

CLI Reference Guide

Cloud Mobility Manager

Release 25.7 SP13

FINAL

DN09131852

Issue 25-0

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Summary of changes

A list of changes between document issues. You can navigate through the respective changed topics.

Changes between issues 24-0 (CMM25.3) and 25-0 (CMM25.7)

Table 1: New and changed commands/parameters

Features	Commands/Parameters
CMM support for third party software updates in CMM25.7 (Feature f02570-70)	- cmm traceData Create Delete List - cmm traceDataStatus Show List
CMM support for CLI / PM changes in CMM25.7 (Feature f02570-80)	- cmm dnnToSnsai Create Show Modify Delete List --dataNetworkName <string> [Changed] - cmm dnnToSnsaiPerPlmn Create Show Modify Delete List --dataNetworkName <string> [Changed] - cmm cmpProc Create --caCertificateFile <string> [New]
AMF support for 3GPP standards Rel17 select Change Requests (CRs) - Phase 10 (Feature f10003-20)	- cmm amfGParms Show Modify List --amfGparmName sendSkipN2PduSessionResRelInd --amfGparmValue <string> [New] - cmm amfLcsGParms Show Modify List --amfGparmName posGpsi --amfGparmValue <string> [New] - cmm networkFunctionInstance Create Show Modify Delete List --pgwFqdnListName <string> [New] - cmm pgwFqdnList Create Show Delete List --name <string> [New] - cmm amfUePlmnServices Create Show Modify Delete List --staticSmf <list-item> [New] - cmm pgwFqdnItem Create Show Delete List --name <string> [New] --fqdn <string> [New]

Table 1: New and changed commands/parameters (continued)

Features	Commands/Parameters
	- cmm pgwFqdnListItem Create Show Delete List --pgwFqdnListName <string> [New] --pgwFqdnItemName <string> [New]
MME support for intelligent PGW selection based on requested PDN type enhancements (Feature f10102-07)	- cmm generalGParams Show Modify List --cmGparamName resendDnsQueryWithoutPdnTypeForTimeout - -cmGparamValue <string> [New]
CMM support for adjustable thresholds for UE paging fanout mitigations (Feature f10737-08)	- cmm mmeUePageFanoutControl Show Modify List --pagingFanoutThreshold <integer> [New] --pageFanoutToTotalMsgThreshold <integer> [New] --msgRateToSafetyNetThreshold <integer> [New] - cmm amfUePageFanoutControl Show Modify List --pagingFanoutThreshold <integer> [New] --pageFanoutToTotalMsgThreshold <integer> [New] --msgRateToSafetyNetThreshold <integer> [New]
MME support for attach rate message throttling for roamers (Feature f10742-01)	- cmm plmn Create Show Modify Delete List --includePlmnAttachRate <list-item> [New] - cmm generalGParams Show Modify List --cmGparamName roamerAttachRateLimitOverride --cmGparamValue <string> [New]
MME support for sending only IMEI portion of IMEISV towards gateways and EIR for all procedures (Feature f10810-01)	- cmm generalGParams Show Modify List --cmGparamName sendOnlyImeiToSgw --cmGparamValue <string> [New] --cmGparamName sendOnlyImeiToEir --cmGparamValue <string> [New]

Table 1: New and changed commands/parameters (continued)

Features	Commands/Parameters
CMM support for RAN serving PLMN in the S1/N2 setup response (Feature f10942-01)	• <code>cmm amfGParms Show Modify List</code> <code>--amfGparmName sendOnlyRanSupportedPlmnsInN2 --amfGparmValue <string> [New]</code> • <code>cmm generalGParms Show Modify List</code> <code>--cmGparmName sendOnlyRanSupportedPlmnsInS1 --cmGparmValue <string> [New]</code>
CMM support for Nokia stacks - SCTP stack enhancements - delivery (Feature f12130-03)	• <code>cmm sctpProfile Create Show Modify Delete List</code> <code>--mtuSize <integer> [Changed]</code> • <code>cmm m3uaProfile Create Show Modify Delete List</code> <code>--mtuSize <integer> [Changed]</code>
MME support for reporting HSS name instead of remote endpoint ID in PM measurement for remote endpoint (Feature f13045-01)	• <code>cmm generalGParms Show Modify List</code> <code>--cmGparmName pmReportHssByName --cmGparmValue <string> [New]</code>
CMM support for automated TLS certificate management - delivery (Feature f13237-03)	• <code>cmm tlsCAcert Create Delete List</code> • <code>cmm tlsCertificate Create Delete List</code> • <code>cmm tlsClientCert Create Delete List</code> • <code>cmm tlsRevocation Create Delete List</code> • <code>cmm oamTlsCAcert Create Delete List</code> <code>--signedCertFile <string> [New]</code> • <code>cmm oamTlsCertificate Create Delete List</code> <code>--privateKeyFile <string> [New]</code> <code>--signedCertFile <string> [New]</code> • <code>cmm oamTlsClientCert Create Delete List</code> <code>--privateKeyFile <string> [New]</code> <code>--signedCertFile <string> [New]</code> • <code>cmm oamTlsRevocation Create Delete List</code>

Table 1: New and changed commands/parameters (continued)

Features	Commands/Parameters
	--revocationListFile <string> [New]
MME support for directed UE move with non-zero relative capacity (Feature f13321-01)	- cmm ueLoadBalance Show Modify List --suppressRcZero <list-item> [New]
CMM support for offload subscribers gracefully when CLI not working (Feature f13322-01)	- cmm aggregateService Show Modify List --lockN2Intf <list-item> [New]
CMM support for NR10 optimization for high runner scenarios - Phase 1 (Feature f14109-15)	- cmm generalGParms Show Modify List --cmGparmName srsTrafficReduction --cmGparmValue <string> [New]
AMF support for OAuth2.0 enhancements phase 3 (Feature f20039-04)	- cmm oauthProfile Create Show Modify Delete List --supportMultiplePlmnsConsumer <list-item> [New] --supportMultiplePlmnsProducer <list-item> [New] --supportModelA <list-item> [New] - cmm consumerOAuthNfType Show Modify List --enable <list-item> [New]
CMM support for using RFSP as selection criterion for access restriction (Feature f20085-10)	- cmm amfGParms Show Modify List --amfGparmName rfspAccessRestriction --amfGparmValue <string> [New] - cmm imsiRangeServices Create Show Modify Delete List --rfspAccessRestrictionListName <string> [New] - cmm amfAccessRestrictionNasMappingProfile Create Show Modify Delete List --rfspAccessRestrictionCc <integer> [New] - cmm restNasMappingProfile Create Show Modify Delete List --rfspAccessRestrictionCc <integer> [New] - cmm amfSupiRangeServices Create Show Modify Delete List

Table 1: New and changed commands/parameters (continued)

Features	Commands/Parameters
	--amfRfspAccessRestrictionListName <string> [New] • cmm generalGParms Show Modify List --cmGparmName rfspAccessRestriction --cmGparmValue <string> [New] • cmm rfspAccessRestrictionList Create Show Delete List --name <string> [New] • cmm uePlmnServices Create Show Modify Delete List --rfspAccessRestrictionListName <string> [New] • cmm amfUePlmnServices Create Show Modify Delete List --amfRfspAccessRestrictionListName <string> [New] • cmm rfspAccessRestrictionListItem Create Show Delete List --rfspAccessRestrictionListName <string> [New] --rfspValue <integer> [New] • cmm amfRfspAccessRestrictionListItem Create Show Delete List --amfRfspAccessRestrictionListName <string> [New] --amfRfspValue <integer> [New] • cmm amfRfspAccessRestrictionList Create Show Delete List --name <string> [New]
AMF support for customer specific DNN conversion enhancement (Feature f20085-11)	• cmm amfAliasDnnConversionProfile Create Show Modify Delete List --replacementDnnListName <string> [New]
AMF support for SMP based priority paging and timer based temporary priority (Feature f20091-06)	• cmm amfGParms Show Modify List --amfGparmName smpBasedPriorityFromSmsf --amfGparm Value <string> [New] • cmm amfSupiRangeServices Create Show Modify Delete List --amfSmpToPagingPriorityProfileListName <string> [New]

Table 1: New and changed commands/parameters (continued)

Features	Commands/Parameters
	• <code>cmm amfSmpToPagingPriorityProfile Create Show Modify Delete List</code> <code>--sbiMessagePriority <integer> [New]</code> <code>--pagingPriLevel <list-item> [New]</code> <code>--amfSmpToPagingPriorityProfileListName <string> [New]</code> <code>--upliftTimer <integer> [New]</code> <code>--prioDuration <list-item> [New]</code> <code>--prioType <list-item> [New]</code> <code>--includePagingPriority <list-item> [New]</code> • <code>cmm amfSmpToPagingPriorityProfileList Create Show Delete List</code> <code>--name <string> [New]</code> • <code>cmm amfUePlmnServices Create Show Modify Delete List</code> <code>--amfSmpToPagingPriorityProfileListName <string> [New]</code> • <code>cmm amfMcsPagingPriProfile Create Show Modify Delete List</code> <code>--includePagingPriority <list-item> [New]</code> • <code>cmm amfPagingPriProfile Create Show Modify Delete List</code> <code>--includePagingPriority <list-item> [New]</code> • <code>cmm amfUePriorityProfile Create Show Modify Delete List</code> <code>--sendPagingForSubsequentPrioN1N2 <list-item> [New]</code>
AMF support for sending PEI with IMEI type towards SMF and EIR for all procedures (Feature f20113-01)	• <code>cmm amfGParms Show Modify List</code> <code>--amfGparmName sendPeiTypeImeiToSmf --amfGparmValue <string> [New]</code> <code>--amfGparmName sendPeiTypeImeiToEir --amfGparmValue <string> [New]</code>
AMF support for network slice specific authentication and authorization - phase 3 (Feature f20606-04)	• <code>cmm amfGParms Show Modify List</code> <code>--amfGparmName sendPendingSnssaiInAllowedNssaiOfIcsr --amfGparmValue <string> [New]</code> • <code>cmm networkFunctionInstance Create Show Modify Delete List</code>

Table 1: New and changed commands/parameters (continued)

Features	Commands/Parameters
	--supiRangeListName <string> [New] - cmm amfUePlmnServices Create Show Modify Delete List --nssaafSelection <list-item> [New] --nssaafStatusSubscribe <list-item> [New] --defaultNssaafSelection <list-item> [New] - cmm callTraceJob Create Show Delete List --n58 <list-item> [New] - cmm callTraceSetting Show Modify List --n58 <list-item> [New]
AMF support for UDM/HSS based P-CSCF restoration procedure - phase 2 (Feature f21302-02)	cmm amfGParms Show Modify List --amfGparmName pageForN1SmlnUpdRsp --amfGparmValue <string> [New]
AMF support for migration of SBI interfaces to HTTPS/TLS (Feature f21510-01)	cmm sbilndirectCommunicationGParms Show Modify List --amfGparmName prefSchemeForCallbacksInModelID --amfGparmValue <string> [New]
AMF support for HLCOM for MICO UEs (Feature f21701-01)	- cmm amfSvcAgreementProfile Create Show Modify Delete List --micoExtendedConnectSource <list-item> [New] --extendedConnectTime <integer> [New] --edrxExtendedConnectSource <list-item> [New] - cmm amfSupiRangeServices Create Show Modify Delete List --hlcomAllowedMico <list-item> [New] - cmm amfUePlmnServices Create Show Modify Delete List --hlcomAllowedMico <list-item> [New] - cmm amfEdrxMicoGParms Show Modify List --amfGparmName hlcomActiveMico --amfGparmValue <string> [New]

Table 1: New and changed commands/parameters (continued)

Features	Commands/Parameters
	--amfGparmName hlcomWaitTimeOffsetMico --amfGparmValue <string> [New]
CMM support for UE nodeB prediction - phase 3 (Feature f21901-05)	- cmm nbNbrGParms Show Modify List --cmGparmName gnbNeighborContUpdate --cmGparmValue <string> [New] --cmGparmName gnbMapLocalRestoration --cmGparmValue <string> [New] --cmGparmName prioritizedUeHistoryCount --cmGparmValue <string> [New] - cmm gnbNbrMapDump List --poolType <list-item> [New] --poolId <integer> [New] --macroGnbId <integer> [New] --gnbMapType <list-item> [New] --weight <integer> [New] --neighborGnbId <integer> [New] --information <string> [New]
MME support for selective RFSP overwrite (Feature f51024-07)	- cmm generalGParms Show Modify List --cmGparmName useLocalRfspHssRfspNotSub --cmGparmValue <string> [New] - cmm svcAgreementProfile Create Show Modify Delete List --useLocalRfspHssRfspNotSub <list-item> [New]
SGSN support for NITZ in PLMN level (Feature f51048-01)	- cmm amfNon3gppTai Create Show Modify Delete List --timeZoneName <list-item> [Changed] - cmm plmnParam Create Show Modify Delete List --timeZone <list-item> [New] --fullNetwName <string> [New] --shortNetwName <string> [New]

Table 1: New and changed commands/parameters (continued)

Features	Commands/Parameters
	• cmm amfPlmn Create Show Modify Delete List --timeZoneName <list-item> [Changed] • cmm enb Create Show Modify Delete List --timeZoneName <list-item> [Changed] • cmm plmn Create Show Modify Delete List --timeZoneName <list-item> [Changed] • cmm tai Create Show Modify Delete List --timeZoneName <list-item> [Changed] • cmm taiDiscoveryDefaults Show Modify List --timeZoneName <list-item> [Changed] • cmm taiRangeDiscoveryProfile Create Show Modify Delete List --timeZoneName <list-item> [Changed] • cmm sgsnGParms Show Modify List --sgsnGparmName networkNamePlmn --sgsnGparmValue <string> [New] • cmm personalize Show Modify List --timeZoneName <list-item> [Changed] • cmm amfMultiTacOperation Create Modify --timeZoneName <list-item> [Changed] • cmm multiTacOperation Modify --timeZoneName <list-item> [Changed] • cmm amfTaiRangeDiscoveryProfile Create Show Modify Delete List --timeZoneName <list-item> [Changed] • cmm amfTai Create Show Modify Delete List --timeZoneName <list-item> [Changed]
CMM support for AI/ML driven overload control and MLIS links	• cmm aiMLControl Show Modify List --aiMLDrivenCpuOverload <list-item> [New]

Table 1: New and changed commands/parameters (continued)

Features	Commands/Parameters
status CLI query - phase 1 (Feature f72013-19)	• cmm aimlMultiLink Show List --aimlLinkIndex <integer> [New] --linkOperationState <list-item> [New] --aimlLinkIndexType <list-item> [New] --localIp <string> [New] --peerNodeIp <string> [New] --peerNode <string> [New] • cmm aiMl5gUeOffloadStatus List --peerNodeIp <string> [New] --ueOffloadStatus <list-item> [New] --peerNode <string> [New] --ownName <string> [New] --localIp <string> [New] --statusStartTime <string> [New] • cmm localEndPtCfg Create Show Modify Delete List
CMM support for paging optimization using AI/ML forecasted UE location/trajectory - PoC phase 2 (Feature f72013-22)	• cmm amfPagingPolicy Create Show Modify Delete List --maxGnbsPaged <integer> [New] --amfMethod <list-item> [Changed] • cmm amfPagingMethodOverride Create Show Modify Delete List --maxGnbsPaged <integer> [New] --nearestNeighborMethod <list-item> [Changed]  NOTE: The parameters introduced in this feature are not supported in CMM25.7.

Table 2: Removed commands/parameters

Type	Command/ Parameter	Reason
Parameter	cmm gParms support5gslwk Flag	The gParm is redundant and does not have any impact on MME behavior with or without N26 in relation to 5GS interworking.

1 CMM command notes

Notes about Cloud Mobility Manager CLI.

Disclaimer notice

The CMM CLI json/yaml/plain output is a reflection of CMM format. This output does not reflect the raw data that is used to generate the output. This format is designed for human readability and not for scripting purposes.

2 Command line interface (CLI)

Run-time management operations can be carried out through the command line interface (CLI) provided by the Network Element Cluster Controller (NECC).

2.1 Authentication, user accounts and user roles

User authentication protects the system from unauthorized users. Preconfigured user accounts can be used when accessing the CMM for the first time. Pre-configured user `cmmsecurity` can be used to create user groups with different access rights.

Authentication

All the defined user accounts need to be found in the LDAP database in the NECC, otherwise, logging to CMM and use of the CLI commands fail.

User accounts and user roles

Table 3: Preconfigured user accounts in the system

Username	Password	Role
root	39pfMBw3W8ec	–
cmm	2pe1qIP4zjXt	sysadmin
cbamuser	KX6GQrNagFSw	sysadmin
sam5620	dipq2zIPLv36	sysadmin
cmm_fallback	–	–
diagnostic	psf45edIKp82	sysadmin
trainee	aple41OKsprq	monitor
rsp	dOje1586Dp78	sysadmin
cgw	pLk5390tGi09	A user for the charging GW, with access only to the data-charging folder in the

Table 3: Preconfigured user accounts in the system (continued)

Username	Password	Role
		NECC through the SFTP with the Bp service IP.
orchest	–	sysadmin This user is present when orchestrating with the CBAM.
dcae-dfc	8u53hgBS53gq	A user for the open network automation platform (ONAP) with access only to the /data-pm/gzip folder in the NECC through the SFTP.
ca4mn	7xrr99Sso05T	A user for the cognitive analytics for mobile networks (CA4MN), with access only to the /data-pcmd folder in the NECC through the SFTP.
cmmsecurity	akflt9nT/pBMdX+V	secadmin
 NOTE: Nokia strongly recommends changing the pre-configured user account password as soon as the application is deployed.		

Restrictions:

- A root user cannot log in through SSH. The only way to log in as root is to use the su command while logged in as cmm user or cmm_fallback user (or some other user in the admin group).
- The cmm_fallback account is not available in the CNF-based deployments.
- A cmm_fallback user can log in only through SSH from the controller node (this user does not have a password). The only thing that the cmm_fallback user can do is to change to root through the su root command. This is a backup user for certain situations, for example, when LDAP is down and prevents login for LDAP users, such as cmm user.
- The Password column is not applicable to CMM CNFs.

**NOTE:**

- Nokia strongly recommends changing the preconfigured user account password as soon as the application is deployed.
- The root account cannot be used when logging in to the NECC directly using its floating IP. Instead, use another account and use `su -` to become the root user.
- A healing/rebuild operation restores the NECC root password to default and password change is required.

User groups and user roles

Preconfigured user roles are:

monitor	The monitor user can only view configuration settings but, cannot change them.
manage	The manage user is an administrative user who can configure and view all settings.
sysadmin	The sysadmin user can view and modify all configurations. The sysadmin user has also superuser (sudo) rights. The superuser is a special user account used for system administration.
lioperator	The lioperator user can configure LIC and LIB network elements and read LI-related logs.
secadmin	The secadmin user can create other users and configure security settings.

Table 4: User group and user role mapping

LDAP group	User role	Tasks
admin	sysadmin	• access to system and security logs • access to almost all Linux administrative components

Table 4: User group and user role mapping (continued)

LDAP group	User role	Tasks
		• configure the CMM parameters through the CLI interface (excluding security-related parameters)
monitor	monitor	read the CMM parameters through the CLI interface (excluding the security-related parameters)
configure	manage	configure the CMM parameters through the CLI interface (excluding the security-related parameters)
li	lioperator	• configure the LI-related parameters through the LI CLI interface • access to LI logs
security	secadmin	• LDAP user provisioning • configure the security-related CMM parameters through the CLI interface

Restrictions:

- Only users in the security group can create, modify or delete the following commands:
 - authServer
 - certificate
 - oamSession
 - oauthAccessTokenKey, oauthAccessTokenKeyList
 - oauthProfile
 - passwdPolicy
 - radiusProxy
 - tacacsAuthRoleMapping
 - tlsCAcert, tlsCAcertList
 - tlsCertificate, tlsCertificateList

- tlsClientCert, tlsClientCertList
- tlsProfile
- tlsRevocation, tlsRevocationList
- secureCopySetup
- extLdapProfile
- oamTlsCAcert, oamTlsCAcertList
- ldapAuthRoleMapping
- oamTlsCertificate, oamTlsCertificateList
- oamTlsClientCert, oamTlsClientCertList
- oamTlsProfile
- oamTlsRevocation, oamTlsRevocationList
- amfMcsPagingPriProfile, amfMcsPagingPriProfileList
- amfPagingPriProfile, amfPagingPriProfileList
- amfUePriorityProfile

- Users in the admin and configuration group can use list and show in the following commands:

- oauthProfile
- oamTlsProfile
- secureCopySetup
- tlsProfile

2.2 Creating user accounts

To create user accounts, use the **fsuseradd** command.

Purpose

In addition to the preconfigured user accounts, a root user and a cmmsecurity user can add new user accounts with the desired user roles. This procedure describes how to create user accounts with command line interface to the internal configuration directory server.

Before you start

Ensure that you have the correct permissions.

Procedure

1. View existing users.

Sub-steps

- a) Log in to the NECC as a cmmsecurity user.
- b) Execute `fsdumpusers`.

Step example

```
[cmmsecurity@necc0 ~]$ fsdumpusers
dn = uid=cmm,ou=People,fsFragmentId=Security,fsClusterId=ClusterRoot
cn: cmm
uid: cmm
objectClass: FSSecAccount posixAccount shadowAccount extensibleObject
uidNumber: 1000
gidNumber: 30500
homeDirectory: /home/cmm
loginShell: /bin/bash
shadowMin: 0
shadowMax: 180
shadowWarning: 7
shadowLastChange: 17114

dn = uid=cbamuser,ou=People,fsFragmentId=Security,fsClusterId=ClusterRoot
cn: cbamuser
uid: cbamuser
objectClass: FSSecAccount posixAccount shadowAccount extensibleObject
uidNumber: 1001
gidNumber: 1001
homeDirectory: /home/cbamuser
loginShell: /bin/bash
shadowMin: 0
shadowMax: 180
shadowWarning: 7
shadowLastChange: 0
```

2. Add a user account.

Sub-steps

- a) Log in to the NECC as a cmmsecurity user.
- b) Create a hash for password using the `slappasswd` command.



NOTE:

A different hash string is used each time.

Step example

{SSHA}Uhy4I9+x7CDLaMQ2SFS8nhFk+IdWXbCP is a hash for Nokia_1234.

```
[cmmsecurity@necc0 ~]$ slappasswd -h {SSHA} -s Nokia_1234

{SSHA}Uhy4I9+x7CDLaMQ2SFS8nhFk+IdWXbCP
```

c) Execute the `fsuseradd` command with the target user group.

Add a new user account with the `fsuseradd` command. The full syntax of the command with all options is the following:

```
fsuseradd [-c cn] [-u uidNumber] [-d homeDirectory] [-p userPassword]
[-s loginShell] [-G group[,...]] [-e expire_date] [-f inactive_time]
[-n mindays] [-x maxdays] [-w warndays] [-l last_change] [--filter
filter_size] [-v] [-h] [--syssec] [-g gidNumber] uid
```

**NOTE:**

A primary group is created automatically with the same textual as that of the user account.

**NOTE:**

If the secondary group does not exist prior to creating the new user account, the secondary group must be created first. It is also possible to create a user account without any secondary groups, but such a user account cannot have any permissions as they are mapped to a user only through secondary groups.

Table 5: Parameters of the `fsuseradd` command

Parameter	Description	Value range
-c cn	Common name. This is the full name of the user.	The default value is an empty string.
-u uidNumber	UID number.	The value is a number in the range 10000 to 60000. The default value is the first available number in this range.

Table 5: Parameters of the `fsuseradd` command (continued)

Parameter	Description	Value range
<code>-d homeDirectory</code>	The home directory of the user.	The default value is <code>/home/<uid></code> .
<code>-e expire_date</code>	The date on which the user account will be disabled.	–
<code>-f inactive_days</code>	The number of days after a password expires until the account is permanently disabled.	The date is specified in the format <code>YYYY-MM-DD</code> or in the number of days from UNIX EPOCH.
<code>-p userPassword</code>	The encrypted password, as generated using SSHA.	The value is the string output of the <code>s1appasswd</code> command. The default value is <code>"*"</code> .
<code>-s loginShell</code>	The default login shell.	The default value is <code>fsclish</code> .
<code>-G group[, ...]</code>	List of secondary groups.	The default value is an empty string.
<code>-n mindays</code>	The number of days from the password change before which the password cannot be changed.	The default value is 0.
<code>-x maxdays</code>	The number of days from the password change after which the password must be changed.	The default value is 180. Recommended value is 90 days or less.
<code>-w warndays</code>	The number of days before the user is notified when the password expires.	The default value is 7.
<code>-l last_change</code>	The date of the last change of the password. The value <code>-l 0</code> forces the user to change the password after the first login.	The date is specified in the format <code>YYYY-MM-DD</code> or in the number of days from UNIX

Table 5: Parameters of the `fsuseradd` command (continued)

Parameter	Description	Value range
		EPOCH. The default value is 2014-12-05 (16409).
<code>--filter filter_size</code>	The initial filter size of the free uidNumber search.	Available range of values is 2 to 2048. The default value is 128.
<code>-v</code>	Verification is asked from user.	–
<code>-h</code>	Help. Prints command help page.	–
<code>--syssec</code>	Add the user under LDAP SystemSecurity fragment (ou=People, fsFragmentId=SystemSecurity, fsClusterId=ClusterRoot) syssec option cannot be combined with usertype option	–
<code>-g gidNumber</code>	Primary group number or name.	–
<code>uid</code>	User's login name.	–

Step example

```
fsuseradd [username] -G [group name] -p [password hash] -l [last_change]
```

```
[cmmsecurity@necc0 ~]$ fsuseradd test_monitor1 -G monitor -p um0D9BIE0Bb+uW+Idp2eg1HvSZaqEVoa -l 0
```

Command executed successfully.

Step example

To create the user account *testuser* with the following attributes:

- Assign the user to the group *testuser*.
- Assign the user to the pre-existing secondary group *admin*

- Set the home directory to /home/testuser.
- Set the inactive time to 14 days.
- Set the expiration date to 31.12.2016.
- Set the login shell to /bin/bash.
- Set the time when the password must be changed to 60 days.
- Set the time to notify the user about the password change to 14 days.
- Set the date of the last change of the password to zero to prompt the user to change the password after the first login.

Enter the following command:

```
fsuseradd testuser -G monitor -d /home/testuser -f 14 -e 2016-12-31 -s /bin/  
bash -x 60 -w 14 -l 0
```

After successful operation, the following message is displayed:

```
Command executed successfully.
```

Step result

Now the added user can log in to the NECC and further authenticate oneself using the created account.



NOTE:

Last login date and time is displayed after a successful login on all the login attempts.

2.3 Deleting user accounts

To delete user accounts, use the `fsuserdel` command. Optionally, the user's home directory can be preserved when the user account is removed.

Before you start

Ensure that you have the correct permissions.

Procedure

1. Log in to the NECC as a `cmmsecurity` user.
2. Delete a user account.

To delete a user account, use the following command:

```
fsuserdel [-v] [-h] [-f] [-p] uid
```

The parameters are described in the following table:

Table 6: Parameters for the `fsuserdel` command

Parameter	Description
-v	Verification is asked from the user.
-h	Print the command help page.
-f	Force to delete entries when the entry is found in both <i>Security</i> and <i>SystemSecurity</i> fragments.
-p	Preserve users home directory.
uid	Specify the username.

Step example

To delete the testuser user, delete also the user's home directory, and ask for verification, enter the following command:

```
fsuserdel -v testuser
```

The following output is displayed:

```
Deleting this from LDAP:

'uid'           = testuser
'uidNumber'     = 1002
'cn'            = testuser
'gidNumber'     = 30500
'groups'        = 1002
'homeDirectory' = /home/testuser
'loginShell'    = /bin/bash
Removing also home directory in the disk: /home/testuser

Delete? y/n
```

Enter y to confirm. The following output is displayed:

```
Command executed successfully.
```

Step example

To delete test_monitor1 user, but preserve the user's home directory (home directory's owner will be changed to root), enter the following command:

```
[cmmsecurity@necc0 ~]$ fsuserdel -p test_monitor1  
Command executed successfully.
```

2.4 Unlocking a user account after failed login attempts

A user is locked out after five unsuccessful login attempts. In addition, if a user fails to log in successfully before the maximum time for login attempts expires, the connection is closed.

Purpose

A locked-out user can attempt to log in after five minutes. Instead of waiting, you can reset (or clear) the number of failed login attempts.

Procedure

- Reset the number of failed attempts for a cmm user.

Enter the command in the NECC:

```
sudo /usr/sbin/faillock --user cmm --reset
```

2.5 Logging in to NECC as the cmm_fallback user

If you cannot log in as a cmm user (and root user login is disabled), you can always log in as the cmm_fallback user from the controller.

Purpose



NOTE:

The cmm_fallback account is not available in the CNF-based deployments.

If the LDAP is down, login as a cmm user does not work. Login as a cmm_fallback user still works because the controller has a public key for the cmm_fallback user and the cmm_fallback user is a normal Linux user and not in LDAP. After login, change to the root user and fix the LDAP issue.



NOTE:

Login banner is not configurable, HTTPs login does not display banner. CMM login using SSH will display banner.

Procedure

1. Log in as the cmm_fallback user from the controller.

```
[IPC-TCL-102@controller-1 ~]$ ssh cmm_fallback@[NECC VIP]
```

2. Change to the root user with the su command.

```
[cmm_fallback@necc0 ~]$ su root
```

2.6 CLI command structure

A CLI command consists of primary keyword (cmm), command group or table name, command, and parameters.

The command format is: cmm <group> <command> [--<parameter1> <value>] [--<parameter2> <value>] ...

cmm

primary keyword

group

parameter group or table name, for example, plmn

command

actions: create, modify, delete, show, list

parameter

parameter types (optional parameters, positional parameters or required named parameters)

Table 7: CLI parameter types

Notation	Description
<parameter value>	Mandatory positional argument. Example: cmm tai show -plmnName <plmnName> --tac <tac>
{value1, value2, value3...}	Mandatory positional argument with one of the values inside the brackets. Example: interfaceName allowed values {S1MME, S6A, S10, S11, GN, SV, SGS, X2, S13, ...}
--parameter <value>	Mandatory required named parameter. Example: cmm tai create --plmnName 310_012 --tac 9999

Table 7: CLI parameter types (continued)

Notation	Description
--parameter {value1, value2, value3...}	Mandatory required named parameter with one of the values inside the brackets. Example: cmm rmtEndPtCfg create --name remoteEndPt_1 --interfaceName {S6A,S13,SGS,SLG,SLS,SBC,S V,SGS_LITE} --ip1 2.2.2.2 --ip2 3.3.3.3
[--parameter <value>]	Optional required named parameter Example: [--multiHomed {true, false}]
[--parameter {value1, value2, value3...}]	Optional required named parameter with one of the values inside the brackets. Example: cmm localEndPtCfg create --interfaceName S1MME --multiHomed true

**NOTE:**

If <value> allows the '-' character in its allowed character set, it cannot be the first character of that string.

The configuration database contains pre-populated tables and defaults, for example, internal subnet, service IPs for internal communication, various profiles, and timers. If the value of a field is not specifically provisioned, the system populates the field with the default value. There are two types of default:

- An absolute value has no dependencies from other field(s) in the same or other tables.
- A relative value is a set of possible default values that are chosen based on the content of other field(s) in the same or other tables.

2.7 CLI command history

CMM supports a dedicated log file for the logging of CLI command history.

For each command entered, the log file captures:

- User ID
- Timestamp

- Command syntax, including all parameters provided on the command line
- Command outcome (for example, success, fail, return code)

There is a log file on each NECC (/var/log/cli_cmd.log).

The file has maximum size and retention policy managed by standard Linux log rotation.

2.8 Executing CLI commands

The command line interface (CLI) can be used in the active NECC.

Procedure

1. Log in to an active NECC VM (NECC-0 or NECC-1) using ssh with a valid user account and authenticate.

User **cmm** is authenticated to the CMM CLI automatically. All other users need to be authenticated before they can enter configuration commands.

The command is `cmm authenticate`.

Step example

```
[root@necc0 ~]# cmm authenticate
Enter Username: john
Enter Password:

Authentication successful.
[root@necc0 ~]#
```

2. Type `cmm -h` or `cmm --help` to see the available commands.

Commands `cmm -h` and `cmm --help` list all commands available for CMM.

3. Use the CLI help for detailed command information (`-h` / `--help`).

Help arguments are `-h` and `--help`.

Step example

```
cmm hssGrp create --help
```

4. Type the command with the required arguments in the command line and press **Enter**.

If you are unsure of the exact command to execute, use the auto-completion feature of CLI. This feature displays the choices allowed for a command or the commands themselves. To use auto-completion, type the command or part of it, and then press **TAB**.



NOTE:

When using the create command, all mandatory parameters must be given at the same time.

**NOTE:**

When the wildcard character is used in a CLI command, enter it in quotation marks (' * '). This will make sure the wildcard character will be expanded as a string instead of the list of files in current directory.

Enter parameter values with word spaces, for example "Home Network", in quotation marks.

Step example

```
cmm clusterMgmt test.ping --target '*'
cmm diamProfile create --name 1 --dwrFreqTimer 10000 -- dwrMsgTimer 2000 --
dwrRetransMax 3 --cerRetransMax 3 -- diamProductName "MME Diameter"
```

Step result

When the command is executed successfully, information is displayed. If no information is available, the command displays nothing or an empty table or a message indicating that there is no information to display.

5. Use output formatters to modify the command output.

All commands include Linux output formatter options:

```
[-f {json,plain,shell,table,value,yaml}]
[-c COLUMN]
[--max-width <integer>]
[--fit-width]
[--print-empty]
[--noindent]
[--prefix PREFIX]
```

Table 8: Output formatter options

Output formatter	Description
-c COLUMN --column COLUMN	Specify the column(s) to include, can be repeated.
-f {csv,json,plain,table,value,yaml} --format {csv,json,plain,table,value,yaml}	Output formatter for list operation, defaults to table.

Table 8: Output formatter options (continued)

Output formatter	Description
<code>-f {json,plain,shell,table,value,yaml}</code> <code>--format {json,plain,shell,table,value,yaml}</code>	Output formatter for show operation, defaults to table.
<code>--max-width <integer></code>	Table formatter. Maximum display width, <1 to disable. You can also use the CLIFF_MAX_TERM_WIDTH environment variable, but the parameter takes precedence.
<code>--fit-width</code>	Table formatter. Fit the table to the display width. Implied if <code>--max-width</code> is greater than 0. Set the environment variable CLIFF_FIT_WIDTH=1 to always enable.
<code>--print-empty</code>	Table formatter. Print empty table if there is no data to show.
<code>--noindent</code>	json formatter: Whether to disable indenting the JSON.
<code>--quote {all,minimal,none,nonnumeric}</code>	csv formatter: When to include quotes, defaults to nonnumeric.
<code>--prefix PREFIX</code>	Shell formatter, a format a UNIX shell can parse (variable="value"). Add a prefix to all variable names.

Step example

Show only the selected column (`-c/--column`):

```
cmm traceDataStatus list
```

```
+-----+-----+-----+-----+-----+-----+-----+
|SUBSCRIBER|MSISDN|TRACE|TRACE|HITS|FIRST|LAST|
|          |      |ID  |ORIGIN|    |    |    |
+-----+-----+-----+-----+-----+-----+-----+
|2622800000000001|918003779680|1|hss|6|2016-11-17|2016-11-17|
```

					T13:19:28.018	T13:19:28.209	
262280000000000001	918003779680		2 hss		15 2016-11-17	2016-11-17	
					T13:19:28.304	T13:19:29.101	
262280000000000002	918003779681		3 hss		20 2016-11-17	2016-11-17	
					T13:19:29.115	T13:19:29.873	
262280000000000002	918003779681		4 hss		20 2016-11-17	2016-11-17	
					T13:19:29.924	T13:19:30.535	
262280000000000002	918003779681		5 hss		20 2016-11-17	2016-11-17	
					T13:19:30.616	T13:19:31.307	
+-----+	+-----+	+-----+	+-----+	+-----+	+-----+	+-----+	+-----+

```
cmm traceDataStatus list --column HITS
```

```
+-----+
| HITS |
+-----+
| 5    |
| 6    |
+-----+
```

```
cmm traceDataStatus list -c HITS
```

```
+-----+
| HITS |
+-----+
| 5    |
| 6    |
+-----+
```

Step example

If the displayed information is too wide and difficult to read, adjust the width of the displayed information. Re-enter the command but use the `--max-width` argument to make the information easier to read.

```
cmm alarm active show 19 --max-width 50
```

6. For debugging, use the `-v` argument.

With the `-v` (`--verbose`) argument, CLI command logs, the REST Request, and the received Response are shown.



TIP:

If a CLI command fails, this may help you identify the cause of the failure.

Step example

```

cmm systemInfo show -v
initialize_app
found extension EntryPoint.parse('plain =
nokia.mmecli.api.plainformatter:PlainFormatter')
found extension EntryPoint.parse('table =
cliff.formatters.table:TableFormatter')
found extension EntryPoint.parse('json =
cliff.formatters.json_format:JSONFormatter')
found extension EntryPoint.parse('shell =
cliff.formatters.shell:ShellFormatter')
found extension EntryPoint.parse('value =
cliff.formatters.value:ValueFormatter')
found extension EntryPoint.parse('yaml =
cliff.formatters.yaml_format:YAMLFormatter')
prepare_to_run_command SysteminfoShow
request header: {'Accept': 'application/json'}
request URL = http://10.10.30.226:8080/api/cm/v1/systemInfos/config
REST server json response = 200, {'_embedded': {'releaseNum':
u'CMM17.0.0', u'application': u'MME',
u'buildNum': u'B1_C1982', u'hwType': u'ATE', u'_links': {'self':
{'href': u'/api/cm/v1/systemInfos/config'}},
u'switchName': u'ns17', u'timeZone': u'CST'}, u'_links': {'self':
{'href': u'/api/cm/v1/systemInfos/config'}}}}
+-----+-----+
| Field | Value |
+-----+-----+
| systemInfo | config |
| switchName | ns17 |
| timeZone | CST |
| application | MME |
| releaseNum | CMM17.0.0 |
| buildNum | B1_C1982 |
| hwType | ATE |
+-----+-----+
clean_up SysteminfoShow

```

7. If you get a timeout when displaying alarm or timer data, try the `--period` argument to select a shorter period.

When using CLI commands to display alarm history or counter data, timeout may occur. Without the `--period` argument, the default period for alarms is 1 d (day) and for counters 15 minutes. Re-enter the command with a shorter period.

Step example

```
cmm alarm history list --period 1h
```

8. If you get output “Command execution not possible yet – try again later” try again later.
CLI cannot send a command forward because the receiving side is not ready. Try again later.

3 CMM CLI commands: cmm aggregateService - cmm amfUePageFanoutControl

CMM CLI command descriptions: syntax, usage, parameters, and user roles.

3.1 cmm aggregateService

Aggregate services. Syntax, description, parameters, and roles for cmm aggregateService command.

Syntax

Description

Administer available services. For QA applications both adminState and amfAdminState must to specified and set to the same state. For AMF-only applications only amfAdminState is applicable. For MME-only and TA applications only adminState is applicable. For TA and QA applications, SGSN PAPSs must be locked (using cmm serviceState modify command with the 'force' option) before locking aggregate service.

Parameters

Name

adminState - Admin state

Syntax

```
--adminState <list-item>
```

Description

Admin state of the services.

Class

Optional.

List

- locked - Locked.
- unlocked - Unlocked.

Name

amfAdminState - AMF Admin state

Syntax

```
--amfAdminState <list-item>
```

Description

Admin state of the AMF services.

Class

Optional.

List

- locked - Locked.
- unlocked - Unlocked.

Name

lockN2Intf - Lock N2 Interface

Syntax

--lockN2Intf <list-item>

Description

Setting this parameter to true will cause the N2 Interface to be locked on application of Aggregate Service Lock for AMF or QA.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

3.2 cmm aiML5gUeOffloadStatus

AIML Driven 5G UE Offload Status. Syntax, description, parameters, and roles for cmm aiML5gUeOffloadStatus command.

Syntax**Description**

AIML Driven 5G UE Offload Status

Parameters**Name**

localIp - local Ip

Syntax

--localIp <string>

Description

Local MLIS external IP

Class

Read only.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

ownName - Own Name

Syntax

--ownName <string>

Description

Local CMM/AMF name

Class

Read only.

Type / Range

Digit string of length 1 to 40 with the allowed pattern of [0-9a-zA-Z\-*]

Name

peerNode - peer Node

Syntax

--peerNode <string>

Description

Remote CMM Name from cmm aiMIPeer table

Class

Read only.

Type / Range

Digit string of length 1 to 40 with the allowed pattern of [0-9a-zA-Z\-*]

Name

peerNodeIp - Peer Node Ip

Syntax

--peerNodeIp <string>

Description

Remote peer CMM MLIS external IP

Class

Read only.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

statusStartTime - offload status start time

Syntax

--statusStartTime <string>

Description

Offload Status Start Time

Class

Read only.

Type / Range

Date and Time Digit string of length 0 to 20 with the allowed pattern of '20[0-9][0-9]/[0-1][0-9]/[0-3][0-9] [0-2][0-9]:[0-5][0-9]:[0-5][0-9]'

Name

ueOffloadStatus - AIML driven UE offload status

Syntax

--ueOffloadStatus <list-item>

Description

AIML driven UE offload status

Class

Read only.

List

- unknown - Offload status unknown
- offloadPendingFromPeer - Offload Pending From Peer - Offload is Pending From Peer CMM
- offloadInProgressFromPeer - Offload is Progress From Peer CMM - Offload in progress from peer CMM
- offloadPendingToPeer - Offload Pending To Peer - Offload is Pending to Peer CMM
- offloadInProgressToPeer - Offload in Progress to Peer CMM - Offload in progress to peer CMM

Roles**List**

sysadmin, manage, monitor

3.3 cmm aiMlAwsParms

Artificial Intelligence and Machine Learning AWS Access Parameters. Syntax, description, parameters, and roles for cmm aiMlAwsParms command.

Syntax

Description

This command is used to provide parameters such as IP address, port, access keys etc in CMM to access AWS cloud to store artificial intelligence and machine learning data.

Parameters

Name

awsAccessKey - Access Key

Syntax

--awsAccessKey <string>

Description

Access Key

Class

Optional.

Type / Range

String of length 0, or 16 to 128, with a pattern of [a-zA-Z0-9]*

Default

"

Name

awsHttpSchema - HTTP schema

Syntax

--awsHttpSchema <list-item>

Description

HTTP schema

Class

Optional.

List

- http - Hypertext Transfer Protocol scheme

- https - Hypertext Transfer Protocol Secure scheme

Default

http

Name

awsIp - IP address

Syntax

--awsIp <string>

Description

IP address

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Default

"

Name

awsPort - Port

Syntax

--awsPort <integer>

Description

Port

Class

Optional.

Type / Range

Integer in the range of 0 or 1,024 to 65,535

Default

8080

Name

awsRegion - Region

Syntax

--awsRegion <string>

Description

Region

Class

Optional.

Type / Range

String of length 0, or 1 to 50, with a pattern of [a-zA-Z0-9]*

Default

"

Name

awsSecretKey - Secret key

Syntax

--awsSecretKey <string>

Description

Secret key

Class

Optional.

Type / Range

Digit string of length 0 to 256 with the allowed pattern of [0-9a-zA-Z\.\-_:]*

Default

"

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

3.4 cmm aiMLControl

Artificial Intelligence and Machine Learning Controls. Syntax, description, parameters, and roles for cmm aiMLControl command.

Syntax**Description**

This command is used to administer Artificial Intelligence and Machine Learning control fields/flags.

Parameters**Name**

aiMLDriven5gUeLocationForecasting - AIML Driven 5g Ue Location Forecasting

Syntax

--aiMLDriven5gUeLocationForecasting <list-item>

Description

Flag to enable or disable AIML driven 5G UE location forecasting.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

aiMLDriven5gUePagingFailureRca - AIML Driven 5g Ue Paging Failure RCA

Syntax

--aiMLDriven5gUePagingFailureRca <list-item>

Description

AIML Driven 5g Ue Paging Failure Root Cause Analysis

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

aiMLDriven5gUeRegistrationFailureRca - AIML Driven 5g Ue Registration Failure RCA

Syntax

--aiMLDriven5gUeRegistrationFailureRca <list-item>

Description

AIML Driven 5g Ue Registration Failure Root Cause Analysis

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

aiMLDriven5gUeServiceFailureRca - AIML Driven 5g Ue Service Failure RCA

Syntax

--aiMLDriven5gUeServiceFailureRca <list-item>

Description

AIML Driven 5g Ue Service Failure Root Cause Analysis

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

aiMLDrivenCpuOverload - AIML Driven CPU Overload detection

Syntax

--aiMLDrivenCpuOverload <list-item>

Description

AIML Driven CPU Overload detection and mitigation

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

aiMLDrivenMsgRateOverload - AIML Driven Message Rate Overload

Syntax

--aiMLDrivenMsgRateOverload <list-item>

Description

AIML Driven Message Rate Overload detection and mitigation

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

analyticsDrivenAmfSetMgmt - Analytics Driven AMF Set Management

Syntax

--analyticsDrivenAmfSetMgmt <list-item>

Description

Flag to enable or disable traffic management across CMM Set or Pool. For AMF, management is across AMF Set. For MME, management is across MME Pool.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

analyticsDrivenAmfSetRebalance - Analytics Driven AMF Set Rebalance

Syntax

--analyticsDrivenAmfSetRebalance <list-item>

Description

Flag to invoke UE rebalance between AMFs within a set based on AI/ML analytics.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

analyticsDrivenAmfSliceMgmt - Analytics Driven AMF Slice Management

Syntax

--analyticsDrivenAmfSliceMgmt <list-item>

Description

Flag to enable or disable AMF Slice management within and across AMF set.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

analyticsDrivenAmfSliceMitigation - Analytics Driven AMF Slice Mitigation

Syntax

--analyticsDrivenAmfSliceMitigation <list-item>

Description

Flag to enable/disable mitigation of AMF slice operations driven by AI/ML models.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

dataLakeType - Data Storage type

Syntax

--dataLakeType <list-item>

Description

Data storage mechanism used to store Artificial Intelligence and Machine Learning data

Class

Optional.

List

- aws - AWS storage - Storage of Artificial Intelligence and Machine Learning data in AWS S3 buckets
- azure - Azure Storage - Storage of Artificial Intelligence and Machine Learning data in MS Azure cloud
- gcp - GCP Storage - Storage of Artificial Intelligence and Machine Learning data in GCP cloud
- local - Local Storage - Local storage of data for Artificial Intelligence and Machine Learning modelling

Default

local

Name

pagingOverloadControl - Flag to control artificial intelligence model to perform paging overload control

Syntax

--pagingOverloadControl <list-item>

Description

Flag to control artificial intelligence model to perform paging overload control

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

trafficLoadForScaling - Flag to enable or disable traffic load modelling for automatic scaling

Syntax

--trafficLoadForScaling <list-item>

Description

Flag to enable or disable traffic load modelling to automatically scaling call processing pods

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

3.5 cmm aimlMultiLink

AIML Multilink Status Query. Syntax, description, parameters, and roles for cmm aimlMultiLink command.

Syntax**Description**

This command is used to query the status of each AIML_DATA_STREAM network interface type external link that the MLIS pod has with MLIS pods in other CMMs within a pool

Parameters**Name**

aimlLinkIndex - AIML Link Index

Syntax

--aimlLinkIndex <integer>

Description

AIML external link index

Class

Required key.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

aimlLinkIndexType - aiml Link Index Type

Syntax

--aimlLinkIndexType <list-item>

Description

Describes if link index is a client or server

Class

Read only.

List

- client - Client socket
- server - Server socket

Name

linkOperationState - AML peer MLIS external interface link status

Syntax

--linkOperationState <list-item>

Description

AML peer MLIS external interface link status

Class

Read only.

List

- disabled - Link down, not established.
- enabled - Link established and enabled.
- unknown - Link status unknown. Link not established.

Default

unknown

Name

localIp - Local MLIS external IP

Syntax

--localIp <string>

Description

Local MLIS external IP

Class

Read only.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

peerNode - Remote CMM Name

Syntax

--peerNode <string>

Description

Remote CMM name from cmm aiMIPeer table

Class

Read only.

Type / Range

Digit string of length 1 to 40 with the allowed pattern of [0-9a-zA-Z\-*]

Name

peerNodeIp - remote peer MLIS IP

Syntax

--peerNodeIp <string>

Description

remote peer MLIS IP

Class

Read only.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Roles**Show**

sysadmin, manage, monitor

List

sysadmin, manage, monitor

3.6 cmm aiMIPeer

Artificial Intelligence and Machine Learning Peer CMM Access Parameters. Syntax, description, parameters, and roles for cmm aiMIPeer command.

Syntax**Description**

This command is used to provide parameters such as IP address, port, access keys etc in CMM to peer CMMs share artificial intelligence and machine learning data.

Parameters

Name

peerNode - Peer CMM Name to Share AIML Data

Syntax

--peerNode <string>

Description

Parameter is used to provision peer CMM name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

peerNodeIp - Remote Node IP address

Syntax

--peerNodeIp <string>

Description

Remote Node (CMM) IP address

Class

Required on create, not modifiable.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

enable - Enable

Syntax

--enable <list-item>

Description

The parameter is used to enable the use of the remote peer.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.7 cmm alarmActive

Alarm Active. Syntax, description, parameters, and roles for cmm alarmActive command.

Syntax

Description

This command has three operations: The cancel operation is used to cancel active alarms. The list operation is used to list active alarms. The show operation is used to show active alarms. Note: Before CMM195M2, the command is cmm alarm active instead of cmm alarmActive. For backward compatibility, both versions of the command are supported.

Parameters

Name

pid - Alarm PID

Syntax

--pid <integer>

Description

Alarm PID(e.g. 42004).

Class

Optional.

Type / Range

Integer in the range of 0 to 2,147,483,647

Name

sourceApp - Source appname

Syntax

--sourceApp <string>

Description

Source appname.

Class

Optional.

Type / Range

Digit string of length

Name

sourceHost - Source hostname

Syntax

--sourceHost <string>

Description

Source hostname.

Class

Optional.

Type / Range

Digit string of length

Name

sourceIp - Source hostip

Syntax

--sourceIp <string>

Description

Source hostip.

Class

Optional.

Type / Range

Digit string of length

Roles**Cancel**

sysadmin, manage

List

sysadmin, manage, monitor

Show

sysadmin, manage, monitor

3.8 cmm alarmDefinition

Alarm Definition. Syntax, description, parameters, and roles for cmm alarmDefinition command.

Syntax

Description

This command has two operations: The list operation is used to list alarm definitions. The show operation is used to show alarm definitions. Note: Before CMM195BR, the command was cmm alarm definition instead of cmm alarmDefinition. For backwards compatibility, both versions of the command are supported.

Parameters

N/A

Roles

List

sysadmin, manage, monitor

Show

sysadmin, manage, monitor

3.9 cmm alarmHistory

Alarm History. Syntax, description, parameters, and roles for cmm alarmHistory command.

Syntax

Description

This command has three operations: The list operation is used to list alarm history. The meta operation is used to provide alarm history meta information. The show operation is used to show alarm history. Note: Before CMM195BR, the command was cmm alarm history instead of cmm alarmHistory. For backwards compatibility, both versions of the command are supported.

Parameters

Name

category - Category

Syntax

--category <string>

Description

Category.

Class

Optional.

Type / Range

Digit string of length

Name

endTime - Ending time

Syntax

--endTime <string>

Description

Ending time('yyyy-mm-dd HH:MM:SS').

Class

Optional.

Type / Range

Digit string of length 19 to 19 with the allowed pattern of (\d{4}-\d{2}-\d{2} \d{2}:\d{2}:\d{2})

Name

limit - Amount limit in response

Syntax

--limit <string>

Description

Amount limit in response.

Class

Optional.

Type / Range

Digit string of length

Name

period - Time period

Syntax

--period <string>

Description

Time period(e.g. 30s,15m,2h,1d ...).

Class

Optional.

Type / Range

Digit string of length

Name

pid - Alarm PID

Syntax

--pid <integer>

Description

Alarm PID(e.g. 42004).

Class

Optional.

Type / Range

Integer in the range of 0 to 2,147,483,647

Name

problem - Problem

Syntax

--problem <string>

Description

Problem.

Class

Optional.

Type / Range

Digit string of length

Name

severity - Severity

Syntax

--severity <string>

Description

Severity(e.g. major,minor,info ...).

Class

Optional.

Type / Range

Digit string of length

Name

sourceApp - Source appname

Syntax

--sourceApp <string>

Description

Source appname.

Class

Optional.

Type / Range

Digit string of length

Name

sourceHost - Source hostname

Syntax

--sourceHost <string>

Description

Source hostname.

Class

Optional.

Type / Range

Digit string of length

Name

sourceIp - Source hostip

Syntax

--sourceIp <string>

Description

Source hostip.

Class

Optional.

Type / Range

Digit string of length

Name

startTime - Starting time

Syntax

--startTime <string>

Description

Starting time('yyyy-mm-dd HH:MM:SS').

Class

Optional.

Type / Range

Digit string of length 19 to 19 with the allowed pattern of `(\d{4}-\d{2}-\d{2} \d{2}:\d{2}:\d{2})`

Name

task - Task

Syntax

`--task <string>`

Description

Task(e.g. raise, cancel, update, expire).

Class

Optional.

Type / Range

Digit string of length

Roles**List**

sysadmin, manage, monitor

Meta

sysadmin, manage

Show

sysadmin, manage, monitor

3.10 cmm alarmIgnore

Alarms ignored. Syntax, description, parameters, and roles for `cmm alarmIgnore` command.

Syntax**Description**

Configure PIDs for alarms to be ignored.

Parameters**Name**

alarmId - Alarm PID

Syntax

`--alarmId <integer>`

Description

PID of the alarm to be ignored.

Class

Required key.

Type / Range

Integer in the range of 0 to 2,147,483,647

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.11 cmm allowedNfDomain

Allowed NF Domain. Syntax, description, parameters, and roles for cmm allowedNfDomain command.

Syntax**Description**

Links a Domain to an allowedNfDomain list. A Domain may be added only once in a list.

Parameters**Name**

allowedNfDomainListName - Allowed NF Domain List Name

Syntax

--allowedNfDomainListName <string>

Description

Allowed NF Domain List Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

domain - NF Domain

Syntax

--domain <string>

Description

NF Domain

Class

Required key.

Type / Range

Digit string of length 1 to 128 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.12 cmm allowedNfDomainList

Allowed NF Domain List. Syntax, description, parameters, and roles for cmm allowedNfDomainList command.

Syntax**Description**

List of allowed NF domains. Up to 8 lists can be created. Each list can have up to 127 domains.

Parameters**Name**

name - Allowed NF Domains List Name

Syntax

--name <string>

Description

Allowed NF Domains List Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.13 cmm allowedNfType

Allowed NF Type. Syntax, description, parameters, and roles for cmm allowedNfType command.

Syntax**Description**

Links a NF Type to an allowed NF Type list. Each NF Type may be added only once in a list.

Parameters**Name**

allowedNfTypeListName - Allowed NF Types List Name

Syntax

--allowedNfTypeListName <string>

Description

Allowed NF Types List Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

nfType - NF Type

Syntax

--nfType <list-item>

Description

NF Type

Class

Required key.

List

- 5GEIR - 5G EIR - The network function type of 5G-Equipment Identity Register.
- AMF - Access and Mobility Manager Function - The network function type of Access and Mobility Manager Function.
- AUSF - Authentication Server Function - The network function type of Authentication Server Function.
- CBCF - Cell Broadcast Centre Function - The network function type of Cell Broadcast Centre Function
- CHF - Charging Function - The network function type of Charging Function
- GMLC - Gateway Mobile Location Center - The network function type of Gateway Mobile Location Center.
- LMF - Location Management Function - The network function type of Location Management Function.
- NRF - Network Repository Function - The network function type of Network Repository Function.
- NSSAAF - Network Slice Specific Authentication and Authorization Function - The network function type of Network Slice Specific Authentication and Authorization Function
- NSSF - Network Slice Selection Function - The network function type of Network Slice Selection Function.
- NWDAF - Network Data Analytics Function - The network function type of NWDAF.
- PCF - Policy Control Function - The network function type of Policy Control Function.
- SMF - Session Management Function - The network function type of Session Management Function.
- SMSF - Short Message Service Function - The network function type of Short Message Service Function.

- UDM - Unified Data Manager - The network function type of Unified Data Manager.
- UDSF - Unstructured Data Storage Function - The network function type of Unstructured Data Storage Function.

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.14 cmm allowedNfTypeList

Allowed NF Type List. Syntax, description, parameters, and roles for cmm allowedNfTypeList command.

Syntax

Description

List of allowed NF types. Up to 8 lists can be created. Each list can have up to 15 NF Types.

Parameters

Name

name - Allowed NF Type List Name

Syntax

--name <string>

Description

Allowed NF Type List Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.15 cmm allowedPlmn

Allowed PLMN. Syntax, description, parameters, and roles for `cmm allowedPlmn` command.

Syntax

Description

Links a PLMN to an allowed PLMN List. Each PLMN can be added only once to a list.

Parameters

Name

allowedPlmnListName - Allowed PLMNs List Name

Syntax

--allowedPlmnListName <string>

Description

Allowed PLMNs List Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

mcc - Mobile Country Code

Syntax

--mcc <digits>

Description

Mobile Country Code

Class

Required key.

Type / Range

Digit string of length 3 to 3 with the allowed pattern of [0-9]*

Name

mnc - Mobile Network Code

Syntax

--mnc <digits>

Description

Mobile Network Code

Class

Required key.

Type / Range

Digit string of length 2 to 3 with the allowed pattern of [0-9]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.16 cmm allowedPlmnList

Allowed PLMN List. Syntax, description, parameters, and roles for cmm allowedPlmnList command.

Syntax**Description**

List of allowed PLMN. Up to 8 lists can be created. Each list can have 31 PLMNs.

Parameters**Name**

name - Allowed PLMNs List Name

Syntax

--name <string>

Description

Allowed PLMNs List Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.17 cmm allowedSnssai

Allowed S-NSSAI. Syntax, description, parameters, and roles for cmm allowedSnssai command.

Syntax**Description**

Create an Allowed S-NSSAI with Name, Service Type and Service Differentiator.

Parameters**Name**

name - Allowed S-NSSAI Name

Syntax

--name <string>

Description

Allowed S-NSSAI Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

sliceServiceType - Slice Service Type

Syntax

--sliceServiceType <integer>

Description

The S-NSSAI consists of a Slice Service Type (SST) and a Slice Differentiator (SD). Each slice must have SST and SD is optional.

Class

Required on create.

Type / Range

5G Slice Service Type

Name

sliceDifferentiator - Slice Differentiator

Syntax

--sliceDifferentiator <string>

Description

The S-NSSAI consists of a Slice Service Type (SST) and a Slice Differentiator (SD). Each slice must have SST and SD is optional. A SST may have multiple SDs. If there is no Differentiator this value shall be set to 'FFFFFF'.

Class

Optional.

Type / Range

Identifies Slice Differentiator (SD)

Default

FFFFFF

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.18 cmm allowedSnssaiList

Allowed S-NSSAI List . Syntax, description, parameters, and roles for cmm allowedSnssaiList command.

Syntax

Description

List with allowed S-NSSAIs. Up to 8 lists can be created. Each list can have up to 255 S-NSSAIs.

Parameters

Name

name - Allowed S-NSSAI List Name

Syntax

--name <string>

Description

Allowed S-NSSAI List Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.19 cmm allowedSnssaiListItem

Allowed SNSSAI List Item. Syntax, description, parameters, and roles for cmm allowedSnssaiListItem command.

Syntax

Description

Adds or removes an Allowed S-NSSAI to an Allowed S-NSSAI List. Each list can have up to 255 S-NSSAIs

Parameters

Name

allowedSnssaiListName - Allowed S-NSSAI List Name

Syntax

--allowedSnssaiListName <string>

Description

Allowed S-NSSAI List Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

allowedSnssaiName - Allowed S-NSSAI Name

Syntax

--allowedSnssaiName <string>

Description

Allowed S-NSSAI Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.20 cmm amfAccessRestrictionNasMappingProfile

Access Restriction To NAS Cause Mapping Profile. Syntax, description, parameters, and roles for cmm amfAccessRestrictionNasMappingProfile command.

Syntax

Description

Command is used to administer the Access Restriction to NAS Cause Mapping Profile. This profile is used to select the right cause code to send to the UE when a UE mobility management request is rejected due to an access restriction. The maximum number of profiles supported is 128. A default profile, named 'default', is provided.

Parameters

Name

name - Name of access restriction to NAS cause code profile name

Syntax

--name <string>

Description

Profile name for access restriction to NAS cause code mapping.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

ausfMsgDecodingFailure - AUSF Message Decoding Failure

Syntax

--ausfMsgDecodingFailure <integer>

Description

NAS cause code sent to the UE when AMF fails in decoding a message from AUSF.

Class

Optional.

Type / Range

5GMM Cause Code.

Default

10

Name

ausfResponseTimeOut - AUSF Response Time Out

Syntax

--ausfResponseTimeOut <integer>

Description

NAS cause code sent to the UE when AUSF times out with no response.

Class

Optional.

Type / Range

5GMM Cause Code.

Default

15

Name

ausfUnavailability - AUSF Unavailability

Syntax

--ausfUnavailability <integer>

Description

NAS cause code sent to the UE when AMF is not able to communicate with the AUSF.

Class

Optional.

Type / Range

5GMM Cause Code.

Default

111

Name

authenticationResponseTimeout - Authentication Response Timeout

Syntax

--authenticationResponseTimeout <integer>

Description

NAS cause code sent to the UE when AMF times out waiting for NAS Authentication Response from the UE.

Class

Optional.

Type / Range

5GMM Cause Code.

Default

10

Name

cagRestrictUeNotSuppCag - CAG Restriction for UE Not Supporting CAG

Syntax

--cagRestrictUeNotSuppCag <integer>

Description

NAS cause code sent when UE is not allowed to access network due to CAG restriction for UE not supporting CAG feature.

Class

Optional.

Type / Range

5GMM Cause Code.

Default

15

Name

cagRestrictUeSuppCag - CAG Restriction for UE Supporting CAG

Syntax

--cagRestrictUeSuppCag <integer>

Description

NAS cause code sent when UE is not allowed to access network due to CAG restriction for UE supporting CAG feature.

Class

Optional.

Type / Range

5GMM Cause Code.

Default

76

Name

coreNetworkTypeRestriction - Core Network Type Restriction

Syntax

--coreNetworkTypeRestriction <integer>

Description

NAS cause code sent when UE is not allowed in 5G Core Network.

Class

Optional.

Type / Range

5GMM Cause Code.

Default

15

Name

forbiddenTaiHplmn - UE is in a forbidden TAI in HPLMN

Syntax

--forbiddenTaiHplmn <integer>

Description

NAS cause code sent when UE is in a forbidden TA in HPLMN.

Class

Optional.

Type / Range

5GMM Cause Code.

Default

15

Name

interPlmnMobRegUpdtRej - Inter-PLMN Mobility Registration Update Reject

Syntax

--interPlmnMobRegUpdtRej <integer>

Description

NAS cause code sent to roamer UE when inter-plmn idle mode Mobility Registration Update is not allowed.

Class

Optional.

Type / Range

5GMM Cause Code.

Default

9

Name

noAllowedNssai - No Allowed NSSAI

Syntax

--noAllowedNssai <integer>

Description

NAS cause code sent to the UE when no allowed NSSAI for the UE.

Class

Optional.

Type / Range

5GMM Cause Code.

Default

10

Name

noCommonSecurityAlgorithm - No Common Security Algorithm

Syntax

--noCommonSecurityAlgorithm <integer>

Description

NAS cause code sent to the UE when there is no common security algorithm between the UE supported algorithms and the AMF configuration.

Class

Optional.

Type / Range

5GMM Cause Code.

Default

10

Name

nonEncryptedSuciRestriction - Non Encrypted SUCI Not Allowed

Syntax

--nonEncryptedSuciRestriction <integer>

Description

NAS cause code sent in Registration Reject message when nonEncryptedSuciNotAllowed is enabled but UE identity received is non-encrypted SUCI.

Class

Optional.

Type / Range

5GMM Cause Code.

Default

3

Name

noSlicesAvailableDueToNssaa - No Network Slices Available Due to NSSAA

Syntax`--noSlicesAvailableDueToNssaa <integer>`**Description**

NAS cause code sent to the UE when there is no Network Slices Available due to NSSAA function. The network shall set the 5GMM cause value to #62 'No network slices available'. If the UE had included requested NSSAI in the REGISTRATION REQUEST message, then the network shall include the rejected S-NSSAI(s) in the Rejected NSSAI IE of the REGISTRATION REJECT message. Otherwise, the network may include the rejected NSSAI

Class

Optional.

Type / Range

5GMM Cause Code.

Default

62

Name

nrPlmnNotAllowed - PLMN of SUPI not allowed in NR

Syntax`--nrPlmnNotAllowed <integer>`**Description**

NAS cause code sent by the AMF to inbound roamer(s) because PLMN of roaming SUPI(s) is not allowed in 5G NR.

Class

Optional.

Type / Range

5GMM Cause Code.

Default

27

Name

nrRatNotAllowedForRoamer - NR RAT restricted for roamer

Syntax`--nrRatNotAllowedForRoamer <integer>`**Description**

NAS cause code sent when a roamer UE is not allowed in the NR RAT based on receipt of RAT_NOT ALLOWED from UDM.

Restrictions

Applies to roamer UEs

Class

Optional.

Type / Range

5GMM Cause Code.

Default

15

Name

nssfMsgDecodingFailure - NSSF Message Decoding Failure

Syntax

--nssfMsgDecodingFailure <integer>

Description

NAS cause code sent to the UE when AMF fails in decoding a message from NSSF.

Class

Optional.

Type / Range

5GMM Cause Code.

Default

10

Name

nssfResponseTimeout - NSSF Response Timeout

Syntax

--nssfResponseTimeout <integer>

Description

NAS cause code sent to the UE when NSSF times out with no response.

Class

Optional.

Type / Range

5GMM Cause Code.

Default

10

Name

nssfUnavailability - NSSF Unavailability

Syntax

--nssfUnavailability <integer>

Description

NAS cause code sent to the UE when AMF is not able to communicate with the NSSF.

Class

Optional.

Type / Range

5GMM Cause Code.

Default

10

Name

ratNotAllowedHplmn - NR restricted in the UE's HPLMN

Syntax

--ratNotAllowedHplmn <integer>

Description

NAS cause code sent when UE is not allowed in the NR RAT.

Class

Optional.

Type / Range

5GMM Cause Code.

Default

15

Name

redcapRatNotAllowedForRoamer - RedCap RAT not allow for Roamer

Syntax

--redcapRatNotAllowedForRoamer <integer>

Description

Defines the cause code to be sent to the UE in case of access restriction for the NR RedCap RAT.

Class

Optional.

Type / Range

5GMM Cause Code.

Default

15

Name

redcapRatNotAllowedHplmn - RedCap RAT not allow for HPLMN

Syntax`--redcapRatNotAllowedHplmn <integer>`**Description**

Defines the cause code to be provided to a home UE that is not allowed to use the NR RedCap RAT.

Class

Optional.

Type / Range

5GMM Cause Code.

Default

15

Name

rfspAccessRestrictionCc - RFSP Access Restriction Cause Code

Syntax`--rfspAccessRestrictionCc <integer>`**Description**

The Cause Code to be sent to a UE if it is restricted access based on RFSP

Class

Optional.

Type / Range

5GMM Cause Code.

Default

15

Name

roam3gppAccessForbiddenTai - Roaming in 5G 3GPP Access in TA not allowed

Syntax`--roam3gppAccessForbiddenTai <integer>`**Description**

NAS cause code sent to roamer UE when roaming is not allowed in 5G 3GPP Access in this Tracking Area based on UDM forbidden TAIs or locally provisioned forbidden TAIs.

Restrictions

Applies to roamer UEs.

Class

Optional.

Type / Range

5GMM Cause Code.

Default

15

Name

roam3gppAccessNotAllowed - Roaming in 5G 3GPP Access not allowed

Syntax

--roam3gppAccessNotAllowed <integer>

Description

NAS cause code sent to roamer UE when roaming is not allowed in 5G 3GPP Access based on receipt of ACCESS_NOT_ALLOWED from UDM or due to local provisioning.

Restrictions

Applies to roamer UEs

Class

Optional.

Type / Range

5GMM Cause Code.

Default

15

Name

roamingSuciRestriction - Roaming UE SUCI Access Restriction

Syntax

--roamingSuciRestriction <integer>

Description

The NAS cause code sent to an inbound roaming UE when UE SUCI has no match in UE Scheme Output allowed list.

Class

Optional.

Type / Range

5GMM Cause Code.

Default

27

Name

securityModeCompleteTimeout - Security Mode Complete Timeout

Syntax

--securityModeCompleteTimeout <integer>

Description

NAS cause code sent to the UE when AMF times out waiting for NAS Security Mode Complete from the UE.

Class

Optional.

Type / Range

5GMM Cause Code.

Default

10

Name

serviceAreaRestriction - Service Area Restriction

Syntax

--serviceAreaRestriction <integer>

Description

NAS cause code sent when UE is not allowed in a Service Area.

Class

Optional.

Type / Range

5GMM Cause Code.

Default

28

Name

snpnNotAllowed - SNPN Not Allowed

Syntax

--snpnNotAllowed <integer>

Description

NAS cause code sent to UE when SNPN service not allowed.

Class

Optional.

Type / Range

5GMM Cause Code.

Default

74

Name

udmImmediateReportUnavailable - UDM Immediate Report Not Available

Syntax

--udmImmediateReportUnavailable <integer>

Description

NAS cause code sent to the UE when UDM indicates Immediate Report feature supported but associated report s not included in the UDM response to the AMF.

Class

Optional.

Type / Range

5GMM Cause Code.

Default

15

Name

udmMsgDecodingFailure - UDM Message Decoding Failure

Syntax

--udmMsgDecodingFailure <integer>

Description

NAS cause code sent to the UE when AMF fails in decoding a message from UDM.

Class

Optional.

Type / Range

5GMM Cause Code.

Default

10

Name

udmOdbAllPacket - UDM ODB All Packet Services

Syntax

--udmOdbAllPacket <integer>

Description

NAS cause code sent to the UE on Registration Reject when both UDM provides ODB All Packet Services restriction for the UE and AMF local provisioning has on service agreement level parameter Restrict Mobility on UDM ODB All Packet Barred set to true.

Class

Optional.

Type / Range

5GMM Cause Code.

Default

15

Name

udmResponseTimeOut - UDM Response Time Out

Syntax

--udmResponseTimeOut <integer>

Description

NAS cause code sent to the UE when UDM times out with no response.

Class

Optional.

Type / Range

5GMM Cause Code.

Default

15

Name

udmUnavailability - UDM Unavailability

Syntax

--udmUnavailability <integer>

Description

NAS cause code sent to the UE when AMF is not able to communicate with the UDM.

Class

Optional.

Type / Range

5GMM Cause Code.

Default

10

Name

wlanRatRestriction - WLAN RAT Restriction HPLMN

Syntax

--wlanRatRestriction <integer>

Description

NAS cause code sent when UE is not allowed in the WLAN RAT in HPLMN.

Class

Optional.

Type / Range

5GMM Cause Code.

Default

72

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.21 cmm amfAdjacentPlmn

Adjacent Plmn. Syntax, description, parameters, and roles for cmm amfAdjacentPlmn command.

Syntax**Description**

Configures an existing network as an adjacent plmn network and includes parameter attributes for adding the adjacent plmn network names.

Parameters**Name**

amfAdjacentPlmnListName - Adjacent Plmn List Name

Syntax`--amfAdjacentPlmnListName <string>`**Description**

Identifies the Adjacent PLMN list name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

amfAdjacentPlmnName - Adjacent Plmn Name

Syntax

--amfAdjacentPlmnName <string>

Description

Identifies the name of the Adjacent PLMN which can be Home, Shared or Other PLMN. There can be up to 5 Adjacent Plmns in the list and each have to exist in the EPLMN List sent to the gNB.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.22 cmm amfAdjacentPlmnList

Adjacent PLMN List. Syntax, description, parameters, and roles for cmm amfAdjacentPlmnList command.

Syntax**Description**

Adjacent PLMN List

Parameters**Name**

name - Adjacent PLMN List Name

Syntax

--name <string>

Description

Identifies the name of the Adjacent PLMN List

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

description - Description of Adjacent PLMN List

Syntax

--description <string>

Description

Describes the use of the Adjacent PLMN List

Class

Optional.

Type / Range

Digit string of length 0 to 160 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.23 cmm amfAliasDnnConversionProfile

AMF Alias DNN Conversion Profile. Syntax, description, parameters, and roles for cmm amfAliasDnnConversionProfile command.

Syntax

Description

This command is used to administer an alias DNN Conversion Profile.

Parameters

Name

name - Name

Syntax

--name <string>

Description

Name that identifies alias Dnn Conversion profile

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

aliasConversionDnnSupport - Alias Conversion DNN Support

Syntax

--aliasConversionDnnSupport <list-item>

Description

Parameter controls the feature activation flag. By default, the flag shall be disabled. When enabled AMF shall support alias Conversion DNN functionality.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

aliasDefaultConversionDnn - Alias Default Conversion DNN

Syntax

--aliasDefaultConversionDnn <string>

Description

Provisioned Alias Default Conversion DNN

Class

Optional.

Type / Range

Data Network Name with Null String.

Default

""

Name

replacementDnnListName - Replacement DNN List Name

Syntax

--replacementDnnListName <string>

Description

Contains list of allowed DNNs that are subjected to conversion. aliasConversionDnnSupport flag need to be enabled to have an entry in replacementDnnListName list.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Default

""

Name

sendAliasConvertedDnnToUe - Send Alias Converted DNN to UE

Syntax

--sendAliasConvertedDnnToUe <list-item>

Description

Parameter controls whether to send the corrected/overridden DNN or the UE requested DNN. When the parameter is false, UE requested DNN shall be sent to UE and SMF. When the parameter is true converted DNN shall be sent to SMF and back to the UE, If the UE did not send a requested DNN, then the AMF shall send the converted DNN to the UE and SMF.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.24 cmm amfAmPolicyProfile

AMF AM Policy Profile. Syntax, description, parameters, and roles for cmm amfAmPolicyProfile command.

Syntax

Description

This command is used to administer an AM Policy Profile. and includes following parameters: RFSP index ServiceAreaRestriction SUPI GPSI URSP location change change of ue presence area

Parameters

Name

name - AM Policy Profile Name

Syntax

--name <string>

Description

A name that identifies AM Policy profile.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

serviceAreaRestriction - Service Area Restriction

Syntax

--serviceAreaRestriction <list-item>

Description

Service Area Restriction specifies the list of allowed or non-allowed and maximum number of allowed TAs

Class

Required on create.

List

- 1 - allowedServiceAreaRestrictionListName - allowed Service Area Restriction List Name
- 2 - notAllowedServiceAreaRestrictionListName - not allowed Service Area Restriction List Name

Name

allowedServiceAreaRestrictionListName - Allowed Service Area Restriction List Name

Syntax

--allowedServiceAreaRestrictionListName <string>

Description

Allowed Service Area Restriction List Name

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

maxNumOfTAs - Maximum number of Tracking Areas

Syntax

--maxNumOfTAs <integer>

Description

Maximum number of Tracking Areas

Class

Optional.

Type / Range

Integer in the range of 1 to 65,535

Default

255

Name

notAllowedServiceAreaRestrictionListName - Not Allowed Service Area Restriction List Name

Syntax

--notAllowedServiceAreaRestrictionListName <string>

Description

Not Allowed Service Area Restriction List Name

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

rfspIndex - RFSP Index Value

Syntax

--rfspIndex <integer>

Description

Value of the RFSP Index

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

v2xSupported - V2X Supported

Syntax

--v2xSupported <list-item>

Description

Specifies whether the PCF supports V2X.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.25 cmm amfAusfHttp2ToNasCcMapping

AMF AUSF HTTP2 To NAS CC mapping. Syntax, description, parameters, and roles for cmm amfAusfHttp2ToNasCcMapping command.

Syntax

Description

Defines AUSF HTTP2 Status Codes Mapping to AMF NAS Cause Codes. The maximum number of profiles supported is 128. A default profile, named 'default', is provided.

Parameters

Name

amfAusfHttp2ToNasCcMappingListName - AMF AUSF HTTP2 to NAS CC mapping list name

Syntax

--amfAusfHttp2ToNasCcMappingListName <string>

Description

The name of the HTTP2 to 5G NAS Cause Code mapping list

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

ausfHttp2StatusCode - AUSF HTTP2 Status Code

Syntax

--ausfHttp2StatusCode <list-item>

Description

Defines the AUSF HTTP2 Status Code to map to 5G NAS Cause Code

Class

Required key.

List

- 400_RESOURCE_CONTEXT_NOT_FOUND - 400 RESOURCE CONTEXT NOT FOUND - The notification request is rejected because the call back URI still exists in the receiver of the notification, but the specific resource context identified within the notification payload is not found in the NF service consumer.
- 400_UNKNOWN - 400 UNKNOWN - The 400_UNKNOWN is used when included problem details is not specified in 3GPP or it is not included.
- 403_CCA_VERIFICATION_FAILURE - 403 CCA VERIFICATION FAILURE - The request is rejected due to a failure to verify the CCA at the receiving entity (AUSF).
- 403_UNKNOWN - 403 UNKNOWN - The 403_UNKNOWN is used when included problem details is not specified in 3GPP or it is not included.
- 404_UNKNOWN - 404 UNKNOWN - The 404_UNKNOWN is used when included problem details is not specified in 3GPP or it is not included.
- 411_UNKNOWN - 411 UNKNOWN - The 411_UNKNOWN is used when included problem details is not specified in 3GPP or it is not included.
- 429_UNKNOWN - 429 UNKNOWN - The 429_UNKNOWN is used when included problem details is not specified in 3GPP or it is not included.
- 500_NF_FAILOVER - 500 NF FAILOVER - The request is rejected due to the unavailability of the NF (AUSF), and the requester may trigger an immediate re-selection of an alternative NF based on this information.
- 500_NF_SERVICE_FAILOVER - 500 NF SERVICE FAILOVER - The request is rejected due to the unavailability of the NF service, and the requester may trigger an immediate re-selection of an alternative NF service based on this information.
- 500_UNKNOWN - 500 UNKNOWN - The 500_UNKNOWN is used when included problem details is not specified in 3GPP or it is not included.
- 501_UNKNOWN - 501 UNKNOWN - The 501_UNKNOWN is used when included problem details is not specified in 3GPP or it is not included.
- 503_UNKNOWN - 503 UNKNOWN - The 503_UNKNOWN is used when included problem details is not specified in 3GPP or it is not included.

- 504_TARGET_NF_NOT_REACHABLE - 504 TARGET NF NOT REACHABLE - The request is not served as the target NF (AUSF) is not reachable.
- 504_TIMED_OUT_REQUEST - 504 TIMED OUT REQUEST - The request is rejected due a request that has timed out at the HTTP client based on origination time stamp
- 504_UNKNOWN - 504 UNKNOWN - The 504_UNKNOWN is used when included problem details is not specified in 3GPP or it is not included.
- 400_INVALID_API - 400 Bad Request Invalid API - The HTTP request contains an unsupported API name or API version in the URI.
- 400_INVALID_MSG_FORMAT - 400 Bad Request Invalid Msg Format - The HTTP request has an invalid format.
- 400_INVALID_QUERY_PARAM - 400 Bad Request Invalid Query Param - The HTTP request contains an unsupported query parameter in the URI.
- 400_MANDATORY_QUERY_PARAM_INCORRECT - 400 Bad Request Mandatory Query Param Incorrect - A mandatory query parameter, or a conditional query parameter but mandatory required, for an HTTP method was received in the URI with semantically incorrect value.
- 400_OPTIONAL_QUERY_PARAM_INCORRECT - 400 Bad Request Optional Query Param Incorrect - An optional query parameter for an HTTP method was received in the URI with a semantically incorrect value that prevents successful processing of the service request.
- 400_MANDATORY_QUERY_PARAM_MISSING - 400 Bad Request Mandatory Query Param Missing - Query parameter which is defined as mandatory, or as conditional but mandatory required, for an HTTP method is not included in the URI of the request.
- 400_MANDATORY_IE_INCORRECT - 400 Bad Request Mandatory IE Incorrect - A mandatory IE or conditional IE in data structure, but mandatory required, for an HTTP method was received with a semantically incorrect value.
- 400_OPTIONAL_IE_INCORRECT - 400 Bad Request Optional IE Incorrect - An optional IE in data structure for an HTTP method was received with a semantically incorrect value that prevents successful processing of the service request.
- 400_MANDATORY_IE_MISSING - 400 Bad Request Mandatory IE Missing - IE which is defined as mandatory or as conditional in data structure, but mandatory required, for an HTTP method is not included in the payload body of the request.
- 400_UNSPECIFIED_MSG_FAILURE - 400 Bad Request Unspecified Msg Failure - The request is rejected due to unspecified client error.
- 403_MODIFICATION_NOT_ALLOWED - 403 Forbidden Modification Not Allowed - The request is rejected because the contained modification instructions attempt to modify IE which is not allowed to be modified.
- 403_SERVING_NETWORK_NOT_AUTHORIZED - 403 Forbidden Serving Network Not Authorized - The serving network is not authorized, e.g. serving PLMN.

- 403_AUTHENTICATION_REJECTED - 403 Forbidden Authentication Rejected - The user is cannot be authenticated with this authentication method e.g. only SIM data available.
- 403_INVALID_HN_PUBLIC_KEY_IDENTIFIER - 403 Forbidden Invalid HN Public Key Identifier - Invalid HN public key identifier received
- 403_INVALID_SCHEME_OUTPUT - 403 Forbidden Invalid Scheme Output - SUCI cannot be decrypted with received data.
- 404_SUBSCRIPTION_NOT_FOUND - 404 Not Found Subscription Not Found - The request for modification or deletion of subscription is rejected because the subscription is not found in the NF.
- 404_CONTEXT_NOT_FOUND - 404 Not Found Context Not Found - The AUSF cannot found the resource corresponding to the URI provided by the NF Service Consumer.
- 404_RESOURCE_URI_STRUCTURE_NOT_FOUND - 404 Not Found Resource URI Structure Not Found - The request is rejected because a fixed part after the first variable part of an apiSpecificResourceUriPart is not found in the NF. This fixed part of the URI may represent a sub-resource collection (e.g. contexts, subscriptions, policies) or a custom operation.
- 404_USER_NOT_FOUND - 404 Not Found User Not Found - The user does not exist in the HPLMN.
- 411_INCORRECT_LENGTH - 411 Length Required Incorrect Length - The request is rejected due to incorrect value of a Content-length header field.
- 429_NF_CONGESTION_RISK - 429 Too Many Requests NF Congestion Risk - The request is rejected due to excessive traffic which, if continued over time, may lead to (or may increase) an overload situation.
- 500_INSUFFICIENT_RESOURCES - 500 Internal Server Error Insufficient Resources - The request is rejected due to insufficient resources.
- 500_UNSPECIFIED_NF_FAILURE - 500 Internal Server Error Unspecified NF Failure - The request is rejected due to unspecified reason at the NF.
- 500_SYSTEM_FAILURE - 500 Internal Server Error System Failure - The request is rejected due to generic error condition in the NF.
- 500_AV_GENERATION_PROBLEM - 500 Internal Server Error AV Generation Problem - The UDM has indicated that it was not able to generate AV.
- 501_UNSUPPORTED_PROTECTION_SCHEME - 501 Not implemented Unsupported Protection Scheme - The received protection scheme is not supported by HPLMN.
- 503_NF_CONGESTION - 503 Service Unavailable NF Congestion - The NF experiences congestion and performs overload control, which does not allow the request to be processed.
- 504_UPSTREAM_SERVER_ERROR - 504 Gateway Timeout Upstream Server Error - No response is received from a remote peer, e.g. from the UDM.

- 504_NETWORK_FAILURE - 504 Gateway Timeout Network Failure - The request is rejected due to a network problem.

Name

amf5gMmCc - 5GMM NAS Cause Code

Syntax

--amf5gMmCc <integer>

Description

Defines the 5GMM NAS cause code to use towards the UE based on the HTTP2 Status Code received from the AUSF

Class

Optional.

Type / Range

5GMM Cause Code.

Default

15

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

3.26 cmm amfAusfHttp2ToNasCcMappingList

AMF AUSF HTTP2 To NAS CC Mapping List. Syntax, description, parameters, and roles for cmm amfAusfHttp2ToNasCcMappingList command.

Syntax**Description**

Specifies AUSF HTTP2 Status Code to 5GMM NAS Cause Code mapping Profile List

Parameters**Name**

name - AMF AUSF HTTP2 To NAS CC Mapping List Name

Syntax

--name <string>

Description

AUSF HTTP2 Status Code to 5GMM NAS Cause Code mapping Profile List name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.27 cmm amfCapacityGParms

AMF Capacity. Syntax, description, parameters, and roles for cmm amfCapacityGParms command.

Syntax**Description**

This command opens the context for configuring capacity related parameters for the AMF network.

Parameters**Name**

amfGparmName - Parameter name

Syntax

--amfGparmName <list-item>

Description

Name of parameter

Class

Required key.

List

- **amfRelativeCapacity** - AMF Relative Capacity - Assigns Relative AMF Capacity for use by the gNB when selecting an AMF from an AMF SET(pool). RAN uses Relative Capacity to load balance across AMFs within an AMF SET. This will only be used if Auto Adjust Capacity is set to false.
- **autoAdjustCap** - Auto Adjust Relative Capacity - Indicates whether or not the AMF automatically adjusts the value of the relative capacity according to an algorithm based upon: 1) The number of active CPPS in the enabled and unlocked states and the number of CPPS considered as 'Protected' (Number Protection CPPS) in an n+k protection or sparing scheme. 2) The number of active IPDS Pools in the enabled and unlocked states.
- **maxTrackingAreaRelativeCapacity** - Maximum Tracking Area Relative Capacity - Selects the maximum provisioned AMF TA relative capacity of the provisioned AMF relative capacities of TAs supported by the gNB when sent to gNB in NGAP SETUP RESPONSE or AMF CONFIGURATION UPDATE messages.
- **numProtCpps** - Total Number of Protection CPPS - This parameter is the number of the CPPS with the AMMS personality in the system that are protection CPPS. This is also considered the k value in the n+k sparing arrangement. This is only used when Auto Adjust Capacity is true. Total number of CPPS minus numProtCpps times relativeCapacityCpps should be less than or equal to 255.
- **relativeCapacityCpps** - Relative Capacity per CPPS - AMF Relative Capacity assigned to each N CPPS in a N + K protection or sparing scheme of CPPS with the AMMS personality. This value will only be used when Auto Adjust Capacity parameter is set to true. This value will be used to calculate the CPPS contribution to the total AMF Relative Capacity based upon the total number of CPPS and the value of K, Number Protection CPPS parameter, which contains the number of Protection CPPS. Total number of CPPS minus numProtCpps times relativeCapacityCpps should be less than or equal to 255.
- **relativeCapacityIPDS0** - Relative Capacity per IPDS Pool 0 - AMF Relative Capacity assigned to the IPDS Pool 0. This value will be used to calculate the IPDS contribution to the total nominal value for the AMF Relative Capacity. $\text{relativeCapacityIPDS0} + (\text{total number of available additional IPDS pools} * \text{relativeCapacityIPDSAdditional})$ should be less than or equal to 255.
- **relativeCapacityIPDSAdditional** - Relative Capacity per IPDS Additional - AMF Relative Capacity assigned per additional IPDS Pool. This value will be used to calculate the IPDS contribution to the total nominal value for the AMF Relative Capacity. $\text{relativeCapacityIPDS0} + (\text{total number of available additional IPDS pools} * \text{relativeCapacityIPDSAdditional})$ should be less than or equal to 255.
- **trackingAreaRelativeCapacity** - Tracking Area Relative Capacity - Sets AMF relative capacity per tracking area

Name

amfGparmValue - Parameter value

Syntax

--amfGparmValue <string>

Description

Value of parameter

Class

Required on create.

Type / Range

Digit string of length 0 to 64

The amfGparmValue string is based on the amfGparmName parameter:

Name

amfRelativeCapacity - AMF Relative Capacity

Syntax

--amfGparmName amfRelativeCapacity --amfGparmValue <string>

Description

Assigns Relative AMF Capacity for use by the gNB when selecting an AMF from an AMF SET(pool). RAN uses Relative Capacity to load balance across AMFs within an AMF SET. This will only be used if Auto Adjust Capacity is set to false.

Type / Range

Integer in the range of 0 to 255

Default

5

Name

autoAdjustCap - Auto Adjust Relative Capacity

Syntax

--amfGparmName autoAdjustCap --amfGparmValue <string>

Description

Indicates whether or not the AMF automatically adjusts the value of the relative capacity according to an algorithm based upon: 1) The number of active CPPS in the enabled and unlocked states and the number of CPPS considered as 'Protected' (Number Protection CPPS) in an n+k protection or sparing scheme. 2) The number of active IPDS Pools in the enabled and unlocked states.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

maxTrackingAreaRelativeCapacity - Maximum Tracking Area Relative Capacity

Syntax

```
--amfGparmName maxTrackingAreaRelativeCapacity --amfGparmValue  
<string>
```

Description

Selects the maximum provisioned AMF TA relative capacity of the provisioned AMF relative capacities of TAIs supported by the gNB when sent to gNB in NGAP SETUP RESPONSE or AMF CONFIGURATION UPDATE messages.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

numProtCpps - Total Number of Protection CPPS

Syntax

```
--amfGparmName numProtCpps --amfGparmValue <string>
```

Description

This parameter is the number of the CPPS with the AMMS personality in the system that are protection CPPS. This is also considered the k value in the n+k sparing arrangement. This is only used when Auto Adjust Capacity is true. Total number of CPPS minus numProtCpps times relativeCapacityCpps should be less than or equal to 255.

Type / Range

Integer in the range of 0 to 8

Default

0

Name

relativeCapacityCpps - Relative Capacity per CPPS

Syntax

```
--amfGparmName relativeCapacityCpps --amfGparmValue <string>
```

Description

AMF Relative Capacity assigned to each N CPPS in a N + K protection or sparing scheme of CPPS with the AMMS personality. This value will only be used when Auto Adjust Capacity parameter is set to true. This value will be used to calculate the CPPS contribution to the total AMF Relative Capacity based upon the total number of CPPS and the value of K, Number

Protection CPPS parameter, which contains the number of Protection CPPS. Total number of CPPS minus numProtCpps times relativeCapacityCpps should be less than or equal to 255.

Type / Range

Integer in the range of 0 to 255

Default

6

Name

relativeCapacityIPDS0 - Relative Capacity per IPDS Pool 0

Syntax

```
--amfGparmName relativeCapacityIPDS0 --amfGparmValue <string>
```

Description

AMF Relative Capacity assigned to the IPDS Pool 0. This value will be used to calculate the IPDS contribution to the total nominal value for the AMF Relative Capacity. $\text{relativeCapacityIPDS0} + (\text{total number of available additional IPDS pools} * \text{relativeCapacityIPDSAdditional})$ should be less than or equal to 255.

Type / Range

Integer in the range of 0 to 255

Default

20

Name

relativeCapacityIPDSAdditional - Relative Capacity per IPDS Additional

Syntax

```
--amfGparmName relativeCapacityIPDSAdditional --amfGparmValue  
<string>
```

Description

AMF Relative Capacity assigned per additional IPDS Pool. This value will be used to calculate the IPDS contribution to the total nominal value for the AMF Relative Capacity. $\text{relativeCapacityIPDS0} + (\text{total number of available additional IPDS pools} * \text{relativeCapacityIPDSAdditional})$ should be less than or equal to 255.

Type / Range

Integer in the range of 0 to 255

Default

20

Name

trackingAreaRelativeCapacity - Tracking Area Relative Capacity

Syntax

```
--amfGparmName trackingAreaRelativeCapacity --amfGparmValue <string>
```

Description

Sets AMF relative capacity per tracking area

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

3.28 cmm amfCellId

Cell Identifiers Subject to Handover. Syntax, description, parameters, and roles for `cmm amfCellId` command.

Syntax**Description**

Administer Cell Identifiers subject to Handover Restriction

Parameters**Name**

name - Name of Cell ID

Syntax

--name <string>

Description

Name of the cell identifier

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

cellIdentifier - Cell Identifier

Syntax

--cellIdentifier <integer>

Description

Identify the cell

Class

Required on create, not modifiable.

Type / Range

Integer in the range of 0..68719476735

Name

plmnName - Mobile network name

Syntax

--plmnName <string>

Description

A name that identifies mobile network

Class

Required on create, not modifiable.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.29 cmm amfCIPHERingAlgorithm

Preferred Ciphering Algorithm. Syntax, description, parameters, and roles for cmm amfCIPHERingAlgorithm command.

Syntax

Description

This command sets the algorithms, in priority order, AMF will use to negotiate encryption of NAS messages with the UE. At UE REGISTRATION, the UE provides, in the UE security capabilities, which encryption algorithms are supported. The AMF will use the algorithm both supported by the UE and highest in priority in the nea ciphering algorithm priority list. - An algorithm must be chosen for 'priority-1' and 'priority-2'. - An additional selection, 'none', can be selected for 'priority-3' and 'priority-4' - effectively truncating the list. - If any ciphering algorithm (including null) is selected for 'priority-4', then 'priority-3' must not be 'none'. - An algorithm cannot be used for more than one priority (except for 'none').

Parameters

Name

cipheringAlgPrio1 - Priority 1 ciphering algorithm

Syntax

--cipheringAlgPrio1 <list-item>

Description

Most preferred ciphering algorithm.

Class

Optional.

List

- aes - 128-NEA2 (AES)
- noCiphering - 128-NEA0 (No ciphering)
- none - No Algorithm Selected
- snow3G - 128-NEA1 (SNOW 3G)
- zuc - 128-NEA3 (ZUC)

Default

snow3G

Name

cipheringAlgPrio2 - Second most preferred ciphering algorithm

Syntax

--cipheringAlgPrio2 <list-item>

Description

Second most preferred ciphering algorithm; used if the user element does not support algorithm(s) of higher priority

Class

Optional.

List

- aes - 128-NEA2 (AES)
- noCiphering - 128-NEA0 (No ciphering)
- none - No Algorithm Selected
- snow3G - 128-NEA1 (SNOW 3G)
- zuc - 128-NEA3 (ZUC)

Default

aes

Name

cipheringAlgPrio3 - Third most preferred ciphering algorithm

Syntax

--cipheringAlgPrio3 <list-item>

Description

Third most preferred ciphering algorithm; used if the user element does not support algorithm(s) of higher priority.

Class

Optional.

List

- aes - 128-NEA2 (AES)
- noCiphering - 128-NEA0 (No ciphering)
- none - No Algorithm Selected
- snow3G - 128-NEA1 (SNOW 3G)
- zuc - 128-NEA3 (ZUC)

Default

none

Name

cipheringAlgPrio4 - Fourth most preferred ciphering algorithm

Syntax

--cipheringAlgPrio4 <list-item>

Description

Fourth most preferred ciphering algorithm; used if the user element does not support algorithm(s) of higher priority.

Class

Optional.

List

- aes - 128-NEA2 (AES)
- noCiphering - 128-NEA0 (No ciphering)
- none - No Algorithm Selected
- snow3G - 128-NEA1 (SNOW 3G)
- zuc - 128-NEA3 (ZUC)

Default

none

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

3.30 cmm amfCmasGParms

AMF CMAS. Syntax, description, parameters, and roles for `cmm amfCmasGParms` command.

Syntax

Description

This command is used to enter the context of CMAS to provision AMF global CMAS parameter

Parameters

Name

amfGparmName - Parameter name

Syntax

--amfGparmName <list-item>

Description

Name of parameter

Class

Required key.

List

- **alertMessageHoldingTime** - Alert Message Holding Time - Alert Message Holding Time timer indicates the time to wait (in seconds) that a N50 NonUeN2MessageTransfer request from the CBCF that will be queued while waiting for a previously received N50 NonUeN2MessageTransfer request completes its broadcast.
- **cmas5GWarningSystemLog** - CMAS 5G WarningSystemLog - 5G CMAS Warning System Log
- **messageReTransmission** - Message Retransmission - This command configures maximum number of times WRITE_REPLACE WARNING REQUEST message to re-transmit if no response is received from the gNB. By default the command sets the re-transmission value to 0.
- **queuingCbcfRequests** - Queuing CBCF Requests - Parameter controls queuing CBCF requests. By default, it is disabled. When enabled, AMF shall support queuing of up to 20 CBCF requests while processing on-going broadcast.
- **stopWarningIndication** - Stop Warning Indication - This command is used to enable sending of STOP WARNING MESSAGE TO the CBCF.
- **warningMessageDelivery** - Warning Message Delivery - This command enables/disables Commercial Mobile Alert System (aka warning message delivery) By default, the warning message delivery is disabled
- **writeReplaceWarningIndication** - Write Replace Warning Indication - This command is used to enable AMF sending of broadcast completed area list to the CBCF. By default this is disabled.

Name

amfGparmValue - Parameter value

Syntax

--amfGparmValue <string>

Description

Value of parameter

Class

Required on create.

Type / Range

Digit string of length 0 to 64

The amfGparmValue string is based on the amfGparmName parameter:

Name

alertMessageHoldingTime - Alert Message Holding Time

Syntax

```
--amfGparmName alertMessageHoldingTime --amfGparmValue <string>
```

Description

Alert Message Holding Time timer indicates the time to wait (in seconds) that a N50 NonUeN2MessageTransfer request from the CBCF that will be queued while waiting for a previously received N50 NonUeN2MessageTransfer request completes its broadcast.

Type / Range

Integer in the range of 0 to 180 in units of Second

Default

0

Name

cmas5GWarningSystemLog - CMAS 5G WarningSystemLog

Syntax

```
--amfGparmName cmas5GWarningSystemLog --amfGparmValue <string>
```

Description

5G CMAS Warning System Log

List

- all - Log both NGAP and CBCF CMAS messages in warnsys log.
- cbcfOnly - CBCF Only - Only log CBCF CMAS messages in warnsys log.
- ngapOnly - NGAP Only - Only log NGAP CMAS messages in warnsys log.
- none - Do not log 5G CMAS messages in warnsys log.

Default

all

Name

messageReTransmission - Message Retransmission

Syntax

```
--amfGparmName messageReTransmission --amfGparmValue <string>
```

Description

This command configures maximum number of times WRITE_REPLACE WARNING REQUEST message to re-transmit if no response is received from the gNB. By default the command sets the re-transmission value to 0.

Type / Range

Integer in the range of 0 to 10

Default

0

Name

queuingCbcfRequests - Queuing CBCF Requests

Syntax

--amfGparmName queuingCbcfRequests --amfGparmValue <string>

Description

Parameter controls queuing CBCF requests. By default, it is disabled. When enabled, AMF shall support queuing of up to 20 CBCF requests while processing on-going broadcast.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

stopWarningIndication - Stop Warning Indication

Syntax

--amfGparmName stopWarningIndication --amfGparmValue <string>

Description

This command is used to enable sending of STOP WARNING MESSAGE TO the CBCF.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

warningMessageDelivery - Warning Message Delivery

Syntax

--amfGparmName warningMessageDelivery --amfGparmValue <string>

Description

This command enables/disables Commercial Mobile Alert System (aka warning message delivery) By default, the warning message delivery is disabled

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

writeReplaceWarningIndication - Write Replace Warning Indication

Syntax

--amfGparmName writeReplaceWarningIndication --amfGparmValue <string>

Description

This command is used to enable AMF sending of broadcast completed area list to the CBCF.
By default this is disabled.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

3.31 cmm amfConsumerLocalEndPointConfig

Consumer AMF Local End Point Configuration. Syntax, description, parameters, and roles for cmm amfConsumerLocalEndPointConfig command.

Syntax**Description**

This command is used to create, modify and delete AMF local endpoint configuration for the AMF in the home network and AMFs in the shared network. An endpoint can be created for AMF supported services and endpoints used by an AMF as a client to setup connections to remote AMF and other NF. Separate endpoints can be configured using this command per a NF for the NF supported NSSAI.

Parameters**Name**

endPointName - AMF Local End Point Interface Name

Syntax

--endPointName <string>

Description

Command to create AMF local end point with parameters such as IPv4 address, IPv6 address, TCP well known port for HTTP and HTTPS, HTTP profile and TCP profile.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

amfServiceIpName - Service IP Name

Syntax

--amfServiceIpName <string>

Description

The parameter points to a set of parameters related with IP address configuration created using the 'serviceIp' command.

Class

Required on create.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

sbiEndpointType - SBI Endpoint Type

Syntax

--sbiEndpointType <list-item>

Description

Specifies endpoint configuration is applied to which SBI.

Class

Required on create, not modifiable.

List

- AUSF - AUSF consumer endpoint - N12 Interface - The endpoint configuration is for AMF as a consumer of AUSF services. The reference point N12 is used between AMF and AUSF.
- CHF - CHF consumer endpoint - N41 interface. - The endpoint configuration is for AMF as a consumer of CHF services. The reference point N41 is used between AMF and CHF

- EIR - EIR consumer endpoint - N17 interface - The endpoint configuration is for as a consumer of EIR services. The reference point N17 is used between AMF and PCF.
- GMLC - GMLC consumer endpoint - NLg interface - The endpoint configuration is for AMF as a consumer of GMLC services. The reference point NLg is used between AMF and GMLC.
- LMF - LMF consumer endpoint - NLs interface - The endpoint configuration is for AMF as a consumer of LMF services. The reference point NLs is used between AMF and LMF.
- NRF - NRF consumer endpoint - The endpoint configuration is for AMF as a consumer of NRF services.
- NSSAAF - NSSAAF consumer endpoint - N58 Interface - The endpoint configuration is for AMF as a consumer of NSSAAF services
- NSSF - NSSF consumer endpoint - N22 interface - The endpoint configuration is for AMF as a consumer of NSSF services. The reference point N22 is used between AMF and AUSF.
- NWDAF - NWDAF consumer endpoint - The endpoint configuration is for an AMF as a consumer of NWDAF services.
- PCF - PCF consumer endpoint - N15 interface - The endpoint configuration is for as a consumer of PCF services. The reference point N15 is used between AMF and PCF.
- RemoteAMF - Remote AMF consumer endpoint - The endpoint configuration is for AMF as a consumer of remote AMF services. The reference point N14 is used between two AMF.
- SMF - SMF consumer endpoint - N11 interface. - The endpoint configuration is for AMF as a consumer of SMF services. The reference point N11 is used between AMF and SMF.
- SMSF - SMSF consumer endpoint - N20 interface - The endpoint configuration is for AMF as a consumer of SMSF services. The reference point N20 is used between AMF and SMSF.
- UDM - UDM consumer endpoint - N13 interface. - The endpoint configuration is for AMF as a consumer of UDM services. The reference point N8 is used between AMF and UDM.
- UDSF - UDSF consumer endpoint - The endpoint configuration is for AMF as a consumer of UDSF services.

Name

dscpCode - DSCP Code

Syntax

--dscpCode <list-item>

Description

Parameter to set DSCP code to be used for an IP packet

Class

Optional.

List

- AF11 - Assured forwarding 11.
- AF12 - Assured forwarding 12.
- AF13 - Assured forwarding 13.
- AF21 - Assured forwarding 21.
- AF22 - Assured forwarding 22.
- AF23 - Assured forwarding 23.
- AF31 - Assured forwarding 31.
- AF32 - Assured forwarding 32.
- AF33 - Assured forwarding 33.
- AF41 - Assured forwarding 41.
- AF42 - Assured forwarding 42.
- AF43 - Assured forwarding 43.
- BE - BE.
- CS1 - Class selector 1.
- CS2 - Class selector 2.
- CS3 - Class selector 3.
- CS4 - Class selector 4.
- CS5 - Class selector 5.
- CS6 - Class selector 6.
- CS7 - Class selector 7.
- EF - Expedited forwarding.
- voiceAdmit - Voice Admit

Default

AF41

Name

httpProfileName - HTTP profile name

Syntax

--httpProfileName <string>

Description

Assign HTTP profile for the end point.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\- _:@]*

Default

default

Name

rmtNfSupportedSnssaiListName - Remote NF Supported S-NSSAI List

Syntax

--rmtNfSupportedSnssaiListName <string>

Description

This parameter only applies to other NF and remote AMF. The parameter specifies the NSSAI supported by the remote NF. This parameter enables AMF to setup separate TCP connection for a NSSAI and NF. This is an optional parameter. If the parameter is not set then the endpoint can be used to connect to any NF irrespective of their support of NSSAI. If the parameter is set then the endpoint is used to setup connection with the NF that supports the provisioned NSSAI.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\- _:@]*

Name

tcpProfileName - TCP Profile Name

Syntax

--tcpProfileName <string>

Description

Configure TCP profile to the endpoint.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\- _:@]*

Default

default

Name

tlsProfileName - Transport Layer Security Certificate

Syntax

```
--tlsProfileName <string>
```

Description

Configure TLS certificate used in establishing secure communication.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.32 cmm amfDeregistrationBehaviorMappingProfile

AMFDeregistration Behavior Mapping Profile. Syntax, description, parameters, and roles for cmm amfDeregistrationBehaviorMappingProfile command.

Syntax**Description**

Administers De-registration Behavior Mapping Profile for various scenarios related to UE de-registration. Profile includes parameters for defining cause codes, de-registration types and option to page UE. The maximum number of profiles supported is 128. A default profile, named 'default', is provided.

Parameters**Name**

name - De-registration behavior mapping profile name

Syntax

```
--name <string>
```

Description

Identifies, via name, a given de-registration behavior mapping.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

adminDeleteUeContextCc - Delete UE context cause code

Syntax

--adminDeleteUeContextCc <integer>

Description

NAS Cause code to include in Deregistration Request when UE context is deleted

Class

Optional.

Type / Range

5GMM Cause Code.

Default

10

Name

adminDeleteUeContextDeregistrationType - Delete UE context deregistration type

Syntax

--adminDeleteUeContextDeregistrationType <list-item>

Description

Deregistration type when UE context is deleted via administrative action e.g. cli command

Class

Optional.

List

- registrationNotRequired - Registration not required - Specifies registration type of registration not required
- registrationRequired - Registration required - Specifies registration type of registration required

Default

registrationNotRequired

Name

adminDeleteUeContextNotifyIdleUe - Delete UE context notify idle UE

Syntax

--adminDeleteUeContextNotifyIdleUe <list-item>

Description

Page UE if in the CM-idle when deleting UE context via administrative action e.g. cli command

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

adminDeleteUeContextSendCc - Delete UE context send cause code

Syntax

--adminDeleteUeContextSendCc <list-item>

Description

Include cause code in deregistration request when deregistration reason is due to deleting UE context

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

udmCoreNetworkTypeDeregistration - UDM Core Network Type Deregistration

Syntax

--udmCoreNetworkTypeDeregistration <list-item>

Description

Deregistration type sent to the UE when UDM Notification of UE subscription change is 5GC is not allowed.

Restrictions

Applies to home or roamer UEs.

Class

Optional.

List

- registrationNotRequired - Registration not required - Specifies registration type of registration not required
- registrationRequired - Registration required - Specifies registration type of registration required

Default

registrationNotRequired

Name

udmCoreNetworkTypeDeregistrationCc - UDM Core Network Type Deregistration cause code mapping

Syntax

--udmCoreNetworkTypeDeregistrationCc <integer>

Description

NAS Cause code to include in Deregistration Request when UDM Notification Change of UE subscription is Core Network Type 5GC not allowed.

Restrictions

Applies to home or roamer UEs.

Class

Optional.

Type / Range

5GMM Cause Code.

Default

27

Name

udmCoreNetworkTypeDeregistrationNotifyIdleUE - Core Network Type Deregistration Notify for idle UE

Syntax

--udmCoreNetworkTypeDeregistrationNotifyIdleUE <list-item>

Description

Page UE if in the CM-idle state when UDM Change Notification is 5GC restricted.

Restrictions

Applies to home to roamer UEs.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

udmCoreNetworkTypeDeregistrationSendCc - UDM Core Network Type Deregistration send cause code to UE

Syntax

--udmCoreNetworkTypeDeregistrationSendCc <list-item>

Description

Include cause code in Deregistration Request when UDM Change Notification is Core Network Type 5GC not allowed.

Restrictions

Applies to home or roamer UEs.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

udmForbiddenTacCc - UDM Forbidden TAC Cause Code

Syntax

--udmForbiddenTacCc <integer>

Description

UDM Forbidden TAC Cause Code

Class

Optional.

Type / Range

Integer in the range of 0 to 63

Default

15

Name

udmForbiddenTacDeregistrationType - UDM Forbidden TAC Deregistration Type

Syntax

--udmForbiddenTacDeregistrationType <list-item>

Description

UDM Forbidden TAC Deregistration Type

Class

Optional.

List

- registrationNotRequired - Registration not required - Specifies registration type of registration not required
- registrationRequired - Registration required - Specifies registration type of registration required

Default

registrationRequired

Name

udmForbiddenTacNotifyIdleUe - UDM Forbidden TAC Notify Idle UE

Syntax

--udmForbiddenTacNotifyIdleUe <list-item>

Description

UDM Forbidden TAC Notify Idle UE

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

udmForbiddenTacSendCc - UDM Forbidden TAC Send Cause Code

Syntax

--udmForbiddenTacSendCc <list-item>

Description

UDM Forbidden TAC Send Cause Code

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

udmNrRatRestrictionDeregistration - UDM NR RAT Restriction Deregistration

Syntax

--udmNrRatRestrictionDeregistration <list-item>

Description

Deregistration type sent to the UE when UDM Notification of UE RAT Restriction in NR is to restrict.

Restrictions

Applies to home or roamer UEs.

Class

Optional.

List

- registrationNotRequired - Registration not required - Specifies registration type of registration not required
- registrationRequired - Registration required - Specifies registration type of registration required

Default

registrationNotRequired

Name

udmNrRatRestrictionDeregistrationCc - UDM NR RAT Restriction Deregistration CC

Syntax

--udmNrRatRestrictionDeregistrationCc <integer>

Description

NAS Cause code to include in Deregistration Request when UDM Notification of UE RAT Restriction in NR is to restrict.

Class

Optional.

Type / Range

5GMM Cause Code.

Default

27

Name

udmNrRatRestrictionDeregistrationNotifyIdleUE - UDM NR RAT Restriction Deregistration Notify UE

Syntax

--udmNrRatRestrictionDeregistrationNotifyIdleUE <list-item>

Description

Page UE if in the CM-idle state when UDM Change Notification is NR RAT restricted.

Restrictions

Applies to home or roamer UEs.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

udmNrRatRestrictionDeregistrationSendCc - UDM NR RAT Restriction Deregistration Send Cause Code to UEC

Syntax

--udmNrRatRestrictionDeregistrationSendCc <list-item>

Description

Include cause code in Deregistration Request when UDM Change Notification NR RAT Restriction is to restrict.

Restrictions

Applies to home or roamer UEs.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

udmOdbAllPacketCc - UDM ODB All Packet Services Cause Code

Syntax

--udmOdbAllPacketCc <integer>

Description

NAS Cause code to include in Deregistration Request when both UDM provides ODB All Packet Services Barred and AMF local provisioning parameter on service agreement level for Restrict Mobility on UDM ODB All Packet Barred is set to true.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

10

Name

udmOdbAllPacketDeregistrationType - UDM ODB All Packet Services Deregistration Type

Syntax

--udmOdbAllPacketDeregistrationType <list-item>

Description

Deregistration type when both UDM provides ODB All Packet Services Barred and AMF local provisioning parameter on service agreement level for Restrict Mobility on UDM ODB All Packet Barred is set to true.

Class

Optional.

List

- registrationNotRequired - Registration not required - Specifies registration type of registration not required
- registrationRequired - Registration required - Specifies registration type of registration required

Default

registrationNotRequired

Name

udmOdbAllPacketNotifyIdleUe - UDM ODB All Packet Services Notify Idle UE

Syntax

--udmOdbAllPacketNotifyIdleUe <list-item>

Description

Defines whether to page the UE for deregistering when both UDM provides ODB All Packet Services Barred and AMF local provisioning parameter on service agreement level for Restrict Mobility on UDM ODB All Packet Barred is set to true.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

udmOdbAllPacketSendCc - UDM ODB All Packet Services Send CC

Syntax

--udmOdbAllPacketSendCc <list-item>

Description

Defines whether the AMF includes the cause code in Deregistration request when both UDM provides ODB All Packet Services Barred and AMF local provisioning parameter on service agreement level for Restrict Mobility on UDM ODB All Packet Barred is set to true.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

udmSubWithdrawnCc - Deregistration Notify with Subscription Withdrawn Cause Code

Syntax

--udmSubWithdrawnCc <integer>

Description

NAS Cause code to include in Deregistration Request when deregistration reason in DeregistrationNotify is subscription withdrawn.

Class

Optional.

Type / Range

5GMM Cause Code.

Default

10

Name

udmSubWithdrawnDeregistrationType - Deregistration Notify with reason subscription
withdrawn deregistration type

Syntax

--udmSubWithdrawnDeregistrationType <list-item>

Description

Deregistration type when UDM deregistration notify reason is subscription withdrawn

Class

Optional.

List

- registrationNotRequired - Registration not required - Specifies registration type of registration not required
- registrationRequired - Registration required - Specifies registration type of registration required

Default

registrationNotRequired

Name

udmSubWithdrawnNotifyIdleUe - Deregistration Notify with Subscription Withdrawn notify
idle UE

Syntax

--udmSubWithdrawnNotifyIdleUe <list-item>

Description

Page UE if in the CM-idle when deregistration reason in UDM deregistration notify is
subscription withdrawn.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

udmSubWithdrawnSendCc - deregistration notify subscription withdrawn send cause code to UE

Syntax

--udmSubWithdrawnSendCc <list-item>

Description

Include cause code in deregistration request when deregistration reason in deregistration notify is subscription withdrawn.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.33 cmm amfDirectedUelb

AMF UE Load Balancing. Syntax, description, parameters, and roles for cmm amfDirectedUelb command.

Syntax**Description**

UE loadbalancing for less than 100% is performed using this CLI command that enables the operator to move a percentage or an absolute number of 5G UEs. If maxOffLoadRate is not specified the If

maxOffLoadRate is not specified the off-load happens as fast as possible.. Source GUAMI defaults to ALL supported GUAMIs (Note: when multiple GUAMIs are supported then a subset may be specified). The radioType defaults to NG-3GPP, in future releases it can be non-3GPP, or ALL. The service lock is not required as this command is never used to empty a CMM. There is no concept of quarantine or filters, once the number of UEs are moved the command shutdowns automatically. The command can be stopped/started using the --action parameter

Parameters

Name

uelbAction - action

Syntax

--uelbAction <list-item>

Description

Allows the operator to turn on or off the UELB. when turning on the profile shall be specified.

Class

Required key.

List

- start - starts the UELB
- stop - Stops the UELB

Name

amfDirectedUelbProfileName - Profile Name

Syntax

--amfDirectedUelbProfileName <string>

Description

a profile created with cmm amfDirectedUeLBProfile

Restrictions

profile name shall be entered only when the action is start.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Modify

sysadmin, manage

3.34 cmm amfDirectedUelbProfile

Directed UELB Profile. Syntax, description, parameters, and roles for cmm amfDirectedUelbProfile command.

Syntax

Description

This sets the parameters that will be used in cmm amfDirectedUelb --amfDirectedUelbProfileName --action.

Parameters

Name

name - Profile Name

Syntax

--name <string>

Description

Profile name that was used to invoke the Directed UELB Command

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

targetGuamiPointer - Target GUAMI

Syntax

--targetGuamiPointer <integer>

Description

The GUAMI where the UEs will go.

Class

Required on create.

Type / Range

5G AMF Pointer

Name

endSupi - End of SUPI Range

Syntax

--endSupi <digits>

Description

The end of the SUPI range.

Class

Optional.

Type / Range

Digit string of length 0 or 9 to 15

Default

""

Name

lbAbsolute - Number of UEs

Syntax

--lbAbsolute <integer>

Description

Target number of UEs to redirect to the target GUAMI. If 0, it is not used, and lbPercentage is used

Restrictions

either --lbAbsolute or --lbPercentage are required but not both

Class

Optional.

Type / Range

Integer in the range of 0 to 4,000,000

Default

0

Name

lbPercentage - Percentage of UEs

Syntax

--lbPercentage <integer>

Description

This specifies a percentage of the Connected UEs. The number is translated into a number of UEs based on the number of connected UEs at the time the command is issued. If 0, it is not used, and lbAbsolute is used.

Class

Optional.

Type / Range

Integer in the range of 0 to 100

Default

100

Name

maxOffloadRate - Off-load Rate

Syntax

--maxOffloadRate <integer>

Description

Target Off-load Rate. The rate at which redirections occur.

Class

Optional.

Type / Range

Integer in the range of 10 to 500

Default

500

Name

offloadGuardTimer - Directed UELB Offload Guard Timer

Syntax

--offloadGuardTimer <integer>

Description

This guard timer is for Directed UELB. If the UE target users moved is not met, the offload will end after this time period.

Class

Optional.

Type / Range

Integer in the range of 1 to 24 in units of Hour

Default

1

Name

startSupi - Start SUPI Range

Syntax

--startSupi <digits>

Description

The start of SUPI range

Class

Optional.

Type / Range

Digit string of length 0 or 9 to 15

Default

""

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.35 cmm amfDirectedUelbStats

Directed UELB Statistics. Syntax, description, parameters, and roles for cmm amfDirectedUelbStats command.

Syntax**Description**

Displays the statistics of the directed UELB Command cmm amfDirectedUelb. This command shows one line for each of the previous 5 invocations of the command(only one amfUeLoadBalance command can be active at any time): The stats include: Target, Radio type Number of UEs moved and Target UE number (derived from percentage if needed), duration, offLoadRate achieved and offLoadRate specified.

Parameters**Name**

startDateTime - Start Date and Time

Syntax

--startDateTime <string>

Description

records the date and time at which the command was entered.

Class

Required key.

Type / Range

Date and Time Digit string of length 0 to 20 with the allowed pattern of '20[0-9][0-9]/[0-1][0-9]/[0-3][0-9] [0-2][0-9]:[0-5][0-9]:[0-5][0-9]'

Name

actualOffloadRate - Actual Off-Load Rate

Syntax

--actualOffloadRate <integer>

Description

the achieved rate of redirections.

Class

Read only.

Type / Range

Integer in the range of 0 to 1,000

Name

amfDirectedUelbProfileName - Profile Name

Syntax

--amfDirectedUelbProfileName <string>

Description

A name.

Class

Read only.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

endSupi - End of the SUPI Range

Syntax

--endSupi <digits>

Description

The end of the SUPI range.

Class

Read only.

Type / Range

Digit string of length 0 or 9 to 15

Default

""

Name

lbAbsolute - Number of UEs

Syntax

--lbAbsolute <integer>

Description

Target number of UEs to redirect to the targetGUAMI. If 0, it is not used, and lbPercentage is used

Restrictions

either --lbAbsolute will be computed from --lbPercentage if the latter is the input

Class

Read only.

Type / Range

Integer in the range of 0 to 4,000,000

Name

lbPercentage - Percentage of UEs

Syntax

--lbPercentage <integer>

Description

shows the percentage of UEs requested. If 0, it is not used, and lbAbsolute is used

Class

Read only.

Type / Range

Integer in the range of 0 to 100

Name

maxOffloadRate - Off-load Rate

Syntax

--maxOffloadRate <integer>

Description

Target Off-load Rate. The requested rate for redirections

Class

Read only.

Type / Range

Integer in the range of 10to 500

Default

500

Name

n1MsgRedirectedUes - N1 message Redirected UEs

Syntax

--n1MsgRedirectedUes <integer>

Description

Count of the number of redirected UEs to the target as a result of the UELB Command

Class

Read only.

Type / Range

Integer in the range of 0 to 268,435,455

Name

n2InfoRedirectedUes - N2InfoRedirectedUes

Syntax

--n2InfoRedirectedUes <integer>

Description

Count of the number of redirected UEs to the target as a result of the UELB Command

Class

Read only.

Type / Range

Integer in the range of 0 to 268,435,455

Name

offloadGuardTimer - Directed UELB Offload Guard Timer

Syntax

--offloadGuardTimer <integer>

Description

The requested guard timer is for Directed UELB.

Class

Read only.

Type / Range

Integer in the range of 1 to 24 in units of Hour

Default

1

Name

radioType5g - Radio Type 5G

Syntax

--radioType5g <list-item>

Description

Indicates the radio type that would be used to select UEs for redirection

Class

Read only.

List

- both - selects both NG-3GPP and non-3GPP radios
- ng3gpp - NG-3GPP - 5G NG RAN 3GPP
- non3gpp - non-3GPP - non-3GPP Radios (via N3IWF)
- unknown - The value is unknown

Default

ng3gpp

Name

redirect308UeCount - UEs 308 redirected during UELB Count

Syntax

--redirect308UeCount <integer>

Description

Count of UEs that have been 308 redirected during a UELB operation

Class

Read only.

Type / Range

Integer in the range of 0 to 268,435,455

Name

runTime - Run Time

Syntax

--runTime <string>

Description

Counts in hh:mm:ss the run time of the command

Class

Read only.

Type / Range

Digit string of length 0 to 8 with the allowed pattern of [0-9a-zA-Z\._:-]*

Name

startSupi - Start of the SUPI Range

Syntax

--startSupi <digits>

Description

The start of the SUPI range.

Class

Read only.

Type / Range

Digit string of length 0 or 9 to 15

Default

""

Name

targetGuamiPointer - Target GUAMI

Syntax

--targetGuamiPointer <integer>

Description

The GUAMI where the UEs are redirected

Class

Read only.

Type / Range

5G AMF Pointer

Name

uelbCommandStatus - UELB Command Status

Syntax

--uelbCommandStatus <list-item>

Description

indicates whether the command is stopped, running, completed

Class

Read only.

List

- completed - command completed the load balancing operation as requested
- errorGuardTimer - error guard timer - The requested guard timer expired without completing the task
- errorSetAmfFailure - error set AMF failure - The UELB was stopped due to a failure triggering restoration in the set

- errorTargetAmfFailure - error target AMF failure - the UELB terminated because the target AMF failed or was unreachable
- running - the UELB command is running
- stopped - The UELB command was stopped.

Roles

List

sysadmin, manage, monitor

3.36 cmm amfDnnOverrideProfile

DNN Override Profile. Syntax, description, parameters, and roles for cmm amfDnnOverrideProfile command.

Syntax

Description

This command is used to administer an DNN Override Profile.

Parameters

Name

name - Name

Syntax

--name <string>

Description

Name that identifies Dnn Override profile

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

defaultOverrideDnn - Default DNN

Syntax

--defaultOverrideDnn <string>

Description

Provisioned Default Override DNN

Class

Optional.

Type / Range

Data Network Name with Null String.

Default

""

Name

sendOverrideDnnToUe - Send Overridden DNN to UE

Syntax

--sendOverrideDnnToUe <list-item>

Description

Parameter controls whether to send the corrected/overridden DNN or the UE requested DNN. When the parameter is false, overridden DNN shall be sent to SMF and back to the UE. When the parameter is true, the UE requested DNN is sent to the UE and SMF. If the UE did not send a requested DNN, then the AMF shall send the overridden DNN to the UE and SMF.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

supportDnnOverride - Support DNN Override

Syntax

--supportDnnOverride <list-item>

Description

Parameter controls the feature activation flag. By default the flag shall be disabled. When enabled AMF shall support DNN Override functionality

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

supportDnnOverrideWithWildcardDnn - Support DNN Override with Wildcard DNN

Syntax

--supportDnnOverrideWithWildcardDnn <list-item>

Description

When this parameter is set to true and when the UE has wildcard DNN subscribed, the AMF uses the UE requested DNN to establish the PDU when the AMF is able to discover an SMF for the UE requested DNN. When the AMF is not able to discover an SMF for the UE requested DNN, the AMF applies DNN override functionality, when the DNN override is enabled. When this parameter is set to false, and the UE has wildcard DNN subscribed, the AMF attempts to discover an SMF for the UE requested DNN, but if the SMF discovery fails, the AMF fails the PDU setup procedure.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.37 cmm amfEdrxMicoGParms

AMF eDRX and MICO Global Parameters. Syntax, description, parameters, and roles for cmm amfEdrxMicoGParms command.

Syntax

Description

Global Parameters for eDRX and MICO.

Parameters

Name

amfGparmName - Parameter name

Syntax

--amfGparmName <list-item>

Description

Name of parameter

Class

Required key.

List

- dnnEdrxRegWithoutPduAllowed - DNN eDRX without PDU on Registration Allowed - For a UE NAS registration at the AMF where a DNN list is to be used for allowing eDRX and the UE has not yet set up PDU(s), this field determines whether the AMF is to allow eDRX.
- edrxActive - eDRX Active - Specifies whether eDRX is allowed to be used on all serving PLMNs within the AMF.
- edrxCarryOverPage - eDRX Carryover Page - In the case when a subsequent eDRX page is to be done and there is not sufficient time remaining the PTW, this parameter specifies whether the page attempt is to be carried over to the next PTW for the UE.
- edrxEpochLeapSeconds - eDRX Epoch Leap Seconds - Number of leap seconds since the eDRX epoch time.
- edrxEpochTime - eDRX Epoch Time - eDRX Epoch Time. Valid range is 1980-01-01 00:00:00 through 2037-12-31 23:59:59. The provisioned time must be older than the current system time.
- edrxNewLeapSecondTime - eDRX new leap second time - The data and time when an additional leap second will occur. Valid range is 1980-01-01 00:00:00 through 2037-12-31 23:59:59 or blank.
- edrxPageStartOfPtw - eDRX Page Start of PTW - For a UE using eDRX, this field specifies whether the first page attempt is to wait until the start of the next paging time window. It applies only for HLCOM (pending downlink data) and only if all potential page attempt timeouts would exceed the end of the UE's paging time window.
- edrxPtwEndOffset - eDRX PTW End Offset - Specifies offset (in milliseconds) that the AMF shall consider the UE PTW to end. This parameter is set to account for hyperframe

time synchronization differences between the AMF and RAN, and to account for transmission delays in paging messages sent to the RAN. This offset should be less than or equal to edrxPtwStartOffset.

- **edrxPtwHoldTime** - eDRX PTW Hold Time - Maximum time (in milliseconds) that the AMF will queue a page prior to the start of a UE's paging time window. If a page request is received prior to the hold time, the request is rejected.
- **edrxPtwStartOffset** - eDRX page start offset - Specifies offset (in milliseconds) that the AMF shall consider the UE PTW to begin. This parameter is set to account for hyperframe time synchronization differences between the AMF and RAN, and to account for transmission delays in paging messages sent to the RAN.
- **hlcomActive** - HLCOM Active - Specifies whether extended buffering (HLCOM) for eDRX UEs is active on the AMF for all serving PLMNs. The default is false.
- **hlcomActiveMico** - HLCOM Active MICO - Specifies whether extended buffering (HLCOM) for MICO UEs is active on the AMF for all serving PLMNs. The default is false.
- **hlcomWaitTimeOffset** - HLCOM Wait Time Offset - Specifies additional time to be added for 'Max Waiting Time' sent to an SMF during a requested eDRX UE page attempt.
- **hlcomWaitTimeOffsetMico** - HLCOM Wait Time Offset MICO - Specifies additional time to be added for 'Max Waiting Time' sent to an SMF supporting HLCOM during a requested MICO UE page attempt.
- **reachableEdrxOffset** - Reachable eDRX Offset - Specifies the amount of time in advance of a UE's Paging Time Window Start that the AMF shall send a notification to the UDM for 'Reachable For DL Traffic'. This applies only if Send Reachable for DL Traffic for eDRX is true.
- **sendReachableForDLTrafficForEdrx** - Send Reachable for DL Traffic for eDRX - Feature flag for the AMF to send a pending 'Reachable for DL Traffic' notification to the UDM if a UE using eDRX enters its paging time window.

Name

amfGparmValue - Parameter value

Syntax

--amfGparmValue <string>

Description

Value of parameter

Class

Optional.

Type / Range

Digit string of length 0 to 64

The amfGparmValue string is based on the amfGparmName parameter:

Name

dnnEdrxRegWithoutPduAllowed - DNN eDRX without PDU on Registration Allowed

Syntax

```
--amfGparmName dnnEdrxRegWithoutPduAllowed --amfGparmValue <string>
```

Description

For a UE NAS registration at the AMF where a DNN list is to be used for allowing eDRX and the UE has not yet set up PDU(s), this field determines whether the AMF is to allow eDRX.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

edrxActive - eDRX Active

Syntax

```
--amfGparmName edrxActive --amfGparmValue <string>
```

Description

Specifies whether eDRX is allowed to be used on all serving PLMNs within the AMF.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

edrxCarryOverPage - eDRX Carryover Page

Syntax

```
--amfGparmName edrxCarryOverPage --amfGparmValue <string>
```

Description

In the case when a subsequent eDRX page is to be done and there is not sufficient time remaining the PTW, this parameter specifies whether the page attempt is to be carried over to the next PTW for the UE.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

edrxEpochLeapSeconds - eDRX Epoch Leap Seconds

Syntax

--amfGparmName edrxEpochLeapSeconds --amfGparmValue <string>

Description

Number of leap seconds since the eDRX epoch time.

Type / Range

Integer in the range of 0 to 100

Default

18

Recommended

18

Name

edrxEpochTime - eDRX Epoch Time

Syntax

--amfGparmName edrxEpochTime --amfGparmValue <string>

Description

eDRX Epoch Time. Valid range is 1980-01-01 00:00:00 through 2037-12-31 23:59:59. The provisioned time must be older than the current system time.

Type / Range

Digit string of length 19 to 19 with the allowed pattern of (\d{4}-\d{2}-\d{2} \d{2}:\d{2}:\d{2})

Default

1980-01-06 00:00:00

Recommended

1980-01-06 00:00:00

Name

edrxNewLeapSecondTime - eDRX new leap second time

Syntax

--amfGparmName edrxNewLeapSecondTime --amfGparmValue <string>

Description

The data and time when an additional leap second will occur. Valid range is 1980-01-01 00:00:00 through 2037-12-31 23:59:59 or blank.

Type / Range

Digit string of length 0 or 19 to 19 with the allowed pattern of (\d{4}-\d{2}-\d{2} \d{2}:\d{2}:\d{2})*

Default

"

Recommended

"

Name

edrxPageStartOfPtw - eDRX Page Start of PTW

Syntax

--amfGparmName edrxPageStartOfPtw --amfGparmValue <string>

Description

For a UE using eDRX, this field specifies whether the first page attempt is to wait until the start of the next paging time window. It applies only for HLCOM (pending downlink data) and only if all potential page attempt timeouts would exceed the end of the UE's paging time window.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Recommended

false

Name

edrxPtwEndOffset - eDRX PTW End Offset

Syntax

--amfGparmName edrxPtwEndOffset --amfGparmValue <string>

Description

Specifies offset (in milliseconds) that the AMF shall consider the UE PTW to end. This parameter is set to account for hyperframe time synchronization differences between the AMF and RAN, and to account for transmission delays in paging messages sent to the RAN. This offset should be less than or equal to edrxPtwStartOffset.

Type / Range

Integer in the range of 100 to 5,000

Default

2000

Recommended

2000

Name

edrxPtwHoldTime - eDRX PTW Hold Time

Syntax

--amfGparmName edrxPtwHoldTime --amfGparmValue <string>

Description

Maximum time (in milliseconds) that the AMF will queue a page prior to the start of a UE's paging time window. if a page request is received prior to the hold time, the request is rejected.

Type / Range

Integer in the range of 3,000 to 15,000

Default

10000

Recommended

10000

Name

edrxPtwStartOffset - eDRX page start offset

Syntax

--amfGparmName edrxPtwStartOffset --amfGparmValue <string>

Description

Specifies offset (in milliseconds) that the AMF shall consider the UE PTW to begin. This parameter is set to account for hyperframe time synchronization differences between the AMF and RAN, and to account for transmission delays in paging messages sent to the RAN.

Type / Range

Integer in the range of 100 to 5,000

Default

2000

Recommended

2000

Name

hlcomActive - HLCOM Active

Syntax

--amfGparmName hlcomActive --amfGparmValue <string>

Description

Specifies whether extended buffering (HLCOM) for eDRX UEs is active on the AMF for all serving PLMNs. The default is false.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

hlcomActiveMico - HLCOM Active MICO

Syntax

```
--amfGparmName hlcomActiveMico --amfGparmValue <string>
```

Description

Specifies whether extended buffering (HLCOM) for MICO UEs is active on the AMF for all serving PLMNs. The default is false.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

hlcomWaitTimeOffset - HLCOM Wait Time Offset

Syntax

```
--amfGparmName hlcomWaitTimeOffset --amfGparmValue <string>
```

Description

Specifies additional time to be added for 'Max Waiting Time' sent to an SMF during a requested eDRX UE page attempt.

Type / Range

Integer in the range of 0 to 10,000 in units of Millisecond

Default

0

Recommended

0

Name

hlcomWaitTimeOffsetMico - HLCOM Wait Time Offset MICO

Syntax

```
--amfGparmName hlcomWaitTimeOffsetMico --amfGparmValue <string>
```

Description

Specifies additional time to be added for 'Max Waiting Time' sent to an SMF supporting HLCOM during a requested MICO UE page attempt.

Type / Range

Integer in the range of 0 to 5,000 in units of Millisecond

Default

0

Name

reachableEdrxOffset - Reachable eDRX Offset

Syntax

```
--amfGparmName reachableEdrxOffset --amfGparmValue <string>
```

Description

Specifies the amount of time in advance of a UE's Paging Time Window Start that the AMF shall send a notification to the UDM for 'Reachable For DL Traffic'. This applies only if Send Reachable for DL Traffic for eDRX is true.

Type / Range

Integer in the range of 0 to 5,000 in units of Millisecond

Default

0

Name

sendReachableForDLTrafficForEdrx - Send Reachable for DL Traffic for eDRX

Syntax

```
--amfGparmName sendReachableForDLTrafficForEdrx --amfGparmValue  
<string>
```

Description

Feature flag for the AMF to send a pending 'Reachable for DL Traffic' notification to the UDM if a UE using eDRX enters its paging time window.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

3.38 cmm amfEirGParms

Equipment Identity Register. Syntax, description, parameters, and roles for cmm amfEirGParms command.

Syntax

Description

This command opens the context for configuring equipment identity register (EIR) related parameters for the network.

Parameters

Name

amfGparmName - Parameter name

Syntax

--amfGparmName <list-item>

Description

Name of parameter

Class

Required key.

List

- allowAttachEirChkFail - Allow Attach EIR Check Fail - Parameter 'allowAttachEirChkFail' controls the registration/attach procedure to continue. By default, this flag shall be disabled. When the flag is enabled, Attach/Registration procedure shall be allowed even if the interaction with EIR fails.
- amfAcceptGreyListedUe - Accept Grey Listed UE - The accept greylisted UE command enables acceptance of UEs that are grey-listed in the EIR repository.
- eirCheckingForInitialRegistrationFromInterAmf - EIR Checking for Initial Registration from Inter AMF - This parameter controls whether the EIR checking is performed for Initial Registration from inter AMF for N14 case.. By default this shall be set to true.
- eirCheckingForInitialRegistrationFromIntraAmf - EIR Checking for Initial Registration from Intra AMF - This parameter controls whether the EIR checking is performed for Initial Registration from intra AMF for N14 case. By default this shall be set to true.

- **eirCheckingForInterAmfMobilityRegistration** - EIR Checking For Inter AMF Mobility Registration - This parameter controls whether the EIR checking is performed for inter AMF Mobility registration. By default this shall be set to true. When parameter is set to false, identity check towards EIR won't be performed by AMF.
- **eirCheckingForMobilityRegistrationFromMme** - EIR Checking For Mobility Registration From MME - This parameter controls whether the EIR checking is performed for registration upon mobility across N26 interface.. For emergency registration, the setting of the amfEsrvcBehavior in the emergency profile is used to control if emergency registration is allowed to continue.
- **eirCheckingForRegistrationWithSuci** - EIR Checking for Registration with SUCI - This parameter controls whether the EIR checking is performed for 5G Registration With SUCI. By default this shall be set to true.
- **rejectRegRequest** - Reject Registration Request - Provisioning control to reject registration request, when AMF initiates EIR checking with 5G-EIR during initial registration detects N17 link is disabled.

Name

amfGparmValue - Parameter value

Syntax`--amfGparmValue <string>`**Description**

Value of parameter

Class

Required on create.

Type / Range

Digit string of length 0 to 64

The amfGparmValue string is based on the amfGparmName parameter:

Name

allowAttachEirChkFail - Allow Attach EIR Check Fail

Syntax`--amfGparmName allowAttachEirChkFail --amfGparmValue <string>`**Description**

Parameter 'allowAttachEirChkFail' controls the registration/attach procedure to continue. By default, this flag shall be disabled. When the flag is enabled, Attach/Registration procedure shall be allowed even if the interaction with EIR fails.

List

- **false** - Boolean value of false.
- **true** - Boolean value of true.

Default

false

Name

amfAcceptGreyListedUe - Accept Grey Listed UE

Syntax

```
--amfGparmName amfAcceptGreyListedUe --amfGparmValue <string>
```

Description

The accept greylisted UE command enables acceptance of UEs that are grey-listed in the EIR repository.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

eirCheckingForInitialRegistrationFromInterAmf - EIR Checking for Initial Registration from Inter AMF

Syntax

```
--amfGparmName eirCheckingForInitialRegistrationFromInterAmf --  
amfGparmValue <string>
```

Description

This parameter controls whether the EIR checking is performed for Initial Registration from inter AMF for N14 case.. By default this shall be set to true.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

eirCheckingForInitialRegistrationFromIntraAmf - EIR Checking for Initial Registration from Intra AMF

Syntax

```
--amfGparmName eirCheckingForInitialRegistrationFromIntraAmf --  
amfGparmValue <string>
```

Description

This parameter controls whether the EIR checking is performed for Initial Registration from intra AMF for N14 case. By default this shall be set to true.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

eirCheckingForInterAmfMobilityRegistration - EIR Checking For Inter AMF Mobility Registration

Syntax

```
--amfGparmName eirCheckingForInterAmfMobilityRegistration --  
amfGparmValue <string>
```

Description

This parameter controls whether the EIR checking is performed for inter AMF Mobility registration. By default this shall be set to true. When parameter is set to false, identity check towards EIR won't be performed by AMF.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

eirCheckingForMobilityRegistrationFromMme - EIR Checking For Mobility Registration From MME

Syntax

```
--amfGparmName eirCheckingForMobilityRegistrationFromMme --  
amfGparmValue <string>
```

Description

This parameter controls whether the EIR checking is performed for registration upon mobility across N26 interface.. For emergency registration, the setting of the amfEsrvcBehavior in the emergency profile is used to control if emergency registration is allowed to continue.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

eirCheckingForRegistrationWithSuci - EIR Checking for Registration with SUCI

Syntax

```
--amfGparmName eirCheckingForRegistrationWithSuci --amfGparmValue  
<string>
```

Description

This parameter controls whether the EIR checking is performed for 5G Registration With SUCI. By default this shall be set to true.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

rejectRegRequest - Reject Registration Request

Syntax

```
--amfGparmName rejectRegRequest --amfGparmValue <string>
```

Description

Provisioning control to reject registration request, when AMF initiates EIR checking with 5G-EIR during initial registration detects N17 link is disabled.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

3.39 cmm amfEirOverrideProfile

AMF EIR Override Profile. Syntax, description, parameters, and roles for cmm amfEirOverrideProfile command.

Syntax

Description

This command opens the context for configuring Equipment Identity Register (EIR) related parameters profile to allow override of AMF EIR global parameters at the UE PLMN Services or SUPI range level.

Parameters

Name

name - EIR Override Profile Name

Syntax

--name <string>

Description

Identifies profile name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

acceptTrackListedUe - Accept Tracklisted UE

Syntax

--acceptTrackListedUe <list-item>

Description

Enables acceptance of UE that are track-listed in the EIR repository.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

allowRegistrationEirCheckFail - Allow Registration EIR Check Failed

Syntax

--allowRegistrationEirCheckFail <list-item>

Description

This parameter controls whether the Registration procedure is allowed to continue if AMF receives a failure error from EIR during EIR check. For emergency registration, the setting of the amfEsrvcBehavior in the emergency profile is used to control if emergency registration is allowed to continue.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

eirCheckingForInitialRegistrationFromInterAmf - EIR Checking for Initial Registration from Inter AMF

Syntax

--eirCheckingForInitialRegistrationFromInterAmf <list-item>

Description

This parameter controls whether the EIR checking is performed for Initial Registration from inter AMF for N14 case. By default this shall be set to false.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

eirCheckingForInitialRegistrationFromMme - EIR Checking for Initial Registration from MME

Syntax

--eirCheckingForInitialRegistrationFromMme <list-item>

Description

This parameter controls whether the EIR checking is performed for Initial Registration from MME (N26) By default this shall be set to false.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

eirCheckingForInterAmfMobilityRegistration - EIR Checking for Inter AMF Mobility Registration

Syntax

```
--eirCheckingForInterAmfMobilityRegistration <list-item>
```

Description

This parameter controls whether the EIR checking is performed for Inter AMF Mobility Registration (N14 and N2). By default this shall be set to false

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

eirCheckingForMobilityRegistrationFromMme - EIR Checking for Mobility Registration from MME

Syntax

```
--eirCheckingForMobilityRegistrationFromMme <list-item>
```

Description

This parameter controls whether the EIR checking is performed for registration upon mobility across N26 interface.. For emergency registration, the setting of the amfEsrvBehavior in the emergency profile is used to control if emergency registration is allowed to continue.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

eirCheckingForRegistrationWithSuci - EIR Checking for Registration with SUCI

Syntax

--eirCheckingForRegistrationWithSuci <list-item>

Description

This parameter controls whether the EIR checking is performed for 5G Registration With SUCI. By default this shall be set to false.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

interAmfMobilityRegistrationPercentage - Inter AMF Mobility Registration Percentage

Syntax

--interAmfMobilityRegistrationPercentage <integer>

Description

Inter AMF Mobility Registration percentage to EIR.

Class

Optional.

Type / Range

Integer in the range of 0 to 100

Default

0

Name

mobilityRegistrationFromMmePercentage - Mobility Registration From MME Percentage

Syntax

--mobilityRegistrationFromMmePercentage <integer>

Description

Mobility Registration from MME Percentage to EIR

Class

Optional.

Type / Range

Integer in the range of 0 to 100

Default

0

Name

rejectRegistrationRequest - Reject Registration Reject

Syntax

--rejectRegistrationRequest <list-item>

Description

This parameter controls whether the Registration procedure is allowed to continue if AMF detects transport error such as N17 link(s) is disabled or EIR check attempts timeout. For emergency registration, the setting of the amfEsrvBehavior in the emergency profile is used to control if emergency registration is allowed to continue.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.40 cmm amfEmergencyNum

Emergency Number List. Syntax, description, parameters, and roles for cmm amfEmergencyNum command.

Syntax

Description

This command is used to administer an Emergency Number List and includes parameters for defining emergency numbers for ambulance, fire brigade, marine guard, mountain rescue and police. Maximum list of up to 6 Emergency Numbers shall be supported.

Parameters

Name

amfEmergencyNumListName - Name

Syntax

--amfEmergencyNumListName <string>

Description

Defines the name of amf emergency number list table

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

emergencyDigits - Emergency Digits

Syntax

--emergencyDigits <string>

Description

Emergency number. Only 6 unique amfEmergencyNum records can be associated with the same amfEmergencyNumList name.

Class

Required key.

Type / Range

Digit string of length 1 to 10 with the allowed pattern of [0-9][0-9A-Ca-c*#]*

Name

ambulance - Ambulance

Syntax

--ambulance <list-item>

Description

If emergency number digits apply to ambulance.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

fireBrigade - Fire brigade

Syntax

--fireBrigade <list-item>

Description

If emergency number digits apply to firebrigade.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

marineGuard - Marine guard

Syntax

--marineGuard <list-item>

Description

If emergency number digits apply to marineguard.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

mountainRescue - Mountain rescue

Syntax

--mountainRescue <list-item>

Description

If emergency number digits apply to mountainrescue.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

police - Police

Syntax

--police <list-item>

Description

If emergency number digits apply to police.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.41 cmm amfEmergencyNumList

Emergency Number List Table. Syntax, description, parameters, and roles for cmm amfEmergencyNumList command.

Syntax

Description

This command is used to administer an Emergency Number List Table.

Parameters

Name

name - Name

Syntax

--name <string>

Description

Defines the name of emergency number list table.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.42 cmm amfEmergencyProfile

AMF Emergency Profile. Syntax, description, parameters, and roles for cmm amfEmergencyProfile command.

Syntax

Description

This command is used to administer an Emergency Profile. AMF allows creation of up to 10 emergency profiles. An emergency profile can be assigned to an AMF home PLMN (for 3GPP access) and/or to a non-3GPP access TA.

Parameters

Name

name - amf Emergency Profile Name

Syntax

--name <string>

Description

Name that identifies amf emergency profile

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

sliceServiceType - Slice Service Type

Syntax

--sliceServiceType <integer>

Description

The S-NSSAI consists of a Slice Service Type (SST) and a Slice Differentiator (SD). Each slice must have SST and SD is optional. SST is mandatory for an sNssai, as such it shall be always provisioned in emergency profile.

Class

Required on create.

Type / Range

5G Slice Service Type

Name

ambrDl - AMBR downlink

Syntax

--ambrDl <integer>

Description

AMBR downlink bandwidth in bits per second

Class

Optional.

Type / Range

Integer in the range of 0 to 4,294,967,295

Default

4096000

Name

ambrUl - AMBR uplink

Syntax

--ambrUl <integer>

Description

AMBR uplink bandwidth in bits per second

Class

Optional.

Type / Range

Integer in the range of 0 to 4,294,967,295

Default

4096000

Name

amfEmerFallback - Emergency Fallback

Syntax

--amfEmerFallback <list-item>

Description

Emergency fall back services shall be provided to a UE to establish emergency call. Not applicable for non-3GPP access TA.

Class

Optional.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Name

amfEmerFallbackNumListIncl - AMF Emergency Fallback Number List Inclusion

Syntax

--amfEmerFallbackNumListIncl <list-item>

Description

Defines whether the AMF includes the UE the Emergency Number List as part of (normal i.e. not emergency) Registration Accept, when the AMF indicates Emergency service support indicator for 3GPP access (EMC) flag as not supported, and Emergency services fallback indicator for 3GPP access (EMF) as supported. Not applicable for non-3GPP access TA.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

amfEmergencyNumListName - Amf Emergency number list table name

Syntax

--amfEmergencyNumListName <string>

Description

Defines the amf emergency number list table name

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

amfEsrvBehavior - Service behavior

Syntax

--amfEsrvBehavior <list-item>

Description

AMF behavior in providing emergency service session support.

Class

Optional.

List

- All UEs are allowed - Specifies AMF emergency support to all UEs are allowed.
- Only UEs that are authenticated are allowed - Specifies AMF emergency support to only UEs that are authenticated are allowed.
- UE SUPI required, authentication optional - Specifies AMF emergency support to UEs SUPI required, authentication optional.
- Valid UEs only - Specifies AMF emergency support to valid UEs only.

Default

All UEs are allowed

Name

amfEsrvcBlackListedUe - Service for black listed UE

Syntax

--amfEsrvcBlackListedUe <list-item>

Description

Emergency services shall be provided to a UE with an PEI that is black listed in the 5G-EIR.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

amfEsrvcT3512Timer - Emergency T3512 Timer In Minutes

Syntax

--amfEsrvcT3512Timer <integer>

Description

Emergency T3512 timer in minutes. If it is 0, this timer is disabled and t3512 timer from AMF NAS Timers table is used. Not applicable for non-3GPP access TA.

Class

Optional.

Type / Range

Integer in the range of 0 to 186

Default

0

Name

dnnNi - DNN (NI)

Syntax

--dnnNi <string>

Description

DNN is a reference to a data network.

Class

Optional.

Type / Range

Digit string of length 1 to 62 with the allowed pattern of [0-9a-zA-Z\.\-]*

Default

sos

Name

indicateEmergencyHandoverAfter5gSrvcc - Indicate Emergency Handover after 5G-SRVCC

Syntax

--indicateEmergencyHandoverAfter5gSrvcc <list-item>

Description

Defines whether the AMF indicates the emergency call handover towards the GMLC after an emergency call has been handed over to UTRAN by 5G-SRVCC, and additionally whether the AMF supports and includes the MSC server address on it. Not applicable for non-3GPP access TA.

Class

Optional.

List

- NoLocationContinuity - No Location Continuity - AMF does not trigger Namf_Location_EventNotify towards the GMLC after 5G-SRVCC handover. (Note: This value should be used when the MSC server is configured to update the GMLC after the 5G-SRVCC completion.
- WithMscServerAddr - With MSC Server Address - AMF triggers Namf_Location_EventNotify towards the GMLC to indicate the emergency call handover, whenever AMF has previously triggered the LCS procedure when the emergency session was started (according to LCS settings). AMF supports receiving the MSC

server address from MME_SRVCC over N26 and includes the MSC server address on Namf_Location_EventNotify towards the GMLC, if received from MME.

- WithoutMscServerAddr - Without MSC Server Address - AMF triggers Namf_Location_EventNotify towards the GMLC to indicate the emergency call handover, whenever AMF has previously triggered the LCS procedure when the emergency session was started (according to LCS settings). AMF does not support receiving the MSC server address from MME_SRVCC over N26 and does not include the MSC server address on Namf_Location_EventNotify towards the GMLC.

Default

WithoutMscServerAddr

Name

networkInducedLcsBehavior - NI-LR Behavior

Syntax

--networkInducedLcsBehavior <list-item>

Description

AMF NI-LR behaviors: E911 without NI-LR, E911 with NI-LR without LMF or E911 with NI-LR and LMF. Not supported on the current product release for non-3GPP access TAI.

Class

Optional.

List

- LMFNI-LR - NI-LR With LMF Contact - NI-LR with LMF contact
- NoLMFNI-LR - NI-LR Without LMF Contact - NI-LR without LMF contact
- NoNI-LR - Do Not Perform NI-LR - Do not perform NI-LR

Default

NoNI-LR

Name

sliceDifferentiator - Slice Differentiator

Syntax

--sliceDifferentiator <string>

Description

The S-NSSAI consists of a Slice Service Type (SST) and a Slice Differentiator (SD). Each slice must have SST and SD is optional. A SST may have multiple SDs.

Class

Optional.

Type / Range

Identifies Slice Differentiator (SD)

Default

FFFFFF

Name

smfEmergencySelection - SMF Emergency Selection

Syntax`--smfEmergencySelection <list-item>`**Description**

The parameter specifies the method of SMF selection used by the AMF for Emergency services. If populated it shall override smf selection in uePlmn services.

Class

Optional.

List

- LocalProvisioning - Local Provisioning - AMF will use locally provisioned NF for the selection of a NF.
- NRF - Network Repository Function - Make use of NRF to select NF.
- NRFLocalProvisioning - NRF with Local Provisioning Override - The AMF will check the locally provisioned selection rules and if there is a match then the AMF will select the NF matching the rule. If there is no matching then AMF will use the NRF. The selection rule by default is that AMF will select a NF supporting a provisioned criterion for the NF.
- NoProvisioning - No Provisioning - Use smfSelection from amfUePlmnServices

Default

NoProvisioning

Name

snssaiName - SNSSAI Name

Syntax`--snssaiName <string>`**Description**

SNSSAI Name

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.43 cmm amfEplmn

Equivalent PLMN . Syntax, description, parameters, and roles for `cmm amfEplmn` command.

Syntax

Description

Configures an existing network as an equivalent network and includes parameter attributes for defining the serving and equivalent network names.

Parameters

Name

amfEplmnListName - Equivalent PLMN List Name

Syntax

--amfEplmnListName <string>

Description

Identifies the Equivalent PLMN list name.

Restrictions

EPLMN List must not include the Served PLMN at list creation or modification.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of `[0-9a-zA-Z\.\-_\:@]*`

Name

amfEquivalentPlmnName - AMF Equivalent PLMN Name

Syntax

--amfEquivalentPlmnName <string>

Description

Identifies the name of the EPLMN which can be Home, Shared or Other PLMN. There can be up to 15 EPLMNs in the list.

Restrictions

EPLMN List must not include the Served PLMN at list creation or modification.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.44 cmm amfEplmnList

EPLMN List. Syntax, description, parameters, and roles for cmm amfEplmnList command.

Syntax**Description**

EPLMN List

Parameters**Name**

name - EPLMN List name

Syntax

--name <string>

Description

Identifies the name of the EPLMN List

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

description - Description of Eplmn List

Syntax

--description <string>

Description

Describes the use of the EPLMN list e.g. for UE or gNB.

Class

Optional.

Type / Range

Digit string of length 0 to 160 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.45 cmm amfEpsCipheringAlgorithm

Preferred EPS Ciphering Algorithm in AMF . Syntax, description, parameters, and roles for cmm amfEpsCipheringAlgorithm command.

Syntax

Description

This command sets EPS algorithms, in priority order, AMF will use it to negotiate encryption of NAS messages with the UE. At UE REGISTRATION, the UE provides, in the UE security capabilities, which encryption algorithms in EPS are supported. The AMF will use the algorithm both supported by the UE and highest in priority in the eea ciphering algorithm priority list. - An algorithm must be chosen for 'priority-1' and 'priority-2'. - An additional selection, 'none', can be selected for 'priority-3' and 'priority-4' - effectively truncating the list. - If any ciphering algorithm (including null) is selected for 'priority-4', then 'priority-3' must not be 'none'. - An algorithm cannot be used for more than one priority (except for 'none')

Parameters

Name

epsCipheringAlgPrio1 - Priority 1 EPS ciphering algorithm

Syntax

--epsCipheringAlgPrio1 <list-item>

Description

Most preferred EPS ciphering algorithm

Class

Optional.

List

- aes - 128-EEA2 (AES) - Specify ciphering of 128-EEA2 (AES).
- noCiphering - 128-EEA0 (No ciphering) - Specify ciphering of 128-EEA0 (No ciphering).
- none - No algorithm selected - Specify ciphering of no algorithm selected.
- snow3G - 128-EEA1 (SNOW 3G) - Specify ciphering of 128-EEA1 (SNOW 3G).
- zuc - 128-EEA3 (ZUC) - Specify ciphering of 128-EEA3 (ZUC).

Default

snow3G

Name

epsCipheringAlgPrio2 - Second most preferred EPS ciphering algorithm

Syntax

--epsCipheringAlgPrio2 <list-item>

Description

Second most preferred EPS ciphering algorithm; used if the user element does not support algorithm(s) of higher priority

Class

Optional.

List

- aes - 128-EEA2 (AES) - Specify ciphering of 128-EEA2 (AES).
- noCiphering - 128-EEA0 (No ciphering) - Specify ciphering of 128-EEA0 (No ciphering).
- none - No algorithm selected - Specify ciphering of no algorithm selected.
- snow3G - 128-EEA1 (SNOW 3G) - Specify ciphering of 128-EEA1 (SNOW 3G).
- zuc - 128-EEA3 (ZUC) - Specify ciphering of 128-EEA3 (ZUC).

Default

aes

Name

epsCipheringAlgPrio3 - Third most preferred ciphering algorithm

Syntax

--epsCipheringAlgPrio3 <list-item>

Description

Third most preferred ciphering algorithm; used if the user element does not support algorithm(s) of higher priority

Class

Optional.

List

- aes - 128-EEA2 (AES) - Specify ciphering of 128-EEA2 (AES).
- noCiphering - 128-EEA0 (No ciphering) - Specify ciphering of 128-EEA0 (No ciphering).
- none - No algorithm selected - Specify ciphering of no algorithm selected.
- snow3G - 128-EEA1 (SNOW 3G) - Specify ciphering of 128-EEA1 (SNOW 3G).
- zuc - 128-EEA3 (ZUC) - Specify ciphering of 128-EEA3 (ZUC).

Default

zuc

Name

epsCipheringAlgPrio4 - Fourth most preferred ciphering algorithm

Syntax

--epsCipheringAlgPrio4 <list-item>

Description

Fourth most preferred ciphering algorithm; used if the user element does not support algorithm(s) of higher priority

Class

Optional.

List

- aes - 128-EEA2 (AES) - Specify ciphering of 128-EEA2 (AES).
- noCiphering - 128-EEA0 (No ciphering) - Specify ciphering of 128-EEA0 (No ciphering).
- none - No algorithm selected - Specify ciphering of no algorithm selected.
- snow3G - 128-EEA1 (SNOW 3G) - Specify ciphering of 128-EEA1 (SNOW 3G).
- zuc - 128-EEA3 (ZUC) - Specify ciphering of 128-EEA3 (ZUC).

Default

none

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

3.46 cmm amfEpsIntegrityAlgorithm

Preferred Integrity EPS Algorithm in AMF . Syntax, description, parameters, and roles for cmm amfEpsIntegrityAlgorithm command.

Syntax**Description**

This command sets the algorithms, in priority order, AMF will use to negotiate integrity of NAS messages with the UE. At REGISTRATION, the UE provides, in the UE security capabilities, which EPS integrity algorithms are supported. The AMF will use the algorithm both supported by the UE and highest in priority in the eia algorithm priority list. - An algorithm must be chosen for 'priority-1' and 'priority-2'. - An additional selection, 'none', can be selected for 'priority-3' and 'priority-4' - effectively truncating the list. - If any integrity algorithm (including null) is selected for 'priority-4', then 'priority-3' must not be 'none'. - An algorithm cannot be used for more than one priority (except for 'none').

Parameters**Name**

epsIntegrityAlgPrio1 - Priority 1 EPS Integrity algorithm

Syntax

--epsIntegrityAlgPrio1 <list-item>

Description

Most preferred EPS Integrity algorithm

Class

Optional.

List

- aes - 128-EIA2 (AES) - Specify integrity algorithm of 128-EIA2 (AES).
- none - No algorithm selected - Specify integrity algorithm of no algorithm selected.
- snow3G - 128-EIA1 (SNOW 3G) - Specify integrity algorithm of 128-EIA1 (SNOW 3G).
- zuc - 128-EIA3 (ZUC) - Specify integrity algorithm of 128-EIA3 (ZUC).

Default

snow3G

Name

epsIntegrityAlgPrio2 - Priority 2 EPS integrity algorithm

Syntax

--epsIntegrityAlgPrio2 <list-item>

Description

Second most preferred integrity algorithm; used if the user element does not support algorithm(s) of higher priority

Class

Optional.

List

- aes - 128-EIA2 (AES) - Specify integrity algorithm of 128-EIA2 (AES).
- none - No algorithm selected - Specify integrity algorithm of no algorithm selected.
- snow3G - 128-EIA1 (SNOW 3G) - Specify integrity algorithm of 128-EIA1 (SNOW 3G).
- zuc - 128-EIA3 (ZUC) - Specify integrity algorithm of 128-EIA3 (ZUC).

Default

aes

Name

epsIntegrityAlgPrio3 - Third most preferred EPS integrity algorithm

Syntax

--epsIntegrityAlgPrio3 <list-item>

Description

Third most preferred integrity algorithm; used if the user element does not support algorithm(s) of higher priority

Class

Optional.

List

- aes - 128-EIA2 (AES) - Specify integrity algorithm of 128-EIA2 (AES).
- none - No algorithm selected - Specify integrity algorithm of no algorithm selected.
- snow3G - 128-EIA1 (SNOW 3G) - Specify integrity algorithm of 128-EIA1 (SNOW 3G).
- zuc - 128-EIA3 (ZUC) - Specify integrity algorithm of 128-EIA3 (ZUC).

Default

zuc

Name

epsIntegrityAlgPrio4 - Priority 4 EPS integrity algorithm

Syntax

--epsIntegrityAlgPrio4 <list-item>

Description

Priority 4 EPS integrity algorithm

Class

Optional.

List

- aes - 128-EIA2 (AES) - Specify integrity algorithm of 128-EIA2 (AES).
- none - No algorithm selected - Specify integrity algorithm of no algorithm selected.
- snow3G - 128-EIA1 (SNOW 3G) - Specify integrity algorithm of 128-EIA1 (SNOW 3G).
- zuc - 128-EIA3 (ZUC) - Specify integrity algorithm of 128-EIA3 (ZUC).

Default

none

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

3.47 cmm amfEquivalentPlmn

AMF Equivalent PLMN. Syntax, description, parameters, and roles for cmm amfEquivalentPlmn command.

Syntax

Description

Configures an Equivalent PLMN network, and includes parameters for defining the Equivalent PLMN network name(s) and the mapping to the Served PLMN(s) Home or Shared. This table allows the provisioning of AMF TAI List for Equivalent PLMN(s).

Parameters

Name

servedPlmnName - Served PLMN Name

Syntax

--servedPlmnName <string>

Description

Plmn Name that identifies the serving network Home or Shared.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

amfEquivalentPlmnName - Equivalent PLMN Name

Syntax

--amfEquivalentPlmnName <string>

Description

Identifies the Equivalent Plmn Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.48 cmm amfEventExposureGParms

AMF Event Exposure Global Parameters. Syntax, description, parameters, and roles for cmm amfEventExposureGParms command.

Syntax

Description

This command is used to enter AMF Event Exposure global parameters.

Parameters

Name

amfGparmName - Parameter name

Syntax

--amfGparmName <list-item>

Description

Name of parameter

Class

Required key.

List

- maxExpiry - Maximum Expiry - Specifies the Maximum Expiration time for a subscribed Event. If an event subscription request has an expiration timer greater than the maximum expiry, the AMF shall set the event expiry to the provisioned value.
- uesInAreaSubscriptionAllowed - UEs In Area Subscription Allowed - Specifies whether the AMF is allowed to accept a subscription to the UEs in Area Event

Name

amfGparmValue - Parameter value

Syntax

--amfGparmValue <string>

Description

Value of parameter

Class

Required on create.

Type / Range

Digit string of length 0 to 64

The amfGparmValue string is based on the amfGparmName parameter:

Name

maxExpiry - Maximum Expiry

Syntax

```
--amfGparmName maxExpiry --amfGparmValue <string>
```

Description

Specifies the Maximum Expiration time for a subscribed Event. If an event subscription request has an expiration timer greater than the maximum expiry, the AMF shall set the event expiry to the provisioned value.

Type / Range

Integer in the range of 10 to 90 in units of Hour

Default

10

Name

uesInAreaSubscriptionAllowed - UEs In Area Subscription Allowed

Syntax

```
--amfGparmName uesInAreaSubscriptionAllowed --amfGparmValue <string>
```

Description

Specifies whether the AMF is allowed to accept a subscription to the UEs in Area Event

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

3.49 cmm amfEventProfile

AMF Event Profile. Syntax, description, parameters, and roles for `cmm amfEventProfile` command.

Syntax**Description**

Profile defines which AMF Event Exposure capabilities are activated at the AMF. A maximum of 16 profiles is supported. The 'default' profile always exists and cannot be deleted.

Parameters**Name**

name - name

Syntax

--name <string>

Description

profile name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

communicationFailure - Communication Failure

Syntax

--communicationFailure <list-item>

Description

Communication Failure Event Feature

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

connectivityStatus - Connectivity Status

Syntax

--connectivityStatus <list-item>

Description

Connectivity Status Event Active

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

groupAllowed - Group Allowed

Syntax

--groupAllowed <list-item>

Description

Specifies whether groupID events are allowed on the AMF

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

location - Location

Syntax

--location <list-item>

Description

Location Reporting Event

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

lossOfConnectivity - Loss Of Connectivity

Syntax

--lossOfConnectivity <list-item>

Description

Loss of Connectivity for the profile

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

maxAvailTimeSupported - Max Avail Time Supported

Syntax

--maxAvailTimeSupported <list-item>

Description

Specifies whether the AMF can include Maximum Availability Time in a UE Reachability Notification for downlink data for a UE currently assigned eDRX active time.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Recommended

false

Name

pageIdleUeForCurrentLocation - Page IDLE UE for Current Location

Syntax

--pageIdleUeForCurrentLocation <list-item>

Description

Defines the AMF behavior for an CM-IDLE UE when the subscribed event type is LOCATION_REPORT and immediateFlag is false or absent. The parameter defines whether the AMF pages the UE to retrieve the current location or, if no paging is done, whether the AMF waits for the UE to establish the signalling connection and only after that the AMF reports the current location towards the service consumer. This setting applies for ONE_TIME reports and for CONTINUOUS reports the first report.

Class

Optional.

List

- AmfNoPaging - AMF No Paging - AMF does not page a CM-IDLE UE for current location.
- AmfPageAlways - AMF Page Always - AMF pages the CM-IDLE UE always no matter the requested location.
- AmfPageForCellOnly - AMF Page For Cell Only - AMF pages the CM-IDLE UE for the cell level only.

Default

AmfNoPaging

Name

presenceInAoI - Presence in Area of Interest

Syntax

--presenceInAoI <list-item>

Description

Presence in Area of Interest.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

reachability - Reachability

Syntax

--reachability <list-item>

Description

UE Reachability Event

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

registrationState - Registration State

Syntax

--registrationState <list-item>

Description

Registration State Event Active

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

typeAllocCode - Type Allocation Code

Syntax

--typeAllocCode <list-item>

Description

Specifies whether the AMF supports the Type Allocation Code Event

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

userState - User State

Syntax

--userState <list-item>

Description

User State Event Report Feature

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.50 cmm amfExtendedEmergencyNum

AMF Extended Emergency Number. Syntax, description, parameters, and roles for cmm amfExtendedEmergencyNum command.

Syntax

Description

This command is used to administer an Extended Emergency Number List Table Maximum of 25 Extended Emergency Number entries can be supported per Extended Emergency Number List.

Parameters

Name

extendedEmergencyNumListName - Extended Emergency Number List Name

Syntax

--extendedEmergencyNumListName <string>

Description

Defines the extended emergency number list table name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

extendedEmergencyDigits - Extended Emergency Digits

Syntax

--extendedEmergencyDigits <string>

Description

Maximum of 25 items can be in a list.

Class

Required key.

Type / Range

Digit string of length 1 to 10 with the allowed pattern of [0-9][0-9A-Ca-c*#]*

Name

subServiceUrn - Sub-service Unified Resource Name

Syntax

--subServiceUrn <string>

Description

Name of the emergency service. If the associated emergency service URN is 'urn:service:sos.gas', there is only one sub-service provided in the sub-services field which is 'gas'. Following are some of the examples of URN services: 'urn:service:sos.ambulance', 'urn:service:sos.police', 'urn:service:sos.fire', 'urn:service:sos.marine', 'urn:service:sos.mountain', 'urn:service:sos.ecall.manual', 'urn:service:sos.ecall.automatic'. If the UE can determine the type of emergency service, the

UE shall use an emergency service URN with a sub-service type corresponding to the type of emergency service. There is no need to enter the prefix: urn:service:

Class

Required key.

Type / Range

Emergency Service Name

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.51 cmm amfExtendedEmergencyNumList

AMF Extended Emergency Number List. Syntax, description, parameters, and roles for cmm amfExtendedEmergencyNumList command.

Syntax**Description**

This command is used to administer an Extended Emergency Number List. Maximum of 10 extended emergency number list shall be supported.

Parameters**Name**

name - Extended Emergency Number List Name

Syntax

--name <string>

Description

Defines the extended emergency number list table name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

extendedEmergencyNumberListValidity - Extended Emergency Number List Validity

Syntax

--extendedEmergencyNumberListValidity <list-item>

Description

Parameter specifies whether extended emergency number list validity is valid in the country or the PLMN from which this IE is received.

Class

Optional.

List

- country - Country Extended Emergency Number List Validity - Extended Local Emergency Numbers List is valid in the country of the PLMN from which this IE is received
- plmn - PLMN Extended Emergency Number List Validity - Extended Local Emergency Numbers List is valid only in the PLMN from which this IE is received.

Default

country

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.52 cmm amfFreqCountMonitoring

AMF Frequency Count Monitoring. Syntax, description, parameters, and roles for cmm amfFreqCountMonitoring command.

Syntax

Description

Manage AMF actions related to frequency count monitoring functionality. The command supports three tasks. A task is created for each action. The Create Deny-listed UE File task dumps/exports currently deny-listed UEs into created file. The Delete Behaviour Data task deletes/purges all behavior contexts i.e. clears any deny-listing related information from AMF internal storage. After performing this, deny-listed UEs can again access the AMF (normal subscriber data is maintained). The Deny-listed UE Count task outputs the number of deny-listed UEs.

Parameters

Name

frequencyMonitoringTask - Frequency Monitoring Task

Syntax

--frequencyMonitoringTask <list-item>

Description

Defines frequency count monitoring task to be executed.

Class

Required key.

List

- denyListedUeCount - Deny-listed UE Count - Request deny-listed UE count.
- createDenyListedUeFile - Create Deny-listed UE File - Request to create a file with deny-listed UEs.
- deleteBehaviourData - Delete Behaviour Data - Request to delete all AMF stored behaviour data (for the UEs that have previously attempted to access the AMF, deny-listed or not yet deny-listed).

Name

denyListedUeCount - Deny-listed UE Count

Syntax

--denyListedUeCount <integer>

Description

The number of deny-listed UEs.

Class

Read only.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

freqMonitoringStatus - Frequency Monitoring Status

Syntax

--freqMonitoringStatus <string>

Description

Shows current status of task.

Class

Read only.

Type / Range

Digit string of length 0 to 64

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

3.53 cmm amfFreqCountMonitoringProfile

AMF Frequency Count Monitoring Profile. Syntax, description, parameters, and roles for cmm amfFreqCountMonitoringProfile command.

Syntax**Description**

This profile defines parameters for frequency count analysis. AMF considers UE attempts for defined procedure(s) abnormally frequent if the number of attempts exceed the provisioned value during a provisioned time period. AMF supports max 128 profiles.

Parameters**Name**

name - Name

Syntax

--name <string>

Description

The name of the AMF frequency count monitoring profile.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

allowed3gppRegistrationAttempts - Allowed 3GPP Registration Attempts

Syntax

--allowed3gppRegistrationAttempts <integer>

Description

Defines the allowed number of NAS Registration attempts over 3GPP access for a single UE within the provisioned time interval before UE is deny-listed.

Class

Optional.

Type / Range

Integer in the range of 1 to 100

Default

5

Name

allowed3gppServiceRequestAttempts - Allowed 3GPP Service Request Attempts

Syntax

--allowed3gppServiceRequestAttempts <integer>

Description

Defines the allowed number of NAS Service Request attempts over 3GPP access for a single UE within the provisioned time interval before UE is deny-listed.

Class

Optional.

Type / Range

Integer in the range of 1 to 100

Default

5

Name

allowed3gppUINasSmAttempts - Allowed 3GPP UL NAS SM Attempts

Syntax

--allowed3gppUINasSmAttempts <integer>

Description

Defines the allowed number of NAS UL NAS Transport attempts over 3GPP access for a single UE, where the Payload Container Type indicates N1 SM information, within the provisioned time interval before UE is deny-listed.

Class

Optional.

Type / Range

Integer in the range of 1 to 100

Default

5

Name

allowed3gppUlNasSmsAttempts - Allowed 3GPP UL NAS SMS Attempts

Syntax

--allowed3gppUlNasSmsAttempts <integer>

Description

Defines the allowed number of NAS UL NAS Transport attempts over 3GPP access for a single UE, where the Payload Container Type indicates SMS, within the provisioned time interval before UE is deny-listed.

Class

Optional.

Type / Range

Integer in the range of 1 to 100

Default

10

Name

allowedNon3gppRegistrationAttempts - Allowed Non-3GPP Registration Attempts

Syntax

--allowedNon3gppRegistrationAttempts <integer>

Description

Defines the allowed number of NAS Registration attempts over non-3GPP access for a single UE within the provisioned time interval before UE is deny-listed.

Class

Optional.

Type / Range

Integer in the range of 1 to 100

Default

5

Name

allowedNon3gppServiceRequestAttempts - Allowed Non-3GPP Service Request Attempts

Syntax

--allowedNon3gppServiceRequestAttempts <integer>

Description

Defines the allowed number of NAS Service Request attempts over non-3GPP access for a single UE within the provisioned time interval before UE is deny-listed.

Class

Optional.

Type / Range

Integer in the range of 1 to 100

Default

5

Name

allowedNon3gppUINasSmAttempts - Allowed Non-3GPP UL NAS SM Attempts

Syntax

--allowedNon3gppUINasSmAttempts <integer>

Description

Defines the allowed number of NAS UL NAS Transport attempts over non-3GPP access for a single UE, where the Payload Container Type indicates N1 SM information, within the provisioned time interval before UE is deny-listed.

Class

Optional.

Type / Range

Integer in the range of 1 to 100

Default

5

Name

allowedNon3gppUINasSmsAttempts - Allowed Non-3GPP UL NAS SMS Attempts

Syntax

--allowedNon3gppUINasSmsAttempts <integer>

Description

Defines the allowed number of NAS UL NAS Transport attempts over non-3GPP access for a single UE, where the Payload Container Type indicates SMS, within the provisioned time interval before UE is deny-listed.

Class

Optional.

Type / Range

Integer in the range of 1 to 100

Default

10

Name

calculateTotalNumberOfAttempts - Calculate Total Number of Attempts

Syntax

--calculateTotalNumberOfAttempts <list-item>

Description

Defines whether the AMF calculates a total number of all monitored procedure attempts per the monitoring period. AMF only calculates into the total number of attempts for the procedures that are provisioned as monitored. When set to true, the AMF performs actions if the complete number of all monitored procedures exceeds the total number of allowed attempts per monitoring period. When set to false, only exceeding the procedure specific attempt(s) causes AMF action.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

initialRegistrationDeregistrationCause - Initial Registration and Deregistration cause

Syntax

--initialRegistrationDeregistrationCause <integer>

Description

Defines the 5GMM cause code to be used on Registration Rejects for Registration Type Initial Registration, and Deregistration Requests from the AMF, when triggered due to frequency count monitoring functionality.

Class

Optional.

Type / Range

5GMM Cause Code.

Default

7

Name

mobilityPeriodicRegistrationServiceRequestCause - Mobility and Periodic Registration Update and Service Request Cause

Syntax

--mobilityPeriodicRegistrationServiceRequestCause <integer>

Description

Defines the 5GMM cause code used on Mobility and periodic registration update procedures or Service request, when the AMF rejects the procedure due to UE procedure frequency count monitoring functionality.

Class

Optional.

Type / Range

5GMM Cause Code that leads on Mobility and Periodic Registration Update the UE into any 5GMM-DEREGISTERED state.

Default

7

Name

monitor3gppRegistrations - Monitor 3GPP Registrations

Syntax

--monitor3gppRegistrations <list-item>

Description

Defines whether the AMF monitors frequency of NAS Registration procedure attempts over 3GPP access.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

monitor3gppServiceRequests - Monitor 3GPP Service Requests

Syntax

--monitor3gppServiceRequests <list-item>

Description

Defines whether the AMF monitors the frequency of NAS Service requests over 3GPP access.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

monitor3gppUINasSm - Monitor 3GPP Uplink NAS SM

Syntax

--monitor3gppUINasSm <list-item>

Description

Defines whether the AMF monitors the frequency of NAS UL NAS transports with payload container type indicating N1 SM information over 3GPP access.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

monitor3gppUINasSms - Monitor 3GPP Uplink NAS SMS

Syntax

--monitor3gppUINasSms <list-item>

Description

Defines whether the AMF monitors the frequency of NAS UL NAS transports with payload container type indicating SMS over 3GPP access.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

monitorNon3gppRegistrations - Monitor Non-3GPP Registrations

Syntax

--monitorNon3gppRegistrations <list-item>

Description

Defines whether the AMF monitors the frequency of NAS Registrations over non-3GPP access.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

monitorNon3gppServiceRequests - Monitor Non-3GPP Service Requests

Syntax

--monitorNon3gppServiceRequests <list-item>

Description

Defines whether the AMF monitors NAS Service requests over non-3GPP access.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

monitorNon3gppULNasSm - Monitor Non-3GPP UL NAS SM

Syntax

--monitorNon3gppULNasSm <list-item>

Description

Defines whether the AMF monitors separately the frequency of NAS UL NAS transports with payload container type indicating N1 SM information over non-3GPP access.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

monitorNon3gppUlnasSms - Monitor Non-3GPP Uplink NAS SMS

Syntax

--monitorNon3gppUlnasSms <list-item>

Description

Defines whether the AMF monitors the frequency of NAS UL NAS transports with payload container type indicating SMS over non-3GPP access.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

timeInterval - Time Interval

Syntax

--timeInterval <integer>

Description

Defines the monitoring interval within which the AMF monitors the number of procedure attempts from a UE.

Class

Optional.

Type / Range

Integer in the range of 1 to 3,600 in units of Second

Default

60

Name

totalAllowedAttempts - Total Allowed Attempts

Syntax

--totalAllowedAttempts <integer>

Description

Defines the total allowed number of all monitored procedure attempts over any access type for a single UE within the provisioned time interval before UE is deny-listed. AMF only calculates into the total number of attempts the attempts for procedures that are provisioned as monitored. This parameter is applied when Calculate Total Number of Attempts is set to true.

Class

Optional.

Type / Range

Integer in the range of 1 to 100

Default

10

Name

ulNasCause - UL NAS Cause

Syntax

--ulNasCause <integer>

Description

Defines the 5GMM cause code used when a NAS UL NAS message is sent back to the UE.

Class

Optional.

Type / Range

5GMM Cause Code.

Default

90

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.54 cmm amfGParms

5G Global Parameters. Syntax, description, parameters, and roles for cmm amfGParms command.

Syntax

Description

AMF global parameters.

Parameters

Name

amfGparmName - Parameter name

Syntax

--amfGparmName <list-item>

Description

Name of parameter.

Class

Required key.

List

- 5g4gRetPrefPlmn - 5g4g Return Preferred PLMN - This parameter enables indicating to serving NG-RAN the last visited EPC PLMN as preferred indication to be used in future 5GS to EPS mobility.
- 5gRanTotalConfigIncompatibilityAlarm - 5G RAN Total Configuration Incompatibility Alarm - When this parameter is enabled, AMF will raise an Alarm when receiving RAN Configuration Update or there is local configuration change (that may or not lead to sending AMF Configuration Update) that results in total configuration incompatibility between the AMF and the RAN.
- acceptVoiceNoCheckResp - Accept Voice No Check Response - Controls whether the AMF allows indicating IMS voice over PS sessions supported towards the UE and the UDM, when UE Radio Capability Check procedure is provisioned to be run towards the NG-RAN from the AMF but the NG-RAN provides no response for NGAP UE Radio Capability Check Request message.

- **accessTypeChangePcfTrigger** - Access Type Change PCF Trigger - Enables the AMF support for Access Type changes triggered from PCF. By default, flag shall be disabled.
- **alwaysSendDnnWithOperatorIdentifier** - Always Send DNN with DNN Operator Identifier - Defines whether AMF will always include DNN-OI in DNNs sent as SBI JSON Body IEs and Query Parameters. When the value is true, the AMF will always include the DNN-OI in DNNs. When value is false, the DNN-OI may not be always present as the AMF transparently conveys the DNN in the non roaming scenarios.
- **amfEirSupport** - AMF EIR Support - Activation of the AMF EIR feature on the AMF. By default this shall be disabled
- **amfHomogeneousSupportTais** - AMF Homogeneous Support TAI - Defines the TAI the AMF considers for IMS voice over PS homogeneous support under the serving AMF. The other access TAI is considered to support all voice options when considered for the other access IMS voice over PS homogeneous support. (Non-3GPP TAI is considered to support VoPS with VoNR and VoLTE Handover and Redirect.)
- **amfN26DirectFwd** - Indicates whether direct data forwarding is enabled or disabled on n26 handover.
- **amfN26IndirectFwd** - Indicates whether indirect data forwarding is enabled or disabled on n26 handover.
- **amfNpli** - AMF NPLI - Enables the support for Network Provided Location Info (NPLI) i.e the Namf_Location_ProvideLocationInfo service for the AMF.
- **amfRanCombiningOption** - AMF RAN Combining Option - Controls how the AMF establishes the user plane sessions during Service Request or Mobility Registration Update for multiple PDU sessions. AMF may either combine/aggregate all SMF responses and formulate a single NGAP message, or formulate separate NGAP messages per each SMF request.
- **amfRequestPsCellInLrc** - AMF Request PS Cell in LRC - When location reporting control procedure is used in AMF, if AMF Request PS Cell in LRC is set, then AMF includes additional location information IE and set to Include PSCell. gNB will include the current PSCell in the report if Dual Connectivity is activated. If AMF Request PS Cell in LRC is not set, AMF will not request additional location information under location reporting procedure as existing function, then gNB will not report PSCell in location report.
- **amfSmsSubscrGet** - AMF SMS Subscription Get - Controls whether the AMF always attempts to get the SMS related subscription data from the UDM, or whether the AMF follows UE indication of support for SMS over NAS when retrieving the SMS related subscription data from the UDM (i.e. gets the subscription data on need only).
- **amfTaiDiscovery** - AMF TAI Discovery - Specifies whether the AMF TAI auto-discovery feature is active. Default is disable.
- **amfTriggeredRrcInactivePaging** - AMF Triggered RRC Inactive Paging - When set to true, the AMF triggers paging of UE(s) in RRC Inactive state but not reachable without network triggered messaging.

- **applProtocolFallbackN26ToS10** - Application Protocol Fallback from N26 to S10 - Current parameter enables fallback to search for 's10' candidate when DNS Query to find target MME during S10 handover using 'n26' fails. This parameter is expected to be used together with parameter Use N26 Service Parameter in 5GS to EPS Handover
- **apra** - Additional Presence Reporting Area - Enables the use of the Additional Presence Reporting Area (APRA) feature on the Namf_EventExposure service. When this parameter is set to true, and when also the service consumer indicates support for the APRA feature, and the Presence Reporting Area (PRA) set is internally provisioned in the AMF, and the service consumer requests reporting for the PRA set, the AMF reports the PRA set id in the 'prald' field and the individual PRA id(s) in the 'additionalPrald' field(s) on the presenceInfo of the areaList (an array of AmfEventAreas) of a reportList (an array of AmfEventReports).
- **asynchrTypeComm** - Asynchronous Type Communication - This parameter defines whether the AMF attempts to contact a CM-IDLE UE when AMF receives signalling message for the UE, e.g. SMF triggers N1/N2 message transfer for PDU session modification. When enabled, the AMF attempts to reach the UE via paging (if the UE is currently considered reachable and would be expected to respond to paging). When disabled, the AMF sends the signalling message to the UE when the UE makes the next contact towards the network.
- **autoAddTaiToTaiList** - Auto Add TAI to TAI List - Auto Add TAI to TAI list, with allowed values are: Off and Update Three.
- **cagSupport** - CAG Support - This parameter enables Closed Access Group feature at the AMF
- **checkTgtMmeN26StatusN26Ho** - Check Target MME N26 Status in N26 Handover - Current parameter enables checking the N26 link/interface status of local target MME node during N26 handover so that if it isn't available to use, then even if the local MME can serve the target RAN, source AMF will select an alternative node to complete the N26 handover.
- **checkUesUsageSetting** - Check UE's Usage Setting - Controls whether the AMF requires UE's Usage Setting IE to be included in NAS Registration Request to indicate IMS voice over PS sessions supported towards the UE and the UDM. If set to false, IMS voice over PS sessions support indication does not depend on whether the UE includes UE's Usage Setting IE. If set to true, AMF indicates IMS voice over PS sessions supported only if UE's Usage Setting IE is included in NAS Registration Request.
- **contactOverNon3gppFor3gpp** - Contact Over Non-3GPP for 3GPP - Defines whether the AMF contacts a UE over non-3GPP access, when the UE is CM-CONNECTED over non-3GPP access and CM-IDLE over 3GPP access, and the AMF gets a 3GPP access related trigger for the UE.
- **cr129ts29500ErrorToken** - CR 129 TS 29.500 Error Token - This parameter controls the activation of CR 0129 of TS29.500. When this parameter is set to true: Removes

condition the access without an OAuth2.0 access token failed for the realm attribute
Adds error code invalid_token

- **defaultHttpPort** - Default HTTP Port - Default port to be used for HTTP connections when it is not explicitly given as part of the the authority of a SBI URI. This parameter applies only to HTTP connections. HTTPS connections always use 443 as the default port (if not explicitly given otherwise in the URI)
- **defErrorCcSnpn** - Default Error Cause for SNPN network - This parameter defines the default error cause to send for UE request towards SNPN network when Access Restriction NAS Mapping profile isn't available to use
- **delayUdmDuringRegistration** - Delay UDM During Registration - When the same AMF still holds the UE data internally after previous N26 mobility to EPS, this parameter enables AMF to successfully complete the registration procedure and suppress the UDM dialogs (Nudm_UECM_Registration, Nudm_SDM_Get and/or Nudm_SDM_Subscribe, depending on other settings) for the enabled UEs. The UDM dialogs will be re-attempted automatically according to provisioning. Note: this functionality only is effective when the AMF global parameter Keep VLR Entry is set to true.
- **deletePduAfterFailedUpAttemptsInIndirectComm** - Delete PDU After Failed User Plane Attempts in Indirect Communication - Enables the possibility to delete a PDU from AMF after provisioned number of failed user plane re-establishment attempts when indirect communication is deployed. When this parameter is set to true, the AMF counts the number of times the AMF receives (after possible SCP reselection(s)) a 504 failure from SCP into the internal counter for failed user plane re-establishment attempts. Once the AMF detects 'nth' time the user plane re-establishment fails (for either direct or indirect attempt towards the SMF), the AMF internally deletes the PDU and indicates towards the UE that the PDU is inactive. The AMF applies the value of the AMF global parameter Failed User Plane Attempts to Delete PDU for the number 'n' above. This parameter is only effective when AMF global parameter Failed User Plane Attempts to Delete PDU is set to a non-zero value. When this parameter is set to false, the AMF does not count 504 response from SCP as SMF not reachable during user plane re-establishment attempts and does not delete PDU(s) due to those.
- **directNasMsgTransfer** - Direct NAS message transfer to the target AMF - If the parameters is set to 'True' then AMF forwards the NAS Registration message to the target AMF over N14 interface if the AMF can determine the target AMF. If the parameters is set to 'False' then AMF forwards the NAS Registration message via 5G RAN.
- **dualFailVnflpds** - Dual Fail IPDS - When enabled, a dual failure of a spread IPDS pool will trigger IPDS pool 0 to dual fail. This is specifically for the two large IPDS configuration in VNF or CNF. It applies to any application type. This dual failure of IPDS pool 0 will result in system outage of the system. When disabled, the propagation of the dual failure will not occur.
- **eaea** - EBI and ARP mapping update in EBI Assignment - Enables supporting EAEA feature

- **ebiAssignReqReject** - EBI Assignment Request Reject - When set to true, the AMF rejects SMF EBI assignment request with an http2 status code associated with the ongoing procedure at the time of the received request, TEMPORARY_REJECT_HANDOVER_ONGOING or TEMPORARY_REJECT_REGISTRATION_ONGOING
- **emcNon3gppN3iwf** - EMC Non-3GPP Access N3IWF - Enables the support for emergency services over untrusted non-3GPP access via N3IWF.
- **emergencyEpslwkWoN26** - Emergency EPS Interworking Without N26 - Enables the possibility to handover an emergency session from EPS to 5GS (3GPP access) without N26 interface on the AMF. Session continuity is ensured by getting the (EPS stored) emergency information from HSS+UDM as part of the subscription profile. Applied when EPS Interworking Without N26 is set to true.
- **emergencyNon3gppEpsPduHo** - Emergency Non-3GPP Access EPS PDU Handover - Enables the AMF support for UE triggered emergency PDU handovers between untrusted non-3GPP via N3IWF/5GS and E-UTRAN/EPS.
- **enhancedPcfSelection3gppAccess** - Enhanced PCF Selection 3GPP Access - When this parameter is enabled, AMF will support the following functionality for 3GPP access:
a) Reselect PCF upon change of Serving PLMN if required. b) Exclude PCFs that do not support both UE and AM Policy Control Services when an association for both Policies is required. c) Reselect PCF during inter-AMF UE mobility if the PCF Instance provided by the old AMF does not meet all the selection criterions. d) Fallback to selection of PCF without considering Slice support, when no valid candidate PCF that serves all slices in the Allowed NSSAI is discovered.
- **epdgEmergencyHo** - ePDG Emergency Handover - This parameter enables session continuity of an emergency session between ePDG/EPC and NG-RAN/5GC.
- **epdgHo** - ePDG Handover - Enables handover between non-3GPP access connected to EPC and 3GPP access (NG-RAN) connected to 5GC
- **epsFallbackNoPageUe** - Page UE during EPS Fallback - Enables suppression of UE paging when gNB releases UE context due to EPS Fallback and EpsFallbackNoPageTimer is running.
- **epslwkWithoutN26** - EPS Interworking Without N26 - Enables the EPS interworking without N26 interface support on the AMF.
- **epsTo5gsHoWithAmfReselect** - EPS To 5GS Handover With AMF Reselect - This parameter enables the capability of AMF reselection during EPS to 5GS handover. It is evaluated at both the initial AMF and target AMF of an EPS to 5GS Handover with AMF reselection procedure
- **epsUpip** - EPS UPIP - Current parameter enables support for the EPS-UPIP (User Plane Integrity Protection)

- **ethernetPduSupportN26** - Ethernet PDU Support N26 - Parameter controls AMF indicating in Context Request if Ethernet PDU is supported for EPS to 5GS idle mobility scenario. By default, the flag shall be disabled.
- **excludeRoutingIndWithSupi** - Exclude Routing Indicator with SUPI - When enabled, the AMF, when SUPI is available, will use SUPI and exclude routing indicator as selection criteria in the query to the NRF to select AUSF and UDM.
- **existingPduNoSmf** - Existing PDU No SMF - Controls whether the AMF sends the SM message back to the UE or drops it, if the AMF receives an SM message with Request Type 'existing PDU session', the AMF does not have a routing context for the PDU session and no SMF can be found on the UDM subscription data for the PDU session id or the DNN.
- **extendedEmergencyNumListAreaCode** - Extended Emergency Number List Area Code - Indicates provisioned extended emergency list area code. When feature activation gParm is enabled and upon receiving AreaCode in UE Subscription data from UDM, if received Area Code matches provisioned area code for this feature, AMF shall send Extended Emergency Numbers List in the REGISTRATION ACCEPT message.
- **failedUserPlaneAttemptsToDeletePdu** - Failed User Plane Attempts to Delete PDU - Defines how many failed UE triggered user plane re-establishment attempts, where the AMF fails to communicate with the SMF due to link issues, cause the AMF to internally delete the requested PDU and to indicate towards the UE that the corresponding requested PDU is inactive. The value 0 stands for the functionality to be disabled. When disabled, the AMF does not delete the PDU due to link related SMF communication failures during user plane re-establishment attempts.
- **freqCountMonitoring** - Frequency Count Monitoring - Defines whether the AMF monitors the UE attempts for different UE triggered procedures, and if too frequent attempts made by the UE during the provisioned monitoring period, treat the UE accordingly, e.g. drop subsequent procedure requests or deregister the UE.
- **getSmfSelSubDataEmptySnssaiInfo** - Get Smf Selection Subscription Data when having empty SnssaiInfo - Enable fetching SMF Selection Subscription Data during PDU Session Establishment (type = initial request) when local UE context indicates that the UE is supposed to have PDU session service subscribed but local subscription data for SMF selection is empty
- **gnbRerouteFallback** - GNB Reroute NAS Fallback - AMF re-routes NAS message to gNB when N14 is not available during UE inter-AMF re-location when source AMF can't service the UE slice.
- **iabEnabled** - IAB Enabled - This gParm enables Integrated Access and Backhauling (IAB) operation at the AMF
- **icsrAloneVoiceRadioCap** - ICSR Alone Voice Radio Capability - Controls whether AMF sends NGAP Initial Context Setup Request alone without NAS Registration Accept, when either -AMF does not hold UE radio capability but needs it for purposes listed further below, and/or -AMF needs to run NGAP UE radio capability check procedure to decide

IMS voice over PS support for the UE. If UE radio capabilities are needed on the AMF (Reject On VoNR Tracking Area and/or EMC Suppress No VoNR and/or IMS VoPS S1 Disabled is true under AMF Voice Profile for the UE on 5G UE PLMN Services), or UE radio capability check procedure result is required to decide for the IMS voice over PS result for the UE, the AMF needs to send NGAP Initial Context Setup Request alone first (without any NAS message) in order to let NG-RAN establish AS security before requesting radio capabilities from the UE. When set to true and when Reject On VoNR Tracking Area and/or EMC Suppress No VoNR and/or IMS VoPS S1 Disabled is true under AMF Voice Profile for the UE on 5G UE PLMN Services, AMF sends NGAP UE Radio Capability Check Request only after receiving Initial Context Setup Response.

- **ignoreAusfFailuresOptionalReAuthentications** - Ignore AUSF Failures on Optional Re-Authentications - This parameter, when set, enables AMF to ignore AUSF dialog related failures during optional re-authentications. The UEs to apply the functionality to are provisioned on the AMF. The functionality is applied when the AMF is locally provisioned to perform an optional re-authentication for a UE that is already registered on the AMF, but the re-authentication fails during the AUSF dialog and the AMF is able to continue serving the UE without the re-authentication.
- **imdRptThirdPartySub** - Immediate Report to Third Party Subscription - Controls whether the AMF supports the 'IERSR' feature negotiation when a client subscribes to an Event at the AMF and requests an immediate report as part of that subscription. When the gparm is set and the client subscription request an immediate report, the AMF optionally can send the first event report in the subscription response, depending on whether the first report is ready by the time the AMF must respond to the subscription. See TS 29.518 CR 0646.
- **imsVopsHomogeneousSupport** - IMS VoPS Homogeneous Support - This parameter controls whether or not the AMF includes the IMS VoPS attribute towards the UDM as part of the Amf3gppAccessRegistration attribute.
- **inhibitInterAmfN2HoSameTa** - Inhibit Inter AMF N2 Handover Source and Target gNBs in same TA - When enabled, source AMF inhibits N2 HO to target AMF if source AMF has no N2 link to target gNB and UE current location (TAC) and the target gNB TAC is the same. Source AMF responds back to source gNB with Handover Preparation Failure message.
- **initialContextSetupTrigger** - Send Initial Context - Controls when the AMF sends NGAP Initial Context Setup Request towards the NG-RAN.
- **interSystemAndEnDcConfTransferOverN26** - Inter-System and EN-DC SON Configuration Transfer Over N26 - Parameter controls feature activation GPARM flag. By default, flag shall be disabled. When enabled, AMF/MME shall transfer inter-system SON configuration information and EN-DC Configuration Transfer between a gNB and eNodeB.
- **intraAmfInterPlmnUdmModRegistration** - Intra AMF Inter PLMN UDM Registration Modification - When set to true and UE moves between Serving PLMNs within the same AMF, the AMF will modify GUAMI IE to the target Serving PLMN GUAMI, within the UE Context Management information in UDM, for given UE.

- **intraAmfInterPlmnUdmModSubscription** - Intra AMF Inter PLMN UDM Subscription Modification - When set to true and UE moves between Serving PLMN within the same AMF, the AMF will modify the PLMN ID to the target Serving PLMN for given UE.
- **ismf** - Intermediate SMF - Enables Support of deployments topologies with specific SMF Service Areas, i.e. the Intermediate SMF (I-SMF) support in the AMF. An Intermediate SMF may be inserted on the signalling path between AMF and the SMF, when the SMF is not able to control the UPF(s) because the UPF(s) belong to a different SMF Service Area. The I-SMF may also be changed or removed due to this during procedures. The I-SMF concept is introduced in 3GPP Release 16.
- **ismfSelectionFailureMaintain** - I-SMF Selection Failure Maintain - Defines whether the AMF maintains PDU session in a situation when 3GPP Support of deployments topologies with specific SMF Service Areas i.e. I-SMF is deployed, and AMF detects the need to select a new I-SMF, but no I-SMF supporting the current UE area is available. If set to 'false', AMF releases the PDU session (i.e. triggers Nsmf_PDUSession_UpdateSMContext with release indication towards the old SMF or I-SMF) if no new I-SMF is available for the current area of the UE.
- **keepUeContextAfterN26Ho** - Keep VLR Entry - When this parameter is set to true, keep VLR entry when UDM DeregistrationReason is 5GS_TO_EPS_MOBILITY
- **ladn** - Local Area Data Network - Local Area Data Network. This parameter activates the LADN functionality support in AMF. With the LADN functionality, access to a provisioned data network is only provided on a restricted service area. The LADN DNN and the associated LADN Service Area (the Tracking Area list) also need to be provisioned.
- **ladnNotifications** - LADN Notifications - Controls whether the AMF triggers Event Notifications on LADN state towards the service consumer.
- **linkAlarmCustomize** - AMF Link Alarm Customize - If the global parameter is set to true, customized severity levels in the AMF Link Alarm Config are used to set the alarm severity for links and interfaces. If the global parameter is set to false, customization of link alarm severity is not supported.
- **locationEvents** - Location Events - Enables the Namf_Event Exposure in the AMF for subscribing to UE location reporting events (direct or continuous location reporting).
- **maskedImeIsV** - Masked IMEISV - Include Masked IMEISV in NGAP messages
- **maxAmfUeRadioCapLength** - Maximum AMF Ue Radio Capability Length - Maximum Ue Radio Capability Length Supported on the AMF. Default is 9216. Setting this value to 24576 enables storage of Radio capability length between 9216 - 24576 bytes in the RCIS if RCIS is available
- **maxNumberOfForbiddenTacs** - Maximum Number of Forbidden TACs - AMF shall support maximum number of 600 forbidden TACs with the default value of 64.
- **mcs** - Mission Critical Services - Enables the support for Mission Critical Services (MCS) on the AMF.

- **memoryRestrictionForbiddenTac** - Memory Restriction Forbidden TAC - Parameter to control CPPS VLR memory allocation for feature f20430-02. **maxNumberOfForbiddenTacs** can be set to larger size (i.e 600) only when this gParm is 'true' A memory restricted gparm that can only be changed, with aggregate service lock. default value is false. **memoryRestrictionForbiddenTac** cannot be changed in CMM24.7 release. It will be deprecated in CMM24.11
- **mps** - Multimedia Priority Services - Enables the support for Multimedia Priority Services (MPS) on the AMF.
- **mrlDINasTransp** - Mobility Restriction List on DL NAS Transport - Controls whether the AMF will only include Mobility Restriction List on DL NAS transport if the list context has been changed, or include even if the content has not changed from previous content.
- **mrlSplmnOnlySending** - Mobility Restriction List Serving PLMN only sending - Controls whether the AMF includes Mobility Restriction List on NGAP messages (Initial Context Setup Request, Handover Request and DL NAS Transport) with Serving PLMN as the only content, or only if also restrictions are to be communicated to the NG-RAN. (Note: DL NAS transport with only serving PLMN included is only applicable if Mobility Restriction List is sent in every DL NAS Transport.
- **multiLevelMps** - Multi-Level MPS - Enables the use of 5 different MPS levels with dedicated SBI Message Priority value for each MPS level. Applies when AMF global parameter Multimedia Priority Services is set to true.
- **multiSimWithSameMsisdnSupport** - Multiple SIM with Same MSISDN Support - Parameter controls feature activation. By default, parameter shall be disabled. When enabled, AMF shall support and allow more than one SUPI to register to the AMF by using the same MSISDN. With the default value setting of false, the AMF shall work as it had before the introduction of this feature.
- **n26iwkSuppt** - N26 interworking Supported - Enables the support for EPS interworking with N26 interface in the AMF
- **n2HoWithAmfReselect** - N2 Handover with AMF Reselection - This parameter enables the capability of target AMF reselection during inter-AMF N2 handover. The reselection is performed by the initial target AMF when it is not able to serve all requested S-NSSAIs
- **n3gppInitialCtxtNas** - Non-3GPP NAS Response with Initial Context - Controls whether AMF sends NAS response on top of the NGAP Initial Context Setup Request for non-3GPP access. When set to false (default), AMF sends NAS response alone via DL NAS Transport after the Initial Context Setup has been completed. When set to true, AMF sends NAS response on top of the NGAP Initial Context Setup Request.
- **ngap3gppRelease** - NGAP asn.1 3GPP Release - Identifies supported NGAP ASN.1 3GPP Releases.
- **non3gppAccessImsVopsHomogeneous** - Non-3GPP Access N3IWF IMS VoPS Homogeneous - This parameter controls whether or not the AMF applies UECM Update service operation towards UDM due to imsVoPs homogeneous supports change for a UE that is registered over untrusted non-3GPP access via N3IWF

- **non3gppAccessN3iwf** - Non-3GPP access N3IWF - Controls the support of non-3GPP access and the Non-3GPP Access Interworking Function (N3IWF) in the AMF.
- **non3gppAccessN3iwfRatType** - AMF Non-3GPP Access N3IWF RAT Types - Defines the RAT type the AMF supports for UEs connected via untrusted non-3GPP access and N3IWF to the AMF.
- **non3gppAccessVoiceN3iwf** - Non-3GPP Access Voice N3IWF - Enables support for IMS Voice over PS sessions over untrusted non-3GPP access via N3IWF.
- **non3gppEpsPduHo** - Non-3GPP Access EPS PDU Handover - Enables the AMF support for UE triggered non-emergency PDU handovers between untrusted non-3GPP via N3IWF/5GS and E-UTRAN/EPS.
- **non3gppNpliN3iwf** - Non-3GPP NPLI N3IWF - Enables the AMF support for Network Provided Location Information (NPLI) for untrusted non-3GPP access via N3IWF.
- **nonIpPdnSupportN26** - Non IP PDN Support N26 - Parameter controls AMF indicating in Context Request if Non IP PDN is supported for EPS to 5GS idle mobility scenario. By default, the flag shall be enabled.
- **nrDualConnectivity** - NR Dual Connectivity - Enables NR Dual Connectivity NR-DC and Secondary RAT Usage Reporting. The NR Dual Connectivity profile, when configured, administers NR-DC support, Secondary RAT Usage Reporting restriction and UDM NR Secondary RAT Access restriction override per UE PLMN Services or SUPI Range Services.
- **nrfDefaultLoadValueForNfs** - Default Load Value for NFs - Determines default 'load' value to be assumed for a Service instance when the 'load' information is not received from NRF. Load is used by the AMF along with Locality, Priority and Capacity in order to select/load balance traffic among multiple NF Service Instances that meet the given criteria.
- **nrfHeartbeatFailures** - NRF Heartbeat Failures - This parameter is used to provision maximum number of consecutive heartbeat failures to NRF. If AMF does not receive responses to heartbeat messages consecutively specified by this parameter then, AMF stops heartbeat messages and tries to re-register to this NRF. If registration is successful, AMF again tries heartbeat messages. If registration is unsuccessful, AMF tries to register to another NRF.
- **nrfUseNFLoadInfo** - NRF Use Network Function Load Information - Determines whether 'load' will be taken into account when applying load balancing among candidate Network Functions / Service Instances. The 'load' indicates the current load percentage of the NF/Service Instance and is acquired from the NRF during NF Discovery and NF Profile Notification Update. Parameter values: - No (default): load is not taken into account - Yes: load is taken into account
- **nssaaEnable** - NSSAA - A control parameter to enable Network Slice-Specific Authentication and Authorization (NSSAA). If the parameter set to true, then AMF shall support NSSAA. Otherwise, AMF will not support it

- **nssfNsSelectionLoadBalancing** - NSSF NSSelection Load Balancing - This parameter controls whether load balancing will be applied among multiple Nssf_NSSelection service instances. When load balancing is enabled, load balancing will be applied among the highest priority available service instances based on their relative capacity.
- **nullSchemeSuciUseSupi** - Null-Scheme SUCI Use SUPI - When enabled, the AMF derives SUPI from Null-Scheme SUCI and include the SUPI: - in the query to the NRF to select AUSF - in the Authentication Request to the AUSF to authenticate the UE
- **nwdafEnable** - NWDAF Enable - A control parameter to enable AMF access to an NWDAF in the home PLMN. If set false, the AMF will not use the SBI towards the NWDAF.
- **oauthSupport** - OAuth Support - Parameter controls the feature activation. If parameter is enabled, OAuth based Authorization of Network Function Service Access shall be supported. By default this parameter shall be disabled.
- **pageForN1SmInUpdRsp** - Page for N1Sm in Update Response - Current parameter enables paging the UE when receive N1Sm in Update Sm Context Response and the UE is in CM-IDLE state.
- **pageOn3gppIfNon3gppFail** - Page UE on 3GPP if non-3GPP fail - Defines AMF action when UE is CM-CONNECTED for non-3GPP access and CM-IDLE on 3GPP access, and AMF has sent Notification on non-3GPP access to re-establish user plane or deliver signaling from SMF for 3GPP access, but AMF timer T3565 supervising the Notification Response, Service Request or Registration Request has expired. When set to 'false', AMF notifies the SMF that the UE is unreachable. When set to 'true', AMF pages the UE on 3GPP access.
- **pageRrcInactiveUe** - Page UE in RRC Inactive state - Enables paging UE in RRC Inactive state feature. When set to true, AMF invokes paging for UEs UE in RRC Inactive state if RAN paging fails.
- **pagingGapFailCountTrigger** - Paging Gap Fail Count Trigger - Parameter AMF Paging Gap Fail Count Trigger indicates the number of failed paging operations that start the AMF Paging Gap Timer.
- **pagingOverloadPrevention** - Paging Overload Prevention - Parameter controls feature activation related to monitoring AMF paging activity and taking action (when necessary) to avoid an overload condition. By default, flag shall be always enabled.
- **peips** - PEIPS Active - Specifies whether the PEIPS feature (Paging Early Indication with Paging Subgroups) is active on the AMF.
- **peiRetrievalMethod** - Query PEI method - Specify method to retrieve PEI from the UE. If the parameter is set to none, PEI is retrieved via identity request procedure if/when needed.
- **pmlInclude5qi** - Include 5QI PM Counters - This parameter controls whether the PM counters relevant to 5QI would be pegged from the system. When this parameter is set, the system is able to extract the 5QI information needed to peg the corresponding counters.

- **praPcfTrigger** - PRA PCF Trigger - Enables the AMF support for Presence Reporting Area changes triggered from PCF (for 3GPP access).
- **presenceInAol** - Presence in Area of Interest - Enables the Presence in Area of Interest in the AMF. The AMF provides this functionality via the Namf_EventExposure service. The service consumer may define an Area of Interest and the AMF reports back the presence of the UE on the area.
- **primaryHttpConnectionWeight** - Primary HTTP connection Weight - Defines how AMF will load balance traffic among 2 HTTP connections to the same SBI Server endpoint. Primary connection will have weight equal to primaryHttpConnectionWeight Secondary connection will have weight equal to 100 - primaryHttpConnectionWeight. Default is 50 which balances UE related traffic over the two connections on a per-UE basis and non-UE related traffic on a random basis.
- **prioritizeLinkStatesForCallbacks** - Prioritize Link States for Callbacks - When this parameter is set to true, when CMM is sending a Request to a callback URI (Subsequent Request or a Notification), it will prefer to use Endpoints with no link (Not Existing) or Endpoints that have links in Enabled and Unlocked state. Endpoints with links Under Creation (UNDEFINED) will be used only if no other valid Endpoint is available.
- **prioritizeLinkStatesForInitialRequests** - Prioritize Link States for Initial Requests - When this parameter is set to true, when CMM is sending an Initial Request in Direct communication, it will prefer to use Endpoints with no link (Not Existing) or Endpoints that have links in Enabled and Unlocked state. Endpoints with links Under Creation (UNDEFINED) will be used only if no other valid Endpoint is available.
- **prioritizeLinkStatesForScps** - Prioritize Link States for SCPs - When this parameter is set to true, when CMM is sending an Request in Indirect Direct communication, it will prefer to use Endpoints with no link (Not Existing) or Endpoints that have links in Enabled and Unlocked state. Endpoints with links Under Creation (UNDEFINED) will be used only if no other valid Endpoint is available.
- **radioCapInfoForPaging** - UE radio capability for paging - Enable/disable inclusion of radio capability for paging IE in the: - NGAP PAGING message - Initial UE Context Setup Request message when RRC Inactive feature is enabled
- **reattemptN26CtxReqCc92** - Reattempt N26 Context Request when receive CC92 - When current parameter is enabled and the AMF receives N26 Context Response with CC92, the AMF verifies UE identity, possibly by doing reauthentication, and attempts again N26 Context Request to source MME
- **reattemptN26CtxReqCc96** - Reattempt N26 Context Request when receive CC96 - When current parameter is enabled and the AMF receives N26 Context Response with CC96, the AMF verifies UE identity, possibly by doing reauthentication, and attempts again N26 Context Request to source MME
- **restrictBroadcastCmmHomePlmn** - Restrict Broadcasting CMM Home PLMN - Current parameter enables restrict sending CMM 'Home' PLMN in NG Setup and NRF Register procedures

- **reuseOldUeContextEpsTo5gs** - Reuse Old UE Context When UE Moves to 5GS - With this parameter, the AMF reuses the subscription data of the old UE Context that were locally preserved when UE moved to EPS and skips SDM GET and SDM Subscribe operations towards UDM
- **reuseRequestApiRootInLocationHeader** - Reuse Request apiRoot in Location Header - When this parameter is set to true and the Response includes a Location header, CMM will construct the apiRoot of the Location URI by reusing apiRoot included in the Request URI. When this parameter is false the apiRoot is deduced from the relevant service Instance provisioning.
- **rfspAccessRestriction** - RFSP Access Restriction - Feature control for RFSP Access Restriction for the AMF.
- **rimInformationTransfer** - RIM Information Transfer - Enables NGAP RIM Information Transfer Procedure. When set to false, AMF silently discards the NGAP Uplink RIM Information Transfer message.
- **rrcInactive** - RRC Inactive - Enables RRC Inactive feature. When enabled, AMF provides Core Network Assistance information to the RAN to assist gNB in moving UE(s) to RRC Inactive state
- **sbiHttp2PingKeepAlive** - SBI HTTP2 Ping Keep Alive - Enables or disables the impact of a received HTTP2 ping upon SBI link availability keep alive. When value is set to false a received HTTP2 ping will have no impact on SBI link availability. When value is set to true a received HTTP2 ping will reset SBI link keep alive, like other traffic messages.
- **sbiQueryConversions** - SBI Query Parameter Conversions - Enables or disables Query Conversion Rules. When value is set to false no Query conversion rule is applied. When value is set to true Query conversion rules are applied according to their Rule Status.
- **selectiveN15UsageControl** - Selective N15 Usage Control - Parameter controls feature activation flag. By default, the flag shall be disabled. When the flag is disabled, for homers AM and UE policy shall be created (old behavior) and for roamers, interPlmnPcfSupport shall be checked to decide if AM/UE policy need to be created or not (old behaviour). When gParm is enabled, for all users, pcfPolicyAssociation of amfSvcAgreement profile shall be checked to decide whether to establish AM and UE policy.
- **send3gppSbiCallbackInAllNotifications** - Send 3gpp-Sbi-Callback in all notifications - When this parameter is false the 3gpp-Sbi-Callback header is included only in Notifications sent via Indirect Communication. When the gParm is true the 3gpp-Sbi-Callback header is included in Notifications for both Direct and Indirect Communication.
- **send3gppSbiOriginatingNetworkId** - Send 3gpp-Sbi-Originating-Network-Id - When parameter is false, AMF does not include the 3gpp-Sbi-Originating-Network-Id header in Requests. When parameter is true, AMF includes 3gpp-Sbi-Originating-Network-Id header in Requests.
- **sendAmfConfigUpdTotalConfigMismatch** - Send AMF Configuration Update when Total Configuration Mismatch with gNB - When this parameter is enabled, AMF will send

AMF Configuration Update to a gNB with which there is total configuration mismatch, following a local configuration update.

- **sendAreaCodeBasedExtEmerNumberInRegistrationAccept** - Send Area Code Based Extended Emergency Number in Registration Accept - Parameter controls feature activation GPARM flag. By default, flag shall be disabled. When enabled, AMF shall send Extended Emergency Numbers List IE in the REGISTRATION ACCEPT messages if one is provisioned on the AMF.
- **sendCagPaging** - Send CAG Information List in Paging - When enabled, the AMF sends the CAG Information List in Paging message
- **sendEmeRegInd** - Send Emergency Registration Indication - The parameter enables sending the emergencyRegistrationInd IE in Amf3GppAccessRegistration data of UECM Registration to indicate emergency Registration to the UDM. This IE is sent in Nudm_UECM SBI v1.2.0-alpha.6 onwards
- **sendEmptyCagUe** - Send Empty CAG Information List to the UE - When enabled, the AMF sends empty CAG Information List in NAS Accept when no PLMN CAG is subscribed for CAG capable UE
- **sendEpslwkUpd** - When this parameter is enabled and the status of interworking with EPS for a PDU session changes due to change of UE 5GMM capability e.g. S1 mode supported, the AMF invokes Nsmf_PDUSession_UpdateSMContext to send updated EPS Interworking Indication to the SMF
- **sendExtEmerNumberInRegistrationAccept** - Send Extended Emergency Number in Registration Accept - Parameter controls feature activation GPARM flag. By default, flag shall be disabled. When enabled, AMF shall send Extended Emergency Numbers List IE in the REGISTRATION ACCEPT messages if one is provisioned on the AMF.
- **sendMappedGummei** - Current parameter enables sending in NGAP Setup Response the mapped GUAMI(s) pointing to the MME GUMMEI(s) for the associated PLMN(s) that coexist in both 4G and 5G at local CMM node
- **sendNsci** - Send NSCI IE - Current parameter enables sending 'New Security Context Indicator' NSCI IE in Handover Request, as part of EPS to 5GS handover
- **sendOnlyCommonlySupportedSlicesToAn** - Send Only Commonly Supported Slices to AN - When this parameter is true, the AMF will send only the intersection of the slices the AMF supports and the slices the gNB/ N3IWF supports for that PLMN (i.e. the union of all slices supported in all TAs the gNB/N3IWF supports per PLMN). Note: In the latter case the gNB/N3IWF will not be aware of all the slices the AMF supports, however it can resolve interworking issues when RAN cannot handle processing the all the slices included in the PLMN support list.
- **sendOnlyRanSupportedPlmnsInN2** - Send Only RAN Supported PLMNs in N2 - Current parameter enables sending in SETUP RSP and AMF Configuration Update only the PLMNs that are supported by the RAN even if additional PLMNs are configured at the AMF.

- **sendPc5ToNgran** - Send PC5 to NG RAN - Specifies whether the AMF is to send the UE PC5 QoS, if available, in applicable 38.413 messages
- **sendPeiTypeImeiToEir** - Send PEI Type IMEI to EIR - Parameter controls the option of sending PEI value type IMEI that indicates IMEI to EIR even though IMEISV is available. By default, flag shall be disabled.
- **sendPeiTypeImeiToSmf** - Send PEI Type IMEI to SMF - Parameter controls the option of sending PEI value type IMEI that indicates IMEI to SMF even though IMEISV is available. By default, flag shall be disabled.
- **sendPendingSnssaiInAllowedNssaiOfIcsr** - Send Pending SNSSAI in Allowed NSSAI of ICSR - Current parameter enables sending a pending SNSSAI (random selection) in Allowed NSSAI IE of the Initial Context Setup Request when latter needs be sent but there is not yet an allowed SNSSAI due to the NSSAA feature.
- **sendRlsSessionInfo** - Send Release Session Info - Current parameter enables sending the toReleaseSessionInfo attribute in Registration Status Update request.
- **sendSkipN2PduSessionResRelInd** - Send Skip N2 PDU Session Resource Release Indication - Current parameter enables sending Update SM Context Request carrying the skipN2PduSessionResRelInd parameter, instead of Release SM Context Request when a PDU Session needs to be released towards the UE at completion of handover e.g. because the associated S-NSSAI isn't supported at target TAI, but not towards the target gNB because a User Plane isn't established at the target gNB.
- **sendSnpnToUnsupportedNgRan** - Send SNPN Info to Unsupported NG-RAN - Current parameter enables sending the supported SNPNs in Setup Response and AMF Configuration Update towards NGRAN nodes that haven't indicated support of SNPN type of network in NG Setup or NG Configuration Update
- **sendTargetNfIdInRedirect** - Send 3gpp-sbi-Target-Nf-Id in Redirect - When this parameter is set to true the AMF adds the target AMF NF Instance and service instance Id in the relevant 3gpp-sbi-Target-Nf-Id HTTP header.
- **sharedPlmnSupport** - Shared PLMN support - Enables the support for Multiple Shared PLMN in the AMF. When the flag is set to TRUE, it shall be possible to set the multiple shared plmn for up to 8 PLMNs.
- **smContextStatusNotifyRequired** - SM Context Status Notify Required - This parameter defines whether the AMF requires an SM context status notify to be triggered from the SMF before the AMF deletes the PDU session. The parameter is applied during -UE/AMF/ SMF triggered PDU session release procedures, -PDU session setup procedures that have been rejected towards the UE after SM context create, and -PDU session setup procedures when the AMF already has a PDU session established for the requested PDU session id. Upon SBI timer pduSmCtxStatusNotify expiry, the AMF behavior is as follows: When set to true, the AMF only clears internal temporary data. AMF maintains the PDU session context. (AMF removes the PDU session only when it receives the SM context status notify with resourceStatusInfo set to 'RELEASED' on statusInfo inside SmContextStatusNotification.) When set to false, the AMF clears internal temporary

data, and also removes the PDU context (without waiting for the SM context status notify.)

- **smfAreaOfValidity** - SMF Area of Validity - Defines how the AMF applies the SMF optionally provided Area of Validity parameter on Namf_Communication_N1N2MessageTransfer request. AMF supports receiving up to 100 TAs.
- **smfRecoveryTime** - SMF Recovery Time - Enables the AMF to use the RecoveryTime attribute related to an SMF (service) instance to detect whether the SMF (service) instance has recently restarted and lost the resources created prior to the indicated recovery timestamp.
- **smpBasedPriorityFromSmsf** - SMP based Priority from SMSF - Enables AMF to apply priority handling for UEs receiving SBI message Priority value matching internal provisioning from SMSF.
- **smsfReselectionAfterUeProcedure** - SMSF Reselection After UE procedure - Enables the AMF to reselect SMSF for all impacted UEs, after AMF has detected an SMSF unavailable. The AMF reselects the SMSF for any impacted UEs after any next procedure for the impacted UEs.
- **smsfReselections** - SMSF Reselections - The number of attempted SMSF reselections per a single UE.
- **smsOverNas** - SMS Over NAS - Activates the SMS over NAS feature. The AMF supports the SMS delivery over NAS signalling and further via the N20 interface towards the SMSF.
- **smsRestrictionVoiceCentricUes** - SMS Restriction Voice Centric UEs - When this parameter is set to true, the AMF restricts the SMS over NAS service for UEs indicating 'voice centric' on UE's usage setting IE on NAS Registration request. Similar restriction also needs to be set per UE PLMN/serving PLMN combination. It is assumed that these UEs will get SMS service via IMS (and do not need SMS over NAS to be allowed). When this parameter is set to false, the SMS over NAS service is controlled in the AMF local provisioning by the AMF global parameter SMS Over NAS, the SMS Over NAS Allowed parameter per UE PLMN/serving PLMN combination and the SMS subscribed indication from UDM.
- **sorForTahs** - SoR for TAHS - Current parameter enables delivering the Steering of Roaming Information to TAHS subscriber during the Registration procedure, when Steering of Roaming subscription data is delivered to the AMF from the UDM.
- **srsInitialRegRelocation** - Initial UE Registration With SRS - Enables new AMF to retrieve UE Context from session restoration server during inter AMF UE initial registration with GUTI.
- **srvccPossibleNormallyRegisteredEmergency** - 5G-SRVCC Possible Normally Registered Emergency - Defines whether the AMF temporarily indicates on 5G-SRVCC Operation Possible indication the value 'Possible' when a normally registered UE supports 5G-SRVCC and establishes an emergency PDU session.

- **standaloneNonPublicNetworkSupport** - Standalone Non-Public Network Support - Parameter controls feature activation flag. By default, flag shall be disabled. When enabled, AMF shall support 3GPP specified Release 16 Stand alone Non Public Network functionality.
- **subscribedEmergencySnssai** - Subscribed Emergency S-NSSAI - Current parameter enables using the S-NSSAI from subscribed emergency DNN if such exists in subscription data
- **subscribedForbiddenTacChangeAction** - Subscribed Forbidden TAC Change Action - Parameter controls behavior invocation whether to release the N2 connection to put UE into idle mode or deregister the UE upon receipt of UDM Notification with updated list of subscribed forbidden Areas/TACs which includes the current UE TAC, and the UE is in connected mode. By default, parameter shall be set to none.
- **support5gAnSliceInconsistencyAlarm** - Support 5G-AN Slice inconsistency Alarm - When this parameter is enabled, AMF will raise a Alarm when multiple 5G-AN network elements (gNB, N3IWF) report different slices for the same Tracking Area.
- **support5gSrvcc** - Support 5G-SRVCC - Enables the AMF support for 5G-SRVCC to hand over regular voice calls from NG-RAN/5GS to 3GPP UTRAN/CS domain.
- **support5gSrvccEmergency** - Support 5G-SRVCC Emergency - Enables the AMF support for 5G-SRVCC to hand over emergency calls from NG-RAN/5GS to 3GPP UTRAN/CS domain.
- **supportAusfUdmGroupId** - Support AUSF UDM GroupId - When this parameter is set to true and the GroupId is included in the Profile of a AUSF or UDM selected for a UE, AMF will store it in the UE's context. The GroupId is accordingly used for subsequent AUSF/UDM selection and conveyed in case of inter AMF mobility.
- **suppressRetryWithNudmSdmSingleGet** - Suppress retry with SDM single Get - When set to true, the AMF will not retry with nudm-sdm single GET to the same UDM on receipt of an error response to a nudm-sdm multi dataset GET from the UDM with status code 404 DATA NOT FOUND or any other status code failure.
- **tappingActivated** - Status of AMF Tapping functionality - This parameter controls the activated/deactivated status of AMF Tapping
- **timeBasedAuthFrequency** - Time Based Authentication Frequency - Parameter controls feature activation global parameter flag. By default, the flag shall be disabled. When enabled, AMF shall support time-based authentication in accordance with new option(s) defined in AMF UE PLMN Services configuration.
- **timerSmfReselection** - Timer SMF Reselection - Enables the functionality to reselect SMF based on a timer during SM context creates. When timer based reselection is set to false, AMF only reselects SMF once per procedure on error cases (i.e. decision to reselect is attempt based). The used timer is deducted from SBI SMF response timer. Only when this parameter is set to 'true', the AMF applies the parameters errorSource, n1Included, reselectSmfVsmf, reselectOnlySmf and reselectRemote on cmm amfSmfNasMapping command.

- **trigUePolicyWhenNoUePolicyContainer** - Trigger UE Policy When No UE Policy Container - Parameter controls establishment of UE Policy Association based on received UE policy container from UE. By default, the flag shall be disabled. When enabled, if no UE Policy Container is received during UE Policy Association Establishment procedure the AMF will establish UE Policy Association.
- **udmBasedPscfRestoration** - UDM based PSCF Restoration - Parameter controls feature activation GPARM flag. By default, flag shall be disabled. When enabled, AMF upon receiving P-CSCF restoration indication from UDM shall initiate a network triggered PDU Session Release procedure of the IMS PDU session with Reactivation indication and accordingly UE shall request to re-establish IMS PDU session and shall discover P-CSCF during IMS PDU session establishment.
- **udmImmediateReport** - UDM Immediate Report - Immediate report indication to UDM
- **udmInitiatedCallTrace** - UDM Initiated Call Trace - Support UDM initiated call trace or not.
- **udmMdt** - UDM MDT - Activation/deactivation minimization of drive tests.
- **ueAmbrChangeFromPcf** - UE AMBR Change From PCF - Parameter controls whether to send UDM AMBR update to PCF in order for PCF to determine if AMBR update should occur. By default, flag shall be disabled. When enabled, AMF shall send UDM AMBR update to PCF.
- **ueRadioCapPagingN14** - Ue Radio Capability for Paging in N14 - Enable sending Ue Radio Capability for Paging IE in N14 interface
- **uniqueIdentifierSupportInSubscriptionId** - Unique Identifier Support In Subscription Id - Parameter controls feature activation flag. By default, flag shall be disabled. When enabled, support for unique identifier support in subscription is provided so the cell broadcast centre function can filter/sort which amf has provided subscription to it.
- **useAdditionalMemory** - Use Additional Memory - When the CPPS has been configured with additional memory, this parameter controls use of the additional memory for selected features. Use of additional memory for a feature must be indicated with this parameter before that feature can be activated.
- **useDnsForN26Ho** - Use Dns for N26 handover - Current parameter enables using DNS for selecting target node as part of N26 handover even when local MME of quad access node serves target RAN.
- **useFqdnAsAuthorityInAllSbiInterfaces** - Use FQDN as Authority in All SBI Interfaces - When this parameter is set to true, and FQDN of the target URI is available, CMM uses the FQDN as the authority in HTTP requests. If the parameter is set to true but there is no FQDN, the IP is used as the authority. When the parameter is set to false the IP is used as the authority in the SBI interfaces with exception of the NRF interface which always uses the FQDN as the preferred authority.
- **useLocationHeaderFromUdmRsp** - Use HTTP Location Header from UDM response - When this parameter is set to true, when CMM receives Response to a UDM Subscribe

the URI in the Location Header is used in subsequent Requests to update (PATCH/PUT) or Delete that Subscription. CMM also stores the Producer ID so as to be able to support OAuth in subsequent requests. Otherwise subsequent Requests are sent to a service instance discovered via NRF. Note: This parameter must be set to true when UDM communication is based on Model D (indirect communication with delegated discovery)

- **useN26SvcParam5gsToEpsHo** - Use N26 Service Parameter in 5GS to EPS Handover - Current parameter enables using N26 service parameter as part of DNS query for target MME selection during 5GS to EPS handover
- **useNssfAvailabilityService** - Use NSSF Availability Service - When this parameter is true, the AMF will use the NSSF Availability Services. When NSSF Availability Services are used, the AMF does not take in account the local provisioned info regarding Slice support per Tracking Area.
- **useNssfAvailabilityServiceSubscription** - Use NSSF Availability Service Subscription - When this parameter is true, the AMF will Subscribe the NSSF Availability Services in order to receive notification from NSSF regarding changes to the NSSAI availability information on a per TA basis. This parameter is applicable only when **amfGParams** **useNssfAvailabilityService** is set to true.
- **useNssfPerPlmnSliceRestrictions** - Use Per PLMN Slice Restrictions - When this parameter is set to true AMF evaluates the **restrictedNssaiList** when included in the **AuthorizedNssaiAvailabilityData** by the NSSF in a **Nnssf_NSSAIAvailability** operation. When set to false the restrictions are ignored.
- **useNssfSelectionServiceForAmf** - Use NSSF Selection Service for AMF - When this parameter is true the AMF uses the NSSF Selection Service to select AMF during UE Registration in case AMF decides that it is unable to serve all the Allowed Slices the UE.
- **useNssfSelectionServiceForSmf** - Use NSSF Selection Service for SMF - When this parameter is true the AMF uses the NSSF Selection Service to select SMF during PDU Session establishment
- **userLocationIgnoreNcgi** - User Location Ignore NCGI - Defines whether the AMF supports **ignoreNcgi** parameter inside the **nrLocation** and further on **userLocation**.
- **v2xAllowed** - V2X Allowed - Feature flag for the v2x feature. When set true, the AMF will request a PCF that supports v2x.
- **vlrCleanupBasedOnOccupancy** - VLR Cleanup Based on Occupancy - Current parameter enables doing VLR cleanup of the VLR Contexts of deregistered UEs based on VLR Occupancy
- **voiceEpsFallback** - Voice with EPS Fallback - This parameter activates the voice functionality using EPS fallback on the AMF.
- **voNR** - Voice over New Radio - Activation of the Voice over New Radio feature on the AMF.
- **vsmfChange** - VSMF Change - Enables VSMF change including selection of a new VSMF when new UE location is not supported by current VSMF.

Name

amfGparmValue - Parameter value

Syntax

--amfGparmValue <string>

Description

Value of parameter.

Class

Required on create.

Type / Range

Digit string of length 0 to 64

The amfGparmValue string is based on the amfGparmName parameter:

Name

5g4gRetPrefPlmn - 5g4g Return Preferred PLMN

Syntax

--amfGparmName 5g4gRetPrefPlmn --amfGparmValue <string>

Description

This parameter enables indicating to serving NG-RAN the last visited EPC PLMN as preferred indication to be used in future 5GS to EPS mobility.

List

- DISABLE - Disable
- ENABLE - Enable

Default

DISABLE

Name

5gRanTotalConfigIncompatibilityAlarm - 5G RAN Total Configuration Incompatibility Alarm

Syntax

--amfGparmName 5gRanTotalConfigIncompatibilityAlarm --amfGparmValue <string>

Description

When this parameter is enabled, AMF will raise an Alarm when receiving RAN Configuration Update or there is local configuration change (that may or not lead to sending AMF Configuration Update) that results in total configuration incompatibility between the AMF and the RAN.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Recommended

true

Name

acceptVoiceNoCheckResp - Accept Voice No Check Response

Syntax

--amfGparmName acceptVoiceNoCheckResp --amfGparmValue <string>

Description

Controls whether the AMF allows indicating IMS voice over PS sessions supported towards the UE and the UDM, when UE Radio Capability Check procedure is provisioned to be run towards the NG-RAN from the AMF but the NG-RAN provides no response for NGAP UE Radio Capability Check Request message.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

accessTypeChangePcfTrigger - Access Type Change PCF Trigger

Syntax

--amfGparmName accessTypeChangePcfTrigger --amfGparmValue <string>

Description

Enables the AMF support for Access Type changes triggered from PCF. By default, flag shall be disabled.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

alwaysSendDnnWithOperatorIdentifier - Always Send DNN with DNN Operator Identifier

Syntax


```
--amfGparmName alwaysSendDnnWithOperatorIdentifier --amfGparmValue <string>
```

Description

Defines whether AMF will always include DNN-OI in DNNs sent as SBI JSON Body IEs and Query Parameters. When the value is true, the AMF will always include the DNN-OI in DNNs. When value is false, the DNN-OI may not be always present as the AMF transparently conveys the DNN in the non roaming scenarios.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

amfEirSupport - AMF EIR Support

Syntax

```
--amfGparmName amfEirSupport --amfGparmValue <string>
```

Description

Activation of the AMF EIR feature on the AMF. By default this shall be disabled

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

amfHomogeneousSupportTais - AMF Homogeneous Support TAIs

Syntax

```
--amfGparmName amfHomogeneousSupportTais --amfGparmValue <string>
```

Description

Defines the TAI the AMF considers for IMS voice over PS homogeneous support under the serving AMF. The other access TAI is considered to support all voice options when considered for the other access IMS voice over PS homogeneous support. (Non-3GPP TAI is considered to support VoPS with VoNR and VoLTE Handover and Redirect.)

List

- AmfHomogSuppAllPlmnsBothAccesses - AMF Homogeneous Support All PLMNs Both Accesses - AMF considers all Tracking Areas of all PLMNs provisioned for both access types when evaluating the IMS voice over PS homogeneous support for the UE.

- AmfHomogSuppAllPlmnsCurrentAccess - AMF Homogeneous Support All PLMNs Current Access - AMF considers all Tracking Areas of all PLMNs provisioned for the current access type when evaluating the IMS voice over PS homogeneous support for the UE.
- AmfHomogSuppServingPlmnBothAccesses - AMF Homogeneous Support Serving PLMN Both Accesses - AMF considers all Tracking Areas of the serving PLMN provisioned for both access types when evaluating the IMS voice over PS homogeneous support for the UE.
- AmfHomogSuppServingPlmnCurrentAccess - AMF Homogeneous Support Serving PLMN Current Access - AMF considers all Tracking Areas of the serving PLMN and the current access type when evaluating the IMS voice over PS homogeneous support for the UE.

Default

AmfHomogSuppAllPlmnsCurrentAccess

Recommended

The recommended value is AmfHomogSuppServingPlmnCurrentAccess.

Name

amfN26DirectFwd - amfN26DirectFwd

Syntax

--amfGparmName amfN26DirectFwd --amfGparmValue <string>

Description

Indicates whether direct data forwarding is enabled or disabled on n26 handover.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

amfN26IndirectFwd - amfN26IndirectFwd

Syntax

--amfGparmName amfN26IndirectFwd --amfGparmValue <string>

Description

Indicates whether indirect data forwarding is enabled or disabled on n26 handover.

List

- disabled - Indirect forwarding is disabled for n26 handover
- enabled - Indirect forwarding is enabled for n26 handover

Default

enabled

Name

amfNpli - AMF NPLI

Syntax

--amfGparmName amfNpli --amfGparmValue <string>

Description

Enables the support for Network Provided Location Info (NPLI) i.e the Namf_Location_ProvideLocationInfo service for the AMF.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

amfRanCombiningOption - AMF RAN Combining Option

Syntax

--amfGparmName amfRanCombiningOption --amfGparmValue <string>

Description

Controls how the AMF establishes the user plane sessions during Service Request or Mobility Registration Update for multiple PDU sessions. AMF may either combine/aggregate all SMF responses and formulate a single NGAP message, or formulate separate NGAP messages per each SMF request.

List

- AMF_COMBINE_RAN_MSGS - Combine RAN messages - The AMF waits for all the SMF responses related to all PDU sessions from all SMFs before sending any NGAP message towards the AN. On Service Request or Mobility Registration Update when user plane connection is established, the AMF formulates a single combined Initial Context Setup Request message on N2 interface with all the N2 resource setup containers included, together with the NAS Service Accept.
- AMF_SEPARATE_RAN_MSGS - Separate RAN Messages - The AMF forwards the SMF responses related to the separate PDU sessions from the SMFs one by one as they arrive at the AMF. On Service Request or Mobility Registration Update when user plane connection is established, the AMF formulates first an Initial Context Setup Request with the first SMF N2 interface container included, and after that formulates subsequent PDU Session Resource Setup Request(s), each request carrying a single N2 setup information container from the corresponding SMF. The last (of the possibly multiple) N2 message(s) contains the NAS Service Accept or Registration Accept message.

Default

AMF_COMBINE_RAN_MSGS

Name

amfRequestPsCellInLrc - AMF Request PS Cell in LRC

Syntax

--amfGparmName amfRequestPsCellInLrc --amfGparmValue <string>

Description

When location reporting control procedure is used in AMF, if AMF Request PS Cell in LRC is set, then AMF includes additional location information IE and set to Include PSCell. gNB will include the current PSCell in the report if Dual Connectivity is activated. If AMF Request PS Cell in LRC is not set, AMF will not request additional location information under location reporting procedure as existing function, then gNB will not report PSCell in location report.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

amfSmsSubscrGet - AMF SMS Subscription Get

Syntax

--amfGparmName amfSmsSubscrGet --amfGparmValue <string>

Description

Controls whether the AMF always attempts to get the SMS related subscription data from the UDM, or whether the AMF follows UE indication of support for SMS over NAS when retrieving the SMS related subscription data from the UDM (i.e. gets the subscription data on need only).

List

- amfSmsGetAlways - AMF SMS Get Always - AMF always attempts to get the SMS Subscription Data and the UE context in SMSF from the UDM during initial registration to the current AMF, if the SMS over NAS feature is enabled both AMF globally and for the UE on UE PLMN services. The AMF attempts to get the subscription data independently of the UE indicated support for SMS over NAS.
- amfSmsGetOnNeed - AMF SMS Get On Need - AMF follows the UE indicated support for SMS over NAS and only attempts to get the SMS Subscription Data and the UE context in SMSF from the UDM during initial registration to the current AMF, if the UE indicates it supports SMS over NAS. The AMF attempts to get the SMS Subscription Data and the UE context in SMSF from the UDM the during a subsequent (periodic or mobility) registration procedure if the UE support for SMS over NAS changes i.e. UE indicates support for SMS over NAS. Further, if AMF does not GET multiple data sets

simultaneously, AMF further does not GET the UE context in SMSF if SMS subscription data is not received.

Default

amfSmsGetAlways

Recommended

The recommended value is amfSmsGetOnNeed.

Name

amfTaiDiscovery - AMF TAI Discovery

Syntax

```
--amfGparmName amfTaiDiscovery --amfGparmValue <string>
```

Description

Specifies whether the AMF TAI auto-discovery feature is active. Default is disable.

List

- DISABLE - Disable
- ENABLE - Enable

Default

DISABLE

Name

amfTriggeredRrcInactivePaging - AMF Triggered RRC Inactive Paging

Syntax

```
--amfGparmName amfTriggeredRrcInactivePaging --amfGparmValue <string>
```

Description

When set to true, the AMF triggers paging of UE(s) in RRC Inactive state but not reachable without network triggered messaging.

Restrictions

This gParm is only taken into consideration if gParm pageRrcInactiveUe is set to true.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Recommended

false

Name

applProtocolFallbackN26ToS10 - Application Protocol Fallback from N26 to S10

Syntax

```
--amfGparmName applProtocolFallbackN26ToS10 --amfGparmValue <string>
```

Description

Current parameter enables fallback to search for 's10' candidate when DNS Query to find target MME during S10 handover using 'n26' fails. This parameter is expected to be used together with parameter Use N26 Service Parameter in 5GS to EPS Handover

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

apra - Additional Presence Reporting Area

Syntax

```
--amfGparmName apra --amfGparmValue <string>
```

Description

Enables the use of the Additional Presence Reporting Area (APRA) feature on the Namf_EventExposure service. When this parameter is set to true, and when also the service consumer indicates support for the APRA feature, and the Presence Reporting Area (PRA) set is internally provisioned in the AMF, and the service consumer requests reporting for the PRA set, the AMF reports the PRA set id in the 'prald' field and the individual PRA id(s) in the 'additionalPrald' field(s) on the presenceInfo of the areaList (an array of AmfEventAreas) of a reportList (an array of AmfEventReports).

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

asynchrTypeComm - Asynchronous Type Communication

Syntax

```
--amfGparmName asynchrTypeComm --amfGparmValue <string>
```

Description

This parameter defines whether the AMF attempts to contact a CM-IDLE UE when AMF receives signalling message for the UE, e.g. SMF triggers N1/N2 message transfer for PDU session modification. When enabled, the AMF attempts to reach the UE via paging (if the UE is currently considered reachable and would be expected to respond to paging). When

disabled, the AMF sends the signalling message to the UE when the UE makes the next contact towards the network.

List

- **ATC_Disabled** - Asynchronous Type Communication Disabled - When AMF receives terminating signalling for a UE that is CM-IDLE, the AMF attempts to reach a CM-IDLE UE via paging (if the UE is currently considered reachable and would be expected to respond to the paging).
- **ATC_Enabled** - Asynchronous Type Communication Enabled - For a CM-IDLE UE for which terminating signalling is received from another network element, the AMF stores the corresponding signalling message and delivers the message towards the UE when the UE makes the next contact towards the network.

Default

ATC_Disabled

Name

autoAddTaiToTaiList - Auto Add TAI to TAI List

Syntax

--amfGparmName autoAddTaiToTaiList --amfGparmValue <string>

Description

Auto Add TAI to TAI list, with allowed values are: Off and Update Three.

List

- **false** - Boolean value of false.
- **true** - Boolean value of true.

Default

false

Name

cagSupport - CAG Support

Syntax

--amfGparmName cagSupport --amfGparmValue <string>

Description

This parameter enables Closed Access Group feature at the AMF

List

- **false** - Boolean value of false.
- **true** - Boolean value of true.

Default

false

Name

checkTgtMmeN26StatusN26Ho - Check Target MME N26 Status in N26 Handover

Syntax

--amfGparmName checkTgtMmeN26StatusN26Ho --amfGparmValue <string>

Description

Current parameter enables checking the N26 link/interface status of local target MME node during N26 handover so that if it isn't available to use, then even if the local MME can serve the target RAN, source AMF will select an alternative node to complete the N26 handover.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Recommended

true

Name

checkUesUsageSetting - Check UE's Usage Setting

Syntax

--amfGparmName checkUesUsageSetting --amfGparmValue <string>

Description

Controls whether the AMF requires UE's Usage Setting IE to be included in NAS Registration Request to indicate IMS voice over PS sessions supported towards the UE and the UDM. If set to false, IMS voice over PS sessions support indication does not depend on whether the UE includes UE's Usage Setting IE. If set to true, AMF indicates IMS voice over PS sessions supported only if UE's Usage Setting IE is included in NAS Registration Request.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Recommended

The recommended value is TRUE.

Name

contactOverNon3gppFor3gpp - Contact Over Non-3GPP for 3GPP

Syntax

--amfGparmName contactOverNon3gppFor3gpp --amfGparmValue <string>

Description

Defines whether the AMF contacts a UE over non-3GPP access, when the UE is CM-CONNECTED over non-3GPP access and CM-IDLE over 3GPP access, and the AMF gets a 3GPP access related trigger for the UE.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

cr129ts29500ErrorToken - CR 129 TS 29.500 Error Token

Syntax

--amfGparmName cr129ts29500ErrorToken --amfGparmValue <string>

Description

This parameter controls the activation of CR 0129 of TS29.500. When this parameter is set to true: Removes condition the access without an OAuth2.0 access token failed for the realm attribute Adds error code invalid_token

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

defaultHttpPort - Default HTTP Port

Syntax

--amfGparmName defaultHttpPort --amfGparmValue <string>

Description

Default port to be used for HTTP connections when it is not explicitly given as part of the the authority of a SBI URI. This parameter applies only to HTTP connections. HTTPS connections always use 443 as the default port (if not explicitly given otherwise in the URI)

Type / Range

Integer in the range of 1 to 65,535

Default

80

Name

defErrorCcSnpn - Default Error Cause for SNPN network

Syntax

```
--amfGparmName defErrorCcSnpn --amfGparmValue <string>
```

Description

This parameter defines the default error cause to send for UE request towards SNPN network when Access Restriction NAS Mapping profile isn't available to use

Type / Range

5GMM Cause Code.

Default

74

Name

delayUdmDuringRegistration - Delay UDM During Registration

Syntax

```
--amfGparmName delayUdmDuringRegistration --amfGparmValue <string>
```

Description

When the same AMF still holds the UE data internally after previous N26 mobility to EPS, this parameter enables AMF to successfully complete the registration procedure and suppress the UDM dialogs (Nudm_UECM_Registration, Nudm_SDM_Get and/or Nudm_SDM_Subscribe, depending on other settings) for the enabled UEs. The UDM dialogs will be re-attempted automatically according to provisioning. Note: this functionality only is effective when the AMF global parameter Keep VLR Entry is set to true.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

deletePduAfterFailedUpAttemptsInIndirectComm - Delete PDU After Failed User Plane Attempts in Indirect Communication

Syntax

```
--amfGparmName deletePduAfterFailedUpAttemptsInIndirectComm --  
amfGparmValue <string>
```

Description

Enables the possibility to delete a PDU from AMF after provisioned number of failed user plane re-establishment attempts when indirect communication is deployed. When this parameter is set to true, the AMF counts the number of times the AMF receives (after possible SCP reselection(s)) a 504 failure from SCP into the internal counter for failed user plane re-establishment attempts. Once the AMF detects 'nth' time the user plane re-establishment fails (for either direct or indirect attempt towards the SMF), the AMF internally

deletes the PDU and indicates towards the UE that the PDU is inactive. The AMF applies the value of the AMF global parameter Failed User Plane Attempts to Delete PDU for the number 'n' above. This parameter is only effective when AMF global parameter Failed User Plane Attempts to Delete PDU is set to a non-zero value. When this parameter is set to false, the AMF does not count 504 response from SCP as SMF not reachable during user plane re-establishment attempts and does not delete PDU(s) due to those.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

directNasMsgTransfer - Direct NAS message transfer to the target AMF

Syntax

--amfGparmName directNasMsgTransfer --amfGparmValue <string>

Description

If the parameters is set to 'True' then AMF forwards the NAS Registration message to the target AMF over N14 interface if the AMF can determine the target AMF. If the parameters is set to 'False' then AMF forwards the NAS Registration message via 5G RAN.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

dualFailVnfIpds - Dual Fail IPDS

Syntax

--amfGparmName dualFailVnfIpds --amfGparmValue <string>

Description

When enabled, a dual failure of a spread IPDS pool will trigger IPDS pool 0 to dual fail. This is specifically for the two large IPDS configuration in VNF or CNF. It applies to any application type. This dual failure of IPDS pool 0 will result in system outage of the system. When disabled, the propagation of the dual failure will not occur.

List

- DISABLE - Disable
- ENABLE - Enable

Default

ENABLE

Name

eaea - EBI and ARP mapping update in EBI Assignment

Syntax

--amfGparmName eaea --amfGparmValue <string>

Description

Enables supporting EAEA feature

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ebiAssignReqReject - EBI Assignment Request Reject

Syntax

--amfGparmName ebiAssignReqReject --amfGparmValue <string>

Description

When set to true, the AMF rejects SMF EBI assignment request with an http2 status code associated with the ongoing procedure at the time of the received request, TEMPORARY_REJECT_HANDOVER_ONGOING or TEMPORARY_REJECT_REGISTRATION_ONGOING

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

emcNon3gppN3iwf - EMC Non-3GPP Access N3IWF

Syntax

--amfGparmName emcNon3gppN3iwf --amfGparmValue <string>

Description

Enables the support for emergency services over untrusted non-3GPP access via N3IWF.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Recommended

Should be set to true when non-3GPP access type is served on 5GS via N3IWF.

Name

emergencyEpsIwkWoN26 - Emergency EPS Interworking Without N26

Syntax

--amfGparmName emergencyEpsIwkWoN26 --amfGparmValue <string>

Description

Enables the possibility to handover an emergency session from EPS to 5GS (3GPP access) without N26 interface on the AMF. Session continuity is ensured by getting the (EPS stored) emergency information from HSS+UDM as part of the subscription profile. Applied when EPS Interworking Without N26 is set to true.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

emergencyNon3gppEpsPduHo - Emergency Non-3GPP Access EPS PDU Handover

Syntax

--amfGparmName emergencyNon3gppEpsPduHo --amfGparmValue <string>

Description

Enables the AMF support for UE triggered emergency PDU handovers between untrusted non-3GPP via N3IWF/5GS and E-UTRAN/EPS.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

enhancedPcfSelection3gppAccess - Enhanced PCF Selection 3GPP Access

Syntax

```
--amfGparmName enhancedPcfSelection3gppAccess --amfGparmValue  
<string>
```

Description

When this parameter is enabled, AMF will support the following functionality for 3GPP access: a) Reselect PCF upon change of Serving PLMN if required. b) Exclude PCFs that do not support both UE and AM Policy Control Services when an association for both Policies is required. c) Reselect PCF during inter-AMF UE mobility if the PCF Instance provided by the old AMF does not meet all the selection criterions. d) Fallback to selection of PCF without considering Slice support, when no valid candidate PCF that serves all slices in the Allowed NSSAI is discovered.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

epdgEmergencyHo - ePDG Emergency Handover

Syntax

```
--amfGparmName epdgEmergencyHo --amfGparmValue <string>
```

Description

This parameter enables session continuity of an emergency session between ePDG/EPC and NG-RAN/5GC.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

epdgHo - ePDG Handover

Syntax

```
--amfGparmName epdgHo --amfGparmValue <string>
```

Description

Enables handover between non-3GPP access connected to EPC and 3GPP access (NG-RAN) connected to 5GC

List

- DISABLE - Disable

- ENABLE - Enable

Default

DISABLE

Name

epsFallbackNoPageUe - Page UE during EPS Fallback

Syntax

--amfGparmName epsFallbackNoPageUe --amfGparmValue <string>

Description

Enables suppression of UE paging when gNB releases UE context due to EPS Fallback and EpsFallbackNoPageTimer is running.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

epsIwkWithoutN26 - EPS Interworking Without N26

Syntax

--amfGparmName epsIwkWithoutN26 --amfGparmValue <string>

Description

Enables the EPS interworking without N26 interface support on the AMF.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

epsTo5gsHoWithAmfReselect - EPS To 5GS Handover With AMF Reselect

Syntax

--amfGparmName epsTo5gsHoWithAmfReselect --amfGparmValue <string>

Description

This parameter enables the capability of AMF reselection during EPS to 5GS handover. It is evaluated at both the initial AMF and target AMF of an EPS to 5GS Handover with AMF reselection procedure

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

epsUpip - EPS UPIP

Syntax

--amfGparmName epsUpip --amfGparmValue <string>

Description

Current parameter enables support for the EPS-UIP (User Plane Integrity Protection)

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ethernetPduSupportN26 - Ethernet PDU Support N26

Syntax

--amfGparmName ethernetPduSupportN26 --amfGparmValue <string>

Description

Parameter controls AMF indicating in Context Request if Ethernet PDU is supported for EPS to 5GS idle mobility scenario. By default, the flag shall be disabled.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

excludeRoutingIndWithSupi - Exclude Routing Indicator with SUPI

Syntax

--amfGparmName excludeRoutingIndWithSupi --amfGparmValue <string>

Description

When enabled, the AMF, when SUPI is available, will use SUPI and exclude routing indicator as selection criteria in the query to the NRF to select AUSF and UDM.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

existingPduNoSmf - Existing PDU No SMF

Syntax

--amfGparmName existingPduNoSmf --amfGparmValue <string>

Description

Controls whether the AMF sends the SM message back to the UE or drops it, if the AMF receives an SM message with Request Type 'existing PDU session', the AMF does not have a routing context for the PDU session and no SMF can be found on the UDM subscription data for the PDU session id or the DNN.

List

- AMF_EXISTING_PDU_NO_SMF_DROP - Existing PDU No SMF Drop - If the AMF receives an SM message with Request Type "existing PDU session", the AMF does not have a routing context for the PDU session and no SMF can be found on the UDM subscription data for the PDU session id or the DNN, the AMF drops the SM message.
- AMF_EXISTING_PDU_NO_SMF_SENDBACK - Existing PDU No SMF Send Back - If the AMF receives an SM message with Request Type "existing PDU session", the AMF does not have a routing context for the PDU session and no SMF can be found on the UDM subscription data for the PDU session id or the DNN, the AMF sends back the SM msg to the UE.

Default

AMF_EXISTING_PDU_NO_SMF_SENDBACK

Name

extendedEmergencyNumListAreaCode - Extended Emergency Number List Area Code

Syntax

--amfGparmName extendedEmergencyNumListAreaCode --amfGparmValue
<string>

Description

Indicates provisioned extended emergency list area code. When feature activation gParm is enabled and upon receiving AreaCode in UE Subscription data from UDM, if received Area Code matches provisioned area code for this feature, AMF shall send Extended Emergency Numbers List in the REGISTRATION ACCEPT message.

Type / Range

Integer in the range of 0 to 65,535

Default

0

Name

failedUserPlaneAttemptsToDeletePdu - Failed User Plane Attempts to Delete PDU

Syntax

```
--amfGparmName failedUserPlaneAttemptsToDeletePdu --amfGparmValue  
<string>
```

Description

Defines how many failed UE triggered user plane re-establishment attempts, where the AMF fails to communicate with the SMF due to link issues, cause the AMF to internally delete the requested PDU and to indicate towards the UE that the corresponding requested PDU is inactive. The value 0 stands for the functionality to be disabled. When disabled, the AMF does not delete the PDU due to link related SMF communication failures during user plane re-establishment attempts.

Type / Range

Integer in the range of 0 to 10

Default

0

Name

freqCountMonitoring - Frequency Count Monitoring

Syntax

```
--amfGparmName freqCountMonitoring --amfGparmValue <string>
```

Description

Defines whether the AMF monitors the UE attempts for different UE triggered procedures, and if too frequent attempts made by the UE during the provisioned monitoring period, treat the UE accordingly, e.g. drop subsequent procedure requests or deregister the UE.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

getSmfSelSubDataEmptySnssaiInfo - Get Smf Selection Subscription Data when having empty SnssaiInfo

Syntax

```
--amfGparmName getSmfSelSubDataEmptySnssaiInfo --amfGparmValue <string>
```

Description

Enable fetching SMF Selection Subscription Data during PDU Session Establishment (type = initial request) when local UE context indicates that the UE is supposed to have PDU session service subscribed but local subscription data for SMF selection is empty

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

gnbRerouteFallback - GNB Reroute NAS Fallback

Syntax

```
--amfGparmName gnbRerouteFallback --amfGparmValue <string>
```

Description

AMF re-routes NAS message to gNB when N14 is not available during UE inter-AMF re-location when source AMF can't service the UE slice.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

iabEnabled - IAB Enabled

Syntax

```
--amfGparmName iabEnabled --amfGparmValue <string>
```

Description

This gParm enables Integrated Access and Backhauling (IAB) operation at the AMF

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

icsrAloneVoiceRadioCap - ICSR Alone Voice Radio Capability

Syntax

```
--amfGparmName icsrAloneVoiceRadioCap --amfGparmValue <string>
```

Description

Controls whether AMF sends NGAP Initial Context Setup Request alone without NAS Registration Accept, when either -AMF does not hold UE radio capability but needs it for purposes listed further below, and/or -AMF needs to run NGAP UE radio capability check procedure to decide IMS voice over PS support for the UE. If UE radio capabilities are needed on the AMF (Reject On VoNR Tracking Area and/or EMC Suppress No VoNR and/or IMS VoPS S1 Disabled is true under AMF Voice Profile for the UE on 5G UE PLMN Services), or UE radio capability check procedure result is required to decide for the IMS voice over PS result for the UE, the AMF needs to send NGAP Initial Context Setup Request alone first (without any NAS message) in order to let NG-RAN establish AS security before requesting radio capabilities from the UE. When set to true and when Reject On VoNR Tracking Area and/or EMC Suppress No VoNR and/or IMS VoPS S1 Disabled is true under AMF Voice Profile for the UE on 5G UE PLMN Services, AMF sends NGAP UE Radio Capability Check Request only after receiving Initial Context Setup Response.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Recommended

The recommended value is TRUE.

Name

ignoreAusfFailuresOptionalReAuthentications - Ignore AUSF Failures on Optional Re-Authentications

Syntax

```
--amfGparmName ignoreAusfFailuresOptionalReAuthentications --  
amfGparmValue <string>
```

Description

This parameter, when set, enables AMF to ignore AUSF dialog related failures during optional re-authentications. The UEs to apply the functionality to are provisioned on the AMF. The functionality is applied when the AMF is locally provisioned to perform an optional re-authentication for a UE that is already registered on the AMF, but the re-authentication fails during the AUSF dialog and the AMF is able to continue serving the UE without the re-authentication.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Name

imdRptThirdPartySub - Immediate Report to Third Party Subscription

Syntax

--amfGparmName imdRptThirdPartySub --amfGparmValue <string>

Description

Controls whether the AMF supports the 'IERSR' feature negotiation when a client subscribes to an Event at the AMF and requests an immediate report as part of that subscription. When the gparm is set and the client subscription request an immediate report, the AMF optionally can send the first event report in the subscription response, depending on whether the first report is ready by the time the AMF must respond to the subscription. See TS 29.518 CR 0646.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

imsVopsHomogeneousSupport - IMS VoPS Homogeneous Support

Syntax

--amfGparmName imsVopsHomogeneousSupport --amfGparmValue <string>

Description

This parameter controls whether or not the AMF includes the IMS VoPS attribute towards the UDM as part of the Amf3gppAccessRegistration attribute.

List

- DISABLE - Disable
- ENABLE - Enable

Default

DISABLE

Name

inhibitInterAmfN2HoSameTa - Inhibit Inter AMF N2 Handover Source and Target gNBs in same TA

Syntax

--amfGparmName inhibitInterAmfN2HoSameTa --amfGparmValue <string>

Description

When enabled, source AMF inhibits N2 HO to target AMF if source AMF has no N2 link to target gNB and UE current location (TAC) and the target gNB TAC is the same. Source AMF responds back to source gNB with Handover Preparation Failure message.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

initialContextSetupTrigger - Send Initial Context

Syntax

--amfGparmName initialContextSetupTrigger --amfGparmValue <string>

Description

Controls when the AMF sends NGAP Initial Context Setup Request towards the NG-RAN.

List

- normal - AMF sends NGAP Initial Context Setup Request when: - The RAN includes the UE Context Request IE in the NGAP Initial UE Message - Mobility Registration Update or Service Request with UL data status IE is set - Session Management procedure is triggered and Initial Context has not yet been established
- registrationAndServiceRequests - Registration and Service Requests - In addition to the "normal" setting triggers, AMF sends Initial Context Setup Request for initial UE Registration or Service Requests.
- serviceRequests - Service Requests - In addition to the "normal" setting triggers, AMF sends NGAP Initial Context Setup Request for initial UE Service Requests (UE in ECM-IDLE or ECM-CONNECTED state).

Default

normal

Recommended

The recommended value is registrationAndServiceRequests.

Name

interSystemAndEnDcConfTransferOverN26 - Inter-System and EN-DC SON Configuration Transfer Over N26

Syntax

--amfGparmName interSystemAndEnDcConfTransferOverN26 --amfGparmValue <string>

Description

Parameter controls feature activation GPARM flag. By default, flag shall be disabled. When enabled, AMF/MME shall transfer inter-system SON configuration information and EN-DC Configuration Transfer between a gNB and eNodeB.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

intraAmfInterPlmnUdmModRegistration - Intra AMF Inter PLMN UDM Registration Modification

Syntax

```
--amfGparmName intraAmfInterPlmnUdmModRegistration --amfGparmValue  
<string>
```

Description

When set to true and UE moves between Serving PLMNs within the same AMF, the AMF will modify GUAMI IE to the target Serving PLMN GUAMI, within the UE Context Management information in UDM, for given UE.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

intraAmfInterPlmnUdmModSubscription - Intra AMF Inter PLMN UDM Subscription Modification

Syntax

```
--amfGparmName intraAmfInterPlmnUdmModSubscription --amfGparmValue  
<string>
```

Description

When set to true and UE moves between Serving PLMN within the same AMF, the AMF will modify the PLMN ID to the target Serving PLMN for given UE.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

ismf - Intermediate SMF

Syntax

--amfGparmName ismf --amfGparmValue <string>

Description

Enables Support of deployments topologies with specific SMF Service Areas, i.e. the Intermediate SMF (I-SMF) support in the AMF. An Intermediate SMF may be inserted on the signalling path between AMF and the SMF, when the SMF is not able to control the UPF(s) because the UPF(s) belong to a different SMF Service Area. The I-SMF may also be changed or removed due to this during procedures. The I-SMF concept is introduced in 3GPP Release 16.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ismfSelectionFailureMaintain - I-SMF Selection Failure Maintain

Syntax

--amfGparmName ismfSelectionFailureMaintain --amfGparmValue <string>

Description

Defines whether the AMF maintains PDU session in a situation when 3GPP Support of deployments topologies with specific SMF Service Areas i.e. I-SMF is deployed, and AMF detects the need to select a new I-SMF, but no I-SMF supporting the current UE area is available. If set to 'false', AMF releases the PDU session (i.e. triggers Nsmf_PDUSession_UpdateSMContext with release indication towards the old SMF or I-SMF) if no new I-SMF is available for the current area of the UE.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

keepUeContextAfterN26Ho - Keep VLR Entry

Syntax

--amfGparmName keepUeContextAfterN26Ho --amfGparmValue <string>

Description

When this parameter is set to true, keep VLR entry when UDM DeregistrationReason is 5GS_TO_EPS_MOBILITY

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ladn - Local Area Data Network

Syntax

```
--amfGparmName ladm --amfGparmValue <string>
```

Description

Local Area Data Network. This parameter activates the LADN functionality support in AMF. With the LADN functionality, access to a provisioned data network is only provided on a restricted service area. The LADN DNN and the associated LADN Service Area (the Tracking Area list) also need to be provisioned.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ladnNotifications - LADN Notifications

Syntax

```
--amfGparmName ladmNotifications --amfGparmValue <string>
```

Description

Controls whether the AMF triggers Event Notifications on LADN state towards the service consumer.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

linkAlarmCustomize - AMF Link Alarm Customize

Syntax

```
--amfGparmName linkAlarmCustomize --amfGparmValue <string>
```

Description

If the global parameter is set to true, customized severity levels in the AMF Link Alarm Config are used to set the alarm severity for links and interfaces. If the global parameter is set to false, customization of link alarm severity is not supported.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

locationEvents - Location Events

Syntax

```
--amfGparmName locationEvents --amfGparmValue <string>
```

Description

Enables the Namf_Event Exposure in the AMF for subscribing to UE location reporting events (direct or continuous location reporting).

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

maskedImeIsV - Masked IMEISV

Syntax

```
--amfGparmName maskedImeIsV --amfGparmValue <string>
```

Description

Include Masked IMEISV in NGAP messages

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

maxAmfUeRadioCapLength - Maximum AMF Ue Radio Capability Length

Syntax

--amfGparmName maxAmfUeRadioCapLength --amfGparmValue <string>

Description

Maximum Ue Radio Capability Length Supported on the AMF. Default is 9216. Setting this value to 24576 enables storage of Radio capability length between 9216 - 24576 bytes in the RCIS if RCIS is available

Type / Range

Integer in the range of 9,216 or 24,576

Default

9216

Name

maxNumberOfForbiddenTacs - Maximum Number of Forbidden TACs

Syntax

--amfGparmName maxNumberOfForbiddenTacs --amfGparmValue <string>

Description

AMF shall support maximum number of 600 forbidden TACs with the default value of 64.

Type / Range

Integer in the range of 64 or 255 or 600

Default

64

Name

mcs - Mission Critical Services

Syntax

--amfGparmName mcs --amfGparmValue <string>

Description

Enables the support for Mission Critical Services (MCS) on the AMF.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

memoryRestrictionForbiddenTac - Memory Restriction Forbidden TAC

Syntax

```
--amfGparmName memoryRestrictionForbiddenTac --amfGparmValue <string>
```

Description

Parameter to control CPPS VLR memory allocation for feature f20430-02. maxNumberOfForbiddenTacs can be set to larger size (i.e 600) only when this gParm is 'true' A memory restricted gparm that can only be changed, with aggregate service lock. default value is false. memoryRestrictionForbiddenTac cannot be changed in CMM24.7 release. It will be deprecated in CMM24.11

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

mps - Multimedia Priority Services

Syntax

```
--amfGparmName mps --amfGparmValue <string>
```

Description

Enables the support for Multimedia Priority Services (MPS) on the AMF.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

mrldINasTransp - Mobility Restriction List on DL NAS Transport

Syntax

```
--amfGparmName mrldINasTransp --amfGparmValue <string>
```

Description

Controls whether the AMF will only include Mobility Restriction List on DL NAS transport if the list context has been changed, or include even if the content has not changed from previous content.

List

- **AMF_MRL_ON_DL_NAS_TRANSP_CHANGES_ONLY** - Mobility Restriction List on DL NAS Transport changes only - AMF will include Mobility Restriction List on DL NAS Transport only if the Mobility Restriction List content has changed.
- **AMF_MRL_ON_DL_NAS_TRANSP_NO_CHANGE** - Mobility Restriction List on DL NAS Transport no changes - AMF will include Mobility Restriction List on DL NAS Transport even if the Mobility Restriction List content has not changed.

Default

AMF_MRL_ON_DL_NAS_TRANSP_CHANGES_ONLY

Name

mrlSplmnOnlySending - Mobility Restriction List Serving PLMN only sending

Syntax

--amfGparmName mrlSplmnOnlySending --amfGparmValue <string>

Description

Controls whether the AMF includes Mobility Restriction List on NGAP messages (Initial Context Setup Request, Handover Request and DL NAS Transport) with Serving PLMN as the only content, or only if also restrictions are to be communicated to the NG-RAN. (Note: DL NAS transport with only serving PLMN included is only applicable if Mobility Restriction List is sent in every DL NAS Transport.

List

- **AMF_MRL_SPLMN_NOT_ALONE** - Mobility Restriction List Serving PLMN not sent alone - AMF includes Mobility Restriction List on NGAP messages Initial Context Setup Request, Handover Request and DL NAS Transport only if also restrictions are to be communicated to the NG-RAN.
- **AMF_MRL_SPLMN_ONLY_SENT** - Mobility Restriction List Serving PLMN only sent - AMF includes Mobility Restriction List on NGAP messages Initial Context Setup Request, Handover Request and DL NAS Transport, even if the Serving PLMN would be the only content included in the list, i.e. no restrictions are to be communicated to the NG-RAN.

Default

AMF_MRL_SPLMN_ONLY_SENT

Name

multiLevelMps - Multi-Level MPS

Syntax

--amfGparmName multiLevelMps --amfGparmValue <string>

Description

Enables the use of 5 different MPS levels with dedicated SBI Message Priority value for each MPS level. Applies when AMF global parameter Multimedia Priority Services is set to true.

List

- **false** - Boolean value of false.

- true - Boolean value of true.

Default

false

Name

multiSimWithSameMsisdnSupport - Multiple SIM with Same MSISDN Support

Syntax

```
--amfGparmName multiSimWithSameMsisdnSupport --amfGparmValue <string>
```

Description

Parameter controls feature activation. By default, parameter shall be disabled. When enabled, AMF shall support and allow more than one SUPI to register to the AMF by using the same MSISDN. With the default value setting of false, the AMF shall work as it had before the introduction of this feature.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

n26iwkSuppt - N26 interworking Supported

Syntax

```
--amfGparmName n26iwkSuppt --amfGparmValue <string>
```

Description

Enables the support for EPS interworking with N26 interface in the AMF

List

- DISABLE - Disable
- ENABLE - Enable

Default

DISABLE

Recommended

Recommended value is ENABLE

Name

n2HoWithAmfReselect - N2 Handover with AMF Reselection

Syntax

```
--amfGparmName n2HoWithAmfReselect --amfGparmValue <string>
```

Description

This parameter enables the capability of target AMF reselection during inter-AMF N2 handover. The reselection is performed by the initial target AMF when it is not able to serve all requested S-NSSAIs

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

n3gppInitialCtxtNas - Non-3GPP NAS Response with Initial Context

Syntax

--amfGparmName n3gppInitialCtxtNas --amfGparmValue <string>

Description

Controls whether AMF sends NAS response on top of the NGAP Initial Context Setup Request for non-3GPP access. When set to false (default), AMF sends NAS response alone via DL NAS Transport after the Initial Context Setup has been completed. When set to true, AMF sends NAS response on top of the NGAP Initial Context Setup Request.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ngap3gppRelease - NGAP asn.1 3GPP Release

Syntax

--amfGparmName ngap3gppRelease --amfGparmValue <string>

Description

Identifies supported NGAP ASN.1 3GPP Releases.

List

- rel16 - 3GPP R16 NGAP asn.1 - Identifies NGAP asn.1 3GPP Release
- rel17 - 3GPP R17 NGAP asn.1 - Identifies NGAP asn.1 3GPP Release
- mar20 - NGAP asn.1 3GPP March 2020 Release - Identifies NGAP asn.1 3GPP Release

Default

mar20

Name

non3gppAccessImsVopsHomogeneous - Non-3GPP Access N3IWF IMS VoPS Homogeneous

Syntax

```
--amfGparmName non3gppAccessImsVopsHomogeneous --amfGparmValue <string>
```

Description

This parameter controls whether or not the AMF applies UECM Update service operation towards UDM due to imsVoPs homogeneous supports change for a UE that is registered over untrusted non-3GPP access via N3IWF

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

non3gppAccessN3iwf - Non-3GPP access N3IWF

Syntax

```
--amfGparmName non3gppAccessN3iwf --amfGparmValue <string>
```

Description

Controls the support of non-3GPP access and the Non-3GPP Access Interworking Function (N3IWF) in the AMF.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

non3gppAccessN3iwfRatType - AMF Non-3GPP Access N3IWF RAT Types

Syntax

```
--amfGparmName non3gppAccessN3iwfRatType --amfGparmValue <string>
```

Description

Defines the RAT type the AMF supports for UEs connected via untrusted non-3GPP access and N3IWF to the AMF.

List

- AmfN3iwfRatTypeVirtual - AMF N3IWF RAT Type VIRTUAL - AMF supports RAT type VIRTUAL for untrusted non-3GPP access for UEs connected via N3IWF to AMF.

- AmfN3iwfRatTypeWlan - AMF N3IWF RAT Type WLAN - AMF supports RAT type WLAN for untrusted non-3GPP access for UEs connected via N3IWF to AMF.

Default

AmfN3iwfRatTypeWlan

Name

non3gppAccessVoiceN3iwf - Non-3GPP Access Voice N3IWF

Syntax

--amfGparmName non3gppAccessVoiceN3iwf --amfGparmValue <string>

Description

Enables support for IMS Voice over PS sessions over untrusted non-3GPP access via N3IWF.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

non3gppEpsPduHo - Non-3GPP Access EPS PDU Handover

Syntax

--amfGparmName non3gppEpsPduHo --amfGparmValue <string>

Description

Enables the AMF support for UE triggered non-emergency PDU handovers between untrusted non-3GPP via N3IWF/5GS and E-UTRAN/EPS.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

non3gppNpliN3iwf - Non-3GPP NPLI N3IWF

Syntax

--amfGparmName non3gppNpliN3iwf --amfGparmValue <string>

Description

Enables the AMF support for Network Provided Location Information (NPLI) for untrusted non-3GPP access via N3IWF.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

nonIpPdnSupportN26 - Non IP PDN Support N26

Syntax

--amfGparmName nonIpPdnSupportN26 --amfGparmValue <string>

Description

Parameter controls AMF indicating in Context Request if Non IP PDN is supported for EPS to 5GS idle mobility scenario. By default, the flag shall be enabled.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

nrDualConnectivity - NR Dual Connectivity

Syntax

--amfGparmName nrDualConnectivity --amfGparmValue <string>

Description

Enables NR Dual Connectivity NR-DC and Secondary RAT Usage Reporting. The NR Dual Connectivity profile, when configured, administers NR-DC support, Secondary RAT Usage Reporting restriction and UDM NR Secondary RAT Access restriction override per UE PLMN Services or SUPI Range Services.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

nrfDefaultLoadValueForNfs - Default Load Value for NFs

Syntax

--amfGparmName nrfDefaultLoadValueForNfs --amfGparmValue <string>

Description

Determines default 'load' value to be assumed for a Service instance when the 'load' information is not received from NRF. Load is used by the AMF along with Locality, Priority and Capacity in order to select/load balance traffic among multiple NF Service Instances that meet the given criteria.

Type / Range

Integer in the range of 0 to 100

Default

0

Name

nrfHeartbeatFailures - NRF Heartbeat Failures

Syntax

--amfGparmName nrfHeartbeatFailures --amfGparmValue <string>

Description

This parameter is used to provision maximum number of consecutive heartbeat failures to NRF. If AMF does not receive responses to heartbeat messages consecutively specified by this parameter then, AMF stops heartbeat messages and tries to re-register to this NRF. If registration is successful, AMF again tries heartbeat messages. If registration is unsuccessful, AMF tries to register to another NRF.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

nrfUseNFLoadInfo - NRF Use Network Function Load Information

Syntax

--amfGparmName nrfUseNFLoadInfo --amfGparmValue <string>

Description

Determines whether 'load' will be taken into account when applying load balancing among candidate Network Functions / Service Instances. The 'load' indicates the current load percentage of the NF/Service Instance and is acquired from the NRF during NF Discovery and NF Profile Notification Update. Parameter values: - No (default): load is not taken into account - Yes: load is taken into account

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

nssaaEnable - NSSAA

Syntax

--amfGparmName nssaaEnable --amfGparmValue <string>

Description

A control parameter to enable Network Slice-Specific Authentication and Authorization (NSSAA). If the parameter set to true, then AMF shall support NSSAA. Otherwise, AMF will not support it

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

nssfNsSelectionLoadBalancing - NSSF NSSelection Load Balancing

Syntax

--amfGparmName nssfNsSelectionLoadBalancing --amfGparmValue <string>

Description

This parameter controls whether load balancing will be applied among multiple Nnssf_NSSelection service instances. When load balancing is enabled, load balancing will be applied among the highest priority available service instances based on their relative capacity.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

nullSchemeSuciUseSupi - Null-Scheme SUCI Use SUPI

Syntax

--amfGparmName nullSchemeSuciUseSupi --amfGparmValue <string>

Description

When enabled, the AMF derives SUPI from Null-Scheme SUCI and include the SUPI: - in the query to the NRF to select AUSF - in the Authentication Request to the AUSF to authenticate the UE

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

nwdafEnable - NWDAF Enable

Syntax

```
--amfGparmName nwdafEnable --amfGparmValue <string>
```

Description

A control parameter to enable AMF access to an NWDAF in the home PLMN. If set false, the AMF will not use the SBI towards the NWDAF.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

oauthSupport - OAuth Support

Syntax

```
--amfGparmName oauthSupport --amfGparmValue <string>
```

Description

Parameter controls the feature activation. If parameter is enabled, OAuth based Authorization of Network Function Service Access shall be supported. By default this parameter shall be disabled.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

pageForN1SmInUpdRsp - Page for N1Sm in Update Response

Syntax

```
--amfGparmName pageForN1SmInUpdRsp --amfGparmValue <string>
```

Description

Current parameter enables paging the UE when receive N1Sm in Update Sm Context Response and the UE is in CM-IDLE state.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

pageOn3gppIfNon3gppFail - Page UE on 3GPP if non-3GPP fail

Syntax

--amfGparmName pageOn3gppIfNon3gppFail --amfGparmValue <string>

Description

Defines AMF action when UE is CM-CONNECTED for non-3GPP access and CM-IDLE on 3GPP access, and AMF has sent Notification on non-3GPP access to re-establish user plane or deliver signaling from SMF for 3GPP access, but AMF timer T3565 supervising the Notification Response, Service Request or Registration Request has expired. When set to 'false', AMF notifies the SMF that the UE is unreachable. When set to 'true', AMF pages the UE on 3GPP access.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

pageRrcInactiveUe - Page UE in RRC Inactive state

Syntax

--amfGparmName pageRrcInactiveUe --amfGparmValue <string>

Description

Enables paging UE in RRC Inactive state feature. When set to true, AMF invokes paging for UEs UE in RRC Inactive state if RAN paging fails.

Restrictions

This gParm can only be set to true if gParm rrcInactive is set to true.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Recommended

false

Name

pagingGapFailCountTrigger - Paging Gap Fail Count Trigger

Syntax

```
--amfGparmName pagingGapFailCountTrigger --amfGparmValue <string>
```

Description

Parameter AMF Paging Gap Fail Count Trigger indicates the number of failed paging operations that start the AMF Paging Gap Timer.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

pagingOverloadPrevention - Paging Overload Prevention

Syntax

```
--amfGparmName pagingOverloadPrevention --amfGparmValue <string>
```

Description

Parameter controls feature activation related to monitoring AMF paging activity and taking action (when necessary) to avoid an overload condition. By default, flag shall be always enabled.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

peips - PEIPS Active

Syntax

```
--amfGparmName peips --amfGparmValue <string>
```

Description

Specifies whether the PEIPS feature (Paging Early Indication with Paging Subgroups) is active on the AMF.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

peiRetrievalMethod - Query PEI method

Syntax

--amfGparmName peiRetrievalMethod --amfGparmValue <string>

Description

Specify method to retrieve PEI from the UE. If the parameter is set to none, PEI is retrieved via identity request procedure if/when needed.

List

- idr - Identity Request - Retrieve PEI via Identity Request
- none - No PEI retrieval
- smc - Security Mode Command - Retrieve PEI via Security Mode Command

Default

smc

Name

pmInclude5qi - Include 5QI PM Counters

Syntax

--amfGparmName pmInclude5qi --amfGparmValue <string>

Description

This parameter controls whether the PM counters relevant to 5QI would be pegged from the system. When this parameter is set, the system is able to extract the 5QI information needed to peg the corresponding counters.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

praPcfTrigger - PRA PCF Trigger

Syntax

--amfGparmName praPcfTrigger --amfGparmValue <string>

Description

Enables the AMF support for Presence Reporting Area changes triggered from PCF (for 3GPP access).

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

presenceInAoI - Presence in Area of Interest

Syntax

--amfGparmName presenceInAoI --amfGparmValue <string>

Description

Enables the Presence in Area of Interest in the AMF. The AMF provides this functionality via the Namf_EventExposure service. The service consumer may define an Area of Interest and the AMF reports back the presence of the UE on the area.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

primaryHttpConnectionWeight - Primary HTTP connection Weight

Syntax

--amfGparmName primaryHttpConnectionWeight --amfGparmValue <string>

Description

Defines how AMF will load balance traffic among 2 HTTP connections to the same SBI Server endpoint. Primary connection will have weight equal to primaryHttpConnectionWeight. Secondary connection will have weight equal to 100 - primaryHttpConnectionWeight. Default is 50 which balances UE related traffic over the two connections on a per-UE basis and non-UE related traffic on a random basis.

Type / Range

Integer in the range of 50 to 100

Default

50

Name

prioritizeLinkStatesForCallbacks - Prioritize Link States for Callbacks

Syntax

```
--amfGparmName prioritizeLinkStatesForCallbacks --amfGparmValue  
<string>
```

Description

When this parameter is set to true, when CMM is sending a Request to a callback URI (Subsequent Request or a Notification), it will prefer to use Endpoints with no link (Not Existing) or Endpoints that have links in Enabled and Unlocked state. Endpoints with links Under Creation (UNDEFINED) will be used only if no other valid Endpoint is available.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

prioritizeLinkStatesForInitialRequests - Prioritize Link States for Initial Requests

Syntax

```
--amfGparmName prioritizeLinkStatesForInitialRequests --amfGparmValue  
<string>
```

Description

When this parameter is set to true, when CMM is sending an Initial Request in Direct communication, it will prefer to use Endpoints with no link (Not Existing) or Endpoints that have links in Enabled and Unlocked state. Endpoints with links Under Creation (UNDEFINED) will be used only if no other valid Endpoint is available.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

prioritizeLinkStatesForScps - Prioritize Link States for SCPs

Syntax

```
--amfGparmName prioritizeLinkStatesForScps --amfGparmValue <string>
```

Description

When this parameter is set to true, when CMM is sending an Request in Indirect Direct communication, it will prefer to use Endpoints with no link (Not Existing) or Endpoints that have links in Enabled and Unlocked state. Endpoints with links Under Creation (UNDEFINED) will be used only if no other valid Endpoint is available.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

radioCapInfoForPaging - UE radio capability for paging

Syntax

--amfGparmName radioCapInfoForPaging --amfGparmValue <string>

Description

Enable/disable inclusion of radio capability for paging IE in the: - NGAP PAGING message - Initial UE Context Setup Request message when RRC Inactive feature is enabled

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

reattemptN26CtxReqCc92 - Reattempt N26 Context Request when receive CC92

Syntax

--amfGparmName reattemptN26CtxReqCc92 --amfGparmValue <string>

Description

When current parameter is enabled and the AMF receives N26 Context Response with CC92, the AMF verifies UE identity, possibly by doing reauthentication, and attempts again N26 Context Request to source MME

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

reattemptN26CtxReqCc96 - Reattempt N26 Context Request when receive CC96

Syntax

--amfGparmName reattemptN26CtxReqCc96 --amfGparmValue <string>

Description

When current parameter is enabled and the AMF receives N26 Context Response with CC96, the AMF verifies UE identity, possibly by doing reauthentication, and attempts again N26 Context Request to source MME

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

restrictBroadcastCmmHomePlmn - Restrict Broadcasting CMM Home PLMN

Syntax

--amfGparmName restrictBroadcastCmmHomePlmn --amfGparmValue <string>

Description

Current parameter enables restrict sending CMM 'Home' PLMN in NG Setup and NRF Register procedures

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

reuseOldUeContextEpsTo5gs - Reuse Old UE Context When UE Moves to 5GS

Syntax

--amfGparmName reuseOldUeContextEpsTo5gs --amfGparmValue <string>

Description

With this parameter, the AMF reuses the subscription data of the old UE Context that were locally preserved when UE moved to EPS and skips SDM GET and SDM Subscribe operations towards UDM

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

reuseRequestApiRootInLocationHeader - Reuse Request apiRoot in Location Header

Syntax

```
--amfGparmName reuseRequestApiRootInLocationHeader --amfGparmValue  
<string>
```

Description

When this parameter is set to true and the Response includes a Location header, CMM will construct the apiRoot of the Location URI by reusing apiRoot included in the Request URI. When this parameter is false the apiRoot is deduced from the relevant service Instance provisioning.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

rfspAccessRestriction - RFSP Access Restriction

Syntax

```
--amfGparmName rfspAccessRestriction --amfGparmValue <string>
```

Description

Feature control for RFSP Access Restriction for the AMF.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

rimInformationTransfer - RIM Information Transfer

Syntax

```
--amfGparmName rimInformationTransfer --amfGparmValue <string>
```

Description

Enables NGAP RIM Information Transfer Procedure. When set to false, AMF silently discards the NGAP Uplink RIM Information Transfer message.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

rrcInactive - RRC Inactive

Syntax

--amfGparmName rrcInactive --amfGparmValue <string>

Description

Enables RRC Inactive feature. When enabled, AMF provides Core Network Assistance information to the RAN to assist gNB in moving UE(s) to RRC Inactive state

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sbiHttp2PingKeepAlive - SBI HTTP2 Ping Keep Alive

Syntax

--amfGparmName sbiHttp2PingKeepAlive --amfGparmValue <string>

Description

Enables or disables the impact of a received HTTP2 ping upon SBI link availability keep alive. When value is set to false a received HTTP2 ping will have no impact on SBI link availability. When value is set to true a received HTTP2 ping will reset SBI link keep alive, like other traffic messages.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sbiQueryConversions - SBI Query Parameter Conversions

Syntax

--amfGparmName sbiQueryConversions --amfGparmValue <string>

Description

Enables or disables Query Conversion Rules. When value is set to false no Query conversion rule is applied. When value is set to true Query conversion rules are applied according to their Rule Status.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

selectiveN15UsageControl - Selective N15 Usage Control

Syntax

```
--amfGparmName selectiveN15UsageControl --amfGparmValue <string>
```

Description

Parameter controls feature activation flag. By default, the flag shall be disabled. When the flag is disabled, for homers AM and UE policy shall be created (old behavior) and for roamers, interPlmnPcfSupport shall be checked to decide if AM/UE policy need to be created or not (old behaviour). When gParm is enabled, for all users, pcfPolicyAssociation of amfSvcAgreement profile shall be checked to decide whether to establish AM and UE policy.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

send3gppSbiCallbackInAllNotifications - Send 3gpp-Sbi-Callback in all notifications

Syntax

```
--amfGparmName send3gppSbiCallbackInAllNotifications --amfGparmValue  
<string>
```

Description

When this parameter is false the 3gpp-Sbi-Callback header is included only in Notifications sent via Indirect Communication. When the gParm is true the 3gpp-Sbi-Callback header is included Notifications for both Direct and Indirect Communication.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Recommended

true

Name

send3gppSbiOriginatingNetworkId - Send 3gpp-Sbi-Originating-Network-Id

Syntax

```
--amfGparmName send3gppSbiOriginatingNetworkId --amfGparmValue  
<string>
```

Description

When parameter is false, AMF does not include the 3gpp-Sbi-Originating-Network-Id header in Requests. When parameter is true, AMF includes 3gpp-Sbi-Originating-Network-Id header in Requests.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sendAmfConfigUpdTotalConfigMismatch - Send AMF Configuration Update when Total Configuration Mismatch with gNB

Syntax

```
--amfGparmName sendAmfConfigUpdTotalConfigMismatch --amfGparmValue  
<string>
```

Description

When this parameter is enabled, AMF will send AMF Configuration Update to a gNB with which there is total configuration mismatch, following a local configuration update.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sendAreaCodeBasedExtEmerNumberInRegistrationAccept - Send Area Code Based Extended Emergency Number in Registration Accept

Syntax

```
--amfGparmName sendAreaCodeBasedExtEmerNumberInRegistrationAccept --  
amfGparmValue <string>
```

Description

Parameter controls feature activation GPARM flag. By default, flag shall be disabled. When enabled, AMF shall send Extended Emergency Numbers List IE in the REGISTRATION ACCEPT messages if one is provisioned on the AMF.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sendCagPaging - Send CAG Information List in Paging

Syntax

```
--amfGparmName sendCagPaging --amfGparmValue <string>
```

Description

When enabled, the AMF sends the CAG Information List in Paging message

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sendEmeRegInd - Send Emergency Registration Indication

Syntax

```
--amfGparmName sendEmeRegInd --amfGparmValue <string>
```

Description

The parameter enables sending the emergencyRegistrationInd IE in Amf3GppAccessRegistration data of UECM Registration to indicate emergency Registration to the UDM. This IE is sent in Nudm_UECM SBI v1.2.0-alpha.6 onwards

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Recommended

true

Name

sendEmptyCagUe - Send Empty CAG Information List to the UE

Syntax

```
--amfGparmName sendEmptyCagUe --amfGparmValue <string>
```

Description

When enabled, the AMF sends empty CAG Information List in NAS Accept when no PLMN CAG is subscribed for CAG capable UE

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sendEpsIwkUpd - sendEpsIwkUpd

Syntax

```
--amfGparmName sendEpsIwkUpd --amfGparmValue <string>
```

Description

When this parameter is enabled and the status of interworking with EPS for a PDU session changes due to change of UE 5GMM capability e.g. S1 mode supported, the AMF invokes Nsmf_PDUSession_UpdateSMContext to send updated EPS Interworking Indication to the SMF

List

- neverSend - If status of interworking with EPS for a PDU session changes due to change of UE 5GMM capability e.g. S1 mode supported, the AMF doesn't invoke Nsmf_PDUSession_UpdateSMContext to send updated EPS Interworking Indication to SMF.
- sendAlways - If status of interworking with EPS for a PDU session changes due to change of UE 5GMM capability e.g. S1 mode supported, the AMF always invokes Nsmf_PDUSession_UpdateSMContext to send updated EPS Interworking Indication to SMF no matter if it is combo SMF or standalone SMF.

Default

neverSend

Name

sendExtEmerNumberInRegistrationAccept - Send Extended Emergency Number in Registration Accept

Syntax

```
--amfGparmName sendExtEmerNumberInRegistrationAccept --amfGparmValue <string>
```

Description

Parameter controls feature activation GPARM flag. By default, flag shall be disabled. When enabled, AMF shall send Extended Emergency Numbers List IE in the REGISTRATION ACCEPT messages if one is provisioned on the AMF.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sendMappedGummei - sendMappedGummei

Syntax

```
--amfGparmName sendMappedGummei --amfGparmValue <string>
```

Description

Current parameter enables sending in NGAP Setup Response the mapped GUAMI(s) pointing to the MME GUMMEI(s) for the associated PLMN(s) that coexist in both 4G and 5G at local CMM node

List

- DISABLE - Disable
- ENABLE - Enable

Default

DISABLE

Name

sendNsci - Send NSCI IE

Syntax

```
--amfGparmName sendNsci --amfGparmValue <string>
```

Description

Current parameter enables sending 'New Security Context Indicator' NSCI IE in Handover Request, as part of EPS to 5GS handover

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sendOnlyCommonlySupportedSlicesToAn - Send Only Commonly Supported Slices to AN

Syntax

```
--amfGparmName sendOnlyCommonlySupportedSlicesToAn --amfGparmValue <string>
```

Description

When this parameter is true, the AMF will send only the intersection of the slices the AMF supports and the slices the gNB/ N3IWF supports for that PLMN (i.e. the union of all slices supported in all TAs the gNB/N3IWF supports per PLMN). Note: In the latter case the gNB/N3IWF will not be aware of all the slices the AMF supports, however it can resolve interworking issues when RAN cannot handle processing the all the slices included in the PLMN support list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sendOnlyRanSupportedPlmnsInN2 - Send Only RAN Supported PLMNs in N2

Syntax

```
--amfGparmName sendOnlyRanSupportedPlmnsInN2 --amfGparmValue <string>
```

Description

Current parameter enables sending in SETUP RSP and AMF Configuration Update only the PLMNs that are supported by the RAN even if additional PLMNs are configured at the AMF.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sendPc5ToNgran - Send PC5 to NG RAN

Syntax

```
--amfGparmName sendPc5ToNgran --amfGparmValue <string>
```

Description

Specifies whether the AMF is to send the UE PC5 QoS, if available, in applicable 38.413 messages

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sendPeiTypeImeiToEir - Send PEI Type IMEI to EIR

Syntax

```
--amfGparmName sendPeiTypeImeiToEir --amfGparmValue <string>
```

Description

Parameter controls the option of sending PEI value type IMEI that indicates IMEI to EIR even though IMEISV is available. By default, flag shall be disabled.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sendPeiTypeImeiToSmf - Send PEI Type IMEI to SMF

Syntax

```
--amfGparmName sendPeiTypeImeiToSmf --amfGparmValue <string>
```

Description

Parameter controls the option of sending PEI value type IMEI that indicates IMEI to SMF even though IMEISV is available. By default, flag shall be disabled.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sendPendingSnssaiInAllowedNssaiOfIcsr - Send Pending SNSSAI in Allowed NSSAI of ICSR

Syntax

```
--amfGparmName sendPendingSnssaiInAllowedNssaiOfIcsr --amfGparmValue  
<string>
```

Description

Current parameter enables sending a pending SNSSAI (random selection) in Allowed NSSAI IE of the Initial Context Setup Request when latter needs be sent but there is not yet an allowed SNSSAI due to the NSSAA feature.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Recommended

true

Name

sendRlsSessionInfo - Send Release Session Info

Syntax

```
--amfGparmName sendRlsSessionInfo --amfGparmValue <string>
```

Description

Current parameter enables sending the toReleaseSessionInfo attribute in Registration Status Update request.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Recommended

true

Name

sendSkipN2PduSessionResRelInd - Send Skip N2 PDU Session Resource Release Indication

Syntax

```
--amfGparmName sendSkipN2PduSessionResRelInd --amfGparmValue <string>
```

Description

Current parameter enables sending Update SM Context Request carrying the skipN2PduSessionResRelInd parameter, instead of Release SM Context Request when a PDU Session needs to be released towards the UE at completion of handover e.g. because the associated S-NSSAI isn't supported at target TAI, but not towards the target gNB because a User Plane isn't established at the target gNB.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Name

sendSnpnToUnsupportedNgRan - Send SNPN Info to Unsupported NG-RAN

Syntax

--amfGparmName sendSnpnToUnsupportedNgRan --amfGparmValue <string>

Description

Current parameter enables sending the supported SNPNs in Setup Response and AMF Configuration Update towards NGRAN nodes that haven't indicated support of SNPN type of network in NG Setup or NG Configuration Update

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sendTargetNfIdInRedirect - Send 3gpp-sbi-Target-Nf-Id in Redirect

Syntax

--amfGparmName sendTargetNfIdInRedirect --amfGparmValue <string>

Description

When this parameter is set to true the AMF adds the target AMF NF Instance and service instance Id in the relevant 3gpp-sbi-Target-Nf-Id HTTP header.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sharedPlmnSupport - Shared PLMN support

Syntax

--amfGparmName sharedPlmnSupport --amfGparmValue <string>

Description

Enables the support for Multiple Shared PLMN in the AMF. When the flag is set to TRUE, it shall be possible to set the multiple shared plmn for up to 8 PLMNs.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

smContextStatusNotifyRequired - SM Context Status Notify Required

Syntax

--amfGparmName smContextStatusNotifyRequired --amfGparmValue <string>

Description

This parameter defines whether the AMF requires an SM context status notify to be triggered from the SMF before the AMF deletes the PDU session. The parameter is applied during -UE/AMF/SMF triggered PDU session release procedures, -PDU session setup procedures that have been rejected towards the UE after SM context create, and -PDU session setup procedures when the AMF already has a PDU session established for the requested PDU session id. Upon SBI timer pduSmCtxStatusNotify expiry, the AMF behavior is as follows: When set to true, the AMF only clears internal temporary data. AMF maintains the PDU session context. (AMF removes the PDU session only when it receives the SM context status notify with resourceStatusInfo set to 'RELEASED' on statusInfo inside SmContextStatusNotification.) When set to false, the AMF clears internal temporary data, and also removes the PDU context (without waiting for the SM context status notify.)

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

smfAreaOfValidity - SMF Area of Validity

Syntax

--amfGparmName smfAreaOfValidity --amfGparmValue <string>

Description

Defines how the AMF applies the SMF optionally provided Area of Validity parameter on Namf_Communication_N1N2MessageTransfer request. AMF supports receiving up to 100 TAs.

List

- SmCtxtUpdateOptimization - SM Context Update Optimization - The AMF does not trigger i.e. skips the SM context update when the Service request is triggered by Paging

or NAS Notification indicating 3GPP access and the UE does not include Uplink data Status (despite of the Area of validity being included or not). Also if the UE was in CM-CONNECTED state on the access related to the PDU session id, the AMF does not trigger SM context update. This solution assumes that the N2 container is always valid and Area of validity is not required to be present on the SMF request.

- **ValidateTheArea** - Validate the Area - When the UE does not include the Uplink Data Status IE on the Service Request that was triggered by Paging or NAS Notification indicating 3GPP access, the AMF triggers the SM Context Update when the Area of Validity is either missing or the UE is outside the SMF provided Area of Validity. Also if the UE was in CM-CONNECTED state on the access related to the PDU session id, the AMF triggers SM context update if the UE either is outside the area of validity or the area of validity is not provided. This solution requires that the SMF includes the Area of validity parameter.

Default

SmCtxtUpdateOptimization

Name

smfRecoveryTime - SMF Recovery Time

Syntax

```
--amfGparmName smfRecoveryTime --amfGparmValue <string>
```

Description

Enables the AMF to use the RecoveryTime attribute related to an SMF (service) instance to detect whether the SMF (service) instance has recently restarted and lost the resources created prior to the indicated recovery timestamp.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

smpBasedPriorityFromSmsf - SMP based Priority from SMSF

Syntax

```
--amfGparmName smpBasedPriorityFromSmsf --amfGparmValue <string>
```

Description

Enables AMF to apply priority handling for UEs receiving SBI message Priority value matching internal provisioning from SMSF.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Name

smsfReselectionAfterUeProcedure - SMSF Reselection After UE procedure

Syntax

```
--amfGparmName smsfReselectionAfterUeProcedure --amfGparmValue  
<string>
```

Description

Enables the AMF to reselect SMSF for all impacted UEs, after AMF has detected an SMSF unavailable. The AMF reselects the SMSF for any impacted UEs after any next procedure for the impacted UEs.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

smsfReselections - SMSF Reselections

Syntax

```
--amfGparmName smsfReselections --amfGparmValue <string>
```

Description

The number of attempted SMSF reselections per a single UE.

Type / Range

Integer in the range of 0 to 5

Default

1

Name

smsOverNas - SMS Over NAS

Syntax

```
--amfGparmName smsOverNas --amfGparmValue <string>
```

Description

Activates the SMS over NAS feature. The AMF supports the SMS delivery over NAS signalling and further via the N20 interface towards the SMSF.

List

- DISABLE - Disable
- ENABLE - Enable

Default

DISABLE

Name

smsRestrictionVoiceCentricUes - SMS Restriction Voice Centric UEs

Syntax`--amfGparmName smsRestrictionVoiceCentricUes --amfGparmValue <string>`**Description**

When this parameter is set to true, the AMF restricts the SMS over NAS service for UEs indicating 'voice centric' on UE's usage setting IE on NAS Registration request. Similar restriction also needs to be set per UE PLMN/serving PLMN combination. It is assumed that these UEs will get SMS service via IMS (and do not need SMS over NAS to be allowed). When this parameter is set to false, the SMS over NAS service is controlled in the AMF local provisioning by the AMF global parameter SMS Over NAS, the SMS Over NAS Allowed parameter per UE PLMN/serving PLMN combination and the SMS subscribed indication from UDM.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sorForTahs - SoR for TAHS

Syntax`--amfGparmName sorForTahs --amfGparmValue <string>`**Description**

Current parameter enables delivering the Steering of Roaming Information to TAHS subscriber during the Registration procedure, when Steering of Roaming subscription data is delivered to the AMF from the UDM.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

srsInitialRegRelocation - Initial UE Registration With SRS

Syntax

```
--amfGparmName srsInitialRegRelocation --amfGparmValue <string>
```

Description

Enables new AMF to retrieve UE Context from session restoration server during inter AMF UE initial registration with GUTI.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

srvccPossibleNormallyRegisteredEmergency - 5G-SRVCC Possible Normally Registered Emergency

Syntax

```
--amfGparmName srvccPossibleNormallyRegisteredEmergency --  
amfGparmValue <string>
```

Description

Defines whether the AMF temporarily indicates on 5G-SRVCC Operation Possible indication the value 'Possible' when a normally registered UE supports 5G-SRVCC and establishes an emergency PDU session.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

standaloneNonPublicNetworkSupport - Standalone Non-Public Network Support

Syntax

```
--amfGparmName standaloneNonPublicNetworkSupport --amfGparmValue  
<string>
```

Description

Parameter controls feature activation flag. By default, flag shall be disabled. When enabled, AMF shall support 3GPP specified Release 16 Stand alone Non Public Network functionality.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

subscribedEmergencySnssai - Subscribed Emergency S-NSSAI

Syntax

```
--amfGparmName subscribedEmergencySnssai --amfGparmValue <string>
```

Description

Current parameter enables using the S-NSSAI from subscribed emergency DNN if such exists in subscription data

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

subscribedForbiddenTacChangeAction - Subscribed Forbidden TAC Change Action

Syntax

```
--amfGparmName subscribedForbiddenTacChangeAction --amfGparmValue  
<string>
```

Description

Parameter controls behavior invocation whether to release the N2 connection to put UE into idle mode or deregister the UE upon receipt of UDM Notification with updated list of subscribed forbidden Areas/TACs which includes the current UE TAC, and the UE is in connected mode. By default, parameter shall be set to none.

List

- deregister - Deregister the UE
- n2Release - N2 release - Release of N2
- none - None

Default

none

Name

support5gAnSliceInconsistencyAlarm - Support 5G-AN Slice inconsistency Alarm

Syntax

```
--amfGparmName support5gAnSliceInconsistencyAlarm --amfGparmValue  
<string>
```

Description

When this parameter is enabled, AMF will raise a Alarm when multiple 5G-AN network elements (gNB, N3IWF) report different slices for the same Tracking Area.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

support5gSrvcc - Support 5G-SRVCC

Syntax

```
--amfGparmName support5gSrvcc --amfGparmValue <string>
```

Description

Enables the AMF support for 5G-SRVCC to hand over regular voice calls from NG-RAN/5GS to 3GPP UTRAN/CS domain.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

support5gSrvccEmergency - Support 5G-SRVCC Emergency

Syntax

```
--amfGparmName support5gSrvccEmergency --amfGparmValue <string>
```

Description

Enables the AMF support for 5G-SRVCC to hand over emergency calls from NG-RAN/5GS to 3GPP UTRAN/CS domain.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

supportAusfUdmGroupId - Support AUSF UDM GroupId

Syntax

```
--amfGparmName supportAusfUdmGroupId --amfGparmValue <string>
```

Description

When this parameter is set to true and the GroupId is included in the Profile of a AUSF or UDM selected for a UE, AMF will store it in the UE's context. The GroupId is accordingly used for subsequent AUSF/UDM selection and conveyed in case of inter AMF mobility.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

suppressRetryWithNudmSdmSingleGet - Suppress retry with SDM single Get

Syntax

```
--amfGparmName suppressRetryWithNudmSdmSingleGet --amfGparmValue  
<string>
```

Description

When set to true, the AMF will not retry with nudm-sdm single GET to the same UDM on receipt of an error response to a nudm-sdm multi dataset GET from the UDM with status code 404 DATA NOT FOUND or any other status code failure.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Recommended

The value should be set to true if the UDM doesn't respond with error response with 404 DATA NOT FOUND to a nudm-sdm multi-data GET request if one of the requested datasets is not available.

Name

tappingActivated - Status of AMF Tapping functionality

Syntax

```
--amfGparmName tappingActivated --amfGparmValue <string>
```

Description

This parameter controls the activated/deactivated status of AMF Tapping

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

timeBasedAuthFrequency - Time Based Authentication Frequency

Syntax

--amfGparmName timeBasedAuthFrequency --amfGparmValue <string>

Description

Parameter controls feature activation global parameter flag. By default, the flag shall be disabled. When enabled, AMF shall support time-based authentication in accordance with new option(s) defined in AMF UE PLMN Services configuration.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

timerSmfReselection - Timer SMF Reselection

Syntax

--amfGparmName timerSmfReselection --amfGparmValue <string>

Description

Enables the functionality to reselect SMF based on a timer during SM context creates. When timer based reselection is set to false, AMF only reselects SMF once per procedure on error cases (i.e. decision to reselect is attempt based). The used timer is deducted from SBI SMF response timer. Only when this parameter is set to 'true', the AMF applies the parameters errorSource, n1Included, reselectIsmfVsmf, reselectOnlySmf and reselectRemote on cmm amfSmfNasMapping command.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

trigUePolicyWhenNoUePolicyContainer - Trigger UE Policy When No UE Policy Container

Syntax

```
--amfGparmName trigUePolicyWhenNoUePolicyContainer --amfGparmValue <string>
```

Description

Parameter controls establishment of UE Policy Association based on received UE policy container from UE. By default, the flag shall be disabled. When enabled, if no UE Policy Container is received during UE Policy Association Establishment procedure the AMF will establish UE Policy Association.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

udmBasedPscfRestoration - UDM based PSCF Restoration

Syntax

```
--amfGparmName udmBasedPscfRestoration --amfGparmValue <string>
```

Description

Parameter controls feature activation GPARM flag. By default, flag shall be disabled. When enabled, AMF upon receiving P-CSCF restoration indication from UDM shall initiate a network triggered PDU Session Release procedure of the IMS PDU session with Reactivation indication and accordingly UE shall request to re-establish IMS PDU session and shall discover P-CSCF during IMS PDU session establishment.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

udmImmediateReport - UDM Immediate Report

Syntax

```
--amfGparmName udmImmediateReport --amfGparmValue <string>
```

Description

Immediate report indication to UDM

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

udmInitiatedCallTrace - UDM Initiated Call Trace

Syntax

--amfGparmName udmInitiatedCallTrace --amfGparmValue <string>

Description

Support UDM initiated call trace or not.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

udmMdt - UDM MDT

Syntax

--amfGparmName udmMdt --amfGparmValue <string>

Description

Activation/deactivation minimization of drive tests.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ueAmbrChangeFromPcf - UE AMBR Change From PCF

Syntax

--amfGparmName ueAmbrChangeFromPcf --amfGparmValue <string>

Description

Parameter controls whether to send UDM AMBR update to PCF in order for PCF to determine if AMBR update should occur. By default, flag shall be disabled. When enabled, AMF shall send UDM AMBR update to PCF.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ueRadioCapPagingN14 - Ue Radio Capability for Paging in N14

Syntax

--amfGparmName ueRadioCapPagingN14 --amfGparmValue <string>

Description

Enable sending Ue Radio Capability for Paging IE in N14 interface

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

uniqueIdentifierSupportInSubscriptionId - Unique Identifier Support In Subscription Id

Syntax

--amfGparmName uniqueIdentifierSupportInSubscriptionId --
amfGparmValue <string>

Description

Parameter controls feature activation flag. By default, flag shall be disabled. When enabled, support for unique identifier support in subscription is provided so the cell broadcast centre function can filter/sort which amf has provided subscription to it.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

useAdditionalMemory - Use Additional Memory

Syntax

--amfGparmName useAdditionalMemory --amfGparmValue <string>

Description

When the CPPS has been configured with additional memory, this parameter controls use of the additional memory for selected features. Use of additional memory for a feature must be indicated with this parameter before that feature can be activated.

List

- frbtac - Forbidden Tac - Use additional memory for Forbidden TAC on CPPS.
- none - No additional memory needed for any features.

Default

none

Name

useDnsForN26Ho - Use Dns for N26 handover

Syntax

```
--amfGparmName useDnsForN26Ho --amfGparmValue <string>
```

Description

Current parameter enables using DNS for selecting target node as part of N26 handover even when local MME of quad access node serves target RAN.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

useFqdnAsAuthorityInAllSbiInterfaces - Use FQDN as Authoruty in All SBI Interfaces

Syntax

```
--amfGparmName useFqdnAsAuthorityInAllSbiInterfaces --amfGparmValue  
<string>
```

Description

When this parameter is set to true, and FQDN of the target URI is available, CMM uses the FQDN as the authority in HTTP requests. If the parameter is set to true but there is no FQDN, the IP is used as the authority. When the parameter is set to false the IP is used as the authority in the SBI interfaces with exception of the NRF interface which always uses the FQDN as the preferred authority.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

useLocationHeaderFromUdmRsp - Use HTTP Location Header from UDM response

Syntax

--amfGparmName useLocationHeaderFromUdmRsp --amfGparmValue <string>

Description

When this parameter is set to true, when CMM receives Response to a UDM Subscribe the URI in the Location Header is used in subsequent Requests to update (PATCH/PUT) or Delete that Subscription. CMM also stores the Producer ID so as to be able to support OAuth in subsequent requests. Otherwise subsequent Requests are sent to a service instance discovered via NRF. Note: This parameter must be set to true when UDM communication is based on Model D (indirect communication with delegated discovery)

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

useN26SvcParam5gsToEpsHo - Use N26 Service Parameter in 5GS to EPS Handover

Syntax

--amfGparmName useN26SvcParam5gsToEpsHo --amfGparmValue <string>

Description

Current parameter enables using N26 service parameter as part of DNS query for target MME selection during 5GS to EPS handover

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

useNssfAvailabilityService - Use NSSF Availability Service

Syntax

--amfGparmName useNssfAvailabilityService --amfGparmValue <string>

Description

When this parameter is true, the AMF will use the NSSF Availability Services. When NSSF Availability Services are used, the AMF does not take in account the local provisioned info regarding Slice support per Tracking Area.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

useNssfAvailabilityServiceSubscription - Use NSSF Availability Service Subscription

Syntax

```
--amfGparmName useNssfAvailabilityServiceSubscription --amfGparmValue  
<string>
```

Description

When this parameter is true, the AMF will Subscribe the NSSF Availability Services in order to receive notification from NSSF regarding changes to the NSSAI availability information on a per TA basis. This parameter is applicable only when amfGParms useNssfAvailabilityService is set to true.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

useNssfPerPlmnSliceRestrictions - Use Per PLMN Slice Restrictions

Syntax

```
--amfGparmName useNssfPerPlmnSliceRestrictions --amfGparmValue  
<string>
```

Description

When this parameter is set to true AMF evaluates the restrictedSnssaiList when included in the AuthorizedNssaiAvailabilityData by the NSSF in a Nnssf_NSSAIAvailability operation. When set to false the restrictions are ignored.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

useNssfSelectionServiceForAmf - Use NSSF Selection Service for AMF

Syntax

```
--amfGparmName useNssfSelectionServiceForAmf --amfGparmValue <string>
```

Description

When this parameter is true the AMF uses the NSSF Selection Service to select AMF during UE Registration in case AMF decides that it is unable to serve all the Allowed Slices the UE.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

useNssfSelectionServiceForSmf - Use NSSF Selection Service for SMF

Syntax

```
--amfGparmName useNssfSelectionServiceForSmf --amfGparmValue <string>
```

Description

When this parameter is true the AMF uses the NSSF Selection Service to select SMF during PDU Session establishment

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

userLocationIgnoreNcgi - User Location Ignore NCGI

Syntax

```
--amfGparmName userLocationIgnoreNcgi --amfGparmValue <string>
```

Description

Defines whether the AMF supports ignoreNcgi parameter inside the nrLocation and further on userLocation.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

v2xAllowed - V2X Allowed

Syntax

--amfGparmName v2xAllowed --amfGparmValue <string>

Description

Feature flag for the v2x feature. When set true, the AMF will request a PCF that supports v2x.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

vlrCleanupBasedOnOccupancy - VLR Cleanup Based on Occupancy

Syntax

--amfGparmName vlrCleanupBasedOnOccupancy --amfGparmValue <string>

Description

Current parameter enables doing VLR cleanup of the VLR Contexts of deregistered UEs based on VLR Occupancy

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Recommended

true

Name

voiceEpsFallback - Voice with EPS Fallback

Syntax

--amfGparmName voiceEpsFallback --amfGparmValue <string>

Description

This parameter activates the voice functionality using EPS fallback on the AMF.

List

- DISABLE - Disable

- ENABLE - Enable

Default

DISABLE

Name

voNR - Voice over New Radio

Syntax

```
--amfGparmName voNR --amfGparmValue <string>
```

Description

Activation of the Voice over New Radio feature on the AMF.

List

- DISABLE - Disable
- ENABLE - Enable

Default

DISABLE

Name

vsmfChange - VSMF Change

Syntax

```
--amfGparmName vsmfChange --amfGparmValue <string>
```

Description

Enables VSMF change including selection of a new VSMF when new UE location is not supported by current VSMF.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

3.55 cmm amfGtpProfile

amf GTP profile. Syntax, description, parameters, and roles for cmm amfGtpProfile command.

Syntax

Description

Administer amf gtp profile

Parameters

Name

name - Amf GTP Profile Name

Syntax

--name <string>

Description

Unique name for this GTP profile

Restrictions

GTP profile with label = default always exists, it cannot be deleted, but can be update.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

contextReqT3 - Context request timeout T3 waiting for response

Syntax

--contextReqT3 <integer>

Description

Max wait time for a response of a context request message

Class

Optional.

Type / Range

Integer in the range of 100 to 60,000 in units of Millisecond

Default

3000

Name

contextRspT3 - Context response timeout T3 waiting for ack

Syntax

--contextRspT3 <integer>

Description

Max wait time for a response to a context response message

Class

Optional.

Type / Range

Integer in the range of 100 to 60,000 in units of Millisecond

Default

3000

Name

echoRequests - Echo requests

Syntax

--echoRequests <integer>

Description

(N3-REQUESTS) the total number of echo request messages to be transmitted without receiving a corresponding echo response message after receiving at least one echo response from an endpoint.

Class

Optional.

Type / Range

Integer in the range of 5 to 20

Default

9

Name

identificationT3 - Identification response timeout T3

Syntax

--identificationT3 <integer>

Description

Max wait time for a response of an identification request message

Class

Optional.

Type / Range

Integer 100...60000, step 100 in units of Millisecond

Default

3000

Name

n3ContextReq - N3 context request attempts

Syntax

--n3ContextReq <integer>

Description

Max number of attempts to send a context request message

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

n3ContextRsp - N3 context response attempts

Syntax

--n3ContextRsp <integer>

Description

Max number of attempts to send a context response message

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

n3Identification - N3 identification request attempts

Syntax

--n3Identification <integer>

Description

Max number of attempts to send an identification request message.

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

n3RelocationCancel - N3 relocation cancel attempts

Syntax`--n3RelocationCancel <integer>`**Description**

Max number of attempts to send a relocation cancel message

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

n3RelocationCmpl - N3 forward relocation complete attempts

Syntax`--n3RelocationCmpl <integer>`**Description**

Max number of attempts to send a relocation complete message

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

n3RelocationReq - N3 forward relocation request attempts

Syntax`--n3RelocationReq <integer>`**Description**

Max number of attempts to send relocation request message

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

relocationCancelT3 - Relocation cancel response timeout T3

Syntax

--relocationCancelT3 <integer>

Description

Max wait time for a response of a relocation cancel message

Class

Optional.

Type / Range

Integer in the range of 100 to 60,000 in units of Millisecond

Default

3000

Name

relocationCmplT3 - Relocation complete ack timeout T3

Syntax

--relocationCmplT3 <integer>

Description

Max wait time for a response to a relocation complete message

Class

Optional.

Type / Range

Integer in the range of 100 to 60,000 in units of Millisecond

Default

3000

Name

relocationReqT3 - Relocation request timeout T3 waiting for response

Syntax

--relocationReqT3 <integer>

Description

Max wait time for a response of a relocation request message

Class

Optional.

Type / Range

Integer in the range of 100 to 60,000 in units of Millisecond

Default

3000

Name

sendEchoOnFailure - Send echo requests on failure

Syntax

--sendEchoOnFailure <list-item>

Description

Controls sending echo requests when no GTP-C non-echo response messages are received for a period of time.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

sendEchoTimerT3 - Send echo request timer T3

Syntax

--sendEchoTimerT3 <integer>

Description

The periodic interval for sending echo requests on interfaces that support GTP communications.

Class

Optional.

Type / Range

Integer value of 60000...10800000, step 100 in units of Millisecond

Default

60000

Name

sendPeriodicEcho - sendPeriodicEcho

Syntax

--sendPeriodicEcho <list-item>

Description

Controls sending of periodic echo requests on enabled links

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

waitEchoTimerT3 - Wait for echo response timer T3

Syntax

--waitEchoTimerT3 <integer>

Description

Specifies the time to wait for an echo response to be received before retransmitting the echo request message. the total time to wait for an echo response (N3-REQUESTS * T3-RESPONSE) should be less than the periodic time (Send echo request timer) used to send the echo request message.

Class

Optional.

Type / Range

Integer 1000...600000, step 100 in units of Millisecond

Default

6000

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.56 cmm amfGuamiStatusAdmin

Update GUAMI Status. Syntax, description, parameters, and roles for cmm amfGuamiStatusAdmin command.

Syntax

Description

cmm amfUpdateGuamiStatus is used to initiate the 100% non-directed off-load. The GUAMI to be moved to either MAINTENANCE STATUS or SUPPORTED STATUS shall be specified in this command. An internalStatus with values shuttingDown, locked and unlocked indicate the following: shuttingDown applies if the GUAMI is in the maintenanceTimer window, locked applies if the GUAMI has entered MAINTENANCE STATUS, this means that is not available and the NRF has been updated that the AMF does not support the GUAMI, unlocked is used when the GUAMI is in service and has been updated in the NRF as SUPPORTED. GUAMIs shall not be removed from the CMM configuration unless they are in internalState == locked. GUAMIs with SUPPORTED STATUS are in Service.To Backout of a Maintenance operation the operator shall move the GUAMI to SUPPORTED STATUS. If the aggregate service lock is applied to the CMM the AMF shall deregister from the NRF. When the aggregate service lock is removed the AMF shall register in the NRF with all GUAMIs that are in SUPPORTED STATUS. The list form of the command displays all the GUAMIs configured in the AMF Instance, requested status (MAINTENANCE or SUPPORTED), maintenanceTimer used and internalStatus. If the RAN supports the 'Timer Approach' the maintenanceTimer shall be set to the RAN configured Timer plus 5 seconds. Note: when a GUAMI is configured with cmm amfInstance, it is created with internalState == unlocked

Parameters

Name

guamiName - GUAMI Name

Syntax

--guamiName <string>

Description

It is the GUAMI name for which the status is going to change.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

guamiStatus - Guami Status

Syntax

--guamiStatus <list-item>

Description

GUAMI Status can be maintenance or supported

Class

Optional.

List

- maintenance - The GUAMI is not available due to maintenance actions.
- supported - The GUAMI is available.

Default

supported

Name

maintenanceTimer - Maintenance Timer

Syntax

--maintenanceTimer <integer>

Description

Specifies the length of the Timer Approach window. This timer shall be set to the timer value configured in the RAN (if supported) + 5 seconds. If the command is run during high traffic time, the value suggested is 0.

Restrictions

May only be provided if the status parameter is set to maintenance

Class

Optional.

Type / Range

Integer in the range of 0 to 100 in units of Second

Default

0

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

3.57 cmm amfGuamiStatusAdminStats

GUAMI Status Administrative Statistics. Syntax, description, parameters, and roles for cmm amfGuamiStatusAdminStats command.

Syntax

Description

This command provides statistics when a GUAMI Status is changed by the operator to 'maintenance'. Changing the status of a GUAMI to 'maintenance' causes the CMM to initiate non-directed off-load for a time interval (--maintenanceTimer) controlled by the operator in the amfGuamiStatusAdmin command. For instance, during the non-directed off-load UEs release and rebind their TNLA Bindings.

Parameters

Name

guamiName - GUAMI Name

Syntax

--guamiName <string>

Description

GUAMI name for which the statistics are provided

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

startDateTime - Starting Date and Time

Syntax

--startDateTime <string>

Description

Date and Time at which the operator issued the status change with the amfGuamiStatusAdmin command

Class

Required key.

Type / Range

Date and Time Digit string of length 0 to 20 with the allowed pattern of '20[0-9][0-9]/[0-1][0-9]/[0-3][0-9] [0-2][0-9]:[0-5][0-9]:[0-5][0-9]'

Name

guamiInternalStatus - GUAMI Internal Status

Syntax

--guamiInternalStatus <list-item>

Description

An internalStatus with values shuttingDown, locked and unlocked indicate the following: shuttingDown applies if the GUAMI is in the maintenanceTimer window, locked applies if the GUAMI has entered MAINTENANCE STATUS, this means that is not available and the NRF has been updated that the AMF does not support the GUAMI, unlocked is used when the GUAMI is in service and has been updated in the NRF as SUPPORTED. GUAMIs shall not be removed from the CMM configuration unless they are in internalState == locked.

Class

Read only.

List

- locked - The GUAMI has entered Maintenance State (i.e., if applicable the maintenanceTimer has run out, the RAN, NRF and (when supported) notifications have been issued.
- shuttingDown - the GUAMI requested status is Maintenance and the maintenanceTimer has not run out
- unlocked - The GUAMI is Supported by the AMF, it is registered as Supported in the NRF, the RAN has been updated too

Default

unlocked

Name

guamiStatus - GUAMI Status

Syntax

--guamiStatus <list-item>

Description

GUAMI external status. GUAMIs with SUPPORTED STATUS are in Service.To Backout of a Maintenance operation the operator shall move the GUAMI to SUPPORTED STATUS. If the aggregate service lock is applied to the CMM the AMF shall deregister from the NRF. When the aggregate service lock is removed the AMF shall register in the NRF with all GUAMIs that are in SUPPORTED STATUS.

Class

Read only.

List

- maintenance - The GUAMI is not available due to maintenance actions.
- supported - The GUAMI is available.

Default

supported

Name

maintenanceTimer - Maintenance Timer

Syntax

--maintenanceTimer <integer>

Description

The maintenance timer value specified in the amfGuamiStatusAdmin command

Class

Read only.

Type / Range

Integer in the range of 0 to 100 in units of Second

Default

0

Name

numUeCpnfProcRedirected - Number of UE Network Procedures Redirected

Syntax

--numUeCpnfProcRedirected <integer>

Description

Provides the number of network initiated procedures that were redirected during the GUAMI status change

Class

Read only.

Type / Range

Integer in the range of 0 to 4,294,967,295

Default

0

Name

numUeTnlaBindingRelease - Number of UE TNLA Binding Releases

Syntax

--numUeTnlaBindingRelease <integer>

Description

Reports the number of UE TNLA Bindings released during the GUAMI status change

Class

Read only.

Type / Range

Integer in the range of 0 to 4,294,967,295

Default

0

Name

numUeTnlaRebinding - Number of UE TNLA Rebindings

Syntax

--numUeTnlaRebinding <integer>

Description

Provides the number of UE TNLA Bindings that resulted in a Rebinding during the GUAMI status change

Class

Read only.

Type / Range

Integer in the range of 0 to 4,294,967,295

Default

0

Roles**List**

sysadmin, manage, monitor

3.58 cmm amfHoCellIdRestrictionList

Handover Cell Identity Restriction List for AMF . Syntax, description, parameters, and roles for cmm amfHoCellIdRestrictionList command.

Syntax**Description**

Administer Handover Cell Identity Restriction List.

Parameters**Name**

name - Cell Identity Restriction List Name

Syntax

--name <string>

Description

Identifies the name of the Cell Identity Restriction List.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

listInterface - List of Interfaces

Syntax

--listInterface <list-item>

Description

Identify the interface used for cell id based handover.

Class

Required on create, not modifiable.

List

- n14 - n14 Interface between AMFs - Cell identifier based handover over N14 interface between AMFs.

Name

listType - List Type

Syntax

--listType <list-item>

Description

Identify if the Cell Identity restriction list is of type allow or deny list.

Class

Optional.

List

- allow - Allow Handover Cell Ids list - Identify the list of Cell Ids from which handover is allowed
- deny - Deny Handover Cell Ids list - Identify the list of Cell Ids from which handover is denied

Default

allow

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.59 cmm amfHoCellIdRestrictionListItem

Handover Cell ID Restriction List. Syntax, description, parameters, and roles for cmm amfHoCellIdRestrictionListItem command.

Syntax

Description

Administers addition (or deletion) of cell id to the handover cell id restriction list. AMF supports up to 16 cell id restriction lists with a maximum number of 10k cell ids. A list can have up to 10k cell ids resulting in one possible list since maximum numbers of cell ids allowed is used.

Parameters

Name

hoCellIdRestrictionListName - Handover Cell ID Restriction List Name

Syntax

--hoCellIdRestrictionListName <string>

Description

Name of the handover cell id restriction list.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

cellIdName - Cell Identifier Name

Syntax

--cellIdName <string>

Description

Name of the cell id.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.60 cmm amfHttp2Ping

AMF Interface HTTP2 Ping. Syntax, description, parameters, and roles for cmm amfHttp2Ping command.

Syntax**Description**

This command is used to determine the status of a TCP/HTTP2 connection in the 5G network

Parameters**Name**

amfClientIntf - Amf Client Interface Ping

Syntax

--amfClientIntf <list-item>

Description

Client Interfaces that can initiate a HTTP2 ping

Class

Required key.

List

- N11 - N11 Interface between AMF and SMF
- N12 - N12 Interface between AMF and AUSF
- N14 - N14 Interface between AMF and AMF
- N15 - N15 Interface between AMF and PCF
- N17 - N17 Interface between AMF and 5G-EIR
- N18 - Nudsf - Nudsf Interface between AMF and UDSF
- N20 - N20 Interface between AMF and SMSF
- N22 - N22 Interface between AMF and NSSF
- N58 - N58 interface between AMF and the NSSAAF
- N8 - N8 Interface between AMF and SDM/UECM
- NLg - NLg Interface between AMF and GMLC
- NLs - NLs Interface between AMF and LMF
- NR10 - Private Interface for Session Restoration
- NamfCommNotification - Namf Communication Notiifcation - Namf Communication Service Notification Interface
- NamfEeNotification - Namf Event Exposure Notiifcation - Namf Event Exposure Service Notification Interface
- NamfLocNotification - Namf Location Notification - Namf Location Notification Interface
- Nnef - Nnef Interface between AMF and NEF
- Nnrf - Nnrf Interface between AMF and NRF
- Nnwdaf - Nnwdaf interface between AMF and NWDAF

Name

amfIntfInstance - 5G Interface Instance

Syntax

--amfIntfInstance <integer>

Description

5G Interface Instance for ping

Class

Required key.

Type / Range

Integer in the range of 1 to 16

Name

amfRmtEndIp - AMF Remote Endpoint IP address

Syntax

--amfRmtEndIp <string>

Description

AMF Remote Endpoint IP address for ping

Class

Required key.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

amfRmtEndIpPort - AMF Remote Endpoint IP Port

Syntax

--amfRmtEndIpPort <integer>

Description

AMF Remote Endpoint IP Port.

Class

Required key.

Type / Range

Integer in the range of 1,024 to 65,535

Name

idx - Index

Syntax

--idx <integer>

Description

Index number for output

Class

Required key.

Type / Range

Integer in the range of 0 to 255

Name

http2PingStatus - HTTP2 Ping Status

Syntax

--http2PingStatus <list-item>

Description

HTTP2 Ping Return Status

Class

Read only.

List

- failure - TCP/HTTP - TCP/HTTP setup failure - The TCP/HTTP2 link establishment failed
- failure - internal error - Ping failure Internal System Error - Internal system error
- failure - time out - Ping Failure Time out - No HTTP2 Ping response received in time interval
- failure - unknown - Ping failure for unknown reason - Ping was unsuccessful - unknown reason
- success - Ping Success - The HTTP2 Ping was successful

Name

http2PingTimeOut - Time out for Ping Operation

Syntax

--http2PingTimeOut <integer>

Description

Time before the Ping will timeout

Class

Optional.

Type / Range

Integer in the range of 1 to 10

Default

5

Name

numHttp2Ping - Number of Pings to Send

Syntax

--numHttp2Ping <integer>

Description

This is the number of Pings that will be send during the operation

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

5

Name

roundTripTimeSec - Ping Round Trip Time In Seconds

Syntax

--roundTripTimeSec <integer>

Description

This is the ping round trip time in seconds

Class

Read only.

Type / Range

Integer in the range of 0 to 100 in units of Second

Name

roundTripTimeUsec - Ping Round Trip Time In NSeconds

Syntax

--roundTripTimeUsec <integer>

Description

This is the ping round trip time in nseconds

Class

Read only.

Type / Range

Integer in the range of 0 to 65,535

Roles**List**

sysadmin, manage, monitor

3.61 cmm amfInstance

AMF Instance. Syntax, description, parameters, and roles for cmm amfInstance command.

Syntax**Description**

This command is used to create a new AMF instance and modify and delete an existing AMF instance. An AMF instance is identified by its Instance ID.

Parameters**Name**

servedPlmnName - Mobile Country Code

Syntax

--servedPlmnName <string>

Description

Set Mobile Country Code of the AMF

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

amfRegionId - AMF Region ID

Syntax

--amfRegionId <integer>

Description

AMF Region Identifier is used to identify an AMF group serving a region. 5G AMF are organized into an AMF region, AMF set and AMF pointer. An AMF is uniquely identified by PLMN ID, AMF Region ID, AMF Set and AMF pointer.

Class

Required key.

Type / Range

Integer in the range of 0 to 255

Name

amfSetId - AMF Set Identifier

Syntax

--amfSetId <integer>

Description

AMF Set Identifier is used to identify a set of AMF within an AMF region. An AMF is uniquely identified by PLMN ID, AMF Region ID, AMF Set and AMF pointer.

Class

Required key.

Type / Range

5G AMF Set ID

Name

amfPointer - AMF Pointer

Syntax

--amfPointer <integer>

Description

The AMF Pointer identifies an AMF in an AMF set. An AMF is uniquely identified by PLMN ID, AMF Region ID, AMF Set and AMF pointer.

Class

Required key.

Type / Range

5G AMF Pointer

Name

amfIdLabels - AMF Identifier labels for AMF name FQDN

Syntax

--amfIdLabels <string>

Description

This parameter is used to construct AMF name FQDN The amfLabel should not include the PLMN. For example, if the amfIdLabel is set to 'amf1.region1' then the AMF Name FQDN for MCC 345 and MNC 12 is 'amf1.region1.amf.5gc.mnc012.mcc345.3gppnetwork.org'. The provisioned part is 'amf1.region1' and the AMF derives the '.amf.5gc.mnc012.mcc345.3gppnetwork.org' part.

Class

Required on create.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

guamiName - GUAMI Name

Syntax

--guamiName <string>

Description

User-provided name for this GUAMI

Class

Required on create, not modifiable.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

nfServicesListName - NF Services List Name

Syntax

--nfServicesListName <string>

Description

Provision services supported by the AMF.

Class

Required on create.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

allowedNfDomainListName - Allowed NF Domain List Name

Syntax

--allowedNfDomainListName <string>

Description

The list with the NF domains the AMF is accessible to. When a list is provisioned, the AMF will add the NF domains included in the list in the the allowedNfDomains IE in the NF Profile Registered to the NRF.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

allowedNfTypeListName - Allowed NF Type List Name

Syntax

--allowedNfTypeListName <string>

Description

The list with the NF types the AMF is accessible to. When a list is provisioned, the AMF will add the NF types included in the list in the the allowedNfTypes IE in the NF Profile Registered to the NRF.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

allowedPlmnListName - Allowed PLMN List Name

Syntax


```
--allowedPlmnListName <string>
```

Description

The list with the PLMNs the AMF is accessible to. When a list is provisioned, the AMF will add the PLMNs included in the list in the the allowedPlmns IE in the NF Profile Registered to the NRF.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

""

Name

allowedSnssaiListName - Allowed S-NSSAI List Name

Syntax

```
--allowedSnssaiListName <string>
```

Description

The list with the S-NSSAIs the AMF is accessible to. When a list is provisioned, the AMF will add the S-NSSAIs included in the list in the the allowedNssais IE in the NF Profile Registered to the NRF.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

""

Name

capacity - AMF Relative Capacity

Syntax

```
--capacity <integer>
```

Description

AMF Relative Capacity. The relative capacity is only used in the other NF in the 5GC. There is a separate parameter for the relative capacity sent to the NG-RAN.

Class

Optional.

Type / Range

Integer in the range of 0 to 65,535

Default

65535

Name

enable - Enable AMF

Syntax`--enable <list-item>`**Description**

The parameter is used to enable use of the provisioned instance of the AMF. If the parameter is set to 'false', the AMF instance will not be used. Once an instance is enabled, the instance can be disabled only under certain TBD conditions.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

instanceId - Instance Identifier

Syntax`--instanceId <string>`**Description**

The AMF Instance ID is a Universally Unique Identifier (UUID) as a Universal Resource Name (URN). The AMF configuration generates the UUID version 4, as described in IETF RFC 4122 [xx].

Class

Read only.

Type / Range

Digit string of length 36 to 36 with the allowed pattern of `[0-9a-fA-F]{8}\-[0-9a-fA-F]{4}\-[0-9a-fA-F]{4}\-[0-9a-fA-F]{4}\-[0-9a-fA-F]{12}`

Name

interPlmnAmfLabel - Identifier labels for AMF Inter PLMN name FQDN

Syntax`--interPlmnAmfLabel <string>`**Description**

This parameter is used to construct AMF Inter PLMN name FQDN. The AMF Label should not include the PLMN ID. For example, if the Label is set to 'amf1.region1' then the AMF Inter PLMN Name FQDN for MCC 345 and MNC 12 is 'amf1.region1.amf.5gc.mnc012.mcc345.3gppnetwork.org'. The provisioned part is 'amf1.region1' and the AMF derives the '.amf.5gc.mnc012.mcc345.3gppnetwork.org' part. If this parameter is null, the AMF uses amfidLabels for the AMF Inter PLMN label.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

locality - Location of the AMF

Syntax

--locality <string>

Description

This parameter is used to configure AMF location like geographical location or data center name.

Class

Optional.

Type / Range

Digit string of length 0 to 64

Default

""

Name

nsiIdListName - NSI ID List Name

Syntax

--nsiIdListName <string>

Description

Provision NSI ID list name. The NSI ID list provisioning is optional. If provisioned AMF will include the list in the profile when the AMF registers with the NRF.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

oauthProfileName - OAuth Access Token Key Name

Syntax

--oauthProfileName <string>

Description

Configure OAuth Access Token Key used for validating OAuth access token received by AMF producer services.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

priority - AMF Priority

Syntax

--priority <integer>

Description

Indicates priority value of the AMF relative to the other AMF within the same AMF set. Lower value indicates higher priority. The capacity and priority parameters, if present, are used for NF selection and load balancing. NFs with a lower priority value is selected before NFs with a higher priority value; for NFs with equal priority value, the selection uses the relative capacity (weight) value of the NFs.

Class

Optional.

Type / Range

Integer in the range of 0 to 65,535

Default

1

Name

restorationEndPtListName - Restoration End Point List Name

Syntax

--restorationEndPtListName <string>

Description

The end point List that is to be used for backing up and restoring this GUAMI

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

scpDomainListName - SCP Domain List Name

Syntax

--scpDomainListName <string>

Description

Name of the list including the SCP Domains CMM belongs to. When a list is provisioned, the AMF will add the NF domains included in the list in the the scpDomains IE in the NF Profile Registered to the NRF.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Default

""

Name

tnlaProfileName - TNLA Profile Name

Syntax

--tnlaProfileName <string>

Description

Provision TNLA profile.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Default

default

Name

userAgentSuffix - User Agent Suffix

Syntax

--userAgentSuffix <string>

Description

The parameter is used to configure the text to be suffixed to User-Agent HTTP header. If the text 'Chicago' is provisioned then the User-Agent HTTP header will be set to 'amf-Chicago'.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.62 cmm amfIntegrityAlgorithm

Preferred Integrity Algorithm. Syntax, description, parameters, and roles for cmm amfIntegrityAlgorithm command.

Syntax**Description**

This command sets the algorithms, in priority order, AMF will use to negotiate integrity of NAS messages with the UE. At REGISTRATION, the UE provides, in the UE security capabilities, which integrity algorithms are supported. The AMF will use the algorithm both supported by the UE and highest in priority in the nia algorithm priority list. - An algorithm must be chosen for 'priority-1' and 'priority-2'. - An additional selection, 'none', can be selected for 'priority-3' and 'priority-4' - effectively truncating the list. - If any integrity algorithm (including null) is selected for 'priority-4', then 'priority-3' must not be 'none'. - An algorithm cannot be used for more than one priority (except for 'none').

Parameters

Name

integrityAlgPrio1 - Most preferred integrity algorithm

Syntax

```
--integrityAlgPrio1 <list-item>
```

Description

The most priority algorithm for integrity of NAS messages with the UE.

Class

Optional.

List

- aes - 128-NIA2 (AES)
- none - No Algorithm Selected
- nullIntegrity - 128-NIA0 (Null integrity)
- snow3G - 128-NIA1 (SNOW 3G)
- zuc - 128-NIA3 (ZUC)

Default

snow3G

Name

integrityAlgPrio2 - Priority 2 integrity algorithm

Syntax

```
--integrityAlgPrio2 <list-item>
```

Description

Second most preferred integrity algorithm; used if the user element does not support algorithm(s) of higher priority

Class

Optional.

List

- aes - 128-NIA2 (AES)
- none - No Algorithm Selected
- nullIntegrity - 128-NIA0 (Null integrity)
- snow3G - 128-NIA1 (SNOW 3G)
- zuc - 128-NIA3 (ZUC)

Default

aes

Name

integrityAlgPrio3 - Priority 3 integrity algorithm

Syntax

--integrityAlgPrio3 <list-item>

Description

Priority 3 integrity algorithm

Class

Optional.

List

- aes - 128-NIA2 (AES)
- none - No Algorithm Selected
- nullIntegrity - 128-NIA0 (Null integrity)
- snow3G - 128-NIA1 (SNOW 3G)
- zuc - 128-NIA3 (ZUC)

Default

none

Name

integrityAlgPrio4 - Priority 4 integrity algorithm

Syntax

--integrityAlgPrio4 <list-item>

Description

Priority 4 integrity algorithm

Class

Optional.

List

- aes - 128-NIA2 (AES)
- none - No Algorithm Selected
- nullIntegrity - 128-NIA0 (Null integrity)
- snow3G - 128-NIA1 (SNOW 3G)
- zuc - 128-NIA3 (ZUC)

Default

none

Roles

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

3.63 cmm amfIntfAdmin

AMF Signalling Interface Administration. Syntax, description, parameters, and roles for cmm amfIntfAdmin command.

Syntax

Description

AMF Signaling Interface Administration - Lock an Interface Locked indicates that no new sessions will be established (Client). UnLocked indicates that the interface is available for use. Blocked indicates that no new HTTP/TCP connections will be allowed into the Interface (Server). When the AMF has multiple Service Instances the command applies to all the Service instances of the Service type.

Parameters

Name

amfIntfType - AMF Interface type

Syntax

--amfIntfType <list-item>

Description

AMF Interface type.

Class

Required key.

List

- N11 - N11 Interface - N11 Interface between AMF and SMF
- N12 - N12 Interface - N12 Interface between AMF and AUSF
- N14 - N14 Interface - N14 Interface between AMF and AMF
- N15 - N15 Interface - N15 Interface between AMF and PCF
- N17 - N17 Interface - N17 Interface between AMF and 5G EIR

- N18 - N18 interface between AMF and UDSF
- N2 - N2 Interface - N2 Interface between AMF and 5gNB
- N20 - N20 Interface - N20 Interface between AMF and SMSF
- N22 - N22 Interface - N22 Interface between AMF and NSSF
- N26 - N26 Interface - N26 Interface between AMF and MME
- N41 - N41 interface - N41 interface between AMF and CHF
- N58 - N58 Interface - N58 interface between AMF and NSSAAF
- N8 - N8 Interface - N8 Interface between AMF and UDM
- NLg - NLg Interface - NLg Interface between AMF and GMLC
- NLs - NLs Interface - NLs Interface between AMF and LMF
- Nnef - Nnef Interface - Nnef Interface between AMF and Network Event Function
- Nnrf - Nnrf Interface - Nnrf Interface between AMF and NRF
- Nnwdaf - Nnwdaf Interface - Nnwdaf interface between AMF and NWDAF
- namf_Communications - AMF Communications Service
- namf_EventExposure - AMF Event Exposure Service
- namf_Location - AMF Location Service
- namf_MT - AMF Mobile Terminated Service
- notification_AMF - AMF Notification - AMF Notification Function
- notification_NRF - NRF Notification - NRF Notification Function
- notification_NSSAAF - NSSAAF Notification - NSSAAF Notification Function
- notification_NSSF - NSSF Notification - NSSF Notification Function
- notification_NWDAF - NWDAF Notification - NWDAF Notification Function
- notification_PCF - PCF Notification - PCF Notification Function
- notification_SMF - SMF Notification - SMF Notification Function
- notification_UDM - UDM Notification - UDM Notification Function

Name

endPointName - AMF Local End Point Interface Name

Syntax

--endPointName <string>

Description

AMF Local End Point Interface Name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

amfIntfAdminState - AMF Interface admin state

Syntax

--amfIntfAdminState <list-item>

Description

AMF Interface admin state.

Class

Required on create.

List

- blocked - Blocked State - only used for SBI_Server Interface types
- blockedDisabled - Blocked (disabled) - Blocked State - only used for SBI_Server Interface types when aggregate service is locked.
- locked - Locked State - only used for SBI_Client, N2 and N26 Interface types
- lockedDisabled - Locked (disabled) - Locked State - only used for SBI_Client, N2 and N26 Interface types when aggregate service is locked.
- unlocked - Unlocked State
- unlockedDisabled - Unlocked (disabled) - Unlocked State when aggregate service is locked.

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

3.64 cmm amfIntfQueryPath

Query sctp path information. Syntax, description, parameters, and roles for cmm amfIntfQueryPath command.

Syntax

Description

Query path information for SCTP link/interface in 5G.

Parameters

Name

amfIntfType - Link interface type

Syntax

--amfIntfType <list-item>

Description

Link interface type. only SCTP interfaces are supported.

Class

Required key.

List

- N2 - N2 Interface - List of (N2, ...)

Name

endLinkIndex - End Link Index

Syntax

--endLinkIndex <integer>

Description

End link index

Class

Optional.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

linkIndex - Link Index

Syntax

--linkIndex <integer>

Description

Link index

Class

Read only.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

maxResponseSize - Max record count

Syntax

--maxResponseSize <integer>

Description

Max number of records to be returned.

Class

Optional.

Type / Range

Integer in the range of 1 to 100

Name

path1Status - Path 1 Status

Syntax

--path1Status <list-item>

Description

Path 1 status

Class

Read only.

List

- not-used - Path status of not used.
- reachable - Path status of reachable.
- unknown - Path status of unknown.
- unreachable - Path status of unreachable.

Name

path2Status - Path 2 Status

Syntax

--path2Status <list-item>

Description

Path 2 status

Class

Read only.

List

- not-used - Path status of not used.
- reachable - Path status of reachable.
- unknown - Path status of unknown.

- unreachable - Path status of unreachable.

Name

startLinkIndex - Start Link Index

Syntax

--startLinkIndex <integer>

Description

Start link index

Class

Optional.

Type / Range

Integer in the range of 0 to 4,294,967,295

Roles**List**

sysadmin, manage, monitor

3.65 cmm amfIntfSummary

AMF Signalling Interface Link Summary. Syntax, description, parameters, and roles for cmm amfIntfSummary command.

Syntax**Description**

AMF Signalling Interface Link Summary Request

Parameters**Name**

amfIntfType - AMF Interface Type

Syntax

--amfIntfType <list-item>

Description

AMF Interface type. Default to N8.

Class

Required key.

List

- N11 - N11 Interface - N11 Interface between AMF and SMF
- N12 - N12 Interface - N12 Interface between AMF and AUSF
- N14 - N14 Interface - N14 Interface between AMF and AMF
- N15 - N15 Interface - N15 Interface between AMF and PCF
- N17 - N17 Interface - N17 Interface between AMF and 5G EIR
- N18 - N18 interface between AMF and UDSF
- N2 - N2 Interface - N2 Interface between AMF and 5gNB
- N20 - N20 Interface - N20 Interface between AMF and SMSF
- N22 - N22 Interface - N22 Interface between AMF and NSSF
- N26 - N26 Interface - N26 Interface between AMF and MME
- N41 - N41 interface - N41 interface between AMF and CHF
- N58 - N58 Interface - N58 interface between AMF and NSSAAF
- N8 - N8 Interface - N8 Interface between AMF and UDM
- NLg - NLg Interface - NLg Interface between AMF and GMLC
- NLs - NLs Interface - NLs Interface between AMF and LMF
- Nnef - Nnef Interface - Nnef Interface between AMF and Network Event Function
- Nnrf - Nnrf Interface - Nnrf Interface between AMF and NRF
- Nnwdaf - Nnwdaf Interface - Nnwdaf interface between AMF and NWDAF
- namf_Communications - AMF Communications Service
- namf_EventExposure - AMF Event Exposure Service
- namf_Location - AMF Location Service
- namf_MT - AMF Mobile Terminated Service
- notification_AMF - AMF Notification - AMF Notification Function
- notification_NRF - NRF Notification - NRF Notification Function
- notification_NSSAAF - NSSAAF Notification - NSSAAF Notification Function
- notification_NSSF - NSSF Notification - NSSF Notification Function
- notification_NWDAF - NWDAF Notification - NWDAF Notification Function
- notification_PCF - PCF Notification - PCF Notification Function
- notification_SMF - SMF Notification - SMF Notification Function
- notification_UDM - UDM Notification - UDM Notification Function

Default

N8

Name

amfLinkAdminState - AMF Link Admin State

Syntax

--amfLinkAdminState <list-item>

Description

AMF Link Admin State

Class

Read only.

List

- blocked - Blocked State - only used for SBI_Server link types
- locked - Locked State - only used for SBI_Client, N2 and N26 links types
- unlocked - Unlocked State

Name

endPointName - AMF Local End Point Interface Name

Syntax

--endPointName <string>

Description

AMF Local End Point Interface Name

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

linkCount - Link count

Syntax

--linkCount <integer>

Description

Link count.

Class

Read only.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

linkOperationState - Link operation state

Syntax

--linkOperationState <list-item>

Description

Link operation state.

Class

Read only.

List

- disabled - Disabled state of the link.
- enabled - Enabled state of the link.
- suspended - Suspended state of the link.
- unknown - Unknown state of the link. this can not be used during query.

Name

poolId - poolId

Syntax

--poolId <integer>

Description

IPDS Pool ID - If chosen only the count of interface links on the specified IPDS will be output. This parameter is only applicable for Multi-IPDS instantiations and requires linkInterfaceType option to display results on any Pool Id greater than 0

Class

Read only.

Type / Range

Integer in the range of 0 to 31

Roles**List**

sysadmin, manage, monitor

3.66 cmm amfLadnMapping

AMF LADN Information. Syntax, description, parameters, and roles for cmm amfLadnMapping command.

Syntax

Description

AMF Local Area Data Network (LADN) mapping. This command provides mapping between LADN service area (Tracking Area list) and the related Data Network Name (DNN). With the LADN functionality, access to a provisioned data network is only provided on a restricted service area. A single DNN can only have a single associated LADN TAI list in the same PLMN. The same LADN TAI list related to a single PLMN may be associated with multiple DNNs. AMF supports up to 200 LADN TAI list-DNN associations for a PLMN.

Parameters

Name

amfLadnTailistName - AMF LADN TAI List Name

Syntax

--amfLadnTailistName <string>

Description

The AMF Tracking Area list associated with a LADN DNN.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

dataNetworkName - Data Network Name

Syntax

--dataNetworkName <string>

Description

The Data Network Name (DNN) associated with the DNN service area (Tracking Area list) which is used for Local Area Data Network (LADN) functionality.

Class

Required key.

Type / Range

Data Network Name is basically APN Ni.

Name

plmnName - Mobile network name

Syntax

--plmnName <string>

Description

A name that identifies mobile network.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.67 cmm amfLadnTai

AMF Local Area Data Network TAI. Syntax, description, parameters, and roles for cmm amfLadnTai command.

Syntax**Description**

Administer Tracking Area (TA) associated with a Local Area Data Network (LADN). A UE can only access a Local Area Data Network when camping inside a specific service area defined by a list of Tracking Area Identifiers. AMF supports 300 TAIs on a single LADN TAI list.

Parameters**Name**

amfLadnTaiListName - AMF Local Area Data Network TAI List Name

Syntax

--amfLadnTaiListName <string>

Description

Name of Local Area Data Network (LADN) associated Tracking Area list.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

servedPlmn - Served PLMN

Syntax

--servedPlmn <string>

Description

Served mobile network name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

amfTac - AMF TAC

Syntax

--amfTac <integer>

Description

The Tracking Area Code in the AMF.

Class

Required key.

Type / Range

Integer in the range of 0 to 16,777,215

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.68 cmm amfLadnTailList

AMF Local Area Data Network TAI List. Syntax, description, parameters, and roles for cmm amfLadnTailList command.

Syntax

Description

Administer AMF Local Area Data Network (LADN) TAI List. A UE can only access a Local Area Data Network when camping inside a specific service area defined by a list of Tracking Area Identifiers. AMF supports 200 LADN TAI lists per PLMN.

Parameters

Name

name - AMF Local Area Data Network TAI List Name

Syntax

--name <string>

Description

The name of the AMF Local Area Data Network TAI List.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.69 cmm amfLcsGParms

Location Service GParms. Syntax, description, parameters, and roles for cmm amfLcsGParms command.

Syntax

Description

This command opens the context for configuring location Service (LCS) related parameters for the network.

Parameters

Name

amfGparmName - Parameter name

Syntax

--amfGparmName <list-item>

Description

Name of parameter

Class

Required key.

List

- activateLcs - Activate LCS - Activate LCS in AMF
- activateLcsClientEmergencyServices - Activate LCS Client Type For Emergency Services - Activate LCS with Client Type is Emergency Services
- activateLcsClientVasService - Activate LCS Client Type For VAS Services - Activated Lcs MT_LR with ClientType= Value added Service. When parameter is set, AMF will accept MT-LR with client type= VAS. If the parameter is false, then AMF will not accept client type= VAS for MT-LR
- clientTypeLmfGmlcSelect - Client Type Based LMF/GMLC Selection - With this parameter set, Client type based GMLC/LMF selection shall be used so that dedicated GMLC/LMF will be selected and used for different client types. If the parameter is not set, existing function will be used -> AMF does NRF query of LMF and GMLC for MT-LR without client type.
- emerMtLrNoLmf - E911 LCS MT LR Without Query LMF - When the parameter is enabled, AMF supports MT-LR during emergency call for clientType emergency without contacting LMF. Instead, AMF reports the current NRCGI to GMLC when UE registers to the 5GC for emergency services
- ePdgEmergencyHoNotificationGmlc - ePDG emergency HO notification to GMLC - Defines the type of procedure the AMF triggers to GMLC when then UE requests an emergency session to be handed over from ePDG/EPC to NG-RAN/5GS. When set to true, the AMF will send Handover notification to GMLC. When set to false, AMF will send LRR with emergency origination. By default, the parameter is set to true.
- includeNrCgiMtlrFailure - Include last seen NR-CGI and timestamp in the ProblemDetails portion of a failed MT-LR - Include last seen NR-CGI and timestamp in the ProblemDetails portion of a failed MT-LR (Provide Position Response) if AMF not able to retrieve latitude/longitude information for UE.
- lawfulInterceptMtLrNoLmf - Lawful Intercept MT_LR No LMF - This is for MT_LR client type lawful Intercept service do not contact LMF for UE location information. If yes, when AMF receives mobile terminated location request with client

type=lawfulInterceptService, and activateLcsClientTypeLawfulInterceptServices is set, the AMF contacts the gNB for UE current cell information. This information is sent back to the GMLC. The AMF does not contact the LMF to obtain any UE location information. If gParms is no, nothing changes and keep existing function (i.e. AMF contacts LMF to obtain UE location)

- lcsClientTypeLawfulInterceptServices - Activate LCS Client Type Lawful Intercept Services
- lcsClientTypePlmnOperatorServices - Activate LCS Client Type PLMN Operator Services
- lcsRetrieveCurrentCellInfo - Retrieve Cell Id of UE - If a UE is 5GMM-CONNECTED, the AMF uses the setting of the lcsRetrieveCurrentCellInfo parameter to determine whether to send NGAP Location Reporting Control to retrieve current cell ID for the UE. If parameter setting is Yes, Location Reporting Control /Location Report exchange with the gNB occurs prior to the AMF contacting the LMF for geographic information
- lengthByteForNrppa - Length Byte For NRPPa - Normally, AMF transparent send NRPPa PDUs received from LMF to gNB without any change. With this parameter set to true, AMF will encode Octet String (with length Byte) towards gNB. If the parameter set to false, then existing function i.e. AMF transparent passes the PDU unchanged. The default value is false.
- lmfgmlcSelectForNiLr - LMF/GMLC Selection For NI-LR - Client type based GMLC/LMF selection has been introduced by 3GPP and Will be used in AMF. However, if Dedicated GMLC/LMF are used for E911 and selection of LMF/GMLC for NI-LR needs to follow the same rule. Thus, with this parameter is set, AMF shall do NRF query of LMF and GMLC for NI-LR. If this parameter is not set, existing function will keep-> AMF does NRF query of LMF and GMLC for NI-LR without client type.
- maxNrCgiAgeEmerMtlrNoLmf - Maximum time interval in seconds for requesting nrCGI for emergency MT-LR - If the parameter is set to 0 with the amfLcsGParms parameter emerMtlrNoLmf sets to yes, AMF page 5GMM-IDLE UE and send nrCGI information or send location-reporting control information for CM. If the parameter is set to a non-zero value, when AMF gets MT-LR request, AMF will check the difference between AMF stored nrCGI timestamp of the UE and time received MT-LR. If the time difference is less than provisioned value, then the AMF reports stored nrCGI information to GMLC. If time difference is greater than the provisioned value, then AMF will request current nrCGI either via paging or location report control depends UE status
- maxNrCgiAgeLawfulInterceptMtlrNoLmf - Maximum Time Interval In Seconds For Requesting nrCGI Lawful Intercept MT-LR - If the parameter is set to 0 with the amfLcsGParms parameter LawfulInterceptMtlrNoLmf sets to yes, AMF page 5GMM-IDLE UE and send nrCGI information or send location-reporting control information for CM. If the parameter is set to a non-zero value, when AMF gets MT-LR request, AMF will check the difference between AMF stored nrCGI timestamp of the UE and time received MT-LR. If the time difference is less than provisioned value, then the AMF reports stored nrCGI information to GMLC. If time difference is greater than the provisioned value (or equal

provisioned value), AMF will request current nrCGI either via paging or location report control depends UE status

- **maxNrCgiAgePlmnOperMtLrNoLmf** - Max nrCGI Age PLMN Operator MT-LR - Maximum time interval in seconds for requesting nrCGI PLMN operator MT-LR. If the parameter is set to 0 with the amfLcsGParms parameter plmnOperMtLrNoLmf sets to yes, AMF page 5GMM-IDLE UE and send nrCGI information or send location-reporting control information. If the parameter is set to a non-zero value, when AMF gets MT-LR request, AMF will check the difference between AMF stored nrCGI timestamp of the UE and time received MT-LR. If the time difference is less than provisioned value, then the AMF reports stored nrCGI information to GMLC. If time difference is greater than the provisioned value (or equal provisioned value), AMF will request current nrCGI either via paging or location report control depends UE status
- **maxNrCgiAgeValueAddedMtLrNoLmf** - Max NrCgi Age Value Added MT-LR - Maximum time interval in seconds for requesting nrCGI for value added MT-LR. If the parameter is set to 0 with the amfLcsGParms parameter valueAddedMtLrNoLmf sets to yes, AMF page 5GMM-IDLE UE and send nrCGI information or send location-reporting control information. If the parameter is set to a non-zero value, when AMF gets MT-LR request, AMF will check the difference between AMF stored nrCGI timestamp of the UE and time received MT-LR. If the time difference is less than provisioned value, then the AMF reports stored nrCGI information to GMLC. If time difference is greater than the provisioned value (or equal provisioned value), AMF will request current nrCGI either via paging or location report control depends UE status
- **nonUeAssociatedNrppa** - Non UE Associated NRPPa - With the parameter set, AMF supports non UE Associated NRPPa transfer between LMF and NG-RAN. If the parameter is not set, AMF will not support non UE Associated NRPPa transfer between LMF and NG-RAN.
- **plmnOperMtLrNoLmf** - PLMN Operator MT_LR No LMF - MT_LR client type PLMN Operator Service do not contact LMF for UE location information. If gParms is 'yes', when AMF receives mobile terminated location request with client type=plmnOperatorServices, and activateLcsClientTypePlmnOperatorServices is set, the AMF contacts the gNB for UE current cell information. This information is sent back to the GMLC. The AMF does not contact the LMF to obtain any UE location information. If gParms is 'no', nothing changes and keep existing function (i.e. AMF contacts LMF to obtain UE location for this MT-LR type) .
- **posGpsi** - POS GPSI - If the control parameter is set, it means that AMF supports the 'POSGPSI' feature, the ueContextId identifies the individual ueContext resource may also be composed by UE's GPSI.
- **valueAddedMtLrNoLmf** - Value Added MT_LR No LMF - MT_LR client type value-added do not contact lmf for UE location information. If gParms is 'yes', when AMF receives mobile terminated location request with client type=valueAdded, and activateLcsClientTypeValueAddedServices is set, the AMF contacts the gNB for UE current cell information. This information is sent back to the GMLC. The AMF does not

contact the LMF to obtain any UE location information. If gParms is 'no', nothing change and keep existing function (i.e. AMF contacts LMF to obtain UE location for this MT-LR type

Name

amfGparmValue - Parameter value

Syntax

```
--amfGparmValue <string>
```

Description

Value of parameter

Class

Required on create.

Type / Range

Digit string of length 0 to 64

The amfGparmValue string is based on the amfGparmName parameter:

Name

activateLcs - Activate LCS

Syntax

```
--amfGparmName activateLcs --amfGparmValue <string>
```

Description

Activate LCS in AMF

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

activateLcsClientEmergencyServices - Activate LCS Client Type For Emergency Services

Syntax

```
--amfGparmName activateLcsClientEmergencyServices --amfGparmValue <string>
```

Description

Activate LCS with Client Type is Emergency Services

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

activateLcsClientVasService - Activate LCS Client Type For VAS Services

Syntax

--amfGparmName activateLcsClientVasService --amfGparmValue <string>

Description

Activated Lcs MT_LR with ClientType= Value added Service. When parameter is set, AMF will accept MT-LR with client type= VAS. If the parameter is false, then AMF will not accept client type= VAS for MT-LR

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

clientTypeLmfGmlcSelect - Client Type Based LMF/GMLC Selection

Syntax

--amfGparmName clientTypeLmfGmlcSelect --amfGparmValue <string>

Description

With this parameter set, Client type based GMLC/LMF selection shall be used so that dedicated GMLC/LMF will be selected and used for different client types. If the parameter is not set, existing function will be used -> AMF does NRF query of LMF and GMLC for MT-LR without client type.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

emerMtLrNoLmf - E911 LCS MT LR Without Query LMF

Syntax

--amfGparmName emerMtLrNoLmf --amfGparmValue <string>

Description

When the parameter is enabled, AMF supports MT-LR during emergency call for clientType emergency without contacting LMF. Instead, AMF reports the current NRCGI to GMLC when UE registers to the 5GC for emergency services

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ePdgEmergencyHoNotificationGmlc - ePDG emergency HO notification to GMLC

Syntax

```
--amfGparmName ePdgEmergencyHoNotificationGmlc --amfGparmValue  
<string>
```

Description

Defines the type of procedure the AMF triggers to GMLC when then UE requests an emergency session to be handed over from ePDG/EPC to NG-RAN/5GS. When set to true, the AMF will send Handover notification to GMLC. When set to false, AMF will send LRR with emergency origination. By default, the parameter is set to true.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

includeNrCgiMtlrFailure - Include last seen NR-CGI and timestamp in the ProblemDetails portion of a failed MT-LR

Syntax

```
--amfGparmName includeNrCgiMtlrFailure --amfGparmValue <string>
```

Description

Include last seen NR-CGI and timestamp in the ProblemDetails portion of a failed MT-LR (Provide Position Response) if AMF not able to retrieve latitude/longitude information for UE.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

lawfulInterceptMtLrNoLmf - Lawful Intercept MT_LR No LMF

Syntax

--amfGparamName lawfulInterceptMtLrNoLmf --amfGparamValue <string>

Description

This is for MT_LR client type lawful Intercept service do not contact LMF for UE location information. If yes, when AMF receives mobile terminated location request with client type=lawfulInterceptService, and activateLcsClientTypeLawfulInterceptServices is set, the AMF contacts the gNB for UE current cell information. This information is sent back to the GMLC. The AMF does not contact the LMF to obtain any UE location information. If gParms is no, nothing changes and keep existing function (i.e. AMF contacts LMF to obtain UE location)

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

lcsClientTypeLawfulInterceptServices - Activate LCS Client Type Lawful Intercept Services

Syntax

--amfGparamName lcsClientTypeLawfulInterceptServices --amfGparamValue <string>

Description

Activate LCS Client Type Lawful Intercept Services

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

lcsClientTypePlmnOperatorServices - Activate LCS Client Type PLMN Operator Services

Syntax

--amfGparamName lcsClientTypePlmnOperatorServices --amfGparamValue <string>

Description

Activate LCS Client Type PLMN Operator Services

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

lcsRetrieveCurrentCellInfo - Retrieve Cell Id of UE

Syntax

```
--amfGparmName lcsRetrieveCurrentCellInfo --amfGparmValue <string>
```

Description

If a UE is 5GMM-CONNECTED, the AMF uses the setting of the lcsRetrieveCurrentCellInfo parameter to determine whether to send NGAP Location Reporting Control to retrieve current cell ID for the UE. If parameter setting is Yes, Location Reporting Control /Location Report exchange with the gNB occurs prior to the AMF contacting the LMF for geographic information

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

lengthByteForNrppa - Length Byte For NRPPa

Syntax

```
--amfGparmName lengthByteForNrppa --amfGparmValue <string>
```

Description

Normally, AMF transparent send NRPPa PDUs received from LMF to gNB without any change. With this parameter set to true, AMF will encode Octet String (with length Byte) towards gNB. If the parameter set to false, then existing function i.e. AMF transparent passes the PDU unchanged. The default value is false.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

lmfGmlcSelectForNiLr - LMF/GMLC Selection For NI-LR

Syntax

```
--amfGparmName lmfGmlcSelectForNiLr --amfGparmValue <string>
```

Description

Client type based GMLC/LMF selection has been introduced by 3GPP and Will be used in AMF. However, if Dedicated GMLC/LMF are used for E911 and selection of LMF/GMLC for NI-LR needs to follow the same rule. Thus, with this parameter is set, AMF shall do NRF query of LMF and GMLC for NI-LR. If this parameter is not set, existing function will keep-> AMF does NRF query of LMF and GMLC for NI-LR without client type.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

maxNrCgiAgeEmerMtLrNoLmf - Maximum time interval in seconds for requesting nrCGI for emergency MT-LR

Syntax

```
--amfGparmName maxNrCgiAgeEmerMtLrNoLmf --amfGparmValue <string>
```

Description

If the parameter is set to 0 with the amfLcsGParms parameter emerMtLrNoLmf sets to yes, AMF page 5GMM-IDLE UE and send nrCGI information or send location-reporting control information for CM. If the parameter is set to a non-zero value, when AMF gets MT-LR request, AMF will check the difference between AMF stored nrCGI timestamp of the UE and time received MT-LR. If the time difference is less than provisioned value, then the AMF reports stored nrCGI information to GMLC. If time difference is greater than the provisioned value, then AMF will request current nrCGI either via paging or location report control depends UE status

Type / Range

Integer in the range of 0 to 180

Default

0

Name

maxNrCgiAgeLawfulInterceptMtLrNoLmf - Maximum Time Interval In Seconds For Requesting nrCGI Lawful Intercept MT-LR

Syntax

```
--amfGparmName maxNrCgiAgeLawfulInterceptMtLrNoLmf --amfGparmValue  
<string>
```

Description

If the parameter is set to 0 with the amfLcsGParms parameter LawfullInterceptMtLrNoLmf sets to yes, AMF page 5GMM-IDLE UE and send nrCGI information or send location-reporting control information for CM. If the parameter is set to a non-zero value, when AMF gets MT-LR request, AMF will check the difference between AMF stored nrCGI timestamp of the UE and time received MT-LR. If the time difference is less than provisioned value, then the AMF reports stored nrCGI information to GMLC. If time difference is greater than the provisioned value (or equal provisioned value), AMF will request current nrCGI either via paging or location report control depends UE status

Type / Range

Integer in the range of 0 to 180 in units of Second

Default

0

Name

maxNrCgiAgePlmnOperMtLrNoLmf - Max nrCGI Age PLMN Operator MT-LR

Syntax

--amfGparmName maxNrCgiAgePlmnOperMtLrNoLmf --amfGparmValue <string>

Description

Maximum time interval in seconds for requesting nrCGI PLMN operator MT-LR. If the parameter is set to 0 with the amfLcsGParms parameter plmnOperMtLrNoLmf sets to yes, AMF page 5GMM-IDLE UE and send nrCGI information or send location-reporting control information. If the parameter is set to a non-zero value, when AMF gets MT-LR request, AMF will check the difference between AMF stored nrCGI timestamp of the UE and time received MT-LR. If the time difference is less than provisioned value, then the AMF reports stored nrCGI information to GMLC. If time difference is greater than the provisioned value (or equal provisioned value), AMF will request current nrCGI either via paging or location report control depends UE status

Type / Range

Integer in the range of 0 to 180 in units of Second

Default

0

Name

maxNrCgiAgeValueAddedMtLrNoLmf - Max NrCgi Age Value Added MT-LR

Syntax

--amfGparmName maxNrCgiAgeValueAddedMtLrNoLmf --amfGparmValue <string>

Description

Maximum time interval in seconds for requesting nrCGI for value added MT-LR. If the parameter is set to 0 with the amfLcsGParms parameter valueAddedMtLrNoLmf sets to yes, AMF page 5GMM-IDLE UE and send nrCGI information or send location-reporting control

information. If the parameter is set to a non-zero value, when AMF gets MT-LR request, AMF will check the difference between AMF stored nrCGI timestamp of the UE and time received MT-LR. If the time difference is less than provisioned value, then the AMF reports stored nrCGI information to GMLC. If time difference is greater than the provisioned value (or equal provisioned value), AMF will request current nrCGI either via paging or location report control depends UE status

Type / Range

Integer in the range of 0 to 180 in units of Second

Default

0

Name

nonUeAssociatedNrppa - Non UE Associated NRPPa

Syntax

--amfGparmName nonUeAssociatedNrppa --amfGparmValue <string>

Description

With the parameter set, AMF supports non UE Associated NRPPa transfer between LMF and NG-RAN. If the parameter is not set, AMF will not support non UE Associated NRPPa transfer between LMF and NG-RAN.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

plmnOperMtLrNoLmf - PLMN Operator MT_LR No LMF

Syntax

--amfGparmName plmnOperMtLrNoLmf --amfGparmValue <string>

Description

MT_LR client type PLMN Operator Service do not contact LMF for UE location information. If gParms is 'yes', when AMF receives mobile terminated location request with client type=plmnOperatorServices, and activateLcsClientTypePlmnOperatorServices is set, the AMF contacts the gNB for UE current cell information. This information is sent back to the GMLC. The AMF does not contact the LMF to obtain any UE location information. If gParms is 'no', nothing changes and keep existing function (i.e. AMF contacts LMF to obtain UE location for this MT-LR type) .

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Name

posGpsi - POS GPSI

Syntax

--amfGparmName posGpsi --amfGparmValue <string>

Description

If the control parameter is set, it means that AMF supports the 'POSGPSI' feature, the ueContextId identifies the individual ueContext resource may also be composed by UE's GPSI.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

valueAddedMtLrNoLmf - Value Added MT_LR No LMF

Syntax

--amfGparmName valueAddedMtLrNoLmf --amfGparmValue <string>

Description

MT_LR client type value-added do not contact lmf for UE location information. If gParms is 'yes', when AMF receives mobile terminated location request with client type=valueAdded, and activateLcsClientTypeValueAddedServices is set, the AMF contacts the gNB for UE current cell information. This information is sent back to the GMLC. The AMF does not contact the LMF to obtain any UE location information. If gParms is 'no', nothing change and keep existing function (i.e. AMF contacts LMF to obtain UE location for this MT-LR type

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

3.70 cmm amfLinkAdmin

AMF Link Administration. Syntax, description, parameters, and roles for cmm amfLinkAdmin command.

Syntax

Description

AMF link administration - Modify/Show the administrative state of a 5G link

Parameters

Name

amfLinkType - AMF Link Type

Syntax

--amfLinkType <list-item>

Description

AMF link type

Class

Required key.

List

- N2 - N2 link - N2 link between AMF and gNB
- N26 - N26 link - N26 link between AMF and MME
- SBI_Client - 5G Client SBI - 5G Client Service Based link
- SBI_Server - 5G Server SBI - 5G Server Service Based link

Name

linkIndex - AMF Link Index

Syntax

--linkIndex <integer>

Description

AMF link index

Class

Required key.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

amfLinkAdminState - AMF Link Admin State

Syntax

--amfLinkAdminState <list-item>

Description

AMF link admin state

Class

Required on create.

List

- blocked - Blocked State - only used for SBI_Server link types
- locked - Locked State - only used for SBI_Client, N2 and N26 links types
- unlocked - Unlocked State

Name

alarmState - Alarm State

Syntax

--alarmState <list-item>

Description

Alarm state

Class

Read only.

List

- critical - Critical alarm.
- major - Major alarm.
- minor - Minor alarm.
- none - No alarm.
- unknown - Unknown.

Name

linkIdentifier - Link identifier

Syntax

--linkIdentifier <string>

Description

Link information.

Class

Read only.

Type / Range

Digit string of length 0 or 1 to 255

Name

linkOperationState - Link Operation State

Syntax

--linkOperationState <list-item>

Description

Link operation state

Class

Read only.

List

- disabled - Disabled state of the link.
- enabled - Enabled state of the link.
- suspended - Suspended state of the link.
- unknown - Unknown state of the link. this can not be used during query.

Name

localIp1 - Local IP1

Syntax

--localIp1 <string>

Description

Local IP1.

Class

Read only.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

localIp2 - Local IP2

Syntax

--localIp2 <string>

Description

Local IP2.

Class

Read only.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

localPort1 - Local Port 1

Syntax

--localPort1 <integer>

Description

Local Port associated with the local IP address 1

Class

Read only.

Type / Range

Integer in the range of 0 to 65,534

Name

localPort2 - Local Port 2

Syntax

--localPort2 <integer>

Description

Local Port associated with the local IP address 2

Class

Read only.

Type / Range

Integer in the range of 0 to 65,534

Name

poolId - poolId

Syntax

--poolId <integer>

Description

IPDS Pool ID - ID will only be 0 for non Multi-IPDS instantiations.

Class

Read only.

Type / Range

Integer in the range of 0 to 31

Name

remotelp1 - Remote IP1

Syntax

--remoteIp1 <string>

Description

Remote IP1.

Class

Read only.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

remotelp2 - Remote IP2

Syntax

--remoteIp2 <string>

Description

Remote IP2.

Class

Read only.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

remotePort1 - Remote Port 1

Syntax

--remotePort1 <integer>

Description

Remote Port associated with the remote IP address 1

Class

Read only.

Type / Range

Integer in the range of 0 to 65,534

Name

remotePort2 - Remote Port 2

Syntax

--remotePort2 <integer>

Description

Remote Port associated with the remote IP address 2

Class

Read only.

Type / Range

Integer in the range of 0 to 65,534

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

3.71 cmm amfLinkAlarmCfg

AMF Link Alarm Config. Syntax, description, parameters, and roles for cmm amfLinkAlarmCfg command.

Syntax**Description**

Administer Link Alarm Severity for AMF.

Parameters**Name**

amfIntfType - AMF Interface Type

Syntax

--amfIntfType <list-item>

Description

AMF interface type.

Class

Required key.

List

- N11 - N11 Interface - N11 Interface between AMF and SMF
- N12 - N12 Interface - N12 Interface between AMF and AUSF
- N14 - N14 Interface - N14 Interface between AMF and AMF
- N15 - N15 Interface - N15 Interface between AMF and PCF
- N17 - N17 Interface - N17 Interface between AMF and 5G EIR

- N18 - N18 interface between AMF and UDSF
- N2 - N2 Interface - N2 Interface between AMF and 5gNB
- N20 - N20 Interface - N20 Interface between AMF and SMSF
- N22 - N22 Interface - N22 Interface between AMF and NSSF
- N26 - N26 Interface - N26 Interface between AMF and MME
- N41 - N41 interface - N41 interface between AMF and CHF
- N58 - N58 Interface - N58 interface between AMF and NSSAAF
- N8 - N8 Interface - N8 Interface between AMF and UDM
- NLg - NLg Interface - NLg Interface between AMF and GMLC
- NLs - NLs Interface - NLs Interface between AMF and LMF
- Nnef - Nnef Interface - Nnef Interface between AMF and Network Event Function
- Nnrf - Nnrf Interface - Nnrf Interface between AMF and NRF
- Nnwdaf - Nnwdaf Interface - Nnwdaf interface between AMF and NWDAF
- namf_Communications - AMF Communications Service
- namf_EventExposure - AMF Event Exposure Service
- namf_Location - AMF Location Service
- namf_MT - AMF Mobile Terminated Service
- notification_AMF - AMF Notification - AMF Notification Function
- notification_NRF - NRF Notification - NRF Notification Function
- notification_NSSAAF - NSSAAF Notification - NSSAAF Notification Function
- notification_NSSF - NSSF Notification - NSSF Notification Function
- notification_NWDAF - NWDAF Notification - NWDAF Notification Function
- notification_PCF - PCF Notification - PCF Notification Function
- notification_SMF - SMF Notification - SMF Notification Function
- notification_UDM - UDM Notification - UDM Notification Function

Name

absoluteLinkLossExcessiveAlarm - Absolute Link Loss Excessive Alarm

Syntax

--absoluteLinkLossExcessiveAlarm <integer>

Description

This parameter indicates the absolute number of unavailable Links necessary to trigger the Excessive Links Lost Alarm. This parameter applies to N2 and N11 only.

Class

Optional.

Type / Range

Integer in the range of 5 to 100

Default

100

Name

aggrSeverity - Aggregate severity

Syntax

--aggrSeverity <list-item>

Description

Aggregate severity.

Class

Optional.

List

- critical - Critical alarm.
- major - Major alarm.
- minor - Minor alarm.
- none - No alarm.

Default

critical

Name

individualSeverity - Individual severity

Syntax

--individualSeverity <list-item>

Description

Individual severity.

Class

Optional.

List

- critical - Critical alarm.
- major - Major alarm.
- minor - Minor alarm.
- none - No alarm.

Default

major

Name

percentLinkLossAlarm - Percentage of Link Loss Alarm

Syntax

--percentLinkLossAlarm <integer>

Description

This parameter specifies the percentage of unavailable links at which to raise a critical interface alarm. This parameter applies to N2 and N11 only.

Class

Optional.

Type / Range

Integer in the range of 5 to 100 in units of %

Default

20

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.72 cmm amfLinkQueryByGnb

5G Link query by gnb. Syntax, description, parameters, and roles for cmm amfLinkQueryByGnb command.

Syntax

Description

5G Link query by gNB ID. When multiple TNLAs exist for a gNB, one is designated the primary TNLA for the purposes of performing non-UE-associated signaling procedures. amfLinkQueryByGnb prints information for the primary gNB TNLA. Use ngRanTnlaQuery for listing gNB information for each TNLA of a gNB.

Parameters

Name

mcc - Mobile Country Code

Syntax

--mcc <digits>

Description

3-digit number used to represent the Mobile Country Code

Class

Required key.

Type / Range

Digit string of length 0 or 3

Name

mnc - Mobile Network Code

Syntax

--mnc <digits>

Description

Digits used to represent Mobile Network Code.

Class

Required key.

Type / Range

Digit string of length 0 or 2 to 3

Name

macroGnbId - Macro gNB ID

Syntax

--macroGnbId <integer>

Description

Macro gNB ID

Class

Required key.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

alarmState - Alarm state

Syntax

--alarmState <list-item>

Description

Alarm state.

Class

Read only.

List

- critical - Critical alarm.
- major - Major alarm.
- minor - Minor alarm.
- none - No alarm.
- unknown - Unknown.

Name

amfLinkAdminState - 5G link admin state

Syntax

--amfLinkAdminState <list-item>

Description

Link admin state.

Class

Read only.

List

- blocked - Blocked State - only used for SBI_Server link types
- locked - Locked State - only used for SBI_Client, N2 and N26 links types
- unlocked - Unlocked State

Name

amfLinkType - 5G Link type

Syntax

--amfLinkType <list-item>

Description

5G Link type.

Class

Read only.

List

- N2 - N2 link - N2 link between AMF and gNB
- N26 - N26 link - N26 link between AMF and MME
- SBI_Client - 5G Client SBI - 5G Client Service Based link
- SBI_Server - 5G Server SBI - 5G Server Service Based link

Name

linkIdentifier - Link identifier

Syntax

--linkIdentifier <string>

Description

Link information.

Class

Read only.

Type / Range

Digit string of length 0 or 1 to 255

Name

linkIndex - Link index

Syntax

--linkIndex <integer>

Description

Link index.

Class

Read only.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

linkOperationState - Link operation state

Syntax

--linkOperationState <list-item>

Description

Link operation state

Class

Read only.

List

- disabled - Disabled state of the link.
- enabled - Enable state of the link.
- unknown - Unknown state of the link. this can not be used during query.

Name

localIp1 - Local IP1

Syntax

--localIp1 <string>

Description

Local IP1.

Class

Read only.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

localIp2 - Local IP2

Syntax

--localIp2 <string>

Description

Local IP2.

Class

Read only.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

localPort1 - Local Port 1

Syntax

--localPort1 <integer>

Description

Local Port associated with the local IP address 1

Class

Read only.

Type / Range

Integer in the range of 0 to 65,534

Name

localPort2 - Local Port 2

Syntax

--localPort2 <integer>

Description

Local Port associated with the local IP address 2

Class

Read only.

Type / Range

Integer in the range of 0 to 65,534

Name

poolId - poolId

Syntax

--poolId <integer>

Description

IPDS Pool ID - ID will only be 0 for non Multi-IPDS instantiations.

Class

Read only.

Type / Range

Integer in the range of 0 to 31

Name

remotep1 - Remote IP1

Syntax

--remoteIp1 <string>

Description

Remote IP1.

Class

Read only.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

remotep2 - Remote IP2

Syntax

--remoteIp2 <string>

Description

Remote IP2.

Class

Read only.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

remotePort1 - Remote Port 1

Syntax

--remotePort1 <integer>

Description

Remote Port associated with the remote IP address 1

Class

Read only.

Type / Range

Integer in the range of 0 to 65,534

Name

remotePort2 - Remote Port 2

Syntax

--remotePort2 <integer>

Description

Remote Port associated with the remote IP address 2

Class

Read only.

Type / Range

Integer in the range of 0 to 65,534

Roles**Show**

sysadmin, manage, monitor

3.73 cmm amfLinkQueryByN3iwf

5G Link query by n3iwf. Syntax, description, parameters, and roles for cmm amfLinkQueryByN3iwf command.

Syntax

Description

5G Link query by N3IWF ID. When multiple TNLAs exist for a N3IWF, one is designated the primary TNLA for the purposes of performing non-UE-associated signaling procedures. amfLinkQueryByN3iwf prints information for the primary N3IWF TNLA. Use ngRanTnlaQuery for listing n3iwf information for each TNLA of a N3IWF.

Parameters

Name

mcc - Mobile Country Code

Syntax

--mcc <digits>

Description

3-digit number used to represent the Mobile Country Code

Class

Required key.

Type / Range

Digit string of length 0 or 3

Name

mnc - Mobile Network Code

Syntax

--mnc <digits>

Description

Digits used to represent Mobile Network Code.

Class

Required key.

Type / Range

Digit string of length 0 or 2 to 3

Name

n3iwfId - N3IWF ID

Syntax

--n3iwfId <integer>

Description

N3IWF ID

Class

Required key.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

alarmState - Alarm state

Syntax

--alarmState <list-item>

Description

Alarm state

Class

Read only.

List

- critical - Critical alarm.
- major - Major alarm.
- minor - Minor alarm.
- none - No alarm.
- unknown - Unknown.

Name

amfLinkAdminState - 5G link admin state

Syntax

--amfLinkAdminState <list-item>

Description

Link admin state.

Class

Read only.

List

- blocked - Blocked State - only used for SBI_Server link types
- locked - Locked State - only used for SBI_Client, N2 and N26 links types
- unlocked - Unlocked State

Name

amfLinkType - 5G Link type

Syntax

--amfLinkType <list-item>

Description

5G Link type.

Class

Read only.

List

- N2 - N2 link - N2 link between AMF and gNB
- N26 - N26 link - N26 link between AMF and MME
- SBI_Client - 5G Client SBI - 5G Client Service Based link
- SBI_Server - 5G Server SBI - 5G Server Service Based link

Name

linkIdentifier - Link identifier

Syntax

--linkIdentifier <string>

Description

Link information

Class

Read only.

Type / Range

Digit string of length 0 or 1 to 255

Name

linkIndex - Link index

Syntax

--linkIndex <integer>

Description

Link index.

Class

Read only.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

linkOperationState - Link operation state

Syntax

--linkOperationState <list-item>

Description

Link operation state

Class

Read only.

List

- disabled - Disabled state of the link.
- enabled - Enable state of the link.
- unknown - Unknown state of the link. this can not be used during query.

Name

localIp1 - Local IP1

Syntax

--localIp1 <string>

Description

Local IP1.

Class

Read only.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

localIp2 - Local IP2

Syntax

--localIp2 <string>

Description

Local IP2

Class

Read only.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

localPort1 - Local Port 1

Syntax

--localPort1 <integer>

Description

Local Port associated with the local IP address 1

Class

Read only.

Type / Range

Integer in the range of 0 to 65,534

Name

localPort2 - Local Port 2

Syntax

--localPort2 <integer>

Description

Local Port associated with the local IP address 2

Class

Read only.

Type / Range

Integer in the range of 0 to 65,534

Name

poolId - poolId

Syntax

--poolId <integer>

Description

IPDS Pool ID - ID will only be 0 for non Multi-IPDS instantiations.

Class

Read only.

Type / Range

Integer in the range of 0 to 31

Name

remotep1 - Remote IP1

Syntax

--remoteIp1 <string>

Description

Remote IP1

Class

Read only.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

remotep2 - Remote IP2

Syntax

--remoteIp2 <string>

Description

Remote IP2

Class

Read only.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

remotePort1 - Remote Port 1

Syntax

--remotePort1 <integer>

Description

Remote Port associated with the remote IP address 1

Class

Read only.

Type / Range

Integer in the range of 0 to 65,534

Name

remotePort2 - Remote Port 2

Syntax

--remotePort2 <integer>

Description

Remote Port associated with the remote IP address 2

Class

Read only.

Type / Range

Integer in the range of 0 to 65,534

Roles**Show**

sysadmin, manage, monitor

3.74 cmm amfLinkQueryDetail

AMF Signalling Link Query Detail. Syntax, description, parameters, and roles for cmm amfLinkQueryDetail command.

Syntax

Description

AMF Signalling Link Show query link detail

Parameters

Name

amfLinkType - AMF Link Type

Syntax

--amfLinkType <list-item>

Description

AMF link type

Class

Required key.

List

- N2 - N2 link - N2 link between AMF and gNB
- N26 - N26 link - N26 link between AMF and MME
- SBI_Client - 5G Client SBI - 5G Client Service Based link
- SBI_Server - 5G Server SBI - 5G Server Service Based link

Name

linkIndex - Link Index

Syntax

--linkIndex <integer>

Description

Link index

Class

Required key.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

linkDetailCode - Link Detail Code

Syntax

--linkDetailCode <integer>

Description

Link detail code. Code 1 has been allocated

Class

Required key.

Type / Range

Integer in the range of 0 to 65,535

Name

detailOutput - Link details

Syntax

--detailOutput <string>

Description

Link details.

Class

Read only.

Type / Range

Digit string of length

Roles**Show**

sysadmin, manage, monitor

3.75 cmm amfLinkQueryPath

Query sctp Path. Syntax, description, parameters, and roles for cmm amfLinkQueryPath command.

Syntax**Description**

Query path information for SCTP links in 5G.

Parameters**Name**

amfIntfType - Link interface type

Syntax

--amfIntfType <list-item>

Description

Link interface type. Only SCTP interfaces are supported. Default to N2

Class

Required key.

List

- N2 - N2 Interface - List of (N2, ...)

Name

linkIndex - Link index

Syntax

--linkIndex <integer>

Description

Link index to read

Class

Required key.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

path1LocalIp - Path 1 Local IP

Syntax

--path1LocalIp <string>

Description

Path 1 Local IP

Class

Read only.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

path1LocalPort - Path 1 Local Port

Syntax

--path1LocalPort <integer>

Description

Path 1 Local Port

Class

Read only.

Type / Range

Integer in the range of 0 to 65,535

Name

path1Remotelp - Path 1 Remote IP

Syntax

--path1RemoteIp <string>

Description

Path 1 Remote IP

Class

Read only.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

path1RemotePort - Path 1 Remote Port

Syntax

--path1RemotePort <integer>

Description

Path 1 Remote Port

Class

Read only.

Type / Range

Integer in the range of 0 to 65,535

Name

path1Status - path 1 status

Syntax

--path1Status <list-item>

Description

Path 1 status

Class

Read only.

List

- not-used - Path status of not used.
- reachable - Path status of reachable.
- unknown - Path status of unknown.
- unreachable - Path status of unreachable.

Name

path2LocalIp - Path 2 Local IP

Syntax

--path2LocalIp <string>

Description

Path 2 Local IP

Class

Read only.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

path2LocalPort - Path 2 Local Port

Syntax

--path2LocalPort <integer>

Description

Path 2 Local Port

Class

Read only.

Type / Range

Integer in the range of 0 to 65,535

Name

path2RemoteIp - Path 2 Remote IP

Syntax

--path2RemoteIp <string>

Description

Path 2 Remote IP

Class

Read only.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

path2RemotePort - Path 2 Remote Port

Syntax

--path2RemotePort <integer>

Description

Path 2 Remote Port

Class

Read only.

Type / Range

Integer in the range of 0 to 65,535

Name

path2Status - Path 2 status

Syntax

--path2Status <list-item>

Description

Path 2 status

Class

Read only.

List

- not-used - Path status of not used.
- reachable - Path status of reachable.
- unknown - Path status of unknown.
- unreachable - Path status of unreachable.

Roles**Show**

sysadmin, manage, monitor

3.76 cmm amfMcsPagingPriProfile

AMF MCS Paging Priority Profile. Syntax, description, parameters, and roles for cmm amfMcsPagingPriProfile command.

Syntax**Description**

This command is used to administer Paging Priority Profiles for Mission Critical Services (MCS/MCX). The profile includes parameters for defining e.g. the ARP Priority level and paging priority levels.

Parameters**Name**

amfMcsPagingPriProfileListName - AMF MCS Paging Priority Profile List Name

Syntax

--amfMcsPagingPriProfileListName <string>

Description

Name that identifies the AMF MCS Paging Priority Profile List.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

arpLevel - Allocation Retention Priority Level

Syntax

--arpLevel <integer>

Description

Allocation Retention Priority (ARP) Priority Level related to MCS specific paging priority profile on AMF. Defines the relative importance of a resource request. Values are ordered in decreasing order of priority, i.e., with 1 as the highest priority and 15 as the lowest priority.

Class

Required key.

Type / Range

Integer in the range of 1 to 15

Name

includePagingPriority - Include Paging Priority

Syntax

--includePagingPriority <list-item>

Description

Defines whether the AMF includes the provisioned Paging Priority on the NGAP Paging message (when set to true) or suppresses it (when set to false).

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

pagingPriorityLevel - Paging Priority Level

Syntax

--pagingPriorityLevel <list-item>

Description

The Priority Level to be used for the paging.

Class

Optional.

List

- PrioLevel0 - Priority level 0 - Priority level 0. No paging priority is set.
- PrioLevel1 - Priority level 1 - Priority level 1. Highest Paging priority.
- PrioLevel2 - Priority level 2 - Priority level 2.
- PrioLevel3 - Priority level 3 - Priority level 3.
- PrioLevel4 - Priority level 4 - Priority level 4.
- PrioLevel5 - Priority level 5 - Priority level 5.
- PrioLevel6 - Priority level 6 - Priority level 6.
- PrioLevel7 - Priority level 7 - Priority level 7.
- PrioLevel8 - Priority level 8 - Priority level 8. Lowest Paging Priority.

Default

PrioLevel0

Roles**Create**

secadmin

Show

sysadmin, manage, secadmin

Modify

secadmin

Delete

secadmin

List

sysadmin, manage, secadmin

3.77 cmm amfMcsPagingPriProfileList

AMF MCS Paging Priority Profile List. Syntax, description, parameters, and roles for cmm amfMcsPagingPriProfileList command.

Syntax

Description

This command is used to administer a Paging Priority Profile List for Mission Critical Services (MCS/MCX).

Parameters

Name

name - AMF MCS Paging Priority Profile Name

Syntax

--name <digits>

Description

Defines the paging priority profile name for Mission Critical Services (MCS/MSX).

Class

Required key.

Type / Range

Digit string of length 1 to 1 with the allowed pattern of [1-8]*

Roles

Create

secadmin

Show

sysadmin, manage, secadmin

Delete

secadmin

List

sysadmin, manage, secadmin

3.78 cmm amfMobilityTimers

AMF Mobility Timers. Syntax, description, parameters, and roles for cmm amfMobilityTimers command.

Syntax

Description

Define AMF Mobility Timers

Parameters

Name

amfTimerName - Timer Name

Syntax

--amfTimerName <list-item>

Description

Name of timer.

Class

Required key.

List

- amfInitDeregistration - AMF initiated de-registration timer - Timer starts when 'Mobile reachable' timer expires and stops when a NAS message is received. UE is implicitly de-registered when timer expires.
- connGutiReallocation - GUTI Reallocation timer - AMF re-allocates a new GUTI, while in CM Connected mode, on timer expiry. AMF restarts the timer when running if AMF reallocates a new GUTI during a NAS procedure (e.g. Service Request in connected mode).
- hoMobilityRegistrationUpdateWait - HO Mobility Registration Update Wait - Configure how long to wait for Mobility Registration Update after Xn HO or N2 HO
- mobilityReachable - Mobility Reachable timer beyond T3512 - Minutes beyond T3512 for taking actions when a UE does not send Periodic Mobility Registration Update. T3512 timer starts when UE enters idle state and stops when a NAS message is received from the UE.
- n14RmIsmfVsmfAtSrcAmf - N14 remove I-SMF or V-SMF at source AMF - Used in N14 interactions to remove SM context from old I-SMF or old V-SMF at the source AMF after UE has moved to a new AMF and new AMF has indicated I-SMF or V-SMF is going to be changed or removed. This timer supersedes timer n14RmUeDataAtSrcAmf, when I-SMF/ V-SMF is to be removed or changed.
- n14RmUeDataAtSrcAmf - N14 remove UE data at source AMF - Used in N14 interactions to remove UE data at the source AMF after UE mobility with AMF relocation.
- n26RmUeDataAtSrcAmf - N26 remove UE data at source AMF - Used in N26 interactions to remove UE data at the source AMF after UE mobility towards MME
- n2HoRanStatusN1N2Transfer - N2 Handover RAN Status N1N2 Transfer - Defines the time the source AMF waits for Namf_N1N2MessageTransfer Response from the target AMF during inter AMF N2 Handover procedure, when the source AMF has received NGAP Uplink RAN Status Transfer from the source NG-RAN and forwards the RAN status to the target AMF.
- purgeTimer - 5G Purge timer - Purge timer starts after the AMF initiated de-registration timer fires. AMF purges UE context data and deletes AMF association with the UDM for this UE after timer expiry.

- rcisAmfPurgeTimer - RCIS AMF Purge Timer - RCIS Purge Timer for AMF UEs
- registrationStatusUpdate - Registration Status Update timer - Supervises UE Context Transfer result during UE inter-AMF Registration or UE Registration with AMF re-allocation.
- secondaryRatUsageReportingDelay - Secondary Rat Usage Reporting Delay during Handover - AMF delays forwarding the Secondary RAT Usage report to the SMF during Handover
- smfAggregate - SMF aggregate timer - Time AMF waits for SMF(s) to respond before aggregating a response back to the RAN during Xn HO. AMF doesn't have to wait for the timer expiry if all expected responses are received before the timer expires.
- srvcc5gRemovePdus - 5G-SRVCC Remove PDUs - Defines the timer on the source AMF to trigger removal of all 3GPP access related PDU sessions of the UE after the 5G-SRVCC handover for the voice call component is complete.
- ueCtxMovedToEpsPurgeTimer - 5G Purge Timer for UE Contexts Moved to EPS - Purge timer starts after the UE moves to EPS. AMF purges UE context data that were preserved when UE moved to EPS and deletes AMF association with the UDM for this UE after timer expiry.
- ueSorAck - UE SoR Acknowledge Timer - Time to wait for receiving SoR Acknowledge from the UE.

Name

amfTimerUnit - Timer Unit

Syntax

--amfTimerUnit <list-item>

Description

Unit of measurement

Class

Required on create, not modifiable.

List

- Millisecond - Specify timer unit of 1 millisecond
- Second - Specify timer unit of 1 second
- 2Second - 2 Second - Specify timer unit of 2 seconds
- 30Second - 30 Second - Specify timer unit of 30 seconds.
- Minute - Specify timer unit of 1 minute
- DeciHour - Specify timer unit of 1 decihour (6 minutes).
- 10Minute - 10 Minute - Specify timer unit of 10 minutes
- Hour - Specify timer unit of 1 hour

- 10Hour - 10 Hour - Specify timer unit of 10 hours.

Name

amfTimerValue - Timer value

Syntax

--amfTimerValue <integer>

Description

Numeric value of timer

Class

Required on create.

Type / Range

Integer in the range of 0 to 65,535

The amfTimerValue string is based on the amfTimerName parameter:

Name

amfInitDeregistration - AMF initiated de-registration timer

Syntax

--amfTimerName amfInitDeregistration --amfTimerValue <string> [--
amfTimerUnit Minute]

Description

Timer starts when 'Mobile reachable' timer expires and stops when a NAS message is received. UE is implicitly de-registered when timer expires.

Type / Range

Integer in the range of 0 to 1,500 in units of Minute

Default

240

Name

connGutiReallocation - GUTI Reallocation timer

Syntax

--amfTimerName connGutiReallocation --amfTimerValue <string> [--
amfTimerUnit 10Minute]

Description

AMF re-allocates a new GUTI, while in CM Connected mode, on timer expiry. AMF restarts the timer when running if AMF reallocates a new GUTI during a NAS procedure (e.g. Service Request in connected mode).

Type / Range

Integer in the range of 0 to 1,000 in units of 10Minute

Default

0

Name

hoMobilityRegistrationUpdateWait - HO Mobility Registration Update Wait

Syntax

```
--amfTimerName hoMobilityRegistrationUpdateWait --amfTimerValue  
<string> [ --amfTimerUnit Millisecond ]
```

Description

Configure how long to wait for Mobility Registration Update after Xn HO or N2 HO

Type / Range

Integer in the range of 100 to 10,000 in units of Millisecond

Default

1000

Name

mobilityReachable - Mobility Reachable timer beyond T3512

Syntax

```
--amfTimerName mobilityReachable --amfTimerValue <string> [ --  
amfTimerUnit Minute ]
```

Description

Minutes beyond T3512 for taking actions when a UE does not send Periodic Mobility Registration Update. T3512 timer starts when UE enters idle state and stops when a NAS message is received from the UE.

Type / Range

Integer in the range of 1 to 60 in units of Minute

Default

4

Name

n14RmIsmfVsmfAtSrcAmf - N14 remove I-SMF or V-SMF at source AMF

Syntax

```
--amfTimerName n14RmIsmfVsmfAtSrcAmf --amfTimerValue <string> [ --  
amfTimerUnit Second ]
```

Description

Used in N14 interactions to remove SM context from old I-SMF or old V-SMF at the source AMF after UE has moved to a new AMF and new AMF has indicated I-SMF or V-SMF is going to be changed or removed. This timer supersedes timer n14RmUeDataAtSrcAmf, when I-SMF/V-SMF is to be removed or changed.

Type / Range

Integer in the range of 0 to 10 in units of Second

Default

2

Name

n14RmUeDataAtSrcAmf - N14 remove UE data at source AMF

Syntax

```
--amfTimerName n14RmUeDataAtSrcAmf --amfTimerValue <string> [ --  
amfTimerUnit Millisecond ]
```

Description

Used in N14 interactions to remove UE data at the source AMF after UE mobility with AMF relocation.

Type / Range

Integer in the range of 0 to 5,000 in units of Millisecond

Default

0

Name

n26RmUeDataAtSrcAmf - N26 remove UE data at source AMF

Syntax

```
--amfTimerName n26RmUeDataAtSrcAmf --amfTimerValue <string> [ --  
amfTimerUnit Millisecond ]
```

Description

Used in N26 interactions to remove UE data at the source AMF after UE mobility towards MME

Type / Range

Integer in the range of 0 to 5,000 in units of Millisecond

Default

0

Name

n2HoRanStatusN1N2Transfer - N2 Handover RAN Status N1N2 Transfer

Syntax

```
--amfTimerName n2HoRanStatusN1N2Transfer --amfTimerValue <string>  
[ --amfTimerUnit Millisecond ]
```

Description

Defines the time the source AMF waits for Namf_N1N2MessageTransfer Response from the target AMF during inter AMF N2 Handover procedure, when the source AMF has received NGAP Uplink RAN Status Transfer from the source NG-RAN and forwards the RAN status to the target AMF.

Type / Range

Integer in the range of 100 to 30,000 in units of Millisecond

Default

5000

Name

purgeTimer - 5G Purge timer

Syntax

```
--amfTimerName purgeTimer --amfTimerValue <string> [ --amfTimerUnit  
Hour ]
```

Description

Purge timer starts after the AMF initiated de-registration timer fires. AMF purges UE context data and deletes AMF association with the UDM for this UE after timer expiry.

Type / Range

Integer in the range of 1 to 120 in units of Hour

Default

48

Name

rcisAmfPurgeTimer - RCIS AMF Purge Timer

Syntax

```
--amfTimerName rcisAmfPurgeTimer --amfTimerValue <string> [ --  
amfTimerUnit Hour ]
```

Description

RCIS Purge Timer for AMF UEs

Type / Range

Integer in the range of 1 to 120 in units of Hour

Default

48

Name

registrationStatusUpdate - Registration Status Update timer

Syntax

```
--amfTimerName registrationStatusUpdate --amfTimerValue <string> [ --  
amfTimerUnit Millisecond ]
```

Description

Supervises UE Context Transfer result during UE inter-AMF Registration or UE Registration with AMF re-allocation.

Type / Range

Integer in the range of 10 to 10,000 in units of Millisecond

Default

8000

Name

secondaryRatUsageReportingDelay - Secondary Rat Usage Reporting Delay during Handover

Syntax

```
--amfTimerName secondaryRatUsageReportingDelay --amfTimerValue  
<string> [ --amfTimerUnit Second ]
```

Description

AMF delays forwarding the Secondary RAT Usage report to the SMF during Handover

Type / Range

Integer in the range of 1 to 30 in units of Second

Default

10

Name

smfAggregate - SMF aggregate timer

Syntax

```
--amfTimerName smfAggregate --amfTimerValue <string> [ --amfTimerUnit  
Millisecond ]
```

Description

Time AMF waits for SMF(s) to respond before aggregating a response back to the RAN during Xn HO. AMF doesn't have to wait for the timer expiry if all expected responses are received before the timer expires.

Type / Range

Integer in the range of 100 to 10,000 in units of Millisecond

Default

500

Name

srvcc5gRemovePdu - 5G-SRVCC Remove PDUs

Syntax

```
--amfTimerName srvcc5gRemovePdu --amfTimerValue <string> [ --  
amfTimerUnit Second ]
```

Description

Defines the timer on the source AMF to trigger removal of all 3GPP access related PDU sessions of the UE after the 5G-SRVCC handover for the voice call component is complete.

Type / Range

Integer in the range of 1 to 10 in units of Second

Default

2

Name

ueCtxMovedToEpsPurgeTimer - 5G Purge Timer for UE Contexts Moved to EPS

Syntax

```
--amfTimerName ueCtxMovedToEpsPurgeTimer --amfTimerValue <string>  
[ --amfTimerUnit Hour ]
```

Description

Purge timer starts after the UE moves to EPS. AMF purges UE context data that were preserved when UE moved to EPS and deletes AMF association with the UDM for this UE after timer expiry.

Type / Range

Integer in the range of 0 to 120 in units of Hour

Default

0

Name

ueSorAck - UE SoR Acknowledge Timer

Syntax

```
--amfTimerName ueSorAck --amfTimerValue <string> [ --amfTimerUnit  
Millisecond ]
```

Description

Time to wait for receiving SoR Acknowledge from the UE.

Type / Range

Integer in the range of 100 to 10,000 in units of Millisecond

Default

5000

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

3.79 cmm amfMpsLevels

AMF MPS Levels. Syntax, description, parameters, and roles for cmm amfMpsLevels command.

Syntax

Description

Defines MPS priority (sub)levels AMF globally. 5 MPS sublevels with specific SBI Message Priority (SMP) value ranges are defined. Applies when global parameter Multilevel MPS is set to true. Default MPS levels are as follows: Level (SMP values): 0 (0..0), 1 (1..4), 2 (5..10), 3 (11..16), 4 (17..23). When MPS level provisioning is modified, all value limits are required to be provided. The SMP value ranges are not allowed to overlap, and all MPS SMP ranges must have higher priority (i.e. lower SMP value) than the 'default' SMP value 24: 0 <= MPS Level 0 SBI Message Priority Max < MPS Level 1 SBI Message Priority Max < MPS Level 2 SBI Message Priority Max < MPS Level 3 SBI Message Priority Max < MPS Level 4 SBI Message Priority Max < 24.

Parameters

Name

mpsLevel0SmpMax - MPS Level 0 SBI Message Priority Max

Syntax

--mpsLevel0SmpMax <integer>

Description

Defines the SBI Message Priority (SMP) max value (lowest priority, highest numerical value) that still is included in the MPS level 0. MPS Level 0 consists of SMP values spanning between SMP values 0 and (MPS Level 0 SBI Message Priority Max).

Class

Optional.

Type / Range

Integer in the range of 0 to 19

Default

0

Name

mpsLevel1SmpMax - MPS Level 1 SBI Message Priority Max

Syntax

--mpsLevel1SmpMax <integer>

Description

Defines the SBI Message Priority (SMP) max value (lowest priority, highest numerical value) that still is included in the MPS level 1. MPS Level 1 consists of SMP values spanning between

SMP values (MPS Level 0 SBI Message Priority Max + 1) and (MPS Level 1 SBI Message Priority Max).

Class

Optional.

Type / Range

Integer in the range of 1 to 20

Default

4

Name

mpsLevel2SmpMax - MPS Level 2 SBI Message Priority Max

Syntax

--mpsLevel2SmpMax <integer>

Description

Defines the SBI Message Priority (SMP) max value (lowest priority, highest numerical value) that still is included in the MPS level 2. MPS Level 2 consists of SMP values spanning between SMP values (MPS Level 1 SBI Message Priority Max + 1) and (MPS Level 2 SBI Message Priority Max).

Class

Optional.

Type / Range

Integer in the range of 2 to 21

Default

10

Name

mpsLevel3SmpMax - MPS Level 3 SBI Message Priority Max

Syntax

--mpsLevel3SmpMax <integer>

Description

Defines the SBI Message Priority (SMP) max value (lowest priority, highest numerical value) that still is included in the MPS level 3. MPS Level 3 consists of SMP values spanning between SMP values (MPS Level 2 SBI Message Priority Max + 1) and (MPS Level 3 SBI Message Priority Max).

Class

Optional.

Type / Range

Integer in the range of 3 to 22

Default

16

Name

mpsLevel4SmpMax - MPS SBI Message Priority Level 4 Max

Syntax`--mpsLevel4SmpMax <integer>`**Description**

Defines the SBI Message Priority (SMP) max value (lowest priority, highest numerical value) that still is included in the MPS level 4. MPS Level 4 consists of SMP values spanning between SMP values (MPS Level 3 SBI Message Priority Max + 1) and (MPS Level 4 SBI Message Priority Max).

Class

Optional.

Type / Range

Integer in the range of 4 to 23

Default

23

Roles**Show**

sysadmin, manage, secadmin

Modify

secadmin

List

sysadmin, manage, secadmin

3.80 cmm amfMultiLink

AMF Signalling Multilink List. Syntax, description, parameters, and roles for `cmm amfMultiLink` command.

Syntax**Description**

AMF signalling multiple link show

Parameters**Name**

amfIntfType - AMF Interface Type

Syntax

--amfIntfType <list-item>

Description

AMF interface type. Default to N8

Class

Required key.

List

- N11 - N11 Interface - N11 Interface between AMF and SMF
- N12 - N12 Interface - N12 Interface between AMF and AUSF
- N14 - N14 Interface - N14 Interface between AMF and AMF
- N15 - N15 Interface - N15 Interface between AMF and PCF
- N17 - N17 Interface - N17 Interface between AMF and 5G EIR
- N18 - N18 interface between AMF and UDSF
- N2 - N2 Interface - N2 Interface between AMF and 5gNB
- N20 - N20 Interface - N20 Interface between AMF and SMSF
- N22 - N22 Interface - N22 Interface between AMF and NSSF
- N26 - N26 Interface - N26 Interface between AMF and MME
- N41 - N41 interface - N41 interface between AMF and CHF
- N58 - N58 Interface - N58 interface between AMF and NSSAAF
- N8 - N8 Interface - N8 Interface between AMF and UDM
- NLg - NLg Interface - NLg Interface between AMF and GMLC
- NLs - NLs Interface - NLs Interface between AMF and LMF
- Nnef - Nnef Interface - Nnef Interface between AMF and Network Event Function
- Nnrf - Nnrf Interface - Nnrf Interface between AMF and NRF
- Nnwdaf - Nnwdaf Interface - Nnwdaf interface between AMF and NWDAF
- namf_Communications - AMF Communications Service
- namf_EventExposure - AMF Event Exposure Service
- namf_Location - AMF Location Service
- namf_MT - AMF Mobile Terminated Service
- notification_AMF - AMF Notification - AMF Notification Function
- notification_NRF - NRF Notification - NRF Notification Function

- notification_NSSAAF - NSSAAF Notification - NSSAAF Notification Function
- notification_NSSF - NSSF Notification - NSSF Notification Function
- notification_NWDAF - NWDAF Notification - NWDAF Notification Function
- notification_PCF - PCF Notification - PCF Notification Function
- notification_SMF - SMF Notification - SMF Notification Function
- notification_UDM - UDM Notification - UDM Notification Function

Default

N8

Name

alarmState - Alarm State

Syntax

--alarmState <list-item>

Description

Alarm state

Class

Read only.

List

- critical - Critical alarm.
- major - Major alarm.
- minor - Minor alarm.
- none - No alarm.
- unknown - Unknown.

Name

amfLinkAdminState - AMF Admin State

Syntax

--amfLinkAdminState <list-item>

Description

AMF admin state

Class

Optional.

List

- blocked - Blocked State - only used for SBI_Server link types
- locked - Locked State - only used for SBI_Client, N2 and N26 links types

- unlocked - Unlocked State

Name

amfLinkType - AMF Link Type

Syntax

--amfLinkType <list-item>

Description

AMF link type

Class

Read only.

List

- N2 - N2 link - N2 link between AMF and gNB
- N26 - N26 link - N26 link between AMF and MME
- SBI_Client - 5G Client SBI - 5G Client Service Based link
- SBI_Server - 5G Server SBI - 5G Server Service Based link

Name

endLinkIndex - End link index

Syntax

--endLinkIndex <integer>

Description

End link index from the read command.

Class

Optional.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

endPointName - AMF Local End Point Interface Name

Syntax

--endPointName <string>

Description

AMF local end point interface name

Class

Read only.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

linkAlarm - Link Alarmed

Syntax

--linkAlarm <list-item>

Description

If true query returns only alarmed links else return all links

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

linkIdentifier - Link identifier

Syntax

--linkIdentifier <string>

Description

Link information.

Class

Read only.

Type / Range

Digit string of length 0 or 1 to 255

Name

linkIndex - Link Index

Syntax

--linkIndex <integer>

Description

Link index

Class

Read only.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

linkOperationState - Link operation state

Syntax

```
--linkOperationState <list-item>
```

Description

Link operation state.

Class

Optional.

List

- disabled - Disabled state of the link.
- enabled - Enabled state of the link.
- suspended - Suspended state of the link.
- unknown - Unknown state of the link. this can not be used during query.

Name

localIp1 - Local IP1

Syntax

```
--localIp1 <string>
```

Description

Local IP1.

Class

Read only.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

localIp2 - Local IP2

Syntax

```
--localIp2 <string>
```

Description

Local IP2.

Class

Read only.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

localPort1 - Local Port 1

Syntax

--localPort1 <integer>

Description

Local Port associated with the local IP address 1

Class

Read only.

Type / Range

Integer in the range of 0 to 65,534

Name

localPort2 - Local Port 2

Syntax

--localPort2 <integer>

Description

Local Port associated with the local IP address 2

Class

Read only.

Type / Range

Integer in the range of 0 to 65,534

Name

maxResponseSize - Max record count

Syntax

--maxResponseSize <integer>

Description

Max number of records to be returned.

Class

Optional.

Type / Range

Integer in the range of 1 to 100

Name

poolId - poolId

Syntax

--poolId <integer>

Description

IPDS Pool ID - If chosen only the links on the specified IPDS will be output. The IPDS Pool ID will only be 0 for non Multi-IPDS instantiations. The default is ALL IPDS pools

Class

Read only.

Type / Range

Integer in the range of 0 to 31

Default

31

Name

previousIndex - Previous index

Syntax

--previousIndex <integer>

Description

Previous index when specified next available index will be returned.

Class

Optional.

Type / Range

Integer in the range of -1 or 0 to 2,147,483,647

Name

queryRmtIp - Query Remote IP address

Syntax

--queryRmtIp <string>

Description

IP address used to query for a link whose remote IP address matches the specified IP.

Class

Optional.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

remotelp1 - Remote IP1

Syntax

--remoteIp1 <string>

Description

Remote IP1.

Class

Read only.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

remoteIp2 - Remote IP2

Syntax

--remoteIp2 <string>

Description

Remote IP2.

Class

Read only.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

remotePort1 - Remote Port 1

Syntax

--remotePort1 <integer>

Description

Remote Port associated with the remote IP address 1

Class

Read only.

Type / Range

Integer in the range of 0 to 65,534

Name

remotePort2 - Remote Port 2

Syntax

--remotePort2 <integer>

Description

Remote Port associated with the remote IP address 2

Class

Read only.

Type / Range

Integer in the range of 0 to 65,534

Name

startLinkIndex - Start Link Index

Syntax

--startLinkIndex <integer>

Description

Start link index

Class

Optional.

Type / Range

Integer in the range of 0 to 4,294,967,295

Roles**List**

sysadmin, manage, monitor

3.81 cmm amfMultipleHomePlmn

Multiple Home Plmn. Syntax, description, parameters, and roles for cmm amfMultipleHomePlmn command.

Syntax**Description**

This command is used to configure an existing network as an multiple home plmn network and includes parameters for defining the serving and multiple home network names

Parameters**Name**

servedPlmnName - Serving Mobile Network

Syntax

--servedPlmnName <string>

Description

Name that identifies serving network.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

multipleHomePlmnName - Multiple Home Plmn Network name

Syntax

--multipleHomePlmnName <string>

Description

Name that identifies equivalent network.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.82 cmm amfMultiTacOperation

Tracking Area ID. Syntax, description, parameters, and roles for cmm amfMultiTacOperation command.

Syntax**Description**

Provision multiple TACs for a serving PLMN. A serving PLMN may be AMF home PLMN or shared PLMN. Maximum number of TACs supported in one operation shall be 100.

Parameters**Name**

plmnName - Serving PLMN Name

Syntax

--plmnName <string>

Description

The serving PLMN name may be the name of the AMF home PLMN or shared PLMN.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

maxTac - Max tracking area code

Syntax

--maxTac <integer>

Description

Maximum TAC value in defining set of TACs to create or update, used with mintac.

Class

Required on create.

Type / Range

Integer in the range of 0 to 16777215

Name

minTac - Min tracking area code

Syntax

--minTac <integer>

Description

Minimum TAC value in defining set of TACs to update or create, used with maxtac.

Class

Required on create.

Type / Range

Integer in the range of 0 to 16777215

Name

allTac - All Tracking Area

Syntax

--allTac <list-item>

Description

Define set of TACs to update to be all TACs in a given mobile network.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

amfRelativeCapacity - AMF Relative Capacity

Syntax

--amfRelativeCapacity <integer>

Description

Configures AMF relative capacity for the tracking area

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

5

Name

emergencyFallback - Emergency Fallback

Syntax

--emergencyFallback <list-item>

Description

Enables Emergency Fallback support in the tracking area. AMF indicates to the UE that emergency fallback is supported by the network if the emergency fallback is enabled in the tracking areas sent to the UE in the tracking area list. By default, the emergency fallback is not supported.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

emergencyServiceTai - Emergency Service TAI

Syntax

--emergencyServiceTai <list-item>

Description

This command is used to enable/disable Emergency service in the tracking area. AMF indicates to the UE that emergency service is supported by the network if the emergency service is enabled in the current tracking area and all other tracking areas sent to the UE in the tracking area neighbor list. By default, the emergency service is disabled.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

imsVopsSupport - IMS VoPS Support

Syntax

--imsVopsSupport <list-item>

Description

This parameter indicates whether IMS VoPS is supported on the Tracking Areas of the AMF. Voice over PS can be supported either via Voice over New Radio connected to 5G Core, or via VoLTE connected to EPS with the NG-RAN supporting handover or redirection to EPS for the UE at QoS Flow establishment for IMS voice.

Class

Optional.

List

- amfVopsNotSupported - AMF VoPS Not Supported - Voice over PS sessions is not supported on this Tracking Area of the AMF.
- amfVopsSupportedOverNrConnectedTo5gc - AMF VoPS Support Over New Radio Connected to 5GC - Voice over PS sessions is supported on this Tracking Area of the AMF by Voice over New Radio connected to 5G Core. Note: this value is only effective when gParm voNR is enabled.
- amfVopsSupportedVoNrVolteHoRe - AMF VoPS Supported VoNR and VoLTE Handover and Redirect - Voice over PS sessions on this Tracking Area of the AMF is supported by - Voice over New Radio connected to 5G Core, -Voice over LTE connected to EPS, and the NG-RAN supports both handover and redirection to EPS at QoS Flow establishment for IMS voice. Note: this value is only effective when gParms voiceEpsFallback and VoNR are enabled.
- amfVopsSupportedVolteHo - AMF VoPS Supported VoLTE Handover - Voice over PS sessions is supported on this Tracking Area of the AMF by Voice over LTE connected to EPS, and the NG-RAN supports handover to EPS at QoS Flow establishment for IMS voice. Note: this value is only effective when gParm voiceEpsFallback is enabled.
- amfVopsSupportedVolteHoRe - AMF VoPS Supported VoLTE Handover and Redirect - Voice over PS sessions on this Tracking Area of the AMF is supported by Voice over LTE connected to EPS, and the NG-RAN supports both handover and redirection to EPS at

QoS Flow establishment for IMS voice. Note: this value is only effective when gParm voiceEpsFallback is enabled.

- amfVopsSupportedVolteRedirect - AMF VoPS Supported VoLTE Redirect - Voice over PS sessions on this Tracking Area of the AMF is supported by Voice over LTE connected to EPS, and the NG-RAN supports redirection to EPS at QoS Flow establishment for IMS voice. Note: this value is only effective when gParm voiceEpsFallback is enabled.
- amfVopsSupportedVonnrVolteHo - AMF VoPS Supported VoNR and VoLTE Handover - Voice over PS sessions on this Tracking Area of the AMF is supported by -Voice over New Radio connected to 5G Core, -Voice over LTE connected to EPS, and the NG-RAN supports handover to EPS at QoS Flow establishment for IMS voice. Note: this value is only effective when gParms voiceEpsFallback and VoNR are enabled.
- amfVopsSupportedVonnrVolteRe - AMF VoPS Supported VoNR and VoLTE Redirect - Voice over PS sessions on this Tracking Area of the AMF is supported by -Voice over New Radio connected to 5G Core, -Voice over LTE connected to EPS, and the NG-RAN supports redirection to EPS at QoS Flow establishment for IMS voice. Note: this value is only effective when gParms voiceEpsFallback and VoNR are enabled.

Default

amfVopsNotSupported

Name

overwriteExistingTaiOnCreate - Overwrite Existing TAIs on Create

Syntax

--overwriteExistingTaiOnCreate <list-item>

Description

When this parameter is set to false (default value) and the given TAC range overlaps with already existing TAIs, the existing TAIs parameter values will be retained (i.e. create command will not overwrite parameter values of an existing amfTai). When this parameter is set to true, and the given TAC range overlaps with already created TAIs, the existing TAIs parameter values will be overwritten. This parameter is not applicable for the amfMultiTacOperation modify operation

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

snssaiListName - SNSSAI List Name

Syntax

```
--snssaiListName <string>
```

Description

This parameter is a pointer to a list of S-NSSAI. The S_NSSAI in the list are supported in all the TACs specified by the amfMultiTacOperation command. If a TAC is not provisioned with SNSSAI list then it is assumed that all SNSSAI are supported in the TAC.

Restrictions

The S_NSSAI lists only apply to NRF and SMF

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

timeZoneName - Time Zone Name

Syntax

```
--timeZoneName <list-item>
```

Description

Provision time zone of the TACs

Class

Optional.

List

- Africa/Abidjan - Timezone for africa/abidjan UTC offset of +00:00.
- Africa/Accra - Timezone for africa/accra UTC offset of +00:00.
- Africa/Addis_Ababa - Timezone for africa/addis_ababa UTC offset of +03:00.
- Africa/Algiers - Timezone for africa/algiers UTC offset of +01:00.
- Africa/Asmara - Timezone for africa/asmara UTC offset of +03:00.
- Africa/Asmera - Timezone for africa/asmera UTC offset of +03:00.
- Africa/Bamako - Timezone for africa/bamako UTC offset of +00:00.
- Africa/Bangui - Timezone for africa/bangui UTC offset of +01:00.
- Africa/Banjul - Timezone for africa/banjul UTC offset of +00:00.
- Africa/Bissau - Timezone for africa/bissau UTC offset of +00:00.
- Africa/Blantyre - Timezone for africa/blantyre UTC offset of +02:00.

- Africa/Brazzaville - Timezone for africa/brazzaville UTC offset of +01:00.
- Africa/Bujumbura - Timezone for africa/bujumbura UTC offset of +02:00.
- Africa/Cairo - Timezone for africa/cairo UTC offset of +02:00.
- Africa/Casablanca - Timezone for africa/casablanca UTC offset of +00:00.
- Africa/Ceuta - Timezone for africa/ceuta UTC offset of +01:00.
- Africa/Conakry - Timezone for africa/conakry UTC offset of +00:00.
- Africa/Dakar - Timezone for africa/dakar UTC offset of +00:00.
- Africa/Dar_es_Salaam - Timezone for africa/dar_es_salaam UTC offset of +03:00.
- Africa/Djibouti - Timezone for africa/djibouti UTC offset of +03:00.
- Africa/Douala - Timezone for africa/douala UTC offset of +01:00.
- Africa/El_Aaiun - Timezone for africa/el_aaiun UTC offset of +00:00.
- Africa/Freetown - Timezone for africa/freetown UTC offset of +00:00.
- Africa/Gaborone - Timezone for africa/gaborone UTC offset of +02:00.
- Africa/Harare - Timezone for africa/harare UTC offset of +02:00.
- Africa/Johannesburg - Timezone for africa/johannesburg UTC offset of +02:00.
- Africa/Juba - Timezone for africa/juba UTC offset of +03:00.
- Africa/Kampala - Timezone for africa/kampala UTC offset of +03:00.
- Africa/Khartoum - Timezone for africa/khartoum UTC offset of +03:00.
- Africa/Kigali - Timezone for africa/kigali UTC offset of +02:00.
- Africa/Kinshasa - Timezone for africa/kinshasa UTC offset of +01:00.
- Africa/Lagos - Timezone for africa/lagos UTC offset of +01:00.
- Africa/Libreville - Timezone for africa/libreville UTC offset of +01:00.
- Africa/Lome - Timezone for africa/lome UTC offset of +00:00.
- Africa/Luanda - Timezone for africa/luanda UTC offset of +01:00.
- Africa/Lubumbashi - Timezone for africa/lubumbashi UTC offset of +02:00.
- Africa/Lusaka - Timezone for africa/lusaka UTC offset of +02:00.
- Africa/Malabo - Timezone for africa/malabo UTC offset of +01:00.
- Africa/Maputo - Timezone for africa/maputo UTC offset of +02:00.
- Africa/Maseru - Timezone for africa/maseru UTC offset of +02:00.
- Africa/Mbabane - Timezone for africa/mbabane UTC offset of +02:00.
- Africa/Mogadishu - Timezone for africa/mogadishu UTC offset of +03:00.

- Africa/Monrovia - Timezone for africa/monrovia UTC offset of +00:00.
- Africa/Nairobi - Timezone for africa/nairobi UTC offset of +03:00.
- Africa/Ndjamena - Timezone for africa/ndjamena UTC offset of +01:00.
- Africa/Niamey - Timezone for africa/niamey UTC offset of +01:00.
- Africa/Nouakchott - Timezone for africa/nouakchott UTC offset of +00:00.
- Africa/Ouagadougou - Timezone for africa/ouagadougou UTC offset of +00:00.
- Africa/Porto-Novo - Timezone for africa/porto-novo UTC offset of +01:00.
- Africa/Sao_Tome - Timezone for africa/sao_tome UTC offset of +00:00.
- Africa/Timbuktu - Timezone for africa/timbuktu UTC offset of +00:00.
- Africa/Tripoli - Timezone for africa/tripoli UTC offset of +02:00.
- Africa/Tunis - Timezone for africa/tunis UTC offset of +01:00.
- Africa/Windhoek - Timezone for africa/windhoek UTC offset of +02:00.
- America/Adak - Timezone for america/adak UTC offset of -10:00.
- America/Anchorage - Timezone for america/anchorage UTC offset of -09:00.
- America/Anguilla - Timezone for america/anguilla UTC offset of -04:00.
- America/Antigua - Timezone for america/antigua UTC offset of -04:00.
- America/Araguaina - Timezone for america/araguaina UTC offset of -03:00.
- America/Argentina/Buenos_Aires - Timezone for america/argentina/buenos_aires UTC offset of -03:00.
- America/Argentina/Catamarca - Timezone for america/argentina/catamarca UTC offset of -03:00.
- America/Argentina/ComodRivadavia - Timezone for america/argentina/comodrivadavia UTC offset of -03:00.
- America/Argentina/Cordoba - Timezone for america/argentina/cordoba UTC offset of -03:00.
- America/Argentina/Jujuy - Timezone for america/argentina/jujuy UTC offset of -03:00.
- America/Argentina/La_Rioja - Timezone for america/argentina/la_rioja UTC offset of -03:00.
- America/Argentina/Mendoza - Timezone for america/argentina/mendoza UTC offset of -03:00.
- America/Argentina/Rio_Gallegos - Timezone for america/argentina/rio_gallegos UTC offset of -03:00.
- America/Argentina/Salta - Timezone for america/argentina/salta UTC offset of -03:00.

- America/Argentina/San_Juan - Timezone for america/argentina/san_juan UTC offset of -03:00.
- America/Argentina/San_Luis - Timezone for america/argentina/san_luis UTC offset of -03:00.
- America/Argentina/Tucuman - Timezone for america/argentina/tucuman UTC offset of -03:00.
- America/Argentina/Ushuaia - Timezone for america/argentina/ushuaia UTC offset of -03:00.
- America/Aruba - Timezone for america/aruba UTC offset of -04:00.
- America/Asuncion - Timezone for america/asuncion UTC offset of -03:00.
- America/Atikokan - Timezone for america/atikokan UTC offset of -05:00.
- America/Atka - Timezone for america/atka UTC offset of -10:00.
- America/Bahia - Timezone for america/bahia UTC offset of -03:00.
- America/Bahia_Banderas - Timezone for america/bahia_banderas UTC offset of -06:00.
- America/Barbados - Timezone for america/barbados UTC offset of -04:00.
- America/Belem - Timezone for america/belem UTC offset of -03:00.
- America/Belize - Timezone for america/belize UTC offset of -06:00.
- America/Blanc-Sablon - Timezone for america/blanc-sablon UTC offset of -04:00.
- America/Boa_Vista - Timezone for america/boa_vista UTC offset of -04:00.
- America/Bogota - Timezone for america/bogota UTC offset of -05:00.
- America/Boise - Timezone for america/boise UTC offset of -07:00.
- America/Buenos_Aires - Timezone for america/buenos_aires UTC offset of -03:00.
- America/Cambridge_Bay - Timezone for america/cambridge_bay UTC offset of -07:00.
- America/Campo_Grande - Timezone for america/campo_grande UTC offset of -03:00.
- America/Cancun - Timezone for america/cancun UTC offset of -05:00.
- America/Caracas - Timezone for america/caracas UTC offset of -04:00.
- America/Catamarca - Timezone for america/catamarca UTC offset of -03:00.
- America/Cayenne - Timezone for america/cayenne UTC offset of -03:00.
- America/Cayman - Timezone for america/cayman UTC offset of -05:00.
- America/Chicago - Timezone for america/chicago UTC offset of -06:00.
- America/Chihuahua - Timezone for america/chihuahua UTC offset of -07:00.
- America/Coral_Harbour - Timezone for america/coral_harbour UTC offset of -05:00.

- America/Cordoba - Timezone for america/cordoba UTC offset of -03:00.
- America/Costa_Rica - Timezone for america/costa_rica UTC offset of -06:00.
- America/Creston - Timezone for america/creston UTC offset of -07:00.
- America/Cuiaba - Timezone for america/cuiaba UTC offset of -03:00.
- America/Curacao - Timezone for america/curacao UTC offset of -04:00.
- America/Danmarkshavn - Timezone for america/danmarkshavn UTC offset of +00:00.
- America/Dawson - Timezone for america/dawson UTC offset of -08:00.
- America/Dawson_Creek - Timezone for america/dawson_creek UTC offset of -07:00.
- America/Denver - Timezone for america/denver UTC offset of -07:00.
- America/Detroit - Timezone for america/detroit UTC offset of -05:00.
- America/Dominica - Timezone for america/dominica UTC offset of -04:00.
- America/Edmonton - Timezone for america/edmonton UTC offset of -07:00.
- America/Eirunepe - Timezone for america/eirunepe UTC offset of -05:00.
- America/El_Salvador - Timezone for america/el_salvador UTC offset of -06:00.
- America/Ensenada - Timezone for america/ensenada UTC offset of -08:00.
- America/Fort_Nelson - Timezone for america/fort_nelson UTC offset of -07:00.
- America/Fort_Wayne - Timezone for america/fort_wayne UTC offset of -05:00.
- America/Fortaleza - Timezone for america/fortaleza UTC offset of -03:00.
- America/Glace_Bay - Timezone for america/glance_bay UTC offset of -04:00.
- America/Godthab - Timezone for america/godthab UTC offset of -03:00.
- America/Goose_Bay - Timezone for america/goose_bay UTC offset of -04:00.
- America/Grand_Turk - Timezone for america/grand_turk UTC offset of -04:00.
- America/Grenada - Timezone for america/grenada UTC offset of -04:00.
- America/Guadeloupe - Timezone for america/guadeloupe UTC offset of -04:00.
- America/Guatemala - Timezone for america/guatemala UTC offset of -06:00.
- America/Guayaquil - Timezone for america/guayaquil UTC offset of -05:00.
- America/Guyana - Timezone for america/guyana UTC offset of -04:00.
- America/Halifax - Timezone for america/halifax UTC offset of -04:00.
- America/Havana - Timezone for america/havana UTC offset of -05:00.
- America/Hermosillo - Timezone for america/hermosillo UTC offset of -07:00.

- America/Indiana/Indianapolis - Timezone for america/indiana/indianapolis UTC offset of -05:00.
- America/Indiana/Knox - Timezone for america/indiana/knox UTC offset of -06:00.
- America/Indiana/Marengo - Timezone for america/indiana/marengo UTC offset of -05:00.
- America/Indiana/Petersburg - Timezone for america/indiana/petersburg UTC offset of -05:00.
- America/Indiana/Tell_City - Timezone for america/indiana/tell_city UTC offset of -06:00.
- America/Indiana/Vevay - Timezone for america/indiana/vevay UTC offset of -05:00.
- America/Indiana/Vincennes - Timezone for america/indiana/vincennes UTC offset of -05:00.
- America/Indiana/Winamac - Timezone for america/indiana/winamac UTC offset of -05:00.
- America/Indianapolis - Timezone for america/indianapolis UTC offset of -05:00.
- America/Inuvik - Timezone for america/inuvik UTC offset of -07:00.
- America/Iqaluit - Timezone for america/iquait UTC offset of -05:00.
- America/Jamaica - Timezone for america/jamaica UTC offset of -05:00.
- America/Jujuy - Timezone for america/jujuy UTC offset of -03:00.
- America/Juneau - Timezone for america/juneau UTC offset of -09:00.
- America/Kentucky/Louisville - Timezone for america/kentucky/louisville UTC offset of -05:00.
- America/Kentucky/Monticello - Timezone for america/kentucky/monticello UTC offset of -05:00.
- America/Knox_IN - Timezone for america/knox_in UTC offset of -06:00.
- America/Kralendijk - Timezone for america/kralendijk UTC offset of -04:00.
- America/La_Paz - Timezone for america/la_paz UTC offset of -04:00.
- America/Lima - Timezone for america/lima UTC offset of -05:00.
- America/Los_Angeles - Timezone for america/los_angeles UTC offset of -08:00.
- America/Louisville - Timezone for america/louisville UTC offset of -05:00.
- America/Lower_Princes - Timezone for america/lower_princes UTC offset of -04:00.
- America/Maceio - Timezone for america/maceio UTC offset of -03:00.
- America/Managua - Timezone for america/managua UTC offset of -06:00.
- America/Manaus - Timezone for america/manaus UTC offset of -04:00.

- America/Marigot - Timezone for america/marigot UTC offset of -04:00.
- America/Martinique - Timezone for america/martinique UTC offset of -04:00.
- America/Matamoros - Timezone for america/matamoros UTC offset of -06:00.
- America/Mazatlan - Timezone for america/mazatlan UTC offset of -07:00.
- America/Mendoza - Timezone for america/mendoza UTC offset of -03:00.
- America/Menominee - Timezone for america/menominee UTC offset of -06:00.
- America/Merida - Timezone for america/merida UTC offset of -06:00.
- America/Metlakatla - Timezone for america/metlakatla UTC offset of -09:00.
- America/Mexico_City - Timezone for america/mexico_city UTC offset of -06:00.
- America/Miquelon - Timezone for america/miquelon UTC offset of -03:00.
- America/Moncton - Timezone for america/moncton UTC offset of -04:00.
- America/Monterrey - Timezone for america/monterrey UTC offset of -06:00.
- America/Montevideo - Timezone for america/montevideo UTC offset of -03:00.
- America/Montreal - Timezone for america/montreal UTC offset of -05:00.
- America/Montserrat - Timezone for america/montserrat UTC offset of -04:00.
- America/Nassau - Timezone for america/nassau UTC offset of -05:00.
- America/New_York - Timezone for america/new_york UTC offset of -05:00.
- America/Nipigon - Timezone for america/nipigon UTC offset of -05:00.
- America/Nome - Timezone for america/nome UTC offset of -09:00.
- America/Noronha - Timezone for america/noronha UTC offset of -02:00.
- America/North_Dakota/Beulah - Timezone for america/north_dakota/beulah UTC offset of -06:00.
- America/North_Dakota/Center - Timezone for america/north_dakota/center UTC offset of -06:00.
- America/North_Dakota/New_Salem - Timezone for america/north_dakota/new_salem UTC offset of -06:00.
- America/Ojinaga - Timezone for america/ojinaga UTC offset of -07:00.
- America/Panama - Timezone for america/panama UTC offset of -05:00.
- America/Pangnirtung - Timezone for america/pangnirtung UTC offset of -05:00.
- America/Paramaribo - Timezone for america/paramaribo UTC offset of -03:00.
- America/Phoenix - Timezone for america/phoenix UTC offset of -07:00.
- America/Port-au-Prince - Timezone for america/port-au-prince UTC offset of -05:00.

- America/Port_of_Spain - Timezone for america/port_of_spain UTC offset of -04:00.
- America/Porto_Acre - Timezone for america/porto_acre UTC offset of -05:00.
- America/Porto_Velho - Timezone for america/porto_velho UTC offset of -04:00.
- America/Puerto_Rico - Timezone for america/puerto_rico UTC offset of -04:00.
- America/Rainy_River - Timezone for america/rainy_river UTC offset of -06:00.
- America/Rankin_Inlet - Timezone for america/rankin_inlet UTC offset of -06:00.
- America/Recife - Timezone for america/recife UTC offset of -03:00.
- America/Regina - Timezone for america/regina UTC offset of -06:00.
- America/Resolute - Timezone for america/resolute UTC offset of -06:00.
- America/Rio_Branco - Timezone for america/rio_branco UTC offset of -05:00.
- America/Rosario - Timezone for america/rosario UTC offset of -03:00.
- America/Santa_Isabel - Timezone for america/santa_isabel UTC offset of -08:00.
- America/Santarem - Timezone for america/santarem UTC offset of -03:00.
- America/Santiago - Timezone for america/santiago UTC offset of -03:00.
- America/Santo_Domingo - Timezone for america/santo_domingo UTC offset of -04:00.
- America/Sao_Paulo - Timezone for america/sao_paulo UTC offset of -02:00.
- America/Scoresbysund - Timezone for america/scoresbysund UTC offset of -01:00.
- America/Shiprock - Timezone for america/shiprock UTC offset of -07:00.
- America/Sitka - Timezone for america/sitka UTC offset of -09:00.
- America/St_Barthelemy - Timezone for america/st_barthelemy UTC offset of -04:00.
- America/St_Johns - Timezone for america/st_johns UTC offset of -03:30.
- America/St_Kitts - Timezone for america/st_kitts UTC offset of -04:00.
- America/St_Lucia - Timezone for america/st_lucia UTC offset of -04:00.
- America/St_Thomas - Timezone for america/st_thomas UTC offset of -04:00.
- America/St_Vincent - Timezone for america/st_vincent UTC offset of -04:00.
- America/Swift_Current - Timezone for america/swift_current UTC offset of -06:00.
- America/Tegucigalpa - Timezone for america/tegucigalpa UTC offset of -06:00.
- America/Thule - Timezone for america/thule UTC offset of -04:00.
- America/Thunder_Bay - Timezone for america/thunder_bay UTC offset of -05:00.
- America/Tijuana - Timezone for america/tijuana UTC offset of -08:00.
- America/Toronto - Timezone for america/toronto UTC offset of -05:00.

- America/Tortola - Timezone for america/tortola UTC offset of -04:00.
- America/Vancouver - Timezone for america/vancouver UTC offset of -08:00.
- America/Virgin - Timezone for america/virgin UTC offset of -04:00.
- America/Whitehorse - Timezone for america/whitehorse UTC offset of -08:00.
- America/Winnipeg - Timezone for america/winnipeg UTC offset of -06:00.
- America/Yakutat - Timezone for america/yakutat UTC offset of -09:00.
- America/Yellowknife - Timezone for america/yellowknife UTC offset of -07:00.
- Antarctica/Casey - Timezone for antarctica/casey UTC offset of +11:00.
- Antarctica/Davis - Timezone for antarctica/davis UTC offset of +07:00.
- Antarctica/DumontDURville - Timezone for antarctica/dumontdurville UTC offset of +10:00.
- Antarctica/Macquarie - Timezone for antarctica/macquarie UTC offset of +11:00.
- Antarctica/Mawson - Timezone for antarctica/mawson UTC offset of +05:00.
- Antarctica/McMurdo - Timezone for antarctica/mcmurdo UTC offset of +13:00.
- Antarctica/Palmer - Timezone for antarctica/palmer UTC offset of -03:00.
- Antarctica/Rothera - Timezone for antarctica/rothera UTC offset of -03:00.
- Antarctica/South_Pole - Timezone for antarctica/south_pole UTC offset of +13:00.
- Antarctica/Syowa - Timezone for antarctica/syowa UTC offset of +03:00.
- Antarctica/Troll - Timezone for antarctica/troll UTC offset of +00:00.
- Antarctica/Vostok - Timezone for antarctica/vostok UTC offset of +06:00.
- Arctic/Longyearbyen - Timezone for arctic/longyearbyen UTC offset of +01:00.
- Asia/Aden - Timezone for asia/aden UTC offset of +03:00.
- Asia/Almaty - Timezone for asia/almaty UTC offset of +06:00.
- Asia/Amman - Timezone for asia/amman UTC offset of +02:00.
- Asia/Anadyr - Timezone for asia/anadyr UTC offset of +12:00.
- Asia/Aqtau - Timezone for asia/aqtau UTC offset of +05:00.
- Asia/Aqtobe - Timezone for asia/aqtobe UTC offset of +05:00.
- Asia/Ashgabat - Timezone for asia/ashgabat UTC offset of +05:00.
- Asia/Ashkhabad - Timezone for asia/ashkhabad UTC offset of +05:00.
- Asia/Baghdad - Timezone for asia/baghdad UTC offset of +03:00.
- Asia/Bahrain - Timezone for asia/bahrain UTC offset of +03:00.

- Asia/Baku - Timezone for asia/baku UTC offset of +04:00.
- Asia/Bangkok - Timezone for asia/bangkok UTC offset of +07:00.
- Asia/Barnaul - Timezone for asia/barnaul UTC offset of +07:00.
- Asia/Beirut - Timezone for asia/beirut UTC offset of +02:00.
- Asia/Bishkek - Timezone for asia/bishkek UTC offset of +06:00.
- Asia/Brunei - Timezone for asia/brunei UTC offset of +08:00.
- Asia/Calcutta - Timezone for asia/calcutta UTC offset of +05:30.
- Asia/Chita - Timezone for asia/chita UTC offset of +09:00.
- Asia/Choibalsan - Timezone for asia/choibalsan UTC offset of +08:00.
- Asia/Chongqing - Timezone for asia/chongqing UTC offset of +08:00.
- Asia/Chungking - Timezone for asia/chungking UTC offset of +08:00.
- Asia/Colombo - Timezone for asia/colombo UTC offset of +05:30.
- Asia/Dacca - Timezone for asia/dacca UTC offset of +06:00.
- Asia/Damascus - Timezone for asia/damascus UTC offset of +02:00.
- Asia/Dhaka - Timezone for asia/dhaka UTC offset of +06:00.
- Asia/Dili - Timezone for asia/dili UTC offset of +09:00.
- Asia/Dubai - Timezone for asia/dubai UTC offset of +04:00.
- Asia/Dushanbe - Timezone for asia/dushanbe UTC offset of +05:00.
- Asia/Gaza - Timezone for asia/gaza UTC offset of +02:00.
- Asia/Harbin - Timezone for asia/harbin UTC offset of +08:00.
- Asia/Hebron - Timezone for asia/hebron UTC offset of +02:00.
- Asia/Ho_Chi_Minh - Timezone for asia/ho_chi_minh UTC offset of +07:00.
- Asia/Hong_Kong - Timezone for asia/hong_kong UTC offset of +08:00.
- Asia/Hovd - Timezone for asia/hovd UTC offset of +07:00.
- Asia/Irkutsk - Timezone for asia/irkutsk UTC offset of +08:00.
- Asia/Istanbul - Timezone for asia/istanbul UTC offset of +03:00.
- Asia/Jakarta - Timezone for asia/jakarta UTC offset of +07:00.
- Asia/Jayapura - Timezone for asia/jayapura UTC offset of +09:00.
- Asia/Jerusalem - Timezone for asia/jerusalem UTC offset of +02:00.
- Asia/Kabul - Timezone for asia/kabul UTC offset of +04:30.
- Asia/Kamchatka - Timezone for asia/kamchatka UTC offset of +12:00.

- Asia/Karachi - Timezone for asia/karachi UTC offset of +05:00.
- Asia/Kashgar - Timezone for asia/kashgar UTC offset of +06:00.
- Asia/Kathmandu - Timezone for asia/kathmandu UTC offset of +05:45.
- Asia/Katmandu - Timezone for asia/katmandu UTC offset of +05:45.
- Asia/Khandyga - Timezone for asia/khandyga UTC offset of +09:00.
- Asia/Kolkata - Timezone for asia/kolkata UTC offset of +05:30.
- Asia/Krasnoyarsk - Timezone for asia/krasnoyarsk UTC offset of +07:00.
- Asia/Kuala_Lumpur - Timezone for asia/kuala_lumpur UTC offset of +08:00.
- Asia/Kuching - Timezone for asia/kuching UTC offset of +08:00.
- Asia/Kuwait - Timezone for asia/kuwait UTC offset of +03:00.
- Asia/Macao - Timezone for asia/macao UTC offset of +08:00.
- Asia/Macau - Timezone for asia/macau UTC offset of +08:00.
- Asia/Magadan - Timezone for asia/magadan UTC offset of +11:00.
- Asia/Makassar - Timezone for asia/makassar UTC offset of +08:00.
- Asia/Manila - Timezone for asia/manila UTC offset of +08:00.
- Asia/Muscat - Timezone for asia/muscat UTC offset of +04:00.
- Asia/Nicosia - Timezone for asia/nicosia UTC offset of +02:00.
- Asia/Novokuznetsk - Timezone for asia/novokuznetsk UTC offset of +07:00.
- Asia/Novosibirsk - Timezone for asia/novosibirsk UTC offset of +07:00.
- Asia/Omsk - Timezone for asia/omsk UTC offset of +06:00.
- Asia/Oral - Timezone for asia/oral UTC offset of +05:00.
- Asia/Phnom_Penh - Timezone for asia/phnom_penh UTC offset of +07:00.
- Asia/Pontianak - Timezone for asia/pontianak UTC offset of +07:00.
- Asia/Pyongyang - Timezone for asia/pyongyang UTC offset of +08:30.
- Asia/Qatar - Timezone for asia/qatar UTC offset of +03:00.
- Asia/Qyzylorda - Timezone for asia/qyzylorda UTC offset of +06:00.
- Asia/Rangoon - Timezone for asia/rangoon UTC offset of +06:30.
- Asia/Riyadh - Timezone for asia/riyadh UTC offset of +03:00.
- Asia/Saigon - Timezone for asia/saigon UTC offset of +07:00.
- Asia/Sakhalin - Timezone for asia/sakhalin UTC offset of +11:00.
- Asia/Samarkand - Timezone for asia/samarkand UTC offset of +05:00.

- Asia/Seoul - Timezone for asia/seoul UTC offset of +09:00.
- Asia/Shanghai - Timezone for asia/shanghai UTC offset of +08:00.
- Asia/Singapore - Timezone for asia/singapore UTC offset of +08:00.
- Asia/Srednekolymsk - Timezone for asia/srednekolymsk UTC offset of +11:00.
- Asia/Taipei - Timezone for asia/taipei UTC offset of +08:00.
- Asia/Tashkent - Timezone for asia/tashkent UTC offset of +05:00.
- Asia/Tbilisi - Timezone for asia/tbilisi UTC offset of +04:00.
- Asia/Tehran - Timezone for asia/tehran UTC offset of +03:30.
- Asia/Tel_Aviv - Timezone for asia/tel_aviv UTC offset of +02:00.
- Asia/Thimbu - Timezone for asia/thimbu UTC offset of +06:00.
- Asia/Thimphu - Timezone for asia/thimphu UTC offset of +06:00.
- Asia/Tokyo - Timezone for asia/tokyo UTC offset of +09:00.
- Asia/Tomsk - Timezone for asia/tomsk UTC offset of +07:00.
- Asia/Ujung_Pandang - Timezone for asia/ujung_pandang UTC offset of +08:00.
- Asia/Ulaanbaatar - Timezone for asia/ulaanbaatar UTC offset of +08:00.
- Asia/Ulan_Bator - Timezone for asia/ulan_bator UTC offset of +08:00.
- Asia/Urumqi - Timezone for asia/urumqi UTC offset of +06:00.
- Asia/Ust-Nera - Timezone for asia/ust-nera UTC offset of +10:00.
- Asia/Vientiane - Timezone for asia/vientiane UTC offset of +07:00.
- Asia/Vladivostok - Timezone for asia/vladivostok UTC offset of +10:00.
- Asia/Yakutsk - Timezone for asia/yakutsk UTC offset of +09:00.
- Asia/Yekaterinburg - Timezone for asia/yekaterinburg UTC offset of +05:00.
- Asia/Yerevan - Timezone for asia/yerevan UTC offset of +04:00.
- Atlantic/Azores - Timezone for atlantic/azores UTC offset of -01:00.
- Atlantic/Bermuda - Timezone for atlantic/bermuda UTC offset of -04:00.
- Atlantic/Canary - Timezone for atlantic/canary UTC offset of +00:00.
- Atlantic/Cape_Verde - Timezone for atlantic/cape_verde UTC offset of -01:00.
- Atlantic/Faeroe - Timezone for atlantic/faeroe UTC offset of +00:00.
- Atlantic/Faroe - Timezone for atlantic/faroe UTC offset of +00:00.
- Atlantic/Jan_Mayen - Timezone for atlantic/jan_mayen UTC offset of +01:00.
- Atlantic/Madeira - Timezone for atlantic/madeira UTC offset of +00:00.

- Atlantic/Reykjavik - Timezone for atlantic/reykjavik UTC offset of +00:00.
- Atlantic/South_Georgia - Timezone for atlantic/south_georgia UTC offset of -02:00.
- Atlantic/St_Helena - Timezone for atlantic/st_helena UTC offset of +00:00.
- Atlantic/Stanley - Timezone for atlantic/stanley UTC offset of -03:00.
- Australia/ACT - Timezone for australia/act UTC offset of +11:00.
- Australia/Adelaide - Timezone for australia/adelaide UTC offset of +10:30.
- Australia/Brisbane - Timezone for australia/brisbane UTC offset of +10:00.
- Australia/Broken_Hill - Timezone for australia/broken_hill UTC offset of +10:30.
- Australia/Canberra - Timezone for australia/canberra UTC offset of +11:00.
- Australia/Currie - Timezone for australia/currie UTC offset of +11:00.
- Australia/Darwin - Timezone for australia/darwin UTC offset of +09:30.
- Australia/Eucla - Timezone for australia/eucla UTC offset of +08:45.
- Australia/Hobart - Timezone for australia/hobart UTC offset of +11:00.
- Australia/LHI - Timezone for australia/lhi UTC offset of +11:00.
- Australia/Lindeman - Timezone for australia/lindeman UTC offset of +10:00.
- Australia/Lord_Howe - Timezone for australia/lord_howe UTC offset of +11:00.
- Australia/Melbourne - Timezone for australia/melbourne UTC offset of +11:00.
- Australia/NSW - Timezone for australia/nsw UTC offset of +11:00.
- Australia/North - Timezone for australia/north UTC offset of +09:30.
- Australia/Perth - Timezone for australia/perth UTC offset of +08:00.
- Australia/Queensland - Timezone for australia/queensland UTC offset of +10:00.
- Australia/South - Timezone for australia/south UTC offset of +10:30.
- Australia/Sydney - Timezone for australia/sydney UTC offset of +11:00.
- Australia/Tasmania - Timezone for australia/tasmania UTC offset of +11:00.
- Australia/Victoria - Timezone for australia/victoria UTC offset of +11:00.
- Australia/West - Timezone for australia/west UTC offset of +08:00.
- Australia/Yancowinna - Timezone for australia/yancowinna UTC offset of +10:30.
- CET - Timezone for CET UTC offset of +01:00.
- CST6CDT - Timezone for CST6CDT UTC offset of -06:00.
- Canada/Atlantic - Timezone for canada/atlantic UTC offset of -04:00.
- Canada/Central - Timezone for canada/central UTC offset of -06:00.

- Canada/East-Saskatchewan - Timezone for canada/east-saskatchewan UTC offset of -06:00.
- Canada/Eastern - Timezone for canada/eastern UTC offset of -05:00.
- Canada/Mountain - Timezone for canada/mountain UTC offset of -07:00.
- Canada/Newfoundland - Timezone for canada/newfoundland UTC offset of -03:30.
- Canada/Pacific - Timezone for canada/pacific UTC offset of -08:00.
- Canada/Saskatchewan - Timezone for canada/saskatchewan UTC offset of -06:00.
- Canada/Yukon - Timezone for canada/yukon UTC offset of -08:00.
- Chile/Continental - Timezone for chile/continental UTC offset of -03:00.
- Chile/EasterIsland - Timezone for chile/easterisland UTC offset of -05:00.
- Cuba - Timezone for cuba UTC offset of -05:00.
- Default - Timezone for system UTC (OS time zone)
- EET - Timezone for EET UTC offset of +02:00.
- EST - Timezone for EST UTC offset of -05:00.
- EST5EDT - Timezone for EST5EDT UTC offset of -05:00.
- Egypt - Timezone for egypt UTC offset of +02:00.
- Eire - Timezone for eire UTC offset of +00:00.
- Etc/Greenwich - Timezone for etc/greenwich UTC offset of +00:00.
- Etc/UCT - Timezone for etc/uct UTC offset of +00:00.
- Etc/UTC - Timezone for etc/utc UTC offset of +00:00.
- Etc/Universal - Timezone for etc/universal UTC offset of +00:00.
- Etc/Zulu - Timezone for etc/zulu UTC offset of +00:00.
- Europe/Amsterdam - Timezone for europe/amsterdam UTC offset of +01:00.
- Europe/Andorra - Timezone for europe/andorra UTC offset of +01:00.
- Europe/Astrakhan - Timezone for europe/astrakhan UTC offset of +04:00.
- Europe/Athens - Timezone for europe/athens UTC offset of +02:00.
- Europe/Belfast - Timezone for europe/belfast UTC offset of +00:00.
- Europe/Belgrade - Timezone for europe/belgrade UTC offset of +01:00.
- Europe/Berlin - Timezone for europe/berlin UTC offset of +01:00.
- Europe/Bratislava - Timezone for europe/bratislava UTC offset of +01:00.
- Europe/Brussels - Timezone for europe/brussels UTC offset of +01:00.

- Europe/Bucharest - Timezone for europe/bucharest UTC offset of +02:00.
- Europe/Budapest - Timezone for europe/budapest UTC offset of +01:00.
- Europe/Busingen - Timezone for europe/busingen UTC offset of +01:00.
- Europe/Chisinau - Timezone for europe/chisinau UTC offset of +02:00.
- Europe/Copenhagen - Timezone for europe/copenhagen UTC offset of +01:00.
- Europe/Dublin - Timezone for europe/dublin UTC offset of +00:00.
- Europe/Gibraltar - Timezone for europe/gibraltar UTC offset of +01:00.
- Europe/Guernsey - Timezone for europe/guernsey UTC offset of +00:00.
- Europe/Helsinki - Timezone for europe/helsinki UTC offset of +02:00.
- Europe/Isle_of_Man - Timezone for europe/isle_of_man UTC offset of +00:00.
- Europe/Istanbul - Timezone for europe/istanbul UTC offset of +03:00.
- Europe/Jersey - Timezone for europe/jersey UTC offset of +00:00.
- Europe/Kaliningrad - Timezone for europe/kaliningrad UTC offset of +02:00.
- Europe/Kiev - Timezone for europe/kyiv UTC offset of +02:00.
- Europe/Kirov - Timezone for europe/kirov UTC offset of +03:00.
- Europe/Lisbon - Timezone for europe/lisbon UTC offset of +00:00.
- Europe/Ljubljana - Timezone for europe/ljubljana UTC offset of +01:00.
- Europe/London - Timezone for europe/london UTC offset of +00:00.
- Europe/Luxembourg - Timezone for europe/luxembourg UTC offset of +01:00.
- Europe/Madrid - Timezone for europe/madrid UTC offset of +01:00.
- Europe/Malta - Timezone for europe/malta UTC offset of +01:00.
- Europe/Mariehamn - Timezone for europe/mariehamn UTC offset of +02:00.
- Europe/Minsk - Timezone for europe/minsk UTC offset of +03:00.
- Europe/Monaco - Timezone for europe/monaco UTC offset of +01:00.
- Europe/Moscow - Timezone for europe/moscow UTC offset of +03:00.
- Europe/Nicosia - Timezone for europe/nicosia UTC offset of +02:00.
- Europe/Oslo - Timezone for europe/oslo UTC offset of +01:00.
- Europe/Paris - Timezone for europe/paris UTC offset of +01:00.
- Europe/Podgorica - Timezone for europe/podgorica UTC offset of +01:00.
- Europe/Prague - Timezone for europe/prague UTC offset of +01:00.
- Europe/Riga - Timezone for europe/riga UTC offset of +02:00.

- Europe/Rome - Timezone for europe/rome UTC offset of +01:00.
- Europe/Samara - Timezone for europe/samara UTC offset of +04:00.
- Europe/San_Marino - Timezone for europe/san_marino UTC offset of +01:00.
- Europe/Sarajevo - Timezone for europe/sarajevo UTC offset of +01:00.
- Europe/Simferopol - Timezone for europe/simferopol UTC offset of +03:00.
- Europe/Skopje - Timezone for europe/skopje UTC offset of +01:00.
- Europe/Sofia - Timezone for europe/sofia UTC offset of +02:00.
- Europe/Stockholm - Timezone for europe/stockholm UTC offset of +01:00.
- Europe/Tallinn - Timezone for europe/tallinn UTC offset of +02:00.
- Europe/Tirane - Timezone for europe/tirane UTC offset of +01:00.
- Europe/Tiraspol - Timezone for europe/tiraspol UTC offset of +02:00.
- Europe/Ulyanovsk - Timezone for europe/ulyanovsk UTC offset of +04:00.
- Europe/Uzhgorod - Timezone for europe/uzhgorod UTC offset of +02:00.
- Europe/Vaduz - Timezone for europe/vaduz UTC offset of +01:00.
- Europe/Vatican - Timezone for europe/vatican UTC offset of +01:00.
- Europe/Vienna - Timezone for europe/vienna UTC offset of +01:00.
- Europe/Vilnius - Timezone for europe/vilnius UTC offset of +02:00.
- Europe/Volgograd - Timezone for europe/volgograd UTC offset of +03:00.
- Europe/Warsaw - Timezone for europe/warsaw UTC offset of +01:00.
- Europe/Zagreb - Timezone for europe/zagreb UTC offset of +01:00.
- Europe/Zaporozhye - Timezone for europe/zaporozhye UTC offset of +02:00.
- Europe/Zurich - Timezone for europe/zurich UTC offset of +01:00.
- GB - Timezone for GB UTC offset of +00:00.
- GB-Eire - Timezone for GB-Eire UTC offset of +00:00.
- GMT - Timezone for GMT UTC offset of +00:00.
- GMT0 - Timezone for GMT0 UTC offset of +00:00.
- Greenwich - Timezone for greenwich UTC offset of +00:00.
- HST - Timezone for HST UTC offset of -10:00.
- Hongkong - Timezone for hongkong UTC offset of +08:00.
- Iceland - Timezone for iceland UTC offset of +00:00.
- Indian/Antananarivo - Timezone for indian/antananarivo UTC offset of +03:00.

- Indian/Chagos - Timezone for indian/chagos UTC offset of +06:00.
- Indian/Christmas - Timezone for indian/christmas UTC offset of +07:00.
- Indian/Cocos - Timezone for indian/cocos UTC offset of +06:30.
- Indian/Comoro - Timezone for indian/comoro UTC offset of +03:00.
- Indian/Kerguelen - Timezone for indian/kerguelen UTC offset of +05:00.
- Indian/Mahe - Timezone for indian/mahe UTC offset of +04:00.
- Indian/Maldives - Timezone for indian/maldives UTC offset of +05:00.
- Indian/Mauritius - Timezone for indian/mauritius UTC offset of +04:00.
- Indian/Mayotte - Timezone for indian/mayotte UTC offset of +03:00.
- Indian/Reunion - Timezone for indian/reunion UTC offset of +04:00.
- Iran - Timezone for iran UTC offset of +03:30.
- Israel - Timezone for israel UTC offset of +02:00.
- Jamaica - Timezone for jamaica UTC offset of -05:00.
- Japan - Timezone for japan UTC offset of +09:00.
- Kwajalein - Timezone for kwajalein UTC offset of +12:00.
- Libya - Timezone for libya UTC offset of +02:00.
- MET - Timezone for MET UTC offset of +01:00.
- MST - Timezone for MST UTC offset of -07:00.
- MST7MDT - Timezone for MST7MDT UTC offset of -07:00.
- Mexico/BajaNorte - Timezone for mexico/bajanorte UTC offset of -08:00.
- Mexico/BajaSur - Timezone for mexico/bajasur UTC offset of -07:00.
- Mexico/General - Timezone for mexico/general UTC offset of -06:00.
- NZ - Timezone for NZ UTC offset of +13:00.
- NZ-CHAT - Timezone for NZ-CHAT UTC offset of +13:45.
- Navajo - Timezone for navajo UTC offset of -07:00.
- PRC - Timezone for PRC UTC offset of +08:00.
- PST8PDT - Timezone for PST8PDT UTC offset of -08:00.
- Pacific/Apia - Timezone for pacific/apia UTC offset of +14:00.
- Pacific/Auckland - Timezone for pacific/auckland UTC offset of +13:00.
- Pacific/Bougainville - Timezone for pacific/bougainville UTC offset of +11:00.
- Pacific/Chatham - Timezone for pacific/chatham UTC offset of +13:45.

- Pacific/Chuuk - Timezone for pacific/chuuk UTC offset of +10:00.
- Pacific/Easter - Timezone for pacific/easter UTC offset of -05:00.
- Pacific/Efate - Timezone for pacific/efate UTC offset of +11:00.
- Pacific/Enderbury - Timezone for pacific/enderbury UTC offset of +13:00.
- Pacific/Fakaofo - Timezone for pacific/fakaofo UTC offset of +13:00.
- Pacific/Fiji - Timezone for pacific/fiji UTC offset of +13:00.
- Pacific/Funafuti - Timezone for pacific/funafuti UTC offset of +12:00.
- Pacific/Galapagos - Timezone for pacific/galapagos UTC offset of -06:00.
- Pacific/Gambier - Timezone for pacific/gambier UTC offset of -09:00.
- Pacific/Guadalcanal - Timezone for pacific/guadalcanal UTC offset of +11:00.
- Pacific/Guam - Timezone for pacific/guam UTC offset of +10:00.
- Pacific/Honolulu - Timezone for pacific/honolulu UTC offset of -10:00.
- Pacific/Johnston - Timezone for pacific/johnston UTC offset of -10:00.
- Pacific/Kiritimati - Timezone for pacific/kiritimati UTC offset of +14:00.
- Pacific/Kosrae - Timezone for pacific/kosrae UTC offset of +11:00.
- Pacific/Kwajalein - Timezone for pacific/kwajalein UTC offset of +12:00.
- Pacific/Majuro - Timezone for pacific/majuro UTC offset of +12:00.
- Pacific/Marquesas - Timezone for pacific/marquesas UTC offset of -09:30.
- Pacific/Midway - Timezone for pacific/midway UTC offset of -11:00.
- Pacific/Nauru - Timezone for pacific/nauru UTC offset of +12:00.
- Pacific/Niue - Timezone for pacific/niue UTC offset of -11:00.
- Pacific/Norfolk - Timezone for pacific/norfolk UTC offset of +11:00.
- Pacific/Noumea - Timezone for pacific/noumea UTC offset of +11:00.
- Pacific/Pago_Pago - Timezone for pacific/pago_pago UTC offset of -11:00.
- Pacific/Palau - Timezone for pacific/palau UTC offset of +09:00.
- Pacific/Pitcairn - Timezone for pacific/pitcairn UTC offset of -08:00.
- Pacific/Pohnpei - Timezone for pacific/pohnpei UTC offset of +11:00.
- Pacific/Ponape - Timezone for pacific/ponape UTC offset of +11:00.
- Pacific/Port_Moresby - Timezone for pacific/port_moresby UTC offset of +10:00.
- Pacific/Rarotonga - Timezone for pacific/rarotonga UTC offset of -10:00.
- Pacific/Saipan - Timezone for pacific/saipan UTC offset of +10:00.

- Pacific/Samoa - Timezone for pacific/samoa UTC offset of -11:00.
- Pacific/Tahiti - Timezone for pacific/tahiti UTC offset of -10:00.
- Pacific/Tarawa - Timezone for pacific/tarawa UTC offset of +12:00.
- Pacific/Tongatapu - Timezone for pacific/tongatapu UTC offset of +14:00.
- Pacific/Truk - Timezone for pacific/truk UTC offset of +10:00.
- Pacific/Wake - Timezone for pacific/wake UTC offset of +12:00.
- Pacific/Wallis - Timezone for pacific/wallis UTC offset of +12:00.
- Pacific/Yap - Timezone for pacific/yap UTC offset of +10:00.
- Poland - Timezone for poland UTC offset of +01:00.
- Portugal - Timezone for portugal UTC offset of +00:00.
- ROC - Timezone for ROC UTC offset of +08:00.
- ROK - Timezone for ROK UTC offset of +09:00.
- Singapore - Timezone for singapore UTC offset of +08:00.
- Turkey - Timezone for turkey UTC offset of +03:00.
- UCT - Timezone for UCT UTC offset of +00:00.
- US/Pacific-New - Timezone for US/Pacific-New UTC offset of -08:00.
- Universal - Timezone for universal UTC offset of +00:00.
- W-SU - Timezone for W-SU UTC offset of +03:00.
- WET - Timezone for WET UTC offset of +00:00.
- Zulu - Timezone for zulu UTC offset of +00:00.
- jdtz - Timezone for jdtz UTC offset of +00:00.

Default

Default

Roles

Create

sysadmin, manage

Modify

sysadmin, manage

3.83 cmm amfN26LocalEndPointConfig

N26 AMF Local End Point Configuration. Syntax, description, parameters, and roles for cmm amfN26LocalEndPointConfig command.

Syntax

Description

This command is used to create, modify and delete AMF local endpoint configuration for the N26 interface.

Parameters

Name

endPointName - End Point Interface Name

Syntax

--endPointName <string>

Description

End Point Interface Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

amfServiceIpName - Service IP Name

Syntax

--amfServiceIpName <string>

Description

The parameter points to a set of parameters related to IP address configuration created using the 'amfServiceIp' command

Class

Required on create.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

enable - Enable

Syntax

--enable <list-item>

Description

The parameter is used to enable use of the provisioned instance of the N26 interface. If the parameter is set to 'false', the N26 instance will not be used. Once an instance is enabled, the instance can be disabled only under certain TBD conditions.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

protocolProfileName - Interface profile name

Syntax

--protocolProfileName <string>

Description

Interface profile name

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

default

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.84 cmm amfNasMsgAttempts

AMF NAS Message Attempts. Syntax, description, parameters, and roles for cmm amfNasMsgAttempts command.

Syntax

Description

This command assigns number of re-transmission attempts for NAS procedure messages

Parameters

Name

msgName - 5G NAS message attempts

Syntax

--msgName <list-item>

Description

This command is used to assign the number of retransmission attempts used for NAS procedure messages

Class

Required key.

List

- authenticationRequest - Authentication Request - Number of AUTHENTICATION REQUEST messages without receiving a corresponding AUTHENTICATION RESPONSE
- identityRequest - Identity Request - Number of IDENTITY REQUEST messages without receiving a corresponding IDENTITY RESPONSE
- netInitDeregistration - Network Initiated Deregistration - Number of network initiated deregistration requests transmitted without receiving a corresponding deregistration accept response
- networkSliceSpecificAuthCommand - Network Slice Specific Authentication Command - Number of NETWORK SLICE-SPECIFIC AUTHENTICATION COMMAND without receiving a corresponding NETWORK SLICE-SPECIFIC AUTHENTICATION COMPLETE
- notification - Number of Notification messages without receiving a corresponding Notification Response, Service Request or Registration Request.
- registrationAccept - Registration Accept - Number of "initial registration" accept messages with assigned GUTI transmitted without receiving a corresponding registration complete response.
- registrationUpdateAccept - Registration Update Accept - Number of registration accept messages with assigned GUTI transmitted, when registration type is mobility or periodic updating, without receiving a corresponding registration complete response.
- securityModeCommand - Security Mode Command - Number of security mode command messages without receiving a corresponding SECURITY MODE COMPLETE

- ueConfigurationUpdate - UE COntfiguration Update - Number of UE configuration update command messages without receiving a corresponding UE configuration update complete response.

Name

msgAttempts - Message retransmission attempts

Syntax

--msgAttempts <integer>

Description

specify number of NAS message retransmission attempts

Class

Optional.

Type / Range

Integer in the range of 0 to 5

Default

3

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

3.85 cmm amfNasTimers

AMF NAS Timers. Syntax, description, parameters, and roles for cmm amfNasTimers command.

Syntax**Description**

This command is to add AMF NAS Timers

Parameters**Name**

amfTimerName - Timer name

Syntax

--amfTimerName <list-item>

Description

Name of timer.

Class

Required key.

List

- dlNasTransportBackoffTimerMaximum - DL NAS Transport Backoff Timer Maximum - Specifies the maximum value to consider when assigning this timer value to be used for a Downlink Nas Transport procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).
- dlNasTransportBackoffTimerMinimum - DL NAS Transport Backoff Timer Minimum - Specifies the minimum value to consider when assigning this timer value to be used for a Downlink Nas Transport procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).
- micoT3324 - MICO T3324 - Value of the T3324 Active Timer to be assigned to a MICO UE. Default is 5 minutes.
- micoT3512 - MICO T3512 - Value of the T3512 Timer to be assigned to a MICO UE. Default is 30 minutes. Maximum supported unit is 'Hours'. Maximum supported timer is 31 Hours.
- non3gppAccessDeregistrationTimer - Non-3GPP Access Deregistration Timer - AMF provides this timer to the UE on Registration Accept when the UE registers over non-3GPP access. Specifies the time after which the UE deregisters after moving to CM-IDLE for non-3GPP access. Maximum value for the timer is 186 Minutes. Default is 9 DeciHour (54 Minutes).
- t3346registerRejectMaximum - T3346 register reject maximum - Specifies the maximum value to consider when randomly assigning the T3346 timer value to be used for an register procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).
- t3346registerRejectMinimum - T3346 register reject minimum - Specifies the minimum value to consider when randomly assigning the T3346 timer value to be used for an register procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).
- t3346srRejectMaximum - T3346 SR reject maximum - Specifies the maximum value to consider when randomly assigning the T3346 timer value to be used for an SR procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).
- t3346srRejectMinimum - T3346 SR reject minimum - Specifies the minimum value to consider when randomly assigning the T3346 timer value to be used for an SR

procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).

- **t3502** - Used in UE Registration Accept and Registration Reject messages. Used by UE to initiate the next registration (initial registration, mobility registration update or periodic registration in case of a registration failure when the registration attempt counter is reached. Default is 12 minutes. Maximum is 186 minutes. (31*DeciHour)
- **t3512** - Supervise the UE registration status when UE is in CM-IDLE and has no other mobility procedure to use. At Expiry of timer T3512, UE sends Mobility Registration Request with 'Periodic Updating' to notify the network it is still registered. If no request is received, AMF starts mobile reachable timer. Default is 12 Minutes. Maximum value is 310 Hours.
- **t3522** - Supervise the re-transmission of the De-Registration Request. At Expiry of timer T3522, AMF retries the associated message and repeats 'n' configured times. On the nth expiry of the timer, AMF considers the UE as 'non-De-Registered'.
- **t3550** - Timer supervising Registration Complete - Supervise the re-transmission of the Registration Accept when the Registration Complete is expected. At Expiry of timer T3550, AMF retries the associated command and repeats 'n' configured times. On the nth expiry of the timer, AMF aborts the associated procedure.
- **t3555** - Supervise the re-transmission of the UE Configuration Update Request when Configuration Update Complete is expected. At Expiry of timer T3555, AMF retries the associated message and repeats 'n' configured times. On the nth expiry of the timer, AMF aborts the procedure.
- **t3560** - Supervise the re-transmission of the Authentication Request/Security mode command. At Expiry of timer T3560, AMF retries the associated command and repeats 'n' configured times. On the nth expiry of the timer, AMF rejects the associated procedure.
- **t3565** - Supervise the re-transmission of the Notification when either Notification Response, Service Request or Registration Request is expected. At expiry of timer T3565, AMF retries the associated message and repeats 'n' configured times. On the nth expiry of the timer, AMF aborts the procedure.
- **t3570** - Supervise the re-transmission of the Identity Request. At Expiry of timer t3570, AMF retries the associated command and repeats 'n' configured times. On the nth expiry of the timer, AMF either continues with the procedure if requesting IMEISV or aborts the associated procedure.
- **t3575** - Supervise transmission of NETWORK SLICE-SPECIFIC AUTHENTICATION COMMAND is expected. At Expiry of timer T3575, AMF retries the associated command and repeats 'n' configured times. On the nth expiry of the timer, AMF aborts the associated procedure.

Name

amfTimerUnit - Timer unit

Syntax

```
--amfTimerUnit <list-item>
```

Description

Unit of measurement. Timer unit is modifiable. Timer unit allowed values are per timer.

Class

Optional.

List

- Millisecond - Specify timer unit of 1 millisecond
- Second - Specify timer unit of 1 second
- 2Second - 2 Second - Specify timer unit of 2 seconds
- 30Second - 30 Second - Specify timer unit of 30 seconds.
- Minute - Specify timer unit of 1 minute
- DeciHour - Specify timer unit of 1 decihour (6 minutes).
- 10Minute - 10 Minute - Specify timer unit of 10 minutes
- Hour - Specify timer unit of 1 hour
- 10Hour - 10 Hour - Specify timer unit of 10 hours.

Name

amfTimerValue - Timer value

Syntax

```
--amfTimerValue <integer>
```

Description

Numeric value of timer.

Class

Optional.

Type / Range

Integer in the range of 0 to 65,535

The amfTimerValue string is based on the amfTimerName parameter:

Name

dlNasTransportBackoffTimerMaximum - DL NAS Transport Backoff Timer Maximum

Syntax

```
--amfTimerName dlNasTransportBackoffTimerMaximum --amfTimerValue  
<string> [ --amfTimerUnit Second ]
```

Description

Specifies the maximum value to consider when assigning this timer value to be used for a Downlink Nas Transport procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).

Type / Range

Integer in the range of 0 to 600 in units of Second

Default

60

Name

dlNasTransportBackoffTimerMinimum - DL NAS Transport Backoff Timer Minimum

Syntax

```
--amfTimerName dlNasTransportBackoffTimerMinimum --amfTimerValue  
<string> [ --amfTimerUnit Second ]
```

Description

Specifies the minimum value to consider when assigning this timer value to be used for a Downlink Nas Transport procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).

Type / Range

Integer in the range of 0 to 600 in units of Second

Default

15

Name

micoT3324 - MICO T3324

Syntax

```
--amfTimerName micoT3324 --amfTimerValue <string> [ --amfTimerUnit  
{10Minute,2Second,30Second,Hour,Minute} ]
```

Description

Value of the T3324 Active Timer to be assigned to a MICO UE. Default is 5 minutes.

Type / Range

Integer in the range of 1 to 31 in units of 10Minute,2Second,30Second,Hour,Minute

Default

5

Name

micoT3512 - MICO T3512

Syntax

```
--amfTimerName micoT3512 --amfTimerValue <string> [ --amfTimerUnit  
{10Minute,2Second,30Second,Hour,Minute} ]
```

Description

Value of the T3512 Timer to be assigned to a MICO UE. Default is 30 minutes. Maximum supported unit is 'Hours'. Maximum supported timer is 31 Hours.

Type / Range

Integer in the range of 1 to 31 in units of 10Minute,2Second,30Second,Hour,Minute

Default

30

Name

non3gppAccessDeregistrationTimer - Non-3GPP Access Deregistration Timer

Syntax

```
--amfTimerName non3gppAccessDeregistrationTimer --amfTimerValue  
<string> [ --amfTimerUnit {2Second,DeciHour,Minute} ]
```

Description

AMF provides this timer to the UE on Registration Accept when the UE registers over non-3GPP access. Specifies the time after which the UE deregisters after moving to CM-IDLE for non-3GPP access. Maximum value for the timer is 186 Minutes. Default is 9 DeciHour (54 Minutes).

Type / Range

Integer in the range of 0 to 31 in units of 2Second,DeciHour,Minute

Default

9

Name

t3346registerRejectMaximum - T3346 register reject maximum

Syntax

```
--amfTimerName t3346registerRejectMaximum --amfTimerValue <string>  
[ --amfTimerUnit Second ]
```

Description

Specifies the maximum value to consider when randomly assigning the T3346 timer value to be used for an register procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).

Type / Range

Integer in the range of 0 to 1800 in units of Second

Default

60

Name

t3346registerRejectMinimum - T3346 register reject minimum

Syntax

```
--amfTimerName t3346registerRejectMinimum --amfTimerValue <string>  
[ --amfTimerUnit Second ]
```

Description

Specifies the minimum value to consider when randomly assigning the T3346 timer value to be used for an register procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).

Type / Range

Integer in the range of 0 to 1800 in units of Second

Default

15

Name

t3346srRejectMaximum - T3346 SR reject maximum

Syntax

```
--amfTimerName t3346srRejectMaximum --amfTimerValue <string> [ --  
amfTimerUnit Second ]
```

Description

Specifies the maximum value to consider when randomly assigning the T3346 timer value to be used for an SR procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).

Type / Range

Integer in the range of 0 to 1800 in units of Second

Default

60

Name

t3346srRejectMinimum - T3346 SR reject minimum

Syntax

```
--amfTimerName t3346srRejectMinimum --amfTimerValue <string> [ --  
amfTimerUnit Second ]
```

Description

Specifies the minimum value to consider when randomly assigning the T3346 timer value to be used for an SR procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).

Type / Range

Integer in the range of 1 to 1800 in units of Second

Default

15

Name

t3502 - t3502

Syntax

```
--amfTimerName t3502 --amfTimerValue <string> [ --amfTimerUnit  
{2Second,DeciHour,Minute} ]
```

Description

Used in UE Registration Accept and Registration Reject messages. Used by UE to initiate the next registration (initial registration, mobility registration update or periodic registration in case of a registration failure when the registration attempt counter is reached. Default is 12 minutes. Maximum is 186 minutes. (31*DeciHour)

Type / Range

Integer in the range of 1 to 31 in units of 2Second,DeciHour,Minute

Default

12

Name

t3512 - t3512

Syntax

```
--amfTimerName t3512 --amfTimerValue <string> [ --amfTimerUnit  
{10Hour,10Minute,2Second,30Second,Hour,Minute} ]
```

Description

Supervise the UE registration status when UE is in CM-IDLE and has no other mobility procedure to use. At Expiry of timer T3512, UE sends Mobility Registration Request with 'Periodic Updating' to notify the network it is still registered. If no request is received, AMF starts mobile reachable timer. Default is 12 Minutes. Maximum value is 310 Hours.

Type / Range

Integer in the range of 0 to 31 in units of 10Hour,10Minute,2Second,30Second,Hour,Minute

Default

12

Name

t3522 - t3522

Syntax

```
--amfTimerName t3522 --amfTimerValue <string> [ --amfTimerUnit  
Second ]
```

Description

Supervise the re-transmission of the De-Registration Request. At Expiry of timer T3522, AMF retries the associated message and repeats 'n' configured times. On the nth expiry of the timer, AMF considers the UE as 'non-De-Registered'.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

6

Name

t3550 - Timer supervising Registration Complete

Syntax

```
--amfTimerName t3550 --amfTimerValue <string> [ --amfTimerUnit  
Second ]
```

Description

Suprvice the re-transmission of the Registration Accept when the Registration Complete is expected. At Expiry of timer T3550, AMF retries the associated command and repeats 'n' configured times. On the nth expiry of the timer, AMF aborts the associated procedure.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

6

Name

t3555 - t3555

Syntax

```
--amfTimerName t3555 --amfTimerValue <string> [ --amfTimerUnit  
Second ]
```

Description

Supervise the re-transmission of the UE Configuration Update Request when Configuration Update Complete is expected. At Expiry of timer T3555, AMF retries the associated message and repeats 'n' configured times. On the nth expiry of the timer, AMF aborts the procedure.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

6

Name

t3560 - t3560

Syntax

```
--amfTimerName t3560 --amfTimerValue <string> [ --amfTimerUnit  
Second ]
```

Description

Supervise the re-transmission of the Authentication Request/Security mode command.
At Expiry of timer T3560, AMF retries the associated command and repeats 'n' configured times. On the nth expiry of the timer, AMF rejects the associated procedure.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

6

Name

t3565 - t3565

Syntax

```
--amfTimerName t3565 --amfTimerValue <string> [ --amfTimerUnit  
Second ]
```

Description

Supervise the re-transmission of the Notification when either Notification Response, Service Request or Registration Request is expected. At expiry of timer T3565, AMF retries the associated message and repeats 'n' configured times. On the nth expiry of the timer, AMF aborts the procedure.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

6

Name

t3570 - t3570

Syntax

```
--amfTimerName t3570 --amfTimerValue <string> [ --amfTimerUnit  
Second ]
```

Description

Supervise the re-transmission of the Identity Request. At Expiry of timer t3570, AMF retries the associated command and repeats 'n' configured times. On the nth expiry of the timer, AMF either continues with the procedure if requesting IMEISV or aborts the associated procedure.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

6

Name

t3575 - t3575

Syntax

```
--amfTimerName t3575 --amfTimerValue <string> [ --amfTimerUnit  
Second ]
```

Description

Supervise transmission of NETWORK SLICE-SPECIFIC AUTHENTICATION COMMAND is expected. At Expiry of timer T3575, AMF retries the associated command and repeats 'n' configured times. On the nth expiry of the timer, AMF aborts the associated procedure.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

15

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

3.86 cmm amfNgRanLocalEndPointConfig

NG Ran Local End Point Configuration. Syntax, description, parameters, and roles for cmm amfNgRanLocalEndPointConfig command.

Syntax**Description**

This command is used to create, modify and delete AMF local endpoint configuration for the NG-RAN N2 interface. The NG-RAN could be gNB, ng-eNB or N3IWF.

Parameters**Name**

endPointName - End Point Interface Name

Syntax

```
--endPointName <string>
```

Description

End Point Interface Name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

amfServiceIpName - Service IP Name

Syntax

--amfServiceIpName <string>

Description

The parameter points to a set of parameters related with IP address configuration created using the 'servicelp' command.

Class

Required on create.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

interfaceType - Interface name

Syntax

--interfaceType <list-item>

Description

Interface name.

Class

Required on create, not modifiable.

List

- N2 - N2 Interface - List of (N2, ...)

Name

amfServiceIpName1 - Service IP Name 1

Syntax

--amfServiceIpName1 <string>

Description

The parameter points to a set of parameters related with IP address configuration created using the 'servicelp' command. When configured, it specifies the second IP Address for N2 SCTP multi-homing.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

""

Name

dscpCode - DSCP code

Syntax

--dscpCode <list-item>

Description

Parameter to set DSCP code to be used for an IP packet

Class

Optional.

List

- AF11 - Assured forwarding 11.
- AF12 - Assured forwarding 12.
- AF13 - Assured forwarding 13.
- AF21 - Assured forwarding 21.
- AF22 - Assured forwarding 22.
- AF23 - Assured forwarding 23.
- AF31 - Assured forwarding 31.
- AF32 - Assured forwarding 32.
- AF33 - Assured forwarding 33.
- AF41 - Assured forwarding 41.
- AF42 - Assured forwarding 42.
- AF43 - Assured forwarding 43.
- BE - BE.
- CS1 - Class selector 1.
- CS2 - Class selector 2.
- CS3 - Class selector 3.
- CS4 - Class selector 4.
- CS5 - Class selector 5.
- CS6 - Class selector 6.

- CS7 - Class selector 7.
- EF - Expedited forwarding.
- voiceAdmit - Voice Admit

Default

AF41

Name

enable - Enable

Syntax`--enable <list-item>`**Description**

The parameter is used to enable use of the provisioned instance of the N2 interface. If the parameter is set to 'false', the N2 instance will not be used. Once an instance is enabled, the instance can be disabled only under certain TBD conditions.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

interfacePort - Interface port

Syntax`--interfacePort <integer>`**Description**

This parameter assigns local sctp port number.

Class

Optional.

Type / Range

Integer in the range of 0 or 1,024 to 65,535

Default

0

Name

multiHomed - Multi home

Syntax

--multiHomed <list-item>

Description

This enables/disables sctp multihomed at the given sctp local termination.

Restrictions

multiHomed is only applicable when interfaceName is N2. multiHomed can be modified only when enable is set to false.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

protocolProfileName - Interface profile name

Syntax

--protocolProfileName <string>

Description

Interface profile name.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

default

Name

tlsProfileName - TLS Profile

Syntax

--tlsProfileName <string>

Description

TLS Profile Name for N1/N2 endpoint

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

..

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.87 cmm amfNon3gppTai

Non-3GPP Tracking Area Identifier. Syntax, description, parameters, and roles for cmm amfNon3gppTai command.

Syntax

Description

Defines the non-3GPP access specific Tracking Area on the AMF. A single TAI dedicated to Non-3GPP access, is defined in a PLMN and applies within the PLMN. When a UE registers with the network over the Non-3GPP access, the AMF allocates a registration area that only includes the Non-3GPP TAI to the UE.

Parameters

Name

plmnName - Serving PLMN Name

Syntax

--plmnName <string>

Description

The serving PLMN name may be the name of the AMF home PLMN or shared PLMN.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

amfTac - AMF Tracking Area Code

Syntax

--amfTac <integer>

Description

Provision 5G TAC for non-3GPP access. 5G TAC is 3 octets long.

Class

Required key.

Type / Range

Integer in the range of 0 to 16777215

Name

amfEmergencyProfileName - AMF Emergency Profile Name

Syntax

--amfEmergencyProfileName <string>

Description

The name of the AMF Emergency profile that applies for the non-3GPP TAI.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

emergencyServiceSupported - Emergency Service Supported

Syntax

--emergencyServiceSupported <list-item>

Description

Defines whether Emergency service is supported in the non-3GPP access tracking area.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

imsVops - IMS Voice Over PS Support

Syntax

--imsVops <list-item>

Description

Indicates whether IMS VoPS is supported on the Non-3GPP Tracking Area of the AMF.

Class

Optional.

List

- AmflmsVopsNon3gppNotSupported - IMS Voice Over PS Non-3GPP Access Not Supported - IMS Voice over PS not supported over Non-3GPP access.
- AmflmsVopsNon3gppSupported - IMS Voice Over PS Non-3GPP Access Supported - IMS Voice over PS supported over Non-3GPP access.

Default

AmflmsVopsNon3gppNotSupported

Name

snssaiListName - SNSSAI List Name

Syntax

--snssaiListName <string>

Description

This parameter is a pointer to the S-NSSAI list. The S-NSSAI in the list are authorized in the TAI. If the TAI is not provisioned with a list then it is assumed that all S-NSSAIs that are supported by AMF and 5G-AN are authorized in the TAI. This parameter is not used if AMF is provisioned to use the NSSF Availability Service as in this case the Slices that are authorized in the TAI are provided by the NSSF.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

""

Name

timeZoneName - Time Zone Name

Syntax

--timeZoneName <list-item>

Description

Time Zone of the Tracking Area.

Class

Optional.

List

- Africa/Abidjan - Timezone for africa/abidjan UTC offset of +00:00.
- Africa/Accra - Timezone for africa/accra UTC offset of +00:00.
- Africa/Addis_Ababa - Timezone for africa/addis_ababa UTC offset of +03:00.
- Africa/Algiers - Timezone for africa/algiers UTC offset of +01:00.
- Africa/Asmara - Timezone for africa/asmara UTC offset of +03:00.
- Africa/Asmera - Timezone for africa/asmera UTC offset of +03:00.
- Africa/Bamako - Timezone for africa/bamako UTC offset of +00:00.
- Africa/Bangui - Timezone for africa/bangui UTC offset of +01:00.
- Africa/Banjul - Timezone for africa/banjul UTC offset of +00:00.
- Africa/Bissau - Timezone for africa/bissau UTC offset of +00:00.
- Africa/Blantyre - Timezone for africa/blantyre UTC offset of +02:00.
- Africa/Brazzaville - Timezone for africa/brazzaville UTC offset of +01:00.
- Africa/Bujumbura - Timezone for africa/bujumbura UTC offset of +02:00.
- Africa/Cairo - Timezone for africa/cairo UTC offset of +02:00.
- Africa/Casablanca - Timezone for africa/casablanca UTC offset of +00:00.
- Africa/Ceuta - Timezone for africa/ceuta UTC offset of +01:00.
- Africa/Conakry - Timezone for africa/conakry UTC offset of +00:00.
- Africa/Dakar - Timezone for africa/dakar UTC offset of +00:00.
- Africa/Dar_es_Salaam - Timezone for africa/dar_es_salaam UTC offset of +03:00.
- Africa/Djibouti - Timezone for africa/djibouti UTC offset of +03:00.
- Africa/Douala - Timezone for africa/douala UTC offset of +01:00.
- Africa/El_Aaiun - Timezone for africa/el_aaiun UTC offset of +00:00.
- Africa/Freetown - Timezone for africa/freetown UTC offset of +00:00.
- Africa/Gaborone - Timezone for africa/gaborone UTC offset of +02:00.
- Africa/Harare - Timezone for africa/harare UTC offset of +02:00.
- Africa/Johannesburg - Timezone for africa/johannesburg UTC offset of +02:00.

- Africa/Juba - Timezone for africa/juba UTC offset of +03:00.
- Africa/Kampala - Timezone for africa/kampala UTC offset of +03:00.
- Africa/Khartoum - Timezone for africa/khartoum UTC offset of +03:00.
- Africa/Kigali - Timezone for africa/kigali UTC offset of +02:00.
- Africa/Kinshasa - Timezone for africa/kinshasa UTC offset of +01:00.
- Africa/Lagos - Timezone for africa/lagos UTC offset of +01:00.
- Africa/Libreville - Timezone for africa/libreville UTC offset of +01:00.
- Africa/Lome - Timezone for africa/lome UTC offset of +00:00.
- Africa/Luanda - Timezone for africa/luanda UTC offset of +01:00.
- Africa/Lubumbashi - Timezone for africa/lubumbashi UTC offset of +02:00.
- Africa/Lusaka - Timezone for africa/lusaka UTC offset of +02:00.
- Africa/Malabo - Timezone for africa/malabo UTC offset of +01:00.
- Africa/Maputo - Timezone for africa/maputo UTC offset of +02:00.
- Africa/Maseru - Timezone for africa/maseru UTC offset of +02:00.
- Africa/Mbabane - Timezone for africa/mbabane UTC offset of +02:00.
- Africa/Mogadishu - Timezone for africa/mogadishu UTC offset of +03:00.
- Africa/Monrovia - Timezone for africa/monrovia UTC offset of +00:00.
- Africa/Nairobi - Timezone for africa/nairobi UTC offset of +03:00.
- Africa/Ndjamena - Timezone for africa/ndjamena UTC offset of +01:00.
- Africa/Niamey - Timezone for africa/niamey UTC offset of +01:00.
- Africa/Nouakchott - Timezone for africa/nouakchott UTC offset of +00:00.
- Africa/Ouagadougou - Timezone for africa/ouagadougou UTC offset of +00:00.
- Africa/Porto-Novo - Timezone for africa/porto-novo UTC offset of +01:00.
- Africa/Sao_Tome - Timezone for africa/sao_tome UTC offset of +00:00.
- Africa/Timbuktu - Timezone for africa/timbuktu UTC offset of +00:00.
- Africa/Tripoli - Timezone for africa/tripoli UTC offset of +02:00.
- Africa/Tunis - Timezone for africa/tunis UTC offset of +01:00.
- Africa/Windhoek - Timezone for africa/windhoek UTC offset of +02:00.
- America/Adak - Timezone for america/adak UTC offset of -10:00.
- America/Anchorage - Timezone for america/anchorage UTC offset of -09:00.
- America/Anguilla - Timezone for america/anguilla UTC offset of -04:00.

- America/Antigua - Timezone for america/antigua UTC offset of -04:00.
- America/Araguaina - Timezone for america/araguaina UTC offset of -03:00.
- America/Argentina/Buenos_Aires - Timezone for america/argentina/buenos_aires UTC offset of -03:00.
- America/Argentina/Catamarca - Timezone for america/argentina/catamarca UTC offset of -03:00.
- America/Argentina/ComodRivadavia - Timezone for america/argentina/comodrivadavia UTC offset of -03:00.
- America/Argentina/Cordoba - Timezone for america/argentina/cordoba UTC offset of -03:00.
- America/Argentina/Jujuy - Timezone for america/argentina/jujuy UTC offset of -03:00.
- America/Argentina/La_Rioja - Timezone for america/argentina/la_rioja UTC offset of -03:00.
- America/Argentina/Mendoza - Timezone for america/argentina/mendoza UTC offset of -03:00.
- America/Argentina/Rio_Gallegos - Timezone for america/argentina/rio_gallegos UTC offset of -03:00.
- America/Argentina/Salta - Timezone for america/argentina/salta UTC offset of -03:00.
- America/Argentina/San_Juan - Timezone for america/argentina/san_juan UTC offset of -03:00.
- America/Argentina/San_Luis - Timezone for america/argentina/san_luis UTC offset of -03:00.
- America/Argentina/Tucuman - Timezone for america/argentina/tucuman UTC offset of -03:00.
- America/Argentina/Ushuaia - Timezone for america/argentina/ushuaia UTC offset of -03:00.
- America/Aruba - Timezone for america/aruba UTC offset of -04:00.
- America/Asuncion - Timezone for america/asuncion UTC offset of -03:00.
- America/Atikokan - Timezone for america/atikokan UTC offset of -05:00.
- America/Atka - Timezone for america/atka UTC offset of -10:00.
- America/Bahia - Timezone for america/bahia UTC offset of -03:00.
- America/Bahia_Banderas - Timezone for america/bahia_banderas UTC offset of -06:00.
- America/Barbados - Timezone for america/barbados UTC offset of -04:00.
- America/Belem - Timezone for america/belem UTC offset of -03:00.

- America/Belize - Timezone for america/belize UTC offset of -06:00.
- America/Blanc-Sablon - Timezone for america/blanc-sablon UTC offset of -04:00.
- America/Boa_Vista - Timezone for america/boa_vista UTC offset of -04:00.
- America/Bogota - Timezone for america/bogota UTC offset of -05:00.
- America/Boise - Timezone for america/boise UTC offset of -07:00.
- America/Buenos_Aires - Timezone for america/buenos_aires UTC offset of -03:00.
- America/Cambridge_Bay - Timezone for america/cambridge_bay UTC offset of -07:00.
- America/Campo_Grande - Timezone for america/campo_grande UTC offset of -03:00.
- America/Cancun - Timezone for america/cancun UTC offset of -05:00.
- America/Caracas - Timezone for america/caracas UTC offset of -04:00.
- America/Catamarca - Timezone for america/catamarca UTC offset of -03:00.
- America/Cayenne - Timezone for america/cayenne UTC offset of -03:00.
- America/Cayman - Timezone for america/cayman UTC offset of -05:00.
- America/Chicago - Timezone for america/chicago UTC offset of -06:00.
- America/Chihuahua - Timezone for america/chihuahua UTC offset of -07:00.
- America/Coral_Harbour - Timezone for america/coral_harbour UTC offset of -05:00.
- America/Cordoba - Timezone for america/cordoba UTC offset of -03:00.
- America/Costa_Rica - Timezone for america/costa_rica UTC offset of -06:00.
- America/Creston - Timezone for america/creston UTC offset of -07:00.
- America/Cuiaba - Timezone for america/cuiaba UTC offset of -03:00.
- America/Curacao - Timezone for america/curacao UTC offset of -04:00.
- America/Danmarkshavn - Timezone for america/danmarkshavn UTC offset of +00:00.
- America/Dawson - Timezone for america/dawson UTC offset of -08:00.
- America/Dawson_Creek - Timezone for america/dawson_creek UTC offset of -07:00.
- America/Denver - Timezone for america/denver UTC offset of -07:00.
- America/Detroit - Timezone for america/detroit UTC offset of -05:00.
- America/Dominica - Timezone for america/dominica UTC offset of -04:00.
- America/Edmonton - Timezone for america/edmonton UTC offset of -07:00.
- America/Eirunepe - Timezone for america/eirunepe UTC offset of -05:00.
- America/El_Salvador - Timezone for america/el_salvador UTC offset of -06:00.
- America/Ensenada - Timezone for america/ensenada UTC offset of -08:00.

- America/Fort_Nelson - Timezone for america/fort_nelson UTC offset of -07:00.
- America/Fort_Wayne - Timezone for america/fort_wayne UTC offset of -05:00.
- America/Fortaleza - Timezone for america/fortaleza UTC offset of -03:00.
- America/Glace_Bay - Timezone for america/glance_bay UTC offset of -04:00.
- America/Godthab - Timezone for america/godthab UTC offset of -03:00.
- America/Goose_Bay - Timezone for america/goose_bay UTC offset of -04:00.
- America/Grand_Turk - Timezone for america/grand_turk UTC offset of -04:00.
- America/Grenada - Timezone for america/grenada UTC offset of -04:00.
- America/Guadeloupe - Timezone for america/guadeloupe UTC offset of -04:00.
- America/Guatemala - Timezone for america/guatemala UTC offset of -06:00.
- America/Guayaquil - Timezone for america/guayaquil UTC offset of -05:00.
- America/Guyana - Timezone for america/guyana UTC offset of -04:00.
- America/Halifax - Timezone for america/halifax UTC offset of -04:00.
- America/Havana - Timezone for america/havana UTC offset of -05:00.
- America/Hermosillo - Timezone for america/hermosillo UTC offset of -07:00.
- America/Indiana/Indianapolis - Timezone for america/indiana/indianapolis UTC offset of -05:00.
- America/Indiana/Knox - Timezone for america/indiana/knox UTC offset of -06:00.
- America/Indiana/Marengo - Timezone for america/indiana/marengo UTC offset of -05:00.
- America/Indiana/Petersburg - Timezone for america/indiana/petersburg UTC offset of -05:00.
- America/Indiana/Tell_City - Timezone for america/indiana/tell_city UTC offset of -06:00.
- America/Indiana/Vevay - Timezone for america/indiana/vevay UTC offset of -05:00.
- America/Indiana/Vincennes - Timezone for america/indiana/vincennes UTC offset of -05:00.
- America/Indiana/Winamac - Timezone for america/indiana/winamac UTC offset of -05:00.
- America/Indianapolis - Timezone for america/indianapolis UTC offset of -05:00.
- America/Inuvik - Timezone for america/inuvik UTC offset of -07:00.
- America/Iqaluit - Timezone for america/iqaluit UTC offset of -05:00.
- America/Jamaica - Timezone for america/jamaica UTC offset of -05:00.

- America/Jujuy - Timezone for america/jujuy UTC offset of -03:00.
- America/Juneau - Timezone for america/juneau UTC offset of -09:00.
- America/Kentucky/Louisville - Timezone for america/kentucky/louisville UTC offset of -05:00.
- America/Kentucky/Monticello - Timezone for america/kentucky/monticello UTC offset of -05:00.
- America/Knox_IN - Timezone for america/knox_in UTC offset of -06:00.
- America/Kralendijk - Timezone for america/kralendijk UTC offset of -04:00.
- America/La_Paz - Timezone for america/la_paz UTC offset of -04:00.
- America/Lima - Timezone for america/lima UTC offset of -05:00.
- America/Los_Angeles - Timezone for america/los_angeles UTC offset of -08:00.
- America/Louisville - Timezone for america/louisville UTC offset of -05:00.
- America/Lower_Princes - Timezone for america/lower_princes UTC offset of -04:00.
- America/Maceio - Timezone for america/maceio UTC offset of -03:00.
- America/Managua - Timezone for america/managua UTC offset of -06:00.
- America/Manaus - Timezone for america/manuel UTC offset of -04:00.
- America/Marigot - Timezone for america/marigot UTC offset of -04:00.
- America/Martinique - Timezone for america/martinique UTC offset of -04:00.
- America/Matamoros - Timezone for america/matamoros UTC offset of -06:00.
- America/Mazatlan - Timezone for america/mazatlan UTC offset of -07:00.
- America/Mendoza - Timezone for america/mendoza UTC offset of -03:00.
- America/Menominee - Timezone for america/menominee UTC offset of -06:00.
- America/Merida - Timezone for america/merida UTC offset of -06:00.
- America/Metlakatla - Timezone for america/metlakatla UTC offset of -09:00.
- America/Mexico_City - Timezone for america/mexico_city UTC offset of -06:00.
- America/Miquelon - Timezone for america/miquelon UTC offset of -03:00.
- America/Moncton - Timezone for america/moncton UTC offset of -04:00.
- America/Monterrey - Timezone for america/monterrey UTC offset of -06:00.
- America/Montevideo - Timezone for america/montevideo UTC offset of -03:00.
- America/Montreal - Timezone for america/montreal UTC offset of -05:00.
- America/Montserrat - Timezone for america/montserrat UTC offset of -04:00.
- America/Nassau - Timezone for america/nassau UTC offset of -05:00.

- America/New_York - Timezone for america/new_york UTC offset of -05:00.
- America/Nipigon - Timezone for america/nipigon UTC offset of -05:00.
- America/Nome - Timezone for america/nome UTC offset of -09:00.
- America/Noronha - Timezone for america/noronha UTC offset of -02:00.
- America/North_Dakota/Beulah - Timezone for america/north_dakota/beulah UTC offset of -06:00.
- America/North_Dakota/Center - Timezone for america/north_dakota/center UTC offset of -06:00.
- America/North_Dakota/New_Salem - Timezone for america/north_dakota/new_salem UTC offset of -06:00.
- America/Ojinaga - Timezone for america/ojinaga UTC offset of -07:00.
- America/Panama - Timezone for america/panama UTC offset of -05:00.
- America/Pangnirtung - Timezone for america/pangnirtung UTC offset of -05:00.
- America/Paramaribo - Timezone for america/paramaribo UTC offset of -03:00.
- America/Phoenix - Timezone for america/phoenix UTC offset of -07:00.
- America/Port-au-Prince - Timezone for america/port-au-prince UTC offset of -05:00.
- America/Port_of_Spain - Timezone for america/port_of_spain UTC offset of -04:00.
- America/Porto_Acre - Timezone for america/porto_acre UTC offset of -05:00.
- America/Porto_Velho - Timezone for america/porto_velho UTC offset of -04:00.
- America/Puerto_Rico - Timezone for america/puerto_rico UTC offset of -04:00.
- America/Rainy_River - Timezone for america/rainy_river UTC offset of -06:00.
- America/Rankin_Inlet - Timezone for america/rankin_inlet UTC offset of -06:00.
- America/Recife - Timezone for america/recife UTC offset of -03:00.
- America/Regina - Timezone for america/regina UTC offset of -06:00.
- America/Resolute - Timezone for america/resolute UTC offset of -06:00.
- America/Rio_Branco - Timezone for america/rio_branco UTC offset of -05:00.
- America/Rosario - Timezone for america/rosario UTC offset of -03:00.
- America/Santa_Isabel - Timezone for america/santa_isabel UTC offset of -08:00.
- America/Santarem - Timezone for america/santarem UTC offset of -03:00.
- America/Santiago - Timezone for america/santiago UTC offset of -03:00.
- America/Santo_Domingo - Timezone for america/santo_domingo UTC offset of -04:00.
- America/Sao_Paulo - Timezone for america/sao_paulo UTC offset of -02:00.

- America/Scoresbysund - Timezone for america/scoresbysund UTC offset of -01:00.
- America/Shiprock - Timezone for america/shiprock UTC offset of -07:00.
- America/Sitka - Timezone for america/sitka UTC offset of -09:00.
- America/St_Barthelemy - Timezone for america/st_barthelemy UTC offset of -04:00.
- America/St_Johns - Timezone for america/st_johns UTC offset of -03:30.
- America/St_Kitts - Timezone for america/st_kitts UTC offset of -04:00.
- America/St_Lucia - Timezone for america/st_lucia UTC offset of -04:00.
- America/St_Thomas - Timezone for america/st_thomas UTC offset of -04:00.
- America/St_Vincent - Timezone for america/st_vincent UTC offset of -04:00.
- America/Swift_Current - Timezone for america/swift_current UTC offset of -06:00.
- America/Tegucigalpa - Timezone for america/tegucigalpa UTC offset of -06:00.
- America/Thule - Timezone for america/thule UTC offset of -04:00.
- America/Thunder_Bay - Timezone for america/thunder_bay UTC offset of -05:00.
- America/Tijuana - Timezone for america/tijuana UTC offset of -08:00.
- America/Toronto - Timezone for america/toronto UTC offset of -05:00.
- America/Tortola - Timezone for america/tortola UTC offset of -04:00.
- America/Vancouver - Timezone for america/vancouver UTC offset of -08:00.
- America/Virgin - Timezone for america/virgin UTC offset of -04:00.
- America/Whitehorse - Timezone for america/whitehorse UTC offset of -08:00.
- America/Winnipeg - Timezone for america/winnipeg UTC offset of -06:00.
- America/Yakutat - Timezone for america/yakutat UTC offset of -09:00.
- America/Yellowknife - Timezone for america/yellowknife UTC offset of -07:00.
- Antarctica/Casey - Timezone for antarctica/casey UTC offset of +11:00.
- Antarctica/Davis - Timezone for antarctica/davis UTC offset of +07:00.
- Antarctica/DumontDURville - Timezone for antarctica/dumontdurville UTC offset of +10:00.
- Antarctica/Macquarie - Timezone for antarctica/macquarie UTC offset of +11:00.
- Antarctica/Mawson - Timezone for antarctica/mawson UTC offset of +05:00.
- Antarctica/McMurdo - Timezone for antarctica/mcmurdo UTC offset of +13:00.
- Antarctica/Palmer - Timezone for antarctica/palmer UTC offset of -03:00.
- Antarctica/Rothera - Timezone for antarctica/rothera UTC offset of -03:00.

- Antarctica/South_Pole - Timezone for antarctica/south_pole UTC offset of +13:00.
- Antarctica/Syowa - Timezone for antarctica/syowa UTC offset of +03:00.
- Antarctica/Troll - Timezone for antarctica/troll UTC offset of +00:00.
- Antarctica/Vostok - Timezone for antarctica/vostok UTC offset of +06:00.
- Arctic/Longyearbyen - Timezone for arctic/longyearbyen UTC offset of +01:00.
- Asia/Aden - Timezone for asia/aden UTC offset of +03:00.
- Asia/Almaty - Timezone for asia/almaty UTC offset of +06:00.
- Asia/Amman - Timezone for asia/amman UTC offset of +02:00.
- Asia/Anadyr - Timezone for asia/anadyr UTC offset of +12:00.
- Asia/Aqtau - Timezone for asia/aqtau UTC offset of +05:00.
- Asia/Aqtobe - Timezone for asia/aqtobe UTC offset of +05:00.
- Asia/Ashgabat - Timezone for asia/ashgabat UTC offset of +05:00.
- Asia/Ashkhabad - Timezone for asia/ashkhabad UTC offset of +05:00.
- Asia/Baghdad - Timezone for asia/baghdad UTC offset of +03:00.
- Asia/Bahrain - Timezone for asia/bahrain UTC offset of +03:00.
- Asia/Baku - Timezone for asia/baku UTC offset of +04:00.
- Asia/Bangkok - Timezone for asia/bangkok UTC offset of +07:00.
- Asia/Barnaul - Timezone for asia/barnaul UTC offset of +07:00.
- Asia/Beirut - Timezone for asia/beirut UTC offset of +02:00.
- Asia/Bishkek - Timezone for asia/bishkek UTC offset of +06:00.
- Asia/Brunei - Timezone for asia/brunei UTC offset of +08:00.
- Asia/Calcutta - Timezone for asia/calcutta UTC offset of +05:30.
- Asia/Chita - Timezone for asia/chita UTC offset of +09:00.
- Asia/Choibalsan - Timezone for asia/choibalsan UTC offset of +08:00.
- Asia/Chongqing - Timezone for asia/chongqing UTC offset of +08:00.
- Asia/Chungking - Timezone for asia/chungking UTC offset of +08:00.
- Asia/Colombo - Timezone for asia/colombo UTC offset of +05:30.
- Asia/Dacca - Timezone for asia/dacca UTC offset of +06:00.
- Asia/Damascus - Timezone for asia/damascus UTC offset of +02:00.
- Asia/Dhaka - Timezone for asia/dhaka UTC offset of +06:00.
- Asia/Dili - Timezone for asia/dili UTC offset of +09:00.

- Asia/Dubai - Timezone for asia/dubai UTC offset of +04:00.
- Asia/Dushanbe - Timezone for asia/dushanbe UTC offset of +05:00.
- Asia/Gaza - Timezone for asia/gaza UTC offset of +02:00.
- Asia/Harbin - Timezone for asia/harbin UTC offset of +08:00.
- Asia/Hebron - Timezone for asia/hebron UTC offset of +02:00.
- Asia/Ho_Chi_Minh - Timezone for asia/ho_chi_minh UTC offset of +07:00.
- Asia/Hong_Kong - Timezone for asia/hong_kong UTC offset of +08:00.
- Asia/Hovd - Timezone for asia/hovd UTC offset of +07:00.
- Asia/Irkutsk - Timezone for asia/irkutsk UTC offset of +08:00.
- Asia/Istanbul - Timezone for asia/istanbul UTC offset of +03:00.
- Asia/Jakarta - Timezone for asia/jakarta UTC offset of +07:00.
- Asia/Jayapura - Timezone for asia/jayapura UTC offset of +09:00.
- Asia/Jerusalem - Timezone for asia/jerusalem UTC offset of +02:00.
- Asia/Kabul - Timezone for asia/kabul UTC offset of +04:30.
- Asia/Kamchatka - Timezone for asia/kamchatka UTC offset of +12:00.
- Asia/Karachi - Timezone for asia/karachi UTC offset of +05:00.
- Asia/Kashgar - Timezone for asia/kashgar UTC offset of +06:00.
- Asia/Kathmandu - Timezone for asia/kathmandu UTC offset of +05:45.
- Asia/Katmandu - Timezone for asia/katmandu UTC offset of +05:45.
- Asia/Khandyga - Timezone for asia/khandyga UTC offset of +09:00.
- Asia/Kolkata - Timezone for asia/kolkata UTC offset of +05:30.
- Asia/Krasnoyarsk - Timezone for asia/krasnoyarsk UTC offset of +07:00.
- Asia/Kuala_Lumpur - Timezone for asia/kuala_lumpur UTC offset of +08:00.
- Asia/Kuching - Timezone for asia/kuching UTC offset of +08:00.
- Asia/Kuwait - Timezone for asia/kuwait UTC offset of +03:00.
- Asia/Macao - Timezone for asia/macao UTC offset of +08:00.
- Asia/Macau - Timezone for asia/macau UTC offset of +08:00.
- Asia/Magadan - Timezone for asia/magadan UTC offset of +11:00.
- Asia/Makassar - Timezone for asia/makassar UTC offset of +08:00.
- Asia/Manila - Timezone for asia/manila UTC offset of +08:00.
- Asia/Muscat - Timezone for asia/muscat UTC offset of +04:00.

- Asia/Nicosia - Timezone for asia/nicosia UTC offset of +02:00.
- Asia/Novokuznetsk - Timezone for asia/novokuznetsk UTC offset of +07:00.
- Asia/Novosibirsk - Timezone for asia/novosibirsk UTC offset of +07:00.
- Asia/Omsk - Timezone for asia/omsk UTC offset of +06:00.
- Asia/Oral - Timezone for asia/oral UTC offset of +05:00.
- Asia/Phnom_Penh - Timezone for asia/phnom_penh UTC offset of +07:00.
- Asia/Pontianak - Timezone for asia/pontianak UTC offset of +07:00.
- Asia/Pyongyang - Timezone for asia/pyongyang UTC offset of +08:30.
- Asia/Qatar - Timezone for asia/qatar UTC offset of +03:00.
- Asia/Qyzylorda - Timezone for asia/qyzylorda UTC offset of +06:00.
- Asia/Rangoon - Timezone for asia/rangoon UTC offset of +06:30.
- Asia/Riyadh - Timezone for asia/riyadh UTC offset of +03:00.
- Asia/Saigon - Timezone for asia/saigon UTC offset of +07:00.
- Asia/Sakhalin - Timezone for asia/sakhalin UTC offset of +11:00.
- Asia/Samarkand - Timezone for asia/samarkand UTC offset of +05:00.
- Asia/Seoul - Timezone for asia/seoul UTC offset of +09:00.
- Asia/Shanghai - Timezone for asia/shanghai UTC offset of +08:00.
- Asia/Singapore - Timezone for asia/singapore UTC offset of +08:00.
- Asia/Srednekolymsk - Timezone for asia/srednekolymsk UTC offset of +11:00.
- Asia/Taipei - Timezone for asia/taipei UTC offset of +08:00.
- Asia/Tashkent - Timezone for asia/tashkent UTC offset of +05:00.
- Asia/Tbilisi - Timezone for asia/tbilisi UTC offset of +04:00.
- Asia/Tehran - Timezone for asia/tehran UTC offset of +03:30.
- Asia/Tel_Aviv - Timezone for asia/tel_aviv UTC offset of +02:00.
- Asia/Thimbu - Timezone for asia/thimbu UTC offset of +06:00.
- Asia/Thimphu - Timezone for asia/thimphu UTC offset of +06:00.
- Asia/Tokyo - Timezone for asia/tokyo UTC offset of +09:00.
- Asia/Tomsk - Timezone for asia/tomsk UTC offset of +07:00.
- Asia/Ujung_Pandang - Timezone for asia/ujung_pandang UTC offset of +08:00.
- Asia/Ulaanbaatar - Timezone for asia/ulaanbaatar UTC offset of +08:00.
- Asia/Ulan_Bator - Timezone for asia/ulan_bator UTC offset of +08:00.

- Asia/Urumqi - Timezone for asia/urumqi UTC offset of +06:00.
- Asia/Ust-Nera - Timezone for asia/ust-nera UTC offset of +10:00.
- Asia/Vientiane - Timezone for asia/vientiane UTC offset of +07:00.
- Asia/Vladivostok - Timezone for asia/vladivostok UTC offset of +10:00.
- Asia/Yakutsk - Timezone for asia/yakutsk UTC offset of +09:00.
- Asia/Yekaterinburg - Timezone for asia/yekaterinburg UTC offset of +05:00.
- Asia/Yerevan - Timezone for asia/yerevan UTC offset of +04:00.
- Atlantic/Azores - Timezone for atlantic/azores UTC offset of -01:00.
- Atlantic/Bermuda - Timezone for atlantic/bermuda UTC offset of -04:00.
- Atlantic/Canary - Timezone for atlantic/canary UTC offset of +00:00.
- Atlantic/Cape_Verde - Timezone for atlantic/cape_verde UTC offset of -01:00.
- Atlantic/Faeroe - Timezone for atlantic/faeroe UTC offset of +00:00.
- Atlantic/Faroe - Timezone for atlantic/faroe UTC offset of +00:00.
- Atlantic/Jan_Mayen - Timezone for atlantic/jan_mayen UTC offset of +01:00.
- Atlantic/Madeira - Timezone for atlantic/madeira UTC offset of +00:00.
- Atlantic/Reykjavik - Timezone for atlantic/reykjavik UTC offset of +00:00.
- Atlantic/South_Georgia - Timezone for atlantic/south_georgia UTC offset of -02:00.
- Atlantic/St_Helena - Timezone for atlantic/st_helena UTC offset of +00:00.
- Atlantic/Stanley - Timezone for atlantic/stanley UTC offset of -03:00.
- Australia/ACT - Timezone for australia/act UTC offset of +11:00.
- Australia/Adelaide - Timezone for australia/adelaide UTC offset of +10:30.
- Australia/Brisbane - Timezone for australia/brisbane UTC offset of +10:00.
- Australia/Broken_Hill - Timezone for australia/broken_hill UTC offset of +10:30.
- Australia/Canberra - Timezone for australia/canberra UTC offset of +11:00.
- Australia/Currie - Timezone for australia/currie UTC offset of +11:00.
- Australia/Darwin - Timezone for australia/darwin UTC offset of +09:30.
- Australia/Eucla - Timezone for australia/eucla UTC offset of +08:45.
- Australia/Hobart - Timezone for australia/hobart UTC offset of +11:00.
- Australia/LHI - Timezone for australia/lhi UTC offset of +11:00.
- Australia/Lindeman - Timezone for australia/lindeman UTC offset of +10:00.
- Australia/Lord_Howe - Timezone for australia/lord_howe UTC offset of +11:00.

- Australia/Melbourne - Timezone for australia/melbourne UTC offset of +11:00.
- Australia/NSW - Timezone for australia/nsw UTC offset of +11:00.
- Australia/North - Timezone for australia/north UTC offset of +09:30.
- Australia/Perth - Timezone for australia/perth UTC offset of +08:00.
- Australia/Queensland - Timezone for australia/queensland UTC offset of +10:00.
- Australia/South - Timezone for australia/south UTC offset of +10:30.
- Australia/Sydney - Timezone for australia/sydney UTC offset of +11:00.
- Australia/Tasmania - Timezone for australia/tasmania UTC offset of +11:00.
- Australia/Victoria - Timezone for australia/victoria UTC offset of +11:00.
- Australia/West - Timezone for australia/west UTC offset of +08:00.
- Australia/Yancowinna - Timezone for australia/yancowinna UTC offset of +10:30.
- CET - Timezone for CET UTC offset of +01:00.
- CST6CDT - Timezone for CST6CDT UTC offset of -06:00.
- Canada/Atlantic - Timezone for canada/atlantic UTC offset of -04:00.
- Canada/Central - Timezone for canada/central UTC offset of -06:00.
- Canada/East-Saskatchewan - Timezone for canada/east-saskatchewan UTC offset of -06:00.
- Canada/Eastern - Timezone for canada/eastern UTC offset of -05:00.
- Canada/Mountain - Timezone for canada/mountain UTC offset of -07:00.
- Canada/Newfoundland - Timezone for canada/newfoundland UTC offset of -03:30.
- Canada/Pacific - Timezone for canada/pacific UTC offset of -08:00.
- Canada/Saskatchewan - Timezone for canada/saskatchewan UTC offset of -06:00.
- Canada/Yukon - Timezone for canada/yukon UTC offset of -08:00.
- Chile/Continental - Timezone for chile/continental UTC offset of -03:00.
- Chile/EasterIsland - Timezone for chile/easterisland UTC offset of -05:00.
- Cuba - Timezone for cuba UTC offset of -05:00.
- Default - Timezone for system UTC (OS time zone)
- EET - Timezone for EET UTC offset of +02:00.
- EST - Timezone for EST UTC offset of -05:00.
- EST5EDT - Timezone for EST5EDT UTC offset of -05:00.
- Egypt - Timezone for egypt UTC offset of +02:00.

- Eire - Timezone for eire UTC offset of +00:00.
- Etc/Greenwich - Timezone for etc/greenwich UTC offset of +00:00.
- Etc/UCT - Timezone for etc/uct UTC offset of +00:00.
- Etc/UTC - Timezone for etc/utc UTC offset of +00:00.
- Etc/Universal - Timezone for etc/universal UTC offset of +00:00.
- Etc/Zulu - Timezone for etc/zulu UTC offset of +00:00.
- Europe/Amsterdam - Timezone for europe/amsterdam UTC offset of +01:00.
- Europe/Andorra - Timezone for europe/andorra UTC offset of +01:00.
- Europe/Astrakhan - Timezone for europe/astrakhan UTC offset of +04:00.
- Europe/Athens - Timezone for europe/athens UTC offset of +02:00.
- Europe/Belfast - Timezone for europe/belfast UTC offset of +00:00.
- Europe/Belgrade - Timezone for europe/belgrade UTC offset of +01:00.
- Europe/Berlin - Timezone for europe/berlin UTC offset of +01:00.
- Europe/Bratislava - Timezone for europe/bratislava UTC offset of +01:00.
- Europe/Brussels - Timezone for europe/brussels UTC offset of +01:00.
- Europe/Bucharest - Timezone for europe/bucharest UTC offset of +02:00.
- Europe/Budapest - Timezone for europe/budapest UTC offset of +01:00.
- Europe/Busingen - Timezone for europe/busingen UTC offset of +01:00.
- Europe/Chisinau - Timezone for europe/chisinau UTC offset of +02:00.
- Europe/Copenhagen - Timezone for europe/copenhagen UTC offset of +01:00.
- Europe/Dublin - Timezone for europe/dublin UTC offset of +00:00.
- Europe/Gibraltar - Timezone for europe/gibraltar UTC offset of +01:00.
- Europe/Guernsey - Timezone for europe/guernsey UTC offset of +00:00.
- Europe/Helsinki - Timezone for europe/helsinki UTC offset of +02:00.
- Europe/Isle_of_Man - Timezone for europe/isle_of_man UTC offset of +00:00.
- Europe/Istanbul - Timezone for europe/istanbul UTC offset of +03:00.
- Europe/Jersey - Timezone for europe/jersey UTC offset of +00:00.
- Europe/Kaliningrad - Timezone for europe/kaliningrad UTC offset of +02:00.
- Europe/Kiev - Timezone for europe/kyiv UTC offset of +02:00.
- Europe/Kirov - Timezone for europe/kirov UTC offset of +03:00.
- Europe/Lisbon - Timezone for europe/lisbon UTC offset of +00:00.

- Europe/Ljubljana - Timezone for europe/ljubljana UTC offset of +01:00.
- Europe/London - Timezone for europe/london UTC offset of +00:00.
- Europe/Luxembourg - Timezone for europe/luxembourg UTC offset of +01:00.
- Europe/Madrid - Timezone for europe/madrid UTC offset of +01:00.
- Europe/Malta - Timezone for europe/malta UTC offset of +01:00.
- Europe/Mariehamn - Timezone for europe/mariehamn UTC offset of +02:00.
- Europe/Minsk - Timezone for europe/minsk UTC offset of +03:00.
- Europe/Monaco - Timezone for europe/monaco UTC offset of +01:00.
- Europe/Moscow - Timezone for europe/moscow UTC offset of +03:00.
- Europe/Nicosia - Timezone for europe/nicosia UTC offset of +02:00.
- Europe/Oslo - Timezone for europe/oslo UTC offset of +01:00.
- Europe/Paris - Timezone for europe/paris UTC offset of +01:00.
- Europe/Podgorica - Timezone for europe/podgorica UTC offset of +01:00.
- Europe/Prague - Timezone for europe/prague UTC offset of +01:00.
- Europe/Riga - Timezone for europe/riga UTC offset of +02:00.
- Europe/Rome - Timezone for europe/rome UTC offset of +01:00.
- Europe/Samara - Timezone for europe/samara UTC offset of +04:00.
- Europe/San_Marino - Timezone for europe/san_marino UTC offset of +01:00.
- Europe/Sarajevo - Timezone for europe/sarajevo UTC offset of +01:00.
- Europe/Simferopol - Timezone for europe/simferopol UTC offset of +03:00.
- Europe/Skopje - Timezone for europe/skopje UTC offset of +01:00.
- Europe/Sofia - Timezone for europe/sofia UTC offset of +02:00.
- Europe/Stockholm - Timezone for europe/stockholm UTC offset of +01:00.
- Europe/Tallinn - Timezone for europe/tallinn UTC offset of +02:00.
- Europe/Tirane - Timezone for europe/tirane UTC offset of +01:00.
- Europe/Tiraspol - Timezone for europe/tiraspol UTC offset of +02:00.
- Europe/Ulyanovsk - Timezone for europe/ulyanovsk UTC offset of +04:00.
- Europe/Uzhgorod - Timezone for europe/uzhgorod UTC offset of +02:00.
- Europe/Vaduz - Timezone for europe/vaduz UTC offset of +01:00.
- Europe/Vatican - Timezone for europe/vatican UTC offset of +01:00.
- Europe/Vienna - Timezone for europe/vienna UTC offset of +01:00.

- Europe/Vilnius - Timezone for europe/vilnius UTC offset of +02:00.
- Europe/Volgograd - Timezone for europe/volgograd UTC offset of +03:00.
- Europe/Warsaw - Timezone for europe/warsaw UTC offset of +01:00.
- Europe/Zagreb - Timezone for europe/zagreb UTC offset of +01:00.
- Europe/Zaporozhye - Timezone for europe/zaporozhye UTC offset of +02:00.
- Europe/Zurich - Timezone for europe/zurich UTC offset of +01:00.
- GB - Timezone for GB UTC offset of +00:00.
- GB-Eire - Timezone for GB-Eire UTC offset of +00:00.
- GMT - Timezone for GMT UTC offset of +00:00.
- GMT0 - Timezone for GMT0 UTC offset of +00:00.
- Greenwich - Timezone for greenwich UTC offset of +00:00.
- HST - Timezone for HST UTC offset of -10:00.
- Hongkong - Timezone for hongkong UTC offset of +08:00.
- Iceland - Timezone for iceland UTC offset of +00:00.
- Indian/Antananarivo - Timezone for indian/antananarivo UTC offset of +03:00.
- Indian/Chagos - Timezone for indian/chagos UTC offset of +06:00.
- Indian/Christmas - Timezone for indian/christmas UTC offset of +07:00.
- Indian/Cocos - Timezone for indian/cocos UTC offset of +06:30.
- Indian/Comoro - Timezone for indian/comoro UTC offset of +03:00.
- Indian/Kerguelen - Timezone for indian/kerguelen UTC offset of +05:00.
- Indian/Mahe - Timezone for indian/mahe UTC offset of +04:00.
- Indian/Maldives - Timezone for indian/maldives UTC offset of +05:00.
- Indian/Mauritius - Timezone for indian/mauritius UTC offset of +04:00.
- Indian/Mayotte - Timezone for indian/mayotte UTC offset of +03:00.
- Indian/Reunion - Timezone for indian/reunion UTC offset of +04:00.
- Iran - Timezone for iran UTC offset of +03:30.
- Israel - Timezone for israel UTC offset of +02:00.
- Jamaica - Timezone for jamaica UTC offset of -05:00.
- Japan - Timezone for japan UTC offset of +09:00.
- Kwajalein - Timezone for kwajalein UTC offset of +12:00.
- Libya - Timezone for libya UTC offset of +02:00.

- MET - Timezone for MET UTC offset of +01:00.
- MST - Timezone for MST UTC offset of -07:00.
- MST7MDT - Timezone for MST7MDT UTC offset of -07:00.
- Mexico/BajaNorte - Timezone for mexico/bajanorte UTC offset of -08:00.
- Mexico/BajaSur - Timezone for mexico/bajasur UTC offset of -07:00.
- Mexico/General - Timezone for mexico/general UTC offset of -06:00.
- NZ - Timezone for NZ UTC offset of +13:00.
- NZ-CHAT - Timezone for NZ-CHAT UTC offset of +13:45.
- Navajo - Timezone for navajo UTC offset of -07:00.
- PRC - Timezone for PRC UTC offset of +08:00.
- PST8PDT - Timezone for PST8PDT UTC offset of -08:00.
- Pacific/Apia - Timezone for pacific/apia UTC offset of +14:00.
- Pacific/Auckland - Timezone for pacific/auckland UTC offset of +13:00.
- Pacific/Bougainville - Timezone for pacific/bougainville UTC offset of +11:00.
- Pacific/Chatham - Timezone for pacific/chatham UTC offset of +13:45.
- Pacific/Chuuk - Timezone for pacific/chuuk UTC offset of +10:00.
- Pacific/Easter - Timezone for pacific/easter UTC offset of -05:00.
- Pacific/Efate - Timezone for pacific/efate UTC offset of +11:00.
- Pacific/Enderbury - Timezone for pacific/enderbury UTC offset of +13:00.
- Pacific/Fakaofu - Timezone for pacific/fakaofu UTC offset of +13:00.
- Pacific/Fiji - Timezone for pacific/fiji UTC offset of +13:00.
- Pacific/Funafuti - Timezone for pacific/funafuti UTC offset of +12:00.
- Pacific/Galapagos - Timezone for pacific/galapagos UTC offset of -06:00.
- Pacific/Gambier - Timezone for pacific/gambier UTC offset of -09:00.
- Pacific/Guadalcanal - Timezone for pacific/guadalcanal UTC offset of +11:00.
- Pacific/Guam - Timezone for pacific/guam UTC offset of +10:00.
- Pacific/Honolulu - Timezone for pacific/honolulu UTC offset of -10:00.
- Pacific/Johnston - Timezone for pacific/johnston UTC offset of -10:00.
- Pacific/Kiritimati - Timezone for pacific/kiritimati UTC offset of +14:00.
- Pacific/Kosrae - Timezone for pacific/kosrae UTC offset of +11:00.
- Pacific/Kwajalein - Timezone for pacific/kwajalein UTC offset of +12:00.

- Pacific/Majuro - Timezone for pacific/majuro UTC offset of +12:00.
- Pacific/Marquesas - Timezone for pacific/marquesas UTC offset of -09:30.
- Pacific/Midway - Timezone for pacific/midway UTC offset of -11:00.
- Pacific/Nauru - Timezone for pacific/nauru UTC offset of +12:00.
- Pacific/Niue - Timezone for pacific/niue UTC offset of -11:00.
- Pacific/Norfolk - Timezone for pacific/norfolk UTC offset of +11:00.
- Pacific/Noumea - Timezone for pacific/noumea UTC offset of +11:00.
- Pacific/Pago_Pago - Timezone for pacific/pago_pago UTC offset of -11:00.
- Pacific/Palau - Timezone for pacific/palau UTC offset of +09:00.
- Pacific/Pitcairn - Timezone for pacific/pitcairn UTC offset of -08:00.
- Pacific/Pohnpei - Timezone for pacific/pohnpei UTC offset of +11:00.
- Pacific/Ponape - Timezone for pacific/ponape UTC offset of +11:00.
- Pacific/Port_Moresby - Timezone for pacific/port_moresby UTC offset of +10:00.
- Pacific/Rarotonga - Timezone for pacific/rarotonga UTC offset of -10:00.
- Pacific/Saipan - Timezone for pacific/saipan UTC offset of +10:00.
- Pacific/Samoa - Timezone for pacific/samoa UTC offset of -11:00.
- Pacific/Tahiti - Timezone for pacific/tahiti UTC offset of -10:00.
- Pacific/Tarawa - Timezone for pacific/tarawa UTC offset of +12:00.
- Pacific/Tongatapu - Timezone for pacific/tongatapu UTC offset of +14:00.
- Pacific/Truk - Timezone for pacific/truk UTC offset of +10:00.
- Pacific/Wake - Timezone for pacific/wake UTC offset of +12:00.
- Pacific/Wallis - Timezone for pacific/wallis UTC offset of +12:00.
- Pacific/Yap - Timezone for pacific/yap UTC offset of +10:00.
- Poland - Timezone for poland UTC offset of +01:00.
- Portugal - Timezone for portugal UTC offset of +00:00.
- ROC - Timezone for ROC UTC offset of +08:00.
- ROK - Timezone for ROK UTC offset of +09:00.
- Singapore - Timezone for singapore UTC offset of +08:00.
- Turkey - Timezone for turkey UTC offset of +03:00.
- UCT - Timezone for UCT UTC offset of +00:00.
- US/Pacific-New - Timezone for US/Pacific-New UTC offset of -08:00.

- Universal - Timezone for universal UTC offset of +00:00.
- W-SU - Timezone for W-SU UTC offset of +03:00.
- WET - Timezone for WET UTC offset of +00:00.
- Zulu - Timezone for zulu UTC offset of +00:00.
- jdj_tz - Timezone for jdj_tz UTC offset of +00:00.

Default

Default

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.88 cmm amfNotificationEndPointConfig

AMF Notification End Point Configuration. Syntax, description, parameters, and roles for cmm amfNotificationEndPointConfig command.

Syntax**Description**

This command is used to create, modify and delete AMF local endpoint configuration for receiving and sending notifications. AMF as a consumer/client of other NF (producer/server) may include callback URI in a request if the AMF expects to receive a notification from the producer NF. The callback URI will be constructed using the parameters of this command and other data such as UE SUPI. The producer will act as a consumer/client when it sends a notification to the AMF and sets up TCP connection to the IP address provisioned to this endpoint. AMF acts as a producer/server to receive the notification.

Parameters**Name**

endPointName - Notification Endpoint Name

Syntax

--endPointName <string>

Description

The name that identifies the Notification Endpoint.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

amfServiceIpName - Service IP

Syntax

--amfServiceIpName <string>

Description

The parameter points to a set of parameters related with IP address configuration created using the 'serviceIp' command.

Class

Required on create.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

nfType - Network Function Type

Syntax

--nfType <list-item>

Description

The endpoint will be used for constructing the callback URI to a NF identified by this parameter. For example, if AMF includes callback URI in a request to SMF then the AMF will construct the callback URI using the parameters of the endpoint for the NF type SMF.

Class

Required on create, not modifiable.

List

- 5GEIR - 5G EIR - The network function type of 5G-Equipment Identity Register.
- AMF - Access and Mobility Manager Function - The network function type of Access and Mobility Manager Function.
- AUSF - Authentication Server Function - The network function type of Authentication Server Function.

- CBCF - Cell Broadcast Centre Function - The network function type of Cell Broadcast Centre Function
- CHF - Charging Function - The network function type of Charging Function
- GMLC - Gateway Mobile Location Center - The network function type of Gateway Mobile Location Center.
- LMF - Location Management Function - The network function type of Location Management Function.
- NRF - Network Repository Function - The network function type of Network Repository Function.
- NSSAAF - Network Slice Specific Authentication and Authorization Function - The network function type of Network Slice Specific Authentication and Authorization Function
- NSSF - Network Slice Selection Function - The network function type of Network Slice Selection Function.
- NWDAF - Network Data Analytics Function - The network function type of NWDAF.
- PCF - Policy Control Function - The network function type of Policy Control Function.
- SMF - Session Management Function - The network function type of Session Management Function.
- SMSF - Short Message Service Function - The network function type of Short Message Service Function.
- UDM - Unified Data Manager - The network function type of Unified Data Manager.
- UDSF - Unstructured Data Storage Function - The network function type of Unstructured Data Storage Function.

Name

callbackUriAuthorityType - Callback URI Authority Type

Syntax

--callbackUriAuthorityType <list-item>

Description

Defines the format of the authority when the Notification Endpoint is provided to a remote NF. -DEFAULT_IP (default): The authority part of the Endpoint is given as an IP. IP Version aligns to the IP version of the interface via which the Request/Response is received/sent (if available) otherwise whatever version is available. -FQDN: The authority part of the Endpoint given as a FQDN. If --fqdn is not provisioned then authority is given as an IP address. IP Version aligns to the IP version of the interface via which the Request/Response is received/sent (if available) otherwise whatever version is available

Class

Optional.

List

- **DEFAULT_IP_VERSION** - Default IP Version - The authority part of the Endpoint is given as an IP. IP Version aligns to the IP version of the interface via which the Request/Response is received/sent (if available) otherwise whatever version is available.
- **FQDN** - The authority part of the Endpoint given as a FQDN. If FQDN is not provisioned then authority is given as an IP address. IP Version aligns to the IP version of the interface via which the Request/Response is received/sent (if available) otherwise whatever version is available.

Default

DEFAULT_IP_VERSION

Name

dscpCode - DSCP Code

Syntax

--dscpCode <list-item>

Description

Parameter used for selecting DSCP code to use for the IP packet.

Class

Optional.

List

- AF11 - Assured forwarding 11.
- AF12 - Assured forwarding 12.
- AF13 - Assured forwarding 13.
- AF21 - Assured forwarding 21.
- AF22 - Assured forwarding 22.
- AF23 - Assured forwarding 23.
- AF31 - Assured forwarding 31.
- AF32 - Assured forwarding 32.
- AF33 - Assured forwarding 33.
- AF41 - Assured forwarding 41.
- AF42 - Assured forwarding 42.
- AF43 - Assured forwarding 43.
- BE - BE.
- CS1 - Class selector 1.
- CS2 - Class selector 2.

- CS3 - Class selector 3.
- CS4 - Class selector 4.
- CS5 - Class selector 5.
- CS6 - Class selector 6.
- CS7 - Class selector 7.
- EF - Expedited forwarding.
- voiceAdmit - Voice Admit

Default

AF41

Name

fqdn - FQDN

Syntax

--fqdn <string>

Description

The FQDN of the endpoint

Class

Optional.

Type / Range

Digit string of length 0 to 96 with the allowed pattern of [0-9a-zA-Z\._:]*

Default

""

Name

httpProfileName - HTTP profile name

Syntax

--httpProfileName <string>

Description

Assign HTTP profile for the end point.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

default

Name

interPlmnFqdn - Inter PLMN FQDN

Syntax

--interPlmnFqdn <string>

Description

The FQDN of the endpoint used in inter PLMN scenarios

Class

Optional.

Type / Range

Digit string of length 0 to 96 with the allowed pattern of [0-9a-zA-Z\._:]*

Default

""

Name

port - TCP Port

Syntax

--port <integer>

Description

TCP Port.

Class

Optional.

Type / Range

TCP ports 0, 443 and 1024 - 65535

Default

8080

Name

tcpProfileName - TCP Profile Name

Syntax

--tcpProfileName <string>

Description

Configure TCP profile to the endpoint.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

default

Name

tlsProfileName - Transport Layer Security Certificate

Syntax

--tlsProfileName <string>

Description

Configure TLS certificate used in establishing secure communication.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.89 cmm amfNrDualConnectivityProfile

AMF NR Dual Connectivity Profile. Syntax, description, parameters, and roles for cmm amfNrDualConnectivityProfile command.

Syntax**Description**

Command used to administer NR Dual Connectivity profile and NR access restrictions per UE PLMN Services or SUPI Range Services when such a profile is configured.

Parameters**Name**

name - Profile Name

Syntax

--name <string>

Description

Name of the NR Dual Connectivity Profile.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

accessRestrictNrSecRat - Access Restriction for NR Secondary RAT

Syntax

--accessRestrictNrSecRat <list-item>

Description

AMF overrides UE subscription access to NR Secondary RAT.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

accessRestrictSubscribedNrSecRatAbsent - Access Restriction for Subscribed NR Secondary RAT Absent

Syntax

--accessRestrictSubscribedNrSecRatAbsent <list-item>

Description

AMF restricts NR Secondary RAT if Subscribed Secondary RAT Restriction IE is absent.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

includeNrsrna - Include NRSRNA over N26

Syntax

--includeNrsrna <list-item>

Description

AMF includes subscribed UDM NR Access Restriction indication in UE Context over N26 during 5GC to EPC Handover.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

nrDc - NR Dual Connectivity

Syntax

--nrDc <list-item>

Description

Enables or disables NR Dual Connectivity at UE PLMN Services or SUPI Range Services if AMF NR-DC support is enabled and NR Dual Connectivity profile is configured.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

secondaryRatUsageReporting - Secondary RAT Usage Reporting

Syntax

--secondaryRatUsageReporting <list-item>

Description

Enables or disables NR Secondary RAT Usage Reporting at UE PLMN Services or SUPI Range Services if AMF NR-DC support is enabled and NR Dual Connectivity profile is configured.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.90 cmm amfNrfGParms

NRF Global Parameters. Syntax, description, parameters, and roles for cmm amfNrfGParms command.

Syntax**Description**

This command opens the context for configuring NRF related parameters for the network.

Parameters**Name**

amfGparmName - Name of parameter

Syntax

--amfGparmName <list-item>

Description

Name of parameter

Class

Required key.

List

- **alwaysAddPlmnInNfDisc** - Always Include Target Plmn In NF Discoveries - When this parameter is enabled AMF will always include the target-plmn-list Query parameter in NF Discoveries for all non-roaming scenarios
- **amfDiscoveryLimit** - AMF Discovery Limit - Value to be used as the 'limit' Query Parameter in the NFDISCOVERY GET Requests for AMFs to the NRF. This parameter limits the Maximum number of NFProfiles to be returned in the response. When value is 0 (default), the AMF will not include the limit Query Parameter in the Request.
- **ausfDiscoveryLimit** - AUSF Discovery Limit - Value to be used as the 'limit' Query Parameter in the NFDISCOVERY GET Requests for AUSFs to the NRF. This parameter limits the Maximum number of NFProfiles to be returned in the response. When value is 0 (default), the AMF will not include the limit Query Parameter in the Request.
- **cbcfDiscoveryLimit** - CBCF Discovery Limit - Value to be used as the 'limit' Query Parameter in the NFDISCOVERY GET Requests for CBCFs to the NRF. This parameter limits the Maximum number of NFProfiles to be returned in the response. When value is 0 (default), the AMF will not include the limit Query Parameter in the Request.
- **checkPgwFqdnMatchInAnySmfInfo** - Check pgwFqdn match in any smfInfo - When this parameter is set to false, all relevant query parameters for SMF discovery must be matched in the same smfInfo IE within the SMF Profile so as to consider the SMF a valid candidate for selection (as per 3GPP specifications). When this parameter is set to true, SMF is considered as a valid candidate even when the pgwFqdn is placed in a different smfInfo IE than the smfInfo which matches all the remaining query parameters.
- **discoveryRateLimit** - Rate Limit for NRF Discoveries - This parameter defines the rate (per second) above which Discovery Messages will be rate limited. When Discovery Requests are load balanced towards multiple NRF (Service) instances the rate is counted separately for each NFDISCOVERY service instance. When this parameter is set to 0 rate limiting is not applied.
- **discoveryRedirectLimit** - Discovery Redirect Limit - When this parameter is set to greater than 0, CMM will perform Redirects.
- **discoveryRetry4xx** - Discovery Retry 4xx - This parameter defines whether NRF Discovery Retry is triggered for 4xx status class Responses.
- **discoveryRetry5xx** - Discovery Retry 5xx - This parameter defines whether NRF Discovery Retry is triggered for 5xx status class Responses.

- **discoveryRetryLimit** - Discovery Retry Limit - When this parameter is set to greater than 0, CMM will perform Retries in case of NRF Discovery errors or timeouts when CMM is provisioned to perform Retry for that type of error.
- **discoveryRetryTimeout** - Discovery Retry Timeout - This parameter defines whether NRF Discovery Retry is triggered for Timeouts.
- **eirDiscoveryLimit** - 5G EIR Discovery Limit - Value to be used as the 'limit' Query Parameter in the NFDISCOVERY GET Requests for 5G-EIRs to the NRF. This parameter limits the Maximum number of NFProfiles to be returned in the response. When value is 0 (default), the AMF will not include the limit Query Parameter in the Request.
- **gmlcDiscoveryLimit** - GMLC Discovery Limit - Value to be used as the 'limit' Query Parameter in the NFDISCOVERY GET Requests for GMLCs to the NRF. This parameter limits the Maximum number of NFProfiles to be returned in the response. When value is 0 (default), the AMF will not include the limit Query Parameter in the Request.
- **includeOauth2RequiredInNfProfile** - Include oauth2required in NfProfile - When this parameter is enabled AMF includes in the NFProfile it registers to NRF information regarding which of its service instances require OAuth2.0. The information is conveyed in the oauth2Required and perPlmnOauth2ReqList attributes of each NFService.
- **interPlmnFqdnNfProfile** - Inter-PLMN FQDN in NF Profile and Service - Support receiving intra-PLMN notification of changes in the NFProfile/NFService containing the interPlmnFqdn attribute
- **lmfDiscoveryLimit** - LMF Discovery Limit - Value to be used as the 'limit' Query Parameter in the NFDISCOVERY GET Requests for LMFs to the NRF. This parameter limits the Maximum number of NFProfiles to be returned in the response. When value is 0 (default), the AMF will not include the limit Query Parameter in the Request.
- **maxPayload** - Max Payload - Defines the Maximum payload size (before compression, if any) of the response, expressed in kiloBytes. When set to value other than zero the relevant Query Parameter will be sent to the NRF which will limit the number of NF profiles returned in the response such as to not exceed the maximum payload size indicated in the request.
- **pcfDiscoveryLimit** - PCF Discovery Limit - Value to be used as the 'limit' Query Parameter in the NFDISCOVERY GET Requests for PCFs to the NRF. This parameter limits the Maximum number of NFProfiles to be returned in the response. When value is 0 (default), the AMF will not include the limit Query Parameter in the Request.
- **prefetchTimerFixedValue** - Prefetch Timer Fixed Value - Prefetch timer fixed value (in sec). Relevant when the NRF NF Profile Prefetch Timer Mode is set to fixed.
- **prefetchTimerMode** - Prefetch Timer Mode - The NRF NFProfile Prefetch timer mode. In the fixed mode, the prefetch timer is set to the value given in the Prefetch Timer Fixed Value parameter and is common for all NFs. In the percent mode, the prefetch timer is calculated per NF as a percent of the Validity timer received in the NF Discovery Response from NRF (Prefetch Timer Percent Value * Validity).

- **prefetchTimerPercentValue** - Prefetch Timer Percent Value - Prefetch timer percent value. Relevant when NRF NF Profile Prefetch Timer Mode is set to percent.
- **sendN2InterfaceAmfInfo** - Send n2InterfaceAmfInfo - When this parameter is set to true AMF will include n2InterfaceAmfInfo in the AmfInfo of the NFProfile it Registers to NRF.
- **sendNfSetIdListToNrf** - Send NfSetIdList to NRF - When this parameter is set to true, AMF includes the NF Sets it belongs to in the nfSetIdList IE of the NFProfile it Registers in NRF.
- **sendOnlyEndpointFqdnInRegistration** - Send Only Endpoint Fqdn In Registration - When this parameter is set to True, the AMF omits the IPs and includes only the FQDN in the Endpoints of the NFServices and defaultNotificationSubscriptions in the Profile Registered to NRF.
- **sendRequesterExtendedSnssais** - Send Requester Extended S-NSSAIS - When this parameter is set to true, the AMF supports populating the supported S-NSSAIs in Discovery Queries and Subscriptions Requests issued to NRF as Extended S-NSSAIs that (may) include SD Ranges (ExtSnssai)
- **sendRequesterNfInstanceId** - Send Requester NF Instance ID - When this parameter is set to true CMM includes its own NF Instance ID when sending a Discovery Query or Subscribing to a NRF
- **sendRequesterPlmnInSubscriptions** - Send Requester PLMN in Subscriptions - When this parameter is set to true, CMM includes all Home and Shared PLMNs in the reqPlmnList IE when subscribing to a NRF for a NF Instance in a remote PLMN
- **sendRequesterSpecificSnssaiList** - Send Requester Specific S-NSSAI List - When this parameter is set to true, the AMF will populate the list of supported S-NSSAIs per Serving PLMN in Discovery Queries issued to NRF when the Home PLMN or Serving PLMNs do not all share the same slice list. Additionally when this parameter is set to true, AMF will send the Requester Specific S-NSSAI List in Subscription Requests when the relevant IE is supported by the NRF API version and when the Home PLMN or Serving PLMNs do not all share the same slice list.
- **sendSubscriptionsToQueryNrf** - Send Subscriptions to Query NRF - Controls the selection of the NRF where the NFStatusSubscribe requests are sent for newly discovered NFs. When the value is false the AMF sends the NFStatusSubscribe to the NRF where the AMF is currently Registered When the value is true the AMF sends the NFStatusSubscribe to the NRF used to discover the NF. Default value is false.
- **smfDiscoveryLimit** - SMF Discovery Limit - Value to be used as the 'limit' Query Parameter in the NFDISCOVERY GET Requests for SMFs to the NRF. This parameter limits the Maximum number of NFProfiles to be returned in the response. When value is 0 (default), the AMF will not include the limit Query Parameter in the Request.
- **smsfDiscoveryLimit** - SMSF Discovery Limit - Value to be used as the 'limit' Query Parameter in the NFDISCOVERY GET Requests for SMSFs to the NRF. This parameter limits the Maximum number of NFProfiles to be returned in the response. When value is 0 (default), the AMF will not include the limit Query Parameter in the Request.

- **subscriptionRateLimit** - Rate Limit for NRF Subscriptions - This parameter defines the rate (per second) above which NRF Subscriptions will be rate limited. Heartbeats and Registrations of the NRF Management API will never be rate limited. When this parameter is set to 0 rate limiting is not applied for Subscription Requests.
- **supportPerPlmnSnssaiListInNfProfiles** - Support per PLMN S-NSSAI List in NF Profiles - When this parameter is set to true, CMM will evaluate the per PLMN S-NSSAI List IE in NF Profile received from NRF during NF Discovery operations. Setting this parameter to false does not block CMM from populating per PLMN S-NSSAI List in its own NF Profile sent to NRF during Registration.
- **supportPerPlmnSnssaiListInSmfServices** - Support per PLMN S-NSSAI List in SMF Services - When this parameter is set to true, CMM will evaluate the per PLMN S-NSSAI List IE in NF Services received from NRF during NF Discovery operations for SMF. This parameter can be activated only when Supporting per PLMN S-NSSAI List on Profile level is activated.
- **suppressIdenticalDiscRequests** - Suppress Identical Discovery Requests - When this parameter is enabled AMF will avoid sending the exact same Discovery Request to NRF multiple times if one is issued and response has not yet been received.
- **udmDiscoveryLimit** - UDM Discovery Limit - Value to be used as the 'limit' Query Parameter in the NFDISCOVERY GET Requests for UDM to the NRF. This parameter limits the Maximum number of NFProfiles to be returned in the response. When value is 0 (default), the AMF will not include the limit Query Parameter in the Request.
- **useExpiredNfProfiles** - Use Expired NF Profiles - When this parameter is set to true, in case CMM failed to find a matching NF profile from the local cache and the NRF Discovery failed, will search whether an expired cached NF Profile matches the Discovery criterions prior to falling back to locally provisioned NFs.
- **useL1L2NrfCache** - Use L1L2 Nrf Cache - This parameter activates the use of the L1L2 Discovery Result Cache for NF selection.
- **useQueryCache** - Use Query Cache - When this parameter is enabled CMM caches the discovery Queries that have been performed towards the NRF and avoids repeating the exact same query (i.e. GET with all query parameters/values exactly the same) to the NRF multiple times. The entry with all the query parameters is maintained in the cache for the period indicated validityPeriod IE in the NRF response. This parameter should be enabled when CMM is provisioned to not use cached NF profiles in certain scenarios (via f20104-01). It may be enabled otherwise.
- **useServiceMaps** - Use Service Maps - When this parameter is enabled CMM will indicate to the NRF that it supports identifying the list of NF Service Instances as a map in Discovery and Subscription Requests, and will Register to NRF its own NF Profile populating its services as a map of NF Service Instances.

Name

amfGparmValue - Value of parameter

Syntax`--amfGparmValue <string>`**Description**

Value of parameter

Class

Required on create.

Type / Range

Digit string of length 0 to 64

The amfGparmValue string is based on the amfGparmName parameter:

Name

alwaysAddPlmnInNfDisc - Always Include Target Plmn In NF Discoveries

Syntax`--amfGparmName alwaysAddPlmnInNfDisc --amfGparmValue <string>`**Description**

When this parameter is enabled AMF will always include the target-plmn-list Query parameter in NF Discoveries for all non-roaming scenarios

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

amfDiscoveryLimit - AMF Discovery Limit

Syntax`--amfGparmName amfDiscoveryLimit --amfGparmValue <string>`**Description**

Value to be used as the 'limit' Query Parameter in the NFDISCOVERY GET Requests for AMFs to the NRF. This parameter limits the Maximum number of NFProfiles to be returned in the response. When value is 0 (default), the AMF will not include the limit Query Parameter in the Request.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

ausfDiscoveryLimit - AUSF Discovery Limit

Syntax

```
--amfGparmName ausfDiscoveryLimit --amfGparmValue <string>
```

Description

Value to be used as the 'limit' Query Parameter in the NFDISCOVERY GET Requests for AUSFs to the NRF. This parameter limits the Maximum number of NFProfiles to be returned in the response. When value is 0 (default), the AMF will not include the limit Query Parameter in the Request.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

cbcfDiscoveryLimit - CBCF Discovery Limit

Syntax

```
--amfGparmName cbcfDiscoveryLimit --amfGparmValue <string>
```

Description

Value to be used as the 'limit' Query Parameter in the NFDISCOVERY GET Requests for CBCFs to the NRF. This parameter limits the Maximum number of NFProfiles to be returned in the response. When value is 0 (default), the AMF will not include the limit Query Parameter in the Request.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

checkPgwfQdnMatchInAnySmfInfo - Check pgwfQdn match in any smfInfo

Syntax

```
--amfGparmName checkPgwfQdnMatchInAnySmfInfo --amfGparmValue <string>
```

Description

When this parameter is set to false, all relevant query parameters for SMF discovery must be matched in the same smfInfo IE within the SMF Profile so as to consider the SMF a valid candidate for selection (as per 3GPP specifications). When this parameter is set to true, SMF is considered as a valid candidate even when the pgwfQdn is placed in a different smfInfo IE than the smfInfo which matches all the remaining query parameters.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Name

discoveryRateLimit - Rate Limit for NRF Discoveries

Syntax

```
--amfGparmName discoveryRateLimit --amfGparmValue <string>
```

Description

This parameter defines the rate (per second) above which Discovery Messages will be rate limited. When Discovery Requests are load balanced towards multiple NRF (Service) instances the rate is counted separately for each NFDISCOVERY service instance. When this parameter is set to 0 rate limiting is not applied.

Type / Range

Limit above which requests will be rate limited in units of Second

Default

0

Name

discoveryRedirectLimit - Discovery Redirect Limit

Syntax

```
--amfGparmName discoveryRedirectLimit --amfGparmValue <string>
```

Description

When this parameter is set to greater than 0, CMM will perform Redirects.

Type / Range

Integer in the range of 0 to 1

Default

0

Name

discoveryRetry4xx - Discovery Retry 4xx

Syntax

```
--amfGparmName discoveryRetry4xx --amfGparmValue <string>
```

Description

This parameter defines whether NRF Discovery Retry is triggered for 4xx status class Responses.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

discoveryRetry5xx - Discovery Retry 5xx

Syntax

--amfGparmName discoveryRetry5xx --amfGparmValue <string>

Description

This parameter defines whether NRF Discovery Retry is triggered for 5xx status class Responses.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

discoveryRetryLimit - Discovery Retry Limit

Syntax

--amfGparmName discoveryRetryLimit --amfGparmValue <string>

Description

When this parameter is set to greater than 0, CMM will perform Retries in case of NRF Discovery errors or timeouts when CMM is provisioned to perform Retry for that type of error.

Type / Range

Integer in the range of 0 to 2

Default

0

Name

discoveryRetryTimeout - Discovery Retry Timeout

Syntax

--amfGparmName discoveryRetryTimeout --amfGparmValue <string>

Description

This parameter defines whether NRF Discovery Retry is triggered for Timeouts.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

eirDiscoveryLimit - 5G EIR Discovery Limit

Syntax

--amfGparmName eirDiscoveryLimit --amfGparmValue <string>

Description

Value to be used as the 'limit' Query Parameter in the NFDISCOVERY GET Requests for 5G-EIRs to the NRF. This parameter limits the Maximum number of NFProfiles to be returned in the response. When value is 0 (default), the AMF will not include the limit Query Parameter in the Request.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

gmlcDiscoveryLimit - GMLC Discovery Limit

Syntax

--amfGparmName gmlcDiscoveryLimit --amfGparmValue <string>

Description

Value to be used as the 'limit' Query Parameter in the NFDISCOVERY GET Requests for GMLCsto the NRF. This parameter limits the Maximum number of NFProfiles to be returned in the response. When value is 0 (default), the AMF will not include the limit Query Parameter in the Request.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

includeOauth2RequiredInNfProfile - Include oauth2required in NfProfile

Syntax

--amfGparmName includeOauth2RequiredInNfProfile --amfGparmValue
<string>

Description

When this parameter is enabled AMF includes in the NFProfile it registers to NRF information regarding which of its service instances require OAuth2.0. The information is conveyed in the oauth2Required and perPlmnOauth2ReqList attributes of each NFService.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Recommended

Should be set to true if OAuth2.0 is enforced in CMM as a Producer.

Name

interPlmnFqdnNfProfile - Inter-PLMN FQDN in NF Profile and Service

Syntax

```
--amfGparmName interPlmnFqdnNfProfile --amfGparmValue <string>
```

Description

Support receiving intra-PLMN notification of changes in the NFProfile/NFService containing the interPlmnFqdn attribute

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Recommended

true

Name

lmfDiscoveryLimit - LMF Discovery Limit

Syntax

```
--amfGparmName lmfDiscoveryLimit --amfGparmValue <string>
```

Description

Value to be used as the 'limit' Query Parameter in the NFDISCOVERY GET Requests for LMFs to the NRF. This parameter limits the Maximum number of NFProfiles to be returned in the response. When value is 0 (default), the AMF will not include the limit Query Parameter in the Request.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

maxPayload - Max Payload

Syntax

```
--amfGparmName maxPayload --amfGparmValue <string>
```

Description

Defines the Maximum payload size (before compression, if any) of the response, expressed in kiloBytes. When set to value other than zero the relevant Query Parameter will be sent to the NRF which will limit the number of NF profiles returned in the response such as to not exceed the maximum payload size indicated in the request.

Type / Range

Integer in the range of 0 or 64 to 2,000

Default

0

Name

pcfDiscoveryLimit - PCF Discovery Limit

Syntax

```
--amfGparmName pcfDiscoveryLimit --amfGparmValue <string>
```

Description

Value to be used as the 'limit' Query Parameter in the NFDISCOVERY GET Requests for PCFs to the NRF. This parameter limits the Maximum number of NFProfiles to be returned in the response. When value is 0 (default), the AMF will not include the limit Query Parameter in the Request.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

prefetchTimerFixedValue - Prefetch Timer Fixed Value

Syntax

```
--amfGparmName prefetchTimerFixedValue --amfGparmValue <string>
```

Description

Prefetch timer fixed value (in sec). Relevant when the NRF NF Profile Prefetch Timer Mode is set to fixed.

Type / Range

Integer in the range of 0 to 300

Default

16

Name

prefetchTimerMode - Prefetch Timer Mode

Syntax

--amfGparmName prefetchTimerMode --amfGparmValue <string>

Description

The NRF NFProfile Prefetch timer mode. In the fixed mode, the prefetch timer is set to the value given in the Prefetch Timer Fixed Value parameter and is common for all NFs. In the percent mode, the prefetch timer is calculated per NF as a percent of the Validity timer received in the NF Discovery Response from NRF (Prefetch Timer Percent Value * Validity).

List

- fixed - Prefetch timer is Fixed and common for all NFs.
- percent - Prefetch timer is calculated per NF as a percent of the validity timer received in the discovery response.

Default

fixed

Name

prefetchTimerPercentValue - Prefetch Timer Percent Value

Syntax

--amfGparmName prefetchTimerPercentValue --amfGparmValue <string>

Description

Prefetch timer percent value. Relevant when NRF NF Profile Prefetch Timer Mode is set to percent.

Type / Range

Integer in the range of 0 to 50

Default

10

Name

sendN2InterfaceAmfInfo - Send n2InterfaceAmfInfo

Syntax

--amfGparmName sendN2InterfaceAmfInfo --amfGparmValue <string>

Description

When this parameter is set to true AMF will include n2InterfaceAmfInfo in the AmfInfo of the NFProfile it Registers to NRF.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sendNfSetIdListToNrf - Send NfSetIdList to NRF

Syntax

--amfGparamName sendNfSetIdListToNrf --amfGparamValue <string>

Description

When this parameter is set to true, AMF includes the NF Sets it belongs to in the nfSetIdList IE of the NFProfile it Registers in NRF.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sendOnlyEndpointFqdnInRegistration - Send Only Endpoint Fqdn In Registration

Syntax

--amfGparamName sendOnlyEndpointFqdnInRegistration --amfGparamValue <string>

Description

When this parameter is set to True, the AMF omits the IPs and includes only the FQDN in the Endpoints of the NFServices and defaultNotificationSubscriptions in the Profile Registered to NRF.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sendRequesterExtendedSnssais - Send Requester Extended S-NSSAIS

Syntax

--amfGparamName sendRequesterExtendedSnssais --amfGparamValue <string>

Description

When this parameter is set to true, the AMF supports populating the supported S-NSSAIs in Discovery Queries and Subscriptions Requests issued to NRF as Extended S-NSSAIs that (may) include SD Ranges (ExtSnssai)

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sendRequesterNfInstanceId - Send Requester NF Instance ID

Syntax

--amfGparmName sendRequesterNfInstanceId --amfGparmValue <string>

Description

When this parameter is set to true CMM includes its own NF Instance ID when sending a Discovery Query or Subscribing to a NRF

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sendRequesterPlmnInSubscriptions - Send Requester PLMN in Subscriptions

Syntax

--amfGparmName sendRequesterPlmnInSubscriptions --amfGparmValue
<string>

Description

When this parameter is set to true, CMM includes all Home and Shared PLMNs in the reqPlmnList IE when subscribing to a NRF for a NF Instance in a remote PLMN

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sendRequesterSpecificSnssaiList - Send Requester Specific S-NSSAI List

Syntax

--amfGparmName sendRequesterSpecificSnssaiList --amfGparmValue
<string>

Description

When this parameter is set to true, the AMF will populate the list of supported S-NSSAIs per Serving PLMN in Discovery Queries issued to NRF when the Home PLMN or Serving PLMNs do not all share the same slice list. Additionally when this parameter is set to true, AMF will send the Requester Specific S-NSSAI List in Subscription Requests when the relevant IE is supported by the NRF API version and when the Home PLMN or Serving PLMNs do not all share the same slice list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

sendSubscriptionsToQueryNrf - Send Subscriptions to Query NRF

Syntax

```
--amfGparamName sendSubscriptionsToQueryNrf --amfGparamValue <string>
```

Description

Controls the selection of the NRF where the NFStatusSubscribe requests are sent for newly discovered NFs. When the value is false the AMF sends the NFStatusSubscribe to the NRF where the AMF is currently Registered When the value is true the AMF sends the NFStatusSubscribe to the NRF used to discover the NF. Default value is false.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

smfDiscoveryLimit - SMF Discovery Limit

Syntax

```
--amfGparamName smfDiscoveryLimit --amfGparamValue <string>
```

Description

Value to be used as the 'limit' Query Parameter in the NFDISCOVERY GET Requests for SMFs to the NRF. This parameter limits the Maximum number of NFProfiles to be returned in the response. When value is 0 (default), the AMF will not include the limit Query Parameter in the Request.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

smsfDiscoveryLimit - SMSF Discovery Limit

Syntax`--amfGparmName smsfDiscoveryLimit --amfGparmValue <string>`**Description**

Value to be used as the 'limit' Query Parameter in the NFDISCOVERY GET Requests for SMSFs to the NRF. This parameter limits the Maximum number of NFProfiles to be returned in the response. When value is 0 (default), the AMF will not include the limit Query Parameter in the Request.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

subscriptionRateLimit - Rate Limit for NRF Subscriptions

Syntax`--amfGparmName subscriptionRateLimit --amfGparmValue <string>`**Description**

This parameter defines the rate (per second) above which NRF Subscriptions will be rate limited. Heartbeats and Registrations of the NRF Management API will never be rate limited. When this parameter is set to 0 rate limiting is not applied for Subscription Requests.

Type / Range

Limit above which requests will be rate limited in units of Second

Default

0

Name

supportPerPlmnSnssaiListInNfProfiles - Support per PLMN S-NSSAI List in NF Profiles

Syntax`--amfGparmName supportPerPlmnSnssaiListInNfProfiles --amfGparmValue <string>`**Description**

When this parameter is set to true, CMM will evaluate the per PLMN S-NSSAI List IE in NF Profile received from NRF during NF Discovery operations. Setting this this parameter to false does not block CMM from populating per PLMN S-NSSAI List in its own NF Profile sent to NRF during Registration.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

supportPerPlmnSnssaiListInSmfServices - Support per PLMN S-NSSAI List in SMF Services

Syntax

```
--amfGparmName supportPerPlmnSnssaiListInSmfServices --amfGparmValue <string>
```

Description

When this parameter is set to true, CMM will evaluate the per PLMN S-NSSAI List IE in NF Services received from NRF during NF Discovery operations for SMF. This parameter can be activated only when Supporting per PLMN S-NSSAI List on Profile level is activated.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

suppressIdenticalDiscRequests - Suppress Identical Discovery Requests

Syntax

```
--amfGparmName suppressIdenticalDiscRequests --amfGparmValue <string>
```

Description

When this parameter is enabled AMF will avoid sending the exact same Discovery Request to NRF multiple times if one is issued and response has not yet been received.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

udmDiscoveryLimit - UDM Discovery Limit

Syntax

```
--amfGparmName udmDiscoveryLimit --amfGparmValue <string>
```

Description

Value to be used as the 'limit' Query Parameter in the NFDISCOVERY GET Requests for UDM to the NRF. This parameter limits the Maximum number of NFProfiles to be returned in the response. When value is 0 (default), the AMF will not include the limit Query Parameter in the Request.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

useExpiredNfProfiles - Use Expired NF Profiles

Syntax

```
--amfGparmName useExpiredNfProfiles --amfGparmValue <string>
```

Description

When this parameter is set to true, in case CMM failed to find a matching NF profile from the local cache and the NRF Discovery failed, will search whether an expired cached NF Profile matches the Discovery criterions prior to falling back to locally provisioned NFs.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

useL1L2NrfCache - Use L1L2 Nrf Cache

Syntax

```
--amfGparmName useL1L2NrfCache --amfGparmValue <string>
```

Description

This parameter activates the use of the L1L2 Discovery Result Cache for NF selection.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

useServiceMaps - Use Service Maps

Syntax

```
--amfGparmName useServiceMaps --amfGparmValue <string>
```

Description

When this parameter is enabled CMM will indicate to the NRF that it supports identifying the list of NF Service Instances as a map in Discovery and Subscription Requests, and will Register to NRF its own NF Profile populating its services as a map of NF Service Instances.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

3.91 cmm amfNrfHttp2ToNasCcMapping

NRF HTTP2 to NAS Cause Codes Mapping. Syntax, description, parameters, and roles for cmm amfNrfHttp2ToNasCcMapping command.

Syntax**Description**

Defines NRF HTTP2 Status Codes Mapping to AMF NAS Cause Codes. The maximum number of profiles supported is 128. A default profile, named 'default', is provided.

Parameters**Name**

amfNrfHttp2ToNasCcMappingListName - NRF HTTP2 to NAS CC mapping list name

Syntax

```
--amfNrfHttp2ToNasCcMappingListName <string>
```

Description

The name of the HTTP2 to 5G NAS Cause Code mapping list to be used for NRF

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

nrfHttp2StatusCode - NRF HTTP2 Status Code

Syntax

--nrfHttp2StatusCode <list-item>

Description

Defines the NRF HTTP2 Status Code to map to 5G NAS Cause Code

Class

Required key.

List

- 200_OK - 200 OK - The NFDDiscover Query is successfully processed by NRF but SearchResult contains no matching NF Profiles.
- 4XX_CLIENT_ERROR - 4xx Client Error - The NFDDiscover Query was rejected by the NRF due to an error in the request
- 5XX_SERVER_ERROR - 5xx Server Error - The NFDDiscover Query was rejected by the NRF as it was unable to process the request due to a Server Error
- DECODING_FAILURE - Decoding Failure - The AMF faces an error when decoding the NRF response.
- LINK_DOWN - Link Down - All TCP connections for the NFDDiscovery Service are unavailable (TCP unable to be setup or Link disabled via link management CLI).
- TIMEOUT - AMF times out when waiting for a response from the NRF.

Name

nfType - NF Type

Syntax

--nfType <list-item>

Description

NF Type that for which the NFDDiscovery Query was sent to the NRF. The Cause code assigned to the NF type 'DEFAULT' is used when no Cause Code is defined for the specific NF Type.

Class

Required key.

List

- 5GEIR - 5G EIR - The network function type of 5G-Equipment Identity Register.
- AMF - Access and Mobility Manager Function - The network function type of Access and Mobility Manager Function.
- AUSF - Authentication Server Function - The network function type of Authentication Server Function.
- NSSF - Network Slice Selection Function - The network function type of Network Slice Selection Function
- SMF - Session Management Function - The network function type of Session Management Function.
- SMSF - Short Message Service Function - The network function type of Short Message Service Function.
- UDM - Unified Data Manager - The network function type of Unified Data Manager.

Name

procedure - Procedure

Syntax

--procedure <list-item>

Description

Defines the procedure to apply the NRF error code into 5GMM cause code mapping.

Class

Required key.

List

- DEFAULT - Default Procedure

Name

amf5gMmCc - 5GMM NAS Cause Code

Syntax

--amf5gMmCc <integer>

Description

Defines the 5GMM NAS cause code to use towards the UE based on either the HTTP2 Status Code received from the NRF, or other error (e.g. timeout). The default value depends on peer NF and nature of the error. For 5G EIR timeout, link down or decoding error, NAS cause code 5 is used by default. For UDM or AUSF timeout, link down or decoding error, NAS cause code 10 is used by default. For SMF 200 OK, NAS cause code 91 is used by default.

Class

Optional.

Type / Range

5GMM Cause Code.

Default

90

Name

useSystemDefaultCC - Use System Default Cause code

Syntax`--useSystemDefaultCC <list-item>`**Description**

This Parameter controls if AMF will use System's default Cause code or the 5GMM Cause Code defined in the amf5gMmCc parameter for the specific combination of Procedure, NfType and HTTP Status Code. Default value is true which means that System's default Cause code will be sent back to the UE.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

3.92 cmm amfNrfHttp2ToNasCcMappingList

NRF HTTP2 to NAS CC Mapping List. Syntax, description, parameters, and roles for cmm amfNrfHttp2ToNasCcMappingList command.

Syntax**Description**

Specifies NRF HTTP2 Status Code to 5GMM NAS Cause Code mapping Profile List

Parameters

Name

name - NRF HTTP2 To NAS CC Mapping List Name

Syntax

--name <string>

Description

NRF HTTP2 Status Code to 5GMM NAS Cause Code mapping Profile List name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.93 cmm amfNrfPreferredParameterOrderProfile

NRF Preferred Parameter Order Profile. Syntax, description, parameters, and roles for cmm amfNrfPreferredParameterOrderProfile command.

Syntax

Description

This command parameter defines the relative order of precedence of the preferred-xxx query parameter when multiple preferred-xxx query parameters are included in the NF discovery and selection. Smaller order value indicates higher precedence. It is not allowed to have multiple preferred-XXX parameters with the same order. A default profile named 'default' always exists and can be modified but not deleted.

Parameters

Name

name - Name

Syntax

--name <string>

Description

Name of the profile

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

preferredFullPlmnOrder - Preferred Full PLMN

Syntax

--preferredFullPlmnOrder <integer>

Description

This parameter defines the order of precedence of the preferred-full-plmn query parameter when multiple preferred-xxx query parameters are included in the NF discovery and selection. Smaller order value indicates higher precedence. It is not allowed to have multiple preferred-XXX parameters with the same order.

Class

Optional.

Type / Range

Precedence order a preferred-xxx query parameter should be evaluated when multiple such query parameters are included for the NF discovery and selection (smaller order value indicates higher precedence).

Default

30

Name

preferredLocalityOrder - Preferred Locality Order

Syntax

--preferredLocalityOrder <integer>

Description

This parameter defines the order of precedence of the preferred-locality query parameter when multiple preferred-xxx query parameters are included in the NF discovery and selection. Smaller order value indicates higher precedence. It is not allowed to have multiple preferred-XXX parameters with the same order.

Class

Optional.

Type / Range

Precedence order a preferred-xxx query parameter should be evaluated when multiple such query parameters are included for the NF discovery and selection (smaller order value indicates higher precedence).

Default

100

Name

preferredNfInstancesOrder - Preferred NF Instances Order

Syntax

--preferredNfInstancesOrder <integer>

Description

This parameter defines the order of precedence the preferred-nf-instances query parameter when multiple preferred-xxx query parameters are included in the NF discovery and selection. Smaller order value indicates higher precedence. It is not allowed to have multiple preferred-XXX parameters with the same order.

Class

Optional.

Type / Range

Precedence order a preferred-xxx query parameter should be evaluated when multiple such query parameters are included for the NF discovery and selection (smaller order value indicates higher precedence).

Default

10

Name

preferredSupiOrder - Preferred SUPI Order

Syntax

--preferredSupiOrder <integer>

Description

This parameter defines the order of precedence of the preferred-supi query parameter when multiple preferred-xxx query parameters are included in the NF discovery and selection. Smaller order value indicates higher precedence. It is not allowed to have multiple preferred-XXX parameters with the same order.

Class

Optional.

Type / Range

Precedence order a preferred-xxx query parameter should be evaluated when multiple such query parameters are included for the NF discovery and selection (smaller order value indicates higher precedence).

Default

20

Name

preferredTaiOrder - Preferred TAI Order

Syntax`--preferredTaiOrder <integer>`**Description**

This parameter defines the order of precedence of the preferred-tai query parameter when multiple preferred-xxx query parameters are included in the NF discovery and selection. Smaller order value indicates higher precedence. It is not allowed to have multiple preferred-XXX parameters with the same order.

Class

Optional.

Type / Range

Precedence order a preferred-xxx query parameter should be evaluated when multiple such query parameters are included for the NF discovery and selection (smaller order value indicates higher precedence).

Default

50

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.94 cmm amfNrRedcapGParms

AMF NR RedCap Global Parameters. Syntax, description, parameters, and roles for cmm amfNrRedcapGParms command.

Syntax

Description

Administer global parameters for NR RedCap.

Parameters

Name

amfGparmName - Parameter name

Syntax

--amfGparmName <list-item>

Description

Name of parameter

Class

Required key.

List

- redcapAsAmfProducer - RedCap As AMF Producer - Defines whether the AMF is to include a redcap RAT, if applicable to a given UE, in communicating as a producer to another core network element when the SBI is R16. Note that RedCap RAT is always included if the SBI version is R17 or greater, regardless of this parameter setting.
- redcapToAmf - RedCap To AMF - Defines whether the AMF is to include a redcap indication, if applicable to a given UE, in communicating to another AMF when the SBI is less than R17. Note that RedCap RAT is always included if the SBI version is R17, regardless of this parameter setting.
- redcapToPcf - RedCap to PCF - Specifies whether the RedCap RAT, if applicable to a given UE, shall be sent to the PCF if an R16 SBI version is used. The default is no. Note that RedCap RAT is always included if the SBI version is R17, regardless of this parameter setting.
- redcapToSmf - RedCap to SMF - Specifies whether the RedCap indication, if applicable to a given UE, is to be sent to an SMF when the SBI is not R17. The default is no. Note that RedCap RAT is always included if the SBI version is R17, regardless of this parameter setting.
- redcapToSmsf - RedCap to SMSF - Defines whether the AMF is to include a RedCap indication, if applicable to a given UE, to the SMSF if the SBI is prior to R17. Note that RedCap RAT is always included if the SBI version is R17, regardless of this parameter setting.
- redcapToUdm - RedCap to UDM - Defines whether the MME is to include a RedCap indication, if applicable to a given UE, to the UDM if the SBI is prior to R17. Note that RedCap RAT is always included if the SBI version is R17, regardless of this parameter setting.

Name

amfGparmValue - Parameter value

Syntax

--amfGparmValue <string>

Description

Value of parameter

Class

Required on create.

Type / Range

Digit string of length 0 to 64

The amfGparmValue string is based on the amfGparmName parameter:

Name

redcapAsAmfProducer - RedCap As AMF Producer

Syntax

--amfGparmName redcapAsAmfProducer --amfGparmValue <string>

Description

Defines whether the AMF is to include a redcap RAT, if applicable to a given UE, in communicating as a producer to another core network element when the SBI is R16. Note that RedCap RAT is always included if the SBI version is R17 or greater, regardless of this parameter setting.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

redcapToAmf - RedCap To AMF

Syntax

--amfGparmName redcapToAmf --amfGparmValue <string>

Description

Defines whether the AMF is to include a redcap indication, if applicable to a given UE, in communicating to another AMF when the SBI is less than R17. Note that RedCap RAT is always included if the SBI version is R17, regardless of this parameter setting.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

redcapToPcf - RedCap to PCF

Syntax

--amfGparmName redcapToPcf --amfGparmValue <string>

Description

Specifies whether the RedCap RAT, if applicable to a given UE, shall be sent to the PCF if an R16 SBI version is used. The default is no. Note that RedCap RAT is always included if the SBI version is R17, regardless of this parameter setting.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

redcapToSmf - RedCap to SMF

Syntax

--amfGparmName redcapToSmf --amfGparmValue <string>

Description

Specifies whether the RedCap indication, if applicable to a given UE, is to be sent to an SMF when the SBI is not R17. The default is no. Note that RedCap RAT is always included if the SBI version is R17, regardless of this parameter setting.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

redcapToSmsf - RedCap to SMSF

Syntax

--amfGparmName redcapToSmsf --amfGparmValue <string>

Description

Defines whether the AMF is to include a RedCap indication, if applicable to a given UE, to the SMSF if the SBI is prior to R17. Note that RedCap RAT is always included if the SBI version is R17, regardless of this parameter setting.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

redcapToUdm - RedCap to UDM

Syntax

--amfGparmName redcapToUdm --amfGparmValue <string>

Description

Defines whether the MME is to include a RedCap indication, if applicable to a given UE, to the UDM if the SBI is prior to R17. Note that RedCap RAT is always included if the SBI version is R17, regardless of this parameter setting.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

3.95 cmm amfNssfHttp2ToNasCcMapping

NSSF HTTP2 To NAS CC mapping. Syntax, description, parameters, and roles for cmm amfNssfHttp2ToNasCcMapping command.

Syntax**Description**

Defines NSSF HTTP2 Status Codes Mapping to AMF NAS Cause Codes.

Parameters

Name

amfNssfHttp2ToNasCcMappingListName - NSSF HTTP2 to NAS CC mapping list name

Syntax

--amfNssfHttp2ToNasCcMappingListName <string>

Description

The name of the NSSF HTTP2 to 5G NAS Cause Code mapping list

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

nssfHttp2StatusCode - NSSF HTTP2 Status Code

Syntax

--nssfHttp2StatusCode <list-item>

Description

NSSF HTTP2 Status Code

Class

Required key.

List

- 403_SNSSAI_NOT_SUPPORTED - 403 SNSSAI NOT SUPPORTED
- OTHER - Used for Status Codes for which mapping for exact Status and Application error is not defined.

Name

procedure - Procedure

Syntax

--procedure <list-item>

Description

Procedure for which the mapping is relevant

Class

Required key.

List

- pduSessionEstablishment - PDU Session Establishment
- ueMobility - UE Mobility

Name

amf5gMmCc - 5GMM NAS Cause Code

Syntax

--amf5gMmCc <integer>

Description

Defines the 5GMM NAS cause code to use towards the UE based on the HTTP2 Status Code received from the NSSF

Class

Optional.

Type / Range

5GMM Cause Code.

Default

62

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

3.96 cmm amfNssfHttp2ToNasCcMappingList

NSSF HTTP2 to NAS CC Mapping List . Syntax, description, parameters, and roles for cmm amfNssfHttp2ToNasCcMappingList command.

Syntax**Description**

Specifies NSSF HTTP2 Status Code to 5GMM NAS Cause Code mapping Profile List

Parameters**Name**

name - NSSF HTTP2 To NAS CC Mapping List Name

Syntax

--name <string>

Description

NSSF HTTP2 Status Code to 5GMM NAS Cause Code mapping Profile List name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.97 cmm amfOlcGParms

AMF OLC. Syntax, description, parameters, and roles for cmm amfOlcGParms command.

Syntax**Description**

This command is used to provision AMF global Overload related parameters

Parameters**Name**

amfGparmName - Parameter Name

Syntax

--amfGparmName <list-item>

Description

Name of parameter

Class

Required key.

List

- **amfOverloadMmRejectionNasCc** - AMF Overload MM Rejection NAS CC - AMM 5GMM Cause Code to be sent in the mobility management reject messages (Registration Reject and Service Reject) due to any AMF overload or throttling of AMM procedures. Valid values are 3,5...7,9...13,15,20...24,26...28,43,65,67,69,71...73,90...92,95...101,111
- **apcmdSheddingDuringOverload** - APCMD Shedding During Overload
- **enableAmfOverloadBackoffTimer** - Enable AMF Overload Backoff Timer - This flag is used to enable the capability of AMF to include a randomly-selected T3346 timer value whenever an initial registration or a service request procedure is rejected due to IPDS CPU overload, CPPS CPU overload, or CXN object overload. no (default) - set the T3346 back-off timer to 0. yes - set the T3346 back-off timer to a randomly-selected value between provisioned minimum and maximum values.
- **memoryAlarmForRestrictedPaging** - Memory Overload Alarm for AMF Restricted Paging - Memory Overload Alarm at which CMM starts AMF Restricted Paging
- **ngapOverloadAction** - NGAP Overload Action - This is the action signaled over NGAP when AMF gets into overload (according to 3GPP 38.413)
- **ocSendOverloadStartAndStopToGnbs** - Overload Control Send Overload Start and Stop Messages to gNBs - A value of true indicates the AMF will send Overload Start and Overload Stop messages to gNBs as part of Overload Control. A value of false indicates the AMF will not send Overload Start and Overload Stop messages.
- **restrictPagingDuringOverload** - Restrict Paging During Overload - Restrict paging during overload.

Name

amfGparmValue - Parameter Value

Syntax

--amfGparmValue <string>

Description

The value of the global parameter

Class

Required on create.

Type / Range

Digit string of length 0 to 64

The amfGparmValue string is based on the amfGparmName parameter:

Name

amfOverloadMmRejectionNasCc - AMF Overload MM Rejection NAS CC

Syntax

--amfGparmName amfOverloadMmRejectionNasCc --amfGparmValue <string>

Description

AMM 5GMM Cause Code to be sent in the mobility management reject messages (Registration Reject and Service Reject) due to any AMF overload or throttling of AMM procedures. Valid values are 3,5...7,9...13,15,20...24,26...28,43,65,67,69,71...73,90...92,95...101,111

Type / Range

5GMM Cause Code.

Default

22

Name

apcmdSheddingDuringOverload - APCMD Shedding During Overload

Syntax

--amfGparmName apcmdSheddingDuringOverload --amfGparmValue <string>

Description

APCMD Shedding During Overload

List

- doNotDropApcmdDuringOverload - Do not drop APCMD during overload
- dropApcmdToNeccDuringMajorOverload - Drop APCMD to NECC during major overload

Default

doNotDropApcmdDuringOverload

Name

enableAmfOverloadBackoffTimer - Enable AMF Overload Backoff Timer

Syntax

--amfGparmName enableAmfOverloadBackoffTimer --amfGparmValue <string>

Description

This flag is used to enable the capability of AMF to include a randomly-selected T3346 timer value whenever an initial registration or a service request procedure is rejected due to IPDS CPU overload, CPPS CPU overload, or CXN object overload. no (default) - set the T3346 back-off timer to 0. yes - set the T3346 back-off timer to a randomly-selected value between provisioned minimum and maximum values.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

memoryAlarmForRestrictedPaging - Memory Overload Alarm for AMF Restricted Paging

Syntax

```
--amfGparmName memoryAlarmForRestrictedPaging --amfGparmValue  
<string>
```

Description

Memory Overload Alarm at which CMM starts AMF Restricted Paging

List

- minor - Minor Memory Alarm
- major - Major Memory Alarm
- critical - Critical Memory Alarm

Default

minor

Name

ngapOverloadAction - NGAP Overload Action

Syntax

```
--amfGparmName ngapOverloadAction --amfGparmValue <string>
```

Description

This is the action signaled over NGAP when AMF gets into overload (according to 3GPP 38.413)

List

- rejRrcForNonEmergMoDt - Reject RRC connection establishments for non-emergency MO DT - Reject RRC connection establishments for non-emergency MO DT (3GPP 38.413)
- rejRrcForSignaling - Reject RRC connection establishments for Signaling - Reject RRC connection establishments for Signaling (3GPP 38.413)
- permitEmergSessAndMtServicesOnly - Permit Emergency Sessions and mobile terminated services only - Permit Emergency Sessions and mobile terminated services only (3GPP 38.413)
- permitHighPrioSessAndMtServicesOnly - Permit High Priority Sessions and mobile terminated services only - Permit High Priority Sessions and mobile terminated services only (3GPP 38.413)

Default

rejRrcForNonEmergMoDt

Name

ocSendOverloadStartAndStopToGnbs - Overload Control Send Overload Start and Stop Messages to gNBs

Syntax


```
--amfGparmName ocSendOverloadStartAndStopToGnbs --amfGparmValue <string>
```

Description

A value of true indicates the AMF will send Overload Start and Overload Stop messages to gNBs as part of Overload Control. A value of false indicates the AMF will not send Overload Start and Overload Stop messages.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

restrictPagingDuringOverload - Restrict Paging During Overload

Syntax

```
--amfGparmName restrictPagingDuringOverload --amfGparmValue <string>
```

Description

Restrict paging during overload.

List

- DISABLE - Disable
- ENABLE - Enable

Default

ENABLE

Roles

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

3.98 cmm amfOlcTimersProfile

AMF Overload Control Timers. Syntax, description, parameters, and roles for cmm amfOlcTimersProfile command.

Syntax

Description

This command is used to provision AMF Overload Control related timers

Parameters

Name

name - AMF OLC Timers Profile Name

Syntax

--name <string>

Description

Name of the profile. Identifies AMF OLC Timers Profile to be used instead of the timer values in amfNasTimers.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

dlNasTransportBackoffTimerMaximum - DL NAS Transport Backoff Timer Maximum

Syntax

--dlNasTransportBackoffTimerMaximum <integer>

Description

Specifies the maximum value to consider when assigning this timer value to be used for a Downlink Nas Transport procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).

Class

Optional.

Type / Range

Integer in the range of 1 to 600

Default

60

Name

dlNasTransportBackoffTimerMinimum - DL NAS Transport Backoff Timer Minimum

Syntax

--dlNasTransportBackoffTimerMinimum <integer>

Description

Specifies the minimum value to consider when assigning this timer value to be used for a Downlink Nas Transport procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).

Class

Optional.

Type / Range

Integer in the range of 1 to 600

Default

15

Name

t3346registerRejectMaximum - T3346 Register Reject Maximum

Syntax

--t3346registerRejectMaximum <integer>

Description

Specifies the maximum value to consider when randomly assigning the T3346 timer value to be used for an register procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion)

Class

Optional.

Type / Range

Integer in the range of 1 to 1800 in units of Second

Default

60

Name

t3346registerRejectMinimum - T3346 Register Reject Minimum

Syntax

--t3346registerRejectMinimum <integer>

Description

Specifies the minimum value to consider when randomly assigning the T3346 timer value to be used for an register procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion)

Class

Optional.

Type / Range

Integer in the range of 1 to 1800 in units of Second

Default

15

Name

t3346srRejectMaximum - T3346 SR Reject Maximum

Syntax

--t3346srRejectMaximum <integer>

Description

Specifies the maximum value to consider when randomly assigning the T3346 timer value to be used for an SR procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).

Class

Optional.

Type / Range

Integer in the range of 1 to 1800 in units of Second

Default

60

Name

t3346srRejectMinimum - T3346 SR Reject Minimum

Syntax

--t3346srRejectMinimum <integer>

Description

Specifies the minimum value to consider when randomly assigning the T3346 timer value to be used for an SR procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).

Class

Optional.

Type / Range

Integer in the range of 1 to 1800 in units of Second

Default

15

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.99 cmm amfPagingMethodOverride

AMF Paging Policy Map. Syntax, description, parameters, and roles for cmm amfPagingMethodOverride command.

Syntax**Description**

Maps an existing paging policy to an override policy based on 5GPaging Type and UE plmn or supi range.

Parameters**Name**

nearestNeighborMethod - Nearest Neighbor Method

Syntax

--nearestNeighborMethod <list-item>

Description

The nearest neighbor paging method that is to be mapped to another paging method at a certain percentage.

Class

Required key.

List

- Level1NearestNeighbor - Level 1 Nearest Neighbor - The set of gNBs immediately adjacent to the UE's last confirmed location.
- Level2NearestNeighbor - Level 2 Nearest Neighbor - The set of gNBs up to two hops away from the UE's last confirmed location.
- Level3NearestNeighbor - Level 3 Nearest Neighbor - The set of gNBs up to three hops away from the UE's last confirmed location.
- AiMIDrivenGnbList - AIML Driven Gnb List - Set of GNBs forecast to be the nearest neighbors of UE's last confirmed location

Name

overridePagingMethod - Override Paging Method

Syntax

--overridePagingMethod <list-item>

Description

The paging method to be used. It is selected at a rate of 100% - percentage.

Class

Required on create.

List

- LastSeenGnb - Last Seen Gnb - The last seen Gnb
- LastSeenGnbList - Last Seen Gnb List
- LastSeenTai - Last Seen TAI - The last seen Tracking Area Identity
- LastSeenTailist - Last Seen TAI List - Last seen tracking area identity neighbor TAI

Name

maxGnbsPaged - Maximum number of gNBs paged

Syntax

--maxGnbsPaged <integer>

Description

Defines the maximum number of gNBs that may be paged for this page attempt. A value of 0 indicates that the default limit of gNBs to be paged applies.

Class

Optional.

Type / Range

Integer in the range of 0 to 30

Default

0

Recommended

10

Name

pctAsHundredths - Percent As Hundredths

Syntax

--pctAsHundredths <list-item>

Description

If set true, the AMF is to interpret the percentage field as hundredths of one percent. The default is false.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Recommended

false

Name

percentage - Percentage

Syntax

--percentage <integer>

Description

The percentage of paging attempts that are to use the specified nearest neighbor paging method. The other paging attempts outside of this percentage will use the specified override paging method.

Class

Optional.

Type / Range

Integer in the range of 0 to 100

Default

0

Recommended

0

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.100 cmm amfPagingPolicy

CMM AMF Paging Policy. Syntax, description, parameters, and roles for cmm amfPagingPolicy command.

Syntax

Description

This command is used to administer a cmm AMF Paging Policy and includes parameters for defining the number of paging attempts, the maximum number of Gnbs paged, paging method, paging type, and paging response timer.

Parameters

Name

plmnName - Mobile network name

Syntax

--plmnName <string>

Description

Name that identifies mobile network.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

amf5GpagingType - AMF 5G Paging type

Syntax

--amf5GpagingType <list-item>

Description

Defines the type of paging to be done, basic, at least one tuple must exist for each paging type.

Class

Required key.

List

- 5QI1 - 5QI-1 - Defines the type of 5QI-1 paging to be done.
- 5QI2 - 5QI-2 - Defines the type of 5QI-2 paging to be done.
- 5QI3 - 5QI-3 - Defines the type of 5QI-3 paging to be done.

- 5QI4 - 5QI-4 - Defines the type of 5QI-1 paging to be done.
- 5QI5 - 5QI-5 - Defines the type of 5QI-1 paging to be done.
- 5QI6 - 5QI-6 - Defines the type of 5QI-1 paging to be done.
- 5QI7 - 5QI-7 - Defines the type of 5QI-7 paging to be done.
- 5QI8 - 5QI-8 - Defines the type of 5QI-8 paging to be done.
- 5QI9 - 5QI-9 - Defines the type of 5QI-9 paging to be done.
- ARP1 - Defines the type of ARP1 paging to be done.
- ARP10 - Defines the type of ARP10 paging to be done.
- ARP11 - Defines the type of ARP11 paging to be done.
- ARP12 - Defines the type of ARP12 paging to be done.
- ARP13 - Defines the type of ARP13 paging to be done.
- ARP14 - Defines the type of ARP14 paging to be done.
- ARP15 - Defines the type of ARP15 paging to be done.
- ARP2 - Defines the type of ARP2 paging to be done.
- ARP3 - Defines the type of ARP3 paging to be done.
- ARP4 - Defines the type of ARP4 paging to be done.
- ARP5 - Defines the type of ARP5 paging to be done.
- ARP6 - Defines the type of ARP6 paging to be done.
- ARP7 - Defines the type of ARP7 paging to be done.
- ARP8 - Defines the type of ARP8 paging to be done.
- ARP9 - Defines the type of ARP9 paging to be done.
- Basic - Defines the type of paging to be done, basic, at least one tuple must exist for each paging type.
- PPI0 - Paging Policy Indication 0 - Paging policy differentiation using Paging Policy Indication value 0.
- PPI1 - Paging Policy Indication 1 - Paging policy differentiation using Paging Policy Indication value 1.
- PPI2 - Paging Policy Indication 2 - Paging policy differentiation using Paging Policy Indication value 2.
- PPI3 - Paging Policy Indication 3 - Paging policy differentiation using Paging Policy Indication value 3.
- PPI4 - Paging Policy Indication 4 - Paging policy differentiation using Paging Policy Indication value 4.

- PPI5 - Paging Policy Indication 5 - Paging policy differentiation using Paging Policy Indication value 5.
- PPI6 - Paging Policy Indication 6 - Paging policy differentiation using Paging Policy Indication value 6.
- PPI7 - Paging Policy Indication 7 - Paging policy differentiation using Paging Policy Indication value 7.
- PtOverride1 - Paging Type Override 1 - The PEIPS feature can map a given paging type (e.g. 5QI5) to this override paging type. If mapped, this paging type can then be used to specify different paging treatment from the mapped (e.g. 5QI5) paging type.
- PtOverride2 - Paging Type Override 2 - The PEIPS feature can map a given paging type (e.g. 5QI5) to this override paging type. If mapped, this paging type can then be used to specify different paging treatment from the mapped (e.g. 5QI5) paging type.
- PtOverride3 - Paging Type Override 3 - The PEIPS feature can map a given paging type (e.g. 5QI5) to this override paging type. If mapped, this paging type can then be used to specify different paging treatment from the mapped (e.g. 5QI5) paging type.
- PtOverride4 - Paging Type Override 4 - The PEIPS feature can map a given paging type (e.g. 5QI5) to this override paging type. If mapped, this paging type can then be used to specify different paging treatment from the mapped (e.g. 5QI5) paging type.
- PtOverride5 - Paging Type Override 5 - The PEIPS feature can map a given paging type (e.g. 5QI5) to this override paging type. If mapped, this paging type can then be used to specify different paging treatment from the mapped (e.g. 5QI5) paging type.
- PtOverride6 - Paging Type Override 6 - The PEIPS feature can map a given paging type (e.g. 5QI5) to this override paging type. If mapped, this paging type can then be used to specify different paging treatment from the mapped (e.g. 5QI5) paging type.
- PtOverride7 - Paging Type Override 7 - The PEIPS feature can map a given paging type (e.g. 5QI5) to this override paging type. If mapped, this paging type can then be used to specify different paging treatment from the mapped (e.g. 5QI5) paging type.
- PtOverride8 - Paging Type Override 8 - The PEIPS feature can map a given paging type (e.g. 5QI5) to this override paging type. If mapped, this paging type can then be used to specify different paging treatment from the mapped (e.g. 5QI5) paging type.
- PtOverride9 - Paging Type Override 9 - The PEIPS feature can map a given paging type (e.g. 5QI5) to this override paging type. If mapped, this paging type can then be used to specify different paging treatment from the mapped (e.g. 5QI5) paging type.
- Restricted - Restricted paging is triggered when OLC determines the CMM is in overload. the restricted paging type is used instead of the normal paging types during an overload condition. the restricted paging type normally has only a single page attempt. The only methods allowed for restricted are LastSeenGnb nad LastSeenGnbList.
- SxnRestoration - Session Restoration Paging - This paging policy is limited to Session Restoration - 1 time only, no second page attempt.

- **currentLocation** - Current Location - The paging is triggered due to consumer NF request for the current location of a CM-IDLE UE: Namf_EventExposure for LOCATION_REPORT or Namf_Location_ProvideLocationInfo.
- **hf1** - For UEs assigned eDRX cycle of one hyperframe (10.24 seconds)
- **hf1024** - Hyperframe 1024 - For UEs assigned eDRX cycle of 1024 hyperframes (10485.76 seconds)
- **hf128** - Hyperframe 128 - For UEs assigned eDRX cycle of 128 hyperframes (1310.72 seconds)
- **hf16** - Hyperframe 16 - For UEs assigned eDRX cycle of 16 hyperframes (163.84 seconds)
- **hf2** - Hyperframe 2 - For UEs assigned eDRX cycle of two hyperframes (20.48 seconds)
- **hf256** - Hyperframe 256 - For UEs assigned eDRX cycle of 256 hyperframes (2621.44 seconds)
- **hf32** - Hyperframe 32 - For UEs assigned eDRX cycle of 32 hyperframes (327.68 seconds)
- **hf4** - Hyperframe 4 - For UEs assigned eDRX cycle of four hyperframes (40.96 seconds)
- **hf512** - Hyperframe 512 - For UEs assigned eDRX cycle of 512 hyperframes (5242.88 seconds)
- **hf64** - Hyperframe 64 - For UEs assigned eDRX cycle of 64 hyperframes (655.36 seconds)
- **hf8** - Hyperframe 8 - For UEs assigned eDRX cycle of 8 hyperframes (81.92 seconds)
- **hfhalf** - Hyperframe Half - For UEs assigned eDRX cycle of a half hyperframe (5.12 seconds)
- **hfquarter** - Hyperframe Quarter - For UEs assigned eDRX cycle of a quarter hyperframe (2.56 seconds)
- **non3gppAccess** - Non-3GPP access - The paging over 3GPP access is triggered by non-3GPP access related signaling.
- **ueReachability** - UE Reachability Request - The paging trigger is Namf_EnableUERachability to make the UE reachable for e.g. MT SMS sending.

Default

Basic

Name

amfAttempt - Attempt

Syntax

--amfAttempt <list-item>

Description

Defines the attempt (1 of 4 possible). attempt numbers must start at 1 and be consecutive (1st attempt, 2nd attempt, and so on).

Class

Required key.

List

- Attempt1 - Attempt 1 - First attempt.
- Attempt2 - Attempt 2 - Second attempt.
- Attempt3 - Attempt 3 - Third attempt.
- Attempt4 - Attempt 4 - 4th attempt.

Name

amf5Gt3513Timer - 5G Paging T3513 timer

Syntax

--amf5Gt3513Timer <integer>

Description

Defines the value for the T3513 timer used for receiving responses to a paging request. When this timer expires, it triggers the AMF to proceed with the next paging attempt as provisioned in the AMF paging policy table. This timer value should be set according to the maximum time it takes a UE to respond to a paging message under normal operating conditions. The T3513 timer interval is in msec. The default timer value varies per the eDRX cycle: Hfquarter 3s; Hfhalf 6s; Hf1 11s; and for all eDRX cycles greater than one hyperframe 2.56s. Timer interval will only have a precision of +/- 100 milliseconds. The CMM timer wheel does not have an accuracy down to the millisecond. This timer also applies for eDRX paging timeout.

Class

Optional.

Type / Range

Integer in the range of 1 to 12,000 in units of Millisecond

Default

6000

Name

amfMethod - Method

Syntax

--amfMethod <list-item>

Description

Method that the designated attempt (1 - 4) should use.

Class

Optional.

List

- LastSeenGnb - Last Seen Gnb - The last seen Gnb
- LastSeenGnbList - Last Seen Gnb List - Last seen Gnb List
- LastSeenTai - Last Seen TAI - The last seen Tracking Area Identity
- LastSeenTaiList - Last Seen TAI List - Last seen tracking area identity neighbor TAI
- Level1NearestNeighbor - Level 1 Nearest Neighbor - The set of gNBs immediately adjacent to the UE's last confirmed location.
- Level2NearestNeighbor - Level 2 Nearest Neighbor - The set of gNBs up to two hops away from the UE's last confirmed location.
- Level3NearestNeighbor - Level 3 Nearest Neighbor - The set of gNBs up to three hops away from the UE's last confirmed location.
- AiMIDrivenGnbList - AIML Driven Gnb List - Set of GNBs forecast to be the nearest neighbors of UE's last confirmed location

Default

LastSeenGnb

Name

maxGnbsPaged - Maximum number of gNBs paged

Syntax

--maxGnbsPaged <integer>

Description

Defines the maximum number of gNBs that may be paged for this page attempt. A value of 0 indicates that the default limit of gNBs to be paged applies. This parameter applies only to the following paging methods: LastSeenGnbList, Level 1 Nearest Neighbor, Level 2 Nearest Neighbor, Level 3 Nearest Neighbor and AiMIDrivenGnbList. For the LastSeenGnbList paging method, a value larger than 5 will be treated as 5.

Class

Optional.

Type / Range

Integer in the range of 0 to 30

Default

0

Name

suppressPagingGap - Suppress Paging Gap

Syntax

--suppressPagingGap <list-item>

Description

Parameter allows the service provider to suppress the AMF Paging Gap functionality.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.101 cmm amfPagingPolicyPeipsMap

AMF Paging Policy PEIPS Map. Syntax, description, parameters, and roles for cmm amfPagingPolicyPeipsMap command.

Syntax**Description**

Applicable when a UE is assigned a paging subgroup per the PEIPS feature, this command allows the AMF to override a Paging Policy with another Policy based on the PEIPS value.

Parameters**Name**

servedPlmnName - PLMN Name

Syntax

--servedPlmnName <string>

Description

Serving PLMN

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

peiSubgroup - PEI Subgroup

Syntax

--peiSubgroup <integer>

Description

The subgroup assigned to a UE per the PEIPS feature.

Class

Required key.

Type / Range

Integer in the range of 0 to 7

Name

pagingTypeMapped - AMF 5G Paging Type

Syntax

--pagingTypeMapped <list-item>

Description

The AMF 5G Paging Type to map. A Paging Override Type is not allowed.

Class

Required key.

List

- 5QI1 - 5QI-1 - Defines the type of 5QI-1 paging to be done.
- 5QI2 - 5QI-2 - Defines the type of 5QI-2 paging to be done.
- 5QI3 - 5QI-3 - Defines the type of 5QI-3 paging to be done.
- 5QI4 - 5QI-4 - Defines the type of 5QI-1 paging to be done.
- 5QI5 - 5QI-5 - Defines the type of 5QI-1 paging to be done.
- 5QI6 - 5QI-6 - Defines the type of 5QI-1 paging to be done.
- 5QI7 - 5QI-7 - Defines the type of 5QI-7 paging to be done.
- 5QI8 - 5QI-8 - Defines the type of 5QI-8 paging to be done.

- 5QI9 - 5QI-9 - Defines the type of 5QI-9 paging to be done.
- ARP1 - Defines the type of ARP1 paging to be done.
- ARP10 - Defines the type of ARP10 paging to be done.
- ARP11 - Defines the type of ARP11 paging to be done.
- ARP12 - Defines the type of ARP12 paging to be done.
- ARP13 - Defines the type of ARP13 paging to be done.
- ARP14 - Defines the type of ARP14 paging to be done.
- ARP15 - Defines the type of ARP15 paging to be done.
- ARP2 - Defines the type of ARP2 paging to be done.
- ARP3 - Defines the type of ARP3 paging to be done.
- ARP4 - Defines the type of ARP4 paging to be done.
- ARP5 - Defines the type of ARP5 paging to be done.
- ARP6 - Defines the type of ARP6 paging to be done.
- ARP7 - Defines the type of ARP7 paging to be done.
- ARP8 - Defines the type of ARP8 paging to be done.
- ARP9 - Defines the type of ARP9 paging to be done.
- Basic - Defines the type of paging to be done, basic, at least one tuple must exist for each paging type.
- PPI0 - Paging Policy Indication 0 - Paging policy differentiation using Paging Policy Indication value 0.
- PPI1 - Paging Policy Indication 1 - Paging policy differentiation using Paging Policy Indication value 1.
- PPI2 - Paging Policy Indication 2 - Paging policy differentiation using Paging Policy Indication value 2.
- PPI3 - Paging Policy Indication 3 - Paging policy differentiation using Paging Policy Indication value 3.
- PPI4 - Paging Policy Indication 4 - Paging policy differentiation using Paging Policy Indication value 4.
- PPI5 - Paging Policy Indication 5 - Paging policy differentiation using Paging Policy Indication value 5.
- PPI6 - Paging Policy Indication 6 - Paging policy differentiation using Paging Policy Indication value 6.
- PPI7 - Paging Policy Indication 7 - Paging policy differentiation using Paging Policy Indication value 7.

- **Restricted** - Restricted paging is triggered when OLC determines the CMM is in overload. the restricted paging type is used instead of the normal paging types during an overload condition. the restricted paging type normally has only a single page attempt. The only methods allowed for restricted are LastSeenGnb nad LastSeenGnbList.
- **SxnRestoration** - Session Restoration Paging - This paging policy is limited to Session Restoration - 1 time only, no second page attempt.
- **currentLocation** - Current Location - The paging is triggered due to consumer NF request for the current location of a CM-IDLE UE: Namf_EventExposure for LOCATION_REPORT or Namf_Location_ProvideLocationInfo.
- **hf1** - For UEs assigned eDRX cycle of one hyperframe (10.24 seconds)
- **hf1024** - Hyperframe 1024 - For UEs assigned eDRX cycle of 1024 hyperframes (10485.76 seconds)
- **hf128** - Hyperframe 128 - For UEs assigned eDRX cycle of 128 hyperframes (1310.72 seconds)
- **hf16** - Hyperframe 16 - For UEs assigned eDRX cycle of 16 hyperframes (163.84 seconds)
- **hf2** - Hyperframe 2 - For UEs assigned eDRX cycle of two hyperframes (20.48 seconds)
- **hf256** - Hyperframe 256 - For UEs assigned eDRX cycle of 256 hyperframes (2621.44 seconds)
- **hf32** - Hyperframe 32 - For UEs assigned eDRX cycle of 32 hyperframes (327.68 seconds)
- **hf4** - Hyperframe 4 - For UEs assigned eDRX cycle of four hyperframes (40.96 seconds)
- **hf512** - Hyperframe 512 - For UEs assigned eDRX cycle of 512 hyperframes (5242.88 seconds)
- **hf64** - Hyperframe 64 - For UEs assigned eDRX cycle of 64 hyperframes (655.36 seconds)
- **hf8** - Hyperframe 8 - For UEs assigned eDRX cycle of 8 hyperframes (81.92 seconds)
- **hfhalf** - Hyperframe Half - For UEs assigned eDRX cycle of a half hyperframe (5.12 seconds)
- **hfquarter** - Hyperframe Quarter - For UEs assigned eDRX cycle of a quarter hyperframe (2.56 seconds)
- **non3gppAccess** - Non-3GPP access - The paging over 3GPP access is triggered by non-3GPP access related signaling.
- **ueReachability** - UE Reachability Request - The paging trigger is Namf_EnableUERachability to make the UE reachable for e.g. MT SMS sending.

Name

peipsPagingType - PEIPS Paging Type

Syntax

--peipsPagingType <list-item>

Description

The override paging type to be used

Class

Required on create.

List

- PtOverride1 - Pagine Type Override 1 - The PEIPS feature can map a given paging type (e.g. 5QI5) to this override paging type. If mapped, this paging type can then be used to specify different paging treatment from the mapped (e.g. 5QI5) paging type.
- PtOverride2 - Pagine Type Override 2 - The PEIPS feature can map a given paging type (e.g. 5QI5) to this override paging type. If mapped, this paging type can then be used to specify different paging treatment from the mapped (e.g. 5QI5) paging type.
- PtOverride3 - Pagine Type Override 3 - The PEIPS feature can map a given paging type (e.g. 5QI5) to this override paging type. If mapped, this paging type can then be used to specify different paging treatment from the mapped (e.g. 5QI5) paging type.
- PtOverride4 - Pagine Type Override 4 - The PEIPS feature can map a given paging type (e.g. 5QI5) to this override paging type. If mapped, this paging type can then be used to specify different paging treatment from the mapped (e.g. 5QI5) paging type.
- PtOverride5 - Pagine Type Override 5 - The PEIPS feature can map a given paging type (e.g. 5QI5) to this override paging type. If mapped, this paging type can then be used to specify different paging treatment from the mapped (e.g. 5QI5) paging type.
- PtOverride6 - Pagine Type Override 6 - The PEIPS feature can map a given paging type (e.g. 5QI5) to this override paging type. If mapped, this paging type can then be used to specify different paging treatment from the mapped (e.g. 5QI5) paging type.
- PtOverride7 - Pagine Type Override 7 - The PEIPS feature can map a given paging type (e.g. 5QI5) to this override paging type. If mapped, this paging type can then be used to specify different paging treatment from the mapped (e.g. 5QI5) paging type.
- PtOverride8 - Pagine Type Override 8 - The PEIPS feature can map a given paging type (e.g. 5QI5) to this override paging type. If mapped, this paging type can then be used to specify different paging treatment from the mapped (e.g. 5QI5) paging type.
- PtOverride9 - Pagine Type Override 9 - The PEIPS feature can map a given paging type (e.g. 5QI5) to this override paging type. If mapped, this paging type can then be used to specify different paging treatment from the mapped (e.g. 5QI5) paging type.

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.102 cmm amfPagingPriProfile

AMF Paging Priority Profile. Syntax, description, parameters, and roles for cmm amfPagingPriProfile command.

Syntax

Description

This command is used to administer a Paging Priority Profile and includes parameters for defining the ARP and paging priority levels

Parameters

Name

pagingPriProfileListName - Paging priority profile name

Syntax

--pagingPriProfileListName <string>

Description

Name that identifies paging priority profile table (alternate key).

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

arpLevel - Allocation Retention Priority level

Syntax

--arpLevel <integer>

Description

Allocation Retention Priority (ARP) priority level related to paging priority profile on AMF. Defines the relative importance of a resource request. Values are ordered in decreasing order of priority, i.e., with 1 as the highest priority and 15 as the lowest priority.

Class

Required key.

Type / Range

Integer in the range of 1 to 15

Name

includePagingPriority - Include Paging Priority

Syntax

```
--includePagingPriority <list-item>
```

Description

Defines whether the AMF includes the provisioned Paging Priority on the NGAP Paging message (when set to true) or suppresses it (when set to false).

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

pagingPriLevel - Paging priority level

Syntax

```
--pagingPriLevel <list-item>
```

Description

Paging priority level

Class

Optional.

List

- PrioLevel0 - Priority level 0 - Priority level 0. No paging priority is set.
- PrioLevel1 - Priority level 1 - Priority level 1. Highest Paging priority.
- PrioLevel2 - Priority level 2 - Priority level 2.
- PrioLevel3 - Priority level 3 - Priority level 3.
- PrioLevel4 - Priority level 4 - Priority level 4.

- PrioLevel5 - Priority level 5 - Priority level 5.
- PrioLevel6 - Priority level 6 - Priority level 6.
- PrioLevel7 - Priority level 7 - Priority level 7.
- PrioLevel8 - Priority level 8 - Priority level 8. Lowest Paging Priority.

Default

PrioLevel0

Name

sbiMessagePriorityForArpPl - SBI Message Priority for ARP PL

Syntax

--sbiMessagePriorityForArpPl <integer>

Description

Defines the SMP value that is used with the related ARP PL. Applies when global parameter Multilevel MPS is set to true and multiple MPS priorities are applied on AMF.

Class

Optional.

Type / Range

Integer in the range of 0 to 23

Default

23

Roles**Create**

secadmin

Show

sysadmin, manage, secadmin

Modify

secadmin

Delete

secadmin

List

sysadmin, manage, secadmin

3.103 cmm amfPagingPriProfileList

Paging Priority Profile List. Syntax, description, parameters, and roles for cmm amfPagingPriProfileList command.

Syntax

Description

This command is used to administer a Paging Priority Profile List.

Parameters

Name

name - Paging priority profile name

Syntax

--name <digits>

Description

Defines the paging priority profile name

Class

Required key.

Type / Range

Digit string of length 1 to 1 with the allowed pattern of [1-8]*

Roles

Create

secadmin

Show

sysadmin, manage, secadmin

Delete

secadmin

List

sysadmin, manage, secadmin

3.104 cmm amfPcfTriggerProfile

AMF PCF Policy Trigger Profile. Syntax, description, parameters, and roles for cmm amfPcfTriggerProfile command.

Syntax

Description

The profile to enable AM and UE policy triggers from PCF. A maximum of 16 profiles is supported. The 'default' profile always exists and cannot be deleted.

Parameters

Name

name - Profile Name

Syntax

--name <string>

Description

Profile name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

amPolicyAccessTypeChange - AM Policy Access Type Change

Syntax

--amPolicyAccessTypeChange <list-item>

Description

Defines whether the PCF can trigger AM policy for Access Type Change towards the AMF

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

amPolicyImmediateReport - AM Policy Immediate Report

Syntax

--amPolicyImmediateReport <list-item>

Description

Enables the AMF support for the immediateReport feature for policy update notification for AM policy.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

amPolicyMultipleAccessTypes - AM Policy Multiple Access Types

Syntax

--amPolicyMultipleAccessTypes <list-item>

Description

Enables the AMF support for the 'multipleAccessTypes' feature for policy update notification for AM policy.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

amPolicyPra - AM Policy Change of UE presence in PRA

Syntax

--amPolicyPra <list-item>

Description

Defines whether the PCF can trigger AM policy for Change of UE presence in PRA towards the AMF.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

pageUeOnPraPolicyNotification - Page UE on PRA Policy Notification

Syntax


```
--pageUeOnPraPolicyNotification <list-item>
```

Description

Indicates whether the AMF triggers paging when it receives UE/AM policy notification with PRA(s). When set to false, AMF waits for next UE activity to analyse whether the UE is in or out of the requested presence reporting area(s).

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

uePolicyPlmnChange - UE Policy PLMN Change

Syntax

```
--uePolicyPlmnChange <list-item>
```

Description

Defines whether the PCF can trigger UE policy for PLMN Change towards the AMF

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

uePolicyPra - UE Policy Change of UE presence in PRA

Syntax

```
--uePolicyPra <list-item>
```

Description

Defines whether the PCF can trigger UE policy for Change of UE presence in PRA towards the AMF.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.105 cmm amfPcmdConfig

AMF PCMD Configuration. Syntax, description, parameters, and roles for cmm amfPcmdConfig command.

Syntax**Description**

This command is used to administer the AMF PCMD configuration.

Parameters**Name**

enableHttpProxyDetails - Enable HTTP Proxy Details

Syntax

--enableHttpProxyDetails <list-item>

Description

Defines whether or not HTTP Proxy Details are included in the PCMD records

Class

Optional.

List

- never - Never store HTTP Proxy details in PCMD data.
- onError - On error - Store HTTP Proxy details in PCMD data only in case of errors.

Default

never

Name

enableSmsReport - Enable SMS Report

Syntax

--enableSmsReport <list-item>

Description

Defines whether or not SMSF interactions are included in the PCMD record

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

enableSubscribeReport - Enable Subscribe Report

Syntax

--enableSubscribeReport <list-item>

Description

Defines whether or not Subscribe/Unsubscribe interactions are included in the PCMD record

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

include5qi - Include 5QI

Syntax

```
--include5qi <list-item>
```

Description

When this is set, the 5QI SM parameter shall be included within the AMF PCMD record.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

includeRequestedDnn - Include Requested DNN

Syntax

```
--includeRequestedDnn <list-item>
```

Description

This parameter controls when to include Requested DNN at AMF PCMD record

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

3.106 cmm amfPduDeletionProfile

AMF PDU Deletion Profile. Syntax, description, parameters, and roles for cmm amfPduDeletionProfile command.

Syntax

Description

Defines options for AMF triggered signalling when PDU session deletion of an existing PDU session needs to be triggered by the AMF. The maximum number of profiles supported is 128. A default profile, named 'default', is provided.

Parameters

Name

name - Name

Syntax

--name <string>

Description

The name of the AMF PDU Deletion Profile.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

causeToSmfDuplicatePdu - Cause to SMF Duplicate PDU

Syntax

--causeToSmfDuplicatePdu <list-item>

Description

The cause on SM context update with release indication to release a PDU session due to duplicate PDU Session.

Class

Optional.

List

- relDueToHo - Release Due to HO
- epsFallback - EPS Fallback
- relDueToUpSec - Release Due to User Plane Security
- dnnCongestion - DNN Congestion
- snssaiCongestion - S-NSSAI Congestion
- relDueToReactivation - Release Due to Reactivation

- 5gAnNotResponding - 5G AN Not Responding
- relDueToSliceNotAvailable - Release Due to Slice Not Available
- relDueToDuplicateSessionId - Release Due to Duplicate Session Id
- pduSessionStatusMismatch - PDU Session Status Mismatch
- hoFailure - HO Failure
- insufficientUpResources - Insufficient UP Resources
- pduSessionHandedOver - PDU Session Handed Over
- pduSessionResumed - PDU Session Resumed
- cnAssistedRanParameterTuning - CN Assisted RAN Parameter Tuning
- ismfContextTransfer - I-SMF Context Transfer
- smfContextTransfer - SMF Context Transfer
- relDueToPsToCsHo - Release Due to PS to CS HO
- relDueToSubscriptionChange - Release Due to Subscription Change
- hoCancel - HO Cancel
- relDueToSliceNotAuthorized - Release Due to Slice Not Authorized
- pduSessionHandoverFailure - PDU Session Handover Failure
- ddnFailureStatus - DDN Failure Status
- relDueToCpOnlyNotApplicable - Release Due to CP Only Not Applicable
- notSupportedWithIsmf - Not supported with I-SMF
- changedAnchorSmf - Changed Anchor SMF
- changedIntermediateSmf - Changed Intermediate SMF
- noCause - No Cause - No cause provided to SMF.

Default

relDueToDuplicateSessionId

Name

causeToSmfHoToCs - Cause to SMF HO To CS

Syntax

--causeToSmfHoToCs <list-item>

Description

The cause on SM context release after 5G-SRVCC Handover to CS domain has been completed.

Class

Optional.

List

- relDueToHo - Release Due to HO
- epsFallback - EPS Fallback
- relDueToUpSec - Release Due to User Plane Security
- dnnCongestion - DNN Congestion
- snssaiCongestion - S-NSSAI Congestion
- relDueToReactivation - Release Due to Reactivation
- 5gAnNotResponding - 5G AN Not Responding
- relDueToSliceNotAvailable - Release Due to Slice Not Available
- relDueToDuplicateSessionId - Release Due to Duplicate Session Id
- pduSessionStatusMismatch - PDU Session Status Mismatch
- hoFailure - HO Failure
- insufficientUpResources - Insufficient UP Resources
- pduSessionHandedOver - PDU Session Handed Over
- pduSessionResumed - PDU Session Resumed
- cnAssistedRanParameterTuning - CN Assisted RAN Parameter Tuning
- ismfContextTransfer - I-SMF Context Transfer
- smfContextTransfer - SMF Context Transfer
- relDueToPsToCsHo - Release Due to PS to CS HO
- relDueToSubscriptionChange - Release Due to Subscription Change
- hoCancel - HO Cancel
- relDueToSliceNotAuthorized - Release Due to Slice Not Authorized
- pduSessionHandoverFailure - PDU Session Handover Failure
- ddnFailureStatus - DDN Failure Status
- relDueToCpOnlyNotApplicable - Release Due to CP Only Not Applicable
- notSupportedWithIsmf - Not supported with I-SMF
- changedAnchorSmf - Changed Anchor SMF
- changedIntermediateSmf - Changed Intermediate SMF
- noCause - No Cause - No cause provided to SMF.

Default

relDueToPsToCsHo

Name

causeToSmfLbo - Cause to SMF LBO

Syntax

--causeToSmfLbo <list-item>

Description

The cause on SM context update with release indication to release a PDU session due to LBO restriction.

Class

Optional.

List

- relDueToHo - Release Due to HO
- epsFallback - EPS Fallback
- relDueToUpSec - Release Due to User Plane Security
- dnnCongestion - DNN Congestion
- snssaiCongestion - S-NSSAI Congestion
- relDueToReactivation - Release Due to Reactivation
- 5gAnNotResponding - 5G AN Not Responding
- relDueToSliceNotAvailable - Release Due to Slice Not Available
- relDueToDuplicateSessionId - Release Due to Duplicate Session Id
- pduSessionStatusMismatch - PDU Session Status Mismatch
- hoFailure - HO Failure
- insufficientUpResources - Insufficient UP Resources
- pduSessionHandedOver - PDU Session Handed Over
- pduSessionResumed - PDU Session Resumed
- cnAssistedRanParameterTuning - CN Assisted RAN Parameter Tuning
- ismfContextTransfer - I-SMF Context Transfer
- smfContextTransfer - SMF Context Transfer
- relDueToPsToCsHo - Release Due to PS to CS HO
- relDueToSubscriptionChange - Release Due to Subscription Change
- hoCancel - HO Cancel
- relDueToSliceNotAuthorized - Release Due to Slice Not Authorized

- pduSessionHandoverFailure - PDU Session Handover Failure
- ddnFailureStatus - DDN Failure Status
- relDueToCpOnlyNotApplicable - Release Due to CP Only Not Applicable
- notSupportedWithIsmf - Not supported with I-SMF
- changedAnchorSmf - Changed Anchor SMF
- changedIntermediateSmf - Changed Intermediate SMF
- noCause - No Cause - No cause provided to SMF.

Default

relDueToSubscriptionChange

Name

causeToSmfOdb - cause to SMF ODB

Syntax

--causeToSmfOdb <list-item>

Description

The cause on SM context update with release indication to release a PDU session due to ODB restriction.

Class

Optional.

List

- relDueToHo - Release Due to HO
- epsFallback - EPS Fallback
- relDueToUpSec - Release Due to User Plane Security
- dnnCongestion - DNN Congestion
- snssaiCongestion - S-NSSAI Congestion
- relDueToReactivation - Release Due to Reactivation
- 5gAnNotResponding - 5G AN Not Responding
- relDueToSliceNotAvailable - Release Due to Slice Not Available
- relDueToDuplicateSessionId - Release Due to Duplicate Session Id
- pduSessionStatusMismatch - PDU Session Status Mismatch
- hoFailure - HO Failure
- insufficientUpResources - Insufficient UP Resources
- pduSessionHandedOver - PDU Session Handed Over

- pduSessionResumed - PDU Session Resumed
- cnAssistedRanParameterTuning - CN Assisted RAN Parameter Tuning
- ismfContextTransfer - I-SMF Context Transfer
- smfContextTransfer - SMF Context Transfer
- relDueToPsToCsHo - Release Due to PS to CS HO
- relDueToSubscriptionChange - Release Due to Subscription Change
- hoCancel - HO Cancel
- relDueToSliceNotAuthorized - Release Due to Slice Not Authorized
- pduSessionHandoverFailure - PDU Session Handover Failure
- ddnFailureStatus - DDN Failure Status
- relDueToCpOnlyNotApplicable - Release Due to CP Only Not Applicable
- notSupportedWithIsmf - Not supported with I-SMF
- changedAnchorSmf - Changed Anchor SMF
- changedIntermediateSmf - Changed Intermediate SMF
- noCause - No Cause - No cause provided to SMF.

Default

relDueToSubscriptionChange

Name

causeToSmfSliceNotAuthorized - Cause to SMF Slice Not Authorized

Syntax

--causeToSmfSliceNotAuthorized <list-item>

Description

The cause on SM context update with release indication to release a PDU session due to slice not authorized.

Class

Optional.

List

- relDueToHo - Release Due to HO
- epsFallback - EPS Fallback
- relDueToUpSec - Release Due to User Plane Security
- dnnCongestion - DNN Congestion
- snssaiCongestion - S-NSSAI Congestion

- relDueToReactivation - Release Due to Reactivation
- 5gAnNotResponding - 5G AN Not Responding
- relDueToSliceNotAvailable - Release Due to Slice Not Available
- relDueToDuplicateSessionId - Release Due to Duplicate Session Id
- pduSessionStatusMismatch - PDU Session Status Mismatch
- hoFailure - HO Failure
- insufficientUpResources - Insufficient UP Resources
- pduSessionHandedOver - PDU Session Handed Over
- pduSessionResumed - PDU Session Resumed
- cnAssistedRanParameterTuning - CN Assisted RAN Parameter Tuning
- ismfContextTransfer - I-SMF Context Transfer
- smfContextTransfer - SMF Context Transfer
- relDueToPsToCsHo - Release Due to PS to CS HO
- relDueToSubscriptionChange - Release Due to Subscription Change
- hoCancel - HO Cancel
- relDueToSliceNotAuthorized - Release Due to Slice Not Authorized
- pduSessionHandoverFailure - PDU Session Handover Failure
- ddnFailureStatus - DDN Failure Status
- relDueToCpOnlyNotApplicable - Release Due to CP Only Not Applicable
- notSupportedWithIsmf - Not supported with I-SMF
- changedAnchorSmf - Changed Anchor SMF
- changedIntermediateSmf - Changed Intermediate SMF
- noCause - No Cause - No cause provided to SMF.

Default

relDueToSliceNotAuthorized

Name

causeToSmfSliceNotAvailable - Cause to SMF Slice Not Available

Syntax

--causeToSmfSliceNotAvailable <list-item>

Description

The cause on SM context update with release indication to release a PDU session due to slice not available.

Class

Optional.

List

- relDueToHo - Release Due to HO
- epsFallback - EPS Fallback
- relDueToUpSec - Release Due to User Plane Security
- dnnCongestion - DNN Congestion
- snssaiCongestion - S-NSSAI Congestion
- relDueToReactivation - Release Due to Reactivation
- 5gAnNotResponding - 5G AN Not Responding
- relDueToSliceNotAvailable - Release Due to Slice Not Available
- relDueToDuplicateSessionId - Release Due to Duplicate Session Id
- pduSessionStatusMismatch - PDU Session Status Mismatch
- hoFailure - HO Failure
- insufficientUpResources - Insufficient UP Resources
- pduSessionHandedOver - PDU Session Handed Over
- pduSessionResumed - PDU Session Resumed
- cnAssistedRanParameterTuning - CN Assisted RAN Parameter Tuning
- ismfContextTransfer - I-SMF Context Transfer
- smfContextTransfer - SMF Context Transfer
- relDueToPsToCsHo - Release Due to PS to CS HO
- relDueToSubscriptionChange - Release Due to Subscription Change
- hoCancel - HO Cancel
- relDueToSliceNotAuthorized - Release Due to Slice Not Authorized
- pduSessionHandoverFailure - PDU Session Handover Failure
- ddnFailureStatus - DDN Failure Status
- relDueToCpOnlyNotApplicable - Release Due to CP Only Not Applicable
- notSupportedWithIsmf - Not supported with I-SMF
- changedAnchorSmf - Changed Anchor SMF
- changedIntermediateSmf - Changed Intermediate SMF
- noCause - No Cause - No cause provided to SMF.

Default

relDueToSliceNotAvailable

Name

causeToSmfStatusMismatch - Cause to SMF Status Mismatch

Syntax

--causeToSmfStatusMismatch <list-item>

Description

The cause on SM context release when releasing a PDU session due to PDU session status mismatch with the UE.

Class

Optional.

List

- relDueToHo - Release Due to HO
- epsFallback - EPS Fallback
- relDueToUpSec - Release Due to User Plane Security
- dnnCongestion - DNN Congestion
- snssaiCongestion - S-NSSAI Congestion
- relDueToReactivation - Release Due to Reactivation
- 5gAnNotResponding - 5G AN Not Responding
- relDueToSliceNotAvailable - Release Due to Slice Not Available
- relDueToDuplicateSessionId - Release Due to Duplicate Session Id
- pduSessionStatusMismatch - PDU Session Status Mismatch
- hoFailure - HO Failure
- insufficientUpResources - Insufficient UP Resources
- pduSessionHandedOver - PDU Session Handed Over
- pduSessionResumed - PDU Session Resumed
- cnAssistedRanParameterTuning - CN Assisted RAN Parameter Tuning
- ismfContextTransfer - I-SMF Context Transfer
- smfContextTransfer - SMF Context Transfer
- relDueToPsToCsHo - Release Due to PS to CS HO
- relDueToSubscriptionChange - Release Due to Subscription Change
- hoCancel - HO Cancel

- relDueToSliceNotAuthorized - Release Due to Slice Not Authorized
- pduSessionHandoverFailure - PDU Session Handover Failure
- ddnFailureStatus - DDN Failure Status
- relDueToCpOnlyNotApplicable - Release Due to CP Only Not Applicable
- notSupportedWithIsmf - Not supported with I-SMF
- changedAnchorSmf - Changed Anchor SMF
- changedIntermediateSmf - Changed Intermediate SMF
- noCause - No Cause - No cause provided to SMF.

Default

pduSessionStatusMismatch

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.107 cmm amfPeipsProfile

AMF PEIPS Profile. Syntax, description, parameters, and roles for `cmm amfPeipsProfile` command.

Syntax**Description**

Mapping of UE characteristics to a paging subgroup to be used with the PEIPS feature.

Parameters**Name**

entryId - Entry ID

Syntax

--entryId <integer>

Description

Numeric identification of the profile. A lower number profile has higher read priority over a higher number profile.

Class

Required key.

Type / Range

Integer in the range of 1 to 1,024

Name

assignedPagingSubgroup - Assigned Paging Subgroup

Syntax

--assignedPagingSubgroup <integer>

Description

The paging subgroup, used with the PEIPS feature, to which a UE matching this profile is to be assigned.

Class

Required on create.

Type / Range

Integer in the range of 0 to 7

Name

amfRat - AMF RAT

Syntax

--amfRat <list-item>

Description

5G RAT

Class

Optional.

List

- any - Any of the supported 5G RATs (NR, NR REDCAP)
- nr - NR
- nrRedcap - NR REDCAP

Default

any

Name

amfTaiListName - TAI List Name

Syntax

--amfTaiListName <string>

Description

Name of a TAI List

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Name

dnnListName - DNN List Name

Syntax

--dnnListName <string>

Description

The name of a DNN List. If this is populated, the UE must have at least one active DNN matching an entry in the list.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Name

edrxCycleLength - f20206-01

Syntax

--edrxCycleLength <list-item>

Description

The 5G eDRX cycle length assigned to the UE.

Class

Optional.

List

- hfquarter - HF Quarter - Quarter of a Hyperframe (2.56 seconds).
- hfhalf - HF Half - Half of a Hyperframe (5.12 seconds)

- hf1 - HF 1 - One Hyperframe (10.24 seconds)
- hf2 - HF 2 - Two Hyperframes (20.48 seconds)
- hf4 - HF 4 - Four Hyperframes (40.96 seconds)
- hf8 - HF 8 - Eight Hyperframes (81.92 seconds)
- hf16 - HF 16 - Sixteen Hyperframes (163.84 seconds)
- hf32 - HF 32 - 32 Hyperframes (327.68 seconds)
- hf64 - HF 64 - 64 Hyperframes (655.36 seconds)
- hf128 - HF 258 - 256 Hyperframes (2621.44 seconds)
- hf256 - HF 258 - 256 Hyperframes (2621.44 seconds)
- hf512 - HF 512 - 512 Hyperframes (5242.88 seconds)
- hf1024 - HF 1024 - 1024 Hyperframes (10485.76 seconds)
- any - Any eDRX Cycle
- none - No eDRX cycle.

Default

none

Name

pagingProbability - Paging Probability

Syntax

--pagingProbability <list-item>

Description

The paging probability provided by the UE during registration

Class

Optional.

List

- any - Any paging probability value from the UE
- none - UE does not provide a paging probability value
- p00 - 0% paging probability value
- p05 - 5% paging probability value
- p10 - 10% paging probability value
- p100 - 100% paging probability value
- p15 - 15% paging probability value
- p20 - 20% paging probability value

- p25 - 25% paging probability value
- p30 - 30% paging probability value
- p35 - 35% paging probability value
- p40 - 40% paging probability value
- p45 - 45% paging probability value
- p50 - 50% paging probability value
- p55 - 55% paging probability value
- p60 - 60% paging probability value
- p65 - 65% paging probability value
- p70 - 70% paging probability value
- p75 - 75% paging probability value
- p80 - 80% paging probability value
- p85 - 85% paging probability value
- p90 - 90% paging probability value
- p95 - 95% paging probability value

Default

none

Name

servedPlmnName - Served PLMN Name

Syntax

--servedPlmnName <string>

Description

The serving PLMN where a UE is registered.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Name

stationaryInd - Stationary Indication

Syntax

--stationaryInd <list-item>

Description

Whether the UDM profile indicates that the UE is stationary.

Class

Optional.

List

- absentOrAny - Absent or Any - The UE is identified as either MOBILE or STATIONARY in the UDM subscription, or the subscription does not include the indication.
- mobile - The UDM identifies the UE as mobile.
- stationary - The UDM identifies the UE as stationary.

Default

absentOrAny

Name

uePlmnName - UE PLMN

Syntax

--uePlmnName <string>

Description

The UE home PLMN

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.108 cmm amfPlmn

Public Land Mobile Networks. Syntax, description, parameters, and roles for cmm amfPlmn command.

Syntax

Description

This command is used to administer a Public Land Mobile Network and includes parameters for defining the network's MCC and MNC, PLMN type, roaming agreements, etc. For amfPlmn record of plmnType='Multiple Home Network' all attributes other than name, mcc, mnc, and plmnType are ignored; the AMF instead applies the attribute values from the amfPlmn record associated with plmnType='Home Network'.

Parameters

Name

name - PLMN name

Syntax

--name <string>

Description

PLMN name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

mcc - Mobile Country Code

Syntax

--mcc <digits>

Description

Mobile Country Code.

Class

Required on create, not modifiable.

Type / Range

Digit string of length 3 to 3 with the allowed pattern of [0-9]*

Name

mnc - Mobile Network Code

Syntax

--mnc <digits>

Description

Mobile Network Code.

Class

Required on create, not modifiable.

Type / Range

Digit string of length 2 to 3 with the allowed pattern of [0-9]*

Name

plmnType - PLMN Type

Syntax

--plmnType <list-item>

Description

Indicates which type of network this is

Class

Required on create, not modifiable.

List

- Home Network - Specify network type of home network.
- Other Network - Specify network type of Other for roaming partner(s)
- Shared Network - Specify network type of Shared Network
- Standalone Non-Public Network - standalone-non-public-network - Specify network type of SNPN

Name

amfEmergencyProfileName - AMF Emergency Profile Name

Syntax

--amfEmergencyProfileName <string>

Description

AMF Emergency profile name

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

assignMode - Assignment Mode

Syntax

--assignMode <integer>

Description

This fields defines the assignment mode for the SNPN

Class

Not modifiable.

Type / Range

Integer in the range of 0 to 2

Default

0

Name

checkImsDnnSubscription - Check IMS DNN subscription

Syntax

--checkImsDnnSubscription <list-item>

Description

Check IMS DNN subscription in determining network IMS voice over PS support.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

cmassEnable - cmass enable

Syntax

--cmassEnable <list-item>

Description

Parameter controls cmass enable with values true or false

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

defaultRacFor5gSrvcc - Default RAC for 5G-SRVCC

Syntax

--defaultRacFor5gSrvcc <integer>

Description

Defines the value for the default RAC used in N26 Forward Relocation Request for 5G-SRVCC from NG-RAN to UTRAN CS. The AMF uses this parameter to fill in the RAC of the Target Identification IE on the N26 Forward Relocation Request. The NGAP Handover Required message with a Target ID of type Target RNC-ID does not contain any RAC.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

255

Name

mncDigits - Number of MNC digits

Syntax

--mncDigits <list-item>

Description

Number of digits used to represent the mobile network code

Class

Not modifiable.

List

- 2/3Digits - 2 or 3 digits - 2 or 3 digits.
- 2Digits - 2 digits - 2 digits.
- 3Digits - 3 digits - 3 digits.

Default

3Digits

Name

nidValue - Nid value

Syntax

--nidValue <integer>

Description

By this field the assigned SNPN Nid is defined

Class

Not modifiable.

Type / Range

Integer in the Range 0 to 1099511627775

Default

0

Name

rerunRadioCapabilityCheck - Rerun Radio Capability Check

Syntax

--rerunRadioCapabilityCheck <list-item>

Description

When the NGAP UE Radio Capability Check is to be run as part of the AMF analysis for IMS voice over PS session support (as controlled by an other parameter), this parameter further controls whether the AMF only runs the NGAP UE Radio Capability Check Request procedure on the initial Registration or handover procedure (when the UE enters the current AMF), or whether the AMF reruns the Radio Capability Check on (each) Mobility Registration procedure, if there is no homogeneous support/homogeneous non-support for voice on the Tracking Areas of the network (the serving PLMN). Note: if there is homogeneous support/non-support for voice, the AMF anyway does not rerun UE Radio Capability Check.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

timeZoneName - Time Zone Name

Syntax

--timeZoneName <list-item>

Description

Provides time zone names that correspond to the Region selected

Class

Optional.

List

- Africa/Abidjan - Timezone for africa/abidjan UTC offset of +00:00.

- Africa/Accra - Timezone for africa/accra UTC offset of +00:00.
- Africa/Addis_Ababa - Timezone for africa/addis_ababa UTC offset of +03:00.
- Africa/Algiers - Timezone for africa/algiers UTC offset of +01:00.
- Africa/Asmara - Timezone for africa/asmara UTC offset of +03:00.
- Africa/Asmera - Timezone for africa/asmera UTC offset of +03:00.
- Africa/Bamako - Timezone for africa/bamako UTC offset of +00:00.
- Africa/Bangui - Timezone for africa/bangui UTC offset of +01:00.
- Africa/Banjul - Timezone for africa/banjul UTC offset of +00:00.
- Africa/Bissau - Timezone for africa/bissau UTC offset of +00:00.
- Africa/Blantyre - Timezone for africa/blantyre UTC offset of +02:00.
- Africa/Brazzaville - Timezone for africa/brazzaville UTC offset of +01:00.
- Africa/Bujumbura - Timezone for africa/bujumbura UTC offset of +02:00.
- Africa/Cairo - Timezone for africa/cairo UTC offset of +02:00.
- Africa/Casablanca - Timezone for africa/casablanca UTC offset of +00:00.
- Africa/Ceuta - Timezone for africa/ceuta UTC offset of +01:00.
- Africa/Conakry - Timezone for africa/conakry UTC offset of +00:00.
- Africa/Dakar - Timezone for africa/dakar UTC offset of +00:00.
- Africa/Dar_es_Salaam - Timezone for africa/dar_es_salaam UTC offset of +03:00.
- Africa/Djibouti - Timezone for africa/djibouti UTC offset of +03:00.
- Africa/Douala - Timezone for africa/douala UTC offset of +01:00.
- Africa/El_Aaiun - Timezone for africa/el_aaiun UTC offset of +00:00.
- Africa/Freetown - Timezone for africa/freetown UTC offset of +00:00.
- Africa/Gaborone - Timezone for africa/gaborone UTC offset of +02:00.
- Africa/Harare - Timezone for africa/harare UTC offset of +02:00.
- Africa/Johannesburg - Timezone for africa/johannesburg UTC offset of +02:00.
- Africa/Juba - Timezone for africa/juba UTC offset of +03:00.
- Africa/Kampala - Timezone for africa/kampala UTC offset of +03:00.
- Africa/Khartoum - Timezone for africa/khartoum UTC offset of +03:00.
- Africa/Kigali - Timezone for africa/kigali UTC offset of +02:00.
- Africa/Kinshasa - Timezone for africa/kinshasa UTC offset of +01:00.
- Africa/Lagos - Timezone for africa/lagos UTC offset of +01:00.

- Africa/Libreville - Timezone for africa/libreville UTC offset of +01:00.
- Africa/Lome - Timezone for africa/lome UTC offset of +00:00.
- Africa/Luanda - Timezone for africa/luanda UTC offset of +01:00.
- Africa/Lubumbashi - Timezone for africa/lubumbashi UTC offset of +02:00.
- Africa/Lusaka - Timezone for africa/lusaka UTC offset of +02:00.
- Africa/Malabo - Timezone for africa/malabo UTC offset of +01:00.
- Africa/Maputo - Timezone for africa/maputo UTC offset of +02:00.
- Africa/Maseru - Timezone for africa/maseru UTC offset of +02:00.
- Africa/Mbabane - Timezone for africa/mbabane UTC offset of +02:00.
- Africa/Mogadishu - Timezone for africa/mogadishu UTC offset of +03:00.
- Africa/Monrovia - Timezone for africa/monrovia UTC offset of +00:00.
- Africa/Nairobi - Timezone for africa/nairobi UTC offset of +03:00.
- Africa/Ndjamena - Timezone for africa/ndjamena UTC offset of +01:00.
- Africa/Niamey - Timezone for africa/niamey UTC offset of +01:00.
- Africa/Nouakchott - Timezone for africa/nouakchott UTC offset of +00:00.
- Africa/Ouagadougou - Timezone for africa/ouagadougou UTC offset of +00:00.
- Africa/Porto-Novo - Timezone for africa/porto-novo UTC offset of +01:00.
- Africa/Sao_Tome - Timezone for africa/sao_tome UTC offset of +00:00.
- Africa/Timbuktu - Timezone for africa/timbuktu UTC offset of +00:00.
- Africa/Tripoli - Timezone for africa/tripoli UTC offset of +02:00.
- Africa/Tunis - Timezone for africa/tunis UTC offset of +01:00.
- Africa/Windhoek - Timezone for africa/windhoek UTC offset of +02:00.
- America/Adak - Timezone for america/adak UTC offset of -10:00.
- America/Anchorage - Timezone for america/anchorage UTC offset of -09:00.
- America/Anguilla - Timezone for america/anguilla UTC offset of -04:00.
- America/Antigua - Timezone for america/antigua UTC offset of -04:00.
- America/Araguaina - Timezone for america/araguaina UTC offset of -03:00.
- America/Argentina/Buenos_Aires - Timezone for america/argentina/buenos_aires UTC offset of -03:00.
- America/Argentina/Catamarca - Timezone for america/argentina/catamarca UTC offset of -03:00.

- America/Argentina/ComodRivadavia - Timezone for america/argentina/comodrivadavia UTC offset of -03:00.
- America/Argentina/Cordoba - Timezone for america/argentina/cordoba UTC offset of -03:00.
- America/Argentina/Jujuy - Timezone for america/argentina/jujuy UTC offset of -03:00.
- America/Argentina/La_Rioja - Timezone for america/argentina/la_rioja UTC offset of -03:00.
- America/Argentina/Mendoza - Timezone for america/argentina/mendoza UTC offset of -03:00.
- America/Argentina/Rio_Gallegos - Timezone for america/argentina/rio_gallegos UTC offset of -03:00.
- America/Argentina/Salta - Timezone for america/argentina/salta UTC offset of -03:00.
- America/Argentina/San_Juan - Timezone for america/argentina/san_juan UTC offset of -03:00.
- America/Argentina/San_Luis - Timezone for america/argentina/san_luis UTC offset of -03:00.
- America/Argentina/Tucuman - Timezone for america/argentina/tucuman UTC offset of -03:00.
- America/Argentina/Ushuaia - Timezone for america/argentina/ushuaia UTC offset of -03:00.
- America/Aruba - Timezone for america/aruba UTC offset of -04:00.
- America/Asuncion - Timezone for america/asuncion UTC offset of -03:00.
- America/Atikokan - Timezone for america/atikokan UTC offset of -05:00.
- America/Atka - Timezone for america/atka UTC offset of -10:00.
- America/Bahia - Timezone for america/bahia UTC offset of -03:00.
- America/Bahia_Banderas - Timezone for america/bahia_banderas UTC offset of -06:00.
- America/Barbados - Timezone for america/barbados UTC offset of -04:00.
- America/Belem - Timezone for america/belem UTC offset of -03:00.
- America/Belize - Timezone for america/belize UTC offset of -06:00.
- America/Blanc-Sablon - Timezone for america/blanc-sablon UTC offset of -04:00.
- America/Boa_Vista - Timezone for america/boa_vista UTC offset of -04:00.
- America/Bogota - Timezone for america/bogota UTC offset of -05:00.
- America/Boise - Timezone for america/boise UTC offset of -07:00.
- America/Buenos_Aires - Timezone for america/buenos_aires UTC offset of -03:00.

- America/Cambridge_Bay - Timezone for america/cambridge_bay UTC offset of -07:00.
- America/Campo_Grande - Timezone for america/campo_grande UTC offset of -03:00.
- America/Cancun - Timezone for america/cancun UTC offset of -05:00.
- America/Caracas - Timezone for america/caracas UTC offset of -04:00.
- America/Catamarca - Timezone for america/catamarca UTC offset of -03:00.
- America/Cayenne - Timezone for america/cayenne UTC offset of -03:00.
- America/Cayman - Timezone for america/cayman UTC offset of -05:00.
- America/Chicago - Timezone for america/chicago UTC offset of -06:00.
- America/Chihuahua - Timezone for america/chihuahua UTC offset of -07:00.
- America/Coral_Harbour - Timezone for america/coral_harbour UTC offset of -05:00.
- America/Cordoba - Timezone for america/cordoba UTC offset of -03:00.
- America/Costa_Rica - Timezone for america/costa_rica UTC offset of -06:00.
- America/Creston - Timezone for america/creston UTC offset of -07:00.
- America/Cuiaba - Timezone for america/cuiaba UTC offset of -03:00.
- America/Curacao - Timezone for america/curacao UTC offset of -04:00.
- America/Danmarkshavn - Timezone for america/danmarkshavn UTC offset of +00:00.
- America/Dawson - Timezone for america/dawson UTC offset of -08:00.
- America/Dawson_Creek - Timezone for america/dawson_creek UTC offset of -07:00.
- America/Denver - Timezone for america/denver UTC offset of -07:00.
- America/Detroit - Timezone for america/detroit UTC offset of -05:00.
- America/Dominica - Timezone for america/dominica UTC offset of -04:00.
- America/Edmonton - Timezone for america/edmonton UTC offset of -07:00.
- America/Eirunepe - Timezone for america/eirunepe UTC offset of -05:00.
- America/El_Salvador - Timezone for america/el_salvador UTC offset of -06:00.
- America/Ensenada - Timezone for america/ensenada UTC offset of -08:00.
- America/Fort_Nelson - Timezone for america/fort_nelson UTC offset of -07:00.
- America/Fort_Wayne - Timezone for america/fort_wayne UTC offset of -05:00.
- America/Fortaleza - Timezone for america/fortaleza UTC offset of -03:00.
- America/Glace_Bay - Timezone for america/glance_bay UTC offset of -04:00.
- America/Godthab - Timezone for america/godthab UTC offset of -03:00.
- America/Goose_Bay - Timezone for america/goose_bay UTC offset of -04:00.

- America/Grand_Turk - Timezone for america/grand_turk UTC offset of -04:00.
- America/Grenada - Timezone for america/grenada UTC offset of -04:00.
- America/Guadeloupe - Timezone for america/guadeloupe UTC offset of -04:00.
- America/Guatemala - Timezone for america/guatemala UTC offset of -06:00.
- America/Guayaquil - Timezone for america/guayaquil UTC offset of -05:00.
- America/Guyana - Timezone for america/guyana UTC offset of -04:00.
- America/Halifax - Timezone for america/halifax UTC offset of -04:00.
- America/Havana - Timezone for america/havana UTC offset of -05:00.
- America/Hermosillo - Timezone for america/hermosillo UTC offset of -07:00.
- America/Indiana/Indianapolis - Timezone for america/indiana/indianapolis UTC offset of -05:00.
- America/Indiana/Knox - Timezone for america/indiana/knox UTC offset of -06:00.
- America/Indiana/Marengo - Timezone for america/indiana/marengo UTC offset of -05:00.
- America/Indiana/Petersburg - Timezone for america/indiana/petersburg UTC offset of -05:00.
- America/Indiana/Tell_City - Timezone for america/indiana/tell_city UTC offset of -06:00.
- America/Indiana/Vevay - Timezone for america/indiana/vevay UTC offset of -05:00.
- America/Indiana/Vincennes - Timezone for america/indiana/vincennes UTC offset of -05:00.
- America/Indiana/Winamac - Timezone for america/indiana/winamac UTC offset of -05:00.
- America/Indianapolis - Timezone for america/indianapolis UTC offset of -05:00.
- America/Inuvik - Timezone for america/inuvik UTC offset of -07:00.
- America/Iqaluit - Timezone for america/iqaluit UTC offset of -05:00.
- America/Jamaica - Timezone for america/jamaica UTC offset of -05:00.
- America/Jujuy - Timezone for america/jujuy UTC offset of -03:00.
- America/Juneau - Timezone for america/juneau UTC offset of -09:00.
- America/Kentucky/Louisville - Timezone for america/kentucky/louisville UTC offset of -05:00.
- America/Kentucky/Monticello - Timezone for america/kentucky/monticello UTC offset of -05:00.
- America/Knox_IN - Timezone for america/knox_in UTC offset of -06:00.

- America/Kralendijk - Timezone for america/kralendijk UTC offset of -04:00.
- America/La_Paz - Timezone for america/la_paz UTC offset of -04:00.
- America/Lima - Timezone for america/lima UTC offset of -05:00.
- America/Los_Angeles - Timezone for america/los_angeles UTC offset of -08:00.
- America/Louisville - Timezone for america/louisville UTC offset of -05:00.
- America/Lower_Princes - Timezone for america/lower_princes UTC offset of -04:00.
- America/Maceio - Timezone for america/maceio UTC offset of -03:00.
- America/Managua - Timezone for america/managua UTC offset of -06:00.
- America/Manaus - Timezone for america/manuel UTC offset of -04:00.
- America/Marigot - Timezone for america/marigot UTC offset of -04:00.
- America/Martinique - Timezone for america/martinique UTC offset of -04:00.
- America/Matamoros - Timezone for america/matamoros UTC offset of -06:00.
- America/Mazatlan - Timezone for america/mazatlan UTC offset of -07:00.
- America/Mendoza - Timezone for america/mendoza UTC offset of -03:00.
- America/Menominee - Timezone for america/menominee UTC offset of -06:00.
- America/Merida - Timezone for america/merida UTC offset of -06:00.
- America/Metlakatla - Timezone for america/metlakatla UTC offset of -09:00.
- America/Mexico_City - Timezone for america/mexico_city UTC offset of -06:00.
- America/Miquelon - Timezone for america/miquelon UTC offset of -03:00.
- America/Moncton - Timezone for america/moncton UTC offset of -04:00.
- America/Monterrey - Timezone for america/monterrey UTC offset of -06:00.
- America/Montevideo - Timezone for america/montevideo UTC offset of -03:00.
- America/Montreal - Timezone for america/montreal UTC offset of -05:00.
- America/Montserrat - Timezone for america/montserrat UTC offset of -04:00.
- America/Nassau - Timezone for america/nassau UTC offset of -05:00.
- America/New_York - Timezone for america/new_york UTC offset of -05:00.
- America/Nipigon - Timezone for america/nipigon UTC offset of -05:00.
- America/Nome - Timezone for america/nome UTC offset of -09:00.
- America/Noronha - Timezone for america/noronha UTC offset of -02:00.
- America/North_Dakota/Beulah - Timezone for america/north_dakota/beulah UTC offset of -06:00.

- America/North_Dakota/Center - Timezone for america/north_dakota/center UTC offset of -06:00.
- America/North_Dakota/New_Salem - Timezone for america/north_dakota/new_salem UTC offset of -06:00.
- America/Ojinaga - Timezone for america/ojinaga UTC offset of -07:00.
- America/Panama - Timezone for america/panama UTC offset of -05:00.
- America/Pangnirtung - Timezone for america/pangnirtung UTC offset of -05:00.
- America/Paramaribo - Timezone for america/paramaribo UTC offset of -03:00.
- America/Phoenix - Timezone for america/phoenix UTC offset of -07:00.
- America/Port-au-Prince - Timezone for america/port-au-prince UTC offset of -05:00.
- America/Port_of_Spain - Timezone for america/port_of_spain UTC offset of -04:00.
- America/Porto_Acre - Timezone for america/porto_acre UTC offset of -05:00.
- America/Porto_Velho - Timezone for america/porto_velho UTC offset of -04:00.
- America/Puerto_Rico - Timezone for america/puerto_rico UTC offset of -04:00.
- America/Rainy_River - Timezone for america/rainy_river UTC offset of -06:00.
- America/Rankin_Inlet - Timezone for america/rankin_inlet UTC offset of -06:00.
- America/Recife - Timezone for america/recife UTC offset of -03:00.
- America/Regina - Timezone for america/regina UTC offset of -06:00.
- America/Resolute - Timezone for america/resolute UTC offset of -06:00.
- America/Rio_Branco - Timezone for america/rio_branco UTC offset of -05:00.
- America/Rosario - Timezone for america/rosario UTC offset of -03:00.
- America/Santa_Isabel - Timezone for america/santa_isabel UTC offset of -08:00.
- America/Santarem - Timezone for america/santarem UTC offset of -03:00.
- America/Santiago - Timezone for america/santiago UTC offset of -03:00.
- America/Santo_Domingo - Timezone for america/santo_domingo UTC offset of -04:00.
- America/Sao_Paulo - Timezone for america/sao_paulo UTC offset of -02:00.
- America/Scoresbysund - Timezone for america/scoresbysund UTC offset of -01:00.
- America/Shiprock - Timezone for america/shiprock UTC offset of -07:00.
- America/Sitka - Timezone for america/sitka UTC offset of -09:00.
- America/St_Barthelemy - Timezone for america/st_barthelemy UTC offset of -04:00.
- America/St_Johns - Timezone for america/st_johns UTC offset of -03:30.
- America/St_Kitts - Timezone for america/st_kitts UTC offset of -04:00.

- America/St_Lucia - Timezone for america/st_lucia UTC offset of -04:00.
- America/St_Thomas - Timezone for america/st_thomas UTC offset of -04:00.
- America/St_Vincent - Timezone for america/st_vincent UTC offset of -04:00.
- America/Swift_Current - Timezone for america/swift_current UTC offset of -06:00.
- America/Tegucigalpa - Timezone for america/tegucigalpa UTC offset of -06:00.
- America/Thule - Timezone for america/thule UTC offset of -04:00.
- America/Thunder_Bay - Timezone for america/thunder_bay UTC offset of -05:00.
- America/Tijuana - Timezone for america/tijuana UTC offset of -08:00.
- America/Toronto - Timezone for america/toronto UTC offset of -05:00.
- America/Tortola - Timezone for america/tortola UTC offset of -04:00.
- America/Vancouver - Timezone for america/vancouver UTC offset of -08:00.
- America/Virgin - Timezone for america/virgin UTC offset of -04:00.
- America/Whitehorse - Timezone for america/whitehorse UTC offset of -08:00.
- America/Winnipeg - Timezone for america/winnipeg UTC offset of -06:00.
- America/Yakutat - Timezone for america/yakutat UTC offset of -09:00.
- America/Yellowknife - Timezone for america/yellowknife UTC offset of -07:00.
- Antarctica/Casey - Timezone for antarctica/casey UTC offset of +11:00.
- Antarctica/Davis - Timezone for antarctica/davis UTC offset of +07:00.
- Antarctica/DumontDURville - Timezone for antarctica/dumontdurville UTC offset of +10:00.
- Antarctica/Macquarie - Timezone for antarctica/macquarie UTC offset of +11:00.
- Antarctica/Mawson - Timezone for antarctica/mawson UTC offset of +05:00.
- Antarctica/McMurdo - Timezone for antarctica/mcmurdo UTC offset of +13:00.
- Antarctica/Palmer - Timezone for antarctica/palmer UTC offset of -03:00.
- Antarctica/Rothera - Timezone for antarctica/rothera UTC offset of -03:00.
- Antarctica/South_Pole - Timezone for antarctica/south_pole UTC offset of +13:00.
- Antarctica/Syowa - Timezone for antarctica/syowa UTC offset of +03:00.
- Antarctica/Troll - Timezone for antarctica/troll UTC offset of +00:00.
- Antarctica/Vostok - Timezone for antarctica/vostok UTC offset of +06:00.
- Arctic/Longyearbyen - Timezone for arctic/longyearbyen UTC offset of +01:00.
- Asia/Aden - Timezone for asia/aden UTC offset of +03:00.

- Asia/Almaty - Timezone for asia/almaty UTC offset of +06:00.
- Asia/Amman - Timezone for asia/amman UTC offset of +02:00.
- Asia/Anadyr - Timezone for asia/anadyr UTC offset of +12:00.
- Asia/Aqtau - Timezone for asia/aqtau UTC offset of +05:00.
- Asia/Aqtobe - Timezone for asia/aqtobe UTC offset of +05:00.
- Asia/Ashgabat - Timezone for asia/ashgabat UTC offset of +05:00.
- Asia/Ashkhabad - Timezone for asia/ashkhabad UTC offset of +05:00.
- Asia/Baghdad - Timezone for asia/baghdad UTC offset of +03:00.
- Asia/Bahrain - Timezone for asia/bahrain UTC offset of +03:00.
- Asia/Baku - Timezone for asia/baku UTC offset of +04:00.
- Asia/Bangkok - Timezone for asia/bangkok UTC offset of +07:00.
- Asia/Barnaul - Timezone for asia/barnaul UTC offset of +07:00.
- Asia/Beirut - Timezone for asia/beirut UTC offset of +02:00.
- Asia/Bishkek - Timezone for asia/bishkek UTC offset of +06:00.
- Asia/Brunei - Timezone for asia/brunei UTC offset of +08:00.
- Asia/Calcutta - Timezone for asia/calcutta UTC offset of +05:30.
- Asia/Chita - Timezone for asia/chita UTC offset of +09:00.
- Asia/Choibalsan - Timezone for asia/choibalsan UTC offset of +08:00.
- Asia/Chongqing - Timezone for asia/chongqing UTC offset of +08:00.
- Asia/Chungking - Timezone for asia/chungking UTC offset of +08:00.
- Asia/Colombo - Timezone for asia/colombo UTC offset of +05:30.
- Asia/Dacca - Timezone for asia/dacca UTC offset of +06:00.
- Asia/Damascus - Timezone for asia/damascus UTC offset of +02:00.
- Asia/Dhaka - Timezone for asia/dhaka UTC offset of +06:00.
- Asia/Dili - Timezone for asia/dili UTC offset of +09:00.
- Asia/Dubai - Timezone for asia/dubai UTC offset of +04:00.
- Asia/Dushanbe - Timezone for asia/dushanbe UTC offset of +05:00.
- Asia/Gaza - Timezone for asia/gaza UTC offset of +02:00.
- Asia/Harbin - Timezone for asia/harbin UTC offset of +08:00.
- Asia/Hebron - Timezone for asia/hebron UTC offset of +02:00.
- Asia/Ho_Chi_Minh - Timezone for asia/ho_chi_minh UTC offset of +07:00.

- Asia/Hong_Kong - Timezone for asia/hong_kong UTC offset of +08:00.
- Asia/Hovd - Timezone for asia/hovd UTC offset of +07:00.
- Asia/Irkutsk - Timezone for asia/irkutsk UTC offset of +08:00.
- Asia/Istanbul - Timezone for asia/istanbul UTC offset of +03:00.
- Asia/Jakarta - Timezone for asia/jakarta UTC offset of +07:00.
- Asia/Jayapura - Timezone for asia/jayapura UTC offset of +09:00.
- Asia/Jerusalem - Timezone for asia/jerusalem UTC offset of +02:00.
- Asia/Kabul - Timezone for asia/kabul UTC offset of +04:30.
- Asia/Kamchatka - Timezone for asia/kamchatka UTC offset of +12:00.
- Asia/Karachi - Timezone for asia/karachi UTC offset of +05:00.
- Asia/Kashgar - Timezone for asia/kashgar UTC offset of +06:00.
- Asia/Kathmandu - Timezone for asia/kathmandu UTC offset of +05:45.
- Asia/Katmandu - Timezone for asia/katmandu UTC offset of +05:45.
- Asia/Khandyga - Timezone for asia/khandyga UTC offset of +09:00.
- Asia/Kolkata - Timezone for asia/kolkata UTC offset of +05:30.
- Asia/Krasnoyarsk - Timezone for asia/krasnoyarsk UTC offset of +07:00.
- Asia/Kuala_Lumpur - Timezone for asia/kuala_lumpur UTC offset of +08:00.
- Asia/Kuching - Timezone for asia/kuching UTC offset of +08:00.
- Asia/Kuwait - Timezone for asia/kuwait UTC offset of +03:00.
- Asia/Macao - Timezone for asia/macao UTC offset of +08:00.
- Asia/Macau - Timezone for asia/macau UTC offset of +08:00.
- Asia/Magadan - Timezone for asia/magadan UTC offset of +11:00.
- Asia/Makassar - Timezone for asia/makassar UTC offset of +08:00.
- Asia/Manila - Timezone for asia/manila UTC offset of +08:00.
- Asia/Muscat - Timezone for asia/muscat UTC offset of +04:00.
- Asia/Nicosia - Timezone for asia/nicosia UTC offset of +02:00.
- Asia/Novokuznetsk - Timezone for asia/novokuznetsk UTC offset of +07:00.
- Asia/Novosibirsk - Timezone for asia/novosibirsk UTC offset of +07:00.
- Asia/Omsk - Timezone for asia/omsk UTC offset of +06:00.
- Asia/Oral - Timezone for asia/oral UTC offset of +05:00.
- Asia/Phnom_Penh - Timezone for asia/phnom_penh UTC offset of +07:00.

- Asia/Pontianak - Timezone for asia/pontianak UTC offset of +07:00.
- Asia/Pyongyang - Timezone for asia/pyongyang UTC offset of +08:30.
- Asia/Qatar - Timezone for asia/qatar UTC offset of +03:00.
- Asia/Qyzylorda - Timezone for asia/qyzylorda UTC offset of +06:00.
- Asia/Rangoon - Timezone for asia/rangoon UTC offset of +06:30.
- Asia/Riyadh - Timezone for asia/riyadh UTC offset of +03:00.
- Asia/Saigon - Timezone for asia/saigon UTC offset of +07:00.
- Asia/Sakhalin - Timezone for asia/sakhalin UTC offset of +11:00.
- Asia/Samarkand - Timezone for asia/samarkand UTC offset of +05:00.
- Asia/Seoul - Timezone for asia/seoul UTC offset of +09:00.
- Asia/Shanghai - Timezone for asia/shanghai UTC offset of +08:00.
- Asia/Singapore - Timezone for asia/singapore UTC offset of +08:00.
- Asia/Srednekolymsk - Timezone for asia/srednekolymsk UTC offset of +11:00.
- Asia/Taipei - Timezone for asia/taipei UTC offset of +08:00.
- Asia/Tashkent - Timezone for asia/tashkent UTC offset of +05:00.
- Asia/Tbilisi - Timezone for asia/tbilisi UTC offset of +04:00.
- Asia/Tehran - Timezone for asia/tehran UTC offset of +03:30.
- Asia/Tel_Aviv - Timezone for asia/tel_aviv UTC offset of +02:00.
- Asia/Thimbu - Timezone for asia/thimbu UTC offset of +06:00.
- Asia/Thimphu - Timezone for asia/thimphu UTC offset of +06:00.
- Asia/Tokyo - Timezone for asia/tokyo UTC offset of +09:00.
- Asia/Tomsk - Timezone for asia/tomsk UTC offset of +07:00.
- Asia/Ujung_Pandang - Timezone for asia/ujung_pandang UTC offset of +08:00.
- Asia/Ulaanbaatar - Timezone for asia/ulaanbaatar UTC offset of +08:00.
- Asia/Ulan_Bator - Timezone for asia/ulan_bator UTC offset of +08:00.
- Asia/Urumqi - Timezone for asia/urumqi UTC offset of +06:00.
- Asia/Ust-Nera - Timezone for asia/ust-nera UTC offset of +10:00.
- Asia/Vientiane - Timezone for asia/vientiane UTC offset of +07:00.
- Asia/Vladivostok - Timezone for asia/vladivostok UTC offset of +10:00.
- Asia/Yakutsk - Timezone for asia/yakutsk UTC offset of +09:00.
- Asia/Yekaterinburg - Timezone for asia/yekaterinburg UTC offset of +05:00.

- Asia/Yerevan - Timezone for asia/yerevan UTC offset of +04:00.
- Atlantic/Azores - Timezone for atlantic/azores UTC offset of -01:00.
- Atlantic/Bermuda - Timezone for atlantic/bermuda UTC offset of -04:00.
- Atlantic/Canary - Timezone for atlantic/canary UTC offset of +00:00.
- Atlantic/Cape_Verde - Timezone for atlantic/cape_verde UTC offset of -01:00.
- Atlantic/Faeroe - Timezone for atlantic/faeroe UTC offset of +00:00.
- Atlantic/Faroe - Timezone for atlantic/faroe UTC offset of +00:00.
- Atlantic/Jan_Mayen - Timezone for atlantic/jan_mayen UTC offset of +01:00.
- Atlantic/Madeira - Timezone for atlantic/madeira UTC offset of +00:00.
- Atlantic/Reykjavik - Timezone for atlantic/reykjavik UTC offset of +00:00.
- Atlantic/South_Georgia - Timezone for atlantic/south_georgia UTC offset of -02:00.
- Atlantic/St_Helena - Timezone for atlantic/st_helena UTC offset of +00:00.
- Atlantic/Stanley - Timezone for atlantic/stanley UTC offset of -03:00.
- Australia/ACT - Timezone for australia/act UTC offset of +11:00.
- Australia/Adelaide - Timezone for australia/adelaide UTC offset of +10:30.
- Australia/Brisbane - Timezone for australia/brisbane UTC offset of +10:00.
- Australia/Broken_Hill - Timezone for australia/broken_hill UTC offset of +10:30.
- Australia/Canberra - Timezone for australia/canberra UTC offset of +11:00.
- Australia/Currie - Timezone for australia/currie UTC offset of +11:00.
- Australia/Darwin - Timezone for australia/darwin UTC offset of +09:30.
- Australia/Eucla - Timezone for australia/eucla UTC offset of +08:45.
- Australia/Hobart - Timezone for australia/hobart UTC offset of +11:00.
- Australia/LHI - Timezone for australia/lhi UTC offset of +11:00.
- Australia/Lindeman - Timezone for australia/lindeman UTC offset of +10:00.
- Australia/Lord_Howe - Timezone for australia/lord_howe UTC offset of +11:00.
- Australia/Melbourne - Timezone for australia/melbourne UTC offset of +11:00.
- Australia/NSW - Timezone for australia/nsw UTC offset of +11:00.
- Australia/North - Timezone for australia/north UTC offset of +09:30.
- Australia/Perth - Timezone for australia/perth UTC offset of +08:00.
- Australia/Queensland - Timezone for australia/queensland UTC offset of +10:00.
- Australia/South - Timezone for australia/south UTC offset of +10:30.

- Australia/Sydney - Timezone for australia/sydney UTC offset of +11:00.
- Australia/Tasmania - Timezone for australia/tasmania UTC offset of +11:00.
- Australia/Victoria - Timezone for australia/victoria UTC offset of +11:00.
- Australia/West - Timezone for australia/west UTC offset of +08:00.
- Australia/Yancowinna - Timezone for australia/yancowinna UTC offset of +10:30.
- CET - Timezone for CET UTC offset of +01:00.
- CST6CDT - Timezone for CST6CDT UTC offset of -06:00.
- Canada/Atlantic - Timezone for canada/atlantic UTC offset of -04:00.
- Canada/Central - Timezone for canada/central UTC offset of -06:00.
- Canada/East-Saskatchewan - Timezone for canada/east-saskatchewan UTC offset of -06:00.
- Canada/Eastern - Timezone for canada/eastern UTC offset of -05:00.
- Canada/Mountain - Timezone for canada/mountain UTC offset of -07:00.
- Canada/Newfoundland - Timezone for canada/newfoundland UTC offset of -03:30.
- Canada/Pacific - Timezone for canada/pacific UTC offset of -08:00.
- Canada/Saskatchewan - Timezone for canada/saskatchewan UTC offset of -06:00.
- Canada/Yukon - Timezone for canada/yukon UTC offset of -08:00.
- Chile/Continental - Timezone for chile/continental UTC offset of -03:00.
- Chile/EasterIsland - Timezone for chile/easterisland UTC offset of -05:00.
- Cuba - Timezone for cuba UTC offset of -05:00.
- Default - Timezone for system UTC (OS time zone)
- EET - Timezone for EET UTC offset of +02:00.
- EST - Timezone for EST UTC offset of -05:00.
- EST5EDT - Timezone for EST5EDT UTC offset of -05:00.
- Egypt - Timezone for egypt UTC offset of +02:00.
- Eire - Timezone for eire UTC offset of +00:00.
- Etc/Greenwich - Timezone for etc/greenwich UTC offset of +00:00.
- Etc/UCT - Timezone for etc/uct UTC offset of +00:00.
- Etc/UTC - Timezone for etc/utc UTC offset of +00:00.
- Etc/Universal - Timezone for etc/universal UTC offset of +00:00.
- Etc/Zulu - Timezone for etc/zulu UTC offset of +00:00.

- Europe/Amsterdam - Timezone for europe/amsterdam UTC offset of +01:00.
- Europe/Andorra - Timezone for europe/andorra UTC offset of +01:00.
- Europe/Astrakhan - Timezone for europe/astrakhan UTC offset of +04:00.
- Europe/Athens - Timezone for europe/athens UTC offset of +02:00.
- Europe/Belfast - Timezone for europe/belfast UTC offset of +00:00.
- Europe/Belgrade - Timezone for europe/belgrade UTC offset of +01:00.
- Europe/Berlin - Timezone for europe/berlin UTC offset of +01:00.
- Europe/Bratislava - Timezone for europe/bratislava UTC offset of +01:00.
- Europe/Brussels - Timezone for europe/brussels UTC offset of +01:00.
- Europe/Bucharest - Timezone for europe/bucharest UTC offset of +02:00.
- Europe/Budapest - Timezone for europe/budapest UTC offset of +01:00.
- Europe/Busingen - Timezone for europe/busingen UTC offset of +01:00.
- Europe/Chisinau - Timezone for europe/chisinau UTC offset of +02:00.
- Europe/Copenhagen - Timezone for europe/copenhagen UTC offset of +01:00.
- Europe/Dublin - Timezone for europe/dublin UTC offset of +00:00.
- Europe/Gibraltar - Timezone for europe/gibraltar UTC offset of +01:00.
- Europe/Guernsey - Timezone for europe/guernsey UTC offset of +00:00.
- Europe/Helsinki - Timezone for europe/helsinki UTC offset of +02:00.
- Europe/Isle_of_Man - Timezone for europe/isle_of_man UTC offset of +00:00.
- Europe/Istanbul - Timezone for europe/istanbul UTC offset of +03:00.
- Europe/Jersey - Timezone for europe/jersey UTC offset of +00:00.
- Europe/Kaliningrad - Timezone for europe/kaliningrad UTC offset of +02:00.
- Europe/Kiev - Timezone for europe/kyiv UTC offset of +02:00.
- Europe/Kirov - Timezone for europe/kirov UTC offset of +03:00.
- Europe/Lisbon - Timezone for europe/lisbon UTC offset of +00:00.
- Europe/Ljubljana - Timezone for europe/ljubljana UTC offset of +01:00.
- Europe/London - Timezone for europe/london UTC offset of +00:00.
- Europe/Luxembourg - Timezone for europe/luxembourg UTC offset of +01:00.
- Europe/Madrid - Timezone for europe/madrid UTC offset of +01:00.
- Europe/Malta - Timezone for europe/malta UTC offset of +01:00.
- Europe/Mariehamn - Timezone for europe/mariehamn UTC offset of +02:00.

- Europe/Minsk - Timezone for europe/minsk UTC offset of +03:00.
- Europe/Monaco - Timezone for europe/monaco UTC offset of +01:00.
- Europe/Moscow - Timezone for europe/moscow UTC offset of +03:00.
- Europe/Nicosia - Timezone for europe/nicosia UTC offset of +02:00.
- Europe/Oslo - Timezone for europe/oslo UTC offset of +01:00.
- Europe/Paris - Timezone for europe/paris UTC offset of +01:00.
- Europe/Podgorica - Timezone for europe/podgorica UTC offset of +01:00.
- Europe/Prague - Timezone for europe/prague UTC offset of +01:00.
- Europe/Riga - Timezone for europe/riga UTC offset of +02:00.
- Europe/Rome - Timezone for europe/rome UTC offset of +01:00.
- Europe/Samara - Timezone for europe/samara UTC offset of +04:00.
- Europe/San_Marino - Timezone for europe/san_marino UTC offset of +01:00.
- Europe/Sarajevo - Timezone for europe/sarajevo UTC offset of +01:00.
- Europe/Simferopol - Timezone for europe/simferopol UTC offset of +03:00.
- Europe/Skopje - Timezone for europe/skopje UTC offset of +01:00.
- Europe/Sofia - Timezone for europe/sofia UTC offset of +02:00.
- Europe/Stockholm - Timezone for europe/stockholm UTC offset of +01:00.
- Europe/Tallinn - Timezone for europe/tallinn UTC offset of +02:00.
- Europe/Tirane - Timezone for europe/tirane UTC offset of +01:00.
- Europe/Tiraspol - Timezone for europe/tiraspol UTC offset of +02:00.
- Europe/Ulyanovsk - Timezone for europe/ulyanovsk UTC offset of +04:00.
- Europe/Uzhgorod - Timezone for europe/uzhgorod UTC offset of +02:00.
- Europe/Vaduz - Timezone for europe/vaduz UTC offset of +01:00.
- Europe/Vatican - Timezone for europe/vatican UTC offset of +01:00.
- Europe/Vienna - Timezone for europe/vienna UTC offset of +01:00.
- Europe/Vilnius - Timezone for europe/vilnius UTC offset of +02:00.
- Europe/Volgograd - Timezone for europe/volgograd UTC offset of +03:00.
- Europe/Warsaw - Timezone for europe/warsaw UTC offset of +01:00.
- Europe/Zagreb - Timezone for europe/zagreb UTC offset of +01:00.
- Europe/Zaporozhye - Timezone for europe/zaporozhye UTC offset of +02:00.
- Europe/Zurich - Timezone for europe/zurich UTC offset of +01:00.

- GB - Timezone for GB UTC offset of +00:00.
- GB-Eire - Timezone for GB-Eire UTC offset of +00:00.
- GMT - Timezone for GMT UTC offset of +00:00.
- GMT0 - Timezone for GMT0 UTC offset of +00:00.
- Greenwich - Timezone for greenwich UTC offset of +00:00.
- HST - Timezone for HST UTC offset of -10:00.
- Hongkong - Timezone for hongkong UTC offset of +08:00.
- Iceland - Timezone for iceland UTC offset of +00:00.
- Indian/Antananarivo - Timezone for indian/antananarivo UTC offset of +03:00.
- Indian/Chagos - Timezone for indian/chagos UTC offset of +06:00.
- Indian/Christmas - Timezone for indian/christmas UTC offset of +07:00.
- Indian/Cocos - Timezone for indian/cocos UTC offset of +06:30.
- Indian/Comoro - Timezone for indian/comoro UTC offset of +03:00.
- Indian/Kerguelen - Timezone for indian/kerguelen UTC offset of +05:00.
- Indian/Mahe - Timezone for indian/mahe UTC offset of +04:00.
- Indian/Maldives - Timezone for indian/maldives UTC offset of +05:00.
- Indian/Mauritius - Timezone for indian/mauritius UTC offset of +04:00.
- Indian/Mayotte - Timezone for indian/mayotte UTC offset of +03:00.
- Indian/Reunion - Timezone for indian/reunion UTC offset of +04:00.
- Iran - Timezone for iran UTC offset of +03:30.
- Israel - Timezone for israel UTC offset of +02:00.
- Jamaica - Timezone for jamaica UTC offset of -05:00.
- Japan - Timezone for japan UTC offset of +09:00.
- Kwajalein - Timezone for kwajalein UTC offset of +12:00.
- Libya - Timezone for libya UTC offset of +02:00.
- MET - Timezone for MET UTC offset of +01:00.
- MST - Timezone for MST UTC offset of -07:00.
- MST7MDT - Timezone for MST7MDT UTC offset of -07:00.
- Mexico/BajaNorte - Timezone for mexico/bajanorte UTC offset of -08:00.
- Mexico/BajaSur - Timezone for mexico/bajasur UTC offset of -07:00.
- Mexico/General - Timezone for mexico/general UTC offset of -06:00.

- NZ - Timezone for NZ UTC offset of +13:00.
- NZ-CHAT - Timezone for NZ-CHAT UTC offset of +13:45.
- Navajo - Timezone for navajo UTC offset of -07:00.
- PRC - Timezone for PRC UTC offset of +08:00.
- PST8PDT - Timezone for PST8PDT UTC offset of -08:00.
- Pacific/Apia - Timezone for pacific/apia UTC offset of +14:00.
- Pacific/Auckland - Timezone for pacific/auckland UTC offset of +13:00.
- Pacific/Bougainville - Timezone for pacific/bougainville UTC offset of +11:00.
- Pacific/Chatham - Timezone for pacific/chatham UTC offset of +13:45.
- Pacific/Chuuk - Timezone for pacific/chuuk UTC offset of +10:00.
- Pacific/Easter - Timezone for pacific/easter UTC offset of -05:00.
- Pacific/Efate - Timezone for pacific/efate UTC offset of +11:00.
- Pacific/Enderbury - Timezone for pacific/enderbury UTC offset of +13:00.
- Pacific/Fakaofu - Timezone for pacific/fakaofu UTC offset of +13:00.
- Pacific/Fiji - Timezone for pacific/fiji UTC offset of +13:00.
- Pacific/Funafuti - Timezone for pacific/funafuti UTC offset of +12:00.
- Pacific/Galapagos - Timezone for pacific/galapagos UTC offset of -06:00.
- Pacific/Gambier - Timezone for pacific/gambier UTC offset of -09:00.
- Pacific/Guadalcanal - Timezone for pacific/guadalcanal UTC offset of +11:00.
- Pacific/Guam - Timezone for pacific/guam UTC offset of +10:00.
- Pacific/Honolulu - Timezone for pacific/honolulu UTC offset of -10:00.
- Pacific/Johnston - Timezone for pacific/johnston UTC offset of -10:00.
- Pacific/Kiritimati - Timezone for pacific/kiritimati UTC offset of +14:00.
- Pacific/Kosrae - Timezone for pacific/kosrae UTC offset of +11:00.
- Pacific/Kwajalein - Timezone for pacific/kwajalein UTC offset of +12:00.
- Pacific/Majuro - Timezone for pacific/majuro UTC offset of +12:00.
- Pacific/Marquesas - Timezone for pacific/marquesas UTC offset of -09:30.
- Pacific/Midway - Timezone for pacific/midway UTC offset of -11:00.
- Pacific/Nauru - Timezone for pacific/nauru UTC offset of +12:00.
- Pacific/Niue - Timezone for pacific/niue UTC offset of -11:00.
- Pacific/Norfolk - Timezone for pacific/norfolk UTC offset of +11:00.

- Pacific/Noumea - Timezone for pacific/noumea UTC offset of +11:00.
- Pacific/Pago_Pago - Timezone for pacific/pago_pago UTC offset of -11:00.
- Pacific/Palau - Timezone for pacific/palau UTC offset of +09:00.
- Pacific/Pitcairn - Timezone for pacific/pitcairn UTC offset of -08:00.
- Pacific/Pohnpei - Timezone for pacific/pohnpei UTC offset of +11:00.
- Pacific/Ponape - Timezone for pacific/ponape UTC offset of +11:00.
- Pacific/Port_Moresby - Timezone for pacific/port_moresby UTC offset of +10:00.
- Pacific/Rarotonga - Timezone for pacific/rarotonga UTC offset of -10:00.
- Pacific/Saipan - Timezone for pacific/saipan UTC offset of +10:00.
- Pacific/Samoa - Timezone for pacific/samoa UTC offset of -11:00.
- Pacific/Tahiti - Timezone for pacific/tahiti UTC offset of -10:00.
- Pacific/Tarawa - Timezone for pacific/tarawa UTC offset of +12:00.
- Pacific/Tongatapu - Timezone for pacific/tongatapu UTC offset of +14:00.
- Pacific/Truk - Timezone for pacific/truk UTC offset of +10:00.
- Pacific/Wake - Timezone for pacific/wake UTC offset of +12:00.
- Pacific/Wallis - Timezone for pacific/wallis UTC offset of +12:00.
- Pacific/Yap - Timezone for pacific/yap UTC offset of +10:00.
- Poland - Timezone for poland UTC offset of +01:00.
- Portugal - Timezone for portugal UTC offset of +00:00.
- ROC - Timezone for ROC UTC offset of +08:00.
- ROK - Timezone for ROK UTC offset of +09:00.
- Singapore - Timezone for singapore UTC offset of +08:00.
- Turkey - Timezone for turkey UTC offset of +03:00.
- UCT - Timezone for UCT UTC offset of +00:00.
- US/Pacific-New - Timezone for US/Pacific-New UTC offset of -08:00.
- Universal - Timezone for universal UTC offset of +00:00.
- W-SU - Timezone for W-SU UTC offset of +03:00.
- WET - Timezone for WET UTC offset of +00:00.
- Zulu - Timezone for zulu UTC offset of +00:00.
- jdj_tz - Timezone for jdj_tz UTC offset of +00:00.

Default

Default

Name

ueRadioCapabilityCheck - UE Radio Capability Check

Syntax

--ueRadioCapabilityCheck <list-item>

Description

This parameter controls whether the AMF uses NGAP UE Radio Check procedure and the result of the procedure to define whether Voice over PS Sessions shall be supported for the UE on this PLMN.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.109 cmm amfPolicySelection

AMF Policy Selection. Syntax, description, parameters, and roles for cmm amfPolicySelection command.

Syntax

Description

AMF Policy Selection

Parameters

Name

servedPlmnName - served plmn name

Syntax

--servedPlmnName <string>

Description

served plmn name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

entryId - Entry ID

Syntax

--entryId <integer>

Description

Entry ID

Class

Required key.

Type / Range

Integer in the range of 1 to 10,000

Name

amfAmPolicyProfileName - AMF AM Policy Profile Name

Syntax

--amfAmPolicyProfileName <string>

Description

AMF AM Policy Profile Name

Class

Required on create.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

gpsiRangeListName - GPSI Range

Syntax

--gpsiRangeListName <string>

Description

Name of the list consisting of one more GPSI ranges.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

shutdownPolicyProfile - Shutdown Policy Profile

Syntax

--shutdownPolicyProfile <list-item>

Description

Parameter shall be used to indicate if a rule is live or not. Shutdown == true implies the rule is not live and should be skipped

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

snssaiListName - SNSSAI List Name

Syntax

--snssaiListName <string>

Description

SNSSAI List Name

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

supiRangeListName - SUPI Range list

Syntax

--supiRangeListName <string>

Description

Name of the list consisting of SUPI ranges

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Default

""

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.110 cmm amfPraGnb

Presence Reporting Area for gNB. Syntax, description, parameters, and roles for cmm amfPraGnb command.

Syntax**Description**

Administer Presence Reporting Area for the gNB area type for AMF. AMF supports up to 64 gNBs per each CN predefined PRA.

Parameters

Name

amfPraId - AMF Presence Reporting Area Identifier

Syntax

--amfPraId <integer>

Description

An integer that identifies the core network predefined presence reporting area in the AMF.

Class

Required key.

Type / Range

Integer in the range of 8,388,608 to 16,777,215

Name

plmnName - Mobile network name

Syntax

--plmnName <string>

Description

A name that identifies mobile network.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

gnbId - Macro gNB Identifier

Syntax

--gnbId <integer>

Description

The identifier of the macro gNB. Equal to the leftmost bits of the NR Cell Identity IE contained in the NR CGI IE of each cell served by the gNB.

Class

Required key.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

gnbIdLength - GNB Identifier Length

Syntax

--gnbIdLength <integer>

Description

The length of the gNB identifier.

Class

Required on create, not modifiable.

Type / Range

Integer in the range of 22 to 32

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.111 cmm amfPraNrCgi

Presence Reporting Area for NR CGI for AMF. Syntax, description, parameters, and roles for cmm amfPraNrCgi command.

Syntax**Description**

Administer Presence Reporting Area for New Radio Cell Global Identifier area type for AMF. AMF supports up to 64 NR CGIs per each CN predefined PRA.

Parameters**Name**

amfPraId - AMF Presence Reporting Area Identifier

Syntax

--amfPraId <integer>

Description

An integer that identifies the core network predefined presence reporting area in the AMF.

Class

Required key.

Type / Range

Integer in the range of 8,388,608 to 16,777,215

Name

plmnName - Mobile network name

Syntax

--plmnName <string>

Description

A name that identifies mobile network.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

nrCgi - New Radio Cell Global Identifier

Syntax

--nrCgi <integer>

Description

New Radio Cell Global Identifier for the NR CGI area type of the Presence Reporting Area.

Class

Required key.

Type / Range

Integer in the range of 0..68719476735

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.112 cmm amfPraSet

AMF Presence Reporting Area Set. Syntax, description, parameters, and roles for cmm amfPraSet command.

Syntax

Description

Administer AMF Presence Reporting Area set. A Core Network predefined PRA id must be unique in the system. A single PRA id may be contained in multiple PRA sets. PRA set id cannot be contained in the same or a different PRA set (i.e. nested PRA sets are not allowed). A PRA set can contain max 9 PRA ids. When PRA sets are used, only a single PRA set may be simultaneously triggered to be reported for a single UE.

Parameters

Name

amfPraSetId - AMF Presence Reporting Area Set Identifier

Syntax

--amfPraSetId <integer>

Description

An integer that identifies the core network predefined presence reporting area set in the AMF.

Class

Required key.

Type / Range

Integer in the range of 8,388,608 to 16,777,215

Name

amfPraId - AMF Presence Reporting Area Identifier

Syntax

--amfPraId <integer>

Description

An integer that identifies the core network predefined presence reporting area in the AMF.

Class

Required key.

Type / Range

Integer in the range of 8,388,608 to 16,777,215

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.113 cmm amfPraTai

Presence Reporting Area for TAI for AMF. Syntax, description, parameters, and roles for cmm amfPraTai command.

Syntax

Description

Administer Presence Reporting Area for Tracking Area Identity area type for the AMF. AMF supports up to 16 TAIs per each CN predefined PRA.

Parameters

Name

amfPraId - AMF Presence Reporting Area Identifier

Syntax

--amfPraId <integer>

Description

An integer that identifies the core network predefined presence reporting area in the AMF.

Class

Required key.

Type / Range

Integer in the range of 8,388,608 to 16,777,215

Name

plmnName - Mobile network name

Syntax

--plmnName <string>

Description

A name that identifies mobile network.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

amfTac - AMF TAC in the TAI of the PRA

Syntax

--amfTac <integer>

Description

AMF Tracking area code in the tracking area identity area type of the presence reporting area.

Class

Required key.

Type / Range

Integer in the range of 0 to 16,777,215

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.114 cmm amfRanTimers

AMF NG-RAN Timers. Syntax, description, parameters, and roles for cmm amfRanTimers command.

Syntax**Description**

Define NG-RAN timers

Parameters

Name

amfTimerName - Timer name

Syntax

--amfTimerName <list-item>

Description

Name of timer.

Class

Required key.

List

- amfCnfgUpdate - NGAP AMF Configuration Update - When the AMF sends an AMF Configuration Update Request message to the gNB, it expects to receive a response message from the gNB (either AMF Configuration Update Acknowledge or AMF Configuration Update Failure). If the timer expires while waiting for the response, the AMF considers this NGAP Class 1 procedure failed.
- followOnRequest - Pending Follow On Request Timer - AMF maintains the UE N2 connection, after sending Registration Accept to the UE or receiving Registration Complete from the UE, for the period of the timer. The AMF starts the timer if 'Follow on Request pending' is set by the UE in the Registration Request.
- handoverRequest - Handover Request - When the AMF sends a handover request message to the NG-RAN, it expects to receive an acknowledgment response message from the NG-RAN. if the timer expires while waiting for this response, the AMF considers that this class 1 procedure failed.
- initCtxtSetup - NGAP Initial Context Setup - When the AMF sends an Initial Context Setup Request message to the gNB, it expects to receive a response message from the gNB (either Initial Context Setup Complete or Initial Context Setup Failure). if the timer expires while waiting for this response, the AMF considers this NGAP Class 1 procedure failed.
- locationRpCtrl - Timer For location reporting control Message - Guard timer for sending Location reporting control message towards NGRAN
- n1InactivityTimer - N1 NAS inactivity timer - When timer is set, AMF maintains the UE N2 connection after sending Registration Accept to the UE or receiving Registration Complete from the UE. The AMF starts the timer if UE didn't indicate 'Follow on Request pending' in the Registration Request and uplinkStatusPending is not set.
- n26HoResourceRelease - N26 Handover Resource Release - Timer starts at 5GS to EPS handover Source AMF after the AMF sends Forward Relocation Complete Notification ACK. On timer expiry, Source AMF releases all UE resources including N2 connection to the Source gNB

- **n2HoComplete** - N2 Handover Complete - Supervises the receipt of either: - a Handover Notify message from the target gNB, in the case of intra-AMF Handover or inter-AMF Handover (target AMF) or N26 EPS to 5GS Handover (target AMF) - an N2InfoNotify from the target AMF in the case of inter-AMF Handover (source AMF) - a Forward Relocation Complete Notification from the target MME, in the case of N26 5GS to EPS Handover (source AMF) The handover is not successful if the expected message is not received before the expiration of the timer.
- **n2HoResourceRelease** - N2 Handover Resource Release - Timer starts on receipt of Handover Notify if no AMF relocation or on receipt of N2InfoNotify if there is AMF relocation. On timer expiry, Source AMF releases all UE resources including N2 connection to the Source gNB
- **n2Reset** - N2 Reset Timer - AMF sends N2 Reset message to the gNB and waits to receive a Reset Acknowledgment response. If timer expires while waiting for this response, the AMF considers this class 1 procedure failed
- **nasNonDelivery** - NAS Non-Delivery - This timer is started whenever any N1 (NAS) message is sent towards the NG-RAN to be forwarded towards the UE. The AMF maintains the callback URI of the service consumer triggering the Namf_Communication_N1N2MessageTransfer when this timer is running.
- **ngapUeContextModify** - UE Context Modification Timer - This command sets the value of the ue-context-modify-timer which is used to supervise the CONTEXT MODIFICATION REQUEST/CONTEXT MODIFICATION RESPONSE exchange between the AMF and gNB
- **pduRsrcHoFailure** - PDU Session Resource Failure For HO Operations - PDU Session Resource Setup and Modify failures for HO operations. Includes XN-handover-triggered an NG-intra-system-handover-triggered.
- **pduRsrcMod** - PDU Session Resource Modify - When the AMF sends a PDU Session Resource Modify Request message to the gNB, it expects to receive a PDU Session Resource Modify Response from the gNB. if the timer expires while waiting for this response, the AMF considers this NGAP Class 1 procedure failed.
- **pduRsrcRel** - PDU Session Resource Release - When the AMF sends a PDU Session Resource Release Command message to the gNB, it expects to receive a PDU Session Resource Release Response from the gNB. if the timer expires while waiting for this response, the AMF considers this NGAP Class 1 procedure failed.
- **pduRsrcSetup** - PDU Session Resource Setup - When the AMF sends a PDU Session Resource Setup Request message to the gNB, it expects to receive a PDU Session Resource Setup Response from the gNB. if the timer expires while waiting for this response, the AMF considers this NGAP Class 1 procedure failed.
- **srvcc5gHoComplete** - SRVCC 5G HO Complete - Defines the time the AMF waits for 5G-SRVCC handover completion i.e. the receipt of N26 Forward Relocation Complete Notification from MME_SRVCC. The 5G-SRVCC handover is not considered successful if this message is not received before the expiration of the timer.

- ueCtxtRel - NGAP UE Context Release - When the AMF sends a UE Context Release Command message to the gNB, it expects to receive a UE Release Complete as a response. If the timer expires while waiting for this response, the AMF considers this NGAP Class 1 procedure failed.
- ueRadioCapabilityCheck - NGAP UE Radio Capability Check Response - Specifies the time the AMF waits for NGAP UE Radio Capability Check Response after it has sent NGAP UE Radio Capability Check Request. If timer expires while waiting for this response, the AMF considers this class 1 procedure failed.
- v2xPc5Guard - V2X PC5 Guard - Specifies how long the AMF is to keep the NAS connection open during N1 registration when the UE is authorized for V2X PC5 service. Values range from 0 to 5 seconds in increments of 100ms.

Name

amfTimerUnit - Timer unit

Syntax

--amfTimerUnit <list-item>

Description

Unit of measurement.

Class

Required on create, not modifiable.

List

- Millisecond - Specify timer unit of 1 millisecond
- Second - Specify timer unit of 1 second
- 2Second - 2 Second - Specify timer unit of 2 seconds
- 30Second - 30 Second - Specify timer unit of 30 seconds.
- Minute - Specify timer unit of 1 minute
- DeciHour - Specify timer unit of 1 decihour (6 minutes).
- 10Minute - 10 Minute - Specify timer unit of 10 minutes
- Hour - Specify timer unit of 1 hour
- 10Hour - 10 Hour - Specify timer unit of 10 hours.

Name

amfTimerValue - Timer value

Syntax

--amfTimerValue <integer>

Description

Numeric value of timer.

Class

Required on create.

Type / Range

Integer in the range of 0 to 65,535

The amfTimerValue string is based on the amfTimerName parameter:

Name

amfCnfgUpdate - NGAP AMF Configuration Update

Syntax

```
--amfTimerName amfCnfgUpdate --amfTimerValue <string> [ --  
amfTimerUnit Second ]
```

Description

When the AMF sends an AMF Configuration Update Request message to the gNB, it expects to receive a response message from the gNB (either AMF Configuration Update Acknowledge or AMF Configuration Update Failure). If the timer expires while waiting for the response, the AMF considers this NGAP Class 1 procedure failed.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

5

Name

followOnRequest - Pending Follow On Request Timer

Syntax

```
--amfTimerName followOnRequest --amfTimerValue <string> [ --  
amfTimerUnit Second ]
```

Description

AMF maintains the UE N2 connection, after sending Registration Accept to the UE or receiving Registration Complete from the UE, for the period of the timer. The AMF starts the timer if 'Follow on Request pending' is set by the UE in the Registration Request.

Type / Range

Integer in the range of 0 to 25 in units of Second

Default

15

Name

handoverRequest - Handover Request

Syntax

```
--amfTimerName handoverRequest --amfTimerValue <string> [ --  
amfTimerUnit Second ]
```


Description

When the AMF sends a handover request message to the NG-RAN, it expects to receive an acknowledgment response message from the NG-RAN. if the timer expires while waiting for this response, the AMF considers that this class 1 procedure failed.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

8

Name

initCtxtSetup - NGAP Initial Context Setup

Syntax

```
--amfTimerName initCtxtSetup --amfTimerValue <string> [ --  
amfTimerUnit Second ]
```

Description

When the AMF sends an Initial Context Setup Request message to the gNB, it expects to receive a response message from the gNB (either Initial Context Setup Complete or Initial Context Setup Failure). if the timer expires while waiting for this response, the AMF considers this NGAP Class 1 procedure failed.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

5

Name

locationRpCtrl - Timer For location reporting control Message

Syntax

```
--amfTimerName locationRpCtrl --amfTimerValue <string> [ --  
amfTimerUnit Second ]
```

Description

Guard timer for sending Location reporting control message towards NGRAN

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

5

Name

n1InactivityTimer - N1 NAS inactivity timer

Syntax

```
--amfTimerName n1InactivityTimer --amfTimerValue <string> [ --  
amfTimerUnit Second ]
```

Description

When timer is set, AMF maintains the UE N2 connection after sending Registration Accept to the UE or receiving Registration Complete from the UE. The AMF starts the timer if UE didn't indicate 'Follow on Request pending' in the Registration Request and uplinkStatusPending is not set.

Type / Range

Integer in the range of 0 to 10 in units of Second

Default

0

Name

n26HoResourceRelease - N26 Handover Resource Release

Syntax

```
--amfTimerName n26HoResourceRelease --amfTimerValue <string> [ --  
amfTimerUnit Millisecond ]
```

Description

Timer starts at 5GS to EPS handover Source AMF after the AMF sends Forward Relocation Complete Notification ACK. On timer expiry, Source AMF releases all UE resources including N2 connection to the Source gNB

Type / Range

Integer in the range of 500 to 15,000 in units of Millisecond

Default

2000

Name

n2HoComplete - N2 Handover Complete

Syntax

```
--amfTimerName n2HoComplete --amfTimerValue <string> [ --amfTimerUnit  
Second ]
```

Description

Supervises the receipt of either: - a Handover Notify message from the target gNB, in the case of intra-AMF Handover or inter-AMF Handover (target AMF) or N26 EPS to 5GS Handover (target AMF) - an N2InfoNotify from the target AMF in the case of inter-AMF Handover (source AMF) - a Forward Relocation Complete Notification from the target MME, in the case of N26 5GS to EPS Handover (source AMF) The handover is not successful if the expected message is not received before the expiration of the timer.

Type / Range

Integer in the range of 1 to 15 in units of Second

Default

5

Name

n2HoResourceRelease - N2 Handover Resource Release

Syntax

```
--amfTimerName n2HoResourceRelease --amfTimerValue <string> [ --  
amfTimerUnit Millisecond ]
```

Description

Timer starts on receipt of Handover Notify if no AMF relocation or on receipt of N2InfoNotify if there is AMF relocation. On timer expiry, Source AMF releases all UE resources including N2 connection to the Source gNB

Type / Range

Integer in the range of 500 to 5,000 in units of Millisecond

Default

2000

Name

n2Reset - N2 Reset Timer

Syntax

```
--amfTimerName n2Reset --amfTimerValue <string> [ --amfTimerUnit  
Second ]
```

Description

AMF sends N2 Reset message to the gNB and waits to receive a Reset Acknowledgment response. If timer expires while waiting for this response, the AMF considers this class 1 procedure failed

Type / Range

Integer in the range of 1 to 180 in units of Second

Default

1

Name

nasNonDelivery - NAS Non-Delivery

Syntax

```
--amfTimerName nasNonDelivery --amfTimerValue <string> [ --  
amfTimerUnit Millisecond ]
```

Description

This timer is started whenever any N1 (NAS) message is sent towards the NG-RAN to be forwarded towards the UE. The AMF maintains the callback URI of the service consumer triggering the Namf_Communication_N1N2MessageTransfer when this timer is running.

Type / Range

Integer in the range of 0 to 10,000 in units of Millisecond

Default

1000

Name

ngapUeContextModify - UE Context Modification Timer

Syntax

```
--amfTimerName ngapUeContextModify --amfTimerValue <string> [ --  
amfTimerUnit Second ]
```

Description

This command sets the value of the ue-context-modify-timer which is used to supervise the CONTEXT MODIFICATION REQUEST/CONTEXT MODIFICATION RESPONSE exchange between the AMF and gNB

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

8

Name

pduRsrcHoFailure - PDU Session Resource Failure For HO Operations

Syntax

```
--amfTimerName pduRsrcHoFailure --amfTimerValue <string> [ --  
amfTimerUnit Millisecond ]
```

Description

PDU Session Resource Setup and Modify failures for HO operations. Includes XN-handover-triggered an NG-intra-system-handover-triggered.

Type / Range

Integer range from 300 to 2000 in units of Millisecond

Default

500

Name

pduRsrcMod - PDU Session Resource Modify

Syntax

```
--amfTimerName pduRsrcMod --amfTimerValue <string> [ --amfTimerUnit  
Second ]
```

Description

When the AMF sends a PDU Session Resource Modify Request message to the gNB, it expects to receive a PDU Session Resource Modify Response from the gNB. If the timer expires while waiting for this response, the AMF considers this NGAP Class 1 procedure failed.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

5

Name

pduRsrcRel - PDU Session Resource Release

Syntax

```
--amfTimerName pduRsrcRel --amfTimerValue <string> [ --amfTimerUnit  
Second ]
```

Description

When the AMF sends a PDU Session Resource Release Command message to the gNB, it expects to receive a PDU Session Resource Release Response from the gNB. If the timer expires while waiting for this response, the AMF considers this NGAP Class 1 procedure failed.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

5

Name

pduRsrcSetup - PDU Session Resource Setup

Syntax

```
--amfTimerName pduRsrcSetup --amfTimerValue <string> [ --amfTimerUnit  
Second ]
```

Description

When the AMF sends a PDU Session Resource Setup Request message to the gNB, it expects to receive a PDU Session Resource Setup Response from the gNB. If the timer expires while waiting for this response, the AMF considers this NGAP Class 1 procedure failed.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

5

Name

srvcc5gHoComplete - SRVCC 5G HO Complete

Syntax

```
--amfTimerName srvcc5gHoComplete --amfTimerValue <string> [ --  
amfTimerUnit Second ]
```

Description

Defines the time the AMF waits for 5G-SRVCC handover completion i.e. the receipt of N26 Forward Relocation Complete Notification from MME_SRVCC. The 5G-SRVCC handover is not considered successful if this message is not received before the expiration of the timer.

Type / Range

Integer in the range of 1 to 15 in units of Second

Default

5

Name

ueCtxtRel - NGAP UE Context Release

Syntax

```
--amfTimerName ueCtxtRel --amfTimerValue <string> [ --amfTimerUnit  
Second ]
```

Description

When the AMF sends a UE Context Release Command message to the gNB, it expects to receive a UE Release Complete as a response. If the timer expires while waiting for this response, the AMF considers this NGAP Class 1 procedure failed.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

5

Name

ueRadioCapabilityCheck - NGAP UE Radio Capability Check Response

Syntax

```
--amfTimerName ueRadioCapabilityCheck --amfTimerValue <string> [ --  
amfTimerUnit Millisecond ]
```

Description

Specifies the time the AMF waits for NGAP UE Radio Capability Check Response after it has sent NGAP UE Radio Capability Check Request. If timer expires while waiting for this response, the AMF considers this class 1 procedure failed.

Type / Range

Integer in the range of 100 to 10,000 in units of Millisecond

Default

8000

Name

v2xPc5Guard - V2X PC5 Guard

Syntax

```
--amfTimerName v2xPc5Guard --amfTimerValue <string> [ --amfTimerUnit  
Millisecond ]
```

Description

Specifies how long the AMF is to keep the NAS connection open during N1 registration when the UE is authorized for V2X PC5 service. Values range from 0 to 5 seconds in increments of 100ms.

Type / Range

Integer in the range of 0 to 5,000 in units of Millisecond

Default

0

Roles

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

3.115 cmm amfRegionServingTai

TAI served by AMF in an AMF Region. Syntax, description, parameters, and roles for cmm amfRegionServingTai command.

Syntax

Description

This command used to provision list of TAI served by AMF in an AMF region.

Parameters

Name

amfRegionId - AMF Region Identifier

Syntax

```
--amfRegionId <integer>
```

Description

AMF Region Identifier.

Class

Required key.

Type / Range

Integer in the range of 0 to 255

Name

plmnName - PLMN Name

Syntax

--plmnName <string>

Description

PLMN name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

tac - Tracking Area Code

Syntax

--tac <integer>

Description

Tracking Area Code.

Class

Required key.

Type / Range

Integer in the range of 0 to 16777215

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.116 cmm amfRfspAccessRestrictionList

RFSP Access Restriction List for the AMF. Syntax, description, parameters, and roles for cmm amfRfspAccessRestrictionList command.

Syntax

Description

Creates list names to be used for an RFSP Access Restriction List

Parameters

Name

name - Name

Syntax

--name <string>

Description

The name of the RFSP Access Restriction List.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.117 cmm amfRfspAccessRestrictionListItem

RFSP AMF Access Restriction List Item. Syntax, description, parameters, and roles for cmm amfRfspAccessRestrictionListItem command.

Syntax

Description

The RFSP value to assign to a RFSP AMF Access Restriction List.

Parameters

Name

amfRfspAccessRestrictionListName - RFSP AMF Access Restriction List Item

Syntax

--amfRfspAccessRestrictionListName <string>

Description

RFSP AMF Access Restriction List Item

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

amfRfspValue - RFSP Value

Syntax

--amfRfspValue <integer>

Description

Value of the RFSP to be assigned to a list.

Class

Required key.

Type / Range

Integer in the range of 1 to 256

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.118 cmm amfRfspLocalOverrideProfile

AMF RFSP Local Override Profile. Syntax, description, parameters, and roles for cmm amfRfspLocalOverrideProfile command.

Syntax

Description

Define local override profile for RAT Frequency Selection Priority (RFSP) index. One or multiple parameters can be defined, either any parameter in the profile defined set is required to match or all parameters in the set are required to match. Parameters that can be defined in the profile include -allowed S-NSSAI list -subscribed RFSP index (i.e. UDM RFSP) -UE's usage setting Max 128 profiles can be created. As a result of create and modify commands, at least one of S-NSSAI list, UDM RFSP and UE's Usage setting must be set, i.e. the result cannot be that the S-NSSAI list name is blank when also UDM RFSP is set to zero and UE's Usage Setting is set to notUsed.

Parameters

Name

name - Name

Syntax

--name <string>

Description

The name of the AMF RFSP Local Override Profile.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

matchLevel - Match Level

Syntax

--matchLevel <list-item>

Description

Whether all of the parameters or any of the provided parameters is/are required to match in order to apply the override profile.

Class

Required on create.

List

- anyMatch - Any Match - Any of the parameters provided in the AMF RFSP Override RFSP profile is required to match to apply the override.
- fullMatch - Full Match - All of the parameters provided in the AMF RFSP Override RFSP profile are required to match to apply the override.

Name

overrideRfsp - Override RFSP

Syntax

--overrideRfsp <integer>

Description

The value of the RFSP used by the AMF towards the other NW elements when the override applies.

Class

Required on create.

Type / Range

Integer in the range of 1 to 256

Name

amfSimpleRfspOverride - AMF Simple RFSP Override

Syntax

--amfSimpleRfspOverride <list-item>

Description

Defines the AMF simple UDM RFSP override policy. This parameter takes precedence over AMF RFSP Local Override Profile included parameters Match Level, Override Level, Allowed S-NSSAI List Name, UDM RFSP and UE's Usage Setting, i.e. the AMF ignores those parameters when this parameter is set to other value than Disabled.

Class

Optional.

List

- disabled - AMF simple RFSP override is disabled.
- amfOverrideUdmAlways - AMF Override UDM Always - AMF applies AMF locally provisioned RFSP index independently of the UDM provided RFSP index.
- amfOverrideNoUdmRfsp - AMF Override No UDM RFSP - AMF uses AMF locally provisioned RFSP index when UDM does not provide any RFSP index value.

Default

disabled

Name

overrideLevel - Override Level

Syntax

--overrideLevel <list-item>

Description

Defines the NW elements from which the AMF may receive the RFSP and override it with the AMF locally provisioned RFSP value.

Class

Optional.

List

- amfRfspOverridePcf - AMF RFSP Override PCF - The AMF overrides PCF provided RFSP value.
- amfRfspOverrideUdm - AMF RFSP Override UDM - The AMF overrides UDM provided RFSP value.
- amfRfspOverrideUdmPcf - AMF RFSP Override UDM PCF - The AMF overrides UDM and PCF provided RFSP values.

Default

amfRfspOverrideUdm

Name

snssaiListName - Allowed S-NSSAI List Name

Syntax

--snssaiListName <string>

Description

The name of the allowed S-NSSAI list to trigger the override. If the list is empty, the S-NSSAI list is not used in the override profile. If any of the allowed S-NSSAIs for the UE is present in the list, the S-NSSAI list is considered to match.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

""

Name

udmRfsp - UDM RFSP

Syntax

--udmRfsp <integer>

Description

The value of the UDM subscribed RFSP index to trigger the AMF override. 0 value stands for not used.

Class

Optional.

Type / Range

Integer in the range of 0 to 256

Default

0

Name

uesUsageSetting - UE's Usage Setting

Syntax

--uesUsageSetting <list-item>

Description

The UE's usage setting.

Class

Optional.

List

- dataCentric - Data Centric - UE's Usage setting is Data Centric.
- none - UE's usage setting is not provided.
- notUsed - Not Used - The UE's usage setting is not used.
- voiceCentric - Voice Centric - UE's Usage setting is Voice Centric.

Default

notUsed

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.119 cmm amfServiceIp

AMF Service IP. Syntax, description, parameters, and roles for cmm amfServiceIp command.

Syntax

Description

This command is used to administer a Service IP and includes parameters for defining AMF N2 and SB interfaces. For Multiple IPDS, the Service IPs have to be defined on all IPDS pools for the following interfaces: - N2 and - SBI and - N26

Parameters

Name

name - End Point Name

Syntax

--name <string>

Description

End point name of N2 interface or Service Based Interface.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

assignmentType - Assignment type

Syntax

--assignmentType <list-item>

Description

Assignment type: fixed or floating

Class

Optional.

List

- fixed - Fixed.
- floating - Floating.

Default

floating

Name

interfaceN2Name - N2 Interface Name

Syntax

--interfaceN2Name <list-item>

Description

This parameter defines the name used for N2 Interfaces (single homed and multi homed) when FDS is used.

Class

Optional.

List

- N2INTF2 - N2 INTF2 - This is used for the N2 Secondary Multi Homed Interface with SFDS
- N2INTF2_2 - N2 INTF2_2 - This is used for the second N2 Secondary Multi Homed Interface with SFDS
- N2INTF3_1 - N2 INTF3_1 - This is used for the third N2 Primary Multi Homed Interface with SFDS
- N2INTF3_2 - N2 INTF3_2 - This is used for the third N2 Secondary Multi Homed Interface with SFDS
- N2SH_2 - N2 SH_2 - This is used for the second N2 Single Homing with SFDS
- N2SH_3 - N2 SH_3 - This is used for the third N2 Single Homing with SFDS
- N2INTF1 - N2 INTF1 - This is used for the N2 Primary Multi Homed Interface with SFDS
- N2INTF2_1 - N2 INTF2_1 - This is used for the second N2 Primary Multi Homed Interface with SFDS
- N2SH - N2 SH - This is used for N2 Single Homing with SFDS
- NONE - When SFDS is not used - Default

Default

NONE

Name

poolId - Pool id

Syntax

--poolId <integer>

Description

Pool identification number.

Class

Optional.

Type / Range

Integer in the range of 0 to 39

Default

0

Name

poolMemberId - Pool member ID

Syntax

--poolMemberId <integer>

Description

Pool member identification number.

Class

Optional.

Type / Range

Integer in the range of 0 to 2

Default

0

Name

poolType - Pool type

Syntax

--poolType <list-item>

Description

Pool (service) type.

Class

Optional.

List

- ALMS - Alarm Service - Alarm Service.
- CPPS - Control plane processing service - Control plane processing service.
- CTCS - Call Trace Collection service - Call Trace Collection service.
- DBS - Database service - Database service.
- EEMS - Event Exposure Management service - Event Exposure Management service.
- IPDS - IP director service - IP director service.
- IPPS - IP packet processing service - IP packet processing service.
- LIHS - Lawful Interception Handling service
- MLIS - Machine Learning Intelligence Service
- NECC - Network element cluster controller - Network element cluster controller.

- NWDS - Network Data Service - NWDAF service
- PAPS - Packet processing service - Packet processing service.
- PCMS - Per Call Measurements Data Service - Per Call Measurements Data (PCMD) Service
- RCIS - Radio Capability Information storage Service
- SFDS - SCTP flow distributor service
- UCDS - User equipment Context Data-dump Service
- UFDS - UDP flow distributor service

Default

IPDS

Name

svclpV4 - Service IP address V4

Syntax

--svcIPv4 <string>

Description

Service IP address V4

Class

Optional.

Type / Range

Digit string of length 7 to 15 with the allowed pattern of [0-9\.]*

Default

0.0.0.0

Name

svclpV6 - Service IP address V6

Syntax

--svcIPv6 <string>

Description

Service IP address V6

Class

Optional.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of '[0-9a-f:]*'

Default

::

Name

useFds - Use Flow Distributor

Syntax

--useFds <list-item>

Description

Indicates whether this service IP shall be used as back-end for single-ip flow distributor

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.120 cmm amfServicesEndPointConfig

AMF Services End Point Configuration. Syntax, description, parameters, and roles for cmm amfServicesEndPointConfig command.

Syntax**Description**

This command is used to create, modify and delete AMF local endpoint configuration for the AMF supported services in the home network and AMFs in the shared network.

Parameters

Name

endPointName - AMF Local End Point Interface Name

Syntax

--endPointName <string>

Description

The parameter points to a set of parameters related with IP address configuration created using the 'endPointName' command.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

amfServiceInstanceName - AMF Service Instance Name

Syntax

--amfServiceInstanceName <string>

Description

Specifies the service instance of the own AMF for which the configuration applies. Each AMF Service Instance can be mapped to only one amfServicesEndPointConfig

Class

Required on create, not modifiable.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

amfServiceIpName - Service IP Name

Syntax

--amfServiceIpName <string>

Description

The parameter points to a set of parameters related with IP address configuration created using the 'serviceIp' command.

Class

Required on create.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

dscpCode - DSCP Code

Syntax

--dscpCode <list-item>

Description

Parameter to set DSCP code to be used for an IP packet.

Class

Optional.

List

- AF11 - Assured forwarding 11.
- AF12 - Assured forwarding 12.
- AF13 - Assured forwarding 13.
- AF21 - Assured forwarding 21.
- AF22 - Assured forwarding 22.
- AF23 - Assured forwarding 23.
- AF31 - Assured forwarding 31.
- AF32 - Assured forwarding 32.
- AF33 - Assured forwarding 33.
- AF41 - Assured forwarding 41.
- AF42 - Assured forwarding 42.
- AF43 - Assured forwarding 43.
- BE - BE.
- CS1 - Class selector 1.
- CS2 - Class selector 2.
- CS3 - Class selector 3.
- CS4 - Class selector 4.
- CS5 - Class selector 5.
- CS6 - Class selector 6.
- CS7 - Class selector 7.
- EF - Expedited forwarding.
- voiceAdmit - Voice Admit

Default

AF41

Name

httpProfileName - HTTP profile name

Syntax

--httpProfileName <string>

Description

Assign HTTP profile for the end point.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

default

Name

port - TCP Port

Syntax

--port <integer>

Description

TCP Port.

Class

Optional.

Type / Range

Integer in the range of 1 to 65,535

Default

8080

Name

tcpProfileName - TCP Profile Name

Syntax

--tcpProfileName <string>

Description

Configure TCP profile to the endpoint.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

default

Name

tlsProfileName - Transport Layer Security Certificate

Syntax

--tlsProfileName <string>

Description

Configure TLS certificate used in establishing secure communication.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.121 cmm amfSetSnssai

AMF set supported S-NSSAI. Syntax, description, parameters, and roles for cmm amfSetSnssai command.

Syntax**Description**

Command to provision list of S-NSSAI supported by an AMF set.

Parameters**Name**

plmnName - PLMN Name

Syntax

--plmnName <string>

Description

PLMN ID of the AMF set.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

amfRegionId - AMF Region Identifier

Syntax

--amfRegionId <integer>

Description

AMF Region ID of the AMF set.

Class

Required key.

Type / Range

Integer in the range of 0 to 255

Name

amfSetId - AMF Set Identifier

Syntax

--amfSetId <integer>

Description

AMF set identifier

Class

Required key.

Type / Range

5G AMF Set ID

Name

snssaiListName - SNSSAI List

Syntax

--snssaiListName <string>

Description

List of S-NSSAI supported by the AMF set.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.122 cmm amfSmfNasMapping

SMF NAS mapping. Syntax, description, parameters, and roles for cmm amfSmfNasMapping command.

Syntax**Description**

Defines mapping from N11 HTTP Status code and protocol/application error code to 5GMM NAS cause code. A default profile, named 'default', is provided.

Parameters**Name**

smfNasMappingListName - SMF Error To NAS 5GMM Mapping Name

Syntax

--smfNasMappingListName <string>

Description

The name of the SMF error to NAS 5GMM mapping.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

procedure - Procedure

Syntax

--procedure <list-item>

Description

Defines the procedure to apply the SMF error code into 5GMM cause code mapping.

Class

Required key.

List

- ULNAS - UL NAS Transport - UL NAS Transport procedure.
- PDU_REACT_RESULT - PDU Reactivation Result - PDU Reactivation Result error code used on Registration Accept and Service Accept.

Name

smfError - SMF Error

Syntax

--smfError <list-item>

Description

Defines the SMF error to map the 5GMM cause into.

Class

Required key.

List

- 400_RESOURCE_CONTEXT_NOT_FOUND - 400 RESOURCE CONTEXT NOT FOUND - The notification request is rejected because the call back URI still exists in the receiver of the notification, but the specific resource context identified within the notification payload is not found in the NF service consumer.
- 400_UNKNOWN - 400 UNKNOWN - The 400_UNKNOWN is used when included problem details is not specified in 3GPP or it is not included.
- 403_CCA_VERIFICATION_FAILURE - 403 CCA VERIFICATION FAILURE - The request is rejected due to a failure to verify the CCA at the receiving entity (SMF).
- 403_DEFAULT_EBI_NOT_TRANSFERRED - 403 DEFAULT EBI NOT TRANSFERRED - It is used during 5GS to EPS mobility if the EBI of the default EPS bearer is included in the notToTransferEbiList attribute.
- 403_LATE_OVERLAPPING_REQUEST - The request is rejected because it collides with an existing SM context or PDU session context with a more recent origination timestamp.
- 403_UNKNOWN - 403 UNKNOWN - The 403_UNKNOWN is used when included problem details is not specified in 3GPP or it is not included.
- 404_UNKNOWN - 404 UNKNOWN - The 404_UNKNOWN is used when included problem details is not specified in 3GPP or it is not included.

- 409_HIGHER_PRIORITY_REQUEST_ONGOING - 409 HIGHER PRIORITY REQUEST ONGOING - The request is rejected temporarily due to procedure for higher priority session in progress.
- 409_UE_IN_CM_IDLE_STATE - UE IN CM IDLE STATE - The request is rejected due to the UE being in CM-IDLE state for the PDU session associated to non-3GPP access.
- 409_UNKNOWN - 409 UNKNOWN - The 409_UNKNOWN is used when included problem details is not specified in 3GPP or it is not included.
- 411_UNKNOWN - 411 UNKNOWN - The 411_UNKNOWN is used when included problem details is not specified in 3GPP or it is not included.
- 429_UNKNOWN - 429 UNKNOWN - The 429_UNKNOWN is used when included problem details is not specified in 3GPP or it is not included.
- 500_NF_FAILOVER - 500 NF FAILOVER - The request is rejected due to the unavailability of the NF (SMF), and the requester may trigger an immediate re-selection of an alternative NF based on this information.
- 500_NF_SERVICE_FAILOVER - The request is rejected due to the unavailability of the NF service, and the requester may trigger an immediate re-selection of an alternative NF service based on this information.
- 500_UNKNOWN - 500 UNKNOWN - The 500_UNKNOWN is used when included problem details is not specified in 3GPP or it is not included.
- 503_UNKNOWN - 503 UNKNOWN - The 503_UNKNOWN is used when included problem details is not specified in 3GPP or it is not included.
- 504_TARGET_NF_NOT_REACHABLE - The request is not served as the target NF (SMF) is not reachable.
- 504_TIMED_OUT_REQUEST - 504 TIMED OUT REQUEST - The request is rejected due a request that has timed out at the HTTP client based on origination timestamp.
- 504_UE_NOT_REACHABLE - 504 UE NOT REACHABLE - The UE is not reachable for service.
- 504_UNKNOWN - 504 UNKNOWN - The 504_UNKNOWN is used when included problem details is not specified in 3GPP or it is not included.
- 504_UPF_NOT_RESPONDING - 504 UPF NOT RESPONDING - The request is rejected due to no response received from the UPF.
- DEFAULT - This item defines the default entry to apply when no specific entry for an HTTP status code and protocol/application error is provisioned.
- 400_INVALID_API - 400 Bad Request Invalid API - HTTP status code 400 Bad Request, protocol or application error INVALID_API.
- 400_INVALID_MSG_FORMAT - 400 Bad Request Invalid Message Format - HTTP status code 400 Bad Request, protocol or application error INVALID_MSG_FORMAT.

- 400_INVALID_QUERY_PARAM - 400 Bad Request Invalid Query Parameter - HTTP status code 400 Bad Request, protocol or application error INVALID_QUERY_PARAM.
- 400_MANDATORY_QUERY_PARAM_INCORRECT - 400 Mandatory Query Param Incorrect - HTTP status code 400 Bad Request, protocol or application error MANDATORY_QUERY_PARAM_INCORRECT
- 400_OPTIONAL_QUERY_PARAM_INCORRECT - 400 Optional Query Param Incorrect - HTTP status code 400 Bad Request, protocol or application error OPTIONAL_QUERY_PARAM_INCORRECT
- 400_MANDATORY_QUERY_PARAM_MISSING - 400 Mandatory Query Param Missing - HTTP status code 400 Bad Request, protocol or application error MANDATORY_QUERY_PARAM_MISSING
- 400_MANDATORY_IE_INCORRECT - 400 Bad Request Mandatory IE Incorrect - HTTP status code 400 Bad Request, protocol or application error MANDATORY_IE_INCORRECT.
- 400_OPTIONAL_IE_INCORRECT - 400 Optional IE Incorrect - HTTP status code 400 Bad Request, protocol or application error OPTIONAL_IE_INCORRECT
- 400_MANDATORY_IE_MISSING - 400 Bad Request Mandatory IE Missing - HTTP status code 400 Bad Request, protocol or application error MANDATORY_IE_MISSING.
- 400_UNSPECIFIED_MSG_FAILURE - 400 Bad Request Unspecified Message Failure - HTTP status code 400 Bad Request, protocol or application error UNSPECIFIED_MSG_FAILURE.
- 403_MODIFICATION_NOT_ALLOWED - 403 Forbidden Modification Not Allowed - HTTP status code 403 Forbidden, protocol or application error MODIFICATION_NOT_ALLOWED.
- 403_N1_SM_ERROR - 403 Forbidden N1 SM Error - HTTP status code 403 Forbidden, application error N1_SM_ERROR.
- 403_SNSSAI_DENIED - 403 Forbidden S-NSSAI Denied - HTTP status code 403 Forbidden, application error SNSSAI_DENIED.
- 403_DNN_DENIED - 403 Forbidden DNN Denied - HTTP status code 403 Forbidden, application error DNN_DENIED.
- 403_PDUTYPE_DENIED - 403 Forbidden PDU Type Denied - HTTP status code 403 Forbidden, application error PDUTYPE_DENIED.
- 403_SSC_DENIED - 403 Forbidden SSC Denied - HTTP status code 403 Forbidden, application error SSC_DENIED.
- 403_SUBSCRIPTION_DENIED - 403 Forbidden Subscription Denied - HTTP status code 403 Forbidden, application error SUBSCRIPTION_DENIED.
- 403_DNN_NOT_SUPPORTED - 403 Forbidden DNN Not Supported - HTTP status code 403 Forbidden, application error DNN_NOT_SUPPORTED.

- 403_PDUTYPE_NOT_SUPPORTED - 403 Forbidden PDU Type Not Supported - HTTP status code 403 Forbidden, application error PDUTYPE_NOT_SUPPORTED.
- 403_SSC_NOT_SUPPORTED - 403 Forbidden SSC Not Supported - HTTP status code 403 Forbidden, application error SSC_NOT_SUPPORTED.
- 403_HOME_ROUTED_ROAMING_REQUIRED - 403 Forbidden Home Routed Roaming Required - HTTP status code 403 Forbidden, application error HOME_ROUTED_ROAMING_REQUIRED.
- 403_OUT_OF_LADN_SERVICE_AREA - 403 Forbidden Out of LADN Service Area - HTTP status code 403 Forbidden, application error OUT_OF_LADN_SERVICE_AREA.
- 403_N2_SM_ERROR - 403 Forbidden N2 SM Error - HTTP status code 403 Forbidden, application error N2_SM_ERROR.
- 403_PRIORITIZED_SERVICES_ONLY - 403 Forbidden Prioritized Services Only - HTTP status code 403 Forbidden, application error PRIORITIZED_SERVICES_ONLY.
- 403_PDU_SESSION_ANCHOR_CHANGE - 403 Forbidden PDU Session Anchor Change - HTTP status code 403 Forbidden, application error PDU_SESSION_ANCHOR_CHANGE.
- 403_TARGET_MME_CAPABILITY - 403 Forbidden Target MME Capability - HTTP status code 403 Forbidden, application error TARGET_MME_CAPABILITY.
- 403_NO_EPS_5GS_CONTINUITY - 403 Forbidden No EPS 5GS Continuity - HTTP status code 403 Forbidden, application error NO_EPS_5GS_CONTINUITY.
- 403_UNABLE_TO_PAGE_UE - 403 Forbidden Unable to Page UE - HTTP status code 403 Forbidden, application error UNABLE_TO_PAGE_UE.
- 403_UE_NOT_RESPONDING - 403 Forbidden UE Not Responding - HTTP status code 403 Forbidden, application error UE_NOT_RESPONDING.
- 403_REJECTED_BY_UE - 403 Forbidden Rejected by UE - HTTP status code 403 Forbidden, application error REJECTED_BY_UE.
- 403_REJECTED_DUE_VPLMN_POLICY - 403 Forbidden Rejected Due VPLMN Policy - HTTP status code 403 Forbidden, application error REJECTED_DUE_VPLMN_POLICY.
- 403_HO_TAU_IN_PROGRESS - 403 Forbidden HO TAU in Progress - HTTP status code 403 Forbidden, application error HO_TAU_IN_PROGRESS.
- 403_INTEGRITY_PROTECTED_MDR_NOT_ACCEPTABLE - 403 Forbidden Integrity Protected MDR not Acceptable - HTTP status code 403 Forbidden, application error INTEGRITY_PROTECTED_MDR_NOT_ACCEPTABLE.
- 403_EBI_EXHAUSTED - 403 Forbidden EBI Exhausted - HTTP status code 403 Forbidden, application error EBI_EXHAUSTED.
- 403_EBI_REJECTED_LOCAL_POLICY - 403 Forbidden EBI Rejected Local Policy - HTTP status code 403 Forbidden, application error EBI_REJECTED_LOCAL_POLICY.

- 403_EBI_REJECTED_NO_N26 - 403 Forbidden EBI Rejected No N26 - HTTP status code 403 Forbidden, application error EBI_REJECTED_NO_N26.
- 403_DEFAULT_EPS_BEARER_INACTIVE - 403 Default EPS Bearer Inactive - HTTP status code 403 Forbidden, application error DEFAULT_EPS_BEARER_INACTIVE.
- 403_HANDOVER_RESOURCE_ALLOCATION_FAILURE - 403 Forbidden Handover Resource Allocation Failure - HTTP status code 403 Forbidden, application error HANDOVER_RESOURCE_ALLOCATION_FAILURE.
- 403_NOT_SUPPORTED_WITH_ISMF - 403 Forbidden Not Supported With I-SMF - HTTP status code 403 Forbidden, application error NOT_SUPPORTED_WITH_ISMF.
- 403_SERVICE_NOT_AUTHORIZED_BY_NEXT_HOP - 403 Forbidden Service Not Authorized By Next Hop - HTTP status code 403 Forbidden, application error SERVICE_NOT_AUTHORIZED_BY_NEXT_HOP.
- 404_SUBSCRIPTION_NOT_FOUND - 404 Not Found Subscription Not Found - HTTP status code 404 Not Found, protocol or application error SUBSCRIPTION_NOT_FOUND.
- 404_RESOURCE_URI_STRUCTURE_NOT_FOUND - 404 Not Found Resource URI Structure Not Found - HTTP status code 404 Not Found, protocol or application error RESOURCE_URI_STRUCTURE_NOT_FOUND.
- 404_CONTEXT_NOT_FOUND - 404 Not Found Context Not Found - HTTP status code 404 Not Found, application error CONTEXT_NOT_FOUND.
- 411_INCORRECT_LENGTH - 411 Length Required Incorrect Length - HTTP status code 411 Length Required, protocol or application error INCORRECT_LENGTH.
- 429_NF_CONGESTION_RISK - 429 Too Many Requests NF Congestion Risk - HTTP status code 429 Too Many Requests, protocol or application error NF_CONGESTION_RISK.
- 500_INSUFFICIENT_RESOURCES - 500 Internal Server Error Insufficient Resources - HTTP status code 500 Internal Server Error, protocol or application error INSUFFICIENT_RESOURCES.
- 500_UNSPECIFIED_NF_FAILURE - 500 Internal Server Error Unspecified NF Failure - HTTP status code 500 Internal Server Error, protocol or application error UNSPECIFIED_NF_FAILURE.
- 500_SYSTEM_FAILURE - 500 Internal Server Error System Failure - HTTP status code 500 Internal Server Error, protocol or application error SYSTEM_FAILURE.
- 500_INSUFFICIENT_RESOURCES_SLICE - 500 Internal Server Error Insufficient Resources Slice - HTTP status code 500 Internal Server Error, application error INSUFFICIENT_RESOURCES_SLICE.
- 500_INSUFFICIENT_RESOURCES_SLICE_DNN - 500 Internal Server Error Insufficient Resources Slice DNN - HTTP status code 500 Internal Server Error, application error INSUFFICIENT_RESOURCES_SLICE_DNN.

- 503_NF_CONGESTION - 503 Service Unavailable NF Congestion - HTTP status code 503 Service Unavailable, protocol or application error NF_CONGESTION.
- 503_DNN_CONGESTION - 503 Service Unavailable DNN Congestion - HTTP status code 503 Service Unavailable, application error DNN_CONGESTION.
- 503_S_NSSAI_CONGESTION - 503 Service Unavailable S-NSSAI Congestion - HTTP status code 503 Service Unavailable, application error S-NSSAI_CONGESTION.
- 504_PEER_NOT_RESPONDING - 504 Gateway Timeout Peer Not Responding - HTTP status code 504 Gateway Timeout, application error PEER_NOT_RESPONDING.
- 504_NETWORK_FAILURE - 504 Gateway Timeout Network Failure - HTTP status code 504 Gateway Timeout, application error NETWORK_FAILURE.

Name

amf5gMmCc - 5GMM Cause Code

Syntax

--amf5gMmCc <integer>

Description

Defines the 5GMM cause code to use towards the UE for the given procedure case and the given SMF error (HTTP status code and application/protocol).

Class

Optional.

Type / Range

5GMM Cause Code. 0 stands for no cause to be provided

Name

errorSource - Error Source

Syntax

--errorSource <list-item>

Description

Defines the source of the SMF error according to AMF analysis of received error based on the Via and the Server headers. Only applied when amfGParm timerSmfReselection is set to 'true'.

Class

Optional.

List

- onlySmf - Only SMF - The SMF is the only used SMF for the PDU.
- ismfVsmf - I-SMF or V-SMF - The SMF is either in the role of an Intermediate SMF (I-SMF) or a visited SMF (V-SMF) for the PDU.

- remoteSmf - Remote SMF - Remote SMF is either the actual SMF (when also I-SMF is included on the path) or the H-SMF of the PDU. Note: remote SMF is only applicable when either I-SMF is added on the path, or home routed roaming is applied for the PDU.

Default

onlySmf

Name

n1Included - N1 Message Included

Syntax

--n1Included <list-item>

Description

Defines whether the SMF provided error response included a NAS Session Management message to be delivered back to the UE. Only applied when amfGParm timerSmfReselection is set to 'true'.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

reselectIsmfVsmf - Reselect I-SMF V-SMF

Syntax

--reselectIsmfVsmf <list-item>

Description

Defines whether the AMF for a UE triggered request overrides the provisioned NAS cause code mapping for the provisioned SMF error and instead of returning a NAS error code towards the UE the AMF attempts to reselect the I-SMF or the V-SMF during the same procedure. This parameter is also used to define whether the AMF reselects an I-SMF or a V-SMF on other procedures after an I-SMF or V-SMF reported error. Only applied when amfGParm timerSmfReselection is set to 'true'. Note: on the current release the AMF only reselects an SMF after receiving an error for an SM context create for an initially selected SMF.

Class

Optional.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Name

reselectOnlySmf - Reselect Only SMF

Syntax

--reselectOnlySmf <list-item>

Description

Defines whether the AMF for a UE triggered request overrides the provisioned NAS cause code mapping for the provisioned SMF error and instead of returning a NAS error code towards the UE the AMF attempts to reselect the only SMF on the path for the PDU during the same procedure. Only applied when amfGParm timerSmfReselection is set to 'true'. Note: on the current release the AMF only reselects an SMF after receiving an error for an SM context create for an initially selected SMF.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

reselectRemote - Reselect Remote

Syntax

--reselectRemote <list-item>

Description

Defines whether the AMF for a UE triggered request overrides the provisioned NAS cause code mapping for the provisioned SMF error and instead of returning a NAS error code towards the UE the AMF attempts to reselect the remote SMF (i.e. the actual SMF when also I-SMF is used for the PDU, or the H-SMF) during the same procedure. Only applied when amfGParm timerSmfReselection is set to 'true'. Note: on the current release the AMF only reselects an SMF after receiving an error for an SM context create for an initially selected SMF.

Class

Optional.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

3.123 cmm amfSmfNasMappingList

SMF Error to NAS Mapping Profile List. Syntax, description, parameters, and roles for cmm amfSmfNasMappingList command.

Syntax**Description**

SMF Error to NAS Mapping Profile List.

Parameters**Name**

name - SMF error to NAS 5GMM mapping list name

Syntax

--name <string>

Description

SMF error to NAS 5GMM mapping list name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.124 cmm amfSmpToPagingPriorityProfile

AMF SMP to Paging Priority Level. Syntax, description, parameters, and roles for cmm amfSmpToPagingPriorityProfile command.

Syntax

Description

Defines the mapping from incoming SBI Message Priority (SMP) to a Paging Priority Level. Altogether 32 profiles may be defined on AMF.

Parameters

Name

amfSmpToPagingPriorityProfileListName - Name

Syntax

--amfSmpToPagingPriorityProfileListName <string>

Description

The name of the SMP to Paging priority level profile.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

sbiMessagePriority - SBI Message Priority

Syntax

--sbiMessagePriority <integer>

Description

Defines the SBI Message priority (SMP) value to map into Paging Priority Level.

Class

Required key.

Type / Range

Integer in the range of 0 to 23

Name

prioType - Priority Type

Syntax

--prioType <list-item>

Description

Defines the priority type.

Class

Required on create.

List

- amfPrioTypeMcs - AMF Priority Type MCS - Priority type Mission Critical Services.
- amfPrioTypeMps - AMF Priority Type MPS - Priority type Multimedia Priority Service.

Name

includePagingPriority - Include Paging Priority

Syntax

--includePagingPriority <list-item>

Description

Defines whether the AMF includes the Paging Priority parameter on the NGAP Paging message. When Paging Priority is not provisioned to be included, AMF applies the associated priority internally but does not include the Paging Priority parameter in the NGAP Paging message.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

pagingPriLevel - Paging Priority Level

Syntax

--pagingPriLevel <list-item>

Description

Defines the Paging Priority Level parameter.

Class

Optional.

List

- PrioLevel0 - Priority level 0 - Priority level 0. No paging priority is set.
- PrioLevel1 - Priority level 1 - Priority level 1. Highest Paging priority.
- PrioLevel2 - Priority level 2 - Priority level 2.
- PrioLevel3 - Priority level 3 - Priority level 3.
- PrioLevel4 - Priority level 4 - Priority level 4.
- PrioLevel5 - Priority level 5 - Priority level 5.
- PrioLevel6 - Priority level 6 - Priority level 6.
- PrioLevel7 - Priority level 7 - Priority level 7.
- PrioLevel8 - Priority level 8 - Priority level 8. Lowest Paging Priority.

Default

PrioLevel8

Name

prioDuration - Priority Duration

Syntax

--prioDuration <list-item>

Description

Defines the time the AMF provides priority treatment for the UE, when a matching SMP is used from other SBI NF.

Class

Optional.

List

- amfPrioDurationReg - AMF Prio Duration Current Registration - AMF provides priority for the UE during the current registration of the UE.
- amfPrioDurationTimer - AMF Priority Duration Timer - AMF provides priority for the UE during the time the uplift timer is running.

Default

amfPrioDurationTimer

Name

upliftTimer - Uplift Timer

Syntax

--upliftTimer <integer>

Description

The timer used to temporarily uplift the priority of the UE to the incoming request SMP value level.

Class

Optional.

Type / Range

Integer in the range of 1 to 900 in units of Second

Default

120

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.125 cmm amfSmpToPagingPriorityProfileList

AMF SMP to Paging Priority Level List. Syntax, description, parameters, and roles for cmm amfSmpToPagingPriorityProfileList command.

Syntax**Description**

Used to administer a AMF SMP to Paging Priority Level List. Altogether 32 lists may be defined on AMF, max 32 profiles per list.

Parameters**Name**

name - Name

Syntax

--name <string>

Description

Defines the SBI Message Priority (SMP) to a Paging Priority Level profile.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.126 cmm amfSmsfNasMapping

AMF SMSF NAS mapping. Syntax, description, parameters, and roles for cmm amfSmsfNasMapping command.

Syntax

Description

Defines mapping from N20 HTTP Status code and SMSF returned protocol/application error code to 5GMM NAS cause code. When no cause is provisioned in the mapping, AMF performs SMSF reselection. A default profile, named 'default', is provided. The maximum number of profiles supported is 128.

Parameters

Name

smsfNasMappingListName - SMSF Error To NAS 5GMM Mapping List Name

Syntax

--smsfNasMappingListName <string>

Description

The name of the SMSF (N20) error to NAS 5GMM mapping.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

smsfError - SMSF Error

Syntax

--smsfError <list-item>

Description

Defines the SMSF error (HTTP status code and protocol/application error) to map the 5GMM cause code into.

Class

Required key.

List

- InvalidApi400 - 400 Bad Request Invalid API - HTTP status code 400 Bad Request, protocol or application error INVALID_API.
- InvalidMessageFormat400 - 400 Bad Request Invalid Message Format - HTTP status code 400 Bad Request, protocol or application error INVALID_MSG_FORMAT.
- InvalidQueryParameter400 - 400 Bad Request Invalid Query Parameter - HTTP status code 400 Bad Request, protocol or application error INVALID_QUERY_PARAM.
- MandatoryQueryParamIncorrect400 - 400 Bad Request Mandatory Query Param Incorrect - HTTP status code 400 Bad Request, protocol or application error MANDATORY_QUERY_PARAM_INCORRECT.
- OptionalQueryParamIncorrect400 - 400 Bad Request Optional Query Param Incorrect - HTTP status code 400 Bad Request, protocol or application error OPTIONAL_QUERY_PARAM_INCORRECT.
- MandatoryQueryParamMissing400 - 400 Bad Request Mandatory Query Param Missing - HTTP status code 400 Bad Request, protocol or application error MANDATORY_QUERY_PARAM_MISSING.
- MandatoryIeIncorrect400 - 400 Bad Request Mandatory IE Incorrect - HTTP status code 400 Bad Request, protocol or application error MANDATORY_IE_INCORRECT.
- OptionalIeIncorrect400 - 400 Bad Request Optional IE Incorrect - HTTP status code 400 Bad Request, protocol or application error OPTIONAL_IE_INCORRECT.
- MandatoryIeMissing400 - 400 Bad Request Mandatory IE Missing - HTTP status code 400 Bad Request, protocol or application error MANDATORY_IE_MISSING.
- UnspecifiedMessageFailure400 - 400 Bad Request Unspecified Message Failure - HTTP status code 400 Bad Request, protocol or application error UNSPECIFIED_MSG_FAILURE.

- **SmsPayloadMissing400** - 400 Bad Request SMS Payload Missing - HTTP status code 400 Bad Request, application error SMS_PAYLOAD_MISSING.
- **SmsPayloadError400** - 400 Bad Request SMS Payload Error - HTTP status code 400 Bad Request, application error SMS_PAYLOAD_ERROR.
- **ModificationNotAllowed403** - 403 Forbidden Modification Not Allowed - HTTP status code 403 Forbidden, protocol or application error MODIFICATION_NOT_ALLOWED.
- **ServiceNotAllowed403** - 403 Forbidden Service Not Allowed - HTTP status code 403 Forbidden, application error SERVICE_NOT_ALLOWED.
- **RoamingNotAllowed403** - 403 Roaming Not Allowed - HTTP status code 403 Forbidden, protocol or application error ROAMING_NOT_ALLOWED.
- **SubscriptionNotFound404** - 404 Not Found Subscription Not Found - HTTP status code 404 Not Found, protocol or application error SUBSCRIPTION_NOT_FOUND.
- **ResourceUriStructureNotFound404** - 404 Resource URI Structure Not Found - HTTP status code 404 Not Found, protocol or application error RESOURCE_URI_STRUCTURE_NOT_FOUND.
- **UserNotFound404** - 404 Not Found User Not Found - HTTP status code 404 Not Found, application error USER_NOT_FOUND.
- **ContextNotFound404** - 404 Not Found Context Not Found - HTTP status code 404 Not Found, application error CONTEXT_NOT_FOUND.
- **IncorrectLength411** - 411 Length Required Incorrect Length - HTTP status code 411 Length Required, protocol or application error INCORRECT_LENGTH.
- **NfCongestionRisk429** - 429 Too Many Requests NF Congestion Risk - HTTP status code 429 Too Many Requests, protocol or application error NF_CONGESTION_RISK.
- **InsufficientResources500** - 500 Internal Server Error Insufficient Resources - HTTP status code 500 Internal Server Error, protocol or application error INSUFFICIENT_RESOURCES.
- **UnspecifiedNfFailure500** - 500 Internal Server Error Unspecified NF Failure - HTTP status code 500 Internal Server Error, protocol or application error UNSPECIFIED_NF_FAILURE.
- **SystemFailure500** - 500 Internal Server Error System Failure - HTTP status code 500 Internal Server Error, protocol or application error SYSTEM_FAILURE.
- **NfCongestion503** - 503 Service Unavailable NF Congestion - HTTP status code 503 Service Unavailable, protocol or application error NF_CONGESTION.
- **CcaVerificationFailure403** - 403 Cca Verification Failure - The request is rejected due to a failure to verify the CCA at the receiving entity (SMSF).
- **NfFailover500** - 500 Nf Failover - The request is rejected due to the unavailability of the NF (SMSF), and the requester may trigger an immediate re-selection of an alternative NF based on this information.

- NfServiceFailover500 - 500 Nf Service Failover - The request is rejected due to the unavailability of the NF service, and the requester may trigger an immediate re-selection of an alternative NF service based on this information.
- ResourceContextNotFound400 - 400 Resource Context Not Found - The notification request is rejected because the call back URI still exists in the receiver of the notification, but the specific resource context identified within the notification payload is not found in the NF service consumer.
- TargetNfNotReachable504 - 504 Target Nf Not Reachable - The request is not served as the target NF (SMSF) is not reachable.
- TimedOutRequest504 - 504 Timed Out Request - The request is rejected due a request that has timed out at the HTTP client based on origination timestamp.
- Unknown400 - 400 Unknown - The 400 Unknown is used when included problem details is not specified in 3GPP or not included.
- Unknown403 - 403 Unknown - The 403 Unknown is used when included problem details is not specified in 3GPP or not included.
- Unknown404 - 404 Unknown - The 404 Unknown is used when included problem details is not specified in 3GPP or not included.
- Unknown411 - 411 Unknown - The 411 Unknown is used when included problem details is not specified in 3GPP or not included.
- Unknown429 - 429 Unknown - The 429 Unknown is used when included problem details is not specified in 3GPP or not included.
- Unknown500 - 500 Unknown - The 500 Unknown is used when included problem details is not specified in 3GPP or not included.
- Unknown503 - 503 Unknown - The 503 Unknown is used when included problem details is not specified in 3GPP or not included.
- Unknown504 - 504 Unknown - The 504 Unknown is used when included problem details is not specified in 3GPP or not included.

Name

amf5gMmCc - 5GMM Cause Code

Syntax

--amf5gMmCc <integer>

Description

Defines the 5GMM cause code to use towards the UE on NAS DL NAS Transport message as the 5GMM cause code for the given SMSF error (HTTP status code and application/protocol). If no 5GMM cause code is defined i.e. 5GMM cause code is 0, the AMF triggers SMSF reselection for the UE.

Class

Optional.

Type / Range

5GMM Cause Code. 0 stands for no cause to be provided

Default

0

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

3.127 cmm amfSmsfNasMappingList

AMF SMSF NAS Mapping List. Syntax, description, parameters, and roles for cmm amfSmsfNasMappingList command.

Syntax**Description**

SMFS (N20) Error to NAS Mapping Profile List.

Parameters**Name**

name - SMSF error to NAS 5GMM mapping list name

Syntax

--name <string>

Description

SMSF (N20) error to NAS 5GMM mapping list name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.128 cmm amfStormLimitsGParms

AMF Storm Limits. Syntax, description, parameters, and roles for cmm amfStormLimitsGParms command.

Syntax

Description

This command is used to administer Global Parameters related to AMF Storm Limits in the CMM.

Parameters

Name

amfGparmName - Parameter Name

Syntax

--amfGparmName <list-item>

Description

Name of parameter.

Class

Required key.

List

- initRegistrationRateLimitOverride - Initial Registrations Rate Limit Override - Number of allowed Initial Registrations per sec per IPDS after which the AMF starts dropping the requests until the rate is below limit. Value 0 means System Calculated, System Default. Non-zero means use this value directly
- initRegistrationRateRejectLimitOverride - Initial Registration Rate Reject Limit Override - Number of allowed Initial Registrations per sec per CPPS after which the AMF starts rejecting the requests until the rate is below limit. Value 0 means System Calculated, System Default. Non-zero means use this value directly
- irat4gTo5gTauRateDropLimitOverride - IRAT 4G to 5G TAU Rate Drop Limit Override - 4G to 5G IRAT TAUs allowed per sec after which AMF start dropping 4G to 5G IRAT TAUs

until rate is below limit. Value of 0 means use system calculated Default value. Non-zero means use this value directly

- **n1n2MessageTransferLimitOverride** - N1N2 Message Transfer Limit Override - Number of allowed N1N2 Message Transfer Requests for Idle UEs that initiate a new paging procedure per sec per CPPS after which the AMF will respond with HTTP/2 status code 503 with a possible retry-after timer. Value 0 means use system calculated Default value. Non-zero means use this value directly.
- **serviceReqRateDropLimitOverride** - Service Request Rate Drop Limit Override - Number of allowed 5G UE initiated Service Requests beyond which AMF/CMM will start to drop them on the IPDS. Value of 0 means use system calculated Default value. Non-zero means use this value directly.
- **serviceReqRateLimitOverride** - Service Request Rate Limit Override - Number of UE initiated service requests allowed per sec per CPPS after which CPPS starts dropping service requests until rate is below limit. Value 0 means System Calculated, System Default. Non-zero means use this value directly.
- **tcpOpenRateLimitOverride** - TCP Open Rate Limit Override - Number of allowed TCP open events per sec per IPDS after which the AMF starts dropping the requests until the rate is below limit. Value 0 means the feature is off. Non-zero value is the limit.

Name

amfGparmValue - Parameter Value

Syntax

```
--amfGparmValue <string>
```

Description

Value of parameter.

Class

Required on create.

Type / Range

Digit string of length 0 to 64

The amfGparmValue string is based on the amfGparmName parameter:

Name

initRegistrationRateLimitOverride - Initial Registrations Rate Limit Override

Syntax

```
--amfGparmName initRegistrationRateLimitOverride --amfGparmValue  
<string>
```

Description

Number of allowed Initial Registrations per sec per IPDS after which the AMF starts dropping the requests until the rate is below limit. Value 0 means System Calculated, System Default. Non-zero means use this value directly

Type / Range

Integer in the range of 0 to 9,999

Default

0

Name

initRegistrationRateRejectLimitOverride - Initial Registration Rate Reject Limit Override

Syntax

```
--amfGparmName initRegistrationRateRejectLimitOverride --  
amfGparmValue <string>
```

Description

Number of allowed Initial Registrations per sec per CPPS after which the AMF starts rejecting the requests until the rate is below limit. Value 0 means System Calculated, System Default. Non-zero means use this value directly

Type / Range

Integer in the range of 0 to 9,999

Default

0

Name

irat4gTo5gTauRateDropLimitOverride - IRAT 4G to 5G TAU Rate Drop Limit Override

Syntax

```
--amfGparmName irat4gTo5gTauRateDropLimitOverride --amfGparmValue  
<string>
```

Description

4G to 5G IRAT TAUs allowed per sec after which AMF start dropping 4G to 5G IRAT TAUs until rate is below limit. Value of 0 means use system calculated Default value. Non-zero means use this value directly

Type / Range

Integer in the range of 0 to 14,999

Default

0

Name

n1n2MessageTransferLimitOverride - N1N2 Message Transfer Limit Override

Syntax

```
--amfGparmName n1n2MessageTransferLimitOverride --amfGparmValue  
<string>
```

Description

Number of allowed N1N2 Message Transfer Requests for Idle UEs that initiate a new paging procedure per sec per CPPS after which the AMF will respond with HTTP/2 status code 503 with a possible retry-after timer. Value 0 means use system calculated Default value. Non-zero means use this value directly.

Type / Range

Integer in the range of 0 to 9,999

Default

0

Name

serviceReqRateDropLimitOverride - Service Request Rate Drop Limit Override

Syntax

```
--amfGparmName serviceReqRateDropLimitOverride --amfGparmValue  
<string>
```

Description

Number of allowed 5G UE initiated Service Requests beyond which AMF/CMM will start to drop them on the IPDS. Value of 0 means use system calculated Default value. Non-zero means use this value directly.

Type / Range

Integer in the range of 0 to 24,999

Default

0

Name

serviceReqRateLimitOverride - Service Request Rate Limit Override

Syntax

```
--amfGparmName serviceReqRateLimitOverride --amfGparmValue <string>
```

Description

Number of UE initiated service requests allowed per sec per CPPS after which CPPS starts dropping service requests until rate is below limit. Value 0 means System Calculated, System Default. Non-zero means use this value directly.

Type / Range

Integer in the range of 0 to 6,999

Default

0

Name

tcpOpenRateLimitOverride - TCP Open Rate Limit Override

Syntax

```
--amfGparmName tcpOpenRateLimitOverride --amfGparmValue <string>
```

Description

Number of allowed TCP open events per sec per IPDS after which the AMF starts dropping the requests until the rate is below limit. Value 0 means the feature is off. Non-zero value is the limit.

Type / Range

Integer in the range of 0, 75 to 150

Default

0

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

3.129 cmm amfSubDataSetsProfile

AMF Subscription DataSets Profile. Syntax, description, parameters, and roles for cmm amfSubDataSetsProfile command.

Syntax**Description**

Defines subscription datasets that amf can choose to combine when retrieving UE subscription data from the UDM. The maximum number of profiles supported is 128. A default profile, named 'default', is provided.

Parameters**Name**

name - Profile name

Syntax

--name <string>

Description

Identifies, via name, a given AMF Subscription DataSets Profile.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

accessMobility - Access Mobility DataSet

Syntax

--accessMobility <list-item>

Description

UE access mobility subscription data

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

amfRetrieveSubMultipleDataSets - AMF Retrieves Subscription Multiple Data

Syntax

--amfRetrieveSubMultipleDataSets <list-item>

Description

Determines whether AMF retrieves combined Datasets or not. When amfRetrieveSubMultipleDataSets is active(true) for each additional parameter (smfSel,smsSub,uecSmf,uecSmsf) will be included in a single combined message if active (set to true) or excluded (if set to false) and sent as a separate query message to UDM

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

smfSel - SMF Selection Subscription Data

Syntax

--smfSel <list-item>

Description

UE SMF Selection Subscription Data

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

smsSub - SMS Subscription Data

Syntax

--smsSub <list-item>

Description

SMS Subscription Data, when SMS feature is enabled

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

traceDataFromUdm - Trace Data From UDM

Syntax

--traceDataFromUdm <list-item>

Description

Trace Data From UDM when UDM Initiated Call Trace or MDT is enabled.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

uecSmf - UE Context in SMF DataSet

Syntax

--uecSmf <list-item>

Description

UE Context in SMF Data

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

uecSmsf - UE Context in SMSF

Syntax

--uecSmsf <list-item>

Description

UE Context in SMSF, when SMS is enabled

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

v2x - V2X

Syntax

--v2x <list-item>

Description

Specifies whether the v2x data type is to be requested from UDM

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.130 cmm amfSupiRangeServices

Services based on Supi Range within UE Network. Syntax, description, parameters, and roles for cmm amfSupiRangeServices command.

Syntax**Description**

This command is used to administer services based on Supi Range within UE Network.

Parameters**Name**

amfUePlmnServicesName - UE PLMN services name

Syntax

--amfUePlmnServicesName <string>

Description

Name of a UE PLMN services record

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

entryId - ENTRY ID

Syntax

--entryId <integer>

Description

ENTRY ID

Class

Required key.

Type / Range

Integer in the range of 1 to 10,000

Name

aliasDnnConversionProfileName - AMF Alias DNN Conversion Profile Name

Syntax

--aliasDnnConversionProfileName <string>

Description

Pointer to the Alias DNN Conversion Profile

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

default

Name

amfAccessRestrictionNasMappingProfileName - Name of access restriction to NAS cause code profile name

Syntax

--amfAccessRestrictionNasMappingProfileName <string>

Description

Profile name for access restriction to NAS cause code mapping.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

default

Name

amfAusfHttp2ToNasCcMappingListName - AUSF HTTP2 Status Code to 5GMM NAS Profile List Name

Syntax

--amfAusfHttp2ToNasCcMappingListName <string>

Description

The name of the AUSF HTTP2 Status Code to 5GMM NAS Cause Code mapping profile list.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

default

Name

amfDnnOverrideProfileName - AMF Dnn Override Profile Name

Syntax

--amfDnnOverrideProfileName <string>

Description

Pointer to the AMF DNN Override Profile

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

default

Name

amfEirOverrideProfileName - AMF EIR Override Profile Name

Syntax

--amfEirOverrideProfileName <string>

Description

Indicates the profile name used during Eir check.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z_\\-:@]*

Default

""

Name

amfEplmnListName - EPLMN list name

Syntax

--amfEplmnListName <string>

Description

Identifies Equivalent PLMN list name

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 31 with the allowed pattern of [0-9a-zA-Z_\\-:@]*

Default

""

Name

amfEplmnListNameGnb - EPLMN list name for gNB

Syntax

--amfEplmnListNameGnb <string>

Description

Identifies Equivalent PLMN list name to be sent to gNB. Both gNB and UE Eplmn lists have to be populated if the list is to be sent to both. Content of both lists could be same or different based on operator choice.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 31 with the allowed pattern of [0-9a-zA-Z_\\-:@]*

Default

""

Name

amfMcsPagingPriProfileListName - AMF MCS Paging Priority Profile List Name

Syntax

--amfMcsPagingPriProfileListName <string>

Description

Name that identifies the AMF MCS Paging Priority Profile List.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

amfNrDualConnectivityProfileName - NR Dual Connectivity Profile Name

Syntax

--amfNrDualConnectivityProfileName <string>

Description

Identifies NR Dual Connectivity Profile Name.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

amfNrfHttp2ToNasCcMappingListName - NRF HTTP2 Status Code to 5GMM NAS Profile List Name

Syntax

--amfNrfHttp2ToNasCcMappingListName <string>

Description

The name of the NRF HTTP2 Status Code to 5GMM NAS Cause Code mapping profile list.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

default

Name

amfRfspAccessRestrictionListName - Access Restriction for RFSP List Name

Syntax`--amfRfspAccessRestrictionListName <string>`**Description**

A reference to a list of RFSP values for which access is to be restricted for the UE.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

amfRfspLocalOverrideProfileName - AMF RFSP Local Override Profile Name

Syntax`--amfRfspLocalOverrideProfileName <string>`**Description**

The name of the AMF RFSP Local Override Profile Name for the UE.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

amfSmpToPagingPriorityProfileListName - AMF SMP to Paging Priority Profile Name

Syntax`--amfSmpToPagingPriorityProfileListName <string>`**Description**

The name of the profile to map SBI message Priority (SMP) to Paging Priority Level.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

amfSubDataSetsProfileName - Subscription Data Sets Profile Name

Syntax

--amfSubDataSetsProfileName <string>

Description

Identifies subscription Data Sets profile name for retrieving UE subscription data.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

default

Name

amfSvcAgreementProfileName - Service agreement profile name

Syntax

--amfSvcAgreementProfileName <string>

Description

Defines the PLMN service agreement profile to be used for service.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

default

Name

amfTaiListName - TAI Area List Name

Syntax

--amfTaiListName <string>

Description

Defines the area in which the service agreement profile applies for the SUPI range.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

""

Name

amfUdmHttp2ToNasCcMappingListName - UDM HTTP2 Status Code to 5GMM NAS Profile List Name

Syntax

--amfUdmHttp2ToNasCcMappingListName <string>

Description

The name of the UDM HTTP2 Status Code to 5GMM NAS Cause Code mapping profile list.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

default

Name

amfUeMrlForbiddenTaiListName - Mobility Restriction TAI List Name

Syntax

--amfUeMrlForbiddenTaiListName <string>

Description

Name of the Forbidden TAI List in the Mobility Restriction List sent to the gNB(s).

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

""

Name

amfUeNitzInfoOverrideProfileName - UE Nitz Information Override Profile Name

Syntax

--amfUeNitzInfoOverrideProfileName <string>

Description

Indicates the profile name used for the UE Nitz information.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

"

Name

amfUePriorityProfileName - AMF UE Priority Profile Name

Syntax

--amfUePriorityProfileName <string>

Description

The name of the UE Priority Profile

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

areaCodeExtendedEmergencyNumListName - Area Code Extended Emergency Number List Name

Syntax

--areaCodeExtendedEmergencyNumListName <string>

Description

Extended Emergency Number List Name

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

cag - Closed Access Group

Syntax

--cag <list-item>

Description

A control parameter to enable Closed Access Group (CAG) feature.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

defaultVsnssaiName - Default Vsnssai Name

Syntax

--defaultVsnssaiName <string>

Description

The default VPLMN Slice used when there is no matching VPLMN slice to the UE requested HPLMN Slice.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

dnnMccOverride - DNN MCC Override

Syntax

--dnnMccOverride <digits>

Description

Override of DNN domain derivation by using this mobility country code and the mobility network code represented by dnnMccOverride.

Class

Optional.

Type / Range

Digit string of length 0 or 3

Default

""

Name

dnnMncOverride - DNN MNC Override

Syntax

--dnnMncOverride <digits>

Description

Override of DNN domain derivation by using this mobility network code and the mobility country code represented by dncMncOverride.

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 3

Default

""

Name

edrxNrAllowed - eDRX Allowed

Syntax

--edrxNrAllowed <list-item>

Description

Specifies whether UEs from this PLMN are allowed to use eDRX.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Recommended

false

Name

edrxNrDnnListName - eDRX DNN List Name

Syntax

--edrxNrDnnListName <string>

Description

Specifies name of a List of DNNs. A UE is allowed to use eDRX only if it is using at least one DNN in the list.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Default

"

Name

edrxNrProfileName - eDRX Profile Name

Syntax

--edrxNrProfileName <string>

Description

Specifies an eDRX profile that is used for negotiation of eDRX parameters with a UE.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Recommended

"

Name

edrxNrRedcapProfileName - eDRX NR Redcap Profile Name

Syntax

--edrxNrRedcapProfileName <string>

Description

Specifies an eDRX profile that is used for negotiation of eDRX parameters with a UE when the UE is using NR Redcap.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Recommended

"

Name

eirCheck - EIR Check

Syntax

--eirCheck <list-item>

Description

This command enables EIR check for subscriber(s) IMEISV based on Supi range local configuration rule(s)

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

emcNon3gppN3iwf - EMC Non-3GPP Access N3IWF

Syntax

--emcNon3gppN3iwf <list-item>

Description

Defines whether Emergency Services are supported for the UE over untrusted non-3GPP access via N3IWF.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

emcSupport - EMC Support

Syntax

--emcSupport <list-item>

Description

Defines whether Emergency Services are supported for the UE. It applies for a home UE or a roamer UE.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

emfSupport - EMF Support

Syntax

--emfSupport <list-item>

Description

Defines whether Emergency Fallback is supported for the UE. It applies for a home UE or a roamer UE.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

enabled - Enabled

Syntax

--enabled <list-item>

Description

Enable entry

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

extendedEmergencyNumListName - Extended Emergency Number List Name

Syntax

--extendedEmergencyNumListName <string>

Description

Extended Emergency Number List Name

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

getAccMobDataCag - Get Access and Mobility Data Due to CAG Enabling

Syntax

--getAccMobDataCag <list-item>

Description

A control parameter to enable requesting again for Access and Mobility Subscription Data in intra-AMF Mobility and Periodic Registration due to CAG feature enabling

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

hlcomAllowed - HLCOM Allowed

Syntax

--hlcomAllowed <list-item>

Description

Specifies whether an eDRX UE from this PLMN is allowed to use High Latency Communication.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Recommended

false

Name

hlcomAllowedMico - HLCOM Allowed MICO

Syntax

--hlcomAllowedMico <list-item>

Description

Specifies whether the AMF will use MICO procedures, if provisioned and requested by the SMF, for UEs from this SUPI Range.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

hplmnToVplmnSnssaiMappingListName - H-PLMN to V-PLMN Snssai Mapping List Name

Syntax

--hplmnToVplmnSnssaiMappingListName <string>

Description

Identifies allowed Home PLMN to Visited PLMN Snssai Mapping list name.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 31 with the allowed pattern of [0-9a-zA-Z_\.\\-:@]*

Default

""

Name

ignoreAusfFailuresOptionalReAuthentications - Ignore AUSF Failures on Optional Re-Authentications

Syntax

--ignoreAusfFailuresOptionalReAuthentications <list-item>

Description

This parameter, when set, enables AMF to ignore AUSF dialog related failures during optional re-authentications. The UEs to apply the functionality to are provisioned on the AMF. The functionality is applied when the AMF is locally provisioned to perform an optional re-

authentication for a UE that is already registered on the AMF, but the re-authentication fails during the AUSF dialog and the AMF is able to continue serving the UE without the re-authentication

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

imsVoiceOverPsEpsFb - IMS Voice over PS EPS Fallback

Syntax

--imsVoiceOverPsEpsFb <list-item>

Description

This parameter controls whether IMS Voice over PS Sessions is supported for the UE SupiRange under the serving PLMN by EPS fallback.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

imsVoiceOverPsVoNR - IMS Voice over PS Voice over New Radio

Syntax

--imsVoiceOverPsVoNR <list-item>

Description

This parameter controls whether IMS Voice over PS Sessions is supported for the UE SupiRange under the serving PLMN by Voice over New Radio.

Class

Optional.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Name

mappingOfAllowedHomeNssaiQuery - Mapping Of Allowed Home Nssai Query

Syntax

--mappingOfAllowedHomeNssaiQuery <list-item>

Description

Enables to query NSSF to map Allowed UE Home PLMN Nssai to Visited PLMN Nssai at UE initial registration or during idle or connected mode mobility. If NSSF query response doesn't include mapped Nssai, returns an error or the setting is set to false, AMF uses the local hplmn to vplmn mapping configuration.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

n26IwkRestrictionTaiListName - N26 IWK Restriction TAI List

Syntax

--n26IwkRestrictionTaiListName <string>

Description

N26 interworking restriction TAI list used by AMF to restrict 5G service when UE(s) move from LTE into the respective TACs in the TAI list.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:~@]*

Default

""

Name

nwdafUeMobilityProfileName - NWDAF Profile Name

Syntax

--nwdafUeMobilityProfileName <string>

Description

Reference to an NWDAF Profile. Default is blank.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Name

pagingPriProfileListName - MPS Paging priority profile name

Syntax

--pagingPriProfileListName <string>

Description

Identifies the paging priority profile for MPS.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

""

Name

pcfPolicyAllowSnssaiListName - PCF Policy Allow S-NSSAI List Name

Syntax

--pcfPolicyAllowSnssaiListName <string>

Description

When a list is provided, and the corresponding Policy Association Profile indicates that UE Policy and/or AM Policy Services should be established, then an additional check is taken in to account so that the the policy establishment is triggered only if at least one of the UE's Subscribed S-NSSAI is also included in PCF Policy Allow S-NSSAI List. If the list is not provided or it is empty then the UEs Subscribed NSSAIs are not taken into account.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

""

Name

sbiTimersProfileListName - SBI Operation Response Timers and Operation Attempts

Syntax

--sbiTimersProfileListName <string>

Description

Profile consisting of SBI Operation Response Timers and Operation Attempts

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

default

Name

smfNasMappingListName - SMF NAS Mapping List Name

Syntax

--smfNasMappingListName <string>

Description

The name of the SMF error to 5GMM NAS mapping list.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

default

Name

smsfNasMappingListName - SMSF NAS Mapping List Name

Syntax

--smsfNasMappingListName <string>

Description

The name of the SMSF (N20) error to 5GMM NAS mapping list.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

default

Name

sor - Steering of Roaming

Syntax

--sor <list-item>

Description

Enables steering of roaming for roamers. The UDM can invoke steering of roaming toward the AMF when UE is in visited PLMN or in home PLMN.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

supiRangeListName - Supi Range List Name

Syntax

--supiRangeListName <string>

Description

Name of the SupiRange list.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

""

Name

support5gSrvcc - Support 5G-SRVCC

Syntax

--support5gSrvcc <list-item>

Description

Enables the 5G-SRVCC support for handing over regular voice calls from NG-RAN/5GS to 3GPP UTRAN/CS domain for the UE.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ueAmbrDownlink - UE AMBR Downlink Bitrate

Syntax

--ueAmbrDownlink <integer>

Description

UE Aggregate Maximum Downlink Bitrate. Allowed for both home and roaming subscribers.

Restrictions

Both uplink/downlink must be 0 or both uplink/downlink must be non-zero.

Class

Optional.

Type / Range

Integer in the range of 0-1,000,000,000,000 bps. in units of Kbps

Default

0

Name

ueAmbrSrc - UE AMBR Source

Syntax

--ueAmbrSrc <list-item>

Description

Identifies UE AMBR source i.e. from UDM or locally provisioned.

Class

Optional.

List

- local - Local value - UE AMBR locally configured value.
- udm - udm value - Subscribed UE AMBR in UDM

Default

udm

Name

ueAmbrUplink - UE AMBR Uplink Bitrate

Syntax

--ueAmbrUplink <integer>

Description

UE Aggregate Maximum Uplink Bitrate. Allowed for both home and roaming subscribers.

Restrictions

Both uplink/downlink must be 0 or both uplink/downlink must be non-zero.

Class

Optional.

Type / Range

Integer in the range of 0-1,000,000,000,000 bps. in units of Kbps

Default

0

Name

useSupiForLocalSmfSelection - Use SUPI for local SMF selection

Syntax

--useSupiForLocalSmfSelection <list-item>

Description

When this parameter is set to true SUPI will be used when selecting SMF from locally provisioned NFs. The SUPI will not be used for SMF discovery/selection from NRF (or when searching for SMF in the NRF cache) even if this parameter is set to true. When SUPI is used for locally provisioned SMF selection the AMF shortlists first SMFs that support the SUPI. The selection parameter shall be handled as preferred-supi: SMFs that have SUPI range that include the specific SUPI and meet the other query criterions are shortlisted first. If no such SMF is found then this selection parameter is omitted (SMF with no SUPI range or with SUPI range that does not include the specific SUPI value) will be selected.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.131 cmm amfSvcAgreementProfile

AMF Service Agreement Profile. Syntax, description, parameters, and roles for cmm amfSvcAgreementProfile command.

Syntax

Description

Administers UE PLMN and Served PLMN Service Agreement Profile and includes parameters to manage HO and service restrictions for roamers.

Parameters

Name

name - Name of Service Agreement Profile

Syntax

--name <string>

Description

Defines the PLMN service agreement profile name to be used for service.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

edrxExtendedConnectSource - eDRX Extended Connect Source

Syntax

--edrxExtendedConnectSource <list-item>

Description

This field specifies the source for the Extended Connect Time.

Class

Optional.

List

- local - The locally provisioned value in the service agreement. A value of 0 means that the AMF is aware of pending traffic but will not provide a specific extended connect time to the RAN.
- none - The Extended Connect Feature is off and the AMF will not provide any extended connect information to the RAN.

Default

none

Name

extendedConnectTime - Extended Connect Time

Syntax

--extendedConnectTime <integer>

Description

Specifies the time in seconds for the AMF to keep a UE in connected mode when the AMF knows of a pending HLCOM DL packet at the SMF/UPF. The default is 10. The AMF supports a maximum value of 60 seconds.

Class

Optional.

Type / Range

Integer in the range of 0 to 60 in units of Second

Default

10

Name

interPlmnOauthSupport - Oauth Support

Syntax

--interPlmnOauthSupport <list-item>

Description

This parameter is deprecated. OAuth is always suppressed in inter-PLMN scenarios.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

mico - MICO Support

Syntax

--mico <list-item>

Description

Enable MICO support.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

micoActiveTime - MICO Active Time

Syntax

--micoActiveTime <list-item>

Description

Defines whether the UE is allowed to use a MICO active time.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

micoActiveTimerSource - MICO Active Time Source

Syntax

--micoActiveTimerSource <list-item>

Description

If a UE requests an active timer as part of MICO negotiation, this field specifies the source that the AMF shall use for the active timer (t3324) negotiation.

Class

Optional.

List

- local - Local Provisioning
- ue - UE

Default

local

Name

micoExtendedConnectSource - MICO Extended Connect Source

Syntax

--micoExtendedConnectSource <list-item>

Description

This field specifies the source for the Extended Connect Time.

Class

Optional.

List

- local - The locally provisioned value in the service agreement. A value of 0 means that the AMF is aware of pending traffic but will not provide a specific extended connect time to the RAN.
- none - The Extended Connect Feature is off and the AMF will not provide any extended connect information to the RAN.

Default

none

Name

micoT3512TimerSource - MICO Registration Timer Source

Syntax

--micoT3512TimerSource <list-item>

Description

Source of the MICO registration timer to be negotiated with the UE.

Class

Optional.

List

- localT3512Timer - Local t3512 timer - The provisioned t3512 timer on the AMF.
- localMicoT3512Timer - Local MICO t3512 timer - The provisioned MICO t3512 timer on the AMF.

- ue - UE

Default

localT3512Timer

Name

micoWithEdrx - MICO with eDRX

Syntax

--micoWithEdrx <list-item>

Description

Specifies whether a UE requesting both MICO with Active Time and eDRX is allowed to use both or one of them.

Class

Optional.

List

- both - A UE requesting both eDRX and MICO can use both.
- mico - A UE requesting both eDRX and MICO can use MICO.
- edrx - A UE requesting both eDRX and MICO can use eDRX.

Default

mico

Name

pc5V2xAllowed - PC5 V2X Allowed

Syntax

--pc5V2xAllowed <list-item>

Description

Specifies whether PC5 V2X is allowed. When yes, the AMF preferentially selects a PCF that supports V2x.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

pcfPolicyAssociation - PCF Policy Association

Syntax

--pcfPolicyAssociation <list-item>

Description

Specifies type of Policy Association to be enabled

Class

Optional.

List

- amPolicy - AM Policy - AM Policy to be enabled
- both - Both UE and AM policy association to be enabled.
- none - None of the policies will be enabled
- uePolicy - UE Policy - UE Policy association to be enabled.

Default

both

Name

restrictMobilityOnUdmOdbAllPacketBarred - Restrict Mobility on UDM ODB All Packet Barred

Syntax

--restrictMobilityOnUdmOdbAllPacketBarred <list-item>

Description

Defines whether the AMF -upon UDM data change notification with ODB all packet services barred, performs deregistration for the UE -upon receiving ODB all packet services barred as part of the UE Access and Mobility subscription profile rejects the UE registration.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

roam3gppAccessAllowed - Roaming allowed in 5G 3GPP Access

Syntax

--roam3gppAccessAllowed <list-item>

Description

Specifies if inbound roamers are allowed to roam in 5G 3GPP Access.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

roam3gppAccessPduSessionOption - Roaming Options for PDU Session Establishment

Syntax

--roam3gppAccessPduSessionOption <list-item>

Description

Specifies roaming options for PDU Session Establishment in the Serving PLMN.

Class

Optional.

List

- both - Local Break Out and Home Routed - Subscribers are allowed to establish PDU Session in V-PLMN (local breakout) if DNN is LBO enabled by UDM subscription or to establish PDU Session in H-PLMN (Home-routed) for non-LBO DNN when roaming outside their home network. If LBO DNN fails to establish in V-PLMN, AMF tries to establish in H-PLMN.
- hr - Home Routed - Subscribers are allowed to establish PDU Session in H-PLMN (Home-routed) when roaming outside their home network.
- lbo - Local Break Out - Subscribers are allowed to establish PDU Session in V-PLMN (local breakout) when roaming outside their home network if DNN is LBO enabled by UDM subscription.
- none - Neither Local Break Out nor Home Routed - Subscribers are not allowed to establish PDU Session in V-PLMN (local breakout) or H-PLMN (Home-routed) when roaming outside their home network. Only services like SMS could be allowed.

Default

none

Name

snpn - Standalone Non-Public Network

Syntax

--snpn <list-item>

Description

Current parameter enables SNPN type of service

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

v2xAllowed - V2X Allowed

Syntax

--v2xAllowed <list-item>

Description

Defines whether V2X is allowed. If so, the AMF shall preferentially select a PCF that supports V2X.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.132 cmm amfTai

Tracking Area ID. Syntax, description, parameters, and roles for cmm amfTai command.

Syntax

Description

Provision TAC for a serving PLMN. A serving PLMN may be AMF home PLMN or shared PLMN.

Parameters

Name

plmnName - Serving PLMN Name

Syntax

--plmnName <string>

Description

The serving PLMN name may be the name of the AMF home PLMN or shared PLMN.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

amfTac - AMF Tracking Area Code

Syntax

--amfTac <integer>

Description

Provision 5G TAC. 5G TAC is 3 octets long.

Class

Required key.

Type / Range

Integer in the range of 0 to 16777215

Name

amfRelativeCapacity - AMF Relative Capacity

Syntax

--amfRelativeCapacity <integer>

Description

Configures AMF relative capacity for the tracking area

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

5

Name

emergencyFallback - Emergency Fallback

Syntax

--emergencyFallback <list-item>

Description

Enables Emergency Fallback support in the tracking area. AMF indicates to the UE that emergency fallback is supported by the network if the emergency fallback is enabled in the tracking areas sent to the UE in the tracking area list. By default, the emergency fallback is not supported.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

emergencyServiceTai - Emergency Service TAI

Syntax

--emergencyServiceTai <list-item>

Description

This command is used to enable/disable Emergency service in the tracking area. AMF indicates to the UE that emergency service is supported by the network if the emergency service is enabled in the current tracking area and all other tracking areas sent to the UE in the tracking area neighbor list. By default, the emergency service is disabled.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

imsVopsSupport - IMS VoPS Support

Syntax

--imsVopsSupport <list-item>

Description

This parameter indicates whether IMS VoPS is supported on the Tracking Areas of the AMF. Voice over PS can be supported either via Voice over New Radio connected to 5G Core, or via VoLTE connected to EPS with the NG-RAN supporting handover or redirection to EPS for the UE at QoS Flow establishment for IMS voice.

Class

Optional.

List

- amfVopsNotSupported - AMF VoPS Not Supported - Voice over PS sessions is not supported on this Tracking Area of the AMF.
- amfVopsSupportedOverNrConnectedTo5gc - AMF VoPS Support Over New Radio Connected to 5GC - Voice over PS sessions is supported on this Tracking Area of the AMF by Voice over New Radio connected to 5G Core. Note: this value is only effective when gParm voNR is enabled.
- amfVopsSupportedVoNrVolteHoRe - AMF VoPS Supported VoNR and VoLTE Handover and Redirect - Voice over PS sessions on this Tracking Area of the AMF is supported by - Voice over New Radio connected to 5G Core, -Voice over LTE connected to EPS, and the NG-RAN supports both handover and redirection to EPS at QoS Flow establishment for IMS voice. Note: this value is only effective when gParms voiceEpsFallback and VoNR are enabled.
- amfVopsSupportedVolteHo - AMF VoPS Supported VoLTE Handover - Voice over PS sessions is supported on this Tracking Area of the AMF by Voice over LTE connected to EPS, and the NG-RAN supports handover to EPS at QoS Flow establishment for IMS voice. Note: this value is only effective when gParm voiceEpsFallback is enabled.
- amfVopsSupportedVolteHoRe - AMF VoPS Supported VoLTE Handover and Redirect - Voice over PS sessions on this Tracking Area of the AMF is supported by Voice over LTE connected to EPS, and the NG-RAN supports both handover and redirection to EPS at QoS Flow establishment for IMS voice. Note: this value is only effective when gParm voiceEpsFallback is enabled.
- amfVopsSupportedVolteRedirect - AMF VoPS Supported VoLTE Redirect - Voice over PS sessions on this Tracking Area of the AMF is supported by Voice over LTE connected to EPS, and the NG-RAN supports redirection to EPS at QoS Flow establishment for IMS voice. Note: this value is only effective when gParm voiceEpsFallback is enabled.

- amfVopsSupportedVonnrVolteHo - AMF VoPS Supported VoNR and VoLTE Handover - Voice over PS sessions on this Tracking Area of the AMF is supported by -Voice over New Radio connected to 5G Core, -Voice over LTE connected to EPS, and the NG-RAN supports handover to EPS at QoS Flow establishment for IMS voice. Note: this value is only effective when gParms voiceEpsFallback and VoNR are enabled.
- amfVopsSupportedVonnrVolteRe - AMF VoPS Supported VoNR and VoLTE Redirect - Voice over PS sessions on this Tracking Area of the AMF is supported by -Voice over New Radio connected to 5G Core, -Voice over LTE connected to EPS, and the NG-RAN supports redirection to EPS at QoS Flow establishment for IMS voice. Note: this value is only effective when gParms voiceEpsFallback and VoNR are enabled.

Default

amfVopsNotSupported

Name

provisionedStatus - Provisioned Status

Syntax

--provisionedStatus <list-item>

Description

Specifies how the amf tai was last updated. Values are provisioned, rangeDiscovered, and provisionedAndRangeDiscovered

Class

Read only.

List

- Provisioned - AMF TAI was explicitly provisioned
- RangeDiscovered - Range Discovered - AMF TAI was discovered matching a range in the amfTaiRangeDiscovery Translation
- RangeDiscoveredAndProvisioned - Range Discovered And Provisioned - AMF TAI was initially range discovered, but additional direct provisioning to the TAI was done.

Name

snssaiListName - SNSSAI List Name

Syntax

--snssaiListName <string>

Description

This parameter is a pointer to the S-NSSAI list. The S-NSSAI in the list are authorized in the TAI. If the TAI is not provisioned with a list or the list is empty then it is assumed that all S-NSSAIs that are supported by AMF and 5G-AN are authorized in the TAI. This parameter is not used if AMF is provisioned to use the NSSF Availability Service as in this case the Slices that are authorized in the TAI are provided by the NSSF.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

timeZoneName - Time Zone Name

Syntax

--timeZoneName <list-item>

Description

Provision time zone of the TAC.

Class

Optional.

List

- Africa/Abidjan - Timezone for africa/abidjan UTC offset of +00:00.
- Africa/Accra - Timezone for africa/accra UTC offset of +00:00.
- Africa/Addis_Ababa - Timezone for africa/addis_ababa UTC offset of +03:00.
- Africa/Algiers - Timezone for africa/algiers UTC offset of +01:00.
- Africa/Asmara - Timezone for africa/asmara UTC offset of +03:00.
- Africa/Asmera - Timezone for africa/asmera UTC offset of +03:00.
- Africa/Bamako - Timezone for africa/bamako UTC offset of +00:00.
- Africa/Bangui - Timezone for africa/bangui UTC offset of +01:00.
- Africa/Banjul - Timezone for africa/banjul UTC offset of +00:00.
- Africa/Bissau - Timezone for africa/bissau UTC offset of +00:00.
- Africa/Blantyre - Timezone for africa/blantyre UTC offset of +02:00.
- Africa/Brazzaville - Timezone for africa/brazzaville UTC offset of +01:00.
- Africa/Bujumbura - Timezone for africa/bujumbura UTC offset of +02:00.
- Africa/Cairo - Timezone for africa/cairo UTC offset of +02:00.
- Africa/Casablanca - Timezone for africa/casablanca UTC offset of +00:00.
- Africa/Ceuta - Timezone for africa/ceuta UTC offset of +01:00.
- Africa/Conakry - Timezone for africa/conakry UTC offset of +00:00.
- Africa/Dakar - Timezone for africa/dakar UTC offset of +00:00.
- Africa/Dar_es_Salaam - Timezone for africa/dar_es_salaam UTC offset of +03:00.

- Africa/Djibouti - Timezone for africa/djibouti UTC offset of +03:00.
- Africa/Douala - Timezone for africa/douala UTC offset of +01:00.
- Africa/El_Aaiun - Timezone for africa/el_aaiun UTC offset of +00:00.
- Africa/Freetown - Timezone for africa/freetown UTC offset of +00:00.
- Africa/Gaborone - Timezone for africa/gaborone UTC offset of +02:00.
- Africa/Harare - Timezone for africa/harare UTC offset of +02:00.
- Africa/Johannesburg - Timezone for africa/johannesburg UTC offset of +02:00.
- Africa/Juba - Timezone for africa/juba UTC offset of +03:00.
- Africa/Kampala - Timezone for africa/kampala UTC offset of +03:00.
- Africa/Khartoum - Timezone for africa/khartoum UTC offset of +03:00.
- Africa/Kigali - Timezone for africa/kigali UTC offset of +02:00.
- Africa/Kinshasa - Timezone for africa/kinshasa UTC offset of +01:00.
- Africa/Lagos - Timezone for africa/lagos UTC offset of +01:00.
- Africa/Libreville - Timezone for africa/libreville UTC offset of +01:00.
- Africa/Lome - Timezone for africa/lome UTC offset of +00:00.
- Africa/Luanda - Timezone for africa/luanda UTC offset of +01:00.
- Africa/Lubumbashi - Timezone for africa/lubumbashi UTC offset of +02:00.
- Africa/Lusaka - Timezone for africa/lusaka UTC offset of +02:00.
- Africa/Malabo - Timezone for africa/malabo UTC offset of +01:00.
- Africa/Maputo - Timezone for africa/maputo UTC offset of +02:00.
- Africa/Maseru - Timezone for africa/maseru UTC offset of +02:00.
- Africa/Mbabane - Timezone for africa/mbabane UTC offset of +02:00.
- Africa/Mogadishu - Timezone for africa/mogadishu UTC offset of +03:00.
- Africa/Monrovia - Timezone for africa/monrovia UTC offset of +00:00.
- Africa/Nairobi - Timezone for africa/nairobi UTC offset of +03:00.
- Africa/Ndjamena - Timezone for africa/ndjamena UTC offset of +01:00.
- Africa/Niamey - Timezone for africa/niamey UTC offset of +01:00.
- Africa/Nouakchott - Timezone for africa/nouakchott UTC offset of +00:00.
- Africa/Ouagadougou - Timezone for africa/ouagadougou UTC offset of +00:00.
- Africa/Porto-Novo - Timezone for africa/porto-novo UTC offset of +01:00.
- Africa/Sao_Tome - Timezone for africa/sao_tome UTC offset of +00:00.

- Africa/Timbuktu - Timezone for africa/timbuktu UTC offset of +00:00.
- Africa/Tripoli - Timezone for africa/tripoli UTC offset of +02:00.
- Africa/Tunis - Timezone for africa/tunis UTC offset of +01:00.
- Africa/Windhoek - Timezone for africa/windhoek UTC offset of +02:00.
- America/Adak - Timezone for america/adak UTC offset of -10:00.
- America/Anchorage - Timezone for america/anchorage UTC offset of -09:00.
- America/Anguilla - Timezone for america/anguilla UTC offset of -04:00.
- America/Antigua - Timezone for america/antigua UTC offset of -04:00.
- America/Araguaina - Timezone for america/araguaina UTC offset of -03:00.
- America/Argentina/Buenos_Aires - Timezone for america/argentina/buenos_aires UTC offset of -03:00.
- America/Argentina/Catamarca - Timezone for america/argentina/catamarca UTC offset of -03:00.
- America/Argentina/ComodRivadavia - Timezone for america/argentina/comodrivadavia UTC offset of -03:00.
- America/Argentina/Cordoba - Timezone for america/argentina/cordoba UTC offset of -03:00.
- America/Argentina/Jujuy - Timezone for america/argentina/jujuy UTC offset of -03:00.
- America/Argentina/La_Rioja - Timezone for america/argentina/la_rioja UTC offset of -03:00.
- America/Argentina/Mendoza - Timezone for america/argentina/mendoza UTC offset of -03:00.
- America/Argentina/Rio_Gallegos - Timezone for america/argentina/rio_gallegos UTC offset of -03:00.
- America/Argentina/Salta - Timezone for america/argentina/salta UTC offset of -03:00.
- America/Argentina/San_Juan - Timezone for america/argentina/san_juan UTC offset of -03:00.
- America/Argentina/San_Luis - Timezone for america/argentina/san_luis UTC offset of -03:00.
- America/Argentina/Tucuman - Timezone for america/argentina/tucuman UTC offset of -03:00.
- America/Argentina/Ushuaia - Timezone for america/argentina/ushuaia UTC offset of -03:00.
- America/Aruba - Timezone for america/aruba UTC offset of -04:00.

- America/Asuncion - Timezone for america/asuncion UTC offset of -03:00.
- America/Atikokan - Timezone for america/atikokan UTC offset of -05:00.
- America/Atka - Timezone for america/atka UTC offset of -10:00.
- America/Bahia - Timezone for america/bahia UTC offset of -03:00.
- America/Bahia_Banderas - Timezone for america/bahia_banderas UTC offset of -06:00.
- America/Barbados - Timezone for america/barbados UTC offset of -04:00.
- America/Belem - Timezone for america/belem UTC offset of -03:00.
- America/Belize - Timezone for america/belize UTC offset of -06:00.
- America/Blanc-Sablon - Timezone for america/blanc-sablon UTC offset of -04:00.
- America/Boa_Vista - Timezone for america/boa_vista UTC offset of -04:00.
- America/Bogota - Timezone for america/bogota UTC offset of -05:00.
- America/Boise - Timezone for america/boise UTC offset of -07:00.
- America/Buenos_Aires - Timezone for america/buenos_aires UTC offset of -03:00.
- America/Cambridge_Bay - Timezone for america/cambridge_bay UTC offset of -07:00.
- America/Campo_Grande - Timezone for america/campo_grande UTC offset of -03:00.
- America/Cancun - Timezone for america/cancun UTC offset of -05:00.
- America/Caracas - Timezone for america/caracas UTC offset of -04:00.
- America/Catamarca - Timezone for america/catamarca UTC offset of -03:00.
- America/Cayenne - Timezone for america/cayenne UTC offset of -03:00.
- America/Cayman - Timezone for america/cayman UTC offset of -05:00.
- America/Chicago - Timezone for america/chicago UTC offset of -06:00.
- America/Chihuahua - Timezone for america/chihuahua UTC offset of -07:00.
- America/Coral_Harbour - Timezone for america/coral_harbour UTC offset of -05:00.
- America/Cordoba - Timezone for america/cordoba UTC offset of -03:00.
- America/Costa_Rica - Timezone for america/costa_rica UTC offset of -06:00.
- America/Creston - Timezone for america/creston UTC offset of -07:00.
- America/Cuiaba - Timezone for america/cuiaba UTC offset of -03:00.
- America/Curacao - Timezone for america/curacao UTC offset of -04:00.
- America/Danmarkshavn - Timezone for america/danmarkshavn UTC offset of +00:00.
- America/Dawson - Timezone for america/dawson UTC offset of -08:00.
- America/Dawson_Creek - Timezone for america/dawson_creek UTC offset of -07:00.

- America/Denver - Timezone for america/denver UTC offset of -07:00.
- America/Detroit - Timezone for america/detroit UTC offset of -05:00.
- America/Dominica - Timezone for america/dominica UTC offset of -04:00.
- America/Edmonton - Timezone for america/edmonton UTC offset of -07:00.
- America/Eirunepe - Timezone for america/eirunepe UTC offset of -05:00.
- America/El_Salvador - Timezone for america/el_salvador UTC offset of -06:00.
- America/Ensenada - Timezone for america/ensenada UTC offset of -08:00.
- America/Fort_Nelson - Timezone for america/fort_nelson UTC offset of -07:00.
- America/Fort_Wayne - Timezone for america/fort_wayne UTC offset of -05:00.
- America/Fortaleza - Timezone for america/fortaleza UTC offset of -03:00.
- America/Glace_Bay - Timezone for america/glance_bay UTC offset of -04:00.
- America/Godthab - Timezone for america/godthab UTC offset of -03:00.
- America/Goose_Bay - Timezone for america/goose_bay UTC offset of -04:00.
- America/Grand_Turk - Timezone for america/grand_turk UTC offset of -04:00.
- America/Grenada - Timezone for america/grenada UTC offset of -04:00.
- America/Guadeloupe - Timezone for america/guadeloupe UTC offset of -04:00.
- America/Guatemala - Timezone for america/guatemala UTC offset of -06:00.
- America/Guayaquil - Timezone for america/guayaquil UTC offset of -05:00.
- America/Guyana - Timezone for america/guyana UTC offset of -04:00.
- America/Halifax - Timezone for america/halifax UTC offset of -04:00.
- America/Havana - Timezone for america/havana UTC offset of -05:00.
- America/Hermosillo - Timezone for america/hermosillo UTC offset of -07:00.
- America/Indiana/Indianapolis - Timezone for america/indiana/indianapolis UTC offset of -05:00.
- America/Indiana/Knox - Timezone for america/indiana/knox UTC offset of -06:00.
- America/Indiana/Marengo - Timezone for america/indiana/marengo UTC offset of -05:00.
- America/Indiana/Petersburg - Timezone for america/indiana/petersburg UTC offset of -05:00.
- America/Indiana/Tell_City - Timezone for america/indiana/tell_city UTC offset of -06:00.
- America/Indiana/Vevay - Timezone for america/indiana/vevay UTC offset of -05:00.

- America/Indiana/Vincennes - Timezone for america/indiana/vincennes UTC offset of -05:00.
- America/Indiana/Winamac - Timezone for america/indiana/winamac UTC offset of -05:00.
- America/Indianapolis - Timezone for america/indianapolis UTC offset of -05:00.
- America/Inuvik - Timezone for america/inuvik UTC offset of -07:00.
- America/Iqaluit - Timezone for america/iqaluit UTC offset of -05:00.
- America/Jamaica - Timezone for america/jamaica UTC offset of -05:00.
- America/Jujuy - Timezone for america/jujuy UTC offset of -03:00.
- America/Juneau - Timezone for america/juneau UTC offset of -09:00.
- America/Kentucky/Louisville - Timezone for america/kentucky/louisville UTC offset of -05:00.
- America/Kentucky/Monticello - Timezone for america/kentucky/monticello UTC offset of -05:00.
- America/Knox_IN - Timezone for america/knox_in UTC offset of -06:00.
- America/Kralendijk - Timezone for america/kralendijk UTC offset of -04:00.
- America/La_Paz - Timezone for america/la_paz UTC offset of -04:00.
- America/Lima - Timezone for america/lima UTC offset of -05:00.
- America/Los_Angeles - Timezone for america/los_angeles UTC offset of -08:00.
- America/Louisville - Timezone for america/louisville UTC offset of -05:00.
- America/Lower_Princes - Timezone for america/lower_princes UTC offset of -04:00.
- America/Maceio - Timezone for america/maceio UTC offset of -03:00.
- America/Managua - Timezone for america/managua UTC offset of -06:00.
- America/Manaus - Timezone for america/manuel UTC offset of -04:00.
- America/Marigot - Timezone for america/marigot UTC offset of -04:00.
- America/Martinique - Timezone for america/martinique UTC offset of -04:00.
- America/Matamoros - Timezone for america/matamoros UTC offset of -06:00.
- America/Mazatlan - Timezone for america/mazatlan UTC offset of -07:00.
- America/Mendoza - Timezone for america/mendoza UTC offset of -03:00.
- America/Menominee - Timezone for america/menominee UTC offset of -06:00.
- America/Merida - Timezone for america/merida UTC offset of -06:00.
- America/Metlakatla - Timezone for america/metlakatla UTC offset of -09:00.

- America/Mexico_City - Timezone for america/mexico_city UTC offset of -06:00.
- America/Miquelon - Timezone for america/miquelon UTC offset of -03:00.
- America/Moncton - Timezone for america/moncton UTC offset of -04:00.
- America/Monterrey - Timezone for america/monterrey UTC offset of -06:00.
- America/Montevideo - Timezone for america/montevideo UTC offset of -03:00.
- America/Montreal - Timezone for america/montreal UTC offset of -05:00.
- America/Montserrat - Timezone for america/montserrat UTC offset of -04:00.
- America/Nassau - Timezone for america/nassau UTC offset of -05:00.
- America/New_York - Timezone for america/new_york UTC offset of -05:00.
- America/Nipigon - Timezone for america/nipigon UTC offset of -05:00.
- America/Nome - Timezone for america/nome UTC offset of -09:00.
- America/Noronha - Timezone for america/noronha UTC offset of -02:00.
- America/North_Dakota/Beulah - Timezone for america/north_dakota/beulah UTC offset of -06:00.
- America/North_Dakota/Center - Timezone for america/north_dakota/center UTC offset of -06:00.
- America/North_Dakota/New_Salem - Timezone for america/north_dakota/new_salem UTC offset of -06:00.
- America/Ojinaga - Timezone for america/ojinaga UTC offset of -07:00.
- America/Panama - Timezone for america/panama UTC offset of -05:00.
- America/Pangnirtung - Timezone for america/pangnirtung UTC offset of -05:00.
- America/Paramaribo - Timezone for america/paramaribo UTC offset of -03:00.
- America/Phoenix - Timezone for america/phoenix UTC offset of -07:00.
- America/Port-au-Prince - Timezone for america/port-au-prince UTC offset of -05:00.
- America/Port_of_Spain - Timezone for america/port_of_spain UTC offset of -04:00.
- America/Porto_Acre - Timezone for america/porto_acre UTC offset of -05:00.
- America/Porto_Velho - Timezone for america/porto_velho UTC offset of -04:00.
- America/Puerto_Rico - Timezone for america/puerto_rico UTC offset of -04:00.
- America/Rainy_River - Timezone for america/rainy_river UTC offset of -06:00.
- America/Rankin_Inlet - Timezone for america/rankin_inlet UTC offset of -06:00.
- America/Recife - Timezone for america/recife UTC offset of -03:00.
- America/Regina - Timezone for america/regina UTC offset of -06:00.

- America/Resolute - Timezone for america/resolute UTC offset of -06:00.
- America/Rio_Branco - Timezone for america/rio_branco UTC offset of -05:00.
- America/Rosario - Timezone for america/rosario UTC offset of -03:00.
- America/Santa_Isabel - Timezone for america/santa_isabel UTC offset of -08:00.
- America/Santarem - Timezone for america/santarem UTC offset of -03:00.
- America/Santiago - Timezone for america/santiago UTC offset of -03:00.
- America/Santo_Domingo - Timezone for america/santo_domingo UTC offset of -04:00.
- America/Sao_Paulo - Timezone for america/sao_paulo UTC offset of -02:00.
- America/Scoresbysund - Timezone for america/scoresbysund UTC offset of -01:00.
- America/Shiprock - Timezone for america/shiprock UTC offset of -07:00.
- America/Sitka - Timezone for america/sitka UTC offset of -09:00.
- America/St_Barthelemy - Timezone for america/st_barthelemy UTC offset of -04:00.
- America/St_Johns - Timezone for america/st_johns UTC offset of -03:30.
- America/St_Kitts - Timezone for america/st_kitts UTC offset of -04:00.
- America/St_Lucia - Timezone for america/st_lucia UTC offset of -04:00.
- America/St_Thomas - Timezone for america/st_thomas UTC offset of -04:00.
- America/St_Vincent - Timezone for america/st_vincent UTC offset of -04:00.
- America/Swift_Current - Timezone for america/swift_current UTC offset of -06:00.
- America/Tegucigalpa - Timezone for america/tegucigalpa UTC offset of -06:00.
- America/Thule - Timezone for america/thule UTC offset of -04:00.
- America/Thunder_Bay - Timezone for america/thunder_bay UTC offset of -05:00.
- America/Tijuana - Timezone for america/tijuana UTC offset of -08:00.
- America/Toronto - Timezone for america/toronto UTC offset of -05:00.
- America/Tortola - Timezone for america/tortola UTC offset of -04:00.
- America/Vancouver - Timezone for america/vancouver UTC offset of -08:00.
- America/Virgin - Timezone for america/virgin UTC offset of -04:00.
- America/Whitehorse - Timezone for america/whitehorse UTC offset of -08:00.
- America/Winnipeg - Timezone for america/winnipeg UTC offset of -06:00.
- America/Yakutat - Timezone for america/yakutat UTC offset of -09:00.
- America/Yellowknife - Timezone for america/yellowknife UTC offset of -07:00.
- Antarctica/Casey - Timezone for antarctica/casey UTC offset of +11:00.

- Antarctica/Davis - Timezone for antarctica/davis UTC offset of +07:00.
- Antarctica/DumontDURville - Timezone for antarctica/dumontdurville UTC offset of +10:00.
- Antarctica/Macquarie - Timezone for antarctica/macquarie UTC offset of +11:00.
- Antarctica/Mawson - Timezone for antarctica/mawson UTC offset of +05:00.
- Antarctica/McMurdo - Timezone for antarctica/mcmurdo UTC offset of +13:00.
- Antarctica/Palmer - Timezone for antarctica/palmer UTC offset of -03:00.
- Antarctica/Rothera - Timezone for antarctica/rothera UTC offset of -03:00.
- Antarctica/South_Pole - Timezone for antarctica/south_pole UTC offset of +13:00.
- Antarctica/Syowa - Timezone for antarctica/syowa UTC offset of +03:00.
- Antarctica/Troll - Timezone for antarctica/troll UTC offset of +00:00.
- Antarctica/Vostok - Timezone for antarctica/vostok UTC offset of +06:00.
- Arctic/Longyearbyen - Timezone for arctic/longyearbyen UTC offset of +01:00.
- Asia/Aden - Timezone for asia/aden UTC offset of +03:00.
- Asia/Almaty - Timezone for asia/almaty UTC offset of +06:00.
- Asia/Amman - Timezone for asia/amman UTC offset of +02:00.
- Asia/Anadyr - Timezone for asia/anadyr UTC offset of +12:00.
- Asia/Aqtau - Timezone for asia/aqtau UTC offset of +05:00.
- Asia/Aqtobe - Timezone for asia/aqtobe UTC offset of +05:00.
- Asia/Ashgabat - Timezone for asia/ashgabat UTC offset of +05:00.
- Asia/Ashkhabad - Timezone for asia/ashkhabad UTC offset of +05:00.
- Asia/Baghdad - Timezone for asia/baghdad UTC offset of +03:00.
- Asia/Bahrain - Timezone for asia/bahrain UTC offset of +03:00.
- Asia/Baku - Timezone for asia/baku UTC offset of +04:00.
- Asia/Bangkok - Timezone for asia/bangkok UTC offset of +07:00.
- Asia/Barnaul - Timezone for asia/barnaul UTC offset of +07:00.
- Asia/Beirut - Timezone for asia/beirut UTC offset of +02:00.
- Asia/Bishkek - Timezone for asia/bishkek UTC offset of +06:00.
- Asia/Brunei - Timezone for asia/brunei UTC offset of +08:00.
- Asia/Calcutta - Timezone for asia/calcutta UTC offset of +05:30.
- Asia/Chita - Timezone for asia/chita UTC offset of +09:00.

- Asia/Choibalsan - Timezone for asia/choibalsan UTC offset of +08:00.
- Asia/Chongqing - Timezone for asia/chongqing UTC offset of +08:00.
- Asia/Chungking - Timezone for asia/chungking UTC offset of +08:00.
- Asia/Colombo - Timezone for asia/colombo UTC offset of +05:30.
- Asia/Dacca - Timezone for asia/dacca UTC offset of +06:00.
- Asia/Damascus - Timezone for asia/damascus UTC offset of +02:00.
- Asia/Dhaka - Timezone for asia/dhaka UTC offset of +06:00.
- Asia/Dili - Timezone for asia/dili UTC offset of +09:00.
- Asia/Dubai - Timezone for asia/dubai UTC offset of +04:00.
- Asia/Dushanbe - Timezone for asia/dushanbe UTC offset of +05:00.
- Asia/Gaza - Timezone for asia/gaza UTC offset of +02:00.
- Asia/Harbin - Timezone for asia/harbin UTC offset of +08:00.
- Asia/Hebron - Timezone for asia/hebron UTC offset of +02:00.
- Asia/Ho_Chi_Minh - Timezone for asia/ho_chi_minh UTC offset of +07:00.
- Asia/Hong_Kong - Timezone for asia/hong_kong UTC offset of +08:00.
- Asia/Hovd - Timezone for asia/hovd UTC offset of +07:00.
- Asia/Irkutsk - Timezone for asia/irkutsk UTC offset of +08:00.
- Asia/Istanbul - Timezone for asia/istanbul UTC offset of +03:00.
- Asia/Jakarta - Timezone for asia/jakarta UTC offset of +07:00.
- Asia/Jayapura - Timezone for asia/jayapura UTC offset of +09:00.
- Asia/Jerusalem - Timezone for asia/jerusalem UTC offset of +02:00.
- Asia/Kabul - Timezone for asia/kabul UTC offset of +04:30.
- Asia/Kamchatka - Timezone for asia/kamchatka UTC offset of +12:00.
- Asia/Karachi - Timezone for asia/karachi UTC offset of +05:00.
- Asia/Kashgar - Timezone for asia/kashgar UTC offset of +06:00.
- Asia/Kathmandu - Timezone for asia/kathmandu UTC offset of +05:45.
- Asia/Katmandu - Timezone for asia/katmandu UTC offset of +05:45.
- Asia/Khandyga - Timezone for asia/khandyga UTC offset of +09:00.
- Asia/Kolkata - Timezone for asia/kolkata UTC offset of +05:30.
- Asia/Krasnoyarsk - Timezone for asia/krasnoyarsk UTC offset of +07:00.
- Asia/Kuala_Lumpur - Timezone for asia/kuala_lumpur UTC offset of +08:00.

- Asia/Kuching - Timezone for asia/kuching UTC offset of +08:00.
- Asia/Kuwait - Timezone for asia/kuwait UTC offset of +03:00.
- Asia/Macao - Timezone for asia/macao UTC offset of +08:00.
- Asia/Macau - Timezone for asia/macau UTC offset of +08:00.
- Asia/Magadan - Timezone for asia/magadan UTC offset of +11:00.
- Asia/Makassar - Timezone for asia/makassar UTC offset of +08:00.
- Asia/Manila - Timezone for asia/manila UTC offset of +08:00.
- Asia/Muscat - Timezone for asia/muscat UTC offset of +04:00.
- Asia/Nicosia - Timezone for asia/nicosia UTC offset of +02:00.
- Asia/Novokuznetsk - Timezone for asia/novokuznetsk UTC offset of +07:00.
- Asia/Novosibirsk - Timezone for asia/novosibirsk UTC offset of +07:00.
- Asia/Omsk - Timezone for asia/omsk UTC offset of +06:00.
- Asia/Oral - Timezone for asia/oral UTC offset of +05:00.
- Asia/Phnom_Penh - Timezone for asia/phnom_penh UTC offset of +07:00.
- Asia/Pontianak - Timezone for asia/pontianak UTC offset of +07:00.
- Asia/Pyongyang - Timezone for asia/pyongyang UTC offset of +08:30.
- Asia/Qatar - Timezone for asia/qatar UTC offset of +03:00.
- Asia/Qyzylorda - Timezone for asia/qyzylorda UTC offset of +06:00.
- Asia/Rangoon - Timezone for asia/rangoon UTC offset of +06:30.
- Asia/Riyadh - Timezone for asia/riyadh UTC offset of +03:00.
- Asia/Saigon - Timezone for asia/saigon UTC offset of +07:00.
- Asia/Sakhalin - Timezone for asia/sakhalin UTC offset of +11:00.
- Asia/Samarkand - Timezone for asia/samarkand UTC offset of +05:00.
- Asia/Seoul - Timezone for asia/seoul UTC offset of +09:00.
- Asia/Shanghai - Timezone for asia/shanghai UTC offset of +08:00.
- Asia/Singapore - Timezone for asia/singapore UTC offset of +08:00.
- Asia/Srednekolymsk - Timezone for asia/srednekolymsk UTC offset of +11:00.
- Asia/Taipei - Timezone for asia/taipei UTC offset of +08:00.
- Asia/Tashkent - Timezone for asia/tashkent UTC offset of +05:00.
- Asia/Tbilisi - Timezone for asia/tbilisi UTC offset of +04:00.
- Asia/Tehran - Timezone for asia/tehran UTC offset of +03:30.

- Asia/Tel_Aviv - Timezone for asia/tel_aviv UTC offset of +02:00.
- Asia/Thimbu - Timezone for asia/thimbu UTC offset of +06:00.
- Asia/Thimphu - Timezone for asia/thimphu UTC offset of +06:00.
- Asia/Tokyo - Timezone for asia/tokyo UTC offset of +09:00.
- Asia/Tomsk - Timezone for asia/tomsk UTC offset of +07:00.
- Asia/Ujung_Pandang - Timezone for asia/ujung_pandang UTC offset of +08:00.
- Asia/Ulaanbaatar - Timezone for asia/ulaanbaatar UTC offset of +08:00.
- Asia/Ulan_Bator - Timezone for asia/ulan_bator UTC offset of +08:00.
- Asia/Urumqi - Timezone for asia/urumqi UTC offset of +06:00.
- Asia/Ust-Nera - Timezone for asia/ust-nera UTC offset of +10:00.
- Asia/Vientiane - Timezone for asia/vientiane UTC offset of +07:00.
- Asia/Vladivostok - Timezone for asia/vladivostok UTC offset of +10:00.
- Asia/Yakutsk - Timezone for asia/yakutsk UTC offset of +09:00.
- Asia/Yekaterinburg - Timezone for asia/yekaterinburg UTC offset of +05:00.
- Asia/Yerevan - Timezone for asia/yerevan UTC offset of +04:00.
- Atlantic/Azores - Timezone for atlantic/azores UTC offset of -01:00.
- Atlantic/Bermuda - Timezone for atlantic/bermuda UTC offset of -04:00.
- Atlantic/Canary - Timezone for atlantic/canary UTC offset of +00:00.
- Atlantic/Cape_Verde - Timezone for atlantic/cape_verde UTC offset of -01:00.
- Atlantic/Faeroe - Timezone for atlantic/faeroe UTC offset of +00:00.
- Atlantic/Faroe - Timezone for atlantic/faroe UTC offset of +00:00.
- Atlantic/Jan_Mayen - Timezone for atlantic/jan_mayen UTC offset of +01:00.
- Atlantic/Madeira - Timezone for atlantic/madeira UTC offset of +00:00.
- Atlantic/Reykjavik - Timezone for atlantic/reykjavik UTC offset of +00:00.
- Atlantic/South_Georgia - Timezone for atlantic/south_georgia UTC offset of -02:00.
- Atlantic/St_Helena - Timezone for atlantic/st_helena UTC offset of +00:00.
- Atlantic/Stanley - Timezone for atlantic/stanley UTC offset of -03:00.
- Australia/ACT - Timezone for australia/act UTC offset of +11:00.
- Australia/Adelaide - Timezone for australia/adelaide UTC offset of +10:30.
- Australia/Brisbane - Timezone for australia/brisbane UTC offset of +10:00.
- Australia/Broken_Hill - Timezone for australia/broken_hill UTC offset of +10:30.

- Australia/Canberra - Timezone for australia/canberra UTC offset of +11:00.
- Australia/Currie - Timezone for australia/currie UTC offset of +11:00.
- Australia/Darwin - Timezone for australia/darwin UTC offset of +09:30.
- Australia/Eucla - Timezone for australia/eucla UTC offset of +08:45.
- Australia/Hobart - Timezone for australia/hobart UTC offset of +11:00.
- Australia/LHI - Timezone for australia/lhi UTC offset of +11:00.
- Australia/Lindeman - Timezone for australia/lindeman UTC offset of +10:00.
- Australia/Lord_Howe - Timezone for australia/lord_howe UTC offset of +11:00.
- Australia/Melbourne - Timezone for australia/melbourne UTC offset of +11:00.
- Australia/NSW - Timezone for australia/nsw UTC offset of +11:00.
- Australia/North - Timezone for australia/north UTC offset of +09:30.
- Australia/Perth - Timezone for australia/perth UTC offset of +08:00.
- Australia/Queensland - Timezone for australia/queensland UTC offset of +10:00.
- Australia/South - Timezone for australia/south UTC offset of +10:30.
- Australia/Sydney - Timezone for australia/sydney UTC offset of +11:00.
- Australia/Tasmania - Timezone for australia/tasmania UTC offset of +11:00.
- Australia/Victoria - Timezone for australia/victoria UTC offset of +11:00.
- Australia/West - Timezone for australia/west UTC offset of +08:00.
- Australia/Yancowinna - Timezone for australia/yancowinna UTC offset of +10:30.
- CET - Timezone for CET UTC offset of +01:00.
- CST6CDT - Timezone for CST6CDT UTC offset of -06:00.
- Canada/Atlantic - Timezone for canada/atlantic UTC offset of -04:00.
- Canada/Central - Timezone for canada/central UTC offset of -06:00.
- Canada/East-Saskatchewan - Timezone for canada/east-saskatchewan UTC offset of -06:00.
- Canada/Eastern - Timezone for canada/eastern UTC offset of -05:00.
- Canada/Mountain - Timezone for canada/mountain UTC offset of -07:00.
- Canada/Newfoundland - Timezone for canada/newfoundland UTC offset of -03:30.
- Canada/Pacific - Timezone for canada/pacific UTC offset of -08:00.
- Canada/Saskatchewan - Timezone for canada/saskatchewan UTC offset of -06:00.
- Canada/Yukon - Timezone for canada/yukon UTC offset of -08:00.

- Chile/Continental - Timezone for chile/continental UTC offset of -03:00.
- Chile/EasterIsland - Timezone for chile/easterisland UTC offset of -05:00.
- Cuba - Timezone for cuba UTC offset of -05:00.
- Default - Timezone for system UTC (OS time zone)
- EET - Timezone for EET UTC offset of +02:00.
- EST - Timezone for EST UTC offset of -05:00.
- EST5EDT - Timezone for EST5EDT UTC offset of -05:00.
- Egypt - Timezone for egypt UTC offset of +02:00.
- Eire - Timezone for eire UTC offset of +00:00.
- Etc/Greenwich - Timezone for etc/greenwich UTC offset of +00:00.
- Etc/UCT - Timezone for etc/uct UTC offset of +00:00.
- Etc/UTC - Timezone for etc/utc UTC offset of +00:00.
- Etc/Universal - Timezone for etc/universal UTC offset of +00:00.
- Etc/Zulu - Timezone for etc/zulu UTC offset of +00:00.
- Europe/Amsterdam - Timezone for europe/amsterdam UTC offset of +01:00.
- Europe/Andorra - Timezone for europe/andorra UTC offset of +01:00.
- Europe/Astrakhan - Timezone for europe/astrakhan UTC offset of +04:00.
- Europe/Athens - Timezone for europe/athens UTC offset of +02:00.
- Europe/Belfast - Timezone for europe/belfast UTC offset of +00:00.
- Europe/Belgrade - Timezone for europe/belgrade UTC offset of +01:00.
- Europe/Berlin - Timezone for europe/berlin UTC offset of +01:00.
- Europe/Bratislava - Timezone for europe/bratislava UTC offset of +01:00.
- Europe/Brussels - Timezone for europe/brussels UTC offset of +01:00.
- Europe/Bucharest - Timezone for europe/bucharest UTC offset of +02:00.
- Europe/Budapest - Timezone for europe/budapest UTC offset of +01:00.
- Europe/Busingen - Timezone for europe/busingen UTC offset of +01:00.
- Europe/Chisinau - Timezone for europe/chisinau UTC offset of +02:00.
- Europe/Copenhagen - Timezone for europe/copenhagen UTC offset of +01:00.
- Europe/Dublin - Timezone for europe/dublin UTC offset of +00:00.
- Europe/Gibraltar - Timezone for europe/gibraltar UTC offset of +01:00.
- Europe/Guernsey - Timezone for europe/guernsey UTC offset of +00:00.

- Europe/Helsinki - Timezone for europe/helsinki UTC offset of +02:00.
- Europe/Isle_of_Man - Timezone for europe/isle_of_man UTC offset of +00:00.
- Europe/Istanbul - Timezone for europe/istanbul UTC offset of +03:00.
- Europe/Jersey - Timezone for europe/jersey UTC offset of +00:00.
- Europe/Kaliningrad - Timezone for europe/kaliningrad UTC offset of +02:00.
- Europe/Kiev - Timezone for europe/kyiv UTC offset of +02:00.
- Europe/Kirov - Timezone for europe/kirov UTC offset of +03:00.
- Europe/Lisbon - Timezone for europe/lisbon UTC offset of +00:00.
- Europe/Ljubljana - Timezone for europe/ljubljana UTC offset of +01:00.
- Europe/London - Timezone for europe/london UTC offset of +00:00.
- Europe/Luxembourg - Timezone for europe/luxembourg UTC offset of +01:00.
- Europe/Madrid - Timezone for europe/madrid UTC offset of +01:00.
- Europe/Malta - Timezone for europe/malta UTC offset of +01:00.
- Europe/Mariehamn - Timezone for europe/mariehamn UTC offset of +02:00.
- Europe/Minsk - Timezone for europe/minsk UTC offset of +03:00.
- Europe/Monaco - Timezone for europe/monaco UTC offset of +01:00.
- Europe/Moscow - Timezone for europe/moscow UTC offset of +03:00.
- Europe/Nicosia - Timezone for europe/nicosia UTC offset of +02:00.
- Europe/Oslo - Timezone for europe/oslo UTC offset of +01:00.
- Europe/Paris - Timezone for europe/paris UTC offset of +01:00.
- Europe/Podgorica - Timezone for europe/podgorica UTC offset of +01:00.
- Europe/Prague - Timezone for europe/prague UTC offset of +01:00.
- Europe/Riga - Timezone for europe/riga UTC offset of +02:00.
- Europe/Rome - Timezone for europe/rome UTC offset of +01:00.
- Europe/Samara - Timezone for europe/samara UTC offset of +04:00.
- Europe/San_Marino - Timezone for europe/san_marino UTC offset of +01:00.
- Europe/Sarajevo - Timezone for europe/sarajevo UTC offset of +01:00.
- Europe/Simferopol - Timezone for europe/simferopol UTC offset of +03:00.
- Europe/Skopje - Timezone for europe/skopje UTC offset of +01:00.
- Europe/Sofia - Timezone for europe/sofia UTC offset of +02:00.
- Europe/Stockholm - Timezone for europe/stockholm UTC offset of +01:00.

- Europe/Tallinn - Timezone for europe/tallinn UTC offset of +02:00.
- Europe/Tirane - Timezone for europe/tirane UTC offset of +01:00.
- Europe/Tiraspol - Timezone for europe/tiraspol UTC offset of +02:00.
- Europe/Ulyanovsk - Timezone for europe/ulyanovsk UTC offset of +04:00.
- Europe/Uzhgorod - Timezone for europe/uzhgorod UTC offset of +02:00.
- Europe/Vaduz - Timezone for europe/vaduz UTC offset of +01:00.
- Europe/Vatican - Timezone for europe/vatican UTC offset of +01:00.
- Europe/Vienna - Timezone for europe/vienna UTC offset of +01:00.
- Europe/Vilnius - Timezone for europe/vilnius UTC offset of +02:00.
- Europe/Volgograd - Timezone for europe/volgograd UTC offset of +03:00.
- Europe/Warsaw - Timezone for europe/warsaw UTC offset of +01:00.
- Europe/Zagreb - Timezone for europe/zagreb UTC offset of +01:00.
- Europe/Zaporozhye - Timezone for europe/zaporozhye UTC offset of +02:00.
- Europe/Zurich - Timezone for europe/zurich UTC offset of +01:00.
- GB - Timezone for GB UTC offset of +00:00.
- GB-Eire - Timezone for GB-Eire UTC offset of +00:00.
- GMT - Timezone for GMT UTC offset of +00:00.
- GMT0 - Timezone for GMT0 UTC offset of +00:00.
- Greenwich - Timezone for greenwich UTC offset of +00:00.
- HST - Timezone for HST UTC offset of -10:00.
- Hongkong - Timezone for hongkong UTC offset of +08:00.
- Iceland - Timezone for iceland UTC offset of +00:00.
- Indian/Antananarivo - Timezone for indian/antananarivo UTC offset of +03:00.
- Indian/Chagos - Timezone for indian/chagos UTC offset of +06:00.
- Indian/Christmas - Timezone for indian/christmas UTC offset of +07:00.
- Indian/Cocos - Timezone for indian/cocos UTC offset of +06:30.
- Indian/Comoro - Timezone for indian/comoro UTC offset of +03:00.
- Indian/Kerguelen - Timezone for indian/kerguelen UTC offset of +05:00.
- Indian/Mahe - Timezone for indian/mahe UTC offset of +04:00.
- Indian/Maldives - Timezone for indian/maldives UTC offset of +05:00.
- Indian/Mauritius - Timezone for indian/mauritius UTC offset of +04:00.

- Indian/Mayotte - Timezone for indian/mayotte UTC offset of +03:00.
- Indian/Reunion - Timezone for indian/reunion UTC offset of +04:00.
- Iran - Timezone for iran UTC offset of +03:30.
- Israel - Timezone for israel UTC offset of +02:00.
- Jamaica - Timezone for jamaica UTC offset of -05:00.
- Japan - Timezone for japan UTC offset of +09:00.
- Kwajalein - Timezone for kwajalein UTC offset of +12:00.
- Libya - Timezone for libya UTC offset of +02:00.
- MET - Timezone for MET UTC offset of +01:00.
- MST - Timezone for MST UTC offset of -07:00.
- MST7MDT - Timezone for MST7MDT UTC offset of -07:00.
- Mexico/BajaNorte - Timezone for mexico/bajanorte UTC offset of -08:00.
- Mexico/BajaSur - Timezone for mexico/bajasur UTC offset of -07:00.
- Mexico/General - Timezone for mexico/general UTC offset of -06:00.
- NZ - Timezone for NZ UTC offset of +13:00.
- NZ-CHAT - Timezone for NZ-CHAT UTC offset of +13:45.
- Navajo - Timezone for navajo UTC offset of -07:00.
- PRC - Timezone for PRC UTC offset of +08:00.
- PST8PDT - Timezone for PST8PDT UTC offset of -08:00.
- Pacific/Apia - Timezone for pacific/apia UTC offset of +14:00.
- Pacific/Auckland - Timezone for pacific/auckland UTC offset of +13:00.
- Pacific/Bougainville - Timezone for pacific/bougainville UTC offset of +11:00.
- Pacific/Chatham - Timezone for pacific/chatham UTC offset of +13:45.
- Pacific/Chuuk - Timezone for pacific/chuuk UTC offset of +10:00.
- Pacific/Easter - Timezone for pacific/easter UTC offset of -05:00.
- Pacific/Efate - Timezone for pacific/efate UTC offset of +11:00.
- Pacific/Enderbury - Timezone for pacific/enderbury UTC offset of +13:00.
- Pacific/Fakaofo - Timezone for pacific/fakaofo UTC offset of +13:00.
- Pacific/Fiji - Timezone for pacific/fiji UTC offset of +13:00.
- Pacific/Funafuti - Timezone for pacific/funafuti UTC offset of +12:00.
- Pacific/Galapagos - Timezone for pacific/galapagos UTC offset of -06:00.

- Pacific/Gambier - Timezone for pacific/gambier UTC offset of -09:00.
- Pacific/Guadalcanal - Timezone for pacific/guadalcanal UTC offset of +11:00.
- Pacific/Guam - Timezone for pacific/guam UTC offset of +10:00.
- Pacific/Honolulu - Timezone for pacific/honolulu UTC offset of -10:00.
- Pacific/Johnston - Timezone for pacific/johnston UTC offset of -10:00.
- Pacific/Kiritimati - Timezone for pacific/kiritimati UTC offset of +14:00.
- Pacific/Kosrae - Timezone for pacific/kosrae UTC offset of +11:00.
- Pacific/Kwajalein - Timezone for pacific/kwajalein UTC offset of +12:00.
- Pacific/Majuro - Timezone for pacific/majuro UTC offset of +12:00.
- Pacific/Marquesas - Timezone for pacific/marquesas UTC offset of -09:30.
- Pacific/Midway - Timezone for pacific/midway UTC offset of -11:00.
- Pacific/Nauru - Timezone for pacific/nauru UTC offset of +12:00.
- Pacific/Niue - Timezone for pacific/niue UTC offset of -11:00.
- Pacific/Norfolk - Timezone for pacific/norfolk UTC offset of +11:00.
- Pacific/Noumea - Timezone for pacific/noumea UTC offset of +11:00.
- Pacific/Pago_Pago - Timezone for pacific/pago_pago UTC offset of -11:00.
- Pacific/Palau - Timezone for pacific/palau UTC offset of +09:00.
- Pacific/Pitcairn - Timezone for pacific/pitcairn UTC offset of -08:00.
- Pacific/Pohnpei - Timezone for pacific/pohnpei UTC offset of +11:00.
- Pacific/Ponape - Timezone for pacific/ponape UTC offset of +11:00.
- Pacific/Port_Moresby - Timezone for pacific/port_moresby UTC offset of +10:00.
- Pacific/Rarotonga - Timezone for pacific/rarotonga UTC offset of -10:00.
- Pacific/Saipan - Timezone for pacific/saipan UTC offset of +10:00.
- Pacific/Samoa - Timezone for pacific/samoa UTC offset of -11:00.
- Pacific/Tahiti - Timezone for pacific/tahiti UTC offset of -10:00.
- Pacific/Tarawa - Timezone for pacific/tarawa UTC offset of +12:00.
- Pacific/Tongatapu - Timezone for pacific/tongatapu UTC offset of +14:00.
- Pacific/Truk - Timezone for pacific/truk UTC offset of +10:00.
- Pacific/Wake - Timezone for pacific/wake UTC offset of +12:00.
- Pacific/Wallis - Timezone for pacific/wallis UTC offset of +12:00.
- Pacific/Yap - Timezone for pacific/yap UTC offset of +10:00.

- Poland - Timezone for poland UTC offset of +01:00.
- Portugal - Timezone for portugal UTC offset of +00:00.
- ROC - Timezone for ROC UTC offset of +08:00.
- ROK - Timezone for ROK UTC offset of +09:00.
- Singapore - Timezone for singapore UTC offset of +08:00.
- Turkey - Timezone for turkey UTC offset of +03:00.
- UCT - Timezone for UCT UTC offset of +00:00.
- US/Pacific-New - Timezone for US/Pacific-New UTC offset of -08:00.
- Universal - Timezone for universal UTC offset of +00:00.
- W-SU - Timezone for W-SU UTC offset of +03:00.
- WET - Timezone for WET UTC offset of +00:00.
- Zulu - Timezone for zulu UTC offset of +00:00.
- jdj_tz - Timezone for jdj_tz UTC offset of +00:00.

Default

Default

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.133 cmm amfTaiDiscoveryLadnTailList

AMF TAI Discovery LADN List. Syntax, description, parameters, and roles for cmm amfTaiDiscoveryLadnTailList command.

Syntax

Description

creates a list name to hold LADN names for auto-discovery. 32 list names are supported.

Parameters

Name

name - Name

Syntax

--name <string>

Description

Name of the AMF TAI Discovery LADN List

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.134 cmm amfTaiDiscoveryLadnTaiListItem

AMF LADN TAI List Names. Syntax, description, parameters, and roles for cmm amfTaiDiscoveryLadnTaiListItem command.

Syntax

Description

This command is used to create a list of AMF LADN TAI List Name items. 16384 List Items are supported.

Parameters

Name

amfTaiDiscoveryLadnTailListName - AMF LADN TAI List Names

Syntax

```
--amfTaiDiscoveryLadnTailListName <string>
```

Description

The name of the list of AMF LADN TAI List items

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

amfLadnTailListName - Name

Syntax

```
--amfLadnTailListName <string>
```

Description

specifies the name for the list of AMF LADN List Name Items

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.135 cmm amfTaiDiscoveryPraId

AMF TAI Discovery PRA Insert. Syntax, description, parameters, and roles for cmm amfTaiDiscoveryPraId command.

Syntax

Description

Specifies pre-existing PRA IDs to which a discovered TAI is to be assigned. 8192 items are supported.

Parameters

Name

amfTaiDiscoveryPraIdListName - TAI Discovery Pra ID List Name

Syntax

--amfTaiDiscoveryPraIdListName <string>

Description

Specifies the name of the list of PRA IDs

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

amfPraId - AMF PRA ID

Syntax

--amfPraId <integer>

Description

Specifies the AMF PRA ID.

Class

Required key.

Type / Range

Integer in the range of 8,388,608 to 16,777,215

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.136 cmm amfTaiDiscoveryPraIdList

AMF TAI Discovery Pra ID List. Syntax, description, parameters, and roles for cmm amfTaiDiscoveryPraIdList command.

Syntax

Description

Specifies the name of a list of AMF PRA IDs. 32 list names are supported.

Parameters

Name

name - Name

Syntax

--name <string>

Description

The name of a PRA ID list for auto-discovery.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.137 cmm amfTaiDiscoveryRegionId

AMF TAI Discovery Region ID Item. Syntax, description, parameters, and roles for cmm amfTaiDiscoveryRegionId command.

Syntax

Description

Creates an item in a amfTaiDiscoveryRegionId list. 8192 List Items are supported.

Parameters

Name

amfTaiDiscoveryRegionIdListName - AMF TAI Discovery Region ID List Name

Syntax

--amfTaiDiscoveryRegionIdListName <string>

Description

Specifies the name of a list of Region IDs

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

amfRegionId - AMF Region ID

Syntax

--amfRegionId <integer>

Description

Specifies the AMF Region ID. Used to identify an AMF group serving a region.

Class

Required key.

Type / Range

Integer in the range of 0 to 255

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.138 cmm amfTaiDiscoveryRegionIdList

AMF TAI Discovery Region Insert. Syntax, description, parameters, and roles for cmm amfTaiDiscoveryRegionIdList command.

Syntax

Description

Specifies pre-existing region Ids to which a discovered TAI is to be assigned. 32 list names are supported.

Parameters

Name

name - name

Syntax

--name <string>

Description

specifies the name of the list of Region IDs

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.139 cmm amfTaiDiscoveryTaiList

AMF Tai Discovery Tai List . Syntax, description, parameters, and roles for cmm amfTaiDiscoveryTaiList command.

Syntax

Description

Specifies the name for a list of amfTaiListname. 32 list names are supported.

Parameters

Name

name - Name

Syntax

--name <string>

Description

The name of a amfTaiDiscoveryTailist

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.140 cmm amfTaiDiscoveryTailistItem

AMF TAI List Names. Syntax, description, parameters, and roles for cmm amfTaiDiscoveryTailistItem command.

Syntax

Description

creates a list of AMF TAI List Name elements for use with amfTaiDiscovery. 8192 List Items are supported.

Parameters

Name

amfTaiDiscoveryTailListName - AMF TAI List Names Name

Syntax

--amfTaiDiscoveryTailListName <string>

Description

Name of the List of AMF TAI List Name items

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

amfTailListName - AMF TAI List Name

Syntax

--amfTailListName <string>

Description

Name of TAI list

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.141 cmm amfTailList

List of TAIs. Syntax, description, parameters, and roles for cmm amfTailList command.

Syntax

Description

Administer List of TAIs

Parameters

Name

name - Name of TAI List

Syntax

--name <string>

Description

Name of TAI list

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

taiListType - AMF TAI List Type

Syntax

--taiListType <list-item>

Description

Identifies the type of the AMF TAI list associated or linked. TAI List Type of NF Instance and SUPI Range Services do not apply to Equivalent PLMN(s).

Class

Required on create, not modifiable.

List

- mrlForbiddenTacs - TAI List for Mobility Restriction List Forbidden Tacs - Identifies TAI List associated or linked to Mobility Restriction List.
- n26lwkRestTacs - TAI List for N26 Restriction - Identifies TAI List associated with N26 Restriction.
- nfInstance - TAI List for Network Function Instance(s) - Identifies TAI List associated or linked to Network Function Instance(s) such as AMF.
- supiRangeServices - TAI List for Supi Range Services - Identifies TAI List associated or linked to Supi Range Services.

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.142 cmm amfTaiRangeDiscoveryProfile

AMF Tai Discovery. Syntax, description, parameters, and roles for cmm amfTaiRangeDiscoveryProfile command.

Syntax

Description

Specifies fields to be entered in the amfTai translation for an auto-discovered AMF TAI. A maximum of 256 profiles is supported.

Parameters

Name

name - name

Syntax

--name <string>

Description

AMF TAI Range Discovery Profile name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

maxTac - AMF TAC Max

Syntax

--maxTac <integer>

Description

The maximum TAC in the range

Class

Required on create.

Type / Range

Integer in the range of 0 to 16777215

Name

minTac - AMF TAC Min

Syntax

--minTac <integer>

Description

Lower Bound of the TAI Range

Class

Required on create.

Type / Range

Integer in the range of 0 to 16777215

Name

plmnName - Serving PLMN Name

Syntax

--plmnName <string>

Description

The serving PLMN name. It may be the name of the AMF home PLMN or shared PLMN

Class

Required on create, not modifiable.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

amfRelativeCapacity - AMF Relative Capacity

Syntax

--amfRelativeCapacity <integer>

Description

Configures AMF relative capacity for the tracking area

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

5

Name

amfTaiDiscoveryLadnTaiListName - AMF LADN TAI List Names

Syntax`--amfTaiDiscoveryLadnTaiListName <string>`**Description**

Name of the List of AMF LADN TAI List Items. This translation is referenced from amfTaiRangeDiscovery and is used for auto-inserting a discovered TAI into the amfTais.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

amfTaiDiscoveryPraIdListName - AMF TAI Discovery PRA Insert Name

Syntax`--amfTaiDiscoveryPraIdListName <string>`**Description**

Specifies the name of a list of PRAs into which a discovered TAI is to be inserted

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

amfTaiDiscoveryRegionIdListName - AMF Discovery Region Insert Name

Syntax`--amfTaiDiscoveryRegionIdListName <string>`**Description**

Specifies the name of the list of Region IDs into which a discovered TAI is to be inserted

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

amfTaiDiscoveryTailListName - AMF TAI List Names

Syntax

--amfTaiDiscoveryTailListName <string>

Description

Name of the List of AMF TAI List Names

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

emergencyFallback - Emergency Fallback

Syntax

--emergencyFallback <list-item>

Description

Enables Emergency Fallback support in the tracking area. AMF indicates to the UE that emergency fallback is supported by the network if the emergency fallback is enabled in the tracking areas sent to the UE in the tracking area list. By default, the emergency fallback is not supported.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

emergencyServiceTai - Emergency Service TAI

Syntax

--emergencyServiceTai <list-item>

Description

This command is used to enable/disable Emergency service in the tracking area. AMF indicates to the UE that emergency service is supported by the network if the emergency service is enabled in the current tracking area and all other tracking areas sent to the UE in the tracking area neighbor list. By default, the emergency service is disabled.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

imsVopsSupport - IMS VoPS Support

Syntax

--imsVopsSupport <list-item>

Description

This parameter indicates whether IMS VoPS is supported on the Tracking Areas of the AMF. Voice over PS can be supported either via Voice over New Radio connected to 5G Core, or via VoLTE connected to EPS with the NG-RAN supporting handover or redirection to EPS for the UE at QoS Flow establishment for IMS voice.

Class

Optional.

List

- amfVopsNotSupported - AMF VoPS Not Supported - Voice over PS sessions is not supported on this Tracking Area of the AMF.
- amfVopsSupportedOverNrConnectedTo5gc - AMF VoPS Support Over New Radio Connected to 5GC - Voice over PS sessions is supported on this Tracking Area of the AMF by Voice over New Radio connected to 5G Core. Note: this value is only effective when gParm voNR is enabled.
- amfVopsSupportedVoNrVolteHoRe - AMF VoPS Supported VoNR and VoLTE Handover and Redirect - Voice over PS sessions on this Tracking Area of the AMF is supported by - Voice over New Radio connected to 5G Core, -Voice over LTE connected to EPS, and the NG-RAN supports both handover and redirection to EPS at QoS Flow establishment for IMS voice. Note: this value is only effective when gParms voiceEpsFallback and VoNR are enabled.
- amfVopsSupportedVolteHo - AMF VoPS Supported VoLTE Handover - Voice over PS sessions is supported on this Tracking Area of the AMF by Voice over LTE connected

to EPS, and the NG-RAN supports handover to EPS at QoS Flow establishment for IMS voice. Note: this value is only effective when gParm voiceEpsFallback is enabled.

- amfVopsSupportedVolteHoRe - AMF VoPS Supported VoLTE Handover and Redirect - Voice over PS sessions on this Tracking Area of the AMF is supported by Voice over LTE connected to EPS, and the NG-RAN supports both handover and redirection to EPS at QoS Flow establishment for IMS voice. Note: this value is only effective when gParm voiceEpsFallback is enabled.
- amfVopsSupportedVolteRedirect - AMF VoPS Supported VoLTE Redirect - Voice over PS sessions on this Tracking Area of the AMF is supported by Voice over LTE connected to EPS, and the NG-RAN supports redirection to EPS at QoS Flow establishment for IMS voice. Note: this value is only effective when gParm voiceEpsFallback is enabled.
- amfVopsSupportedVonnrVolteHo - AMF VoPS Supported VoNR and VoLTE Handover - Voice over PS sessions on this Tracking Area of the AMF is supported by -Voice over New Radio connected to 5G Core, -Voice over LTE connected to EPS, and the NG-RAN supports handover to EPS at QoS Flow establishment for IMS voice. Note: this value is only effective when gParms voiceEpsFallback and VoNR are enabled.
- amfVopsSupportedVonnrVolteRe - AMF VoPS Supported VoNR and VoLTE Redirect - Voice over PS sessions on this Tracking Area of the AMF is supported by -Voice over New Radio connected to 5G Core, -Voice over LTE connected to EPS, and the NG-RAN supports redirection to EPS at QoS Flow establishment for IMS voice. Note: this value is only effective when gParms voiceEpsFallback and VoNR are enabled.

Default

amfVopsNotSupported

Name

snssaiListName - SNSSAI List Name

Syntax

--snssaiListName <string>

Description

The S-NSSAI in the list are authorized in the TAI. If TAI is not provisioned with a list then it is assumed that all S-NSSAIs that are supported by AMF and 5G-AN are authorized in the TAI. This parameter is not used if AMF is provisioned to use the NSSF Availability Service as in this case the Slices that are authorized in the TAI are provided by the NSSF.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

timeZoneName - Time Zone Name

Syntax

--timeZoneName <list-item>

Description

Provision time zone of the TAC.

Class

Optional.

List

- Africa/Abidjan - Timezone for africa/abidjan UTC offset of +00:00.
- Africa/Accra - Timezone for africa/accra UTC offset of +00:00.
- Africa/Addis_Ababa - Timezone for africa/addis_ababa UTC offset of +03:00.
- Africa/Algiers - Timezone for africa/algiers UTC offset of +01:00.
- Africa/Asmara - Timezone for africa/asmara UTC offset of +03:00.
- Africa/Asmera - Timezone for africa/asmera UTC offset of +03:00.
- Africa/Bamako - Timezone for africa/bamako UTC offset of +00:00.
- Africa/Bangui - Timezone for africa/bangui UTC offset of +01:00.
- Africa/Banjul - Timezone for africa/banjul UTC offset of +00:00.
- Africa/Bissau - Timezone for africa/bissau UTC offset of +00:00.
- Africa/Blantyre - Timezone for africa/blantyre UTC offset of +02:00.
- Africa/Brazzaville - Timezone for africa/brazzaville UTC offset of +01:00.
- Africa/Bujumbura - Timezone for africa/bujumbura UTC offset of +02:00.
- Africa/Cairo - Timezone for africa/cairo UTC offset of +02:00.
- Africa/Casablanca - Timezone for africa/casablanca UTC offset of +00:00.
- Africa/Ceuta - Timezone for africa/ceuta UTC offset of +01:00.
- Africa/Conakry - Timezone for africa/conakry UTC offset of +00:00.
- Africa/Dakar - Timezone for africa/dakar UTC offset of +00:00.
- Africa/Dar_es_Salaam - Timezone for africa/dar_es_salaam UTC offset of +03:00.
- Africa/Djibouti - Timezone for africa/djibouti UTC offset of +03:00.
- Africa/Douala - Timezone for africa/douala UTC offset of +01:00.
- Africa/El_Aaiun - Timezone for africa/el_aaiun UTC offset of +00:00.
- Africa/Freetown - Timezone for africa/freetown UTC offset of +00:00.
- Africa/Gaborone - Timezone for africa/gaborone UTC offset of +02:00.

- Africa/Harare - Timezone for africa/harare UTC offset of +02:00.
- Africa/Johannesburg - Timezone for africa/johannesburg UTC offset of +02:00.
- Africa/Juba - Timezone for africa/juba UTC offset of +03:00.
- Africa/Kampala - Timezone for africa/kampala UTC offset of +03:00.
- Africa/Khartoum - Timezone for africa/khartoum UTC offset of +03:00.
- Africa/Kigali - Timezone for africa/kigali UTC offset of +02:00.
- Africa/Kinshasa - Timezone for africa/kinshasa UTC offset of +01:00.
- Africa/Lagos - Timezone for africa/lagos UTC offset of +01:00.
- Africa/Libreville - Timezone for africa/libreville UTC offset of +01:00.
- Africa/Lome - Timezone for africa/lome UTC offset of +00:00.
- Africa/Luanda - Timezone for africa/luanda UTC offset of +01:00.
- Africa/Lubumbashi - Timezone for africa/lubumbashi UTC offset of +02:00.
- Africa/Lusaka - Timezone for africa/lusaka UTC offset of +02:00.
- Africa/Malabo - Timezone for africa/malabo UTC offset of +01:00.
- Africa/Maputo - Timezone for africa/maputo UTC offset of +02:00.
- Africa/Maseru - Timezone for africa/maseru UTC offset of +02:00.
- Africa/Mbabane - Timezone for africa/mbabane UTC offset of +02:00.
- Africa/Mogadishu - Timezone for africa/mogadishu UTC offset of +03:00.
- Africa/Monrovia - Timezone for africa/monrovia UTC offset of +00:00.
- Africa/Nairobi - Timezone for africa/nairobi UTC offset of +03:00.
- Africa/Ndjamena - Timezone for africa/ndjamena UTC offset of +01:00.
- Africa/Niamey - Timezone for africa/niamey UTC offset of +01:00.
- Africa/Nouakchott - Timezone for africa/nouakchott UTC offset of +00:00.
- Africa/Ouagadougou - Timezone for africa/ouagadougou UTC offset of +00:00.
- Africa/Porto-Novo - Timezone for africa/porto-novo UTC offset of +01:00.
- Africa/Sao_Tome - Timezone for africa/sao_tome UTC offset of +00:00.
- Africa/Timbuktu - Timezone for africa/timbuktu UTC offset of +00:00.
- Africa/Tripoli - Timezone for africa/tripoli UTC offset of +02:00.
- Africa/Tunis - Timezone for africa/tunis UTC offset of +01:00.
- Africa/Windhoek - Timezone for africa/windhoek UTC offset of +02:00.
- America/Adak - Timezone for america/adak UTC offset of -10:00.

- America/Anchorage - Timezone for america/anchorage UTC offset of -09:00.
- America/Anguilla - Timezone for america/anguilla UTC offset of -04:00.
- America/Antigua - Timezone for america/antigua UTC offset of -04:00.
- America/Araguaina - Timezone for america/araguaina UTC offset of -03:00.
- America/Argentina/Buenos_Aires - Timezone for america/argentina/buenos_aires UTC offset of -03:00.
- America/Argentina/Catamarca - Timezone for america/argentina/catamarca UTC offset of -03:00.
- America/Argentina/ComodRivadavia - Timezone for america/argentina/comodrivadavia UTC offset of -03:00.
- America/Argentina/Cordoba - Timezone for america/argentina/cordoba UTC offset of -03:00.
- America/Argentina/Jujuy - Timezone for america/argentina/jujuy UTC offset of -03:00.
- America/Argentina/La_Rioja - Timezone for america/argentina/la_rioja UTC offset of -03:00.
- America/Argentina/Mendoza - Timezone for america/argentina/mendoza UTC offset of -03:00.
- America/Argentina/Rio_Gallegos - Timezone for america/argentina/rio_gallegos UTC offset of -03:00.
- America/Argentina/Salta - Timezone for america/argentina/salta UTC offset of -03:00.
- America/Argentina/San_Juan - Timezone for america/argentina/san_juan UTC offset of -03:00.
- America/Argentina/San_Luis - Timezone for america/argentina/san_luis UTC offset of -03:00.
- America/Argentina/Tucuman - Timezone for america/argentina/tucuman UTC offset of -03:00.
- America/Argentina/Ushuaia - Timezone for america/argentina/ushuaia UTC offset of -03:00.
- America/Aruba - Timezone for america/aruba UTC offset of -04:00.
- America/Asuncion - Timezone for america/asuncion UTC offset of -03:00.
- America/Atikokan - Timezone for america/atikokan UTC offset of -05:00.
- America/Atka - Timezone for america/atka UTC offset of -10:00.
- America/Bahia - Timezone for america/bahia UTC offset of -03:00.
- America/Bahia_Banderas - Timezone for america/bahia_banderas UTC offset of -06:00.

- America/Barbados - Timezone for america/barbados UTC offset of -04:00.
- America/Belem - Timezone for america/belem UTC offset of -03:00.
- America/Belize - Timezone for america/belize UTC offset of -06:00.
- America/Blanc-Sablon - Timezone for america/blanc-sablon UTC offset of -04:00.
- America/Boa_Vista - Timezone for america/boa_vista UTC offset of -04:00.
- America/Bogota - Timezone for america/bogota UTC offset of -05:00.
- America/Boise - Timezone for america/boise UTC offset of -07:00.
- America/Buenos_Aires - Timezone for america/buenos_aires UTC offset of -03:00.
- America/Cambridge_Bay - Timezone for america/cambridge_bay UTC offset of -07:00.
- America/Campo_Grande - Timezone for america/campo_grande UTC offset of -03:00.
- America/Cancun - Timezone for america/cancun UTC offset of -05:00.
- America/Caracas - Timezone for america/caracas UTC offset of -04:00.
- America/Catamarca - Timezone for america/catamarca UTC offset of -03:00.
- America/Cayenne - Timezone for america/cayenne UTC offset of -03:00.
- America/Cayman - Timezone for america/cayman UTC offset of -05:00.
- America/Chicago - Timezone for america/chicago UTC offset of -06:00.
- America/Chihuahua - Timezone for america/chihuahua UTC offset of -07:00.
- America/Coral_Harbour - Timezone for america/coral_harbour UTC offset of -05:00.
- America/Cordoba - Timezone for america/cordoba UTC offset of -03:00.
- America/Costa_Rica - Timezone for america/costa_rica UTC offset of -06:00.
- America/Creston - Timezone for america/creston UTC offset of -07:00.
- America/Cuiaba - Timezone for america/cuiaba UTC offset of -03:00.
- America/Curacao - Timezone for america/curacao UTC offset of -04:00.
- America/Danmarkshavn - Timezone for america/danmarkshavn UTC offset of +00:00.
- America/Dawson - Timezone for america/dawson UTC offset of -08:00.
- America/Dawson_Creek - Timezone for america/dawson_creek UTC offset of -07:00.
- America/Denver - Timezone for america/denver UTC offset of -07:00.
- America/Detroit - Timezone for america/detroit UTC offset of -05:00.
- America/Dominica - Timezone for america/dominica UTC offset of -04:00.
- America/Edmonton - Timezone for america/edmonton UTC offset of -07:00.
- America/Eirunepe - Timezone for america/eirunepe UTC offset of -05:00.

- America/El_Salvador - Timezone for america/el_salvador UTC offset of -06:00.
- America/Ensenada - Timezone for america/ensenada UTC offset of -08:00.
- America/Fort_Nelson - Timezone for america/fort_nelson UTC offset of -07:00.
- America/Fort_Wayne - Timezone for america/fort_wayne UTC offset of -05:00.
- America/Fortaleza - Timezone for america/fortaleza UTC offset of -03:00.
- America/Glace_Bay - Timezone for america/glance_bay UTC offset of -04:00.
- America/Godthab - Timezone for america/godthab UTC offset of -03:00.
- America/Goose_Bay - Timezone for america/goose_bay UTC offset of -04:00.
- America/Grand_Turk - Timezone for america/grand_turk UTC offset of -04:00.
- America/Grenada - Timezone for america/grenada UTC offset of -04:00.
- America/Guadeloupe - Timezone for america/guadeloupe UTC offset of -04:00.
- America/Guatemala - Timezone for america/guatemala UTC offset of -06:00.
- America/Guayaquil - Timezone for america/guayaquil UTC offset of -05:00.
- America/Guyana - Timezone for america/guyana UTC offset of -04:00.
- America/Halifax - Timezone for america/halifax UTC offset of -04:00.
- America/Havana - Timezone for america/havana UTC offset of -05:00.
- America/Hermosillo - Timezone for america/hermosillo UTC offset of -07:00.
- America/Indiana/Indianapolis - Timezone for america/indiana/indianapolis UTC offset of -05:00.
- America/Indiana/Knox - Timezone for america/indiana/knox UTC offset of -06:00.
- America/Indiana/Marengo - Timezone for america/indiana/marengo UTC offset of -05:00.
- America/Indiana/Petersburg - Timezone for america/indiana/petersburg UTC offset of -05:00.
- America/Indiana/Tell_City - Timezone for america/indiana/tell_city UTC offset of -06:00.
- America/Indiana/Vevay - Timezone for america/indiana/vevay UTC offset of -05:00.
- America/Indiana/Vincennes - Timezone for america/indiana/vincennes UTC offset of -05:00.
- America/Indiana/Winamac - Timezone for america/indiana/winamac UTC offset of -05:00.
- America/Indianapolis - Timezone for america/indianapolis UTC offset of -05:00.
- America/Inuvik - Timezone for america/inuvik UTC offset of -07:00.

- America/Iqaluit - Timezone for america/iqaluit UTC offset of -05:00.
- America/Jamaica - Timezone for america/jamaica UTC offset of -05:00.
- America/Jujuy - Timezone for america/jujuy UTC offset of -03:00.
- America/Juneau - Timezone for america/juneau UTC offset of -09:00.
- America/Kentucky/Louisville - Timezone for america/kentucky/louisville UTC offset of -05:00.
- America/Kentucky/Monticello - Timezone for america/kentucky/monticello UTC offset of -05:00.
- America/Knox_IN - Timezone for america/knox_in UTC offset of -06:00.
- America/Kralendijk - Timezone for america/kralendijk UTC offset of -04:00.
- America/La_Paz - Timezone for america/la_paz UTC offset of -04:00.
- America/Lima - Timezone for america/lima UTC offset of -05:00.
- America/Los_Angeles - Timezone for america/los_angeles UTC offset of -08:00.
- America/Louisville - Timezone for america/louisville UTC offset of -05:00.
- America/Lower_Princes - Timezone for america/lower_princes UTC offset of -04:00.
- America/Maceio - Timezone for america/maceio UTC offset of -03:00.
- America/Managua - Timezone for america/managua UTC offset of -06:00.
- America/Manaus - Timezone for america/manuel UTC offset of -04:00.
- America/Marigot - Timezone for america/marigot UTC offset of -04:00.
- America/Martinique - Timezone for america/martinique UTC offset of -04:00.
- America/Matamoros - Timezone for america/matamoros UTC offset of -06:00.
- America/Mazatlan - Timezone for america/mazatlan UTC offset of -07:00.
- America/Mendoza - Timezone for america/mendoza UTC offset of -03:00.
- America/Menominee - Timezone for america/menominee UTC offset of -06:00.
- America/Merida - Timezone for america/merida UTC offset of -06:00.
- America/Metlakatla - Timezone for america/metlakatla UTC offset of -09:00.
- America/Mexico_City - Timezone for america/mexico_city UTC offset of -06:00.
- America/Miquelon - Timezone for america/miquelon UTC offset of -03:00.
- America/Moncton - Timezone for america/moncton UTC offset of -04:00.
- America/Monterrey - Timezone for america/monterrey UTC offset of -06:00.
- America/Montevideo - Timezone for america/montevideo UTC offset of -03:00.
- America/Montreal - Timezone for america/montreal UTC offset of -05:00.

- America/Montserrat - Timezone for america/montserrat UTC offset of -04:00.
- America/Nassau - Timezone for america/nassau UTC offset of -05:00.
- America/New_York - Timezone for america/new_york UTC offset of -05:00.
- America/Nipigon - Timezone for america/nipigon UTC offset of -05:00.
- America/Nome - Timezone for america/nome UTC offset of -09:00.
- America/Noronha - Timezone for america/noronha UTC offset of -02:00.
- America/North_Dakota/Beulah - Timezone for america/north_dakota/beulah UTC offset of -06:00.
- America/North_Dakota/Center - Timezone for america/north_dakota/center UTC offset of -06:00.
- America/North_Dakota/New_Salem - Timezone for america/north_dakota/new_salem UTC offset of -06:00.
- America/Ojinaga - Timezone for america/ojinaga UTC offset of -07:00.
- America/Panama - Timezone for america/panama UTC offset of -05:00.
- America/Pangnirtung - Timezone for america/pangnirtung UTC offset of -05:00.
- America/Paramaribo - Timezone for america/paramaribo UTC offset of -03:00.
- America/Phoenix - Timezone for america/phoenix UTC offset of -07:00.
- America/Port-au-Prince - Timezone for america/port-au-prince UTC offset of -05:00.
- America/Port_of_Spain - Timezone for america/port_of_spain UTC offset of -04:00.
- America/Porto_Acre - Timezone for america/porto_acre UTC offset of -05:00.
- America/Porto_Velho - Timezone for america/porto_velho UTC offset of -04:00.
- America/Puerto_Rico - Timezone for america/puerto_rico UTC offset of -04:00.
- America/Rainy_River - Timezone for america/rainy_river UTC offset of -06:00.
- America/Rankin_Inlet - Timezone for america/rankin_inlet UTC offset of -06:00.
- America/Recife - Timezone for america/recife UTC offset of -03:00.
- America/Regina - Timezone for america/regina UTC offset of -06:00.
- America/Resolute - Timezone for america/resolute UTC offset of -06:00.
- America/Rio_Branco - Timezone for america/rio_branco UTC offset of -05:00.
- America/Rosario - Timezone for america/rosario UTC offset of -03:00.
- America/Santa_Isabel - Timezone for america/santa_isabel UTC offset of -08:00.
- America/Santarem - Timezone for america/santarem UTC offset of -03:00.
- America/Santiago - Timezone for america/santiago UTC offset of -03:00.

- America/Santo_Domingo - Timezone for america/santo_domingo UTC offset of -04:00.
- America/Sao_Paulo - Timezone for america/sao_paulo UTC offset of -02:00.
- America/Scoresbysund - Timezone for america/scoresbysund UTC offset of -01:00.
- America/Shiprock - Timezone for america/shiprock UTC offset of -07:00.
- America/Sitka - Timezone for america/sitka UTC offset of -09:00.
- America/St_Barthelemy - Timezone for america/st_barthelemy UTC offset of -04:00.
- America/St_Johns - Timezone for america/st_johns UTC offset of -03:30.
- America/St_Kitts - Timezone for america/st_kitts UTC offset of -04:00.
- America/St_Lucia - Timezone for america/st_lucia UTC offset of -04:00.
- America/St_Thomas - Timezone for america/st_thomas UTC offset of -04:00.
- America/St_Vincent - Timezone for america/st_vincent UTC offset of -04:00.
- America/Swift_Current - Timezone for america/swift_current UTC offset of -06:00.
- America/Tegucigalpa - Timezone for america/tegucigalpa UTC offset of -06:00.
- America/Thule - Timezone for america/thule UTC offset of -04:00.
- America/Thunder_Bay - Timezone for america/thunder_bay UTC offset of -05:00.
- America/Tijuana - Timezone for america/tijuana UTC offset of -08:00.
- America/Toronto - Timezone for america/toronto UTC offset of -05:00.
- America/Tortola - Timezone for america/tortola UTC offset of -04:00.
- America/Vancouver - Timezone for america/vancouver UTC offset of -08:00.
- America/Virgin - Timezone for america/virgin UTC offset of -04:00.
- America/Whitehorse - Timezone for america/whitehorse UTC offset of -08:00.
- America/Winnipeg - Timezone for america/winnipeg UTC offset of -06:00.
- America/Yakutat - Timezone for america/yakutat UTC offset of -09:00.
- America/Yellowknife - Timezone for america/yellowknife UTC offset of -07:00.
- Antarctica/Casey - Timezone for antarctica/casey UTC offset of +11:00.
- Antarctica/Davis - Timezone for antarctica/davis UTC offset of +07:00.
- Antarctica/DumontDURville - Timezone for antarctica/dumontdurville UTC offset of +10:00.
- Antarctica/Macquarie - Timezone for antarctica/macquarie UTC offset of +11:00.
- Antarctica/Mawson - Timezone for antarctica/mawson UTC offset of +05:00.
- Antarctica/McMurdo - Timezone for antarctica/mcmurdo UTC offset of +13:00.

- Antarctica/Palmer - Timezone for antarctica/palmer UTC offset of -03:00.
- Antarctica/Rothera - Timezone for antarctica/rothera UTC offset of -03:00.
- Antarctica/South_Pole - Timezone for antarctica/south_pole UTC offset of +13:00.
- Antarctica/Syowa - Timezone for antarctica/syowa UTC offset of +03:00.
- Antarctica/Troll - Timezone for antarctica/troll UTC offset of +00:00.
- Antarctica/Vostok - Timezone for antarctica/vostok UTC offset of +06:00.
- Arctic/Longyearbyen - Timezone for arctic/longyearbyen UTC offset of +01:00.
- Asia/Aden - Timezone for asia/aden UTC offset of +03:00.
- Asia/Almaty - Timezone for asia/almaty UTC offset of +06:00.
- Asia/Amman - Timezone for asia/amman UTC offset of +02:00.
- Asia/Anadyr - Timezone for asia/anadyr UTC offset of +12:00.
- Asia/Aqtau - Timezone for asia/aqtau UTC offset of +05:00.
- Asia/Aqtobe - Timezone for asia/aqtobe UTC offset of +05:00.
- Asia/Ashgabat - Timezone for asia/ashgabat UTC offset of +05:00.
- Asia/Ashkhabad - Timezone for asia/ashkhabad UTC offset of +05:00.
- Asia/Baghdad - Timezone for asia/baghdad UTC offset of +03:00.
- Asia/Bahrain - Timezone for asia/bahrain UTC offset of +03:00.
- Asia/Baku - Timezone for asia/baku UTC offset of +04:00.
- Asia/Bangkok - Timezone for asia/bangkok UTC offset of +07:00.
- Asia/Barnaul - Timezone for asia/barnaul UTC offset of +07:00.
- Asia/Beirut - Timezone for asia/beirut UTC offset of +02:00.
- Asia/Bishkek - Timezone for asia/bishkek UTC offset of +06:00.
- Asia/Brunei - Timezone for asia/brunei UTC offset of +08:00.
- Asia/Calcutta - Timezone for asia/calcutta UTC offset of +05:30.
- Asia/Chita - Timezone for asia/chita UTC offset of +09:00.
- Asia/Choibalsan - Timezone for asia/choibalsan UTC offset of +08:00.
- Asia/Chongqing - Timezone for asia/chongqing UTC offset of +08:00.
- Asia/Chungking - Timezone for asia/chungking UTC offset of +08:00.
- Asia/Colombo - Timezone for asia/colombo UTC offset of +05:30.
- Asia/Dacca - Timezone for asia/dacca UTC offset of +06:00.
- Asia/Damascus - Timezone for asia/damascus UTC offset of +02:00.

- Asia/Dhaka - Timezone for asia/dhaka UTC offset of +06:00.
- Asia/Dili - Timezone for asia/dili UTC offset of +09:00.
- Asia/Dubai - Timezone for asia/dubai UTC offset of +04:00.
- Asia/Dushanbe - Timezone for asia/dushanbe UTC offset of +05:00.
- Asia/Gaza - Timezone for asia/gaza UTC offset of +02:00.
- Asia/Harbin - Timezone for asia/harbin UTC offset of +08:00.
- Asia/Hebron - Timezone for asia/hebron UTC offset of +02:00.
- Asia/Ho_Chi_Minh - Timezone for asia/ho_chi_minh UTC offset of +07:00.
- Asia/Hong_Kong - Timezone for asia/hong_kong UTC offset of +08:00.
- Asia/Hovd - Timezone for asia/hovd UTC offset of +07:00.
- Asia/Irkutsk - Timezone for asia/irkutsk UTC offset of +08:00.
- Asia/Istanbul - Timezone for asia/istanbul UTC offset of +03:00.
- Asia/Jakarta - Timezone for asia/jakarta UTC offset of +07:00.
- Asia/Jayapura - Timezone for asia/jayapura UTC offset of +09:00.
- Asia/Jerusalem - Timezone for asia/jerusalem UTC offset of +02:00.
- Asia/Kabul - Timezone for asia/kabul UTC offset of +04:30.
- Asia/Kamchatka - Timezone for asia/kamchatka UTC offset of +12:00.
- Asia/Karachi - Timezone for asia/karachi UTC offset of +05:00.
- Asia/Kashgar - Timezone for asia/kashgar UTC offset of +06:00.
- Asia/Kathmandu - Timezone for asia/kathmandu UTC offset of +05:45.
- Asia/Katmandu - Timezone for asia/katmandu UTC offset of +05:45.
- Asia/Khandyga - Timezone for asia/khandyga UTC offset of +09:00.
- Asia/Kolkata - Timezone for asia/kolkata UTC offset of +05:30.
- Asia/Krasnoyarsk - Timezone for asia/krasnoyarsk UTC offset of +07:00.
- Asia/Kuala_Lumpur - Timezone for asia/kuala_lumpur UTC offset of +08:00.
- Asia/Kuching - Timezone for asia/kuching UTC offset of +08:00.
- Asia/Kuwait - Timezone for asia/kuwait UTC offset of +03:00.
- Asia/Macao - Timezone for asia/macao UTC offset of +08:00.
- Asia/Macau - Timezone for asia/macau UTC offset of +08:00.
- Asia/Magadan - Timezone for asia/magadan UTC offset of +11:00.
- Asia/Makassar - Timezone for asia/makassar UTC offset of +08:00.

- Asia/Manila - Timezone for asia/manila UTC offset of +08:00.
- Asia/Muscat - Timezone for asia/muscat UTC offset of +04:00.
- Asia/Nicosia - Timezone for asia/nicosia UTC offset of +02:00.
- Asia/Novokuznetsk - Timezone for asia/novokuznetsk UTC offset of +07:00.
- Asia/Novosibirsk - Timezone for asia/novosibirsk UTC offset of +07:00.
- Asia/Omsk - Timezone for asia/omsk UTC offset of +06:00.
- Asia/Oral - Timezone for asia/oral UTC offset of +05:00.
- Asia/Phnom_Penh - Timezone for asia/phnom_penh UTC offset of +07:00.
- Asia/Pontianak - Timezone for asia/pontianak UTC offset of +07:00.
- Asia/Pyongyang - Timezone for asia/pyongyang UTC offset of +08:30.
- Asia/Qatar - Timezone for asia/qatar UTC offset of +03:00.
- Asia/Qyzylorda - Timezone for asia/qyzylorda UTC offset of +06:00.
- Asia/Rangoon - Timezone for asia/rangoon UTC offset of +06:30.
- Asia/Riyadh - Timezone for asia/riyadh UTC offset of +03:00.
- Asia/Saigon - Timezone for asia/saigon UTC offset of +07:00.
- Asia/Sakhalin - Timezone for asia/sakhalin UTC offset of +11:00.
- Asia/Samarkand - Timezone for asia/samarkand UTC offset of +05:00.
- Asia/Seoul - Timezone for asia/seoul UTC offset of +09:00.
- Asia/Shanghai - Timezone for asia/shanghai UTC offset of +08:00.
- Asia/Singapore - Timezone for asia/singapore UTC offset of +08:00.
- Asia/Srednekolymsk - Timezone for asia/srednekolymsk UTC offset of +11:00.
- Asia/Taipei - Timezone for asia/taipei UTC offset of +08:00.
- Asia/Tashkent - Timezone for asia/tashkent UTC offset of +05:00.
- Asia/Tbilisi - Timezone for asia/tbilisi UTC offset of +04:00.
- Asia/Tehran - Timezone for asia/tehran UTC offset of +03:30.
- Asia/Tel_Aviv - Timezone for asia/tel_aviv UTC offset of +02:00.
- Asia/Thimbu - Timezone for asia/thimbu UTC offset of +06:00.
- Asia/Thimphu - Timezone for asia/thimphu UTC offset of +06:00.
- Asia/Tokyo - Timezone for asia/tokyo UTC offset of +09:00.
- Asia/Tomsk - Timezone for asia/tomsk UTC offset of +07:00.
- Asia/Ujung_Pandang - Timezone for asia/ujung_pandang UTC offset of +08:00.

- Asia/Ulaanbaatar - Timezone for asia/ulaanbaatar UTC offset of +08:00.
- Asia/Ulan_Bator - Timezone for asia/ulan_bator UTC offset of +08:00.
- Asia/Urumqi - Timezone for asia/urumqi UTC offset of +06:00.
- Asia/Ust-Nera - Timezone for asia/ust-nera UTC offset of +10:00.
- Asia/Vientiane - Timezone for asia/vientiane UTC offset of +07:00.
- Asia/Vladivostok - Timezone for asia/vladivostok UTC offset of +10:00.
- Asia/Yakutsk - Timezone for asia/yakutsk UTC offset of +09:00.
- Asia/Yekaterinburg - Timezone for asia/yekaterinburg UTC offset of +05:00.
- Asia/Yerevan - Timezone for asia/yerevan UTC offset of +04:00.
- Atlantic/Azores - Timezone for atlantic/azores UTC offset of -01:00.
- Atlantic/Bermuda - Timezone for atlantic/bermuda UTC offset of -04:00.
- Atlantic/Canary - Timezone for atlantic/canary UTC offset of +00:00.
- Atlantic/Cape_Verde - Timezone for atlantic/cape_verde UTC offset of -01:00.
- Atlantic/Faeroe - Timezone for atlantic/faeroe UTC offset of +00:00.
- Atlantic/Faroe - Timezone for atlantic/faroe UTC offset of +00:00.
- Atlantic/Jan_Mayen - Timezone for atlantic/jan_mayen UTC offset of +01:00.
- Atlantic/Madeira - Timezone for atlantic/madeira UTC offset of +00:00.
- Atlantic/Reykjavik - Timezone for atlantic/reykjavik UTC offset of +00:00.
- Atlantic/South_Georgia - Timezone for atlantic/south_georgia UTC offset of -02:00.
- Atlantic/St_Helena - Timezone for atlantic/st_helena UTC offset of +00:00.
- Atlantic/Stanley - Timezone for atlantic/stanley UTC offset of -03:00.
- Australia/ACT - Timezone for australia/act UTC offset of +11:00.
- Australia/Adelaide - Timezone for australia/adelaide UTC offset of +10:30.
- Australia/Brisbane - Timezone for australia/brisbane UTC offset of +10:00.
- Australia/Broken_Hill - Timezone for australia/broken_hill UTC offset of +10:30.
- Australia/Canberra - Timezone for australia/canberra UTC offset of +11:00.
- Australia/Currie - Timezone for australia/currie UTC offset of +11:00.
- Australia/Darwin - Timezone for australia/darwin UTC offset of +09:30.
- Australia/Eucla - Timezone for australia/eucla UTC offset of +08:45.
- Australia/Hobart - Timezone for australia/hobart UTC offset of +11:00.
- Australia/LHI - Timezone for australia/lhi UTC offset of +11:00.

- Australia/Lindeman - Timezone for australia/lindeman UTC offset of +10:00.
- Australia/Lord_Howe - Timezone for australia/lord_howe UTC offset of +11:00.
- Australia/Melbourne - Timezone for australia/melbourne UTC offset of +11:00.
- Australia/NSW - Timezone for australia/nsw UTC offset of +11:00.
- Australia/North - Timezone for australia/north UTC offset of +09:30.
- Australia/Perth - Timezone for australia/perth UTC offset of +08:00.
- Australia/Queensland - Timezone for australia/queensland UTC offset of +10:00.
- Australia/South - Timezone for australia/south UTC offset of +10:30.
- Australia/Sydney - Timezone for australia/sydney UTC offset of +11:00.
- Australia/Tasmania - Timezone for australia/tasmania UTC offset of +11:00.
- Australia/Victoria - Timezone for australia/victoria UTC offset of +11:00.
- Australia/West - Timezone for australia/west UTC offset of +08:00.
- Australia/Yancowinna - Timezone for australia/yancowinna UTC offset of +10:30.
- CET - Timezone for CET UTC offset of +01:00.
- CST6CDT - Timezone for CST6CDT UTC offset of -06:00.
- Canada/Atlantic - Timezone for canada/atlantic UTC offset of -04:00.
- Canada/Central - Timezone for canada/central UTC offset of -06:00.
- Canada/East-Saskatchewan - Timezone for canada/east-saskatchewan UTC offset of -06:00.
- Canada/Eastern - Timezone for canada/eastern UTC offset of -05:00.
- Canada/Mountain - Timezone for canada/mountain UTC offset of -07:00.
- Canada/Newfoundland - Timezone for canada/newfoundland UTC offset of -03:30.
- Canada/Pacific - Timezone for canada/pacific UTC offset of -08:00.
- Canada/Saskatchewan - Timezone for canada/saskatchewan UTC offset of -06:00.
- Canada/Yukon - Timezone for canada/yukon UTC offset of -08:00.
- Chile/Continental - Timezone for chile/continental UTC offset of -03:00.
- Chile/EasterIsland - Timezone for chile/easterisland UTC offset of -05:00.
- Cuba - Timezone for cuba UTC offset of -05:00.
- Default - Timezone for system UTC (OS time zone)
- EET - Timezone for EET UTC offset of +02:00.
- EST - Timezone for EST UTC offset of -05:00.

- EST5EDT - Timezone for EST5EDT UTC offset of -05:00.
- Egypt - Timezone for egypt UTC offset of +02:00.
- Eire - Timezone for eire UTC offset of +00:00.
- Etc/Greenwich - Timezone for etc/greenwich UTC offset of +00:00.
- Etc/UCT - Timezone for etc/uct UTC offset of +00:00.
- Etc/UTC - Timezone for etc/utc UTC offset of +00:00.
- Etc/Universal - Timezone for etc/universal UTC offset of +00:00.
- Etc/Zulu - Timezone for etc/zulu UTC offset of +00:00.
- Europe/Amsterdam - Timezone for europe/amsterdam UTC offset of +01:00.
- Europe/Andorra - Timezone for europe/andorra UTC offset of +01:00.
- Europe/Astrakhan - Timezone for europe/astrakhan UTC offset of +04:00.
- Europe/Athens - Timezone for europe/athens UTC offset of +02:00.
- Europe/Belfast - Timezone for europe/belfast UTC offset of +00:00.
- Europe/Belgrade - Timezone for europe/belgrade UTC offset of +01:00.
- Europe/Berlin - Timezone for europe/berlin UTC offset of +01:00.
- Europe/Bratislava - Timezone for europe/bratislava UTC offset of +01:00.
- Europe/Brussels - Timezone for europe/brussels UTC offset of +01:00.
- Europe/Bucharest - Timezone for europe/bucharest UTC offset of +02:00.
- Europe/Budapest - Timezone for europe/budapest UTC offset of +01:00.
- Europe/Busingen - Timezone for europe/busingen UTC offset of +01:00.
- Europe/Chisinau - Timezone for europe/chisinau UTC offset of +02:00.
- Europe/Copenhagen - Timezone for europe/copenhagen UTC offset of +01:00.
- Europe/Dublin - Timezone for europe/dublin UTC offset of +00:00.
- Europe/Gibraltar - Timezone for europe/gibraltar UTC offset of +01:00.
- Europe/Guernsey - Timezone for europe/guernsey UTC offset of +00:00.
- Europe/Helsinki - Timezone for europe/helsinki UTC offset of +02:00.
- Europe/Isle_of_Man - Timezone for europe/isle_of_man UTC offset of +00:00.
- Europe/Istanbul - Timezone for europe/istanbul UTC offset of +03:00.
- Europe/Jersey - Timezone for europe/jersey UTC offset of +00:00.
- Europe/Kaliningrad - Timezone for europe/kaliningrad UTC offset of +02:00.
- Europe/Kiev - Timezone for europe/kyiv UTC offset of +02:00.

- Europe/Kirov - Timezone for europe/kirov UTC offset of +03:00.
- Europe/Lisbon - Timezone for europe/lisbon UTC offset of +00:00.
- Europe/Ljubljana - Timezone for europe/ljubljana UTC offset of +01:00.
- Europe/London - Timezone for europe/london UTC offset of +00:00.
- Europe/Luxembourg - Timezone for europe/luxembourg UTC offset of +01:00.
- Europe/Madrid - Timezone for europe/madrid UTC offset of +01:00.
- Europe/Malta - Timezone for europe/malta UTC offset of +01:00.
- Europe/Mariehamn - Timezone for europe/mariehamn UTC offset of +02:00.
- Europe/Minsk - Timezone for europe/minsk UTC offset of +03:00.
- Europe/Monaco - Timezone for europe/monaco UTC offset of +01:00.
- Europe/Moscow - Timezone for europe/moscow UTC offset of +03:00.
- Europe/Nicosia - Timezone for europe/nicosia UTC offset of +02:00.
- Europe/Oslo - Timezone for europe/oslo UTC offset of +01:00.
- Europe/Paris - Timezone for europe/paris UTC offset of +01:00.
- Europe/Podgorica - Timezone for europe/podgorica UTC offset of +01:00.
- Europe/Prague - Timezone for europe/prague UTC offset of +01:00.
- Europe/Riga - Timezone for europe/riga UTC offset of +02:00.
- Europe/Rome - Timezone for europe/rome UTC offset of +01:00.
- Europe/Samara - Timezone for europe/samara UTC offset of +04:00.
- Europe/San_Marino - Timezone for europe/san_marino UTC offset of +01:00.
- Europe/Sarajevo - Timezone for europe/sarajevo UTC offset of +01:00.
- Europe/Simferopol - Timezone for europe/simferopol UTC offset of +03:00.
- Europe/Skopje - Timezone for europe/skopje UTC offset of +01:00.
- Europe/Sofia - Timezone for europe/sofia UTC offset of +02:00.
- Europe/Stockholm - Timezone for europe/stockholm UTC offset of +01:00.
- Europe/Tallinn - Timezone for europe/tallinn UTC offset of +02:00.
- Europe/Tirane - Timezone for europe/tirane UTC offset of +01:00.
- Europe/Tiraspol - Timezone for europe/tiraspol UTC offset of +02:00.
- Europe/Ulyanovsk - Timezone for europe/ulyanovsk UTC offset of +04:00.
- Europe/Uzhgorod - Timezone for europe/uzhgorod UTC offset of +02:00.
- Europe/Vaduz - Timezone for europe/vaduz UTC offset of +01:00.

- Europe/Vatican - Timezone for europe/vatican UTC offset of +01:00.
- Europe/Vienna - Timezone for europe/vienna UTC offset of +01:00.
- Europe/Vilnius - Timezone for europe/vilnius UTC offset of +02:00.
- Europe/Volgograd - Timezone for europe/volgograd UTC offset of +03:00.
- Europe/Warsaw - Timezone for europe/warsaw UTC offset of +01:00.
- Europe/Zagreb - Timezone for europe/zagreb UTC offset of +01:00.
- Europe/Zaporozhye - Timezone for europe/zaporozhye UTC offset of +02:00.
- Europe/Zurich - Timezone for europe/zurich UTC offset of +01:00.
- GB - Timezone for GB UTC offset of +00:00.
- GB-Eire - Timezone for GB-Eire UTC offset of +00:00.
- GMT - Timezone for GMT UTC offset of +00:00.
- GMT0 - Timezone for GMT0 UTC offset of +00:00.
- Greenwich - Timezone for greenwich UTC offset of +00:00.
- HST - Timezone for HST UTC offset of -10:00.
- Hongkong - Timezone for hongkong UTC offset of +08:00.
- Iceland - Timezone for iceland UTC offset of +00:00.
- Indian/Antananarivo - Timezone for indian/antananarivo UTC offset of +03:00.
- Indian/Chagos - Timezone for indian/chagos UTC offset of +06:00.
- Indian/Christmas - Timezone for indian/christmas UTC offset of +07:00.
- Indian/Cocos - Timezone for indian/cocos UTC offset of +06:30.
- Indian/Comoro - Timezone for indian/comoro UTC offset of +03:00.
- Indian/Kerguelen - Timezone for indian/kerguelen UTC offset of +05:00.
- Indian/Mahe - Timezone for indian/mahe UTC offset of +04:00.
- Indian/Maldives - Timezone for indian/maldives UTC offset of +05:00.
- Indian/Mauritius - Timezone for indian/mauritius UTC offset of +04:00.
- Indian/Mayotte - Timezone for indian/mayotte UTC offset of +03:00.
- Indian/Reunion - Timezone for indian/reunion UTC offset of +04:00.
- Iran - Timezone for iran UTC offset of +03:30.
- Israel - Timezone for israel UTC offset of +02:00.
- Jamaica - Timezone for jamaica UTC offset of -05:00.
- Japan - Timezone for japan UTC offset of +09:00.

- Kwajalein - Timezone for kwajalein UTC offset of +12:00.
- Libya - Timezone for libya UTC offset of +02:00.
- MET - Timezone for MET UTC offset of +01:00.
- MST - Timezone for MST UTC offset of -07:00.
- MST7MDT - Timezone for MST7MDT UTC offset of -07:00.
- Mexico/BajaNorte - Timezone for mexico/bajanorte UTC offset of -08:00.
- Mexico/BajaSur - Timezone for mexico/bajasur UTC offset of -07:00.
- Mexico/General - Timezone for mexico/general UTC offset of -06:00.
- NZ - Timezone for NZ UTC offset of +13:00.
- NZ-CHAT - Timezone for NZ-CHAT UTC offset of +13:45.
- Navajo - Timezone for navajo UTC offset of -07:00.
- PRC - Timezone for PRC UTC offset of +08:00.
- PST8PDT - Timezone for PST8PDT UTC offset of -08:00.
- Pacific/Apia - Timezone for pacific/apia UTC offset of +14:00.
- Pacific/Auckland - Timezone for pacific/auckland UTC offset of +13:00.
- Pacific/Bougainville - Timezone for pacific/bougainville UTC offset of +11:00.
- Pacific/Chatham - Timezone for pacific/chatham UTC offset of +13:45.
- Pacific/Chuuk - Timezone for pacific/chuuk UTC offset of +10:00.
- Pacific/Easter - Timezone for pacific/easter UTC offset of -05:00.
- Pacific/Efate - Timezone for pacific/efate UTC offset of +11:00.
- Pacific/Enderbury - Timezone for pacific/enderbury UTC offset of +13:00.
- Pacific/Fakaofo - Timezone for pacific/fakaofo UTC offset of +13:00.
- Pacific/Fiji - Timezone for pacific/fiji UTC offset of +13:00.
- Pacific/Funafuti - Timezone for pacific/funafuti UTC offset of +12:00.
- Pacific/Galapagos - Timezone for pacific/galapagos UTC offset of -06:00.
- Pacific/Gambier - Timezone for pacific/gambier UTC offset of -09:00.
- Pacific/Guadalcanal - Timezone for pacific/guadalcanal UTC offset of +11:00.
- Pacific/Guam - Timezone for pacific/guam UTC offset of +10:00.
- Pacific/Honolulu - Timezone for pacific/honolulu UTC offset of -10:00.
- Pacific/Johnston - Timezone for pacific/johnston UTC offset of -10:00.
- Pacific/Kiritimati - Timezone for pacific/kiritimati UTC offset of +14:00.

- Pacific/Kosrae - Timezone for pacific/kosrae UTC offset of +11:00.
- Pacific/Kwajalein - Timezone for pacific/kwajalein UTC offset of +12:00.
- Pacific/Majuro - Timezone for pacific/majuro UTC offset of +12:00.
- Pacific/Marquesas - Timezone for pacific/marquesas UTC offset of -09:30.
- Pacific/Midway - Timezone for pacific/midway UTC offset of -11:00.
- Pacific/Nauru - Timezone for pacific/nauru UTC offset of +12:00.
- Pacific/Niue - Timezone for pacific/niue UTC offset of -11:00.
- Pacific/Norfolk - Timezone for pacific/norfolk UTC offset of +11:00.
- Pacific/Noumea - Timezone for pacific/noumea UTC offset of +11:00.
- Pacific/Pago_Pago - Timezone for pacific/pago_pago UTC offset of -11:00.
- Pacific/Palau - Timezone for pacific/palau UTC offset of +09:00.
- Pacific/Pitcairn - Timezone for pacific/pitcairn UTC offset of -08:00.
- Pacific/Pohnpei - Timezone for pacific/pohnpei UTC offset of +11:00.
- Pacific/Ponape - Timezone for pacific/ponape UTC offset of +11:00.
- Pacific/Port_Moresby - Timezone for pacific/port_moresby UTC offset of +10:00.
- Pacific/Rarotonga - Timezone for pacific/rarotonga UTC offset of -10:00.
- Pacific/Saipan - Timezone for pacific/saipan UTC offset of +10:00.
- Pacific/Samoa - Timezone for pacific/samoa UTC offset of -11:00.
- Pacific/Tahiti - Timezone for pacific/tahiti UTC offset of -10:00.
- Pacific/Tarawa - Timezone for pacific/tarawa UTC offset of +12:00.
- Pacific/Tongatapu - Timezone for pacific/tongatapu UTC offset of +14:00.
- Pacific/Truk - Timezone for pacific/truk UTC offset of +10:00.
- Pacific/Wake - Timezone for pacific/wake UTC offset of +12:00.
- Pacific/Wallis - Timezone for pacific/wallis UTC offset of +12:00.
- Pacific/Yap - Timezone for pacific/yap UTC offset of +10:00.
- Poland - Timezone for poland UTC offset of +01:00.
- Portugal - Timezone for portugal UTC offset of +00:00.
- ROC - Timezone for ROC UTC offset of +08:00.
- ROK - Timezone for ROK UTC offset of +09:00.
- Singapore - Timezone for singapore UTC offset of +08:00.
- Turkey - Timezone for turkey UTC offset of +03:00.

- UCT - Timezone for UCT UTC offset of +00:00.
- US/Pacific-New - Timezone for US/Pacific-New UTC offset of -08:00.
- Universal - Timezone for universal UTC offset of +00:00.
- W-SU - Timezone for W-SU UTC offset of +03:00.
- WET - Timezone for WET UTC offset of +00:00.
- Zulu - Timezone for zulu UTC offset of +00:00.
- jdj_tz - Timezone for jdj_tz UTC offset of +00:00.

Default

Default

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.143 cmm amfTais

AMF TAI List. Syntax, description, parameters, and roles for `cmm amfTais` command.

Syntax**Description**

This command creates a list of 5G TAI.

Parameters**Name**

amfTaiListName - Area List Name

Syntax

--amfTaiListName <string>

Description

Name of TAI list

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

servingPlmnName - Serving PLMN Name

Syntax

--servingPlmnName <string>

Description

Serving PLMN Name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

amfTac - AMF Tracking Area Code

Syntax

--amfTac <integer>

Description

AMF Tracking Area Code

Restrictions

Each List can have a maximum of 1024 TACs. Maximum number of TAI lists supported is 256.

Class

Required key.

Type / Range

Integer in the range of 0 to 16,777,215

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.144 cmm amfTimers

5G Timers. Syntax, description, parameters, and roles for cmm amfTimers command.

Syntax

Description

5G timers like NG-AP, mobility management, etc.

Parameters

Name

amfTimerName - Timer name

Syntax

--amfTimerName <list-item>

Description

Name of timer.

Class

Required key.

List

- amfLoadUpdate - AMF Load Update - Defines the period after which the AMF re-checks the AMF load info and updates NRF if needed.
- amfSbiDisabledLinkDel - SBI Link Inactive Timer for Disabled links - This parameter specifies the Link Inactive Timer for SBI Disabled Links
- amfSbiInactiveLinkDel - SBI Link Inactive Timer - This parameter specifies the Link Delete Timeout for SBI links
- ausfEapAuthOngoing - EAP AKA' Is Still Ongoing - A guard timer for receiving response from AUSF for EAP authentication ongoing message, 6000 ms (T3560) * 4 (retries) +ausf Gurad timer 3000=27000. Note: this timer value and/or its default should be updated whenever T3560 timer value and/or its default is changed. The max range value is capped at: 6000 ms (T3560) * 6 (retries) +ausf Gurad timer 3000+1000=40000.

- **cmassWarningMsgTimer** - CMAS Warning Message Timer - The warning message timer is an NG-AP interface timer. It indicates the time to wait (in seconds) for the Warning Message Delivery Service. Failure to receive a Write-Replace Warning Response from the gNB before the timer expires which indicates that the broadcast or broadcast cancellation is unsuccessful in that gNB.
- **epsFallbackNoPageTimer** - EPS Fallback no page timer - AMF suppresses UE paging while timer is running and uePageEpsFallback is set to true.
- **nssaafSliceAuthorizationOngoing** - NSSAA Is Still Ongoing - A guard timer for all pending S-NSSAIs. For the case the NSSAA is done in sequence and not in parallel for the 'pending' S-NSSAIs, the AMF shall check remaining time before starting NSSAA for next slice, so when there isn't enough time or timer expires the AMF shall update the UE with the S-NSSAI status up to that point and then resume NSSAA in next mobility procedure. $N(\text{NAS retries})=1$ as default value used here. So, $15000 \text{ ms (T3575)} + \text{nssaaf Guard timer } 5000 = 20000$. The max range value +retry in late phase could be capped at: $15000 \text{ ms (T3575)} * 3 (\text{retries}) + \text{nssaaf Guard timer } 5000 = 50000 \text{ ms}$. Default = $15000 \text{ ms} * 1 + 5000 = 20000 \text{ ms}$
- **pagingGapTimer** - Paging Gap Timer - This timer indicates the interval in which the AMF will suppress paging for a UE after multiple consecutive page response failures have occurred for the UE. a value of 0 indicates that paging should not be suppressed due to previous page response failures. Paging Gap timer will be stopped if the UE reconnects to the network with AMF call processing procedure such as Service Request. If the Paging Gap timer is stopped or expires, the AMF will once again trigger paging for that UE. If the UE fails to respond after the AMF paged the UE after the Paging Gap timer expired (i.e., the UE has still not connected to the network since Paging Gap timer was started), the Paging Gap timer will be started again. The trigger for starting suppression of AMF paging is 3 consecutive failed procedures due to an inability to reach a UE with paging.
- **radioCapInfoWait** - Radio Capability Info Waiting - Defines the time waiting for NGAP UE Radio Capability Info Indication after receiving Initial Context Setup Response and possibly UE Radio Capability Check Response. The timer is only started when AMF does not hold UE radio capability and UE radio capabilities are needed for either -enhanced voice rejection (registration needs to be rejected if UE does not support VoNR on VoNR only TA), or -EMC needs to be suppressed if UE does not support VoNR (and additionally optionally EMF needs to be suppressed if VoNR is supported), or -IMS VoPS needs to be suppressed if UE has disabled S1 support.
- **rrcInactiveTimerIncr** - RRC inactive timer increment - This timer is added to 24.501 and 38.413 timers when RRC Inactive feature is enabled.
- **smsNoLastMsgIndication** - SMS No Last Message Indication - Defines the timer value the AMF uses to release AMF reserved internal resources, if the SMSF does not provide the lastMsgIndication during Namf_Communication_N1N2MessageTransfer.
- **udmSubscriptionExpires** - UDM Subscription Expires timer - During a UDM subscribe to notification of data change procedure, the AMF can specify a data/time that the

subscription expires. The expiration date/time used will be the current data/time when subscribe is requested plus udmSubExpires. Note: On expiry, AMF sends a subscribe message with previously subscribed resources to UDM.

- vlrCleanupIratUeCtxt - VLR Cleanup IRAT UE Contexts - This parameter specifies the time for deleting from the VLR the preserved UE Contexts of the deregistered UEs that have moved to EPC. Value 0 means that the UE Contexts are not deleted from the VLR
- waitReleaseUeCtxtSrcAmf - Wait Release UE Context from Source AMF - Current timer enables maintaining the UE context as initial target AMF after receiving Create UE Context Response (success) from final target AMF in N2 Handover with AMF reselection scenario so that a Release UE Context Request due to handover Cancellation may still be sent to final target AMF. When the timer expires, initial target AMF shall release local resources. When timer is zero, the initial target AMF releases the UE Context immediately after sending Create UE Context Response to source AMF.
- waitRelocateCancelSrcMme - Wait Relocate Cancel from Source MME - Supervises receipt of Relocate Cancel Request from source MME after receiving Relocate UE Context Response from target AMF in EPS to 5GS with AMF relocation scenario. When timer expires, initial AMF shall release local resources

Name

amfTimerUnit - Timer unit

Syntax

--amfTimerUnit <list-item>

Description

Unit of measurement.

Class

Required on create, not modifiable.

List

- 320Hour - 320 hour - Specify timer unit of 320 hours.
- Millisecond - Specify timer unit of 1 millisecond.
- Second - Specify timer unit of 1 second.
- 2Second - 2 second - Specify timer unit of 2 seconds.
- 30Second - 30 second - Specify timer unit of 30 seconds.
- Minute - Specify timer unit of 1 minute.
- DeciHour - Specify timer unit of 1 decihour (6 minutes).
- 10Minute - 10 minute - Specify timer unit of 10 minutes.
- Hour - Specify timer unit of 1 hour.
- 10Hour - 10 hour - Specify timer unit of 10 hours.

Name

amfTimerValue - Timer value

Syntax

```
--amfTimerValue <integer>
```

Description

Numeric value of timer.

Class

Required on create.

Type / Range

Integer in the range of 0 to 65,535

The amfTimerValue string is based on the amfTimerName parameter:

Name

amfLoadUpdate - AMF Load Update

Syntax

```
--amfTimerName amfLoadUpdate --amfTimerValue <string> [ --  
amfTimerUnit Second ]
```

Description

Defines the period after which the AMF re-checks the AMF load info and updates NRF if needed.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

10

Name

amfSbiDisabledLinkDel - SBI Link Inactive Timer for Disabled links

Syntax

```
--amfTimerName amfSbiDisabledLinkDel --amfTimerValue <string> [ --  
amfTimerUnit 10Minute ]
```

Description

This parameter specifies the Link Inactive Timer for SBI Disabled Links

Type / Range

Integer in the range of 1 to 140 in units of 10Minute

Default

6

Name

amfSbiInactiveLinkDel - SBI Link Inactive Timer

Syntax

```
--amfTimerName amfSbiInactiveLinkDel --amfTimerValue <string> [ --  
amfTimerUnit 10Minute ]
```

Description

This parameter specifies the Link Delete Timeout for SBI links

Type / Range

Integer in the range of 1 to 140 in units of 10Minute

Default

6

Name

ausfEapAuthOngoing - EAP AKA' Is Still Ongoing

Syntax

```
--amfTimerName ausfEapAuthOngoing --amfTimerValue <string> [ --  
amfTimerUnit Millisecond ]
```

Description

A guard timer for receiving response from AUSF for EAP authentication ongoing message, 6000 ms (T3560) * 4 (retries) +ausf Gurad timer 3000=27000. Note: this timer value and/or its default should be updated whenever T3560 timer value and/or its default is changed. The max range value is capped at: 6000 ms (T3560) * 6 (retries) +ausf Gurad timer 3000+1000=40000.

Type / Range

Integer in the range of 10 to 40,000 in units of Millisecond

Default

27000

Name

cmassWarningMsgTimer - CMAS Warning Message Timer

Syntax

```
--amfTimerName cmassWarningMsgTimer --amfTimerValue <string> [ --  
amfTimerUnit Second ]
```

Description

The warning message timer is an NG-AP interface timer. It indicates the time to wait (in seconds) for the Warning Message Delivery Service. Failure to receive a Write-Replace Warning Response from the gNB before the timer expires which indicates that the broadcast or broadcast cancellation is unsuccessful in that gNB.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

8

Name

epsFallbackNoPageTimer - EPS Fallback no page timer

Syntax

```
--amfTimerName epsFallbackNoPageTimer --amfTimerValue <string> [ --  
amfTimerUnit Millisecond ]
```

Description

AMF suppresses UE paging while timer is running and uePageEpsFallback is set to true.

Type / Range

Integer in the range of 10 to 2000 in units of Millisecond

Default

1000

Name

nssaafSliceAuthorizationOngoing - NSSAA Is Still Ongoing

Syntax

```
--amfTimerName nssaafSliceAuthorizationOngoing --amfTimerValue  
<string> [ --amfTimerUnit Millisecond ]
```

Description

A guard timer for all pending S-NSSAIs. For the case the NSSAA is done in sequence and not in parallel for the 'pending' S-NSSAIs, the AMF shall check remaining time before starting NSSAA for next slice, so when there isn't enough time or timer expires the AMF shall update the UE with the S-NSSAI status up to that point and then resume NSSAA in next mobility procedure. N(NAS retries)=1 as default value used here. So, 15000 ms (T3575) +nssaaf Guard timer 5000=20000. The max range value +retry in late phase could be capped at: 15000 ms (T3575) * 3 (retries) +nssaaf Guard timer 5000 =50000 ms. Default=15000 ms*1+5000=20000 ms

Type / Range

Integer in the range of 10 to 60,000 in units of Millisecond

Default

20000

Name

pagingGapTimer - Paging Gap Timer

Syntax

```
--amfTimerName pagingGapTimer --amfTimerValue <string> [ --  
amfTimerUnit Minute ]
```

Description

This timer indicates the interval in which the AMF will suppress paging for a UE after multiple consecutive page response failures have occurred for the UE. a value of 0 indicates that

paging should not be suppressed due to previous page response failures. Paging Gap timer will be stopped if the UE reconnects to the network with AMF call processing procedure such as Service Request. If the Paging Gap timer is stopped or expires, the AMF will once again trigger paging for that UE. If the UE fails to respond after the AMF paged the UE after the Paging Gap timer expired (i.e., the UE has still not connected to the network since Paging Gap timer was started), the Paging Gap timer will be started again. The trigger for starting suppression of AMF paging is 3 consecutive failed procedures due to an inability to reach a UE with paging.

Type / Range

Integer in the range of 0 to 1,440 in units of Minute

Default

0

Name

radioCapInfoWait - Radio Capability Info Waiting

Syntax

```
--amfTimerName radioCapInfoWait --amfTimerValue <string> [ --  
amfTimerUnit Millisecond ]
```

Description

Defines the time waiting for NGAP UE Radio Capability Info Indication after receiving Initial Context Setup Response and possibly UE Radio Capability Check Response. The timer is only started when AMF does not hold UE radio capability and UE radio capabilities are needed for either -enhanced voice rejection (registration needs to be rejected if UE does not support VoNR on VoNR only TA), or -EMC needs to be suppressed if UE does not support VoNR (and additionally optionally EMF needs to be suppressed if VoNR is supported), or -IMS VoPS needs to be suppressed if UE has disabled S1 support.

Type / Range

Integer in the range of 0 to 3,000 in units of Millisecond

Default

500

Name

rrcInactiveTimerIncr - RRC inactive timer increment

Syntax

```
--amfTimerName rrcInactiveTimerIncr --amfTimerValue <string> [ --  
amfTimerUnit Second ]
```

Description

This timer is added to 24.501 and 38.413 timers when RRC Inactive feature is enabled.

Type / Range

Integer in the range of 0 to 3 in units of Second

Default

0

Name

smsNoLastMsgIndication - SMS No Last Message Indication

Syntax

```
--amfTimerName smsNoLastMsgIndication --amfTimerValue <string> [ --  
amfTimerUnit Second ]
```

Description

Defines the timer value the AMF uses to release AMF reserved internal resources, if the SMSF does not provide the lastMsgIndication during Namf_Communication_N1N2MessageTransfer.

Type / Range

Integer in the range of 1 to 10 in units of Second

Default

3

Name

udmSubscriptionExpires - UDM Subscription Expires timer

Syntax

```
--amfTimerName udmSubscriptionExpires --amfTimerValue <string> [ --  
amfTimerUnit Hour ]
```

Description

During a UDM subscribe to notification of data change procedure, the AMF can specify a data/time that the subscription expires. The expiration date/time used will be the current data/time when subscribe is requested plus udmSubExpires. Note: On expiry, AMF sends a subscribe message with previously subscribed resources to UDM.

Type / Range

Integer in the range of 1 to 700 in units of Hour

Default

168

Name

vlrCleanupIratUeCtxt - VLR Cleanup IRAT UE Contexts

Syntax

```
--amfTimerName vlrCleanupIratUeCtxt --amfTimerValue <string> [ --  
amfTimerUnit Minute ]
```

Description

This parameter specifies the time for deleting from the VLR the preserved UE Contexts of the deregistered UEs that have moved to EPC. Value 0 means that the UE Contexts are not deleted from the VLR

Type / Range

Integer in the range of 5 to 120, allowed 0 value also. in units of Minute

Default

5

Recommended

5

Name

waitReleaseUeCtxtSrcAmf - Wait Release UE Context from Source AMF

Syntax

```
--amfTimerName waitReleaseUeCtxtSrcAmf --amfTimerValue <string> [ --  
amfTimerUnit Millisecond ]
```

Description

Current timer enables maintaining the UE context as initial target AMF after receiving Create UE Context Response (success) from final target AMF in N2 Handover with AMF reselection scenario so that a Release UE Context Request due to handover Cancellation may still be sent to final target AMF. When the timer expires, initial target AMF shall release local resources. When timer is zero, the initial target AMF releases the UE Context immediately after sending Create UE Context Response to source AMF.

Type / Range

Integer in the range of 0 to 1,000 in units of Millisecond

Default

500

Name

waitRelocateCancelSrcMme - Wait Relocate Cancel from Source MME

Syntax

```
--amfTimerName waitRelocateCancelSrcMme --amfTimerValue <string> [ --  
amfTimerUnit Millisecond ]
```

Description

Supervises receipt of Relocate Cancel Request from source MME after receiving Relocate UE Context Response from target AMF in EPS to 5GS with AMF relocation scenario. When timer expires, initial AMF shall release local resources

Type / Range

Integer in the range of 0 to 1,000 in units of Millisecond

Default

500

Roles

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

3.145 cmm amfTnlaQuery

AMF TNLA Query. Syntax, description, parameters, and roles for cmm amfTnlaQuery command.

Syntax

Description

Outputs for a given AMF N2 IP address a list of NG-RAN gNBs that use the AMF N2 endpoint for a TNLA or as the initial AMF N2 endpoint used to perform the first N2 Setup procedure with the AMF.

Parameters

Name

amfIpAddress - AMF N2 IP address

Syntax

--amfIpAddress <string>

Description

Specifies AMF N2 local endpoint IP address

Class

Required key.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

assignMode - Assignment Mode

Syntax

--assignMode <integer>

Description

This field depicts the assignment mode for the SNPN

Class

Optional.

Type / Range

Integer in the range of 0 to 2

Default

0

Name

information - Information

Syntax

--information <string>

Description

Information

Class

Read only.

Type / Range

Digit string of length 0 to 256 with the allowed pattern of [0-9a-zA-Z\._:-]*

Name

initialTnla - Initial TNLA

Syntax

--initialTnla <list-item>

Description

Indicates if gNB used the AMF N2 local endpoint IP address to send its initial NG Setup Request to the AMF.

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Name

linkIndex - AMF Link Index

Syntax

--linkIndex <integer>

Description

AMF link index

Class

Read only.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

mcc - Mobile country code

Syntax

--mcc <digits>

Description

3-digit number used to represent the mobile country code.

Class

Read only.

Type / Range

Digit string of length 3

Name

mnc - Mobile network code

Syntax

--mnc <digits>

Description

Digits used to represent mobile network code.

Class

Read only.

Type / Range

Digit string of length 2 to 3

Name

ngRanId - Global RAN Node ID

Syntax

--ngRanId <integer>

Description

Specifies Global RAN Node identifier as specified in TS 38.413

Class

Read only.

Type / Range

Integer in the range of 0 to 268,435,455

Name

ngRanIpAddress - TNLA RAN IP address

Syntax


```
--ngRanIpAddress <string>
```

Description

Specifies TNLA RAN IP address

Class

Read only.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

ngRanPort - TNLA RAN port number

Syntax

```
--ngRanPort <integer>
```

Description

Specifies the TNLA RAN port number

Class

Read only.

Type / Range

Integer in the range of 0 to 65,535

Name

nidValue - Nid value

Syntax

```
--nidValue <integer>
```

Description

This field depicts the Nid value of the NID for the SNPN

Class

Optional.

Type / Range

Integer in the Range 0 to 1099511627775

Default

0

Roles**List**

sysadmin, manage, monitor

3.146 cmm amfUdmHttp2ToNasCcMapping

AMF Udm Http2 To Nas Cc Mapping. Syntax, description, parameters, and roles for cmm amfUdmHttp2ToNasCcMapping command.

Syntax

Description

Defines UDM HTTP2 Status Codes Mapping to AMF NAS Cause Codes. The maximum number of profiles supported is 128. A default profile, named 'default', is provided.

Parameters

Name

amfUdmHttp2ToNasCcMappingListName - AMF UDM HTTP2 to NAS CC mapping list name

Syntax

--amfUdmHttp2ToNasCcMappingListName <string>

Description

The name of the HTTP2 to 5G NAS Cause Code mapping list

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

udmHttp2StatusCode - UDM HTTP2 Status Code

Syntax

--udmHttp2StatusCode <list-item>

Description

Defines the UDM HTTP2 Status Code to map to 5G NAS Cause Code

Class

Required key.

List

- 307_UNKNOWN - 307 UNKNOWN - The 307_UNKNOWN is used when included problem details is not specified in 3GPP or it is not included.
- 308_UNKNOWN - 308 UNKNOWN - The 308_UNKNOWN is used when included problem details is not specified in 3GPP or it is not included.

- **400_RESOURCE_CONTEXT_NOT_FOUND - 400 RESOURCE CONTEXT NOT FOUND** - The notification request is rejected because the call back URI still exists in the receiver of the notification, but the specific resource context identified within the notification payload is not found in the NF service consumer
- **400_UNKNOWN - 400 UNKNOWN** - The 400_UNKNOWN is used when included problem details is not specified in 3GPP or it is not included.
- **403_CCA_VERIFICATION_FAILURE - 403 CCA VERIFICATION FAILURE** - The request is rejected due to a failure to verify the CCA at the receiving entity (UDM).
- **403_SERVICE_NOT_ALLOWED - 403 SERVICE NOT ALLOWED** - The request is related to a service that is not allowed for the user in the 5GS subscription data (e.g. SMS is barred). Note: This Error Status is specific for Nudm_UEContextManagement service.
- **403_SERVICE_NOT_PROVISIONED - 403 SERVICE NOT PROVISIONED** - The request is related to a service that is not provisioned for the user in the 5GS subscription data (e.g. SMS teleservice not provisioned). Note: This Error Status is specific for Nudm_UEContextManagement service.
- **403_UNKNOWN - 403 UNKNOWN** - The 403_UNKNOWN is used when included problem details is not specified in 3GPP or it is not included.
- **404_UNKNOWN - 404 UNKNOWN** - The 404_UNKNOWN is used when included problem details is not specified in 3GPP or it is not included.
- **409_TEMPORARY_REJECT_HANDOVER_ONGOING - 409 TEMPORARY REJECT HANDOVER ONGOING** - The request cannot be processed due to an ongoing N2 handover procedure. Note: This Error Status is specific for Nudm_UEContextManagement service.
- **409_TEMPORARY_REJECT_REGISTRATION_ONGOING - 409 TEMPORARY REJECT REGISTRATION ONGOING** - The request cannot be processed due to an ongoing registration procedure. Note: This Error Status is specific for Nudm_UEContextManagement service.
- **409_UNKNOWN - 409 UNKNOWN** - The 409_UNKNOWN is used when included problem details is not specified in 3GPP or it is not included.
- **411_UNKNOWN - 411 UNKNOWN** - The 411_UNKNOWN is used when included problem details is not specified in 3GPP or it is not included.
- **422_UNKNOWN - 422 UNKNOWN** - The 422_UNKNOWN is used when included problem details is not specified in 3GPP or it is not included.
- **429_UNKNOWN - 429 UNKNOWN** - The 429_UNKNOWN is used when included problem details is not specified in 3GPP or it is not included.
- **500_NF_FAILOVER - 500 NF FAILOVER** - The request is rejected due to the unavailability of the NF (UDM), and the requester may trigger an immediate re-selection of an alternative NF based on this information.

- 500_NF_SERVICE_FAILOVER - 500 NF SERVICE FAILOVER - The request is rejected due to the unavailability of the NF service, and the requester may trigger an immediate re-selection of an alternative NF service based on this information.
- 500_UNKNOWN - 500 UNKNOWN - The 500_UNKNOWN is used when included problem details is not specified in 3GPP or it is not included.
- 501_UNKNOWN - 501 UNKNOWN - The 501_UNKNOWN is used when included problem details is not specified in 3GPP or it is not included.
- 503_UNKNOWN - 503 UNKNOWN - The 503_UNKNOWN is used when included problem details is not specified in 3GPP or it is not included.
- 504_TARGET_NF_NOT_REACHABLE - 504 TARGET NF NOT REACHABLE - The request is not served as the target NF (UDM) is not reachable.
- 504_TIMED_OUT_REQUEST - 504 TIMED OUT REQUEST - The request is rejected due a request that has timed out at the HTTP client based on origination stamp.
- 504_UNKNOWN - 504 UNKNOWN - The 504_UNKNOWN is used when included problem details is not specified in 3GPP or it is not included.
- 307_NF_CONSUMER_REDIRECT_ONE_TXN - 307 Temporary Redirect NF Consumer Redirect One TXN - The request has been asked to be redirected to a specified target for one transaction. Note: This Error Code is specific for Nudm_SubscriberDataManagement service.
- 308_CONTEXT_NOT_FOUND - 308 Permanent Redirect Context Not Found - The request has been asked to be redirected to a specified target. Note: This Error Status is specific for Nudm_SubscriberDataManagement service.
- 400_INVALID_API - 400 Bad Request Invalid API - The HTTP request contains an unsupported API name or API version in the URI.
- 400_INVALID_MSG_FORMAT - 400 Bad Request Invalid Msg Format - The HTTP request has an invalid format.
- 400_INVALID_QUERY_PARAM - 400 Bad Request Invalid Query Param - The HTTP request contains an unsupported query parameter in the URI.
- 400_MANDATORY_QUERY_PARAM_INCORRECT - 400 Bad Request Mandatory Query Param Incorrect - A mandatory query parameter, or a conditional query parameter but mandatory required, for an HTTP method was received in the URI with semantically incorrect value.
- 400_OPTIONAL_QUERY_PARAM_INCORRECT - 400 Bad Request Optional Query Param Incorrect - An optional query parameter for an HTTP method was received in the URI with a semantically incorrect value that prevents successful processing of the service request.
- 400_MANDATORY_QUERY_PARAM_MISSING - 400 Mandatory Query Param Missing - Query parameter which is defined as mandatory, or as conditional but mandatory required, for an HTTP method is not included in the URI of the request.

- **400_MANDATORY_IE_INCORRECT** - 400 Bad Request Mandatory IE Incorrect - A mandatory IE or conditional IE in data structure, but mandatory required, for an HTTP method was received with a semantically incorrect value.
- **400_OPTIONAL_IE_INCORRECT** - 400 Bad Request Optional IE Incorrect - An optional IE in data structure for an HTTP method was received with a semantically incorrect value that prevents successful processing of the service request.
- **400_MANDATORY_IE_MISSING** - 400 Bad Request Mandatory IE Missing - IE which is defined as mandatory or as conditional in data structure, but mandatory required, for an HTTP method is not included in the payload body of the request.
- **400_UNSPECIFIED_MSG_FAILURE** - 400 Bad Request Unspecified Msg Failure - The request is rejected due to unspecified client error.
- **403_MODIFICATION_NOT_ALLOWED** - 403 Forbidden Modification Not Allowed - The request is rejected because the contained modification instructions attempt to modify IE which is not allowed to be modified.
- **403_UNKNOWN_5GS_SUBSCRIPTION** - 403 Forbidden Unknown 5GS Subscription - No 5GS subscription is associated with the user. Note: This Error Status is specific for Nudm_UEContextManagement service
- **403_NO_PS_SUBSCRIPTION** - 403 Forbidden No PS Subscription - No PS (5GS, EPS, GPRS) subscription is associated with the user. Note: This Error Status is specific for Nudm_UEContextManagement service
- **403_ROAMING_NOT_ALLOWED** - 403 Forbidden Roaming Not Allowed - The subscriber is not allowed to roam within that PLMN. Note: This Error Status is specific for Nudm_UEContextManagement service.
- **403_ACCESS_NOT_ALLOWED** - 403 Forbidden Access Not Allowed - Access type not allowed for the user. Note: This Error Status is specific for Nudm_UEContextManagement service
- **403_RAT_NOT_ALLOWED** - 403 Forbidden RAT Not Allowed - RAT is not allowed for the user. Note: This Error Status is specific for Nudm_UEContextManagement service.
- **403_DNN_NOT_ALLOWED** - 403 Forbidden DNN Not Allowed - DNN not authorized for the user. Note: This Error Status is specific for Nudm_UEContextManagement service.
- **403_REAUTHENTICATION_REQUIRED** - 403 Forbidden Reauthentication Required - Due to operator policies the user needs to be re-authenticated, e.g. last valid authentication is considered obsolete Note: This Error Status is specific for Nudm_UEContextManagement service.
- **403_INVALID_GUAMI** - 403 Forbidden Invalid Guami - The AMF is not allowed to modify the registration information stored in the UDM, as it is not the registered AMF. Note: This Error Status is specific for Nudm_UEContextManagement service.

- **404_SUBSCRIPTION_NOT_FOUND** - 404 Not Found Subscription Not Found - The request for modification or deletion of subscription is rejected because the subscription is not found in the NF
- **404_RESOURCE_URI_STRUCTURE_NOT_FOUND** - 404 Not Found Resource Structure Not Found - The request is rejected because a fixed part after the first variable part of an `apiSpecificResourceUriPart` is not found in the NF. This fixed part of the URI may represent a sub-resource collection (e.g. contexts, subscriptions, policies) or a custom operation.
- **404_DATA_NOT_FOUND** - 404 Not Found Data Not Found - The requested UE subscription data is not found/does not exist. This error is applicable to all `Nudm_SDM` GET operations.
- **404_USER_NOT_FOUND** - 404 Not Found User Not Found - The user does not exist in the HPLMN when `Nudm_UEContextManagement` service is used. The user does not exist when `Nudm_SubscriberDataManagement` service is used. This error is applicable to all `Nudm_SDM` GET operations.
- **404_CONTEXT_NOT_FOUND** - 404 Not Found Context Not Found - It is used during the modification of an existing subscription when no corresponding context exists when `Nudm_SubscriberDataManagement` service is used. It is used when no corresponding context exists when `Nudm_UEContextManagement` service is used.
- **404_GROUP_IDENTIFIER_NOT_FOUND** - 404 Not Found Group Identifier Not Found - The requested Group Identifier does not exist. Note: This Error Status is specific for `Nudm_SubscriberDataManagement` service
- **411_INCORRECT_LENGTH** - 411 Length Required Incorrect Length - The request is rejected due to incorrect value of a Content-length header field.
- **422_UNPROCESSABLE_REQUEST** - 422 Unprocessable Entity Unprocessable Request - The request can not be processed due to semantic errors when trying to process a patch method. Note: This Error Status is specific for `Nudm_UEContextManagement` service.
- **429_NF_CONGESTION_RISK** - 429 Too Many Requests NF Congestion Risk - The request is rejected due to excessive traffic which, if continued over time, may lead to (or may increase) an overload situation.
- **500_INSUFFICIENT_RESOURCES** - 500 Internal Server Error Insufficient Resources - The request is rejected due to insufficient resources.
- **500_UNSPECIFIED_NF_FAILURE** - 501 Internal Server Error Unspecified NF Failure - The request is rejected due to unspecified reason at the NF.
- **500_SYSTEM_FAILURE** - 501 Internal Server Error System Failure - The request is rejected due to generic error condition in the NF.
- **501_UNSUPPORTED_RESOURCE_URI** - 501 Not Implemented Unsupported Resource URI - The `SDM` Subscription contains unsupported resource URI to be monitored when

Nudm_SubscriberDataManagement service is used. Note: This Error Status is specific for Nudm_SubscriberDataManagement service

- 503_NF_CONGESTION - 503 Service Unavailable NF Congestion - The NF experiences congestion and performs overload control, which does not allow the request to be processed.

Name

amf5gMmCc - 5GMM NAS Cause Code

Syntax

--amf5gMmCc <integer>

Description

Defines the 5GMM NAS cause code to use towards the UE based on the HTTP2 Status Code received from the UDM

Class

Optional.

Type / Range

5GMM Cause Code.

Default

15

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

3.147 cmm amfUdmHttp2ToNasCcMappingList

AMF UDM HTTP2 To NAS CC Mapping List. Syntax, description, parameters, and roles for cmm amfUdmHttp2ToNasCcMappingList command.

Syntax**Description**

UDM HTTP2 status code to 5GMM NAS Cause Code mapping profile list

Parameters

Name

name - AMF UDM HTTP2 To NAS CC Mapping List Name

Syntax

--name <string>

Description

UDM HTTP2 Status Code to 5GMM NAS Cause Code mapping Profile List name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.148 cmm amfUeNitzInfo

Ue Nitz Information. Syntax, description, parameters, and roles for cmm amfUeNitzInfo command.

Syntax

Description

Specifies parameters to be sent in NAS UE Configuration Procedure to the UE, e.g. during UE Registration procedure.

Parameters

Name

plmnName - Mobile network name

Syntax

--plmnName <string>

Description

Name that identifies mobile network (alternate key).

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

encodingName - Encoding name

Syntax

--encodingName <list-item>

Description

Indicates the encoding type to use for name fields.

Class

Optional.

List

- GSM default alphabet - Specifies GSM default alphabet encoding name.
- UCS2 - USC2 - Specifies USC2 encoding name.

Default

GSM default alphabet

Name

networkName - Network name

Syntax

--networkName <string>

Description

Unique network name.

Class

Optional.

Type / Range

The Network Name for the PLMN. The following characters in the GSM character set per 3GPP TS 23.038 Section 6.2.1.

Default

""

Name

networkShortName - Network short name

Syntax

--networkShortName <string>

Description

Unique short network name.

Class

Optional.

Type / Range

The Network Short Name for the PLMN. The following characters in the GSM character set per 3GPP TS 23.038 Section 6.2.1.

Default

"

Name

sendCountryInit - Send country init

Syntax

--sendCountryInit <list-item>

Description

Indicates whether or not the UE should display or not display the country initials.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sendNetworkName - Send network name

Syntax

--sendNetworkName <list-item>

Description

Indicates whether or not to send the network names.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sendTzOffSet - Send time zone offset

Syntax

--sendTzOffSet <list-item>

Description

Indicates whether or not to send the time zone offset.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.149 cmm amfUeNitzInfoOverrideProfile

UE Nitz Information Override Profile. Syntax, description, parameters, and roles for cmm amfUeNitzInfoOverrideProfile command.

Syntax

Description

This command is used to administer UE Nitz Information Override Profile.

Parameters

Name

name - UE Nitz Info Override Profile Name

Syntax

```
--name <string>
```

Description

Profile name for UE Nitz Information.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

encodingName - Encoding Name

Syntax

```
--encodingName <list-item>
```

Description

Indicates the encoding type to use for the network name fields

Class

Optional.

List

- GSM default alphabet - Specifies GSM default alphabet encoding name.
- UCS2 - USC2 - Specifies USC2 encoding name.

Default

GSM default alphabet

Name

networkName - Network Name

Syntax

```
--networkName <string>
```

Description

Unique network name

Class

Optional.

Type / Range

The Network Name for the PLMN. The following characters in the GSM character set per 3GPP TS 23.038 Section 6.2.1.

Default

"

Name

networkShortName - Network Short Name

Syntax

--networkShortName <string>

Description

Unique short network name.

Class

Optional.

Type / Range

The Network Short Name for the PLMN. The following characters in the GSM character set per 3GPP TS 23.038 Section 6.2.1.

Default

"

Name

sendCountryInit - Send Country Initials

Syntax

--sendCountryInit <list-item>

Description

Indicates whether or not the UE should display or not display the country initials.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sendNetworkName - Send Network Name

Syntax

--sendNetworkName <list-item>

Description

Indicates whether or not to send the network names.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sendTzOffSet - Send Time Zone Offset

Syntax

--sendTzOffSet <list-item>

Description

Indicates whether or not to send the time zone offset.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

3.150 cmm amfUePageFanoutControl

AMF 5G UE paging fanout overload control parameters. Syntax, description, parameters, and roles for `cmm amfUePageFanoutControl` command.

Syntax

Description

This command is used to set thresholds for 5G UE paging fan-out overload control

Parameters

Name

msgRateToSafetyNetThreshold - Message rate to total allowed message rate (Safety Net) threshold

Syntax

--msgRateToSafetyNetThreshold <integer>

Description

Threshold used for CMM total message to total allowed message rate (Safety Net), in conjunction with other parameters under this command above which to trigger paging restriction

Class

Optional.

Type / Range

Integer in the range of 0 to 100

Default

60

Name

pageFanoutToTotalMsgThreshold - UE paging fanout to total message rate threshold

Syntax

--pageFanoutToTotalMsgThreshold <integer>

Description

Threshold used for UE paging fan-out (number of GNBs contacted) contribution to total CMM message rate, in conjunction with other parameters under this command above which to trigger paging restriction

Class

Optional.

Type / Range

Integer in the range of 0 to 100

Default

50

Name

pagingFanoutThreshold - Paging fanout threshold

Syntax

--pagingFanoutThreshold <integer>

Description

Threshold used for UE paging fan-out (number of GNBs contacted) in conjunction with other parameters under this command above which to trigger paging restriction

Class

Optional.

Type / Range

Integer in the range of 0 to 100

Default

25

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

4 CMM CLI commands: cmm amfUePlmnServices - cmm genCdr

CMM CLI command descriptions: syntax, usage, parameters, and user roles.

4.1 cmm amfUePlmnServices

5G UE PLMN Services. Syntax, description, parameters, and roles for cmm amfUePlmnServices command.

Syntax

Description

This command is used to administer services to a UE PLMN based on UE PLMN and serving PLMN. Records should not be created for 'Multiple Home Network' UE PLMNs; Instead, the AMF uses the amfUePlmnServices record attributes associated with the AMF Home PLMN (where UE and Serving PLMN are set to the Home Network PLMN).

Parameters

Name

name - Name of the UE PLMN services for a serving PLMN

Syntax

```
--name <string>
```

Description

Name of a list of UE PLMN services for a UE PLMN and serving PLMN combination.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

servedPlmnName - Served Mobile Network Name

Syntax

```
--servedPlmnName <string>
```

Description

Name that identifies the serving network.

Class

Required on create, not modifiable.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

uePlmnName - UE Mobile Network Name

Syntax

--uePlmnName <string>

Description

PLMN name identifying the mobile network of the UE.

Class

Required on create, not modifiable.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

aasn - Alternative Anchor SMF Notification

Syntax

--aasn <list-item>

Description

Defines whether the AMF supports Alternative Anchor SMF Notification (AASN) on SM context create for the V-SMF to indicate alternatively selected anchor SMF. Note: on the current product release this is only supported for V-SMF.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

acscr - Absence of smfUri and hSmfUri in Create SM Context Request

Syntax

--acscr <list-item>

Description

Enables the feature Absence of smfUri and hSmfUri attributes in Create SM Context Request for procedures with I-SMF/V-SMF insertion/change other than PDU session establishment and EPS to 5GS mobility procedures. The support for this feature in AMF may remove the need for the AMF to fetch the smfUri or hsmfUri from the NRF where the anchor SMF profile is registered to fasten the mobility scenarios.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

additionalHsmfUri - Additional H-SMF URIs

Syntax

--additionalHsmfUri <integer>

Description

Identifies additional H-SMF URIs included by the AMF in the Create SM Context to the V-SMF for home routed traffic. V-SMF can use the additional H-SMF URIs if included primary H-SMF is not reachable or failed to establish the PDU Session. Default of 0 means no additional H-SMF URIs included.

Class

Optional.

Type / Range

Integer in the range of 0 to 6

Default

0

Name

aliasDnnConversionProfileName - AMF Alias DNN Conversion Profile Name

Syntax

--aliasDnnConversionProfileName <string>

Description

Pointer to the Alias DNN Conversion Profile

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

default

Name

allowEthernetPdnTypeN26 - Allow Ethernet PDN Type N26

Syntax

--allowEthernetPdnTypeN26 <list-item>

Description

Parameter controls sending in Forward Relocation Request allowing Ethernet PDN Type across N26 interface. By default this shall be set to false.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

allowNonIpPdnTypeN26 - Allow Non IP PDN Type N26

Syntax

--allowNonIpPdnTypeN26 <list-item>

Description

Parameter controls sending in Forward Relocation Request Non IP PDN Type across N26. By default this shall be set to Yes.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

alwaysQueryNssfForAmfSelection - Always Query NSSF for AMF Selection

Syntax

--alwaysQueryNssfForAmfSelection <list-item>

Description

This parameter determines whether AMF will query NSSF for AMF selection during UE Registration even when the AMF can serve all Requested and Valid -according to the UE subscription- Slices. Default value is 'false', which means the AMF will Query NSSF for AMF selection only when: - it does not support at least one of the valid Requested Slices. - UE is using non-standard slices in VPLMN and mapped HPLMN list is not provided by the UE.

Restrictions

Applies to home to roamer UEs.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

amfAccessRestrictionNasMappingProfileName - Name of access restriction to NAS cause
code profile name

Syntax

--amfAccessRestrictionNasMappingProfileName <string>

Description

Profile name for access restriction to NAS cause code mapping.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

default

Name

amfAdjacentPlmnListName - adjacent Plmn List Name

Syntax

--amfAdjacentPlmnListName <string>

Description

Identifies the Adjacent PLMN list name

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

""

Name

amfAusfHttp2ToNasCcMappingListName - AUSF HTTP2 Status Code to 5GMM NAS Profile
List Name

Syntax

--amfAusfHttp2ToNasCcMappingListName <string>

Description

The name of the AUSF HTTP2 Status Code to 5GMM NAS Cause Code mapping profile list

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

default

Name

amfDeregistrationBehaviorMappingProfileName - De-registration behavior mapping profile
name

Syntax

--amfDeregistrationBehaviorMappingProfileName <string>

Description

Identifies, via name, a given de-registration behavior mapping.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

default

Name

amfDnnOverrideProfileName - AMF Dnn Override Profile Name

Syntax

--amfDnnOverrideProfileName <string>

Description

Pointer to the AMF DNN Override Profile

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

default

Name

amfEirOverrideProfileName - AMF EIR Override Profile Name

Syntax

--amfEirOverrideProfileName <string>

Description

It indicates the profile name used during Eir check.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

""

Name

amfEplmnListName - EPLMN list name

Syntax

--amfEplmnListName <string>

Description

Identifies Equivalent PLMN list name

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

""

Name

amfEplmnListNameGnb - EPLMN list name for gNB

Syntax

--amfEplmnListNameGnb <string>

Description

Identifies Equivalent PLMN list name to be sent to gNB. Both gNB and UE Eplmn lists have to be populated if the list is to be sent to both. Content of both lists could be same or different based on operator choice.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

""

Name

amfEventProfileName - AMF Event Profile Name

Syntax

--amfEventProfileName <string>

Description

Pointer to the AMF Event Profile

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

default

Name

amfFreqCountMonitoringProfileName - AMF Frequency Count Monitoring Profile Name

Syntax

--amfFreqCountMonitoringProfileName <string>

Description

Defines the frequency count monitoring profile name applied for the UE.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

""

Name

amfMcsPagingPriProfileListName - AMF MCS Paging Priority Profile List Name

Syntax

--amfMcsPagingPriProfileListName <string>

Description

Name that identifies the AMF MCS Paging Priority Profile List.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

amfNrDualConnectivityProfileName - NR Dual Connectivity Profile Name

Syntax

--amfNrDualConnectivityProfileName <string>

Description

Identifies NR Dual Connectivity Profile Name.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

amfNrfHttp2ToNasCcMappingListName - NRF HTTP2 Status Code to 5GMM NAS Profile List Name

Syntax

--amfNrfHttp2ToNasCcMappingListName <string>

Description

The name of the NRF HTTP2 Status Code to 5GMM NAS Cause Code mapping profile list

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

default

Name

amfNssfHttp2ToNasCcMappingListName - NSSF HTTP2 Status Code to 5GMM NAS Profile List Name

Syntax

--amfNssfHttp2ToNasCcMappingListName <string>

Description

The name of the NSSF HTTP2 Status Code to 5GMM NAS Cause Code mapping profile list

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

default

Name

amfOlcTimersProfileName - AMF OLC Timers Profile Name

Syntax

--amfOlcTimersProfileName <string>

Description

Identifies AMF OLC Timers Profile Name.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

amfPcfTriggerProfileName - AMF PCF Policy Trigger Profile Name

Syntax

--amfPcfTriggerProfileName <string>

Description

The name of the AMF PCF Policy Trigger Profile.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

default

Name

amfPduDeletionProfileName - AMF PDU Deletion Profile Name

Syntax

--amfPduDeletionProfileName <string>

Description

The name of the AMF PDU Deletion Profile that applies for the UE.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

default

Name

amfRfspAccessRestrictionListName - Access Restriction for RFSP List Name

Syntax

--amfRfspAccessRestrictionListName <string>

Description

A reference to a list of RFSP values for which access is to be restricted for the UE.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

amfRfspLocalOverrideProfileName - AMF RFSP Local Override Profile Name

Syntax

--amfRfspLocalOverrideProfileName <string>

Description

The name of the AMF RFSP Local Override Profile Name for the UE.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

amfSelection - AMF selection

Syntax

--amfSelection <list-item>

Description

The parameter specifies the method of AMF selection used by the AMF for UE. If there are multiple AMF matches then the AMF will use the capacity parameter to load balance the UE across the AMFs.

Class

Optional.

List

- LocalProvisioning - Local Provisioning - AMF will use locally provisioned NF for the selection of a NF.
- NRF - Network Repository Function - Make use of NRF to select NF.
- NRFLocalProvisioning - NRF with Local Provisioning Override - The AMF will check the locally provisioned selection rules and if there is a match then the AMF will select the NF matching the rule. If there is no matching then AMF will use the NRF. The selection rule by default is that AMF will select a NF supporting a provisioned criterion for the NF

Default

LocalProvisioning

Name

amfSmpToPagingPriorityProfileListName - AMF SMP to Paging Priority Profile Name

Syntax

--amfSmpToPagingPriorityProfileListName <string>

Description

The name of the profile to map SBI message Priority (SMP) to Paging Priority Level.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

amfStatusSubscribe - AMF Status Subscribe

Syntax

--amfStatusSubscribe <list-item>

Description

Subscribe NRF for remote AMF status change event notifications.

Class

Optional.

List

- byNfType - All NF of a given NF type - AMF subscribes to get status of by NF type to get status of all NF registered with NRF whether AMF use them or not.
- noSubscription - Do not Subscribe to NF Status - AMF does not subscribe to get NF status .
- selectedOnly - Selected NFs only - AMF subscribes to get status of NF by instance ID for the NFs that are discovered via the NRF

Default

noSubscription

Name

amfSubDataSetsProfileName - Profile name

Syntax

- -amfSubDataSetsProfileName <string>

Description

Identifies, via name, a given AMF Subscription DataSets Profile.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Default

default

Name

amfSvcAgreementProfileName - Service agreement profile name

Syntax

- -amfSvcAgreementProfileName <string>

Description

Defines the PLMN service agreement profile to be used for service.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Default

default

NameamfUdmHttp2ToNasCcMappingListName - UDM HTTP2 Status Code to 5GMM NAS Profile
List Name

Syntax

--amfUdmHttp2ToNasCcMappingListName <string>

Description

The name of the UDM HTTP2 Status Code to 5GMM NAS Cause Code mapping profile list

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

default

Name

amfUeMrlForbiddenTaiListName - Mobility Restriction TAI List Name

Syntax

--amfUeMrlForbiddenTaiListName <string>

Description

Name of the Forbidden TAI List in the Mobility Restriction List sent to the gNB(s)

Restrictions

Applies to home or roamer UEs.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

amfUeNitzInfoOverrideProfileName - UE Nitz Information Override Profile Name

Syntax

--amfUeNitzInfoOverrideProfileName <string>

Description

Indicates the profile name used for the UE Nitz information.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

amfUePriorityProfileName - AMF UE Priority Profile Name

Syntax

--amfUePriorityProfileName <string>

Description

The name of the UE Priority Profile.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

amfUINasUnableToForwardProfileName - AMF Uplink NAS Unable to Forward Profile Name

Syntax

--amfUINasUnableToForwardProfileName <string>

Description

Defines the name of the AMF Uplink NAS Unable to Forward Profile that applies for the UE.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

default

Name

amfVoiceProfileName - AMf Voice Profile Name

Syntax

--amfVoiceProfileName <string>

Description

The name of the profile for voice related handling.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

default

Name

amPolicySelection - AM Policy Selection

Syntax

--amPolicySelection <list-item>

Description

if the parameter is set to 'true' then the AMF will always select an provisioned AM Policy for specific SUP/GPSI ranges.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

applProtFallbackN26ToS10 - Application Protocol Fallback from N26 to S10

Syntax

--applProtFallbackN26ToS10 <list-item>

Description

Current parameter enables fallback to search for 's10' candidate when DNS Query to find target MME during S10 handover using 'n26' fails. This parameter is expected to be used together with parameter Use N26 Service Parameter in 5GS to EPS Handover

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

areaCodeExtendedEmergencyNumListName - Area Code Extended Emergency Number List
Name

Syntax

--areaCodeExtendedEmergencyNumListName <string>

Description

Area Code Extended Emergency Number List Name

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

ausfSelection - AUSF Selection

Syntax

--ausfSelection <list-item>

Description

The parameter specifies the method of AUSF selection used by the AMF for UE. If there are multiple AUSF matches then the AMF will use the capacity parameter to load balance the UE across the AUSFs.

Class

Optional.

List

- LocalProvisioning - Local Provisioning - AMF will use locally provisioned NF for the selection of a NF.
- NRF - Network Repository Function - Make use of NRF to select NF.
- NRFLocalProvisioning - NRF with Local Provisioning Override - The AMF will check the locally provisioned selection rules and if there is a match then the AMF will select the NF matching the rule. If there is no matching then AMF will use the NRF. The selection rule by default is that AMF will select a NF supporting a provisioned criterion for the NF

Default

LocalProvisioning

Name

ausfStatusSubscribe - AUSF Status Subscribe

Syntax

--ausfStatusSubscribe <list-item>

Description

Subscribe NRF for AUSF status change event notifications.

Class

Optional.

List

- **byNfType** - All NF of a given NF type - AMF subscribes to get status of by NF type to get status of all NF registered with NRF whether AMF use them or not.
- **noSubscription** - Do not Subscribe to NF Status - AMF does not subscribe to get NF status .
- **selectedOnly** - Selected NFs only - AMF subscribes to get status of NF by instance ID for the NFs that are discovered via the NRF

Default

noSubscription

Name

cag - Closed Access Group

Syntax

--cag <list-item>

Description

A control parameter to enable Closed Access Group (CAG) feature.

Class

Optional.

List

- **false** - Boolean value of false.
- **true** - Boolean value of true.

Default

false

Name

checkImsDnnSubscriptionVops - Check IMS DNN Subscription IMS Voice Over PS

Syntax

--checkImsDnnSubscriptionVops <list-item>

Description

Defines whether the AMF checks IMS DNN subscription in determining network IMS voice over PS support for the UE.

Class

Optional.

List

- **false** - Boolean value of false.
- **true** - Boolean value of true.

Default

false

Name

default5GEIRSelection - Default 5GEIR Selection

Syntax

--default5GEIRSelection <list-item>

Description

if the parameter is set to 'true' then the AMF will select an 5GEIR provisioned as a default 5GEIR if the AMF fails to select 5GEIR supporting using NRF or a locally provisioned 5GEIR that meets all the selection criterion.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

defaultAmPolicySelection - Default AM Policy Selection

Syntax

--defaultAmPolicySelection <list-item>

Description

if the parameter is set to 'true' then the AMF will select locally provisioned AM Policy if the AMF fails to receive AM policy from PCF..

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

defaultAusfSelection - Default AUSF Selection

Syntax

--defaultAusfSelection <list-item>

Description

if the parameter is set to 'true' then the AMF will select an AUSF provisioned as a default AUSF if the AMF fails to select an AUSF supporting using NRF or a locally provisioned AUSF that meets all the selection criterion.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

defaultGmlcSelection - Default Gmlc Selection

Syntax

--defaultGmlcSelection <list-item>

Description

If the parameter is set to 'true' then the AMF will select an GMLC provisioned as a default GMLC if the AMF fails to select an GMLC using NRF.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

defaultLmfSelection - Default LMF Selection

Syntax

--defaultLmfSelection <list-item>

Description

If the parameter is set to 'true' then the AMF will select an LMF provisioned as a default LMF if the AMF fails to select an LMF using NRF or a locally provisioned LMF that meets all the selection criterion.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

defaultNssaafSelection - Default Nssaaf Selection

Syntax

--defaultNssaafSelection <list-item>

Description

if the parameter is set to 'true' then the AMF will select an NSSAAF provisioned as a default NSSAAF if the AMF fails to select an NSSAAF supporting using NRF or a locally provisioned NSSAAF that meets all the selection criterion.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

defaultPcfSelection - Default PCF Selection

Syntax

--defaultPcfSelection <list-item>

Description

if the parameter is set to 'true' then the AMF will select a PCF provisioned as a default PCF if the AMF fails to select a PCF using NRF or a locally provisioned PCF that meets all the selection criterion.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

defaultSmfSelection - Default SMF selection

Syntax

--defaultSmfSelection <list-item>

Description

If the parameter is set to 'true' then the AMF will select an SMF provisioned as a default SMF if the AMF fails to select SMF using NRF or a locally provisioned SMF that meets all the selection criterion.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

defaultSmsfSelection - Default SMSF Selection

Syntax

--defaultSmsfSelection <list-item>

Description

If the parameter is set to 'true' then the AMF will select an SMSF provisioned as a default SMSF if the AMF fails to select an SMSF using NRF or a locally provisioned SMSF that meets all the selection criterion.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

defaultSnssaiListName - Default S-NSSAI List Name

Syntax

--defaultSnssaiListName <string>

Description

Configure default S-NSSAI list to be used as UE default S-NSSAI. The list must have only eight S-NSSAI. If the list has more than 8, the AMF shall only use the first eight SNSSAI.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Default

"

Name

defaultSnssaiSelection - Default S-NSSAI Selection Options

Syntax

--defaultSnssaiSelection <list-item>

Description

This parameter is deprecated. Value should be set to UDM. Value LOCAL is only for test purposes and does not interwork properly with other slicing features. Configuration parameter was used to choose the locally provisioned S-NSSAI list or UDM provided.

Class

Optional.

List

- LOCAL - Locally provisioned. For roamers locally provisioned is used when none of the UE subscribed SNSSAI(s) is marked as "default" and the locally provisioned SNSSAI(s) is standardized 3GPP SNSSAI(s).
- UDM - UDM provided default S-NSSAI list.

Default

UDM

Name

defaultUdmSelection - Default UDM Selection

Syntax

--defaultUdmSelection <list-item>

Description

If the parameter is set to 'true' then the AMF will select a UDM th provisioned as a default UDM if the AMF fails to select a UDM using NRF or a locally provisioned UDM that meets all the selection criterion.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

defaultVsnssaiName - Default Vsnssai Name

Syntax

--defaultVsnssaiName <string>

Description

The default VPLMN Slice used when there is no matching VPLMN slice to the UE requested HPLMN Slice.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

delayUdmDuringRegistrationProfileName - Delay UDM During Registration Profile Name

Syntax

--delayUdmDuringRegistrationProfileName <string>

Description

The name of the Delay UDM During Registration Profile.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

default

Name

dnnMccOverride - DNN MCC Override

Syntax

--dnnMccOverride <digits>

Description

Override of DNN domain derivation by using this mobility country code and the mobility network code represented by dnnMccOverride.

Class

Optional.

Type / Range

Digit string of length 0 or 3

Default

""

Name

dnnMncOverride - DNN MNC Override

Syntax

--dnnMncOverride <digits>

Description

Override of DNN domain derivation by using this mobility network code and the mobility country code represented by dncMncOverride

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 3

Default

""

Name

dnnToSnssaiListName - DNN to S-NSSAI List Name

Syntax

--dnnToSnssaiListName <string>

Description

Configure DNN to S-NSSAI list name.The AMF will use the list in n26 interworking scenarios

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

dnnToSnssaiPerPlmnListName - DNN to SNSSAI per PLMN List Name

Syntax

```
--dnnToSnssaiPerPlmnListName <string>
```

Description

Configure DNN to SNSSAI per PLMN List Name. The AMF will use the list in n26 inter-working scenarios for associated DNN and serving TAL.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Default

""

Name

edrxNrAllowed - eDRX Allowed

Syntax

```
--edrxNrAllowed <list-item>
```

Description

Specifies whether UEs from this PLMN are allowed to use eDRX.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Recommended

false

Name

edrxNrDnnListName - eDRX DNN List Name

Syntax

```
--edrxNrDnnListName <string>
```

Description

Specifies name of a List of DNNs. A UE is allowed to use eDRX only if it is using at least one DNN in the list.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Name

edrxNrProfileName - eDRX Profile Name

Syntax

--edrxNrProfileName <string>

Description

Specifies an eDRX profile that is used for negotiation of eDRX parameters with a UE.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Recommended

"

Name

edrxNrRedcapProfileName - eDRX NR Redcap Profile Name

Syntax

--edrxNrRedcapProfileName <string>

Description

Specifies an eDRX profile that is used for negotiation of eDRX parameters with a UE when the UE is using NR Redcap.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Recommended

"

Name

eir5gStatusSubscribe - 5G EIR Status Subscribe

Syntax

--eir5gStatusSubscribe <list-item>

Description

Subscribe NRF for 5G EIR status change event notifications.

Class

Optional.

List

- byNfType - All NF of a given NF type - AMF subscribes to get status of by NF type to get status of all NF registered with NRF whether AMF use them or not.
- noSubscription - Do not Subscribe to NF Status - AMF does not subscribe to get NF status .
- selectedOnly - Selected NFs only - AMF subscribes to get status of NF by instance ID for the NFs that are discovered via the NRF

Default

noSubscription

Name

eirCheck - EIR Check

Syntax

--eirCheck <list-item>

Description

This command enables EIR check for subscriber(s) IMEISV.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

eirSelection - EIR Selection

Syntax

--eirSelection <list-item>

Description

The parameter specifies the method of EIR selection used by the AMF for UE. If there are multiple EIR matches then the AMF will the capacity parameter to load balance the UE across the EIRs.

Class

Optional.

List

- LocalProvisioning - Local Provisioning - AMF will use locally provisioned NF for the selection of a NF.
- NRF - Network Repository Function - Make use of NRF to select NF.
- NRFLocalProvisioning - NRF with Local Provisioning Override - The AMF will check the locally provisioned selection rules and if there is a match then the AMF will select the NF matching the rule. If there is no matching then AMF will use the NRF. The selection rule by default is that AMF will select a NF supporting a provisioned criterion for the NF

Default

LocalProvisioning

Name

emcNon3gppN3iwf - EMC Non-3GPP Access N3IWF

Syntax

--emcNon3gppN3iwf <list-item>

Description

Defines whether Emergency Services are supported for the UE over untrusted non-3GPP access via N3IWF.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

emcSupport - EMC Support

Syntax

--emcSupport <list-item>

Description

Defines whether Emergency Services are supported for the UE. It applies for a home UE or a roamer UE.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

emergencyNon3gppEpsPduHo - Emergency Non-3GPP Access EPS PDU Handover

Syntax

--emergencyNon3gppEpsPduHo <list-item>

Description

Enables the support for UE triggered emergency PDU handovers between untrusted non-3GPP via N3IWF/5GS and E-UTRAN/EPS for the UE.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

emfSupport - EMF Support

Syntax

--emfSupport <list-item>

Description

Defines whether Emergency Fallback is supported for the UE. It applies for a home UE or a roamer UE.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

epslwkWithoutN26 - EPS Interworking Without N26

Syntax

```
--epsIwkWithoutN26 <list-item>
```

Description

Enables the EPS interworking without N26 interface support for the UE.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

epsTo5gsSmfReselectFailMaintain - EPS to 5GS SMF Reselection Failure Maintain

Syntax

```
--epsTo5gsSmfReselectFailMaintain <list-item>
```

Description

Defines whether the AMF maintains PDU session in a situation where during EPS to 5GS mobility, the AMF detects the need to select a new I-SMF (or V-SMF) due to actual S-NSSAI not served by default I-SMF (or V-SMF) but SMF reselection fails. If set to 'false', AMF releases the PDU session (i.e. triggers Nsmf_PDUSession_UpdateSMContext with release indication towards the default V-SMF or I-SMF), otherwise the PDU is served by default V-SMF (or I-SMF) and default SNSSAI

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

epsUpip - EPS UPIP

Syntax

```
--epsUpip <list-item>
```

Description

Current parameter enables support for the EPS-UIPI (User Plane Integrity Protection)

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

extendedEmergencyNumListName - Extended Emergency Number List Name

Syntax

--extendedEmergencyNumListName <string>

Description

Extended Emergency Number List Name

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Default

''

Name

getAccMobDataCag - Get Access and Mobility Data Due to CAG Enabling

Syntax

--getAccMobDataCag <list-item>

Description

A control parameter to enable requesting again for Access and Mobility Subscription Data in intra-AMF Mobility and Periodic Registration due to CAG feature enabling

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

gmlcSelection - GMLC selection

Syntax

--gmlcSelection <list-item>

Description

The parameter specifies the method of GMLC selection used by the AMF. If there are multiple GMLC matches then the AMF will use the capacity parameter to load balance the UE across the GMLC.

Class

Optional.

List

- LocalProvisioning - Local Provisioning - AMF will use locally provisioned NF for the selection of a NF.
- NRF - Network Repository Function - Make use of NRF to select NF.
- NRFLocalProvisioning - NRF with Local Provisioning Override - The AMF will check the locally provisioned selection rules and if there is a match then the AMF will select the NF matching the rule. If there is no matching then AMF will use the NRF. The selection rule by default is that AMF will select a NF supporting a provisioned criterion for the NF

Default

LocalProvisioning

Name

gmlcStatusSubscribe - GMLC Status Subscribe

Syntax

--gmlcStatusSubscribe <list-item>

Description

Subscribe NRF for remote LMF status change event notifications.

Class

Optional.

List

- byNfType - All NF of a given NF type - AMF subscribes to get status of by NF type to get status of all NF registered with NRF whether AMF use them or not.
- noSubscription - Do not Subscribe to NF Status - AMF does not subscribe to get NF status .
- selectedOnly - Selected NFs only - AMF subscribes to get status of NF by instance ID for the NFs that are discovered via the NRF

Default

noSubscription

Name

hlcomAllowed - HLCOM Allowed

Syntax

--hlcomAllowed <list-item>

Description

Specifies whether an eDRX UE from this PLMN is allowed to use High Latency Communication.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Recommended

false

Name

hlcomAllowedMico - HLCOM Allowed MICO

Syntax

--hlcomAllowedMico <list-item>

Description

Specifies whether the AMF will use MICO procedures, if provisioned and requested by the SMF, for UEs from this PLMN.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

hoCellIdRestrictionListNameN14 - NR Cells Lists for Handover Restriction over N14

Syntax

--hoCellIdRestrictionListNameN14 <string>

Description

Identifies the list of source NR Cells that the UE is either allowed or restricted from being handover over between AMFs. The target AMF uses the list to either receive or reject the Create UE Context Request from the source AMF.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

""

Name

hplmnToVplmnSnssaiMappingListName - H-PLMN to V-PLMN Snssai Mapping List Name

Syntax

--hplmnToVplmnSnssaiMappingListName <string>

Description

Identifies allowed Home PLMN to Visited PLMN Snssai Mapping list name.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

""

Name

ignoreAusfFailuresOptionalReAuthentications - Ignore AUSF Failures on Optional Re-Authentications

Syntax

--ignoreAusfFailuresOptionalReAuthentications <list-item>

Description

This parameter, when set, enables AMF to ignore AUSF dialog related failures during optional re-authentications. The UEs to apply the functionality to are provisioned on the AMF. The functionality is applied when the AMF is locally provisioned to perform an optional re-authentication for a UE that is already registered on the AMF, but the re-authentication fails during the AUSF dialog and the AMF is able to continue serving the UE without the re-authentication

Class

Optional.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Name

imsReestablishmentAfter5gSrvccCancel - IMS Re-Establishment After 5G-SRVCC Cancelled

Syntax

--imsReestablishmentAfter5gSrvccCancel <list-item>

Description

Enables the AMF support for NAS Event Notification for 5G-SRVCC handover cancelled, IMS re-establishment required indication towards the UE.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

imsVoiceOverPsEpsFb - IMS Voice over PS EPS Fallback

Syntax

--imsVoiceOverPsEpsFb <list-item>

Description

This parameter controls whether IMS Voice over PS Sessions is supported for the UE with this PLMN under the serving PLMN by EPS fallback.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

imsVoiceOverPsVoNR - IMS Voice over PS Voice over New Radio

Syntax

--imsVoiceOverPsVoNR <list-item>

Description

This parameter controls whether IMS Voice over PS Sessions is supported for the UE with this PLMN under the serving PLMN by Voice over New Radio.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

includeSelectedDnn - Include Selected DNN

Syntax

--includeSelectedDnn <list-item>

Description

Defines whether the AMF includes the selectedDnn attribute inside -SmContextCreateData towards the SMF/I-SMF/V-SMF and/or -PduSessionContext between old and new AMF during inter AMF mobility procedures, when a different DNN than the UE requested DNN has been selected for the PDU.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Recommended

The recommended value is TRUE.

Name

informUeEmergencyRegisteredOnly - Inform UE Emergency Registered Only

Syntax

--informUeEmergencyRegisteredOnly <list-item>

Description

AMF enables the option to inform the UE that it is emergency registered. The AMF uses this option if the AMF for a normally registered UE having an emergency session ongoing receives a trigger for network triggered deregistration. After the AMF in such a case has indicated to

SMF(s) that non-emergency PDU(s) are released, the AMF indicates on Registration Accept or Configuration Update Command that the UE is registered for emergency services.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

interPlmnApiRoot - Inter PLMN API Root

Syntax

--interPlmnApiRoot <list-item>

Description

When set to true the AMF supports receiving interPlmnApiRoot from SMF storing it in the UE Context and transferring to new AMF during UE mobility.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

interPlmnEmergencyHoSourceNode - Inter Plmn Emergency Call Handover Source Node

Syntax

--interPlmnEmergencyHoSourceNode <list-item>

Description

Identifies AMF source node behaviour (allow or reject) for inter plmn 5G to 5G handover of emergency calls triggered by the source gNB.

Class

Optional.

List

- both - Both LBO and Home Routed emergency calls are allowed by the source node to be handed over.

- hrt - Home Routed Traffic - when UE originates emergency call in UE H-PLMN, the call is allowed by source node to be handed over to V-PLMN as home routed call.
- lbo - Local Break Out - When roaming UE originates LBO emergency call in V-PLMN, the call is allowed by source node to be handed over to another PLMN (another V-PLMN or UE H-PLMN).
- none - Neither LBO nor Home Routed emergency calls are allowed by the source node to be handed over.

Default

both

Name

interPlmnEmergencyHoTargetNode - Inter PLMN Emergency Call Handover Target Node

Syntax

--interPlmnEmergencyHoTargetNode <list-item>

Description

Identifies AMF target node behaviour (allow or reject) for inter plmn 5G to 5G handover of emergency calls from AMF source node.

Class

Optional.

List

- both - LBO and HRT - Both LBO and Home Routed emergency calls are allowed by the target node to be handed in.
- hrt - Home Routed Traffic - When UE originates emergency call in UE H-PLMN, the call is allowed by V-PLMN target node to be handed in.
- lbo - Local Break Out - When roaming UE originates emergency call in V-PLMN, the call is allowed by target node in another Plmn (V-PLMN or H-PLMN) to be handed in.
- none - Neither LBO nor Home Routed emergency calls are allowed by the target node to be handed in.

Default

both

Name

interPlmnHandover - Inter-PLMN Handover

Syntax

--interPlmnHandover <list-item>

Description

Allows or restricts gNB triggered Handover at source AMF for intra 5GS Handover or 5GS to EPS Handover.

Class

Optional.

List

- 5gsToEps - 5G to EPS Handover Allowed - Source AMF allows inter-plmn Handover to LTE (target MME)
- both - 5GS to 5GS and 5GS to EPS Handover Allowed - Source AMF allows inter-plmn Handover to LTE (target MME) or to 5GS (target AMF)
- intra5gs - 5GS to 5GS Handover Allowed - Source AMF allows inter-plmn Handover to 5GS (target AMF)
- none - No Handover Allowed - Source AMF restricts inter-plmn Handover to LTE (target MME) or to 5GS (target AMF)

Default

none

Name

interPlmnHandoverTargetNodeBehavior - Inter PLMN Handover Target Node Behavior

Syntax

--interPlmnHandoverTargetNodeBehavior <list-item>

Description

Sets target AMF behaviour over N14 or N26 during inter-plmn Handover.

Class

Optional.

List

- both - Both N14 and N26 Interfaces - The target AMF allows inter-plmn Create UE Context procedure from source AMF or Forward Relocation Request procedure from source MME to proceed.
- n14 - N14 Interface - The target AMF allows inter-plmn Create UE Context procedure from source AMF to proceed.
- n26 - N26 Interface - The target AMF allows inter-plmn Forward Relocation Request procedure from source MME to proceed.
- none - The target AMF rejects inter-plmn Create UE Context from source AMF or Forward Relocation Request from source MME due to Handover not allowed.

Default

both

Name

interPlmnIdleModeMobSrcNodeBehavior - Inter PLMN Idle Mode Mobility Source Node Behavior

Syntax

--interPlmnIdleModeMobSrcNodeBehavior <list-item>

Description

Sets source or old AMF behaviour over N14 or N26 during inter-plmn UE Idle Mode Mobility.

Class

Optional.

List

- both - Both N14 and N26 - Source or old AMF allows UE Context Transfer procedure from new or target AMF or Context Request procedure from new or target MME to proceed based on Serving-Target PLMN ID combination during inter-plmn UE Idle Mode Mobility.
- n14 - N14 Interface - Source or old AMF allows UE Context Transfer procedure from new or target AMF to proceed based on Serving-Target PLMN ID combination during inter-plmn UE Mobility Registration Update.
- n26 - N26 Interface - Source or old AMF allows Context Request procedure from new or target MME to proceed based on Serving-Target PLMN ID combination during inter-plmn UE Tracking Area Update.
- none - Source or old AMF rejects UE Context Transfer procedure from new or target AMF or Context Request procedure from new or target MME based on Serving-Target PLMN ID combination during inter-plmn UE Idle Mode Mobility.

Default

both

Name

interPlmnInitRegAttachSrcNodeBehavior - Inter PLMN Initial Registration or Attach Source Node Behavior

Syntax

--interPlmnInitRegAttachSrcNodeBehavior <list-item>

Description

Sets source or old AMF behaviour over N14 or N26 during inter-plmn UE Initial Registration or Attach.

Class

Optional.

List

- both - Both N14 and N26 Interfaces - Source or old AMF allows both UE Context Transfer procedure from new or target AMF or Identification Request procedure from new or target MME to proceed based on Serving-Target PLMN ID combination during inter-plmn UE Initial Registration or Initial Attach procedure.

- n14 - n14 Interface - Source or old AMF allows UE Context Transfer procedure from new or target AMF to proceed based on Serving-Target PLMN ID combination during inter-plmn UE Initial Registration procedure.
- n26 - n26 Interface - Source or old AMF allows Identification Request procedure from new or target MME to proceed based on Serving-Target PLMN ID combination during inter-plmn UE Initial Attach procedure.
- none - Source or old AMF rejects UE Context Transfer procedure from new or target AMF or Identification Request procedure from new or target MME to proceed based on Serving-Target PLMN ID combination during inter-plmn UE Initial Registration or Initial Attach procedure.

Default

both

Name

interPlmnInitRegReq - Inter-PLMN Initial Registration Request

Syntax

--interPlmnInitRegReq <list-item>

Description

AMF allows or restricts inbound inter-plmn UE Initial Registration Request roaming from 5G NR or LTE based on UE GUTI PLMN ID. When restricted, AMF invokes NAS Identity Request toward the UE to retrieve UE SUCI and proceed with the initial registration.

Class

Optional.

List

- both - Both nr and lte - AMF allows inbound inter-plmn UE Initial Registration Request roaming from 5G NR or LTE.
- lte - AMF allows inbound inter-plmn UE Initial Registration Request roaming from LTE
- none - AMF restricts inbound inter-plmn UE Initial Registration Request roaming from 5G NR or LTE
- nr - 5G New Radio - AMF allows inbound inter-plmn UE Initial Registration Request roaming from 5G NR

Default

none

Name

interPlmnMobRegUpdtReq - Inter-PLMN Mobility Registration Update Request

Syntax

--interPlmnMobRegUpdtReq <list-item>

Description

Allows or restricts inbound inter-plmn idle Mobility Registration Update (MRU) from NR over N14, or from LTE over N26, or both based on UE GUTI PLMN ID. When restricted, AMF rejects the MRU with cause code #9, which triggers UE to initiate Initial Registration with SUCI.

Class

Optional.

List

- both - 4G LTE and 5G New Radio - Allows inbound inter-plmn idle Mobility Registration Update Request from source AMF or source MME
- lte - 4G LTE - Allows inbound inter-plmn idle Mobility Registration Update Request from source MME
- none - Neither 4G LTE nor 5G New Radio - Restricts inbound inter-plmn idle Mobility Registration Update Request from source AMF or source MME
- nr - 5G New Radio - Allows inbound inter-plmn idle Mobility Registration Update Request from source AMF

Default

none

Name

interPlmnN26EmergencyHoSourceNode - Inter Plmn N26 Emergency Call Handover Source Node

Syntax

--interPlmnN26EmergencyHoSourceNode <list-item>

Description

Identifies AMF source node behaviour (allow or reject) for inter plmn 5G to 4G handover of emergency calls triggered by the source gNB.

Class

Optional.

List

- both - Both LBO and Home Routed emergency calls are allowed by the source node to be handed over.
- hrt - Home Routed Traffic - when UE originates emergency call in UE H-PLMN, the call is allowed by source node to be handed over to V-PLMN as home routed call.
- lbo - Local Break Out - When roaming UE originates LBO emergency call in V-PLMN, the call is allowed by source node to be handed over to another PLMN (another V-PLMN or UE H-PLMN).
- none - Neither LBO nor Home Routed emergency calls are allowed by the source node to be handed over.

Default

both

Name

interPlmnN26EmergencyHoTargetNode - Inter PLMN N26 Emergency Call Handover Target Node

Syntax

--interPlmnN26EmergencyHoTargetNode <list-item>

Description

Identifies AMF target node behaviour (allow or reject) for inter plmn 4G to 5G handover of emergency calls from MME source node.

Class

Optional.

List

- both - LBO and HRT - Both LBO and Home Routed emergency calls are allowed by the target node to be handed in.
- hrt - Home Routed Traffic - When UE originates emergency call in UE H-PLMN, the call is allowed by V-PLMN target node to be handed in.
- lbo - Local Break Out - When roaming UE originates emergency call in V-PLMN, the call is allowed by target node in another Plmn (V-PLMN or H-PLMN) to be handed in.
- none - Neither LBO nor Home Routed emergency calls are allowed by the target node to be handed in.

Default

both

Name

interPlmnPcfSupport - Inter-PLMN PCF Support

Syntax

--interPlmnPcfSupport <list-item>

Description

Allows or restricts the use of PCF for Access Mobility and UE Policy Rules, based on roaming SLAs between operators.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ladn - Local Area Data Network

Syntax

--ladn <list-item>

Description

Local Area Data Network. This parameter activates the LADN functionality support for the UE. With the LADN functionality, access to a provisioned data network is only provided on a restricted service area.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

lmfSelection - LMF Selection

Syntax

--lmfSelection <list-item>

Description

The parameter specifies the method of LMF selection used by the AMF. If there are multiple LMF matches then the AMF will use the capacity parameter to load balance the UE across the LMFs.

Class

Optional.

List

- LocalProvisioning - Local Provisioning - AMF will use locally provisioned NF for the selection of a NF.
- NRF - Network Repository Function - Make use of NRF to select NF.
- NRFLocalProvisioning - NRF with Local Provisioning Override - The AMF will check the locally provisioned selection rules and if there is a match then the AMF will select the NF matching the rule. If there is no matching then AMF will use the NRF. The selection rule by default is that AMF will select a NF supporting a provisioned criterion for the NF

Default

LocalProvisioning

Name

ImfStatusSubscribe - LMF Status Subscribe

Syntax

```
--lmfStatusSubscribe <list-item>
```

Description

Subscribe NRF for remote LMF status change event notifications.

Class

Optional.

List

- **byNfType** - All NF of a given NF type - AMF subscribes to get status of by NF type to get status of all NF registered with NRF whether AMF use them or not.
- **noSubscription** - Do not Subscribe to NF Status - AMF does not subscribe to get NF status .
- **selectedOnly** - Selected NFs only - AMF subscribes to get status of NF by instance ID for the NFs that are discovered via the NRF

Default

noSubscription

Name

mappingOfAllowedHomeNssaiQuery - Mapping Of Allowed Home Nssai Query

Syntax

```
--mappingOfAllowedHomeNssaiQuery <list-item>
```

Description

Enables to query NSSF to map Allowed UE Home PLMN Nssai to Visited PLMN Nssai at UE initial registration or during idle or connected mode mobility. If NSSF query response doesn't include mapped Nssai, returns an error or the setting is set to false, AMF uses the local hplmn to vplmn mapping configuration.

Class

Optional.

List

- **false** - Boolean value of false.
- **true** - Boolean value of true.

Default

false

Name

mergeUeAllowedNssaiAndSubInSliceSubChange - Merge UE Allowed NSSAI and Subscribed in Slice Subscription Change

Syntax

--mergeUeAllowedNssaiAndSubInSliceSubChange <list-item>

Description

When this parameter is enabled and the UDM send a Notification with slice change in the UE subscription the AMF will use the union the existing Allowed NSSAI with the Slices of the Subscribed NSSAI in replacement of the original Allowed NSSAI and use it to determine the final Allowed NSSAI and include it in the Configuration Update sent to the UE

Class

Optional.

List

- none - None
- allSubscribed - All Subscribed - All Subscribed S-NSSAIs

Default

none

Name

mergeUeReqNssaiAndSubInFirstReg - Merge UE Requested NSSAI and Subscribed in First Registration

Syntax

--mergeUeReqNssaiAndSubInFirstReg <list-item>

Description

When this parameter is enabled AMF will use the union the received Requested NSSAI with the Slices of the Subscribed NSSAI as the Requested NSSAI in Initial Registrations, Mobility Registration from different AMF or Mobility Registration from 4G.

Class

Optional.

List

- none - None
- allSubscribed - All Subscribed - All Subscribed S-NSSAIs

Default

none

Name

n26lwk - N26 interworking enabled

Syntax

--n26lwk <list-item>

Description

When set, N26 interworking is enabled

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

n26IwkRestrictionTaiListName - N26 IWK Restriction TAI List

Syntax

--n26IwkRestrictionTaiListName <string>

Description

N26 interworking restriction TAI list used by AMF to restrict 5G service when UE(s) move from LTE into the respective TACs in the TAI list.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 31 with the allowed pattern of [0-9a-zA-Z_\.\\-:@]*

Default

""

Name

non3gppAccessN3iwf - Non-3GPP access N3IWF

Syntax

--non3gppAccessN3iwf <list-item>

Description

Controls the support of non-3GPP access and the Non-3GPP Access Interworking Function (N3IWF) for the UE.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

non3gppAccessN3iwfImsVops - Non-3GPP Access N3IWF IMS Voice Over PS

Syntax

--non3gppAccessN3iwfImsVops <list-item>

Description

Controls whether IMS Voice over PS Sessions over untrusted non-3GPP access over N3IWF is supported for the UE with this PLMN under the serving PLMN.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

non3gppEpsPduHo - Non-3GPP Access EPS PDU Handover

Syntax

--non3gppEpsPduHo <list-item>

Description

Enables the support for UE triggered non-emergency PDU handovers between untrusted non-3GPP via N3IWF/5GS and E-UTRAN/EPS for the UE.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

non3gppPduHo - Non-3GPP PDU Handover

Syntax

--non3gppPduHo <list-item>

Description

Enables the support for UE triggered non-emergency PDU handovers between 3GPP and untrusted non-3GPP access via N3IWF for the UE.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

non3gppPduHoSmfHoFail - Non-3GPP PDU Handover SMF HO Failure

Syntax

- non3gppPduHoSmfHoFail <list-item>

Description

Defines whether the AMF supports the Nsmf_PDUSession service feature HOFail for the UE to indicate a failure of a PDU session handover between 3GPP and non-3GPP access types via SM context status notify.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

non3gppVopsN3iwfCheckImsDnn - Non-3GPP IMS VoPS N3IWF Check IMS DNN

Syntax

- non3gppVopsN3iwfCheckImsDnn <list-item>

Description

Defines whether the AMF checks IMS DNN subscription in determining network IMS voice over PS support over non-3GPP access for the UE, when the UE registers over untrusted non-3GPP access via N3IWF.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

non3gppVopsN3iwfCheckUeUsageSetting - Non-3GPP VoPS N3IWF Check UE Usage Setting

Syntax

--non3gppVopsN3iwfCheckUeUsageSetting <list-item>

Description

Controls whether the AMF requires UE's Usage Setting IE to be included in NAS Registration Request to indicate IMS voice over PS session over non-3GPP access as supported towards the UE and the UDM. If set to false, IMS voice over PS session support over non-3GPP access indication does not depend on whether the UE includes UE's Usage Setting IE. If set to true, AMF indicates IMS voice over PS session supported over non-3GPP access only if UE's Usage Setting IE is included in NAS Registration Request.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

nonEncryptedSuciNotAllowed - Non Encrypted SUCI Not Allowed

Syntax

--nonEncryptedSuciNotAllowed <list-item>

Description

When set, AMF rejects UE Registration if UE provided a non-encrypted SUCI in the Registration Request.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

npIiNon3gppAddUeLocation - NPLI Non-3GPP Additional UE Location

Syntax

```
--npliNon3gppAddUeLocation <list-item>
```

Description

Defines whether the AMF provides the additional non-3GPP access location as part of Network Provided Location Information (NPLI), when both 3GPP access location information and non-3GPP access location information are available for the UE.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

nssaa - NSSAA

Syntax

```
--nssaa <list-item>
```

Description

A control parameter to enable Network Slice-Specific Authentication and Authorization (NSSAA). If the parameter set to true, then AMF support NSSAA for this ue serving PLMN. Otherwise, AMF will not support it

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

nssaafSelection - Nssaaf Selection

Syntax

```
--nssaafSelection <list-item>
```

Description

This parameter specifies the method of NSSAAF selection used by the AMF for the NSSAA Authenticate svc operation.

Class

Optional.

List

- LocalProvisioning - Local Provisioning - AMF will use locally provisioned NF for the selection of a NF.
- NRF - Network Repository Function - Make use of NRF to select NF.
- NRFLocalProvisioning - NRF with Local Provisioning Override - The AMF will check the locally provisioned selection rules and if there is a match then the AMF will select the NF matching the rule. If there is no matching then AMF will use the NRF. The selection rule by default is that AMF will select a NF supporting a provisioned criterion for the NF

Default

LocalProvisioning

Name

nssaafStatusSubscribe - NSSAAF Status Subscribe

Syntax

--nssaafStatusSubscribe <list-item>

Description

Subscribe NRF for NSSAAF status change event notifications.

Class

Optional.

List

- byNfType - All NF of a given NF type - AMF subscribes to get status of by NF type to get status of all NF registered with NRF whether AMF use them or not.
- noSubscription - Do not Subscribe to NF Status - AMF does not subscribe to get NF status .
- selectedOnly - Selected NFs only - AMF subscribes to get status of NF by instance ID for the NFs that are discovered via the NRF

Default

noSubscription

Name

nssaiInclusionMode - NSSAI Inclusion Mode

Syntax

--nssaiInclusionMode <list-item>

Description

This Parameter control of the NSSAI inclusion mode is sent to the UE. Can have one of the following values: - None: Inclusion mode is not sent in the Registration Accept no matter if UE subscription allows inclusion modes. - Mode A: Inclusion mode A is sent in the

Registration Accept if UE subscription allows NSSAI Inclusion mode. Otherwise, no NSSAI Inclusion mode sent. - Mode B: Inclusion mode B is sent in the Registration Accept if UE subscription allows NSSAI Inclusion mode. Otherwise, no NSSAI Inclusion mode sent. - Mode C: Inclusion mode C is sent in the Registration Accept if UE subscription allows NSSAI Inclusion mode. Otherwise, no NSSAI Inclusion mode sent. - Mode D: Inclusion mode D is sent no matter if UE subscription allows inclusion modes.

Class

Optional.

List

- modeA - Mode A - NSSAI Inclusion mode A
- modeB - Mode B - NSSAI Inclusion mode B
- modeC - Mode C - NSSAI Inclusion mode C
- modeD - Mode D - NSSAI Inclusion mode D
- none - No NSSAI Inclusion mode

Default

none

Name

pagingPriProfileListName - Paging priority profile name

Syntax

--pagingPriProfileListName <string>

Description

Identifies the paging priority profile.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

pcfPolicyAllowSnssaiListName - PCF Policy Allow S-NSSAI List Name

Syntax

--pcfPolicyAllowSnssaiListName <string>

Description

When a list is provided, and the corresponding Policy Association Profile indicates that UE Policy and/or AM Policy Services should be established, then an additional check is taken in to account so that the the policy establishment is triggered only if at least one of the UE's Subscribed S-NSSAI is also included in PCF Policy Allow S-NSSAI List. If the list is not provided or it is empty then the UEs Subscribed NSSAIs are not taken into account.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

pcfSelection - PCF Selection

Syntax

--pcfSelection <list-item>

Description

The parameter specifies the method of PCF selection used by the AMF for UE. If there are multiple PCF matches then the AMF will use the capacity parameter to load balance the UE across the PCFs.

Class

Optional.

List

- LocalProvisioning - Local Provisioning - AMF will use locally provisioned NF for the selection of a NF.
- NRF - Network Repository Function - Make use of NRF to select NF.
- NRFLocalProvisioning - NRF with Local Provisioning Override - The AMF will check the locally provisioned selection rules and if there is a match then the AMF will select the NF matching the rule. If there is no matching then AMF will use the NRF. The selection rule by default is that AMF will select a NF supporting a provisioned criterion for the NF

Default

LocalProvisioning

Name

pcfStatusSubscribe - PCF Status Subscribe

Syntax

--pcfStatusSubscribe <list-item>

Description

Subscribe NRF for PCF status change event notifications.

Class

Optional.

List

- **byNfType** - All NF of a given NF type - AMF subscribes to get status of by NF type to get status of all NF registered with NRF whether AMF use them or not.
- **noSubscription** - Do not Subscribe to NF Status - AMF does not subscribe to get NF status .
- **selectedOnly** - Selected NFs only - AMF subscribes to get status of NF by instance ID for the NFs that are discovered via the NRF

Default

noSubscription

Name

preferredFullPlmn - Preferred Full PLMN

Syntax

--preferredFullPlmn <list-item>

Description

When set to true, the AMF includes Query-Params-Ext2 with preferred-full-plmn in the Get method.

Class

Optional.

List

- **false** - Boolean value of false.
- **true** - Boolean value of true.

Default

true

Name

preferredNfInstances - Preferred NF Instances

Syntax

--preferredNfInstances <list-item>

Description

Effective when ismf global parameter is set to 'true'. Controls whether the target AMF provides the anchor SMF and the I-SMF instance id(s) towards the NRF when performing I-SMF selection/checking the need to insert/change/remove I-SMF at AMF change.

Class

Optional.

List

- **false** - Boolean value of false.
- **true** - Boolean value of true.

Default

false

Name

preferredTaiSmfSelection - Preferred TAI SMF Selection

Syntax

--preferredTaiSmfSelection <list-item>

Description

Effective when ismf global parameter is set to 'true'. Controls the option to include 'preferred TAI' as part of the SMF selection criteria. Applicable to SMF selection based on both local provisioning and the NRF.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

preferSameSmfForAllPdus - Prefer Same SMF for All UE PDUs

Syntax

--preferSameSmfForAllPdus <list-item>

Description

When this parameter is set to true the AMF will select the same SMF for all PDU sessions of a UE when feasible. This can optimize the SGW/PGW collocation when the UE moves to RAT LTE in order to minimize the number of session on SGW and PGW not collocated.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

preferSameVsmfForAllPdus - Prefer Same V-SMF for All UE PDUs

Syntax

--preferSameVsmfForAllPduSes <list-item>

Description

When this parameter is set to true the AMF will select the same V-SMF for all PDU sessions of a UE when feasible. This can minimize the probability of collisions

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

queryNssfForConfigurationUpdate - Query NSSF For Configuration Update

Syntax

--queryNssfForConfigurationUpdate <list-item>

Description

This parameter defines where the AMF triggers the NSSF Get service operation of Nnssf_NSSelection service during UE Configuration Update procedure.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

reAuthTimePeriod - Reauthentication Time Period

Syntax

--reAuthTimePeriod <integer>

Description

This field defines the time period between successive UE authentications forced by the AMF. When value equals zero then the feature is disabled.

Class

Optional.

Type / Range

Integer in the range of 60 to 1440 including value zero in units of Minute

Default

0

Name

requiredFeaturesNrfDtssaSmf - Required-features DTSSA to NRF for SMF

Syntax

--requiredFeaturesNrfDtssaSmf <list-item>

Description

Defines whether the AMF includes the Deployments Topologies with specific SMF Service Areas (DTSSA) feature id inside required-features on NRF query when discovering the actual SMF, and I-SMF feature is enabled.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

reregisterAfterSliceModification - Reregister UE after Slice Modification

Syntax

--reregisterAfterSliceModification <list-item>

Description

Controls when the AMF indicates to the UE the need for the UE to perform Registration procedure without including the GUAMI or 5G-S-TMSI after a Slice modification.

Class

Optional.

List

- allPotentiallyAffectedUes - All Potentially Affected UEs - The AMF indicates to all potentially affected UE the need for the UE to perform Registration procedure without including the GUAMI or 5G-S-TMSI if any slice included the UEs configured NSSAI and UE's current TAI has been added irrelevant if the Allowed NSSAI has been affected.
- allowedNssaiChange - Allowed NSSAI - The AMF indicates to the UE the need for the UE to perform Registration procedure without including the GUAMI or 5G-S-TMSI when the UE's Allowed NSSAI has been changed irrelevant if existing Slice connectivity is affected.

- always - The AMF always indicates to all UE the need for the UE to perform Registration procedure without including the GUAMI or 5G-S-TMSI whenever there is a Slice change in the UEs Subscribed NSSAI or a Network Slice Change.
- existingSliceConnectivityAffected - Existing Slice Connectivity Affected - The AMF indicates to all UE the need for the UE to perform Registration procedure without including the GUAMI or 5G-S-TMSI when the UE's Allowed NSSAI has been changed only if existing connectivity is affected.

Default

allowedNssaiChange

Name

reselectPcfOnRegistration - Reselect PCF on Registration

Syntax

--reselectPcfOnRegistration <list-item>

Description

When this parameter is enabled AMF will (re)check that the existing PCF supports all mandatory criterions it would use to select a PCF when the UE performs a Registration (Periodic / Mobility). If the existing PCF meets all the criterions the existing PCF will be maintained. If not, a new PCF will be selected.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

reuseOldUeContextEpsTo5gs - Reuse Old UE Context When UE Moves to 5GS

Syntax

--reuseOldUeContextEpsTo5gs <list-item>

Description

With this parameter, the AMF reuses the subscription data of the old UE Context that were locally preserved when UE moved to EPS and skips SDM GET and SDM Subscribe operations towards UDM

Class

Optional.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Name

roamDiscoverPcf - Discover PCF in HPLMN

Syntax

--roamDiscoverPcf <list-item>

Description

Specifies the option to discover H-PCF in the UE HPLMN to manage UE Policy rules based on roaming agreements between operators for Local Break Out roaming. When set, the AMF discovers H-PCH and includes in the UE Policy Association Creation request to the V-PCF which it uses to reach to the H-PCF to manage UE Policy Rules.

Restrictions

Applies to roamer UEs.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sameSmfForMultiplePdus - Same SMF for Multiple PDUs

Syntax

--sameSmfForMultiplePdus <list-item>

Description

Defines whether the AMF supports the functionality to select same SMF for multiple PDUs per same (S-NSSAI, DNN) as indicated by UDM.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sbiCommunicationProfileName - SBI Communication Profile Name

Syntax

--sbiCommunicationProfileName <string>

Description

The name of the sbiCommunicationProfile that is relevant in case of Roaming NFs

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

default

Name

sbiTimersProfileListName - SBI Operation Response Timers and Operation Attempts

Syntax

--sbiTimersProfileListName <string>

Description

Profile consisting of SBI Operation Response Timers and Operation Attempts.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

default

Name

sbiVersionControlProfileListName - SBI Version Control Profile List Name

Syntax

--sbiVersionControlProfileListName <string>

Description

Name of the SBI version control Profile. Refers to the 'default' profile a SBI Version Control Profile created via sbiVersionControlProfile CLI

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

default

Name

sbiVersionProducerControlProfileListName - SBI Version Producer Control Profile List Name

Syntax

--sbiVersionProducerControlProfileListName <string>

Description

Name of the SBI version producer control Profile. Refers to the 'default' profile a SBI Version Producer Control Profile created via sbiVersionProducerControlProfile CLI.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

default

Name

sendConfiguredNssaiInFirstReg - Send Configured NSSAI in First Registration

Syntax

--sendConfiguredNssaiInFirstReg <list-item>

Description

This parameter controls if AMF will always send the Configured NSSAI during Initial UE Registration, Mobility Registration from different AMF or Mobility Registration from 4G

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

smCtxtCreateSbiOriginationTimestamp - SM Context Create SBI Origination Timestamp

Syntax

--smCtxtCreateSbiOriginationTimestamp <list-item>

Description

Defines whether the AMF include the 3gpp-Sbi-Origination-Timestamp HTTP header on SM context create towards the SMF/I-SMF/V-SMF on SM context create request. This is aimed to help the SMF help handling colliding and late arriving requests.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

smfAddUeLocation - SMF Additional UE Location

Syntax

--smfAddUeLocation <list-item>

Description

AMF supports providing Additional UE Location as part of SM context create, update and delete towards the SMF, when additional location is available on AMF. The additional UE Location provides the 3GPP location information of a UE, when the corresponding PDU is related to non-3GPP access (i.e. access type for the SM context create/update/delete equals to non-3GPP access) and the AMF additionally holds valid 3GPP access location.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

smfExactDnnSelectionMatch - SMF Exact DNN Selection Match

Syntax

--smfExactDnnSelectionMatch <list-item>

Description

This parameter is applicable only for selection from locally provisioned SMFs. When the SMF Exact DNN Selection Match is set to true: If a DNN list is not provisioned with the Network Function Instances command, or the DNN list is empty, the defined NF (SMF) shall not be used if an explicit matching DNN is available on another SMF that matches all other selection

parameters. When the SMF Selection Match is set to false, the SMF is selected according to SMF selection and Default SMF selection parameters.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

smfNasMappingListName - SMF NAS Mapping List Name

Syntax

--smfNasMappingListName <string>

Description

The name of the SMF error to 5GMM NAS mapping list.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

default

Name

smfSelection - SMF Selection

Syntax

--smfSelection <list-item>

Description

The parameter specifies the method of SMF selection used by the AMF for UE requested PDU session establishment. If there are multiple SMF matches then the AMF will the capacity parameter to load balance the UE across the SMFs.

Class

Optional.

List

- LocalProvisioning - Local Provisioning - AMF will use locally provisioned NF for the selection of a NF.
- NRF - Network Repository Function - Make use of NRF to select NF.

- **NRFLocalProvisioning** - NRF with Local Provisioning Override - The AMF will check the locally provisioned selection rules and if there is a match then the AMF will select the NF matching the rule. If there is no matching then AMF will use the NRF. The selection rule by default is that AMF will select a NF supporting a provisioned criterion for the NF

Default

LocalProvisioning

Name

smfStatusSubscribe - SMF Status Subscribe

Syntax

--smfStatusSubscribe <list-item>

Description

Subscribe NRF for SMF status change event notifications.

Class

Optional.

List

- **byNfType** - All NF of a given NF type - AMF subscribes to get status of by NF type to get status of all NF registered with NRF whether AMF use them or not.
- **noSubscription** - Do not Subscribe to NF Status - AMF does not subscribe to get NF status .
- **selectedOnly** - Selected NFs only - AMF subscribes to get status of NF by instance ID for the NFs that are discovered via the NRF

Default

noSubscription

Name

smsfNasMappingListName - SMSF NAS Mapping List Name

Syntax

--smsfNasMappingListName <string>

Description

The name of the SMSF (N20) error to 5GMM NAS mapping list.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

default

Name

smsfSelection - SMSF Selection

Syntax

--smsfSelection <list-item>

Description

The parameter specifies the method of SMSF selection used by the AMF for delivering SMS over NAS. If there are multiple SMSF matches then the AMF will use the capacity parameter to load balance the UE across the SMSFs.

Class

Optional.

List

- LocalProvisioning - Local Provisioning - AMF will use locally provisioned NF for the selection of a NF.
- NRF - Network Repository Function - Make use of NRF to select NF.
- NRFLocalProvisioning - NRF with Local Provisioning Override - The AMF will check the locally provisioned selection rules and if there is a match then the AMF will select the NF matching the rule. If there is no matching then AMF will use the NRF. The selection rule by default is that AMF will select a NF supporting a provisioned criterion for the NF

Default

LocalProvisioning

Name

smsfStatusSubscribe - Subscribe to SMSF status

Syntax

--smsfStatusSubscribe <list-item>

Description

Subscribe NRF for SMSF status change event notifications

Class

Optional.

List

- byNfType - All NF of a given NF type - AMF subscribes to get status of by NF type to get status of all NF registered with NRF whether AMF use them or not.
- noSubscription - Do not Subscribe to NF Status - AMF does not subscribe to get NF status .
- selectedOnly - Selected NFs only - AMF subscribes to get status of NF by instance ID for the NFs that are discovered via the NRF

Default

noSubscription

Name

smsOverNasAllowed - SMS Over NAS Allowed

Syntax

--smsOverNasAllowed <list-item>

Description

Defines whether SMS over NAS is allowed. The Short Message is delivered via the NAS signalling from the UE to/from the AMF and further via the N20 interface towards the SMSF.

Class

Optional.

List

- ALLOWED - Allowed.
- DENIED - Denied.

Default

DENIED

Name

smsOverNasForImsCapableVoiceCentricUEs - SMS Over NAS for IMS Capable Voice Centric UEs

Syntax

--smsOverNasForImsCapableVoiceCentricUEs <list-item>

Description

Defines whether SMS over NAS is allowed for UEs indicating 'voice centric' on UE's usage setting IE on NAS Registration request, assuming that these IMS capable, voice centric UEs will get SMS service via IMS (and do not need SMS over NAS to be allowed).

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

smsPreferredAccess - SMS Preferred Access

Syntax

--smsPreferredAccess <list-item>

Description

Defines the preferred access for SMS over NAS delivery of MT SMS, when the UE is RM-REGISTERED and CM-CONNECTED via the same AMF over both 3GPP access and non-3GPP access types.

Class

Optional.

List

- amf3gppAccess - 3GPP Access - 3GPP Access type.
- amfNon3gppAccess - Non-3GPP access - Non-3GPP Access type.

Default

amfNon3gppAccess

Name

smsReattemptViaOtherAccess - SMS Reattempt via Other Access

Syntax

--smsReattemptViaOtherAccess <list-item>

Description

Defines whether the AMF reattempts the delivery of an MT SMS via the other access, when the UE is RM-REGISTERED and CM-CONNECTED via both 3GPP and non-3GPP access types and the initial SMS delivery has failed via the preferred access type.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

snssaiToDnnListName - S-NSSAI to Default DNN List Name

Syntax

--snssaiToDnnListName <string>

Description

Configure S-NSSAI to default DNN list. The AMF will use the list if the AMF failed to select a DNN for the UE requested S-NSSAI PDU connection.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

sor - Steering of Roaming

Syntax

--sor <list-item>

Description

Enables steering of roaming for roamers. The UDM can invoke steering of roaming toward the AMF when UE is in visited PLMN or in home PLMN.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

staticSmf - Static SMF

Syntax

--staticSmf <list-item>

Description

Enables the functionality to select static SMF for a PDU connection of a UE. The static SMF address is delivered from UDM inside SmfSelectionSubscriptionData per DNN. AMF primarily attempts to interpret the received SMF address as an instance id. If no match then the AMF interprets it as an FQDN.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

stdStaticSmf - Standard Static SMF

Syntax

--stdStaticSmf <list-item>

Description

Defines whether 3GPP standard based static SMF selection functionality is enabled for the UE. If static SMF list is defined for the DNN on UDM and AMF is not able to select any of the UDM provided SMF(s), the AMF will fail the PDU session establishment procedure.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

subscribedEmergencySnssai - Subscribed Emergency S-NSSAI

Syntax

--subscribedEmergencySnssai <list-item>

Description

Current parameter enables using the S-NSSAI from subscribed emergency DNN if such exists in subscription data

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

suciSchemeAllowedListName - UEs SUCI Scheme Output Allow List Name

Syntax

--suciSchemeAllowedListName <string>

Description

Identifies the name of the SUCI Scheme Output allowed list. The AMF will immediately reject the UE registration, without attempting to authenticate the UE, if the UE SUCI Scheme Output is not in the allowed list. The registration reject NAS cause code is configurable with default value set to 27.

Restrictions

This applies only to inbound roaming UEs.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

support5gSrvcc - Support 5G-SRVCC

Syntax

--support5gSrvcc <list-item>

Description

Enables the 5G-SRVCC support for handing over regular voice calls from NG-RAN/5GS to 3GPP UTRAN/CS domain for the UE.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

supportAnLevelSliceValidation - Support AN Level Slice Validation

Syntax

--supportAnLevelSliceValidation <list-item>

Description

When this feature is enabled, AMF will consider whether a Slice is actually supported by the specific 5G-AN Network Element (gNBs, N3IWF) the UE is currently connected to (i.e. not just the Tracking area level status) and act accordingly in the next UE Registration, Service Request, or Configuration Update. Note: This functionality is available only for non-roaming UEs.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

t3512Src - t3512 Source

Syntax

--t3512Src <list-item>

Description

Determines how the T3512 timer is set. this field provides the ability to override the AMF locally provisioned T3512 timer with the subscribed-periodic timer from UE subscription data in UDM.

Class

Optional.

List

- Provisioned - Use AMF locally provisioned T3512 periodic timer
- Subscribed - Use subscribed-periodic-timer value from UE subscription data in UDM

Default

Provisioned

Name

treatAsHomeSubscriber - FiveG Treat As Home Subscriber

Syntax

--treatAsHomeSubscriber <list-item>

Description

When set, treat a UE as if it were a home subscriber of the serving PLMN. Note: Intra-AMF inter-PLMN mobility is not supported among Serving PLMNs if the UEs roaming status is not uniform (i.e. the value of this parameter changes). In other words a UE must either be treated as Roamer in both Serving networks or as Home Subscriber in both Serving networks. Furthermore treating UEs belonging to a Home or a Shared PLMN (i.e. PLMN Type is Home or Shared Network) as Roamers is not supported (i.e. value should be set to true).

Restrictions

This element is not applicable when the serving and UE PLMNs are the same.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

trigUePolicyWhenNoUePolicyContainer - Trigger UE Policy When No UE Policy Container

Syntax

--trigUePolicyWhenNoUePolicyContainer <list-item>

Description

Parameter controls establishment of UE Policy Association based on received UE policy container from UE. By default, the flag shall be disabled.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

udmImmediateReport - UDM Immediate Report

Syntax

--udmImmediateReport <list-item>

Description

The AMF uses the Nudm SDM Subscribe service operation with the immediateReport indication to retrieve the UE Subscription Data and Subscribe for Data Change Notification in one combined operation.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

udmSelection - UDM Selection

Syntax

--udmSelection <list-item>

Description

The parameter specifies the method of UDM selection used by the AMF for UE. If there are multiple UDM matches then the AMF will use the capacity parameter to load balance the UE across the UDMs.

Class

Optional.

List

- LocalProvisioning - Local Provisioning - AMF will use locally provisioned NF for the selection of a NF.
- NRF - Network Repository Function - Make use of NRF to select NF.
- NRFLocalProvisioning - NRF with Local Provisioning Override - The AMF will check the locally provisioned selection rules and if there is a match then the AMF will select the NF matching the rule. If there is no matching then AMF will use the NRF. The selection rule by default is that AMF will select a NF supporting a provisioned criterion for the NF

Default

LocalProvisioning

Name

udmStatusSubscribe - UDM Status Subscribe

Syntax

--udmStatusSubscribe <list-item>

Description

Subscribe NRF for UDM status change event notifications.

Class

Optional.

List

- byNfType - All NF of a given NF type - AMF subscribes to get status of by NF type to get status of all NF registered with NRF whether AMF use them or not.
- noSubscription - Do not Subscribe to NF Status - AMF does not subscribe to get NF status .
- selectedOnly - Selected NFs only - AMF subscribes to get status of NF by instance ID for the NFs that are discovered via the NRF

Default

noSubscription

Name

udmSubscribeOptions - UDM Subscribe Options

Syntax

--udmSubscribeOptions <list-item>

Description

Options for AMF, in subscribe to notification of data change procedure, to specify the expiration of a subscription.

Class

Optional.

List

- both - ImplicitUnsubscribe And Expires - AMF sets implicitUnsubscribe to true and also includes "expires" IE/attribute. Upon expiry of "expires" time and UE Context still exists in the AMF, the AMF sends a Subscribe request to UDM (with same resource as previous subscribe). When AMF sends Deregistration request to UDM, AMF does not send Unsubscribe request to UDM. When UDM send DeregistrationNotificaton to AMF, AMF does not send Unsubscribe request to UDM.
- expires - Subscribing AMF Expires Timer - Upon expiry of "expires" time and UE Context still exists in the AMF, the AMF sends a Subscribe request (with same resources as previous subscribe). AMF sends Unsubscribe request to UDM when UE Context is removed from AMF.
- implicitUnsubscribe - Implicit Unsubscribe From UDM - When present and set to true, AMF subscription for UE is terminated at the UDM when the UE Context in the AMF no longer exists. When implicitUnsubscribe is not set to true or not present, AMF must send "expires" IE/attribute to the UDM. Note: When implicitUnsubscribe is set to true:
 - When AMF sends Deregistration request to UDM, AMF does not send Unsubscribe request to UDM.
 - When UDM send DeregistrationNotificaton to AMF, AMF does not send Unsubscribe request to UDM.

Default

expires

Name

ueAmbrDownlink - UE AMBR Downlink Bitrate

Syntax

--ueAmbrDownlink <integer>

Description

UE Aggregate Maximum Downlink Bitrate. Allowed for both home and roaming subscribers.

Restrictions

Both uplink/downlink must be 0 or both uplink/downlink must be non-zero.

Class

Optional.

Type / Range

Integer in the range of 0-1,000,000,000,000 bps. in units of Kbps

Default

0

Name

ueAmbrSrc - UE AMBR Source

Syntax`--ueAmbrSrc <list-item>`**Description**

Identifies UE AMBR source i.e. from UDM or locally provisioned.

Restrictions

Applies to home or roamer UEs.

Class

Optional.

List

- local - Local value - UE AMBR locally configured value.
- udm - udm value - Subscribed UE AMBR in UDM

Default

udm

Name

ueAmbrUplink - UE AMBR Uplink Bitrate

Syntax`--ueAmbrUplink <integer>`**Description**

UE Aggregate Maximum Uplink Bitrate. Allowed for both home and roaming subscribers.

Restrictions

Both uplink/downlink must be 0 or both uplink/downlink must be non-zero.

Class

Optional.

Type / Range

Integer in the range of 0-1,000,000,000,000 bps. in units of Kbps

Default

0

Name

ueRadioCapabilityCheck - UE Radio Capability Check

Syntax`--ueRadioCapabilityCheck <list-item>`

Description

This parameter controls whether the AMF uses NGAP UE Radio Check procedure and the result of the procedure to define whether Voice over PS Sessions shall be supported for the UE on this PLMN.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

unsupportedSliceHandlingInReg - Unsupported Slice Handling In Registration

Syntax

--unsupportedSliceHandlingInReg <list-item>

Description

When the parameter is set to rejectReg, AMF sends Registration Reject when AMF cannot serve all the UE Candidate slices and cannot discover the Target AMF for reallocating the UE Registration. Otherwise, when the parameter is set to rejectSlicePlmn, or rejectSliceRA, when at least one or more slices are allowed, the AMF sends Registration Accept and includes the unsupported slices in the Rejected NSSAI with configurable cause as SNSSAI not available in the current PLMN or SNPN, or SNSSAI not available in the current registration area respectively. This parameter is not applicable for Roaming subscribers.

Class

Optional.

List

- rejectReg - Reject Registration - When the parameter is set to RejectRegistration, AMF sends Registration Reject when AMF cannot serve all the UE Candidate slices and cannot discover the Target AMF for reallocating the UE Registration
- rejectSlicePlmn - Reject Slice Plmn - when the parameter is set to RejectSlicePlmn, when at least one or more slices are allowed, the AMF sends Registration Accept and includes the unsupported slices in the Rejected NSSAI with configurable cause as SNSSAI not available in the current PLMN or SNPN
- rejectSliceRA - Reject Slice Registration Area - when the parameter is set to RejectSliceRA, when at least one or more slices are allowed, the AMF sends Registration Accept and includes the unsupported slices in the Rejected NSSAI with configurable cause as SNSSAI not available in the current registration area

Default

rejectReg

Recommended

It is recommended to use the value "rejectSlicePlmn" assuming the UE Subscription errors are more probable than the AMF provisioning errors.

Name

useAllowedNssaiForPcfSelection - Use Allowed NSSAI for PCF Selection

Syntax

--useAllowedNssaiForPcfSelection <list-item>

Description

This parameter defines whether the AMF adds the UE's allowed NSSAI as a NF_Discovery GET Query Parameter for PCF selection. When the value is true, the Allowed NSSAI is added in the snnsais query Parameter for PCF selection. When the value is false the AMF does not include snnsais as a query Parameter for PCF selection. Default Value is false.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

useGpsiForPcfSelection - Use GPSI for PCF Selection

Syntax

--useGpsiForPcfSelection <list-item>

Description

This parameter defines whether the AMF adds the UE's GPSI as a query Parameter when discovering PCF from NRF, or when searching for matching PCF in the locally provisioned NFs list. When the value is true, the GPSI is used for PCF selection. When the value is false, the GPSI is not used for PCF selection. Default Value is false.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

useN26SvcParam5gsToEpsHo - Use N26 Service Parameter in 5GS to EPS Handover

Syntax

--useN26SvcParam5gsToEpsHo <list-item>

Description

Current parameter enables using N26 service parameter as part of DNS query for target MME selection during 5GS to EPS handover

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

useNssfSelectionServiceForSmf - Use NSSF Selection Service for Slice Info for SMF

Syntax

--useNssfSelectionServiceForSmf <list-item>

Description

Specifies options to use NSSF Selection Service for Slice Info during PDU Session Establishment.

Restrictions

This applies to roamer UEs.

Class

Optional.

List

- both - Local Break Out and Home Routed - AMF invokes NSSF Service for Slice Information during PDU Session Establishment when roaming subscriber is allowed PDU Session Service for Local Break Out and Home Routed.
- hr - Home Routed - AMF invokes NSSF Service for Slice Information during PDU Session Establishment when roaming subscriber is allowed PDU Session Service for Home Routed.
- lbo - Local Break OUT - AMF invokes NSSF Service for Slice Information during PDU Session Establishment when roaming subscriber is allowed PDU Session Service for Local Break Out.

- none - AMF doesn't invoke NSSF Service for Slice Information during PDU Session Establishment when roaming.

Default

none

Name

useTaiForSmfSelection - Use TAI for SMF Selection

Syntax

--useTaiForSmfSelection <list-item>

Description

Controls usage of TAI as a SMF selection criterion: - 'No' (default): TAI IE is not added in NRF Discovery Query for SMF selection and not used as a criterion when searching for SMF in local or external NRF Cache. - 'Yes' TAI IE is added in NRF Discovery Query for SMF selection and used as a criterion when searching for SMF from local or external NRF Cache.

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

vsmfChange - VSMF Change

Syntax

--vsmfChange <list-item>

Description

Enables VSMF change including selection of a new VSMF when new UE location is not supported by current VSMF.

Restrictions

Requires vsmfChange gparm to be set to true to enable this parameter at the UE PLMN Services level.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

vsmfSelectionFailureMaintain - VSMF Selection Failure Maintain

Syntax

--vsmfSelectionFailureMaintain <list-item>

Description

Defines whether the AMF maintains PDU session when V-SMF change is enabled, and AMF detects the need to select a new V-SMF, but no V-SMF supporting the current UE service area is available. The parameter is similarly used when V-SMF needs to be changed but H-SMF does not support DTSSA. If set to false, AMF releases the PDU session (i.e. triggers Nsmf PDUSession UpdateSMContext with release indication towards the old SMF or V-SMF) if no new V-SMF is available for the current service area of the UE or H-SMF does not support DTSSA

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

vsmfSupportInd - V-SMF Support Indication to NRF for SMF

Syntax

--vsmfSupportInd <list-item>

Description

AMF includes V-SMF Indication Support in R15 API version when discovering an SMF in the V-PLMN for home routed traffic.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.2 cmm amfUePlmnServicesSecurity

amf UE Plmn Services Security. Syntax, description, parameters, and roles for **cmm amfUePlmnServicesSecurity** command.

Syntax

Description

This command is used to administer UE PLMN Services Security for AMF and includes parameters for defining the authentication interaction for AMF procedure(s). For UE Plmns of plmnType "Multiple Home Network", the security parameters associated with the Home Network amfUePlmnServicesName will be used.

Parameters

Name

amfUePlmnServicesName - UE PLMN services name

Syntax

--amfUePlmnServicesName <string>

Description

Name of a UE PLMN services record

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

amfProcedureName - AMF procedure name

Syntax

--amfProcedureName <list-item>

Description

Name of AMF procedure

Class

Required key.

List

- initialRegistration - Initial Registration - UE Initial Registration
- mobilityRegistrationUpdate - Mobility Registration Update - UE Mobility Registration Update
- periodicRegistration - Mobility Registration Update with type "Periodic Registration" - UE Periodic Mobility Registration
- deregistration - De-registration - UE Initiated De-registration
- serviceRequest - Service Request Procedure - Service Request Procedure Note: For GUTI Reallocation this represents mobile originated service request
- iratChange - IRAT Change
- amfChange - AMF Change
- nlServiceRequest - NI Service Request - Network-initiated Service Request Procedure

Name

authInteraction - Authentication Interaction

Syntax

--authInteraction <integer>

Description

The authentication repetition rate defines how many times a particular event occurs before an optional authentication needs to be performed. If the value is 0, no optional authentication is needed. If the value is N, the authentication is done in connection with every Nth event. The parameter values range from 0 to 255 except for inter-AMF and IRAT relocations the range is 0-1.

Restrictions

Default values: initialRegistration = 1 mobilityRegistrationUpdate = 1 periodicRegistration = 0 serviceRequest = 0 iratChange = 0 (range is 0-1) amfChange = 0 (range is 0-1) deregistration = 0

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

gutiReallocation - 5G-GUTI Re-allocation

Syntax

--gutiReallocation <integer>

Description

The GUTI Re-allocation defines how many times a particular event occurs before a GUTI Re-allocation is performed. If the value is 0, no GUTI Re-allocation is performed. If the value is N, the GUTI Re-allocation is performed with every nth event. GUTI Re-allocation applies to Initial UE Registration, Mobility Registration Update, Periodic Mobility Registration and Service Request events. For inter-AMF and I-RAT relocation GUTI re-allocation is mandatory i.e. value can only be "1". For Deregistration value can only be "0". Default value can be different for each event

Restrictions

GUTI Re-allocation is restricted to Initial UE Registration, Mobility Registration Update and Service Request events. Default values: initialRegistration = 5 mobilityRegistrationUpdate = 5 periodicRegistration = 0 serviceRequest = 0 iratChange = 1 (range is 1) amfChange = 1 (range is 1) deregistration = 0 (range is 0)

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

horizDerivatAtSourceAmf - Horizontal Derivation Required At Source AMF

Syntax

--horizDerivatAtSourceAmf <list-item>

Description

horizontal derivation is required at source AMF

Restrictions

only apply for the inter AMFs

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

horizDerivatAtTargetAmf - Horizontal Derivation Kamf At Target AMF

Syntax

--horizDerivatAtTargetAmf <list-item>

Description

if target AMF decide to use horizontal derivation Kamf received from source AMF

Restrictions

only apply for inter AMF

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

4.3 cmm amfUePriorityProfile

AMF UE Priority Profile. Syntax, description, parameters, and roles for cmm amfUePriorityProfile command.

Syntax

Description

Manage UE priority profile; control settings to allow priority services for the UE. AMF supports up to 64 UE priority profiles.

Parameters

Name

name - Name

Syntax

--name <string>

Description

The name of the UE Priority Profile.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

gtpMpN26Mcs - GTP Message Priority Over N26 MCS

Syntax

--gtpMpN26Mcs <list-item>

Description

Enables AMF to use GTP Message Priority value 0 over N26 interface for MCS priority UEs.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

gtpMpN26Mps - GTP Message Priority Over N26 MPS

Syntax

--gtpMpN26Mps <list-item>

Description

Enables AMF to use GTP Message Priority value 0 over N26 interface for MPS priority UEs.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

hpaInitialPriorityMpsMcs - High Priority Access Initial Priority MPS/MCS

Syntax

--hpaInitialPriorityMpsMcs <list-item>

Description

Defines whether the AMF provides initial priority for UEs indicating RRC establishment cause highPriorityAccess. The UE needs to have MPS or MCS subscribed, or have PDU with priority mapped ARP in order to be continued to be treated with priority.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

hpaInitialSbiMessagePriority - HPA Initial SBI Message Priority

Syntax

--hpaInitialSbiMessagePriority <integer>

Description

Defines the SBI Message Priority (SMP) that is used for UEs indicating RRC establishment cause highPriorityAccess. This SMP value is used only when enabled by the separate parameter High Priority Access Initial Priority as part of the UE priority profile. The AMF uses this SMP initially on AMF triggered request messages towards other NFs, prior to the AMF being able to validate the priority of the UE.

Class

Optional.

Type / Range

Integer in the range of 0 to 23

Default

23

Name

mcs - Mission Critical Services

Syntax`--mcs <list-item>`**Description**

Enables the use of Mission Critical Services (MCS) for the UE.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

mcsSbiMessagePriority - MCS SBI Message Priority

Syntax`--mcsSbiMessagePriority <integer>`**Description**

Defines the SBI Message Priority (SMP) that is used for MCS-detected UEs on AMF triggered request messages towards other NFs.

Class

Optional.

Type / Range

Integer in the range of 0 to 23

Default

23

Name

mps - Multimedia Priority Services

Syntax`--mps <list-item>`**Description**

Enables the use of Multimedia Priority Services (MPS) for the UE.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

mpsSbiMessagePriority - MPS SBI Message Priority

Syntax

--mpsSbiMessagePriority <integer>

Description

Defines the SBI Message Priority (SMP) that is used for MPS-detected UEs on AMF triggered request messages towards other NFs. AMF call processing treats any value defined between 24..31 according to the value of 23. This is to ensure all MPS calls will have higher priority than regular calls.

Class

Optional.

Type / Range

Integer in the range of 0 to 31

Default

23

Name

n1n2RejPrioIncludeHighestArp - N1 N2 Rejection due to Priority Include Highest ARP

Syntax

--n1n2RejPrioIncludeHighestArp <list-item>

Description

Defines whether AMF, when rejecting N1N2 Message Transfer request due to higher priority request ongoing, includes 'highestPrioArp' (Highest Priority ARP) into the reject message.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sendPagingForSubsequentPrioN1N2 - Send paging for subsequent Priority N1N2 or EnableUeReachability request

Syntax

--sendPagingForSubsequentPrioN1N2 <list-item>

Description

Defines whether AMF sends Paging message with suitable Paging Priority, when AMF receives subsequent N1N2 Message Transfer request including priority defined ARP or Enable UE Reachability request with priority defined SMP. When set to false, AMF does not send any paging message when subsequent priority request(s) come before paging timer for the already started paging expires.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Create**

secadmin

Show

sysadmin, manage, secadmin

Modify

secadmin

Delete

secadmin

List

sysadmin, manage, secadmin

4.4 cmm amfUINasUnableToForwardProfile

AMF Uplink NAS Unable to Forward Profile. Syntax, description, parameters, and roles for cmm amfUINasUnableToForwardProfile command.

Syntax

Description

Defines parameters when AMF is unable to forward an Uplink NAS message to the target NF. The maximum number of profiles supported is 128. A default profile, named 'default', is provided.

Parameters

Name

name - Name

Syntax

--name <string>

Description

Defines the name of the AMF Uplink NAS Unable to Forward Profile.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

smAmfLockedLinkLocked - SM AMF Locked Link Locked

Syntax

--smAmfLockedLinkLocked <integer>

Description

Defines NAS 5GMM cause that the AMF uses when the NAS: Uplink NAS carries an SM type Payload Container and the message cannot be forwarded to SMF due to AMF being locked or the S11 link is locked. Defining 0 stands for no cause code included on AMF triggered NAS: DL NAS message.

Class

Optional.

Type / Range

5GMM Cause Code. 0 stands for no cause to be provided

Default

90

Name

smDnnNotSubscribed - SM DNN Not Subscribed

Syntax

--smDnnNotSubscribed <integer>

Description

Defines NAS 5GMM cause that the AMF uses when the NAS: Uplink NAS carries an SM type Payload Container and the message cannot be forwarded to SMF due to DNN not subscribed in the slice. Defining 0 stands for no cause code included on AMF triggered NAS: DL NAS message.

Class

Optional.

Type / Range

5GMM Cause Code. 0 stands for no cause to be provided

Default

91

Name

smEmergencyNoNid - SM Emergency No NID

Syntax

--smEmergencyNoNid <integer>

Description

Defines NAS 5GMM cause that the AMF uses when the NAS: Uplink NAS carries an SM type Payload Container and the message cannot be forwarded to SMF due to NID not present for SNPN. Defining 0 stands for no cause code included on AMF triggered NAS: DL NAS message.

Class

Optional.

Type / Range

5GMM Cause Code. 0 stands for no cause to be provided

Default

90

Name

smEmergencyRegistered - SM Emergency Registered

Syntax

--smEmergencyRegistered <integer>

Description

Defines NAS 5GMM cause that the AMF uses when the NAS: Uplink NAS carries an SM type Payload Container and the message cannot be forwarded to SMF due to the Request Type not concerning an emergency session and the UE being emergency registered. Defining 0 stands for no cause code included on AMF triggered NAS: DL NAS message.

Class

Optional.

Type / Range

5GMM Cause Code. 0 stands for no cause to be provided

Default

90

Name

smInternalSwError - SM Internal SW Error

Syntax`--smInternalSwError <integer>`**Description**

Defines NAS 5GMM cause that the AMF uses when the NAS: Uplink NAS carries an SM type Payload Container and the message cannot be forwarded to SMF due to internal SW error. Defining 0 stands for no cause code included on AMF triggered NAS: DL NAS message.

Class

Optional.

Type / Range

5GMM Cause Code. 0 stands for no cause to be provided

Default

90

Name

smInvalidRequestTypeNoSmf - SM Invalid Request Type No SMF

Syntax`--smInvalidRequestTypeNoSmf <integer>`**Description**

Defines NAS 5GMM cause that the AMF uses when the NAS: Uplink NAS carries an SM type Payload Container and the message cannot be forwarded to SMF due to Request Type is not included or the Request Type value is unknown to AMF and the AMF does not have a PDU session routing context for the PDU session ID and the UE. Defining 0 stands for no cause code included on AMF triggered NAS: DL NAS message.

Class

Optional.

Type / Range

5GMM Cause Code. 0 stands for no cause to be provided

Default

90

Name

smPduSessionIdInvalidMissing - SM PDU Session Id Invalid or Missing

Syntax`--smPduSessionIdInvalidMissing <integer>`**Description**

Defines NAS 5GMM cause that the AMF uses when the NAS: Uplink NAS carries an SM type Payload Container and the message cannot be forwarded to SMF due to the PDU Session Id being invalid or missing. Defining 0 stands for no cause code included on AMF triggered NAS: DL NAS message.

Class

Optional.

Type / Range

5GMM Cause Code. 0 stands for no cause to be provided

Default

96

Name

smProcedureCollision - SM Procedure Collision

Syntax

--smProcedureCollision <integer>

Description

Defines NAS 5GMM cause that the AMF uses when the NAS: Uplink NAS carries an SM type Payload Container and the message cannot be forwarded to SMF due to procedure collision. Defining 0 stands for no cause code included on AMF triggered NAS: DL NAS message.

Class

Optional.

Type / Range

5GMM Cause Code. 0 stands for no cause to be provided

Default

90

Name

smRestrictedServiceArea - SM Restricted Service Area

Syntax

--smRestrictedServiceArea <integer>

Description

Defines NAS 5GMM cause that the AMF uses when the NAS: Uplink NAS carries an SM type Payload Container and the message cannot be forwarded to SMF due to restricted service area. Defining 0 stands for no cause code included on AMF triggered NAS: DL NAS message.

Class

Optional.

Type / Range

5GMM Cause Code. 0 stands for no cause to be provided

Default

28

Name

smSmfSelectionFails - SM SMF Selection Fails

Syntax`--smSmfSelectionFails <integer>`**Description**

Defines NAS 5GMM cause that the AMF uses when the NAS: Uplink NAS carries an SM type Payload Container and the message cannot be forwarded to SMF due to SMF selection failing. Defining 0 stands for no cause code included on AMF triggered NAS: DL NAS message.

Class

Optional.

Type / Range

5GMM Cause Code. 0 stands for no cause to be provided

Default

90

Name

smSnssaiNotAllowed - SM S-NSSAI Not Allowed

Syntax`--smSnssaiNotAllowed <integer>`**Description**

Defines NAS 5GMM cause that the AMF uses when the NAS: Uplink NAS carries an SM type Payload Container, and the message cannot be forwarded due to S-NSSAI not allowed. Defining 0 stands for no cause code included on AMF triggered NAS: DL NAS message.

Class

Optional.

Type / Range

5GMM Cause Code. 0 stands for no cause to be provided

Default

91

Name

smTimeout - SM Timeout

Syntax`--smTimeout <integer>`**Description**

Defines NAS 5GMM cause that the AMF uses when the NAS: Uplink NAS carries an SM type Payload Container and the message cannot be forwarded to SMF due to SMF timeout. Defining 0 stands for no cause code included on AMF triggered NAS: DL NAS message.

Class

Optional.

Type / Range

5GMM Cause Code. 0 stands for no cause to be provided

Default

90

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.5 cmm amfVoiceProfile

AMF Voice Profile. Syntax, description, parameters, and roles for cmm amfVoiceProfile command.

Syntax

Description

Controls 5GS voice service related parameters and AMF actions related to voice, emergency and emergency fallback services. The maximum number of profiles supported is 128. A default profile, named 'default', is provided.

Parameters

Name

name - AMF Voice Profile Name

Syntax

--name <string>

Description

The name of the profile for voice related handling.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

acceptVoiceNoUeCapCheck - Accept Voice No UE Capability Check

Syntax

--acceptVoiceNoUeCapCheck <list-item>

Description

Defines whether the AMF skips UE radio capability check during IMS voice over PS evaluation, when the UE radio capability check is provisioned to be run as part of the IMS voice over PS evaluation, but the UE radio capability check cannot be run due to Initial Context Setup Failure or timeout. When this parameter is set to true, the AMF then also skips the UE radio capability check on the IMS voice over PS evaluation. I.e. If the only reason for the AMF to not indicate IMS voice over PS as supported was the UE radio capability check not being possible to be run, the AMF indicates IMS voice over PS as supported for the UE. When this parameter is set to false, AMF considers the lack of the possibility to run the UE radio capability check (due to Initial Context Setup Failure) to cause IMS voice over PS as not supported for the UE, when provisioned to be run but cannot be run due to Initial context setup failure or timeout.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

emcSuppressNoVonnr - EMC Suppress No VoNR

Syntax

--emcSuppressNoVonnr <list-item>

Description

The AMF suppresses EMC support i.e. indicates 'Emergency services not supported' on 5GS Network Feature Support IE on NAS Registration Accept for the UE when UE radio capabilities indicate that the UE does not support VoNR.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

emfSuppressS1Disabled - EMF Suppress S1 Disabled

Syntax

--emfSuppressS1Disabled <list-item>

Description

Defines whether the AMF suppresses Emergency Services Fallback (EMF) when the UE has disabled S1 mode.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

emfSuppressVonn - EMF Suppress VoNR

Syntax

--emfSuppressVonn <list-item>

Description

Controls whether the AMF suppresses EMF when the UE supports VoNR. When set to true, and the EMC suppression (EMC Suppress No VoNR) is set to true for the UE (on the AMF Voice Profile on 5G UE PLMN Services), and the UE radio capabilities indicate that the UE supports VoNR, the AMF suppresses EMF support i.e. indicates 'Emergency services fallback not supported' on 5GS Network Feature Support IE on NAS Registration Accept for the UE. When set to false, or EMC suppression (EMC Suppress No VoNR) is set to false for the UE (on the AMF Voice Profile on 5G UE PLMN Services), or the UE does not support VoNR according to the UE radio capabilities, the AMF sets the EMF according to other EMF settings. The other settings for the EMF include -Emergency Fallback inside the AMF Emergency Profile from the AMF PLMN -Emergency Fallback under the AMF Tracking Area -EMF Support inside the AMF Voice Profile from the 5G UE PLMN Services

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

imsVopsNonVonrUeVonrTa - IMS Voice Over PS Non-VoNR UE on VoNR TA

Syntax

--imsVopsNonVonrUeVonrTa <list-item>

Description

Defines whether to set IMS voice over PS as not supported for UEs not supporting VoNR according to UE radio capabilities when camping on a VoNR only TA.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

imsVopsS1Disabled - IMS VoPS S1 Disabled

Syntax

--imsVopsS1Disabled <list-item>

Description

Enables AMF to adjust the IMS voice over PS sessions support when the UE disabled S1 mode. When IMS voice over PS is enabled by both VoNR and EPS FB for the UE and on the TA, but the UE has disabled the S1 mode support, the AMF checks the UE VoNR support from the UE radio capabilities. If the UE does not support VoNR (or the UE VoNR support is unknown), the AMF indicates IMS VoPS not supported for the UE.

Class

Optional.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Name

rejectionCause - Rejection Cause

Syntax

--rejectionCause <integer>

Description

Indicates the 5GMM cause used on Registration Reject.

Class

Optional.

Type / Range

5GMM Cause Code.

Default

15

Name

rejectNoVoice - Reject No Voice

Syntax

--rejectNoVoice <list-item>

Description

Enables the functionality to reject 5GS Registration for any UE indicating 'voice centric' UE's Usage setting and not getting voice service ('IMS voice over PS session supported over 3GPP access') on the current Tracking Area.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

rejectOnVoNrTa - Reject On VoNR Tracking Area

Syntax

--rejectOnVoNrTa <list-item>

Description

Enables the functionality to investigate UE Radio Capability IE from NGAP UE Radio Capability Info Indication message in order to find out the UE support for VoNR. Further, when -this control parameter is set to 'true', -the UE does not support VoNR according to UE radio capabilities, -the current TA is VoNR only TA, -the UE indicates 'voice centric' in UE's usage setting, the AMF rejects 5GS Registration for such UE. The AMF uses provisioned 5GMM cause on the rejection.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.6 cmm apnAdministration

APN Administration. Syntax, description, parameters, and roles for cmm apnAdministration command.

Syntax**Description**

APN Administration. Parameters to administer the PDNs towards the APN(s) defined per serving PLMN. Maximum 10 PLMNs can simultaneously have APN administration list provisioned. Parameter backOffTimerMin value must be specified smaller than parameter backOffTimerMax value. Parameter t3396Min value must be specified smaller than parameter t3396Max value. When the functionality is

currently not active (APN Administration Statistics Actual Running Level indicates Not Running, Stopped During Deactivation, Stopped During Rejection, Timed Out During Deactivation, Timed Out During Rejection or the initial state when APN Administration Statistics is not present), the Action Level is not allowed to be set to Not Running. When the functionality is currently active (APN Administration Statistics Actual Running Level indicates Deactivating and Rejecting, Deactivation Aborted Rejecting Continues, Deactivation Successful Rejecting or Rejecting), the operation must first be stopped (i.e. set to Not Running) before changing to same or different action level.

Parameters

Name

actionLevel - Action Level

Syntax

```
--actionLevel <list-item>
```

Description

The action level the MME is operating on. The process to reject and/or deactivate the PDNs is started and stopped by operating this parameter.

Class

Required key.

List

- deactivatingRejecting - Deactivating and Rejecting - MME both rejects incoming PDN connectivity requests and also deactivates previously existing PDN connections towards the defined APN(s).
- notRunning - Not Running - APN administration process is currently not running.
- rejecting - MME rejects incoming PDN connectivity requests towards the defined APN(s).

Default

notRunning

Name

backOffTimerMax - Back-off Timer Max Value

Syntax

```
--backOffTimerMax <integer>
```

Description

Defines the maximum of the back-off timer value in seconds. The back-off timer is used on PDN Connectivity Reject messages. The MME generates the back-off timer value for each UE based on the provisioned minimum and maximum values.

Class

Optional.

Type / Range

Integer in the range of 1 to 600 in units of Second

Default

60

Name

backOffTimerMin - Back-Off Timer Min Value

Syntax

--backOffTimerMin <integer>

Description

Defines the minimum of the back-off timer value in seconds. The back-off timer is used on PDN Connectivity Reject messages. The MME generates the back-off timer value for each UE based on the provisioned minimum and maximum values.

Class

Optional.

Type / Range

Integer in the range of 0 to 599 in units of Second

Default

15

Name

deactivationCause - Deactivation Cause

Syntax

--deactivationCause <integer>

Description

ESM cause sent on Deactivate EPS Bearer Context Request, when the MME deactivates a previously existing default bearer towards the defined APN due to APN administration functionality.

Class

Optional.

Type / Range

Integer in the range of 8, 26-39, 41-47, 49-60, 65-66, 81, 95-101, 111-113

Default

26

Name

deactivationRate - Deactivation Rate

Syntax

--deactivationRate <integer>

Description

Defines the rate (in UEs/second) the MME performs the PDN deactivations/detaches when the MME is provisioned to deactivate previously existing PDN connections towards the defined APN(s). Must be a multiple of 10.

Class

Optional.

Type / Range

Integer in the range of 10 to 100

Default

100

Name

includeBackOffTimer - Include Back-Off Timer

Syntax

--includeBackOffTimer <list-item>

Description

Defines whether back-off timer is included when PDN rejection or deactivation happens due to APN administration. When set to true, the back-off timer IE is included in PDN Connectivity Reject messages (when configured ESM cause code is NOT one of the following: 50 'PDN type IPv4 only allowed', 51 'PDN type IPv6 only allowed', 54 'PDN connection does not exist', 57 'PDN type IPv4v6 only allowed', 58 'PDN type non IP only allowed', 65 'maximum number of EPS bearers reached'). When set to false, the back-off timer is not included in the above cases.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

includeT3396 - Include T3396

Syntax

--includeT3396 <list-item>

Description

Defines whether T3396 is included when PDN deactivation happens due to APN administration. When set to true, the T3396 IE is included in Deactivate EPS Bearer Context Request messages (when configured ESM cause code is 26 'insufficient resources'). When set to false, the T3396 is not included in the above case.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

maxDuration - Max Duration

Syntax

--maxDuration <integer>

Description

Defines the maximum time of the APN disabling state in seconds.

Class

Optional.

Type / Range

Integer in the range of 900 to 54,000 in units of Second

Default

5400

Name

pdnConnectivityRejectCause - PDN Connectivity Reject Cause

Syntax

--pdnConnectivityRejectCause <integer>

Description

ESM cause sent on PDN Connectivity Reject, when the MME rejects the PDN establishment during an Attach or standalone PDN connectivity request procedure due to APN administration functionality.

Class

Optional.

Type / Range

Integer in the range of 8, 26-39, 41-47, 49-60, 65-66, 81, 95-101, 111-113

Default

26

Name

t3396Max - T3396 Max Value

Syntax

--t3396Max <integer>

Description

Defines the maximum of the timer T3396 value in seconds. The timer T3396 is used on Deactivate EPS Bearer Context Request messages. The MME generates the timer T3396 value for each UE based on the provisioned minimum and maximum values.

Class

Optional.

Type / Range

Integer in the range of 1 to 600 in units of Second

Default

60

Name

t3396Min - T3396 Min Value

Syntax

--t3396Min <integer>

Description

Defines the minimum of the timer T3396 value in seconds. The timer T3396 is used on Deactivate EPS Bearer Context Request messages. The MME generates the timer T3396 value for each UE based on the provisioned minimum and maximum values.

Class

Optional.

Type / Range

Integer in the range of 0 to 599 in units of Second

Default

15

Roles**Modify**

sysadmin, manage

4.7 cmm apnAdministrationStats

APN Administration Statistics. Syntax, description, parameters, and roles for cmm apnAdministrationStats command.

Syntax

Description

Statistics of the APN Administration process.

Parameters

Name

startDateTime - Start Date and Time

Syntax

--startDateTime <string>

Description

Records the date and time at which the command was entered.

Class

Required key.

Type / Range

Date and Time Digit string of length 0 to 20 with the allowed pattern of '20[0-9][0-9]/[0-1][0-9]/[0-3][0-9] [0-2][0-9]:[0-5][0-9]:[0-5][0-9]'

Name

actualRunningLevel - Actual Running Level

Syntax

--actualRunningLevel <list-item>

Description

The action level the MME is/was last operating on.

Class

Read only.

List

- deactivatingRejecting - Deactivating and Rejecting - MME both deactivates previously existing PDN connections and rejects incoming PDN connectivity requests towards the defined APN(s).
- deactivationAbortedRejectingContinues - Deactivation Aborted Rejecting Continues - The APN administration deactivation process was aborted due to e.g. other scanning loop started during the deactivation process. MME action level is downgraded to incoming PDN connectivity request rejection.
- deactivationSuccessfulRejecting - Deactivation Successful Rejecting - Deactivation process was successfully finished for all impacted UEs. MME action level is downgraded to incoming PDN connectivity request rejection.
- notRunning - Not Running - APN administration is currently not running.
- rejecting - MME rejects incoming PDN connectivity requests.

- stoppedDuringDeactivation - Stopped During Deactivation - MME APN administration process was stopped during PDN deactivation process.
- stoppedDuringRejection - Stopped During Rejection - MME APN administration process was stopped during PDN rejection.
- timedOutDuringDeactivation - Timed Out During Deactivation - MME timed out (max duration reached) during deactivation process before handling all impacted UEs.
- timedOutDuringRejection - Timed Out During Rejection - MME timed out (max duration reached) during PDN connectivity rejection.

Name

backOffTimerMax - Back-off Timer Max Value

Syntax

--backOffTimerMax <integer>

Description

Defines the maximum of the back-off timer value in seconds. The back-off timer is used on PDN Connectivity Reject messages. The MME generates the back-off timer value for each UE based on the provisioned minimum and maximum values.

Class

Read only.

Type / Range

Integer in the range of 1 to 600

Name

backOffTimerMin - Back-Off Timer Min Value

Syntax

--backOffTimerMin <integer>

Description

Defines the minimum of the back-off timer value in seconds. The back-off timer is used on PDN Connectivity Reject messages. The MME generates the back-off timer value for each UE based on the provisioned minimum and maximum values.

Class

Read only.

Type / Range

Integer in the range of 0 to 599

Name

deactivationCause - Deactivation Cause

Syntax

--deactivationCause <integer>

Description

ESM cause sent on Deactivate EPS Bearer Context Request, when the MME deactivates a previously existing default bearer towards the defined APN due to APN administration functionality.

Class

Read only.

Type / Range

Integer in the range of 8, 26-39, 41-47, 49-60, 65-66, 81, 95-101, 111-113

Name

deactivationRate - Deactivation Rate

Syntax

--deactivationRate <integer>

Description

Defines the rate (in UEs/second) the MME performs the PDN deactivations/detaches when the MME is provisioned to deactivate previously existing PDN connections towards the defined APN(s).

Class

Read only.

Type / Range

Integer in the range of 10 to 100

Name

includeBackOffTimer - Include Back-Off Timer

Syntax

--includeBackOffTimer <list-item>

Description

Defines whether back-off timer is included when PDN rejection happens due to APN administration. When set to true, the back-off timer IE is included in PDN Connectivity Reject messages (when configured ESM cause code is NOT one of the following: 50 'PDN type IPv4 only allowed', 51 'PDN type IPv6 only allowed', 54 'PDN connection does not exist', 57 'PDN type IPv4v6 only allowed', 58 'PDN type non IP only allowed', 65 'maximum number of EPS bearers reached'). When set to false, the back-off timer is not included in the above cases.

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Name

includeT3396 - Include T3396

Syntax

--includeT3396 <list-item>

Description

Defines whether T3396 is included when PDN deactivation happens due to APN administration. When set to true, the T3396 IE is included in Deactivate EPS Bearer Context Request messages (when configured ESM cause code is 26 'insufficient resources'). When set to false, the T3396 is not included in the above case.

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Name

maxDuration - Max Duration

Syntax

--maxDuration <integer>

Description

Defines the maximum time of the APN disabling state in seconds.

Class

Read only.

Type / Range

Integer in the range of 900 to 54,000 in units of Second

Name

pdnConnectivityRejectCause - PDN Connectivity Reject Cause

Syntax

--pdnConnectivityRejectCause <integer>

Description

ESM cause sent on PDN Connectivity Reject, when the MME rejects the PDN establishment during an Attach or standalone PDN connectivity request procedure due to APN administration functionality.

Class

Read only.

Type / Range

Integer in the range of 8, 26-39, 41-47, 49-60, 65-66, 81, 95-101, 111-113

Name

requestedActionLevel - Requested Action Level

Syntax

--requestedActionLevel <list-item>

Description

The action level the MME is/was requested to operate on.

Class

Read only.

List

- deactivatingRejecting - Deactivating and Rejecting - MME both rejects incoming PDN connectivity requests and also deactivates previously existing PDN connections towards the defined APN(s).
- notRunning - Not Running - APN administration process is currently not running.
- rejecting - MME rejects incoming PDN connectivity requests towards the defined APN(s).

Name

t3396Max - T3396 Max Value

Syntax

--t3396Max <integer>

Description

Defines the maximum of the timer T3396 value in seconds. The timer T3396 is used on Deactivate EPS Bearer Context Request messages. The MME generates the timer T3396 value for each UE based on the provisioned minimum and maximum values.

Class

Read only.

Type / Range

Integer in the range of 1 to 600

Name

t3396Min - T3396 Min Value

Syntax

--t3396Min <integer>

Description

Defines the minimum of the timer T3396 value in seconds. The timer T3396 is used on Deactivate EPS Bearer Context Request messages. The MME generates the timer T3396 value for each UE based on the provisioned minimum and maximum values.

Class

Read only.

Type / Range

Integer in the range of 0 to 599 in units of Second

Roles

List

sysadmin, manage, monitor

4.8 cmm apnAdminList

APN Administration List. Syntax, description, parameters, and roles for cmm apnAdminList command.

Syntax

Description

APN (Access Point Name) Network Identifier (NI) list for administrative actions. Max 10 APN NIs may be included on a list. Max 10 APN NI lists may be defined.

Parameters

Name

name - APN Administration List

Syntax

--name <string>

Description

The name of the APN Administration List.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.9 cmm apnAdminListNi

APN Administration List Network Identifier. Syntax, description, parameters, and roles for cmm apnAdminListNi command.

Syntax

Description

Administer the APN Network Identifiers in the APN Administration list.

Parameters

Name

apnAdminListName - APN Administration List

Syntax

--apnAdminListName <string>

Description

The name of the APN Administration List.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

apnNi - APN Network Identifier

Syntax

--apnNi <string>

Description

The Network Identifier (NI) of the Access Point Name (APN). Note: 'sos' APN is not allowed to be provisioned to ensure emergency service is not impacted.

Class

Required key.

Type / Range

Digit string of length 1 to 62 with the allowed pattern of [0-9a-zA-Z\._]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.10 cmm apnConversion

apnConversion. Syntax, description, parameters, and roles for cmm apnConversion command.

Syntax

Description

Configure, list and show APN conversion.

Parameters

Name

apnConversionId - ApnConversionId

Syntax

--apnConversionId <integer>

Description

Primary key of APN_CONVERSION table, numeric id for the table.

Class

Required key.

Type / Range

Integer in the range of 0 to 255

Name

plmnParamId - PLMN parameter set name

Syntax

--plmnParamId <string>

Description

PLMN parameter set name where this APN conversion belongs to. (corresponds to plmnparamid of plmnparam table. you must create that plmnparam first).

Class

Required key.

Type / Range

Digit string of length 1 to 15 with the allowed pattern of [0-9a-zA-Z\._:]*

Name

convertedApnIndex - Converted APN index

Syntax

--convertedApnIndex <integer>

Description

This parameter defines the index for APN name used as converted APN in APN conversion.

Class

Required on create.

Type / Range

Integer in the range of 0 to 1022

Name

requestedApnIndex - Requested APN index

Syntax

--requestedApnIndex <integer>

Description

This parameter defines the index for APN name used as requested APN in APN conversion.

Class

Required on create.

Type / Range

Integer in the range of 0 to 1022

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.11 cmm apnConversionList

APN Conversion List. Syntax, description, parameters, and roles for **cmm apnConversionList** command.

Syntax

Description

Administer APN Conversion List

Parameters

Name

name - APN Conversion List Name

Syntax

--name <string>

Description

APN conversion list name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.12 cmm apnConversionListNi

APN Conversion List NI. Syntax, description, parameters, and roles for **cmm apnConversionListNi** command.

Syntax

Description

Administer APN Conversion List NI.

Parameters

Name

apnConversionListName - APN Conversion List Name

Syntax

--apnConversionListName <string>

Description

APN Conversion List Name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

apnNi - APN Network Indicator

Syntax

--apnNi <string>

Description

APN NI for Conversion List

Class

Required key.

Type / Range

Digit string of length 1 to 62 with the allowed pattern of [0-9a-zA-Z\._]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.13 cmm apnImsiRangeServices

APN IMSI Range Services. Syntax, description, parameters, and roles for `cmm apnImsiRangeServices` command.

Syntax

Description

This command is used to administer APN IMSI Range Services. The 'shutdown' parameter is used to admin lock the APN Replace list for an IMSI range.

Parameters

Name

uePlmnServicesName - Roaming network name

Syntax

--uePlmnServicesName <string>

Description

Name that identifies the assigned network.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

entryId - Entry ID

Syntax

--entryId <integer>

Description

Entry ID.

Class

Required key.

Type / Range

Integer in the range of 1 to 10,000

Name

allowAutonomousLbo - Allow autonomous LBO

Syntax

--allowAutonomousLbo <list-item>

Description

Allows MME to support local breakout for a roamer PDN connection irrespective of setting of VPLMN dynamic address allowed AVP in UE subscription data.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

apnConversionListName - APN Conversion List Name

Syntax

--apnConversionListName <string>

Description

Used to assign a converted APN list to an IMSI range. APN Conversion List Name shall be used only when gParms aliasApnConversionPerImsi is enabled. Either apnReplaceListName or apnConversionListName must be set when shutdown is false. Both cannot be set at the same time.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

apnReplaceListName - APN replace list name

Syntax

--apnReplaceListName <string>

Description

It is used to assign a replacement APN list to an IMSI range.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

apn-all

Name

description - IMSIRange entry description

Syntax

--description <string>

Description

Description for IMSI range services entry.

Class

Optional.

Type / Range

Digit string of length 0 to 160 with the allowed pattern of [0-9a-zA-Z\.\- _:@]*

Default

"

Name

maxImei - Maximum IMEISV

Syntax

--maxImei <string>

Description

The maximum masked IMEISV in range.

Class

Optional.

Type / Range

Digit string of length 15 to 16 with the allowed pattern of [0-9*]*

Default

Name

maxMsin - Maximum MSIN

Syntax

--maxMsin <digits>

Description

The maximum IMSI MSIN value.

Class

Optional.

Type / Range

Digit string of length 0 or 8 to 10

Default

9999999999

Name

minImei - Minimum IMEISV

Syntax

--minImei <string>

Description

The minimum masked IMEISV in range.

Class

Optional.

Type / Range

Digit string of length 15 to 16 with the allowed pattern of [0-9\]*

Default

Name

minMsin - Minimum MSIN

Syntax

--minMsin <digits>

Description

The minimum IMSI MSIN value.

Class

Optional.

Type / Range

Digit string of length 0 or 8 to 10

Default

000000000

Name

pdpTypeMappingOverride - PDP Type Mapping Override

Syntax

--pdpTypeMappingOverride <list-item>

Description

Parameter controls PDP Type Mapping which when set to Unsubscribed APN will cause MME to map a UE request PDP type of IPV4 to IPV6 if HSS subscription is IPV6, and to map UE requested PDP type of IPV6 to IPV4 if HSS Subscription is IPV4 when UE requests an APN that is not subscribed in the HSS or the UE did not provide a valid APN. PDN Type Mapping refers

to the UE requested PDN Type IE as per TS 3GPP 23.401 and the PDN Type AVP as per TS 3GPP 29.272. Default value shall be set to never, in which case MME will follow standards-based PDP Type conversion rules.

Class

Optional.

List

- never - Never
- unsubscribedApn - Unsubscribed APN

Default

never

Name

replacementAction - Replacement Action

Syntax

--replacementAction <list-item>

Description

Replacement Action field in the existing IMSI/IMEI table controls APN Override with the following values: - UseDefault: Means use the default APN in the HSS if it exists as correction APN (if the default APN does not exist, reject the attach or PDN connection). Any provisioned APN in the IMSI/IMEI table is ignored. - Reject: Means reject the attach/PDN connection regardless of any provisioned APN in the IMSI/IMEI table - UseMap: Use the APN in the IMSI/IMEI map.

Class

Optional.

List

- Reject - Reject means reject the attach/PDN connection regardless of any provisioned APN in the IMSI/IMEI table
- UseDefault - Use Default - UseDefault means use the default APN in the HSS if it exists as correction APN. Any provisioned APN in the IMSI/IMEI table is ignored.
- UseMap - Use Map - Use the APN in the IMSI/IMEI map.

Default

UseMap

Name

shutdown - Shutdown

Syntax

--shutdown <list-item>

Description

It is used to admin lock the APN replace list for an IMSI range.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

ueRequestedApnNi - UE requested APN NI

Syntax

--ueRequestedApnNi <string>

Description

It is used to assign a UE requested APN NI. apn-all is a special case value that will match any non-empty APN of 1 to 63 octets in length. apn-none is a special case value that will match a requested APN that is too long (more than 63 octets), or when no APN is requested. apn-all and apn-none have special meaning only when the APN Conversion Enhanced Map Treatment global parameter is enabled, else they are treated as a literal APN name for matching.

Class

Optional.

Type / Range

Digit string of length 1 to 62 with the allowed pattern of [0-9a-zA-Z\.\-]*

Default

all

Name

useMapTreatmentForSubscribedApn - Use Map Treatment For Subscribed APN

Syntax

--useMapTreatmentForSubscribedApn <list-item>

Description

Map Treatment shall be applied for Subscribed APNs with values: - Always Convert: the converted APN is used and the procedure will continue. - No Convert And Continue: no conversion is done and the procedure will continue with the requested APN

Class

Optional.

List

- alwaysConvert - Always Convert

- noConvertAndContinue - No Convert And Continue

Default

alwaysConvert

Name

useMapTreatmentForUnsubscribedApn - Use Map Treatment For Unsubscribed APN

Syntax

--useMapTreatmentForUnsubscribedApn <list-item>

Description

Map Treatment shall be applied for Unsubscribed APNs with values: - Apply Subscribed Default Then Convert (default): apply the treatment from feature f10107-12 - MME support for Custom Specific APN Correction - No Convert And Continue: no conversion is done and the procedure will continue with the requested APN - No Convert And Reject: no conversion is done and the procedure is rejected - Convert If Local Subscribed Else Reject: the APN in the APN Conversion List is used if the UE has a subscription. Otherwise the procedure is rejected. - Convert If Local Subscribed Else Lowest Context Id: the APN in the APN Conversion List is used if the UE has a subscription. Otherwise the APN with the lowest context Id in the HSS will be used and the procedure will continue. - No Convert and Use Lowest Context Id: no conversion is done and the lowest context Id in the HSS will be used and the procedure will continue. If APN Conversion List Name is not set, then it is considered no match with subscription and the subsequent treatment applies (Use Lowest Subscription ID or Reject, depending on setting).

Class

Optional.

List

- applySubscribedDefaultThenConvert - Apply Subscribed Default Then Convert
- convertIfLocalSubscribedElseLowestContextId - Convert If Local Subscribed Else Lowest Context Id
- convertIfLocalSubscribedElseReject - Convert If Local Subscribed Else Reject
- noConvertAndContinue - No Convert and Continue
- noConvertAndReject - No Convert and Reject
- noConvertAndUseLowestContextId - No Convert and Use Lowest Context Id

Default

applySubscribedDefaultThenConvert

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.14 cmm apnList

apnLists. Syntax, description, parameters, and roles for cmm apnList command.

Syntax

Description

Configure, list and show APN lists.

Parameters

Name

apnConversion - APN conversion list

Syntax

--apnConversion <list-item>

Description

APN conversion list.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ccPgw - Charging Characteristics for PGW selection

Syntax

--ccPgw <list-item>

Description

Defines whether Charging characteristics are appended to the APN FQDN during rel-8 DNS query for P-GW selection. When value is false (default) charging characteristics are not appended. When value is true charging characteristics are appended.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

homeGgsn - Intelligent GGSN selection

Syntax

--homeGgsn <list-item>

Description

Intelligent GGSN selection (HGGSN).

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

hpGwCc - home PGW selection based on CC

Syntax

--hpGwCc <list-item>

Description

home PGW selection based on CC

Class

Optional, not filterable on list.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Name

locationGw - Location Based PGW/GGSN Selection

Syntax

--locationGw <list-item>

Description

Location Based PGW/GGSN selection

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

msisdnPgw - MSISDN based PGW selection

Syntax

--msisdnPgw <list-item>

Description

MSISDN based PGW selection.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

oneGtpTunnel - Direct Tunnel

Syntax

--oneGtpTunnel <list-item>

Description

Direct tunnel (DT).

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

pgw - APN based P-GW selection

Syntax

--pgw <list-item>

Description

APN based P-GW selection (P-GW).

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

id - Apn for the special handling

Syntax

<string>

Description

This parameter defines the external network to which the SGSN or MME is connected. In the GPRS backbone, the APN is a reference to the gateway GPRS support node (GGSN) to be used. The APN string is a set of labels separated by dots, for example, WWW.COMPANY.COM. The length of the identifier ranges from 2 to 100 alphanumeric and special characters. For the APN Conversion List the maximum length of the network identifier is 63 characters

Class

Required value.

Type / Range

Digit string of length 2 to 100 with the allowed pattern of [0-9a-zA-Z\.\-]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.15 cmm apnPagingTypeMapping

APN Paging Policy. Syntax, description, parameters, and roles for cmm apnPagingTypeMapping command.

Syntax

Description

This command is used to administer a APN based Paging Policy and includes parameters for defining the number of paging attempts, paging method, paging type, and paging response timer. Note, Coverage Enhancement paging and eDRX paging types would have higher precedence than APN paging.

Parameters

Name

plmnName - Serving PLMN Name

Syntax

--plmnName <string>

Description

The serving PLMN name may be the name of the MME home PLMN or shared PLMN.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

apnNi - APN network identifier

Syntax

--apnNi <string>

Description

APN network identifier

Class

Required key.

Type / Range

Digit string of length 1 to 62 with the allowed pattern of [0-9a-zA-Z\.\-]*

Name

apnPagingType - APN Paging Type

Syntax

--apnPagingType <list-item>

Description

Names the APN-related MME paging type for this mapping record.

Class

Required on create.

List

- APN1 - Paging Type for APN1
- APN10 - Paging Type for APN10
- APN2 - Paging Type for APN2
- APN3 - Paging type for APN3
- APN4 - Paging Type for APN4
- APN5 - Paging Type for APN5
- APN6 - Paging Type for APN6
- APN7 - Paging Type for APN7
- APN8 - Paging type for APN8
- APN9 - Paging Type for APN9

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.16 cmm apnPgwSelection

APN PGW selection. Syntax, description, parameters, and roles for `cmm apnPgwSelection` command.

Syntax

Description

APNs for PGW selection.

Parameters

Name

apnPgwSelectionListName - APN PGW selection list name

Syntax

--apnPgwSelectionListName <string>

Description

APN PGW selection list name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of `[0-9a-zA-Z\.\_:@]*`

Name

apnNi - APN network identifier

Syntax

--apnNi <string>

Description

APN network identifier.

Class

Required key.

Type / Range

Digit string of length 1 to 62 with the allowed pattern of `[0-9a-zA-Z\.\_]*`

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.17 cmm apnPgwSelectionList

APN PGW selection list. Syntax, description, parameters, and roles for `cmm apnPgwSelectionList` command.

Syntax

Description

List of APNs for PGW selections.

Parameters

Name

apnPgwSelectionListName - APN PGW selection list name

Syntax

--apnPgwSelectionListName <string>

Description

APN PGW selection list name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of `[0-9a-zA-Z\.\-_:@]*`

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.18 cmm apnPriority

APN Priority. Syntax, description, parameters, and roles for **cmm apnPriority** command.

Syntax

Description

This command is used to add an APN to the list of APNs whose PDN connections will be recovered for SGW enhanced recovery and/or PGW restart.

Parameters

Name

apnPriorityListName - APN priority list name

Syntax

--apnPriorityListName <string>

Description

APN priority list name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

apnPriorityNi - APN NI

Syntax

--apnPriorityNi <string>

Description

APN NI.

Class

Required key.

Type / Range

Digit string of length 1 to 62 with the allowed pattern of [0-9a-zA-Z\._]*

Name

apnPriorityValue - APN priority value

Syntax

--apnPriorityValue <integer>

Description

APN priority value (1 is the highest value).

Class

Optional.

Type / Range

Integer in the range of 1 to 8

Default

8

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.19 cmm apnPriorityList

APN Priority List. Syntax, description, parameters, and roles for **cmm apnPriorityList** command.

Syntax**Description**

This command defines an APN priority list for prioritizing UE recovery upon SGW failure/restart or UE recovery of PDN connections upon restart of a PGW based on the UE PDN connection to the APN in the list at the time of the failure. An APN priority list can only be deleted if it is not assigned to a network. The maximum number of APNs allowed is 8.

Parameters

Name

name - APN priority list name

Syntax

--name <string>

Description

APN priority list name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.20 cmm apnQciPpiMap

APN QciPpiMap. Syntax, description, parameters, and roles for **cmm apnQciPpiMap** command.

Syntax

Description

This command is used to define an APN-NI and includes parameters for defining Paging Policy Indication (PPI) and QoS Class Identifier (QCI) values.

Parameters

Name

apnQciPpiMapListName - APN QCI PPI mapping name

Syntax


```
--apnQciPpiMapListName <string>
```

Description

APN QCI PPI mapping name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

apnNi - Access point name - network indicator

Syntax

```
--apnNi <string>
```

Description

APN NI for this roaming QoS profile.

Class

Required key.

Type / Range

Digit string of length 1 to 62 with the allowed pattern of [0-9a-zA-Z\._]*

Name

qci - QoS class identifier

Syntax

```
--qci <integer>
```

Description

QoS class identifier number.

Class

Required key.

Type / Range

Integer in the range of 1-9,65-67,69-70

Default

5

Name

ppi - Paging policy indication

Syntax

```
--ppi <integer>
```

Description

Paging policy indication value.

Class

Required key.

Type / Range

Integer in the range of 0 to 63

Default

0

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.21 cmm apnQciPpiMapList

APN QCI PPI Mapping List. Syntax, description, parameters, and roles for `cmm apnQciPpiMapList` command.

Syntax**Description**

This command creates an APN to QCI to PPI mapping profile for use when MME determines whether to use Priority Paging Indicator value in DDN to initiate differentiated paging. For the selected HPLMN, the combination of APN/QCI/PPI defined by this command's mapping will be compared to the: - PPI value received in the DDN Paging Service Information IE - APN and QCI in the UE Context Database mapped to the EBI (EPS Bearer Identity) in the DDN Paging Service Information IE. If the values in/derived from the DDN Paging Service Information IE do not match the values provisioned in the APN QCI PPI mapping profile, differentiated paging service will not be provided for the DDN. The maximum number of APN mapping profiles supported is 8. The maximum number of combinations of QCI/PPI per APN is limited to 3.

Parameters**Name**

name - APN QCI PPI mapping name

Syntax

--name <string>

Description

APN QCI PPI mapping name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.22 cmm apnReplaceList

APN Replace List. Syntax, description, parameters, and roles for cmm apnReplaceList command.

Syntax**Description**

Administer APN Replace List.

Parameters**Name**

apnReplaceListName - APN replace list name

Syntax

--apnReplaceListName <string>

Description

APN replace list name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.23 cmm apnReplaceListNi

APN Replace List NI. Syntax, description, parameters, and roles for cmm apnReplaceListNi command.

Syntax

Description

Administer APN Replace List NI.

Parameters

Name

apnReplaceListName - APN replace list name

Syntax

--apnReplaceListName <string>

Description

APN replace list name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

apnNi - Access point name - network indicator

Syntax

--apnNi <string>

Description

APN NI for replace list.

Class

Required key.

Type / Range

Digit string of length 1 to 62 with the allowed pattern of [0-9a-zA-Z\.\-]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.24 cmm apnWhiteList

APN White List. Syntax, description, parameters, and roles for cmm apnWhiteList command.

Syntax**Description**

This command is used to create an APN White List.

Parameters**Name**

name - APN white list name

Syntax

--name <string>

Description

APN white list name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.25 cmm apnWhiteListNi

APN White List NI. Syntax, description, parameters, and roles for `cmm apnWhiteListNi` command.

Syntax

Description

This command is used to administer an APN White List and includes a parameter for defining an APN-NI.

Parameters

Name

apnWhiteListName - APN white list name

Syntax

--apnWhiteListName <string>

Description

APN white list name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

apnNi - Access point name - network indicator

Syntax

--apnNi <string>

Description

APN NI for white list.

Class

Required key.

Type / Range

Digit string of length 1 to 62 with the allowed pattern of [0-9a-zA-Z\.\-]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.26 cmm areaList

Area List. Syntax, description, parameters, and roles for cmm areaList command.

Syntax**Description**

This command is used to administer an Area List and includes parameters for defining the serving network name and list type.

Parameters**Name**

name - Area list name

Syntax

--name <string>

Description

Name associated with this list, provided as a way to have a brief but meaningful description of the list.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

plmnName - Served mobile network name

Syntax

--plmnName <string>

Description

Name that identifies serving network.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

listType - List type

Syntax

--listType <list-item>

Description

Indicates the type of list being provisioned: TAI List, LAI List, Forbidden TAI List, LTE HO Restrict List.

Class

Required on create, not modifiable.

List

- Forbidden TAI List - Forbidden tracking area identifier list.
- Forbidden TAI Range List - Forbidden Tracking Area List specified by a range.
- LAI List - Location area identifier list.
- LTE HO Restrict List - LTE handoff restriction list.
- TAI List - Tracking area identifier list.
- TAI Range List - Tracking Area List specified by a range.
- n26lwkRestTailList - TAI List for N26 Restriction - Identifies TAI List associated with N26 Restriction.

Name

areaListId - Associated table ID

Syntax

--areaListId <integer>

Description

ID associated with this list in the table containing the TAIs or LAIs themselves.

Class

Read only.

Type / Range

Integer in the range of 1 to 5,000

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.27 cmm arpPara

ARP Parameter. Syntax, description, parameters, and roles for `cmm arpPara` command.

Syntax**Description**

ARP parameters in the R8 mapping table.

Parameters**Name**

serviceLevel - Service Level

Syntax

`--serviceLevel <list-item>`

Description

This parameter specifies the service level. A service level is determined based on the traffic class, downlink guaranteed bit rate (GBR), and traffic handling priority (THP) in the QoS profile contained in the PDP context of a UE.

Class

Required key.

List

- BACKGROUND - BACKGROUND(Background): indicates a service
- CONVERSATION - indicates a service.
- INTERACTIVETRAFFICPRI1 - INTERACTIVETRAFFICPRI1(InteractiveTrafficPri1): indicates a service level whose traffic class is interactive and whose THP is 1.
- INTERACTIVETRAFFICPRI2 - INTERACTIVETRAFFICPRI2(InteractiveTrafficPri2): indicates a service level whose traffic class is interactive and whose THP is 2.
- INTERACTIVETRAFFICPRI3 - INTERACTIVETRAFFICPRI3(InteractiveTrafficPri3): indicates a service level whose traffic class is interactive and whose THP is 3
- STREAMINGGBRLESS24KBPS - streaming and whose downlink GBR is less than 25
- STREAMINGGBRMORE25KBPS - streaming and whose downlink GBR is greater

Name

userLevel - User Level

Syntax

--userLevel <list-item>

Description

The different user priority level based on received pre-R8 ARP under subscription data: ARP 1->high level; ARP 2-> Normal level; ARP 3-low level

Class

Required key.

List

- HighLevel - High Level User with ARP 1
- LowLevel - Low Level User with ARP 3
- NormalLevel - Normal Level User with ARP 2

Name

preLevel - Pre-emption Level

Syntax

--preLevel <integer>

Description

This parameter specifies the pre-emption capability of the requested radio bearer. This capability corresponds to the Pre-emption Capability IE in the ARP attribute of the radio bearer (see 3GPP TS 25.413). The default value for PCI shall be 1 (may trigger pre-emption). If it's configured as 0, SGSN Will send it to radio as 0 (shall not trigger pre-emption).

Class

Optional.

Type / Range

Integer in the range of 0 to 1

Default

1

Name

preVulnerability - Pre-emption Vulnerability

Syntax

--preVulnerability <integer>

Description

This parameter specifies the pre-emption vulnerability of the requested radio bearer. This vulnerability corresponds to the Pre-emption Vulnerability IE in the ARP attribute of the radio bearer. For details, see 3GPP TS 25.413. The default value of PVI shall be 0 (not pre-emptable). If it's configured as 1, SGSN will send it to radio as 1 (pre-emptable)

Class

Optional.

Type / Range

Integer in the range of 0 to 1

Default

0

Name

priorityLevel - Priority Level

Syntax

--priorityLevel <integer>

Description

This parameter specifies the bearer priority used in the resource allocation by the RNC. This priority corresponds to the Priority Level IE. The values 1 to 14 are arranged in descending order of priority, with 1 indicating the highest priority and 14 indicating the lowest priority. The value 15 indicates shall not trigger pre-emption

Class

Optional.

Type / Range

Integer in the range of 1 to 15

Default

7

Name

queue - Queue

Syntax

--queue <integer>

Description

Queuing Allowed. This parameter specifies whether a radio bearer setup request can be included in the resource allocation queue. This parameter corresponds to the Queuing Allowed IE in the ARP attribute of the radio bearer. For details, see 3GPP TS 25.413. The default value shall be 1 (queuing allowed). if it's configured to 0, SGSN Will send 0 to RNC (queuing not allowed).

Class

Optional.

Type / Range

Integer in the range of 0 to 1

Default

1

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

4.28 cmm assignEmergencyPgw

Assign Emergency PGW. Syntax, description, parameters, and roles for `cmm assignEmergencyPgw` command.

Syntax**Description**

Allows an operator to assign a specific emergency PGW to a specific group of subscribers based on the PDN type requested by the UE in the emergency PDN connection request

Parameters**Name**

uePlmnServicesName - Name of Services based on UE Network and Served Network

Syntax

```
--uePlmnServicesName <string>
```

Description

Name that identifies roaming agreement

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

pgwTypeReceived - PGW Type Received From UE

Syntax

```
--pgwTypeReceived <list-item>
```

Description

PGW Type Received From UE

Class

Required key.

List

- IPv4 - IPv4
- IPv4IPv6 - IPv4IPv6
- IPv6 - IPv6

Name

fqdnTag - FQDN Tag

Syntax

```
--fqdnTag <string>
```

Description

Tag (String to represent a FQDN label. All the rules of FQDN label apply. Provisioning of no string is valid.)

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 128

Default

""

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.29 cmm authenticate

Authenticate. Syntax, description, parameters, and roles for `cmm authenticate` command.

Syntax**Description**

This command is used for authentication. It prompts the user for a Username and Password.

Parameters

N/A

Roles

sysadmin, manage

4.30 cmm authServer

authServer. Syntax, description, parameters, and roles for `cmm authServer` command.

Syntax**Description**

external authentication server IP/identity.

Parameters**Name**

priority - Server priority

Syntax

--priority <integer>

Description

External authentication server priority.

Class

Required key.

Type / Range

Integer in the range of 1 to 3

Name

ip - IP address

Syntax

--ip <string>

Description

IP of external authentication server.

Class

Required on create.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

extLdapProfileName - External LDAP Profile Name

Syntax

--extLdapProfileName <string>

Description

External LDAP Profile Name. When type is extldap this field must be entered.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

secret - Secret

Syntax

--secret <string>

Description

Shared secret for authentication between client and external authentication server. This field must be entered when Type is RADIUS or TACACS.

Restrictions

This field must be entered when Type is RADIUS or TACACS. Check must be in the backend.
This field is optional when type is externalLdap.

Class

Optional.

Type / Range

Authentication server secret for RADIUS/TACACS+. Following characters are supported: 0-9a-zA-Z.-_ and space.

Name

timeout - IP timeout

Syntax

--timeout <integer>

Description

External authentication server timeout.

Class

Optional.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

type - Type of external AAA server

Syntax

--type <list-item>

Description

Type of external AAA server

Class

Not modifiable.

List

- RADIUS - RADIUS authentication server
- TACACS - TACACS+ - TACACS+ authentication server
- externalLdap - External LDAP

Default

RADIUS

Roles

Create

secadmin

Show

secadmin

Modify

secadmin

Delete

secadmin

List

secadmin

4.31 cmm cacheAgeOut

Cache Ageout Quiet Timer configuration. Syntax, description, parameters, and roles for cmm cacheAgeOut command.

Syntax

Description

This command is used to administer the Quiet Timer configurations per UE class. The Quiet Timer defines the time of inactivity for a UE of a given time to be eligible for eviction from the CPPS cache.

Parameters

Name

iotQuietTimer - IoT UE Quiet Timer

Syntax

--iotQuietTimer <integer>

Description

This parameter defines the value of the Quiet Timer for IoT UEs. An IoT UE are identified as those 4G UEs that: a) Use NBIoT RAT, b) use S11U, c) use T6a, d) are registered without PDN connection, e) are catm1-cemodeb. Caching is controlled by the IoT Quiet timer. The value is by default zero. If the value is zero it means that UEs of the IoT type are not evicted from the cache. Minimum value is 1 minute , maximum value is 120 minutes.

Restrictions

If the parameter is zero then no cache evictions take place. the minimum value is 1 minute and the maximum value is 120.

Class

Optional.

Type / Range

Integer in the range of 0 to 120 in units of Minute

Default

0

Name

retainImsiGroupIdListName - Retain IMSI Group ID List Name

Syntax

--retainImsiGroupIdListName <string>

Description

List of IMSI Group IDs that are to be excluded from age-out

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

''

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

4.32 cmm callTraceAllowedRange

Call Trace Allowed Range. Syntax, description, parameters, and roles for cmm callTraceAllowedRange command.

Syntax**Description**

This command is used to create, delete, show and list the allowed call trace range.

Parameters

Name

name - Allowed List Name

Syntax

--name <string>

Description

Name of the allowed range list.

Class

Required key.

Type / Range

Digit string of length 1 to 32 with the allowed pattern of [0-9a-zA-Z\.\-:]*

Name

userIdentity - User identity

Syntax

--userIdentity <list-item>

Description

Call trace user identity.

Class

Required on create, not modifiable.

List

- imsi - Specify user identity by international mobile station identify.
- msisdn - Specify user identify by mobile station international subscriber directory number.

Name

maxImsi - Max Imsi

Syntax

--maxImsi <digits>

Description

Maximum IMSI for IMSI range.

Class

Optional.

Type / Range

Digit string of length 0 or 6 to 15 with the allowed pattern of [0-9]*

Default

""

Name

maxMsisdn - Max Msisdn

Syntax

--maxMsisdn <digits>

Description

Maximum MSISDN for MSISDN range.

Class

Optional.

Type / Range

Digit string of length 0 or 6 to 15 with the allowed pattern of [0-9]*

Default

""

Name

minImsi - Min Imsi

Syntax

--minImsi <digits>

Description

Minimum IMSI for IMSI range.

Class

Optional.

Type / Range

Digit string of length 0 or 6 to 15 with the allowed pattern of [0-9]*

Default

""

Name

minMsisdn - Min Msisdn

Syntax

--minMsisdn <digits>

Description

Minimum MSISDN for MSISDN range.

Class

Optional.

Type / Range

Digit string of length 0 or 6 to 15 with the allowed pattern of [0-9]*

Default

""

Roles

Create

secadmin

Show

sysadmin, manage, secadmin

Delete

secadmin

List

sysadmin, manage, secadmin

4.33 cmm callTraceJob

Call Trace Job. Syntax, description, parameters, and roles for `cmm callTraceJob` command.

Syntax

Description

Configuration context for specifying which UE the call-trace packets should be collected for.

Parameters

Name

userIdentity - User identity

Syntax

--userIdentity <list-item>

Description

Call trace user identity.

Class

Required key.

List

- imsi - Specify user identity by international mobile station identify.
- msisdn - Specify user identify by mobile station international subscriber directory number.

Name

digits - Digits

Syntax

--digits <digits>

Description

IMSI or MSISDN digits.

Class

Required key.

Type / Range

Digit string of length 6 to 15 with the allowed pattern of [0-9]*

Name

dns - DNS

Syntax

--dns <list-item>

Description

The call trace collection from DNS interface.

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

dnssgsn - DNS for SGSN

Syntax

--dnssgsn <list-item>

Description

The call trace collection from SGSN DNS interface

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

gb - Gb Interface

Syntax

--gb <list-item>

Description

The call trace collection from GB interface

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

gf - Gf Interface

Syntax

--gf <list-item>

Description

The call trace collection from GF interface

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

gn - Gn

Syntax

--gn <list-item>

Description

The call trace collection from Gn interface.

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

gnsgsn - Gn Interface for SGSN

Syntax

--gnsgsn <list-item>

Description

The call trace collection from SGSN GN interface

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

gr - Gr Interface

Syntax

--gr <list-item>

Description

The call trace collection from GR interface

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

gs - Gs Interface

Syntax

--gs <list-item>

Description

The call trace collection from GS interface

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

iucp - lu-C Interface

Syntax

--iucp <list-item>

Description

The call trace collection from IUCP interface

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

mode - Debug mode

Syntax

--mode <list-item>

Description

Call trace debug mode. Ten IMSI/MSISDN may be under debug mode at one time.

Class

Optional.

List

- debug - Debug call trace mode.
- default - Default call trace mode.

Default

default

Name

n11 - N11 Session Management Function

Syntax

--n11 <list-item>

Description

The call trace collection from N11 interface.

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

n12 - N12 Authentication Server Function

Syntax

--n12 <list-item>

Description

The call trace collection from N12 interface.

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

n14 - N14 Access and MM Function

Syntax

--n14 <list-item>

Description

The call trace collection from N14 interface.

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

n15 - N15 Policy Control Function

Syntax

--n15 <list-item>

Description

The call trace collection from N15 interface.

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

n17 - N17 5G EIR

Syntax

--n17 <list-item>

Description

The call trace collection from N17 interface.

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

n2 - N2 New Radio

Syntax

--n2 <list-item>

Description

The call trace collection from N2 interface.

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

n20 - N20 SMS Function

Syntax

--n20 <list-item>

Description

The call trace collection from Nsms interface.

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

n22 - N22 Network Slice Function

Syntax

--n22 <list-item>

Description

The call trace collection from N22 interface.

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

n26 - N26 Mobility Management Entity

Syntax

--n26 <list-item>

Description

The call trace collection from N26 interface.

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

n26mme - n26 MME

Syntax

--n26mme <list-item>

Description

N26 - Interface to AMF from MME

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

n58 - N58

Syntax

--n58 <list-item>

Description

The call trace collection from N58 interface

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

n8 - N8 Unified Data Management

Syntax

--n8 <list-item>

Description

The call trace collection from N8 interface.

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

namfEvts - Namf Event Exposure Service

Syntax

--namfEvts <list-item>

Description

Namf Event Exposure Service

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

namfServer - AMF Server

Syntax

--namfServer <list-item>

Description

The call trace collection from COMM, LOC on AMF

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

nlg - NLg interface

Syntax

--nlg <list-item>

Description

The call trace collection from Nlg interface

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

nls - NLS interface

Syntax

--nls <list-item>

Description

The call trace collection from NLS interface

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

nnrf - Nnrf Network Resource Function

Syntax

--nnrf <list-item>

Description

The call trace collection from Nnrf interface.

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

notificationService - AMF Notification Service

Syntax

--notificationService <list-item>

Description

This parameter represents the Notification Services on the AMF. Any NF that wants to notify the AMF will use one of these services.

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

nr10 - NR10

Syntax

--nr10 <list-item>

Description

The call trace collection from NR10 interface

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

rs10 - RS10

Syntax

--rs10 <list-item>

Description

The call trace collection from RS10 interface.

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

s10 - S10

Syntax

--s10 <list-item>

Description

The call trace collection from S10 interface.

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

s102 - S102

Syntax

--s102 <list-item>

Description

The call trace collection from S102 interface.

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

s11 - S11

Syntax

--s11 <list-item>

Description

The call trace collection from S11 interface.

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

s11u - S11U

Syntax

--s11u <list-item>

Description

The call trace collection from S11U interface.

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

s13 - S13

Syntax

--s13 <list-item>

Description

The call trace collection from S13 interface.

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

s13sgsn - S13 Interface for SGSN

Syntax

--s13sgsn <list-item>

Description

The call trace collection from S13sgsn interface

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

s16 - S16 Interface

Syntax

--s16 <list-item>

Description

The call trace collection from S16 interface

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

s1mme - S1MME

Syntax

--s1mme <list-item>

Description

The call trace collection from S1MME interface.

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

s3 - S3

Syntax

--s3 <list-item>

Description

The call trace collection from S3 interface.

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

s3sgsn - S3 Interface for SGSN

Syntax

--s3sgsn <list-item>

Description

The call trace collection from SGSN S3 interface

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

s4 - S4 Interface

Syntax

--s4 <list-item>

Description

The call trace collection from S4 interface

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

s4gn2gu - 2G Data

Syntax

--s4gn2gu <list-item>

Description

The call trace collection for 2G Data(Gn-U and S4-U)

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

s4gn3gu - 3G Data

Syntax

--s4gn3gu <list-item>

Description

The call trace collection for 3G Data (Gn-U and S4-U).

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

s6ad - S6ad

Syntax

--s6ad <list-item>

Description

The call trace collection from S6ad interface.

Class

Read only.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

true

Name

s6c - S6c

Syntax

--s6c <list-item>

Description

The call trace collection from S6c interface.

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

s6d - S6d

Syntax

--s6d <list-item>

Description

The call trace collection from S6D interface

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

sgd - SGd

Syntax

--sgd <list-item>

Description

The call trace collection from SGd interface.

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

sgs - SGs

Syntax

--sgs <list-item>

Description

The call trace collection from SGs interface.

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

sgslite - SGSLite

Syntax

--sgslite <list-item>

Description

The call trace collection from SGSLite interface.

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

slg - Slg

Syntax

--slg <list-item>

Description

The call trace collection from Slg interface.

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

sls - Sls

Syntax

--sls <list-item>

Description

The call trace collection from Sls interface.

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

sv - Sv

Syntax

--sv <list-item>

Description

The call trace collection from Sv interface.

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

t6a - T6A

Syntax

--t6a <list-item>

Description

The call trace collection from T6A interface.

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

t6aU - T6AU

Syntax

--t6aU <list-item>

Description

The call trace collection from T6AU interface.

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

traceld - Call trace job ID

Syntax

--traceId <integer>

Description

Call trace job ID.

Class

Read only.

Type / Range

Integer in the range of 1 to 65,535

Default

1

Name

traceSrc - Trace source

Syntax

--traceSrc <list-item>

Description

Call trace job source.

Class

Read only.

List

- hss - Home subscriber server - Specify trace source of home subscriber server.
- mgt - Management - Specify trace source of management.
- udm - United Data Management Server - Specify trace source of United Data Management Server

Default

mgt

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.34 cmm callTraceJobStream

Streaming for call trace Job. Syntax, description, parameters, and roles for cmm callTraceJobStream command.

Syntax

Description

Associate a call trace Job with a streaming destination, i.e. define a remote call trace streaming server to receive call trace records related to a specific call trace job

Parameters

Name

traceJobId - Trace Job Identifier

Syntax

--traceJobId <integer>

Description

Trace job identifier produced by Call Trace Job command which identifies a specific call trace job.

Class

Required key.

Type / Range

Integer in the range of 1 to 65,535

Name

callTraceStreamingName - Name of the Streaming Destination

Syntax

--callTraceStreamingName <string>

Description

This is the name of the streaming destination (i.e stream profile) that will be used to associate a streaming remote endpoint to a call trace job.

Restrictions

This should be an existing streaming name defined earlier using the Call Trace Streaming command.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.35 cmm callTraceSetting

Call Trace Interface Setting. Syntax, description, parameters, and roles for cmm callTraceSetting command.

Syntax

Description

Configuration Call Trace Default Interface Setting.

Parameters

Name

dns - DNS

Syntax

--dns <list-item>

Description

The call trace collection from DNS interface.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

dnssgsn - DNS Interface for SGSN

Syntax

--dnssgsn <list-item>

Description

The call trace collection from SGSN DNS interface

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

gb - Gb interface

Syntax

--gb <list-item>

Description

The Call Trace collection from Gb interface

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

gf - Gf Interface

Syntax

--gf <list-item>

Description

The call trace collection from GF interface

Class

Optional.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

true

Name

gn - Gn

Syntax

--gn <list-item>

Description

The call trace collection from Gn interface.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

gnsgsn - Gn Interface for SGSN

Syntax

--gnsgsn <list-item>

Description

The call trace collection from SGSN GN interface

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

gr - Gr Interface

Syntax

--gr <list-item>

Description

The call trace collection from GR interface

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

gs - Gs Interface

Syntax

--gs <list-item>

Description

The call trace collection from GS interface

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

iucp - lu-C interface

Syntax

--iucp <list-item>

Description

The call trace collection from IU-CP interface

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

n11 - N11 Session Management Function

Syntax

--n11 <list-item>

Description

The call trace collection from N11 interface.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

n12 - N12 Authentication Server Function

Syntax

--n12 <list-item>

Description

The call trace collection from N12 interface.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

n14 - N14 Access and MM function

Syntax

--n14 <list-item>

Description

The call trace collection from N14 interface.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

n15 - N15 Policy Control Function

Syntax

--n15 <list-item>

Description

The call trace collection from N15 interface.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

n17 - N17 5G EIR

Syntax

--n17 <list-item>

Description

The call trace collection from N17 interface.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

n2 - N2 New Radio

Syntax

--n2 <list-item>

Description

The call trace collection from N2 interface.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

n20 - N20 SMS Function

Syntax

--n20 <list-item>

Description

The call trace collection from Nsms interface.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

n22 - N22 Network Slice Selection Function

Syntax

--n22 <list-item>

Description

The call trace collection from N22 interface.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

n26 - N26 Mobility Management Function

Syntax

--n26 <list-item>

Description

The call trace collection from N26 interface.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

n26mme - n26 MME

Syntax

--n26mme <list-item>

Description

N26 - Interface to AMF from MME

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

n58 - N58 NSSAA Server Function

Syntax

--n58 <list-item>

Description

The call trace collection from N58 interface

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

n8 - N8 Unified Data Management

Syntax

--n8 <list-item>

Description

The call trace collection from N8 interface.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

namfEvts - Namf Event Exposure Service

Syntax

--namfEvts <list-item>

Description

Call Trace for Namf Event Exposure Service

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

namfServer - AMF Server

Syntax

--namfServer <list-item>

Description

The call trace collection from COMM, LOC on AMF

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

nlg - Nlg interface

Syntax

--nlg <list-item>

Description

The call trace collection from Nlg interface

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

nls - Nls interface

Syntax

--nls <list-item>

Description

The call trace collection from Nls interface

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

nnrf - Nnrf Network Resource Function

Syntax

--nnrf <list-item>

Description

The call trace collection from Nnrf interface.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

notificationService - AMF Notification Service

Syntax

--notificationService <list-item>

Description

This parameter represents the Notification Services on the AMF. Any NF that wants to notify the AMF will use one of these services.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

nr10 - NR10

Syntax

```
--nr10 <list-item>
```

Description

The call trace collection from NR10 interface

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

rs10 - RS10

Syntax

```
--rs10 <list-item>
```

Description

The call trace collection from RS10 interface.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

s10 - S10

Syntax

```
--s10 <list-item>
```

Description

The call trace collection from S10 interface.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

s102 - S102

Syntax

--s102 <list-item>

Description

The call trace collection from S102 interface.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

s11 - S11

Syntax

--s11 <list-item>

Description

The call trace collection from S11 interface.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

s11u - S11U

Syntax

--s11u <list-item>

Description

The call trace collection from S11U interface.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

s13 - S13

Syntax

--s13 <list-item>

Description

The call trace collection from S13 interface.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

s13sgsn - S13 Interface for SGSN

Syntax

--s13sgsn <list-item>

Description

The call trace collection from S13sgsn interface

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

s16 - S16 Interface

Syntax

--s16 <list-item>

Description

The call trace collection from S16 interface

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

s1mme - S1MME

Syntax

--s1mme <list-item>

Description

The call trace collection from S1MME interface.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

s3 - S3

Syntax

--s3 <list-item>

Description

The call trace collection from S3 interface.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

s3sgsn - S3 Interface for SGSN

Syntax

--s3sgsn <list-item>

Description

The call trace collection from SGSN S3 interface

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

s4 - S4 Interface

Syntax

--s4 <list-item>

Description

The call trace collection from S4 interface

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

s4gn2gu - 2G Data

Syntax

--s4gn2gu <list-item>

Description

The call trace collection for 2G Data(Gn-U and S4-U)

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

s4gn3gu - 3G Data

Syntax

--s4gn3gu <list-item>

Description

The call trace collection for 3G Data (Gn-U and S4-U).

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

s6ad - S6ad

Syntax

--s6ad <list-item>

Description

The call trace collection from S6ad interface.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

s6c - S6c

Syntax

--s6c <list-item>

Description

The call trace collection from S6c interface.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

s6d - S6d Interface

Syntax

--s6d <list-item>

Description

The call trace collection from S6D interface

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

sgd - SGd

Syntax

--sgd <list-item>

Description

The call trace collection from SGd interface.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

sgs - SGs

Syntax

--sgs <list-item>

Description

The call trace collection from SGs interface.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

sgslite - SGSLite

Syntax

--sgslite <list-item>

Description

The call trace collection from SGSLite interface.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

slg - Slg

Syntax

--slg <list-item>

Description

The call trace collection from Slg interface.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

sls - Sls

Syntax

--sls <list-item>

Description

The call trace collection from Sls interface.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

sv - Sv

Syntax

--sv <list-item>

Description

The call trace collection from Sv interface.

Class

Optional.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

true

Name

t6a - T6A

Syntax

--t6a <list-item>

Description

The call trace collection from T6A interface.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

t6aU - T6AU

Syntax

--t6aU <list-item>

Description

The call trace collection from T6AU interface.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

4.36 cmm callTraceStreaming

Call Trace streaming destination. Syntax, description, parameters, and roles for cmm callTraceStreaming command.

Syntax

Description

With this command, the Call Trace Streaming function of CMM is configured

Parameters

Name

name - Streaming Destination Name

Syntax

--name <string>

Description

This is the name of the streaming destination (i.e stream profile) that will be used to associate a streaming remote endpoint to a call trace job.

Restrictions

Only up to 5 streaming destination can be defined, so in maximum up to 5 records.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

port - Streaming Server Port

Syntax

--port <integer>

Description

Port of the streaming destination, i.e. this is the (remote) Port of the streaming server (e.g. LTMT).

Class

Required on create.

Type / Range

Integer in the range of 1,024 to 65,535

Name

streamIp - Streaming Server IP

Syntax

--streamIp <string>

Description

IP of the streaming destination, i.e. this is the (remote) IP of the streaming server (e.g. LTMT).

Class

Required on create.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

oamTlsProfileName - OAM TLS Profile

Syntax

--oamTlsProfileName <string>

Description

This parameter contains the profile for TLS. It contains the certificate names needed for establishing secure communication.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

operationStateAct - Operation State Active

Syntax

--operationStateAct <list-item>

Description

Operation State of the particular call trace streaming destination

Class

Optional.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Name

type - Trace Type

Syntax

--type <list-item>

Description

This parameter indicates the type of data in the streaming connection. For call trace the type is pcap, for Minimization of Drive Testing the type is mdt

Class

Optional.

List

- mdt - The value of mdt indicates that the data is for Minimization of Drive Testing.
- pcap - The value of pcap indicates that the data is for Call Trace.

Default

pcap

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.37 cmm causeCodeMappingSets

causeCodeSetsNew. Syntax, description, parameters, and roles for cmm causeCodeMappingSets command.

Syntax

Description

Configure, list and show cause code sets name. Each cause code set creates the respective `causeCodeMappingSetsProcs` and `causeCodeMappingSetsCauses`. The `causeCodeMappingSetsCauses` maps the internal cause codes to external ones as defined in this group. New cause code sets need to be created to be able to define the values for the mappings. Cause code mappings can be defined per NAS message (procedure).

Parameters

Name

name - name

Syntax

--name <string>

Description

Cause code set name

Class

Required key.

Type / Range

Digit string of length 1 to 8 with the allowed pattern of `[0-9a-zA-Z\.\-_:]*`

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.38 cmm causeCodeMappingSetsCauses

causeCodeMappingSetsCauses. Syntax, description, parameters, and roles for **cmm causeCodeMappingSetsCauses** command.

Syntax

Description

Update, list and show cause code sets causes. Mapping of internal cause codes to external ones are defined in this group. New cause code sets need to be created to be able to define the values for the mappings. Cause code mappings can be defined per NAS message (procedure).

Parameters

Name

name - name

Syntax

--name <string>

Description

Cause code set name

Class

Required key.

Type / Range

Digit string of length 1 to 8 with the allowed pattern of [0-9a-zA-Z\.\-_:]*

Name

proc - proc

Syntax

--proc <list-item>

Description

The message used for the cause codes. This parameter identifies the procedure. Allowed values, attRej ... attach reject causes attAcc ... attach accept causes rauRej ... RAU reject causes rauAcc ... RAU accept causes det ... NW initiated detach causes pdpActRej ... PDP context activation reject causes pdpModRej ... PDP context modification reject causes pdpDeact ... PDP context deactivation causes servRej ... service request reject causes.

Class

Required key.

List

- attAcc - Attach accept - Attach accept causes.
- attRej - Attach reject - Attach reject causes.
- det - Detach - Network initiated detach causes.
- pdpActRej - PDP activation reject - PDP context activation reject causes.
- pdpDeact - PDP deactivation - PDP context deactivation causes.
- pdpModRej - PDP modification reject - PDP context modification reject causes.

- rauAcc - RAU accept - RAU accept causes.
- rauRej - RAU reject - RAU reject causes.
- servRej - Service request reject - Service request reject causes.

Name

intCause - internal cause

Syntax

--intCause <integer>

Description

internal cause to be mapped to an external cause

Class

Required key.

Type / Range

Integer in the range of 0 to 499

Name

extCause - external cause

Syntax

--extCause <integer>

Description

external cause mapped from an internal cause

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

4.39 cmm causeCodeMappingSetsProcs

causeCodeMappingSetsProcs. Syntax, description, parameters, and roles for cmm causeCodeMappingSetsProcs command.

Syntax

Description

List and show cause code sets procedures. Cause code mappings of internal cause codes to external ones are defined per NAS message (procedure).

Parameters

Name

name - name

Syntax

--name <string>

Description

Cause code set name

Class

Required key.

Type / Range

Digit string of length 1 to 8 with the allowed pattern of [0-9a-zA-Z\._]*

Name

proc - proc

Syntax

--proc <list-item>

Description

the message used for the cause code procedures. This parameter identifies the procedure. Allowed values, attRej ... Attach reject causes attAcc ... Attach accept causes rauRej ... RAU reject causes rauAcc ... RAU accept causes det ... NW initiated detach causes pdpActRej ... PDP context activation reject causes pdpModRej ... PDP context modification reject causes pdpDeact ... PDP context deactivation causes servRej ... Service request reject causes

Class

Required key.

List

- attAcc - Attach accept - Attach accept causes.
- attRej - Attach reject - Attach reject causes.
- det - Detach - Network initiated detach causes.
- pdpActRej - PDP activation reject - PDP context activation reject causes.
- pdpDeact - PDP deactivation - PDP context deactivation causes.

- pdpModRej - PDP modification reject - PDP context modification reject causes.
- rauAcc - RAU accept - RAU accept causes.
- rauRej - RAU reject - RAU reject causes.
- servRej - Service request reject - Service request reject causes.

Roles

Show

sysadmin, manage, monitor

List

sysadmin, manage, monitor

4.40 cmm cdrEncryption

CDR Encryption. Syntax, description, parameters, and roles for `cmm cdrEncryption` command.

Syntax

Description

This command is used to enable or disable the encryption in SGSN CDR files and set the encryption key.

Parameters

Name

enable - Enable

Syntax

--enable <list-item>

Description

Enables or disable CDR File encryption

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

encryptionKey - Encryption Key

Syntax

--encryptionKey <string>

Description

Key used to encrypt CDR files

Class

Optional.

Type / Range

Digit string of length 0 or 32 with the allowed pattern of [A-Za-z0-9+/=]*

Default

""

Roles**Show**

secadmin

Modify

secadmin

List

secadmin

4.41 cmm cdrFiles

CDR Files. Syntax, description, parameters, and roles for cmm cdrFiles command.

Syntax**Description**

Configure, modify and show CDR Files.

Parameters**Name**

cdrFilesId - Id for cdrfile

Syntax

--cdrFilesId <integer>

Description

Id for cdrfile - just set it to 1

Class

Required key.

Type / Range

Integer in the range of 1 to 1

Name

amountOfFiles - AmountOfFiles

Syntax

--amountOfFiles <integer>

Description

The total amount of CDR files/indexes to use. The resulting value when multiplied to the sizeperfile parameter, should be below 200GB.

Class

Optional.

Type / Range

Integer in the range of 0 to 68,000

Default

20000

Name

compression - Compression

Syntax

--compression <list-item>

Description

Set this to 'true' when cdr files should be compressed

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

noOfDigitsInFileName - noOfDigitsInFileName

Syntax

--noOfDigitsInFileName <integer>

Description

The number of digits in the filename of CDR file to use

Class

Optional.

Type / Range

Integer in the range of 4 to 5

Default

5

Name

overwrite - Overwrite

Syntax

--overwrite <list-item>

Description

When there is no available/free CDR file index to use, overwrite could be an option

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sizePerFile - SizePerFile

Syntax

--sizePerFile <integer>

Description

The size per CDR file (in KBs) to use. The transmission of large file sizes could be more error prone.

Class

Optional.

Type / Range

Integer in the range of 3,072 to 10,240

Default

5120

Name

useControlfiles - UseControlfiles

Syntax

--useControlfiles <list-item>

Description

The operator has the possibility to select whether the cdrfiles are transferred towards the billing domain in a controlled way

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.42 cmm cdrPapsCount

cdrPapsCount. Syntax, description, parameters, and roles for cmm cdrPapsCount command.

Syntax**Description**

cdr count of paps.

Parameters**Name**

papsId - Paps index

Syntax

--papsId <string>

Description

Index of paps.

Class

Optional.

Type / Range

Digit string of length 1 to 4 with the allowed pattern of [0-9a-zA-Z\._:-]*

Roles**List**

sysadmin, manage, monitor

4.43 cmm cellId

Cell Identifiers subject to Handover. Syntax, description, parameters, and roles for cmm cellId command.

Syntax**Description**

Administer Cell Identifiers subject to Handover Restriction

Parameters**Name**

name - Name of the Cell ID

Syntax

--name <string>

Description

Name of the cell identifier.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:-@]*

Name

cellIdentifier - Cell Identifier

Syntax

--cellIdentifier <integer>

Description

Identify the cell.

Class

Required on create, not modifiable.

Type / Range

Integer in the range of 0..68719476735

Name

plmnName - Mobile network name

Syntax

--plmnName <string>

Description

A name that identifies mobile network.

Class

Required on create, not modifiable.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.44 cmm certificate

Certificate. Syntax, description, parameters, and roles for **cmm certificate** command.

Syntax

Description

Generate signed certificate.

Parameters

Name

privateKeyFile - Private key file

Syntax

--privateKeyFile <string>

Description

Private key file. Type it in format <path/yourPrivateKeyFileName>

Class

Optional, not filterable on list.

Type / Range

Digit string of length

Name

serverCertificate - Server Certificate

Syntax

--serverCertificate <string>

Description

Certificate which is created and stored

Class

Output only.

Type / Range

Digit string of length

Name

signedCertFile - Signed certificate file

Syntax

--signedCertFile <string>

Description

Signed certificate file. Type it in format <path/yourSignedCertFileName>

Class

Optional, not filterable on list.

Type / Range

Digit string of length

Name

id - Id

Syntax

<list-item>

Description

Just type 'cert' here.

Class

Required value.

List

- cert - cert

Roles**Create**

secadmin

Show

secadmin

Delete

secadmin

List

secadmin

4.45 cmm cgInfo

CG status information. Syntax, description, parameters, and roles for cmm cgInfo command.

Syntax**Description**

Charging Gateway runtime information.

Parameters**Name**

cgOrder - Order of charging gateways in the CG table

Syntax

--cgOrder <integer>

Description

Order of the Charging GW to be displayed. If cgOrder is specified, the status information for the specific CG will be displayed, else the status information for all CGs in the CG table will be displayed.

Class

Optional.

Type / Range

Integer in the range of 1 to 10

Roles**List**

sysadmin, manage, monitor

4.46 cmm cgw

Charging Gateways. Syntax, description, parameters, and roles for cmm cgw command.

Syntax**Description**

Management of the list of known Charging Gateways used by Ga interface.

Parameters**Name**

cgOrder - CG order

Syntax

--cgOrder <integer>

Description

Order of the CG in the CG table.

Restrictions

- Order field is a primary key for the CG table.
- There can only be one record in the CG table with a specific order value.
- There cannot be two records with different order value but the same (IPv4,remPort) pair values.

Class

Required key.

Type / Range

Order of charging gateways in the CG table

Name

adminState - Administrative state

Syntax

--adminState <list-item>

Description

Administrative state of the charging gateway

Class

Required on create.

List

- locked - Locked.
- unlocked - Unlocked.

Name

cglp - Charging Gateway IP

Syntax

--cgIp <string>

Description

Charging Gateway IP

Class

Required on create.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

operState - Operational state

Syntax

--operState <integer>

Description

Operational state of the charging gateway. This is set only from call processing

Class

Read only.

Type / Range

Integer in the range of 0 to 13

Default

9

Name

redirectInd - Redirect Indicator

Syntax

--redirectInd <integer>

Description

Index of a CG where CDRs are redirected to. This is set only from call processing

Class

Read only.

Type / Range

Integer in the range of 0 to 255

Default

255

Name

remPort - Remote UDP port

Syntax

--remPort <integer>

Description

Remote UDP port

Class

Optional.

Type / Range

Remote Port

Default

3386

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.47 cmm chargingCdr

Charging CDR. Syntax, description, parameters, and roles for cmm chargingCdr command.

Syntax

Description

Show and list subscriber CDR.

Parameters

Name

imsi - IMSI

Syntax

--imsi <digits>

Description

IMSI.

Class

Required on create.

Type / Range

Digit string of length 5 to 15

Roles

Show

sysadmin, manage, monitor

List

sysadmin, manage, monitor

4.48 cmm chargingChangeGroup

Charging Change Group. Syntax, description, parameters, and roles for cmm chargingChangeGroup command.

Syntax

Description

Configure, list and show Charging Change Group.

Parameters

Name

chargingChangeGroupId - ChargingChangeGroup id

Syntax

--chargingChangeGroupId <integer>

Description

Charging change group index.

Class

Required key.

Type / Range

Integer in the range of 1 to 50

Name

chargeRateIndex - Charge rate index

Syntax

--chargeRateIndex <string>

Description

Defines the charge rates that are used within a day. the character & is permissible. this parameter is given as a decimal number with values 1 to 4. a maximum of eight indices may be defined and there must be one extra index defined compared to the number of charge rate change times. the indices must be given in a chronological order.

Class

Required on create.

Type / Range

Digit string of length 0 to 16 with the allowed pattern of ([1-4])(&[1-4]){,7}

Name

changeTime - ChangeTime

Syntax

--changeTime <string>

Description

The change time of the charge rate (hours-minutes) within a day. the character & is permissible. a maximum of seven change times may be defined. the change times must be given in a chronological order.

Class

Optional.

Type / Range

Digit string of length 0 to 42 with the allowed pattern of `(([0-1][0-9]|2[0-3])-(00|15|30|45))`
`(&([0-1][0-9]|2[0-3])-(00|15|30|45))){{,6}}`

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.49 cmm chargingCharAccessRestrictionProfile

Charging Characteristics Access Restriction Profile. Syntax, description, parameters, and roles for `cmm chargingCharAccessRestrictionProfile` command.

Syntax

Description

Defines parameters for charging characteristics based access restriction. Max 64 profiles are supported. The value of the T3346 Minimum must be set smaller than the value of the T3346 Maximum.

Parameters

Name

name - Name

Syntax

`--name <string>`

Description

The name of the profile.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of `[0-9a-zA-Z\.\_:@]*`

Name

chargingCharValue1 - Charging Characteristics Value 1

Syntax

--chargingCharValue1 <string>

Description

Defines the charging characteristics value 1 that triggers the access restriction.

Class

Required on create.

Type / Range

Digit string of length 4 with the allowed pattern of [0-9a-fA-F]*

Name

attachRejectCause - Attach Reject Cause

Syntax

--attachRejectCause <integer>

Description

Defines the Attach reject cause the MME uses.

Class

Optional.

Type / Range

Integer in the range of 3,6-8,11-15,22,25,35,42

Default

15

Name

chargingCharValue2 - Charging Characteristics Value 2

Syntax

--chargingCharValue2 <string>

Description

Defines the charging characteristics value 2 that triggers the access restriction. An empty value stands for this value being not in use.

Class

Optional.

Type / Range

Digit string of length 0 or 4 with the allowed pattern of [0-9a-fA-F]*

Default

""

Name

chargingCharValue3 - Charging Characteristics Value 3

Syntax

--chargingCharValue3 <string>

Description

Defines the charging characteristics value 3 that triggers the access restriction. An empty value stands for this value being not in use.

Class

Optional.

Type / Range

Digit string of length 0 or 4 with the allowed pattern of [0-9a-fA-F]*

Default

""

Name

chargingCharValue4 - Charging Characteristics Value 4

Syntax

--chargingCharValue4 <string>

Description

Defines the charging characteristics value 4 that triggers the access restriction. An empty value stands for this value being not in use.

Class

Optional.

Type / Range

Digit string of length 0 or 4 with the allowed pattern of [0-9a-fA-F]*

Default

""

Name

includeT3346 - Include T3346

Syntax

--includeT3346 <list-item>

Description

Defines whether MME includes T3346 value on the NAS rejection message. The MME only includes the T3346 in the rejection message when both this parameter is set to true and the Rejection cause is provisioned as 22 (congestion).

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

restrictAccess - Restrict Access

Syntax

--restrictAccess <list-item>

Description

Defines whether access is restricted based on provisioned charging characteristics value.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

t3346max - T3346 Maximum

Syntax

--t3346max <integer>

Description

Defines the maximum value for T3346.

Class

Optional.

Type / Range

Integer in the range of 0 to 11,160 in units of Second

Default

60

Name

t3346min - T3346 Minimum

Syntax

--t3346min <integer>

Description

Defines the minimum value for T3346.

Class

Optional.

Type / Range

Integer in the range of 0 to 11,159 in units of Second

Default

15

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.50 cmm chargingParams

Charging Parameters. Syntax, description, parameters, and roles for `cmm chargingParams` command.

Syntax**Description**

Configure, list and show Charging Parameters.

Parameters**Name**

chargingParamsId - ChargingParamsId

Syntax

--chargingParamsId <integer>

Description

ChargingParamsId - just set it to 1.

Class

Required key.

Type / Range

Integer in the range of 1 to 1

Name

cdrCreateDt - CdrCreateDt

Syntax

--cdrCreateDt <list-item>

Description

Direct tunnel S-CDRs generation mode off - S-CDRs for direct tunnel PDPs not generated on
- S-CDRs for direct tunnel PDPs generated.

Class

Optional.

List

- off - off.
- on - on.

Default

off

Name

cdrCreateGb - CdrCreateGb

Syntax

--cdrCreateGb <list-item>

Description

CDR generation mode for gb subscribers - all - S-CDR opening for roaming and home 2G
subscribers - home - S-CDR opening for home 2G subscribers - roam - S-CDR opening for
roaming 2G subscribers - none - no S-CDR opening for 2G subscribers.

Class

Optional.

List

- all - Specify CDR for both home and roaming subscribers.
- home - Specify CDR for home subscribers.
- none - Specify CDR for no subscribers.
- roam - Specify CDR for roaming subscribers.

Default

all

Name

cdrCreatelhspa - CdrCreatelhspa

Syntax

--cdrCreateIhspa <list-item>

Description

CDR generation mode for ihspa 3G subscribers - all - S-CDR opening for roaming and home ihspa 3G subscribers - home - S-CDR opening for home ihspa 3G subscribers - roam - S-CDR opening for roaming ihspa 3G subscribers - none - no S-CDR opening for ihspa 3G subscribers.

Class

Optional.

List

- all - Specify CDR for both home and roaming subscribers.
- home - Specify CDR for home subscribers.
- none - Specify CDR for no subscribers.
- roam - Specify CDR for roaming subscribers.

Default

all

Name

cdrCreatelu - CdrCreatelu

Syntax

--cdrCreateIu <list-item>

Description

CDR generation mode for iu subscribers - all - S-CDR opening for roaming and home 3G subscribers - home - S-CDR opening for home 3G subscribers - roam - S-CDR opening for roaming 3G subscribers - none - no S-CDR opening for 3G subscribers.

Class

Optional.

List

- all - Specify CDR for both home and roaming subscribers.
- home - Specify CDR for home subscribers.
- none - Specify CDR for no subscribers.
- roam - Specify CDR for roaming subscribers.

Default

all

Name

ctmNorm - Continuous trigger mode

Syntax

--ctmNorm <list-item>

Description

Continuous trigger mode - off - continuous trigger mode off for S-CDR triggers - timetr
- continuous trigger mode on for S-CDR time trigger the time trigger expires always as defined, that is, if parameter normal S-CDR time trigger has been set to 00-15 - 15 minutes, an intermediate S-CDR is generated in every 15 minutes with the closing cause - time limit
- even if another intermediate S-CDRs has been generated during this 15 minute period - with the closing cause - volume limit - or - max changes. - volumetr - continuous trigger mode on for 2G and 3G S-CDR vol trigger the volume trigger expires always as defined, that is, if parameter normal 2G data volume trigger is set to 500 KB, an intermediate S-CDR is generated after every 500 KB of data volumes with the closing cause - volume limit - even if another intermediate S-CDRs has been generated during this period - with the closing cause time limit or max changes. - both - continuous trigger mode on for S-CDR time and vol triggers continuous time and continuous volume triggers can be activated to be used simultaneously. basically this means that an intermediate S-CDR is generated every time when either time trigger or volume trigger expires even if there other intermediate S-CDRs have been generated during the collection period with closing causes max changes - volume limit - for continuous time trigger - and max changes and time limit - for continuous volume trigger.

Class

Optional.

List

- off - Continuous trigger mode off for S-CDR triggers.
- both - Both time and volume trigger - Continuous trigger mode on for S-CDR time and volume triggers. continuous time and continuous volume triggers can be activated to be used simultaneously. basically this means that an intermediate S-CDR is generated every time when either time trigger or volume trigger expires even if there other intermediate S-CDRs have been generated during the collection period with closing causes max changes - volume limit - for continuous time trigger - and max changes and time limit - for continuous volume trigger.
- timeTr - Time trigger - Continuous trigger mode on for S-CDR time trigger. the time trigger expires always as defined, that is, if parameter normal S-CDR time trigger has been set to 00-15 - 15 minutes, an intermediate S-CDR is generated in every 15 minutes with the closing cause - time limit - even if another intermediate S-CDRs has been generated during this 15 minute period - with the closing cause - volume limit - or - max changes. - .

- volumeTr - Volume trigger - Continuous trigger mode on for 2G and 3G S-CDR volume trigger. the volume trigger expires always as defined, that is, if parameter normal 2G data volume trigger is set to 500 KB, an intermediate S-CDR is generated after every 500 KB of data volumes with the closing cause - volume limit - even if another intermediate S-CDRs has been generated during this period - with the closing cause time limit or max changes.

Default

off

Name

direction - Direction

Syntax

--direction <list-item>

Description

CDR direction: off - Charging function disabled and no S-CDR generation bp - S-CDR generation with Bp interface on ga - S-CDR generation with Ga interface on

Restrictions

From ga or bp it is only possible to switch to off. From off, it is possible to switch to ga or bp.

Class

Optional.

List

- ga - S-CDR generation with bp interface on.
- off - Charging function disabled and no S-CDR generation.
- bp - S-CDR generation with bp interface on.

Default

off

Name

flatRate - Flat rate

Syntax

--flatRate <list-item>

Description

Flat rate S-CDR generation: off - no flat rate S-CDR generation on - flat rate S-CDR generation on.

Class

Optional.

List

- off - off.

- on - on.

Default

off

Name

minVol2g - Min volume trigger for normal 2G S-CDR

Syntax

--minVol2g <integer>

Description

Min volume trigger for normal 2G S-CDR - decimal value in KB, min 0 in 100 KB steps. normal vol2g value must be divisible by normal minvol2g value, if the normal minvol2g value is not 0 KB.

Class

Optional.

Type / Range

Integer in the range of 0 to 4,294,967,294

Default

3000

Name

minVol3g - Min volume trigger for normal 3G S-CDR

Syntax

--minVol3g <integer>

Description

Min volume trigger for normal 3G S-CDR - decimal value in KB, min 0 in 100 KB steps. normal vol3g value must be divisible by normal minvol3g value, if the normal minvol3g value is not 0 KB.

Class

Optional.

Type / Range

Integer in the range of 0 to 4,294,967,294

Default

15000

Name

periodicFileCloseTimer - Periodic File Close Timer

Syntax

--periodicFileCloseTimer <integer>

Description

This timer defines the period (in minutes) after which the SGSN Charging file will be closed and stored for retrieval even if the files provisioned size limit has not been reached. When the value is set to 0 (default) the files are not periodically closed.

Class

Optional.

Type / Range

Integer in the range of 0 to 1,440 in units of Minute

Default

0

Name

purgeCDRs - Purge S-CDRs

Syntax

--purgeCDRs <list-item>

Description

Purge and close all non-updated S-CDRs from database: - off - purge functionality off - 1day - purge S-CDRs not updated since last 24 h - 2day - purge S-CDRs not updated since last 48 h - 3day - purge S-CDRs not updated since last 72 h - 4day - purge S-CDRs not updated since last 96 h - 5day - purge S-CDRs not updated since last 120 h - 6day - purge S-CDRs not updated since last 144 h - 7day - purge S-CDRs not updated since last 168 h.

Class

Optional.

List

- off - Purge functionality off.
- 1day - 1 day - Purge S-CDRs not updated since last 24 hours.
- 2day - 2 days - Purge S-CDRs not updated since last 48 hours.
- 3day - 3 days - Purge S-CDRs not updated since last 72 hours.
- 4day - 4 days - Purge S-CDRs not updated since last 96 hours.
- 5day - 5 days - Purge S-CDRs not updated since last 120 hours.
- 6day - 6 days - Purge S-CDRs not updated since last 144 hours.
- 7day - 7 days - Purge S-CDRs not updated since last 168 hours.

Default

off

Name

sCdrDiscard - sCdr discard

Syntax

--sCdrDiscard <list-item>

Description

With this parameter operator can block the disk storing of S-CDRs for the charging parameter modification for direction, cdrCreateGb, cdrCreatelu or cdrCreatelhspa: off - the S-CDRs generated due to parameter modification are stored to disk on - the S-CDRs generated due to parameter modification are not stored to disk

Class

Optional.

List

- off - off.
- on - on.

Default

off

Name

t24h - Trigger for normal S-CDR, 24 hours

Syntax

--t24h <list-item>

Description

Trigger for normal S-CDR, 24 hours off - intermediate S-CDR is not generated due to T24H trigger at all on - intermediate S-CDR is generated due to T24H trigger if an S-CDR is open and its intermediate S-CDR has not been generated in the last 24 hours.

Class

Optional.

List

- off - off.
- on - on.

Default

off

Name

time - Time trigger for normal S-CDR

Syntax

--time <string>

Description

time trigger for normal S-CDR: hh-mm - minimum value 00-10, in one minute steps

Class

Optional.

Type / Range

Digit string of length 5 to 5 with the allowed pattern of (([0-2][0-9])-[0-5][0-9]))

Default

04-00

Name

ttcMode - Tunnel type change related S-CDR handling

Syntax

--ttcMode <list-item>

Description

Tunnel type change related S-CDR handling - blockingoncont - tunnel type change related container changes are blocked, no container change in S-CDR due to tunnel type change. - blockingoffcont - tunnel type change related container changes are not blocked, tunnel type change will cause container change in S-CDR. - blockingoncdr - tunnel type change related S-CDR generation is blocked, no S-CDR generation due to tunnel type change. - blockingoffcdr - tunnel type change related S-CDR generation is not blocked, tunnel type change will cause S-CDR generation. - blockingoncdrwodatavol - tunnel type change related S-CDR generation is blocked for S-CDRs without data volumes, only 2T S-CDRs with data volumes are generated due to tunnel type change.

Class

Optional.

List

- blockingOffCdr - Tunnel type change related S-CDR handling.
- blockingOffCont - Tunnel type change related container changes are not blocked, tunnel type change will cause container change in S-CDR. -.
- blockingOnCdr - Tunnel type change related S-CDR generation is blocked, no S-CDR generation due to tunnel type change.
- blockingOnCdrWoDatavol - Blocking on CDR without data volumes - Tunnel type change related S-CDR generation is blocked for S-CDRs without data volumes, only 2T S-CDRs with data volumes are generated due to tunnel type change.
- blockingOnCont - Blocking on container - Tunnel type change related container changes are blocked, no container change in S-CDR due to tunnel type change.

Default

blockingOnCont

Name

ttcTso - Tunnel type change related CDR timestamp override

Syntax

--ttcTso <list-item>

Description

Tunnel type change related CDR timestamp override off - timestamp override mechanism not in use on - timestamp override mechanism in use.

Class

Optional.

List

- off - off.
- on - on.

Default

off

Name

vol2g - Volume trigger for normal 2G S-CDR

Syntax

--vol2g <integer>

Description

Volume trigger for normal 2G S-CDR - decimal value in KB, min 500 in 100 KB steps. normal vol2g value must be divisible by normal minvol2g value, if the normal minvol2g value is not 0 KB.

Class

Optional.

Type / Range

Integer in the range of 500 to 4,294,967,294

Default

12000

Name

vol3g - Volume trigger for normal 3G S-CDR

Syntax

--vol3g <integer>

Description

Volume trigger for normal 3G S-CDR - decimal value in KB, min 1000 in 100 KB steps. normal vol3g value must be divisible by normal minvol3g value, if the normal minvol3g value is not 0 KB.

Class

Optional.

Type / Range

Integer in the range of 1,000 to 4,294,967,294

Default

60000

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.51 cmm chargingSpecialday

Charging Special day. Syntax, description, parameters, and roles for **cmm chargingSpecialday** command.

Syntax**Description**

Configure, list and show Charging Special day.

Parameters**Name**

date - Charging special day

Syntax

--date <string>

Description

Charging special day.

Class

Required key.

Type / Range

Digit string of length 0 to 10 with the allowed pattern of ((20[2-9][0-9])-(0[1-9])(1[0-2]))-(0[1-9]|[1-2][0-9]|3[0-1])), representing YYYY-MM-DD (year-month-day)

Name

changeGroupIndex - ChangeGroupIndex

Syntax

--changeGroupIndex <integer>

Description

Index to chargingchangeGroup.

Class

Required on create.

Type / Range

Integer in the range of 1 to 50

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.52 cmm chargingWeekday

Charging Weekday. Syntax, description, parameters, and roles for cmm chargingWeekday command.

Syntax**Description**

Configure, list and show Charging Weekday.

Parameters**Name**

weekday - Weekday

Syntax

--weekday <list-item>

Description

Is the day of the week and what changegroupindex may be linked to this day of the week.

Class

Required key.

List

- MON - Specify weekday of monday.
- TUE - Specify weekday of tuesday.
- WED - Specify weekday of wednesday.
- THU - Specify weekday of thursday.
- FRI - Specify weekday of friday.
- SAT - Specify weekday of saturday.
- SUN - Specify weekday of sunday.

Name

changeGroupIndex - ChangeGroupIndex

Syntax

--changeGroupIndex <integer>

Description

Index to chargingchangegroup.

Class

Optional.

Type / Range

Integer in the range of 0 to 50

Default

0

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

4.53 cmm ciotProfile

CloT Profile. Syntax, description, parameters, and roles for cmm ciotProfile command.

Syntax

Description

Defines profile for CloT settings to be used in IMSI Range Service

Parameters

Name

name - name

Syntax

--name <string>

Description

Profile Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

allowNonIpOverS1u - allow non IP over S1

Syntax

--allowNonIpOverS1u <list-item>

Description

specifies whether a non-IP connection is allowed over S1.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ciotEmmRegisterdWithoutPdn - Enable EMM-REGISTERED without PDN connection for C-IOT UE.
Enable EMM-REGISTERED without PDN connection for C-IOT UE

Syntax

--ciotEmmRegisterdWithoutPdn <list-item>

Description

Enable EMM-REGISTERED without PDN connection for C-IOT UE.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

cpCioTDeliveryMechanism - CP CIOT delivery mechanism

Syntax

--cpCioTDeliveryMechanism <list-item>

Description

CP CIOT delivery mechanism

Class

Optional.

List

- all - Delivery using both S11-u and T6a is supported.
- scef - Delivery using T6a interface.
- sgi - Delivery using S11-u interface.

Default

sgi

Name

ePcoSupport - Extended protocol configuration option support

Syntax

--ePcoSupport <list-item>

Description

Extended protocol configuration option support

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

hlcomDdnAckWithDlBuffering - HLCOM ddn ack with downlink dl buffering

Syntax

--hlcomDdnAckWithDlBuffering <list-item>

Description

DDN ACK with downlink buffering for HLCOM.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

hlcomEnabled - HLCOM enabled

Syntax

--hlcomEnabled <list-item>

Description

HLCOM enabled

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

hlcomPsmDlBufferingPacketCount - HLCOM/PSM downlink buffering packet count

Syntax

--hlcomPsmDlBufferingPacketCount <integer>

Description

PSM downlink buffering packet count for HLCOM and PSM.

Class

Optional.

Type / Range

Integer in the range of 0 to 65,535

Default

1

Name

hlcomUseLocalDlBufferingPacketCount - HLCOM use local downlink buffering packet count

Syntax

--hlcomUseLocalDlBufferingPacketCount <list-item>

Description

Use local downlink buffering packet count for HLCOM.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

mmeNbIotCptoUpSwitch - Enable MME triggered control plane bearers to user plane bearers switch for CIOT UE

Syntax

--mmeNbIotCptoUpSwitch <list-item>

Description

This parameters enables MME triggered control plane (CP) bearers to use user plane (UP) i.e. S1-U bearers for CIOT UE. if this parameter is set then MME triggers CP bearers to UP. By default the parameter is not set.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

mmeWbEutranCpToUpSwitch - Enable MME triggered control plane bearers to user plane bearers switch for WB-EUTRAN UE

Syntax

--mmeWbEutranCpToUpSwitch <list-item>

Description

This parameters enables MME triggered control plane (CP) bearers to use user plane (UP) i.e. S1-U bearers for WB-EUTRAN UE. if this parameter is set then MME triggers CP bearers to UP. By default the parameter is not set.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

moEdtCp - Mobile originated Early Data Transfer over Control Plane

Syntax

--moEdtCp <list-item>

Description

Mobile Originated Early Data Transfer for Control Plane for S11u and T6a

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

mtEdtCp - Mobile Terminated Early Data Transfer over Control Plane

Syntax

--mtEdtCp <list-item>

Description

Mobile Terminated Early Data Transmission for Control Plane

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

mtEdtT6a - Mobile Terminated Early Data Transmission for T6a

Syntax

--mtEdtT6a <list-item>

Description

Mobile Terminated Early Data Transmission for T6a

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

mtEdtUp - Mobile Terminated Early Data Transfer over User Plane

Syntax

--mtEdtUp <list-item>

Description

Mobile Terminated Early Data Transmission for User Plane

Class

Optional.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Name

mtSmsOpt2ForUeInPowerSavings - MT SMS option 2 deployment for UE in power savings

Syntax

--mtSmsOpt2ForUeInPowerSavings <list-item>

Description

Use deployment option 2 specified in TS 23.272 for handling mobile terminated SMS for UE in power savings (idle mode DRX and power savings mode). if this parameter is false then MME uses the option 1 specified in TS 23.272.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

nbIotDlDataSizeForCpToUpSwitch - Downlink data packet size threshold for CP bearers to UP bearers switch

Syntax

--nbIotDlDataSizeForCpToUpSwitch <integer>

Description

Provision the size of CIOT UE downlink control plane data packet size threshold in octets for MME to trigger control plane bearers to switch to use user plane bearers i.e. S1-U bearers. By default the data packet size is set to 120 octets. the range of allowed values are 20 to 1024 octets.

Class

Optional.

Type / Range

Integer in the range of 20 to 1,024

Default

120

Name

nbIotSmsWithoutCombinedAttach - Enable SMS without combined Attach For NB-IOT UE

Syntax

--nbiotSmsWithoutCombinedAttach <list-item>

Description

Enable SMS without combined Attach For NB-IOT UE

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

nBlotUeCpToUpSwitch - CIOT UE control plane bearer to user plane bearer switching

Syntax

--nBIotUeCpToUpSwitch <list-item>

Description

Parameter to allow/not allow CIOT UE control plane traffic to user plane i.e. setup radio bearers and transport user data over S1-U.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

nblotUlDataSizeForCpToUpSwitch - Upnlk data packet size threshold for CP bearers to UP bearers switch

Syntax

--nbIotUlDataSizeForCpToUpSwitch <integer>

Description

Provision the size of CIOT UE uplink control plane data packet size threshold in octets for MME to trigger control plane bearers to switch to use user plane bearers i.e. S1-U bearers. By default the data packet size is set to 120 octets. the range of allowed values are 20 to 1024 octets.

Class

Optional.

Type / Range

Integer in the range of 20 to 1,024

Default

120

Name

nonIpDataTrans - Non-IP data transport

Syntax

--nonIpDataTrans <list-item>

Description

Enable/disable non-ip data transport.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

rohc - ROHC

Syntax

--rohc <list-item>

Description

ROHC

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

s1uDataTrans - S1U data transfer

Syntax

```
--s1uDataTrans <list-item>
```

Description

Enable S1-U data transfer.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

serviceGapControlActive - Service Gap Control Active

Syntax

```
--serviceGapControlActive <list-item>
```

Description

Service Gap Control is active for UEs from the PLMN

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

serviceGapTimerSource - Service Gap Timer Source

Syntax

```
--serviceGapTimerSource <list-item>
```

Description

Specifies whether to use Service Gap Timer from HSS or Local provisioning

Class

Optional.

List

- Local - Use locally-provisioned Service Gap Timer

- HSS - Use HSS-provided Service Gap Timer

Default

HSS

Name

supportGrpSvcProvision - Support group service provision

Syntax

--supportGrpSvcProvision <list-item>

Description

Support group service provision

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

upCloT - UP CloT

Syntax

--upCIoT <list-item>

Description

Enable CP CIOT optimization.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

wbEutranDlDataSizeForCpToUpSwitch - Downlink data packet size threshold for CP bearers to UP bearers switch

Syntax

--wbEutranDlDataSizeForCpToUpSwitch <integer>

Description

Provision the size of WB-EUTRAN UE downlink control plane data packet size threshold in octets for MME to trigger control plane bearers to switch to use user plane bearers i.e. S1-U bearers. By default the data packet size is set to 120 octets. The range of allowed values are 20 to 1024 octets.

Class

Optional.

Type / Range

Integer in the range of 20 to 1,024

Default

120

Name

wbEutranUeCpToUpSwitch - WB-EUTRAN UE control plane bearers to user plane switching

Syntax

- -wbEutranUeCpToUpSwitch <list-item>

Description

Parameter to allow WB-EUTRAN UE control plane bearers switching to user plane i.e. set up radio bearers and transport user data over S1-U.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

wbEutranUlDataSizeForCpToUpSwitch - Uplink data packet size threshold for CP bearers to UP bearers switch

Syntax

- -wbEutranUlDataSizeForCpToUpSwitch <integer>

Description

Provision the size of WB-EUTRAN UE uplink control plane data packet size threshold in octets for MME to trigger control plane bearers to switch to use user plane bearers i.e. S1-U bearers. By default the data packet size is set to 120 octets. The range of allowed values are 20 to 1024 octets.

Class

Optional.

Type / Range

Integer in the range of 20 to 1,024

Default

120

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.54 cmm cipheringAlgorithm

Ciphering Algorithm. Syntax, description, parameters, and roles for cmm cipheringAlgorithm command.

Syntax

Description

This command sets the algorithms, in priority order, MME will use to negotiate encryption of NAS messages with the UE. At attach and TAU, the UE provides, in the UE EPS security capabilities, which encryption algorithms are supported. The MME will use the algorithm both supported by the UE and highest in priority in the eea ciphering algorithm priority list. - An algorithm must be chosen for 'priority-1' and 'priority-2'. - An additional selection, 'none', can be selected for 'priority-3' and 'priority-4' - effectively truncating the list. - If any ciphering algorithm (including null) is selected for 'priority-4', then 'priority-3' must not be 'none'. - An algorithm cannot be used for more than one priority (except for 'none'). Note: A pre-requisite for changing values in the cipheringAlgorithm is that MME aggregate service must be locked.

Parameters

Name

cipheringAlgPrio1 - Priority 1 ciphering algorithm

Syntax

--cipheringAlgPrio1 <list-item>

Description

Most preferred ciphering algorithm.

Class

Optional.

List

- aes - 128-EEA2 (AES) - Specify ciphering of 128-EEA2 (AES).
- noCiphering - 128-EEA0 (No ciphering) - Specify ciphering of 128-EEA0 (No ciphering).
- none - No algorithm selected - Specify ciphering of no algorithm selected.
- snow3G - 128-EEA1 (SNOW 3G) - Specify ciphering of 128-EEA1 (SNOW 3G).
- zuc - 128-EEA3 (ZUC) - Specify ciphering of 128-EEA3 (ZUC).

Default

snow3G

Name

cipheringAlgPrio2 - Priority 2 ciphering algorithm

Syntax

--cipheringAlgPrio2 <list-item>

Description

Second most preferred ciphering algorithm; used if the user element does not support algorithm(s) of higher priority.

Class

Optional.

List

- aes - 128-EEA2 (AES) - Specify ciphering of 128-EEA2 (AES).
- noCiphering - 128-EEA0 (No ciphering) - Specify ciphering of 128-EEA0 (No ciphering).
- none - No algorithm selected - Specify ciphering of no algorithm selected.
- snow3G - 128-EEA1 (SNOW 3G) - Specify ciphering of 128-EEA1 (SNOW 3G).
- zuc - 128-EEA3 (ZUC) - Specify ciphering of 128-EEA3 (ZUC).

Default

aes

Name

cipheringAlgPrio3 - Priority 3 ciphering algorithm

Syntax

--cipheringAlgPrio3 <list-item>

Description

Third most preferred ciphering algorithm; used if the user element does not support algorithm(s) of higher priority.

Class

Optional.

List

- aes - 128-EEA2 (AES) - Specify ciphering of 128-EEA2 (AES).
- noCiphering - 128-EEA0 (No ciphering) - Specify ciphering of 128-EEA0 (No ciphering).
- none - No algorithm selected - Specify ciphering of no algorithm selected.
- snow3G - 128-EEA1 (SNOW 3G) - Specify ciphering of 128-EEA1 (SNOW 3G).
- zuc - 128-EEA3 (ZUC) - Specify ciphering of 128-EEA3 (ZUC).

Default

none

Name

cipheringAlgPrio4 - Priority 4 ciphering algorithm

Syntax

--cipheringAlgPrio4 <list-item>

Description

Fourth most preferred ciphering algorithm; used if the user element does not support algorithm(s) of higher priority.

Class

Optional.

List

- aes - 128-EEA2 (AES) - Specify ciphering of 128-EEA2 (AES).
- noCiphering - 128-EEA0 (No ciphering) - Specify ciphering of 128-EEA0 (No ciphering).
- none - No algorithm selected - Specify ciphering of no algorithm selected.
- snow3G - 128-EEA1 (SNOW 3G) - Specify ciphering of 128-EEA1 (SNOW 3G).
- zuc - 128-EEA3 (ZUC) - Specify ciphering of 128-EEA3 (ZUC).

Default

none

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

4.55 cmm closeCdr

close CDR. Syntax, description, parameters, and roles for cmm c loseCdr command.

Syntax

Description

Close CDR

Parameters

Name

transferType - transferType

Syntax

--transferType <list-item>

Description

Transfer type of the S-CDRs to close.Offline means that Charging will go into offline mode afterwards.PAPS restart is then needed

Class

Required on create.

List

- currentFile - Current File - Specify transfer type of Current File.
- offline - Specify transfer type of offline.

Roles

Create

sysadmin, manage

4.56 cmm clusterMgmt

Cluster Management. Syntax, description, parameters, and roles for cmm c lusterMgmt command.

Syntax

Description

This command is used to manage clusters.

Parameters

Name

args - Arguments

Syntax

--args <string>

Description

Argument for the specified command that takes parameter.

Class

Optional.

Type / Range

Digit string of length

Name

target - Target

Syntax

--target <string>

Description

Target host. Example: ipds0 . Pattern matching is supported with pattern value surrounded by double quotes. Pattern value examples: *, necc*, ?p?s*

Class

Required.

Type / Range

Digit string of length

Roles

sysadmin, manage

4.57 cmm cmpPki

CMP PKI Connection. Syntax, description, parameters, and roles for cmm cmpPki command.

Syntax

Description

This command will set the parameters for a PKI connection using CMPv2.

Parameters

Name

pkild - PKI CMP Name

Syntax

--pkiId <integer>

Description

PKI CMP Name

Class

Required key.

Type / Range

Integer in the range of 1 to 10

Name

referenceNumber - PKI Reference Number

Syntax

--referenceNumber <string>

Description

Reference Number for PKI CMP interactions

Class

Required on create.

Type / Range

Digit string of length 1 to 64 with the allowed pattern of [0-9a-zA-Z\._:]*

Name

remotelp - Remote IP Address

Syntax

--remoteIp <string>

Description

Remote IP-address.

Class

Required on create.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

sharedKey - Shared Key

Syntax

--sharedKey <string>

Description

Shared key.

Restrictions

CMM shall encrypt the sharedKey in the Database

Class

Required on create.

Type / Range

String length of 1...256

Name

cmmDn - CMM distinguished name

Syntax

--cmmDn <string>

Description

Distinguished Name for CMM CMP

Class

Optional.

Type / Range

Distinguished name in CMPv2

Default

""

Name

pkiDn - Distinguished Name

Syntax

--pkiDn <string>

Description

Distinguished Name for PKI CMP

Class

Optional.

Type / Range

Distinguished name in CMPv2

Default

""

Name

remotePath - Remote path

Syntax

--remotePath <string>

Description

Remote path of the CMPv2 server

Class

Optional.

Type / Range

Remote path of the CMPv2 server

Default

/

Name

remotePort - CMPv2 Remote Port

Syntax

--remotePort <integer>

Description

CMPv2 Remote Port

Class

Optional.

Type / Range

PKI CMP remote port

Default

80

Roles**Create**

secadmin

Show

secadmin

Modify

secadmin

Delete

secadmin

List

secadmin

4.58 cmm cmpProc

CMPv2 Procedure. Syntax, description, parameters, and roles for `cmm cmpProc` command.

Syntax

Description

This command specifies the CMPv2 procedure to be executed.

Parameters

Name

pkild - PKI CMP Name

Syntax

--pkid <integer>

Description

PKI CMP Name

Class

Required key.

Type / Range

Integer in the range of 1 to 10

Name

cmpProcedureType - CMP Procedure Type

Syntax

--cmpProcedureType <list-item>

Description

This parameter specifies the CMPv2 Procedure to execute.

Class

Required on create, not modifiable.

List

- cr - Certificate Update - CMPv2 Certificate Update
- crl - Certificate Revocation List - CMPv2 Certificate Revocation List Request
- ir - Initial Request - CMPv2 Initial Request
- kur - Key Pair Update - CMPv2 Key Pair Update

Name

privateKeyFile - Private Key File Name

Syntax

--privateKeyFile <string>

Description

Absolute path of the private key file name. Example: /tmp/privkey.pem

Class

Required on create.

Type / Range

Digit string of length 1 to 100 with the allowed pattern of [0-9a-zA-Z\.\-_:&(),/]*

Name

signedCertFile - Signed Certificate File Name

Syntax

--signedCertFile <string>

Description

Absolute path of the signed certificate file name. Example: /tmp/certificate

Class

Required on create.

Type / Range

Digit string of length 1 to 100 with the allowed pattern of [0-9a-zA-Z\.\-_:&(),/]*

Name

tlsCertType - TLS Certificate Type

Syntax

--tlsCertType <list-item>

Description

Type of TLS Certificate to be generated with CMPv2

Class

Required on create.

List

- client - Client TLS certificate
- server - Server TLS certificate

Name

caCertificateFile - Certified Authority Certificate File

Syntax

--caCertificateFile <string>

Description

Absolute path of the signed certificate file name. Example: '/tmp/certificate'

Class

Optional.

Type / Range

Digit string of length 0 to 100 with the allowed pattern of [0-9a-zA-Z\.\-_:&(),/]*

Default

""

Name

cmpProcInfo - CMP Proc Info

Syntax

--cmpProcInfo <string>

Description

Information about command execution

Class

Read only.

Type / Range

Digit string of length 0 to 128

Roles**Create**

secadmin

4.59 cmm cnfLcmGParms

CNF CMM global parameters for LCM operations. Syntax, description, parameters, and roles for cmm cnfLcmGParms command.

Syntax**Description**

This command is used to modify global parameters that control the behavior of CNF CMM's operator pod in applicable LCM operations.

Parameters**Name**

cmGparmName - Parameter Name

Syntax

--cmGparmName <list-item>

Description

Name of the global parameter

Class

Required key.

List

- mmelssuWithoutSrs - MME ISSU Without SRS - When set to true, this parameter instructs the CMM operator pod to assume that SRS for MME is not available and make it allow a predefined amount of time between the update of EMMS and DBS layers for UE activity that will cause the 4G UEs to be assigned again on a CPPS. The waiting time is defined via the MME ISSU Without SRS Time for Sync global CNF LCM parameter.
- mmelssuWithoutSrsTimeForSync - MME ISSU Without SRS Waiting Time Between DBS and EMMS Upgrade - This parameter defines the amount of time in seconds that the CMM operator pod will allow between the update of EMMS and DBS layers for UE activity that will cause the 4G UEs to be assigned again on a CPPS. This parameter is only effective if MME ISSU Without SRS is true.

Name

cmGparmValue - Parameter Value

Syntax

--cmGparmValue <string>

Description

Value of parameter

Class

Required on create.

Type / Range

Digit string of length 0 to 64

The cmGparmValue string is based on the cmGparmName parameter:

Name

mmelssuWithoutSrs - MME ISSU Without SRS

Syntax

--cmGparmName mmeIssuWithoutSrs --cmGparmValue <string>

Description

When set to true, this parameter instructs the CMM operator pod to assume that SRS for MME is not available and make it allow a predefined amount of time between the update of EMMS and DBS layers for UE activity that will cause the 4G UEs to be assigned again on a

CPPS. The waiting time is defined via the MME ISSU Without SRS Time for Sync global CNF LCM parameter.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

mmeIssuWithoutSrsTimeForSync - MME ISSU Without SRS Waiting Time Between DBS and EMMS Upgrade

Syntax

```
--cmGparmName mmeIssuWithoutSrsTimeForSync --cmGparmValue <string>
```

Description

This parameter defines the amount of time in seconds that the CMM operator pod will allow between the update of EMMS and DBS layers for UE activity that will cause the 4G UEs to be assigned again on a CPPS. This parameter is only effective if MME ISSU Without SRS is true.

Type / Range

Integer in the range of 1 to 3,600

Default

900

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

4.60 cmm combinedAttachWithoutCsProfile

Combined Attach without CS Profile. Syntax, description, parameters, and roles for cmm combinedAttachWithoutCsProfile command.

Syntax

Description

Define the parameters for supporting combined attach without CS domain. Maximum number of profiles is 128.

Parameters

Name

name - Name

Syntax

--name <string>

Description

The name of the profile defining parameters for combined attached without CS.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

checkNetworkAccessMode - Check Network Access Mode

Syntax

--checkNetworkAccessMode <list-item>

Description

Defines whether the MME checks the Network Access Mode value from the HSS. When set to false, MME is able to accept combined attach without CS despite of the Network Access Mode parameter value from HSS. When set to true, MME only accepts combined attach without CS when Network Access Mode equals to PACKET_AND_CIRCUIT.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

combinedAttachWithoutCsEnabled - Combined Attach without CS Enabled

Syntax

--combinedAttachWithoutCsEnabled <list-item>

Description

Defines whether Combined Attach without CS is enabled for the UE.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

extendedServiceRequestRejectCc - Extended Service Request Reject CC

Syntax

--extendedServiceRequestRejectCc <integer>

Description

Defines the EMM cause code used to reject Extended Service Request when Combined Attach without CS is enabled.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

18

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.61 cmm connectedEnGnbQuery

Connected en-gNB Query. Syntax, description, parameters, and roles for cmm connectedEnGnbQuery command.

Syntax

Description

Dump command displays connected en-gNB List IE included in the S1 SETUP REQUEST message

Parameters

Name

enbId - engNB ID

Syntax

--enbId <integer>

Description

engNB identifier as specified in TS 36.413 v9.2.1.37

Class

Required key.

Type / Range

Integer in the range of 0 to 268,435,455

Name

mcc - Mobile country code

Syntax

--mcc <digits>

Description

3-digit number used to represent the mobile country code.

Class

Required key.

Type / Range

Digit string of length 3

Name

mnc - Mobile network code

Syntax

--mnc <digits>

Description

Digits used to represent mobile network code.

Class

Required key.

Type / Range

Digit string of length 2 to 3

Name

enGnbid - en Gnb Id

Syntax

--enGnbid <integer>

Description

engNB identifier

Class

Read only.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

enGnbMcc - en Gnb MCC

Syntax

--enGnbMcc <digits>

Description

3-digit number used to represent the mobile country code.

Class

Read only.

Type / Range

Digit string of length 3

Name

enGnbMnc - Mobile network code

Syntax

--enGnbMnc <digits>

Description

Digits used to represent mobile network code.

Class

Read only.

Type / Range

Digit string of length 2 to 3

Roles

List

sysadmin, manage, monitor

4.62 cmm consumerOAuthNfType

Consumer OAuth NF Type. Syntax, description, parameters, and roles for cmm consumerOAuthNfType command.

Syntax

Description

List of the NFTypes for which AMF supports OAuth as a Consumer.

Parameters

Name

consumerOAuthNfTypeListName - List Name

Syntax

--consumerOAuthNfTypeListName <string>

Description

Name of the list for which the NFType is relevant

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

nfType - NF Type

Syntax

--nfType <list-item>

Description

NFType of the consumer interface

Class

Required key.

List

- 5GEIR - 5G EIR - The network function type of 5G-Equipment Identity Register.

- AMF - Access and Mobility Manager Function - The network function type of Access and Mobility Manager Function.
- AUSF - Authentication Server Function - The network function type of Authentication Server Function.
- LMF - Location Management Function - The network function type of Location Management Function.
- PCF - Policy Control Function - The network function type of Policy Control Function.
- SMF - Session Management Function - The network function type of Session Management Function.
- SMSF - Short Message Service Function - The network function type of Short Message Service Function.
- UDM - Unified Data Manager - The network function type of Unified Data Manager.

Name

enable - Enable

Syntax

--enable <list-item>

Description

Enables OAuth2.0 as a Consumer for a NFType

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

4.63 cmm consumerOAuthNfTypeList

Consumer OAuth 2.0 NfType List. Syntax, description, parameters, and roles for cmm consumerOAuthNfTypeList command.

Syntax

Description

List of NfTypes for which AMF shall support OAuth as a Consumer.

Parameters

Name

name - Name

Syntax

--name <string>

Description

Name of the list

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.64 cmm counter

Counter List. Syntax, description, parameters, and roles for cmm counter command.

Syntax

Description

This command is used to list counters, provide counter meta information, or show counters.

Parameters

Name

endTime - Ending time

Syntax

--endTime <string>

Description

Ending time('yyyy-mm-dd HH:MM:SS') Only the data of maximum last 15 minutes will be returned.

Class

Optional.

Type / Range

Digit string of length 19 to 19 with the allowed pattern of (\d{4}-\d{2}-\d{2} \d{2}:\d{2}:\d{2})

Name

entity - Entity filter

Syntax

--entity <string>

Description

Entity filter.

Class

Optional.

Type / Range

Digit string of length

Name

entity_actc - Entity ACTC

Syntax

--entity_actc <string>

Description

Entity ACTC.

Class

Optional.

Type / Range

Digit string of length

Name

entity_bearerclass - Entity bearer class

Syntax

--entity_bearerclass <string>

Description

Entity bearer class.

Class

Optional.

Type / Range

Digit string of length

Name

entity_cellid - Entity cell ID

Syntax

--entity_cellid <string>

Description

Entity cell ID.

Class

Optional.

Type / Range

Digit string of length

Name

entity_core - Entity core

Syntax

--entity_core <string>

Description

Entity core.

Class

Optional.

Type / Range

Digit string of length

Name

entity_disk - Entity disk

Syntax

--entity_disk <string>

Description

Entity disk.

Class

Optional.

Type / Range

Digit string of length

Name

entity_filesystem - Entity filesystem

Syntax

--entity_filesystem <string>

Description

Entity filesystem.

Class

Optional.

Type / Range

Digit string of length

Name

entity_hss - Entity HSS

Syntax

--entity_hss <string>

Description

Entity HSS.

Class

Optional.

Type / Range

Digit string of length

Name

entity_interface - Entity interface

Syntax

--entity_interface <string>

Description

Entity interface.

Class

Optional.

Type / Range

Digit string of length

Name

entity_interfacetype - Entity interface type

Syntax

--entity_interfacetype <string>

Description

Entity interface type.

Class

Optional.

Type / Range

Digit string of length

Name

entity_iulac - Entity IU LAC

Syntax

--entity_iulac <string>

Description

Entity IU LAC.

Class

Optional.

Type / Range

Digit string of length

Name

entity_iuraplmn - Entity IU RA PLMN

Syntax

--entity_iuraplmn <string>

Description

Entity IU RA PLMN.

Class

Optional.

Type / Range

Digit string of length

Name

entity_iurarac - Entity IU RA RAC

Syntax

--entity_iurarac <string>

Description

Entity IU RA RAC.

Class

Optional.

Type / Range

Digit string of length

Name

entity_iurarnc - Entity IU RA RNC

Syntax

--entity_iurarnc <string>

Description

Entity IU RA RNC.

Class

Optional.

Type / Range

Digit string of length

Name

entity_iusalac - Entity IU SA LAC

Syntax

--entity_iusalac <string>

Description

Entity IU SA LAC.

Class

Optional.

Type / Range

Digit string of length

Name

entity_iusaplmn - Entity IU SA PLMN

Syntax

--entity_iusaplmn <string>

Description

Entity IU SA PLMN.

Class

Optional.

Type / Range

Digit string of length

Name

entity_lac - Entity LAC

Syntax

--entity_lac <string>

Description

Entity LAC.

Class

Optional.

Type / Range

Digit string of length

Name

entity_lip - Entity local IP

Syntax

--entity_lip <string>

Description

Entity local IP.

Class

Optional.

Type / Range

Digit string of length

Name

entity_liptp - Entity local IP type

Syntax

--entity_liptp <string>

Description

Entity local IP type.

Class

Optional.

Type / Range

Digit string of length

Name

entity_ludp - Entity local UDP port

Syntax

--entity_ludp <string>

Description

Entity local UDP port.

Class

Optional.

Type / Range

Digit string of length

Name

entity_nsei - Entity NSEI

Syntax

--entity_nsei <string>

Description

Entity NSEI.

Class

Optional.

Type / Range

Digit string of length

Name

entity_nsvci - Entity NSVCI

Syntax

--entity_nsvci <string>

Description

Entity NSVCI.

Class

Optional.

Type / Range

Digit string of length

Name

entity_olac - Entity old LAC

Syntax

--entity_olac <string>

Description

Entity old LAC.

Class

Optional.

Type / Range

Digit string of length

Name

entity_operdefqci - Entity operdefqci

Syntax

--entity_operdefqci <string>

Description

Entity operdefqci.

Class

Optional.

Type / Range

Digit string of length

Name

entity_oplmn - Entity old PLMN

Syntax

--entity_oplmn <string>

Description

Entity old PLMN.

Class

Optional.

Type / Range

Digit string of length

Name

entity_orac - Entity old RAC

Syntax

--entity_orac <string>

Description

Entity old RAC.

Class

Optional.

Type / Range

Digit string of length

Name

entity_ornc - Entity old RNC

Syntax

--entity_ornc <string>

Description

Entity old RNC.

Class

Optional.

Type / Range

Digit string of length

Name

entity_pagingtype - Entity pagingtype

Syntax

--entity_pagingtype <string>

Description

Entity pagingtype.

Class

Optional.

Type / Range

Digit string of length

Name

entity_partner - Entity partner

Syntax

--entity_partner <string>

Description

Entity partner.

Class

Optional.

Type / Range

Digit string of length

Name

entity_plmn - Entity PLMN

Syntax

--entity_plmn <string>

Description

Entity PLMN.

Class

Optional.

Type / Range

Digit string of length

Name

entity_pltc - Entity PLTC

Syntax

--entity_pltc <string>

Description

Entity PLTC.

Class

Optional.

Type / Range

Digit string of length

Name

entity_rac - Entity RAC

Syntax

--entity_rac <string>

Description

Entity RAC.

Class

Optional.

Type / Range

Digit string of length

Name

entity_rautype - Entity RAU type

Syntax

--entity_rautype <string>

Description

Entity RAU type.

Class

Optional.

Type / Range

Digit string of length

Name

entity_reselection - Entity reselection

Syntax

--entity_reselection <string>

Description

Entity reselection.

Class

Optional.

Type / Range

Digit string of length

Name

entity_rip - Entity remote IP

Syntax

--entity_rip <string>

Description

Entity remote IP.

Class

Optional.

Type / Range

Digit string of length

Name

entity_riotp - Entity remote IP type

Syntax

--entity_riotp <string>

Description

Entity remote IP type.

Class

Optional.

Type / Range

Digit string of length

Name

entity_rncid - Entity RNC ID

Syntax

--entity_rncid <string>

Description

Entity RNC ID.

Class

Optional.

Type / Range

Digit string of length

Name

entity_rudp - Entity remote UDP port

Syntax

--entity_rudp <string>

Description

Entity remote UDP port.

Class

Optional.

Type / Range

Digit string of length

Name

entity_sac - Entity SAC

Syntax

--entity_sac <string>

Description

Entity SAC.

Class

Optional.

Type / Range

Digit string of length

Name

entity_snet - Entity signaling network

Syntax

--entity_snet <string>

Description

Entity signaling network.

Class

Optional.

Type / Range

Digit string of length

Name

entity_spc - Entity signaling point code

Syntax

--entity_spc <string>

Description

Entity signaling point code.

Class

Optional.

Type / Range

Digit string of length

Name

entity_ssname - Entity subsystem name

Syntax

--entity_ssname <string>

Description

Entity subsystem name.

Class

Optional.

Type / Range

Digit string of length

Name

entity_ssnumber - Entity subsystem number

Syntax

--entity_ssnumber <string>

Description

Entity subsystem number.

Class

Optional.

Type / Range

Digit string of length

Name

entity_tac - Entity TAC

Syntax

--entity_tac <string>

Description

Entity TAC.

Class

Optional.

Type / Range

Digit string of length

Name

group - CounterGroup PID

Syntax

--group <string>

Description

Counter Group PID(e.g. m128).

Class

Optional.

Type / Range

Digit string of length

Name

mode - Mode

Syntax

--mode <string>

Description

When set to 'detailed', show the counter definition when a counter value is unavailable.

Class

Optional.

Type / Range

Digit string of length

Name

period - Time period

Syntax

--period <string>

Description

Time period only within the last 15 minutes (e.g. 2m, 5m, 10m)

Class

Optional.

Type / Range

Digit string of length

Name

pid - Counter PID

Syntax

--pid <string>

Description

Counter PID(e.g. m101c001).

Class

Optional.

Type / Range

Digit string of length

Name

source - Source filter

Syntax

--source <string>

Description

Source filter.

Class

Optional.

Type / Range

Digit string of length

Name

sourceHost - Source hostname

Syntax

--sourceHost <string>

Description

Source hostname.

Class

Optional.

Type / Range

Digit string of length

Name

sourceIp - Source hostip

Syntax

--sourceIp <string>

Description

Source hostip.

Class

Optional.

Type / Range

Digit string of length

Name

startTime - Starting time

Syntax

--startTime <string>

Description

Starting time('yyyy-mm-dd HH:MM:SS') Only the data of maximum last 15 minutes will be returned.

Class

Optional.

Type / Range

Digit string of length 19 to 19 with the allowed pattern of (\d{4}-\d{2}-\d{2} \d{2}:\d{2}:\d{2})

Roles**List**

sysadmin, manage, monitor

Meta

sysadmin, manage

Show

sysadmin, manage, monitor

4.65 cmm counterDefinition

Counter Definition. Syntax, description, parameters, and roles for cmm counterDefinition command.

Syntax**Description**

This command has two operations: The list operation is used to list counter definitions. The show operation is used to show counter definitions. Note: Before CMM195BR, the command was cmm counter definition instead of cmm counterDefinition. For backwards compatibility, both versions of the command are supported.

Parameters**Name**

group - Counter group

Syntax

--group <string>

Description

CounterGroup PID(e.g. m101c0).

Class

Optional.

Type / Range

Digit string of length

Roles**List**

sysadmin, manage, monitor

Show

sysadmin, manage, monitor

4.66 cmm counterGroupDefinition

Counter Group Definition List. Syntax, description, parameters, and roles for cmm counterGroupDefinition command.

Syntax**Description**

This command is used to list or show counter group definitions. Note: Before CMM195BR, the command was cmm counterGroup definition instead of cmm counterGroupDefinition. For backwards compatibility, both versions of the command are supported.

Parameters

N/A

Roles**List**

sysadmin, manage, monitor

Show

sysadmin, manage, monitor

4.67 cmm cpi

Critical Performance Indicator Threshold. Syntax, description, parameters, and roles for cmm cpi command.

Syntax

Description

Administer the Critical Performance Indicator Threshold

Parameters

Name

cpiService - CPI service

Syntax

--cpiService <list-item>

Description

CPI service.

Class

Required key.

List

- CPPS - Control plane processing service - Specify service of control plane processing service.
- IPDS - Internet protocol director service - Specify service of internet protocol director service.
- PAPS - Packet processing service - Specify service of packet processing service.

Name

cpiElement - CPI element

Syntax

--cpiElement <list-item>

Description

CPI element name.

Class

Required key.

List

- AttachFailures - Failure rate for attach procedures.

- AttachFailuresSysRelated - Failure rate for system-related attach procedures.
- AttachWithPGWreselection - PGW reselection rate during attach.
- AttachWithSGWreselection - SGW reselection rate during attach.
- CreateDedicatedBearerFailures - Failure rate for create dedicated bearer procedure.
- DeactivateDedBearerFailures - Failure rate for deactivate dedicated bearer.
- EIRfailuresS13 - Failure rate for unsuccessful equipmentcheckrequests (ECRs) to the number of ECRs attempted.
- ExtServiceReqFailuresSysRelated - System related extended service request failures.
- ExtServiceRequestFailures - Extended service request failures.
- FailuresOverSGs - Failure rate for procedures between the MME and the UE context using SGs.
- GTPcResponseTOS3 - Failure rate for GTPc response to S3.
- GTPcResponseTO_Gn - Failure rate for GTPc expected response to messages timed out (Gn interface).
- GTPcResponseTO_S10 - Failure rate for GTP-C expected response messages timed out (S10 interface).
- GTPcResponseTO_S11 - Failure rate for GTP-C expected response messages timed out (S11 interface).
- GTPcResponseTO_Sv - Failure rate for GTPc expected response to messages timed out (Sv interface).
- HOFailuresTo3G2GOverGn - Failure rate for handover to UTRAN/GERAN scenarios over gn.
- HOFailuresToLTEOverGn - Failure rate for handover from UTRAN/GERAN scenarios over gn.
- HOfailuresFromGERANoverS3 - Handover from GERAN over S3.
- HOfailuresFromUTRANoverS3 - Handover failures from UTRAN over S3.
- HOfailuresRAUto2G3GOverS3 - Handover failures RAU to 2G3G over S3.
- HOfailuresRAUto2G3GnewSgwOverS3 - HOfailuresRAUto2G3GnewSgwOverS3 over S3 with SGW relocation.
- HOfailuresRAUto2G3GsameSgwOverS3 - RAU-Based handover to UTRAN/GERAN over S3 without SGW relocation.
- HOfailuresToGERANoverS3 - Handover to GERAN over S3.
- HOfailurestoUTRANoverS3 - Handover failures to UTRAN over S3.

- HOwMMERelocFailures_atSource - Failure rate for handover with MME relocation scenario (at a source MME).
- HOwMMERelocFailures_atTarget - Failure rate for handover with MME relocation scenario (at a target MME).
- HOwNoRelocFailures - Failure rate for handover without MME relocation.
- HOwSGWrelocFailures - Failure rate for handover without MME relocation and with SGW relocation scenario.
- HSSauthFailures - Failure rate for HSS authentication.
- HSSreselection - HSS reselection rate.
- MBMSSessionStartM3FailureRate - MBMS session start M3 failure rate.
- MBMSSessionStartSmFailureRate - MBMS session start sm failure rate.
- MBMSSessionStopM3FailureRate - MBMS session stop M3 failure rate.
- MBMSSessionStopSmFailureRate - MBMS session stop sm failure rate.
- MBMSSessionUpdateM3FailureRate - MBMS session update M3 failure rate.
- MBMSSessionUpdateSmFailureRate - MBMS session update sm failure rate.
- MMEUEcapacity - MME capacity thresholds.
- MobileTermLocRequestFailures - Mobile termination location request failures.
- NetwrkInducedLocRequestFailures - Network induced location request failures.
- NoPSHOFailuresOverSv - Failure rate for hand down to UTRAN/GERAN over Sv without PSHO.
- PDNconnWithPGWreselection - PGW reselection rate during PDN connectivity procedures.
- PSHOFailuresOverSv - Failure rate for hand down to UTRAN/GERAN over Sv with PSHO.
- S1MMEconnFailures - Failure rate for S1-MME failed connection attempts.
- S3TauFailures - S3 TAU failures.
- S3TauFailuresInterSgw - S3 TAU intersgw failures.
- S3TauFailuresIntraSGW - S3 TAU intrasgw failures.
- ServiceReqFailuresSysRelated - Failure rate for system-related UE service request procedures.
- ServiceRequestFailures - Failure rate for UE service request procedures.
- Sgsn2GattachFailures - GPRS attach failure in GERAN.
- Sgsn2GpagingFailures - PS paging failure in GERAN.
- Sgsn2GpdpActFailures - MS initiated PDP context activation failure in GERAN.

- Sgsn2GpdpDeactmsFailures - MS initiated PDP CONTEXT deactivation failure in GERAN.
- Sgsn2GrauFailures - RAU failure rate in GERAN.
- Sgsn3GattachFailures - GPRS attach failure in UTRAN.
- Sgsn3GpagingFailures - PS paging failure in UTRAN.
- Sgsn3GpdpActFailures - MS initiated PDP context activation failure in UTRAN.
- Sgsn3GpdpDeactmsFailures - MS initiated PDP context deactivation failure in UTRAN.
- Sgsn3GrauFailures - RAU failure rate in UTRAN.
- SgsnGtpcResponseTO_S16 - SGSN GTP-C expected response time out for S16.
- SgsnGtpcResponseTO_S3 - SGSN GTP-C expected response time out for S3.
- SgsnGtpcResponseTO_S4 - SGSN GTP-C expected response time out for S4.
- StopWarnMsgDeliveryS1MMEFailureRate - Stop warning message delivery S1MME failure rate.
- StopWarnMsgDeliverySBcFailureRate - Stop warning message delivery SBc failure rate.
- TauFailures - Failure rate for TAU scenario.
- TauFailuresInterMme - Failure rate for TAU inter-mme scenario.
- TauFailuresInterMmeInterSgw - Failure rate for TAU procedures for inter MME and inter SGW relocation.
- TauFailuresInterSgw - Failure rate for TAU inter-sgw scenario.
- UEauthFailures - UE authentication failures.
- UpdateBearerFailures - Failure rate for update bearer.
- UpdateDedicatedBearerFailures - Failure rate for update dedicated bearer.
- WarnMsgDeliveryS1MMEFailureRate - Warning message delivery S1MME failure rate.
- WarnMsgDeliverySBcFailureRate - Warning message delivery SBc failure rate.

Name

cpiCritical - Critical alarm

Syntax

--cpiCritical <integer>

Description

CPI threshold for critical alarm.

Class

Optional.

Type / Range

Integer in the range of -1 or 1 to 99 in units of %

Name

cpiEnabled - CPI enable

Syntax

--cpiEnabled <list-item>

Description

When checked (true), defines that CPIs are enabled.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Name

cpiMajor - Major alarm

Syntax

--cpiMajor <integer>

Description

CPI threshold for major alarm.

Class

Optional.

Type / Range

Integer in the range of -1 or 1 to 99 in units of %

Name

cpiMinAttempts - Min number of attempts

Syntax

--cpiMinAttempts <integer>

Description

CPI minimum attempt limit.

Class

Optional.

Type / Range

Integer in the range of 0 to 500

Default

100

Name

cpiMinor - Minor alarm

Syntax

--cpiMinor <integer>

Description

CPI threshold for minor alarm.

Class

Optional.

Type / Range

Integer in the range of -1 or 1 to 99 in units of %

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

4.68 cmm csfbEmlppPagPriProfile

CSFB eMLPP Paging Priority Profile. Syntax, description, parameters, and roles for cmm csfbEmlppPagPriProfile command.

Syntax**Description**

This command is used to administer a CSFB eMLPP Paging Priority Profile and includes a parameter for defining the paging priority level.

Parameters**Name**

csfbPagingPriProfileName - CSFB paging priority profile name

Syntax

--csfbPagingPriProfileName <string>

Description

Defines the CSFB paging priority profile name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

emlpp - eMLPP

Syntax

--emlpp <integer>

Description

eMLPP.

Class

Required key.

Type / Range

Integer in the range of 0 to 7

Name

pagingPriLevel - Paging priority level

Syntax

--pagingPriLevel <list-item>

Description

Paging priority level.

Class

Optional.

List

- PrioLevel0 - Priority level 0 - Priority level 0. No paging priority is set.
- PrioLevel1 - Priority level 1 - Priority level 1. Highest Paging priority.
- PrioLevel2 - Priority level 2 - Priority level 2.
- PrioLevel3 - Priority level 3 - Priority level 3.
- PrioLevel4 - Priority level 4 - Priority level 4.
- PrioLevel5 - Priority level 5 - Priority level 5.
- PrioLevel6 - Priority level 6 - Priority level 6.
- PrioLevel7 - Priority level 7 - Priority level 7.
- PrioLevel8 - Priority level 8 - Priority level 8. Lowest Paging Priority.

Default

PrioLevel0

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.69 cmm csfbPagingPriProfile

CSFB Paging Priority Profile. Syntax, description, parameters, and roles for cmm csfbPagingPriProfile command.

Syntax

Description

This command is used to administer a CSFB Paging Priority Profile and includes a parameter for defining the No-eMLPP paging priority level.

Parameters

Name

name - CSFB paging priority profile name

Syntax

--name <digits>

Description

Defines the CSFB paging priority profile name.

Class

Required key.

Type / Range

Digit string of length 1 to 1 with the allowed pattern of [1-8]*

Name

description - Description

Syntax

--description <string>

Description

Profile description.

Class

Optional.

Type / Range

Digit string of length 0 to 160 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

''

Name

noeMlppPagingPriLevel - No-eMLPP paging priority level

Syntax

--noeMlppPagingPriLevel <list-item>

Description

No-eMLPP paging priority level.

Class

Optional.

List

- PrioLevel0 - Priority level 0 - Priority level 0. No paging priority is set.
- PrioLevel1 - Priority level 1 - Priority level 1. Highest Paging priority.
- PrioLevel2 - Priority level 2 - Priority level 2.
- PrioLevel3 - Priority level 3 - Priority level 3.
- PrioLevel4 - Priority level 4 - Priority level 4.
- PrioLevel5 - Priority level 5 - Priority level 5.
- PrioLevel6 - Priority level 6 - Priority level 6.
- PrioLevel7 - Priority level 7 - Priority level 7.
- PrioLevel8 - Priority level 8 - Priority level 8. Lowest Paging Priority.

Default

PrioLevel0

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.70 cmm dataTrafficList

Data Traffic List. Syntax, description, parameters, and roles for cmm dataTrafficList command.

Syntax

Description

Data Traffic List.

Parameters

Name

name - Data traffic list name

Syntax

--name <string>

Description

Data traffic list name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.71 cmm dataTrafficListNi

Data Traffic List NI. Syntax, description, parameters, and roles for cmm dataTrafficListNi command.

Syntax**Description**

Administer Data Traffic List NI.

Parameters**Name**

dataTrafficListName - Data traffic list name

Syntax

--dataTrafficListName <string>

Description

Data traffic list name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

apnNi - Network indicator

Syntax

--apnNi <string>

Description

Access point name - network indicator.

Class

Required key.

Type / Range

Digit string of length 1 to 62 with the allowed pattern of [0-9a-zA-Z\._]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.72 cmm dcaeServer

DCAE Server Interface. Syntax, description, parameters, and roles for **cmm dcaeServer** command.

Syntax**Description**

Provision the address of the ONAP Server for DCAE

Parameters**Name**

dcaeReportId - ONAP report ID

Syntax

--dcaeReportId <list-item>

Description

Type of report for the ONAP server.

Class

Required key.

List

- amfRTPMReport - AMF Real Time PM Report - AMF Real Time PM Data
- mmeRTPMReport - MME Real Time PM Report
- nonRTPMReport - Non Real Time PM Report - Non Real Time PM Data for AMF, MME, SGSN
- sgsnRTPMReport - SGSN Real Time PM Report

Name

primaryIp - ONAP server Primary IP

Syntax

--primaryIp <string>

Description

Primary IP address of the ONAP server.

Class

Required on create.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

primaryPort - Primary Port

Syntax

--primaryPort <integer>

Description

Primary port address of service.

Class

Required on create.

Type / Range

Integer in the range of 1,024 to 65,535

Name

dcaeConnName - Connection Name for DCAE

Syntax

--dcaeConnName <string>

Description

Name used to identify the AMF on DCAE connection

Restrictions

This parameter is required only if the dcaeReportId = NonRTPMReport

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 64 with the allowed pattern of [0-9a-zA-Z\._:-]*

Default

"

Name

dcaeConnWord - DCAE Connection word

Syntax

--dcaeConnWord <string>

Description

Connection word for DCAE

Restrictions

This parameter is required only if the dcaeReportId = NonRTPMReport

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 64 with the allowed pattern of [0-9a-zA-Z\.\-_:]*

Default

"

Name

dscpCode - DSCP Code

Syntax

--dscpCode <list-item>

Description

Parameter to set DSCP code to be used for an IP packet.

Class

Optional.

List

- AF11 - Assured forwarding 11.
- AF12 - Assured forwarding 12.
- AF13 - Assured forwarding 13.
- AF21 - Assured forwarding 21.
- AF22 - Assured forwarding 22.
- AF23 - Assured forwarding 23.
- AF31 - Assured forwarding 31.
- AF32 - Assured forwarding 32.
- AF33 - Assured forwarding 33.
- AF41 - Assured forwarding 41.
- AF42 - Assured forwarding 42.
- AF43 - Assured forwarding 43.
- BE - BE.
- CS1 - Class selector 1.
- CS2 - Class selector 2.

- CS3 - Class selector 3.
- CS4 - Class selector 4.
- CS5 - Class selector 5.
- CS6 - Class selector 6.
- CS7 - Class selector 7.
- EF - Expedited forwarding.
- voiceAdmit - Voice Admit

Default

AF41

Name

oamTlsProfileName - OAM TLS Profile

Syntax

--oamTlsProfileName <string>

Description

This parameter contains the profile for TLS. It contains the certificate names needed for establishing secure communication.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

secondaryIp - ONAP server Secondary IP

Syntax

--secondaryIp <string>

Description

Secondary IP address of the ONAP server.

Class

Optional.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Default

0.0.0.0

Name

secondaryPort - Secondary Port

Syntax

--secondaryPort <integer>

Description

Secondary port address of service

Class

Optional.

Type / Range

Integer in the range of 0 or 1,024 to 65,535

Default

0

Name

tcpProfileName - TCP Profile Name

Syntax

--tcpProfileName <string>

Description

Configure TCP profile to the endpoint.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

default

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.73 cmm dcnIdUeUsageType

DCN ID supported UE Usage Types. Syntax, description, parameters, and roles for cmm dcnIdUeUsageType command.

Syntax

Description

This command is used to provision UE Usage Types for a DCN ID.

Parameters

Name

dcnId - Default Dedicated Core Network ID

Syntax

--dcnId <integer>

Description

DCN ID to which UE Usage Types are provisioned.

Class

Required key.

Type / Range

Integer in the range of 1 to 126 or 128 to 65,535

Name

ueUsageType - UE Usage Type

Syntax

--ueUsageType <integer>

Description

Provision UE Usage Types per DCN ID. A maximum of 32 UE Usage Types can be provisioned to a DCN ID. The UE Usage Types must be unique across DCN IDs

Class

Required key.

Type / Range

Integer in the range of 1 to 126 or 128 to 255

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.74 cmm dcnIdUutRoamers

DCN ID UE Usage TType for Roamers. Syntax, description, parameters, and roles for cmm dcnIdUutRoamers command.

Syntax

Description

This command maps an inbound roamer UUT, if present in subscription data, to an internal Dedicated Core Network ID.

Parameters

Name

uut - UE Usage Type

Syntax

--uut <integer>

Description

The UE Usage Type as specified in the profile from the HSS for the inbound roamer UE.

Class

Required key.

Type / Range

Integer in the range of 1 to 126 or 128 to 255

Name

dcnId - DCN ID

Syntax

--dcnId <integer>

Description

Dedicated Core Network Identifier

Class

Required on create.

Type / Range

Integer in the range of 1 to 126 or 128 to 65,535

Name

ueUsageTypeForGwSelection - UE Usage Type for GW Selection

Syntax

--ueUsageTypeForGwSelection <integer>

Description

UE Usage Type to be used solely for DNS selection queries for SGW and PGW.

Class

Required on create.

Type / Range

Integer in the range of 1 to 126 or 128 to 255

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.75 cmm dcnMmeGrp

MME Group for DCN. Syntax, description, parameters, and roles for cmm dcnMmeGrp command.

Syntax**Description**

MME Group provision for DCN

Parameters

Name

name - Name of the MME Group

Syntax

--name <string>

Description

Name of the MME group.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

mmeGi - MME Group Identifier

Syntax

--mmeGi <integer>

Description

MME Group Identifier

Class

Required on create, not modifiable.

Type / Range

Integer in the range of 0 to 65,535

Name

servedPlmnName - Served PLMN Name

Syntax

--servedPlmnName <string>

Description

Name of the PLMN to which the MME group belongs to.

Class

Required on create, not modifiable.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.76 cmm defaultNotifications

Default Notifications. Syntax, description, parameters, and roles for cmm defaultNotifications command.

Syntax

Description

List containing pointers to default notification subscriptions. A maximum of 16 default notification subscription pointers can be added.

Parameters

Name

defaultNotificationSubscriptionListName - Default Notification Subscription List Name

Syntax

```
--defaultNotificationSubscriptionListName <string>
```

Description

Default Notification Subscription List Name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

defaultNotificationSubscriptionName - Default Notification Subscription Name

Syntax

```
--defaultNotificationSubscriptionName <string>
```

Description

Default Notification Subscription Name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.77 cmm defaultNotificationSubscription

Notification endpoints for different notification types. Syntax, description, parameters, and roles for cmm defaultNotificationSubscription command.

Syntax**Description**

Default notification endpoints for different notifications. Other NF use this information to send notifications.

Parameters**Name**

name - Notification name

Syntax

--name <string>

Description

Notification name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

callbackUri - Callback URI

Syntax

--callbackUri <string>

Description

The callback URI used for the notification.

Class

Required on create.

Type / Range

Digit string of length 512 with the allowed pattern of (http:\\|https:\\)[0-9a-zA-Z\\.\\-\\~\$+* _;&=#?\\[\\]@,()\\! '%]*

Name

n1MessageClass - N1 Message Class

Syntax

--n1MessageClass <list-item>

Description

If the notification type is N1_MESSAGES, this IE must be provisioned to identify class of N1 messages.

Class

Optional.

List

- 5GMM - 5G Mobility Management N1 messages - The whole NAS message as received (for e.g. used in forwarding the Registration message to target AMF during Registration procedure with AMF redirection).
- LPP - LPP N1 message
- SM - Session Management N1 messages
- SMS - SMS N1 message - The N1 message of SMS type.

Default

5GMM

Name

n2InformationClass - N2 Information Class

Syntax

--n2InformationClass <list-item>

Description

If the notification type is N2_INFORMATION, this parameter must be provisioned to identify N2 Information Class.

Class

Optional.

List

- NRPPa - N2 NRRPa information
- PWS - N2 PWS Information
- PWS_BCAL - N2 Broadcast Completed Area List or the Broadcast Cancelled Area List.
- PWS_RF - N2 Restart Indication or Failure Indication
- RAN - N2 RAN related information
- SM - N2 SM Information

Default

NRPPa

Name

notificationType - Notification Type

Syntax

--notificationType <list-item>

Description

Type of notification for which the corresponding callback URI is provided.

Class

Optional.

List

- LOCATION_NOTIFICATION - Location Notification - Notification of Location Information by AMF towards NF Service Consumers (e.g GMLC).
- N1_MESSAGES - N1 Messages - Notification of N1 messages
- N2_INFORMATION - N2 Information - Notification of N2 information

Default

N1_MESSAGES

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.78 cmm defaultNotificationSubscriptionList

Default Notification Subscription List. Syntax, description, parameters, and roles for cmm defaultNotificationSubscriptionList command.

Syntax

Description

Create a Default Notification Subscription list.

Parameters

Name

name - List Name

Syntax

--name <string>

Description

List name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.79 cmm delayUdmDuringRegistrationProfile

Delay UDM During Registration Profile. Syntax, description, parameters, and roles for cmm delayUdmDuringRegistrationProfile command.

Syntax

Description

Define the parameters for delayed UDM dialogs during Registration, and for re-attempting the delayed UDM dialogs during subsequent procedures. Maximum number of profiles is 128.

Parameters

Name

name - Name

Syntax

--name <string>

Description

The name of the Delay UDM During Registration Profile.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

automaticReattempts - Automatic Reattempts

Syntax

--automaticReattempts <list-item>

Description

Enables the AMF to automatically reattempt any delayed UDM dialogs of the UE during subsequent UE triggered Registration/Service request procedures.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

delayUdmDuringRegistration - Delay UDM During Registration

Syntax

--delayUdmDuringRegistration <list-item>

Description

When the same AMF still holds the UE data internally after previous N26 mobility to EPS, this parameter enables AMF to successfully complete the registration procedure and suppress the UDM dialogs (Nudm_UECM_Registration, Nudm_SDM_Get and/or Nudm_SDM_Subscribe, depending on other settings) for the enabled UEs. The UDM dialogs will be re-attempted automatically according to provisioning. Note: this functionality only is effective when the AMF global parameter Keep VLR Entry is set to true.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.80 cmm detachBehaviorMappingProfile

Detach Behavior Mapping Profile. Syntax, description, parameters, and roles for cmm detachBehaviorMappingProfile command.

Syntax

Description

This command is used to administer a Detach Behavior Mapping Profile and includes parameters for defining cause codes and detach types for various scenarios related to UE detach. A default profile, named 'default' is provided.

Parameters

Name

name - Detach behavior mapping profile name

Syntax

--name <string>

Description

Identifies, via name, a given detach behavior mapping.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

adminDeleteUeContextCc - Delete UE context cause code

Syntax

--adminDeleteUeContextCc <integer>

Description

NAS Cause code to include in Detach Request when UE context is deleted

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

10

Name

adminDeleteUeContextDetachType - Delete UE context detach type

Syntax

--adminDeleteUeContextDetachType <list-item>

Description

Detach type when UE context is deleted via administrative action e.g. cli command

Class

Optional.

List

- Reattach not required - Reattached not required - Specifies detach type of reattach not required.
- Reattach required - Specifies detach type of reattach required.

Default

Reattach required

Name

adminDeleteUeContextNotifyIdleUe - Delete UE context notify idle UE

Syntax

--adminDeleteUeContextNotifyIdleUe <list-item>

Description

Page UE if in the CM-idle when deleting UE context via administrative action e.g. cli command

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

adminDeleteUeContextSendCc - Delete UE context send cause code

Syntax

--adminDeleteUeContextSendCc <list-item>

Description

Include cause code in detach request when detach reason is due to deleting UE context

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

apnAdminCc - APN Administration Cause Code

Syntax

--apnAdminCc <integer>

Description

Configure the NAS EMM cause code to be used on the Detach Request as part of the APN administration functionality. MME performs detach for a UE when the PDNs towards the provisioned APN(s) are commanded to be deactivated from MME, and the UE only has PDN(s) towards the defined APN(s).

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

10

Name

apnAdminDetachType - APN Administration Detach Type

Syntax

--apnAdminDetachType <list-item>

Description

Configure the detach type to be used for APN administration functionality. The MME detaches the UE when the PDNs towards defined APN(s) are to be deactivated and the UE only has PDN(s) towards the defined APN(s).

Class

Optional.

List

- Reattach not required - Reattached not required - Specifies detach type of reattach not required.
- Reattach required - Specifies detach type of reattach required.

Default

Reattach not required

Name

apnAdminNotifyIdleUe - APN Administration Notify Idle UE

Syntax

--apnAdminNotifyIdleUe <list-item>

Description

Define whether an ECM-IDLE UE will be paged for detaching due to the APN Administration.
By default the detaching will be implicit without contacting an ECM-IDLE UE.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

apnAdminSendCc - APN Administration Send CC

Syntax

--apnAdminSendCc <list-item>

Description

Define whether to include NAS EMM cause code on Detach Request when the MME detaches the UE due to APN Administration functionality.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

clrSubWithdrawalCc - CLR with cancellation type subscription withdrawal cause code

Syntax

--clrSubWithdrawalCc <integer>

Description

Cause code to send when detach reason is CLR with cancellation type subscription withdrawal.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

10

Name

clrSubWithdrawalDetachType - CLR with cancellation type subscription withdrawal detach type

Syntax

--clrSubWithdrawalDetachType <list-item>

Description

Detach type when detach reason is CLR with cancellation type subscription withdrawal.

Class

Optional.

List

- Reattach not required - Reattached not required - Specifies detach type of reattach not required.
- Reattach required - Specifies detach type of reattach required.

Default

Reattach not required

Name

clrSubWithdrawalNotifyIdleUe - CLR with cancellation type subscription withdrawal notify idle UE

Syntax

--clrSubWithdrawalNotifyIdleUe <list-item>

Description

Page a UE, if it is in the ECM idle state, when detach reason is CLR with cancellation type subscription withdrawal.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

clrSubWithdrawalSendCc - CLR with cancellation type subscription withdrawal send cause code

Syntax

--clrSubWithdrawalSendCc <list-item>

Description

Include cause code in detach when detach reason is CLR with cancellation type subscription withdrawal.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

delBearerReqCcCc - Delete bearer request with CC cause code

Syntax

--delBearerReqCcCc <integer>

Description

Cause code to send when detach reason is delete bearer request with cause code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

10

Name

delBearerReqCcDetachType - Delete bearer request with CC detach type

Syntax

--delBearerReqCcDetachType <list-item>

Description

Detach type when detach reason is delete bearer request with cause code.

Class

Optional.

List

- Reattach not required - Reattached not required - Specifies detach type of reattach not required.
- Reattach required - Specifies detach type of reattach required.

Default

Reattach not required

Name

delBearerReqCcImsCc - IMS delete bearer request with CC cause code

Syntax

--delBearerReqCcImsCc <integer>

Description

Cause code to send for IMS when detach reason is delete bearer request with cause code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

10

Name

delBearerReqCcImsDetachType - IMS delete bearer request with CC detach type

Syntax

--delBearerReqCcImsDetachType <list-item>

Description

Detach type for IMS when detach reason is delete bearer request with cause code.

Class

Optional.

List

- Reattach not required - Reattached not required - Specifies detach type of reattach not required.
- Reattach required - Specifies detach type of reattach required.

Default

Reattach not required

Name

delBearerReqCcImsNotifyIdleUe - IMS delete bearer request with CC notify idle UE

Syntax

--delBearerReqCcImsNotifyIdleUe <list-item>

Description

Page a UE for IMS, if it is in the ECM idle state, when detach reason is delete bearer request with cause code.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

delBearerReqCcImsSendCc - IMS delete bearer request with CC send cause code

Syntax

--delBearerReqCcImsSendCc <list-item>

Description

Include cause code for IMS in detach when detach reason is delete bearer request with cause code.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

delBearerReqCcNotifyIdleUe - Delete bearer request with CC notify idle UE

Syntax

--delBearerReqCcNotifyIdleUe <list-item>

Description

Page a UE, if it is in the ECM idle state, when detach reason is delete bearer request with cause code.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

delBearerReqCcSendCc - Delete bearer request with CC send cause code

Syntax

--delBearerReqCcSendCc <list-item>

Description

Include cause code in detach when detach reason is delete bearer request with cause code.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

delBearerReqNoCcCc - Delete bearer request with no CC cause code

Syntax

--delBearerReqNoCcCc <integer>

Description

Cause code to send when detach reason is delete bearer request with no cause code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

10

Name

delBearerReqNoCcDetachType - Delete bearer request with no CC detach type

Syntax

--delBearerReqNoCcDetachType <list-item>

Description

Detach type when detach reason is delete bearer request with no cause code.

Class

Optional.

List

- Reattach not required - Reattached not required - Specifies detach type of reattach not required.
- Reattach required - Specifies detach type of reattach required.

Default

Reattach not required

Name

delBearerReqNoCclmsCc - IMS delete bearer request with no CC cause code

Syntax

--delBearerReqNoCcImsCc <integer>

Description

Cause code to send for IMS when detach reason is delete bearer request with no cause code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

10

Name

delBearerReqNoCclmsDetachType - IMS delete bearer request with no CC detach type

Syntax

--delBearerReqNoCcImsDetachType <list-item>

Description

Detach type for IMS when detach reason is delete bearer request with no cause code.

Class

Optional.

List

- Reattach not required - Reattached not required - Specifies detach type of reattach not required.
- Reattach required - Specifies detach type of reattach required.

Default

Reattach not required

Name

delBearerReqNoCclmsNotifyIdleUe - IMS delete bearer request with no CC notify idle UE

Syntax

--delBearerReqNoCcImsNotifyIdleUe <list-item>

Description

Page a UE for IMS, if it is in the ECM idle state, when detach reason is delete bearer request with no cause code.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

delBearerReqNoCcImsSendCc - IMS delete bearer request with no CC send cause code

Syntax

--delBearerReqNoCcImsSendCc <list-item>

Description

Include cause code for IMS in detach when detach reason is delete bearer request with no cause code.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

delBearerReqNoCcNotifyIdleUe - Delete bearer request with no CC notify idle UE

Syntax

--delBearerReqNoCcNotifyIdleUe <list-item>

Description

Page a UE, if it is in the ECM idle state, when detach reason is delete bearer request with no cause code.

Class

Optional.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Name

delBearerReqNoCcSendCc - Delete bearer request with no CC send cause code

Syntax

--delBearerReqNoCcSendCc <list-item>

Description

Include cause code in detach when detach reason is delete bearer request with no cause code.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

description - Profile description

Syntax

--description <string>

Description

Profile description.

Class

Optional.

Type / Range

Digit string of length 0 to 160 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

failopenRestorationCc - Failopen Restoration Cause Code

Syntax

--failopenRestorationCc <integer>

Description

Define the NAS EMM cause code on Detach Request due to HSS unavailability manual restoration, when the MME detaches a UE that has been attached with the help of the failopen profile.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

10

Name

failopenRestorationDetachType - Failopen Restoration Detach Type

Syntax

--failopenRestorationDetachType <list-item>

Description

Configure the Detach Type when the MME due to HSS unavailability manual restoration detaches a UE that has been attached with the help of the failopen profile.

Class

Optional.

List

- Reattach not required - Reattached not required - Specifies detach type of reattach not required.
- Reattach required - Specifies detach type of reattach required.

Default

Reattach required

Name

failopenRestorationNotifyIdleUe - Failopen Restoration Notify Idle UE

Syntax

--failopenRestorationNotifyIdleUe <list-item>

Description

Define whether an ECM-IDLE UE will be paged for detaching, when the MME due to HSS unavailability manual restoration detaches a UE that has been attached with the help of the failopen profile.

Class

Optional.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Name

failopenRestorationSendCc - Failopen Restoration Send CC

Syntax

--failopenRestorationSendCc <list-item>

Description

Define whether to include NAS EMM cause code on Detach Request when the MME detaches the UE due to manual HSS unavailability restoration, and the UE has been attached with the help of the failopen profile. By default no EMM cause code is included.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

imsPdnActivationTimerExpiredCc - IMS PDN Activation Timer Expired CC

Syntax

--imsPdnActivationTimerExpiredCc <integer>

Description

Configure the NAS EMM cause code to be used on the Detach Request as part of the IMS supervision functionality. MME performs detach for a UE when the IMS PDN Activation Timer expires and IMS PDN is not activated.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

10

Name

imsPdnActivationTimerExpiredDetachType - IMS PDN Activation Timer Expired Detach Type

Syntax

--imsPdnActivationTimerExpiredDetachType <list-item>

Description

Configure the detach type to be used for IMS supervision functionality. The MME detaches the UE when the IMS PDN Activation Timer expires and IMS PDN is not activated.

Class

Optional.

List

- Reattach not required - Reattached not required - Specifies detach type of reattach not required.
- Reattach required - Specifies detach type of reattach required.

Default

Reattach not required

Name

imsPdnActivationTimerExpiredNotifyIdleUe - IMS PDN Activation Timer Expired Notify Idle UE

Syntax

```
--imsPdnActivationTimerExpiredNotifyIdleUe <list-item>
```

Description

Define whether an ECM-IDLE UE will be paged for detaching after the IMS PDN Activation Timer has expired and IMS PDN is not activated.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

imsPdnActivationTimerExpiredSendCc - IMS PDN Activation Timer Expired Send CC

Syntax

```
--imsPdnActivationTimerExpiredSendCc <list-item>
```

Description

Define whether to include NAS EMM cause code on Detach Request when the MME detaches the UE due to IMS PDN Activation Timer expiry and the IMS PDN is not activated. By default no EMM cause code is included.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

imsRegistrationFailedCc - IMS Registration Failed CC

Syntax

--imsRegistrationFailedCc <integer>

Description

Configure the NAS EMM cause code used on Detach Request when the IMS Registration Timer expires and the UE has not performed successful IMS registration.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

10

Name

imsRegistrationFailedDetachType - IMS Registration Failed Detach Type

Syntax

--imsRegistrationFailedDetachType <list-item>

Description

Configure the Detach Type when the UE is detached due to IMS Registration Timer expiry and the UE has not performed successful IMS registration.

Class

Optional.

List

- Reattach not required - Reattached not required - Specifies detach type of reattach not required.
- Reattach required - Specifies detach type of reattach required.

Default

Reattach required

Name

imsRegistrationFailedNotifyIdleUe - IMS Registration Failed Notify Idle UE

Syntax

```
--imsRegistrationFailedNotifyIdleUe <list-item>
```

Description

Define whether an ECM-IDLE UE will be paged for detaching when the IMS Registration Timer expires and the UE has not performed successful IMS registration.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

imsRegistrationFailedSendCc - IMS Registration Failed Send CC

Syntax

```
--imsRegistrationFailedSendCc <list-item>
```

Description

Define whether the NAS EMM cause is included in the Detach Request when the UE is detached due to IMS Registration Timer expiry and the UE has not performed successful IMS registration.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

noaUeCc - NOA user unknown NAS cause code

Syntax

```
--noaUeCc <integer>
```

Description

Configure NAS cause code sent when detach reason is user unknown received in notify response from HSS.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

8

Name

noaUeDetachType - NOA user unknown detach type

Syntax

--noaUeDetachType <list-item>

Description

Specify the detach type "Reattach required" or "Reattach not required".

Class

Optional.

List

- Reattach not required - Reattached not required - Specifies detach type of reattach not required.
- Reattach required - Specifies detach type of reattach required.

Default

Reattach not required

Name

noaUeNotifyIdleUe - NOA user unknown notify idle UE

Syntax

--noaUeNotifyIdleUe <list-item>

Description

Enable/disable explicit detach of an ECM-IDLE UE.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

noaUeSendCc - NOA user unknown send NAS cause code

Syntax

--noaUeSendCc <list-item>

Description

Enable/disable inclusion of EMM cause in detach request.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

pcscfBrrReactCc - PCSCF detach bearer request NAS cause code

Syntax

--pcscfBrrReactCc <integer>

Description

Configure NAS cause code sent when detach BR with reactivation requested.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

10

Name

pcscfBrrReactDetachType - PCSCF detach bearer request detach type

Syntax

--pcscfBrrReactDetachType <list-item>

Description

Configure reattach option for det BR with reactivation requested.

Class

Optional.

List

- Reattach not required - Reattached not required - Specifies detach type of reattach not required.
- Reattach required - Specifies detach type of reattach required.

Default

Reattach not required

Name

pcscfBrrReactNotifyIdleUe - PCSCF detach bearer request notify idle UE

Syntax

--pcscfBrrReactNotifyIdleUe <list-item>

Description

Enable/disable paging of idle UE when detach reason is det BR with reactivation requested.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

pcscfBrrReactSendCc - PCSCF detach bearer request send NAS cause code

Syntax

--pcscfBrrReactSendCc <list-item>

Description

Enable/disable inclusion of cause code when det BR with reactivation requested.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

pgwRstCc - PGW restart NAS cause code

Syntax

--pgwRstCc <integer>

Description

NAS cause code to be sent to UE if its detached.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

10

Name

pgwRstDetachType - PGW restart detach type

Syntax

--pgwRstDetachType <list-item>

Description

Specify the detach type "Reattach required" or "Reattach not required".

Class

Optional.

List

- Reattach not required - Reattached not required - Specifies detach type of reattach not required.
- Reattach required - Specifies detach type of reattach required.

Default

Reattach not required

Name

pgwRstNotifyIdleUe - PGW restart notify idle UE

Syntax

--pgwRstNotifyIdleUe <list-item>

Description

Enable/disable explicit detach of an ECM-IDLE UE.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

pgwRstSendCc - PGW restart send NAS cause code

Syntax

--pgwRstSendCc <list-item>

Description

Enable/disable inclusion of EMM cause in detach request.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

sgwRestartCc - SGW restart cause code

Syntax

--sgwRestartCc <integer>

Description

Cause code to send when detach reason is SGW restart.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

10

Name

sgwRestartDetachType - SGW restart detach type

Syntax

--sgwRestartDetachType <list-item>

Description

Detach type when detach reason is SGW restart.

Class

Optional.

List

- Reattach not required - Reattached not required - Specifies detach type of reattach not required.
- Reattach required - Specifies detach type of reattach required.

Default

Reattach required

Name

sgwRestartNotifyIdleUe - SGW restart notify idle UE

Syntax

--sgwRestartNotifyIdleUe <list-item>

Description

Page a UE, if it is in the ECM idle state, when detach reason is SGW restart.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sgwRestartSendCc - SGW restart send cause code

Syntax

--sgwRestartSendCc <list-item>

Description

Include cause code in detach when detach reason is SGW restart.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.81 cmm dfltGateway

Default gateway. Syntax, description, parameters, and roles for cmm dfltGateway command.

Syntax

Description

This command is used to administer a default gateway and includes parameters for defining the pool type, ID, and member ID, assignment type, gateway IP address, and source IP address.

Parameters

Name

poolType - Pool type

Syntax

--poolType <list-item>

Description

Pool (service) type.

Class

Required key.

List

- ALMS - Alarm Service - Alarm Service.
- CPPS - Control plane processing service - Control plane processing service.
- CTCS - Call Trace Collection service - Call Trace Collection service.
- DBS - Database service - Database service.
- EEMS - Event Exposure Management service - Event Exposure Management service.
- IPDS - IP director service - IP director service.
- IPPS - IP packet processing service - IP packet processing service.
- LIHS - Lawful Interception Handling service

- MLIS - Machine Learning Intelligence Service
- NECC - Network element cluster controller - Network element cluster controller.
- NWDS - Network Data Service - NWDAF service
- PAPS - Packet processing service - Packet processing service.
- PCMS - Per Call Measurements Data Service - Per Call Measurements Data (PCMD) Service
- RCIS - Radio Capability Information storage Service
- SFDS - SCTP flow distributor service
- UCDS - User equipment Context Data-dump Service
- UFDS - UDP flow distributor service

Name

poolId - Pool id

Syntax

--poolId <integer>

Description

Pool identification number.

Class

Required key.

Type / Range

Integer in the range of 0 to 39

Name

poolMemberId - Pool member ID

Syntax

--poolMemberId <integer>

Description

Pool member identification number.

Class

Required key.

Type / Range

Integer in the range of 0 to 2

Name

assignmentType - Assignment type

Syntax

--assignmentType <list-item>

Description

Assignment type: fixed or floating

Class

Required key.

List

- fixed - Fixed.
- floating - Floating.

Name

ipv - IP version

Syntax

--ipv <integer>

Description

Internet protocol version number.

Class

Required key.

Type / Range

Integer in the range of 4 or 6

Name

gateway - Gateway IP

Syntax

--gateway <string>

Description

Gateway IP address.

Class

Required on create.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

srcIp - Source IP

Syntax

--srcIp <string>

Description

Source IP address.

Class

Optional.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.82 cmm diamAppProfile

diamAppProfile. Syntax, description, parameters, and roles for **cmm diamAppProfile** command.

Syntax**Description**

This command is used to administer an operator defined diameter application profile.

Parameters**Name**

diamAppProfileListName - Diameter Application Profile List Name

Syntax

--diamAppProfileListName <string>

Description

Diameter application profile name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

diamAppType - Diameter Application Type

Syntax

--diamApplType <list-item>

Description

Diameter Application Types

Class

Required key.

List

- S13 - S13 diam application type.
- S6A - S6AD - S6AD diam application type.
- SGD - SGd diam application type.
- SLG - Slg diam application type.
- T6A - T6A diam application type.

Name

diamApplTimeOut - Diameter Application Timeout

Syntax

--diamApplTimeOut <integer>

Description

Diameter Application Timeout

Class

Optional.

Type / Range

Integer in the range of 1 to 10

Default

5

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.83 cmm diamApplProfileList

Diameter Application Profile List. Syntax, description, parameters, and roles for cmm diamApplProfileList command.

Syntax

Description

This command is used to administer the Diameter Application Profile List

Parameters

Name

name - Diameter Application Profile Name

Syntax

--name <string>

Description

Diameter Application Profile Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.84 cmm diamConnection

Diameter Connections. Syntax, description, parameters, and roles for cmm diamConnection command.

Syntax

Description

This command is used to administer a Diameter connection and includes parameters for defining Diameter request message timeout, application type, minimum number of active connections, origin host and realm names (FQDN), and endpoint names.

Parameters

Name

diamApplType - Application type

Syntax

--diamApplType <list-item>

Description

Diameter application type.

Class

Required key.

List

- S13 - S13 diam application type.
- S6A - S6AD - S6AD diam application type.
- SGD - SGd diam application type.
- SLG - Slg diam application type.
- T6A - T6A diam application type.

Default

S6A

Name

combineApplType - Combine application type

Syntax

--combineApplType <list-item>

Description

Combine application type.

Restrictions

This element must be 'Empty' when diamApplType is not S6A.

Class

Optional.

List

- S6A/S13/SGD - Combined S6a, S13, and SGd interfaces.
- S6A/S13/T6A - Combined S6a, S13, and T6a interfaces.
- S6A/S13/T6A/SGD - Combined S6a, S13, T6a, and SGd interfaces.
- S6A/T6A/SGD - Combined S6a, T6a, and SGd interfaces.
- " - Empty - Interfaces not combined.
- S6A/S13/SLG - Combined S6a, S13 and SLg interfaces.
- S6A/S13/SLG/T6A - Combined S6a, S13, SLg, and T6a interfaces.
- S6A/S13/SLG/SGD - Combined S6a, S13, SLg, and SGd interfaces.
- S6A/S13/SLG/T6A/SGD - Combined S6a, S13, SLg, T6a, and SGd interfaces.
- S6A/S13 - Combined S6a and S13 interfaces.
- S6A/SLG - Combined S6a and SLg interfaces.
- S6A/SGD - Combined S6a and SGd interfaces.
- S6A/SLG/SGD - Combined S6a, SLg, and SGd interfaces.
- S6A/SLG/T6A - Combined S6a, SLg, and T6a interfaces.
- S6A/SLG/T6A/SGD - Combined S6a, SLg, T6a, and SGd interfaces.
- S6A/T6A - Combined S6a, and T6a interfaces.

Default

""

Name

diamApplTimeOut - Application timeout

Syntax

--diamApplTimeOut <integer>

Description

Diameter request message timeout.

Class

Optional.

Type / Range

Integer in the range of 1 to 10

Default

5

Name

diamOrgHost - Origin host

Syntax

--diamOrgHost <string>

Description

FQDN defined for this MME.

Class

Optional.

Type / Range

Digit string of length 0 to 96 with the allowed pattern of [0-9a-zA-Z\.\-_:]*

Name

diamOrgRealm - Origin realm

Syntax

--diamOrgRealm <string>

Description

FQDN for all the MME nodes that can access the HSS host.

Class

Optional.

Type / Range

Digit string of length 0 to 96 with the allowed pattern of [0-9a-zA-Z\.\-_:]*

Name

diamProfileName - Diam profile name

Syntax

--diamProfileName <string>

Description

Diam profile name.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

default

Name

diamUseSctp - Use SCTP

Syntax

--diamUseSctp <list-item>

Description

When checked (true), indicates that SCTP is used to support the diameter sessions.

Restrictions

diamUseSctp must be true.

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

endpoint1Name - Endpoint 1 name

Syntax

--endpoint1Name <string>

Description

Long name for endpoint 1 id matching the label specified in remote end point configuration.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

endpoint2Name - Endpoint 2 name

Syntax

--endpoint2Name <string>

Description

Long name for endpoint 2 id matching the label specified in remote end point configuration.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

endpoint3Name - Endpoint 3 name

Syntax

--endpoint3Name <string>

Description

Long name for endpoint 3 id matching the label specified in remote end point configuration.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

endpoint4Name - Endpoint 4 name

Syntax

--endpoint4Name <string>

Description

Long name for endpoint 4 id matching the label specified in remote end point configuration.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

endpoint5Name - Endpoint 5 name

Syntax

--endpoint5Name <string>

Description

Long name for endpoint 5 id matching the label specified in remote end point configuration.

Restrictions

This element only applies to diamApplType SLG.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

endpoint6Name - Endpoint 6 name

Syntax

--endpoint6Name <string>

Description

Long name for endpoint 6 id matching the label specified in remote end point configuration.

Restrictions

This element only applies to diamApplType SLG.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

endpoint7Name - Endpoint 7 name

Syntax

--endpoint7Name <string>

Description

Long name for endpoint 7 id matching the label specified in remote end point configuration.

Restrictions

This element only applies to diamApplType SLG.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

endpoint8Name - Endpoint 8 name

Syntax

--endpoint8Name <string>

Description

Long name for endpoint 8 id matching the label specified in remote end point configuration.

Restrictions

This element only applies to diamApplType SLG.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

minActiveConn - Min active connections

Syntax

--minActiveConn <integer>

Description

Percent of active connections dropping before switching to an alternate HSS host.

Class

Optional.

Type / Range

Integer in the range of 10 to 90

Default

50

Name

sgsnNumberUlr - SGSN Number In Ulr

Syntax

--sgsnNumberUlr <digits>

Description

MME supports to send SGSN number in ULR for combined MME/SGSN.

Class

Optional.

Type / Range

Digit string of length 0 to 18

Default

""

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

4.85 cmm diamNasMappingProfile

Diameter Code to NAS Cause Code Mapping. Syntax, description, parameters, and roles for cmm diamNasMappingProfile command.

Syntax

Description

This command is used to administer a Diameter Code to NAS Cause Code (CC) mapping. The MME uses this profile to select the right CC to send to the UE when MME rejects a UE mobility management request due to an error received from the HSS. The maximum number of profiles supported is 1024. A default profile, named 'default' is provided.

Parameters

Name

name - Diameter to NAS cause code mapping profile name

Syntax

--name <string>

Description

Defines all diameter code to NAS cause code mapping.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

appUnsupported - Application unsupported

Syntax

--appUnsupported <integer>

Description

Mapping for application unsupported to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

17

Name

authenticationDataUnavailable - Authentication data unavailable

Syntax

--authenticationDataUnavailable <integer>

Description

Mapping for authentication data unavailable to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

17

Name

authenticationRejected - Authentication rejected

Syntax

--authenticationRejected <integer>

Description

Mapping for authentication rejected to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

17

Name

authorizationRejected - Authorization rejected

Syntax

--authorizationRejected <integer>

Description

Mapping for authorization rejected to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

15

Name

avpNotAllowed - AVP not allowed

Syntax

--avpNotAllowed <integer>

Description

Mapping for AVP not allowed to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

17

Name

avpOccursTooManyTimes - AVP occurs too many times

Syntax

--avpOccursTooManyTimes <integer>

Description

Mapping for AVP occurs too many times to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

17

Name

avpUnsupported - AVP unsupported

Syntax

--avpUnsupported <integer>

Description

Mapping for AVP unsupported to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

17

Name

commandUnsupported - Command unsupported

Syntax

--commandUnsupported <integer>

Description

Mapping for command unsupported to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

17

Name

contradictingAvps - Contradicting AVPs

Syntax

--contradictingAvps <integer>

Description

Mapping for contradicting AVPs to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

17

Name

description - Description

Syntax

--description <string>

Description

Profile description.

Class

Optional.

Type / Range

Digit string of length 0 to 160 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Name

electionLost - Election lost

Syntax

--electionLost <integer>

Description

Mapping for election lost to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

17

Name

equipmentUnknown - Equipment unknown

Syntax

--equipmentUnknown <integer>

Description

Mapping for equipment unknown to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

6

Name

featureUnsupported - Feature unsupported

Syntax

--featureUnsupported <integer>

Description

Mapping for feature unsupported to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

17

Name

hssOrEirDown - HSS or EIR down

Syntax

```
--hssOrEirDown <integer>
```

Description

Mapping for HSS or EIR down to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

15

Name

invalidAvpBitCombination - Invalid AVP bit combination

Syntax

```
--invalidAvpBitCombination <integer>
```

Description

Mapping for invalid AVP bit combination to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

17

Name

invalidAvpBits - Invalid AVP bits

Syntax

```
--invalidAvpBits <integer>
```

Description

Mapping for invalid AVP bits to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

17

Name

invalidAvpLength - Invalid AVP length

Syntax

--invalidAvpLength <integer>

Description

Mapping for invalid AVP length to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

17

Name

invalidAvpValue - Invalid AVP value

Syntax

--invalidAvpValue <integer>

Description

Mapping for invalid AVP value to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

17

Name

invalidBitHeader - Invalid bit in header

Syntax

--invalidBitHeader <integer>

Description

Mapping for invalid bit in header to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

17

Name

invalidHeaderBits - Invalid header bits

Syntax

--invalidHeaderBits <integer>

Description

Mapping for invalid header bits to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

17

Name

invalidMessageLength - Invalid message length

Syntax

--invalidMessageLength <integer>

Description

Mapping for invalid message length to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

17

Name

loopDetected - Loop detected

Syntax

--loopDetected <integer>

Description

Mapping for loop detected to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

17

Name

missingAvp - Missing AVP

Syntax

--missingAvp <integer>

Description

Mapping for missing AVP to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

17

Name

multiRoundAuth - Multi round authorization

Syntax

--multiRoundAuth <integer>

Description

Mapping for multi round authorization to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

17

Name

noCommonApplication - No common application

Syntax

--noCommonApplication <integer>

Description

Mapping for no common application to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

17

Name

noCommonSecurity - No common security

Syntax`--noCommonSecurity <integer>`**Description**

Mapping for no common security to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

17

Name

noProvisionedHss - No provisioned HSS

Syntax`--noProvisionedHss <integer>`**Description**

Mapping for no provisioned HSS to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

14

Name

noResponseFromHss - No Response from HSS

Syntax`--noResponseFromHss <integer>`**Description**

Mapping for no response from HSS to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

17

Name

outOfSpace - Out of space

Syntax`--outOfSpace <integer>`**Description**

Mapping for out of space to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

17

Name

overloadRetryNotAllowedAny - Overload retry not allowed to any

Syntax`--overloadRetryNotAllowedAny <integer>`**Description**

Mapping from diameter overload retry not allowed to any to NAS cause code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

42

Name

ratNotAllowed - RAT not allowed

Syntax`--ratNotAllowed <integer>`**Description**

Mapping for RAT not allowed to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

15

Name

realmNotServed - Realm not served

Syntax

--realmNotServed <integer>

Description

Mapping for realm not served to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

17

Name

redirectionIndication - Redirection indication

Syntax

--redirectionIndication <integer>

Description

Mapping for redirection indication to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

17

Name

resourcesExceeded - Resources exceeded

Syntax

--resourcesExceeded <integer>

Description

Mapping for resources exceeded to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

17

Name

roamingNotAllowed - Roaming not allowed

Syntax

--roamingNotAllowed <integer>

Description

Mapping for roaming not allowed to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

11

Name

roamingNotAllowedAllApn - Roaming not allowed all APN

Syntax

--roamingNotAllowedAllApn <integer>

Description

Mapping for roaming not allowed all APN to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

19

Name

roamingNotAllowedHplmnApn - Roaming not allowed HPLMN APN

Syntax

--roamingNotAllowedHplmnApn <integer>

Description

Mapping for roaming not allowed HPLMN APN to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

14

Name

roamingNotAllowedOther - Roaming not allowed, other

Syntax

--roamingNotAllowedOther <integer>

Description

Mapping for roaming not allowed, other to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

17

Name

roamingNotAllowedVplmnApn - Roaming not allowed VPLMN APN

Syntax

--roamingNotAllowedVplmnApn <integer>

Description

Mapping for roaming not allowed VPLMN APN to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

14

Name

tooBusy - Too busy

Syntax

--tooBusy <integer>

Description

Mapping for too busy to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

17

Name

unableToComply - Unable to comply

Syntax

--unableToComply <integer>

Description

Mapping for unable to comply to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

17

Name

unableToDeliver - Unable to deliver

Syntax

--unableToDeliver <integer>

Description

Mapping for unable to deliver to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

17

Name

unknownEpsGprsData - Unknown EPS GPRS DATA

Syntax

--unknownEpsGprsData <integer>

Description

Mapping for unknown EPS GPRS DATA to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

15

Name

unknownEpsNoGprsData - Unknown EPS no GPRS DATA

Syntax

--unknownEpsNoGprsData <integer>

Description

Mapping for unknown EPS no GPRS DATA to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

7

Name

unknownEpsSubscription - Unknown EPS subscription

Syntax

--unknownEpsSubscription <integer>

Description

Mapping for unknown EPS subscription to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

15

Name

unknownEpsSubscriptionOther - Unknown EPS subscription, other

Syntax

--unknownEpsSubscriptionOther <integer>

Description

Mapping for unknown EPS subscription, other to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

15

Name

unknownPeer - Unknown peer

Syntax

--unknownPeer <integer>

Description

Mapping for unknown peer to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

17

Name

unknownServingNode - Unknown serving node

Syntax

--unknownServingNode <integer>

Description

Mapping for unknown serving node to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

2

Name

unknownSessionId - Unknown session ID

Syntax

--unknownSessionId <integer>

Description

Mapping for unknown session ID to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

17

Name

unsupportedVersion - Unsupported version

Syntax

--unsupportedVersion <integer>

Description

Mapping for unsupported version to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

17

Name

userUnknown - User unknown

Syntax

--userUnknown <integer>

Description

Mapping for user unknown to NAS code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

8

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.86 cmm diamProfile

Diameter Profile. Syntax, description, parameters, and roles for `cmm diamProfile` command.

Syntax

Description

This command is used to administer a Diameter Profile and includes parameters for defining setup, connect, reconnect, frequency, and retransmission timers, and maximum number of retransmissions. A default profile, named 'default' is provided.

Parameters

Name

name - Diameter profile name

Syntax

--name <string>

Description

Profile name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

cerMsgTimer - CER message timer

Syntax

--cerMsgTimer <integer>

Description

Capability exchange request (Diameter setup) timer value.

Class

Optional.

Type / Range

Integer in the range of 100 to 10,000 in units of Millisecond

Default

3000

Name

cerRetransMax - CER max retransmissions

Syntax

--cerRetransMax <integer>

Description

Capability exchange request maximum retransmissions. when set to 0 if an MME initiated CER is lost or CEA is lost/delayed beyond timeout (CER message timer 3 sec) the connection shall be closed, it shall not be retransmitted, only connection timer shall be used to schedule the next re-connection.

Class

Optional.

Type / Range

Integer in the range of 0 to 5

Default

3

Name

connectTimer - Connect timer

Syntax

--connectTimer <integer>

Description

Connect timer in 100 msec increments.

Class

Optional.

Type / Range

Integer in the range of 100 to 10,000 in units of MilliSecond

Default

3000

Name

description - Description

Syntax

--description <string>

Description

Profile description.

Class

Optional.

Type / Range

Digit string of length 0 to 160 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Name

diamProductName - Diameter product name

Syntax

--diamProductName <string>

Description

Unique diameter product name.

Class

Optional.

Type / Range

Digit string of length 1 to 100 with the allowed pattern of [0-9a-zA-Z\.\-_:]*

Default

LTE MME Diameter

Name

dprMsgTimer - DPR message timer

Syntax

--dprMsgTimer <integer>

Description

DPR timeout message timer value in 100 msec increments.

Class

Optional.

Type / Range

Integer in the range of 100 to 30,000 in units of MilliSecond

Default

6000

Name

dwrFreqTimer - DWR frequency timer

Syntax

--dwrFreqTimer <integer>

Description

DeviceWatchDog request frequency timer in 100 msec increments.

Class

Optional.

Type / Range

Integer in the range of 100 to 30,000 in units of Millisecond

Default

1000

Name

dwrMsgTimer - DWR message timer

Syntax

--dwrMsgTimer <integer>

Description

DeviceWatchDog request (heartbeat) message timer value in 100 msec increments.

Class

Optional.

Type / Range

Integer in the range of 100 to 30,000 in units of Millisecond

Default

200

Name

dwrRetransMax - DWR max retransmissions

Syntax

--dwrRetransMax <integer>

Description

DeviceWatchDog request maximum retransmissions. when set to 0 the first time a DWR is not responded or DWA is rejected the connection shall be closed. when set to 0, DWR frequency timer shall be still used to send DWR.

Class

Optional.

Type / Range

Integer in the range of 0 to 5

Default

3

Name

reconnectTimer - Reconnect timer

Syntax

--reconnectTimer <integer>

Description

Reconnect timer in 100 msec increments.

Class

Optional.

Type / Range

Integer in the range of 100 to 10,000 in units of MilliSecond

Default

3000

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.87 cmm district

district name. Syntax, description, parameters, and roles for cmm district command.

Syntax**Description**

create or delete a district

Parameters

Name

districtName - district Name

Syntax

--districtName <string>

Description

This parameter defines the name of district.

Class

Required key.

Type / Range

District type

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.88 cmm districtRai

RAI list under a district. Syntax, description, parameters, and roles for cmm districtRai command.

Syntax

Description

a RAI or more RAI defined under a district

Parameters

Name

mcc - mobile country code

Syntax

--mcc <digits>

Description

Mobile Country Code.

Class

Required key.

Type / Range

Digit string of length 3 to 3 with the allowed pattern of [0-9]*

Name

mnc - mobile network code

Syntax

--mnc <digits>

Description

Mobile Network Code.

Class

Required key.

Type / Range

Digit string of length 2 to 3 with the allowed pattern of [0-9]*

Name

rac - routing area code

Syntax

--rac <integer>

Description

Routing Area Code

Class

Required key.

Type / Range

Integer in the range of 0 to 255

Name

lac - location area code

Syntax

--lac <integer>

Description

Location Area Code

Class

Required key.

Type / Range

Integer in the range of 1 to 65,533

Name

districtName - name of district

Syntax

--districtName <string>

Description

This parameter defines the name of district. The parameter is obligatory when you have defined RAllist

Class

Required on create.

Type / Range

District type

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.89 cmm dnn

Data Network Name. Syntax, description, parameters, and roles for cmm dnn command.

Syntax**Description**

Add a DNN to a DNN list specified by a list name.

Parameters**Name**

dnnListName - List Name

Syntax

--dnnListName <string>

Description

Name of the DNN list.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

dataNetworkName - Data Network Name (DNN)

Syntax

--dataNetworkName <string>

Description

Add a DNN to the DNN list.

Class

Required key.

Type / Range

Data Network Name is basically APN Ni.

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.90 cmm dnnList

DNN List. Syntax, description, parameters, and roles for cmm dnnList command.

Syntax**Description**

Creates IDNN list.

Parameters

Name

name - List Name

Syntax

--name <string>

Description

Name of the list.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.91 cmm dnnToSnssai

dnnToSnssai. Syntax, description, parameters, and roles for **cmm dnnToSnssai** command.

Syntax

Description

DNN to S-NSSAI mapping that is used in n26 interworking scenarios

Parameters

Name

dnnToSnssaiListName - dnnToSnssaiListName

Syntax

--dnnToSnssaiListName <string>

Description

DNN to S-NSSAI List Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

dataNetworkName - dataNetworkName

Syntax

--dataNetworkName <string>

Description

Data Network Name

Class

Required key.

Type / Range

DataNetworkName is typically APN Ni, but also allows wildcard APN

Name

snssaiName - S-NSSAI Name

Syntax

--snssaiName <string>

Description

Name of the S-NSSAI

Class

Required on create.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.92 cmm dnnToSnssaiList

dnnToSnssaiList. Syntax, description, parameters, and roles for **cmm dnnToSnssaiList** command.

Syntax

Description

DNN to SNSSAI mapping list name

Parameters

Name

name - name

Syntax

--name <string>

Description

Name of the DNN to SNSSAI mapping list name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.93 cmm dnnToSnssaiPerPlmn

dnnToSnssaiPerPlmn. Syntax, description, parameters, and roles for cmm dnnToSnssaiPerPlmn command.

Syntax

Description

This commands enables creating map of {DNN, TAI List} to SNSSAI that is used in n26 interworking scenarios as part of default V-SMF selection

Parameters

Name

dnnToSnssaiPerPlmnListName - DNN to SNSSAI per PLMN List Name

Syntax

```
--dnnToSnssaiPerPlmnListName <string>
```

Description

Associated list name for {DNN, TAI} to SNSSAI mapping

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

dataNetworkName - Data Network Name

Syntax

```
--dataNetworkName <string>
```

Description

Data Network Name

Class

Required key.

Type / Range

DataNetworkName is typically APN Ni, but also allows wildcard APN

Name

servingPlmn - Serving PLMN

Syntax

```
--servingPlmn <string>
```

Description

Serving PLMN Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

amfTac - AMF TAC

Syntax

--amfTac <integer>

Description

AMF Tracking Area Code

Class

Required key.

Type / Range

Integer in the range of 0 to 16,777,215

Name

snssaiName - SNSSAI Name

Syntax

--snssaiName <string>

Description

Name of the S-NSSAI

Class

Required on create.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.94 cmm dnnToSnssaiPerPlmnList

DNN to SNSSAI per PLMN List. Syntax, description, parameters, and roles for cmm dnnToSnssaiPerPlmnList command.

Syntax

Description

This commands enables creating list name for {DNN, TAI List} to SNSSAI mapping

Parameters

Name

name - Name

Syntax

--name <string>

Description

Name of the DNN to SNSSAI per PLMN mapping list name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.95 cmm dns

DNS configurations. Syntax, description, parameters, and roles for cmm dns command.

Syntax

Description

This command is used to administer DNS configurations and includes parameters for defining the external DNS IP addresses. The IP addresses must be added in order and no holes can be left during deletions.

Parameters

Name

dnsConfigName - DNS ID

Syntax

--dnsConfigName <list-item>

Description

Name of the DNS configuration.

Class

Required key.

List

- config - Config.

Name

priDnsIp - DNS primary ID

Syntax

--priDnsIp <string>

Description

The IP address of the primary external DNS.

Class

Optional.

Type / Range

Digit string of length 3 to 39 with valid IPv4 or IPv6 address

Name

quatDnsIp - DNS quaternary IP

Syntax

--quatDnsIp <string>

Description

The IP address of the quaternary external DNS.

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Default

""

Name

secDnsIp - DNS secondary ID

Syntax

--secDnsIp <string>

Description

The IP address of the secondary external DNS.

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

terDnsIp - DNS tertiary IP

Syntax

--terDnsIp <string>

Description

The IP address of the tertiary external DNS.

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Default

""

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

4.96 cmm draDestFqdn

DRA Destination FQDN. Syntax, description, parameters, and roles for cmm draDestFqdn command.

Syntax**Description**

This command is used to administer a DRA Destination FQDN and includes parameters for defining the FQDN name and string.

Parameters**Name**

draDestFqdnName - DRA destination FQDN name

Syntax

--draDestFqdnName <string>

Description

DRA destination FQDN name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

draDestFqdnStr - DRA destination FQDN string

Syntax

--draDestFqdnStr <string>

Description

DRA destination FQDN string.

Class

Optional.

Type / Range

Digit string of length 1 to 128 with the allowed pattern of [0-9a-zA-Z\.\-:{}]@*

Default

epc.mnc{ueMnc}.mcc{ueMcc}.3gppnetwork.org

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.97 cmm drmpCiotProfile

DRMP CIOT profile. Syntax, description, parameters, and roles for `cmm drmpCiotProfile` command.

Syntax

Description

Diameter Routing Message Priority cellular internet of things profile.

Parameters

Name

name - DRMP CIOT profile name

Syntax

--name <string>

Description

DRMP cellular internet of things profile name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

attachWithoutPdnAir - Attach without PDN, AIR

Syntax

--attachWithoutPdnAir <integer>

Description

Attach without PDN, AIR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

14

Name

attachWithoutPdnUlr - Attach without PDN, ULR

Syntax

--attachWithoutPdnUlr <integer>

Description

Attach without PDN, ULR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

12

Name

cirAuthorizationToIwkScef - CIR Authorization to IWK-SCEF

Syntax

--cirAuthorizationToIwkScef <integer>

Description

Specifies the message priority of the T6ai:CIR used to authorize an inbound roaming UE's monitoring events

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

12

Name

connectionUpdateCmr - Connection update, CMR

Syntax

--connectionUpdateCmr <integer>

Description

Connection update (update SCEF of node change), CMR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

8

Name

connectModeTauMmeChangeUlr - Connect mode TAU MME change, ULR

Syntax

--connectModeTauMmeChangeUlr <integer>

Description

Connected mode TAU with MME change, ULR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

10

Name

description - Description

Syntax

--description <string>

Description

Profile description.

Class

Optional.

Type / Range

Digit string of length 0 to 160 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

""

Name

eUtranAttachImsiGutiAir - eUTRAN attach IMSI GUTI, AIR

Syntax

--eUtranAttachImsiGutiAir <integer>

Description

eUTRAN attach by IMSI/by GUTI, AIR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

12

Name

eUtranAttachImsiGutiNor - eUTRAN attach IMSI GUTI, NOR

Syntax

--eUtranAttachImsiGutiNor <integer>

Description

eUTRAN attach by IMSI/by GUTI, NOR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

14

Name

eUtranAttachImsiGutiUlr - eUTRAN attach IMSI GUTI, ULR

Syntax

--eUtranAttachImsiGutiUlr <integer>

Description

eUTRAN attach by IMSI/by GUTI, ULR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

10

Name

hssRestorationUlr - HSS restoration, ULR

Syntax

--hssRestorationUlr <integer>

Description

HSS restoration, ULR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

14

Name

idleModeTauMmeChangeAir - Idle mode TAU MME change, AIR

Syntax

--idleModeTauMmeChangeAir <integer>

Description

Idle mode TAU with MME change (no SGW change), AIR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

8

Name

idleModeTauMmeChangeUlr - Idle mode TAU MME change, ULR

Syntax

--idleModeTauMmeChangeUlr <integer>

Description

Idle mode TAU with MME change (no SGW change), ULR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

8

Name

idleModeTauMmeSgwChangeAir - Idle mode TAU MME SGW change, AIR

Syntax`--idleModeTauMmeSgwChangeAir <integer>`**Description**

Idle mode TAU with MME change and SGW change, AIR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

10

Name

idleModeTauMmeSgwChangeUlr - Idle mode TAU MME SGW change, ULR

Syntax`--idleModeTauMmeSgwChangeUlr <integer>`**Description**

Idle mode TAU with MME change and SGW change, ULR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

8

Name

idleModeTauSgwChangeAir - Idle mode TAU SGW change, AIR

Syntax`--idleModeTauSgwChangeAir <integer>`**Description**

Idle mode TAU with SGW change, AIR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

10

Name

initAttachMmeChangeAir - Initial attach MME change, AIR

Syntax`--initAttachMmeChangeAir <integer>`**Description**

Initial attach (usable GUTI) where MME change occurs, AIR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

10

Name

initAttachMmeChangeNor - Initial attach MME change, NOR

Syntax`--initAttachMmeChangeNor <integer>`**Description**

Initial attach (usable GUTI) where MME change occurs, NOR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

10

Name

initAttachMmeChangeUlr - Initial attach MME change, ULR

Syntax`--initAttachMmeChangeUlr <integer>`**Description**

Initial attach (usable GUTI) where MME change occurs, ULR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

10

Name

interMmeInterPlmnHoTauAir - Inter MME inter PLMN HO TAU, AIR

Syntax

--interMmeInterPlmnHoTauAir <integer>

Description

Inter MME inter PLMN HO/TAU, AIR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

8

Name

interMmeInterPlmnHoTauUlr - Inter MME inter PLMN HO TAU, ULR

Syntax

--interMmeInterPlmnHoTauUlr <integer>

Description

Inter MME inter PLMN HO/TAU, ULR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

8

Name

lteTauNoSgwMmeChangesAir - LTE TAU no SGW MME changes, AIR

Syntax

--lteTauNoSgwMmeChangesAir <integer>

Description

LTE TAU without SGW/MME changes, AIR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

10

Name

midCallUeAvailFailedDataAttemptCmr - Mid call UE availability failed data attempt, CMR

Syntax

--midCallUeAvailFailedDataAttemptCmr <integer>

Description

Mid call procedure - UE availability after failed SCEF initiated MT data attempt, CMR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

8

Name

mmeFailureNewMmeUlr - MME failure new MME, ULR

Syntax

--mmeFailureNewMmeUlr <integer>

Description

MME failure (we are new MME), ULR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

10

Name

mmeRestartUeSameMmeUlr - MME restart UE same MME, ULR

Syntax

--mmeRestartUeSameMmeUlr <integer>

Description

MME restart (UE comes back to same MME), ULR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

10

Name

moDataOdr - MO data, ODR

Syntax

--moDataOdr <integer>

Description

MO data, ODR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

8

Name

monitorEventReportRir - Monitoring event report, RIR

Syntax

--monitorEventReportRir <integer>

Description

Monitoring event report, RIR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

10

Name

nonIpPdnActivationCmr - Non IP PDN activation, CMR

Syntax

--nonIpPdnActivationCmr <integer>

Description

Non-IP PDN activation, CMR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

12

Name

roamerAir - Roamer, AIR

Syntax

--roamerAir <integer>

Description

Roamer, AIR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

12

Name

roamerUlr - Roamer, ULR

Syntax

--roamerUlr <integer>

Description

Roamer, ULR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

12

Name

s1HoMmeRelocDirectDataUlr - S1 HO MME relocation direct data, ULR

Syntax

--s1HoMmeRelocDirectDataUlr <integer>

Description

S1-based HO with MME relocation (Direct data tunnel), ULR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

8

Name

s1HoMmeRelocIndirectDataUlr - S1 HO MME relocation indirect data, ULR

Syntax

--s1HoMmeRelocIndirectDataUlr <integer>

Description

S1-based HO with MME relocation (Indirect data tunnel), ULR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

8

Name

s1HoMmeSgwRelocDirectDataUlr - S1 HO MME SGW relocation direct data, ULR

Syntax

--s1HoMmeSgwRelocDirectDataUlr <integer>

Description

S1-based HO with MME/SGW relocation (Direct data tunnel), ULR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

8

Name

s1HoMmeSgwRelocIndirectDataUlr - S1 HO MME SGW relocation indirect data, ULR

Syntax

--s1HoMmeSgwRelocIndirectDataUlr <integer>

Description

S1-based HO with MME/SGW relocation (Indirect data tunnel), ULR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

8

Name

smsAlr - SMS Alert, ALR

Syntax

--smsAlr <integer>

Description

SGd interface Alert Service Center Request (ALR) message. The MME uses the alert when either the UE has moved to another MME or SGSN area or the UE is available for MT SMS (after previously not being available for SMS).

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

12

Name

smsOfr - MO SMS delivery, OFR

Syntax

--smsOfr <integer>

Description

MO SMS delivery towards SMS router, SGd interface MO Forward Short Message Request (OFR) message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

12

Name

ueInitNonIpPdnDiscCmr - UE initiated non IP PDN disconnect, CMR

Syntax

--ueInitNonIpPdnDiscCmr <integer>

Description

UE initiated non-ip PDN disconnection, CMR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

6

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.98 cmm drmpProfile

DRMP profile. Syntax, description, parameters, and roles for cmm drmpProfile command.

Syntax**Description**

Diameter Routing Message Priority profile.

Parameters**Name**

name - DRMP profile name

Syntax

--name <string>

Description

DRMP profile name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

connectModeTauMmeChangeEmpsUlr - Connect mode TAU MME change eMPS, ULR

Syntax

--connectModeTauMmeChangeEmpsUlr <integer>

Description

Connected mode TAU with MME change - while eMPS call is going on, ULR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

0

Name

connectModeTauMmeChangeUlr - Connect mode TAU MME change, ULR

Syntax

--connectModeTauMmeChangeUlr <integer>

Description

Connected mode TAU with MME change, ULR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

8

Name

description - Description

Syntax

--description <string>

Description

Profile description.

Class

Optional.

Type / Range

Digit string of length 0 to 160 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

ehrpdlteHandoverAir - eHRPD LTE handover, AIR

Syntax

--ehrpdlteHandoverAir <integer>

Description

eHRPD - LTE handover, AIR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

6

Name

ehrpdlteHandoverUlr - eHRPD LTE handover, ULR

Syntax

--ehrpdlteHandoverUlr <integer>

Description

eHRPD - LTE handover, ULR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

6

Name

emerPdnInitAttachAir - Emergency PDN initial attach, AIR

Syntax

--emerPdnInitAttachAir <integer>

Description

Emergency PDN initial attach, AIR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

4

Name

emerPdnInitAttachUlr - Emergency PDN initial attach, ULR

Syntax

--emerPdnInitAttachUlr <integer>

Description

Emergency PDN initial attach, ULR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

4

Name

empsAir - eMPS AIR

Syntax

--empsAir <integer>

Description

Effective when gParm empsPrioToHss is set to Yes. DRMP value on S6a AIR for user indicating priority on RRC establishment cause.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

0

Name

empsUlr - eMPS ULR

Syntax

--empsUlr <integer>

Description

Effective when gParm empsPrioToHss is set to Yes. DRMP value on S6a ULR for user indicating priority on RRC establishment cause.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

0

Name

eUtranAttachImsiGutiAir - eUTRAN attach IMSI GUTI, AIR

Syntax

--eUtranAttachImsiGutiAir <integer>

Description

eUTRAN attach by IMSI/by GUTI, AIR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

10

Name

eUtranAttachImsiGutiNor - eUTRAN attach IMSI GUTI, NOR

Syntax

--eUtranAttachImsiGutiNor <integer>

Description

eUTRAN attach by IMSI/by GUTI, NOR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

12

Name

eUtranAttachImsiGutiUlr - eUTRAN attach IMSI GUTI, ULR

Syntax`--eUtranAttachImsiGutiUlr <integer>`**Description**

eUTRAN attach by IMSI/by GUTI, ULR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

8

Name

handoverAttachAir - Handover attach, AIR

Syntax`--handoverAttachAir <integer>`**Description**

Handover attach (non-3gpp to LTE usable GUTI), AIR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

6

Name

handoverAttachUlr - Handover attach, ULR

Syntax`--handoverAttachUlr <integer>`**Description**

Handover attach (non-3gpp to LTE usable GUTI), ULR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

6

Name

hssRestorationUlr - HSS restoration, ULR

Syntax

--hssRestorationUlr <integer>

Description

HSS restoration, ULR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

12

Name

idleModeTauMmeChangeAir - Idle mode TAU MME change, AIR

Syntax

--idleModeTauMmeChangeAir <integer>

Description

Idle mode TAU with MME change (no SGW change), AIR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

6

Name

idleModeTauMmeChangeUlr - Idle mode TAU MME change, ULR

Syntax

--idleModeTauMmeChangeUlr <integer>

Description

Idle mode TAU with MME change (no SGW change), ULR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

6

Name

idleModeTauMmeSgwChangeAir - Idle mode TAU MME SGW change, AIR

Syntax

--idleModeTauMmeSgwChangeAir <integer>

Description

Idle mode TAU with MME change and SGW change, AIR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

8

Name

idleModeTauMmeSgwChangeUlr - Idle mode TAU MME SGW change, ULR

Syntax

--idleModeTauMmeSgwChangeUlr <integer>

Description

Idle mode TAU with MME change and SGW change, ULR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

6

Name

idleModeTauSgwChangeAir - Idle mode TAU SGW change, AIR

Syntax

--idleModeTauSgwChangeAir <integer>

Description

Idle mode TAU with SGW change, AIR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

8

Name

initAttachMmeChangeAir - Initial attach MME change, ULR

Syntax

--initAttachMmeChangeAir <integer>

Description

Initial attach (usable GUTI) where MME change occurs, AIR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

8

Name

initAttachMmeChangeNor - Initial attach MME change, ULR

Syntax

--initAttachMmeChangeNor <integer>

Description

Initial attach (usable GUTI) where MME change occurs, NOR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

10

Name

initAttachMmeChangeUlr - Initial attach MME change, ULR

Syntax

--initAttachMmeChangeUlr <integer>

Description

Initial attach (usable GUTI) where MME change occurs, ULR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

8

Name

interMmeInterPlmnHoTauAir - Inter MME inter PLMN HO TAU, AIR

Syntax

```
--interMmeInterPlmnHoTauAir <integer>
```

Description

Inter MME inter PLMN HO/TAU, AIR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

6

Name

interMmeInterPlmnHoTauUlr - Inter MME inter PLMN HO TAU, ULR

Syntax

```
--interMmeInterPlmnHoTauUlr <integer>
```

Description

Inter MME inter PLMN HO/TAU, ULR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

6

Name

lteEhrpdHoAddPdnsPur - LTE eHRPD HO additional PDNs, PUR

Syntax

```
--lteEhrpdHoAddPdnsPur <integer>
```

Description

LTE-eHRPD HO additional PDNs, PUR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

8

Name

lteTauNoSgwMmeChangesAir - LTE TAU no SGW MME changes, AIR

Syntax

--lteTauNoSgwMmeChangesAir <integer>

Description

LTE TAU without SGW/MME changes, AIR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

8

Name

mmeFailureNewMmeUlr - MME failure new MME, ULR

Syntax

--mmeFailureNewMmeUlr <integer>

Description

MME failure (we are new MME), ULR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

8

Name

mmeRestartUeSameMmeUlr - MME restart UE same MME, ULR

Syntax

--mmeRestartUeSameMmeUlr <integer>

Description

MME restart (UE comes back to same MME), ULR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

8

Name

roamerAir - Roamer, AIR

Syntax

--roamerAir <integer>

Description

Roamer, AIR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

10

Name

roamerEmerPdnInitAttachAir - Roamer emergency PDN initial attach, AIR

Syntax

--roamerEmerPdnInitAttachAir <integer>

Description

Roamer emergency PDN initial attach, AIR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

4

Name

roamerEmerPdnInitAttachUlr - Roamer emergency PDN initial attach, ULR

Syntax

--roamerEmerPdnInitAttachUlr <integer>

Description

Roamer emergency PDN initial attach, ULR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

4

Name

roamerUlr - Roamer, ULR

Syntax

--roamerUlr <integer>

Description

Roamer, ULR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

10

Name

s1HoMmeRelocDirectDataUlr - S1 HO MME relocation direct data, ULR

Syntax

--s1HoMmeRelocDirectDataUlr <integer>

Description

S1-based HO with MME relocation (Direct data tunnel), ULR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

6

Name

s1HoMmeRelocIndirectDataUlr - S1 HO MME relocation indirect data, ULR

Syntax

```
--s1HoMmeRelocIndirectDataUlr <integer>
```

Description

S1-based HO with MME relocation (Indirect data tunnel), ULR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

6

Name

s1HoMmeSgwRelocDirectDataUlr - S1 HO MME SGW relocation direct data, ULR

Syntax

```
--s1HoMmeSgwRelocDirectDataUlr <integer>
```

Description

S1-based HO with MME/SGW relocation (Direct data tunnel), ULR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

6

Name

s1HoMmeSgwRelocIndirectDataUlr - S1 HO MME SGW relocation indirect data, ULR

Syntax

```
--s1HoMmeSgwRelocIndirectDataUlr <integer>
```

Description

S1-based HO with MME/SGW relocation (Indirect data tunnel), ULR message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

6

Name

smsAlr - SMS Alert

Syntax

--smsAlr <integer>

Description

SGd interface Alert Service Center Request (ALR) message. The MME uses the alert when either the UE has moved to another MME or SGSN area or the UE is available for MT SMS (after previously not being available for SMS).

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

12

Name

smsOfr - SMS OFR

Syntax

--smsOfr <integer>

Description

MO SMS delivery towards SMS router, SGd interface MO Forward Short Message Request (OFR) message.

Class

Optional.

Type / Range

Integer in the range of 0, 2, 4, 6, 8, 10, 12, or 14

Default

12

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.99 cmm edrxApn

eDRX APN. Syntax, description, parameters, and roles for **cmm edrxApn** command.

Syntax**Description**

Administer eDRX for APN.

Parameters**Name**

edrxApnListName - eDRX APN list name

Syntax

--edrxApnListName <string>

Description

eDRX APN list name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

apnNi - APN NI

Syntax

--apnNi <string>

Description

APN NI.

Class

Required key.

Type / Range

Digit string of length 1 to 62 with the allowed pattern of [0-9a-zA-Z_\. \-]*

Name

edrxEnabled - eDRX enabled

Syntax

--edrxEnabled <list-item>

Description

eDRX enabled.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

nbEdrxCycleLength - NB-S1 mode eDRX cycle length

Syntax

--nbEdrxCycleLength <list-item>

Description

eDRX cycle value for NB-S1 UE.

Class

Optional.

List

- 20.48 - Specifies eDRX narrow band cycle length of 20.48 seconds.
- 40.96 - Specifies eDRX narrow band cycle length 40.96 seconds.
- 81.92 - Specifies eDRX narrow band cycle length 81.92 seconds.
- 163.84 - Specifies eDRX narrow band cycle length of 163.84 seconds.
- 327.68 - Specifies eDRX narrow band cycle length of 327.68 seconds.
- 655.36 - Specifies eDRX narrow band cycle length of 655.36 seconds.
- 1310.72 - Specifies eDRX narrow band cycle length of 1310.72 seconds.
- 2621.44 - Specifies eDRX narrow band cycle length of 2621.44 seconds.
- 5242.88 - Specifies eDRX narrow band cycle length of 5242.88 seconds.
- 10485.76 - Specifies eDRX narrow band cycle length of 10485.76 seconds.

Default

20.48

Name

nbEdrxPtw - NB-S1 mode PTW

Syntax

--nbEdrxPtw <list-item>

Description

PTW value for NB-S1 UE. PTW value must be less than the eDRX cycle value.

Class

Optional.

List

- 2.56 - Specifies eDRX narrow band paging time window of 2.56 seconds.
- 5.12 - Specifies eDRX narrow band paging time window of 5.12 seconds.
- 7.68 - Specifies eDRX narrow band paging time window of 7.68 seconds.
- 10.24 - Specifies eDRX narrow band paging time window of 10.24 seconds.
- 12.80 - Specifies eDRX narrow band paging time window of 12.80 seconds.
- 15.36 - Specifies eDRX narrow band paging time window of 15.36 seconds.
- 17.92 - Specifies eDRX narrow band paging time window of 17.92 seconds.
- 20.48 - Specifies eDRX narrow band paging time window of 20.48 seconds.
- 23.04 - Specifies eDRX narrow band paging time window of 23.04 seconds.
- 25.60 - Specifies eDRX narrow band paging time window of 25.60 seconds.
- 28.16 - Specifies eDRX narrow band paging time window of 28.16 seconds.
- 30.72 - Specifies eDRX narrow band paging time window of 30.72 seconds.
- 33.28 - Specifies eDRX narrow band paging time window of 33.28 seconds.
- 35.84 - Specifies eDRX narrow band paging time window of 35.84 seconds.
- 38.40 - Specifies eDRX narrow band paging time window of 38.40 seconds.
- 40.96 - Specifies eDRX narrow band paging time window of 40.96 seconds.

Default

10.24

Name

source - source

Syntax

--source <list-item>

Description

source.

Class

Optional.

List

- UE - description: Specifies eDRX/PSM source as UE.
- local - description: local means Specifies eDRX/PSM source as APN .

Default

local

Name

wbEdrxCycleLength - WB-S1 mode eDRX cycle length

Syntax

--wbEdrxCycleLength <list-item>

Description

eDRX cycle value for WB-S1 UE.

Class

Optional.

List

- 10.24 - Specifies eDRX wide band cycle length of 20.48 seconds.
- 20.48 - 10.48 - Specifies eDRX wide band cycle length of 20.48 seconds.
- 40.96 - Specifies eDRX wide band cycle length of 40.96 seconds.
- 61.44 - Specifies eDRX wide band cycle length of 61.44 seconds.
- 81.92 - Specifies eDRX wide band cycle length of 81.92 seconds.
- 102.40 - Specifies eDRX wide band cycle length of 102.40 seconds.
- 122.88 - Specifies eDRX wide band cycle length of 122.88 seconds.
- 143.36 - Specifies eDRX wide band cycle length of 143.36 seconds.
- 163.84 - Specifies eDRX wide band cycle length of 163.84 seconds.
- 327.68 - Specifies eDRX wide band cycle length of 327.68 seconds.
- 655.36 - Specifies eDRX wide band cycle length of 655.36 seconds.
- 1310.72 - Specifies eDRX wide band cycle length of 1310.72 seconds.
- 2621.44 - Specifies eDRX wide band cycle length of 2621.44 seconds.

Default

20.48

Name

wbEdrxPtw - WB-S1 mode PTW

Syntax

--wbEdrxPtw <list-item>

Description

PTW value for WB-S1 UE. PTW value must be less than the eDRX cycle value.

Class

Optional.

List

- 1.28 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 2.56 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 3.84 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 5.12 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 6.40 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 7.68 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 8.96 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 10.24 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 11.52 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 12.80 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 14.08 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 15.36 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 16.64 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 17.92 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 19.20 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 20.48 - Specifies eDRX wide band paging time window length of 1.28 seconds.

Default

10.24

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.100 cmm edrxApnList

eDRX APN List. Syntax, description, parameters, and roles for cmm edrxApnList command.

Syntax

Description

Administer List of APNs for eDRX.

Parameters

Name

name - eDRX APN list name

Syntax

--name <string>

Description

eDRX APN list name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

description - Description

Syntax

--description <string>

Description

Profile description.

Class

Optional.

Type / Range

Digit string of length 0 to 160 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

"

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.101 cmm edrxHssPtwProfile

HSS PTW eDRX Profile. Syntax, description, parameters, and roles for cmm edrxHssPtwProfile command.

Syntax

Description

eDRX HSS PTW Profile used by eDRX and HLCOM.

Parameters

Name

name - Profile name

Syntax

--name <string>

Description

eDRX HSS PTW profile name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

description - Profile description

Syntax

--description <string>

Description

eDRX HSS PTW profile description.

Class

Optional.

Type / Range

Digit string of length 0 to 160 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Name

nbEdrx10Ptw - NB-S1 mode eDRX-10 PTW length

Syntax

--nbEdrx10Ptw <list-item>

Description

PTW length for eDRX cycle length of 327.68 seconds for NB-S1 UE.

Class

Optional.

List

- 2.56 - Specifies eDRX narrow band paging time window of 2.56 seconds.
- 5.12 - Specifies eDRX narrow band paging time window of 5.12 seconds.
- 7.68 - Specifies eDRX narrow band paging time window of 7.68 seconds.
- 10.24 - Specifies eDRX narrow band paging time window of 10.24 seconds.
- 12.80 - Specifies eDRX narrow band paging time window of 12.80 seconds.
- 15.36 - Specifies eDRX narrow band paging time window of 15.36 seconds.
- 17.92 - Specifies eDRX narrow band paging time window of 17.92 seconds.
- 20.48 - Specifies eDRX narrow band paging time window of 20.48 seconds.
- 23.04 - Specifies eDRX narrow band paging time window of 23.04 seconds.
- 25.60 - Specifies eDRX narrow band paging time window of 25.60 seconds.
- 28.16 - Specifies eDRX narrow band paging time window of 28.16 seconds.
- 30.72 - Specifies eDRX narrow band paging time window of 30.72 seconds.
- 33.28 - Specifies eDRX narrow band paging time window of 33.28 seconds.
- 35.84 - Specifies eDRX narrow band paging time window of 35.84 seconds.
- 38.40 - Specifies eDRX narrow band paging time window of 38.40 seconds.
- 40.96 - Specifies eDRX narrow band paging time window of 40.96 seconds.

Default

28.16

Name

nbEdrx11Ptw - NB-S1 mode eDRX-11 PTW length

Syntax`--nbEdrx11Ptw <list-item>`**Description**

PTW length for eDRX cycle length of 655.36 seconds for NB-S1 UE.

Class

Optional.

List

- 2.56 - Specifies eDRX narrow band paging time window of 2.56 seconds.
- 5.12 - Specifies eDRX narrow band paging time window of 5.12 seconds.
- 7.68 - Specifies eDRX narrow band paging time window of 7.68 seconds.
- 10.24 - Specifies eDRX narrow band paging time window of 10.24 seconds.
- 12.80 - Specifies eDRX narrow band paging time window of 12.80 seconds.
- 15.36 - Specifies eDRX narrow band paging time window of 15.36 seconds.
- 17.92 - Specifies eDRX narrow band paging time window of 17.92 seconds.
- 20.48 - Specifies eDRX narrow band paging time window of 20.48 seconds.
- 23.04 - Specifies eDRX narrow band paging time window of 23.04 seconds.
- 25.60 - Specifies eDRX narrow band paging time window of 25.60 seconds.
- 28.16 - Specifies eDRX narrow band paging time window of 28.16 seconds.
- 30.72 - Specifies eDRX narrow band paging time window of 30.72 seconds.
- 33.28 - Specifies eDRX narrow band paging time window of 33.28 seconds.
- 35.84 - Specifies eDRX narrow band paging time window of 35.84 seconds.
- 38.40 - Specifies eDRX narrow band paging time window of 38.40 seconds.
- 40.96 - Specifies eDRX narrow band paging time window of 40.96 seconds.

Default

28.16

Name

nbEdrx12Ptw - NB-S1 mode eDRX-12 PTW length

Syntax`--nbEdrx12Ptw <list-item>`

Description

PTW length for eDRX cycle length of 1310.72 seconds for NB-S1 UE.

Class

Optional.

List

- 2.56 - Specifies eDRX narrow band paging time window of 2.56 seconds.
- 5.12 - Specifies eDRX narrow band paging time window of 5.12 seconds.
- 7.68 - Specifies eDRX narrow band paging time window of 7.68 seconds.
- 10.24 - Specifies eDRX narrow band paging time window of 10.24 seconds.
- 12.80 - Specifies eDRX narrow band paging time window of 12.80 seconds.
- 15.36 - Specifies eDRX narrow band paging time window of 15.36 seconds.
- 17.92 - Specifies eDRX narrow band paging time window of 17.92 seconds.
- 20.48 - Specifies eDRX narrow band paging time window of 20.48 seconds.
- 23.04 - Specifies eDRX narrow band paging time window of 23.04 seconds.
- 25.60 - Specifies eDRX narrow band paging time window of 25.60 seconds.
- 28.16 - Specifies eDRX narrow band paging time window of 28.16 seconds.
- 30.72 - Specifies eDRX narrow band paging time window of 30.72 seconds.
- 33.28 - Specifies eDRX narrow band paging time window of 33.28 seconds.
- 35.84 - Specifies eDRX narrow band paging time window of 35.84 seconds.
- 38.40 - Specifies eDRX narrow band paging time window of 38.40 seconds.
- 40.96 - Specifies eDRX narrow band paging time window of 40.96 seconds.

Default

28.16

Name

nbEdrx13Ptw - NB-S1 mode eDRX-13 PTW length

Syntax

--nbEdrx13Ptw <list-item>

Description

PTW length for eDRX cycle length of 2621.44 seconds for NB-S1 UE.

Class

Optional.

List

- 2.56 - Specifies eDRX narrow band paging time window of 2.56 seconds.

- 5.12 - Specifies eDRX narrow band paging time window of 5.12 seconds.
- 7.68 - Specifies eDRX narrow band paging time window of 7.68 seconds.
- 10.24 - Specifies eDRX narrow band paging time window of 10.24 seconds.
- 12.80 - Specifies eDRX narrow band paging time window of 12.80 seconds.
- 15.36 - Specifies eDRX narrow band paging time window of 15.36 seconds.
- 17.92 - Specifies eDRX narrow band paging time window of 17.92 seconds.
- 20.48 - Specifies eDRX narrow band paging time window of 20.48 seconds.
- 23.04 - Specifies eDRX narrow band paging time window of 23.04 seconds.
- 25.60 - Specifies eDRX narrow band paging time window of 25.60 seconds.
- 28.16 - Specifies eDRX narrow band paging time window of 28.16 seconds.
- 30.72 - Specifies eDRX narrow band paging time window of 30.72 seconds.
- 33.28 - Specifies eDRX narrow band paging time window of 33.28 seconds.
- 35.84 - Specifies eDRX narrow band paging time window of 35.84 seconds.
- 38.40 - Specifies eDRX narrow band paging time window of 38.40 seconds.
- 40.96 - Specifies eDRX narrow band paging time window of 40.96 seconds.

Default

28.16

Name

nbEdrx14Ptw - NB-S1 mode eDRX-14 PTW length

Syntax

--nbEdrx14Ptw <list-item>

Description

PTW length for eDRX cycle length of 5242.88 seconds for NB-S1 UE.

Class

Optional.

List

- 2.56 - Specifies eDRX narrow band paging time window of 2.56 seconds.
- 5.12 - Specifies eDRX narrow band paging time window of 5.12 seconds.
- 7.68 - Specifies eDRX narrow band paging time window of 7.68 seconds.
- 10.24 - Specifies eDRX narrow band paging time window of 10.24 seconds.
- 12.80 - Specifies eDRX narrow band paging time window of 12.80 seconds.
- 15.36 - Specifies eDRX narrow band paging time window of 15.36 seconds.

- 17.92 - Specifies eDRX narrow band paging time window of 17.92 seconds.
- 20.48 - Specifies eDRX narrow band paging time window of 20.48 seconds.
- 23.04 - Specifies eDRX narrow band paging time window of 23.04 seconds.
- 25.60 - Specifies eDRX narrow band paging time window of 25.60 seconds.
- 28.16 - Specifies eDRX narrow band paging time window of 28.16 seconds.
- 30.72 - Specifies eDRX narrow band paging time window of 30.72 seconds.
- 33.28 - Specifies eDRX narrow band paging time window of 33.28 seconds.
- 35.84 - Specifies eDRX narrow band paging time window of 35.84 seconds.
- 38.40 - Specifies eDRX narrow band paging time window of 38.40 seconds.
- 40.96 - Specifies eDRX narrow band paging time window of 40.96 seconds.

Default

28.16

Name

nbEdrx15Ptw - NB-S1 mode eDRX-15 PTW length

Syntax

--nbEdrx15Ptw <list-item>

Description

PTW length for eDRX cycle length of 10485.76 seconds for NB-S1 UE.

Class

Optional.

List

- 2.56 - Specifies eDRX narrow band paging time window of 2.56 seconds.
- 5.12 - Specifies eDRX narrow band paging time window of 5.12 seconds.
- 7.68 - Specifies eDRX narrow band paging time window of 7.68 seconds.
- 10.24 - Specifies eDRX narrow band paging time window of 10.24 seconds.
- 12.80 - Specifies eDRX narrow band paging time window of 12.80 seconds.
- 15.36 - Specifies eDRX narrow band paging time window of 15.36 seconds.
- 17.92 - Specifies eDRX narrow band paging time window of 17.92 seconds.
- 20.48 - Specifies eDRX narrow band paging time window of 20.48 seconds.
- 23.04 - Specifies eDRX narrow band paging time window of 23.04 seconds.
- 25.60 - Specifies eDRX narrow band paging time window of 25.60 seconds.
- 28.16 - Specifies eDRX narrow band paging time window of 28.16 seconds.

- 30.72 - Specifies eDRX narrow band paging time window of 30.72 seconds.
- 33.28 - Specifies eDRX narrow band paging time window of 33.28 seconds.
- 35.84 - Specifies eDRX narrow band paging time window of 35.84 seconds.
- 38.40 - Specifies eDRX narrow band paging time window of 38.40 seconds.
- 40.96 - Specifies eDRX narrow band paging time window of 40.96 seconds.

Default

28.16

Name

nbEdrx2Ptw - NB-S1 mode eDRX-2 PTW length

Syntax

--nbEdrx2Ptw <list-item>

Description

PTW length for eDRX cycle length of 20.48 seconds for NB-S1 UE.

Class

Optional.

List

- 2.56 - Specifies eDRX narrow band paging time window of 2.56 seconds.
- 5.12 - Specifies eDRX narrow band paging time window of 5.12 seconds.
- 7.68 - Specifies eDRX narrow band paging time window of 7.68 seconds.
- 10.24 - Specifies eDRX narrow band paging time window of 10.24 seconds.
- 12.80 - Specifies eDRX narrow band paging time window of 12.80 seconds.
- 15.36 - Specifies eDRX narrow band paging time window of 15.36 seconds.
- 17.92 - Specifies eDRX narrow band paging time window of 17.92 seconds.

Default

10.24

Name

nbEdrx3Ptw - NB-S1 mode eDRX-3 PTW length

Syntax

--nbEdrx3Ptw <list-item>

Description

PTW length for eDRX cycle length of 40.96 seconds for NB-S1 UE.

Class

Optional.

List

- 2.56 - Specifies EDRX narrow band paging time window of 2.56 seconds.
- 5.12 - Specifies EDRX narrow band paging time window of 5.12 seconds.
- 7.68 - Specifies EDRX narrow band paging time window of 7.68 seconds.
- 10.24 - Specifies EDRX narrow band paging time window of 10.24 seconds.
- 12.80 - Specifies EDRX narrow band paging time window of 12.80 seconds.
- 15.36 - Specifies EDRX narrow band paging time window of 15.36 seconds.
- 17.92 - Specifies EDRX narrow band paging time window of 17.92 seconds.
- 20.48 - Specifies EDRX narrow band paging time window of 20.48 seconds.
- 23.04 - Specifies EDRX narrow band paging time window of 23.04 seconds.
- 25.60 - Specifies EDRX narrow band paging time window of 25.60 seconds.
- 28.16 - Specifies EDRX narrow band paging time window of 28.16 seconds.
- 30.72 - Specifies EDRX narrow band paging time window of 30.72 seconds.
- 33.28 - Specifies EDRX narrow band paging time window of 33.28 seconds.
- 35.84 - Specifies EDRX narrow band paging time window of 35.84 seconds.
- 38.40 - Specifies EDRX narrow band paging time window of 38.40 seconds.

Default

10.24

Name

nbEdrx5Ptw - NB-S1 mode eDRX-5 PTW length

Syntax`--nbEdrx5Ptw <list-item>`**Description**

PTW length for eDRX cycle length of 81.92 seconds for NB-S1 UE.

Class

Optional.

List

- 2.56 - Specifies eDRX narrow band paging time window of 2.56 seconds.
- 5.12 - Specifies eDRX narrow band paging time window of 5.12 seconds.
- 7.68 - Specifies eDRX narrow band paging time window of 7.68 seconds.
- 10.24 - Specifies eDRX narrow band paging time window of 10.24 seconds.
- 12.80 - Specifies eDRX narrow band paging time window of 12.80 seconds.

- 15.36 - Specifies eDRX narrow band paging time window of 15.36 seconds.
- 17.92 - Specifies eDRX narrow band paging time window of 17.92 seconds.
- 20.48 - Specifies eDRX narrow band paging time window of 20.48 seconds.
- 23.04 - Specifies eDRX narrow band paging time window of 23.04 seconds.
- 25.60 - Specifies eDRX narrow band paging time window of 25.60 seconds.
- 28.16 - Specifies eDRX narrow band paging time window of 28.16 seconds.
- 30.72 - Specifies eDRX narrow band paging time window of 30.72 seconds.
- 33.28 - Specifies eDRX narrow band paging time window of 33.28 seconds.
- 35.84 - Specifies eDRX narrow band paging time window of 35.84 seconds.
- 38.40 - Specifies eDRX narrow band paging time window of 38.40 seconds.
- 40.96 - Specifies eDRX narrow band paging time window of 40.96 seconds.

Default

10.24

Name

nbEdrx9Ptw - NB-S1 mode eDRX-9 PTW length

Syntax

--nbEdrx9Ptw <list-item>

Description

PTW length for eDRX cycle length of 163.84 seconds for NB-S1 UE.

Class

Optional.

List

- 2.56 - Specifies eDRX narrow band paging time window of 2.56 seconds.
- 5.12 - Specifies eDRX narrow band paging time window of 5.12 seconds.
- 7.68 - Specifies eDRX narrow band paging time window of 7.68 seconds.
- 10.24 - Specifies eDRX narrow band paging time window of 10.24 seconds.
- 12.80 - Specifies eDRX narrow band paging time window of 12.80 seconds.
- 15.36 - Specifies eDRX narrow band paging time window of 15.36 seconds.
- 17.92 - Specifies eDRX narrow band paging time window of 17.92 seconds.
- 20.48 - Specifies eDRX narrow band paging time window of 20.48 seconds.
- 23.04 - Specifies eDRX narrow band paging time window of 23.04 seconds.
- 25.60 - Specifies eDRX narrow band paging time window of 25.60 seconds.

- 28.16 - Specifies eDRX narrow band paging time window of 28.16 seconds.
- 30.72 - Specifies eDRX narrow band paging time window of 30.72 seconds.
- 33.28 - Specifies eDRX narrow band paging time window of 33.28 seconds.
- 35.84 - Specifies eDRX narrow band paging time window of 35.84 seconds.
- 38.40 - Specifies eDRX narrow band paging time window of 38.40 seconds.
- 40.96 - Specifies eDRX narrow band paging time window of 40.96 seconds.

Default

17.92

Name

wbEdrx10Ptw - WB-S1 mode eDRX-10 PTW length

Syntax

--wbEdrx10Ptw <list-item>

Description

PTW length for eDRX cycle length of 327.68 seconds for WB-S1 UE.

Class

Optional.

List

- 1.28 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 2.56 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 3.84 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 5.12 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 6.40 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 7.68 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 8.96 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 10.24 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 11.52 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 12.80 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 14.08 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 15.36 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 16.64 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 17.92 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 19.20 - Specifies eDRX wide band paging time window length of 1.28 seconds.

- 20.48 - Specifies eDRX wide band paging time window length of 1.28 seconds.

Default

20.48

Name

wbEdrx11Ptw - WB-S1 mode eDRX-11 PTW length

Syntax

--wbEdrx11Ptw <list-item>

Description

PTW length for eDRX cycle length of 655.36 seconds for WB-S1 UE.

Class

Optional.

List

- 1.28 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 2.56 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 3.84 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 5.12 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 6.40 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 7.68 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 8.96 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 10.24 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 11.52 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 12.80 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 14.08 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 15.36 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 16.64 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 17.92 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 19.20 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 20.48 - Specifies eDRX wide band paging time window length of 1.28 seconds.

Default

20.48

Name

wbEdrx12Ptw - WB-S1 mode eDRX-12 PTW length

Syntax

--wbEdrx12Ptw <list-item>

Description

PTW length for eDRX cycle length of 1310.72 seconds for WB-S1 UE.

Class

Optional.

List

- 1.28 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 2.56 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 3.84 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 5.12 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 6.40 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 7.68 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 8.96 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 10.24 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 11.52 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 12.80 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 14.08 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 15.36 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 16.64 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 17.92 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 19.20 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 20.48 - Specifies eDRX wide band paging time window length of 1.28 seconds.

Default

20.48

Name

wbEdrx13Ptw - WB-S1 mode eDRX-13 PTW length

Syntax

--wbEdrx13Ptw <list-item>

Description

PTW length for eDRX cycle length of 2621.44 seconds for WB-S1 UE.

Class

Optional.

List

- 1.28 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 2.56 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 3.84 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 5.12 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 6.40 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 7.68 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 8.96 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 10.24 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 11.52 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 12.80 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 14.08 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 15.36 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 16.64 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 17.92 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 19.20 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 20.48 - Specifies eDRX wide band paging time window length of 1.28 seconds.

Default

20.48

Name

wbEdrx1Ptw - WB-S1 mode eDRX-1 PTW length

Syntax

--wbEdrx1Ptw <list-item>

Description

PTW length for eDRX cycle length of 10.24 seconds for WB-S1 UE.

Class

Optional.

List

- 1.28 - Specifies eDRX wide band paging time window of 1.28 seconds.
- 2.56 - Specifies eDRX wide band paging time window of 2.56 seconds.
- 3.84 - Specifies eDRX wide band paging time window of 3.84 seconds.
- 5.12 - Specifies eDRX wide band paging time window of 5.12 seconds.
- 6.40 - Specifies eDRX wide band paging time window of 6.40 seconds.

- 7.68 - Specifies eDRX wide band paging time window of 7.68 seconds.
- 8.96 - Specifies eDRX wide band paging time window of 8.96 seconds.

Default

6.40

Name

wbEdrx2Ptw - WB-S1 mode eDRX-2 PTW length

Syntax

--wbEdrx2Ptw <list-item>

Description

PTW length for eDRX cycle length of 20.48 seconds for WB-S1 UE.

Class

Optional.

List

- 1.28 - Specifies EDRX wide band paging time window of 1.28 seconds.
- 2.56 - Specifies EDRX wide band paging time window of 2.56 seconds.
- 3.84 - Specifies EDRX wide band paging time window of 3.84 seconds.
- 5.12 - Specifies EDRX wide band paging time window of 5.12 seconds.
- 6.40 - Specifies EDRX wide band paging time window of 6.40 seconds.
- 7.68 - Specifies EDRX wide band paging time window of 7.68 seconds.
- 8.96 - Specifies EDRX wide band paging time window of 8.96 seconds.
- 10.24 - Specifies EDRX wide band paging time window of 10.24 seconds.
- 11.52 - Specifies EDRX wide band paging time window of 11.52seconds.
- 12.80 - Specifies EDRX wide band paging time window of 12.80 seconds.
- 14.08 - Specifies EDRX wide band paging time window of 14.08 seconds.
- 15.36 - Specifies EDRX wide band paging time window of 15.36 seconds.
- 16.64 - Specifies EDRX wide band paging time window of 16.64 seconds.
- 17.92 - Specifies EDRX wide band paging time window of 17.92 seconds.
- 19.20 - Specifies EDRX wide band paging time window of 19.20 seconds.

Default

6.40

Name

wbEdrx3Ptw - WB-S1 mode eDRX-3 PTW length

Syntax

--wbEdrx3Ptw <list-item>

Description

PTW length for eDRX cycle length of 40.96 seconds for WB-S1 UE.

Class

Optional.

List

- 1.28 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 2.56 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 3.84 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 5.12 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 6.40 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 7.68 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 8.96 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 10.24 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 11.52 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 12.80 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 14.08 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 15.36 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 16.64 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 17.92 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 19.20 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 20.48 - Specifies eDRX wide band paging time window length of 1.28 seconds.

Default

6.40

Name

wbEdrx4Ptw - WB-S1 mode eDRX-4 PTW length

Syntax

--wbEdrx4Ptw <list-item>

Description

PTW length for eDRX cycle length of 61.44 seconds for WB-S1 UE.

Class

Optional.

List

- 1.28 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 2.56 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 3.84 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 5.12 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 6.40 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 7.68 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 8.96 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 10.24 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 11.52 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 12.80 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 14.08 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 15.36 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 16.64 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 17.92 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 19.20 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 20.48 - Specifies eDRX wide band paging time window length of 1.28 seconds.

Default

6.40

Name

wbEdrx5Ptw - WB-S1 mode eDRX-5 PTW length

Syntax

--wbEdrx5Ptw <list-item>

Description

PTW length for eDRX cycle length of 81.92 seconds for WB-S1 UE.

Class

Optional.

List

- 1.28 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 2.56 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 3.84 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 5.12 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 6.40 - Specifies eDRX wide band paging time window length of 1.28 seconds.

- 7.68 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 8.96 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 10.24 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 11.52 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 12.80 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 14.08 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 15.36 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 16.64 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 17.92 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 19.20 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 20.48 - Specifies eDRX wide band paging time window length of 1.28 seconds.

Default

8.96

Name

wbEdrx6Ptw - WB-S1 mode eDRX-6 PTW length

Syntax

--wbEdrx6Ptw <list-item>

Description

PTW length for eDRX cycle length of 102.40 seconds for WB-S1 UE.

Class

Optional.

List

- 1.28 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 2.56 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 3.84 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 5.12 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 6.40 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 7.68 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 8.96 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 10.24 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 11.52 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 12.80 - Specifies eDRX wide band paging time window length of 1.28 seconds.

- 14.08 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 15.36 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 16.64 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 17.92 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 19.20 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 20.48 - Specifies eDRX wide band paging time window length of 1.28 seconds.

Default

10.24

Name

wbEdrx7Ptw - WB-S1 mode eDRX-7 PTW length

Syntax

--wbEdrx7Ptw <list-item>

Description

PTW length for eDRX cycle length of 122.88 seconds for WB-S1 UE.

Class

Optional.

List

- 1.28 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 2.56 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 3.84 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 5.12 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 6.40 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 7.68 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 8.96 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 10.24 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 11.52 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 12.80 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 14.08 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 15.36 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 16.64 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 17.92 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 19.20 - Specifies eDRX wide band paging time window length of 1.28 seconds.

- 20.48 - Specifies eDRX wide band paging time window length of 1.28 seconds.

Default

12.80

Name

wbEdrx8Ptw - WB-S1 mode eDRX-8 PTW length

Syntax

--wbEdrx8Ptw <list-item>

Description

PTW length for eDRX cycle length of 143.36 seconds for WB-S1 UE.

Class

Optional.

List

- 1.28 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 2.56 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 3.84 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 5.12 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 6.40 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 7.68 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 8.96 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 10.24 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 11.52 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 12.80 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 14.08 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 15.36 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 16.64 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 17.92 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 19.20 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 20.48 - Specifies eDRX wide band paging time window length of 1.28 seconds.

Default

14.08

Name

wbEdrx9Ptw - WB-S1 mode eDRX-9 PTW length

Syntax

--wbEdrx9Ptw <list-item>

Description

PTW length for eDRX cycle length of 163.84 seconds for WB-S1 UE.

Class

Optional.

List

- 1.28 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 2.56 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 3.84 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 5.12 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 6.40 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 7.68 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 8.96 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 10.24 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 11.52 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 12.80 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 14.08 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 15.36 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 16.64 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 17.92 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 19.20 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 20.48 - Specifies eDRX wide band paging time window length of 1.28 seconds.

Default

16.64

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.102 cmm edrxNrProfile

eDRX NR Profile. Syntax, description, parameters, and roles for `cmm edrxNrProfile` command.

Syntax

Description

eDRX profile for UEs using NR, NR_Redcap, or both access types in the 5G RAN. 32 profiles are supported.

Parameters

Name

name - name

Syntax

--name <string>

Description

Name of the eDRX NR Profile

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

edrxHf1024Override - eDRX HF1024 Override

Syntax

--edrxHf1024override <list-item>

Description

Override value for the HF1024 cycle. Default is hf1024 (i.e. no override).

Class

Optional.

List

- hfquarter - HF Quarter - Quarter of a Hyperframe (2.56 seconds).
- hfhalf - HF Half - Half of a Hyperframe (5.12 seconds)
- hf1 - HF 1 - One Hyperframe (10.24 seconds)

- hf2 - HF 2 - Two Hyperframes (20.48 seconds)
- hf4 - HF 4 - Four Hyperframes (40.96 seconds)
- hf8 - HF 8 - Eight Hyperframes (81.92 seconds)
- hf16 - HF 16 - Sixteen Hyperframes (163.84 seconds)
- hf32 - HF 32 - 32 Hyperframes (327.68 seconds)
- hf64 - HF 64 - 64 Hyperframes (655.36 seconds)
- hf128 - HF 128 - 128 Hyperframes (1310.72 seconds)
- hf256 - HF 256 - 256 Hyperframes (2621.44 seconds)
- hf512 - 512 Hyperframes (5242.88 seconds)
- hf1024 - HF 1024 - 1024 Hyperframes (10485.76 seconds)

Default

hf1024

Recommended

hf1024

Name

edrxHf128Override - eDRX HF128 Override

Syntax

--edrxHf128Override <list-item>

Description

Override value for the HF128 cycle. Default is hf128 (i.e. no override).

Class

Optional.

List

- hfquarter - HF Quarter - Quarter of a Hyperframe (2.56 seconds).
- hfhalf - HF Half - Half of a Hyperframe (5.12 seconds)
- hf1 - HF 1 - One Hyperframe (10.24 seconds)
- hf2 - HF 2 - Two Hyperframes (20.48 seconds)
- hf4 - HF 4 - Four Hyperframes (40.96 seconds)
- hf8 - HF 8 - Eight Hyperframes (81.92 seconds)
- hf16 - HF 16 - Sixteen Hyperframes (163.84 seconds)
- hf32 - HF 32 - 32 Hyperframes (327.68 seconds)
- hf64 - HF 64 - 64 Hyperframes (655.36 seconds)

- hf128 - HF 128 - 128 Hypterframes (1310.72 seconds)
- hf256 - HF 258 - 256 Hyperframes (2621.44 seconds)
- hf512 - 512 Hyperframes (5242.88 seconds)
- hf1024 - HF 1024 - 1024 Hyperframes (10485.76 seconds)

Default

hf128

Recommended

hf128

Name

edrxHf16Override - eDRX HF16 Override

Syntax

--edrxHf16Override <list-item>

Description

Override value for the HF16 cycle. Default is hf16 (i.e. no override).

Class

Optional.

List

- hfquarter - HF Quarter - Quarter of a Hyperframe (2.56 seconds).
- hfhalf - HF Half - Half of a Hyperframe (5.12 seconds)
- hf1 - HF 1 - One Hyperframe (10.24 seconds)
- hf2 - HF 2 - Two Hyperframes (20.48 seconds)
- hf4 - HF 4 - Four Hyperframes (40.96 seconds)
- hf8 - HF 8 - Eight Hyperframes (81.92 seconds)
- hf16 - HF 16 - Sixteen Hyperframes (163.84 seconds)
- hf32 - HF 32 - 32 Hyperframes (327.68 seconds)
- hf64 - HF 64 - 64 Hyperframes (655.36 seconds)
- hf128 - HF 128 - 128 Hypterframes (1310.72 seconds)
- hf256 - HF 258 - 256 Hyperframes (2621.44 seconds)
- hf512 - 512 Hyperframes (5242.88 seconds)
- hf1024 - HF 1024 - 1024 Hyperframes (10485.76 seconds)

Default

hf16

Recommended

hf16

Name

edrxHf1Override - eDRX HF1 Override

Syntax

--edrxHf10override <list-item>

Description

Override value for the HF1 cycle. Default is HF1 (i.e. no override).

Class

Optional.

List

- hfquarter - HF Quarter - Quarter of a Hyperframe (2.56 seconds).
- hfhalf - HF Half - Half of a Hyperframe (5.12 seconds)
- hf1 - HF 1 - One Hyperframe (10.24 seconds)
- hf2 - HF 2 - Two Hyperframes (20.48 seconds)
- hf4 - HF 4 - Four Hyperframes (40.96 seconds)
- hf8 - HF 8 - Eight Hyperframes (81.92 seconds)
- hf16 - HF 16 - Sixteen Hyperframes (163.84 seconds)
- hf32 - HF 32 - 32 Hyperframes (327.68 seconds)
- hf64 - HF 64 - 64 Hyperframes (655.36 seconds)
- hf128 - HF 128 - 128 Hypterframes (1310.72 seconds)
- hf256 - HF 258 - 256 Hyperframes (2621.44 seconds)
- hf512 - 512 Hyperframes (5242.88 seconds)
- hf1024 - HF 1024 - 1024 Hyperframes (10485.76 seconds)

Default

hf1

Recommended

hf1

Name

edrxHf256Override - eDRX HF256 Override

Syntax

--edrxHf2560override <list-item>

Description

Override value for the HF256 cycle. Default is hf256 (i.e. no override).

Class

Optional.

List

- hfquarter - HF Quarter - Quarter of a Hyperframe (2.56 seconds).
- hfhalf - HF Half - Half of a Hyperframe (5.12 seconds)
- hf1 - HF 1 - One Hyperframe (10.24 seconds)
- hf2 - HF 2 - Two Hyperframes (20.48 seconds)
- hf4 - HF 4 - Four Hyperframes (40.96 seconds)
- hf8 - HF 8 - Eight Hyperframes (81.92 seconds)
- hf16 - HF 16 - Sixteen Hyperframes (163.84 seconds)
- hf32 - HF 32 - 32 Hyperframes (327.68 seconds)
- hf64 - HF 64 - 64 Hyperframes (655.36 seconds)
- hf128 - HF 128 - 128 Hypterframes (1310.72 seconds)
- hf256 - HF 258 - 256 Hyperframes (2621.44 seconds)
- hf512 - 512 Hyperframes (5242.88 seconds)
- hf1024 - HF 1024 - 1024 Hyperframes (10485.76 seconds)

Default

hf256

Recommended

hf256

Name

edrxHf2Override - eDRX HF2 Override

Syntax

--edrxHf2Override <list-item>

Description

Override value for the HF2 cycle. Default is hf2 (i.e. no override).

Class

Optional.

List

- hfquarter - HF Quarter - Quarter of a Hyperframe (2.56 seconds).
- hfhalf - HF Half - Half of a Hyperframe (5.12 seconds)
- hf1 - HF 1 - One Hyperframe (10.24 seconds)
- hf2 - HF 2 - Two Hyperframes (20.48 seconds)

- hf4 - HF 4 - Four Hyperframes (40.96 seconds)
- hf8 - HF 8 - Eight Hyperframes (81.92 seconds)
- hf16 - HF 16 - Sixteen Hyperframes (163.84 seconds)
- hf32 - HF 32 - 32 Hyperframes (327.68 seconds)
- hf64 - HF 64 - 64 Hyperframes (655.36 seconds)
- hf128 - HF 128 - 128 Hypterframes (1310.72 seconds)
- hf256 - HF 258 - 256 Hyperframes (2621.44 seconds)
- hf512 - 512 Hyperframes (5242.88 seconds)
- hf1024 - HF 1024 - 1024 Hyperframes (10485.76 seconds)

Default

hf2

Recommended

hf2

Name

edrxHf32Override - eDRX HF32 Override

Syntax

--edrxHf32override <list-item>

Description

Override value for the HF32 cycle. Default is hf32 (i.e. no override).

Class

Optional.

List

- hfquarter - HF Quarter - Quarter of a Hyperframe (2.56 seconds).
- hfhalf - HF Half - Half of a Hyperframe (5.12 seconds)
- hf1 - HF 1 - One Hyperframe (10.24 seconds)
- hf2 - HF 2 - Two Hyperframes (20.48 seconds)
- hf4 - HF 4 - Four Hyperframes (40.96 seconds)
- hf8 - HF 8 - Eight Hyperframes (81.92 seconds)
- hf16 - HF 16 - Sixteen Hyperframes (163.84 seconds)
- hf32 - HF 32 - 32 Hyperframes (327.68 seconds)
- hf64 - HF 64 - 64 Hyperframes (655.36 seconds)
- hf128 - HF 128 - 128 Hypterframes (1310.72 seconds)

- hf256 - HF 258 - 256 Hyperframes (2621.44 seconds)
- hf512 - 512 Hyperframes (5242.88 seconds)
- hf1024 - HF 1024 - 1024 Hyperframes (10485.76 seconds)

Default

hf32

Recommended

hf32

Name

edrxHf4Override - eDRX HF4 Override

Syntax

--edrxHf4Override <list-item>

Description

Override value for the HF4 cycle. Default is hf4 (i.e. no override).

Class

Optional.

List

- hfquarter - HF Quarter - Quarter of a Hyperframe (2.56 seconds).
- hfhalf - HF Half - Half of a Hyperframe (5.12 seconds)
- hf1 - HF 1 - One Hyperframe (10.24 seconds)
- hf2 - HF 2 - Two Hyperframes (20.48 seconds)
- hf4 - HF 4 - Four Hyperframes (40.96 seconds)
- hf8 - HF 8 - Eight Hyperframes (81.92 seconds)
- hf16 - HF 16 - Sixteen Hyperframes (163.84 seconds)
- hf32 - HF 32 - 32 Hyperframes (327.68 seconds)
- hf64 - HF 64 - 64 Hyperframes (655.36 seconds)
- hf128 - HF 128 - 128 Hypterframes (1310.72 seconds)
- hf256 - HF 258 - 256 Hyperframes (2621.44 seconds)
- hf512 - 512 Hyperframes (5242.88 seconds)
- hf1024 - HF 1024 - 1024 Hyperframes (10485.76 seconds)

Default

hf4

Recommended

hf4

Name

edrxHf512Override - eDRX HF512 Override

Syntax

--edrxHf512Override <list-item>

Description

Override value for the HF512 cycle. Default is hf512 (i.e. no override).

Class

Optional.

List

- hfquarter - HF Quarter - Quarter of a Hyperframe (2.56 seconds)
- hfhalf - HF Half - Half of a Hyperframe (5.12 seconds)
- hf1 - HF 1 - One Hyperframe (10.24 seconds)
- hf2 - HF 2 - Two Hyperframes (20.48 seconds)
- hf4 - HF 4 - Four Hyperframes (40.96 seconds)
- hf8 - HF 8 - Eight Hyperframes (81.92 seconds)
- hf16 - HF 16 - Sixteen Hyperframes (163.84 seconds)
- hf32 - HF 32 - 32 Hyperframes (327.68 seconds)
- hf64 - HF 64 - 64 Hyperframes (655.36 seconds)
- hf128 - HF 128 - 128 Hyperframes (1310.72 seconds)
- hf256 - HF 256 - 256 Hyperframes (2621.44 seconds)
- hf512 - 512 Hyperframes (5242.88 seconds)
- hf1024 - HF 1024 - 1024 Hyperframes (10485.76 seconds)

Default

hf512

Recommended

hf512

Name

edrxHf64Override - eDRX HF64 Override

Syntax

--edrxHf64Override <list-item>

Description

Override value for the HF64 cycle. Default is hf64 (i.e. no override).

Class

Optional.

List

- hfquarter - HF Quarter - Quarter of a Hyperframe (2.56 seconds).
- hfhalf - HF Half - Half of a Hyperframe (5.12 seconds)
- hf1 - HF 1 - One Hyperframe (10.24 seconds)
- hf2 - HF 2 - Two Hyperframes (20.48 seconds)
- hf4 - HF 4 - Four Hyperframes (40.96 seconds)
- hf8 - HF 8 - Eight Hyperframes (81.92 seconds)
- hf16 - HF 16 - Sixteen Hyperframes (163.84 seconds)
- hf32 - HF 32 - 32 Hyperframes (327.68 seconds)
- hf64 - HF 64 - 64 Hyperframes (655.36 seconds)
- hf128 - HF 128 - 128 Hyperframes (1310.72 seconds)
- hf256 - HF 256 - 256 Hyperframes (2621.44 seconds)
- hf512 - 512 Hyperframes (5242.88 seconds)
- hf1024 - HF 1024 - 1024 Hyperframes (10485.76 seconds)

Default

hf64

Recommended

hf64

Name

edrxHf8Override - eDRX HF8 Override

Syntax

--edrxHf8override <list-item>

Description

Override value for the HF8 cycle. Default is hf8 (i.e. no override).

Class

Optional.

List

- hfquarter - HF Quarter - Quarter of a Hyperframe (2.56 seconds).
- hfhalf - HF Half - Half of a Hyperframe (5.12 seconds)
- hf1 - HF 1 - One Hyperframe (10.24 seconds)
- hf2 - HF 2 - Two Hyperframes (20.48 seconds)
- hf4 - HF 4 - Four Hyperframes (40.96 seconds)

- hf8 - HF 8 - Eight Hyperframes (81.92 seconds)
- hf16 - HF 16 - Sixteen Hyperframes (163.84 seconds)
- hf32 - HF 32 - 32 Hyperframes (327.68 seconds)
- hf64 - HF 64 - 64 Hyperframes (655.36 seconds)
- hf128 - HF 128 - 128 Hypterframes (1310.72 seconds)
- hf256 - HF 258 - 256 Hyperframes (2621.44 seconds)
- hf512 - 512 Hyperframes (5242.88 seconds)
- hf1024 - HF 1024 - 1024 Hyperframes (10485.76 seconds)

Default

hf8

Recommended

hf8

Name

edrxHfhalfOverride - eDRX HF Half Override

Syntax

--edrxHfhalfOverride <list-item>

Description

Override value for the HF half cycle. Default is hfhalf (i.e. no override).

Class

Optional.

List

- hfquarter - HF Quarter - Quarter of a Hyperframe (2.56 seconds).
- hfhalf - HF Half - Half of a Hyperframe (5.12 seconds)
- hf1 - HF 1 - One Hyperframe (10.24 seconds)
- hf2 - HF 2 - Two Hyperframes (20.48 seconds)
- hf4 - HF 4 - Four Hyperframes (40.96 seconds)
- hf8 - HF 8 - Eight Hyperframes (81.92 seconds)
- hf16 - HF 16 - Sixteen Hyperframes (163.84 seconds)
- hf32 - HF 32 - 32 Hyperframes (327.68 seconds)
- hf64 - HF 64 - 64 Hyperframes (655.36 seconds)
- hf128 - HF 128 - 128 Hypterframes (1310.72 seconds)
- hf256 - HF 258 - 256 Hyperframes (2621.44 seconds)

- hf512 - 512 Hyperframes (5242.88 seconds)
- hf1024 - HF 1024 - 1024 Hyperframes (10485.76 seconds)

Default

hfhalf

Recommended

hfhalf

Name

edrxHfquarterOverride - eDRX HF Quarter Override

Syntax

--edrxHfquarterOverride <list-item>

Description

Override value for the HF quarter cycle. Default is hfquarter (i.e. no override).

Class

Optional.

List

- hfquarter - HF Quarter - Quarter of a Hyperframe (2.56 seconds).
- hfhalf - HF Half - Half of a Hyperframe (5.12 seconds)
- hf1 - HF 1 - One Hyperframe (10.24 seconds)
- hf2 - HF 2 - Two Hyperframes (20.48 seconds)
- hf4 - HF 4 - Four Hyperframes (40.96 seconds)
- hf8 - HF 8 - Eight Hyperframes (81.92 seconds)
- hf16 - HF 16 - Sixteen Hyperframes (163.84 seconds)
- hf32 - HF 32 - 32 Hyperframes (327.68 seconds)
- hf64 - HF 64 - 64 Hyperframes (655.36 seconds)
- hf128 - HF 128 - 128 Hypterframes (1310.72 seconds)
- hf256 - HF 258 - 256 Hyperframes (2621.44 seconds)
- hf512 - 512 Hyperframes (5242.88 seconds)
- hf1024 - HF 1024 - 1024 Hyperframes (10485.76 seconds)

Default

hfquarter

Recommended

hfquarter

Name

ptwHf1024Default - PTW HF1024 Default

Syntax

--ptwHf1024Default <list-item>

Description

Default PTW to use for HF1024 if user requests value outside of the low-high range. The default must be within the low-high range.

Class

Optional.

List

- s1 - 1.28 Seconds
- s2 - 2.56 Seconds
- s3 - 3.84 Seconds
- s4 - 5.12 Seconds
- s5 - 6.40 Seconds
- s6 - 7.68 Seconds
- s7 - 8.96 Seconds
- s8 - 10.24 Seconds
- s9 - 11.52 Seconds
- s10 - 12.80 Seconds
- s11 - 14.08 Seconds
- s12 - 15.36 Seconds
- s13 - 16.64 Seconds
- s14 - 17.92 Seconds
- s15 - 19.20 Seconds
- s16 - 20.48 Seconds
- s17 - 21.76 Seconds
- s18 - 23.04 Seconds
- s19 - 24.32 Seconds
- s20 - 25.60 Seconds
- s21 - 26.88 Seconds
- s22 - 28.16 Seconds
- s23 - 29.44 Seconds

- s24 - 30.72 Seconds
- s25 - 32.00 Seconds
- s26 - 33.28 Seconds
- s27 - 34.56 Seconds
- s28 - 35.84 Seconds
- s29 - 37.12 Seconds
- s30 - 38.40 Seconds
- s31 - 39.68 Seconds
- s32 - 40.96 Seconds

Default

s32

Recommended

s32

Name

ptwHf1024High - PTW HF1024 high

Syntax

--ptwHf1024High <list-item>

Description

Highest allowable PTW to be used with HF1024.

Class

Optional.

List

- s1 - 1.28 Seconds
- s2 - 2.56 Seconds
- s3 - 3.84 Seconds
- s4 - 5.12 Seconds
- s5 - 6.40 Seconds
- s6 - 7.68 Seconds
- s7 - 8.96 Seconds
- s8 - 10.24 Seconds
- s9 - 11.52 Seconds
- s10 - 12.80 Seconds

- s11 - 14.08 Seconds
- s12 - 15.36 Seconds
- s13 - 16.64 Seconds
- s14 - 17.92 Seconds
- s15 - 19.20 Seconds
- s16 - 20.48 Seconds
- s17 - 21.76 Seconds
- s18 - 23.04 Seconds
- s19 - 24.32 Seconds
- s20 - 25.60 Seconds
- s21 - 26.88 Seconds
- s22 - 28.16 Seconds
- s23 - 29.44 Seconds
- s24 - 30.72 Seconds
- s25 - 32.00 Seconds
- s26 - 33.28 Seconds
- s27 - 34.56 Seconds
- s28 - 35.84 Seconds
- s29 - 37.12 Seconds
- s30 - 38.40 Seconds
- s31 - 39.68 Seconds
- s32 - 40.96 Seconds

Default

s32

Recommended

s32

Name

ptwHf1024Low - PTW HF1024 low

Syntax

--ptwHf1024Low <list-item>

Description

Lowest allowable PTW to be used with HF1024. It is recommended that this time be at least as long as the RAN DRX timer.

Class

Optional.

List

- s1 - 1.28 Seconds
- s2 - 2.56 Seconds
- s3 - 3.84 Seconds
- s4 - 5.12 Seconds
- s5 - 6.40 Seconds
- s6 - 7.68 Seconds
- s7 - 8.96 Seconds
- s8 - 10.24 Seconds
- s9 - 11.52 Seconds
- s10 - 12.80 Seconds
- s11 - 14.08 Seconds
- s12 - 15.36 Seconds
- s13 - 16.64 Seconds
- s14 - 17.92 Seconds
- s15 - 19.20 Seconds
- s16 - 20.48 Seconds
- s17 - 21.76 Seconds
- s18 - 23.04 Seconds
- s19 - 24.32 Seconds
- s20 - 25.60 Seconds
- s21 - 26.88 Seconds
- s22 - 28.16 Seconds
- s23 - 29.44 Seconds
- s24 - 30.72 Seconds
- s25 - 32.00 Seconds
- s26 - 33.28 Seconds
- s27 - 34.56 Seconds

- s28 - 35.84 Seconds
- s29 - 37.12 Seconds
- s30 - 38.40 Seconds
- s31 - 39.68 Seconds
- s32 - 40.96 Seconds

Default

s1

Recommended

s1

Name

ptwHf128Default - PTW HF2 Default

Syntax

--ptwHf128Default <list-item>

Description

Default PTW to use for HF128 if user requests value outside of the low-high range. The default must be within the low-high range.

Class

Optional.

List

- s1 - 1.28 Seconds
- s2 - 2.56 Seconds
- s3 - 3.84 Seconds
- s4 - 5.12 Seconds
- s5 - 6.40 Seconds
- s6 - 7.68 Seconds
- s7 - 8.96 Seconds
- s8 - 10.24 Seconds
- s9 - 11.52 Seconds
- s10 - 12.80 Seconds
- s11 - 14.08 Seconds
- s12 - 15.36 Seconds
- s13 - 16.64 Seconds
- s14 - 17.92 Seconds

- s15 - 19.20 Seconds
- s16 - 20.48 Seconds
- s17 - 21.76 Seconds
- s18 - 23.04 Seconds
- s19 - 24.32 Seconds
- s20 - 25.60 Seconds
- s21 - 26.88 Seconds
- s22 - 28.16 Seconds
- s23 - 29.44 Seconds
- s24 - 30.72 Seconds
- s25 - 32.00 Seconds
- s26 - 33.28 Seconds
- s27 - 34.56 Seconds
- s28 - 35.84 Seconds
- s29 - 37.12 Seconds
- s30 - 38.40 Seconds
- s31 - 39.68 Seconds
- s32 - 40.96 Seconds

Default

s32

Recommended

s32

Name

ptwHf128High - PTW HF2 high

Syntax

--ptwHf128High <list-item>

Description

Highest allowable PTW to be used with HF128.

Class

Optional.

List

- s1 - 1.28 Seconds

- s2 - 2.56 Seconds
- s3 - 3.84 Seconds
- s4 - 5.12 Seconds
- s5 - 6.40 Seconds
- s6 - 7.68 Seconds
- s7 - 8.96 Seconds
- s8 - 10.24 Seconds
- s9 - 11.52 Seconds
- s10 - 12.80 Seconds
- s11 - 14.08 Seconds
- s12 - 15.36 Seconds
- s13 - 16.64 Seconds
- s14 - 17.92 Seconds
- s15 - 19.20 Seconds
- s16 - 20.48 Seconds
- s17 - 21.76 Seconds
- s18 - 23.04 Seconds
- s19 - 24.32 Seconds
- s20 - 25.60 Seconds
- s21 - 26.88 Seconds
- s22 - 28.16 Seconds
- s23 - 29.44 Seconds
- s24 - 30.72 Seconds
- s25 - 32.00 Seconds
- s26 - 33.28 Seconds
- s27 - 34.56 Seconds
- s28 - 35.84 Seconds
- s29 - 37.12 Seconds
- s30 - 38.40 Seconds
- s31 - 39.68 Seconds
- s32 - 40.96 Seconds

Default

s32

Recommended

s32

Name

ptwHf128Low - PTW HF128 low

Syntax`--ptwHf128Low <list-item>`**Description**

Lowest allowable PTW to be used with HF128. It is recommended that this time be at least as long as the RAN DRX timer.

Class

Optional.

List

- s1 - 1.28 Seconds
- s2 - 2.56 Seconds
- s3 - 3.84 Seconds
- s4 - 5.12 Seconds
- s5 - 6.40 Seconds
- s6 - 7.68 Seconds
- s7 - 8.96 Seconds
- s8 - 10.24 Seconds
- s9 - 11.52 Seconds
- s10 - 12.80 Seconds
- s11 - 14.08 Seconds
- s12 - 15.36 Seconds
- s13 - 16.64 Seconds
- s14 - 17.92 Seconds
- s15 - 19.20 Seconds
- s16 - 20.48 Seconds
- s17 - 21.76 Seconds
- s18 - 23.04 Seconds
- s19 - 24.32 Seconds

- s20 - 25.60 Seconds
- s21 - 26.88 Seconds
- s22 - 28.16 Seconds
- s23 - 29.44 Seconds
- s24 - 30.72 Seconds
- s25 - 32.00 Seconds
- s26 - 33.28 Seconds
- s27 - 34.56 Seconds
- s28 - 35.84 Seconds
- s29 - 37.12 Seconds
- s30 - 38.40 Seconds
- s31 - 39.68 Seconds
- s32 - 40.96 Seconds

Default

s1

Recommended

s1

Name

ptwHf16Default - PTW HF16 Default

Syntax

--ptwHf16Default <list-item>

Description

Default PTW to use for HF16 if user requests value outside of the low-high range. The default must be within the low-high range.

Class

Optional.

List

- s1 - 1.28 Seconds
- s2 - 2.56 Seconds
- s3 - 3.84 Seconds
- s4 - 5.12 Seconds
- s5 - 6.40 Seconds
- s6 - 7.68 Seconds

- s7 - 8.96 Seconds
- s8 - 10.24 Seconds
- s9 - 11.52 Seconds
- s10 - 12.80 Seconds
- s11 - 14.08 Seconds
- s12 - 15.36 Seconds
- s13 - 16.64 Seconds
- s14 - 17.92 Seconds
- s15 - 19.20 Seconds
- s16 - 20.48 Seconds
- s17 - 21.76 Seconds
- s18 - 23.04 Seconds
- s19 - 24.32 Seconds
- s20 - 25.60 Seconds
- s21 - 26.88 Seconds
- s22 - 28.16 Seconds
- s23 - 29.44 Seconds
- s24 - 30.72 Seconds
- s25 - 32.00 Seconds
- s26 - 33.28 Seconds
- s27 - 34.56 Seconds
- s28 - 35.84 Seconds
- s29 - 37.12 Seconds
- s30 - 38.40 Seconds
- s31 - 39.68 Seconds
- s32 - 40.96 Seconds

Default

s32

Recommended

s32

Name

ptwHf16High - PTW HF16 high

Syntax

--ptwHf16High <list-item>

Description

Highest allowable PTW to be used with HF16.

Class

Optional.

List

- s1 - 1.28 Seconds
- s2 - 2.56 Seconds
- s3 - 3.84 Seconds
- s4 - 5.12 Seconds
- s5 - 6.40 Seconds
- s6 - 7.68 Seconds
- s7 - 8.96 Seconds
- s8 - 10.24 Seconds
- s9 - 11.52 Seconds
- s10 - 12.80 Seconds
- s11 - 14.08 Seconds
- s12 - 15.36 Seconds
- s13 - 16.64 Seconds
- s14 - 17.92 Seconds
- s15 - 19.20 Seconds
- s16 - 20.48 Seconds
- s17 - 21.76 Seconds
- s18 - 23.04 Seconds
- s19 - 24.32 Seconds
- s20 - 25.60 Seconds
- s21 - 26.88 Seconds
- s22 - 28.16 Seconds
- s23 - 29.44 Seconds
- s24 - 30.72 Seconds
- s25 - 32.00 Seconds

- s26 - 33.28 Seconds
- s27 - 34.56 Seconds
- s28 - 35.84 Seconds
- s29 - 37.12 Seconds
- s30 - 38.40 Seconds
- s31 - 39.68 Seconds
- s32 - 40.96 Seconds

Default

s32

Recommended

s32

Name

ptwHf16Low - PTW HF16 low

Syntax

--ptwHf16Low <list-item>

Description

Lowest allowable PTW to be used with HF16. It is recommended that this time be at least as long as the RAN DRX timer.

Class

Optional.

List

- s1 - 1.28 Seconds
- s2 - 2.56 Seconds
- s3 - 3.84 Seconds
- s4 - 5.12 Seconds
- s5 - 6.40 Seconds
- s6 - 7.68 Seconds
- s7 - 8.96 Seconds
- s8 - 10.24 Seconds
- s9 - 11.52 Seconds
- s10 - 12.80 Seconds
- s11 - 14.08 Seconds
- s12 - 15.36 Seconds

- s13 - 16.64 Seconds
- s14 - 17.92 Seconds
- s15 - 19.20 Seconds
- s16 - 20.48 Seconds
- s17 - 21.76 Seconds
- s18 - 23.04 Seconds
- s19 - 24.32 Seconds
- s20 - 25.60 Seconds
- s21 - 26.88 Seconds
- s22 - 28.16 Seconds
- s23 - 29.44 Seconds
- s24 - 30.72 Seconds
- s25 - 32.00 Seconds
- s26 - 33.28 Seconds
- s27 - 34.56 Seconds
- s28 - 35.84 Seconds
- s29 - 37.12 Seconds
- s30 - 38.40 Seconds
- s31 - 39.68 Seconds
- s32 - 40.96 Seconds

Default

s1

Recommended

s1

Name

ptwHf256Default - PTW HF256 Default

Syntax

--ptwHf256Default <list-item>

Description

Default PTW to use for HF256 if user requests value outside of the low-high range. The default must be within the low-high range.

Class

Optional.

List

- s1 - 1.28 Seconds
- s2 - 2.56 Seconds
- s3 - 3.84 Seconds
- s4 - 5.12 Seconds
- s5 - 6.40 Seconds
- s6 - 7.68 Seconds
- s7 - 8.96 Seconds
- s8 - 10.24 Seconds
- s9 - 11.52 Seconds
- s10 - 12.80 Seconds
- s11 - 14.08 Seconds
- s12 - 15.36 Seconds
- s13 - 16.64 Seconds
- s14 - 17.92 Seconds
- s15 - 19.20 Seconds
- s16 - 20.48 Seconds
- s17 - 21.76 Seconds
- s18 - 23.04 Seconds
- s19 - 24.32 Seconds
- s20 - 25.60 Seconds
- s21 - 26.88 Seconds
- s22 - 28.16 Seconds
- s23 - 29.44 Seconds
- s24 - 30.72 Seconds
- s25 - 32.00 Seconds
- s26 - 33.28 Seconds
- s27 - 34.56 Seconds
- s28 - 35.84 Seconds
- s29 - 37.12 Seconds
- s30 - 38.40 Seconds

- s31 - 39.68 Seconds
- s32 - 40.96 Seconds

Default

s32

Recommended

s32

Name

ptwHf256High - PTW HF256 high

Syntax

--ptwHf256High <list-item>

Description

Highest allowable PTW to be used with HF256.

Class

Optional.

List

- s1 - 1.28 Seconds
- s2 - 2.56 Seconds
- s3 - 3.84 Seconds
- s4 - 5.12 Seconds
- s5 - 6.40 Seconds
- s6 - 7.68 Seconds
- s7 - 8.96 Seconds
- s8 - 10.24 Seconds
- s9 - 11.52 Seconds
- s10 - 12.80 Seconds
- s11 - 14.08 Seconds
- s12 - 15.36 Seconds
- s13 - 16.64 Seconds
- s14 - 17.92 Seconds
- s15 - 19.20 Seconds
- s16 - 20.48 Seconds
- s17 - 21.76 Seconds

- s18 - 23.04 Seconds
- s19 - 24.32 Seconds
- s20 - 25.60 Seconds
- s21 - 26.88 Seconds
- s22 - 28.16 Seconds
- s23 - 29.44 Seconds
- s24 - 30.72 Seconds
- s25 - 32.00 Seconds
- s26 - 33.28 Seconds
- s27 - 34.56 Seconds
- s28 - 35.84 Seconds
- s29 - 37.12 Seconds
- s30 - 38.40 Seconds
- s31 - 39.68 Seconds
- s32 - 40.96 Seconds

Default

s32

Recommended

s32

Name

ptwHf256Low - PTW HF256 low

Syntax`--ptwHf256Low <list-item>`**Description**

Lowest allowable PTW to be used with HF256. It is recommended that this time be at least as long as the RAN DRX timer.

Class

Optional.

List

- s1 - 1.28 Seconds
- s2 - 2.56 Seconds
- s3 - 3.84 Seconds
- s4 - 5.12 Seconds

- s5 - 6.40 Seconds
- s6 - 7.68 Seconds
- s7 - 8.96 Seconds
- s8 - 10.24 Seconds
- s9 - 11.52 Seconds
- s10 - 12.80 Seconds
- s11 - 14.08 Seconds
- s12 - 15.36 Seconds
- s13 - 16.64 Seconds
- s14 - 17.92 Seconds
- s15 - 19.20 Seconds
- s16 - 20.48 Seconds
- s17 - 21.76 Seconds
- s18 - 23.04 Seconds
- s19 - 24.32 Seconds
- s20 - 25.60 Seconds
- s21 - 26.88 Seconds
- s22 - 28.16 Seconds
- s23 - 29.44 Seconds
- s24 - 30.72 Seconds
- s25 - 32.00 Seconds
- s26 - 33.28 Seconds
- s27 - 34.56 Seconds
- s28 - 35.84 Seconds
- s29 - 37.12 Seconds
- s30 - 38.40 Seconds
- s31 - 39.68 Seconds
- s32 - 40.96 Seconds

Default

s1

Recommended

s1

Name

ptwHf2Default - PTW HF2 Default

Syntax

--ptwHf2Default <list-item>

Description

Default PTW to use for HF2 if user requests value outside of the low-high range. The default must be within the low-high range.

Class

Optional.

List

- s1 - 1.28 Seconds
- s2 - 2.56 Seconds
- s3 - 3.84 Seconds
- s4 - 5.12 Seconds
- s5 - 6.40 Seconds
- s6 - 7.68 Seconds
- s7 - 8.96 Seconds
- s8 - 10.24 Seconds
- s9 - 11.52 Seconds
- s10 - 12.80 Seconds
- s11 - 14.08 Seconds
- s12 - 15.36 Seconds
- s13 - 16.64 Seconds
- s14 - 17.92 Seconds
- s15 - 19.20 Seconds
- s16 - 20.48 Seconds
- s17 - 21.76 Seconds
- s18 - 23.04 Seconds
- s19 - 24.32 Seconds
- s20 - 25.60 Seconds
- s21 - 26.88 Seconds

- s22 - 28.16 Seconds
- s23 - 29.44 Seconds
- s24 - 30.72 Seconds
- s25 - 32.00 Seconds
- s26 - 33.28 Seconds
- s27 - 34.56 Seconds
- s28 - 35.84 Seconds
- s29 - 37.12 Seconds
- s30 - 38.40 Seconds
- s31 - 39.68 Seconds
- s32 - 40.96 Seconds

Default

s5

Recommended

s5

Name

ptwHf2High - PTW HF2 high

Syntax

--ptwHf2High <list-item>

Description

Highest allowable PTW to be used with HF2.

Class

Optional.

List

- s1 - 1.28 Seconds
- s2 - 2.56 Seconds
- s3 - 3.84 Seconds
- s4 - 5.12 Seconds
- s5 - 6.40 Seconds
- s6 - 7.68 Seconds
- s7 - 8.96 Seconds
- s8 - 10.24 Seconds

- s9 - 11.52 Seconds
- s10 - 12.80 Seconds
- s11 - 14.08 Seconds
- s12 - 15.36 Seconds
- s13 - 16.64 Seconds
- s14 - 17.92 Seconds
- s15 - 19.20 Seconds
- s16 - 20.48 Seconds
- s17 - 21.76 Seconds
- s18 - 23.04 Seconds
- s19 - 24.32 Seconds
- s20 - 25.60 Seconds
- s21 - 26.88 Seconds
- s22 - 28.16 Seconds
- s23 - 29.44 Seconds
- s24 - 30.72 Seconds
- s25 - 32.00 Seconds
- s26 - 33.28 Seconds
- s27 - 34.56 Seconds
- s28 - 35.84 Seconds
- s29 - 37.12 Seconds
- s30 - 38.40 Seconds
- s31 - 39.68 Seconds
- s32 - 40.96 Seconds

Default

s8

Recommended

s8

Name

ptwHf2Low - PTW HF2 low

Syntax`--ptwHf2Low <list-item>`

Description

Lowest allowable PTW to be used with HF2. It is recommended that this time be at least as long as the RAN DRX timer.

Class

Optional.

List

- s1 - 1.28 Seconds
- s2 - 2.56 Seconds
- s3 - 3.84 Seconds
- s4 - 5.12 Seconds
- s5 - 6.40 Seconds
- s6 - 7.68 Seconds
- s7 - 8.96 Seconds
- s8 - 10.24 Seconds
- s9 - 11.52 Seconds
- s10 - 12.80 Seconds
- s11 - 14.08 Seconds
- s12 - 15.36 Seconds
- s13 - 16.64 Seconds
- s14 - 17.92 Seconds
- s15 - 19.20 Seconds
- s16 - 20.48 Seconds
- s17 - 21.76 Seconds
- s18 - 23.04 Seconds
- s19 - 24.32 Seconds
- s20 - 25.60 Seconds
- s21 - 26.88 Seconds
- s22 - 28.16 Seconds
- s23 - 29.44 Seconds
- s24 - 30.72 Seconds
- s25 - 32.00 Seconds
- s26 - 33.28 Seconds

- s27 - 34.56 Seconds
- s28 - 35.84 Seconds
- s29 - 37.12 Seconds
- s30 - 38.40 Seconds
- s31 - 39.68 Seconds
- s32 - 40.96 Seconds

Default

s1

Recommended

s1

Name

ptwHf32Default - PTW HF32 Default

Syntax

--ptwHf32Default <list-item>

Description

Default PTW to use for HF32 if user requests value outside of the low-high range. The default must be within the low-high range.

Class

Optional.

List

- s1 - 1.28 Seconds
- s2 - 2.56 Seconds
- s3 - 3.84 Seconds
- s4 - 5.12 Seconds
- s5 - 6.40 Seconds
- s6 - 7.68 Seconds
- s7 - 8.96 Seconds
- s8 - 10.24 Seconds
- s9 - 11.52 Seconds
- s10 - 12.80 Seconds
- s11 - 14.08 Seconds
- s12 - 15.36 Seconds
- s13 - 16.64 Seconds

- s14 - 17.92 Seconds
- s15 - 19.20 Seconds
- s16 - 20.48 Seconds
- s17 - 21.76 Seconds
- s18 - 23.04 Seconds
- s19 - 24.32 Seconds
- s20 - 25.60 Seconds
- s21 - 26.88 Seconds
- s22 - 28.16 Seconds
- s23 - 29.44 Seconds
- s24 - 30.72 Seconds
- s25 - 32.00 Seconds
- s26 - 33.28 Seconds
- s27 - 34.56 Seconds
- s28 - 35.84 Seconds
- s29 - 37.12 Seconds
- s30 - 38.40 Seconds
- s31 - 39.68 Seconds
- s32 - 40.96 Seconds

Default

s32

Recommended

s32

Name

ptwHf32High - PTW HF32 high

Syntax

--ptwHf32High <list-item>

Description

Highest allowable PTW to be used with HF32.

Class

Optional.

List

- s1 - 1.28 Seconds
- s2 - 2.56 Seconds
- s3 - 3.84 Seconds
- s4 - 5.12 Seconds
- s5 - 6.40 Seconds
- s6 - 7.68 Seconds
- s7 - 8.96 Seconds
- s8 - 10.24 Seconds
- s9 - 11.52 Seconds
- s10 - 12.80 Seconds
- s11 - 14.08 Seconds
- s12 - 15.36 Seconds
- s13 - 16.64 Seconds
- s14 - 17.92 Seconds
- s15 - 19.20 Seconds
- s16 - 20.48 Seconds
- s17 - 21.76 Seconds
- s18 - 23.04 Seconds
- s19 - 24.32 Seconds
- s20 - 25.60 Seconds
- s21 - 26.88 Seconds
- s22 - 28.16 Seconds
- s23 - 29.44 Seconds
- s24 - 30.72 Seconds
- s25 - 32.00 Seconds
- s26 - 33.28 Seconds
- s27 - 34.56 Seconds
- s28 - 35.84 Seconds
- s29 - 37.12 Seconds
- s30 - 38.40 Seconds
- s31 - 39.68 Seconds

- s32 - 40.96 Seconds

Default

s32

Recommended

s32

Name

ptwHf32Low - PTW HF32 low

Syntax

--ptwHf32Low <list-item>

Description

Lowest allowable PTW to be used with HF32. It is recommended that this time be at least as long as the RAN DRX timer.

Class

Optional.

List

- s1 - 1.28 Seconds
- s2 - 2.56 Seconds
- s3 - 3.84 Seconds
- s4 - 5.12 Seconds
- s5 - 6.40 Seconds
- s6 - 7.68 Seconds
- s7 - 8.96 Seconds
- s8 - 10.24 Seconds
- s9 - 11.52 Seconds
- s10 - 12.80 Seconds
- s11 - 14.08 Seconds
- s12 - 15.36 Seconds
- s13 - 16.64 Seconds
- s14 - 17.92 Seconds
- s15 - 19.20 Seconds
- s16 - 20.48 Seconds
- s17 - 21.76 Seconds
- s18 - 23.04 Seconds

- s19 - 24.32 Seconds
- s20 - 25.60 Seconds
- s21 - 26.88 Seconds
- s22 - 28.16 Seconds
- s23 - 29.44 Seconds
- s24 - 30.72 Seconds
- s25 - 32.00 Seconds
- s26 - 33.28 Seconds
- s27 - 34.56 Seconds
- s28 - 35.84 Seconds
- s29 - 37.12 Seconds
- s30 - 38.40 Seconds
- s31 - 39.68 Seconds
- s32 - 40.96 Seconds

Default

s1

Recommended

s1

Name

ptwHf4Default - PTW HF4 Default

Syntax

--ptwHf4Default <list-item>

Description

Default PTW to use for HF4 if user requests value outside of the low-high range. The default must be within the low-high range.

Class

Optional.

List

- s1 - 1.28 Seconds
- s2 - 2.56 Seconds
- s3 - 3.84 Seconds
- s4 - 5.12 Seconds
- s5 - 6.40 Seconds

- s6 - 7.68 Seconds
- s7 - 8.96 Seconds
- s8 - 10.24 Seconds
- s9 - 11.52 Seconds
- s10 - 12.80 Seconds
- s11 - 14.08 Seconds
- s12 - 15.36 Seconds
- s13 - 16.64 Seconds
- s14 - 17.92 Seconds
- s15 - 19.20 Seconds
- s16 - 20.48 Seconds
- s17 - 21.76 Seconds
- s18 - 23.04 Seconds
- s19 - 24.32 Seconds
- s20 - 25.60 Seconds
- s21 - 26.88 Seconds
- s22 - 28.16 Seconds
- s23 - 29.44 Seconds
- s24 - 30.72 Seconds
- s25 - 32.00 Seconds
- s26 - 33.28 Seconds
- s27 - 34.56 Seconds
- s28 - 35.84 Seconds
- s29 - 37.12 Seconds
- s30 - 38.40 Seconds
- s31 - 39.68 Seconds
- s32 - 40.96 Seconds

Default

s16

Recommended

s16

Name

ptwHf4High - ptwHf4High

Syntax

--ptwHf4High <list-item>

Description

Highest allowable PTW to be used with HF4.

Class

Optional.

List

- s1 - 1.28 Seconds
- s2 - 2.56 Seconds
- s3 - 3.84 Seconds
- s4 - 5.12 Seconds
- s5 - 6.40 Seconds
- s6 - 7.68 Seconds
- s7 - 8.96 Seconds
- s8 - 10.24 Seconds
- s9 - 11.52 Seconds
- s10 - 12.80 Seconds
- s11 - 14.08 Seconds
- s12 - 15.36 Seconds
- s13 - 16.64 Seconds
- s14 - 17.92 Seconds
- s15 - 19.20 Seconds
- s16 - 20.48 Seconds
- s17 - 21.76 Seconds
- s18 - 23.04 Seconds
- s19 - 24.32 Seconds
- s20 - 25.60 Seconds
- s21 - 26.88 Seconds
- s22 - 28.16 Seconds
- s23 - 29.44 Seconds
- s24 - 30.72 Seconds

- s25 - 32.00 Seconds
- s26 - 33.28 Seconds
- s27 - 34.56 Seconds
- s28 - 35.84 Seconds
- s29 - 37.12 Seconds
- s30 - 38.40 Seconds
- s31 - 39.68 Seconds
- s32 - 40.96 Seconds

Default

s16

Recommended

s16

Name

ptwHf4Low - PTW HF4 low

Syntax

--ptwHf4Low <list-item>

Description

Lowest allowable PTW to be used with HF4. It is recommended that this time be at least as long as the RAN DRX timer.

Class

Optional.

List

- s1 - 1.28 Seconds
- s2 - 2.56 Seconds
- s3 - 3.84 Seconds
- s4 - 5.12 Seconds
- s5 - 6.40 Seconds
- s6 - 7.68 Seconds
- s7 - 8.96 Seconds
- s8 - 10.24 Seconds
- s9 - 11.52 Seconds
- s10 - 12.80 Seconds
- s11 - 14.08 Seconds

- s12 - 15.36 Seconds
- s13 - 16.64 Seconds
- s14 - 17.92 Seconds
- s15 - 19.20 Seconds
- s16 - 20.48 Seconds
- s17 - 21.76 Seconds
- s18 - 23.04 Seconds
- s19 - 24.32 Seconds
- s20 - 25.60 Seconds
- s21 - 26.88 Seconds
- s22 - 28.16 Seconds
- s23 - 29.44 Seconds
- s24 - 30.72 Seconds
- s25 - 32.00 Seconds
- s26 - 33.28 Seconds
- s27 - 34.56 Seconds
- s28 - 35.84 Seconds
- s29 - 37.12 Seconds
- s30 - 38.40 Seconds
- s31 - 39.68 Seconds
- s32 - 40.96 Seconds

Default

s1

Recommended

s1

Name

ptwHf512Default - PTW HF512 Default

Syntax

--ptwHf512Default <list-item>

Description

Default PTW to use for HF512 if user requests value outside of the low-high range. The default must be within the low-high range.

Class

Optional.

List

- s1 - 1.28 Seconds
- s2 - 2.56 Seconds
- s3 - 3.84 Seconds
- s4 - 5.12 Seconds
- s5 - 6.40 Seconds
- s6 - 7.68 Seconds
- s7 - 8.96 Seconds
- s8 - 10.24 Seconds
- s9 - 11.52 Seconds
- s10 - 12.80 Seconds
- s11 - 14.08 Seconds
- s12 - 15.36 Seconds
- s13 - 16.64 Seconds
- s14 - 17.92 Seconds
- s15 - 19.20 Seconds
- s16 - 20.48 Seconds
- s17 - 21.76 Seconds
- s18 - 23.04 Seconds
- s19 - 24.32 Seconds
- s20 - 25.60 Seconds
- s21 - 26.88 Seconds
- s22 - 28.16 Seconds
- s23 - 29.44 Seconds
- s24 - 30.72 Seconds
- s25 - 32.00 Seconds
- s26 - 33.28 Seconds
- s27 - 34.56 Seconds
- s28 - 35.84 Seconds

- s29 - 37.12 Seconds
- s30 - 38.40 Seconds
- s31 - 39.68 Seconds
- s32 - 40.96 Seconds

Default

s32

Recommended

s32

Name

ptwHf512High - PTW HF512 high

Syntax

--ptwHf512High <list-item>

Description

Highest allowable PTW to be used with HF512.

Class

Optional.

List

- s1 - 1.28 Seconds
- s2 - 2.56 Seconds
- s3 - 3.84 Seconds
- s4 - 5.12 Seconds
- s5 - 6.40 Seconds
- s6 - 7.68 Seconds
- s7 - 8.96 Seconds
- s8 - 10.24 Seconds
- s9 - 11.52 Seconds
- s10 - 12.80 Seconds
- s11 - 14.08 Seconds
- s12 - 15.36 Seconds
- s13 - 16.64 Seconds
- s14 - 17.92 Seconds
- s15 - 19.20 Seconds

- s16 - 20.48 Seconds
- s17 - 21.76 Seconds
- s18 - 23.04 Seconds
- s19 - 24.32 Seconds
- s20 - 25.60 Seconds
- s21 - 26.88 Seconds
- s22 - 28.16 Seconds
- s23 - 29.44 Seconds
- s24 - 30.72 Seconds
- s25 - 32.00 Seconds
- s26 - 33.28 Seconds
- s27 - 34.56 Seconds
- s28 - 35.84 Seconds
- s29 - 37.12 Seconds
- s30 - 38.40 Seconds
- s31 - 39.68 Seconds
- s32 - 40.96 Seconds

Default

s32

Recommended

s32

Name

ptwHf512Low - PTW HF512 low

Syntax

--ptwHf512Low <list-item>

Description

Lowest allowable PTW to be used with HF512. It is recommended that this time be at least as long as the RAN DRX timer.

Class

Optional.

List

- s1 - 1.28 Seconds
- s2 - 2.56 Seconds

- s3 - 3.84 Seconds
- s4 - 5.12 Seconds
- s5 - 6.40 Seconds
- s6 - 7.68 Seconds
- s7 - 8.96 Seconds
- s8 - 10.24 Seconds
- s9 - 11.52 Seconds
- s10 - 12.80 Seconds
- s11 - 14.08 Seconds
- s12 - 15.36 Seconds
- s13 - 16.64 Seconds
- s14 - 17.92 Seconds
- s15 - 19.20 Seconds
- s16 - 20.48 Seconds
- s17 - 21.76 Seconds
- s18 - 23.04 Seconds
- s19 - 24.32 Seconds
- s20 - 25.60 Seconds
- s21 - 26.88 Seconds
- s22 - 28.16 Seconds
- s23 - 29.44 Seconds
- s24 - 30.72 Seconds
- s25 - 32.00 Seconds
- s26 - 33.28 Seconds
- s27 - 34.56 Seconds
- s28 - 35.84 Seconds
- s29 - 37.12 Seconds
- s30 - 38.40 Seconds
- s31 - 39.68 Seconds
- s32 - 40.96 Seconds

Default

s1

Recommended

s1

Name

ptwHf64Default - PTW HF64 Default

Syntax

--ptwHf64Default <list-item>

Description

Default PTW to use for HF64 if user requests value outside of the low-high range. The default must be within the low-high range.

Class

Optional.

List

- s1 - 1.28 Seconds
- s2 - 2.56 Seconds
- s3 - 3.84 Seconds
- s4 - 5.12 Seconds
- s5 - 6.40 Seconds
- s6 - 7.68 Seconds
- s7 - 8.96 Seconds
- s8 - 10.24 Seconds
- s9 - 11.52 Seconds
- s10 - 12.80 Seconds
- s11 - 14.08 Seconds
- s12 - 15.36 Seconds
- s13 - 16.64 Seconds
- s14 - 17.92 Seconds
- s15 - 19.20 Seconds
- s16 - 20.48 Seconds
- s17 - 21.76 Seconds
- s18 - 23.04 Seconds
- s19 - 24.32 Seconds
- s20 - 25.60 Seconds

- s21 - 26.88 Seconds
- s22 - 28.16 Seconds
- s23 - 29.44 Seconds
- s24 - 30.72 Seconds
- s25 - 32.00 Seconds
- s26 - 33.28 Seconds
- s27 - 34.56 Seconds
- s28 - 35.84 Seconds
- s29 - 37.12 Seconds
- s30 - 38.40 Seconds
- s31 - 39.68 Seconds
- s32 - 40.96 Seconds

Default

s32

Recommended

s32

Name

ptwHf64High - PTW HF64 high

Syntax

--ptwHf64High <list-item>

Description

Highest allowable PTW to be used with HF64.

Class

Optional.

List

- s1 - 1.28 Seconds
- s2 - 2.56 Seconds
- s3 - 3.84 Seconds
- s4 - 5.12 Seconds
- s5 - 6.40 Seconds
- s6 - 7.68 Seconds
- s7 - 8.96 Seconds

- s8 - 10.24 Seconds
- s9 - 11.52 Seconds
- s10 - 12.80 Seconds
- s11 - 14.08 Seconds
- s12 - 15.36 Seconds
- s13 - 16.64 Seconds
- s14 - 17.92 Seconds
- s15 - 19.20 Seconds
- s16 - 20.48 Seconds
- s17 - 21.76 Seconds
- s18 - 23.04 Seconds
- s19 - 24.32 Seconds
- s20 - 25.60 Seconds
- s21 - 26.88 Seconds
- s22 - 28.16 Seconds
- s23 - 29.44 Seconds
- s24 - 30.72 Seconds
- s25 - 32.00 Seconds
- s26 - 33.28 Seconds
- s27 - 34.56 Seconds
- s28 - 35.84 Seconds
- s29 - 37.12 Seconds
- s30 - 38.40 Seconds
- s31 - 39.68 Seconds
- s32 - 40.96 Seconds

Default

s32

Recommended

s32

Name

ptwHf64Low - PTW HF64 low

Syntax

--ptwHf64Low <list-item>

Description

Lowest allowable PTW to be used with HF64. It is recommended that this time be at least as long as the RAN DRX timer.

Class

Optional.

List

- s1 - 1.28 Seconds
- s2 - 2.56 Seconds
- s3 - 3.84 Seconds
- s4 - 5.12 Seconds
- s5 - 6.40 Seconds
- s6 - 7.68 Seconds
- s7 - 8.96 Seconds
- s8 - 10.24 Seconds
- s9 - 11.52 Seconds
- s10 - 12.80 Seconds
- s11 - 14.08 Seconds
- s12 - 15.36 Seconds
- s13 - 16.64 Seconds
- s14 - 17.92 Seconds
- s15 - 19.20 Seconds
- s16 - 20.48 Seconds
- s17 - 21.76 Seconds
- s18 - 23.04 Seconds
- s19 - 24.32 Seconds
- s20 - 25.60 Seconds
- s21 - 26.88 Seconds
- s22 - 28.16 Seconds
- s23 - 29.44 Seconds
- s24 - 30.72 Seconds
- s25 - 32.00 Seconds

- s26 - 33.28 Seconds
- s27 - 34.56 Seconds
- s28 - 35.84 Seconds
- s29 - 37.12 Seconds
- s30 - 38.40 Seconds
- s31 - 39.68 Seconds
- s32 - 40.96 Seconds

Default

s1

Recommended

s1

Name

ptwHf8Default - PTW HF8 Default

Syntax

--ptwHf8Default <list-item>

Description

Default PTW to use for HF8 if user requests value outside of the low-high range. The default must be within the low-high range.

Class

Optional.

List

- s1 - 1.28 Seconds
- s2 - 2.56 Seconds
- s3 - 3.84 Seconds
- s4 - 5.12 Seconds
- s5 - 6.40 Seconds
- s6 - 7.68 Seconds
- s7 - 8.96 Seconds
- s8 - 10.24 Seconds
- s9 - 11.52 Seconds
- s10 - 12.80 Seconds
- s11 - 14.08 Seconds
- s12 - 15.36 Seconds

- s13 - 16.64 Seconds
- s14 - 17.92 Seconds
- s15 - 19.20 Seconds
- s16 - 20.48 Seconds
- s17 - 21.76 Seconds
- s18 - 23.04 Seconds
- s19 - 24.32 Seconds
- s20 - 25.60 Seconds
- s21 - 26.88 Seconds
- s22 - 28.16 Seconds
- s23 - 29.44 Seconds
- s24 - 30.72 Seconds
- s25 - 32.00 Seconds
- s26 - 33.28 Seconds
- s27 - 34.56 Seconds
- s28 - 35.84 Seconds
- s29 - 37.12 Seconds
- s30 - 38.40 Seconds
- s31 - 39.68 Seconds
- s32 - 40.96 Seconds

Default

s32

Recommended

s32

Name

ptwHf8High - PTW HF8 high

Syntax

--ptwHf8High <list-item>

Description

Highest allowable PTW to be used with HF8.

Class

Optional.

List

- s1 - 1.28 Seconds
- s2 - 2.56 Seconds
- s3 - 3.84 Seconds
- s4 - 5.12 Seconds
- s5 - 6.40 Seconds
- s6 - 7.68 Seconds
- s7 - 8.96 Seconds
- s8 - 10.24 Seconds
- s9 - 11.52 Seconds
- s10 - 12.80 Seconds
- s11 - 14.08 Seconds
- s12 - 15.36 Seconds
- s13 - 16.64 Seconds
- s14 - 17.92 Seconds
- s15 - 19.20 Seconds
- s16 - 20.48 Seconds
- s17 - 21.76 Seconds
- s18 - 23.04 Seconds
- s19 - 24.32 Seconds
- s20 - 25.60 Seconds
- s21 - 26.88 Seconds
- s22 - 28.16 Seconds
- s23 - 29.44 Seconds
- s24 - 30.72 Seconds
- s25 - 32.00 Seconds
- s26 - 33.28 Seconds
- s27 - 34.56 Seconds
- s28 - 35.84 Seconds
- s29 - 37.12 Seconds
- s30 - 38.40 Seconds

- s31 - 39.68 Seconds
- s32 - 40.96 Seconds

Default

s32

Recommended

s32

Name

ptwHf8Low - PTW HF8 low

Syntax

--ptwHf8Low <list-item>

Description

Lowest allowable PTW to be used with HF8. It is recommended that this time be at least as long as the RAN DRX timer.

Class

Optional.

List

- s1 - 1.28 Seconds
- s2 - 2.56 Seconds
- s3 - 3.84 Seconds
- s4 - 5.12 Seconds
- s5 - 6.40 Seconds
- s6 - 7.68 Seconds
- s7 - 8.96 Seconds
- s8 - 10.24 Seconds
- s9 - 11.52 Seconds
- s10 - 12.80 Seconds
- s11 - 14.08 Seconds
- s12 - 15.36 Seconds
- s13 - 16.64 Seconds
- s14 - 17.92 Seconds
- s15 - 19.20 Seconds
- s16 - 20.48 Seconds
- s17 - 21.76 Seconds

- s18 - 23.04 Seconds
- s19 - 24.32 Seconds
- s20 - 25.60 Seconds
- s21 - 26.88 Seconds
- s22 - 28.16 Seconds
- s23 - 29.44 Seconds
- s24 - 30.72 Seconds
- s25 - 32.00 Seconds
- s26 - 33.28 Seconds
- s27 - 34.56 Seconds
- s28 - 35.84 Seconds
- s29 - 37.12 Seconds
- s30 - 38.40 Seconds
- s31 - 39.68 Seconds
- s32 - 40.96 Seconds

Default

s1

Recommended

s1

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.103 cmm edrxProfile

eDRX Profile. Syntax, description, parameters, and roles for `cmm edrxProfile` command.

Syntax

Description

eDRX Profile used by eDRX and HLCOM.

Parameters

Name

name - Profile name

Syntax

--name <string>

Description

eDRX profile name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\.\-:@]*

Name

description - Profile description

Syntax

--description <string>

Description

eDRX profile description.

Class

Optional.

Type / Range

Digit string of length 0 to 160 with the allowed pattern of [0-9a-zA-Z_\.\-:@]*

Default

"

Name

edrx0CycleOverride - edrx-0 Override

Syntax

--edrx0CycleOverride <list-item>

Description

Enable parameter override for eDRX cycle length of 5.12 seconds

Class

Optional.

List

- 5.12 - Specifies EDRX cycle length of 5.12 seconds
- 10.24 - Specifies EDRX cycle length of 10.24 seconds.
- 20.48 - Specifies EDRX cycle length of 20.48 seconds.
- 40.96 - Specifies EDRX cycle length of 40.96 seconds.
- 61.44 - Specifies EDRX cycle length of 61.44 seconds.
- 81.92 - Specifies EDRX cycle length of 81.92 seconds.
- 102.40 - Specifies EDRX cycle length of 102.40 seconds.
- 122.88 - Specifies EDRX cycle length of 122.88 seconds.
- 143.36 - Specifies EDRX cycle length of 143.36 seconds.
- 163.84 - Specifies EDRX cycle length of 163.84 seconds.
- 327.68 - Specifies EDRX cycle length of 327.68 seconds.
- 655.36 - Specifies EDRX cycle length of 655.36 seconds.
- 1310.72 - Specifies EDRX cycle length of 1310.72 seconds.
- 2621.44 - Specifies EDRX cycle length of 2621.44 seconds.
- 5242.88 - Specifies EDRX cycle length of 5242.88 seconds.
- 10485.76 - Specifies EDRX cycle length of 10485.76 seconds.

Default

5.12

Name

edrx00override - eDRX-0 override

Syntax

--edrx00override <list-item>

Description

Enable parameter override for eDRX cycle length of 5.12 seconds

Class

Optional.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Name

edrx10CycleOverride - eDRX-10 cycle override

Syntax

--edrx10CycleOverride <list-item>

Description

Override eDRX cycle selection length of 327.68 seconds.

Class

Optional.

List

- 5.12 - Specifies EDRX cycle length of 5.12 seconds
- 10.24 - Specifies EDRX cycle length of 10.24 seconds.
- 20.48 - Specifies EDRX cycle length of 20.48 seconds.
- 40.96 - Specifies EDRX cycle length of 40.96 seconds.
- 61.44 - Specifies EDRX cycle length of 61.44 seconds.
- 81.92 - Specifies EDRX cycle length of 81.92 seconds.
- 102.40 - Specifies EDRX cycle length of 102.40 seconds.
- 122.88 - Specifies EDRX cycle length of 122.88 seconds.
- 143.36 - Specifies EDRX cycle length of 143.36 seconds.
- 163.84 - Specifies EDRX cycle length of 163.84 seconds.
- 327.68 - Specifies EDRX cycle length of 327.68 seconds.
- 655.36 - Specifies EDRX cycle length of 655.36 seconds.
- 1310.72 - Specifies EDRX cycle length of 1310.72 seconds.
- 2621.44 - Specifies EDRX cycle length of 2621.44 seconds.
- 5242.88 - Specifies EDRX cycle length of 5242.88 seconds.
- 10485.76 - Specifies EDRX cycle length of 10485.76 seconds.

Default

327.68

Name

edrx10DataBufferCount - eDRX-10 data buffer count

Syntax

--edrx10DataBufferCount <integer>

Description

Suggested maximum number of DL packets buffered in the SGW for eDRX cycle length of 327.68 seconds.

Class

Optional.

Type / Range

Integer in the range of 1 to 65,535

Default

1

Name

edrx10Override - eDRX-10 override

Syntax

--edrx10override <list-item>

Description

Enable parameter override for eDRX cycle length of 327.68 seconds.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

edrx11CycleOverride - eDRX-11 cycle override

Syntax

--edrx11cycle0override <list-item>

Description

Override eDRX cycle selection length of 655.36 seconds.

Class

Optional.

List

- 5.12 - Specifies EDRX cycle length of 5.12 seconds
- 10.24 - Specifies EDRX cycle length of 10.24 seconds.

- 20.48 - Specifies EDRX cycle length of 20.48 seconds.
- 40.96 - Specifies EDRX cycle length of 40.96 seconds.
- 61.44 - Specifies EDRX cycle length of 61.44 seconds.
- 81.92 - Specifies EDRX cycle length of 81.92 seconds.
- 102.40 - Specifies EDRX cycle length of 102.40 seconds.
- 122.88 - Specifies EDRX cycle length of 122.88 seconds.
- 143.36 - Specifies EDRX cycle length of 143.36 seconds.
- 163.84 - Specifies EDRX cycle length of 163.84 seconds.
- 327.68 - Specifies EDRX cycle length of 327.68 seconds.
- 655.36 - Specifies EDRX cycle length of 655.36 seconds.
- 1310.72 - Specifies EDRX cycle length of 1310.72 seconds.
- 2621.44 - Specifies EDRX cycle length of 2621.44 seconds.
- 5242.88 - Specifies EDRX cycle length of 5242.88 seconds.
- 10485.76 - Specifies EDRX cycle length of 10485.76 seconds.

Default

655.36

Name

edrx11DataBufferCount - eDRX-11 data buffer count

Syntax

--edrx11DataBufferCount <integer>

Description

Suggested maximum number of DL packets buffered in the SGW for eDRX cycle length of 655.36 seconds.

Class

Optional.

Type / Range

Integer in the range of 1 to 65,535

Default

1

Name

edrx11Override - eDRX-11 override

Syntax

--edrx11override <list-item>

Description

Enable parameter override for eDRX cycle length of 655.36 seconds.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

edrx12CycleOverride - eDRX-12 cycle override

Syntax

--edrx12CycleOverride <list-item>

Description

Override eDRX cycle selection length of 1310.72 seconds.

Class

Optional.

List

- 5.12 - Specifies EDRX cycle length of 5.12 seconds
- 10.24 - Specifies EDRX cycle length of 10.24 seconds.
- 20.48 - Specifies EDRX cycle length of 20.48 seconds.
- 40.96 - Specifies EDRX cycle length of 40.96 seconds.
- 61.44 - Specifies EDRX cycle length of 61.44 seconds.
- 81.92 - Specifies EDRX cycle length of 81.92 seconds.
- 102.40 - Specifies EDRX cycle length of 102.40 seconds.
- 122.88 - Specifies EDRX cycle length of 122.88 seconds.
- 143.36 - Specifies EDRX cycle length of 143.36 seconds.
- 163.84 - Specifies EDRX cycle length of 163.84 seconds.
- 327.68 - Specifies EDRX cycle length of 327.68 seconds.
- 655.36 - Specifies EDRX cycle length of 655.36 seconds.
- 1310.72 - Specifies EDRX cycle length of 1310.72 seconds.
- 2621.44 - Specifies EDRX cycle length of 2621.44 seconds.
- 5242.88 - Specifies EDRX cycle length of 5242.88 seconds.

- 10485.76 - Specifies EDRX cycle length of 10485.76 seconds.

Default

1310.72

Name

edrx12DataBufferCount - eDRX-12 data buffer count

Syntax

--edrx12DataBufferCount <integer>

Description

Suggested maximum number of DL packets buffered in the SGW for eDRX cycle length of 1310.72 seconds.

Class

Optional.

Type / Range

Integer in the range of 1 to 65,535

Default

1

Name

edrx12Override - eDRX-12 override

Syntax

--edrx12override <list-item>

Description

Enable parameter override for eDRX cycle length of 1310.72 seconds.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

edrx13CycleOverride - eDRX-13 cycle override

Syntax

--edrx13CycleOverride <list-item>

Description

Override eDRX cycle selection length of 2621.44 seconds.

Class

Optional.

List

- 5.12 - Specifies EDRX cycle length of 5.12 seconds
- 10.24 - Specifies EDRX cycle length of 10.24 seconds.
- 20.48 - Specifies EDRX cycle length of 20.48 seconds.
- 40.96 - Specifies EDRX cycle length of 40.96 seconds.
- 61.44 - Specifies EDRX cycle length of 61.44 seconds.
- 81.92 - Specifies EDRX cycle length of 81.92 seconds.
- 102.40 - Specifies EDRX cycle length of 102.40 seconds.
- 122.88 - Specifies EDRX cycle length of 122.88 seconds.
- 143.36 - Specifies EDRX cycle length of 143.36 seconds.
- 163.84 - Specifies EDRX cycle length of 163.84 seconds.
- 327.68 - Specifies EDRX cycle length of 327.68 seconds.
- 655.36 - Specifies EDRX cycle length of 655.36 seconds.
- 1310.72 - Specifies EDRX cycle length of 1310.72 seconds.
- 2621.44 - Specifies EDRX cycle length of 2621.44 seconds.
- 5242.88 - Specifies EDRX cycle length of 5242.88 seconds.
- 10485.76 - Specifies EDRX cycle length of 10485.76 seconds.

Default

2621.44

Name

edrx13DataBufferCount - eDRX-13 data buffer count

Syntax

--edrx13DataBufferCount <integer>

Description

Suggested maximum number of DL packets buffered in the SGW for eDRX cycle length of 2621.44 seconds.

Class

Optional.

Type / Range

Integer in the range of 1 to 65,535

Default

1

Name

edrx13Override - eDRX-13 override

Syntax

--edrx13override <list-item>

Description

Enable parameter override for eDRX cycle length of 2621.44 seconds.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

edrx14CycleOverride - eDRX-14 cycle override

Syntax

--edrx14CycleOverride <list-item>

Description

Override eDRX cycle selection length of 5242.88 seconds.

Class

Optional.

List

- 5.12 - Specifies EDRX cycle length of 5.12 seconds
- 10.24 - Specifies EDRX cycle length of 10.24 seconds.
- 20.48 - Specifies EDRX cycle length of 20.48 seconds.
- 40.96 - Specifies EDRX cycle length of 40.96 seconds.
- 61.44 - Specifies EDRX cycle length of 61.44 seconds.
- 81.92 - Specifies EDRX cycle length of 81.92 seconds.
- 102.40 - Specifies EDRX cycle length of 102.40 seconds.
- 122.88 - Specifies EDRX cycle length of 122.88 seconds.
- 143.36 - Specifies EDRX cycle length of 143.36 seconds.
- 163.84 - Specifies EDRX cycle length of 163.84 seconds.

- 327.68 - Specifies EDRX cycle length of 327.68 seconds.
- 655.36 - Specifies EDRX cycle length of 655.36 seconds.
- 1310.72 - Specifies EDRX cycle length of 1310.72 seconds.
- 2621.44 - Specifies EDRX cycle length of 2621.44 seconds.
- 5242.88 - Specifies EDRX cycle length of 5242.88 seconds.
- 10485.76 - Specifies EDRX cycle length of 10485.76 seconds.

Default

5242.88

Name

edrx14DataBufferCount - eDRX-14 data buffer count

Syntax

--edrx14DataBufferCount <integer>

Description

Suggested maximum number of DL packets buffered in the SGW for eDRX cycle length of 5242.88 seconds.

Class

Optional.

Type / Range

Integer in the range of 1 to 65,535

Default

1

Name

edrx14Override - eDRX-14 override

Syntax

--edrx140verrride <list-item>

Description

Enable parameter override for eDRX cycle length of 5242.88 seconds.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

edrx15CycleOverride - eDRX-15 cycle override

Syntax

--edrx15CycleOverride <list-item>

Description

Override eDRX cycle selection length of 10485.76 seconds.

Class

Optional.

List

- 5.12 - Specifies EDRX cycle length of 5.12 seconds
- 10.24 - Specifies EDRX cycle length of 10.24 seconds.
- 20.48 - Specifies EDRX cycle length of 20.48 seconds.
- 40.96 - Specifies EDRX cycle length of 40.96 seconds.
- 61.44 - Specifies EDRX cycle length of 61.44 seconds.
- 81.92 - Specifies EDRX cycle length of 81.92 seconds.
- 102.40 - Specifies EDRX cycle length of 102.40 seconds.
- 122.88 - Specifies EDRX cycle length of 122.88 seconds.
- 143.36 - Specifies EDRX cycle length of 143.36 seconds.
- 163.84 - Specifies EDRX cycle length of 163.84 seconds.
- 327.68 - Specifies EDRX cycle length of 327.68 seconds.
- 655.36 - Specifies EDRX cycle length of 655.36 seconds.
- 1310.72 - Specifies EDRX cycle length of 1310.72 seconds.
- 2621.44 - Specifies EDRX cycle length of 2621.44 seconds.
- 5242.88 - Specifies EDRX cycle length of 5242.88 seconds.
- 10485.76 - Specifies EDRX cycle length of 10485.76 seconds.

Default

10485.76

Name

edrx15DataBufferCount - eDRX-15 data buffer count

Syntax

--edrx15DataBufferCount <integer>

Description

Suggested maximum number of DL packets buffered in the SGW for eDRX cycle length of 10485.76 seconds.

Class

Optional.

Type / Range

Integer in the range of 1 to 65,535

Default

1

Name

edrx15Override - eDRX-15 override

Syntax

--edrx15override <list-item>

Description

Enable parameter override for eDRX cycle length of 10485.76 seconds.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

edrx1CycleOverride - eDRX-1 cycle override

Syntax

--edrx1CycleOverride <list-item>

Description

Override eDRX cycle selection length of 10.24 seconds.

Class

Optional.

List

- 5.12 - Specifies EDRX cycle length of 5.12 seconds
- 10.24 - Specifies EDRX cycle length of 10.24 seconds.
- 20.48 - Specifies EDRX cycle length of 20.48 seconds.
- 40.96 - Specifies EDRX cycle length of 40.96 seconds.

- 61.44 - Specifies EDRX cycle length of 61.44 seconds.
- 81.92 - Specifies EDRX cycle length of 81.92 seconds.
- 102.40 - Specifies EDRX cycle length of 102.40 seconds.
- 122.88 - Specifies EDRX cycle length of 122.88 seconds.
- 143.36 - Specifies EDRX cycle length of 143.36 seconds.
- 163.84 - Specifies EDRX cycle length of 163.84 seconds.
- 327.68 - Specifies EDRX cycle length of 327.68 seconds.
- 655.36 - Specifies EDRX cycle length of 655.36 seconds.
- 1310.72 - Specifies EDRX cycle length of 1310.72 seconds.
- 2621.44 - Specifies EDRX cycle length of 2621.44 seconds.
- 5242.88 - Specifies EDRX cycle length of 5242.88 seconds.
- 10485.76 - Specifies EDRX cycle length of 10485.76 seconds.

Default

10.24

Name

edrx1DataBufferCount - eDRX-1 data buffer count

Syntax

--edrx1DataBufferCount <integer>

Description

Suggested maximum number of DL packets buffered in the SGW for eDRX cycle length of 10.24 seconds.

Class

Optional.

Type / Range

Integer in the range of 1 to 65,535

Default

1

Name

edrx1Override - eDRX-1 override

Syntax

--edrx10override <list-item>

Description

Enable parameter override for eDRX cycle length of 10.24 seconds.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

edrx2CycleOverride - eDRX-2 cycle override

Syntax

--edrx2CycleOverride <list-item>

Description

Override eDRX cycle selection length of 20.48 seconds.

Class

Optional.

List

- 5.12 - Specifies EDRX cycle length of 5.12 seconds
- 10.24 - Specifies EDRX cycle length of 10.24 seconds.
- 20.48 - Specifies EDRX cycle length of 20.48 seconds.
- 40.96 - Specifies EDRX cycle length of 40.96 seconds.
- 61.44 - Specifies EDRX cycle length of 61.44 seconds.
- 81.92 - Specifies EDRX cycle length of 81.92 seconds.
- 102.40 - Specifies EDRX cycle length of 102.40 seconds.
- 122.88 - Specifies EDRX cycle length of 122.88 seconds.
- 143.36 - Specifies EDRX cycle length of 143.36 seconds.
- 163.84 - Specifies EDRX cycle length of 163.84 seconds.
- 327.68 - Specifies EDRX cycle length of 327.68 seconds.
- 655.36 - Specifies EDRX cycle length of 655.36 seconds.
- 1310.72 - Specifies EDRX cycle length of 1310.72 seconds.
- 2621.44 - Specifies EDRX cycle length of 2621.44 seconds.
- 5242.88 - Specifies EDRX cycle length of 5242.88 seconds.
- 10485.76 - Specifies EDRX cycle length of 10485.76 seconds.

Default

20.48

Name

edrx2DataBufferCount - eDRX-2 data buffer count

Syntax

--edrx2DataBufferCount <integer>

Description

Suggested maximum number of DL packets buffered in the SGW for eDRX cycle length of 20.48 seconds.

Class

Optional.

Type / Range

Integer in the range of 1 to 65,535

Default

1

Name

edrx2Override - eDRX-2 override

Syntax

--edrx2Override <list-item>

Description

Enable parameter override for eDRX cycle length of 20.48 seconds.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

edrx3CycleOverride - eDRX-3 cycle override

Syntax

--edrx3CycleOverride <list-item>

Description

Override eDRX cycle selection length of 40.96 seconds.

Class

Optional.

List

- 5.12 - Specifies EDRX cycle length of 5.12 seconds
- 10.24 - Specifies EDRX cycle length of 10.24 seconds.
- 20.48 - Specifies EDRX cycle length of 20.48 seconds.
- 40.96 - Specifies EDRX cycle length of 40.96 seconds.
- 61.44 - Specifies EDRX cycle length of 61.44 seconds.
- 81.92 - Specifies EDRX cycle length of 81.92 seconds.
- 102.40 - Specifies EDRX cycle length of 102.40 seconds.
- 122.88 - Specifies EDRX cycle length of 122.88 seconds.
- 143.36 - Specifies EDRX cycle length of 143.36 seconds.
- 163.84 - Specifies EDRX cycle length of 163.84 seconds.
- 327.68 - Specifies EDRX cycle length of 327.68 seconds.
- 655.36 - Specifies EDRX cycle length of 655.36 seconds.
- 1310.72 - Specifies EDRX cycle length of 1310.72 seconds.
- 2621.44 - Specifies EDRX cycle length of 2621.44 seconds.
- 5242.88 - Specifies EDRX cycle length of 5242.88 seconds.
- 10485.76 - Specifies EDRX cycle length of 10485.76 seconds.

Default

40.96

Name

edrx3DataBufferCount - eDRX-3 data buffer count

Syntax

- edrx3DataBufferCount <integer>

Description

Suggested maximum number of DL packets buffered in the SGW for eDRX cycle length of 40.96 seconds.

Class

Optional.

Type / Range

Integer in the range of 1 to 65,535

Default

1

Name

edrx3Override - eDRX-3 override

Syntax

--edrx3override <list-item>

Description

Enable parameter override for eDRX cycle length of 40.96 seconds.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

edrx4CycleOverride - eDRX-4 cycle override

Syntax

--edrx4CycleOverride <list-item>

Description

Override eDRX cycle selection length of 61.44 seconds.

Class

Optional.

List

- 5.12 - Specifies EDRX cycle length of 5.12 seconds
- 10.24 - Specifies EDRX cycle length of 10.24 seconds.
- 20.48 - Specifies EDRX cycle length of 20.48 seconds.
- 40.96 - Specifies EDRX cycle length of 40.96 seconds.
- 61.44 - Specifies EDRX cycle length of 61.44 seconds.
- 81.92 - Specifies EDRX cycle length of 81.92 seconds.
- 102.40 - Specifies EDRX cycle length of 102.40 seconds.
- 122.88 - Specifies EDRX cycle length of 122.88 seconds.
- 143.36 - Specifies EDRX cycle length of 143.36 seconds.
- 163.84 - Specifies EDRX cycle length of 163.84 seconds.
- 327.68 - Specifies EDRX cycle length of 327.68 seconds.
- 655.36 - Specifies EDRX cycle length of 655.36 seconds.

- 1310.72 - Specifies EDRX cycle length of 1310.72 seconds.
- 2621.44 - Specifies EDRX cycle length of 2621.44 seconds.
- 5242.88 - Specifies EDRX cycle length of 5242.88 seconds.
- 10485.76 - Specifies EDRX cycle length of 10485.76 seconds.

Default

61.44

Name

edrx4DataBufferCount - eDRX-4 data buffer count

Syntax

--edrx4DataBufferCount <integer>

Description

Suggested maximum number of DL packets buffered in the SGW for eDRX cycle length of 61.44 seconds.

Class

Optional.

Type / Range

Integer in the range of 1 to 65,535

Default

1

Name

edrx4Override - eDRX-4 override

Syntax

--edrx4Override <list-item>

Description

Enable parameter override for eDRX cycle length of 61.44 seconds.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

edrx5CycleOverride - eDRX-5 cycle override

Syntax

--edrx5CycleOverride <list-item>

Description

Override eDRX cycle selection length of 81.92 seconds.

Class

Optional.

List

- 5.12 - Specifies EDRX cycle length of 5.12 seconds
- 10.24 - Specifies EDRX cycle length of 10.24 seconds.
- 20.48 - Specifies EDRX cycle length of 20.48 seconds.
- 40.96 - Specifies EDRX cycle length of 40.96 seconds.
- 61.44 - Specifies EDRX cycle length of 61.44 seconds.
- 81.92 - Specifies EDRX cycle length of 81.92 seconds.
- 102.40 - Specifies EDRX cycle length of 102.40 seconds.
- 122.88 - Specifies EDRX cycle length of 122.88 seconds.
- 143.36 - Specifies EDRX cycle length of 143.36 seconds.
- 163.84 - Specifies EDRX cycle length of 163.84 seconds.
- 327.68 - Specifies EDRX cycle length of 327.68 seconds.
- 655.36 - Specifies EDRX cycle length of 655.36 seconds.
- 1310.72 - Specifies EDRX cycle length of 1310.72 seconds.
- 2621.44 - Specifies EDRX cycle length of 2621.44 seconds.
- 5242.88 - Specifies EDRX cycle length of 5242.88 seconds.
- 10485.76 - Specifies EDRX cycle length of 10485.76 seconds.

Default

81.92

Name

edrx5DataBufferCount - eDRX-5 data buffer count

Syntax

--edrx5DataBufferCount <integer>

Description

Suggested maximum number of DL packets buffered in the SGW for eDRX cycle length of 81.92 seconds.

Class

Optional.

Type / Range

Integer in the range of 1 to 65,535

Default

1

Name

edrx5Override - eDRX-5 override

Syntax

--edrx5override <list-item>

Description

Enable parameter override for eDRX cycle length of 81.92 seconds.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

edrx6CycleOverride - eDRX-6 cycle override

Syntax

--edrx6CycleOverride <list-item>

Description

Override eDRX cycle selection length of 102.40 seconds.

Class

Optional.

List

- 5.12 - Specifies EDRX cycle length of 5.12 seconds
- 10.24 - Specifies EDRX cycle length of 10.24 seconds.
- 20.48 - Specifies EDRX cycle length of 20.48 seconds.
- 40.96 - Specifies EDRX cycle length of 40.96 seconds.
- 61.44 - Specifies EDRX cycle length of 61.44 seconds.
- 81.92 - Specifies EDRX cycle length of 81.92 seconds.
- 102.40 - Specifies EDRX cycle length of 102.40 seconds.

- 122.88 - Specifies EDRX cycle length of 122.88 seconds.
- 143.36 - Specifies EDRX cycle length of 143.36 seconds.
- 163.84 - Specifies EDRX cycle length of 163.84 seconds.
- 327.68 - Specifies EDRX cycle length of 327.68 seconds.
- 655.36 - Specifies EDRX cycle length of 655.36 seconds.
- 1310.72 - Specifies EDRX cycle length of 1310.72 seconds.
- 2621.44 - Specifies EDRX cycle length of 2621.44 seconds.
- 5242.88 - Specifies EDRX cycle length of 5242.88 seconds.
- 10485.76 - Specifies EDRX cycle length of 10485.76 seconds.

Default

102.40

Name

edrx6DataBufferCount - eDRX-6 data buffer count

Syntax

--edrx6DataBufferCount <integer>

Description

Suggested maximum number of DL packets buffered in the SGW for eDRX cycle length of 102.40 seconds.

Class

Optional.

Type / Range

Integer in the range of 1 to 65,535

Default

1

Name

edrx6Override - eDRX-6 override

Syntax

--edrx6Override <list-item>

Description

Enable parameter override for eDRX cycle length of 102.40 seconds.

Class

Optional.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Name

edrx7CycleOverride - eDRX-7 cycle override

Syntax

--edrx7CycleOverride <list-item>

Description

Override eDRX cycle selection length of 122.88 seconds.

Class

Optional.

List

- 5.12 - Specifies EDRX cycle length of 5.12 seconds
- 10.24 - Specifies EDRX cycle length of 10.24 seconds.
- 20.48 - Specifies EDRX cycle length of 20.48 seconds.
- 40.96 - Specifies EDRX cycle length of 40.96 seconds.
- 61.44 - Specifies EDRX cycle length of 61.44 seconds.
- 81.92 - Specifies EDRX cycle length of 81.92 seconds.
- 102.40 - Specifies EDRX cycle length of 102.40 seconds.
- 122.88 - Specifies EDRX cycle length of 122.88 seconds.
- 143.36 - Specifies EDRX cycle length of 143.36 seconds.
- 163.84 - Specifies EDRX cycle length of 163.84 seconds.
- 327.68 - Specifies EDRX cycle length of 327.68 seconds.
- 655.36 - Specifies EDRX cycle length of 655.36 seconds.
- 1310.72 - Specifies EDRX cycle length of 1310.72 seconds.
- 2621.44 - Specifies EDRX cycle length of 2621.44 seconds.
- 5242.88 - Specifies EDRX cycle length of 5242.88 seconds.
- 10485.76 - Specifies EDRX cycle length of 10485.76 seconds.

Default

122.88

Name

edrx7DataBufferCount - eDRX-7 data buffer count

Syntax

--edrx7DataBufferCount <integer>

Description

Suggested maximum number of DL packets buffered in the SGW for eDRX cycle length of 122.88 seconds.

Class

Optional.

Type / Range

Integer in the range of 1 to 65,535

Default

1

Name

edrx7Override - eDRX-7 override

Syntax

--edrx70override <list-item>

Description

Enable parameter override for eDRX cycle length of 122.88 seconds.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

edrx8CycleOverride - eDRX-8 cycle override

Syntax

--edrx8CycleOverride <list-item>

Description

Override eDRX cycle selection length of 143.36 seconds.

Class

Optional.

List

- 5.12 - Specifies EDRX cycle length of 5.12 seconds
- 10.24 - Specifies EDRX cycle length of 10.24 seconds.

- 20.48 - Specifies EDRX cycle length of 20.48 seconds.
- 40.96 - Specifies EDRX cycle length of 40.96 seconds.
- 61.44 - Specifies EDRX cycle length of 61.44 seconds.
- 81.92 - Specifies EDRX cycle length of 81.92 seconds.
- 102.40 - Specifies EDRX cycle length of 102.40 seconds.
- 122.88 - Specifies EDRX cycle length of 122.88 seconds.
- 143.36 - Specifies EDRX cycle length of 143.36 seconds.
- 163.84 - Specifies EDRX cycle length of 163.84 seconds.
- 327.68 - Specifies EDRX cycle length of 327.68 seconds.
- 655.36 - Specifies EDRX cycle length of 655.36 seconds.
- 1310.72 - Specifies EDRX cycle length of 1310.72 seconds.
- 2621.44 - Specifies EDRX cycle length of 2621.44 seconds.
- 5242.88 - Specifies EDRX cycle length of 5242.88 seconds.
- 10485.76 - Specifies EDRX cycle length of 10485.76 seconds.

Default

143.36

Name

edrx8DataBufferCount - eDRX-8 data buffer count

Syntax

--edrx8DataBufferCount <integer>

Description

Suggested maximum number of DL packets buffered in the SGW for eDRX cycle length of 143.36 seconds.

Class

Optional.

Type / Range

Integer in the range of 1 to 65,535

Default

1

Name

edrx8Override - eDRX-8 override

Syntax

--edrx8Override <list-item>

Description

Enable parameter override for eDRX cycle length of 143.36 seconds.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

edrx9CycleOverride - eDRX-9 cycle override

Syntax

--edrx9CycleOverride <list-item>

Description

Override eDRX cycle selection length of 163.84 seconds.

Class

Optional.

List

- 5.12 - Specifies EDRX cycle length of 5.12 seconds
- 10.24 - Specifies EDRX cycle length of 10.24 seconds.
- 20.48 - Specifies EDRX cycle length of 20.48 seconds.
- 40.96 - Specifies EDRX cycle length of 40.96 seconds.
- 61.44 - Specifies EDRX cycle length of 61.44 seconds.
- 81.92 - Specifies EDRX cycle length of 81.92 seconds.
- 102.40 - Specifies EDRX cycle length of 102.40 seconds.
- 122.88 - Specifies EDRX cycle length of 122.88 seconds.
- 143.36 - Specifies EDRX cycle length of 143.36 seconds.
- 163.84 - Specifies EDRX cycle length of 163.84 seconds.
- 327.68 - Specifies EDRX cycle length of 327.68 seconds.
- 655.36 - Specifies EDRX cycle length of 655.36 seconds.
- 1310.72 - Specifies EDRX cycle length of 1310.72 seconds.
- 2621.44 - Specifies EDRX cycle length of 2621.44 seconds.
- 5242.88 - Specifies EDRX cycle length of 5242.88 seconds.

- 10485.76 - Specifies EDRX cycle length of 10485.76 seconds.

Default

163.84

Name

edrx9DataBufferCount - eDRX-9 data buffer count

Syntax

--edrx9DataBufferCount <integer>

Description

Suggested maximum number of DL packets buffered in the SGW for eDRX cycle length of 163.84 seconds.

Class

Optional.

Type / Range

Integer in the range of 1 to 65,535

Default

1

Name

edrx9Override - eDRX-9 override

Syntax

--edrx90override <list-item>

Description

Enable parameter override for eDRX cycle length of 163.84 seconds.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

nbEdrx10PtwOverride - NB-S1 mode eDRX-10 PTW length override

Syntax

--nbEdrx10PtwOverride <list-item>

Description

Override PTW length for eDRX cycle length of 327.68 seconds for NB-S1 UE.

Class

Optional.

List

- 2.56 - Specifies eDRX narrow band paging time window of 2.56 seconds.
- 5.12 - Specifies eDRX narrow band paging time window of 5.12 seconds.
- 7.68 - Specifies eDRX narrow band paging time window of 7.68 seconds.
- 10.24 - Specifies eDRX narrow band paging time window of 10.24 seconds.
- 12.80 - Specifies eDRX narrow band paging time window of 12.80 seconds.
- 15.36 - Specifies eDRX narrow band paging time window of 15.36 seconds.
- 17.92 - Specifies eDRX narrow band paging time window of 17.92 seconds.
- 20.48 - Specifies eDRX narrow band paging time window of 20.48 seconds.
- 23.04 - Specifies eDRX narrow band paging time window of 23.04 seconds.
- 25.60 - Specifies eDRX narrow band paging time window of 25.60 seconds.
- 28.16 - Specifies eDRX narrow band paging time window of 28.16 seconds.
- 30.72 - Specifies eDRX narrow band paging time window of 30.72 seconds.
- 33.28 - Specifies eDRX narrow band paging time window of 33.28 seconds.
- 35.84 - Specifies eDRX narrow band paging time window of 35.84 seconds.
- 38.40 - Specifies eDRX narrow band paging time window of 38.40 seconds.
- 40.96 - Specifies eDRX narrow band paging time window of 40.96 seconds.

Default

28.16

Name

nbEdrx11PtwOverride - NB-S1 mode eDRX-11 PTW length override

Syntax

--nbEdrx11PtwOverride <list-item>

Description

Override PTW length for eDRX cycle length of 655.36 seconds for NB-S1 UE.

Class

Optional.

List

- 2.56 - Specifies eDRX narrow band paging time window of 2.56 seconds.
- 5.12 - Specifies eDRX narrow band paging time window of 5.12 seconds.

- 7.68 - Specifies eDRX narrow band paging time window of 7.68 seconds.
- 10.24 - Specifies eDRX narrow band paging time window of 10.24 seconds.
- 12.80 - Specifies eDRX narrow band paging time window of 12.80 seconds.
- 15.36 - Specifies eDRX narrow band paging time window of 15.36 seconds.
- 17.92 - Specifies eDRX narrow band paging time window of 17.92 seconds.
- 20.48 - Specifies eDRX narrow band paging time window of 20.48 seconds.
- 23.04 - Specifies eDRX narrow band paging time window of 23.04 seconds.
- 25.60 - Specifies eDRX narrow band paging time window of 25.60 seconds.
- 28.16 - Specifies eDRX narrow band paging time window of 28.16 seconds.
- 30.72 - Specifies eDRX narrow band paging time window of 30.72 seconds.
- 33.28 - Specifies eDRX narrow band paging time window of 33.28 seconds.
- 35.84 - Specifies eDRX narrow band paging time window of 35.84 seconds.
- 38.40 - Specifies eDRX narrow band paging time window of 38.40 seconds.
- 40.96 - Specifies eDRX narrow band paging time window of 40.96 seconds.

Default

28.16

Name

nbEdrx12PtwOverride - NB-S1 mode eDRX-12 PTW length override

Syntax

--nbEdrx12PtwOverride <list-item>

Description

Override PTW length for eDRX cycle length of 1310.72 seconds for NB-S1 UE.

Class

Optional.

List

- 2.56 - Specifies eDRX narrow band paging time window of 2.56 seconds.
- 5.12 - Specifies eDRX narrow band paging time window of 5.12 seconds.
- 7.68 - Specifies eDRX narrow band paging time window of 7.68 seconds.
- 10.24 - Specifies eDRX narrow band paging time window of 10.24 seconds.
- 12.80 - Specifies eDRX narrow band paging time window of 12.80 seconds.
- 15.36 - Specifies eDRX narrow band paging time window of 15.36 seconds.
- 17.92 - Specifies eDRX narrow band paging time window of 17.92 seconds.

- 20.48 - Specifies eDRX narrow band paging time window of 20.48 seconds.
- 23.04 - Specifies eDRX narrow band paging time window of 23.04 seconds.
- 25.60 - Specifies eDRX narrow band paging time window of 25.60 seconds.
- 28.16 - Specifies eDRX narrow band paging time window of 28.16 seconds.
- 30.72 - Specifies eDRX narrow band paging time window of 30.72 seconds.
- 33.28 - Specifies eDRX narrow band paging time window of 33.28 seconds.
- 35.84 - Specifies eDRX narrow band paging time window of 35.84 seconds.
- 38.40 - Specifies eDRX narrow band paging time window of 38.40 seconds.
- 40.96 - Specifies eDRX narrow band paging time window of 40.96 seconds.

Default

28.16

Name

nbEdrx13PtwOverride - NB-S1 mode eDRX-13 PTW length override

Syntax

--nbEdrx13PtwOverride <list-item>

Description

Override PTW length for eDRX cycle length of 2621.44 seconds for NB-S1 UE.

Class

Optional.

List

- 2.56 - Specifies eDRX narrow band paging time window of 2.56 seconds.
- 5.12 - Specifies eDRX narrow band paging time window of 5.12 seconds.
- 7.68 - Specifies eDRX narrow band paging time window of 7.68 seconds.
- 10.24 - Specifies eDRX narrow band paging time window of 10.24 seconds.
- 12.80 - Specifies eDRX narrow band paging time window of 12.80 seconds.
- 15.36 - Specifies eDRX narrow band paging time window of 15.36 seconds.
- 17.92 - Specifies eDRX narrow band paging time window of 17.92 seconds.
- 20.48 - Specifies eDRX narrow band paging time window of 20.48 seconds.
- 23.04 - Specifies eDRX narrow band paging time window of 23.04 seconds.
- 25.60 - Specifies eDRX narrow band paging time window of 25.60 seconds.
- 28.16 - Specifies eDRX narrow band paging time window of 28.16 seconds.
- 30.72 - Specifies eDRX narrow band paging time window of 30.72 seconds.

- 33.28 - Specifies eDRX narrow band paging time window of 33.28 seconds.
- 35.84 - Specifies eDRX narrow band paging time window of 35.84 seconds.
- 38.40 - Specifies eDRX narrow band paging time window of 38.40 seconds.
- 40.96 - Specifies eDRX narrow band paging time window of 40.96 seconds.

Default

28.16

Name

nbEdrx14PtwOverride - NB-S1 mode eDRX-14 PTW length override

Syntax

--nbEdrx14PtwOverride <list-item>

Description

Override PTW length for eDRX cycle length of 5242.88 seconds for NB-S1 UE.

Class

Optional.

List

- 2.56 - Specifies eDRX narrow band paging time window of 2.56 seconds.
- 5.12 - Specifies eDRX narrow band paging time window of 5.12 seconds.
- 7.68 - Specifies eDRX narrow band paging time window of 7.68 seconds.
- 10.24 - Specifies eDRX narrow band paging time window of 10.24 seconds.
- 12.80 - Specifies eDRX narrow band paging time window of 12.80 seconds.
- 15.36 - Specifies eDRX narrow band paging time window of 15.36 seconds.
- 17.92 - Specifies eDRX narrow band paging time window of 17.92 seconds.
- 20.48 - Specifies eDRX narrow band paging time window of 20.48 seconds.
- 23.04 - Specifies eDRX narrow band paging time window of 23.04 seconds.
- 25.60 - Specifies eDRX narrow band paging time window of 25.60 seconds.
- 28.16 - Specifies eDRX narrow band paging time window of 28.16 seconds.
- 30.72 - Specifies eDRX narrow band paging time window of 30.72 seconds.
- 33.28 - Specifies eDRX narrow band paging time window of 33.28 seconds.
- 35.84 - Specifies eDRX narrow band paging time window of 35.84 seconds.
- 38.40 - Specifies eDRX narrow band paging time window of 38.40 seconds.
- 40.96 - Specifies eDRX narrow band paging time window of 40.96 seconds.

Default

28.16

Name

nbEdrx15PtwOverride - NB-S1 mode eDRX-15 PTW length override

Syntax`--nbEdrx15PtwOverride <list-item>`**Description**

Override PTW length for eDRX cycle length of 10485.76 seconds for NB-S1 UE.

Class

Optional.

List

- 2.56 - Specifies eDRX narrow band paging time window of 2.56 seconds.
- 5.12 - Specifies eDRX narrow band paging time window of 5.12 seconds.
- 7.68 - Specifies eDRX narrow band paging time window of 7.68 seconds.
- 10.24 - Specifies eDRX narrow band paging time window of 10.24 seconds.
- 12.80 - Specifies eDRX narrow band paging time window of 12.80 seconds.
- 15.36 - Specifies eDRX narrow band paging time window of 15.36 seconds.
- 17.92 - Specifies eDRX narrow band paging time window of 17.92 seconds.
- 20.48 - Specifies eDRX narrow band paging time window of 20.48 seconds.
- 23.04 - Specifies eDRX narrow band paging time window of 23.04 seconds.
- 25.60 - Specifies eDRX narrow band paging time window of 25.60 seconds.
- 28.16 - Specifies eDRX narrow band paging time window of 28.16 seconds.
- 30.72 - Specifies eDRX narrow band paging time window of 30.72 seconds.
- 33.28 - Specifies eDRX narrow band paging time window of 33.28 seconds.
- 35.84 - Specifies eDRX narrow band paging time window of 35.84 seconds.
- 38.40 - Specifies eDRX narrow band paging time window of 38.40 seconds.
- 40.96 - Specifies eDRX narrow band paging time window of 40.96 seconds.

Default

28.16

Name

nbEdrx2PtwOverride - NB-S1 mode eDRX-2 PTW length override

Syntax`--nbEdrx2PtwOverride <list-item>`**Description**

Override PTW length for eDRX cycle length of 20.48 seconds for NB-S1 UE.

Class

Optional.

List

- 2.56 - Specifies eDRX narrow band paging time window of 2.56 seconds.
- 5.12 - Specifies eDRX narrow band paging time window of 5.12 seconds.
- 7.68 - Specifies eDRX narrow band paging time window of 7.68 seconds.
- 10.24 - Specifies eDRX narrow band paging time window of 10.24 seconds.
- 12.80 - Specifies eDRX narrow band paging time window of 12.80 seconds.
- 15.36 - Specifies eDRX narrow band paging time window of 15.36 seconds.
- 17.92 - Specifies eDRX narrow band paging time window of 17.92 seconds.

Default

10.24

Name

nbEdrx3PtwOverride - NB-S1 mode eDRX-3 PTW length override

Syntax

--nbEdrx3PtwOverride <list-item>

Description

Override PTW length for eDRX cycle length of 40.96 seconds for NB-S1 UE.

Class

Optional.

List

- 2.56 - Specifies eDRX narrow band paging time window of 2.56 seconds.
- 5.12 - Specifies eDRX narrow band paging time window of 5.12 seconds.
- 7.68 - Specifies eDRX narrow band paging time window of 7.68 seconds.
- 10.24 - Specifies eDRX narrow band paging time window of 10.24 seconds.
- 12.80 - Specifies eDRX narrow band paging time window of 12.80 seconds.
- 15.36 - Specifies eDRX narrow band paging time window of 15.36 seconds.
- 17.92 - Specifies eDRX narrow band paging time window of 17.92 seconds.
- 20.48 - Specifies eDRX narrow band paging time window of 20.48 seconds.
- 23.04 - Specifies eDRX narrow band paging time window of 23.04 seconds.
- 25.60 - Specifies eDRX narrow band paging time window of 25.60 seconds.

- 28.16 - Specifies eDRX narrow band paging time window of 28.16 seconds.
- 30.72 - Specifies eDRX narrow band paging time window of 30.72 seconds.
- 33.28 - Specifies eDRX narrow band paging time window of 33.28 seconds.
- 35.84 - Specifies eDRX narrow band paging time window of 35.84 seconds.
- 38.40 - Specifies eDRX narrow band paging time window of 38.40 seconds.
- 40.96 - Specifies eDRX narrow band paging time window of 40.96 seconds.

Default

10.24

Name

nbEdrx5PtwOverride - NB-S1 mode eDRX-5 PTW length override

Syntax

--nbEdrx5PtwOverride <list-item>

Description

Override PTW length for eDRX cycle length of 81.92 seconds for NB-S1 UE.

Class

Optional.

List

- 2.56 - Specifies eDRX narrow band paging time window of 2.56 seconds.
- 5.12 - Specifies eDRX narrow band paging time window of 5.12 seconds.
- 7.68 - Specifies eDRX narrow band paging time window of 7.68 seconds.
- 10.24 - Specifies eDRX narrow band paging time window of 10.24 seconds.
- 12.80 - Specifies eDRX narrow band paging time window of 12.80 seconds.
- 15.36 - Specifies eDRX narrow band paging time window of 15.36 seconds.
- 17.92 - Specifies eDRX narrow band paging time window of 17.92 seconds.
- 20.48 - Specifies eDRX narrow band paging time window of 20.48 seconds.
- 23.04 - Specifies eDRX narrow band paging time window of 23.04 seconds.
- 25.60 - Specifies eDRX narrow band paging time window of 25.60 seconds.
- 28.16 - Specifies eDRX narrow band paging time window of 28.16 seconds.
- 30.72 - Specifies eDRX narrow band paging time window of 30.72 seconds.
- 33.28 - Specifies eDRX narrow band paging time window of 33.28 seconds.
- 35.84 - Specifies eDRX narrow band paging time window of 35.84 seconds.
- 38.40 - Specifies eDRX narrow band paging time window of 38.40 seconds.

- 40.96 - Specifies eDRX narrow band paging time window of 40.96 seconds.

Default

10.24

Name

nbEdrx9PtwOverride - NB-S1 mode eDRX-9 PTW length override

Syntax

--nbEdrx9PtwOverride <list-item>

Description

Override PTW length for eDRX cycle length of 163.84 seconds for NB-S1 UE.

Class

Optional.

List

- 2.56 - Specifies eDRX narrow band paging time window of 2.56 seconds.
- 5.12 - Specifies eDRX narrow band paging time window of 5.12 seconds.
- 7.68 - Specifies eDRX narrow band paging time window of 7.68 seconds.
- 10.24 - Specifies eDRX narrow band paging time window of 10.24 seconds.
- 12.80 - Specifies eDRX narrow band paging time window of 12.80 seconds.
- 15.36 - Specifies eDRX narrow band paging time window of 15.36 seconds.
- 17.92 - Specifies eDRX narrow band paging time window of 17.92 seconds.
- 20.48 - Specifies eDRX narrow band paging time window of 20.48 seconds.
- 23.04 - Specifies eDRX narrow band paging time window of 23.04 seconds.
- 25.60 - Specifies eDRX narrow band paging time window of 25.60 seconds.
- 28.16 - Specifies eDRX narrow band paging time window of 28.16 seconds.
- 30.72 - Specifies eDRX narrow band paging time window of 30.72 seconds.
- 33.28 - Specifies eDRX narrow band paging time window of 33.28 seconds.
- 35.84 - Specifies eDRX narrow band paging time window of 35.84 seconds.
- 38.40 - Specifies eDRX narrow band paging time window of 38.40 seconds.
- 40.96 - Specifies eDRX narrow band paging time window of 40.96 seconds.

Default

17.92

Name

ptwNbCarryover - PTW NB Carryover

Syntax

--ptwNbCarryover <list-item>

Description

If the MME is provisioned to allow paging to extend to another PTW, this field specifies the maximum WB eDRX cycle value that a UE is allowed to have for it to be eligible for paging carry over. The default is 20.48.

Class

Optional.

List

- 20.48 - Specifies eDRX narrow band cycle length of 20.48 seconds.
- 40.96 - Specifies eDRX narrow band cycle length 40.96 seconds.
- 81.92 - Specifies eDRX narrow band cycle length 81.92 seconds.
- 163.84 - Specifies eDRX narrow band cycle length of 163.84 seconds.
- 327.68 - Specifies eDRX narrow band cycle length of 327.68 seconds.
- 655.36 - Specifies eDRX narrow band cycle length of 655.36 seconds.
- 1310.72 - Specifies eDRX narrow band cycle length of 1310.72 seconds.
- 2621.44 - Specifies eDRX narrow band cycle length of 2621.44 seconds.
- 5242.88 - Specifies eDRX narrow band cycle length of 5242.88 seconds.
- 10485.76 - Specifies eDRX narrow band cycle length of 10485.76 seconds.

Default

20.48

Name

ptwWbCarryover - PTW WB Carryover

Syntax

--ptwWbCarryover <list-item>

Description

If the MME is provisioned to allow paging to extend to another PTW, this field specifies the maximum WB eDRX cycle value that a UE is allowed to have for it to be eligible for paging carry over. The default is 10.24.

Class

Optional.

List

- 10.24 - Specifies eDRX wide band cycle length of 20.48 seconds.
- 20.48 - 10.48 - Specifies eDRX wide band cycle length of 20.48 seconds.
- 40.96 - Specifies eDRX wide band cycle length of 40.96 seconds.

- 61.44 - Specifies eDRX wide band cycle length of 61.44 seconds.
- 81.92 - Specifies eDRX wide band cycle length of 81.92 seconds.
- 102.40 - Specifies eDRX wide band cycle length of 102.40 seconds.
- 122.88 - Specifies eDRX wide band cycle length of 122.88 seconds.
- 143.36 - Specifies eDRX wide band cycle length of 143.36 seconds.
- 163.84 - Specifies eDRX wide band cycle length of 163.84 seconds.
- 327.68 - Specifies eDRX wide band cycle length of 327.68 seconds.
- 655.36 - Specifies eDRX wide band cycle length of 655.36 seconds.
- 1310.72 - Specifies eDRX wide band cycle length of 1310.72 seconds.
- 2621.44 - Specifies eDRX wide band cycle length of 2621.44 seconds.

Default

10.24

Name

wbEdrx10PtwOverride - WB-S1 mode eDRX-10 PTW length override

Syntax

--wbEdrx10PtwOverride <list-item>

Description

Override PTW length for eDRX cycle length of 327.68 seconds for WB-S1 UE.

Class

Optional.

List

- 1.28 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 2.56 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 3.84 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 5.12 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 6.40 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 7.68 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 8.96 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 10.24 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 11.52 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 12.80 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 14.08 - Specifies eDRX wide band paging time window length of 1.28 seconds.

- 15.36 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 16.64 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 17.92 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 19.20 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 20.48 - Specifies eDRX wide band paging time window length of 1.28 seconds.

Default

20.48

Name

wbEdrx11PtwOverride - WB-S1 mode eDRX-11 PTW length override

Syntax

--wbEdrx11PtwOverride <list-item>

Description

Override PTW length for eDRX cycle length of 655.36 seconds for WB-S1 UE.

Class

Optional.

List

- 1.28 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 2.56 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 3.84 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 5.12 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 6.40 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 7.68 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 8.96 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 10.24 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 11.52 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 12.80 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 14.08 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 15.36 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 16.64 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 17.92 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 19.20 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 20.48 - Specifies eDRX wide band paging time window length of 1.28 seconds.

Default

20.48

Name

wbEdrx12PtwOverride - WB-S1 mode eDRX-12 PTW length override

Syntax`--wbEdrx12PtwOverride <list-item>`**Description**

Override PTW length for eDRX cycle length of 1310.72 seconds for WB-S1 UE.

Class

Optional.

List

- 1.28 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 2.56 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 3.84 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 5.12 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 6.40 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 7.68 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 8.96 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 10.24 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 11.52 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 12.80 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 14.08 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 15.36 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 16.64 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 17.92 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 19.20 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 20.48 - Specifies eDRX wide band paging time window length of 1.28 seconds.

Default

20.48

Name

wbEdrx13PtwOverride - WB-S1 mode eDRX-13 PTW length override

Syntax`--wbEdrx13PtwOverride <list-item>`

Description

Override PTW length for eDRX cycle length of 2621.44 seconds for WB-S1 UE.

Class

Optional.

List

- 1.28 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 2.56 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 3.84 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 5.12 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 6.40 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 7.68 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 8.96 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 10.24 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 11.52 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 12.80 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 14.08 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 15.36 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 16.64 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 17.92 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 19.20 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 20.48 - Specifies eDRX wide band paging time window length of 1.28 seconds.

Default

20.48

Name

wbEdrx1PtwOverride - WB-S1 mode eDRX-1 PTW length override

Syntax

--wbEdrx1PtwOverride <list-item>

Description

Override PTW length for eDRX cycle length of 10.24 seconds for WB-S1 UE.

Class

Optional.

List

- 1.28 - Specifies eDRX wide band paging time window of 1.28 seconds.

- 2.56 - Specifies eDRX wide band paging time window of 2.56 seconds.
- 3.84 - Specifies eDRX wide band paging time window of 3.84 seconds.
- 5.12 - Specifies eDRX wide band paging time window of 5.12 seconds.
- 6.40 - Specifies eDRX wide band paging time window of 6.40 seconds.
- 7.68 - Specifies eDRX wide band paging time window of 7.68 seconds.
- 8.96 - Specifies eDRX wide band paging time window of 8.96 seconds.

Default

6.40

Name

wbEdrx2PtwOverride - WB-S1 mode eDRX-2 PTW length override

Syntax

--wbEdrx2PtwOverride <list-item>

Description

Override PTW length for eDRX cycle length of 20.48 seconds for WB-S1 UE.

Class

Optional.

List

- 1.28 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 2.56 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 3.84 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 5.12 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 6.40 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 7.68 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 8.96 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 10.24 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 11.52 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 12.80 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 14.08 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 15.36 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 16.64 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 17.92 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 19.20 - Specifies eDRX wide band paging time window length of 1.28 seconds.

- 20.48 - Specifies eDRX wide band paging time window length of 1.28 seconds.

Default

6.40

Name

wbEdrx3PtwOverride - WB-S1 mode eDRX-3 PTW length override

Syntax

--wbEdrx3PtwOverride <list-item>

Description

Override PTW length for eDRX cycle length of 40.96 seconds for WB-S1 UE.

Class

Optional.

List

- 1.28 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 2.56 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 3.84 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 5.12 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 6.40 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 7.68 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 8.96 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 10.24 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 11.52 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 12.80 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 14.08 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 15.36 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 16.64 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 17.92 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 19.20 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 20.48 - Specifies eDRX wide band paging time window length of 1.28 seconds.

Default

6.40

Name

wbEdrx4PtwOverride - WB-S1 mode eDRX-4 PTW length override

Syntax

--wbEdrx4PtwOverride <list-item>

Description

Override PTW length for eDRX cycle length of 61.44 seconds for WB-S1 UE.

Class

Optional.

List

- 1.28 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 2.56 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 3.84 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 5.12 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 6.40 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 7.68 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 8.96 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 10.24 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 11.52 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 12.80 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 14.08 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 15.36 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 16.64 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 17.92 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 19.20 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 20.48 - Specifies eDRX wide band paging time window length of 1.28 seconds.

Default

6.40

Name

wbEdrx5PtwOverride - WB-S1 mode eDRX-5 PTW length override

Syntax

--wbEdrx5PtwOverride <list-item>

Description

Override PTW length for eDRX cycle length of 81.92 seconds for WB-S1 UE.

Class

Optional.

List

- 1.28 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 2.56 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 3.84 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 5.12 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 6.40 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 7.68 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 8.96 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 10.24 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 11.52 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 12.80 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 14.08 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 15.36 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 16.64 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 17.92 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 19.20 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 20.48 - Specifies eDRX wide band paging time window length of 1.28 seconds.

Default

8.96

Name

wbEdrx6PtwOverride - WB-S1 mode eDRX-6 PTW length override

Syntax

--wbEdrx6PtwOverride <list-item>

Description

Override PTW length for eDRX cycle length of 102.40 seconds for WB-S1 UE.

Class

Optional.

List

- 1.28 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 2.56 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 3.84 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 5.12 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 6.40 - Specifies eDRX wide band paging time window length of 1.28 seconds.

- 7.68 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 8.96 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 10.24 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 11.52 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 12.80 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 14.08 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 15.36 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 16.64 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 17.92 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 19.20 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 20.48 - Specifies eDRX wide band paging time window length of 1.28 seconds.

Default

10.24

Name

wbEdrx7PtwOverride - WB-S1 mode eDRX-7 PTW length override

Syntax

-wbEdrx7PtwOverride <list-item>

Description

Override PTW length for eDRX cycle length of 122.88 seconds for WB-S1 UE.

Class

Optional.

List

- 1.28 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 2.56 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 3.84 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 5.12 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 6.40 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 7.68 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 8.96 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 10.24 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 11.52 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 12.80 - Specifies eDRX wide band paging time window length of 1.28 seconds.

- 14.08 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 15.36 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 16.64 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 17.92 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 19.20 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 20.48 - Specifies eDRX wide band paging time window length of 1.28 seconds.

Default

12.80

Name

wbEdrx8PtwOverride - WB-S1 mode eDRX-8 PTW length override

Syntax

--wbEdrx8PtwOverride <list-item>

Description

Override PTW length for eDRX cycle length of 143.36 seconds for WB-S1 UE.

Class

Optional.

List

- 1.28 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 2.56 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 3.84 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 5.12 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 6.40 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 7.68 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 8.96 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 10.24 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 11.52 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 12.80 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 14.08 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 15.36 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 16.64 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 17.92 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 19.20 - Specifies eDRX wide band paging time window length of 1.28 seconds.

- 20.48 - Specifies eDRX wide band paging time window length of 1.28 seconds.

Default

14.08

Name

wbEdrx9PtwOverride - WB-S1 mode eDRX-9 PTW length override

Syntax

--wbEdrx9PtwOverride <list-item>

Description

Override PTW length for eDRX cycle length of 163.84 seconds for WB-S1 UE.

Class

Optional.

List

- 1.28 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 2.56 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 3.84 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 5.12 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 6.40 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 7.68 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 8.96 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 10.24 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 11.52 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 12.80 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 14.08 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 15.36 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 16.64 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 17.92 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 19.20 - Specifies eDRX wide band paging time window length of 1.28 seconds.
- 20.48 - Specifies eDRX wide band paging time window length of 1.28 seconds.

Default

16.64

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.104 cmm emergencyNum

Emergency Number List. Syntax, description, parameters, and roles for cmm emergencyNum command.

Syntax

Description

This command is used to administer an Emergency Number List and includes parameters for defining emergency numbers for ambulance, fire brigade, marine guard, mountain rescue, and police.

Parameters

Name

emergencyNumListName - Emergency number list table name

Syntax

--emergencyNumListName <string>

Description

Defines the emergency number list table name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

emergencyDigits - Emergency digits

Syntax

--emergencyDigits <string>

Description

Emergency number. only 8 unique numbers per emergency numbertableid.

Class

Required key.

Type / Range

Digit string of length 1 to 10 with the allowed pattern of [0-9][0-9A-Ca-c*#]*

Name

ambulance - Ambulance

Syntax

--ambulance <list-item>

Description

If emergency number digits apply to ambulance.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

fireBrigade - Fire brigade

Syntax

--fireBrigade <list-item>

Description

If Emergency Number Digits apply to Fire Brigade.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

marineGuard - Marine guard

Syntax

--marineGuard <list-item>

Description

If Emergency Number Digits apply to Marine Guard.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

mountainRescue - Mountain rescue

Syntax

--mountainRescue <list-item>

Description

If Emergency Number Digits apply to Mountain Rescue.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

police - Police

Syntax

--police <list-item>

Description

If emergency number digits apply to police.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.105 cmm emergencyNumList

Emergency Number List Table. Syntax, description, parameters, and roles for cmm emergencyNumList command.

Syntax

Description

This command is used to administer an Emergency Number List Table

Parameters

Name

name - Name

Syntax

--name <string>

Description

Defines the name of emergency number list table.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

description - Description

Syntax

--description <string>

Description

Description.

Class

Optional.

Type / Range

Digit string of length 0 to 160 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.106 cmm emergencyProfile

Emergency Profile. Syntax, description, parameters, and roles for cmm emergencyProfile command.

Syntax**Description**

This command is used to administer an Emergency Profile. MME allows creation of up to 10 emergency profiles for the shared network. An emergency profile can be assigned to an MME home PLMN or any one of the 8 shared PLMNs.

Parameters**Name**

name - Emergency profile name

Syntax

--name <string>

Description

Name that identifies emergency profile.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

ambrDL - AMBR downlink in bps

Syntax

--ambrDL <integer>

Description

AMBR downlink bandwidth in bits per second.

Class

Optional.

Type / Range

Integer in the range of 0 to 4,294,967,295

Default

4096000

Name

ambrUI - AMBR uplink in bps

Syntax

--ambrUI <integer>

Description

AMBR uplink bandwidth in bits per second.

Class

Optional.

Type / Range

Integer in the range of 0 to 4,294,967,295

Default

4096000

Name

apnNi - APN (NI)

Syntax

--apnNi <string>

Description

Access point name network identifier part.

Class

Optional.

Type / Range

Digit string of length 1 to 62 with the allowed pattern of [0-9a-zA-Z\.\-]*

Default

sos

Name

description - Description

Syntax

--description <string>

Description

Description.

Class

Optional.

Type / Range

Digit string of length 0 to 160 with the allowed pattern of [0-9a-zA-Z\.\- _:@]*

Default

"

Name

emerArp - Emergency ARP

Syntax

--emerArp <integer>

Description

Allocation and retention priority for emergency service default bearers.

Class

Optional.

Type / Range

Integer in the range of 1 to 15

Default

1

Name

emergencyGmlcAlternateName - Emergency GMLC alternate name

Syntax

--emergencyGmlcAlternateName <string>

Description

Emergency gateway mobile location center alternate name is the name associated with the IP address of Slg interface on remote end point configuration prior to entering the name in this field. this name must be provisioned on the diamconnection with Slg interface. this should be different from all other GMLCs in this profile table.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

emergencyGmlcPrimaryName - Emergency GMLC primary name

Syntax

--emergencyGmlcPrimaryName <string>

Description

Emergency gateway mobile location center primary name is the name associated with the IP address of Slg interface on remote end point configuration prior to entering the name in this field. this name must be provisioned on the diamconnection with Slg interface. this should be different from all other GMLCs in this profile table.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

emergencyNumListName - Emergency number list table name

Syntax

--emergencyNumListName <string>

Description

Defines the emergency number list table name.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

emerQci - Emergency QCI

Syntax

--emerQci <integer>

Description

QoS class identifier for emergency service default bearers.

Class

Optional.

Type / Range

Integer in the range of 5 to 9

Default

5

Name

esrvcBehavior - Service behavior

Syntax

--esrvcBehavior <list-item>

Description

MME behavior in providing emergency bearer support.

Class

Optional.

List

- All UEs are allowed - Specifies MME emergency bearer support to all UEs are allowed.
- Only UEs that are authenticated are allowed - Specifies MME emergency bearer support to only UEs that are authenticated are allowed.
- UE IMSI required, authentication optional - Specifies MME emergency bearer support to UE IMSI required, authentication optional.
- Valid UEs only - Specifies MME emergency bearer support to valid UEs only.

Default

All UEs are allowed

Name

esrvcBlackListedUe - Service for black listed UE

Syntax

--esrvcBlackListedUe <list-item>

Description

Emergency services shall be provided to a UE with an IMEI that is black listed in the EIR.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

esrvct3412Timer - Emergency T3412 timer in minutes

Syntax

--esrvct3412Timer <integer>

Description

Emergency T3412 timer in minutes. if it is 0, this timer is disabled and t3412 timer from timer table is used.

Class

Optional.

Type / Range

Integer in the range of 0 to 186 in units of Minute

Default

0

Name

horizontalAccuracy - Horizontal accuracy

Syntax

--horizontalAccuracy <integer>

Description

Horizontal accuracy parameter in the location request.

Class

Optional.

Type / Range

Integer in the range of 0 to 127

Default

20

Name

initiateLcsRequest - Initiate LCS request

Syntax

--initiateLcsRequest <list-item>

Description

Initiate LCS request indicates whether or not the MME shall initiate a location request for the UE. valid values are do not initiate, send SLR to GMLC with estimated UE location, send SLR without location estimate to GMLC.

Class

Optional.

List

- Do not initiate - Do not initiate.
- Send SLR to GMLC with estimated UE location - Send SLR to GMLC with estimated UE location.
- Send SLR without location estimate to GMLC - Send SLR without location estimate to GMLC.

Default

Do not initiate

Name

nilrLcsClientName - NI LR LCS EPS Client Name

Syntax

--nilrLcsClientName <string>

Description

NI-LR LCS EPS Client Name

Class

Optional.

Type / Range

Digit string of length 0 to 62 with the allowed pattern of [A-Za-z0-9._-]*

Default

""

Name

pdnGwFqdn - PDNGW FQDN

Syntax

--pdnGwFqdn <string>

Description

Emergency packet data network gateway fully qualified domain name.

Class

Optional.

Type / Range

Digit string of length 0 to 96 with the allowed pattern of [0-9a-zA-Z\._:\]*

Name

pdnGwIpV4 - PDNGW IPv4

Syntax

--pdnGwIpV4 <string>

Description

Emergency packet data network gateway IP V4 address.

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Default

0.0.0.0

Name

pdnGwIpV6 - PDNGW IPv6

Syntax

--pdnGwIpV6 <string>

Description

Emergency packet data network gateway IP V6 address.

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Default

::

Name

pdnTypeWhenv4v6Requested - PDN type when IPv4v6 requested

Syntax

--pdnTypeWhenv4v6Requested <list-item>

Description

PDN type used for the emergency session when the UE requests PDN type IPv4v6.

Class

Optional.

List

- v4 - IPv4 - Specify IPv4.

- v4v6 - IPv4 / IPv6 - Specify IPv4 / IPv6.
- v6 - IPv6 - Specify IPv6.

Default

v4v6

Name

pgwcSmfGwFqdn - PGW-C+SMF GW FQDN

Syntax

--pgwcSmfGwFqdn <string>

Description

Emergency PGW-C+SMF gateway fully qualified domain name.

Class

Optional.

Type / Range

FQDN string of length 0 to 96 with the allowed pattern of [0-9a-zA-Z.-]* where each label (separated by dot) must start with a letter and a single label must not exceed 63 characters in length

Default

""

Name

pgwcSmfGwIpv4 - PGW-C+SMF GW IPv4

Syntax

--pgwcSmfGwIpv4 <string>

Description

Emergency PGW-C+SMF gateway IP V4 address.

Class

Optional.

Type / Range

Digit string of length 7 to 15 with the allowed pattern of [0-9\.]*

Default

0.0.0.0

Name

pgwcSmfGwIpv6 - PGW-C+SMF GW IPv6

Syntax

--pgwcSmfGwIpv6 <string>

Description

Emergency PGW-C+SMF gateway IPv6 address

Class

Optional.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of '[0-9a-f:]*'

Default

::

Name

pgwcSmfGwPdnTypeWhenv4v6Requested - PGW-C+SMF GW PDN Type When v4v6 Requested

Syntax

- pgwcSmfGwPdnTypeWhenv4v6Requested <list-item>

Description

PDN type used for the emergency session when the UE requests PDN type IPv4v6

Class

Optional.

List

- v4 - IPv4 - Specify IPv4.
- v4v6 - IPv4 / IPv6 - Specify IPv4 / IPv6.
- v6 - IPv6 - Specify IPv6.

Default

v4v6

Name

preemptionCapability - Preemption capability

Syntax

--preemptionCapability <list-item>

Description

Preemption capability used for emergency service default bearers.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

preemptionVulnerability - Preemption vulnerability

Syntax

--preemptionVulnerability <list-item>

Description

Preemption vulnerability for emergency service default bearers.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

responseTime - Response time

Syntax

--responseTime <list-item>

Description

Response time parameter in the location request.

Class

Optional.

List

- Delay Tolerant - Specify response time of delay tolerant.
- Low Delay - Specify response time of low delay.

Default

Low Delay

Name

verticalAccuracy - Vertical accuracy

Syntax

--verticalAccuracy <integer>

Description

Vertical accuracy parameter in the location request.

Class

Optional.

Type / Range

Integer in the range of 0 to 127

Default

20

Name

verticalRequested - Vertical requested

Syntax

--verticalRequested <list-item>

Description

Vertical requested parameter in the location request.

Class

Optional.

List

- Requested - Specify vertical requested.
- Not Requested - Specify vertical not requested.

Default

Not Requested

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.107 cmm emmCauseCodeMapping

Configurable EMM causes mapping table. Syntax, description, parameters, and roles for cmm emmCauseCodeMapping command.

Syntax

Description

Configurable EMM causes mapping table.

Parameters

Name

emmCauseCodeMappingId - Configurable EMM causes mapping table ID

Syntax

--emmCauseCodeMappingId <integer>

Description

Configurable EMM causes mapping table ID - just type 1.

Class

Required key.

Type / Range

Integer in the range of 1 to 1

Name

intErrorBearerSetupFailed - EMM cause code for bearer setup failed

Syntax

--intErrorBearerSetupFailed <integer>

Description

EMM cause code to be sent if bearer setup failed.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

19

Name

intErrorCreateSessRespUserAuthFail - Create Session Response User Authentication Fail

Syntax

--intErrorCreateSessRespUserAuthFail <integer>

Description

EMM cause code sent to the UE upon receiving create session reject from SGW for attach request.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

17

Name

intErrorIpv4OnlyAllowed - EMM cause code for IPv4 only allowed

Syntax

--intErrorIpv4OnlyAllowed <integer>

Description

EMM cause code to be sent if IPv4 only allowed.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

19

Name

intErrorIpv6OnlyAllowed - EMM cause code for IPv6 only allowed

Syntax

--intErrorIpv6OnlyAllowed <integer>

Description

EMM cause code to be sent if IPv6 only allowed.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

19

Name

intErrorIpVerMismatchAtt - EMM cause code for IP version mismatch in attach

Syntax

--intErrorIpVerMismatchAtt <integer>

Description

EMM cause code to be sent if IP version mismatch in attach.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

19

Name

intErrorIpVerMismatchTau - EMM cause code for IP version mismatch in TAU

Syntax

--intErrorIpVerMismatchTau <integer>

Description

EMM cause code to be sent if IP version mismatch tau.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

17

Name

intErrorMandIeInc - EMM cause code for mandatory IE included

Syntax

--intErrorMandIeInc <integer>

Description

EMM cause code to be sent if mandatory IE included.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

19

Name

intErrorNoApnAllowedByHss - EMM cause code for no APN allowed by HSS

Syntax

--intErrorNoApnAllowedByHss <integer>

Description

EMM cause code to be sent if no apn allowed by HSS.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

19

Name

intErrorNoRespFromSgwAttPdn - EMM cause code for no resp from SGW attach PDN

Syntax

--intErrorNoRespFromSgwAttPdn <integer>

Description

EMM cause code to be sent if no response from SGW attach pdn.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

19

Name

intErrorNoRespFromSgwTau - EMM cause code for no resp from SGW TAU

Syntax

--intErrorNoRespFromSgwTau <integer>

Description

EMM cause code to be sent if no response from SGW tau.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

10

Name

intErrorOptIeInc - EMM cause code for optional IE included

Syntax

--intErrorOptIeInc <integer>

Description

EMM cause code to be sent if optional IE included.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

19

Name

intErrorServiceDenied - EMM cause code for servicedenied

Syntax

--intErrorServiceDenied <integer>

Description

EMM cause code to be sent if service denied.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

19

Name

intErrorSmcFailed - EMM cause code for SMC failed

Syntax

--intErrorSmcFailed <integer>

Description

EMM cause code to be sent if smc failed.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

3

Name

intErrorSystemFailure - EMM cause code for system failure

Syntax

--intErrorSystemFailure <integer>

Description

EMM cause code to be sent in system failure.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

19

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

4.108 cmm emmInfo

EMM Information. Syntax, description, parameters, and roles for `cmm emmInfo` command.

Syntax**Description**

This command is used to administer EMM information and configure parameters to be sent in NAS EMM INFORMATION message and also equivalent PLMNs to be sent in the NAS ATTACH ACCEPT and TAU ACCEPT messages.

Parameters**Name**

plmnName - Mobile network name

Syntax

--plmnName <string>

Description

Name that identifies mobile network (alternate key).

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

encodingName - Encoding name

Syntax

--encodingName <list-item>

Description

Indicates the encoding type to use for name fields.

Class

Optional.

List

- GSM default alphabet - Specifies GSM default alphabet encoding name.
- UCS2 - USC2 - Specifies USC2 encoding name.

Default

GSM default alphabet

Name

networkName - Network name

Syntax

--networkName <string>

Description

Unique network name.

Class

Optional.

Type / Range

The Network Name for the PLMN. The following characters in the GSM character set per 3GPP TS 23.038 Section 6.2.1.

Default

"

Name

networkShortName - Network short name

Syntax

--networkShortName <string>

Description

Unique short network name.

Class

Optional.

Type / Range

The Network Short Name for the PLMN. The following characters in the GSM character set per 3GPP TS 23.038 Section 6.2.1.

Default

"

Name

sendCountryInit - Send country init

Syntax

--sendCountryInit <list-item>

Description

Indicates whether or not the UE should display or not display the country initials.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sendNetworkName - Send network name

Syntax

--sendNetworkName <list-item>

Description

Indicates whether or not to send the network names.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sendOnAttach - Send On Attach

Syntax

```
--sendOnAttach <list-item>
```

Description

Defines whether the EMM information message is sent to the UE after the Attach procedure.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

sendOnInterMmeTau - Send On Inter MME TAU

Syntax

```
--sendOnInterMmeTau <list-item>
```

Description

Defines whether the EMM information message is sent to the UE after the Inter MME TAU procedure.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

sendOnIratTau - Send On IRAT TAU

Syntax

```
--sendOnIratTau <list-item>
```

Description

Defines whether the EMM information message is sent to the UE after the Inter RAT TAU procedure.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

sendOnNonPeriodicIntraMmeTau - Send On Non Periodic Intra MME TAU

Syntax

--sendOnNonPeriodicIntraMmeTau <list-item>

Description

Defines whether the EMM information message is sent to the UE after the Non Periodic Intra MME TAU procedure

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sendOnPeriodicTau - Send On Periodic TAU

Syntax

--sendOnPeriodicTau <list-item>

Description

Defines whether the EMM information message is sent to the UE after the Periodic TAU procedure.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sendOnSrsRecovery - Send On SRS Recovery

Syntax

--sendOnSrsRecovery <list-item>

Description

Defines whether the EMM information message is sent to the UE after UE data recovery with the help of the Session Restoration Server.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

sendOnTzChange - Send On Time Zone Change

Syntax

--sendOnTzChange <list-item>

Description

Defines whether the EMM information message is sent to the UE after Time Zone has changed after the last provided Network Identity and Time Zone delivery towards the UE.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

sendTzOffset - Send time zone offset

Syntax

--sendTzOffset <list-item>

Description

Indicates whether or not to send the time zone offset.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

4.109 cmm emmInfoOverrideProfile

Override of EMM Info for IMEI/SV. Syntax, description, parameters, and roles for cmm emmInfoOverrideProfile command.

Syntax**Description**

override information for TAC/SV based emmInfo

Parameters**Name**

name - Override of EMM Info by IMEI/SV

Syntax

--name <string>

Description

Override of EMM Info by IMEI/SV

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

encodingName - Encoding Name

Syntax

--encodingName <list-item>

Description

Indicates the encoding type to use for the name fields

Class

Optional.

List

- GSM default alphabet - Specifies GSM default alphabet encoding name.
- UCS2 - USC2 - Specifies USC2 encoding name.

Default

GSM default alphabet

Name

networkName - Network Name

Syntax

--networkName <string>

Description

Network Name

Restrictions

Digit string of length 0 or 3 to 64 with the allowed pattern of [0-9a-zA-Z\._&:]*

Class

Optional.

Type / Range

The Network Name for the PLMN. The following characters in the GSM character set per 3GPP TS 23.038 Section 6.2.1.

Default

""

Name

sendCountryInit - Send Country Initials

Syntax

--sendCountryInit <list-item>

Description

Indicates whether the UE should display the country initials

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

shortNetworkNameOverride - Short Network Name Override

Syntax

--shortNetworkNameOverride <string>

Description

Defines the network short name value that is to be used to activate a special icon in the UE

Class

Optional.

Type / Range

The Network Short Name for the PLMN. The following characters in the GSM character set per 3GPP TS 23.038 Section 6.2.1.

Default

""

Name

shortNetworkNameOverrideDeactivate - Short Network Name Override Deactivate

Syntax

--shortNetworkNameOverrideDeactivate <string>

Description

Defines the network short name value that is to be used to deactivate a special icon in the UE

Class

Optional.

Type / Range

The Network Short Name for the PLMN. The following characters in the GSM character set per 3GPP TS 23.038 Section 6.2.1.

Default

""

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.110 cmm emmInfoOverrideTacSv

EMM Info Override TAC SV. Syntax, description, parameters, and roles for cmm emmInfoOverrideTacSv command.

Syntax

Description

List of UE Type Approval Code (TAC) and Software Version (SV) that should have a specific EmmInfo profile instead of the system default.

Parameters

Name

imeiTac - IMEI TAC

Syntax

--imeiTac <digits>

Description

Type Approval Code (TAC) of the IMEI

Class

Required key.

Type / Range

Digit string of length 8 to 8

Name

svRange - SV Range

Syntax

--svRange <string>

Description

Software Version (SV) Range

Class

Required key.

Type / Range

Software Version Range, where each member of the range must be two digits (0..9), separated by a dash, and where the first range member must be less than or equal to the second range number.

Name

emmInfoOverrideProfileName - EMM Info Override Profile Name

Syntax

--emmInfoOverrideProfileName <string>

Description

EMM Info Override Profile to be used for this TAC/SV range

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\- _:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.111 cmm empsDscpMarking

eMPS DSCP Marking. Syntax, description, parameters, and roles for **cmm empsDscpMarking** command.

Syntax**Description**

Administer a list of derived DSCP code point to QCI-ARP mapping. The combination of a unique QCI-ARP pair defines the DSCP value derived from that pair. QCI may range from 1-9, 65-67, 69-70. ARP may range

from 1-15. Example: list administration can create individual entries such as (QCI = 1, ARP = 1), (QCI = 1, ARP = 2), and (QCI = 1, ARP = 3) where each may be marked with VOICE-ADMIT (DSCP44)

Parameters

Name

arp - ARP Priority Level

Syntax

--arp <integer>

Description

ARP (Allocation and Retention Priority) An integer ranging from 1 to 15, with 1 being the highest level of priority

Class

Required key.

Type / Range

Integer in the range of 1 to 15

Name

qci - QoS Class Identifier

Syntax

--qci <integer>

Description

QoS class identifier number.

Class

Required key.

Type / Range

Integer in the range of 1-9, 65-67, 69-70. Integer value represent the 3GPP QCI values, including mission critical value.

Name

dscpCode - DSCP Code

Syntax

--dscpCode <list-item>

Description

Differentiated Services Code Point (DSCP). A value utilized as a means of classifying and managing network traffic and of providing quality of service (QoS)

Class

Required on create.

List

- AF11 - Assured forwarding 11.
- AF12 - Assured forwarding 12.
- AF13 - Assured forwarding 13.
- AF21 - Assured forwarding 21.
- AF22 - Assured forwarding 22.
- AF23 - Assured forwarding 23.
- AF31 - Assured forwarding 31.
- AF32 - Assured forwarding 32.
- AF33 - Assured forwarding 33.
- AF41 - Assured forwarding 41.
- AF42 - Assured forwarding 42.
- AF43 - Assured forwarding 43.
- BE - BE.
- CS1 - Class selector 1.
- CS2 - Class selector 2.
- CS3 - Class selector 3.
- CS4 - Class selector 4.
- CS5 - Class selector 5.
- CS6 - Class selector 6.
- CS7 - Class selector 7.
- EF - Expedited forwarding.
- voiceAdmit - Voice Admit

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.112 cmm emsTrapNotification

EMS Trap Notification. Syntax, description, parameters, and roles for cmm emsTrapNotification command.

Syntax**Description**

This command is used to define a list of Element Management Systems for delivery of Northbound SNMP information and includes parameters for defining the EMS host, port, and port type, and SNMP user name, context name, row status, and user-based security level and model.

Parameters**Name**

emsHost - EMS host

Syntax

--emsHost <string>

Description

The IP address where the EMS is listening for notifications.

Class

Required key.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

emsPort - EMS port

Syntax

--emsPort <integer>

Description

The UDP port at which the EMS is listening for notifications.

Class

Required key.

Type / Range

Integer in the range of 1 to 65,535

Name

contextName - Context name

Syntax

--contextName <string>

Description

SNMP context name. this context name is same as in vacmaccessstable of VACM MIB RFC2575.

Class

Optional.

Type / Range

Digit string of length 0 to 64 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

NoAuth

Name

emsHostType - EMS host type

Syntax

--emsHostType <integer>

Description

The InetAddressType for the manager host. 0:unknown, 1:ipv4, 2:ipv6

Restrictions

Value of emsHostType is based on value of emsHost

Class

Read only.

Type / Range

Integer in the range of 0 to 2

Default

0

Name

rowStatus - Row status

Syntax

--rowStatus <integer>

Description

SNMP row status. used to manage insert/delete of SNMP conceptual row.

Class

Optional.

Type / Range

Integer in the range of 1 to 6

Default

4

Name

userName - User name

Syntax

--userName <string>

Description

SNMP user name.

Class

Optional.

Type / Range

Digit string of length 0 to 64 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

userSecLevel - User security level

Syntax

--userSecLevel <integer>

Description

SNMP user-based security level. 1:noAuthNoPriv, 2:authNoPriv, 3:authPriv.

Class

Optional.

Type / Range

Integer in the range of 1 to 3

Default

1

Name

userSecModel - User security model

Syntax

--userSecModel <integer>

Description

SNMP user-based security model.

Restrictions

CMM only supports User-based Security model(USM).

Class

Read only.

Type / Range

Integer in the range of 0 to 2,147,483,647

Default

3

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.113 cmm enb

eNB. Syntax, description, parameters, and roles for cmm enb command.

Syntax

Description

This command is used to administer eNB time zone information for a mobile network and includes parameters for defining the macro eNB ID and region time zone.

Parameters

Name

plmnName - Mobile network name

Syntax

--plmnName <string>

Description

Name that identifies the mobile network.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

macroEnbId - Macro eNB id

Syntax

--macroEnbId <integer>

Description

Macro eNB identifier as specified in TS 36.413 v8.6.1.

Class

Required key.

Type / Range

Integer in the range of 0 to 1,048,575

Name

timeZoneName - Time zone name

Syntax

--timeZoneName <list-item>

Description

Provides time zone names that correspond to the region selected.

Restrictions

See comments on TIME_ZONE type in CfgDataType.xsd.

Class

Optional.

List

- Africa/Abidjan - Timezone for africa/abidjan UTC offset of +00:00.
- Africa/Accra - Timezone for africa/accra UTC offset of +00:00.
- Africa/Addis_Ababa - Timezone for africa/addis_ababa UTC offset of +03:00.
- Africa/Algiers - Timezone for africa/algiers UTC offset of +01:00.
- Africa/Asmara - Timezone for africa/asmara UTC offset of +03:00.
- Africa/Asmera - Timezone for africa/asmera UTC offset of +03:00.
- Africa/Bamako - Timezone for africa/bamako UTC offset of +00:00.
- Africa/Bangui - Timezone for africa/bangui UTC offset of +01:00.
- Africa/Banjul - Timezone for africa/banjul UTC offset of +00:00.
- Africa/Bissau - Timezone for africa/bissau UTC offset of +00:00.

- Africa/Blantyre - Timezone for africa/blantyre UTC offset of +02:00.
- Africa/Brazzaville - Timezone for africa/brazzaville UTC offset of +01:00.
- Africa/Bujumbura - Timezone for africa/bujumbura UTC offset of +02:00.
- Africa/Cairo - Timezone for africa/cairo UTC offset of +02:00.
- Africa/Casablanca - Timezone for africa/casablanca UTC offset of +00:00.
- Africa/Ceuta - Timezone for africa/ceuta UTC offset of +01:00.
- Africa/Conakry - Timezone for africa/conakry UTC offset of +00:00.
- Africa/Dakar - Timezone for africa/dakar UTC offset of +00:00.
- Africa/Dar_es_Salaam - Timezone for africa/dar_es_salaam UTC offset of +03:00.
- Africa/Djibouti - Timezone for africa/djibouti UTC offset of +03:00.
- Africa/Douala - Timezone for africa/douala UTC offset of +01:00.
- Africa/El_Aaiun - Timezone for africa/el_aaiun UTC offset of +00:00.
- Africa/Freetown - Timezone for africa/freetown UTC offset of +00:00.
- Africa/Gaborone - Timezone for africa/gaborone UTC offset of +02:00.
- Africa/Harare - Timezone for africa/harare UTC offset of +02:00.
- Africa/Johannesburg - Timezone for africa/johannesburg UTC offset of +02:00.
- Africa/Juba - Timezone for africa/juba UTC offset of +03:00.
- Africa/Kampala - Timezone for africa/kampala UTC offset of +03:00.
- Africa/Khartoum - Timezone for africa/khartoum UTC offset of +03:00.
- Africa/Kigali - Timezone for africa/kigali UTC offset of +02:00.
- Africa/Kinshasa - Timezone for africa/kinshasa UTC offset of +01:00.
- Africa/Lagos - Timezone for africa/lagos UTC offset of +01:00.
- Africa/Libreville - Timezone for africa/libreville UTC offset of +01:00.
- Africa/Lome - Timezone for africa/lome UTC offset of +00:00.
- Africa/Luanda - Timezone for africa/luanda UTC offset of +01:00.
- Africa/Lubumbashi - Timezone for africa/lubumbashi UTC offset of +02:00.
- Africa/Lusaka - Timezone for africa/lusaka UTC offset of +02:00.
- Africa/Malabo - Timezone for africa/malabo UTC offset of +01:00.
- Africa/Maputo - Timezone for africa/maputo UTC offset of +02:00.
- Africa/Maseru - Timezone for africa/maseru UTC offset of +02:00.
- Africa/Mbabane - Timezone for africa/mbabane UTC offset of +02:00.

- Africa/Mogadishu - Timezone for africa/mogadishu UTC offset of +03:00.
- Africa/Monrovia - Timezone for africa/monrovia UTC offset of +00:00.
- Africa/Nairobi - Timezone for africa/nairobi UTC offset of +03:00.
- Africa/Ndjamena - Timezone for africa/ndjamena UTC offset of +01:00.
- Africa/Niamey - Timezone for africa/niamey UTC offset of +01:00.
- Africa/Nouakchott - Timezone for africa/nouakchott UTC offset of +00:00.
- Africa/Ouagadougou - Timezone for africa/ouagadougou UTC offset of +00:00.
- Africa/Porto-Novo - Timezone for africa/porto-novo UTC offset of +01:00.
- Africa/Sao_Tome - Timezone for africa/sao_tome UTC offset of +00:00.
- Africa/Timbuktu - Timezone for africa/timbuktu UTC offset of +00:00.
- Africa/Tripoli - Timezone for africa/tripoli UTC offset of +02:00.
- Africa/Tunis - Timezone for africa/tunis UTC offset of +01:00.
- Africa/Windhoek - Timezone for africa/windhoek UTC offset of +02:00.
- America/Adak - Timezone for america/adak UTC offset of -10:00.
- America/Anchorage - Timezone for america/anchorage UTC offset of -09:00.
- America/Anguilla - Timezone for america/anguilla UTC offset of -04:00.
- America/Antigua - Timezone for america/antigua UTC offset of -04:00.
- America/Araguaina - Timezone for america/araguaina UTC offset of -03:00.
- America/Argentina/Buenos_Aires - Timezone for america/argentina/buenos_aires UTC offset of -03:00.
- America/Argentina/Catamarca - Timezone for america/argentina/catamarca UTC offset of -03:00.
- America/Argentina/ComodRivadavia - Timezone for america/argentina/comodrivadavia UTC offset of -03:00.
- America/Argentina/Cordoba - Timezone for america/argentina/cordoba UTC offset of -03:00.
- America/Argentina/Jujuy - Timezone for america/argentina/jujuy UTC offset of -03:00.
- America/Argentina/La_Rioja - Timezone for america/argentina/la_rioja UTC offset of -03:00.
- America/Argentina/Mendoza - Timezone for america/argentina/mendoza UTC offset of -03:00.
- America/Argentina/Rio_Gallegos - Timezone for america/argentina/rio_gallegos UTC offset of -03:00.

- America/Argentina/Salta - Timezone for america/argentina/salta UTC offset of -03:00.
- America/Argentina/San_Juan - Timezone for america/argentina/san_juan UTC offset of -03:00.
- America/Argentina/San_Luis - Timezone for america/argentina/san_luis UTC offset of -03:00.
- America/Argentina/Tucuman - Timezone for america/argentina/tucuman UTC offset of -03:00.
- America/Argentina/Ushuaia - Timezone for america/argentina/ushuaia UTC offset of -03:00.
- America/Aruba - Timezone for america/aruba UTC offset of -04:00.
- America/Asuncion - Timezone for america/asuncion UTC offset of -03:00.
- America/Atikokan - Timezone for america/atikokan UTC offset of -05:00.
- America/Atka - Timezone for america/atka UTC offset of -10:00.
- America/Bahia - Timezone for america/bahia UTC offset of -03:00.
- America/Bahia_Banderas - Timezone for america/bahia_banderas UTC offset of -06:00.
- America/Barbados - Timezone for america/barbados UTC offset of -04:00.
- America/Belem - Timezone for america/belem UTC offset of -03:00.
- America/Belize - Timezone for america/belize UTC offset of -06:00.
- America/Blanc-Sablon - Timezone for america/blanc-sablon UTC offset of -04:00.
- America/Boa_Vista - Timezone for america/boa_vista UTC offset of -04:00.
- America/Bogota - Timezone for america/bogota UTC offset of -05:00.
- America/Boise - Timezone for america/boise UTC offset of -07:00.
- America/Buenos_Aires - Timezone for america/buenos_aires UTC offset of -03:00.
- America/Cambridge_Bay - Timezone for america/cambridge_bay UTC offset of -07:00.
- America/Campo_Grande - Timezone for america/campo_grande UTC offset of -03:00.
- America/Cancun - Timezone for america/cancun UTC offset of -05:00.
- America/Caracas - Timezone for america/caracas UTC offset of -04:00.
- America/Catamarca - Timezone for america/catamarca UTC offset of -03:00.
- America/Cayenne - Timezone for america/cayenne UTC offset of -03:00.
- America/Cayman - Timezone for america/cayman UTC offset of -05:00.
- America/Chicago - Timezone for america/chicago UTC offset of -06:00.
- America/Chihuahua - Timezone for america/chihuahua UTC offset of -07:00.

- America/Coral_Harbour - Timezone for america/coral_harbour UTC offset of -05:00.
- America/Cordoba - Timezone for america/cordoba UTC offset of -03:00.
- America/Costa_Rica - Timezone for america/costa_rica UTC offset of -06:00.
- America/Creston - Timezone for america/creston UTC offset of -07:00.
- America/Cuiaba - Timezone for america/cuiaba UTC offset of -03:00.
- America/Curacao - Timezone for america/curacao UTC offset of -04:00.
- America/Danmarkshavn - Timezone for america/danmarkshavn UTC offset of +00:00.
- America/Dawson - Timezone for america/dawson UTC offset of -08:00.
- America/Dawson_Creek - Timezone for america/dawson_creek UTC offset of -07:00.
- America/Denver - Timezone for america/denver UTC offset of -07:00.
- America/Detroit - Timezone for america/detroit UTC offset of -05:00.
- America/Dominica - Timezone for america/dominica UTC offset of -04:00.
- America/Edmonton - Timezone for america/edmonton UTC offset of -07:00.
- America/Eirunepe - Timezone for america/eirunepe UTC offset of -05:00.
- America/El_Salvador - Timezone for america/el_salvador UTC offset of -06:00.
- America/Ensenada - Timezone for america/ensenada UTC offset of -08:00.
- America/Fort_Nelson - Timezone for america/fort_nelson UTC offset of -07:00.
- America/Fort_Wayne - Timezone for america/fort_wayne UTC offset of -05:00.
- America/Fortaleza - Timezone for america/fortaleza UTC offset of -03:00.
- America/Glace_Bay - Timezone for america/glance_bay UTC offset of -04:00.
- America/Godthab - Timezone for america/godthab UTC offset of -03:00.
- America/Goose_Bay - Timezone for america/goose_bay UTC offset of -04:00.
- America/Grand_Turk - Timezone for america/grand_turk UTC offset of -04:00.
- America/Grenada - Timezone for america/grenada UTC offset of -04:00.
- America/Guadeloupe - Timezone for america/guadeloupe UTC offset of -04:00.
- America/Guatemala - Timezone for america/guatemala UTC offset of -06:00.
- America/Guayaquil - Timezone for america/guayaquil UTC offset of -05:00.
- America/Guyana - Timezone for america/guyana UTC offset of -04:00.
- America/Halifax - Timezone for america/halifax UTC offset of -04:00.
- America/Havana - Timezone for america/havana UTC offset of -05:00.
- America/Hermosillo - Timezone for america/hermosillo UTC offset of -07:00.

- America/Indiana/Indianapolis - Timezone for america/indiana/indianapolis UTC offset of -05:00.
- America/Indiana/Knox - Timezone for america/indiana/knox UTC offset of -06:00.
- America/Indiana/Marengo - Timezone for america/indiana/marengo UTC offset of -05:00.
- America/Indiana/Petersburg - Timezone for america/indiana/petersburg UTC offset of -05:00.
- America/Indiana/Tell_City - Timezone for america/indiana/tell_city UTC offset of -06:00.
- America/Indiana/Vevay - Timezone for america/indiana/vevay UTC offset of -05:00.
- America/Indiana/Vincennes - Timezone for america/indiana/vincennes UTC offset of -05:00.
- America/Indiana/Winamac - Timezone for america/indiana/winamac UTC offset of -05:00.
- America/Indianapolis - Timezone for america/indianapolis UTC offset of -05:00.
- America/Inuvik - Timezone for america/inuvik UTC offset of -07:00.
- America/Iqaluit - Timezone for america/iquait UTC offset of -05:00.
- America/Jamaica - Timezone for america/jamaica UTC offset of -05:00.
- America/Jujuy - Timezone for america/jujuy UTC offset of -03:00.
- America/Juneau - Timezone for america/juneau UTC offset of -09:00.
- America/Kentucky/Louisville - Timezone for america/kentucky/louisville UTC offset of -05:00.
- America/Kentucky/Monticello - Timezone for america/kentucky/monticello UTC offset of -05:00.
- America/Knox_IN - Timezone for america/knox_in UTC offset of -06:00.
- America/Kralendijk - Timezone for america/kralendijk UTC offset of -04:00.
- America/La_Paz - Timezone for america/la_paz UTC offset of -04:00.
- America/Lima - Timezone for america/lima UTC offset of -05:00.
- America/Los_Angeles - Timezone for america/los_angeles UTC offset of -08:00.
- America/Louisville - Timezone for america/louisville UTC offset of -05:00.
- America/Lower_Princes - Timezone for america/lower_princes UTC offset of -04:00.
- America/Maceio - Timezone for america/maceio UTC offset of -03:00.
- America/Managua - Timezone for america/managua UTC offset of -06:00.
- America/Manaus - Timezone for america/manaus UTC offset of -04:00.

- America/Marigot - Timezone for america/marigot UTC offset of -04:00.
- America/Martinique - Timezone for america/martinique UTC offset of -04:00.
- America/Matamoros - Timezone for america/matamoros UTC offset of -06:00.
- America/Mazatlan - Timezone for america/mazatlan UTC offset of -07:00.
- America/Mendoza - Timezone for america/mendoza UTC offset of -03:00.
- America/Menominee - Timezone for america/menominee UTC offset of -06:00.
- America/Merida - Timezone for america/merida UTC offset of -06:00.
- America/Metlakatla - Timezone for america/metlakatla UTC offset of -09:00.
- America/Mexico_City - Timezone for america/mexico_city UTC offset of -06:00.
- America/Miquelon - Timezone for america/miquelon UTC offset of -03:00.
- America/Moncton - Timezone for america/moncton UTC offset of -04:00.
- America/Monterrey - Timezone for america/monterrey UTC offset of -06:00.
- America/Montevideo - Timezone for america/montevideo UTC offset of -03:00.
- America/Montreal - Timezone for america/montreal UTC offset of -05:00.
- America/Montserrat - Timezone for america/montserrat UTC offset of -04:00.
- America/Nassau - Timezone for america/nassau UTC offset of -05:00.
- America/New_York - Timezone for america/new_york UTC offset of -05:00.
- America/Nipigon - Timezone for america/nipigon UTC offset of -05:00.
- America/Nome - Timezone for america/nome UTC offset of -09:00.
- America/Noronha - Timezone for america/noronha UTC offset of -02:00.
- America/North_Dakota/Beulah - Timezone for america/north_dakota/beulah UTC offset of -06:00.
- America/North_Dakota/Center - Timezone for america/north_dakota/center UTC offset of -06:00.
- America/North_Dakota/New_Salem - Timezone for america/north_dakota/new_salem UTC offset of -06:00.
- America/Ojinaga - Timezone for america/ojinaga UTC offset of -07:00.
- America/Panama - Timezone for america/panama UTC offset of -05:00.
- America/Pangnirtung - Timezone for america/pangnirtung UTC offset of -05:00.
- America/Paramaribo - Timezone for america/paramaribo UTC offset of -03:00.
- America/Phoenix - Timezone for america/phoenix UTC offset of -07:00.
- America/Port-au-Prince - Timezone for america/port-au-prince UTC offset of -05:00.

- America/Port_of_Spain - Timezone for america/port_of_spain UTC offset of -04:00.
- America/Porto_Acre - Timezone for america/porto_acre UTC offset of -05:00.
- America/Porto_Velho - Timezone for america/porto_velho UTC offset of -04:00.
- America/Puerto_Rico - Timezone for america/puerto_rico UTC offset of -04:00.
- America/Rainy_River - Timezone for america/rainy_river UTC offset of -06:00.
- America/Rankin_Inlet - Timezone for america/rankin_inlet UTC offset of -06:00.
- America/Recife - Timezone for america/recife UTC offset of -03:00.
- America/Regina - Timezone for america/regina UTC offset of -06:00.
- America/Resolute - Timezone for america/resolute UTC offset of -06:00.
- America/Rio_Branco - Timezone for america/rio_branco UTC offset of -05:00.
- America/Rosario - Timezone for america/rosario UTC offset of -03:00.
- America/Santa_Isabel - Timezone for america/santa_isabel UTC offset of -08:00.
- America/Santarem - Timezone for america/santarem UTC offset of -03:00.
- America/Santiago - Timezone for america/santiago UTC offset of -03:00.
- America/Santo_Domingo - Timezone for america/santo_domingo UTC offset of -04:00.
- America/Sao_Paulo - Timezone for america/sao_paulo UTC offset of -02:00.
- America/Scoresbysund - Timezone for america/scoresbysund UTC offset of -01:00.
- America/Shiprock - Timezone for america/shiprock UTC offset of -07:00.
- America/Sitka - Timezone for america/sitka UTC offset of -09:00.
- America/St_Barthelemy - Timezone for america/st_barthelemy UTC offset of -04:00.
- America/St_Johns - Timezone for america/st_johns UTC offset of -03:30.
- America/St_Kitts - Timezone for america/st_kitts UTC offset of -04:00.
- America/St_Lucia - Timezone for america/st_lucia UTC offset of -04:00.
- America/St_Thomas - Timezone for america/st_thomas UTC offset of -04:00.
- America/St_Vincent - Timezone for america/st_vincent UTC offset of -04:00.
- America/Swift_Current - Timezone for america/swift_current UTC offset of -06:00.
- America/Tegucigalpa - Timezone for america/tegucigalpa UTC offset of -06:00.
- America/Thule - Timezone for america/thule UTC offset of -04:00.
- America/Thunder_Bay - Timezone for america/thunder_bay UTC offset of -05:00.
- America/Tijuana - Timezone for america/tijuana UTC offset of -08:00.
- America/Toronto - Timezone for america/toronto UTC offset of -05:00.

- America/Tortola - Timezone for america/tortola UTC offset of -04:00.
- America/Vancouver - Timezone for america/vancouver UTC offset of -08:00.
- America/Virgin - Timezone for america/virgin UTC offset of -04:00.
- America/Whitehorse - Timezone for america/whitehorse UTC offset of -08:00.
- America/Winnipeg - Timezone for america/winnipeg UTC offset of -06:00.
- America/Yakutat - Timezone for america/yakutat UTC offset of -09:00.
- America/Yellowknife - Timezone for america/yellowknife UTC offset of -07:00.
- Antarctica/Casey - Timezone for antarctica/casey UTC offset of +11:00.
- Antarctica/Davis - Timezone for antarctica/davis UTC offset of +07:00.
- Antarctica/DumontDUrville - Timezone for antarctica/dumontdurville UTC offset of +10:00.
- Antarctica/Macquarie - Timezone for antarctica/macquarie UTC offset of +11:00.
- Antarctica/Mawson - Timezone for antarctica/mawson UTC offset of +05:00.
- Antarctica/McMurdo - Timezone for antarctica/mcmurdo UTC offset of +13:00.
- Antarctica/Palmer - Timezone for antarctica/palmer UTC offset of -03:00.
- Antarctica/Rothera - Timezone for antarctica/rothera UTC offset of -03:00.
- Antarctica/South_Pole - Timezone for antarctica/south_pole UTC offset of +13:00.
- Antarctica/Syowa - Timezone for antarctica/syowa UTC offset of +03:00.
- Antarctica/Troll - Timezone for antarctica/troll UTC offset of +00:00.
- Antarctica/Vostok - Timezone for antarctica/vostok UTC offset of +06:00.
- Arctic/Longyearbyen - Timezone for arctic/longyearbyen UTC offset of +01:00.
- Asia/Aden - Timezone for asia/aden UTC offset of +03:00.
- Asia/Almaty - Timezone for asia/almaty UTC offset of +06:00.
- Asia/Amman - Timezone for asia/amman UTC offset of +02:00.
- Asia/Anadyr - Timezone for asia/anadyr UTC offset of +12:00.
- Asia/Aqtau - Timezone for asia/aqtau UTC offset of +05:00.
- Asia/Aqtobe - Timezone for asia/aqtobe UTC offset of +05:00.
- Asia/Ashgabat - Timezone for asia/ashgabat UTC offset of +05:00.
- Asia/Ashkhabad - Timezone for asia/ashkhabad UTC offset of +05:00.
- Asia/Baghdad - Timezone for asia/baghdad UTC offset of +03:00.
- Asia/Bahrain - Timezone for asia/bahrain UTC offset of +03:00.

- Asia/Baku - Timezone for asia/baku UTC offset of +04:00.
- Asia/Bangkok - Timezone for asia/bangkok UTC offset of +07:00.
- Asia/Barnaul - Timezone for asia/barnaul UTC offset of +07:00.
- Asia/Beirut - Timezone for asia/beirut UTC offset of +02:00.
- Asia/Bishkek - Timezone for asia/bishkek UTC offset of +06:00.
- Asia/Brunei - Timezone for asia/brunei UTC offset of +08:00.
- Asia/Calcutta - Timezone for asia/calcutta UTC offset of +05:30.
- Asia/Chita - Timezone for asia/chita UTC offset of +09:00.
- Asia/Choibalsan - Timezone for asia/choibalsan UTC offset of +08:00.
- Asia/Chongqing - Timezone for asia/chongqing UTC offset of +08:00.
- Asia/Chungking - Timezone for asia/chungking UTC offset of +08:00.
- Asia/Colombo - Timezone for asia/colombo UTC offset of +05:30.
- Asia/Dacca - Timezone for asia/dacca UTC offset of +06:00.
- Asia/Damascus - Timezone for asia/damascus UTC offset of +02:00.
- Asia/Dhaka - Timezone for asia/dhaka UTC offset of +06:00.
- Asia/Dili - Timezone for asia/dili UTC offset of +09:00.
- Asia/Dubai - Timezone for asia/dubai UTC offset of +04:00.
- Asia/Dushanbe - Timezone for asia/dushanbe UTC offset of +05:00.
- Asia/Gaza - Timezone for asia/gaza UTC offset of +02:00.
- Asia/Harbin - Timezone for asia/harbin UTC offset of +08:00.
- Asia/Hebron - Timezone for asia/hebron UTC offset of +02:00.
- Asia/Ho_Chi_Minh - Timezone for asia/ho_chi_minh UTC offset of +07:00.
- Asia/Hong_Kong - Timezone for asia/hong_kong UTC offset of +08:00.
- Asia/Hovd - Timezone for asia/hovd UTC offset of +07:00.
- Asia/Irkutsk - Timezone for asia/irkutsk UTC offset of +08:00.
- Asia/Istanbul - Timezone for asia/istanbul UTC offset of +03:00.
- Asia/Jakarta - Timezone for asia/jakarta UTC offset of +07:00.
- Asia/Jayapura - Timezone for asia/jayapura UTC offset of +09:00.
- Asia/Jerusalem - Timezone for asia/jerusalem UTC offset of +02:00.
- Asia/Kabul - Timezone for asia/kabul UTC offset of +04:30.
- Asia/Kamchatka - Timezone for asia/kamchatka UTC offset of +12:00.

- Asia/Karachi - Timezone for asia/karachi UTC offset of +05:00.
- Asia/Kashgar - Timezone for asia/kashgar UTC offset of +06:00.
- Asia/Kathmandu - Timezone for asia/kathmandu UTC offset of +05:45.
- Asia/Katmandu - Timezone for asia/katmandu UTC offset of +05:45.
- Asia/Khandyga - Timezone for asia/khandyga UTC offset of +09:00.
- Asia/Kolkata - Timezone for asia/kolkata UTC offset of +05:30.
- Asia/Krasnoyarsk - Timezone for asia/krasnoyarsk UTC offset of +07:00.
- Asia/Kuala_Lumpur - Timezone for asia/kuala_lumpur UTC offset of +08:00.
- Asia/Kuching - Timezone for asia/kuching UTC offset of +08:00.
- Asia/Kuwait - Timezone for asia/kuwait UTC offset of +03:00.
- Asia/Macao - Timezone for asia/macao UTC offset of +08:00.
- Asia/Macau - Timezone for asia/macau UTC offset of +08:00.
- Asia/Magadan - Timezone for asia/magadan UTC offset of +11:00.
- Asia/Makassar - Timezone for asia/makassar UTC offset of +08:00.
- Asia/Manila - Timezone for asia/manila UTC offset of +08:00.
- Asia/Muscat - Timezone for asia/muscat UTC offset of +04:00.
- Asia/Nicosia - Timezone for asia/nicosia UTC offset of +02:00.
- Asia/Novokuznetsk - Timezone for asia/novokuznetsk UTC offset of +07:00.
- Asia/Novosibirsk - Timezone for asia/novosibirsk UTC offset of +07:00.
- Asia/Omsk - Timezone for asia/omsk UTC offset of +06:00.
- Asia/Oral - Timezone for asia/oral UTC offset of +05:00.
- Asia/Phnom_Penh - Timezone for asia/phnom_penh UTC offset of +07:00.
- Asia/Pontianak - Timezone for asia/pontianak UTC offset of +07:00.
- Asia/Pyongyang - Timezone for asia/pyongyang UTC offset of +08:30.
- Asia/Qatar - Timezone for asia/qatar UTC offset of +03:00.
- Asia/Qyzylorda - Timezone for asia/qyzylorda UTC offset of +06:00.
- Asia/Rangoon - Timezone for asia/rangoon UTC offset of +06:30.
- Asia/Riyadh - Timezone for asia/riyadh UTC offset of +03:00.
- Asia/Saigon - Timezone for asia/saigon UTC offset of +07:00.
- Asia/Sakhalin - Timezone for asia/sakhalin UTC offset of +11:00.
- Asia/Samarkand - Timezone for asia/samarkand UTC offset of +05:00.

- Asia/Seoul - Timezone for asia/seoul UTC offset of +09:00.
- Asia/Shanghai - Timezone for asia/shanghai UTC offset of +08:00.
- Asia/Singapore - Timezone for asia/singapore UTC offset of +08:00.
- Asia/Srednekolymsk - Timezone for asia/srednekolymsk UTC offset of +11:00.
- Asia/Taipei - Timezone for asia/taipei UTC offset of +08:00.
- Asia/Tashkent - Timezone for asia/tashkent UTC offset of +05:00.
- Asia/Tbilisi - Timezone for asia/tbilisi UTC offset of +04:00.
- Asia/Tehran - Timezone for asia/tehran UTC offset of +03:30.
- Asia/Tel_Aviv - Timezone for asia/tel_aviv UTC offset of +02:00.
- Asia/Thimbu - Timezone for asia/thimbu UTC offset of +06:00.
- Asia/Thimphu - Timezone for asia/thimphu UTC offset of +06:00.
- Asia/Tokyo - Timezone for asia/tokyo UTC offset of +09:00.
- Asia/Tomsk - Timezone for asia/tomsk UTC offset of +07:00.
- Asia/Ujung_Pandang - Timezone for asia/ujung_pandang UTC offset of +08:00.
- Asia/Ulaanbaatar - Timezone for asia/ulaanbaatar UTC offset of +08:00.
- Asia/Ulan_Bator - Timezone for asia/ulan_bator UTC offset of +08:00.
- Asia/Urumqi - Timezone for asia/urumqi UTC offset of +06:00.
- Asia/Ust-Nera - Timezone for asia/ust-nera UTC offset of +10:00.
- Asia/Vientiane - Timezone for asia/vientiane UTC offset of +07:00.
- Asia/Vladivostok - Timezone for asia/vladivostok UTC offset of +10:00.
- Asia/Yakutsk - Timezone for asia/yakutsk UTC offset of +09:00.
- Asia/Yekaterinburg - Timezone for asia/yekaterinburg UTC offset of +05:00.
- Asia/Yerevan - Timezone for asia/yerevan UTC offset of +04:00.
- Atlantic/Azores - Timezone for atlantic/azores UTC offset of -01:00.
- Atlantic/Bermuda - Timezone for atlantic/bermuda UTC offset of -04:00.
- Atlantic/Canary - Timezone for atlantic/canary UTC offset of +00:00.
- Atlantic/Cape_Verde - Timezone for atlantic/cape_verde UTC offset of -01:00.
- Atlantic/Faeroe - Timezone for atlantic/faeroe UTC offset of +00:00.
- Atlantic/Faroe - Timezone for atlantic/faroe UTC offset of +00:00.
- Atlantic/Jan_Mayen - Timezone for atlantic/jan_mayen UTC offset of +01:00.
- Atlantic/Madeira - Timezone for atlantic/madeira UTC offset of +00:00.

- Atlantic/Reykjavik - Timezone for atlantic/reykjavik UTC offset of +00:00.
- Atlantic/South_Georgia - Timezone for atlantic/south_georgia UTC offset of -02:00.
- Atlantic/St_Helena - Timezone for atlantic/st_helena UTC offset of +00:00.
- Atlantic/Stanley - Timezone for atlantic/stanley UTC offset of -03:00.
- Australia/ACT - Timezone for australia/act UTC offset of +11:00.
- Australia/Adelaide - Timezone for australia/adelaide UTC offset of +10:30.
- Australia/Brisbane - Timezone for australia/brisbane UTC offset of +10:00.
- Australia/Broken_Hill - Timezone for australia/broken_hill UTC offset of +10:30.
- Australia/Canberra - Timezone for australia/canberra UTC offset of +11:00.
- Australia/Currie - Timezone for australia/currie UTC offset of +11:00.
- Australia/Darwin - Timezone for australia/darwin UTC offset of +09:30.
- Australia/Eucla - Timezone for australia/eucla UTC offset of +08:45.
- Australia/Hobart - Timezone for australia/hobart UTC offset of +11:00.
- Australia/LHI - Timezone for australia/lhi UTC offset of +11:00.
- Australia/Lindeman - Timezone for australia/lindeman UTC offset of +10:00.
- Australia/Lord_Howe - Timezone for australia/lord_howe UTC offset of +11:00.
- Australia/Melbourne - Timezone for australia/melbourne UTC offset of +11:00.
- Australia/NSW - Timezone for australia/nsw UTC offset of +11:00.
- Australia/North - Timezone for australia/north UTC offset of +09:30.
- Australia/Perth - Timezone for australia/perth UTC offset of +08:00.
- Australia/Queensland - Timezone for australia/queensland UTC offset of +10:00.
- Australia/South - Timezone for australia/south UTC offset of +10:30.
- Australia/Sydney - Timezone for australia/sydney UTC offset of +11:00.
- Australia/Tasmania - Timezone for australia/tasmania UTC offset of +11:00.
- Australia/Victoria - Timezone for australia/victoria UTC offset of +11:00.
- Australia/West - Timezone for australia/west UTC offset of +08:00.
- Australia/Yancowinna - Timezone for australia/yancowinna UTC offset of +10:30.
- CET - Timezone for CET UTC offset of +01:00.
- CST6CDT - Timezone for CST6CDT UTC offset of -06:00.
- Canada/Atlantic - Timezone for canada/atlantic UTC offset of -04:00.
- Canada/Central - Timezone for canada/central UTC offset of -06:00.

- Canada/East-Saskatchewan - Timezone for canada/east-saskatchewan UTC offset of -06:00.
- Canada/Eastern - Timezone for canada/eastern UTC offset of -05:00.
- Canada/Mountain - Timezone for canada/mountain UTC offset of -07:00.
- Canada/Newfoundland - Timezone for canada/newfoundland UTC offset of -03:30.
- Canada/Pacific - Timezone for canada/pacific UTC offset of -08:00.
- Canada/Saskatchewan - Timezone for canada/saskatchewan UTC offset of -06:00.
- Canada/Yukon - Timezone for canada/yukon UTC offset of -08:00.
- Chile/Continental - Timezone for chile/continental UTC offset of -03:00.
- Chile/EasterIsland - Timezone for chile/easterisland UTC offset of -05:00.
- Cuba - Timezone for cuba UTC offset of -05:00.
- Default - Timezone for system UTC (OS time zone)
- EET - Timezone for EET UTC offset of +02:00.
- EST - Timezone for EST UTC offset of -05:00.
- EST5EDT - Timezone for EST5EDT UTC offset of -05:00.
- Egypt - Timezone for egypt UTC offset of +02:00.
- Eire - Timezone for eire UTC offset of +00:00.
- Etc/Greenwich - Timezone for etc/greenwich UTC offset of +00:00.
- Etc/UCT - Timezone for etc/uct UTC offset of +00:00.
- Etc/UTC - Timezone for etc/utc UTC offset of +00:00.
- Etc/Universal - Timezone for etc/universal UTC offset of +00:00.
- Etc/Zulu - Timezone for etc/zulu UTC offset of +00:00.
- Europe/Amsterdam - Timezone for europe/amsterdam UTC offset of +01:00.
- Europe/Andorra - Timezone for europe/andorra UTC offset of +01:00.
- Europe/Astrakhan - Timezone for europe/astrakhan UTC offset of +04:00.
- Europe/Athens - Timezone for europe/athens UTC offset of +02:00.
- Europe/Belfast - Timezone for europe/belfast UTC offset of +00:00.
- Europe/Belgrade - Timezone for europe/belgrade UTC offset of +01:00.
- Europe/Berlin - Timezone for europe/berlin UTC offset of +01:00.
- Europe/Bratislava - Timezone for europe/bratislava UTC offset of +01:00.
- Europe/Brussels - Timezone for europe/brussels UTC offset of +01:00.

- Europe/Bucharest - Timezone for europe/bucharest UTC offset of +02:00.
- Europe/Budapest - Timezone for europe/budapest UTC offset of +01:00.
- Europe/Busingen - Timezone for europe/busingen UTC offset of +01:00.
- Europe/Chisinau - Timezone for europe/chisinau UTC offset of +02:00.
- Europe/Copenhagen - Timezone for europe/copenhagen UTC offset of +01:00.
- Europe/Dublin - Timezone for europe/dublin UTC offset of +00:00.
- Europe/Gibraltar - Timezone for europe/gibraltar UTC offset of +01:00.
- Europe/Guernsey - Timezone for europe/guernsey UTC offset of +00:00.
- Europe/Helsinki - Timezone for europe/helsinki UTC offset of +02:00.
- Europe/Isle_of_Man - Timezone for europe/isle_of_man UTC offset of +00:00.
- Europe/Istanbul - Timezone for europe/istanbul UTC offset of +03:00.
- Europe/Jersey - Timezone for europe/jersey UTC offset of +00:00.
- Europe/Kaliningrad - Timezone for europe/kaliningrad UTC offset of +02:00.
- Europe/Kiev - Timezone for europe/kyiv UTC offset of +02:00.
- Europe/Kirov - Timezone for europe/kirov UTC offset of +03:00.
- Europe/Lisbon - Timezone for europe/lisbon UTC offset of +00:00.
- Europe/Ljubljana - Timezone for europe/ljubljana UTC offset of +01:00.
- Europe/London - Timezone for europe/london UTC offset of +00:00.
- Europe/Luxembourg - Timezone for europe/luxembourg UTC offset of +01:00.
- Europe/Madrid - Timezone for europe/madrid UTC offset of +01:00.
- Europe/Malta - Timezone for europe/malta UTC offset of +01:00.
- Europe/Mariehamn - Timezone for europe/mariehamn UTC offset of +02:00.
- Europe/Minsk - Timezone for europe/minsk UTC offset of +03:00.
- Europe/Monaco - Timezone for europe/monaco UTC offset of +01:00.
- Europe/Moscow - Timezone for europe/moscow UTC offset of +03:00.
- Europe/Nicosia - Timezone for europe/nicosia UTC offset of +02:00.
- Europe/Oslo - Timezone for europe/oslo UTC offset of +01:00.
- Europe/Paris - Timezone for europe/paris UTC offset of +01:00.
- Europe/Podgorica - Timezone for europe/podgorica UTC offset of +01:00.
- Europe/Prague - Timezone for europe/prague UTC offset of +01:00.
- Europe/Riga - Timezone for europe/riga UTC offset of +02:00.

- Europe/Rome - Timezone for europe/rome UTC offset of +01:00.
- Europe/Samara - Timezone for europe/samara UTC offset of +04:00.
- Europe/San_Marino - Timezone for europe/san_marino UTC offset of +01:00.
- Europe/Sarajevo - Timezone for europe/sarajevo UTC offset of +01:00.
- Europe/Simferopol - Timezone for europe/simferopol UTC offset of +03:00.
- Europe/Skopje - Timezone for europe/skopje UTC offset of +01:00.
- Europe/Sofia - Timezone for europe/sofia UTC offset of +02:00.
- Europe/Stockholm - Timezone for europe/stockholm UTC offset of +01:00.
- Europe/Tallinn - Timezone for europe/tallinn UTC offset of +02:00.
- Europe/Tirane - Timezone for europe/tirane UTC offset of +01:00.
- Europe/Tiraspol - Timezone for europe/tiraspol UTC offset of +02:00.
- Europe/Ulyanovsk - Timezone for europe/ulyanovsk UTC offset of +04:00.
- Europe/Uzhgorod - Timezone for europe/uzhgorod UTC offset of +02:00.
- Europe/Vaduz - Timezone for europe/vaduz UTC offset of +01:00.
- Europe/Vatican - Timezone for europe/vatican UTC offset of +01:00.
- Europe/Vienna - Timezone for europe/vienna UTC offset of +01:00.
- Europe/Vilnius - Timezone for europe/vilnius UTC offset of +02:00.
- Europe/Volgograd - Timezone for europe/volgograd UTC offset of +03:00.
- Europe/Warsaw - Timezone for europe/warsaw UTC offset of +01:00.
- Europe/Zagreb - Timezone for europe/zagreb UTC offset of +01:00.
- Europe/Zaporozhye - Timezone for europe/zaporozhye UTC offset of +02:00.
- Europe/Zurich - Timezone for europe/zurich UTC offset of +01:00.
- GB - Timezone for GB UTC offset of +00:00.
- GB-Eire - Timezone for GB-Eire UTC offset of +00:00.
- GMT - Timezone for GMT UTC offset of +00:00.
- GMT0 - Timezone for GMT0 UTC offset of +00:00.
- Greenwich - Timezone for greenwich UTC offset of +00:00.
- HST - Timezone for HST UTC offset of -10:00.
- Hongkong - Timezone for hongkong UTC offset of +08:00.
- Iceland - Timezone for iceland UTC offset of +00:00.
- Indian/Antananarivo - Timezone for indian/antananarivo UTC offset of +03:00.

- Indian/Chagos - Timezone for indian/chagos UTC offset of +06:00.
- Indian/Christmas - Timezone for indian/christmas UTC offset of +07:00.
- Indian/Cocos - Timezone for indian/cocos UTC offset of +06:30.
- Indian/Comoro - Timezone for indian/comoro UTC offset of +03:00.
- Indian/Kerguelen - Timezone for indian/kerguelen UTC offset of +05:00.
- Indian/Mahe - Timezone for indian/mahe UTC offset of +04:00.
- Indian/Maldives - Timezone for indian/maldives UTC offset of +05:00.
- Indian/Mauritius - Timezone for indian/mauritius UTC offset of +04:00.
- Indian/Mayotte - Timezone for indian/mayotte UTC offset of +03:00.
- Indian/Reunion - Timezone for indian/reunion UTC offset of +04:00.
- Iran - Timezone for iran UTC offset of +03:30.
- Israel - Timezone for israel UTC offset of +02:00.
- Jamaica - Timezone for jamaica UTC offset of -05:00.
- Japan - Timezone for japan UTC offset of +09:00.
- Kwajalein - Timezone for kwajalein UTC offset of +12:00.
- Libya - Timezone for libya UTC offset of +02:00.
- MET - Timezone for MET UTC offset of +01:00.
- MST - Timezone for MST UTC offset of -07:00.
- MST7MDT - Timezone for MST7MDT UTC offset of -07:00.
- Mexico/BajaNorte - Timezone for mexico/bajanorte UTC offset of -08:00.
- Mexico/BajaSur - Timezone for mexico/bajasur UTC offset of -07:00.
- Mexico/General - Timezone for mexico/general UTC offset of -06:00.
- NZ - Timezone for NZ UTC offset of +13:00.
- NZ-CHAT - Timezone for NZ-CHAT UTC offset of +13:45.
- Navajo - Timezone for navajo UTC offset of -07:00.
- PRC - Timezone for PRC UTC offset of +08:00.
- PST8PDT - Timezone for PST8PDT UTC offset of -08:00.
- Pacific/Apia - Timezone for pacific/apia UTC offset of +14:00.
- Pacific/Auckland - Timezone for pacific/auckland UTC offset of +13:00.
- Pacific/Bougainville - Timezone for pacific/bougainville UTC offset of +11:00.
- Pacific/Chatham - Timezone for pacific/chatham UTC offset of +13:45.

- Pacific/Chuuk - Timezone for pacific/chuuk UTC offset of +10:00.
- Pacific/Easter - Timezone for pacific/easter UTC offset of -05:00.
- Pacific/Efate - Timezone for pacific/efate UTC offset of +11:00.
- Pacific/Enderbury - Timezone for pacific/enderbury UTC offset of +13:00.
- Pacific/Fakaofo - Timezone for pacific/fakaofo UTC offset of +13:00.
- Pacific/Fiji - Timezone for pacific/fiji UTC offset of +13:00.
- Pacific/Funafuti - Timezone for pacific/funafuti UTC offset of +12:00.
- Pacific/Galapagos - Timezone for pacific/galapagos UTC offset of -06:00.
- Pacific/Gambier - Timezone for pacific/gambier UTC offset of -09:00.
- Pacific/Guadalcanal - Timezone for pacific/guadalcanal UTC offset of +11:00.
- Pacific/Guam - Timezone for pacific/guam UTC offset of +10:00.
- Pacific/Honolulu - Timezone for pacific/honolulu UTC offset of -10:00.
- Pacific/Johnston - Timezone for pacific/johnston UTC offset of -10:00.
- Pacific/Kiritimati - Timezone for pacific/kiritimati UTC offset of +14:00.
- Pacific/Kosrae - Timezone for pacific/kosrae UTC offset of +11:00.
- Pacific/Kwajalein - Timezone for pacific/kwajalein UTC offset of +12:00.
- Pacific/Majuro - Timezone for pacific/majuro UTC offset of +12:00.
- Pacific/Marquesas - Timezone for pacific/marquesas UTC offset of -09:30.
- Pacific/Midway - Timezone for pacific/midway UTC offset of -11:00.
- Pacific/Nauru - Timezone for pacific/nauru UTC offset of +12:00.
- Pacific/Niue - Timezone for pacific/niue UTC offset of -11:00.
- Pacific/Norfolk - Timezone for pacific/norfolk UTC offset of +11:00.
- Pacific/Noumea - Timezone for pacific/noumea UTC offset of +11:00.
- Pacific/Pago_Pago - Timezone for pacific/pago_pago UTC offset of -11:00.
- Pacific/Palau - Timezone for pacific/palau UTC offset of +09:00.
- Pacific/Pitcairn - Timezone for pacific/pitcairn UTC offset of -08:00.
- Pacific/Pohnpei - Timezone for pacific/pohnpei UTC offset of +11:00.
- Pacific/Ponape - Timezone for pacific/ponape UTC offset of +11:00.
- Pacific/Port_Moresby - Timezone for pacific/port_moresby UTC offset of +10:00.
- Pacific/Rarotonga - Timezone for pacific/rarotonga UTC offset of -10:00.
- Pacific/Saipan - Timezone for pacific/saipan UTC offset of +10:00.

- Pacific/Samoa - Timezone for pacific/samoa UTC offset of -11:00.
- Pacific/Tahiti - Timezone for pacific/tahiti UTC offset of -10:00.
- Pacific/Tarawa - Timezone for pacific/tarawa UTC offset of +12:00.
- Pacific/Tongatapu - Timezone for pacific/tongatapu UTC offset of +14:00.
- Pacific/Truk - Timezone for pacific/truk UTC offset of +10:00.
- Pacific/Wake - Timezone for pacific/wake UTC offset of +12:00.
- Pacific/Wallis - Timezone for pacific/wallis UTC offset of +12:00.
- Pacific/Yap - Timezone for pacific/yap UTC offset of +10:00.
- Poland - Timezone for poland UTC offset of +01:00.
- Portugal - Timezone for portugal UTC offset of +00:00.
- ROC - Timezone for ROC UTC offset of +08:00.
- ROK - Timezone for ROK UTC offset of +09:00.
- Singapore - Timezone for singapore UTC offset of +08:00.
- Turkey - Timezone for turkey UTC offset of +03:00.
- UCT - Timezone for UCT UTC offset of +00:00.
- US/Pacific-New - Timezone for US/Pacific-New UTC offset of -08:00.
- Universal - Timezone for universal UTC offset of +00:00.
- W-SU - Timezone for W-SU UTC offset of +03:00.
- WET - Timezone for WET UTC offset of +00:00.
- Zulu - Timezone for zulu UTC offset of +00:00.
- jdj_tz - Timezone for jdj_tz UTC offset of +00:00.

Default

Default

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.114 cmm enbInfoQuery

eNB Info Query. Syntax, description, parameters, and roles for `cmm enbInfoQuery` command.

Syntax**Description**

List the info of the eNB(s) corresponding to the typed IP address

Parameters**Name**

enbId - eNB id

Syntax

--enbId <integer>

Description

eNB identifier as specified in TS 36.413

Class

Optional.

Type / Range

Integer in the range of 0 to 268,435,455

Name

enbIpAddress - enbIpAddress

Syntax

--enbIpAddress <string>

Description

eNB IP address that is used as key in query

Class

Optional.

Type / Range

Digit string of length 3 to 39 with valid IPv4 or IPv6 address

Name

enbIpAddress2 - enbIpAddress2

Syntax

--enbIpAddress2 <string>

Description

eNB IP address 2

Class

Optional.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

enbName - eNB name

Syntax

--enbName <string>

Description

eNB name

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 255

Name

enbType - ENodeB type

Syntax

--enbType <list-item>

Description

Specifies which enodeb type of the enodeb

Class

Read only.

List

- home - Specify home eNodeB.
- macro - Specify macro eNodeB.

Name

information - information

Syntax

--information <string>

Description

Information about outcome of command

Class

Read only.

Type / Range

Digit string of length 0 to 256 with the allowed pattern of [0-9a-zA-Z\.\-_:]*

Name

linkAdminState - Link admin state

Syntax

--linkAdminState <list-item>

Description

Link admin state

Class

Read only.

List

- locked - Locked.
- unlocked - Unlocked.

Name

linkIndex - Link index

Syntax

--linkIndex <integer>

Description

Link index.

Class

Read only.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

linkOperationState - Link operation state

Syntax

--linkOperationState <list-item>

Description

Link operation state

Class

Read only.

List

- disabled - Disabled state.

- enabled - Enabled state.

Name

remotePort - Remote Port

Syntax

--remotePort <integer>

Description

Remote Port associated with the remote IP address

Class

Read only.

Type / Range

Integer in the range of 0 to 65,534

Name

tac - Tracking area code

Syntax

--tac <integer>

Description

Tracking area Code

Class

Read only.

Type / Range

Integer in the range of 1 to 65,535

Name

taiMcc - TAI MCC

Syntax

--taiMcc <digits>

Description

3-digit number used to represent the mobile country code for TAI.

Class

Read only.

Type / Range

Digit string of length 3

Name

taiMnc - TAI MNC

Syntax

--taiMnc <digits>

Description

Digits used to represent mobile network code for TAI.

Class

Read only.

Type / Range

Digit string of length 2 to 3

Roles**List**

sysadmin, manage, monitor

4.115 cmm enbTaiQuery

eNB TAI Query. Syntax, description, parameters, and roles for `cmm enbTaiQuery` command.

Syntax**Description**

List TAIs of eNB. By default, only TAIs with enbType 'macro' are shown

Parameters**Name**

mcc - Mobile country code

Syntax

--mcc <digits>

Description

3-digit number used to represent the mobile country code.

Class

Required key.

Type / Range

Digit string of length 3

Name

mnc - Mobile network code

Syntax

--mnc <digits>

Description

Digits used to represent mobile network code.

Class

Required key.

Type / Range

Digit string of length 2 to 3

Name

enbId - eNB id

Syntax

--enbId <integer>

Description

eNB identifier as specified in TS 36.413 v8.6.1.

Class

Required key.

Type / Range

Integer in the range of 0 to 268,435,455

Name

enbIpAddress - eNB IP address

Syntax

--enbIpAddress <string>

Description

eNB IP address.

Class

Read only.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

enbName - eNB name

Syntax

--enbName <string>

Description

eNB name.

Class

Read only.

Type / Range

Digit string of length 0 or 1 to 255

Name

enbType - ENodeB type

Syntax

--enbType <list-item>

Description

Specifies which ENodeB Type of the ENodeB. If not given then enbType 'macro' is assumed

Class

Optional.

List

- home - Specify home eNodeB.
- macro - Specify macro eNodeB.

Name

linkAdminState - Link admin state

Syntax

--linkAdminState <list-item>

Description

Link admin state.

Class

Read only.

List

- locked - Locked.
- unlocked - Unlocked.

Name

linkOperationState - Link operation state

Syntax

--linkOperationState <list-item>

Description

Link operation state.

Class

Read only.

List

- disabled - Disabled state.
- enabled - Enabled state.

Name

tac - Tracking area code

Syntax

--tac <integer>

Description

Tracking area code.

Class

Read only.

Type / Range

Integer in the range of 1 to 65,535

Name

taiMcc - TAI MCC

Syntax

--taiMcc <digits>

Description

3-digit number used to represent the mobile country code for TAI.

Class

Read only.

Type / Range

Digit string of length 3

Name

taiMnc - TAI MNC

Syntax

--taiMnc <digits>

Description

Digits used to represent mobile network code for TAI.

Class

Read only.

Type / Range

Digit string of length 2 to 3

Roles**List**

sysadmin, manage, monitor

4.116 cmm enbUeQuery

enb Ue Query. Syntax, description, parameters, and roles for cmm enbUeQuery command.

Syntax

Description

Provides information about number of UEs per connected eNB

Parameters

Name

disconnectTime - Disconnect Time

Syntax

--disconnectTime <string>

Description

This is the time that an enb is disconnected

Class

Read only.

Type / Range

Date and Time Digit string of length 0 or 19 with the allowed pattern of (20[0-9][0-9]/[0-1][0-9]/[0-3][0-9] [0-2][0-9]:[0-5][0-9]:[0-5][0-9])*

Name

enbId - eNB Id

Syntax

--enbId <integer>

Description

eNB identifier as specified in TS 36.413

Class

Optional.

Type / Range

Integer in the range of 0 to 268,435,455

Name

enbIpAddress - eNB IP Address

Syntax

--enbIpAddress <string>

Description

eNB IP address

Class

Optional.

Type / Range

Digit string of length 3 to 39 with valid IPv4 or IPv6 address

Name

enbIpAddress2 - eNB IP Address 2

Syntax

--enbIpAddress2 <string>

Description

eNB IP address 2

Class

Optional.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

enbName - eNB Name

Syntax

--enbName <string>

Description

eNB name

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 255

Name

enbType - eNB Type

Syntax

--enbType <list-item>

Description

Specifies which enodeb type of the enodeb

Class

Optional.

List

- home - Specify home eNodeB.

- macro - Specify macro eNodeB.

Name

information - Information

Syntax

--information <string>

Description

Information about outcome of command

Class

Read only.

Type / Range

Digit string of length 0 to 256 with the allowed pattern of [0-9a-zA-Z\._:-]*

Name

mcc - Mobile Country Code

Syntax

--mcc <digits>

Description

Mobile Country Code

Class

Optional.

Type / Range

Digit string of length 3 to 3 with the allowed pattern of [0-9]*

Name

mnc - Mobile Network Code

Syntax

--mnc <digits>

Description

Mobile Network Code

Class

Optional.

Type / Range

Digit string of length 2 to 3 with the allowed pattern of [0-9]*

Name

s1ConnectionAmount - S1 Connection Amount

Syntax

--s1ConnectionAmount <integer>

Description

Number of UEs connected to a specific eNodeB

Class

Read only.

Type / Range

Integer in the range of 0 to 4,294,967,295

Roles**List**

sysadmin, manage, monitor

4.117 cmm enbUeQuerySum

enb Ue Query Sum. Syntax, description, parameters, and roles for cmm enbUeQuerySum command.

Syntax**Description**

Provides overview information about the total number of enodeBs, total number of S1 Connections and total number of disconnected enodeBs.

Parameters**Name**

information - Information

Syntax

--information <string>

Description

Information about outcome of command

Class

Read only.

Type / Range

Digit string of length 0 to 256 with the allowed pattern of [0-9a-zA-Z\._:\]*

Name

totalNumConnectedEnbs - Total Number eNBs

Syntax

--totalNumConnectedEnbs <integer>

Description

Provides information about the total number of connected eNBs

Class

Read only.

Type / Range

Integer in the range of 0 to 300000

Name

totalNumDisconnectedEnbs - Total Number Disconnected eNBs

Syntax

--totalNumDisconnectedEnbs <integer>

Description

Provides information about the total number of disconnected eNBs

Class

Read only.

Type / Range

Integer in the range of 0 to 300000

Name

totalNumS1Connections - Total Number S1 Connections

Syntax

--totalNumS1Connections <integer>

Description

Provides information about the total number of S1 connections

Class

Read only.

Type / Range

Integer in the range of 0 to 4,294,967,295

Roles**List**

sysadmin, manage, monitor

4.118 cmm enhancedDualConnectivityProfile

Enhanced Dual Connectivity Profile. Syntax, description, parameters, and roles for cmm enhancedDualConnectivityProfile command.

Syntax

Description

This profile lists the NR access restriction options, GW selection with NR, and Secondary Rat Usage Reporting for NSA Option 3A/3X.

Parameters

Name

name - Enhanced Dual Connectivity Profile Name

Syntax

--name <string>

Description

Identifies the name of the Enhanced Dual Connectivity Profile for NSA Option 3A/3X

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

accRestAvpAbsent - Access Restriction AVP Absent

Syntax

--accRestAvpAbsent <list-item>

Description

Identifies whether to allow or restrict a UE New Radio (NR) access when HSS doesn't include Access Restriction AVP and HSS Feature Support AVP for NR over S6a.

Class

Optional.

List

- ALLOW - Allow
- RESTRICT - Restrict

Default

RESTRICT

Name

accRestAvpNrAllow - UE NR Access Allowed in HSS

Syntax

--accRestAvpNrAllow <list-item>

Description

Identifies whether to allow or restrict a UE New Radio (NR) access when HSS indicates UE NR access is allowed in Access Restriction AVP but HSS Feature Support AVP for NR is absent over S6a.

Class

Optional.

List

- ALLOW - Allow
- RESTRICT - Restrict

Default

RESTRICT

Name

accRestAvpNrRestrict - UE NR Access Restricted in HSS

Syntax

--accRestAvpNrRestrict <list-item>

Description

Identifies whether to allow or restrict a UE New Radio (NR) access when HSS indicates UE NR access is Restricted in Access Restriction AVP but HSS Feature Support AVP for NR is absent over S6a.

Class

Optional.

List

- ALLOW - Allow
- RESTRICT - Restrict

Default

RESTRICT

Name

accRestNrAllow - UE NR Access & NR Feature support in HSS

Syntax

--accRestNrAllow <list-item>

Description

Identifies whether to allow or restrict a UE New Radio (NR) access when HSS indicates UE NR access is allowed or absent in Access Restriction AVP and HSS Feature Support AVP for NR is set.

Class

Optional.

List

- ALLOW - Allow
- RESTRICT - Restrict

Default

ALLOW

Name

accRestNrAllowGwNotNrCapable - UE NR Access Allow when GW not NR capable

Syntax

--accRestNrAllowGwNotNrCapable <list-item>

Description

When set to true, MME will not indicate NR Restriction in Handover Restriction List (HRL) to eNB or RestrictDCNR in the TAU Accept to NR Capable UE if returned SNAPTR SGW/PGW records (SGW, PGW or both SGW/PGW records) do not indicate NR capable SGW/PGW. This parameter applies to only idle Gn-TAU (with or without active flag).

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

dcnrIndToUe - DCNR support indication to UE

Syntax

--dcnrIndToUe <list-item>

Description

MME indicates DCNR support (RestrictDCNR=0) to the UE in Attach or TAU Accept message even if MME determined UE is DCNR Access Restricted and regardless of UE DCNR support indication (or lack of) in Attach or TAU Request. MME will still indicate UE DCNR restricted to the eNB.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

gwSelectionFallbackWithoutNr - S/PGW Selection Without NR

Syntax

--gwSelectionFallbackWithoutNr <list-item>

Description

Enables the MME to select S/PGW with no New Radio (NR) capability for NR capable UEs if SNAPTR query returned gateway candidate list has no S/PGW candidate nodes with service parameter '+nc-nr'.

Class

Optional.

List

- DISABLE - Disable
- ENABLE - Enable

Default

ENABLE

Name

iratNrGwReselection - IRAT TAU NR GW reselection

Syntax

--iratNrGwReselection <list-item>

Description

enables/disables PGW re-selection with +nc-nr when UE moves in idle mode from Gn-SGSN to MME.

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

irPgw - PGW is intended recipient of Secondary RAT Usage Reports

Syntax

--irPgw <list-item>

Description

Identifies whether the Secondary RAT Usage Report(s) are intended (or not) for PGW.
irPgw=0, SGW does not forward report(s) to PGW irPgw=1, SGW forwards report(s) to PGW
irPgw applies only when Secondary RAT Usage Data Reporting is enabled

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

irSgw - SGW is intended recipient of Secondary RAT Usage Reports

Syntax

--irSgw <list-item>

Description

Identifies whether the Secondary RAT Usage Report(s) are intended (or not) for SGW.
irSgw=0, SGW ignores the report(s) and forward to PGW if irPgw=1 irSgw=1, SGW stores/
process the report(s) and also forward to PGW if irPGW=1 irSgw applies only when Secondary
RAT Usage Data Reporting is enabled

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

restrictDcnrToUePerTaiDcnrSupported - Restrict DCNR to UE Per TAI DCNR Supported

Syntax

--restrictDcnrToUePerTaiDcnrSupported <list-item>

Description

Specifies whether the MME is to restrict the UE's use of DC-NR via NAS messaging if the TAI
is locally provisioned as not allowing DC-NR.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

secondaryRatUsageDataReporting - Secondary RAT Usage Data Reporting for NSA Option 3A/3X

Syntax

--secondaryRatUsageDataReporting <list-item>

Description

Enables Secondary RAT Usage Data Reporting for NSA Option 3X. MME forwards usage data reports, if/when received from the eNB over S1AP, to S/PGW over S11.

Class

Optional.

List

- DISABLE - Disable
- ENABLE - Enable

Default

DISABLE

Name

triggerSgwChangePerTaiDcnrSupported - Trigger SGW Change Per TAI DCNR Supported

Syntax

--triggerSgwChangePerTaiDcnrSupported <list-item>

Description

This field controls whether the MME will attempt to change SGW if the UE moves from a TAI provisioned as supporting DCNR to a TAI marked as not supporting DCNR, as well as when the UE move from a TAI provisioned as not supporting DCNR to one that supports DCNR. For this field to be set true, useTaiDcnrSupportedInCsrToSgw must also be true.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

useTaiDcnrSupportedInCsrToSgw - Use TAI DCNR Supported

Syntax

--useTaiDcnrSupportedInCsrToSgw <list-item>

Description

This field controls whether the MME uses the TAI 'DCNR Supported' field when determining how to set the 'DCNR' bit in a Create Session Request message to the SGW.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.119 cmm ePlmnPerImsiTai

Equivalent PLMN per IMSI and TAI. Syntax, description, parameters, and roles for cmm ePlmnPerImsiTai command.

Syntax

Description

Equivalent PLMN List. The list can be provisioned to an IMSI series.

Parameters

Name

ePlmnListName - EPLMN list name

Syntax

--ePlmnListName <string>

Description

Equivalent PLMN list name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

equivalentPlmnName - Equivalent PLMN

Syntax

--equivalentPlmnName <string>

Description

Name of the PLMN to be treated as an equivalent PLMN.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.120 cmm ePlmnPerImsiTaiList

EPLMN per IMSI and TAI List. Syntax, description, parameters, and roles for cmm ePlmnPerImsiTaiList command.

Syntax

Description

EPLMN List

Parameters

Name

name - Name of the list

Syntax

```
--name <string>
```

Description

Name of the list

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.121 cmm eqPlmn

equivalent plmn. Syntax, description, parameters, and roles for cmm eqPlmn command.

Syntax

Description

Configure, list and show equivalent PLMN.

Parameters

Name

eqPlmnId - Primary key of EQPLMN table

Syntax

--eqPlmnId <integer>

Description

Primary key of EQPLMN table, numeric id for the table.

Class

Required key.

Type / Range

Integer in the range of 0 to 255

Name

plmnParamId - PLMN parameter set name

Syntax

--plmnParamId <string>

Description

PLMN parameter set name where this EQPLMN belongs to. (corresponds to plmnparamid of plmnparam table. you must create that plmnparam first).

Class

Required key.

Type / Range

Digit string of length 1 to 15 with the allowed pattern of [0-9a-zA-Z\._:-]*

Name

eplmnIndex - EplmnIndex

Syntax

--eplmnIndex <string>

Description

All index combinations from 1 to 5 is possible, some examples: 00000 - NO INDEX IN USE
00001 - INDEX 1 IN USE 00011 - INDEX 1 AND INDEX 2 IN USE 01010 - INDEX 2 AND INDEX 4
IN USE 01110 - INDEX 2, INDEX 3 AND INDEX 4 IN USE 11111 - ALL 5 INDEXES IN USE.

Class

Optional.

Type / Range

Digit string of length 5 to 5 with the allowed pattern of [0-9a-zA-Z\._ :]*

Default

00000

Name

mcc - MCC of equivalent PLMN

Syntax

--mcc <digits>

Description

This parameter defines the MCC of equivalent PLMN.

Class

Optional.

Type / Range

Digit string of length 3

Name

mnc - MNC of equivalent PLMN

Syntax

--mnc <digits>

Description

This parameter defines the MNC of equivalent PLMN.

Class

Optional.

Type / Range

Digit string of length 2 to 3

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.122 cmm equivalentPlmn

Equivalent PLMN. Syntax, description, parameters, and roles for cmm equivalentPlmn command.

Syntax

Description

This command is used to configure an existing network as an equivalent network and includes parameters for defining the serving and equivalent network names.

Parameters

Name

servedPlmnName - Serving mobile network

Syntax

--servedPlmnName <string>

Description

Name that identifies serving network.

Restrictions

plmnType of servedPlmnName must be Home or Shared Network

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

equivalentPlmnName - Equivalent network name

Syntax

--equivalentPlmnName <string>

Description

Name that identifies equivalent network.

Restrictions

plmnType of equivalentPlmnName must be Shared or Other Network

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.123 cmm esmCauseCodeMapping

Configurable ESM causes mapping table. Syntax, description, parameters, and roles for cmm esmCauseCodeMapping command.

Syntax

Description

Configurable ESM causes mapping table.

Parameters

Name

esmCauseCodeMappingId - Configurable ESM causes mapping table ID

Syntax

--esmCauseCodeMappingId <integer>

Description

Configurable ESM causes mapping table ID - just type 1.

Class

Required key.

Type / Range

Integer in the range of 1 to 1

Name

intErrorBearerSetupFailed - ESM cause code for bearer setup failed

Syntax

--intErrorBearerSetupFailed <integer>

Description

ESM cause code to be sent if bearer setup failed.

Class

Optional.

Type / Range

Integer in the range of 8, 26-39, 41-47, 49-60, 65-66, 81, 95-101, 111-113

Default

112

Name

intErrorIpv4OnlyAllowed - IntErrorIpv4OnlyAllowed

Syntax

--intErrorIpv4OnlyAllowed <integer>

Description

ESM cause code to be sent if IPv4 only allowed.

Class

Optional.

Type / Range

Integer in the range of 8, 26-39, 41-47, 49-60, 65-66, 81, 95-101, 111-113

Default

50

Name

intErrorIpv6OnlyAllowed - ESM cause code for IPv6 only allowed

Syntax

--intErrorIpv6OnlyAllowed <integer>

Description

ESM cause code to be sent if IPv6 only allowed.

Class

Optional.

Type / Range

Integer in the range of 8, 26-39, 41-47, 49-60, 65-66, 81, 95-101, 111-113

Default

51

Name

intErrorIpVerMismatchAtt - ESM cause code for IP version mismatch in attach

Syntax

--intErrorIpVerMismatchAtt <integer>

Description

ESM cause code to be sent if IP version mismatch in attach.

Class

Optional.

Type / Range

Integer in the range of 8, 26-39, 41-47, 49-60, 65-66, 81, 95-101, 111-113

Default

38

Name

intErrorMandleInc - ESM cause code for mandatory IE included

Syntax

--intErrorMandIeInc <integer>

Description

ESM cause code to be sent if mandatory IE included.

Class

Optional.

Type / Range

Integer in the range of 8, 26-39, 41-47, 49-60, 65-66, 81, 95-101, 111-113

Default

96

Name

intErrorMaxNumOfPdnPerApn - ESM cause code for max num of PDN per apn exceeded

Syntax

--intErrorMaxNumOfPdnPerApn <integer>

Description

ESM cause code to be sent if maximum number of PDN per apn.

Class

Optional.

Type / Range

Integer in the range of 8, 26-39, 41-47, 49-60, 65-66, 81, 95-101, 111-113

Default

55

Name

intErrorNoRespFromSgwAttPdn - ESM cause code for no resp from SGW attach PDN

Syntax

--intErrorNoRespFromSgwAttPdn <integer>

Description

ESM cause code to be sent if no response from SGW attach pdn.

Class

Optional.

Type / Range

Integer in the range of 8, 26-39, 41-47, 49-60, 65-66, 81, 95-101, 111-113

Default

26

Name

intErrorNoRespFromSgwBrrAlloc - ESM cause code for no resp from SGW on bearer allocation

Syntax

--intErrorNoRespFromSgwBrrAlloc <integer>

Description

ESM cause code to be sent if no response from SGW brr allocation.

Class

Optional.

Type / Range

Integer in the range of 8, 26-39, 41-47, 49-60, 65-66, 81, 95-101, 111-113

Default

30

Name

intErrorNoRespFromSgwBrrMod - ESM cause code for no resp from SGW on bearer modification

Syntax

--intErrorNoRespFromSgwBrrMod <integer>

Description

ESM cause code to be sent if no response from SGW brr modification.

Class

Optional.

Type / Range

Integer in the range of 8, 26-39, 41-47, 49-60, 65-66, 81, 95-101, 111-113

Default

31

Name

intErrorNoRespFromSgwSvcReq - ESM cause code for no resp from SGW service request

Syntax

--intErrorNoRespFromSgwSvcReq <integer>

Description

ESM cause code to be sent if no response from SGW svc request.

Class

Optional.

Type / Range

Integer in the range of 8, 26-39, 41-47, 49-60, 65-66, 81, 95-101, 111-113

Default

38

Name

intErrorOptIeInc - ESM cause code for optional IE included

Syntax

--intErrorOptIeInc <integer>

Description

ESM cause code to be sent if optional IE included.

Class

Optional.

Type / Range

Integer in the range of 8, 26-39, 41-47, 49-60, 65-66, 81, 95-101, 111-113

Default

100

Name

intErrorPdnRejectDueToNoApnSuppEscCc - ESM cause code for PDN reject due to no APN support

Syntax

--intErrorPdnRejectDueToNoApnSuppEscCc <integer>

Description

ESM cause code to be sent if PDN rejected due to no APN support for IMS roaming control.

Class

Optional.

Type / Range

Integer in the range of 8, 26-39, 41-47, 49-60, 65-66, 81, 95-101, 111-113

Default

66

Name

intErrorReqPdnNotAllowed - ESM cause code for req PDN not allowed

Syntax`--intErrorReqPdnNotAllowed <integer>`**Description**

ESM cause code to be sent if requested PDN not allowed.

Class

Optional.

Type / Range

Integer in the range of 8, 26-39, 41-47, 49-60, 65-66, 81, 95-101, 111-113

Default

112

Name

intErrorServiceDenied - ESM cause code for service denied

Syntax`--intErrorServiceDenied <integer>`**Description**

ESM cause code to be sent if service denied.

Class

Optional.

Type / Range

Integer in the range of 8, 26-39, 41-47, 49-60, 65-66, 81, 95-101, 111-113

Default

31

Name

intErrorSystemFailure - ESM cause code for system failure

Syntax`--intErrorSystemFailure <integer>`**Description**

ESM cause code to be sent in system failure.

Class

Optional.

Type / Range

Integer in the range of 8, 26-39, 41-47, 49-60, 65-66, 81, 95-101, 111-113

Default

38

Name

intErrorUserAuthenticationFailure - ESM cause code for user authentication failed from SGW

Syntax`--intErrorUserAuthenticationFailure <integer>`**Description**

ESM cause code to be sent if user authentication failed response received from SGW for standalone pdn connectivity request.

Class

Optional.

Type / Range

Integer in the range of 8, 26-39, 41-47, 49-60, 65-66, 81, 95-101, 111-113

Default

33

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

4.124 cmm esmlc

ESMLC. Syntax, description, parameters, and roles for `cmm esmlc` command.

Syntax**Description**

This command is used to create an Evolved Serving Mobile Location Center (E-SMLC) and to configure E-SMLC endpoint parameters.

Parameters**Name**

esmlcIdentity - ESMLC identifier

Syntax

--esmlcIdentity <integer>

Description

Evolved serving mobile location center identifier.

Class

Required key.

Type / Range

Integer in the range of 0 to 255

Name

esmlcEndPoint1Name - ESMC end point 1 name

Syntax

--esmlcEndPoint1Name <string>

Description

Name of primary evolved serving mobile location center end point. corresponds to the label field of the rmtendptcfg configuration table. name must be unique across all ESMC.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

esmlcEndPoint2Name - ESMC end point 2 name

Syntax

--esmlcEndPoint2Name <string>

Description

Name of secondary evolved serving mobile location center end point. corresponds to the label field of the rmtendptcfg configuration table. name must be unique across all ESMC.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

name - ESMC name

Syntax

--name <string>

Description

Evolved serving mobile location center identifier name.

Class

Optional.

Type / Range

Digit string of length 0 to 32 with the allowed pattern of [0-9a-zA-Z\.\-:]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.125 cmm extendedEmergencyNum

Extended Emergency Number List. Syntax, description, parameters, and roles for cmm extendedEmergencyNum command.

Syntax**Description**

This command is used to administer an Extended Emergency Number List Table Maximum of 25 Extended Emergency Number entries can be supported per extended emergency number list.

Parameters**Name**

extendedEmergencyNumListName - Extended Emergency Number List Name

Syntax

--extendedEmergencyNumListName <string>

Description

Defines the extended emergency number list table name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

extendedEmergencyDigits - Extended Emergency Digits

Syntax

--extendedEmergencyDigits <string>

Description

Maximum of 25 items can be in a list.

Class

Required key.

Type / Range

Digit string of length 1 to 10 with the allowed pattern of [0-9][0-9A-Ca-c*#]*

Name

subServiceUrn - Sub-service URN

Syntax

--subServiceUrn <string>

Description

Name of the emergency service. If the associated emergency service URN is 'urn:service:sos.gas', there is only one sub-service provided in the sub-services field which is 'gas'. Following are some of the examples of URN services: 'urn:service:sos.ambulance', 'urn:service:sos.police', 'urn:service:sos.fire', 'urn:service:sos.marine', 'urn:service:sos.mountain', 'urn:service:sos.ecall.manual', 'urn:service:sos.ecall.automatic'. If the UE can determine the type of emergency service the UE shall use an emergency service URN with a sub-service type corresponding to the type of emergency service. There is no need to enter the prefix: urn:service:

Class

Required key.

Type / Range

Emergency Service Name

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.126 cmm extendedEmergencyNumList

Extended Emergency Number List. Syntax, description, parameters, and roles for cmm extendedEmergencyNumList command.

Syntax

Description

This command is used to administer an Extended Emergency Number List. Maximum of 10 extended emergency number lists shall be supported.

Parameters

Name

name - Name

Syntax

--name <string>

Description

Defines the name of extended emergency number list table.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

extendedEmergencyNumberListValidity - Extended Emergency Number List Validity

Syntax

--extendedEmergencyNumberListValidity <list-item>

Description

Parameter specifies whether extended emergency number list validity is valid in the country or the PLMN from which this IE is received.

Class

Optional.

List

- country - Country Extended Emergency Number List Validity - Extended Local Emergency Numbers List is valid in the country of the PLMN from which this IE is received.
- plmn - PLMN Extended Emergency Number List Validity - Extended Local Emergency Numbers List is valid only in the PLMN from which this IE is received.

Default

country

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.127 cmm extLdapProfile

External LDAP Profile. Syntax, description, parameters, and roles for `cmm extLdapProfile` command.

Syntax

Description

This command is used to administer an External LDAP Profile. Maximum of 3 profiles are supported. This profile can be assigned on authServer

Parameters

Name

name - External LDAP Profile Name

Syntax

--name <string>

Description

External LDAP Profile Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

bindCredential - Bind Credential

Syntax

--bindCredential <string>

Description

Bind Credential used on the External ldap server for the Bind DN.

Class

Optional.

Type / Range

Digit string of length 0 to 128 with the allowed pattern of [\]0-9a-zA-Z\.\~\$+* _.;&=#?\[@,()
/!"%]* in units of Byte

Default

""

Name

bindDn - Bind DN

Syntax

--bindDn <string>

Description

DN on the external LDAP server used to use for performing External LDAP operations eg: ,
cn=admin, dc=example, dc=org

Class

Optional.

Type / Range

Digit string of length 0 to 128

Default

""

Name

connectionExpireTimeout - Connection Expire Timeout

Syntax

--connectionExpireTimeout <integer>

Description

Specifies a timeout (in seconds) that a connection to an LDAP server will be maintained. After this time, the connection will be re-established.

Class

Optional.

Type / Range

Integer in the range of 1 to 3,600

Default

900

Name

insecure - Insecure

Syntax

--insecure <list-item>

Description

When set to false, ldaps protocol is used using port 636 When set to true, ldap protocol is used using port 389. Insecure value of true is not recommended

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

kerberosRealm - Kerberos Realm

Syntax

--kerberosRealm <string>

Description

Kerberos Realm. This field is optional.

Class

Optional.

Type / Range

Digit string of length 96 with the allowed pattern of [0-9a-zA-Z.-]*

Name

kerberosServerAddress - Kerberos Server Address

Syntax

--kerberosServerAddress <string>

Description

Kerberos Server Address. This field is optional. Can be IPv4 or IPv6

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

ldapSearchBase - LDAP Search Base

Syntax

--ldapSearchBase <string>

Description

LDAP Search Base. The default base DN to use for performing LDAP user operations. eg
dc=example,dc=com

Class

Optional.

Type / Range

Digit string of length 0 to 128 with the allowed pattern of [\\]0-9a-zA-Z\\.\\-~\$+*_.:;&=#?\\[\\@,
(/!'%"*]*

Name

oamTlsProfileName - OAM TLS Profile

Syntax

--oamTlsProfileName <string>

Description

This parameter contains the profile for TLS. It contains the certificate names needed for
establishing secure communication.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\\.\\-_:@]*

Default

""

Name

port - External LDAP Port

Syntax

--port <integer>

Description

External LDAP Port

Class

Optional.

Type / Range

Integer in the range of 1 to 65,535

Default

636

Name

rdnLdapAttribute - RDN LDAP Attribute Name

Syntax

--rdnLdapAttribute <string>

Description

External LDAP DN attribute Name

Class

Optional.

Type / Range

Digit string of length 0 to 128

Default

""

Name

schemaType - Schema Type

Syntax

--schemaType <list-item>

Description

Specifies the Schema Type in use on the External LDAP server.

Class

Optional.

List

- rfc2307bis - With rfc2307bis group members are listed by DN and stored in the member attribute.
- rfc2307 - With rfc2307, group members are listed by name in the memberUid attribute.
- ad - Active Directory

Default

rfc2307bis

Name

userNameLdapAttribute - User Name LDAP Attribute

Syntax

--userNameLdapAttribute <string>

Description

Name of the Remote LDAP Attribute which is mapped to local user name eg CN.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z]*

Default

""

Roles**Create**

secadmin

Show

secadmin

Modify

secadmin

Delete

secadmin

List

secadmin

4.128 cmm failopenProfile

Failopen profile. Syntax, description, parameters, and roles for cmm failopenProfile command.

Syntax**Description**

Failopen profile for HSS unavailability.

Parameters

Name

chargingCharacteristics - Charging characteristics

Syntax

--chargingCharacteristics <string>

Description

Charging characteristics.

Class

Optional.

Type / Range

Digit string of length 1 to 4 with the allowed pattern of [0-9a-fA-F]*

Default

0000

Name

defaultApnNi - APN NI type

Syntax

--defaultApnNi <string>

Description

Access point name network indicator type.

Class

Optional.

Type / Range

Digit string of length 0 to 62 with the allowed pattern of ([0-9a-zA-Z\.\-]*|*)

Default

''

Name

esmPdnConnRejectionCode - ESM PDN connection rejection code

Syntax

--esmPdnConnRejectionCode <integer>

Description

EPS session management packet data network connection rejection code.

Class

Optional.

Type / Range

Integer in the range of 8, 26-35, 38, 50-51, 53-55, 65-66, 95-112

Default

34

Name

networkAccessMode - Network access mode

Syntax`--networkAccessMode <list-item>`**Description**

Network access mode.

Class

Optional.

List

- Packet and Circuit - Specifies network access mode of packet and circuit.
- Packet only - Specifies network access mode of packet only.

Default

Packet and Circuit

Name

rfspId - RFSP ID

Syntax`--rfspId <integer>`**Description**

RAT/frequency selection priority identification.

Class

Optional.

Type / Range

Integer in the range of 1 to 256

Default

256

Name

ueAmbrDL - UE AMBR downlink

Syntax`--ueAmbrDL <integer>`**Description**

User equipment aggregate maximum bit rate downlink.

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64- 568, 576-8640, 8700-16000, 17000-128000,
130000-4000000000

Default

128000

Recommended

The recommended value is 100000000.

Name

ueAmbrUl - UE AMBR uplink

Syntax

--ueAmbrUl <integer>

Description

User equipment aggregate maximum bit rate uplink.

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64- 568, 576-8640, 8700-16000, 17000-128000,
130000-4000000000

Default

128000

Recommended

The recommended value is 100000000.

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

4.129 cmm failopenProfileApn

Failopen Profile APN. Syntax, description, parameters, and roles for cmm failopenProfileApn command.

Syntax

Description

Define a failopen APN for HSS unavailability. The failopen APN is used instead of the HSS subscription profile when HSS is unavailable and the MME does not hold the UE subscription profile. MME supports up to 50 failopen profile APN lists and 50 APNs (i.e. APN NIs) altogether on all failopen profile APN lists.

Parameters

Name

failopenProfileApnListName - Failopen Profile APN List Name

Syntax

--failopenProfileApnListName <string>

Description

The name of the failopen profile associated APN list.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

apnNi - APN NI

Syntax

--apnNi <string>

Description

Access point name network indicator.

Class

Required key.

Type / Range

Digit string of length 1 to 62 with the allowed pattern of ([0-9a-zA-Z\._]*|*)

Name

apnAmbrDl - APN AMBR Downlink

Syntax

--apnAmbrDl <integer>

Description

Access point name aggregate maximum bit rate downlink.

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64- 568, 576-8640, 8700-16000, 17000-128000,
130000-4000000000

Default

128000

Recommended

The recommended value is 100000000.

Name

apnAmbrUl - APN AMBR Uplink

Syntax

--apnAmbrUl <integer>

Description

Access point name aggregate maximum bit rate uplink.

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64- 568, 576-8640, 8700-16000, 17000-128000,
130000-4000000000

Default

128000

Recommended

The recommended value is 100000000.

Name

arpPriorityLevel - ARP PL

Syntax

--arpPriorityLevel <integer>

Description

Allocation and retention priority, priority level.

Class

Optional.

Type / Range

Integer in the range of 1 to 15

Default

15

Name

interworking5gsIndicator - Interworking 5GS Indicator

Syntax

--interworking5gsIndicator <list-item>

Description

Indicates whether the interworking between 5GS and EPS is subscribed for the APN.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

pdnAddrType - PDN Address Type

Syntax

--pdnAddrType <list-item>

Description

PDN address type.

Class

Optional.

List

- ipv4 - Specify PDN address type of IPv4.
- ipv4OrIpv6 - Specify PDN address type of IPv4 or IPv6.
- ipv4v6 - Specify PDN address type of IPv4 and IPv6.
- ipv6 - Specify PDN address type of IPv6.

Default

ipv4

Name

qci - QCI

Syntax

--qci <integer>

Description

Quality of Service Class Identifier.

Class

Optional.

Type / Range

Integer in the range of 5 to 9

Default

9

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.130 cmm failopenProfileApnConfig

Failopen profile APN configuration. Syntax, description, parameters, and roles for cmm failopenProfileApnConfig command.

Syntax**Description**

Failopen profile APN configuration for HSS unavailability.

Parameters**Name**

apnNi - APN NI type

Syntax

--apnNi <string>

Description

Access point name network indicator type.

Class

Required key.

Type / Range

Digit string of length 1 to 62 with the allowed pattern of ([0-9a-zA-Z\.\-]*\|*)

Name

apnAmbrDL - APN AMBR downlink

Syntax

--apnAmbrDL <integer>

Description

Access point name aggregate maximum bit rate downlink.

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64- 568, 576-8640, 8700-16000, 17000-128000,
130000-4000000000

Default

128000

Recommended

The recommended value is 100000000.

Name

apnAmbrUL - APN AMBR uplink

Syntax

--apnAmbrUL <integer>

Description

Access point name aggregate maximum bit rate uplink.

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64- 568, 576-8640, 8700-16000, 17000-128000,
130000-4000000000

Default

128000

Recommended

The recommended value is 100000000.

Name

arpPL - ARP PL

Syntax

--arpPL <integer>

Description

Allocation and retention priority PL.

Class

Optional.

Type / Range

Integer in the range of 1 to 15

Default

15

Name

pdnAddrType - Charging characteristics

Syntax

--pdnAddrType <list-item>

Description

Charging characteristics.

Class

Optional.

List

- ipv4 - Specify PDN address type of IPv4.
- ipv4OrIpv6 - Specify PDN address type of IPv4 or IPv6.
- ipv4v6 - Specify PDN address type of IPv4 and IPv6.
- ipv6 - Specify PDN address type of IPv6.

Default

ipv4

Name

qci - QCI

Syntax

--qci <integer>

Description

Quality of service class identifier.

Class

Optional.

Type / Range

Integer in the range of 5 to 9

Default

9

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.131 cmm failopenProfileApnList

Failopen profile APN configuration list. Syntax, description, parameters, and roles for cmm failopenProfileApnList command.

Syntax

Description

Define the list of APN configuration profiles for a failopen profile. MME supports up to 16 APNs per APN configuration list. MME supports up to 50 APNs in all APN configuration lists. Applicable when enhanced failopen profile is enabled.

Parameters

Name

name - Failopen Profile APN Configuration List Name

Syntax

--name <string>

Description

The name of the failopen profile APN configuration list.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.132 cmm features

features. Syntax, description, parameters, and roles for **cmm features** command.

Syntax

Description

Configure, list and show the states of features.

Parameters

Name

parameter - Parameter

Syntax

--parameter <string>

Description

Name of the parameter.

Class

Output only.

Type / Range

Digit string of length 1 to 100 with the allowed pattern of [0-9a-zA-Z\._&(),/]*

Name

status - Status

Syntax

--status <list-item>

Description

Is feature active or not.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

id - FeatureId

Syntax

<list-item>

Description

Feature for modification.

Class

Required value.

List

- 2gMocn - 2G multiple operator core network - 2G multiple operator core network functionality.
- 2gSubsOffloading - 2G subscriber offloading - 2G subscriber offloading.
- abnUeDetection - Abn UE detection - Abnormal signaling reduction functionality.
- addSupport - Add support - Add support.
- advancedCdr - Advanced CDR - Advanced CDR (PM) functionality.
- advancedData - Advanced data - Advanced data (PM) functionality.
- advancedData2 - Advanced data 2 - Advanced data 2 (PM) functionality.
- advancedMobilityMan - Advanced mobility management - Advanced mobility management (PM) functionality.
- advancedSessionMan - Advanced session management - Advanced session management (PM) functionality.
- advancedUser - Advanced user - Advanced user (PM) functionality.
- apnConv - APN conversion - APN conversion and correction functionality.
- apnDedicatedPrepaid - APN dedicated prepaid - APN dedicated prepaid functionality.
- attachCapacity - Attach capacity - Attach capacity. Warning: For Nokia personnel only.
- causeCodeFrwdS4 - Cause code forward S4 - Cause code forwarding over S4 interface functionality.

- ccGgsn - CC GGSN - CC GGSN.
- ccMapping - CC mapping - Cause code mapping functionality.
- cellSaStat - Cell SA statistics - Cell SA statistics (PM) functionality.
- cipheringGea3 - Ciphering GEA3 - Ciphering GEA3.
- configCharging - Configuration charging - Configurable charging functionality. Warning: For Nokia personnel only.
- controlledRoaming - Controlled roaming - Controlled roaming functionality.
- conversationalQoS - Conversational QoS - Conversational QoS functionality.
- csServices - CC services - CC services.
- directTunnelLicence - Direct tunnel licence - Direct tunnel functionality.
- dualStackPdp - Dual stack PDP - Dual stack PDP context support functionality.
- edgeSupport - Edge support - Edge support functionality.
- equivalentPlmn - Equivalent PLMN - Equivalent PLMN network selection services functionality.
- extdGtpTmrRoaming - Extend GTP timer roaming - Extend GTP timer roaming.
- externalNaccSupport - External NACC support - RIM procedures - external NACC support.
- faultyImeisv - Faulty IMEI Sv - Enable faulty IMEISV UE functionality.
- gbAttachCapacity - Gb attach capacity - Gb attach capacity. Warning: For Nokia personnel only.
- gbOverIp - Gb overlap - Gb over IP functionality.
- ggsnImei - GGSN IMEI - IMEI based GGSN selection functionality.
- gtpInfoSending - GTP information sending - GTP information sending.
- gwIpAddrSv - Gateway IP address Sv - Gateway IP address Sv. Warning: For Nokia personnel only.
- hlr2g3gAccCtr - HLR 2G/3G access control - HLR 2G/3G access control.
- homeGgsnSolution - Home GGSN solution - Home GGSN solution functionality.
- hsdpa168m - HSDPA 168M - HSDPA 168M functionality.
- hsdpa21m - HSDPA 21M - HSDPA 21M functionality.
- hsdpa28m - HSDPA 28M - HSDPA 28M functionality.
- hsdpa42m - HSDPA 42M - HSDPA 42M functionality.
- hsdpa84m - HSDPA 84M - HSDPA 84M functionality.
- hsdpaSupport - HSPDA support - HSPDA support functionality.

- hsdpaSupport12M - HSDPA support 12M - HSDPA support 12M functionality.
- hsdpaSupport16M - HSDPA support 16M - HSDPA support 16M functionality.
- hsdpaSupport4M - HSDPA support 4M - HSDPA support 4M functionality.
- hspaStatistics - HSPA statistics - HSPA statistics functionality.
- hsupa11m - HSUPA 11M - HSUPA 11M functionality.
- hsupa23m - HSUPA 23M - HSUPA 23M functionality.
- hsupa46m - HSUPA 46M - HSUPA 46M functionality.
- hsupaSupport1500k - HSUPA support 1500K - HSUPA support 1500K functionality.
- hsupaSupport2M - HSUPA support 2M - HSUPA support 2M functionality.
- hsupaSupport6M - HSUPA support 6M - HSUPA support 6M functionality.
- hsupaSupport8M - HSUPA support 8M - HSUPA support 8M functionality.
- iHspaEnabler - iHSPA enabler - I-HSPA enabler functionality.
- imeiRestrDetTimer - IMEI restr det timer - IMEI restricted detach timer functionality.
- imeisvInCdr - IMEISlvn CDR - IMEISV in CDR functionality.
- implmeiChk - Imp IMEI check - Enhanced IMEI check control functionality.
- imsiBasedHomeGgsnS - IMSI based home GGSNs - IMSI based home GGSNs selection functionality.
- imsiBasedOffload - IMSI Based Offload
- intGwSelRoamSubs - Int gateway selection roam subscribers - Int gateway selection roam subscribers.
- ipMultimediaSupport - IP multimedia support - IP multimedia support.
- ipv6UserLayer - IPv6 user layer - IPv6 user layer.
- iuAttachCapacity - Iu attach capacity - Iu attach capacity. Warning: For Nokia personnel only.
- lacBasedEplmn - LAC based equivalent PLMN - LAC based equivalent PLMN functionality.
- largeRaSupport - Large RA support - Large RA support.
- lcs - Location services - Location services. Warning: For Nokia personnel only.
- locInfoInScdr - Location information in S-CDR - Location information in S-CDR. Warning: For Nokia personnel only.
- localBreakout - Local breakout - Local breakout.
- localUiIntoPm - Local ui inform PM - Local ui inform PM. Warning: For Nokia personnel only.
- lrasEnhancement - LRAS enhancement - LRAS enhancement functionality.

- lteRoamQosControl - LTE roaming QoS control - LTE roaming QoS control.
- m3uaLoadMonitoring - M3UA load monitoring - M3UA load monitoring. Warning: For Nokia personnel only.
- mmCause2 - MM cause 2 - MM cause 2 (PM) functionality.
- mobilityManCause - Mobility management cause - Mobility management cause (PM) functionality.
- mocn - Multiple operator core network - Multiple operator core network functionality.
- mpAForMsc - Multipoint A support for MSC - SGSN Multipoint A support for MSC
- mpGb - MP gb - Multipoint gb functionality.
- mplu - MPLU - Multipoint iu functionality.
- mpluForMsc - Multipoint lu support for MSC - SGSN Multipoint lu support for MSC
- msisdNBasedGwCc - MSISDN based GW selection with CC
- msisdNBasedPgwSel - MSISDN based PQW selection - SGSN MSISDN based PGW selection.
- multiPdpCtxSubs - Multiple PDP context subscribers - Multiple PDP context for one subscriber functionality.
- multiSim - Multi-SIM - Multi-SIM.
- multipGbNri - Multip gb NRI - Multip gb NRI.
- multipointDns - Multipoint DNS - DNS query optimization for SGSN pool functionality.
- neLevelIpv6 - Network element level IPv6 - Network element level IPv6.
- nearestGgsn - Nearest GGSN - Nearest GGSN selection functionality.
- networkIdTimeZone - Network id time zone - Network id time zone.
- noCsDet - No CS determination - SGSN enabling blacklisted mobile to have CS service.
- nonStdGbp - Non-Standard gb OLP - Non-Standard gb over IP functionality.
- nseAmountInc2 - NSE amount inc 2 - NSE amount increase II functionality.
- numberOf2GCells - Number of 2G cells - Number of 2G cells.
- numberOfRncs - Number of RNCs - Number of RNCs.
- pagingOpt - Paging opt - Paging optimizations functionality.
- papuBloMaint - PAPU blo maint - PAPS blocking for maintenance functionality. Warning: For Nokia personnel only.
- pdpContextCapacity - PDP context capacity - PDP context capacity. Warning: For Nokia personnel only.

- pdpCtxMassDea - PDP context mass detach - PDP context mass deactivation protection functionality.
- plmnStatistics - PLMN level statistics - Home and foreign PLMN level statistics functionality.
- preventScdrForDt - Prevent S-CDR for direct tunnel - Prevent S-CDR for direct tunnel.
- psOperBarring - PS oper barring - PS oper barring.
- qosCtrRoamSub - QoS ctr roam subscriber - QoS control for roaming subscribers functionality.
- qqMessagingSupport - QQ message support - QQ message support. Warning: For Nokia personnel only.
- r8DnsForR7EpcSubs - R8 DNS for R7 EPC subscribers - R8 DNS query support for gn-sgsn functionality.
- r9RimFor2g3g - R9 RIM for 2G/3G - R9 NACC/RIM for 2G/3G support functionality.
- ranapCause - RANAP cause - RANAP cause (PM) functionality.
- roundTripTime - Round trip time - Round trip time counter (Advanced MM/SM) functionality.
- s13ImeiChecking - S13 IMEI checking - S13 IMEI checking.
- sccpLoadshAcross128 - SCCP load share across 128 - SCCP load share across 128. Warning: For Nokia personnel only.
- security - Security.
- selDetachTimer - Sel detach timer - Selective detach timer.
- sessionManCause - Session management cause - Session management cause (PM) functionality.
- sessionManCause2 - Session management cause 2 - Session management cause 2 (PM) functionality.
- sg01074OverReqApn - SG01074 over req APN - Override of requested APN functionality.
- sg02019OverReqApn - SG02019 over req APN - IMEI based APN override functionality.
- sgsnDetachTimer - SGSN detach timer - SGSN detach timer.
- signalReductM2m - Signaling reduction mobile to mobile - Signaling reduction for stationary devices (M2M) functionality.
- ss7NetworkSharing - SS7 network sharing - SS7 network sharing functionality.
- streamQualOpt - Stream qual opt - Streaming QoS optimizations functionality.
- streamingQoS - Streaming QoS - Streaming QoS functionality.

- subsOffloading - Subscriber offloading - Subscriber offloading.
- termManagSupp - Term management support - Term management support.
- throughputCap - Throughput capacity - Throughput capacity. Warning: For Nokia personnel only.
- totalAttachCapacitylocalCgf - Total attach capacity local CGF - Total attach capacity local CGF. Warning: For Nokia personnel only.
- ueCapBasedGwSel - UE capability based gateway selection - UE capability based gateway selection.
- updQosAboveHlr - Update QoS above HLR - Update QoS above HLR.

Roles

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

4.133 cmm firewallRule

Firewall Rules Configuration. Syntax, description, parameters, and roles for cmm firewallRule command.

Syntax

Description

This cli will give operator a alternative way for defining Dynamic firewall rule. Before, IP table is automatically formed by system

Parameters

Name

ruleName - firewall rule name

Syntax

--ruleName <string>

Description

Customer defined firewall rule name

Class

Required key.

Type / Range

Digit string of length 1 to 50 with the allowed pattern of [0-9a-zA-Z\.\-_:]*

Name

dport - Destination Port(s)

Syntax

--dport <string>

Description

destination ports

Class

Optional.

Type / Range

Digit string of length 0 to 11 with the allowed pattern of ((([0-9]{1,4})[1-5][0-9]{4}|6[0-4][0-9]{3}|65[0-4][0-9]{2}|655[0-2][0-9]|6553[0-5])|((([0-9]{1,4})[1-5][0-9]{4}|6[0-4][0-9]{3}|65[0-4][0-9]{2}|655[0-2][0-9]|6553[0-5])-([0-9]{1,4})[1-5][0-9]{4}|6[0-4][0-9]{3}|65[0-4][0-9]{2}|655[0-2][0-9]|6553[0-5])))

Name

externalIp - Source IP address

Syntax

--externalIp <string>

Description

Source IP address on which the firewall rule will apply (external IP)

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 43 with the allowed pattern of [0-9a-fA-F\.:/*]

Name

firewallProtocol - Protocol

Syntax

--firewallProtocol <list-item>

Description

the protocol of rule apply

Class

Optional.

List

- all - All

- tcp - TCP
- udp - UDP

Default

all

Name

ruleResult - firewall rule effect

Syntax

--ruleResult <list-item>

Description

Determine the effect type of the firewall rule applied

Class

Optional.

List

- allow - Allow.
- block - Block.

Default

block

Name

sport - Source Port(s)

Syntax

--sport <string>

Description

source port

Class

Optional.

Type / Range

Digit string of length 0 to 11 with the allowed pattern of ((([0-9]{1,4})[1-5][0-9]{4})6[0-4][0-9]{3}|65[0-4][0-9]{2}|655[0-2][0-9]|6553[0-5])((([0-9]{1,4})[1-5][0-9]{4})6[0-4][0-9]{3}|65[0-4][0-9]{2}|655[0-2][0-9]|6553[0-5])-(|[0-9]{1,4})[1-5][0-9]{4}|6[0-4][0-9]{3}|65[0-4][0-9]{2}|655[0-2][0-9]|6553[0-5]))

Roles**Create**

sysadmin

Show

sysadmin, manage, monitor

Modify

sysadmin

Delete

sysadmin

List

sysadmin, manage, monitor

4.134 cmm fluentdTcpConfig

PM FluentD TCP Streaming Option Configuration. Syntax, description, parameters, and roles for cmm fluentdTcpConfig command.

Syntax

Description

This command is used to add or delete IPs and Ports for the tcp input fluentD plugin. In addition the security aspects are also controlled via this command.

Parameters

Name

name - tcp socket name

Syntax

--name <string>

Description

This is the name of the tcp socket for the fluentD.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

fluentdIp - FluentD IP

Syntax

--fluentdIp <string>

Description

This parameter defines the IP to be used for the tcp input fluentD plugin. The number of the IPs could be up to 1.

Class

Required on create.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

fluentdPort - FluentD port

Syntax

--fluentdPort <integer>

Description

This parameter defines the port to be used for the tcp input fluentD plugin

Class

Required on create.

Type / Range

Integer in the range of 1,024 to 65,535

Name

oamTlsProfileName - OAM TLS Profile

Syntax

--oamTlsProfileName <string>

Description

This parameter contains the profile for TLS. It contains the certificate names needed for establishing secure communication.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

securityProtocol - Security Protocol

Syntax

--securityProtocol <list-item>

Description

This parameter defines the security protocol for the communication between CMM and tcp input fluentD plugin.

Class

Optional.

List

- plaintext - Plain Text - Plain text would be the protocol to communicate with tcp input fluentD plugin
- ssl - SSL would be the protocol to communicate with tcp input fluentD plugi

Default

plaintext

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.135 cmm forbiddenRszc

Forbidden Zone Code PLMN List. Syntax, description, parameters, and roles for cmm forbiddenRszc command.

Syntax**Description**

This command is used to administer a Forbidden Zone Code PLMN list and includes parameters for defining the MCC, MNC, and regional subscription zone code for the forbidden network.

Parameters**Name**

plmnName - Mobile network name

Syntax

--plmnName <string>

Description

Name that identifies served mobile network.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

zcId - Zone code

Syntax

--zcId <integer>

Description

Regional subscription zone code.

Class

Required key.

Type / Range

Integer in the range of 0 to 127

Name

mcc - Mobile country code

Syntax

--mcc <digits>

Description

Mobile country code of forbidden network.

Class

Required key.

Type / Range

Digit string of length 3

Name

mnc - Mobile network code

Syntax

--mnc <digits>

Description

Mobile network code of forbidden network.

Class

Required key.

Type / Range

Digit string of length 2 to 3

Name

mncDigits - Number of MNC digits

Syntax

--mncDigits <list-item>

Description

Number of MNC digits of forbidden network.

Class

Optional.

List

- 2/3Digits - 2 or 3 digits - 2 or 3 digits.
- 2Digits - 2 digits - 2 digits.
- 3Digits - 3 digits - 3 digits.

Default

3Digits

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.136 cmm forbiddenRszcList

Restricted Zone Code per Served PLMN. Syntax, description, parameters, and roles for cmm forbiddenRszcList command.

Syntax

Description

This command is used to administer a Restricted Zone Code per Served PLMN.

Parameters

Name

plmnName - Mobile network name

Syntax

--plmnName <string>

Description

Name that identifies served mobile network.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

zcId - ZoneCode

Syntax

--zcId <integer>

Description

Regional subscription zone code.

Class

Required key.

Type / Range

Integer in the range of 0 to 127

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.137 cmm forbiddenTailList

List of forbidden TAls. Syntax, description, parameters, and roles for cmm forbiddenTailList command.

Syntax

Description

Administer List of forbidden TAIs.

Parameters

Name

areaListName - Name of TAI list

Syntax

--areaListName <string>

Description

Name of forbidden TAI list.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

servedPlmnName - Served mobile network name

Syntax

--servedPlmnName <string>

Description

Served mobile network name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

tac - Tracking area code

Syntax

--tac <integer>

Description

Tracking area code.

Class

Required key.

Type / Range

Integer in the range of 1 to 65,535

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.138 cmm forcelpChange

force IP address Change. Syntax, description, parameters, and roles for **cmm forceIpChange** command.

Syntax

Description

Use this cli to manage the forced UE IP change operations in RA under a specific district (for example, initiate forced IP change, terminate forced IP change and interrogate the number of subscribers in RA) etc.

Parameters

Name

rac - rac

Syntax

--rac <integer>

Description

Routing Area Code

Class

Required key.

Type / Range

Integer in the range of 0 to 255

Name

lac - lac

Syntax

--lac <integer>

Description

Location Area Code

Class

Required key.

Type / Range

Integer in the range of 1 to 65,533

Name

mcc - mcc

Syntax

--mcc <digits>

Description

Mobile Country Code digits

Class

Required key.

Type / Range

Digit string of length 3 to 3 with the allowed pattern of [0-9]*

Name

mnc - mnc

Syntax

--mnc <digits>

Description

Mobile Network Code digits

Class

Required key.

Type / Range

Digit string of length 2 to 3 with the allowed pattern of [0-9]*

Name

papsId - papsid

Syntax

--papsId <integer>

Description

PAPS id

Class

Read only.

Type / Range

Integer in the range of 0 to 19

Name

progressSubs - forced IP count progress

Syntax

--progressSubs <integer>

Description

forced ms/ue count per paps

Class

Read only.

Type / Range

Integer in the range of 0 to 1,000,000

Default

0

Name

statusCode - forced IP change status

Syntax

--statusCode <string>

Description

RAI forced IP change status reported by each PAPS. Use modify command to set this parameter to STOP, if you want to terminate an ongoing RAI forced IP change command

Class

Optional.

Type / Range

Digit string of length

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.139 cmm fqdnLabelMatchProfile

labelMatchProfile. Syntax, description, parameters, and roles for **cmm fqdnLabelMatchProfile** command.

Syntax

Description

Provision the Service Combinations for which fqdn label matching is enabled to when searching for SPGW pair . It is also used to provision the number of fqdn labels to match when searching for SGW-PGW pair. The number of labels to match applies commonly for all Service Combinations for which fqdn label matching is enabled. Value zero (0) for minLabelMatchValue means that minimum label matching is disabled

Parameters

Name

name - name

Syntax

--name <string>

Description

Identifies the name of the fqdn label match profile

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

minLabelMatchValue - minLabelMatchValue

Syntax

--minLabelMatchValue <integer>

Description

Minimum label match Value to be used across all Service Combinations when filtering out SPGW pairs Value 0 (default) means that min fqdn label matching is disabled

Class

Optional.

Type / Range

Integer in the range of 0 to 10

Default

0

Name

ncNrMinLabelMatch - nc nr Minimum Label Match

Syntax

--ncNrMinLabelMatch <list-item>

Description

This parameter defines the minimum matching fqdn labels between SGW and PGW fqdn(s) that is used for filtering out the candidate SPGW pairs found by DNS query when requested nc.nr service parameters. (Default) Value false means that minimum label matching is disabled for this combination of Service Parameters

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ncNrSmfMinLabelMatch - ncNrSmfMinLabelMatch

Syntax

--ncNrSmfMinLabelMatch <list-item>

Description

This parameter defines the minimum matching fqdn labels between SGW and PGW fqdn(s) that is used for filtering out the candidate SPGW pairs found by DNS query when requested nc.nr.smf service parameters. (Default) Value false means that minimum label matching is disabled for this combination of Service Parameters

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ncNrSmfUutMinLabelMatch - ncNrSmfUutMinLabelMatch

Syntax

--ncNrSmfUutMinLabelMatch <list-item>

Description

This parameter defines the minimum matching fqdn labels between SGW and PGW fqdn(s) that is used for filtering out the candidate SPGW pairs found by DNS query when requested nc.nr.smf+ue-<uut> service parameters. (Default) Value false means that minimum label matching is disabled for this combination of Service Parameters

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ncNrUutMinLabelMatch - nc nr uut Minimum Label Match

Syntax

--ncNrUutMinLabelMatch <list-item>

Description

This parameter defines the minimum matching fqdn labels between SGW and PGW fqdn(s) that is used for filtering out the candidate SPGW pairs found by DNS query when requested nc.nr.+ue- service parameters. (Default) Value false means that minimum label matching is disabled for this combination of Service Parameters

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ncSmfMinLabelMatch - ncSmfMinLabelMatch

Syntax

--ncSmfMinLabelMatch <list-item>

Description

This parameter defines the minimum matching fqdn labels between SGW and PGW fqdn(s) that is used for filtering out the candidate SPGW pairs found by DNS query when requested

nc.smf service parameter. (Default) Value false means that minimum label matching is disabled for this combination of Service Parameters

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ncSmfUutMinLabelMatch - ncSmfUutMinLabelMatch

Syntax

--ncSmfUutMinLabelMatch <list-item>

Description

This parameter defines the minimum matching fqdn labels between SGW and PGW fqdn(s) that is used for filtering out the candidate SPGW pairs found by DNS query when requested nc.smf+Uut service parameters. (Default) Value false means that minimum label matching is disabled for this combination of Service Parameters

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

noModifiers - No Modifiers

Syntax

--noModifiers <list-item>

Description

This parameter is used for filtering out the candidate SPGW pairs found by DNS query when requested +noModifiers- service parameters. (Default) Value false means that minimum label matching is disabled for this combination of Service Parameters

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ueLabelMatch - UE Label Match

Syntax

--ueLabelMatch <list-item>

Description

This parameter defines the minimum matching fqdn labels between SGW and PGW fqdn(s) that is used for filtering out the candidate SPGW pairs found by DNS query when requested +ue- service parameters. (Default) Value false means that minimum label matching is disabled for this combination of Service Parameters

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.140 cmm fqdnLabelMatchUut

fqdnLabelMatchUeUsageType. Syntax, description, parameters, and roles for cmm fqdnLabelMatchUut command.

Syntax

Description

Provision the UE Usage Type values for which min fqdn label matching applies. When minFqdnLabelMatch = false for specific UE Usage Type then label matching doesn't apply for that UE Usage Type

Parameters

Name

ueUsageType - UE Usage Type

Syntax

--ueUsageType <integer>

Description

This parameter defines the UE Usage Type associated with the configured min fqdn label matching parameter

Class

Required key.

Type / Range

Integer in the range of 1 to 126 or 128 to 255

Name

minFqdnLabelMatch - minfqdnLabelMatch

Syntax

--minFqdnLabelMatch <list-item>

Description

This parameter defines if min fqdn label matching needs be applied for associated UE Usage Type. (Default) Value false means that minimum label matching is disabled for the associated Ue Usage Type

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.141 cmm freqCountMonitoring

Frequency Count Monitoring. Syntax, description, parameters, and roles for cmm freqCountMonitoring command.

Syntax

Description

Manage MME actions related to frequency count monitoring functionality. The command supports three tasks. A task is created for each action. The create task dumps/exports currently blacklisted UEs into created file. The delete task deletes/purges all behavior contexts i.e. clears any blacklisting related information from MME internal storage. After performing this, blacklisted UEs can again access the MME (normal subscriber data is maintained). The count task outputs the number of blacklisted UEs.

Parameters

Name

freqMonTask - Frequency Monitoring Task

Syntax

--freqMonTask <list-item>

Description

Defines frequency count monitoring tasks to be executed

Class

Required key.

List

- blacklistedUeCount - Blacklisted Ue Count - Request blacklisted Ue count

- createBlacklistedUeFile - Create blacklisted UE file - Request to create a file with blacklisted UEs
- deleteBehaviorData - Request to delete all MME stored behavior data (for the UEs that have previously attempted to access the MME, blacklisted or not yet blacklisted)

Name

blacklistedUeCount - Blacklisted UE count

Syntax

--blacklistedUeCount <integer>

Description

The number of blacklisted UEs

Class

Read only.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

freqMonStatus - Frequency Monitoring Status

Syntax

--freqMonStatus <string>

Description

Shows current status of task

Class

Read only.

Type / Range

Digit string of length 0 to 64

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

4.142 cmm freqCountMonitoringProfile

Frequency Count Monitoring Profile. Syntax, description, parameters, and roles for cmm freqCountMonitoringProfile command.

Syntax

Description

This profile defines parameters for frequency count analysis. MME considers UE attempts for defined procedure(s) abnormally frequent if the number of attempts exceed the provisioned value during a provisioned time period. MME supports max 128 profiles.

Parameters

Name

name - Name

Syntax

--name <string>

Description

The name of the frequency count monitoring profile.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

attempts - Attempts

Syntax

--attempts <integer>

Description

Defines the number of ALL the procedure(s) attempts allowed for a single UE within the defined time interval before UE is blacklisted.

Class

Optional.

Type / Range

Integer in the range of 1 to 1000

Default

5

Name

emmCc - EMM Cause Code

Syntax

--emmCc <integer>

Description

The EMM cause code used on Attach Rejects and Detach Requests.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

7

Name

esmCc - ESM cause Code

Syntax

--esmCc <integer>

Description

The ESM Cause Code used on PDN Connectivity Reject.

Class

Optional.

Type / Range

Integer in the range of 8, 26-39, 41-47, 49-60, 65-66, 81, 95-101, 111-113

Default

55

Name

interval - Interval

Syntax

--interval <integer>

Description

Defines the monitoring interval within which the MME monitors the number of procedure attempts from a UE.

Class

Optional.

Type / Range

Integer in the range of 1 to 3,600 in units of Second

Default

60

Name

monitorAttaches - Monitor Attaches

Syntax

--monitorAttaches <list-item>

Description

Enables Attach procedure monitoring.

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

monitorPdnConnReqs - Monitor PDN Connectedness Requests

Syntax

--monitorPdnConnReqs <list-item>

Description

Enables monitoring of Standalone PDN Connectivity Requests.

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

monitorServiceReqs - Monitor Service Requests

Syntax

--monitorServiceReqs <list-item>

Description

Enables monitoring of Service Requests (including Control Plane Service Requests and Extended Service Requests).

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

serviceRejectCause - Service Reject Cause

Syntax

--serviceRejectCause <integer>

Description

The EMM cause used for Service Rejects.

Class

Optional.

Type / Range

Integer in the range of 3,6-12,35,40,42

Default

7

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.143 cmm gaParams

Ga interface parameters. Syntax, description, parameters, and roles for cmm gaParams command.

Syntax**Description**

Use this command to configure parameters specific to the Ga interface.

Parameters

Name

gaParamsId - Ga Parameters set ID

Syntax

--gaParamsId <integer>

Description

Ga Parameters set ID. Exactly one Parameter set is supported.

Class

Required key.

Type / Range

Integer in the range of 1 to 1

Name

cdrResendDelay - CDR resend delay

Syntax

--cdrResendDelay <integer>

Description

CDR resend delay in seconds

Class

Optional.

Type / Range

CDR resend delay in seconds

Default

5

Name

cdrRetransmissions - CDR retransmission count

Syntax

--cdrRetransmissions <integer>

Description

CDR retransmission count on Ga interface

Class

Optional.

Type / Range

CDR retransmission count

Default

5

Name

dcpmException - DCPM exception handling

Syntax

--dcpmException <list-item>

Description

CDR exception handling (default instruction to alternative CGs on what to do in case of potentially duplicated CDR packets: values R (release) or C (cancel). Default R).

Class

Optional.

List

- cancel - DCPM on Ga sends CANCEL command in case of permanent CG failure
- release - DCPM on Ga sends RELEASE command in case of permanent CG failure

Default

release

Name

gaBackupOnDisk - Ga backup on Disk

Syntax

--gaBackupOnDisk <list-item>

Description

Ga backup on disk in case of risk to lose CDRs.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

gaEchoInt - Ga ECHO Interval

Syntax

--gaEchoInt <integer>

Description

Ga ECHO interval in seconds.

Class

Optional.

Type / Range

Ga echo interval

Default

120

Name

gaT3res - Ga T3 ECHO response

Syntax

--gaT3res <integer>

Description

Ga T3 ECHO response in seconds.

Class

Optional.

Type / Range

Ga Echo T3 Response

Default

3

Name

gaTick - Ga transmission period

Syntax

--gaTick <integer>

Description

Ga transmission period in steps of 0.01 seconds.

Class

Optional.

Type / Range

CDR transmission period on Ga interface in steps of 0.01 seconds

Default

1

Name

gaTickVolume - Ga transmission volume

Syntax

--gaTickVolume <integer>

Description

Ga CDR packets per tick transmission volume.

Class

Optional.

Type / Range

Transmitted CDR packets per tick, on Ga interface

Default

10

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.144 cmm gbBvcReset

Gb BVC reset. Syntax, description, parameters, and roles for cmm gbBvcReset command.

Syntax**Description**

The purpose is to trigger the BVCRESET procedure in order to reset the traffic to the sleeping cells.

Parameters**Name**

bvci - BVC index

Syntax

--bvci <integer>

Description

BSSGP virtual connection id (requires NSEI).

Class

Required on create.

Type / Range

Integer in the range of 2 to 65,535

Name

nsei - NSEI 0-65535

Syntax

--nsei <string>

Description

Network service entity identifier(s).

Class

Required on create.

Type / Range

Digit string of length 0 to 30 with the allowed pattern of ((([0-9]|[1-8][0-9]|9[0-9]|[1-8][0-9]{2}|9[0-8][0-9]|99[0-9]|[1-8][0-9]{3}|9[0-8][0-9]{2}|99[0-8][0-9]|999[0-9]|[1-5][0-9]{4}|6[0-4][0-9]{3}|65[0-4][0-9]{2}|655[0-2][0-9]|6553[0-5]))([,-]([0-9]|[1-8][0-9]|9[0-9]|[1-8][0-9]{2}|9[0-8][0-9]|99[0-9]|[1-8][0-9]{3}|9[0-8][0-9]{2}|99[0-8][0-9]|999[0-9]|[1-5][0-9]{4}|6[0-4][0-9]{3}|65[0-4][0-9]{2}|655[0-2][0-9]|6553[0-5]))))*

Name

cause - Cause

Syntax

--cause <string>

Description

Cause successful/unsuccessful.

Class

Optional.

Type / Range

Digit string of length

Roles**Create**

sysadmin, manage

4.145 cmm gbLinkInfo

GB Link Info. Syntax, description, parameters, and roles for cmm gbLinkInfo command.

Syntax

Description

GB Link Info.

Parameters

Name

adminState - Admin state

Syntax

--adminState <string>

Description

Administrative state.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

ldw - Local data weight

Syntax

--ldw <integer>

Description

Local data weight.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Name

lsw - Local sign weight

Syntax

--lsw <integer>

Description

Local sign weight.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Name

nsei - NSEI

Syntax

--nsei <string>

Description

Network service entity identifier.

Class

Optional.

Type / Range

Digit string of length 0 to 30 with the allowed pattern of ((([0-9]|[1-8][0-9]|9[0-9]|[1-8][0-9]{2}|9[0-8][0-9]|99[0-9]|[1-8][0-9]{3}|9[0-8][0-9]{2}|99[0-8][0-9]|999[0-9]|[1-5][0-9]{4}|6[0-4][0-9]{3}|65[0-4][0-9]{2}|655[0-2][0-9]|6553[0-5]))([, -]((([0-9]|[1-8][0-9]|9[0-9]|[1-8][0-9]{2}|9[0-8][0-9]|99[0-9]|[1-8][0-9]{3}|9[0-8][0-9]{2}|99[0-8][0-9]|999[0-9]|[1-5][0-9]{4}|6[0-4][0-9]{3}|65[0-4][0-9]{2}|655[0-2][0-9]|6553[0-5]))))*

Name

nsvci - NSVC

Syntax

--nsvci <integer>

Description

Network service virtual connection identifier.

Class

Optional.

Type / Range

Integer in the range of 0 to 65,535

Name

opState - Operational state

Syntax

--opState <integer>

Description

Operational state.

Class

Optional.

Type / Range

Integer in the range of 1 to 4

Name

papsId - PapsId

Syntax

--papsId <integer>

Description

PAPS index.

Class

Optional.

Type / Range

Integer in the range of 0 to 19

Name

rdw - Remote data weight

Syntax

--rdw <integer>

Description

Remote data weight.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Name

remotelp - Remote IP endpoint

Syntax

--remoteIp <string>

Description

Remote IP endpoint.

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

remotePort - Remote udp port

Syntax

--remotePort <integer>

Description

Remote UDP port number.

Class

Optional.

Type / Range

Integer in the range of 0 to 65,535

Name

rsw - Remote sign weight

Syntax

--rsw <integer>

Description

Remote sign weight.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Roles**List**

sysadmin, manage, monitor

4.146 cmm gbNetworkInfo

Gb Network Info. Syntax, description, parameters, and roles for cmm gbNetworkInfo command.

Syntax**Description**

BVCIs of LAI, nse_bvci_pair_of_cell_req_s = 0xB7C3.

Parameters**Name**

bvci - BVC index

Syntax

--bvci <integer>

Description

BSSGP virtual connection id (requires NSEI).

Class

Optional.

Type / Range

Integer in the range of 2 to 65,535

Name

ci - Cell id

Syntax

--ci <integer>

Description

Cell identity, 0-65535.

Class

Optional.

Type / Range

Integer in the range of 0 to 65,535

Name

epind - Equivalent PLMN list index

Syntax

--epind <integer>

Description

.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Name

lac - LAC

Syntax

--lac <string>

Description

Location area code(s), use with MCC and MNC.

Class

Optional.

Type / Range

Digit string of length with the allowed pattern of ([1-9][1-8][0-9]9[0-9][1-8][0-9]{2}9[0-8][0-9]99[0-9][1-8][0-9]{3}9[0-8][0-9]{2}99[0-8][0-9]999[0-9][1-5][0-9]{4}6[0-4][0-9]{3}65[0-4][0-9]{2}655[0-2][0-9]6553[0-3])([,-]([1-9][1-8][0-9]9[0-9][1-8][0-9]{2}9[0-8][0-9]99[0-9][1-8][0-9]{3}9[0-8][0-9]{2}99[0-8][0-9]999[0-9][1-5][0-9]{4}6[0-4][0-9]{3}65[0-4][0-9]{2}655[0-2][0-9]6553[0-3]))*

Name

mcc - MCC

Syntax

--mcc <digits>

Description

MCC digits of broadcasted PLMN.

Class

Optional.

Type / Range

Digit string of length 3

Name

mnc - MNC

Syntax

--mnc <digits>

Description

MNC digits of broadcasted PLMN.

Class

Optional.

Type / Range

Digit string of length 2 to 3

Name

nsei - NSEI 0-65535

Syntax

--nsei <string>

Description

Network service entity identifier(s).

Class

Optional.

Type / Range

Digit string of length 0 to 30 with the allowed pattern of ((([0-9][1-8][0-9]9[0-9][1-8][0-9]{2}9[0-8][0-9]99[0-9][1-8][0-9]{3}9[0-8][0-9]{2}99[0-8][0-9]999[0-9][1-5][0-9]{4}6[0-4][0-9]{3}65[0-4][0-9]{2}655[0-2][0-9]6553[0-5]))([,-]([0-9][1-8][0-9]9[0-9][1-8][0-9]{2}9[0-8][0-9]99[0-9][1-8][0-9]{3}9[0-8][0-9]{2}99[0-8][0-9]999[0-9][1-5][0-9]{4}6[0-4][0-9]{3}65[0-4][0-9]{2}655[0-2][0-9]6553[0-5])))*

Name

papsId - PAPS index

Syntax

--papsId <integer>

Description

PAPS index (given as the only parameter).

Class

Optional.

Type / Range

Integer in the range of 0 to 19

Name

rac - RoutingAreaCode

Syntax

--rac <string>

Description

Routing area code(s), use with MCC, MNC and LAC.

Class

Optional.

Type / Range

Digit string of length with the allowed pattern of ((([0-9]|[1-8][0-9]|9[0-9]|1[0-9]{2}|2[0-4][0-9]|25[0-5]))(,[-]([0-9]|[1-8][0-9]|9[0-9]|1[0-9]{2}|2[0-4][0-9]|25[0-5])))*

Roles**List**

sysadmin, manage, monitor

4.147 cmm gbrQciQosProfile

GBR Dedicated Bearer per QCI QoS Profile. Syntax, description, parameters, and roles for cmm gbrQciQosProfile command.

Syntax**Description**

This command is used to administer a QCI QoS Profile for GBR Dedicated Bearers and includes parameters for defining Allocation and Retention Priority (ARP), uplink and downlink bitrates, Pre-emption Capability Indicator (PCI), and Pre-emption Vulnerability Indicator (PVI) values. The 'shutdown' parameter is used to enable/disable the profile.

Parameters

Name

qosProfileName - QoS profile name

Syntax

--qosProfileName <string>

Description

Identifies the enhanced QoS profile.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

gbrDedBrrQci - Guaranteed dedicated bearer QCI

Syntax

--gbrDedBrrQci <integer>

Description

Guaranteed dedicated bearer QoS class identifier.

Class

Required key.

Type / Range

Integer in the range of 1 to 4

Name

gbrDedBrrArp - Guaranteed dedicated bearer ARP

Syntax

--gbrDedBrrArp <integer>

Description

Guaranteed dedicated bearer allocation and retention priority.

Class

Optional.

Type / Range

Integer in the range of 1 to 15

Default

15

Name

gbrDedBrrBitrateDownlink - Guaranteed dedicated bearer downlink bitrate

Syntax

--gbrDedBrrBitrateDownlink <integer>

Description

Guaranteed dedicated bearer downlink bitrate.

Restrictions

The granularity is 8 when the value is between 64 and 568. The granularity is 64 when the value is between 576 and 8640. The granularity is 100 when the value is between 8700 and 16000. The granularity is 1000 when the value is between 17000 and 128000. The granularity is 2000 when the value is between 130000 and 256000.

Class

Optional.

Type / Range

Integer in the range of 1-63, 64-568, 576-8640, 8700-16000, 17000-128000, 130000-256000, 260000-500000, 510000-1500000, 1600000-4000000000 in units of Kbps

Default

128000

Name

gbrDedBrrBitrateUplink - Guaranteed dedicated bearer uplink bitrate

Syntax

--gbrDedBrrBitrateUplink <integer>

Description

Guaranteed dedicated bearer uplink bitrate.

Restrictions

The granularity is 8 when the value is between 64 and 568. The granularity is 64 when the value is between 576 and 8640. The granularity is 100 when the value is between 8700 and 16000. The granularity is 1000 when the value is between 17000 and 128000. The granularity is 2000 when the value is between 130000 and 256000.

Class

Optional.

Type / Range

Integer in the range of 1-63, 64-568, 576-8640, 8700-16000, 17000-128000, 130000-256000, 260000-500000, 510000-1500000, 1600000-4000000000 in units of Kbps

Default

128000

Name

gbrDedBrrMaxBitrateDownlink - Guaranteed dedicated bearer maximum downlink bitrate

Syntax

--gbrDedBrrMaxBitrateDownlink <integer>

Description

Guaranteed dedicated bearer maximum downlink bitrate.

Restrictions

The granularity is 8 when the value is between 64 and 568. The granularity is 64 when the value is between 576 and 8640. The granularity is 100 when the value is between 8700 and 16000. The granularity is 1000 when the value is between 17000 and 128000. The granularity is 2000 when the value is between 130000 and 256000.

Class

Optional.

Type / Range

Integer in the range of 1-63, 64-568, 576-8640, 8700-16000, 17000-128000, 130000-256000, 260000-500000, 510000-1500000, 1600000-4000000000 in units of Kbps

Default

128000

Name

gbrDedBrrMaxBitrateUplink - Guaranteed dedicated bearer maximum uplink bitrate

Syntax

--gbrDedBrrMaxBitrateUplink <integer>

Description

Guaranteed dedicated bearer maximum uplink bitrate.

Restrictions

The granularity is 8 when the value is between 64 and 568. The granularity is 64 when the value is between 576 and 8640. The granularity is 100 when the value is between 8700 and 16000. The granularity is 1000 when the value is between 17000 and 128000. The granularity is 2000 when the value is between 130000 and 256000.

Class

Optional.

Type / Range

Integer in the range of 1-63, 64-568, 576-8640, 8700-16000, 17000-128000, 130000-256000, 260000-500000, 510000-1500000, 1600000-4000000000 in units of Kbps

Default

128000

Name

gbrDedBrrPci - Guaranteed dedicated bearer PCI

Syntax

--gbrDedBrrPci <list-item>

Description

Guaranteed dedicated bearer pre-emption capability indicator.

Class

Optional.

List

- Disabled - PCI disabled - Pre-emption capability disabled.
- Enabled - PCI enabled - Pre-emption capability enabled.

Default

Disabled

Name

gbrDedBrrPvi - Guaranteed dedicated bearer PVI

Syntax

--gbrDedBrrPvi <list-item>

Description

Guaranteed dedicated bearer pre-emption vulnerability indicator.

Class

Optional.

List

- Disabled - PVI disabled - Pre-emption vulnerability disabled.
- Enabled - PVI enabled - Pre-emption vulnerability enabled.

Default

Disabled

Name

shutdown - Shutdown

Syntax

--shutdown <list-item>

Description

Indicates if the profile is active.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

4.148 cmm genCdr

generate CDR. Syntax, description, parameters, and roles for cmm genCdr command.

Syntax**Description**

Generate CDR.

Parameters**Name**

imsi - IMSI

Syntax

--imsi <digits>

Description

IMSI for the forced S-CDR closure.

Class

Required on create.

Type / Range

Digit string of length 5 to 15

Name

nsapi - NSAPI

Syntax

--nsapi <integer>

Description

NSAPI for the forced S-CDR closure.

Class

Optional.

Type / Range

Integer in the range of 5 to 15

Roles**Create**

sysadmin, manage

5 CMM CLI commands: cmm generalGParms - cmm ngReset

CMM CLI command descriptions: syntax, usage, parameters, and user roles.

5.1 cmm generalGParms

General Global Parameters. Syntax, description, parameters, and roles for cmm generalGParms command.

Syntax

Description

This command is used to administer General Global Parameters defined to configure global office data in CMM

Parameters

Name

cmGparmName - Parameter Name

Syntax

--cmGparmName <list-item>

Description

Name of the global parameter

Class

Required key.

List

- adaptiveSafetyNet - Adaptive Safety Net Control - Parameter to control adaptive safety net functionality for CMM. This global parameter controls allows operator to enable or disable a functionality that allows CMM to dynamically calculate the message rate safety net applicable for a CMM deployment. This applies to both CNF and VNF deployments. The dynamic safety net functionality applies to the following: 1. Per IPDS message rate safety net value 2. Total CMM message rate safety net value based on number of IPDSs present 3. Per AMMS and EMMS message rate safety net value
- allowEdrxCarryoverPaging - Allow EDRX Carryover Paging - Specifies whether the MME is allowed to implement carryover paging for an eDRX UE. If set, the settings in the corresponding edrxProfile define the eligible eDRX cycle values for carryover paging
- allowExtScefRefIdS10 - Allow Extended SCEF Reference ID Over S10 - Specifies whether a source MME is allowed to send the Extended SCEF Reference ID over S10 for MONTE event transfer. Also specifies whether a target MME will use the extended Reference ID information if present over S10. The default is false.

- **apnConversionEnhancedMapTreatment** - APN Conversion Enhanced Map Treatment - Parameter controls feature activation GPARAM flag. By default, flag shall be disabled. When enabled, MME shall support following list of enhancements/exceptions to existing custom APN override feature: 1) Support for rejecting with EMM cause code #19 once APN conversion is done upon requested APN does not match HSS subscription profile. 2) Support for handling special APN names (e.g. prepaid.customer.br) and IMS APN (ims.customer.com.br). 3) Support for lowest context id APN in HSS upon requested APN does not exist in HSS. Additionally, two Map Treatment fields under APN IMSI Range Services shall be used to determine the behavior of either rejecting, converting, or using lowest context id APN for various scenarios. This parameter can be enabled only when the Alias APN Conversion Per IMSI global parameter is enabled.
- **applProtocolFallbackN26ToS10** - Application Protocol Fallback from N26 to S10 - Current parameter enables fallback to search for 's10' candidate when DNS Query to find target MME during S10 handover using 'n26' fails. This parameter is expected to be used together with parameter Use N26 Service Parameter in S10 Handover
- **attachEirPercentageChecking** - Attach EIR Percentage Checking - Parameter controls feature activation GPARAM flag which shall be used or activated only to perform IMSI and GUTI initial attach EIR percentage. By default, flag shall be disabled.
- **automatedTlsCertManagement** - Automated TLS Certificate Management - This parameter controls the usage of Automated TLS Certificate Management using cert-manager and Kubernetes Secrets. If ENABLE automated TLS Certificate Management is active.
- **cc100SgwIsolate** - CC100 SGW Isolate - Defines whether the MME isolates SGW when the MME receives CC100/CS=0 in Create Session Response of Attach (including handover Attach). Current parameter takes effect only when parameter lteSgwIsolation is enabled
- **cc100SgwReselect** - CC100 SGW Reselection - Defines whether the MME does SGW reselection when the MME receives CC100/CS=0 in Create Session Response of Attach (including handover Attach). Because PGW reselection happens by default in such case, when current parameter is enabled, the MME does SPGW pair reselection
- **chargingCharAccessRestriction** - Charging Characteristics Access Restriction - Enables the MME to restrict access of UEs based on provisioned charging characteristics value.
- **checkTgtAmfN26StatusN26Ho** - Check Target AMF N26 Status in N26 Handover - Current parameter enables checking the N26 link/interface status of local target AMF node during N26 handover so that if it isn't available to use, then even if the local AMF can serve the target RAN, source MME will select an alternative node to complete the N26 handover.
- **doubleS1N26Mob** - Handle Double S1 at N26 Mobility - Enable processing of double S1 IRAT scenario where UE abandons N26 handover procedure and restarts idle mode IRAT TAU possibly at the same MME that served N26 handover so that the Create Session Request towards the new serving SGW, as part of the idle mode TAU, is sent before

Delete Session Request (Scope Indication) that is sent towards old SGW as part of resource cleanup of N26 handover context. The old and new SGW can be the same.

- **edrxExtendStopPageTime** - eDRX Extend Stop Page Time - Specifies whether the eDRX Stop Page Timer should be extended to be the true end of the paging time window for the UE
- **effectiveLoadForScaling** - Effective Load For Scaling - This parameter is to control effective load scaling feature for CNF Horizontal Auto Scaler. When set to false the feature to compute effective load is disabled. To enable Effective load computation set the parameter to true. This parameter is CNF specific.
- **empsGtpMessagePriorityN26** - MME eMPS GTP Message Priority N26 - Enables MME to use GTP Message Priority value 0 during N26 mobility procedures for priority UEs.
- **empsGtpMpOnlyWhenDscpMapped** - eMPS GTP Message Priority Only When DSCP Mapped - When this parameter is set to true, the MME will indicate the GTP Message Priority only when the GTP message involves a bearer whose QCI and ARP PL values map to a DSCP value on the eMPS DSCP Marking provisioning command. When set to false, GTP Message Priority is indicated for all GTP messages of the priority UE.
- **epsUip** - EPS UPIP - Current parameter enables support for the EPS-UIPI (User Plane Integrity Protection)
- **extendedEmergencyNumberListZoneCode** - Extended Emergency Number List Zone Code - Indicates provisioned extended emergency list zone code. When feature activation gParm is enabled and upon receiving Regional-Subscription-Zone-Code in UE Subscription data from HSS in ULA/IDR message, if received Zone Code matches provisioned zone code for this feature, MME shall send Extended Emergency Numbers List in the ATTACH and TAU ACCEPT message
- **fallbackFromLocationGatewaySelectionToNormalDns** - Fallback from Location Gateway Selection to Normal DNS - Parameter controls fallback to normal DNS operation. When the parameter is set to true, fallback to normal DNS query without tag is supported. If the parameter is not set (default) fallback is not supported.
- **gtpMpForEmergencyFallbackFrom5g** - GTP MP for Emergency Service Fallback from 5G - Defines whether the MME uses GTP Message Priority (MP) on GTP N26 Context Request towards the AMF, when the UE uses RRC establishment cause 'emergency' on TAU request and the MME sends N26 Context Request to the AMF as part of the procedure.
- **gtpMpValueForEmergency** - GTP MP Value for Emergency - Defines the GTP Message Priority (MP) value the MME uses, when GTP MP is provisioned to be used for emergency services.
- **healthCheckCompare** - Health Check Compare - This parameter is used to control when a comparison between the health check functionality results shall be executed.
- **inclBackoffTimerForImsPdnReqRej** - Include Backoff Timer For IMS PDN Request Reject - Parameter controls feature activation flag. By default, flag shall be disabled. When enabled, MME shall send T3396 back-off timer IE when rejecting IMS PDN Connectivity

Request that is sent either as standalone or as part of Attach procedure due to not subscribed IMS APN with cause code #27.

- **includeNwPolicyInAttachAndTauAccept** - Include Network Policy in Attach and TAU Accept - Parameter controls inclusion of Network Policy IE in Attach Accept and TAU Accept messages. By default, it shall be disabled. When enabled, MME shall include Network Policy IE in Attach Accept and TAU accept messages sent to the UE.
- **includeUliForMmeTauWithoutSgwChange** - Include ULI for MME TAU without SGW Change - Parameter controls inclusion of ULI in S11 message for the following procedure upon enabling the gparm flag. By default, flag shall be disabled. 1) TAU MME no change and no SGW change
- **includeUliForX2AndS1HoWithoutSgwChange** - Include ULI for X2 and S1 HO without SGW Change - Parameter controls inclusion of ULI in S11 messages for following procedures upon enabling the gparm flag. By default, the flag shall be disabled. 1) X2 HO with no SGW change 2) S1 HO with MME change and no SGW change 3) S1 HO with no MME change and no SGW change
- **keepUeCtxtN26Mobility** - Keep UE Context After N26 Mobility - Current parameter enables keeping the UE Context after UE has moved to 5G with N26 mobility (idle or connected mode). When the gParm is disabled then the UE Context is deleted at the end of the mobility procedure to 5G (default behavior).
- **locallyProvisionedImsLbo** - Locally Provisioned IMS LBO - Enables the locally provisioned IMS LBO functionality in the MME.
- **locationGatewaySelection** - Location Gateway Selection. - Parameter controls location based gateway selection. By default, it shall be set to false. When the parameter is set to true, MME will insert a tag (either zone name or TAC) to the DNS query for certain APNs.
- **memoryAlarmForRestrictedPaging** - Memory Overload Alarm for MME Restricted Paging - Memory Overload Alarm at which CMM starts MME Restricted Paging
- **mmeRateControlT6aDI** - MME Rate Control T6a Downlink - Specifies whether the MME is to enforce downlink Serving PLMN Rate Control for T6a bearers. If not set, the SCEF is expected to carry out the rate control. The default is for the MME not to enforce.
- **pdnReactivateN1Enable** - PDN Reactivation when N1 mode gets enabled - Current parameter enables the MME performing PDN deactivation (reactivation requested) for the non-emergency PDNs after TAU completes for which the UE re-enabled N1 mode for those PDN sessions that are established with standalone PGWs but the associated APNs allow 5GS IWK and UE is not 5G Restricted.
- **pdnTypeIpv6Restriction** - PDN Type IPv6 Restriction - Enables MME to restrict the use of PDN Type IPv6 and IPv4v6 for the provisioned UEs.
- **pdnTypeProtocolTypeGwReselection** - PDN Type or Protocol Type GW Reselection - Defines whether the MME reselects an alternate GW during Attach and PDN Connectivity procedures when SGW returns cause code 83 (Preferred PDN type not supported) or 86

(Protocol type not supported). MME reselects either SGW or PGW depending on which one (SGW or PGW) the cause source indicates. Setting this parameter to true requires that both global parameters Support alternate gw reselection feature and Enhanced Alternate GW Reselection are also set to Yes.

- **pmReportHssByName** - PM Report HSS by Name - Specifies whether PM counts reported by HSS are to use the name field from the remote endpoint configuration for that S6a interface. If false, the PM HSS name continues to use the endpoint ID from the remote endpoint configuration.
- **proactiveAdaptiveSafetyNet** - Proactive Adaptive Safety Net - Enables or disables proactive adaptive safety net enhancements, in particular the automatic, proactive lowering of the safety net threshold if it is predicted that the busiest IPDS will reach 80% CPU occupancy before the threshold would have been reached.
- **resendDnsQueryWithoutPdnTypeForTimeout** - Resend DNS Query Without PDN Type for Timeout - Parameter controls feature activation GPARAM flag which by default shall be disabled. When enabled, MME shall upon receiving no response from DNS/timeout shall resend the DNS query by excluding the PDN type information from APN FQDN.
- **resilientIpdsFdsFailure** - System Resilience on IPDS/FDS dual fail - This parameter specifies that system recovery is using the enhanced recovery method for dual IPDS or dual xFDS failure. This method will lock the CMM Aggregate Service if IPDSs and FDSs are not available after dual failure. The system will be unlocked when the normal state is recovered.
- **restorationWithFullBackup** - SRS Restoration with Full Backup - This parameter allows SRS restoration (MME failure) with MME Full Backup of UE Contexts. Previously full backup only allowed restoring Users during ISSU. This feature supports restoral of basic procedures on any MME in the pool, using a Full Context backup. The following global parameters must be enabled prior to enabling Restoration With Full Backup: Support Ue Restoration Context Deletion, Inhibit Peer Srs Responses, Ue Ctx Update In Patch Load Support, Session Ddn Restoration Std Based, and Support Full Ue Restoration.
- **rfspAccessRestriction** - RFSP Access Restriction - Feature control for RFSP Access Restriction for the MME.
- **roamerAttachRateLimitOverride** - Roamer Attach Rate Limit Override - IMSI and foreign GUTI Attaches allowed per second after which CMM starts dropping attaches for roamers until the rate is below the limit. If the Roamer Attach Rate Limit exceeds the general Attach Rate Limit for any reason, the Roamer Attach Rate Limit Override is effectively disabled. This is because the general Attach Rate Limit will be reached before the Roamer Attach Rate Limit. If the value is set to 0, it means the Roamer Attach Rate Limit is disabled. Value needs to be set to non-zero for the feature to be enabled. Refer to the description of the Attach Rate Limit Override global parameter to understand how this value is used in different CMM environments. Roamer Attach Rate Limit Override configuration is for both IPDS and mIPDS.

- **safetyNetVariation** - Safety Net Variation - Allows for variation in safety net in special circumstances. It is ordinarily not changed and should be changed only at the direction of Nokia personnel.
- **sendExtEmerNumberInAttachAndTauAccept** - Send Extended Emergency Number in Attach and TAU Accept - Parameter controls feature activation global parameter flag. By default, flag shall be disabled. When enabled, MME shall send Extended Emergency Numbers List IE in the ATTACH and TAU ACCEPT messages if one is provisioned on the MME.
- **sendOnlyImeiToEir** - Send Only IMEI to EIR - When this parameter is set to true MME will include only the IMEI part towards EIR for authentication even if the IMEI SV is available in the UEs MM context. When set to false the MME will include the IMEISV when this is available.
- **sendOnlyImeiToSgw** - Send Only IMEI to SGW - When this parameter is set to true MME will include only the IMEI towards SGW even if the IMEI SV is available in the UEs MM context. When set to false the MME will include the IMEISV when this is available.
- **sendOnlyRanSupportedPlmnsInS1** - Send Only RAN Supported PLMNs in S1 - Current parameter enables sending in SETUP RSP and MME Configuration Update only the PLMNs that are supported by the RAN even if additional PLMNs are configured at the MME
- **sendSgsMmeIdentityS6a** - Send SGs-MME-Identity AVP on S6a - Specifies whether the MME is to include the SGs-MME-Identity AVP in S6a messaging for a UE that is being registered over SGs. The default is false.
- **separateEplmnListNasS1ap** - Separate EPLMN List on NAS and S1AP - Enable sending separate Equivalent PLMN Lists to NAS and S1AP. The EPLMN Lists applied to NAS and S1AP are copied from EPLMN List Name and EPLMN List Name for eNB, respectively, that are provisioned in IMSI Range Services or UE PLMN Services.
- **serviceReqRateDropLimitOverride** - Service Request Rate Drop Limit Override - Number of allowed 4G UE initiated Service Requests beyond which MME/CMM will start to drop them on the IPDS. Value of 0 means use system calculated Default value. Non-zero means use this value directly.
- **sessionDdnRestorationStdBased** - Session DDN Restoration Standards Based - MME Session DDN Restoration based on Standards
- **sgsLocUpdCombTauAfterCsfb** - SGs Location Update for Combined TAU After CSFB - Current parameter enables the MME doing SGs Location Update when the UE does intra-MME Combined TAU after CSFB even if Old and New LAI are the same.
- **srStormInternalMonitoring** - Service Request Storm Internal Monitoring - This parameter enables internal software resource monitoring to further protect the CMM during Service Request Storms, taking action as necessary to prevent internal resource exhaustion and overloads, including in scenarios such as when SGW overloads result in delays in SGW responses to the CMM, which in turn could result in CMM overloads.

- **srsTrafficReduction** - SRS traffic reduction - This parameter enables using new format (compressed) to send NR10 update requests and NR10 fetch responses. For AMF, all the NR10 Update requests and Fetch responses will follow the new compressed format if the value of this parameter is set to true. For MME, Update Requests and Fetch Responses will follow the new compressed format when the value of this parameter is set to true as well as Support Full UE Restoration is enabled. If there is a need to change the value of this gParm please contact Nokia support team
- **staticPgwSelectionAllPlmns** - Static PGW selection for defined IMSIs on all serving PLMNs - MME supports static PGW selection for defined IMSIs on all available serving PLMNs.
- **tauRateDropLimitOverride** - IRAT tau Rate Drop Limit Override - 5G to 4G IRAT TAUs allowed per sec after which MME start dropping 5G to 4G IRAT TAUs until rate is below limit. Value of 0 means use system calculated Default value. Non-zero means use this value directly.
- **transferTo5gPdnWo5gslwk** - Transfer to 5GS PDN Session without 5GS IWK - Current parameter enables transferring to 5GS a PDN Session that is known that 5GSIWK flag were not sent in Create Session Request during PDN Session Establishment procedure of type handover.
- **ueRadioCapPagingS10** - UE Radio Capability Paging S10 - This gparm specifies whether the MME will include the 'UE Radio Capability for Paging' AVP, if available, in S10 messaging. The default is false. The same gparm specifies whether the MME will accept and store 'UE Radio Capability for Paging' AVP, if available in S10 messaging, in the UE context.
- **ueTzInCreateAndUpdateBearerResp** - UE Time Zone In Create And Update Bearer Response - Parameter controls feature activation GPARM flag. By default, flag shall be disabled. When enabled, MME will include UE Time Zone in S11 Create Bearer and Update Bearer Response messages for both homers and roamers.
- **useApnOiToDetermineHrt** - Use APN-OI to Determine HRT - Current parameter enables using the APN-OI instead of the PGW FQDN to determine if a PDN Session is HRT for a roaming subscriber when applying QoS modification for a PDN session in accordance to local configuration and when sending the UE Time Zone IE in UBR and CBR Response messages.
- **useDnsForN26Ho** - Use Dns for N26 handover - Current parameter enables using DNS for selecting target node as part of N26 handover even when local AMF of quad access node serves target RAN.
- **useLocalRfsspHssRfsspNotSub** - Use Local RFSP if RFSP isn't subscribed at the HSS - Current parameter enables using the locally provisioned RFSP value when local configuration at the uePlmnServices or imsiRangeServices indicates applying the HSS subscribed value but no RFSP value is received from the HSS

- useN26SvcParamS10Ho - Use N26 Service Parameter in S10 Handover - Current parameter enables using N26 service parameter as part of DNS query for target MME selection during S10 handover
- useSameIpDynStaticS11Link - Use Same IP Address for Dynamic and Static S11 Links - Enable using same IP address for dynamic and static S11 links.
- vlrCleanupBasedOnOccupancy - VLR Cleanup Based on Occupancy - Current parameter enables doing VLR cleanup of the VLR Contexts of deregistered UEs based on VLR Occupancy
- wifiHoConnectedUe - Wifi Handover for Connected Ue - When a Delete Bearer Request is received for a default bearer with the cause 'RAT changed from 3GPP to non-3GPP' for an ECM-CONNECTED UE and this is not the last bearer for the UE, instead of doing S1 Release and Release Access Bearer, the MME sends an S1AP ERAB Release Command for all bearers associated with the PDN session, but without including a NAS EPS Deactivate Bearer Context Request in the E-RAB Release.
- zeroOiCsrS10Tau - Set OI flag to Zero in CSR of S10 TAU - Current parameter enables setting OI flag to value zero in Create Session Request that MME sends as part of s10 Idle mobility. When this parameter is disabled, the MME sets OI flag to 1
- zoneCodeExtendedEmergencyNumberList - Zone Code Extended Emergency Number List - Parameter controls feature activation GPARM flag. By default, flag shall be disabled. When enabled, MME shall use the received regional zone code information from HSS to determine whether to send extended emergency number list to UE.
- zoneNameSwitch - Zone Name Switch - Parameter controls whether the mapping between zone names and TAC is enabled (switch on). If the parameter is set, zone name will be used as tag in the APN query. Otherwise, TAI will be used as tag in the APN query.

Name

cmGparmValue - Parameter Value

Syntax

--cmGparmValue <string>

Description

Value of parameter

Class

Required on create.

Type / Range

Digit string of length 0 to 64

The cmGparmValue string is based on the cmGparmName parameter:

Name

adaptiveSafetyNet - Adaptive Safety Net Control

Syntax

```
--cmGparmName adaptiveSafetyNet --cmGparmValue <string>
```

Description

Parameter to control adaptive safety net functionality for CMM. This global parameter controls allows operator to enable or disable a functionality that allows CMM to dynamically calculate the message rate safety net applicable for a CMM deployment. This applies to both CNF and VNF deployments. The dynamic safety net functionality applies to the following: 1. Per IPDS message rate safety net value 2. Total CMM message rate safety net value based on number of IPDSs present 3. Per AMMS and EMMS message rate safety net value

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

allowEdrxCarryoverPaging - Allow EDRX Carryover Paging

Syntax

```
--cmGparmName allowEdrxCarryoverPaging --cmGparmValue <string>
```

Description

Specifies whether the MME is allowed to implement carryover paging for an eDRX UE. If set, the settings in the corresponding edrxProfile define the eligible eDRX cycle values for carryover paging

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

allowExtScefRefIdS10 - Allow Extended SCEF Reference ID Over S10

Syntax

```
--cmGparmName allowExtScefRefIdS10 --cmGparmValue <string>
```

Description

Specifies whether a source MME is allowed to send the Extended SCEF Reference ID over S10 for MONTE event transfer. Also specifies whether a target MME will use the extended Reference ID information if present over S10. The default is false.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Recommended

true

Name

apnConversionEnhancedMapTreatment - APN Conversion Enhanced Map Treatment

Syntax

--cmGparmName apnConversionEnhancedMapTreatment --cmGparmValue
<string>

Description

Parameter controls feature activation GPARM flag. By default, flag shall be disabled. When enabled, MME shall support following list of enhancements/exceptions to existing custom APN override feature: 1) Support for rejecting with EMM cause code #19 once APN conversion is done upon requested APN does not match HSS subscription profile. 2) Support for handling special APN names (e.g. prepaid.customer.br) and IMS APN (ims.customer.com.br). 3) Support for lowest context id APN in HSS upon requested APN does not exist in HSS. Additionally, two Map Treatment fields under APN IMSI Range Services shall be used to determine the behavior of either rejecting, converting, or using lowest context id APN for various scenarios. This parameter can be enabled only when the Alias APN Conversion Per IMSI global parameter is enabled.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

applProtocolFallbackN26ToS10 - Application Protocol Fallback from N26 to S10

Syntax

--cmGparmName applProtocolFallbackN26ToS10 --cmGparmValue <string>

Description

Current parameter enables fallback to search for 's10' candidate when DNS Query to find target MME during S10 handover using 'n26' fails. This parameter is expected to be used together with parameter Use N26 Service Parameter in S10 Handover

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

attachEirPercentageChecking - Attach EIR Percentage Checking

Syntax

--cmGparmName attachEirPercentageChecking --cmGparmValue <string>

Description

Parameter controls feature activation GPARM flag which shall be used or activated only to perform IMSI and GUTI initial attach EIR percentage. By default, flag shall be disabled.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

automatedTlsCertManagement - Automated TLS Certificate Management

Syntax

--cmGparmName automatedTlsCertManagement --cmGparmValue <string>

Description

This parameter controls the usage of Automated TLS Certificate Management using cert-manager and Kubernetes Secrets. If ENABLE automated TLS Certificate Management is active.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

cc100SgwIsolate - CC100 SGW Isolate

Syntax

--cmGparmName cc100SgwIsolate --cmGparmValue <string>

Description

Defines whether the MME isolates SGW when the MME receives CC100/CS=0 in Create Session Response of Attach (including handover Attach). Current parameter takes effect only when parameter lteSgwIsolation is enabled

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

cc100SgwReselect - CC100 SGW Reselection

Syntax

```
--cmGparmName cc100SgwReselect --cmGparmValue <string>
```

Description

Defines whether the MME does SGW reselection when the MME receives CC100/CS=0 in Create Session Response of Attach (including handover Attach). Because PGW reselection happens by default in such case, when current parameter is enabled, the MME does SPGW pair reselection

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

chargingCharAccessRestriction - Charging Characteristics Access Restriction

Syntax

```
--cmGparmName chargingCharAccessRestriction --cmGparmValue <string>
```

Description

Enables the MME to restrict access of UEs based on provisioned charging characteristics value.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

checkTgtAmfN26StatusN26Ho - Check Target AMF N26 Status in N26 Handover

Syntax

```
--cmGparmName checkTgtAmfN26StatusN26Ho --cmGparmValue <string>
```

Description

Current parameter enables checking the N26 link/interface status of local target AMF node during N26 handover so that if it isn't available to use, then even if the local AMF can serve the target RAN, source MME will select an alternative node to complete the N26 handover.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Recommended

true

Name

doubleS1N26Mob - Handle Double S1 at N26 Mobility

Syntax

```
--cmGparamName doubleS1N26Mob --cmGparamValue <string>
```

Description

Enable processing of double S1 IRAT scenario where UE abandons N26 handover procedure and restarts idle mode IRAT TAU possibly at the same MME that served N26 handover so that the Create Session Request towards the new serving SGW, as part of the idle mode TAU, is sent before Delete Session Request (Scope Indication) that is sent towards old SGW as part of resource cleanup of N26 handover context. The old and new SGW can be the same.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

edrxExtendStopPageTime - eDRX Extend Stop Page Time

Syntax

```
--cmGparamName edrxExtendStopPageTime --cmGparamValue <string>
```

Description

Specifies whether the eDRX Stop Page Timer should be extended to be the true end of the paging time window for the UE

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Name

effectiveLoadForScaling - Effective Load For Scaling

Syntax

--cmGparmName effectiveLoadForScaling --cmGparmValue <string>

Description

This parameter is to control effective load scaling feature for CNF Horizontal Auto Scaler. When set to false the feature to compute effective load is disabled. To enable Effective load computation set the parameter to true. This parameter is CNF specific.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

empsGtpMessagePriorityN26 - MME eMPS GTP Message Priority N26

Syntax

--cmGparmName empsGtpMessagePriorityN26 --cmGparmValue <string>

Description

Enables MME to use GTP Message Priority value 0 during N26 mobility procedures for priority UEs.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

empsGtpMpOnlyWhenDscpMapped - eMPS GTP Message Priority Only When DSCP Mapped

Syntax

--cmGparmName empsGtpMpOnlyWhenDscpMapped --cmGparmValue <string>

Description

When this parameter is set to true, the MME will indicate the GTP Message Priority only when the GTP message involves a bearer whose QCI and ARP PL values map to a DSCP value on

the eMPS DSCP Marking provisioning command. When set to false, GTP Message Priority is indicated for all GTP messages of the priority UE.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

epsUpip - EPS UPIP

Syntax

--cmGparmName epsUpip --cmGparmValue <string>

Description

Current parameter enables support for the EPS-UIP (User Plane Integrity Protection)

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

extendedEmergencyNumberListZoneCode - Extended Emergency Number List Zone Code

Syntax

--cmGparmName extendedEmergencyNumberListZoneCode --cmGparmValue
<string>

Description

Indicates provisioned extended emergency list zone code. When feature activation gParm is enabled and upon receiving Regional-Subscription-Zone-Code in UE Subscription data from HSS in ULA/IDR message, if received Zone Code matches provisioned zone code for this feature, MME shall send Extended Emergency Numbers List in the ATTACH and TAU ACCEPT message

Type / Range

Integer in the range of 0 to 65,535

Default

0

Name

fallbackFromLocationGatewaySelectionToNormalDns - Fallback from Location Gateway Selection to Normal DNS

Syntax

```
--cmGparamName fallbackFromLocationGatewaySelectionToNormalDns --  
cmGparamValue <string>
```

Description

Parameter controls fallback to normal DNS operation. When the parameter is set to true, fallback to normal DNS query without tag is supported. If the parameter is not set (default) fallback is not supported.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

gtpMpForEmergencyFallbackFrom5g - GTP MP for Emergency Service Fallback from 5G

Syntax

```
--cmGparamName gtpMpForEmergencyFallbackFrom5g --cmGparamValue <string>
```

Description

Defines whether the MME uses GTP Message Priority (MP) on GTP N26 Context Request towards the AMF, when the UE uses RRC establishment cause 'emergency' on TAU request and the MME sends N26 Context Request to the AMF as part of the procedure.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

gtpMpValueForEmergency - GTP MP Value for Emergency

Syntax

```
--cmGparamName gtpMpValueForEmergency --cmGparamValue <string>
```

Description

Defines the GTP Message Priority (MP) value the MME uses, when GTP MP is provisioned to be used for emergency services.

Type / Range

Integer in the range of 0 to 15

Default

0

Name

healthCheckCompare - Health Check Compare

Syntax

--cmGparamName healthCheckCompare --cmGparamValue <string>

Description

This parameter is used to control when a comparison between the health check functionality results shall be executed.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Recommended

true

Name

inclBackoffTimerForImsPdnReqRej - Include Backoff Timer For IMS PDN Request Reject

Syntax

--cmGparamName inclBackoffTimerForImsPdnReqRej --cmGparamValue <string>

Description

Parameter controls feature activation flag. By default, flag shall be disabled. When enabled, MME shall send T3396 back-off timer IE when rejecting IMS PDN Connectivity Request that is sent either as standalone or as part of Attach procedure due to not subscribed IMS APN with cause code #27.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

includeNwPolicyInAttachAndTauAccept - Include Network Policy in Attach and TAU Accept

Syntax

--cmGparamName includeNwPolicyInAttachAndTauAccept --cmGparamValue
<string>

Description

Parameter controls inclusion of Network Policy IE in Attach Accept and TAU Accept messages. By default, it shall be disabled. When enabled, MME shall include Network Policy IE in Attach Accept and TAU accept messages sent to the UE.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

includeUliForMmeTauWithoutSgwChange - Include ULI for MME TAU without SGW Change

Syntax

```
--cmGparamName includeUliForMmeTauWithoutSgwChange --cmGparamValue  
<string>
```

Description

Parameter controls inclusion of ULI in S11 message for the following procedure upon enabling the gparam flag. By default, flag shall be disabled. 1) TAU MME no change and no SGW change

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

includeUliForX2AndS1HoWithoutSgwChange - Include ULI for X2 and S1 HO without SGW Change

Syntax

```
--cmGparamName includeUliForX2AndS1HoWithoutSgwChange --cmGparamValue  
<string>
```

Description

Parameter controls inclusion of ULI in S11 messages for following procedures upon enabling the gparam flag. By default, the flag shall be disabled. 1) X2 HO with no SGW change 2) S1 HO with MME change and no SGW change 3) S1 HO with no MME change and no SGW change

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

keepUeCtxtN26Mobility - Keep UE Context After N26 Mobility

Syntax

--cmGparmName keepUeCtxtN26Mobility --cmGparmValue <string>

Description

Current parameter enables keeping the UE Context after UE has moved to 5G with N26 mobility (idle or connected mode). When the gParm is disabled then the UE Context is deleted at the end of the mobility procedure to 5G (default behavior).

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

locallyProvisionedImsLbo - Locally Provisioned IMS LBO

Syntax

--cmGparmName locallyProvisionedImsLbo --cmGparmValue <string>

Description

Enables the locally provisioned IMS LBO functionality in the MME.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

locationGatewaySelection - Location Gateway Selection.

Syntax

--cmGparmName locationGatewaySelection --cmGparmValue <string>

Description

Parameter controls location based gateway selection. By default, it shall be set to false. When the parameter is set to true, MME will insert a tag (either zone name or TAC) to the DNS query for certain APNs.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Name

memoryAlarmForRestrictedPaging - Memory Overload Alarm for MME Restricted Paging

Syntax

--cmGparmName memoryAlarmForRestrictedPaging --cmGparmValue <string>

Description

Memory Overload Alarm at which CMM starts MME Restricted Paging

List

- minor - Minor Memory Alarm
- major - Major Memory Alarm
- critical - Critical Memory Alarm

Default

minor

Name

mmeRateControlT6aDI - MME Rate Control T6a Downlink

Syntax

--cmGparmName mmeRateControlT6aDI --cmGparmValue <string>

Description

Specifies whether the MME is to enforce downlink Serving PLMN Rate Control for T6a bearers. If not set, the SCEF is expected to carry out the rate control. The default is for the MME not to enforce.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

pdnReactivateN1Enable - PDN Reactivation when N1 mode gets enabled

Syntax

--cmGparmName pdnReactivateN1Enable --cmGparmValue <string>

Description

Current parameter enables the MME performing PDN deactivation (reactivation requested) for the non-emergency PDNs after TAU completes for which the UE re-enabled N1 mode

for those PDN sessions that are established with standalone PGWs but the associated APNs allow 5GS IWK and UE is not 5G Restricted.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

pdnTypeIpv6Restriction - PDN Type IPv6 Restriction

Syntax

```
--cmGparamName pdnTypeIpv6Restriction --cmGparamValue <string>
```

Description

Enables MME to restrict the use of PDN Type IPv6 and IPv4v6 for the provisioned UEs.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

pdnTypeProtocolTypeGwReselection - PDN Type or Protocol Type GW Reselection

Syntax

```
--cmGparamName pdnTypeProtocolTypeGwReselection --cmGparamValue  
<string>
```

Description

Defines whether the MME reselects an alternate GW during Attach and PDN Connectivity procedures when SGW returns cause code 83 (Preferred PDN type not supported) or 86 (Protocol type not supported). MME reselects either SGW or PGW depending on which one (SGW or PGW) the cause source indicates. Setting this parameter to true requires that both global parameters Support alternate gw reselection feature and Enhanced Alternate GW Reselection are also set to Yes.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

pmReportHssByName - PM Report HSS by Name

Syntax

--cmGparmName pmReportHssByName --cmGparmValue <string>

Description

Specifies whether PM counts reported by HSS are to use the name field from the remote endpoint configuration for that S6a interface. If false, the PM HSS name continues to use the endpoint ID from the remote endpoint configuration.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

proactiveAdaptiveSafetyNet - Proactive Adaptive Safety Net

Syntax

--cmGparmName proactiveAdaptiveSafetyNet --cmGparmValue <string>

Description

Enables or disables proactive adaptive safety net enhancements, in particular the automatic, proactive lowering of the safety net threshold if it is predicted that the busiest IPDS will reach 80% CPU occupancy before the threshold would have been reached.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

resendDnsQueryWithoutPdnTypeForTimeout - Resend DNS Query Without PDN Type for Timeout

Syntax

--cmGparmName resendDnsQueryWithoutPdnTypeForTimeout --cmGparmValue <string>

Description

Parameter controls feature activation GPARM flag which by default shall be disabled. When enabled, MME shall upon receiving no response from DNS/timeout shall resend the DNS query by excluding the PDN type information from APN FQDN.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

resilientIpdsFdsFailure - System Resilience on IPDS/FDS dual fail

Syntax

--cmGparmName resilientIpdsFdsFailure --cmGparmValue <string>

Description

This parameter specifies that system recovery is using the enhanced recovery method for dual IPDS or dual xFDS failure. This method will lock the CMM Aggregate Service if IPDSs and FDSs are not available after dual failure. The system will be unlocked when the normal state is recovered.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

restorationWithFullBackup - SRS Restoration with Full Backup

Syntax

--cmGparmName restorationWithFullBackup --cmGparmValue <string>

Description

This parameter allows SRS restoration (MME failure) with MME Full Backup of UE Contexts. Previously full backup only allowed restoring Users during ISSU. This feature supports restoral of basic procedures on any MME in the pool, using a Full Context backup. The following global parameters must be enabled prior to enabling Restoration With Full Backup: Support Ue Restoration Context Deletion, Inhibit Peer Srs Responses, Ue Ctx Update In Patch Load Support, Session Ddn Restoration Std Based, and Support Full Ue Restoration.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Recommended

true

Name

rfspAccessRestriction - RFSP Access Restriction

Syntax

--cmGparmName rfspAccessRestriction --cmGparmValue <string>

Description

Feature control for RFSP Access Restriction for the MME.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

roamerAttachRateLimitOverride - Roamer Attach Rate Limit Override

Syntax

--cmGparmName roamerAttachRateLimitOverride --cmGparmValue <string>

Description

IMSI and foreign GUTI Attaches allowed per second after which CMM starts dropping attaches for roamers until the rate is below the limit. If the Roamer Attach Rate Limit exceeds the general Attach Rate Limit for any reason, the Roamer Attach Rate Limit Override is effectively disabled. This is because the general Attach Rate Limit will be reached before the Roamer Attach Rate Limit. If the value is set to 0, it means the Roamer Attach Rate Limit is disabled. Value needs to be set to non-zero for the feature to be enabled. Refer to the description of the Attach Rate Limit Override global parameter to understand how this value is used in different CMM environments. Roamer Attach Rate Limit Override configuration is for both IPDS and mIPDS.

Type / Range

Integer in the range of 0 to 9,999

Default

0

Name

safetyNetVariation - Safety Net Variation

Syntax

--cmGparmName safetyNetVariation --cmGparmValue <string>

Description

Allows for variation in safety net in special circumstances. It is ordinarily not changed and should be changed only at the direction of Nokia personnel.

Restrictions

This parameter should be changed only at the direction of Nokia personnel.

List

- loadAlpha - This value should be used always unless directed otherwise by Nokia personnel.
- loadBeta - This value should be used only at the direction of Nokia personnel.
- loadDelta - This value should be used only at the direction of Nokia personnel.
- loadGamma - This value should be used only at the direction of Nokia personnel.

Default

loadAlpha

Name

sendExtEmerNumberInAttachAndTauAccept - Send Extended Emergency Number in Attach and TAU Accept

Syntax

```
--cmGparmName sendExtEmerNumberInAttachAndTauAccept --cmGparmValue <string>
```

Description

Parameter controls feature activation global parameter flag. By default, flag shall be disabled. When enabled, MME shall send Extended Emergency Numbers List IE in the ATTACH and TAU ACCEPT messages if one is provisioned on the MME.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sendOnlyImeiToEir - Send Only IMEI to EIR

Syntax

```
--cmGparmName sendOnlyImeiToEir --cmGparmValue <string>
```

Description

When this parameter is set to true MME will include only the IMEI part towards EIR for authentication even if the IMEI SV is available in the UEs MM context. When set to false the MME will include the IMEISV when this is available.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Name

sendOnlyImeiToSgw - Send Only IMEI to SGW

Syntax

--cmGparmName sendOnlyImeiToSgw --cmGparmValue <string>

Description

When this parameter is set to true MME will include only the IMEI towards SGW even if the IMEI SV is available in the UEs MM context. When set to false the MME will include the IMEISV when this is available.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sendOnlyRanSupportedPlmnsInS1 - Send Only RAN Supported PLMNs in S1

Syntax

--cmGparmName sendOnlyRanSupportedPlmnsInS1 --cmGparmValue <string>

Description

Current parameter enables sending in SETUP RSP and MME Configuration Update only the PLMNs that are supported by the RAN even if additional PLMNs are configured at the MME

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sendSgsMmeIdentityS6a - Send SGs-MME-Identity AVP on S6a

Syntax

--cmGparmName sendSgsMmeIdentityS6a --cmGparmValue <string>

Description

Specifies whether the MME is to include the SGs-MME-Identity AVP in S6a messaging for a UE that is being registered over SGs. The default is false.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

separateEplmnListNasS1ap - Separate EPLMN List on NAS and S1AP

Syntax

--cmGparmName separateEplmnListNasS1ap --cmGparmValue <string>

Description

Enable sending separate Equivalent PLMN Lists to NAS and S1AP. The EPLMN Lists applied to NAS and S1AP are copied from EPLMN List Name and EPLMN List Name for eNB, respectively, that are provisioned in IMSI Range Services or UE PLMN Services.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

serviceReqRateDropLimitOverride - Service Request Rate Drop Limit Override

Syntax

--cmGparmName serviceReqRateDropLimitOverride --cmGparmValue <string>

Description

Number of allowed 4G UE initiated Service Requests beyond which MME/CMM will start to drop them on the IPDS. Value of 0 means use system calculated Default value. Non-zero means use this value directly.

Type / Range

Integer in the range of 0 to 24,999

Default

0

Name

sessionDdnRestorationStdBased - Session DDN Restoration Standards Based

Syntax

--cmGparmName sessionDdnRestorationStdBased --cmGparmValue <string>

Description

MME Session DDN Restoration based on Standards

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sgsLocUpdCombTauAfterCsfb - SGs Location Update for Combined TAU After CSFB

Syntax

```
--cmGparmName sgsLocUpdCombTauAfterCsfb --cmGparmValue <string>
```

Description

Current parameter enables the MME doing SGs Location Update when the UE does intra-MME Combined TAU after CSFB even if Old and New LAI are the same.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

srStormInternalMonitoring - Service Request Storm Internal Monitoring

Syntax

```
--cmGparmName srStormInternalMonitoring --cmGparmValue <string>
```

Description

This parameter enables internal software resource monitoring to further protect the CMM during Service Request Storms, taking action as necessary to prevent internal resource exhaustion and overloads, including in scenarios such as when SGW overloads result in delays in SGW responses to the CMM, which in turn could result in CMM overloads.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Recommended

true

Name

srsTrafficReduction - SRS traffic reduction

Syntax

--cmGparamName srsTrafficReduction --cmGparamValue <string>

Description

This parameter enables using new format (compressed) to send NR10 update requests and NR10 fetch responses. For AMF, all the NR10 Update requests and Fetch responses will follow the new compressed format if the value of this parameter is set to true. For MME, Update Requests and Fetch Responses will follow the new compressed format when the value of this parameter is set to true as well as Support Full UE Restoration is enabled. If there is a need to change the value of this gParam please contact Nokia support team

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

staticPgwSelectionAllPlmns - Static PGW selection for defined IMSIs on all serving PLMNs

Syntax

--cmGparamName staticPgwSelectionAllPlmns --cmGparamValue <string>

Description

MME supports static PGW selection for defined IMSIs on all available serving PLMNs.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

tauRateDropLimitOverride - IRAT tau Rate Drop Limit Override

Syntax

--cmGparamName tauRateDropLimitOverride --cmGparamValue <string>

Description

5G to 4G IRAT TAUs allowed per sec after which MME start dropping 5G to 4G IRAT TAUs until rate is below limit. Value of 0 means use system calculated Default value. Non-zero means use this value directly.

Type / Range

Integer in the range of 0 to 14,999

Default

0

Name

transferTo5gPdnWo5gsIwk - Transfer to 5GS PDN Session without 5GS IWK

Syntax`--cmGparamName transferTo5gPdnWo5gsIwk --cmGparamValue <string>`**Description**

Current parameter enables transferring to 5GS a PDN Session that is known that 5GSIWK flag were not sent in Create Session Request during PDN Session Establishment procedure of type handover.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Recommended

false

Name

ueRadioCapPagingS10 - UE Radio Capability Paging S10

Syntax`--cmGparamName ueRadioCapPagingS10 --cmGparamValue <string>`**Description**

This gparam specifies whether the MME will include the 'UE Radio Capability for Paging' AVP, if available, in S10 messaging. The default is false. The same gparam specifies whether the MME will accept and store 'UE Radio Capability for Paging' AVP, if available in S10 messaging, in the UE context.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ueTzInCreateAndUpdateBearerResp - UE Time Zone In Create And Update Bearer Response

Syntax`--cmGparamName ueTzInCreateAndUpdateBearerResp --cmGparamValue <string>`

Description

Parameter controls feature activation GPARM flag. By default, flag shall be disabled. When enabled, MME will include UE Time Zone in S11 Create Bearer and Update Bearer Response messages for both homers and roamers.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

useApnOiToDetermineHrt - Use APN-OI to Determine HRT

Syntax

--cmGparmName useApnOiToDetermineHrt --cmGparmValue <string>

Description

Current parameter enables using the APN-OI instead of the PGW FQDN to determine if a PDN Session is HRT for a roaming subscriber when applying QoS modification for a PDN session in accordance to local configuration and when sending the UE Time Zone IE in UBR and CBR Response messages.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

useDnsForN26Ho - Use Dns for N26 handover

Syntax

--cmGparmName useDnsForN26Ho --cmGparmValue <string>

Description

Current parameter enables using DNS for selecting target node as part of N26 handover even when local AMF of quad access node serves target RAN.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

useLocalRfspHssRfspNotSub - Use Local RFSP if RFSP isn't subscribed at the HSS

Syntax

```
--cmGparmName useLocalRfspHssRfspNotSub --cmGparmValue <string>
```

Description

Current parameter enables using the locally provisioned RFSP value when local configuration at the uePlmnServices or imsiRangeServices indicates applying the HSS subscribed value but no RFSP value is received from the HSS

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

useN26SvcParamS10Ho - Use N26 Service Parameter in S10 Handover

Syntax

```
--cmGparmName useN26SvcParamS10Ho --cmGparmValue <string>
```

Description

Current parameter enables using N26 service parameter as part of DNS query for target MME selection during S10 handover

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

useSameIpDynStaticS11Link - Use Same IP Address for Dynamic and Static S11 Links

Syntax

```
--cmGparmName useSameIpDynStaticS11Link --cmGparmValue <string>
```

Description

Enable using same IP address for dynamic and static S11 links.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

vlrCleanupBasedOnOccupancy - VLR Cleanup Based on Occupancy

Syntax

--cmGparmName vlrCleanupBasedOnOccupancy --cmGparmValue <string>

Description

Current parameter enables doing VLR cleanup of the VLR Contexts of deregistered UEs based on VLR Occupancy

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Recommended

true

Name

wifiHoConnectedUe - Wifi Handover for Connected Ue

Syntax

--cmGparmName wifiHoConnectedUe --cmGparmValue <string>

Description

When a Delete Bearer Request is received for a default bearer with the cause 'RAT changed from 3GPP to non-3GPP' for an ECM-CONNECTED UE and this is not the last bearer for the UE, instead of doing S1 Release and Release Access Bearer, the MME sends an S1AP ERAB Release Command for all bearers associated with the PDN session, but without including a NAS EPS Deactivate Bearer Context Request in the E-RAB Release.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

zeroOiCsrS10Tau - Set OI flag to Zero in CSR of S10 TAU

Syntax

--cmGparmName zeroOiCsrS10Tau --cmGparmValue <string>

Description

Current parameter enables setting OI flag to value zero in Create Session Request that MME sends as part of s10 Idle mobility. When this parameter is disabled, the MME sets OI flag to 1

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

zoneCodeExtendedEmergencyNumberList - Zone Code Extended Emergency Number List

Syntax

```
--cmGparmName zoneCodeExtendedEmergencyNumberList --cmGparmValue <string>
```

Description

Parameter controls feature activation GPARM flag. By default, flag shall be disabled. When enabled, MME shall use the received regional zone code information from HSS to determine whether to send extended emergency number list to UE.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

zoneNameSwitch - Zone Name Switch

Syntax

```
--cmGparmName zoneNameSwitch --cmGparmValue <string>
```

Description

Parameter controls whether the mapping between zone names and TAC is enabled (switch on). If the parameter is set, zone name will be used as tag in the APN query. Otherwise, TAI will be used as tag in the APN query.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

5.2 cmm ggsnIpsWeight

Different weights on GGSN IP addresses for the host file. Syntax, description, parameters, and roles for cmm ggsnIpsWeight command.

Syntax

Description

This command is used in order to create/list/show different weights for the available GGSN IP addresses used in the host file

Parameters

Name

apnNI - APN Domain Name

Syntax

--apnNI <string>

Description

Use this parameter to specify the APN Domain Name

Class

Required key.

Type / Range

Use this parameter to specify the APN Domain Name

Name

mcc - mcc

Syntax

--mcc <digits>

Description

Mobile Country Code digits of broadcasted PLMN

Class

Required key.

Type / Range

"Mobile Country Code digits of broadcasted PLMN"

Name

mnc - mnc

Syntax

--mnc <digits>

Description

Mobile Network Code digits of broadcasted PLMN

Class

Required key.

Type / Range

Mobile Network Code digits of broadcasted PLMN

Name

ggsnIpAddress - GGSN IP Address

Syntax

--ggsnIpAddress <string>

Description

Use this parameter to specify the GGSN IP Address.

Class

Required key.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

fullApn - Full APN Domain Name

Syntax

--fullApn <string>

Description

'This field contains the full APN Domain Name'

Class

Read only.

Type / Range

Digit string of length 1 to 100 with the allowed pattern of [0-9a-z\.\-]*

Name

ipAddressType - IP Address Type

Syntax

--ipAddressType <list-item>

Description

Use this parameter to specify the IP Address Type. Allowed values IPV4, IPV6

Class

Read only.

List

- IPV4 - Specify trace entity of IPv4.
- IPV6 - Specify trace entity of IPv6.

Default

IPV4

Name

weight - GGSN IP weight

Syntax

--weight <integer>

Description

Use this parameter to specify the weight for the GGSN IP address. Allowed values 0 to 10

Class

Optional.

Type / Range

Integer in the range of 0 to 10

Default

1

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.3 cmm gnbNbrMapDump

gNB Neighbor Map Dump. Syntax, description, parameters, and roles for cmm gnbNbrMapDump command.

Syntax

Description

This command outputs to a file the gNB Neighbor Map residing in the specified node.

Parameters

Name

gnbMapType - Map

Syntax

--gnbMapType <list-item>

Description

The requested map to be retrieved.

Class

Optional.

List

- active - The active gNB map in the requested node.
- updating - The updating gNB map in the requested node.

Default

active

Name

information - Information

Syntax

--information <string>

Description

Output information for the request.

Class

Read only.

Type / Range

Digit string of length 0 to 256 with the allowed pattern of [0-9a-zA-Z\._:\-]*

Name

macroGnbId - Macro gNB ID

Syntax

--macroGnbId <integer>

Description

Macro gNB ID

Class

Optional.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

neighborGnbId - Neighbor gNB ID

Syntax

--neighborGnbId <integer>

Description

Neighbor gNB ID

Class

Optional.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

poolId - Pool ID

Syntax

--poolId <integer>

Description

Pool ID

Class

Optional.

Type / Range

Integer in the range of 0 to 39

Name

poolType - Pool Type

Syntax

--poolType <list-item>

Description

Pool Type

Class

Optional.

List

- ALMS - Alarm Service - Alarm Service.
- CPPS - Control plane processing service - Control plane processing service.
- CTCS - Call Trace Collection service - Call Trace Collection service.
- DBS - Database service - Database service.
- EEMS - Event Exposure Management service - Event Exposure Management service.
- IPDS - IP director service - IP director service.
- IPPS - IP packet processing service - IP packet processing service.
- LIHS - Lawful Interception Handling service
- MLIS - Machine Learning Intelligence Service
- NECC - Network element cluster controller - Network element cluster controller.
- NWDS - Network Data Service - NWDAF service
- PAPS - Packet processing service - Packet processing service.
- PCMS - Per Call Measurements Data Service - Per Call Measurements Data (PCMD) Service
- RCIS - Radio Capability Information storage Service
- SFDS - SCTP flow distributor service
- UCDS - User equipment Context Data-dump Service
- UFDS - UDP flow distributor service

Name

weight - Weight

Syntax

--weight <integer>

Description

Weight

Class

Optional.

Type / Range

Integer in the range of 0 to 4,294,967,295

Roles

List

sysadmin, manage, monitor

5.4 cmm gParms

Global Parameters. Syntax, description, parameters, and roles for cmm gParms command.

Syntax

Description

This command is used to administer Global Parameters defined to configure global office data in CMM.

Parameters

Name

gParmName - Parameter name

Syntax

--gParmName <list-item>

Description

Name of parameter.

Class

Required key.

List

- 20bitHenbId - 20 bit home eNodeB identifier - Control flag for routing of S1AP HO messages based on target TAI when the target eNB with 20-bit eNB ID is not directly connected with the MME. no (default) - disable this routing. yes - enable this routing.
- 2g3gPgwlIsolation - 2g3g PGW isolation - Setting to yes enables PGW isolation feature on SGSN. setting to no disables PGW isolation feature on SGSN.
- 2g3gSgwlIsolation - 2g3g SGW isolation - Setting to yes enables SGW isolation feature on SGSN. setting to no disables SGW isolation feature on SGSN.
- 4g5gRetPrefPlmn - 4g5g Return Preferred Plmn - This parameter enables indicating to serving eNB the last visited 5GS PLMN as preferred indication to be used in future EPS to 5GS mobility.
- acceptActFlagTauCpsrForT6aonly - Accept active flag TAU CPSR for T6A ONLY - Enable acceptance of TAU with active flag and CPSR to establish S1-U bearers for a UE with SCEF PDN connections only.

- **acceptScefRelForPsmEdrx** - Accept SCEF Release for PSM eDRX - This gparm specifies whether the MME is to return diameter Success if the SCEF attempts to release a T6a bearer for an Idle UE in PSM or eDRX mode and unavailable for paging.
- **acceptSgsPageFromNonRegMsc** - Accept an SGs Page from non-registered MSC - This gparm specifies whether the CMM should accept and process an SGs page request from an MSC other than the currently registered MSC.
- **accessRestrictionToLte** - support ARD from s6d and pass it to Radio - This parameter allows SGSN to support ARD from s6d and pass it to Radio. This parameter also allows MME in combi node during attach or TAU to store 'WB-E-UTRAN Not Allowed' in the DB if receives DIAMETER_ERROR_USER_UNKNOWN during location updating. If the condition such user is still in the DB e.g. the subscriber was attached in the SGSN part before, purge is not done yet. MME will send #15 CC towards UE in this case. Default value is No, it means that both SGSN/MME will keep current behavior, no changes. If the parameter is set to Yes, above function will take in place.
- **activateHssUnavailBlacklisting** - Activate HSS unavailable blacklisting - Enable the extended HSS unavailability detection and blacklisting.
- **activateLcs** - Activate LCS - Indicates whether or not to activate LCS.
- **activateLcsClientTypeEmergencyServices** - Activate LCS client type emergency services - Activate LCS client type emergency services.
- **activateLcsClientTypeLawfulInterceptServices** - Activate LCS client type Lawful Intercept Services - Activate LCS client type Lawful Intercept Services.
- **activateLcsClientTypePlmnOperatorServices** - Activate LCS client type PLMN operator services - Activate LCS client type PLMN operator services.
- **activateLcsClientTypeValueAddedServices** - Activate LCS client type Value Added Services
- **aerialSubscriptionActive** - Aerial Subscription Feature Active - Specifies whether the MME is to include aerial subscription information (if available) in S1 messaging to the eNodeB.
- **alarmCmasFailure** - Alarm CMAS failure - Alarm CMAS failure.
- **alarmFailToReceiveCmasResponse** - Alarm fail to receive CMAS response - Alarm fail to receive CMAS response.
- **alarmFailToSendCmasMessage** - Alarm fail to send CMAS message - Alarm fail to send CMAS message.
- **aliasApnConversionPerImsi** - Alias APN Conversion Per IMSI - Parameter controls feature activation flag. By default, flag shall be disabled. When enabled, MME shall support converting the UE requested APN with the locally configured APN.
- **allApns** - All APNS - MME to store context of all PDN connections of UE on SRS.

- **allowBrrDeactvtRglrDeactvt** - Allow bearer deactivate regular deactivate - Yes - enables the MME, upon receiving delete bearer request from the network during a PGW initiated PDN disconnection scenario with cause of 'Reactivation requested', to send deactivate EPS bearer context request message to the UE overriding the standard cause with 'regular deactivation'. upon receiving DEACTIVATE EPS BEARER CONTEXT ACCEPT message from the UE, the MME will send S11 delete bearer response message to the SGW. no (default) - in the scenario above, the MME will send the standard cause, 'reactivation requested', towards the UE.
- **allowEdrxValue0** - support eDRX cycle 5.12s - support eDRX cycle 5.12s. Default value is not set. In this case, MME will keep current behavior: a UE requests eDRX 0, MME responds with an ATTACH/TAU RESPONSE message with the eDRX IE not present. Since EMM currently rejects the use of eDRX 0, the UE will be paged the same as if the UE didn't request eDRX 0. When this gParm is set, then this feature function will be enable. MME will accept eDRX0, and a new paging type WB_eDRX_0 will be used
- **allowEnGnbRouting** - Allow En-Gnb Routing - Allow the use of the en-Gnb mapping to be used in routing the Enb Configuration Transfer message received at MME. When false, the en-gNB ID and TAC will be used in message routing. When true, message routing occurs using only the en-gNB ID.
- **allowExtendedImsiRangeServices** - Allow Extended Number of IMSIRangeServices - This parameter determines the use of an extended range of IMSIRangeServices. When set to no the number of IMSIRangeServices is limited to 4K. When set to yes, the number of MSIRangeServices is limited to 10K.
- **allowLIPAPDNWhenLITarget** - allow LIPA PDN Connections - Use this parameter to allow checks on LIPA PDN Connections.
- **allowMixedPdnAddressAllocationTypes** - Allow mixed PDN address allocation types - When set to yes PDN type is IPv4v6, if only one of the IP versions is statically provisioned in the HSS, set the other IP version to all zeros thus also allowing for dynamic PDN address allocation.
- **allowNonEmerAttachEmerArpProvCc** - Allow non-emergency Attach with emergency ARP when Provisionable Cause Code feature is enabled - When this parameter is enabled, the MME allows non-emergency Attach for which PDN session is associated with emergency ARP, as defined in local emergency profile, that is otherwise by default rejected when the Provisionable Cause Code feature is enabled.
- **allowProcEirFailure** - Allow Procedure EIR Failure - Feature introduces support for enhancing the EIR support by having MME reject a user's Attach when the IMEI is explicitly blacklisted in EIR. In all other cases attach should succeed, if for example - the AVP value is different from BLACKLISTED (ie UNKNOWN, WHITELISTED or GREYLISTED) - any other Diameter Error Code received on S13 '3GPP-ME-Identity_Check' - No response received from EIR - Any other diameter error cause codes received from EIR Feature shall be controlled by GPARM flag

- **allowR8ArpGnSgsn** - Allow R8 ARP Gn SGSN - Gn SGSN is pre-rel8 element and it connects/receives pre-rel8 ARP from HLR. With this parameter set to true, Gn SGSN allows customer configured mapping table and transfer rel8 ARP towards RNC. With this parameter set, a new configured Group IMSIs Will be given ARP 1 towards RNC. If the parameter is not set, then normal function will take place. Rel8 ARP will not allow to send to RNC.
- **allowScefMtEdt** - Allow SCEF MT-EDT - Allows MT-EDT to be used with SCEFs
- **allowTransferOfEmergencyBearersToGnSgsn** - Allow transfer of emergency bearers to Gn SGSN - This global parameter determines whether emergency bearers can be transferred to a Gn SGSN. yes: the MME allows transfer of emergency bearers to a Gn SGSN. no: the MME denies transfer of emergency bearers to a Gn SGSN.
- **allowTransferOfEmergencyBearersToS3sgsn** - Allow transfer of emergency bearers to S3 SGSN - This global parameter determines whether emergency bearers can be transferred to a S3 SGSN. yes: the MME allows transfer of emergency bearers to a S3 SGSN. no: the MME denies transfer of emergency bearers to a S3 SGSN.
- **allowUeInitiatedBearerResourceAllocationRequest** - Allow UE initiated bearer resource allocation request - If the bearer resource allocation request parameter is set to yes, the request shall be allowed. if the request parameter is set to FALSE, the request shall be rejected by the MME. MME shall send BEARER RESOURCE ALLOCATION REJECT message with the ESM cause value #97 message type non-existent or nor implemented to the UE.
- **allowUeOffloadOnMmeDuringMafCpuOverload** - Allow UE offload on MME during CPPS CPU overload - Enable UE offloading on the MME during CPPS CPU overload when lightly loaded CPPSs are available on the MME.
- **allowUnSupportDcnReroute** - Allow UnSupported DCN Reroute - Indicates to allow an unsupported Dcn Rerouted request to proceed on a non-default DCN MME
- **allowUnSupportDcnRerouteRoamer** - Allow UnSupported DCN Reroute for Roamer - This field specifies whether to allow an unsupported DCN Rerouted request to proceed on a non-default DCN at the MME. The default is No.
- **alwaysSndEpsNwFeatSuppt** - Always Send EPS Network Feature Support - Always send EPS Network Feature Support IE in attach and tau accept message
- **anyUeRequestedApn** - Any UE Requested APN - GPARM introduces APN override support for rogue APNs. GPARM allows a match on 'any' UE requested APN. There will be an apnImsiRangeServcies entry with IMSI or IMEI range and a UE requested apn of 'any'. If this gparm is enabled it shall replace the UE requested APN for the search in apnImsiRangeServices with our own string 'any'. By default this shall be disabled.
- **apnRateControlStatus** - APN Rate Control Status - When this gparm is set to yes, the MME will include any saved APN rate control status for the first newly created PDN for a given APN. The PGW and SCEF can provide the rate control status to the MME when the last PDN in a given APN is deleted.

- **apply2g3gApnWhiteList** - Apply 2g3g APN white list - It is used to enable/disable the LBO APN white list checks globally for 2G3G. If it is no, then SGSN will not apply LBO APN white list checks irrespective of the UE PLMN LBO APN white list provisioning.
- **applyDeleteGbrToVolteBearersOnly** - Apply Delete GBR timer only to VOLTE bearers - Parameter controls the usage of existing Delete GBR timer. When this parameter is enabled, then Delete GBR timer applies only for VOLTE bearers. When this parameter is disabled, then Delete GBR timer applies for all GBR beares.
- **applyLocallyProvQosValuesForHrt** - Apply Locally Provisioned QoS Values For HRT - Parameter controls the behaviour of MME whether to apply QCI/ARP/PCI/PVI values from the locally provisioned QoS values. If the parameter is enabled and if MME receives QCI/ARP/PCI/PVI values from gateways that are different than locally configured qos values, MME shall apply locally configured values for HRT scenarios. By default the flag shall be disabled.
- **applyLteApnWhiteList** - Apply LTE APN white list - It is used to enable/disable the LBO APN white list checks globally for LTE. If it is no, then MME will not apply LBO APN white list checks irrespective of the UE PLMN LBO APN white list provisioning.
- **arpHighPriorityAccessLevel** - ARP high priority access level - Allocation Retention Priority (ARP) priority level value indicating high priority access in MME. ARP priority level indicates the priority of allocation and retention and can take values between 1 and 15. Value 15 means 'no priority'. Values between 1 and 14 are ordered in decreasing order of priority, i.e., 1 indicates highest priority and 14 indicates lowest priority. The MME grants high priority access for a UE when SGW/PGW provides an ARP PL value equal to this MME provisioned value for a bearer of the UE.
- **arpLowAccessPriorityLevel** - ARP low access priority level - Allocation Retention Priority (ARP) priority level value indicating low priority traffic in MME. ARP priority level indicates the priority of allocation and retention and can take values between 1 and 15. Value 15 means 'no priority'. Values between 1 and 14 are ordered in decreasing order of priority, i.e., 1 indicates highest priority and 14 indicates lowest priority. MME detects low priority traffic based on SGW/PGW providing an ARP PL value equal to this MME provisioned value for a bearer of the UE.
- **attachRateLimitOverride** - Attaches allowed per sec after which CMM starts dropping attaches until the rate is below the limit. The limit value is specified as either 'systemDefault', which means the CMM determines what the value is, or a user provisioned number of attaches per second. If the user provisioned value is greater than the maximum effective value listed below, the limit value that the CMM actually uses will be the maximum effective value. For 10 vcpu EMMS in large CMM, systemDefault is calculated as 400 attaches per second times the number of CPPS VMs with a maximum value of 10000 attaches per second. If the value is provisioned, the maximum effective value is the lesser of 1000 attaches per second times the number of CPPS VMs or 10000 attaches per second. For 12 vcpu EMMS in large CMM, Attach storm rate is 480 attaches per second per cpps. For an openstack small CMM, systemDefault is calculated as 400 attaches per second times the number of CPPS VMs with a maximum value of

1600 attaches per second. If the value is provisioned, the maximum effective value is the lesser of 1000 attaches per second times the number of CPPS VMs or 1600 attaches per second. For an A8 appliance CMM, systemDefault is calculated as 400 attaches per second times the number of CPPS VMs with a maximum value of 10000 attaches per second. If the value is provisioned, the maximum effective value is the lesser of 1000 attaches per second times the number of CPPS VMs or 10000 attaches per second. For an A2 appliance CMM, systemDefault is calculated as 200 attaches per second times the number of CPPS VMs with a maximum value of 1750 attaches per second. If the value is provisioned, the maximum effective value is the lesser of 400 attaches per second times the number of CPPS VMs or 1750 attaches per second. The implementation of the systemDefault is done at each IPDS and not factored for a m-IPDS CMM. For a m-IPDS CMM with 3 IPDS pairs for example, the actual limit is effectively tripled. We strongly suggest customer to overwrite the systemDefault with an adjusted value.

- **attachRateRejectLimitOverride** - Attaches allowed per sec after which CMM starts rejecting attaches until the rate is below the limit. The limit value is specified as either 'systemDefault', which means the CMM determines what the value is, or a user provisioned number of attaches per second. If the user provisioned value is greater than the maximum effective value listed below, the limit value that the CMM actually uses will be the maximum effective value. For an openstack large CMM, systemDefault is 10000 attaches per second. If the value is provisioned, the maximum effective value is 10000 attaches per second. For an openstack small CMM, systemDefault is 1600 attaches per second. If the value is provisioned, the maximum effective value is 1600 attaches per second. For an A8 appliance CMM, systemDefault is 10000 attaches per second. If the value is provisioned, the maximum effective value is 10000 attaches per second. For an A2 appliance CMM, systemDefault is 1750 attaches per second. If the value is provisioned, the maximum effective value is 1750 attaches per second. The implementation of the systemDefault is done at each IPDS and not factored for a m-IPDS CMM. For a m-IPDS CMM with 3 IPDS pairs for example, the actual limit is effectively tripled. We strongly suggest customer to overwrite the systemDefault with an adjusted value.
- **attachRejForUnsublmsApn** - Attach Reject for Unsubscribed APN - Global Parameter controls the feature activation. By default the parameter shall be disabled. When enabled, MME shall reject with Cause Value #33 if IMS APN is not subscribed or if IMS APN is subscribed only via Wild Card APN.
- **attachRejForUnsublmsApnWithLogging** - MME Track VoLTE Fail - Parameter controls the feature activation. By default the parameter shall be disabled. When the parameter is enabled MME shall log PDN Connectivity Failures (standalone or part of the Attach) with ESM cause code 27 for IMS APN only. IMSI, MSISDN and IMEI shall be captured as part of the log.
- **attachSgsUnknownLaiEmmCause** - Attach SGs Unknown LAI EMM Cause - The parameter is used to provision EMM cause code to be sent to UE combined attach request if MME fails to select MSC due to LAI to TAI to MSC mapping provisioning errors.

- **authntDtchIntgChkFail** - Authenticate UE for Detach Request message Integrity Checks Fail - Parameter is used to activate MME to authenticate a UE if the UE sent Detach Request fails message integrity checks. If authentication fails or if MME is unable to authenticate the UE then MME does not detach the UE.
- **autoAddTaiToTaiList** - Auto add TAI to TAI list - Indicates how the 'Automatically add TAI to the TAI list' should be handled.
- **backoffTimerImspdnReqRej** - Back-off Timer Inclusion - Global Parameter controls the inclusion of back-off timer IE. By default this parameter shall be disabled. When this parameter is enabled MME shall send T3396 back-off timer IE when rejecting IMS PDN Connectivity Request that is sent either as standalone or as part of Attach procedure due to not subscribed IMS APN or IMS APN is subscribed via the wildcard APN. Particularly for the case where IMS APN is subscribed via wildcard APN, the MME shall reject Attach/PDN Connectivity procedure(s) towards the IMS APN when GPARM is enabled.
- **bandwidthRoundup** - Bandwidth roundup - Round up or round down AMBR, GBR and MBR sent to UE.
- **bssapPlusSsn** - BSSAP plus SSN - BSSAP+ protocol sub system number.
- **bufferCbrDuringHo** - Buffer Create Bearer Request During HO - When this parameter is enabled, then if the MME receives CBR, UBR or DBR while processing 5G to 4G HO or inter-MME handover, it places the CBR (or UBR, DBR) on hold that is processed after completion of connected mode TAU that follows handover procedure.
- **bufferCreateBearerRequestDuringTau** - Buffer Create Bearer Request During TAU - Parameter controls handling collision scenario. By default, the flag shall be disabled. When enabled, MME while processing Modify bearer Request during TAU procedure, receives Create Bearer Request from SGW, it shall be buffered and processed only upon completion of TAU procedure.
- **canonicalNodeNameLabel** - Canonical node name label - Comparison of S-GW and P-GW network names in order to decide whether they are far enough to trigger a PDN connection re-establishment.
- **catM1bPagingTimerIncr** - Category M1 CE mode b paging timer increment - This value is added to paging response timer for CAT-M1 CE mode b UE.
- **cc15eUtranAllowed** - CC 15 E-UTRAN allowed - Configures the E-UTRAN allowed bit of the extended EMM cause.
- **cgfAlwaysSetSequenceNumberToOne** - CFG Always Set Sequence Number to One - When this parameter is set to true SGSN support the optional customization to overwrite the 'recordSequenceNumber' field for each unique CDR with value 1
- **cgfServingPlmnSupport** - CFG Serving PLMN Support - When this parameter is set to true, SGSN stores in the S-CDR servingNodePLMNIdentifier field the PLMN of the RA the UE is in. When set to false the Serving PLMN will not be added in the CDR.

- **checkDcnUeUsageTypeSgsnToMme** - Check DCN UE Usage Type SGSN to MME - Specifies whether the MME shall use UE Usage Type during 2G/3G->4G Attach and TAU procedure in order to perform Decor functionality as part of the procedure.
- **checkDcnUeUsageTypeSgsnToMmeRoamer** - Check DCN UE Usage Type SGSN to MME for Roamer - Specifies whether the MME shall use UE Usage Type during 2G/3G to 4G Attach and TAU procedure in order to perform Decor functionality as part of the procedure. This setting applies only for inbound roamers. The default is No.
- **checkMatchApnConfig** - check matchApn configuration - Check matchApn configuration to decide on IMS APN. If matchApn is not configured and checkMatchApnConfig set to No - consider apn with QCI 5 as IMS APN Yes - consider apn with QCI 5 as non IMS APN
- **ciotS1APMsgTunneling** - CMM Support for Tunnelling of S1AP Messages Associated with CIOT or DCNR Attach/TAU Requests - Enables CMM message tunnelling support over S10 of S1AP messages associated with CIOT (f11728-01) or DCNR Attach/TAU Requests.
- **clrHandlingWithoutClrFlags** - CLR Handling Without Clr Flags - Some functions are missing when combined MME/Gn SGSN (S6d use) handles CLR without CLR flag. With this parameter enabled, missing function should be supported. By default, the missing function is not supported.
- **collDbreqUbreq** - Collision of DL and UL bearer request - Enable/Disable support for collision scenario.
- **colocatedGwSearchForNoMatch** - Co-located Gateway Search for No Match - Parameter controls the support for searching co-located gateways for UEs when the served PLMN does not match the PGW FQDN PLMN for gateway mode selection 2. By default the flag shall be disabled.
- **combinedAttachWithoutCs** - Combined Attach Without CS - Enables the support for combined attach without actual CS domain in the MME.
- **comboPgwcRetries3XUe** - Combo PGW-C Retries for 3X capable UE - Number of retries selecting combo PGW-C(s) during PDN Connectivity procedure before switching selecting standalone PGW-C for 3X capable UE(s)
- **comboPgwcRetriesN1Ue** - Combo PGW-C Retries for N1 capable UE - Number of retries selecting combo PGW-C(s) during PDN Connectivity procedure before switching selecting standalone PGW-C for N1 capable UE(s)
- **coreNetworkRestrictSupport** - Core Network Restrictions Support - Current parameter enables sending Core Network Restrictions field and NRNA bit of MM Context IE in s10 interface. Note that when n26 interworking is enabled, it is expected that this gParm is also enabled by the operator
- **coreNetworkSharing** - Core network sharing - Indicates whether core network sharing enabled or not the default is no, indicating core network sharing is not enabled.
- **coverageEnhancement** - Coverage enhancement - Enable the MME support for coverage enhancement level paging feature.

- cpCiotReloc - CP CloT Reloc - Feature activation for supporting NB-IoT Control Plane CloT Relocation
- cpDataPktSizeToSwitchCpToUp - CP Data Packet Size to Switch CP to UP - CP data packet size threshold in octets for MME to trigger CP to UP switch. The same parameters is used for both UL and DL. If UL or DL CP data packet size exceeds the provisioned threshold then MME shall switch CP bearers to UP. MME shall not perform CP to UP switch if MME has knowledge that no more UL or DL packets are expected or a single DL packet is expected. MME will have this knowledge from the UE sent ESM DATA TRANSFER message IE Release Assistance Indication.
- crsi - Change reporting support indication - Indicates whether to set the CRSI flag in MME GTPv2 messages.
- csfbReturnLastPlmn - CSFB return last PLMN - Configure MME to indicate MSC to return UE to the last PLMN after CSFB.
- csRecoveryIndication - MSC CS recovery indication support - When set to yes enables MME to accept SGsAP-PAGING REQUEST message with CS restoration indicator and paging of the UE in the location information provided by the MSC. when set to no disables the feature.
- ctrlRoamerQos - Control QoS for roaming subscribers - ctrlRoamerQos controls below functionality only for roaming subscribers: If it sets to YES: SGSN will 1 Sending of HSS/HLR subscribed QoS values to S/P-GW or GGSN. 2 Ignoring QoS changes (upgrade/downgrade) from S/P-GW or GGSN. 3 Sending local configured QoS parameters towards RAN in case of: - Inter RAT handover MME to S4/Gn-SGSN - RAU MME to S4/Gn-SGSN - Inter S4/Gn-SGSN RAU If it sets to No, SGSN has no changes. Default value is NO
- dcnrGwSellte - DCNR capable PGW-C Selection for LTE only UE - Enable selecting DCNR capable PGW-C for LTE only UE (i.e. without EN-DC support) during PDN Connectivity procedure
- dcnrOverrideLipa - DCNR overrides Lipa service - Allow MME to override local PGW selection for Lipa enabled PDN with mobile network PGW selection for DCNR capable UE(s) with allowed NR as a Secondary RAT service.
- dcPraActionAddProcs - DC PRA action additional procedures - Enables MME to react upon received PRA action for dual connectivity PRA, when MME receives new PRA action from SGW/PGW during -Dedicated bearer activation -Default/dedicated bearer modification -intra LTE Mobility procedures -PDN setup for a PDN being handed over from ePDG. -Change notification response Applied when either -PRA Dual connectivity segment reporting is set to other than DC Segment Reporting Disabled, or -Presence Reporting Area Dual Connectivity IPv4 is set to other than DC PRA IPv4 Reporting Disabled, or -PRA DC IPv6 Prefix is set to other than DC PRA IPv6 Prefix Disabled.
- dcPraInterMme - DC PRA Inter MME Mobility - Enables MME to hand over already started PRA action for DC PRA between MMEs during inter MME mobility. Applied when either -PRA Dual connectivity segment reporting is set to other than DC Segment Reporting Disabled, or -Presence Reporting Area Dual Connectivity IPv4 is set to other than DC

PRA IPv4 Reporting Disabled, or -PRA DC IPv6 Prefix is set to other than DC PRA IPv6 Prefix Disabled.

- **dcPraStart5gsToEps** - DC PRA action start 5GS to EPS Mobility - Enables MME to react upon received PRA action for dual connectivity PRA, when MME receives new PRA action for start reporting during 5GS to EPS mobility. Applied when either -PRA Dual connectivity segment reporting is set to other than DC Segment Reporting Disabled, or -Presence Reporting Area Dual Connectivity IPv4 is set to other than DC PRA IPv4 Reporting Disabled, or -PRA DC IPv6 Prefix is set to other than DC PRA IPv6 Prefix Disabled.
- **ddnAckWithDLBuffering** - DDN ack with DL buffering - Enable sending of DDN ack with downlink buffering duration set to number of page attempts times (T3413 timer + provisioned increment for ESM messages) seconds and DL buffering suggested packet count.
- **ddnRateLimitOverride** - Number of DDNs allowed per sec after which MME start dropping DDNs until rate is below limit.
- **decorRerouteSr** - Decor Reroute on SR - This parameter indicates whether MME is enabled to reroute a UE upon UE SR or CPSR to a MME group supporting the UE DCN ID if the current MME does not support the DCN ID. By default the parameter is set to False.
- **decorRerouteSrRoamer** - Decor Reroute on SR for Inbound Roamer - This parameter indicates whether the MME is enabled to reroute an inbound roamer UE upon UE SR or CPSR to an MME group supporting the UE DCN ID if the current MME does not support the DCN ID. By default the parameter is set to False.
- **dedicatedBearerPiggybackSupport** - Dedicated bearer piggyback support - This parameter indicates whether the MME should send piggyback support for dedicated bearer setup to the SGW in the create session message. if the parameter is yes the message will indicate MME support, if it is no the message will indicate non support.
- **defaultIpv4ForPdnTypeIpv4OrIpv6** - Default Ipv4 for Pdn Type Ipv4OrIpv6 - GPARM controls to choose between IPv4 and IPv6 upon receiving from HSS. When GPARM is set to Yes default is IPv4 for pdnType IPv4 or IPv6. When gparm is set to 'No' it means ipv6. By default this parameter shall be set to 'No'.
- **defaultPreferredDataMode** - Default preferred data mode - Select default preferred network behavior when preferred-data-mode AVP is not received for a PDN connection.
- **defaultRacForSrvccMscSelection** - Default RAC for SRVCC MSC selection - Default RAC for SRVCC MSC selection.
- **delbrrRateLimitOverride** - Number of delete bearers allowed per sec after which MME start dropping delete bearers until rate is below limit.
- **deletePdnSessionS11u** - Delete PDN session S11U - Delete PDN sessions using S11-U when UE moves in to TAI that does not support CP CIOT optimization.

- deletePdnSessionS1u - Delete PDN session S1U - Delete PDN sessions using S1-U when UE moves in to TAI that does not support S1-U data transport.
- deletePdnSessionT6a - Delete PDN session T6A - Delete PDN sessions using T6A when UE moves in to TAI that does not support T6A data transport.
- depthOfSgwPgwRetry - Depth of SGW PGW retry - Defines the number of retries MME takes when there is no response from SGW on CSR.
- detachUeMoveRegions2g3gEnable - Detach UE move regions 2G3G enable - Yes - feature 'detach ue move regions is enabled for 2g3g. no (default) - feature 'detach ue move regions is disabled for 2g3g.
- detachUeMoveRegionsLteApnSustOrDel - Detach UE move regions LTE APNs sustained or deleted with delay - MME feature with enhanced APN specific functionality for detaching UEs when moving between regions in LTE. The enhanced functionality enables defining two specific APN lists. One list defining APN(s) or which the PDN(s) are sustained despite of the UE moving between regions (=detaching is disabled), and the other list defining APN(s) for which the PDN(s) are deleted only when no dedicated bearer(s) exist (=detaching is delayed). This feature is only functional when the basic feature for detach UE move regions is enabled for LTE (detachUeMoveRegionsLteEnable).
- detachUeMoveRegionsLteEnable - Detach UE move regions LTE enable - Yes - feature 'detach ue move regions is enabled for lte. no (default) - feature 'detach ue move regions is disabled for lte.
- detachUeMoveRegionsMonitorTime - Detach UE move regions monitor time - Time slice for monitoring the number of detaches per UE due to move across regions. if the number of detaches for a given UE exceed 'Detach UE Move Regions Threshold' during this time slice, the CMM will NOT detach the UE for 'Detach UE Move Regions Suppress Time' minutes.
- detachUeMoveRegionsSuppressTime - Detach UE move regions suppress time - Time duration in which the WMM will suppress a UE detach due to move across regions. this is in order to prevent frequent detaches of a UE that is ping-ponging between two regions.
- detachUeMoveRegionsThreshold - Detach UE move regions threshold - Threshold for number of UE detaches due to move across regions. if threshold is met within 'DETACH UE MOVE REGIONS MONITOR TIME' then detach suppression kicks in for a given UE.
- detachUponSrMbrTimeout - Detach upon SR MBR timeout - Detach immediately upon service request modify bearer response timeout.
- diameterIgnoreUnknownAvp - Diameter ignore unknown AVP - Ignore unknown AVPs from HSS?.
- diameterMsgCiotPriority - Diameter CIOT message priority - Use message priority when handling cellular internet of things diameter messages.
- diameterMsgPriority - Diameter message priority - Use message priority when handling diameter messages.

- **disableMmeRateControlExecute** - disable ratue control execute/drop in MME - There are be different count between MME and GW for the packet for the downlink rate control. This has been noticed from testing with CMG for IPv6. This parameter is introduced to give operator choice, to disable MME drop more packets than already did by GW.
- **discoverMme** - Discover MME - Indicates whether or not (YES or NO) to use a DNS query to obtain a list of MME IP addresses for handling a given TAI. if set to yes, the S10 interface managed objects (MOs) are created dynamically. if set to 'no', the S10 interface MOs are created from provisioned data and are deleted when this provisioning data is removed from the system.
- **discoverMmeEnh** - Discover MME Enhancement - Parameter controls reselection of another MME if the response from the target MME times out or the first choice per DNS records is unavailable. By default the flag shall be set to No. When flag is set to Yes, MME shall select 2nd S10 link if the first choice (per DNS records) is unavailable or if the response from the target MME times out.
- **discoverSgw** - Discover SGW - Indicates whether or not (YES or NO) to use a DNS query to obtain a list of SGW IP addresses for handling a given TAI. if set to yes, the S11 interface managed objects (MOs) are created dynamically. if set to 'no', the S11 interface MOs are created from provisioned data and are deleted when this provisioning data is removed from the system.
- **discoverSrvccMsc** - Discover SRVCC MSC - Indicates whether to discover the MSC's SRVCC IP address via DNS. the default is no, indicating that the IP address must be provisioned.
- **discoverTaiDefaults** - Discover TAI Defaults - Feature activation for default TAC Mapping to TA Discovery Provisioned Parameters
- **discoverTais** - Discover TAIs - Enable WMM to discover non-provisioned TAIs.
- **dlBufferingPacketCount** - DL buffering suggested packet count - Downlink buffering suggested packet count.
- **dltPrevPdnConnOnNewReq** - Delete previous PDN connection on new request - This flag determines the call flow of a PDN connectivity request where a UE already has PDN connection. no (default) - the current call flow is used. when an additional PDN connection request is received after attach the MME checks to see whether there is already a PDN connection to the network specified by the APN in the request. when the received APN is the same as the APN of a PDN connection that has already been established for the given UE, the MME sends a PDN connectivity reject with cause code 55. yes - instead of sending a PDN connectivity reject the MME will first delete the previously established PDN connection and then honor the subsequent PDN connection request. note that if the PDN connectivity request with APN=x is for a UE with only one already established PDN connection and that connection is for APN x, the MME would have to reject the PDN connectivity request as is currently done.
- **dualConnectivity** - Dual connectivity - Enable/disable dual connectivity feature.

- **e911apnNiForMode1pgwSelection** - E911 APN NI for mode 1 PGW selection - If set to yes, MME will support the use of the E911 APN NI for the mode 1 selection of PGW for emergency attach and also PDN connection request to E911. if set to no, MME will default to using the S-NAPTR procedure to select a PGW.
- **edrxEpochLeapSeconds** - eDRX epoch leap second - Number of leap seconds that have occurred since the eDRX epoch time.
- **edrxEpochTime** - eDRX epoch time - Starting point in time of counting H-SFN for calculation of eDRX paging opportunities. Any valid past date (YYYY-MM-DD) and time (HH:MM:SS) in GMT can be entered.
- **edrxMaxPagingHoldTime** - eDRX maximum paging hold time - Maximum time interval (in milliseconds) that the MME will hold a call processing procedure.
- **edrxNewLeapSecondTime** - eDRX new leap second time - The date and time when an additional leap second will occur.
- **edrxNotifyHssTime** - eDRX notify HSS time - Time interval (in milliseconds) prior to the eDRX paging send time that the MME should send the notify command to HSS when handling a pending SMS request for an idle UE with eDRX enabled. this parameter is set to account for the time needed for the SMS service center to be informed of the UE's availability and the resulting SGS paging request to be sent to the MME.
- **edrxPagingSendTime** - eDRX paging send time - Time interval (in milliseconds) before paging time window (PTW) start time that the MME should start sending the S1AP-PAGING messages to the eNB. this parameter is set to account for small time synchronization differences between the MME and the eNB.
- **emergency5gslwkWoN26** - Emergency 5GS Interworking Without N26 - Enables emergency session to be handed over between 5GS and EPS without N26 interface in the MME. Applied when EPS-5GS Interworking without N26 is set to Yes for the MME.
- **emergencySimlessUeOverride** - Emergency SIMless UE override - Emergency SIMless UE override when set to yes MME will accept emergency attach from SIMless UEs from any TAI served by the MME irrespective of emergency provisioning per TAI.
- **emergencyWlanHo** - Emergency WLAN Handover - This parameter controls the support for IMS emergency session continuity for handovers between LTE access and WLAN for authenticated home subscribers. When supported, the MME receives WLAN access selected PGW information from HSS, and on handover to LTE the same PGW can be selected. When an emergency call is initially started on LTE access, the MME also registers via NOR the selected PGW to HSS to be possibly handed over to WLAN access.
- **emerMtLrNoEsmLc** - Emergency MT-LR do not contact E-SMLC for UE location information - If yes, when the MME receives mobile terminated location request with emergency client type or MT-LR with any client type while an emergency call is ongoing, the MME contacts the eNB for UE current E-CGI information. this information is sent back to the GMLC. the MME does not contact the E-SMLC to obtain any UE location information.

- **empsAdvancePriority** - EMPS advance priority - Enable the detection of advance priority eMPS users and eMPS IMS priority users (Non advance priority users).
- **empsDscp44Marking** - EMPS DSCP 44 Marking - Enable the ability to mark eMPS S11 and S10 messages with a provisionable DSCP value. Enabling the eMPS DSCP 44 Marking global parameter shall be denied if the eMPS Advance Priority global parameter is disabled. Disabling the eMPS Advance Priority global parameter shall be denied if the eMPS DSCP 44 Marking global parameter is enabled
- **empsPriorityOnlySubscriptionBased** - eMPS Priority Only Subscription Based - When set to Yes, enables MME to provide eMPS priority for UEs that have HSS subscription indicating MPS-EPS-Priority bit set on the MPS-Priority AVP. MME does not check other parameters for UEs that have MPS EPS functionality enabled on (on MME local provisioning on service agreement profile) and MPS-EPS-Priority bit set on the HSS subscription.
- **empsPrioToHss** - eMPS Priority to HSS - MME treats with priority any signal it receives with RRC establishment cause indicating priority. If set to no, MME only treats with priority a UE that additionally either has priority HSS subscription or a bearer is established with a priority ARP value.
- **enable2gBvccountIndb** - Ability to store BVC counts DB.
- **enable2GdataJumboFrame** - Enable 2G data Jumbo frame - This parameter enables the 2G data Jumbo Frame
- **enable2gEnhancesgblb** - Ability to load PAPSS using the number of BVCs assigned to NSEs.
- **enable2gUsrdataSafetynet** - Ability for WMM to detect 2G user data throughput safety net breach and throttle 2G attaches and PDP activations. 2G allowed user data throughput is 145000000 bits/sec.
- **enable3GdataJumboFrame** - Enable 3G data Jumbo frame - This parameter enables the 3G data Jumbo Frame
- **enableAlarmEventsStream** - Enables Alarm Events streaming via fluentd for CNF deployment
- **enableCacheAgeOut** - enable Cache Age Out feature - If the enableCacheAgeOut value is set to Yes the Cache Age Out feature uses defined Quiet-period-timers per UE Class. These timers are used to manage the CPPS cache. By default the enableCacheAgeOut is set to NO, the feature is by default disabled.
- **enableCallTraceBuffering** - Enable Call Trace Buffering - Enables the buffering of Call Trace messaging before IMSI/SUPI is known.
- **enableCapdegradeOn2gciphering** - Ability to downgrade WMM 2G capacity by 20% when 2G ciphering is on.
- **enableCpuBasedControlOnly** - Whether or not WMM performs overload control based on CPU only or take into account message rate and storm rate for controls.

- enableEffectiveloadSnalgo - Enable effective load snalgo.
- enableLocalRoamerQciArpOvr - Enable local roamer QCI ARP override - Enable local roamer QCI ARP override whereby skipping QCI and ARP override parameters for dedicated bearers in case of LBO traffic.
- enableLocalRoamerQosPolicy - Enable local roamer QoS policy - Enable local roamer QoS policy.
- enableLogEventsStream - Enables Log Events streaming via fluentd for CNF deployment
- enableMmeOverloadBackoffTimer - Enable MME overload backoff timer - This flag is used to enable the capability of MME to include a randomly-selected T3346 timer value whenever an attach request, TAU request, service request, or extended service request procedure is rejected due to IPDS CPU overload, CPPS CPU overload, or CXN object overload. no (default) - set the T3346 back-off timer to 0. yes - set the T3346 back-off timer to a randomly-selected value between provisioned minimum and maximum values.
- enableOLCContextDump - Whether or not WMM dumps service context data during overload report.
- enableOlcSchddelayDegrade - Whether or not VMM capacity should be degraded on scheduling delay.
- enableOldGutiPaging - Enable old GUTI paging per MME - Enable old GUTI paging per MME.
- enablePra - Enable presence reporting area - Enable MME support for presence reporting of an UE entering or leaving an area.
- enablePrometheusGW - Enable Prometheus Functionality - This parameter controls whether Prometheus functionality is enabled in the system. This means that this gParm does not control the way the PM data are sent/retrieved from Prometheus (i.e. sent via Prometheus Push GW or retrieved directly from Prometheus). This would be specified via a another gParm that is named Prometheus Mode.
- enableProvCc - Enable provisional cause code - Enable MME support for configurable or provisional cause code functionality.
- enablePsmT3412MinTime - Enable use of PSN T3412 Min Time - This parameter enables/disables the feature CMM support for Minimum T3412 Timer for UE Supporting Power Savings Mode'. By default the feature is disabled.
- enableRejectIdrepslocinfoOnol - Enable/Disable reject IDR with EPS location info request during overload.
- enbToSgwIpv4v6TransportLayerAddress - eNB/gNB IPv4v6 Transport Layer Address IP Type - MME passes eNB/gNB S1-u IPv4v6 Transport Layer Address to the SGW if it received this IP Type from eNB/gNB and this parameter is set to YES. Otherwise MME matches S1-C IP Address Type and passes it to the SGW.

- **enhancedAlternateGwReselection** - Enhanced Alternate GW Reselection - Defines whether the MME reselects an alternate SGW or PGW during the same procedure in defined error cases. Setting this parameter to 'Yes' requires that gParm supportAlternateGwReselection is also set to 'Yes'. MME reselects an alternate PGW during Attach and PDN Connectivity procedures when SGW returns cause code 72 (System failure), 78 (Missing or unknown APN), 84 (All dynamic addresses are occupied) or 91 (No memory available) and the cause source indicates PGW. MME also reselects an alternate PGW when SGW returns cause code 100 (Remote peer not responding) and cause source indicates SGW. MME reselects an alternate SGW during Attach and mobility procedures when SGW returns cause code 72 (System failure) or 91 (No memory available) and cause source indicates SGW.
- **enhancedPgwlIsolation** - Enhanced PGW Isolation - Defines whether the MME isolates PGW during additional cases. Enabling this parameter requires gParm ltePgwlIsolation to be set to 'Yes'. MME considers PGW related errors for PGW isolation functionality additionally to the legacy functionality when SGW returns error code 73 (No resources available), 91 (No memory available) or 100 (Remote peer not responding) and cause source indicates PGW. MME also considers SGW returned error for PGW isolation functionality when SGW returns error code 100 (Remote peer not responding) and cause source indicates SGW.
- **enhancedSgwIsolation** - Enhanced SGW Isolation - Defines whether the MME isolates SGW during additional cases. Enabling this parameter requires gParm lteSgwIsolation to be set to 'Yes'. MME considers SGW related errors for SGW isolation functionality additionally to the legacy functionality when SGW returns error code 73 (No resources available) or 91 (No memory available) and cause source indicates SGW.
- **enhancedSgwRestorationProcedures** - Enhanced SGW restoration procedures - Support enhanced SGW restoration procedures or not.
- **enhancedSGWselectionIRAT** - Enhanced SGW selection for IRAT mobility - Enable/Disable support enhanced SGW selection for IRAT mobility per MME.
- **enhApnCorrectionPerImsi** - Enhanced APN Correction per Imsi - Parameter controls when the rejection cause code shall be sent by MME. By default the flag shall be disabled, when enabled MME during an Attach or PDN Connectivity Request, if the APN received from the UE, is not an HSS provisioned APN and is not in locally configured in the MME correction/replacement table, MME shall reject the procedure with cause code 'Requested service option not subscribed'. In addition if the APN received from the UE, is locally configured in MME APN correction/replacement table but correction comes back with APN which is not an HSS provisioned APN. (In other words, even after correction the APN must be subscribed or it will be rejected) Feature shall be enabled when apnlist table is enabled.
- **ePlmnPerImsiTaiRange** - EPLMN per IMSI Range - MME uses the EPLMN list provisioned per IMSI and TAI range if the parameter is set to 'True'.
- **eps5gslwkWithoutN26** - EPS-5GS Interworking without N26 - Enables the EPS-5GS interworking without N26 interface support in the MME.

- **epsInfoInUlr** - Support of EPS info in ULR - When parameter is YES, EPS info is supported in updateGprsLocation message, otherwise it is not supported
- **epsNasSecurityContextInclusion** - EPS NAS Security Context Inclusion - Parameter controls inclusion of NAS security context at the end of MM Context IE. By default, flag shall be disabled. When enabled, MME will include NAS security context at the end of MM Context IE to indicate the mapped or native security context during the MME-1 to MME-2 reallocation
- **eutranSupportedRatType** - Eutran supported RAT type - Configure eutran supported RAT type. values are both, wbeutran and Clot.
- **exclDelayDlPktNotifReq** - Exclude delay downlink packet notification request - Control the inclusion of the delay downlink packet notification request.
- **excludeDestinationHost** - Exclude destination host - Enable or disable exclusion of destination host for S6ad and S13 interfaces.
- **excludeNbiotDrbCheckForCpCiot** - Exclude NB-IoT Check on DRB for CP Clot - The MME will not consider the NB-IOT UE support of multiple DRBs when the UE requests to set up a subsequent CP Clot bearer.
- **excludeUIExceptionReports** - exception reports are excluded from rate limit - By default MME rate control applies all packets. With this parameter is set, Mo exception data packet will be excluded in the rate limited (not count)
- **excludeVendorSpecificApplicationId** - Exclude vendor specific application id - Enable or disable exclusion of exclusion of vendor specific application id for S13 interface.
- **extendEmmCauseIe** - Extend EMM cause IE - Enable inclusion of extended cause IE in attach reject and TAU reject messages with EMM cause #15.
- **fallbackToDefaultApnOi** - Fallback to default APN-OI - Activate fallback to use of default APN-OI in NAPTR query if NAPTR query with APN-OI replacement fails.
- **fiveGcNotRestrictedIndicationSupport** - 5GC Not Restricted Indication Support - Parameter controls the support for 5GCNRI and 5GCNRS indication flags in create session request message. By default, the flag shall be disabled.
- **fmKafkaStream** - Kafka Stream for the Alarm Functionality - This parameter controls the option to stream the Alarms to an external KAFKA cluster. When this parameter is set, the Alarms would be streamed to a specific topic and broker of an external KAFKA cluster.
- **fnsBckpSrvrHbNoRspns** - FNS Backup Server No Responses to heartbeat message - If MME does not receive responses to the heartbeat messages for consecutive number of times specified by this parameter then MME declares the link is not available and generates a major alarm.
- **fnsBckupServer** - FNS Backup Server - Global parameter to enable support of FNS backup server. If the parameter is set to 'true' then CMM establishes TCP connection to the provisioned backup server and start updating the UE context.

- **fnsBckupSrvrHbTimer** - FNS Backup Server Heart Beat Timer - Parameter specifies the heart beat interval to the FNS backup server.
- **fnsBckupSrvrRspTimer** - FNS Backup Server Response Timer - Time to wait for FNS backup server response after sending a request to the server.
- **fnsBckupSrvrUpdateType** - FNS Backup Server Update Type - Parameter to specify which UE context needs to be stored on the FNS backup server.
- **fnsInMmePool** - FNS IN MME Pool - This parameter is used to indicate whether FNS MME and CMM MME are in the same pool. The parameter will be used by CMM uelb command directed move to FNS to make the CMM UELB is compatible with FNS directed move.
- **forcePgwReselectionOnCsRequestTimeout** - Force PGW reselection on CS request timeout - If set to yes, mme shall select an alternate PGW if there is no response to the create session request. the number of attempts to select PGW remains the same. the feature only applies to GW selection mode 2. the applicable procedures are attach and subsequent PDN connectivity request.
- **fourDigitMncMccSuppt** - Four Digit MNC MCC Support - The parameter controls whether four digit MNC and MCC is used in RAI and RNCid pre-REL8 DNS queries for Gn-SGSN. By default the parameter shall be disabled.
- **freqCountMonitoring** - Frequency Count Monitoring - Enables the MME functionality to monitor the frequency count of UE attempts for defined procedures.
- **freqCountMonitoringMtAccess** - Frequency Count Monitoring MT Access - Defines whether the MME, while frequency count monitoring functionality is activated, allows the Mobile Terminated Access for the UEs. If the UE is blacklisted but not yet implicitly detached, the network may page the UE due to e.g. incoming downlink data and the UE may trigger Service Request/Extended Service Request/Control Plane Service Request due to that. When MT access is allowed, MME does not increment procedure attempt counter for MT access attempts.
- **freqCountMonitoringSecurity** - Frequency Count Monitoring Security - Defines whether the MME increments procedure frequency count before or after running any of the security procedures (especially integrity check).
- **fullPgwApnGtpOvlControlSupport** - Full PGW APN or Node Level Overload Control Support - Defines whether the MME supports PGW indicated overload. -PGW Overload Control Information IE containing APN list on GTPv2 messages, when APN is congested on a PGW. -Back-off Time IE on Create Session Response when PGW rejects the request due to APN congestion.
- **gea1Degradation** - GEA1 degradation - GEA1 degradation percent. Warning: For Nokia personnel only.
- **gea2Degradation** - GEA2 degradation - GEA2 degradation percent. Warning: For Nokia personnel only.
- **gea3Degradation** - GEA3 degradation - GEA3 Degradation percent. Warning: For Nokia personnel only.

- **ggsnLbMode** - GGSN Load Balancing mode - Set the corresponding mode for GGSN Load Balancing
- **gmlcOverlappingLocationRequests** - GMLC overlapping location requests - Enable/disable GMLC overlapping UE location requests.
- **gnSgsnSnaptr** - Gn SGSN SNAPTR - None: gn-sgsn will not use S-NAPTR. pgw-ggsn: gn-sgsn will use S-NAPTR only for allocation of a co-located pgw-ggsn. all: gn-sgsn will use S-NAPTR for all FQDN resolutions.
- **gnSgsnSrv** - Gn SGSN SRV - None SRV RR only for pgw-ggsn selection.
- **gnTargetIdFormat** - Gn target id format - Determines how it encodes and decodes the target identification IE when the MME is sending a GTPv1 (Gn) message.
- **gprsSubscriptionData** - GPRS subscription data - Request the HSS to include GPRS subscription data in ULA if available.
- **gtpLoadControlSupport** - GTP load control support - GTP load control support.
- **gtpOvlControlSupport** - GTP OVL control support - Support for GTP overload control.
- **gwGtpOvlControlSupport** - GW GTP OVL control support - Support for GTP overload control from gateways to MME.
- **gwSelectionMode** - Gateway selection mode - Select PGW topologically close to selected SGW or topologically close SGW and PGW pair.
- **gwSelectionModeForSgsn** - SGW PGW Selection Enhancement for SGSN - This parameter controls the S/P -GW pair selection for S4-SGSN. When the parameter is GW Selection Mode 1 then the SGW priorities are taken into consideration for selecting the appropriate S/P -GW pair. When the parameter is GW Selection Mode 2 then the PGW priorities are taken into consideration for selecting the appropriate S/P -GW pair.
- **gxInterfacesNetworkAnsiSs7** - GX interfaces network ANSI SS7 - This flag configures the SS7 protocol type 'ANSI' or 'ITU' for the m3ua-network(s) of the gx (i.e. gr, gf, ge, gd) interfaces. when multiple m3ua-networks are defined for the gx interfaces, the protocol type must be the same across all the m3ua-networks.
- **highCapacityIHspa** - High Capacity iHSPA - When set to 'No' it is possible to create a total sum of 512 iHSPAs and RNCs When set to 'Yes' it is possible to create a total sum of 8192 iHSPAs and RNCs (of which only up to 512 may be RNCs)
- **homeDomainName** - Home domain name - Home domain name.
- **homogeneousIMSSupportAVP** - Whether or not WMM includes homogeneous IMS support AVP in ULR and NOR.
- **homogeneousSupportNblotTas** - Homogeneous support NB-IoT TAs - Controls whether the MME includes Tracking Areas supporting NB-IoT Radio Access Technology when evaluating Homogeneous support of Voice over PS sessions towards the HSS. Yes - NB-IoT TAs are included when evaluating Homogeneous support of IMS Voice over PS

sessions No - NB-IoT TAs are not included when evaluating Homogeneous support of IMS Voice over PS sessions

- homogeneousSupportTais - Homogeneous Support TAs - Defines whether MME includes IMS Voice over PS of the serving PLMN TAs or all TAs MME globally (serving PLMN and shared PLMN TAs) when evaluating homogeneous IMS voice over PS support towards the HSS.
- homogeneousVopsSrvccCapa - Homogeneous Voice over PS SRVCC Capability - MME provides Homogeneous-Support-of-IMS-Voice-Over-PS-Sessions towards HSS according to specific requirements: TAs support for IMS voice over PS (separate activation parameter controlling whether to include all TAs MME globally, or all TAs on the serving PLMN), voice over PS support for the UE (on svcAgreementProfile) and checking or not checking the UE SRVCC capability (separate parameter on svcAgreementProfile).
- hssInitiatedCallTrace - HSS initiated call trace - Support HSS initiated call trace or not.
- hssNorReduction - HSS NOR reduction - When set to yes: if MME selects the same PGW as in ULA/IDR from HSS, MME does not send NOR to HSS. when set to no: MME sends NOR to HSS for this case.
- hssRestorationMode - HSS Restoration Mode - Defines how HSS restoration is performed after HSS unavailability. The HSS restoration should only be enabled i.e. set to values other than 'HssRestorationNone' with MME only configurations.
- hssUlrReduction - HSS ULR reduction - When set to yes: if UE attaches with valid local GUTI and MME has IMSI and subscription data, MME does not send ULR to HSS. when set to no: MME sends ULR to HSS for this case.
- hssUnavailabilityContinueN26 - HSS Unavailability Continue N26 - Defines whether the MME continues the procedure when HSS is unavailable and the UE moves from 5GS via N26 to the MME, and the MME has available security context or mapped security context and thus the MME is able to continue serving the UE without new authentication on EPS.
- hssUnavailFailedPercentTriggerBlacklisting - HSS unavailable failed percentage triggering blacklisting - Percentage of failed MME attempted dialogs triggering blacklisting.
- ignoreSGSapMMInfo - IgnoreSGSapMMInfo.
- imeiOperatorPolicyAccessRestriction - IMEI Operator Policy Access Restriction - Enables IMEI based operator policy for access restrictions based on provisioned criteria.
- imeiOperatorPolicyAuthenticationDisabled - imeiTac Based Authentication - authentication to be enabled/disabled for a particular TAC, whereas entire network it might be running. Similarly based on TAC, IMSI Combination.
- imeiOperatorPolicyEirCheckDisabled - IMEI Operator Policy EIR Check Disabled - Enables IMEI based operator policy to disable EIR check of UEs matching the provisioned criteria.

- **immediateLrrDuringMtLr** - Immediate LRR During MT-LR - If the MME needs to send a LRR related to an emergency call event and there is an ongoing MT-LR procedure, this parameter controls whether the LRR is to be sent immediately when the event occurs or later after the MT-LR procedure completes. When enabled, the LRR is sent immediately.
- **imsEmergencySupportIndication** - IMS emergency support indication - Enable/Disable IMS voice enabled check for IMS emergency support indication.
- **imsilmeiLookupFail** - IMSI IMEI Lookup Fail - Parameter **imsilmeiLookupFail** controls APN Override feature with the values 'UseDefault and Reject. By default this parameter shall be disabled.
- **imsiPagingInAllMmeServingAreas** - IMSI paging in all MME serving areas - Allow IMSI paging in all MME serving areas?.
- **inclEmerNumListImsAllowed** - Include Emergency Number List When IMS Allowed - Parameter controls whether MME includes the UE the Emergency Number List for a UE that has EPS-Network-Feature-Support with IMS-voice-over-PS allowed, but EPS-Network-Feature-Support emergency call is not allowed on the MME. When Emergency Number list is included it uses the Serving PLMN Emergency Profile information for the Emergency Number List.
- **inclExt2InEPSQoSIEs** - Include extended 2 in EPS QoS IEs - Enable/Disable inclusion of extended-2 octets in EPS QoS IEs.
- **inclExt2InNegotiatedQoSIEs** - Include extended 2 in negotiated QoS IEs - Enable/Disable inclusion of extended-2-octets in R99 negotiated QoS IE.
- **includeActiveApnAvpInUlrAfterRsr** - Include active-apn AVP in ULR after RSR - Include active-apn AVP with PGW information in ULR after RSR.
- **includeEcgidPlainFailure** - Include ECGID in PLA in failure - Supports inclusion of ECGID in PSLR even if MME as not able to get UE location.
- **includeExtUeIndexInPagingMsg** - Include Extended UE Identity Index in PAGING message - If this parameter is set to 'Yes' then the MME shall include the Extended UE Identity Index IE in PAGING messages.
- **includeHrlAlways** - Include Handover Restriction List Always - When enabled, MME shall include the Handover Restriction List IE in S1AP messages irrespective of EPLMN, forbidden TA or forbidden LA provisioning. If EPLMN and forbidden TA and LA are not provisioned then MME shall only include the serving PLMN IE in the HRL IE.
- **includeMapped3gppQciToUe** - Include mapped 3GPP QCI value to UE - When set to yes, the 3GPP QCI value mapped from the operator defined QCI value will be included in the activate default bearer context request, activate dedicated bearer context request, and modify bearer context request messages sent to the UE. this applies to both default and dedicated bearers.
- **includeMappedGummei** - Include mapped GUMMEI - This command is only applicable for WMM deployed as a combo. this command enables/disables inclusion of the mapped GUMMEI in S1-AP S1 SETUP RESPONSE message and S1-MME CONFIGURATION UPDATE

message. the inclusion of the mapped GUMMEI facilitates eNB to select the same physical node supporting the combo MME/SGSN when UE is moving between 2G/3G and LTE area served by the same combo MME/SGSN.

- **includeNblotSmsServiceStatus** - Include CIOT SMS service status - Include SMS services status IE in attach accept and TAU accept message if MME fails to set up SMS for CIOT UEs and MME only accepts EPS attach only.
- **includeNeighborListInTaiList** - Include neighbor list in TAI list - Indicates whether the MME includes (Yes) the provisioned set of neighbor TAIs in the list of TAIs that is sent to the UE when the UE registers at the MME (Attach or TAU). if this parameter is set to 'No', the MME only includes the last seen TAI in the list of TAIs sent to the UE.
- **includeRanapTli** - Include RANAP TLI - Enables RANAP TLI in RAB assignment request to RNC triggered by GGSN PDP context modification.
- **includeServingNetworkS11ie** - Include serving network S11 IE - BSM to control whether the serving network IE is included in the modify bearer request when the PLMN has changed.
- **includeServingPlmnRateControl** - support Serving Plmn Rate Control function - With this parameter enable, Serving PLMN Rate Control will be supported
- **includeT3448Timer** - include T3448 timer in overload situation - support T3448 timer for overload control with limited data transfer. T3448 used for limited UE transfer data over CIOT Control Plane Optimization during timer run. With this parameter, T3448 is allowed to be used in CMM.
- **includeUliCrSessModBrr** - ULI Inclusion in Create Session And Mod Brr Msg - Parameter controls inclusion of 'ULI for SGW' IE in Create Session and Modify Bearer Request message that is sent to SGW. By default, the flag shall be set to Yes. When flag is set to 'No', MME will not include the IE in Create Session and Modify Bearer Request message that is sent to SGW.
- **includeUliForSgsn** - Include ULI for SGSN - This flag is used to enable/disable inclusion of the user location information IE in associated GTP messages over the Gn interface to the GGSN. no - indicates that the ULI IE is not included. yes (default) - indicates that the ULI IE is included.
- **inhibitPeerSrsResponses** - Inhibit Peer SRS Responses - This parameter provides the ability on the NR10 client side to inhibit NR10 responses from being sent by the peer NR10 server. When set to true, responses to update requests and delete requests are inhibited. This parameter applies to both AMF and MME NR10 functionality.
- **initialAttachIndicationInUlr** - Initial attach indication in ULR - Initial attach indication flag.
- **interPlmnTauAttach** - Inter PLMN TAU attach - If it is set to yes then MME shall check per PLMN provisioning whether inter-plmn S10/S3/Gn message can be sent. by default, the feature is disabled.

- **ipAddrInclusion** - IP Address Inclusion - Parameter controls inclusion of same IP address type as the destination address used in the IP header in the Create Session Request message. By default, the flag shall be disabled.
- **ipv4v6sgsn** - IPv4v6 SGSN - A value of yes indicates release 8 SGSN and above. a value of no indicates pre-release 8 SGSN.
- **ipv6SelectionAllowedOnGn** - IPv6 Selection Allowed on Gn - Use this parameter to allow IPV6 support on Gn-C.
- **iratGnMobilityQoSParametersUpdate** - IRAT Gn mobility QoS parameters update - Indicates whether MME sends modifybearercmd to SGW at the end of TAU when QoS parameters in S6ad ULA are different than those in Gn contextresponse.
- **iratMobilityWithOtherPLMNAllowed** - IRAT mobility with other PLMN allowed - IRAT mobility with other PLMN allowed.
- **iratS3mobilityQoSParametersUpdate** - IRAT S3 mobility QoS parameters update - Indicates whether MME sends modifybearercmd to SGW at the end of TAU when QoS parameters in S6ad ULA are different than those in S3 contextresponse.
- **iuAlgsSelection** - new algs support in lu interface - Customer is claiming that UIA1 and UEA1 is compromised already, thus, supporting UIA2 and UEA2 in 3G is needed. The parameter will control to use a configured preference list of algs or not. If the parameter is setting to Yes, then a preference list of algs needed to be configured by customer, SGSN will based on both UE capability and configured preference list to inform RNC the selected algs. If the parameter is set to No, then existing function is unchanged i.e. continue with UIA1/UEA1.
- **keepGbrWhenEnbFails** - Keep GBR when eNB fails - Indicates whether (Yes), or not (No), all the GBR bearers associated with an eNB are deactivated when the S1-MME link to that eNB fails.
- **keepGbrWhenPagingFails** - Keep GBR when paging fails - Indicates whether (Yes), or not (No), the GBR bearers of the UE are deactivated if paging fails to locate the UE.
- **keepSameSgw** - Keep the Same SGW - Keep the Same SGW during IRAT Mobility from 4G (in case of SGSN)
- **keepUpClotUeS1Bearer** - Keep UP ClOT UE S1-U bearer - Enable keeping of a S1-U bearer of a UE supporting UP ClOT moves into a TAI that does not support UP ClOT optimization and S1-U data transfer.
- **lawfulInterceptMtLrNoEsmlc** - lawful Intercept MT_LR no E-SMLC - This is for MT_LR client type lawful Intercept service do not contact E-SMLC for UE location information. If yes, when MME receives mobile terminated location request with client type=lawfulInterceptService, and activateLcsClientTypeLawfulInterceptServices is set, the MME contacts the eNB for UE current E-CGI information. This information is sent back to the GMLC. The MME does not contact the E-SMLC to obtain any UE location information. If no, nothing changes and keep existing function (i.e. MME contacts E-SMLC to obtain UE location)

- **lcsapNumberOfResetsWithoutAckForAlarm** - LCSAP number of resets without ack for alarm - LCSAP number of resets without ack for alarm.
- **limitS1apConnEst** - Limit S1AP conn est - The parameter is used to limit the sending of the S1AP CONNECTION ESTABLISHMENT INDICATION message to establish UE-associated logical S1-connection if the logical connection is already established. the UE-associated logical S1-connection would be established by the MME if the MME sends S1AP DOWNLINK NAS TRANSPORT before sending the S1AP CONNECTION ESTABLISHMENT INDICATION message.
- **linkAlarmCustomize** - Link alarm customize - If the global parameter is set to 'Yes', customized severity levels in the linkalarmcfg is used to set the alarm severity for links and interfaces. if the global parameter is set to 'No', customization of link alarm severity is not supported.
- **localImeiTacRejectionList** - Local IMEI-TAC Rejection List - Enables the functionality to provision IMEI-TAC rejection list locally in the MME. When a UE whose IMEI-TAC is included in the list attempts to attach in the MME, the MME rejects the attempt with a provisioned EMM cause code.
- **locallyProvisionedSgwPgw** - Locally provisioned SGW or PGW - Test switch to provision sgw idx in range from 241 to 256.
- **locallyProvisionedSgwPgwTopon** - Locally provisioned S/PGW Topological Selection - Enables topological S/PGW selection for locally provisioned S/PGW.
- **locationReportingForUeWithLppsSupportOnly** - Location reporting for UE with LPP support only - This flag indicates whether LCS will be supported for UEs with or without LPP capability. when flag is enabled only UEs that indicated LPP support can perform LCS procedure. this flag is used in conjunction with the other LCS feature activation fields. those fields apply as well. UE indicates support of LPP capability in the UE network capability IE of the attach request and/or TAU request message.
- **logEcgiReceivedInWriteReplaceWarningMessage** - Log ECGI Received In Write Replace Warning Message - Parameter controls feature activation GPARM flag. By default, flag shall be disabled. When enabled, MME shall log eCGI that is received in Write-Replace-Warning-Response Message from eNB.
- **longestGtDigitsMatch** - Longest GT Digits Match - This parameter allows or denies whether the SCCP is able to route on the longest GT digits match
- **lowPriorityAccess** - Low priority access - Activate the low access priority capabilities on MME.
- **lowPriorityTrafficDdnThrottling** - Low priority traffic DDN throttling - Throttling of DDN requests for the low priority traffic.
- **lraUli** - LRA ULI - Enable/Disable LRA-ULI.
- **ltePgwIsolation** - LTE PGW isolation - Setting to yes enables PGW isolation feature on MME. setting to no disables PGW isolation feature on MME.

- lteSgwIsolation - LTE SGW isolation - Setting to yes enables SGW isolation feature on MME. setting to no disables SGW isolation feature on MME.
- maintainErab - Maintain ERAB - When set to yes, MME will maintain E-RAB that are not successfully switched in the core network during the E-RAB modification indication procedure. when set to no, MME will release/delete e-rab that are not successfully switched in the core network during the E-RAB modification indication procedure.
- maskedImeIsV - Masked IMEISV - Masked IMEISV in S1AP messages.
- maxCid - Max CID - Max CID.
- maxEcgiAgeEmerMtLrNoEsmlc - Maximum time interval in seconds for requesting eCGI for emergency MT-LR - If the parameter is set to 0 with the global feature activation parameter emermtlrnoesmlc is yes, MME page ECM-IDLE UE and send eCGI information or send location-reporting control information for ECM. if the parameter is set to a non-zero value, when MME gets MT-LR request, MME will check the difference between MME stored eCGI timestamp of the UE and time received MT-LR. if the time difference is less than provisioned value, then the MME reports stored eCGI information to GMLC. if time difference is greater than the provisioned value, then MME will request current eCGI either via paging or location report control depends UE status.
- maxEcgiAgeLawfulInterceptMtLrNoEsmlc - Max eCGI age lawful intercept MT_LR_no E-SMLC - Maximum time interval in seconds for requesting eCGI for lawful intercept MT-LR. If the parameter is set to 0 with the global feature activation parameter lawfulInterceptMtLrNoEsmlc is yes, MME page ECM-IDLE UE and send eCGI information or send location-reporting control information for ECM_connected UE. if the parameter is set to a non-zero value, when MME gets MT-LR request, MME will check the difference between MME stored eCGI timestamp of the UE and time received MT-LR. if the time difference is less than provisioned value, then the MME reports stored eCGI information to GMLC. if time difference is greater than the provisioned value (or equal provisioned value), then MME will request current eCGI either via paging or location report control depends UE status
- maxEcgiAgePlmnOperMtLrNoEsmlc - Max eCGI age PLMN operator MT_LR no E-SMLC - Maximum time interval in seconds for requesting eCGI for PLMN Operator MT-LR. If the parameter is set to 0 with the global feature activation parameter plmnOperatorMtLrNoEsmlc is yes, MME page ECM-IDLE UE and send eCGI information or send location-reporting control information for ECM_connected UE. if the parameter is set to a non-zero value, when MME gets MT-LR request, MME will check the difference between MME stored eCGI timestamp of the UE and time received MT-LR. if the time difference is less than provisioned value, then the MME reports stored eCGI information to GMLC. if time difference is greater than the provisioned value ((or equal provisioned value), then MME will request current eCGI either via paging or location report control depends UE status
- maxEcgiAgeValueAddedMtLrNoEsmlc - Max eCGI age value added MT_LR no E-SMLC - Maximum time interval in seconds for requesting eCGI for value Added service MT-LR. If the parameter is set to 0 with the global feature activation parameter

valueAddedMtLrNoEsmIc is yes, MME page ECM-IDLE UE and send eCGI information or send location-reporting control information for ECM_connected UE. if the parameter is set to a non-zero value, when MME gets MT-LR request, MME will check the difference between MME stored eCGI timestamp of the UE and time received MT-LR. if the time difference is less than provisioned value, then the MME reports stored eCGI information to GMLC. if time difference is greater than the provisioned value (or equal provisioned value), then MME will request current eCGI either via paging or location report control depends UE status

- **maxEnbsPagedDuringMinorOverload** - Maximum eNBs paged during minor overload - The parameter specifies the maximum number of eNBs paged for a single page attempt when a minor overload condition is present.
- **maxMmeUeRadioCapLength** - Maximum MME Ue Radio Capability Length - Maximum Ue Radio Capability Length Supported on the MME. Default is 8192. Setting this value to 24576 or 49152 enables storage of Radio capability length greater than 8192 bytes in the RCIS if RCIS is available
- **maxNumberWarningAreasEcgi** - Max number warning areas ecgi - Maximum number of warning areas (ecgi) that can be includes in S1AP WRITE-REPLACE WARNING REQUEST or S1AP KIL-REQUEST messages sent to eNB.
- **maxTaRelativeCapacity** - Max TA relative capacity - Used to select to send to the eNB the maximum or minimum MME relative capacity of provisioned MME relative capacities of TAIs supported by an eNB to the eNB in S1-AP SETUP RESPONSE message and in MME CONFIGURATION UPDATE message.
- **mdt** - Minimization of drive tests - Activation/deactivation minimization of drive tests.
- **measInfoldUsage** - measInfold use in PM - This parameter describes the type of measInfold parameter to add to PM files.
- **minFqdnLabelMatching** - Enable applying minimum matching fqdn labels between SGW and PGW fqdn(s) during SPGW pair selection. When gParm is enabled, the MME use configuration from fqdnLabelMatchProfile and fqdnLabelMatchUut to further resolve min fqdn label matching per Service Combination and subscribed UE Usage Type
- **mmeAdditionalMsisdn** - MME Additional MSISDN - GPARM parameter
mmeAdditionalMsisdn controls the support for Additional MSISDN by MME. By default the parameter shall be no. When GPARM parameter mmeAdditionalMsisdn is yes, the MME shall inform HSS that it supports the A-MSISDN feature by setting the corresponding feature bit of the Feature List AVP in the Update Location Request, Insert Subscriber Data Answer and Delete Subscriber Data Answer commands.
- **mmeApnNiExtension** - MME APN NI extension - No (default) - for MME, disable the extension of the APN NI with configurable length of MSISDN digits. yes - for MME, enable the extension of the APN NI with configurable length of MSISDN digits.
- **mmeCpToUpBearerSwitch** - Enable MME Triggered CP to UP Bearer Switch - If this parameter is set, the MME shall proceed to switch CP bearers to UP if DL or UL CP data

packet size exceeds the provisioned threshold (cpDataPktSizeToSwitchCpToUp). If the parameter is not set, the MME shall not switch CP bearers to UP.

- **mmeOverloadMmRejectionNasCc** - MME overload MM rejection NAS CC - EMM NAS cause code to be sent in the mobility management reject messages (Attach reject, TAU reject and service reject) due to any MME overload or throttling of MM procedures. valid values are 2-3,5-26,35,39-40,42,95-101,111 per annex a of TS 24.301.
- **mmeOverloadPlaResult** - MME overload PLA result - Select diameter result to be sent in Slg provide subscriber location answer when MME is in major or critical overload.
- **mmePgwQueryEmptyflagRr** - MME PGW QUERY EMPTYFLAG RR - No (default) - disable the MME to support NAPTR RR with '' (empty) flag for PGW selection. yes - enable the MME to support NAPTR RR with '' (empty) flag for PGW selection.
- **mmeProseAuthorizationSupport** - Public safety proximity services - Public safety proximity services.
- **mmeSgwQueryEmptyflagRr** - MME SGW Query Empty Flag - This parameter is used to enable MME handling of reception of S-NAPTR RR with empty flag. By default, the capability disabled. If the parameter is set to 'Yes' then the capability is enabled.
- **mocnAccurateLoc** - P-GW/GGSN enhancement with 3G MOCN setup for location update. - In 3G MOCN setup, during SRNS relocation procedure or EUTRAN-to-UTRAN PS handover, it is possible that the PLMN ID received in RAU after relocation is not identical to the PLMN ID that the target SGSN learns with the GTP:FORWARD_RELOCATION_REQ message. Activate this parameter to force a GTP:UPDATE_PDP_CONTEXT message that updates PGW/GGSN with the correct PLMN ID. SGSN Multi Operator Core Network (3G MOCN) needs to be activated for this functionality.
- **monitoringConfigFailureSendNor** - Monitoring Config Failure Send Nor - Specifies whether the MME is to send a Notification Report to the HSS if a monitoring event cannot be configured at the MME (3GPP R14 functionality).
- **monitoringEventsDeletionByUla** - Monitoring Events Deletion by ULA - Parameter controls detection and deletion of monitoring event configuration AVPs received in ULA. When parameter is enabled, upon receiving one or more Monitoring-Event-Configuration AVP(s) in the ULA, the MME shall start the detection of the Monitoring events indicated in those AVP(s), if not already started, and shall stop the detection and delete the previous monitoring events (if any) which are not indicated in those AVP(s). If there is a failure when starting the detection (e.g. maximum resources exceeded), the MME shall not store the failed configuration(s) and shall send a notification of those events whose configuration have failed, If the Subscription-Data AVP is received in the ULA but it does not contain any Monitoring-Event-Configuration AVP(s), the MME shall stop the detection and delete all stored monitoring event configurations (if any).
- **monteCommFailRanUeInact** - Specifies whether the MME is to send a T6a:RIR Communication Failure Event Report if the RAN requests UE S1 release due to UE Inactivity.
- **mscLoadBalance** - MSC load balance - Configure MSC load balance algorithm.

- **mscSelMultipleCsCoreSuppt** - MSC selection of Multiple CS Core network - GPARM will control the feature flag. By default the global parameter shall be false/disabled.
- **mscVlrReselection** - MSC VLR reselection - Enable/disable alternate MSC selection if location update times out or MSC rejects the request with reject cause 'congestion.
- **msisdNHomeGwWithCc** - MSISDN Based Home GW selection with CC - MSISDN based home GW selection with CC(charging characteristic)
- **multipleIwsSupport** - Multiple IWS support - CLI command to control the support for multiple IWSs between ALU MME and 3rd party IWS.
- **multiPra** - Multiple PRAs - Enables support for -multiple Presence Reporting Areas (PRAs) per PDN connection in the MME, -multiple IP address segments per PRA identifier for Dual connectivity segment based reporting in the MME.
- **multiPraOperationEnbsEcgis** - Multiple PRA Operation eNBs and ECGIs - Enables the functionality to enter multiple eNBs and ECGIs with a single command addressing a Presence Reporting Area.
- **multiPraOperationTacsEnabled** - Multiple PRA Operation with TACs Enabled - Enables the functionality to enter multiple Tracking Areas with a single command addressing a Presence Reporting Area.
- **multiSimSupport** - Multi SIM Support - GPARM controls support for MultiSim with YES or NO flag with a default value of NO. When the parameter is set to YES the MME will drop an MT-LR if it only contains an MSISDN for the location query. With the default value setting of NO, the CMM will allow an MT-LR that contains only an MSISDN as the identifier
- **n26DirectFwd** - Indicates whether direct data forwarding is enabled or disabled on n26 handover.
- **n26IndirectFwd** - Indicates whether indirect data forwarding is enabled or disabled on n26 handover.
- **n26iwk** - N26iwk - N26 interworking - Enables N26 interworking.
- **nasNonDeliveryIndicationEsmDataTransfer** - Enable NAS Non Delivery Indication for ESM Data Transfer
- **nasTokensToCompare** - NAS tokens to compare - The number of NAS tokens to compare in determining whether a UE is authenticated in a handover scenario from UTRAN/GERAN to LTE.
- **nblotEmmTimerIncr** - CIOT EMM timer increment - This value is added to the timers in 10.2.2 of 24.301 for CIOT UE.
- **nblotEsmTimerIncr** - CIOT ESM timer increment - This value is added to the timers in 10.3.2 of 24.301 for CIOT UE.
- **nblotOlAtMinorOverload** - NB-IOT Overload Control At Minor Overload - If the parameter is Yes, the MME will perform NB-IOT overload control when the MME starts

experiencing minor overload. If the parameter is No, the MME would not take any NB-IOT overload control action until the MME starts experiencing major overload.

- **nblotPagingTimerIncr** - CIOT paging timer increment - This value is added to paging response timer for CIOT UE.
- **nbiotRatTypeToHss** - NBIOT RAT Type to HSS - If the global parameter is set to 'Yes' then MME shall send NB-IoT RAT Type to HSS. If the global parameter is set to 'No', then MME shall send EUTRAN RAT Type to HSS.
- **noClrRejForMissingClrFlags** - No CLR Reject for Missing CLR-Flags - Parameter controls rejecting CLR for Missing CLR-Flags AVP. By default the flag shall be disabled. When enabled MME will not reject CLR with Diameter Unable to Comply if CLR-Flags AVP is missing.
- **nonDdnPagingTauCollision** - Non DDN paging TAU collision - Enable MME to treat TAU request without active flag as a paging response and re-establish bearers if TAU is received while MME is paging a UE.
- **nonEmergencyPdnConnAsNewPdnConnAtAttach** - Non-emergency PDN connection as new PDN connection at attach - MME support to treat non-emergency PDN connection as a new PDN connection at attach.
- **nrBasedGwSelectionSgsn** - NR Capability based GW selection - This parameter enables SGSN selecting GW using NR capability of the UE. When this parameter is enabled, SGSN selects NR capable GW if NR capability is supported at the UE. If the parameter is setting to Yes, then a fixed tag=5gc will be added to pre-rel8 DNS query or rel8 DNS with service +nc-nr. If the parameter is set to No, then existing function is unchanged i.e. no changes for DNS query of 5G NSA UEs.
- **nrCcDoubleS1PsAck** - Normal release cause for double S1 connections and path switch request ack - If this parameter is set to 'Yes' then the cause IE of S1AP UE CONTEXT RELEASE COMMAND is set to 'Normal release' for the following cases.
- **nriLength** - Length of Network Resource Identifier, LEN - This parameter defines the length of Network Resource Identifier (NRI) in the CMM SGSN. Possible values are 0, 5, 6 and 7.
- **nrPriorityOverSmf** - NC-NR priority over SMF - When this parameter is enabled, then the MME gives priority to NC-NR capable gw against combo PGW-C capable gw when multiple candidates, each serving only one of these service parameters are retrieved by DNS query
- **nullNri** - Null network resource identifier - This parameter defines the null network resource identifier, 255 ... not defined (default) 0 to 31 ... valid values.
- **numIpsPerPgwFqdn** - Supported number of IPs per PGW FQDN - Sets the maximum number of IPv4 and IPv6 addresses per PGW FQDN. Only 2 values are supported (2 or 8). When set to '2' or '8', as many as 2 or 8 IPv4 and 2 or 8 IPv6 addresses are supported per PGW FQDN.

- **obtainPgwFqdnEps5gsMob** - Current parameter enables performing PGW FQDN lookup when PGW FQDN isn't available during EPS to 5GS mobility
- **ocCallTraceContinue** - OC call trace continue - Indicates whether or not call trace will continue (YES), or be suspended (NO), on a CPPS when the CPPS is in a major or critical overload condition.
- **ocSendOverloadStartAndStopToEnbs** - Overload control send overload start and stop messages to eNBs - A value of yes indicates the MME will send overload start and overload stop messages to eNBs as part of overload control. a value of no indicates the MME will not send overload start and overload stop messages.
- **octetEncodingSupport** - Octet Encoding Support - Parameter controls support for Octet Encoding for the Indication IE contained within messages sent over the S11 interface. By default the flag shall be disabled. When the flag is enabled, MME shall only encode the first three bytes of indication flags.
- **odbBitsSupport** - ODB Bits Support - Parameter controls feature activation. By default the GPARAM flag is off. When the feature is activated, MME upon receiving ODB AVP in ULA/IDR from HSS, shall set the following 2 bits: `hssPxy_SupportedFeatureList_ODBHplmnApn` and `hssPxy_SupportedFeatureList_ODBVplmnApn` regardless of `homer` vs. `homer`, and regardless of value of `odb_all_apn` and `odb_hplmn_apn` bits in `ServiceAgreementProfile`
- **overrideEnhEmmCcCreateSessRespUserAuthFail** - Override Enh EMM Cause Code Create Session Response User Auth Fail - This parameter controls whether to use `restNasMappingProfile` provisioning for Create Session Request cause code #72 or #92. By default, flag shall be disabled. When this parameter is enabled, `restNasMappingProfile` shall be used.
- **overrideVaa** - APN Override regardless the VAA parameter value - This parameter controls the APN override functionality of network identifier for roamer subscribers. When this parameter is ON the relevant functionality is decoupled from the value of the VAA parameter. When this value is OFF the relevant functionality continues to work as currently (i.e. coupled with the value of the VAA parameter)
- **pagingAfterInterRatRedirection** - Paging After Inter Rat Redirection - Parameter controls MME to allow paging to continue upon receiving a DDN after handling of an UE context release request with cause inter-rat-redirection, By default the flag shall be disabled. When enabled, MME upon receiving a DDN after handling of an UE context release request with cause inter-rat-redirection, MME shall allow paging to continue.
- **pagingGapFailCountTrigger** - Paging Gap Fail Count Trigger - Parameter Paging Gap Fail Count Trigger indicates the number of failed paging operations that start the paging gap timer.
- **pagingOptimizationInHenb** - Paging optimization in home eNB - Is paging optimization supported for the home eNB?
- **pagingOptimizationInMme** - Paging optimization in MME - A value of y indicates that paging optimization is supported for the MME.

- **pagingOverloadPrevention** - Paging Overload Prevention - Parameter controls feature activation. By default the flag shall be always enabled.
- **pagingUponTadsReceived** - Paging Upon T-ADS Received - Parameter controls paging upon receiving T-ADS from HSS in IDR message. By default, flag shall be disabled. When enabled, MME upon receiving T-ADS in IDR message from HSS when UE is in either registered or in idle state, shall respond with Insert Subscriber-Data Answer (with IMS-Voice-Over-PS Sessions-Supported (1) and shall immediately starts paging the UE.
- **passiveOffload** - Passive offload - MME is in passive offload mode or not.
- **passiveUelbOfLoadRate** - UELB Off-Load Rate - For Passive UELB this parameter specifies the per CPPS of load rate
- **pcmdSheddingDuringOverload** - PCMD shedding during overload - Control flag for PCMD shedding during overload. this flag can be used to determine MME behavior related to PCMD during CPU overload.
- **pcscfRecovery** - PCSCF recovery - Configure P-CSCF recovery or not.
- **pdnConnEstEnhSupport** - PDN Connection Re-establishment after SGW Change with Roamers and Voice Enhancements - Parameter controls enabling or disabling the support for enhancements to PDN Connection Re-establishment after SGW change with Roamers and Voice feature. By default this parameter shall be disabled.
- **pdnDifferentApnOi** - PDN connection for a different APN OI - This parameter is used to enable/disable MME to set up a second PDN connection with a different APN OI for APN when UE already has a PDN connection for the same APN.
- **pdnHoS11ApnFormulation** - PDN Handover S11 APN Formulation - Defines how the MME forms the APN-OI of the APN IE on S11 Create Session Request towards the SGW, when the MME receives a 'Handover' PDN Connectivity Request from the UE. When set to 'Yes', the MME uses enhanced logic to utilise either the possibly provisioned override PLMN id, the PLMN id obtained from the UE's IMSI or the UE's requested APN-OI. When set to 'No' (default), the MME uses the Destination-Realm as received from the HSS indicating the selected PGW of the PDN to set the APN-OI on S11 CSR.
- **pdnReestablishSgwChange** - PDN reestablish SGW change - All PDN connections for each UE shall be always examined for PDN-reestablishment after every SGW change.
- **pdnRejectDueToSubscriptionEsmCc** - ESM cause for PDN rejected due to subscription - ESM cause code in PDN connection reject message when PDN connection request is rejected due to subscription.
- **pdnReleaseNrSecRatRestriction** - Release PDN connection due to NR as Secondary RAT restriction - Allow MME to release a PDN Connection for a DCNR capable UE after MME determines UE is restricted access for NR as a Secondary RAT.
- **pgwAddrOnS6aAndPgwFinding** - PGW Address on S6a and PGW Finding - **pgwAddrOnS6aAndPgwFinding** parameter has two purposes for e.g. VoLTE - VoWifi handovers: it -controls which received PGW IP address(es) the MME uses, when the MME sends the selected PGW information inside the MIP6-Agent-Info AVP on the S6a NOR

and the S6a ULR (Active-APN AVP(s)). -controls how the MME finds the PGW for the PDN when the MME receives Attach Request or PDN Connectivity Request with Request type Handover and establishes the PDN with Create Session Request.

- **pgwApnGtpOvlControlSupport** - APN Level Overload Control Support - When enabled the MME shall support APN level overload control functionality, whenever overload information is provided on APN level from the PGW.
- **pgwIsolationFaultThreshold** - PGW isolation fault threshold - This command is used to provision PGW fault threshold for the PGW isolation. if the number of the PGW faults (remote peer not responding cause received in the S11/S4 create session response) exceed the provisioned threshold in a provisioned duration timer then WMM will isolate the PGW from the selection to new PDN connection for provisionable duration.
- **pgwPlmnIdExclusionSupport** - PGW plmn ID exclusion support - Enable/Disable exclusion of PGW PLMN ID AVP when PGW FQDN is present in NOR.
- **pgwRestartNotification** - PGW restart notification - If it is yes, then MME/S4-SGSN sends support of PGW restart notification indication in node feature IE of GTPv2 echo request and echo response. if it is no, then MME will not indicate the support of PGW restart notification to the SGW.
- **pgwSelectionRestrictedToColocatedGgsnPgW** - PGW Selection Restricted To Colocated GGSN/PGW - Parameter controls MME support for Configuration Control for Restriction of PGW Selection to Colocated GGSN/PGW during IRAT Mobility scenario. By default, flag shall be disabled. When enabled, restriction for gp-service for roamer and gn-service for homer shall be removed. This shall apply to Gateway Mode Selection 2.
- **pgwSelectionWithNonNcSmf** - PGW Selection With Non nc-smf - Parameter controls feature activation. By default the flag shall be disabled. When enabled, MME shall support preferring PGW(s) which specifically have a non nc-smf tagged service type for non N1 mode UEs. In other words, with the new gParm enabled, MME shall enforce associating LTE only UE with non-combo PGW. This feature shall be applied only for gateway mode selection 2.
- **pgwSelectionWithSnaptr** - PWG selection with snaptr - Enable/Disable support for PGW selection with snaptr.
- **plmnCtrlPreferGw** - plmn level control of preferred Gw - PLMN Level Control of Preferred Gateway. Customer preferred control of GW type selection. This gParm is to control this feature. When this plmnCtrlPreferGw parameter is set, then PLMN level Control of Preferred Gateway will be enable. If the parameter is false (default value), then S4 SGSN keeps existing behaviour.
- **plmnOperMtlrNoEsmIc** - PLMN operator MT_LR no E-SMLC - MT_LR client type PLMN Operator Service do not contact E-SMLC for UE location information. If yes, when MME receives mobile terminated location request with client type=plmnOperatorServices, and activateLcsClientTypePlmnOperatorServices is set, the MME contacts the eNB for UE current E-CGI information. This information is sent back to the GMLC. The MME does not contact the E-SMLC to obtain any UE location information. If no, nothing changes

and keep existing function (i.e. MME contacts E-SMLC to obtain UE location for this MT-LR type) .

- **plmnS6dSelection** - PLMN level s6d&Gr selection - When rel8 architecture enable, S4-SGSN always selects HSS. With this feature enable, PLMN level control to select either S6d/Gr will be controlled by operators. Also fallback from s6d to Gr could be also controlled by operators. When this parameter set to true (yes), then above mentioned two parameters can be configured by operator/PLMN level. If the parameter set to false (no, default), then SGSN keeps the same as current. When this feature is enable, then SG01236 is not used.
- **pmFileToFluentd** - Send PM File to Fluentd - This parameter controls the functionality for sending PM XML file to fluentd.
- **pMtuDiscoveryS1IpV6** - Path MTU Discovery for IPv6 S1/M3 links - Controls Path MTU Discovery for IPv6 S1/M3 links. no (default) - disable Path MTU Discovery yes - enable Path MTU Discovery
- **powerSavingsMode** - Power savings mode - Support for UE power saving mode.
- **praDclpv4** - Presence Reporting Area Dual Connectivity IPv4 - Activates customized Presence Reporting Area (PRA) functionality for dual connectivity. When dual connectivity is applied for a UE, presence reporting is used to inform the SGW/PGW/PCRF (upon request) whether the UE currently uses a 5G gNB (UE is inside) or not (UE is outside). The gNBs use IPv4 addressing. MME uses a customized schema in analyzing whether a provided RAN IPv4 user plane endpoint belongs to an eNB or a gNB.
- **praDclpv6Prefix** - PRA DC IPv6 Prefix - Activate the Presence Reporting Area functionality to report on dual connectivity with IPv6 prefixes. MME detects the dual connectivity usage based on provisioned IPv6 address prefixes. The functionality uses CN predefined PRAs. The MME reports via PRA reporting that a UE is inside the area when the MME detects that the gNB user plane IP address is covered by the provisioned IPv6 address prefix i.e. UE is using dual connectivity with a 5G gNB. The UE is outside the area when the gNB user plane IP address is not covered by the provisioned IPv6 address prefix i.e. UE is not using dual connectivity with a 5G gNB.
- **praDcSegment** - PRA Dual connectivity segment reporting - Activate the Presence Reporting Area functionality to report on dual connectivity utilizing a provisioned IPv6 address segment. The functionality uses CN predefined PRAs. The MME reports via PRA reporting that a UE is inside the area when the MME detects that a 5G gNB having a provisioned IPv6 address segment is added with dual connectivity for the UE. The UE is outside the area when the corresponding 5G gNB is removed for the UE.
- **praDcSingleTimerAllowedRestartCount** - PRA DC Single Timer Allowed Restart Count - Defines the number of times the MME is sequentially allowed to restart the single DC PRA optimisation timer. When the MME has already previously restarted the single optimisation timer the times allowed by this parameter and the MME still receives new RAN endpoints causing DC PRA status change (and the MME would in this case again restart the single optimisation timer without this limit), the MME stops the single

optimisation timer and sends out the current status of the DC PRAs that is not yet reported towards the SGW/PGW. This parameter is only applied when global parameter PRA Dual connectivity segment reporting is set to DC Segment Optimized Single Timer.

- **praDormantState** - PRA Dormant State Enabled - Enables the functionality to change the state of the CN predefined Presence Reporting Area (PRA) between active (i.e. normal) and dormant. In dormant PRA state, the MME suppresses any PRA status (inside/outside) reports towards the SGW/PGW. In dormant PRA state the MME still accepts and stores PRA action to request PRA status reporting. Dormant PRA state is applicable for both 3GPP standard based PRA functionality as well as the dual connectivity segment based reporting functionality.
- **praForRoamers** - PRA for roamers - When set to Yes, this parameter allows MME to support Presence Reporting Area (PRA) functionality for roaming UEs.
- **praldleState** - PRA Idle State - Enables Presence Reporting Area (PRA) functionality clarification change. If the UE is in ECM-IDLE state, the MME may either bring the UE into ECM-CONNECTED state, or, report based on the UE's last known location and when the UE was there.
- **praModifyAction** - PRA Modify Action - Controls whether the MME reports immediately 'inside'/'outside' when receiving PRA action with action value 'modify' independently of the previous report. When set to 'yes', MME always reports immediately 'inside'/'outside' even if previous report was the same. When set to 'no', MME does not report immediately 'inside'/'outside' if previous report was the same.
- **praModifyEnabling** - PRA Modify Enabling - Enables Presence Reporting Area (PRA) modify action support in the MME.
- **praPace** - Presence Reporting Area Pace - The pace to send Presence Reporting Area Information towards the SGW/PGW with the standard based Presence Reporting Area functionality. This setting is applicable for both dual connectivity segment based PRA and standard based PRA functionalities. The pace is defined in messages/second per CPPS.
- **praPacing** - PRA Pacing - Enables Presence Reporting Area pacing functionality to ensure the surrounding network elements are not impacted by bursts of Presence Area Reporting Information.
- **praRel145** - Presence Reporting Area 3GPP Rel 14-15 - Enables 3GPP Release 14-15 enhancements for Presence Reporting Area functionalities: e.g. multiple PRA ids per PDN, PRA sets and PRA modify action.
- **praSets** - Enables support of Presence Reporting Area (PRA) sets in MME.
- **praStdSuppressInitialOut** - PRA Standard Suppress Initial Out - Controls whether on 3GPP Standard Presence Reporting Area (PRA) functionality the first/initial outside report is suppressed. If set to Yes and MME receives PRA action Start from SGW/PGW, MME only sends initial PRA report (PRA Information) to SGW/PGW when the UE is/has arrived inside the corresponding area. If set to No, the MME will send the initial report

(PRA Information) immediately after receiving PRA action Start from SGW/PGW despite of UE being inside or outside of the corresponding area initially.

- **praStdSuppression** - PRA Standard Suppression - Enables MME suppression of too frequent PRA reports for 3GPP standard based PRA (Presence Reporting Area) functionality. If set to Yes, the MME suppresses a subsequent PRA report inside the PRA suppress time period, unless the PRA status stays the same for the PRA suppress cancel defined time period. If set to No, MME sends subsequent PRA reports according to UE mobility based on 3GPP standard definitions with no restriction to reporting frequency.
- **pRau2g** - PRAU timer in 2G - This parameter controls the usage of independent 2G and 3G pRau timers. If the parameter is set, then SGSN is allowed to configure pRau timer differently in 2G and 3G. If the parameter is not set, existing functionality will be used (2G and 3G with the same value for pRau timer).
- **preferDataMode** - Prefer data mode - Selection of preferred data mode when a UE subscription has APN configuration.
- **preferredPgwId** - Preferred PGW Identifier - Preferred PGW identifier to determine PGW identity
- **prefGnToGGSNIpVer** - IP address version-preference for transport layer - Use this parameter to control the IP address version-preference for transport layer.
- **preRel8DomainName** - Pre release 8 domain name - Pre release 8 domain name.
- **prometheusFormat** - Prometheus Format - This option controls the format of the data model and notation used to report Prometheus Counters. This option is applicable in all cases i.e. either when CMM is configured to push the Prometheus Counters to PrometheusPushGW or when Prometheus Server directly scrapes the Prometheus Counters from CMM.
- **prometheusMode** - Prometheus Mode - There are two functionalities supported for Prometheus Server in order to retrieve the relevant PM Data. The first functionality is when CMM pushes the relevant PM Counters to PrometheusPushGW and then Prometheus Serves scrapes the relevant PM Counters from PrometheusPushGW. The second functionality is when Prometheus Server directly scrapes the relevant PM Counters from CMM. The scope of this parameter is control which functionality will be enabled from CMM.
- **prometheusRequests** - Prometheus Requests - This parameter defines the maximum allowed number of Prometheus Requests, in the specific Prometheus Sliding Window (defined with parameter **prometheusSlidingWindow**) , that could in parallel directly scrape the relevant PM Counters from CMM. This parameter is applicable only when the functionality/option to allow Prometheus Server to directly scrape the relevant PM Counters from CMM is configured. Warning: For Nokia personnel only.
- **prometheusSlidingWindow** - Prometheus Sliding Window - This parameter defines the CMM sliding window (in seconds) in which Requests from Prometheus Server to directly scrape the data from CMM are accepted. The number of parallel requests in this sliding window are defined with parameter **prometheusRequests**. This parameter is applicable

only when the functionality/option to allow Prometheus Server to directly scrape the relevant PM Counters from CMM is configured. Warning: For Nokia personnel only.

- provPciPvi - Provisionable PCI and PVI value supported - Supports provisionable PCI and PVI values.
- psCsCoordination - PS CS coordination - PS CS coordination capability for MOCN configuration.
- pseudoRandomTlliCollisionHandling - Pseudo-random Tlli Collision Handling - When parameter is set to No existing collision handling mechanism is maintained regarding 2G Attach with random TLLI. When status is set to Yes collision handling will not be applied for a 2G Attach with Random TLLI even if is already used by an ongoing Attach procedure, as long as Attach message was received via different nsei/bvci pair. Furthermore the SGSN will silently drop a new Attach request with a random TLLI when the included random TLLI in the Attach request is already used by ongoing Attach from the same nsei/bvci pair.
- pseudoSpgwIpv4address - Pseudo SPGW IPv4 address - The SGW/PGW IPv4 address used in creating the S1-U SGW tunnel IDs to be sent to eNB in S1AP INITIAL UE CONTEXT SETUP message when MME does not use SGW or PGW for bearer creation during SDS UE attach request.
- pseudoSpgwIpv6address - Pseudo SPGW IPv6 address - The SGW/PGW IPv6 address used in creating the S1-U SGW tunnel IDs to be sent to eNB in S1AP INITIAL UE CONTEXT SETUP message when MME does not use SGW or PGW for bearer creation during SDS UE attach request.
- pseudoUeIpv4address - Pseudo UE IPv4 address - The UE IPv4 address to be used for an SDS UE when MME does not use SGW and PGW for bearer session setup during attach procedure.
- pseudoUeIpv6address - Pseudo UE IPv6 address - The UE IPv6 address to be used for an SDS UE when MME does not use SGW and PGW for bearer session setup during attach procedure.
- purFlagsAvp - PUR-Flags AVP - PUR-Flags.
- purgeUeWhenHandoffToEhrpd - Purge UE when UE hands off to eHRPD - When UE hands off to eHRPD, if this global parameter is yes, MME deletes VLR and sends PUR to HSS.
- pwsRestartProcedure - PWS restart procedure - Forwarding of PWS RESTART INDICATION message to CBC.
- qci1HoRejectionGeran - QCI1 HO Rejection GERAN - When set to 'Yes', MME rejects incoming S1AP Handover Required attempts when -the handover is attempted towards 2G i.e. Handover Type is set to 'LTEtoGERAN' -the Handover Required message does NOT include SRVCC HO Indication IE -the UE currently has a dedicated bearer with QCI1 established. The MME sends Handover Preparation Failure with cause 'Handover Failure In Target EPC/eNB Or Target System' and does not proceed further with the handover. When set to 'No', MME proceeds with the handover normally.

- **qci1HoRejectionUtran** - QCI1 HO Rejection UTRAN - When set to 'Yes', MME rejects incoming S1AP Handover Required attempts when -the handover is attempted towards 3G i.e. Handover Type is set to 'LTEtoUTRAN' -the Handover Required message does NOT include SRVCC HO Indication IE -the UE currently has a dedicated bearer with QCI1 established. The MME sends Handover Preparation Failure with cause 'Handover Failure In Target EPC/eNB Or Target System' and does not proceed further with the handover. When set to 'No', MME proceeds with the handover normally.
- **qosModCollEnh** - QoS Modification Collision Enhancements - This parameter controls the feature activation. By default this parameter shall be set to value 'No'.
- **queueUeCtxtRelUserInactivityErabModIndication** - Queue UE Context Release Request User Inactivity During E-RAB Modification Indication - Defines whether the MME queues an incoming S1AP UE Context Release Request with radio network cause 'user inactivity' from eNB when the MME has received S1AP E-RAB Modification Indication from the eNB and has not yet sent E-RAB Modification Confirm as response. When set to Yes, MME triggers UE Context Release Command after sending out E-RAB Modification Confirm. When set to No (default), MME drops the incoming UE Context Release Request with radio network cause 'user inactivity'.
- **raiBasedDeactPdpRate** - deactivation PDP rate - This parameter determines with what rate (reactivations per second) PDP context reactivations should be done during RAI-based context reactivation.
- **raiBasedUeHandlingEnabled** - RAI based UE handling - RAI/RAI group based UE handling
- **ranapResetAfterSysRestart** - Enables performing SGSN init RANAP RESET after system restart/upgrade
- **rangeDiscoverTais** - Range Discover TAIs - Feature activation for Enhanced TA Discovery Auto-Provisioning
- **rateLimitPerApn** - Rate Limit Per APN - Whether the MME should enforce PLMN Rate Control per APN/PDN per 3GPP TS 23.401 CR 3486 or per PDN. The default is no rate control per PDN.
- **rcvryDlSdsMaxLimitOverride** - Maximum number of recovery DL SDS messages allowed to protect SRS from MME.
- **rcvrySrUlrLimitOverride** - Recovery SR ULRs allowed per second after which CPPS will start dropping recovery ULRs until rate is below limit.
- **rcvrySrUlrMaxdeferCountOverride** - Maximum number of times recovery SR ULRs can be deferred to protect MME and HSS during recovery ULR storm.
- **redirectionCommandRate** - Redirection command rate - Maximum allowed rate of S102 redirection command messages.
- **regionalSubscription** - Regional subscription - Setting to yes activates regional subscription on the gn-sgsn. setting to no deactivates regional subscription on the gn-sgsn.

- **rejDtrExtendedServiceRequest** - Reject Dual Transceiver Receiver Extended Service Request - MME shall be provisioned with GPARAM flag 'rejDtrExtendedServiceRequest' to control sending UE Context Modification Request message to E-UTRAN upon receiving Extended Service Request that is attached to E-UTRAN. By default this flag shall be disabled. When the flag is enabled, MME shall send cause code #18 (CS Domain not available) in service reject to eNodeB. When the flag is disabled, MME shall act as it did before introduction of this functionality, i.e. MME upon receiving Extended Service Request shall send UE Context Modification Request message to E-UTRAN.
- **rejectAlgorithmDuringOverload** - Reject algorithm during overload - Overload control logic with a different set of messages/procedures to shed at major and critical overload thresholds.
- **rejectCcForGmmErrorCode11** - Reject Cause Code for GMM Error Code 11 - This parameter defines which Reject Cause code will be used for GMM Error Code 11.
- **rejectCcForGmmErrorCode12** - Reject Cause Code for GMM Error Code 12 - This parameter defines which Reject Cause code will be used for GMM Error Code 12.
- **rejectCcForGmmErrorCode13** - Reject Cause Code for GMM Error Code 13 - This parameter defines which Reject Cause code will be used for GMM Error Code 13.
- **rejectCcForGmmErrorCode14** - Reject Cause Code for GMM Error Code 14 - This parameter defines which Reject Cause code will be used for GMM Error Code 14.
- **rejectCcForGmmErrorCode15** - Reject Cause Code for GMM Error Code 15 - This parameter defines which Reject Cause code will be used for GMM Error Code 15.
- **rejectEsrNoSrvcc** - Reject ESR no SRVCC - Reject ESR requested by an ECM-CONNECTED UE with IMS voice bearer.
- **rejectNodeRelocatingPeriodicTau** - Reject node relocating periodic TAU - Enable/disable MME support for enhanced handling of intermme TAU with update type periodic.
- **requestPsCellInLrc** - Request PS Cell in LRC - When location reporting control procedure is used in MME, if Request PS Cell in LRC is set, then MME includes additional location information towards eNB. eNB will include the current PSCell in the report if EN-DC is activated. If Request PS Cell in LRC is not set, MME will not request additional location information under location reporting procedure as existing function, then eNB will not report PSCell in location report.
- **reservedBoolParameter1** - Reserved bool parameter 1 - This parameter is reserved for future use.
- **reservedBoolParameter2** - Reserved bool parameter 2 - This parameter is reserved for future use.
- **reservedBoolParameter3** - Reserved bool parameter 3 - This parameter is reserved for future use.
- **reservedBoolParameter4** - Reserved bool parameter 4 - This parameter is reserved for future use.

- reservedBoolParameter5 - Reserved bool parameter 5 - This parameter is reserved for future use.
- reservedIntParameter1 - Reserved int parameter 1 - This parameter is reserved for future use.
- reservedIntParameter2 - Reserved int parameter 2 - This parameter is reserved for future use.
- reservedIntParameter3 - Reserved int parameter 3 - This parameter is reserved for future use.
- reservedIntParameter4 - Reserved int parameter 4 - This parameter is reserved for future use.
- reservedIntParameter5 - Reserved int parameter 5 - This parameter is reserved for future use.
- restrictPagingDuringOverload - Restrict paging during overload - Restrict paging during overload.
- restrictS10linksToNeighborMme - Restrict S10 links to neighbor MME - MME support for restricting S10 to neighboring MME.
- restrictSubExtQoS - Restrict Subscribed Extended QoS - CMM MME restricts subscribed extended QoS when UE is 5G DCNR restricted or 5G DCNR not capable. MME limits QoS to 4.29 Gbps or less when setting up or modifying default, GBR, or non-GBR bearer(s).
- retransCsSvcNotifyOnNasNonDeliveryRadioConnLost - Retransmit CS Service Notify NAS Non-Delivery Radio Conn Lost - Current gParm enables treating NAS Non-Delivery Indication msg carrying cause 'radio connection with ue lost' the same as if cause were 'S1 intra system Handover triggered', or 'X2 Handover triggered' for SGs downlink service operation. When parameter is enabled, then when receiving NAS Non Delivery Indication (cause=radio connection with ue lost) the MME will expect to receive X2 Handover safeguarded by timer, at expiration of which the CS Service Notification will be resent
- retriesForDcNrUes - Retries for DCNR Capable UE - Number of retries selecting PGW-C(s) during PDN Connectivity procedure upon receiving GTP error or timeout from gateway for DCNR Capable UEs.. For example, MME will send its first attempt to the PGW, if it receives no response within a certain period of time, the MME will retransmit the same request to the SGW/PGW and if/when it gets an error back from the SGW then it will move on to select the next best gateway to try. Assuming there is only one PGW in the sub-area, this means that the MME will perform round 2 (look for +ue entries but not find any) and then round 3 (look for _nc-nr entries, where it will find many to choose from). The next MME attempt will then be to a nearby +nc-nr PGW.
- retriesForUutUes - Retries For UUT UEs - Number of retries selecting PGW-C(s) during PDN Connectivity procedure upon receiving GTP error or timeout from the gateway for UUT UE(s) For example, MME will send its first attempt to the PGW, if it receives no response within a certain period of time, the MME will retransmit the same request to

the SGW/PGW and if/when it gets an error back from the SGW then it will move on to select the next best gateway to try. Assuming there is only one PGW in the sub-area, this means that the MME will perform round 2 (look for +ue<uut> entries but not find any) and then round 3 (look for _nc-nr entries, where it will find many to choose from). The next MME attempt will then be to a nearby +nc-nr PGW.

- **retrieveCellId** - Retrieve cell id - Yes - enable retrieval of ECGI from the eNB to be included in the LSC-AP location request message sent to the E-SMLC when MME is triggered either by GMLC or by HSS to get UE location. the retrieval of the ECGI applies to all client types. no (default) - disable the retrieval and MME sends the ECGI that was received in the most recent S1-AP message.
- **retrieveUeLocation** - Retrieve UE location - This flag determines whether UE location information is retrieved from the eNB. no (default) - indicates that the operator policy is to use the last known user location information. yes - indicates that the WMM operator policy is to use the location reporting procedure (23.401 section 5.9.1) to retrieve the ECGI from the eNB.
- **retryPagingForSgsPageRequest** - Retry paging for SGs page request - A value of yes indicates that the MME will perform additional page attempts for each SGs page request received until the UE responds or all page attempts in the paging policy have been exhausted. a value of no indicates that the MME will perform only one page attempt for each SGs page request received. if needed, the MSC must trigger additional page attempts with new SGs page requests.
- **rfspIndexSelectionOption** - RFSP Index Selection Option - When this parameter is enabled, then the MME enables using rfspSelOption option, available in SvcAgreementProfile, to decide the way for assigning RFSP Index for the UE
- **rfspIndexVallmeiTac** - Support RFSP Index Value - GPARM controls the feature activation. By default the parameter shall be disabled. When parameter is enabled, MME shall validate the TAC received in IMEI against provisioned TAC entry and if there is a match, MME shall apply the provisioned RFSP value.
- **rfspInS4Sgsn** - support RFSP in S4 SGSN
- **rfspOverridelmeilmsi** - RFSP Override by IMEI TAC and IMSI - RFSP Override by IMEI TAC and IMSI, giving priority to IMEI TAC
- **roamerQosForWpsUserSupport** - Roamer QoS Profile for WPS User Support - Parameter controls feature activation flag. By default, flag shall be disabled, when enabled MME shall support provisioned Roamer QoS Profile for WPS user.
- **roamQciSuppt** - Roam QCI Support - Upon setting the parameter to TRUE, MME shall check the received inbound roamer GBR QCI value and if the received GBR QCI value is 2, it shall be rejected. With the Default value setting of FALSE, MME shall work as it had before the introduction of this feature.
- **roamQosOvdForRan** - Roamer QoS Override for RAN - When this parameter is on, locally configured QoS for Roamers is sent only towards RAN. The default value shall be off.

- **rsrUlrPercentForCtrlPlaneServiceReq** - Percentage of control-plane service req that MME sends ULR - Percentage of control-plane service request that MME sends ULR due to HSS reset.
- **rsrUlrPercentForDdnServiceRequest** - Percentage of ddn service req that MME sends ULR - Parameter controls percentage of ddn service req that MME sends ULR
- **rsrUlrPercentForServiceReq** - Percentage of MO Service Req that MME sends ULR and UE-Act-Ind - Percentage of MO Service Request that MME sends S6a ULR to HSS, and sends SGs UE-Activity-Indication to MSC/VLR due to HSS Reset; MT Service Request (following Paging) does not trigger ULR due to HSS reset.
- **rsrVlrScanPercentWithoutOverload** - (Reset Subscriber Request) Vlr Scan Percentage Without Overload - When Reset Subscriber Request(RSR)including Subscription-Data or Subscription-Data-Deletion AVP is received in CPPS, VLR scan is triggered in CPPS to find VLR entries that match the RSR user-ids and/or reset-ids AVPs. This VLR scan is triggered periodically to avoid system overload. When the system is not in overload, the number of scanned entries per period is defined in this gParm as percentage of 50000(which is hardcoded). The allowed percentages are [10, 20, ... , 90, 100]. Warning: For Nokia personnel only.
- **rsrVlrScanPercentWithOverload** - (Reset Subscriber Request) Vlr Scan Percentage With Overload - When Reset Subscriber Request(RSR)including Subscription-Data or Subscription-Data-Deletion AVP is received in CPPS, VLR scan is triggered in CPPS to find VLR entries that match the RSR user-ids and/or reset-ids AVPs. This VLR scan is triggered periodically to avoid system overload. If the system is in overload, VLR scan is postponed for the next period. If VLR scan is postponed for 5 consecutive times, VLR scan is triggered in the next period, but the number of scanned entries is reduced. The percentage of the scanned entries is defined in this gParm as percentage of 25000(which is hardcoded). The allowed percentages are [10, 20, ... , 90, 100]. Note that 'rsrVlrScanPercentWithOverload * 25000' must be smaller than 'rsrVlrScanPercentWithoutOverload * 50000'. Warning: For Nokia personnel only.
- **s102** - MME to send S102 redirection command to the IWS obtained from the SRS in case of UE recovery if the UE previously registered with CIOT IWS.
- **s1021xRttRcidInSlr** - S102 CIOT RCID in SLR - Include CIOT reference cell ID in Slg subscriber location report message.
- **s10InterPlmnRelocationQci1Only** - S10 Inter PLMN Relocation QCI-1 Only - Parameter controls feature activation GPARM flag. By default, flag shall be disabled. When enabled, source MME upon receiving inter plmn handover request from eNB will find target MME from TAC reported by eNB, will send forward relocation request to target MME. Target MME upon receiving forward location request will check for the presence of QCI-1 bearer and if QCI-1 bearer is not present target MME shall reject the handover with cause value handover reject target not allowed. Additionally, for TAU, upon receiving context request without QCI-1, Context Response reject with a cause code Target access restricted for the subscriber shall be sent.

- **s10mobilityQosParametersUpdate** - Inter MME S10 mobility QoS parameters update - Indicates whether MME sends modifybearercmd to SGW at the end of TAU when QoS parameters in S6ad ULA are different than those in S10 Context Response or Forward Relocation
- **s11CauseRsrcUnavailPdnDelInSr** - S11 Cause Resource Unavailable PDN Delete Session Request in Service Request - Parameter controls MME sending Delete Session Request to SGW upon receiving Modify Bearer Response with 'Resources not available'. By default the flag shall be disabled.
- **s11ipAddressLoadBalancer** - S11 IP Address Load Balancer - Parameter when enabled prevents MME from creating an S11 link/MO directly to an S11 load balancer. An IP is determined to be a load balancer if the MME has received a Create Session Response with a sender F-TEID for ctrl plane specifying a different IP address than the IP header. If we subsequently receive a Create Session Response from an LB IP and FTEID matches the IP header, an S11 link/MO will not be created. By default, the flag shall be disabled.
- **s11sendCidfDuringS1ho** - S11 send CIDF during S1HO - To enable/disable sending of the S11 create indirect forwarding tunnel request message during an S1HO.
- **s11SuspendNotificationSentPerLbi** - S11 suspend notification sent per LBI - Setting to yes enables S11 suspend notification sent per LBI feature. setting to no disables S11 suspend notification sent per LBI feature.
- **s11uBearerSetUp** - S11-U Bearer Set Up - Enable MME to setup S11-U bearer for a UE supporting both CP and UP optimization indicates preference for UP CloT optimization but the MME is not provisioned to support UP CloT optimization.
- **s11uDIDataBuffering** - S11-U bearer downlink data buffering - Enable S11-U downlink data buffering to handle collision of downlink data with ESM procedures.
- **s13retryDifferentEir** - S13 retry different EIR - When MME gets some errors from EIR (too busy, out of space, etc) or EIR does not respond, MME sends the same request to another EIR in the same PLMN or IMSI range.
- **s1ResumeSecurityContextUpdate** - S1 Resume Security Context Update - If the parameter is set to 'Yes' then MME includes the Security Context ID in the S1AP UE CONTEXT RESUME RESPONSE message.
- **s1SendAllServedPlmnsToEnb** - Send all Served Plmns to Enb - When enabled, MME includes all configured home and shared PLMN IDs in the MME Served PLMNs list when MME sends S1AP Configuration Update, or in S1AP Setup Response when MME responds to eNB S1AP Setup Request.
- **s1SuspendSecurityContextUpdate** - S1 Suspend Security Context Update - If the parameter is set to 'Yes' then MME includes the Security Context ID in the S1AP UE CONTEXT SUSPEND RESPONSE message.
- **s3GnIndirectForwarding** - S3 Gn indirect forwarding - Indicates whether inter-rat handover using Gn or S3 always employs indirect data forwarding (ALWAYS), does not

employ indirect data forwarding (NEVER), or employs indirect 4290 data forwarding for inter-plmn inter-rat handover (INTER-PLMN/RAT ONLY).

- **s4Func** - allow SGSN S4 functionality (Gr/S6d) - Use this parameter to allow SGSN S4 functionality (Gr/S6d).
- **s4SgsnLbo** - S4 SGSN support LBO - S4 SGSN support local breakout function. With this parameter set, the feature will be enable.
- **s6aDraRoamingSupported** - S6ad DRA roaming supported - When set to yes: use provisioned destinationrealm in remoteendpoint table only for homer. for roamer, use hard-coded 'epc.mnc###.mcc###.3gppnetwork.org' instead of provisioned destinationrealm when set to no: regular S6ad DRA.
- **s6adUeTimeZoneRetrieval** - S6ad UE time zone retrieval - This flag is used to enable the retrieval of UE time zone information. no (default) - indicates that UE time zone retrieval is not enabled. yes - indicates that UE time zone retrieval is enabled.
- **s6aFaultContinueProcedureWithAkaOptional** - S6A fault continue procedure with AKA optional - This flag determines whether processing will continue without authentication when there are no active enabled S6a links for the home PLMN and it has been determined that authentication is optional. no (default) - processing will not continue without authentication. yes - processing will continue without authentication.
- **s6aHssUpdateEnbCapacity** - S6A HSS update eNB capacity - This flag identifies how HSS links will be monitored.
- **s6aReportPreviousHssSelectionForHssDown** - S6A Report Previous HSS Selection For HSS Down - Parameter controls feature activation. By default, flag parameter shall be disabled. When enabled, if an HSS selection rule is found (matched) and no HSS available (all links down for hssGrp selection) report 'hssOrEirDown' even when no other HSS selection (fall-back) rule is provisioned. Else continue reporting 'noProvisionedHss' when no HSS can be found and no other available HSS selection rule is provisioned. In other words, when a single no IMSI group does not have (imsiToHss) is found with an HSS association provisioned, MME shall consider the 'noProvisionedHss'. Otherwise existing functionality shall be used, which looks at all IMSI group availability/provisioning before declaring unavailable or down.
- **s6aRetryDifferentHss** - S6AD retry different HSS - When WMM receives certain errors from HSS (too busy, out of space, etc) or HSS does not respond, WMM sends the same request to another HSS in the same PLMN or IMSI range.
- **s6dDiameter** - allow SGSN S6d/Gr Selection - Use this parameter to allow SGSN S6d/Gr Selection.
- **scefHighLatencySupport** - SCEF high latency communication support - This parameter indicates to the MME whether SCEF supports buffering of MT data if UE is not reachable due to power savings.

- **secureHandlingOfAttachTau** - Secure Handling of Attach and TAU Requests - The parameter is used to enable the secure handling of Attach and TAU Requests if these messages fail integrity checks.
- **selectedPgwFormat** - Selected PGW Format - selectedPgwFormat parameter controls how the MME formulates the selected PGW information inside the MIP6-Agent-Info AVP, which the MME sends on the S6a NOR and the S6a ULR (Active-APN AVP(s)). When 'selectedPgwFormat' is set to value
- **selectiveVolteEmerAllowed** - Selective volte enablement feature emergency call handling - If selectivevolteenabled is enabled, allow/disallow IMS emergency call.
- **selectiveVolteEnabled** - Selective volte enablement feature - Enable/Disable selective volte enablement to deny certain types of UEs access to LTE.
- **sendCallRelOnSlrEmerOrigFailure** - Send Call Release On SLR Emergency Call Origination Failure - When the parameter is Yes, if the MME sent LRR Emergency Origination to the GMLC and did not receive a successful LRR ACK, the MME will send LRR Emergency Release when the Emergency bearer is removed or deleted. When the parameter is No, LRR Emergency Release is sent after receiving a successful LRR ACK for the LRR Emergency Origination.
- **sendDdnAckWithContextNotFoundForAbnormalSgwSwitchover** - Send DDN Ack with Context Not Found for Abnormal SGW Switchover - Parameter controls feature activation. By default, flag shall be disabled. When enabled, MME upon getting a DDN with a CV of 254, and if the TEID is recognized but the DDN is from the wrong SGW, the MME shall respond with a DDN Ack, with a TEID of 0 (zero), with the same sequence number as in the DDN, and with a CV of 64 (context not found), back to the sending (wrong) SGW, so that SGW will know to clean up automatically.
- **sendDeleteSessionReqForIntersystemRau** - Send Delete Session Request for Intersystem RAU - Parameter controls sending Delete Session Request to SGW upon receiving Cancel Location Request from HSS with cancellation Type=MME Update Procedure during Gn based Inter System RAU and S3 based Inter System RAU with SGW relocation. By default the flag shall be disabled.
- **sendEquivalentPlmnListOnlytoEnbInHandoverRestrictionList** - Send Equivalent Plmn List Only to Enb in Handover Restriction List - Parameter controls feature activation. By default, the flag shall be disabled. When enabled, MME shall send Equivalent Plmn List only to eNB in Handover Restriction List. Sending to the UE shall be suppressed even if ePLMN List Name is provisioned. When gparm is disabled, if ePLMN List Enb is empty, ePLMN List shall be copied.
- **sendEsmDummyAttachPdnRejected** - Send ESM Dummy Attach PDN Rejected - MME sends Attach Accept with ESM Dummy esmMsgContainer, if PDN is Rejected and Attach is accepted, for Attach Request with PDN Connectivity Request, from an UE supporting and allowed for, registration without PDN Values: Yes - Send Attach Accept with esmMsgContainer -> ESM DUMMY message No - Send Attach Accept with esmMsgContainer -> PDN CONNECTIVITY REJECT message

- **sendForbiddenTaiListDCsOnly** - Send forbidden TAI list for data centric CS voice only - UE usage setting is data centric and voice domain preference is CS only. if **sendforbiddentailistdcsonly** is yes, send the forbidden TAI list.
- **sendForbiddenTaiListDCsPrefer** - Send forbidden TAI list for data centric CS voice preferred - UE usage setting is data centric and voice domain preference is CS preferred, IMS secondary. if **sendforbiddentailistdcprefer** is yes, send the forbidden TAI list.
- **sendForbiddenTaiListDlmsOnly** - Send forbidden TAI list for data centric IMS PS voice only - UE usage setting is data centric and voice domain preference is IMS only. if **sendforbiddentailistdimsonly** is yes, send the forbidden TAI list.
- **sendForbiddenTaiListDlmsPrefer** - Send forbidden TAI list for data centric IMS PS voice preferred - UE usage setting is data centric and voice domain preference is IMS preferred, CS secondary. if **sendforbiddentailistdimsprefer** is yes, send the forbidden TAI list.
- **sendForbiddenTaiListNone** - Selective volte enablement feature send forbidden TAI list none - UE does not indicate both UE usage setting and voice domain preference. if **sendforbiddentailistnone** is yes, send the forbidden TAI list.
- **sendForbiddenTaiListVCsOnly** - Send forbidden TAI list for voice centric CS voice only - UE usage setting is voice centric and voice domain preference is CS only. if **sendforbiddentailistvcsonly** is yes, send the forbidden TAI list.
- **sendForbiddenTaiListVCsPrefer** - Send forbidden TAI list for voice centric CS voice preferred - UE usage setting is voice centric and voice domain preference is CS preferred, IMS secondary. if **sendforbiddentailistvcprefer** is yes, send the forbidden TAI list.
- **sendForbiddenTaiListVlmsOnly** - Send forbidden TAI list for voice centric IMS PS voice only - UE usage setting is voice centric and voice domain preference is IMS only. if **sendforbiddentailistvimsonly** is yes, send the forbidden TAI list.
- **sendForbiddenTaiListVlmsPrefer** - Send forbidden TAI list for voice centric IMS PS voice preferred - UE usage setting is voice centric and voice domain preference is IMS preferred, CS secondary. if **sendforbiddentailistvimsprefer** is yes, send the forbidden TAI list.
- **sendFwdRelCancelAndHoPrepFailure** - Send Forward Relocation Cancel and HO Preparation Failure - Parameter controls sending Forward Relocation Cancel and HO preparation failure message upon indirect forwarding failure at SGW. By default the flag shall be disabled
- **sendingMBR** - Sending MBR - Sending modify bearer request.
- **sendLtemRatTypeToScef** - Specifies whether the MME can send the 'LTE-M' RAT to the SCEF in T6a CMR message.
- **sendLtemRatTypeToSgw** - Specifies whether the MME is to send the 'LTE-M' RAT Type to SGWs for UEs that are identified as LTE-M..

- **sendMappedGuami** - Current parameter enables sending in S1 Setup Response mapped GUMMEI(s) pointing to the AMF GUAMI(s) for the associated PLMN(s) that coexist in both 4G and 5G at local CMM node
- **sendMaximumUeAvailabilityTimeInNor** - Send Maximum UE Availability Time in NOR - Parameter controls support for sending maximum ue availability time to hss in NOR message. By default the flag shall be disabled.
- **sendNasRanCauseOnS11** - Send NAS RAN cause on S11 - Indicates whether (Yes) or not (No) to send the NAS or RAN failure indication code on the S11 interface to the SGW.
- **sendNorWithPgwInfoAfterTauUla** - Send NOR with PGW information after TAU ULA - Send NOR with PGW information after TAU ULA.
- **sendPdnConnRejWithReactReq** - Send PDN Connectivity Reject With CC#54 (PDN connection does not exist) - Parameter controls sending PDN Connectivity Reject with cause code #54 (PDN connection does not exist).. By default the flag shall be No.
- **sendRabrDuringDoubleS1ServiceRequest** - Send Release Access Bearer Request During Double S1 Service Request - When enabled, upon receiving Double S1 Service Request, MME shall release the old S1 connection before sending any Modify Bearer Request to SGW. MME shall send Release Access Bearer Request to SGW to release the old S1U connection on SGW for the UE. As a result SGW shall not send data over the old S1U to the enB during the Double S1 procedure. By default, flag shall be disabled.
- **sendRabUponEcmConnectedDdn** - Send RAB upon ECM-CONNECTED DDN - Specifies whether MME shall send DDN ack with cause 'Unable to page UE', followed by a release access bearer (RAB) to the SGW upon the receipt of a DDN for a ECM-CONNECTED UE.
- **sendRequestedApn** - Send requested APN - This parameter indicates to send either the UE requested APN-NI or the replacement APN-NI.
- **sendSgw3gppNasRanCause** - Send SGW 3GPP NAS ran cause - Enable/disable inclusion of NAS or RAN failure indication cause code on S11 interface to SGW.
- **sendSgw3gppS11cc** - Send SGW 3GPP S11 cause code - Sent in GTPv2 message to SGW across S11 interface. a value of 'Yes' indicates that MME shall support and send cause code 110 to SGW.
- **sendUeUnreachable** - Send UE UN_REACHABLE - Parameter controls MME behavior related to SGs Paging and specifically controlling if/when an SGs Paging Reject is sent (with UE_UNREACHABLE). Parameter allows customization of when an SGs Paging Reject is sent with a cause of UE_UNREACHABLE. When the global parameter flag is disabled, an SGs Paging Reject with the UE_UNREACHABLE cause is sent when the MME is unable to page the UE (i.e., due to UE inactivity or MME paging suppression). When the global parameter flag is enabled, the MME will also send an SGs Paging Reject with the UE_UNREACHABLE when MME paging times out trying to reach the UE. By default, flag shall be disabled.
- **sendUpdatedRfspReceivedInIdr** - Send Updated RFSP Received in IDR - Parameter controls sending updated Subscriber Profile ID IE to eNb in UE Context Modification

message. By default, the flag shall be disabled. When enabled, MME shall send updated Subscriber Profile ID IE that it receives in IDR message from HSS to eNb in UE Context Modification message.

- **serveS6dClrBySgsnOnly** - Serve S6d CLR by SGSN Only - Current parameter controls the handling of s6d CLR (cancellation-type= initial attach procedure) with the 'S6a/S6d-Indicator' flag indicating the SGSN as the affected part of combo node. When the parameter is disabled and the UE isn't found in SGSN side, then the CLR is forwarded to the MME side for 4G detach. When the parameter is enabled then the CLR isn't forwarded to the MME side for 4G detach
- **serviceGapControlActive** - Service Gap Control Active - Global Feature Flag for the Service Gap Control Feature
- **serviceGapControlSendBackoffTimer** - Service Gap Control Backoff Timer - Specifies whether the MME is to provide the UE with a backoff timer in case the UE access is blocked due to the Service Gap Control Feature
- **serviceModifiersWithLabelMatching** - Service Modifiers with Label Matching - Parameter controls feature activation flag. By default, flag shall be disabled. When enabled, MME shall support the following four service combinations: +nc-nr+ue, +ue, +nc-nr, and no modifiers.
- **serviceReqRateLimitOverride** - Number of UE initiated service requests allowed per sec per CPPS after which CPPS starts dropping service requests until rate is below limit.
- **serviceReqRateLimitOverrideSRS** - Number of service requests (that requiring SRS) allowed per sec per CPPS, after which CPPS starts rejecting service request until rate is below limit.
- **servingNodeInLRR** - Serving node in LRR - If this parameter is set to 'Yes' then then MME shall include the serving node AVP in LRR of all location scenarios.
- **sessionRestoration** - Session Restoration - Enable Session Restoration
- **setCapdegradeOn2gcipherGea1** - Set 2G capacity degrade percentage for GEA1 ciphering algorithm.
- **setCapdegradeOn2gcipherGea2** - Set 2G capacity degrade percentage for GEA2 ciphering algorithm.
- **setCapdegradeOn2gcipherGea3** - Set 2G capacity degrade percentage for GEA3 ciphering algorithm.
- **setlaiForGutiAttach** - Set IAI for GUTI attach - Set initialattachindication in ULR for GUTI attach when UE IMSI is not received directly from the UE during the ongoing attach procedure.
- **sgs** - Recovery of 3GPP CSFB services of UE using SRS.
- **sgsAssStateCsfbEmergencyCall** - SGS ass state CSFB emergency call - Configure to change the SGs association state to SGS-NULL.

- **sgsLinkLock** - SGS link lock - Enable/Disable support of MME to change SCTP state when locking SGs links.
- **sgsLiteDetachIndicationAttempts** - Detach indication attempts - Configure maximum number of transmit attempts for SGsLite-IMSI-DETACH-INDICATION message.
- **sgsLiteLocationUpdateAttempts** - Location update attempts - Configure maximum number of transmit attempts for SGsLite-LOCATION-UPDATE message.
- **sgsLiteUplinkDataAttempts** - Uplink data attempts - Configure maximum number of transmit attempts for SGsLite-UPLINK-UNITDATA message.
- **sgsnCsfbReturnLastPlmn** - SGSN CSFB return last PLMN - Configure SGSN to support return UE to last PLMN by including the last used LTE PLMN ID into the equivalent PLMN list provided to the UE in RAU accept message.
- **sgsnDiscoveryMethod** - SGSN discovery method - Method used to discover SGSN IP address for inter-rat handover to UTRAN/GERAN.
- **sgsnId** - SGSN ID - This parameter defines the SGSN identifier, SGID. Possible identifier values range from 1 to 31. 0 means not used.
- **sgsnInformImplicitlyDetachedUEs** - SGSN informs Implicitly Detached UEs - This parameter controls if the SGSN shall take action to inform UEs that have been implicitly detached due to PAPS reset that their MM context has been lost.
- **sgsnIntPreluGbSessionResiliency** - SGSN support for Internal Preparation for Lu Gb Session Resiliency - Enable SGSN support for Internal Preparation for Lu Gb Session Resiliency. Warning: For Nokia personnel only.
- **sgsnIpAddrCmpSgwSel** - SGSN IP Address Comparison During SGW Selection - Current gParm enables S4 SGSN recognizing that same SGW has been selected during inter-PLMN mobility (RAU and handover) by comparing the IP addresses of the source and target SGW(s).
- **sgsnIpv4AddressForUserTraffic** - SGSN IPv4 address for user traffic - SGSN IPv4 address for user traffic to support the SGSN context acknowledge message sent from the MME during a gn/gp SGSN to MME TAU
- **sgsnMonLogSheddingDuringOverload** - Monitor Logs Shedding During Overload - This parameter handles shedding of SGSN Monitoring Logs during overload in PAPS. Warning: For Nokia personnel only.
- **sgsnNumberSending** - SGSN Number Sending - This parameter is to control sending SGSN number to HSS via diameter to HSS. By default, this parameter is set to No, which means Not Send.
- **sgsnPoolId** - SGSN pool ID - This parameter defines the SGSN pool identifier. Possible values range from 0 to 3.
- **sgsnUseCclnApnForR8DnsPgW** - SGSN use of Charging Characteristics in APN for R8 DNS query for P-GW selection - Defines if SGSN will add Charging Characteristics in APN for

Rel-8 DNS query for P-GW selection. - 'No' (default) CC is not added - 'Yes' CC added when APN is in the apnList and marked to use CC for Rel-8 DNS P-GW selection

- sgsSvMscPreference - SGS Sv MSC preference - Configure UE assignment preference for MSC supporting both SGs and Sv interface.
- sgwIsolationFaultThreshold - SGW isolation fault threshold - This command is used to provision SGW fault threshold for the SGW isolation. If the number of the SGW faults (number of S11/S4 create session response timeouts) exceed the provisioned threshold in a provisioned duration timer then WMM will isolate the SGW from the selection for provisionable duration.
- sgwRelocationOnMbrFailure - SGW relocation on MBR failure - Enable SGW relocation if MME times out waiting for SGW response to modify bearer requests or modify access bearer request.
- sgwRelocationRestriction - SGW relocation restriction - If the feature is activated then, MME shall not perform SGW relocation for TAU request, X2 HO, S1HO and MME relocations.
- sgwSelectionWithSnaptr - SGW selection with snaptr - Enable/Disable support for SGW selection with snaptr.
- sgwSelForHomeRoutedGwSelMode - SGW Selection for GW Selection Mode 1 - Parameter controls feature activation flag. By default the flag shall be disabled. When enabled MME shall then if the UE is a home-routed roamer, the s8-gtp order and preference for each of the SGW candidates will be used to determine which SGW to select instead of the order and preference for s5-gtp, and also the selection of an SGW that supports only s8-gtp and not s5-gtp will be supported.
- sgwToEnblpv4v6TransportLayerAddress - SGW IPv4v6 Transport Layer Address IP Type - MME passes SGW S1-u IPv4v6 Transport Layer Address to the eNB/gNB if it received this IP Type from the SGW and this parameter is set to YES. Otherwise MME matches S1-C IP Address Type and passes it to the eNB/gNB.
- shortDataService - Short data service - Enable/disable short data service at MME level.
- signalingOnlyUe - Signaling only UE - Enable/disable feature 'MME support for UE with signalling connections only.
- singlePdpPerApnSgsn - SGSN allows one PDP Per APN - This parameter enables restriction the number of active PDPs towards same APN to one. When the parameter is not set (by default), the UE may have multiple active PDP(s) towards same APN. When the parameter is set, then UE is allowed having only one active PDP towards same APN.
- singleThreadOlcSheddingEffectiveness - Single Thread OLC Shedding Effectiveness - When enabled, OLC will determine the effectiveness of shedding during a single thread CPU overload. If shedding is ineffective, then OLC will turn shedding off, else the shedding will continue. This feature applies to IPDS and CPPS.
- skipSubscriptionData - Skip subscription data - HSS may skip subscription data in the ULA if subscription data has not changed since last successful update.

- skipUlrForSgsnToMmeTau - Skip ULR for SGSN to MME Tracking Area Update - Parameter controls sending NOR message to HSS at the end of SGSN to MME TAU procedure (applies to both Gn and S3 SGSN) instead of ULR, for the case where subscription data is already available in MME. By default the flag shall be disabled
- slgRetryLrrEmergencySameDra - SLG Retry LRR Emergency Same DRA - This parameter will control MME to retry the same DRA/GMLC when MME receives Diameter error 3002 (DIAMETER_UNABLE_TO_DELIVER). If the parameter is enabled, MME will retry the diameter message in SLG the same DRA for LRR related with emergency. If the parameter is disabled, MME will keep existing function (i.e. no retry). By default, the parameter is disabled.
- slrEmergencyInitiationDedicatedBearer - SLR emergency initiation based on dedicated bearer - SLR emergency initiation based on dedicated bearer.
- slrSenderDuringHandover - SLR sender during handover - Specifies whether the source or target MME sends the SLR to the GMLC during handover. the default is the target MME.
- smfPgwcSelSupport - SMF Combo PGW-C Selection support - Selection of combo PGW-C is done using only HSS subscription information as criteria, without considering UE N1 mode support
- smsInMme - SMS in MME - Activate the SMS in MME feature. This feature enables the possibility to deploy the SMS service without the use of the CS domain.
- sourceMmeRelocSgsDetach - Source MME reloc send SGs detach - This flag controls whether the source MME for a SGS associated UE (i.e. a combined attach UE) will send an SGs detach message to the MSC/VLR following a successful relocation. yes - source MME sends a SGs detach message to the MSC/VLR following a successful relocation no - source MME do not sends a SGs detach message to the MSC/VLR following a successful relocation.
- spcmdSheddingDuringOverload - SPCMD Shedding During Overload - This parameter handles shedding of SGSN PCMD records during overload in PAPS
- srvcc - MME support of SRVCC with the parameters obtained from the SRS.
- srvccCollSendDbrCc110 - SRVCC Collision Send Del brr response - Global Parameter flag 'srvccCollSendDbrCc110' controls the feature activation. By default the flag shall be disabled. When enabled, MME shall support sending Delete Bearer Response with Cause Code 110 during SRVCC collision.
- srvccForAllImsEme - SRVCC for all IMS EME - Use the srvccforallimseme to ensure all UEs making an IMS emergency call (and supporting SRVCC) will be allowed to perform SRVCC.
- staticGwSelection - STATIC GGSN/GW SELECTION - StaticGwSelection shall be used to control static GGSN/GW selection for SGSN. When it's set, then the new function f52070-01 (configured GWs) will apply. Otherwise, exist function will be used.
- stdPraActionAddProcs - Standard PRA Action Additional Procedures - Enables MME to react upon received PRA action for standard PRA, when MME receives new PRA action

from SGW/PGW during 5GS to EPS mobility, or during PDN setup for a PDN being handed over from ePDG. Applied when Enable Presence Reporting Area functionality is set to Yes. For modify action also the PRA Modify Enabling parameter must be set to Yes.

- **stdsNotifyResponse** - Stds notify response - Standards specified MME behavior upon receiving notify response from HSS.
- **stopWarningIndication** - Stop warning indication - Sending of STOP WARNING MESSAGE TO the CBC.
- **subscriberDataExtraction** - Subscriber Data Extraction - Enable/disable periodic subscriber data extraction
- **supportAlternateGwReselection** - Support alternate gw reselection feature - If enabled, the feature does not check the source bit of cause IE received in create session response for reselection of GW on failures instead MME uses the cause code to determine GW reselection.
- **supportApnCorrection** - Support APN correction - Configure whether or not APN correction is enabled.
- **supportAsstData** - Support Assistance Data - Parameter controls the support for Assistance Data for Recommended Cells info that MME receives info in the S1 CONTEXT RELEASE COMPLETE message shall store it in the VLR and shall include the IE in the paging message. The IE in UE CONTEXT RELEASE COMPLETE that contains the info is 'Information on Recommended Cells' and eNBs for Paging. The info from this IE shall be stored in VLR until a new UE CONTEXT RELEASE COMPLETE message is received. The contents of this IE shall be used to populate Assistance Data for Recommended Cells IE of the Assistance Data for Paging of S1AP PAGING. By default the flag shall be disabled.
- **supportCatM1b** - Support category M1 CE mode b UE - Support category M1 CE mode b UE.
- **supportCmasRestriction** - Support cmas restriction - Feature activation per MME.
- **supportCppsAutoPersonalityChange** - Support CPPS Auto Personality Change - Parameter controls whether CMM changes CPPS personality between EMMS and AMMS based on 4G and 5G traffic load.
- **supportCsgChangeReporting** - Support CSG change reporting - Is support for CSG change reporting to be provided?.
- **supportDedicatedBearerUli** - Support dedicated bearer ULI - Enable or disable ULI enhancement for dedicated bearer activation, update and delete bearer response messages.
- **supportDelayedNor** - Support delayed NOR - Delay NOR when the HSS is unavailable.
- **supportDelayedUlr** - Support delayed ULR - Delay ULR when the HSS is unavailable.
- **supportDfltBearerOperDefQci** - Support Default Bearer Operator Defined Qci - Enable or Disable the support for Default Bearers to use Operator Defined QCIs

- supportDnsSrv - Support DNS SRV - MME support for service (SRV) resource records (RR) for MME DNS discovery of MME/PGW/SGW/SGSN.
- supportDupAppProtDnsRec - Support Duplicate 'app-protocol' DNS Records - When this parameter is enabled, the MME does not discard DNS records carrying multiple app-protocol entries of the same app-protocol value, and the MME assumes that the record supports the superset of the supported services across the multiple app-protocol entries.
- supportEdrxRelatedRat - Support eDRX Related RAT - Parameter controls eDRX related RAT support by MME. By default the flag shall be disabled. When the gparm is off existing behavior (i.e. if eDRX-Cycle-Length-Withdrawal is provided the edrx cycle length is removed for both nb-iot and e-utran). If the gparm is enabled, then the only edrx-cycle length deleted will be what the AVP specified. By default, the flag shall be disabled.
- supportForAiMlFunctionalities - Support for AI/ML functionalities in CMM - This global parameter can be used to enable or disable Artificial Intelligence and Machine Learning (AIML) capabilities in a CMM. By default the global parameter should be disabled (set to No). The AIML functionalities are only applicable to CNF based CMMs, hence, this parameter should always be disabled in VNF based CMMs. When enabled, the Machine Learning Intelligence Service (MLIS) will setup client-server connections with call processing pods for data streaming. When enabled, MLIS pod will collect necessary data to serve AIML algorithms/models in the CMM.
- supportFullUeRestoration - Full 4G SRS Restoration - If this gparm is enabled full UE Contexts will be stored in the Backup. Restorations will take place on the Full UE Contents. This feature allows rolling update of DBS for 4G in Containers
- supportImeiDigitValidation - Support IMEI digit validation - This command enables/disables IMEI validation.
- supportImsIimeiApnEnh - Support IMS IIMEI APN enhancement - Parameter controls feature activation. By default the parameter shall be disabled. When enabled, MME shall use the replacement APN NI in the DNS query. Replacement APN NI shall be sent in the Create Session Request towards SGW. QoS and other parameters shall be used from the subscription data. In addition, MME shall send the NOR message with default APN context identifier
- supportMmeSrvcc - MME Supporting 5G-SRVCC - Enables the MME functionality to act in the intermediary role of the MME_SRVCC (MME supporting 5G-SRVCC) to perform the 5G-SRVCC handover for regular voice calls from NG-RAN/5GS to 3GPP UTRAN/CS domain.
- supportMmeSrvccEmergency - Support MME_SRVCC Emergency - Enables the MME functionality to act in the intermediary role of the MME_SRVCC (MME supporting 5G-SRVCC) to perform the 5G-SRVCC handover for emergency calls from NG-RAN/5GS to 3GPP UTRAN/CS domain.
- supportModifyAccessBearerRequest - Support modify access bearer request - Support for S11 modify access bearer request message.

- supportMsisdnInEcrMsg - Support MSISDN in ECR message - When set to yes, MME shall populate MSISDN AVP in the ECR message towards EIR that is sent across S13/S13' interface as part of ME identity check procedure.
- supportNonhomeTaiLaiMapping - Support Non-home TAI/LAI Mapping - Feature activation for Non-home TAI/LAI Mapping
- supportOrigTimeStampMaxWait - Support Origination Time Stamp and Maximum Wait Time IE - Parameter controls support for Origination Time stamp and Maximum Wait Time IE inclusion in Create Session Request message. By default this flag shall be disabled. When enabled, MME shall include Origination Time stamp and Maximum Wait Time IEs in Create Session Request that is sent on S11 message to gateways.
- supportQoSOverride - Support QoS Override - Parameter controls the feature activation. By default this shall be disabled. When enabled, MME shall support HSS/P-GW QoS Parameter Override for home subscribers
- supportReattachRequired - Support reattach required - Enables/Disables the ability to detach the UE based on the bit 'Reattach-Required' bit received from HSS.
- supportSbcAccessControl - Support SBC access control - When support SBC access control set to yes, access control list for SBC provisioned on the MMERmtEndPtCfgr is used for access control of SBC.
- supportSdsUeRestoration - Support SDS UE restoration - When enabled, MME supports sending data for SDS UEs to SRS.
- supportSessionRestoration - Support session restoration - Support session restoration or not.
- supportSgsnOperDefQci - Support SGSN Operator Defined QCI - Defines if SGSN supports the provisioned Operator Defined QCI. Up to 32 SGSN supported Operator Defined QCI can be defined via CLI oprDefQciToRel99QoSMapping
- supportSignalingOnlyUeRestoration - Support signaling only UE restoration - When enabled, MME supports sending data for signaling-only UEs to SRS.
- supportTwoUeBearersForDualPdnType - Support two UE bearers for dual PDN type - Support two PDN connections to the same APN if the value is 'false', then 2 PDN connections to the same APN shall not be allowed.
- supportUeRestorationContextDeletion - Support UE restoration context deletion - Enables/Disables UE restoration context deletion functionality on MME.
- suppressRegisteredLai - Suppress Registered LAI - If the feature is enabled (Yes) then MME shall not include the Registered LAI IE in the S1AP INITIAL CONTEXT SETUP and UE CONTEXT MODIFICATION REQUEST messages
- suppressSendingAltGnGgsnAddress - Suppress Sending Alternate Gn GGSN Address - MME shall be provisioned with a global parameter flag that controls sending two IEs (Alternative GGSN Address for control Plane and the Alternative GGSN Address for

user traffic IEs) in forward relocation message and context response message during handover scenario. By default, the flag shall be disabled.

- **supptIMSIIMEIAPNconv** - Support IMSI IMEI APN conversion - Support IMSI IMEI APN conversion.
- **switchPdnToLipa** - Switch PDN to LIPA - Enables MME to dis-connect a LIPA enabled PDN Connection and to re-establish it when UE in idle mode performs a TAU in a LIPA area. MME detaches UE with 're-attach required' to trigger UE to re-establish the PDN connection in the LIPA area.
- **t6aDiamOverloadControl** - T6A diam overload control - Enable(Yes)/disable(No) T6A overload control.
- **t6aRetryDifferentScef** - T6A retry different SCEF - When MME gets some errors from SCEF (too busy, out of space, etc) or SCEF does not respond, MME sends the same request to another SCEF in the same PLMN.
- **tappingActivated** - Status of MME/SGSN Tapping functionality - This parameter controls the activated/deactivated status of MME/SGSN Tapping
- **tauDdnCollision** - TAU DDN collision - Configure MME handling TAU request collision with an ongoing DDN procedure.
- **tauRateLimitOverride** - SGSN to MME IRAT TAUs allowed per sec after which MME start dropping SGSN to MME IRAT TAUs until rate is below limit.
- **tauSgsUnknownLaiEmmCause** - TAU SGs Unknown LAI EMM Cause - The parameter is used to provision EMM cause code to be sent to UE combined TAU request if MME fails to select MSC due to LAI to TAI to MSC mapping provisioning errors.
- **teidToSgsn** - Tunnel endpoint identifier data to SGSN - Tunnel endpoint identifier data II information element parameters to support the SGSN context acknowledge message sent from the MME during a gn/gp SGSN to MME TAU enum takes all zero and all ones.
- **threegAttachRateLimitOverride** - 3G attaches allowed per sec per PAPS after which PAPS start dropping attaches until rate is below limit.
- **threegAttachRateLimitOverrideAtlpds** - 3G Attach Rate Limit Override - 3G Attaches allowed per sec per PAPS after which PAPS start dropping attaches until rate is below limit at IPDS
- **threegControlMsgLimit** - 3G control message limit - 3G control messages per second without ciphering. Warning: For Nokia personnel only.
- **threeglsauRateLimitOverride** - Number of 3G ISRAUs allowed per sec per PAPS, after which PAPS starts dropping 3G ISRAUs until rate is below limit.
- **threeglsauRateLimitOverrideAtlpds** - 3G Isau Rate Limit Override - Number of 3G ISRAUs allowed per sec per CPPS, after which CPPS starts dropping 3G ISRAUs until rate is below limit at IPDS

- **timeBasedAuthFrequency** - Time Based Authentication Frequency - Parameter controls feature activation. By default, the flag shall be disabled. When enabled, MME shall support time-based authentication in accordance with new option(s) defined in UE PLMN Services and IMSI Range Services configuration.
- **timeZoneUpdateRate** - Time zone update rate - Configure time zone update rate.
- **timeZoneUpdateThrottling** - Time zone update throttling - Activate controlling the rate of time zone updates to PGW due to DST change.
- **tmsiBasedNriMscSelection** - TMSI based NRI MSC selection - When set to yes enables the capability to map NRI in TMSI received from the UE to a MSC so that SGs update location is sent to the same MSC as UE was registered with previously to avoid dual VLR registrations. enabling of this feature also requires an operator to provision a non-broadcasting LAC and also NRI for each provisioned MSC. when set to no disables the feature.
- **topologicalLabelMatchingForS8pgwSelection** - Topological label matching for S8 PGW selection - This parameter controls whether or not topological label matching is used to compare the selected SGW FQDN labels with the candidate S8 PGW FQDN labels when selecting a PGW candidate for roaming. when the topological label matching for S8 PGW selection parameter is set to false, the MME will not attempt to use label matching to select the PGW but instead use just the DNS order and preference values. when the parameter is set to true, the MME will use the same label matching algorithm used for S5 PGW selection for S8 PGW selection. when this parameter is set to true, it is up to the service provider to ensure that the labels provided in the SGW and PGW FQDNs are consistent so that label matching will be effective. in particular, the SGW and PGW FQDN values provisioned in the DNS server for MNC and MCC must match.
- **topologicalSGWselectionIRAT** - Topological SGW Selection IRAT - Performs topological matching of SGW to PGW for iRAT TAU and GW Selection Mode 2, when the PGW FQDN had been derived from the PGW IP address, even when pre-existing APNs for the bearers have PLMNs that don't match the UE's selected PLMN
- **tracAreaRelativeCapacity** - Trac area relative capacity - Enable/disable feature to MME support of configuration of MME relative capacity.
- **trafloadCombPgw3XUe** - Traffic Load to Combo PGW-C for 3X capable UE - This parameter controls percentage (0% - 100%) of traffic that shall be sent to combo PGW-C for 3X UE(s) eligible using combo PGW-C. If parameter is set to 0%, then MME shall not send any traffic to combo PGW-C.
- **trafloadCombPgwN1Ue** - Traffic Load to Combo PGW-C for N1 capable UE - This parameter controls percentage (0% - 100%) of traffic that shall be sent to combo PGW-C for N1 UE(s) eligible using combo PGW-C. If parameter is set to 0%, then MME shall not send any traffic to combo PGW-C.
- **treatGbrTauActiveTau** - Treat GBR TAU active TAU - Enable(Yes)/disable(No) the feature during TAU request with GBR bearers present regardless of active flag value. if set to yes, then MME shall check to determine whether an ECM-IDLE UE has any GBR bearers

when TAU request is received. if yes, MME shall treat the TAU request as though TAU request is received with active flags set regardless of active flags setting in the received TAU request and bring up all the bearers. this feature applies to both relocating and non-relocating TAU.

- **treatIratTauActiveTau** - Treat inter TAU active TAU - Controls to determine whether or not the active flag is treated as if it were set whenever interrta TAU requests are processed.
- **treatMip6AgentInfoInIidrAsOneEntity** - Treat MIP6Agent-Info in IDR as one Entity - GPARAM controls the support for treating the MIP6-Agent-Info in IDR as one entity rather than individually modifiable elements. When the GPARAM is enabled MME will treat MIP6-Agent-Info in IDR as one entity, By default this shall be disabled.
- **twogAttachRateLimitOverride** - 2G attaches allowed per sec per PAPS after which PAPS start dropping attaches until rate is below limit.
- **twogControlMsgLimit** - 2G control message limit - 2G control messages per second without ciphering. Warning: For Nokia personnel only.
- **twogDataMsgLimit** - 2G data message limit - 2G data messages per second without ciphering. Warning: For Nokia personnel only.
- **twogGnThroughput** - 2G Gn throughput - 2G Gn throughput (Mbps). Warning: For Nokia personnel only.
- **twogImpactOfCiphering** - 2G impact of ciphering - 2G impact of ciphering (%). Warning: For Nokia personnel only.
- **twogIsauRateLimitOverride** - Number of 2G ISRAUs allowed per sec per PAPS, after which PAPS starts dropping 2G ISRAUs until rate is below limit.
- **twogMaxPktsConcurrentDataPresence** - 2G max packets concurrent data presence - Max 2G control packets due to concurrent 2G data presence. Warning: For Nokia personnel only.
- **twogUeThroughput** - 2G UE throughput - 2G UE throughput (Mbps). Warning: For Nokia personnel only.
- **twothreegAttachRateLimitOverride** - Number of 2G and 3G ISRAUs allowed per sec per CPPS, after which CPPS starts dropping 2G and 3G attaches until rate is below limit.
- **ubrRejCauseDuringIratRedirect** - UBR Reject Cause During Interrat-Redirection - A user specified GTPv2 cause to be used within the Update Bearer Response sent back to the SGW, when the Update Bearer Request is received during an interrta-redirection scenario.
- **ueCpToUpBearerSwitch** - Enable UE Triggered CP to UP Bearer Switch - If this parameter is set, the MME shall proceed with the UE request to switch CP bearers to UP. If the parameter is not set, the MME shall reject the UE request to switch CP bearers to UP.
- **ueCtxUpdateInPatchLoadSupport** - Support for UE Context Update Over NR10 in Patch Loads - Support for UE Context Update over NR10 in patch loads

- **ueEnbServingCellChangeReporting** - UE eNB serving cell change reporting - Indicates whether UE eNB serving cell change reporting is enabled or not the default is no, indicating UE eNB serving cell change reporting is not enabled.
- **ueReachTAdsInSupportedFeaturesAvp** - UE reach T-ADS in supported features AVP - A value of yes indicates that MME shall send UE-reachability-Notification and T-ADS data retrieval in the in the supported-features AVP, in the feature-list AVP, by setting bits 26 and 27 of the feature-list flags.
- **ueSrvccCapabilityHss** - UE SRVCC Capability HSS - This parameter controls the value of the UE-SRVCC-Capability AVP the MME sends towards the HSS. UE-SRVCC-Capability AVP is absent when the UE does not include MS network capability IE. MME sends UE-SRVCC-Capability AVP as 'UE-SRVCC-NOT-SUPPORTED' when the UE does not support 'SRVCC to GERAN/UTRAN capability' on MS network capability, or MME does not support SRVCC for the UE according to provisioning (as part of svcAgreementProfile). MME sends UE-SRVCC-Capability AVP as 'UE-SRVCC-SUPPORTED' if the MME supports SRVCC for the UE according to provisioning, and the UE supports 'SRVCC to GERAN/UTRAN capability' on MS network capability.
- **ueUnreachableMsgSgsEdrxPsmCause** - SGs cause code UE unreachable - Use 'UE unreachable' instead of the default 'UE temporarily unreachable' as cause code with SGs cause IE in SGs interface
- **uliEnhancement** - ULI enhancement - Indicates whether ULI enhancement is enabled or not the default is no, indicating ULI enhancement is not enabled.
- **uliHomeRoutedUe** - ULI home routed UE - This flag determines whether ULI will be included for home routed UEs. no - ULI will not be included for home routed UEs. yes (default) - ULI will be included for home routed UEs.
- **uliInS11messagesWithSgwRelocation** - Enable ULI in S11 messages with SGW relocation - A value of yes for the parameter shall indicate that the WMM will include the ULI (ECGI + TAI) for S11 messages with SGW relocation. a value of no for the parameter shall indicate that existing, 29.274 standards based IE inclusion policy shall apply.
- **unconditionallyAddHrIe** - Unconditionally Add HRL IE - When enabled, MME shall unconditionally include the Handover Restriction List IE in all S1AP messages as described within 3GPP specifications.
- **upper20BitHenbGwRouting** - Upper 20-Bit henb GW routing - Enable/disable non-standard usage of upper 20 bits of the received 28-bit henb ID to route to the henb GW.
- **use3gpplmsiHashForSgs** - Use 3GPP IMSI Hash for SGs - Specifies whether the CMM is to use the 3GPP IMSI Hash Algorithm specified in TS 23.236 for UE distribution to MSCs via the SGs interface. The default is no. When this field is set to no, the CMM shall use its proprietary IMSI hash algorithm.
- **useExplicitDetach** - Use explicit detach - Indicates whether explicit detach is used (and the MME pages the UE as part of the explicit detach procedure), or whether an implicit detach is used (and the MME does not page the UE).

- **useFailopenProfile** - Use failopen profile - Enable the use of the failopen profile for attach and PDN connectivity procedures, used for home subscribers and roamers considered as homers.
- **useHplmnforSgsInterface** - Use Home PLMN For SGs Interface - MME shall be provisioned with GPARM 'useHplmnforSgsInterface' flag. By default, this flag shall be disabled. When the flag is enabled, MME shall always use home plmn of the specific UE towards MSS across SGs interface.
- **useLocalDnsHost** - Use Local DNS Host - Use local DNS host.
- **useModifiedSbcAsn1** - Use Modified 29.168 SBC ASN.1 - Parameter controls the GPARM flag. By default, this flag shall be disabled. When the flag is enabled, MME shall use the modified version of 29.168 ASN.1.
- **useMTUpathdiscoveryforDFflag** - use MTU path discovery for DF flag - This parameter indicates whether to use MTU path discovery to set the DF flag for IP for S1mme. If it is set to yes, then DF flag will be set if MTU path discovery is enabled. If it is set to no, then the DF flag will not be set, even if MTU path discovery is enabled.
- **useNon3gppEmergencyPolicy** - Use Non-3GPP NW provided Policies for emergency calls - This parameter is used to enable MME to include the Non-3GPP NW provided policies IE in the Attach Request and TAU Request message. By default the IE is not included.
- **usePcmdZstdCompression** - Use PCMD zstd compression - When this is set to Yes, zstd compression is applied at PCMD files under /data-pcmd/ directory. When this is No, gzip compression is applied instead.
- **useTopologicalLabelMatchingAlways** - Use Topological Label Matching Always - Provisioned Parameter 'UseTopologicalLabelMatchingAlways' for Gateway selection controls whether topological matching shall be used to select SGW and PGW, regardless of whether SGW/PGW FQDN MNC/MCC identifies UE PLMN or Serving PLMN or MME Home PLMN or some other PLMN, during Attach Request procedures and to select a PGW during PDN Connectivity Request procedures. By default, GPARM shall be disabled. GPARM is valid only for GW selection mode 2 and shall not have any effect if GW selection Mode 1 is enabled.
- **valueAddedMtLrNoEsmlc** - Value added MT_LR no E-SMLC - MT_LR client type value-added do not contact E-SMLC for UE location information. If yes, when MME receives mobile terminated location request with client type=valueAdded, and activateLcsClientTypeValueAddedServices is set, the MME contacts the eNB for UE current E-CGI information. This information is sent back to the GMLC. The MME does not contact the E-SMLC to obtain any UE location information. If no, nothing change and keep existing function (i.e. MME contacts E-SMLC to obtain UE location for this MT-LR type).
- **vdpForbiddenTaiListOnVdpProfile** - VDP Forbidden TAI List on VDP Profile - Defines whether the VDP Profile associated Forbidden TAI List is sent according to global parameters or VDP Profile level parameters. When set to Yes, and MME finds a matching VDP Profile for the UE, the MME sends VDP profile associated Forbidden TAI List for

UEs according to VDP Profile level settings for UE's usage type and Voice Domain Preference, i.e. parameters Data Centric CS Voice Only Send Forbidden TAI List, Data Centric CS Voice Preferred Send Forbidden TAI List, etc. MME ignores global parameters Send forbidden TAI list for data centric CS voice only, Send forbidden TAI list for data centric CS voice preferred, etc. When a) set to Yes and MME does not find a matching VDP Profile for the UE, or b) when set to No, MME sends VDP profile associated Forbidden TAI List for UEs according to global parameters Send forbidden TAI list for data centric CS voice only, Send forbidden TAI list for data centric CS voice preferred, etc. MME ignores VDP profile level parameters Data Centric CS Voice Only Send Forbidden TAI List, Data Centric CS Voice Preferred Send Forbidden TAI List, etc.

- vlrAutoRecoveryPurgeCount - VLR auto recovery purge count - VLR auto recovery purge count when set to 0, the counter is disabled.
- warningRetransmissions - Warning retransmissions - The number of times the MME will attempt to send a S1 CMAS message to a eNB.
- warningSystemLog - Warning system log - Warning system log.
- weightedSgslmsiHash - Weighted SGs IMSI Hash - Specifies whether the MME is to use a weighted IMSI hash distribution for assigning UEs to an SGs MSC.
- writeReplaceWarningIndication - Write replace warning indication - Enable MME sending of broadcast completed area list to the CBC.
- wusProfilesEnabled - WUS Profiles Enabled - Specifies whether a WUS profile name referenced in UE PLMN Services and IMSI Range Services can be used. Default is No.
- x2hoInclSgwUlteid - X2 HO include SGW ul - Enable the E-RAB to BE switched in uplink list IE in the S1AP PATH SWITCH REQUEST ACKNOWLEDGE message irrespective of SGW relocation.
- zeroOiCsrN26Tau - Set OI flag to Zero in CSR of N26 TAU - Current parameter enables setting OI flag to value zero in Create Session Request that MME sends as part of n26 Idle mobility. When this parameter is disabled, the MME sets OI flag to 1

Name

gParmValue - Parameter value

Syntax

--gParmValue <string>

Description

Value of parameter.

Class

Optional.

Type / Range

Digit string of length 0 to 64

The gParmValue string is based on the gParmName parameter:

Name

20bitHenbld - 20 bit home eNodeB identifier

Syntax

--gParmName 20bitHenbId --gParmValue <string>

Description

Control flag for routing of S1AP HO messages based on target TAI when the target eNB with 20-bit eNB ID is not directly connected with the MME. no (default) - disable this routing. yes - enable this routing.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

2g3gPgwlIsolation - 2g3g PGW isolation

Syntax

--gParmName 2g3gPgwlIsolation --gParmValue <string>

Description

Setting to yes enables PGW isolation feature on SGSN. setting to no disables PGW isolation feature on SGSN.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

2g3gSgwlIsolation - 2g3g SGW isolation

Syntax

--gParmName 2g3gSgwlIsolation --gParmValue <string>

Description

Setting to yes enables SGW isolation feature on SGSN. setting to no disables SGW isolation feature on SGSN.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

4g5gRetPrefPlmn - 4g5g Return Preferred Plmn

Syntax

--gParmName 4g5gRetPrefPlmn --gParmValue <string>

Description

This parameter enables indicating to serving eNB the last visited 5GS PLMN as preferred indication to be used in future EPS to 5GS mobility.

List

- OFF - OFF
- ON - ON

Default

OFF

Name

acceptActFlagTauCpsrForT6aonly - Accept active flag TAU CPSR for T6A ONLY

Syntax

--gParmName acceptActFlagTauCpsrForT6aonly --gParmValue <string>

Description

Enable acceptance of TAU with active flag and CPSR to establish S1-U bearers for a UE with SCEF PDN connections only.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

acceptScefRelForPsmEdrx - Accept SCEF Release for PSM eDRX

Syntax

--gParmName acceptScefRelForPsmEdrx --gParmValue <string>

Description

This gparm specifies whether the MME is to return diameter Success if the SCEF attempts to release a T6a bearer for an Idle UE in PSM or eDRX mode and unavailable for paging.

List

- No - Boolean value of no.

- Yes - Boolean value of yes.

Default

No

Name

acceptSgsPageFromNonRegMsc - Accept an SGs Page from non-registered MSC

Syntax`--gParmName acceptSgsPageFromNonRegMsc --gParmValue <string>`**Description**

This gparm specifies whether the CMM should accept and process an SGs page request from an MSC other than the currently registered MSC.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

accessRestrictionToLte - support ARD from s6d and pass it to Radio

Syntax`--gParmName accessRestrictionToLte --gParmValue <string>`**Description**

This parameter allows SGSN to support ARD from s6d and pass it to Radio. This parameter also allows MME in combi node during attach or TAU to store 'WB-E-UTRAN Not Allowed' in the DB if receives DIAMETER_ERROR_USER_UNKNOWN during location updating. If the condition such user is still in the DB e.g. the subscriber was attached in the SGSN part before, purge is not done yet. MME will send #15 CC towards UE in this case. Default value is No, it means that both SGSN/MME will keep current behavior, no changes. If the parameter is set to Yes, above function will take in place.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

activateHssUnavailBlacklisting - Activate HSS unavailable blacklisting

Syntax`--gParmName activateHssUnavailBlacklisting --gParmValue <string>`

Description

Enable the extended HSS unavailability detection and blacklisting.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

activateLcs - Activate LCS

Syntax

```
--gParmName activateLcs --gParmValue <string>
```

Description

Indicates whether or not to activate LCS.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

activateLcsClientTypeEmergencyServices - Activate LCS client type emergency services

Syntax

```
--gParmName activateLcsClientTypeEmergencyServices --gParmValue  
<string>
```

Description

Activate LCS client type emergency services.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

activateLcsClientTypeLawfulInterceptServices - Activate LCS client type Lawful Intercept
Services

Syntax

```
--gParmName activateLcsClientTypeLawfulInterceptServices --gParmValue  
<string>
```

Description

Activate LCS client type Lawful Intercept Services.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

activateLcsClientTypePlmnOperatorServices - Activate LCS client type PLMN operator services

Syntax

```
--gParmName activateLcsClientTypePlmnOperatorServices --gParmValue  
<string>
```

Description

Activate LCS client type PLMN operator services.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

activateLcsClientTypeValueAddedServices - Activate LCS client type Value Added Services

Syntax

```
--gParmName activateLcsClientTypeValueAddedServices --gParmValue  
<string>
```

Description

Activate LCS client type Value Added Services

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

aerialSubscriptionActive - Aerial Subscription Feature Active

Syntax

```
--gParmName aerialSubscriptionActive --gParmValue <string>
```

Description

Specifies whether the MME is to include aerial subscription information (if available) in S1 messaging to the eNodeB.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

alarmCmasFailure - Alarm CMAS failure

Syntax

```
--gParmName alarmCmasFailure --gParmValue <string>
```

Description

Alarm CMAS failure.

List

- Critical - Alarm level to be used for this message (Critical).
- Major - Alarm level to be used for this message (Major).
- Minor - Alarm level to be used for this message (Minor).
- None - Alarm level to be used for this message (None).

Default

None

Name

alarmFailToReceiveCmasResponse - Alarm fail to receive CMAS response

Syntax

```
--gParmName alarmFailToReceiveCmasResponse --gParmValue <string>
```

Description

Alarm fail to receive CMAS response.

List

- Critical - Alarm level to be used for this message (Critical).
- Major - Alarm level to be used for this message (Major).
- Minor - Alarm level to be used for this message (Minor).

- None - Alarm level to be used for this message (None).

Default

None

Name

alarmFailToSendCmasMessage - Alarm fail to send CMAS message

Syntax

```
--gParmName alarmFailToSendCmasMessage --gParmValue <string>
```

Description

Alarm fail to send CMAS message.

List

- Critical - Alarm level to be used for this message (Critical).
- Major - Alarm level to be used for this message (Major).
- Minor - Alarm level to be used for this message (Minor).
- None - Alarm level to be used for this message (None).

Default

None

Name

aliasApnConversionPerImsi - Alias APN Conversion Per IMSI

Syntax

```
--gParmName aliasApnConversionPerImsi --gParmValue <string>
```

Description

Parameter controls feature activation flag. By default, flag shall be disabled. When enabled, MME shall support converting the UE requested APN with the locally configured APN.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

allApns - All APNS

Syntax

```
--gParmName allApns --gParmValue <string>
```

Description

MME to store context of all PDN connections of UE on SRS.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

allowBrrDeactvtRglrDeactvt - Allow bearer deactivate regular deactivate

Syntax

--gParmName allowBrrDeactvtRglrDeactvt --gParmValue <string>

Description

Yes - enables the MME, upon receiving delete bearer request from the network during a PGW initiated PDN disconnection scenario with cause of 'Reactivation requested', to send deactivate EPS bearer context request message to the UE overriding the standard cause with 'regular deactivation'. upon receiving DEACTIVATE EPS BEARER CONTEXT ACCEPT message from the UE, the MME will send S11 delete bearer response message to the SGW. no (default) - in the scenario above, the MME will send the standard cause, 'reactivation requested', towards the UE.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

allowEdrxValue0 - support eDRX cycle 5.12s

Syntax

--gParmName allowEdrxValue0 --gParmValue <string>

Description

support eDRX cycle 5.12s. Default value is not set. In this case, MME will keep current behavior: a UE requests eDRX 0, MME responds with an ATTACH/TAU RESPONSE message with the eDRX IE not present. Since EMM currently rejects the use of eDRX 0, the UE will be paged the same as if the UE didn not request eDRX 0. When this gParm is set, then this feature function will be enable. MME will accept eDRX0, and a new paging type WB_eDRX_0 will be used

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

allowEnGnbRouting - Allow En-Gnb Routing

Syntax

--gParmName allowEnGnbRouting --gParmValue <string>

Description

Allow the use of the en-Gnb mapping to be used in routing the Enb Configuration Transfer message received at MME. When false, the en-gNB ID and TAC will be used in message routing. When true, message routing occurs using only the en-gNB ID.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

allowExtendedImsiRangeServices - Allow Extended Number of IMSIRangeServices

Syntax

--gParmName allowExtendedImsiRangeServices --gParmValue <string>

Description

This parameter determines the use of an extended range of IMSIRangeServices. When set to no the number of IMSIRangeServices is limited to 4K. When set to yes, the number of MSIRangeServices is limited to 10K.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

allowLIPAPDNWhenLITarget - allow LIPA PDN Connections

Syntax

--gParmName allowLIPAPDNWhenLITarget --gParmValue <string>

Description

Use this parameter to allow checks on LIPA PDN Connections.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

allowMixedPdnAddressAllocationTypes - Allow mixed PDN address allocation types

Syntax

--gParmName allowMixedPdnAddressAllocationTypes --gParmValue <string>

Description

When set to yes PDN type is IPv4v6, if only one of the IP versions is statically provisioned in the HSS, set the other IP version to all zeros thus also allowing for dynamic PDN address allocation.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

allowNonEmerAttachEmerArpProvCc - Allow non-emergency Attach with emergency ARP when Provisionable Cause Code feature is enabled

Syntax

--gParmName allowNonEmerAttachEmerArpProvCc --gParmValue <string>

Description

When this parameter is enabled, the MME allows non-emergency Attach for which PDN session is associated with emergency ARP, as defined in local emergency profile, that is otherwise by default rejected when the Provisionable Cause Code feature is enabled.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

allowProcEirFailure - Allow Procedure EIR Failure

Syntax

--gParmName allowProcEirFailure --gParmValue <string>

Description

Feature introduces support for enhancing the EIR support by having MME reject a user's Attach when the IMEI is explicitly blacklisted in EIR. In all other cases attach should succeed, if for example - the AVP value is different from BLACKLISTED (ie UNKNOWN, WHITELISTED or GREYLISTED) - any other Diameter Error Code received on S13 '3GPP-ME-Identity_Check' - No response received from EIR - Any other diameter error cause codes received from EIR Feature shall be controlled by GPARM flag

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

allowR8ArpGnSgsn - Allow R8 ARP Gn SGSN

Syntax

```
--gParmName allowR8ArpGnSgsn --gParmValue <string>
```

Description

Gn SGSN is pre-rel8 element and it connects/receives pre-rel8 ARP from HLR. With this parameter set to true, Gn SGSN allows customer configured mapping table and transfer rel8 ARP towards RNC. With this parameter set, a new configured Group IMSIs Will be given ARP 1 towards RNC. If the parameter is not set, then normal function will take place. Rel8 ARP will not allow to send to RNC.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

allowScefMtEdt - Allow SCEF MT-EDT

Syntax

```
--gParmName allowScefMtEdt --gParmValue <string>
```

Description

Allows MT-EDT to be used with SCEFs

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

allowTransferOfEmergencyBearersToGnSgsn - Allow transfer of emergency bearers to Gn SGSN

Syntax

```
--gParmName allowTransferOfEmergencyBearersToGnSgsn --gParmValue  
<string>
```

Description

This global parameter determines whether emergency bearers can be transferred to a Gn SGSN. yes: the MME allows transfer of emergency bearers to a Gn SGSN. no: the MME denies transfer of emergency bearers to a Gn SGSN.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

allowTransferOfEmergencyBearersToS3sgsn - Allow transfer of emergency bearers to S3 SGSN

Syntax

```
--gParmName allowTransferOfEmergencyBearersToS3sgsn --gParmValue  
<string>
```

Description

This global parameter determines whether emergency bearers can be transferred to a S3 SGSN. yes: the MME allows transfer of emergency bearers to a S3 SGSN. no: the MME denies transfer of emergency bearers to a S3 SGSN.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

allowUeInitiatedBearerResourceAllocationRequest - Allow UE initiated bearer resource allocation request

Syntax

```
--gParmName allowUeInitiatedBearerResourceAllocationRequest --  
gParmValue <string>
```

Description

If the bearer resource allocation request parameter is set to yes, the request shall be allowed. if the request parameter is set to FALSE, the request shall be rejected by the MME. MME shall send BEARER RESOURCE ALLOCATION REJECT message with the ESM cause value #97 message type non-existent or nor implemented to the UE.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

allowUeOffloadOnMmeDuringMafCpuOverload - Allow UE offload on MME during CPPS CPU overload

Syntax

```
--gParmName allowUeOffloadOnMmeDuringMafCpuOverload --gParmValue  
<string>
```

Description

Enable UE offloading on the MME during CPPS CPU overload when lightly loaded CPPSs are available on the MME.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

allowUnSupportDcnReroute - Allow UnSupported DCN Reroute

Syntax

```
--gParmName allowUnSupportDcnReroute --gParmValue <string>
```

Description

Indicates to allow an unsupported Dcn Rerouted request to proceed on a non-default DCN MME

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

allowUnSupportDcnRerouteRoamer - Allow Unsupported DCN Reroute for Roamer

Syntax

--gParmName allowUnSupportDcnRerouteRoamer --gParmValue <string>

Description

This field specifies whether to allow an unsupported DCN Rerouted request to proceed on a non-default DCN at the MME. The default is No.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

alwaysSndEpsNwFeatSuppt - Always Send EPS Network Feature Support

Syntax

--gParmName alwaysSndEpsNwFeatSuppt --gParmValue <string>

Description

Always send EPS Network Feature Support IE in attach and tau accept message

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

anyUeRequestedApn - Any UE Requested APN

Syntax

--gParmName anyUeRequestedApn --gParmValue <string>

Description

GPARAM introduces APN override support for rogue APNs. GPARAM allows a match on 'any' UE requested APN. There will be an apnImsiRangeServcies entry with IMSI or IMEI range and a UE requested apn of 'any'. If this gparm is enabled it shall replace the UE requested APN for the search in apnImsiRangeServices with our own string 'any'. By default this shall be disabled.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

apnRateControlStatus - APN Rate Control Status

Syntax

```
--gParmName apnRateControlStatus --gParmValue <string>
```

Description

When this gparm is set to yes, the MME will include any saved APN rate control status for the first newly created PDN for a given APN. The PGW and SCEF can provide the rate control status to the MME when the last PDN in a given APN is deleted.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

apply2g3gApnWhiteList - Apply 2g3g APN white list

Syntax

```
--gParmName apply2g3gApnWhiteList --gParmValue <string>
```

Description

It is used to enable/disable the LBO APN white list checks globally for 2G3G. if it is no, then SGSN will not apply LBO APN white list checks irrespective of the UE PLMN LBO APN white list provisioning.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

applyDeleteGbrToVolteBearersOnly - Apply Delete GBR timer only to VOLTE bearers

Syntax

```
--gParmName applyDeleteGbrToVolteBearersOnly --gParmValue <string>
```

Description

Parameter controls the usage of existing Delete GBR timer. When this parameter is enabled, then Delete GBR timer applies only for VOLTE bearers. When this parameter is disabled, then Delete GBR timer applies for all GBR beares.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

applyLocallyProvQosValuesForHrt - Apply Locally Provisioned QoS Values For HRT

Syntax

```
--gParmName applyLocallyProvQosValuesForHrt --gParmValue <string>
```

Description

Parameter controls the behaviour of MME whether to apply QCI/ARP/PCI/PVI values from the locally provisioned QoS values. If the parameter is enabled and if MME receives QCI/ARP/PCI/PVI values from gateways that are different than locally configured qos values, MME shall apply locally configured values for HRT scenarios. By default the flag shall be disabled.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

applyLteApnWhiteList - Apply LTE APN white list

Syntax

```
--gParmName applyLteApnWhiteList --gParmValue <string>
```

Description

It is used to enable/disable the LBO APN white list checks globally for LTE. if it is no, then MME will not apply LBO APN white list checks irrespective of the UE PLMN LBO APN white list provisioning.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

arpHighPriorityAccessLevel - ARP high priority access level

Syntax

```
--gParmName arpHighPriorityAccessLevel --gParmValue <string>
```

Description

Allocation Retention Priority (ARP) priority level value indicating high priority access in MME. ARP priority level indicates the priority of allocation and retention and can take values between 1 and 15. Value 15 means 'no priority'. Values between 1 and 14 are ordered in decreasing order of priority, i.e., 1 indicates highest priority and 14 indicates lowest priority. The MME grants high priority access for a UE when SGW/PGW provides an ARP PL value equal to this MME provisioned value for a bearer of the UE.

Type / Range

Integer in the range of 1 to 15

Default

3

Name

arpLowAccessPriorityLevel - ARP low access priority level

Syntax

```
--gParmName arpLowAccessPriorityLevel --gParmValue <string>
```

Description

Allocation Retention Priority (ARP) priority level value indicating low priority traffic in MME. ARP priority level indicates the priority of allocation and retention and can take values between 1 and 15. Value 15 means 'no priority'. Values between 1 and 14 are ordered in decreasing order of priority, i.e., 1 indicates highest priority and 14 indicates lowest priority. MME detects low priority traffic based on SGW/PGW providing an ARP PL value equal to this MME provisioned value for a bearer of the UE.

Type / Range

Integer in the range of 1 to 15

Default

15

Name

attachRateLimitOverride - AttachRateLimitOverride

Syntax

```
--gParmName attachRateLimitOverride --gParmValue <string>
```

Description

Attaches allowed per sec after which CMM starts dropping attaches until the rate is below the limit. The limit value is specified as either 'systemDefault', which means the CMM determines what the value is, or a user provisioned number of attaches per second. If the

user provisioned value is greater than the maximum effective value listed below, the limit value that the CMM actually uses will be the maximum effective value. For 10 vcpu EMMS in large CMM, systemDefault is calculated as 400 attaches per second times the number of CPPS VMs with a maximum value of 10000 attaches per second. If the value is provisioned, the maximum effective value is the lesser of 1000 attaches per second times the number of CPPS VMs or 10000 attaches per second. For 12 vcpu EMMS in large CMM, Attach storm rate is 480 attaches per second per cpps. For an openstack small CMM, systemDefault is calculated as 400 attaches per second times the number of CPPS VMs with a maximum value of 1600 attaches per second. If the value is provisioned, the maximum effective value is the lesser of 1000 attaches per second times the number of CPPS VMs or 1600 attaches per second. For an A8 appliance CMM, systemDefault is calculated as 400 attaches per second times the number of CPPS VMs with a maximum value of 10000 attaches per second. If the value is provisioned, the maximum effective value is the lesser of 1000 attaches per second times the number of CPPS VMs or 10000 attaches per second. For an A2 appliance CMM, systemDefault is calculated as 200 attaches per second times the number of CPPS VMs with a maximum value of 1750 attaches per second. If the value is provisioned, the maximum effective value is the lesser of 400 attaches per second times the number of CPPS VMs or 1750 attaches per second. The implementation of the systemDefault is done at each IPDS and not factored for a m-IPDS CMM. For a m-IPDS CMM with 3 IPDS pairs for example, the actual limit is effectively tripled. We strongly suggest customer to overwrite the systemDefault with an adjusted value.

Type / Range

Integer in the range of systemDefault or 0 to 9,999

Default

systemDefault

Name

attachRateRejectLimitOverride - AttachRateRejectLimitOverride

Syntax

--gParmName attachRateRejectLimitOverride --gParmValue <string>

Description

Attaches allowed per sec after which CMM starts rejecting attaches until the rate is below the limit. The limit value is specified as either 'systemDefault', which means the CMM determines what the value is, or a user provisioned number of attaches per second. If the user provisioned value is greater than the maximum effective value listed below, the limit value that the CMM actually uses will be the maximum effective value. For an openstack large CMM, systemDefault is 10000 attaches per second. If the value is provisioned, the maximum effective value is 10000 attaches per second. For an openstack small CMM, systemDefault is 1600 attaches per second. If the value is provisioned, the maximum effective value is 1600 attaches per second. For an A8 appliance CMM, systemDefault is 10000 attaches per second. If the value is provisioned, the maximum effective value is 10000 attaches per second. For an A2 appliance CMM, systemDefault is 1750 attaches per second. If the value is provisioned, the maximum effective value is 1750 attaches per second. The implementation

of the systemDefault is done at each IPDS and not factored for a m-IPDS CMM. For a m-IPDS CMM with 3 IPDS pairs for example, the actual limit is effectively tripled. We strongly suggest customer to overwrite the systemDefault with an adjusted value.

Type / Range

Integer in the range of systemDefault or 0 to 9,999

Default

systemDefault

Name

attachRejForUnsubImsApn - Attach Reject for Unsubscribed APN

Syntax

```
--gParmName attachRejForUnsubImsApn --gParmValue <string>
```

Description

Global Parameter controls the feature activation. By default the parameter shall be disabled. When enabled, MME shall reject with Cause Value #33 if IMS APN is not subscribed or if IMS APN is subscribed only via Wild Card APN.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

attachRejForUnsubImsApnWithLogging - MME Track VoLTE Fail

Syntax

```
--gParmName attachRejForUnsubImsApnWithLogging --gParmValue <string>
```

Description

Parameter controls the feature activation. By default the parameter shall be disabled. When the parameter is enabled MME shall log PDN Connectivity Failures (standalone or part of the Attach) with ESM cause code 27 for IMS APN only. IMSI, MSISDN and IMEI shall be captured as part of the log.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

attachSgsUnknownLaiEmmCause - Attach SGs Unknown LAI EMM Cause

Syntax

```
--gParmName attachSgsUnknownLaiEmmCause --gParmValue <string>
```

Description

The parameter is used to provision EMM cause code to be sent to UE combined attach request if MME fails to select MSC due to LAI to TAI to MSC mapping provisioning errors.

Type / Range

Integer in the range of 2 or 16 to 18 or 22

Default

16

Name

authntDtchIntgChkFail - Authenticate UE for Detach Request message Integrity Checks Fail

Syntax

```
--gParmName authntDtchIntgChkFail --gParmValue <string>
```

Description

Parameter is used to activate MME to authenticate a UE if the UE sent Detach Request fails message integrity checks. If authentication fails or if MME is unable to authenticate the UE then MME does not detach the UE.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

autoAddTaiToTaiList - Auto add TAI to TAI list

Syntax

```
--gParmName autoAddTaiToTaiList --gParmValue <string>
```

Description

Indicates how the 'Automatically add TAI to the TAI list' should be handled.

List

- Basic - MME adds the old last seen TAI to the TAI list determined by the provisioning of "Neighbor list in the registration TAI list", and the enhanced list is sent to the UE (note: only a unique set of TAIs is sent).
- Enhanced - MME adds the previous old last seen TAI and the old last seen TAI to the TAI list determined by the provisioning of "Neighbor list in the registration TAI list", and the enhanced list is sent to the UE (note: only a unique set of TAIs is sent).

- Off - MME uses the list determined by the provisioning of 'Neighbor list in the registration TAI list' to determine the set of TAIs that is sent to the UE.
- UpdateThree - Update three - No cyclical behavior is required for update-three option.

Default

Basic

Name

backoffTimerImspdnReqRej - Back-off Timer Inclusion

Syntax

--gParmName backoffTimerImspdnReqRej --gParmValue <string>

Description

Global Parameter controls the inclusion of back-off timer IE. By default this parameter shall be disabled. When this parameter is enabled MME shall send T3396 back-off timer IE when rejecting IMS PDN Connectivity Request that is sent either as standalone or as part of Attach procedure due to not subscribed IMS APN or IMS APN is subscribed via the wildcard APN. Particularly for the case where IMS APN is subscribed via wildcard APN, the MME shall reject Attach/PDN Connectivity procedure(s) towards the IMS APN when GPARAM is enabled.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

bandwidthRoundup - Bandwidth roundup

Syntax

--gParmName bandwidthRoundup --gParmValue <string>

Description

Round up or round down AMBR, GBR and MBR sent to UE.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

bssapPlusSsn - BSSAP plus SSN

Syntax

```
--gParmName  bssapPlusSsn --gParmValue <string>
```

Description

BSSAP+ protocol sub system number.

Type / Range

Integer in the range of 0 to 254

Default

192

Name

bufferCbrDuringHo - Buffer Create Bearer Request During HO

Syntax

```
--gParmName  bufferCbrDuringHo --gParmValue <string>
```

Description

When this parameter is enabled, then if the MME receives CBR, UBR or DBR while processing 5G to 4G HO or inter-MME handover, it places the CBR (or UBR, DBR) on hold that is processed after completion of connected mode TAU that follows handover procedure.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

bufferCreateBearerRequestDuringTau - Buffer Create Bearer Request During TAU

Syntax

```
--gParmName  bufferCreateBearerRequestDuringTau --gParmValue <string>
```

Description

Parameter controls handling collision scenario. By default, the flag shall be disabled. When enabled, MME while processing Modify bearer Request during TAU procedure, receives Create Bearer Request from SGW, it shall be buffered and processed only upon completion of TAU procedure.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

canonicalNodeNameLabel - Canonical node name label

Syntax

```
--gParmName canonicalNodeNameLabel --gParmValue <string>
```

Description

Comparison of S-GW and P-GW network names in order to decide whether they are far enough to trigger a PDN connection re-establishment.

Type / Range

Integer in the range of 2 to 4

Default

2

Name

catM1bPagingTimerIncr - Category M1 CE mode b paging timer increment

Syntax

```
--gParmName catM1bPagingTimerIncr --gParmValue <string>
```

Description

This value is added to paging response timer for CAT-M1 CE mode b UE.

Type / Range

Integer in the range of 0 to 100

Default

0

Name

cc15eUtranAllowed - CC 15 E-UTRAN allowed

Syntax

```
--gParmName cc15eUtranAllowed --gParmValue <string>
```

Description

Configures the E-UTRAN allowed bit of the extended EMM cause.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

cgfAlwaysSetSequenceNumberToOne - CFG Always Set Sequence Number to One

Syntax

```
--gParmName cgfAlwaysSetSequenceNumberToOne --gParmValue <string>
```

Description

When this parameter is set to true SGSN support the optional customization to overwrite the 'recordSequenceNumber' field for each unique CDR with value 1

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

cgfServingPlmnSupport - CFG Serving PLMN Support

Syntax

```
--gParmName cgfServingPlmnSupport --gParmValue <string>
```

Description

When this parameter is set to true, SGSN stores in the S-CDR servingNodePLMNIdentifier field the PLMN of the RA the UE is in. When set to false the Serving PLMN will not be added in the CDR.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

checkDcnUeUsageTypeSgsnToMme - Check DCN UE Usage Type SGSN to MME

Syntax

```
--gParmName checkDcnUeUsageTypeSgsnToMme --gParmValue <string>
```

Description

Specifies whether the MME shall use UE Usage Type during 2G/3G->4G Attach and TAU procedure in order to perform Decor functionality as part of the procedure.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

checkDcnUeUsageTypeSgsnToMmeRoamer - Check DCN UE Usage Type SGSN to MME for Roamer

Syntax

--gParmName checkDcnUeUsageTypeSgsnToMmeRoamer --gParmValue <string>

Description

Specifies whether the MME shall use UE Usage Type during 2G/3G to 4G Attach and TAU procedure in order to perform Decor functionality as part of the procedure. This setting applies only for inbound roamers. The default is No.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

checkMatchApnConfig - check matchApn configuration

Syntax

--gParmName checkMatchApnConfig --gParmValue <string>

Description

Check matchApn configuration to decide on IMS APN. If matchApn is not configured and checkMatchApnConfig set to No - consider apn with QCI 5 as IMS APN Yes - consider apn with QCI 5 as non IMS APN

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

ciotS1ApMsgTunneling - CMM Support for Tunnelling of S1AP Messages Associated with CIOT or DCNR Attach/TAU Requests

Syntax

--gParmName ciotS1ApMsgTunneling --gParmValue <string>

Description

Enables CMM message tunnelling support over S10 of S1AP messages associated with CIOT (f11728-01) or DCNR Attach/TAU Requests.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

clrHandlingWithoutClrFlags - CLR Handling Without Clr Flags

Syntax

```
--gParmName clrHandlingWithoutClrFlags --gParmValue <string>
```

Description

Some functions are missing when combined MME/Gn SGSN (S6d use) handles CLR without CLR flag. With this parameter enabled, missing function should be supported. By default, the missing function is not supported.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

colldbreqUbreq - Collision of DL and UL bearer request

Syntax

```
--gParmName colldbreqUbreq --gParmValue <string>
```

Description

Enable/Disable support for collision scenario.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

colocatedGwSearchForNoMatch - Co-located Gateway Search for No Match

Syntax

```
--gParmName colocatedGwSearchForNoMatch --gParmValue <string>
```

Description

Parameter controls the support for searching co-located gateways for UEs when the served PLMN does not match the PGW FQDN PLMN for gateway mode selection 2. By default the flag shall be disabled.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

combinedAttachWithoutCs - Combined Attach Without CS

Syntax

```
--gParmName combinedAttachWithoutCs --gParmValue <string>
```

Description

Enables the support for combined attach without actual CS domain in the MME.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

comboPgwcRetries3XUe - Combo PGW-C Retries for 3X capable UE

Syntax

```
--gParmName comboPgwcRetries3XUe --gParmValue <string>
```

Description

Number of retries selecting combo PGW-C(s) during PDN Connectivity procedure before switching selecting standalone PGW-C for 3X capable UE(s)

Type / Range

Integer in the range of 0 to 4

Default

0

Name

comboPgwcRetriesN1Ue - Combo PGW-C Retries for N1 capable UE

Syntax

```
--gParmName comboPgwcRetriesN1Ue --gParmValue <string>
```

Description

Number of retries selecting combo PGW-C(s) during PDN Connectivity procedure before switching selecting standalone PGW-C for N1 capable UE(s)

Type / Range

Integer in the range of 0 to 4

Default

0

Name

coreNetworkRestrictSupport - Core Network Restrictions Support

Syntax

```
--gParmName coreNetworkRestrictSupport --gParmValue <string>
```

Description

Current parameter enables sending Core Network Restrictions field and NRNA bit of MM Context IE in s10 interface. Note that when n26 interworking is enabled, it is expected that this gParm is also enabled by the operator

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

coreNetworkSharing - Core network sharing

Syntax

```
--gParmName coreNetworkSharing --gParmValue <string>
```

Description

Indicates whether core network sharing enabled or not the default is no, indicating core network sharing is not enabled.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

coverageEnhancement - Coverage enhancement

Syntax

```
--gParmName coverageEnhancement --gParmValue <string>
```

Description

Enable the MME support for coverage enhancement level paging feature.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

cpCiotReloc - CP CloT Reloc

Syntax

```
--gParmName cpCiotReloc --gParmValue <string>
```

Description

Feature activation for supporting NB-IoT Control Plane CloT Relocation

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

crsi - Change reporting support indication

Syntax

```
--gParmName crsi --gParmValue <string>
```

Description

Indicates whether to set the CRSI flag in MME GTPv2 messages.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

csfbReturnLastPlmn - CSFB return last PLMN

Syntax

```
--gParmName csfbReturnLastPlmn --gParmValue <string>
```

Description

Configure MME to indicate MSC to return UE to the last PLMN after CSFB.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

csRecoveryIndication - MSC CS recovery indication support

Syntax

```
--gParmName csRecoveryIndication --gParmValue <string>
```

Description

When set to yes enables MME to accept SGsAP-PAGING REQUEST message with CS restoration indicator and paging of the UE in the location information provided by the MSC. when set to no disables the feature.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

ctrlRoamerQos - Control QoS for roaming subscribers

Syntax

```
--gParmName ctrlRoamerQos --gParmValue <string>
```

Description

ctrlRoamerQos controls below functionality only for roaming subscribers: If it sets to YES: SGSN will 1 Sending of HSS/HLR subscribed QoS values to S/P-GW or GGSN. 2 Ignoring QoS changes (upgrade/downgrade) from S/P-GW or GGSN. 3 Sending local configured QoS parameters towards RAN in case of: - Inter RAT handover MME to S4/Gn-SGSN - RAU MME to S4/Gn-SGSN - Inter S4/Gn-SGSN RAU If it sets to No, SGSN has no changes. Default value is NO

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

dcnrGwSelLte - DCNR capable PGW-C Selection for LTE only UE

Syntax

--gParmName dcnrGwSelLte --gParmValue <string>

Description

Enable selecting DCNR capable PGW-C for LTE only UE (i.e. without EN-DC support) during PDN Connectivity procedure

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

dcnrOverrideLipa - DCNR overrides Lipa service

Syntax

--gParmName dcnrOverrideLipa --gParmValue <string>

Description

Allow MME to override local PGW selection for Lipa enabled PDN with mobile network PGW selection for DCNR capable UE(s) with allowed NR as a Secondary RAT service.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

dcPraActionAddProcs - DC PRA action additional procedures

Syntax

--gParmName dcPraActionAddProcs --gParmValue <string>

Description

Enables MME to react upon received PRA action for dual connectivity PRA, when MME receives new PRA action from SGW/PGW during -Dedicated bearer activation -Default/dedicated bearer modification -intra LTE Mobility procedures -PDN setup for a PDN being handed over from ePDG. -Change notification response Applied when either -PRA Dual connectivity segment reporting is set to other than DC Segment Reporting Disabled, or -Presence Reporting Area Dual Connectivity IPv4 is set to other than DC PRA IPv4 Reporting Disabled, or -PRA DC IPv6 Prefix is set to other than DC PRA IPv6 Prefix Disabled.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

dcPraInterMme - DC PRA Inter MME Mobility

Syntax

--gParmName dcPraInterMme --gParmValue <string>

Description

Enables MME to hand over already started PRA action for DC PRA between MMEs during inter MME mobility. Applied when either -PRA Dual connectivity segment reporting is set to other than DC Segment Reporting Disabled, or -Presence Reporting Area Dual Connectivity IPv4 is set to other than DC PRA IPv4 Reporting Disabled, or -PRA DC IPv6 Prefix is set to other than DC PRA IPv6 Prefix Disabled.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

dcPraStart5gsToEps - DC PRA action start 5GS to EPS Mobility

Syntax

--gParmName dcPraStart5gsToEps --gParmValue <string>

Description

Enables MME to react upon received PRA action for dual connectivity PRA, when MME receives new PRA action for start reporting during 5GS to EPS mobility. Applied when either -PRA Dual connectivity segment reporting is set to other than DC Segment Reporting Disabled, or -Presence Reporting Area Dual Connectivity IPv4 is set to other than DC PRA IPv4 Reporting Disabled, or -PRA DC IPv6 Prefix is set to other than DC PRA IPv6 Prefix Disabled.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

ddnAckWithDLBuffering - DDN ack with DL buffering

Syntax

```
--gParmName ddnAckWithDLBuffering --gParmValue <string>
```

Description

Enable sending of DDN ack with downlink buffering duration set to number of page attempts times (T3413 timer + provisioned increment for ESM messages) seconds and DL buffering suggested packet count.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

ddnRateLimitOverride - DdnRateLimitOverride

Syntax

```
--gParmName ddnRateLimitOverride --gParmValue <string>
```

Description

Number of DDNs allowed per sec after which MME start dropping DDNs until rate is below limit.

Type / Range

Integer in the range of systemDefault or 0 to 31,999

Default

systemDefault

Name

decorRerouteSr - Decor Reroute on SR

Syntax

```
--gParmName decorRerouteSr --gParmValue <string>
```

Description

This parameter indicates whether MME is enabled to reroute a UE upon UE SR or CPSR to a MME group supporting the UE DCN ID if the current MME does not support the DCN ID. By default the parameter is set to False.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

decorRerouteSrRoamer - Decor Reroute on SR for Inbound Roamer

Syntax

```
--gParmName decorRerouteSrRoamer --gParmValue <string>
```

Description

This parameter indicates whether the MME is enabled to reroute an inbound roamer UE upon UE SR or CPSR to an MME group supporting the UE DCN ID if the current MME does not support the DCN ID. By default the parameter is set to False.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

dedicatedBearerPiggybackSupport - Dedicated bearer piggyback support

Syntax

```
--gParmName dedicatedBearerPiggybackSupport --gParmValue <string>
```

Description

This parameter indicates whether the MME should send piggyback support for dedicated bearer setup to the SGW in the create session message. if the parameter is yes the message will indicate MME support, if it is no the message will indicate non support.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

defaultIpv4ForPdnTypeIpv4OrIv6 - Default Ipv4 for Pdn Type IPv4OrIv6

Syntax

```
--gParmName defaultIpv4ForPdnTypeIpv4OrIv6 --gParmValue <string>
```

Description

GPARAM controls to choose between IPv4 and IPv6 upon receiving from HSS. When GPARAM is set to Yes default is IPv4 for pdnType IPv4 or IPv6. When gparm is set to 'No' it means ipv6. By default this parameter shall be set to 'No'.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

defaultPreferredDataMode - Default preferred data mode

Syntax

```
--gParmName defaultPreferredDataMode --gParmValue <string>
```

Description

Select default preferred network behavior when preferred-data-mode AVP is not received for a PDN connection.

List

- Control Plane - Default preferred data mode type of control plane.
- User Plane - Default preferred data mode type of user plane.

Default

User Plane

Name

defaultRacForSrvccMscSelection - Default RAC for SRVCC MSC selection

Syntax

```
--gParmName defaultRacForSrvccMscSelection --gParmValue <string>
```

Description

Default RAC for SRVCC MSC selection.

Type / Range

Integer in the range of 0 to 255

Default

255

Name

delbrrRateLimitOverride - DelbrrRateLimitOverride

Syntax

```
--gParmName delbrrRateLimitOverride --gParmValue <string>
```

Description

Number of delete bearers allowed per sec after which MME start dropping delete bearers until rate is below limit.

Type / Range

Integer in the range of systemDefault or 0 to 23,999

Default

systemDefault

Name

deletePdnSessionS11u - Delete PDN session S11U

Syntax

```
--gParmName deletePdnSessionS11u --gParmValue <string>
```

Description

Delete PDN sessions using S11-U when UE moves in to TAI that does not support CP CIOT optimization.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

deletePdnSessionS1u - Delete PDN session S1U

Syntax

```
--gParmName deletePdnSessionS1u --gParmValue <string>
```

Description

Delete PDN sessions using S1-U when UE moves in to TAI that does not support S1-U data transport.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

deletePdnSessionT6a - Delete PDN session T6A

Syntax

```
--gParmName deletePdnSessionT6a --gParmValue <string>
```

Description

Delete PDN sessions using T6A when UE moves in to TAI that does not support T6A data transport.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

depthOfSgwPgwrtry - Depth of SGW PGW retry

Syntax

```
--gParmName depthOfSgwPgwrtry --gParmValue <string>
```

Description

Defines the number of retries MME takes when there is no response from SGW on CSR.

Type / Range

Integer in the range of 1 to 2

Default

2

Name

detachUeMoveRegions2g3gEnable - Detach UE move regions 2G3G enable

Syntax

```
--gParmName detachUeMoveRegions2g3gEnable --gParmValue <string>
```

Description

Yes - feature 'detach ue move regions is enabled for 2g3g. no (default) - feature 'detach ue move regions is disabled for 2g3g.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

detachUeMoveRegionsLteApnSustOrDel - Detach UE move regions LTE APNs sustained or deleted with delay

Syntax

```
--gParmName detachUeMoveRegionsLteApnSustOrDel --gParmValue <string>
```

Description

MME feature with enhanced APN specific functionality for detaching UEs when moving between regions in LTE. The enhanced functionality enables defining two specific APN lists. One list defining APN(s) or which the PDN(s) are sustained despite of the UE moving between regions (=detaching is disabled), and the other list defining APN(s) for which the PDN(s)

are deleted only when no dedicated bearer(s) exist (=detaching is delayed). This feature is only functional when the basic feature for detach UE move regions is enabled for LTE (detachUeMoveRegionsLteEnable).

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

detachUeMoveRegionsLteEnable - Detach UE move regions LTE enable

Syntax

```
--gParmName detachUeMoveRegionsLteEnable --gParmValue <string>
```

Description

Yes - feature 'detach ue move regions is enabled for lte. no (default) - feature 'detach ue move regions is disabled for lte.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

detachUeMoveRegionsMonitorTime - Detach UE move regions monitor time

Syntax

```
--gParmName detachUeMoveRegionsMonitorTime --gParmValue <string>
```

Description

Time slice for monitoring the number of detaches per UE due to move across regions. if the number of detaches for a given UE exceed 'Detach UE Move Regions Threshold' during this time slice, the CMM will NOT detach the UE for 'Detach UE Move Regions Suppress Time' minutes.

Type / Range

Integer in the range of 1 to 3,600

Default

300

Name

detachUeMoveRegionsSuppressTime - Detach UE move regions suppress time

Syntax

```
--gParmName detachUeMoveRegionsSuppresTime --gParmValue <string>
```

Description

Time duration in which the WMM will suppress a UE detach due to move across regions. this is in order to prevent frequent detaches of a UE that is ping-ponging between two regions.

Type / Range

Integer in the range of 1 to 60

Default

10

Name

detachUeMoveRegionsThreshold - Detach UE move regions threshold

Syntax

```
--gParmName detachUeMoveRegionsThreshold --gParmValue <string>
```

Description

Threshold for number of UE detaches due to move across regions. if threshold is met within 'DETACH UE MOVE REGIONS MONITOR TIME' then detach suppression kicks in for a given UE.

Type / Range

Integer in the range of 1 to 16

Default

5

Name

detachUponSrMbrTimeout - Detach upon SR MBR timeout

Syntax

```
--gParmName detachUponSrMbrTimeout --gParmValue <string>
```

Description

Detach immediately upon service request modify bearer response timeout.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

diameterIgnoreUnknownAvp - Diameter ignore unknown AVP

Syntax

```
--gParmName diameterIgnoreUnknownAvp --gParmValue <string>
```

Description

Ignore unknown AVPs from HSS?.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

diameterMsgCiotPriority - Diameter CIOT message priority

Syntax

```
--gParmName diameterMsgCiotPriority --gParmValue <string>
```

Description

Use message priority when handling cellular internet of things diameter messages.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

diameterMsgPriority - Diameter message priority

Syntax

```
--gParmName diameterMsgPriority --gParmValue <string>
```

Description

Use message priority when handling diameter messages.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

disableMmeRateControlExecute - disable ratue control execute/drop in MME

Syntax

```
--gParmName disableMmeRateControlExecute --gParmValue <string>
```

Description

There are be different count between MME and GW for the packet for the downlink rate control. This has been noticed from testing with CMG for IPv6. This parameter is introduced to give operator choice, to disable MME drop more packets than already did by GW.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

discoverMme - Discover MME

Syntax

```
--gParmName discoverMme --gParmValue <string>
```

Description

Indicates whether or not (YES or NO) to use a DNS query to obtain a list of MME IP addresses for handling a given TAI. if set to yes, the S10 interface managed objects (MOs) are created dynamically. if set to 'no', the S10 interface MOs are created from provisioned data and are deleted when this provisioning data is removed from the system.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

discoverMmeEnh - Discover MME Enhancement

Syntax

```
--gParmName discoverMmeEnh --gParmValue <string>
```

Description

Parameter controls reselection of another MME if the response from the target MME times out or the first choice per DNS records is unavailable. By default the flag shall be set to No. When flag is set to Yes, MME shall select 2nd S10 link if the first choice (per DNS records) is unavailable or if the response from the target MME times out.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

discoverSgw - Discover SGW

Syntax

--gParmName discoverSgw --gParmValue <string>

Description

Indicates whether or not (YES or NO) to use a DNS query to obtain a list of SGW IP addresses for handling a given TAI. if set to yes, the S11 interface managed objects (MOs) are created dynamically. if set to 'no', the S11 interface MOs are created from provisioned data and are deleted when this provisioning data is removed from the system.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

discoverSrvccMsc - Discover SRVCC MSC

Syntax

--gParmName discoverSrvccMsc --gParmValue <string>

Description

Indicates whether to discover the MSC's SRVCC IP address via DNS. the default is no, indicating that the IP address must be provisioned.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

discoverTaiDefaults - Discover TAI Defaults

Syntax

--gParmName discoverTaiDefaults --gParmValue <string>

Description

Feature activation for default TAC Mapping to TA Discovery Provisioned Parameters

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

discoverTais - Discover TAIs

Syntax

```
--gParmName discoverTais --gParmValue <string>
```

Description

Enable WMM to discover non-provisioned TAIs.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

dlBufferingPacketCount - DL buffering suggested packet count

Syntax

```
--gParmName dlBufferingPacketCount --gParmValue <string>
```

Description

Downlink buffering suggested packet count.

Type / Range

Integer in the range of 0 to 10

Default

2

Name

dltPrevPdnConnOnNewReq - Delete previous PDN connection on new request

Syntax

```
--gParmName dltPrevPdnConnOnNewReq --gParmValue <string>
```

Description

This flag determines the call flow of a PDN connectivity request where a UE already has PDN connection. no (default) - the current call flow is used. when an additional PDN connection request is received after attach the MME checks to see whether there is already a PDN connection to the network specified by the APN in the request. when the received APN is the same as the APN of a PDN connection that has already been established for the given UE, the MME sends a PDN connectivity reject with cause code 55. yes - instead of sending a PDN connectivity reject the MME will first delete the previously established PDN connection and then honor the subsequent PDN connection request. note that if the PDN connectivity

request with APN=x is for a UE with only one already established PDN connection and that connection is for APN x, the MME would have to reject the PDN connectivity request as is currently done.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

dualConnectivity - Dual connectivity

Syntax

```
--gParmName dualConnectivity --gParmValue <string>
```

Description

Enable/disable dual connectivity feature.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

e911apnNiForMode1pgwSelection - E911 APN NI for mode 1 PGW selection

Syntax

```
--gParmName e911apnNiForMode1pgwSelection --gParmValue <string>
```

Description

If set to yes, MME will support the use of the E911 APN NI for the mode 1 selection of PGW for emergency attach and also PDN connection request to E911. if set to no, MME will default to using the S-NAPTR procedure to select a PGW.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

edrxEpochLeapSeconds - eDRX epoch leap second

Syntax

```
--gParmName edrxEpochLeapSeconds --gParmValue <string>
```

Description

Number of leap seconds that have occurred since the eDRX epoch time.

Type / Range

Integer in the range of 0 to 100

Default

18

Name

edrxEpochTime - eDRX epoch time

Syntax

```
--gParmName edrxEpochTime --gParmValue <string>
```

Description

Starting point in time of counting H-SFN for calculation of eDRX paging opportunities. Any valid past date (YYYY-MM-DD) and time (HH:MM:SS) in GMT can be entered.

Type / Range

Digit string of length 19 to 19 with the allowed pattern of (\d{4}-\d{2}-\d{2} \d{2}:\d{2}:\d{2})

Default

1980-01-06 00:00:00

Name

edrxMaxPagingHoldTime - eDRX maximum paging hold time

Syntax

```
--gParmName edrxMaxPagingHoldTime --gParmValue <string>
```

Description

Maximum time interval (in milliseconds) that the MME will hold a call processing procedure.

Type / Range

Integer in the range of 3,000 to 15,000

Default

10000

Name

edrxNewLeapSecondTime - eDRX new leap second time

Syntax

```
--gParmName edrxNewLeapSecondTime --gParmValue <string>
```

Description

The date and time when an additional leap second will occur.

Type / Range

Digit string of length 0 or 19 to 19 with the allowed pattern of (\d{4}-\d{2}-\d{2} \d{2}:\d{2}:\d{2})*\d{2})*

Name

edrxNotifyHssTime - eDRX notify HSS time

Syntax

```
--gParmName edrxNotifyHssTime --gParmValue <string>
```

Description

Time interval (in milliseconds) prior to the eDRX paging send time that the MME should send the notify command to HSS when handling a pending SMS request for an idle UE with eDRX enabled. this parameter is set to account for the time needed for the SMS service center to be informed of the UE's availability and the resulting SGS paging request to be sent to the MME.

Type / Range

Integer in the range of 100 to 3,000

Default

500

Name

edrxPagingSendTime - eDRX paging send time

Syntax

```
--gParmName edrxPagingSendTime --gParmValue <string>
```

Description

Time interval (in milliseconds) before paging time window (PTW) start time that the MME should start sending the S1AP-PAGING messages to the eNB. this parameter is set to account for small time synchronization differences between the MME and the eNB.

Type / Range

Integer in the range of 100 to 5,000

Default

2000

Name

emergency5gsIwkWoN26 - Emergency 5GS Interworking Without N26

Syntax

```
--gParmName emergency5gsIwkWoN26 --gParmValue <string>
```

Description

Enables emergency session to be handed over between 5GS and EPS without N26 interface in the MME. Applied when EPS-5GS Interworking without N26 is set to Yes for the MME.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

emergencySimlessUeOverride - Emergency SIMless UE override

Syntax

--gParmName emergencySimlessUeOverride --gParmValue <string>

Description

Emergency SIMless UE override when set to yes MME will accept emergency attach from SIMless UEs from any TAI served by the MME irrespective of emergency provisioning per TAI.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

emergencyWlanHo - Emergency WLAN Handover

Syntax

--gParmName emergencyWlanHo --gParmValue <string>

Description

This parameter controls the support for IMS emergency session continuity for handovers between LTE access and WLAN for authenticated home subscribers. When supported, the MME receives WLAN access selected PGW information from HSS, and on handover to LTE the same PGW can be selected. When an emergency call is initially started on LTE access, the MME also registers via NOR the selected PGW to HSS to be possibly handed over to WLAN access.

List

- ewLanHoDisabled - Emergency WLAN Handover Disabled - Feature is disabled.
- ewLanHoSendHost - Emergency WLAN Handover Send Host - Feature is enabled and when MME sends NOR to HSS for emergency WLAN Handover session continuity, the MME sends the PGW information on MIP6-Agent-Info AVP as MIP-Home-Agent-Host, if it is available. If the hostname is not available, the MME sends the IP address(es).
- ewLanHoSendIp - Emergency WLAN Handover Send Ip - Feature is enabled and when MME sends NOR to HSS for emergency WLAN Handover session continuity, the MME sends the PGW information on MIP6-Agent-Info AVP as MIP-Home-Agent-Address(es).

Default

ewLanHoDisabled

Name

emerMtLrNoEsmLc - Emergency MT-LR do not contact E-SMLC for UE location information

Syntax

--gParmName emerMtLrNoEsmLc --gParmValue <string>

Description

If yes, when the MME receives mobile terminated location request with emergency client type or MT-LR with any client type while an emergency call is ongoing, the MME contacts the eNB for UE current E-CGI information. this information is sent back to the GMLC. the MME does not contact the E-SMLC to obtain any UE location information.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

empsAdvancePriority - EMPS advance priority

Syntax

--gParmName empsAdvancePriority --gParmValue <string>

Description

Enable the detection of advance priority eMPS users and eMPS IMS priority users (Non advance priority users).

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

empsDscp44Marking - EMPS DSCP 44 Marking

Syntax

--gParmName empsDscp44Marking --gParmValue <string>

Description

Enable the ability to mark eMPS S11 and S10 messages with a provisionable DSCP value. Enabling the eMPS DSCP 44 Marking global parameter shall be denied if the eMPS Advance

Priority global parameter is disabled. Disabling the eMPS Advance Priority global parameter shall be denied if the eMPS DSCP 44 Marking global parameter is enabled

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

empsPriorityOnlySubscriptionBased - eMPS Priority Only Subscription Based

Syntax

--gParmName empsPriorityOnlySubscriptionBased --gParmValue <string>

Description

When set to Yes, enables MME to provide eMPS priority for UEs that have HSS subscription indicating MPS-EPS-Priority bit set on the MPS-Priority AVP. MME does not check other parameters for UEs that have MPS EPS functionality enabled on (on MME local provisioning on service agreement profile) and MPS-EPS-Priority bit set on the HSS subscription.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

empsPrioToHss - eMPS Priority to HSS

Syntax

--gParmName empsPrioToHss --gParmValue <string>

Description

MME treats with priority any signal it receives with RRC establishment cause indicating priority. If set to no, MME only treats with priority a UE that additionally either has priority HSS subscription or a bearer is established with a priority ARP value.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

enable2gBvccountIndb - Enable2gBvccountIndb

Syntax

--gParmName enable2gBvccountIndb --gParmValue <string>

Description

Ability to store BVC counts DB.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

enable2GdataJumboFrame - Enable 2G data Jumbo frame

Syntax

--gParmName enable2GdataJumboFrame --gParmValue <string>

Description

This parameter enables the 2G data Jumbo Frame

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

enable2gEnhancesgbaLb - Enable2gEnhancesgbaLb

Syntax

--gParmName enable2gEnhancesgbaLb --gParmValue <string>

Description

Ability to load PAPSs using the number of BVCs assigned to NSEs.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

enable2gUsrdataSafetynet - Enable2gUsrdataSafetynet

Syntax

```
--gParmName enable2gUsrdataSafetynet --gParmValue <string>
```

Description

Ability for WMM to detect 2G user data throughput safety net breach and throttle 2G attaches and PDP activations. 2G allowed user data throughput is 145000000 bits/sec.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

enable3GdataJumboFrame - Enable 3G data Jumbo frame

Syntax

```
--gParmName enable3GdataJumboFrame --gParmValue <string>
```

Description

This parameter enables the 3G data Jumbo Frame

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

enableAlarmEventsStream - enableAlarmEventsStream

Syntax

```
--gParmName enableAlarmEventsStream --gParmValue <string>
```

Description

Enables Alarm Events streaming via fluentd for CNF deployment

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

enableCacheAgeOut - enable Cache Age Out feature

Syntax

```
--gParmName enableCacheAgeOut --gParmValue <string>
```

Description

If the enableCacheAgeOut value is set to Yes the Cache Age Out feature uses defined Quiet-period-timers per UE Class. These timers are used to manage the CPPS cache. By default the enableCacheAgeOut is set to NO, the feature is by default disabled.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

enableCallTraceBuffering - Enable Call Trace Buffering

Syntax

```
--gParmName enableCallTraceBuffering --gParmValue <string>
```

Description

Enables the buffering of Call Trace messaging before IMSI/SUPI is known.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Recommended

Recommended value is Yes

Name

enableCapdegradeOn2gciphering - EnableCapdegradeOn2gciphering

Syntax

```
--gParmName enableCapdegradeOn2gciphering --gParmValue <string>
```

Description

Ability to downgrade WMM 2G capacity by 20% when 2G ciphering is on.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

enableCpuBasedControlOnly - EnableCpuBasedControlOnly

Syntax

--gParmName enableCpuBasedControlOnly --gParmValue <string>

Description

Whether or not WMM performs overload control based on CPU only or take into account message rate and storm rate for controls.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

enableEffectiveloadSnalgo - EnableEffectiveloadSnalgo

Syntax

--gParmName enableEffectiveloadSnalgo --gParmValue <string>

Description

Enable effective load snalgo.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

enableLocalRoamerQciArpOvrd - Enable local roamer QCI ARP override

Syntax

--gParmName enableLocalRoamerQciArpOvrd --gParmValue <string>

Description

Enable local roamer QCI ARP override whereby skipping QCI and ARP override parameters for dedicated bearers in case of LBO traffic.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

enableLocalRoamerQoSPolicy - Enable local roamer QoS policy

Syntax

```
--gParmName enableLocalRoamerQoSPolicy --gParmValue <string>
```

Description

Enable local roamer QoS policy.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

enableLogEventsStream - enableLogEventsStream

Syntax

```
--gParmName enableLogEventsStream --gParmValue <string>
```

Description

Enables Log Events streaming via fluentd for CNF deployment

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

enableMmeOverloadBackoffTimer - Enable MME overload backoff timer

Syntax

```
--gParmName enableMmeOverloadBackoffTimer --gParmValue <string>
```

Description

This flag is used to enable the capability of MME to include a randomly-selected T3346 timer value whenever an attach request, TAU request, service request, or extended service request procedure is rejected due to IPDS CPU overload, CPPS CPU overload, or CXN object overload. no (default) - set the T3346 back-off timer to 0. yes - set the T3346 back-off timer to a randomly-selected value between provisioned minimum and maximum values.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

enableOLCContextDump - EnableOLCContextDump

Syntax

```
--gParmName enableOLCContextDump --gParmValue <string>
```

Description

Whether or not WMM dumps service context data during overload report.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

enableOlcSchddelayDegrade - EnableOlcSchddelayDegrade

Syntax

```
--gParmName enableOlcSchddelayDegrade --gParmValue <string>
```

Description

Whether or not VMM capacity should be degraded on scheduling delay.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

enableOldGutiPaging - Enable old GUTI paging per MME

Syntax

```
--gParmName enableOldGutiPaging --gParmValue <string>
```

Description

Enable old GUTI paging per MME.

List

- No - Boolean value of no.

- Yes - Boolean value of yes.

Default

No

Name

enablePra - Enable presence reporting area

Syntax

--gParmName enablePra --gParmValue <string>

Description

Enable MME support for presence reporting of an UE entering or leaving an area.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

enablePrometheusGW - Enable Prometheus Functionality

Syntax

--gParmName enablePrometheusGW --gParmValue <string>

Description

This parameter controls whether Prometheus functionality is enabled in the system. This means that this gParm does not control the way the PM data are sent/retrieved from Prometheus (i.e. sent via Prometheus Push GW or retrieved directly from Prometheus). This would be specified via a another gParm that is named Prometheus Mode.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

enableProvCc - Enable provisional cause code

Syntax

--gParmName enableProvCc --gParmValue <string>

Description

Enable MME support for configurable or provisional cause code functionality.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

enablePsmT3412MinTime - Enable use of PSN T3412 Min Time

Syntax

```
--gParmName enablePsmT3412MinTime --gParmValue <string>
```

Description

This parameter enables/disables the feature CMM support for Minimum T3412 Timer for UE Supporting Power Savings Mode'. By default the feature is disabled.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

enableRejectIdrepslocinfoOnol - EnableRejectIdrepslocinfoOnol

Syntax

```
--gParmName enableRejectIdrepslocinfoOnol --gParmValue <string>
```

Description

Enable/Disable reject IDR with EPS location info request during overload.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

enbToSgwIpv4v6TransportLayerAddress - eNB/gNB IPv4v6 Transport Layer Address IP Type

Syntax

```
--gParmName enbToSgwIpv4v6TransportLayerAddress --gParmValue <string>
```

Description

MME passes eNB/gNB S1-u IPv4v6 Transport Layer Address to the SGW if it received this IP Type from eNB/gNB and this parameter is set to YES. Otherwise MME matches S1-C IP Address Type and passes it to the SGW.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

enhancedAlternateGwReselection - Enhanced Alternate GW Reselection

Syntax

--gParmName enhancedAlternateGwReselection --gParmValue <string>

Description

Defines whether the MME reselects an alternate SGW or PGW during the same procedure in defined error cases. Setting this parameter to 'Yes' requires that gParm supportAlternateGwReselection is also set to 'Yes'. MME reselects an alternate PGW during Attach and PDN Connectivity procedures when SGW returns cause code 72 (System failure), 78 (Missing or unknown APN), 84 (All dynamic addresses are occupied) or 91 (No memory available) and the cause source indicates PGW. MME also reselects an alternate PGW when SGW returns cause code 100 (Remote peer not responding) and cause source indicates SGW. MME reselects an alternate SGW during Attach and mobility procedures when SGW returns cause code 72 (System failure) or 91 (No memory available) and cause source indicates SGW.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

enhancedPgwlIsolation - Enhanced PGW Isolation

Syntax

--gParmName enhancedPgwlIsolation --gParmValue <string>

Description

Defines whether the MME isolates PGW during additional cases. Enabling this parameter requires gParm ltePgwlIsolation to be set to 'Yes'. MME considers PGW related errors for PGW isolation functionality additionally to the legacy functionality when SGW returns error code 73 (No resources available), 91 (No memory available) or 100 (Remote peer not responding) and cause source indicates PGW. MME also considers SGW returned error for PGW isolation functionality when SGW returns error code 100 (Remote peer not responding) and cause source indicates SGW.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

enhancedSgwIsolation - Enhanced SGW Isolation

Syntax

```
--gParmName enhancedSgwIsolation --gParmValue <string>
```

Description

Defines whether the MME isolates SGW during additional cases. Enabling this parameter requires gParm lteSgwIsolation to be set to 'Yes'. MME considers SGW related errors for SGW isolation functionality additionally to the legacy functionality when SGW returns error code 73 (No resources available) or 91 (No memory available) and cause source indicates SGW.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

enhancedSgwRestorationProcedures - Enhanced SGW restoration procedures

Syntax

```
--gParmName enhancedSgwRestorationProcedures --gParmValue <string>
```

Description

Support enhanced SGW restoration procedures or not.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

enhancedSGWselectionIRAT - Enhanced SGW selection for IRAT mobility

Syntax

```
--gParmName enhancedSGWselectionIRAT --gParmValue <string>
```

Description

Enable/Disable support enhanced SGW selection for IRAT mobility per MME.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

enhApnCorrectionPerImsi - Enhanced APN Correction per Imsi

Syntax`--gParmName enhApnCorrectionPerImsi --gParmValue <string>`**Description**

Parameter controls when the rejection cause code shall be sent by MME. By default the flag shall be disabled, when enabled MME during an Attach or PDN Connectivity Request, if the APN received from the UE, is not an HSS provisioned APN and is not in locally configured in the MME correction/replacement table, MME shall reject the procedure with cause code 'Requested service option not subscribed'. In addition if the APN received from the UE, is locally configured in MME APN correction/replacement table but correction comes back with APN which is not an HSS provisioned APN. (In other words, even after correction the APN must be subscribed or it will be rejected) Feature shall be enabled when apnlist table is enabled.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

ePlmnPerImsiTaiRange - EPLMN per IMSI Range

Syntax`--gParmName ePlmnPerImsiTaiRange --gParmValue <string>`**Description**

MME uses the EPLMN list provisioned per IMSI and TAI range if the parameter is set to 'True'.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

eps5gslwkWithoutN26 - EPS-5GS Interworking without N26

Syntax

```
--gParmName eps5gsIwkWithoutN26 --gParmValue <string>
```

Description

Enables the EPS-5GS interworking without N26 interface support in the MME.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

epsInfoInUlr - Support of EPS info in ULR

Syntax

```
--gParmName epsInfoInUlr --gParmValue <string>
```

Description

When parameter is YES, EPS info is supported in updateGprsLocation message, otherwise it is not supported

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

epsNasSecurityContextInclusion - EPS NAS Security Context Inclusion

Syntax

```
--gParmName epsNasSecurityContextInclusion --gParmValue <string>
```

Description

Parameter controls inclusion of NAS security context at the end of MM Context IE. By default, flag shall be disabled. When enabled, MME will include NAS security context at the end of MM Context IE to indicate the mapped or native security context during the MME-1 to MME-2 reallocation

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

eutranSupportedRatType - Eutran supported RAT type

Syntax

--gParmName eutranSupportedRatType --gParmValue <string>

Description

Configure eutran supported RAT type. values are both, wbeutran and ClOT.

List

- both - Both wide band EUTRAN and CIOT.
- nbiot - CIOT - CIOT.
- wbeutran - Wide band EUTRAN - Wide band EUTRAN.

Default

wbeutran

Name

exclDelayDlPktNotifReq - Exclude delay downlink packet notification request

Syntax

--gParmName exclDelayDlPktNotifReq --gParmValue <string>

Description

Control the inclusion of the delay downlink packet notification request.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

excludeDestinationHost - Exclude destination host

Syntax

--gParmName excludeDestinationHost --gParmValue <string>

Description

Enable or disable exclusion of destination host for S6ad and S13 interfaces.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

excludeNbiotDrbCheckForCpCiot - Exclude NB-IoT Check on DRB for CP CIoT

Syntax

```
--gParmName excludeNbiotDrbCheckForCpCiot --gParmValue <string>
```

Description

The MME will not consider the NB-IOT UE support of multiple DRBs when the UE requests to set up a subsequent CP CIoT bearer.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

excludeUIExceptionReports - exception reports are excluded from rate limit

Syntax

```
--gParmName excludeUIExceptionReports --gParmValue <string>
```

Description

By default MME rate control applies all packets. With this parameter is set, Mo exception data packet will be excluded in the rate limited (not count)

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

excludeVendorSpecificApplicationId - Exclude vendor specific application id

Syntax

```
--gParmName excludeVendorSpecificApplicationId --gParmValue <string>
```

Description

Enable or disable exclusion of exclusion of vendor specific application id for S13 interface.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

extendEmmCauseIe - Extend EMM cause IE

Syntax

```
--gParmName extendEmmCauseIe --gParmValue <string>
```

Description

Enable inclusion of extended cause IE in attach reject and TAU reject messages with EMM cause #15.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

fallbackToDefaultApnOi - Fallback to default APN-OI

Syntax

```
--gParmName fallbackToDefaultApnOi --gParmValue <string>
```

Description

Activate fallback to use of default APN-OI in NAPTR query if NAPTR query with APN-OI replacement fails.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

fivegcNotRestrictedIndicationSupport - 5GC Not Restricted Indication Support

Syntax

```
--gParmName fivegcNotRestrictedIndicationSupport --gParmValue  
<string>
```

Description

Parameter controls the support for 5GCNRI and 5GCNRS indication flags in create session request message. By default, the flag shall be disabled.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

fmKafkaStream - Kafka Stream for the Alarm Functionality

Syntax

--gParmName fmKafkaStream --gParmValue <string>

Description

This parameter controls the option to stream the Alarms to an external KAFKA cluster. When this parameter is set, the Alarms would be streamed to a specific topic and broker of an external KAFKA cluster.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

fnsBckpSrvrHbNoRspns - FNS Backup Server No Responses to heartbeat message

Syntax

--gParmName fnsBckpSrvrHbNoRspns --gParmValue <string>

Description

If MME does not receive responses to the heartbeat messages for consecutive number of times specified by this parameter then MME declares the link is not available and generates a major alarm.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

fnsBckupServer - FNS Backup Server

Syntax

--gParmName fnsBckupServer --gParmValue <string>

Description

Global parameter to enable support of FNS backup server. If the parameter is set to 'true' then CMM establishes TCP connection to the provisioned backup server and start updating the UE context.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

fnsBckupSrvrHbTimer - FNS Backup Server Heart Beat Timer

Syntax

--gParmName fnsBckupSrvrHbTimer --gParmValue <string>

Description

Parameter specifies the heart beat interval to the FNS backup server.

Type / Range

Integer in the range of 1 to 600 in units of Second

Default

10

Name

fnsBckupSrvrRspTimer - FNS Backup Server Response Timer

Syntax

--gParmName fnsBckupSrvrRspTimer --gParmValue <string>

Description

Time to wait for FNS backup server response after sending a request to the server.

Type / Range

Integer in the range of 1 to 10 in units of Second

Default

2

Name

fnsBckupSrvrUpdateType - FNS Backup Server Update Type

Syntax

--gParmName fnsBckupSrvrUpdateType --gParmValue <string>

Description

Parameter to specify which UE context needs to be stored on the FNS backup server.

List

- All - ALL UE - Store IMSI and last visited TA of all UE.
- IMS - Store IMSI and last visited TA of all UE with IMS PDN connection.
- IMS_AND_CSFBCK - IMS and CS Fallback - Store IMSI and last visited TA of all UE with IMS PDN connection and UE combined attach with CSFB. Combined attach UE with SMS only service are not stored for this option.

Default

All

Name

fnsInMmePool - FNS IN MME Pool

Syntax`--gParmName fnsInMmePool --gParmValue <string>`**Description**

This parameter is used to indicate whether FNS MME and CMM MME are in the same pool. The parameter will be used by CMM uelb command directed move to FNS to make the CMM UELB is compatible with FNS directed move.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

forcePgwReselectionOnCsRequestTimeout - Force PGW reselection on CS request timeout

Syntax`--gParmName forcePgwReselectionOnCsRequestTimeout --gParmValue
<string>`**Description**

If set to yes, mme shall select an alternate PGW if there is no response to the create session request. the number of attempts to select PGW remains the same. the feature only applies to GW selection mode 2. the applicable procedures are attach and subsequent PDN connectivity request.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

fourDigitMncMccSuppt - Four Digit MNC MCC Support

Syntax

```
--gParmName fourDigitMncMccSuppt --gParmValue <string>
```

Description

The parameter controls whether four digit MNC and MCC is used in RAI and RNCid pre-REL8 DNS queries for Gn-SGSN. By default the parameter shall be disabled.

List

- OFF - OFF
- ON - ON

Default

OFF

Name

freqCountMonitoring - Frequency Count Monitoring

Syntax

```
--gParmName freqCountMonitoring --gParmValue <string>
```

Description

Enables the MME functionality to monitor the frequency count of UE attempts for defined procedures.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

freqCountMonitoringMtAccess - Frequency Count Monitoring MT Access

Syntax

```
--gParmName freqCountMonitoringMtAccess --gParmValue <string>
```

Description

Defines whether the MME, while frequency count monitoring functionality is activated, allows the Mobile Terminated Access for the UEs. If the UE is blacklisted but not yet implicitly detached, the network may page the UE due to e.g. incoming downlink data and the UE may trigger Service Request/Extended Service Request/Control Plane Service Request due to that. When MT access is allowed, MME does not increment procedure attempt counter for MT access attempts.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

freqCountMonitoringSecurity - Frequency Count Monitoring Security

Syntax

--gParmName freqCountMonitoringSecurity --gParmValue <string>

Description

Defines whether the MME increments procedure frequency count before or after running any of the security procedures (especially integrity check).

List

- AfterSecurity - After Security - MME increments frequency count monitoring procedure attempt counter after running any of the security procedures (e.g. integrity check).
- BeforeSecurity - Before Security - MME increments frequency count monitoring procedure attempt counter before running any of the security procedures (e.g. integrity check).

Default

AfterSecurity

Name

fullPgwApnGtpOvlControlSupport - Full PGW APN or Node Level Overload Control Support

Syntax

--gParmName fullPgwApnGtpOvlControlSupport --gParmValue <string>

Description

Defines whether the MME supports PGW indicated overload. -PGW Overload Control Information IE containing APN list on GTPv2 messages, when APN is congested on a PGW. -Back-off Time IE on Create Session Response when PGW rejects the request due to APN congestion.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

gea1Degradation - GEA1 degradation

Syntax

```
--gParmName gea1Degradation --gParmValue <string>
```

Description

GEA1 degradation percent. Warning: For Nokia personnel only.

Type / Range

Integer in the range of 0 to 100

Default

52

Name

gea2Degradation - GEA2 degradation

Syntax

```
--gParmName gea2Degradation --gParmValue <string>
```

Description

GEA2 degradation percent. Warning: For Nokia personnel only.

Type / Range

Integer in the range of 0 to 100

Default

64

Name

gea3Degradation - GEA3 degradation

Syntax

```
--gParmName gea3Degradation --gParmValue <string>
```

Description

GEA3 Degradation percent. Warning: For Nokia personnel only.

Type / Range

Integer in the range of 0 to 100

Default

28

Name

ggsnLbMode - GGSN Load Balancing mode

Syntax

```
--gParmName ggsnLbMode --gParmValue <string>
```

Description

Set the corresponding mode for GGSN Load Balancing

List

- DNS - Weighted round robin (DNS)

- HOST - Weighted round robin (HOST)
- RROFF - Round robin is not in use
- RRON - Round Robin is in use

Default

RROFF

Name

gmlcOverlappingLocationRequests - GMLC overlapping location requests

Syntax`--gParmName gmlcOverlappingLocationRequests --gParmValue <string>`**Description**

Enable/disable GMLC overlapping UE location requests.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

gnSgsnSnaptr - Gn SGSN SNAPTR

Syntax`--gParmName gnSgsnSnaptr --gParmValue <string>`**Description**

None: gn-sgsn will not use S-NAPTR. pgw-ggsn: gn-sgsn will use S-NAPTR only for allocation of a co-located pgw-ggsn. all: gn-sgsn will use S-NAPTR for all FQDN resolutions.

List

- all - Specify S-NAPTR type of all.
- none - Specify S-NAPTR type of none.
- pgw-ggsn - Specify S-NAPTR type of PGW / GGSN.

Default

none

Name

gnSgsnSrv - Gn SGSN SRV

Syntax`--gParmName gnSgsnSrv --gParmValue <string>`**Description**

None SRV RR only for pgw-ggsn selection.

List

- none - Specify SRV type of none.
- pgw-ggsn - Specify SRV type of PGW / GGSN.

Default

none

Name

gnTargetIdFormat - Gn target id format

Syntax

--gParmName gnTargetIdFormat --gParmValue <string>

Description

Determines how it encodes and decodes the target identification IE when the MME is sending a GTPv1 (Gn) message.

List

- ASN.1 With Choice/Options Octet - Encode according to type 1 in SEI-d59695-Gn_SEI-100.
- ASN.1 Without Choice/Options Octet - Encode according to type 2 in SEI-d59695-Gn_SEI-100.
- GTP with CN-order of PLMN Digits - Encode according to type 3 in SEI-d59695-Gn_SEI-100.
- GTP with RANAP-order of PLMN Digits - Encode according to type 4 in SEI-d59695-Gn_SEI-100.

Default

ASN.1 With Choice/Options Octet

Name

gprsSubscriptionData - GPRS subscription data

Syntax

--gParmName gprsSubscriptionData --gParmValue <string>

Description

Request the HSS to include GPRS subscription data in ULA if available.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

gtpLoadControlSupport - GTP load control support

Syntax

```
--gParmName gtpLoadControlSupport --gParmValue <string>
```

Description

GTP load control support.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

gtpOvlControlSupport - GTP OVL control support

Syntax

```
--gParmName gtpOvlControlSupport --gParmValue <string>
```

Description

Support for GTP overload control.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

gwGtpOvlControlSupport - GW GTP OVL control support

Syntax

```
--gParmName gwGtpOvlControlSupport --gParmValue <string>
```

Description

Support for GTP overload control from gateways to MME.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

gwSelectionMode - Gateway selection mode

Syntax

--gParmName gwSelectionMode --gParmValue <string>

Description

Select PGW topologically close to selected SGW or topologically close SGW and PGW pair.

List

- GW Selection Mode 1 - MME will select a PGW topologically close to the selected SGW.
- GW Selection Mode 2 - MME will select topologically close SGW and PGW pair.

Default

GW Selection Mode 1

Name

gwSelectionModeForSgsn - SGW PGW Selection Enhancement for SGSN

Syntax

--gParmName gwSelectionModeForSgsn --gParmValue <string>

Description

This parameter controls the S/P -GW pair selection for S4-SGSN. When the parameter is GW Selection Mode 1 then the SGW priorities are taken into consideration for selecting the appropriate S/P -GW pair. When the parameter is GW Selection Mode 2 then the PGW priorities are taken into consideration for selecting the appropriate S/P -GW pair.

List

- GW Selection Mode 1 - SGSN will use as a reference the S-GW priorities in order to select a S/P-GW pair
- GW Selection Mode 2 - SGSN will use as a reference the P-GW priorities in order to select a S/P-GW pair

Default

GW Selection Mode 1

Name

gxInterfacesNetworkAnsiSs7 - GX interfaces network ANSI SS7

Syntax

--gParmName gxInterfacesNetworkAnsiSs7 --gParmValue <string>

Description

This flag configures the SS7 protocol type 'ANSI' or 'ITU' for the m3ua-network(s) of the gx (i.e. gr, gf, ge, gd) interfaces. when multiple m3ua-networks are defined for the gx interfaces, the protocol type must be the same across all the m3ua-networks.

List

- ANSI - Specify SS7 variant of american national standards institute.
- ITU - Specify SS7 variant of international telecommunication union - telecommunications.

Default

ITU

Name

highCapacityIHspa - High Capacity iHSPA

Syntax

--gParmName highCapacityIHspa --gParmValue <string>

Description

When set to 'No' it is possible to create a total sum of 512 iHSPAs and RNCs When set to 'Yes' it is possible to create a total sum of 8192 iHSPAs and RNCs (of which only up to 512 may be RNCs)

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

homeDomainName - Home domain name

Syntax

--gParmName homeDomainName --gParmValue <string>

Description

Home domain name.

Type / Range

Digit string of length 0 to 62 with the allowed pattern of [0-9a-zA-Z\.\-]*

Name

homogeneousIMSSupportAVP - HomogeneousIMSSupportAVP

Syntax

--gParmName homogeneousIMSSupportAVP --gParmValue <string>

Description

Whether or not WMM includes homogeneous IMS support AVP in ULR and NOR.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

homogeneousSupportNblotTas - Homogeneous support NB-IoT TAs

Syntax

```
--gParmName homogeneousSupportNbIotTas --gParmValue <string>
```

Description

Controls whether the MME includes Tracking Areas supporting NB-IoT Radio Access Technology when evaluating Homogeneous support of Voice over PS sessions towards the HSS. Yes - NB-IoT TAs are included when evaluating Homogeneous support of IMS Voice over PS sessions No - NB-IoT TAs are not included when evaluating Homogeneous support of IMS Voice over PS sessions

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

homogeneousSupportTais - Homogeneous Support TAs

Syntax

```
--gParmName homogeneousSupportTais --gParmValue <string>
```

Description

Defines whether MME includes IMS Voice over PS of the serving PLMN TAs or all TAs MME globally (serving PLMN and shared PLMN TAs) when evaluating homogeneous IMS voice over PS support towards the HSS.

List

- HomogeneousSupportAllTais - Homogeneous Support All TAs - MME includes IMS Voice over PS support of all TAs MME globally (TAs of the serving PLMN as well as TAs of shared PLMNs) when evaluating Homogeneous voice over PS support.
- HomogeneousSupportServingPlmn - Homogeneous Support Serving PLMN - MME includes IMS Voice over PS support of TAs of the serving PLMN when evaluating Homogeneous voice over PS support.

Default

HomogeneousSupportServingPlmn

Name

homogeneousVopsSrvccCapa - Homogeneous Voice over PS SRVCC Capability

Syntax

```
--gParmName homogeneousVopsSrvccCapa --gParmValue <string>
```

Description

MME provides Homogeneous-Support-of-IMS-Voice-Over-PS-Sessions towards HSS according to specific requirements: TAs support for IMS voice over PS (separate activation parameter controlling whether to include all TAIs MME globally, or all TAIs on the serving PLMN), voice over PS support for the UE (on svcAgreementProfile) and checking or not checking the UE SRVCC capability (separate parameter on svcAgreementProfile).

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

hssInitiatedCallTrace - HSS initiated call trace

Syntax

```
--gParmName hssInitiatedCallTrace --gParmValue <string>
```

Description

Support HSS initiated call trace or not.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

hssNorReduction - HSS NOR reduction

Syntax

```
--gParmName hssNorReduction --gParmValue <string>
```

Description

When set to yes: if MME selects the same PGW as in ULA/IDR from HSS, MME does not send NOR to HSS. when set to no: MME sends NOR to HSS for this case.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

hssRestorationMode - HSS Restoration Mode

Syntax

--gParmName hssRestorationMode --gParmValue <string>

Description

Defines how HSS restoration is performed after HSS unavailability. The HSS restoration should only be enabled i.e. set to values other than 'HssRestorationNone' with MME only configurations.

List

- HssRestorationAutomatic - HSS Restoration Automatic - HSS restoration is performed automatically when MME detects available HSS link.
- HssRestorationManual - HSS Restoration Manual - HSS restoration is performed manually by the operator.
- HssRestorationNone - HSS Restoration None - HSS restoration is not performed.

Default

HssRestorationNone

Name

hssUlrReduction - HSS ULR reduction

Syntax

--gParmName hssUlrReduction --gParmValue <string>

Description

When set to yes: if UE attaches with valid local GUTI and MME has IMSI and subscription data, MME does not send ULR to HSS. when set to no: MME sends ULR to HSS for this case.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

hssUnavailabilityContinueN26 - HSS Unavailability Continue N26

Syntax

--gParmName hssUnavailabilityContinueN26 --gParmValue <string>

Description

Defines whether the MME continues the procedure when HSS is unavailable and the UE moves from 5GS via N26 to the MME, and the MME has available security context or

mapped security context and thus the MME is able to continue serving the UE without new authentication on EPS.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

hssUnavailFailedPercentTriggerBlacklisting - HSS unavailable failed percentage triggering blacklisting

Syntax

```
--gParmName hssUnavailFailedPercentTriggerBlacklisting --gParmValue <string>
```

Description

Percentage of failed MME attempted dialogs triggering blacklisting.

Type / Range

Integer in the range of 1 to 100

Default

50

Name

ignoreSGSapMMInfo - IgnoreSGSapMMInfo

Syntax

```
--gParmName ignoreSGSapMMInfo --gParmValue <string>
```

Description

IgnoreSGSapMMInfo.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

imeiOperatorPolicyAccessRestriction - IMEI Operator Policy Access Restriction

Syntax

```
--gParmName imeiOperatorPolicyAccessRestriction --gParmValue <string>
```

Description

Enables IMEI based operator policy for access restrictions based on provisioned criteria.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

imeiOperatorPolicyAuthenticationDisabled - imeiTac Based Authentication

Syntax

```
--gParmName imeiOperatorPolicyAuthenticationDisabled --gParmValue  
<string>
```

Description

authentication to be enabled/disabled for a particular TAC, whereas entire network it might be running. Similarly based on TAC, IMSI Combination.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

imeiOperatorPolicyEirCheckDisabled - IMEI Operator Policy EIR Check Disabled

Syntax

```
--gParmName imeiOperatorPolicyEirCheckDisabled --gParmValue <string>
```

Description

Enables IMEI based operator policy to disable EIR check of UEs matching the provisioned criteria.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

immediateLrrDuringMtLr - Immediate LRR During MT-LR

Syntax

```
--gParmName immediateLrrDuringMtLr --gParmValue <string>
```

Description

If the MME needs to send a LRR related to an emergency call event and there is an ongoing MT-LR procedure, this parameter controls whether the LRR is to be sent immediately when the event occurs or later after the MT-LR procedure completes. When enabled, the LRR is sent immediately.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

imsEmergencySupportIndication - IMS emergency support indication

Syntax

```
--gParmName imsEmergencySupportIndication --gParmValue <string>
```

Description

Enable/Disable IMS voice enabled check for IMS emergency support indication.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

imsilmeiLookupFail - IMSI IMEI Lookup Fail

Syntax

```
--gParmName imsiImeiLookupFail --gParmValue <string>
```

Description

Parameter imsilmeiLookupFail controls APN Override feature with the values 'UseDefault and Reject. By default this parameter shall be disabled.

List

- Reject - Reject value means reject the attach/PDN connection regardless of any provisioned APN in the IMSI/IMEI table
- UseDefault - Use Default - Use Default value means if there is a wildcard APN provided by the HSS then it will be used with the UE-requested APN if applicable, else use the default APN in the HSS if it exists as correction APN.

Default

UseDefault

Name

imsiPagingInAllMmeServingAreas - IMSI paging in all MME serving areas

Syntax

--gParmName imsiPagingInAllMmeServingAreas --gParmValue <string>

Description

Allow IMSI paging in all MME serving areas?.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

inclEmerNumListImsAllowed - Include Emergency Number List When IMS Allowed

Syntax

--gParmName inclEmerNumListImsAllowed --gParmValue <string>

Description

Parameter controls whether MME includes the UE the Emergency Number List for a UE that has EPS-Network-Feature-Support with IMS-voice-over-PS allowed, but EPS-Network-Feature-Support emergency call is not allowed on the MME. When Emergency Number list is included it uses the Serving PLMN Emergency Profile information for the Emergency Number List.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

inclExt2InEPSQoSIEs - Include extended 2 in EPS QoS IEs

Syntax

--gParmName inclExt2InEPSQoSIEs --gParmValue <string>

Description

Enable/Disable inclusion of extended-2 octets in EPS QoS IEs.

List

- No - Boolean value of no.

- Yes - Boolean value of yes.

Default

No

Name

inclExt2InNegotiatedQoSIEs - Include extended 2 in negotiated QoS IEs

Syntax

```
--gParmName inclExt2InNegotiatedQoSIEs --gParmValue <string>
```

Description

Enable/Disable inclusion of extended-2-octets in R99 negotiated QoS IE.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

includeActiveApnAvpInUlrAfterRsr - Include active-apn AVP in ULR after RSR

Syntax

```
--gParmName includeActiveApnAvpInUlrAfterRsr --gParmValue <string>
```

Description

Include active-apn AVP with PGW information in ULR after RSR.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

includeEcgidPlaInFailure - Include ECGID in PLA in failure

Syntax

```
--gParmName includeEcgidPlaInFailure --gParmValue <string>
```

Description

Supports inclusion of ECGID in PSLR even if MME as not able to get UE location.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

includeExtUeIndexInPagingMsg - Include Extended UE Identity Index in PAGING message

Syntax

--gParmName includeExtUeIndexInPagingMsg --gParmValue <string>

Description

If this parameter is set to 'Yes' then the MME shall include the Extended UE Identity Index IE in PAGING messages.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

includeHrIAAlways - Include Handover Restriction List Always

Syntax

--gParmName includeHrIAAlways --gParmValue <string>

Description

When enabled, MME shall include the Handover Restriction List IE in S1AP messages irrespective of EPLMN, forbidden TA or forbidden LA provisioning. If EPLMN and forbidden TA and LA are not provisioned then MME shall only include the serving PLMN IE in the HRL IE.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

includeMapped3gppQciToUe - Include mapped 3GPP QCI value to UE

Syntax

--gParmName includeMapped3gppQciToUe --gParmValue <string>

Description

When set to yes, the 3GPP QCI value mapped from the operator defined QCI value will be included in the activate default bearer context request, activate dedicated bearer context request, and modify bearer context request messages sent to the UE. this applies to both default and dedicated bearers.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

includeMappedGummei - Include mapped GUMMEI

Syntax

--gParmName includeMappedGummei --gParmValue <string>

Description

This command is only applicable for WMM deployed as a combo. this command enables/disables inclusion of the mapped GUMMEI in S1-AP S1 SETUP RESPONSE message and S1-MME CONFIGURATION UPDATE message. the inclusion of the mapped GUMMEI facilitates eNB to select the same physical node supporting the combo MME/SGSN when UE is moving between 2G/3G and LTE area served by the same combo MME/SGSN.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

includeNblotSmsServiceStatus - Include CIOT SMS service status

Syntax

--gParmName includeNbIotSmsServiceStatus --gParmValue <string>

Description

Include SMS services status IE in attach accept and TAU accept message if MME fails to set up SMS for CIOT UEs and MME only accepts EPS attach only.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

includeNeighborListInTailList - Include neighbor list in TAI list

Syntax

```
--gParmName includeNeighborListInTailList --gParmValue <string>
```

Description

Indicates whether the MME includes (Yes) the provisioned set of neighbor TAIs in the list of TAIs that is sent to the UE when the UE registers at the MME (Attach or TAU). If this parameter is set to 'No', the MME only includes the last seen TAI in the list of TAIs sent to the UE.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

includeRanapTli - Include RANAP TLI

Syntax

```
--gParmName includeRanapTli --gParmValue <string>
```

Description

Enables RANAP TLI in RAB assignment request to RNC triggered by GGSN PDP context modification.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

includeServingNetworkS11ie - Include serving network S11 IE

Syntax

```
--gParmName includeServingNetworkS11ie --gParmValue <string>
```

Description

BSM to control whether the serving network IE is included in the modify bearer request when the PLMN has changed.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

includeServingPlmnRateControl - support Serving Plmn Rate Control function

Syntax

--gParmName includeServingPlmnRateControl --gParmValue <string>

Description

With this parameter enable, Serving PLMN Rate Control will be supported

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

includeT3448Timer - include T3448 timer in overload situation

Syntax

--gParmName includeT3448Timer --gParmValue <string>

Description

support T3448 timer for overload control with limited data transfer. T3448 used for limited UE transfer data over CIOT Control Plane Optimization during timer run. With this parameter, T3448 is allowed to be used in CMM.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

includeUliCrSessModBrr - ULI Inclusion in Create Session And Mod Brr Msg

Syntax

--gParmName includeUliCrSessModBrr --gParmValue <string>

Description

Parameter controls inclusion of 'ULI for SGW' IE in Create Session and Modify Bearer Request message that is sent to SGW. By default, the flag shall be set to Yes. When flag is set to 'No', MME will not include the IE in Create Session and Modify Bearer Request message that is sent to SGW.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

includeUliForSgsn - Include ULI for SGSN

Syntax

```
--gParmName includeUliForSgsn --gParmValue <string>
```

Description

This flag is used to enable/disable inclusion of the user location information IE in associated GTP messages over the Gn interface to the GGSN. no - indicates that the ULI IE is not included. yes (default) - indicates that the ULI IE is included.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

inhibitPeerSrsResponses - Inhibit Peer SRS Responses

Syntax

```
--gParmName inhibitPeerSrsResponses --gParmValue <string>
```

Description

This parameter provides the ability on the NR10 client side to inhibit NR10 responses from being sent by the peer NR10 server. When set to true, responses to update requests and delete requests are inhibited. This parameter applies to both AMF and MME NR10 functionality.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

initialAttachIndicationInUlr - Initial attach indication in ULR

Syntax

```
--gParmName initialAttachIndicationInUlr --gParmValue <string>
```

Description

Initial attach indication flag.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

interPlmnTauAttach - Inter PLMN TAU attach

Syntax

```
--gParmName interPlmnTauAttach --gParmValue <string>
```

Description

If it is set to yes then MME shall check per PLMN provisioning whether inter-plmn S10/S3/Gn message can be sent. by default, the feature is disabled.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

ipAddrInclusion - IP Address Inclusion

Syntax

```
--gParmName ipAddrInclusion --gParmValue <string>
```

Description

Parameter controls inclusion of same IP address type as the destination address used in the IP header in the Create Session Request message. By default, the flag shall be disabled.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

ipv4v6sgsn - IPv4v6 SGSN

Syntax

```
--gParmName ipv4v6sgsn --gParmValue <string>
```

Description

A value of yes indicates release 8 SGSN and above. a value of no indicates pre-release 8 SGSN.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

ipv6SelectionAllowedOnGn - IPv6 Selection Allowed on Gn

Syntax

```
--gParmName ipv6SelectionAllowedOnGn --gParmValue <string>
```

Description

Use this parameter to allow IPV6 support on Gn-C.

List

- OFF - OFF
- ON - ON

Default

OFF

Name

iratGnMobilityQoSParametersUpdate - IRAT Gn mobility QoS parameters update

Syntax

```
--gParmName iratGnMobilityQoSParametersUpdate --gParmValue <string>
```

Description

Indicates whether MME sends modifybearercmd to SGW at the end of TAU when QoS parameters in S6ad ULA are different than those in Gn contextresponse.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

iratMobilityWithOtherPlmnAllowed - IRAT mobility with other PLMN allowed

Syntax

```
--gParmName iratMobilityWithOtherPlmnAllowed --gParmValue <string>
```


Description

IRAT mobility with other PLMN allowed.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

iratS3mobilityQosParametersUpdate - IRAT S3 mobility QoS parameters update

Syntax

--gParmName iratS3mobilityQosParametersUpdate --gParmValue <string>

Description

Indicates whether MME sends modifybearercmd to SGW at the end of TAU when QoS parameters in S6ad ULA are different than those in S3 contextresponse.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

iuAlgsSelection - new algs support in lu interface

Syntax

--gParmName iuAlgsSelection --gParmValue <string>

Description

Customer is claiming that UIA1 and UEA1 is compromised already, thus, supporting UIA2 and UEA2 in 3G is needed. The parameter will control to use a configured preference list of algs or not. If the parameter is setting to Yes, then a preference list of algs needed to be configured by customer, SGSN will based on both UE capability and configured preference list to inform RNC the selected algs. If the parameter is set to No, then existing function is unchanged i.e. continue with UIA1/UEA1.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

keepGbrWhenEnbFails - Keep GBR when eNB fails

Syntax

```
--gParmName keepGbrWhenEnbFails --gParmValue <string>
```

Description

Indicates whether (Yes), or not (No), all the GBR bearers associated with an eNB are deactivated when the S1-MME link to that eNB fails.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

keepGbrWhenPagingFails - Keep GBR when paging fails

Syntax

```
--gParmName keepGbrWhenPagingFails --gParmValue <string>
```

Description

Indicates whether (Yes), or not (No), the GBR bearers of the UE are deactivated if paging fails to locate the UE.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

keepSameSgw - Keep the Same SGW

Syntax

```
--gParmName keepSameSgw --gParmValue <string>
```

Description

Keep the Same SGW during IRAT Mobility from 4G (in case of SGSN)

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

keepUpCioTUEs1Bearer - Keep UP CIOT UE S1-U bearer

Syntax

--gParmName keepUpCioTUEs1Bearer --gParmValue <string>

Description

Enable keeping of a S1-U bearer of a UE supporting UP CloT moves into a TAI that does not support UP CloT optimization and S1-U data transfer.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

lawfulInterceptMtLrNoEsmlc - lawful Intercept MT_LR no E-SMLC

Syntax

--gParmName lawfulInterceptMtLrNoEsmlc --gParmValue <string>

Description

This is for MT_LR client type lawful Intercept service do not contact E-SMLC for UE location information. If yes, when MME receives mobile terminated location request with client type=lawfulInterceptService, and activateLcsClientTypeLawfullInterceptServices is set, the MME contacts the eNB for UE current E-CGI information. This information is sent back to the GMLC. The MME does not contact the E-SMLC to obtain any UE location information. If no, nothing changes and keep existing function (i.e. MME contacts E-SMLC to obtain UE location)

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

lcsapNumberOfResetsWithoutAckForAlarm - LCSAP number of resets without ack for alarm

Syntax

--gParmName lcsapNumberOfResetsWithoutAckForAlarm --gParmValue
<string>

Description

LCSAP number of resets without ack for alarm.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

limitS1apConnEst - Limit S1AP conn est

Syntax

```
--gParmName limitS1apConnEst --gParmValue <string>
```

Description

The parameter is used to limit the sending of the S1AP CONNECTION ESTABLISHMENT INDICATION message to establish UE-associated logical S1-connection if the logical connection is already established. the UE-associated logical S1-connection would be established by the MME if the MME sends S1AP DOWNLINK NAS TRANSPORT before sending the S1AP CONNECTION ESTABLISHMENT INDICATION message.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

linkAlarmCustomize - Link alarm customize

Syntax

```
--gParmName linkAlarmCustomize --gParmValue <string>
```

Description

If the global parameter is set to 'Yes', customized severity levels in the linkalarmcfg is used to set the alarm severity for links and interfaces. if the global parameter is set to 'No', customization of link alarm severity is not supported.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

localIimeiTacRejectionList - Local IMEI-TAC Rejection List

Syntax

```
--gParmName localIimeiTacRejectionList --gParmValue <string>
```

Description

Enables the functionality to provision IMEI-TAC rejection list locally in the MME. When a UE whose IMEI-TAC is included in the list attempts to attach in the MME, the MME rejects the attempt with a provisioned EMM cause code.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

locallyProvisionedSgwPgw - Locally provisioned SGW or PGW

Syntax

```
--gParmName locallyProvisionedSgwPgw --gParmValue <string>
```

Description

Test switch to provision sgw idx in range from 241 to 256.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

locallyProvisionedSgwPgwTopon - Locally provisioned S/PGW Topological Selection

Syntax

```
--gParmName locallyProvisionedSgwPgwTopon --gParmValue <string>
```

Description

Enables topological S/PGW selection for locally provisioned S/PGW.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

locationReportingForUeWithLppsupportOnly - Location reporting for UE with LPP support only

Syntax

```
--gParmName locationReportingForUeWithLppsSupportOnly --gParmValue  
<string>
```

Description

This flag indicates whether LCS will be supported for UEs with or without LPP capability. when flag is enabled only UEs that indicated LPP support can perform LCS procedure. this flag is used in conjunction with the other LCS feature activation fields. those fields apply as well. UE indicates support of LPP capability in the UE network capability IE of the attach request and/or TAU request message.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

logEcgiReceivedInWriteReplaceWarningMessage - Log ECGI Received In Write Replace
Warning Message

Syntax

```
--gParmName logEcgiReceivedInWriteReplaceWarningMessage --gParmValue  
<string>
```

Description

Parameter controls feature activation GPARAM flag. By default, flag shall be disabled. When enabled, MME shall log eCGI that is received in Write-Replace-Warning-Response Message from eNB.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

longestGtDigitsMatch - Longest GT Digits Match

Syntax

```
--gParmName longestGtDigitsMatch --gParmValue <string>
```

Description

This parameter allows or denies whether the SCCP is able to route on the longest GT digits match

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

lowPriorityAccess - Low priority access

Syntax

```
--gParmName lowPriorityAccess --gParmValue <string>
```

Description

Activate the low access priority capabilities on MME.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

lowPriorityTrafficDdnThrottling - Low priority traffic DDN throttling

Syntax

```
--gParmName lowPriorityTrafficDdnThrottling --gParmValue <string>
```

Description

Throttling of DDN requests for the low priority traffic.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

lraUli - LRA ULI

Syntax

```
--gParmName lraUli --gParmValue <string>
```

Description

Enable/Disable LRA-ULI.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

ltePgwlIsolation - LTE PGW isolation

Syntax

```
--gParmName ltePgwlIsolation --gParmValue <string>
```

Description

Setting to yes enables PGW isolation feature on MME. setting to no disables PGW isolation feature on MME.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

lteSgwIsolation - LTE SGW isolation

Syntax

```
--gParmName lteSgwIsolation --gParmValue <string>
```

Description

Setting to yes enables SGW isolation feature on MME. setting to no disables SGW isolation feature on MME.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

maintainErab - Maintain ERAB

Syntax

```
--gParmName maintainErab --gParmValue <string>
```

Description

When set to yes, MME will maintain E-RAB that are not successfully switched in the core network during the E-RAB modification indication procedure. when set to no, MME will

release/deletee-rab that are not successfully switched in the core network during the E-RAB modification indication procedure.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

maskedImeIsV - Masked IMEISV

Syntax

```
--gParmName maskedImeIsV --gParmValue <string>
```

Description

Masked IMEISV in S1AP messages.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

maxCid - Max CID

Syntax

```
--gParmName maxCid --gParmValue <string>
```

Description

Max CID.

Type / Range

Integer in the range of 1 to 16,383

Default

15

Name

maxEcgiAgeEmerMtLrNoEsmlc - Maximum time interval in seconds for requesting eCGI for emergency MT-LR

Syntax

```
--gParmName maxEcgiAgeEmerMtLrNoEsmlc --gParmValue <string>
```

Description

If the parameter is set to 0 with the global feature activation parameter `emermtlrnoesmlc` is yes, MME page ECM-IDLE UE and send eCGI information or send location-reporting control information for ECM. if the parameter is set to a non-zero value, when MME gets MT-LR request, MME will check the difference between MME stored eCGI timestamp of the UE and time received MT-LR. if the time difference is less than provisioned value, then the MME reports stored eCGI information to GMLC. if time difference is greater than the provisioned value, then MME will request current eCGI either via paging or location report control depends UE status.

Type / Range

Integer in the range of 0 to 180

Default

0

Name

`maxEcgiAgeLawfulInterceptMtLrNoEsmc` - Max ECGI age lawful intercept MT_LR_no E-SMLC

Syntax

```
--gParmName maxEcgiAgeLawfulInterceptMtLrNoEsmc --gParmValue <string>
```

Description

Maximum time interval in seconds for requesting eCGI for lawful intercept MT-LR. If the parameter is set to 0 with the global feature activation parameter `lawfulInterceptMtLrNoEsmc` is yes, MME page ECM-IDLE UE and send eCGI information or send location-reporting control information for ECM_connected UE. if the parameter is set to a non-zero value, when MME gets MT-LR request, MME will check the difference between MME stored eCGI timestamp of the UE and time received MT-LR. if the time difference is less than provisioned value, then the MME reports stored eCGI information to GMLC. if time difference is greater than the provisioned value (or equal provisioned value), then MME will request current eCGI either via paging or location report control depends UE status

Type / Range

Integer in the range of 0 to 180 in units of Second

Default

0

Name

`maxEcgiAgePlmnOperMtLrNoEsmc` - Max ECGI age PLMN operator MT_LR no E-SMLC

Syntax

```
--gParmName maxEcgiAgePlmnOperMtLrNoEsmc --gParmValue <string>
```

Description

Maximum time interval in seconds for requesting eCGI for PLMN Operator MT-LR. If the parameter is set to 0 with the global feature activation parameter `plmnOperatorMtLrNoEsmc` is yes, MME page ECM-IDLE UE and send eCGI information or

send location-reporting control information for ECM_connected UE. if the parameter is set to a non-zero value, when MME gets MT-LR request, MME will check the difference between MME stored eCGI timestamp of the UE and time received MT-LR. if the time difference is less than provisioned value, then the MME reports stored eCGI information to GMLC. if time difference is greater than the provisioned value ((or equal provisioned value), then MME will request current eCGI either via paging or location report control depends UE status

Type / Range

Integer in the range of 0 to 180 in units of Second

Default

0

Name

maxEcgiAgeValueAddedMtLrNoEsmlc - Max ECgi age value added MT_LR no E-SMLC

Syntax

--gParmName maxEcgiAgeValueAddedMtLrNoEsmlc --gParmValue <string>

Description

Maximum time interval in seconds for requesting eCGI for value Added service MT-LR. If the parameter is set to 0 with the global feature activation parameter valueAddedMtLrNoEsmlc is yes, MME page ECM-IDLE UE and send eCGI information or send location-reporting control information for ECM_connected UE. if the parameter is set to a non-zero value, when MME gets MT-LR request, MME will check the difference between MME stored eCGI timestamp of the UE and time received MT-LR. if the time difference is less than provisioned value, then the MME reports stored eCGI information to GMLC. if time difference is greater than the provisioned value (or equal provisioned value), then MME will request current eCGI either via paging or location report control depends UE status

Type / Range

Integer in the range of 0 to 180 in units of Second

Default

0

Name

maxEnbsPagedDuringMinorOverload - Maximum eNBs paged during minor overload

Syntax

--gParmName maxEnbsPagedDuringMinorOverload --gParmValue <string>

Description

The parameter specifies the maximum number of eNBs paged for a single page attempt when a minor overload condition is present.

Type / Range

Used for maximum number of eNBs paged for a single page attempt.

Default

600

Name

maxMmeUeRadioCapLength - Maximum MME Ue Radio Capability Length

Syntax

--gParmName maxMmeUeRadioCapLength --gParmValue <string>

Description

Maximum Ue Radio Capability Length Supported on the MME. Default is 8192. Setting this value to 24576 or 49152 enables storage of Radio capability length greater than 8192 bytes in the RCIS if RCIS is available

Type / Range

Integer in the range of 8,192 or 24,576 or 49,152

Default

8192

Name

maxNumberWarningAreasEcgi - Max number warning areas ecgi

Syntax

--gParmName maxNumberWarningAreasEcgi --gParmValue <string>

Description

Maximum number of warning areas (ecgi) that can be includes in S1AP WRITE-REPLACE WARNING REQUEST or S1AP KIL-REQUEST messages sent to eNB.

Type / Range

Integer in the range of 1 to 9,000

Default

4000

Name

maxTaRelativeCapacity - Max TA relative capacity

Syntax

--gParmName maxTaRelativeCapacity --gParmValue <string>

Description

Used to select to send to the eNB the maximum or minimum MME relative capacity of provisioned MME relative capacities of TAs supported by an eNB to the eNB in S1-AP SETUP RESPONSE message and in MME CONFIGURATION UPDATE message.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

mdt - Minimization of drive tests

Syntax

--gParmName mdt --gParmValue <string>

Description

Activation/deactivation minimization of drive tests.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

measInfoldUsage - measInfold use in PM

Syntax

--gParmName measInfoIdUsage --gParmValue <string>

Description

This parameter describes the type of measInfold parameter to add to PM files.

List

- none - No measInfold - The parameter is not included in PM file.
- serviceAndCategory - Service and Category - The parameter is set to Service and Category in the PM file.

Default

none

Name

minFqdnLabelMatching - minFqdnLabelMatching

Syntax

--gParmName minFqdnLabelMatching --gParmValue <string>

Description

Enable applying minimum matching fqdn labels between SGW and PGW fqdn(s) during SPGW pair selection. When gParm is enabled, the MME use configuration from fqdnLabelMatchProfile and fqdnLabelMatchUut to further resolve min fqdn label matching per Service Combination and subscribed UE Usage Type

List

- No - Boolean value of no.

- Yes - Boolean value of yes.

Default

No

Name

mmeAdditionalMsisdn - MME Additional MSISDN

Syntax

```
--gParmName mmeAdditionalMsisdn --gParmValue <string>
```

Description

GPARM parameter mmeAdditionalMsisdn controls the support for Additional MSISDN by MME. By default the parameter shall be no. When GPARM parameter mmeAdditionalMsisdn is yes, the MME shall inform HSS that it supports the A-MSISDN feature by setting the corresponding feature bit of the Feature List AVP in the Update Location Request, Insert Subscriber Data Answer and Delete Subscriber Data Answer commands.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

mmeApnNiExtension - MME APN NI extension

Syntax

```
--gParmName mmeApnNiExtension --gParmValue <string>
```

Description

No (default) - for MME, disable the extension of the APN NI with configurable length of MSISDN digits. yes - for MME, enable the extension of the APN NI with configurable length of MSISDN digits.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

mmeOverloadMmRejectionNasCc - MME overload MM rejection NAS CC

Syntax

```
--gParmName mmeOverloadMmRejectionNasCc --gParmValue <string>
```

Description

EMM NAS cause code to be sent in the mobility management reject messages (Attach reject, TAU reject and service reject) due to any MME overload or throttling of MM procedures. valid values are 2-3,5-26,35,39-40,42,95-101,111 per annex a of TS 24.301.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

22

Name

mmeOverloadPlaResult - MME overload PLA result

Syntax

--gParmName mmeOverloadPlaResult --gParmValue <string>

Description

Select diameter result to be sent in Slg provide subscriber location answer when MME is in major or critical overload.

List

- 3004 - Too Busy - 3004 - DIAMETER_TOO_BUSY.
- 4002 - Out of Space - 4002 - DIAMETER_OUT_OF_SPACE.
- 4224 - Positioning Denied - 4224 - DIAMETER_ERROR_POSITIONING_DENIED.
- 4225 - Positioning Failed - 4225 - DIAMETER_ERROR_POSITIONING_FAILED.
- 5006 - Resources Exceeded - 5006 - DIAMETER_RESOURCES_EXCEEDED.

Default

4224 - Positioning Denied

Name

mmePgwQueryEmptyflagRr - MME PGW QUERY EMPTYFLAG RR

Syntax

--gParmName mmePgwQueryEmptyflagRr --gParmValue <string>

Description

No (default) - disable the MME to support NAPTR RR with " (empty) flag for PGW selection.
yes - enable the MME to support NAPTR RR with " (empty) flag for PGW selection.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

mmeProseAuthorizationSupport - Public safety proximity services

Syntax

--gParmName mmeProseAuthorizationSupport --gParmValue <string>

Description

Public safety proximity services.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

mmeSgwQueryEmptyflagRr - MME SGW Query Empty Flag

Syntax

--gParmName mmeSgwQueryEmptyflagRr --gParmValue <string>

Description

This parameter is used to enable MME handling of reception of S-NAPTR RR with empty flag. By default, the capability disabled. If the parameter is set to 'Yes' then the capability is enabled.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

mocnAccurateLoc - P-GW/GGSN enhancement with 3G MOCN setup for location update.

Syntax

--gParmName mocnAccurateLoc --gParmValue <string>

Description

In 3G MOCN setup, during SRNS relocation procedure or EUTRAN-to-UTRAN PS handover, it is possible that the PLMN ID received in RAU after relocation is not identical to the PLMN ID that the target SGSN learns with the GTP:FORWARD_RELOCATION_REQ message. Activate this parameter to force a GTP:UPDATE_PDP_CONTEXT message that updates PGW/GGSN with the correct PLMN ID. SGSN Multi Operator Core Network (3G MOCN) needs to be activated for this functionality.

List

- OFF - OFF
- ON - ON

Default

OFF

Name

monitoringConfigFailureSendNor - Monitoring Config Failure Send Nor

Syntax

```
--gParmName monitoringConfigFailureSendNor --gParmValue <string>
```

Description

Specifies whether the MME is to send a Notification Report to the HSS if a monitoring event cannot be configured at the MME (3GPP R14 functionality).

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

monitoringEventsDeletionByUla - Monitoring Events Deletion by ULA

Syntax

```
--gParmName monitoringEventsDeletionByUla --gParmValue <string>
```

Description

Parameter controls detection and deletion of monitoring event configuration AVPs received in ULA. When parameter is enabled, upon receiving one or more Monitoring-Event-Configuration AVP(s) in the ULA, the MME shall start the detection of the Monitoring events indicated in those AVP(s), if not already started, and shall stop the detection and delete the previous monitoring events (if any) which are not indicated in those AVP(s). If there is a failure when starting the detection (e.g. maximum resources exceeded), the MME shall not store the failed configuration(s) and shall send a notification of those events whose configuration have failed. If the Subscription-Data AVP is received in the ULA but it does not contain any Monitoring-Event-Configuration AVP(s), the MME shall stop the detection and delete all stored monitoring event configurations (if any).

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

monteCommFailRanUeInact - monteCommFailRanUeInact

Syntax

--gParmName monteCommFailRanUeInact --gParmValue <string>

Description

Specifies whether the MME is to send a T6a:RIR Communication Failure Event Report if the RAN requests UE S1 release due to UE Inactivity.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

mscLoadBalance - MSC load balance

Syntax

--gParmName mscLoadBalance --gParmValue <string>

Description

Configure MSC load balance algorithm.

List

- IMSI Hash - Assign UEs based on hash derived from the UE IMSI.
- Round Robin - Round-Robin - Use round robin assignment of UEs.

Default

Round Robin

Name

mscSelMultipleCsCoreSuppt - MSC selection of Multiple CS Core network

Syntax

--gParmName mscSelMultipleCsCoreSuppt --gParmValue <string>

Description

GPARAM will control the feature flag. By default the global parameter shall be false/disabled.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

mScVlrReselection - MSC VLR reselection

Syntax

--gParmName mScVlrReselection --gParmValue <string>

Description

Enable/disable alternate MSC selection if location update times out or MSC rejects the request with reject cause 'congestion'.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

mSisdNHomeGwWithCc - MSISDN Based Home GW selection with CC

Syntax

--gParmName mSisdNHomeGwWithCc --gParmValue <string>

Description

MSISDN based home GW selection with CC(charging characteristic)

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

mUltipleIwsSupport - Multiple IWS support

Syntax

--gParmName mUltipleIwsSupport --gParmValue <string>

Description

CLI command to control the support for multiple IWSs between ALU MME and 3rd party IWS.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

multiPra - Multiple PRAs

Syntax

```
--gParmName multiPra --gParmValue <string>
```

Description

Enables support for -multiple Presence Reporting Areas (PRAs) per PDN connection in the MME, -multiple IP address segments per PRA identifier for Dual connectivity segment based reporting in the MME.

Restrictions

gParm praRel145 required to be set to 'Yes'.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

multiPraOperationEnbsEcgis - Multiple PRA Operation eNBs and ECGIs

Syntax

```
--gParmName multiPraOperationEnbsEcgis --gParmValue <string>
```

Description

Enables the functionality to enter multiple eNBs and ECGIs with a single command addressing a Presence Reporting Area.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

multiPraOperationTacsEnabled - Multiple PRA Operation with TACs Enabled

Syntax

```
--gParmName multiPraOperationTacsEnabled --gParmValue <string>
```

Description

Enables the functionality to enter multiple Tracking Areas with a single command addressing a Presence Reporting Area.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

multiSimSupport - Multi SIM Support

Syntax

```
--gParmName multiSimSupport --gParmValue <string>
```

Description

GPARM controls support for MultiSim with YES or NO flag with a default value of NO. When the parameter is set to YES the MME will drop an MT-LR if it only contains an MSISDN for the location query. With the default value setting of NO, the CMM will allow an MT-LR that contains only an MSISDN as the identifier

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

n26DirectFwd - n26DirectFwd

Syntax

```
--gParmName n26DirectFwd --gParmValue <string>
```

Description

Indicates whether direct data forwarding is enabled or disabled on n26 handover.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

n26IndirectFwd - n26IndirectFwd

Syntax

```
--gParmName n26IndirectFwd --gParmValue <string>
```

Description

Indicates whether indirect data forwarding is enabled or disabled on n26 handover.

List

- disabled - Indirect forwarding is disabled for n26 handover
- enabled - Indirect forwarding is enabled for n26 handover

Default

enabled

Name

n26iwk - N26iwk - N26 interworking

Syntax

--gParmName n26iwk --gParmValue <string>

Description

Enables N26 interworking.

List

- OFF - OFF
- ON - ON

Default

OFF

Name

nasNonDeliveryIndicationEsmDataTransfer - Enable NAS Non Delivery Indication for ESM Data Transfer

Syntax

--gParmName nasNonDeliveryIndicationEsmDataTransfer --gParmValue <string>

Description

Enable NAS Non Delivery Indication for ESM Data Transfer

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

nasTokensToCompare - NAS tokens to compare

Syntax

--gParmName nasTokensToCompare --gParmValue <string>

Description

The number of NAS tokens to compare in determining whether a UE is authenticated in a handover scenario from UTRAN/GERAN to LTE.

Type / Range

Integer in the range of 2 to 5

Default

3

Name

nbIotEmmTimerIncr - CIOT EMM timer increment

Syntax

```
--gParmName nbIotEmmTimerIncr --gParmValue <string>
```

Description

This value is added to the timers in 10.2.2 of 24.301 for CIOT UE.

Type / Range

Integer in the range of 0 to 240

Default

0

Name

nbIotEsmTimerIncr - CIOT ESM timer increment

Syntax

```
--gParmName nbIotEsmTimerIncr --gParmValue <string>
```

Description

This value is added to the timers in 10.3.2 of 24.301 for CIOT UE.

Type / Range

Integer in the range of 0 to 180

Default

0

Name

nbIotOlAtMinorOverload - NB-IOT Overload Control At Minor Overload

Syntax

```
--gParmName nbIotOlAtMinorOverload --gParmValue <string>
```

Description

If the parameter is Yes, the MME will perform NB-IOT overload control when the MME starts experiencing minor overload. If the parameter is No, the MME would not take any NB-IOT overload control action until the MME starts experiencing major overload.

List

- No - Boolean value of no.

- Yes - Boolean value of yes.

Default

No

Name

nbIotPagingTimerIncr - CIOT paging timer increment

Syntax

```
--gParmName nbIotPagingTimerIncr --gParmValue <string>
```

Description

This value is added to paging response timer for CIOT UE.

Type / Range

Integer in the range of 0 to 240

Default

0

Name

nbIotRatTypeToHss - NBIOT RAT Type to HSS

Syntax

```
--gParmName nbIotRatTypeToHss --gParmValue <string>
```

Description

If the global parameter is set to 'Yes' then MME shall send NB-IoT RAT Type to HSS. If the global parameter is set to 'No', then MME shall send EUTRAN RAT Type to HSS.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

noClrRejForMissingClrFlags - No CLR Reject for Missing CLR-Flags

Syntax

```
--gParmName noClrRejForMissingClrFlags --gParmValue <string>
```

Description

Parameter controls rejecting CLR for Missing CLR-Flags AVP. By default the flag shall be disabled. When enabled MME will not reject CLR with Diameter Unable to Comply if CLR-Flags AVP is missing.

List

- No - Boolean value of no.

- Yes - Boolean value of yes.

Default

No

Name

nonDdnPagingTauCollision - Non DDN paging TAU collision

Syntax

--gParmName nonDdnPagingTauCollision --gParmValue <string>

Description

Enable MME to treat TAU request without active flag as a paging response and re-establish bearers if TAU is received while MME is paging a UE.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

nonEmergencyPdnConnAsNewPdnConnAtAttach - Non-emergency PDN connection as new PDN connection at attach

Syntax

--gParmName nonEmergencyPdnConnAsNewPdnConnAtAttach --gParmValue <string>

Description

MME support to treat non-emergency PDN connection as a new PDN connection at attach.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

nrBasedGwSelectionSgsn - NR Capability based GW selection

Syntax

--gParmName nrBasedGwSelectionSgsn --gParmValue <string>

Description

This parameter enables SGSN selecting GW using NR capability of the UE. When this parameter is enabled, SGSN selects NR capable GW if NR capability is supported at the

UE. If the parameter is setting to Yes, then a fixed tag=5gc will be added to pre-rel8 DNS query or rel8 DNS with service +nc-nr. If the parameter is set to No, then existing function is unchanged i.e. no changes for DNS query of 5G NSA UEs.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

nrCcDoubleS1PsAck - Normal release cause for double S1 connections and path switch request ack

Syntax

--gParmName nrCcDoubleS1PsAck --gParmValue <string>

Description

If this parameter is set to 'Yes' then the cause IE of S1AP UE CONTEXT RELEASE COMMAND is set to 'Normal release' for the following cases.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

nriLength - Length of Network Resource Identifier, LEN

Syntax

--gParmName nriLength --gParmValue <string>

Description

This parameter defines the length of Network Resource Identifier (NRI) in the CMM SGSN. Possible values are 0, 5, 6 and 7.

Type / Range

Integer in the range of 0 or 5 to 7

Default

0

Name

nrPriorityOverSmf - NC-NR priority over SMF

Syntax

```
--gParmName nrPriorityOverSmf --gParmValue <string>
```

Description

When this parameter is enabled, then the MME gives priority to NC-NR capable gw against combo PGW-C capable gw when multiple candidates, each serving only one of these service parameters are retrieved by DNS query

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

nullNri - Null network resource identifier

Syntax

```
--gParmName nullNri --gParmValue <string>
```

Description

This parameter defines the null network resource identifier, 255 ... not defined (default) 0 to 31 ... valid values.

Type / Range

Integer in the range of 0 to 31 or 255

Default

255

Name

numIpsPerPgwFqdn - Supported number of IPs per PGW FQDN

Syntax

```
--gParmName numIpsPerPgwFqdn --gParmValue <string>
```

Description

Sets the maximum number of IPv4 and IPv6 addresses per PGW FQDN. Only 2 values are supported (2 or 8). When set to '2' or '8', as many as 2 or 8 IPv4 and 2 or 8 IPv6 addresses are supported per PGW FQDN.

Type / Range

Indicates maximum number of IPv4s and IPv6s supported per PGW FQDN: 2 or 8.

Default

2

Name

obtainPgwFqdnEps5gsMob - obtainPgwFqdnEps5gsMob

Syntax

```
--gParmName obtainPgwFqdnEps5gsMob --gParmValue <string>
```

Description

Current parameter enables performing PGW FQDN lookup when PGW FQDN isn't available during EPS to 5GS mobility

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

ocCallTraceContinue - OC call trace continue

Syntax

```
--gParmName ocCallTraceContinue --gParmValue <string>
```

Description

Indicates whether or not call trace will continue (YES), or be suspended (NO), on a CPPS when the CPPS is in a major or critical overload condition.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

ocSendOverloadStartAndStopToEnbs - Overload control send overload start and stop messages to eNBs

Syntax

```
--gParmName ocSendOverloadStartAndStopToEnbs --gParmValue <string>
```

Description

A value of yes indicates the MME will send overload start and overload stop messages to eNBs as part of overload control. a value of no indicates the MME will not send overload start and overload stop messages.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

octetEncodingSupport - Octet Encoding Support

Syntax

```
--gParmName octetEncodingSupport --gParmValue <string>
```

Description

Parameter controls support for Octet Encoding for the Indication IE contained within messages sent over the S11 interface. By default the flag shall be disabled. When the flag is enabled, MME shall only encode the first three bytes of indication flags.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

odbBitsSupport - ODB Bits Support

Syntax

```
--gParmName odbBitsSupport --gParmValue <string>
```

Description

Parameter controls feature activation. By default the GPARM flag is off. When the feature is activated, MME upon receiving ODB AVP in ULA/IDR from HSS, shall set the following 2 bits: hssPxy_SupportedFeatureList_ODBHplmnApn and hssPxy_SupportedFeatureList_ODBVplmnApn regardless of homer vs. homer, and regardless of value of odb_all_apn and odb_hplmn_apn bits in ServiceAgreementProfile

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

overrideEnhEmmCcCreateSessRespUserAuthFail - Override Enh EMM Cause Code Create Session Response User Auth Fail

Syntax

```
--gParmName overrideEnhEmmCcCreateSessRespUserAuthFail --gParmValue  
<string>
```

Description

This parameter controls whether to use restNasMappingProfile provisioning for Create Session Request cause code #72 or #92. By default, flag shall be disabled. When this parameter is enabled, restNasMappingProfile shall be used.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

overrideVaa - APN Override regardless the VAA parameter value

Syntax

```
--gParmName overrideVaa --gParmValue <string>
```

Description

This parameter controls the APN override functionality of network identifier for roamer subscribers. When this parameter is ON the relevant functionality is decoupled from the value of the VAA parameter. When this value is OFF the relevant functionality continues to work as currently (i.e. coupled with the value of the VAA parameter)

List

- OFF - OFF
- ON - ON

Default

OFF

Name

pagingAfterInterRatRedirection - Paging After Inter Rat Redirection

Syntax

```
--gParmName pagingAfterInterRatRedirection --gParmValue <string>
```

Description

Parameter controls MME to allow paging to continue upon receiving a DDN after handling of an UE context release request with cause inter-rat-redirection, By default the flag shall be disabled. When enabled, MME upon receiving a DDN after handling of an UE context release request with cause inter-rat-redirection, MME shall allow paging to continue.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

pagingGapFailCountTrigger - Paging Gap Fail Count Trigger

Syntax

```
--gParmName pagingGapFailCountTrigger --gParmValue <string>
```

Description

Parameter Paging Gap Fail Count Trigger indicates the number of failed paging operations that start the paging gap timer.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

pagingOptimizationInHenb - Paging optimization in home eNB

Syntax

```
--gParmName pagingOptimizationInHenb --gParmValue <string>
```

Description

Is paging optimization supported for the home eNB?.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

pagingOptimizationInMme - Paging optimization in MME

Syntax

```
--gParmName pagingOptimizationInMme --gParmValue <string>
```

Description

A value of y indicates that paging optimization is supported for the MME.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

pagingOverloadPrevention - Paging Overload Prevention

Syntax

```
--gParmName pagingOverloadPrevention --gParmValue <string>
```

Description

Parameter controls feature activation. By default the flag shall be always enabled.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

pagingUponTadsReceived - Paging Upon T-ADS Received

Syntax

```
--gParmName pagingUponTadsReceived --gParmValue <string>
```

Description

Parameter controls paging upon receiving T-ADS from HSS in IDR message. By default, flag shall be disabled. When enabled, MME upon receiving T-ADS in IDR message from HSS when UE is in either registered or in idle state, shall respond with Insert Subscriber-Data Answer (with IMS-Voice-Over-PS Sessions-Supported (1) and shall immediately starts paging the UE.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

passiveOffload - Passive offload

Syntax

```
--gParmName passiveOffload --gParmValue <string>
```

Description

MME is in passive offload mode or not.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

passiveUelbOfLoadRate - UELB Off-Load Rate

Syntax

--gParmName passiveUelbOfLoadRate --gParmValue <string>

Description

For Passive UELB this parameter specifies the per CPPS of load rate

Type / Range

Integer in the range of 10 to 1000

Default

80

Name

pcmdSheddingDuringOverload - PCMD shedding during overload

Syntax

--gParmName pcmdSheddingDuringOverload --gParmValue <string>

Description

Control flag for PCMD shedding during overload. this flag can be used to determine MME behavior related to PCMD during CPU overload.

List

- Do not drop PCMD during overload - Do not drop PCMD during overload.
- Drop PCMD to OAM during overload - Drop PCMD to OAM during overload.
- Drop all PCMD in major overload - Drop all PCMD in major overload (includes eNB PCMD and PCMD to OAM).
- Drop all PCMD in minor overload - Drop all PCMD in minor overload (includes eNB PCMD and PCMD to OAM).

Default

Do not drop PCMD during overload

Name

pcscfRecovery - PCSCF recovery

Syntax

--gParmName pcscfRecovery --gParmValue <string>

Description

Configure P-CSCF recovery or not.

List

- mode1 - With mode 1 option, MME will utilize bit 31 in IDR flags that shall be used for p-cscf recovery, and it shall be carried in private extension IE. in addition, with this option MME shall send modify bearer request to SGW.
- mode2 - IMME shall utilize bit 8 in IDR flags AVP for p-cscf recovery and MME shall use feature bit 16 in supported features AVP. MME with this option shall disconnect PDN connection with NAS cause CNode reactivation required and shall send DSReq to SGW.
- mode3 - Supports 3GPP specified PCO based P-CSCF restoration.
- none - With none option, if MME receives bit in IDR flags AVP for p-cscf recovery it shall be ignored by MME and will continue with the ongoing procedure.

Default

none

Name

pdnConnEstEnhSupport - PDN Connection Re-establishment after SGW Change with Roamers and Voice Enhancements

Syntax

```
--gParmName pdnConnEstEnhSupport --gParmValue <string>
```

Description

Parameter controls enabling or disabling the support for enhancements to PDN Connection Re-establishment after SGW change with Roamers and Voice feature. By default this parameter shall be disabled.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

pdnDifferentApnOi - PDN connection for a different APN OI

Syntax

```
--gParmName pdnDifferentApnOi --gParmValue <string>
```

Description

This parameter is used to enable/disable MME to set up a second PDN connection with a different APN OI for APN when UE already has a PDN connection for the same APN.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

pdnHoS11ApnFormulation - PDN Handover S11 APN Formulation

Syntax

--gParmName pdnHoS11ApnFormulation --gParmValue <string>

Description

Defines how the MME forms the APN-OI of the APN IE on S11 Create Session Request towards the SGW, when the MME receives a 'Handover' PDN Connectivity Request from the UE. When set to 'Yes', the MME uses enhanced logic to utilise either the possibly provisioned override PLMN id, the PLMN id obtained from the UE's IMSI or the UE's requested APN-OI. When set to 'No' (default), the MME uses the Destination-Realm as received from the HSS indicating the selected PGW of the PDN to set the APN-OI on S11 CSR.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

pdnReestablishSgwChange - PDN reestablish SGW change

Syntax

--gParmName pdnReestablishSgwChange --gParmValue <string>

Description

All PDN connections for each UE shall be always examined for PDN-reestablishment after every SGW change.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

pdnRejectDueToSubscriptionEsmCc - ESM cause for PDN rejected due to subscription

Syntax

--gParmName pdnRejectDueToSubscriptionEsmCc --gParmValue <string>

Description

ESM cause code in PDN connection reject message when PDN connection request is rejected due to subscription.

Type / Range

Integer in the range of 8, 26-39, 41-47, 49-60, 65-66, 81, 95-101, 111-113

Default

32

Name

pdnReleaseNrSecRatRestriction - Release PDN connection due to NR as Secondary RAT restriction

Syntax

--gParmName pdnReleaseNrSecRatRestriction --gParmValue <string>

Description

Allow MME to release a PDN Connection for a DCNR capable UE after MME determines UE is restricted access for NR as a Secondary RAT.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

pgwAddrOnS6aAndPgwFinding - PGW Address on S6a and PGW Finding

Syntax

--gParmName pgwAddrOnS6aAndPgwFinding --gParmValue <string>

Description

pgwAddrOnS6aAndPgwFinding parameter has two purposes for e.g. VoLTE - VoWifi handovers: it -controls which received PGW IP address(es) the MME uses, when the MME sends the selected PGW information inside the MIP6-Agent-Info AVP on the S6a NOR and the S6a ULR (Active-APN AVP(s)). -controls how the MME finds the PGW for the PDN when the MME receives Attach Request or PDN Connectivity Request with Request type Handover and establishes the PDN with Create Session Request.

List

- csrHss - Create Session Response returned IP(s) or HSS given PGW IP(s) - On S6a NOR/ULR, the MME uses the IP address(es) returned by the SGW on Create Session Response. When sending Create Session Request for a PDN after receiving Attach Request or PDN Connectivity Request with Request type "Handover", the MME uses the PGW IP address(es) from the HSS for the PDN and does not perform any DNS query.
- dns - DNS given PGW IP(s) - On S6a NOR/ULR, the MME uses the PGW IP address(es) received from the DNS. When sending Create Session Request for a PDN after receiving Attach Request or PDN Connectivity Request with Request type "Handover", the MME uses the DNS to find the PGW address to use for the PDN.

Default

dns

Name

pgwApnGtpOvlControlSupport - APN Level Overload Control Support

Syntax

--gParmName pgwApnGtpOvlControlSupport --gParmValue <string>

Description

When enabled the MME shall support APN level overload control functionality, whenever overload information is provided on APN level from the PGW.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

pgwIsolationFaultThreshold - PGW isolation fault threshold

Syntax

--gParmName pgwIsolationFaultThreshold --gParmValue <string>

Description

This command is used to provision PGW fault threshold for the PGW isolation. if the number of the PGW faults (remote peer not responding cause received in the S11/S4 create session response) exceed the provisioned threshold in a provisioned duration timer then WMM will isolate the PGW from the selection to new PDN connection for provisionable duration.

Type / Range

Integer in the range of 1 to 1000

Default

50

Name

pgwPlmnIdExclusionSupport - PGW plmn ID exclusion support

Syntax

--gParmName pgwPlmnIdExclusionSupport --gParmValue <string>

Description

Enable/Disable exclusion of PGW PLMN ID AVP when PGW FQDN is present in NOR.

List

- No - Boolean value of no.

- Yes - Boolean value of yes.

Default

No

Name

pgwRestartNotification - PGW restart notification

Syntax

```
--gParmName pgwRestartNotification --gParmValue <string>
```

Description

If it is yes, then MME/S4-SGSN sends support of PGW restart notification indication in node feature IE of GTPv2 echo request and echo response. if it is no, then MME will not indicate the support of PGW restart notification to the SGW.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

pgwSelectionRestrictedToCollocatedGgsnPgw - PGW Selection Restricted To Collocated GGSN/PGW

Syntax

```
--gParmName pgwSelectionRestrictedToCollocatedGgsnPgw --gParmValue <string>
```

Description

Parameter controls MME support for Configuration Control for Restriction of PGW Selection to Collocated GGSN/PGW during IRAT Mobility scenario. By default, flag shall be disabled. When enabled, restriction for gp-service for roamer and gn-service for homer shall be removed. This shall apply to Gateway Mode Selection 2.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

pgwSelectionWithNonNcSmf - PGW Selection With Non nc-smf

Syntax

```
--gParmName pgwSelectionWithNonNcSmf --gParmValue <string>
```

Description

Parameter controls feature activation. By default the flag shall be disabled. When enabled, MME shall support preferring PGW(s) which specifically have a non nc-smf tagged service type for non N1 mode UEs. In other words, with the new gParm enabled, MME shall enforce associating LTE only UE with non-combo PGW. This feature shall be applied only for gateway mode selection 2.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

pgwSelectionWithSnaptr - PWG selection with snaptr

Syntax

--gParmName pgwSelectionWithSnaptr --gParmValue <string>

Description

Enable/Disable support for PGW selection with snaptr.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

plmnCtrlPreferGw - plmn level control of preferred Gw

Syntax

--gParmName plmnCtrlPreferGw --gParmValue <string>

Description

PLMN Level Control of Preferred Gateway. Customer preferred control of GW type selection. This gParm is to control this feature. When this plmnCtrlpreferGw parameter is set, then PLMN level Control of Preferred Gateway will be enable. If the parameter is false (default value), then S4 SGSN keeps existing behaviour.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

plmnOperMtLrNoEsmc - PLMN operator MT_LR no E-SMLC

Syntax

--gParmName plmnOperMtLrNoEsmc --gParmValue <string>

Description

MT_LR client type PLMN Operator Service do not contact E-SMLC for UE location information. If yes, when MME receives mobile terminated location request with client type=plmnOperatorServices, and activateLcsClientTypePlmnOperatorServices is set, the MME contacts the eNB for UE current E-CGI information. This information is sent back to the GMLC. The MME does not contact the E-SMLC to obtain any UE location information. If no, nothing changes and keep existing function (i.e. MME contacts E-SMLC to obtain UE location for this MT-LR type) .

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

plmnS6dSelection - PLMN level s6d&Gr selection

Syntax

--gParmName plmnS6dSelection --gParmValue <string>

Description

When rel8 architecture enable, S4-SGSN always selects HSS. With this feature enable, PLMN level control to select either S6d/Gr will be controlled by operators. Also fallback from s6d to Gr could be also controlled by operators. When this parameter set to true (yes), then above mentioned two parameters can be configured by operator/PLMN level. If the parameter set to false (no, default), then SGSN keeps the same as current. When this feature is enable, then SG01236 is not used.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

pmFileToFluentd - Send PM File to Fluentd

Syntax


```
--gParmName pmFileToFluentd --gParmValue <string>
```

Description

This parameter controls the functionality for sending PM XML file to fluentd.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

pMtuDiscoveryS1IPv6 - Path MTU Discovery for IPv6 S1/M3 links

Syntax

```
--gParmName pMtuDiscoveryS1IPv6 --gParmValue <string>
```

Description

Controls Path MTU Discovery for IPv6 S1/M3 links. no (default) - disable Path MTU Discovery
yes - enable Path MTU Discovery

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

powerSavingsMode - Power savings mode

Syntax

```
--gParmName powerSavingsMode --gParmValue <string>
```

Description

Support for UE power saving mode.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

pradClpv4 - Presence Reporting Area Dual Connectivity IPv4

Syntax

```
--gParmName praDcIpv4 --gParmValue <string>
```

Description

Activates customized Presence Reporting Area (PRA) functionality for dual connectivity. When dual connectivity is applied for a UE, presence reporting is used to inform the SGW/PGW/PCRF (upon request) whether the UE currently uses a 5G gNB (UE is inside) or not (UE is outside). The gNBs use IPv4 addressing. MME uses a customized schema in analyzing whether a provided RAN IPv4 user plane endpoint belongs to an eNB or a gNB.

List

- dcPralpv4Disabled - DC PRA IPv4 Reporting Disabled - Dual Connectivity Presence Reporting for IPv4 addresses is disabled.
- dcPralpv4Basic - DC PRA IPv4 Reporting Basic - MME sends all Dual Connectivity IPv4 address based reports. MME sends initially what is the UE status when the PRA action is requested from the SGW/PGW/PCRF, and also all changes on the status. "Inside" means that 5G gNB is currently used by the UE, and "outside" means that 5G gNB is currently not used.
- dcPralpv4OptIdleConn - DC PRA IPv4 Reporting Optimized Idle Connected - MME does not send Dual Connectivity IPv4 address based initial report, if the initial status is 'outside'. MME sends initially what is the UE status only if the status is initially 'inside' when the PRA action is requested from the SGW/PGW/PCRF. If the initial status is 'outside', MME does not report that and only reports the very first 'inside'. After the initial status the MME sends changes on the status. 'Inside' means that 5G gNB is currently used by the UE, and 'outside' means that 5G gNB is currently not used. MME supports three separate timers for optimizing PRA reporting. MME starts timer whenever it first receives new RAN endpoints, no timer is previously running and PRA status would be changed due to the new endpoints. Timer is stopped when MME receives different new RAN endpoints or UE context is released. MME sends the PRA report towards the SGW/PGW when timer expires or is stopped if reporting is still required i.e. status differs from previous reported and the UE is still ECM-CONNECTED. Note: when PRA action is received from another MME on Forward Relocation Request/ Context Response, the MME always sends the first report under the new MME even with this setting selected.

Default

dcPralpv4Disabled

Name

praDcIpv6Prefix - PRA DC IPv6 Prefix

Syntax

```
--gParmName praDcIpv6Prefix --gParmValue <string>
```

Description

Activate the Presence Reporting Area functionality to report on dual connectivity with IPv6 prefixes. MME detects the dual connectivity usage based on provisioned IPv6 address

prefixes. The functionality uses CN predefined PRAs. The MME reports via PRA reporting that a UE is inside the area when the MME detects that the gNB user plane IP address is covered by the provisioned IPv6 address prefix i.e. UE is using dual connectivity with a 5G gNB. The UE is outside the area when the gNB user plane IP address is not covered by the provisioned IPv6 address prefix i.e. UE is not using dual connectivity with a 5G gNB.

List

- **dcPralpv6PrefixBasic** - DC PRA IPv6 Prefix Basic - MME sends all Dual Connectivity IPv6 prefix based reports. MME sends initially what is the UE status after the PRA action has been requested from the SGW/PGW/PCRF, and also all changes after that on the status. 'Inside' means that 5G gNB is currently used by the UE, and 'outside' means that 5G gNB is currently not used.
- **dcPralpv6PrefixDisabled** - DC PRA IPv6 Prefix Disabled - Dual Connectivity Presence Reporting Area based on IPv6 prefixes is disabled.
- **dcPralpv6PrefixSingleTimer** - DC PRA IPv6 Prefix Single Timer - MME does not send Dual Connectivity IPv6 prefix based initial report, if the initial status is 'outside'. MME sends initially what is the UE status only if the status is initially 'inside' when the PRA action is requested from the SGW/PGW/PCRF. If the initial status is 'outside', MME does not report that and only reports the very first 'inside'. After the initial status the MME sends changes on the status. 'Inside' means that 5G gNB is currently used by the UE, and 'outside' means that 5G gNB is currently not used. MME supports a single timer for optimizing PRA reporting for all PRA ids of the UE. MME starts the timer whenever it first receives new RAN endpoints, no timer is previously running and PRA status would be changed due to the new endpoints. Timer is stopped when MME receives different new RAN endpoints to update the status back to previous or UE context is released. The timer is restarted if it was running and the MME receives new endpoints changing the status of a different PRA id of the same UE. MME sends the PRA report towards the SGW/PGW when timer expires or is stopped if reporting is still required i.e. status of any PRA differs from previous reported. Note: when PRA action is received from another MME on Forward Relocation Request/Context Response, the MME always sends the first report under the new MME even with this setting selected.

Default

dcPralpv6PrefixDisabled

Name

praDcSegment - PRA Dual connectivity segment reporting

Syntax

```
--gParmName praDcSegment --gParmValue <string>
```

Description

Activate the Presence Reporting Area functionality to report on dual connectivity utilizing a provisioned IPv6 address segment. The functionality uses CN predefined PRAs. The MME reports via PRA reporting that a UE is inside the area when the MME detects that a 5G gNB

having a provisioned IPv6 address segment is added with dual connectivity for the UE. The UE is outside the area when the corresponding 5G gNB is removed for the UE.

List

- **dcSegmentReportingDisabled** - DC Segment Reporting Disabled - Dual Connectivity Presence Area Reporting feature is disabled.
- **dcSegmentBasic** - DC Segment Basic - MME sends all Dual Connectivity IPv6 segment based reports. MME sends initially what is the UE status when the PRA action is requested from the SGW/PGW/PCRF, and also all changes on the status. 'inside' means that 5G gNB is currently used by the UE, and 'outside' means that 5G gNB is currently not used.
- **dcSegmentOpt** - DC Segment Optimized - MME does not send Dual Connectivity IPv6 segment based initial report, if the initial status is 'outside'. MME sends initially what is the UE status only if the status is initially 'inside' when the PRA action is requested from the SGW/PGW/PCRF. If the initial status is 'outside', MME does not report that and only reports the very first 'inside'. After the initial status the MME sends changes on the status. 'Inside' means that 5G gNB is currently used by the UE, and 'outside' means that 5G gNB is currently not used. On Service request the MME supports a provisioned timer to wait for any incoming E-RAB Modification Indication for a subsequent status change, and only reports the possible status change after the timer expiry. Note: when PRA action is received from another MME on Forward Relocation Request/Context Response, the MME always sends the first report under the new MME even with this setting selected.
- **dcSegmentOptIdleConn** - DC Segment Optimized Idle Connected - MME does not send Dual Connectivity IPv6 segment based initial report, if the initial status is 'outside'. MME sends initially what is the UE status only if the status is initially 'inside' when the PRA action is requested from the SGW/PGW/PCRF. If the initial status is 'outside', MME does not report that and only reports the very first 'inside'. After the initial status the MME sends changes on the status. 'Inside' means that 5G gNB is currently used by the UE, and 'outside' means that 5G gNB is currently not used. MME supports three separate timers for optimizing PRA reporting. MME starts timer whenever it first receives new RAN endpoints, no timer is previously running and PRA status would be changed due to the new endpoints. Timer is stopped when MME receives different new RAN endpoints or UE context is released. MME sends the PRA report towards the SGW/PGW when timer expires or is stopped if reporting is still required i.e. status differs from previous reported and the UE is still ECM-CONNECTED. Note: when PRA action is received from another MME on Forward Relocation Request/Context Response, the MME always sends the first report under the new MME even with this setting selected.
- **dcSegmentOptSingleTimer** - DC Segment Optimized Single Timer - MME does not send Dual Connectivity IPv6 segment based initial report, if the initial status is 'outside'. MME sends initially what is the UE status only if the status is initially 'inside' when the PRA action is requested from the SGW/PGW/PCRF. If the initial status is 'outside', MME does not report that and only reports the very first 'inside'. After the initial status the MME

sends changes on the status. 'Inside' means that 5G gNB is currently used by the UE, and 'outside' means that 5G gNB is currently not used. MME supports a single timer for optimizing PRA reporting for all PRA ids of the UE. MME starts the timer whenever it first receives new RAN endpoints, no timer is previously running and PRA status would be changed due to the new endpoints. Timer is stopped when MME receives different new RAN endpoints to update the status back to previous or UE context is released. The timer is restarted if it was running and the MME receives new endpoints changing the status of a different PRA id of the same UE. MME sends the PRA report towards the SGW/PGW when timer expires or is stopped if reporting is still required i.e. status of any PRA differs from previous reported. Note: when PRA action is received from another MME on Forward Relocation Request/Context Response, the MME always sends the first report under the new MME even with this setting selected.

Default

dcSegmentReportingDisabled

Name

praDcSingleTimerAllowedRestartCount - PRA DC Single Timer Allowed Restart Count

Syntax

--gParmName praDcSingleTimerAllowedRestartCount --gParmValue <string>

Description

Defines the number of times the MME is sequentially allowed to restart the single DC PRA optimisation timer. When the MME has already previously restarted the single optimisation timer the times allowed by this parameter and the MME still receives new RAN endpoints causing DC PRA status change (and the MME would in this case again restart the single optimisation timer without this limit), the MME stops the single optimisation timer and sends out the current status of the DC PRAs that is not yet reported towards the SGW/PGW. This parameter is only applied when global parameter PRA Dual connectivity segment reporting is set to DC Segment Optimized Single Timer.

Type / Range

Integer in the range of 0 to 50

Default

5

Name

praDormantState - PRA Dormant State Enabled

Syntax

--gParmName praDormantState --gParmValue <string>

Description

Enables the functionality to change the state of the CN predefined Presence Reporting Area (PRA) between active (i.e. normal) and dormant. In dormant PRA state, the MME suppresses any PRA status (inside/outside) reports towards the SGW/PGW. In dormant PRA state the MME still accepts and stores PRA action to request PRA status reporting. Dormant PRA state

is applicable for both 3GPP standard based PRA functionality as well as the dual connectivity segment based reporting functionality.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

praForRoamers - PRA for roamers

Syntax

--gParmName praForRoamers --gParmValue <string>

Description

When set to Yes, this parameter allows MME to support Presence Reporting Area (PRA) functionality for roaming UEs.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

praidleState - PRA Idle State

Syntax

--gParmName praIdleState --gParmValue <string>

Description

Enables Presence Reporting Area (PRA) functionality clarification change. If the UE is in ECM-IDLE state, the MME may either bring the UE into ECM-CONNECTED state, or, report based on the UE's last known location and when the UE was there.

Restrictions

gParm praRel145 required to be set to 'Yes'.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

praModifyAction - PRA Modify Action

Syntax

--gParmName praModifyAction --gParmValue <string>

Description

Controls whether the MME reports immediately 'inside'/'outside' when receiving PRA action with action value 'modify' independently of the previous report. When set to 'yes', MME always reports immediately 'inside'/'outside' even if previous report was the same. When set to 'no', MME does not report immediately 'inside'/'outside' if previous report was the same.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

praModifyEnabling - PRA Modify Enabling

Syntax

--gParmName praModifyEnabling --gParmValue <string>

Description

Enables Presence Reporting Area (PRA) modify action support in the MME.

Restrictions

gParm praRel145 required to be set to 'Yes'.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

praPace - Presence Reporting Area Pace

Syntax

--gParmName praPace --gParmValue <string>

Description

The pace to send Presence Reporting Area Information towards the SGW/PGW with the standard based Presence Reporting Area functionality. This setting is applicable for both dual

connectivity segment based PRA and standard based PRA functionalities. The pace is defined in messages/second per CPPS.

Type / Range

Integer in the range of 10 to 2000

Default

50

Name

praPacing - PRA Pacing

Syntax

```
--gParmName praPacing --gParmValue <string>
```

Description

Enables Presence Reporting Area pacing functionality to ensure the surrounding network elements are not impacted by bursts of Presence Area Reporting Information.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

praRel145 - Presence Reporting Area 3GPP Rel 14-15

Syntax

```
--gParmName praRel145 --gParmValue <string>
```

Description

Enables 3GPP Release 14-15 enhancements for Presence Reporting Area functionalities: e.g. multiple PRA ids per PDN, PRA sets and PRA modify action.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

praSets - praSets

Syntax

```
--gParmName praSets --gParmValue <string>
```

Description

Enables support of Presence Reporting Area (PRA) sets in MME.

Restrictions

gParm praRel145 required to be set to 'Yes'.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

praStdSuppressInitialOut - PRA Standard Suppress Initial Out

Syntax

```
--gParmName praStdSuppressInitialOut --gParmValue <string>
```

Description

Controls whether on 3GPP Standard Presence Reporting Area (PRA) functionality the first/initial outside report is suppressed. If set to Yes and MME receives PRA action Start from SGW/PGW, MME only sends initial PRA report (PRA Information) to SGW/PGW when the UE is/has arrived inside the corresponding area. If set to No, the MME will send the initial report (PRA Information) immediately after receiving PRA action Start from SGW/PGW despite of UE being inside or outside of the corresponding area initially.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

praStdSuppression - PRA Standard Suppression

Syntax

```
--gParmName praStdSuppression --gParmValue <string>
```

Description

Enables MME suppression of too frequent PRA reports for 3GPP standard based PRA (Presence Reporting Area) functionality. If set to Yes, the MME suppresses a subsequent PRA report inside the PRA suppress time period, unless the PRA status stays the same for the PRA suppress cancel defined time period. If set to No, MME sends subsequent PRA reports according to UE mobility based on 3GPP standard definitions with no restriction to reporting frequency.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

pRau2g - PRAU timer in 2G

Syntax

--gParmName pRau2g --gParmValue <string>

Description

This parameter controls the usage of independent 2G and 3G pRau timers. If the parameter is set, then SGSN is allowed to configure pRau timer differently in 2G and 3G. If the parameter is not set, existing functionality will be used (2G and 3G with the same value for pRau timer).

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

preferDataMode - Prefer data mode

Syntax

--gParmName preferDataMode --gParmValue <string>

Description

Selection of preferred data mode when a UE subscription has APN configuration.

List

- Data over CP Preferred - Data over CP preferred.
- Data over UP Preferred - Data over UP preferred.

Default

Data over UP Preferred

Name

preferredPgwId - Preferred PGW Identifier

Syntax

--gParmName preferredPgwId --gParmValue <string>

Description

Preferred PGW identifier to determine PGW identity

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

prefGnToGGSNIpVer - IP address version-preference for transport layer

Syntax

--gParmName prefGnToGGSNIpVer --gParmValue <string>

Description

Use this parameter to control the IP address version-preference for transport layer.

List

- IPV4 - Specify trace entity of IPv4.
- IPV6 - Specify trace entity of IPv6.

Default

IPV4

Name

preRel8DomainName - Pre release 8 domain name

Syntax

--gParmName preRel8DomainName --gParmValue <string>

Description

Pre release 8 domain name.

Type / Range

Digit string of length 0 to 62 with the allowed pattern of [0-9a-zA-Z\.\-]*

Name

prometheusFormat - Prometheus Format

Syntax

--gParmName prometheusFormat --gParmValue <string>

Description

This option controls the format of the data model and notation used to report Prometheus Counters. This option is applicable in all cases i.e. either when CMM is configured to push the Prometheus Counters to PrometheusPushGW or when Prometheus Server directly scrapes the Prometheus Counters from CMM.

List

- **basicFormat** - Basic Format - The Basic Format supports the data model and notation that was initially introduced in CMM20.0 when Prometheus Server functionality was implemented.
- **advancedFormat** - Advanced Format - The Advanced Format is based on the Basic format with the following changes: 1. Instead of using Counter PIDs, the 3gpp counter name is used. 2. the **elementType** is added.

Default

basicFormat

Name

prometheusMode - Prometheus Mode

Syntax

--gParmName prometheusMode --gParmValue <string>

Description

There are two functionalities supported for Prometheus Server in order to retrieve the relevant PM Data. The first functionality is when CMM pushes the relevant PM Counters to PrometheusPushGW and then Prometheus Server scrapes the relevant PM Counters from PrometheusPushGW. The second functionality is when Prometheus Server directly scrapes the relevant PM Counters from CMM. The scope of this parameter is control which functionality will be enabled from CMM.

List

- **push** - This value is used when the Prometheus counters are pushed from CMM towards PrometheusPushGW. Prometheus Server will then scrape the PM Data from PrometheusPushGW.
- **pull** - This value is used when the Prometheus Server directly scrapes from CMM the Prometheus Counters (i.e. PrometheusPushGW is not part of the solution).

Default

push

Name

prometheusRequests - Prometheus Requests

Syntax

--gParmName prometheusRequests --gParmValue <string>

Description

This parameter defines the maximum allowed number of Prometheus Requests, in the specific Prometheus Sliding Window (defined with parameter **prometheusSlidingWindow**), that could in parallel directly scrape the relevant PM Counters from CMM. This parameter is applicable only when the functionality/option to allow Prometheus Server to directly scrape the relevant PM Counters from CMM is configured. Warning: For Nokia personnel only.

Type / Range

Integer in the range of 0 to 11

Default

10

Name

prometheusSlidingWindow - Prometheus Sliding Window

Syntax

```
--gParmName prometheusSlidingWindow --gParmValue <string>
```

Description

This parameter defines the CMM sliding window (in seconds) in which Requests from Prometheus Server to directly scrape the data from CMM are accepted. The number of parallel requests in this sliding window are defined with parameter prometheusRequests. This parameter is applicable only when the functionality/option to allow Prometheus Server to directly scrape the relevant PM Counters from CMM is configured. Warning: For Nokia personnel only.

Type / Range

Prometheus Sliding Window in seconds

Default

60

Name

provPciPvi - Provisionable PCI and PVI value supported

Syntax

```
--gParmName provPciPvi --gParmValue <string>
```

Description

Supports provisionable PCI and PVI values.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

psCsCoordination - PS CS coordination

Syntax

```
--gParmName psCsCoordination --gParmValue <string>
```

Description

PS CS coordination capability for MOCN configuration.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

pseudoRandomTlliCollisionHandling - Pseudo-random Tlli Collision Handling

Syntax

--gParmName pseudoRandomTlliCollisionHandling --gParmValue <string>

Description

When parameter is set to No existing collision handling mechanism is maintained regarding 2G Attach with random TLLI. When status is set to Yes collision handling will not be applied for a 2G Attach with Random TLLI even if is already used by an ongoing Attach procedure, as long as Attach message was received via different nsei/bvci pair. Furthermore the SGSN will silently drop a new Attach request with a random TLLI when the included random TLLI in the Attach request is already used by ongoing Attach from the same nsei/bvci pair.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

pseudoSpgwIpv4address - Pseudo SPGW IPv4 address

Syntax

--gParmName pseudoSpgwIpv4address --gParmValue <string>

Description

The SGW/PGW IPv4 address used in creating the S1-U SGW tunnel IDs to be sent to eNB in S1AP INITIAL UE CONTEXT SETUP message when MME does not use SGW or PGW for bearer creation during SDS UE attach request.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Default

0.0.0.0

Name

pseudoSpgwIpv6address - Pseudo SPGW IPv6 address

Syntax

--gParmName pseudoSpgwIpv6address --gParmValue <string>

Description

The SGW/PGW IPv6 address used in creating the S1-U SGW tunnel IDs to be sent to eNB in S1AP INITIAL UE CONTEXT SETUP message when MME does not use SGW or PGW for bearer creation during SDS UE attach request.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Default

::

Name

pseudoUeIpv4address - Pseudo UE IPv4 address

Syntax

--gParmName pseudoUeIpv4address --gParmValue <string>

Description

The UE IPv4 address to be used for an SDS UE when MME does not use SGW and PGW for bearer session setup during attach procedure.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Default

0.0.0.0

Name

pseudoUeIpv6address - Pseudo UE IPv6 address

Syntax

--gParmName pseudoUeIpv6address --gParmValue <string>

Description

The UE IPv6 address to be used for an SDS UE when MME does not use SGW and PGW for bearer session setup during attach procedure.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Default

::

Name

purFlagsAvp - PUR-Flags AVP

Syntax

--gParmName purFlagsAvp --gParmValue <string>

Description

PUR-Flags.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

purgeUeWhenHandoffToEhrpd - Purge UE when UE hands off to eHRPD

Syntax

--gParmName purgeUeWhenHandoffToEhrpd --gParmValue <string>

Description

When UE hands off to eHRPD, if this global parameter is yes, MME deletes VLR and sends PUR to HSS.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

pwsRestartProcedure - PWS restart procedure

Syntax

--gParmName pwsRestartProcedure --gParmValue <string>

Description

Forwarding of PWS RESTART INDICATION message to CBC.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

qci1HoRejectionGeran - QCI1 HO Rejection GERAN

Syntax

--gParmName qci1HoRejectionGeran --gParmValue <string>

Description

When set to 'Yes', MME rejects incoming S1AP Handover Required attempts when -the handover is attempted towards 2G i.e. Handover Type is set to 'LTEtoGERAN' -the Handover Required message does NOT include SRVCC HO Indication IE -the UE currently has a dedicated bearer with QCI1 established. The MME sends Handover Preparation Failure with cause 'Handover Failure In Target EPC/eNB Or Target System' and does not proceed further with the handover. When set to 'No', MME proceeds with the handover normally.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

qci1HoRejectionUtran - QCI1 HO Rejection UTRAN

Syntax

```
--gParmName qci1HoRejectionUtran --gParmValue <string>
```

Description

When set to 'Yes', MME rejects incoming S1AP Handover Required attempts when -the handover is attempted towards 3G i.e. Handover Type is set to 'LTEtoUTRAN' -the Handover Required message does NOT include SRVCC HO Indication IE -the UE currently has a dedicated bearer with QCI1 established. The MME sends Handover Preparation Failure with cause 'Handover Failure In Target EPC/eNB Or Target System' and does not proceed further with the handover. When set to 'No', MME proceeds with the handover normally.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

qosModCollEnh - QoS Modification Collision Enhancements

Syntax

```
--gParmName qosModCollEnh --gParmValue <string>
```

Description

This parameter controls the feature activation. By default this parameter shall be set to value 'No'.

List

- No - Boolean value of no.

- Yes - Boolean value of yes.

Default

No

Name

queueUeCtxtRelUserInactivityErabModIndication - Queue UE Context Release Request User Inactivity During E-RAB Modification Indication

Syntax

```
--gParmName queueUeCtxtRelUserInactivityErabModIndication --  
gParmValue <string>
```

Description

Defines whether the MME queues an incoming S1AP UE Context Release Request with radio network cause 'user inactivity' from eNB when the MME has received S1AP E-RAB Modification Indication from the eNB and has not yet sent E-RAB Modification Confirm as response. When set to Yes, MME triggers UE Context Release Command after sending out E-RAB Modification Confirm. When set to No (default), MME drops the incoming UE Context Release Request with radio network cause 'user inactivity'.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

raiBasedDeactPdpRate - deactivation PDP rate

Syntax

```
--gParmName raiBasedDeactPdpRate --gParmValue <string>
```

Description

This parameter determines with what rate (reactivations per second) PDP context reactivations should be done during RAI-based context reactivation.

Restrictions

FNS has limitation 1-25 after PET testing

Type / Range

Integer in the range of 1 to 25

Default

18

Name

raiBasedUeHandlingEnabled - RAI based UE handling

Syntax

```
--gParmName raiBasedUeHandlingEnabled --gParmValue <string>
```

Description

RAI/RAI group based UE handling

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

ranapResetAfterSysRestart - ranapResetAfterSysRestart

Syntax

```
--gParmName ranapResetAfterSysRestart --gParmValue <string>
```

Description

Enables performing SGSN init RANAP RESET after system restart/upgrade

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

rangeDiscoverTais - Range Discover TAIs

Syntax

```
--gParmName rangeDiscoverTais --gParmValue <string>
```

Description

Feature activation for Enhanced TA Discovery Auto-Provisioning

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

rateLimitPerApn - Rate Limit Per APN

Syntax

```
--gParmName rateLimitPerApn --gParmValue <string>
```

Description

Whether the MME should enforce PLMN Rate Control per APN/PDN per 3GPP TS 23.401 CR 3486 or per PDN. The default is no rate control per PDN.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

rcvryDlsdsMaxLimitOverride - RcvryDlsdsMaxLimitOverride

Syntax

```
--gParmName rcvryDlsdsMaxLimitOverride --gParmValue <string>
```

Description

Maximum number of recovery DL SDS messages allowed to protect SRS from MME.

Type / Range

Integer in the range of systemDefault or 0 to 50,000

Default

systemDefault

Name

rcvrySrulrLimitOverride - RcvrySrulrLimitOverride

Syntax

```
--gParmName rcvrySrulrLimitOverride --gParmValue <string>
```

Description

Recovery SR ULRs allowed per second after which CPPS will start droppping recovery ULRs until rate is below limit.

Type / Range

Integer in the range of systemDefault or 0 to 1,000,000

Default

systemDefault

Name

rcvrySrulrMaxdeferCountOverride - RcvrySrulrMaxdeferCountOverride

Syntax

```
--gParmName rcvrySrulrMaxdeferCountOverride --gParmValue <string>
```

Description

Maximum number of times recovery SR ULRs can be deferred to protect MME and HSS during recovery ULR storm.

Type / Range

Integer in the range of systemDefault or 1 to 20

Default

systemDefault

Name

redirectionCommandRate - Redirection command rate

Syntax

```
--gParmName redirectionCommandRate --gParmValue <string>
```

Description

Maximum allowed rate of S102 redirection command messages.

Type / Range

Integer in the range of 0 to 50,000

Default

5000

Name

regionalSubscription - Regional subscription

Syntax

```
--gParmName regionalSubscription --gParmValue <string>
```

Description

Setting to yes activates regional subscription on the gn-sgsn. setting to no deactivates regional subscription on the gn-sgsn.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

rejDtrExtendedServiceRequest - Reject Dual Transceiver Receiver Extended Service Request

Syntax

```
--gParmName rejDtrExtendedServiceRequest --gParmValue <string>
```

Description

MME shall be provisioned with GPARM flag 'rejDtrExtendedServiceRequest' to control sending UE Context Modification Request message to E-UTRAN upon receiving Extended Service Request that is attached to E-UTRAN. By default this flag shall be disabled. When

the flag is enabled, MME shall send cause code #18 (CS Domain not available) in service reject to eNodeB. When the flag is disabled, MME shall act as it did before introduction of this functionality, i.e. MME upon receiving Extended Service Request shall send UE Context Modification Request message to E-UTRAN.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

rejectAlgorithmDuringOverload - Reject algorithm during overload

Syntax

--gParmName rejectAlgorithmDuringOverload --gParmValue <string>

Description

Overload control logic with a different set of messages/procedures to shed at major and critical overload thresholds.

List

- Reject Idle User First - Overload control reject idle user initiated procedure first.
- Reject Max Signaling Load First - Overload control reject highest signaling load generating procedures first.

Default

Reject Max Signaling Load First

Name

rejectCcForGmmErrorCode11 - Reject Cause Code for GMM Error Code 11

Syntax

--gParmName rejectCcForGmmErrorCode11 --gParmValue <string>

Description

This parameter defines which Reject Cause code will be used for GMM Error Code 11.

List

- gprsServicesNotAllowed - GPRS services not allowed in this PLMN - Reject Cause: GPRS services not allowed in this PLMN
- locationAreaNotAllowed - Location Area not allowed - Reject Cause: Location Area not allowed
- noSuitableCellInLocationArea - No Suitable Cell in Location Area - Reject Cause: No Suitable Cell in Location Area

- plmnNotAllowed - PLMN not allowed - Reject Cause: PLMN not allowed
- psCsCoordinationRequired - PS CS Coordination Required - Reject Cause: ps-cs-coordination-required
- roamingNotAllowedInLocationArea - Roaming not allowed in Location Area - Reject Cause: Roaming-not-allowed-in- location -area

Default

psCsCoordinationRequired

Name

rejectCcForGmmErrorCode12 - Reject Cause Code for GMM Error Code 12

Syntax

--gParmName rejectCcForGmmErrorCode12 --gParmValue <string>

Description

This parameter defines which Reject Cause code will be used for GMM Error Code 12.

List

- gprsServicesNotAllowed - GPRS services not allowed in this PLMN - Reject Cause: GPRS services not allowed in this PLMN
- locationAreaNotAllowed - Location Area not allowed - Reject Cause: Location Area not allowed
- noSuitableCellInLocationArea - No Suitable Cell in Location Area - Reject Cause: No Suitable Cell in Location Area
- plmnNotAllowed - PLMN not allowed - Reject Cause: PLMN not allowed
- psCsCoordinationRequired - PS CS Coordination Required - Reject Cause: ps-cs-coordination-required
- roamingNotAllowedInLocationArea - Roaming not allowed in Location Area - Reject Cause: Roaming-not-allowed-in- location -area

Default

locationAreaNotAllowed

Name

rejectCcForGmmErrorCode13 - Reject Cause Code for GMM Error Code 13

Syntax

--gParmName rejectCcForGmmErrorCode13 --gParmValue <string>

Description

This parameter defines which Reject Cause code will be used for GMM Error Code 13.

List

- gprsServicesNotAllowed - GPRS services not allowed in this PLMN - Reject Cause: GPRS services not allowed in this PLMN
- locationAreaNotAllowed - Location Area not allowed - Reject Cause: Location Area not allowed
- noSuitableCellInLocationArea - No Suitable Cell in Location Area - Reject Cause: No Suitable Cell in Location Area
- plmnNotAllowed - PLMN not allowed - Reject Cause: PLMN not allowed
- psCsCoordinationRequired - PS CS Coordination Required - Reject Cause: ps-cs-coordination-required
- roamingNotAllowedInLocationArea - Roaming not allowed in Location Area - Reject Cause: Roaming-not-allowed-in- location -area

Default

roamingNotAllowedInLocationArea

Name

rejectCcForGmmErrorCode14 - Reject Cause Code for GMM Error Code 14

Syntax

--gParmName rejectCcForGmmErrorCode14 --gParmValue <string>

Description

This parameter defines which Reject Cause code will be used for GMM Error Code 14

List

- gprsServicesNotAllowed - GPRS services not allowed in this PLMN - Reject Cause: GPRS services not allowed in this PLMN
- locationAreaNotAllowed - Location Area not allowed - Reject Cause: Location Area not allowed
- noSuitableCellInLocationArea - No Suitable Cell in Location Area - Reject Cause: No Suitable Cell in Location Area
- plmnNotAllowed - PLMN not allowed - Reject Cause: PLMN not allowed
- psCsCoordinationRequired - PS CS Coordination Required - Reject Cause: ps-cs-coordination-required
- roamingNotAllowedInLocationArea - Roaming not allowed in Location Area - Reject Cause: Roaming-not-allowed-in- location -area

Default

gprsServicesNotAllowed

Name

rejectCcForGmmErrorCode15 - Reject Cause Code for GMM Error Code 15

Syntax


```
--gParmName rejectCcForGmmErrorCode15 --gParmValue <string>
```

Description

This parameter defines which Reject Cause code will be used for GMM Error Code 15.

List

- gprsServicesNotAllowed - GPRS services not allowed in this PLMN - Reject Cause: GPRS services not allowed in this PLMN
- locationAreaNotAllowed - Location Area not allowed - Reject Cause: Location Area not allowed
- noSuitableCellInLocationArea - No Suitable Cell in Location Area - Reject Cause: No Suitable Cell in Location Area
- plmnNotAllowed - PLMN not allowed - Reject Cause: PLMN not allowed
- psCsCoordinationRequired - PS CS Coordination Required - Reject Cause: ps-cs-coordination-required
- roamingNotAllowedInLocationArea - Roaming not allowed in Location Area - Reject Cause: Roaming-not-allowed-in- location -area

Default

noSuitableCellInLocationArea

Name

rejectEsrNoSrvcc - Reject ESR no SRVCC

Syntax

```
--gParmName rejectEsrNoSrvcc --gParmValue <string>
```

Description

Reject ESR requested by an ECM-CONNECTED UE with IMS voice bearer.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

rejectNodeRelocatingPeriodicTau - Reject node relocating periodic TAU

Syntax

```
--gParmName rejectNodeRelocatingPeriodicTau --gParmValue <string>
```

Description

Enable/disable MME support for enhanced handling of intermme TAU with update type periodic.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

requestPsCellInLrc - Request PS Cell in LRC

Syntax

```
--gParmName requestPsCellInLrc --gParmValue <string>
```

Description

When location reporting control procedure is used in MME, if Request PS Cell in LRC is set, then MME includes additional location information towards eNB. eNB will include the current PSCell in the report if EN-DC is activated. If Request PS Cell in LRC is not set, MME will not request additional location information under location reporting procedure as existing function, then eNB will not report PSCell in location report.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

reservedBoolParameter1 - Reserved bool parameter 1

Syntax

```
--gParmName reservedBoolParameter1 --gParmValue <string>
```

Description

This parameter is reserved for future use.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

reservedBoolParameter2 - Reserved bool parameter 2

Syntax

```
--gParmName reservedBoolParameter2 --gParmValue <string>
```

Description

This parameter is reserved for future use.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

reservedBoolParameter3 - Reserved bool parameter 3

Syntax

```
--gParmName reservedBoolParameter3 --gParmValue <string>
```

Description

This parameter is reserved for future use.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

reservedBoolParameter4 - Reserved bool parameter 4

Syntax

```
--gParmName reservedBoolParameter4 --gParmValue <string>
```

Description

This parameter is reserved for future use.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

reservedBoolParameter5 - Reserved bool parameter 5

Syntax

```
--gParmName reservedBoolParameter5 --gParmValue <string>
```

Description

This parameter is reserved for future use.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

reservedIntParameter1 - Reserved int parameter 1

Syntax

```
--gParmName reservedIntParameter1 --gParmValue <string>
```

Description

This parameter is reserved for future use.

Type / Range

Digit string of length 1 to 5

Default

0

Name

reservedIntParameter2 - Reserved int parameter 2

Syntax

```
--gParmName reservedIntParameter2 --gParmValue <string>
```

Description

This parameter is reserved for future use.

Type / Range

Digit string of length 1 to 5

Default

0

Name

reservedIntParameter3 - Reserved int parameter 3

Syntax

```
--gParmName reservedIntParameter3 --gParmValue <string>
```

Description

This parameter is reserved for future use.

Type / Range

Digit string of length 1 to 5

Default

0

Name

reservedIntParameter4 - Reserved int parameter 4

Syntax

--gParmName reservedIntParameter4 --gParmValue <string>

Description

This parameter is reserved for future use.

Type / Range

Digit string of length 1 to 5

Default

0

Name

reservedIntParameter5 - Reserved int parameter 5

Syntax

--gParmName reservedIntParameter5 --gParmValue <string>

Description

This parameter is reserved for future use.

Type / Range

Digit string of length 1 to 5

Default

0

Name

restrictPagingDuringOverload - Restrict paging during overload

Syntax

--gParmName restrictPagingDuringOverload --gParmValue <string>

Description

Restrict paging during overload.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

restrictS10linksToNeighborMme - Restrict S10 links to neighbor MME

Syntax

```
--gParmName restrictS10linksToNeighborMme --gParmValue <string>
```

Description

MME support for restricting S10 to neighboring MME.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

restrictSubExtQoS - Restrict Subscribed Extended QoS

Syntax

```
--gParmName restrictSubExtQoS --gParmValue <string>
```

Description

CMM MME restricts subscribed extended QoS when UE is 5G DCNR restricted or 5G DCNR not capable. MME limits QoS to 4.29 Gbps or less when setting up or modifying default, GBR, or non-GBR bearer(s).

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

retransCsSvcNotifyOnNasNonDeliveryRadioConnLost - Retransmit CS Service Notify NAS Non-Delivery Radio Conn Lost

Syntax

```
--gParmName retransCsSvcNotifyOnNasNonDeliveryRadioConnLost --  
gParmValue <string>
```

Description

Current gParm enables treating NAS Non-Delivery Indication msg carrying cause 'radio connection with ue lost' the same as if cause were 'S1 intra system Handover triggered', or 'X2 Handover triggered' for SGs downlink service operation. When parameter is enabled, then when receiving NAS Non Delivery Indication (cause=radio connection with ue lost) the MME will expect to receive X2 Handover safeguarded by timer, at expiration of which the CS Service Notification will be resent

List

- No - Boolean value of no.

- Yes - Boolean value of yes.

Default

No

Name

retriesForDcNrUes - Retries for DCNR Capable UE

Syntax

--gParmName retriesForDcNrUes --gParmValue <string>

Description

Number of retries selecting PGW-C(s) during PDN Connectivity procedure upon receiving GTP error or timeout from gateway for DCNR Capable UEs.. For example, MME will send its first attempt to the PGW, if it receives no response within a certain period of time, the MME will retransmit the same request to the SGW/PGW and if/when it gets an error back from the SGW then it will move on to select the next best gateway to try. Assuming there is only one PGW in the sub-area, this means that the MME will perform round 2 (look for +ue entries but not find any) and then round 3 (look for _nc-nr entries, where it will find many to choose from). The next MME attempt will then be to a nearby +nc-nr PGW.

Type / Range

Integer in the range of 0 to 4

Default

0

Name

retriesForUutUes - Retries For UUT UEs

Syntax

--gParmName retriesForUutUes --gParmValue <string>

Description

Number of retries selecting PGW-C(s) during PDN Connectivity procedure upon receiving GTP error or timeout from the gateway for UUT UE(s) For example, MME will send its first attempt to the PGW, if it receives no response within a certain period of time, the MME will retransmit the same request to the SGW/PGW and if/when it gets an error back from the SGW then it will move on to select the next best gateway to try. Assuming there is only one PGW in the sub-area, this means that the MME will perform round 2 (look for +ue<uut> entries but not find any) and then round 3 (look for _nc-nr entries, where it will find many to choose from). The next MME attempt will then be to a nearby +nc-nr PGW.

Type / Range

Integer in the range of 0 to 4

Default

0

Name

retrieveCellId - Retrieve cell id

Syntax

--gParmName retrieveCellId --gParmValue <string>

Description

Yes - enable retrieval of ECGI from the eNB to be included in the LSC-AP location request message sent to the E-SMLC when MME is triggered either by GMLC or by HSS to get UE location. the retrieval of the ECGI applies to all client types. no (default) - disable the retrieval and MME sends the ECGI that was received in the most recent S1-AP message.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

retrieveUeLocation - Retrieve UE location

Syntax

--gParmName retrieveUeLocation --gParmValue <string>

Description

This flag determines whether UE location information is retrieved from the eNB. no (default) - indicates that the operator policy is to use the last known user location information. yes - indicates that the WMM operator policy is to use the location reporting procedure (23.401 section 5.9.1) to retrieve the ECGI from the eNB.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

retryPagingForSgsPageRequest - Retry paging for SGs page request

Syntax

--gParmName retryPagingForSgsPageRequest --gParmValue <string>

Description

A value of yes indicates that the MME will perform additional page attempts for each SGs page request received until the UE responds or all page attempts in the paging policy have been exhausted. a value of no indicates that the MME will perform only one page attempt for

each SGs page request received. if needed, the MSC must trigger additional page attempts with new SGs page requests.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

rfspIndexSelectionOption - RFSP Index Selection Option

Syntax

```
--gParmName rfspIndexSelectionOption --gParmValue <string>
```

Description

When this parameter is enabled, then the MME enables using rfspSelOption option, available in SvcAgreementProfile, to decide the way for assigning RFSP Index for the UE

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

rfspIndexValImeiTac - Support RFSP Index Value

Syntax

```
--gParmName rfspIndexValImeiTac --gParmValue <string>
```

Description

GPARM controls the feature activation. By default the parameter shall be disabled. When parameter is enabled, MME shall validate the TAC received in IMEI against provisioned TAC entry and if there is a match, MME shall apply the provisioned RFSP value.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

rfspInS4Sgsn - support RFSP in S4 SGSN

Syntax

```
--gParmName rfspInS4Sgsn --gParmValue <string>
```

Description

support RFSP in S4 SGSN

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

rfspOverridelmeilmsi - RFSP Override by IMEI TAC and IMSI

Syntax

```
--gParmName rfspOverrideImeiImsi --gParmValue <string>
```

Description

RFSP Override by IMEI TAC and IMSI, giving priority to IMEI TAC

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

roamerQosForWpsUserSupport - Roamer QoS Profile for WPS User Support

Syntax

```
--gParmName roamerQosForWpsUserSupport --gParmValue <string>
```

Description

Parameter controls feature activation flag. By default, flag shall be disabled, when enabled MME shall support provisioned Roamer QoS Profile for WPS user.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

roamQciSuppt - Roam QCI Support

Syntax

```
--gParmName roamQciSuppt --gParmValue <string>
```

Description

Upon setting the parameter to TRUE, MME shall check the received inbound roamer GBR QCI value and if the received GBR QCI value is 2, it shall be rejected. With the Default value setting of FALSE, MME shall work as it had before the introduction of this feature.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

roamQosOvdForRan - Roamer QoS Override for RAN

Syntax

```
--gParmName roamQosOvdForRan --gParmValue <string>
```

Description

When this parameter is on, locally configured QoS for Roamers is sent only towards RAN. The default value shall be off.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

rsrUlrPercentForCtrlPlaneServiceReq - Percentage of control-plane service req that MME sends ULR

Syntax

```
--gParmName rsrUlrPercentForCtrlPlaneServiceReq --gParmValue <string>
```

Description

Percentage of control-plane service request that MME sends ULR due to HSS reset.

Type / Range

Integer in the range of 0 to 100

Default

100

Name

rsrUlrPercentForDdnServiceRequest - Percentage of ddn service req that MME sends ULR

Syntax

```
--gParmName rsrUlrPercentForDdnServiceRequest --gParmValue <string>
```

Description

Parameter controls percentage of ddn service req that MME sends ULR

Type / Range

Integer in the range of 0 to 100

Default

100

Name

rsrUlrPercentForServiceReq - Percentage of MO Service Req that MME sends ULR and UE-Act-Ind

Syntax

```
--gParmName rsrUlrPercentForServiceReq --gParmValue <string>
```

Description

Percentage of MO Service Request that MME sends S6a ULR to HSS, and sends SGs UE-Activity-Indication to MSC/VLR due to HSS Reset; MT Service Request (following Paging) does not trigger ULR due to HSS reset.

Type / Range

Integer in the range of 0 to 100

Default

0

Name

rsrVlrScanPercentWithoutOverload - (Reset Subscriber Request) Vlr Scan Percentage Without Overload

Syntax

```
--gParmName rsrVlrScanPercentWithoutOverload --gParmValue <string>
```

Description

When Reset Subscriber Request(RSR)including Subscription-Data or Subscription-Data-Deletion AVP is received in CPPS, VLR scan is triggered in CPPS to find VLR entries that match the RSR user-ids and/or reset-ids AVPs. This VLR scan is triggered periodically to avoid system overload. When the system is not in overload, the number of scanned entries per period is defined in this gParm as percentage of 50000(which is hardcoded). The allowed percentages are [10, 20, ... , 90, 100]. Warning: For Nokia personnel only.

Type / Range

Integer in the range of 10 or 20 or 30 or 40 or 50 or 60 or 70 or 80 or 90 or 100

Default

100

Name

rsrVlrScanPercentWithOverload - (Reset Subscriber Request) Vlr Scan Percentage With Overload

Syntax

--gParmName rsrVlrScanPercentWithOverload --gParmValue <string>

Description

When Reset Subscriber Request(RSR)including Subscription-Data or Subscription-Data-Deletion AVP is received in CPPS, VLR scan is triggered in CPPS to find VLR entries that match the RSR user-ids and/or reset-ids AVPs. This VLR scan is triggered periodically to avoid system overload. If the system is in overload, VLR scan is postponed for the next period. If VLR scan is postponed for 5 consecutive times, VLR scan is triggered in the next period, but the number of scanned entries is reduced. The percentage of the scanned entries is defined in this gParm as percentage of 25000(which is hardcoded). The allowed percentages are [10, 20, ... , 90, 100]. Note that 'rsrVlrScanPercentWithOverload * 25000' must be smaller than 'rsrVlrScanPercentWithoutOverload * 50000'. Warning: For Nokia personnel only.

Type / Range

Integer in the range of 10 or 20 or 30 or 40 or 50 or 60 or 70 or 80 or 90 or 100

Default

100

Name

s102 - S102

Syntax

--gParmName s102 --gParmValue <string>

Description

MME to send S102 redirection command to the IWS obtained from the SRS in case of UE recovery if the UE previously registered with CIOT IWS.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

s1021xRttRcidInSlr - S102 CIOT RCID in SLR

Syntax

--gParmName s1021xRttRcidInSlr --gParmValue <string>

Description

Include CIOT reference cell ID in Slg subscriber location report message.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

s10InterPlmnRelocationQci1Only - S10 Inter PLMN Relocation QCI-1 Only

Syntax

--gParmName s10InterPlmnRelocationQci1Only --gParmValue <string>

Description

Parameter controls feature activation GPARM flag. By default, flag shall be disabled. When enabled, source MME upon receiving inter plmn handover request from eNB will find target MME from TAC reported by eNB, will send forward relocation request to target MME. Target MME upon receiving forward location request will check for the presence of QCI-1 bearer and if QCI-1 bearer is not present target MME shall reject the handover with cause value handover reject target not allowed. Additionally, for TAU, upon receiving context request without QCI-1, Context Response reject with a cause code Target access restricted for the subscriber shall be sent.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

s10mobilityQosParametersUpdate - Inter MME S10 mobility QoS parameters update

Syntax

--gParmName s10mobilityQosParametersUpdate --gParmValue <string>

Description

Indicates whether MME sends modifybearercmd to SGW at the end of TAU when QoS parameters in S6ad ULA are different than those in S10 Context Response or Forward Relocation

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

s11CauseRsrcUnavailPdnDelInSr - S11 Cause Resource Unavailable PDN Delete Session Request in Service Request

Syntax

--gParmName s11CauseRsrcUnavailPdnDelInSr --gParmValue <string>

Description

Parameter controls MME sending Delete Session Request to SGW upon receiving Modify Bearer Response with 'Resources not available'. By default the flag shall be disabled.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

s11ipAddressLoadBalancer - S11 IP Address Load Balancer

Syntax

--gParmName s11ipAddressLoadBalancer --gParmValue <string>

Description

Parameter when enabled prevents MME from creating an S11 link/MO directly to an S11 load balancer. An IP is determined to be a load balancer if the MME has received a Create Session Response with a sender F-TEID for ctrl plane specifying a different IP address than the IP header. If we subsequently receive a Create Session Response from an LB IP and FTEID matches the IP header, an S11 link/MO will not be created. By default, the flag shall be disabled.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

s11sendCidfDuringS1ho - S11 send CIDF during S1HO

Syntax

--gParmName s11sendCidfDuringS1ho --gParmValue <string>

Description

To enable/disable sending of the S11 create indirect forwarding tunnel request message during an S1HO.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

s11SuspendNotificationSentPerLbi - S11 suspend notification sent per LBI

Syntax

```
--gParmName s11SuspendNotificationSentPerLbi --gParmValue <string>
```

Description

Setting to yes enables S11 suspend notification sent per LBI feature. setting to no disables S11 suspend notification sent per LBI feature.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

s11uBearerSetUp - S11-U Bearer Set Up

Syntax

```
--gParmName s11uBearerSetUp --gParmValue <string>
```

Description

Enable MME to setup S11-U bearer for a UE supporting both CP and UP optimization indicates preference for UP Clot optimization but the MME is not provisioned to support UP Clot optimization.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

s11uDlDataBuffering - S11-U bearer downlink data buffering

Syntax

```
--gParmName s11uDlDataBuffering --gParmValue <string>
```

Description

Enable S11-U downlink data buffering to handle collision of downlink data with ESM procedures.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

s13retryDifferentEir - S13 retry different EIR

Syntax

```
--gParmName s13retryDifferentEir --gParmValue <string>
```

Description

When MME gets some errors from EIR (too busy, out of space, etc) or EIR does not respond, MME sends the same request to another EIR in the same PLMN or IMSI range.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

s1ResumeSecurityContextUpdate - S1 Resume Security Context Update

Syntax

```
--gParmName s1ResumeSecurityContextUpdate --gParmValue <string>
```

Description

If the parameter is set to 'Yes' then MME includes the Security Context ID in the S1AP UE CONTEXT RESUME RESPONSE message.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

s1SendAllServedPlmnsToEnb - Send all Served Plmns to Enb

Syntax

```
--gParmName s1SendAllServedPlmnsToEnb --gParmValue <string>
```

Description

When enabled, MME includes all configured home and shared PLMN IDs in the MME Served PLMNs list when MME sends S1AP Configuration Update, or in S1AP Setup Response when MME responds to eNB S1AP Setup Request.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

s1SuspendSecurityContextUpdate - S1 Suspend Security Context Update

Syntax

```
--gParmName s1SuspendSecurityContextUpdate --gParmValue <string>
```

Description

If the parameter is set to 'Yes' then MME includes the Security Context ID in the S1AP UE CONTEXT SUSPEND RESPONSE message.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

s3GnIndirectForwarding - S3 Gn indirect forwarding

Syntax

```
--gParmName s3GnIndirectForwarding --gParmValue <string>
```

Description

Indicates whether inter-rat handover using Gn or S3 always employs indirect data forwarding (ALWAYS), does not employ indirect data forwarding (NEVER), or employs indirect 4290 data forwarding for inter-plmn inter-rat handover (INTER-PLMN/RAT ONLY).

List

- Always - Indicates whether inter-rat handover using Gn or S3 always employs indirect data forwarding.
- Inter-PLMN/RAT Only - Indicates whether inter-rat handover using Gn or S3 employs indirect 4290 data forwarding for inter-plmn inter-rat handover only.

- Never - Indicates whether inter-rat handover using Gn or S3 does not employ indirect data forwarding.

Default

Always

Name

s4Func - allow SGSN S4 functionality (Gr/S6d)

Syntax

--gParmName s4Func --gParmValue <string>

Description

Use this parameter to allow SGSN S4 functionality (Gr/S6d).

List

- OFF - OFF
- ON - ON

Default

OFF

Name

s4SgsnLbo - S4 SGSN support LBO

Syntax

--gParmName s4SgsnLbo --gParmValue <string>

Description

S4 SGSN support local breakout function. With this parameter set, the feature will be enable.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

s6aDraRoamingSupported - S6ad DRA roaming supported

Syntax

--gParmName s6aDraRoamingSupported --gParmValue <string>

Description

When set to yes: use provisioned destinationrealm in remoteendpoint table only for homer. for roamer, use hard-coded 'epc.mnc###.mcc###.3gppnetwork.org' instead of provisioned destinationrealm when set to no: regular S6ad DRA.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

s6adUeTimeZoneRetrieval - S6ad UE time zone retrieval

Syntax

```
--gParmName s6adUeTimeZoneRetrieval --gParmValue <string>
```

Description

This flag is used to enable the retrieval of UE time zone information. no (default) - indicates that UE time zone retrieval is not enabled. yes - indicates that UE time zone retrieval is enabled.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

s6aFaultContinueProcedureWithAkaOptional - S6A fault continue procedure with AKA optional

Syntax

```
--gParmName s6aFaultContinueProcedureWithAkaOptional --gParmValue <string>
```

Description

This flag determines whether processing will continue without authentication when there are no active enabled S6a links for the home PLMN and it has been determined that authentication is optional. no (default) - processing will not continue without authentication. yes - processing will continue without authentication.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

s6aHssUpdateEnbCapacity - S6A HSS update eNB capacity

Syntax

```
--gParmName s6aHssUpdateEnbCapacity --gParmValue <string>
```

Description

This flag identifies how HSS links will be monitored.

List

- AllHSS - All HSS links in use are monitored.
- HomeHSS - HSS links in the home PLMN all rule are monitored.
- NoHSS - No HSS links are monitored.

Default

NoHSS

Name

s6aReportPreviousHssSelectionForHssDown - S6A Report Previous HSS Selection For HSS Down

Syntax

```
--gParmName s6aReportPreviousHssSelectionForHssDown --gParmValue  
<string>
```

Description

Parameter controls feature activation. By default, flag parameter shall be disabled. When enabled, if an HSS selection rule is found (matched) and no HSS available (all links down for hssGrp selection) report 'hssOrEirDown' even when no other HSS selection (fall-back) rule is provisioned. Else continue reporting 'noProvisionedHss' when no HSS can be found and no other available HSS selection rule is provisioned. In other words, when a single no IMSI group does not have (imsiToHss) is found with an HSS association provisioned, MME shall consider the 'noProvisionedHss'. Otherwise existing functionality shall be used, which looks at all IMSI group availability/provisioning before declaring unavailable or down.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

s6aRetryDifferentHss - S6AD retry different HSS

Syntax

```
--gParmName s6aRetryDifferentHss --gParmValue <string>
```

Description

When WMM receives certain errors from HSS (too busy, out of space, etc) or HSS does not respond, WMM sends the same request to another HSS in the same PLMN or IMSI range.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

s6dDiameter - allow SGSN S6d/Gr Selection

Syntax

--gParmName s6dDiameter --gParmValue <string>

Description

Use this parameter to allow SGSN S6d/Gr Selection.

List

- OFF - OFF
- ON - ON

Default

OFF

Name

scefHighLatencySupport - SCEF high latency communication support

Syntax

--gParmName scefHighLatencySupport --gParmValue <string>

Description

This parameter indicates to the MME whether SCEF supports buffering of MT data if UE is not reachable due to power savings.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

secureHandlingOfAttachTau - Secure Handling of Attach and TAU Requests

Syntax

--gParmName secureHandlingOfAttachTau --gParmValue <string>

Description

The parameter is used to enable the secure handling of Attach and TAU Requests if these messages fail integrity checks.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

selectedPgwFormat - Selected PGW Format

Syntax

--gParmName selectedPgwFormat --gParmValue <string>

Description

selectedPgwFormat parameter controls how the MME formulates the selected PGW information inside the MIP6-Agent-Info AVP, which the MME sends on the S6a NOR and the S6a ULR (Active-APN AVP(s)). When 'selectedPgwFormat' is set to value

List

- includePgwFqdn - Include PGW FQDN - MME includes the PGW FQDN on MIP-Home-Agent-Host AVP always when available in the MME. in case of LIPA, as DNS is not applicable i.e. FQDN is not available, the MME includes the PGW IP address(es) on MIP-Home-Agent-Address AVP.
- includePgwIp - Include PGW IP - MME includes (only) the PGW IP address(es) on MIP-Home-Agent-Address AVP.
- includePgwIpFqdn - Include PGW IP FQDN - MME includes the PGW IP address(es) on MIP-Home-Agent-Address AVP. MME additionally also includes the PGW FQDN on MIP-Home-Agent-Host AVP, when the FQDN is available in the MME. in case of LIPA, as DNS is not applicable i.e. the FQDN is not available, the MME includes only the PGW IP address(es) on MIP-Home-Agent-Address AVP.

Default

includePgwFqdn

Name

selectiveVoLteEmerAllowed - Selective volte enablement feature emergency call handling

Syntax

--gParmName selectiveVoLteEmerAllowed --gParmValue <string>

Description

If selectivevolteenabled is enabled, allow/disallow IMS emergency call.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

selectiveVoLteEnabled - Selective volte enablement feature

Syntax

```
--gParmName selectiveVoLteEnabled --gParmValue <string>
```

Description

Enable/Disable selective volte enablement to deny certain types of UEs access to LTE.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sendCallRelOnSlrEmerOrigFailure - Send Call Release On SLR Emergency Call Origination Failure

Syntax

```
--gParmName sendCallRelOnSlrEmerOrigFailure --gParmValue <string>
```

Description

When the parameter is Yes, if the MME sent LRR Emergency Origination to the GMLC and did not receive a successful LRR ACK, the MME will send LRR Emergency Release when the Emergency bearer is removed or deleted. When the parameter is No, LRR Emergency Release is sent after receiving a successful LRR ACK for the LRR Emergency Origination.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sendDdnAckWithContextNotFoundForAbnormalSgwSwitchover - Send DDN Ack with Context Not Found for Abnormal SGW Switchover

Syntax


```
--gParmName sendDdnAckWithContextNotFoundForAbnormalSgwSwitchover --  
gParmValue <string>
```

Description

Parameter controls feature activation. By default, flag shall be disabled. When enabled, MME upon getting a DDN with a CV of 254, and if the TEID is recognized but the DDN is from the wrong SGW, the MME shall respond with a DDN Ack, with a TEID of 0 (zero), with the same sequence number as in the DDN, and with a CV of 64 (context not found), back to the sending (wrong) SGW, so that SGW will know to clean up automatically.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sendDeleteSessionReqForIntersystemRau - Send Delete Session Request for Intersystem RAU

Syntax

```
--gParmName sendDeleteSessionReqForIntersystemRau --gParmValue  
<string>
```

Description

Parameter controls sending Delete Session Request to SGW upon receiving Cancel Location Request from HSS with cancellation Type=MME Update Procedure during Gn based Inter System RAU and S3 based Inter System RAU with SGW relocation. By default the flag shall be disabled.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sendEquivalentPlmnListOnlytoEnbInHandoverRestrictionList - Send Equivalent Plmn List Only to Enb in Handover Restriction List

Syntax

```
--gParmName sendEquivalentPlmnListOnlytoEnbInHandoverRestrictionList  
--gParmValue <string>
```

Description

Parameter controls feature activation. By default, the flag shall be disabled. When enabled, MME shall send Equivalent Plmn List only to eNB in Handover Restriction List. Sending to the UE shall be suppressed even if ePLMN List Name is provisioned. When gparm is disabled, if ePLMN List Enb is empty, ePLMN List shall be copied.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sendEsmDummyAttachPdnRejected - Send ESM Dummy Attach PDN Rejected

Syntax

--gParmName sendEsmDummyAttachPdnRejected --gParmValue <string>

Description

MME sends Attach Accept with ESM Dummy esmMsgContainer, if PDN is Rejected and Attach is accepted, for Attach Request with PDN Connectivity Request, from an UE supporting and allowed for, registration without PDN Values: Yes - Send Attach Accept with esmMsgContainer -> ESM DUMMY message No - Send Attach Accept with esmMsgContainer -> PDN CONNECTIVITY REJECT message

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

sendForbiddenTailistDCsOnly - Send forbidden TAI list for data centric CS voice only

Syntax

--gParmName sendForbiddenTailistDCsOnly --gParmValue <string>

Description

UE usage setting is data centric and voice domain preference is CS only. if sendforbiddentailistdcsonly is yes, send the forbidden TAI list.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sendForbiddenTailListDCsPrefer - Send forbidden TAI list for data centric CS voice preferred

Syntax

--gParmName sendForbiddenTailListDCsPrefer --gParmValue <string>

Description

UE usage setting is data centric and voice domain preference is CS preferred, IMS secondary. if sendforbiddentailstdcsprefer is yes, send the forbidden TAI list.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sendForbiddenTailListDImSOnly - Send forbidden TAI list for data centric IMS PS voice only

Syntax

--gParmName sendForbiddenTailListDImSOnly --gParmValue <string>

Description

UE usage setting is data centric and voice domain preference is IMS only. if sendforbiddentailstdimonly is yes, send the forbidden TAI list.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sendForbiddenTailListDImSPrefer - Send forbidden TAI list for data centric IMS PS voice preferred

Syntax

--gParmName sendForbiddenTailListDImSPrefer --gParmValue <string>

Description

UE usage setting is data centric and voice domain preference is IMS preferred, CS secondary. if sendforbiddentailstdimprefer is yes, send the forbidden TAI list.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sendForbiddenTailListNone - Selective volte enablement feature send forbidden TAI list none

Syntax

```
--gParmName sendForbiddenTailListNone --gParmValue <string>
```

Description

UE does not indicate both UE usage setting and voice domain preference. if sendforbiddentailistnone is yes, send the forbidden TAI list.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sendForbiddenTailListVCsOnly - Send forbidden TAI list for voice centric CS voice only

Syntax

```
--gParmName sendForbiddenTailListVCsOnly --gParmValue <string>
```

Description

UE usage setting is voice centric and voice domain preference is CS only. if sendforbiddentailistvcsonly is yes, send the forbidden TAI list.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sendForbiddenTailListVCsPrefer - Send forbidden TAI list for voice centric CS voice preferred

Syntax

```
--gParmName sendForbiddenTailListVCsPrefer --gParmValue <string>
```

Description

UE usage setting is voice centric and voice domain preference is CS preferred, IMS secondary. if sendforbiddentailistvcprefer is yes, send the forbidden TAI list.

List

- No - Boolean value of no.

- Yes - Boolean value of yes.

Default

No

Name

sendForbiddenTailListVImOnly - Send forbidden TAI list for voice centric IMS PS voice only

Syntax

```
--gParmName sendForbiddenTailListVImOnly --gParmValue <string>
```

Description

UE usage setting is voice centric and voice domain preference is IMS only. if
sendforbiddentailvimsonly is yes, send the forbidden TAI list.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sendForbiddenTailListVImPrefer - Send forbidden TAI list for voice centric IMS PS voice
preferred

Syntax

```
--gParmName sendForbiddenTailListVImPrefer --gParmValue <string>
```

Description

UE usage setting is voice centric and voice domain preference is IMS preferred, CS secondary.
if sendforbiddentailvimprefer is yes, send the forbidden TAI list.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sendFwdRelCancelAndHoPrepFailure - Send Forward Relocation Cancel and HO Preparation
Failure

Syntax

```
--gParmName sendFwdRelCancelAndHoPrepFailure --gParmValue <string>
```

Description

Parameter controls sending Forward Relocation Cancel and HO preparation failure message upon indirect forwarding failure at SGW. By default the flag shall be disabled

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sendingMBR - Sending MBR

Syntax

--gParmName sendingMBR --gParmValue <string>

Description

Sending modify bearer request.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

sendLtemRatTypeToScef - sendLtemRatTypeToScef

Syntax

--gParmName sendLtemRatTypeToScef --gParmValue <string>

Description

Specifies whether the MME can send the 'LTE-M' RAT to the SCEF in T6a CMR message.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sendLtemRatTypeToSgw - sendLtemRatTypeToSgw

Syntax

--gParmName sendLtemRatTypeToSgw --gParmValue <string>

Description

Specifies whether the MME is to send the 'LTE-M' RAT Type to SGWs for UEs that are identified as LTE-M..

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sendMappedGuami - sendMappedGuami

Syntax

```
--gParmName sendMappedGuami --gParmValue <string>
```

Description

Current parameter enables sending in S1 Setup Response mapped GUMMEI(s) pointing to the AMF GUAMI(s) for the associated PLMN(s) that coexist in both 4G and 5G at local CMM node

List

- DISABLE - Disable
- ENABLE - Enable

Default

DISABLE

Name

sendMaximumUeAvailabilityTimeInNor - Send Maximum UE Availability Time in NOR

Syntax

```
--gParmName sendMaximumUeAvailabilityTimeInNor --gParmValue <string>
```

Description

Parameter controls support for sending maximum ue availability time to hss in NOR message. By default the flag shall be disabled.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sendNasRanCauseOnS11 - Send NAS RAN cause on S11

Syntax

```
--gParmName sendNasRanCauseOnS11 --gParmValue <string>
```

Description

Indicates whether (Yes) or not (No) to send the NAS or RAN failure indication code on the S11 interface to the SGW.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sendNorWithPgwInfoAfterTauUla - Send NOR with PGW information after TAU ULA

Syntax

```
--gParmName sendNorWithPgwInfoAfterTauUla --gParmValue <string>
```

Description

Send NOR with PGW information after TAU ULA.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sendPdnConnRejWithReactReq - Send PDN Connectivity Reject With CC#54 (PDN connection does not exist)

Syntax

```
--gParmName sendPdnConnRejWithReactReq --gParmValue <string>
```

Description

Parameter controls sending PDN Connectivity Reject with cause code #54 (PDN connection does not exist).. By default the flag shall be No.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sendRabrDuringDoubleS1ServiceRequest - Send Release Access Bearer Request During Double S1 Service Request

Syntax

```
--gParmName sendRabrDuringDoubleS1ServiceRequest --gParmValue  
<string>
```

Description

When enabled, upon receiving Double S1 Service Request, MME shall release the old S1 connection before sending any Modify Bearer Request to SGW. MME shall send Release Access Bearer Request to SGW to release the old S1U connection on SGW for the UE. As a result SGW shall not send data over the old S1U to the enB during the Double S1 procedure. By default, flag shall be disabled.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sendRabUponEcmConnectedDdn - Send RAB upon ECM-CONNECTED DDN

Syntax

```
--gParmName sendRabUponEcmConnectedDdn --gParmValue <string>
```

Description

Specifies whether MME shall send DDN ack with cause 'Unable to page UE', followed by a release access bearer (RAB) to the SGW upon the receipt of a DDN for a ECM-CONNECTED UE.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sendRequestedApn - Send requested APN

Syntax

```
--gParmName sendRequestedApn --gParmValue <string>
```

Description

This parameter indicates to send either the UE requested APN-NI or the replacement APN-NI.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sendSgw3gppNasRanCause - Send SGW 3GPP NAS ran cause

Syntax

--gParmName sendSgw3gppNasRanCause --gParmValue <string>

Description

Enable/disable inclusion of NAS or RAN failure indication cause code on S11 interface to SGW.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sendSgw3gppS11cc - Send SGW 3GPP S11 cause code

Syntax

--gParmName sendSgw3gppS11cc --gParmValue <string>

Description

Sent in GTPv2 message to SGW across S11 interface. a value of 'Yes' indicates that MME shall support and send cause code 110 to SGW.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sendUeUnreachable - Send UE UN_REACHABLE

Syntax

--gParmName sendUeUnreachable --gParmValue <string>

Description

Parameter controls MME behavior related to SGs Paging and specifically controlling if/when an SGs Paging Reject is sent (with UE_UNREACHABLE). Parameter allows customization of when an SGs Paging Reject is sent with a cause of UE_UNREACHABLE. When the global parameter flag is disabled, an SGs Paging Reject with the UE_UNREACHABLE cause is sent when the MME is unable to page the UE (i.e., due to UE inactivity or MME paging suppression).

When the global parameter flag is enabled, the MME will also send an SGs Paging Reject with the UE_UNREACHABLE when MME paging times out trying to reach the UE. By default, flag shall be disabled.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sendUpdatedRfspReceivedInIdr - Send Updated RFSP Received in IDR

Syntax

```
--gParmName sendUpdatedRfspReceivedInIdr --gParmValue <string>
```

Description

Parameter controls sending updated Subscriber Profile ID IE to eNb in UE Context Modification message. By default, the flag shall be disabled. When enabled, MME shall send updated Subscriber Profile ID IE that it receives in IDR message from HSS to eNb in UE Context Modification message.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

serveS6dClrBySgsnOnly - Serve S6d CLR by SGSN Only

Syntax

```
--gParmName serveS6dClrBySgsnOnly --gParmValue <string>
```

Description

Current parameter controls the handling of s6d CLR (cancellation-type= initial attach procedure) with the 'S6a/S6d-Indicator' flag indicating the SGSN as the affected part of combo node. When the parameter is disabled and the UE isn't found in SGSN side, then the CLR is forwarded to the MME side for 4G detach. When the parameter is enabled then the CLR isn't forwarded to the MME side for 4G detach

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

serviceGapControlActive - Service Gap Control Active

Syntax

--gParmName serviceGapControlActive --gParmValue <string>

Description

Global Feature Flag for the Service Gap Control Feature

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

serviceGapControlSendBackoffTimer - Service Gap Control Backoff Timer

Syntax

--gParmName serviceGapControlSendBackoffTimer --gParmValue <string>

Description

Specifies whether the MME is to provide the UE with a backoff timer in case the UE access is blocked due to the Service Gap Control Feature

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

serviceModifiersWithLabelMatching - Service Modifiers with Label Matching

Syntax

--gParmName serviceModifiersWithLabelMatching --gParmValue <string>

Description

Parameter controls feature activation flag. By default, flag shall be disabled. When enabled, MME shall support the following four service combinations: +nc-nr+ue, +ue, +nc-nr, and no modifiers.

List

- No - Boolean value of no.

- Yes - Boolean value of yes.

Default

No

Name

serviceReqRateLimitOverride - ServiceReqRateLimitOverride

Syntax

```
--gParmName serviceReqRateLimitOverride --gParmValue <string>
```

Description

Number of UE initiated service requests allowed per sec per CPPS after which CPPS starts dropping service requests until rate is below limit.

Type / Range

Integer in the range of systemDefault or 0 to 6,999

Default

systemDefault

Name

serviceReqRateLimitOverrideSRS - ServiceReqRateLimitOverrideSRS

Syntax

```
--gParmName serviceReqRateLimitOverrideSRS --gParmValue <string>
```

Description

Number of service requests (that requiring SRS) allowed per sec per CPPS, after which CPPS starts rejecting service request until rate is below limit.

Type / Range

Integer in the range of systemDefault or 0 to 999

Default

systemDefault

Name

servingNodeInLRR - Serving node in LRR

Syntax

```
--gParmName servingNodeInLRR --gParmValue <string>
```

Description

If this parameter is set to 'Yes' then then MME shall include the serving node AVP in LRR of all location scenarios.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sessionRestoration - Session Restoration

Syntax

```
--gParmName sessionRestoration --gParmValue <string>
```

Description

Enable Session Restoration

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

setCapdegradeOn2gcipherGea1 - SetCapdegradeOn2gcipherGea1

Syntax

```
--gParmName setCapdegradeOn2gcipherGea1 --gParmValue <string>
```

Description

Set 2G capacity degrade percentage for GEA1 ciphering algorithm.

Type / Range

Integer in the range of systemDefault or 0 to 100

Default

systemDefault

Name

setCapdegradeOn2gcipherGea2 - SetCapdegradeOn2gcipherGea2

Syntax

```
--gParmName setCapdegradeOn2gcipherGea2 --gParmValue <string>
```

Description

Set 2G capacity degrade percentage for GEA2 ciphering algorithm.

Type / Range

Integer in the range of systemDefault or 0 to 100

Default

systemDefault

Name

setCapdegradeOn2gcipherGea3 - SetCapdegradeOn2gcipherGea3

Syntax

```
--gParmName setCapdegradeOn2gcipherGea3 --gParmValue <string>
```

Description

Set 2G capacity degrade percentage for GEA3 ciphering algorithm.

Type / Range

Integer in the range of systemDefault or 0 to 100

Default

systemDefault

Name

setIaiForGutiAttach - Set IAI for GUTI attach

Syntax

```
--gParmName setIaiForGutiAttach --gParmValue <string>
```

Description

Set initialattachindication in ULR for GUTI attach when UE IMSI is not received directly from the UE during the ongoing attach procedure.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sgs - SGS

Syntax

```
--gParmName sgs --gParmValue <string>
```

Description

Recovery of 3GPP CSFB services of UE using SRS.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sgsAssStateCsfbEmergencyCall - SGS ass state CSFB emergency call

Syntax

```
--gParmName sgsAssStateCsfbEmergencyCall --gParmValue <string>
```

Description

Configure to change the SGs association state to SGS-NULL.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sgsLinkLock - SGS link lock

Syntax

```
--gParmName sgsLinkLock --gParmValue <string>
```

Description

Enable/Disable support of MME to change SCTP state when locking SGs links.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sgsLiteDetachIndicationAttempts - Detach indication attempts

Syntax

```
--gParmName sgsLiteDetachIndicationAttempts --gParmValue <string>
```

Description

Configure maximum number of transmit attempts for SGsLite-IMSI-DETACH-INDICATION message.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

sgsLiteLocationUpdateAttempts - Location update attempts

Syntax

```
--gParmName sgsLiteLocationUpdateAttempts --gParmValue <string>
```


Description

Configure maximum number of transmit attempts for SGsLite-LOCATION-UPDATE message.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

sgsLiteUplinkDataAttempts - Uplink data attempts

Syntax

```
--gParmName sgsLiteUplinkDataAttempts --gParmValue <string>
```

Description

Configure maximum number of transmit attempts for SGsLite-UPLINK-UNITDATA message.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

sgsnCsfbReturnLastPlmn - SGSN CSFB return last PLMN

Syntax

```
--gParmName sgsnCsfbReturnLastPlmn --gParmValue <string>
```

Description

Configure SGSN to support return UE to last PLMN by including the last used LTE PLMN ID into the equivalent PLMN list provided to the UE in RAU accept message.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sgsnDiscoveryMethod - SGSN discovery method

Syntax

```
--gParmName sgsnDiscoveryMethod --gParmValue <string>
```

Description

Method used to discover SGSN IP address for inter-rat handover to UTRAN/GERAN.

List

- Use S-NAPTR Procedure - The MME shall perform a DNS S-NAPTR query to determine the IP address of the SGSN.
- Use S-NAPTR and fallback - MME falls back to pre-rel8 query if MME receives NAPTR records with missing interfaces.
- Use pre-Rel8 FQDN - The MME shall perform a DNS query with pre-rel 8 FQDN determine the IP address of the SGSN.

Default

Use S-NAPTR Procedure

Name

sgsnId - SGSN ID

Syntax

```
--gParmName sgsnId --gParmValue <string>
```

Description

This parameter defines the SGSN identifier, SGID. Possible identifier values range from 1 to 31. 0 means not used.

Type / Range

Integer in the range of 0 to 31

Default

15

Name

sgsnInformImplicitlyDetachedUEs - SGSN informs Implicitly Detached UEs

Syntax

```
--gParmName sgsnInformImplicitlyDetachedUEs --gParmValue <string>
```

Description

This parameter controls if the SGSN shall take action to inform UEs that have been implicitly detached due to PAPS reset that their MM context has been lost.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sgsnIntPreIuGbSessionResiliency - SGSN support for Internal Preparation for Iu Gb Session Resiliency

Syntax

```
--gParmName sgsnIntPreIuGbSessionResiliency --gParmValue <string>
```

Description

Enable SGSN support for Internal Preparation for Lu Gb Session Resiliency. Warning: For Nokia personnel only.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sgsnIpAddrCmpSgwSel - SGSN IP Address Comparison During SGW Selection

Syntax

```
--gParmName sgsnIpAddrCmpSgwSel --gParmValue <string>
```

Description

Current gParm enables S4 SGSN recognizing that same SGW has been selected during inter-PLMN mobility (RAU and handover) by comparing the IP addresses of the source and target SGW(s).

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sgsnIpv4AddressForUserTraffic - SGSN IPv4 address for user traffic

Syntax

```
--gParmName sgsnIpv4AddressForUserTraffic --gParmValue <string>
```

Description

SGSN IPv4 address for user traffic to support the SGSN context acknowledge message sent from the MME during a gn/gp SGSN to MME TAU

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Default

127.0.0.1

Name

sgsnMonLogSheddingDuringOverload - Monitor Logs Shedding During Overload

Syntax

```
--gParmName sgsnMonLogSheddingDuringOverload --gParmValue <string>
```

Description

This parameter handles shedding of SGSN Monitoring Logs during overload in PAPS. Warning: For Nokia personnel only.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

sgsnNumberSending - SGSN Number Sending

Syntax

```
--gParmName sgsnNumberSending --gParmValue <string>
```

Description

This parameter is to control sending SGSN number to HSS via diameter to HSS. By default, this parameter is set to No, which means Not Send.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sgsnPoolId - SGSN pool ID

Syntax

```
--gParmName sgsnPoolId --gParmValue <string>
```

Description

This parameter defines the SGSN pool identifier. Possible values range from 0 to 3.

Type / Range

Integer in the range of 0 to 3

Default

0

Name

sgsnUseCclnApnForR8DnsPgw - SGSN use of Charging Characteristics in APN for R8 DNS query for P-GW selection

Syntax

```
--gParmName sgsnUseCcInApnForR8DnsPgw --gParmValue <string>
```

Description

Defines if SGSN will add Charging Characteristics in APN for Rel-8 DNS query for P-GW selection. - 'No' (default) CC is not added - 'Yes' CC added when APN is in the apnList and marked to use CC for Rel-8 DNS P-GW selection

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sgsSvMscPreference - SGS Sv MSC preference

Syntax

```
--gParmName sgsSvMscPreference --gParmValue <string>
```

Description

Configure UE assignment preference for MSC supporting both SGs and Sv interface.

List

- All UE - MME uses the MSCs supporting SGs and Sv interface only for the SRVCC capable UEs and these MSC are excluded for the UEs that do not support SRVCC.
- SRVCC UE Only - MME will use these MSC for both SRVCC capable and SRVCC not capable UEs.

Default

All UE

Name

sgwIsolationFaultThreshold - SGW isolation fault threshold

Syntax

```
--gParmName sgwIsolationFaultThreshold --gParmValue <string>
```

Description

This command is used to provision SGW fault threshold for the SGW isolation. if the number of the SGW faults (number of S11/S4 create session response timeouts) exceed the provisioned threshold in a provisioned duration timer then WMM will isolate the SGW from the selection for provisionable duration.

Type / Range

Integer in the range of 1 to 1000

Default

50

Name

sgwRelocationOnMbrFailure - SGW relocation on MBR failure

Syntax

```
--gParmName sgwRelocationOnMbrFailure --gParmValue <string>
```

Description

Enable SGW relocation if MME times out waiting for SGW response to modify bearer requests or modify access bearer request.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sgwRelocationRestriction - SGW relocation restriction

Syntax

```
--gParmName sgwRelocationRestriction --gParmValue <string>
```

Description

If the feature is activated then, MME shall not perform SGW relocation for TAU request, X2 HO, S1HO and MME relocations.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sgwSelectionWithSnaptr - SGW selection with snaptr

Syntax

```
--gParmName sgwSelectionWithSnaptr --gParmValue <string>
```

Description

Enable/Disable support for SGW selection with snaptr.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sgwSelForHomeRoutedGwSelMode - SGW Selection for GW Selection Mode 1

Syntax

--gParmName sgwSelForHomeRoutedGwSelMode --gParmValue <string>

Description

Parameter controls feature activation flag. By default the flag shall be disabled. When enabled MME shall then if the UE is a home-routed roamer, the s8-gtp order and preference for each of the SGW candidates will be used to determine which SGW to select instead of the order and preference for s5-gtp, and also the selection of an SGW that supports only s8-gtp and not s5-gtp will be supported.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sgwToEnblpv4v6TransportLayerAddress - SGW IPv4v6 Transport Layer Address IP Type

Syntax

--gParmName sgwToEnblpv4v6TransportLayerAddress --gParmValue <string>

Description

MME passes SGW S1-u IPv4v6 Transport Layer Address to the eNB/gNB if it received this IP Type from the SGW and this parameter is set to YES. Otherwise MME matches S1-C IP Address Type and passes it to the eNB/gNB.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

shortDataService - Short data service

Syntax

--gParmName shortDataService --gParmValue <string>

Description

Enable/disable short data service at MME level.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

signalingOnlyUe - Signaling only UE

Syntax

--gParmName signalingOnlyUe --gParmValue <string>

Description

Enable/disable feature 'MME support for UE with signalling connections only.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

singlePdpPerApnSgsn - SGSN allows one PDP Per APN

Syntax

--gParmName singlePdpPerApnSgsn --gParmValue <string>

Description

This parameter enables restriction the number of active PDPs towards same APN to one. When the parameter is not set (by default), the UE may have multiple active PDP(s) towards same APN. When the parameter is set, then UE is allowed having only one active PDP towards same APN.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

singleThreadOlcSheddingEffectiveness - Single Thread OLC Shedding Effectiveness

Syntax

--gParmName singleThreadOlcSheddingEffectiveness --gParmValue
<string>

Description

When enabled, OLC will determine the effectiveness of shedding during a single thread CPU overload. If shedding is ineffective, then OLC will turn shedding off, else the shedding will continue. This feature applies to IPDS and CPPS.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

skipSubscriptionData - Skip subscription data

Syntax

```
--gParmName skipSubscriptionData --gParmValue <string>
```

Description

HSS may skip subscription data in the ULA if subscription data has not changed since last successful update.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

skipUlrForSgsnToMmeTau - Skip ULR for SGSN to MME Tracking Area Update

Syntax

```
--gParmName skipUlrForSgsnToMmeTau --gParmValue <string>
```

Description

Parameter controls sending NOR message to HSS at the end of SGSN to MME TAU procedure (applies to both Gn and S3 SGSN) instead of ULR, for the case where subscription data is already available in MME. By default the flag shall be disabled

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

slgRetryLrrEmergencySameDra - SLG Retry LRR Emergency Same DRA

Syntax

```
--gParmName slgRetryLrrEmergencySameDra --gParmValue <string>
```

Description

This parameter will control MME to retry the same DRA/GMLC when MME receives Diameter error 3002 (DIAMETER_UNABLE_TO_DELIVER). If the parameter is enabled, MME will retry the diameter message in SLG the same DRA for LRR related with emergency. If the parameter is disabled, MME will keep existing function (i.e. no retry). By default, the parameter is disabled.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

slrEmergencyInitiationDedicatedBearer - SLR emergency initiation based on dedicated bearer

Syntax

```
--gParmName slrEmergencyInitiationDedicatedBearer --gParmValue  
<string>
```

Description

SLR emergency initiation based on dedicated bearer.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

slrSenderDuringHandover - SLR sender during handover

Syntax

```
--gParmName slrSenderDuringHandover --gParmValue <string>
```

Description

Specifies whether the source or target MME sends the SLR to the GMLC during handover. the default is the target MME.

List

- Source MME - The source MME shall send SLR during handover.

- Target MME - The target MME shall send SLR during handover.

Default

Target MME

Name

smfPgwSelSupport - SMF Combo PGW-C Selection support

Syntax

--gParmName smfPgwSelSupport --gParmValue <string>

Description

Selection of combo PGW-C is done using only HSS subscription information as criteria, without considering UE N1 mode support

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

smsInMme - SMS in MME

Syntax

--gParmName smsInMme --gParmValue <string>

Description

Activate the SMS in MME feature. This feature enables the possibility to deploy the SMS service without the use of the CS domain.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sourceMmeRelocSgsDetach - Source MME reloc send SGs detach

Syntax

--gParmName sourceMmeRelocSgsDetach --gParmValue <string>

Description

This flag controls whether the source MME for a SGS associated UE (i.e. a combined attach UE) will send an SGs detach message to the MSC/VLR following a successful relocation. yes - source MME sends a SGs detach message to the MSC/VLR following a successful relocation

no - source MME do not sends a SGs detach message to the MSC/VLR following a successful relocation.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

spcmdSheddingDuringOverload - SPCMD Shedding During Overload

Syntax

--gParmName spcmdSheddingDuringOverload --gParmValue <string>

Description

This parameter handles shedding of SGSN PCMD records during overload in PAPS

List

- Do not drop SPCMD during overload - Do not drop SPCMD during overload
- Drop SM SPCMD in major overload - Drop SM related SPCMD records in major overload
- Drop all SPCMD in critical overload - Drop all SPCMD records in critical overload (SM and MM related)

Default

Do not drop SPCMD during overload

Name

srvcc - SRVCC

Syntax

--gParmName srvcc --gParmValue <string>

Description

MME support of SRVCC with the parameters obtained from the SRS.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

srvccCollSendDbrCc110 - SRVCC Collision Send Del brr response

Syntax

```
--gParmName srvccCollSendDbrCc110 --gParmValue <string>
```

Description

Global Parameter flag 'srvccCollSendDbrCc110' controls the feature activation. By default the flag shall be disabled. When enabled, MME shall support sending Delete Bearer Response with Cause Code 110 during SRVCC collision.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

srvccForAllImsEme - SRVCC for all IMS EME

Syntax

```
--gParmName srvccForAllImsEme --gParmValue <string>
```

Description

Use the srvccforallimseme to ensure all UEs making an IMS emergency call (and supporting SRVCC) will be allowed to perform SRVCC.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

staticGwSelection - STATIC GGSN/GW SELECTION

Syntax

```
--gParmName staticGwSelection --gParmValue <string>
```

Description

StaticGwSelection shall be used to control static GGSN/GW selection for SGSN. When it's set, then the new function f52070-01 (configured GWs) will apply. Otherwise, exist function will be used.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

stdPraActionAddProcs - Standard PRA Action Additional Procedures

Syntax

```
--gParmName stdPraActionAddProcs --gParmValue <string>
```

Description

Enables MME to react upon received PRA action for standard PRA, when MME receives new PRA action from SGW/PGW during 5GS to EPS mobility, or during PDN setup for a PDN being handed over from ePDG. Applied when Enable Presence Reporting Area functionality is set to Yes. For modify action also the PRA Modify Enabling parameter must be set to Yes.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

stdsNotifyResponse - Stds notify response

Syntax

```
--gParmName stdsNotifyResponse --gParmValue <string>
```

Description

Standards specified MME behavior upon receiving notify response from HSS.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

stopWarningIndication - Stop warning indication

Syntax

```
--gParmName stopWarningIndication --gParmValue <string>
```

Description

Sending of STOP WARNING MESSAGE TO the CBC.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

subscriberDataExtraction - Subscriber Data Extraction

Syntax

```
--gParmName subscriberDataExtraction --gParmValue <string>
```

Description

Enable/disable periodic subscriber data extraction

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

supportAlternateGwReselection - Support alternate gw reselection feature

Syntax

```
--gParmName supportAlternateGwReselection --gParmValue <string>
```

Description

If enabled, the feature does not check the source bit of cause IE received in create session response for reselection of GW on failures instead MME uses the cause code to determine GW reselection.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

supportApnCorrection - Support APN correction

Syntax

```
--gParmName supportApnCorrection --gParmValue <string>
```

Description

Configure whether or not APN correction is enabled.

List

- No - Boolean value of no.

- Yes - Boolean value of yes.

Default

No

Name

supportAsstData - Support Assistance Data

Syntax

--gParmName supportAsstData --gParmValue <string>

Description

Parameter controls the support for Assistance Data for Recommended Cells info that MME receives info in the S1 CONTEXT RELEASE COMPLETE message shall store it in the VLR and shall include the IE in the paging message. The IE in UE CONTEXT RELEASE COMPLETE that contains the info is 'Information on Recommended Cells' and eNBs for Paging. The info from this IE shall be stored in VLR until a new UE CONTEXT RELEASE COMPLETE message is received. The contents of this IE shall be used to populate Assistance Data for Recommended Cells IE of the Assistance Data for Paging of S1AP PAGING. By default the flag shall be disabled.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

supportCatM1b - Support category M1 CE mode b UE

Syntax

--gParmName supportCatM1b --gParmValue <string>

Description

Support category M1 CE mode b UE.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

supportCmasRestriction - Support cmas restriction

Syntax

--gParmName supportCmasRestriction --gParmValue <string>

Description

Feature activation per MME.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

supportCppsAutoPersonalityChange - Support CPPS Auto Personality Change

Syntax

```
--gParmName supportCppsAutoPersonalityChange --gParmValue <string>
```

Description

Parameter controls whether CMM changes CPPS personality between EMMS and AMMS based on 4G and 5G traffic load.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

supportCsgChangeReporting - Support CSG change reporting

Syntax

```
--gParmName supportCsgChangeReporting --gParmValue <string>
```

Description

Is support for CSG change reporting to be provided?.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

supportDedicatedBearerUli - Support dedicated bearer ULI

Syntax

```
--gParmName supportDedicatedBearerUli --gParmValue <string>
```

Description

Enable or disable ULI enhancement for dedicated bearer activation, update and delete bearer response messages.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

supportDelayedNor - Support delayed NOR

Syntax

```
--gParmName supportDelayedNor --gParmValue <string>
```

Description

Delay NOR when the HSS is unavailable.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

supportDelayedUlr - Support delayed ULR

Syntax

```
--gParmName supportDelayedUlr --gParmValue <string>
```

Description

Delay ULR when the HSS is unavailable.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

supportDfltBearerOperDefQci - Support Default Bearer Operator Defined Qci

Syntax

```
--gParmName supportDfltBearerOperDefQci --gParmValue <string>
```

Description

Enable or Disable the support for Default Bearers to use Operator Defined QClis

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

supportDnsSrv - Support DNS SRV

Syntax

--gParmName supportDnsSrv --gParmValue <string>

Description

MME support for service (SRV) resource records (RR) for MME DNS discovery of MME/PGW/SGW/SGSN.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

supportDupAppProtDnsRec - Support Duplicate 'app-protocol' DNS Records

Syntax

--gParmName supportDupAppProtDnsRec --gParmValue <string>

Description

When this parameter is enabled, the MME does not discard DNS records carrying multiple app-protocol entries of the same app-protocol value, and the MME assumes that the record supports the superset of the supported services across the multiple app-protocol entries.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

supportEdrxRelatedRat - Support eDRX Related RAT

Syntax

```
--gParmName supportEdrxRelatedRat --gParmValue <string>
```

Description

Parameter controls eDRX related RAT support by MME. By default the flag shall be disabled. When the gparm is off existing behavior (i.e. if eDRX-Cycle-Length-Withdrawal is provided the edrx cycle length is removed for both nb-iot and e-utran). If the gparm is enabled, then the only edrx-cycle length deleted will be what the AVP specified. By default, the flag shall be disabled.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

supportForAiMLFunctionalities - Support for AI/ML functionalities in CMM

Syntax

```
--gParmName supportForAiMLFunctionalities --gParmValue <string>
```

Description

This global parameter can be used to enable or disable Artificial Intelligence and Machine Learning (AIML) capabilities in a CMM. By default the global parameter should be disabled (set to No). The AIML functionalities are only applicable to CNF based CMMs, hence, this parameter should always be disabled in VNF based CMMs. When enabled, the Machine Learning Intelligence Service (MLIS) will setup client-server connections with call processing pods for data streaming. When enabled, MLIS pod will collect necessary data to serve AIML algorithms/models in the CMM.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

supportFullUeRestoration - Full 4G SRS Restoration

Syntax

```
--gParmName supportFullUeRestoration --gParmValue <string>
```

Description

If this gparm is enabled full UE Contexts will be stored in the Backup. Restorations will take place on the Full UE Contents. This feature allows rolling update of DBS for 4G in Containers

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

supportIimeiDigitValidation - Support IMEI digit validation

Syntax

--gParmName supportIimeiDigitValidation --gParmValue <string>

Description

This command enables/disables IMEI validation.

List

- allow - Validate IMEI and if invalid do not store in VLR and continue request (or in case IMEI is required procedure is rejected).
- no - IMEI validation is not done.
- reject - Validate IMEI and if invalid reject request.

Default

no

Name

supportImsIimeiApnEnh - Support IMS IIMEI APN enhancement

Syntax

--gParmName supportImsIimeiApnEnh --gParmValue <string>

Description

Parameter controls feature activation. By default the parameter shall be disabled. When enabled, MME shall use the replacement APN NI in the DNS query. Replacement APN NI shall be sent in the Create Session Request towards SGW. QoS and other parameters shall be used from the subscription data. In addition, MME shall send the NOR message with default APN context identifier

List

- EnhDisable - Disabled - Parameter controls feature activation. When EnhDisable is enabled, feature shall be disabled.
- EnhEnableDefault - Enable and use default APN context ID - When EnhEnableDefault is enabled, MME shall send NOR message with default APN context identifier
- EnhEnableSuppress - Enable and Suppress NOR - When EnhEnableSuppress is enabled, MME shall suppress the NOR

Default

EnhDisable

Name

supportMmeSrvcc - MME Supporting 5G-SRVCC

Syntax

--gParmName supportMmeSrvcc --gParmValue <string>

Description

Enables the MME functionality to act in the intermediary role of the MME_SRVCC (MME supporting 5G-SRVCC) to perform the 5G-SRVCC handover for regular voice calls from NG-RAN/5GS to 3GPP UTRAN/CS domain.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

supportMmeSrvccEmergency - Support MME_SRVCC Emergency

Syntax

--gParmName supportMmeSrvccEmergency --gParmValue <string>

Description

Enables the MME functionality to act in the intermediary role of the MME_SRVCC (MME supporting 5G-SRVCC) to perform the 5G-SRVCC handover for emergency calls from NG-RAN/5GS to 3GPP UTRAN/CS domain.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

supportModifyAccessBearerRequest - Support modify access bearer request

Syntax

--gParmName supportModifyAccessBearerRequest --gParmValue <string>

Description

Support for S11 modify access bearer request message.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

supportMsisdnInEcrMsg - Support MSISDN in ECR message

Syntax

```
--gParmName supportMsisdnInEcrMsg --gParmValue <string>
```

Description

When set to yes, MME shall populate MSISDN AVP in the ECR message towards EIR that is sent across S13/S13' interface as part of ME identity check procedure.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

supportNonhomeTaiLaiMapping - Support Non-home TAI/LAI Mapping

Syntax

```
--gParmName supportNonhomeTaiLaiMapping --gParmValue <string>
```

Description

Feature activation for Non-home TAI/LAI Mapping

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

supportOrigTimeStampMaxWait - Support Origination Time Stamp and Maximum Wait Time IE

Syntax

```
--gParmName supportOrigTimeStampMaxWait --gParmValue <string>
```

Description

Parameter controls support for Origination Time stamp and Maximum Wait Time IE inclusion in Create Session Request message. By default this flag shall be disabled. When enabled,

MME shall include Origination Time stamp and Maximum Wait Time IEs in Create Session Request that is sent on S11 message to gateways.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

supportQoSOverride - Support QoS Override

Syntax

--gParmName supportQoSOverride --gParmValue <string>

Description

Parameter controls the feature activation. By default this shall be disabled. When enabled, MME shall support HSS/P-GW QoS Parameter Override for home subscribers

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

supportReattachRequired - Support reattach required

Syntax

--gParmName supportReattachRequired --gParmValue <string>

Description

Enables/Disables the ability to detach the UE based on the bit 'Reattach-Required' bit received from HSS.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

supportSbcAccessControl - Support SBC access control

Syntax

--gParmName supportSbcAccessControl --gParmValue <string>

Description

When support SBc access control set to yes, access control list for SBc provisioned on the MMERmtEndPtCfg is used for access control of SBc.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

supportSdsUeRestoration - Support SDS UE restoration

Syntax

```
--gParmName supportSdsUeRestoration --gParmValue <string>
```

Description

When enabled, MME supports sending data for SDS UEs to SRS.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

supportSessionRestoration - Support session restoration

Syntax

```
--gParmName supportSessionRestoration --gParmValue <string>
```

Description

Support session restoration or not.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

supportSgsnOperDefQci - Support SGSN Operator Defined QCI

Syntax

```
--gParmName supportSgsnOperDefQci --gParmValue <string>
```

Description

Defines if SGSN supports the provisioned Operator Defined QCI. Up to 32 SGSN supported Operator Defined QCI can be defined via CLI oprDefQciToRel99QoSMapping

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

supportSignalingOnlyUeRestoration - Support signaling only UE restoration

Syntax

```
--gParmName supportSignalingOnlyUeRestoration --gParmValue <string>
```

Description

When enabled, MME supports sending data for signaling-only UEs to SRS.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

supportTwoUeBearersForDualPdnType - Support two UE bearers for dual PDN type

Syntax

```
--gParmName supportTwoUeBearersForDualPdnType --gParmValue <string>
```

Description

Support two PDN connections to the same APN if the value is 'false', then 2 PDN connections to the same APN shall not be allowed.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

supportUeRestorationContextDeletion - Support UE restoration context deletion

Syntax

```
--gParmName supportUeRestorationContextDeletion --gParmValue <string>
```

Description

Enables/Disables UE restoration context deletion functionality on MME.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

suppressRegisteredLai - Suppress Registered LAI

Syntax

```
--gParmName suppressRegisteredLai --gParmValue <string>
```

Description

If the feature is enabled (Yes) then MME shall not include the Registered LAI IE in the S1AP INITIAL CONTEXT SETUP and UE CONTEXT MODIFICATION REQUEST messages

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

suppressSendingAltGnGgsnAddress - Suppress Sending Alternate Gn GGSN Address

Syntax

```
--gParmName suppressSendingAltGnGgsnAddress --gParmValue <string>
```

Description

MME shall be provisioned with a global parameter flag that controls sending two IEs (Alternative GGSN Address for control Plane and the Alternative GGSN Address for user traffic IEs) in forward relocation message and context response message during handover scenario. By default, the flag shall be disabled.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

supptIMSIIMEIAPNconv - Support IMSI IMEI APN conversion

Syntax

--gParmName supptIMSIIMEIAPNconv --gParmValue <string>

Description

Support IMSI IMEI APN conversion.

List

- deActivated - Specify support IMSI / IMEI APN conversion type of deactivate.
- imeiPriority - IMEI priority - Specify support IMSI / IMEI APN conversion type of IMEI priority.
- imsiPriority - IMSI priority - Specify support IMSI / IMEI APN conversion type of IMSI priority.

Default

deActivated

Name

switchPdnToLipa - Switch PDN to LIPA

Syntax

--gParmName switchPdnToLipa --gParmValue <string>

Description

Enables MME to dis-connect a LIPA enabled PDN Connection and to re-establish it when UE in idle mode performs a TAU in a LIPA area. MME detaches UE with 're-attach required' to trigger UE to re-establish the PDN connection in the LIPA area.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

t6aDiamOverloadControl - T6A diam overload control

Syntax

--gParmName t6aDiamOverloadControl --gParmValue <string>

Description

Enable(Yes)/disable(No) T6A overload control.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

t6aRetryDifferentScef - T6A retry different SCEF

Syntax

```
--gParmName t6aRetryDifferentScef --gParmValue <string>
```

Description

When MME gets some errors from SCEF (too busy, out of space, etc) or SCEF does not respond, MME sends the same request to another SCEF in the same PLMN.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

tappingActivated - Status of MME/SGSN Tapping functionality

Syntax

```
--gParmName tappingActivated --gParmValue <string>
```

Description

This parameter controls the activated/deactivated status of MME/SGSN Tapping

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

tauDdnCollision - TAU DDN collision

Syntax

```
--gParmName tauDdnCollision --gParmValue <string>
```

Description

Configure MME handling TAU request collision with an ongoing DDN procedure.

List

- Fail DDN - This option preserves the current behavior (i.e. behavior supported until WM8.1.0)). MME sends DDN failure with cause #89 and proceed with TAU.

- Re Page - MME stops paging and proceeds with the TAU request. after TAU is complete and S1 release done, MME pages the UE. MME uses the provisioned paging procedure just as original page used.
- Treat as SR - MME treats the TAU request without the "Active flag" set in the EPS update IE as a TAU request with the "Active flag" set and sends S1AP INITIAL UE CONTEXT REQUEST to set up the bearers. this option basically eliminates the need to page the UE.

Default

Fail DDN

Name

tauRateLimitOverride - TauRateLimitOverride

Syntax

--gParmName tauRateLimitOverride --gParmValue <string>

Description

SGSN to MME IRAT TAUs allowed per sec after which MME start dropping SGSN to MME IRAT TAUs until rate is below limit.

Type / Range

Integer in the range of systemDefault or 0 to 14,999

Default

systemDefault

Name

tauSgsUnknownLaiEmmCause - TAU SGs Unknown LAI EMM Cause

Syntax

--gParmName tauSgsUnknownLaiEmmCause --gParmValue <string>

Description

The parameter is used to provision EMM cause code to be sent to UE combined TAU request if MME fails to select MSC due to LAI to TAI to MSC mapping provisioning errors.

Type / Range

Integer in the range of 2 or 16 to 18 or 22

Default

16

Name

teidToSgsn - Tunnel endpoint identifier data to SGSN

Syntax

--gParmName teidToSgsn --gParmValue <string>

Description

Tunnel endpoint identifier data II information element parameters to support the SGSN context acknowledge message sent from the MME during a gn/gp SGSN to MME TAU enum takes all zero and all ones.

List

- All Ones - Specify all ones.
- All Zeros - Specify all zeros.

Default

All Ones

Name

threegAttachRateLimitOverride - ThreegAttachRateLimitOverride

Syntax

```
--gParmName threegAttachRateLimitOverride --gParmValue <string>
```

Description

3G attaches allowed per sec per PAPS after which PAPS start dropping attaches until rate is below limit.

Type / Range

Integer in the range of 0 to 5,000

Default

900

Name

threegAttachRateLimitOverrideAtIpds - 3G Attach Rate Limit Override

Syntax

```
--gParmName threegAttachRateLimitOverrideAtIpds --gParmValue <string>
```

Description

3G Attaches allowed per sec per PAPS after which PAPS start dropping attaches until rate is below limit at IPDS

Type / Range

Integer in the range of 0 to 15,000

Default

10000

Name

threegControlMsgLimit - 3G control message limit

Syntax

```
--gParmName threegControlMsgLimit --gParmValue <string>
```

Description

3G control messages per second without ciphering. Warning: For Nokia personnel only.

Type / Range

Integer in the range of 1 to 30,000

Default

15700

Name

threegIsauRateLimitOverride - ThreegIsauRateLimitOverride

Syntax

```
--gParmName threegIsauRateLimitOverride --gParmValue <string>
```

Description

Number of 3G ISRAUs allowed per sec per PAPS, after which PAPS starts dropping 3G ISRAUs until rate is below limit.

Type / Range

Integer in the range of 0 to 5,000

Default

1500

Name

threegIsauRateLimitOverrideAtIpds - 3G Isau Rate Limit Override

Syntax

```
--gParmName threegIsauRateLimitOverrideAtIpds --gParmValue <string>
```

Description

Number of 3G ISRAUs allowed per sec per CPPS, after which CPPS starts dropping 3G ISRAUs until rate is below limit at IPDS

Type / Range

Integer in the range of 0 to 20,000

Default

15000

Name

timeBasedAuthFrequency - Time Based Authentication Frequency

Syntax

```
--gParmName timeBasedAuthFrequency --gParmValue <string>
```

Description

Parameter controls feature activation. By default, the flag shall be disabled. When enabled, MME shall support time-based authentication in accordance with new option(s) defined in UE PLMN Services and IMSI Range Services configuration.

List

- No - Boolean value of no.

- Yes - Boolean value of yes.

Default

No

Name

timeZoneUpdateRate - Time zone update rate

Syntax

```
--gParmName timeZoneUpdateRate --gParmValue <string>
```

Description

Configure time zone update rate.

Type / Range

Integer in the range of 30 to 7,200

Default

50

Name

timeZoneUpdateThrottling - Time zone update throttling

Syntax

```
--gParmName timeZoneUpdateThrottling --gParmValue <string>
```

Description

Activate controlling the rate of time zone updates to PGW due to DST change.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

tmsiBasedNriMscSelection - TMSI based NRI MSC selection

Syntax

```
--gParmName tmsiBasedNriMscSelection --gParmValue <string>
```

Description

When set to yes enables the capability to map NRI in TMSI received from the UE to a MSC so that SGs update location is sent to the same MSC as UE was registered with previously to avoid dual VLR registrations. enabling of this feature also requires an operator to provision a non-broadcasting LAC and also NRI for each provisioned MSC. when set to no disables the feature.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

topologicalLabelMatchingForS8pgwSelection - Topological label matching for S8 PGW selection

Syntax

```
--gParmName topologicalLabelMatchingForS8pgwSelection --gParmValue <string>
```

Description

This parameter controls whether or not topological label matching is used to compare the selected SGW FQDN labels with the candidate S8 PGW FQDN labels when selecting a PGW candidate for roaming. when the topological label matching for S8 PGW selection parameter is set to false, the MME will not attempt to use label matching to select the PGW but instead use just the DNS order and preference values. when the parameter is set to true, the MME will use the same label matching algorithm used for S5 PGW selection for S8 PGW selection. when this parameter is set to true, it is up to the service provider to ensure that the labels provided in the SGW and PGW FQDNs are consistent so that label matching will be effective. in particular, the SGW and PGW FQDN values provisioned in the DNS server for MNC and MCC must match.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

topologicalSGWselectionIRAT - Topological SGW Selection IRAT

Syntax

```
--gParmName topologicalSGWselectionIRAT --gParmValue <string>
```

Description

Performs topological matching of SGW to PGW for iRAT TAU and GW Selection Mode 2, when the PGW FQDN had been derived from the PGW IP address, even when pre-existing APNs for the bearers have PLMNs that don't match the UE's selected PLMN

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

tracAreaRelativeCapacity - Trac area relative capacity

Syntax

```
--gParmName tracAreaRelativeCapacity --gParmValue <string>
```

Description

Enable/disable feature to MME support of configuration of MME relative capacity.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

trafloadCombPgw3XUe - Traffic Load to Combo PGW-C for 3X capable UE

Syntax

```
--gParmName trafloadCombPgw3XUe --gParmValue <string>
```

Description

This parameter controls percentage (0% - 100%) of traffic that shall be sent to combo PGW-C for 3X UE(s) eligible using combo PGW-C. If parameter is set to 0%, then MME shall not send any traffic to combo PGW-C.

Type / Range

Integer in the range of 0 to 100

Default

0

Name

trafloadCombPgwN1Ue - Traffic Load to Combo PGW-C for N1 capable UE

Syntax

```
--gParmName trafloadCombPgwN1Ue --gParmValue <string>
```

Description

This parameter controls percentage (0% - 100%) of traffic that shall be sent to combo PGW-C for N1 UE(s) eligible using combo PGW-C. If parameter is set to 0%, then MME shall not send any traffic to combo PGW-C.

Type / Range

Integer in the range of 0 to 100

Default

0

Name

treatGbrTauActiveTau - Treat GBR TAU active TAU

Syntax`--gParmName treatGbrTauActiveTau --gParmValue <string>`**Description**

Enable(Yes)/disable(No) the feature during TAU request with GBR bearers present regardless of active flag value. if set to yes, then MME shall check to determine whether an ECM-IDLE UE has any GBR bearers when TAU request is received. if yes, MME shall treat the TAU request as though TAU request is received with active flags set regardless of active flags setting in the received TAU request and bring up all the bearers. this feature applies to both relocating and non-relocating TAU.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

treatIratTauActiveTau - Treat inter TAU active TAU

Syntax`--gParmName treatIratTauActiveTau --gParmValue <string>`**Description**

Controls to determine whether or not the active flag is treated as if it were set whenever interrta TAU requests are processed.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

treatMip6AgentInfoInIdrAsOneEntity - Treat MIP6Agent-Info in IDR as one Entity

Syntax`--gParmName treatMip6AgentInfoInIdrAsOneEntity --gParmValue <string>`**Description**

GPARM controls the support for treating the MIP6-Agent-Info in IDR as one entity rather than individually modifiable elements. When the GPARM is enabled MME will treat MIP6-Agent-Info in IDR as one entity, By default this shall be disabled.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

twogAttachRateLimitOverride - TwogAttachRateLimitOverride

Syntax

```
--gParmName twogAttachRateLimitOverride --gParmValue <string>
```

Description

2G attaches allowed per sec per PAPS after which PAPS start dropping attaches until rate is below limit.

Type / Range

Integer in the range of 0 to 5,000

Default

1000

Name

twogControlMsgLimit - 2G control message limit

Syntax

```
--gParmName twogControlMsgLimit --gParmValue <string>
```

Description

2G control messages per second without ciphering. Warning: For Nokia personnel only.

Type / Range

Integer in the range of 1 to 30,000

Default

14000

Name

twogDataMsgLimit - 2G data message limit

Syntax

```
--gParmName twogDataMsgLimit --gParmValue <string>
```

Description

2G data messages per second without ciphering. Warning: For Nokia personnel only.

Type / Range

Integer in the range of 1 to 120,000

Default

55000

Name

twogGnThroughput - 2G Gn throughput

Syntax

```
--gParmName twogGnThroughput --gParmValue <string>
```

Description

2G Gn throughput (Mbps). Warning: For Nokia personnel only.

Type / Range

Integer in the range of 1 to 600

Default

220

Name

twogImpactOfCipherring - 2G impact of cipherring

Syntax

```
--gParmName twogImpactOfCipherring --gParmValue <string>
```

Description

2G impact of cipherring (%). Warning: For Nokia personnel only.

Type / Range

Integer in the range of 0 to 100

Default

20

Name

twogIsauRateLimitOverride - TwogIsauRateLimitOverride

Syntax

```
--gParmName twogIsauRateLimitOverride --gParmValue <string>
```

Description

Number of 2G ISRAUs allowed per sec per PAPS, after which PAPS starts dropping 2G ISRAUs until rate is below limit.

Type / Range

Integer in the range of 0 to 5,000

Default

1800

Name

twogMaxPktsConcurrentDataPresence - 2G max packets concurrent data presence

Syntax

--gParmName twogMaxPktsConcurrentDataPresence --gParmValue <string>

Description

Max 2G control packets due to concurrent 2G data presence. Warning: For Nokia personnel only.

Type / Range

Integer in the range of 0 to 50,000

Default

4900

Name

twogUeThroughput - 2G UE throughput

Syntax

--gParmName twogUeThroughput --gParmValue <string>

Description

2G UE throughput (Mbps). Warning: For Nokia personnel only.

Type / Range

Integer in the range of 1 to 600

Default

220

Name

twothreegAttachRateLimitOverride - TwothreegAttachRateLimitOverride

Syntax

--gParmName twothreegAttachRateLimitOverride --gParmValue <string>

Description

Number of 2G and 3G ISRAUs allowed per sec per CPPS, after which CPPS starts dropping 2G and 3G attaches until rate is below limit.

Type / Range

Integer in the range of systemDefault or 0 to 699

Default

systemDefault

Name

ubrRejCauseDuringIratRedirect - UBR Reject Cause During Interrat-Redirection

Syntax

--gParmName ubrRejCauseDuringIratRedirect --gParmValue <string>

Description

A user specified GTPv2 cause to be used within the Update Bearer Response sent back to the SGW, when the Update Bearer Request is received during an interratt-redirection scenario.

Type / Range

29.274 reject Cause value. Integer value in the range of 64 to 127.

Default

94

Name

ueCtxUpdateInPatchLoadSupport - Support for UE Context Update Over NR10 in Patch Loads

Syntax

--gParmName ueCtxUpdateInPatchLoadSupport --gParmValue <string>

Description

Support for UE Context Update over NR10 in patch loads

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

ueEnbServingCellChangeReporting - UE eNB serving cell change reporting

Syntax

--gParmName ueEnbServingCellChangeReporting --gParmValue <string>

Description

Indicates whether UE eNB serving cell change reporting is enabled or not the default is no, indicating UE eNB serving cell change reporting is not enabled.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

ueReachTAdsInSupportedFeaturesAvp - UE reach T-ADS in supported features AVP

Syntax

--gParmName ueReachTAdsInSupportedFeaturesAvp --gParmValue <string>

Description

A value of yes indicates that MME shall send UE-reachability-Notification and T-ADS data retrieval in the in the supported-features AVP, in the feature-list AVP, by setting bits 26 and 27 of the feature-list flags.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

ueSrvccCapabilityHss - UE SRVCC Capability HSS

Syntax

--gParmName ueSrvccCapabilityHss --gParmValue <string>

Description

This parameter controls the value of the UE-SRVCC-Capability AVP the MME sends towards the HSS. UE-SRVCC-Capability AVP is absent when the UE does not include MS network capability IE. MME sends UE-SRVCC-Capability AVP as 'UE-SRVCC-NOT-SUPPORTED' when the UE does not support 'SRVCC to GERAN/UTRAN capability' on MS network capability, or MME does not support SRVCC for the UE according to provisioning (as part of svcAgreementProfile). MME sends UE-SRVCC-Capability AVP as 'UE-SRVCC-SUPPORTED' if the MME supports SRVCC for the UE according to provisioning, and the UE supports 'SRVCC to GERAN/UTRAN capability' on MS network capability.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

ueUnreachableMsgSgsEdrxPsmCause - SGs cause code UE unreachable

Syntax

--gParmName ueUnreachableMsgSgsEdrxPsmCause --gParmValue <string>

Description

Use 'UE unreachable' instead of the default 'UE temporarily unreachable' as cause code with SGs cause IE in SGs interface

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

uliEnhancement - ULI enhancement

Syntax

--gParmName uliEnhancement --gParmValue <string>

Description

Indicates whether ULI enhancement is enabled or not the default is no, indicating ULI enhancement is not enabled.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

uliHomeRoutedUe - ULI home routed UE

Syntax

--gParmName uliHomeRoutedUe --gParmValue <string>

Description

This flag determines whether ULI will be included for home routed UEs. no - ULI will not be included for home routed UEs. yes (default) - ULI will be included for home routed UEs.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

uliInS11messagesWithSgwRelocation - Enable ULI in S11 messages with SGW relocation

Syntax

--gParmName uliInS11messagesWithSgwRelocation --gParmValue <string>

Description

A value of yes for the parameter shall indicate that the WMM will include the ULI (ECGI+TAI) for S11 messages with SGW relocation. a value of no for the parameter shall indicate that existing, 29.274 standards based IE inclusion policy shall apply.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

unconditionallyAddHrIe - Unconditionally Add HRL IE

Syntax

```
--gParmName unconditionallyAddHrIe --gParmValue <string>
```

Description

When enabled, MME shall unconditionally include the Handover Restriction List IE in all S1AP messages as described within 3GPP specifications.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

upper20BitHenbGwRouting - Upper 20-Bit henb GW routing

Syntax

```
--gParmName upper20BitHenbGwRouting --gParmValue <string>
```

Description

Enable/disable non-standard usage of upper 20 bits of the received 28-bit henb ID to route to the henb GW.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

use3gppImsiHashForSgs - Use 3GPP IMSI Hash for SGs

Syntax

```
--gParmName use3gppImsiHashForSgs --gParmValue <string>
```

Description

Specifies whether the CMM is to use the 3GPP IMSI Hash Algorithm specified in TS 23.236 for UE distribution to MSCs via the SGs interface. The default is no. When this field is set to no, the CMM shall use its proprietary IMSI hash algorithm.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

useExplicitDetach - Use explicit detach

Syntax

```
--gParmName useExplicitDetach --gParmValue <string>
```

Description

Indicates whether explicit detach is used (and the MME pages the UE as part of the explicit detach procedure), or whether an implicit detach is used (and the MME does not page the UE).

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

useFailopenProfile - Use failopen profile

Syntax

```
--gParmName useFailopenProfile --gParmValue <string>
```

Description

Enable the use of the failopen profile for attach and PDN connectivity procedures, used for home subscribers and roamers considered as homers.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

useHplmnforSgsInterface - Use Home PLMN For SGs Interface

Syntax

```
--gParmName useHplmnforSgsInterface --gParmValue <string>
```

Description

MME shall be provisioned with GPARM 'useHplmnforSgsInterface' flag. By default, this flag shall be disabled. When the flag is enabled, MME shall always use home plmn of the specific UE towards MSS across SGs interface.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

useLocalDnsHost - Use Local DNS Host

Syntax

```
--gParmName useLocalDnsHost --gParmValue <string>
```

Description

Use local DNS host.

List

- mme - MME will use local DNS file
- mmeS4sgsn - MME,S4SGSN - MME and S4 SGSN will use local DNS file
- none - No elements will use local DNS file
- s4sgsn - S4 SGSN - S4 SGSN will use DNS file

Default

none

Name

useModifiedSbcAsn1 - Use Modified 29.168 SBC ASN.1

Syntax

```
--gParmName useModifiedSbcAsn1 --gParmValue <string>
```

Description

Parameter controls the GPARM flag. By default, this flag shall be disabled. When the flag is enabled, MME shall use the modified version of 29.168 ASN.1.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

useMTUpathdiscoveryforDFflag - use MTU path discovery for DF flag

Syntax

--gParmName useMTUpathdiscoveryforDFflag --gParmValue <string>

Description

This parameter indicates whether to use MTU path discovery to set the DF flag for IP for S1-MME. If it is set to yes, then DF flag will be set if MTU path discovery is enabled. If it is set to no, then the DF flag will not be set, even if MTU path discovery is enabled.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

useNon3gppEmergencyPolicy - Use Non-3GPP NW provided Policies for emergency calls

Syntax

--gParmName useNon3gppEmergencyPolicy --gParmValue <string>

Description

This parameter is used to enable MME to include the Non-3GPP NW provided policies IE in the Attach Request and TAU Request message. By default the IE is not included.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

usePcmdZstdCompression - Use PCMD zstd compression

Syntax

--gParmName usePcmdZstdCompression --gParmValue <string>

Description

When this is set to Yes, zstd compression is applied at PCMD files under /data-pcmd/ directory. When this is No, gzip compression is applied instead.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

useTopologicalLabelMatchingAlways - Use Topological Label Matching Always

Syntax

```
--gParmName useTopologicalLabelMatchingAlways --gParmValue <string>
```

Description

Provisioned Parameter 'UseTopologicalLabelMatchingAlways' for Gateway selection controls whether topological matching shall be used to select SGW and PGW, regardless of whether SGW/PGW FQDN MNC/MCC identifies UE PLMN or Serving PLMN or MME Home PLMN or some other PLMN, during Attach Request procedures and to select a PGW during PDN Connectivity Request procedures. By default, GPARAM shall be disabled. GPARAM is valid only for GW selection mode 2 and shall not have any effect if GW selection Mode 1 is enabled.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

valueAddedMtLrNoEsmLc - Value added MT_LR no E-SMLC

Syntax

```
--gParmName valueAddedMtLrNoEsmLc --gParmValue <string>
```

Description

MT_LR client type value-added do not contact E-SMLC for UE location information. If yes, when MME receives mobile terminated location request with client type=valueAdded, and activateLcsClientTypeValueAddedServices is set, the MME contacts the eNB for UE current E-CGI information. This information is sent back to the GMLC. The MME does not contact the E-SMLC to obtain any UE location information. If no, nothing change and keep existing function (i.e. MME contacts E-SMLC to obtain UE location for this MT-LR type).

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

vdpForbiddenTailListOnVdpProfile - VDP Forbidden TAI List on VDP Profile

Syntax

--gParmName vdpForbiddenTailListOnVdpProfile --gParmValue <string>

Description

Defines whether the VDP Profile associated Forbidden TAI List is sent according to global parameters or VDP Profile level parameters. When set to Yes, and MME finds a matching VDP Profile for the UE, the MME sends VDP profile associated Forbidden TAI List for UEs according to VDP Profile level settings for UE's usage type and Voice Domain Preference, i.e. parameters Data Centric CS Voice Only Send Forbidden TAI List, Data Centric CS Voice Preferred Send Forbidden TAI List, etc. MME ignores global parameters Send forbidden TAI list for data centric CS voice only, Send forbidden TAI list for data centric CS voice preferred, etc. When a) set to Yes and MME does not find a matching VDP Profile for the UE, or b) when set to No, MME sends VDP profile associated Forbidden TAI List for UEs according to global parameters Send forbidden TAI list for data centric CS voice only, Send forbidden TAI list for data centric CS voice preferred, etc. MME ignores VDP profile level parameters Data Centric CS Voice Only Send Forbidden TAI List, Data Centric CS Voice Preferred Send Forbidden TAI List, etc.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

vlrAutoRecoveryPurgeCount - VLR auto recovery purge count

Syntax

--gParmName vlrAutoRecoveryPurgeCount --gParmValue <string>

Description

VLR auto recovery purge count when set to 0, the counter is disabled.

Type / Range

Integer in the range of 0 or 6 to 10

Default

0

Name

warningRetransmissions - Warning retransmissions

Syntax

--gParmName warningRetransmissions --gParmValue <string>

Description

The number of times the MME will attempt to send a S1 CMAS message to a eNB.

Type / Range

Integer in the range of 0 to 10

Default

0

Name

warningSystemLog - Warning system log

Syntax

```
--gParmName warningSystemLog --gParmValue <string>
```

Description

Warning system log.

List

- All - Log both S1 and SBC CMAS messages in warnsys log.
- None - Do not log CMAS messages in warnsys log.
- S1 Only - Only log S1 CMAS messages in warnsys log.
- SBC Only - Only log SBC CMAS messages in warnsys log.

Default

All

Name

weightedSgsImsiHash - Weighted SGs IMSI Hash

Syntax

```
--gParmName weightedSgsImsiHash --gParmValue <string>
```

Description

Specifies whether the MME is to use a weighted IMSI hash distribution for assigning UEs to an SGs MSC.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

writeReplaceWarningIndication - Write replace warning indication

Syntax

```
--gParmName writeReplaceWarningIndication --gParmValue <string>
```

Description

Enable MME sending of broadcast completed area list to the CBC.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

wusProfilesEnabled - WUS Profiles Enabled

Syntax

```
--gParmName wusProfilesEnabled --gParmValue <string>
```

Description

Specifies whether a WUS profile name referenced in UE PLMN Services and IMSI Range Services can be used. Default is No.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

x2hoInclSgwUlTeid - X2 HO include SGW ul

Syntax

```
--gParmName x2hoInclSgwUlTeid --gParmValue <string>
```

Description

Enable E-RAB to BE switched in uplink list IE in the S1AP PATH SWITCH REQUEST ACKNOWLEDGE message irrespective of SGW relocation.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

zeroOiCsrN26Tau - Set OI flag to Zero in CSR of N26 TAU

Syntax

```
--gParmName zeroOiCsrN26Tau --gParmValue <string>
```

Description

Current parameter enables setting OI flag to value zero in Create Session Request that MME sends as part of n26 Idle mobility. When this parameter is disabled, the MME sets OI flag to 1

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

5.5 cmm gpsiRange

GPSI Range. Syntax, description, parameters, and roles for cmm gpsiRange command.

Syntax**Description**

Manages a GPSI range. Up to 1024 GPSI ranges can be defined.

Parameters**Name**

gpsiRangeListName - GPSI Range

Syntax

```
--gpsiRangeListName <string>
```

Description

Name of the list consisting of one more GPSI ranges.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

startGpsi - Start of the GPSI range

Syntax

--startGpsi <digits>

Description

Start of the GPSI range

Class

Required key.

Type / Range

Digit string of length 5 to 15

Name

endGpsi - End GPSI

Syntax

--endGpsi <digits>

Description

End of the GPSI Range

Class

Required key.

Type / Range

Digit string of length 5 to 15

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.6 cmm gpsiRangeList

GPSI Range List . Syntax, description, parameters, and roles for cmm gpsiRangeList command.

Syntax

Description

List for GPSI ranges. Up to 32 GPSI Range List may be created. Each list may contain up to 1024 GPSI Ranges.

Parameters

Name

name - List Name

Syntax

--name <string>

Description

GPSI ranges list name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.7 cmm gtAnalysis

GT analysis. Syntax, description, parameters, and roles for cmm gtAnalysis command.

Syntax

Description

Use this command to configure, list and show SCCP Global Title (GT) analysis.

Parameters

Name

gtald - GT analysis digits identifier

Syntax

--gtald <digits>

Description

Use this parameter to specify GT analysis,digits. Give as many digits from the beginning of the global title as are needed to find the analysis result.In case that you are trying to add an analysis, with the same numbering plan, the same nature of address and the same prefix of digits with an already existing analysis, please check that longestGtDigitsMatch is set to Yes (through cmm gParm show longestGtDigitsMatch).If not, please set it, and audit mechanism will heal the issue.

Class

Required key.

Type / Range

Digit string of length 1 to 24 with the allowed pattern of [0-9]*

Name

numberingPlan - Numbering plan

Syntax

--numberingPlan <list-item>

Description

Use this parameter to specify the numbering plan for the global title. Cases, numberingPlan 0 unknown numberingPlan 1 ISDN,Telephony (Rec. E.164,E.163) ISDN,telephone network numberingPlan 2 spare numberingPlan 3 Data (Rec. X.121) , data network numberingPlan 4 Telex (Rec. F.69) , telex network numberingPlan 5 Maritime Mobile (Rec. E.210, E.211) , maritime mobile telephone network numberingPlan 6 Land Mobile (Rec. E.212) , mobile telephone network (Mobile Subscriber Identity) numberingPlan 7 ISDN,Mobile (Rec. E.214) , ISDN,mobile telephone network (Mobile Global Title) numberingPlan 8 to 14 spare numberingPlan 15 reserved

Class

Required key.

List

- 1 - Specifies the global title numbering plan of 1.
- 6 - Specifies the global title numbering plan of 6.

- 7 - Specifies the global title numbering plan of 7.

Name

gtResultId - Identifier to GT analysis result

Syntax

--gtResultId <integer>

Description

Use this parameter to specify the record number of the Global Title Translation Result File (GTRFIL). It can get values from the range presented in the dynamic parameter guides. The analysis result of the global title has to be created before it can be referred to from the analysis tree. So, check the available records through cmm gtAnalysisResult list command.

Class

Required on create.

Type / Range

Integer in the range of 1 to 10,239

Name

atf - Analysis type filter

Syntax

--atf <list-item>

Description

output GT analyses that have the same analysis type as is mentioned in the analysis result

Class

Optional.

List

- ALL - All.
- TRUNCATED - Truncated.

Default

ALL

Name

gti - Gti

Syntax

--gti <integer>

Description

Use this parameter to specify the global title indicator for the specific ss7 standard. ITU Cases, gti 0 no global title is included in the address gti 1 the global title only includes the nature of address indicator gti 2 the global title only includes the translation type gti 3 the global title includes the translation type, numbering plan and encoding scheme gti 4 the

global title includes the translation type, numbering plan, encoding scheme and nature of address indicator gti 5 to 7 spare (international) gti 8 to 14 spare (national) gti 15 reserved for expansion ANSI Cases, gti 0 no global title is included in the address gti 1 the global title includes the translation type, numbering plan and encoding scheme gti 2 the global title includes the translation type gti 3 to 4 not specified in the USA gti 5 to 7 spare (international) gti 8 to 14 spare (national) gti 15 reserved for expansion

Class

Optional.

Type / Range

Integer in the range of 1 to 15

Default

4

Name

last - Last global title element to be analysed

Syntax

--last <list-item>

Description

Use this parameter to specify the last element of the called global title that you want to be analysed. use this parameter when all necessary information can not be given for some reason. cases, last GTI the analysis tree only contains the global title indicator values. numbers are not given, and they are not analysed. last TT the analysis tree contains the global title indicator values and the translation type values. other data is not analysed. last NP the analysis tree contains the global title indicator values, the translation type values and the numbering plan. other data is not analysed. last NAI the analysis tree contains all parameters of the root. last DIG the analysis tree contains all parameters and numbers of the root.

Class

Optional.

List

- DIG - Specify DIG as last global title element to be analyzed.
- GTI - Specify GTI as last global title element to be analyzed.
- NAI - Specify NAI as last global title element to be analyzed.
- NP - Specify NP as last global title element to be analyzed.
- TT - Specify TT as last global title element to be analyzed.

Default

DIG

Name

nai - Nature of address indicator

Syntax

--nai <integer>

Description

Use this parameter to specify the nature of address indicator. Cases: nai 0 spare nai 1 subscriber number nai 2 reserved for national use nai 3 national significant number nai 4 international number nai 5 to 127 spare

Class

Optional.

Type / Range

Integer in the range of 0 to 127

Default

4

Name

nle - Number length

Syntax

--nle <integer>

Description

output GT analyses which has the same digit amount with the set number length

Class

Optional.

Type / Range

Integer in the range of 0 to 24

Default

0

Name

ss7Standard - Ss7Standard

Syntax

--ss7Standard <list-item>

Description

Use this parameter to specify the global title standard (ss7 standard). Allowed values, ANSI,ITU

Class

Optional.

List

- ANSI - Specify SS7 variant of american national standards institute.

- ITU - Specify SS7 variant of international telecommunication union - telecommunications.

Default

ITU

Name

translationType - Translation type

Syntax`--translationType <integer>`**Description**

Use this parameter to specify the translation type. The recommended parameter values depend on the standard and on the values of GTI you give. ITU Cases, translationType 0 translation type is not used. In the SGSN, this is the only supported value. translationType 1 to 127 can be defined for use between networks translationType 128 to 254 can be defined for internal use in the network translationType 255 reserved for expansion ANSI Cases, translationType 0 reserved translationType 1 981 telecommunications credit cards translationType 2 14 digit calling card translationType 3 cellular nationwide roaming service translationType 4 global title point code translationType 5 calling name delivery translationType 6 to 31 can be defined for use between networks translationType 32 to 222 spare translationType 223 to 252 can be defined for internal use in the network translationType 253 14 digit calling card translationType 254 can be defined for internal use in the network

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.8 cmm gtAnalysisResult

GT analysis result. Syntax, description, parameters, and roles for `cmm gtAnalysisResult` command.

Syntax

Description

Use this command to configure, list and show (GT) global title translation result records.

Parameters

Name

gtResultId - GT analysis result identifier

Syntax

--gtResultId <integer>

Description

Use this parameter to specify the global title translation result record to be created with the parameter value, which is a decimal numeral greater than 0. you can specify a given number value for the GT result index with the GT result number parameter. this parameter supports creating similar translation results in different signalling points in any order if external tools are used for data generating and maintaining.it will be mapped to GTR file.

Class

Required key.

Type / Range

Integer in the range of 1 to 10,239

Name

primaryDpc - Primary destination point code

Syntax

--primaryDpc <integer>

Description

Use this parameter to specify the primary destination point code. please, check that this destination point code has been configured through `cmm sccpsignallingpoint` list, and that the signallingnetwork is the same as the given primarynetwork. you must give the primarynetwork parameter before the primarydpc parameter.

Class

Required on create.

Type / Range

Integer in the range of 1 to 16,777,215

Name

primaryNetwork - Primary signalling network

Syntax

--primaryNetwork <list-item>

Description

Use this parameter to specify the primary signalling network. cases, IN0 international network number 0 IN1 international network number 1 NA0 national network number 0 NA1 national network number 1.

Class

Required on create.

List

- IN0 - International network number 0.
- IN1 - International network number 1.
- NA0 - National network number 0.
- NA1 - National network number 1.

Name

backup - Backup subsystem indication

Syntax

--backup <list-item>

Description

Use this parameter to specify the existence of the subsystems individual backups in the global title translation result. Allowed values: true, false

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

issn - Input subsystem for backup

Syntax

--issn <integer>

Description

Use this parameter to specify the subsystem for which the individual backup is created. please, check that this subsystem number has been configured through cmm sccpsubsystem list.

Class

Optional.

Type / Range

Integer in the range of 0 to 254

Default

0

Name

loadSharing - Load sharing

Syntax

--loadSharing <list-item>

Description

Use this parameter to specify the use of load sharing. you can add records for load sharing results through cmm gtloadsharingnet create command. Allowed values: true, false

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

overwr - Overwrite method

Syntax

--overwr <list-item>

Description

Use this parameter to specify the overwrite method. you can use this parameter with the GT result number parameter for handling the situations, where you have defined the GT result index value of an existing record for the global title translation result record to be created. you can overwrite the global title translation result record if the record is not associated with any global title analysis. with this parameter you can define a predetermined way for handling any possible overwriting situation. Allowed values: true, false

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

priGtmd - Primary GT modification data id for called

Syntax

--priGtmd <integer>

Description

Use this parameter to specify the primary modification data record, which you use to modify or replace the existing global title in the called address. used in load sharing case to modify or replace called GT address. please, check that this GT modification data record has been configured through cmm gtmodificationdata list. you cannot give this parameter unless the value of the RI primary parameter is GT (route on gt). 0 default, id to primary GT modification called data not exists.

Class

Optional.

Type / Range

Integer in the range of 0 to 256

Default

0

Name

priGtmg - Primary GT modification data id for calling

Syntax

--priGtmg <integer>

Description

Use this parameter to specify the primary modification data record, which you use to modify or replace the existing global title in the calling address. used in load sharing case to modify or replace calling GT address. please, check that this GT modification data record has been configured through cmm gtmodificationdata list. 0 default, id to primary GT modification calling data not exists.

Class

Optional.

Type / Range

Integer in the range of 0 to 256

Default

0

Name

primaryPrio - Primary priority for load sharing

Syntax

--primaryPrio <list-item>

Description

Use this parameter to specify the primary priority for a load sharing destination. if you give the primaryprio parameter, you must indicate the load sharing later by setting loadsharing as true.

Class

Optional.

List

- 1 - Specify load sharing priority of 1.
- 2 - Specify load sharing priority of 2.

Default

1

Name

primaryRi - Primary routing indicator

Syntax

--primaryRi <list-item>

Description

Use this parameter to specify the primary routing indicator. Allowed values: SSN (route on ssn) , GT (route on gt)

Class

Optional.

List

- GT - Global title - Specify routing indicator of global title.
- SSN - Subsystem number - Specify routing indicator of subsystem number.

Default

SSN

Name

primarySsn - Primary subsystem number

Syntax

--primarySsn <integer>

Description

Use this parameter to specify the primary subsystem number. please, check that this subsystem number has been configured through cmm sccpsubsystem list. you must give the primarynetwork and primarydpc parameters before giving the primaryssn parameter.

Class

Optional.

Type / Range

Integer in the range of 0 to 254

Default

0

Name

primaryWeight - Primary Weight

Syntax

--primaryWeight <integer>

Description

Use this parameter to specify the weight for the Primary destination

Class

Optional.

Type / Range

Integer in the range of 1 to 255

Default

1

Name

rdpc - Replicative destination point code

Syntax

--rdpc <integer>

Description

Use this parameter to specify the replicative destination point code. please, check that this destination point code has been configured through cmm sccpsignallingpoint list, and that the signallingnetwork is the same as the given rnet. you must give the rnet parameter before the rdpc parameter.

Class

Optional.

Type / Range

Integer in the range of 0 to 16,777,215

Default

0

Name

rnet - Replicative network

Syntax

--rnet <list-item>

Description

Use this parameter to specify the replicative signalling network. cases, IN0 international network number 0 IN1 international network number 1 NAO national network number 0 NA1 national network number 1.

Class

Optional.

List

- " - No signalling network - No signalling network.
- IN0 - Signalling network IN0 - Signalling network IN0.
- IN1 - Signalling network IN1 - Signalling network IN1.
- NAO - Signalling network NAO - Signalling network NAO.
- NA1 - Signalling network NA1 - Signalling network NA1.

Default

""

Name

rssn - Replicative subsystem

Syntax

--rssn <integer>

Description

Use this parameter to specify the replicative subsystem number. please, check that this subsystem number has been configured through cmm sccpsubsystem list. you must give the rnet and rdpc parameters before giving the rssn parameter.

Class

Optional.

Type / Range

Integer in the range of 0 to 254

Default

0

Name

secGtmd - Secondary GT modification data id for called

Syntax

--secGtmd <integer>

Description

Use this parameter to specify the secondary modification data record, which you use to modify or replace the existing global title in the called address. used in load sharing case to modify or replace called GT address. please, check that this GT modification data record has been configured through cmm gtmodificationdata list. you cannot give this parameter unless the value of the RI secondary parameter is GT (route on gt). 0 default, id to secondary GT modification called data not exists.

Class

Optional.

Type / Range

Integer in the range of 0 to 256

Default

0

Name

secGtmg - Secondary GT modification data id for calling

Syntax

--secGtmg <integer>

Description

Use this parameter to specify the secondary modification data record, which you use to modify or replace the existing global title in the calling address. used in load sharing case to modify or replace calling GT address. please, check that this GT modification data record has been configured through cmm gtmodificationdata list. 0 default, id to secondary GT modification calling data not exists.

Class

Optional.

Type / Range

Integer in the range of 0 to 256

Default

0

Name

secondaryDpc - Secondary destination point code

Syntax

--secondaryDpc <integer>

Description

Use this parameter to specify the secondary destination point code. please, check that this destination point code has been configured through cmm sccpsignallingpoint list, and

that the signallingnetwork is the same as the given secondarynetwork. you must give the secondarynetwork parameter before the secondarydpc parameter.

Class

Optional.

Type / Range

Integer in the range of 0 to 16,777,215

Default

0

Name

secondaryNetwork - Secondary signalling network

Syntax

--secondaryNetwork <list-item>

Description

Use this parameter to specify the secondary signalling network. cases, IN0 international network number 0 IN1 international network number 1 NAO national network number 0 NA1 national network number 1.

Class

Optional.

List

- " - No signalling network - No signalling network.
- IN0 - Signalling network IN0 - Signalling network IN0.
- IN1 - Signalling network IN1 - Signalling network IN1.
- NAO - Signalling network NAO - Signalling network NAO.
- NA1 - Signalling network NA1 - Signalling network NA1.

Default

"

Name

secondaryPrio - Secondary priority for load sharing

Syntax

--secondaryPrio <list-item>

Description

Use this parameter to specify the secondary priority for a load sharing destination. if you give the secondaryprio parameter, you must indicate the load sharing later by setting loadsharing as true.

Class

Optional.

List

- 0 - No load sharing priority - No load sharing priority.
- 1 - Load sharing priority 1 - Load sharing priority 1.
- 2 - Load sharing priority 2 - Load sharing priority 2.

Default

0

Name

secondaryRi - Secondary routing indicator

Syntax

--secondaryRi <list-item>

Description

Use this parameter to specify the secondary routing indicator. Allowed values: SSN (route on ssn), GT (route on gt)

Class

Optional.

List

- " - No routing indicator - No routing indicator.
- GT - Routing indicator GT - Routing indicator GT.
- SSN - Routing indicator SSN - Routing indicator SSN.

Default

""

Name

secondarySsn - Secondary subsystem number

Syntax

--secondarySsn <integer>

Description

Use this parameter to specify the secondary subsystem number. please, check that this subsystem number has been configured through cmm sccpsubsystem list. you must give the secondarynetwork and secondarydpc parameters before giving the secondaryssn parameter.

Class

Optional.

Type / Range

Integer in the range of 0 to 254

Default

0

Name

secondaryWeight - Secondary Weight

Syntax

--secondaryWeight <integer>

Description

Use this parameter to specify the weight for the Secondary destination

Class

Optional.

Type / Range

Integer in the range of 1 to 255

Default

1

Name

special - Special indication

Syntax

--special <list-item>

Description

Use this parameter to specify whether the global title translation result contains a special treatment for the global title translated parameter. Allowed values: true, false

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.9 cmm gtLoadSharingNet

GT analysis Load Sharing Net. Syntax, description, parameters, and roles for cmm gtLoadSharingNet command.

Syntax**Description**

Use this command to configure, list and show Load Sharing Net records for a specific (GT) global title translation result record.

Parameters**Name**

gtLoadSharingNetId - GT analysis load sharing net identifier

Syntax

--gtLoadSharingNetId <integer>

Description

Use this parameter to specify the load sharing net record number for the specific (GT) global title translation result record.

Class

Required key.

Type / Range

Integer in the range of 3 to 32

Name

gtResultId - GT analysis result identifier

Syntax

--gtResultId <integer>

Description

Use this parameter to specify the (GT) global title translation result record. please check that this record exists through cmm gtanalysisresult list command.

Class

Required key.

Type / Range

Integer in the range of 1 to 10,239

Name

ldpc - Destination point code for load sharing

Syntax

--ldpc <integer>

Description

Use this parameter to specify the destination point code used in the load sharing result. please, check that this destination point code has been configured through cmm sccpsignallingpoint list, and that the signallingnetwork is the same as the given lnet. you must give the lnet parameter before the ldpc parameter.

Class

Required on create.

Type / Range

Integer in the range of 1 to 16,777,215

Name

lnet - Network for load sharing

Syntax

--lnet <list-item>

Description

Use this parameter to specify the signalling network used in the load sharing result cases, IN0 international network number 0 IN1 international network number 1 NAO national network number 0 NA1 national network number 1.

Class

Required on create.

List

- IN0 - International network number 0.
- IN1 - International network number 1.
- NAO - National network number 0.
- NA1 - National network number 1.

Name

lgtmd - GT modification data id for called

Syntax

--lgtmd <integer>

Description

Use this parameter to specify the modification data record, which you use to modify or replace the existing global title in the called address. used in load sharing case to modify or replace called GT address. please, check that this GT modification data record has been configured through cmm gtmodificationdata list. you cannot give this parameter unless the value of the RI parameter is GT (route on gt). 0 default, id to GT modification called data not exists.

Class

Optional.

Type / Range

Integer in the range of 0 to 256

Default

0

Name

lgtmg - GT modification data id for calling

Syntax

--lgtmg <integer>

Description

Use this parameter to specify the modification data record, which you use to modify or replace the existing global title in the calling address. used in load sharing case to modify or replace calling GT address. please, check that this GT modification data record has been configured through cmm gtmodificationdata list. you cannot give this parameter unless the value of the RI parameter is GT (route on gt). 0 default, id to GT modification calling data not exists.

Class

Optional.

Type / Range

Integer in the range of 0 to 256

Default

0

Name

lprio - Priority for load sharing

Syntax

--lprio <list-item>

Description

Use this parameter to specify the priority level for the load sharing destination.

Class

Optional.

List

- 0 - No load sharing priority - No load sharing priority.
- 1 - Load sharing priority 1 - Load sharing priority 1.
- 2 - Load sharing priority 2 - Load sharing priority 2.

Default

0

Name

lri - Routing indicator for load sharing

Syntax`--lri <list-item>`**Description**

Use this parameter to specify the routing indicator used in the load sharing result. Allowed values: SSN (route on ssn) , GT (route on gt)

Class

Optional.

List

- " - No routing indicator - No routing indicator.
- GT - Routing indicator GT - Routing indicator GT.
- SSN - Routing indicator SSN - Routing indicator SSN.

Default

""

Name

lssn - Subsystem number for load sharing

Syntax`--lssn <integer>`**Description**

Use this parameter to specify the new subsystem, which is the primary result of the global title translation. please, check that this subsystem number has been configured through cmm sccpsubsystem list. you must give the lnet and ldpc parameters before giving the lssn parameter.

Class

Optional.

Type / Range

Integer in the range of 0 to 254

Default

0

Name

lweight - Weight for load sharing

Syntax

--lweight <integer>

Description

Use this parameter to specify the weight for the load sharing destination.

Class

Optional.

Type / Range

Integer in the range of 1 to 255

Default

1

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.10 cmm gtModificationData

GT modification data. Syntax, description, parameters, and roles for **cmm gtModificationData** command.

Syntax**Description**

Use this command to configure, list and show (GT) global title modification data records. Note: Record 0 is a dummy record and it cannot be modified or removed.

Parameters

Name

gtModificationDataId - GT modification data id

Syntax

--gtModificationDataId <integer>

Description

Use this parameter to specify the id to the GT modification data record. it will be mapped to GTM file.

Class

Required key.

Type / Range

Integer in the range of 0 to 256

Name

digits - Digits

Syntax

--digits <digits>

Description

Use this parameter to specify the digit string to be added to the original global title.

Class

Required on create.

Type / Range

Digit string of length 1 to 24 with the allowed pattern of [0-9]*

Name

adp - Add digits pointer

Syntax

--adp <integer>

Description

Use this parameter to specify the digits pointer to be added.

Class

Optional.

Type / Range

Integer in the range of 0 to 24

Default

0

Name

ddc - Delete digits count

Syntax

--ddc <integer>

Description

Use this parameter to specify the number of digits to be deleted from an existing global title digit string.

Class

Optional.

Type / Range

Integer in the range of 0 to 24

Default

0

Name

ddp - Delete digits pointer

Syntax

--ddp <integer>

Description

Use this parameter to specify the digits pointer to be deleted.

Class

Optional.

Type / Range

Integer in the range of 0 to 24

Default

0

Name

gti - Gti

Syntax

--gti <integer>

Description

Use this parameter to specify the global title indicator for the specific SS7 standard. ITU cases, gti 0 no global title is included in the address gti 1 the global title only includes the nature of address indicator gti 2 the global title only includes the translation type gti 3 the global title includes the translation type, numbering plan and encoding scheme gti 4 the global title includes the translation type, numbering plan, encoding scheme and nature of address indicator gti 5 to 7 reserved for international use gti 8 to 14 reserved for national use gti 15 reserved for extension ANSI cases, gti 0 no global title is included in the address gti 1 the global title includes the translation type, numbering plan and encoding scheme

gti 2 the global title includes the translation type gti 3 to 4 not specified in the USA gti 5 to 7 reserved for international use gti 8 to 14 reserved for national use gti 15 reserved for extension.

Class

Optional.

Type / Range

Integer in the range of 1 to 4

Default

4

Name

nai - Nature of address indicator

Syntax

--nai <integer>

Description

Use this parameter to specify the nature of address indicator. cases, nai 0 spare nai 1 subscriber number nai 2 reserved for national use nai 3 national significant number nai 4 international number nai 5 to 127 spare.

Class

Optional.

Type / Range

Integer in the range of 0 to 127

Default

4

Name

numberingPlan - Numbering plan

Syntax

--numberingPlan <integer>

Description

Use this parameter to specify the numbering plan for the global title. cases, numberingplan 0 unknown numberingplan 1 ISDN,Telephony (Rec. E.164,E.163) , ISDN,telephone network numberingplan 2 spare numberingplan 3 data (Rec. X.121) , data network numberingplan 4 telex (Rec. F.69) , telex network numberingplan 5 maritime mobile (Rec. E.210, E.211) , maritime mobile telephone network numberingplan 6 land mobile (Rec. E.212) , mobile telephone network (Mobile subscriber identity) numberingplan 7 ISDN,Mobile (Rec. E.214) , ISDN, mobile telephone network (Mobile global title) numberingplan 8 to 14 spare numberingplan 15 reserved.

Class

Optional.

Type / Range

Integer in the range of 0 to 15

Default

1

Name

ss7Standard - Ss7Standard

Syntax

--ss7Standard <list-item>

Description

Use this parameter to specify the global title standard (ss7 standard). allowed values, ANSI , ITU.

Class

Optional.

List

- ANSI - Specify SS7 variant of american national standards institute.
- ITU - Specify SS7 variant of international telecommunication union - telecommunications.

Default

ITU

Name

translationType - Translation type

Syntax

--translationType <integer>

Description

Use this parameter to specify the translation type. the recommended parameter values depend on the standard and on the values of GTI you give. ITU cases, translationtype 0 translation type is not used. in the SGSN, this is the only supported value. translationtype 1 to 127 can be defined for use between networks translationtype 128 to 254 can be defined for internal use in the network translationtype 255 reserved for expansion ANSI cases, translationtype 0 reserved translationtype 1 981 telecommunications credit cards translationtype 2 14 digit calling card translationtype 3 cellular nationwide roaming service translationtype 4 global title point code translationtype 5 calling name delivery translationtype 6 to 31 can be defined for use between networks translationtype 32 to 222 spare translationtype 223 to 252 can be defined for internal use in the network translationtype 253 14 digit calling card translationtype 254 can be defined for internal use in the network.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.11 cmm gtpProfile

GTP Profile. Syntax, description, parameters, and roles for `cmm gtpProfile` command.

Syntax**Description**

This command is used to administer a GTP Profile and includes parameters for defining message attempts and timers.

Parameters**Name**

name - GTP profile name

Syntax

--name <string>

Description

Unique name for this GTP profile.

Restrictions

GTP profile with label = default always exists, it cannot be deleted, but can be update.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

bearerResourceCommandT3 - Bearer resource command timeout T3

Syntax

--bearerResourceCommandT3 <integer>

Description

Max wait time for bearer resource command.

Class

Optional.

Type / Range

Integer 100...60000, step 100 in units of Millisecond

Default

3000

Name

changeNotificationResponseT3 - Change notification response timeout T3

Syntax

--changeNotificationResponseT3 <integer>

Description

Max wait time for change notification response.

Class

Optional.

Type / Range

Integer 100...60000, step 100 in units of Millisecond

Default

3000

Name

ciotDIS1APMsgTunnelingN3 - N3 Downlink S1APMessage Tunneling message

Syntax

--ciotDlS1APMsgTunnelingN3 <integer>

Description

Maximum number of attempts to send the Downlink S1APMessage Tunneling message.

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

2

Name

ciotDIS1APMsgTunnelingT3 - DL S1AP message tunneling timeout waiting for ACK message

Syntax

--ciotDlS1APMsgTunnelingT3 <integer>

Description

Max wait time for an ACK message to an Downlink S1APMessage Tunneling message.

Class

Optional.

Type / Range

Integer in the range of 100 to 3,000 in units of Millisecond

Default

500

Name

contextReqT3 - Context request timeout T3 waiting for response

Syntax

--contextReqT3 <integer>

Description

Max wait time for a response of a context request message (29.060).

Class

Optional.

Type / Range

Integer 100...60000, step 100 in units of Millisecond

Default

3000

Name

contextRspT3 - Context response timeout T3 waiting for ack

Syntax

--contextRspT3 <integer>

Description

Max wait time for a response to a context response message (29.060).

Class

Optional.

Type / Range

Integer 100...60000, step 100 in units of Millisecond

Default

3000

Name

createFwdTunnelResponseT3 - Create forward tunnel response timeout T3

Syntax

--createFwdTunnelResponseT3 <integer>

Description

Max wait time for create forward tunnel response.

Class

Optional.

Type / Range

Integer 100...60000, step 100 in units of Millisecond

Default

3000

Name

createIndirectDataFwdTunnelT3 - Create indirect data forward tunnel timeout T3

Syntax

--createIndirectDataFwdTunnelT3 <integer>

Description

Max wait time for create indirect data forward tunnel response.

Class

Optional.

Type / Range

Integer 100...60000, step 100 in units of Millisecond

Default

3000

Name

createSessResponseT3 - Create session response timeout T3

Syntax

--createSessResponseT3 <integer>

Description

Max wait time for create session response.

Class

Optional.

Type / Range

Integer 100...60000, step 100 in units of Millisecond

Default

3000

Name

createT3 - Create response timeout T3

Syntax

--createT3 <integer>

Description

Max wait time for a response of a create PDP context request message (29.060).

Class

Optional.

Type / Range

Integer 100...60000, step 100 in units of Millisecond

Default

3000

Name

deleteBearerCommandT3 - Delete bearer command timeout T3

Syntax

--deleteBearerCommandT3 <integer>

Description

Max wait time for delete bearer command.

Class

Optional.

Type / Range

Integer 100...60000, step 100 in units of Millisecond

Default

3000

Name

deleteIndirectDataFwdTunnelT3 - Delete indirect data forward tunnel timeout T3

Syntax

--deleteIndirectDataFwdTunnelT3 <integer>

Description

Max wait time for delete indirect data forward tunnel response.

Class

Optional.

Type / Range

Integer 100...60000, step 100 in units of Millisecond

Default

3000

Name

deletePdnConnSetResponseT3 - Delete PDN connection set response timeout T3

Syntax

--deletePdnConnSetResponseT3 <integer>

Description

Max wait time for delete PDN connection set response.

Class

Optional.

Type / Range

Integer 100...60000, step 100 in units of Millisecond

Default

3000

Name

deleteSessResponseT3 - Delete session response timeout T3

Syntax

--deleteSessResponseT3 <integer>

Description

Max wait time for delete session response.

Class

Optional.

Type / Range

Integer 100...60000, step 100 in units of Millisecond

Default

3000

Name

deleteT3 - Delete response timeout T3

Syntax

--deleteT3 <integer>

Description

Max wait time for a response to a delete PDP context request message (29.060).

Class

Optional.

Type / Range

Integer 100...60000, step 100 in units of Millisecond

Default

3000

Name

description - GTP profile description

Syntax

--description <string>

Description

GTP profile description.

Class

Optional.

Type / Range

Digit string of length 0 to 80 with the allowed pattern of [0-9a-zA-Z\.\- _:@]*

Default

"

Name

echoRequests - Echo requests

Syntax

--echoRequests <integer>

Description

(N3-REQUESTS) the total number of echo request messages to be transmitted without receiving a corresponding echo response message after receiving at least one echo response from an endpoint.

Class

Optional.

Type / Range

Integer in the range of 5 to 20

Default

9

Name

fwdContextT3 - Forward context ack timeout T3

Syntax

--fwdContextT3 <integer>

Description

Max wait time for a response to a forward context message (29.060).

Class

Optional.

Type / Range

Integer 100...60000, step 100 in units of Millisecond

Default

3000

Name

identificationT3 - Identification response timeout T3

Syntax

--identificationT3 <integer>

Description

Max wait time for a response of an identification request message (29.060).

Class

Optional.

Type / Range

Integer 100...60000, step 100 in units of Millisecond

Default

3000

Name

modifyAccessBearersResponseT3 - Modify access bearers response timeout T3

Syntax

--modifyAccessBearersResponseT3 <integer>

Description

Max wait time for modify access bearers response.

Class

Optional.

Type / Range

Integer 100...60000, step 100 in units of Millisecond

Default

3000

Name

modifyBearerCommandT3 - Modify bearer command timeout T3

Syntax

--modifyBearerCommandT3 <integer>

Description

Max wait time for modify bearer command.

Class

Optional.

Type / Range

Integer 100...60000, step 100 in units of Millisecond

Default

3000

Name

modifyBearerResponseT3 - Modify bearer response timeout T3

Syntax

--modifyBearerResponseT3 <integer>

Description

Max wait time for modify bearer response.

Class

Optional.

Type / Range

Integer 100...60000, step 100 in units of Millisecond

Default

3000

Name

n3BearerResourceCommand - N3 bearer resource command attempts

Syntax

--n3BearerResourceCommand <integer>

Description

Max number of attempts to send bearer resource command.

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

n3ChangeNotificationRequest - N3 change notification request attempts

Syntax

--n3ChangeNotificationRequest <integer>

Description

Max number of attempts to send change notification request.

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

n3ContextReq - N3 context request attempts

Syntax

--n3ContextReq <integer>

Description

Max number of attempts to send a context request message (29.060).

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

n3ContextRsp - N3 context response attempts

Syntax

--n3ContextRsp <integer>

Description

Max number of attempts to send a context response message (29.060).

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

n3Create - N3 create request attempts

Syntax`--n3Create <integer>`**Description**

Max number of attempts to send a create request message (29.060).

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

n3CreateFwdTunnelRequest - N3 create forward tunnel request attempts

Syntax`--n3CreateFwdTunnelRequest <integer>`**Description**

Max number of attempts to send create forward tunnel request.

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

n3CreateIndirectDataFwdTunnel - N3 create indirect data forward tunnel attempts

Syntax`--n3CreateIndirectDataFwdTunnel <integer>`**Description**

Max number of attempts to send create indirect data forward tunnel request.

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

n3CreateSessRequest - N3 create session request attempts

Syntax

--n3CreateSessRequest <integer>

Description

Max number of attempts to send create session request.

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

n3Delete - N3 delete request attempts

Syntax

--n3Delete <integer>

Description

Max number of attempts to send a delete request message (29.060).

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

n3DeleteBearerCommand - N3 delete bearer command attempts

Syntax

--n3DeleteBearerCommand <integer>

Description

Max number of attempts to send delete bearer command.

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

n3DeleteIndirectDataFwdTunnel - N3 delete indirect data forward tunnel attempts

Syntax

--n3DeleteIndirectDataFwdTunnel <integer>

Description

Max number of attempts to send delete indirect data forward tunnel request.

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

n3DeletePdnConnSetResponse - N3 delete PDN conn set req attempts

Syntax

--n3DeletePdnConnSetResponse <integer>

Description

Max number of attempts to send delete PDN connection set request.

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

n3DeleteSessRequest - N3 delete session request attempts

Syntax

--n3DeleteSessRequest <integer>

Description

Max number of attempts to send delete session request.

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

n3FwdContext - N3 forward context attempts

Syntax

--n3FwdContext <integer>

Description

Max number of attempts to send a forward context message (29.060).

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

n3Identification - N3 identification request attempts

Syntax

--n3Identification <integer>

Description

Max number of attempts to send an identification request message.

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

n3ModifyAccessBearersRequest - N3 modify access bearers request attempts

Syntax

--n3ModifyAccessBearersRequest <integer>

Description

Max number of attempts to send modify access bearers request.

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

n3ModifyBearerCommand - N3 modify bearer command attempts

Syntax

--n3ModifyBearerCommand <integer>

Description

Max number of attempts to send modify bearer command.

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

n3ModifyBearerRequest - N3 modify bearer request attempts

Syntax

--n3ModifyBearerRequest <integer>

Description

Max number of attempts to send modify bearer request.

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

n3ReleaseAccessBearersRequest - N3 release access bearers request attempts

Syntax

--n3ReleaseAccessBearersRequest <integer>

Description

Max number of attempts to send release access bearers request.

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

n3RelocationCancel - N3 relocation cancel attempts

Syntax

--n3RelocationCancel <integer>

Description

Max number of attempts to send a relocation cancel message (29.060).

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

n3RelocationCmpl - N3 forward relocation complete attempts

Syntax

--n3RelocationCmpl <integer>

Description

Max number of attempts to send a relocation complete message (29.060).

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

n3RelocationReq - N3 forward relocation request attempts

Syntax

--n3RelocationReq <integer>

Description

Max number of attempts to send relocation request message (29.060).

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

n3ResumeNotification - N3 resume notification attempts

Syntax

--n3ResumeNotification <integer>

Description

Max number of attempts to send resume notification.

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

n3SrvccCancelNotification - N3 srvcc cancel notification attempts

Syntax

--n3SrvccCancelNotification <integer>

Description

Max number of attempts to send srvcc cancel notification.

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

n3SrvccRequest - N3 srvcc request attempts

Syntax

--n3SrvccRequest <integer>

Description

Max number of attempts to send srvcc request.

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

n3SuspendNotification - N3 suspend notification attempts

Syntax

--n3SuspendNotification <integer>

Description

Max number of attempts to send suspend notification.

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

n3Update - N3 update request attempts

Syntax

--n3Update <integer>

Description

Max number of attempts made by GTP-C to send a update request message (29.060).

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

numTransmits - Number of retransmits

Syntax

--numTransmits <integer>

Description

Number of transmits. if an acknowledgment is not sent in the time frame specified in the GTP message response timer, this value is the total number of times to transmit the message.

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

releaseAccessBearersResponseT3 - Release access bearers response timeout T3

Syntax

--releaseAccessBearersResponseT3 <integer>

Description

Max wait time for release access bearers response.

Class

Optional.

Type / Range

Integer 100...60000, step 100 in units of Millisecond

Default

3000

Name

relocationCancelT3 - Relocation cancel response timeout T3

Syntax

--relocationCancelT3 <integer>

Description

Max wait time for a response of a relocation cancel message (29.060).

Class

Optional.

Type / Range

Integer 100...60000, step 100 in units of Millisecond

Default

3000

Name

relocationCmplT3 - Relocation complete ack timeout T3

Syntax

--relocationCmplT3 <integer>

Description

Max wait time for a response to a relocation complete message (29.060).

Class

Optional.

Type / Range

Integer 100...60000, step 100 in units of Millisecond

Default

3000

Name

relocationReqT3 - Relocation request timeout T3 waiting for response

Syntax

--relocationReqT3 <integer>

Description

Max wait time for a response of a relocation request message (29.060).

Class

Optional.

Type / Range

Integer 100...60000, step 100 in units of Millisecond

Default

3000

Name

resumeNotificationAckT3 - Resume notification acknowledge timeout T3

Syntax

--resumeNotificationAckT3 <integer>

Description

Max wait time for resume notification acknowledge.

Class

Optional.

Type / Range

Integer 100...60000, step 100 in units of Millisecond

Default

3000

Name

retransTimerT3 - Retransmit timer T3

Syntax

--retransTimerT3 <integer>

Description

The time to wait for acknowledgment of a message.

Class

Optional.

Type / Range

Integer 100...60000, step 100 in units of Millisecond

Default

3000

Name

sendEchoOnFailure - Send echo requests on failure

Syntax

--sendEchoOnFailure <list-item>

Description

Controls sending echo requests when no GTP-C non-echo response messages are received for a period of time.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

sendEchoTimerT3 - Send echo request timer T3

Syntax

--sendEchoTimerT3 <integer>

Description

The periodic interval for sending echo requests on interfaces that support GTP communications.

Class

Optional.

Type / Range

Integer value of 60000...10800000, step 100 in units of Millisecond

Default

60000

Name

sendPeriodicEcho - Send periodic echo requests

Syntax

--sendPeriodicEcho <list-item>

Description

Controls sending of periodic echo requests on enabled links.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

srvccCancelAckT3 - Srvcc cancel acknowledge timeout T3

Syntax

--srvccCancelAckT3 <integer>

Description

Max wait time for srvcc cancel acknowledge.

Class

Optional.

Type / Range

Integer 100...60000, step 100 in units of Millisecond

Default

3000

Name

srvccResponseT3 - Srvcc response timeout T3

Syntax

--srvccResponseT3 <integer>

Description

Max wait time for srvc response.

Class

Optional.

Type / Range

Integer 100...60000, step 100 in units of Millisecond

Default

3000

Name

suspendAckT3 - Suspend acknowledge timeout T3

Syntax

--suspendAckT3 <integer>

Description

Max wait time for suspend acknowledge.

Class

Optional.

Type / Range

Integer 100...60000, step 100 in units of Millisecond

Default

3000

Name

updateT3 - Update response timeout T3

Syntax

--updateT3 <integer>

Description

Max wait time for a response to a update PDP context request message (29.060).

Class

Optional.

Type / Range

Integer 100...60000, step 100 in units of Millisecond

Default

3000

Name

waitEchoTimerT3 - Wait for echo response timer T3

Syntax

--waitEchoTimerT3 <integer>

Description

Specifies the time to wait for an echo response to be received before retransmitting the echo request message. the total time to wait for an echo response (N3-REQUESTS * T3-RESPONSE) should be less than the periodic time (Send echo request timer) used to send the echo request message.

Class

Optional.

Type / Range

Integer 1000...600000, step 100 in units of Millisecond

Default

6000

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.12 cmm gtpRestartCount

GTP Restart Count. Syntax, description, parameters, and roles for cmm gtpRestartCount command.

Syntax**Description**

Command to modify GTP link Restart Counter information. The GTP Restart Count Status command can used list and show the MME restart counter, and peer restart counter data.

Parameters**Name**

mmeRestart - MME Restart Counter

Syntax

--mmeRestart <integer>

Description

The local MME restart counter, as described in 29.274. MME restart counter applies to all GTP interfaces, not a specific linkIndex.

Class

Required on create.

Type / Range

Integer in the range of 0 to 255

Roles**Modify**

sysadmin, manage

5.13 cmm gtpRestartCountStatus

GTP Restart Count Status. Syntax, description, parameters, and roles for cmm gtpRestartCountStatus command.

Syntax**Description**

This command can be used to list the link Index; the MME restart counter; and peer restart counter for all the GTP links of a specified link Interface Type. This command can be used to show the the MME restart counter and peer restart counter for a specific GTP link Index. The GTP Restart Count command is used to modify the MME restart counter.

Parameters**Name**

linkInterfaceType - Link Interface type

Syntax

--linkInterfaceType <list-item>

Description

Link Interface Type applies only to the list and show operations. When included in the list operation, all links of this interface type will be listed. When included in the show operation, a link Index is expected so only the specific link Index for that interface type will be shown.

Class

Required key.

List

- s11 - Specify the S11 interface to an SGW.

Default

s11

Name

linkIndex - Link Index

Syntax

--linkIndex <integer>

Description

Link Index from the list operation, or from cmm multiLink or cmm linkAdmin commands

Class

Required key.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

gtpRestart - GTP Restart Counter

Syntax

--gtpRestart <integer>

Description

Peer GTP Restart Recovery Counter. Peer node type relates to the link Interface Type (e.g. SGW is the peer node for the S11interfaceType)

Class

Read only.

Type / Range

Integer in the range of 0 to 255

Name

mmeRestart - MME Restart

Syntax

--mmeRestart <integer>

Description

MME GTP Restart Recovery Counter

Class

Read only.

Type / Range

Integer in the range of 0 to 255

Roles

Show

sysadmin, manage, monitor

List

sysadmin, manage, monitor

5.14 cmm gtpTimer

gtpTimer. Syntax, description, parameters, and roles for cmm gtpTimer command.

Syntax

Description

Configure, list and show GTP timers.

Parameters

Name

gtptmrInUse - IN USE flag for GTP timer profile, INUSE

Syntax

--gtptmrInUse <list-item>

Description

This parameter specifies whether a specific GTP timer profile is currently in use. it is used in cases where operator wishes to disable a specific timer profile without deactivating the whole feature.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

gtptmrN3CreSesReqCounter - N3-response timer of create session response, N3CS

Syntax

--gtptmrN3CreSesReqCounter <integer>

Description

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunneling protocol V2 (GTPv2) to send a create session request message. the values range from 1 to 7 seconds in 1 second steps.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

gtptmrN3ModBeaReqCounter - N3-request counter for GTPv2 modify bearer request, N3MB

Syntax

--gtptmrN3ModBeaReqCounter <integer>

Description

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunneling protocol v2 (GTPv2) to send a modify bearer request message. the values range from 1 to 7 seconds in 1 second steps.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

gtptmrN3PdpCreReqCounter - N3-requests counter for create PDP context requests, N3CRE

Syntax

--gtptmrN3PdpCreReqCounter <integer>

Description

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunnelling protocol (GTP) to send a create PDP context request message. the values range from 1 to 7 seconds in 1 second steps.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

gtptmrN3PdpUpdReqCounter - N3-request counter for update PDP context requests, N3UPD

Syntax

--gtptmrN3PdpUpdReqCounter <integer>

Description

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunnelling protocol (GTP) to send an update PDP context request message. the values range from 1 to 7 seconds in 1 second steps.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

gtptmrPrName - GTP timer profile name, TMRPR

Syntax

--gtptmrPrName <string>

Description

This parameter defines a short name for a GTP timer profile.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 to 6 with the allowed pattern of [0-9a-zA-Z\._:-]*

Default

""

Name

gtptmrT3CreSesRespTimer - T3-response timer of create session response in seconds, T3CS

Syntax

--gtptmrT3CreSesRespTimer <integer>

Description

This parameter defines the value for the timer that holds the maximum waiting time for a create session response message to a create session request message. the values range from 1 to 10 seconds in 1 second steps.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

gtptmrT3ModBeaRespTimer - T3-response timer of modify bearer response in seconds, T3MB

Syntax

--gtptmrT3ModBeaRespTimer <integer>

Description

This parameter defines the value for the timer that holds the maximum waiting time for a modify bearer response message to a modify bearer request message. the values range from 1 to 10 seconds in 1 second steps.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

gtptmrT3PdpCreRespTimer - N3-requests counter for create PDP context requests, T3CRE

Syntax

--gtptmrT3PdpCreRespTimer <integer>

Description

This parameter defines the value for the timer that holds the maximum waiting time for a create PDP context response message to a create PDP context request message. the values range from 1 to 10 seconds in 1 second steps.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

gtptmrT3PdpUpdRespTimer - T3-response timer of update PDP context response, T3UPD

Syntax

```
--gtptmrT3PdpUpdRespTimer <integer>
```

Description

This parameter defines the value for the timer that holds the maximum waiting time for an update PDP context response message to an update PDP context request message. the values range from 1 to 10 seconds in 1 second steps.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

id - Id, TBLIND

Syntax

<integer>

Description

This parameter defines an id of GTP timer profile table to hold n3/t3 values.

Class

Required value.

Type / Range

Integer in the range of 1 to 6

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.15 cmm gwNodeAdmin

Signaling Node Admin. Syntax, description, parameters, and roles for cmm gwNodeAdmin command.

Syntax

Description

Signaling Node Admin. This command is used to (1) suspend a node from node selection, (2) resume selection of a node that was previously suspended. (3) to delete a node by giving node_index. (4) when reading pgw , PGW* and SGW* fields will be filled with data (5) when reading sgw , SGW* fields will be filled with data.

Parameters

Name

sigNodeType - Node type

Syntax

--sigNodeType <list-item>

Description

Node type specifies the type of node. the nodes types supported are pgw and sgw.

Class

Required key.

List

- pgw - PDN gateway - Specify node type of packet data network gateway.
- sgw - Serving gateway - Specify node type of serving gateway.

Name

networkTechType - Network technology type

Syntax

--networkTechType <list-item>

Description

Network technology type of node.

Class

Required key.

List

- lte - This selection provides node status maintained by the MME.

Name

nodeIndex - Node index

Syntax

--nodeIndex <integer>

Description

Node index.

Class

Required key.

Type / Range

Integer in the range of 1 to 10,000

Name

nodeAdminState - Node admin state

Syntax

--nodeAdminState <list-item>

Description

Specified current admin state.

Class

Optional.

List

- delete - To delete the entity. this is a special case as it is a command.
- resume - To resume the entity.
- suspend - To suspend the entity.
- unknown - This is not a valid value during update.

Name

numberOfTimeoutFailure - Number of timeout failure

Syntax

--numberOfTimeoutFailure <integer>

Description

Number of timeout failure.

Class

Read only.

Type / Range

Integer in the range of 0 to 65,535

Name

pgwApnNi - PGW APN NI

Syntax

--pgwApnNi <string>

Description

APN of a node.

Class

Read only.

Type / Range

Digit string of length 0 or 1 to 62

Name

pgwFqdn - PGW FQDN

Syntax

--pgwFqdn <string>

Description

FQDN of the node.

Class

Read only.

Type / Range

Digit string of length 0 or 1 to 128

Name

pgwNodeIpAddress1 - PGW IP address 1

Syntax

--pgwNodeIpAddress1 <string>

Description

IP address 1 of a node.

Class

Read only.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

pgwNodeIpAddress2 - PGW IP address 2

Syntax

--pgwNodeIpAddress2 <string>

Description

IP address 2 of a node.

Class

Read only.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

remainingTimeSuspendState - Remaining time in suspend state

Syntax

--remainingTimeSuspendState <string>

Description

Remaining Time in Suspend state. If manually suspended it will indicate Manually Suspended. If in resumed state it will be 0. If automatically suspended it will indicate the remaining time in minutes

Class

Read only.

Type / Range

Digit string of length 0 or 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

sgwFqdn - SGW FQDN

Syntax

--sgwFqdn <string>

Description

FQDN of the node of SGW.

Class

Read only.

Type / Range

Digit string of length 0 or 1 to 128

Name

sgwNodeIpAddress1 - SGW IP address 1

Syntax

--sgwNodeIpAddress1 <string>

Description

IP address 1 of a node.

Class

Read only.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

sgwNodeIpAddress2 - SGW IP address 2

Syntax

--sgwNodeIpAddress2 <string>

Description

IP address 2 of a node.

Class

Read only.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

5.16 cmm gwPool

GW pool. Syntax, description, parameters, and roles for cmm gwPool command.

Syntax**Description**

GW pool.

Parameters**Name**

gwPoolName - GW pool name

Syntax

--gwPoolName <string>

Description

GW pool name.

Class

Required key.

Type / Range

Digit string of length 1 to 20 with the allowed pattern of [0-9a-zA-Z\._ :]*

Name

gwType - PGW or SGW

Syntax

--gwType <list-item>

Description

PGW or SGW.

Class

Required on create.

List

- pgw - PDN gateway - Specify node type of packet data network gateway.
- sgw - Serving gateway - Specify node type of serving gateway.

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.17 cmm gwPoolMap

GW pool map. Syntax, description, parameters, and roles for cmm gwPoolMap command.

Syntax**Description**

GW pool map.

Parameters**Name**

gwPoolName - GW pool name

Syntax

--gwPoolName <string>

Description

GW pool name.

Class

Required key.

Type / Range

Digit string of length 1 to 20 with the allowed pattern of [0-9a-zA-Z\._ :]*

Name

ip - GW IP

Syntax

--ip <string>

Description

GW ip.

Class

Required key.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.18 cmm gwts

GW-TS. Syntax, description, parameters, and roles for cmm gwts command.

Syntax**Description**

This command is used to administer a GW-TS and includes parameters for defining the GW-TS ID, minimum number of TCP connections, and the GW-TS preference.

Parameters

Name

name - GW-TS name

Syntax

--name <string>

Description

GW-TS name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

gwtsId - GW-TS ID

Syntax

--gwtsId <integer>

Description

GW-TS ID.

Class

Required on create, not modifiable.

Type / Range

Integer in the range of 1 to 128

Name

minTcpConns - Minimum TCP connections

Syntax

--minTcpConns <integer>

Description

Minimum number of TCP connections required on a GW-TS for normal operation.

Class

Optional.

Type / Range

Integer in the range of 1 to 8

Default

2

Name

preference - Preference

Syntax

--preference <integer>

Description

Preference for a GW-TS.

Class

Optional.

Type / Range

Integer in the range of 1 to 1000

Default

1

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.19 cmm gwtsNodeAdmin

GWTS Node Admin. Syntax, description, parameters, and roles for cmm gwtsNodeAdmin command.

Syntax**Description**

GWTS Node Administration. This command is used to lock/unlock the GWTS node.

Parameters**Name**

gwtsName - GWTS node name

Syntax

--gwtsName <string>

Description

GWTS node name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

gwtsId - GWTS ID

Syntax

--gwtsId <integer>

Description

GWTS ID.

Class

Read only.

Type / Range

Integer in the range of 1 to 128

Default

1

Name

nodeAdminState - Node admin state

Syntax

--nodeAdminState <list-item>

Description

Specified current admin state.

Class

Optional.

List

- lock - To lock the entity.
- shutdown - To shutdown the entity.
- unlock - To unlock the entity.

Default

unlock

Name

nodeOperState - Node operation state

Syntax

--nodeOperState <list-item>

Description

Node operation state.

Class

Read only.

List

- disabled - Specify operational state of disabled.
- enabled - Specify operational state of enabled.
- unknown - Specify operational state of unknown.

Default

unknown

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

5.20 cmm gwtsRmtEndPtMap

GW-TS to Remote End Point. Syntax, description, parameters, and roles for cmm gwtsRmtEndPtMap command.

Syntax**Description**

This command is used to associate a GW-TS with a Remote End Point.

Parameters**Name**

gwtsName - GW-TS name

Syntax

--gwtsName <string>

Description

GW-TS name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

sgsLtRmtEndPtName - SGS lite remote end point name

Syntax

--sgsLtRmtEndPtName <string>

Description

Remote end point name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.21 cmm heartbeat

Heartbeat. Syntax, description, parameters, and roles for cmm heartbeat command.

Syntax**Description**

Heartbeat.

Parameters

Name

label - Label

Syntax

--label <string>

Description

Just need a parameter for the operation.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Modify

sysadmin, manage

5.22 cmm highPriArpGroup

High Priority ARP Group. Syntax, description, parameters, and roles for cmm highPriArpGroup command.

Syntax

Description

Customized Group users (IMSI) have higher ARP priorities than the default all users. If an IMSI is configured under this cli. SGSN will assign ARP 1 to it, also PVI, PCI and Queue should be configured.

Parameters

Name

entryId - Entry ID

Syntax

--entryId <integer>

Description

Defines the order i.e. precedence of the entry

Class

Required key.

Type / Range

Integer in the range of 1 to 1000

Name

plmnName - Home PLMN Name

Syntax

--plmnName <string>

Description

Home PLMN parameter set connected to the IMSI digits

Class

Required key.

Type / Range

Digit string of length 1 to 15 with the allowed pattern of [0-9a-zA-Z\.\-_:]*

Name

maxMsin - Maximum MSIN

Syntax

--maxMsin <digits>

Description

The maximum IMSI MSIN value.

Class

Optional.

Type / Range

Digit string of length 0 or 9 to 10

Default

9999999999

Name

minMsin - Minimum MSIN

Syntax

--minMsin <digits>

Description

The minimum IMSI MSIN value.

Class

Optional.

Type / Range

Digit string of length 0 or 9 to 10

Default

0000000000

Name

preLevel - Pre-emption Level

Syntax

--preLevel <integer>

Description

This parameter specifies the pre-emption capability of the requested radio bearer. This capability corresponds to the Pre-emption Capability IE in the ARP attribute of the radio bearer (see 3GPP TS 25.413). The default value for PCI shall be 1 (may trigger pre-emption). If it's configured as 0, SGSN Will send it to radio as 0 (shall not trigger pre-emption).

Class

Optional.

Type / Range

Integer in the range of 0 to 1

Default

1

Name

preVulnerability - Pre-emption Vulnerability

Syntax

--preVulnerability <integer>

Description

This parameter specifies the pre-emption vulnerability of the requested radio bearer. This vulnerability corresponds to the Pre-emption Vulnerability IE in the ARP attribute of the radio bearer. For details, see 3GPP TS 25.413. The default value of PVI shall be 0 (not pre-emptable). If it's configured as 1, SGSN will send it to radio as 1 (pre-emptable).

Class

Optional.

Type / Range

Integer in the range of 0 to 1

Default

0

Name

queue - Queue

Syntax

--queue <integer>

Description

Queuing Allowed. This parameter specifies whether a radio bearer setup request can be included in the resource allocation queue. This parameter corresponds to the Queuing

Allowed IE in the ARP attribute of the radio bearer. For details, see 3GPP TS 25.413. The default value shall be 1 (queuing allowed). if it's configured to 0, SGSN Will send 0 to RNC (queuing not allowed).

Class

Optional.

Type / Range

Integer in the range of 0 to 1

Default

1

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.23 cmm highTrafficCell

highTrafficCell. Syntax, description, parameters, and roles for **cmm highTrafficCell** command.

Syntax

Description

Configure, list and show distribution of cells. Use this command to add, interrogate or remove high traffic cell location in the PAPS. The count of possible high traffic cells is limited, that is, only the most important cells can be defined to be the high traffic cells.

Parameters

Name

cellIdentity - Cell identity

Syntax

--cellIdentity <integer>

Description

This parameter defines the cell number.

Class

Required on create, not filterable on list.

Type / Range

Integer in the range of 0 to 65,535

Name

lac - Location area code

Syntax

--lac <integer>

Description

This parameter defines the LAC number.

Class

Required on create, not filterable on list.

Type / Range

Integer in the range of 1 to 65,533

Name

mcc - Mobile country code

Syntax

--mcc <digits>

Description

This parameter defines the MCC number.

Class

Required on create, not filterable on list.

Type / Range

Digit string of length 3

Name

mnc - Mobile network code

Syntax

--mnc <digits>

Description

This parameter defines the MNC number.

Class

Required on create, not filterable on list.

Type / Range

Digit string of length 2 to 3

Name

papsIndex - PAPS index

Syntax

--papsIndex <integer>

Description

This parameter defines the PAPS to which the high traffic cell is located.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 19, 255 for invalid

Default

255

Name

id - Id

Syntax

<integer>

Description

High traffic cell id.

Class

Required value.

Type / Range

Integer in the range of 1 to 100

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.24 cmm hoCellIdRestrictionList

Handover Cell Identity Restriction List for MME. Syntax, description, parameters, and roles for cmm hoCellIdRestrictionList command.

Syntax

Description

Administer Handover Cell Identity Restriction List

Parameters

Name

name - Cell Identity Restriction List name

Syntax

--name <string>

Description

Identifies the name of the Cell Identity Restriction List.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

listInterface - List of Interfaces

Syntax

--listInterface <list-item>

Description

Identify the interface used for cell id based handover.

Class

Required on create, not modifiable.

List

- n26 - n26 Interface between MME and AMF - Cell identifier based handover over N26 interface between target MME and Source AMF

Name

listType - List Type

Syntax

--listType <list-item>

Description

Identify if the Cell Identity restriction list is of type allow or deny list.

Class

Optional.

List

- allow - Allow Handover Cell Ids list - Identify the list of Cell Ids from which handover is allowed
- deny - Deny Handover Cell Ids list - Identify the list of Cell Ids from which handover is denied

Default

allow

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.25 cmm hoCellIdRestrictionListItem

Handover Cell ID Restriction List. Syntax, description, parameters, and roles for cmm hoCellIdRestrictionListItem command.

Syntax**Description**

Administers addition (or deletion) of cell id to the handover cell id restriction list. MME supports up to 16 cell id restriction lists with a maximum number of 10k cell ids. A list can have up to 10k cell ids resulting in one possible list since maximum numbers of cell ids allowed is used.

Parameters

Name

hoCellIdRestrictionListName - Handover Cell ID Restriction List Name

Syntax

--hoCellIdRestrictionListName <string>

Description

Name of the handover cell id restriction list.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

cellIdName - Cell Identifier Name

Syntax

--cellIdName <string>

Description

Name of the cell id.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.26 cmm homeAccessList

home access list. Syntax, description, parameters, and roles for cmm homeAccessList command.

Syntax

Description

Configure, list and show home access list.

Parameters

Name

homeAccessListId - Home access list id

Syntax

--homeAccessListId <integer>

Description

Home access list identifier.

Class

Required key.

Type / Range

Integer in the range of 1 to 1000

Name

alName - Access list name

Syntax

--alName <string>

Description

Access list name.

Class

Optional.

Type / Range

Digit string of length 0 to 15 with the allowed pattern of [0-9a-zA-Z\.\-:]*

Default

""

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.27 cmm homeSubscriberGroup

home subscriber group. Syntax, description, parameters, and roles for cmm homeSubscriberGroup command.

Syntax

Description

Configure, list and show SGSN home subscribers.

Parameters

Name

homeSubscriberGroupId - Home subscriber group id

Syntax

--homeSubscriberGroupId <integer>

Description

Home subscriber group identifier.

Class

Required key.

Type / Range

Integer in the range of 1 to 5,000

Name

homeAccessListId - Home access list id

Syntax

--homeAccessListId <integer>

Description

Home access list identifier.

Class

Required on create, not modifiable.

Type / Range

Integer in the range of 1 to 1000

Name

accessRights - Access rights

Syntax

--accessRights <list-item>

Description

Access rights of subscriber group. possible values, norights ... no rights
2groamgeranandutran ... 2G roaming allowed with GERAN and UTRAN
3groamgeranandutran ... 3G roaming allowed with GERAN and UTRAN
2gand3groamgeranandutran ... 2G and 3G roaming allowed with GERAN and UTRAN
2groamgeran ... 2G roaming allowed with GERAN 3groamgeran ... 3G roaming allowed
with GERAN 2groamutran ... 2G roaming allowed with UTRAN 3groamutran ... 3G roaming
allowed with UTRAN 2gand3groamgeran ... 2G and 3G roaming allowed with GERAN
2gand3groamutran ... 2G and 3G roaming allowed with UTRAN.

Class

Optional.

List

- 2gAnd3gRoamGeran - 2G and 3G roaming allowed with GERAN - 2G and 3G roaming allowed with GERAN.
- 2gAnd3gRoamGeranAndUtran - 2G and 3G roaming allowed with GERAN and UTRAN - 2G and 3G roaming allowed with GERAN and UTRAN.
- 2gAnd3gRoamUtran - 2G and 3G roaming allowed with UTRAN - 2G and 3G roaming allowed with UTRAN.
- 2gRoamGeran - 2G roaming allowed with GERAN - 2G roaming allowed with GERAN.
- 2gRoamGeranAndUtran - 2G roaming allowed with GERAN and UTRAN - 2G roaming allowed with GERAN and UTRAN.
- 2gRoamUtran - 2G roaming allowed with UTRAN - 2G roaming allowed with UTRAN.
- 3gRoamGeran - 3G roaming allowed with GERAN - 3G roaming allowed with GERAN.
- 3gRoamGeranAndUtran - 3G roaming allowed with GERAN and UTRAN - 3G roaming allowed with GERAN and UTRAN.
- 3gRoamUtran - 3G roaming allowed with UTRAN - 3G roaming allowed with UTRAN.
- noRights - No rights - No rights.

Default

noRights

Name

imsi - IMSI

Syntax

--imsi <digits>

Description

Imsi.

Class

Optional.

Type / Range

Digit string of length 0 to 15

Default

""

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.28 cmm hpGwCcApn

Add APNNI (the hpwGwCc) to list. Syntax, description, parameters, and roles for cmm hpGwCcApn command.

Syntax

Description

Add APNNI (the hpwGwCc) to list

Parameters

Name

hpGwCcApnListName - Access point name - network identifier

Syntax

--hpGwCcApnListName <string>

Description

Access point name - network identifier

Class

Required key.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

apnNi - Access point name - network identifier

Syntax

--apnNi <string>

Description

Access point name - network identifier

Class

Required key.

Type / Range

Digit string of length 1 to 62 with the allowed pattern of [0-9a-zA-Z\._]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.29 cmm hpGwCcApnList

MSISDN based home GW selection with CC list. Syntax, description, parameters, and roles for cmm hpGwCcApnList command.

Syntax**Description**

MSISDN based home GW selection with CC list

Parameters

Name

name - create apnNI list

Syntax

--name <string>

Description

create apnNI list

Class

Required key.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.30 cmm hplmnToVplmnSnssaiMap

Home PLMN to Visited PLMN Snssai Map. Syntax, description, parameters, and roles for cmm hp_lmnToVp_lmnSnssaiMap command.

Syntax

Description

UE Home PLMN Slices mapping to Visited PLMN Slices used at UE initial registration or during idle or connected mode mobility 5G to 5G or 4G to 5G, when NSSF is not deployed or NSSF is deployed but operator setting is to use local configuration or NSSF is deployed but does not return Nssai mapping or returns an error.

Parameters

Name

hplmnToVplmnSnssaiMappingListName - Home PLMN to Visited PLMN SNSSAI Mapping List Name

Syntax

--hplmnToVplmnSnssaiMappingListName <string>

Description

Identifies the Home PLMN to Visited PLMN Snssai Mapping list name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

hSliceServiceType - Slice Service Type

Syntax

--hSliceServiceType <integer>

Description

The UE home PLMN S-NSSAI Slice Service Type (SST).

Class

Required key.

Type / Range

5G Slice Service Type

Name

hSliceDifferentiator - Slice Differentiator

Syntax

--hSliceDifferentiator <string>

Description

The UE home PLMN S-NSSAI Slice Differentiator (SD). Each slice must have SST and SD is optional. A SST may have multiple SDs.

Class

Required key.

Type / Range

Identifies Slice Differentiator (SD)

Name

vsnsaiName - Visited PLMN SNSSAI

Syntax

--vsnsaiName <string>

Description

Identifies the name of the Visited PLMN S-NSSAI that maps to the allowed Home PLMN S-NSSAI.

Class

Required on create.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.31 cmm hplmnToVplmnSnssaiMappingList

Home PLMN to Visited PLMN Snssai Mapping List Name. Syntax, description, parameters, and roles for cmm hplmnToVplmnSnssaiMappingList command.

Syntax**Description**

Identifies the name of the Home PLMN to Visited PLMN Snssai Mapping List

Parameters**Name**

name - Name of Snssai mapping list

Syntax

--name <string>

Description

Identifies the name of the Home PLMN to Visited PLMN Snssai Mapping List

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.32 cmm hssGrp

HSS Group Profile. Syntax, description, parameters, and roles for cmm hssGrp command.

Syntax

Description

This command is used to administer an HSS Group Profile and includes parameters for defining the HSS name, the HSS order in the group, and the weight used in the load balancing algorithm.

Parameters

Name

hssGrpListName - HSS group profile name

Syntax

--hssGrpListName <string>

Description

HSS group profile name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

hssOrderId - HSS order in group

Syntax

--hssOrderId <integer>

Description

HSS order in group.

Class

Required key.

Type / Range

Integer in the range of 1 to 16

Name

hssName - HSS name

Syntax

--hssName <string>

Description

HSS name specified in remote end point configuration.

Class

Required on create, not modifiable.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

lbWeight - Weight for load balancing

Syntax

--lbWeight <integer>

Description

Weight for load balancing algorithm.

Class

Optional.

Type / Range

Integer in the range of 0 to 100

Default

0

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.33 cmm hssGrpList

HSS Group Profile List. Syntax, description, parameters, and roles for cmm hssGrpList command.

Syntax

Description

This command is used to administer an HSS Group Profile List and includes the definition of the HSS Group Profile Name. This name is used to define an HSS/DRA set used in load balancing or in default primary/secondary/n-ary selection.

Parameters

Name

name - HSS group profile name

Syntax

--name <string>

Description

HSS group profile name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.34 cmm hssRestoration

HSS Restoration. Syntax, description, parameters, and roles for cmm hssRestoration command.

Syntax

Description

Trigger manually the process to restore normal services for UEs impacted by HSS unavailability period. A single restoration process may be ongoing at a time. Restoration Start action restarts the restoration process with new parameters, if the Start action is performed during an already ongoing process. By default an action is targeted to all impacted UEs. The command is blocked during ISSU.

Parameters

Name

action - Action

Syntax

--action <list-item>

Description

Define the HSS restoration process action.

Class

Required key.

List

- start - Starts the HSS restoration process.
- stop - Stops the HSS restoration process.

Name

maxDuration - Maximum Duration

Syntax

--maxDuration <integer>

Description

Defines the maximum duration of the HSS restoration process. After reaching the maximum duration of time before having all targeted UEs status restored back to normal, the MME will stop the restoration process. The remaining UEs will remain in their non-restored state.

Class

Optional.

Type / Range

Integer in the range of 10 to 300 in units of Minute

Default

60

Name

rate - Rate

Syntax

--rate <integer>

Description

Defines the rate of the restoration (UEs/sec).

Class

Optional.

Type / Range

Integer in the range of 10 to 1000

Default

100

Name

targetUes - Target UEs

Syntax

--targetUes <list-item>

Description

Defines the target UEs for the HSS restoration process in the MME.

Class

Optional.

List

- all - Triggers the restoration process for all impacted UEs: -detaches of UEs that are attached with the help of the failopen profile and -deactivation of failopen profile APN configuration based PDN for UEs that also have PDNs that are not marked as failopen PDNs.
- failopenDetach - Failopen Detach - Triggers the detaches of UEs that are attached with the help of the failopen profile.
- failopenPdnDeactivation - Failopen PDN Deactivation - Triggers the deactivation of failopen profile APN configuration based PDN for UEs that also have PDNs that are not marked as failopen PDNs.

Default

all

Roles

Modify

sysadmin, manage

5.35 cmm hssRestorationStats

HSS Restoration Statistics. Syntax, description, parameters, and roles for cmm hssRestorationStats command.

Syntax

Description

Displays the statistics of the HSS Restoration process.

Parameters

Name

action - Action

Syntax

--action <list-item>

Description

The HSS restoration process action.

Class

Read only.

List

- start - Starts the HSS restoration process.
- stop - Stops the HSS restoration process.

Name

additionalStatus - Additional Status

Syntax

--additionalStatus <string>

Description

Provides additional status information for a failure.

Class

Read only.

Type / Range

Digit string of length 0 to 255

Name

maxDuration - Maximum Duration

Syntax

--maxDuration <integer>

Description

The maximum duration of the HSS restoration process.

Class

Read only.

Type / Range

Integer in the range of 10 to 300 in units of Minute

Name

rate - Rate

Syntax

--rate <integer>

Description

The rate of the HS restoration process (UEs/sec).

Class

Read only.

Type / Range

Integer in the range of 10 to 1000

Name

status - Status

Syntax

--status <list-item>

Description

The status of the HSS restoration.

Class

Read only.

List

- completed - Restoration process is completed.
- running - Restoration process is running.
- stopped - Restoration process is stopped.
- maxDurationReachedNotFinished - Maximum duration reached not finished - The defined maximum duration time has been reached without finishing the HSS restoration process.

- failed - Restoration process has failed to complete due to an error.

Name

targetUes - Target UEs

Syntax

--targetUes <list-item>

Description

Defines the target UEs for the HSS restoration process in the MME.

Class

Read only.

List

- all - Triggers the restoration process for all impacted UEs: -detaches of UEs that are attached with the help of the failopen profile and -deactivation of failopen profile APN configuration based PDN for UEs that also have PDNs that are not marked as failopen PDNs.
- failopenDetach - Failopen Detach - Triggers the detaches of UEs that are attached with the help of the failopen profile.
- failopenPdnDeactivation - Failopen PDN Deactivation - Triggers the deactivation of failopen profile APN configuration based PDN for UEs that also have PDNs that are not marked as failopen PDNs.

Roles**List**

sysadmin, manage, monitor

5.36 cmm httpProfile

HTTP parameters profile. Syntax, description, parameters, and roles for cmm httpProfile command.

Syntax**Description**

Command to create, modify and delete HTTP profile.

Parameters**Name**

name - HTTP profile name for SBI

Syntax

--name <string>

Description

HTTP profile name for SBI.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

clientAutoRedirect - HTTP/2 Client Automatic Redirect Enable

Syntax

--clientAutoRedirect <list-item>

Description

This flag enables and disables HTTP/2 client automatic redirection upon receiving an HTTP/2 response with status code 307 Temporary Redirect. When enabled, automatic redirection triggered by status code 307 is performed. When disabled, automatic redirection is not performed, and the HTTP/2 transaction is considered failed.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

clientRestartTimer - Client Restart Timer

Syntax

--clientRestartTimer <integer>

Description

This parameter provisions a restart timer (in seconds) for SBI connections where CMM is the HTTP Client. When the timer is set and expires CMM will gracefully shutdown the connection. Setting this parameter to 0 disables this functionality. The parameter does not apply to NR10 interface connections.

Class

Optional.

Type / Range

Integer in the range of 0 or 300 to 3,600 in units of Second

Default

0

Name

headerTableSize - Header Table Size

Syntax`--headerTableSize <integer>`**Description**

Header table size parameter in the HTTP connection setup. A value of 0 means that HPACK Dynamic Compression is OFF.

Class

Optional.

Type / Range

Integer in the range of 0 to 32,768

Default

4096

Name

initialWindowSize - HTTP Initial Window Size

Syntax`--initialWindowSize <integer>`**Description**

Value of the Initial Window Size parameter in the HTTP connection setup.

Class

Optional.

Type / Range

Initial HTTP window size

Default

16384

Name

inputRateLimiting - HTTP/2 connection level input rate limiting enable

Syntax`--inputRateLimiting <list-item>`**Description**

This flag enables and disables HTTP/2 connection level input (ingress) rate limiting. When enabled, input rate limiting using HTTP/2 response status code 429 is performed. When disabled, connection level input rate limiting is not performed.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

maxConcurrentStreams - HTTP Maximum Concurrent Streams

Syntax

--maxConcurrentStreams <integer>

Description

Value of the Maximum Concurrent Streams parameter in the HTTP connection setup.

Class

Optional.

Type / Range

Integer in the range of 1 to 32,768

Default

100

Name

maxHeaderListSize - HTTP Maximum Header List Size

Syntax

--maxHeaderListSize <integer>

Description

Value of the Maximum Header List Size parameter in the HTTP connection setup.

Class

Optional.

Type / Range

Integer in the range of 1 to 32,768

Default

2000

Name

refusedStreamHandling - Refused Stream Handling

Syntax

--refusedStreamHandling <list-item>

Description

This parameter defines how AMF handles the reception of a HTTP2 Stream Reset (RST_STREAM) with Error REFUSED_STREAM (0x7) in the SBI interfaces when acting as a HTTP Client. The parameter can have following values: None: Reset messages for Refused Streams are ignored; Link Error: Reset messages for Refused Streams are handled as link error for the specific Stream.

Class

Optional.

List

- linkError - Link Error - Stream Reset is handled as Link Error
- none - Stream Reset is not handled

Default

none

Name

serverRestartTimer - Server Restart Timer

Syntax

--serverRestartTimer <integer>

Description

This parameter provisions a restart timer (in seconds) for SBI connections where CMM is the HTTP Server. When the timer is set and expires CMM will gracefully shutdown the connection. Setting this parameter to 0 disables this functionality. The parameter does not apply to NR10 interface connections.

Class

Optional.

Type / Range

Integer in the range of 0 or 300 to 3,600 in units of Second

Default

0

Name

serverStreamLifetime - Server Stream Lifetime

Syntax

--serverStreamLifetime <integer>

Description

Maximum default Lifetime of a HTTP/2 Server stream. Note that in certain occasions such as namf-mt ue-reachind and namf-loc provide-pos-info/provide-loc-info the timer may be overwritten/extended by call processing.

Class

Optional.

Type / Range

Integer in the range of 6 to 30 in units of Second

Default

6

Name

timerToWaitForSettingsFrameWithAck - Timer to wait for SETTINGS frame with ack

Syntax

--timerToWaitForSettingsFrameWithAck <integer>

Description

Timer started after sending SETTINGS frame for the reception of SETTINGS frame with ack.
Time unit millisecond with 100 as granularity.

Class

Optional.

Type / Range

Integer in the range of 100 to 60,000 in units of Millisecond

Default

2000

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.37 cmm httpProxy

HTTP Proxy. Syntax, description, parameters, and roles for cmm httpProxy command.

Syntax

Description

This command is used to manage HTTP Proxies. Up to 64 Proxies can be created.

Parameters

Name

name - Name

Syntax

--name <string>

Description

Name of the HTTP Proxy.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

reportedName - Reported Name

Syntax

--reportedName <string>

Description

The name used as the HTTP Proxy identity in PM and PCMD reports.

Class

Required on create.

Type / Range

Digit string of length 1 to 96 with the allowed pattern of [0-9a-zA-Z\.\-_:]*

Name

capacity - Capacity

Syntax

--capacity <integer>

Description

Relative capacity of the HTTP Proxy.

Class

Optional.

Type / Range

Integer in the range of 0 to 65,535

Default

65535

Name

enable - Enable

Syntax

--enable <list-item>

Description

When HTTP Proxy is enabled it is considered candidate for new Requests When disabled only ongoing sessions are served until completion

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

fqdn - Fully Qualified Domain Name

Syntax

--fqdn <string>

Description

FQDN of HTTP Proxy. If HTTP Proxy is provisioned with FQDN and nfEndpointListName is not provisioned then DNS A/AAAA query will be triggered for IP resolution and standard ports will be used (80 for HTTP and 443 for HTTPS) Up to 2 IPv6 addresses will be selected with higher priority (as long as AMF interface also supports IPv6). Otherwise up to 2 IPv4 endpoints will be selected.

Class

Optional.

Type / Range

Digit string of length 0 to 96 with the allowed pattern of [0-9a-zA-Z\.\-_:]*

Default

""

Name

fqdnPort - FQDN Port

Syntax

--fqdnPort <integer>

Description

Port to be used when HTTP proxy address is given as FQDN and resolved via DNS A/AAAA query. When set to '0' default ports are used (80 for scheme http and 443 for https)

Class

Optional.

Type / Range

Integer in the range of 0 to 65,535

Default

0

Name

httpScheme - HTTP Scheme

Syntax

--httpScheme <list-item>

Description

Defines use of HTTP or HTTPS towards HTTP Proxy. When HTTPS is selected Proxy's FQDN must be provided

Class

Optional.

List

- http - The schema of Hypertext Transfer Protocol.
- https - The schema of Hypertext Transfer Protocol Secure.

Default

https

Name

nfIpEndPointListName - NF End Point IP List Name

Syntax

--nfIpEndPointListName <string>

Description

HTTP Proxy's End Point IP List Name. Up to 2 Endpoints are supported towards a HTTP Proxy which must be of same IP version. IPv6 addresses will be selected with higher priority as long as AMF interface also supports IPv6. Otherwise IPv4 endpoints will be selected. If Endpoint list is not provided FQDN must be provisioned for DNS resolution via A/AAAA query.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"""

Name

proxyPrefix - Proxy Prefix

Syntax`--proxyPrefix <string>`**Description**

Optional deployment-specific string to be pretended to the ':path' HTTP/2 pseudo-header when sending messages to the HTTP Proxy. This string is removed by the Proxy when the message is forwarded to the next hop.

Class

Optional.

Type / Range

API Prefix as defined in 29.500. May be null otherwise shall have at least allowed character, shall not start with / and shall not end with /. When not-null, the leading / is implied and shall not be included.

Default

"""

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.38 cmm httpProxyIsolationPerCauseCodeSettings

HTTP Proxy Isolation per Status Code Settings. Syntax, description, parameters, and roles for cmm httpProxyIsolationPerCauseCodeSettings command.

Syntax

Description

Includes action for specific Cause/Status codes. Used when the action for a Causes/Status codes combination for a service needs to differentiate from the default behaviour provisioned for the relevant status class within the sbilndirectCommunicationGParams CLI.

Parameters

Name

statusCode - Status Code

Syntax

--statusCode <list-item>

Description

The HTTP Status Code

Class

Required key.

List

- 400_Bad_Request - 400 Bad Request
- 401_Unauthorized - 401 Unauthorized
- 403_Forbidden - 403 Forbidden
- 404_Not_Found - 404 Not Found
- 405_Method_Not_Allowed - 405 Method Not Allowed
- 406_Not_Acceptable - 406 Not Acceptable
- 408_Request_Timeout - 408 Request Timeout
- 409_Conflict - 409 Conflict
- 410_Gone - 410 Gone
- 411_Length_Required - 411 Length Required
- 412_Precondition_Failed - 412 Precondition Failed
- 413_Payload_Too_Large - 413 Payload Too Large
- 414_URI_Too_Long - 414 URI Too Long
- 415_Unsupported_Media_Type - 415 Unsupported Media Type
- 429_Too_Many_Requests - 429 Too Many Requests
- 500_Internal_Server_Error - 500 Internal Server Error
- 501_Not_Implemented - 501 Not Implemented
- 502_Bad_Gateway - 502 Bad Gateway

- 503_Service_Unavailable - 503 Service Unavailable
- 504_Gateway_Timeout - 504 Gateway Timeout

Name

errorCause - Error Cause

Syntax

--errorCause <string>

Description

Application Error Cause value. Used to applies string match with the Cause in the response. 'NULL' string literal has special meaning and is used to match cases where Response did not include Application Error Cause

Class

Required key.

Type / Range

SBI Application Error Cause value. Digit string of length 1 to 64 with the allowed pattern of [A-Z_]*

Name

action - Action

Syntax

--action <list-item>

Description

Action to be applied for specific status code and Error cause

Class

Optional.

List

- Isolate - Isolate the Service Instance
- KPI_Fail - KPI Fail - Increase the Failure Counter used for the Service Instance KPI
- KPI_Success - KPI Success - Increase the Success counter used for the Service Instance KPI

Default

KPI_Fail

Name

enable - Enable

Syntax

--enable <list-item>

Description

Controls whether the action is enabled or disabled. User created and System inserted actions (i.e. actions automatically created by the system during initial deployment) may be enabled or disabled but only user created actions may be deleted.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.39 cmm httpProxyList

HTTP Proxy List. Syntax, description, parameters, and roles for `cmm httpProxyList` command.

Syntax**Description**

List of HTTP proxies. Up to 16 HTTP Proxies lists may be defined.

Parameters**Name**

name - Name

Syntax

--name <string>

Description

Name of list

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.40 cmm httpProxyListItem

HTTP Proxies. Syntax, description, parameters, and roles for cmm httpProxyListItem command.

Syntax**Description**

Adds or removes a HTTP Proxy from a HTTP proxy list with a given priority. Each list may have up to 16 HTTP Proxies. The same Proxy may be part of multiple lists. The same HTTP Proxy have different priority in different lists.

Parameters**Name**

httpProxyListName - HTTP Proxy List Name

Syntax

--httpProxyListName <string>

Description

Name of HTTP Proxy List

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

httpProxyName - HTTP Proxy Name

Syntax

--httpProxyName <string>

Description

Name of HTTP Proxy.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

priority - Priority

Syntax

--priority <integer>

Description

Priority of HTTP Proxy. Priority is only relevant for the specific HTTP Proxy list.

Class

Optional.

Type / Range

Integer in the range of 0 to 15

Default

0

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.41 cmm httpProxyListItemStatus

HTTP Proxy List Item Status. Syntax, description, parameters, and roles for cmm httpProxyListItemStatus command.

Syntax

Description

List the status of the HTTP Proxies included in a Proxy list

Parameters

Name

blacklisted - Blacklisted

Syntax

--blacklisted <list-item>

Description

Blacklist status

Class

Optional.

List

- blacklisted - Status Blacklisted
- kpi-blacklisted - KPI Blacklisted - Status Blacklisted due to low KPI
- nonblacklisted - Non Blacklisted - Status not Blacklisted
- unknown - Status Unknown

Name

httpProxyListName - HTTP Proxy List Name

Syntax

--httpProxyListName <string>

Description

Name of the HTTP Proxy List

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-V:@]*

Name

httpProxyName - HTTP Proxy Name

Syntax

--httpProxyName <string>

Description

Name of the HTTP Proxy

Class

Optional.

Type / Range

Digit string of length 0 to 128

Roles**List**

sysadmin, manage, monitor

5.42 cmm imeiAnalysis

imeiAnalysis. Syntax, description, parameters, and roles for cmm imeiAnalysis command.

Syntax**Description**

Configure, list and show IMEI analysis.

Parameters**Name**

apno - Feature name of APN not overridden

Syntax

--apno <list-item>

Description

Feature name of APN not overridden.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ccgb - Feature name of GB cause code set name

Syntax

--ccgb <list-item>

Description

Feature name of GB cause code set name.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

cciu - Feature name of IU cause code set name

Syntax

--cciu <list-item>

Description

Feature name of IU cause code set name.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

gbset - GB cause code set name for GB cause code set name feature

Syntax

--gbset <string>

Description

GB cause code set name for GB cause code set name feature.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 1 to 10 with the allowed pattern of [0-9a-zA-Z\._ :]*

Default

0

Name

ggsn - Feature name of IMEI based GGSN selection

Syntax

--ggsn <list-item>

Description

Feature name of IMEI based GGSN selection.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ggsnTag - TAG identifier for new IMEI based GGSN selection for IMEI based GGSN selection feature

Syntax

--ggsnTag <string>

Description

TAG identifier for new IMEI based GGSN selection for IMEI based GGSN selection feature.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 1 to 10 with the allowed pattern of [0-9a-zA-Z\._ :]*

Default

''

Name

iuset - IU cause code set name for IU cause code set name feature

Syntax

```
--iuset <string>
```

Description

IU cause code set name for IU cause code set name feature.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 1 to 10 with the allowed pattern of [0-9a-zA-Z\._ :]*

Default

0

Name

pgw - Feature name of IMEI based P-GW selection

Syntax

```
--pgw <list-item>
```

Description

Feature name of IMEI based P-GW selection.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

pgwTag - TAG identifier for new IMEI based P-GW selection for IMEI based P-GW selection feature

Syntax

```
--pgwTag <string>
```

Description

TAG identifier for new IMEI based P-GW selection for IMEI based P-GW selection feature.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 1 to 10 with the allowed pattern of [0-9a-zA-Z\._ :]*

Default

''

Name

id - International mobile equipment identity id

Syntax

<digits>

Description

International mobile equipment identity id.

Class

Required value.

Type / Range

Digit string of length 1 to 14

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.43 cmm imeiOperatorPolicy

IMEI Operator Policy. Syntax, description, parameters, and roles for cmm imeiOperatorPolicy command.

Syntax**Description**

Define IMEI operator policy. Maximum number of operator policies is 32.

Parameters**Name**

name - Name

Syntax

--name <string>

Description

The name pointing out the operator policy criteria.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

accessRestricted - Access Restricted

Syntax

--accessRestricted <list-item>

Description

Defines whether access is restricted for UEs matching the provisioned criteria.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

accessRestrictedAttachRejectCause - Access Restricted Attach Reject Cause

Syntax

--accessRestrictedAttachRejectCause <integer>

Description

Defines the Attach Reject EMM cause when IMEI based operator policy to restrict access is applied for the UE.

Class

Optional.

Type / Range

Integer in the range of 3,6-8,11-15,22,25,35,42

Default

15

Name

accessRestrictedServiceRejectCause - Access Restricted Service Reject Cause

Syntax

--accessRestrictedServiceRejectCause <integer>

Description

Defines the Service Reject EMM cause when IMEI based operator policy to restrict access is applied for the UE.

Class

Optional.

Type / Range

Integer in the range of 3,6-12,35,40,42

Default

7

Name

accessRestrictedTauRejectCause - Access Restricted TAU Reject cause

Syntax

--accessRestrictedTauRejectCause <integer>

Description

Defines the TAU Reject EMM cause when IMEI based operator policy to restrict access is applied for the UE.

Class

Optional.

Type / Range

Integer in the range of 3,6,7,9,10-12,14,15,35,40,42

Default

7

Name

authenticationDisabledPolicy - authentication disabled policy

Syntax

--authenticationDisabledPolicy <list-item>

Description

Defines whether authentication shall be disabled for the UEs matching the provisioned criteria. If policy is set to true, then authentication should be disabled for matching UE. If policy is set to false, then authentication should follow plmnSecurity. Note: disable authentication applies for optional part authentication only.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

eirCheckDisabledPolicy - EIR check disabled policy

Syntax

--eirCheckDisabledPolicy <list-item>

Description

Defines whether EIR check shall be disabled for the UEs matching the provisioned criteria.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.44 cmm imeiOperatorPolicyCriteria

Operator Policy Criteria. Syntax, description, parameters, and roles for cmm imeiOperatorPolicyCriteria command.

Syntax

Description

Define operator policy criteria to find UEs to apply operator policies. MME supports altogether up to 25 000 IMEI Type Approval Codes (TACs) on IMEI TAC lists. IMEI TAC ranges and lists of IMEI TACs (or any other parameters) may overlap. Both IMEI TAC range and IMEI TAC list may be included on the same search criteria. An IMEI TAC matches if the IMEI TAC can be found in either the IMEI TAC list or the IMEI TAC range. At minimum either the IMEI TAC list or the IMEI TAC range must be specified. At maximum 10000 operator policy criteria can be defined.

Parameters

Name

entryId - Entry ID

Syntax

--entryId <integer>

Description

Defines the order i.e. precedence of the entry.

Class

Required key.

Type / Range

Integer in the range of 1 to 10,000

Name

imeiOperatorPolicyName - IMEI Operator Policy Name

Syntax

--imeiOperatorPolicyName <string>

Description

The name of the operator policy to apply to the UEs matching the criteria.

Class

Required on create.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

adminState - Admin state

Syntax

--adminState <list-item>

Description

Administrative state.

Class

Optional.

List

- locked - Locked.
- unlocked - Unlocked.

Default

locked

Name

imeiTacSvListName - IMEI TAC SV List Name

Syntax

--imeiTacSvListName <string>

Description

The IMEI TAC SV List name the operator policy applies to.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

maxImeiSv - Maximum IMEI SV

Syntax

--maxImeiSv <digits>

Description

The maximum IMEI SV in range.

Class

Optional.

Type / Range

Digit string of length 2 to 2

Default

99

Name

maxImeiTac - Max IMEI TAC

Syntax

--maxImeiTac <digits>

Description

Maximum IMEI TAC for IMEI TAC range to apply operator policy to.

Class

Optional.

Type / Range

Digit string of length 0 or 8 to 8

Default

"

Name

maxMsin - Maximum MSIN

Syntax

--maxMsin <digits>

Description

The maximum IMSI MSIN value to apply operator policy to.

Class

Optional.

Type / Range

Digit string of length 0 or 8 to 10

Default

"

Name

minImeiSv - Minimum IMEI SV

Syntax

--minImeiSv <digits>

Description

The minimum IMEI SV in range.

Class

Optional.

Type / Range

Digit string of length 2 to 2

Default

00

Name

minImeiTac - Min IMEI TAC

Syntax

--minImeiTac <digits>

Description

Minimum IMEI TAC value of IMEI TAC range to apply operator policy to.

Class

Optional.

Type / Range

Digit string of length 0 or 8 to 8

Default

''

Name

minMsin - Minimum MSIN

Syntax

--minMsin <digits>

Description

The minimum IMSI MSIN value to apply operator policy to.

Class

Optional.

Type / Range

Digit string of length 0 or 8 to 10

Default

''

Name

servingPlmnName - Serving Mobile Network Name

Syntax

--servingPlmnName <string>

Description

Name that identifies the serving mobile network.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Default

''

Name

uePlmnName - UE Mobile Network Name

Syntax

```
--uePlmnName <string>
```

Description

Name that identifies the mobile network of the UE.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.45 cmm imeiTacRfsp

IMEI TAC RFSP. Syntax, description, parameters, and roles for cmm imeiTacRfsp command.

Syntax**Description**

This command is used to administer the TAC (Type Approval Code) and RFSP(RAT/Frequency Selection Priority) table to activate TAC/RFSP based features.

Parameters**Name**

imeiTac - Type approval code

Syntax

```
--imeiTac <digits>
```

Description

Type Approval Code of the IMEI

Class

Required key.

Type / Range

Digit string of length 8 to 8

Name

rfsp - IMEI TAC RFSP

Syntax

--rfsp <integer>

Description

IMEI TAC with associated RFSP value. Indicate to use index value for RAT/Frequency selection priority configured with the associated TAC locally in the UE home PLMN should be used

Class

Optional.

Type / Range

Integer in the range of 0 to 256

Default

1

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.46 cmm imeiTacSv

IMEI TAC SV. Syntax, description, parameters, and roles for cmm imeiTacSv command.

Syntax

Description

This command is used to administer the TAC (Type Approval Code) and SV (Software Version) table to activate TAC/SV based features. Complete maximum number of IMEI TAC SV entries on all lists is 25 000. Maximum number of IMEI TAC SV lists is 1000.

Parameters

Name

imeiTacSvListName - IMEI TAC SV List Name

Syntax

--imeiTacSvListName <string>

Description

name of the IMEI TAC SV List.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

imeiTac - Type approval code

Syntax

--imeiTac <digits>

Description

Type Approval Code of the IMEI

Class

Required key.

Type / Range

Digit string of length 8 to 8

Name

maxImeiSv - Max IMEI Software Version

Syntax

--maxImeiSv <digits>

Description

The maximum IMEI SV in range.

Class

Optional.

Type / Range

Digit string of length 2 to 2

Default

99

Name

minImeiSv - Min IMEI Software Version

Syntax

--minImeiSv <digits>

Description

The minimum IMEI SV in range.

Class

Optional.

Type / Range

Digit string of length 2 to 2

Default

00

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.47 cmm imeiTacSvList

IMEI Type Allocation Code Software Version List. Syntax, description, parameters, and roles for cmm imeiTacSvList command.

Syntax

Description

Define IMEI TAC SV list. Complete maximum number of IMEI TAC SV entries on all lists is 25 000. Maximum number of IMEI TAC SV lists is 1000.

Parameters

Name

name - IMEI TAC SV List Name

Syntax

--name <string>

Description

IMEI TAC SV List Name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.48 cmm imeiTacSvRejectionProfile

IMEI TAC SV Rejection Profile. Syntax, description, parameters, and roles for cmm imeiTacSvRejectionProfile command.

Syntax

Description

Define IMEI TAC SV Rejection profile.

Parameters

Name

name - IMEI TAC SV Rejection Profile Name

Syntax

--name <string>

Description

Profile name defining the IMEI-TAC MME local rejection parameters.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

attachRejectCc - Attach Reject Cause Code

Syntax

--attachRejectCc <integer>

Description

Define the EMM cause code when a UE with IMEI TAC to be rejected attempts to attach.

Class

Optional.

Type / Range

Integer in the range of 3,6-8,11-15,22,25,35,42

Default

15

Name

extendedEmmCauseCurrentRatNotAllowed - Extended EMM cause IE Current RAT Not Allowed

Syntax

--extendedEmmCauseCurrentRatNotAllowed <list-item>

Description

Defines whether the MME includes the Extended EMM cause IE on the Attach Rejects and/or TAU rejects, when the Attach reject and/or TAU reject cause is set to EMM cause code 15 (No suitable cells in tracking area). When the MME includes the Extended EMM Cause IE, the MME sets it to either 'E-UTRAN not allowed' (if the UE attempts to access WB-E-UTRAN) or 'NB-IoT not allowed' (if the UE attempts to access NB-IoT).

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

imeiTacSvListName - IMEI TAC SV List Name

Syntax

--imeiTacSvListName <string>

Description

Define the IMEI TAC SV List Name for MME local rejection.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

serviceRejectCc - Service Reject Cause Code

Syntax

--serviceRejectCc <integer>

Description

Define the EMM cause code when a UE with IMEI TAC to be rejected attempts Service Request.

Class

Optional.

Type / Range

Integer in the range of 3,6-12,35,40,42

Default

7

Name

tauRejectCc - TAU Reject Cause Code

Syntax

--tauRejectCc <integer>

Description

Define the EMM cause code when a UE with IMEI TAC to be rejected attempts Tracking Area Update.

Class

Optional.

Type / Range

Integer in the range of 3,6,7,9,10-12,14,15,35,40,42

Default

7

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.49 cmm imsiAnalysis

imsiAnalysis. Syntax, description, parameters, and roles for **cmm imsiAnalysis** command.

Syntax

Description

Configure, list and show IMSI analysis.

Parameters

Name

plmnName - Name of the PLMN parameter set connected to the IMSI digits

Syntax

--plmnName <string>

Description

This parameter indicates the name of the home PLMN of the IMSI number.

Class

Required on create, not filterable on list.

Type / Range

Digit string of length 1 to 15 with the allowed pattern of [0-9a-zA-Z\._ :]*

Name

dia - Digits to be added

Syntax

--dia <digits>

Description

Use this parameter to specify the digits to be added to the IMSI number. use this parameter only when the TYPE OF ADDRESS parameter has the GT value. this parameter is obligatory if the START POINT OF DIGITS TO BE ADDED parameter has a value other than 0.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 to 15 with the allowed pattern of [0-9]*

Default

0

Name

ni - Network indicator

Syntax

--ni <list-item>

Description

Use this parameter to identify the type of the network. the possible values are, IN0 ... international network IN1 ... spare NAO ... national network NA1 ... reserved for national use.

Class

Optional, not filterable on list.

List

- IN0 - International network number 0.
- IN1 - International network number 1.
- NAO - National network number 0.
- NA1 - National network number 1.

Default

NA0

Name

np - Numbering plan

Syntax

--np <list-item>

Description

The numbering plan refers to the standardised numbering instructions described by the ITU-T. use this parameter only when the TYPE OF ADDRESS parameter gets the GT value. the possible values are, E164 ... the result GT address follows the ITU-T recommendation E.163/E.164 (the numbering plan for the international telephone service/the numbering plan for ISDN era). the result GT address is in international ISDN number or national (significant) number format, depending on the value of the TYPE OF NUMBER parameter. E212 ... the result GT address follows the ITU-T recommendation E.212 (the identification plan for land mobile stations). the result GT address is in IMSI or NMSI format, depending on the value of the TYPE OF NUMBER parameter. E214 ... the result GT address follows the ITU-T recommendation E.214 (the structure of the land mobile global title for the signalling connection control part). the result GT address is in mobile global title format if the value of the TYPE OF NUMBER parameter is INT. it is in NC+MSIN format if the value of the TYPE OF NUMBER parameter is NAT.

Class

Optional, not filterable on list.

List

- E164 - E.164 - Specify numbering plan of E.164.
- E212 - E.212 - Specify numbering plan of E.212.
- E214 - E.214 - Specify numbering plan of E.214.

Default

E212

Name

nrd - Number of digits to be removed

Syntax

--nrd <integer>

Description

Use this parameter to specify the number of digits to be removed from the IMSI number. Use this parameter only when the TYPE OF ADDRESS parameter has the GT value. This parameter must have the value 0 (default) if the START POINT OF DIGITS TO BE REMOVED parameter has the value 0. This parameter must have a value other than 0 if the START POINT OF DIGITS TO BE REMOVED parameter has a value other than 0. The values of START POINT OF DIGITS TO BE REMOVED and NUMBER OF DIGITS TO BE REMOVED are calculated together. As the IMSI has 15 digits the sum of these parameters should not extend 16 as it will be bigger than the length of IMSI

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 15

Default

0

Name

sad - Start point of digits to be added

Syntax

--sad <integer>

Description

Use this parameter to indicate the start point of digits to be added to the IMSI number. use this parameter only when the TYPE OF ADDRESS parameter has the GT value. if you use the DIGITS TO BE ADDED parameter, give this parameter a value other than 0.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 15

Default

0

Name

spc - Signalling point code

Syntax

--spc <integer>

Description

Use this parameter to identify a signalling point code in the signalling network.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 16,383

Default

0

Name

srd - Start point of digits to be removed

Syntax

--srd <integer>

Description

Use this parameter to define the start point of the digits to be removed from the IMSI number. use this parameter only when the TYPE OF ADDRESS parameter gets the GT value. this parameter must have a value other than 0 if the NUMBERING PLAN parameter has any other value than E212. this parameter must have the value 0 if the NUMBER OF DIGITS TO BE REMOVED parameter has the value 0. this parameter must have a value other than 0 if the NUMBER OF DIGITS TO BE REMOVED parameter has any other value than 0.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 15

Default

0

Name

toA - Identify type of RNC address

Syntax

--toA <list-item>

Description

Use this parameter to define the type of address given as the result of the analysis. the possible values are, GT ... global title SPC ... signalling point code.

Class

Optional, not filterable on list.

List

- GT - Global title address.
- SPC - Destination point code (ni + spc).

Default

GT

Name

ton - Type of number

Syntax

--ton <list-item>

Description

Use this parameter to identify the type of number for the result address. use this parameter only when the TYPE OF ADDRESS parameter gets the GT value. the possible values are, INT ... international number NAT ... national number.

Class

Optional, not filterable on list.

List

- INT - International - Specify address format as international.
- NAT - National - Specify address format as national.

Default

INT

Name

id - IMSI digits for the analysis

Syntax

<digits>

Description

Use this parameter to indicate the IMSI number for which you create the analysis. the IMSI number uniquely identifies the mobile subscriber and consists of the national mobile subscriber identifier and the mobile country code.

Class

Required value.

Type / Range

Digit string of length 1 to 15

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.50 cmm imsiGroupId

IMSI Group Identifier. Syntax, description, parameters, and roles for cmm imsiGroupId command.

Syntax

Description

Specifies an IMSI Group Identifier according to 3GPP 23.003 Section 19.9

Parameters

Name

imsiGroupIdListName - Name

Syntax

```
--imsiGroupIdListName <string>
```

Description

Internal Name for IMSI Group ID. This is used to assign the IMSI Group ID to a list.

Class

Required key.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

groupServiceId - Group Service ID

Syntax

```
--groupServiceId <integer>
```

Description

Specifies the Group Service ID portion of the IMSI Group ID. 15 Group Service IDs are supported.

Class

Required key.

Type / Range

Integer in the range of 0 to 4,294,967,294

Name

localGroup - Local Group

Syntax

```
--localGroup <string>
```

Description

Local Group portion of the IMSI Group ID

Restrictions

String length of 10 octets, where each octet is composed of two hex digits each with the allowed pattern of [0-9a-fA-F\.:/*]

Class

Required key.

Type / Range

Hexadecimal value ranging from 0 to 10 octets in length.

Name

plmnName - PLMN Name

Syntax

--plmnName <string>

Description

UE's Home PLMN

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.51 cmm imsiGroupIdList

IMSI Group ID List. Syntax, description, parameters, and roles for cmm imsiGroupIdList command.

Syntax**Description**

List of IMSI Group Identifiers

Parameters

Name

name - IMSI Group ID Name

Syntax

--name <string>

Description

Name of IMSI Group ID to be added to the list.

Class

Required key.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

''

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.52 cmm imsiList

IMSI List. Syntax, description, parameters, and roles for cmm **imsiList** command.

Syntax

Description

Define list of individual IMSIs. The IMSIs inside an IMSI list may be related to different PLMNs. A single IMSI list may contain up to 1000 IMSIs. The IMSI lists altogether may contain up to 1000 individual IMSIs. Up to 64 IMSI lists may be created.

Parameters

Name

name - Name

Syntax

--name <string>

Description

The name of the IMSI list.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.53 cmm imsiRangeServices

Services based on IMSI Range within UE Network. Syntax, description, parameters, and roles for cmm imsiRangeServices command.

Syntax**Description**

This command is used to administer services based on IMSI Range within UE Network and includes parameters for defining APN QOS profiles, NAS mapping profiles, and AMBR values.

Parameters**Name**

uePlmnServicesName - Name of services based on UE network and served network

Syntax

--uePlmnServicesName <string>

Description

Name that identifies roaming agreement.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

entryId - Entry ID

Syntax

--entryId <integer>

Description

Entry ID.

Class

Required key.

Type / Range

Integer in the range of 1 to 10,000

Name

adminState - Administrative state

Syntax

--adminState <list-item>

Description

Administrative state.

Class

Optional.

List

- locked - Locked.
- unlocked - Unlocked.

Default

locked

Name

apnCorrectionAttach - APN correction during attach

Syntax

--apnCorrectionAttach <list-item>

Description

APN correction that is done during attach

Class

Optional.

List

- featureDisabled - Feature disabled - This feature is disabled.
- locallyConfigured - Locally configured - Locally configured.
- receivedFromHss - Received from HSS - Received from HSS.

Default

featureDisabled

Name

apnCorrectionStandalone - APN correction during standalone PDN connect

Syntax

--apnCorrectionStandalone <list-item>

Description

APN correction that is done during standalone PDN connectivity request.

Class

Optional.

List

- featureDisabled - Feature disabled - This feature is disabled.
- locallyConfigured - Locally configured - Locally configured.
- receivedFromHss - Received from HSS - Received from HSS.

Default

featureDisabled

Name

ciotProfileName - IoT Profile Name

Syntax

--ciotProfileName <string>

Description

Profile for IoT settings to be used in IMSI Range Service

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Name

combinedAttachWithoutCsProfileName - Combined Attach without CS Profile Name

Syntax

--combinedAttachWithoutCsProfileName <string>

Description

The profile name to include parameters for combined attach without CS domain.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

cpCIoT - CP CIOT

Syntax

--cpCIoT <list-item>

Description

Specifies whether CP CIoT is allowed for the IMSI range

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

custApn1QosprofileName - Custom APN QoS profile for roamers - 1

Syntax

--custApn1QosprofileName <string>

Description

Points to the first custom APN QoS profile for roamers.

Restrictions

This element is not applicable when the serving and UE PLMNs are the same. This element applies only to 4G.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

custApn1WpsQosprofileName - Custom APN WPS QoS profile for roamers - 1

Syntax

--custApn1WpsQosprofileName <string>

Description

Points to the first custom APN WPS QoS profile for roamers.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

custApn2QosprofileName - Custom APN QoS profile for roamers - 2

Syntax

--custApn2QosprofileName <string>

Description

Points to the second custom APN QoS profile for roamers.

Restrictions

This element is not applicable when the serving and UE PLMNs are the same. This element applies only to 4G.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

custApn2WpsQosprofileName - Custom APN WPS QoS profile for roamers - 2

Syntax

--custApn2WpsQosprofileName <string>

Description

Points to the second custom APN WPS QoS profile for roamers.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Name

custApn3QosprofileName - Custom APN QoS profile for roamers - 3

Syntax

--custApn3QosprofileName <string>

Description

Points to the third custom APN QoS profile for roamers.

Restrictions

This element is not applicable when the serving and UE PLMNs are the same. This element applies only to 4G.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Name

custApn3WpsQosprofileName - Custom APN WPS QoS profile for roamers - 3

Syntax

--custApn3WpsQosprofileName <string>

Description

Points to the third custom APN WPS QoS profile for roamers.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Name

description - IMSIRange entry description

Syntax

--description <string>

Description

Description for IMSI range services entry.

Class

Optional.

Type / Range

Digit string of length 0 to 160 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

diamNasMappingProfileName - Diameter cause code to NAS cause code profile name

Syntax

--diamNasMappingProfileName <string>

Description

Defines all diameter cause code to NAS cause code mapping.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

disableUeMonitoring - Disable UE Monitoring

Syntax

--disableUeMonitoring <list-item>

Description

This field specifies whether the UEs in this IMSI Range Service should not be allowed to use monitoring events.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

edrxApnListName - eDRX APN list name

Syntax

--edrxApnListName <string>

Description

eDRX APN list name.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

"

Name

edrxEnabled - eDRX enabled

Syntax

--edrxEnabled <list-item>

Description

Indicates if eDRX is enabled.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

edrxHssPtwProfileName - eDRX HSS PTW profile name

Syntax

--edrxHssPtwProfileName <string>

Description

eDRX HSS PTW profile name.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

default

Name

edrxModeWithPsm - eDRX mode with PSM

Syntax

--edrxModeWithPsm <list-item>

Description

eDRX mode with PSM.

Class

Optional.

List

- Both - Specifies both eDRX and PSM.
- PSM - Specifies PSM.
- eDRX - Specifies eDRX.

Default

eDRX

Name

edrxProfileName - eDRX profile name

Syntax

--edrxProfileName <string>

Description

eDRX profile name.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

default

Name

edrxSource - eDRX source

Syntax

--edrxSource <list-item>

Description

eDRX source.

Class

Optional.

List

- HSS - Specifies eDRX source as HSS.
- Local - Specifies eDRX source as local.

Default

Local

Name

emmInfoOverrideProfileName - EMM Info Override Profile Name

Syntax

--emmInfoOverrideProfileName <string>

Description

EMM Info Override Profile to be used for this IMSI range

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

''

Name

ePlmnListName - EPLMN list name

Syntax

--ePlmnListName <string>

Description

EPLMN list name.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

ePlmnListNameEnb - EPLMN List Name for eNB

Syntax

--ePlmnListNameEnb <string>

Description

Identifies Equivalent PLMN list name to be sent to eNB

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Name

extendedEmergencyNumListName - Extended Emergency Number List Name

Syntax

--extendedEmergencyNumListName <string>

Description

Extended Emergency Number List Name

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Name

externalIdSupported - External ID Supported

Syntax

--externalIdSupported <list-item>

Description

Specifies whether the MME advertises support of the External ID to the HSS.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

hoRestrictAreaListName - Handover restrict area list name

Syntax

--hoRestrictAreaListName <string>

Description

Defines the UE roaming restrict profile to be used for the service

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Name

imeiTacSvListName - imeiTacSvListName

Syntax

--imeiTacSvListName <string>

Description

List containing IMEI Type Approval Codes (TAC) and Software Versions (SV) to be used for record matching.

Restrictions

An IMEI TAC/SV list name may not be populated if an MSIN subrange or a TAI List subrange is populated.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Name

imsApnQosprofileName - Default IMS APN QoS profile for roamers

Syntax

--imsApnQosprofileName <string>

Description

Points to the roaming QoS profile for IMS APNs. IMS APN QoS profile name is not used when served PLMN and UE PLMN are the same. Otherwise, the default is Def-Roam-IMS-APN-QoS-Prfl.

Restrictions

This element is not applicable when the serving and UE PLMNs are the same. This element applies only to 4G.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

imsApnWpsQosprofileName - Default IMS APN WPS QoS profile for roamers

Syntax

--imsApnWpsQosprofileName <string>

Description

Points to the roaming WPS QoS profile for IMS APNs. IMS APN WPS QoS profile name is not used when Serving PLMN equals UE PLMN or when Treat As Home Subscriber is true.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

imsLboProfileName - IMS LBO Profile Name

Syntax

--imsLboProfileName <string>

Description

The name of the IMS LBO profile.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

initialGutiAttachEirPercentage - Initial GUTI Attach EIR Percentage

Syntax

--initialGutiAttachEirPercentage <integer>

Description

Parameter controls percentage EIR checking for Initial GUTI Attach Procedure. Default value of 0 indicates no EIR checking. This field is valid when the Attach EIR Percentage Checking General Global Parameter is enabled.

Class

Optional.

Type / Range

Integer in the range of 0 to 100

Default

0

Name

initialImsiAttachEirPercentage - Initial IMSI Attach EIR Percentage

Syntax

--initialImsiAttachEirPercentage <integer>

Description

Parameter controls percentage EIR checking for Initial IMSI Attach Procedure. Default value of 0 indicates no EIR checking. This field is valid when the Attach EIR Percentage Checking General Global Parameter is enabled.

Class

Optional.

Type / Range

Integer in the range of 0 to 100

Default

0

Name

lcsOptionsProfileName - Location services options profile name

Syntax

--lcsOptionsProfileName <string>

Description

Defines location services options.

Restrictions

This element applies only to 4G.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

default

Name

locationBasedGatewaySelection - Location Based Gateway Selection.

Syntax

--locationBasedGatewaySelection <list-item>

Description

Specifies whether location based gateway selection is allowed for IMSI range.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

maxMsin - Maximum MSIN

Syntax

--maxMsin <digits>

Description

The maximum IMSI MSIN value.

Class

Optional.

Type / Range

Digit string of length 0 or 8 to 10

Default

"

Name

mccHmRtPgWdnsDmOvrd - MCC home routed PGW DNS domain override

Syntax

--mccHmRtPgWdnsDmOvrd <digits>

Description

Override the normal PGW domain derivation by using the given MCC and MNC values for a home routed APN.

Class

Optional.

Type / Range

Digit string of length 0 or 3

Name

minMsin - Minimum MSIN

Syntax

--minMsin <digits>

Description

The minimum IMSI MSIN value.

Class

Optional.

Type / Range

Digit string of length 0 or 8 to 10

Default

"

Name

mncHmRtPgWdnsDmOvrd - MNC home routed PGW DNS domain override

Syntax

--mncHmRtPgWdnsDmOvrd <digits>

Description

Override the normal PGW domain derivation by using the given MCC and MNC values for a home routed APN.

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 3

Default

"

Name

n26IwkMmeRestrictionTaiListName - MME N26 IWK Restriction TAI List

Syntax

--n26IwkMmeRestrictionTaiListName <string>

Description

N26 interworking restriction TAI list used by MME to restrict 4G service when UE(s) move from 5G into the respective TACs in the TAI list.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 31 with the allowed pattern of [0-9a-zA-Z_\.\\-:@]*

Default

""

Name

nonImsApnQosprofileName - Default non-ims APN QoS profile for roamers

Syntax

--nonImsApnQosprofileName <string>

Description

Points to the roaming QoS profile for non IMS APNs. Non IMS APN QoS profile name is not used when served PLMN and UE PLMN are the same. Otherwise, the default is Def-Roam-NonIMS-APN-QoS-Prfl.

Restrictions

This element is not applicable when the serving and UE PLMNs are the same. This element applies only to 4G.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

nonImsApnWpsQosprofileName - Default non-ims APN WPS QoS profile for roamers

Syntax

--nonImsApnWpsQosprofileName <string>

Description

Points to the roaming QoS profile for non IMS APNs. Non IMS APN WPS QoS profile name is not used when Serving PLMN equals UE PLMN or when Treat As Home Subscriber is true.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

pgwDomainMccOverride - PGW domain name MCC override

Syntax

--pgwDomainMccOverride <digits>

Description

Override of normal PGW DNS domain derivation by using this mobility country code and the mobility network code represented by pgwdomainmncoveride.

Restrictions

1. This element is not applicable when the serving and UE PLMNs are the same. 2. If this element is present so must be pgwDomainMncOverride

Class

Optional.

Type / Range

Digit string of length 0 or 3

Name

pgwDomainMncOverride - PGW domain name MNC override

Syntax

--pgwDomainMncOverride <digits>

Description

Override of normal PGW DNS domain derivation by using this mobility network code and the mobility country code represented by pgwdomainmccoverride.

Restrictions

1. This element is not applicable when the serving and UE PLMNs are the same. 2. If this element is present so must be pgwDomainMccOverride

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 3

Name

plmnSecurityProfileListName - Name of identifies profile

Syntax

--plmnSecurityProfileListName <string>

Description

Name of identifies plmnSecurityProfile, which will be used for IMSI Range

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Default

""

Name

psmApnListName - PSM APN list name

Syntax

--psmApnListName <string>

Description

PSM APN list name.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 31 with the allowed pattern of [0-9a-zA-Z_\.\\-:@]*

Default

""

Name

psmEnabled - PSM enabled

Syntax

--psmEnabled <list-item>

Description

Indicates if PSM is enabled. This field is only used if the Power Saving Mode global parameter is enabled.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

reAuthTimePeriod - Reauthentication Time Period

Syntax

--reAuthTimePeriod <integer>

Description

This field defines the time period between successive UE authentications forced by the MME. When value equals zero then the feature is disabled

Class

Optional.

Type / Range

Integer in the range of 60 to 1440 including value zero in units of Minute

Default

0

Name

restNasMappingProfileName - Access restriction to NAS cause code profile name

Syntax

--restNasMappingProfileName <string>

Description

Defines all access restriction to NAS cause code mapping.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

default

Name

rfspAccessRestrictionListName - Access Restriction for RFSP List Name

Syntax

--rfspAccessRestrictionListName <string>

Description

A reference to a list of RFSP values for which access is to be restricted for the UE.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

''

Name

separateEplmnListNasS1ap - Separate EPLMN List on NAS and S1AP

Syntax

--separateEplmnListNasS1ap <list-item>

Description

Enable sending separate Equivalent PLMN Lists to NAS and S1AP. The EPLMN Lists applied to NAS and S1AP are copied from EPLMN List Name and EPLMN List Name for eNB, respectively.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

shortDataService - Short data service

Syntax

--shortDataService <list-item>

Description

Enable SDS for an MSIN range.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

support5gSrvcc - Support 5G-SRVCC

Syntax

--support5gSrvcc <list-item>

Description

Enables the intermediary role of the MME_SRVCC (MME supporting 5G-SRVCC) for the UE for regular voice calls. The MME_SRVCC passes messages between the AMF and MSC server during 5G-SRVCC handover of voice calls from NG-RAN/5GS to 3GPP UTRAN/CS domain.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

svcAgreementProfileName - Service agreement profile name

Syntax

```
--svcAgreementProfileName <string>
```

Description

Defines the PLMN service agreement profile to be used for service.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

default

Name

taiLaiArealistName - TAI/LAI area list name

Syntax

```
--taiLaiArealistName <string>
```

Description

Defines the area in which the service agreement profile applies for the IMSI range.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

all

Name

taiListPref - TAI List Preference

Syntax

```
--taiListPref <list-item>
```

Description

Specifies which TAI list to use for the imsi Range Services record match.

Class

Optional.

List

- TAI List - TAI List.
- TAI Range List - TAI Range List

Default

TAI List

Name

taiRangeListName - TAI Range List Name

Syntax

--taiRangeListName <string>

Description

Reference to a TAI Range List

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

""

Name

taiZoneMappingListName - Zone TAI Mapping List Name

Syntax

--taiZoneMappingListName <string>

Description

TAI Zone Mapping List Name

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

""

Name

treatAsHomeSubscriber - Treat roamer as home subscriber

Syntax

--treatAsHomeSubscriber <list-item>

Description

When true, treat a roaming UE as if it were a home subscriber of the serving PLMN. This overrides the LTE Treat As Home Subscriber setting in UE PLMN Services for the given IMSI range. If Treat As Home Subscriber is true for a roaming UE, the IMS APN QoS Profile and Non-IMS APN QoS Profile fields do not take effect even if populated.

Restrictions

This element is not applicable when the serving and UE PLMNs are the same.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ueAmbrDownLink - UE AMBR downlink bitrate

Syntax

--ueAmbrDownLink <integer>

Description

UE aggregate maximum downlink bitrate.

Restrictions

The granularity is 8 when the value is between 64 and 568. The granularity is 64 when the value is between 576 and 8640. The granularity is 100 when the value is between 8700 and 16000. The granularity is 1000 when the value is between 17000 and 128000. The granularity is 2000 when the value is between 130000 and 256000. This element is not applicable when the serving and UE PLMNs are the same.

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64- 568, 576-8640, 8700-16000, 17000-128000, 130000-4000000000 in units of kbps

Default

0

Name

ueAmbrUpLink - UE AMBR uplink bitrate

Syntax

--ueAmbrUpLink <integer>

Description

UE aggregate maximum uplink bitrate.

Restrictions

The granularity is 8 when the value is between 64 and 568. The granularity is 64 when the value is between 576 and 8640. The granularity is 100 when the value is between 8700 and 16000. The granularity is 1000 when the value is between 17000 and 128000. The granularity is 2000 when the value is between 130000 and 256000. This element is not applicable when the serving and UE PLMNs are the same.

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64- 568, 576-8640, 8700-16000, 17000-128000, 130000-4000000000 in units of kbps

Default

0

Name

ueAmbrUseHssValue - UE AMBR use HSS value

Syntax

--ueAmbrUseHssValue <list-item>

Description

Enable or disable locally configured uplink UE-AMBR and downlink UE-AMBR values with subscribed uplink UE-AMBR and downlink UE-AMBR values in HSS for IMSI NS.

Restrictions

This element is not applicable when the serving and UE PLMNs are the same.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Name

ueUsageType - ueUsageType

Syntax

--ueUsageType <integer>

Description

UE usage type to be assigned to UE. This field is applicable only if ueUsageSourceImsiRange in uePlmnServices is set appropriately. A value of 0 means that no UUT is assigned to the UE via IMSI Range Services.

Class

Optional.

Type / Range

UE Usage type in the range of 1 to 126 or 128 to 255, or 0 if not used.

Default

0

Name

useApnOiToDetermineHrt - Use APN-OI to Determine HRT

Syntax

--useApnOiToDetermineHrt <list-item>

Description

Current parameter enables using the APN-OI instead of the PGW FQDN to determine if a PDN Session is HRT for a roaming subscriber when applying QoS modification for a PDN session in accordance to local configuration and when sending the UE Time Zone IE in UBR and CBR Response messages. The parameter is effective only when the gParm useApnOiToDetermineHrt is also enabled.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

uutOverrideProfileName - UE Usage Type Override Profile Name

Syntax

--uutOverrideProfileName <string>

Description

Specifies profile name for assigning local UE Usage Type override

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

"

Name

wusProfileName - Wake Up Signal Profile Name

Syntax

--wusProfileName <string>

Description

Specifies WUS profile to use for UEs from this IMSI Range

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

zoneCodeExtendedEmergencyNumListName - Zone Code Extended Emergency Number List
Name

Syntax

--zoneCodeExtendedEmergencyNumListName <string>

Description

Zone Code Extended Emergency Number List Name

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.54 cmm imsiRangeServicesPlmn

PLMN based IMSI Range Services. Syntax, description, parameters, and roles for cmm
imsiRangeServicesPlmn command.

Syntax

Description

This identifies a set of imsi-ranges for a PLMN covering all TACs

Parameters

Name

uePlmnServicesName - Services defined for an imsi-range

Syntax

--uePlmnServicesName <string>

Description

PLMN based imsi-range defined services/capabilities

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

entryId - imsi-range Entry ID

Syntax

--entryId <integer>

Description

imsi-range Entry ID

Class

Required key.

Type / Range

Integer in the range of 1 to 1000

Name

description - imsi-range Entry description

Syntax

--description <string>

Description

describes offered services

Class

Optional.

Type / Range

Digit string of length 0 to 160 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Default

""

Name

enhancedDualConnectivity - 5G NSA Option 3X

Syntax

--enhancedDualConnectivity <list-item>

Description

This parameter enables, for an imsi-range, 5G NSA Option 3x with enhanced capabilities which include higher bit rates, NR access restriction and S/PGW selection with NR capability

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

enhancedDualConnectivityProfileName - Enhanced Dual Connectivity Profile Name

Syntax

--enhancedDualConnectivityProfileName <string>

Description

Identifies the Enhanced Dual Connectivity Profile for NSA Option 3A/3X

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

""

Name

maxMsin - Maximum MSIN

Syntax

--maxMsin <digits>

Description

The maximum IMSI MSIN value

Class

Optional.

Type / Range

Digit string of length 8 to 10

Default

999999999

Name

minMsin - Minimum MSIN

Syntax

--minMsin <digits>

Description

The minimum IMSI MSIN value

Class

Optional.

Type / Range

Digit string of length 8 to 10

Default

000000000

Name

shutdown - Shutdown

Syntax

--shutdown <list-item>

Description

When shutdown is true, the imsi-range rule is not active

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.55 cmm imsiToHss

IMSI to HSS Routing. Syntax, description, parameters, and roles for cmm imsiToHss command.

Syntax

Description

This command is used to administer IMSI to HSS Routing and includes parameters for defining FQDN names, load balancing, PLMN name, subscriber database type, and IMSI MSIN values. The 'shutdown' parameter is used to enable/disable the mapping rule.

Parameters

Name

mappingId - Mapping identifier

Syntax

--mappingId <integer>

Description

Unique identifier for this mapping rule.

Class

Required key.

Type / Range

Integer in the range of 1 to 8,192

Name

mappingType - Mapping type

Syntax

--mappingType <list-item>

Description

Designates whether even IMSI values, odd IMSI values, or an optional range (minimum/maximum IMSI values) should be used.

Class

Required on create, not modifiable.

List

- ALL - All.
- EVEN - Even.
- ODD - Odd.

Name

plmnName - Mobile network name

Syntax

--plmnName <string>

Description

Name that identifies mobile network.

Class

Required on create, not modifiable.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

draDestHostFqdnName - DRA destination host name

Syntax

--draDestHostFqdnName <string>

Description

If MME does not have destination host in VLR (UE shows up for the first time), and the destination is a DRA, MME uses the value of this new field to populate destination host AVP in AIR and ULR.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

draDestRealmFqdnName - DRA destination realm name

Syntax

--draDestRealmFqdnName <string>

Description

If MME does not have destination realm in VLR (UE shows up for the first time), and the destination is a DRA, MME uses the value of this new field to populate destination realm AVP in AIR and ULR.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Name

hssGrpListName - Hss group profile name

Syntax

--hssGrpListName <string>

Description

HSS group profile name.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Name

loadBalance - Load balance

Syntax

--loadBalance <list-item>

Description

Indicates load balancing options for S6ad.

Class

Optional.

List

- all - Balance load on all S6ad links based on weight.
- none - No load balancing between S65ad links.
- priority group - Balance load on all S6ad links based on group priority. the primary group is the set of HSS/DRA that have the same highest weight. the secondary group is the set of HSS/DRA that have the same second highest weight value, etc. the last N-ary group is always the group with 0 weight (if used).

Default

none

Name

maxMsin - Maximum MSIN

Syntax

--maxMsin <digits>

Description

This field is used to enter the maximum IMSI MSIN value. Note: minimum/maximum ranges must be non-overlapping.

Class

Optional.

Type / Range

Digit string of length 0 to 10

Name

minMsin - Minimum MSIN

Syntax

--minMsin <digits>

Description

This field is used to enter the minimum IMSI MSIN value. Note: minimum/maximum ranges must be non-overlapping.

Class

Optional.

Type / Range

Digit string of length 0 to 10

Name

shutdown - Shutdown

Syntax

--shutdown <list-item>

Description

Indicates if the mapping rule is active. if shutdown is true, the rule behaves as if it is deleted.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

subscriberDbType - Subscriber database type

Syntax

--subscriberDbType <list-item>

Description

Subscriber database type.

Class

Optional.

List

- EPC-Capability - Specify EPC capability.
- HLR - Specify HLR database.
- HSS - Specify HSS database.

Default

EPC-Capability

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.56 cmm imsiToList

IMSI to List. Syntax, description, parameters, and roles for cmm imsiToList command.

Syntax

Description

Add a new IMSI to the IMSI list.

Parameters

Name

imsiListName - IMSI List Name

Syntax

```
--imsiListName <string>
```

Description

The name of the IMSI List.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

imsi - IMSI

Syntax

```
--imsi <digits>
```

Description

The full IMSI including the MNC and MCC.

Class

Required key.

Type / Range

Digit string of length 9 to 15

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.57 cmm imsiUseLocalDns

IMSI Use Local DNS. Syntax, description, parameters, and roles for cmm imsiUseLocalDns command.

Syntax

Description

This command will be used for configuring the IMSIs which will be allowed to use local DNS for resolution.

Parameters

Name

entryId - Entry Id

Syntax

--entryId <integer>

Description

Entry Id

Class

Required key.

Type / Range

Integer in the range of 1 to 100

Name

maxMsin - Maximum MSIN

Syntax

--maxMsin <digits>

Description

The maximum IMSI MSIN value. For an IMSI the allowed length is 8-10. Under each entryId, the range of MSIN (between maxMsin and minMsin) has to be smaller than 100. The total number of MSINs under all configured entryIds can not exceed 100.

Class

Required on create.

Type / Range

Digit string of length 8 to 10

Name

mcc - Mobile Country Code

Syntax

--mcc <digits>

Description

Mobile Country Code

Class

Required on create, not modifiable.

Type / Range

Digit string of length 3 to 3 with the allowed pattern of [0-9]*

Name

minMsin - Minimum MSIN

Syntax

--minMsin <digits>

Description

The minimum IMSI MSIN. For an IMSI the allowed length is 8-10. Under each entryId, the range of MSIN (between maxMsin and minMsin) has to be smaller than 100. The total number of MSINs under all configured entryIds can not exceed 100.

Class

Required on create.

Type / Range

Digit string of length 8 to 10

Name

mnc - Mobile Network Code

Syntax

--mnc <digits>

Description

Mobile Network Code

Class

Required on create, not modifiable.

Type / Range

Digit string of length 2 to 3 with the allowed pattern of [0-9]*

Name

applicationInd - IMSI For S4 SGSN indicator

Syntax

--applicationInd <list-item>

Description

This parameter is to indicate entryId IMSI is for S4 SGSN or MME or MME and S4SGSN.

Class

Optional.

List

- mme - MME will use local DNS file
- mmeS4sgsn - MME and S4 SGSN - MME and S4 SGSN will use local DNS file
- s4sgsn - S4 SGSN - S4 SGSN will use DNS file

Default

mmeS4sgsn

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.58 cmm imsLboProfile

IMS LBO Profile. Syntax, description, parameters, and roles for cmm `imsLboProfile` command.

Syntax

Description

The profile that MME uses when the MME is locally provisioned to force local breakout for the PDN towards the IMS APN. Max 64 profiles can be created.

Parameters

Name

name - Name

Syntax

--name <string>

Description

The name of the IMS LBO profile.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

apnAmbrDI - APN AMBR Downlink

Syntax

--apnAmbrDI <integer>

Description

Aggregate maximum bit rate for downlink for the IMS APN.

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64- 568, 576-8640, 8700-16000, 17000-128000,
130000-4000000000

Default

100000000

Name

apnAmbrUI - APN AMBR Uplink

Syntax

--apnAmbrUI <integer>

Description

Aggregate maximum bit rate for uplink for the IMS APN.

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64- 568, 576-8640, 8700-16000, 17000-128000,
130000-4000000000

Default

100000000

Name

applyImsApnSpecificChargingCharacteristics - Apply IMS APN Specific Charging
Characteristics

Syntax

--applyImsApnSpecificChargingCharacteristics <list-item>

Description

Defines whether IMS APN Specific Charging Characteristics based on the provisioned profile is applied.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

applyLocalImsNotSubscribed - Apply Local IMS Not Subscribed

Syntax

--applyLocalImsNotSubscribed <list-item>

Description

Defines whether the MME applies the local IMS profile for the UE when the APN configuration profile on the UE HSS subscription does not include explicit IMS APN configuration.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

applyLocalImsSubscribedLboAllowed - Apply Local IMS Subscribed LBO Allowed

Syntax

--applyLocalImsSubscribedLboAllowed <list-item>

Description

Defines whether the MME applies the local IMS profile for the UE when the APN configuration profile on the UE HSS subscription includes explicit IMS APN configuration, and the VPLMN-Dynamic-Address-Allowed AVP is set to ALLOWED on the IMS APN configuration.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

applyLocalImsSubscribedLboNotAllowed - Apply Local IMS Subscribed LBO Not Allowed

Syntax

--applyLocalImsSubscribedLboNotAllowed <list-item>

Description

Defines whether the MME applies the local IMS profile for the UE when the APN configuration profile on the UE HSS subscription includes explicit IMS APN configuration, and the VPLMN-Dynamic-Address-Allowed AVP is either absent or set to NOTALLOWED on the IMS APN configuration.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

arpPriorityLevel - ARP Priority Level

Syntax

--arpPriorityLevel <integer>

Description

Allocation and retention priority, priority level for the IMS APN.

Class

Optional.

Type / Range

Integer in the range of 1 to 15

Default

15

Name

chargingCharacteristics - Charging Characteristics

Syntax

```
--chargingCharacteristics <string>
```

Description

Charging Characteristics for the IMS APN.

Class

Optional.

Type / Range

Digit string of length 4 with the allowed pattern of [0-9a-fA-F]*

Default

0000

Name

interworking5gsIndicator - Interworking 5GS Indicator

Syntax

```
--interworking5gsIndicator <list-item>
```

Description

Indicates whether the interworking between 5GS and EPS is subscribed for the IMS APN.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

pdnAddrType - PDN Address Type

Syntax

```
--pdnAddrType <list-item>
```

Description

The PDN address type for the IMS APN.

Class

Optional.

List

- ipv4 - Specify PDN address type of IPv4.
- ipv4OrIpv6 - Specify PDN address type of IPv4 or IPv6.
- ipv4v6 - Specify PDN address type of IPv4 and IPv6.

- ipv6 - Specify PDN address type of IPv6.

Default

ipv6

Name

preemptionCapability - Pre-emption Capability

Syntax

--preemptionCapability <list-item>

Description

Allocation and retention priority, pre-emption capability for the IMS APN.

Class

Optional.

List

- DISABLE - Disable
- ENABLE - Enable

Default

DISABLE

Name

preemptionVulnerability - Pre-emption Vulnerability

Syntax

--preemptionVulnerability <list-item>

Description

Allocation and retention priority, pre-emption vulnerability for the IMS APN.

Class

Optional.

List

- DISABLE - Disable
- ENABLE - Enable

Default

ENABLE

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.59 cmm integrityAlgorithm

Integrity Algorithm. Syntax, description, parameters, and roles for cmm integrityAlgorithm command.

Syntax

Description

This command sets the algorithms, in priority order, MME will use to negotiate Integrity Protection of NAS messages with the UE. At attach and TAU, the UE provides, in the UE EPS security capabilities, which integrity protection algorithms are supported. The MME will use the algorithm both supported by the UE and highest in priority in the eia integrity protection algorithm priority list. Note: per 3GPP EIA0/null Integrity Protection is not allowed - all NAS messages after authentication must be Integrity Protected - An algorithm must be chosen for 'priority-1' and 'priority-2' ('none' option is not supported) - An additional selection, 'none', can be selected for 'priority-3' - effectively truncating the list. - An algorithm cannot be used for more than one priority Note: A pre-requisite for changing values in the eea-integrity-alg is that MME aggregate service must be locked.

Parameters

Name

integrityAlgPrio1 - Priority 1 integrity algorithm

Syntax

--integrityAlgPrio1 <list-item>

Description

Most preferred integrity algorithm.

Class

Optional.

List

- aes - 128-EIA2 (AES) - Specify integrity algorithm of 128-EIA2 (AES).
- none - No algorithm selected - Specify integrity algorithm of no algorithm selected.

- snow3G - 128-EIA1 (SNOW 3G) - Specify integrity algorithm of 128-EIA1 (SNOW 3G).
- zuc - 128-EIA3 (ZUC) - Specify integrity algorithm of 128-EIA3 (ZUC).

Default

snow3G

Name

integrityAlgPrio2 - Priority 2 integrity algorithm

Syntax

--integrityAlgPrio2 <list-item>

Description

Second most preferred integrity algorithm; used if the user element does not support algorithm(s) of higher priority.

Class

Optional.

List

- aes - 128-EIA2 (AES) - Specify integrity algorithm of 128-EIA2 (AES).
- none - No algorithm selected - Specify integrity algorithm of no algorithm selected.
- snow3G - 128-EIA1 (SNOW 3G) - Specify integrity algorithm of 128-EIA1 (SNOW 3G).
- zuc - 128-EIA3 (ZUC) - Specify integrity algorithm of 128-EIA3 (ZUC).

Default

aes

Name

integrityAlgPrio3 - Priority 3 integrity algorithm

Syntax

--integrityAlgPrio3 <list-item>

Description

Third most preferred integrity algorithm; used if the user element does not support algorithm(s) of higher priority.

Class

Optional.

List

- aes - 128-EIA2 (AES) - Specify integrity algorithm of 128-EIA2 (AES).
- none - No algorithm selected - Specify integrity algorithm of no algorithm selected.
- snow3G - 128-EIA1 (SNOW 3G) - Specify integrity algorithm of 128-EIA1 (SNOW 3G).
- zuc - 128-EIA3 (ZUC) - Specify integrity algorithm of 128-EIA3 (ZUC).

Default

none

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

5.60 cmm intfAdmin

Signaling Interface Administration. Syntax, description, parameters, and roles for cmm intfAdmin command.

Syntax**Description**

Signaling Interface Administration.

Parameters**Name**

linkInterfaceType - Link interface type

Syntax

--linkInterfaceType <list-item>

Description

Link interface type. not all interfaces are supported. s1mme is supported in this release.

Class

Required key.

List

- gn - Specify the Gn interface to an SGSN.
- m3 - Specify the M3 interface to an eNodeB.
- n26 - Specify n26 interface to an AMF
- nr10 - Specify nr10 interface.
- rs10 - Specify the RS10 interface to an SRS.

- s10 - Specify the S10 interface to another MME.
- s102 - Specify the S102 interface to an IWF.
- s11 - Specify the S11 interface to an SGW.
- s11u - Specify S11-u interface to an SGW.
- s13 - Specify the S13 interface to an EIR.
- s16 - Specify S16 interface loop around to this SGSN.
- s1mme - Specify the S1-MME interface to an eNodeB.
- s3 - Specify the S3 interface to an SGSN.
- s4 - Specify S4 interface to an SGW.
- s6ad - Specify the S6ad interface to an HSS.
- sbc - Specify the SBc interface to a CBC..
- sgd - Specify SGd interface to an SMSC/SMS router.
- sgs - Specify the SGs interface to a GSM/UMTS MSC.
- sgslite - Specify SGs lite interface to an IWF.
- slg - Specify the Slg interface to a GMLC.
- sls - Specify the Sls interface to an E-SMLC.
- sm - Specify the sm interface to an MBMS gateway.
- sv - Specify the Sv interface to a GSM/UMTS MSC.
- t6a - Specify T6a interface to an SCEF.

Name

linkAdminState - Link admin state

Syntax

--linkAdminState <list-item>

Description

Specified current link admin state.

Class

Optional.

List

- blocked - Blocked state. this applies only to s6ad, s13, slg, sgd and t6a.
- draining - Draining state. only applies to s11.
- locked - Locked state.

- unknown - Used in special cases to report state. a link cannot be updated to unknown state.
- unlocked - UnLocked state.

Name

terminationName - TerminationName

Syntax

--terminationName <string>

Description

Termination name. this is used for s1mme alone.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Roles**Modify**

sysadmin, manage

5.61 cmm intfQueryPath

Signaling single link Query Path. Syntax, description, parameters, and roles for cmm intfQueryPath command.

Syntax**Description**

Signaling Link Show query sctp path.

Parameters**Name**

linkInterfaceType - Link interface type

Syntax

--linkInterfaceType <list-item>

Description

Link interface type. only SCTP interfaces are supported.

Class

Required key.

List

- m3 - Specify the M3 interface to an eNodeB.
- m3ua - Specify an M3UA interface.
- s13 - Specify the S13 interface to an EIR.
- s1mme - S1-MME - Specify the S1-MME interface to an eNodeB.
- s6ad - Specify the S6ad interface to an HSS.
- sbc - Specify the SBc interface to a CBC.
- sgd - Specify SGd interface to an SMSC/SMS router.
- sgs - Specify the SGs interface to a GSM/UMTS MSC.
- slg - Specify the SLg interface to a GMLC.
- sls - Specify the SLs interface to an E-SMLC.
- t6a - Specify T6a interface to an SCEF.

Name

linkIndex - Link index

Syntax

--linkIndex <integer>

Description

Link index from the read command.

Class

Read only.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

maxResponseSize - Max record count

Syntax

--maxResponseSize <integer>

Description

Max number of records to be returned.

Class

Optional.

Type / Range

Integer in the range of 1 to 2,000

Name

pathStat1 - Path 1 loc-ip0 rem-ip0

Syntax

--pathStat1 <list-item>

Description

Path status between local IP0 and remote IP0.

Class

Read only.

List

- not-used - Path status of not used.
- reachable - Path status of reachable.
- unknown - Path status of unknown.
- unreachable - Path status of unreachable.

Name

pathStat2 - Path 2 loc-ip0 rem-ip1

Syntax

--pathStat2 <list-item>

Description

Path status between local IP0 and remote IP1.

Class

Read only.

List

- not-used - Path status of not used.
- reachable - Path status of reachable.
- unknown - Path status of unknown.
- unreachable - Path status of unreachable.

Name

pathStat3 - Path 3 loc-ip1 rem-ip0

Syntax

--pathStat3 <list-item>

Description

Path status between local IP1 and remote IP0.

Class

Read only.

List

- not-used - Path status of not used.
- reachable - Path status of reachable.
- unknown - Path status of unknown.
- unreachable - Path status of unreachable.

Name

pathStat4 - Path 4 loc-ip1 rem-ip1

Syntax

--pathStat4 <list-item>

Description

Path status between local IP1 and remote IP1.

Class

Read only.

List

- not-used - Path status of not used.
- reachable - Path status of reachable.
- unknown - Path status of unknown.
- unreachable - Path status of unreachable.

Name

previousIndex - Previous index

Syntax

--previousIndex <integer>

Description

Previous index when specified, next available index will be returned.

Class

Optional.

Type / Range

Integer in the range of -1 or 0 to 2,147,483,647

Name

startLinkIndex - Start link index

Syntax

--startLinkIndex <integer>

Description

Start link index from the read command.

Class

Optional.

Type / Range

Integer in the range of 0 to 4,294,967,295

Roles**List**

sysadmin, manage, monitor

5.62 cmm intfSummary

Signaling Interface Link Summary Request. Syntax, description, parameters, and roles for cmm intfSummary command.

Syntax**Description**

Signaling Interface Link Summary Request.

Parameters**Name**

linkInterfaceType - Link interface type

Syntax

--linkInterfaceType <list-item>

Description

Link interface type.

Class

Required key.

List

- gn - Specify the Gn interface to an SGSN.
- m3 - Specify the M3 interface to an eNodeB.
- n26 - Specify n26 interface to an AMF
- nr10 - Specify nr10 interface.
- rs10 - Specify the RS10 interface to an SRS.
- s10 - Specify the S10 interface to another MME.
- s102 - Specify the S102 interface to an IWF.

- s11 - Specify the S11 interface to an SGW.
- s11u - Specify S11-u interface to an SGW.
- s13 - Specify the S13 interface to an EIR.
- s16 - Specify S16 interface loop around to this SGSN.
- s1mme - Specify the S1-MME interface to an eNodeB.
- s3 - Specify the S3 interface to an SGSN.
- s4 - Specify S4 interface to an SGW.
- s6ad - Specify the S6ad interface to an HSS.
- sbc - Specify the SBc interface to a CBC..
- sgd - Specify SGd interface to an SMSC/SMS router.
- sgs - Specify the SGs interface to a GSM/UMTS MSC.
- sgslite - Specify SGs lite interface to an IWF.
- slg - Specify the Slg interface to a GMLC.
- sls - Specify the Sls interface to an E-SMLC.
- sm - Specify the sm interface to an MBMS gateway.
- sv - Specify the Sv interface to a GSM/UMTS MSC.
- t6a - Specify T6a interface to an SCEF.

Name

linkAdminState - Link admin state

Syntax

--linkAdminState <list-item>

Description

Link admin state.

Class

Read only.

List

- blocked - Blocked state. this applies only to s6ad, s13, slg, sgd and t6a.
- draining - Draining state. only applies to s11.
- locked - Locked state.
- unknown - Used in special cases to report state. a link cannot be updated to unknown state.
- unlocked - UnLocked state.

Name

linkCount - Link count

Syntax

--linkCount <integer>

Description

Link count.

Class

Read only.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

linkOperationState - Link operation state

Syntax

--linkOperationState <list-item>

Description

Link operation state.

Class

Read only.

List

- disabled - Disabled state of the link.
- enabled - Enable state of the link.
- unknown - Unknown state of the link. this can not be used during query.

Name

poolId - poolId

Syntax

--poolId <integer>

Description

IPDS Pool ID - If chosen only the count of interface links on the specified IPDS will be output. This parameter is only applicable for Multi-IPDS instantiations and requires linkInterfaceType option to display results on any Pool Id greater than 0

Class

Read only.

Type / Range

Integer in the range of 0 to 31

Name

terminationName - TerminationName

Syntax

--terminationName <string>

Description

Termination name. it is used for s1mme only.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Roles**List**

sysadmin, manage, monitor

5.63 cmm issuCanaryConfig

ISSU Canary Configuration. Syntax, description, parameters, and roles for cmm issuCanaryConfig command.

Syntax**Description**

Use this command to define the canary configuration to be used when running a canary test for each supported POD type.

Parameters**Name**

poolType - pool type

Syntax

--poolType <list-item>

Description

Pool Type - Only CPPS pool type is allowed

Class

Required key.

List

- CPPS - Control Plane Processing Service - Control Plane Processing Service.

- DBS - Database service - Database service.
- IPDS - IP director service - IP director service.
- NECC - Network element cluster controller - Network element cluster controller.

Name

personality - personality

Syntax

--personality <list-item>

Description

This field determines whether the CPPS serves 4g or 5g traffic.

Class

Required key.

List

- - - None Personality
- EMMS - EMMS Personality
- AMMS - AMMS Personality

Default

-

Name

imsiRangeCount - Count of Test IMSI Range

Syntax

--imsiRangeCount <integer>

Description

The count of test IMSI range. When setting this parameter to non-zero value, IMSI Range Start must be also set.

Restrictions

When setting this parameter to non-zero value, IMSI Range Start must be also set. To set this parameter to 0, IMSI Range Start must be also set to empty ('').

Class

Optional.

Type / Range

Integer in the range of 0 to 128

Default

0

Name

imsiRangeStart - IMSI Range Start for Canary Test IMSIs

Syntax

--imsiRangeStart <digits>

Description

First IMSI of test IMSI range. When setting this parameter, IMSI Range Count must be also set to a valid value.

Restrictions

To set this parameter, IMSI Range Count must be also set to a valid value. To clear this parameter (i.e. set it to empty), IMSI Range Count must be also set to 0.

Class

Optional.

Type / Range

Digit string of length 0 or 15

Default

""

Name

max4gUeCnt - Max 4G UE count

Syntax

--max4gUeCnt <integer>

Description

Maximum 4G UE count for this canary instance (where applicable).

Class

Optional.

Type / Range

Integer in the range of 0 to 50,000

Default

1000

Name

max5gUeCnt - Max 5G UE count

Syntax

--max5gUeCnt <integer>

Description

Maximum 5G UE count for this canary instance (where applicable).

Class

Optional.

Type / Range

Integer in the range of 0 to 50,000

Default

1000

Name

mode - mode

Syntax**Description**

This field determines the CPPS mode

Class

Optional.

Type / Range

Digit string of length 1 to 255

Name

poolId - Pool ID

Syntax**Description**

Pool ID of the canary instance

Class

Optional.

Type / Range

Digit string of length 1 to 255

Name

poolMemberId - Pool Member Id

Syntax**Description**

Pool member id of the canary instance'

Class

Optional.

Type / Range

Digit string of length 1 to 255

Name

softwareVersion - Software version

Syntax**Description**

Software version of the canary instance

Class

Optional.

Type / Range

Digit string of length 1 to 255

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.64 cmm issuCanaryStatus

ISSU Canary status. Syntax, description, parameters, and roles for `cmm issuCanaryStatus` command.

Syntax**Description**

Interrogate the system for any services running in canary mode and display information.

Parameters**Name**

personality - personality

Syntax

--personality <list-item>

Description

This field determines whether the CPPS serves 4g or 5g traffic.

Class

Read only.

List

- - - None Personality
- EMMS - EMMS Personality
- AMMS - AMMS Personality

Default

-

Name

poolId - Pool ID

Syntax

--poolId <integer>

Description

Pool ID of the canary instance

Class

Read only.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

poolMemberId - Pool Member Id

Syntax

--poolMemberId <integer>

Description

Pool member id of the canary instance

Class

Read only.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

poolType - Pool Type

Syntax

--poolType <list-item>

Description

AMF/MME pod type

Class

Read only.

List

- CPPS - Control Plane Processing Service - Control Plane Processing Service.
- DBS - Database service - Database service.
- IPDS - IP director service - IP director service.
- NECC - Network element cluster controller - Network element cluster controller.

Name

softwareVersion - Software version

Syntax

--softwareVersion <string>

Description

Software version of the canary instance

Class

Read only.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles**List**

sysadmin, manage, monitor

5.65 cmm iuCipheringAlg

Preferred lu Ciphering Algorithm . Syntax, description, parameters, and roles for cmm iuCipheringAlg command.

Syntax**Description**

This command sets lu algorithms in priority order, SGSN will use it to negotiate encryption alg with the UE. At UE attach/Rau, the UE provides, in the UE network capability IE, which encryption algorithms in 3G are supported. SGSN will send the selected algs to RNC in priority order. The algorithm should be both supported by the UE and highest priority in the configured UEA encryption algorithm priority list

Parameters**Name**

iuCipheringAlgPrio1 - Priority 1 lu ciphering algorithm

Syntax

--iuCipheringAlgPrio1 <list-item>

Description

Most preferred lu ciphering algorithm

Class

Optional.

List

- kasumi - UEA1 (Kasumi) - Specify ciphering of UEA1 (Kasumi).
- noCiphering - UEA0(no ciphering) - Specify ciphering of UEA0 (no encryption)
- none - No Algorithm Selected
- snow3G - UEA2 (SNOW 3G) - Specify ciphering of UEA2 (SNOW 3G).

Default

snow3G

Name

iuCipheringAlgPrio2 - Priority 2 lu ciphering algorithm

Syntax

--iuCipheringAlgPrio2 <list-item>

Description

Second preferred lu ciphering algorithm

Class

Optional.

List

- kasumi - UEA1 (Kasumi) - Specify ciphering of UEA1 (Kasumi).
- noCiphering - UEA0(no ciphering) - Specify ciphering of UEA0 (no encryption)
- none - No Algorithm Selected
- snow3G - UEA2 (SNOW 3G) - Specify ciphering of UEA2 (SNOW 3G).

Default

kasumi

Name

iuCipheringAlgPrio3 - Priority 3 lu ciphering algorithm

Syntax

--iuCipheringAlgPrio3 <list-item>

Description

3rd preferred lu ciphering algorithm, used if the user element does not support algorithm(s) of higher priority

Class

Optional.

List

- kasumi - UEA1 (Kasumi) - Specify ciphering of UEA1 (Kasumi).
- noCiphering - UEA0(no ciphering) - Specify ciphering of UEA0 (no encryption)
- none - No Algorithm Selected
- snow3G - UEA2 (SNOW 3G) - Specify ciphering of UEA2 (SNOW 3G).

Default

none

Roles

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

5.66 cmm iuIntegrityAlg

Preferred lu Integrity Algorithm. Syntax, description, parameters, and roles for cmm iuIntegrityAlg command.

Syntax

Description

This command sets lu algorithms in priority order, SGSN will use it to negotiate Integrity alg with the UE. At UE attach/Rau, the UE provides, in the UE network capability IE, which integrity algorithms in 3G are supported. The SGSN will send the algs to RNC in priority order. The algorithm should be both supported by the UE and highest priority in the configured UIA integrity algorithm priority list

Parameters

Name

iuIntegrityAlgPrio1 - Priority 1 lu integrity algorithm

Syntax

--iuIntegrityAlgPrio1 <list-item>

Description

Most preferred lu integrity algorithm

Class

Optional.

List

- kasumi - UIA1 (Kasumi) - Specify Integrity of UIA1 (Kasumi).
- none - No Algorithm Selected
- snow3G - UIA2 (SNOW 3G) - Specify Integrity of UIA2 (SNOW 3G).

Default

snow3G

Name

iuIntegrityAlgPrio2 - Priority 2 lu integrity algorithm

Syntax

--iuIntegrityAlgPrio2 <list-item>

Description

Second preferred lu integrity algorithm

Class

Optional.

List

- kasumi - UIA1 (Kasumi) - Specify Integrity of UIA1 (Kasumi).
- none - No Algorithm Selected
- snow3G - UIA2 (SNOW 3G) - Specify Integrity of UIA2 (SNOW 3G).

Default

kasumi

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

5.67 cmm kafkaBroker

Kafka Broker. Syntax, description, parameters, and roles for cmm kafkaBroker command.

Syntax

Description

This command is used to add or delete IPs or FQDNs for a Kafka Broker.

Parameters

Name

name - Broker Name

Syntax

--name <string>

Description

This is the name of the Kafka Broker

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

brokerPort - Kafka Broker Port

Syntax

--brokerPort <integer>

Description

The parameter defines the Kafka Broker Port.

Class

Required on create.

Type / Range

Integer in the range of 1,024 to 65,535

Name

brokerFqdn - Kafka Broker FQDN

Syntax

--brokerFqdn <string>

Description

The parameter defines the Kafka Broker FQDN. The number of the Kafka Broker FQDNs could be up to 18.

Restrictions

Either the Broker FQDN or the Broker IP must be set

Class

Optional.

Type / Range

Digit string of length 0 to 255 with the allowed pattern of [A-Za-z0-9.-]

Default

""

Name

brokerIp - Kafka Broker IP

Syntax

--brokerIp <string>

Description

The parameter defines the Kafka Broker IP. The number of the Kafka Broker IPs could be up to 18.

Restrictions

Either the Broker FQDN or the Broker IP must be set

Class

Optional.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Default

0.0.0.0

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.68 cmm kafkaBrokerList

Kafka Broker List. Syntax, description, parameters, and roles for cmm kafkaBrokerList command.

Syntax

Description

This command is used to create, modify and delete the Kafka Broker List so as to send the relevant data to an external Kafka cluster. The functional area in scope as well as the Kafka topic needs to be provided. The security parameters are used to define the level of the communication.

Parameters

Name

area - Functional Area

Syntax

--area <list-item>

Description

This parameter defines the functional area for the communication between Kafka Producer and Kafka Broker

Class

Required key.

List

- FM - FM defines the Alarm area
- PM - PM defines the Performance Management (counters) area
- LOG - LOG defines the Logging area

Name

topic - Kafka Topic Name

Syntax

--topic <string>

Description

This parameter defines the name of the Kafka Topic

Restrictions

logical key, shall be unique

Class

Required on create.

Type / Range

Digit String of length 1 to 255 with the allowed pattern [a-zA-Z0-9._-]

Name

oamTlsProfileName - OAM TLS Profile

Syntax

```
--oamTlsProfileName <string>
```

Description

This parameter contains the profile for TLS. It contains the certificate names needed for establishing secure communication.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Name

securityProtocol - Security Protocol with Kafka broker

Syntax

```
--securityProtocol <list-item>
```

Description

This parameter defines the security protocol for the communication between Kafka Producer and Kafka Broker

Class

Optional.

List

- PLAINTEXT - PLAINTEXT would be the protocol to communicate with brokers
- SSL - SSL would be the protocol to communicate with brokers

Default

PLAINTEXT

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.69 cmm kafkaBrokerListItem

Kafka Broker List Item. Syntax, description, parameters, and roles for cmm kafkaBrokerListItem command.

Syntax**Description**

Adds or removes a Kafka Broker to a Kafka Broker List

Parameters**Name**

kafkaBrokerListArea - Kafka Broker List Area

Syntax

--kafkaBrokerListArea <list-item>

Description

Kafka Broker List Area

Class

Required key.

List

- FM - FM defines the Alarm area
- PM - PM defines the Performance Management (counters) area
- LOG - LOG defines the Logging area

Name

kafkaBrokerName - Kafka Broker Name

Syntax

--kafkaBrokerName <string>

Description

Kafka Broker Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.70 cmm lai

Location Area Identifier. Syntax, description, parameters, and roles for cmm lai command.

Syntax

Description

This command is used to define a location area in a given mobile network and includes a parameter to indicate whether CSFB is supported for the associated LAI.

Parameters

Name

plmnName - Mobile network name

Syntax

--plmnName <string>

Description

Name that identifies mobile network.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

lac - Location area code

Syntax

--lac <integer>

Description

Location area code.

Class

Required key.

Type / Range

Integer in the range of 1 to 65,535

Name

csfbSupported - CSFB supported

Syntax

--csfbSupported <list-item>

Description

Boolean flag to indication whether or not CSFB is supported for this LAI.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.71 cmm laiList

List of LAIs. Syntax, description, parameters, and roles for cmm laiList command.

Syntax

Description

This command is used to administer a list of LAIs and includes parameters for defining the Location Area Code and Served Mobile Network Name.

Parameters

Name

areaListName - Name of LAI list

Syntax

--areaListName <string>

Description

Name of LAI list.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

servedPlmnName - Served mobile network name

Syntax

--servedPlmnName <string>

Description

Label for served PLMN.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

lac - Location area code

Syntax

--lac <integer>

Description

Location area code.

Class

Required key.

Type / Range

Integer in the range of 1 to 65,535

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.72 cmm laiMscMapping

LAI to MSC Mapping. Syntax, description, parameters, and roles for `cmm laiMscMapping` command.

Syntax

Description

This command is used to administer an LAI to MSC Mapping and includes parameters for defining the Mobile Network Name, Location Area Code, and MSC Server ID.

Parameters

Name

plmnName - Mobile network name

Syntax

--plmnName <string>

Description

Name that identifies mobile network.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

lac - Location area code

Syntax

--lac <integer>

Description

Location area code.

Class

Required key.

Type / Range

Integer in the range of 1 to 65,535

Name

mscSrvId - MSC ID

Syntax

--mscSrvId <integer>

Description

The ID of the MSC server this LAI maps to.

Class

Required key.

Type / Range

Integer in the range of 1 to 128

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.73 cmm laiOfHomeAccessList

LAI of home access list. Syntax, description, parameters, and roles for cmm laiOfHomeAccessList command.

Syntax**Description**

LAI home access list relation.

Parameters

Name

mcc - MCC

Syntax

--mcc <digits>

Description

MCC digits of broadcasted PLMN.

Class

Required key.

Type / Range

Digit string of length 3

Name

mnc - MNC

Syntax

--mnc <digits>

Description

MNC digits of broadcasted PLMN.

Class

Required key.

Type / Range

Digit string of length 2 to 3

Name

lac - LAC

Syntax

--lac <integer>

Description

Location area code.

Class

Required key.

Type / Range

Integer in the range of 1 to 65,533

Name

homeAccessListId - Home access list id

Syntax

--homeAccessListId <integer>

Description

Home access list identifier. this parameter should match parameter homeaccesslistid of table homeaccesslist.

Class

Required key.

Type / Range

Integer in the range of 1 to 1000

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.74 cmm laiOfVisitorAccessList

SGSN LAI of visitor access list. Syntax, description, parameters, and roles for cmm laiOfVisitorAccessList command.

Syntax**Description**

SGSN LAI visitor access list relation.

Parameters**Name**

mcc - MCC

Syntax

--mcc <digits>

Description

MCC digits of broadcasted PLMN.

Class

Required key.

Type / Range

Digit string of length 3

Name

mnc - MNC

Syntax

--mnc <digits>

Description

MNC digits of broadcasted PLMN.

Class

Required key.

Type / Range

Digit string of length 2 to 3

Name

lac - LAC

Syntax

--lac <integer>

Description

Location area code.

Class

Required key.

Type / Range

Integer in the range of 1 to 65,533

Name

visitorAccessListId - Visitor access list id

Syntax

--visitorAccessListId <integer>

Description

Visitor access list identifier. this parameter should match parameter visitoraccesslistid of table visitoraccesslist.

Class

Required key.

Type / Range

Integer in the range of 1 to 1000

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.75 cmm laiOfVlr

SGSN LAI of VLR. Syntax, description, parameters, and roles for cmm laiOfVlr command.

Syntax

Description

SGSN LAI VLR relation.

Parameters

Name

mcc - MCC

Syntax

--mcc <digits>

Description

MCC digits of broadcasted PLMN.

Class

Required key.

Type / Range

Digit string of length 3

Name

mnc - MNC

Syntax

--mnc <digits>

Description

MNC digits of broadcasted PLMN.

Class

Required key.

Type / Range

Digit string of length 2 to 3

Name

lac - LAC

Syntax

--lac <integer>

Description

Location area code.

Class

Required key.

Type / Range

Integer in the range of 1 to 65,533

Name

vlrId - VlrId

Syntax

--vlrId <integer>

Description

VLR index.

Class

Required key.

Type / Range

Integer in the range of 0 to 199

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.76 cmm lapPagingPriProfile

Low Access Priority Paging Priority Profile. Syntax, description, parameters, and roles for cmm lapPagingPriProfile command.

Syntax

Description

Administer Low Access Priority Paging Priority Profile.

Parameters

Name

name - Low access priority paging profile name

Syntax

--name <digits>

Description

Defines the LAP paging profile name.

Class

Required key.

Type / Range

Digit string of length 1 to 1 with the allowed pattern of [1-8]*

Name

description - Description

Syntax

--description <string>

Description

Profile description.

Class

Optional.

Type / Range

Digit string of length 0 to 160 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Name

lapPagingPriLevel - LAP paging priority level

Syntax

--lapPagingPriLevel <list-item>

Description

LAP paging priority level.

Class

Optional.

List

- PrioLevel0 - Priority level 0 - Priority level 0. No paging priority is set.
- PrioLevel1 - Priority level 1 - Priority level 1. Highest Paging priority.
- PrioLevel2 - Priority level 2 - Priority level 2.
- PrioLevel3 - Priority level 3 - Priority level 3.
- PrioLevel4 - Priority level 4 - Priority level 4.
- PrioLevel5 - Priority level 5 - Priority level 5.
- PrioLevel6 - Priority level 6 - Priority level 6.
- PrioLevel7 - Priority level 7 - Priority level 7.
- PrioLevel8 - Priority level 8 - Priority level 8. Lowest Paging Priority.

Default

PrioLevel0

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.77 cmm lboImsiRangeServices

LBO IMSI Range Services. Syntax, description, parameters, and roles for cmm lboImsiRangeServices command.

Syntax

Description

This command is used to administer Local Breakout IMSI Range Services and includes parameters for defining the roaming network name, an LBO APN white list, and IMSI MSIN values. The 'shutdown' parameter is used to enable/disable the APN white list for an IMSI series.

Parameters

Name

uePlmnServicesName - Name of services based on UE network and served network

Syntax

--uePlmnServicesName <string>

Description

Name that identifies roaming network.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

entryId - Entry ID

Syntax

--entryId <integer>

Description

Entry ID.

Class

Required key.

Type / Range

Integer in the range of 1 to 10,000

Name

apnWhiteListName - APN white list name

Syntax

--apnWhiteListName <string>

Description

It is used to assign a LBO APN white list to a roaming partner network.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

apn-all

Name

description - IMSIRange entry description

Syntax

--description <string>

Description

Description for IMSI range services entry.

Class

Optional.

Type / Range

Digit string of length 0 to 160 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

maxMsin - Maximum MSIN

Syntax

--maxMsin <digits>

Description

The maximum IMSI MSIN value.

Class

Optional.

Type / Range

Digit string of length 0 or 8 to 10

Default

"

Name

minMsin - Minimum MSIN

Syntax

--minMsin <digits>

Description

The minimum IMSI MSIN value.

Class

Optional.

Type / Range

Digit string of length 0 or 8 to 10

Default

..

Name

shutdown - Shutdown

Syntax

--shutdown <list-item>

Description

It is used to disables the APN white list for an IMSI series.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.78 cmm lboPgWSelectionApn

LBO PGW Selection Supplementary APN. Syntax, description, parameters, and roles for cmm lboPgWSelectionApn command.

Syntax

Description

This command is used to provision a tag to be suffixed to an APN FQDN for the selection of PGW for a roamer UE LBO PDN connection based on the PDN type requested.

Parameters

Name

uePlmnServicesName - Name of Services based on UE Network and Served Network

Syntax

--uePlmnServicesName <string>

Description

Name that identifies roaming network

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

pdnType - Pdn Type

Syntax

--pdnType <list-item>

Description

Pdn Type

Class

Required key.

List

- ipv4 - Specify PDN type of IPv4.
- ipv4v6 - Specify PDN type of IPv4 / IPv6.
- ipv6 - Specify PDN type of IPv6.

Name

apnNi - Supplementary Access Point Name

Syntax

--apnNi <string>

Description

The parameter configures the tag to be added to the APN NI as a suffix in the FQDN used in the S-NAPTR query for the selection of roamer UE LBO PDN connection.

Class

Required on create.

Type / Range

Digit string of length 1 to 62 with the allowed pattern of [0-9a-zA-Z\.\-]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.79 cmm lcsOptionsProfile

Location Services Options Profile. Syntax, description, parameters, and roles for cmm lcsOptionsProfile command.

Syntax**Description**

This command is used to administer a Location Services Options Profile and includes parameters for defining options for handling HSS EPS Location Data Requests in IDR, horizontal and vertical accuracy, response time, and operator and value added services. A default profile, named 'default', is provided.

Parameters**Name**

name - Profile name

Syntax

--name <string>

Description

Location services options profile name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\.\-:@]*

Name

description - Description

Syntax

--description <string>

Description

Profile description.

Class

Optional.

Type / Range

Digit string of length 0 to 160 with the allowed pattern of [0-9a-zA-Z_\.\-:@]*

Default

"

Name

edrxPsmDelayedLocReqAllow - Mobile-Terminated location request delayed location request allowed

Syntax

--edrxPsmDelayedLocReqAllow <list-item>

Description

Allow LCS mobile terminated location request for delayed location reporting when a UE is unreachable due to eDRX or PSM.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

epsStateLocInfoReq - Report UE state/location information if requested

Syntax

--epsStateLocInfoReq <list-item>

Description

Select options for handling HSS EPS location data request in IDR.

Class

Optional.

List

- ECGI and TAI Only - MME will indicate to the HSS in ULR and IDA that MME supports the EPS state/location. when HSS sends IDR requesting EPS location information then MME only sends the current ECGI and TAI information to the HSS. in this case MME does not run LCSAP procedures to obtain UE location.
- Not Supported - If this option is selected then MME will not indicate to the HSS that MME supports EPS state/location retrieval in ULR and IDA.
- UE Geolocation - MME will indicate to the HSS in ULR and IDA that MME supports the EPS state/location. when HSS sends IDR requesting EPS location information then MME runs the LCSAP procedures to obtain UE location and sends the obtained UE geo-location in IDA along with the most recent ECGI and TAI.

Default

Not Supported

Name

horizontalAccuracy - Horizontal accuracy

Syntax

--horizontalAccuracy <integer>

Description

Horizontal accuracy parameter in the location request.

Class

Optional.

Type / Range

Integer in the range of 0 to 127

Default

20

Name

lawfulInterceptServices - Lawful Intercept location services

Syntax

--lawfulInterceptServices <list-item>

Description

Allow/reject a location request by LCS client of type lawful intercept services.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

operatorServices - Operator services LCS client type

Syntax

--operatorServices <list-item>

Description

Allow/reject a location request by LCS client of type operator services.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

responseTime - Response time

Syntax

--responseTime <list-item>

Description

Response time parameter in the location request.

Class

Optional.

List

- Delay Tolerant - Specify response time of delay tolerant.
- Low Delay - Specify response time of low delay.

Default

Low Delay

Name

valueAddedServices - Value added services LCS client type

Syntax

--valueAddedServices <list-item>

Description

Allow/reject a location request by LCS client of type value added services.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

verticalAccuracy - Vertical accuracy

Syntax

--verticalAccuracy <integer>

Description

Vertical accuracy parameter in the location request.

Class

Optional.

Type / Range

Integer in the range of 0 to 127

Default

20

Name

verticalRequested - Vertical requested

Syntax

--verticalRequested <list-item>

Description

Vertical requested parameter in the location request.

Class

Optional.

List

- Requested - Specify vertical requested.
- Not Requested - Specify vertical not requested.

Default

Not Requested

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.80 cmm ldapAuthRoleMapping

LDAP Authorizaiton Role Mapping. Syntax, description, parameters, and roles for cmm
ldapAuthRoLeMapping command.

Syntax

Description

This command allows mapping of External LDAP Authorization Roles to the Internal CMM Roles. Maximum number of entries supported shall be 50.

Parameters

Name

name - Name

Syntax

--name <string>

Description

Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

ldapExternalRole - LDAP External Role Name

Syntax

```
--ldapExternalRole <string>
```

Description

LDAP External Role Name

Class

Required key.

Type / Range

Digit string of length 1 to 100 with the allowed pattern of [0-9a-zA-Z\._-\$]*

Name

rbacRole - RBAC Role

Syntax

```
--rbacRole <list-item>
```

Description

The internal RBAC Role Allowed

Class

Required on create.

List

- Admin - Admin Role
- Configurator - Configurator Role
- LI - Lawful Intercept - Lawful Intercept Role
- Monitor - Monitor Role
- SecAdmin - Security Admin - Security Admin Role

Roles**Create**

secadmin

Show

secadmin

Modify

secadmin

Delete

secadmin

List

secadmin

5.81 cmm linkAdmin

Signaling Link Admin. Syntax, description, parameters, and roles for cmm linkAdmin command.

Syntax

Description

Signaling Link Admin.

Parameters

Name

linkInterfaceType - Link interface type

Syntax

```
--linkInterfaceType <list-item>
```

Description

Link interface type.

Class

Required key.

List

- gn - Specify the Gn interface to an SGSN.
- m3 - Specify the M3 interface to an eNodeB.
- n26 - Specify n26 interface to an AMF
- nr10 - Specify nr10 interface.
- rs10 - Specify the RS10 interface to an SRS.
- s10 - Specify the S10 interface to another MME.
- s102 - Specify the S102 interface to an IWF.
- s11 - Specify the S11 interface to an SGW.
- s11u - Specify S11-u interface to an SGW.
- s13 - Specify the S13 interface to an EIR.
- s16 - Specify S16 interface loop around to this SGSN.
- s1mme - Specify the S1-MME interface to an eNodeB.
- s3 - Specify the S3 interface to an SGSN.
- s4 - Specify S4 interface to an SGW.

- s6ad - Specify the S6ad interface to an HSS.
- sbc - Specify the SBc interface to a CBC..
- sgd - Specify SGd interface to an SMSC/SMS router.
- sgs - Specify the SGs interface to a GSM/UMTS MSC.
- sgslite - Specify SGs lite interface to an IWF.
- slg - Specify the Slg interface to a GMLC.
- sls - Specify the Sls interface to an E-SMLC.
- sm - Specify the sm interface to an MBMS gateway.
- sv - Specify the Sv interface to a GSM/UMTS MSC.
- t6a - Specify T6a interface to an SCEF.

Name

linkIndex - Link index

Syntax

--linkIndex <integer>

Description

Link index from the read command

Class

Required key.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

linkAdminState - Link admin state

Syntax

--linkAdminState <list-item>

Description

Specified current link admin state.

Class

Required on create.

List

- blocked - Blocked state. this applies only to s6ad, s13, slg, sgd and t6a.
- draining - Draining state. only applies to s11.
- locked - Locked state.

- unknown - Used in special cases to report state. a link cannot be updated to unknown state.
- unlocked - UnLocked state.

Name

alarmState - Alarm state

Syntax

--alarmState <list-item>

Description

Specifies alarm state.

Class

Read only.

List

- critical - Critical alarm.
- major - Major alarm.
- minor - Minor alarm.
- none - No alarm.
- unknown - Unknown.

Name

forceFlag - Force flag

Syntax

--forceFlag <list-item>

Description

When you are locking last link on S6ad, use this flag to be true to lock or block the S6ad link unconditionally. linkadminstate must be entered when force is entered.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Name

linkAvailState - Link available state

Syntax

--linkAvailState <list-item>

Description

Specifies link available state.

Class

Read only.

List

- degraded - Degraded.
- dependency - Dependency.
- none - None.
- unknown - Unknown.

Name

linkIdentifier - Link identifier

Syntax

```
--linkIdentifier <string>
```

Description

Link information.

Class

Read only.

Type / Range

Digit string of length 0 or 1 to 255

Name

linkInUseState - Link inuse state

Syntax

```
--linkInUseState <list-item>
```

Description

Link inuse state.

Class

Read only.

List

- in-use - Specify link in use state of in use.
- not-applicable - Specify link in use state of not application.
- not-used - Specify link in use state of not used.

Name

linkOperationState - Link operation state

Syntax

```
--linkOperationState <list-item>
```

Description

Link operation state requested. valid values are locked, unlocked, draining or blocked.

Class

Read only.

List

- disabled - Disabled state of the link.
- enabled - Enable state of the link.
- unknown - Unknown state of the link. this can not be used during query.

Name

linkPlmnType - PLMN type of the link

Syntax

--linkPlmnType <list-item>

Description

Specifies PLMN type of the link. only applies to S6ad.

Class

Read only.

List

- home - Specify link PLMN type of home.
- not-applicable - Specify link PLMN type of not applicable.
- not-home - Specify link PLMN type of not home.

Name

linkProcState - Link procedural state

Syntax

--linkProcState <list-item>

Description

Specifies link procedural state.

Class

Read only.

List

- init req - Init request - Init request.
- initing - Initiating.
- none - None.
- not init - Not init.

- not-applicable - Not-applicable.
- report - Report.
- term - Term.

Name

localIp1 - Local IP1

Syntax

--localIp1 <string>

Description

Local IP1.

Class

Read only.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

localIp2 - Local IP2

Syntax

--localIp2 <string>

Description

Local IP2.

Class

Read only.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

poolId - poolId

Syntax

--poolId <integer>

Description

IPDS Pool ID - ID will only be 0 for non Multi-IPDS instantiations.

Class

Read only.

Type / Range

Integer in the range of 0 to 31

Name

remotelp1 - Remote IP1

Syntax

```
--remoteIp1 <string>
```

Description

Remote IP1.

Class

Read only.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

remotelp2 - Remote IP2

Syntax

```
--remoteIp2 <string>
```

Description

Remote IP2.

Class

Read only.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

terminationName - TerminationName

Syntax

```
--terminationName <string>
```

Description

Termination name. this is used for s1mme alone. this field is part of input for optional key fields. this field cannot be updated.

Class

Read only.

Type / Range

Digit string of length 0 or 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

5.82 cmm linkAlarmCfg

linkAlarmCfg. Syntax, description, parameters, and roles for cmm linkAlarmCfg command.

Syntax

Description

Administer Link Alarm Severity for MME.

Parameters

Name

linkInterfaceType - Link interface type

Syntax

```
--linkInterfaceType <list-item>
```

Description

Link interface type.

Class

Required key.

List

- gn - Specify the Gn interface to an SGSN.
- m3 - Specify the M3 interface to an eNodeB.
- n26 - Specify n26 interface to an AMF
- nr10 - Specify nr10 interface.
- rs10 - Specify the RS10 interface to an SRS.
- s10 - Specify the S10 interface to another MME.
- s102 - Specify the S102 interface to an IWF.
- s11 - Specify the S11 interface to an SGW.
- s11u - Specify S11-u interface to an SGW.
- s13 - Specify the S13 interface to an EIR.
- s16 - Specify S16 interface loop around to this SGSN.
- s1mme - Specify the S1-MME interface to an eNodeB.
- s3 - Specify the S3 interface to an SGSN.
- s4 - Specify S4 interface to an SGW.

- s6ad - Specify the S6ad interface to an HSS.
- sbc - Specify the SBc interface to a CBC..
- sgd - Specify SGd interface to an SMSC/SMS router.
- sgs - Specify the SGs interface to a GSM/UMTS MSC.
- sgslite - Specify SGs lite interface to an IWF.
- slg - Specify the Slg interface to a GMLC.
- sls - Specify the Sls interface to an E-SMLC.
- sm - Specify the sm interface to an MBMS gateway.
- sv - Specify the Sv interface to a GSM/UMTS MSC.
- t6a - Specify T6a interface to an SCEF.

Name

absoluteLinkLossExcessiveAlarm - Absolute Link Loss Excessive Alarm

Syntax

--absoluteLinkLossExcessiveAlarm <integer>

Description

This parameter indicates the absolute number of unavailable Links necessary to trigger the Excessive Links Lost Alarm. This parameter applies to S1, S11-U and S11 only.

Class

Optional.

Type / Range

Integer in the range of 5 to 100

Default

100

Name

aggrSeverity - Aggregate severity

Syntax

--aggrSeverity <list-item>

Description

Aggregate severity.

Class

Optional.

List

- critical - Critical alarm.
- major - Major alarm.

- minor - Minor alarm.
- none - No alarm.

Default

critical

Name

individualSeverity - Individual severity

Syntax

--individualSeverity <list-item>

Description

Individual severity.

Class

Optional.

List

- critical - Critical alarm.
- major - Major alarm.
- minor - Minor alarm.
- none - No alarm.

Default

major

Name

percentLinkLossAlarm - Percentage of Link Loss Alarm

Syntax

--percentLinkLossAlarm <integer>

Description

This parameter specifies the percentage of unavailable links at which to raise a critical interface alarm. This parameter applies to S1, S11-U and S11 only.

Class

Optional.

Type / Range

Integer in the range of 5 to 100 in units of %

Default

20

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.83 cmm linkQueryDetail

Signaling single link Query Detail. Syntax, description, parameters, and roles for cmm linkQueryDetail command.

Syntax

Description

Signaling Link Show query link detail.

Parameters

Name

linkInterfaceType - Link interface type

Syntax

--linkInterfaceType <list-item>

Description

Link interface type. default to s6ad.

Class

Required key.

List

- m3 - Specify the M3 interface to an eNodeB.
- m3ua - Specify an M3UA interface.
- s13 - Specify the S13 interface to an EIR.
- s1mme - S1-MME - Specify the S1-MME interface to an eNodeB.
- s6ad - Specify the S6ad interface to an HSS.
- sbc - Specify the SBc interface to a CBC.

- sgd - Specify SGd interface to an SMSC/SMS router.
- sgs - Specify the SGs interface to a GSM/UMTS MSC.
- slg - Specify the SLg interface to a GMLC.
- sls - Specify the SLs interface to an E-SMLC.
- t6a - Specify T6a interface to an SCEF.

Name

linkIndex - Link index

Syntax

--linkIndex <integer>

Description

Link index from the read command. default to first enabled link.

Class

Required key.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

linkDetailCode - Link detail

Syntax

--linkDetailCode <integer>

Description

Link detail code from the read command. default to 1.

Class

Required key.

Type / Range

Integer in the range of 0 to 65,535

Name

detailOutputLineNumber - Detail output line number

Syntax

--detailOutputLineNumber <string>

Description

Detail output line number as output.

Class

Required key.

Type / Range

Digit string of length

Name

detailOutput - Link details

Syntax

--detailOutput <string>

Description

Link details.

Class

Read only.

Type / Range

Digit string of length

Roles**List**

sysadmin, manage, monitor

5.84 cmm linkQueryPath

Signaling single link Query Path. Syntax, description, parameters, and roles for cmm linkQueryPath command.

Syntax**Description**

Signaling Link Show query sctp path.

Parameters**Name**

linkInterfaceType - Link interface type

Syntax

--linkInterfaceType <list-item>

Description

Link interface type. only SCTP interfaces are supported. default to s6ad.

Class

Required key.

List

- m3 - Specify the M3 interface to an eNodeB.
- m3ua - Specify an M3UA interface.
- s13 - Specify the S13 interface to an EIR.
- s1mme - S1-MME - Specify the S1-MME interface to an eNodeB.
- s6ad - Specify the S6ad interface to an HSS.
- sbc - Specify the SBc interface to a CBC.
- sgd - Specify SGd interface to an SMSC/SMS router.
- sgs - Specify the SGs interface to a GSM/UMTS MSC.
- slg - Specify the SLg interface to a GMLC.
- sls - Specify the SLs interface to an E-SMLC.
- t6a - Specify T6a interface to an SCEF.

Name

linkIndex - Link index

Syntax

--linkIndex <integer>

Description

Link index to read. default to first enabled link.

Class

Required key.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

localIp1 - Local IP1

Syntax

--localIp1 <string>

Description

Local IP1.

Class

Read only.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

localPort1 - LocalPort1

Syntax


```
--localPort1 <integer>
```

Description

Local port 1.

Class

Read only.

Type / Range

Integer in the range of 0 to 65,535

Name

pathIndex - Path number

Syntax

```
--pathIndex <integer>
```

Description

Path index.

Class

Read only.

Type / Range

Integer in the range of 0 to 3

Name

remotelp1 - Remote IP1

Syntax

```
--remoteIp1 <string>
```

Description

Remote IP1.

Class

Read only.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

remotePort1 - RemotePort1

Syntax

```
--remotePort1 <integer>
```

Description

Remote port 1.

Class

Read only.

Type / Range

Integer in the range of 0 to 65,535

Roles**List**

sysadmin, manage, monitor

5.85 cmm localDnsHostFile

Local DNS Host File. Syntax, description, parameters, and roles for cmm localDnsHostFile command.

Syntax**Description**

This command creates a DB entry of local DNS host file to be loaded into openssl. The local DNS host file is used as resolution for S-GW (TAI), P-GW (APN), and MME/S4 SGSN

Parameters**Name**

name - Local DNS Host File Name

Syntax

--name <string>

Description

Name of the DB entry. Not applicable for the list operation

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

hostFile - Local DNS Host File

Syntax

--hostFile <string>

Description

Absolute path of local DNS host file. Example: /tmp/localdnshostfile

Class

Required on create.

Type / Range

Digit string of length 1 to 100 with the allowed pattern of [0-9a-zA-Z\._&(),/]*

Name

content - DNS Local Host File Content

Syntax

--content <string>

Description

DNS Local Host File Content

Class

Read only.

Type / Range

String of length 0 to 512

Name

lineNumber - Content Line Number

Syntax

--lineNumber <integer>

Description

Content Line Number

Class

Read only.

Type / Range

Integer in the range of 0 to 1,000

Roles**Create**

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.86 cmm localEndPtCfg

Local End Point Configuration. Syntax, description, parameters, and roles for cmm localEndPtCfg command.

Syntax

Description

This command is used to administer a Local End Point Configuration and includes parameters for defining the interface name, label, and port, DSCP code, and SCTP inbound and outbound streams. The 'shutdown' parameter is used to enable/shutdown the interface. The 'shutdown' parameter does not apply to AIML_DATA_STREAM - AIML Data Stream interface

Parameters

Name

interfaceName - Interface name

Syntax

--interfaceName <list-item>

Description

Interface name.

Class

Required key.

List

- GB - Specify gb interface to a BSC.
- GN - Specify the Gn interface to an SGSN.
- GNGGSN - Specify Gn interface to a GGSN.
- M3 - Specify the M3 interface to an eNodeB.
- N26 - Specify N26 interface
- RS10 - Specify RS10 interface to an SRS.
- S10 - Specify the S10 interface to another MME.
- S102 - Specify S102 interface to an IWF.
- S11 - Specify the S11 interface to an SGW.
- S11U - Specify S11u interface to a SGW.
- S13 - Specify the S13 interface to an EIR.
- S16 - Specify S16 interface loop around to this SGSN.
- S1MME - Specify the S1-MME interface to an eNodeB.
- S1MME_2 - Specify 2nd S1MME interface to an eNodeB.
- S1MME_3 - Specify 3rd S1MME interface to an eNodeB.
- S1MME_4 - Specify 4th S1MME interface to an eNodeB.
- S1MME_5 - Specify 5th S1MME interface to an eNodeB.

- S1MME_6 - Specify 6th S1MME interface to an eNodeB.
- S1MME_7 - Specify 7th S1MME interface to an eNodeB.
- S1MME_8 - Specify 8th S1MME interface to an eNodeB.
- S1MME_9 - Specify 9th S1MME interface to an eNodeB.
- S3 - Specify S3 interface from the MME to an SGSN.
- S3SGSN - Specify S3 interface from the SGSN to an SGSN.
- S4 - Specify the S4 interface to an SGW.
- S6A - Specify the S6a interface to an HSS.
- SBC - Specify the SBc interface to a CBC.
- SGD - Specify the SGd interface to an SMSC/SMS router.
- SGS - Specify the SGs interface to a GSM/UMTS MSC.
- SGS_LITE - Specify SGs lite interface to an IWF.
- SLG - Specify the SLg interface to a GMLC.
- SLS - Specify the SLs interface to an E-SMLC.
- SV - Specify the Sv interface to a GSM/UMTS MSC.
- Sm - Specify the sm interface to an MBMS gateway.
- T6A - Specify T6a interface to an SCEF.
- X1_1 - Specify X1_1 interface to an ADMF.
- X2 - Specify X2 interface to a DF2.
- SBI_NR10 - Specify SBI_NR10 interface.
- SBI_NR10_2 - SBI_NR_10_2 - Specify SBI_NR10_2 interface.
- SBI_NR10_3 - Specify the SBI_NR10_3 interface.
- SBI_NR10_4 - Specify the SBI_NR10_4 interface.
- SBI_NR10_5 - Specify the SBI_NR10_5 interface.
- SBI_NR10_6 - Specify the SBI_NR10_6 interface.
- SBI_NR10_7 - Specify the SBI_NR10_7 interface.
- SBI_NR10_8 - Specify the SBI_NR10_8 interface.
- SBI_NR10_9 - Specify the SBI_NR10_9 interface.
- SBI_NR10_10 - Specify the SBI_NR10_10 interface.
- SBI_NR10_11 - Specify the SBI_NR10_11 interface.
- SBI_NR10_12 - Specify the SBI_NR10_12 interface.

- SBI_NR10_13 - Specify the SBI_NR10_13 interface.
- SBI_NR10_14 - Specify the SBI_NR10_14 interface.
- SBI_NR10_15 - Specify the SBI_NR10_15 interface.
- SBI_NR10_16 - Specify the SBI_NR10_16 interface.
- AIML_DATA_STREAM - AIML Data Stream - Specify the AIML interface.

Name

avRequested - Number of authentication vectors

Syntax

--avRequested <integer>

Description

Authentication vector requested. this field is only used with the S6ad interface.

Restrictions

This element is only applicable when interfaceName is S6A.

Class

Optional.

Type / Range

Integer in the range of 0 to 5

Default

3

Name

comboIndication - Combo indication

Syntax

--comboIndication <list-item>

Description

This parameter enables system when running as a combo MME/SGSN configuration to use s6ad to connect to the HSS for both MME and SGSN.

Restrictions

1. This element is only applicable when interfaceName is S6A. 2. comboIndication is only valid when the system is running as a combo MME/SGSN and SGSN is running as S4-SGSN or both Gn/S4-SGSN. When enabled, MME or SGSN shall set the combo flag when communicating with the HSS. 3. comboIndication can be modified only when shutdown is set to true.

Class

Optional.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Name

description - Interface description

Syntax

--description <string>

Description

Description for interface.

Class

Optional.

Type / Range

Digit string of length 0 to 160 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Name

dscpCode - DSCP code

Syntax

--dscpCode <list-item>

Description

DSCP code.

Class

Optional.

List

- AF11 - Assured forwarding 11.
- AF12 - Assured forwarding 12.
- AF13 - Assured forwarding 13.
- AF21 - Assured forwarding 21.
- AF22 - Assured forwarding 22.
- AF23 - Assured forwarding 23.
- AF31 - Assured forwarding 31.
- AF32 - Assured forwarding 32.
- AF33 - Assured forwarding 33.
- AF41 - Assured forwarding 41.

- AF42 - Assured forwarding 42.
- AF43 - Assured forwarding 43.
- BE - BE.
- CS1 - Class selector 1.
- CS2 - Class selector 2.
- CS3 - Class selector 3.
- CS4 - Class selector 4.
- CS5 - Class selector 5.
- CS6 - Class selector 6.
- CS7 - Class selector 7.
- EF - Expedited forwarding.
- voiceAdmit - Voice Admit

Default

AF41

Name

httpProfileName - HTTP Profile Name for SBI

Syntax

--httpProfileName <string>

Description

HTTP Profile Name for SBI local endpoint

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

immRspPref - Immediate response preferred

Syntax

--immRspPref <list-item>

Description

Whether to include "Immediate response preferred" AVP in AIR.

Restrictions

This element is only applicable when interfaceName is S6A.

Class

Optional.

List

- Dynamic - Include when WMM needs authentication vector for current procedure.
- No - Always do not include "Immediate response preferred.
- Yes - Always include "Immediate response preferred.

Default

No

Name

interfaceLabel - Interface label

Syntax

--interfaceLabel <string>

Description

Interface label.

Class

Read only.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

interfacePort - Interface port

Syntax

--interfacePort <integer>

Description

This parameter assigns local sctp port number, and local tcp port for SBI interfaces. For SBI interfaces, if port is 0 and tlsEncrypted is true, port 443 is used. if port is 0 and tlsEncrypted is false, port 80 is used

Class

Optional.

Type / Range

Integer in the range of 0 or 1,024 to 65,535

Default

0

Name

multiHomed - Multi home

Syntax

--multiHomed <list-item>

Description

This enables/disables sctp multihomed at the given local termination.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

protocolProfileName - Interface profile name

Syntax

--protocolProfileName <string>

Description

Interface profile name.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Name

sctpInboundStreams - SCTP inbound streams

Syntax

--sctpInboundStreams <integer>

Description

SCTP inbound streams. this field is only used with the S6ad interface.

Restrictions

This element is only applicable when interfaceName is S6A.

Class

Optional.

Type / Range

Integer in the range of 1 to 10

Default

1

Name

sctpOutboundStreams - SCTP outbound streams

Syntax

--sctpOutboundStreams <integer>

Description

SCTP outbound streams. this field is only used with the S6ad interface.

Restrictions

This element is only applicable when interfaceName is S6A.

Class

Optional.

Type / Range

Integer in the range of 1 to 10

Default

1

Name

shutdown - Shutdown

Syntax

--shutdown <list-item>

Description

When it is true, indicates that interface is shutdown.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

tlsProfileName - TLS Profile Name for SBI

Syntax

--tlsProfileName <string>

Description

TLS Profile Name for SBI local endpoint

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.87 cmm locationGatewayApn

Location Gateway APN List. Syntax, description, parameters, and roles for cmm locationGatewayApn command.

Syntax**Description**

Configure, list and show APN lists.

Parameters**Name**

apnNi - APN Network Identifier

Syntax

--apnNi <string>

Description

APN Network Identifier

Class

Required key.

Type / Range

Digit string of length 1 to 62 with the allowed pattern of [0-9a-zA-Z\.\-]*

Name

locationGw - Location Based PGW Selection

Syntax

--locationGw <list-item>

Description

Location Based PGW Selection

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.88 cmm lteQosPolicySet

LTE QoS Policy set configuration. Syntax, description, parameters, and roles for cmm lteQosPolicySet command.

Syntax

Description

Configure, list and show LTE QoS Policy set parameters.

Parameters

Name

apnAmbrDL - APN aggregate maximum bit rate for downlink in kbps

Syntax

--apnAmbrDL <integer>

Description

This parameter defines the aggregated maximum bit rate for non-guaranteed bit rate for downlink per APN in kbps. the values range from 0 to 168000 with variable increment.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 168,000

Default

50

Name

apnAmbrUL - APN aggregate maximum bit rate for uplink in kbps

Syntax

--apnAmbrUL <integer>

Description

This parameter defines the aggregated maximum bit rate for non-guaranteed bit rate for uplink per APN in kbps. the values range from 0 to 46000 with variable increment.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 46,000

Default

50

Name

arpGbr - Allocation/retention priority for GBR

Syntax

--arpGbr <integer>

Description

This parameter defines allocation and retention priority for guaranteed bit rate. the value range from 1 to 15.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 15

Default

14

Name

arpNGbr - Allocation/retention for non-gbr

Syntax

--arpNGbr <integer>

Description

This parameter defines allocation and retention priority for non-guaranteed bit rate. the value range from 1 to 15.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 15

Default

14

Name

gbrDl - Guaranteed bit rate for downlink in kbps

Syntax

--gbrDl <integer>

Description

This parameter defines the guaranteed bit rate for downlink in kbps. the values range from 0 to 168000 with variable increment.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 168,000

Default

120

Name

gbrUL - Guaranteed bit rate for uplink in kbps

Syntax

--gbrUL <integer>

Description

This parameter defines the guaranteed bit rate for uplink in kbps. the values range from 0 to 46000 with variable increment.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 46,000

Default

50

Name

mbrDL - Maximum bit rate for downlink in kbps

Syntax

--mbrDL <integer>

Description

This parameter defines the maximum bit rate for downlink in kbps. the values range from 0 to 168000 with variable increment.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 168,000

Default

120

Name

mbrUL - Maximum bit rate for uplink in kbps

Syntax

--mbrUL <integer>

Description

This parameter defines the maximum bit rate for uplink in kbps. the values range from 0 to 46000 with variable increment.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 46,000

Default

50

Name

pciGbr - ARP pre-emption capability information for GBR

Syntax

--pciGbr <list-item>

Description

This parameter defines arp pre-emption capability information for guaranteed bit rate.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

pciNGbr - ARP pre-emption capability information for non-gbr

Syntax

--pciNGbr <list-item>

Description

This parameter defines arp pre-emption capability information for non-guaranteed bit rate.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

pviGbr - ARP pre-emption vulnerability information for GBR

Syntax

--pviGbr <list-item>

Description

This parameter defines arp pre-emption vulnerability information for guaranteed bit rate.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

pviNGbr - ARP pre-emption vulnerability information for non-gbr

Syntax

--pviNGbr <list-item>

Description

This parameter defines arp pre-emption vulnerability information for non-guaranteed bit rate.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

qci1 - LTE QoS class identifier 1

Syntax

--qci1 <list-item>

Description

This parameter defines the QoS class identifier 1.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

qci2 - LTE QoS class identifier 1

Syntax

--qci2 <list-item>

Description

This parameter defines the QoS class identifier 2.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

qci3 - LTE QoS class identifier 3

Syntax

--qci3 <list-item>

Description

This parameter defines the QoS class identifier 3.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

qci4 - LTE QoS class identifier 4

Syntax

--qci4 <list-item>

Description

This parameter defines the QoS class identifier 4.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

qci5 - LTE QoS class identifier 5

Syntax

--qci5 <list-item>

Description

This parameter defines the QoS class identifier 5.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

qci6 - LTE QoS class identifier 6

Syntax

--qci6 <list-item>

Description

This parameter defines the QoS class identifier 6.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

qci7 - LTE QoS class identifier 7

Syntax

```
--qci7 <list-item>
```

Description

This parameter defines the QoS class identifier 7.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

qci8 - LTE QoS class identifier 8

Syntax

```
--qci8 <list-item>
```

Description

This parameter defines the QoS class identifier 8.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

qci9 - LTE QoS class identifier 9

Syntax

```
--qci9 <list-item>
```

Description

This parameter defines the QoS class identifier 9.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Name

ueAmbrDl - UE aggregate maximum bit rate for downlink in kpbs

Syntax

--ueAmbrDl <integer>

Description

This parameter defines the UE-aggregated maximum bit rate for non-guaranteed bit rate for downlink. the values range from 0 to 168000 with variable increment.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 168,000

Default

120

Name

ueAmbrUl - UE aggregate maximum bit rate for uplink in kpbs

Syntax

--ueAmbrUl <integer>

Description

This parameter defines the UE-aggregated maximum bit rate for non-guaranteed bit rate for uplink. the values range from 0 to 46000 with variable increment.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 46,000

Default

50

Name

id - Id of LTE QoS policy set number

Syntax

<integer>

Description

Id of LTE QoS policy set number.

Class

Required value.

Type / Range

Integer in the range of 1 to 6

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.89 cmm lteQosPolicySetDefault

LTE QoS Policy default set configuration. Syntax, description, parameters, and roles for cmm lteQosPolicySetDefault command.

Syntax**Description**

Configure and show LTE QoS Policy default set parameters.

Parameters**Name**

apnAmbrDL - APN aggregate maximum bit rate for downlink in kbps

Syntax

--apnAmbrDL <integer>

Description

This parameter defines the aggregated maximum bit rate for non-guaranteed bit rate for downlink per APN in kbps. the values range from 0 to 168000 with variable increment.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 168,000

Default

50

Name

apnAmbrUl - APN aggregate maximum bit rate for uplink in kbps

Syntax

--apnAmbrUl <integer>

Description

This parameter defines the aggregated maximum bit rate for non-guaranteed bit rate for uplink per APN in kbps. the values range from 0 to 46000 with variable increment.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 46,000

Default

50

Name

arpGbr - Allocation/retention priority for GBR

Syntax

--arpGbr <integer>

Description

This parameter defines allocation and retention priority for guaranteed bit rate. the value range from 1 to 15.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 15

Default

14

Name

arpNGbr - Allocation/retention for non-gbr

Syntax

--arpNGbr <integer>

Description

This parameter defines allocation and retention priority for non-guaranteed bit rate. the value range from 1 to 15.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 15

Default

14

Name

gbrDL - Guaranteed bit rate for downlink in kbps

Syntax

--gbrDL <integer>

Description

This parameter defines the guaranteed bit rate for downlink in kbps. the values range from 0 to 168000 with variable increment.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 168,000

Default

120

Name

gbrUL - Guaranteed bit rate for uplink in kbps

Syntax

--gbrUL <integer>

Description

This parameter defines the guaranteed bit rate for uplink in kbps. the values range from 0 to 46000 with variable increment.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 46,000

Default

50

Name

mbrDL - Maximum bit rate for downlink in kbps

Syntax

```
--mbrDl <integer>
```

Description

This parameter defines the maximum bit rate for downlink in kbps. the values range from 0 to 168000 with variable increment.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 168,000

Default

120

Name

mbrUl - Maximum bit rate for uplink in kbps

Syntax

```
--mbrUl <integer>
```

Description

This parameter defines the maximum bit rate for uplink in kbps. the values range from 0 to 46000 with variable increment.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 46,000

Default

50

Name

pciGbr - ARP pre-emption capability information for GBR

Syntax

```
--pciGbr <list-item>
```

Description

This parameter defines arp pre-emption capability information for guaranteed bit rate.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

pciNGbr - ARP pre-emption capability information for non-gbr

Syntax

--pciNGbr <list-item>

Description

This parameter defines arp pre-emption capability information for non-guaranteed bit rate.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

pviGbr - ARP pre-emption vulnerability information for GBR

Syntax

--pviGbr <list-item>

Description

This parameter defines arp pre-emption vulnerability information for guaranteed bit rate.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

pviNGbr - ARP pre-emption vulnerability information for non-gbr

Syntax

--pviNGbr <list-item>

Description

This parameter defines arp pre-emption vulnerability information for non-guaranteed bit rate.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

qci1 - LTE QoS class identifier 1

Syntax

--qci1 <list-item>

Description

This parameter defines the QoS class identifier 1.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

qci2 - LTE QoS class identifier 1

Syntax

--qci2 <list-item>

Description

This parameter defines the QoS class identifier 2.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

qci3 - LTE QoS class identifier 3

Syntax

--qci3 <list-item>

Description

This parameter defines the QoS class identifier 3.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

qci4 - LTE QoS class identifier 4

Syntax

--qci4 <list-item>

Description

This parameter defines the QoS class identifier 4.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

qci5 - LTE QoS class identifier 5

Syntax

--qci5 <list-item>

Description

This parameter defines the QoS class identifier 5.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

qci6 - LTE QoS class identifier 6

Syntax

--qci6 <list-item>

Description

This parameter defines the QoS class identifier 6.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

qci7 - LTE QoS class identifier 7

Syntax

--qci7 <list-item>

Description

This parameter defines the QoS class identifier 7.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

qci8 - LTE QoS class identifier 8

Syntax

--qci8 <list-item>

Description

This parameter defines the QoS class identifier 8.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

qci9 - LTE QoS class identifier 9

Syntax

--qci9 <list-item>

Description

This parameter defines the QoS class identifier 9.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ueAmbrDL - UE aggregate maximum bit rate for downlink in kpbs

Syntax

--ueAmbrDL <integer>

Description

This parameter defines the UE-aggregated maximum bit rate for non-guaranteed bit rate for downlink. the values range from 0 to 168000 with variable increment.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 168,000

Default

120

Name

ueAmbrUl - UE aggregate maximum bit rate for uplink in kpbs

Syntax

--ueAmbrUl <integer>

Description

This parameter defines the UE-aggregated maximum bit rate for non-guaranteed bit rate for uplink. the values range from 0 to 46000 with variable increment.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 46,000

Default

50

Name

id - Id of default LTE QoS policy set number

Syntax

config

Description

Just type 'config' here.

Class

Required value.

List

- config - Specify LTE QoS policy default of config.

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

5.90 cmm lteTo2G3GQosMapping

LTE 2G3G QoS Mapping. Syntax, description, parameters, and roles for cmm lteTo2G3GQosMapping command.

Syntax

Description

This command is used to administer LTE 2G3G QoS Mapping and includes parameters for defining AMBR and ARP values, and traffic classes and QCI values for background, conversational, interactive, and streaming. A default profile is provided.

Parameters

Name

name - LTE 2G3G QoS mapping name

Syntax

--name <string>

Description

LTE 2G3G QoS mapping name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

apnAmbrDownLink - APN AMBR downlink

Syntax

--apnAmbrDownLink <integer>

Description

Configures apn AMBR for downlink rate in kilobits per second.

Restrictions

Valid bit rates are as follows... if in range 0-63 multiple of 1 kbps, if in the range 64..568 must be a multiple of 8 kbps, if in the range 576..8640 must be a multiple of 64 kbps, if in the range 8700..16000 must be a multiple of 100 kbps, if in the range 17000..128000 must be a multiple of 1 Mbps, if in the range 130000..256000 must be a multiple of 2 Mbps

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64-568, 576-8640, 8700-16000, 17000-128000, 130000-256000, 260000-500000, 510000-1500000, 1600000-4000000000

Default

256000

Name

apnAmbrUpLink - APN AMBR uplink

Syntax

--apnAmbrUpLink <integer>

Description

Configures apn AMBR for uplink rate in kilobits per second.

Restrictions

Valid bit rates are as follows... if in range 0-63 multiple of 1 kbps, if in the range 64..568 must be a multiple of 8 kbps, if in the range 576..8640 must be a multiple of 64 kbps, if in the range 8700..16000 must be a multiple of 100 kbps, if in the range 17000..128000 must be a multiple of 1 Mbps, if in the range 130000..256000 must be a multiple of 2 Mbps

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64-568, 576-8640, 8700-16000, 17000-128000, 130000-256000, 260000-500000, 510000-1500000, 1600000-4000000000

Default

256000

Name

arpHValue - ArpH value

Syntax

--arpHValue <integer>

Description

ArpH value indicates a high priority (H) value to use for ARP translations.

Class

Optional.

Type / Range

Integer in the range of 1 to 13

Default

1

Name

arpMValue - ArpM value

Syntax

--arpMValue <integer>

Description

ArpM value indicates a medium priority (M) value to use for ARP translations.

Class

Optional.

Type / Range

Integer in the range of 2 to 14

Default

2

Name

backgroundQci - Background QCI

Syntax

--backgroundQci <integer>

Description

Configures background qci.

Class

Optional.

Type / Range

Integer in the range of 1 to 9

Default

9

Name

conversationalSpeechQci - Conversational speech QCI

Syntax

--conversationalSpeechQci <integer>

Description

Configures conversationalspeech qci.

Class

Optional.

Type / Range

Integer in the range of 1 to 9

Default

1

Name

conversationalUnknownHighDelayQci - ConversationalUnknownHighDelay QCI

Syntax

--conversationalUnknownHighDelayQci <integer>

Description

Configures conversationalunknownhighdelay qci.

Class

Optional.

Type / Range

Integer in the range of 1 to 9

Default

2

Name

conversationalUnknownLowDelayQci - ConversationalUnknownLowDelay QCI

Syntax

--conversationalUnknownLowDelayQci <integer>

Description

Configures conversationalunknownlowdelay qci.

Class

Optional.

Type / Range

Integer in the range of 1 to 9

Default

3

Name

description - Description

Syntax

--description <string>

Description

Description.

Class

Optional.

Type / Range

Digit string of length 0 to 80 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

includeR11QosExtensionsGn - Include R11 QoS extensions Gn

Syntax

--includeR11QosExtensionsGn <list-item>

Description

Configures in the MME to include/exclude r1 QoS in SGSN context response or forward relocation request over the Gn interface..

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

includeR7QosExtensionsGn - Include R7 QoS extensions Gn

Syntax

```
--includeR7QosExtensionsGn <list-item>
```

Description

Configures in the MME to include/exclude r7 QoS in SGSN context response or forward relocation request over the Gn interface..

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

interactiveThp1NonSignalingQci - InteractiveTHP1NonSignaling QCI

Syntax

```
--interactiveThp1NonSignalingQci <integer>
```

Description

Configures interactivethp1nonsignaling qci.

Class

Optional.

Type / Range

Integer in the range of 1 to 9

Default

6

Name

interactiveThp1SignalingQci - InteractiveTHP1Signaling QCI

Syntax`--interactiveThp1SignalingQci <integer>`**Description**

Configures interactivethp1signaling qci.

Class

Optional.

Type / Range

Integer in the range of 1 to 9

Default

5

Name

interactiveThp2SignalingQci - InteractiveTHP2Signaling QCI

Syntax`--interactiveThp2SignalingQci <integer>`**Description**

Configures interactivethp2signaling qci.

Class

Optional.

Type / Range

Integer in the range of 1 to 9

Default

7

Name

interactiveThp3SignalingQci - InteractiveTHP3Signaling QCI

Syntax`--interactiveThp3SignalingQci <integer>`**Description**

Configures interactivethp3signaling qci.

Class

Optional.

Type / Range

Integer in the range of 1 to 9

Default

8

Name

obtainUeAmbr - Obtain UE-AMBR

Syntax`--obtainUeAmbr <integer>`**Description**

Obtain UE-AMBR indicates the rule to use to obtain the local UE-AMBR. entering "0", you designate the "calculate rule" which means that the MME software implements the algorithm specified in annex e of TS 23.401 to derive a local value for the parameter 5361 until the subscribed value can be obtained from the HSS. by entering a number between 1 and 50,000, the MME software uses the provisioned value until the subscribed value can be obtained from the HSS.

Class

Optional.

Type / Range

Integer in the range of 0 to 50,000

Default

0

Name

qci1QosProfileName - Qci1 QoS profile name

Syntax`--qci1QosProfileName <string>`**Description**

This is the traffic class profile name for QCI value 1. it must be entered together with qci1trafficclass.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

qci1-default

Name

qci1TrafficClass - Qci1 trafficclass

Syntax`--qci1TrafficClass <list-item>`**Description**

Qci1 trafficclass.

Class

Optional.

List

- Background - Traffic class type of background.
- Conversational - Traffic class type of conversational.
- Interactive - Traffic class type of interactive.
- Streaming - Traffic class type of streaming.

Default

Conversational

Name

qci2QosProfileName - Qci2 QoS profile name

Syntax

--qci2QosProfileName <string>

Description

This is the traffic class profile name for QCI value 2. it must be entered together with qci2trafficclass.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

qci2-default

Name

qci2TrafficClass - Qci2 trafficclass

Syntax

--qci2TrafficClass <list-item>

Description

Qci2 trafficclass.

Class

Optional.

List

- Background - Traffic class type of background.
- Conversational - Traffic class type of conversational.

- Interactive - Traffic class type of interactive.
- Streaming - Traffic class type of streaming.

Default

Conversational

Name

qci3QosProfileName - Qci3 QoS profile name

Syntax

--qci3QosProfileName <string>

Description

This is the traffic class profile name for QCI value 3. it must be entered together with qci3trafficclass.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

qci3-default

Name

qci3TrafficClass - Qci3 trafficclass

Syntax

--qci3TrafficClass <list-item>

Description

Qci3 trafficclass.

Class

Optional.

List

- Background - Traffic class type of background.
- Conversational - Traffic class type of conversational.
- Interactive - Traffic class type of interactive.
- Streaming - Traffic class type of streaming.

Default

Conversational

Name

qci4QosProfileName - Qci4 QoS profile name

Syntax

```
--qci4QosProfileName <string>
```

Description

This is the traffic class profile name for QCI value 4. it must be entered together with qci4trafficclass.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

qci4-default

Name

qci4TrafficClass - Qci4 trafficclass

Syntax

```
--qci4TrafficClass <list-item>
```

Description

Qci4 trafficclass.

Class

Optional.

List

- Background - Traffic class type of background.
- Conversational - Traffic class type of conversational.
- Interactive - Traffic class type of interactive.
- Streaming - Traffic class type of streaming.

Default

Streaming

Name

qci5QosProfileName - Qci5 QoS profile name

Syntax

```
--qci5QosProfileName <string>
```

Description

This is the traffic class profile name for QCI value 5. it must be entered together with qci5trafficclass.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

qci5-default

Name

qci5TrafficClass - Qci5 trafficclass

Syntax

--qci5TrafficClass <list-item>

Description

Qci5 trafficclass.

Class

Optional.

List

- Background - Traffic class type of background.
- Conversational - Traffic class type of conversational.
- Interactive - Traffic class type of interactive.
- Streaming - Traffic class type of streaming.

Default

Interactive

Name

qci6QosProfileName - Qci6 QoS profile name

Syntax

--qci6QosProfileName <string>

Description

This is the traffic class profile name for QCI value 6. it must be entered together with qci6trafficclass.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

qci6-default

Name

qci6TrafficClass - Qci6 trafficclass

Syntax

```
--qci6TrafficClass <list-item>
```

Description

Qci6 trafficclass.

Class

Optional.

List

- Background - Traffic class type of background.
- Conversational - Traffic class type of conversational.
- Interactive - Traffic class type of interactive.
- Streaming - Traffic class type of streaming.

Default

Interactive

Name

qci7QosProfileName - Qci7 QoS profile name

Syntax

```
--qci7QosProfileName <string>
```

Description

This is the traffic class profile name for QCI value 7. it must be entered together with qci7trafficclass.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

qci7-default

Name

qci7TrafficClass - Qci7 trafficclass

Syntax

```
--qci7TrafficClass <list-item>
```

Description

Qci7 trafficclass.

Class

Optional.

List

- Background - Traffic class type of background.
- Conversational - Traffic class type of conversational.
- Interactive - Traffic class type of interactive.
- Streaming - Traffic class type of streaming.

Default

Interactive

Name

qci8QosProfileName - Qci8 QoS profile name

Syntax

--qci8QosProfileName <string>

Description

This is the traffic class profile name for QCI value 8. it must be entered together with qci8trafficclass.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Default

qci8-default

Name

qci8TrafficClass - Qci8 trafficclass

Syntax

--qci8TrafficClass <list-item>

Description

Qci8 trafficclass.

Class

Optional.

List

- Background - Traffic class type of background.
- Conversational - Traffic class type of conversational.
- Interactive - Traffic class type of interactive.
- Streaming - Traffic class type of streaming.

Default

Interactive

Name

qci9QosProfileName - Qci9 QoS profile name

Syntax

--qci9QosProfileName <string>

Description

This is the traffic class profile name for QCI value 9. it must be entered together with qci9trafficclass.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

qci9-default

Name

qci9TrafficClass - Qci9 trafficclass

Syntax

--qci9TrafficClass <list-item>

Description

Qci9 trafficclass.

Class

Optional.

List

- Background - Traffic class type of background.
- Conversational - Traffic class type of conversational.
- Interactive - Traffic class type of interactive.
- Streaming - Traffic class type of streaming.

Default

Background

Name

qosMappingHighDelayThr - QoS mapping highdelay threshold

Syntax

--qosMappingHighDelayThr <integer>

Description

QoS mapping highdelay threshold is used to distinguish between the conversational_unknown_highdelay and conversational_unknown_lowdelay cases. the R99

QoS transfer delay parameter is compared to this threshold value. if it is equal to or greater than this threshold then it is considered "high delay". otherwise it is "low delay". range is 10 - 400 in millisec.

Class

Optional.

Type / Range

Integer in the range of 10 to 400

Default

150

Name

streamingSpeechQci - StreamingSpeech QCI

Syntax

--streamingSpeechQci <integer>

Description

Configures streamingspeech qci.

Class

Optional.

Type / Range

Integer in the range of 1 to 9

Default

4

Name

streamingUnknownQci - StreamingUnknown QCI

Syntax

--streamingUnknownQci <integer>

Description

Configures streamingunknown qci.

Class

Optional.

Type / Range

Integer in the range of 1 to 9

Default

4

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.91 cmm m3uaLinkInfo

M3UA Link Info. Syntax, description, parameters, and roles for cmm m3uaLinkInfo command.

Syntax

Description

M3UA Link Info.

Parameters

Name

link - M3UA link

Syntax

--link <string>

Description

Use this parameter to specify the signalling links you wish to interrogate. if you wish to interrogate all existing signalling links, leave this parameter out.

Class

Optional.

Type / Range

Digit string of length 0 to 30 with the allowed pattern of `^(1203[0-1]|120[0-2][0-9]|1[0-1][0-9][0-9][0-9][1-9][0-9][0-9][0-9][1-9][0-9][0-9][1-9][0-9][0-9])([,-](1203[0-1]|120[0-2][0-9]|1[0-1][0-9][0-9][0-9][1-9][0-9][0-9][0-9][1-9][0-9][0-9][1-9][0-9][0-9]))*`
\$

Name

state - Link state queried

Syntax

--state <list-item>

Description

Use this parameter to specify the state of the signalling links you wish to interrogate. allowed values unavailable, available. only links with the specified state are displayed.

Class

Optional.

List

- available - Available.
- unavailable - Unavailable.

Roles

List

sysadmin, manage, monitor

5.92 cmm m3uaLinkSetInfo

M3UA Link Set Info. Syntax, description, parameters, and roles for cmm m3uaLinkSetInfo command.

Syntax

Description

M3UA Link Set Info.

Parameters

Name

network - Signalling network

Syntax

--network <list-item>

Description

Use this parameter to specify the symbolic name of the signalling network. allowed values IN0,IN1,NA0,NA1.

Class

Required on create.

List

- IN0 - International network number 0.
- IN1 - International network number 1.

- NAO - National network number 0.
- NA1 - National network number 1.

Name

spc - Signalling point code

Syntax

--spc <string>

Description

Use this parameter to specify the signalling point codes of those signalling points where the signalling link sets, whose state data you wish to output, lead to.

Class

Optional.

Type / Range

Digit string of length 0 to 30 with the allowed pattern of (1203[0-1]|120[0-2][0-9]|1[0-1][0-9][0-9][0-9]|1[1-9][0-9][0-9][0-9]|1[1-9][0-9][0-9]|1[1-9][0-9][0-9]|1[1-9][0-9][0-9])([,,-](1203[0-1]|120[0-2][0-9]|1[0-1][0-9][0-9][0-9]|1[1-9][0-9][0-9][0-9]|1[1-9][0-9][0-9]|1[1-9][0-9][0-9]))*

Roles**List**

sysadmin, manage, monitor

5.93 cmm m3uaProfile

M3UA Profile. Syntax, description, parameters, and roles for cmm m3uaProfile command.

Syntax**Description**

Administer M3UA Profile.

Parameters**Name**

name - M3UA profile name

Syntax

--name <string>

Description

M3UA profile name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

assoMaxRetrans - Max association retrans

Syntax

--assoMaxRetrans <integer>

Description

Maximum association retransmissions.

Class

Optional.

Type / Range

Integer in the range of 1 to 50

Default

10

Name

cookieLife - Cookie life

Syntax

--cookieLife <integer>

Description

Valid cookie life time frame in units of seconds.

Class

Optional.

Type / Range

Integer in the range of 5 to 120 in units of Second

Default

60

Name

description - Description

Syntax

--description <string>

Description

Profile description.

Class

Optional.

Type / Range

Digit string of length 0 to 160 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

hbInterval - Heartbeat interval

Syntax

--hbInterval <integer>

Description

Heartbeat interval in units of seconds.

Class

Optional.

Type / Range

Integer in the range of 1 to 300 in units of Second

Default

30

Name

maxInitRetrans - Max init retrans

Syntax

--maxInitRetrans <integer>

Description

Maximum initial retransmissions.

Class

Optional.

Type / Range

Integer in the range of 1 to 25

Default

8

Name

mtuSize - MTU Size

Syntax

--mtuSize <integer>

Description

Maximum transmission unit size in bytes.

Class

Optional.

Type / Range

Integer in the range of 512 to 9000 in units of Byte

Default

1500

Name

pathMaxRetrans - Max path retrans

Syntax

--pathMaxRetrans <integer>

Description

Maximum path retransmissions.

Class

Optional.

Type / Range

Integer in the range of 1 to 25

Default

5

Name

rtoInitValid - RTO initial value

Syntax

--rtoInitValid <integer>

Description

Initial retransmission timeout in units of ms.

Class

Optional.

Type / Range

Integer in the range of 10 to 10,000 in units of Millisecond

Default

300

Name

rtoMax - RTO maximum

Syntax

--rtoMax <integer>

Description

Maximum retransmission timeout in units of ms.

Class

Optional.

Type / Range

Integer in the range of 20 to 60,000 in units of Millisecond

Default

400

Name

rtoMin - RTO minimum

Syntax

--rtoMin <integer>

Description

Minimum retransmission timeout in units of ms.

Class

Optional.

Type / Range

Integer in the range of 10 to 5,000 in units of Millisecond

Default

50

Name

sackFreq - SACK frequency

Syntax

--sackFreq <integer>

Description

SCTP acknowledgment frequency.

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

2

Name

sackPeriod - SACK period

Syntax

--sackPeriod <integer>

Description

SCTP acknowledgment period in units of ms.

Class

Optional.

Type / Range

Integer in the range of 0 to 500 in units of Millisecond

Default

200

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.94 cmm m3uaRouteSetInfo

M3UA Route Set Info. Syntax, description, parameters, and roles for cmm m3uaRouteSetInfo command.

Syntax**Description**

M3UA Route Set Info.

Parameters**Name**

network - Signalling network

Syntax

--network <list-item>

Description

Use this parameter to specify the symbolic name of the signalling network. allowed values IN0,IN1,NA0,NA1.

Class

Required on create.

List

- IN0 - International network number 0.
- IN1 - International network number 1.
- NA0 - National network number 0.
- NA1 - National network number 1.

Name

spc - Signalling point code

Syntax

--spc <string>

Description

Use this parameter to specify the signalling point codes of the signalling points where the signalling route sets, whose state data you wish to output, lead to. if you wish to interrogate the state data on the signalling route sets leading to all existing signalling points, leave this parameter out.

Class

Optional.

Type / Range

Digit string of length 0 to 30 with the allowed pattern of (1203[0-1]|120[0-2][0-9]|1[0-1][0-9][0-9][0-9][1-9][0-9][0-9][0-9][1-9][0-9][0-9][1-9][0-9][0-9])([,-](1203[0-1]|120[0-2][0-9]|1[0-1][0-9][0-9][0-9][1-9][0-9][0-9][0-9][1-9][0-9][0-9][1-9][0-9][0-9]))*

Roles

List

sysadmin, manage, monitor

5.95 cmm m3uaSpLoadSharingInfo

M3UA Signalling Point Load Sharing info. Syntax, description, parameters, and roles for cmm m3uaSpLoadSharingInfo command.

Syntax

Description

M3UA Signalling Point Load Sharing info.

Parameters

Name

network - Signalling network

Syntax

--network <list-item>

Description

Use this parameter to specify the symbolic name of the signalling network. allowed values IN0,IN1,NA0,NA1.

Class

Required on create.

List

- IN0 - International network number 0.
- IN1 - International network number 1.
- NA0 - National network number 0.
- NA1 - National network number 1.

Name

opc - Originator point code

Syntax

--opc <string>

Description

Use this parameter to specify the originator point of STP messages.

Class

Optional.

Type / Range

Digit string of length 0 to 30 with the allowed pattern of (1203[0-1]|120[0-2][0-9]|1[0-1][0-9][0-9][0-9][1-9][0-9][0-9][0-9][1-9][0-9][0-9][1-9][0-9][0-9])([,-](1203[0-1]|120[0-2][0-9]|1[0-1][0-9][0-9][0-9][1-9][0-9][0-9][0-9][1-9][0-9][0-9][1-9][0-9][0-9]))*

Name

spc - Signalling point code

Syntax

--spc <string>

Description

Use this parameter to specify the signalling point code of the signalling point the load sharing of which you wish to interrogate.

Class

Optional.

Type / Range

Digit string of length 0 to 30 with the allowed pattern of (1203[0-1]|120[0-2][0-9]|1[0-1][0-9][0-9][0-9]|1[1-9][0-9][0-9][0-9]|1[1-9][0-9][0-9][0-9]|1[1-9][0-9][0-9][0-9])([,,-](1203[0-1]|120[0-2][0-9]|1[0-1][0-9][0-9][0-9]|1[1-9][0-9][0-9][0-9]|1[1-9][0-9][0-9][0-9]))*

Roles

List

sysadmin, manage, monitor

5.96 cmm maintenance

maintenance. Syntax, description, parameters, and roles for cmm maintenance command.

Syntax

Description

Configure and list maintenance.

Parameters

Name

offload - Offload

Syntax

--offload <list-item>

Description

With this parameter, you can activate the phase of the multipoint offloading process manually or start the automated offloading process. Possible values are: 1 ... activate offloading phase 1. 2 ... activate offloading phase 2. AUTO ... start automated offloading.

Class

Optional, not filterable on list.

List

- 1 - Phase 1 - Activate offloading phase 1.
- 2 - Phase 2 - Activate offloading phase 2.
- AUTO - Automated - Start automated offloading.

Default

AUTO

Name

rat - Radio access type

Syntax`--rat <list-item>`**Description**

This parameter defines the access type (automated or manual) multipoint offloading procedure. possible values are GB ... GB interface IU ... IU interface ALL ... all interfaces.

Class

Optional, not filterable on list.

List

- ALL - All interfaces.
- GB - Gb interface.
- IU - Iu interface.

Default

ALL

Name

type - Type

Syntax`--type <list-item>`**Description**

This parameter defines whether offload subscribers or Remove SGSN for maintenance (as described in f50106-01 SGSN support for Remove SGSN for Maintenance. Possible values are: OFFL ... Offload Subscribers. MAINT ... Remove SGSN for Maintenance (default). IOFFL ... IMSI Based Offloading

Class

Optional, not filterable on list.

List

- IOFFL - Offload specific subscribers specified with IMSI
- MAINT - Offload all SUBS.
- OFFL - Offload number of subscribers specified in OFFLCOUNT.

Default

MAINT

Name

id - Maintenance id

Syntax

<list-item>

Description

Just type 'SGSN' here.

Class

Required value.

List

- SGSN - Specify maintenance identifier of SGSN.

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.97 cmm maintForPaps

maintForPaps. Syntax, description, parameters, and roles for **cmm maintForPaps** command.

Syntax**Description**

Configure and show PAPS maintenance.

Parameters**Name**

blockForMaint - Blocking for maintenance

Syntax

--blockForMaint <list-item>

Description

With this parameter you can set all PAPS to the blocked state or unblock all PAPS for normal use. Cannot be issued with massDetach. Possible values are: ACT ... Block all PAPS for maintenance DACT ... Unblock all PAPS for normal use OFF ... Not used parameter

Class

Optional, not filterable on list.

List

- OFF - Not used parameter.
- ACT - Active - Block all PAPS for maintenance.
- DACT - Deactive - Unblock all PAPS for normal use.

Default

OFF

Name

massDetach - Mass detach

Syntax

--massDetach <list-item>

Description

With this parameter you can start or cancel the execution of mass detach. cannot be issued with blockformaint. possible values are START ... start mass detach CANCEL ... cancel mass detach OFF ... not used parameter.

Class

Optional, not filterable on list.

List

- OFF - Not used parameter.
- CANCEL - Cancel mass detach.
- START - Start mass detach.

Default

OFF

Name

rat - Radio access type

Syntax

--rat <list-item>

Description

This parameter defines the access type for the mass detach or (automated or manual) multipoint offloading procedure. possible values are GB ... GB interface IU ... IU interface ALL ... all interfaces NA ... not applicable parameter.

Class

Optional, not filterable on list.

List

- ALL - All interfaces.
- GB - Gb interface.
- IU - Iu interface.
- NA - Not applicable - Not applicable parameter.

Default

NA

Name

id - Maintforpaps id

Syntax

<integer>

Description

PAPS unit index.

Class

Required value.

Type / Range

Integer in the range of 0 to 19

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.98 cmm mapAddrAnalysis

MAP Address Analysis. Syntax, description, parameters, and roles for cmm mapAddrAnalysis command.

Syntax

Description

Use this command to configure, list and show network entity address analysis records for determining the best MAP version and virtual MSC,VLR addresses (optional) as well as the standard of the TCAP (optional), the standard of the SCCP address (optional), and the standard of the object identifier (optional).

Parameters

Name

gtDigits - GT digits

Syntax

--gtDigits <digits>

Description

Use this parameter to specify the global title address of the network entity.

Class

Required key.

Type / Range

Digit string of length 1 to 15

Name

spc - Spc

Syntax

--spc <integer>

Description

Use this parameter to specify the signalling point code of the network entity.you must give the ni parameter before the spc parameter.

Class

Required key.

Type / Range

Integer in the range of 1 to 16,777,215

Name

ni - Network indicator

Syntax

--ni <list-item>

Description

Use this parameter to specify the value that identifies the type of the network. cases, IN0 international network number 0 IN1 international network number 1 NAO national network number 0 NA1 national network number 1.

Class

Required key.

List

- IN0 - International network number 0.
- IN1 - International network number 1.
- NAO - National network number 0.
- NA1 - National network number 1.

Name

resAddressIndex - Result address index

Syntax

--resAddressIndex <integer>

Description

Use this parameter to specify the result address pair index, on the basis of which the analysis is to be carried out. please, check that this result record has been configured through cmm mapaddranalysisresaddr list.

Class

Required on create.

Type / Range

Integer in the range of 0 to 9

Name

resNumber - Result record number

Syntax

--resNumber <integer>

Description

Use this parameter to specify the result record number, on the basis of which the analysis is to be carried out. please, check that this result record has been configured through cmm mapaddranalysisresult list.

Class

Required on create.

Type / Range

Integer in the range of 0 to 254

Name

np - Numbering plan

Syntax

--np <list-item>

Description

Use this parameter to specify the numbering plan for the global title address. cases, E164 ISDN,telephony E212 land mobile E214 ISDN,mobile.

Class

Optional.

List

- E164 - E.164 - Specify numbering plan of E.164.
- E212 - E.212 - Specify numbering plan of E.212.
- E214 - E.214 - Specify numbering plan of E.214.

Default

E164

Name

obid - Standard of object identifier

Syntax

--obid <list-item>

Description

Use this parameter to specify the standard that is used in the object identifiers in the result of the current analysis. allowed values, ANSI , ITU.

Class

Optional.

List

- ANSI - Specify standards type of american national standards institute.
- ITU - Specify standards type of international telecommunication union.

Default

ITU

Name

resAddressName - Result address name

Syntax

--resAddressName <string>

Description

Use this parameter to specify the result address pair name, on the basis of which the analysis is to be carried out. you do not have to specify this, specify just resaddressindex.

Class

Read only.

Type / Range

Digit string of length 1 to 15 with the allowed pattern of [0-9a-zA-Z\._ :]*

Name

resName - Result record name

Syntax

--resName <string>

Description

Use this parameter to specify the result record name, on the basis of which the analysis is to be carried out. you do not have to specify this, specify just resnumber.

Class

Read only.

Type / Range

Digit string of length 1 to 24 with the allowed pattern of [0-9a-zA-Z\._ :]*

Name

sccp - Standard of SCCP addresses

Syntax

--sccp <list-item>

Description

Use this parameter to specify the standard that is used in the SCCP addresses in the result of the current analysis. allowed values, ANSI , ITU.

Class

Optional.

List

- ANSI - Specify standards type of american national standards institute.
- ITU - Specify standards type of international telecommunication union.

Default

ITU

Name

tcap - Standard of TCAP

Syntax

--tcap <list-item>

Description

Use this parameter to specify the standard that is used in the TCAP in the result of the current analysis. allowed values, ANSI , ITU.

Class

Optional.

List

- ANSI - Specify standards type of american national standards institute.
- ITU - Specify standards type of international telecommunication union.

Default

ITU

Name

toA - Type of address

Syntax

--toA <list-item>

Description

Use this parameter to specify the type of address. allowed values, GT (global title address) , DPC (destination point code).

Class

Optional.

List

- DPC - Destination point code - Specify address type of destination point code.
- GT - Global title - Specify address type of global title.

Default

GT

Name

toN - Type of number

Syntax

--toN <list-item>

Description

Use this parameter to specify the country format for the global title address. cases, INT international NAT national.

Class

Optional.

List

- INT - International - Specify address format as international.
- NAT - National - Specify address format as national.

Default

INT

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.99 cmm mapAddrAnalysisResAddr

MAP Address Analysis Result Address. Syntax, description, parameters, and roles for cmm mapAddrAnalysisResAddr command.

Syntax

Description

Use this command to configure, list and show the result address pairs for the network entity address analysis.

Parameters

Name

resAddressIndex - Result address index

Syntax

--resAddressIndex <integer>

Description

Use this parameter to specify the result address pair index, on the basis of which the analysis is to be carried out.

Class

Required key.

Type / Range

Integer in the range of 0 to 9

Name

resAddressName - Result address name

Syntax

--resAddressName <string>

Description

Use this parameter to specify the result address pair name, on the basis of which the analysis is to be carried out.

Class

Required on create.

Type / Range

Digit string of length 1 to 15 with the allowed pattern of [0-9a-zA-Z\._ :]*

Name

mscDigits - MSC address

Syntax

--mscDigits <digits>

Description

Use this parameter to specify the virtual MSC address in E164 format.

Class

Optional.

Type / Range

Digit string of length 1 to 18

Default

0

Name

mscTon - Type of MSC address

Syntax

--mscTon <list-item>

Description

Use this parameter to specify the country format of virtual MSC address.

Class

Optional.

List

- INT - International - Specify address format as international.
- NAT - National - Specify address format as national.

Default

INT

Name

vlrDigits - VLR address

Syntax

--vlrDigits <digits>

Description

Use this parameter to specify the virtual VLR address in E164 format.

Class

Optional.

Type / Range

Digit string of length 1 to 18

Default

0

Name

vlrTon - Type of VLR address

Syntax

--vlrTon <list-item>

Description

Use this parameter to specify the country format of virtual VLR address.

Class

Optional.

List

- INT - International - Specify address format as international.
- NAT - National - Specify address format as national.

Default

INT

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.100 cmm mapAddrAnalysisResult

MAP Address Analysis Result. Syntax, description, parameters, and roles for cmm mapAddrAnalysisResult command.

Syntax

Description

Use this command to configure, list and show the result records of the network entity address analysis.

Parameters

Name

result - Result record number

Syntax

--result <integer>

Description

Use this parameter to specify the result record number.

Class

Required key.

Type / Range

Integer in the range of 0 to 254

Name

name - Result record name

Syntax

--name <string>

Description

Use this parameter to specify the name for the result record.

Class

Required on create.

Type / Range

Digit string of length 1 to 24 with the allowed pattern of [0-9a-zA-Z\._:-]*

Name

extension - Extensions in use

Syntax

--extension <list-item>

Description

Use this parameter to specify the extension status in use. allowed values, y (Yes) , n (No).

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.101 cmm mapApplicationContextDefault

MAP application context default. Syntax, description, parameters, and roles for cmm mapApplicationContextDefault command.

Syntax**Description**

Use this command to modify or show the application-context-specific default MAP version, the default use of the 0x1647 alarm, and the default use of the 0x2646 alarm.

Parameters**Name**

applicationContext - MAP application context id

Syntax

--applicationContext <list-item>

Description

Use this parameter to specify the application context. this parameter is associated with each dialogue established by a MAP user, that is, it identifies the reason for the MAP communication between two network elements.

Class

Required key.

List

- anyTimeEnquiry - Any time enquiry - Any time enquiry.
- anyTimeInfoHandling - Any time information handling - Any time information handling.
- authenticationFailureReport - Authentication failure report - Authentication failure report.
- callCompletion - Call completion - Call completion.
- equipmentMngt - Equipment management - Equipment management.
- gprsLocationUpdate - GRPS location update - GRPS location update.
- handoverControl - Handover control - Handover control.
- imsiRetrieval - IMSI retrieval - IMSI retrieval.
- infoRetrieval - Information retrieval - Information retrieval.
- interVlrInfoRetrieval - Inter VLR information retrieval - Inter VLR information retrieval.
- locationCancellation - Location cancellation - Location cancellation.
- locationInfoRetrieval - Location information retrieval - Location information retrieval.
- locationSvcEnquiry - Location svc enquiry - Location svc enquiry.
- locationSvcGateway - Location svc gateway - Location svc gateway.
- mmEventReporting - MM event reporting - MM event reporting.
- msPurging - MS purging - MS purging.
- mwdMngt - MWD management - MWD management.
- networkFunctionalSs - Network functional ss - Network functional ss.
- networkLocUp - Network loc up - Network loc up.
- networkUnstructuredSs - Network unstructured ss - Network unstructured ss.
- reporting - Reporting.
- reset - Reset.
- roamingNumberEnquiry - Roaming number enquiry - Roaming number enquiry.

- shortMsgAlert - Short message alert - Short message alert.
- shortMsgGateway - Short message gateway - Short message gateway.
- shortMsgMORelay - Short message mobile origination relay - Short message mobile origination relay.
- shortMsgMTRelay - Short message mobile termination relay - Short message mobile termination relay.
- sslInvocationNotification - SS invocation notification - SS invocation notification.
- subscriberDataMngt - Subscriber data management - Subscriber data management.
- subscriberDataModificationNotif - Subscriber data modification - Subscriber data modification.
- subscriberInfoEnquiry - Subscriber information enquiry - Subscriber information enquiry.
- tracing - Tracing.

Name

alarm1 - Alarm 1

Syntax

--alarm1 <list-item>

Description

Use this parameter to specify the OPERATION TIMER EXPIRY (0x1647) alarm status. allowed values, ALLOWED , DENIED.

Class

Optional.

List

- ALLOWED - Specify default MAP alarm of allowed.
- DENIED - Specify default MAP alarm of denied.

Default

ALLOWED

Name

alarm2 - Alarm 2

Syntax

--alarm2 <list-item>

Description

Use this parameter to specify the PROTOCOL ROUTING FAILURE (0x2646) alarm status. allowed values, ALLOWED , DENIED.

Class

Optional.

List

- ALLOWED - Specify default MAP alarm of allowed.
- DENIED - Specify default MAP alarm of denied.

Default

ALLOWED

Name

extensionInUse - Extension in use

Syntax

--extensionInUse <list-item>

Description

Use this parameter to specify the extension status in use.

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

mapVersion - MAP version

Syntax

--mapVersion <list-item>

Description

Use this parameter to specify the MAP version used. each version is used for communicating with a particular network in the chosen application context. cases, 1 2 3 4 DENIED.

Class

Optional.

List

- 1 - Specify MAP default version of 1.
- 2 - Specify MAP default version of 2.
- 3 - Specify MAP default version of 3.
- 4 - Specify MAP default version of 4.
- DENIED - Specify MAP default version of denied.

Default

3

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

5.102 cmm mapApplicationContextResult

MAP Application Context Result. Syntax, description, parameters, and roles for cmm mapApplicationContextResult command.

Syntax**Description**

Use this command to configure, list and show the application context data for the result records of the network entity address analysis.

Parameters**Name**

result - Result record number

Syntax`--result <integer>`**Description**

Use this parameter to specify the result record number of the specific application context.

Class

Required key.

Type / Range

Integer in the range of 1 to 254

Name

applicationContext - Application context

Syntax`--applicationContext <list-item>`

Description

Use this parameter to specify the application context. this parameter is associated with each dialogue established by a MAP user, that is, it identifies the reason for the MAP communication between two network elements.

Class

Required key.

List

- anyTimeEnquiry - Any time enquiry - Any time enquiry.
- anyTimeInfoHandling - Any time information handling - Any time information handling.
- authenticationFailureReport - Authentication failure report - Authentication failure report.
- callCompletion - Call completion - Call completion.
- equipmentMngt - Equipment management - Equipment management.
- gprsLocationUpdate - GRPS location update - GRPS location update.
- handoverControl - Handover control - Handover control.
- imsiRetrieval - IMSI retrieval - IMSI retrieval.
- infoRetrieval - Information retrieval - Information retrieval.
- interVlrInfoRetrieval - Inter VLR information retrieval - Inter VLR information retrieval.
- locationCancellation - Location cancellation - Location cancellation.
- locationInfoRetrieval - Location information retrieval - Location information retrieval.
- locationSvcEnquiry - Location svc enquiry - Location svc enquiry.
- locationSvcGateway - Location svc gateway - Location svc gateway.
- mmEventReporting - MM event reporting - MM event reporting.
- msPurging - MS purging - MS purging.
- mwdMngt - MWD management - MWD management.
- networkFunctionalSs - Network functional ss - Network functional ss.
- networkLocUp - Network loc up - Network loc up.
- networkUnstructuredSs - Network unstructured ss - Network unstructured ss.
- reporting - Reporting.
- reset - Reset.
- roamingNumberEnquiry - Roaming number enquiry - Roaming number enquiry.
- shortMsgAlert - Short message alert - Short message alert.

- shortMsgGateway - Short message gateway - Short message gateway.
- shortMsgMORElay - Short message mobile origination relay - Short message mobile origination relay.
- shortMsgMTRelay - Short message mobile termination relay - Short message mobile termination relay.
- sslInvocationNotification - SS invocation notification - SS invocation notification.
- subscriberDataMngt - Subscriber data management - Subscriber data management.
- subscriberDataModificationNotif - Subscriber data modification - Subscriber data modification.
- subscriberInfoEnquiry - Subscriber information enquiry - Subscriber information enquiry.
- tracing - Tracing.

Name

alarm1 - Alarm 0x1647

Syntax

--alarm1 <list-item>

Description

Use this parameter to specify the OPERATION TIMER EXPIRY (0x1647) alarm status. allowed values, ALLOWED , DENIED.

Class

Optional.

List

- ALLOWED - Specify MAP alarm of allowed.
- DEF - Default - Specify MAP alarm of default.
- DENIED - Specify MAP alarm of denied.

Default

ALLOWED

Name

alarm2 - Alarm 0x2646

Syntax

--alarm2 <list-item>

Description

Use this parameter to specify the PROTOCOL ROUTING FAILURE (0x2646) alarm status. allowed values, ALLOWED , DENIED.

Class

Optional.

List

- ALLOWED - Specify MAP alarm of allowed.
- DEF - Default - Specify MAP alarm of default.
- DENIED - Specify MAP alarm of denied.

Default

ALLOWED

Name

mapVersion - MAP version

Syntax

--mapVersion <list-item>

Description

Use this parameter to specify the MAP version used. each version is used for communicating with a particular network in the chosen application context. cases, 1 2 3 4 DENIED.

Class

Optional.

List

- 1 - Specify MAP version of 1.
- 2 - Specify MAP version of 2.
- 3 - Specify MAP version of 3.
- 4 - Specify MAP version of 4.
- DEF - Default - Specify MAP version of default.
- DENIED - Specify MAP version of denied.

Default

3

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.103 cmm mapDefaultParameter

MAP default Parameters. Syntax, description, parameters, and roles for cmm mapDefaultParameter command.

Syntax

Description

Use this command to modify, list and show the SCCP return option, the abort in overload control, the standard of the TCAP (optional), the standard of the SCCP address (optional), the standard of the object identifier (optional), and the number of authentication sets 1 and 2.

Parameters

Name

parmName - Parameter Name

Syntax

--parmName <list-item>

Description

'Use this parameter to specify the name of parameter. Cases, SCCP, When SCCP return is activated, messages sent to the network are returned in case of a routing failure. That is, if the SCCP message cannot be routed further, it is returned to the sender. A message can be sent back from the SCCP layer of any signalling transfer point or destination. Messages can also be sent back to TCAP users from the SCCP layer of the originating network element (local return). Local return is always active. The parmValue of this parameter is either Yes (SCCP return is on) or No (SCCP return is off). OLC, If an overload condition is detected while handling the TC begin message received from the network (that is, a MAP dialogue is to be initiated), the dialogue is not accepted and nothing is sent back to the network. This means that the remote network element is not informed by default about the rejection of the dialogue. Use this parameter to force the network element to send a notification (TC U Abort) to the initiating network element in the case of an overload situation. The parmValue of this parameter is either Yes (notification sending is on) or No (notification sending is off). TCAP, This parameter indicates the standard used in the TCAP as default if no network entity address analysis is created, or if the standard is given as default in the network entity address analysis. The specific parmValue can have the following values, ANSI (ANSI standard) , ITU (ITU standard). ADDR, This parameter indicates the standard used in the SCCP addresses as default if no network entity address analysis is created, or if the standard is given as default in the network entity address analysis. The specific parmValue can have the

following values, ANSI (ANSI standard) , ITU (ITU standard). OBID, This parameter indicates the standard used in the object identifiers as default if no network entity address analysis is created, or if the standard is given as default in the network entity address analysis. The specific parmValue can have the following values: ANSI (ANSI standard) , ITU (ITU standard). NAS1, Use this parameter to define the number of authentication sets in the case of MAP version 1. The specific parmValue can have range from 1 to 5. NAS2 , Use this parameter to define the number of authentication sets in the case of MAP version 2. The specific parmValue can have range from 1 to 4.

Class

Required key.

List

- addr - standard of SCCP addresses - Standard that is used in the SCCP addresses in the result of the current analysis.
- nas1 - number of authentication sets 1 - Define the number of authentication sets in the case of MAP version 1.
- nas2 - number of authentication sets 2 - Define the number of authentication sets in the case of MAP version 2.
- obid - standard of object identifier - Standard that is used in the object identifiers in the result of the current analysis.
- olc - abort in overload control - Send a notification (TC-U-Abort) to the initiating network element in the case of an overload situation.
- sccp - SCCP return option - When SCCP return is activated, messages sent to the network are returned in case of a routing failure.
- tcap - standard of TCAP - Standard that is used in the TCAP in the result of the current analysis.

Name

parmValue - Parameter Value

Syntax

--parmValue <string>

Description

Use this parameter to specify the value of parameter, according to the name of the parameter.

Class

Optional.

Type / Range

Digit string of length 1 to 15 with the allowed pattern of [0-9a-zA-Z\._:-]*

The parmValue string is based on the parmName parameter:

Name

addr - standard of SCCP addresses

Syntax

--parmName addr --parmValue <string>

Description

Standard that is used in the SCCP addresses in the result of the current analysis.

List

- ANSI - Specify standards type of american national standards institute.
- ITU - Specify standards type of international telecommunication union.

Default

ITU

Name

nas1 - number of authentication sets 1

Syntax

--parmName nas1 --parmValue <string>

Description

Define the number of authentication sets in the case of MAP version 1.

Type / Range

Integer in the range of 1 to 5

Default

4

Name

nas2 - number of authentication sets 2

Syntax

--parmName nas2 --parmValue <string>

Description

Define the number of authentication sets in the case of MAP version 2.

Type / Range

Integer in the range of 1 to 5

Default

4

Name

obid - standard of object identifier

Syntax

--parmName obid --parmValue <string>

Description

Standard that is used in the object identifiers in the result of the current analysis.

List

- ANSI - Specify standards type of american national standards institute.
- ITU - Specify standards type of international telecommunication union.

Default

ITU

Name

olc - abort in overload control

Syntax

--parmName olc --parmValue <string>

Description

Send a notification (TC-U-Abort) to the initiating network element in the case of an overload situation.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sccp - SCCP return option

Syntax

--parmName sccp --parmValue <string>

Description

When SCCP return is activated, messages sent to the network are returned in case of a routing failure.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

tcap - standard of TCAP

Syntax

```
--parmName tcap --parmValue <string>
```

Description

Standard that is used in the TCAP in the result of the current analysis.

List

- ANSI - Specify standards type of american national standards institute.
- ITU - Specify standards type of international telecommunication union.

Default

ITU

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

5.104 cmm matchApn

Match APN List. Syntax, description, parameters, and roles for cmm matchApn command.

Syntax**Description**

This command is used to administer a Match APN List, assigning a match method to an APN NI.

Parameters**Name**

matchApnNi - APN NI

Syntax

```
--matchApnNi <string>
```

Description

APN NI to be matched.

Class

Required key.

Type / Range

Digit string of length 1 to 62 with the allowed pattern of [0-9a-zA-Z\.\-]*

Name

matchMethod - Match method

Syntax

--matchMethod <list-item>

Description

Configure IMS alias selection that applies and covers ims prefix, ims suffix, ims anywhere and ims match in an APN NI.

Class

Optional.

List

- AnyWhere - Match any where - Match IMS anywhere also covers IMS prefix and IMS suffix in an APN NI.
- Exact - Match exact - Match exact IMS covers exact IMS match without any additional characters in an APN NI.
- Prefix - Match prefix - Match prefix IMS covers IMS prefix in an APN NI.
- Suffix - Match suffix - Match suffix IMS covers IMS suffix in an APN NI.

Default

AnyWhere

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.105 cmm mbmsAdmin

mbmsAdmin. Syntax, description, parameters, and roles for cmm mbmsAdmin command.

Syntax

Description

MBMS Administration. Only supported operation is Modify. Output goes to the master.log.

Parameters

Name

mbmsType - MBMS type

Syntax

--mbmsType <list-item>

Description

MbmsType.

Class

Required key.

List

- mce - Specify MBMS type of mce.
- session - Specify MBMS type of session.
- table - Specify MBMS type of table.
- tmgi - Specify MBMS type of tmgi.

Name

mbmsOp - MBMS operation

Syntax

--mbmsOp <list-item>

Description

Mbms operation to perform.

Class

Required key.

List

- delete - Specify MBMS operation of list.
- dump - Specify MBMS operation of list.
- list - Specify MBMS operation of list.

Name

linkIndex - Link index

Syntax

--linkIndex <integer>

Description

Link index from the read command

Class

Required on create.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

flowId - FlowId

Syntax

--flowId <integer>

Description

Flow identifier.

Class

Optional.

Type / Range

Integer in the range of 0 to 65,535

Name

mbmsTable - MBMS table identifier

Syntax

--mbmsTable <list-item>

Description

MBMS table identifier.

Class

Optional.

List

- all - Specify MBMS table type of all.
- m3conn - Specify MBMS table type of m3conn.
- sessdb - Specify MBMS table type of sessdb.

Name

sessionId - MBMS session identifier

Syntax

--sessionId <integer>

Description

MBMS session identifier.

Class

Optional.

Type / Range

Integer in the range of 0 to 65,535

Name

tmgi - TMGI

Syntax

--tmgi <string>

Description

TMGI value as string of digits. TMGI must be 12 digits long with the format SSSSSSPPPPPP. SSSSSS is a six-digit hexadecimal service ID between 000000 and FFFFFFFF. PPPPPP is six-digit PLMN (MCC+MNC). for two-digit MNC, use f for the last digit.

Class

Optional.

Type / Range

Digit string of length 12 to 12 with the allowed pattern of [0-9a-fA-F]{12}

Roles**Modify**

sysadmin, manage

5.106 cmm mmeforcelpChange

MME force IP address Change. Syntax, description, parameters, and roles for cmm mmeforceIpChange command.

Syntax**Description**

Use this cli to manage the MME forced UE IP change operations in TA region (for example, initiate MME forced IP change, terminate forced IP change and interrogate the status of forced UE IP change in specific TA region) . Note: list command shows the status of the last mass detach operation. It is proposed to use list command without filtering.

Parameters**Name**

regionListName - regionListName

Syntax

--regionListName <string>

Description

Name that identifies region

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

forceIpTask - Force IP Change Task

Syntax

--forceIpTask <list-item>

Description

Force IP Change Task. Show is used only when performing a query of the Force IP Change procedure.

Class

Required key.

List

- start - Start MME Force IP Change
- stop - Stop MME Force IP Change
- show - Show MME Force IP Change

Name

detachStatus - Detach Status

Syntax

--detachStatus <string>

Description

Shows current status of the mass detach operation

Class

Read only.

Type / Range

Digit string of length 0 to 64

Roles**Create**

sysadmin, manage

List

sysadmin, manage, monitor

5.107 cmm mmeGrpDcnId

Dedicated Core Network Identifier. Syntax, description, parameters, and roles for cmm mmeGrpDcnId command.

Syntax**Description**

This command is used to provision DCN ID to a MME group. All MMEs in a MME group will support the same set of DNC IDs.

Parameters**Name**

mmeGrpName - MME Group Name

Syntax

--mmeGrpName <string>

Description

MME group to whom the DCN IDs are provisioned.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

dcnId - Dedicated Core Network Identifier

Syntax

--dcnId <integer>

Description

Provision DCN ID supported by the MME group. A maximum of 32 DCN IDs can be provisioned for a MME group.

Class

Required key.

Type / Range

Integer in the range of 1 to 126 or 128 to 65,535

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.108 cmm mmeGrpNode

MME Group Node. Syntax, description, parameters, and roles for cmm mmeGrpNode command.

Syntax

Description

This command is used to administer an MME Group Node and includes parameters for defining the relative MME capacity for use in the eNB MME selection algorithm, automatically adjusting the relative capacity, the MME code, the MME group identifier, and node type.

Parameters

Name

plmnName - Mobile network name

Syntax

--plmnName <string>

Description

Name that identifies mobile network.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

mmeGi - MME group ID

Syntax

--mmeGi <integer>

Description

The MME group identifier.

Class

Required key.

Type / Range

Integer in the range of 0 to 65,535

Name

mmeC - MME CODE

Syntax

--mmeC <integer>

Description

The MME code (that is, the node ID) of the MME within the group.

Class

Required key.

Type / Range

Integer in the range of 0 to 255

Name

autoAdjustCap - Auto adjusts relative capacity

Syntax

--autoAdjustCap <list-item>

Description

Indicates whether or not the MME automatically adjusts the value of the relative capacity according to the number of active MAF blades in the enabled and unlocked states. this field can only be provisioned if the home MME field is checked (true).

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

capacity - Relative capacity

Syntax

--capacity <integer>

Description

Value for the relative MME capacity for use in the eNB MME selection algorithm. this field is only significant if the auto adjusts relative capacity field is unchecked. this field can also only be provisioned if home MME is checked (true).

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

5

Name

localName - Local name

Syntax

--localName <string>

Description

A name that identifies this node. when this is the home MME, this is a required field and it is highly recommended that the local name match the switch name.

Class

Optional.

Type / Range

Digit string of length 0 to 32 with the allowed pattern of [0-9a-zA-Z\.\-:]*

Default

"

Name

nodeType - Node type

Syntax

--nodeType <list-item>

Description

Indicates for the node type.

Class

Not modifiable.

List

- Home Network - Specify node type of home network.
- Other Network - Specify node type of other network.
- Shared Network - Specify node type of shared network.

Default

Other Network

Name

numProtCPPS - Total number of protection CPPS

Syntax

--numProtCPPS <integer>

Description

This parameter is the number of the CPPS in the system that are protection CPPS. This is also considered the k value in the n+k sparing arrangement. This is only used when autoAdjustCap is TRUE.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

s10Ip - S10 IP

Syntax

--s10Ip <string>

Description

S10 IP address.

Class

Not modifiable.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Default

0.0.0.0

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.109 cmm mmeGrpTai

MME Group to TAI List. Syntax, description, parameters, and roles for cmm mmeGrpTai command.

Syntax

Description

This command is used to administer the MME Group to TAI List and includes parameters to define the MME group ID and Tracking Area Code for the associated network.

Parameters

Name

servedPlmnName - Mobile network name

Syntax

--servedPlmnName <string>

Description

Name that identifies mobile network (alternate key).

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

mmeGi - MME group ID

Syntax

--mmeGi <integer>

Description

MME group ID.

Class

Required key.

Type / Range

Integer in the range of 0 to 65,535

Name

tac - Tracking area code

Syntax

--tac <integer>

Description

Tracking area code.

Class

Required key.

Type / Range

Integer in the range of 1 to 65,535

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.110 cmm mmePlmnStatistic

MME PLMN Statistic. Syntax, description, parameters, and roles for cmm mmePlmnStatistic command.

Syntax**Description**

This command would be used to enable/disable as well to configure the functionality for the M169 Measurement Group.

Parameters**Name**

entryId - Entry Id

Syntax

--entryId <integer>

Description

PlmnStatistic identifier.

Class

Required key.

Type / Range

Integer in the range of 1 to 24

Name

plmnType - PLMN Type

Syntax

--plmnType <list-item>

Description

PLMN Type allowed values. Includes ALL PLMNs (Home, Roaming, Specific PLMNs), HOME (home plmns), ROAMING (roaming PLMNs), Specific PLMN IDs

Class

Required on create.

List

- ALL - All PLMNs. Includes HOME and ROAMING PLMNs
- HOME - Home PLMNs. This is the serving PLMNs (i.e. home + share)
- ROAMING - Roaming PLMNs
- PLMNID - Specific PLMN id. Includes only one specific PLMN id (MCC+MNC).
- EMERGENCY - Emergency cases

Name

mcc - MCC

Syntax

--mcc <digits>

Description

Mobile country code.

Class

Optional.

Type / Range

Digit string of length 0 or 3

Default

''

Name

mnc - MNC

Syntax

--mnc <digits>

Description

Mobile network code.

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 3

Default

""

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.111 cmm mmeUePageFanoutControl

MME 4G UE paging fanout overload control parameters. Syntax, description, parameters, and roles for cmm mmeUePageFanoutControl command.

Syntax**Description**

This command is used to set thresholds for 4G UE paging fan-out overload control

Parameters**Name**

msgRateToSafetyNetThreshold - Message rate to total allowed message rate (Safety Net) threshold

Syntax

--msgRateToSafetyNetThreshold <integer>

Description

Threshold used for CMM total message to total allowed message rate (Safety Net), in conjunction with other parameters under this command above which to trigger paging restriction

Class

Optional.

Type / Range

Integer in the range of 0 to 100

Default

60

Name

pageFanoutToTotalMsgThreshold - UE paging fanout to total message rate threshold

Syntax

--pageFanoutToTotalMsgThreshold <integer>

Description

Threshold used for UE paging fan-out (number of ENBs contacted) contribution to total CMM message rate, in conjunction with other parameters under this command above which to trigger paging restriction

Class

Optional.

Type / Range

Integer in the range of 0 to 100

Default

50

Name

pagingFanoutThreshold - Paging fanout threshold

Syntax

--pagingFanoutThreshold <integer>

Description

Threshold used for UE paging fan-out (number of ENBs contacted) in conjunction with other parameters under this command above which to trigger paging restriction

Class

Optional.

Type / Range

Integer in the range of 0 to 100

Default

25

Roles

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

5.112 cmm mscNri

MSC NRI. Syntax, description, parameters, and roles for `cmm mscNri` command.

Syntax

Description

This command is used to associate a Network Resource Identifier with a Mobile Switching Center.

Parameters

Name

mscSrvId - MSC ID

Syntax

--mscSrvId <integer>

Description

Mobile switching center identifier.

Class

Required key.

Type / Range

Integer in the range of 1 to 128

Name

nri - Network resource identifier for the MSC

Syntax

--nri <integer>

Description

Network resource identifier for the MSC.

Restrictions

There can be a maximum of 8 NRI values per MSC.

Class

Required key.

Type / Range

Integer in the range of 0 to 1,023

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.113 cmm mscServer

MSC Server. Syntax, description, parameters, and roles for `cmm mscServer` command.

Syntax

Description

This command is used to administer a Mobile Switching Center (MSC) Server and includes parameters for defining MSC offload, remote endpoint configurations, and to designate an MSC for maintenance mode.

Parameters

Name

mscSrvId - MSC ID

Syntax

--mscSrvId <integer>

Description

Unique MSC identifier.

Class

Required key.

Type / Range

Integer in the range of 1 to 128

Name

currentOffloadPercentage - Percentage of UEs offloaded

Syntax

--currentOffloadPercentage <integer>

Description

Percentage of UEs offloaded.

Class

Read only.

Type / Range

Integer in the range of 0 to 100

Name

lpaSmsOnlyUe - LPA SMS only UE

Syntax

--lpaSmsOnlyUe <list-item>

Description

Designate a MSC for the SMS only LPA UE.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

maintenanceMode - Maintenance mode

Syntax

--maintenanceMode <list-item>

Description

Designate a MSC into maintenance mode.

Restrictions

This field cannot be true during insert

Class

Optional.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Name

mscOffload - MSC offload start or cancel

Syntax

--mscOffload <list-item>

Description

Start or cancel an MSC offload.

Class

Optional.

List

- offloadCancel - Offload cancel - Offload cancel.
- offloadNotIssued - Offload not issued - Offload not issued.
- offloadStart - Offload start - Offload start.

Default

offloadNotIssued

Name

mscSrvE164Address - MSC E.164 address

Syntax

--mscSrvE164Address <digits>

Description

The MSC's E.164 address.

Class

Optional.

Type / Range

Digit string of length 0 to 15

Default

"

Name

mscVlrRegSubs - Current VLR registered subscribers in this MSC

Syntax

--mscVlrRegSubs <integer>

Description

Current VLR registered subscribers. this is only valid during maintenance mode is true.

Class

Read only.

Type / Range

Integer in the range of 0 to 2,147,483,647

Name

mscVlrRegSubsAtOffloadStart - VLR registered subscribers when offload started

Syntax

--mscVlrRegSubsAtOffloadStart <integer>

Description

VLR registered subscribers when offload started.

Class

Read only.

Type / Range

Integer in the range of 0 to 2,147,483,647

Name

name - MSC name

Syntax

--name <string>

Description

Unique MSC name.

Class

Optional.

Type / Range

Digit string of length 1 to 64 with the allowed pattern of [0-9a-zA-Z\.\-:]*

Name

nonBroadCastLac - Non broadcast LAC

Syntax

--nonBroadCastLac <integer>

Description

The command is used to provision a location area code (LAC) for non-broadcasting LAI that is need to be included in SGsAP-LOCATION-UPDATE-REQUEST if MME determines that the serving MSC/VLR is changed. the MCC and MNC of the LAI is set to the MCC and MNC of the MSC/VLR. when the MSC/VLR receives SGsAP-LOCATION-UPDATE-REQUEST message with the specific LAI value, the MSC/VLR may retrieve the MM context from the HLR.

Class

Optional.

Type / Range

Integer in the range of 0 to 65,535

Default

65534

Name

offloadDuration - Duration of offload in minutes

Syntax

--offloadDuration <integer>

Description

Duration of offload in minutes.

Class

Optional.

Type / Range

Integer in the range of 10 to 300 in units of Minute

Default

60

Name

offloadEndTime - Offload end time

Syntax

--offloadEndTime <string>

Description

Offload end time.

Class

Read only.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

offloadPercentage - Percentage of UEs to be offloaded

Syntax

--offloadPercentage <integer>

Description

Percentage of UEs to be offloaded.

Class

Optional.

Type / Range

Integer in the range of 10 to 100

Default

100

Name

offloadRate - Offload rate

Syntax

--offloadRate <integer>

Description

Offload rate.

Class

Optional.

Type / Range

Integer in the range of 10 to 5,000

Default

50

Name

offloadStartTime - Offload start time

Syntax

--offloadStartTime <string>

Description

Offload start time.

Class

Read only.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

offloadStatus - Offload status

Syntax

--offloadStatus <string>

Description

Offload status.

Class

Read only.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

offloadStatusString - Offload status string

Syntax

--offloadStatusString <string>

Description

Offload status string.

Class

Read only.

Type / Range

Digit string of length 0 to 64

Name

sctpProfileName - SCTP profile name

Syntax

--sctpProfileName <string>

Description

Empty string or valid sctpprofile.name.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

''

Name

sgsEndPt1 - SGS endpoint 1

Syntax

--sgsEndPt1 <string>

Description

SGS name specified in remote end point configuration.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

''

Name

sgsEndPt10 - SGS Endpoint 10

Syntax

--sgsEndPt10 <string>

Description

SGS name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\._:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

sgsEndPt11 - SGS Endpoint 11

Syntax

--sgsEndPt11 <string>

Description

SGS name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\._:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

sgsEndPt12 - SGS Endpoint 12

Syntax

--sgsEndPt12 <string>

Description

SGS name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\._:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

sgsEndPt13 - SGS Endpoint 13

Syntax

--sgsEndPt13 <string>

Description

SGS name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\.\-_:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

sgsEndPt14 - SGS Endpoint 14

Syntax

--sgsEndPt14 <string>

Description

SGS name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\.\-_:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

sgsEndPt15 - SGS Endpoint 15

Syntax

--sgsEndPt15 <string>

Description

SGS name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\.\-_:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

sgsEndPt16 - SGS Endpoint 16

Syntax

--sgsEndPt16 <string>

Description

SGS name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\._:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

sgsEndPt17 - SGs Endpoint 17

Syntax

--sgsEndPt17 <string>

Description

SGS name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\._:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

sgsEndPt18 - SGS Endpoint 18

Syntax

--sgsEndPt18 <string>

Description

SGS name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\._:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

sgsEndPt19 - SGS Endpoint 19

Syntax

--sgsEndPt19 <string>

Description

SGS name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\._:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

sgsEndPt2 - SGS endpoint 2

Syntax

--sgsEndPt2 <string>

Description

SGS name specified in remote end point configuration.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

sgsEndPt20 - SGS Endpoint 20

Syntax

--sgsEndPt20 <string>

Description

SGS name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\._:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

sgsEndPt21 - SGS Endpoint 21

Syntax

--sgsEndPt21 <string>

Description

SGS name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\._:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

sgsEndPt22 - SGS Endpoint 22

Syntax

--sgsEndPt22 <string>

Description

SGS name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\._:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

sgsEndPt23 - SGS Endpoint 23

Syntax

--sgsEndPt23 <string>

Description

SGS name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\._:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

sgsEndPt24 - SGS Endpoint 24

Syntax

--sgsEndPt24 <string>

Description

SGS name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\._:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

sgsEndPt25 - SGS Endpoint 25

Syntax

--sgsEndPt25 <string>

Description

SGS name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\._:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

sgsEndPt26 - SGS Endpoint 26

Syntax

--sgsEndPt26 <string>

Description

SGS name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\._:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

sgsEndPt27 - SGS Endpoint 27

Syntax

--sgsEndPt27 <string>

Description

SGS name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\._:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

sgsEndPt28 - SGS Endpoint 28

Syntax

--sgsEndPt28 <string>

Description

SGS name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\._:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

sgsEndPt29 - SGS Endpoint 29

Syntax

--sgsEndPt29 <string>

Description

SGS name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\._:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

sgsEndPt3 - SGS endpoint 3

Syntax

--sgsEndPt3 <string>

Description

SGS name specified in remote end point configuration.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

sgsEndPt30 - SGS Endpoint 30

Syntax

--sgsEndPt30 <string>

Description

SGS name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\._:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

sgsEndPt31 - SGS Endpoint 31

Syntax

--sgsEndPt31 <string>

Description

SGS name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\.\-_:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

sgsEndPt32 - SGS Endpoint 32

Syntax

--sgsEndPt32 <string>

Description

SGS name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\.\-_:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

sgsEndPt4 - SGS endpoint 4

Syntax

--sgsEndPt4 <string>

Description

SGS name specified in remote end point configuration.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

''

Name

sgsEndPt5 - SGS Endpoint 5

Syntax

--sgsEndPt5 <string>

Description

SGS name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\.\-_:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

sgsEndPt6 - SGS Endpoint 6

Syntax

--sgsEndPt6 <string>

Description

SGS name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\.\-_:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

sgsEndPt7 - SGS Endpoint 7

Syntax

--sgsEndPt7 <string>

Description

SGS name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\.\-_:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

sgsEndPt8 - SGS Endpoint 8

Syntax

--sgsEndPt8 <string>

Description

SGS name specified in remote end point configuration

Restrictions

[0-9a-zA-Z\._:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

sgsEndPt9 - SGS Endpoint 9

Syntax

--sgsEndPt9 <string>

Description

SGS name specified in remote end point configuration

Restrictions

[0-9a-zA-Z\._:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

svEndPt1 - Sv endpoint 1

Syntax

--svEndPt1 <string>

Description

Sv name specified in remote end point configuration.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

svEndPt10 - Sv endpoint 10

Syntax

--svEndPt10 <string>

Description

Sv name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\._:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

svEndPt11 - Sv endpoint 11

Syntax

--svEndPt11 <string>

Description

Sv name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\._:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

svEndPt12 - Sv endpoint 12

Syntax

--svEndPt12 <string>

Description

Sv name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\._:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

svEndPt13 - Sv endpoint 13

Syntax

--svEndPt13 <string>

Description

Sv name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\._:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

svEndPt14 - Sv endpoint 14

Syntax

--svEndPt14 <string>

Description

Sv name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\._:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

svEndPt15 - Sv endpoint 15

Syntax

--svEndPt15 <string>

Description

Sv name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\._:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

svEndPt16 - Sv endpoint 16

Syntax

--svEndPt16 <string>

Description

Sv name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\._:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

svEndPt17 - Sv endpoint 17

Syntax

--svEndPt17 <string>

Description

Sv name specified in remote end point configuration.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

svEndPt18 - Sv endpoint 18

Syntax

--svEndPt18 <string>

Description

Sv name specified in remote end point configuration.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

svEndPt19 - Sv endpoint 19

Syntax

--svEndPt19 <string>

Description

Sv name specified in remote end point configuration.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

svEndPt2 - Sv endpoint 2

Syntax

--svEndPt2 <string>

Description

Sv name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\._:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

svEndPt20 - Sv endpoint 20

Syntax

--svEndPt20 <string>

Description

Sv name specified in remote end point configuration.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

svEndPt21 - Sv endpoint 21

Syntax

--svEndPt21 <string>

Description

Sv name specified in remote end point configuration.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

svEndPt22 - Sv endpoint 22

Syntax

--svEndPt22 <string>

Description

Sv name specified in remote end point configuration.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

svEndPt23 - Sv endpoint 23

Syntax

--svEndPt23 <string>

Description

Sv name specified in remote end point configuration.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

svEndPt24 - Sv endpoint 24

Syntax

--svEndPt24 <string>

Description

Sv name specified in remote end point configuration.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

svEndPt25 - Sv endpoint 26

Syntax

--svEndPt25 <string>

Description

Sv name specified in remote end point configuration.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

svEndPt26 - Sv endpoint 26

Syntax

--svEndPt26 <string>

Description

Sv name specified in remote end point configuration.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

svEndPt27 - Sv endpoint 27

Syntax

--svEndPt27 <string>

Description

Sv name specified in remote end point configuration.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

svEndPt28 - Sv endpoint 28

Syntax

--svEndPt28 <string>

Description

Sv name specified in remote end point configuration.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

svEndPt29 - Sv endpoint 29

Syntax

--svEndPt29 <string>

Description

Sv name specified in remote end point configuration.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

svEndPt3 - Sv endpoint 3

Syntax

--svEndPt3 <string>

Description

Sv name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\.\-_:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

svEndPt30 - Sv endpoint 30

Syntax

--svEndPt30 <string>

Description

Sv name specified in remote end point configuration.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

svEndPt31 - Sv endpoint 31

Syntax

--svEndPt31 <string>

Description

Sv name specified in remote end point configuration.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

svEndPt32 - Sv endpoint 32

Syntax

--svEndPt32 <string>

Description

Sv name specified in remote end point configuration.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

svEndPt4 - Sv endpoint 4

Syntax

--svEndPt4 <string>

Description

Sv name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\._:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

svEndPt5 - Sv endpoint 5

Syntax

--svEndPt5 <string>

Description

Sv name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\.\-_:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

svEndPt6 - Sv endpoint 6

Syntax

--svEndPt6 <string>

Description

Sv name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\.\-_:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

svEndPt7 - Sv endpoint 7

Syntax

--svEndPt7 <string>

Description

Sv name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\.\-_:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

svEndPt8 - Sv endpoint 8

Syntax

--svEndPt8 <string>

Description

Sv name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\._:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

svEndPt9 - Sv endpoint 9

Syntax

--svEndPt9 <string>

Description

Sv name specified in remote end point configuration.

Restrictions

[0-9a-zA-Z\._:@]

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

weight - weight

Syntax

--weight <integer>

Description

Relative MSC weight compared to other MSCs defined in the mscServer translation. This value applies only if cmm gparmWeightedSgslmsiHash is true.

Class

Optional.

Type / Range

Integer in the range of 1 to 100

Default

50

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.114 cmm mscToIws

MSC to IWS. Syntax, description, parameters, and roles for cmm mscToIws command.

Syntax**Description**

This command is used to administer an IWS Mobile Switching Center (MSC) and includes parameters for defining heartbeat, IP address, and ISDN number.

Parameters**Name**

name - MSC to IWS name

Syntax

--name <string>

Description

Unique MSC to IWS name.

Class

Required key.

Type / Range

Digit string of length 1 to 64 with the allowed pattern of [0-9a-zA-Z\.\-:]*

Name

msclwsSrvID - MSC server ID

Syntax

--mscIwsSrvID <integer>

Description

MSC to IWS server ID.

Class

Required on create, not modifiable.

Type / Range

Integer in the range of 0 to 16,777,215

Name

htBt - Heart beat

Syntax

--htBt <list-item>

Description

Indicates whether or not to do heart beat.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

htBtTmOut - Heart beat time out

Syntax

--htBtTmOut <integer>

Description

Heart beat time out value.

Class

Optional.

Type / Range

Integer in the range of 1 to 300

Default

15

Name

ip1XCslws - 1x CS IWS IP

Syntax

--ip1XCslws <string>

Description

1x CS IWS IP address.

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Default

0.0.0.0

Name

isdnNumber - ISDN number

Syntax

--isdnNumber <digits>

Description

ISDN number.

Class

Optional.

Type / Range

Digit string of length 0 to 15

Default

''

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.115 cmm msgRetries

Number of Message Retransmissions. Syntax, description, parameters, and roles for cmm msgRetries command.

Syntax

Description

This command is used to assign the number of retransmissions used for NAS procedure messages.

Parameters

Name

msgName - Message name

Syntax

--msgName <list-item>

Description

NAS procedure name.

Class

Required key.

List

- ActDedBrr - Number of activate dedicated bearer requests transmitted without receiving a corresponding response.
- ActDefBrr - Number of activate default bearer requests transmitted without receiving a corresponding response.
- Attach - Number of attach procedures transmitted without receiving a corresponding response.
- AuthRquest - Number of authentication requests transmitted without receiving a corresponding response.
- DactBrr - Number of deactivate bearer requests transmitted without receiving a corresponding response.
- ESMInfo - Number of ESM information requests transmitted without receiving a corresponding response.
- GUTIRelocation - Number of globally unique temporary identity relocation procedures transmitted without receiving a corresponding response.
- IdentityCom - Number of identity procedure requests transmitted without receiving a corresponding response.

- ModifyBrr - Number of modify bearer requests transmitted without receiving a corresponding response.
- NetDetach - Number of network initiated detach procedures transmitted without receiving a corresponding response.
- Ns10 - Number of retries to deliver messages across the SGs interface to the designated end point. the total number of attempts to deliver a message is the number of retries plus 1.
- Ns102 - Ns102.
- Ns12 - Number of retries to deliver messages across the SGs interface to the designated end point. the total number of attempts to deliver a message is the number of retries plus 1.
- Ns8 - Number of retries to deliver messages across the SGs interface to the designated end point. the total number of attempts to deliver a message is the number of retries plus 1.
- Ns9 - Number of retries to deliver messages across the SGs interface to the designated end point. the total number of attempts to deliver a message is the number of retries plus 1.
- SGSN Attach Accept - Reserved for future use.
- SGSN Auth Ciph Request - Reserved for future use.
- SGSN Deactivate PDP Cxt - Reserved for future use.
- SGSN Detach Request - Reserved for future use.
- SGSN Identity Request - Reserved for future use.
- SGSN Rau Accept - Reserved for future use.
- SecurityMode - Number of security mode requests transmitted without receiving a corresponding response.
- TAU - Number of tracking area update procedure transmitted without receiving a corresponding response.

Name

msgNumRetries - Number of retransmissions

Syntax

--msgNumRetries <integer>

Description

Number of retries.

Class

Optional.

Type / Range

Integer in the range of 1 to 4

Roles

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

5.116 cmm muaAssociation

M3UA Association. Syntax, description, parameters, and roles for `cmm muaAssociation` command.

Syntax

Description

Use this command to configure, list and show M3UA association that can be added to an association set.

Parameters

Name

`muaAssociationId` - M3UA association identifier

Syntax

```
--muaAssociationId <integer>
```

Description

M3UA association identifier.

Class

Required key.

Type / Range

Integer in the range of 0 to 27168

Name

`associationInUnitIndex` - Association in unit index

Syntax

```
--associationInUnitIndex <integer>
```

Description

Use this parameter to specify the index of the VM in which the association resides: decimal number 0 ... 1

Class

Required on create, not modifiable.

Type / Range

Integer in the range of 0 to 1

Name

associationInUnitType - Association in unit type

Syntax

--associationInUnitType <list-item>

Description

Use this parameter to specify the type of the VM in which the association resides: IPDS .. IP director unit.

Class

Not modifiable.

List

- IPDS - Internet protocol director service.

Default

IPDS

Name

destinationPort - Destination port

Syntax

--destinationPort <integer>

Description

identify destination SCTP port number decimal number 1 ... 65535 default for client is IANA specified port (2905) default for server is IANA specified port (2905)

Class

Optional.

Type / Range

Integer in the range of 1 to 65,535

Default

2905

Name

m3uaProfileName - M3UA profile name

Syntax

--m3uaProfileName <string>

Description

Specify M3UA profile name.

Restrictions

There must exist a record in m3uaProfile with this name

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

default

Name

primaryDestinationIpv4v6Address - primary Destination Ipv4/Ipv6 Address

Syntax

```
--primaryDestinationIpv4v6Address <string>
```

Description

identify primary destination IP address: IPV4 address: dotted decimal integer (##.##.##) IPV6 address: colon hexadecimal integer (##::##)

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Default

0.0.0.0

Name

primaryDestinationIpv4v6Netmask - primary Destination Ipv4/Ipv6 Netmask

Syntax

```
--primaryDestinationIpv4v6Netmask <integer>
```

Description

identify netmask length: decimal number 1 ... 128 default is 8

Class

Optional.

Type / Range

Integer in the range of 1 to 128

Default

8

Name

primarySourceIpv4v6Address - primary Source Ipv4/Ipv6 Address

Syntax

--primarySourceIpv4v6Address <string>

Description

identify primary source IP address: IPv4 address: dotted decimal integer (#.#.#.#) IPv6
address: colon hexadecimal integer (#: #:: #: #)

Restrictions

There must exist a record in servicelp with this source IP and pool_type of
associationInUnitType

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Default

0.0.0.0

Name

secondaryDestinationIpv4v6Address - secondary Destination Ipv4/Ipv6 Address

Syntax

--secondaryDestinationIpv4v6Address <string>

Description

identify secondary destination IP address: IPv4 address: dotted decimal integer (#.#.#.#)
IPv6 address: colon hexadecimal integer (#: #:: #: #)

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Default

0.0.0.0

Name

secondaryDestinationIpv4v6Netmask - secondary Destination Ipv4/Ipv6 Netmask

Syntax

--secondaryDestinationIpv4v6Netmask <integer>

Description

identify netmask length: decimal number 1 ... 128 default is 8

Class

Optional.

Type / Range

Integer in the range of 1 to 128

Default

8

Name

secondarySourceIpv4v6Address - secondary Source Ipv4/Ipv6 Address

Syntax

--secondarySourceIpv4v6Address <string>

Description

identify secondary source IP address: IPV4 address: dotted decimal integer (###.###) IPV6 address: colon hexadecimal integer (###:###:###)

Restrictions

There must exist a record in servicelp with this source IP and pool_type of associationInUnitType

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Default

0.0.0.0

Name

sourcePort - Source port

Syntax

--sourcePort <integer>

Description

identify source SCTP port number decimal number 1 ... 65535 default for client is IANA specified port (2905) default for server is IANA specified port (2905)

Class

Optional.

Type / Range

Integer in the range of 1 to 65,535

Default

2905

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.117 cmm muaAssociationList

M3UA Association List. Syntax, description, parameters, and roles for cmm muaAssociationList command.

Syntax

Description

Configure, list and show M3UA association list.

Parameters

Name

muaAssociationSetId - M3UA association set identifier

Syntax

--muaAssociationSetId <integer>

Description

M3UA association set identifier the number is the record number of a certain association set in the association set configuration file (S3FILE).

Restrictions

record with this id must exist in muaAssociationSet. Max number of list entries per muaAssociationSet is 32.

Class

Required key.

Type / Range

Integer in the range of 0 to 12000

Name

muaAssociationId - M3UA association identifier

Syntax

--muaAssociationId <integer>

Description

M3UA association identifier specifies a single association within an association set.

Restrictions

record with this id must exist in muaAssociation

Class

Required key.

Type / Range

Integer in the range of 0 to 27168

Name

associationState - Association state

Syntax

--associationState <list-item>

Description

Specifies the state of the association. the following abbreviations are used to describe the state of the association: DOWN the SCTP association is set down by user INACT ... the remote peer at the ASP is available but application traffic is stopped (only management messages are allowed) ACT the remote peer at the ASP is available and application traffic is active.

Class

Optional.

List

- ACT - Active - Active.
- DOWN - Down.
- INACT - Inactive - Inactive.

Default

ACT

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.118 cmm muaAssociationSet

M3UA Association Set. Syntax, description, parameters, and roles for cmm muaAssociationSet command.

Syntax

Description

Configure, list and show M3UA association set.

Parameters

Name

muaAssociationSetId - M3UA association set identifier

Syntax

--muaAssociationSetId <integer>

Description

M3UA association set identifier.

Class

Required key.

Type / Range

Integer in the range of 0 to 12000

Name

associationSetName - Identify association set name

Syntax

--associationSetName <string>

Description

Use this parameter to specify the name of the association set that you want to create: 1 ...
16 characters.

Restrictions

associationSetName must be unique within muaAssociationSet. When modifying associationSetName it must be the only parameter to be modified during that operation.

Class

Required on create.

Type / Range

Digit string of length 1 to 16 with the allowed pattern of [0-9a-zA-Z\.\-_:]*

Name

sctpRole - Identify role

Syntax

--sctpRole <list-item>

Description

Use this parameter to specify the role of the association set: c ... client s ... server.

Class

Required on create.

List

- C - Client - Specify SCTP role of client.
- S - Server - Specify SCTP role of server.

Name

activationDelay - Activation delay

Syntax

--activationDelay <integer>

Description

Activation delay is used in overload situations to delay the association in getting to the traffic handling state. It starts whenever there is overload in activation. 30 successive ASPUP messages can be received in less than one second. Configuring value 0 means there is no delay, and the activation of the association does not need to compete with activation of other associations that have delay more than 0. The load of a VM resulting from activations can be smoothed by having different values in different association sets.

Class

Optional.

Type / Range

Integer in the range of 0 to 30

Default

0

Name

aspMsgsInIpsp - ASP messages in IPSP

Syntax

--aspMsgsInIpsp <list-item>

Description

The parameter defines a sending of ASP messages in IPSP connection for an association set.

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

aspMsgsInUse - ASP messages in use

Syntax

--aspMsgsInUse <list-item>

Description

The parameter defines application server process (ASP) messages in use for an association set.

Restrictions

If trafficMode is 1 (OVER-RIDE) then aspMsgsInUse must be Yes.

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

broadcastSsnmMsg - Broadcast SSNM messages

Syntax

--broadcastSsnmMsg <list-item>

Description

The parameter defines SS7 signalling network management (SSNM) messages broadcast to all application server processes (ASP) for an association set.

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

firstDataStreamNum - First data stream number

Syntax

--firstDataStreamNum <integer>

Description

This parameter is defined as the number of the first data stream.

Restrictions

The value of firstDataStreamNum must be 1 if specificationVersion is 16.

Class

Optional.

Type / Range

Integer in the range of 0 to 1

Default

1

Name

hbInterval - M3UA heartbeat interval in seconds

Syntax

--hbInterval <integer>

Description

This parameter defines the time that M3UA waits until it sends the next M3UA BEAT after receiving BEAT ACK to the previous BEAT. value 0 means that the M3UA does not send M3UA heartbeat.

Class

Optional.

Type / Range

Integer in the range of 0 to 30

Default

15

Name

networkAppearance - Network appearance

Syntax

--networkAppearance <string>

Description

This parameter identifies the network appearance for a given association set [value = 0...4294967294 or '-' if networkappearance is not supported].

Class

Optional.

Type / Range

Digit string of length 1 to 10 with the allowed pattern of ([-]|[0-9]|[1-8][0-9]|9[0-9]|[1-8][0-9]{2}|9[0-8][0-9]|99[0-9]|[1-8][0-9]{3}|9[0-8][0-9]{2}|99[0-8][0-9]|999[0-9]|[1-8][0-9]{4}|9[0-8][0-9]{3}|99[0-8][0-9]{2}|999[0-8][0-9]|9999[0-9]|[1-8][0-9]{5}|9[0-8][0-9]{4}|99[0-8][0-9]{3}|999[0-8][0-9]{2}|9999[0-8][0-9]|99999[0-9]|[1-8][0-9]{6}|9[0-8][0-9]{5}|99[0-8][0-9]{4}|999[0-8][0-9]{3}|9999[0-8][0-9]{2}|99999[0-8][0-9]|999999[0-9]|[1-8][0-9]{7}|9[0-8][0-9]{6}|99[0-8][0-9]{5}|999[0-8][0-9]{4}|9999[0-8][0-9]{3}|99999[0-8][0-9]{2}|999999[0-8][0-9]|9999999[0-9]|[1-8][0-9]{8}|9[0-8][0-9]{7}|99[0-8][0-9]{6}|999[0-8][0-9]{5}|9999[0-8][0-9]{4}|99999[0-8][0-9]{3}|999999[0-8][0-9]{2}|9999999[0-8][0-9]|99999999[0-9]|[1-3][0-9]{9}|4[01][0-9]{8}|42[0-8][0-9]{7}|429[0-3][0-9]{6}|4294[0-8][0-9]{5}|42949[0-5][0-9]{4}|429496[0-6][0-9]{3}|4294967[01][0-9]{2}|42949672[0-8][0-9]|429496729[0-4])

Default

0

Name

registrationReqInUse - Registration request in use

Syntax

--registrationReqInUse <list-item>

Description

This parameter controls if dynamic routing key registration is in use.

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

routingContext - Routing context

Syntax

--routingContext <string>

Description

This parameter is the defined value of the routing context [value = 0...4294967294 or '-' if registrationreqinuse is yes].

Restrictions

The value of routingContext must be '-' if registrationReqInUse is Yes. The value of routingContext must be 0...4294967294 if registrationReqInUse is No.

Class

Optional.

Type / Range

Digit string of length 1 to 10 with the allowed pattern of ([-]|[0-9]|[1-8][0-9]|9[0-9]|[1-8][0-9]{2}|9[0-8][0-9]|99[0-9]|[1-8][0-9]{3}|9[0-8][0-9]{2}|99[0-8][0-9]|999[0-9]|[1-8][0-9]{4}|9[0-8][0-9]{3}|99[0-8][0-9]{2}|999[0-8][0-9]|9999[0-9]|[1-8][0-9]{5}|9[0-8][0-9]{4}|99[0-8][0-9]{3}|999[0-8][0-9]{2}|9999[0-8][0-9]|99999[0-9]|[1-8][0-9]{6}|9[0-8][0-9]{5}|99[0-8][0-9]{4}|999[0-8][0-9]{3}|9999[0-8][0-9]{2}|99999[0-8][0-9]|999999[0-9]|[1-8][0-9]{7}|9[0-8][0-9]{6}|99[0-8][0-9]{5}|999[0-8][0-9]{4}|9999[0-8][0-9]{3}|99999[0-8][0-9]{2}|999999[0-8][0-9]|9999999[0-9]|[1-3][0-9]{9}|4[01][0-9]{8}|42[0-8][0-9]{7}|429[0-3][0-9]{6}|4294[0-8][0-9]{5}|42949[0-5][0-9]{4}|429496[0-6][0-9]{3}|4294967[01][0-9]{2}|42949672[0-8][0-9]|429496729[0-4])

Default

-

Name

sctpUser - Identify user

Syntax

--sctpUser <list-item>

Description

Use this parameter to specify the SCTP user of the association set. the default value is M3UA (MTP3 USER ADAPTION LAYER) M3UA ... MTP3 user adaptation layer.

Class

Not modifiable.

List

- IUA - Specify SCTP user of IUA.
- M3UA - Specify SCTP user of M3UA.

- SGSAP - Specify SCTP user of SGSAP.

Default

M3UA

Name

specificationVersion - M3UA specification version

Syntax

--specificationVersion <integer>

Description

This parameter specifies the version of the M3UA specification. The possible values are: 5 (M3UA SPECIFICATION VERSION 05), 7 (M3UA SPECIFICATION VERSION 07), 9 (M3UA SPECIFICATION VERSION 09), 16 (M3UA SPECIFICATION VERSION 1.0 (RFC))

Restrictions

If other peers support a different version of the M3UA, then the version parameter should be changed to the same version as that of the remote peer.

Class

Optional.

Type / Range

Integer in the range of 5 or 7 or 9 or 16

Default

16

Name

tAck - Timed acknowledgement for ASPSM and ASPTM in seconds

Syntax

--tAck <integer>

Description

This parameter defines the time that M3UA waits for a response to the sent ASPSM and ASPTM message, (for instance, ASPUP ACK for the sent ASPUP, or ASPAC ACK for the sent ASPAC) before resending the message.

Class

Optional.

Type / Range

Integer in the range of 1 to 20

Default

2

Name

trafficMode - M3UA traffic mode

Syntax

--trafficMode <integer>

Description

This parameter controls how traffic is routed between SCTP associations into the association set. Values (decimal): 1 (OVER-RIDE), 2 (LOAD-SHARE)

Class

Optional.

Type / Range

Integer in the range of 1 to 2

Default

2

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.119 cmm muaLink

M3UA Signalling Link. Syntax, description, parameters, and roles for cmm muaLink command.

Syntax**Description**

Configure, list and show M3UA signalling link.

Parameters**Name**

muaLinkId - MUA signalling link id

Syntax

--muaLinkId <integer>

Description

M3UA signalling link identifier.

Class

Required key.

Type / Range

Integer in the range of 0 to 12000

Name

associationSetName - Association set name

Syntax

--associationSetName <string>

Description

Name of the M3UA association set. reference to the muaassociationset table. you can check the created association sets with the cmm muaassociationset list command.

Class

Required on create, not modifiable.

Type / Range

Digit string of length 1 to 16 with the allowed pattern of [0-9a-zA-Z\.\-_:]*

Name

linkSignallingNetwork - Link signalling network

Syntax

--linkSignallingNetwork <list-item>

Description

Use this parameter to define the signalling network.

Class

Required on create, not modifiable.

List

- IN0 - International network number 0.
- IN1 - International network number 1.
- NA0 - National network number 0.
- NA1 - National network number 1.

Name

linkSignallingPointCode - Link signalling point code

Syntax

--linkSignallingPointCode <integer>

Description

Use this parameter to define the signalling point code.

Class

Required on create, not modifiable.

Type / Range

Integer in the range of 0 to 16,777,215

Name

signallingLinkNumber - Signalling link number

Syntax

```
--signallingLinkNumber <integer>
```

Description

Use this parameter to define the signalling link number.

Class

Required on create, not modifiable.

Type / Range

Integer in the range of 0 to 12000

Name

signallingLinkSetName - Signalling link set name

Syntax

```
--signallingLinkSetName <string>
```

Description

Use this parameter to define the signalling link set name.

Class

Required on create.

Type / Range

Digit string of length 1 to 8 with the allowed pattern of [0-9a-zA-Z\._:-]*

Name

signallingLinkState - Signalling link state

Syntax

```
--signallingLinkState <list-item>
```

Description

Use this parameter to define the signalling link state.

Class

Optional.

List

- ACT - Active - Specify signaling link state of active.
- INA - Inactive - Specify signaling link state of inactive.

Default

INA

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.120 cmm muaOwnSignallingPoint

M3UA Own Signalling Point. Syntax, description, parameters, and roles for cmm muaOwnSignallingPoint command.

Syntax**Description**

Configure, list and show M3UA own signalling point.

Parameters**Name**

muaOwnSignallingPointId - MUA own signalling point id

Syntax

--muaOwnSignallingPointId <integer>

Description

Use this parameter to define M3UA own signalling point identifier.

Class

Required key.

Type / Range

Integer in the range of 0 to 12000

Name

signallingNetwork - Signalling network

Syntax

--signallingNetwork <list-item>

Description

Use this parameter to define signalling network of the own signalling point.

Class

Required on create, not modifiable.

List

- IN0 - International network number 0.
- IN1 - International network number 1.
- NA0 - National network number 0.
- NA1 - National network number 1.

Name

signallingPointCode - Signalling point code

Syntax

--signallingPointCode <integer>

Description

Use this parameter to define signalling point code of the own signalling point.

Class

Required on create, not modifiable.

Type / Range

Integer in the range of 0 to 16,777,215

Name

signallingPointName - Signalling point name

Syntax

--signallingPointName <string>

Description

Use this parameter to define the name of the own signalling point.

Class

Required on create, not modifiable.

Type / Range

Digit string of length 1 to 5 with the allowed pattern of [0-9a-zA-Z\._ :]*

Name

adpc - Additional signalling destination point code

Syntax

--adpc <integer>

Description

Use this parameter to define the additional signalling destination point code. only used when signalling point type is "PRV".

Class

Optional.

Type / Range

Integer in the range of 0 to 16,777,215

Default

0

Name

dpc - Signalling destination point code

Syntax

--dpc <integer>

Description

Use this parameter to define the signalling destination point code. only used when signalling point type is "PRV".

Class

Optional.

Type / Range

Integer in the range of 0 to 16,777,215

Default

0

Name

signallingPointType - Signalling point type

Syntax

--signallingPointType <list-item>

Description

Use this parameter to define the type of the own signalling point - SEP : the own signalling point is defined to be signalling end point, STP : the own signalling point is defined to be signalling transfer point, REC : creation of an additional signalling point that is recognized as the own point in incoming signalling messages, USP : creation of an additional signalling

point that can be set as an originating point code in user part messages, IWF : creation of an additional signalling point that supports shared use, PRV : private additional signalling point code

Class

Optional.

List

- DEL - The own signalling point is deleted.
- IWF - Creation of an additional signaling point that supports shared use.
- PRV - Private additional signaling point code.
- REC - Creation of an additional signaling point that is recognized as the own point in incoming signaling messages.
- SEP - The own signaling point is changed to or created as a signaling end point.
- STP - The own signaling point is changed to or created as a signaling transfer point.
- USP - Creation of an additional signaling point that can be set as an originating point code in user part messages.

Default

STP

Name

spcSubfields - SPC subfields

Syntax

--spcSubfields <integer>

Description

Use this parameter to define the number of subfields under the signalling point code. 1 means no subfields.

Class

Optional.

Type / Range

Integer in the range of 1 to 4

Default

1

Name

spcSubfieldsLength1 - SPC subfields length 1

Syntax

--spcSubfieldsLength1 <integer>

Description

Use this parameter to define the length of the 1st subfield in bits.

Class

Optional.

Type / Range

Integer in the range of 1 to 24

Default

14

Name

spcSubfieldsLength2 - SPC subfields length 2

Syntax

--spcSubfieldsLength2 <integer>

Description

Use this parameter to define the length of the 2nd subfield in bits.

Class

Optional.

Type / Range

Integer in the range of 0 to 24

Default

0

Name

spcSubfieldsLength3 - SPC subfields length 3

Syntax

--spcSubfieldsLength3 <integer>

Description

Use this parameter to define the length of the 3rd subfield in bits.

Class

Optional.

Type / Range

Integer in the range of 0 to 24

Default

0

Name

spcSubfieldsLength4 - SPC subfields length 4

Syntax

--spcSubfieldsLength4 <integer>

Description

Use this parameter to define the length of the 4th subfield in bits.

Class

Optional.

Type / Range

Integer in the range of 0 to 24

Default

0

Name

ss7Standard - SS7 standard

Syntax

--ss7Standard <list-item>

Description

Use this parameter to define what kind of Signalling System No.7 (SS7) is used in the own signalling point, for example the width of the signalling point code in bits and the standard that the signalling network operates according to. The possible values of this parameter are: ITU-T : for 14 bit SPC code of ITU-T SS7 standard, JAP16 : for 16 bit SPC code of Japan SS7 standard, CHI24 : for 24 bit SPC code of China SS7 standard, ANSI : for 24 bit SPC code of ANSI SS7 standard.

Class

Optional.

List

- ANSI - Specify SS7 standard of american national standards institute.
- CHI24 - Specify SS7 standard of china 24 bit point code.
- ITUT - Specify SS7 standard of international telecommunication union - telecommunication.
- JAP16 - Specify SS7 standard of japan 16 bit point code.

Default

ITUT

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.121 cmm muaRoute

M3UA route. Syntax, description, parameters, and roles for `cmm muaRoute` command.

Syntax

Description

Configure, list and show M3UA route.

Parameters

Name

`muaRouteId` - M3UA route identifier

Syntax

`--muaRouteId <integer>`

Description

M3UA route identifier.

Class

Required key.

Type / Range

Integer in the range of 0 to 12000

Name

`destinationSignallingNetwork` - Destination signalling network

Syntax

`--destinationSignallingNetwork <list-item>`

Description

Identify signalling network IN0 ... international network number 0 IN1 ... international network number 1 NAO ... national network number 0 NA1 ... national network number 1.

Class

Required on create, not modifiable.

List

- IN0 - International network number 0.
- IN1 - International network number 1.
- NA0 - National network number 0.
- NA1 - National network number 1.

Name

destinationSignallingPointCode - Destination signalling point code

Syntax

--destinationSignallingPointCode <integer>

Description

Identify the destination signalling point code: decimal number 0 ... 16777215 default is given signalling point code.

Class

Required on create, not modifiable.

Type / Range

Integer in the range of 0 to 16,777,215

Name

destinationSignallingPointName - Destination signalling point name

Syntax

--destinationSignallingPointName <string>

Description

Signalling point name: identifier 1 ... 8 characters. If signalling point code and network are already defined in muaLink, destinationSignallingPointName must be the same with signallingLinkSetName of the related muaLink.

Class

Required on create, not modifiable.

Type / Range

Digit string of length 1 to 8 with the allowed pattern of [0-9a-zA-Z\.\-_:]*

Name

loadSharingStatus - Load sharing status

Syntax

--loadSharingStatus <list-item>

Description

Identify load sharing status: A: load is shared between all available routes having the highest equal priority D: traffic is routed to only one available route having the highest priority (note that if there is more than one route with same priority it is unpredictable which one will carry the load) U: load is shared evenly between all available links of the maximum two available

routes (combined link set) having the highest equal priority default is deny load sharing between signalling routes.

Class

Optional.

List

- A - Load is shared between all available routes having the highest equal priority.
- D - Traffic is routed to only one available route having the highest priority (note that if there is more than one route with same priority it is unpredictable which one will carry the load).
- U - Load is shared evenly between all available links of the maximum two available routes (combined link set) having the highest equal priority .

Default

D

Name

restrictionStatus - Restriction status

Syntax

--restrictionStatus <list-item>

Description

Identify restriction status: n ... no restrictions p ... use as STP toward cluster denied s ... use as STP denied r ... a interface a subgroup of an MTP function, where, for example, the STP function has been prevented over interface a to both directions. t ... test routing via level 3 l ... test routing via transmission loop i ... inverted view of AOPC this restricted route has to connect to the own additional signalling point. m ... management cluster default is no restrictions.

Class

Optional.

List

- I - Inverted view of AOPC.
- L - Test routing via transmission loop.
- M - Management cluster.
- N - No restrictions.
- P - Use as STP toward cluster denied.
- R - A interface.
- S - Use as STP denied.
- T - Test routing via level 3.

Default

N

Name

routeParamSetName - M3UA route parameter set name

Syntax`--routeParamSetName <string>`**Description**

Identify signalling route set parameter set name: itut ... ITU-T for signalling parameters defined by ITU (International telecommunications union) in Q.703 and Q.704 ainterface ... a INTERFACE for a interface e.g. between MSC and BSC btnr146 ... BTNR 146 for signalling parameters defined by BTNR (British telecommunications network requirement) ansit111 ... ANSI T111 for signalling parameters defined by ANSI (American national standards institute) in T1.111 and T1.114 japanrtc ... JAPAN TTC for signalling parameters defined by TTC (Telecommunication technology committee) in JT.Q703 and JT.Q704 japanrtt ... JAPAN NTT for signalling parameters defined by NTT (Nippon telegraph and telecommunication corporation) in JT.Q703 and JT.Q704 ietfm3ua ... IETF M3UA for routes using IP type signalling links defined by IETF (Internet engineering task force) or ... use own created set name (in routeparamset).

Restrictions

a record must exist in routeParamSet with name of routeParamSetName

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

itut

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.122 cmm muaService

M3UA Service. Syntax, description, parameters, and roles for cmm muaService command.

Syntax

Description

Configure, list and show M3UA service.

Parameters

Name

muaServiceId - MUA service id

Syntax

```
--muaServiceId <integer>
```

Description

M3UA service identifier.

Class

Required key.

Type / Range

Integer in the range of 0 to 255

Name

serviceForOwnUserPart - Service for own user part

Syntax

```
--serviceForOwnUserPart <list-item>
```

Description

Use this parameter to indicate whether service exists for the user parts of own signalling point.

Class

Required on create.

List

- false - Boolean value of false.
- true - Boolean value of true.

Name

serviceForStpMessage - Service for STP message

Syntax

--serviceForStpMessage <list-item>

Description

Use this parameter to indicate whether service exists for STP messages.

Class

Required on create.

List

- false - Boolean value of false.
- true - Boolean value of true.

Name

serviceIndicatorIndex - Service indicator index

Syntax

--serviceIndicatorIndex <integer>

Description

Use this parameter to define an index for the service you want to create: 0 SNM ... signalling network management messages, 1 SNT ... signalling network testing and maintenance messages, 2 SNTA ... ANSI - signalling network testing and maintenance messages, 3 SCCP ... signalling connection control part.

Class

Required on create, not modifiable.

Type / Range

Integer in the range of 0 to 3

Name

serviceIndicatorName - Service indicator name

Syntax

--serviceIndicatorName <list-item>

Description

Use this parameter to define a name for the service you want to create: recommended names for some indexes 0 SNM ... signalling network management messages, 1 SNT ... signalling network testing and maintenance messages, 2 SNTA ... ANSI - signalling network testing and maintenance messages, 3 SCCP ... signalling connection control part.

Class

Required on create.

List

- SNM - Signaling network management messages - recommended index 0.
- SNT - Signaling network testing and maintenance messages. recommended index 1.

- SNTT1 - Signaling network testing and maintenance messages. recommended index 2.
- SCCP - Signaling connection control part. recommended index 3.

Name

serviceSignallingNetwork - Service signalling network

Syntax

--serviceSignallingNetwork <list-item>

Description

Use this parameter to define the signalling network where the service you want to create, will exist.

Class

Required on create, not modifiable.

List

- IN0 - International network number 0.
- IN1 - International network number 1.
- NA0 - National network number 0.
- NA1 - National network number 1.

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.123 cmm muaSignallingRoute

M3UA signalling route. Syntax, description, parameters, and roles for `cmm muaSignallingRoute` command.

Syntax

Description

Configure, list and show M3UA signalling route.

Parameters

Name

muaSignallingRouteId - Id

Syntax

--muaSignallingRouteId <integer>

Description

M3UA signalling route identifier.

Class

Required key.

Type / Range

Integer in the range of 0 to 12000

Name

signallingRoutePriority - Signalling route priority

Syntax

--signallingRoutePriority <integer>

Description

Identify signalling route priority: decimal number 0 ... 7 priority 7 is highest (for primary route) and 0 lowest. load is shared between routes with highest equal priority.

Class

Required on create.

Type / Range

Integer in the range of 0 to 7

Name

signallingTransferNetwork - Signalling transfer network

Syntax

--signallingTransferNetwork <list-item>

Description

Identify signalling network IN0 ... international network number 0 IN1 ... international network number 1 NAO ... national network number 0 NA1 ... national network number 1 if you give some other value than given signalling network, it means that you are creating signalling route set to the adjacent SP, utilizing direct route belonging to some other signalling route set leading to the same physical SP. if signalling route set is leading to the

adjacent SP and there is not any other signalling route sets leading to the same SP, signalling transfer point network must be same as given signalling point network.

Class

Required on create, not modifiable.

List

- IN0 - International network number 0.
- IN1 - International network number 1.
- NA0 - National network number 0.
- NA1 - National network number 1.

Name

signallingTransferPointCode - Signalling transfer point code

Syntax

--signallingTransferPointCode <integer>

Description

Identify signalling point code: decimal number 0 ... 16777215 default is given signalling point code.

Class

Required on create, not modifiable.

Type / Range

Integer in the range of 0 to 16,777,215

Name

signallingTransferPointName - Signalling transfer point name

Syntax

--signallingTransferPointName <string>

Description

identify signalling point name: identifier 1 ... 8 characters Default is given signalling point name.

Class

Read only.

Type / Range

Digit string of length 1 to 8 with the allowed pattern of [0-9a-zA-Z\.\-_:]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.124 cmm muaSignallingRouteList

M3UA Signalling Route List. Syntax, description, parameters, and roles for cmm muaSignallingRouteList command.

Syntax

Description

Use this command to configure, list and show M3UA signalling route set. If the signalling point code and network of muaSignallingRouteId are the same as those of the muaRouteId, the created route is a direct route. Else, the created route is an indirect route.

Parameters

Name

muaRouteId - M3UA route identifier

Syntax

- -muaRouteId <integer>

Description

M3UA route identifier muaRoute identifier where the created signalling route set is leading.

Restrictions

record with this id must exist in muaRoute. Max number of list entries per muaRoute is 8.

Class

Required key.

Type / Range

Integer in the range of 0 to 12000

Name

muaSignallingRouteId - M3UA signalling route identifier

Syntax

--muaSignallingRouteId <integer>

Description

M3UA signalling route identifier.

Restrictions

record with this id must exist in muaSignallingRoute

Class

Required key.

Type / Range

Integer in the range of 0 to 12000

Name

signallingRouteActivationAllowed - Signalling route activation allowed

Syntax

--signallingRouteActivationAllowed <list-item>

Description

Allow the activation of a signalling route, i.e. used to remove the state ACTIVATION DENIED from the signalling route. this command is needed, for example, in situation when the user wants to be able to activate a new signalling route that has been automatically set on state UA-AD (Unavailable-Activation denied).

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

signallingRouteState - Signalling route state

Syntax

--signallingRouteState <list-item>

Description

Identify state change: ACT ... activate INA ... inactivate with the state change INA, a signalling route which is in the state 'executing', 'spare' or 'unavailable' can be inactivated. with the state change ACT, an inactivated signalling route can be activated. on the basis of the signalling route priority, the signalling route is set in the 'executing', 'spare', or 'unavailable' state. if load sharing is prohibited, the route is only activated if its priority is higher than that of the signalling route which is currently in use.

Class

Optional.

List

- ACT - Activate - Specify state change activate.
- INA - Inactivate - Specify state change inactivate.

Default

ACT

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.125 cmm multiLink

Signaling Multiple Link Show. Syntax, description, parameters, and roles for cmm multiLink command.

Syntax**Description**

Signaling Multiple Link Show.

Parameters**Name**

linkInterfaceType - Link interface type

Syntax

--linkInterfaceType <list-item>

Description

Link interface type. default to s6ad.

Class

Required key.

List

- gn - Specify the Gn interface to an SGSN.
- m3 - Specify the M3 interface to an eNodeB.
- n26 - Specify n26 interface to an AMF
- nr10 - Specify nr10 interface.
- rs10 - Specify the RS10 interface to an SRS.
- s10 - Specify the S10 interface to another MME.
- s102 - Specify the S102 interface to an IWF.
- s11 - Specify the S11 interface to an SGW.
- s11u - Specify S11-u interface to an SGW.
- s13 - Specify the S13 interface to an EIR.
- s16 - Specify S16 interface loop around to this SGSN.
- s1mme - Specify the S1-MME interface to an eNodeB.
- s3 - Specify the S3 interface to an SGSN.
- s4 - Specify S4 interface to an SGW.
- s6ad - Specify the S6ad interface to an HSS.
- sbc - Specify the SBc interface to a CBC..
- sgd - Specify SGd interface to an SMSC/SMS router.
- sgs - Specify the SGs interface to a GSM/UMTS MSC.
- sgslite - Specify SGs lite interface to an IWF.
- slg - Specify the Slg interface to a GMLC.
- sls - Specify the Sls interface to an E-SMLC.
- sm - Specify the sm interface to an MBMS gateway.
- sv - Specify the Sv interface to a GSM/UMTS MSC.
- t6a - Specify T6a interface to an SCEF.

Name

alarmState - Alarm state

Syntax

--alarmState <list-item>

Description

Specifies alarm state.

Class

Read only.

List

- critical - Critical alarm.
- major - Major alarm.
- minor - Minor alarm.
- none - No alarm.
- unknown - Unknown.

Name

linkAdminState - Link admin state

Syntax

--linkAdminState <list-item>

Description

Specified current link admin state.

Class

Optional.

List

- blocked - Blocked state. this applies only to s6ad, s13, slg, sgd and t6a.
- draining - Draining state. only applies to s11.
- locked - Locked state.
- unknown - Used in special cases to report state. a link cannot be updated to unknown state.
- unlocked - UnLocked state.

Name

linkAlarm - Link alarmed or not

Syntax

--linkAlarm <list-item>

Description

If false query returns all links. if true query returns only alarmed links.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Name

linkAvailState - Link available state

Syntax

```
--linkAvailState <list-item>
```

Description

Specifies link available state.

Class

Read only.

List

- degraded - Degraded.
- dependency - Dependency.
- none - None.
- unknown - Unknown.

Name

linkIdentifier - Link identifier

Syntax

```
--linkIdentifier <string>
```

Description

Link information.

Class

Read only.

Type / Range

Digit string of length 0 or 1 to 255

Name

linkInUseState - Link inuse state

Syntax

```
--linkInUseState <list-item>
```

Description

Link inuse state.

Class

Read only.

List

- in-use - Specify link in use state of in use.
- not-applicable - Specify link in use state of not application.
- not-used - Specify link in use state of not used.

Name

linkOperationState - Link operation state

Syntax

--linkOperationState <list-item>

Description

Link operation state.

Class

Optional.

List

- disabled - Disabled state of the link.
- enabled - Enable state of the link.
- unknown - Unknown state of the link. this can not be used during query.

Name

linkPlmnType - PLMN type of the link

Syntax

--linkPlmnType <list-item>

Description

Specifies PLMN type of the link. only applies to S6ad.

Class

Read only.

List

- home - Specify link PLMN type of home.
- not-applicable - Specify link PLMN type of not applicable.
- not-home - Specify link PLMN type of not home.

Name

linkProcState - Link procedural state

Syntax

--linkProcState <list-item>

Description

Specifies link procedural state.

Class

Read only.

List

- init req - Init request - Init request.
- initing - Initiating.
- none - None.
- not init - Not init.
- not-applicable - Not-applicable.
- report - Report.
- term - Term.

Name

localIp1 - Local IP1

Syntax

--localIp1 <string>

Description

Local IP1.

Class

Read only.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

localIp2 - Local IP2

Syntax

--localIp2 <string>

Description

Local IP2.

Class

Read only.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

macroEnbId - Macro eNB id

Syntax

--macroEnbId <integer>

Description

Macro eNB identifier as specified in TS 36.413 v8.6.1.

Class

Optional.

Type / Range

Integer in the range of 0 to 1,048,575

Name

maxResponseSize - Max record count

Syntax

--maxResponseSize <integer>

Description

Max number of records to be returned.

Class

Optional.

Type / Range

Integer in the range of 1 to 100

Name

mcc - Mobile country code

Syntax

--mcc <digits>

Description

3-digit number used to represent the mobile country code.

Class

Optional.

Type / Range

Digit string of length 0 or 3

Name

mnc - Mobile network code

Syntax

--mnc <digits>

Description

Digits used to represent mobile network code.

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 3

Name

poolId - IPDS Pool ID

Syntax

--poolId <integer>

Description

IPDS Pool ID - If chosen only the links on the specified IPDS will be output. The IPDS Pool ID will only be 0 for non Multi-IPDS instantiations. The default is ALL IPDS pools

Class

Read only.

Type / Range

Integer in the range of 0 to 31

Name

previousIndex - Previous index

Syntax

--previousIndex <integer>

Description

Previous index when specified next available index will be returned.

Class

Optional.

Type / Range

Integer in the range of -1 or 0 to 2,147,483,647

Name

queryRmtIp - Query Remote IP address

Syntax

--queryRmtIp <string>

Description

IP address used to query for a link whose remote IP address matches the specified IP for non-diameter and non-sgslite links

Class

Read only.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

remotelp1 - Remote IP1

Syntax

--remoteIp1 <string>

Description

Remote IP1.

Class

Read only.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

remotelp2 - Remote IP2

Syntax

--remoteIp2 <string>

Description

Remote IP2.

Class

Read only.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

startLinkIndex - Start link index

Syntax

--startLinkIndex <integer>

Description

Start link index from the read command.

Class

Optional.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

terminationName - TerminationName

Syntax

--terminationName <string>

Description

Termination name. this is used for s1mme alone.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles**List**

sysadmin, manage, monitor

5.126 cmm multipleHomePlmn

Multiple Home Plmn. Syntax, description, parameters, and roles for cmm multipleHomePlmn command.

Syntax**Description**

This command is used to configure an existing network as an multiple home plmn network and includes parameters for defining the serving and multiple home network names

Parameters**Name**

servedPlmnName - Serving Mobile Network

Syntax

--servedPlmnName <string>

Description

Name that identifies serving network.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

multipleHomePlmnName - Multiple Home Plmn Network name

Syntax

--multipleHomePlmnName <string>

Description

Name that identifies equivalent network.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.127 cmm multiPraOperation

Multi PRA Operation. Syntax, description, parameters, and roles for `cmm multiPraOperation` command.

Syntax**Description**

Manage multiple areas associated with a Presence Reporting Area Identifier. The ranges inside the list cannot overlap. Exclude ranges must be inside an include range and cannot overlap with an include range, i.e. lower (or upper) bound cannot be the same on both include and exclude range. MME supports up to 1500 TAs, 3500 macro eNBs, 3500 home eNBs and 10500 ECGIs per each CN predefined PRA, with the limit of 50000 of the combined number of TAs, eNBs, and ECGIs.

Parameters**Name**

praid - Presence Reporting Area Identifier

Syntax

--praid <integer>

Description

Presence Reporting Area Identifier associated with the area.

Class

Required key.

Type / Range

Integer in the range of 8,388,608 to 16,777,215

Name

plmnName - Mobile Network name

Syntax

--plmnName <string>

Description

A name that identifies mobile network.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

areaType - Area Type

Syntax

--areaType <list-item>

Description

The area type to be administered related to the Presence Reporting Area.

Class

Required key.

List

- ecgi - E-UTRAN Cell Global Identifier area type.
- homeENbId - Home eNB Identifier - Home eNB Identifier area type.
- macroENbId - Macro eNB Identifier - Macro eNB Identifier area type.
- tac - Tracking Area Code - Tracking Area Code area type.

Default

tac

Name

includeList - Include List

Syntax

--includeList <string>

Description

The list of area identifier ranges in decimal or hexadecimal format) to include in the Presence Reporting Area. The ranges inside the list cannot overlap. Exclude ranges must be inside an include range and cannot overlap with an include range, i.e. lower (or upper) bound cannot be the same on both include and exclude range. Single numbers are allowed. Lower bound and upper bound are inclusive in the range(s). Ranges are separated by comma. Lower bound and upper bound is delimited by a dash. Hexadecimal must be prefix by 0x or 0X.

Class

Required on create.

Type / Range

Digit string of length 0 to 100 with the allowed pattern of '[0-9a-fA-FxX,-]+'

Name

action - Action

Syntax

--action <list-item>

Description

Defines the action to perform on the PRA area lists.

Class

Optional.

List

- delete - Delete of items from PRA areas.
- insert - Insert of items into PRA areas.

Default

insert

Name

excludeList - Exclude List

Syntax

--excludeList <string>

Description

The list of area identifier ranges (in decimal or hexadecimal format) to exclude from the Presence Reporting Area. The ranges inside the list cannot overlap. Exclude ranges must be inside an include range and cannot overlap with an include range, i.e. lower (or upper) bound cannot be the same on both include and exclude range. Single numbers are allowed. Lower bound and upper bound are inclusive in the range(s). Ranges are separated by comma. Lower bound and upper bound is delimited by a dash. Hexadecimal must be prefix by 0x or 0X.

Class

Optional.

Type / Range

Digit string of length 0 to 100 with the allowed pattern of '[0-9a-fA-FxX,-]+'

Roles**Modify**

sysadmin, manage

5.128 cmm multiTacOperation

Multi TAC Operation. Syntax, description, parameters, and roles for cmm multiTacOperation command.

Syntax

Description

Update an attribute(s) on a set of TACs. Define the set of TACs by plmnName,minTac,maxTac OR plmnName,tacListName OR plmnName,allTac. Maximum number of TACs in the set shall be limited to 2000.

Parameters

Name

plmnName - Mobile network name

Syntax

--plmnName <string>

Description

Mobile network in defining set of TACs to update.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

allTac - All tracking area

Syntax

--allTac <list-item>

Description

Define set of TACs to update to be all TACs in a given mobile network.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Name

ciotEmmRegisteredWithoutPdn - Enable EMM-REGISTERED without PDN connection for C-IOT UE

Syntax

--ciotEmmRegisteredWithoutPdn <list-item>

Description

Enable EMM-REGISTERED without PDN connection for C-IOT UE.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

dcnrSupported - DCNR Supported

Syntax

--dcnrSupported <list-item>

Description

Whether DCNR Supported is set.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

emergencySupported - Emergency supported

Syntax

--emergencySupported <list-item>

Description

Indicates if emergency is supported. only allowed if IMS supported is true.

Class

Optional.

List

- false - Boolean value of false.

- true - Boolean value of true.

Name

emmlInfoOverride - EMM info override

Syntax

--emmlInfoOverride <list-item>

Description

Flag defining whether the emmlInfo message should be overridden with profile information in the emmlInfoOverrideTacSv table.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

esmlcIdentity1 - ESMC identity 1

Syntax

--esmlcIdentity1 <digits>

Description

First TAI to E-SMLC (Evolved serving mobile location center) mapping.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 3

Name

esmlcIdentity2 - ESMC identity 2

Syntax

--esmlcIdentity2 <digits>

Description

Second TAI to E-SMLC (Evolved serving mobile location center) mapping.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 3

Name

esmlcIdentity3 - ESMC identity 3

Syntax

--esmlcIdentity3 <digits>

Description

Third TAI to E-SMLC (Evolved serving mobile location center) mapping. The E-SMLC is used for emergency services.

Class

Optional.

Type / Range

Digit string of length 0 to 3

Name

esmlcIdentity4 - ESMC identity 4

Syntax

--esmlcIdentity4 <digits>

Description

Fourth TAI to E-SMLC (Evolved serving mobile location center) mapping. The set of 4 ESMC are used for emergency services.

Class

Optional.

Type / Range

Digit string of length 0 to 3

Name

imsSupported - IMS supported

Syntax

--imsSupported <list-item>

Description

Indicates if IMS is supported.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Name

lcsEntryId - Location services entry id

Syntax

--lcsEntryId <integer>

Description

Unique id for this association of E-SMLC to tracking area and location services type. this field is used as the priority when MME TAI selection algorithm matching location services type is primarysecondary.

Class

Optional.

Type / Range

Integer in the range of 1 to 6

Name

lcsType - Location services type

Syntax

--lcsType <list-item>

Description

Specifies the location services type associated with ESMLC for this tracking area.

Class

Optional.

List

- LawfullIntercept - Lawful interception location services - Lawful interception location services.
- Operator - Operator location services - Operator location services.
- Value Added - Value added location services - Value added location services.

Name

liSelectionAlgorithm - Lawful Intercept location services selection algorithm

Syntax

--liSelectionAlgorithm <list-item>

Description

Indicates for the set of lawful intercept location services ESMLCs (Evolved serving mobile location center) whether they will load share or function as primary/secondary.

Class

Optional.

List

- LoadShare - Load share - Load share.
- PrimarySecondary - Primary / secondary - Function as primary / secondary.

Default

PrimarySecondary

Name

maxTac - Max tracking area code

Syntax

--maxTac <integer>

Description

Maximum TAC value in defining set of TACs to update, used with mintac.

Class

Optional.

Type / Range

Integer in the range of 1 to 65,535

Name

minTac - Min tracking area code

Syntax

--minTac <integer>

Description

Minimum TAC value in defining set of TACs to update, used with maxtac.

Class

Optional.

Type / Range

Integer in the range of 1 to 65,535

Name

mmeRelCapacity - MME relative capacity

Syntax

--mmeRelCapacity <integer>

Description

Provision MME relative capacity per tracking area.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Name

mtEdtCp - mtEdtCp

Syntax

```
--mtEdtCp <list-item>
```

Description

Mobile-Terminated Early Data Transfer over Control Plan

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

mtEdtUp - MT EDT User Plane

Syntax

```
--mtEdtUp <list-item>
```

Description

Mobile Terminated Early Data Transfer over User Plane

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

nbiotSmsWithoutCombinedAttach - Enable SMS without combined Attach For NB-IOT UE

Syntax

```
--nbiotSmsWithoutCombinedAttach <list-item>
```

Description

Enable SMS without combined Attach For NB-IOT UE

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

nonEmerEsmlcIdentity - Non-Emergency ESMLC identifier

Syntax

--nonEmerEsmlcIdentity <digits>

Description

Indicates the ESMLC being provisioned for a given tracking area and non-emergency location services type.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 3

Name

operSelectionAlgorithm - Operator location services selection algorithm

Syntax

--operSelectionAlgorithm <list-item>

Description

Indicates for the set of operator location services ESMLCs (Evolved serving mobile location center) whether they will load share or function as primary/secondary.

Class

Optional.

List

- LoadShare - Load share - Load share.
- PrimarySecondary - Primary / secondary - Function as primary / secondary.

Name

selectionAlgorithm - Selection algorithm

Syntax

--selectionAlgorithm <list-item>

Description

Indicates for the set of ESMLCs, (Evolved serving mobile location center) whether they will load share or function as primary/secondary.

Class

Optional.

List

- LoadShare - Load share - Load share.
- PrimarySecondary - Primary / secondary - Function as primary / secondary.

Name

sgwPoolId - SGW pool ID

Syntax

--sgwPoolId <string>

Description

ID for a serving gateway pool.

Class

Optional.

Type / Range

Digit string of length 0 to 20

Name

supportedRatType - Supported RAT type for this TAC

Syntax

--supportedRatType <list-item>

Description

Supported RAT type for this TAC. it can only be wbeutran or nbiot.

Class

Optional.

List

- nbiot - Narrow band internet of things - Specify type of narrow band internet of things.
- wbeutran - Wide band EUTRAN - Specify type of wide band EUTRAN.

Default

wbeutran

Name

tacListName - Tracking area list

Syntax

--tacListName <string>

Description

Define set of TACs to update to be all TACs in given tracking area list.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

timeZoneName - Time zone name

Syntax

--timeZoneName <list-item>

Description

Provides time zone names that correspond to the region selected.

Class

Optional.

List

- Africa/Abidjan - Timezone for africa/abidjan UTC offset of +00:00.
- Africa/Accra - Timezone for africa/accra UTC offset of +00:00.
- Africa/Addis_Ababa - Timezone for africa/addis_ababa UTC offset of +03:00.
- Africa/Algiers - Timezone for africa/algiers UTC offset of +01:00.
- Africa/Asmara - Timezone for africa/asmara UTC offset of +03:00.
- Africa/Asmera - Timezone for africa/asmera UTC offset of +03:00.
- Africa/Bamako - Timezone for africa/bamako UTC offset of +00:00.
- Africa/Bangui - Timezone for africa/bangui UTC offset of +01:00.
- Africa/Banjul - Timezone for africa/banjul UTC offset of +00:00.
- Africa/Bissau - Timezone for africa/bissau UTC offset of +00:00.
- Africa/Blantyre - Timezone for africa/blantyre UTC offset of +02:00.
- Africa/Brazzaville - Timezone for africa/brazzaville UTC offset of +01:00.
- Africa/Bujumbura - Timezone for africa/bujumbura UTC offset of +02:00.
- Africa/Cairo - Timezone for africa/cairo UTC offset of +02:00.
- Africa/Casablanca - Timezone for africa/casablanca UTC offset of +00:00.
- Africa/Ceuta - Timezone for africa/ceuta UTC offset of +01:00.
- Africa/Conakry - Timezone for africa/conakry UTC offset of +00:00.
- Africa/Dakar - Timezone for africa/dakar UTC offset of +00:00.
- Africa/Dar_es_Salaam - Timezone for africa/dar_es_salaam UTC offset of +03:00.
- Africa/Djibouti - Timezone for africa/djibouti UTC offset of +03:00.
- Africa/Douala - Timezone for africa/douala UTC offset of +01:00.
- Africa/El_Aaiun - Timezone for africa/el_aaiun UTC offset of +00:00.

- Africa/Freetown - Timezone for africa/freetown UTC offset of +00:00.
- Africa/Gaborone - Timezone for africa/gaborone UTC offset of +02:00.
- Africa/Harare - Timezone for africa/harare UTC offset of +02:00.
- Africa/Johannesburg - Timezone for africa/johannesburg UTC offset of +02:00.
- Africa/Juba - Timezone for africa/juba UTC offset of +03:00.
- Africa/Kampala - Timezone for africa/kampala UTC offset of +03:00.
- Africa/Khartoum - Timezone for africa/khartoum UTC offset of +03:00.
- Africa/Kigali - Timezone for africa/kigali UTC offset of +02:00.
- Africa/Kinshasa - Timezone for africa/kinshasa UTC offset of +01:00.
- Africa/Lagos - Timezone for africa/lagos UTC offset of +01:00.
- Africa/Libreville - Timezone for africa/libreville UTC offset of +01:00.
- Africa/Lome - Timezone for africa/lome UTC offset of +00:00.
- Africa/Luanda - Timezone for africa/luanda UTC offset of +01:00.
- Africa/Lubumbashi - Timezone for africa/lubumbashi UTC offset of +02:00.
- Africa/Lusaka - Timezone for africa/lusaka UTC offset of +02:00.
- Africa/Malabo - Timezone for africa/malabo UTC offset of +01:00.
- Africa/Maputo - Timezone for africa/maputo UTC offset of +02:00.
- Africa/Maseru - Timezone for africa/maseru UTC offset of +02:00.
- Africa/Mbabane - Timezone for africa/mbabane UTC offset of +02:00.
- Africa/Mogadishu - Timezone for africa/mogadishu UTC offset of +03:00.
- Africa/Monrovia - Timezone for africa/monrovia UTC offset of +00:00.
- Africa/Nairobi - Timezone for africa/nairobi UTC offset of +03:00.
- Africa/Ndjamena - Timezone for africa/ndjamena UTC offset of +01:00.
- Africa/Niamey - Timezone for africa/niamey UTC offset of +01:00.
- Africa/Nouakchott - Timezone for africa/nouakchott UTC offset of +00:00.
- Africa/Ouagadougou - Timezone for africa/ouagadougou UTC offset of +00:00.
- Africa/Porto-Novo - Timezone for africa/porto-novo UTC offset of +01:00.
- Africa/Sao_Tome - Timezone for africa/sao_tome UTC offset of +00:00.
- Africa/Timbuktu - Timezone for africa/timbuktu UTC offset of +00:00.
- Africa/Tripoli - Timezone for africa/tripoli UTC offset of +02:00.
- Africa/Tunis - Timezone for africa/tunis UTC offset of +01:00.

- Africa/Windhoek - Timezone for africa/windhoek UTC offset of +02:00.
- America/Adak - Timezone for america/adak UTC offset of -10:00.
- America/Anchorage - Timezone for america/anchorage UTC offset of -09:00.
- America/Anguilla - Timezone for america/anguilla UTC offset of -04:00.
- America/Antigua - Timezone for america/antigua UTC offset of -04:00.
- America/Araguaina - Timezone for america/araguaina UTC offset of -03:00.
- America/Argentina/Buenos_Aires - Timezone for america/argentina/buenos_aires UTC offset of -03:00.
- America/Argentina/Catamarca - Timezone for america/argentina/catamarca UTC offset of -03:00.
- America/Argentina/ComodRivadavia - Timezone for america/argentina/comodrivadavia UTC offset of -03:00.
- America/Argentina/Cordoba - Timezone for america/argentina/cordoba UTC offset of -03:00.
- America/Argentina/Jujuy - Timezone for america/argentina/jujuy UTC offset of -03:00.
- America/Argentina/La_Rioja - Timezone for america/argentina/la_rioja UTC offset of -03:00.
- America/Argentina/Mendoza - Timezone for america/argentina/mendoza UTC offset of -03:00.
- America/Argentina/Rio_Gallegos - Timezone for america/argentina/rio_gallegos UTC offset of -03:00.
- America/Argentina/Salta - Timezone for america/argentina/salta UTC offset of -03:00.
- America/Argentina/San_Juan - Timezone for america/argentina/san_juan UTC offset of -03:00.
- America/Argentina/San_Luis - Timezone for america/argentina/san_luis UTC offset of -03:00.
- America/Argentina/Tucuman - Timezone for america/argentina/tucuman UTC offset of -03:00.
- America/Argentina/Ushuaia - Timezone for america/argentina/ushuaia UTC offset of -03:00.
- America/Aruba - Timezone for america/aruba UTC offset of -04:00.
- America/Asuncion - Timezone for america/asuncion UTC offset of -03:00.
- America/Atikokan - Timezone for america/atikokan UTC offset of -05:00.
- America/Atka - Timezone for america/atka UTC offset of -10:00.

- America/Bahia - Timezone for america/bahia UTC offset of -03:00.
- America/Bahia_Banderas - Timezone for america/bahia_banderas UTC offset of -06:00.
- America/Barbados - Timezone for america/barbados UTC offset of -04:00.
- America/Belem - Timezone for america/belem UTC offset of -03:00.
- America/Belize - Timezone for america/belize UTC offset of -06:00.
- America/Blanc-Sablon - Timezone for america/blanc-sablon UTC offset of -04:00.
- America/Boa_Vista - Timezone for america/boa_vista UTC offset of -04:00.
- America/Bogota - Timezone for america/bogota UTC offset of -05:00.
- America/Boise - Timezone for america/boise UTC offset of -07:00.
- America/Buenos_Aires - Timezone for america/buenos_aires UTC offset of -03:00.
- America/Cambridge_Bay - Timezone for america/cambridge_bay UTC offset of -07:00.
- America/Campo_Grande - Timezone for america/campo_grande UTC offset of -03:00.
- America/Cancun - Timezone for america/cancun UTC offset of -05:00.
- America/Caracas - Timezone for america/caracas UTC offset of -04:00.
- America/Catamarca - Timezone for america/catamarca UTC offset of -03:00.
- America/Cayenne - Timezone for america/cayenne UTC offset of -03:00.
- America/Cayman - Timezone for america/cayman UTC offset of -05:00.
- America/Chicago - Timezone for america/chicago UTC offset of -06:00.
- America/Chihuahua - Timezone for america/chihuahua UTC offset of -07:00.
- America/Coral_Harbour - Timezone for america/coral_harbour UTC offset of -05:00.
- America/Cordoba - Timezone for america/cordoba UTC offset of -03:00.
- America/Costa_Rica - Timezone for america/costa_rica UTC offset of -06:00.
- America/Creston - Timezone for america/creston UTC offset of -07:00.
- America/Cuiaba - Timezone for america/cuiaba UTC offset of -03:00.
- America/Curacao - Timezone for america/curacao UTC offset of -04:00.
- America/Danmarkshavn - Timezone for america/danmarkshavn UTC offset of +00:00.
- America/Dawson - Timezone for america/dawson UTC offset of -08:00.
- America/Dawson_Creek - Timezone for america/dawson_creek UTC offset of -07:00.
- America/Denver - Timezone for america/denver UTC offset of -07:00.
- America/Detroit - Timezone for america/detroit UTC offset of -05:00.
- America/Dominica - Timezone for america/dominica UTC offset of -04:00.

- America/Edmonton - Timezone for america/edmonton UTC offset of -07:00.
- America/Eirunepe - Timezone for america/eirunepe UTC offset of -05:00.
- America/El_Salvador - Timezone for america/el_salvador UTC offset of -06:00.
- America/Ensenada - Timezone for america/ensenada UTC offset of -08:00.
- America/Fort_Nelson - Timezone for america/fort_nelson UTC offset of -07:00.
- America/Fort_Wayne - Timezone for america/fort_wayne UTC offset of -05:00.
- America/Fortaleza - Timezone for america/fortaleza UTC offset of -03:00.
- America/Glace_Bay - Timezone for america/glance_bay UTC offset of -04:00.
- America/Godthab - Timezone for america/godthab UTC offset of -03:00.
- America/Goose_Bay - Timezone for america/goose_bay UTC offset of -04:00.
- America/Grand_Turk - Timezone for america/grand_turk UTC offset of -04:00.
- America/Grenada - Timezone for america/grenada UTC offset of -04:00.
- America/Guadeloupe - Timezone for america/guadeloupe UTC offset of -04:00.
- America/Guatemala - Timezone for america/guatemala UTC offset of -06:00.
- America/Guayaquil - Timezone for america/guayaquil UTC offset of -05:00.
- America/Guyana - Timezone for america/guyana UTC offset of -04:00.
- America/Halifax - Timezone for america/halifax UTC offset of -04:00.
- America/Havana - Timezone for america/havana UTC offset of -05:00.
- America/Hermosillo - Timezone for america/hermosillo UTC offset of -07:00.
- America/Indiana/Indianapolis - Timezone for america/indiana/indianapolis UTC offset of -05:00.
- America/Indiana/Knox - Timezone for america/indiana/knox UTC offset of -06:00.
- America/Indiana/Marengo - Timezone for america/indiana/marengo UTC offset of -05:00.
- America/Indiana/Petersburg - Timezone for america/indiana/petersburg UTC offset of -05:00.
- America/Indiana/Tell_City - Timezone for america/indiana/tell_city UTC offset of -06:00.
- America/Indiana/Vevay - Timezone for america/indiana/vevay UTC offset of -05:00.
- America/Indiana/Vincennes - Timezone for america/indiana/vincennes UTC offset of -05:00.
- America/Indiana/Winamac - Timezone for america/indiana/winamac UTC offset of -05:00.

- America/Indianapolis - Timezone for america/indianapolis UTC offset of -05:00.
- America/Inuvik - Timezone for america/inuvik UTC offset of -07:00.
- America/Iqaluit - Timezone for america/iqaluit UTC offset of -05:00.
- America/Jamaica - Timezone for america/jamaica UTC offset of -05:00.
- America/Jujuy - Timezone for america/jujuy UTC offset of -03:00.
- America/Juneau - Timezone for america/juneau UTC offset of -09:00.
- America/Kentucky/Louisville - Timezone for america/kentucky/louisville UTC offset of -05:00.
- America/Kentucky/Monticello - Timezone for america/kentucky/monticello UTC offset of -05:00.
- America/Knox_IN - Timezone for america/knox_in UTC offset of -06:00.
- America/Kralendijk - Timezone for america/kralendijk UTC offset of -04:00.
- America/La_Paz - Timezone for america/la_paz UTC offset of -04:00.
- America/Lima - Timezone for america/lima UTC offset of -05:00.
- America/Los_Angeles - Timezone for america/los_angeles UTC offset of -08:00.
- America/Louisville - Timezone for america/louisville UTC offset of -05:00.
- America/Lower_Princes - Timezone for america/lower_princes UTC offset of -04:00.
- America/Maceio - Timezone for america/maceio UTC offset of -03:00.
- America/Managua - Timezone for america/managua UTC offset of -06:00.
- America/Manaus - Timezone for america/manuel UTC offset of -04:00.
- America/Marigot - Timezone for america/marigot UTC offset of -04:00.
- America/Martinique - Timezone for america/martinique UTC offset of -04:00.
- America/Matamoros - Timezone for america/matamoros UTC offset of -06:00.
- America/Mazatlan - Timezone for america/mazatlan UTC offset of -07:00.
- America/Mendoza - Timezone for america/mendoza UTC offset of -03:00.
- America/Menominee - Timezone for america/menominee UTC offset of -06:00.
- America/Merida - Timezone for america/merida UTC offset of -06:00.
- America/Metlakatla - Timezone for america/metlakatla UTC offset of -09:00.
- America/Mexico_City - Timezone for america/mexico_city UTC offset of -06:00.
- America/Miquelon - Timezone for america/miquelon UTC offset of -03:00.
- America/Moncton - Timezone for america/moncton UTC offset of -04:00.
- America/Monterrey - Timezone for america/monterrey UTC offset of -06:00.

- America/Montevideo - Timezone for america/montevideo UTC offset of -03:00.
- America/Montreal - Timezone for america/montreal UTC offset of -05:00.
- America/Montserrat - Timezone for america/montserrat UTC offset of -04:00.
- America/Nassau - Timezone for america/nassau UTC offset of -05:00.
- America/New_York - Timezone for america/new_york UTC offset of -05:00.
- America/Nipigon - Timezone for america/nipigon UTC offset of -05:00.
- America/Nome - Timezone for america/nome UTC offset of -09:00.
- America/Noronha - Timezone for america/noronha UTC offset of -02:00.
- America/North_Dakota/Beulah - Timezone for america/north_dakota/beulah UTC offset of -06:00.
- America/North_Dakota/Center - Timezone for america/north_dakota/center UTC offset of -06:00.
- America/North_Dakota/New_Salem - Timezone for america/north_dakota/new_salem UTC offset of -06:00.
- America/Ojinaga - Timezone for america/ojinaga UTC offset of -07:00.
- America/Panama - Timezone for america/panama UTC offset of -05:00.
- America/Pangnirtung - Timezone for america/pangnirtung UTC offset of -05:00.
- America/Paramaribo - Timezone for america/paramaribo UTC offset of -03:00.
- America/Phoenix - Timezone for america/phoenix UTC offset of -07:00.
- America/Port-au-Prince - Timezone for america/port-au-prince UTC offset of -05:00.
- America/Port_of_Spain - Timezone for america/port_of_spain UTC offset of -04:00.
- America/Porto_Acre - Timezone for america/porto_acre UTC offset of -05:00.
- America/Porto_Velho - Timezone for america/porto_velho UTC offset of -04:00.
- America/Puerto_Rico - Timezone for america/puerto_rico UTC offset of -04:00.
- America/Rainy_River - Timezone for america/rainy_river UTC offset of -06:00.
- America/Rankin_Inlet - Timezone for america/rankin_inlet UTC offset of -06:00.
- America/Recife - Timezone for america/recife UTC offset of -03:00.
- America/Regina - Timezone for america/regina UTC offset of -06:00.
- America/Resolute - Timezone for america/resolute UTC offset of -06:00.
- America/Rio_Branco - Timezone for america/rio_branco UTC offset of -05:00.
- America/Rosario - Timezone for america/rosario UTC offset of -03:00.
- America/Santa_Isabel - Timezone for america/santa_isabel UTC offset of -08:00.

- America/Santarem - Timezone for america/santarem UTC offset of -03:00.
- America/Santiago - Timezone for america/santiago UTC offset of -03:00.
- America/Santo_Domingo - Timezone for america/santo_domingo UTC offset of -04:00.
- America/Sao_Paulo - Timezone for america/sao_paulo UTC offset of -02:00.
- America/Scoresbysund - Timezone for america/scoresbysund UTC offset of -01:00.
- America/Shiprock - Timezone for america/shiprock UTC offset of -07:00.
- America/Sitka - Timezone for america/sitka UTC offset of -09:00.
- America/St_Barthelemy - Timezone for america/st_barthelemy UTC offset of -04:00.
- America/St_Johns - Timezone for america/st_johns UTC offset of -03:30.
- America/St_Kitts - Timezone for america/st_kitts UTC offset of -04:00.
- America/St_Lucia - Timezone for america/st_lucia UTC offset of -04:00.
- America/St_Thomas - Timezone for america/st_thomas UTC offset of -04:00.
- America/St_Vincent - Timezone for america/st_vincent UTC offset of -04:00.
- America/Swift_Current - Timezone for america/swift_current UTC offset of -06:00.
- America/Tegucigalpa - Timezone for america/tegucigalpa UTC offset of -06:00.
- America/Thule - Timezone for america/thule UTC offset of -04:00.
- America/Thunder_Bay - Timezone for america/thunder_bay UTC offset of -05:00.
- America/Tijuana - Timezone for america/tijuana UTC offset of -08:00.
- America/Toronto - Timezone for america/toronto UTC offset of -05:00.
- America/Tortola - Timezone for america/tortola UTC offset of -04:00.
- America/Vancouver - Timezone for america/vancouver UTC offset of -08:00.
- America/Virgin - Timezone for america/virgin UTC offset of -04:00.
- America/Whitehorse - Timezone for america/whitehorse UTC offset of -08:00.
- America/Winnipeg - Timezone for america/winnipeg UTC offset of -06:00.
- America/Yakutat - Timezone for america/yakutat UTC offset of -09:00.
- America/Yellowknife - Timezone for america/yellowknife UTC offset of -07:00.
- Antarctica/Casey - Timezone for antarctica/casey UTC offset of +11:00.
- Antarctica/Davis - Timezone for antarctica/davis UTC offset of +07:00.
- Antarctica/DumontDURville - Timezone for antarctica/dumontdurville UTC offset of +10:00.
- Antarctica/Macquarie - Timezone for antarctica/macquarie UTC offset of +11:00.

- Antarctica/Mawson - Timezone for antarctica/mawson UTC offset of +05:00.
- Antarctica/McMurdo - Timezone for antarctica/mcmurdo UTC offset of +13:00.
- Antarctica/Palmer - Timezone for antarctica/palmer UTC offset of -03:00.
- Antarctica/Rothera - Timezone for antarctica/rothera UTC offset of -03:00.
- Antarctica/South_Pole - Timezone for antarctica/south_pole UTC offset of +13:00.
- Antarctica/Syowa - Timezone for antarctica/syowa UTC offset of +03:00.
- Antarctica/Troll - Timezone for antarctica/troll UTC offset of +00:00.
- Antarctica/Vostok - Timezone for antarctica/vostok UTC offset of +06:00.
- Arctic/Longyearbyen - Timezone for arctic/longyearbyen UTC offset of +01:00.
- Asia/Aden - Timezone for asia/aden UTC offset of +03:00.
- Asia/Almaty - Timezone for asia/almaty UTC offset of +06:00.
- Asia/Amman - Timezone for asia/amman UTC offset of +02:00.
- Asia/Anadyr - Timezone for asia/anadyr UTC offset of +12:00.
- Asia/Aqtau - Timezone for asia/aqtau UTC offset of +05:00.
- Asia/Aqtobe - Timezone for asia/aqtobe UTC offset of +05:00.
- Asia/Ashgabat - Timezone for asia/ashgabat UTC offset of +05:00.
- Asia/Ashkhabad - Timezone for asia/ashkhabad UTC offset of +05:00.
- Asia/Baghdad - Timezone for asia/baghdad UTC offset of +03:00.
- Asia/Bahrain - Timezone for asia/bahrain UTC offset of +03:00.
- Asia/Baku - Timezone for asia/baku UTC offset of +04:00.
- Asia/Bangkok - Timezone for asia/bangkok UTC offset of +07:00.
- Asia/Barnaul - Timezone for asia/barnaul UTC offset of +07:00.
- Asia/Beirut - Timezone for asia/beirut UTC offset of +02:00.
- Asia/Bishkek - Timezone for asia/bishkek UTC offset of +06:00.
- Asia/Brunei - Timezone for asia/brunei UTC offset of +08:00.
- Asia/Calcutta - Timezone for asia/calcutta UTC offset of +05:30.
- Asia/Chita - Timezone for asia/chita UTC offset of +09:00.
- Asia/Choibalsan - Timezone for asia/choibalsan UTC offset of +08:00.
- Asia/Chongqing - Timezone for asia/chongqing UTC offset of +08:00.
- Asia/Chungking - Timezone for asia/chungking UTC offset of +08:00.
- Asia/Colombo - Timezone for asia/colombo UTC offset of +05:30.

- Asia/Dacca - Timezone for asia/dacca UTC offset of +06:00.
- Asia/Damascus - Timezone for asia/damascus UTC offset of +02:00.
- Asia/Dhaka - Timezone for asia/dhaka UTC offset of +06:00.
- Asia/Dili - Timezone for asia/dili UTC offset of +09:00.
- Asia/Dubai - Timezone for asia/dubai UTC offset of +04:00.
- Asia/Dushanbe - Timezone for asia/dushanbe UTC offset of +05:00.
- Asia/Gaza - Timezone for asia/gaza UTC offset of +02:00.
- Asia/Harbin - Timezone for asia/harbin UTC offset of +08:00.
- Asia/Hebron - Timezone for asia/hebron UTC offset of +02:00.
- Asia/Ho_Chi_Minh - Timezone for asia/ho_chi_minh UTC offset of +07:00.
- Asia/Hong_Kong - Timezone for asia/hong_kong UTC offset of +08:00.
- Asia/Hovd - Timezone for asia/hovd UTC offset of +07:00.
- Asia/Irkutsk - Timezone for asia/irkutsk UTC offset of +08:00.
- Asia/Istanbul - Timezone for asia/istanbul UTC offset of +03:00.
- Asia/Jakarta - Timezone for asia/jakarta UTC offset of +07:00.
- Asia/Jayapura - Timezone for asia/jayapura UTC offset of +09:00.
- Asia/Jerusalem - Timezone for asia/jerusalem UTC offset of +02:00.
- Asia/Kabul - Timezone for asia/kabul UTC offset of +04:30.
- Asia/Kamchatka - Timezone for asia/kamchatka UTC offset of +12:00.
- Asia/Karachi - Timezone for asia/karachi UTC offset of +05:00.
- Asia/Kashgar - Timezone for asia/kashgar UTC offset of +06:00.
- Asia/Kathmandu - Timezone for asia/kathmandu UTC offset of +05:45.
- Asia/Katmandu - Timezone for asia/katmandu UTC offset of +05:45.
- Asia/Khandyga - Timezone for asia/khandyga UTC offset of +09:00.
- Asia/Kolkata - Timezone for asia/kolkata UTC offset of +05:30.
- Asia/Krasnoyarsk - Timezone for asia/krasnoyarsk UTC offset of +07:00.
- Asia/Kuala_Lumpur - Timezone for asia/kuala_lumpur UTC offset of +08:00.
- Asia/Kuching - Timezone for asia/kuching UTC offset of +08:00.
- Asia/Kuwait - Timezone for asia/kuwait UTC offset of +03:00.
- Asia/Macao - Timezone for asia/macao UTC offset of +08:00.
- Asia/Macau - Timezone for asia/macau UTC offset of +08:00.

- Asia/Magadan - Timezone for asia/magadan UTC offset of +11:00.
- Asia/Makassar - Timezone for asia/makassar UTC offset of +08:00.
- Asia/Manila - Timezone for asia/manila UTC offset of +08:00.
- Asia/Muscat - Timezone for asia/muscat UTC offset of +04:00.
- Asia/Nicosia - Timezone for asia/nicosia UTC offset of +02:00.
- Asia/Novokuznetsk - Timezone for asia/novokuznetsk UTC offset of +07:00.
- Asia/Novosibirsk - Timezone for asia/novosibirsk UTC offset of +07:00.
- Asia/Omsk - Timezone for asia/omsk UTC offset of +06:00.
- Asia/Oral - Timezone for asia/oral UTC offset of +05:00.
- Asia/Phnom_Penh - Timezone for asia/phnom_penh UTC offset of +07:00.
- Asia/Pontianak - Timezone for asia/pontianak UTC offset of +07:00.
- Asia/Pyongyang - Timezone for asia/pyongyang UTC offset of +08:30.
- Asia/Qatar - Timezone for asia/qatar UTC offset of +03:00.
- Asia/Qyzylorda - Timezone for asia/qyzylorda UTC offset of +06:00.
- Asia/Rangoon - Timezone for asia/rangoon UTC offset of +06:30.
- Asia/Riyadh - Timezone for asia/riyadh UTC offset of +03:00.
- Asia/Saigon - Timezone for asia/saigon UTC offset of +07:00.
- Asia/Sakhalin - Timezone for asia/sakhalin UTC offset of +11:00.
- Asia/Samarkand - Timezone for asia/samarkand UTC offset of +05:00.
- Asia/Seoul - Timezone for asia/seoul UTC offset of +09:00.
- Asia/Shanghai - Timezone for asia/shanghai UTC offset of +08:00.
- Asia/Singapore - Timezone for asia/singapore UTC offset of +08:00.
- Asia/Srednekolymsk - Timezone for asia/srednekolymsk UTC offset of +11:00.
- Asia/Taipei - Timezone for asia/taipei UTC offset of +08:00.
- Asia/Tashkent - Timezone for asia/tashkent UTC offset of +05:00.
- Asia/Tbilisi - Timezone for asia/tbilisi UTC offset of +04:00.
- Asia/Tehran - Timezone for asia/tehran UTC offset of +03:30.
- Asia/Tel_Aviv - Timezone for asia/tel_aviv UTC offset of +02:00.
- Asia/Thimbu - Timezone for asia/thimbu UTC offset of +06:00.
- Asia/Thimphu - Timezone for asia/thimphu UTC offset of +06:00.
- Asia/Tokyo - Timezone for asia/tokyo UTC offset of +09:00.

- Asia/Tomsk - Timezone for asia/tomsk UTC offset of +07:00.
- Asia/Ujung_Pandang - Timezone for asia/ujung_pandang UTC offset of +08:00.
- Asia/Ulaanbaatar - Timezone for asia/ulaanbaatar UTC offset of +08:00.
- Asia/Ulan_Bator - Timezone for asia/ulan_bator UTC offset of +08:00.
- Asia/Urumqi - Timezone for asia/urumqi UTC offset of +06:00.
- Asia/Ust-Nera - Timezone for asia/ust-nera UTC offset of +10:00.
- Asia/Vientiane - Timezone for asia/vientiane UTC offset of +07:00.
- Asia/Vladivostok - Timezone for asia/vladivostok UTC offset of +10:00.
- Asia/Yakutsk - Timezone for asia/yakutsk UTC offset of +09:00.
- Asia/Yekaterinburg - Timezone for asia/yekaterinburg UTC offset of +05:00.
- Asia/Yerevan - Timezone for asia/yerevan UTC offset of +04:00.
- Atlantic/Azores - Timezone for atlantic/azores UTC offset of -01:00.
- Atlantic/Bermuda - Timezone for atlantic/bermuda UTC offset of -04:00.
- Atlantic/Canary - Timezone for atlantic/canary UTC offset of +00:00.
- Atlantic/Cape_Verde - Timezone for atlantic/cape_verde UTC offset of -01:00.
- Atlantic/Faeroe - Timezone for atlantic/faeroe UTC offset of +00:00.
- Atlantic/Faroe - Timezone for atlantic/faroe UTC offset of +00:00.
- Atlantic/Jan_Mayen - Timezone for atlantic/jan_mayen UTC offset of +01:00.
- Atlantic/Madeira - Timezone for atlantic/madeira UTC offset of +00:00.
- Atlantic/Reykjavik - Timezone for atlantic/reykjavik UTC offset of +00:00.
- Atlantic/South_Georgia - Timezone for atlantic/south_georgia UTC offset of -02:00.
- Atlantic/St_Helena - Timezone for atlantic/st_helena UTC offset of +00:00.
- Atlantic/Stanley - Timezone for atlantic/stanley UTC offset of -03:00.
- Australia/ACT - Timezone for australia/act UTC offset of +11:00.
- Australia/Adelaide - Timezone for australia/adelaide UTC offset of +10:30.
- Australia/Brisbane - Timezone for australia/brisbane UTC offset of +10:00.
- Australia/Broken_Hill - Timezone for australia/broken_hill UTC offset of +10:30.
- Australia/Canberra - Timezone for australia/canberra UTC offset of +11:00.
- Australia/Currie - Timezone for australia/currie UTC offset of +11:00.
- Australia/Darwin - Timezone for australia/darwin UTC offset of +09:30.
- Australia/Eucla - Timezone for australia/eucla UTC offset of +08:45.

- Australia/Hobart - Timezone for australia/hobart UTC offset of +11:00.
- Australia/LHI - Timezone for australia/lhi UTC offset of +11:00.
- Australia/Lindeman - Timezone for australia/lindeman UTC offset of +10:00.
- Australia/Lord_Howe - Timezone for australia/lord_howe UTC offset of +11:00.
- Australia/Melbourne - Timezone for australia/melbourne UTC offset of +11:00.
- Australia/NSW - Timezone for australia/nsw UTC offset of +11:00.
- Australia/North - Timezone for australia/north UTC offset of +09:30.
- Australia/Perth - Timezone for australia/perth UTC offset of +08:00.
- Australia/Queensland - Timezone for australia/queensland UTC offset of +10:00.
- Australia/South - Timezone for australia/south UTC offset of +10:30.
- Australia/Sydney - Timezone for australia/sydney UTC offset of +11:00.
- Australia/Tasmania - Timezone for australia/tasmania UTC offset of +11:00.
- Australia/Victoria - Timezone for australia/victoria UTC offset of +11:00.
- Australia/West - Timezone for australia/west UTC offset of +08:00.
- Australia/Yancowinna - Timezone for australia/yancowinna UTC offset of +10:30.
- CET - Timezone for CET UTC offset of +01:00.
- CST6CDT - Timezone for CST6CDT UTC offset of -06:00.
- Canada/Atlantic - Timezone for canada/atlantic UTC offset of -04:00.
- Canada/Central - Timezone for canada/central UTC offset of -06:00.
- Canada/East-Saskatchewan - Timezone for canada/east-saskatchewan UTC offset of -06:00.
- Canada/Eastern - Timezone for canada/eastern UTC offset of -05:00.
- Canada/Mountain - Timezone for canada/mountain UTC offset of -07:00.
- Canada/Newfoundland - Timezone for canada/newfoundland UTC offset of -03:30.
- Canada/Pacific - Timezone for canada/pacific UTC offset of -08:00.
- Canada/Saskatchewan - Timezone for canada/saskatchewan UTC offset of -06:00.
- Canada/Yukon - Timezone for canada/yukon UTC offset of -08:00.
- Chile/Continental - Timezone for chile/continental UTC offset of -03:00.
- Chile/EasterIsland - Timezone for chile/easterisland UTC offset of -05:00.
- Cuba - Timezone for cuba UTC offset of -05:00.
- Default - Timezone for system UTC (OS time zone)

- EET - Timezone for EET UTC offset of +02:00.
- EST - Timezone for EST UTC offset of -05:00.
- EST5EDT - Timezone for EST5EDT UTC offset of -05:00.
- Egypt - Timezone for egypt UTC offset of +02:00.
- Eire - Timezone for eire UTC offset of +00:00.
- Etc/Greenwich - Timezone for etc/greenwich UTC offset of +00:00.
- Etc/UCT - Timezone for etc/uct UTC offset of +00:00.
- Etc/UTC - Timezone for etc/utc UTC offset of +00:00.
- Etc/Universal - Timezone for etc/universal UTC offset of +00:00.
- Etc/Zulu - Timezone for etc/zulu UTC offset of +00:00.
- Europe/Amsterdam - Timezone for europe/amsterdam UTC offset of +01:00.
- Europe/Andorra - Timezone for europe/andorra UTC offset of +01:00.
- Europe/Astrakhan - Timezone for europe/astrakhan UTC offset of +04:00.
- Europe/Athens - Timezone for europe/athens UTC offset of +02:00.
- Europe/Belfast - Timezone for europe/belfast UTC offset of +00:00.
- Europe/Belgrade - Timezone for europe/belgrade UTC offset of +01:00.
- Europe/Berlin - Timezone for europe/berlin UTC offset of +01:00.
- Europe/Bratislava - Timezone for europe/bratislava UTC offset of +01:00.
- Europe/Brussels - Timezone for europe/brussels UTC offset of +01:00.
- Europe/Bucharest - Timezone for europe/bucharest UTC offset of +02:00.
- Europe/Budapest - Timezone for europe/budapest UTC offset of +01:00.
- Europe/Busingen - Timezone for europe/busingen UTC offset of +01:00.
- Europe/Chisinau - Timezone for europe/chisinau UTC offset of +02:00.
- Europe/Copenhagen - Timezone for europe/copenhagen UTC offset of +01:00.
- Europe/Dublin - Timezone for europe/dublin UTC offset of +00:00.
- Europe/Gibraltar - Timezone for europe/gibraltar UTC offset of +01:00.
- Europe/Guernsey - Timezone for europe/guernsey UTC offset of +00:00.
- Europe/Helsinki - Timezone for europe/helsinki UTC offset of +02:00.
- Europe/Isle_of_Man - Timezone for europe/isle_of_man UTC offset of +00:00.
- Europe/Istanbul - Timezone for europe/istanbul UTC offset of +03:00.
- Europe/Jersey - Timezone for europe/jersey UTC offset of +00:00.

- Europe/Kaliningrad - Timezone for europe/kaliningrad UTC offset of +02:00.
- Europe/Kiev - Timezone for europe/kyiv UTC offset of +02:00.
- Europe/Kirov - Timezone for europe/kirov UTC offset of +03:00.
- Europe/Lisbon - Timezone for europe/lisbon UTC offset of +00:00.
- Europe/Ljubljana - Timezone for europe/ljubljana UTC offset of +01:00.
- Europe/London - Timezone for europe/london UTC offset of +00:00.
- Europe/Luxembourg - Timezone for europe/luxembourg UTC offset of +01:00.
- Europe/Madrid - Timezone for europe/madrid UTC offset of +01:00.
- Europe/Malta - Timezone for europe/malta UTC offset of +01:00.
- Europe/Mariehamn - Timezone for europe/mariehamn UTC offset of +02:00.
- Europe/Minsk - Timezone for europe/minsk UTC offset of +03:00.
- Europe/Monaco - Timezone for europe/monaco UTC offset of +01:00.
- Europe/Moscow - Timezone for europe/moscow UTC offset of +03:00.
- Europe/Nicosia - Timezone for europe/nicosia UTC offset of +02:00.
- Europe/Oslo - Timezone for europe/oslo UTC offset of +01:00.
- Europe/Paris - Timezone for europe/paris UTC offset of +01:00.
- Europe/Podgorica - Timezone for europe/podgorica UTC offset of +01:00.
- Europe/Prague - Timezone for europe/prague UTC offset of +01:00.
- Europe/Riga - Timezone for europe/riga UTC offset of +02:00.
- Europe/Rome - Timezone for europe/rome UTC offset of +01:00.
- Europe/Samara - Timezone for europe/samara UTC offset of +04:00.
- Europe/San_Marino - Timezone for europe/san_marino UTC offset of +01:00.
- Europe/Sarajevo - Timezone for europe/sarajevo UTC offset of +01:00.
- Europe/Simferopol - Timezone for europe/simferopol UTC offset of +03:00.
- Europe/Skopje - Timezone for europe/skopje UTC offset of +01:00.
- Europe/Sofia - Timezone for europe/sofia UTC offset of +02:00.
- Europe/Stockholm - Timezone for europe/stockholm UTC offset of +01:00.
- Europe/Tallinn - Timezone for europe/tallinn UTC offset of +02:00.
- Europe/Tirane - Timezone for europe/tirane UTC offset of +01:00.
- Europe/Tiraspol - Timezone for europe/tiraspol UTC offset of +02:00.
- Europe/Ulyanovsk - Timezone for europe/ulyanovsk UTC offset of +04:00.

- Europe/Uzhgorod - Timezone for europe/uzhgorod UTC offset of +02:00.
- Europe/Vaduz - Timezone for europe/vaduz UTC offset of +01:00.
- Europe/Vatican - Timezone for europe/vatican UTC offset of +01:00.
- Europe/Vienna - Timezone for europe/vienna UTC offset of +01:00.
- Europe/Vilnius - Timezone for europe/vilnius UTC offset of +02:00.
- Europe/Volgograd - Timezone for europe/volgograd UTC offset of +03:00.
- Europe/Warsaw - Timezone for europe/warsaw UTC offset of +01:00.
- Europe/Zagreb - Timezone for europe/zagreb UTC offset of +01:00.
- Europe/Zaporozhye - Timezone for europe/zaporozhye UTC offset of +02:00.
- Europe/Zurich - Timezone for europe/zurich UTC offset of +01:00.
- GB - Timezone for GB UTC offset of +00:00.
- GB-Eire - Timezone for GB-Eire UTC offset of +00:00.
- GMT - Timezone for GMT UTC offset of +00:00.
- GMT0 - Timezone for GMT0 UTC offset of +00:00.
- Greenwich - Timezone for greenwich UTC offset of +00:00.
- HST - Timezone for HST UTC offset of -10:00.
- Hongkong - Timezone for hongkong UTC offset of +08:00.
- Iceland - Timezone for iceland UTC offset of +00:00.
- Indian/Antananarivo - Timezone for indian/antananarivo UTC offset of +03:00.
- Indian/Chagos - Timezone for indian/chagos UTC offset of +06:00.
- Indian/Christmas - Timezone for indian/christmas UTC offset of +07:00.
- Indian/Cocos - Timezone for indian/cocos UTC offset of +06:30.
- Indian/Comoro - Timezone for indian/comoro UTC offset of +03:00.
- Indian/Kerguelen - Timezone for indian/kerguelen UTC offset of +05:00.
- Indian/Mahe - Timezone for indian/mahe UTC offset of +04:00.
- Indian/Maldives - Timezone for indian/maldives UTC offset of +05:00.
- Indian/Mauritius - Timezone for indian/mauritius UTC offset of +04:00.
- Indian/Mayotte - Timezone for indian/mayotte UTC offset of +03:00.
- Indian/Reunion - Timezone for indian/reunion UTC offset of +04:00.
- Iran - Timezone for iran UTC offset of +03:30.
- Israel - Timezone for israel UTC offset of +02:00.

- Jamaica - Timezone for jamaica UTC offset of -05:00.
- Japan - Timezone for japan UTC offset of +09:00.
- Kwajalein - Timezone for kwajalein UTC offset of +12:00.
- Libya - Timezone for libya UTC offset of +02:00.
- MET - Timezone for MET UTC offset of +01:00.
- MST - Timezone for MST UTC offset of -07:00.
- MST7MDT - Timezone for MST7MDT UTC offset of -07:00.
- Mexico/BajaNorte - Timezone for mexico/bajanorte UTC offset of -08:00.
- Mexico/BajaSur - Timezone for mexico/bajasur UTC offset of -07:00.
- Mexico/General - Timezone for mexico/general UTC offset of -06:00.
- NZ - Timezone for NZ UTC offset of +13:00.
- NZ-CHAT - Timezone for NZ-CHAT UTC offset of +13:45.
- Navajo - Timezone for navajo UTC offset of -07:00.
- PRC - Timezone for PRC UTC offset of +08:00.
- PST8PDT - Timezone for PST8PDT UTC offset of -08:00.
- Pacific/Apia - Timezone for pacific/apia UTC offset of +14:00.
- Pacific/Auckland - Timezone for pacific/auckland UTC offset of +13:00.
- Pacific/Bougainville - Timezone for pacific/bougainville UTC offset of +11:00.
- Pacific/Chatham - Timezone for pacific/chatham UTC offset of +13:45.
- Pacific/Chuuk - Timezone for pacific/chuuk UTC offset of +10:00.
- Pacific/Easter - Timezone for pacific/easter UTC offset of -05:00.
- Pacific/Efate - Timezone for pacific/efate UTC offset of +11:00.
- Pacific/Enderbury - Timezone for pacific/enderbury UTC offset of +13:00.
- Pacific/Fakaofo - Timezone for pacific/fakaofo UTC offset of +13:00.
- Pacific/Fiji - Timezone for pacific/fiji UTC offset of +13:00.
- Pacific/Funafuti - Timezone for pacific/funafuti UTC offset of +12:00.
- Pacific/Galapagos - Timezone for pacific/galapagos UTC offset of -06:00.
- Pacific/Gambier - Timezone for pacific/gambier UTC offset of -09:00.
- Pacific/Guadalcanal - Timezone for pacific/guadalcanal UTC offset of +11:00.
- Pacific/Guam - Timezone for pacific/guam UTC offset of +10:00.
- Pacific/Honolulu - Timezone for pacific/honolulu UTC offset of -10:00.

- Pacific/Johnston - Timezone for pacific/johnston UTC offset of -10:00.
- Pacific/Kiritimati - Timezone for pacific/kiritimati UTC offset of +14:00.
- Pacific/Kosrae - Timezone for pacific/kosrae UTC offset of +11:00.
- Pacific/Kwajalein - Timezone for pacific/kwajalein UTC offset of +12:00.
- Pacific/Majuro - Timezone for pacific/majuro UTC offset of +12:00.
- Pacific/Marquesas - Timezone for pacific/marquesas UTC offset of -09:30.
- Pacific/Midway - Timezone for pacific/midway UTC offset of -11:00.
- Pacific/Nauru - Timezone for pacific/nauru UTC offset of +12:00.
- Pacific/Niue - Timezone for pacific/niue UTC offset of -11:00.
- Pacific/Norfolk - Timezone for pacific/norfolk UTC offset of +11:00.
- Pacific/Noumea - Timezone for pacific/noumea UTC offset of +11:00.
- Pacific/Pago_Pago - Timezone for pacific/pago_pago UTC offset of -11:00.
- Pacific/Palau - Timezone for pacific/palau UTC offset of +09:00.
- Pacific/Pitcairn - Timezone for pacific/pitcairn UTC offset of -08:00.
- Pacific/Pohnpei - Timezone for pacific/pohnpei UTC offset of +11:00.
- Pacific/Ponape - Timezone for pacific/ponape UTC offset of +11:00.
- Pacific/Port_Moresby - Timezone for pacific/port_moresby UTC offset of +10:00.
- Pacific/Rarotonga - Timezone for pacific/rarotonga UTC offset of -10:00.
- Pacific/Saipan - Timezone for pacific/saipan UTC offset of +10:00.
- Pacific/Samoa - Timezone for pacific/samoa UTC offset of -11:00.
- Pacific/Tahiti - Timezone for pacific/tahiti UTC offset of -10:00.
- Pacific/Tarawa - Timezone for pacific/tarawa UTC offset of +12:00.
- Pacific/Tongatapu - Timezone for pacific/tongatapu UTC offset of +14:00.
- Pacific/Truk - Timezone for pacific/truk UTC offset of +10:00.
- Pacific/Wake - Timezone for pacific/wake UTC offset of +12:00.
- Pacific/Wallis - Timezone for pacific/wallis UTC offset of +12:00.
- Pacific/Yap - Timezone for pacific/yap UTC offset of +10:00.
- Poland - Timezone for poland UTC offset of +01:00.
- Portugal - Timezone for portugal UTC offset of +00:00.
- ROC - Timezone for ROC UTC offset of +08:00.
- ROK - Timezone for ROK UTC offset of +09:00.

- Singapore - Timezone for singapore UTC offset of +08:00.
- Turkey - Timezone for turkey UTC offset of +03:00.
- UCT - Timezone for UCT UTC offset of +00:00.
- US/Pacific-New - Timezone for US/Pacific-New UTC offset of -08:00.
- Universal - Timezone for universal UTC offset of +00:00.
- W-SU - Timezone for W-SU UTC offset of +03:00.
- WET - Timezone for WET UTC offset of +00:00.
- Zulu - Timezone for zulu UTC offset of +00:00.
- jdj_tz - Timezone for jdj_tz UTC offset of +00:00.

Name

vaSelectionAlgorithm - Value added location services selection algorithm

Syntax

--vaSelectionAlgorithm <list-item>

Description

Indicates for the set of value added location services ESMLCs (Evolved serving mobile location center) whether they will load share or function as primary/secondary.

Class

Optional.

List

- LoadShare - Load share - Load share.
- PrimarySecondary - Primary / secondary - Function as primary / secondary.

Roles**Modify**

sysadmin, manage

5.129 cmm nationalApn

National APN. Syntax, description, parameters, and roles for cmm nationalApn command.

Syntax**Description**

National APN.

Parameters

Name

plmnName - Mobile network name

Syntax

--plmnName <string>

Description

Name that identifies served mobile network.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

apnNi - Access point name - network identifier

Syntax

--apnNi <string>

Description

Access point name.

Class

Required key.

Type / Range

Digit string of length 1 to 62 with the allowed pattern of [0-9a-zA-Z\._]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.130 cmm nbiPing

nbiPing. Syntax, description, parameters, and roles for cmm nbiPing command.

Syntax

Description

Ping Command.

Parameters

Name

pingTarget - Ping target

Syntax

--pingTarget <string>

Description

Ping target can be IP address or DNS name.

Class

Required key.

Type / Range

Digit string of length 1 to 128 with the allowed pattern of [0-9a-zA-Z\.\-_:]*

Name

idx - Index

Syntax

--idx <integer>

Description

Index number for output

Class

Required key.

Type / Range

Integer in the range of 0 to 255

Name

byPassRouting - By pass routing

Syntax

--byPassRouting <list-item>

Description

Send the ping request to a host on a directly attached network by passing routing table.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

doNotFragment - Do not fragment

Syntax

--doNotFragment <list-item>

Description

Do not fragment specifies that request should not be fragmented.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

pingCount - Count

Syntax

--pingCount <integer>

Description

Number of ping requests to the host.

Class

Optional.

Type / Range

Integer in the range of 1 to 10,000

Default

5

Name

pingInterval - Interval

Syntax

--pingInterval <integer>

Description

Interval in seconds between ping.

Class

Optional.

Type / Range

Integer in the range of 1 to 10,000

Default

1

Name

pingOutput - Ping output

Syntax

--pingOutput <string>

Description

Ping output.

Class

Read only.

Type / Range

Digit string of length

Name

pingPattern - Pattern

Syntax

--pingPattern <string>

Description

16 byte string to be included in the ping.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 16 with the allowed pattern of [0-9a-fA-F]*

Default

0

Name

pingRptType - Ping report type

Syntax

--pingRptType <list-item>

Description

Valid values are rapid and normal.

Class

Optional.

List

- normal - Specify ping report type of normal.
- rapid - Specify ping report type of rapid.

Default

normal

Name

pingSize - Size

Syntax

--pingSize <integer>

Description

Size in bytes of ping requests.

Class

Optional.

Type / Range

Integer in the range of 1 to 65,507

Default

56

Name

pingSourceIp - Source IP address

Syntax

--pingSourceIp <string>

Description

Source IP address to be used for ping request.

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Default

0.0.0.0

Name

pingTimeOut - Timeout

Syntax

--pingTimeOut <integer>

Description

Timeout in sec.

Class

Optional.

Type / Range

Integer in the range of 1 to 100

Default

5

Name

timeToLive - Time to live

Syntax

--timeToLive <integer>

Description

Time to live TTL value in ping request.

Class

Optional.

Type / Range

Integer in the range of 1 to 255

Default

64

Name

typeOfService - Type of service

Syntax

--typeOfService <integer>

Description

Type of service bits in the IP header.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Roles**List**

sysadmin, manage, monitor

5.131 cmm nbiTraceroute

nbiTraceroute. Syntax, description, parameters, and roles for cmm nbiTraceroute command.

Syntax

Description

Traceroute Command.

Parameters

Name

trTarget - Traceroute target

Syntax

--trTarget <string>

Description

Traceroute target can be IP address or DNS name.

Class

Required key.

Type / Range

Digit string of length 1 to 128 with the allowed pattern of [0-9a-zA-Z\.\-_:]*

Name

idx - Index

Syntax

--idx <integer>

Description

Index number for output

Class

Required key.

Type / Range

Integer in the range of 0 to 255

Name

trNoDns - DNS reverse lookup allowed

Syntax

--trNoDns <list-item>

Description

Do not perform a DNS reverse-lookup name. by default a DNS lookup is performed.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

trOutput - Traceroute output

Syntax

--trOutput <string>

Description

Traceroute output.

Class

Read only.

Type / Range

Digit string of length

Name

trSourceIp - Source IP address

Syntax

--trSourceIp <string>

Description

Source IP address to be used for traceroute request.

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

trTos - Type of service

Syntax

--trTos <integer>

Description

Type of service bits in the IP header.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

trTtl - Time to live

Syntax

--trTtl <integer>

Description

Time to live TTL value in tr request.

Class

Optional.

Type / Range

Integer in the range of 1 to 255

Default

15

Name

trWaitTime - Wait time

Syntax

--trWaitTime <integer>

Description

Interval in seconds between tr.

Class

Optional.

Type / Range

Integer in the range of 1 to 60,000

Default

5000

Roles**List**

sysadmin, manage, monitor

5.132 cmm nbNbrGParams

NodeB Neighbor GParams. Syntax, description, parameters, and roles for cmm nbNbrGParams command.

Syntax

Description

Global parameters for creating and maintaining a NodeB neighbor map

Parameters

Name

cmGparmName - Parameter Name

Syntax

--cmGparmName <list-item>

Description

Global parameters for creating and maintaining a NodeB neighbor map

Class

Required key.

List

- gnbMapLocalRestoration - gNB Map Local Restoration - Specifies whether AMMS utilizes a locally stored copy of the neighbor gNB map rather than regenerating it following a restart or scale-out.
- gnbNeighborContUpdate - gNB Neighbor Continual Update - Specifies whether the gNB neighbor map shall continue to be updated after initial generation has completed.
- gnbNeighborGenInterval - gNB Neighbor Generation Interval - gNB neighbor map generation interval in days. 0 means the map is not created.
- gnbNeighborMinWeightCutoff - gNB Neighbor Minimum Weight Cutoff - Specifies the minimum weight that a system-discovered gNB pair must have to be considered as neighboring. The weight cutoff ranges from 0% to 100% by 1% increments.
- gnbNeighborPairMaxInterval - gNB Neighbor Pair Maximum Interval - Defines the maximum interval between successive idle UE access to the AMF to be used for the identification of neighboring gNB pairs.
- gnbNeighborSampleRate - gNB Neighbor Sample Rate - When creating or updating the gNB neighbor map, this value defines the percentage of UE accesses that are used.
- prioritizedUeHistoryCount - Prioritize UE History Count - Specifies the number of gNB from the UE history that are prioritized to the head of the paging set for Neighbor Level Paging Method. After the prioritized gNB(s), the remainder of the paging set is

Name

cmGparmValue - Parameter Value

Syntax

--cmGparmValue <string>

Description

Value of parameter

Class

Required on create.

Type / Range

Digit string of length 0 to 64

The cmGparmValue string is based on the cmGparmName parameter:

Name

gnbMapLocalRestoration - gNB Map Local Restoration

Syntax

--cmGparmName gnbMapLocalRestoration --cmGparmValue <string>

Description

Specifies whether AMMS utilizes a locally stored copy of the neighbor gNB map rather than regenerating it following a restart or scale-out.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

gnbNeighborContUpdate - gNB Neighbor Continual Update

Syntax

--cmGparmName gnbNeighborContUpdate --cmGparmValue <string>

Description

Specifies whether the gNB neighbor map shall continue to be updated after initial generation has completed.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

gnbNeighborGenInterval - gNB Neighbor Generation Interval

Syntax

```
--cmGparmName gnbNeighborGenInterval --cmGparmValue <string>
```

Description

gNB neighbor map generation interval in days. 0 means the map is not created.

Type / Range

Integer in the range of 0 to 100

Default

0

Name

gnbNeighborMinWeightCutoff - gNB Neighbor Minimum Weight Cutoff

Syntax

```
--cmGparmName gnbNeighborMinWeightCutoff --cmGparmValue <string>
```

Description

Specifies the minimum weight that a system-discovered gNB pair must have to be considered as neighboring. The weight cutoff ranges from 0% to 100% by 1% increments.

Type / Range

Integer in the range of 0 to 100

Default

0

Name

gnbNeighborPairMaxInterval - gNB Neighbor Pair Maximum Interval

Syntax

```
--cmGparmName gnbNeighborPairMaxInterval --cmGparmValue <string>
```

Description

Defines the maximum interval between successive idle UE access to the AMF to be used for the identification of neighboring gNB pairs.

Type / Range

Integer in the range of 0 to 30,000 in units of Second

Default

60

Name

gnbNeighborSampleRate - gNB Neighbor Sample Rate

Syntax

```
--cmGparmName gnbNeighborSampleRate --cmGparmValue <string>
```

Description

When creating or updating the gNB neighbor map, this value defines the percentage of UE accesses that are used.

Type / Range

Integer in the range of 0 to 100

Default

0

Name

prioritizedUeHistoryCount - Prioritize UE History Count

Syntax

```
--cmGparamName prioritizedUeHistoryCount --cmGparamValue <string>
```

Description

Specifies the number of gNB from the UE history that are prioritized to the head of the paging set for Neighbor Level Paging Method. After the prioritized gNB(s), the remainder of the paging set is

Type / Range

Integer in the range of 1 to 5

Default

5

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

5.133 cmm networkFunctionInstance

Network Function Instance. Syntax, description, parameters, and roles for cmm networkFunctionInstance command.

Syntax

Description

This command is used to create a new NF instance other than AMF and modify and delete an existing NF instance.

Parameters

Name

instanceId - Network Function Instance Identifier

Syntax

--instanceId <string>

Description

Uniquely identifies a network function. The instance ID is a UUID represented as a URN (string format). The operator must obtain the UUID from the NF configuration data. It is used for NF selection when NF discovery is based on target-nf-instance-id or preferred-nf-instances. Also may be used by AMF call processing when NF is selected with different criterions but NF Instance Id has to be stored for further usage. For example, a SMF may be selected based on DNN and S-NSSAI but the selected SMF instance Id for each PDU must be stored in UE Context as defined in 3GPP TS 29.518 (UeContext-->sessionContextList[x]-->hsmfId) The UUID must have the following string format: '8 hex digits-4 hex digits-4 hex digits-4 hex digits-12 hex digits'. Example: '4947a69a-f61b-4bc1-b9da-47c9c5d14b64'

Class

Required key.

Type / Range

Digit string of length 36 to 36 with the allowed pattern of [0-9a-fA-F]{8}\-[0-9a-fA-F]{4}\-[0-9a-fA-F]{4}\-[0-9a-fA-F]{4}\-[0-9a-fA-F]{12}

Default

No default

Name

nfType - Network Function Type

Syntax

--nfType <list-item>

Description

Specifies the type of Network Function. Used to match the NF discovery 'target-nf-type' Parameter. Note: while the NFType CBCF and AMF is available in the CLI they are not relevant. A CBCF it is never Discovered from AMF thus not related to local provisioning, while locally provisioned remote AMFs are created via rmtAmfInstance command.

Class

Required on create, not modifiable.

List

- 5GEIR - 5G EIR - The network function type of 5G-Equipment Identity Register.
- AMF - Access and Mobility Manager Function - The network function type of Access and Mobility Manager Function.
- AUSF - Authentication Server Function - The network function type of Authentication Server Function.
- CBCF - Cell Broadcast Centre Function - The network function type of Cell Broadcast Centre Function
- CHF - Charging Function - The network function type of Charging Function
- GMLC - Gateway Mobile Location Center - The network function type of Gateway Mobile Location Center.
- LMF - Location Management Function - The network function type of Location Management Function.
- NRF - Network Repository Function - The network function type of Network Repository Function.
- NSSAAF - Network Slice Specific Authentication and Authorization Function - The network function type of Network Slice Specific Authentication and Authorization Function
- NSSF - Network Slice Selection Function - The network function type of Network Slice Selection Function.
- NWDAF - Network Data Analytics Function - The network function type of NWDAF.
- PCF - Policy Control Function - The network function type of Policy Control Function.
- SMF - Session Management Function - The network function type of Session Management Function.
- SMSF - Short Message Service Function - The network function type of Short Message Service Function.
- UDM - Unified Data Manager - The network function type of Unified Data Manager.
- UDSF - Unstructured Data Storage Function - The network function type of Unstructured Data Storage Function.

Name

plmnName - PLMN Name

Syntax

--plmnName <string>

Description

Name of the PLMN. Must be set to CMM's own Home PLMN. Currently it is not allowed to be set to a Shared PLMN. Also not allowed to be set to Roaming PLMN (PLMN type 'Other') as local provisioning is not relevant for NF in different PLMNs

Class

Required on create.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

amfTaiListName - AMF TAI List Name

Syntax

--amfTaiListName <string>

Description

This parameter is pointer to a list of TAIs served by the SMF or the I-SMF. Used for NF selection when SMF discovery is based on tai or preferred-tai AMF may use TAI list (if provisioned for UE PLMN/serving PLMN combination) to select an SMF serving a set of TAIs. If the list is not provisioned then it is assumed that SMF serves all the TAIs in the serving PLMN. AMF uses TAI list to select an I-SMF serving a set of TAIs. If the list is not provisioned then it is assumed that I-SMF serves all the TAI in the serving PLMN. AMF also uses TAI list in SMF selection when I-SMF functionality is activated in the AMF and the preferred TAI is used for SMF selection for UE PLMN/serving PLMN combination. Note: The parameter is relevant only for SMF but named amfTaiListName to indicate that this list relevant for CMM AMF functions and distinguish from taiList command which is used for CMM's AMF functions

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

capacity - Relative capacity of the NF

Syntax

--capacity <integer>

Description

Relative capacity of the NF as defined in NFProfile level in 29.510. It is used for load balancing among NFs with equal Priority. As per 29.510 this parameter is used to deduce the NF Service instance Capacity when -and only when- the relevant information is not available on Service Instance level. However since locally provisioned Service instances always have a default capacity value this parameter has no effect for load balancing Service Requests (only relevant for Default Notifications)

Class

Optional.

Type / Range

Integer in the range of 0 to 65,535

Default

65535

Name

defaultNetworkFunction - Default Network Function

Syntax`--defaultNetworkFunction <list-item>`**Description**

The parameter is used to identify a NF treated as a default NF. If the AMF can not select a NF for a selection rule then the AMF will select the NF if provisioning allows the selection of a default NF (controlled via amfUePlmnServices defaultXXXSelection where 'XXX' is the relevant NF Type)

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

defaultNotificationSubscriptionListName - Default Notification Subscription List Name

Syntax`--defaultNotificationSubscriptionListName <string>`**Description**

References a list which links to NFs Default notification subscription Endpoints. Refers to the 3GPP TS 29.510 NFProfile-->defaultNotificationSubscriptions. This is relevant for GMLC and LMF. The list can be added either in NF Instance level or Service instance level. For GMLC only NF Instance level is relevant as GMLC has no services.

Class

Optional.

Type / RangeDigit string of length 0 to 31 with the allowed pattern of `[0-9a-zA-Z\.\_:@]*`**Default**

""

Name

dnnListName - Data Network List Name

Syntax

--dnnListName <string>

Description

The list consists of the DNN supported by the SMF. It is used for SMF selection based on dnn query parameter. All DNNs included in this list are assumed to support all the S-NSSAIs the SMF supports (i.e. not possible to define DNN per S-NSSAI). If a DNN list is not provisioned then it is assumed that the SMF supports all the DNNs. Note: DNN should be provisioned in the complete format (i.e. including the DNN-OI) when amfGParams --amfGparamName alwaysSendDnnWithOperatorIdentifier is enabled

Restrictions

This parameter is only applicable to SMF. All other NF types ignore this parameter if provisioned.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

emergencyServices - Emergency Services

Syntax

--emergencyServices <list-item>

Description

This parameter defines if the NF supports emergency services. This parameter is relevant only for GMLC and LMF.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

enable - Enable NF

Syntax

--enable <list-item>

Description

The parameter is used to enable use of the provisioned instance of a NF. If not enabled the NF will not be considered as candidate for selection.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

fqdn - FQDN

Syntax

--fqdn <string>

Description

The NF's FQDN. This is relevant only if all services of the NF use the same port/fqdn and the FQDN or nfEndPointListName information is not given in NF Service instance level. If both the FQDN and the Endpoints are provided, the Request URI is constructed with the FQDN but the IPs/port are deduced by the Endpoint list (i.e. no DNS query is issued) unless the endpoints include only Port in which case the port is deduced from the endpoint list and the IP is resolved via DNS query. Note: DNS resolution is not supported for NSSF and NNS resolution is not performed in case of Indirect communication with SCP. In that case the SCP is expected to resolve the IP of the FQDN

Restrictions

The FQDN parameter is for future use. The FID f20031-01 ignores this parameter.

Class

Optional.

Type / Range

Digit string of length 96 with the allowed pattern of [0-9a-zA-Z.-]*

Name

gpsiRangeListName - GPSI Range List

Syntax

--gpsiRangeListName <string>

Description

Name of the list consisting of GPSI ranges supported by NF. A NF without GPSI Range List is assumed to support all GPSIs. Relevant for PCF discovery based on gpsi when amfUePlmnServices - useGpsiForPcfSelection is true

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

groupId - UDM or AUSF Group ID

Syntax

--groupId <string>

Description

The GroupId the NF belongs to. Used for UDM and AUSF selection based on the Group ID (e.g. inter-AMF mobility). AMF stores the GroupId of the selected UDM or AUSF in the UE's context (e.g. when selection is based on SUPI due to absence of the GroupId) Note: This parameter is supported for AUSF and UDM starting CMM 22.0. In prior releases this parameter is not used. 3GPP also has defined groupId for PCF however CMM currently does not support using Group ID for PCF selection.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Default

""

Name

locality - Location of the Network Function

Syntax

--locality <string>

Description

This parameter is used to indicate location of the NF like geographical location, data center name, etc. When set, if the locality of the NF matches the locality of the own AMF (amfInstance --locality) the NF is selected with highest priority (if all other criterions are also met). If own AMF has not be provisioned with locality this parameter has no effect.

Class

Optional.

Type / Range

Digit string of length 0 to 64

Default

""

Name

nfIpEndPointListName - NF End Point IP List Name

Syntax

```
--nfIpEndPointListName <string>
```

Description

References a list which links to NFs endpoints (i.e. IP port combination). This parameter is relevant only if all services of the NF use the same port/fqdn and the fqdn or nfEndPointListName information is not given in NF Service instance level. Up to two endpoints per IP version can be provisioned. If both the FQDN and the Endpoints List are provided, the Request URI is constructed including the FQDN but the IPs and port are deduced by the Endpoint list (i.e. no DNS is query) unless the endpoints include only Port in which case the port is deduced from the endpoint list and the IP is resolved via DNS query.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

nfServicePersistence - NF Service Persistence

Syntax

```
--nfServicePersistence <list-item>
```

Description

When this value is True, it indicates that the different service instances of a same NF Service in this NF instance, supporting a same API version, are capable to persist their resource state in shared storage and therefore these resources are available after a new NF service instance supporting the same API version is selected by a NF Service Consumer. Otherwise, it indicates that the NF Service Instances of a same NF Service are not capable to share resource state inside the NF Instance. Currently this parameter is supported only for NRF ignored for all other NF Types. When set to True in NRF CMM will not create a new registration but will update existing when selecting new Management Service within the same NRF

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

nfServicesListName - NF Service List

Syntax

--nfServicesListName <string>

Description

The lists consists of the services supported by the NF. References a list which links to the NFs service instances the NF has. A NF may be provisioned to have multiple service instances for same or different service type.

Restrictions

This list must contain the services of the NF type.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

""

Name

nrfHeartBeatTimer - Heartbeat Timer Proposed From AMF To The NRF

Syntax

--nrfHeartBeatTimer <integer>

Description

Time in seconds expected between 2 consecutive heart-beat messages from the AMF to the NRF. When present the AMF will use this value in the registration request. NRF is allowed to override this value in the response.

Restrictions

This field is relevant only for NRF networkFunctionInstances

Class

Optional.

Type / Range

Integer in the range of 0 or 10 to 3,600 in units of Second

Default

60

Name

nsildListName - NSI ID List Name

Syntax

--nsiIdListName <string>

Description

References the list with the NSIs the NF belongs to. If not given NF is expected to belong to all NSIs. Only relevant for SMF selection from locally provisioned NFs when NSI is provided by NSSF.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

pgwFqdn - PGW FQDN

Syntax

--pgwFqdn <string>

Description

FQDN of PGW-C for combo SMF+PGW-C. Used for SMF selection during 4G-->5G mobility.

Class

Optional.

Type / Range

Digit string of length 96 with the allowed pattern of [0-9a-zA-Z.-]*

Default

"

Name

pgwFqdnListName - PGW FQDN List Name

Syntax

--pgwFqdnListName <string>

Description

Name of list of FQDNs of PGW-C for combo SMF+PGW-C. Used for SMF selection during 4G to 5G mobility.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

preferredNfTypeListName - Preferred NF Type List Name

Syntax

--preferredNfTypeListName <string>

Description

Name of the list that includes the NFs Types for which this NRF is preferred . When selecting NRF, CMM first checks if a list of preferred NF Types exists and includes the target NF. If so the corresponding NRF(s) are shortlisted first . Otherwise all NRF are considered valid candidates and selected based on priority, capacity etc.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

""

Name

priority - NF Priority

Syntax

--priority <integer>

Description

Relative Priority of the NF as defined in NFProfile level in 29.510. It is used for shortlisting candidate NFs. As per 29.510 this parameter is used to deduce the NF Service instance Priority when -and only when- the relevant information is not available on Service Instance level. However since locally provisioned Service instances always have a default priority value this parameter has no effect for shortlisting load balancing Service Requests (only relevant for Default Notifications)

Class

Optional.

Type / Range

Integer in the range of 0 to 65,535

Default

1

Name

routingIndicator - Routing Indicator

Syntax

--routingIndicator <digits>

Description

Routing Indicator, consisting of 1 to 4 decimal digits assigned by the home network operator and provisioned in the USIM, that allow together with the Home Network Identifier to route

network signalling with SUCI to AUSF and UDM instances capable to serve the subscriber. Each decimal digit present in the Routing Indicator shall be regarded as meaningful (e.g. value 012 is not the same as value 12). If a NF supports multiple Routing Indicators then they can be added in the Routing Indicator list referenced in the routingIndicatorListName parameter. If this parameter is set to one zero decimal digit (0) or four zero decimal digits (0000) then the NF supports all routing indicators.

Class

Optional.

Type / Range

Used for Routing Indicator which consists of 1 to 4 digits.

Default

0000

Name

routingIndicatorListName - Routing Indicator List Name

Syntax

```
--routingIndicatorListName <string>
```

Description

The Name of the Routing Indicator List. This list includes the Routing Indicators the NF serves. If the list is not provided or is empty and the routingIndicator is also set to digit 0 then the NF supports all routing indicators.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

snssaiListName - S-NSSAI List

Syntax

```
--snssaiListName <string>
```

Description

This list consists of the S-NSSAI the NF belongs to. Relevant only for slice specific NFs (NRF, PCF, SMF, NSSAAF). If a list is not provisioned then it is assumed that the NF supports all the S-NSSAI. When assigned to NRF this parameter is used to select Slice Specific NRF for SMF selection when NSSF is not used.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

supiRangeListName - SUPI Range list

Syntax

--supiRangeListName <string>

Description

Name of the list consisting of SUPI ranges supported by NF such as UDM, AUSF, PCF, NSSAAF and SMF (the later is Nokia Proprietary). The AMF selects NF based on the SUPI of the UE and SUPI range supported by the NF. Note that there is a difference in the logical interpretation of the absence of this list: in local UDM and AUSF selection absence of SUPI ranges list indicates that all SUPIs are supported while in SMF selection absence of SUPI ranges list indicates that this SMF does not pertain to any specific SUPI.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

v2xAllowed - V2X Allowed

Syntax

--v2xAllowed <list-item>

Description

Specifies whether the Network Function Instance supports v2x. This currently is applicable only for a PCF.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.134 cmm networkTrace

Network Trace. Syntax, description, parameters, and roles for cmm networkTrace command.

Syntax

Description

Specifies Network Trace (Tapping) specifications. This data is to assign local and remote addresses for the Network Trace agent.

Parameters

Name

hostName - Internal host name

Syntax

--hostName <string>

Description

Customer defined VM identifier.

Class

Required key.

Type / Range

Digit string of length 1 to 255

Name

localNtwkTracelp - Local trace IP address

Syntax

```
--localNtwkTraceIp <string>
```

Description

Local Network Trace address.

Class

Required on create, not modifiable.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

remoteNtwkTraceIp - Remote Trace IP address

Syntax

```
--remoteNtwkTraceIp <string>
```

Description

Remote Network Trace address.

Class

Required on create, not modifiable.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

destinationPort - Destination port

Syntax

```
--destinationPort <integer>
```

Description

Destination port

Class

Optional.

Type / Range

Integer in the range of 1,024 to 65,535

Default

5000

Name

endPoint - Remote endpoint

Syntax

```
--endPoint <string>
```

Description

Remote endpoint for the http connection

Class

Optional.

Type / Range

Digit string of length 0 to 255

Name

hep3MsgBundling - HEP3 Message Bundling

Syntax

--hep3MsgBundling <list-item>

Description

Choose if multiple HEP3 msgs need to be bundled into TCP/HTTP msgs

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

noTLS - Transport Layer Security

Syntax

--noTLS <list-item>

Description

Disables TLS

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

oamTlsProfileName - OAM TLS Profile

Syntax

--oamTlsProfileName <string>

Description

This parameter contains the profile for TLS. It contains the certificate names needed for establishing secure communication.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

sourcePort - Source port

Syntax

--sourcePort <integer>

Description

Source port

Class

Optional.

Type / Range

Integer in the range of 1,024 to 65,535

Default

5000

Roles**Create**

secadmin

Show

sysadmin, manage, secadmin

Modify

secadmin

Delete

secadmin

List

sysadmin, manage, secadmin

5.135 cmm nfIpEndPoint

NF IP Endpoint. Syntax, description, parameters, and roles for cmm nfIpEndPoint command.

Syntax

Description

This command is used to create an IP endpoint. The created IP endpoint can be provisioned to a NF instance or NF service instance other than AMF

Parameters

Name

name - Name

Syntax

--name <string>

Description

Name of the nflpEndpoint.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

ipAddress - IP Address

Syntax

--ipAddress <string>

Description

Provision IPv4 or IPv6 address.

Class

Optional.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Default

0.0.0.0

Name

port - TCP Port

Syntax

--port <integer>

Description

TCP port to be used for HTTP/TCP connection. Note: The port is not applicable if nflpEndpoint is configured to a networkFunctionInstance.

Class

Optional.

Type / Range

Integer in the range of 1 to 65,535

Default

8080

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.136 cmm nflpEndpointList

NF IP Endpoint List. Syntax, description, parameters, and roles for cmm nflpEndpointList command.

Syntax**Description**

Create a list for nflpEndpoints. A maximum of 256 lists can be created. Each list can consist of a maximum of 16 nflpEndpoints

Parameters**Name**

name - List Name

Syntax

--name <string>

Description

List name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.137 cmm nfIpEndpoints

NF IP Endpoints. Syntax, description, parameters, and roles for `cmm nfIpEndpoints` command.

Syntax**Description**

Add or delete a nfIpEndPoint to the named list. A maximum of 4 IPv4 and/or IPv6 nfIpEndpoints can be added.

Parameters**Name**

nfIpEndPointListName - Name of the List

Syntax

--nfIpEndPointListName <string>

Description

Specify name of the NF IP endpoint list.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

nfIpEndPointName - NF IP Endpoint Name

Syntax

--nfIpEndPointName <string>

Description

NF IP Endpoint Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.138 cmm nfServiceInstance

Network Service Instance Identifier.. Syntax, description, parameters, and roles for cmm nfServiceInstance command.

Syntax**Description**

This command is used to create a NF service instance and to modify and delete an existing NF service instance. For nfServiceInstances referring to own AMF, up to 16 service instances of the same service Type may be defined.

Parameters**Name**

name - Service Instance Identifier

Syntax

--name <string>

Description

The name of the service instance. It must be unique within CMM. Note: This is the key identifier of the Service instance used to reference the service when e.g. to adding to list. This is not the NFProfile-->nfServices [x]-->serviceInstanceId. This information, which may be used in various cases when NF is discovered via NRF (e.g. stored UeContext-->sessionContextList[x]-->smfServiceInstanceId) is currently not stored or provisioned for locally configured NF and thus the relevant use cases are not supported for locally configured NFs. This parameter is neither the nfProfile-->nfServices [x]-->serviceName Enumeration (e.g. nrrf-nfm, nausf-auth) which defines the type of the service. The Service type is deduced from the service parameter

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

service - Service Name

Syntax

--service <list-item>

Description

Name of the service offered by a NF. Note: Nudm_EventExposure and Nsmf_EventExposure may be provisioned but are currently not used by AMF. Also Namf_EventExposure, Namf_Location, Namf_MT and not relevant when provisioning remote AMF as CMM is the producer but never the consumer of these services.

Class

Required on create, not modifiable.

List

- N5g_eir_EquipmentIdentityCheck - 5GEIR Equipment Identity Check Service - The network function service type of 5GEIR Equipment Identity Check Service.
- Namf_Communication - AMF Communication Service - The network function service type of AMF Communication Service.
- Namf_EventExposure - AMF Event Exposure Service - The network function service type of AMF Event Exposure Service.
- Namf_Location - AMF Location Service - The network function service type of AMF Location Service.
- Namf_MT - AMF Mobile Terminated Service - The network function service type of AMF Mobile Terminated Service.
- Nausf_UEAuthentication - AUSF Authentication Service - The network function service type of AUSF Authentication Service.

- Nchf_ConvergedCharging service - The network function service type of Converged Charging Service.
- Nlmf_Location - LMF Location Service - The network function service type of LMF Location Service.
- Nnrf_AccessToken - NRF Access Token Request Service - The network function service type of NRF Access Token Request Service.
- Nnrf_NFDiscovery - NRF Network Function Discovery Service - The network function service type of NRF Network Function Discovery Service.
- Nnrf_NFManagement - NRF Network Function Management Service - The network function service type of NRF Network Function Management Service
- Nnssaaf_NSSAA - Nssaaf_Network Slice Specific Authentication and Authorization services - The network function service type of NSSAAF Network slice Specific Authentication and Authorization Service.
- Nnssf_NSSAIAvailability - NSSF Network Slice Availability Update Service - The network function service type of NSSF Network Slice Availability Update Service
- Nnssf_NSSelection - NSF Network Slice Selection Service - The network function service type of NSF Network Slice Selection Service.
- Nnwdaf_AnalyticsInfo - NWDAF Analytics Info Service - The Network Data Analytics Function Analytics Info Service
- Nnwdaf_EventsSubscription - NWDAF Events Subscription Service - The Network Data Analytics Function Events Subscription Service
- Npcf_AMPolicyControl - AM Policy Control service - The network function service type of AM Policy Control service.
- Npcf_UEPolicyControl - PCF UE Policy Control Service - The network function service type of PCF UE Policy Control Service.
- Nsmf_EventExposure - Nsmf Event Exposure - he network function service type of Nsmf Event Exposure Service
- Nsmf_PDUSession - SMF PDU Session Service - The network function service type of SMF PDU Session Service.
- Nsmsf_SMSservice - SMSF SM Service - The network function service type of SMSF SM Service.
- Nudm_EventExposure - UDM Event Exposure Services - The network function service type of UDM Event Exposure Services.
- Nudm_SubscriberDataManagement - UDM subscriber Data Management Service - The network function service type of UDM subscriber Data Management Service.
- Nudm_UEContextManagement - UDM UE Context Management Service - The network function service type of UDM UE Context Management Service.

- Nudsf_DataRepository - UDSF Data Repository Service - The network function service type of UDSF Data Repository Service.

Name

allowedNfDomainListName - Allowed NF Domain List Name

Syntax

--allowedNfDomainListName <string>

Description

The list with the NF domains the Service Instance is accessible to. When a list is provisioned, the AMF will add the NF domains included in the list in the the allowedNfDomains IE in the NF Service Registered to the NRF.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Default

""

Name

allowedNfTypeListName - Allowed NF Type List Name

Syntax

--allowedNfTypeListName <string>

Description

The list with the NF types the Service Instance is accessible to. When a list is provisioned, the AMF will add the NF types included in the list in the the allowedNfTypes IE in the NF Service Registered to the NRF.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Default

""

Name

allowedPlmnListName - Allowed PLMN List Name

Syntax

--allowedPlmnListName <string>

Description

The list with the PLMNs the Service Instance is accessible to. When a list is provisioned, the AMF will add the PLMNs included in the list in the the allowedPlmns IE in the NF Service Registered to the NRF.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

allowedSnssaiListName - Allowed S-NSSAI List Name

Syntax

--allowedSnssaiListName <string>

Description

The list with the S-NSSAIs the Service Instance is accessible to. When a list is provisioned, the AMF will add the S-NSSAIs included in the list in the the allowedNssais IE in the NF Service Registered to the NRF.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

apiMinorVersion - API Minor Version

Syntax

--apiMinorVersion <integer>

Description

For Service Instances belonging to remote Network functions this parameter defines the minor API version the AMF will use to encode messages as a Consumer. This parameter has no usage for Service Instances belonging to own AMF as the version for own Services is defined via sbiVersionProducerControlProfile CLI. Default value is 0

Class

Optional.

Type / Range

Integer in the range of 0 to 127

Default

0

Name

apiPatchVersion - API Patch Version

Syntax

--apiPatchVersion <integer>

Description

For Service Instances belonging to remote Network functions this parameter defines the Patch API version the AMF will use to encode messages as a Consumer. This parameter has no usage for Service Instances belonging to own AMF as the version for own Services is defined via sbiVersionProducerControlProfile CLI. Default value is 0

Class

Optional.

Type / Range

Integer in the range of 0 to 127

Default

0

Name

apiPrefix - API Prefix

Syntax

--apiPrefix <string>

Description

Optional path segment(s) used to construct the {apiRoot} variable of the different API URIs.

Class

Optional.

Type / Range

API Prefix as defined in 29.500. May be null otherwise shall have at least allowed character, shall not start with / and shall not end with /. When not-null, the leading / is implied and shall not be included.

Default

""

Name

apiVersionInUri - Service version

Syntax

--apiVersionInUri <list-item>

Description

Version of the service instance to be used in the URI for accessing the API. For Service Instances belonging to remote Network functions this parameter defines the Major API version the AMF will use to encode messages as a Consumer. For Service Instances belonging to own AMF this parameter defines the Major API version the AMF will use to encode messages as a Producer. Default value is v1

Class

Optional.

List

- v1 - Version 1
- v2 - Version 2
- v3 - Version 3
- v4 - Version 4

Default

v1

Name

capacity - Capacity

Syntax

--capacity <integer>

Description

Relative capacity of the NF as defined in NFService level in 3GPP TS 29.510. It is used for load balancing among NF Service instances with equal Priority. As per TS 29.510 this parameter supersedes the capacity given in NF level.

Class

Optional.

Type / Range

Integer in the range of 0 to 65,535

Default

65535

Name

defaultNotificationSubscriptionListName - Default Notification Subscription Name

Syntax

--defaultNotificationSubscriptionListName <string>

Description

Parameter to configure pointer to the Default Notification Subscriptions Endpoints. Refers to the 3GPP TS 29.510 NFProfile nfServices[x] defaultNotificationSubscriptions. The list can be added either in NF Instance level and/or Service instance level. For remote AMFs it must

be set in Service Instance as NF level is not supported. For GMLC only NF Instance level is relevant as GMLC has no services. If the list is given in both NF and service Instance level they are considered complementary (i.e. one does not supersede the other). Note: Though Default Notification Subscription can be placed within Service instance they do not relate to the service offered by the service instance as this is a Consumer interface while service instances represent Producer interfaces and attributes.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

fqdn - FQDN

Syntax

--fqdn <string>

Description

The NF Service Instance fqdn. When set, it supersedes the FQDN or nfEndPointListName given in NF level (networkFunctionInstance). If both the FQDN and the Endpoints are provided, the Request URI is constructed with the FQDN but the IPs/port are deduced by the Endpoint list (i.e. no DNS query issued) unless the endpoints include only Port in which case the port is deduced from the endpoint list and the IP is resolved via DNS query. Note: DNS resolution is not supported for NSSF and DNS resolution is not performed in case of Indirect communication with SCP. In that case the SCP is expected to resolve the IP of the FQDN. In case the NF Service Instance belongs to the own AMF, this FQDN is used as the fqdn of the relevant NFService of the NFProfile registered to NRF.

Class

Optional.

Type / Range

Digit string of length 96 with the allowed pattern of [0-9a-zA-Z.-]*

Name

httpScheme - HTTP Scheme

Syntax

--httpScheme <list-item>

Description

The scheme the Service instance supports (equivalent to 29.510 NFProfile-->nfServices[x]-->scheme) Note: In case of direct communication (model A), the scheme must match the scheme supported from the relevant consumer interface(s) in CMM. In case of indirect communication (model C), it must align to the setting of the relevant

sbiHttpProxySettingsProfile --targetNfScheme. Otherwise this service instance will never be considered as candidate.

Class

Optional.

List

- http - The schema of Hypertext Transfer Protocol.
- https - The schema of Hypertext Transfer Protocol Secure.

Default

https

Name

interPlmnFqdn - Inter PLMN FQDN

Syntax

--interPlmnFqdn <string>

Description

Inter PLMN FQDN to get the listening IP address of the service. This parameter does not have any use for Service instances related to remote NFs. It is only relevant for service instances belonging to CMM's AMF own Service Instances

Class

Optional.

Type / Range

Digit string of length 96 with the allowed pattern of [0-9a-zA-Z.-]*

Name

nfIpEndPointListName - NF End Point IP List Name

Syntax

--nfIpEndPointListName <string>

Description

This parameter points to list of pointers to IPv4 and/or IPv6 endpoints (i.e. IP port combination) of the NF or NF service instance. The list may contain 1 to 4 endpoints. When set, it supersedes the FQDN or nfEndPointListName given in NF level via networkFunctionInstance command. If both the FQDN and the Endpoints List is provided, the Request URI is constructed including the FQDN but the IPs and port are deduced by the Endpoint list (i.e. no DNS query issued) unless the endpoints include only Port in which case the port is deduced from the endpoint list and the IP is resolved via DNS query.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

""

Name

oauthCanary - OAuth Canary

Syntax

--oauthCanary <list-item>

Description

When this parameter is enabled and oauthRequired is also enabled the service instance will check the validity of the Access Tokens received in Requests and will produce OAuth related counters but will accept Requests that do not include Tokens or have invalid Tokens.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

oauthRequired - OAuth2.0 Required

Syntax

--oauthRequired <list-item>

Description

When this parameter is enabled oauthRequired OAuth based authorization is enforced in the Service Requests received by this service instance when general support of OAuth as a producer is also enabled in AMF.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

priority - Service Priority

Syntax

--priority <integer>

Description

Indicates priority relative to other services of the same type on a NF. Lower values indicate a higher priority. Relative Priority of the Service Instance as defined in NFService level in 3GPP TS 29.510. It is used for shortlisting NFs. As per TS 29.510 this parameter supersedes the priority given in NF level.

Class

Optional.

Type / Range

Integer in the range of 0 to 65,535

Default

1

Name

supportedFeatures - Supported Features

Syntax

--supportedFeatures <string>

Description

A string used to indicate the features supported by an API. Each API defines the bitmask supported by an API as defined in 3GPP TS 29.510 NFProfile--> nfServices[x]-->supportedFeatures

Class

Optional.

Type / Range

Digit string of length 0 to 10 with the allowed pattern of [0-9a-fA-F]*

Default

""

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.139 cmm nfServices

NF Services instances ID list. Syntax, description, parameters, and roles for **cmm nfServices** command.

Syntax**Description**

Command to add or delete a NF Services Instance ID to the specified list. This list is provisioned to a NF.

Parameters**Name**

nfServicesListName - List Name

Syntax

--nfServicesListName <string>

Description

Specifies the name of the list consisting of NF service instance IDs. The maximum number of services instances in a list is TBD. The list name has to created by using the 5GListName command.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

nfServiceInstanceName - NF Service Instance ID

Syntax

--nfServiceInstanceName <string>

Description

Assign a NF Service Instance ID

Restrictions

The list must have at least one NF service instance ID.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.140 cmm nfServicesList

NF Services List. Syntax, description, parameters, and roles for cmm nfServicesList command.

Syntax

Description

Create, modify and delete a NF services list.

Parameters

Name

name - List Name

Syntax

--name <string>

Description

Name of the NF services list.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

5.141 cmm ngRanInfoQuery

ngRan Info Query. Syntax, description, parameters, and roles for cmm ngRanInfoQuery command.

Syntax

Description

List the info of the ngRAN(s) corresponding to the typed IP address

Parameters

Name

amfLinkAdminState - AMF Link Admin State

Syntax

--amfLinkAdminState <list-item>

Description

Link admin state locked or unlocked

Class

Read only.

List

- blocked - Blocked State - only used for SBI_Server link types
- locked - Locked State - only used for SBI_Client, N2 and N26 links types
- unlocked - Unlocked State

Name

assignMode - Assignment Mode

Syntax

--assignMode <integer>

Description

This field depicts the assignment mode for the SNPN

Class

Optional.

Type / Range

Integer in the range of 0 to 2

Default

0

Name

information - information

Syntax

--information <string>

Description

Information about outcome of command

Class

Read only.

Type / Range

Digit string of length 0 to 256 with the allowed pattern of [0-9a-zA-Z\._:_]*

Name

linkIndex - Link index

Syntax

--linkIndex <integer>

Description

Link index.

Class

Read only.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

linkOperationState - Link Operation State

Syntax

--linkOperationState <list-item>

Description

Link operation state enabled or disabled.

Class

Read only.

List

- disabled - Disabled state of the link.

- enabled - Enable state of the link.
- unknown - Unknown state of the link. this can not be used during query.

Name

ngRanId - Global RAN Node ID

Syntax

--ngRanId <integer>

Description

Specifies Global RAN Node identifier as specified in TS 38.413

Class

Optional.

Type / Range

Integer in the range of 0 to 268,435,455

Name

ngRanIpAddress - ngRanIpAddress

Syntax

--ngRanIpAddress <string>

Description

Specifies TNLA RAN IP address

Class

Optional.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

ngRanIpAddress2 - ngRanIpAddress2

Syntax

--ngRanIpAddress2 <string>

Description

ngRan IP address 2

Class

Optional.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

ngRanName - ngRan node name

Syntax

--ngRanName <string>

Description

Specifies RAN node name

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 255

Name

ngRanType - NG-RAN Type

Syntax

--ngRanType <list-item>

Description

Specifies the NG-RAN type

Class

Read only.

List

- gnb - GNB
- n3iwf - N3IWF

Name

nidValue - Nid value

Syntax

--nidValue <integer>

Description

This field depicts the Nid value of the NID for the SNPN

Class

Optional.

Type / Range

Integer in the Range 0 to 1099511627775

Default

0

Name

remotePort - Remote Port

Syntax

--remotePort <integer>

Description

Remote Port associated with the remote IP address

Class

Read only.

Type / Range

Integer in the range of 0 to 65,534

Name

sliceDifferentiator - Slice Differentiator

Syntax

--sliceDifferentiator <string>

Description

The Slice Differentiator (SD) of the S-NSSAI supported in associated TAI

Class

Read only.

Type / Range

Digit string of length 1 or 6 with the allowed pattern of \-?[0-9A-F]*

Name

sliceServiceType - Slice Service Type

Syntax

--sliceServiceType <integer>

Description

The Slice Service Type (SST) of the S-NSSAI supported in associated TAI

Class

Read only.

Type / Range

5G Slice Service Type

Name

tac - Tracking area code

Syntax

--tac <integer>

Description

Tracking area code

Class

Read only.

Type / Range

Integer in the range of 1 to 16,777,215

Name

taiMcc - TAI MCC

Syntax

--taiMcc <digits>

Description

3-digit number used to represent the mobile country code for TAI

Class

Read only.

Type / Range

Digit string of length 3

Name

taiMnc - TAI MNC

Syntax

--taiMnc <digits>

Description

Digits used to represent mobile network code for TAI

Class

Read only.

Type / Range

Digit string of length 2 to 3

Roles**List**

sysadmin, manage, monitor

5.142 cmm ngRanTaiQuery

ng-eNB TAI Query. Syntax, description, parameters, and roles for cmm ngRanTaiQuery command.

Syntax**Description**

Lists TAls of ng RAN (without slice specific info per gNB/ng-eNB) 'mcc', 'mnc' and 'ngRanId' are required input filters. If any of these are not given query will return empty list.

Parameters

Name

mcc - Mobile country code

Syntax

--mcc <digits>

Description

3-digit number used to represent the mobile country code.

Class

Required key.

Type / Range

Digit string of length 3

Name

mnc - Mobile network code

Syntax

--mnc <digits>

Description

Digits used to represent mobile network code.

Class

Required key.

Type / Range

Digit string of length 2 to 3

Name

ngRanId - Global RAN Node ID

Syntax

--ngRanId <integer>

Description

Specifies Global RAN Node identifier as specified in TS 38.413

Class

Required key.

Type / Range

Integer in the range of 0 to 268,435,455

Name

assignMode - Assignment Mode

Syntax

--assignMode <integer>

Description

This field depicts the assignment mode for the SNPN

Class

Optional.

Type / Range

Integer in the range of 0 to 2

Default

0

Name

ngRanIpAddress - RAN IP address

Syntax

--ngRanIpAddress <string>

Description

Specifies RAN IP address

Class

Read only.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

ngRanName - ngRan node name

Syntax

--ngRanName <string>

Description

Specifies RAN node name

Class

Read only.

Type / Range

Digit string of length 0 or 1 to 255

Name

ngRanType - NG-RAN Type

Syntax

--ngRanType <list-item>

Description

Specifies the NG-RAN type.

Class

Optional.

List

- gnb - GNB
- n3iwf - N3IWF

Name

nidValue - Nid value

Syntax

--nidValue <integer>

Description

This field depicts the Nid value of the NID for the SNPN

Class

Optional.

Type / Range

Integer in the Range 0 to 1099511627775

Default

0

Name

tac - Tracking area code

Syntax

--tac <integer>

Description

Tracking area code.

Class

Read only.

Type / Range

Integer in the range of 1 to 16,777,215

Name

taiMcc - TAI MCC

Syntax

--taiMcc <digits>

Description

3-digit number used to represent the mobile country code for TAI.

Class

Read only.

Type / Range

Digit string of length 3

Name

taiMnc - TAI MNC

Syntax

--taiMnc <digits>

Description

Digits used to represent mobile network code for TAI.

Class

Read only.

Type / Range

Digit string of length 2 to 3

Roles**List**

sysadmin, manage, monitor

5.143 cmm ngRanTaiSnssaiQuery

ng-eNB TAI/Snssai Query. Syntax, description, parameters, and roles for cmm ngRanTaiSnssaiQuery command.

Syntax**Description**

Displays all S-NSSAIs per TAIs of a specific ng RAN gNB/ng-eNB 'mcc', 'mnc' and 'ngRanId' are required input filters. If any of these are not given query will return empty list.

Parameters**Name**

mcc - Mobile country code

Syntax

--mcc <digits>

Description

3-digit number used to represent the mobile country code.

Class

Required key.

Type / Range

Digit string of length 3

Name

mnc - Mobile network code

Syntax

--mnc <digits>

Description

Digits used to represent mobile network code.

Class

Required key.

Type / Range

Digit string of length 2 to 3

Name

ngRanId - Global RAN Node ID

Syntax

--ngRanId <integer>

Description

Specifies Global RAN Node identifier as specified in TS 38.413

Class

Required key.

Type / Range

Integer in the range of 0 to 268,435,455

Name

assignMode - Assignment Mode

Syntax

--assignMode <integer>

Description

This field depicts the assignment mode for the SNPN

Class

Optional.

Type / Range

Integer in the range of 0 to 2

Default

0

Name

ngRanIpAddress - RAN IP address

Syntax

--ngRanIpAddress <string>

Description

Specifies RAN IP address

Class

Read only.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

ngRanName - ngRan node name

Syntax

--ngRanName <string>

Description

Specifies RAN node name

Class

Read only.

Type / Range

Digit string of length 0 or 1 to 255

Name

ngRanType - NG-RAN Type

Syntax

--ngRanType <list-item>

Description

Specifies the NG-RAN type.

Class

Optional.

List

- gnb - GNB
- n3iwf - N3IWF

Name

nidValue - Nid value

Syntax

--nidValue <integer>

Description

This field depicts the Nid value of the NID for the SNPN

Class

Optional.

Type / Range

Integer in the Range 0 to 1099511627775

Default

0

Name

sliceDifferentiator - Slice Differentiator

Syntax

--sliceDifferentiator <string>

Description

The S-NSSAI consists of a Slice Service Type (SST) and a Slice Differentiator (SD). Each slice must have SST and SD is optional. A SST may have multiple SDs.

Class

Read only.

Type / Range

Identifies Slice Differentiator (SD)

Default

FFFFFF

Name

sliceServiceType - Slice Service Type

Syntax

--sliceServiceType <integer>

Description

The S-NSSAI consists of a Slice Service Type (SST) and a Slice Differentiator (SST). Each slice must have SST and SD is optional.

Class

Read only.

Type / Range

5G Slice Service Type

Name

tac - Tracking Area Code

Syntax

--tac <integer>

Description

5G TAC. 5G TAC is 3 octets long.

Class

Read only.

Type / Range

Integer in the range of 0 to 16777215

Name

taiMcc - TAI MCC

Syntax

--taiMcc <digits>

Description

3-digit number used to represent the mobile country code for TAI.

Class

Read only.

Type / Range

Digit string of length 3

Name

taiMnc - TAI MNC

Syntax

--taiMnc <digits>

Description

Digits used to represent mobile network code for TAI.

Class

Read only.

Type / Range

Digit string of length 2 to 3

Roles**List**

sysadmin, manage, monitor

5.144 cmm ngRanTnlaQuery

NG-RAN gNB TNLA Query. Syntax, description, parameters, and roles for cmm ngRanTnlaQuery command.

Syntax

Description

Lists TNLAs of an NG-RAN gNB. 'mcc', 'mnc' and 'ngRanId' are required input filters. If any of these are not given query will return empty list.

Parameters

Name

mcc - Mobile country code

Syntax

--mcc <digits>

Description

3-digit number used to represent the mobile country code.

Class

Required key.

Type / Range

Digit string of length 3

Name

mnc - Mobile network code

Syntax

--mnc <digits>

Description

Digits used to represent mobile network code.

Class

Required key.

Type / Range

Digit string of length 2 to 3

Name

ngRanId - Global RAN Node ID

Syntax

--ngRanId <integer>

Description

Specifies Global RAN Node identifier as specified in TS 38.413

Class

Required key.

Type / Range

Integer in the range of 0 to 268,435,455

Name

alarmState - Alarm State

Syntax

--alarmState <list-item>

Description

Link alarm state.

Class

Read only.

List

- critical - Critical alarm.
- major - Major alarm.
- minor - Minor alarm.
- none - No alarm.
- unknown - Unknown.

Name

amfIpAddress - TNLA AMF IP address

Syntax

--amfIpAddress <string>

Description

Specifies TNLA AMF IP address

Class

Read only.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

amfPort - TNLA AMF port number

Syntax

--amfPort <integer>

Description

Specifies the TNLA AMF port number

Class

Read only.

Type / Range

Integer in the range of 0 to 65,535

Name

assignMode - Assignment Mode

Syntax

--assignMode <integer>

Description

This field depicts the assignment mode for the SNPN

Class

Optional.

Type / Range

Integer in the range of 0 to 2

Default

0

Name

information - Information

Syntax

--information <string>

Description

Information

Class

Read only.

Type / Range

Digit string of length 0 to 256 with the allowed pattern of [0-9a-zA-Z\._:-]*

Name

linkAdminState - Link Admin State

Syntax

--linkAdminState <list-item>

Description

Link admin state.

Class

Read only.

List

- blocked - Blocked State - only used for SBI_Server link types
- locked - Locked State - only used for SBI_Client, N2 and N26 links types
- unlocked - Unlocked State

Name

linkIndex - AMF Link Index

Syntax

--linkIndex <integer>

Description

AMF link index

Class

Read only.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

linkOperationState - Link Operation State

Syntax

--linkOperationState <list-item>

Description

Link operation state.

Class

Read only.

List

- disabled - Disabled state of the link.
- enabled - Enable state of the link.
- unknown - Unknown state of the link. this can not be used during query.

Name

ngRanIpAddress - TNLA RAN IP address

Syntax

--ngRanIpAddress <string>

Description

Specifies TNLA RAN IP address

Class

Read only.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

ngRanPort - TNLA RAN port number

Syntax

--ngRanPort <integer>

Description

Specifies the TNLA RAN port number

Class

Read only.

Type / Range

Integer in the range of 0 to 65,535

Name

ngRanType - NG-RAN Type

Syntax

--ngRanType <list-item>

Description

Specifies the NG-RAN type.

Class

Read only.

List

- gnb - GNB
- n3iwf - N3IWF

Name

nidValue - Nid value

Syntax

--nidValue <integer>

Description

This field depicts the Nid value of the NID for the SNPN

Class

Optional.

Type / Range

Integer in the Range 0 to 1099511627775

Default

0

Name

poolId - Pool Id

Syntax

--poolId <integer>

Description

IPDS Pool ID - ID will only be 0 for non Multi-IPDS instantiations.

Class

Read only.

Type / Range

Integer in the range of 0 to 31

Roles**List**

sysadmin, manage, monitor

5.145 cmm ngReset

NG Reset. Syntax, description, parameters, and roles for cmm ngReset command.

Syntax**Description**

This command will send an NG Reset with 'Reset All' message to the specified gNB to reset all the NGconnections from that gNB to the AMF.

Parameters**Name**

linkIndex - Link Index

Syntax

--linkIndex <integer>

Description

This parameter is the link index of the NG link to be Reset

Class

Required.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

ranNodeName - RAN Node Name

Syntax

--ranNodeName <string>

Description

RAN Node Name received by the AMF in the NG Setup Message. Used to identify a unique gNB.

Class

Optional.

Type / Range

ASN.1 Printable String

Roles

sysadmin, manage

6 CMM CLI commands: cmm nrfCacheAdmin - cmm sbiVersionControlProfileList

CMM CLI command descriptions: syntax, usage, parameters, and user roles.

6.1 cmm nrfCacheAdmin

NRF Cache Administration. Syntax, description, parameters, and roles for `cmm nrfCacheAdmin` command.

Syntax

Description

Command to administer NRF cache. AMF stores the NF profiles obtained from the NRF cache. The command can be used to perform one of the following: - Output NRF cache. - Flush NRF cache. The flush operation will result in deleting all the NF profile in the cache and forces AMF to query NRF to get fresh NF profile data. - Delete a NF profile from the cache. This operation requires NF instance to be specified. - Delete NF profiles of all NFs of a type. This operation requires NF instance to be specified. - Display NF profile of a NF Instance. This operation requires NF instance to be specified. - Display Isolated (a.k.a. 'blacklisted') Service Instances. This operation requires both NF instance and NF Type to be specified.

Parameters

Name

instanceId - Network Function Instance Identifier

Syntax

--instanceId <string>

Description

Uniquely identifies a network function. The instance ID has to be specified for operations to display or delete NF profile of a NF instance

Class

Optional.

Type / Range

Digit string of length 36 to 36 with the allowed pattern of [0-9a-fA-F]{8}\-[0-9a-fA-F]{4}\-[0-9a-fA-F]{4}\-[0-9a-fA-F]{4}\-[0-9a-fA-F]{12}

Name

nfType - Network Function Type

Syntax

--nfType <list-item>

Description

NF Type has to be specified for Display NF Type and Delete NF Type operations.

Class

Optional.

List

- 5GEIR - 5G EIR - The network function type of 5G-Equipment Identity Register.
- AMF - Access and Mobility Manager Function - The network function type of Access and Mobility Manager Function.
- AUSF - Authentication Server Function - The network function type of Authentication Server Function.
- GMLC - Gateway Mobile Location Center - The network function type of Gateway Mobile Location Center.
- LMF - Location Management Function - The network function type of Location Management Function.
- NRF - Network Repository Function - The network function type of Network Repository Function.
- NSSAAF - Network Slice Specific Authentication and Authorization Function - The network function type of Network Slice Specific Authentication and Authorization Function.
- PCF - Policy Control Function - The network function type of Policy Control Function.
- SEPP - Security Edge Protection Proxy - The network function type of Security Edge Protection Proxy.
- SMF - Session Management Function - The network function type of Session Management Function.
- SMSF - Short Message Service Function - The network function type of Short Message Service Function.
- UDM - Unified Data Manager - The network function type of Unified Data Manager.
- UDSF - Unstructured Data Storage Function - The network function type of Unstructured Data Storage Function.

Name

nrfBlacklist - Blacklisted services

Syntax

--nrfBlacklist <string>

Description

Blacklisted services

Class

Read only.

Type / Range

Digit string of length

Name

nrfFileName - Data File Name

Syntax

--nrfFileName <string>

Description

The data file name from the operation

Class

Read only.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Roles**Modify**

sysadmin, manage

List

sysadmin, manage, monitor

6.2 cmm nrfIsolationGeneralSettings

NRF Isolation Settings. Syntax, description, parameters, and roles for cmm nrfIsolationGeneralSettings command.

Syntax**Description**

Includes general setting related to NRF Isolation.

Parameters**Name**

service - Service

Syntax

--service <list-item>

Description

NRF service for which the settings in the other parameters apply.

Class

Required key.

List

- Nnrf_NFDiscovery - NRF Network Function Discovery Service - The network function service type of NRF Network Function Discovery Service.

Default

Nnrf_NFDiscovery

Name

action3xx - Action 3XX

Syntax

--action3xx <list-item>

Description

Default action for 3xx Status class Responses

Class

Optional.

List

- Isolate - Isolate the Service Instance
- KPI_Fail - KPI Fail - Increase the Failure Counter used for the Service Instance KPI
- KPI_Success - KPI Success - Increase the Success counter used for the Service Instance KPI

Default

KPI_Fail

Name

action4xx - Action 4XX

Syntax

--action4xx <list-item>

Description

Default action for 4XX Status class Responses

Class

Optional.

List

- Isolate - Isolate the Service Instance
- KPI_Fail - KPI Fail - Increase the Failure Counter used for the Service Instance KPI
- KPI_Success - KPI Success - Increase the Success counter used for the Service Instance KPI

Default

KPI_Success

Name

action5xx - Action 5XX

Syntax

--action5xx <list-item>

Description

Default action for 5XX Status class Responses

Class

Optional.

List

- Isolate - Isolate the Service Instance
- KPI_Fail - KPI Fail - Increase the Failure Counter used for the Service Instance KPI
- KPI_Success - KPI Success - Increase the Success counter used for the Service Instance KPI

Default

KPI_Fail

Name

actionTimeout - Action Timeout

Syntax

--actionTimeout <list-item>

Description

Action for Timeouts

Class

Optional.

List

- Isolate - Isolate the Service Instance
- KPI_Fail - KPI Fail - Increase the Failure Counter used for the Service Instance KPI
- KPI_Success - KPI Success - Increase the Success counter used for the Service Instance KPI

Default

KPI_Fail

Name

enable - Enable

Syntax


```
--enable <list-item>
```

Description

Enables/disables enhanced Service Instance Isolation functionality

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

isolationDuration - Isolation Duration

Syntax

```
--isolationDuration <integer>
```

Description

Duration (in seconds) the Isolation lasts. 0 effectively means no Isolation takes place

Class

Optional.

Type / Range

Integer in the range of 0 to 3,600

Default

300

Name

kpiIsolationThreshold - KPI Isolation Threshold

Syntax

```
--kpiIsolationThreshold <integer>
```

Description

Threshold of Success/Attempts percent under which a Service Instance will be isolated. When value is set to 0 the Service Instance is not isolated based on KPIs.

Class

Optional.

Type / Range

Integer in the range of 0 to 100

Default

90

Name

kpiMinSamples - KPI Minimum Samples

Syntax

--kpiMinSamples <integer>

Description

Minimum number of Requests that are required in order to consider Isolation for a Service Instance based on KPIs

Class

Optional.

Type / Range

Integer in the range of 1 to 1,000,000

Default

100

Name

kpiSamplingPeriod - KPI Sampling Period

Syntax

--kpiSamplingPeriod <integer>

Description

Sampling Period (in seconds) for which response success and failures are aggregated to produce KPI for specific Service Instance.

Class

Optional.

Type / Range

Integer in the range of 5 to 300

Default

30

Name

totalIsolationDuration - Total Isolation Duration

Syntax

--totalIsolationDuration <integer>

Description

Duration (in seconds) the Isolation will last when all Service Instances are isolated. When set to 0 Isolation is effectively removed from all services when all relevant Services are isolated

Class

Optional.

Type / Range

Integer in the range of 0 to 3,600

Default

60

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

6.3 cmm nrflsolationPerCauseCodeSettings

NRF Isolation Per Cause Code Settings . Syntax, description, parameters, and roles for cmm nrflsolationPerCauseCodeSettings command.

Syntax**Description**

Includes action for specific Cause/Status codes. Used when the action for a Causes/Status codes combination for a service needs to differentiate from the default behaviour provisioned for the relevant status class within the nrflsolationGeneralSettings CLI.

Parameters**Name**

service - Service

Syntax

--service <list-item>

Description

NRF service for which the settings in the other parameters apply.

Class

Required key.

List

- Nnrf_NFDDiscovery - NRF Network Function Discovery Service - The network function service type of NRF Network Function Discovery Service.

Default

Nnrf_NFDiscovery

Name

statusCode - Status Code

Syntax

--statusCode <list-item>

Description

The HTTP Status Code

Class

Required key.

List

- 400_Bad_Request - 400 Bad Request
- 401_Unauthorized - 401 Unauthorized
- 403_Forbidden - 403 Forbidden
- 404_Not_Found - 404 Not Found
- 405_Method_Not_Allowed - 405 Method Not Allowed
- 406_Not_Acceptable - 406 Not Acceptable
- 408_Request_Timeout - 408 Request Timeout
- 409_Conflict - 409 Conflict
- 410_Gone - 410 Gone
- 411_Length_Required - 411 Length Required
- 412_Precondition_Failed - 412 Precondition Failed
- 413_Payload_Too_Large - 413 Payload Too Large
- 414_URI_Too_Long - 414 URI Too Long
- 415_Unsupported_Media_Type - 415 Unsupported Media Type
- 429_Too_Many_Requests - 429 Too Many Requests
- 500_Internal_Server_Error - 500 Internal Server Error
- 501_Not_Implemented - 501 Not Implemented
- 502_Bad_Gateway - 502 Bad Gateway
- 503_Service_Unavailable - 503 Service Unavailable
- 504_Gateway_Timeout - 504 Gateway Timeout

Name

errorCause - Error Cause

Syntax

--errorCause <string>

Description

Application Error Cause value. Used to applies string match with the Cause in the response. 'NULL' string literal has special meaning and is used to match cases where Response did not include Application Error Cause

Class

Required key.

Type / Range

SBI Application Error Cause value. Digit string of length 1 to 64 with the allowed pattern of [A-Z_]*

Default

"NULL"

Name

action - Action

Syntax

--action <list-item>

Description

Action to be applied for specific status code and Error cause.

Class

Optional.

List

- Isolate - Isolate the Service Instance
- KPI_Fail - KPI Fail - Increase the Failure Counter used for the Service Instance KPI
- KPI_Success - KPI Success - Increase the Success counter used for the Service Instance KPI

Default

KPI_Fail

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.4 cmm nriConfig

nriConfig. Syntax, description, parameters, and roles for **cmm nriConfig** command.

Syntax**Description**

Show network resource information (NRI) configuration

Parameters

N/A

Roles**List**

sysadmin, manage, monitor

6.5 cmm nse

Network Service Entity. Syntax, description, parameters, and roles for **cmm nse** command.

Syntax**Description**

Configure, list and show Network Service Entity.

Parameters**Name**

nseld - Network service entity identifier

Syntax

--nseId <integer>

Description

Network service entity identifier. this parameter defines the NSE. the NSEI provides the network management functionality required for the operation of the gb interface. the NSEI is used in the SGSN and in the BSS to determine the NS-VCs that provide services to the base station system GPRS protocol (BSSGP) and the virtual connection identifiers (BVCI). the value must be the same both in the BSS and in the SGSN, and it must be unique in the SGSN. the value can range from 0 to 65535.

Class

Required key.

Type / Range

Integer in the range of 0 to 65,535

Name

localUdpPort - Local udp port

Syntax

--localUdpPort <integer>

Description

Local UDP port number the parameter is obligatory in the IP configuration and unique.

Class

Required on create.

Type / Range

Integer in the range of 50,000 to 65,535

Name

nseName - NSE name

Syntax

--nseName <string>

Description

Network service entity name use this parameter to assign a name containing 1 to 10 characters to the NSE. the first character must be a letter. the name must be unique in the SGSN. by default, the NSENAME parameter does not have a name.

Class

Required on create.

Type / Range

Digit string of length 1 to 10 with the allowed pattern of [0-9a-zA-Z\._:-]*

Name

connectionType - Connection type

Syntax

--connectionType <list-item>

Description

Connection type DYN ... dynamic IP configuration STA ... static IP configuration.

Class

Not modifiable.

List

- DYN - Dynamic - Dynamic IP configuration.
- STA - Static - Static IP configuration.

Default

DYN

Name

nseIpType - NSE IP type

Syntax

--nseIpType <list-item>

Description

Allowed IP type of the configured NSE

Class

Not modifiable.

List

- IPV4 - Allowed IP type IPv4
- IPV6 - Allowed IP type IPv6

Default

IPV4

Name

papsEndPts - Paps end points

Syntax

--papsEndPts <string>

Description

This parameter is used to define to which PAPS it is needed to create the endpoint. the value can range from 0 to 20. It is possible to enter the values of PAPS by using the ',' and '-' in order to list ranges of PAPS. If this parameter hasn't taken any value, the system shares the capacity between all the active paps. the parameter is given only in the dynamic IP configuration. all the PAPS to which it is needed to create an endpoint should be given at the same time, because it is not allowed to add individual paps to the system.

Class

Optional.

Type / Range

Digit string of length 0 to 86 with the allowed pattern of `^(?=[0-9]|1[0-9])([0-9]|1[0-9])({,19}$0?,?-?1?,?-?2?,?-?3?,?-?4?,?-?5?,?-?6?,?-?7?,?-?8?,?-?9?,?-(10)?,?-(11)?,?-(12)?,?-(13)?,?-(14)?,?-(15)?,?-(16)?,?-(17)?,?-(18)?,?-(19)?$`

Default

""

Name

sfunc - Sfunc

Syntax

--sfunc <integer>

Description

NSE special functionality 0 ... general 1 ... no RESUME-NACK bit for this NSE is on --> send RESUME-ACK.

Class

Optional.

Type / Range

Integer in the range of 0 to 1

Default

0

Name

weightMode - Weight mode

Syntax

--weightMode <integer>

Description

Configuration mode 0 ... auto balance 1 ... operator configurable.

Class

Optional.

Type / Range

Integer in the range of 0 to 1

Default

0

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.6 cmm nseDynEp

NSE dynamic endpoint. Syntax, description, parameters, and roles for cmm nseDynEp command.

Syntax

Description

Configure, list and show NSE dynamic endpoint. The purpose of nseDynEp is to modify the weights(data/ signalling) of an already configured endpoint.

Parameters

Name

nseId - NSE id

Syntax

--nseId <integer>

Description

Network service virtual connection identifier.

Class

Required key.

Type / Range

Integer in the range of 0 to 163,840

Name

papsId - PAPS id

Syntax

--papsId <integer>

Description

This parameter defines the paps id for which the weights should be configured.

Class

Required key.

Type / Range

Integer in the range of 0 to 19

Name

weightData - Local data weight

Syntax

--weightData <integer>

Description

Local data weight for scheduling priority.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Name

weightSign - Local sign weight

Syntax

--weightSign <integer>

Description

Local sign weight for scheduling priority.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

6.7 cmm nsildList

NSI ID List. Syntax, description, parameters, and roles for cmm nsiIdList command.

Syntax

Description

Create a list for NSI ID. Only a single list can be created.

Parameters

Name

name - Name of the list

Syntax

--name <string>

Description

Provide name of the NSSI ID list.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.8 cmm nsilds

NSI ID. Syntax, description, parameters, and roles for cmm nsiIds command.

Syntax

Description

Add a NSI ID to a NSI ID list specified by a list name.

Parameters

Name

nsildListName - NSI ID List Name

Syntax

--nsiIdListName <string>

Description

Name of the NSI ID list

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

nsild - NSI ID

Syntax

--nsiId <string>

Description

NSI ID

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.9 cmm nssfCacheAdmin

NSSF Cache Administration. Syntax, description, parameters, and roles for cmm nssfCacheAdmin command.

Syntax

Description

Command to administer NSSF cache. The command can be used to perform one of the following: - List NSSF cache. - Modify Cache. Either NSSF or local provisioning cache may be flushed: --> NSSF: This command is used to delete the cache and query NSSF for fresh S-NSSAI per TA data for own AMF. The AMF will issue Nnssf_NSSAIAvailability Update request immediately. -->Local provisioning: This command will recreate the local provisioned cache from the provisioning database. While cache is automatically updated whenever provisioning changes, this command may be used as a heal mechanism in case data is inconsistent.

Parameters

Name

cacheSource - Cache Source

Syntax

--cacheSource <list-item>

Description

The cache source for which the operation is applicable. Required only for cache flush operations. Not relevant for listoperation.

Class

Optional.

List

- nssf - Apply cache operation on content which is formed by information received from NSSF
- prov - Local Provisioning - Apply operation on cache content which is formed by local provisioning

Name

nssfCacheOperation - NSSF Cache Operation

Syntax

--nssfCacheOperation <list-item>

Description

Select operation of be performed on the NSSF cache.

Class

Optional.

List

- flushCache - Flush NSSF Cache - Deletes NSSF cache contents for the selected source.

Default

flushCache

Name

nssfFileName - Data File Name

Syntax

--nssfFileName <string>

Description

The data file name from the operation

Class

Read only.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Roles**Modify**

sysadmin, manage

List

sysadmin, manage, monitor

6.10 cmm nsvc

network service virtual connection. Syntax, description, parameters, and roles for cmm nsvc command.

Syntax**Description**

Configure, list and show NSVC - network service virtual connection to the network service entity.

Parameters**Name**

nsvcid - NSVC id

Syntax

--nsvcId <integer>

Description

Network service virtual connection identifier.

Class

Required key.

Type / Range

Integer in the range of 0 to 65,535

Name

nseld - NSE id

Syntax

--nseId <integer>

Description

Network service entity identifier.

Class

Required on create, not modifiable.

Type / Range

Integer in the range of 0 to 65,535

Name

nsvcName - NSVC name

Syntax

--nsvcName <string>

Description

Network service virtual connection name.

Class

Required on create.

Type / Range

Digit string of length 1 to 10 with the allowed pattern of [0-9a-zA-Z\._ :]*

Name

papsId - Unit log index

Syntax

--papsId <integer>

Description

Unit log index.

Class

Required on create, not modifiable.

Type / Range

Integer in the range of 0 to 19

Name

remoteDataWeight - Remote data weight

Syntax

--remoteDataWeight <integer>

Description

Remote data weight.

Class

Required on create.

Type / Range

Integer in the range of 0 to 255

Name

remotelpEndpoint - Remote IP endpoint

Syntax

--remoteIpEndpoint <string>

Description

Remote IP endpoint.

Class

Required on create.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

remoteSignWeight - Remote sign weight

Syntax

--remoteSignWeight <integer>

Description

Remote sign weight.

Class

Required on create.

Type / Range

Integer in the range of 0 to 255

Name

remoteUdpPort - Remote udp port

Syntax

--remoteUdpPort <integer>

Description

Remote UDP port number.

Class

Required on create.

Type / Range

Integer in the range of 0 to 65,535

Name

localUdpPort - Local udp port

Syntax

--localUdpPort <integer>

Description

Local UDP port number.

Class

Read only.

Type / Range

Integer in the range of 50,000 to 65,535

Name

nsvcStatus - User status

Syntax

--nsvcStatus <list-item>

Description

Network service virtual connection state.

Class

Optional.

List

- locked - Locked.
- unlocked - Unlocked.

Default

locked

Name

remoteIpEndpointVersion - Remote IP endpoint version

Syntax

--remoteIpEndpointVersion <list-item>

Description

Remote IP endpoint version.

Class

Read only.

List

- IPV4 - Specify trace entity of IPv4.

- IPV6 - Specify trace entity of IPv6.

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.11 cmm nsvcAdmin

network service virtual connection administration. Syntax, description, parameters, and roles for cmm nsvcAdmin command.

Syntax

Description

NSVC administration Change state of NSVCs.

Parameters

Name

state - Administrative state

Syntax

--state <list-item>

Description

Network service virtual connection administrative state.

Class

Required key.

List

- locked - Locked.
- unlocked - Unlocked.

Default

locked

Name

nseld - NSE

Syntax

--nseId <integer>

Description

Network service entity identifier.

Class

Optional.

Type / Range

Integer in the range of 0 to 65,535

Default

0

Name

papsId - PAPS identifier

Syntax

--papsId <integer>

Description

PAPS which NSVCs administrative state is changed.

Restrictions

0..19

Class

Optional.

Type / Range

Integer in the range of 0 to 19

Default

0

Roles**Modify**

sysadmin, manage

6.12 cmm nsvcEndps

Control NSVC endpoints. Syntax, description, parameters, and roles for cmm nsvcEndps command.

Syntax

Description

Control all dynamic NSVC endpoints from a certain PAPS

Parameters

Name

papsId - PAPS Identifier

Syntax

--papsId <integer>

Description

It identifies the PAPS Id for which this command should run for

Class

Required on create.

Type / Range

Integer in the range of 0 to 19

Roles

Add

sysadmin, manage

Remove

sysadmin, manage

6.13 cmm ntpAuthenticate

NTP Authenticate. Syntax, description, parameters, and roles for **cmm ntpAuthenticate** command.

Syntax

Description

This command is used to configure NTP authentication

Parameters

Name

ntpServer - NTP Server

Syntax

```
--ntpServer <string>
```

Description

This parameter specifies the NTP Server that given NTP authentication mechanism (symmetric key or NTS) will be used.

Restrictions

Given address should be already configured as NTP Server.

Class

Required key.

Type / Range

Digit string of length 1 to 128 with the allowed pattern of [0-9a-zA-Z\.\-:~@]*

Name

keyFormat - key format

Syntax

```
--keyFormat <list-item>
```

Description

This parameter specifies key's format.

Class

Optional.

List

- ASCII - ASCII format
- HEX - HEX format

Default

HEX

Name

keyId - key ID

Syntax

```
--keyId <integer>
```

Description

This parameter defines the key's ID that will be used to the relevant NTP server. Key ID 0 is used when no key has been defined. If NTS is enabled (Yes), keyId has no effect.

Class

Optional.

Type / Range

Integer in the range of 0 to 65,535

Default

0

Name

keyType - key type

Syntax

--keyType <list-item>

Description

This parameter specifies key's type. The type is a name of a cryptographic hash function or cipher which is used to generate and verify the MAC.

Class

Optional.

List

- AES128 - AES128 function is used for authentication.
- AES256 - AES256 function is used for authentication.
- SHA1 - SHA1 function is used for authentication.
- SHA256 - SHA256 function is used for authentication.
- SHA384 - SHA384 function is used for authentication.
- SHA512 - SHA512 function is used for authentication.

Default

SHA1

Name

keyValue - key value

Syntax

--keyValue <string>

Description

This parameter specifies key's actual value.

Class

Optional.

Type / Range

Digit string of length 0, 20 to 1024

Default

"

Name

nts - NTS

Syntax

--nts <list-item>

Description

This parameter enables Network Time Security (NTS) authentication to the relevant NTP server. If NTS is enabled (Yes), keyId has no effect.

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

oamTlsProfileName - OAM TLS Profile

Syntax

--oamTlsProfileName <string>

Description

This parameter contains the profile for TLS used if nts is yes. It contains the certificate names needed for establishing secure communication.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Roles

Create

secadmin

Show

secadmin

Modify

secadmin

Delete

secadmin

List

secadmin

6.14 cmm ntpUpdate

NTP Update. Syntax, description, parameters, and roles for cmm ntpUpdate command.

Syntax

Description

This command allows the update of the CMM NTP server(s). This command is only applicable to VNF and appliance. It is not allowed for CNF.

Parameters

Name

updateId - Id

Syntax

--updateId <string>

Description

Just type config here.

Class

Required key.

Type / Range

Digit string of length 6 to 6 with the allowed pattern of [0-9a-zA-Z\._ :]*

Name

ntpPrimaryIp - NTP primary IP

Syntax

--ntpPrimaryIp <string>

Description

NTP primary IP.

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

ntpSecondaryIp - NTP secondary IP

Syntax

--ntpSecondaryIp <string>

Description

NTP secondary IP.

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Roles**Modify**

secadmin

Show

secadmin

6.15 cmm nwdafUeMobilityProfile

NWDAF Profile. Syntax, description, parameters, and roles for cmm nwdafUeMobilityProfile command.

Syntax**Description**

Specifies the communication parameters with an NWDAF

Parameters**Name**

name - Name

Syntax

--name <string>

Description

Name of the NWDAF Profile

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

accuracy - Accuracy

Syntax

--accuracy <list-item>

Description

Requested Accuracy for NWDAF Predictions

Class

Optional.

List

- high - High Accuracy requested
- low - Low Accuracy Requested

Default

high

Name

endTs - End Time

Syntax

--endTs <string>

Description

Ending Time of the window for which predictions are requested. . The default is blank (no time specified). Valid range is 1980-01-01 00:00:00 through 2037-12-31 23:59:59 or blank. This value, if non-blank, must be greater than startTs.

Class

Optional.

Type / Range

Digit string of length 0 or 19 to 19 with the allowed pattern of (\d{4}-\d{2}-\d{2} \d{2}:\d{2}:\d{2})*\d{2})*

Default

"

Name

maxReports - Maximum Number of Reports

Syntax

--maxReports <integer>

Description

Specifies the maximum number of reports, after which the subscription ceases to exist. If blank, the subscription does not request maximum reports. See TS 29.523 Section 5.6.2.4.

Class

Optional.

Type / Range

Integer in the range of 1 to 25,000

Default

100

Name

monitoringDuration - Monitoring Duration

Syntax

--monitoringDuration <string>

Description

The time at which a subscription is requested to expire. The default is blank (no time specified). Valid range is 1980-01-01 00:00:00 through 2037-12-31 23:59:59 or blank. If blank, there is no time limit. See TS 29.523 Section 5.6.2.4.

Class

Optional.

Type / Range

Digit string of length 0 or 19 to 19 with the allowed pattern of (\d{4}-\d{2}-\d{2} \d{2}:\d{2}:\d{2})*\d{2})*

Default

"

Name

repetitionPeriod - Repetition Period

Syntax

--repetitionPeriod <integer>

Description

The requested interval between periodic reports received from the NWDAF.

Class

Optional.

Type / Range

Integer in the range of 1 to 60,000 in units of Second

Default

600

Name

serviceApi - Service API

Syntax

--serviceApi <list-item>

Description

Specifies the Service API to be used for requesting UE_MOBILITY predictive analytics.

Class

Optional.

List

- Nnwdaf_AnalyticsInfo - The Nnwdaf_AnalyticsInfo Service API.
- Nnwdaf_EventsSubscription - The Nnwdaf_EventsSubscription Service API.

Default

Nnwdaf_EventsSubscription

Name

startTs - Start Time

Syntax`--startTs <string>`**Description**

Starting Time of the window for which predictions are requested. The default is blank (no time specified). Valid range is 1980-01-01 00:00:00 through 2037-12-31 23:59:59 or blank.

Class

Optional.

Type / Range

Digit string of length 0 or 19 to 19 with the allowed pattern of `(\d{4}-\d{2}-\d{2} \d{2}:\d{2}:\d{2})*\d{2}`

Default

"

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.16 cmm oamDns

OAM DNS Configurations . Syntax, description, parameters, and roles for cmm oamDns command.

Syntax

Description

This command is used to administer OAM DNS configurations and includes parameters for defining the external DNS IP addresses. The IP addresses must be added in order and no holes can be left during deletions.

Parameters

Name

dnsProfileName - OAM DNS ID

Syntax

--dnsProfileName <string>

Description

Name of the OAM DNS configuration.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

priDnsIp - DNS primary ID

Syntax

--priDnsIp <string>

Description

The IP address of the primary external DNS.

Class

Required on create.

Type / Range

Digit string of length 3 to 39 with valid IPv4 or IPv6 address

Name

secDnsIp - DNS secondary ID

Syntax

--secDnsIp <string>

Description

The IP address of the secondary external DNS.

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Default

""

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.17 cmm oamLogins

OAM Logins. Syntax, description, parameters, and roles for `cmm oamLogins` command.

Syntax**Description**

This command is used to configure the maximum logins/sessions per OAM user.

Parameters**Name**

user - User

Syntax

--user <string>

Description

OAM user name.

Class

Required key.

Type / Range

User name in Linux format. Username must start with a letter, a digit, an underscore or a full stop. The last character must be a letter, a digit, an underscore, a hyphen, a dollar sign or a full stop. The other characters must be letters, digits, underscores, hyphens or full stops. The length must be between 1 and 32 characters.

Name

dualFactorAuthentication - Flag to enable or disable Dual-Factor Authentication for user accounts

Syntax

--dualFactorAuthentication <list-item>

Description

This parameter activates the Dual-Factor Authentication for the user accounts, using PASSWORD and CRYPTOGRAPHIC KEYS simultaneously. Activation can be done after the user has ssh public keys installed. This will force the user to enter password once ssh key matches.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

maxLogins - Maximum logins

Syntax

--maxLogins <integer>

Description

Maximum logins/sessions per OAM user.

Class

Optional.

Type / Range

Integer in the range of 1 to 20

Default

20

Roles

Create

secadmin

Show

secadmin

Modify

secadmin

Delete

secadmin

List

secadmin

6.18 cmm oamSession

OAM Session. Syntax, description, parameters, and roles for cmm oamSession command.

Syntax

Description

This command is used to configure the OAM users interactive session time-out period.

Parameters

Name

userGroup - User Group

Syntax

--userGroup <list-item>

Description

User Group

Class

Required key.

List

- security - security user group
- admin - admin user group
- manage - manage (/configurator) user group
- monitor - monitor user group

- ligroup - lawful interception (li) user group

Name

timeout - time out

Syntax

--timeout <integer>

Description

Inactivity OAM session time out

Class

Optional.

Type / Range

Integer in the range of 0 or 30 to 28,800 in units of Second

Default

3600

Roles**Show**

secadmin

Modify

secadmin

List

secadmin

6.19 cmm oamTlsCAcert

OAM TLSCAcert. Syntax, description, parameters, and roles for cmm oamTlsCAcert command.

Syntax**Description**

This command creates the CA certificate to be loaded into openssl for OAM users. The Certificate is used to validate certificates received from other tls entities.

Parameters**Name**

name - OAM TLS Certificate Name

Syntax

--name <string>

Description

OAM TLS CA Certificate Name. This is the pem file for the CA authority.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles**Create**

secadmin

Delete

secadmin

List

secadmin

6.20 cmm oamTlsCAcertList

OAM TLSCAcert List. Syntax, description, parameters, and roles for `cmm oamTlsCAcertList` command.

Syntax**Description**

This command is to create/delete OAM TLS Certified Authority certificate name.

Parameters**Name**

name - OAM TLS CA Cert list

Syntax

--name <string>

Description

OAM TLS Certified Authority Certificate

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

secadmin

Show

secadmin

Delete

secadmin

List

secadmin

6.21 cmm oamTlsCertificate

OAM tlsCertificate. Syntax, description, parameters, and roles for cmm oamTlsCertificate command.

Syntax

Description

OAM TLS Server Certificate.

Parameters

Name

name - OAM TLS Certificate Name

Syntax

--name <string>

Description

OAM TLS Certificate Name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

type - certificate type

Syntax

--type <string>

Description

certificate type

Class

Read only.

Type / Range

Digit string of length 0 to 20

Roles**Create**

secadmin

Delete

secadmin

List

secadmin

6.22 cmm oamTlsCertificateList

OAM tlsCertificateList. Syntax, description, parameters, and roles for cmm oamTlsCertificateList command.

Syntax**Description**

This command is to create/delete TLS server certificate name.

Parameters**Name**

name - OAM TLS Certificate Name

Syntax

--name <string>

Description

OAM TLS Certificate Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles**Create**

secaadmin

Show

secaadmin

Delete

secaadmin

List

secaadmin

6.23 cmm oamTlsClientCert

OAM tlsClientCert. Syntax, description, parameters, and roles for cmm oamTlsClientCert command.

Syntax

Description

OAM TLS Client Certificate.

Parameters

Name

name - OAM TLS Client Certificate Name

Syntax

--name <string>

Description

OAM TLS Client Certificate Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

type - client certificate type

Syntax

--type <string>

Description

client certificate type

Class

Read only.

Type / Range

Digit string of length 0 to 20

Roles**Create**

secadmin

Delete

secadmin

List

secadmin

6.24 cmm oamTlsClientCertList

OAM TlsClientCertList. Syntax, description, parameters, and roles for `cmm oamTlsClientCertList` command.

Syntax**Description**

This command is to create/delete OAM TLS client certificate name.

Parameters**Name**

name - OAM TLS Client Certificate Name

Syntax

--name <string>

Description

OAM TLS Client Certificate Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of `[0-9a-zA-Z\.\_:@]*`

Roles**Create**

secadmin

Show

seadmin

Delete

seadmin

List

seadmin

6.25 cmm oamTlsProfile

OAM Tls Profile. Syntax, description, parameters, and roles for cmm oamTlsProfile command.

Syntax

Description

Create, modify and delete a Transport Layer Security profile for OAM services. The CA certificate must be provisioned. Server and Client certificates may be provisioned based on the service need.

Parameters

Name

name - OAM TLS Profile Name

Syntax

--name <string>

Description

OAM TLS Profile Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

mutualTlsAuth - Mutual TLS Authentication

Syntax

--mutualTlsAuth <list-item>

Description

This parameter indicates if the Service should ask to Authenticate the Client through mutual Authentication. If the EndPoint associated with this Profile is a Client this parameter is ignored.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

tlsCACertificateName - TLS CA Certificate Name

Syntax

--tlsCACertificateName <string>

Description

Configure TLS CA Root certificate used in establishing secure communication.

Restrictions

This element is only applicable when interfaceName is a SBI interface.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

tlsCAIMCertName - TLS CA IM Cert Name

Syntax

--tlsCAIMCertName <string>

Description

Configure TLS CA IM certificate used in establishing secure communication.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

tlsClientCertName - TLS Client Certificate Name

Syntax

--tlsClientCertName <string>

Description

Configure TLS Client certificate used in establishing secure communication.

Restrictions

This element is only applicable when interfaceName is a SBI interface.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

tlsServerCertName - TLS Server Certificate Name

Syntax

--tlsServerCertName <string>

Description

Configure TLS Server certificate used in establishing secure communication.

Restrictions

This element is only applicable when interfaceName is a SBI interface.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

tlsVersion - TLS Version

Syntax

--tlsVersion <list-item>

Description

TLS version used to set up secure TCP connection

Class

Optional.

List

- 1.2 - Version 1.2

- 1.3 - Version 1.3
- negotiation - Negotiation of TLS 1.3 OR 1.2

Default

1.2

Roles**Create**

secadmin

Show

sysadmin, manage, secadmin

Modify

secadmin

Delete

secadmin

List

sysadmin, manage, secadmin

6.26 cmm oamTlsRevocation

oam TLS Revocation. Syntax, description, parameters, and roles for `cmm oamTlsRevocation` command.

Syntax**Description**

OAM TLS Revocation CRL - A file of Revoke Certificate Serial Numbers. This is a complete list of Revoked Certificates not a delta.

Parameters**Name**

name - OAM TLS Revocation Name

Syntax

--name <string>

Description

OAM TLS CA Revocation Name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles**Create**

secadmin

Delete

secadmin

List

secadmin

6.27 cmm oamTlsRevocationList

OAM tls Revocation List. Syntax, description, parameters, and roles for cmm oamTlsRevocationList command.

Syntax**Description**

This command is to create/delete OAM TLS Revocation List name.

Parameters**Name**

name - OAM TLS Revocation List

Syntax

--name <string>

Description

OAM TLS Revocation List Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles**Create**

secadmin

Show

secadmin

Delete

secadmin

List

secadmin

6.28 cmm oauthAccessTokenKey

OAuth 2.0 Access Token Key. Syntax, description, parameters, and roles for cmm oauthAccessTokenKey command.

Syntax

Description

OAuth 2.0 Access Token Key. Syntax, description, parameters, and roles for the cmm oauth2AccessTokenKey command.

Parameters

Name

oauthAccessTokenKeyListName - OAuth Access Token Key Name

Syntax

--oauthAccessTokenKeyListName <string>

Description

OAuth Access Token Key. Not applicable for the list operation.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

keyFile - Key File Name

Syntax

--keyFile <string>

Description

Absolute path of the key file name. Example: /tmp/accesstoken.key

Class

Required on create, not modifiable.

Type / Range

Digit string of length 1 to 100 with the allowed pattern of [0-9a-zA-Z\.\-_:&()/]*

Name

content - Key Content

Syntax

--content <string>

Description

Key content

Class

Optional.

Type / Range

Digit string of length 1 to 100 with the allowed pattern of [0-9a-zA-Z\.\-_:]*

Name

lineNumber - Content Line Number

Syntax

--lineNumber <integer>

Description

Line number of the key content

Class

Read only.

Type / Range

Integer in the range of 1 to 100

Roles**Create**

secadmin

Delete

secadmin

List

secadmin

6.29 cmm oauthAccessTokenKeyList

OAuth Access Token Key List. Syntax, description, parameters, and roles for cmm oauthAccessTokenKeyList command.

Syntax

Description

OAuth access token key names

Parameters

Name

name - OAuth Access Token Key Name

Syntax

--name <string>

Description

Name of the OAuth access token key

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

secadmin

Show

secadmin

Delete

secadmin

List

secadmin

6.30 cmm oauthNrfCertificate

OAuth2.0 NRF Certificates. Syntax, description, parameters, and roles for cmm oauthNrfCertificate command.

Syntax

Description

This command manages certificates of the NRFs acting as the Authorization Server for AMF. The certificate that matches the UUID and algorithm indicated in the the OAuth2.0 Token is used for the signature validation.

Parameters

Name

name - Name

Syntax

--name <string>

Description

Name of the certificate

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

signedCertFile - Signed Certificate File Name

Syntax

--signedCertFile <string>

Description

Absolute path of the signed certificate file name. Example: /tmp/certificate/nrf1.crt

Class

Required on create, not modifiable.

Type / Range

Digit string of length 1 to 100 with the allowed pattern of [0-9a-zA-Z\._&(),/]*

Name

content - certificate content

Syntax

--content <string>

Description

certificate content

Class

Optional.

Type / Range

String of length 0 to 120

Name

kid - Key Identifier

Syntax

--kid <string>

Description

The key associated with the certificate. It is used along with the Access Token issuer and the algorithm when selecting the locally provisioned NRF Public Keys to be used for signature validation.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Default

""

Name

lineNumber - content line number

Syntax

--lineNumber <integer>

Description

line number of the certificate content

Class

Optional.

Type / Range

Integer in the range of 0 to 65,535

Roles**Create**

secadmin

Delete

secadmin

List

secadmin

6.31 cmm oauthProfile

OAuth Profile. Syntax, description, parameters, and roles for cmm oauthProfile command.

Syntax**Description**

Administer oauth Profile.

Parameters

Name

name - OAuth Profile Name

Syntax

--name <string>

Description

OAuth profile name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

asNrfConnectionAuditTimer - AS-NRF Connection Audit Timer

Syntax

--asNrfConnectionAuditTimer <integer>

Description

This parameter controls how often CMM periodically audits the state of the link to the AS-NRFs. If the link does not exist CMM will try to create one so as to acquire the Public Key. The audit is relevant only when CMM is provisioned to support multiple Public Keys for OAuth token validation (i.e. use of public keys of AS-NRFs which CMM is not currently registered to)

Class

Optional.

Type / Range

Integer in the range of 1 to 60 in units of Minute

Default

30

Name

consumerOAuthEnable - Consumer OAuth Enable

Syntax

--consumerOAuthEnable <list-item>

Description

Parameter controls if the AMF as a consumer uses OAuth for the authorization of the API access. When the parameter is set to 'true', the AMF consumer shall support OAuth for authorization of the API access. When the parameter is set to 'false', the AMF consumer does not support OAuth for authorization. By default, this parameter shall be set to 'true'.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

consumerTokenPrefetch - Consumer Access Token Prefetch Period

Syntax

--consumerTokenPrefetch <integer>

Description

This parameter defines the period (in seconds) within which AMF will try to prefetch a new Access Token which is about to expire. If the Token to be used has expiration within the prefetch period AMF will issue a background Request to NRF to get a new one.

Class

Optional.

Type / Range

Integer in the range of 0 to 3,600 in units of Second

Default

0

Recommended

It is recommended to set the prefetch period to 60 seconds or more.

Name

oauthAccessTokenKeyListName - OAuth Access Token Key List Name

Syntax

--oauthAccessTokenKeyListName <string>

Description

OAuth Access Token Key List Name

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

oauthVersion - OAuth Version

Syntax

--oauthVersion <list-item>

Description

Version of OAuth

Class

Optional.

List

- 2.0.0 - V2.0.0 - Version 2.0.0

Default

2.0.0

Name

producerOAuthEnable - Producer OAuth Enable

Syntax

--producerOAuthEnable <list-item>

Description

Parameter controls if the AMF as producer requires an API request to have an OAuth access token in the Authorization HTTP request header. When the parameter is set to 'true', the AMF producer requires an access token to be provided by the consumer. When the parameter is set to 'false', the AMF producer does not perform OAuth authorization upon receiving an API request. By default, this parameter shall be set to 'true'

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

producerTokenExpGracePeriod - Producer Token Expiration Grace Period

Syntax

--producerTokenExpGracePeriod <integer>

Description

This parameter defines the period (in seconds) for which OAuth Access tokens that have recently expired as still considered valid.

Class

Optional.

Type / Range

Integer in the range of 0 to 3,600 in units of Second

Default

0

Recommended

Unless there are security concerns value should be set to 60 sec or longer. This can improve robustness during NRF failures as it allows Consumer use (recently) expired tokens.

Name

requestOAuthAccessTokenOnChallenge - Request OAuth Access Token On Challenge

Syntax

--requestOAuthAccessTokenOnChallenge <list-item>

Description

Parameter shall be used to control whether the AMF service consumer (a) requests an OAuth access token only after receiving challenge from the NF service producer, or (b) requests an OAuth access token prior to accessing an NF service producer. When the parameter is set to 'true', AMF requests an OAuth access token only when it receives a challenge from an NF service producer. When the parameter is set to 'false', AMF requests an OAuth access token prior to accessing an NF service producer. By default this shall be set to 'false'.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

supportModelA - Support Model A

Syntax

--supportModelA <list-item>

Description

When this parameter is enabled AMF will use OAuth tokens in model A and model C when producer is selected from local provisioning.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

supportMultiplePlmnsConsumer - Support Multiple Serving PLMNs as a Consumer

Syntax

--supportMultiplePlmnsConsumer <list-item>

Description

This parameter enhances OAuth functionality by supporting attributes that are required when AMF is the Consumer and the AMF and/or the Peers NFs serve multiple PLMNs.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Recommended

Should be set to true when AMF or Peers serve multiple PLMNs

Name

supportMultiplePlmnsProducer - Support Multiple Serving PLMNs as a Producer

Syntax

--supportMultiplePlmnsProducer <list-item>

Description

This parameter enhances OAuth functionality by supporting attributes that are required when AMF is the Producer and the AMF and/or the Peers NFs serve multiple PLMNs.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Recommended

Should be set to true when AMF or Peers serve multiple PLMNs

Name

supportMultiplePublicKeys - Support Multiple Public Keys

Syntax

--supportMultiplePublicKeys <list-item>

Description

When this parameter is set to true, the validation of the issuer of an OAuth Token can use the Public Key of any locally Provisioned NRF. When this parameter is set to false, the validation of the issuer of an OAuth Token done only with the Public Key of the NRF the AMF is currently registered.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

supportSliceInformation - Support Slice Information

Syntax

--supportSliceInformation <list-item>

Description

Defines if AMF supports Slice Information in Access Token Claim when a access token is requested for an NF type.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

tokenRequestByNFType - Token Request By NF Type

Syntax

--tokenRequestByNfType <list-item>

Description

Parameter controls whether the AMF consumer requests access tokens for an NF type or requests access token for an NF service instance. When the parameter is set to 'true', the AMF consumer will request access token for an NF type. When the parameter is set to 'false', the AMF consumer will request access token for an NR service instance. The only value supported is 'false'.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

useLocalNrfCertificatesForValidation - Use local NRF Certificates for Validation

Syntax

--useLocalNrfCertificatesForValidation <list-item>

Description

When this parameter is enabled, and AMF receives a Request including an OAuth Token, it searches for the local NRF certificate that matches the algorithm, and Instance ID (as extracted from the JWS) and uses the relevant algorithm / public Key to validate the signature. When this parameter is disabled AMF uses the NRF Certificate(s) acquired via the TLS handshake to the NRF(s).

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Recommended

This parameter should be set to true if AMF is provisioned to supports OAuth2.0 as a Producer there is no direct TLS connection to the NRF that issues the OAuth Tokens.

Roles

Create

secadmin

Show

sysadmin, manage, secadmin

Modify

secadmin

Delete

secadmin

List

sysadmin, manage, secadmin

6.32 cmm openCdr

Open CDR. Syntax, description, parameters, and roles for `cmm openCdr` command.

Syntax

Description

Show open CDRs.

Parameters

Name

access - Access

Syntax

--access <list-item>

Description

Radio access type of the S-CDRs to be shown.

Class

Required on create.

List

- ALL - Specify charging access of all subscribers.
- DTIHSPA - Direct tunnel I-HSPA - Specify charging access of direct tunnel I-HSPA subscribers.
- DTIU - Direct tunnel iu interface - Specify charging access of direct tunnel iu interface subscribers.

- GB - Gb interface - Specify charging access of gb interface subscribers.
- I-HSPA - I-HSPA - Specify charging access of I-HSPA subscribers.
- IU - Iu interface - Specify charging access of Iu interface subscribers.

Name

sub - Subscriber

Syntax

--sub <list-item>

Description

Roaming status of the S-CDRs to be shown.

Class

Required on create.

List

- ALL - All.
- HOME - Home.
- ROAM - Roam.

Roles**Show**

sysadmin, manage, monitor

List

sysadmin, manage, monitor

6.33 cmm oprDefQciProfile

Operator Defined QCI Profile. Syntax, description, parameters, and roles for `cmm oprDefQciProfile` command.

Syntax**Description**

This command is used to administer an Operator Defined QCI Profile and includes parameters for defining the operator defined QCI value, the mapped QCI value (from 1 - 9), and the associated paging type.

Parameters**Name**

oprDefQciProfileListName - Operator defined QCI profile name

Syntax

--oprDefQciProfileListName <string>

Description

Operator defined QCI profile name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

oprDefQci - Operator defined QCI

Syntax

--oprDefQci <integer>

Description

Operator defined QCI.

Class

Required key.

Type / Range

Integer in the range of 128 to 254

Name

mappedQci - Mapped QCI

Syntax

--mappedQci <integer>

Description

Mapped QCI value from 1 - 9 specifies QCI value.

Class

Optional.

Type / Range

Integer in the range of 1 to 9

Default

1

Name

pagingType - Paging type

Syntax

--pagingType <list-item>

Description

Paging type for the operator defined QCI value.

Class

Optional.

List

- APN1 - paging type for APN1
- APN10 - paging type for APN10
- APN2 - paging type for APN2
- APN3 - paging type for APN3
- APN4 - paging type for APN4
- APN5 - paging type for APN5
- APN6 - paging type for APN6
- APN7 - paging type for APN7
- APN8 - paging type for APN8
- APN9 - paging type for APN9
- ARP15 - Paging policy ARP15.
- Basic - Default paging policy for UE reactivation.
- Basic_QCI_1 - Optional paging policy for bearers with QCI 1.
- Basic_QCI_2 - Optional paging policy for bearers with QCI 2.
- Basic_QCI_3 - Optional paging policy for bearers with QCI 3.
- Basic_QCI_4 - Optional paging policy for bearers with QCI 4.
- Basic_QCI_5 - Optional paging policy for bearers with QCI 5.
- Basic_QCI_6 - Optional paging policy for bearers with QCI 6.
- Basic_QCI_7 - Optional paging policy for bearers with QCI 7.
- Basic_QCI_8 - Optional paging policy for bearers with QCI 8.
- Basic_QCI_9 - Optional paging policy for bearers with QCI 9.
- CoverageEnhancement - Coverage enhancement information policy.
- LPA - Low priority access UEs.
- NB_EDRX_10 - Paging type for NB-S1 UE with eDRX cycle length of 327.68 seconds.
- NB_EDRX_11 - Paging type for NB-S1 UE with eDRX cycle length of 655.36 seconds.
- NB_EDRX_12 - Paging type for NB-S1 UE with eDRX cycle length of 1310.72 seconds.
- NB_EDRX_13 - Paging type for NB-S1 UE with eDRX cycle length of 2621.44 seconds.

- NB_EDRX_14 - Paging type for NB-S1 UE with eDRX cycle length of 5242.88 seconds.
- NB_EDRX_15 - Paging type for NB-S1 UE with eDRX cycle length of 10485.76 seconds.
- NB_EDRX_2 - Paging type for NB-S1 UE with eDRX cycle length of 20.48 seconds.
- NB_EDRX_3 - Paging type for NB-S1 UE with eDRX cycle length of 40.96 seconds.
- NB_EDRX_5 - Paging type for NB-S1 UE with eDRX cycle length of 81.92 seconds.
- NB_EDRX_9 - Paging type for NB-S1 UE with eDRX cycle length of 163.84 seconds.
- PGWRestart - PGW restart paging policy for the mobile network for a paging attempt.
- PPI0 - Paging policy differentiation using paging policy indication flag.
- PPI1 - Paging policy differentiation using paging policy indication flag.
- PPI10 - Paging policy differentiation using paging policy indication flag.
- PPI11 - Paging policy differentiation using paging policy indication flag.
- PPI12 - Paging policy differentiation using paging policy indication flag.
- PPI13 - Paging policy differentiation using paging policy indication flag.
- PPI14 - Paging policy differentiation using paging policy indication flag.
- PPI15 - Paging policy differentiation using paging policy indication flag.
- PPI16 - Paging policy differentiation using paging policy indication flag.
- PPI17 - Paging policy differentiation using paging policy indication flag.
- PPI18 - Paging policy differentiation using paging policy indication flag.
- PPI19 - Paging policy differentiation using paging policy indication flag.
- PPI2 - Paging policy differentiation using paging policy indication flag.
- PPI20 - Paging policy differentiation using paging policy indication flag.
- PPI21 - Paging policy differentiation using paging policy indication flag.
- PPI22 - Paging policy differentiation using paging policy indication flag.
- PPI23 - Paging policy differentiation using paging policy indication flag.
- PPI24 - Paging policy differentiation using paging policy indication flag.
- PPI25 - Paging policy differentiation using paging policy indication flag.
- PPI26 - Paging policy differentiation using paging policy indication flag.
- PPI27 - Paging policy differentiation using paging policy indication flag.
- PPI28 - Paging policy differentiation using paging policy indication flag.
- PPI29 - Paging policy differentiation using paging policy indication flag.
- PPI3 - Paging policy differentiation using paging policy indication flag.

- PPI30 - Paging policy differentiation using paging policy indication flag.
- PPI31 - Paging policy differentiation using paging policy indication flag.
- PPI32 - Paging policy differentiation using paging policy indication flag.
- PPI33 - Paging policy differentiation using paging policy indication flag.
- PPI34 - Paging policy differentiation using paging policy indication flag.
- PPI35 - Paging policy differentiation using paging policy indication flag.
- PPI36 - Paging policy differentiation using paging policy indication flag.
- PPI37 - Paging policy differentiation using paging policy indication flag.
- PPI38 - Paging policy differentiation using paging policy indication flag.
- PPI39 - Paging policy differentiation using paging policy indication flag.
- PPI4 - Paging policy differentiation using paging policy indication flag.
- PPI40 - Paging policy differentiation using paging policy indication flag.
- PPI41 - Paging policy differentiation using paging policy indication flag.
- PPI42 - Paging policy differentiation using paging policy indication flag.
- PPI43 - Paging policy differentiation using paging policy indication flag.
- PPI44 - Paging policy differentiation using paging policy indication flag.
- PPI45 - Paging policy differentiation using paging policy indication flag.
- PPI46 - Paging policy differentiation using paging policy indication flag.
- PPI47 - Paging policy differentiation using paging policy indication flag.
- PPI48 - Paging policy differentiation using paging policy indication flag.
- PPI49 - Paging policy differentiation using paging policy indication flag.
- PPI5 - Paging policy differentiation using paging policy indication flag.
- PPI50 - Paging policy differentiation using paging policy indication flag.
- PPI51 - Paging policy differentiation using paging policy indication flag.
- PPI52 - Paging policy differentiation using paging policy indication flag.
- PPI53 - Paging policy differentiation using paging policy indication flag.
- PPI54 - Paging policy differentiation using paging policy indication flag.
- PPI55 - Paging policy differentiation using paging policy indication flag.
- PPI56 - Paging policy differentiation using paging policy indication flag.
- PPI57 - Paging policy differentiation using paging policy indication flag.
- PPI58 - Paging policy differentiation using paging policy indication flag.

- PPI59 - Paging policy differentiation using paging policy indication flag.
- PPI6 - Paging policy differentiation using paging policy indication flag.
- PPI60 - Paging policy differentiation using paging policy indication flag.
- PPI61 - Paging policy differentiation using paging policy indication flag.
- PPI62 - Paging policy differentiation using paging policy indication flag.
- PPI63 - Paging policy differentiation using paging policy indication flag.
- PPI7 - Paging policy differentiation using paging policy indication flag.
- PPI8 - Paging policy differentiation using paging policy indication flag.
- PPI9 - Paging policy differentiation using paging policy indication flag.
- Restricted - Restricted paging is triggered when OLC determines the CMM/WMM is in overload. the restricted paging type is used instead of the normal paging types during an overload condition. the restricted paging type normally has only a single page attempt.
- S102 - S102 is the paging type when in the "PS" mode and the request was received over the S102 interface.
- S102_CS - S102 is the paging type when in the "CS" mode and the request was received over the S102 interface.
- SGD_PAGING - SGD - Paging policy for SGd triggered paging requests.
- SGS_CS - Paging policy for SGs paging requests in circuit-switch mode.
- SGS_PS - Paging policy for SGs paging requests in packet-switch mode.
- SxnRestoration - SRS restoration paging.
- UELB_PAGING - Paging type for UE Load Balancing pagings.
- UeMonitLocationPaging - UE monitoring location paging - This is for Ue Monitoring feature with monitor type location. If UE is in idle and location trigger from SCEF via HSS to MME, then Page is requested
- UserDefPaging1 - Operator-defined QCI policy.
- UserDefPaging2 - Operator-defined QCI policy.
- UserDefPaging3 - Operator-defined QCI policy.
- UserDefPaging4 - Operator-defined QCI policy.
- UserDefPaging5 - Operator-defined QCI policy.
- UserDefPaging6 - Operator-defined QCI policy.
- WB_EDRX_0 - Paging type for WB-S1 UE with eDRX cycle length of 5.12 seconds
- WB_EDRX_1 - Paging type for WB-S1 UE with eDRX cycle length of 10.24 seconds.

- WB_EDRX_10 - Paging type for WB-S1 UE with eDRX cycle length of 327.68 seconds.
- WB_EDRX_11 - Paging type for WB-S1 UE with eDRX cycle length of 655.36 seconds.
- WB_EDRX_12 - Paging type for WB-S1 UE with eDRX cycle length of 1310.72 seconds.
- WB_EDRX_13 - Paging type for WB-S1 UE with eDRX cycle length of 2621.44 seconds.
- WB_EDRX_2 - Paging type for WB-S1 UE with eDRX cycle length of 20.48 seconds.
- WB_EDRX_3 - Paging type for WB-S1 UE with eDRX cycle length of 40.96 seconds.
- WB_EDRX_4 - Paging type for WB-S1 UE with eDRX cycle length of 61.44 seconds.
- WB_EDRX_5 - Paging type for WB-S1 UE with eDRX cycle length of 81.92 seconds.
- WB_EDRX_6 - Paging type for WB-S1 UE with eDRX cycle length of 102.40 seconds.
- WB_EDRX_7 - Paging type for WB-S1 UE with eDRX cycle length of 122.88 seconds.
- WB_EDRX_8 - Paging type for WB-S1 UE with eDRX cycle length of 143.36 seconds.
- WB_EDRX_9 - Paging type for WB-S1 UE with eDRX cycle length of 163.84 seconds.

Default

Basic

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.34 cmm oprDefQciProfileList

Operator Defined QCI Profile List. Syntax, description, parameters, and roles for cmm oprDefQciProfileList command.

Syntax

Description

This command is used to administer the Operator Defined QCI Profile List.

Parameters

Name

name - Operator defined QCI profile name

Syntax

--name <string>

Description

Operator defined QCI profile name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.35 cmm oprDefQciToRel99QosMapping

Operator Defined QCI to Rel99 QoS Mapping. Syntax, description, parameters, and roles for cmm oprDefQciToRel99QosMapping command.

Syntax

Description

Used by the SGSN to defines the Mapping between operator Defined QCIs and Release 99 QoS parameter values. Up to 32 operator Defined QCIs may be defined

Parameters

Name

qci - QCI

Syntax

--qci <integer>

Description

Operator Defined QoS Class Identifier

Class

Required key.

Type / Range

Integer in the range of 128 to 254

Name

deliveryOfErroneousSDUs - Delivery Of Erroneous SDUs

Syntax

--deliveryOfErroneousSDUs <list-item>

Description

Indicates whether SDUs detected as erroneous shall be delivered or discarded. 'accept' implies that error detection is employed and that erroneous SDUs are delivered together with an error indication, 'discard' implies that error detection is employed and that erroneous SDUs are discarded, and 'do-not-detect' implies that SDUs are delivered without considering error detection. Values: accept, discard, do-not-detect Default: discard

Class

Optional.

List

- accept - Specifies detect and accept erroneous data packets.
- discard - Specifies detect and discard erroneous data packets.
- do-not-detect - Do not detect - Specifies do not detect erroneous data packets.

Default

discard

Name

deliveryOrder - Delivery Order

Syntax

--deliveryOrder <list-item>

Description

Definition: indicates whether the UMTS bearer shall provide in-sequence SDU delivery or not.
Default: No

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

dscp - Differentiated Services Code Point

Syntax

--dscp <integer>

Description

DSCP value (in decimal) to be applied in the user plane traffic for the specific QCI. Values 0..64 Default: 0

Class

Optional.

Type / Range

Integer in the range of 0 to 63

Default

0

Name

maxSduSize - Maximum SDU Size

Syntax

--maxSduSize <integer>

Description

The maximum SDU size for which the network shall satisfy the negotiated QoS

Class

Optional.

Type / Range

Integer in the range of 0 to 1,500

Default

1500

Name

residualBitErrorRatioNonRealTime - Residual Bit Error Ratio Non Real Time

Syntax

--residualBitErrorRatioNonRealTime <list-item>

Description

Error Ratio the undetected bit error ratio in the delivered SDUs for Non Real Time traffic (i.e Interactive, Background). If no error detection is requested, Residual bit error ratio indicates the bit error ratio in the delivered SDUs Default: 1×10^{-5}

Class

Optional.

List

- 1×10^{-5} - 0.00001 - Specify residual bit error ratio 0.00001.
- 4×10^{-3} - 0.004 - Specify residual bit error ratio of 0.004.
- 6×10^{-8} - 0.00000006 - Specify residual bit error ratio of 0.00000006.

Default

1×10^{-5}

Name

residualBitErrorRatioRealTime - Residual Bit Error Real Time

Syntax

--residualBitErrorRatioRealTime <list-item>

Description

Error Ratio the undetected bit error ratio in the delivered SDUs for Real Time traffic (i.e Conversational, Streaming). If no error detection is requested, Residual bit error ratio indicates the bit error ratio in the delivered SDUs Default: 1×10^{-5}

Class

Optional.

List

- 1×10^{-2} - 0.01 - Specify residual bit error ratio of 0.01.
- 1×10^{-3} - 0.001 - Specify residual bit error ratio of 0.001.
- 1×10^{-4} - 0.0001 - Specify residual bit error ratio of 0.0001.
- 1×10^{-5} - 0.00001 - Specify residual bit error ratio of 0.00001.
- 1×10^{-6} - 0.000001 - Specify residual bit error ratio of 0.000001.
- 5×10^{-2} - 0.05 - Specify residual bit error ratio of 0.05.
- 5×10^{-3} - 0.005 - Specify residual bit error ratio of 0.005.

Default

1*10⁻⁵

Name

sduErrorRatioConversational - SDU Error Ratio Conversational

Syntax

--sduErrorRatioConversational <list-item>

Description

Ratio of SDUs lost or detected as erroneous. Relevant only for Conversational traffic.
Default: 1*10⁻³

Class

Optional.

List

- 1*10⁻² - 0.01 - Specify service data unit error ratio of 0.01.
- 1*10⁻³ - 0.001 - Specify service data unit error ratio of 0.001.
- 1*10⁻⁴ - 0.0001 - Specify service data unit error ratio of 0.0001.
- 1*10⁻⁵ - 0.00001 - Specify service data unit error ratio of 0.00001.
- 7*10⁻³ - 0.007 - Specify service data unit error ratio of 0.007.

Default

1*10⁻³

Name

sduErrorRatioNonRealTime - SDU Error Ratio Non Real Time

Syntax

--sduErrorRatioNonRealTime <list-item>

Description

Ratio of SDUs lost or detected as erroneous. SDU error ratio for non Real Time traffic is defined only for conforming traffic (i.e. Background and Interactive) . Default: 1*10⁻⁴

Class

Optional.

List

- 1*10⁻³ - 0.001 - Specify service data unit error ratio of 0.001.
- 1*10⁻⁴ - 0.0001 - Specify service data unit error ratio of 0.0001.
- 1*10⁻⁶ - 0.000001 - Specify service data unit error ratio of 0.000001.

Default

1*10⁻⁴

Name

sduErrorRatioStreaming - SDU Error Ratio Streaming

Syntax

--sduErrorRatioStreaming <list-item>

Description

Ratio of SDUs lost or detected as erroneous. Relevant only for Streaming traffic. Default: 1×10^{-4}

Class

Optional.

List

- 1×10^{-1} - 0.1 - Specify service data unit error ratio of 0.1.
- 1×10^{-2} - 0.01 - Specify service data unit error ratio of 0.01.
- 1×10^{-3} - 0.001 - Specify service data unit error ratio of 0.001.
- 1×10^{-4} - 0.0001 - Specify service data unit error ratio of 0.0001.
- 1×10^{-5} - 0.00001 - Specify service data unit error ratio of 0.00001.
- 7×10^{-3} - 0.007 - Specify service data unit error ratio of 0.007.

Default

1×10^{-4}

Name

signalingIndication - Signaling Indication

Syntax

--signalingIndication <list-item>

Description

Definition: Indicates the signalling nature of the submitted SDUs. This attribute is additional to the other QoS attributes and does not over-ride them. This attribute is only defined for the interactive traffic class. If signalling indication is set to 'Yes', the UE should set the traffic handling priority to '1'. Default: No

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

sourceStatisticsDescriptorSpeech - Source Statistics Descriptor Speech

Syntax

--sourceStatisticsDescriptorSpeech <list-item>

Description

Source statistics descriptor Definition: specifies characteristics of the source of submitted SDUs. Conversational speech has a well-known statistical behaviour (or the discontinuous transmission (DTX) factor). By being informed that the SDUs for a RAB are generated by a speech source, RAN may, based on experience, calculate a statistical multiplex gain for use in admission control on the radio and RAN Access interfaces. Values: False (Unknown), True (Speech) Default: False

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

trafficClass - Traffic Class

Syntax

--trafficClass <list-item>

Description

Type of application for which the bearer service is optimised Default: Background

Class

Optional.

List

- Background - Traffic class type of background.
- Conversational - Traffic class type of conversational.
- Interactive - Traffic class type of interactive.
- Streaming - Traffic class type of streaming.

Default

Background

Name

trafficHandlingPriority - Traffic Handling Priority

Syntax

--trafficHandlingPriority <integer>

Description

Specifies the relative importance for handling of all SDUs belonging to the bearer compared to the SDUs of other bearers. Possible values are 0 (N/A,) 1,2 or 4. Default: 0 (Not Applicable)

Class

Optional.

Type / Range

Integer in the range of 0 to 3

Default

0

Name

transferDelay - Transfer Delay

Syntax

--transferDelay <integer>

Description

Indicates maximum delay in ms for 95th percentile of the distribution of delay for all delivered SDUs during the lifetime of a bearer service, where delay for an SDU is defined as the time from a request to transfer an SDU at one SAP to its delivery at the other SAP. Values: 10..4000 Default: 10

Class

Optional.

Type / Range

Integer in the range of 10 to 4,000

Default

10

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.36 cmm outputPapsStatus

outputPapsStatus. Syntax, description, parameters, and roles for cmm outputPapsStatus command.

Syntax

Description

Show maintenance status of PAPS.

Parameters

Name

active - Activation state of PAPS

Syntax

--active <list-item>

Description

Activation state of PAPS.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

bfMRespCode - Response codes for blocked for maintenance feature

Syntax

--bfMRespCode <string>

Description

Response codes for PAPS maintenance block feature.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 to 10 with the allowed pattern of [0-9a-zA-Z\.\-:]*

Name

bfMRespCodeTimestamp - Timestamp of last response for block PAPS feature

Syntax

--bfMRespCodeTimestamp <string>

Description

Timestamp of last response for block PAPU feature.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

blockForMaint - Blocked for maintenance feature status

Syntax

--blockForMaint <list-item>

Description

Status code for PAPS maintenance block feature.

Class

Optional, not filterable on list.

List

- Blocked - Specify block status of blocked.
- Started - Specify block status of started.
- Unblocked - Specify block status of unblocked.

Default

Unblocked

Name

massDetach - Mass detach status

Syntax

--massDetach <list-item>

Description

Status code for mass detach feature.

Class

Optional, not filterable on list.

List

- Cancelled - Specify mass detach status of cancelled.
- Cancelling - Specify mass detach status of cancelling.
- Idle - Specify mass detach status of idle.

- Started - Specify mass detach status of started.

Default

Idle

Name

mDRespCode - Response codes for mass detach feature

Syntax

--mDRespCode <string>

Description

Response codes for mass detach feature.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 to 10 with the allowed pattern of [0-9a-zA-Z\.\-:]*

Name

mDRespCodeTimestamp - Timestamp of last response for mass detach feature

Syntax

--mDRespCodeTimestamp <string>

Description

Timestamp of last response for mass detach feature.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

offload - Status for offload feature

Syntax

--offload <list-item>

Description

Status code for offload feature.

Class

Optional, not filterable on list.

List

- AUTO completed - PAPS offload - AUTO completed.
- Cancelled - PAPS offload - cancelled.
- Cancelling - PAPS offload - cancelling.

- Idle - PAPS offload - idle.
- Phase1 - PAPS offload - phase 1.
- Phase1 completed - PAPS offload - phase 1 completed.
- Phase2 - PAPS offload - phase 2.
- Phase2 completed - PAPS offload - phase 2 completed.
- Phase3 - PAPS offload - phase 3.
- Phase3 completed - PAPS offload - phase 3 completed.
- Started - PAPS offload - started.

Default

Idle

Name

offlRespCode - Response codes for offload feature

Syntax

--offlRespCode <string>

Description

Response codes for offload feature.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 to 10 with the allowed pattern of [0-9a-zA-Z\.\-:]*

Name

offlRespCodeTimestamp - Timestamp of last response for offload feature

Syntax

--offlRespCodeTimestamp <string>

Description

Timestamp of last response for offload feature.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-:@]*

Name

id - PAPS unit index

Syntax

<list-item>

Description

This parameter defines the PAPU unit index. the value can range from 0 to 19.

Class

Required value.

List

- paps0 - Specify packet processing service unit id 0.
- paps1 - Specify packet processing service unit id 1.
- paps2 - Specify packet processing service unit id 2.
- paps3 - Specify packet processing service unit id 3.
- paps4 - Specify packet processing service unit id 4.
- paps5 - Specify packet processing service unit id 5.
- paps6 - Specify packet processing service unit id 6.
- paps7 - Specify packet processing service unit id 7.
- paps8 - Specify packet processing service unit id 8.
- paps9 - Specify packet processing service unit id 9.
- paps10 - Specify packet processing service unit id 10.
- paps11 - Specify packet processing service unit id 11.
- paps12 - Specify packet processing service unit id 12.
- paps13 - Specify packet processing service unit id 13.
- paps14 - Specify packet processing service unit id 14.
- paps15 - Specify packet processing service 15.
- paps16 - Specify packet processing service unit id 16.
- paps17 - Specify packet processing service unit id 17.
- paps18 - Specify packet processing service unit id 18.
- paps19 - Specify packet processing service unit id 19.

Roles

Show

sysadmin, manage, monitor

List

sysadmin, manage, monitor

6.37 cmm pagingPolicy

Paging Policy. Syntax, description, parameters, and roles for cmm pagingPolicy command.

Syntax

Description

This command is used to administer a Paging Policy and includes parameters for defining the number of paging attempts, the maximum number of eNBs paged, paging method, paging type, and paging response timer. Note, for provisioned SGd paging type, if also eDRX is provisioned, the eDRX value takes priority

Parameters

Name

plmnName - Mobile network name

Syntax

--plmnName <string>

Description

Name that identifies mobile network.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

pagingType - Paging type

Syntax

--pagingType <list-item>

Description

Defines the type of paging to be done, basic, SGS_CS, or SGS_PS at least one tuple must exist for each paging type.

Class

Required key.

List

- APN1 - paging type for APN1
- APN10 - paging type for APN10
- APN2 - paging type for APN2
- APN3 - paging type for APN3

- APN4 - paging type for APN4
- APN5 - paging type for APN5
- APN6 - paging type for APN6
- APN7 - paging type for APN7
- APN8 - paging type for APN8
- APN9 - paging type for APN9
- ARP15 - Paging policy ARP15.
- Basic - Default paging policy for UE reactivation.
- Basic_QCI_1 - Optional paging policy for bearers with QCI 1.
- Basic_QCI_2 - Optional paging policy for bearers with QCI 2.
- Basic_QCI_3 - Optional paging policy for bearers with QCI 3.
- Basic_QCI_4 - Optional paging policy for bearers with QCI 4.
- Basic_QCI_5 - Optional paging policy for bearers with QCI 5.
- Basic_QCI_6 - Optional paging policy for bearers with QCI 6.
- Basic_QCI_7 - Optional paging policy for bearers with QCI 7.
- Basic_QCI_8 - Optional paging policy for bearers with QCI 8.
- Basic_QCI_9 - Optional paging policy for bearers with QCI 9.
- CoverageEnhancement - Coverage enhancement information policy.
- LPA - Low priority access UEs.
- NB_EDRX_10 - Paging type for NB-S1 UE with eDRX cycle length of 327.68 seconds.
- NB_EDRX_11 - Paging type for NB-S1 UE with eDRX cycle length of 655.36 seconds.
- NB_EDRX_12 - Paging type for NB-S1 UE with eDRX cycle length of 1310.72 seconds.
- NB_EDRX_13 - Paging type for NB-S1 UE with eDRX cycle length of 2621.44 seconds.
- NB_EDRX_14 - Paging type for NB-S1 UE with eDRX cycle length of 5242.88 seconds.
- NB_EDRX_15 - Paging type for NB-S1 UE with eDRX cycle length of 10485.76 seconds.
- NB_EDRX_2 - Paging type for NB-S1 UE with eDRX cycle length of 20.48 seconds.
- NB_EDRX_3 - Paging type for NB-S1 UE with eDRX cycle length of 40.96 seconds.
- NB_EDRX_5 - Paging type for NB-S1 UE with eDRX cycle length of 81.92 seconds.
- NB_EDRX_9 - Paging type for NB-S1 UE with eDRX cycle length of 163.84 seconds.
- PGWRestart - PGW restart paging policy for the mobile network for a paging attempt.
- PPI0 - Paging policy differentiation using paging policy indication flag.

- PPI1 - Paging policy differentiation using paging policy indication flag.
- PPI10 - Paging policy differentiation using paging policy indication flag.
- PPI11 - Paging policy differentiation using paging policy indication flag.
- PPI12 - Paging policy differentiation using paging policy indication flag.
- PPI13 - Paging policy differentiation using paging policy indication flag.
- PPI14 - Paging policy differentiation using paging policy indication flag.
- PPI15 - Paging policy differentiation using paging policy indication flag.
- PPI16 - Paging policy differentiation using paging policy indication flag.
- PPI17 - Paging policy differentiation using paging policy indication flag.
- PPI18 - Paging policy differentiation using paging policy indication flag.
- PPI19 - Paging policy differentiation using paging policy indication flag.
- PPI2 - Paging policy differentiation using paging policy indication flag.
- PPI20 - Paging policy differentiation using paging policy indication flag.
- PPI21 - Paging policy differentiation using paging policy indication flag.
- PPI22 - Paging policy differentiation using paging policy indication flag.
- PPI23 - Paging policy differentiation using paging policy indication flag.
- PPI24 - Paging policy differentiation using paging policy indication flag.
- PPI25 - Paging policy differentiation using paging policy indication flag.
- PPI26 - Paging policy differentiation using paging policy indication flag.
- PPI27 - Paging policy differentiation using paging policy indication flag.
- PPI28 - Paging policy differentiation using paging policy indication flag.
- PPI29 - Paging policy differentiation using paging policy indication flag.
- PPI3 - Paging policy differentiation using paging policy indication flag.
- PPI30 - Paging policy differentiation using paging policy indication flag.
- PPI31 - Paging policy differentiation using paging policy indication flag.
- PPI32 - Paging policy differentiation using paging policy indication flag.
- PPI33 - Paging policy differentiation using paging policy indication flag.
- PPI34 - Paging policy differentiation using paging policy indication flag.
- PPI35 - Paging policy differentiation using paging policy indication flag.
- PPI36 - Paging policy differentiation using paging policy indication flag.
- PPI37 - Paging policy differentiation using paging policy indication flag.

- PPI38 - Paging policy differentiation using paging policy indication flag.
- PPI39 - Paging policy differentiation using paging policy indication flag.
- PPI4 - Paging policy differentiation using paging policy indication flag.
- PPI40 - Paging policy differentiation using paging policy indication flag.
- PPI41 - Paging policy differentiation using paging policy indication flag.
- PPI42 - Paging policy differentiation using paging policy indication flag.
- PPI43 - Paging policy differentiation using paging policy indication flag.
- PPI44 - Paging policy differentiation using paging policy indication flag.
- PPI45 - Paging policy differentiation using paging policy indication flag.
- PPI46 - Paging policy differentiation using paging policy indication flag.
- PPI47 - Paging policy differentiation using paging policy indication flag.
- PPI48 - Paging policy differentiation using paging policy indication flag.
- PPI49 - Paging policy differentiation using paging policy indication flag.
- PPI5 - Paging policy differentiation using paging policy indication flag.
- PPI50 - Paging policy differentiation using paging policy indication flag.
- PPI51 - Paging policy differentiation using paging policy indication flag.
- PPI52 - Paging policy differentiation using paging policy indication flag.
- PPI53 - Paging policy differentiation using paging policy indication flag.
- PPI54 - Paging policy differentiation using paging policy indication flag.
- PPI55 - Paging policy differentiation using paging policy indication flag.
- PPI56 - Paging policy differentiation using paging policy indication flag.
- PPI57 - Paging policy differentiation using paging policy indication flag.
- PPI58 - Paging policy differentiation using paging policy indication flag.
- PPI59 - Paging policy differentiation using paging policy indication flag.
- PPI6 - Paging policy differentiation using paging policy indication flag.
- PPI60 - Paging policy differentiation using paging policy indication flag.
- PPI61 - Paging policy differentiation using paging policy indication flag.
- PPI62 - Paging policy differentiation using paging policy indication flag.
- PPI63 - Paging policy differentiation using paging policy indication flag.
- PPI7 - Paging policy differentiation using paging policy indication flag.
- PPI8 - Paging policy differentiation using paging policy indication flag.

- PPI9 - Paging policy differentiation using paging policy indication flag.
- Restricted - Restricted paging is triggered when OLC determines the CMM/WMM is in overload. the restricted paging type is used instead of the normal paging types during an overload condition. the restricted paging type normally has only a single page attempt.
- S102 - S102 is the paging type when in the "PS" mode and the request was received over the S102 interface.
- S102_CS - S102 is the paging type when in the "CS" mode and the request was received over the S102 interface.
- SGD_PAGING - SGD - Paging policy for SGd triggered paging requests.
- SGS_CS - Paging policy for SGs paging requests in circuit-switch mode.
- SGS_PS - Paging policy for SGs paging requests in packet-switch mode.
- SxnRestoration - SRS restoration paging.
- UELB_PAGING - Paging type for UE Load Balancing pagings.
- UeMonitLocationPaging - UE monitoring location paging - This is for Ue Monitoring feature with monitor type location. If UE is in idle and location trigger from SCEF via HSS to MME, then Page is requested
- UserDefPaging1 - Operator-defined QCI policy.
- UserDefPaging2 - Operator-defined QCI policy.
- UserDefPaging3 - Operator-defined QCI policy.
- UserDefPaging4 - Operator-defined QCI policy.
- UserDefPaging5 - Operator-defined QCI policy.
- UserDefPaging6 - Operator-defined QCI policy.
- WB_EDRX_0 - Paging type for WB-S1 UE with eDRX cycle length of 5.12 seconds
- WB_EDRX_1 - Paging type for WB-S1 UE with eDRX cycle length of 10.24 seconds.
- WB_EDRX_10 - Paging type for WB-S1 UE with eDRX cycle length of 327.68 seconds.
- WB_EDRX_11 - Paging type for WB-S1 UE with eDRX cycle length of 655.36 seconds.
- WB_EDRX_12 - Paging type for WB-S1 UE with eDRX cycle length of 1310.72 seconds.
- WB_EDRX_13 - Paging type for WB-S1 UE with eDRX cycle length of 2621.44 seconds.
- WB_EDRX_2 - Paging type for WB-S1 UE with eDRX cycle length of 20.48 seconds.
- WB_EDRX_3 - Paging type for WB-S1 UE with eDRX cycle length of 40.96 seconds.
- WB_EDRX_4 - Paging type for WB-S1 UE with eDRX cycle length of 61.44 seconds.
- WB_EDRX_5 - Paging type for WB-S1 UE with eDRX cycle length of 81.92 seconds.

- WB_EDRX_6 - Paging type for WB-S1 UE with eDRX cycle length of 102.40 seconds.
- WB_EDRX_7 - Paging type for WB-S1 UE with eDRX cycle length of 122.88 seconds.
- WB_EDRX_8 - Paging type for WB-S1 UE with eDRX cycle length of 143.36 seconds.
- WB_EDRX_9 - Paging type for WB-S1 UE with eDRX cycle length of 163.84 seconds.

Name

attempt - Attempt

Syntax

--attempt <list-item>

Description

Defines the attempt (1 of 4 possible). attempt numbers must start at 1 and be consecutive (1st attempt, 2nd attempt, and so on).

Class

Required key.

List

- Attempt1 - Attempt 1 - First attempt.
- Attempt2 - Attempt 2 - Second attempt.
- Attempt3 - Attempt 3 - Third attempt.
- Attempt4 - Attempt 4 - 4th attempt.

Name

extRangeFlag - Extended range flag

Syntax

--extRangeFlag <list-item>

Description

Extended range flag for a given paging mode.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

maxEnbsPaged - Maximum Number of eNBs Paged

Syntax

--maxEnbsPaged <integer>

Description

Defines the maximum number of eNBs that may be paged for this page attempt. This parameter is only applicable to the Last Seen eNB List paging method. The default value of 0 is used to indicate that no maximum limit of eNBs paged is specified. If the default value of 0 is used with the Last Seen eNB List paging method, MME paging will assume a default value of 3.

Class

Optional.

Type / Range

Integer in the range of 0 to 16

Default

0

Name

method - Method

Syntax

--method <list-item>

Description

Method that the designated attempt (1 - 4) should use.

Class

Optional.

List

- LastSeenENB - Last seen eNB - Last seen eNB.
- LastSeenENBList - Last seen eNB list - Last seen eNB list.
- LastSeenTAI - Last seen TAI - Last seen TAI.
- LastSeenTAINBTAI - Last seen TAINBTAI - Last seen TAINBTAI.

Default

LastSeenENB

Name

overridePagingGapTimer - Override Paging Gap Timer Value

Syntax

--overridePagingGapTimer <integer>

Description

Parameter would allow the Paging Gap feature to behave differently for different paging types. For example, one could make the Paging Gap timer longer (or shorter) when an eDRX paging type is used. The time unit is in minutes.

Class

Optional.

Type / Range

Integer in the range of 0 to 1,440

Default

0

Name

preemptionLevel - Paging Pre-Emption Level

Syntax

--preemptionLevel <integer>

Description

Defines the pre-emption level for a paging type. The pre-emption level determines whether a new DDN request will pre-empt an ongoing DDN procedure. A lower non-zero numeric value represents a higher (i.e., stronger) pre-emption level. For example, a DDN with a preemptionLevel value of 1 will pre-empt a DDN with a preemptionLevel value of 2. However, a DDN using Multimedia Priority Services (MPS) will be considered to have the highest pre-emption level. The value of '0' (default) indicates that no pre-emption level is associated with that paging type. This parameter is only relevant to pagingPolicy records for Attempt1. In addition, this parameter does not apply to the following paging types, SxnRestoration, SGS_CS, SGS_PS, S102, and S102_CS.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

radioCapInfo - UE radio capability for paging

Syntax

--radioCapInfo <list-item>

Description

Enable/disable inclusion of radio capability for paging IE in the S1AP PAGING message.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

suppressPagingGap - Suppress Paging Gap

Syntax

--suppressPagingGap <list-item>

Description

Parameter allows the service provider to suppress the Paging Gap functionality for any MME paging type that they desire.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

t3413Timer - T3413 timer

Syntax

--t3413Timer <integer>

Description

Defines the value for the T3413 timer used for receiving responses to a paging request. when this timer expires, it triggers the MME to proceed with the next paging attempt as provisioned in the MME pagingpolicy table. this timer value should be set according to the maximum time it takes a UE to respond to a paging message under normal operating conditions. indicates the T3413 timer interval in seconds.

Class

Optional.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

6

Name

t3415Timer - T3415 timer

Syntax

--t3415Timer <integer>

Description

Defines the value for the T3415 timer used for receiving responses to a paging request. when this timer expires, it triggers the MME to proceed with the next paging attempt as provisioned in the MME pagingpolicy table. this timer value should be set according to the maximum time it takes an eDRX UE to respond to a paging message under normal operating conditions. indicates the T3415 timer interval in milliseconds. Default value is 0 for non eDRX paging types and 6000 ms for WB_EDRX_0 per f11603-06.

Class

Optional.

Type / Range

Integer in the range of 0 or 100 to 60,000 in units of Millisecond

Default

0

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.38 cmm pagingPriProfile

Paging Priority Profile. Syntax, description, parameters, and roles for cmm pagingPriProfile command.

Syntax**Description**

This command is used to administer a Paging Priority Profile and includes parameters for defining the ARP and paging priority levels.

Parameters

Name

pagingPriProfileListName - Paging priority profile name

Syntax

--pagingPriProfileListName <string>

Description

Name that identifies paging priority profile table (alternate key).

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

arpLevel - Allocation Retention Priority level

Syntax

--arpLevel <integer>

Description

Allocation Retention Priority (ARP) priority level related to paging priority profile. ARP priority level indicates the priority of allocation and retention and can take values between 1 and 15. Value 15 means 'no priority'. Values between 1 and 14 are ordered in decreasing order of priority, i.e., 1 indicates highest priority and 14 indicates lowest priority.

Class

Required key.

Type / Range

Integer in the range of 1 to 15

Name

pagingPriLevel - Paging priority level

Syntax

--pagingPriLevel <list-item>

Description

Paging priority level.

Class

Optional.

List

- PrioLevel0 - Priority level 0 - Priority level 0. No paging priority is set.
- PrioLevel1 - Priority level 1 - Priority level 1. Highest Paging priority.

- PrioLevel2 - Priority level 2 - Priority level 2.
- PrioLevel3 - Priority level 3 - Priority level 3.
- PrioLevel4 - Priority level 4 - Priority level 4.
- PrioLevel5 - Priority level 5 - Priority level 5.
- PrioLevel6 - Priority level 6 - Priority level 6.
- PrioLevel7 - Priority level 7 - Priority level 7.
- PrioLevel8 - Priority level 8 - Priority level 8. Lowest Paging Priority.

Default

PrioLevel0

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.39 cmm pagingPriProfileList

Paging Priority Profile List. Syntax, description, parameters, and roles for cmm pagingPriProfileList command.

Syntax**Description**

This command is used to administer a Paging Priority Profile List.

Parameters**Name**

name - Paging priority profile name

Syntax

--name <digits>

Description

Defines the paging priority profile name.

Class

Required key.

Type / Range

Digit string of length 1 to 1 with the allowed pattern of [1-8]*

Name

description - Description

Syntax

--description <string>

Description

Profile description.

Class

Optional.

Type / Range

Digit string of length 0 to 160 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

..

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.40 cmm passwdPolicy

Minimum password length. Syntax, description, parameters, and roles for cmm passwdPolicy command.

Syntax

Description

Change minimum password length for CMM user accounts.

Parameters

Name

passwdPolicyId - Id

Syntax

--passwdPolicyId <string>

Description

Just type 'config' here.

Class

Required key.

Type / Range

Digit string of length 6 to 6 with the allowed pattern of [0-9a-zA-Z\.\-_:]*

Name

minLen - Minimum password length

Syntax

--minLen <integer>

Description

Minimum password length for CMM user accounts.

Class

Optional.

Type / Range

Integer in the range of 10 to 127

Default

10

Roles

Show

secadmin

Modify

secadmin

List

seadmin

6.41 cmm patchParameter

patchparameters. Syntax, description, parameters, and roles for cmm patchParameter command.

Syntax

Description

Configure, list and show patched parameters (former utpfil).

Parameters

Name

appld - Identifier for application process

Syntax

--appId <string>

Description

Identifier for application process.

Class

Required on create, not filterable on list.

Type / Range

Digit string of length 1 to 10 with the allowed pattern of [0-9a-zA-Z\._ :]*

Name

paramId - Identifier for parameter

Syntax

--paramId <string>

Description

Identifier for parameter.

Class

Required on create, not filterable on list.

Type / Range

Digit string of length 1 to 10 with the allowed pattern of [0-9a-zA-Z\._ :]*

Name

value - Value for parameter

Syntax

--value <string>

Description

Value for parameter.

Class

Required on create, not filterable on list.

Type / Range

Digit string of length 1 to 10 with the allowed pattern of [0-9a-zA-Z\.\-_:]*

Name

id - Identifier for instance

Syntax

<integer>

Description

Identifier for instance.

Class

Required value.

Type / Range

Integer in the range of 1 to 200

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.42 cmm pcmdConfig

PCMD Configuration. Syntax, description, parameters, and roles for cmm pcmdConfig command.

Syntax

Description

This command is used to administer the PCMD configuration and includes a parameter to enable/disable inclusion of eNB per-cell measurement reports in PCMD records.

Parameters

Name

enableBothIpv4v6TeidAddresses - Enable Both IPv4 IPv6 Teid Addresses

Syntax

--enableBothIpv4v6TeidAddresses <list-item>

Description

When both IPv4 and IPv6 Teid addresses are available and this parameter is set, then peg both addresses at pcmd record. If not set, then include only the IPv6, if available. If not available then include the IPv4 address instead.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

enableCmReport - Enable CM report

Syntax

--enableCmReport <list-item>

Description

Defines whether or not eNB per-cell measurement reports are included in the PCMD records.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

enableTeidParams - Enable TEID Parameters

Syntax

--enableTeidParams <list-item>

Description

This parameter enables TEID parameters (PdnType, Endpoint Id, TEID IP address) at MME PCMD records. If disabled then these parameters are all removed from the PCMD record to minimize the PCMD record length

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

includeRequestedApn - Include Requested APN

Syntax

--includeRequestedApn <list-item>

Description

When this parameter is enabled, the Requested APN may be pegged at the secondary record, if available. When disabled, the Requested APN is not pegged even if it is available.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

includeS1ulpAddress - Include RAN S1-u IP Address

Syntax

--includeS1uIpAddress <list-item>

Description

When enabled, MME populates RAN IPv4 or IPv6 S1-U IP address in Secondary Record (default bearer) or Tertiary Record (dedicated bearer). Note: RAN S1-U IP address could be that of eNB or gNB for 3X split bearer.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

6.43 cmm pcmdStreaming

PCMD Streaming. Syntax, description, parameters, and roles for cmm pcmdStreaming command.

Syntax**Description**

With this command, the PCMD Streaming function of CMM is configured

Parameters**Name**

streamingType - PCMD Streaming Type

Syntax

--streamingType <list-item>

Description

Configure the respective PCMD Streaming Type

Restrictions

Only one Streaming Type may be created per application. So, in maximum up to 3 records.
One for AMF, one for MME and one for SGSN

Class

Required key.

List

- AMF - Configure PCMD streaming type for AMF
- MME - Configure PCMD streaming type for MME
- SGSN - Configure PCMD streaming type for SGSN

Name

primaryPort - Primary Port

Syntax

--primaryPort <integer>

Description

Primary port address of service.

Class

Required on create.

Type / Range

Integer in the range of 1,024 to 65,535

Name

dscpCode - DSCP Code

Syntax

--dscpCode <list-item>

Description

Parameter to set DSCP code to be used for an IP packet.

Class

Optional.

List

- AF11 - Assured forwarding 11.
- AF12 - Assured forwarding 12.
- AF13 - Assured forwarding 13.
- AF21 - Assured forwarding 21.
- AF22 - Assured forwarding 22.
- AF23 - Assured forwarding 23.
- AF31 - Assured forwarding 31.
- AF32 - Assured forwarding 32.
- AF33 - Assured forwarding 33.

- AF41 - Assured forwarding 41.
- AF42 - Assured forwarding 42.
- AF43 - Assured forwarding 43.
- BE - BE.
- CS1 - Class selector 1.
- CS2 - Class selector 2.
- CS3 - Class selector 3.
- CS4 - Class selector 4.
- CS5 - Class selector 5.
- CS6 - Class selector 6.
- CS7 - Class selector 7.
- EF - Expedited forwarding.
- voiceAdmit - Voice Admit

Default

AF41

Name

oamTlsProfileName - OAM TLS Profile

Syntax

--oamTlsProfileName <string>

Description

This parameter contains the profile for TLS. It contains the certificate names needed for establishing secure communication.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 31 with the allowed pattern of [0-9a-zA-Z_\.\\-:@]*

Name

operationStateAct - Operation State Active

Syntax

--operationStateAct <list-item>

Description

Operation State of the particular streaming type

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

pcmdFqdn - PCMD FQDN

Syntax

--pcmdFqdn <string>

Description

The parameter defines the PCMD Analytics Server FQDN.

Restrictions

Either the PCMD Analytics FQDN or the Primary IP must be set

Class

Optional.

Type / Range

Digit string of length 0 to 255 with the allowed pattern of [A-Za-z0-9.-]

Default

""

Name

primaryIp - Analytics server Primary IP

Syntax

--primaryIp <string>

Description

Primary IP address of the Analytics server.

Restrictions

Either the PCMD Analytics FQDN or the Primary IP must be set

Class

Optional.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Default

0.0.0.0

Name

runOnNecc - Run On NECC

Syntax

--runOnNecc <list-item>

Description

Run On Active or Standby NECC node. This parameter is not applicable either on CNF or VNF based deployments.

Class

Optional.

List

- active - Refer to NECC active node
- default - Software decides automatically based on predefined rules
- standby - Refer to NECC standby node

Default

default

Name

secondaryIp - Analytics server Secondary IP

Syntax

--secondaryIp <string>

Description

Secondary IP address of the Analytics server.

Class

Optional.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Default

0.0.0.0

Name

secondaryPort - Secondary Port

Syntax

--secondaryPort <integer>

Description

Secondary port address of service

Class

Optional.

Type / Range

Integer in the range of 0 or 1,024 to 65,535

Default

0

Name

tcpProfileName - TCP Profile Name

Syntax`--tcpProfileName <string>`**Description**

Configure TCP profile to the endpoint.

Class

Optional.

Type / RangeDigit string of length 1 to 31 with the allowed pattern of `[0-9a-zA-Z\.\-_\:@]*`**Default**

default

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.44 cmm pcsCluster

pcs command wrapper. Syntax, description, parameters, and roles for **cmm pcsCluster** command.

Syntax**Description**

This command is used to take an action on the PCS cluster. The actions are config - show PCS cluster configuration details, status - show PCS cluster status, standby - place a cluster in standby, unstandby - restore a PCS cluster from standby.

Parameters

Name

operation - pcs operation

Syntax

--operation <list-item>

Description

Operation to perform

Class

Required key.

List

- config - PCS operation of config.
- standby - PCS operation of standby.
- status - PCS operation of status.
- unstandby - PCS operation of unstandby.

Name

data - response field

Syntax

--data <string>

Description

response field

Class

Read only.

Type / Range

Digit string of length

Name

node - Node

Syntax

--node <integer>

Description

Node.

Class

Optional.

Type / Range

Integer in the range of 0 to 2

Roles

Create

sysadmin, manage

List

sysadmin, manage, monitor

6.45 cmm perfParams

perf tool parameters . Syntax, description, parameters, and roles for cmm perfParams command.

Syntax

Description

Parameters needed for configuring perf tool for automatic perf collection and recovery actions. This command should only be used by Nokia support team.

Parameters

Name

poolType - Pool type

Syntax

--poolType <list-item>

Description

Target sgsn pool type.

Class

Required key.

List

- IPPS - IP packet processing service

Name

allowedDuration - Allowed duration

Syntax

--allowedDuration <integer>

Description

Allowed active duration in seconds of the configured overload before restarting the process.
0 should indicate that we will never restart the process.

Class

Optional.

Type / Range

Integer in the range of 0 to 600

Default

0

Name

callGraph - Call graph during perf collection

Syntax

--callGraph <list-item>

Description

If call graph will be enabled during perf log collection

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

enable - State of the mechanism

Syntax

--enable <list-item>

Description

State of the whole mechanism, if its active or not

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

logCollectionAlarmSeverity - Severity of the overload

Syntax

--logCollectionAlarmSeverity <list-item>

Description

Severity of the overload the automatic log collection / recovery actions will start.

Class

Optional.

List

- critical - Critical alarm severity
- major - Major alarm severity
- minor - Minor alarm severity

Default

critical

Name

perfDuration - Duration of perf collection

Syntax

--perfDuration <integer>

Description

Duration of perf collection in seconds

Class

Optional.

Type / Range

Integer in the range of 1 to 10

Default

5

Name

samplesToCollect - Perf samples

Syntax

--samplesToCollect <integer>

Description

Perf samples we want to collect in each occurrence

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

1

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.46 cmm personalize

Personalize. Syntax, description, parameters, and roles for cmm personalize command.

Syntax

Description

Personalize locale, timezone, naming and ntp parameters.

Parameters

Name

personalizeId - Id

Syntax

--personalizeId <string>

Description

Just type 'config' here.

Class

Required key.

Type / Range

Digit string of length 6 to 6 with the allowed pattern of [0-9a-zA-Z\._:\]*

Name

ntpPrimaryIp - NTP primary IP

Syntax

--ntpPrimaryIp <string>

Description

NTP primary IP.

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

ntpSecondaryIp - NTP secondary IP

Syntax

--ntpSecondaryIp <string>

Description

NTP secondary IP.

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

timeZoneName - Time zone

Syntax

--timeZoneName <list-item>

Description

Time zone.

Class

Optional.

List

- Africa/Abidjan - Timezone for africa/abidjan UTC offset of +00:00.
- Africa/Accra - Timezone for africa/accra UTC offset of +00:00.
- Africa/Addis_Ababa - Timezone for africa/addis_ababa UTC offset of +03:00.
- Africa/Algiers - Timezone for africa/algiers UTC offset of +01:00.
- Africa/Asmara - Timezone for africa/asmara UTC offset of +03:00.
- Africa/Asmera - Timezone for africa/asmera UTC offset of +03:00.
- Africa/Bamako - Timezone for africa/bamako UTC offset of +00:00.
- Africa/Bangui - Timezone for africa/bangui UTC offset of +01:00.
- Africa/Banjul - Timezone for africa/banjul UTC offset of +00:00.

- Africa/Bissau - Timezone for africa/bissau UTC offset of +00:00.
- Africa/Blantyre - Timezone for africa/blantyre UTC offset of +02:00.
- Africa/Brazzaville - Timezone for africa/brazzaville UTC offset of +01:00.
- Africa/Bujumbura - Timezone for africa/bujumbura UTC offset of +02:00.
- Africa/Cairo - Timezone for africa/cairo UTC offset of +02:00.
- Africa/Casablanca - Timezone for africa/casablanca UTC offset of +00:00.
- Africa/Ceuta - Timezone for africa/ceuta UTC offset of +01:00.
- Africa/Conakry - Timezone for africa/conakry UTC offset of +00:00.
- Africa/Dakar - Timezone for africa/dakar UTC offset of +00:00.
- Africa/Dar_es_Salaam - Timezone for africa/dar_es_salaam UTC offset of +03:00.
- Africa/Djibouti - Timezone for africa/djibouti UTC offset of +03:00.
- Africa/Douala - Timezone for africa/douala UTC offset of +01:00.
- Africa/El_Aaiun - Timezone for africa/el_aaiun UTC offset of +00:00.
- Africa/Freetown - Timezone for africa/freetown UTC offset of +00:00.
- Africa/Gaborone - Timezone for africa/gaborone UTC offset of +02:00.
- Africa/Harare - Timezone for africa/harare UTC offset of +02:00.
- Africa/Johannesburg - Timezone for africa/johannesburg UTC offset of +02:00.
- Africa/Juba - Timezone for africa/juba UTC offset of +03:00.
- Africa/Kampala - Timezone for africa/kampala UTC offset of +03:00.
- Africa/Khartoum - Timezone for africa/khartoum UTC offset of +03:00.
- Africa/Kigali - Timezone for africa/kigali UTC offset of +02:00.
- Africa/Kinshasa - Timezone for africa/kinshasa UTC offset of +01:00.
- Africa/Lagos - Timezone for africa/lagos UTC offset of +01:00.
- Africa/Libreville - Timezone for africa/libreville UTC offset of +01:00.
- Africa/Lome - Timezone for africa/lome UTC offset of +00:00.
- Africa/Luanda - Timezone for africa/luanda UTC offset of +01:00.
- Africa/Lubumbashi - Timezone for africa/lubumbashi UTC offset of +02:00.
- Africa/Lusaka - Timezone for africa/lusaka UTC offset of +02:00.
- Africa/Malabo - Timezone for africa/malabo UTC offset of +01:00.
- Africa/Maputo - Timezone for africa/maputo UTC offset of +02:00.
- Africa/Maseru - Timezone for africa/maseru UTC offset of +02:00.

- Africa/Mbabane - Timezone for africa/mbabane UTC offset of +02:00.
- Africa/Mogadishu - Timezone for africa/mogadishu UTC offset of +03:00.
- Africa/Monrovia - Timezone for africa/monrovia UTC offset of +00:00.
- Africa/Nairobi - Timezone for africa/nairobi UTC offset of +03:00.
- Africa/Ndjamena - Timezone for africa/ndjamena UTC offset of +01:00.
- Africa/Niamey - Timezone for africa/niamey UTC offset of +01:00.
- Africa/Nouakchott - Timezone for africa/nouakchott UTC offset of +00:00.
- Africa/Ouagadougou - Timezone for africa/ouagadougou UTC offset of +00:00.
- Africa/Porto-Novo - Timezone for africa/porto-novo UTC offset of +01:00.
- Africa/Sao_Tome - Timezone for africa/sao_tome UTC offset of +00:00.
- Africa/Timbuktu - Timezone for africa/timbuktu UTC offset of +00:00.
- Africa/Tripoli - Timezone for africa/tripoli UTC offset of +02:00.
- Africa/Tunis - Timezone for africa/tunis UTC offset of +01:00.
- Africa/Windhoek - Timezone for africa/windhoek UTC offset of +02:00.
- America/Adak - Timezone for america/adak UTC offset of -10:00.
- America/Anchorage - Timezone for america/anchorage UTC offset of -09:00.
- America/Anguilla - Timezone for america/anguilla UTC offset of -04:00.
- America/Antigua - Timezone for america/antigua UTC offset of -04:00.
- America/Araguaina - Timezone for america/araguaina UTC offset of -03:00.
- America/Argentina/Buenos_Aires - Timezone for america/argentina/buenos_aires UTC offset of -03:00.
- America/Argentina/Catamarca - Timezone for america/argentina/catamarca UTC offset of -03:00.
- America/Argentina/ComodRivadavia - Timezone for america/argentina/comodrivadavia UTC offset of -03:00.
- America/Argentina/Cordoba - Timezone for america/argentina/cordoba UTC offset of -03:00.
- America/Argentina/Jujuy - Timezone for america/argentina/jujuy UTC offset of -03:00.
- America/Argentina/La_Rioja - Timezone for america/argentina/la_rioja UTC offset of -03:00.
- America/Argentina/Mendoza - Timezone for america/argentina/mendoza UTC offset of -03:00.

- America/Argentina/Rio_Gallegos - Timezone for america/argentina/rio_gallegos UTC offset of -03:00.
- America/Argentina/Salta - Timezone for america/argentina/salta UTC offset of -03:00.
- America/Argentina/San_Juan - Timezone for america/argentina/san_juan UTC offset of -03:00.
- America/Argentina/San_Luis - Timezone for america/argentina/san_luis UTC offset of -03:00.
- America/Argentina/Tucuman - Timezone for america/argentina/tucuman UTC offset of -03:00.
- America/Argentina/Ushuaia - Timezone for america/argentina/ushuaia UTC offset of -03:00.
- America/Aruba - Timezone for america/aruba UTC offset of -04:00.
- America/Asuncion - Timezone for america/asuncion UTC offset of -03:00.
- America/Atikokan - Timezone for america/atikokan UTC offset of -05:00.
- America/Atka - Timezone for america/atka UTC offset of -10:00.
- America/Bahia - Timezone for america/bahia UTC offset of -03:00.
- America/Bahia_Banderas - Timezone for america/bahia_banderas UTC offset of -06:00.
- America/Barbados - Timezone for america/barbados UTC offset of -04:00.
- America/Belem - Timezone for america/belem UTC offset of -03:00.
- America/Belize - Timezone for america/belize UTC offset of -06:00.
- America/Blanc-Sablon - Timezone for america/blanc-sablon UTC offset of -04:00.
- America/Boa_Vista - Timezone for america/boa_vista UTC offset of -04:00.
- America/Bogota - Timezone for america/bogota UTC offset of -05:00.
- America/Boise - Timezone for america/boise UTC offset of -07:00.
- America/Buenos_Aires - Timezone for america/buenos_aires UTC offset of -03:00.
- America/Cambridge_Bay - Timezone for america/cambridge_bay UTC offset of -07:00.
- America/Campo_Grande - Timezone for america/campo_grande UTC offset of -03:00.
- America/Cancun - Timezone for america/cancun UTC offset of -05:00.
- America/Caracas - Timezone for america/caracas UTC offset of -04:00.
- America/Catamarca - Timezone for america/catamarca UTC offset of -03:00.
- America/Cayenne - Timezone for america/cayenne UTC offset of -03:00.
- America/Cayman - Timezone for america/cayman UTC offset of -05:00.
- America/Chicago - Timezone for america/chicago UTC offset of -06:00.

- America/Chihuahua - Timezone for america/chihuahua UTC offset of -07:00.
- America/Coral_Harbour - Timezone for america/coral_harbour UTC offset of -05:00.
- America/Cordoba - Timezone for america/cordoba UTC offset of -03:00.
- America/Costa_Rica - Timezone for america/costa_rica UTC offset of -06:00.
- America/Creston - Timezone for america/creston UTC offset of -07:00.
- America/Cuiaba - Timezone for america/cuiaba UTC offset of -03:00.
- America/Curacao - Timezone for america/curacao UTC offset of -04:00.
- America/Danmarkshavn - Timezone for america/danmarkshavn UTC offset of +00:00.
- America/Dawson - Timezone for america/dawson UTC offset of -08:00.
- America/Dawson_Creek - Timezone for america/dawson_creek UTC offset of -07:00.
- America/Denver - Timezone for america/denver UTC offset of -07:00.
- America/Detroit - Timezone for america/detroit UTC offset of -05:00.
- America/Dominica - Timezone for america/dominica UTC offset of -04:00.
- America/Edmonton - Timezone for america/edmonton UTC offset of -07:00.
- America/Eirunepe - Timezone for america/eirunepe UTC offset of -05:00.
- America/El_Salvador - Timezone for america/el_salvador UTC offset of -06:00.
- America/Ensenada - Timezone for america/ensenada UTC offset of -08:00.
- America/Fort_Nelson - Timezone for america/fort_nelson UTC offset of -07:00.
- America/Fort_Wayne - Timezone for america/fort_wayne UTC offset of -05:00.
- America/Fortaleza - Timezone for america/fortaleza UTC offset of -03:00.
- America/Glace_Bay - Timezone for america/glance_bay UTC offset of -04:00.
- America/Godthab - Timezone for america/godthab UTC offset of -03:00.
- America/Goose_Bay - Timezone for america/goose_bay UTC offset of -04:00.
- America/Grand_Turk - Timezone for america/grand_turk UTC offset of -04:00.
- America/Grenada - Timezone for america/grenada UTC offset of -04:00.
- America/Guadeloupe - Timezone for america/guadeloupe UTC offset of -04:00.
- America/Guatemala - Timezone for america/guatemala UTC offset of -06:00.
- America/Guayaquil - Timezone for america/guayaquil UTC offset of -05:00.
- America/Guyana - Timezone for america/guyana UTC offset of -04:00.
- America/Halifax - Timezone for america/halifax UTC offset of -04:00.
- America/Havana - Timezone for america/havana UTC offset of -05:00.

- America/Hermosillo - Timezone for america/hermosillo UTC offset of -07:00.
- America/Indiana/Indianapolis - Timezone for america/indiana/indianapolis UTC offset of -05:00.
- America/Indiana/Knox - Timezone for america/indiana/knox UTC offset of -06:00.
- America/Indiana/Marengo - Timezone for america/indiana/marengo UTC offset of -05:00.
- America/Indiana/Petersburg - Timezone for america/indiana/petersburg UTC offset of -05:00.
- America/Indiana/Tell_City - Timezone for america/indiana/tell_city UTC offset of -06:00.
- America/Indiana/Vevay - Timezone for america/indiana/vevay UTC offset of -05:00.
- America/Indiana/Vincennes - Timezone for america/indiana/vincennes UTC offset of -05:00.
- America/Indiana/Winamac - Timezone for america/indiana/winamac UTC offset of -05:00.
- America/Indianapolis - Timezone for america/indianapolis UTC offset of -05:00.
- America/Inuvik - Timezone for america/inuvik UTC offset of -07:00.
- America/Iqaluit - Timezone for america/iquait UTC offset of -05:00.
- America/Jamaica - Timezone for america/jamaica UTC offset of -05:00.
- America/Jujuy - Timezone for america/jujuy UTC offset of -03:00.
- America/Juneau - Timezone for america/juneau UTC offset of -09:00.
- America/Kentucky/Louisville - Timezone for america/kentucky/louisville UTC offset of -05:00.
- America/Kentucky/Monticello - Timezone for america/kentucky/monticello UTC offset of -05:00.
- America/Knox_IN - Timezone for america/knox_in UTC offset of -06:00.
- America/Kralendijk - Timezone for america/kralendijk UTC offset of -04:00.
- America/La_Paz - Timezone for america/la_paz UTC offset of -04:00.
- America/Lima - Timezone for america/lima UTC offset of -05:00.
- America/Los_Angeles - Timezone for america/los_angeles UTC offset of -08:00.
- America/Louisville - Timezone for america/louisville UTC offset of -05:00.
- America/Lower_Princes - Timezone for america/lower_princes UTC offset of -04:00.
- America/Maceio - Timezone for america/maceio UTC offset of -03:00.
- America/Managua - Timezone for america/managua UTC offset of -06:00.

- America/Manaus - Timezone for america/manaus UTC offset of -04:00.
- America/Marigot - Timezone for america/marigot UTC offset of -04:00.
- America/Martinique - Timezone for america/martinique UTC offset of -04:00.
- America/Matamoros - Timezone for america/matamoros UTC offset of -06:00.
- America/Mazatlan - Timezone for america/mazatlan UTC offset of -07:00.
- America/Mendoza - Timezone for america/mendoza UTC offset of -03:00.
- America/Menominee - Timezone for america/menominee UTC offset of -06:00.
- America/Merida - Timezone for america/merida UTC offset of -06:00.
- America/Metlakatla - Timezone for america/metlakatla UTC offset of -09:00.
- America/Mexico_City - Timezone for america/mexico_city UTC offset of -06:00.
- America/Miquelon - Timezone for america/miquelon UTC offset of -03:00.
- America/Moncton - Timezone for america/moncton UTC offset of -04:00.
- America/Monterrey - Timezone for america/monterrey UTC offset of -06:00.
- America/Montevideo - Timezone for america/montevideo UTC offset of -03:00.
- America/Montreal - Timezone for america/montreal UTC offset of -05:00.
- America/Montserrat - Timezone for america/montserrat UTC offset of -04:00.
- America/Nassau - Timezone for america/nassau UTC offset of -05:00.
- America/New_York - Timezone for america/new_york UTC offset of -05:00.
- America/Nipigon - Timezone for america/nipigon UTC offset of -05:00.
- America/Nome - Timezone for america/nome UTC offset of -09:00.
- America/Noronha - Timezone for america/noronha UTC offset of -02:00.
- America/North_Dakota/Beulah - Timezone for america/north_dakota/beulah UTC offset of -06:00.
- America/North_Dakota/Center - Timezone for america/north_dakota/center UTC offset of -06:00.
- America/North_Dakota/New_Salem - Timezone for america/north_dakota/new_salem UTC offset of -06:00.
- America/Ojinaga - Timezone for america/ojinaga UTC offset of -07:00.
- America/Panama - Timezone for america/panama UTC offset of -05:00.
- America/Pangnirtung - Timezone for america/pangnirtung UTC offset of -05:00.
- America/Paramaribo - Timezone for america/paramaribo UTC offset of -03:00.
- America/Phoenix - Timezone for america/phoenix UTC offset of -07:00.

- America/Port-au-Prince - Timezone for america/port-au-prince UTC offset of -05:00.
- America/Port_of_Spain - Timezone for america/port_of_spain UTC offset of -04:00.
- America/Porto_Acre - Timezone for america/porto_acre UTC offset of -05:00.
- America/Porto_Velho - Timezone for america/porto_velho UTC offset of -04:00.
- America/Puerto_Rico - Timezone for america/puerto_rico UTC offset of -04:00.
- America/Rainy_River - Timezone for america/rainy_river UTC offset of -06:00.
- America/Rankin_Inlet - Timezone for america/rankin_inlet UTC offset of -06:00.
- America/Recife - Timezone for america/recife UTC offset of -03:00.
- America/Regina - Timezone for america/regina UTC offset of -06:00.
- America/Resolute - Timezone for america/resolute UTC offset of -06:00.
- America/Rio_Branco - Timezone for america/rio_branco UTC offset of -05:00.
- America/Rosario - Timezone for america/rosario UTC offset of -03:00.
- America/Santa_Isabel - Timezone for america/santa_isabel UTC offset of -08:00.
- America/Santarem - Timezone for america/santarem UTC offset of -03:00.
- America/Santiago - Timezone for america/santiago UTC offset of -03:00.
- America/Santo_Domingo - Timezone for america/santo_domingo UTC offset of -04:00.
- America/Sao_Paulo - Timezone for america/sao_paulo UTC offset of -02:00.
- America/Scoresbysund - Timezone for america/scoresbysund UTC offset of -01:00.
- America/Shiprock - Timezone for america/shiprock UTC offset of -07:00.
- America/Sitka - Timezone for america/sitka UTC offset of -09:00.
- America/St_Barthelemy - Timezone for america/st_barthelemy UTC offset of -04:00.
- America/St_Johns - Timezone for america/st_johns UTC offset of -03:30.
- America/St_Kitts - Timezone for america/st_kitts UTC offset of -04:00.
- America/St_Lucia - Timezone for america/st_lucia UTC offset of -04:00.
- America/St_Thomas - Timezone for america/st_thomas UTC offset of -04:00.
- America/St_Vincent - Timezone for america/st_vincent UTC offset of -04:00.
- America/Swift_Current - Timezone for america/swift_current UTC offset of -06:00.
- America/Tegucigalpa - Timezone for america/tegucigalpa UTC offset of -06:00.
- America/Thule - Timezone for america/thule UTC offset of -04:00.
- America/Thunder_Bay - Timezone for america/thunder_bay UTC offset of -05:00.
- America/Tijuana - Timezone for america/tijuana UTC offset of -08:00.

- America/Toronto - Timezone for america/toronto UTC offset of -05:00.
- America/Tortola - Timezone for america/tortola UTC offset of -04:00.
- America/Vancouver - Timezone for america/vancouver UTC offset of -08:00.
- America/Virgin - Timezone for america/virgin UTC offset of -04:00.
- America/Whitehorse - Timezone for america/whitehorse UTC offset of -08:00.
- America/Winnipeg - Timezone for america/winnipeg UTC offset of -06:00.
- America/Yakutat - Timezone for america/yakutat UTC offset of -09:00.
- America/Yellowknife - Timezone for america/yellowknife UTC offset of -07:00.
- Antarctica/Casey - Timezone for antarctica/casey UTC offset of +11:00.
- Antarctica/Davis - Timezone for antarctica/davis UTC offset of +07:00.
- Antarctica/DumontDURville - Timezone for antarctica/dumontdurville UTC offset of +10:00.
- Antarctica/Macquarie - Timezone for antarctica/macquarie UTC offset of +11:00.
- Antarctica/Mawson - Timezone for antarctica/mawson UTC offset of +05:00.
- Antarctica/McMurdo - Timezone for antarctica/mcmurdo UTC offset of +13:00.
- Antarctica/Palmer - Timezone for antarctica/palmer UTC offset of -03:00.
- Antarctica/Rothera - Timezone for antarctica/rothera UTC offset of -03:00.
- Antarctica/South_Pole - Timezone for antarctica/south_pole UTC offset of +13:00.
- Antarctica/Syowa - Timezone for antarctica/syowa UTC offset of +03:00.
- Antarctica/Troll - Timezone for antarctica/troll UTC offset of +00:00.
- Antarctica/Vostok - Timezone for antarctica/vostok UTC offset of +06:00.
- Arctic/Longyearbyen - Timezone for arctic/longyearbyen UTC offset of +01:00.
- Asia/Aden - Timezone for asia/aden UTC offset of +03:00.
- Asia/Almaty - Timezone for asia/almaty UTC offset of +06:00.
- Asia/Amman - Timezone for asia/amman UTC offset of +02:00.
- Asia/Anadyr - Timezone for asia/anadyr UTC offset of +12:00.
- Asia/Aqtau - Timezone for asia/aqtau UTC offset of +05:00.
- Asia/Aqtobe - Timezone for asia/aqtobe UTC offset of +05:00.
- Asia/Ashgabat - Timezone for asia/ashgabat UTC offset of +05:00.
- Asia/Ashkhabad - Timezone for asia/ashkhabad UTC offset of +05:00.
- Asia/Baghdad - Timezone for asia/baghdad UTC offset of +03:00.

- Asia/Bahrain - Timezone for asia/bahrain UTC offset of +03:00.
- Asia/Baku - Timezone for asia/baku UTC offset of +04:00.
- Asia/Bangkok - Timezone for asia/bangkok UTC offset of +07:00.
- Asia/Barnaul - Timezone for asia/barnaul UTC offset of +07:00.
- Asia/Beirut - Timezone for asia/beirut UTC offset of +02:00.
- Asia/Bishkek - Timezone for asia/bishkek UTC offset of +06:00.
- Asia/Brunei - Timezone for asia/brunei UTC offset of +08:00.
- Asia/Calcutta - Timezone for asia/calcutta UTC offset of +05:30.
- Asia/Chita - Timezone for asia/chita UTC offset of +09:00.
- Asia/Choibalsan - Timezone for asia/choibalsan UTC offset of +08:00.
- Asia/Chongqing - Timezone for asia/chongqing UTC offset of +08:00.
- Asia/Chungking - Timezone for asia/chungking UTC offset of +08:00.
- Asia/Colombo - Timezone for asia/colombo UTC offset of +05:30.
- Asia/Dacca - Timezone for asia/dacca UTC offset of +06:00.
- Asia/Damascus - Timezone for asia/damascus UTC offset of +02:00.
- Asia/Dhaka - Timezone for asia/dhaka UTC offset of +06:00.
- Asia/Dili - Timezone for asia/dili UTC offset of +09:00.
- Asia/Dubai - Timezone for asia/dubai UTC offset of +04:00.
- Asia/Dushanbe - Timezone for asia/dushanbe UTC offset of +05:00.
- Asia/Gaza - Timezone for asia/gaza UTC offset of +02:00.
- Asia/Harbin - Timezone for asia/harbin UTC offset of +08:00.
- Asia/Hebron - Timezone for asia/hebron UTC offset of +02:00.
- Asia/Ho_Chi_Minh - Timezone for asia/ho_chi_minh UTC offset of +07:00.
- Asia/Hong_Kong - Timezone for asia/hong_kong UTC offset of +08:00.
- Asia/Hovd - Timezone for asia/hovd UTC offset of +07:00.
- Asia/Irkutsk - Timezone for asia/irkutsk UTC offset of +08:00.
- Asia/Istanbul - Timezone for asia/istanbul UTC offset of +03:00.
- Asia/Jakarta - Timezone for asia/jakarta UTC offset of +07:00.
- Asia/Jayapura - Timezone for asia/jayapura UTC offset of +09:00.
- Asia/Jerusalem - Timezone for asia/jerusalem UTC offset of +02:00.
- Asia/Kabul - Timezone for asia/kabul UTC offset of +04:30.

- Asia/Kamchatka - Timezone for asia/kamchatka UTC offset of +12:00.
- Asia/Karachi - Timezone for asia/karachi UTC offset of +05:00.
- Asia/Kashgar - Timezone for asia/kashgar UTC offset of +06:00.
- Asia/Kathmandu - Timezone for asia/kathmandu UTC offset of +05:45.
- Asia/Katmandu - Timezone for asia/katmandu UTC offset of +05:45.
- Asia/Khandyga - Timezone for asia/khandyga UTC offset of +09:00.
- Asia/Kolkata - Timezone for asia/kolkata UTC offset of +05:30.
- Asia/Krasnoyarsk - Timezone for asia/krasnoyarsk UTC offset of +07:00.
- Asia/Kuala_Lumpur - Timezone for asia/kuala_lumpur UTC offset of +08:00.
- Asia/Kuching - Timezone for asia/kuching UTC offset of +08:00.
- Asia/Kuwait - Timezone for asia/kuwait UTC offset of +03:00.
- Asia/Macao - Timezone for asia/macao UTC offset of +08:00.
- Asia/Macau - Timezone for asia/macau UTC offset of +08:00.
- Asia/Magadan - Timezone for asia/magadan UTC offset of +11:00.
- Asia/Makassar - Timezone for asia/makassar UTC offset of +08:00.
- Asia/Manila - Timezone for asia/manila UTC offset of +08:00.
- Asia/Muscat - Timezone for asia/muscat UTC offset of +04:00.
- Asia/Nicosia - Timezone for asia/nicosia UTC offset of +02:00.
- Asia/Novokuznetsk - Timezone for asia/novokuznetsk UTC offset of +07:00.
- Asia/Novosibirsk - Timezone for asia/novosibirsk UTC offset of +07:00.
- Asia/Omsk - Timezone for asia/omsk UTC offset of +06:00.
- Asia/Oral - Timezone for asia/oral UTC offset of +05:00.
- Asia/Phnom_Penh - Timezone for asia/phnom_penh UTC offset of +07:00.
- Asia/Pontianak - Timezone for asia/pontianak UTC offset of +07:00.
- Asia/Pyongyang - Timezone for asia/pyongyang UTC offset of +08:30.
- Asia/Qatar - Timezone for asia/qatar UTC offset of +03:00.
- Asia/Qyzylorda - Timezone for asia/qyzylorda UTC offset of +06:00.
- Asia/Rangoon - Timezone for asia/rangoon UTC offset of +06:30.
- Asia/Riyadh - Timezone for asia/riyadh UTC offset of +03:00.
- Asia/Saigon - Timezone for asia/saigon UTC offset of +07:00.
- Asia/Sakhalin - Timezone for asia/sakhalin UTC offset of +11:00.

- Asia/Samarkand - Timezone for asia/samarkand UTC offset of +05:00.
- Asia/Seoul - Timezone for asia/seoul UTC offset of +09:00.
- Asia/Shanghai - Timezone for asia/shanghai UTC offset of +08:00.
- Asia/Singapore - Timezone for asia/singapore UTC offset of +08:00.
- Asia/Srednekolymsk - Timezone for asia/srednekolymsk UTC offset of +11:00.
- Asia/Taipei - Timezone for asia/taipei UTC offset of +08:00.
- Asia/Tashkent - Timezone for asia/tashkent UTC offset of +05:00.
- Asia/Tbilisi - Timezone for asia/tbilisi UTC offset of +04:00.
- Asia/Tehran - Timezone for asia/tehran UTC offset of +03:30.
- Asia/Tel_Aviv - Timezone for asia/tel_aviv UTC offset of +02:00.
- Asia/Thimbu - Timezone for asia/thimbu UTC offset of +06:00.
- Asia/Thimphu - Timezone for asia/thimphu UTC offset of +06:00.
- Asia/Tokyo - Timezone for asia/tokyo UTC offset of +09:00.
- Asia/Tomsk - Timezone for asia/tomsk UTC offset of +07:00.
- Asia/Ujung_Pandang - Timezone for asia/ujung_pandang UTC offset of +08:00.
- Asia/Ulaanbaatar - Timezone for asia/ulaanbaatar UTC offset of +08:00.
- Asia/Ulan_Bator - Timezone for asia/ulan_bator UTC offset of +08:00.
- Asia/Urumqi - Timezone for asia/urumqi UTC offset of +06:00.
- Asia/Ust-Nera - Timezone for asia/ust-nera UTC offset of +10:00.
- Asia/Vientiane - Timezone for asia/vientiane UTC offset of +07:00.
- Asia/Vladivostok - Timezone for asia/vladivostok UTC offset of +10:00.
- Asia/Yakutsk - Timezone for asia/yakutsk UTC offset of +09:00.
- Asia/Yekaterinburg - Timezone for asia/yekaterinburg UTC offset of +05:00.
- Asia/Yerevan - Timezone for asia/yerevan UTC offset of +04:00.
- Atlantic/Azores - Timezone for atlantic/azores UTC offset of -01:00.
- Atlantic/Bermuda - Timezone for atlantic/bermuda UTC offset of -04:00.
- Atlantic/Canary - Timezone for atlantic/canary UTC offset of +00:00.
- Atlantic/Cape_Verde - Timezone for atlantic/cape_verde UTC offset of -01:00.
- Atlantic/Faeroe - Timezone for atlantic/faeroe UTC offset of +00:00.
- Atlantic/Faroe - Timezone for atlantic/faroe UTC offset of +00:00.
- Atlantic/Jan_Mayen - Timezone for atlantic/jan_mayen UTC offset of +01:00.

- Atlantic/Madeira - Timezone for atlantic/madeira UTC offset of +00:00.
- Atlantic/Reykjavik - Timezone for atlantic/reykjavik UTC offset of +00:00.
- Atlantic/South_Georgia - Timezone for atlantic/south_georgia UTC offset of -02:00.
- Atlantic/St_Helena - Timezone for atlantic/st_helena UTC offset of +00:00.
- Atlantic/Stanley - Timezone for atlantic/stanley UTC offset of -03:00.
- Australia/ACT - Timezone for australia/act UTC offset of +11:00.
- Australia/Adelaide - Timezone for australia/adelaide UTC offset of +10:30.
- Australia/Brisbane - Timezone for australia/brisbane UTC offset of +10:00.
- Australia/Broken_Hill - Timezone for australia/broken_hill UTC offset of +10:30.
- Australia/Canberra - Timezone for australia/canberra UTC offset of +11:00.
- Australia/Currie - Timezone for australia/currie UTC offset of +11:00.
- Australia/Darwin - Timezone for australia/darwin UTC offset of +09:30.
- Australia/Eucla - Timezone for australia/eucla UTC offset of +08:45.
- Australia/Hobart - Timezone for australia/hobart UTC offset of +11:00.
- Australia/LHI - Timezone for australia/lhi UTC offset of +11:00.
- Australia/Lindeman - Timezone for australia/lindeman UTC offset of +10:00.
- Australia/Lord_Howe - Timezone for australia/lord_howe UTC offset of +11:00.
- Australia/Melbourne - Timezone for australia/melbourne UTC offset of +11:00.
- Australia/NSW - Timezone for australia/nsw UTC offset of +11:00.
- Australia/North - Timezone for australia/north UTC offset of +09:30.
- Australia/Perth - Timezone for australia/perth UTC offset of +08:00.
- Australia/Queensland - Timezone for australia/queensland UTC offset of +10:00.
- Australia/South - Timezone for australia/south UTC offset of +10:30.
- Australia/Sydney - Timezone for australia/sydney UTC offset of +11:00.
- Australia/Tasmania - Timezone for australia/tasmania UTC offset of +11:00.
- Australia/Victoria - Timezone for australia/victoria UTC offset of +11:00.
- Australia/West - Timezone for australia/west UTC offset of +08:00.
- Australia/Yancowinna - Timezone for australia/yancowinna UTC offset of +10:30.
- CET - Timezone for CET UTC offset of +01:00.
- CST6CDT - Timezone for CST6CDT UTC offset of -06:00.
- Canada/Atlantic - Timezone for canada/atlantic UTC offset of -04:00.

- Canada/Central - Timezone for canada/central UTC offset of -06:00.
- Canada/East-Saskatchewan - Timezone for canada/east-saskatchewan UTC offset of -06:00.
- Canada/Eastern - Timezone for canada/eastern UTC offset of -05:00.
- Canada/Mountain - Timezone for canada/mountain UTC offset of -07:00.
- Canada/Newfoundland - Timezone for canada/newfoundland UTC offset of -03:30.
- Canada/Pacific - Timezone for canada/pacific UTC offset of -08:00.
- Canada/Saskatchewan - Timezone for canada/saskatchewan UTC offset of -06:00.
- Canada/Yukon - Timezone for canada/yukon UTC offset of -08:00.
- Chile/Continental - Timezone for chile/continental UTC offset of -03:00.
- Chile/EasterIsland - Timezone for chile/easterisland UTC offset of -05:00.
- Cuba - Timezone for cuba UTC offset of -05:00.
- Default - Timezone for system UTC (OS time zone)
- EET - Timezone for EET UTC offset of +02:00.
- EST - Timezone for EST UTC offset of -05:00.
- EST5EDT - Timezone for EST5EDT UTC offset of -05:00.
- Egypt - Timezone for egypt UTC offset of +02:00.
- Eire - Timezone for eire UTC offset of +00:00.
- Etc/Greenwich - Timezone for etc/greenwich UTC offset of +00:00.
- Etc/UCT - Timezone for etc/uct UTC offset of +00:00.
- Etc/UTC - Timezone for etc/utc UTC offset of +00:00.
- Etc/Universal - Timezone for etc/universal UTC offset of +00:00.
- Etc/Zulu - Timezone for etc/zulu UTC offset of +00:00.
- Europe/Amsterdam - Timezone for europe/amsterdam UTC offset of +01:00.
- Europe/Andorra - Timezone for europe/andorra UTC offset of +01:00.
- Europe/Astrakhan - Timezone for europe/astrakhan UTC offset of +04:00.
- Europe/Athens - Timezone for europe/athens UTC offset of +02:00.
- Europe/Belfast - Timezone for europe/belfast UTC offset of +00:00.
- Europe/Belgrade - Timezone for europe/belgrade UTC offset of +01:00.
- Europe/Berlin - Timezone for europe/berlin UTC offset of +01:00.
- Europe/Bratislava - Timezone for europe/bratislava UTC offset of +01:00.

- Europe/Brussels - Timezone for europe/brussels UTC offset of +01:00.
- Europe/Bucharest - Timezone for europe/bucharest UTC offset of +02:00.
- Europe/Budapest - Timezone for europe/budapest UTC offset of +01:00.
- Europe/Busingen - Timezone for europe/busingen UTC offset of +01:00.
- Europe/Chisinau - Timezone for europe/chisinau UTC offset of +02:00.
- Europe/Copenhagen - Timezone for europe/copenhagen UTC offset of +01:00.
- Europe/Dublin - Timezone for europe/dublin UTC offset of +00:00.
- Europe/Gibraltar - Timezone for europe/gibraltar UTC offset of +01:00.
- Europe/Guernsey - Timezone for europe/guernsey UTC offset of +00:00.
- Europe/Helsinki - Timezone for europe/helsinki UTC offset of +02:00.
- Europe/Isle_of_Man - Timezone for europe/isle_of_man UTC offset of +00:00.
- Europe/Istanbul - Timezone for europe/istanbul UTC offset of +03:00.
- Europe/Jersey - Timezone for europe/jersey UTC offset of +00:00.
- Europe/Kaliningrad - Timezone for europe/kaliningrad UTC offset of +02:00.
- Europe/Kiev - Timezone for europe/kyiv UTC offset of +02:00.
- Europe/Kirov - Timezone for europe/kirov UTC offset of +03:00.
- Europe/Lisbon - Timezone for europe/lisbon UTC offset of +00:00.
- Europe/Ljubljana - Timezone for europe/ljubljana UTC offset of +01:00.
- Europe/London - Timezone for europe/london UTC offset of +00:00.
- Europe/Luxembourg - Timezone for europe/luxembourg UTC offset of +01:00.
- Europe/Madrid - Timezone for europe/madrid UTC offset of +01:00.
- Europe/Malta - Timezone for europe/malta UTC offset of +01:00.
- Europe/Mariehamn - Timezone for europe/mariehamn UTC offset of +02:00.
- Europe/Minsk - Timezone for europe/minsk UTC offset of +03:00.
- Europe/Monaco - Timezone for europe/monaco UTC offset of +01:00.
- Europe/Moscow - Timezone for europe/moscow UTC offset of +03:00.
- Europe/Nicosia - Timezone for europe/nicosia UTC offset of +02:00.
- Europe/Oslo - Timezone for europe/oslo UTC offset of +01:00.
- Europe/Paris - Timezone for europe/paris UTC offset of +01:00.
- Europe/Podgorica - Timezone for europe/podgorica UTC offset of +01:00.
- Europe/Prague - Timezone for europe/prague UTC offset of +01:00.

- Europe/Riga - Timezone for europe/riga UTC offset of +02:00.
- Europe/Rome - Timezone for europe/rome UTC offset of +01:00.
- Europe/Samara - Timezone for europe/samara UTC offset of +04:00.
- Europe/San_Marino - Timezone for europe/san_marino UTC offset of +01:00.
- Europe/Sarajevo - Timezone for europe/sarajevo UTC offset of +01:00.
- Europe/Simferopol - Timezone for europe/simferopol UTC offset of +03:00.
- Europe/Skopje - Timezone for europe/skopje UTC offset of +01:00.
- Europe/Sofia - Timezone for europe/sofia UTC offset of +02:00.
- Europe/Stockholm - Timezone for europe/stockholm UTC offset of +01:00.
- Europe/Tallinn - Timezone for europe/tallinn UTC offset of +02:00.
- Europe/Tirane - Timezone for europe/tirane UTC offset of +01:00.
- Europe/Tiraspol - Timezone for europe/tiraspol UTC offset of +02:00.
- Europe/Ulyanovsk - Timezone for europe/ulyanovsk UTC offset of +04:00.
- Europe/Uzhgorod - Timezone for europe/uzhgorod UTC offset of +02:00.
- Europe/Vaduz - Timezone for europe/vaduz UTC offset of +01:00.
- Europe/Vatican - Timezone for europe/vatican UTC offset of +01:00.
- Europe/Vienna - Timezone for europe/vienna UTC offset of +01:00.
- Europe/Vilnius - Timezone for europe/vilnius UTC offset of +02:00.
- Europe/Volgograd - Timezone for europe/volgograd UTC offset of +03:00.
- Europe/Warsaw - Timezone for europe/warsaw UTC offset of +01:00.
- Europe/Zagreb - Timezone for europe/zagreb UTC offset of +01:00.
- Europe/Zaporozhye - Timezone for europe/zaporozhye UTC offset of +02:00.
- Europe/Zurich - Timezone for europe/zurich UTC offset of +01:00.
- GB - Timezone for GB UTC offset of +00:00.
- GB-Eire - Timezone for GB-Eire UTC offset of +00:00.
- GMT - Timezone for GMT UTC offset of +00:00.
- GMT0 - Timezone for GMT0 UTC offset of +00:00.
- Greenwich - Timezone for greenwich UTC offset of +00:00.
- HST - Timezone for HST UTC offset of -10:00.
- Hongkong - Timezone for hongkong UTC offset of +08:00.
- Iceland - Timezone for iceland UTC offset of +00:00.

- Indian/Antananarivo - Timezone for indian/antananarivo UTC offset of +03:00.
- Indian/Chagos - Timezone for indian/chagos UTC offset of +06:00.
- Indian/Christmas - Timezone for indian/christmas UTC offset of +07:00.
- Indian/Cocos - Timezone for indian/cocos UTC offset of +06:30.
- Indian/Comoro - Timezone for indian/comoro UTC offset of +03:00.
- Indian/Kerguelen - Timezone for indian/kerguelen UTC offset of +05:00.
- Indian/Mahe - Timezone for indian/mahe UTC offset of +04:00.
- Indian/Maldives - Timezone for indian/maldives UTC offset of +05:00.
- Indian/Mauritius - Timezone for indian/mauritius UTC offset of +04:00.
- Indian/Mayotte - Timezone for indian/mayotte UTC offset of +03:00.
- Indian/Reunion - Timezone for indian/reunion UTC offset of +04:00.
- Iran - Timezone for iran UTC offset of +03:30.
- Israel - Timezone for israel UTC offset of +02:00.
- Jamaica - Timezone for jamaica UTC offset of -05:00.
- Japan - Timezone for japan UTC offset of +09:00.
- Kwajalein - Timezone for kwajalein UTC offset of +12:00.
- Libya - Timezone for libya UTC offset of +02:00.
- MET - Timezone for MET UTC offset of +01:00.
- MST - Timezone for MST UTC offset of -07:00.
- MST7MDT - Timezone for MST7MDT UTC offset of -07:00.
- Mexico/BajaNorte - Timezone for mexico/bajanorte UTC offset of -08:00.
- Mexico/BajaSur - Timezone for mexico/bajasur UTC offset of -07:00.
- Mexico/General - Timezone for mexico/general UTC offset of -06:00.
- NZ - Timezone for NZ UTC offset of +13:00.
- NZ-CHAT - Timezone for NZ-CHAT UTC offset of +13:45.
- Navajo - Timezone for navajo UTC offset of -07:00.
- PRC - Timezone for PRC UTC offset of +08:00.
- PST8PDT - Timezone for PST8PDT UTC offset of -08:00.
- Pacific/Apia - Timezone for pacific/apia UTC offset of +14:00.
- Pacific/Auckland - Timezone for pacific/auckland UTC offset of +13:00.
- Pacific/Bougainville - Timezone for pacific/bougainville UTC offset of +11:00.

- Pacific/Chatham - Timezone for pacific/chatham UTC offset of +13:45.
- Pacific/Chuuk - Timezone for pacific/chuuk UTC offset of +10:00.
- Pacific/Easter - Timezone for pacific/easter UTC offset of -05:00.
- Pacific/Efate - Timezone for pacific/efate UTC offset of +11:00.
- Pacific/Enderbury - Timezone for pacific/enderbury UTC offset of +13:00.
- Pacific/Fakaofo - Timezone for pacific/fakaofo UTC offset of +13:00.
- Pacific/Fiji - Timezone for pacific/fiji UTC offset of +13:00.
- Pacific/Funafuti - Timezone for pacific/funafuti UTC offset of +12:00.
- Pacific/Galapagos - Timezone for pacific/galapagos UTC offset of -06:00.
- Pacific/Gambier - Timezone for pacific/gambier UTC offset of -09:00.
- Pacific/Guadalcanal - Timezone for pacific/guadalcanal UTC offset of +11:00.
- Pacific/Guam - Timezone for pacific/guam UTC offset of +10:00.
- Pacific/Honolulu - Timezone for pacific/honolulu UTC offset of -10:00.
- Pacific/Johnston - Timezone for pacific/johnston UTC offset of -10:00.
- Pacific/Kiritimati - Timezone for pacific/kiritimati UTC offset of +14:00.
- Pacific/Kosrae - Timezone for pacific/kosrae UTC offset of +11:00.
- Pacific/Kwajalein - Timezone for pacific/kwajalein UTC offset of +12:00.
- Pacific/Majuro - Timezone for pacific/majuro UTC offset of +12:00.
- Pacific/Marquesas - Timezone for pacific/marquesas UTC offset of -09:30.
- Pacific/Midway - Timezone for pacific/midway UTC offset of -11:00.
- Pacific/Nauru - Timezone for pacific/nauru UTC offset of +12:00.
- Pacific/Niue - Timezone for pacific/niue UTC offset of -11:00.
- Pacific/Norfolk - Timezone for pacific/norfolk UTC offset of +11:00.
- Pacific/Noumea - Timezone for pacific/noumea UTC offset of +11:00.
- Pacific/Pago_Pago - Timezone for pacific/pago_pago UTC offset of -11:00.
- Pacific/Palau - Timezone for pacific/palau UTC offset of +09:00.
- Pacific/Pitcairn - Timezone for pacific/pitcairn UTC offset of -08:00.
- Pacific/Pohnpei - Timezone for pacific/pohnpei UTC offset of +11:00.
- Pacific/Ponape - Timezone for pacific/ponape UTC offset of +11:00.
- Pacific/Port_Moresby - Timezone for pacific/port_moresby UTC offset of +10:00.
- Pacific/Rarotonga - Timezone for pacific/rarotonga UTC offset of -10:00.

- Pacific/Saipan - Timezone for pacific/saipan UTC offset of +10:00.
- Pacific/Samoa - Timezone for pacific/samoa UTC offset of -11:00.
- Pacific/Tahiti - Timezone for pacific/tahiti UTC offset of -10:00.
- Pacific/Tarawa - Timezone for pacific/tarawa UTC offset of +12:00.
- Pacific/Tongatapu - Timezone for pacific/tongatapu UTC offset of +14:00.
- Pacific/Truk - Timezone for pacific/truk UTC offset of +10:00.
- Pacific/Wake - Timezone for pacific/wake UTC offset of +12:00.
- Pacific/Wallis - Timezone for pacific/wallis UTC offset of +12:00.
- Pacific/Yap - Timezone for pacific/yap UTC offset of +10:00.
- Poland - Timezone for poland UTC offset of +01:00.
- Portugal - Timezone for portugal UTC offset of +00:00.
- ROC - Timezone for ROC UTC offset of +08:00.
- ROK - Timezone for ROK UTC offset of +09:00.
- Singapore - Timezone for singapore UTC offset of +08:00.
- Turkey - Timezone for turkey UTC offset of +03:00.
- UCT - Timezone for UCT UTC offset of +00:00.
- US/Pacific-New - Timezone for US/Pacific-New UTC offset of -08:00.
- Universal - Timezone for universal UTC offset of +00:00.
- W-SU - Timezone for W-SU UTC offset of +03:00.
- WET - Timezone for WET UTC offset of +00:00.
- Zulu - Timezone for zulu UTC offset of +00:00.
- jdj_tz - Timezone for jdj_tz UTC offset of +00:00.

Name

vnfcCpps - VNFC CPPSO

Syntax

--vnfcCpps <string>

Description

VNFC CPPSO.

Class

Optional.

Type / Range

Digit string of length 1 to 15 with the allowed pattern of [0-9a-zA-Z\-*]

Name

vnfcDbcs - VNFC DBS0

Syntax

--vnfcDbcs <string>

Description

VNFC DBS0.

Class

Optional.

Type / Range

Digit string of length 1 to 15 with the allowed pattern of [0-9a-zA-Z\-*]

Name

vnfcIpds - VNFC IPDS0

Syntax

--vnfcIpds <string>

Description

VNFC IPDS0.

Class

Optional.

Type / Range

Digit string of length 1 to 15 with the allowed pattern of [0-9a-zA-Z\-*]

Name

vnfcIpps - VNFC IPPS0

Syntax

--vnfcIpps <string>

Description

VNFC IPPS0.

Class

Optional.

Type / Range

Digit string of length 1 to 15 with the allowed pattern of [0-9a-zA-Z\-*]

Name

vnfcNecc - VNFC NECC0

Syntax

--vnfcNecc <string>

Description

VNFC NECC0.

Class

Optional.

Type / Range

Digit string of length 1 to 15 with the allowed pattern of [0-9a-zA-Z\-*]

Name

vnfcPaps - VNFC PAPSO

Syntax

--vnfcPaps <string>

Description

VNFC PAPSO.

Class

Optional.

Type / Range

Digit string of length 1 to 15 with the allowed pattern of [0-9a-zA-Z\-*]

Name

vnfName - VNF name

Syntax

--vnfName <string>

Description

VNF name.

Class

Optional.

Type / Range

Digit string of length 1 to 40 with the allowed pattern of [0-9a-zA-Z\-*]

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

6.47 cmm pgw

PGW. Syntax, description, parameters, and roles for cmm pgw command.

Syntax

Description

PGW.

Parameters

Name

ip - PGW IP address

Syntax

--ip <string>

Description

PGW IP address.

Class

Required key.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

hostName - PGW host name

Syntax

--hostName <string>

Description

PGW host name.

Class

Required on create.

Type / Range

Digit string of length 1 to 255

Name

combinedSmfPgwc - Combined SMF and PGW-c

Syntax

--combinedSmfPgwc <list-item>

Description

Identifies a combined SMF and PGW-c node

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.48 cmm pgwFqdnItem

PGW FQDN. Syntax, description, parameters, and roles for cmm pgwFqdnItem command.

Syntax**Description**

Define an FQDN to be added to a PGW FQDN List.

Parameters**Name**

name - PGW FQDN Name

Syntax

--name <string>

Description

PGW FQDN Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

fqdn - FQDN

Syntax

--fqdn <string>

Description

FQDN

Class

Required on create, not modifiable.

Type / Range

Digit string of length 1 to 96 with the allowed pattern of [0-9a-zA-Z.-]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.49 cmm pgwFqdnList

PGW FQDN List. Syntax, description, parameters, and roles for cmm pgwFqdnList command.

Syntax**Description**

List of PGW FQDNs. Up to 1024 lists can be created. Each list can have up to 16 FQDNs.

Parameters**Name**

name - Name

Syntax

--name <string>

Description

PGW FQDN List Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.50 cmm pgwFqdnListItem

PGW FQDN List Item. Syntax, description, parameters, and roles for cmm pgwFqdnListItem command.

Syntax**Description**

Link a PGW FQDN to a PGW FQDN List. An FQDN may be added only once in a list.

Parameters**Name**

pgwFqdnListName - PGW FQDN List Name

Syntax

--pgwFqdnListName <string>

Description

PGW FQDN List Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

pgwFqdnItemName - PGW FQDN Item Name

Syntax

--pgwFqdnItemName <string>

Description

PGW FQDN Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.51 cmm pgwSelectionAllServingPlmns

PGW selection All Serving PLMNs. Syntax, description, parameters, and roles for cmm pgwSelectionAllServingPlmns command.

Syntax**Description**

MME global static PGW selection service for defined full IMSIs on all available serving PLMNs. An entry identified by an entry id includes the match criteria: -IMSI list or IMSI range (minImsi, maxImsi) -APN list - for roamers, the APN list type (LBO/HRT/both), separate list required for different roaming types. The APN list type only applies for roaming. When a UE PDN connectivity request (type 'Initial Request') matches the above criteria, the entry defines the GW pool with the static PGW(s) that shall be used for these IMSI(s)

towards the indicated APN(s) (per roaming type). Either IMSI list or IMSI range can be set for a single entry id: If IMSI list is provided, both min IMSI and max IMSI must not be set. For IMSI range, both min IMSI and max IMSI need to be provided. Min IMSI must have smaller value than max IMSI. If min IMSI and max IMSI are provided, IMSI list must not be set. If neither IMSI list nor IMSI range (min IMSI, max IMSI) is provided, the entry id is valid for all IMSIs. If no APN list is provided, the entry id applies for any APN for the indicated IMSI(s) (or all IMSIs, if neither IMSI list nor IMSI range is provided as above).

Parameters

Name

entryId - Entry identifier

Syntax

--entryId <integer>

Description

Entry identifier.

Class

Required key.

Type / Range

Integer in the range of 1 to 1000

Name

gwPoolName - GW Pool name

Syntax

--gwPoolName <string>

Description

The GW pool name.

Class

Required on create.

Type / Range

Digit string of length 1 to 20 with the allowed pattern of [0-9a-zA-Z\._:-]*

Name

apnListType - APN List Type

Syntax

--apnListType <list-item>

Description

Identifies the roaming usage type of the APN list, applied for rules defined for PGW pools.

Class

Optional.

List

- both - Local Breakout and Home Routed Traffic - APN List is used for roaming with Local Breakout and Home Routed Traffic
- hrt - Home Routed Traffic - APN List is used for roaming with Home Routed Traffic
- lbo - Local Break Out - APN List is used for roaming with Local Break Out

Default

both

Name

apnPgwSelectionListName - APN PGW selection list name

Syntax

--apnPgwSelectionListName <string>

Description

The name of APN PGW selection list indicating the APN(s).

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Name

imsiListName - IMSI List Name

Syntax

--imsiListName <string>

Description

The name of the IMSI list to apply the selection.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Name

maxImsi - Maximum IMSI

Syntax

--maxImsi <digits>

Description

The Maximum full IMSI value of an IMSI range.

Class

Optional.

Type / Range

Digit string of length 0 or 9 to 15

Default

"

Name

minImsi - Minimum IMSI

Syntax

--minImsi <digits>

Description

The Minimum full IMSI value of an IMSI range.

Class

Optional.

Type / Range

Digit string of length 0 or 9 to 15

Default

"

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.52 cmm pgwSelectionApn

PGW Selection Supplementary APN. Syntax, description, parameters, and roles for cmm pgwSelectionApn command.

Syntax

Description

This command is used to administer a PGW Selection Supplementary APN and includes parameters for defining the supplementary APN name, served mobile network name, and PDN type.

Parameters

Name

plmnName - Mobile network name

Syntax

--plmnName <string>

Description

Name that identifies served mobile network.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

pdnType - PDN type

Syntax

--pdnType <list-item>

Description

PDN type.

Class

Required key.

List

- ipv4 - Specify PDN type of IPv4.
- ipv4v6 - Specify PDN type of IPv4 / IPv6.
- ipv6 - Specify PDN type of IPv6.

Name

apnNi - Supplementary access point name

Syntax

--apnNi <string>

Description

Supplementary access point name.

Class

Required on create.

Type / Range

Digit string of length 1 to 62 with the allowed pattern of [0-9a-zA-Z\.\-]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.53 cmm pgwSelectionCcApn

PGW selection charging characteristics per APN. Syntax, description, parameters, and roles for cmm pgwSelectionCcApn command.

Syntax**Description**

charging characteristics per APN for selecting P gateway.

Parameters**Name**

listName - List name

Syntax

--listName <string>

Description

Name of the list of APNs and other attributes.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

apnNi - Access point name - network indicator

Syntax

--apnNi <string>

Description

APN NI to match on.

Class

Required key.

Type / Range

Digit string of length 1 to 62 with the allowed pattern of [0-9a-zA-Z\.-]*

Name

ccLength - Charging characteristics length

Syntax

--ccLength <integer>

Description

Length of CC to be used for P-GW selection.

Class

Optional.

Type / Range

Integer in the range of 0 to 16

Default

4

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.54 cmm pgwSelectionRangeServices

Services based on PGW selection range. Syntax, description, parameters, and roles for cmm pgwSelectionRangeServices command.

Syntax

Description

Services based on PGW selection range.

Parameters

Name

uePlmnServicesName - Name of services based on UE network and served network

Syntax

--uePlmnServicesName <string>

Description

Name that identifies roaming agreement.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

entryId - Entry ID

Syntax

--entryId <integer>

Description

Entry ID.

Class

Required key.

Type / Range

Integer in the range of 1 to 10,000

Name

apnPgwSelectionListName - Name of APN PGW selection list

Syntax

--apnPgwSelectionListName <string>

Description

Name of APN PGW selection list.

Class

Required on create.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

gwPoolName - PGW pool name

Syntax

--gwPoolName <string>

Description

Gateway pool name this gateway is assigned to.

Class

Required on create.

Type / Range

Digit string of length 1 to 20 with the allowed pattern of [0-9a-zA-Z\.\-_:]*

Name

maxMsin - Maximum MSIN

Syntax

--maxMsin <digits>

Description

The maximum IMSI MSIN value.

Restrictions

The length 2-15 is only allowed for an MSISDN type. For an IMSI the allowed length is 8-10.

Class

Required on create.

Type / Range

Digit string of length 0 or 2 to 15

Name

minMsin - Minimum MSIN

Syntax

--minMsin <digits>

Description

The minimum IMSI MSIN value.

Restrictions

The length 2-15 is only allowed for an MSISDN type. For an IMSI the allowed length is 8-10.

Class

Required on create.

Type / Range

Digit string of length 0 or 2 to 15

Name

rangeServiceType - Range service type

Syntax

--rangeServiceType <list-item>

Description

Range service type.

Class

Required on create, not modifiable.

List

- imsi - Specify range service type of IMSI.
- msisdn - Specify range service type of MSISDN.

Name

apnListType - APN List Type

Syntax

--apnListType <list-item>

Description

Identifies the roaming usage type of the APN list.

Class

Optional.

List

- both - Local Breakout and Home Routed Traffic - APN List is used for roaming with Local Breakout and Home Routed Traffic
- hrt - Home Routed Traffic - APN List is used for roaming with Home Routed Traffic
- lbo - Local Break Out - APN List is used for roaming with Local Break Out

Default

both

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.55 cmm pgwSelectionService

Service based on PGW selection. Syntax, description, parameters, and roles for cmm pgwSelectionService command.

Syntax**Description**

Service based on PGW selection.

Parameters**Name**

apnPgwSelectionListName - Name of APN PGW selection list

Syntax

--apnPgwSelectionListName <string>

Description

Name of APN PGW selection list.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

gwPoolName - PGW pool name

Syntax

--gwPoolName <string>

Description

Gateway pool name this gateway is assigned to.

Class

Required key.

Type / Range

Digit string of length 1 to 20 with the allowed pattern of [0-9a-zA-Z\._ :]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.56 cmm plmn

Public Land Mobile Networks. Syntax, description, parameters, and roles for cmm plmn command.

Syntax**Description**

This command is used to administer a Public Land Mobile Network and includes parameters for defining the network's MCC and MNC, PLMN type, roaming agreements, ARP level, QCI value, and APN-NI.

Parameters**Name**

name - Mobile network name

Syntax

--name <string>

Description

Name that identifies mobile network (alternate key).

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

mcc - Mobile country code

Syntax

--mcc <digits>

Description

3-digit number used to represent the mobile country code.

Class

Required on create, not modifiable.

Type / Range

Digit string of length 3

Name

mnc - Mobile network code

Syntax

--mnc <digits>

Description

Digits used to represent mobile network code.

Class

Required on create, not modifiable.

Type / Range

Digit string of length 2 to 3

Name

allowUnknEquipEirRsp - Allow Unknown Equipment Eir Rsp

Syntax

--allowUnknEquipEirRsp <list-item>

Description

Allow EIR Check supports a provisioning option per PLMN to allow UEs with a EIR check result of 'Equipment Unknown' to attach successfully.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

apnAdminListName - APN Administration List Name

Syntax

--apnAdminListName <string>

Description

The list for the APN administration actions.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

apnQciPpiMapListName - APN QCI PPI mapping list name

Syntax

--apnQciPpiMapListName <string>

Description

Assign an APN to QCI to PPI map previously created to home network.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

arp - Allocation retention priority

Syntax

--arp <integer>

Description

Allocation retention priority, priority-level.

Restrictions

This element is not applicable when plmnType is Other Network.

Class

Optional.

Type / Range

Integer in the range of 1 to 15

Default

1

Name

arpHiPriorityLevel - ARP high priority level

Syntax`--arpHiPriorityLevel <integer>`**Description**

Indicate the largest ARP value that can be assigned to a high priority access user.

Restrictions

This element is not applicable when plmnType is Other Network.

Class

Optional.

Type / Range

Integer in the range of 0 to 15

Default

0

Name

arpPciDisabled - ARP PCI always disabled

Syntax`--arpPciDisabled <list-item>`**Description**

ARP PCI always disabled.

Restrictions

This element is not applicable when plmnType is Other Network.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

camelAllowed - Camel subscriber roaming

Syntax`--camelAllowed <list-item>`

Description

Indicates whether roaming of camel subscribers is allowed or not.

Restrictions

This element is not applicable when plmnType is Other Network.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

checkImsApnSubscription - Check IMS APN subscription

Syntax

--checkImsApnSubscription <list-item>

Description

Check IMS APN subscription in determining network IMS voice over PS support.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

checkStnSrAvpAvailability - Check availability of STN-SR

Syntax

--checkStnSrAvpAvailability <list-item>

Description

Check STN-SR AVP availability in determining network IMS voice over PS support.

Class

Optional.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Name

decorAllowed - Decor Allowed

Syntax

--decorAllowed <list-item>

Description

Specifies whether Decor is allowed to be used in this serving PLMN.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

defaultDcnId - Default Dedicated Core Network ID

Syntax

--defaultDcnId <integer>

Description

This parameter indicates the DCN ID for the default dedicated core. Operators must choose the same default DCN ID for the MME group designated as default MME group. The MME supporting the will ignore the UE Usage Types assigned to the DCN ID as default DCN supports all UE Usage Types and UE with no UE Usage Type parameter.

Class

Optional.

Type / Range

Integer in the range of 1 to 126 or 128 to 65,535

Default

65535

Name

defaultDcnIdRoamer - Default Dedicated Core Network ID for Roamer

Syntax

--defaultDcnIdRoamer <integer>

Description

This parameter indicates the DCN ID for the default dedicated core for an inbound roamer to the PLMN. Operators must choose the same default DCN ID for the MME group designated as default MME group. The MME supporting the will ignore the UE Usage Types assigned to the DCN ID as default DCN supports all UE Usage Types and UE with no UE Usage Type parameter.

Class

Optional.

Type / Range

Integer in the range of 1 to 126 or 128 to 65,535

Default

65535

Name

defaultMappingPlmnName - Default Mapping PLMN Name

Syntax

--defaultMappingPlmnName <string>

Description

Default PLMN ID for LAI for TAI-to-LAI Mapping

Restrictions

Can be set only in home PLMN

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Name

defaultRacFor5gSrvccMme - Default RAC for 5G-SRVCC on MME_SRVCC

Syntax

--defaultRacFor5gSrvccMme <integer>

Description

Defines the value for the default RAC used by the MME_SRVCC during 5G-SRVCC for MSC server discovery. The MME_SRVCC ignores the AMF N26 provided RAC to always discover the MSC server with same (internally provisioned) RAC to interwork with any AMF.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

255

Name

description - Mobile PLMN description

Syntax

--description <string>

Description

Mobile PLMN description.

Class

Optional.

Type / Range

Digit string of length 0 to 160 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

directTunnelOn - Direct tunnel enabled

Syntax

--directTunnelOn <list-item>

Description

Indicates whether direct tunnel is enabled or not.

Restrictions

This element is not applicable when plmnType is Other Network.

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

emergencyProfileName - Emergency profile name

Syntax

--emergencyProfileName <string>

Description

Emergency profile name.

Restrictions

This element is not applicable when plmnType is Other Network.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Name

grAvRequested - Gr authentication vectors requested

Syntax

--grAvRequested <integer>

Description

2g3g authentication vectors requested over gr interface.

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

grImmRspPref - Gr immediate response preferred

Syntax

--grImmRspPref <list-item>

Description

Indicates whether to include "Immediate response preferred" parameter in map send auth info message over gr interface.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

homeApnNi - Home APN NI

Syntax

--homeApnNi <string>

Description

Home access point name (APN) network identifier (NI).

Restrictions

This element is not applicable when plmnType is Other Network.

Class

Optional.

Type / Range

Digit string of length 0 to 62 with the allowed pattern of [0-9a-zA-Z\.\-]*

Default

''

Name

hpGwCcApnListName - msisdn based home GW selection with CC ListName

Syntax

--hpGwCcApnListName <string>

Description

msisdn based home GW selection with CC ListName. This list applies to UEs whose IMSI is served by the Home/Shared PLMN, and also to UEs whose IMSI has a Treat-as-Home-Subscriber Roaming Agreement with this serving PLMN

Restrictions

This element is not applicable when plmnType is Other Network. For Other Network, plmnType hpGwCcApnListName must be left empty

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

hplmnCcLength - ISDN country code length

Syntax

--hplmnCcLength <integer>

Description

Returning to the home province GGSN feature. value specifies MSISDN offset from which digits will be taken and added to APN. length is specified by number of characters added to APN parameter. This attribute applies to UEs whose IMSI is served by the Home/Shared

PLMN, and also to UEs whose IMSI has a Treat-as-Home-Subscriber Roaming Agreement with this serving PLMN.

Restrictions

This element is not applicable when plmnType is Other Network.

Class

Optional.

Type / Range

Integer in the range of 1 to 3

Default

2

Name

immediateCancelLocationAck - Immediate cancel location ack

Syntax

--immediateCancelLocationAck <list-item>

Description

Cancel location acknowledgment is sent immediately.

Restrictions

This element is not applicable when plmnType is Other Network.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

includePlmnAttachRate - Include Plmn Attach Rate

Syntax

--includePlmnAttachRate <list-item>

Description

When enabled, UE attaches with this PLMN are counted towards the Roamer Attach Rate calculation when the Roamer Attach Rate Limit global parameter is enabled.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

initiateLcsEmergency - Initiate LCS emergency

Syntax

--initiateLcsEmergency <list-item>

Description

Indicates whether the MME should initiate a location request for a UE sending an emergency attach request or an emergency PDN connectivity request valid values are do not initiate, send SLR to GMLC with estimated UE location, send SLR without location estimate to GMLC.

Class

Optional.

List

- Do not initiate - Do not initiate.
- Send SLR to GMLC with estimated UE location - Send SLR to GMLC with estimated UE location.
- Send SLR without location estimate to GMLC - Send SLR without location estimate to GMLC.

Default

Do not initiate

Name

interMmeTauEcrPercentage - Inter MME TAU ECR percentage

Syntax

--interMmeTauEcrPercentage <integer>

Description

Inter MME TAU ECR percentage to EIR.

Class

Optional.

Type / Range

Integer in the range of 0 to 100

Default

0

Name

ipV6SupportForNoDab - IPv6 PDN type used when DAB is not supported

Syntax

--ipV6SupportForNoDab <list-item>

Description

Indicates whether IPv6 PDN type is used when PLMN does not support DAB (Dual access bearer) otherwise IPv4 PDN type is used.

Restrictions

This element is not applicable when plmnType is of type Other Network.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

iratTauEcrPercentage - IRAT TAU ECR percentage

Syntax

--iratTauEcrPercentage <integer>

Description

IRAT TAU ECR percentage.

Class

Optional.

Type / Range

Integer in the range of 0 to 100

Default

0

Name

isdnCc - ISDN country code

Syntax

--isdnCc <digits>

Description

Digits use to represent the ISDN country code. used when validating zone codes received from the HSS. zone codes include a country code identifier.

Class

Optional.

Type / Range

Digit string of length 0 to 3

Name

lteTo2G3GQosMappingName - LTE to 2G3G QoS mapping profile name

Syntax

--lteTo2G3GQosMappingName <string>

Description

Name that identifies the LTE-2G3G QoS mapping profile, whose parameters are used during handover between 2G/3G and 4G networks.

Restrictions

This element is not applicable when plmnType is Other Network.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

default

Name

maxBitRateDown - Max UE AMBR downlink rate

Syntax

--maxBitRateDown <integer>

Description

Max UE AMBR downlink rate.

Restrictions

This element is not applicable when plmnType is Other Network.

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64- 568, 576-8640, 8700-16000, 17000-128000, 130000-4000000000 in units of kbps

Default

256000

Name

maxBitRateUp - Max UE AMBR uplink rate

Syntax

--maxBitRateUp <integer>

Description

Max UE AMBR uplink rate.

Restrictions

This element is not applicable when plmnType is Other Network.

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64- 568, 576-8640, 8700-16000, 17000-128000, 130000-4000000000 in units of kbps

Default

256000

Name

mbmsEnabled - MBMS enabled

Syntax

--mbmsEnabled <list-item>

Description

Indicates whether multimedia broadcast/multicast service is enabled.

Restrictions

This element is not applicable when plmnType is Other Network.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

mncDigits - Number of MNC digits

Syntax

--mncDigits <list-item>

Description

Number of digits used to represent the mobile network code.

Class

Not modifiable.

List

- 2/3Digits - 2 or 3 digits - 2 or 3 digits.
- 2Digits - 2 digits - 2 digits.
- 3Digits - 3 digits - 3 digits.

Default

3Digits

Name

mnSubscriberDbType - Mobile network subscriber database type

Syntax

--mnSubscriberDbType <list-item>

Description

Mobile network subscriber database type.

Restrictions

Home and Shared PLMN must not be None. Other PLMN must be set to None. This element is not applicable when plmnType is Other Network.

Class

Optional.

List

- " - Empty - Empty.
- EPC-Capability - EPC-Capability.
- HLR - HLR.
- HLR-HSS - HLR-HSS.
- HSS - HSS.
- None - None.

Default

"

Name

nonBroadcastLaiLac - Non-Broadcast LAI LAC

Syntax

--nonBroadcastLaiLac <integer>

Description

The non-broadcast LAI is used for two different purposes: in the SMS in MME and combined attach without CS domain functionalities. This parameter defines the LAC part of it. The MME may provide a Location Area Identification (LAI) for the UE on Combined Attach/TAU Accept to indicate the CS registration area. When either the SMS in MME or the combined attach without CS domain feature is used, the CS domain is not involved and thus the MME needs to provide the UE a LAI that is not associated with any VLR. The MME provides this provisioned LAI in order not to mix with any real CS NW used LAI.

Class

Optional.

Type / Range

Integer in the range of 1 to 65,533

Default

65533

Name

nonBroadcastTac - Non Broadcast TAC

Syntax

--nonBroadcastTac <integer>

Description

MME uses this TAC in GUTI allocation request to force UE to send TAU request

Class

Optional.

Type / Range

Integer in the range of 0 to 65,535

Default

65534

Name

nriLength - NRI length

Syntax

--nriLength <integer>

Description

The number of bits within a P-TMSI that are used as the network resource identifier.

Class

Optional.

Type / Range

Integer in the range of 0 to 8

Default

0

Name

numberOfAddedDigits - Number of digits added to APN

Syntax

--numberOfAddedDigits <integer>

Description

Returning to the home province GGSN (0=feature is off). value specifies number of digits taken from MSISDN (starting after ISDN country code length) and added to APN. This attribute applies to UEs whose IMSI is served by the Home/Shared PLMN, and also to UEs whose IMSI has a Treat-as-Home-Subscriber Roaming Agreement with this serving PLMN.

Restrictions

This element is not applicable when plmnType is Other Network.

Class

Optional.

Type / Range

Integer in the range of 0 to 15

Default

0

Name

obtainImeiSv - Obtain IMEISV

Syntax

--obtainImeiSv <list-item>

Description

Obtain International Mobile Equipment Identifier. Defines whether the MME obtains the IMEI from the UE, or not. If this field is set to "None" then Validate IMEI with EIR must be set to false when plmnType is either Home Network or Shared Network

Restrictions

This element is not applicable when plmnType is Other Network.

Class

Optional.

List

- IDR - Use NAS IDENTITY REQUEST.
- None - Do not obtain IMEI.
- SMC - Use NAS security mode control procedure.

Default

SMC

Name

oprDefQciProfileListName - Operator defined QCI profile name

Syntax

--oprDefQciProfileListName <string>

Description

Operator defined QCI profile name.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

plmnNetProtocol - Inter-Gateway protocol

Syntax

--plmnNetProtocol <list-item>

Description

PLMN inter-gateway protocol type used by this MME.

Restrictions

This element is not applicable when plmnType is Other Network.

Class

Optional.

List

- GTP - Specify SGW supports S5/S8 based on GTP.
- PMIP - Specify SGW supports S5/S8 based on PMIP.

Default

GTP

Name

plmnType - PLMN type

Syntax

--plmnType <list-item>

Description

Indicates for the PLMN type.

Class

Not modifiable.

List

- Home Network - Specify network type of home network.
- Other Network - Specify network type of other network (not home or shared).
- Shared Network - Specify network type of shared network.

Default

Other Network

Name

qci - QoS class identifier

Syntax

--qci <integer>

Description

Indicates a cap (best possible QoS value allowed) for roamer UEs.

Restrictions

This element is not applicable when plmnType is Other Network.

Class

Optional.

Type / Range

Integer in the range of 6 to 9

Default

6

Name

ratDefaultRoamingHandling - Roaming agreement default handling

Syntax

--ratDefaultRoamingHandling <list-item>

Description

Roaming agreement default handling - allows / restricts access for those PLMNs not explicitly defined in the roaming networks.

Restrictions

This element is not applicable when plmnType is Other Network.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

regionalSubscriptionForRoamers - Regional subscription for roamers

Syntax

--regionalSubscriptionForRoamers <list-item>

Description

When set to true enables regional subscription for roamers in SGSN.

Restrictions

This element is not applicable when plmnType is of type Other Network.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

reservedTmsi - Reserved TMSI

Syntax

--reservedTmsi <integer>

Description

The reserved TMSI is used for two functionalities: the SMS in MME and the combined attach without CS domain. The MME may provide a Temporary Mobile Subscriber Identity (TMSI) for the UE on Combined Attach/TAU Accept for the CS registration. When either the SMS in MME or combined attach without CS domain feature is used, the CS domain is not involved and thus the MME needs to provide the UE a reserved TMSI value that cannot cause any ambiguities with other TMSIs (e.g. when the UE changes to an area where it needs to derive SMS services from a GERAN/UTRAN MSC). The MME provides this provisioned reserved TMSI value in order not to mix with any real CS NW used TMSIs.

Class

Optional.

Type / Range

Integer in the range of 0 to 4,294,967,295

Default

0

Name

sbcEnabled - SBc enabled type

Syntax

--sbcEnabled <list-item>

Description

Indicates whether the CBC feature shall be enabled or not.

Restrictions

This element is not applicable when plmnType is Other Network.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

tauEcrPercentage - TAU ECR percentage

Syntax

--tauEcrPercentage <integer>

Description

TAU ECR percentage to EIR.

Class

Optional.

Type / Range

Integer in the range of 0 to 100

Default

0

Name

timeZoneName - Time zone name

Syntax

--timeZoneName <list-item>

Description

Provides time zone names that correspond to the region selected.

Restrictions

See comments on TIME_ZONE type in CfgDataType.xsd. This element is not applicable when plmnType is Other Network.

Class

Optional.

List

- Africa/Abidjan - Timezone for africa/abidjan UTC offset of +00:00.
- Africa/Accra - Timezone for africa/accra UTC offset of +00:00.
- Africa/Addis_Ababa - Timezone for africa/addis_ababa UTC offset of +03:00.
- Africa/Algiers - Timezone for africa/algiers UTC offset of +01:00.
- Africa/Asmara - Timezone for africa/asmara UTC offset of +03:00.
- Africa/Asmera - Timezone for africa/asmera UTC offset of +03:00.
- Africa/Bamako - Timezone for africa/bamako UTC offset of +00:00.
- Africa/Bangui - Timezone for africa/bangui UTC offset of +01:00.
- Africa/Banjul - Timezone for africa/banjul UTC offset of +00:00.
- Africa/Bissau - Timezone for africa/bissau UTC offset of +00:00.
- Africa/Blantyre - Timezone for africa/blantyre UTC offset of +02:00.
- Africa/Brazzaville - Timezone for africa/brazzaville UTC offset of +01:00.
- Africa/Bujumbura - Timezone for africa/bujumbura UTC offset of +02:00.
- Africa/Cairo - Timezone for africa/cairo UTC offset of +02:00.
- Africa/Casablanca - Timezone for africa/casablanca UTC offset of +00:00.
- Africa/Ceuta - Timezone for africa/ceuta UTC offset of +01:00.
- Africa/Conakry - Timezone for africa/conakry UTC offset of +00:00.
- Africa/Dakar - Timezone for africa/dakar UTC offset of +00:00.
- Africa/Dar_es_Salaam - Timezone for africa/dar_es_salaam UTC offset of +03:00.
- Africa/Djibouti - Timezone for africa/djibouti UTC offset of +03:00.
- Africa/Douala - Timezone for africa/douala UTC offset of +01:00.
- Africa/El_Aaiun - Timezone for africa/el_aaiun UTC offset of +00:00.
- Africa/Freetown - Timezone for africa/freetown UTC offset of +00:00.
- Africa/Gaborone - Timezone for africa/gaborone UTC offset of +02:00.
- Africa/Harare - Timezone for africa/harare UTC offset of +02:00.
- Africa/Johannesburg - Timezone for africa/johannesburg UTC offset of +02:00.
- Africa/Juba - Timezone for africa/juba UTC offset of +03:00.

- Africa/Kampala - Timezone for africa/kampala UTC offset of +03:00.
- Africa/Khartoum - Timezone for africa/khartoum UTC offset of +03:00.
- Africa/Kigali - Timezone for africa/kigali UTC offset of +02:00.
- Africa/Kinshasa - Timezone for africa/kinshasa UTC offset of +01:00.
- Africa/Lagos - Timezone for africa/lagos UTC offset of +01:00.
- Africa/Libreville - Timezone for africa/libreville UTC offset of +01:00.
- Africa/Lome - Timezone for africa/lome UTC offset of +00:00.
- Africa/Luanda - Timezone for africa/luanda UTC offset of +01:00.
- Africa/Lubumbashi - Timezone for africa/lubumbashi UTC offset of +02:00.
- Africa/Lusaka - Timezone for africa/lusaka UTC offset of +02:00.
- Africa/Malabo - Timezone for africa/malabo UTC offset of +01:00.
- Africa/Maputo - Timezone for africa/maputo UTC offset of +02:00.
- Africa/Maseru - Timezone for africa/maseru UTC offset of +02:00.
- Africa/Mbabane - Timezone for africa/mbabane UTC offset of +02:00.
- Africa/Mogadishu - Timezone for africa/mogadishu UTC offset of +03:00.
- Africa/Monrovia - Timezone for africa/monrovia UTC offset of +00:00.
- Africa/Nairobi - Timezone for africa/nairobi UTC offset of +03:00.
- Africa/Ndjamena - Timezone for africa/ndjamena UTC offset of +01:00.
- Africa/Niamey - Timezone for africa/niamey UTC offset of +01:00.
- Africa/Nouakchott - Timezone for africa/nouakchott UTC offset of +00:00.
- Africa/Ouagadougou - Timezone for africa/ouagadougou UTC offset of +00:00.
- Africa/Porto-Novo - Timezone for africa/porto-novo UTC offset of +01:00.
- Africa/Sao_Tome - Timezone for africa/sao_tome UTC offset of +00:00.
- Africa/Timbuktu - Timezone for africa/timbuktu UTC offset of +00:00.
- Africa/Tripoli - Timezone for africa/tripoli UTC offset of +02:00.
- Africa/Tunis - Timezone for africa/tunis UTC offset of +01:00.
- Africa/Windhoek - Timezone for africa/windhoek UTC offset of +02:00.
- America/Adak - Timezone for america/adak UTC offset of -10:00.
- America/Anchorage - Timezone for america/anchorage UTC offset of -09:00.
- America/Anguilla - Timezone for america/anguilla UTC offset of -04:00.
- America/Antigua - Timezone for america/antigua UTC offset of -04:00.

- America/Araguaina - Timezone for america/araguaina UTC offset of -03:00.
- America/Argentina/Buenos_Aires - Timezone for america/argentina/buenos_aires UTC offset of -03:00.
- America/Argentina/Catamarca - Timezone for america/argentina/catamarca UTC offset of -03:00.
- America/Argentina/ComodRivadavia - Timezone for america/argentina/comodrivadavia UTC offset of -03:00.
- America/Argentina/Cordoba - Timezone for america/argentina/cordoba UTC offset of -03:00.
- America/Argentina/Jujuy - Timezone for america/argentina/jujuy UTC offset of -03:00.
- America/Argentina/La_Rioja - Timezone for america/argentina/la_rioja UTC offset of -03:00.
- America/Argentina/Mendoza - Timezone for america/argentina/mendoza UTC offset of -03:00.
- America/Argentina/Rio_Gallegos - Timezone for america/argentina/rio_gallegos UTC offset of -03:00.
- America/Argentina/Salta - Timezone for america/argentina/salta UTC offset of -03:00.
- America/Argentina/San_Juan - Timezone for america/argentina/san_juan UTC offset of -03:00.
- America/Argentina/San_Luis - Timezone for america/argentina/san_luis UTC offset of -03:00.
- America/Argentina/Tucuman - Timezone for america/argentina/tucuman UTC offset of -03:00.
- America/Argentina/Ushuaia - Timezone for america/argentina/ushuaia UTC offset of -03:00.
- America/Aruba - Timezone for america/aruba UTC offset of -04:00.
- America/Asuncion - Timezone for america/asuncion UTC offset of -03:00.
- America/Atikokan - Timezone for america/atikokan UTC offset of -05:00.
- America/Atka - Timezone for america/atka UTC offset of -10:00.
- America/Bahia - Timezone for america/bahia UTC offset of -03:00.
- America/Bahia_Banderas - Timezone for america/bahia_banderas UTC offset of -06:00.
- America/Barbados - Timezone for america/barbados UTC offset of -04:00.
- America/Belem - Timezone for america/belem UTC offset of -03:00.
- America/Belize - Timezone for america/belize UTC offset of -06:00.

- America/Blanc-Sablon - Timezone for america/blanc-sablon UTC offset of -04:00.
- America/Boa_Vista - Timezone for america/boa_vista UTC offset of -04:00.
- America/Bogota - Timezone for america/bogota UTC offset of -05:00.
- America/Boise - Timezone for america/boise UTC offset of -07:00.
- America/Buenos_Aires - Timezone for america/buenos_aires UTC offset of -03:00.
- America/Cambridge_Bay - Timezone for america/cambridge_bay UTC offset of -07:00.
- America/Campo_Grande - Timezone for america/campo_grande UTC offset of -03:00.
- America/Cancun - Timezone for america/cancun UTC offset of -05:00.
- America/Caracas - Timezone for america/caracas UTC offset of -04:00.
- America/Catamarca - Timezone for america/catamarca UTC offset of -03:00.
- America/Cayenne - Timezone for america/cayenne UTC offset of -03:00.
- America/Cayman - Timezone for america/cayman UTC offset of -05:00.
- America/Chicago - Timezone for america/chicago UTC offset of -06:00.
- America/Chihuahua - Timezone for america/chihuahua UTC offset of -07:00.
- America/Coral_Harbour - Timezone for america/coral_harbour UTC offset of -05:00.
- America/Cordoba - Timezone for america/cordoba UTC offset of -03:00.
- America/Costa_Rica - Timezone for america/costa_rica UTC offset of -06:00.
- America/Creston - Timezone for america/creston UTC offset of -07:00.
- America/Cuiaba - Timezone for america/cuiaba UTC offset of -03:00.
- America/Curacao - Timezone for america/curacao UTC offset of -04:00.
- America/Danmarkshavn - Timezone for america/danmarkshavn UTC offset of +00:00.
- America/Dawson - Timezone for america/dawson UTC offset of -08:00.
- America/Dawson_Creek - Timezone for america/dawson_creek UTC offset of -07:00.
- America/Denver - Timezone for america/denver UTC offset of -07:00.
- America/Detroit - Timezone for america/detroit UTC offset of -05:00.
- America/Dominica - Timezone for america/dominica UTC offset of -04:00.
- America/Edmonton - Timezone for america/edmonton UTC offset of -07:00.
- America/Eirunepe - Timezone for america/eirunepe UTC offset of -05:00.
- America/El_Salvador - Timezone for america/el_salvador UTC offset of -06:00.
- America/Ensenada - Timezone for america/ensenada UTC offset of -08:00.
- America/Fort_Nelson - Timezone for america/fort_nelson UTC offset of -07:00.

- America/Fort_Wayne - Timezone for america/fort_wayne UTC offset of -05:00.
- America/Fortaleza - Timezone for america/fortaleza UTC offset of -03:00.
- America/Glace_Bay - Timezone for america/glance_bay UTC offset of -04:00.
- America/Godthab - Timezone for america/godthab UTC offset of -03:00.
- America/Goose_Bay - Timezone for america/goose_bay UTC offset of -04:00.
- America/Grand_Turk - Timezone for america/grand_turk UTC offset of -04:00.
- America/Grenada - Timezone for america/grenada UTC offset of -04:00.
- America/Guadeloupe - Timezone for america/guadeloupe UTC offset of -04:00.
- America/Guatemala - Timezone for america/guatemala UTC offset of -06:00.
- America/Guayaquil - Timezone for america/guayaquil UTC offset of -05:00.
- America/Guyana - Timezone for america/guyana UTC offset of -04:00.
- America/Halifax - Timezone for america/halifax UTC offset of -04:00.
- America/Havana - Timezone for america/havana UTC offset of -05:00.
- America/Hermosillo - Timezone for america/hermosillo UTC offset of -07:00.
- America/Indiana/Indianapolis - Timezone for america/indiana/indianapolis UTC offset of -05:00.
- America/Indiana/Knox - Timezone for america/indiana/knox UTC offset of -06:00.
- America/Indiana/Marengo - Timezone for america/indiana/marengo UTC offset of -05:00.
- America/Indiana/Petersburg - Timezone for america/indiana/petersburg UTC offset of -05:00.
- America/Indiana/Tell_City - Timezone for america/indiana/tell_city UTC offset of -06:00.
- America/Indiana/Vevay - Timezone for america/indiana/vevay UTC offset of -05:00.
- America/Indiana/Vincennes - Timezone for america/indiana/vincennes UTC offset of -05:00.
- America/Indiana/Winamac - Timezone for america/indiana/winamac UTC offset of -05:00.
- America/Indianapolis - Timezone for america/indianapolis UTC offset of -05:00.
- America/Inuvik - Timezone for america/inuvik UTC offset of -07:00.
- America/Iqaluit - Timezone for america/iqaluit UTC offset of -05:00.
- America/Jamaica - Timezone for america/jamaica UTC offset of -05:00.
- America/Jujuy - Timezone for america/jujuy UTC offset of -03:00.

- America/Juneau - Timezone for america/juneau UTC offset of -09:00.
- America/Kentucky/Louisville - Timezone for america/kentucky/louisville UTC offset of -05:00.
- America/Kentucky/Monticello - Timezone for america/kentucky/monticello UTC offset of -05:00.
- America/Knox_IN - Timezone for america/knox_in UTC offset of -06:00.
- America/Kralendijk - Timezone for america/kralendijk UTC offset of -04:00.
- America/La_Paz - Timezone for america/la_paz UTC offset of -04:00.
- America/Lima - Timezone for america/lima UTC offset of -05:00.
- America/Los_Angeles - Timezone for america/los_angeles UTC offset of -08:00.
- America/Louisville - Timezone for america/louisville UTC offset of -05:00.
- America/Lower_Princes - Timezone for america/lower_princes UTC offset of -04:00.
- America/Maceio - Timezone for america/maceio UTC offset of -03:00.
- America/Managua - Timezone for america/managua UTC offset of -06:00.
- America/Manaus - Timezone for america/manaus UTC offset of -04:00.
- America/Marigot - Timezone for america/marigot UTC offset of -04:00.
- America/Martinique - Timezone for america/martinique UTC offset of -04:00.
- America/Matamoros - Timezone for america/matamoros UTC offset of -06:00.
- America/Mazatlan - Timezone for america/mazatlan UTC offset of -07:00.
- America/Mendoza - Timezone for america/mendoza UTC offset of -03:00.
- America/Menominee - Timezone for america/menominee UTC offset of -06:00.
- America/Merida - Timezone for america/merida UTC offset of -06:00.
- America/Metlakatla - Timezone for america/metlakatla UTC offset of -09:00.
- America/Mexico_City - Timezone for america/mexico_city UTC offset of -06:00.
- America/Miquelon - Timezone for america/miquelon UTC offset of -03:00.
- America/Moncton - Timezone for america/moncton UTC offset of -04:00.
- America/Monterrey - Timezone for america/monterrey UTC offset of -06:00.
- America/Montevideo - Timezone for america/montevideo UTC offset of -03:00.
- America/Montreal - Timezone for america/montreal UTC offset of -05:00.
- America/Montserrat - Timezone for america/montserrat UTC offset of -04:00.
- America/Nassau - Timezone for america/nassau UTC offset of -05:00.
- America/New_York - Timezone for america/new_york UTC offset of -05:00.

- America/Nipigon - Timezone for america/nipigon UTC offset of -05:00.
- America/Nome - Timezone for america/nome UTC offset of -09:00.
- America/Noronha - Timezone for america/noronha UTC offset of -02:00.
- America/North_Dakota/Beulah - Timezone for america/north_dakota/beulah UTC offset of -06:00.
- America/North_Dakota/Center - Timezone for america/north_dakota/center UTC offset of -06:00.
- America/North_Dakota/New_Salem - Timezone for america/north_dakota/new_salem UTC offset of -06:00.
- America/Ojinaga - Timezone for america/ojinaga UTC offset of -07:00.
- America/Panama - Timezone for america/panama UTC offset of -05:00.
- America/Pangnirtung - Timezone for america/pangnirtung UTC offset of -05:00.
- America/Paramaribo - Timezone for america/paramaribo UTC offset of -03:00.
- America/Phoenix - Timezone for america/phoenix UTC offset of -07:00.
- America/Port-au-Prince - Timezone for america/port-au-prince UTC offset of -05:00.
- America/Port_of_Spain - Timezone for america/port_of_spain UTC offset of -04:00.
- America/Porto_Acre - Timezone for america/porto_acre UTC offset of -05:00.
- America/Porto_Velho - Timezone for america/porto_velho UTC offset of -04:00.
- America/Puerto_Rico - Timezone for america/puerto_rico UTC offset of -04:00.
- America/Rainy_River - Timezone for america/rainy_river UTC offset of -06:00.
- America/Rankin_Inlet - Timezone for america/rankin_inlet UTC offset of -06:00.
- America/Recife - Timezone for america/recife UTC offset of -03:00.
- America/Regina - Timezone for america/regina UTC offset of -06:00.
- America/Resolute - Timezone for america/resolute UTC offset of -06:00.
- America/Rio_Branco - Timezone for america/rio_branco UTC offset of -05:00.
- America/Rosario - Timezone for america/rosario UTC offset of -03:00.
- America/Santa_Isabel - Timezone for america/santa_isabel UTC offset of -08:00.
- America/Santarem - Timezone for america/santarem UTC offset of -03:00.
- America/Santiago - Timezone for america/santiago UTC offset of -03:00.
- America/Santo_Domingo - Timezone for america/santo_domingo UTC offset of -04:00.
- America/Sao_Paulo - Timezone for america/sao_paulo UTC offset of -02:00.
- America/Scoresbysund - Timezone for america/scoresbysund UTC offset of -01:00.

- America/Shiprock - Timezone for america/shiprock UTC offset of -07:00.
- America/Sitka - Timezone for america/sitka UTC offset of -09:00.
- America/St_Barthelemy - Timezone for america/st_barthelemy UTC offset of -04:00.
- America/St_Johns - Timezone for america/st_johns UTC offset of -03:30.
- America/St_Kitts - Timezone for america/st_kitts UTC offset of -04:00.
- America/St_Lucia - Timezone for america/st_lucia UTC offset of -04:00.
- America/St_Thomas - Timezone for america/st_thomas UTC offset of -04:00.
- America/St_Vincent - Timezone for america/st_vincent UTC offset of -04:00.
- America/Swift_Current - Timezone for america/swift_current UTC offset of -06:00.
- America/Tegucigalpa - Timezone for america/tegucigalpa UTC offset of -06:00.
- America/Thule - Timezone for america/thule UTC offset of -04:00.
- America/Thunder_Bay - Timezone for america/thunder_bay UTC offset of -05:00.
- America/Tijuana - Timezone for america/tijuana UTC offset of -08:00.
- America/Toronto - Timezone for america/toronto UTC offset of -05:00.
- America/Tortola - Timezone for america/tortola UTC offset of -04:00.
- America/Vancouver - Timezone for america/vancouver UTC offset of -08:00.
- America/Virgin - Timezone for america/virgin UTC offset of -04:00.
- America/Whitehorse - Timezone for america/whitehorse UTC offset of -08:00.
- America/Winnipeg - Timezone for america/winnipeg UTC offset of -06:00.
- America/Yakutat - Timezone for america/yakutat UTC offset of -09:00.
- America/Yellowknife - Timezone for america/yellowknife UTC offset of -07:00.
- Antarctica/Casey - Timezone for antarctica/casey UTC offset of +11:00.
- Antarctica/Davis - Timezone for antarctica/davis UTC offset of +07:00.
- Antarctica/DumontDURville - Timezone for antarctica/dumontdurville UTC offset of +10:00.
- Antarctica/Macquarie - Timezone for antarctica/macquarie UTC offset of +11:00.
- Antarctica/Mawson - Timezone for antarctica/mawson UTC offset of +05:00.
- Antarctica/McMurdo - Timezone for antarctica/mcmurdo UTC offset of +13:00.
- Antarctica/Palmer - Timezone for antarctica/palmer UTC offset of -03:00.
- Antarctica/Rothera - Timezone for antarctica/rothera UTC offset of -03:00.
- Antarctica/South_Pole - Timezone for antarctica/south_pole UTC offset of +13:00.

- Antarctica/Syowa - Timezone for antarctica/syowa UTC offset of +03:00.
- Antarctica/Troll - Timezone for antarctica/troll UTC offset of +00:00.
- Antarctica/Vostok - Timezone for antarctica/vostok UTC offset of +06:00.
- Arctic/Longyearbyen - Timezone for arctic/longyearbyen UTC offset of +01:00.
- Asia/Aden - Timezone for asia/aden UTC offset of +03:00.
- Asia/Almaty - Timezone for asia/almaty UTC offset of +06:00.
- Asia/Amman - Timezone for asia/amman UTC offset of +02:00.
- Asia/Anadyr - Timezone for asia/anadyr UTC offset of +12:00.
- Asia/Aqtau - Timezone for asia/aqtau UTC offset of +05:00.
- Asia/Aqtobe - Timezone for asia/aqtobe UTC offset of +05:00.
- Asia/Ashgabat - Timezone for asia/ashgabat UTC offset of +05:00.
- Asia/Ashkhabad - Timezone for asia/ashkhabad UTC offset of +05:00.
- Asia/Baghdad - Timezone for asia/baghdad UTC offset of +03:00.
- Asia/Bahrain - Timezone for asia/bahrain UTC offset of +03:00.
- Asia/Baku - Timezone for asia/baku UTC offset of +04:00.
- Asia/Bangkok - Timezone for asia/bangkok UTC offset of +07:00.
- Asia/Barnaul - Timezone for asia/barnaul UTC offset of +07:00.
- Asia/Beirut - Timezone for asia/beirut UTC offset of +02:00.
- Asia/Bishkek - Timezone for asia/bishkek UTC offset of +06:00.
- Asia/Brunei - Timezone for asia/brunei UTC offset of +08:00.
- Asia/Calcutta - Timezone for asia/calcutta UTC offset of +05:30.
- Asia/Chita - Timezone for asia/chita UTC offset of +09:00.
- Asia/Choibalsan - Timezone for asia/choibalsan UTC offset of +08:00.
- Asia/Chongqing - Timezone for asia/chongqing UTC offset of +08:00.
- Asia/Chungking - Timezone for asia/chungking UTC offset of +08:00.
- Asia/Colombo - Timezone for asia/colombo UTC offset of +05:30.
- Asia/Dacca - Timezone for asia/dacca UTC offset of +06:00.
- Asia/Damascus - Timezone for asia/damascus UTC offset of +02:00.
- Asia/Dhaka - Timezone for asia/dhaka UTC offset of +06:00.
- Asia/Dili - Timezone for asia/dili UTC offset of +09:00.
- Asia/Dubai - Timezone for asia/dubai UTC offset of +04:00.

- Asia/Dushanbe - Timezone for asia/dushanbe UTC offset of +05:00.
- Asia/Gaza - Timezone for asia/gaza UTC offset of +02:00.
- Asia/Harbin - Timezone for asia/harbin UTC offset of +08:00.
- Asia/Hebron - Timezone for asia/hebron UTC offset of +02:00.
- Asia/Ho_Chi_Minh - Timezone for asia/ho_chi_minh UTC offset of +07:00.
- Asia/Hong_Kong - Timezone for asia/hong_kong UTC offset of +08:00.
- Asia/Hovd - Timezone for asia/hovd UTC offset of +07:00.
- Asia/Irkutsk - Timezone for asia/irkutsk UTC offset of +08:00.
- Asia/Istanbul - Timezone for asia/istanbul UTC offset of +03:00.
- Asia/Jakarta - Timezone for asia/jakarta UTC offset of +07:00.
- Asia/Jayapura - Timezone for asia/jayapura UTC offset of +09:00.
- Asia/Jerusalem - Timezone for asia/jerusalem UTC offset of +02:00.
- Asia/Kabul - Timezone for asia/kabul UTC offset of +04:30.
- Asia/Kamchatka - Timezone for asia/kamchatka UTC offset of +12:00.
- Asia/Karachi - Timezone for asia/karachi UTC offset of +05:00.
- Asia/Kashgar - Timezone for asia/kashgar UTC offset of +06:00.
- Asia/Kathmandu - Timezone for asia/kathmandu UTC offset of +05:45.
- Asia/Katmandu - Timezone for asia/katmandu UTC offset of +05:45.
- Asia/Khandyga - Timezone for asia/khandyga UTC offset of +09:00.
- Asia/Kolkata - Timezone for asia/kolkata UTC offset of +05:30.
- Asia/Krasnoyarsk - Timezone for asia/krasnoyarsk UTC offset of +07:00.
- Asia/Kuala_Lumpur - Timezone for asia/kuala_lumpur UTC offset of +08:00.
- Asia/Kuching - Timezone for asia/kuching UTC offset of +08:00.
- Asia/Kuwait - Timezone for asia/kuwait UTC offset of +03:00.
- Asia/Macao - Timezone for asia/macao UTC offset of +08:00.
- Asia/Macau - Timezone for asia/macau UTC offset of +08:00.
- Asia/Magadan - Timezone for asia/magadan UTC offset of +11:00.
- Asia/Makassar - Timezone for asia/makassar UTC offset of +08:00.
- Asia/Manila - Timezone for asia/manila UTC offset of +08:00.
- Asia/Muscat - Timezone for asia/muscat UTC offset of +04:00.
- Asia/Nicosia - Timezone for asia/nicosia UTC offset of +02:00.

- Asia/Novokuznetsk - Timezone for asia/novokuznetsk UTC offset of +07:00.
- Asia/Novosibirsk - Timezone for asia/novosibirsk UTC offset of +07:00.
- Asia/Omsk - Timezone for asia/omsk UTC offset of +06:00.
- Asia/Oral - Timezone for asia/oral UTC offset of +05:00.
- Asia/Phnom_Penh - Timezone for asia/phnom_penh UTC offset of +07:00.
- Asia/Pontianak - Timezone for asia/pontianak UTC offset of +07:00.
- Asia/Pyongyang - Timezone for asia/pyongyang UTC offset of +08:30.
- Asia/Qatar - Timezone for asia/qatar UTC offset of +03:00.
- Asia/Qyzylorda - Timezone for asia/qyzylorda UTC offset of +06:00.
- Asia/Rangoon - Timezone for asia/rangoon UTC offset of +06:30.
- Asia/Riyadh - Timezone for asia/riyadh UTC offset of +03:00.
- Asia/Saigon - Timezone for asia/saigon UTC offset of +07:00.
- Asia/Sakhalin - Timezone for asia/sakhalin UTC offset of +11:00.
- Asia/Samarkand - Timezone for asia/samarkand UTC offset of +05:00.
- Asia/Seoul - Timezone for asia/seoul UTC offset of +09:00.
- Asia/Shanghai - Timezone for asia/shanghai UTC offset of +08:00.
- Asia/Singapore - Timezone for asia/singapore UTC offset of +08:00.
- Asia/Srednekolymsk - Timezone for asia/srednekolymsk UTC offset of +11:00.
- Asia/Taipei - Timezone for asia/taipei UTC offset of +08:00.
- Asia/Tashkent - Timezone for asia/tashkent UTC offset of +05:00.
- Asia/Tbilisi - Timezone for asia/tbilisi UTC offset of +04:00.
- Asia/Tehran - Timezone for asia/tehran UTC offset of +03:30.
- Asia/Tel_Aviv - Timezone for asia/tel_aviv UTC offset of +02:00.
- Asia/Thimbu - Timezone for asia/thimbu UTC offset of +06:00.
- Asia/Thimphu - Timezone for asia/thimphu UTC offset of +06:00.
- Asia/Tokyo - Timezone for asia/tokyo UTC offset of +09:00.
- Asia/Tomsk - Timezone for asia/tomsk UTC offset of +07:00.
- Asia/Ujung_Pandang - Timezone for asia/ujung_pandang UTC offset of +08:00.
- Asia/Ulaanbaatar - Timezone for asia/ulaanbaatar UTC offset of +08:00.
- Asia/Ulan_Bator - Timezone for asia/ulan_bator UTC offset of +08:00.
- Asia/Urumqi - Timezone for asia/urumqi UTC offset of +06:00.

- Asia/Ust-Nera - Timezone for asia/ust-nera UTC offset of +10:00.
- Asia/Vientiane - Timezone for asia/vientiane UTC offset of +07:00.
- Asia/Vladivostok - Timezone for asia/vladivostok UTC offset of +10:00.
- Asia/Yakutsk - Timezone for asia/yakutsk UTC offset of +09:00.
- Asia/Yekaterinburg - Timezone for asia/yekaterinburg UTC offset of +05:00.
- Asia/Yerevan - Timezone for asia/yerevan UTC offset of +04:00.
- Atlantic/Azores - Timezone for atlantic/azores UTC offset of -01:00.
- Atlantic/Bermuda - Timezone for atlantic/bermuda UTC offset of -04:00.
- Atlantic/Canary - Timezone for atlantic/canary UTC offset of +00:00.
- Atlantic/Cape_Verde - Timezone for atlantic/cape_verde UTC offset of -01:00.
- Atlantic/Faeroe - Timezone for atlantic/faeroe UTC offset of +00:00.
- Atlantic/Faroe - Timezone for atlantic/faroe UTC offset of +00:00.
- Atlantic/Jan_Mayen - Timezone for atlantic/jan_mayen UTC offset of +01:00.
- Atlantic/Madeira - Timezone for atlantic/madeira UTC offset of +00:00.
- Atlantic/Reykjavik - Timezone for atlantic/reykjavik UTC offset of +00:00.
- Atlantic/South_Georgia - Timezone for atlantic/south_georgia UTC offset of -02:00.
- Atlantic/St_Helena - Timezone for atlantic/st_helena UTC offset of +00:00.
- Atlantic/Stanley - Timezone for atlantic/stanley UTC offset of -03:00.
- Australia/ACT - Timezone for australia/act UTC offset of +11:00.
- Australia/Adelaide - Timezone for australia/adelaide UTC offset of +10:30.
- Australia/Brisbane - Timezone for australia/brisbane UTC offset of +10:00.
- Australia/Broken_Hill - Timezone for australia/broken_hill UTC offset of +10:30.
- Australia/Canberra - Timezone for australia/canberra UTC offset of +11:00.
- Australia/Currie - Timezone for australia/currie UTC offset of +11:00.
- Australia/Darwin - Timezone for australia/darwin UTC offset of +09:30.
- Australia/Eucla - Timezone for australia/eucla UTC offset of +08:45.
- Australia/Hobart - Timezone for australia/hobart UTC offset of +11:00.
- Australia/LHI - Timezone for australia/lhi UTC offset of +11:00.
- Australia/Lindeman - Timezone for australia/lindeman UTC offset of +10:00.
- Australia/Lord_Howe - Timezone for australia/lord_howe UTC offset of +11:00.
- Australia/Melbourne - Timezone for australia/melbourne UTC offset of +11:00.

- Australia/NSW - Timezone for australia/nsw UTC offset of +11:00.
- Australia/North - Timezone for australia/north UTC offset of +09:30.
- Australia/Perth - Timezone for australia/perth UTC offset of +08:00.
- Australia/Queensland - Timezone for australia/queensland UTC offset of +10:00.
- Australia/South - Timezone for australia/south UTC offset of +10:30.
- Australia/Sydney - Timezone for australia/sydney UTC offset of +11:00.
- Australia/Tasmania - Timezone for australia/tasmania UTC offset of +11:00.
- Australia/Victoria - Timezone for australia/victoria UTC offset of +11:00.
- Australia/West - Timezone for australia/west UTC offset of +08:00.
- Australia/Yancowinna - Timezone for australia/yancowinna UTC offset of +10:30.
- CET - Timezone for CET UTC offset of +01:00.
- CST6CDT - Timezone for CST6CDT UTC offset of -06:00.
- Canada/Atlantic - Timezone for canada/atlantic UTC offset of -04:00.
- Canada/Central - Timezone for canada/central UTC offset of -06:00.
- Canada/East-Saskatchewan - Timezone for canada/east-saskatchewan UTC offset of -06:00.
- Canada/Eastern - Timezone for canada/eastern UTC offset of -05:00.
- Canada/Mountain - Timezone for canada/mountain UTC offset of -07:00.
- Canada/Newfoundland - Timezone for canada/newfoundland UTC offset of -03:30.
- Canada/Pacific - Timezone for canada/pacific UTC offset of -08:00.
- Canada/Saskatchewan - Timezone for canada/saskatchewan UTC offset of -06:00.
- Canada/Yukon - Timezone for canada/yukon UTC offset of -08:00.
- Chile/Continental - Timezone for chile/continental UTC offset of -03:00.
- Chile/EasterIsland - Timezone for chile/easterisland UTC offset of -05:00.
- Cuba - Timezone for cuba UTC offset of -05:00.
- Default - Timezone for system UTC (OS time zone)
- EET - Timezone for EET UTC offset of +02:00.
- EST - Timezone for EST UTC offset of -05:00.
- EST5EDT - Timezone for EST5EDT UTC offset of -05:00.
- Egypt - Timezone for egypt UTC offset of +02:00.
- Eire - Timezone for eire UTC offset of +00:00.

- Etc/Greenwich - Timezone for etc/greenwich UTC offset of +00:00.
- Etc/UCT - Timezone for etc/uct UTC offset of +00:00.
- Etc/UTC - Timezone for etc/utc UTC offset of +00:00.
- Etc/Universal - Timezone for etc/universal UTC offset of +00:00.
- Etc/Zulu - Timezone for etc/zulu UTC offset of +00:00.
- Europe/Amsterdam - Timezone for europe/amsterdam UTC offset of +01:00.
- Europe/Andorra - Timezone for europe/andorra UTC offset of +01:00.
- Europe/Astrakhan - Timezone for europe/astrakhan UTC offset of +04:00.
- Europe/Athens - Timezone for europe/athens UTC offset of +02:00.
- Europe/Belfast - Timezone for europe/belfast UTC offset of +00:00.
- Europe/Belgrade - Timezone for europe/belgrade UTC offset of +01:00.
- Europe/Berlin - Timezone for europe/berlin UTC offset of +01:00.
- Europe/Bratislava - Timezone for europe/bratislava UTC offset of +01:00.
- Europe/Brussels - Timezone for europe/brussels UTC offset of +01:00.
- Europe/Bucharest - Timezone for europe/bucharest UTC offset of +02:00.
- Europe/Budapest - Timezone for europe/budapest UTC offset of +01:00.
- Europe/Busingen - Timezone for europe/busingen UTC offset of +01:00.
- Europe/Chisinau - Timezone for europe/chisinau UTC offset of +02:00.
- Europe/Copenhagen - Timezone for europe/copenhagen UTC offset of +01:00.
- Europe/Dublin - Timezone for europe/dublin UTC offset of +00:00.
- Europe/Gibraltar - Timezone for europe/gibraltar UTC offset of +01:00.
- Europe/Guernsey - Timezone for europe/guernsey UTC offset of +00:00.
- Europe/Helsinki - Timezone for europe/helsinki UTC offset of +02:00.
- Europe/Isle_of_Man - Timezone for europe/isle_of_man UTC offset of +00:00.
- Europe/Istanbul - Timezone for europe/istanbul UTC offset of +03:00.
- Europe/Jersey - Timezone for europe/jersey UTC offset of +00:00.
- Europe/Kaliningrad - Timezone for europe/kaliningrad UTC offset of +02:00.
- Europe/Kiev - Timezone for europe/kiev UTC offset of +02:00.
- Europe/Kirov - Timezone for europe/kirov UTC offset of +03:00.
- Europe/Lisbon - Timezone for europe/lisbon UTC offset of +00:00.
- Europe/Ljubljana - Timezone for europe/ljubljana UTC offset of +01:00.

- Europe/London - Timezone for europe/london UTC offset of +00:00.
- Europe/Luxembourg - Timezone for europe/luxembourg UTC offset of +01:00.
- Europe/Madrid - Timezone for europe/madrid UTC offset of +01:00.
- Europe/Malta - Timezone for europe/malta UTC offset of +01:00.
- Europe/Mariehamn - Timezone for europe/mariehamn UTC offset of +02:00.
- Europe/Minsk - Timezone for europe/minsk UTC offset of +03:00.
- Europe/Monaco - Timezone for europe/monaco UTC offset of +01:00.
- Europe/Moscow - Timezone for europe/moscow UTC offset of +03:00.
- Europe/Nicosia - Timezone for europe/nicosia UTC offset of +02:00.
- Europe/Oslo - Timezone for europe/oslo UTC offset of +01:00.
- Europe/Paris - Timezone for europe/paris UTC offset of +01:00.
- Europe/Podgorica - Timezone for europe/podgorica UTC offset of +01:00.
- Europe/Prague - Timezone for europe/prague UTC offset of +01:00.
- Europe/Riga - Timezone for europe/riga UTC offset of +02:00.
- Europe/Rome - Timezone for europe/rome UTC offset of +01:00.
- Europe/Samara - Timezone for europe/samara UTC offset of +04:00.
- Europe/San_Marino - Timezone for europe/san_marino UTC offset of +01:00.
- Europe/Sarajevo - Timezone for europe/sarajevo UTC offset of +01:00.
- Europe/Simferopol - Timezone for europe/simferopol UTC offset of +03:00.
- Europe/Skopje - Timezone for europe/skopje UTC offset of +01:00.
- Europe/Sofia - Timezone for europe/sofia UTC offset of +02:00.
- Europe/Stockholm - Timezone for europe/stockholm UTC offset of +01:00.
- Europe/Tallinn - Timezone for europe/tallinn UTC offset of +02:00.
- Europe/Tirane - Timezone for europe/tirane UTC offset of +01:00.
- Europe/Tiraspol - Timezone for europe/tiraspol UTC offset of +02:00.
- Europe/Ulyanovsk - Timezone for europe/ulyanovsk UTC offset of +04:00.
- Europe/Uzhgorod - Timezone for europe/uzhgorod UTC offset of +02:00.
- Europe/Vaduz - Timezone for europe/vaduz UTC offset of +01:00.
- Europe/Vatican - Timezone for europe/vatican UTC offset of +01:00.
- Europe/Vienna - Timezone for europe/vienna UTC offset of +01:00.
- Europe/Vilnius - Timezone for europe/vilnius UTC offset of +02:00.

- Europe/Volgograd - Timezone for europe/volgograd UTC offset of +03:00.
- Europe/Warsaw - Timezone for europe/warsaw UTC offset of +01:00.
- Europe/Zagreb - Timezone for europe/zagreb UTC offset of +01:00.
- Europe/Zaporozhye - Timezone for europe/zaporozhye UTC offset of +02:00.
- Europe/Zurich - Timezone for europe/zurich UTC offset of +01:00.
- GB - Timezone for GB UTC offset of +00:00.
- GB-Eire - Timezone for GB-Eire UTC offset of +00:00.
- GMT - Timezone for GMT UTC offset of +00:00.
- GMT0 - Timezone for GMT0 UTC offset of +00:00.
- Greenwich - Timezone for greenwich UTC offset of +00:00.
- HST - Timezone for HST UTC offset of -10:00.
- Hongkong - Timezone for hongkong UTC offset of +08:00.
- Iceland - Timezone for iceland UTC offset of +00:00.
- Indian/Antananarivo - Timezone for indian/antananarivo UTC offset of +03:00.
- Indian/Chagos - Timezone for indian/chagos UTC offset of +06:00.
- Indian/Christmas - Timezone for indian/christmas UTC offset of +07:00.
- Indian/Cocos - Timezone for indian/cocos UTC offset of +06:30.
- Indian/Comoro - Timezone for indian/comoro UTC offset of +03:00.
- Indian/Kerguelen - Timezone for indian/kerguelen UTC offset of +05:00.
- Indian/Mahe - Timezone for indian/mahe UTC offset of +04:00.
- Indian/Maldives - Timezone for indian/maldives UTC offset of +05:00.
- Indian/Mauritius - Timezone for indian/mauritius UTC offset of +04:00.
- Indian/Mayotte - Timezone for indian/mayotte UTC offset of +03:00.
- Indian/Reunion - Timezone for indian/reunion UTC offset of +04:00.
- Iran - Timezone for iran UTC offset of +03:30.
- Israel - Timezone for israel UTC offset of +02:00.
- Jamaica - Timezone for jamaica UTC offset of -05:00.
- Japan - Timezone for japan UTC offset of +09:00.
- Kwajalein - Timezone for kwajalein UTC offset of +12:00.
- Libya - Timezone for libya UTC offset of +02:00.
- MET - Timezone for MET UTC offset of +01:00.

- MST - Timezone for MST UTC offset of -07:00.
- MST7MDT - Timezone for MST7MDT UTC offset of -07:00.
- Mexico/BajaNorte - Timezone for mexico/bajanorte UTC offset of -08:00.
- Mexico/BajaSur - Timezone for mexico/bajasur UTC offset of -07:00.
- Mexico/General - Timezone for mexico/general UTC offset of -06:00.
- NZ - Timezone for NZ UTC offset of +13:00.
- NZ-CHAT - Timezone for NZ-CHAT UTC offset of +13:45.
- Navajo - Timezone for navajo UTC offset of -07:00.
- PRC - Timezone for PRC UTC offset of +08:00.
- PST8PDT - Timezone for PST8PDT UTC offset of -08:00.
- Pacific/Apia - Timezone for pacific/apia UTC offset of +14:00.
- Pacific/Auckland - Timezone for pacific/auckland UTC offset of +13:00.
- Pacific/Bougainville - Timezone for pacific/bougainville UTC offset of +11:00.
- Pacific/Chatham - Timezone for pacific/chatham UTC offset of +13:45.
- Pacific/Chuuk - Timezone for pacific/chuuk UTC offset of +10:00.
- Pacific/Easter - Timezone for pacific/easter UTC offset of -05:00.
- Pacific/Efate - Timezone for pacific/efate UTC offset of +11:00.
- Pacific/Enderbury - Timezone for pacific/enderbury UTC offset of +13:00.
- Pacific/Fakaofo - Timezone for pacific/fakaofo UTC offset of +13:00.
- Pacific/Fiji - Timezone for pacific/fiji UTC offset of +13:00.
- Pacific/Funafuti - Timezone for pacific/funafuti UTC offset of +12:00.
- Pacific/Galapagos - Timezone for pacific/galapagos UTC offset of -06:00.
- Pacific/Gambier - Timezone for pacific/gambier UTC offset of -09:00.
- Pacific/Guadalcanal - Timezone for pacific/guadalcanal UTC offset of +11:00.
- Pacific/Guam - Timezone for pacific/guam UTC offset of +10:00.
- Pacific/Honolulu - Timezone for pacific/honolulu UTC offset of -10:00.
- Pacific/Johnston - Timezone for pacific/johnston UTC offset of -10:00.
- Pacific/Kiritimati - Timezone for pacific/kiritimati UTC offset of +14:00.
- Pacific/Kosrae - Timezone for pacific/kosrae UTC offset of +11:00.
- Pacific/Kwajalein - Timezone for pacific/kwajalein UTC offset of +12:00.
- Pacific/Majuro - Timezone for pacific/majuro UTC offset of +12:00.

- Pacific/Marquesas - Timezone for pacific/marquesas UTC offset of -09:30.
- Pacific/Midway - Timezone for pacific/midway UTC offset of -11:00.
- Pacific/Nauru - Timezone for pacific/nauru UTC offset of +12:00.
- Pacific/Niue - Timezone for pacific/niue UTC offset of -11:00.
- Pacific/Norfolk - Timezone for pacific/norfolk UTC offset of +11:00.
- Pacific/Noumea - Timezone for pacific/noumea UTC offset of +11:00.
- Pacific/Pago_Pago - Timezone for pacific/pago_pago UTC offset of -11:00.
- Pacific/Palau - Timezone for pacific/palau UTC offset of +09:00.
- Pacific/Pitcairn - Timezone for pacific/pitcairn UTC offset of -08:00.
- Pacific/Pohnpei - Timezone for pacific/pohnpei UTC offset of +11:00.
- Pacific/Ponape - Timezone for pacific/ponape UTC offset of +11:00.
- Pacific/Port_Moresby - Timezone for pacific/port_moresby UTC offset of +10:00.
- Pacific/Rarotonga - Timezone for pacific/rarotonga UTC offset of -10:00.
- Pacific/Saipan - Timezone for pacific/saipan UTC offset of +10:00.
- Pacific/Samoa - Timezone for pacific/samoa UTC offset of -11:00.
- Pacific/Tahiti - Timezone for pacific/tahiti UTC offset of -10:00.
- Pacific/Tarawa - Timezone for pacific/tarawa UTC offset of +12:00.
- Pacific/Tongatapu - Timezone for pacific/tongatapu UTC offset of +14:00.
- Pacific/Truk - Timezone for pacific/truk UTC offset of +10:00.
- Pacific/Wake - Timezone for pacific/wake UTC offset of +12:00.
- Pacific/Wallis - Timezone for pacific/wallis UTC offset of +12:00.
- Pacific/Yap - Timezone for pacific/yap UTC offset of +10:00.
- Poland - Timezone for poland UTC offset of +01:00.
- Portugal - Timezone for portugal UTC offset of +00:00.
- ROC - Timezone for ROC UTC offset of +08:00.
- ROK - Timezone for ROK UTC offset of +09:00.
- Singapore - Timezone for singapore UTC offset of +08:00.
- Turkey - Timezone for turkey UTC offset of +03:00.
- UCT - Timezone for UCT UTC offset of +00:00.
- US/Pacific-New - Timezone for US/Pacific-New UTC offset of -08:00.
- Universal - Timezone for universal UTC offset of +00:00.

- W-SU - Timezone for W-SU UTC offset of +03:00.
- WET - Timezone for WET UTC offset of +00:00.
- Zulu - Timezone for zulu UTC offset of +00:00.
- jdj_tz - Timezone for jdj_tz UTC offset of +00:00.

Default

Default

Name

ueRadioCapabilityMatch - UE radio capability match

Syntax

--ueRadioCapabilityMatch <list-item>

Description

When set to true enables UE radio capability match procedure context feature and also to provision whether to use SRVCC capability in determining the support of IMS voice.

Restrictions

This element is not applicable when plmnType is of type Other Network.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ueSrvccCapability - UE SRVCC capability

Syntax

--ueSrvccCapability <list-item>

Description

When set to true enables the use of UE SRVCC capability and also local SRVCC enable/disable provisioning (Refer to service profile for the SRVCC command) in determining IMS voice over PS indication to a UE. in some cases, operators may like to force a UE to perform CSFB rather than using IMS voice over PS if the UE is not capable of SRVCC to prevent call drops when UE moves from an area where IMS voice is supported to an area IMS voice is not supported. the no form of the command disables the capability. this provisioning is only applicable if the UE radio capability match feature is enabled.

Restrictions

This element is not applicable when plmnType is of type Other Network.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

validateImeiEir - Validate IMEI with EIR

Syntax

```
--validateImeiEir <list-item>
```

Description

Indicates whether the MME performs a validation of the IMEI in the EIR. This field can only be set to "checked" if field Obtain IMEISV is different than "None" when plmnType is either Home Network or Shared Network

Restrictions

This element is not applicable when plmnType is Other Network.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.57 cmm plmnParam

plmnParam. Syntax, description, parameters, and roles for **cmm plmnParam** command.

Syntax**Description**

Configure, list and show PLMN parameter sets.

Parameters**Name**

plmnParamId - PLMN parameter set name

Syntax

--plmnParamId <string>

Description

PLMN parameter set name. Note, authentication repetition parameters are taken into use when authenticationMode (cmm securityParam) is set equal to ACT.

Class

Required key.

Type / Range

Digit string of length 1 to 15 with the allowed pattern of [0-9a-zA-Z\._ :]*

Name

homePlmn - Home plmn

Syntax

--homePlmn <list-item>

Description

Home PLMN or not.

Class

Required on create.

List

- false - Boolean value of false.
- true - Boolean value of true.

Name

accRestrCauCnt - Reject cause code for subscriber count

Syntax

--accRestrCauCnt <list-item>

Description

Reject cause code for subscriber count la - location area not allowed plmn - PLMN not allowed roamLa - roaming not allowed in this LA gprsPlmn - GPRS services not allowed in this PLMN cellLa - no suitable cells in LA notUsed - not set.

Class

Optional.

List

- cellLa - No suitable cells in LA.
- gprsPlmn - GPRS services not allowed in this PLMN.
- la - Location area not allowed.
- notUsed - Not set.
- plmn - PLMN not allowed.
- roamLa - Roaming not allowed in this LA.

Default

notUsed

Name

accRestrCauGtoG - Reject cause code for a GSM subscriber to the GSM network

Syntax

--accRestrCauGtoG <list-item>

Description

Reject cause code for a GSM subscriber to the GSM network la - location area not allowed plmn - PLMN not allowed roamLa - roaming not allowed in this LA gprsPlmn - GPRS services not allowed in this PLMN cellLa - no suitable cells in LA notUsed - not set.

Class

Optional.

List

- cellLa - No suitable cells in LA.
- gprsPlmn - GPRS services not allowed in this PLMN.
- la - Location area not allowed.
- notUsed - Not set.
- plmn - PLMN not allowed.

- roamLa - Roaming not allowed in this LA.

Default

notUsed

Name

accRestrCauGtoU - Reject cause code for a GSM subscriber to the UMTS network

Syntax

--accRestrCauGtoU <list-item>

Description

Reject cause code for a GSM subscriber to the UMTS network la - location area not allowed
plmn - PLMN not allowed roamLa - roaming not allowed in this LA gprsPlmn - GPRS services
not allowed in this PLMN cellLa - no suitable cells in LA notUsed - not set.

Class

Optional.

List

- cellLa - No suitable cells in LA.
- gprsPlmn - GPRS services not allowed in this PLMN.
- la - Location area not allowed.
- notUsed - Not set.
- plmn - PLMN not allowed.
- roamLa - Roaming not allowed in this LA.

Default

notUsed

Name

accRestrCauR97 - Reject cause code for a R97/98 terminal to the GSM network

Syntax

--accRestrCauR97 <list-item>

Description

Reject cause code for a R97/98 terminal to the GSM network la - location area not allowed
plmn - PLMN not allowed roamLa - roaming not allowed in this LA gprsPlmn - GPRS services
not allowed in this PLMN cellLa - no suitable cells in LA notUsed - not set.

Class

Optional.

List

- cellLa - No suitable cells in LA.
- gprsPlmn - GPRS services not allowed in this PLMN.

- la - Location area not allowed.
- notUsed - Not set.
- plmn - PLMN not allowed.
- roamLa - Roaming not allowed in this LA.

Default

notUsed

Name

accRestrCauUtoG - Reject cause code for a UMTS subscriber to the GSM network

Syntax

--accRestrCauUtoG <list-item>

Description

Reject cause code for a UMTS subscriber to the GSM network
la - location/ area not allowed
plmn - PLMN not allowed
roamLa - roaming not allowed in this LA
gprsPlmn - GPRS services not allowed in this PLMN
cellLa - no suitable cells in LA
notUsed - not set.

Class

Optional.

List

- cellLa - No suitable cells in LA.
- gprsPlmn - GPRS services not allowed in this PLMN.
- la - Location area not allowed.
- notUsed - Not set.
- plmn - PLMN not allowed.
- roamLa - Roaming not allowed in this LA.

Default

notUsed

Name

accRestrCauUtoU - Reject cause code for a UMTS subscriber to the UMTS network

Syntax

--accRestrCauUtoU <list-item>

Description

Reject cause code for a UMTS subscriber to the UMTS network
la - location area not allowed
plmn - PLMN not allowed
roamLa - roaming not allowed in this LA
gprsPlmn - GPRS services not allowed in this PLMN
cellLa - no suitable cells in LA
notUsed - not set.

Class

Optional.

List

- cellLa - No suitable cells in LA.
- gprsPlmn - GPRS services not allowed in this PLMN.
- la - Location area not allowed.
- notUsed - Not set.
- plmn - PLMN not allowed.
- roamLa - Roaming not allowed in this LA.

Default

notUsed

Name

accRestrGtoG - Access right for a GSM subscriber to the GSM network

Syntax

--accRestrGtoG <list-item>

Description

Access right for a GSM subscriber to the GSM network notused - the access right parameters are ignored allowed - the access right is granted notallowed - the access right is denied.

Class

Optional.

List

- allowed - Allowed.
- notAllowed - Not allowed - Not allowed.
- notUsed - Not used - Not used.

Default

notUsed

Name

accRestrGtoU - Access right for a GSM subscriber to the UMTS network

Syntax

--accRestrGtoU <list-item>

Description

Access right for a GSM subscriber to the UMTS network notused - the access right parameters are ignored allowed - the access right is granted notallowed - the access right is denied.

Class

Optional.

List

- allowed - Allowed.
- notAllowed - Not allowed - Not allowed.
- notUsed - Not used - Not used.

Default

notUsed

Name

accRestrUtoG - Access right for a UMTS subscriber to the GSM network

Syntax

--accRestrUtoG <list-item>

Description

Access right for a UMTS subscriber to the GSM network notused - the access right parameters are ignored allowed - the access right is granted notallowed - the access right is denied.

Class

Optional.

List

- allowed - Allowed.
- notAllowed - Not allowed - Not allowed.
- notUsed - Not used - Not used.

Default

notUsed

Name

accRestrUtoU - Access right for a UMTS subscriber to the UMTS network

Syntax

--accRestrUtoU <list-item>

Description

Access right for a UMTS subscriber to the UMTS network notused - the access right parameters are ignored allowed - the access right is granted notallowed - the access right is denied.

Class

Optional.

List

- allowed - Allowed.
- notAllowed - Not allowed - Not allowed.

- notUsed - Not used - Not used.

Default

notUsed

Name

allocRetention - Allocation/retention priority for roaming subscribers

Syntax

--allocRetention <integer>

Description

Allocation/retention priority for roaming subscribers. this parameter defines the allocation/retention priority which can be used by roaming subscriber at most.

Class

Optional.

Type / Range

Integer in the range of 1 to 3

Default

3

Name

blackListEffect - Black list effect

Syntax

--blackListEffect <list-item>

Description

The black list effect. With the black list effect you can define how to handle mobile stations on black list allow - the action is allowed block - the action is blocked note, when feature SG02220 S13 IMEI checking is activated BLE is not visible.

Class

Optional.

List

- allow - Allow.
- block - Block.

Default

block

Name

blacklistImeiEffect - Allow blacklisted equipment

Syntax

--blacklistImeiEffect <list-item>

Description

Allow blacklisted equipment allow - is used for allowing blacklisted IMEI block - is used for not allowing blacklisted IMEI.

Class

Optional.

List

- allow - Allow.
- block - Block.

Default

block

Name

cipherMode - Ciphering mode

Syntax

--cipherMode <list-item>

Description

Ciphering mode. with the ciphering mode you can define the usage of ciphering true - the ciphering is in use false - the ciphering is not in use.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

combAttIimeiChk - IMEI status check from EIR for combined IMSI and GPRS attach

Syntax

--combAttIimeiChk <list-item>

Description

IMEI status check from EIR for combined IMSI and GPRS attach.

Class

Optional.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Name

combRaulmeiChk - IMEI status check from EIR for combined RA and LA update

Syntax

--combRauIimeiChk <list-item>

Description

IMEI status check from EIR for combined RA and LA update.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

csPsCoord - CS/PS coordination

Syntax

--csPsCoord <list-item>

Description

CS/PS coordination.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ctrlR7R8DnsQuery - control R8-R7 DNS query/PLMN

Syntax

--ctrlR7R8DnsQuery <list-item>

Description

With this parameter, the operator can control on PLMN level whether Rel-8 DNS query for S/P-GW selection or Rel-7 query DNS for GGSN selection will be preferred during PDP context creation. The provided options are: - Default: based on existing selection criteria such as EPC capability, EPS subscription and APN. - S4 preferred: First attempt is with R8 query for S/P-GW selection even if UE is non-EPC capable nor has EPS subscription. If no valid entry is found fallback to R7 for GGSN selection is triggered. - Gn only : In this case only Rel7 DNS query to GGSN is made

Class

Optional.

List

- ExistFunc - Existing R8 GW selection - operator control on PLMN level whether Rel-8 APN DNS query for S/P-GW selection or Rel-7 query DNS for GGSN/GW selection will be preferred during 1st PDP context creation. This item is to select based on existing selection criteria such as EPC, EPS subscription and APN.
- Gn - Gn is preferred - Gn only : In this case only Rel7 DNS query to GGSN is made
- S4 - s4 is preferred - S4 preferred: First attempt is with R8 query for S/P-GW selection even if UE is non-EPC capable nor has EPS subscription. If no valid entry is found fallback to R7 for GGSN selection is triggered.

Default

ExistFunc

Name

defaultApnOi - Define APN operator identifier

Syntax

--defaultApnOi <string>

Description

Define APN operator identifier: maximum of 10 unique default APN operator identifier allowed in database. contains 6...37 characters ('A'...'Z', 'a'...'z', '0'...'9', '.') last part of defapn has to be '.GPRS' remove default APN operator identifier using value 0.

Class

Optional.

Type / Range

Digit string of length 1 to 37 with the allowed pattern of [0-9a-zA-Z\._:-]*

Default

0

Name

defaultApnOi2 - Secondary default APN operator identifier

Syntax

--defaultApnOi2 <string>

Description

Secondary default APN operator identifier.

Class

Optional.

Type / Range

Digit string of length 1 to 37 with the allowed pattern of [0-9a-zA-Z\._ :]*

Default

0

Name

detachImeiChk - IMEI status check from EIR for MO GPRS detach

Syntax

--detachImeiChk <list-item>

Description

IMEI status check from EIR for MO GPRS detach.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

dnsMcc - Define LTE inbound roaming subscribers default DNS domain mobile country code

Syntax

--dnsMcc <string>

Description

Define LTE inbound roaming subscribers default DNS domain mobile country code.

Class

Optional.

Type / Range

Digit string of length 3 to 3 with the allowed pattern of [0-9a-zA-Z\._ :]*

Name

dnsMnc - Define LTE inbound roaming subscribers default DNS domain mobile network code

Syntax

--dnsMnc <string>

Description

Define LTE inbound roaming subscribers default DNS domain mobile network code.

Class

Optional.

Type / Range

Digit string of length 2 to 3 with the allowed pattern of [0-9a-zA-Z\.\-_:]*

Name

earp - Evolved allocation/retention priority

Syntax

--earp <integer>

Description

Evolved allocation/retention priority.

Class

Optional.

Type / Range

Integer in the range of 1 to 15

Default

14

Name

fallbackGr - fall back to Gr interface

Syntax

--fallbackGr <list-item>

Description

This parameter is needed also/PLMN(IMSI NS). For most PLMN (IMSI NS), if S6d is set, then by default, fallback to Gr is also required unless this parameter is set to false->no need to support fall back. fallbackGr=true, fallback to Gr will be supported fallbackGr=false, fallback to Gr will be not supported default is true

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

fullNetwName - Full network name

Syntax

--fullNetwName <string>

Description

This parameter defines the full network name of the network operator (PLMN level). This is for the case MVNO used, different PLMN number is used for different operators.

Class

Optional.

Type / Range

Digit string of length 0 to 64

Default

""

Name

gbCauseCodeRec - Cause code set for gb interface

Syntax

--gbCauseCodeRec <string>

Description

Cause code set for gb interface. with this parameter you can define the name for cause code set, which will be used for actual PLMN on the gb interface. default values can be used if 0 is given as the cause code set name.

Class

Optional.

Type / Range

Digit string of length 1 to 8 with the allowed pattern of [0-9a-zA-Z\._:-]*

Default

0

Name

gbCombAttAuth - Authentication repetition rate for combined IMSI and GPRS attach

Syntax

--gbCombAttAuth <integer>

Description

Authentication repetition rate for combined IMSI and GPRS attach.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

1

Name

gbCombAttImei - IMEI check repetition rate for combined gb IMSI and GPRS attach

Syntax`--gbCombAttImei <integer>`**Description**

IMEI check repetition rate for combined gb IMSI and GPRS attach.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

gbComRaLaAuth - Authentication repetition rate for combined gb RA and LA update

Syntax`--gbComRaLaAuth <integer>`**Description**

Authentication repetition rate for combined gb RA and LA update.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

gbComRaLaImei - IMEI check repetition rate for combined gb RA and LA update

Syntax`--gbComRaLaImei <integer>`**Description**

IMEI check repetition rate for combined gb RA and LA update.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

gbGprsAttAuth - Authentication repetition rate for gb GPRS attach

Syntax

--gbGprsAttAuth <integer>

Description

Authentication repetition rate for gb GPRS attach.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

1

Name

gbGprsAttImei - IMEI check repetition rate for gb GPRS attach

Syntax

--gbGprsAttImei <integer>

Description

IMEI check repetition rate for gb GPRS attach.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

gbGprsAttPtmsiSign - PTMSI signature repetition rate for gb GPRS attach

Syntax

--gbGprsAttPtmsiSign <integer>

Description

PTMSI signature repetition rate for gb GPRS attach.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

gbImsiAttAuth - Authentication repetition rate for gb IMSI attach

Syntax

--gbImsiAttAuth <integer>

Description

Authentication repetition rate for gb IMSI attach.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

1

Name

gbImsiAttImei - IMEI check repetition rate for gb IMSI attach

Syntax

--gbImsiAttImei <integer>

Description

IMEI check repetition rate for gb IMSI attach.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

gbMncl - Gb MNC length

Syntax

--gbMncl <integer>

Description

Gb MNC length.

Class

Optional.

Type / Range

Integer in the range of 2 to 3

Default

2

Name

gbMoGprsDeaAuth - Authentication repetition rate for gb MO GPRS detach

Syntax

--gbMoGprsDeaAuth <integer>

Description

Authentication repetition rate for gb MO GPRS detach.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

gbMoGprsDeaImei - IMEI check repetition rate for gb MO GPRS detach

Syntax

--gbMoGprsDeaImei <integer>

Description

IMEI check repetition rate for gb MO GPRS detach.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

gbMoPdpActAuth - Authentication repetition rate for gb MO PDP context activation

Syntax

--gbMoPdpActAuth <integer>

Description

Authentication repetition rate for gb MO PDP context activation.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

gbMoPdpActImei - IMEI check repetition rate for gb MO PDP context activation

Syntax

--gbMoPdpActImei <integer>

Description

IMEI check repetition rate for gb MO PDP context activation.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

gbMoPdpDeaAuth - Authentication repetition rate for MO PDP context deactivation

Syntax

--gbMoPdpDeaAuth <integer>

Description

Authentication repetition rate for MO PDP context deactivation.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

gbMoPdpDeaImei - IMEI check repetition rate for MO PDP context deactivation

Syntax

--gbMoPdpDeaImei <integer>

Description

IMEI check repetition rate for MO PDP context deactivation.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

gbNormalRaPtmsiSign - PTMSI signature repetition rate for normal gb RA update

Syntax

--gbNormalRaPtmsiSign <integer>

Description

PTMSI signature repetition rate for normal gb RA update.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

gbNormalRaPtmsiSignVisitor - PTMSI signature repetition rate for normal gb RA update, new visitor

Syntax

--gbNormalRaPtmsiSignVisitor <integer>

Description

PTMSI signature repetition rate for normal gb RA update, new visitor.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

gbNormRaAuth - Authentication repetition rate for normal gb RA update

Syntax

--gbNormRaAuth <integer>

Description

Authentication repetition rate for normal gb RA update.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

gbNormRaAuthVisitor - Authentication repetition rate for normal gb RA update, new visitor

Syntax

--gbNormRaAuthVisitor <integer>

Description

Authentication repetition rate for normal gb RA update, new visitor.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

gbNormRaImei - IMEI check repetition rate for normal gb RA update

Syntax

--gbNormRaImei <integer>

Description

IMEI check repetition rate for normal gb RA update.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

gbNormRaImeiVisitor - IMEI check repetition rate for normal gb RA update, new visitor

Syntax

```
--gbNormRaImeiVisitor <integer>
```

Description

IMEI check repetition rate for normal gb RA update, new visitor.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

gbPeriodicRaAuth - Authentication repetition rate for periodic gb RA update

Syntax

```
--gbPeriodicRaAuth <integer>
```

Description

Authentication repetition rate for periodic gb RA update.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

gbPeriodicRaImei - IMEI check repetition rate for periodic RA update

Syntax

```
--gbPeriodicRaImei <integer>
```

Description

IMEI check repetition rate for periodic RA update.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

gbPeriodicRaPtmsiAlloc - PTMSI allocation repetition rate for periodic gb RA update

Syntax

--gbPeriodicRaPtmsiAlloc <integer>

Description

PTMSI allocation repetition rate for periodic gb RA update.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

1

Name

gbPeriodicRaPtmsiSign - PTMSI signature repetition rate for periodic gb RA update

Syntax

--gbPeriodicRaPtmsiSign <integer>

Description

PTMSI signature repetition rate for periodic gb RA update.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

genericTreatmentForUnsubscribedApn - Generic Treatment For Unsubscribed APN

Syntax

--genericTreatmentForUnsubscribedApn <list-item>

Description

Generic Treatment For Unsubscribed APN

Class

Optional.

List

- existSystemBehavior - Existing System Behavior - existing system behavior as a default action
- convertIfLocalSubscribedElseLowestContextId - Convert If Local Subscribed Else Lowest Context Id - convert If Local and Subscribed Else uses Lowest Context Id

- convertIfLocalSubscribedElseReject - Convert If Local Subscribed Else Reject - Convert If it matches Local and Subscribed Else Reject
- noConvertAndReject - No Convert And Reject - No convert and reject PDP

Default

existSystemBehavior

Name

gprsAttImeiChk - IMEI status check from EIR for GPRS attach

Syntax

--gprsAttImeiChk <list-item>

Description

IMEI status check from EIR for GPRS attach.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

gtpTmrprId - GTP timer profile id

Syntax

--gtpTmrprId <integer>

Description

GTP timer profile id.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

255

Name

guaranBitRateDl - Guaranteed bit rate for downlink for roaming subscribers

Syntax

--guaranBitRateDl <integer>

Description

This parameter defines the guaranteed bit rate for downlink for roaming subscribers.

Class

Optional.

Type / Range

Integer in the range of 0 to 28,000

Default

0

Name

guaranBitRateUl - Guaranteed bit rate for uplink for roaming subscribers

Syntax

--guaranBitRateUl <integer>

Description

This parameter defines the guaranteed bit rate for uplink for roaming subscribers.

Class

Optional.

Type / Range

Integer in the range of 0 to 11,500

Default

0

Name

imeiValidityTimer - IMEI check validity timer in hours

Syntax

--imeiValidityTimer <integer>

Description

IMEI check validity timer in hours.

Class

Optional.

Type / Range

Integer in the range of 0 to 743

Default

0

Name

imsApn - IMS APN allowed

Syntax

--imsApn <list-item>

Description

Specifies whether IMS APN is allowed in SGSN or not.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

imsiAttIimeiChk - IMEI status check from EIR for IMSI attach

Syntax

```
--imsiAttIimeiChk <list-item>
```

Description

IMEI status check from EIR for IMSI attach.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

iuCauseCodeRec - Cause code set for iu interface

Syntax

```
--iuCauseCodeRec <string>
```

Description

Cause code set for iu interface. with this parameter you can define the name for cause code set, which will be used for actual PLMN on the iu interface. default values can be used if 0 is given as the cause code set name.

Class

Optional.

Type / Range

Digit string of length 1 to 8 with the allowed pattern of [0-9a-zA-Z\._:-]*

Default

0

Name

iuCombAttImei - IMEI check repetition rate for combined iu IMSI and GPRS attach

Syntax

--iuCombAttImei <integer>

Description

IMEI check repetition rate for combined iu IMSI and GPRS attach.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

iuComRaLaAuth - Authentication repetition rate for combined iu RA and LA update

Syntax

--iuComRaLaAuth <integer>

Description

Authentication repetition rate for combined iu RA and LA update.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

iuComRaLaImei - IMEI check repetition rate for combined iu RA and LA update

Syntax

--iuComRaLaImei <integer>

Description

IMEI check repetition rate for combined iu RA and LA update.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

iuGprsAttImei - IMEI check repetition rate for iu GPRS attach

Syntax`--iuGprsAttImei <integer>`**Description**

IMEI check repetition rate for iu GPRS attach.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

iuGprsAttPtmsiSign - PTMSI signature repetition rate for iu GPRS attach

Syntax`--iuGprsAttPtmsiSign <integer>`**Description**

PTMSI signature repetition rate for iu GPRS attach.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

iuImsiAttImei - IMEI check repetition rate for iu IMSI attach

Syntax`--iuImsiAttImei <integer>`**Description**

IMEI check repetition rate for iu IMSI attach.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

iuMncl - Iu MNC length

Syntax

--iuMncl <integer>

Description

Iu MNC length.

Class

Optional.

Type / Range

Integer in the range of 2 to 3

Default

2

Name

iuMoGprsDeaAuth - Authentication repetition rate for Iu MO GPRS detach

Syntax

--iuMoGprsDeaAuth <integer>

Description

Authentication repetition rate for Iu MO GPRS detach.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

iuMoGprsDeaImei - IMEI check repetition rate for Iu MO GPRS detach

Syntax

--iuMoGprsDeaImei <integer>

Description

IMEI check repetition rate for Iu MO GPRS detach.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

iuMoPdpActAuth - Authentication repetition rate for iu MO PDP context activation

Syntax

--iuMoPdpActAuth <integer>

Description

Authentication repetition rate for iu MO PDP context activation.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

iuMoPdpActImei - IMEI check repetition rate for iu MO PDP context activation

Syntax

--iuMoPdpActImei <integer>

Description

IMEI check repetition rate for iu MO PDP context activation.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

iuMoPdpDeaAuth - Authentication repetition rate for iu MO PDP context deactivation

Syntax

--iuMoPdpDeaAuth <integer>

Description

Authentication repetition rate for iu MO PDP context deactivation.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

iuMoPdpDeaImei - IMEI check repetition rate for iu MO PDP context deactivation

Syntax

--iuMoPdpDeaImei <integer>

Description

IMEI check repetition rate for iu MO PDP context deactivation.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

iuNormalRaPtmsiSign - PTMSI signature repetition rate for normal iu RA update

Syntax

--iuNormalRaPtmsiSign <integer>

Description

PTMSI signature repetition rate for normal iu RA update.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

iuNormalRaPtmsiSignVisitor - PTMSI signature repetition rate for normal iu RA update, new visitor

Syntax

--iuNormalRaPtmsiSignVisitor <integer>

Description

PTMSI signature repetition rate for normal iu RA update, new visitor.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

iuNormRaAuth - Authentication repetition rate for normal iu RA update

Syntax

--iuNormRaAuth <integer>

Description

Authentication repetition rate for normal iu RA update.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

iuNormRaAuthVisitor - Authentication repetition rate for normal iu RA update, new visitor

Syntax

--iuNormRaAuthVisitor <integer>

Description

Authentication repetition rate for normal iu RA update, new visitor.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

iuNormRaImei - IMEI check repetition rate for normal iu RA update

Syntax

--iuNormRaImei <integer>

Description

IMEI check repetition rate for normal iu RA update.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

iuNormRalmeiVisitor - IMEI check repetition rate for normal iu RA update, new visitor

Syntax

```
--iuNormRaImeiVisitor <integer>
```

Description

IMEI check repetition rate for normal iu RA update, new visitor.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

iuPeriodicRaAuth - Authentication repetition rate for periodic iu RA update

Syntax

```
--iuPeriodicRaAuth <integer>
```

Description

Authentication repetition rate for periodic iu RA update.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

iuPeriodicRalmei - IMEI check repetition rate for periodic iu RA update

Syntax


```
--iuPeriodicRaImei <integer>
```

Description

IMEI check repetition rate for periodic iu RA update.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

iuPeriodicRaPtmsiAlloc - PTMSI allocation repetition rate for periodic iu RA update

Syntax

```
--iuPeriodicRaPtmsiAlloc <integer>
```

Description

PTMSI allocation repetition rate for periodic iu RA update.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

1

Name

iuPeriodicRaPtmsiSign - PTMSI signature repetition rate for periodic iu RA update

Syntax

```
--iuPeriodicRaPtmsiSign <integer>
```

Description

PTMSI signature repetition rate for periodic iu RA update.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

iuServReqAuth - Authentication repetition rate for iu service request

Syntax

--iuServReqAuth <integer>

Description

Authentication repetition rate for iu service request.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

iuServReqImei - IMEI check repetition rate for iu service request

Syntax

--iuServReqImei <integer>

Description

IMEI check repetition rate for iu service request.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

iuServReqPtmsiAlloc - PTMSI allocation repetition rate for iu service request

Syntax

--iuServReqPtmsiAlloc <integer>

Description

PTMSI allocation repetition rate for iu service request.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

lb - Enable or disable the local breakout functionality

Syntax

--lb <list-item>

Description

Enable or disable the local breakout functionality.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

lbo - Specifies local break out feature status

Syntax

--lbo <list-item>

Description

Specifies local break out feature status.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

localApnPlmn - Local APN PLMN

Syntax

--localApnPlmn <string>

Description

Local configured APN/PLMN in SGSN for plmn level override function (v4)

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 62

Default

""

Name

localApnPlmnV4V6 - Local APN PLMN for IPv4v6

Syntax

--localApnPlmnV4V6 <string>

Description

Local configured APN/PLMN in SGSN for IPv4v6 for plmn level override function

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 62

Default

""

Name

localApnPlmnV6 - Local APN PLMN for IPv6

Syntax

--localApnPlmnV6 <string>

Description

Local configured APN/PLMN in SGSN for IPv6 type for plmn level override function

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 62

Default

""

Name

locationBasedGwForRoamer - Location Based Gateway Selection For Roamer

Syntax

--locationBasedGwForRoamer <list-item>

Description

With this parameter set to true, SGSN will allow the roamers under this PLMN (national roamer) to do DNS query for gateway selection with current roamer's location info, thus to select closest gateway

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

maxBitRateForDL - Maximum bit rate for downlink for roaming subscribers

Syntax

--maxBitRateForDL <integer>

Description

This parameter defines the maximum bit rate for downlink for roaming subscribers.

Class

Optional.

Type / Range

Integer in the range of 0 to 28,000

Default

50

Name

maxBitRateForUL - Maximum bit rate for uplink for roaming subscribers

Syntax

--maxBitRateForUL <integer>

Description

This parameter defines the maximum bit rate for uplink for roaming subscribers.

Class

Optional.

Type / Range

Integer in the range of 0 to 11,500

Default

50

Name

normalRaulmeiChk - IMEI status check from EIR for normal RA update

Syntax

--normalRauImeiChk <list-item>

Description

IMEI status check from EIR for normal RA update.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

normalRaulmeiChkVisitor - IMEI status check from EIR for normal RA update, new visitor

Syntax

```
--normalRauImeiChkVisitor <list-item>
```

Description

IMEI status check from EIR for normal RA update, new visitor.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

overrideApn - Override APN Allowed

Syntax

```
--overrideApn <list-item>
```

Description

This PLMN parameter will be used to control whether Override APN is allowed for this PLMN. If this parameter sets to true, then override APN feature will apply for the roamer from the PLMN. Otherwise, override APN feature will not work/allow for roamers. By default, this parameter is set to false. This parameter applies to roamers only, for home PLMNs, no impacts. This parameter can be used only when sgsnGParam overridePlmn set to True

Class

Optional.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Name

overrideRfspIndex - Override RFSP Index

Syntax

--overrideRfspIndex <integer>

Description

When sgsnGParam rfspIndexSelectionOption is enabled and rfspIndexSelectionOption equals overrideRfspIndex then the SGSN sends locally provisioned RFSP value, as defined in current field.

Class

Optional.

Type / Range

Integer in the range of 0 to 256

Default

0

Name

pdpActImeiChk - IMEI status check from EIR for MO PDP context activation

Syntax

--pdpActImeiChk <list-item>

Description

IMEI status check from EIR for MO PDP context activation.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

pdpDeactImeiChk - IMEI status check from EIR for MO PDP context deactivation

Syntax

--pdpDeactImeiChk <list-item>

Description

IMEI status check from EIR for MO PDP context deactivation.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

periodRaulmeiChk - IMEI status check from EIR for periodic RA update

Syntax

--periodRauImeiChk <list-item>

Description

IMEI status check from EIR for periodic RA update.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

plmnEARpHPriority - ARP priority value for high priority

Syntax

--plmnEARpHPriority <integer>

Description

ARP priority value for high priority. plmnearphpriority should be less than plmnearpmpriority.

Class

Optional.

Type / Range

Integer in the range of 1 to 13

Default

5

Name

plmnEARpMPriority - ARP priority value for medium priority

Syntax

--plmnEARpMPriority <integer>

Description

ARP priority value for medium priority.plmnearpmpriority should be less than plmnearphpriority.

Class

Optional.

Type / Range

Integer in the range of 2 to 14

Default

10

Name

plmnEARpPCap - Pre-emption capability

Syntax

--plmnEARpPCap <list-item>

Description

Pre-emption capability true - enable false - disable.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

plmnEARpPVul - Pre-emption vulnerability

Syntax

--plmnEARpPVul <list-item>

Description

Pre-emption vulnerability true - enable false - disable.

Class

Optional.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Name

plmnSpecQosRParInUse - PLMN specific QoS roaming parameters in use

Syntax

--plmnSpecQosRParInUse <list-item>

Description

PLMN specific QoS roaming parameters in use true - PLMN specific QoS roaming parameters are in use. Roaming check is performed according to PLMN specific parameters, which can be defined with the help of MAXQOS parameter group. false - PLMN specific QoS roaming parameters are not in use. Roaming check is performed according to SGSN specific QoS roaming subscribers parameters. Note: this parameter has effect only if the parameter qosRoamingAllowed is defined as false.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

pon - LTE QoS policy set id

Syntax

--pon <integer>

Description

LTE QoS policy set id.

Class

Optional.

Type / Range

Integer in the range of 0 to 6

Default

0

Name

qosRoamingAllowed - QoS based roaming agreement

Syntax

--qosRoamingAllowed <list-item>

Description

QoS based roaming agreement. with the QoS based roaming agreement you can define to allow QoS roaming. true - QoS roaming is allowed. roaming check is not performed, and QoS parameter downgrade, according to QoS roaming list, is not possible false - QoS roaming is not allowed. roaming check must be performed according to QOSPLMN parameter value.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

rfspIndexSelectionOption - RFSP Selection Option

Syntax

--rfspIndexSelectionOption <list-item>

Description

This field defines the way that RFSP Index is selected for the UE. Options are (1) (feature) disabled (2) apply local value as configured in overrideRfspIndex field

Class

Optional.

List

- disabled - Feature is disabled at the rfspProfile
- overrideRfspIndex - Override RFSP Index - SGSN assigns to the UE the RFSP Index as defined in Override RFSP Index field in rfspProfile

Default

disabled

Name

roamingControl - Roaming control

Syntax

--roamingControl <list-item>

Description

Roaming control parameter LBO - only local breakout is allowed. HRT - only home routed traffic is allowed. BOTH - both home routed traffic and local breakout are allowed. NOT - roaming not allowed. Default value is HRT

Class

Optional.

List

- BOTH - Both home routed traffic and local breakout are allowed.
- HRT - Home routed traffic - Only home routed traffic is allowed.
- LBO - Local breakout - Only local breakout is allowed.
- NOT - Not.

Default

HRT

Name

s4SgsnRoamingControl - Roaming control for S4 SGSN

Syntax

--s4SgsnRoamingControl <list-item>

Description

S4 SGSN Roaming control parameter LBO - only local breakout is allowed. HRT - only home routed traffic is allowed. BOTH - both home routed traffic and local breakout are allowed. NOT - roaming not allowed. Default value is HRT

Class

Optional.

List

- BOTH - Both home routed traffic and local breakout are allowed.
- HRT - Home routed traffic - Only home routed traffic is allowed.
- LBO - Local breakout - Only local breakout is allowed.
- NOT - Not.

Default

HRT

Name

s6dSelection - s6dGr selection per PLMN

Syntax

--s6dSelection <list-item>

Description

Operator can control on PLMN level whether S6d interface to HSS or Gr interface to HLR will be selected to acquire the UE subscription. The options are: - true, SGSN initially attempts to get EPS Subscription from HSS. Fallback to Gr is triggered if subscription is not acquired

from HSS unless Gr fallback parameter is set to not support fallback. -false, SGSN will get GPRS Subscription from HLR. Fallback to HSS is not supported

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

serviceReqImeiChk - IMEI status check from EIR for service request

Syntax

--serviceReqImeiChk <list-item>

Description

IMEI status check from EIR for service request.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sgsnLboAllowedApnListName - name of sgsn LBO Allowed Apn List

Syntax

--sgsnLboAllowedApnListName <string>

Description

name of sgsn LBO Allowed Apn List

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

shortNetwName - Short network name

Syntax

--shortNetwName <string>

Description

This parameter defines the short network name of the network operator (PLMN level). This is for the case MVNO used, different PLMN number is used for different operators.

Class

Optional.

Type / Range

Digit string of length 0 to 64

Default

""

Name

specialApnListName - Special Apn List Name

Syntax

--specialApnListName <string>

Description

Special Apn List Name

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 31 with the allowed pattern of [0-9a-zA-Z_\.\\-:@]*

Default

""

Name

tcForQci1 - Traffic class for QCI

Syntax

--tcForQci1 <list-item>

Description

Select any traffic class based on the QCI selected possible values if QCI 1 to 4 traffic class 1 to 4 - 1 - .convesational speech - 2 - .convesational with time delay 150ms - 3 - .convesational with time delay 80ms - 4 - .streaming - 10 - default which maps to standard mapping.

Class

Optional.

List

- 1 - Conversational speech.
- 2 - Conversational with time delay 150ms.

- 3 - Conversational with time delay 80ms.
- 4 - Streaming.
- 10 - Default which maps to standard mapping.

Default

10

Name

tcForQci2 - Traffic class for QCI

Syntax`--tcForQci2 <list-item>`**Description**

Select any traffic class based on the QCI selected possible values if QCI 1 to 4 traffic class 1 to 4 - 1 - .convesational speech - 2 - .convesational with time delay 150ms - 3 - .convesational with time delay 80ms - 4 - .streaming - 10 - default which maps to standard mapping.

Class

Optional.

List

- 1 - Conversational speech.
- 2 - Conversational with time delay 150ms.
- 3 - Conversational with time delay 80ms.
- 4 - Streaming.
- 10 - Default which maps to standard mapping.

Default

10

Name

tcForQci3 - Traffic class for QCI

Syntax`--tcForQci3 <list-item>`**Description**

Select any traffic class based on the QCI selected possible values if QCI 1 to 4 traffic class 1 to 4 - 1 - .convesational speech - 2 - .convesational with time delay 150ms - 3 - .convesational with time delay 80ms - 4 - .streaming - 10 - default which maps to standard mapping.

Class

Optional.

List

- 1 - Conversational speech.
- 2 - Conversational with time delay 150ms.
- 3 - Conversational with time delay 80ms.
- 4 - Streaming.
- 10 - Default which maps to standard mapping.

Default

10

Name

tcForQci4 - Traffic class for QCI

Syntax`--tcForQci4 <list-item>`**Description**

Select any traffic class based on the QCI selected possible values if QCI 1 to 4 traffic class 1 to 4 - 1 - .convesational speech - 2 - .convesational with time delay 150ms - 3 - .convesational with time delay 80ms - 4 - .streaming - 10 - default which maps to standard mapping.

Class

Optional.

List

- 1 - Conversational speech.
- 2 - Conversational with time delay 150ms.
- 3 - Conversational with time delay 80ms.
- 4 - Streaming.
- 10 - Default which maps to standard mapping.

Default

10

Name

tcForQci5 - Traffic class for QCI

Syntax`--tcForQci5 <list-item>`**Description**

Select any traffic class based on the QCI selected possible values if QCI 5 to 9 traffic class 5 to 9 - 5 - interactive priority level 1 with signalling indication enabled - 6 - interactive priority

level 1 - 7 - interactive priority level 2 - 8 - interactive priority level 3 - 9 - background - 10 - default which maps to standard mapping.

Class

Optional.

List

- 5 - Interactive priority level 1 with signalling indication enabled.
- 6 - Interactive priority level 1.
- 7 - Interactive priority level 2.
- 8 - Interactive priority level 3.
- 9 - Background.
- 10 - Default which maps to standard mapping.

Default

10

Name

tcForQci6 - Traffic class for QCI

Syntax

--tcForQci6 <list-item>

Description

Select any traffic class based on the QCI selected possible values if QCI 5 to 9 traffic class 5 to 9 - 5 - interactive priority level 1 with signalling indication enabled - 6 - interactive priority level 1 - 7 - interactive priority level 2 - 8 - interactive priority level 3 - 9 - background - 10 - default which maps to standard mapping.

Class

Optional.

List

- 5 - Interactive priority level 1 with signalling indication enabled.
- 6 - Interactive priority level 1.
- 7 - Interactive priority level 2.
- 8 - Interactive priority level 3.
- 9 - Background.
- 10 - Default which maps to standard mapping.

Default

10

Name

tcForQci7 - Traffic class for QCI

Syntax

--tcForQci7 <list-item>

Description

Select any traffic class based on the QCI selected possible values if QCI 5 to 9 traffic class 5 to 9 - 5 - interactive priority level 1 with signalling indication enabled - 6 - interactive priority level 1 - 7 - interactive priority level 2 - 8 - interactive priority level 3 - 9 - background - 10 - default which maps to standard mapping.

Class

Optional.

List

- 5 - Interactive priority level 1 with signalling indication enabled.
- 6 - Interactive priority level 1.
- 7 - Interactive priority level 2.
- 8 - Interactive priority level 3.
- 9 - Background.
- 10 - Default which maps to standard mapping.

Default

10

Name

tcForQci8 - Traffic class for QCI

Syntax

--tcForQci8 <list-item>

Description

Select any traffic class based on the QCI selected possible values if QCI 5 to 9 traffic class 5 to 9 - 5 - interactive priority level 1 with signalling indication enabled - 6 - interactive priority level 1 - 7 - interactive priority level 2 - 8 - interactive priority level 3 - 9 - background - 10 - default which maps to standard mapping.

Class

Optional.

List

- 5 - Interactive priority level 1 with signalling indication enabled.
- 6 - Interactive priority level 1.
- 7 - Interactive priority level 2.
- 8 - Interactive priority level 3.

- 9 - Background.
- 10 - Default which maps to standard mapping.

Default

10

Name

tcForQci9 - Traffic class for QCI

Syntax

--tcForQci9 <list-item>

Description

Select any traffic class based on the QCI selected possible values if QCI 5 to 9 traffic class 5 to 9 - 5 - interactive priority level 1 with signalling indication enabled - 6 - interactive priority level 1 - 7 - interactive priority level 2 - 8 - interactive priority level 3 - 9 - background - 10 - default which maps to standard mapping.

Class

Optional.

List

- 5 - Interactive priority level 1 with signalling indication enabled.
- 6 - Interactive priority level 1.
- 7 - Interactive priority level 2.
- 8 - Interactive priority level 3.
- 9 - Background.
- 10 - Default which maps to standard mapping.

Default

10

Name

timeZone - TIME ZONE

Syntax

--timeZone <list-item>

Description

Time Zone per PLMN

Class

Optional.

List

- Africa/Abidjan - Timezone for africa/abidjan UTC offset of +00:00.

- Africa/Accra - Timezone for africa/accra UTC offset of +00:00.
- Africa/Addis_Ababa - Timezone for africa/addis_ababa UTC offset of +03:00.
- Africa/Algiers - Timezone for africa/algiers UTC offset of +01:00.
- Africa/Asmara - Timezone for africa/asmara UTC offset of +03:00.
- Africa/Asmera - Timezone for africa/asmera UTC offset of +03:00.
- Africa/Bamako - Timezone for africa/bamako UTC offset of +00:00.
- Africa/Bangui - Timezone for africa/bangui UTC offset of +01:00.
- Africa/Banjul - Timezone for africa/banjul UTC offset of +00:00.
- Africa/Bissau - Timezone for africa/bissau UTC offset of +00:00.
- Africa/Blantyre - Timezone for africa/blantyre UTC offset of +02:00.
- Africa/Brazzaville - Timezone for africa/brazzaville UTC offset of +01:00.
- Africa/Bujumbura - Timezone for africa/bujumbura UTC offset of +02:00.
- Africa/Cairo - Timezone for africa/cairo UTC offset of +02:00.
- Africa/Casablanca - Timezone for africa/casablanca UTC offset of +00:00.
- Africa/Ceuta - Timezone for africa/ceuta UTC offset of +01:00.
- Africa/Conakry - Timezone for africa/conakry UTC offset of +00:00.
- Africa/Dakar - Timezone for africa/dakar UTC offset of +00:00.
- Africa/Dar_es_Salaam - Timezone for africa/dar_es_salaam UTC offset of +03:00.
- Africa/Djibouti - Timezone for africa/djibouti UTC offset of +03:00.
- Africa/Douala - Timezone for africa/douala UTC offset of +01:00.
- Africa/El_Aaiun - Timezone for africa/el_aaiun UTC offset of +00:00.
- Africa/Freetown - Timezone for africa/freetown UTC offset of +00:00.
- Africa/Gaborone - Timezone for africa/gaborone UTC offset of +02:00.
- Africa/Harare - Timezone for africa/harare UTC offset of +02:00.
- Africa/Johannesburg - Timezone for africa/johannesburg UTC offset of +02:00.
- Africa/Juba - Timezone for africa/juba UTC offset of +03:00.
- Africa/Kampala - Timezone for africa/kampala UTC offset of +03:00.
- Africa/Khartoum - Timezone for africa/khartoum UTC offset of +03:00.
- Africa/Kigali - Timezone for africa/kigali UTC offset of +02:00.
- Africa/Kinshasa - Timezone for africa/kinshasa UTC offset of +01:00.
- Africa/Lagos - Timezone for africa/lagos UTC offset of +01:00.

- Africa/Libreville - Timezone for africa/libreville UTC offset of +01:00.
- Africa/Lome - Timezone for africa/lome UTC offset of +00:00.
- Africa/Luanda - Timezone for africa/luanda UTC offset of +01:00.
- Africa/Lubumbashi - Timezone for africa/lubumbashi UTC offset of +02:00.
- Africa/Lusaka - Timezone for africa/lusaka UTC offset of +02:00.
- Africa/Malabo - Timezone for africa/malabo UTC offset of +01:00.
- Africa/Maputo - Timezone for africa/maputo UTC offset of +02:00.
- Africa/Maseru - Timezone for africa/maseru UTC offset of +02:00.
- Africa/Mbabane - Timezone for africa/mbabane UTC offset of +02:00.
- Africa/Mogadishu - Timezone for africa/mogadishu UTC offset of +03:00.
- Africa/Monrovia - Timezone for africa/monrovia UTC offset of +00:00.
- Africa/Nairobi - Timezone for africa/nairobi UTC offset of +03:00.
- Africa/Ndjamena - Timezone for africa/ndjamena UTC offset of +01:00.
- Africa/Niamey - Timezone for africa/niamey UTC offset of +01:00.
- Africa/Nouakchott - Timezone for africa/nouakchott UTC offset of +00:00.
- Africa/Ouagadougou - Timezone for africa/ouagadougou UTC offset of +00:00.
- Africa/Porto-Novo - Timezone for africa/porto-novo UTC offset of +01:00.
- Africa/Sao_Tome - Timezone for africa/sao_tome UTC offset of +00:00.
- Africa/Timbuktu - Timezone for africa/timbuktu UTC offset of +00:00.
- Africa/Tripoli - Timezone for africa/tripoli UTC offset of +02:00.
- Africa/Tunis - Timezone for africa/tunis UTC offset of +01:00.
- Africa/Windhoek - Timezone for africa/windhoek UTC offset of +02:00.
- America/Adak - Timezone for america/adak UTC offset of -10:00.
- America/Anchorage - Timezone for america/anchorage UTC offset of -09:00.
- America/Anguilla - Timezone for america/anguilla UTC offset of -04:00.
- America/Antigua - Timezone for america/antigua UTC offset of -04:00.
- America/Araguaina - Timezone for america/araguaina UTC offset of -03:00.
- America/Argentina/Buenos_Aires - Timezone for america/argentina/buenos_aires UTC offset of -03:00.
- America/Argentina/Catamarca - Timezone for america/argentina/catamarca UTC offset of -03:00.

- America/Argentina/ComodRivadavia - Timezone for america/argentina/comodrivadavia UTC offset of -03:00.
- America/Argentina/Cordoba - Timezone for america/argentina/cordoba UTC offset of -03:00.
- America/Argentina/Jujuy - Timezone for america/argentina/jujuy UTC offset of -03:00.
- America/Argentina/La_Rioja - Timezone for america/argentina/la_rioja UTC offset of -03:00.
- America/Argentina/Mendoza - Timezone for america/argentina/mendoza UTC offset of -03:00.
- America/Argentina/Rio_Gallegos - Timezone for america/argentina/rio_gallegos UTC offset of -03:00.
- America/Argentina/Salta - Timezone for america/argentina/salta UTC offset of -03:00.
- America/Argentina/San_Juan - Timezone for america/argentina/san_juan UTC offset of -03:00.
- America/Argentina/San_Luis - Timezone for america/argentina/san_luis UTC offset of -03:00.
- America/Argentina/Tucuman - Timezone for america/argentina/tucuman UTC offset of -03:00.
- America/Argentina/Ushuaia - Timezone for america/argentina/ushuaia UTC offset of -03:00.
- America/Aruba - Timezone for america/aruba UTC offset of -04:00.
- America/Asuncion - Timezone for america/asuncion UTC offset of -03:00.
- America/Atikokan - Timezone for america/atikokan UTC offset of -05:00.
- America/Atka - Timezone for america/atka UTC offset of -10:00.
- America/Bahia - Timezone for america/bahia UTC offset of -03:00.
- America/Bahia_Banderas - Timezone for america/bahia_banderas UTC offset of -06:00.
- America/Barbados - Timezone for america/barbados UTC offset of -04:00.
- America/Belem - Timezone for america/belem UTC offset of -03:00.
- America/Belize - Timezone for america/belize UTC offset of -06:00.
- America/Blanc-Sablon - Timezone for america/blanc-sablon UTC offset of -04:00.
- America/Boa_Vista - Timezone for america/boa_vista UTC offset of -04:00.
- America/Bogota - Timezone for america/bogota UTC offset of -05:00.
- America/Boise - Timezone for america/boise UTC offset of -07:00.
- America/Buenos_Aires - Timezone for america/buenos_aires UTC offset of -03:00.

- America/Cambridge_Bay - Timezone for america/cambridge_bay UTC offset of -07:00.
- America/Campo_Grande - Timezone for america/campo_grande UTC offset of -03:00.
- America/Cancun - Timezone for america/cancun UTC offset of -05:00.
- America/Caracas - Timezone for america/caracas UTC offset of -04:00.
- America/Catamarca - Timezone for america/catamarca UTC offset of -03:00.
- America/Cayenne - Timezone for america/cayenne UTC offset of -03:00.
- America/Cayman - Timezone for america/cayman UTC offset of -05:00.
- America/Chicago - Timezone for america/chicago UTC offset of -06:00.
- America/Chihuahua - Timezone for america/chihuahua UTC offset of -07:00.
- America/Coral_Harbour - Timezone for america/coral_harbour UTC offset of -05:00.
- America/Cordoba - Timezone for america/cordoba UTC offset of -03:00.
- America/Costa_Rica - Timezone for america/costa_rica UTC offset of -06:00.
- America/Creston - Timezone for america/creston UTC offset of -07:00.
- America/Cuiaba - Timezone for america/cuiaba UTC offset of -03:00.
- America/Curacao - Timezone for america/curacao UTC offset of -04:00.
- America/Danmarkshavn - Timezone for america/danmarkshavn UTC offset of +00:00.
- America/Dawson - Timezone for america/dawson UTC offset of -08:00.
- America/Dawson_Creek - Timezone for america/dawson_creek UTC offset of -07:00.
- America/Denver - Timezone for america/denver UTC offset of -07:00.
- America/Detroit - Timezone for america/detroit UTC offset of -05:00.
- America/Dominica - Timezone for america/dominica UTC offset of -04:00.
- America/Edmonton - Timezone for america/edmonton UTC offset of -07:00.
- America/Eirunepe - Timezone for america/eirunepe UTC offset of -05:00.
- America/El_Salvador - Timezone for america/el_salvador UTC offset of -06:00.
- America/Ensenada - Timezone for america/ensenada UTC offset of -08:00.
- America/Fort_Nelson - Timezone for america/fort_nelson UTC offset of -07:00.
- America/Fort_Wayne - Timezone for america/fort_wayne UTC offset of -05:00.
- America/Fortaleza - Timezone for america/fortaleza UTC offset of -03:00.
- America/Glace_Bay - Timezone for america/glance_bay UTC offset of -04:00.
- America/Godthab - Timezone for america/godthab UTC offset of -03:00.
- America/Goose_Bay - Timezone for america/goose_bay UTC offset of -04:00.

- America/Grand_Turk - Timezone for america/grand_turk UTC offset of -04:00.
- America/Grenada - Timezone for america/grenada UTC offset of -04:00.
- America/Guadeloupe - Timezone for america/guadeloupe UTC offset of -04:00.
- America/Guatemala - Timezone for america/guatemala UTC offset of -06:00.
- America/Guayaquil - Timezone for america/guayaquil UTC offset of -05:00.
- America/Guyana - Timezone for america/guyana UTC offset of -04:00.
- America/Halifax - Timezone for america/halifax UTC offset of -04:00.
- America/Havana - Timezone for america/havana UTC offset of -05:00.
- America/Hermosillo - Timezone for america/hermosillo UTC offset of -07:00.
- America/Indiana/Indianapolis - Timezone for america/indiana/indianapolis UTC offset of -05:00.
- America/Indiana/Knox - Timezone for america/indiana/knox UTC offset of -06:00.
- America/Indiana/Marengo - Timezone for america/indiana/marengo UTC offset of -05:00.
- America/Indiana/Petersburg - Timezone for america/indiana/petersburg UTC offset of -05:00.
- America/Indiana/Tell_City - Timezone for america/indiana/tell_city UTC offset of -06:00.
- America/Indiana/Vevay - Timezone for america/indiana/vevay UTC offset of -05:00.
- America/Indiana/Vincennes - Timezone for america/indiana/vincennes UTC offset of -05:00.
- America/Indiana/Winamac - Timezone for america/indiana/winamac UTC offset of -05:00.
- America/Indianapolis - Timezone for america/indianapolis UTC offset of -05:00.
- America/Inuvik - Timezone for america/inuvik UTC offset of -07:00.
- America/Iqaluit - Timezone for america/iqaluit UTC offset of -05:00.
- America/Jamaica - Timezone for america/jamaica UTC offset of -05:00.
- America/Jujuy - Timezone for america/jujuy UTC offset of -03:00.
- America/Juneau - Timezone for america/juneau UTC offset of -09:00.
- America/Kentucky/Louisville - Timezone for america/kentucky/louisville UTC offset of -05:00.
- America/Kentucky/Monticello - Timezone for america/kentucky/monticello UTC offset of -05:00.
- America/Knox_IN - Timezone for america/knox_in UTC offset of -06:00.

- America/Kralendijk - Timezone for america/kralendijk UTC offset of -04:00.
- America/La_Paz - Timezone for america/la_paz UTC offset of -04:00.
- America/Lima - Timezone for america/lima UTC offset of -05:00.
- America/Los_Angeles - Timezone for america/los_angeles UTC offset of -08:00.
- America/Louisville - Timezone for america/louisville UTC offset of -05:00.
- America/Lower_Princes - Timezone for america/lower_princes UTC offset of -04:00.
- America/Maceio - Timezone for america/maceio UTC offset of -03:00.
- America/Managua - Timezone for america/managua UTC offset of -06:00.
- America/Manaus - Timezone for america/manaus UTC offset of -04:00.
- America/Marigot - Timezone for america/marigot UTC offset of -04:00.
- America/Martinique - Timezone for america/martinique UTC offset of -04:00.
- America/Matamoros - Timezone for america/matamoros UTC offset of -06:00.
- America/Mazatlan - Timezone for america/mazatlan UTC offset of -07:00.
- America/Mendoza - Timezone for america/mendoza UTC offset of -03:00.
- America/Menominee - Timezone for america/menominee UTC offset of -06:00.
- America/Merida - Timezone for america/merida UTC offset of -06:00.
- America/Metlakatla - Timezone for america/metlakatla UTC offset of -09:00.
- America/Mexico_City - Timezone for america/mexico_city UTC offset of -06:00.
- America/Miquelon - Timezone for america/miquelon UTC offset of -03:00.
- America/Moncton - Timezone for america/moncton UTC offset of -04:00.
- America/Monterrey - Timezone for america/monterrey UTC offset of -06:00.
- America/Montevideo - Timezone for america/montevideo UTC offset of -03:00.
- America/Montreal - Timezone for america/montreal UTC offset of -05:00.
- America/Montserrat - Timezone for america/montserrat UTC offset of -04:00.
- America/Nassau - Timezone for america/nassau UTC offset of -05:00.
- America/New_York - Timezone for america/new_york UTC offset of -05:00.
- America/Nipigon - Timezone for america/nipigon UTC offset of -05:00.
- America/Nome - Timezone for america/nome UTC offset of -09:00.
- America/Noronha - Timezone for america/noronha UTC offset of -02:00.
- America/North_Dakota/Beulah - Timezone for america/north_dakota/beulah UTC offset of -06:00.

- America/North_Dakota/Center - Timezone for america/north_dakota/center UTC offset of -06:00.
- America/North_Dakota/New_Salem - Timezone for america/north_dakota/new_salem UTC offset of -06:00.
- America/Ojinaga - Timezone for america/ojinaga UTC offset of -07:00.
- America/Panama - Timezone for america/panama UTC offset of -05:00.
- America/Pangnirtung - Timezone for america/pangnirtung UTC offset of -05:00.
- America/Paramaribo - Timezone for america/paramaribo UTC offset of -03:00.
- America/Phoenix - Timezone for america/phoenix UTC offset of -07:00.
- America/Port-au-Prince - Timezone for america/port-au-prince UTC offset of -05:00.
- America/Port_of_Spain - Timezone for america/port_of_spain UTC offset of -04:00.
- America/Porto_Acre - Timezone for america/porto_acre UTC offset of -05:00.
- America/Porto_Velho - Timezone for america/porto_velho UTC offset of -04:00.
- America/Puerto_Rico - Timezone for america/puerto_rico UTC offset of -04:00.
- America/Rainy_River - Timezone for america/rainy_river UTC offset of -06:00.
- America/Rankin_Inlet - Timezone for america/rankin_inlet UTC offset of -06:00.
- America/Recife - Timezone for america/recife UTC offset of -03:00.
- America/Regina - Timezone for america/regina UTC offset of -06:00.
- America/Resolute - Timezone for america/resolute UTC offset of -06:00.
- America/Rio_Branco - Timezone for america/rio_branco UTC offset of -05:00.
- America/Rosario - Timezone for america/rosario UTC offset of -03:00.
- America/Santa_Isabel - Timezone for america/santa_isabel UTC offset of -08:00.
- America/Santarem - Timezone for america/santarem UTC offset of -03:00.
- America/Santiago - Timezone for america/santiago UTC offset of -03:00.
- America/Santo_Domingo - Timezone for america/santo_domingo UTC offset of -04:00.
- America/Sao_Paulo - Timezone for america/sao_paulo UTC offset of -02:00.
- America/Scoresbysund - Timezone for america/scoresbysund UTC offset of -01:00.
- America/Shiprock - Timezone for america/shiprock UTC offset of -07:00.
- America/Sitka - Timezone for america/sitka UTC offset of -09:00.
- America/St_Barthelemy - Timezone for america/st_barthelemy UTC offset of -04:00.
- America/St_Johns - Timezone for america/st_johns UTC offset of -03:30.
- America/St_Kitts - Timezone for america/st_kitts UTC offset of -04:00.

- America/St_Lucia - Timezone for america/st_lucia UTC offset of -04:00.
- America/St_Thomas - Timezone for america/st_thomas UTC offset of -04:00.
- America/St_Vincent - Timezone for america/st_vincent UTC offset of -04:00.
- America/Swift_Current - Timezone for america/swift_current UTC offset of -06:00.
- America/Tegucigalpa - Timezone for america/tegucigalpa UTC offset of -06:00.
- America/Thule - Timezone for america/thule UTC offset of -04:00.
- America/Thunder_Bay - Timezone for america/thunder_bay UTC offset of -05:00.
- America/Tijuana - Timezone for america/tijuana UTC offset of -08:00.
- America/Toronto - Timezone for america/toronto UTC offset of -05:00.
- America/Tortola - Timezone for america/tortola UTC offset of -04:00.
- America/Vancouver - Timezone for america/vancouver UTC offset of -08:00.
- America/Virgin - Timezone for america/virgin UTC offset of -04:00.
- America/Whitehorse - Timezone for america/whitehorse UTC offset of -08:00.
- America/Winnipeg - Timezone for america/winnipeg UTC offset of -06:00.
- America/Yakutat - Timezone for america/yakutat UTC offset of -09:00.
- America/Yellowknife - Timezone for america/yellowknife UTC offset of -07:00.
- Antarctica/Casey - Timezone for antarctica/casey UTC offset of +11:00.
- Antarctica/Davis - Timezone for antarctica/davis UTC offset of +07:00.
- Antarctica/DumontDURville - Timezone for antarctica/dumontdurville UTC offset of +10:00.
- Antarctica/Macquarie - Timezone for antarctica/macquarie UTC offset of +11:00.
- Antarctica/Mawson - Timezone for antarctica/mawson UTC offset of +05:00.
- Antarctica/McMurdo - Timezone for antarctica/mcmurdo UTC offset of +13:00.
- Antarctica/Palmer - Timezone for antarctica/palmer UTC offset of -03:00.
- Antarctica/Rothera - Timezone for antarctica/rothera UTC offset of -03:00.
- Antarctica/South_Pole - Timezone for antarctica/south_pole UTC offset of +13:00.
- Antarctica/Syowa - Timezone for antarctica/syowa UTC offset of +03:00.
- Antarctica/Troll - Timezone for antarctica/troll UTC offset of +00:00.
- Antarctica/Vostok - Timezone for antarctica/vostok UTC offset of +06:00.
- Arctic/Longyearbyen - Timezone for arctic/longyearbyen UTC offset of +01:00.
- Asia/Aden - Timezone for asia/aden UTC offset of +03:00.

- Asia/Almaty - Timezone for asia/almaty UTC offset of +06:00.
- Asia/Amman - Timezone for asia/amman UTC offset of +02:00.
- Asia/Anadyr - Timezone for asia/anadyr UTC offset of +12:00.
- Asia/Aqtau - Timezone for asia/aqtau UTC offset of +05:00.
- Asia/Aqtobe - Timezone for asia/aqtobe UTC offset of +05:00.
- Asia/Ashgabat - Timezone for asia/ashgabat UTC offset of +05:00.
- Asia/Ashkhabad - Timezone for asia/ashkhabad UTC offset of +05:00.
- Asia/Baghdad - Timezone for asia/baghdad UTC offset of +03:00.
- Asia/Bahrain - Timezone for asia/bahrain UTC offset of +03:00.
- Asia/Baku - Timezone for asia/baku UTC offset of +04:00.
- Asia/Bangkok - Timezone for asia/bangkok UTC offset of +07:00.
- Asia/Barnaul - Timezone for asia/barnaul UTC offset of +07:00.
- Asia/Beirut - Timezone for asia/beirut UTC offset of +02:00.
- Asia/Bishkek - Timezone for asia/bishkek UTC offset of +06:00.
- Asia/Brunei - Timezone for asia/brunei UTC offset of +08:00.
- Asia/Calcutta - Timezone for asia/calcutta UTC offset of +05:30.
- Asia/Chita - Timezone for asia/chita UTC offset of +09:00.
- Asia/Choibalsan - Timezone for asia/choibalsan UTC offset of +08:00.
- Asia/Chongqing - Timezone for asia/chongqing UTC offset of +08:00.
- Asia/Chungking - Timezone for asia/chungking UTC offset of +08:00.
- Asia/Colombo - Timezone for asia/colombo UTC offset of +05:30.
- Asia/Dacca - Timezone for asia/dacca UTC offset of +06:00.
- Asia/Damascus - Timezone for asia/damascus UTC offset of +02:00.
- Asia/Dhaka - Timezone for asia/dhaka UTC offset of +06:00.
- Asia/Dili - Timezone for asia/dili UTC offset of +09:00.
- Asia/Dubai - Timezone for asia/dubai UTC offset of +04:00.
- Asia/Dushanbe - Timezone for asia/dushanbe UTC offset of +05:00.
- Asia/Gaza - Timezone for asia/gaza UTC offset of +02:00.
- Asia/Harbin - Timezone for asia/harbin UTC offset of +08:00.
- Asia/Hebron - Timezone for asia/hebron UTC offset of +02:00.
- Asia/Ho_Chi_Minh - Timezone for asia/ho_chi_minh UTC offset of +07:00.

- Asia/Hong_Kong - Timezone for asia/hong_kong UTC offset of +08:00.
- Asia/Hovd - Timezone for asia/hovd UTC offset of +07:00.
- Asia/Irkutsk - Timezone for asia/irkutsk UTC offset of +08:00.
- Asia/Istanbul - Timezone for asia/istanbul UTC offset of +03:00.
- Asia/Jakarta - Timezone for asia/jakarta UTC offset of +07:00.
- Asia/Jayapura - Timezone for asia/jayapura UTC offset of +09:00.
- Asia/Jerusalem - Timezone for asia/jerusalem UTC offset of +02:00.
- Asia/Kabul - Timezone for asia/kabul UTC offset of +04:30.
- Asia/Kamchatka - Timezone for asia/kamchatka UTC offset of +12:00.
- Asia/Karachi - Timezone for asia/karachi UTC offset of +05:00.
- Asia/Kashgar - Timezone for asia/kashgar UTC offset of +06:00.
- Asia/Kathmandu - Timezone for asia/kathmandu UTC offset of +05:45.
- Asia/Katmandu - Timezone for asia/katmandu UTC offset of +05:45.
- Asia/Khandyga - Timezone for asia/khandyga UTC offset of +09:00.
- Asia/Kolkata - Timezone for asia/kolkata UTC offset of +05:30.
- Asia/Krasnoyarsk - Timezone for asia/krasnoyarsk UTC offset of +07:00.
- Asia/Kuala_Lumpur - Timezone for asia/kuala_lumpur UTC offset of +08:00.
- Asia/Kuching - Timezone for asia/kuching UTC offset of +08:00.
- Asia/Kuwait - Timezone for asia/kuwait UTC offset of +03:00.
- Asia/Macao - Timezone for asia/macao UTC offset of +08:00.
- Asia/Macau - Timezone for asia/macau UTC offset of +08:00.
- Asia/Magadan - Timezone for asia/magadan UTC offset of +11:00.
- Asia/Makassar - Timezone for asia/makassar UTC offset of +08:00.
- Asia/Manila - Timezone for asia/manila UTC offset of +08:00.
- Asia/Muscat - Timezone for asia/muscat UTC offset of +04:00.
- Asia/Nicosia - Timezone for asia/nicosia UTC offset of +02:00.
- Asia/Novokuznetsk - Timezone for asia/novokuznetsk UTC offset of +07:00.
- Asia/Novosibirsk - Timezone for asia/novosibirsk UTC offset of +07:00.
- Asia/Omsk - Timezone for asia/omsk UTC offset of +06:00.
- Asia/Oral - Timezone for asia/oral UTC offset of +05:00.
- Asia/Phnom_Penh - Timezone for asia/phnom_penh UTC offset of +07:00.

- Asia/Pontianak - Timezone for asia/pontianak UTC offset of +07:00.
- Asia/Pyongyang - Timezone for asia/pyongyang UTC offset of +08:30.
- Asia/Qatar - Timezone for asia/qatar UTC offset of +03:00.
- Asia/Qyzylorda - Timezone for asia/qyzylorda UTC offset of +06:00.
- Asia/Rangoon - Timezone for asia/rangoon UTC offset of +06:30.
- Asia/Riyadh - Timezone for asia/riyadh UTC offset of +03:00.
- Asia/Saigon - Timezone for asia/saigon UTC offset of +07:00.
- Asia/Sakhalin - Timezone for asia/sakhalin UTC offset of +11:00.
- Asia/Samarkand - Timezone for asia/samarkand UTC offset of +05:00.
- Asia/Seoul - Timezone for asia/seoul UTC offset of +09:00.
- Asia/Shanghai - Timezone for asia/shanghai UTC offset of +08:00.
- Asia/Singapore - Timezone for asia/singapore UTC offset of +08:00.
- Asia/Srednekolymsk - Timezone for asia/srednekolymsk UTC offset of +11:00.
- Asia/Taipei - Timezone for asia/taipei UTC offset of +08:00.
- Asia/Tashkent - Timezone for asia/tashkent UTC offset of +05:00.
- Asia/Tbilisi - Timezone for asia/tbilisi UTC offset of +04:00.
- Asia/Tehran - Timezone for asia/tehran UTC offset of +03:30.
- Asia/Tel_Aviv - Timezone for asia/tel_aviv UTC offset of +02:00.
- Asia/Thimbu - Timezone for asia/thimbu UTC offset of +06:00.
- Asia/Thimphu - Timezone for asia/thimphu UTC offset of +06:00.
- Asia/Tokyo - Timezone for asia/tokyo UTC offset of +09:00.
- Asia/Tomsk - Timezone for asia/tomsk UTC offset of +07:00.
- Asia/Ujung_Pandang - Timezone for asia/ujung_pandang UTC offset of +08:00.
- Asia/Ulaanbaatar - Timezone for asia/ulaanbaatar UTC offset of +08:00.
- Asia/Ulan_Bator - Timezone for asia/ulan_bator UTC offset of +08:00.
- Asia/Urumqi - Timezone for asia/urumqi UTC offset of +06:00.
- Asia/Ust-Nera - Timezone for asia/ust-nera UTC offset of +10:00.
- Asia/Vientiane - Timezone for asia/vientiane UTC offset of +07:00.
- Asia/Vladivostok - Timezone for asia/vladivostok UTC offset of +10:00.
- Asia/Yakutsk - Timezone for asia/yakutsk UTC offset of +09:00.
- Asia/Yekaterinburg - Timezone for asia/yekaterinburg UTC offset of +05:00.

- Asia/Yerevan - Timezone for asia/yerevan UTC offset of +04:00.
- Atlantic/Azores - Timezone for atlantic/azores UTC offset of -01:00.
- Atlantic/Bermuda - Timezone for atlantic/bermuda UTC offset of -04:00.
- Atlantic/Canary - Timezone for atlantic/canary UTC offset of +00:00.
- Atlantic/Cape_Verde - Timezone for atlantic/cape_verde UTC offset of -01:00.
- Atlantic/Faeroe - Timezone for atlantic/faeroe UTC offset of +00:00.
- Atlantic/Faroe - Timezone for atlantic/faroe UTC offset of +00:00.
- Atlantic/Jan_Mayen - Timezone for atlantic/jan_mayen UTC offset of +01:00.
- Atlantic/Madeira - Timezone for atlantic/madeira UTC offset of +00:00.
- Atlantic/Reykjavik - Timezone for atlantic/reykjavik UTC offset of +00:00.
- Atlantic/South_Georgia - Timezone for atlantic/south_georgia UTC offset of -02:00.
- Atlantic/St_Helena - Timezone for atlantic/st_helena UTC offset of +00:00.
- Atlantic/Stanley - Timezone for atlantic/stanley UTC offset of -03:00.
- Australia/ACT - Timezone for australia/act UTC offset of +11:00.
- Australia/Adelaide - Timezone for australia/adelaide UTC offset of +10:30.
- Australia/Brisbane - Timezone for australia/brisbane UTC offset of +10:00.
- Australia/Broken_Hill - Timezone for australia/broken_hill UTC offset of +10:30.
- Australia/Canberra - Timezone for australia/canberra UTC offset of +11:00.
- Australia/Currie - Timezone for australia/currie UTC offset of +11:00.
- Australia/Darwin - Timezone for australia/darwin UTC offset of +09:30.
- Australia/Eucla - Timezone for australia/eucla UTC offset of +08:45.
- Australia/Hobart - Timezone for australia/hobart UTC offset of +11:00.
- Australia/LHI - Timezone for australia/lhi UTC offset of +11:00.
- Australia/Lindeman - Timezone for australia/lindeman UTC offset of +10:00.
- Australia/Lord_Howe - Timezone for australia/lord_howe UTC offset of +11:00.
- Australia/Melbourne - Timezone for australia/melbourne UTC offset of +11:00.
- Australia/NSW - Timezone for australia/nsw UTC offset of +11:00.
- Australia/North - Timezone for australia/north UTC offset of +09:30.
- Australia/Perth - Timezone for australia/perth UTC offset of +08:00.
- Australia/Queensland - Timezone for australia/queensland UTC offset of +10:00.
- Australia/South - Timezone for australia/south UTC offset of +10:30.

- Australia/Sydney - Timezone for australia/sydney UTC offset of +11:00.
- Australia/Tasmania - Timezone for australia/tasmania UTC offset of +11:00.
- Australia/Victoria - Timezone for australia/victoria UTC offset of +11:00.
- Australia/West - Timezone for australia/west UTC offset of +08:00.
- Australia/Yancowinna - Timezone for australia/yancowinna UTC offset of +10:30.
- CET - Timezone for CET UTC offset of +01:00.
- CST6CDT - Timezone for CST6CDT UTC offset of -06:00.
- Canada/Atlantic - Timezone for canada/atlantic UTC offset of -04:00.
- Canada/Central - Timezone for canada/central UTC offset of -06:00.
- Canada/East-Saskatchewan - Timezone for canada/east-saskatchewan UTC offset of -06:00.
- Canada/Eastern - Timezone for canada/eastern UTC offset of -05:00.
- Canada/Mountain - Timezone for canada/mountain UTC offset of -07:00.
- Canada/Newfoundland - Timezone for canada/newfoundland UTC offset of -03:30.
- Canada/Pacific - Timezone for canada/pacific UTC offset of -08:00.
- Canada/Saskatchewan - Timezone for canada/saskatchewan UTC offset of -06:00.
- Canada/Yukon - Timezone for canada/yukon UTC offset of -08:00.
- Chile/Continental - Timezone for chile/continental UTC offset of -03:00.
- Chile/EasterIsland - Timezone for chile/easterisland UTC offset of -05:00.
- Cuba - Timezone for cuba UTC offset of -05:00.
- Default - Timezone for system UTC (OS time zone)
- EET - Timezone for EET UTC offset of +02:00.
- EST - Timezone for EST UTC offset of -05:00.
- EST5EDT - Timezone for EST5EDT UTC offset of -05:00.
- Egypt - Timezone for egypt UTC offset of +02:00.
- Eire - Timezone for eire UTC offset of +00:00.
- Etc/Greenwich - Timezone for etc/greenwich UTC offset of +00:00.
- Etc/UCT - Timezone for etc/uct UTC offset of +00:00.
- Etc/UTC - Timezone for etc/utc UTC offset of +00:00.
- Etc/Universal - Timezone for etc/universal UTC offset of +00:00.
- Etc/Zulu - Timezone for etc/zulu UTC offset of +00:00.

- Europe/Amsterdam - Timezone for europe/amsterdam UTC offset of +01:00.
- Europe/Andorra - Timezone for europe/andorra UTC offset of +01:00.
- Europe/Astrakhan - Timezone for europe/astrakhan UTC offset of +04:00.
- Europe/Athens - Timezone for europe/athens UTC offset of +02:00.
- Europe/Belfast - Timezone for europe/belfast UTC offset of +00:00.
- Europe/Belgrade - Timezone for europe/belgrade UTC offset of +01:00.
- Europe/Berlin - Timezone for europe/berlin UTC offset of +01:00.
- Europe/Bratislava - Timezone for europe/bratislava UTC offset of +01:00.
- Europe/Brussels - Timezone for europe/brussels UTC offset of +01:00.
- Europe/Bucharest - Timezone for europe/bucharest UTC offset of +02:00.
- Europe/Budapest - Timezone for europe/budapest UTC offset of +01:00.
- Europe/Busingen - Timezone for europe/busingen UTC offset of +01:00.
- Europe/Chisinau - Timezone for europe/chisinau UTC offset of +02:00.
- Europe/Copenhagen - Timezone for europe/copenhagen UTC offset of +01:00.
- Europe/Dublin - Timezone for europe/dublin UTC offset of +00:00.
- Europe/Gibraltar - Timezone for europe/gibraltar UTC offset of +01:00.
- Europe/Guernsey - Timezone for europe/guernsey UTC offset of +00:00.
- Europe/Helsinki - Timezone for europe/helsinki UTC offset of +02:00.
- Europe/Isle_of_Man - Timezone for europe/isle_of_man UTC offset of +00:00.
- Europe/Istanbul - Timezone for europe/istanbul UTC offset of +03:00.
- Europe/Jersey - Timezone for europe/jersey UTC offset of +00:00.
- Europe/Kaliningrad - Timezone for europe/kaliningrad UTC offset of +02:00.
- Europe/Kiev - Timezone for europe/kyiv UTC offset of +02:00.
- Europe/Kirov - Timezone for europe/kirov UTC offset of +03:00.
- Europe/Lisbon - Timezone for europe/lisbon UTC offset of +00:00.
- Europe/Ljubljana - Timezone for europe/ljubljana UTC offset of +01:00.
- Europe/London - Timezone for europe/london UTC offset of +00:00.
- Europe/Luxembourg - Timezone for europe/luxembourg UTC offset of +01:00.
- Europe/Madrid - Timezone for europe/madrid UTC offset of +01:00.
- Europe/Malta - Timezone for europe/malta UTC offset of +01:00.
- Europe/Mariehamn - Timezone for europe/mariehamn UTC offset of +02:00.

- Europe/Minsk - Timezone for europe/minsk UTC offset of +03:00.
- Europe/Monaco - Timezone for europe/monaco UTC offset of +01:00.
- Europe/Moscow - Timezone for europe/moscow UTC offset of +03:00.
- Europe/Nicosia - Timezone for europe/nicosia UTC offset of +02:00.
- Europe/Oslo - Timezone for europe/oslo UTC offset of +01:00.
- Europe/Paris - Timezone for europe/paris UTC offset of +01:00.
- Europe/Podgorica - Timezone for europe/podgorica UTC offset of +01:00.
- Europe/Prague - Timezone for europe/prague UTC offset of +01:00.
- Europe/Riga - Timezone for europe/riga UTC offset of +02:00.
- Europe/Rome - Timezone for europe/rome UTC offset of +01:00.
- Europe/Samara - Timezone for europe/samara UTC offset of +04:00.
- Europe/San_Marino - Timezone for europe/san_marino UTC offset of +01:00.
- Europe/Sarajevo - Timezone for europe/sarajevo UTC offset of +01:00.
- Europe/Simferopol - Timezone for europe/simferopol UTC offset of +03:00.
- Europe/Skopje - Timezone for europe/skopje UTC offset of +01:00.
- Europe/Sofia - Timezone for europe/sofia UTC offset of +02:00.
- Europe/Stockholm - Timezone for europe/stockholm UTC offset of +01:00.
- Europe/Tallinn - Timezone for europe/tallinn UTC offset of +02:00.
- Europe/Tirane - Timezone for europe/tirane UTC offset of +01:00.
- Europe/Tiraspol - Timezone for europe/tiraspol UTC offset of +02:00.
- Europe/Ulyanovsk - Timezone for europe/ulyanovsk UTC offset of +04:00.
- Europe/Uzhgorod - Timezone for europe/uzhgorod UTC offset of +02:00.
- Europe/Vaduz - Timezone for europe/vaduz UTC offset of +01:00.
- Europe/Vatican - Timezone for europe/vatican UTC offset of +01:00.
- Europe/Vienna - Timezone for europe/vienna UTC offset of +01:00.
- Europe/Vilnius - Timezone for europe/vilnius UTC offset of +02:00.
- Europe/Volgograd - Timezone for europe/volgograd UTC offset of +03:00.
- Europe/Warsaw - Timezone for europe/warsaw UTC offset of +01:00.
- Europe/Zagreb - Timezone for europe/zagreb UTC offset of +01:00.
- Europe/Zaporozhye - Timezone for europe/zaporozhye UTC offset of +02:00.
- Europe/Zurich - Timezone for europe/zurich UTC offset of +01:00.

- GB - Timezone for GB UTC offset of +00:00.
- GB-Eire - Timezone for GB-Eire UTC offset of +00:00.
- GMT - Timezone for GMT UTC offset of +00:00.
- GMT0 - Timezone for GMT0 UTC offset of +00:00.
- Greenwich - Timezone for greenwich UTC offset of +00:00.
- HST - Timezone for HST UTC offset of -10:00.
- Hongkong - Timezone for hongkong UTC offset of +08:00.
- Iceland - Timezone for iceland UTC offset of +00:00.
- Indian/Antananarivo - Timezone for indian/antananarivo UTC offset of +03:00.
- Indian/Chagos - Timezone for indian/chagos UTC offset of +06:00.
- Indian/Christmas - Timezone for indian/christmas UTC offset of +07:00.
- Indian/Cocos - Timezone for indian/cocos UTC offset of +06:30.
- Indian/Comoro - Timezone for indian/comoro UTC offset of +03:00.
- Indian/Kerguelen - Timezone for indian/kerguelen UTC offset of +05:00.
- Indian/Mahe - Timezone for indian/mahe UTC offset of +04:00.
- Indian/Maldives - Timezone for indian/maldives UTC offset of +05:00.
- Indian/Mauritius - Timezone for indian/mauritius UTC offset of +04:00.
- Indian/Mayotte - Timezone for indian/mayotte UTC offset of +03:00.
- Indian/Reunion - Timezone for indian/reunion UTC offset of +04:00.
- Iran - Timezone for iran UTC offset of +03:30.
- Israel - Timezone for israel UTC offset of +02:00.
- Jamaica - Timezone for jamaica UTC offset of -05:00.
- Japan - Timezone for japan UTC offset of +09:00.
- Kwajalein - Timezone for kwajalein UTC offset of +12:00.
- Libya - Timezone for libya UTC offset of +02:00.
- MET - Timezone for MET UTC offset of +01:00.
- MST - Timezone for MST UTC offset of -07:00.
- MST7MDT - Timezone for MST7MDT UTC offset of -07:00.
- Mexico/BajaNorte - Timezone for mexico/bajanorte UTC offset of -08:00.
- Mexico/BajaSur - Timezone for mexico/bajasur UTC offset of -07:00.
- Mexico/General - Timezone for mexico/general UTC offset of -06:00.

- NZ - Timezone for NZ UTC offset of +13:00.
- NZ-CHAT - Timezone for NZ-CHAT UTC offset of +13:45.
- Navajo - Timezone for navajo UTC offset of -07:00.
- PRC - Timezone for PRC UTC offset of +08:00.
- PST8PDT - Timezone for PST8PDT UTC offset of -08:00.
- Pacific/Apia - Timezone for pacific/apia UTC offset of +14:00.
- Pacific/Auckland - Timezone for pacific/auckland UTC offset of +13:00.
- Pacific/Bougainville - Timezone for pacific/bougainville UTC offset of +11:00.
- Pacific/Chatham - Timezone for pacific/chatham UTC offset of +13:45.
- Pacific/Chuuk - Timezone for pacific/chuuk UTC offset of +10:00.
- Pacific/Easter - Timezone for pacific/easter UTC offset of -05:00.
- Pacific/Efate - Timezone for pacific/efate UTC offset of +11:00.
- Pacific/Enderbury - Timezone for pacific/enderbury UTC offset of +13:00.
- Pacific/Fakaofu - Timezone for pacific/fakaofu UTC offset of +13:00.
- Pacific/Fiji - Timezone for pacific/fiji UTC offset of +13:00.
- Pacific/Funafuti - Timezone for pacific/funafuti UTC offset of +12:00.
- Pacific/Galapagos - Timezone for pacific/galapagos UTC offset of -06:00.
- Pacific/Gambier - Timezone for pacific/gambier UTC offset of -09:00.
- Pacific/Guadalcanal - Timezone for pacific/guadalcanal UTC offset of +11:00.
- Pacific/Guam - Timezone for pacific/guam UTC offset of +10:00.
- Pacific/Honolulu - Timezone for pacific/honolulu UTC offset of -10:00.
- Pacific/Johnston - Timezone for pacific/johnston UTC offset of -10:00.
- Pacific/Kiritimati - Timezone for pacific/kiritimati UTC offset of +14:00.
- Pacific/Kosrae - Timezone for pacific/kosrae UTC offset of +11:00.
- Pacific/Kwajalein - Timezone for pacific/kwajalein UTC offset of +12:00.
- Pacific/Majuro - Timezone for pacific/majuro UTC offset of +12:00.
- Pacific/Marquesas - Timezone for pacific/marquesas UTC offset of -09:30.
- Pacific/Midway - Timezone for pacific/midway UTC offset of -11:00.
- Pacific/Nauru - Timezone for pacific/nauru UTC offset of +12:00.
- Pacific/Niue - Timezone for pacific/niue UTC offset of -11:00.
- Pacific/Norfolk - Timezone for pacific/norfolk UTC offset of +11:00.

- Pacific/Noumea - Timezone for pacific/noumea UTC offset of +11:00.
- Pacific/Pago_Pago - Timezone for pacific/pago_pago UTC offset of -11:00.
- Pacific/Palau - Timezone for pacific/palau UTC offset of +09:00.
- Pacific/Pitcairn - Timezone for pacific/pitcairn UTC offset of -08:00.
- Pacific/Pohnpei - Timezone for pacific/pohnpei UTC offset of +11:00.
- Pacific/Ponape - Timezone for pacific/ponape UTC offset of +11:00.
- Pacific/Port_Moresby - Timezone for pacific/port_moresby UTC offset of +10:00.
- Pacific/Rarotonga - Timezone for pacific/rarotonga UTC offset of -10:00.
- Pacific/Saipan - Timezone for pacific/saipan UTC offset of +10:00.
- Pacific/Samoa - Timezone for pacific/samoa UTC offset of -11:00.
- Pacific/Tahiti - Timezone for pacific/tahiti UTC offset of -10:00.
- Pacific/Tarawa - Timezone for pacific/tarawa UTC offset of +12:00.
- Pacific/Tongatapu - Timezone for pacific/tongatapu UTC offset of +14:00.
- Pacific/Truk - Timezone for pacific/truk UTC offset of +10:00.
- Pacific/Wake - Timezone for pacific/wake UTC offset of +12:00.
- Pacific/Wallis - Timezone for pacific/wallis UTC offset of +12:00.
- Pacific/Yap - Timezone for pacific/yap UTC offset of +10:00.
- Poland - Timezone for poland UTC offset of +01:00.
- Portugal - Timezone for portugal UTC offset of +00:00.
- ROC - Timezone for ROC UTC offset of +08:00.
- ROK - Timezone for ROK UTC offset of +09:00.
- Singapore - Timezone for singapore UTC offset of +08:00.
- Turkey - Timezone for turkey UTC offset of +03:00.
- UCT - Timezone for UCT UTC offset of +00:00.
- US/Pacific-New - Timezone for US/Pacific-New UTC offset of -08:00.
- Universal - Timezone for universal UTC offset of +00:00.
- W-SU - Timezone for W-SU UTC offset of +03:00.
- WET - Timezone for WET UTC offset of +00:00.
- Zulu - Timezone for zulu UTC offset of +00:00.
- jdj_tz - Timezone for jdj_tz UTC offset of +00:00.

Default

Default

Name

traffHandlingPrio - Traffic handling priority for roaming subscribers

Syntax

--traffHandlingPrio <integer>

Description

Traffic handling priority for roaming subscribers. this parameter defines the traffic handling priority which can be used by roaming subscriber at most.

Class

Optional.

Type / Range

Integer in the range of 1 to 3

Default

3

Name

trafficClass - Traffic class for roaming subscribers

Syntax

--trafficClass <list-item>

Description

Traffic class for roaming subscribers. this parameter defines the traffic class which can be used by roaming subscriber at most conversational - traffic class type of conversational streaming - traffic class type of streaming interactive - traffic class type of interactive background - traffic class type of background.

Class

Optional.

List

- Background - Traffic class type of background.
- Conversational - Traffic class type of conversational.
- Interactive - Traffic class type of interactive.
- Streaming - Traffic class type of streaming.

Default

Interactive

Name

unknownImeiEffect - Allow unknown equipment

Syntax

--unknownImeiEffect <list-item>

Description

Allow unknown equipment allow - is used for allowing unknown IMEI block - is used for not allowing unknown IMEI.

Class

Optional.

List

- allow - Allow.
- block - Block.

Default

block

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.58 cmm plmnSecurity

PLMN Security. Syntax, description, parameters, and roles for cmm plmnSecurity command.

Syntax

Description

This command is used to administer PLMN Security and includes parameters for defining the authentication interaction and GUTI reallocation percentages for an MME procedure. GUTI reallocation is only applicable to the following procedures: SubAttach TAUUpdate

Parameters

Name

plmnName - Mobile network name

Syntax

--plmnName <string>

Description

Name that identifies mobile network (alternate key).

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

procedureName - Procedure name

Syntax

--procedureName <list-item>

Description

Name of MME procedure.

Class

Required key.

List

- CPSRequest - CPS request - CPS request.
- IPLMNTAUUpdate - Inter PLMN TA update - Inter-PLMN TA update.
- IRATTAUUpdate - IRAT TAU - Authentication and GUTI reallocation frequency for IRAT TAU procedure.
- ServiceReq - Service request - Service request.
- SubAttach - Sub attach - Sub attach.
- TAUUpdate - TAU update - TAU update.
- UEInitDetach - UE init detach - UE init detach.
- UEInitExtSrvReq - UE init extended service request - UE init extended service request.

Name

authInteraction - Authentication interaction

Syntax

--authInteraction <integer>

Description

Indicates the percentage of cases in which an authentication interaction is performed for the specified procedure.

Class

Optional.

Type / Range

Integer in the range of 0 to 100

Default

20

Name

gutiReallocation - GUTI reallocation

Syntax

--gutiReallocation <integer>

Description

Indicates the percentage of cases in which a GUTI (Globally unique temporary identity) reallocation is performed for the specified procedure.

Class

Optional.

Type / Range

Integer in the range of 0 to 100

Default

20

Name

irat5gAuth - Authentication when UE is moving from 5G to 4G

Syntax

--irat5gAuth <list-item>

Description

Enable authentication when UE is moving from 5G to 4G

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

6.59 cmm plmnSecurityProfile

plmn Security Profile. Syntax, description, parameters, and roles for cmm plmnSecurityProfile command.

Syntax

Description

plmnSecurity is being created whenever an MME PLMN is created. So it is tied to the PLMN. This profile intents for a given range of IMSIs that can point to a different plmnSecurity. This new profile is used as an override to plmnSecurity - if it is provisioned and linked.

Parameters

Name

plmnSecurityProfileListName - Name of identifies profile

Syntax

--plmnSecurityProfileListName <string>

Description

Name that identifies a profile

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

procedureName - Procedure name

Syntax

--procedureName <list-item>

Description

Name of MME procedure.

Class

Required key.

List

- CPSRequest - CPS request - CPS request.
- IPLMNTAUUpdate - Inter PLMN TA update - Inter-PLMN TA update.
- IRATTAUUpdate - IRAT TAU - Authentication and GUTI reallocation frequency for IRAT TAU procedure.
- ServiceReq - Service request - Service request.
- SubAttach - Sub attach - Sub attach.
- TAUUpdate - TAU update - TAU update.
- UEInitDetach - UE init detach - UE init detach.
- UEInitExtSrvReq - UE init extended service request - UE init extended service request.

Name

authInteraction - Authentication interaction

Syntax

--authInteraction <integer>

Description

Indicates the percentage of cases in which an authentication interaction is performed for the specified procedure. Note: default value should align with cli plmnSecurity. The default values in plmnSecurity for each procedure set by call processing.

Class

Optional.

Type / Range

Integer in the range of 0 to 100

Default

20

Name

gutiReallocation - GUTI reallocation

Syntax

--gutiReallocation <integer>

Description

Indicates the percentage of cases in which a GUTI (Globally unique temporary identity) reallocation is performed for the specified procedure. Note: default value should align with cli plmnSecurity. The default values in plmnSecurity for each procedure set by call processing.

Class

Optional.

Type / Range

Integer in the range of 0 to 100

Default

20

Name

irat5gAuth - Authentication when UE is moving from 5G to 4G

Syntax

--irat5gAuth <list-item>

Description

Enable authentication when UE is moving from 5G to 4G

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

6.60 cmm plmnSecurityProfileList

Plmn Security Profile List. Syntax, description, parameters, and roles for cmm plmnSecurityProfileList command.

Syntax**Description**

This command is used to administer a Plmn Security Profile List

Parameters

Name

name - Name of Plmn Security Profile List

Syntax

--name <string>

Description

Name of Plmn Security Profile List

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.61 cmm plmnStatistic

plmn Statistic. Syntax, description, parameters, and roles for cmm plmnStatistic command.

Syntax

Description

Home/Foreign PLMN statistics file.

Parameters

Name

mcc - MCC

Syntax

--mcc <digits>

Description

Mobile country code.

Class

Required on create, not filterable on list.

Type / Range

Digit string of length 3

Name

mnc - MNC

Syntax

--mnc <digits>

Description

Mobile network code.

Class

Required on create, not filterable on list.

Type / Range

Digit string of length 2 to 3

Name

plmnType - Plmn type

Syntax

--plmnType <list-item>

Description

Plmn type. allowed values, ALL ... all PLMNs. includes all PLMNs (home, roaming and specific PLMNs) HOME ... home PLMNs. includes operator's home PLMNs ROAMING ... roaming PLMNs. includes roaming subscribers PLMNs PLMNID ... specific PLMN id. includes only one specific PLMN id (MCC+MNC) INVALID ... invalidate entry.

Class

Required on create, not filterable on list.

List

- ALL - All PLMNs. includes all PLMNs (home, roaming and specific PLMNs).
- HOME - Home PLMNs. includes operator's home PLMNs.
- INVALID - Invalidate entry.
- PLMNID - Specific PLMN id. includes only one specific PLMN id (MCC+MNC).
- ROAMING - Roaming PLMNs. includes roaming subscribers PLMNs.

Name

id - Id

Syntax

<integer>

Description

PlmnStatistic identifier.

Class

Required value.

Type / Range

Integer in the range of 1 to 24

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.62 cmm pmFluentdConfig

PM FluentD Configuration. Syntax, description, parameters, and roles for cmm pmFluentdConfig command.

Syntax**Description**

This command is used in order to configure the various capabilities for streaming PM information towards fluentD.

Parameters**Name**

name - Name

Syntax

--name <string>

Description

Fluentd configuration name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

pmJobName - PM Job Name

Syntax

--pmJobName <string>

Description

This parameter defines which PM job name would be used in order to stream its content towards fluentD.

Class

Required on create.

Type / Range

Digit string of length 1 to 50 with the allowed pattern of [0-9a-zA-Z_\.\-:@]*

Name

streamingMethod - Streaming method

Syntax

--streamingMethod <list-item>

Description

This parameter defines the streaming method to be used from CMM towards fluentD.

Class

Optional.

List

- stderr - The PM info would be sent to stderr
- tcp - tcp socket - The PM info would be sent to the tcp socket
- none - To disable the job (no streaming)

Default

stderr

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

6.63 cmm pmJob

Performance Measurement Job Entry. Syntax, description, parameters, and roles for cmm pmJob command.

Syntax**Description**

This command is used to create a Performance Measurements (PM) job and includes parameters for defining start and end dates, granularity period, and to activate/suspend the job.

Parameters**Name**

jobName - Job name

Syntax

--jobName <string>

Description

Identifies the name of a performance measurement job.

Class

Required key.

Type / Range

Digit string of length 1 to 50 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

active - Job status

Syntax

--active <list-item>

Description

Identifies whether a performance measurement job is active or suspended.

Class

Optional.

List

- Active - Specify performance measurements job status of active.
- Suspended - Specify performance measurements job status of suspended.

Default

Suspended

Name

endDate - End date

Syntax

--endDate <string>

Description

Identifies the end date of a performance measurement job.

Class

Optional.

Type / Range

Digit string of length 8 to 11 with the allowed pattern of (\d{2}:\d{2}:\d{2})|(No_End_Date)

Default

No_End_Date

Name

granuPeriod - Granularity period

Syntax

--granuPeriod <list-item>

Description

Identifies the performance measurement job interval.

Class

Optional.

List

- 5 - Specify performance measurements period of 5 minutes.
- 15 - Specify performance measurements period of 15 minutes.

Default

15

Name

longTerm - Long term

Syntax

--longTerm <list-item>

Description

Identifies if a performance measurement job is defined as long term.

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

startDate - Start date

Syntax

--startDate <string>

Description

Identifies the start date of a performance measurement job.

Class

Optional.

Type / Range

Digit string of length 8 to 13 with the allowed pattern of (\d{2}:\d{2}:\d{2})|(No_Start_Date)

Default

No_Start_Date

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.64 cmm pmJobMeas

Performance Measurement Job Measurements. Syntax, description, parameters, and roles for cmm pmJobMeas command.

Syntax

Description

This command is used to add a measurement group or type to a PM job. Groups are either system defined or custom defined. The group named 'allMts' is reserved and contains all measurements available in the system.

Parameters

Name

jobName - Job name

Syntax

--jobName <string>

Description

Identifies the name of a performance measurement job.

Class

Required key.

Type / Range

Digit string of length 1 to 50 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

mts - Measurement name

Syntax

--mts <string>

Description

Identifies a performance measurement name.

Class

Required key.

Type / Range

Digit string of length 1 to 70 with the allowed pattern of ([a-zA-Z0-9_\. \-:@]*)

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.65 cmm pmJobSched

Performance Measurement Job Schedules. Syntax, description, parameters, and roles for cmm pmJobSched command.

Syntax

Description

This command is used to define the day of week and time of day values for a PM job. Special day (spd) schedules can be configured for specific dates within the start-date and end-date. A job can have multiple schedules for a given day-of-week or spd; this capability enables the operator to collect information for several time periods in a day. A PM job can have an unlimited number of schedules, and schedules can overlap. Each time a schedule is added, the system assigns an integer value identifier to the schedule. Assignment and reassignment of IDs: Schedule IDs are assigned and reassigned (upon deletion) in a contiguous manner. The system automatically reassigns schedule IDs upon deletion of a schedule. If no schedule is specified, a job runs uninterrupted between the start-date and end-date settings.

Parameters

Name

jobName - Job name

Syntax

--jobName <string>

Description

Identifies the name of a performance measurement job.

Class

Required key.

Type / Range

Digit string of length 1 to 50 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

schedId - Schedule ID

Syntax

--schedId <integer>

Description

Identifies the ID of a performance measurement job schedule.

Class

Required key.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

colDate - Collection date

Syntax

--colDate <string>

Description

Identifies the data collection date for a special performance measurement job.

Class

Optional.

Type / Range

Digit string of length 8 to 11 with the allowed pattern of (\d{2}:\d{2}:\d{2})|(No_Spd_Date)

Default

No_Spd_Date

Name

endTime - End time

Syntax

--endTime <string>

Description

Identifies the end time of a scheduled performance measurement job.

Class

Optional.

Type / Range

Digit string of length 5 to 5 with the allowed pattern of (\d{2}:\d{2})

Default

24:00

Name

schedType - Schedule type

Syntax

--schedType <list-item>

Description

Schedule-type.

Class

Optional.

List

- dly - Daily - Performance measurement job runs every day.
- fri - Friday - Performance measurement job runs every friday.

- mon - Monday - Performance measurement job runs every monday.
- sat - Saturday - Performance measurement job runs every saturday.
- spd - Specific date - Performance measurement job runs on a specific date (e.g. new year's day).
- sun - Sunday - Performance measurement job runs every sunday.
- thu - Thursday - Performance measurement job runs every thursday.
- tue - Tuesday - Performance measurement job runs every tuesday.
- wed - Wednesday - Performance measurement job runs every wednesday.

Default

dly

Name

startTime - Start time

Syntax

--startTime <string>

Description

Identifies the start time of a scheduled performance measurement job.

Class

Optional.

Type / Range

Digit string of length 5 to 5 with the allowed pattern of (\d{2}:\d{2})

Default

00:00

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.66 cmm pmMeasGroupName

Performance Measurement Group Name. Syntax, description, parameters, and roles for cmm pmMeasGroupName command.

Syntax

Description

This command is used to administer a Performance Measurement Group Name and includes a parameter for defining the group level (the relative location of a PM group within a hierarchy of performance measurements).

Parameters

Name

groupName - Measurement group name

Syntax

--groupName <string>

Description

Identifies the name of a performance measurement group.

Class

Required key.

Type / Range

Digit string of length 1 to 50 with the allowed pattern of [0-9a-zA-Z_\.\-:@]*

Name

groupLevel - Measurement group level

Syntax

--groupLevel <list-item>

Description

Identifies the relative location of a performance measurement group within a hierarchy of performance measurements. Group Level can be: 1) System_Leaf - a system-defined group that contains only measurement types, 2) System_Non_Leaf - a system-defined group that can contain measurement groups and measurement types, 3) Custom - a user-defined group.

Class

Not modifiable.

List

- Custom - A user-defined group.

- **System_Leaf** - System leaf - A system-defined group that contains only measurement types.
- **System_Non_Leaf** - System non-leaf - A system-defined group that can contain measurement groups and measurement types.

Default

Custom

Name

measGroupDesc - Measurement group description

Syntax

--measGroupDesc <string>

Description

Identifies the description of a performance measurement group.

Class

Optional.

Type / Range

Digit string of length 0 to 94 with the allowed pattern of [0-9a-zA-Z.\-/_:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.67 cmm pmMeasGroups

Performance Measurement Groups. Syntax, description, parameters, and roles for cmm pmMeasGroups command.

Syntax

Description

This command is used to add/remove a measurement group or measurement type to a custom measurement group.

Parameters

Name

groupName - Measurement group name

Syntax

--groupName <string>

Description

Identifies the name of a performance measurement group.

Class

Required key.

Type / Range

Digit string of length 1 to 50 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

measName - Measurement name

Syntax

--measName <string>

Description

Identifies the name of a performance measurement.

Class

Required key.

Type / Range

Digit string of length 1 to 70 with the allowed pattern of ([a-zA-Z0-9_\. \-:@]*)

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.68 cmm pool

Pools. Syntax, description, parameters, and roles for `cmm pool` command.

Syntax

Description

Configure, list and show Pools.

Parameters

Name

ipAddress - ipAddress

Syntax

--ipAddress <string>

Description

IPv4-IPv6 address of the target SGSN over Gn or S16 interface

Class

Required on create, not filterable on list.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

poolnri - Pool nri

Syntax

--poolnri <integer>

Description

Network resource identifier (SGSN).

Class

Required on create, not filterable on list.

Type / Range

Integer in the range of 0 to 127

Name

interface - Interface

Syntax

--interface <list-item>

Description

Interface type.

Class

Optional, not filterable on list.

List

- GN - Specify pool interface type of Gn.
- S16 - Specify pool interface type of S16.

Default

GN

Name

id - Id

Syntax

<integer>

Description

Index of the respective unit record.

Class

Required value.

Type / Range

Integer in the range of 1 to 64

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.69 cmm praAttributes

PRA Attributes. Syntax, description, parameters, and roles for cmm praAttributes command.

Syntax**Description**

Manage the Presence Reporting Area.

Parameters

Name

prald - PRA Identifier

Syntax

--prald <integer>

Description

The identifier of the Presence Reporting Area.

Class

Required key.

Type / Range

Integer in the range of 8,388,608 to 16,777,215

Name

praState - PRA Identifier State

Syntax

--praState <list-item>

Description

Presence Reporting Area identifier state.

Class

Optional.

List

- praActive - PRA Active - Presence Reporting Area identifier is active. MME reports UE presence on the associated CN predefined area normally.
- praDormant - PRA Dormant - Presence Reporting Area identifier is dormant. MME suppresses UE presence reports on the associated CN predefined area.

Default

praActive

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.70 cmm praDclpv4List

PRA Dual Connectivity IPv4 List. Syntax, description, parameters, and roles for cmm praDcIpv4List command.

Syntax

Description

PRA Dual Connectivity IPv4 List. The Dual connectivity Presence Reporting Area functionality updating based on provisioned IPv4 address range utilizes the IPv4 address definitions. The functionality uses CN predefined PRAs. The MME reports via PRA reporting that a UE is inside the area when a 5G gNB having an IPv4 address belonging to the provisioned range is used for this UE with dual connectivity. The UE is outside the area when the corresponding 5G gNB is not used for the UE. A single list (that is mapped to a single PRA identifier) can contain 10 address ranges. MME supports max or altogether 128 segment lists (or PRA identifiers).

Parameters

Name

name - PRA Dual Connectivity IPv4 Mapping List Name

Syntax

--name <string>

Description

The name of the PRA Dual Connectivity IPv4 Mapping List.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.71 cmm praDclpv4Range

PRA Dual Connectivity IPv4 Address Range. Syntax, description, parameters, and roles for cmm praDcIpv4Range command.

Syntax

Description

Manage a 5G gNB IPv4 address range list. The IPv4 address ranges belong to 5G gNBs that are used for dual connectivity and dual connectivity based reporting of 5G usage. The MME reports via PRA reporting that a UE is inside the area when a 5G gNB having an IPv4 address belonging to the provisioned range is used for this UE with dual connectivity. The UE is outside the area when the corresponding 5G gNB is not used for the UE. An IPv4 address range list can contain 10 ranges. The list maps to a single PRA identifier. MME supports max 128 PRA identifiers and thus max 128 segment lists.

Parameters

Name

praDclpv4ListName - PRA Dual Connectivity IPv4 List Name

Syntax

--praDcIpv4ListName <string>

Description

The name of the PRA Dual Connectivity IPv4 List.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

ipv4Base - IPv4 Subnet Base

Syntax

--ipv4Base <string>

Description

Defines the IPv4 subnet used to differentiate gNBs (5G) from eNBs (4G) in dual connectivity PRA reporting.

Class

Required key.

Type / Range

Digit string of length 7 to 15 with the allowed pattern of [0-9\.*]

Name

ipv4RangeMask - IPv4 Range Mask

Syntax

--ipv4RangeMask <string>

Description

Defines the IPv4 range mask used to differentiate gNBs (5G) from eNBs (4G) in dual connectivity PRA reporting. This parameter defines the range of bits used for the low/high limits.

Class

Required key.

Type / Range

Digit string of length 7 to 15 with the allowed pattern of [0-9\.*]

Name

ipv4RangeHighLimit - IPv4 Range High Limit

Syntax

--ipv4RangeHighLimit <string>

Description

Defines the high limit of the ipv4 address range to differentiate gNBs (5G) from eNBs (4G) in dual connectivity PRA reporting. The high limit value is included in the range.

Class

Required on create.

Type / Range

Digit string of length 7 to 15 with the allowed pattern of [0-9\.*]

Name

ipv4RangeLowLimit - IPv4 Range Low Limit

Syntax

--ipv4RangeLowLimit <string>

Description

Defines the low limit of the ipv4 address range to differentiate gNBs (5G) from eNBs (4G) in dual connectivity PRA reporting. The low limit is included in the range.

Class

Required on create.

Type / Range

Digit string of length 7 to 15 with the allowed pattern of [0-9\.*]

Name

ipv4SubnetMask - IPv4 Subnet Mask

Syntax

--ipv4SubnetMask <string>

Description

Defines the IPv4 mask used to differentiate gNBs (5G) from eNBs (4G) in dual connectivity PRA reporting. This parameter defines the base and the mask.

Class

Required on create.

Type / Range

Digit string of length 7 to 15 with the allowed pattern of [0-9\.*]

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.72 cmm praDcIpv6Prefix

PRA Dual Connectivity IPv6 Address Prefix. Syntax, description, parameters, and roles for cmm praDcIpv6Prefix command.

Syntax**Description**

Manage a 5G gNB IPv6 address prefix list. The defined IPv6 addresses belong to 5G gNBs that are used for dual connectivity and dual connectivity based reporting of 5G usage. The MME reports via PRA reporting that a UE is inside the area when a 5G gNB having an IPv6 address covered by the provisioned prefix is used for this UE with dual connectivity. The UE is outside the area when a 5G gNB is not used for the UE. An IPv6

address prefix list can contain up to 25 prefixes. MME supports max 128 PRA identifiers and thus max 128 prefix lists.

Parameters

Name

praDcIpv6PrefixListName - PRA DC IPv6 Prefix List Name

Syntax

--praDcIpv6PrefixListName <string>

Description

The name of the IPv6 address prefix list.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

baseIpv6Address - IPv6 Prefix

Syntax

--baseIpv6Address <string>

Description

The base IPv6 address of the IPv6 prefix identifying the 5G gNBs.

Class

Required key.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of '[0-9a-f:]*'

Name

prefixLength - Prefix Length

Syntax

--prefixLength <integer>

Description

The length of the IPv6 prefix.

Class

Required on create, not modifiable.

Type / Range

Integer in the range of 0 to 128

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.73 cmm praDclpv6PrefixList

PRA Dual Connectivity IPv6 Address Prefix List. Syntax, description, parameters, and roles for cmm praDcIpv6PrefixList command.

Syntax

Description

PRA Dual Connectivity IPv6 Address Prefix List. The Dual Connectivity Presence Reporting Area functionality updating based on provisioned IPv6 Address Prefix List defines the gNB addresses with IPv6 prefixes. The functionality uses CN predefined PRAs. The MME reports via PRA reporting that a UE is inside the area when a 5G gNB having an IPv6 address covered by the provisioned prefix is used for this UE with dual connectivity. The UE is outside the area when a 5G gNB is not used for the UE. A single list (that is mapped to a single PRA identifier) can contain 25 IPv6 address prefixes. MME supports max or altogether 128 IPv6 address prefix lists (or PRA identifiers).

Parameters

Name

name - Name

Syntax

--name <string>

Description

The name of the IPv6 address prefix list.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.74 cmm praDcSegment

PRA Dual connectivity segment. Syntax, description, parameters, and roles for cmm praDcSegment command.

Syntax

Description

Manage a 5G gNB IPv6 address segments. The IPv6 address segments belong to 5G gNBs that are used for dual connectivity and segment based reporting of 5G usage. The MME reports via PRA reporting that a UE is inside the area when a 5G gNB having a provisioned IPv6 address segment is used for this UE with dual connectivity. The UE is outside the area when the corresponding 5G gNB is not used for the UE. An IPv6 segment list can contain 10 segments. The segment list maps to a single PRA identifier. MME supports max 128 PRA identifiers and thus max 128 segment lists.

Parameters

Name

praDcSegmentListName - PRA Dual Connectivity Segment List Name

Syntax

--praDcSegmentListName <string>

Description

The name of the PRA Dual Connectivity Segment List.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

ipv6SegmentContent - IPv6 address segment content

Syntax

--ipv6SegmentContent <string>

Description

The full content of the special IPv6 address segment.

Class

Required key.

Type / Range

Digit string of length 1 to 4 with the allowed pattern of [0-9a-fA-F]*

Name

ipv6SegmentNumber - IPv6 Address Segment Number

Syntax

--ipv6SegmentNumber <integer>

Description

This field specifies which segment number is specified in the Segment Content field.

Class

Required key.

Type / Range

Integer in the range of 1 to 8

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.75 cmm praDcSegmentList

PRA Dual Connectivity Segment List. Syntax, description, parameters, and roles for cmm praDcSegmentList command.

Syntax

Description

PRA Dual Connectivity Segment List. The Dual connectivity Presence Reporting Area functionality updating based on provisioned IPv6 address segment utilizes the segment lists. The functionality uses CN predefined PRAs. The MME reports via PRA reporting that a UE is inside the area when a 5G gNB having a provisioned IPv6 address segment is used for this UE with dual connectivity. The UE is outside the area when the corresponding 5G gNB is not used for the UE. A single list (that is mapped to a single PRA identifier) can contain 10 segments. MME supports max/altogether 128 segment lists (or PRA identifiers).

Parameters

Name

name - PRA Dual Connectivity Segment Mapping List Name

Syntax

--name <string>

Description

The name of the PRA Dual Connectivity Segment Mapping List.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.76 cmm praDcSegmentMapping

PRA Dual connectivity segment mapping. Syntax, description, parameters, and roles for cmm praDcSegmentMapping command.

Syntax

Description

Map a CN predefined PRA id to the specific 5G gNB IP address segment/range or a segment/range list. The Dual connectivity Presence Reporting Area functionality updating based on provisioned IP address segment/range utilizes this mapping. The functionality uses CN predefined PRAs. The MME reports via PRA reporting that a UE is inside the area when a 5G gNB having a provisioned IPv6 address segment or belonging to the provisioned IPv4 address range is used for this UE with dual connectivity. The UE is outside the area when the corresponding 5G gNB is not used for the UE. MME supports max 128 PRA identifiers and max 128 segment/range lists. Each PRA identifier can map to a single segment list and/or a single range list. Each segment/range list may contain max 10 segments.

Parameters

Name

dcSegmentPraId - Dual Connectivity Segment Presence Reporting Area Identifier

Syntax

--dcSegmentPraId <integer>

Description

An integer that identifies the Core Network predefined Presence Reporting Area that is used for the Dual Connectivity segment based reporting.

Restrictions

In the current product release only a single PRA id may be provisioned.

Class

Required key.

Type / Range

Integer in the range of 8,388,608 to 16,777,215

Name

ipv6SegmentContent - IPv6 address segment content

Syntax

--ipv6SegmentContent <string>

Description

The full content of the special IPv6 address segment. Note: if singleSegmentMapping is set to false, ipv6SegmentContent will show default value 0.

Class

Optional.

Type / Range

Digit string of length 1 to 4 with the allowed pattern of [0-9a-fA-F]*

Default

0

Name

ipv6SegmentNumber - IPv6 Address Segment Number

Syntax

--ipv6SegmentNumber <integer>

Description

This field specifies which segment number is specified in the Segment Content field. Note: if singleSegmentMapping is set to false, ipv6SegmentNumber will show default value 6.

Class

Optional.

Type / Range

Integer in the range of 1 to 8

Default

6

Name

praDclpv4ListName - PRA Dual Connectivity IPv4 List Name

Syntax

--praDcIpv4ListName <string>

Description

The name of the PRA Dual Connectivity IPv4 List.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

praDclpv6PrefixListName - PRA DC IPv6 Prefix List Name

Syntax

--praDcIpv6PrefixListName <string>

Description

The name of the presence reporting area dual connectivity IPv6 address prefix list to define the gNBs.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

praDcSegmentListName - PRA Dual Connectivity Segment List Name

Syntax

--praDcSegmentListName <string>

Description

The name of the PRA Dual Connectivity Segment List including the IPv6 segments associated with the Presence Reporting Area identifier.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

""

Name

singleSegmentMapping - Single Segment Mapping

Syntax

--singleSegmentMapping <list-item>

Description

Controls whether the IPv6 segment(s) is(are) defined as a single segment (value true) or as a list of IPv6 segments (value false).

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.77 cmm praEcgi

Presence Reporting Area for ECGI. Syntax, description, parameters, and roles for cmm praEcgi command.

Syntax

Description

Administer Presence Reporting Area for E-UTRAN Cell Global Identifier area type. MME supports up to 10500 ECGIs per each CN predefined PRA, with a limit of 50000 of the combined number of TACs, eNBs, and ECGIs.

Parameters

Name

prald - Presence reporting area identifier

Syntax

--praId <integer>

Description

An integer that identifies the core network predefined presence reporting area.

Class

Required key.

Type / Range

Integer in the range of 8,388,608 to 16,777,215

Name

plmnName - Mobile network name

Syntax

--plmnName <string>

Description

A name that identifies mobile network.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

eci - ECI of the ECGI in the PRA

Syntax

--eci <integer>

Description

E-UTRAN cell identifier for the ECGI area type of the presence reporting area.

Class

Required key.

Type / Range

Integer in the range of 0 to 268,435,455

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.78 cmm praEnb

Presence Reporting Area for ENB. Syntax, description, parameters, and roles for cmm praEnb command.

Syntax**Description**

Administer Presence Reporting Area for the ENB area type. MME supports up to 3500 home eNBs and 3500 macro eNBs per each CN predefined PRA, with the limit of 50000 of the combined number of TACs, eNBs, and ECGIs.

Parameters**Name**

prald - Presence reporting area identifier

Syntax

--praId <integer>

Description

An integer that identifies the core network predefined presence reporting area.

Class

Required key.

Type / Range

Integer in the range of 8,388,608 to 16,777,215

Name

plmnName - Mobile network name

Syntax

--plmnName <string>

Description

A name that identifies mobile network.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

enbType - Presence reporting area eNB type

Syntax

--enbType <list-item>

Description

Indicates the eNB type for the presence reporting area.

Class

Required key.

List

- homeEnb - Home eNodeB - Specify eNodeB type of home eNodeB.
- macroEnb - Macro eNodeB - Specify eNodeB type of macro eNodeB.

Name

enbId - eNB identifier of the eNB area type

Syntax

--enbId <integer>

Description

eNB identifier of the eNB area type for the presence reporting area .

Class

Required key.

Type / Range

Integer in the range of 0 to 268,435,455

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.79 cmm praSet

Presence Reporting Area Set. Syntax, description, parameters, and roles for cmm praSet command.

Syntax**Description**

Administer Presence Reporting Area set. PRA set can only consist of PRA ids related to either standard based PRA functionality, or related to segment based PRA functionality. Both PRA ids related to segment based functionality and PRA ids related to standard based are not allowed on the same PRA set. A Core Network predefined PRA id must be unique in the system. A single PRA id may be contained in multiple PRA sets. PRA set id cannot be contained in the same or a different PRA set (i.e. nested PRA sets are not allowed). A PRA set can contain max 9 PRA ids. Only a single PRA set can be requested to be reported at a time for a single UE. Another PRA set id reporting request for the same UE and PDN (or the same PRA set id reporting request for a different or same PDN of the same UE) will be ignored in the MME.

Parameters**Name**

praSetId - Presence Reporting Area Set Identifier

Syntax

--praSetId <integer>

Description

An integer that identifies the core network predefined presence reporting area set.

Class

Required key.

Type / Range

Integer in the range of 8,388,608 to 16,777,215

Name

prald - Presence Reporting Area Identifier

Syntax

--praId <integer>

Description

An integer that identifies the core network predefined presence reporting area.

Class

Required key.

Type / Range

Integer in the range of 8,388,608 to 16,777,215

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.80 cmm praTai

Presence Reporting Area for TAI. Syntax, description, parameters, and roles for `cmm praTai` command.

Syntax**Description**

Administer Presence Reporting Area for Tracking Area Identity area type. MME supports up to 1500 TAIs per each CN predefined PRA, with the limit of 50000 of the combined number of TACs, eNBs, and ECGIs.

Parameters

Name

prald - Presence reporting area identifier

Syntax

--praId <integer>

Description

An integer that identifies the core network predefined presence reporting area.

Class

Required key.

Type / Range

Integer in the range of 8,388,608 to 16,777,215

Name

plmnName - Mobile network name

Syntax

--plmnName <string>

Description

A name that identifies mobile network.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

tac - TAC in the TAI of the PRA

Syntax

--tac <integer>

Description

Tracking area code in the tracking area identity area type of the presence reporting area.

Class

Required key.

Type / Range

Integer in the range of 1 to 65,535

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.81 cmm preferredNfTypeList

Preferred NF Type List. Syntax, description, parameters, and roles for cmm preferredNfTypeList command.

Syntax

Description

List of Preferred NF Types. Each NF Type can be added only once

Parameters

Name

name - Preferred NF Type List Name

Syntax`--name <string>`**Description**

Preferred NF Type List Name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.82 cmm preferredNfTypes

Preferred NF Type. Syntax, description, parameters, and roles for cmm preferredNfTypes command.

Syntax

Description

Adds a preferred NFType to a preferred NFType list

Parameters

Name

preferredNfTypeListName - Preferred NF Type List Name

Syntax

--preferredNfTypeListName <string>

Description

Preferred NF Type List Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

preferredNfType - Preferred NF Type

Syntax

--preferredNfType <list-item>

Description

Preferred NF Type

Class

Required key.

List

- hAusf - Home Network AUSF
- hPcf - Home Network PCF
- hSmf - Home Network SMF
- hUdm - Home Network UDM

- vPcf - Visiting Network PCF
- vSmf - Visiting Network SMF

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.83 cmm prometheusGW

Prometheus pushGateway. Syntax, description, parameters, and roles for cmm prometheusGW command.

Syntax

Description

Provision the address and characteristics of the Prometheus pushGateway

Parameters

Name

prometheusIndex - Prometheus GW Index

Syntax

--prometheusIndex <integer>

Description

Index for the Prometheus GW

Class

Required key.

Type / Range

Integer in the range of 1 to 1

Name

primaryIp - Prometheus GW Primary IP

Syntax

--primaryIp <string>

Description

Primary IP address of the Prometheus GW

Class

Required on create.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

primaryPort - Primary Port

Syntax

--primaryPort <integer>

Description

Primary port address of Prometheus GW.

Class

Required on create.

Type / Range

Integer in the range of 1,024 to 65,535

Name

prometheusConnName - Connection Name for Prometheus

Syntax

--prometheusConnName <string>

Description

Name used to identify the AMF on Prometheus connection

Class

Required on create.

Type / Range

Digit string of length 0 or 1 to 64 with the allowed pattern of [0-9a-zA-Z\._:-]*

Name

prometheusConnWord - Prometheus Connection word

Syntax

--prometheusConnWord <string>

Description

Connection word for Prometheus

Class

Required on create.

Type / Range

Digit string of length 0 or 1 to 64 with the allowed pattern of [0-9a-zA-Z\._:\]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.84 cmm prometheusScrape

Prometheus Server Interface. Syntax, description, parameters, and roles for **cmm prometheusScrape** command.

Syntax**Description**

This command is used to provision the configuration needed so that Prometheus Server can directly scrape the relevant metrics from NECC (prometheusClient).

Parameters**Name**

dscpCode - DSCP Code

Syntax

--dscpCode <list-item>

Description

Parameter to set DSCP code to be used for an IP packet

Class

Optional.

List

- AF11 - Assured forwarding 11.

- AF12 - Assured forwarding 12.
- AF13 - Assured forwarding 13.
- AF21 - Assured forwarding 21.
- AF22 - Assured forwarding 22.
- AF23 - Assured forwarding 23.
- AF31 - Assured forwarding 31.
- AF32 - Assured forwarding 32.
- AF33 - Assured forwarding 33.
- AF41 - Assured forwarding 41.
- AF42 - Assured forwarding 42.
- AF43 - Assured forwarding 43.
- BE - BE.
- CS1 - Class selector 1.
- CS2 - Class selector 2.
- CS3 - Class selector 3.
- CS4 - Class selector 4.
- CS5 - Class selector 5.
- CS6 - Class selector 6.
- CS7 - Class selector 7.
- EF - Expedited forwarding.
- voiceAdmit - Voice Admit

Default

AF41

Name

httpProfileName - HTTP Profile Name

Syntax`--httpProfileName <string>`**Description**

Assign HTTP profile for the end point.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

default

Name

oamTlsProfileName - OAM TLS Profile

Syntax

--oamTlsProfileName <string>

Description

This parameter contains the profile for TLS. It contains the certificate names needed for establishing secure communication.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

tcpProfileName - TCP Profile Name

Syntax

--tcpProfileName <string>

Description

Configure TCP profile to the endpoint.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

default

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

6.85 cmm psmApn

PSM APN. Syntax, description, parameters, and roles for cmm psmApn command.

Syntax

Description

Administer PSM for APN.

Parameters

Name

psmApnListName - PSM APN list name

Syntax

--psmApnListName <string>

Description

PSM APN list name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

apnNi - APN NI

Syntax

--apnNi <string>

Description

APN NI.

Class

Required key.

Type / Range

Digit string of length 1 to 62 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

nbPsmT3412Timer - NB-S1 mode power savings mode T3412 timer

Syntax

--nbPsmT3412Timer <integer>

Description

Configure T3412 timer to be used for a NB-S1 UE that supports PSM.

Class

Optional.

Type / Range

Integer in the range of 1 to 31

Default

31

Name

nbPsmT3412TimerUnit - NB-S1 mode power savings mode T3412 timer unit

Syntax

--nbPsmT3412TimerUnit <list-item>

Description

Power savings mode T3412 timer unit.

Class

Optional.

List

- 10Hours - Specify unit of T3412 timer of 10 hours.
- 10Minutes - Specify unit of T3412 timer of 10 minutes.
- 2Seconds - Specify unit of T3412 timer of 2 seconds.
- 30Seconds - Specify unit of T3412 timer of 30 seconds.
- Hour - Specify unit of T3412 timer of hours.
- Minute - Specify unit of T3412 timer of minutes.
- 320Hours - Specify unit of T3412 timer of 320 hours.

Default

10Minutes

Name

nbT3324Timer - NB-S1 mode T3324 timer

Syntax

--nbT3324Timer <integer>

Description

Active timer for NB-S1 UE in seconds.

Class

Optional.

Type / Range

Integer in the range of 0 to 11,160

Default

300

Name

psmEnabled - PSM enabled

Syntax

--psmEnabled <list-item>

Description

PSM enabled.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

source - source

Syntax

--source <list-item>

Description

source

Class

Optional.

List

- UE - description: Specifies eDRX/PSM source as UE.
- local - description: local means Specifies eDRX/PSM source as APN .

Default

local

Name

wbPsmT3412Timer - WB-S1 mode power savings mode T3412 timer

Syntax

--wbPsmT3412Timer <integer>

Description

Configure T3412 timer to be used for a WB-S1 UE that supports PSM.

Class

Optional.

Type / Range

Integer in the range of 1 to 31

Default

31

Name

wbPsmT3412TimerUnit - WB-S1 mode power savings mode T3412 timer unit

Syntax

--wbPsmT3412TimerUnit <list-item>

Description

Power savings mode T3412 timer unit.

Class

Optional.

List

- 10Hours - Specify unit of T3412 timer of 10 hours.
- 10Minutes - Specify unit of T3412 timer of 10 minutes.
- 2Seconds - Specify unit of T3412 timer of 2 seconds.
- 30Seconds - Specify unit of T3412 timer of 30 seconds.
- Hour - Specify unit of T3412 timer of hours.
- Minute - Specify unit of T3412 timer of minutes.
- 320Hours - Specify unit of T3412 timer of 320 hours.

Default

10Minutes

Name

wbT3324Timer - WB-S1 mode T3324 timer

Syntax

--wbT3324Timer <integer>

Description

Active timer for WB-S1 UE in seconds.

Class

Optional.

Type / Range

Integer in the range of 0 to 11,160

Default

300

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.86 cmm psmApnList

PSM APN List. Syntax, description, parameters, and roles for cmm psmApnList command.

Syntax**Description**

Administer List of APNs for PSM.

Parameters**Name**

name - PSM APN list name

Syntax

--name <string>

Description

PSM APN list name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

description - Description

Syntax

--description <string>

Description

Profile description.

Class

Optional.

Type / Range

Digit string of length 0 to 160 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.87 cmm qosConf

QoS configuration. Syntax, description, parameters, and roles for cmm qosConf command.

Syntax**Description**

Configure and show QoS parameters.

Parameters**Name**

arpRoam - Allocation/retention priority for roaming

Syntax`--arpRoam <integer>`**Description**

This parameter defines the allocation/retention priority which can be used by a roaming subscriber at most. possible values range from 1 to 3.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 3

Default

3

Name

downMaxRoam - Maximum bit rate for downlink for roaming

Syntax`--downMaxRoam <integer>`**Description**

This parameter defines the maximum bit rate for downlink traffic for a roaming subscriber. the values range from 0 to 63 in steps of 1, from 64 to 568 in steps of 8, and from 576 to 8640 in steps of 64. optionally, there are also the range of values from 8700 to 16000 in steps of 100, from 17000 to 128000 in steps of 1000 and from 130000 to 168000 in steps of 2000.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 168,000

Default

50

Name

dscpBackground - Differentiated services codepoints background

Syntax`--dscpBackground <list-item>`**Description**

Differentiated services codepoints background, 000000 ... BEST EFFORT 001010 ... AF CLASS 1 LOW DROP PRECEDENCE 001100 ... AF CLASS 1 MEDIUM DROP PRECEDENCE 001110 ... AF CLASS 1 HIGH DROP PRECEDENCE 010010 ... AF CLASS 2 LOW DROP PRECEDENCE 010100 ... AF CLASS 2 MEDIUM DROP PRECEDENCE 010110 ... AF CLASS 2 HIGH DROP PRECEDENCE 011010 ... AF CLASS 3 LOW DROP PRECEDENCE 011100 ... AF CLASS 3 MEDIUM DROP

PRECEDENCE 011110 ... AF CLASS 3 HIGH DROP PRECEDENCE 100010 ... AF CLASS 4 LOW
DROP PRECEDENCE 100100 ... AF CLASS 4 MEDIUM DROP PRECEDENCE 100110 ... AF CLASS
4 HIGH DROP PRECEDENCE 101110 ... EXPEDITED FORWARDING.

Class

Output only.

List

- 000000 - BEST EFFORT.
- 001010 - AF CLASS 1 LOW DROP PRECEDENCE.
- 001100 - AF CLASS 1 MEDIUM DROP PRECEDENCE.
- 001110 - AF CLASS 1 HIGH DROP PRECEDENCE.
- 010010 - AF CLASS 2 LOW DROP PRECEDENCE.
- 010100 - AF CLASS 2 MEDIUM DROP PRECEDENCE.
- 010110 - AF CLASS 2 HIGH DROP PRECEDENCE.
- 011010 - AF CLASS 3 LOW DROP PRECEDENCE.
- 011100 - AF CLASS 3 MEDIUM DROP PRECEDENCE.
- 011110 - AF CLASS 3 HIGH DROP PRECEDENCE.
- 100010 - AF CLASS 4 LOW DROP PRECEDENCE.
- 100100 - AF CLASS 4 MEDIUM DROP PRECEDENCE.
- 100110 - AF CLASS 4 HIGH DROP PRECEDENCE.
- 101110 - EXPEDITED FORWARDING.

Default

000000

Name

dscpConversional - Differentiated services codepoints conversional

Syntax

--dscpConversional <list-item>

Description

Differentiated services codepoints conversional, 000000 ... BEST EFFORT 001010 ... AF
CLASS 1 LOW DROP PRECEDENCE 001100 ... AF CLASS 1 MEDIUM DROP PRECEDENCE
001110 ... AF CLASS 1 HIGH DROP PRECEDENCE 010010 ... AF CLASS 2 LOW DROP
PRECEDENCE 010100 ... AF CLASS 2 MEDIUM DROP PRECEDENCE 010110 ... AF CLASS 2 HIGH
DROP PRECEDENCE 011010 ... AF CLASS 3 LOW DROP PRECEDENCE 011100 ... AF CLASS
3 MEDIUM DROP PRECEDENCE 011110 ... AF CLASS 3 HIGH DROP PRECEDENCE 100010 ...
AF CLASS 4 LOW DROP PRECEDENCE 100100 ... AF CLASS 4 MEDIUM DROP PRECEDENCE
100110 ... AF CLASS 4 HIGH DROP PRECEDENCE 101110 ... EXPEDITED FORWARDING.

Class

Output only.

List

- 000000 - BEST EFFORT.
- 001010 - AF CLASS 1 LOW DROP PRECEDENCE.
- 001100 - AF CLASS 1 MEDIUM DROP PRECEDENCE.
- 001110 - AF CLASS 1 HIGH DROP PRECEDENCE.
- 010010 - AF CLASS 2 LOW DROP PRECEDENCE.
- 010100 - AF CLASS 2 MEDIUM DROP PRECEDENCE.
- 010110 - AF CLASS 2 HIGH DROP PRECEDENCE.
- 011010 - AF CLASS 3 LOW DROP PRECEDENCE.
- 011100 - AF CLASS 3 MEDIUM DROP PRECEDENCE.
- 011110 - AF CLASS 3 HIGH DROP PRECEDENCE.
- 100010 - AF CLASS 4 LOW DROP PRECEDENCE.
- 100100 - AF CLASS 4 MEDIUM DROP PRECEDENCE.
- 100110 - AF CLASS 4 HIGH DROP PRECEDENCE.
- 101110 - EXPEDITED FORWARDING.

Default

101110

Name

dscpGbSignalling - Differentiated services codepoints for gb signalling

Syntax**Description**

Differentiated services codepoints for gb signalling, 000000 ... BEST EFFORT 001010 ... AF CLASS 1 LOW DROP PRECEDENCE 001100 ... AF CLASS 1 MEDIUM DROP PRECEDENCE 001110 ... AF CLASS 1 HIGH DROP PRECEDENCE 010010 ... AF CLASS 2 LOW DROP PRECEDENCE 010100 ... AF CLASS 2 MEDIUM DROP PRECEDENCE 010110 ... AF CLASS 2 HIGH DROP PRECEDENCE 011010 ... AF CLASS 3 LOW DROP PRECEDENCE 011100 ... AF CLASS 3 MEDIUM DROP PRECEDENCE 011110 ... AF CLASS 3 HIGH DROP PRECEDENCE 100010 ... AF CLASS 4 LOW DROP PRECEDENCE 100100 ... AF CLASS 4 MEDIUM DROP PRECEDENCE 100110 ... AF CLASS 4 HIGH DROP PRECEDENCE 101110 ... EXPEDITED FORWARDING.

Class

Output only.

Type / Range

Digit string of length 1 to 255

Default

101110

Name

dscpLevel1 - Differentiated services codepoints interactive priority level 1

Syntax`--dscpLevel1 <list-item>`**Description**

Differentiated services codepoints interactive priority level 1, 000000 ... BEST EFFORT 001010 ... AF CLASS 1 LOW DROP PRECEDENCE 001100 ... AF CLASS 1 MEDIUM DROP PRECEDENCE 001110 ... AF CLASS 1 HIGH DROP PRECEDENCE 010010 ... AF CLASS 2 LOW DROP PRECEDENCE 010100 ... AF CLASS 2 MEDIUM DROP PRECEDENCE 010110 ... AF CLASS 2 HIGH DROP PRECEDENCE 011010 ... AF CLASS 3 LOW DROP PRECEDENCE 011100 ... AF CLASS 3 MEDIUM DROP PRECEDENCE 011110 ... AF CLASS 3 HIGH DROP PRECEDENCE 100010 ... AF CLASS 4 LOW DROP PRECEDENCE 100100 ... AF CLASS 4 MEDIUM DROP PRECEDENCE 100110 ... AF CLASS 4 HIGH DROP PRECEDENCE 101110 ... EXPEDITED FORWARDING.

Class

Output only.

List

- 000000 - BEST EFFORT.
- 001010 - AF CLASS 1 LOW DROP PRECEDENCE.
- 001100 - AF CLASS 1 MEDIUM DROP PRECEDENCE.
- 001110 - AF CLASS 1 HIGH DROP PRECEDENCE.
- 010010 - AF CLASS 2 LOW DROP PRECEDENCE.
- 010100 - AF CLASS 2 MEDIUM DROP PRECEDENCE.
- 010110 - AF CLASS 2 HIGH DROP PRECEDENCE.
- 011010 - AF CLASS 3 LOW DROP PRECEDENCE.
- 011100 - AF CLASS 3 MEDIUM DROP PRECEDENCE.
- 011110 - AF CLASS 3 HIGH DROP PRECEDENCE.
- 100010 - AF CLASS 4 LOW DROP PRECEDENCE.
- 100100 - AF CLASS 4 MEDIUM DROP PRECEDENCE.
- 100110 - AF CLASS 4 HIGH DROP PRECEDENCE.
- 101110 - EXPEDITED FORWARDING.

Default

011010

Name

dscpLevel2 - Differentiated services codepoints interactive priority level 2

Syntax

--dscpLevel2 <list-item>

Description

Differentiated services codepoints interactive priority level 2, 000000 ... BEST EFFORT
001010 ... AF CLASS 1 LOW DROP PRECEDENCE 001100 ... AF CLASS 1 MEDIUM DROP
PRECEDENCE 001110 ... AF CLASS 1 HIGH DROP PRECEDENCE 010010 ... AF CLASS 2 LOW
DROP PRECEDENCE 010100 ... AF CLASS 2 MEDIUM DROP PRECEDENCE 010110 ... AF CLASS
2 HIGH DROP PRECEDENCE 011010 ... AF CLASS 3 LOW DROP PRECEDENCE 011100 ...
AF CLASS 3 MEDIUM DROP PRECEDENCE 011110 ... AF CLASS 3 HIGH DROP PRECEDENCE
100010 ... AF CLASS 4 LOW DROP PRECEDENCE 100100 ... AF CLASS 4 MEDIUM DROP
PRECEDENCE 100110 ... AF CLASS 4 HIGH DROP PRECEDENCE 101110 ... EXPEDITED
FORWARDING.

Class

Output only.

List

- 000000 - BEST EFFORT.
- 001010 - AF CLASS 1 LOW DROP PRECEDENCE.
- 001100 - AF CLASS 1 MEDIUM DROP PRECEDENCE.
- 001110 - AF CLASS 1 HIGH DROP PRECEDENCE.
- 010010 - AF CLASS 2 LOW DROP PRECEDENCE.
- 010100 - AF CLASS 2 MEDIUM DROP PRECEDENCE.
- 010110 - AF CLASS 2 HIGH DROP PRECEDENCE.
- 011010 - AF CLASS 3 LOW DROP PRECEDENCE.
- 011100 - AF CLASS 3 MEDIUM DROP PRECEDENCE.
- 011110 - AF CLASS 3 HIGH DROP PRECEDENCE.
- 100010 - AF CLASS 4 LOW DROP PRECEDENCE.
- 100100 - AF CLASS 4 MEDIUM DROP PRECEDENCE.
- 100110 - AF CLASS 4 HIGH DROP PRECEDENCE.
- 101110 - EXPEDITED FORWARDING.

Default

010010

Name

dscpLevel3 - Differentiated services codepoints interactive priority level 3

Syntax

--dscpLevel3 <list-item>

Description

Differentiated services codepoints interactive priority level 3, 000000 ... BEST EFFORT 001010 ... AF CLASS 1 LOW DROP PRECEDENCE 001100 ... AF CLASS 1 MEDIUM DROP PRECEDENCE 001110 ... AF CLASS 1 HIGH DROP PRECEDENCE 010010 ... AF CLASS 2 LOW DROP PRECEDENCE 010100 ... AF CLASS 2 MEDIUM DROP PRECEDENCE 010110 ... AF CLASS 2 HIGH DROP PRECEDENCE 011010 ... AF CLASS 3 LOW DROP PRECEDENCE 011100 ... AF CLASS 3 MEDIUM DROP PRECEDENCE 011110 ... AF CLASS 3 HIGH DROP PRECEDENCE 100010 ... AF CLASS 4 LOW DROP PRECEDENCE 100100 ... AF CLASS 4 MEDIUM DROP PRECEDENCE 100110 ... AF CLASS 4 HIGH DROP PRECEDENCE 101110 ... EXPEDITED FORWARDING.

Class

Output only.

List

- 000000 - BEST EFFORT.
- 001010 - AF CLASS 1 LOW DROP PRECEDENCE.
- 001100 - AF CLASS 1 MEDIUM DROP PRECEDENCE.
- 001110 - AF CLASS 1 HIGH DROP PRECEDENCE.
- 010010 - AF CLASS 2 LOW DROP PRECEDENCE.
- 010100 - AF CLASS 2 MEDIUM DROP PRECEDENCE.
- 010110 - AF CLASS 2 HIGH DROP PRECEDENCE.
- 011010 - AF CLASS 3 LOW DROP PRECEDENCE.
- 011100 - AF CLASS 3 MEDIUM DROP PRECEDENCE.
- 011110 - AF CLASS 3 HIGH DROP PRECEDENCE.
- 100010 - AF CLASS 4 LOW DROP PRECEDENCE.
- 100100 - AF CLASS 4 MEDIUM DROP PRECEDENCE.
- 100110 - AF CLASS 4 HIGH DROP PRECEDENCE.
- 101110 - EXPEDITED FORWARDING.

Default

001010

Name

dscpStreaming - Differentiated services codepoints streaming

Syntax

--dscpStreaming <list-item>

Description

Differentiated services codepoints streaming, 000000 ... BEST EFFORT 001010 ... AF CLASS 1 LOW DROP PRECEDENCE 001100 ... AF CLASS 1 MEDIUM DROP PRECEDENCE 001110 ... AF CLASS 1 HIGH DROP PRECEDENCE 010010 ... AF CLASS 2 LOW DROP PRECEDENCE 010100 ... AF CLASS 2 MEDIUM DROP PRECEDENCE 010110 ... AF CLASS 2 HIGH DROP PRECEDENCE 011010 ... AF CLASS 3 LOW DROP PRECEDENCE 011100 ... AF CLASS 3 MEDIUM DROP PRECEDENCE 011110 ... AF CLASS 3 HIGH DROP PRECEDENCE 100010 ... AF CLASS 4 LOW DROP PRECEDENCE 100100 ... AF CLASS 4 MEDIUM DROP PRECEDENCE 100110 ... AF CLASS 4 HIGH DROP PRECEDENCE 101110 ... EXPEDITED FORWARDING.

Class

Output only.

List

- 000000 - BEST EFFORT.
- 001010 - AF CLASS 1 LOW DROP PRECEDENCE.
- 001100 - AF CLASS 1 MEDIUM DROP PRECEDENCE.
- 001110 - AF CLASS 1 HIGH DROP PRECEDENCE.
- 010010 - AF CLASS 2 LOW DROP PRECEDENCE.
- 010100 - AF CLASS 2 MEDIUM DROP PRECEDENCE.
- 010110 - AF CLASS 2 HIGH DROP PRECEDENCE.
- 011010 - AF CLASS 3 LOW DROP PRECEDENCE.
- 011100 - AF CLASS 3 MEDIUM DROP PRECEDENCE.
- 011110 - AF CLASS 3 HIGH DROP PRECEDENCE.
- 100010 - AF CLASS 4 LOW DROP PRECEDENCE.
- 100100 - AF CLASS 4 MEDIUM DROP PRECEDENCE.
- 100110 - AF CLASS 4 HIGH DROP PRECEDENCE.
- 101110 - EXPEDITED FORWARDING.

Default

100010

Name

dwGbrRoam - Guaranteed bit rate for downlink for roaming

Syntax

--dwGbrRoam <integer>

Description

This parameter defines the guaranteed bit rate for downlink traffic for a roaming subscriber. the values range from 0 to 63 in steps of 1, from 64 to 568 in steps of 8, and from 576 to

8640 in steps of 64. optionally, there is also the range of values from 8700 to 16000 in steps of 100, from 17000 to 128000 in steps of 1000 and from 130000 to 168000 in steps of 2000.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 168,000

Default

0

Name

earpRoam - Evolved allocation/retention priority

Syntax

--earpRoam <integer>

Description

This parameter indicates the evolved ARP. the values range from 1 to 15.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 15

Default

14

Name

epsHighPrioRoam - EPS bearer ARP high prior for roaming subscriber

Syntax

--epsHighPrioRoam <integer>

Description

EPS bearer ARP high prior for roaming subscriber. the values range from 1 to 13.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 13

Default

5

Name

epsMedPrioRoam - EPS bearer ARP medium prior for roaming subscriber

Syntax

--epsMedPrioRoam <integer>

Description

EPS bearer ARP medium prior for roaming subscriber. the values range from 2 to 14.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 2 to 14

Default

10

Name

gbWredSwitch - Enhanced random early detection in gb buffer, QGB

Syntax

--gbWredSwitch <list-item>

Description

This parameter defines whether the enhanced random early detection (ERED) algorithm is used in the gb buffer.

Class

Optional, not filterable on list.

List

- inUse - In use - Specify QoS switch type of in use.
- notInUse - Not in use - Specify QoS switch type of not in use.

Default

inUse

Name

gtpWredSwitch - Random early detection in GTP buffer, QGTP

Syntax

--gtpWredSwitch <list-item>

Description

This parameter defines whether the random early detection (RED) algorithm is used in the GPRS tunnelling protocol (GTP) buffer.

Class

Optional, not filterable on list.

List

- inUse - In use - Specify QoS switch type of in use.
- notInUse - Not in use - Specify QoS switch type of not in use.

Default

inUse

Name

hPriority - EPS bearer ARP high priority

Syntax

--hPriority <integer>

Description

EPS bearer ARP high priority. possible weight values range from 1 to 13.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 13

Default

5

Name

igbr - Guaranteed bit rate for UTRAN to GERAN in ISHO

Syntax

--igbr <integer>

Description

This parameter defines the guaranteed bit rate for UTRAN to GERAN in ISHO. the values range from 0 to 63 kbit/s in steps of 1 and from 64 to 472 kbit/s in steps of 8. IGBR has to be smaller or same than IMBR.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 472

Default

472

Name

imbr - Maximum bit rate for UTRAN to GERAN in ISHO

Syntax

--imbr <integer>

Description

This parameter defines the maximum bit rate for UTRAN to GERAN in ISHO. the values range from 0 to 63 kbit/s in steps of 1 and from 64 to 472 kbit/s in steps of 8. IMBR has to be greater or same than IGBR.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 472

Default

472

Name

mPriority - EPS bearer ARP medium priority

Syntax

--mPriority <integer>

Description

EPS bearer ARP medium priority. possible weight values range from 2 to 14.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 2 to 14

Default

10

Name

pcap - Pre-emption capability

Syntax

--pcap <list-item>

Description

Pre-emption capability. enable or disable.

Class

Optional, not filterable on list.

List

- disable - Specify preempt value of disable.
- enable - Specify preempt value of enable.

Default

disable

Name

pcapRoam - Pre-emption capability for roaming subscriber

Syntax

--pcapRoam <list-item>

Description

Pre-emption capability for roaming subscriber. enable or disable.

Class

Optional, not filterable on list.

List

- disable - Specify preempt value of disable.
- enable - Specify preempt value of enable.

Default

disable

Name

pvul - Pre-emption vulnerability

Syntax

--pvul <list-item>

Description

Pre-emption vulnerability. enable or disable.

Class

Optional, not filterable on list.

List

- disable - Specify preempt value of disable.
- enable - Specify preempt value of enable.

Default

enable

Name

pvulRoam - Pre-emption vulnerability for roaming subscriber

Syntax

--pvulRoam <list-item>

Description

Pre-emption vulnerability for roaming subscriber. enable or disable.

Class

Optional, not filterable on list.

List

- disable - Specify preempt value of disable.
- enable - Specify preempt value of enable.

Default

disable

Name

scheduleSwitch - Scheduling in gb buffer, QSGB

Syntax

--scheduleSwitch <list-item>

Description

This parameter defines whether the scheduling algorithm is used.

Class

Optional, not filterable on list.

List

- inUse - In use - Specify QoS switch type of in use.
- notInUse - Not in use - Specify QoS switch type of not in use.

Default

inUse

Name

trafClassRoam - Traffic class for roaming

Syntax

--trafClassRoam <list-item>

Description

This parameter defines the traffic class which can be used by a roaming subscriber at most.

Class

Optional, not filterable on list.

List

- Background - Traffic class type of background.
- Conversational - Traffic class type of conversational.
- Interactive - Traffic class type of interactive.
- Streaming - Traffic class type of streaming.

Default

Interactive

Name

trafHandPrioRoam - Traffic handling priority for roaming

Syntax

--trafHandPrioRoam <integer>

Description

This parameter defines the traffic handling priority which can be used by a roaming subscriber at most. possible values range from 1 to 3.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 3

Default

3

Name

upGbrRoam - Guaranteed bit rate for uplink for roaming

Syntax

--upGbrRoam <integer>

Description

This parameter defines the guaranteed bit rate for uplink traffic for a roaming subscriber. the values range from 0 to 63 in steps of 1, from 64 to 568 in steps of 8, and from 576 to 8640 in steps of 64. optionally, there is also the range of values from 8700 to 16000 in steps of 100 and from 17000 to 46000 in steps of 1000.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 46,000

Default

0

Name

upMaxRoam - Maximum bit rate for uplink for roaming

Syntax

--upMaxRoam <integer>

Description

This parameter defines the maximum bit rate for uplink traffic for a roaming subscriber. the values range from 0 to 63 in steps of 1, from 64 to 568 in steps of 8, and from 576 to 8640 in steps of 64. optionally, there is also the range of values from 8700 to 16000 in steps of 100 and from 17000 to 46000 in steps of 1000.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 46,000

Default

50

Name

weightPri1 - Weight for scheduling priority one, QP1

Syntax

--weightPri1 <integer>

Description

This parameter defines the relative capacity of delay class 1 compared to the other delay classes. possible weight values range from 25 to 90 in steps of 5. if you enter a value for the weight for scheduling priority one parameter, you must enter values for the other weightpri parameters as well.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 25 to 90

Default

50

Name

weightPri2 - Weight for scheduling priority two, QP2

Syntax

--weightPri2 <integer>

Description

This parameter defines the relative capacity of delay class 2 compared to the other delay classes. possible weight values range from 5 to 45 in steps of 5. if you enter a value for the weight for scheduling priority one parameter, you must enter values for the other weightpri parameters as well.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 5 to 45

Default

30

Name

weightPri3 - Weight for scheduling priority three, QP3

Syntax

--weightPri3 <integer>

Description

This parameter defines the relative capacity of delay class 3 compared to the other delay classes. possible weight values range from 5 to 30 in steps of 5. if you enter a value for the weight for scheduling priority one parameter, you must enter values for the other weightpri parameters as well.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 5 to 30

Default

15

Name

weightPri4 - Weight for scheduling priority four, QP4

Syntax

--weightPri4 <integer>

Description

This parameter defines the relative capacity of delay class 4 compared to the other delay classes. possible weight values range from 0 to 25 in steps of 5. if you enter a value for the weight for scheduling priority one parameter, you must enter values for the other weightpri parameters as well.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 25

Default

5

Name

weightPri5 - Non-guaranteed streaming traffic queue priority

Syntax

--weightPri5 <integer>

Description

This parameter defines the weight value for non-guaranteed streaming traffic queue. the priority of streaming queue is set to be guaranteed by default. if other queues start to fill up, the streaming queue weight is downgraded to operatordefined non-guaranteed weight value until traffic is balanced, and the streaming queue can start to use higher priority again. possible values range from 0 to 200.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 200

Default

200

Name

id - Id

Syntax

config

Description

Just type 'config' here.

Class

Required value.

List

- config - Specify LTE QoS policy default of config.

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

6.88 cmm qosProfile

Enhanced QoS Profile for Roaming Subscribers. Syntax, description, parameters, and roles for cmm qosProfile command.

Syntax**Description**

This command is used to administer an Enhanced QoS Profile for Roaming Subscribers and includes parameters for defining the APN-NI, APN AMBR bitrates, default bearer ARP, QCI, PCI and PVI values, and guaranteed and non-guaranteed dedicated bearer ARP, bitrate, QCI, PCI and PVI values.

Parameters**Name**

name - Roaming QoS profile

Syntax

--name <string>

Description

A name that identifies this roaming QoS profile.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

profileType - Enhanced roaming QoS profile type

Syntax

--profileType <list-item>

Description

Profile Type. Uniquely identifies the various types of profiles we support: IMS, Non-IMS, Custom APN 1, 2 or 3

Class

Required on create, not modifiable.

List

- CustomAPN2gPrfl - 2G custom APN profile - 2G S4 QoS profile for custom APNs.
- CustomAPN3gPrfl - 3G custom APN profile - 3G S4 QoS profile for custom APNs.
- CustomAPNPrfl1 - Custom APN profile - 1 - Roaming QoS profile for custom APN 1.
- CustomAPNPrfl2 - Custom APN profile - 2 - Roaming QoS profile for custom APN 2.
- CustomAPNPrfl3 - Custom APN profile - 3 - Roaming QoS profile for custom APN 3.
- IMSAPN2gPrfl - 2G S4 IMS APN profile - 2G S4 QoS profile IMS APNs.
- IMSAPN3gPrfl - 3G S4 IMS APN profile - 3G S4 QoS profile IMS APNs.
- IMSAPNPrfl - IMS APN profile - Roaming QoS profile for IMS APNs.
- NonIMSAPN2gPrfl - 2G non-ims APN profile - 2G S4 QoS profile for non-ims APNs.
- NonIMSAPN3gPrfl - 3G non-ims APN profile - 3G S4 QoS profile for non-ims APNs.
- NonIMSAPNPrfl - Non-IMS APN profile - Roaming QoS profile for non-ims APNs.

Name

apnAmbrDownLink - APN AMBR downlink for default bearer

Syntax

--apnAmbrDownLink <integer>

Description

APN aggregated maximum bitrate downlink for default bearer.

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64- 568, 576-8640, 8700-16000, 17000-128000,
130000-4000000000 in units of kbps

Default

128000

Name

apnAmbrUpLink - APN AMBR uplink for default bearer

Syntax

--apnAmbrUpLink <integer>

Description

APN aggregated maximum bitrate uplink for default bearer.

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64- 568, 576-8640, 8700-16000, 17000-128000,
130000-4000000000 in units of kbps

Default

128000

Name

apnAmbrUseHssValue - APN AMBR use HSS value

Syntax

--apnAmbrUseHssValue <list-item>

Description

Indicates whether to override both of the local QoS profile uplink APN-AMBR and downlink APN-AMBR values with subscribed uplink APN-AMBR and downlink APN-AMBR values in HSS.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Name

apnNi - Access point name - network indicator

Syntax

--apnNi <string>

Description

APN NI for this roaming QoS profile.

Restrictions

apnNi must be a null string if profileType is one of IMSAPNPrfl or NonIMSAPNPrfl apnNi must NOT be a null string IF (profileType is one of CustomAPNPrfl1 or CustomAPNPrfl2 or CustomAPNPrfl3) AND (given qosProfile is referenced by another table)

Class

Optional.

Type / Range

Digit string of length 0 to 62 with the allowed pattern of [0-9a-zA-Z\.\-]*

Default

"

Name

defaultBrrArp - Default bearer ARP

Syntax

--defaultBrrArp <integer>

Description

Default bearer allocation and retention priority.

Class

Optional.

Type / Range

Integer in the range of 1 to 15

Default

15

Name

defaultBrrQci - Default bearer QCI

Syntax

--defaultBrrQci <integer>

Description

QoS class identifier for default bearer.

Class

Optional.

Type / Range

Integer in the range of 5 to 9

Default

9

Name

defaultBrrUseHssValue - Default bearer use HSS value

Syntax

--defaultBrrUseHssValue <list-item>

Description

Indicates whether to override the default bearer attributes (arp-priority-level, pci, pvi) values in the local QoS profile with subscribed values in HSS.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Name

defBrrPci - Default bearer PCI

Syntax

--defBrrPci <list-item>

Description

Default bearer pre-emption capability indicator.

Class

Optional.

List

- Disabled - PCI disabled - Pre-emption capability disabled.
- Enabled - PCI enabled - Pre-emption capability enabled.

Default

Enabled

Name

defBrrPvi - Default bearer PVI

Syntax

--defBrrPvi <list-item>

Description

Default bearer pre-emption vulnerability indicator.

Class

Optional.

List

- Disabled - PVI disabled - Pre-emption vulnerability disabled.
- Enabled - PVI enabled - Pre-emption vulnerability enabled.

Default

Enabled

Name

description - Description

Syntax

--description <string>

Description

Profile description.

Class

Optional.

Type / Range

Digit string of length 0 to 160 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Name

gbrDedBrrArp - Guaranteed dedicated bearer ARP

Syntax

--gbrDedBrrArp <integer>

Description

Guaranteed dedicated bearer allocation and retention priority.

Class

Optional.

Type / Range

Integer in the range of 0 to 15

Default

15

Name

gbrDedBrrBitrateDownlink - Guaranteed dedicated bearer downlink bitrate

Syntax

--gbrDedBrrBitrateDownlink <integer>

Description

Guaranteed dedicated bearer downlink bitrate.

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64-568, 576-8640, 8700-16000, 17000-128000, 130000-256000, 260000-500000, 510000-1500000, 1600000-4000000000 in units of kbps

Default

128000

Name

gbrDedBrrBitrateUplink - Guaranteed dedicated bearer uplink bitrate

Syntax

--gbrDedBrrBitrateUplink <integer>

Description

Guaranteed dedicated bearer uplink bitrate.

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64-568, 576-8640, 8700-16000, 17000-128000, 130000-256000, 260000-500000, 510000-1500000, 1600000-4000000000 in units of kbps

Default

128000

Name

gbrDedBrrMaxBitrateDownlink - Guaranteed dedicated bearer maximum downlink bitrate

Syntax

--gbrDedBrrMaxBitrateDownlink <integer>

Description

Guaranteed dedicated bearer maximum downlink bitrate.

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64-568, 576-8640, 8700-16000, 17000-128000, 130000-256000, 260000-500000, 510000-1500000, 1600000-4000000000 in units of kbps

Default

128000

Name

gbrDedBrrMaxBitrateUplink - Guaranteed dedicated bearer maximum uplink bitrate

Syntax

--gbrDedBrrMaxBitrateUplink <integer>

Description

Guaranteed dedicated bearer maximum uplink bitrate.

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64-568, 576-8640, 8700-16000, 17000-128000, 130000-256000, 260000-500000, 510000-1500000, 1600000-4000000000 in units of kbps

Default

128000

Name

gbrDedBrrPci - Guaranteed dedicated bearer PCI

Syntax

--gbrDedBrrPci <list-item>

Description

Guaranteed dedicated bearer pre-emption capability indicator.

Class

Optional.

List

- Disabled - PCI disabled - Pre-emption capability disabled.
- Enabled - PCI enabled - Pre-emption capability enabled.

Default

Disabled

Name

gbrDedBrrPvi - Guaranteed dedicated bearer PVI

Syntax

--gbrDedBrrPvi <list-item>

Description

Guaranteed dedicated bearer pre-emption vulnerability indicator.

Class

Optional.

List

- Disabled - PVI disabled - Pre-emption vulnerability disabled.
- Enabled - PVI enabled - Pre-emption vulnerability enabled.

Default

Disabled

Name

gbrDedBrrQci - Guaranteed dedicated bearer QCI

Syntax

--gbrDedBrrQci <integer>

Description

Guaranteed dedicated bearer QoS class identifier.

Class

Optional.

Type / Range

Integer in the range of 0 to 4

Default

4

Name

nonGbrDedBrrArp - Non guaranteed dedicated bearer ARP

Syntax

--nonGbrDedBrrArp <integer>

Description

Non guaranteed dedicated bearer allocation and retention priority.

Class

Optional.

Type / Range

Integer in the range of 0 to 15

Default

15

Name

nonGbrDedBrrPci - Non guaranteed dedicated bearer PCI

Syntax

--nonGbrDedBrrPci <list-item>

Description

Non guaranteed dedicated bearer pre-emption capability indicator.

Class

Optional.

List

- Disabled - PCI disabled - Pre-emption capability disabled.
- Enabled - PCI enabled - Pre-emption capability enabled.

Default

Disabled

Name

nonGbrDedBrrPvi - Non guaranteed dedicated bearer PVI

Syntax

--nonGbrDedBrrPvi <list-item>

Description

Non guaranteed dedicated bearer pre-emption vulnerability indicator.

Class

Optional.

List

- Disabled - PVI disabled - Pre-emption vulnerability disabled.
- Enabled - PVI enabled - Pre-emption vulnerability enabled.

Default

Disabled

Name

nonGbrDedBrrQci - Non guaranteed dedicated bearer QCI

Syntax

--nonGbrDedBrrQci <integer>

Description

Non guaranteed dedicated bearer QoS class identifier.

Class

Optional.

Type / Range

Integer in the range of 0 or 6 to 9

Default

9

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.89 cmm qosProfileBackTc

Background Traffic Class QoS Profile. Syntax, description, parameters, and roles for cmm qosProfileBackTc command.

Syntax

Description

This command is used to administer a Background Traffic Class QoS Profile and includes parameters for defining the Allocation Retention Priority (ARP), QoS maximum bitrates, priority level, error ratios, and PCI and PVI values.

Parameters

Name

name - Background traffic class QoS profile name

Syntax

--name <string>

Description

Name that identifies the background traffic class quality of service (QoS) profile.

Restrictions

default QoS profiles cannot be deleted, but can be updated.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

arp - ARP

Syntax

--arp <list-item>

Description

Allocation retention priority (ARP) for background traffic class.

Class

Optional.

List

- high - Specify priority of high.
- low - Specify priority of low.
- normal - Specify priority of normal.

Default

low

Name

deliveryErroneousSdu - Delivery erroneous SDU

Syntax

--deliveryErroneousSdu <list-item>

Description

Indicates whether to detect erroneous service data units (SDUs) for background traffic class, and if so, whether to accept or discard them; the default is do-not-detect.

Class

Optional.

List

- accept - Specifies detect and accept erroneous data packets.
- discard - Specifies detect and discard erroneous data packets.
- do-not-detect - Do not detect - Specifies do not detect erroneous data packets.

Default

do-not-detect

Name

deliveryOrder - Delivery order

Syntax

--deliveryOrder <list-item>

Description

Enable (with-order) or disable (without-order) delivery order for background traffic class.

Class

Optional.

List

- with-order - With order - Specifies delivery with order.
- without-order - Without order - Specifies delivery without order.

Default

without-order

Name

description - Background traffic class QoS profile description

Syntax

--description <string>

Description

Background traffic class quality of service (QoS) profile description.

Class

Optional.

Type / Range

Digit string of length 0 to 80 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

maxBitRateDownlink - Max bit rate for downlink

Syntax

--maxBitRateDownlink <integer>

Description

Quality of service (QoS) default maximum bit rate for downlink for background class, in kbps.

Restrictions

if in the range 64..568, must be a multiple of 8 kbps if in the range 576..8640, must be a multiple of 64 kbps if in the range 8700..16000, must be a multiple of 100 kbps if in the range 17000..128000, must be a multiple of 1 Mbps if in the range 130000..256000, must be a multiple of 2 Mbps

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64-568, 576-8640, 8700-16000, 17000-128000, 130000-256000, 260000-500000, 510000-1500000, 1600000-4000000000 in units of Kbps

Default

384

Name

maxBitRateUplink - Max bit rate for uplink

Syntax

--maxBitRateUplink <integer>

Description

Quality of service (QoS) default maximum bit rate for uplink for background class, in kbps.

Restrictions

if in the range 64..568, must be a multiple of 8 kbps if in the range 576..8640, must be a multiple of 64 kbps if in the range 8700..16000, must be a multiple of 100 kbps if in the range 17000..128000, must be a multiple of 1 Mbps if in the range 130000..256000, must be a multiple of 2 Mbps

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64-568, 576-8640, 8700-16000, 17000-128000, 130000-256000, 260000-500000, 510000-1500000, 1600000-4000000000 in units of Kbps

Default

384

Name

maxSduSize - Max SDU size

Syntax

--maxSduSize <integer>

Description

Maximum service data unit (SDU) size for background traffic class, in bytes.

Restrictions

if in the range 10..1500, must be a multiple of 10 bytes if greater than 1500, can only be 1502

Class

Optional.

Type / Range

Integer in the range of 10 to 1,502 in units of Byte

Default

1502

Name

pci - Pre-emption capability indicator

Syntax`--pci <list-item>`**Description**

Indicates whether a bearer request can preempt resources in a limited resource condition as part of the allocation and retention policy (ARP) aspect of quality of service (QoS) information.

Class

Optional.

List

- disabled - Specify type of disabled.
- enabled - Specify type of enabled.

Default

disabled

Name

prioLevel - Priority level

Syntax`--prioLevel <integer>`**Description**

The priority level value of EARP used to override the value received from the HSS/HLR or to use as a default value if no value is received.

Class

Optional.

Type / Range

Integer in the range of 1 to 15

Default

15

Name

pvi - Pre-emption vulnerability indicator

Syntax`--pvi <list-item>`**Description**

Indicates whether an active bearer can be preempted by a preemption capable high priority bearer in a limited resource condition as part of the allocation and retention policy (ARP) aspect of quality of service (QoS) information.

Class

Optional.

List

- disabled - Specify type of disabled.
- enabled - Specify type of enabled.

Default

enabled

Name

residualBerNrt - Residual BER

Syntax

--residualBerNrt <list-item>

Description

Residual bit error ratio (BER) for background traffic class.

Class

Optional.

List

- $1 \cdot 10^{-5}$ - 0.00001 - Specify residual bit error ratio 0.00001.
- $4 \cdot 10^{-3}$ - 0.004 - Specify residual bit error ratio of 0.004.
- $6 \cdot 10^{-8}$ - 0.00000006 - Specify residual bit error ratio of 0.00000006.

Default

$6 \cdot 10^{-8}$

Name

sduErrorRatioNrt - SDU error ratio

Syntax

--sduErrorRatioNrt <list-item>

Description

Service data unit (SDU) error ratio for background traffic class.

Class

Optional.

List

- $1 \cdot 10^{-3}$ - 0.001 - Specify service data unit error ratio of 0.001.

- 1×10^{-4} - 0.0001 - Specify service data unit error ratio of 0.0001.
- 1×10^{-6} - 0.000001 - Specify service data unit error ratio of 0.000001.

Default

1×10^{-6}

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.90 cmm qosProfileConvTc

Conversational Traffic Class QoS Profile. Syntax, description, parameters, and roles for `cmm qosProfileConvTc` command.

Syntax**Description**

This command is used to administer a Conversational Traffic Class QoS Profile and includes parameters for defining the Allocation Retention Priority (ARP), QoS maximum and guaranteed bitrates, priority level, error ratios, and PCI and PVI values.

Parameters**Name**

name - Conversational traffic class QoS profile name

Syntax

--name <string>

Description

Name that identifies the conversational traffic class quality of service (QoS) profile.

Restrictions

default QoS profiles cannot be deleted, but can be updated.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

arp - ARP

Syntax

--arp <list-item>

Description

Allocation retention priority (ARP) for conversational traffic class.

Class

Optional.

List

- high - Specify priority of high.
- low - Specify priority of low.
- normal - Specify priority of normal.

Default

high

Name

deliveryErroneousSdu - Delivery erroneous SDU

Syntax

--deliveryErroneousSdu <list-item>

Description

Indicates whether to detect erroneous Service Data Units (SDUs) for Conversational traffic class, and if so, whether to accept or discard them; the default is do-not-detect.

Class

Optional.

List

- accept - Specifies detect and accept erroneous data packets.
- discard - Specifies detect and discard erroneous data packets.
- do-not-detect - Do not detect - Specifies do not detect erroneous data packets.

Default

do-not-detect

Name

deliveryOrder - Delivery order

Syntax

--deliveryOrder <list-item>

Description

Enable (with-order) or disable (without-order) delivery order for conversational traffic class.

Class

Optional.

List

- with-order - With order - Specifies delivery with order.
- without-order - Without order - Specifies delivery without order.

Default

without-order

Name

description - Conversational traffic class QoS profile description

Syntax

--description <string>

Description

Conversational traffic class quality of service (QoS) profile description.

Class

Optional.

Type / Range

Digit string of length 0 to 80 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

guaranteedBitRateDownlink - Guaranteed bit rate for downlink

Syntax

--guaranteedBitRateDownlink <integer>

Description

Quality of service (QoS) default guaranteed bit rate for downlink for conversational traffic class, in kbps.

Restrictions

if in the range 64..568, must be a multiple of 8 kbps if in the range 576..8640, must be a multiple of 64 kbps if in the range 8700..16000, must be a multiple of 100 kbps if in the

range 17000..128000, must be a multiple of 1 Mbps if in the range 130000..256000, must be a multiple of 2 Mbps

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64-568, 576-8640, 8700-16000, 17000-128000, 130000-256000, 260000-500000, 510000-1500000, 1600000-4000000000 in units of Kbps

Default

384

Name

guaranteedBitRateUplink - Guaranteed bit rate for uplink

Syntax

--guaranteedBitRateUplink <integer>

Description

Quality of service (QoS) default guaranteed bit rate for uplink for conversational traffic class, in kbps.

Restrictions

if in the range 64..568, must be a multiple of 8 kbps if in the range 576..8640, must be a multiple of 64 kbps if in the range 8700..16000, must be a multiple of 100 kbps if in the range 17000..128000, must be a multiple of 1 Mbps if in the range 130000..256000, must be a multiple of 2 Mbps

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64-568, 576-8640, 8700-16000, 17000-128000, 130000-256000, 260000-500000, 510000-1500000, 1600000-4000000000 in units of Kbps

Default

384

Name

maxBitRateDownlink - Max bit rate for downlink

Syntax

--maxBitRateDownlink <integer>

Description

Quality of service (QoS) default maximum bit rate for downlink for conversational traffic class, in kbps.

Restrictions

if in the range 64..568, must be a multiple of 8 kbps if in the range 576..8640, must be a multiple of 64 kbps if in the range 8700..16000, must be a multiple of 100 kbps if in the range 17000..128000, must be a multiple of 1 Mbps if in the range 130000..256000, must be a multiple of 2 Mbps

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64-568, 576-8640, 8700-16000, 17000-128000, 130000-256000, 260000-500000, 510000-1500000, 1600000-4000000000 in units of Kbps

Default

384

Name

maxBitRateUplink - Max bit rate for uplink

Syntax

--maxBitRateUplink <integer>

Description

Quality of service (QoS) default maximum bit rate for uplink for conversational traffic class, in kbps.

Restrictions

if in the range 64..568, must be a multiple of 8 kbps if in the range 576..8640, must be a multiple of 64 kbps if in the range 8700..16000, must be a multiple of 100 kbps if in the range 17000..128000, must be a multiple of 1 Mbps if in the range 130000..256000, must be a multiple of 2 Mbps

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64-568, 576-8640, 8700-16000, 17000-128000, 130000-256000, 260000-500000, 510000-1500000, 1600000-4000000000 in units of Kbps

Default

384

Name

maxSduSize - Max SDU size

Syntax

--maxSduSize <integer>

Description

Maximum service data unit (SDU) size for conversational traffic class, in bytes.

Restrictions

if in the range 10..1500, must be a multiple of 10 bytes if greater than 1500, can only be 1502

Class

Optional.

Type / Range

Integer in the range of 10 to 1,502 in units of Byte

Default

1502

Name

pci - Pre-emption capability indicator

Syntax

--pci <list-item>

Description

Indicates whether a bearer request can preempt resources in a limited resource condition as part of the allocation and retention policy (ARP) aspect of quality of service (QoS) information.

Class

Optional.

List

- disabled - Specify type of disabled.
- enabled - Specify type of enabled.

Default

disabled

Name

prioLevel - Priority level

Syntax

--prioLevel <integer>

Description

The priority level value of EARP used to override the value received from the HSS/HLR or to use as a default value if no value is received.

Class

Optional.

Type / Range

Integer in the range of 1 to 15

Default

15

Name

pvi - Pre-emption vulnerability indicator

Syntax

--pvi <list-item>

Description

Indicates whether an active bearer can be preempted by a preemption capable high priority bearer in a limited resource condition as part of the allocation and retention policy (ARP) aspect of quality of service (QoS) information.

Class

Optional.

List

- disabled - Specify type of disabled.
- enabled - Specify type of enabled.

Default

enabled

Name

residualBerRt - Residual BER

Syntax

--residualBerRt <list-item>

Description

Residual bit error ratio (BER) for conversational traffic class.

Class

Optional.

List

- 1×10^{-2} - 0.01 - Specify residual bit error ratio of 0.01.
- 1×10^{-3} - 0.001 - Specify residual bit error ratio of 0.001.
- 1×10^{-4} - 0.0001 - Specify residual bit error ratio of 0.0001.
- 1×10^{-5} - 0.00001 - Specify residual bit error ratio of 0.00001.
- 1×10^{-6} - 0.000001 - Specify residual bit error ratio of 0.000001.
- 5×10^{-2} - 0.05 - Specify residual bit error ratio of 0.05.

- 5×10^{-3} - 0.005 - Specify residual bit error ratio of 0.005.

Default

1×10^{-6}

Name

sduErrorRatioConv - SDU error ratio

Syntax

--sduErrorRatioConv <list-item>

Description

Service data unit (SDU) error ratio for conversational traffic class.

Class

Optional.

List

- 1×10^{-2} - 0.01 - Specify service data unit error ratio of 0.01.
- 1×10^{-3} - 0.001 - Specify service data unit error ratio of 0.001.
- 1×10^{-4} - 0.0001 - Specify service data unit error ratio of 0.0001.
- 1×10^{-5} - 0.00001 - Specify service data unit error ratio of 0.00001.
- 7×10^{-3} - 0.007 - Specify service data unit error ratio of 0.007.

Default

1×10^{-5}

Name

transferDelay - Transfer delay

Syntax

--transferDelay <integer>

Description

Transfer delay for conversational traffic class.

Restrictions

if in the range 10..150, must be a multiple of 10 ms if in the range 200..950, must be a multiple of 50 ms if in the range 1000..4000, must be a multiple of 100 ms

Class

Optional.

Type / Range

Integer in the range of 10 to 4,000 in units of Millisecond

Default

100

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.91 cmm qosProfileInteractTc

Interactive Traffic Class QoS Profile. Syntax, description, parameters, and roles for `cmm qosProfileInteractTc` command.

Syntax

Description

This command is used to administer an Interactive Traffic Class QoS Profile and includes parameters for defining the Allocation Retention Priority (ARP), QoS maximum bitrates, priority level, error ratios, and PCI and PVI values.

Parameters

Name

name - Interactive traffic class QoS profile name

Syntax

--name <string>

Description

Name that identifies the interactive traffic class quality of service (QoS) profile.

Restrictions

default QoS profiles cannot be deleted, but can be updated.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of `[0-9a-zA-Z\.\-_:@]*`

Name

arp - ARP

Syntax

--arp <list-item>

Description

Allocation retention priority (ARP) for interactive traffic class.

Class

Optional.

List

- high - Specify priority of high.
- low - Specify priority of low.
- normal - Specify priority of normal.

Default

normal

Name

deliveryErroneousSdu - Delivery erroneous SDU

Syntax

--deliveryErroneousSdu <list-item>

Description

Indicates whether to detect erroneous service data units (SDUs) for interactive traffic class, and if so, whether to accept or discard them; the default is do-not-detect.

Class

Optional.

List

- accept - Specifies detect and accept erroneous data packets.
- discard - Specifies detect and discard erroneous data packets.
- do-not-detect - Do not detect - Specifies do not detect erroneous data packets.

Default

do-not-detect

Name

deliveryOrder - Delivery order

Syntax

--deliveryOrder <list-item>

Description

Enable (with-order) or disable (without-order) delivery order for interactive traffic class.

Class

Optional.

List

- with-order - With order - Specifies delivery with order.
- without-order - Without order - Specifies delivery without order.

Default

without-order

Name

description - Interactive traffic class QoS profile description

Syntax

--description <string>

Description

Interactive traffic class quality of service (QoS) profile description.

Class

Optional.

Type / Range

Digit string of length 0 to 80 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Name

handlingPriority - Traffic handling priority

Syntax

--handlingPriority <list-item>

Description

Traffic handling priority for interactive traffic class.

Class

Optional.

List

- high - Specify priority of high.
- low - Specify priority of low.
- normal - Specify priority of normal.

Default

high

Name

maxBitRateDownlink - Max bit rate for downlink

Syntax

--maxBitRateDownlink <integer>

Description

Quality of service (QoS) default maximum bit rate for downlink for interactive traffic class, in kbps.

Restrictions

if in the range 64..568, must be a multiple of 8 kbps if in the range 576..8640, must be a multiple of 64 kbps if in the range 8700..16000, must be a multiple of 100 kbps if in the range 17000..128000, must be a multiple of 1 Mbps if in the range 130000..256000, must be a multiple of 2 Mbps

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64-568, 576-8640, 8700-16000, 17000-128000, 130000-256000, 260000-500000, 510000-1500000, 1600000-4000000000 in units of Kbps

Default

384

Name

maxBitRateUplink - Max bit rate for uplink

Syntax

--maxBitRateUplink <integer>

Description

Quality of service (QoS) default maximum bit rate for uplink for interactive traffic class, in kbps.

Restrictions

if in the range 64..568, must be a multiple of 8 kbps if in the range 576..8640, must be a multiple of 64 kbps if in the range 8700..16000, must be a multiple of 100 kbps if in the range 17000..128000, must be a multiple of 1 Mbps if in the range 130000..256000, must be a multiple of 2 Mbps

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64-568, 576-8640, 8700-16000, 17000-128000, 130000-256000, 260000-500000, 510000-1500000, 1600000-4000000000 in units of Kbps

Default

384

Name

maxSduSize - Max SDU size

Syntax`--maxSduSize <integer>`**Description**

Maximum service data unit (SDU) size for streaming traffic class, in bytes.

Restrictionsif in the range 10..1500, must be a multiple of 10 bytes if greater than 1500, can only be
1502**Class**

Optional.

Type / Range

Integer in the range of 10 to 1,502 in units of Byte

Default

1502

Name

pci - Pre-emption capability indicator

Syntax`--pci <list-item>`**Description**Indicates whether a bearer request can preempt resources in a limited resource condition
as part of the allocation and retention policy (ARP) aspect of quality of service (QoS)
information.**Class**

Optional.

List

- disabled - Specify type of disabled.
- enabled - Specify type of enabled.

Default

disabled

Name

prioLevel - Priority level

Syntax

--prioLevel <integer>

Description

The priority level value of EARP used to override the value received from the HSS/HLR or to use as a default value if no value is received.

Class

Optional.

Type / Range

Integer in the range of 1 to 15

Default

15

Name

pvi - Pre-emption vulnerability indicator

Syntax

--pvi <list-item>

Description

Indicates whether an active bearer can be preempted by a preemption capable high priority bearer in a limited resource condition as part of the allocation and retention policy (ARP) aspect of quality of service (QoS) information.

Class

Optional.

List

- disabled - Specify type of disabled.
- enabled - Specify type of enabled.

Default

enabled

Name

residualBerNrt - Residual BER

Syntax

--residualBerNrt <list-item>

Description

Residual bit error ratio (BER) for interactive traffic class.

Class

Optional.

List

- 1×10^{-5} - 0.00001 - Specify residual bit error ratio 0.00001.

- 4×10^{-3} - 0.004 - Specify residual bit error ratio of 0.004.
- 6×10^{-8} - 0.00000006 - Specify residual bit error ratio of 0.00000006.

Default

6×10^{-8}

Name

sduErrorRatioNrt - SDU error ratio

Syntax

--sduErrorRatioNrt <list-item>

Description

Service data unit (SDU) error ratio for interactive traffic class.

Class

Optional.

List

- 1×10^{-3} - 0.001 - Specify service data unit error ratio of 0.001.
- 1×10^{-4} - 0.0001 - Specify service data unit error ratio of 0.0001.
- 1×10^{-6} - 0.000001 - Specify service data unit error ratio of 0.000001.

Default

1×10^{-6}

Name

signallingIndication3G - Signaling indication for 3G

Syntax

--signallingIndication3G <list-item>

Description

Indicates whether to optimize for signaling traffic for interactive traffic class.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.92 cmm qosProfileStreamTc

Streaming Traffic Class QoS Profile. Syntax, description, parameters, and roles for cmm qosProfileStreamTc command.

Syntax

Description

This command is used to administer a Streaming Traffic Class QoS Profile and includes parameters for defining the Allocation Retention Priority (ARP), QoS maximum and guaranteed bitrates, priority level, error ratios, and PCI and PVI values.

Parameters

Name

name - Streaming traffic class QoS profile name

Syntax

--name <string>

Description

Name that identifies the streaming traffic class quality of service (QoS) profile.

Restrictions

default QoS profiles cannot be deleted, but can be updated.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

arp - ARP

Syntax

--arp <list-item>

Description

Allocation retention priority (ARP) for streaming traffic class.

Class

Optional.

List

- high - Specify priority of high.
- low - Specify priority of low.
- normal - Specify priority of normal.

Default

normal

Name

deliveryErroneousSdu - Delivery erroneous SDU

Syntax

--deliveryErroneousSdu <list-item>

Description

Indicates whether to detect erroneous service data units (SDUs) for streaming traffic class, and if so, whether to accept or discard them; the default is do-not-detect.

Class

Optional.

List

- accept - Specifies detect and accept erroneous data packets.
- discard - Specifies detect and discard erroneous data packets.
- do-not-detect - Do not detect - Specifies do not detect erroneous data packets.

Default

do-not-detect

Name

deliveryOrder - Delivery order

Syntax

--deliveryOrder <list-item>

Description

Enable (with-order) or disable (without-order) delivery order for streaming traffic class.

Class

Optional.

List

- with-order - With order - Specifies delivery with order.
- without-order - Without order - Specifies delivery without order.

Default

without-order

Name

description - Streaming traffic class QoS profile description

Syntax

--description <string>

Description

Streaming traffic class quality of service (QoS) profile description.

Class

Optional.

Type / Range

Digit string of length 0 to 80 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

''

Name

guaranteedBitRateDownlink - Guaranteed bit rate for downlink

Syntax

--guaranteedBitRateDownlink <integer>

Description

Quality of service (QoS) default guaranteed bit rate for downlink for streaming traffic class, in kbps.

Restrictions

if in the range 64..568, must be a multiple of 8 kbps if in the range 576..8640, must be a multiple of 64 kbps if in the range 8700..16000, must be a multiple of 100 kbps if in the range 17000..128000, must be a multiple of 1 Mbps if in the range 130000..256000, must be a multiple of 2 Mbps

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64-568, 576-8640, 8700-16000, 17000-128000, 130000-256000, 260000-500000, 510000-1500000, 1600000-4000000000 in units of Kbps

Default

384

Name

guaranteedBitRateUplink - Guaranteed bit rate for uplink

Syntax`--guaranteedBitRateUplink <integer>`**Description**

Quality of service (QoS) default guaranteed bit rate for uplink for streaming traffic class, in kbps.

Restrictions

if in the range 64..568, must be a multiple of 8 kbps if in the range 576..8640, must be a multiple of 64 kbps if in the range 8700..16000, must be a multiple of 100 kbps if in the range 17000..128000, must be a multiple of 1 Mbps if in the range 130000..256000, must be a multiple of 2 Mbps

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64-568, 576-8640, 8700-16000, 17000-128000, 130000-256000, 260000-500000, 510000-1500000, 1600000-4000000000 in units of Kbps

Default

384

Name

maxBitRateDownlink - Max bit rate for downlink

Syntax`--maxBitRateDownlink <integer>`**Description**

Quality of service (QoS) default maximum bit rate for downlink for streaming traffic class, in kbps.

Restrictions

if in the range 64..568, must be a multiple of 8 kbps if in the range 576..8640, must be a multiple of 64 kbps if in the range 8700..16000, must be a multiple of 100 kbps if in the range 17000..128000, must be a multiple of 1 Mbps if in the range 130000..256000, must be a multiple of 2 Mbps

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64-568, 576-8640, 8700-16000, 17000-128000, 130000-256000, 260000-500000, 510000-1500000, 1600000-4000000000 in units of Kbps

Default

384

Name

maxBitRateUplink - Max bit rate for uplink

Syntax

--maxBitRateUplink <integer>

Description

Quality of service (QoS) default maximum bit rate for uplink for streaming traffic class, in kbps.

Restrictions

if in the range 64..568, must be a multiple of 8 kbps if in the range 576..8640, must be a multiple of 64 kbps if in the range 8700..16000, must be a multiple of 100 kbps if in the range 17000..128000, must be a multiple of 1 Mbps if in the range 130000..256000, must be a multiple of 2 Mbps

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64-568, 576-8640, 8700-16000, 17000-128000, 130000-256000, 260000-500000, 510000-1500000, 1600000-4000000000 in units of Kbps

Default

384

Name

maxSduSize - Max SDU size

Syntax

--maxSduSize <integer>

Description

Maximum service data unit (SDU) size for streaming traffic class, in bytes.

Restrictions

if in the range 10..1500, must be a multiple of 10 bytes if greater than 1500, can only be 1502

Class

Optional.

Type / Range

Integer in the range of 10 to 1,502 in units of Byte

Default

1502

Name

pci - Pre-emption capability indicator

Syntax

--pci <list-item>

Description

Indicates whether a bearer request can preempt resources in a limited resource condition as part of the allocation and retention policy (ARP) aspect of quality of service (QoS) information.

Class

Optional.

List

- disabled - Specify type of disabled.
- enabled - Specify type of enabled.

Default

disabled

Name

prioLevel - Priority level

Syntax

--prioLevel <integer>

Description

The priority level value of EARP used to override the value received from the HSS/HLR or to use as a default value if no value is received.

Class

Optional.

Type / Range

Integer in the range of 1 to 15

Default

15

Name

pvi - Pre-emption vulnerability indicator

Syntax

--pvi <list-item>

Description

Indicates whether an active bearer can be preempted by a preemption capable high priority bearer in a limited resource condition as part of the allocation and retention policy (ARP) aspect of quality of service (QoS) information.

Class

Optional.

List

- disabled - Specify type of disabled.
- enabled - Specify type of enabled.

Default

enabled

Name

residualBerRt - Residual BER

Syntax

--residualBerRt <list-item>

Description

Residual bit error ratio (BER) for streaming traffic class.

Class

Optional.

List

- 1×10^{-2} - 0.01 - Specify residual bit error ratio of 0.01.
- 1×10^{-3} - 0.001 - Specify residual bit error ratio of 0.001.
- 1×10^{-4} - 0.0001 - Specify residual bit error ratio of 0.0001.
- 1×10^{-5} - 0.00001 - Specify residual bit error ratio of 0.00001.
- 1×10^{-6} - 0.000001 - Specify residual bit error ratio of 0.000001.
- 5×10^{-2} - 0.05 - Specify residual bit error ratio of 0.05.
- 5×10^{-3} - 0.005 - Specify residual bit error ratio of 0.005.

Default

1×10^{-6}

Name

sduErrorRatioStream - SDU error ratio

Syntax

--sduErrorRatioStream <list-item>

Description

Service data unit (SDU) error ratio for streaming traffic class.

Class

Optional.

List

- $1 \cdot 10^{-1}$ - 0.1 - Specify service data unit error ratio of 0.1.
- $1 \cdot 10^{-2}$ - 0.01 - Specify service data unit error ratio of 0.01.
- $1 \cdot 10^{-3}$ - 0.001 - Specify service data unit error ratio of 0.001.
- $1 \cdot 10^{-4}$ - 0.0001 - Specify service data unit error ratio of 0.0001.
- $1 \cdot 10^{-5}$ - 0.00001 - Specify service data unit error ratio of 0.00001.
- $7 \cdot 10^{-3}$ - 0.007 - Specify service data unit error ratio of 0.007.

Default

$1 \cdot 10^{-5}$

Name

transferDelay - Transfer delay

Syntax

--transferDelay <integer>

Description

Transfer delay for streaming traffic class.

Restrictions

if in the range 10..150, must be a multiple of 10 ms if in the range 200..950, must be a multiple of 50 ms if in the range 1000..4000, must be a multiple of 100 ms

Class

Optional.

Type / Range

Integer in the range of 10 to 4,000 in units of Millisecond

Default

300

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.93 cmm racOfNse

racs of nse. Syntax, description, parameters, and roles for cmm racOfNse command.

Syntax

Description

RAC and NSE linking.

Parameters

Name

mcc - MCC

Syntax

--mcc <digits>

Description

Mobile country code.

Class

Required key.

Type / Range

Digit string of length 3

Name

mnc - MNC

Syntax

--mnc <digits>

Description

Mobile network code.

Class

Required key.

Type / Range

Digit string of length 2 to 3

Name

lac - LAC

Syntax

--lac <integer>

Description

Location area code.

Class

Required key.

Type / Range

Integer in the range of 1 to 65,533

Name

rac - RAC id

Syntax

--rac <integer>

Description

RAC identifier.

Class

Required key.

Type / Range

Integer in the range of 0 to 255

Name

nsId - NSE id

Syntax

--nseId <integer>

Description

Network service entity identifier.

Class

Required key.

Type / Range

Integer in the range of 0 to 65,535

Roles**Show**

sysadmin, manage, monitor

List

sysadmin, manage, monitor

6.94 cmm radiusProxy

radiusProxy. Syntax, description, parameters, and roles for cmm radiusProxy command.

Syntax

Description

Radius Proxy configuration

Parameters

Name

proxyId - Proxy ID

Syntax

--proxyId <integer>

Description

Radius Proxy config identification (1)

Class

Required key.

Type / Range

Integer in the range of 1 to 1

Name

nasIp - NAS-IP-address

Syntax

--nasIp <string>

Description

CMM External address (NAS-IP-Address)

Class

Required on create.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Roles

Create

secadmin

Show

seadmin

Modify

seadmin

Delete

seadmin

List

seadmin

6.95 cmm rait

Routing Area Identification. Syntax, description, parameters, and roles for cmm rait command.

Syntax

Description

Configure, list and show Routing Area Identification.

Parameters

Name

raitId - Rai id

Syntax`--raitId <integer>`**Description**

RAI identifier.

Class

Required key.

Type / Range

Integer in the range of 1 to 3,000

Name

lac - LAC

Syntax`--lac <integer>`**Description**

Location area code.

Class

Required on create.

Type / Range

Integer in the range of 1 to 65,533

Name

mcc - MCC

Syntax

--mcc <digits>

Description

Mobile country code digits of broadcasted PLMN.

Class

Required on create.

Type / Range

Digit string of length 3 to 3 with the allowed pattern of [0-9]*

Name

mnc - MNC

Syntax

--mnc <digits>

Description

Mobile network code digits of broadcasted PLMN.

Class

Required on create.

Type / Range

Digit string of length 2 to 3 with the allowed pattern of [0-9]*

Name

rac - RAC

Syntax

--rac <integer>

Description

Routing area code.

Class

Required on create.

Type / Range

Integer in the range of 0 to 255

Name

epind - Epind

Syntax

--epind <integer>

Description

Identify equivalent PLMN list index: decimal number 0 ... 5 or 255 0 ... all equivalent PLMN are used 255 ... no equivalent PLMN are used.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.96 cmm raitOfRnc

rait Of Rnc. Syntax, description, parameters, and roles for cmm raitOfRnc command.

Syntax**Description**

Configure, list and show RAI of RNC.

Parameters**Name**

raitId - Rai id

Syntax

--raitId <integer>

Description

RAI identifier. this parameter should match parameter raitid of table rait.

Class

Required key.

Type / Range

Integer in the range of 1 to 3,000

Name

rncName - RNC name

Syntax

--rncName <string>

Description

RNC name. this parameter should match parameter rncname of table rnc.

Class

Required key.

Type / Range

Digit string of length 1 to 10 with the allowed pattern of [0-9a-zA-Z\._:-]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.97 cmm regionLac

Region LA Lists. Syntax, description, parameters, and roles for cmm regionLac command.

Syntax**Description**

This command is used to administer a Region LA List and includes parameters for defining the mobile network name and Location Area Codes (LACs). The maximum number of LACs per region is 1024.

Parameters

Name

regionListName - Region list name

Syntax

--regionListName <string>

Description

Name that identifies region.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

plmnName - Mobile network name

Syntax

--plmnName <string>

Description

Name that identifies mobile network containing the given region.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

lac - Location area code

Syntax

--lac <integer>

Description

This element represents a LAC (Location area code) to be added to the LA list of a given region.

Class

Required key.

Type / Range

Integer in the range of 1 to 65,535

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.98 cmm regionList

LA TA Region. Syntax, description, parameters, and roles for cmm regionList command.

Syntax

Description

This command is used to administer an LA TA Region and includes parameters for defining the region and mobile network names and whether the region will be used by the Regional UE Detach application. The maximum number of regions is 3000.

Parameters

Name

name - Region name

Syntax

--name <string>

Description

Name that identifies region.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

plmnName - Mobile network name

Syntax

--plmnName <string>

Description

Name that identifies mobile network containing the given region.

Restrictions

plmnType of plmnName must not be Other Network

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

description - Region description

Syntax

--description <string>

Description

Region description.

Class

Optional.

Type / Range

Digit string of length 0 to 160 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Name

regionalUeDetach - Regional UE detach

Syntax

--regionalUeDetach <list-item>

Description

If set to true, the regional UE detach application will use this region when determining if UE has transitioned from one region to another.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.99 cmm regionTac

Region TA Lists. Syntax, description, parameters, and roles for cmm regionTac command.

Syntax

Description

This command is used to administer a Region TA List and includes parameters for defining the mobile network name and Tracking Area Codes (TACs). The maximum number of TACs per region is 1024.

Parameters

Name

regionListName - Region label

Syntax

--regionListName <string>

Description

Name that identifies region.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

plmnName - Mobile network name

Syntax

--plmnName <string>

Description

Name that identifies mobile network containing the given region.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

tac - Tracking area code

Syntax

--tac <integer>

Description

This element represents a TAC (Tracking area code) to be added to the TA list of a given region.

Class

Required key.

Type / Range

Integer in the range of 1 to 65,535

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.100 cmm registeredUePerTa

Registered UE per TA. Syntax, description, parameters, and roles for cmm registeredUePerTa command.

Syntax**Description**

This command creates a list of TA which can be used to assign maximum number of UEs allowed per TA in the list.

Parameters

Name

registeredUePerTaListName - Name of the list

Syntax

--registeredUePerTaListName <string>

Description

Name of the list.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

servingPlmnName - Serving PLMN Name

Syntax

--servingPlmnName <string>

Description

The serving PLMN is either MME home network or shared network.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

tac - Tracking Area Code

Syntax

--tac <integer>

Description

Tracking Area Code

Class

Required key.

Type / Range

Integer in the range of 1 to 65,535

Name

maxUeAllowedPerTac - Maximum UE Allowed PER Tracking Area Code

Syntax

--maxUeAllowedPerTac <integer>

Description

Provision maximum number of UE allowed in the TAC of the serving PLMN.

Class

Optional.

Type / Range

Integer in the range of 0 to 4,294,967,294

Default

0

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.101 cmm registeredUePerTaList

Registered UE per TA List. Syntax, description, parameters, and roles for cmm registeredUePerTaList command.

Syntax**Description**

Add or delete a TAC to the TAI list used for restricting max number of roamer UE.

Parameters**Name**

name - Registered UE per TA List Name

Syntax

--name <string>

Description

Name of the registered UE per TA List.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.102 cmm registeredUePerTaRange

Registered UE per TA List. Syntax, description, parameters, and roles for cmm registeredUePerTaRange command.

Syntax**Description**

This command creates a list of TA which can be used to assign maximum number of UEs allowed per TA in the list.

Parameters**Name**

registeredUePerTaListName - Name of the list

Syntax

--registeredUePerTaListName <string>

Description

Name of the list.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

servingPlmnName - Serving PLMN Name

Syntax

--servingPlmnName <string>

Description

The serving PLMN is either MME home network or shared network.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

allTac - All Tracking Area Codes

Syntax

--allTac <list-item>

Description

This parameter can be used to configure Max Allowed UE per TAC for all the TACs in the PLMN. If the parameter is set to False then the 'minTac' and 'maxTac' parameters are used.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

maxTac - Max Tracking Area Code

Syntax

--maxTac <integer>

Description

Ending TAC of a TA range.

Class

Optional.

Type / Range

Integer in the range of 0 to 65,535

Default

0

Name

maxUeAllowedPerTac - Maximum UE Allowed PER Tracking Area Code

Syntax`--maxUeAllowedPerTac <integer>`**Description**

Provision maximum number of UE allowed in the TAC of the serving PLMN.

Class

Optional.

Type / Range

Integer in the range of 0 to 4,294,967,294

Default

0

Name

minTac - Min Tracking Area Code

Syntax`--minTac <integer>`**Description**

The starting Tracking Area Code of a TA range.

Class

Optional.

Type / Range

Integer in the range of 0 to 65,535

Default

0

Roles**Modify**

sysadmin, manage

6.103 cmm remoteCopySetup

remote-copy parameters. Syntax, description, parameters, and roles for `cmm remoteCopySetup` command.

Syntax

Description

Create, list, modify, delete and show remote-copy parameters.

Parameters

Name

remoteId - Remote Copy Name

Syntax

--remoteId <list-item>

Description

Remote-copy name.

Class

Required key.

List

- remote-config - Specify remote copy of remote-config.

Name

remoteDir - Remote directory

Syntax

--remoteDir <string>

Description

Remote-copy directory.

Class

Required on create.

Type / Range

Digit string of length 1 to 512 with the allowed pattern of `^/(/?[\w\-\~\.\,]+)/?$`

Name

remoteLogin - Remote login name

Syntax

--remoteLogin <string>

Description

Remote login name.

Class

Required on create.

Type / Range

User name in Linux format. Username must start with a letter, a digit, an underscore or a full stop. The last character must be a letter, a digit, an underscore, a hyphen, a dollar sign or a

full stop. The other characters must be letters, digits, underscores, hyphens or full stops.
The length must be between 1 and 32 characters.

Name

remoteCopyFqdn - Remote Copy FQDN

Syntax

--remoteCopyFqdn <string>

Description

The parameter defines the Remote Copy FQDN.

Restrictions

Either the Remote Copy FQDN or the remote IP must be set

Class

Optional.

Type / Range

Digit string of length 0 to 255 with the allowed pattern of [A-Za-z0-9.-]

Default

""

Name

remotelp - Remote IP Address

Syntax

--remoteIp <string>

Description

Remote IP-address.

Restrictions

Either the PCMD Analytics FQDN or the Primary IP must be set

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Default

""

Name

secondaryRemotelp - Secondary Remote IP Address

Syntax

--secondaryRemoteIp <string>

Description

Secondary Remote IP Address. If backup to the first remoteIP fails, this IP will be attempted. If not populated there will be no secondary attempt.

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.104 cmm restNasMappingProfile

Access Restriction To NAS Cause Mapping Profile. Syntax, description, parameters, and roles for cmm restNasMappingProfile command.

Syntax**Description**

This command is used to administer the Access Restriction To NAS Cause Mapping Profile. This profile is used to select the right cause code to send to the UE when a UE mobility management request is rejected due to an access restriction. The maximum number of profiles supported is 1024. A default profile, named is provided.

Parameters**Name**

name - Name of access restriction to NAS cause code profile name

Syntax

--name <string>

Description

Profile name for access restriction to NAS cause code mapping.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

attachRequestWithoutSmsGrpId - Attach request without SMS group ID

Syntax

--attachRequestWithoutSmsGrpId <integer>

Description

Selection EMM CC to be used when MME rejects attach request without SMS indication when bit 0 of local-group-id AVP is set to indicate that UE must indicate SMS request.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

14

Name

csgRestricted - CSG restricted

Syntax

--csgRestricted <integer>

Description

NAS cause code to be sent if UE MME is rejected because of CSG restrictions.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

25

Name

description - Description of profile

Syntax

--description <string>

Description

Description of profile.

Class

Optional.

Type / Range

Digit string of length 0 to 160 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

""

Name

emmCcForEsmCc29 - EMM CC for ESM CC29

Syntax

--emmCcForEsmCc29 <integer>

Description

Configure NAS MM CC for attach reject when PDN connectivity request is rejected with cause code #29.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

15

Name

emmCcForEsmCc72 - EMM CC for ESM CC72

Syntax

--emmCcForEsmCc72 <integer>

Description

Configure NAS MM CC for attach reject when MME receives Create Session Response with cause code #72 for homers or treat roamer as homer.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

19

Name

emmCcForEsmCc92 - EMM CC for ESM CC92

Syntax

--emmCcForEsmCc92 <integer>

Description

Configure NAS MM CC for attach reject when MME receives Create Session Response with cause code #92 for homers or treat roamer as homer. If gParms enableProvCc is enabled, this field is not used unless generalGParms overrideEnhEmmCcCreateSessRespUserAuthFail is enabled.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

19

Name

emmRegisteredWithoutPdnNotAllowed - EMM-REGISTERED without PDN not allowed

Syntax

--emmRegisteredWithoutPdnNotAllowed <integer>

Description

NAS cause code to be sent if attach or TAU request is rejected if EMM-REGISTERED without PDN connection feature support is not enabled.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

15

Name

eutranNotAllowedHplmn - EUTRAN restricted in the UE's HPLMN

Syntax

--eutranNotAllowedHplmn <integer>

Description

NAS cause code to be sent if UE MME is rejected because a roaming UE is not allowed to access LTE.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

7

Name

imsApnRoamingFailure - IMS APN roaming failure

Syntax`--imsApnRoamingFailure <integer>`**Description**

NAS cause code to be sent if attach is rejected because of IMS APN roaming failure.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

19

Name

interPlmnTauRejDuelmsRoamControl - Inter plmn TAU reject due to IMS roaming control

Syntax`--interPlmnTauRejDueImsRoamControl <integer>`**Description**

NAS cause code to be sent if inter PLMN TAU is rejected because of IMS APN roaming control.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

40

Name

interPlmnTauReject - Inter PLMN TAU reject

Syntax`--interPlmnTauReject <integer>`**Description**

NAS cause code to be used by MME when inter-plmn TAU request is rejected.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

9

Name

lteBarAllPacketOdb - ODB - all packet oriented services barred

Syntax

--lteBarAllPacketOdb <integer>

Description

NAS cause code to be sent if UE MME is rejected because LTE all packet oriented services barred.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

8

Name

ItemNotAllowedHplmn - LTE-M Not Allowed in HPLMN

Syntax

--ItemNotAllowedHplmn <integer>

Description

NAS cause code to be sent to UE if it is not allowed LTE-M access

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

15

Name

ltePlmnNotAllowed - PLMN of IMSI is not allowed

Syntax

--ltePlmnNotAllowed <integer>

Description

NAS cause code to be sent if UE MME request is rejected because PLMN access is not allowed.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

11

Name

nbiotAttachSgsFail - NB-IOT Attach Sgs Fail

Syntax

--nbiotAttachSgsFail <integer>

Description

NAS cause code to be sent in attach reject if fails to set up SGs association.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

15

Name

nbiotBarAllPacketOdb - NB-IOT Bar All Packet ODB

Syntax

--nbiotBarAllPacketOdb <integer>

Description

Specifies the cause code to send to a NB-IoT UE that has ODB.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

8

Name

nbiotNotAllowed - NB-IOT Not Allowed

Syntax

--nbiotNotAllowed <integer>

Description

NAS cause code to be sent if CIOT is not allowed.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

15

Name

nbiotTauSgsFail - NB-IOT TAU Sgs Fail

Syntax

--nbiotTauSgsFail <integer>

Description

NAS cause code to be sent in TAU reject if fails to set up SGs association.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

15

Name

notAllowedTaiHplmn - UE is not allowed in a TAI in HPLMN

Syntax

--notAllowedTaiHplmn <integer>

Description

NAS cause code to be sent if UE MME is rejected because UE is in a restricted TA where UE is not allowed to access services.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

12

Name

pdnTypeIpv6RestrictionTauRejectionCause - PDN Type IPv6 Restriction TAU Rejection Cause

Syntax

--pdnTypeIpv6RestrictionTauRejectionCause <integer>

Description

Defines the EMM cause value when MME rejects the inter MME or inter RAT TAU due to PDN Type IPv6 being restricted, and the UE not having any PDN that can be maintained at the new MME.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

15

Name

rejectionS1u - Rejection S1U

Syntax

--rejectionS1u <integer>

Description

Rejection cause code when UE requests to establish S1-U by sending TAU with active flag, SR or CPSR when a UE has all PDN connections data transport over CP.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

35

Name

rfspAccessRestrictionCc - RFSP Access Restriction Cause Code

Syntax

--rfspAccessRestrictionCc <integer>

Description

The Cause Code to be sent to a UE if it is restricted access based on RFSP

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

15

Name

roamingEmmCcForEsmCc72 - Roaming EMM CC for ESM CC72

Syntax

```
--roamingEmmCcForEsmCc72 <integer>
```

Description

Configure NAS MM CC for attach reject when MME receives Create Session Response with cause code #72 for roamers.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

19

Name

roamingEmmCcForEsmCc92 - Roaming EMM CC for ESM CC92

Syntax

```
--roamingEmmCcForEsmCc92 <integer>
```

Description

Configure NAS MM CC for attach reject when MME receives Create Session Response with cause code #92 for roamers. If gParms enableProvCc is enabled, this field is not used unless generalGParms overrideEnhEmmCcCreateSessRespUserAuthFail is enabled.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

19

Name

roamingLtemNotAllowed - Roaming LTE-M Not Allowed

Syntax

```
--roamingLtemNotAllowed <integer>
```

Description

NAS cause code to be sent if inbound roaming UE is not allowed access to LTE-M

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

15

Name

roamingNbiotNotAllowed - Roaming NB-IoT Not Allowed

Syntax`--roamingNbiotNotAllowed <integer>`**Description**

NAS cause code to be sent to UE if it is not allowed NB-IoT access.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

15

Name

roamingUeEutranNotAllowedVplmn - Roaming UE EUTRAN restricted in a VPLMN

Syntax`--roamingUeEutranNotAllowedVplmn <integer>`**Description**

NAS cause code to be sent if UE MME is rejected because a roaming UE is not allowed to roam in the VPLMN.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

14

Name

roamingUeNotAllowedTai - Roaming UE is not allowed in the TAI

Syntax`--roamingUeNotAllowedTai <integer>`**Description**

NAS cause code to be sent if UE MME is rejected because UE is in a restricted TA where UE is not allowed to roam.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

12

Name

roamingWbEutranExceptLtemNotAllowed - Roaming WB-UTRAN Except LTE-M Not Allowed

Syntax

--roamingWbEutranExceptLtemNotAllowed <integer>

Description

NAS cause code to be sent to UE if it is not allowed WB EUTRAN except LTE-M access.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

15

Name

targetNodeLte - LTE target access restricted by node

Syntax

--targetNodeLte <integer>

Description

NAS cause code to send to the UE when we receive GTPv2 context response with cause code "Target access restricted for the subscriber".

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

15

Name

tauRequestWithoutSmsGrpId - TAU request without SMS group ID

Syntax

--tauRequestWithoutSmsGrpId <integer>

Description

Selection EMM CC to be used when MME rejects TAU request without SMS indication when bit 0 of local-group-id AVP is set to indicate that UE must indicate SMS request.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

14

Name

unsupportedFeatureRestriction - Roaming restricted due to unsupported features

Syntax

--unsupportedFeatureRestriction <integer>

Description

NAS cause code to be sent if UE MME is rejected because of unsupported feature restrictions.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

14

Name

wbEutranExceptLtemNotAllowedHplmn - WB-EUTRAN Except LTE-M Not Allowed in HPLMN

Syntax

--wbEutranExceptLtemNotAllowedHplmn <integer>

Description

NAS cause code to be sent to inbound roaming UE if it is not allowed WB EUTRAN except LTE-M access.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

15

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.105 cmm restorationAcc

Restoration Access. Syntax, description, parameters, and roles for `cmm restorationAcc` command.

Syntax

Description

This command is used to administer restoration access control and includes parameters for defining the remote node IP address and whether it is a primary access record. The 'shutdown' parameter is used to enable/shutdown the interface.

Parameters

Name

rmtNodeName - Remote node name

Syntax

--rmtNodeName <string>

Description

Remote node name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

rmtNodeIp - Remote node IP address

Syntax

--rmtNodeIp <string>

Description

Remote Node IP address

Class

Optional.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

shutdown - Shutdown

Syntax

--shutdown <list-item>

Description

When it is true, indicates that interface is shut down.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Name

srsPrimaryAcc - SRS primary access

Syntax

--srsPrimaryAcc <list-item>

Description

Indicates whether it is a primary access record or not.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.106 cmm restorationApn

Restoration APN. Syntax, description, parameters, and roles for `cmm restorationApn` command.

Syntax

Description

This command is used to administer a Restoration APN Network Identifier.

Parameters

Name

srsApn - APN network identifier

Syntax

--srsApn <string>

Description

APN network identifier.

Class

Required key.

Type / Range

Digit string of length 1 to 62 with the allowed pattern of [0-9a-zA-Z\.\-]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.107 cmm restorationApnList

Restoration APN List. Syntax, description, parameters, and roles for cmm restorationApnList command.

Syntax

Description

This command is used to administer a Restoration APN List. The APN list consists of APNs whose bearer context needs to be stored on the Session Restoration Server (SRS) for restoration in case of MME failure/restart. MME does not save any bearer context if there is no APN list or if the APN list has no APN-NI. Only one APN list can be created. Each APN list can have up to three APN-NI.

Parameters

Name

name - Restoration APN list name

Syntax

--name <string>

Description

Restoration APN list name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.108 cmm restorationConfig

Session Restoration Configuration. Syntax, description, parameters, and roles for cmm restorationConfig command.

Syntax

Description

This command configures the inter-AMF session restoration architecture. A default config named Default is auto created and it cannot be deleted.

Parameters

Name

name - Name

Syntax

--name <string>

Description

Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"default"

Name

realmId - Realm ID

Syntax

--realmId <string>

Description

The realm Id under which the records are stored in UDSF. Applicable only when restoration mode is UDSF.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

""

Name

restorationMode - Restoration Mode

Syntax

--restorationMode <list-item>

Description

UE Context Restoration Mode. SRS: UE Context is stored in backup CMM UDSF: UE Context is stored in UDSF. when mode is UDSF StorageId and realmId must not be Null.

Class

Optional.

List

- SRS - Inter-CMM based UE context backup storage
- UDSF - UDSF based UE context backup storage

Default

SRS

Name

storageId - Storage ID

Syntax

--storageId <string>

Description

The Storage Id under which the records are stored in UDSF. Applicable only when restoration mode is UDSF.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.109 cmm restorationEndPt

Restoration End Point. Syntax, description, parameters, and roles for cmm restorationEndPt command.

Syntax

Description

This command is used to configure the CMMs in a pool that are Restoration Endpoints. A Restoration Endpoint is a CMM that is available to be used as 1st session restoration server (i.e., this endpoint is used by the CMM to store session restoration contexts), or as a source of restoration contexts (i.e., the CMM uses it to retrieve a session restoration context). A CMM is typically configured with restoration endpoints to all other CMMs in the pool, however, only one endpoint is configured as session restoration server (--srsServer is true).

Parameters

Name

srsName - SRS Name

Syntax

--srsName <string>

Description

It is a local string that identifies another CMM as a restoration endpoint

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

interfaceName - Interface Name

Syntax

--interfaceName <list-item>

Description

Interface Name

Class

Required on create, not modifiable.

List

- SBI_NR10 - Specify SBI_NR10 interface.
- SBI_NR10_2 - SBI_NR_10_2 - Specify SBI_NR10_2 interface.
- SBI_NR10_3 - Specify the SBI_NR10_3 interface.
- SBI_NR10_4 - Specify the SBI_NR10_4 interface.
- SBI_NR10_5 - Specify the SBI_NR10_5 interface.
- SBI_NR10_6 - Specify the SBI_NR10_6 interface.
- SBI_NR10_7 - Specify the SBI_NR10_7 interface.
- SBI_NR10_8 - Specify the SBI_NR10_8 interface.
- SBI_NR10_9 - Specify the SBI_NR10_9 interface.
- SBI_NR10_10 - Specify the SBI_NR10_10 interface.
- SBI_NR10_11 - Specify the SBI_NR10_11 interface.
- SBI_NR10_12 - Specify the SBI_NR10_12 interface.
- SBI_NR10_13 - Specify the SBI_NR10_13 interface.
- SBI_NR10_14 - Specify the SBI_NR10_14 interface.
- SBI_NR10_15 - Specify the SBI_NR10_15 interface.
- SBI_NR10_16 - Specify the SBI_NR10_16 interface.

Name

rmtNodeIp - Remote Node Ip

Syntax

--rmtNodeIp <string>

Description

IP address of a CMM that is uses in the pool restoration feature

Class

Required on create.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

sysType - System Type

Syntax

--sysType <list-item>

Description

System type (List of MME, AMF)

Class

Required on create, not modifiable.

List

- MME - MME
- AMF - AMF

Name

state - Administrative state

Syntax

--state <list-item>

Description

Administrative state.

Class

Read only.

List

- locked - Locked.
- unlocked - Unlocked.

Default

locked

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.110 cmm restorationEndPtList

Restoration End Point List. Syntax, description, parameters, and roles for cmm restorationEndPtList command.

Syntax

Description

List of restoration End Points - Contains more than one entry when using multiple IPDSs

Parameters

Name

name - List Name

Syntax

--name <string>

Description

Restoration End Point List Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

sysType - System Type

Syntax

--sysType <list-item>

Description

System Type (List of MME, AMF)

Class

Required on create, not modifiable.

List

- MME - MME
- AMF - AMF

Name

srsServer - SRS Server

Syntax

--srsServer <list-item>

Description

If this parameter is set to true it indicates that the MME identified by the corresponding restoration end point list item(s) are used to store this MME restoration contexts.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.111 cmm restorationEndPtListItem

Restoration End Point List Item. Syntax, description, parameters, and roles for cmm restorationEndPtListItem command.

Syntax**Description**

Add or delete a restoration End Point to the named list.

Parameters**Name**

restorationEndPtListName - Reatoration End Point List Name

Syntax

--restorationEndPtListName <string>

Description

List of Restoration End Points

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

restorationEndPtName - Restoration End Point Name

Syntax

--restorationEndPtName <string>

Description

Name of the Restoration End Point

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.112 cmm restorationSync

Restoration Synchronization. Syntax, description, parameters, and roles for cmm restorationSync command.

Syntax

Description

This command synchronizes the SRS restoration server contexts after an SU.

Parameters

Name

primaryReSync - Primary Resynchronization

Syntax

--primaryReSync <list-item>

Description

This parameter indicates the CMM SRS server should resync the UE contexts with the Primary Server

Class

Optional.

List

- 4g - 4G contexts - This indicates to rsync 4G contexts
- 4g5g - 5G and 4G contexts - This indicates to rsync both 5G and contexts
- 5g - 5G contexts - This indicates to rsync 5G contexts
- none - Not Applicable

Default

none

Name

srsReSync - SRS Resynchronization

Syntax

--srsReSync <list-item>

Description

This parameter indicates the CMM Primary server should resync the UE contexts with the SRS Server

Class

Optional.

List

- 4g - 4G contexts - This indicates to rsync 4G contexts
- 4g5g - 5G and 4G contexts - This indicates to rsync both 5G and contexts
- 5g - 5G contexts - This indicates to rsync 5G contexts
- none - Not Applicable

Default

none

Roles

Create

sysadmin, manage

6.113 cmm restorationSyncStatus

Restoration Synchronization Status. Syntax, description, parameters, and roles for cmm restorationSyncStatus command.

Syntax

Description

Restoration Synchronization Status Status of the last or running restoration command

Parameters

Name

primaryReSync - Primary Resynchronization

Syntax

--primaryReSync <list-item>

Description

This parameter indicates the CMM SRS server should resync the UE contexts with the Primary Server

Class

Optional.

List

- 4g - 4G contexts - This indicates to rsync 4G contexts
- 4g5g - 5G and 4G contexts - This indicates to rsync both 5G and contexts
- 5g - 5G contexts - This indicates to rsync 5G contexts
- none - Not Applicable

Default

none

Name

primaryReSyncStatus - Primary Resynchronization Status

Syntax

--primaryReSyncStatus <list-item>

Description

Status of the last or running Primary Recsync command

Class

Optional.

List

- Aborted - Aborted - This state is entered when nr10 links are down or administratively locked. It can also be entered when 10 nr10 heartbeats are missed.
- Completed - Completed
- Inprogress - In Progress
- None - None
- Pending - Pending - This state is a transient state when the paired CMM has not replied to the resync start.

Name

srsReSync - SRS Resynchronization

Syntax

--srsReSync <list-item>

Description

This parameter indicates the CMM Primary server should resync the UE contexts with the SRS Server

Class

Optional.

List

- 4g - 4G contexts - This indicates to rsync 4G contexts
- 4g5g - 5G and 4G contexts - This indicates to rsync both 5G and contexts
- 5g - 5G contexts - This indicates to rsync 5G contexts
- none - Not Applicable

Default

none

Name

srsReSyncStatus - SRS Resynchronization Status

Syntax

--srsReSyncStatus <list-item>

Description

Status of the last or running SRS resync command

Class

Optional.

List

- Aborted - Aborted - This state is entered when nr10 links are down or administratively locked. It can also be entered when 10 nr10 heartbeats are missed.
- Completed - Completed
- Inprogress - In Progress
- None - None
- Pending - Pending - This state is a transient state when the paired CMM has not replied to the resync start.

Roles

List

sysadmin, manage, monitor

6.114 cmm rfspAccessRestrictionList

RFSP Access Restriction List. Syntax, description, parameters, and roles for cmm rfspAccessRestrictionList command.

Syntax

Description

Creates list names to be used for an RFSP Access Restriction List

Parameters

Name

name - Name

Syntax

--name <string>

Description

The name of the RFSP Access Restriction List

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.115 cmm rfspAccessRestrictionListItem

RFSP Access Restriction List Item. Syntax, description, parameters, and roles for cmm rfspAccessRestrictionListItem command.

Syntax

Description

The RFSP value to assign to a RFSP Access Restriction List.

Parameters

Name

rfspAccessRestrictionListName - RFSP Access Restriction List Item

Syntax

--rfspAccessRestrictionListName <string>

Description

RFSP value to be added to a list.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

rfspValue - RFSP Value

Syntax

--rfspValue <integer>

Description

Value of the RFSP to be assigned to a list

Class

Required key.

Type / Range

Integer in the range of 1 to 256

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.116 cmm rmtAmfInstance

Remote AMF Instance. Syntax, description, parameters, and roles for `cmm rmtAmfInstance` command.

Syntax**Description**

This command is used to create a new AMF instance and modify and delete an existing AMF instance. An AMF instance is identified by its Instance ID.

Parameters**Name**

plmnName - PLMN Name

Syntax

--plmnName <string>

Description

Name of the PLMN of the AMF. Must be set to own AMF's Home PLMN. Currently not allowed to set to Shared PLMN (restriction is not enforced by CLI but impact is undefined). Also not allowed to set to Roaming PLMN as local provisioning is not relevant for NF in different PLMNs

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

amfRegionId - AMF Region ID

Syntax

--amfRegionId <integer>

Description

AMF Region Identifier is used to identify an AMF group serving a region. 5G AMF are organized into an AMF region, AMF set and AMF pointer. An AMF is uniquely identified by PLMN ID, AMF Region ID, AMF Set and AMF pointer. The PLMN/Region/AMF Set combination may point to a same or different AMF Set than the one CMM belongs. Region Id is given in decimal format.

Class

Required key.

Type / Range

Integer in the range of 0 to 255

Name

amfSetId - AMF Set Identifier

Syntax

--amfSetId <integer>

Description

AMF Set Identifier is used to identify a set of AMF within an AMF region. An AMF is uniquely identified by PLMN ID, AMF Region ID, AMF Set and AMF pointer.

Class

Required key.

Type / Range

5G AMF Set ID

Name

amfPointer - AMF Pointer

Syntax

--amfPointer <integer>

Description

The AMF Pointer identifies an AMF in an AMF set. An AMF is uniquely identified by PLMN ID, AMF Region ID, AMF Set and AMF pointer.

Class

Required key.

Type / Range

5G AMF Pointer

Name

instanceId - Instance Identifier

Syntax

--instanceId <string>

Description

AMF Instance Identifier is a UUID. The operator must obtain the UUID from the AMF configuration data. The UUID must have the following string format: 8 hex digits-4 hex digts-4hex digits-4 hex digits-12hex digits. Example: 4947a69a-f61b-4bc1-b9da-47c9c5d14b64. Used for AMF selection when AMF discovery is based on target-nf-instance-id

Class

Required on create, not modifiable.

Type / Range

Digit string of length 36 to 36 with the allowed pattern of [0-9a-fA-F]{8}\-[0-9a-fA-F]{4}\-[0-9a-fA-F]{4}\-[0-9a-fA-F]{4}\-[0-9a-fA-F]{12}

Name

capacity - AMF Relative Capacity

Syntax

--capacity <integer>

Description

Relative capacity of the AMF as defined in NFProfile level in 29.510. It is used for load balancing among NFs with equal Priority. As per 29.510 this parameter is used to deduce the NF Service instance Capacity when -and only when- the relevant information is not available on Service Instance level. However since locally provisioned Service instances always have a default capacity value this parameter has no effect for load balancing Service Request (only relevant for Default Notifications)

Class

Optional.

Type / Range

Integer in the range of 0 to 65,535

Default

65535

Name

enable - Enable AMF

Syntax

--enable <list-item>

Description

The parameter is used to enable use of the provisioned instance of the AMF. If not enabled the NF will not be considered as candidate for selection,

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

fqdn - AMF FQDN

Syntax

--fqdn <string>

Description

AMF FQDN. This parameter is currently not used by CMM

Restrictions

FQDN is an optional parameter in CMM19.0 and it is not used even it is provisioned. The AMF instance must be provisioned either with IPv4 or IPv6 address.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 128

Name

ipAddress - IP V4 Address

Syntax

--ipAddress <string>

Description

An IPv4 or IPv6 address needs to be provisioned if an IP address and the FQDN for an AMF service Instance is not provisioned.

Class

Optional.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Default

0.0.0.0

Name

locality - Location of the AMF

Syntax

--locality <string>

Description

This parameter is used to configure AMF location like geographical location or data center name. Locality of the NF as defined per NF Profile in 29.510. When set, if the locality of the NF matches the locality of the own AMF (amfInstance --locality) the NF is selected with highest priority (if all other criterions are also met). If own AMF has not be provisioned with locality this parameter has no effect.

Class

Optional.

Type / Range

Digit string of length 0 to 64

Default

""

Name

nfServicesListName - NF Service List Name

Syntax

--nfServicesListName <string>

Description

Name of the AMF services list. This list consists of services instances supported by the remote AMF. References a list which links to the NFs service instances. A NF may be provisioned to have multiple service instances for the same service type. For more details on the Services refer to nfServiceInstance command

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

priority - AMF Priority

Syntax

--priority <integer>

Description

Indicates priority value relative to the other AMFs in the same AMF set. Relative Priority of the NF as defined in NFProfile level in 29.510. It is used for shortlisting NFs As per 29.510 this parameter is used to deduce the NF Service instance priority when -and only when- the relevant information is not available on Service Instance level. However since locally provisioned Service instances always have a default priority value this parameter has no effect for shortlisting load balancing Service Requests (only relevant for Default Notifications)

Class

Optional.

Type / Range

Integer in the range of 0 to 65,535

Default

1

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.117 cmm rmtEndPtCfg

Remote End Point Configuration. Syntax, description, parameters, and roles for cmm rmtEndPtCfg command.

Syntax

Description

This command is used to administer a Remote End Point Configuration and includes parameters for defining the interface type, DPR disconnect cause, destination host and realm AVP values, IP addresses, end point ID, and port number. The 'shutdown' parameter is used to enable/shutdown the remote endpoint.

Parameters

Name

name - Name

Syntax

--name <string>

Description

Label.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

interfaceName - Interface name

Syntax

--interfaceName <list-item>

Description

Type of interfaces.

Class

Required on create, not modifiable.

List

- S13 - Specify S13 interface to an EIR.
- S6A - Specify S6a interface to an HSS.
- SBC - Specify the SBc interface to a CBC.
- SGD - Specify SGd interface to an SMSC/SMS router.
- SGS - Specify the SGs interface to a GSM/UMTS MSC.
- SGS_LITE - Specify SGs lite interface to an IWF.
- SLG - Specify the Slg interface to a GMLC.
- SLS - Specify the Sls interface to an E-SMLC.
- SV - Specify the Sv interface to a GSM/UMTS MSC.
- T6A - Specify T6a interface to a SCEF.

Name

dprDisconnectCause - DPR disconnect cause

Syntax

--dprDisconnectCause <list-item>

Description

DPR disconnect cause.

Restrictions

dprDisconnectCause parameter is only applicable when interfaceName is S6A, S13, SLG or SGD.

Class

Optional.

List

- Busy - The peer's internal resources are constrained, and it has determined that the transport connection needs to be closed. a receiver of a DPR with above result code SHOULD NOT attempt reconnection.
- DoNotWantToTalkToYou - Do not want to talk to you - The peer has determined that it does not see a need for the transport connection to exist, since it does not expect any messages to be exchanged in the near future. a receiver of a DPR with above result code SHOULD NOT attempt reconnection.
- Reboot - A scheduled reboot is imminent. a receiver of a DPR with above result code MAY attempt reconnection.

Default

DoNotWantToTalkToYou

Name

draDestHostFqdnName - DRA destination host name

Syntax

--draDestHostFqdnName <string>

Description

If MME does not have destination host in VLR (UE shows up for the first time), and the destination is a DRA, MME uses the value of this new field to populate destination host AVP in AIR and ULR.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

''

Name

draDestRealmFqdnName - DRA destination realm name

Syntax

--draDestRealmFqdnName <string>

Description

If MME does not have destination realm in VLR (UE shows up for the first time), and the destination is a DRA, MME uses the value of this new field to populate destination realm AVP in AIR and ULR.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

draSupported - DRA supported

Syntax

--draSupported <list-item>

Description

When the drasupported parameter is true, if MME does not have destination-host/destination-realm in VLR (UE shows up for the first time), MME uses the value of the field draDestRealmFqdnName to populate destination-realm AVP in AIR and ULR.

Restrictions

draSupported parameter is only applicable when interfaceName is S6A, S13, SLG or SGD.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

endPtId - End point ID

Syntax

--endPtId <integer>

Description

Unique numeric ID for this remote end point IP address(es). If the maximum number of HSS and EIRs are provisioned, it is recommended to place the MSC IP id in the higher range of

100-200 to leave enough entries available for HSS and EIR reconfiguration procedures. in particular, when converting HSS / EIR from IPv4 to IPv6, twice the number of S6ad and S13 remote endpoints may be defined during the conversion, and the remote endpoint IP ID (1-100 range) will be filled by the SGa/S13 interfaces, requiring the SGs interfaces to utilize the higher range of 100-200.

Class

Optional.

Type / Range

Integer in the range of 1 to 1,024

Name

ip1 - IP address 1

Syntax

--ip1 <string>

Description

First IP address.

Class

Optional.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Default

0.0.0.0

Name

ip2 - IP address 2

Syntax

--ip2 <string>

Description

Secondary IP address.

Restrictions

ip2 parameter is only applicable when interfaceName is S6A, S13, SLG, SBC, SGS, SLS or SGD.

Class

Optional.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Default

0.0.0.0

Name

port - Port

Syntax

--port <integer>

Description

Port number.

Restrictions

0 forces the default port to be used. The default port values are as follows: S6A:3868, S13:3868, SBC:29168, SLG:3868, SGS:29118, SLS:9082, SV:2123, SGD 3868

Class

Optional.

Type / Range

Integer in the range of 0 or 1,024 to 65,535

Default

0

Name

scefProfileName - SCEF Profile Name

Syntax

--scefProfileName <string>

Description

SCEF Profile

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Name

shutdown - Shutdown

Syntax

--shutdown <list-item>

Description

Indicates if the remote endpoint is shutdown.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

shutdownReconnectTimer - Shutdown reconnect timer

Syntax

--shutdownReconnectTimer <integer>

Description

Shutdown reconnect timer.

Restrictions

shutdownReconnectTimer is only applicable when interfaceName is S6A, S13, SLG or SGD. It must be 0 for other interface types. shutdownReconnectTimer is entered in units of milliseconds and must be a multiple of 100.

Class

Optional.

Type / Range

Integer in the range of 0 to 10,000 in units of Millisecond

Default

0

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.118 cmm rnc

radio network controller. Syntax, description, parameters, and roles for cmm rnc command.

Syntax

Description

Configure, list and show Radio Network controllers.

Parameters

Name

rncName - RNC name

Syntax

--rncName <string>

Description

RNC name.

Class

Required key.

Type / Range

Digit string of length 1 to 10 with the allowed pattern of [0-9a-zA-Z\.\-_:]*

Name

mcc - MCC

Syntax

--mcc <digits>

Description

Mobile country code digits of global RNC identifier.

Class

Required on create.

Type / Range

Digit string of length 3

Name

mnc - MNC

Syntax

--mnc <digits>

Description

Mobile network code digits of global RNC identifier.

Class

Required on create.

Type / Range

Digit string of length 2 to 3

Name

rnclId - RnclId

Syntax

--rncId <integer>

Description

RNC identifier part of global RNC identifier.

Class

Required on create.

Type / Range

Integer in the range of 1 to 65,534

Name

dig - Digits

Syntax

--dig <digits>

Description

Global title address.

Class

Optional.

Type / Range

Digit string of length 0 to 18

Default

""

Name

gipm - Gipm

Syntax

--gipm <integer>

Description

Gateway IP address netmask length (GGSN/S-GW).

Class

Optional.

Type / Range

Integer in the range of 0 to 127

Default

0

Name

ni - Network indicator

Syntax

--ni <list-item>

Description

Network indicator. this parameter defines the value that represents the type of network where the RNC is located. allowed values, IN0 ... international network IN1 ... reserved for international use NAO ... national network NA1 ... reserved for national use.

Class

Optional.

List

- " - No signalling network - No signalling network.
- IN0 - Signalling network IN0 - Signalling network IN0.
- IN1 - Signalling network IN1 - Signalling network IN1.
- NAO - Signalling network NAO - Signalling network NAO.
- NA1 - Signalling network NA1 - Signalling network NA1.

Default

""

Name

operState - operational state

Syntax

--operState <string>

Description

Radio network controller operational state. allowed values, unlocked ... unlocked (RESET can be started by RNC or SGSN, after reset operState is changed to available) locked ... locked (RNC is locked) blocked ... blocked (RNC blocked, no new IU connections, existing remains) available ... available (SGSN is connected to RNC) resetbysgsn ... reset by sgsn (note this state is visible in database only when sgsninitresetprocedure is ongoing). after resetprocedure, iucontrol process change state to available. resetbyrnc ... reset by RNC (note this state is visible in database only when rncinitresetprocedure is ongoing). after resetprocedure, iucontrol process change state to available outofservice ... out of service.

Class

Read only.

Type / Range

Digit string of length 1 to 12 with the allowed pattern of [0-9a-zA-Z\._:]*

Name

pSetId - P set id

Syntax

--pSetId <integer>

Description

Radio network controller parameter set index. this parameter should match parameter rncparamsetid of table rncparamset. when pSetId is 0, rncdefaultparamset is used.

Class

Optional.

Type / Range

Integer in the range of 0 to 10

Default

0

Name

pSetName - P set name

Syntax

--pSetName <string>

Description

Radio network controller parameter set name.

Class

Read only.

Type / Range

Digit string of length 0 to 15 with the allowed pattern of [0-9a-zA-Z\.\-:]*

Name

spc - Spc

Syntax

--spc <integer>

Description

Signalling point code.

Class

Optional.

Type / Range

Integer in the range of 0 to 16,777,215

Default

0

Name

state - RNC state

Syntax

--state <list-item>

Description

Radio network controller administrative state. allowed values, unlocked ... unlocked (RESET can be started by RNC or SGSN, after reset state is changed to available) locked ... locked (RNC is locked) blocked ... blocked (RNC blocked, no new IU connections, existing remains) resetbysgsn ... reset by sgsn (note this state is visible in database only when sgsninitresetprocedure is ongoing). after resetprocedure, iucontrol process change state to available in operState parameter

Class

Optional.

List

- blocked - Blocked (RNC blocked, no new IU connections, existing remains).
- locked - Locked (RNC is locked).
- resetbysgsn - Reset by SGSN - Reset by SGSN (note this state is visible in redis only when sgsninitresetprocedure is ongoing). after resetprocedure, iucontrol process change state to available in redis.
- unlocked - Unlocked (RESET can be started by RNC or SGSN, after reset state is changed to available).

Default

locked

Name

toA - Type of RNC address

Syntax

--toA <list-item>

Description

Identify type of RNC address. allowed values. GT ... global title address. SPC ... destination point code (ni + spc) if GT is given, the dig parameter should be given. if SPC is given, the ni and the spc parameters should be given.

Class

Optional.

List

- GT - Global title address.
- SPC - Destination point code (ni + spc).

Default

GT

Name

type - RNC type

Syntax

--type <list-item>

Description

Type of RNC. this parameter defines the radio network controller type. allowed values, RNC ... radio network controller (in traditional UTRAN architecture) IHSPA ... I-HSPA nodeb (in flat UTRAN architecture).

Class

Optional.

List

- IHSPA - I-HSPA nodeb (in flat UTRAN architecture).
- RNC - Radio network controller (in traditional UTRAN architecture).

Default

RNC

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.119 cmm rncGwlp

RNC Gateway IPs. Syntax, description, parameters, and roles for `cmm rncGwIp` command.

Syntax

Description

RNC Gateway IPs.

Parameters

Name

rncName - RNC name

Syntax

--rncName <string>

Description

RNC name. this parameter should match parameter rncname of table rnc.

Class

Required key.

Type / Range

Digit string of length 1 to 10 with the allowed pattern of [0-9a-zA-Z\.\-_:]*

Name

ipAddress - RNC gateway IP address

Syntax

--ipAddress <string>

Description

RNC gateway IP address.

Class

Required key.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.120 cmm rncParamSet

RNC parameter set. Syntax, description, parameters, and roles for cmm rncParamSet command.

Syntax

Description

Configure, list and show Radio Network controllers parameter set.

Parameters

Name

rncParamSetId - RNC parameter set id

Syntax

--rncParamSetId <integer>

Description

Radio network controller parameter set index, this is pSetId parameter in rncs.yaml.

Class

Required key.

Type / Range

Integer in the range of 0 to 10

Name

cspsCoordinationSupported - SGSN requests CS/PS coordination from the RNC

Syntax

--cspsCoordinationSupported <list-item>

Description

SGSN requests CS/PS coordination from the RNC.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

epsBearerPresSupported - EPS bearer preservation is supported

Syntax

--epsBearerPresSupported <list-item>

Description

EPS bearer preservation is supported.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

guardPeriodAfteSubsProhib - Timer defines the guard period (in seconds) after the subsystem prohibited

Syntax

--guardPeriodAfteSubsProhib <integer>

Description

Timer defines the guard period (in seconds) after the subsystem prohibited.

Class

Optional.

Type / Range

Integer in the range of 1 to 150

Default

29

Name

guardPeriodBeforeResetAck - This timer defines a guard period (in seconds) in the CN before sending a reset acknowledge message

Syntax

--guardPeriodBeforeResetAck <integer>

Description

This timer defines a guard period (in seconds) in the CN before sending a reset acknowledge message.

Class

Optional.

Type / Range

Integer in the range of 3 to 15

Default

5

Name

ipv6AddressSupported - RNC supports IPv6 addresses

Syntax

--ipv6AddressSupported <list-item>

Description

RNC supports IPv6 addresses.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

maxDlBitrateForRab - Maximum downlink value (kbps) for the RAB maximum bit rate (MBR) supported in the RNC

Syntax

--maxDlBitrateForRab <list-item>

Description

Maximum downlink value (kbps) for the RAB maximum bit rate (MBR) supported in the RNC.

Class

Optional.

List

- 0 - Specify maximum bit rate of 0 kbps.
- 16 - Specify maximum bit rate of 16 kbps.
- 32 - Specify maximum bit rate of 32 kbps.
- 64 - Specify maximum bit rate of 64 kbps.
- 128 - Specify maximum bit rate of 128 kbps.
- 256 - Specify maximum bit rate of 256 kbps.
- 512 - Specify maximum bit rate of 512 kbps.
- 1024 - Specify maximum bit rate of 1024 kbps.

- 2048 - Specify maximum bit rate of 2048 kbps.
- 4096 - Specify maximum bit rate of 4096 kbps.
- 8192 - Specify maximum bit rate of 8192 kbps.
- 11500 - 1150 - Specify maximum bit rate of 11500 kbps.
- 16000 - Specify maximum bit rate of 16000 kbps.
- 21000 - Specify maximum bit rate of 21000 kbps.
- 28000 - Specify maximum bit rate of 28000 kbps.
- 42000 - Specify maximum bit rate of 42000 kbps.
- 84000 - Specify maximum bit rate of 84000 kbps.
- 168000 - Specify maximum bit rate of 168000 kbps.

Default

0

Name

maxGuaranteedUlBitrateForRab - Maximum uplink value (kbps) for the RAB guaranteed bit rate (GBR) supported in the RNC

Syntax

- maxGuaranteedUlBitrateForRab <list-item>

Description

Maximum uplink value (kbps) for the RAB guaranteed bit rate (GBR) supported in the RNC.

Class

Optional.

List

- 0 - Specify maximum bit rate of 0 kbps.
- 16 - Specify maximum bit rate of 16 kbps.
- 32 - Specify maximum bit rate of 32 kbps.
- 64 - Specify maximum bit rate of 64 kbps.
- 128 - Specify maximum bit rate of 128 kbps.
- 256 - Specify maximum bit rate of 256 kbps.
- 512 - Specify maximum bit rate of 512 kbps.
- 1024 - Specify maximum bit rate of 1024 kbps.
- 2048 - Specify maximum bit rate of 2048 kbps.
- 4096 - Specify maximum bit rate of 4096 kbps.

- 6144 - Specify maximum bit rate of 6144 kbps.
- 8192 - Specify maximum bit rate of 8192 kbps.
- 11500 - Specify maximum bit rate of 11500 kbps.
- 23000 - Specify maximum bit rate of 23000 kbps.
- 46000 - Specify maximum bit rate of 46000 kbps.

Default

0

Name

maxQuaranteedDLBitRateForRab - Maximum downlink value (kbps) for the RAB guaranteed bit rate (GBR) supported in the RNC

Syntax

--maxQuaranteedDLBitRateForRab <list-item>

Description

Maximum downlink value (kbps) for the RAB guaranteed bit rate (GBR) supported in the RNC.

Class

Optional.

List

- 0 - Specify maximum bit rate of 0 kbps.
- 16 - Specify maximum bit rate of 16 kbps.
- 32 - Specify maximum bit rate of 32 kbps.
- 64 - Specify maximum bit rate of 64 kbps.
- 128 - Specify maximum bit rate of 128 kbps.
- 256 - Specify maximum bit rate of 256 kbps.
- 512 - Specify maximum bit rate of 512 kbps.
- 1024 - Specify maximum bit rate of 1024 kbps.
- 2048 - Specify maximum bit rate of 2048 kbps.
- 4096 - Specify maximum bit rate of 4096 kbps.
- 8192 - Specify maximum bit rate of 8192 kbps.
- 11500 - 1150 - Specify maximum bit rate of 11500 kbps.
- 16000 - Specify maximum bit rate of 16000 kbps.
- 21000 - Specify maximum bit rate of 21000 kbps.
- 28000 - Specify maximum bit rate of 28000 kbps.

- 42000 - Specify maximum bit rate of 42000 kbps.
- 84000 - Specify maximum bit rate of 84000 kbps.
- 168000 - Specify maximum bit rate of 168000 kbps.

Default

0

Name

maxRepeatCntOfRncGlobRst - Maximum repeat count of the RNC global reset message

Syntax`--maxRepeatCntOfRncGlobRst <integer>`**Description**

Maximum repeat count of the RNC global reset message.

Class

Optional.

Type / Range

Integer in the range of 0 to 256

Default

5

Name

maxRepeatForResourceReset - Maximum repeat count of the RNC global reset message

Syntax`--maxRepeatForResourceReset <integer>`**Description**

Maximum repeat count of the resource reset.

Class

Optional.

Type / Range

Integer in the range of 0 to 16

Default

1

Name

maxTimeForIuRelease - Maximum time in seconds to wait for activity in the iu connection

Syntax`--maxTimeForIuRelease <integer>`**Description**

Maximum time in seconds to wait for activity in the iu connection.

Class

Optional.

Type / Range

Integer in the range of 0 to 60

Default

10

Name

maxTimeForIuReleaseCompl - Maximum time in seconds to wait for iu release complete after the iu release command has been sent

Syntax

--maxTimeForIuReleaseCompl <integer>

Description

Maximum time in seconds to wait for iu release complete after the iu release command has been sent.

Class

Optional.

Type / Range

Integer in the range of 1 to 15

Default

5

Name

maxTimeForRabAssignment - Maximum time in seconds for the whole radio access bearers (RABs) assignment procedure

Syntax

--maxTimeForRabAssignment <integer>

Description

Maximum time in seconds for the whole radio access bearers (RABs) assignment procedure.

Class

Optional.

Type / Range

Integer in the range of 10 to 30

Default

20

Name

maxTimeForRelocResAlloc - Maximum time in seconds for the relocation resource allocation procedure

Syntax

--maxTimeForRelocResAlloc <integer>

Description

Maximum time in seconds for the relocation resource allocation procedure.

Class

Optional.

Type / Range

Integer in the range of 2 to 20

Default

10

Name

maxTimeForRelocResCompl - Maximum time in seconds for relocation resource completion

Syntax

--maxTimeForRelocResCompl <integer>

Description

Maximum time in seconds for relocation resource completion.

Class

Optional.

Type / Range

Integer in the range of 2 to 30

Default

15

Name

maxTimeForResetProcedure - Maximum time in seconds for the reset procedure

Syntax

--maxTimeForResetProcedure <integer>

Description

Maximum time in seconds for the reset procedure.

Class

Optional.

Type / Range

Integer in the range of 30 to 120

Default

60

Name

maxTimeForSecModeAck - Maximum time in seconds for security mode complete/reject after the security mode command has been sent

Syntax

--maxTimeForSecModeAck <integer>

Description

Maximum time in seconds for security mode complete/reject after the security mode command has been sent.

Class

Optional.

Type / Range

Integer in the range of 1 to 16

Default

5

Name

maxTimeForSrnsContextResp - Maximum time in seconds to wait for SRNS context response after the SRNS context request has been sent

Syntax

--maxTimeForSrnsContextResp <integer>

Description

Maximum time in seconds to wait for SRNS context response after the SRNS context request has been sent.

Class

Optional.

Type / Range

Integer in the range of 1 to 16

Default

8

Name

maxTimeUntilLocationReport - Maximum time (in seconds) the SGSN is allowed to wait for the radio access network application part RANAP LOCATION_REPORT after the reloc, handover command is sent to the source access

Syntax

--maxTimeUntilLocationReport <integer>

Description

Maximum time (in seconds) the SGSN is allowed to wait for the radio access network application part RANAP LOCATION_REPORT after the reloc, handover command is sent to the source access.

Class

Optional.

Type / Range

Integer in the range of 1 to 30

Default

10

Name

maxUlBitrateForRab - Maximum uplink value (kbps) for the RAB maximum bit rate (MBR) supported in the RNC

Syntax

-maxUlBitrateForRab <list-item>

Description

Maximum uplink value (kbps) for the RAB maximum bit rate (MBR) supported in the RNC.

Class

Optional.

List

- 0 - Specify maximum bit rate of 0 kbps.
- 16 - Specify maximum bit rate of 16 kbps.
- 32 - Specify maximum bit rate of 32 kbps.
- 64 - Specify maximum bit rate of 64 kbps.
- 128 - Specify maximum bit rate of 128 kbps.
- 256 - Specify maximum bit rate of 256 kbps.
- 512 - Specify maximum bit rate of 512 kbps.
- 1024 - Specify maximum bit rate of 1024 kbps.
- 2048 - Specify maximum bit rate of 2048 kbps.
- 4096 - Specify maximum bit rate of 4096 kbps.
- 6144 - Specify maximum bit rate of 6144 kbps.
- 8192 - Specify maximum bit rate of 8192 kbps.
- 11500 - Specify maximum bit rate of 11500 kbps.
- 23000 - Specify maximum bit rate of 23000 kbps.
- 46000 - Specify maximum bit rate of 46000 kbps.

Default

0

Name

overloadLongTimerInSgsn - While this timer is running (in seconds), the CN is not allowed to increase traffic

Syntax

--overloadLongTimerInSgsn <integer>

Description

While this timer is running (in seconds), the CN is not allowed to increase traffic.

Class

Optional.

Type / Range

Integer in the range of 2 to 120

Default

30

Name

overloadShortTimerInSgsn - While this timer is running (in seconds), overload messages and information on congested signalling points in the CN are ignored

Syntax

--overloadShortTimerInSgsn <integer>

Description

While this timer is running (in seconds), overload messages and information on congested signalling points in the CN are ignored.

Class

Optional.

Type / Range

Integer in the range of 1 to 120

Default

10

Name

pSetName - Radio network controller parameter set name

Syntax

--pSetName <string>

Description

Radio network controller parameter set name.

Class

Optional.

Type / Range

Digit string of length 0 to 15 with the allowed pattern of [0-9a-zA-Z\.\-:]*

Default

""

Name

rabModifySupported - RAB modification is supported

Syntax

--rabModifySupported <list-item>

Description

RAB modification is supported.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

rel8RncSupported - RNC supports REL 8

Syntax

--rel8RncSupported <list-item>

Description

RNC supports REL 8.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

resetResourceRepeatInterval - Timer defines the period (in seconds) between reset resource messages if reset acknowledgement is not received

Syntax

--resetResourceRepeatInterval <integer>

Description

Timer defines the period (in seconds) between reset resource messages if reset acknowledgement is not received.

Class

Optional.

Type / Range

Integer in the range of 1 to 30

Default

10

Name

rncSupportForDirectTunnel - RNC supports the direct tunnel

Syntax

--rncSupportForDirectTunnel <list-item>

Description

RNC supports the direct tunnel.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

specialCollisionHandlingDuringHo - Special collision handling during handover

Syntax

--specialCollisionHandlingDuringHo <list-item>

Description

Special handling of GGSN-initiated PDP deactivation collision with SRNS relocation.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ueAmbrSupported - UE AMBR support in RNC parameter set

Syntax

--ueAmbrSupported <list-item>

Description

UE AMBR support in RNC parameter set.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.121 cmm rohcProfile

ROHC Profile. Syntax, description, parameters, and roles for cmm rohcProfile command.

Syntax

Description

Administer ROHC Profile.

Parameters

Name

name - ROHC profile name

Syntax

--name <string>

Description

ROHC profile name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

maxCidNum - Max CID

Syntax

--maxCidNum <integer>

Description

Max CID.

Class

Optional.

Type / Range

Integer in the range of 1 to 16,383

Default

15

Name

prfl0002 - Profile 0x0002

Syntax

--prfl0002 <list-item>

Description

UDP/IP (Profile 0x0002) (RFC 3095).

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

prfl0003 - Profile 0x0003

Syntax

--prfl0003 <list-item>

Description

ESP/IP (Encapsulating security payload) (Profile 0x0003) (RFC 3095).

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

prfl0004 - Profile 0x0004

Syntax

--prfl0004 <list-item>

Description

IP (Profile 0x0004) (RFC 3843 and RFC 4815).

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

prfl0006 - Profile 0x0006

Syntax

--prfl0006 <list-item>

Description

TCP/IP (Profile 0x0006) (RFC 4996).

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

6.122 cmm routeParamSet

M3UA route parameter set. Syntax, description, parameters, and roles for `cmm routeParamSet` command.

Syntax**Description**

Configure, list and show M3UA route parameter set.

Parameters**Name**

name - M3UA route parameter set name

Syntax

--name <string>

Description

M3UA route parameter set name.

Class

Required key.

Type / Range

Digit string of length 1 to 12 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

circRouPrevInUse - Circ rou prev in use

Syntax

--circRouPrevInUse <list-item>

Description

D1. control of sending additional transfer prohibited messages concerning this destination in order to limit occurrence of circular routing.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

confusionMsgDenied - Confusion msg denied

Syntax

--confusionMsgDenied <list-item>

Description

D4. denial of sending of MTP confusion messages.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

congLevelSupport - Cong level support

Syntax

--congLevelSupport <list-item>

Description

D3. control of coding congestion level to signalling traffic flow control messages.

Class

Optional.

List

- NONE - No congestion level control.
- TFC - TFC congestion control.
- TFCANDRCT - TFC and RCT congestion control.

Default

TFC

Name

indirectRoutesDefault - Indirect routes default

Syntax

--indirectRoutesDefault <list-item>

Description

C3. action in signalling link set recovery when adjacent signalling point restart procedure is not started.

Class

Optional.

List

- AVAILABLE - Specify route default available.
- RESTRICTED - Specify route default restricted.
- TFMBASED - TFM based - Specify route default TFM based.
- UNAVAILABLE - Specify route default unavailable.

Default

AVAILABLE

Name

jtQ707T10 - JtQ707T10

Syntax

--jtQ707T10 <integer>

Description

A7. supervision timer for signalling routing test ack relevant only in japanese network, in 0.1 seconds.

Class

Optional.

Type / Range

Integer in the range of 20 to 150

Default

100

Name

pstnImplication - Pstn implication

Syntax

--pstnImplication <list-item>

Description

C7. alignment of PSTN link sets when designated M3UA link sets are unavailable.

Class

Optional.

List

- ALIGN - Specify PSTN implication of align.
- FAIL - Specify PSTN implication of fail.

Default

ALIGN

Name

q704T10 - q704T10

Syntax

--q704T10 <integer>

Description

A2. waiting to repeat signalling route set test message, in 1 second.

Class

Optional.

Type / Range

Integer in the range of 10 to 120

Default

31

Name

q704T11 - q704T11

Syntax

--q704T11 <integer>

Description

A3. transfer restricted timer, in 1 second.

Class

Optional.

Type / Range

Integer in the range of 30 to 90

Default

60

Name

q704T15 - q704T15

Syntax

--q704T15 <integer>

Description

A4. waiting to start signalling route set congestion test, in 0.1 seconds.

Class

Optional.

Type / Range

Integer in the range of 20 to 30

Default

25

Name

q704T16JtQ704Tc - q704T16JtQ704Tc

Syntax

--q704T16JtQ704Tc <integer>

Description

A5. waiting for signalling route set congestion status allowed values for Q704_T16 are 14...25 in japan the allowed values for JT_Q704_TC are 26...250, in 0.1 seconds.

Class

Optional.

Type / Range

Integer in the range of 14 to 250

Default

15

Name

q704T19WhiteT111T29 - q704T19WhiteT111T29

Syntax

--q704T19WhiteT111T29 <integer>

Description

B2. waiting to avoid TFP, TFR and TRA messages being bounced back and forth, in 1 second.

Class

Optional.

Type / Range

Integer in the range of 60 to 80

Default

68

Name

q704T21T111T25 - q704T21T111T25

Syntax

--q704T21T111T25 <integer>

Description

B0. waiting to restart traffic through adjacent SP, in 1 second.

Class

Optional.

Type / Range

Integer in the range of 20 to 70

Default

64

Name

q704T6 - q704T6

Syntax

--q704T6 <integer>

Description

A0. delay to avoid MIS-sequencing on controlled rerouting, in 0.1 seconds.

Class

Optional.

Type / Range

Integer in the range of 5 to 20

Default

8

Name

q704T8 - q704T8

Syntax

--q704T8 <integer>

Description

A1. transfer prohibited inhibition timer, in 0.1 seconds.

Class

Optional.

Type / Range

Integer in the range of 5 to 20

Default

10

Name

respTfmControl - Resp tfm control

Syntax

--respTfmControl <list-item>

Description

C5. control of sending transfer prohibited and transfer restricted messages based on response method.

Class

Optional.

List

- TFMDENIED - Transfer denied - Specify traffic management control of transfer denied.
- TFPALLOWED - Transfer prohibited allowed - Specify traffic management control of transfer prohibited allowed.
- TFPFORKNOWN - Transfer prohibited for unknown - Specify traffic management control of transfer prohibited for unknown.
- TFRALLOWED - Transfer restricted allowed - Specify traffic management control of transfer restricted allowed.

Default

TFPALLOWED

Name

rstOnTfpAllowed - Rst on tfp allowed

Syntax

--rstOnTfpAllowed <list-item>

Description

D6. indicates whether sending route set test message is allowed on reception of transfer prohibited message.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

sorRefWithNodeInfo - Sor re fWith node info

Syntax

--sorRefWithNodeInfo <list-item>

Description

C8. indicates whether local reference identifiers of co SCCP messages include also SP info.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

spRestartType - Sp restart type

Syntax

--spRestartType <list-item>

Description

C2. type of signalling point restart procedure.

Class

Optional.

List

- ANSI - ANSI for restart type from american national standards institute.
- BLUE - BLUE for restart type defined by international telecommunications union (ITU) in Q.700 - Q.716 blue book.
- NONE - NONE for restart type of none.
- WHITE - WHITE for restart type defined by international telecommunications union (ITU) in Q.700 - Q.716 3/93 white book.

Default

BLUE

Name

srtDenied - Srt denied

Syntax

--srtDenied <list-item>

Description

D9. control of signal routing test.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

supportOfM3uaSnm - Support of M3UA snm

Syntax

--supportOfM3uaSnm <list-item>

Description

D8. control of sending M3UA signalling network management messages.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

t111T18 - t111T18

Syntax

--t111T18 <integer>

Description

A6. transfer restricted inhibition timer, in 1 second.

Class

Optional.

Type / Range

Integer in the range of 2 to 20

Default

3

Name

t111T28 - t111T28

Syntax

--t111T28 <integer>

Description

B1. waiting for traffic restart waiting message, in 1 second.

Class

Optional.

Type / Range

Integer in the range of 3 to 35

Default

30

Name

t111T30 - t111T30

Syntax

--t111T30 <integer>

Description

B3. waiting to limit sending of TFP and TFR messages in response to unexpected TRA or TRW messages, in 1 second.

Class

Optional.

Type / Range

Integer in the range of 20 to 40

Default

30

Name

tfcDenied - Tfc denied

Syntax

--tfcDenied <list-item>

Description

D2. denial of sending of transfer controlled messages.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

tfmControl - Tfm control

Syntax

--tfmControl <list-item>

Description

C4. control of sending transfer prohibited, transfer restricted and transfer allowed messages.

Class

Optional.

List

- ALLALLOWED - All allowed - Specify TFM control of all allowed.
- ALLDENIED - All denied - Specify TFM control of all denied.
- BROADCASTDENIED - Broadcast denied - Specify TFM control broadcast denied.

Default

ALLALLOWED

Name

tfrDenied - Tfr denied

Syntax

--tfrDenied <list-item>

Description

C6. denial of transfer restricted procedure.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

tfrSendingBasis - Tfr sending basis

Syntax

--tfrSendingBasis <list-item>

Description

D0. method for calculating right time to broadcast transfer restricted messages concerning this destination.

Class

Optional.

List

- ANSI - Specify TFR sending basis of american national standards institute.
- ITU - Specify TFR sending basis of international telecommunication union.
- NONE - Specify TFR sending basis of none.

Default

NONE

Name

trmDenied - Trm denied

Syntax

--trmDenied <list-item>

Description

C0. denial of sending of traffic restart messages.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

trmExpected - Trm expected

Syntax

--trmExpected <list-item>

Description

C1. indicates whether reception of traffic restart messages is expected.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

upuAllowed - Upu allowed

Syntax

--upuAllowed <list-item>

Description

D5. indicates whether the sending of user part unavailable message is allowed.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

useOfM3uaNwApp - Use of M3UA nw app

Syntax

--useOfM3uaNwApp <list-item>

Description

D7. control of using network appearance parameter in the M3UA signalling network management messages.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

usnDenied - Usn denied

Syntax

--usnDenied <list-item>

Description

D10. control of unallocated signalling point number message.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.123 cmm routingIndicatorList

Routing Indicator List. Syntax, description, parameters, and roles for cmm routingIndicatorList command.

Syntax**Description**

List of Routing Indicators

Parameters**Name**

name - Name

Syntax

--name <string>

Description

Routing Indicator List Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.124 cmm routingIndicators

Routing Indicator. Syntax, description, parameters, and roles for **cmm routingIndicators** command.

Syntax**Description**

Links a Routing Indicator to a Routing Indicator List.

Parameters**Name**

routingIndicatorListName - Routing Indicator List Name

Syntax

--routingIndicatorListName <string>

Description

The name of the Routing Indicator

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

routingIndicator - Routing Indicator

Syntax

--routingIndicator <digits>

Description

Routing Indicator, consisting of 1 to 4 decimal digits assigned by the home network operator and provisioned in the USIM, that allow together with the Home Network Identifier to route network signalling with SUCI to AUSF and UDM instances capable to serve the subscriber. Each decimal digit present in the Routing Indicator shall be regarded as meaningful (e.g. value 012 is not the same as value 12).

Class

Required key.

Type / Range

Used for Routing Indicator which consists of 1 to 4 digits.

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.125 cmm s102CallPriPagingProfile

S102 Call Priority Paging Profile. Syntax, description, parameters, and roles for cmm s102CallPriPagingProfile command.

Syntax

Description

This command is used to administer a S102 Call Priority Paging Profile and includes parameters for defining the S102 call priority and S1AP paging priority level.

Parameters

Name

s102PagingPriProfileName - S102 paging priority profile name

Syntax

--s102PagingPriProfileName <string>

Description

Identifies the S102 paging priority profile label (alternate key).

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

s102CallPriority - S102 call priority

Syntax

--s102CallPriority <integer>

Description

Identifies the S102 call priority.

Class

Required key.

Type / Range

Integer in the range of 0 to 15

Name

s1ApPagingPriLevel - S1AP paging priority level

Syntax

--s1ApPagingPriLevel <list-item>

Description

Identifies the S1AP paging priority level.

Class

Optional.

List

- PrioLevel0 - Priority level 0 - Priority level 0. No paging priority is set.

- PrioLevel1 - Priority level 1 - Priority level 1. Highest Paging priority.
- PrioLevel2 - Priority level 2 - Priority level 2.
- PrioLevel3 - Priority level 3 - Priority level 3.
- PrioLevel4 - Priority level 4 - Priority level 4.
- PrioLevel5 - Priority level 5 - Priority level 5.
- PrioLevel6 - Priority level 6 - Priority level 6.
- PrioLevel7 - Priority level 7 - Priority level 7.
- PrioLevel8 - Priority level 8 - Priority level 8. Lowest Paging Priority.

Default

PrioLevel0

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.126 cmm s102PagingPriProfile

S102 Paging Priority Profile. Syntax, description, parameters, and roles for cmm s102PagingPriProfile command.

Syntax**Description**

This command is used to administer a S102 Paging Priority Profile and includes parameters for defining the S102 paging priority profile name, no call priority paging priority level, and GEC emergency paging priority level.

Parameters

Name

name - S102 paging priority profile name

Syntax

--name <digits>

Description

Defines the S102 paging priority profile name.

Class

Required key.

Type / Range

Digit string of length 1 to 1 with the allowed pattern of [1-8]*

Name

description - Description

Syntax

--description <string>

Description

Profile description.

Class

Optional.

Type / Range

Digit string of length 0 to 160 with the allowed pattern of [0-9a-zA-Z\.\- _:@]*

Default

"

Name

gecEmerPagingPriLevel - GEC emergency paging priority level

Syntax

--gecEmerPagingPriLevel <list-item>

Description

Identifies the GEC emergency paging priority level.

Class

Optional.

List

- PrioLevel0 - Priority level 0 - Priority level 0. No paging priority is set.
- PrioLevel1 - Priority level 1 - Priority level 1. Highest Paging priority.

- PrioLevel2 - Priority level 2 - Priority level 2.
- PrioLevel3 - Priority level 3 - Priority level 3.
- PrioLevel4 - Priority level 4 - Priority level 4.
- PrioLevel5 - Priority level 5 - Priority level 5.
- PrioLevel6 - Priority level 6 - Priority level 6.
- PrioLevel7 - Priority level 7 - Priority level 7.
- PrioLevel8 - Priority level 8 - Priority level 8. Lowest Paging Priority.

Default

PrioLevel0

Name

noCallPriPagingPriLevel - No call priority paging priority level

Syntax

--noCallPriPagingPriLevel <list-item>

Description

Identifies the no call priority paging priority level.

Class

Optional.

List

- PrioLevel0 - Priority level 0 - Priority level 0. No paging priority is set.
- PrioLevel1 - Priority level 1 - Priority level 1. Highest Paging priority.
- PrioLevel2 - Priority level 2 - Priority level 2.
- PrioLevel3 - Priority level 3 - Priority level 3.
- PrioLevel4 - Priority level 4 - Priority level 4.
- PrioLevel5 - Priority level 5 - Priority level 5.
- PrioLevel6 - Priority level 6 - Priority level 6.
- PrioLevel7 - Priority level 7 - Priority level 7.
- PrioLevel8 - Priority level 8 - Priority level 8. Lowest Paging Priority.

Default

PrioLevel0

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.127 cmm s1apCauseCodeMappingProfile

S1AP Cause Code Mapping Profile. Syntax, description, parameters, and roles for cmm s1apCauseCodeMappingProfile command.

Syntax

Description

Define S1AP cause code mappings.

Parameters

Name

name - S1AP Cause Code Mapping Profile Name

Syntax

--name <string>

Description

The name of the S1AP cause code mapping profile.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

eRabRelApnAdministration - E-RAB Release Command APN Administration

Syntax

--eRabRelApnAdministration <list-item>

Description

The S1AP cause the MME includes on E-RAB release command related to an E-RAB as part of the E-RAB to be released list, when the MME releases the E-RAB due to APN Administration.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.

- `RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId` - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- `RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId` - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- `RadioNetworkLayer_HandoverDesirableForRadioReasons` - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- `RadioNetworkLayer_TimeCriticalHandover` - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- `RadioNetworkLayer_ResourceOptimisationHandover` - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- `RadioNetworkLayer_ReduceLoadInServingCell` - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- `RadioNetworkLayer_UserInactivity` - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- `RadioNetworkLayer_RadioConnectionWithUeLost` - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- `RadioNetworkLayer_LoadBalancingTauRequired` - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- `RadioNetworkLayer_CsFallbackTriggered` - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- `RadioNetworkLayer_UeNotAvailableForPsService` - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- `RadioNetworkLayer_RadioResourcesNotAvailable` - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- `RadioNetworkLayer_FailureInTheRadioInterfaceProcedure` - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- `RadioNetworkLayer_InvalidQosCombination` - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- `RadioNetworkLayer_InterRatRedirection` - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.

- RadioNetworkLayer_InteractionWithOtherProcedure - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- RadioNetworkLayer_UnknownErabId - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- RadioNetworkLayer_MultipleErabIdInstances - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- RadioNetworkLayer_S1IntraSystemHandoverTriggered - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- RadioNetworkLayer_S1InterSystemHandoverTriggered - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- RadioNetworkLayer_x2HandoverTriggered - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- RadioNetworkLayer_redirectionTowards1xRtt - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- RadioNetworkLayer_NotSupportedQciValue - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- RadioNetworkLayer_InvalidCsgId - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- RadioNetworkLayer_ReleaseDueToPreEmption - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- RadioNetworkLayer_N26InterfaceNotAvailable - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- TransportLayer_TransportResourceUnavailable - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- TransportLayer_Unspecified - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- Nas_NormalRelease - NAS Normal Release - S1AP NAS cause Normal Release.
- Nas_AuthenticationFailure - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- Nas_Detach - NAS Detach - S1AP NAS cause Detach.

- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Nas_NormalRelease

Name

eRabRelMmeDeactFailureHo - E-RAB Release Command MME Deactivation Handover

Syntax

--eRabRelMmeDeactFailureHo <list-item>

Description

The S1AP cause the MME includes on E-RAB release command related to an E-RAB as part of the E-RAB to be released list, when the MME releases the E-RAB due to MME initiated PDN connection deactivation when PDN(s) have failed during TAU/handover.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.

- `RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId` - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- `RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId` - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- `RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId` - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- `RadioNetworkLayer_HandoverDesirableForRadioReasons` - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- `RadioNetworkLayer_TimeCriticalHandover` - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- `RadioNetworkLayer_ResourceOptimisationHandover` - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- `RadioNetworkLayer_ReduceLoadInServingCell` - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- `RadioNetworkLayer_UserInactivity` - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- `RadioNetworkLayer_RadioConnectionWithUeLost` - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- `RadioNetworkLayer_LoadBalancingTauRequired` - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- `RadioNetworkLayer_CsFallbackTriggered` - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- `RadioNetworkLayer_UeNotAvailableForPsService` - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- `RadioNetworkLayer_RadioResourcesNotAvailable` - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- `RadioNetworkLayer_FailureInTheRadioInterfaceProcedure` - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- `RadioNetworkLayer_InvalidQosCombination` - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.

- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.

- Nas_Detach - NAS Detach - S1AP NAS cause Detach.
- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

eRabRelPdnDisconnect - E-RAB Release Command PDN Disconnect

Syntax

--eRabRelPdnDisconnect <list-item>

Description

The S1AP cause the MME includes on E-RAB release command related to an E-RAB as part of the E-RAB to be released list, when the MME releases the E-RAB due to PDN Disconnect Request from the UE.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.

- **RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell** - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- **RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId** - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- **RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId** - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- **RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId** - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- **RadioNetworkLayer_HandoverDesirableForRadioReasons** - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- **RadioNetworkLayer_TimeCriticalHandover** - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- **RadioNetworkLayer_ResourceOptimisationHandover** - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- **RadioNetworkLayer_ReduceLoadInServingCell** - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- **RadioNetworkLayer_UserInactivity** - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- **RadioNetworkLayer_RadioConnectionWithUeLost** - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- **RadioNetworkLayer_LoadBalancingTauRequired** - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- **RadioNetworkLayer_CsFallbackTriggered** - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- **RadioNetworkLayer_UeNotAvailableForPsService** - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- **RadioNetworkLayer_RadioResourcesNotAvailable** - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- **RadioNetworkLayer_FailureInTheRadioInterfaceProcedure** - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.

- RadioNetworkLayer_InvalidQosCombination - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- RadioNetworkLayer_InterRatRedirection - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- RadioNetworkLayer_InteractionWithOtherProcedure - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- RadioNetworkLayer_UnknownErabId - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- RadioNetworkLayer_MultipleErabIdInstances - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- RadioNetworkLayer_S1IntraSystemhandoverTriggered - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- RadioNetworkLayer_S1InterSystemHandoverTriggered - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- RadioNetworkLayer_x2HandoverTriggered - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- RadioNetworkLayer_redirectionTowards1xRtt - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- RadioNetworkLayer_NotSupportedQciValue - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- RadioNetworkLayer_InvalidCsgId - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- RadioNetworkLayer_ReleaseDueToPreEmption - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- RadioNetworkLayer_N26InterfaceNotAvailable - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- TransportLayer_TransportResourceUnavailable - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- TransportLayer_Unspecified - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- Nas_NormalRelease - NAS Normal Release - S1AP NAS cause Normal Release.

- Nas_AuthenticationFailure - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- Nas_Detach - NAS Detach - S1AP NAS cause Detach.
- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

eRabRelPdnSetupFailure - E-RAB Release Command PDN Setup Failure

Syntax

--eRabRelPdnSetupFailure <list-item>

Description

The S1AP cause the MME includes on E-RAB release command related to an E-RAB as part of the E-RAB to be released list, when PDN Connectivity Request fails due to either collision with handover, or no response or rejection from UE.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.

- **RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell** - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- **RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId** - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- **RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId** - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- **RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId** - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- **RadioNetworkLayer_HandoverDesirableForRadioReasons** - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- **RadioNetworkLayer_TimeCriticalHandover** - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- **RadioNetworkLayer_ResourceOptimisationHandover** - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- **RadioNetworkLayer_ReduceLoadInServingCell** - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- **RadioNetworkLayer_UserInactivity** - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- **RadioNetworkLayer_RadioConnectionWithUeLost** - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- **RadioNetworkLayer_LoadBalancingTauRequired** - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- **RadioNetworkLayer_CsFallbackTriggered** - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- **RadioNetworkLayer_UeNotAvailableForPsService** - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- **RadioNetworkLayer_RadioResourcesNotAvailable** - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- **RadioNetworkLayer_FailureInTheRadioInterfaceProcedure** - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.

- **RadioNetworkLayer_InvalidQosCombination** - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemhandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.

- Nas_AuthenticationFailure - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- Nas_Detach - NAS Detach - S1AP NAS cause Detach.
- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

eRabRelPgwActivationCollisionUe - E-RAB Release Command PGW Activation Collision UE

Syntax

--eRabRelPgwActivationCollisionUe <list-item>

Description

The S1AP cause the MME includes on E-RAB release command related to an E-RAB as part of the E-RAB to be released list, when the MME terminates PGW initiated dedicated bearer activation due to collision after E-RAB setup has been sent, or the UE does not respond or rejects the request.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.

- **RadioNetworkLayer_UnknownTargetId** - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- **RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell** - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- **RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId** - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- **RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId** - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- **RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId** - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- **RadioNetworkLayer_HandoverDesirableForRadioReasons** - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- **RadioNetworkLayer_TimeCriticalHandover** - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- **RadioNetworkLayer_ResourceOptimisationHandover** - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- **RadioNetworkLayer_ReduceLoadInServingCell** - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- **RadioNetworkLayer_UserInactivity** - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- **RadioNetworkLayer_RadioConnectionWithUeLost** - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- **RadioNetworkLayer_LoadBalancingTauRequired** - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- **RadioNetworkLayer_CsFallbackTriggered** - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- **RadioNetworkLayer_UeNotAvailableForPsService** - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- **RadioNetworkLayer_RadioResourcesNotAvailable** - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.

- **RadioNetworkLayer_FailureInTheRadioInterfaceProcedure** - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- **RadioNetworkLayer_InvalidQosCombination** - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownEranId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleEranIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.

- TransportLayer_Unspecified - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- Nas_NormalRelease - NAS Normal Release - S1AP NAS cause Normal Release.
- Nas_AuthenticationFailure - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- Nas_Detach - NAS Detach - S1AP NAS cause Detach.
- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

eRabRelPgwDeactivation - E-RAB Release Command PGW Deactivation

Syntax

--eRabRelPgwDeactivation <list-item>

Description

The S1AP cause the MME includes on E-RAB release command related to an E-RAB as part of the E-RAB to be released list, when the PGW has triggered dedicated bearer deactivation.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.

- **RadioNetworkLayer_UnknownTargetId** - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- **RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell** - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- **RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId** - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- **RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId** - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- **RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId** - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- **RadioNetworkLayer_HandoverDesirableForRadioReasons** - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- **RadioNetworkLayer_TimeCriticalHandover** - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- **RadioNetworkLayer_ResourceOptimisationHandover** - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- **RadioNetworkLayer_ReduceLoadInServingCell** - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- **RadioNetworkLayer_UserInactivity** - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- **RadioNetworkLayer_RadioConnectionWithUeLost** - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- **RadioNetworkLayer_LoadBalancingTauRequired** - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- **RadioNetworkLayer_CsFallbackTriggered** - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- **RadioNetworkLayer_UeNotAvailableForPsService** - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- **RadioNetworkLayer_RadioResourcesNotAvailable** - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.

- **RadioNetworkLayer_FailureInTheRadioInterfaceProcedure** - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- **RadioNetworkLayer_InvalidQosCombination** - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownErbId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErbIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.

- TransportLayer_Unspecified - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- Nas_NormalRelease - NAS Normal Release - S1AP NAS cause Normal Release.
- Nas_AuthenticationFailure - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- Nas_Detach - NAS Detach - S1AP NAS cause Detach.
- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

eRabRelReEstablishment - E-RAB Release Command Re-establishment

Syntax

--eRabRelReEstablishment <list-item>

Description

The S1AP cause the MME includes on E-RAB release command related to an E-RAB as part of the E-RAB to be released list, when the MME releases the E-RAB (and the PDN) due to PDN connection re-establishment after SGW relocation functionality.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.

- **RadioNetworkLayer_CellNotAvailable** - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- **RadioNetworkLayer_UnknownTargetId** - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- **RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell** - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- **RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId** - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- **RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId** - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- **RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId** - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- **RadioNetworkLayer_HandoverDesirableForRadioReasons** - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- **RadioNetworkLayer_TimeCriticalHandover** - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- **RadioNetworkLayer_ResourceOptimisationHandover** - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- **RadioNetworkLayer_ReduceLoadInServingCell** - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- **RadioNetworkLayer_UserInactivity** - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- **RadioNetworkLayer_RadioConnectionWithUeLost** - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- **RadioNetworkLayer_LoadBalancingTauRequired** - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- **RadioNetworkLayer_CsFallbackTriggered** - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- **RadioNetworkLayer_UeNotAvailableForPsService** - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.

- **RadioNetworkLayer_RadioResourcesNotAvailable** - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- **RadioNetworkLayer_FailureInTheRadioInterfaceProcedure** - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- **RadioNetworkLayer_InvalidQoSCombination** - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.

- RadioNetworkLayer_N26InterfaceNotAvailable - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- TransportLayer_TransportResourceUnavailable - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- TransportLayer_Unspecified - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- Nas_NormalRelease - NAS Normal Release - S1AP NAS cause Normal Release.
- Nas_AuthenticationFailure - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- Nas_Detach - NAS Detach - S1AP NAS cause Detach.
- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.

- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

eRabRelUeNoBearer - E-RAB Release Command UE No Bearer

Syntax

--eRabRelUeNoBearer <list-item>

Description

The S1AP cause the MME includes on E-RAB release command related to an E-RAB as part of the E-RAB to be released list, when the MME releases the E-RAB when the bearer was successfully setup on E-UTRAN but the UE indicates a corresponding bearer inactive during TAU.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.

- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- RadioNetworkLayer_HandoverDesirableForRadioReasons - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- RadioNetworkLayer_TimeCriticalHandover - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- RadioNetworkLayer_ResourceOptimisationHandover - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- RadioNetworkLayer_ReduceLoadInServingCell - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- RadioNetworkLayer_UserInactivity - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- RadioNetworkLayer_RadioConnectionWithUeLost - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.

- **RadioNetworkLayer_LoadBalancingTauRequired** - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- **RadioNetworkLayer_CsFallbackTriggered** - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- **RadioNetworkLayer_UeNotAvailableForPsService** - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- **RadioNetworkLayer_RadioResourcesNotAvailable** - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- **RadioNetworkLayer_FailureInTheRadioInterfaceProcedure** - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- **RadioNetworkLayer_InvalidQosCombination** - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.

- RadioNetworkLayer_NotSupportedQciValue - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- RadioNetworkLayer_InvalidCsgId - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- RadioNetworkLayer_ReleaseDueToPreEmption - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- RadioNetworkLayer_N26InterfaceNotAvailable - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- TransportLayer_TransportResourceUnavailable - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- TransportLayer_Unspecified - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- Nas_NormalRelease - NAS Normal Release - S1AP NAS cause Normal Release.
- Nas_AuthenticationFailure - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- Nas_Detach - NAS Detach - S1AP NAS cause Detach.
- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.

- **Miscellaneous_NotEnoughUserPlaneProcessingResources** - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- **Miscellaneous_HardwareFailure** - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- **Miscellaneous_OmIntervention** - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- **Miscellaneous_Unspecified** - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- **Miscellaneous_UnknownPlmn** - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

pathSwReqAckCtxtNotFoundERab - Path Switch Request Acknowledge Context Not Found E-RAB

Syntax

--pathSwReqAckCtxtNotFoundERab <list-item>

Description

The S1AP cause the MME includes on Path Switch Request Acknowledge related to an E-RAB as part of the E-RAB to be released list, when the MME releases the E-RAB due to receiving Modify Bearer Response from the SGW indicating 'Context not found' GTP cause.

Class

Optional.

List

- **Disabled** - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- **RadioNetworkLayer_Unspecified** - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- **RadioNetworkLayer_TX2RelocOverallExpiry** - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- **RadioNetworkLayer_SuccessfulHandover** - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- **RadioNetworkLayer_ReleaseDueToEutranGeneratedReason** - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.

- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- RadioNetworkLayer_HandoverDesirableForRadioReasons - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- RadioNetworkLayer_TimeCriticalHandover - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- RadioNetworkLayer_ResourceOptimisationHandover - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.

- **RadioNetworkLayer_ReduceLoadInServingCell** - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- **RadioNetworkLayer_UserInactivity** - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- **RadioNetworkLayer_RadioConnectionWithUeLost** - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- **RadioNetworkLayer_LoadBalancingTauRequired** - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- **RadioNetworkLayer_CsFallbackTriggered** - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- **RadioNetworkLayer_UeNotAvailableForPsService** - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- **RadioNetworkLayer_RadioResourcesNotAvailable** - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- **RadioNetworkLayer_FailureInTheRadioInterfaceProcedure** - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- **RadioNetworkLayer_InvalidQosCombination** - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemhandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.

- RadioNetworkLayer_S1InterSystemHandoverTriggered - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- RadioNetworkLayer_x2HandoverTriggered - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- RadioNetworkLayer_redirectionTowards1xRtt - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- RadioNetworkLayer_NotSupportedQciValue - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- RadioNetworkLayer_InvalidCsgId - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- RadioNetworkLayer_ReleaseDueToPreEmption - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- RadioNetworkLayer_N26InterfaceNotAvailable - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- TransportLayer_TransportResourceUnavailable - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- TransportLayer_Unspecified - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- Nas_NormalRelease - NAS Normal Release - S1AP NAS cause Normal Release.
- Nas_AuthenticationFailure - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- Nas_Detach - NAS Detach - S1AP NAS cause Detach.
- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.

- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

pathSwReqAckFailToSwitchBearer - Path Switch Request Acknowledge Fail to Switch Bearer

Syntax

--pathSwReqAckFailToSwitchBearer <list-item>

Description

The S1AP cause the MME includes on Path Switch Request Acknowledge related to an E-RAB as part of the E-RAB to be released list, when the MME releases the E-RAB due to failing to switch the requested bearer during the procedure.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.

- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- RadioNetworkLayer_HandoverDesirableForRadioReasons - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.

- RadioNetworkLayer_TimeCriticalHandover - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- RadioNetworkLayer_ResourceOptimisationHandover - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- RadioNetworkLayer_ReduceLoadInServingCell - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- RadioNetworkLayer_UserInactivity - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- RadioNetworkLayer_RadioConnectionWithUeLost - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- RadioNetworkLayer_LoadBalancingTauRequired - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- RadioNetworkLayer_CsFallbackTriggered - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- RadioNetworkLayer_UeNotAvailableForPsService - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- RadioNetworkLayer_RadioResourcesNotAvailable - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- RadioNetworkLayer_FailureInTheRadioInterfaceProcedure - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- RadioNetworkLayer_InvalidQosCombination - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- RadioNetworkLayer_InterRatRedirection - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- RadioNetworkLayer_InteractionWithOtherProcedure - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- RadioNetworkLayer_UnknownErabId - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- RadioNetworkLayer_MultipleErabIdInstances - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported -

S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.

- RadioNetworkLayer_S1IntraSystemHandoverTriggered - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- RadioNetworkLayer_S1InterSystemHandoverTriggered - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- RadioNetworkLayer_x2HandoverTriggered - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- RadioNetworkLayer_redirectionTowards1xRtt - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- RadioNetworkLayer_NotSupportedQciValue - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- RadioNetworkLayer_InvalidCsgId - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- RadioNetworkLayer_ReleaseDueToPreEmption - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- RadioNetworkLayer_N26InterfaceNotAvailable - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- TransportLayer_TransportResourceUnavailable - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- TransportLayer_Unspecified - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- Nas_NormalRelease - NAS Normal Release - S1AP NAS cause Normal Release.
- Nas_AuthenticationFailure - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- Nas_Detach - NAS Detach - S1AP NAS cause Detach.
- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).

- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

pathSwReqAckUnknownERab - Path Switch Request Acknowledge Unknown E-RAB

Syntax

--pathSwReqAckUnknownERab <list-item>

Description

The S1AP cause the MME includes on Path Switch Request Acknowledge related to an E-RAB as part of the E-RAB to be released list, when the MME releases the E-RAB due to receiving an unknown E-RAB on the Path Switch Request.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.

- **RadioNetworkLayer_TX2RelocOverallExpiry** - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- **RadioNetworkLayer_SuccessfulHandover** - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- **RadioNetworkLayer_ReleaseDueToEutranGeneratedReason** - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- **RadioNetworkLayer_HandoverCancelled** - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- **RadioNetworkLayer_PartialHandover** - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- **RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem** - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- **RadioNetworkLayer_HandoverTargetNotAllowed** - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- **RadioNetworkLayer_TS1RelocOverallExpiry** - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- **RadioNetworkLayer_TS1RelocPrepExpiry** - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- **RadioNetworkLayer_CellNotAvailable** - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- **RadioNetworkLayer_UnknownTargetId** - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- **RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell** - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- **RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId** - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- **RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId** - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- **RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId** - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.

- **RadioNetworkLayer_HandoverDesirableForRadioReasons** - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- **RadioNetworkLayer_TimeCriticalHandover** - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- **RadioNetworkLayer_ResourceOptimisationHandover** - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- **RadioNetworkLayer_ReduceLoadInServingCell** - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- **RadioNetworkLayer_UserInactivity** - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- **RadioNetworkLayer_RadioConnectionWithUeLost** - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- **RadioNetworkLayer_LoadBalancingTauRequired** - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- **RadioNetworkLayer_CsFallbackTriggered** - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- **RadioNetworkLayer_UeNotAvailableForPsService** - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- **RadioNetworkLayer_RadioResourcesNotAvailable** - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- **RadioNetworkLayer_FailureInTheRadioInterfaceProcedure** - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- **RadioNetworkLayer_InvalidQosCombination** - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.

- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- **Nas_Detach** - NAS Detach - S1AP NAS cause Detach.
- **Nas_Unspecified** - NAS Unspecified - S1AP NAS cause Unspecified.
- **Nas_CsgSubscriptionExpiry** - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- **Protocol_TransferSyntaxError** - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- **Protocol_AbstractSyntaxErrorReject** - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).

- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelAdminDelete - UE Context Release Command Admin Delete

Syntax

--ueCtxtRelAdminDelete <list-item>

Description

S1AP cause code to include on UE context release command when UE context is deleted by network administrator.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.

- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.

- **RadioNetworkLayer_HandoverDesirableForRadioReasons** - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- **RadioNetworkLayer_TimeCriticalHandover** - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- **RadioNetworkLayer_ResourceOptimisationHandover** - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- **RadioNetworkLayer_ReduceLoadInServingCell** - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- **RadioNetworkLayer_UserInactivity** - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- **RadioNetworkLayer_RadioConnectionWithUeLost** - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- **RadioNetworkLayer_LoadBalancingTauRequired** - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- **RadioNetworkLayer_CsFallbackTriggered** - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- **RadioNetworkLayer_UeNotAvailableForPsService** - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- **RadioNetworkLayer_RadioResourcesNotAvailable** - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- **RadioNetworkLayer_FailureInTheRadioInterfaceProcedure** - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- **RadioNetworkLayer_InvalidQosCombination** - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.

- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- **Nas_Detach** - NAS Detach - S1AP NAS cause Detach.
- **Nas_Unspecified** - NAS Unspecified - S1AP NAS cause Unspecified.
- **Nas_CsgSubscriptionExpiry** - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- **Protocol_TransferSyntaxError** - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- **Protocol_AbstractSyntaxErrorReject** - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).

- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelApnAdministration - UE Context Release APN Administration

Syntax

--ueCtxtRelApnAdministration <list-item>

Description

The S1AP cause the MME includes on UE context release command on message level, when the MME detaches the UE due to APN administration.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.

- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.

- **RadioNetworkLayer_HandoverDesirableForRadioReasons** - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- **RadioNetworkLayer_TimeCriticalHandover** - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- **RadioNetworkLayer_ResourceOptimisationHandover** - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- **RadioNetworkLayer_ReduceLoadInServingCell** - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- **RadioNetworkLayer_UserInactivity** - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- **RadioNetworkLayer_RadioConnectionWithUeLost** - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- **RadioNetworkLayer_LoadBalancingTauRequired** - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- **RadioNetworkLayer_CsFallbackTriggered** - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- **RadioNetworkLayer_UeNotAvailableForPsService** - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- **RadioNetworkLayer_RadioResourcesNotAvailable** - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- **RadioNetworkLayer_FailureInTheRadioInterfaceProcedure** - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- **RadioNetworkLayer_InvalidQosCombination** - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownEranId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleEranIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.

- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- **Nas_Detach** - NAS Detach - S1AP NAS cause Detach.
- **Nas_Unspecified** - NAS Unspecified - S1AP NAS cause Unspecified.
- **Nas_CsgSubscriptionExpiry** - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- **Protocol_TransferSyntaxError** - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- **Protocol_AbstractSyntaxErrorReject** - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).

- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Nas_Detach

Name

ueCtxtRelBearerSetupFailed - UE Context Release Command Bearer Setup Failed

Syntax

--ueCtxtRelBearerSetupFailed <list-item>

Description

S1AP cause code to include on UE context release command when bearer setup failed.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.

- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.

- **RadioNetworkLayer_HandoverDesirableForRadioReasons** - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- **RadioNetworkLayer_TimeCriticalHandover** - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- **RadioNetworkLayer_ResourceOptimisationHandover** - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- **RadioNetworkLayer_ReduceLoadInServingCell** - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- **RadioNetworkLayer_UserInactivity** - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- **RadioNetworkLayer_RadioConnectionWithUeLost** - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- **RadioNetworkLayer_LoadBalancingTauRequired** - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- **RadioNetworkLayer_CsFallbackTriggered** - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- **RadioNetworkLayer_UeNotAvailableForPsService** - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- **RadioNetworkLayer_RadioResourcesNotAvailable** - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- **RadioNetworkLayer_FailureInTheRadioInterfaceProcedure** - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- **RadioNetworkLayer_InvalidQosCombination** - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.

- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- **Nas_Detach** - NAS Detach - S1AP NAS cause Detach.
- **Nas_Unspecified** - NAS Unspecified - S1AP NAS cause Unspecified.
- **Nas_CsgSubscriptionExpiry** - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- **Protocol_TransferSyntaxError** - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- **Protocol_AbstractSyntaxErrorReject** - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).

- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelClrSubWithdrawal - UE Context Release Command CLR Subscriber Withdrawal

Syntax

--ueCtxtRelClrSubWithdrawal <list-item>

Description

S1AP cause code to include on UE context release command when detach reason is CLR with cancellation type subscription withdrawal.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.

- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.

- RadioNetworkLayer_HandoverDesirableForRadioReasons - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- RadioNetworkLayer_TimeCriticalHandover - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- RadioNetworkLayer_ResourceOptimisationHandover - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- RadioNetworkLayer_ReduceLoadInServingCell - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- RadioNetworkLayer_UserInactivity - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- RadioNetworkLayer_RadioConnectionWithUeLost - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- RadioNetworkLayer_LoadBalancingTauRequired - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- RadioNetworkLayer_CsFallbackTriggered - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- RadioNetworkLayer_UeNotAvailableForPsService - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- RadioNetworkLayer_RadioResourcesNotAvailable - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- RadioNetworkLayer_FailureInTheRadioInterfaceProcedure - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- RadioNetworkLayer_InvalidQosCombination - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- RadioNetworkLayer_InterRatRedirection - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- RadioNetworkLayer_InteractionWithOtherProcedure - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- RadioNetworkLayer_UnknownErabId - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- RadioNetworkLayer_MultipleErabIdInstances - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.

- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- **Nas_Detach** - NAS Detach - S1AP NAS cause Detach.
- **Nas_Unspecified** - NAS Unspecified - S1AP NAS cause Unspecified.
- **Nas_CsgSubscriptionExpiry** - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- **Protocol_TransferSyntaxError** - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- **Protocol_AbstractSyntaxErrorReject** - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).

- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelCongestion - UE Context Release Command Congestion

Syntax

--ueCtxtRelCongestion <list-item>

Description

The S1AP cause the MME includes on UE context release command on message level, when the UE context release is following Attach/TAU/Service Reject with NAS EMM cause code 22 'congestion' when MME is overloaded and T3346 sending is enabled.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.

- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.

- **RadioNetworkLayer_HandoverDesirableForRadioReasons** - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- **RadioNetworkLayer_TimeCriticalHandover** - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- **RadioNetworkLayer_ResourceOptimisationHandover** - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- **RadioNetworkLayer_ReduceLoadInServingCell** - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- **RadioNetworkLayer_UserInactivity** - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- **RadioNetworkLayer_RadioConnectionWithUeLost** - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- **RadioNetworkLayer_LoadBalancingTauRequired** - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- **RadioNetworkLayer_CsFallbackTriggered** - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- **RadioNetworkLayer_UeNotAvailableForPsService** - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- **RadioNetworkLayer_RadioResourcesNotAvailable** - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- **RadioNetworkLayer_FailureInTheRadioInterfaceProcedure** - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- **RadioNetworkLayer_InvalidQosCombination** - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.

- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- **Nas_Detach** - NAS Detach - S1AP NAS cause Detach.
- **Nas_Unspecified** - NAS Unspecified - S1AP NAS cause Unspecified.
- **Nas_CsgSubscriptionExpiry** - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- **Protocol_TransferSyntaxError** - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- **Protocol_AbstractSyntaxErrorReject** - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).

- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelCreateSessRespUserAuthFail - UE Context Release Command Create Session Response User Authentication Fail

Syntax

- ueCtxtRelCreateSessRespUserAuthFail <list-item>

Description

S1AP release cause code to include on UE context release command upon receiving create session reject from SGW for attach request.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.

- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.

- RadioNetworkLayer_HandoverDesirableForRadioReasons - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- RadioNetworkLayer_TimeCriticalHandover - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- RadioNetworkLayer_ResourceOptimisationHandover - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- RadioNetworkLayer_ReduceLoadInServingCell - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- RadioNetworkLayer_UserInactivity - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- RadioNetworkLayer_RadioConnectionWithUeLost - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- RadioNetworkLayer_LoadBalancingTauRequired - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- RadioNetworkLayer_CsFallbackTriggered - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- RadioNetworkLayer_UeNotAvailableForPsService - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- RadioNetworkLayer_RadioResourcesNotAvailable - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- RadioNetworkLayer_FailureInTheRadioInterfaceProcedure - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- RadioNetworkLayer_InvalidQosCombination - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- RadioNetworkLayer_InterRatRedirection - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- RadioNetworkLayer_InteractionWithOtherProcedure - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- RadioNetworkLayer_UnknownEranId - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- RadioNetworkLayer_MultipleEranIdInstances - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.

- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- **Nas_Detach** - NAS Detach - S1AP NAS cause Detach.
- **Nas_Unspecified** - NAS Unspecified - S1AP NAS cause Unspecified.
- **Nas_CsgSubscriptionExpiry** - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- **Protocol_TransferSyntaxError** - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- **Protocol_AbstractSyntaxErrorReject** - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).

- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelCsfBMo - UE Context Release Command CSFB MO

Syntax

--ueCtxtRelCsfBMo <list-item>

Description

S1AP cause code to include on UE context release command due to Mobile Originated CSFB.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.

- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.

- **RadioNetworkLayer_HandoverDesirableForRadioReasons** - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- **RadioNetworkLayer_TimeCriticalHandover** - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- **RadioNetworkLayer_ResourceOptimisationHandover** - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- **RadioNetworkLayer_ReduceLoadInServingCell** - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- **RadioNetworkLayer_UserInactivity** - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- **RadioNetworkLayer_RadioConnectionWithUeLost** - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- **RadioNetworkLayer_LoadBalancingTauRequired** - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- **RadioNetworkLayer_CsFallbackTriggered** - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- **RadioNetworkLayer_UeNotAvailableForPsService** - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- **RadioNetworkLayer_RadioResourcesNotAvailable** - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- **RadioNetworkLayer_FailureInTheRadioInterfaceProcedure** - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- **RadioNetworkLayer_InvalidQosCombination** - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.

- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- **Nas_Detach** - NAS Detach - S1AP NAS cause Detach.
- **Nas_Unspecified** - NAS Unspecified - S1AP NAS cause Unspecified.
- **Nas_CsgSubscriptionExpiry** - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- **Protocol_TransferSyntaxError** - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- **Protocol_AbstractSyntaxErrorReject** - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).

- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelCsfBmt - UE Context Release Command CSFB MT

Syntax

--ueCtxtRelCsfBmt <list-item>

Description

S1AP cause code to include on UE context release command due to Mobile Terminated CSFB.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.

- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.

- RadioNetworkLayer_HandoverDesirableForRadioReasons - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- RadioNetworkLayer_TimeCriticalHandover - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- RadioNetworkLayer_ResourceOptimisationHandover - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- RadioNetworkLayer_ReduceLoadInServingCell - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- RadioNetworkLayer_UserInactivity - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- RadioNetworkLayer_RadioConnectionWithUeLost - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- RadioNetworkLayer_LoadBalancingTauRequired - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- RadioNetworkLayer_CsFallbackTriggered - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- RadioNetworkLayer_UeNotAvailableForPsService - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- RadioNetworkLayer_RadioResourcesNotAvailable - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- RadioNetworkLayer_FailureInTheRadioInterfaceProcedure - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- RadioNetworkLayer_InvalidQosCombination - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- RadioNetworkLayer_InterRatRedirection - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- RadioNetworkLayer_InteractionWithOtherProcedure - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- RadioNetworkLayer_UnknownErabId - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- RadioNetworkLayer_MultipleErabIdInstances - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.

- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- **Nas_Detach** - NAS Detach - S1AP NAS cause Detach.
- **Nas_Unspecified** - NAS Unspecified - S1AP NAS cause Unspecified.
- **Nas_CsgSubscriptionExpiry** - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- **Protocol_TransferSyntaxError** - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- **Protocol_AbstractSyntaxErrorReject** - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).

- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelCsgRestricted - UE Context Release Command CSG Restricted

Syntax

--ueCtxtRelCsgRestricted <list-item>

Description

S1AP cause code to include on UE context release command when the procedure is rejected because of CSG restrictions.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.

- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.

- **RadioNetworkLayer_HandoverDesirableForRadioReasons** - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- **RadioNetworkLayer_TimeCriticalHandover** - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- **RadioNetworkLayer_ResourceOptimisationHandover** - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- **RadioNetworkLayer_ReduceLoadInServingCell** - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- **RadioNetworkLayer_UserInactivity** - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- **RadioNetworkLayer_RadioConnectionWithUeLost** - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- **RadioNetworkLayer_LoadBalancingTauRequired** - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- **RadioNetworkLayer_CsFallbackTriggered** - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- **RadioNetworkLayer_UeNotAvailableForPsService** - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- **RadioNetworkLayer_RadioResourcesNotAvailable** - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- **RadioNetworkLayer_FailureInTheRadioInterfaceProcedure** - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- **RadioNetworkLayer_InvalidQosCombination** - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownEranId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleEranIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.

- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- **Nas_Detach** - NAS Detach - S1AP NAS cause Detach.
- **Nas_Unspecified** - NAS Unspecified - S1AP NAS cause Unspecified.
- **Nas_CsgSubscriptionExpiry** - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- **Protocol_TransferSyntaxError** - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- **Protocol_AbstractSyntaxErrorReject** - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).

- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelDelBearerReq - UE Context Release Command Delete Bearer Request

Syntax

--ueCtxtRelDelBearerReq <list-item>

Description

S1AP cause code to include on UE context release command when detach reason is delete bearer request.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.

- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.

- **RadioNetworkLayer_HandoverDesirableForRadioReasons** - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- **RadioNetworkLayer_TimeCriticalHandover** - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- **RadioNetworkLayer_ResourceOptimisationHandover** - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- **RadioNetworkLayer_ReduceLoadInServingCell** - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- **RadioNetworkLayer_UserInactivity** - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- **RadioNetworkLayer_RadioConnectionWithUeLost** - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- **RadioNetworkLayer_LoadBalancingTauRequired** - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- **RadioNetworkLayer_CsFallbackTriggered** - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- **RadioNetworkLayer_UeNotAvailableForPsService** - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- **RadioNetworkLayer_RadioResourcesNotAvailable** - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- **RadioNetworkLayer_FailureInTheRadioInterfaceProcedure** - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- **RadioNetworkLayer_InvalidQosCombination** - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownEranId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleEranIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.

- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- **Nas_Detach** - NAS Detach - S1AP NAS cause Detach.
- **Nas_Unspecified** - NAS Unspecified - S1AP NAS cause Unspecified.
- **Nas_CsgSubscriptionExpiry** - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- **Protocol_TransferSyntaxError** - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- **Protocol_AbstractSyntaxErrorReject** - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).

- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelDiamError - UE Context Release Command Diameter Error

Syntax

--ueCtxtRelDiamError <list-item>

Description

Mapping for incoming Diameter error to S1 release cause code.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.

- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.

- **RadioNetworkLayer_HandoverDesirableForRadioReasons** - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- **RadioNetworkLayer_TimeCriticalHandover** - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- **RadioNetworkLayer_ResourceOptimisationHandover** - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- **RadioNetworkLayer_ReduceLoadInServingCell** - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- **RadioNetworkLayer_UserInactivity** - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- **RadioNetworkLayer_RadioConnectionWithUeLost** - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- **RadioNetworkLayer_LoadBalancingTauRequired** - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- **RadioNetworkLayer_CsFallbackTriggered** - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- **RadioNetworkLayer_UeNotAvailableForPsService** - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- **RadioNetworkLayer_RadioResourcesNotAvailable** - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- **RadioNetworkLayer_FailureInTheRadioInterfaceProcedure** - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- **RadioNetworkLayer_InvalidQosCombination** - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownEranId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleEranIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.

- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- **Nas_Detach** - NAS Detach - S1AP NAS cause Detach.
- **Nas_Unspecified** - NAS Unspecified - S1AP NAS cause Unspecified.
- **Nas_CsgSubscriptionExpiry** - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- **Protocol_TransferSyntaxError** - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- **Protocol_AbstractSyntaxErrorReject** - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).

- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelDoubleS1 - UE Context Release Command Double S1

Syntax

--ueCtxtRelDoubleS1 <list-item>

Description

The S1AP cause the MME includes on UE context release command on message level, when the MME releases the old S1 connection due to receiving an S1AP Initial UE Message for an ECM-CONNECTED UE.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.

- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.

- **RadioNetworkLayer_HandoverDesirableForRadioReasons** - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- **RadioNetworkLayer_TimeCriticalHandover** - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- **RadioNetworkLayer_ResourceOptimisationHandover** - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- **RadioNetworkLayer_ReduceLoadInServingCell** - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- **RadioNetworkLayer_UserInactivity** - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- **RadioNetworkLayer_RadioConnectionWithUeLost** - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- **RadioNetworkLayer_LoadBalancingTauRequired** - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- **RadioNetworkLayer_CsFallbackTriggered** - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- **RadioNetworkLayer_UeNotAvailableForPsService** - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- **RadioNetworkLayer_RadioResourcesNotAvailable** - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- **RadioNetworkLayer_FailureInTheRadioInterfaceProcedure** - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- **RadioNetworkLayer_InvalidQosCombination** - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.

- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- **Nas_Detach** - NAS Detach - S1AP NAS cause Detach.
- **Nas_Unspecified** - NAS Unspecified - S1AP NAS cause Unspecified.
- **Nas_CsgSubscriptionExpiry** - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- **Protocol_TransferSyntaxError** - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- **Protocol_AbstractSyntaxErrorReject** - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).

- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelDoubleS1Ddn - UE Context Release Command Double S1 DDN

Syntax

--ueCtxtRelDoubleS1Ddn <list-item>

Description

The S1AP cause the MME includes on UE context release command on message level, when the MME releases the old S1 connection when receiving Downlink Data Notification for an ECM-CONNECTED UE.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.

- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.

- **RadioNetworkLayer_HandoverDesirableForRadioReasons** - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- **RadioNetworkLayer_TimeCriticalHandover** - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- **RadioNetworkLayer_ResourceOptimisationHandover** - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- **RadioNetworkLayer_ReduceLoadInServingCell** - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- **RadioNetworkLayer_UserInactivity** - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- **RadioNetworkLayer_RadioConnectionWithUeLost** - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- **RadioNetworkLayer_LoadBalancingTauRequired** - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- **RadioNetworkLayer_CsFallbackTriggered** - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- **RadioNetworkLayer_UeNotAvailableForPsService** - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- **RadioNetworkLayer_RadioResourcesNotAvailable** - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- **RadioNetworkLayer_FailureInTheRadioInterfaceProcedure** - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- **RadioNetworkLayer_InvalidQosCombination** - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.

- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- **Nas_Detach** - NAS Detach - S1AP NAS cause Detach.
- **Nas_Unspecified** - NAS Unspecified - S1AP NAS cause Unspecified.
- **Nas_CsgSubscriptionExpiry** - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- **Protocol_TransferSyntaxError** - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- **Protocol_AbstractSyntaxErrorReject** - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).

- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelEmmRegisteredWithoutPdnNotAllowed - UE Context Release Command EMM
Registered Without PDN Not Allowed

Syntax

- ueCtxtRelEmmRegisteredWithoutPdnNotAllowed <list-item>

Description

S1AP cause code to include on UE context release command when attach or TAU is rejected due to EMM-REGISTERED without PDN connection feature support is not enabled.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.

- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.

- **RadioNetworkLayer_HandoverDesirableForRadioReasons** - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- **RadioNetworkLayer_TimeCriticalHandover** - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- **RadioNetworkLayer_ResourceOptimisationHandover** - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- **RadioNetworkLayer_ReduceLoadInServingCell** - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- **RadioNetworkLayer_UserInactivity** - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- **RadioNetworkLayer_RadioConnectionWithUeLost** - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- **RadioNetworkLayer_LoadBalancingTauRequired** - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- **RadioNetworkLayer_CsFallbackTriggered** - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- **RadioNetworkLayer_UeNotAvailableForPsService** - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- **RadioNetworkLayer_RadioResourcesNotAvailable** - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- **RadioNetworkLayer_FailureInTheRadioInterfaceProcedure** - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- **RadioNetworkLayer_InvalidQosCombination** - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownEranId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleEranIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.

- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- **Nas_Detach** - NAS Detach - S1AP NAS cause Detach.
- **Nas_Unspecified** - NAS Unspecified - S1AP NAS cause Unspecified.
- **Nas_CsgSubscriptionExpiry** - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- **Protocol_TransferSyntaxError** - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- **Protocol_AbstractSyntaxErrorReject** - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).

- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelEsmCc29 - UE Context Release Command ESM CC 29

Syntax

--ueCtxtRelEsmCc29 <list-item>

Description

S1AP cause code to include on UE context release command when attach is rejected due to PDN connectivity request is rejected with cause code 29.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.

- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.

- **RadioNetworkLayer_HandoverDesirableForRadioReasons** - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- **RadioNetworkLayer_TimeCriticalHandover** - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- **RadioNetworkLayer_ResourceOptimisationHandover** - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- **RadioNetworkLayer_ReduceLoadInServingCell** - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- **RadioNetworkLayer_UserInactivity** - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- **RadioNetworkLayer_RadioConnectionWithUeLost** - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- **RadioNetworkLayer_LoadBalancingTauRequired** - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- **RadioNetworkLayer_CsFallbackTriggered** - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- **RadioNetworkLayer_UeNotAvailableForPsService** - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- **RadioNetworkLayer_RadioResourcesNotAvailable** - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- **RadioNetworkLayer_FailureInTheRadioInterfaceProcedure** - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- **RadioNetworkLayer_InvalidQosCombination** - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.

- RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- RadioNetworkLayer_S1IntraSystemHandoverTriggered - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- RadioNetworkLayer_S1InterSystemHandoverTriggered - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- RadioNetworkLayer_x2HandoverTriggered - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- RadioNetworkLayer_redirectionTowards1xRtt - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- RadioNetworkLayer_NotSupportedQciValue - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- RadioNetworkLayer_InvalidCsgId - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- RadioNetworkLayer_ReleaseDueToPreEmption - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- RadioNetworkLayer_N26InterfaceNotAvailable - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- TransportLayer_TransportResourceUnavailable - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- TransportLayer_Unspecified - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- Nas_NormalRelease - NAS Normal Release - S1AP NAS cause Normal Release.
- Nas_AuthenticationFailure - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- Nas_Detach - NAS Detach - S1AP NAS cause Detach.
- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).

- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelEutranNotAllowed - UE Context Release Command E-UTRAN Not Allowed

Syntax

--ueCtxtRelEutranNotAllowed <list-item>

Description

S1AP cause code to include on UE context release command when the procedure is rejected because UE is not allowed to access LTE.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.

- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.

- RadioNetworkLayer_HandoverDesirableForRadioReasons - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- RadioNetworkLayer_TimeCriticalHandover - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- RadioNetworkLayer_ResourceOptimisationHandover - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- RadioNetworkLayer_ReduceLoadInServingCell - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- RadioNetworkLayer_UserInactivity - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- RadioNetworkLayer_RadioConnectionWithUeLost - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- RadioNetworkLayer_LoadBalancingTauRequired - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- RadioNetworkLayer_CsFallbackTriggered - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- RadioNetworkLayer_UeNotAvailableForPsService - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- RadioNetworkLayer_RadioResourcesNotAvailable - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- RadioNetworkLayer_FailureInTheRadioInterfaceProcedure - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- RadioNetworkLayer_InvalidQosCombination - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- RadioNetworkLayer_InterRatRedirection - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- RadioNetworkLayer_InteractionWithOtherProcedure - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- RadioNetworkLayer_UnknownErabId - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- RadioNetworkLayer_MultipleErabIdInstances - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.

- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- **Nas_Detach** - NAS Detach - S1AP NAS cause Detach.
- **Nas_Unspecified** - NAS Unspecified - S1AP NAS cause Unspecified.
- **Nas_CsgSubscriptionExpiry** - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- **Protocol_TransferSyntaxError** - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- **Protocol_AbstractSyntaxErrorReject** - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).

- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelFailopenRestoration - UE Context Release Command Failopen Restoration

Syntax

--ueCtxtRelFailopenRestoration <list-item>

Description

Define the S1AP cause code to include on UE context release command, when the MME detaches a UE that has been attached with the help of the failopen profile due to HSS unavailability manual restoration.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.

- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.

- `RadioNetworkLayer_HandoverDesirableForRadioReasons` - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- `RadioNetworkLayer_TimeCriticalHandover` - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- `RadioNetworkLayer_ResourceOptimisationHandover` - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- `RadioNetworkLayer_ReduceLoadInServingCell` - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- `RadioNetworkLayer_UserInactivity` - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- `RadioNetworkLayer_RadioConnectionWithUeLost` - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- `RadioNetworkLayer_LoadBalancingTauRequired` - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- `RadioNetworkLayer_CsFallbackTriggered` - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- `RadioNetworkLayer_UeNotAvailableForPsService` - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- `RadioNetworkLayer_RadioResourcesNotAvailable` - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- `RadioNetworkLayer_FailureInTheRadioInterfaceProcedure` - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- `RadioNetworkLayer_InvalidQosCombination` - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- `RadioNetworkLayer_InterRatRedirection` - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- `RadioNetworkLayer_InteractionWithOtherProcedure` - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- `RadioNetworkLayer_UnknownEranId` - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- `RadioNetworkLayer_MultipleEranIdInstances` - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.

- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- **Nas_Detach** - NAS Detach - S1AP NAS cause Detach.
- **Nas_Unspecified** - NAS Unspecified - S1AP NAS cause Unspecified.
- **Nas_CsgSubscriptionExpiry** - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- **Protocol_TransferSyntaxError** - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- **Protocol_AbstractSyntaxErrorReject** - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).

- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelHssOrEirDown - UE Context Release Command HSS or EIR Down

Syntax

--ueCtxtRelHssOrEirDown <list-item>

Description

Mapping for HSS or EIR down to S1 release cause code.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.

- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.

- **RadioNetworkLayer_HandoverDesirableForRadioReasons** - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- **RadioNetworkLayer_TimeCriticalHandover** - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- **RadioNetworkLayer_ResourceOptimisationHandover** - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- **RadioNetworkLayer_ReduceLoadInServingCell** - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- **RadioNetworkLayer_UserInactivity** - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- **RadioNetworkLayer_RadioConnectionWithUeLost** - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- **RadioNetworkLayer_LoadBalancingTauRequired** - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- **RadioNetworkLayer_CsFallbackTriggered** - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- **RadioNetworkLayer_UeNotAvailableForPsService** - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- **RadioNetworkLayer_RadioResourcesNotAvailable** - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- **RadioNetworkLayer_FailureInTheRadioInterfaceProcedure** - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- **RadioNetworkLayer_InvalidQosCombination** - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownEranId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleEranIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.

- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- **Nas_Detach** - NAS Detach - S1AP NAS cause Detach.
- **Nas_Unspecified** - NAS Unspecified - S1AP NAS cause Unspecified.
- **Nas_CsgSubscriptionExpiry** - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- **Protocol_TransferSyntaxError** - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- **Protocol_AbstractSyntaxErrorReject** - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).

- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelImsApnRoamingFailure - UE Context Release Command IMS APN Roaming Failure

Syntax

--ueCtxtRelImsApnRoamingFailure <list-item>

Description

S1AP cause code to include on UE context release command when attach is rejected because of IMS APN roaming failure.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.

- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.

- **RadioNetworkLayer_HandoverDesirableForRadioReasons** - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- **RadioNetworkLayer_TimeCriticalHandover** - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- **RadioNetworkLayer_ResourceOptimisationHandover** - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- **RadioNetworkLayer_ReduceLoadInServingCell** - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- **RadioNetworkLayer_UserInactivity** - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- **RadioNetworkLayer_RadioConnectionWithUeLost** - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- **RadioNetworkLayer_LoadBalancingTauRequired** - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- **RadioNetworkLayer_CsFallbackTriggered** - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- **RadioNetworkLayer_UeNotAvailableForPsService** - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- **RadioNetworkLayer_RadioResourcesNotAvailable** - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- **RadioNetworkLayer_FailureInTheRadioInterfaceProcedure** - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- **RadioNetworkLayer_InvalidQosCombination** - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownEranId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleEranIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.

- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- **Nas_Detach** - NAS Detach - S1AP NAS cause Detach.
- **Nas_Unspecified** - NAS Unspecified - S1AP NAS cause Unspecified.
- **Nas_CsgSubscriptionExpiry** - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- **Protocol_TransferSyntaxError** - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- **Protocol_AbstractSyntaxErrorReject** - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).

- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelImsPdnActivationTimerExpired - UE Context Release Command IMS PDN Activation Timer Expired

Syntax

- ueCtxtRelImsPdnActivationTimerExpired <list-item>

Description

The S1AP Cause code to include on UE context release command when the MME detaches the UE as part of the IMS supervision functionality. MME performs detach for a UE when the IMS PDN Activation Timer expires and IMS PDN is not activated.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.

- RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- RadioNetworkLayer_HandoverDesirableForRadioReasons - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- RadioNetworkLayer_TimeCriticalHandover - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- RadioNetworkLayer_ResourceOptimisationHandover - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- RadioNetworkLayer_ReduceLoadInServingCell - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- RadioNetworkLayer_UserInactivity - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- RadioNetworkLayer_RadioConnectionWithUeLost - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- RadioNetworkLayer_LoadBalancingTauRequired - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- RadioNetworkLayer_CsFallbackTriggered - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- RadioNetworkLayer_UeNotAvailableForPsService - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- RadioNetworkLayer_RadioResourcesNotAvailable - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- RadioNetworkLayer_FailureInTheRadioInterfaceProcedure - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- RadioNetworkLayer_InvalidQosCombination - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- RadioNetworkLayer_InterRatRedirection - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- RadioNetworkLayer_InteractionWithOtherProcedure - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.

- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- **Nas_Detach** - NAS Detach - S1AP NAS cause Detach.
- **Nas_Unspecified** - NAS Unspecified - S1AP NAS cause Unspecified.
- **Nas_CsgSubscriptionExpiry** - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.

- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelImsRegistrationFailed - UE Context Release Command IMS Registration Failed

Syntax

--ueCtxtRelImsRegistrationFailed <list-item>

Description

The S1AP cause code to include on UE context release command when the IMS Registration Timer expires and the UE has not performed successful IMS registration.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.

- RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- RadioNetworkLayer_HandoverDesirableForRadioReasons - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- RadioNetworkLayer_TimeCriticalHandover - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- RadioNetworkLayer_ResourceOptimisationHandover - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- RadioNetworkLayer_ReduceLoadInServingCell - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- RadioNetworkLayer_UserInactivity - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- RadioNetworkLayer_RadioConnectionWithUeLost - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- RadioNetworkLayer_LoadBalancingTauRequired - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- RadioNetworkLayer_CsFallbackTriggered - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- RadioNetworkLayer_UeNotAvailableForPsService - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- RadioNetworkLayer_RadioResourcesNotAvailable - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- RadioNetworkLayer_FailureInTheRadioInterfaceProcedure - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- RadioNetworkLayer_InvalidQosCombination - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- RadioNetworkLayer_InterRatRedirection - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- RadioNetworkLayer_InteractionWithOtherProcedure - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.

- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- **Nas_Detach** - NAS Detach - S1AP NAS cause Detach.
- **Nas_Unspecified** - NAS Unspecified - S1AP NAS cause Unspecified.
- **Nas_CsgSubscriptionExpiry** - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.

- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelInterPlmnTauReject - UE Context Release Command Inter PLMN TAU Reject

Syntax

--ueCtxtRelInterPlmnTauReject <list-item>

Description

S1AP cause code to include on UE context release command when inter PLMN TAU request is rejected.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.

- **RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId** - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- **RadioNetworkLayer_HandoverDesirableForRadioReasons** - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- **RadioNetworkLayer_TimeCriticalHandover** - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- **RadioNetworkLayer_ResourceOptimisationHandover** - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- **RadioNetworkLayer_ReduceLoadInServingCell** - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- **RadioNetworkLayer_UserInactivity** - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- **RadioNetworkLayer_RadioConnectionWithUeLost** - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- **RadioNetworkLayer_LoadBalancingTauRequired** - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- **RadioNetworkLayer_CsFallbackTriggered** - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- **RadioNetworkLayer_UeNotAvailableForPsService** - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- **RadioNetworkLayer_RadioResourcesNotAvailable** - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- **RadioNetworkLayer_FailureInTheRadioInterfaceProcedure** - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- **RadioNetworkLayer_InvalidQosCombination** - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.

- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- **Nas_Detach** - NAS Detach - S1AP NAS cause Detach.
- **Nas_Unspecified** - NAS Unspecified - S1AP NAS cause Unspecified.
- **Nas_CsgSubscriptionExpiry** - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.

- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelIntraMmeHoFailure - UE Context Release Command Intra MME Handover Failure

Syntax

--ueCtxtRelIntraMmeHoFailure <list-item>

Description

The S1AP cause the MME includes on UE context release command on message level, when the S1 is released due to a failure during intra MME handover.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.

- RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- RadioNetworkLayer_HandoverDesirableForRadioReasons - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- RadioNetworkLayer_TimeCriticalHandover - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- RadioNetworkLayer_ResourceOptimisationHandover - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- RadioNetworkLayer_ReduceLoadInServingCell - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- RadioNetworkLayer_UserInactivity - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- RadioNetworkLayer_RadioConnectionWithUeLost - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- RadioNetworkLayer_LoadBalancingTauRequired - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- RadioNetworkLayer_CsFallbackTriggered - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- RadioNetworkLayer_UeNotAvailableForPsService - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- RadioNetworkLayer_RadioResourcesNotAvailable - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- RadioNetworkLayer_FailureInTheRadioInterfaceProcedure - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- RadioNetworkLayer_InvalidQosCombination - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- RadioNetworkLayer_InterRatRedirection - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- RadioNetworkLayer_InteractionWithOtherProcedure - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.

- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- **Nas_Detach** - NAS Detach - S1AP NAS cause Detach.
- **Nas_Unspecified** - NAS Unspecified - S1AP NAS cause Unspecified.
- **Nas_CsgSubscriptionExpiry** - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.

- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelIpVerMismatch - UE Context Release Command IP Version Mismatch

Syntax

--ueCtxtRelIpVerMismatch <list-item>

Description

S1AP cause code to include on UE context release command when IPv4 only allowed, IPv6 only allowed or IP version mismatch during Attach, Service request or TAU.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.

- RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- RadioNetworkLayer_HandoverDesirableForRadioReasons - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- RadioNetworkLayer_TimeCriticalHandover - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- RadioNetworkLayer_ResourceOptimisationHandover - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- RadioNetworkLayer_ReduceLoadInServingCell - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- RadioNetworkLayer_UserInactivity - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- RadioNetworkLayer_RadioConnectionWithUeLost - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- RadioNetworkLayer_LoadBalancingTauRequired - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- RadioNetworkLayer_CsFallbackTriggered - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- RadioNetworkLayer_UeNotAvailableForPsService - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- RadioNetworkLayer_RadioResourcesNotAvailable - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- RadioNetworkLayer_FailureInTheRadioInterfaceProcedure - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- RadioNetworkLayer_InvalidQosCombination - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- RadioNetworkLayer_InterRatRedirection - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- RadioNetworkLayer_InteractionWithOtherProcedure - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.

- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- **Nas_Detach** - NAS Detach - S1AP NAS cause Detach.
- **Nas_Unspecified** - NAS Unspecified - S1AP NAS cause Unspecified.
- **Nas_CsgSubscriptionExpiry** - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.

- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelLteBarAllPacketOdb - UE Context Release Command ODB All Packet Oriented Services Barred

Syntax

--ueCtxtRelLteBarAllPacketOdb <list-item>

Description

S1AP cause code to include on UE context release command when in LTE all packet oriented services are barred.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.

- **RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId** - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- **RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId** - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- **RadioNetworkLayer_HandoverDesirableForRadioReasons** - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- **RadioNetworkLayer_TimeCriticalHandover** - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- **RadioNetworkLayer_ResourceOptimisationHandover** - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- **RadioNetworkLayer_ReduceLoadInServingCell** - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- **RadioNetworkLayer_UserInactivity** - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- **RadioNetworkLayer_RadioConnectionWithUeLost** - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- **RadioNetworkLayer_LoadBalancingTauRequired** - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- **RadioNetworkLayer_CsFallbackTriggered** - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- **RadioNetworkLayer_UeNotAvailableForPsService** - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- **RadioNetworkLayer_RadioResourcesNotAvailable** - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- **RadioNetworkLayer_FailureInTheRadioInterfaceProcedure** - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- **RadioNetworkLayer_InvalidQosCombination** - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.

- RadioNetworkLayer_InteractionWithOtherProcedure - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- RadioNetworkLayer_UnknownErbId - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- RadioNetworkLayer_MultipleErbIdInstances - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- RadioNetworkLayer_S1IntraSystemHandoverTriggered - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- RadioNetworkLayer_S1InterSystemHandoverTriggered - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- RadioNetworkLayer_x2HandoverTriggered - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- RadioNetworkLayer_redirectionTowards1xRtt - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- RadioNetworkLayer_NotSupportedQciValue - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- RadioNetworkLayer_InvalidCsgId - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- RadioNetworkLayer_ReleaseDueToPreEmption - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- RadioNetworkLayer_N26InterfaceNotAvailable - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- TransportLayer_TransportResourceUnavailable - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- TransportLayer_Unspecified - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- Nas_NormalRelease - NAS Normal Release - S1AP NAS cause Normal Release.
- Nas_AuthenticationFailure - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- Nas_Detach - NAS Detach - S1AP NAS cause Detach.

- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelLtemNotAllowed - UE Context Release Command LTE-M Not Allowed

Syntax

--ueCtxtRelLtemNotAllowed <list-item>

Description

S1AP cause code to include on UE context release command when the UE is not allowed in LTE-M access.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.

- `RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId` - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- `RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId` - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- `RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId` - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- `RadioNetworkLayer_HandoverDesirableForRadioReasons` - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- `RadioNetworkLayer_TimeCriticalHandover` - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- `RadioNetworkLayer_ResourceOptimisationHandover` - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- `RadioNetworkLayer_ReduceLoadInServingCell` - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- `RadioNetworkLayer_UserInactivity` - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- `RadioNetworkLayer_RadioConnectionWithUeLost` - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- `RadioNetworkLayer_LoadBalancingTauRequired` - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- `RadioNetworkLayer_CsFallbackTriggered` - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- `RadioNetworkLayer_UeNotAvailableForPsService` - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- `RadioNetworkLayer_RadioResourcesNotAvailable` - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- `RadioNetworkLayer_FailureInTheRadioInterfaceProcedure` - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- `RadioNetworkLayer_InvalidQosCombination` - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.

- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.

- Nas_Detach - NAS Detach - S1AP NAS cause Detach.
- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelLtePlmnNotAllowed - UE Context Release Command PLMN of IMSI Is Not Allowed

Syntax

--ueCtxtRelLtePlmnNotAllowed <list-item>

Description

S1AP cause code to include on UE context release command when PLMN access is not allowed.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.

- RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- RadioNetworkLayer_HandoverDesirableForRadioReasons - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- RadioNetworkLayer_TimeCriticalHandover - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- RadioNetworkLayer_ResourceOptimisationHandover - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- RadioNetworkLayer_ReduceLoadInServingCell - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- RadioNetworkLayer_UserInactivity - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- RadioNetworkLayer_RadioConnectionWithUeLost - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- RadioNetworkLayer_LoadBalancingTauRequired - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- RadioNetworkLayer_CsFallbackTriggered - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- RadioNetworkLayer_UeNotAvailableForPsService - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- RadioNetworkLayer_RadioResourcesNotAvailable - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- RadioNetworkLayer_FailureInTheRadioInterfaceProcedure - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- RadioNetworkLayer_InvalidQosCombination - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.

- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownEranId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleEranIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.

- Nas_Detach - NAS Detach - S1AP NAS cause Detach.
- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelNasProtocolError - UE Context Release Command NAS Protocol Error

Syntax

--ueCtxtRelNasProtocolError <list-item>

Description

S1AP cause code to include on UE context release command when NAS protocol error is met:
-mandatory IE incorrect -optional IE incorrect

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.

- RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- RadioNetworkLayer_HandoverDesirableForRadioReasons - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- RadioNetworkLayer_TimeCriticalHandover - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- RadioNetworkLayer_ResourceOptimisationHandover - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- RadioNetworkLayer_ReduceLoadInServingCell - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- RadioNetworkLayer_UserInactivity - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- RadioNetworkLayer_RadioConnectionWithUeLost - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- RadioNetworkLayer_LoadBalancingTauRequired - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- RadioNetworkLayer_CsFallbackTriggered - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- RadioNetworkLayer_UeNotAvailableForPsService - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- RadioNetworkLayer_RadioResourcesNotAvailable - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- RadioNetworkLayer_FailureInTheRadioInterfaceProcedure - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- RadioNetworkLayer_InvalidQosCombination - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.

- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.

- Nas_Detach - NAS Detach - S1AP NAS cause Detach.
- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelNasRejectApnAdministration - UE Context Release NAS Reject APN Administration

Syntax

--ueCtxtRelNasRejectApnAdministration <list-item>

Description

The S1AP cause the MME includes on UE context release command on message level, when the MME rejects the Attach due to APN administration.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.

- RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- RadioNetworkLayer_HandoverDesirableForRadioReasons - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- RadioNetworkLayer_TimeCriticalHandover - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- RadioNetworkLayer_ResourceOptimisationHandover - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- RadioNetworkLayer_ReduceLoadInServingCell - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- RadioNetworkLayer_UserInactivity - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- RadioNetworkLayer_RadioConnectionWithUeLost - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- RadioNetworkLayer_LoadBalancingTauRequired - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- RadioNetworkLayer_CsFallbackTriggered - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- RadioNetworkLayer_UeNotAvailableForPsService - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- RadioNetworkLayer_RadioResourcesNotAvailable - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- RadioNetworkLayer_FailureInTheRadioInterfaceProcedure - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- RadioNetworkLayer_InvalidQosCombination - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.

- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemhandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.

- Nas_Detach - NAS Detach - S1AP NAS cause Detach.
- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Nas_NormalRelease

Name

ueCtxtRelNbiotBarAllPacketOdb - UE Context Release Command NB-IOT Bar All Packet ODB

Syntax

--ueCtxtRelNbiotBarAllPacketOdb <list-item>

Description

S1AP cause code to include on UE context release command when NB-IoT UE that has ODB.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.

- RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- RadioNetworkLayer_HandoverDesirableForRadioReasons - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- RadioNetworkLayer_TimeCriticalHandover - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- RadioNetworkLayer_ResourceOptimisationHandover - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- RadioNetworkLayer_ReduceLoadInServingCell - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- RadioNetworkLayer_UserInactivity - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- RadioNetworkLayer_RadioConnectionWithUeLost - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- RadioNetworkLayer_LoadBalancingTauRequired - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- RadioNetworkLayer_CsFallbackTriggered - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- RadioNetworkLayer_UeNotAvailableForPsService - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- RadioNetworkLayer_RadioResourcesNotAvailable - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- RadioNetworkLayer_FailureInTheRadioInterfaceProcedure - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- RadioNetworkLayer_InvalidQosCombination - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.

- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.

- Nas_Detach - NAS Detach - S1AP NAS cause Detach.
- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelNbIotNotAllowed - UE Context Release Command NB-IOT Not Allowed

Syntax

--ueCtxtRelNbIotNotAllowed <list-item>

Description

S1AP cause code to include on UE context release command when NB-IoT is not allowed.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.

- RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- RadioNetworkLayer_HandoverDesirableForRadioReasons - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- RadioNetworkLayer_TimeCriticalHandover - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- RadioNetworkLayer_ResourceOptimisationHandover - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- RadioNetworkLayer_ReduceLoadInServingCell - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- RadioNetworkLayer_UserInactivity - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- RadioNetworkLayer_RadioConnectionWithUeLost - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- RadioNetworkLayer_LoadBalancingTauRequired - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- RadioNetworkLayer_CsFallbackTriggered - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- RadioNetworkLayer_UeNotAvailableForPsService - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- RadioNetworkLayer_RadioResourcesNotAvailable - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- RadioNetworkLayer_FailureInTheRadioInterfaceProcedure - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- RadioNetworkLayer_InvalidQosCombination - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.

- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemhandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.

- Nas_Detach - NAS Detach - S1AP NAS cause Detach.
- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelNbiotSgsFail - UE Context Release Command NB-IOT SGs Fail

Syntax

--ueCtxtRelNbiotSgsFail <list-item>

Description

S1AP cause code to include on UE context release command when attach or TAU rejected if MME fails to set up SGs association.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.

- `RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId` - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- `RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId` - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- `RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId` - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- `RadioNetworkLayer_HandoverDesirableForRadioReasons` - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- `RadioNetworkLayer_TimeCriticalHandover` - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- `RadioNetworkLayer_ResourceOptimisationHandover` - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- `RadioNetworkLayer_ReduceLoadInServingCell` - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- `RadioNetworkLayer_UserInactivity` - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- `RadioNetworkLayer_RadioConnectionWithUeLost` - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- `RadioNetworkLayer_LoadBalancingTauRequired` - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- `RadioNetworkLayer_CsFallbackTriggered` - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- `RadioNetworkLayer_UeNotAvailableForPsService` - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- `RadioNetworkLayer_RadioResourcesNotAvailable` - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- `RadioNetworkLayer_FailureInTheRadioInterfaceProcedure` - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- `RadioNetworkLayer_InvalidQosCombination` - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.

- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.

- Nas_Detach - NAS Detach - S1AP NAS cause Detach.
- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelNoApnAllowedByHss - UE Context Release Command No APN Allowed by HSS

Syntax`--ueCtxtRelNoApnAllowedByHss <list-item>`

Description

S1AP cause code to include on UE context release command when no APN is allowed by HSS:

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.

- `RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId` - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- `RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId` - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- `RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId` - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- `RadioNetworkLayer_HandoverDesirableForRadioReasons` - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- `RadioNetworkLayer_TimeCriticalHandover` - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- `RadioNetworkLayer_ResourceOptimisationHandover` - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- `RadioNetworkLayer_ReduceLoadInServingCell` - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- `RadioNetworkLayer_UserInactivity` - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- `RadioNetworkLayer_RadioConnectionWithUeLost` - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- `RadioNetworkLayer_LoadBalancingTauRequired` - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- `RadioNetworkLayer_CsFallbackTriggered` - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- `RadioNetworkLayer_UeNotAvailableForPsService` - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- `RadioNetworkLayer_RadioResourcesNotAvailable` - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- `RadioNetworkLayer_FailureInTheRadioInterfaceProcedure` - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- `RadioNetworkLayer_InvalidQosCombination` - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.

- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.

- Nas_Detach - NAS Detach - S1AP NAS cause Detach.
- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelNoaUe - UE Context Release Command NOA User Unknown

Syntax`--ueCtxtRelNoaUe <list-item>`

Description

The S1AP cause code to include on UE context release command when detach reason is user unknown received in Notify Answer from HSS.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.

- RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- RadioNetworkLayer_HandoverDesirableForRadioReasons - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- RadioNetworkLayer_TimeCriticalHandover - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- RadioNetworkLayer_ResourceOptimisationHandover - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- RadioNetworkLayer_ReduceLoadInServingCell - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- RadioNetworkLayer_UserInactivity - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- RadioNetworkLayer_RadioConnectionWithUeLost - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- RadioNetworkLayer_LoadBalancingTauRequired - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- RadioNetworkLayer_CsFallbackTriggered - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- RadioNetworkLayer_UeNotAvailableForPsService - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- RadioNetworkLayer_RadioResourcesNotAvailable - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- RadioNetworkLayer_FailureInTheRadioInterfaceProcedure - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- RadioNetworkLayer_InvalidQosCombination - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.

- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.

- Nas_Detach - NAS Detach - S1AP NAS cause Detach.
- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelNoProvisionedHss - UE Context Release Command No Provisioned HSS

Syntax

--ueCtxtRelNoProvisionedHss <list-item>

Description

Mapping for no provisioned HSS to S1 release cause code.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.

- `RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId` - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- `RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId` - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- `RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId` - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- `RadioNetworkLayer_HandoverDesirableForRadioReasons` - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- `RadioNetworkLayer_TimeCriticalHandover` - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- `RadioNetworkLayer_ResourceOptimisationHandover` - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- `RadioNetworkLayer_ReduceLoadInServingCell` - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- `RadioNetworkLayer_UserInactivity` - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- `RadioNetworkLayer_RadioConnectionWithUeLost` - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- `RadioNetworkLayer_LoadBalancingTauRequired` - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- `RadioNetworkLayer_CsFallbackTriggered` - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- `RadioNetworkLayer_UeNotAvailableForPsService` - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- `RadioNetworkLayer_RadioResourcesNotAvailable` - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- `RadioNetworkLayer_FailureInTheRadioInterfaceProcedure` - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- `RadioNetworkLayer_InvalidQosCombination` - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.

- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.

- Nas_Detach - NAS Detach - S1AP NAS cause Detach.
- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelNoResponseFromHss - UE Context Release Command No Response from HSS

Syntax

--ueCtxtRelNoResponseFromHss <list-item>

Description

Mapping for no response from HSS to S1 release cause code.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.

- `RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId` - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- `RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId` - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- `RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId` - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- `RadioNetworkLayer_HandoverDesirableForRadioReasons` - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- `RadioNetworkLayer_TimeCriticalHandover` - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- `RadioNetworkLayer_ResourceOptimisationHandover` - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- `RadioNetworkLayer_ReduceLoadInServingCell` - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- `RadioNetworkLayer_UserInactivity` - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- `RadioNetworkLayer_RadioConnectionWithUeLost` - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- `RadioNetworkLayer_LoadBalancingTauRequired` - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- `RadioNetworkLayer_CsFallbackTriggered` - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- `RadioNetworkLayer_UeNotAvailableForPsService` - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- `RadioNetworkLayer_RadioResourcesNotAvailable` - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- `RadioNetworkLayer_FailureInTheRadioInterfaceProcedure` - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- `RadioNetworkLayer_InvalidQosCombination` - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.

- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownEranId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleEranIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.

- Nas_Detach - NAS Detach - S1AP NAS cause Detach.
- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelNoResponseFromSgw - UE Context Release Command No Response from SGW

Syntax

--ueCtxtRelNoResponseFromSgw <list-item>

Description

S1AP cause code to include on UE context release command when there is no response from SGW.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.

- RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- RadioNetworkLayer_HandoverDesirableForRadioReasons - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- RadioNetworkLayer_TimeCriticalHandover - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- RadioNetworkLayer_ResourceOptimisationHandover - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- RadioNetworkLayer_ReduceLoadInServingCell - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- RadioNetworkLayer_UserInactivity - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- RadioNetworkLayer_RadioConnectionWithUeLost - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- RadioNetworkLayer_LoadBalancingTauRequired - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- RadioNetworkLayer_CsFallbackTriggered - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- RadioNetworkLayer_UeNotAvailableForPsService - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- RadioNetworkLayer_RadioResourcesNotAvailable - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- RadioNetworkLayer_FailureInTheRadioInterfaceProcedure - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- RadioNetworkLayer_InvalidQosCombination - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.

- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.

- Nas_Detach - NAS Detach - S1AP NAS cause Detach.
- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelNotAllowedTai - UE Context Release Command Not Allowed in a TAI

Syntax`--ueCtxtRelNotAllowedTai <list-item>`

Description

S1AP cause code to include on UE context release command when UE is in a restricted TA where UE is not allowed to access services.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.

- RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- RadioNetworkLayer_HandoverDesirableForRadioReasons - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- RadioNetworkLayer_TimeCriticalHandover - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- RadioNetworkLayer_ResourceOptimisationHandover - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- RadioNetworkLayer_ReduceLoadInServingCell - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- RadioNetworkLayer_UserInactivity - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- RadioNetworkLayer_RadioConnectionWithUeLost - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- RadioNetworkLayer_LoadBalancingTauRequired - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- RadioNetworkLayer_CsFallbackTriggered - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- RadioNetworkLayer_UeNotAvailableForPsService - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- RadioNetworkLayer_RadioResourcesNotAvailable - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- RadioNetworkLayer_FailureInTheRadioInterfaceProcedure - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- RadioNetworkLayer_InvalidQosCombination - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.

- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.

- Nas_Detach - NAS Detach - S1AP NAS cause Detach.
- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelPcscfBrrReact - UE Context Release Command P-CSCF Detach Bearer Request

Syntax`--ueCtxtRelPcscfBrrReact <list-item>`

Description

The S1AP cause code include on UE context release command when P-CSCF detach bearer with reactivation requested.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.

- `RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId` - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- `RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId` - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- `RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId` - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- `RadioNetworkLayer_HandoverDesirableForRadioReasons` - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- `RadioNetworkLayer_TimeCriticalHandover` - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- `RadioNetworkLayer_ResourceOptimisationHandover` - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- `RadioNetworkLayer_ReduceLoadInServingCell` - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- `RadioNetworkLayer_UserInactivity` - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- `RadioNetworkLayer_RadioConnectionWithUeLost` - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- `RadioNetworkLayer_LoadBalancingTauRequired` - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- `RadioNetworkLayer_CsFallbackTriggered` - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- `RadioNetworkLayer_UeNotAvailableForPsService` - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- `RadioNetworkLayer_RadioResourcesNotAvailable` - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- `RadioNetworkLayer_FailureInTheRadioInterfaceProcedure` - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- `RadioNetworkLayer_InvalidQosCombination` - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.

- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.
- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.

- Nas_Detach - NAS Detach - S1AP NAS cause Detach.
- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelPdnReEstDetach - UE Context Release Command PDN Re-establishment Detach

Syntax

--ueCtxtRelPdnReEstDetach <list-item>

Description

The S1AP cause the MME includes on UE context release command on message level, when the UE context release is performed as part of a Detach procedure due to PDN connection re-establishment after SGW relocation functionality.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.

- **RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell** - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- **RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId** - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- **RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId** - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- **RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId** - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- **RadioNetworkLayer_HandoverDesirableForRadioReasons** - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- **RadioNetworkLayer_TimeCriticalHandover** - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- **RadioNetworkLayer_ResourceOptimisationHandover** - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- **RadioNetworkLayer_ReduceLoadInServingCell** - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- **RadioNetworkLayer_UserInactivity** - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- **RadioNetworkLayer_RadioConnectionWithUeLost** - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- **RadioNetworkLayer_LoadBalancingTauRequired** - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- **RadioNetworkLayer_CsFallbackTriggered** - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- **RadioNetworkLayer_UeNotAvailableForPsService** - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- **RadioNetworkLayer_RadioResourcesNotAvailable** - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- **RadioNetworkLayer_FailureInTheRadioInterfaceProcedure** - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.

- **RadioNetworkLayer_InvalidQosCombination** - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemhandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.

- Nas_AuthenticationFailure - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- Nas_Detach - NAS Detach - S1AP NAS cause Detach.
- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxRelPgwrRestart - UE Context Release Command PGW Restart

Syntax

--ueCtxtRelPgwRestart <list-item>

Description

The S1AP cause code to include on UE context release command sent to UE if its detached due to PGW restart.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.

- **RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell** - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- **RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId** - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- **RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId** - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- **RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId** - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- **RadioNetworkLayer_HandoverDesirableForRadioReasons** - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- **RadioNetworkLayer_TimeCriticalHandover** - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- **RadioNetworkLayer_ResourceOptimisationHandover** - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- **RadioNetworkLayer_ReduceLoadInServingCell** - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- **RadioNetworkLayer_UserInactivity** - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- **RadioNetworkLayer_RadioConnectionWithUeLost** - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- **RadioNetworkLayer_LoadBalancingTauRequired** - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- **RadioNetworkLayer_CsFallbackTriggered** - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- **RadioNetworkLayer_UeNotAvailableForPsService** - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- **RadioNetworkLayer_RadioResourcesNotAvailable** - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- **RadioNetworkLayer_FailureInTheRadioInterfaceProcedure** - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.

- **RadioNetworkLayer_InvalidQosCombination** - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemhandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.

- Nas_AuthenticationFailure - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- Nas_Detach - NAS Detach - S1AP NAS cause Detach.
- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxRelRejectionS1u - UE Context Release Command Rejection S1-U

Syntax

--ueCtxtRelRejectionS1u <list-item>

Description

S1AP cause code to include on UE context release command when UE requests to establish S1-U by sending TAU with active flag, SR or CPSR when a UE has all PDN connections data transport over CP.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.

- **RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell** - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- **RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId** - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- **RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId** - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- **RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId** - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- **RadioNetworkLayer_HandoverDesirableForRadioReasons** - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- **RadioNetworkLayer_TimeCriticalHandover** - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- **RadioNetworkLayer_ResourceOptimisationHandover** - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- **RadioNetworkLayer_ReduceLoadInServingCell** - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- **RadioNetworkLayer_UserInactivity** - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- **RadioNetworkLayer_RadioConnectionWithUeLost** - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- **RadioNetworkLayer_LoadBalancingTauRequired** - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- **RadioNetworkLayer_CsFallbackTriggered** - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- **RadioNetworkLayer_UeNotAvailableForPsService** - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- **RadioNetworkLayer_RadioResourcesNotAvailable** - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- **RadioNetworkLayer_FailureInTheRadioInterfaceProcedure** - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.

- **RadioNetworkLayer_InvalidQosCombination** - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemhandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.

- Nas_AuthenticationFailure - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- Nas_Detach - NAS Detach - S1AP NAS cause Detach.
- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxRelServiceDenied - UE Context Release Command Service Denied

Syntax

--ueCtxRelServiceDenied <list-item>

Description

S1AP cause code to include on UE context release command when service denied.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.

- **RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell** - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- **RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId** - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- **RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId** - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- **RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId** - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- **RadioNetworkLayer_HandoverDesirableForRadioReasons** - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- **RadioNetworkLayer_TimeCriticalHandover** - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- **RadioNetworkLayer_ResourceOptimisationHandover** - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- **RadioNetworkLayer_ReduceLoadInServingCell** - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- **RadioNetworkLayer_UserInactivity** - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- **RadioNetworkLayer_RadioConnectionWithUeLost** - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- **RadioNetworkLayer_LoadBalancingTauRequired** - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- **RadioNetworkLayer_CsFallbackTriggered** - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- **RadioNetworkLayer_UeNotAvailableForPsService** - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- **RadioNetworkLayer_RadioResourcesNotAvailable** - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- **RadioNetworkLayer_FailureInTheRadioInterfaceProcedure** - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.

- **RadioNetworkLayer_InvalidQosCombination** - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.

- Nas_AuthenticationFailure - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- Nas_Detach - NAS Detach - S1AP NAS cause Detach.
- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxRelSgwRestart - UE Context Release Command SGW Restart

Syntax

--ueCtxtRelSgwRestart <list-item>

Description

S1AP cause code to include on UE context release command when detach reason is SGW restart.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.

- **RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell** - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- **RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId** - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- **RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId** - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- **RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId** - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- **RadioNetworkLayer_HandoverDesirableForRadioReasons** - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- **RadioNetworkLayer_TimeCriticalHandover** - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- **RadioNetworkLayer_ResourceOptimisationHandover** - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- **RadioNetworkLayer_ReduceLoadInServingCell** - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- **RadioNetworkLayer_UserInactivity** - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- **RadioNetworkLayer_RadioConnectionWithUeLost** - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- **RadioNetworkLayer_LoadBalancingTauRequired** - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- **RadioNetworkLayer_CsFallbackTriggered** - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- **RadioNetworkLayer_UeNotAvailableForPsService** - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- **RadioNetworkLayer_RadioResourcesNotAvailable** - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- **RadioNetworkLayer_FailureInTheRadioInterfaceProcedure** - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.

- **RadioNetworkLayer_InvalidQosCombination** - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.

- **Nas_AuthenticationFailure** - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- **Nas_Detach** - NAS Detach - S1AP NAS cause Detach.
- **Nas_Unspecified** - NAS Unspecified - S1AP NAS cause Unspecified.
- **Nas_CsgSubscriptionExpiry** - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- **Protocol_TransferSyntaxError** - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- **Protocol_AbstractSyntaxErrorReject** - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- **Protocol_AbstractSyntaxErrorIgnoreAndNotify** - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- **Protocol_MessageNotCompatibleWithReceiverState** - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- **Protocol_SemanticError** - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- **Protocol_AbstractSyntaxErrorFalselyConstructedMessage** - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- **Protocol_Unspecified** - Protocol Unspecified - S1AP Protocol cause Unspecified.
- **Miscellaneous_ControlProcessingOverload** - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- **Miscellaneous_NotEnoughUserPlaneProcessingResources** - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- **Miscellaneous_HardwareFailure** - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- **Miscellaneous_OmIntervention** - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- **Miscellaneous_Unspecified** - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- **Miscellaneous_UnknownPlmn** - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxRelSmcFailed - UE Context Release Command SMC Failed

Syntax

--ueCtxRelSmcFailed <list-item>

Description

S1AP cause code to include on UE context release command when SMC failed.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.

- RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- RadioNetworkLayer_HandoverDesirableForRadioReasons - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- RadioNetworkLayer_TimeCriticalHandover - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- RadioNetworkLayer_ResourceOptimisationHandover - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- RadioNetworkLayer_ReduceLoadInServingCell - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- RadioNetworkLayer_UserInactivity - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- RadioNetworkLayer_RadioConnectionWithUeLost - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- RadioNetworkLayer_LoadBalancingTauRequired - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- RadioNetworkLayer_CsFallbackTriggered - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- RadioNetworkLayer_UeNotAvailableForPsService - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- RadioNetworkLayer_RadioResourcesNotAvailable - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- RadioNetworkLayer_FailureInTheRadioInterfaceProcedure - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.

- **RadioNetworkLayer_InvalidQosCombination** - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemhandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.

- Nas_AuthenticationFailure - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- Nas_Detach - NAS Detach - S1AP NAS cause Detach.
- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelSrvccError - UE Context Release Command SRVCC Error

Syntax

--ueCtxtRelSrvccError <list-item>

Description

The S1AP cause the MME includes on UE context release command on message level, when the MME releases the S1 after an error indicated on SRVCC PS to CS handover completion.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.

- **RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell** - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- **RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId** - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- **RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId** - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- **RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId** - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- **RadioNetworkLayer_HandoverDesirableForRadioReasons** - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- **RadioNetworkLayer_TimeCriticalHandover** - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- **RadioNetworkLayer_ResourceOptimisationHandover** - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- **RadioNetworkLayer_ReduceLoadInServingCell** - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- **RadioNetworkLayer_UserInactivity** - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- **RadioNetworkLayer_RadioConnectionWithUeLost** - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- **RadioNetworkLayer_LoadBalancingTauRequired** - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- **RadioNetworkLayer_CsFallbackTriggered** - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- **RadioNetworkLayer_UeNotAvailableForPsService** - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- **RadioNetworkLayer_RadioResourcesNotAvailable** - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- **RadioNetworkLayer_FailureInTheRadioInterfaceProcedure** - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.

- **RadioNetworkLayer_InvalidQosCombination** - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.

- Nas_AuthenticationFailure - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- Nas_Detach - NAS Detach - S1AP NAS cause Detach.
- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxRelSystemFailure - UE Context Release Command System Failure

Syntax

--ueCtxtRelSystemFailure <list-item>

Description

S1AP cause code to include on UE context release command on System Failure.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.

- **RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell** - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- **RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId** - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- **RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId** - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- **RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId** - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- **RadioNetworkLayer_HandoverDesirableForRadioReasons** - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- **RadioNetworkLayer_TimeCriticalHandover** - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- **RadioNetworkLayer_ResourceOptimisationHandover** - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- **RadioNetworkLayer_ReduceLoadInServingCell** - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- **RadioNetworkLayer_UserInactivity** - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- **RadioNetworkLayer_RadioConnectionWithUeLost** - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- **RadioNetworkLayer_LoadBalancingTauRequired** - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- **RadioNetworkLayer_CsFallbackTriggered** - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- **RadioNetworkLayer_UeNotAvailableForPsService** - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- **RadioNetworkLayer_RadioResourcesNotAvailable** - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- **RadioNetworkLayer_FailureInTheRadioInterfaceProcedure** - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.

- **RadioNetworkLayer_InvalidQosCombination** - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.

- Nas_AuthenticationFailure - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- Nas_Detach - NAS Detach - S1AP NAS cause Detach.
- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelT3TunnelExpiry - UE Context Release Command T3 Tunnel Expiry

Syntax

--ueCtxtRelT3TunnelExpiry <list-item>

Description

The S1AP cause the MME includes on UE context release command on message level, when the S1 is released due to T3 tunnel timer expiry after IRAT RAU or successful handover.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.

- **RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell** - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- **RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId** - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- **RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId** - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- **RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId** - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- **RadioNetworkLayer_HandoverDesirableForRadioReasons** - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- **RadioNetworkLayer_TimeCriticalHandover** - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- **RadioNetworkLayer_ResourceOptimisationHandover** - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- **RadioNetworkLayer_ReduceLoadInServingCell** - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- **RadioNetworkLayer_UserInactivity** - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- **RadioNetworkLayer_RadioConnectionWithUeLost** - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- **RadioNetworkLayer_LoadBalancingTauRequired** - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- **RadioNetworkLayer_CsFallbackTriggered** - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- **RadioNetworkLayer_UeNotAvailableForPsService** - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- **RadioNetworkLayer_RadioResourcesNotAvailable** - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- **RadioNetworkLayer_FailureInTheRadioInterfaceProcedure** - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.

- **RadioNetworkLayer_InvalidQoSCombination** - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.

- Nas_AuthenticationFailure - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- Nas_Detach - NAS Detach - S1AP NAS cause Detach.
- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxRelTargetNodeLte - UE Context Release Command LTE Target Access Restricted

Syntax

--ueCtxtRelTargetNodeLte <list-item>

Description

S1AP cause code to include on UE context release command when MME receives GTPv2 context response with cause code 'Target access restricted for the subscriber'.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.

- **RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell** - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- **RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId** - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- **RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId** - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- **RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId** - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- **RadioNetworkLayer_HandoverDesirableForRadioReasons** - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- **RadioNetworkLayer_TimeCriticalHandover** - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- **RadioNetworkLayer_ResourceOptimisationHandover** - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- **RadioNetworkLayer_ReduceLoadInServingCell** - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- **RadioNetworkLayer_UserInactivity** - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- **RadioNetworkLayer_RadioConnectionWithUeLost** - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- **RadioNetworkLayer_LoadBalancingTauRequired** - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- **RadioNetworkLayer_CsFallbackTriggered** - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- **RadioNetworkLayer_UeNotAvailableForPsService** - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- **RadioNetworkLayer_RadioResourcesNotAvailable** - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- **RadioNetworkLayer_FailureInTheRadioInterfaceProcedure** - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.

- **RadioNetworkLayer_InvalidQosCombination** - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemhandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.

- Nas_AuthenticationFailure - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- Nas_Detach - NAS Detach - S1AP NAS cause Detach.
- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

NameueCtxRelUnsupportedFeatureRestriction - UE Context Release Command Roaming
Restricted Due to Unsupported Features

Syntax

--ueCtxRelUnsupportedFeatureRestriction <list-item>

Description

S1AP cause code to include on UE context release command when the procedure is rejected because of unsupported feature restrictions.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- RadioNetworkLayer_UnknownTargetId - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.

- **RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell** - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- **RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId** - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- **RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId** - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- **RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId** - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- **RadioNetworkLayer_HandoverDesirableForRadioReasons** - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- **RadioNetworkLayer_TimeCriticalHandover** - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- **RadioNetworkLayer_ResourceOptimisationHandover** - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- **RadioNetworkLayer_ReduceLoadInServingCell** - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- **RadioNetworkLayer_UserInactivity** - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- **RadioNetworkLayer_RadioConnectionWithUeLost** - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- **RadioNetworkLayer_LoadBalancingTauRequired** - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- **RadioNetworkLayer_CsFallbackTriggered** - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- **RadioNetworkLayer_UeNotAvailableForPsService** - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- **RadioNetworkLayer_RadioResourcesNotAvailable** - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- **RadioNetworkLayer_FailureInTheRadioInterfaceProcedure** - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.

- **RadioNetworkLayer_InvalidQoSCombination** - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownEranId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleEranIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- **TransportLayer_Unspecified** - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- **Nas_NormalRelease** - NAS Normal Release - S1AP NAS cause Normal Release.

- Nas_AuthenticationFailure - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- Nas_Detach - NAS Detach - S1AP NAS cause Detach.
- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxRelVdpForbiddenArea - UE Context Release Command VDP Forbidden Area

Syntax

--ueCtxtRelVdpForbiddenArea <list-item>

Description

S1AP cause code to include on UE context release command when the MME releases the S1 connection after the MME has rejected the UE Attach, TAU or X2 Handover attempt due to the UE/eNB attempting to access a restricted area defined as part of the VDP Profile or VDP Range Services provisioning for this UE.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- RadioNetworkLayer_CellNotAvailable - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.

- **RadioNetworkLayer_UnknownTargetId** - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- **RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell** - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- **RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId** - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- **RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId** - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- **RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId** - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- **RadioNetworkLayer_HandoverDesirableForRadioReasons** - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- **RadioNetworkLayer_TimeCriticalHandover** - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- **RadioNetworkLayer_ResourceOptimisationHandover** - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- **RadioNetworkLayer_ReduceLoadInServingCell** - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- **RadioNetworkLayer_UserInactivity** - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- **RadioNetworkLayer_RadioConnectionWithUeLost** - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- **RadioNetworkLayer_LoadBalancingTauRequired** - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- **RadioNetworkLayer_CsFallbackTriggered** - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- **RadioNetworkLayer_UeNotAvailableForPsService** - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- **RadioNetworkLayer_RadioResourcesNotAvailable** - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.

- **RadioNetworkLayer_FailureInTheRadioInterfaceProcedure** - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- **RadioNetworkLayer_InvalidQosCombination** - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownErbId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErbIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemHandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- **RadioNetworkLayer_N26InterfaceNotAvailable** - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- **TransportLayer_TransportResourceUnavailable** - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.

- TransportLayer_Unspecified - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- Nas_NormalRelease - NAS Normal Release - S1AP NAS cause Normal Release.
- Nas_AuthenticationFailure - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- Nas_Detach - NAS Detach - S1AP NAS cause Detach.
- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelWbEutranExceptLtemNotAllowed - UE Context Release Command WB-EUTRAN
Except LTE-M Not Allowed

Syntax

--ueCtxtRelWbEutranExceptLtemNotAllowed <list-item>

Description

S1AP cause code to include on UE context release command when UE is not allowed WB
EUTRAN except LTE-M access.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.
- RadioNetworkLayer_TS1RelocOverallExpiry - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- RadioNetworkLayer_TS1RelocPrepExpiry - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.

- **RadioNetworkLayer_CellNotAvailable** - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- **RadioNetworkLayer_UnknownTargetId** - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- **RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell** - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- **RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId** - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- **RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId** - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- **RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId** - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- **RadioNetworkLayer_HandoverDesirableForRadioReasons** - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- **RadioNetworkLayer_TimeCriticalHandover** - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- **RadioNetworkLayer_ResourceOptimisationHandover** - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- **RadioNetworkLayer_ReduceLoadInServingCell** - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- **RadioNetworkLayer_UserInactivity** - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- **RadioNetworkLayer_RadioConnectionWithUeLost** - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- **RadioNetworkLayer_LoadBalancingTauRequired** - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.
- **RadioNetworkLayer_CsFallbackTriggered** - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- **RadioNetworkLayer_UeNotAvailableForPsService** - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.

- **RadioNetworkLayer_RadioResourcesNotAvailable** - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- **RadioNetworkLayer_FailureInTheRadioInterfaceProcedure** - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- **RadioNetworkLayer_InvalidQoSCombination** - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- **RadioNetworkLayer_InterRatRedirection** - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- **RadioNetworkLayer_InteractionWithOtherProcedure** - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- **RadioNetworkLayer_UnknownErabId** - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- **RadioNetworkLayer_MultipleErabIdInstances** - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- **RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported** - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- **RadioNetworkLayer_S1IntraSystemhandoverTriggered** - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- **RadioNetworkLayer_S1InterSystemHandoverTriggered** - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- **RadioNetworkLayer_x2HandoverTriggered** - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- **RadioNetworkLayer_redirectionTowards1xRtt** - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- **RadioNetworkLayer_NotSupportedQciValue** - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.
- **RadioNetworkLayer_InvalidCsgId** - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- **RadioNetworkLayer_ReleaseDueToPreEmption** - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.

- RadioNetworkLayer_N26InterfaceNotAvailable - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- TransportLayer_TransportResourceUnavailable - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- TransportLayer_Unspecified - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- Nas_NormalRelease - NAS Normal Release - S1AP NAS cause Normal Release.
- Nas_AuthenticationFailure - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- Nas_Detach - NAS Detach - S1AP NAS cause Detach.
- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.
- Miscellaneous_HardwareFailure - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- Miscellaneous_OmIntervention - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.

- Miscellaneous_Unspecified - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- Miscellaneous_UnknownPlmn - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Name

ueCtxtRelWithoutSmsGrpId - UE Context Release Command Without SMS Group ID

Syntax

- ueCtxtRelWithoutSmsGrpId <list-item>

Description

S1AP cause code to include on UE context release command when MME rejects attach or TAU request without SMS indication when bit 0 of local-group-id AVP is set to indicate that UE must indicate SMS request.

Class

Optional.

List

- Disabled - The feature is disabled in the corresponding scenario. The MME does not override the S1AP cause code.
- RadioNetworkLayer_Unspecified - Radio Network Layer Unspecified - S1AP Radio Network Layer cause Unspecified.
- RadioNetworkLayer_TX2RelocOverallExpiry - Radio Network Layer TX2 Reloc Overall Expiry - S1AP Radio Network Layer cause TX2 Reloc Overall Expiry.
- RadioNetworkLayer_SuccessfulHandover - Radio Network Layer Successful Handover - S1AP Radio Network Layer cause Successful Handover.
- RadioNetworkLayer_ReleaseDueToEutranGeneratedReason - Radio Network Layer Release Due to E-UTRAN Generated Reason - S1AP Radio Network Layer cause Release Due to E-UTRAN Generated Reason.
- RadioNetworkLayer_HandoverCancelled - Radio Network Layer Handover Cancelled - S1AP Radio Network Layer cause Handover Cancelled.
- RadioNetworkLayer_PartialHandover - Radio Network Layer Partial Handover - S1AP Radio Network Layer cause Partial Handover.
- RadioNetworkLayer_HandoverFailureInTargetEpcEnbOrTargetSystem - Radio Network Layer Handover Failure in Target EPC/eNB or Target System - S1AP Radio Network Layer cause Handover Failure In Target EPC/eNB or Target System.
- RadioNetworkLayer_HandoverTargetNotAllowed - Radio Network Layer Handover Target Not Allowed - S1AP Radio Network Layer cause Handover Target not allowed.

- **RadioNetworkLayer_TS1RelocOverallExpiry** - Radio Network Layer TS1 Reloc Overall Expiry - S1AP Radio Network Layer cause TS1Relocoverall Expiry.
- **RadioNetworkLayer_TS1RelocPrepExpiry** - Radio Network Layer TS1 Reloc Prep Expiry - S1AP Radio Network Layer cause TS1RelocPrep Expiry.
- **RadioNetworkLayer_CellNotAvailable** - Radio Network Layer Cell Not Available - S1AP Radio Network Layer cause Cell Not Available.
- **RadioNetworkLayer_UnknownTargetId** - Radio Network Layer Unknown Target Id - S1AP Radio Network Layer cause Unknown Target Id.
- **RadioNetworkLayer_NoRadioResourcesAvailableInTargetCell** - Radio Network Layer No Resources Available in Target Cell - S1AP Radio Network Layer cause No Resources Available in Target Cell.
- **RadioNetworkLayer_UnknownOrAlreadyAllocatedMmeUeS1apId** - Radio Network Layer Unknown or Already Allocated MME UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated MME UE S1AP ID.
- **RadioNetworkLayer_UnknownOrAlreadyAllocatedEnbUeS1apId** - Radio Network Layer Unknown or Already Allocated eNB UE S1AP ID - S1AP Radio Network Layer cause Unknown or already allocated eNB UE S1AP ID.
- **RadioNetworkLayer_UnknownOrInconsistentPairOfUeS1apId** - Radio Network Layer Unknown or Inconsistent Pair of S1AP UE ID - S1AP Radio Network Layer cause Unknown or inconsistent pair of S1AP UE ID.
- **RadioNetworkLayer_HandoverDesirableForRadioReasons** - Radio Network Layer Handover Desirable for Radio Reasons - S1AP Radio Network Layer cause Handover desirable for radio reasons.
- **RadioNetworkLayer_TimeCriticalHandover** - Radio Network Layer Time Critical Handover - S1AP Radio Network Layer cause Time critical handover.
- **RadioNetworkLayer_ResourceOptimisationHandover** - Radio Network Layer Resource Optimisation Handover - S1AP Radio Network Layer cause Resource optimisation handover.
- **RadioNetworkLayer_ReduceLoadInServingCell** - Radio Network Layer Reduce Load in Serving Cell - S1AP Radio Network Layer cause Reduce load in serving cell.
- **RadioNetworkLayer_UserInactivity** - Radio Network Layer User Inactivity - S1AP Radio Network Layer cause User Inactivity.
- **RadioNetworkLayer_RadioConnectionWithUeLost** - Radio Network Layer Radio Connection with UE Lost - S1AP Radio Network Layer cause Radio Connection With UE Lost.
- **RadioNetworkLayer_LoadBalancingTauRequired** - Radio Network Layer Load Balancing TAU Required - S1AP Radio Network Layer cause Load Balancing TAU Required.

- RadioNetworkLayer_CsFallbackTriggered - Radio Network Layer CS Fallback Triggered - S1AP Radio Network Layer cause CS Fallback Triggered.
- RadioNetworkLayer_UeNotAvailableForPsService - Radio Network Layer UE Not Available for PS Service - S1AP Radio Network Layer cause UE Not Available For PS Service.
- RadioNetworkLayer_RadioResourcesNotAvailable - Radio Network Layer Radio Resources Not Available - S1AP Radio Network Layer cause Radio resources not available.
- RadioNetworkLayer_FailureInTheRadioInterfaceProcedure - Radio Network Layer Failure in the Radio Interface Procedure - S1AP Radio Network Layer cause Failure in the Radio Interface Procedure.
- RadioNetworkLayer_InvalidQosCombination - Radio Network Layer Invalid QoS Combination - S1AP Radio Network Layer cause Invalid QoS combination.
- RadioNetworkLayer_InterRatRedirection - Radio Network Layer Inter-RAT Redirection - S1AP Radio Network Layer cause Inter-RAT redirection.
- RadioNetworkLayer_InteractionWithOtherProcedure - Radio Network Layer Interaction with Other Procedure - S1AP Radio Network Layer cause Interaction with other procedure.
- RadioNetworkLayer_UnknownErabId - Radio Network Layer Unknown E-RAB ID - S1AP Radio Network Layer cause Unknown E-RAB ID.
- RadioNetworkLayer_MultipleErabIdInstances - Radio Network Layer Multiple E-RAB ID Instances - S1AP Radio Network Layer cause Multiple E-RAB ID instances.
- RadioNetworkLayer_EncryptionAndOrIntegrityProtectionAlgorithmsNotSupported - Radio Network Layer Encryption and/or Integrity Protection Algorithms Not Supported - S1AP Radio Network Layer cause Encryption and/or Integrity protection algorithms not supported.
- RadioNetworkLayer_S1IntraSystemhandoverTriggered - Radio Network Layer S1 Intra System Handover Triggered - S1AP Radio Network Layer cause S1 intra system Handover triggered.
- RadioNetworkLayer_S1InterSystemHandoverTriggered - Radio Network Layer S1 Inter System Handover Triggered - S1AP Radio Network Layer cause S1 inter system Handover triggered.
- RadioNetworkLayer_x2HandoverTriggered - Radio Network Layer X2 Handover Triggered - S1AP Radio Network Layer cause X2 Handover triggered.
- RadioNetworkLayer_redirectionTowards1xRtt - Radio Network Layer Redirection Towards 1xRTT - S1AP Radio Network Layer cause Redirection towards 1xRTT.
- RadioNetworkLayer_NotSupportedQciValue - Radio Network Layer Not Supported QCI Value - S1AP Radio Network Layer cause Not Supported QCI Value.

- RadioNetworkLayer_InvalidCsgId - Radio Network Layer Invalid CSG Id - S1AP Radio Network Layer cause Invalid CSG Id.
- RadioNetworkLayer_ReleaseDueToPreEmption - Radio Network Layer Release Due to Pre-Emption - S1AP Radio Network Layer cause Release due to Pre-Emption.
- RadioNetworkLayer_N26InterfaceNotAvailable - Radio Network Layer N26 Interface Not Available - S1AP Radio Network Layer cause N26 interface not available.
- TransportLayer_TransportResourceUnavailable - Transport Layer Transport Resource Unavailable - S1AP Transport Layer cause Transport Resource Unavailable.
- TransportLayer_Unspecified - Transport Layer Unspecified - S1AP Transport Layer cause Unspecified.
- Nas_NormalRelease - NAS Normal Release - S1AP NAS cause Normal Release.
- Nas_AuthenticationFailure - NAS Authentication Failure - S1AP NAS cause Authentication failure.
- Nas_Detach - NAS Detach - S1AP NAS cause Detach.
- Nas_Unspecified - NAS Unspecified - S1AP NAS cause Unspecified.
- Nas_CsgSubscriptionExpiry - NAS CSG Subscription Expiry - S1AP NAS cause CSG Subscription Expiry.
- Protocol_TransferSyntaxError - Protocol Transfer Syntax Error - S1AP Protocol cause Transfer Syntax Error.
- Protocol_AbstractSyntaxErrorReject - Protocol Abstract Syntax Error (Reject) - S1AP Protocol cause Abstract Syntax Error (Reject).
- Protocol_AbstractSyntaxErrorIgnoreAndNotify - Protocol Abstract Syntax Error (Ignore and Notify) - S1AP Protocol cause Abstract Syntax Error (Ignore and Notify).
- Protocol_MessageNotCompatibleWithReceiverState - Protocol Message Not Compatible with Receiver State - S1AP Protocol cause Message not Compatible with Receiver State.
- Protocol_SemanticError - Protocol Semantic Error - S1AP Protocol cause Semantic Error.
- Protocol_AbstractSyntaxErrorFalselyConstructedMessage - Protocol Abstract Syntax Error (Falsely Constructed Message) - S1AP Protocol cause Abstract Syntax Error (Falsely Constructed Message).
- Protocol_Unspecified - Protocol Unspecified - S1AP Protocol cause Unspecified.
- Miscellaneous_ControlProcessingOverload - Miscellaneous Control Processing Overload - S1AP Miscellaneous cause Control Processing Overload.
- Miscellaneous_NotEnoughUserPlaneProcessingResources - Miscellaneous Not Enough User Plane Processing Resources - S1AP Miscellaneous cause Not enough User Plane Processing Resources.

- **Miscellaneous_HardwareFailure** - Miscellaneous Hardware Failure - S1AP Miscellaneous cause Hardware Failure.
- **Miscellaneous_OmIntervention** - Miscellaneous O&M Intervention - S1AP Miscellaneous cause O&M Intervention.
- **Miscellaneous_Unspecified** - Miscellaneous Unspecified - S1AP Miscellaneous cause Unspecified.
- **Miscellaneous_UnknownPlmn** - Miscellaneous Unknown PLMN - S1AP Miscellaneous cause Unknown PLMN.

Default

Disabled

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

6.128 cmm s1ConfigUpdate

S1-MME Configuration Update. Syntax, description, parameters, and roles for **cmm s1ConfigUpdate** command.

Syntax**Description**

This command will send an S1 MME Configuration Update to single eNB when non-zero linkindex value is specified or to all eNBs when linkindex=0.

Parameters**Name**

linkIndex - Link Index

Syntax

--linkIndex <integer>

Description

This parameter is the link index of the S1mme link to the eNB that MME will send an s1ap Configuration Update message. A link index of value '0' indicates all S1mme links resulting in a Configuration Update message to all eNBs

Class

Required.

Type / Range

Integer in the range of 0 to 4,294,967,295

Roles

sysadmin, manage

6.129 cmm s1Reset

S1 Reset. Syntax, description, parameters, and roles for cmm s1Reset command.

Syntax**Description**

This command will send an S1 Reset ALL message to the specified eNB to reset all the S1 connections from that eNB to the MME.

Parameters**Name**

linkIndex - Link Index

Syntax

--linkIndex <integer>

Description

This parameter is the link index of the S1mme link to be Reset

Class

Required.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

terminationName - TerminationName

Syntax

--terminationName <string>

Description

Termination Name. This is used for s1mme alone. This field is part of input for optional key fields. This field cannot be updated.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

sysadmin, manage

6.130 cmm sbiChainedQueryConversionRule

SBI Chained Query Conversion Rule. Syntax, description, parameters, and roles for cmm sbiChainedQueryConversionRule command.

Syntax**Description**

Chained Query Conversion Rules are referenced by a single 'Operation only' and possibly multiple 'Condition only' Conversion Rules to create a Conversion rule with multiple Conditions

Parameters**Name**

name - Name

Syntax

--name <string>

Description

Chained Query Conversion Rule Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.131 cmm sbiCommunicationProfile

SBI Communication Profile. Syntax, description, parameters, and roles for cmm sbiCommunicationProfile command.

Syntax

Description

Defines the Communication Model per SBI Interface

Parameters

Name

name - Name

Syntax

--name <string>

Description

Profile name for SBI communication profile

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

amfConsumer - AMF Consumer

Syntax

--amfConsumer <list-item>

Description

SBI Communication Model to be used in the AMF Consumer interface (N14)

Class

Optional.

List

- DIRECT - Direct Communication without SCP
- INDIRECT_WITHOUT_DELEGATED_DISCOVERY - Indirect without Delegated Discovery - Indirect Communication without Delegated Discovery (Model C) where AMF selects the target Service Instance
- INDIRECT_DELEGATED_DISCOVERY - Indirect Delegated Discovery - Indirect Communication with Delegated Discovery (Model D)

Default

DIRECT

Name

ausfConsumer - AUSF Consumer

Syntax

--ausfConsumer <list-item>

Description

SBI Communication Model to be used in the AUSF Consumer interface (N12)

Class

Optional.

List

- DIRECT - Direct Communication without SCP
- INDIRECT_WITHOUT_DELEGATED_DISCOVERY - Indirect without Delegated Discovery - Indirect Communication without Delegated Discovery (Model C) where AMF selects the target Service Instance
- INDIRECT_DELEGATED_DISCOVERY - Indirect Delegated Discovery - Indirect Communication with Delegated Discovery (Model D)

Default

DIRECT

Name

communicationNotifications - Communication Notifications

Syntax

--communicationNotifications <list-item>

Description

SBI Communication Model to be used for AMF Communication Notifications.

Class

Optional.

List

- DIRECT - Direct Communication without SCP

- **INDIRECT_WITHOUT_DELEGATED_DISCOVERY** - Indirect without Delegated Discovery - Indirect Communication without Delegated Discovery (Model C) where AMF selects the target Service Instance
- **INDIRECT_DELEGATED_DISCOVERY** - Indirect Delegated Discovery - Indirect Communication with Delegated Discovery (Model D)

Default

DIRECT

Name

eir5gConsumer - EIR 5G Consumer

Syntax

--eir5gConsumer <list-item>

Description

SBI Communication Model to be used in the 5G EIR Consumer interface (N17)

Class

Optional.

List

- **DIRECT** - Direct Communication without SCP
- **INDIRECT_WITHOUT_DELEGATED_DISCOVERY** - Indirect without Delegated Discovery - Indirect Communication without Delegated Discovery (Model C) where AMF selects the target Service Instance
- **INDIRECT_DELEGATED_DISCOVERY** - Indirect Delegated Discovery - Indirect Communication with Delegated Discovery (Model D)

Default

DIRECT

Name

eventExposureNotifications - Event Exposure Notifications

Syntax

--eventExposureNotifications <list-item>

Description

SBI Communication Model to be used for AMF Event Exposure Notifications.

Class

Optional.

List

- **DIRECT** - Direct Communication without SCP

- **INDIRECT_WITHOUT_DELEGATED_DISCOVERY** - Indirect without Delegated Discovery - Indirect Communication without Delegated Discovery (Model C) where AMF selects the target Service Instance

Default

DIRECT

Name

lmfConsumer - LMF Consumer

Syntax

--lmfConsumer <list-item>

Description

SBI Communication Model to be used in the LMF Consumer interface (NL1)

Class

Optional.

List

- **DIRECT** - Direct Communication without SCP
- **INDIRECT_WITHOUT_DELEGATED_DISCOVERY** - Indirect without Delegated Discovery - Indirect Communication without Delegated Discovery (Model C) where AMF selects the target Service Instance
- **INDIRECT_DELEGATED_DISCOVERY** - Indirect Delegated Discovery - Indirect Communication with Delegated Discovery (Model D)

Default

DIRECT

Name

locationNotifications - Location Notifications

Syntax

--locationNotifications <list-item>

Description

SBI Communication Model to be used for AMF Location Notifications.

Class

Optional.

List

- **DIRECT** - Direct Communication without SCP
- **INDIRECT_WITHOUT_DELEGATED_DISCOVERY** - Indirect without Delegated Discovery - Indirect Communication without Delegated Discovery (Model C) where AMF selects the target Service Instance

- INDIRECT_DELEGATED_DISCOVERY - Indirect Delegated Discovery - Indirect Communication with Delegated Discovery (Model D)

Default

DIRECT

Name

nrfConsumer - NRF Consumer

Syntax

--nrfConsumer <list-item>

Description

SBI Communication Model to be used in the NRF Consumer interface

Class

Optional.

List

- DIRECT - Direct Communication without SCP
- INDIRECT_WITHOUT_DELEGATED_DISCOVERY - Indirect without Delegated Discovery - Indirect Communication without Delegated Discovery (Model C) where AMF selects the target Service Instance

Default

DIRECT

Name

nssaafConsumer - NSSAAF Consumer

Syntax

--nssaafConsumer <list-item>

Description

SBI Communication Model to be used in the NSSAAF Consumer interface

Class

Optional.

List

- DIRECT - Direct Communication without SCP
- INDIRECT_WITHOUT_DELEGATED_DISCOVERY - Indirect without Delegated Discovery - Indirect Communication without Delegated Discovery (Model C) where AMF selects the target Service Instance
- INDIRECT_DELEGATED_DISCOVERY - Indirect Delegated Discovery - Indirect Communication with Delegated Discovery (Model D)

Default

DIRECT

Name

nssfConsumer - NSSF Consumer

Syntax

--nssfConsumer <list-item>

Description

SBI Communication Model to be used in the NSSF Consumer interface (N22)

Class

Optional.

List

- DIRECT - Direct Communication without SCP
- INDIRECT_WITHOUT_DELEGATED_DISCOVERY - Indirect without Delegated Discovery - Indirect Communication without Delegated Discovery (Model C) where AMF selects the target Service Instance

Default

DIRECT

Name

pcfConsumer - PCF Consumer

Syntax

--pcfConsumer <list-item>

Description

SBI Communication Model to be used in the PCF Consumer interface (N15)

Class

Optional.

List

- DIRECT - Direct Communication without SCP
- INDIRECT_WITHOUT_DELEGATED_DISCOVERY - Indirect without Delegated Discovery - Indirect Communication without Delegated Discovery (Model C) where AMF selects the target Service Instance
- INDIRECT_DELEGATED_DISCOVERY - Indirect Delegated Discovery - Indirect Communication with Delegated Discovery (Model D)

Default

DIRECT

Name

sbiHttpProxySettingsProfileName - SBI HTTP Proxy Settings Profile Name

Syntax

--sbiHttpProxySettingsProfileName <string>

Description

Name of SBI HTTP Proxy Settings Profile that is applicable to the Communication Profile.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

default

Name

smfConsumer - SMF Consumer

Syntax

--smfConsumer <list-item>

Description

SBI Communication Model to be used in the SMF Consumer interface (N11)

Class

Optional.

List

- DIRECT - Direct Communication without SCP
- INDIRECT_WITHOUT_DELEGATED_DISCOVERY - Indirect without Delegated Discovery - Indirect Communication without Delegated Discovery (Model C) where AMF selects the target Service Instance
- INDIRECT_DELEGATED_DISCOVERY - Indirect Delegated Discovery - Indirect Communication with Delegated Discovery (Model D)

Default

DIRECT

Name

smsfConsumer - SMSF Consumer

Syntax

--smsfConsumer <list-item>

Description

SBI Communication Model to be used in the SMSF Consumer interface (N20)

Class

Optional.

List

- DIRECT - Direct Communication without SCP
- INDIRECT_WITHOUT_DELEGATED_DISCOVERY - Indirect without Delegated Discovery - Indirect Communication without Delegated Discovery (Model C) where AMF selects the target Service Instance
- INDIRECT_DELEGATED_DISCOVERY - Indirect Delegated Discovery - Indirect Communication with Delegated Discovery (Model D)

Default

DIRECT

Name

udmConsumer - UDM Consumer

Syntax

--udmConsumer <list-item>

Description

SBI Communication Model to be used in the UDM Consumer interface (N8)

Class

Optional.

List

- DIRECT - Direct Communication without SCP
- INDIRECT_WITHOUT_DELEGATED_DISCOVERY - Indirect without Delegated Discovery - Indirect Communication without Delegated Discovery (Model C) where AMF selects the target Service Instance
- INDIRECT_DELEGATED_DISCOVERY - Indirect Delegated Discovery - Indirect Communication with Delegated Discovery (Model D)

Default

DIRECT

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.132 cmm sbiConnectionTestUri

SBI Connection Test to URI. Syntax, description, parameters, and roles for cmm sbiConnectionTestUri command.

Syntax**Description**

This command performs a connection test by sending an HTTP Request to the target URI from the given HTTP Client Endpoint. A httpProxy may optionally be given as input parameter in order to issue the Request via indirect communication. The Response received from the remote peer is displayed in the command output.

Parameters**Name**

endPointName - Endpoint Name

Syntax

--endPointName <string>

Description

The amfConsumerLocalEndPointConfig or amfServicesEndPointConfig endPointName from which the test HTTP Request will be issued. Note: The referenced endPointName cannot be a NR10 interface.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

targetUri - Target URI

Syntax

--targetUri <string>

Description

The URI to which the HTTP Request message will be issued

Class

Required key.

Type / Range

Digit string of length 512 with the allowed pattern of (http:\\|https:\\)[0-9a-zA-Z\\.\\-\\~\$+*_;&=#?\\[\\]@,()\\!'%%]*

Name

connectionTestStatus - Connection Test Status

Syntax

--connectionTestStatus <string>

Description

HTTP2 status returned as response of the connection Test

Class

Read only.

Type / Range

Digit string of length 0 to 255

Default

""

Name

httpProxyName - HTTP Proxy Name

Syntax

--httpProxyName <string>

Description

When this parameter is not NULL it refers to a name of a provisioned HTTP proxy. This triggers the request to be send via Indirect communication

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\\.\\-_:@]*

Default

""

Name

requestMethod - Method

Syntax

--requestMethod <list-item>

Description

The HTTP Method of the Test Request

Class

Optional.

List

- DELETE - HTTP Method DELETE
- GET - HTTP Method GET
- OPTIONS - HTTP Method OPTIONS
- POST - HTTP Method POST
- PUT - HTTP Method PUT

Default

OPTIONS

Roles**List**

sysadmin, manage, monitor

6.133 cmm sbiCorrelationInfoGParms

SBI Correlation Info General Parameters. Syntax, description, parameters, and roles for cmm sbiCorrelationInfoGParms command.

Syntax**Description**

Configuration of 3gpp-Sbi-Correlation-Info HTTP Header handling

Parameters**Name**

amfGparmName - Parameter name

Syntax

--amfGparmName <list-item>

Description

Name of parameter

Class

Required key.

List

- includeImsi - Include IMSI - This parameter controls if AMF will add the IMSI in the 3gpp-Sbi-Correlation-Info in Requests and Responses sent by the AMF when the relevant feature is enabled

- mirrorCorrelationInfo - Mirror Correlation Info - This parameter controls if AMF will mirror all Correlation Info received in Service Requests and Subscriptions when the relevant feature is enabled
- sbiCorrelationInfoEnable - Enable - This parameter enables or disables sending 3gpp-Sbi-Correlation-Info in Requests and Responses

Name

amfGparmValue - Parameter value

Syntax

--amfGparmValue <string>

Description

Value of parameter

Class

Required on create.

Type / Range

Digit string of length 0 to 64

The amfGparmValue string is based on the amfGparmName parameter:

Name

includeImsi - Include IMSI

Syntax

--amfGparmName includeImsi --amfGparmValue <string>

Description

This parameter controls if AMF will add the IMSI in the 3gpp-Sbi-Correlation-Info in Requests and Responses sent by the AMF when the relevant feature is enabled

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

mirrorCorrelationInfo - Mirror Correlation Info

Syntax

--amfGparmName mirrorCorrelationInfo --amfGparmValue <string>

Description

This parameter controls if AMF will mirror all Correlation Info received in Service Requests and Subscriptions when the relevant feature is enabled

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sbiCorrelationInfoEnable - Enable

Syntax

--amfGparmName sbiCorrelationInfoEnable --amfGparmValue <string>

Description

This parameter enables or disables sending 3gpp-Sbi-Correlation-Info in Requests and Responses

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

6.134 cmm sbiEnhancedLoadOverloadControl

Enhanced Load and Overload Control. Syntax, description, parameters, and roles for cmm sbiEnhancedLoadOverloadControl command.

Syntax**Description**

This command configures Enhanced Load and Overload Control related parameters related to 3gpp-sbi-oci 3gpp-sbi-lci HTTP headers

Parameters

Name

defaultLoadValueForScps - Default Load value for SCP/SEPPs

Syntax

--defaultLoadValueForScps <integer>

Description

This parameter determines the default load that is assumed for SCP/SEPPs in case no load information has previously been received via the 3gpp-sbi-lci header.

Class

Optional.

Type / Range

Integer in the range of 0 to 100

Default

0

Name

enable - Enable

Syntax

--enable <list-item>

Description

Enables or disables the Enhanced Load and Overload control functionality

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

supportNfLciRsp - Support NF LCI in Responses

Syntax

--supportNfLciRsp <list-item>

Description

This parameter controls whether 3gpp-sbi-lci header is evaluated in responses for remote NFs .

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

supportNfOciRsp - Support NF Oci in Responses

Syntax

--supportNfOciRsp <list-item>

Description

This parameter controls whether 3gpp-sbi-Oci header is evaluated in responses for remote NFs or not

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

supportScpLci - Support SCP/SEPP LCI

Syntax

--supportScpLci <list-item>

Description

This parameter controls whether 3gpp-sbi-Lci header is taken into account for SCP/SEPP Load control

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

6.135 cmm sbiHttpProxySettingsProfile

SBI HTTP Proxy Settings. Syntax, description, parameters, and roles for cmm sbiHttpProxySettingsProfile command.

Syntax

Description

This command opens the context for creating a profile for HTTP Proxy related parameters for the network. A default profile named 'default', is provided and can not be deleted.

Parameters

Name

name - Name

Syntax

--name <string>

Description

Name of the HTTP Proxy Setting Profile. A default profile named 'default' is provided

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

defaultScpFqdn - Default SCP FQDN

Syntax

--defaultScpFqdn <string>

Description

FQDN used to discover SCPs via SRV DNS Query. When the SRV FQDN is provided (i.e. is a non Null String) it replaces the SCP selection based on sbiHttpProxySettingsProfile

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 128

Default

""

Name

defaultScpFqdnPrefix - defaultScpFqdnPrefix

Syntax

--defaultScpFqdnPrefix <string>

Description

Optional deployment-specific string to be pretended to the ':path' HTTP/2 pseudo-header when sending messages to a SCP discovered via SRV DNS query with the provisioned FQDN. This string is removed by the Proxy when the message is forwarded to the next hop. It is assumed that all SCP servers discovered in the profile use the same prefix

Class

Optional.

Type / Range

API Prefix as defined in 29.500. May be null otherwise shall have at least allowed character, shall not start with / and shall not end with /. When not-null, the leading / is implied and shall not be included.

Default

""

Name

defaultScpFqdnScheme - Default SCP FQDN Scheme

Syntax

--defaultScpFqdnScheme <list-item>

Description

Defines the scheme to be used when SCPs are discovered via SRV DNS query.

Class

Optional.

List

- http - The schema of Hypertext Transfer Protocol.
- https - The schema of Hypertext Transfer Protocol Secure.

Default

https

Name

defaultScpList - Default SCP HTTP Proxy List

Syntax

--defaultScpList <string>

Description

Name of the list with the http Proxies to be used by default when sending HTTP Request to target NF in own network. When the list name is provided, the AMF uses the Indirect Communication Model via SCP. When the list name is empty, the Direct Communication Model is followed.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

""

Name

defaultSeppList - Default SEPP Proxy List

Syntax

--defaultSeppList <string>

Description

Name of of the list with the http Proxies to be used when sending HTTP Request to target NF in Roamer's Home network. List must be set when Roaming is supported.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

""

Name

maxRetries - Max Retries

Syntax

--maxRetries <integer>

Description

Max number of retries to use a HTTP Proxy for indirect communication in a specific Request.

Class

Optional.

Type / Range

Integer in the range of 0 to 3

Default

0

Name

targetNfScheme - Target NF Scheme

Syntax

--targetNfScheme <list-item>

Description

HTTP scheme that should be supported by the target NF when communication is via an HTTP Proxy. Only Service instances that support the target Scheme are considered as valid candidates.

Class

Optional.

List

- both - Both HTTP/HTTPS schemes
- http - Hypertext Transfer Protocol scheme.
- https - Hypertext Transfer Protocol Secure scheme.

Default

https

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.136 cmm sbiIndirectCommunicationErrorRule

SBI Indirect Communication Error Rule. Syntax, description, parameters, and roles for cmm sbiIndirectCommunicationErrorRule command.

Syntax

Description

Defines rules for deducing the error response originator based on the received HTTP/2 status code - application error combination.

Parameters

Name

statusCode - Status Code

Syntax

--statusCode <list-item>

Description

The HTTP Status Code

Class

Required key.

List

- 400_Bad_Request - 400 Bad Request
- 401_Unauthorized - 401 Unauthorized
- 403_Forbidden - 403 Forbidden
- 404_Not_Found - 404 Not Found
- 405_Method_Not_Allowed - 405 Method Not Allowed
- 406_Not_Acceptable - 406 Not Acceptable
- 408_Request_Timeout - 408 Request Timeout
- 409_Conflict - 409 Conflict
- 410_Gone - 410 Gone
- 411_Length_Required - 411 Length Required
- 412_Precondition_Failed - 412 Precondition Failed
- 413_Payload_Too_Large - 413 Payload Too Large
- 414_URI_Too_Long - 414 URI Too Long
- 415_Unsupported_Media_Type - 415 Unsupported Media Type
- 429_Too_Many_Requests - 429 Too Many Requests
- 500_Internal_Server_Error - 500 Internal Server Error
- 501_Not_Implemented - 501 Not Implemented

- 502_Bad_Gateway - 502 Bad Gateway
- 503_Service_Unavailable - 503 Service Unavailable
- 504_Gateway_Timeout - 504 Gateway Timeout

Name

errorCause - Error Cause

Syntax

--errorCause <string>

Description

Application Error Cause value. Used to applies string match with the Cause in the response. 'NULL' string literal has special meaning and is used to match cases where Response did not include Application Error Cause

Class

Required key.

Type / Range

SBI Application Error Cause value. Digit string of length 1 to 64 with the allowed pattern of [A-Z_]*

Default

"NULL"

Name

errorOriginator - Error Originator

Syntax

--errorOriginator <list-item>

Description

The entity that produced the received SBI error response

Class

Required on create.

List

- NF - Error response originator is the NF that received the request by the AMF
- HTTP_PROXY - HTTP Proxy - Error response originator is the HTTP Proxy

Name

enable - Enable

Syntax

--enable <list-item>

Description

Controls whether the rule is enabled or disabled. User created and System inserted rules (i.e. rules automatically created by the system during initial deployment) may be enabled or disabled but only user created rules may be deleted.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.137 cmm sbiIndirectCommunicationGParams

SBI Indirect Communication Global Params. Syntax, description, parameters, and roles for cmm sbiIndirectCommunicationGParams command.

Syntax**Description**

This command configures Indirect Communication related parameters for AMF

Parameters**Name**

amfGparmName - Parameter Name

Syntax

--amfGparmName <list-item>

Description

Name of parameter

Class

Required key.

List

- action429 - Action 429 - Default action for 429 Status Responses
- action4xx - Action 4xx - Default action for 4XX Status class Responses
- action503 - Action 503 - Default action for 503 Status Responses
- action5xx - Action 5xx - Default action for 5XX Status class Responses
- actionTimeout - Action Timeout - Action for Timeouts
- conveyAuthDiscoveryParams - Convey oAuth Discovery Params - This parameters controls whether the AMF sends the Discovery '3gpp-Sbi-Discovery-*' headers related to oAuth in an HTTP/2 request targeting an existing resource context in the NF service producer to enable the SCP to perform the Service access authorization procedures.
- discoverSeppViaNrf - Discover SEPP Via NRF - When this parameter is set to true CMM will select the SEPP via issuing a Discovery query to the NRF. This functionality is for TEST ONLY purposes.
- enhancedErrorDetection - Enhanced Error Detection - This Parameter enables Enhanced Error detection for Indirect communication refer to the following aspects: - Support of handling Single Server header with multiple Producers. For example, Server Header may include both NF-instance and SCP-FQDN or multiple SCP-FQDNs. - Overruling the Server header logic when specific Error/application code combinations are received (e.g. NF_SERVICE_FAILOVER should be attributed to NF even if Server header includes the SCP)
- fallbackToDirectCommunication - Fallback to Direct Communication - This parameter controls if CMM will fallback to direct communication if no SCP is available. Fallback to Direct communication is not applicable in Roaming scenarios where SEPP must be used.
- fallbackToLocalSeppList - Fallback To Local Sepp List - This parameter controls if CMM will fallback to locally provisioned SEPP list in case NRF-based SEPP discovery fails.
- includeHttpServerHeaderInErrorResponses - Include HTTP Server Header In Error Responses - When this parameter is set to true, CMM inserts the HTTP Server header in Error Responses (4xx, 5xx)
- includeHttpServerHeaderInSuccessResponses - Include HTTP Server Header In Success Responses - When this parameter is set to true, CMM inserts the HTTP Server header in Success Responses (2xx)

- **isolateLastHttpProxy** - Isolate last HTTP proxy - When this parameter is set to true CMM will isolate the last available HTTP proxy even if **fallbackToDirectCommunication** parameter is disabled.
- **kpiIsolationDuration** - KPI Isolation Duration - Duration (in sec) a HTTP Proxy is Isolated due to low KPI or other reasons such as a 503 Response that indicates congestion.
- **kpiIsolationThreshold** - KPI isolation threshold - Threshold of Success/Attempts percent under which a HTTP proxy will be Isolated. Only Proxy related failures and timeouts are considered as HTTP proxy failures. Failures attributed to NFs are considered as Successes from HTTP Proxy perspective. Default value is 0 which means that Proxy are not isolated based on KPIs.
- **kpiMinSamples** - KPI Minimum Samples - Minimum number of Requests that are required in order to consider blacklisting a httpProxy based on KPI. Sample window is 10 seconds.
- **kpiSamplingPeriod** - KPI Sampling Period - Sampling Period for which response success and failures are aggregated to produce KPI for specific httpProxy.
- **nrfSubscriptionsInModelD** - Support NRF Subscriptions in Model D - When this parameter is enabled, CMM will support Subscribing to NRF for NF Profiles learned in Model D via the 3gpp-Sbi-Producer-Id
- **prefSchemeForCallbacksInModelD** - Preferred Scheme for Callbacks in Model D - When Model D is used and the SBI request contains a Callback URI for which both schemes are supported, this parameter defines which one is preferred. If only one scheme is supported it is used irrelevant of the value of this parameter
- **supportScpManagedServiceRedirects** - Support SCP Managed Service Redirects - This parameter controls if AMF evaluates the 3gpp-sbi-target-apiRoute and 3gpp-Producer-id HTTP headers added from SCP and altered HTTP Location header in subsequent Responses.
- **supportTokenCachingInModelD** - Support Token Caching in Model D - This parameter controls whether AMF supports Caching Authentication Token when conveyed back to CMM in a response via the 3gpp-Sbi-Access-Scope and 3gpp-Sbi-Access-Token Headers and sending them in relevant subsequent requests
- **unblockKpiBlockedScps** - Unblock KPI Blocked SCPs - This parameter Controls if SCPs that were isolated due to low KPI will be unblocked when no SCP is available.

Name

amfGparmValue - Parameter Value

Syntax

--amfGparmValue <string>

Description

Value of parameter

Class

Required on create.

Type / Range

Digit string of length 0 to 64

The amfGparmValue string is based on the amfGparmName parameter:

Name

action429 - Action 429

Syntax

```
--amfGparmName action429 --amfGparmValue <string>
```

Description

Default action for 429 Status Responses

List

- Isolate - Isolate the Service Instance
- KPI_Fail - KPI Fail - Increase the Failure Counter used for the Service Instance KPI
- KPI_Success - KPI Success - Increase the Success counter used for the Service Instance KPI

Default

Isolate

Name

action4xx - Action 4xx

Syntax

```
--amfGparmName action4xx --amfGparmValue <string>
```

Description

Default action for 4XX Status class Responses

List

- Isolate - Isolate the Service Instance
- KPI_Fail - KPI Fail - Increase the Failure Counter used for the Service Instance KPI
- KPI_Success - KPI Success - Increase the Success counter used for the Service Instance KPI

Default

KPI_Fail

Name

action503 - Action 503

Syntax

```
--amfGparmName action503 --amfGparmValue <string>
```

Description

Default action for 503 Status Responses

List

- Isolate - Isolate the Service Instance
- KPI_Fail - KPI Fail - Increase the Failure Counter used for the Service Instance KPI
- KPI_Success - KPI Success - Increase the Success counter used for the Service Instance KPI

Default

Isolate

Name

action5xx - Action 5xx

Syntax

--amfGparmName action5xx --amfGparmValue <string>

Description

Default action for 5XX Status class Responses

List

- Isolate - Isolate the Service Instance
- KPI_Fail - KPI Fail - Increase the Failure Counter used for the Service Instance KPI
- KPI_Success - KPI Success - Increase the Success counter used for the Service Instance KPI

Default

KPI_Fail

Name

actionTimeout - Action Timeout

Syntax

--amfGparmName actionTimeout --amfGparmValue <string>

Description

Action for Timeouts

List

- Isolate - Isolate the Service Instance
- KPI_Fail - KPI Fail - Increase the Failure Counter used for the Service Instance KPI
- KPI_Success - KPI Success - Increase the Success counter used for the Service Instance KPI

Default

KPI_Fail

Name

conveyAuthDiscoveryParams - Convey oAuth Discovery Params

Syntax

--amfGparmName conveyAuthDiscoveryParams --amfGparmValue <string>

Description

This parameters controls whether the AMF sends the Discovery '3gpp-Sbi-Discovery-*' headers related to oAuth in an HTTP/2 request targeting an existing resource context in the NF service producer to enable the SCP to perform the Service access authorization procedures.

List

- always - Always add oAuth related Discovery Parameters
- never - Never add oAuth related Discovery Parameters
- noCachedToken - No Cached Token - Add oAuth related Discovery Parameters only when there is no cached oAuth token to convey

Default

never

Name

discoverSeppViaNrf - Discover SEPP Via NRF

Syntax

--amfGparmName discoverSeppViaNrf --amfGparmValue <string>

Description

When this parameter is set to true CMM will select the SEPP via issuing a Discovery query to the NRF. This functionality is for TEST ONLY purposes.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

enhancedErrorDetection - Enhanced Error Detection

Syntax

--amfGparmName enhancedErrorDetection --amfGparmValue <string>

Description

This Parameter enables Enhanced Error detection for Indirect communication refer to the following aspects: - Support of handling Single Server header with multiple Producers. For example, Server Header may include both NF-instance and SCP-FQDN or multiple SCP-FQDNs. - Overruling the Server header logic when specific Error/application code combinations are received (e.g. NF_SERVICE_FAILOVER should be attributed to NF even if Server header includes the SCP)

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

fallbackToDirectCommunication - Fallback to Direct Communication

Syntax

```
--amfGparmName fallbackToDirectCommunication --amfGparmValue <string>
```

Description

This parameter controls if CMM will fallback to direct communication if no SCP is available. Fallback to Direct communication is not applicable in Roaming scenarios where SEPP must be used.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

fallbackToLocalSepplList - Fallback To Local Sepp List

Syntax

```
--amfGparmName fallbackToLocalSepplList --amfGparmValue <string>
```

Description

This parameter controls if CMM will fallback to locally provisioned SEPP list in case NRF-based SEPP discovery fails.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

includeHttpServerHeaderInErrorResponses - Include HTTP Server Header In Error Responses

Syntax

```
--amfGparmName includeHttpServerHeaderInErrorResponses --  
amfGparmValue <string>
```

Description

When this parameter is set to true, CMM inserts the HTTP Server header in Error Responses (4xx, 5xx)

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

includeHttpServerHeaderInSuccessResponses - Include HTTP Server Header In Success Responses

Syntax

```
--amfGparmName includeHttpServerHeaderInSuccessResponses --  
amfGparmValue <string>
```

Description

When this parameter is set to true, CMM inserts the HTTP Server header in Success Responses (2xx)

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

isolateLastHttpProxy - Isolate last HTTP proxy

Syntax

```
--amfGparmName isolateLastHttpProxy --amfGparmValue <string>
```

Description

When this parameter is set to true CMM will isolate the last available HTTP proxy even if fallbackToDirectCommunication parameter is disabled.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

kpiIsolationDuration - KPI Isolation Duration

Syntax

```
--amfGparmName kpiIsolationDuration --amfGparmValue <string>
```

Description

Duration (in sec) a HTTP Proxy is Isolated due to low KPI or other reasons such as a 503 Response that indicates congestion.

Type / Range

Integer in the range of 0 to 3,600

Default

30

Name

kpiIsolationThreshold - KPI isolation threshold

Syntax

```
--amfGparmName kpiIsolationThreshold --amfGparmValue <string>
```

Description

Threshold of Success/Attempts percent under which a HTTP proxy will be Isolated. Only Proxy related failures and timeouts are considered as HTTP proxy failures. Failures attributed to NFs are considered as Successes from HTTP Proxy perspective. Default value is 0 which means that Proxy are not isolated based on KPIs.

Type / Range

Integer in the range of 0 to 100

Default

0

Name

kpiMinSamples - KPI Minimum Samples

Syntax

```
--amfGparmName kpiMinSamples --amfGparmValue <string>
```

Description

Minimum number of Requests that are required in order to consider blacklisting a httpProxy based on KPI. Sample window is 10 seconds.

Type / Range

Integer in the range of 1 to 100000

Default

100

Name

kpiSamplingPeriod - KPI Sampling Period

Syntax

```
--amfGparmName kpiSamplingPeriod --amfGparmValue <string>
```

Description

Sampling Period for which response success and failures are aggregated to produce KPI for specific httpProxy.

Type / Range

Integer in the range of 5 to 300

Default

30

Name

nrfSubscriptionsInModelD - Support NRF Subscriptions in Model D

Syntax

```
--amfGparmName nrfSubscriptionsInModelD --amfGparmValue <string>
```

Description

When this parameter is enabled, CMM will support Subscribing to NRF for NF Profiles learned in Model D via the 3gpp-Sbi-Producer-Id

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

prefSchemeForCallbacksInModelD - Preferred Scheme for Callbacks in Model D

Syntax

```
--amfGparmName prefSchemeForCallbacksInModelD --amfGparmValue  
<string>
```

Description

When Model D is used and the SBI request contains a Callback URI for which both schemes are supported, this parameter defines which one is preferred. If only one scheme is supported it is used irrelevant of the value of this parameter

List

- http - The schema of Hypertext Transfer Protocol.
- https - The schema of Hypertext Transfer Protocol Secure.

Default

http

Name

supportScpManagedServiceRedirects - Support SCP Managed Service Redirects

Syntax

```
--amfGparmName supportScpManagedServiceRedirects --amfGparmValue  
<string>
```

Description

This parameter controls if AMF evaluates the 3gpp-sbi-target-apiRoute and 3gpp-Producer-id HTTP headers added from SCP and altered HTTP Location header in subsequent Responses.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

supportTokenCachingInModelD - Support Token Caching in Model D

Syntax

```
--amfGparmName supportTokenCachingInModelD --amfGparmValue <string>
```

Description

This parameter controls whether AMF supports Caching Authentication Token when conveyed back to CMM in a response via the 3gpp-Sbi-Access-Scope and 3gpp-Sbi-Access-Token Headers and sending them in relevant subsequent requests

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

unlockKpiBlockedScps - Unblock KPI Blocked SCPs

Syntax

```
--amfGparmName unlockKpiBlockedScps --amfGparmValue <string>
```


Description

This parameter Controls if SCPs that were isolated due to low KPI will be unblocked when no SCP is available.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

6.138 cmm sbiInitialRequestReselectionProfile

SBI Initial Request Reselection Profile. Syntax, description, parameters, and roles for cmm sbiInitialRequestReselectionProfile command.

Syntax**Description**

Used to enable or disable reselection for Initial Requests based on Rules for the interfaces where this feature is supported. Only one Profile exists named 'default' . The default profile can be modified but cannot be deleted nor new one can be created.

Parameters**Name**

name - Name

Syntax

--name <string>

Description

Profile Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

ausfConsumer - AUSF Consumer Enable

Syntax

--ausfConsumer <list-item>

Description

Enables Reselection for AUSF consumer interface

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

enable - Enable

Syntax

--enable <list-item>

Description

When this parameter is false the feature is turned off for all interfaces (no rule is checked, retries are not supported). Otherwise enabling is per interface.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

udmConsumer - UDM Consumer Enable

Syntax

--udmConsumer <list-item>

Description

Enables Reselection for UDM consumer interface

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

6.139 cmm sbiInitialRequestReselectionRule

SBI Initial Request Reselection Rule. Syntax, description, parameters, and roles for cmm sbiInitialRequestReselectionRule command.

Syntax**Description**

This command provisions the Reselection logic for Initial Requests .The rule with the lowest numbered rule order that matches all conditions defines the action. Rules with a higher numbered rule order are not checked once a match is found. A Default Rule that matches any condition is always placed last and cannot be deleted nor modified except for the action.

Parameters**Name**

sbiInitialRequestReselectionRuleListName - SBI Subsequent Request Reselection Rule List
Name

Syntax

--sbiInitialRequestReselectionRuleListName <string>

Description

The List the rule belongs

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

ruleOrder - Rule Order

Syntax

--ruleOrder <integer>

Description

The order the rule is evaluated.

Class

Required key.

Type / Range

Integer in the range of 1 to 65,535

Name

action - Action

Syntax

--action <list-item>

Description

Reselection action.

Class

Optional.

List

- Abort - Abort
- Reselect_Furthest - Reselect Furthest

Default

Abort

Name

communicationModel - Communication Model

Syntax

--communicationModel <list-item>

Description

Communication Model.

Class

Optional.

List

- ANY - Any
- DIRECT - Direct Communication (Model A/B)

Default

ANY

Name

description - Description

Syntax

--description <string>

Description

Descriptive name of the Rule

Class

Optional.

Type / Range

Digit string of length 0 to 128

Default

""

Name

error - Error

Syntax

--error <list-item>

Description

Error that triggered the reselection

Class

Optional.

List

- Any - Any
- 400_Bad_Request - 400 Bad Request
- 403_Forbidden - 403 Forbidden
- 404_Not_Found - 404 Not Found
- 405_Method_Not_Allowed - 405 Method Not Allowed
- 406_Not_Acceptable - 406 Not Acceptable
- 408_Request_Timeout - 408 Request Timeout

- 409_Conflict - 409 Conflict
- 410_Gone - 410 Gone
- 412_Precondition_Failed - 412 Precondition Failed
- 413_Payload_Too_Large - 413 Payload Too Large
- 414_URI_Too_Long - 414 URI Too Long
- 415_Unsupported_Media_Type - 415 Unsupported Media Type
- 429_Too_Many_Requests - 429 Too Many Requests
- 4xx_Client_Error - 4xx Client Error
- 500_Internal_Server_Error - 500 Internal Server Error
- 501_Not_Implemented - 501 Not Implemented
- 502_Bad_Gateway - 502 Bad Gateway
- 503_Service_Unavailable - 503 Service Unavailable
- 504_Gateway_Timeout - 504 Gateway Timeout
- 5xx_Server_Error - 5xx Server Error

Default

Any

Name

errorCause - Error Cause

Syntax

--errorCause <string>

Description

Application Error Cause value. String literal 'ANY' has special meaning and is applies to any Error Cause (including no Error Cause Value). When not set it matches the cases where no Error Cause Value is provided in the Response.

Class

Optional.

Type / Range

SBI Application Error Cause value

Default

"ANY"

Name

nfServiceOperation - NF Service Operation

Syntax

--nfServiceOperation <list-item>

Description

The NF Service Operation

Class

Optional.

List

- ANY - Any
- Nausf_UEAuthentication_Authenticate - Nausf UEAuthentication Authenticate
- Nudm_SubscriberDataManagement_Get - Nudm SubscriberDataManagement Get
- Nudm_SubscriberDataManagement_Subscribe - Nudm SubscriberDataManagement Subscribe
- Nudm_UEContextManagement_Registration - Nudm UEContextManagement Registration

Default

ANY

Name

nfType - NF Type

Syntax

--nfType <list-item>

Description

Type of the NF

Class

Optional.

List

- ANY - Any
- AUSF - AUSF
- UDM - UDM

Default

ANY

Name

reselectAttempt - Reselect Attempt

Syntax

--reselectAttempt <list-item>

Description

Reselect Attempt

Class

Optional.

List

- ANY - Any
- First - First

Default

ANY

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.140 cmm sbiInitialRequestReselectionRuleList

SBI Initial Request Reselection Rule List. Syntax, description, parameters, and roles for cmm sbiInitialRequestReselectionRuleList command.

Syntax

Description

List of Initial Request Reselection Rules

Parameters

Name

name - Name

Syntax

--name <string>

Description

List name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Roles**Show**

sysadmin, manage, monitor

List

sysadmin, manage, monitor

6.141 cmm sbiNfSetSupportProfile

SBI NF Set Support Profile. Syntax, description, parameters, and roles for cmm sbiNfSetSupportProfile command.

Syntax**Description**

SBI NF Set Support Profile

Parameters**Name**

name - Name

Syntax

--name <string>

Description

Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

addSuffixInCallbackUris - Add suffix in Callback URIs

Syntax

--addSuffixInCallbackUris <list-item>

Description

When this parameter is enabled, AMF will add pt appended with AMF Pointer ID as extra suffix in the paths in all Callback URIs it sends as extra suffix in the paths in all Callback URIs it sends. This suffix may be used by the AMF to distinguish between Notifications originally sent to this AMF vs. notifications received after redirection within AMFs in the Set.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

amfSetRegionSeparator - AMF Set Region Separator

Syntax

--amfSetRegionSeparator <list-item>

Description

Separator used to construct the Set ID(i.e. '<AMF Set ID>-region<AMF Region ID>' instead of '<AMF Set ID>.region<AMF Region ID>').

Class

Optional.

List

- hyphen - hyphen (-)
- period - period (.)

Default

hyphen

Name

indirectCommunicationBindingSupport - Binding Support for Indirect Communication

Syntax

--indirectCommunicationBindingSupport <list-item>

Description

This parameter controls whether AMF supports binding for Indirect Communication. When enabled, AMF will store the binding information send by a Producer UDM, AUSF, SMSF and PCF and convey it as 3gpp-Sbi-Routing-Binding HTTP Header in subsequent Requests sent via indirect communication. AMF will also store the binding information Consumers send when subscribing to amf-comm and amf-event services and send it as the 3gpp-Sbi-Routing-

Binding HTTP Header in Notifications sent via Indirect Communication. (Note: SMF binding is supported via separate parameters)

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

rel16OwnAmfSetConsumer - Rel-16 Own AMS-Set as Consumer

Syntax

--rel16OwnAmfSetConsumer <list-item>

Description

When this parameter is set to true, sends binding information in the relevant Service Requests and Notification Responses.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

rel16OwnAmfSetProducer - Rel-16 Own AMS-Set as Producer

Syntax

--rel16OwnAmfSetProducer <list-item>

Description

When this parameter is set to true, AMF includes the NF Sets it belongs to in the nfSetIdList IE of the NFProfile it Registers in NRF. Furthermore, it sends binding information in the Service Producer and Subscription Responses and Notification Requests

Class

Optional.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Name

smfBindingSupportMode - SMF Binding Support Mode

Syntax

--smfBindingSupportMode <list-item>

Description

The Binding support mode for SMF. Possible values are: - None: Binding Headers/IEs are ignored and not conveyed. - NFSet_Instance_ServiceInstance: Binding Headers/IEs are used and conveyed.

Class

Optional.

List

- NFSet_Instance_ServiceInstance - NFSet NFInstance NF Service Instance - NFSet, NFInstance and NF Service Instance
- None - None

Default

None

Name

smfConveyNfSetAsDiscoveryHeaderToScp - SMF Convey NF Set as Discovery Header to SCP

Syntax

--smfConveyNfSetAsDiscoveryHeaderToScp <list-item>

Description

This parameter controls if AMF will convey the NF Set via the 3gpp-sbi-Discovery-target-nf-set HTTP header for subsequent requests. This allows the SCP perform reselection in case of failures. This Header will be included only if target NF belongs to a NF Set and Route-Binding information is not available. Recommended value is Always as AMF based Reselection in case of Indirect Communication is supported only for Test purposes.

Class

Optional.

List

- Always - Always
- Never - Never

Default

Always

Name

smfNfSetSupportMode - SMF NF Set Support Mode

Syntax

--smfNfSetSupportMode <list-item>

Description

SMF NF Set Support Mode. Possible values are: - None: NFSets are not stored nor conveyed in any API. NFServicePersistence in the NFProfile is not evaluated. - NF_Instance: NFSets are not stored and not conveyed in any API but NFServicePersistence is evaluated. - NF_Set: NFSets are stored and conveyed in all API and NFServicePersistence is evaluated.

Class

Optional.

List

- NF_Instance - NF Instance - NF Instance.
- NF_Set - NF set - NF Set
- None - None

Default

None

Name

smfSupportRedirectForSubsequentServiceRequests - Support Redirect for Subsequent Requests

Syntax

--smfSupportRedirectForSubsequentServiceRequests <list-item>

Description

Defines if AMF will support redirect for Subsequent Requests towards SMF.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.142 cmm sbiPeerInfoGParms

SBI Peer Info General Parameters. Syntax, description, parameters, and roles for cmm sbiPeerInfoGParms command.

Syntax

Description

Configuration of 3gpp-Sbi-Peer-Info HTTP Header handling

Parameters

Name

amfGparmName - Parameter name

Syntax

--amfGparmName <list-item>

Description

Name of Parameter

Class

Required key.

List

- enable - This parameter enables or disables sending 3gpp-Sbi-Peer-Info in Requests and Responses. When enabled AMF will include own NF (Service) Instance Info as the Source NF
- includeDestinationNfInfo - Include Destination NF Info - This parameter controls if AMF will add the Destination NF (Service) Instance in Requests and Responses sent by the AMF when the relevant feature is enabled.
- includeScpSeppInfo - Include SCP/SEPP Info - This parameter controls if AMF will add the SCP/SEPP Info in Requests and Responses sent by the AMF when the relevant feature is enabled.

Name

amfGparmValue - Parameter value

Syntax

--amfGparmValue <string>

Description

Value of the Parameter

Class

Required on create.

Type / Range

Digit string of length 0 to 64

The amfGparmValue string is based on the amfGparmName parameter:

Name

enable - Enable

Syntax

--amfGparmName enable --amfGparmValue <string>

Description

This parameter enables or disables sending 3gpp-Sbi-Peer-Info in Requests and Responses. When enabled AMF will include own NF (Service) Instance Info as the Source NF

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

includeDestinationNfInfo - Include Destination NF Info

Syntax

--amfGparmName includeDestinationNfInfo --amfGparmValue <string>

Description

This parameter controls if AMF will add the Destination NF (Service) Instance in Requests and Responses sent by the AMF when the relevant feature is enabled.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

includeScpSeppInfo - Include SCP/SEPP Info

Syntax

--amfGparmName includeScpSeppInfo --amfGparmValue <string>

Description

This parameter controls if AMF will add the SCP/SEPP Info in Requests and Responses sent by the AMF when the relevant feature is enabled.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

6.143 cmm sbiQueryConversionRule

SBI Query Conversion Rule. Syntax, description, parameters, and roles for cmm sbiQueryConversionRule command.

Syntax**Description**

Manages the Conversion Rules applied in Query Parameters of SBI APIs

Parameters**Name**

ruleName - Rule Name

Syntax

--ruleName <string>

Description

Query Conversion Rule Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

queryServiceName - NF Service Name

Syntax

--queryServiceName <list-item>

Description

NF service for which the conversion Rule is relevant

Class

Required on create, not modifiable.

List

- nnrfNFDiscovery - Nnrf NF Discovery - NRF Network Function Discovery Service

Name

resource - Resource

Syntax

--resource <string>

Description

Resource for which the conversion Rule is relevant. Must start with '/' and must not end with '/'. E.g. '/nf-instances'

Class

Required on create, not modifiable.

Type / Range

Digit string of length 2 to 254

Name

ruleOrder - Rule order

Syntax

--ruleOrder <integer>

Description

Order with which the rule will be executed compared to other rules.

Class

Required on create.

Type / Range

Integer in the range of 1 to 10,000

Name

chainedQueryConversionRuleName - Chained Query Conversion Rule Name

Syntax

--chainedQueryConversionRuleName <string>

Description

This parameter is used exclusively by rules of type Only Condition and Only Operation. When Multiple Only Condition type rules are linked to the same Chained rule they are combined with logical AND. Each Chained Query Conversion Rule can have max one Only Operation Rule

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

comparison - Comparison

Syntax

--comparison <list-item>

Description

Used to compare a Query Parameter or Information Element to a Value

Class

Optional.

List

- eq - Equal
- gt - Greater Than
- gte - Greater Than or Equal to
- lt - Less Than
- lte - Less Than or Equal to
- neq - Not Equal
- na - Not applicable

Default

na

Name

comparisonValue - Comparison Value

Syntax

```
--comparisonValue <string>
```

Description

Value against which the query parameter is compared in a condition. Can be of any type but Format /structure must match the structure of the referenced Parameter

Class

Optional.

Type / Range

Digit string of length 0 to 352

Default

""

Name

conditionQueryParam - Condition Query Param

Syntax

```
--conditionQueryParam <string>
```

Description

Query Param used in the condition of a Rule. May be empty in which case the Rule's operation is applied unconditionally.

Class

Optional.

Type / Range

Digit string of length 0 to 128

Default

""

Name

conditionQueryParamOperator - Condition Query Param Operator

Syntax

```
--conditionQueryParamOperator <list-item>
```

Description

Operator applied to the Information Element. It is not applicable when and only when there is not an IE in the condition. Otherwise it must be one of the following types: 1) val: can be used for any type of primitive or object. 2) length: can be applied only to strings. 3) itemCount: can be applied only to arrays. 4) exists: any type of primitive or object. 5) notExists: any type of primitive or object.

Class

Optional.

List

- exists - Exists. Can be used for Any type of primitive or object. Condition is met if the Query Parameter or Information Element exists.
- itemCount - Item Count - Item Count. Can be applied only if the Query Parameter or Information Element is an array.
- length - Length. Can be applied only if the the Query Parameter or Information Element is a string.
- notExists - Not Exists - Not Exists. Can be used for any type of primitive or object. Condition is met if the Query Parameter or Information Element does not exist.
- val - Value - Value. The value of the Query Parameter or Information Element. Can be used for any type of primitive or object Parameter.
- na - Not applicable - Operator is not applicable. This value is used when and only when no Query parameter or Information Element is referenced in the condition.

Default

na

Name

httpMessageType - HTTP Message Type

Syntax

--httpMessageType <list-item>

Description

HTTP Message Type (Request or Response) for which the conversion Rule is relevant

Class

Optional.

List

- request - HTTP Request

Default

request

Name

httpMethod - HTTP Method

Syntax

--httpMethod <list-item>

Description

HTTP Method for which the conversion Rule is relevant

Class

Optional.

List

- get - HTTP method GET

Default

get

Name

operationAction - Operation Action

Syntax

--operationAction <list-item>

Description

Action that will be applied to the Operation Query Parameter.

Class

Optional.

List

- copy - Set the Operation Query Parameter or Information Element to the value of the Query Parameter or Information Element indicated in the operation Value
- remove - Removes the operation Query Parameter. or Information Element
- rename - Rename the Operation Query Parameter or Information Element to the Name indicated in the operation Value
- set - Set the Set the Operation Query Parameter or Information Element to the Operation value. If the Query Parameter or Information Element does not exists it is added. Otherwise its value is replaced.
- truncate - Can be used only for for arrays. The value refers to the max number of array elements the Operation Query Parameter of Information Element will contain.
- na - Not applicable - Not applicable

Default

na

Name

operationQueryParam - Operation Query Param

Syntax

--operationQueryParam <string>

Description

Query Param (optional) to which the operation of a Rule applies

Class

Optional.

Type / Range

Digit string of length 0 to 128

Default

""

Name

operationValue - Operation Value

Syntax

--operationValue <string>

Description

Usage of Value depends on the operation action: 1) set: can be of any type but format / structure must match the structure of the referenced Parameter.2)Copy: value indicates the Query Parameter from which the value will be copied 3)Truncate: refers to the max number of array elements operation Query Param will contain 4) Rename: refers to the new name of the operation Parameter. 5)It is not applicable in other cases

Class

Optional.

Type / Range

Digit string of length 0 to 352

Default

""

Name

queryApiMajorVersion - Major Version

Syntax

--queryApiMajorVersion <list-item>

Description

API Major Version for which the conversion Rule is relevant

Class

Optional.

List

- v1 - Version 1 - Major Version 1

Default

v1

Name

ruleDescription - Description

Syntax

--ruleDescription <string>

Description

Description of the Rule

Class

Optional.

Type / Range

Digit string of length 0 to 160 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

""

Name

ruleState - Rule State

Syntax

--ruleState <list-item>

Description

State of the Rule (enabled. disabled, test only)

Class

Optional.

List

- disabled - Rule is Disabled.
- enabled - Rule is Enabled.
- testOnly - Test Only - Rule is not applied messages sent to external NF. However conversion input and outcome is reported to testlog when rule condition is met.

Default

disabled

Name

ruleType - Rule Type

Syntax

--ruleType <list-item>

Description

Type of Rule. There are three types of rules. 1) Complete conversion rules: They are not associated to other conversion Rules (however condition is optional). 2. Condition only rules. These are Rules which include only a Condition. Must be associated with an Operation only Rule. If multiple condition rules point to the same Operation only Rule they are combined with logical 'AND'. 3) Operation only Rules which include only an Operation. Used by only Condition Rules.

Class

Not modifiable.

List

- **complete** - Complete/ conversion Rule which is autonomously executed. Not associated to other conversion Rules.
- **onlyCondition** - Only Condition - Rule which includes only a Condition. Must be associated with an only Operation Rule
- **onlyOperation** - Only Operation - Rule which includes only an Operation. Used by only Condition Rules

Default

complete

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.144 cmm sbiSubsequentRequestReselectionRule

SBI Subsequent Request Reselection Profile. Syntax, description, parameters, and roles for cmm sbiSubsequentRequestReselectionRule command.

Syntax**Description**

This command provisions the Reselection logic for Subsequent Requests .The rule with the lowest numbered rule order that matches all conditions defines the action. Rules with a higher numbered rule order are not checked once a match is found. A Default Rule that matches any condition is always placed last and cannot be deleted nor modified except for the action.

Parameters**Name**

sbiSubsequentRequestReselectionRuleListName - SBI Subsequent Request Reselection Rule List Name

Syntax

--sbiSubsequentRequestReselectionRuleListName <string>

Description

The List the rule belongs

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

ruleOrder - Rule Order

Syntax

--ruleOrder <integer>

Description

The order the rule is evaluated.

Class

Required key.

Type / Range

Integer in the range of 1 to 65,535

Name

action - Action

Syntax

--action <list-item>

Description

Reselection action.

Class

Optional.

List

- Abort - Abort
- Reselect_Binding_Closest - Reselect Binding Closest - Reselect First based on Binding. If Binding does not exist reselect Closest
- Reselect_Binding_Furthest - Reselect Binding Furthest - Reselect First based on Binding. If Binding does not exist reselect Furthest
- Reselect_Closest - Reselect Closest - Reselect Closest (ignore binding level priority)
- Reselect_Furthest - Reselect Furthest - Reselect Furthest (ignore binding level priority)

Default

Abort

Name

communicationModel - Communication Model

Syntax

--communicationModel <list-item>

Description

Communication Model. Only Direct officially Supported. Any and Indirect is only for test purposes.

Class

Optional.

List

- ANY - Any (test only) - Any - For Test Purposes Only
- DIRECT - Direct communication
- INDIRECT - Indirect (Test only) - Indirect communication - For Test Purposes Only

Default

DIRECT

Name

description - Description

Syntax

--description <string>

Description

Descriptive name of the Rule

Class

Optional.

Type / Range

Digit string of length 0 to 128

Default

""

Name

error - Error

Syntax

--error <list-item>

Description

Error that triggered the reselection

Class

Optional.

List

- 400_Bad_Request - 400 Bad Request
- 403_Forbidden - 403 Forbidden
- 404_Not_Found - 404 Not Found
- 405_Method_Not_Allowed - 405 Method Not Allowed
- 406_Not_Acceptable - 406 Not Acceptable
- 408_Request_Timeout - 408 Request Timeout
- 409_Conflict - 409 Conflict
- 410_Gone - 410 Gone
- 411_Length_Required - 411 Length Required
- 412_Precondition_Failed - 412 Precondition Failed
- 413_Payload_Too_Large - 413 Payload Too Large
- 414_URI_Too_Long - 414 URI Too Long
- 415_Unsupported_Media_Type - 415 Unsupported Media Type
- 429_Too_Many_Requests - 429 Too Many Requests
- 4XX_Client_Error - 4xx Client Error
- 500_Internal_Server_Error - 500 Internal Server Error
- 501_Not_Implemented - 501 Not Implemented
- 502_Bad_Gateway - 502 Bad Gateway
- 503_Service_Unavailable - 503 Service Unavailable
- 504_Gateway_Timeout - 504 Gateway Timeout
- 5xx_Server_Error - 5xx ServerError - 5xx Server Error
- Any - Any
- Link_Error - Link Error - Link Error (TCP/HTTP connection failure)
- Timeout - Timeout

Default

Any

Name

errorCause - Error Cause

Syntax

--errorCause <string>

Description

Application Error Cause value. String literal 'ANY' has special meaning and is applies to any Error Cause (including no Error Cause Value). When not set it matches the cases where no Error Cause Value is provided in the Response.

Class

Optional.

Type / Range

SBI Application Error Cause value

Default

"ANY"

Name

nfServiceOperation - NF Service Operation

Syntax

--nfServiceOperation <list-item>

Description

The NF Service Operation

Class

Optional.

List

- ANY - Any
- Nsmf_PDUSession_ReleaseSMContext - Nsmf PDUSession ReleaseSMContext
- Nsmf_PDUSession_RetrieveSMContext - Nsmf PDUSession RetrieveSMContext
- Nsmf_PDUSession_UpdateSMContext - Nsmf PDUSession UpdateSMContext - Nsmf_PDUSession UpdateSMContext

Default

ANY

Name

nfType - NF Type

Syntax

--nfType <list-item>

Description

Type of the NF

Class

Optional.

List

- ANY - Any
- SMF - SMF

Default

ANY

Name

reselectAttempt - Reselect Attempt

Syntax

--reselectAttempt <list-item>

Description

Reselect Attempt

Class

Optional.

List

- ANY - Any
- First - First

Default

ANY

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.145 cmm sbiSubsequentRequestReselectionRuleList

SBI Subsequent Request Reselection Rule List. Syntax, description, parameters, and roles for cmm sbiSubsequentRequestReselectionRuleList command.

Syntax

Description

List of Subsequent Request Reselection Rule

Parameters

Name

name - Name

Syntax

--name <string>

Description

Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Roles

Show

sysadmin, manage, monitor

List

sysadmin, manage, monitor

6.146 cmm sbiTimersProfile

Service Based Interface Message Response Wait Timer. Syntax, description, parameters, and roles for cmm sbiTimersProfile command.

Syntax

Description

This profile consists of message wait response timer for all other NF services operation message responses used by the AMF as a consumer. The profile is provisioned to a local consumer AMF endpoint.

Parameters

Name

sbiTimersProfileListName - Profile Name

Syntax

--sbiTimersProfileListName <string>

Description

Name of the profile. The AMF will provide a default profile named `sbiTimersAttemptsDefaultProfile`. By default all UE PLMNs will use the default profile.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of `[0-9a-zA-Z\.\_:@]*`

Name

`sbiTimerName` - Timer name

Syntax

```
--sbiTimerName <list-item>
```

Description

Name of timer.

Class

Required key.

List

- `ausfResponse` - Guard timer for receiving response from AUSF - When the AMF invokes any AUSF service operation, this parameter defines the default timer to be used unless a dedicated timer is defined for the particular service or service operation procedure. The AMF expects to receive a response from the AUSF. If the timer expires while waiting for this response, the AMF considers that the service or service operation utilizing the timer has failed
- `cancelRelocateUeContext` - Namf Cancel Relocate UE Context Request - Supervises receipt of Cancel Relocate UE Context Response from the target AMF. If no response is received, the service operation is considered as failed.
- `cbcfResponse` - Guard timer for receiving response from CBCF - When the AMF invokes any CBCF or CBCF IWF service operation, this parameter defines the default timer to be used unless a dedicated timer is defined for the particular service or service operation procedure. The AMF expects to receive a response from the CBCF. If the timer expires while waiting for this response, the AMF considers that the service or service operation utilising the timer has failed.
- `createUeContext` - Namf Create UE Context Request - Supervises receipt of Create UE Context Response from the target AMF. If no response is received, the service operation is considered as failed.
- `defaultBlacklist` - Default Blacklist Timer - The timer defines the duration that a NF service instance or remote endpoint is blacklisted when considered unavailable.
- `eirResponse` - EIR Response Waiting Timer - When the AMF invokes any EIR service operation, this parameter defines the default timer to be used unless a dedicated timer

is defined for the particular service or service operation procedure. The AMF expects to receive a response from the EIR. If the timer expires while waiting for this response, the AMF considers that the service or service operation utilizing the timer has failed.

- **gmlcResponse** - Guard timer for receiving response from GMLC - When the AMF invokes any GMLC service operation, this parameter defines the default timer to be used unless a dedicated timer is defined for the particular service or service operation procedure. The AMF expects to receive a response from the GMLC. If the timer expires while waiting for this response, the AMF considers that the service or service operation utilizing the timer has failed.
- **lmfResponse** - Guard timer for receiving response from LMF - When the AMF invokes any LMF service operation, this parameter defines the default timer to be used unless a dedicated timer is defined for the particular service or service operation procedure. The AMF expects to receive a response from the LMF. If the timer expires while waiting for this response, the AMF considers that the service or service operation utilizing the timer has failed.
- **n1MessageNotify** - N1 Message Notify - Supervises receipt of N1 Message Notify Response from the destination NF. If no response is received, the service operation is considered as failed.
- **n2InfoNotify** - Namf n2InfoNotify - Supervises receipt of n2InfoNoity Ack from the source AMF. If no response is received, the service operation is considered as failed.
- **nrfAccessTokenBlacklist** - NRF Access Token Service Blacklist Timer - The timer defines the duration that an AccessToken service instance is blacklisted when considered unavailable.
- **nrfDeRegister** - NRF NF Deregister Timer - The NRF NF De-register operation response timer.
- **nrfDiscoveryBlacklist** - NRF NF Discovery Service Blacklist Timer - The timer defines the duration that an NfDiscovery service instance is blacklisted when considered unavailable.
- **nrfDiscoveryResponse** - NRF Discovery Response - When the AMF invokes the NRF Discovery service operation, this parameter defines the default timer to be used unless a dedicated timer is defined for the particular NF Type (e.g SMF Discovery). The AMF expects to receive a response from the NRF. If the timer expires while waiting for this response, the AMF considers that the service operation utilizing the timer has failed.
- **nrfManagementBlacklist** - NRF NF Management Service Blacklist Timer - The timer defines the duration that an NfManagement service instance is blacklisted when considered unavailable.
- **nrfPurgeExpiredNfTimer** - Purge Expired NF Timer - This timer defines how long expired Entries continue to be selected for use by Connection management after the availability time has expired. These entries can be used up to 1 day for cases where the NRF is not responding.

- **nrfRegister** - NRF Register Operation Wait Timer - The timer is used to wait for response from NRF for NFRegister operation.
- **nrfRegisterBlacklist** - NRF Register Operation Wait Timer - The timer defines the duration a NRF Management Service Instance is blacklisted during NF Register failure.
- **nrfRegisterRetry** - NRF NFRegister Retry Wait Timer - The timer defines the wait period between the a failed NRF NFRegister operation and its retry. When value is set to 0 the AMF does not retry the operation
- **nrfResponse** - NRF Response - When the AMF invokes any NRF service operation, this parameter defines the default timer to be used unless a dedicated timer is defined for the particular service operation. The AMF expects to receive a response from the NRF. If the timer expires while waiting for this response, the AMF considers that the service operation utilizing the timer has failed.
- **nrfUpdateRetry** - NRF NFUpdate Retry Wait Timer - The timer defines the wait period between the a failed NRF NFUpdate via PATCH operation and its retry. When value is set to 0 the AMF does not retry the operation
- **nssaafResponse** - Guard Timer For Receiving Response From NSSAAF - When the AMF invokes any NSSAAF service operation, this parameter defines the default timer to be used unless a dedicated timer is defined for the particular service or service operation procedure. The AMF expects to receive a response from the NSSAAF. If the timer expires while waiting for this response, the AMF considers that the service or service operation utilising the timer has failed.
- **nssfAvailabilityPeriodicUpdate** - NSSF Availability Periodic Update - Time in minutes between 2 consecutive Nnssf_NSSAIAvailability Updates from AMF to the NSSF. Upon timer expiration, the AMF will send an Update to the NSSF irrelevant if there are any changes in the AMF Slice/TA information. When value is 0 (default) this mechanism is disabled.
- **nssfAvailabilityRetry** - NSSF Availability Retry Wait Timer - The timer defines the wait period between the a failed NSSF Availability update operation and its retry. When value is set to 0 the AMF does not retry the operation
- **nssfAvailabilityUpdateWait** - NSSF Availability Periodic Update Timer - Defines the period the AMF waits since last change regarding supported Slices/Tracking areas before it sends Nnssf_NSSAIAvailabilityUpdate. This timer is added in order to avoid excessive signalling to NSSF during network re-planning.
- **nssfBlacklist** - NSSF Blacklist Timer - The timer defines the duration a NSSF Instance is blacklisted when considered unavailable.
- **nssfResponse** - NSSF Response Timer - The timer is used to monitor responses from NSSF to the AMF sent requests.
- **nssfSubscribeExpiration** - NSSF Subscribe Expiration - Used to define the suggested time after which the AMF subscription to NSSF becomes invalid. The time is added in

the 'expires' IE during the Subscription towards the NSSF. When this value is set to 0, the AMF will omit the 'expires' IE.

- **nwdafResponse** - NWDAF Response - Guard Timer For Receiving Response From NWDAF. When the AMF invokes any NWDAF service operation, this parameter defines the default timer to be used unless a dedicated timer is defined for the particular service or service operation procedure. The AMF expects to receive a response from the NWDAF. If the timer expires while waiting for this response, the AMF considers that the service or service operation has failed.
- **pcfResponse** - Guard timer for receiving response from PCF - When the AMF invokes any PCF service operation, this parameter defines the default timer to be used unless a dedicated timer is defined for the particular service or service operation procedure. The AMF expects to receive a response from the PCF. If the timer expires while waiting for this response, the AMF considers that the service or service operation utilizing the timer has failed.
- **pduSessEstWt4N1N2Msg** - PDU Session establishment Wait for N1N2 Message Transfer - Defines the time the AMF waits for N1N2 Message Transfer trigger from SMF during PDU session setup procedure, when AMF has received Create SM context response from the SMF.
- **pduSmCtxStatusNotify** - PDU SM Context Status Notify - Defines the time the AMF waits for SM context status notify when the AMF expects to receive SM context status notify. AMF expects to receive SM context status notify during -UE/AMF/SMF triggered PDU session release procedures, -PDU session setup procedures that have been rejected towards the UE after SM context create, and -PDU session setup procedures when the AMF already has a PDU session established for the requested PDU session id. The cmm amfGParam defined setting smContextStatusNotifyRequired further defines the AMF behaviour after this timer expires.
- **releaseUeContext** - Namf Release UE Context Request - Supervises receipt of Release UE Context Response from the target AMF. If no response is received, the service operation is considered as successful
- **relocateUeContext** - Namf Relocate UE Context Request - Supervises receipt of Relocate UE Context Response from the target AMF. If no response is received, the service operation is considered as failed.
- **retrieveSmCtx5g4gHo** - Retrieve SM Context 5G to 4G Handover - When the AMF invokes Retrieve Sm Context service operation, as part of 5G to 4G handover, this parameter defines the default timer to be used for the particular service operation and its dedicated use case. If the timer expires while waiting for this response, the AMF considers that the service operation utilizing the timer has failed i.e. Sm Context Retrieval for the associated PDU Session has failed.
- **retrieveSmCtx5g4gIdleMob** - Retrieve SM Context 5G to 4G Idle Mobility - When the AMF invokes Retrieve Sm Context service operation, as part of 5G to 4G Idle mobility, this parameter defines the default timer to be used for the particular service operation

and its dedicated use case. If the timer expires while waiting for this response, the AMF considers that the service operation utilizing the timer has failed i.e. Sm Context Retrieval for the associated PDU Session has failed.

- smfDisc - SMF Discovery Operation Wait Timer - The timer is used to wait for response from NRF for SMF Discovery operation.
- smfIsolation - SMF Isolation Timer - The timer defines the duration that an SMF service instance is isolated when considered unavailable.
- smfResponse - SMF Response Timer - When the AMF invokes any SMF service operation, this parameter defines the default timer to be used unless a dedicated timer is defined for the particular service operation and its dedicated use case. The AMF expects to receive a response from the SMF. If the timer expires while waiting for this response, the AMF considers that the service operation utilizing the timer has failed.
- smfReuse - SMF Reuse - Defines time to wait for PDU session re-establishment request when SMF has triggered N1N2 Message Transfer service of AMF, and provided SMF Reallocation indication attribute and set this indication as true.
- smsfResponse - Guard timer for receiving response from SMSF - When the AMF invokes any SMSF service operation, this parameter defines the default timer to be used unless a dedicated timer is defined for the particular service or service operation procedure. The AMF expects to receive a response from the SMSF. If the timer expires while waiting for this response, the AMF considers that the service or service operation utilizing the timer has failed.
- udmResponse - Guard timer for receiving response from UDM - When the AMF invokes any UDM service operation, this parameter defines the default timer to be used unless a dedicated timer is defined for the particular service or service operation procedure. The AMF expects to receive a response from the UDM. If the timer expires while waiting for this response, the AMF considers that the service or service operation utilizing the timer has failed.
- ueContextTransfer - UE Context Transfer - Supervises the receipt of the UE Context Transfer response from the old AMF. If no response is received, the operation is considered as failed.
- waitRelocateCancelSrcMme - Wait Relocate Cancel from Source MME - Supervises receipt of Relocate Cancel Request from source MME after receiving Relocate UE Context Response from target AMF in EPS to 5GS with AMF relocation scenario. When timer expires, initial AMF shall release local resources

Name

sbiTimerUnit - Timer unit

Syntax

--sbiTimerUnit <list-item>

Description

Unit of measurement.

Class

Required on create, not modifiable.

List

- 320Hour - 320 hour - Specify timer unit of 320 hours.
- Millisecond - Specify timer unit of 1 millisecond.
- Second - Specify timer unit of 1 second.
- 2Second - 2 second - Specify timer unit of 2 seconds.
- 30Second - 30 second - Specify timer unit of 30 seconds.
- Minute - Specify timer unit of 1 minute.
- DeciHour - Specify timer unit of 1 decihour (6 minutes).
- 10Minute - 10 minute - Specify timer unit of 10 minutes.
- Hour - Specify timer unit of 1 hour.
- 10Hour - 10 hour - Specify timer unit of 10 hours.

Name

sbiTimerValue - Timer value

Syntax

```
--sbiTimerValue <integer>
```

Description

Numeric value of timer.

Class

Optional.

Type / Range

Integer in the range of 0 to 4,294,967,295

The sbiTimerValue string is based on the sbiTimerName parameter:

Name

ausfResponse - Guard timer for receiving response from AUSF

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName ausfResponse --  
sbiTimerValue <string> [ --sbiTimerUnit Millisecond ]
```

Description

When the AMF invokes any AUSF service operation, this parameter defines the default timer to be used unless a dedicated timer is defined for the particular service or service operation procedure. The AMF expects to receive a response from the AUSF. If the timer expires while waiting for this response, the AMF considers that the service or service operation utilizing the timer has failed

Type / Range

Integer in the range of 10 to 10,000 in units of Millisecond

Default

5000

Name

cancelRelocateUeContext - Namf Cancel Relocate UE Context Request

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName  
cancelRelocateUeContext --sbiTimerValue <string> [ --sbiTimerUnit  
Millisecond ]
```

Description

Supervises receipt of Cancel Relocate UE Context Response from the target AMF. If no response is received, the service operation is considered as failed.

Type / Range

Integer in the range of 100 to 30,000 in units of Millisecond

Default

2000

Name

cbcfResponse - Guard timer for receiving response from CBCF

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName cbcfResponse --  
sbiTimerValue <string> [ --sbiTimerUnit Millisecond ]
```

Description

When the AMF invokes any CBCF or CBCF IWF service operation, this parameter defines the default timer to be used unless a dedicated timer is defined for the particular service or service operation procedure. The AMF expects to receive a response from the CBCF. If the timer expires while waiting for this response, the AMF considers that the service or service operation utilising the timer has failed.

Type / Range

Integer in the range of 100 to 30,000 in units of Millisecond

Default

5000

Name

createUeContext - Namf Create UE Context Request

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName createUeContext --  
sbiTimerValue <string> [ --sbiTimerUnit Millisecond ]
```

Description

Supervises receipt of Create UE Context Response from the target AMF. If no response is received, the service operation is considered as failed.

Type / Range

Integer in the range of 100 to 30,000 in units of Millisecond

Default

10000

Name

defaultBlacklist - Default Blacklist Timer

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName defaultBlacklist  
--sbiTimerValue <string> [ --sbiTimerUnit Second ]
```

Description

The timer defines the duration that a NF service instance or remote endpoint is blacklisted when considered unavailable.

Type / Range

Integer in the range of 0 to 3,600 in units of Second

Default

5

Name

eirResponse - EIR Response Waiting Timer

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName eirResponse --  
sbiTimerValue <string> [ --sbiTimerUnit Millisecond ]
```

Description

When the AMF invokes any EIR service operation, this parameter defines the default timer to be used unless a dedicated timer is defined for the particular service or service operation procedure. The AMF expects to receive a response from the EIR. If the timer expires while waiting for this response, the AMF considers that the service or service operation utilizing the timer has failed.

Type / Range

Integer in the range of 100 to 30,000 in units of Millisecond

Default

5000

Name

gmlcResponse - Guard timer for receiving response from GMLC

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName gmlcResponse --  
sbiTimerValue <string> [ --sbiTimerUnit Millisecond ]
```

Description

When the AMF invokes any GMLC service operation, this parameter defines the default timer to be used unless a dedicated timer is defined for the particular service or service operation procedure. The AMF expects to receive a response from the GMLC. If the timer expires while waiting for this response, the AMF considers that the service or service operation utilising the timer has failed.

Type / Range

Integer in the range of 100 to 30,000 in units of Millisecond

Default

5000

Name

lmfResponse - Guard timer for receiving response from LMF

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName lmfResponse --  
sbiTimerValue <string> [ --sbiTimerUnit Millisecond ]
```

Description

When the AMF invokes any LMF service operation, this parameter defines the default timer to be used unless a dedicated timer is defined for the particular service or service operation procedure. The AMF expects to receive a response from the LMF. If the timer expires while waiting for this response, the AMF considers that the service or service operation utilizing the timer has failed.

Type / Range

Integer in the range of 100 to 180000 in units of Millisecond

Default

5000

Name

n1MessageNotify - N1 Message Notify

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName n1MessageNotify --  
sbiTimerValue <string> [ --sbiTimerUnit Millisecond ]
```

Description

Supervises receipt of N1 Message Notify Response from the destination NF. If no response is received, the service operation is considered as failed.

Type / Range

Integer in the range of 100 to 30,000 in units of Millisecond

Default

5000

Name

n2InfoNotify - Namf n2InfoNotify

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName n2InfoNotify --  
sbiTimerValue <string> [ --sbiTimerUnit Millisecond ]
```

Description

Supervises receipt of n2InfoNoity Ack from the source AMF. If no response is received, the service operation is considered as failed.

Type / Range

Integer in the range of 100 to 30,000 in units of Millisecond

Default

5000

Name

nrfAccessTokenBlacklist - NRF Access Token Service Blacklist Timer

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName  
nrfAccessTokenBlacklist --sbiTimerValue <string> [ --sbiTimerUnit  
Second ]
```

Description

The timer defines the duration that an AccessToken service instance is blacklisted when considered unavailable.

Type / Range

Integer in the range of 0 to 3,600 in units of Second

Default

5

Name

nrfDeRegister - NRF NF Deregister Timer

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName nrfDeRegister --  
sbiTimerValue <string> [ --sbiTimerUnit Millisecond ]
```

Description

The NRF NF De-register operation response timer.

Type / Range

Integer in the range of 100 to 30,000 in units of Millisecond

Default

5000

Name

nrfDiscoveryBlacklist - NRF NF Discovery Service Blacklist Timer

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName  
nrfDiscoveryBlacklist --sbiTimerValue <string> [ --sbiTimerUnit  
Second ]
```

Description

The timer defines the duration that an NFDISCOVERY service instance is blacklisted when considered unavailable.

Type / Range

Integer in the range of 0 to 3,600 in units of Second

Default

5

Name

nrfDiscoveryResponse - NRF Discovery Response

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName  
nrfDiscoveryResponse --sbiTimerValue <string> [ --sbiTimerUnit  
Millisecond ]
```

Description

When the AMF invokes the NRF Discovery service operation, this parameter defines the default timer to be used unless a dedicated timer is defined for the particular NF Type (e.g SMF Discovery). The AMF expects to receive a response from the NRF. If the timer expires while waiting for this response, the AMF considers that the service operation utilizing the timer has failed.

Type / Range

Integer in the range of 100 to 30,000 in units of Millisecond

Default

5000

Name

nrfManagementBlacklist - NRF NF Management Service Blacklist Timer

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName  
nrfManagementBlacklist --sbiTimerValue <string> [ --sbiTimerUnit  
Second ]
```

Description

The timer defines the duration that an NFManagement service instance is blacklisted when considered unavailable.

Type / Range

Integer in the range of 0 to 3,600 in units of Second

Default

5

Name

nrfPurgeExpiredNfTimer - Purge Expired NF Timer

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName  
nrfPurgeExpiredNfTimer --sbiTimerValue <string> [ --sbiTimerUnit  
Second ]
```

Description

This timer defines how long expired Entries continue to be selected for use by Connection management after the availability time has expired. These entries can be used up to 1 day for cases where the NRF is not responding.

Type / Range

Integer in the range of 0 or 30 to 86,400 in units of Second

Default

60

Name

nrfRegister - NRF Register Operation Wait Timer

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName nrfRegister --  
sbiTimerValue <string> [ --sbiTimerUnit Millisecond ]
```

Description

The timer is used to wait for response from NRF for NFRegister operation.

Type / Range

Integer in the range of 100 to 30,000 in units of Millisecond

Default

5000

Name

nrfRegisterBlacklist - NRF Register Operation Wait Timer

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName  
nrfRegisterBlacklist --sbiTimerValue <string> [ --sbiTimerUnit  
Second ]
```

Description

The timer defines the duration a NRF Management Service Instance is blacklisted during NF Register failure.

Type / Range

Integer in the range of 0 to 3,600 in units of Second

Default

300

Name

nrfRegisterRetry - NRF NFRegister Retry Wait Timer

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName nrfRegisterRetry  
--sbiTimerValue <string> [ --sbiTimerUnit Second ]
```

Description

The timer defines the wait period between the a failed NRF NFRegister peration and its retry. When value is set to 0 the AMF does not retry the operation

Type / Range

Integer in the range of 0 to 300 in units of Second

Default

10

Name

nrfResponse - NRF Response

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName nrfResponse --  
sbiTimerValue <string> [ --sbiTimerUnit Millisecond ]
```

Description

When the AMF invokes any NRF service operation, this parameter defines the default timer to be used unless a dedicated timer is defined for the particular service operation. The AMF expects to receive a response from the NRF. If the timer expires while waiting for this response, the AMF considers that the service operation utilizing the timer has failed.

Type / Range

Integer in the range of 100 to 30,000 in units of Millisecond

Default

5000

Name

nrfUpdateRetry - NRF NFUpdate Retry Wait Timer

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName nrfUpdateRetry --  
sbiTimerValue <string> [ --sbiTimerUnit Second ]
```

Description

The timer defines the wait period between the a failed NRF NFUpdate via PATCH operation and its retry. When value is set to 0 the AMF does not retry the operation

Type / Range

Integer in the range of 0 to 60 in units of Second

Default

10

Name

nssaafResponse - Guard Timer For Receiving Response From NSSAAF

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName nssaafResponse --  
sbiTimerValue <string> [ --sbiTimerUnit Millisecond ]
```

Description

When the AMF invokes any NSSAAF service operation, this parameter defines the default timer to be used unless a dedicated timer is defined for the particular service or service operation procedure. The AMF expects to receive a response from the NSSAAF. If the timer expires while waiting for this response, the AMF considers that the service or service operation utilising the timer has failed.

Type / Range

Integer in the range of 100 to 60,000 in units of Millisecond

Default

5000

Name

nssfAvailabilityPeriodicUpdate - NSSF Availability Periodic Update

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName  
nssfAvailabilityPeriodicUpdate --sbiTimerValue <string> [ --  
sbiTimerUnit Minute ]
```

Description

Time in minutes between 2 consecutive Nnssf_NSSAIAvailability Updates from AMF to the NSSF. Upon timer expiration, the AMF will sent an Update to the NSSF irrelevant if there are any changes in the AMF Slice/TA information. When value is 0 (default) this mechanism is disabled.

Type / Range

Integer in the range of 0 or 10 to 14400 in units of Minute

Default

0

Name

nssfAvailabilityRetry - NSSF Availaiblity Retry Wait Timer

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName
nssfAvailabilityRetry --sbiTimerValue <string> [ --sbiTimerUnit
Second ]
```

Description

The timer defines the wait period between the a failed NSSF Availaiblity update operation and its retry. When value is set to 0 the AMF does not retry the operation

Type / Range

Integer in the range of 0 to 300 in units of Second

Default

10

Name

nssfAvailabilityUpdateWait - NSSF Availability Periodic Update Timer

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName
nssfAvailabilityUpdateWait --sbiTimerValue <string> [ --sbiTimerUnit
Second ]
```

Description

Defines the period the AMF waits since last change regarding supported Slices/Tracking areas before it sends Nnssf_NSSAIAvailabilityUpdate. This timer is added in order to avoid excessive signalling to NSSF during network re-planning.

Type / Range

Integer in the range of 10 to 300 in units of Second

Default

60

Name

nssfBlacklist - NSSF Blacksit Timer

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName nssfBlacklist --
sbiTimerValue <string> [ --sbiTimerUnit Second ]
```

Description

The timer defines the duuration a NSSF Instance is blacklisted when considered unavailable.

Type / Range

Integer in the range of 0 to 3,600 in units of Second

Default

300

Name

nssfResponse - NSSF Response Timer

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName nssfResponse --  
sbiTimerValue <string> [ --sbiTimerUnit Millisecond ]
```

Description

The timer is used to monitor responses from NSSF to the AMF sent requests.

Type / Range

Integer in the range of 100 to 30,000 in units of Millisecond

Default

5000

Name

nssfSubscribeExpiration - NSSF Subscribe Expiration

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName  
nssfSubscribeExpiration --sbiTimerValue <string> [ --sbiTimerUnit  
Minute ]
```

Description

Used to define the suggested time after which the AMF subscription to NSSF becomes invalid. The time is added in the 'expires' IE during the Subscription towards the NSSF. When this value is set to 0, the AMF will omit the 'expires' IE.

Type / Range

Integer in the range of 0 to 14,400 in units of Minute

Default

1440

Name

nwdafResponse - NWDAF Response

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName nwdafResponse --  
sbiTimerValue <string> [ --sbiTimerUnit Millisecond ]
```

Description

Guard Timer For Receiving Response From NWDAF. When the AMF invokes any NWDAF service operation, this parameter defines the default timer to be used unless a dedicated timer is defined for the particular service or service operation procedure. The AMF expects to receive

a response from the NWDAF. If the timer expires while waiting for this response, the AMF considers that the service or service operation has failed.

Type / Range

Integer in the range of 100 to 30,000 in units of Millisecond

Default

5000

Name

pcfResponse - Gurad timer for receiving response from PCF

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName pcfResponse --  
sbiTimerValue <string> [ --sbiTimerUnit Millisecond ]
```

Description

When the AMF invokes any PCF service operation, this parameter defines the default timer to be used unless a dedicated timer is defined for the particular service or service operation procedure. The AMF expects to receive a response from the PCF. If the timer expires while waiting for this response, the AMF considers that the service or service operation utilizing the timer has failed.

Type / Range

Integer in the range of 10 to 10,000 in units of Millisecond

Default

5000

Name

pduSessEstWt4N1N2Msg - PDU Session establishment Wait for N1N1 Message Transfer

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName  
pduSessEstWt4N1N2Msg --sbiTimerValue <string> [ --sbiTimerUnit  
Second ]
```

Description

Defines the time the AMF waits for N1N2 Message Transfer trigger from SMF during PDU session setup procedure, when AMF has received Create SM context response from the SMF.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

5

Name

pduSmCtxStatusNotify - PDU SM Context Status Notify

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName  
pduSmCtxStatusNotify --sbiTimerValue <string> [ --sbiTimerUnit  
Second ]
```

Description

Defines the time the AMF waits for SM context status notify when the AMF expects to receive SM context status notify. AMF expects to receive SM context status notify during -UE/AMF/ SMF triggered PDU session release procedures, -PDU session setup procedures that have been rejected towards the UE after SM context create, and -PDU session setup procedures when the AMF already has a PDU session established for the requested PDU session id. The cmm amfGParm defined setting smContextStatusNotifyRequired further defines the AMF behaviour after this timer expires.

Type / Range

Integer in the range of 1 to 10 in units of Second

Default

8

Name

releaseUeContext - Namf Release UE Context Request

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName releaseUeContext  
--sbiTimerValue <string> [ --sbiTimerUnit Millisecond ]
```

Description

Supervises receipt of Release UE Context Response from the target AMF. If no response is received, the service operation is considered as successful

Type / Range

Integer in the range of 100 to 30,000 in units of Millisecond

Default

10000

Name

relocateUeContext - Namf Relocate UE Context Request

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName relocateUeContext  
--sbiTimerValue <string> [ --sbiTimerUnit Millisecond ]
```

Description

Supervises receipt of Relocate UE Context Response from the target AMF. If no response is received, the service operation is considered as failed.

Type / Range

Integer in the range of 100 to 30,000 in units of Millisecond

Default

2000

Name

retrieveSmCtx5g4gHo - Retrieve SM Context 5G to 4G Handover

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName
retrieveSmCtx5g4gHo --sbiTimerValue <string> [ --sbiTimerUnit
Millisecond ]
```

Description

When the AMF invokes Retrieve Sm Context service operation, as part of 5G to 4G handover, this parameter defines the default timer to be used for the particular service operation and its dedicated use case. If the timer expires while waiting for this response, the AMF considers that the service operation utilizing the timer has failed i.e. Sm Context Retrieval for the associated PDU Session has failed.

Type / Range

Integer in the range of 100 to 30,000 in units of Millisecond

Default

5000

Name

retrieveSmCtx5g4gIdleMob - Retrieve SM Context 5G to 4G Idle Mobility

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName
retrieveSmCtx5g4gIdleMob --sbiTimerValue <string> [ --sbiTimerUnit
Millisecond ]
```

Description

When the AMF invokes Retrieve Sm Context service operation, as part of 5G to 4G Idle mobility, this parameter defines the default timer to be used for the particular service operation and its dedicated use case. If the timer expires while waiting for this response, the AMF considers that the service operation utilizing the timer has failed i.e. Sm Context Retrieval for the associated PDU Session has failed.

Type / Range

Integer in the range of 100 to 30,000 in units of Millisecond

Default

5000

Name

smfDisc - SMF Discovery Operation Wait Timer

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName smfDisc --
sbiTimerValue <string> [ --sbiTimerUnit Second ]
```

Description

The timer is used to wait for response from NRF for SMF Discovery operation.

Type / Range

Integer in the range of 1 to 30 in units of Second

Default

5

Name

smfIsolation - SMF Isolation Timer

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName smfIsolation --  
sbiTimerValue <string> [ --sbiTimerUnit Second ]
```

Description

The timer defines the duration that an SMF service instance is isolated when considered unavailable.

Type / Range

Integer in the range of 0 to 3,600 in units of Second

Default

5

Name

smfResponse - SMF Response Timer

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName smfResponse --  
sbiTimerValue <string> [ --sbiTimerUnit Millisecond ]
```

Description

When the AMF invokes any SMF service operation, this parameter defines the default timer to be used unless a dedicated timer is defined for the particular service operation and its dedicated use case. The AMF expects to receive a response from the SMF. If the timer expires while waiting for this response, the AMF considers that the service operation utilizing the timer has failed.

Type / Range

Integer in the range of 100 to 30,000 in units of Millisecond

Default

5000

Name

smfReuse - SMF Reuse

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName smfReuse --  
sbiTimerValue <string> [ --sbiTimerUnit Second ]
```

Description

Defines time to wait for PDU session re-establishment request when SMF has triggered N1N2 Message Transfer service of AMF, and provided SMF Reallocation indication attribute and set this indication as true.

Type / Range

Integer in the range of 1 to 10 in units of Second

Default

2

Name

smsfResponse - Guard timer for receiving response from SMSF

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName smsfResponse --  
sbiTimerValue <string> [ --sbiTimerUnit Millisecond ]
```

Description

When the AMF invokes any SMSF service operation, this parameter defines the default timer to be used unless a dedicated timer is defined for the particular service or service operation procedure. The AMF expects to receive a response from the SMSF. If the timer expires while waiting for this response, the AMF considers that the service or service operation utilising the timer has failed.

Type / Range

Integer in the range of 100 to 30,000 in units of Millisecond

Default

5000

Name

udmResponse - Guard timer for receiving response from UDM

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName udmResponse --  
sbiTimerValue <string> [ --sbiTimerUnit Millisecond ]
```

Description

When the AMF invokes any UDM service operation, this parameter defines the default timer to be used unless a dedicated timer is defined for the particular service or service operation procedure. The AMF expects to receive a response from the UDM. If the timer expires while waiting for this response, the AMF considers that the service or service operation utilizing the timer has failed.

Type / Range

Integer in the range of 100 to 30,000 in units of Millisecond

Default

5000

Name

ueContextTransfer - UE Context Transfer

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName ueContextTransfer  
--sbiTimerValue <string> [ --sbiTimerUnit Millisecond ]
```

Description

Supervises the receipt of the UE Context Transfer response from the old AMF. If no response is received, the operation is considered as failed.

Type / Range

Integer in the range of 100 to 30,000 in units of Millisecond

Default

5000

Name

waitRelocateCancelSrcMme - Wait Relocate Cancel from Source MME

Syntax

```
--sbiTimersProfileListName <string> --sbiTimerName  
waitRelocateCancelSrcMme --sbiTimerValue <string> [ --sbiTimerUnit  
Millisecond ]
```

Description

Supervises receipt of Relocate Cancel Request from source MME after receiving Relocate UE Context Response from target AMF in EPS to 5GS with AMF relocation scenario. When timer expires, initial AMF shall release local resources

Type / Range

Integer in the range of 10 to 1000 in units of Millisecond

Default

500

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

6.147 cmm sbiTimersProfileList

SBI Timer Profile List. Syntax, description, parameters, and roles for cmm sbiTimersProfileList command.

Syntax

Description

SBI Timer profile list.

Parameters

Name

name - Name of the SBI Timer Profile List

Syntax

```
--name <string>
```

Description

Name of the SBI Timer Profile List.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.148 cmm sbiVersionControlProfile

SBI Version Control Profile. Syntax, description, parameters, and roles for cmm sbiVersionControlProfile command.

Syntax

Description

Defines the allowed major API versions for each of the API services the AMF consumes in the given Profile. At least one API version must be defined for each API. An API may have more than one allowed full version. When a new profile is created by default for all APIs are assigned to v1.0.0.

Parameters

Name

sbiVersionControlProfileListName - SBI Version Control Profile List Name

Syntax

--sbiVersionControlProfileListName <string>

Description

Name of SBI version control profile list

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

nfServiceName - Network Function Service Name

Syntax

--nfServiceName <list-item>

Description

Network Function Service Name

Class

Required key.

List

- n5gEirEquipmentIdentityCheck - N5G EIR EquipmentIdentity Check - 5GEIR Equipment Identity Check Service
- namfCommunication - Namf Communication - AMF Communication Service
- nausfUEAuthentication - Nausf UEAuthentication - AUSF Authentication Service
- nchfConvergedCharging - Nchf Converged Charging - CHF ConvergedCharging service
- nlmfLocation - Nlmf Location - LMF Location Service
- nnrfAccessToken - Nnrf Access Token Service - NRF Access Token Service

- nrfNFDiscovery - Nrf NF Discovery - NRF Network Function Discovery Service
- nrfNFManagement - Nrf NF Management - NRF Network Function Management Service
- nssaafNssaaService - Nssaaf NSSAA Service - NSSAAF NSSAA service
- nssfNSSAIAvailability - Nssf NSSAI Availability - NSSF Network Slice Availability Update Service
- nssfNSSelection - Nssf NS Selection - NSSF Network Slice Selection Service
- nwdafAnalyticsInfo - NWDAF Analytics Info - NWDAF Analytics Information Service
- nwdafEventsSubscription - NWDAF Events Subscription - NWDAF Events Subscription Service
- npcAMPolicyControl - Npcf AM Policy Control - PCF AM Policy Control Service
- npcUEPolicyControl - Npcf UE Policy Control - PCF UE Policy Control Service
- nsmfPDUSession - Nsmf PDU Session - SMF PDU Session Service
- nsmsfSMSservice - Nsmf SM Service - SMSF SM Service
- nudmSubscriberDataManagement - Nudm Subscriber Data Management - UDM Subscriber Data Management Service
- nudmUEContextManagement - Nudm UE Context Management - UDM UE Context Management Service
- nudsfDataRepository - Nudsf Data Repository - UDSF Data Repository Service

Name

fullVersion - Version

Syntax

--fullVersion <list-item>

Description

The full version of the SBI API.

Class

Required key.

List

- 1.0.0 - v1.0.0 - API version v1.0.0
- 1.0.1 - v1.0.1 - API version v1.0.1
- 1.0.2 - v1.0.2 - API version v1.0.2
- 1.0.3 - v1.0.3 - API version v1.0.3
- 1.0.4 - v1.0.4 - API version v1.0.4

- 1.0.5 - v1.0.5 - API version v1.0.5
- 1.0.6 - v1.0.6 - API version v1.0.6
- 1.0.8 - v1.0.8 - API version v1.0.8
- 1.1.2 - v1.1.2 - API version v1.1.2
- 1.1.3 - v1.1.3 - API version v1.1.3
- 1.1.4 - v1.1.4 - API version v1.1.4
- 1.1.5 - v1.1.5 - API version v1.1.5
- 1.1.6 - v1.1.6 - API version v1.1.6
- 1.1.7 - v1.1.7 - API version v1.1.7
- 1.1.8 - v1.1.8 - API version v1.1.8
- 1.1.10 - v1.1.10 - API version v1.1.10
- 1.1.0 - v1.1.0 - API version 1.1.0
- 1.1.1 - v1.1.1 - API version v1.1.1
- 1.1.11 - v1.1.11 - API version v1.1.11
- 1.2.0 - v1.2.0 - API version v1.2.0
- 1.2.1 - v1.2.1 - API version v1.2.1
- 1.2.2 - v1.2.2 - API version v1.2.2
- 1.2.3 - v1.2.3 - API version v1.2.3
- 1.2.4 - v1.2.4 - API v1.2.4
- 1.2.5 - v1.2.5 - API version v1.2.5
- 1.2.6 - v1.2.6 - API version v1.2.6
- 1.2.7 - v1.2.7 - API version v1.2.7
- 2.0.0 - v2.0.0 - API version v2.0.0
- 2.0.1 - v2.0.1 - API version v2.0.1
- 2.0.2 - v2.0.2 - API version v2.0.2
- 2.0.3 - v2.0.3 - API version v2.0.3
- 2.0.4 - v2.0.4 - API version v2.0.4
- 2.0.5 - v2.0.5 - API version v2.0.5
- 2.1.1 - v2.1.1 - API version v2.1.1
- 2.1.2 - v2.1.2 - API version v2.1.1
- 2.1.3 - v2.1.3 - API version v2.1.3

- 2.1.4 - v2.1.4 - API version v2.1.4
- 2.1.6 - v2.1.6 - API version v2.1.6
- 2.1.7 - v2.1.7 - API version v2.1.7
- 2.2.0 - v2.2.0 - API version v2.2.0
- 2.2.1 - v2.2.1 - API version v2.2.1
- 2.2.2 - v2.2.2 - API version v2.2.2
- 2.2.3 - v2.2.3 - API version v2.2.3
- 2.2.5 - v2.2.5 - API version v2.2.5
- 3.0.0 - v3.0.0 - API version v3.0.0
- 3.1.0 - v3.1.0 - API version v3.1.0

Name

applyVersionControlInDecode - Apply Version Control In Decode

Syntax

--applyVersionControlInDecode <list-item>

Description

This parameter defines if the Full API version is taken into account when Decoding the JSON body of received messages (either Requests or Responses). If set to true IEs that are not supported in that Full API version are ignored.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

enable - Enable

Syntax

--enable <list-item>

Description

This parameter is used to enable or disable sbiVersionControlProfile tuples. User created and System inserted tuples (i.e. tuples automatically created by the system during initial deployment) may be enabled or disabled but only user created tuples may be deleted. At any time at least one enabled sbiVersionControlProfile per nf service must exist.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

6.149 cmm sbiVersionControlProfileList

SBI Version Control Profile List. Syntax, description, parameters, and roles for cmm sbiVersionControlProfileList command.

Syntax**Description**

SBI Version Control Profile List

Parameters**Name**

name - SBI Version Control Profile Name

Syntax

--name <string>

Description

Name of SBI version control profile name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

useMinVersionAsDefault - Use Minimum Version as Default

Syntax

--useMinVersionAsDefault <list-item>

Description

Defines whether the the highest or lowest supported minor/patch Version that matches the Producer major version will be used when the full version of is unknown.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7 CMM CLI commands: cmm sbiVersionProducerControlProfile - cmm zoneCode

CMM CLI command descriptions: syntax, usage, parameters, and user roles.

7.1 cmm sbiVersionProducerControlProfile

SBI Version Producer Control Profile. Syntax, description, parameters, and roles for cmm sbiVersionProducerControlProfile command.

Syntax

Description

Defines the allowed API versions for each of the API services the own AMF supports as a Producer. Only one Profile List ('default') is supported.

Parameters

Name

sbiVersionProducerControlProfileListName - SBI Version Producer Control Profile List Name

Syntax

```
--sbiVersionProducerControlProfileListName <string>
```

Description

Name of SBI version producer control profile list.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

nfServiceName - Network Function Service Name

Syntax

```
--nfServiceName <list-item>
```

Description

Name of the AMF producer service.

Class

Required key.

List

- namfCommunication - Namf Communication - Namf Communication Service

- namfEventExposure - Namf Event Exposure - Namf Event Exposure Service
- namfLocation - Namf Location - Namf Location Service
- namfMT - Namf MT - Namf Mobile Terminated Service

Name

majorVersion - Major Version

Syntax

--majorVersion <list-item>

Description

The major version of the SBI API.

Class

Required key.

List

- v1 - Version 1

Name

fullVersion - Full Version

Syntax

--fullVersion <list-item>

Description

The full version to be used as the default for the specified AMF producer service and major version

Class

Required on create.

List

- 1.0.1 - v1.0.1 - API version v1.0.1
- 1.0.2 - v1.0.2 - API version v1.0.2
- 1.0.3 - v1.0.3 - API version v1.0.3
- 1.0.4 - v1.0.4 - API version v1.0.4
- 1.0.5 - v1.0.5 - API version v1.0.5
- 1.0.6 - v1.0.6 - API version v1.0.6
- 1.0.8 - v1.0.8 - API version v1.0.8
- 1.1.2 - v1.1.2 - API version v1.1.2
- 1.1.3 - v1.1.3 - API version v1.1.3
- 1.1.4 - v1.1.4 - API version v1.1.4

- 1.1.7 - v1.1.7 - API version v1.1.7
- 1.1.8 - v1.1.8 - API version v1.1.8
- 1.1.10 - v1.1.10 - API version v1.1.10
- 1.1.11 - v1.1.11 - API version v1.1.11
- 1.2.1 - v1.2.1 - API version v1.2.1
- 1.2.2 - v1.2.2 - API version v1.2.2
- 1.2.3 - v1.2.3 - API version v1.2.3
- 1.2.4 - v1.2.4 - API version v1.2.4

Name

applyVersionControlInDecode - Apply Version Control In Decode

Syntax

--applyVersionControlInDecode <list-item>

Description

This parameter defines if the Full API version is taken into account when Decoding the JSON body of received messages (either Requests or Responses). If set to true IEs that are not supported in that Full API version are ignored.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

fourthVersionField - Fourth Version Field

Syntax

--fourthVersionField <string>

Description

Additional optional version field in the API version of the Service Instance. When set this string field is appended to the full major.minor.patch API version registered to the NRF using a '.' as a delimiter.

Class

Optional.

Type / Range

Default

""

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

7.2 cmm sbiVersionProducerControlProfileList

SBI Version Producer Control Profile List. Syntax, description, parameters, and roles for cmm sbiVersionProducerControlProfileList command.

Syntax**Description**

SBI Version Producer Control Profile List. Only one list 'default' is supported.

Parameters**Name**

name - SBI Version Producer Control Profile Name

Syntax

--name <string>

Description

Name of SBI version producer control profile name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles**Show**

sysadmin, manage, monitor

List

sysadmin, manage, monitor

7.3 cmm sccpSignallingPoint

SCCP Signalling Point. Syntax, description, parameters, and roles for cmm sccpSignallingPoint command.

Syntax

Description

Configure, list and show SCCP signalling point.

Parameters

Name

signallingPointId - SCCP signalling point identifier

Syntax

--signallingPointId <integer>

Description

SCCP signalling point identifier.

Class

Required key.

Type / Range

Integer in the range of 0 to 12000

Name

signallingNetwork - Signalling network

Syntax

--signallingNetwork <list-item>

Description

Identify signalling network IN0 ... international network number 0 IN1 ... international network number 1 NAO ... national network number 0 NA1 ... national network number 1.

Class

Required on create, not modifiable.

List

- IN0 - International network number 0.
- IN1 - International network number 1.
- NAO - National network number 0.

- NA1 - National network number 1.

Name

signallingPointCode - Signalling point code

Syntax

--signallingPointCode <integer>

Description

Identify signalling point code decimal number 0 ... 16777215.

Class

Required on create, not modifiable.

Type / Range

Integer in the range of 0 to 16,777,215

Name

spcState - Signalling point code state

Syntax

--spcState <list-item>

Description

Identify state change: ACT ... activate INA ... inactivate.

Class

Required on create.

List

- ACT - Activate - Specify state change activate.
- INA - Inactivate - Specify state change inactivate.

Name

csbg - Concerned signalling points broadcast group

Syntax

--csbg <integer>

Description

Broadcast group of concerned signalling points for SP broadcast.

Class

Optional.

Type / Range

Integer in the range of 0 to 65,535

Default

0

Name

spcParamSetName - Signalling point code parameter set name

Syntax

--spcParamSetName <string>

Description

Identify signalling point parameter set name The following parameter sets are predefined:
blue ... BLUE for signalling parameters defined by International Telecommunications Union (ITU) in Q.700 - Q.716, Blue Book
aint ... A-INT for A interface e.g. between MSC and BSC
white ... WHITE for signalling parameters defined by International Telecommunications Union (ITU) in Q.700 - Q.716 3/93, White Book or ... use own created set name (in spcParamSet)

Restrictions

a record must exist in spcParamSet with name of spcParamSetName

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

blue

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.4 cmm sccpSubsystem

SCCP Subsystem. Syntax, description, parameters, and roles for cmm sccpSubsystem command.

Syntax

Description

Configure, list and show SCCP signalling point subsystem.

Parameters

Name

subsystemId - SCCP subsystem identifier

Syntax

--subsystemId <integer>

Description

SCCP subsystem identifier.

Class

Required key.

Type / Range

Integer in the range of 0 to 12000

Name

subsystemName - SCCP subsystem name

Syntax

--subsystemName <string>

Description

Identify subsystem name: identifier 1 ... 5 character.

Restrictions

subsystemName must be unique within sccpSubsystem

Class

Required on create, not modifiable.

Type / Range

Digit string of length 1 to 5 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

subsystemNumber - SCCP subsystem number

Syntax

--subsystemNumber <string>

Description

Identify subsystem number: hexadecimal number 2 ... FE.

Restrictions

subsystemNumber must be unique within sccpSubsystem

Class

Required on create, not modifiable.

Type / Range

Digit string of length 1 to 2 with the allowed pattern of ([2-9a-fA-F][0][2-9a-fA-F][fF][0-9a-eA-E][1-9a-eA-E][0-9a-fA-F])

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.5 cmm sccpSubsystemInfo

SCCP Subsystem Info. Syntax, description, parameters, and roles for cmm sccpSubsystemInfo command.

Syntax**Description**

Display the SCCP subsystems configured for a specific point code and display them together with the currently known status.

Parameters**Name**

network - Signalling Network

Syntax

--network <list-item>

Description

Use this parameter to specify the symbolic name of the signalling network. Allowed values IN0,IN1,NA0,NA1

Restrictions

This parameter is required if --spc has been also used.

Class

Required on create.

List

- IN0 - International network number 0.
- IN1 - International network number 1.
- NA0 - National network number 0.
- NA1 - National network number 1.

Name

spc - Signalling Point Code

Syntax

--spc <integer>

Description

Use this parameter to specify the signalling point codes of those signalling points where the signalling link sets, whose state data you wish to output, lead to

Restrictions

If this parameter is specified, then also --network parameter is required.

Class

Optional.

Type / Range

Integer in the range of 0 to 16,777,215

Roles**List**

sysadmin, manage, monitor

7.6 cmm sccpSubsystemList

SCCP Subsystem List. Syntax, description, parameters, and roles for cmm sccpSubsystemList command.

Syntax**Description**

Configure, list and show SCCP subsystem list.

Parameters

Name

signallingPointId - SCCP signalling point identifier

Syntax

```
--signallingPointId <integer>
```

Description

SCCP signalling point identifier.

Restrictions

record with this id must exist in sccpSignallingPoint. Max number of list entries per sccpSignallingPoint is 20.

Class

Required key.

Type / Range

Integer in the range of 0 to 12000

Name

subsystemId - SCCP subsystem identifier

Syntax

```
--subsystemId <integer>
```

Description

SCCP subsystem identifier.

Restrictions

record with this id must exist in sccpSubsystem

Class

Required key.

Type / Range

Integer in the range of 0 to 12000

Name

albfg - Broadcast group of affected subsystems for local broadcast

Syntax

```
--albfg <integer>
```

Description

Broadcast group of affected subsystems for local broadcast.

Class

Optional.

Type / Range

Integer in the range of 0 to 65,535

Default

0

Name

asbg - Broadcast group of affected subsystems for sp broadcast

Syntax

--asbg <integer>

Description

Broadcast group of affected subsystems for sp broadcast.

Class

Optional.

Type / Range

Integer in the range of 0 to 65,535

Default

0

Name

subParamSetName - SCCP subsystem parameter set name

Syntax

--subParamSetName <string>

Description

identify subsystem parameter set name The following parameter sets are predefined (in sccpsubsystemparamsetdefault): gener ... GENER for general use in other than with A interface aint ... A-INT for A interface, for example, between MSC and BSC or ... use own created set name (in subParamSet)

Restrictions

a record must exist in subParamSet with name of subParamSetName

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

gener

Name

subStatusTest - SCCP subsystem test

Syntax

--subStatusTest <list-item>

Description

Identify subsystem status test: yes ... subsystem test is allowed no ... subsystem test is not allowed parameter is valid, only if the white book management is used in signalling point parameter set. default is yes.

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

Yes

Name

subsystemState - SCCP subsystem state

Syntax

--subsystemState <list-item>

Description

Identify state change: ACT ... active INA ... inactivate CINA ... inactivate coordinated way.

Class

Optional.

List

- ACT - Active - Active.
- CINA - Inactive coordinated - Inactive coordinated.
- INA - Inactive - Inactive.

Default

ACT

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.7 cmm scefGrp

SCEF Group Profile. Syntax, description, parameters, and roles for cmm scefGrp command.

Syntax**Description**

Administer SCEF Group Profile.

Parameters**Name**

scefGrpListName - SCEF group profile name

Syntax

--scefGrpListName <string>

Description

SCEF group profile name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

scefOrderId - SCEF order in group

Syntax

--scefOrderId <integer>

Description

SCEF order in group.

Class

Required key.

Type / Range

Integer in the range of 1 to 16

Name

scefName - SCEF name

Syntax

--scefName <string>

Description

SCEF name specified in remote end point configuration.

Class

Required on create, not modifiable.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

lbWeight - Weight for load balancing

Syntax

--lbWeight <integer>

Description

Weight for load balancing algorithm.

Class

Optional.

Type / Range

Integer in the range of 0 to 100

Default

0

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.8 cmm scefGrpList

SCEF Group Profile List. Syntax, description, parameters, and roles for cmm scefGrpList command.

Syntax

Description

Administer SCEF Group Profile List.

Parameters

Name

name - SCEF group profile name

Syntax

--name <string>

Description

SCEF group profile name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.9 cmm sceflnputSet

SCEF Input Set. Syntax, description, parameters, and roles for cmm sceflnputSet command.

Syntax

Description

SCEF input set.

Parameters

Name

scefInputSetListName - SCEF input set list name

Syntax

--scefInputSetListName <string>

Description

SCEF input set list name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

scefInputFqdnName - SCEF input set FQDN name

Syntax

--scefInputFqdnName <string>

Description

List name of SCEF input set FQDN.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.10 cmm scefInputSetList

SCEF Input Set List. Syntax, description, parameters, and roles for cmm scefInputSetList command.

Syntax

Description

SCEF Input Set List.

Parameters

Name

name - SCEF input set list name

Syntax

--name <string>

Description

Name of SCEF input set list.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.11 cmm scefProfile

SCEF Profile. Syntax, description, parameters, and roles for `cmm scefProfile` command.

Syntax

Description

Defines profile for SCEF settings

Parameters

Name

name - Profile Name

Syntax

--name <string>

Description

Profile Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

allowExtScefRefId - Allow Extended SCEF Reference ID

Syntax

--allowExtScefRefId <list-item>

Description

Specifies whether the MME is to allow negotiation of the use of extended SCEF reference IDs for the UEs-in-Area Monitoring Event configured by the SCEF directly to the MME. The default is false.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Recommended

true

Name

enbTargetIdWithoutHeader - eNodeB-ID Target Without Header

Syntax

--enbTargetIdWithoutHeader <list-item>

Description

When the MME receives an event configuration for 'UEs in Area' that contains eNodeB-ID, and when the MME sends an event report for 'UEs in Area' that contains eNodeB-ID, this

flag specifies whether the leading octets 1-4 for the eNodeB-ID AVP are to be excluded. See 29.274 Section 8.51 for encoding details. The default is false.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

imsiGroupSupported - IMSI Group Supported

Syntax

--imsiGroupSupported <list-item>

Description

Specifies whether the IMSI Group Identifier is supported. This is used for filtering the Number of UEs Present in a Geographical Area Monitoring Event Report.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

includeServingPlmnRateControl - Include Serving PLMN Rate Control

Syntax

--includeServingPlmnRateControl <list-item>

Description

Specifies whether the MME is to include serving PLMN rate control in the T6a:CMR message to the SCEF, if applicable for the UE. The default is false.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

locationReportTargetIdWithoutHeader - Location Report Target ID Without Header

Syntax

--locationReportTargetIdWithoutHeader <list-item>

Description

When the MME sends a Location Event Report containing an ENodeB-Identifier, this flag specifies whether the eNodeB-ID should be sent without leading octets 1-4. See 29.274 Section 8.51 for the message layout. The default is False.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

monNumberUesInGeographicArea - Number of UEs in Geographic Area Monitoring

Syntax

--monNumberUesInGeographicArea <list-item>

Description

Specifies whether a SCEF is allowed to request the Number of UEs in a Geographic Location Monitoring Event.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.12 cmm scefToGrp

SCEF To Group. Syntax, description, parameters, and roles for `cmm scefToGrp` command.

Syntax

Description

This command is used to administer the SCEF To Group Mapping. The 'shutdown' parameter is used to enable/disable the mapping rule.

Parameters

Name

mappingId - Mapping ID

Syntax`--mappingId <integer>`**Description**

Mapping ID.

Class

Required key.

Type / Range

Integer in the range of 1 to 128

Name

cirDestHostFqdnName - CIR Destination Host FQDN Name

Syntax`--cirDestHostFqdnName <string>`**Description**

Specifies the Destination Host FQDN to be used when routing a CIR to an IWK-SCEF

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

cirDestRealmFqdnName - CIR Destination Realm FQDN Name

Syntax

--cirDestRealmFqdnName <string>

Description

Specifies the Destination Realm FQDN to be used when routing a CIR to an IWK-SCEF.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

draDestHostFqdnName - DRA destination host name

Syntax

--draDestHostFqdnName <string>

Description

If MME does not have destination host in VLR (UE shows up for the first time), and the destination is a DRA, MME uses the value of this new field to populate destination host AVP in AIR and ULR.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

draDestRealmFqdnName - DRA destination realm name

Syntax

--draDestRealmFqdnName <string>

Description

If MME does not have destination realm in VLR (UE shows up for the first time), and the destination is a DRA, MME uses the value of this new field to populate destination realm AVP in AIR and ULR.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

hssGrpListName - HSS group list name

Syntax

--hssGrpListName <string>

Description

Name of HSS group list.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

iwkScef - IWK-SCEF

Syntax

--iwkScef <list-item>

Description

Specifies whether the peer supports the 3GPP IWK-SCEF function.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

loadBalance - Load balance

Syntax

--loadBalance <list-item>

Description

Indicates load balancing options.

Class

Optional.

List

- all - Balance load on all S6ad links based on weight.
- none - No load balancing between S65ad links.
- priority group - Balance load on all S6ad links based on group priority. the primary group is the set of HSS/DRA that have the same highest weight. the secondary group is the set of HSS/DRA that have the same second highest weight value, etc. the last N-ary group is always the group with 0 weight (if used).

Default

none

Name

plmnName - Mobile network name

Syntax

--plmnName <string>

Description

Name that identifies the mobile network.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

scefGrpListName - SCEF group list name

Syntax

--scefGrpListName <string>

Description

Name of SCEF group list.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

scefInputSetListName - SCEF input set list name

Syntax

--scefInputSetListName <string>

Description

SCEF input set list name.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

scefMappingType - SCEF mapping type

Syntax

--scefMappingType <list-item>

Description

SCEF mapping type.

Class

Optional.

List

- all - Specify SCEF mapping type of set.
- set - Specify SCEF mapping type of set.

Default

all

Name

shutdown - Shutdown

Syntax

--shutdown <list-item>

Description

Indicates if the mapping rule is active. if shutdown is true, the rule behaves as if it is deleted.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.13 cmm schedule

schedule. Syntax, description, parameters, and roles for **cmm schedule** command.

Syntax**Description**

Create, list, modify, delete and show schedule jobs.

Parameters**Name**

schedJobName - Job name

Syntax

--schedJobName <list-item>

Description

Schedule-job name.

Class

Required key.

List

- backup - Specify schedule job of backup.
- remote-copy - Specify schedule job of remote copy.
- rsp - Specify schedule job of RSP.
- secure-copy - Specify schedule job of secure copy.

Name

dayOfMonth - Day of month

Syntax

--dayOfMonth <string>

Description

Schedule-job day of month.

Class

Required on create.

Type / Range

Digit string of length 1 to 50 with the allowed pattern of `(\*(?:\d+)?(?:3[01][12]\d|0?[1-9])(?:-(?:3[01][12]\d|0?[1-9])(?:\d+)?(?:,|(?:3[01][12]\d|0?[1-9])(?:-(?:3[01][12]\d|0?[1-9])(?:\d+)?)?*))`

Name

dayOfWeek - Day of week

Syntax

--dayOfWeek <string>

Description

Schedule-job day of week.

Class

Required on create.

Type / Range

Digit string of length 1 to 50 with the allowed pattern of `(\*(?:\d+)?(?:[0-6])(?:-(?:[0-6])(?:\d+)?(?:,|(?:[0-6])(?:-(?:[0-6])(?:\d+)?)*((?:Mon|Tue|Wed|Thu|Fri|Sat|Sun)(?:-(?:Mon|Tue|Wed|Thu|Fri|Sat|Sun)(?:\d+)?(?:,|(?:Mon|Tue|Wed|Thu|Fri|Sat|Sun)(?:-(?:Mon|Tue|Wed|Thu|Fri|Sat|Sun)(?:\d+)?)*))`

Name

hour - Hour

Syntax

--hour <string>

Description

Schedule-job hour.

Class

Required on create.

Type / Range

Digit string of length 1 to 50 with the allowed pattern of (*(?:\V\d+)?(?:2[0-3][01]?d)(?:- (?:2[0-3][01]?d)(?:\V\d+)?(?:, (?:2[0-3][01]?d)(?:-(?:2[0-3][01]?d)(?:\V\d+)?))*))

Name

minute - Minute

Syntax

--minute <string>

Description

Schedule-job minute.

Class

Required on create.

Type / Range

Digit string of length 1 to 50 with the allowed pattern of (*(?:\V\d+)?(?:[0-5]?d)(?:-(?: [0-5]?d)(?:\V\d+)?(?:, (?:[0-5]?d)(?:-(?:[0-5]?d)(?:\V\d+)?))*))

Name

month - Month

Syntax

--month <string>

Description

Schedule-job month.

Class

Required on create.

Type / Range

Digit string of length 1 to 50 with the allowed pattern of (*(?:\V\d+)?(?:1[012][1-9])(?:- (?:1[012][1-9])(?:\V\d+)?(?:, (?:1[012][1-9])(?:-(?:1[012][1-9])(?:\V\d+)?))**(?:Jan|Feb| Mar|Apr|May|Jun|Jul|Aug|Sep|Oct|Nov|Dec)(?:-(?:Jan|Feb|Mar|Apr|May|Jun|Jul|Aug|Sep|Oct| Nov|Dec)(?:\V\d+)?(?:, (?:Jan|Feb|Mar|Apr|May|Jun|Jul|Aug|Sep|Oct|Nov|Dec)(?:-(?:Jan|Feb| Mar|Apr|May|Jun|Jul|Aug|Sep|Oct|Nov|Dec)(?:\V\d+)?))*))

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.14 cmm scpDomain

SCP Domain. Syntax, description, parameters, and roles for `cmm scpDomain` command.

Syntax

Description

Links a Domain to the CMM scpDomain list. A SCP Domain may be added only once in a list.

Parameters

Name

scpDomainListName - SCP Domain List Name

Syntax

--scpDomainListName <string>

Description

SCP Domain List Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of `[0-9a-zA-Z\.\_:@]*`

Name

domain - SCP Domain

Syntax

--domain <string>

Description

SCP Domain

Class

Required key.

Type / Range

Digit string of length 1 to 100 with the allowed pattern of [0-9a-zA-Z.-]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.15 cmm scpDomainList

SCP Domain List. Syntax, description, parameters, and roles for cmm scpDomainList command.

Syntax**Description**

The list of SCP Domains CMM belongs to. Only one default List named 'default' exists. The list can have up to 127 domains.

Parameters**Name**

name - Name

Syntax

--name <string>

Description

Name of the list. Only one list exists

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"default"

Roles**Show**

sysadmin, manage, monitor

List

sysadmin, manage, monitor

7.16 cmm sctpAssociationInfo

M3UA/SCTP Association Information. Syntax, description, parameters, and roles for cmm sctpAssociationInfo command.

Syntax**Description**

Displays the status of the SCTP associations that are used for M3UA signaling. Shows all the SCTP associations used by M3UA along with their respective status.

Parameters

N/A

Roles**List**

sysadmin, manage, monitor

7.17 cmm sctpProfile

SCTP Profile. Syntax, description, parameters, and roles for cmm sctpProfile command.

Syntax

Description

This command is used to administer an SCTP Profile and includes parameters for defining the maximum number of retransmissions, retransmission timeouts, SCTP acknowledgement frequency and period, and heartbeat interval.

Parameters

Name

name - SCTP profile name

Syntax

--name <string>

Description

SCTP profile name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

assoMaxRetrans - Max association retrans

Syntax

--assoMaxRetrans <integer>

Description

Maximum association retransmissions.

Class

Optional.

Type / Range

Integer in the range of 1 to 50

Default

10

Name

cookieLife - Cookie life

Syntax

--cookieLife <integer>

Description

Valid cookie life time frame in units of seconds.

Class

Optional.

Type / Range

Integer in the range of 5 to 120 in units of Second

Default

60

Name

description - Description

Syntax

--description <string>

Description

Profile description.

Class

Optional.

Type / Range

Digit string of length 0 to 160 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

hbInterval - Heartbeat interval

Syntax

--hbInterval <integer>

Description

Heartbeat interval in units of seconds.

Class

Optional.

Type / Range

Integer in the range of 1 to 300 in units of Second

Default

30

Name

maxInitRetrans - Max init retrans

Syntax

--maxInitRetrans <integer>

Description

Maximum initial retransmissions.

Class

Optional.

Type / Range

Integer in the range of 1 to 25

Default

8

Name

mtuSize - MTU Size

Syntax

--mtuSize <integer>

Description

Maximum transmission unit size in bytes.

Class

Optional.

Type / Range

Integer in the range of 512 to 9000 in units of Byte

Default

1500

Name

pathMaxRetrans - Max path retrans

Syntax

--pathMaxRetrans <integer>

Description

Maximum path retransmissions.

Class

Optional.

Type / Range

Integer in the range of 1 to 25

Default

5

Name

rtoInitValid - RTO initial value

Syntax

--rtoInitValid <integer>

Description

Initial retransmission timeout in units of ms.

Class

Optional.

Type / Range

Integer in the range of 10 to 10,000 in units of Millisecond

Default

300

Name

rtoMax - RTO maximum

Syntax

--rtoMax <integer>

Description

Maximum retransmission timeout in units of ms.

Class

Optional.

Type / Range

Integer in the range of 20 to 10,000 in units of Millisecond

Default

400

Name

rtoMin - RTO minimum

Syntax

--rtoMin <integer>

Description

Minimum retransmission timeout in units of ms.

Class

Optional.

Type / Range

Integer in the range of 10 to 5,000 in units of Millisecond

Default

50

Name

sackFreq - SACK frequency

Syntax

--sackFreq <integer>

Description

SCTP acknowledgment frequency.

Class

Optional.

Type / Range

Integer in the range of 1 to 5

Default

2

Name

sackPeriod - SACK period

Syntax

--sackPeriod <integer>

Description

SCTP acknowledgment period in units of ms.

Class

Optional.

Type / Range

Integer in the range of 0 to 500 in units of Millisecond

Default

200

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.18 cmm secureCopySetup

Secure Copy Setup. Syntax, description, parameters, and roles for cmm secureCopySetup command.

Syntax

Description

Create, list, modify, delete and show secure-copy parameters.

Parameters

Name

secureId - Secure Copy Name

Syntax

--secureId <list-item>

Description

Secure Copy Name

Class

Required key.

List

- remote-config - Specify remote copy of remote-config.

Name

privateKeyFile - Private key file name

Syntax

--privateKeyFile <string>

Description

Absolute path of the private key file name. Example: /tmp/privkey.pem

Class

Required on create.

Type / Range

Digit string of length 1 to 100 with the allowed pattern of [0-9a-zA-Z\._&(),/]*

Name

signedCertFile - Signed certificate file name

Syntax

--signedCertFile <string>

Description

Absolute path of the signed certificate file name. Example: /tmp/certificate

Class

Required on create.

Type / Range

Digit string of length 1 to 100 with the allowed pattern of [0-9a-zA-Z\._&(),/]*

Name

remoteDir - Remote directory

Syntax

--remoteDir <string>

Description

Remote-copy directory.

Class

Optional.

Type / Range

Digit string of length 1 to 512 with the allowed pattern of ^/(/?[\w\-\~\.\,])+/?\$

Name

remotelp - Remote IP Address

Syntax

--remoteIp <string>

Description

Remote IP-address.

Class

Optional.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

remoteLogin - Remote login name

Syntax

--remoteLogin <string>

Description

Remote login name.

Class

Optional.

Type / Range

User name in Linux format. Username must start with a letter, a digit, an underscore or a full stop. The last character must be a letter, a digit, an underscore, a hyphen, a dollar sign or a full stop. The other characters must be letters, digits, underscores, hyphens or full stops. The length must be between 1 and 32 characters.

Name

secondaryRemotelp - Secondary Remote IP Address

Syntax

--secondaryRemoteIp <string>

Description

Secondary Remote IP Address. If backup to the first remotelP fails, this IP will be attempted. If not populated there will be no secondary attempt.

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Roles**Create**

secadmin

Show

sysadmin, manage, secadmin

Modify

secadmin

Delete

secadmin

List

sysadmin, manage, secadmin

7.19 cmm securityParam

securityParam. Syntax, description, parameters, and roles for **cmm securityParam** command.

Syntax**Description**

Configure and show security parameters.

Parameters**Name**

attachRestriction - Attach restriction

Syntax

--attachRestriction <list-item>

Description

This parameter defines the non ciphering subscriber attach restriction in flexi NS SGSN in the 2G access, true ... attach restriction is in use false ... attach restriction is not in use.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

authenticationMode - Authentication mode

Syntax

```
--authenticationMode <list-item>
```

Description

This parameter defines the authentication mode, act ... Authentication is in use deact ... Authentication is not in use whenNeeded ... Authentication when needed Note, when act is in use, SGSN follows repetition rates configured under plmnParam like iuPeriodicRaAuth etc

Class

Optional, not filterable on list.

List

- act - Active - Active.
- deact - Deactive - Deactive.
- whenNeeded - When needed - When needed.

Default

whenNeeded

Name

cipheringMode - Ciphering mode

Syntax

```
--cipheringMode <list-item>
```

Description

This parameter defines the ciphering mode in flexi NS SGSN in 2G access mode. the parameter deals with non-standard A5 algorithms, true ... ciphering mode is in use false ... ciphering mode is not in use.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

imeiCheckMode - IMEI check mode

Syntax

--imeiCheckMode <list-item>

Description

This parameter defines the IMEI check mode, true ... IMEI check is in use false ... IMEI check is not in use.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

imeiRequestMode - IMEI request mode

Syntax

--imeiRequestMode <list-item>

Description

This parameter defines the IMEI request mode, true ... IMEI request optimization mode is in use. when IMEI is already known, SGSN does not request the IMEI from ME false ... IMEI request optimization mode is not in use. SGSN requests the IMEI from ME, even if IMEI is already known.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ptmsiSignatureMode - PTMSI signature mode

Syntax

--ptmsiSignatureMode <list-item>

Description

This parameter defines the packet temporary mobile subscriber identity (PTMSI) signature reallocation mode, true ... PTMSI signature is in use false ... PTMSI signature is not in use.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

umtsAttachRestriction - UMTS attach restriction

Syntax

--umtsAttachRestriction <list-item>

Description

This parameter defines the usage of non ciphering subscriber UMTS attach restriction, true ... UMTS attach restriction is in use false ... UMTS attach restriction is not in use.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

umtsCipheringMode - UMTS ciphering mode

Syntax

--umtsCipheringMode <list-item>

Description

This parameter defines the UMTS ciphering mode in flexi NS SGSN. the parameter deals with non-standard A5 algorithms, true ... UMTS ciphering mode is in use false ... UMTS ciphering mode is not in use.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

id - SecurityParams id

Syntax

config

Description

SecurityParams ID. just type 'config' here.

Class

Required value.

List

- config - Specify security parameters identifier of config.

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

7.20 cmm serviceAreaRestrictionItem

Service Area Restriction Item. Syntax, description, parameters, and roles for cmm serviceAreaRestrictionItem command.

Syntax

Description

Service Area Restriction Item

Parameters

Name

serviceAreaRestrictionListName - Service Area Restriction List Name

Syntax

--serviceAreaRestrictionListName <string>

Description

Service Area Restriction List Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

minTac - Min Tracking Area Code

Syntax

--minTac <integer>

Description

Minimum TAC value in defining set of TACs to update, used with maxtac. TAC range can only be up to 16 and the max number of serviceAreaRestrictionItem is 2.

Class

Required key.

Type / Range

Integer in the range of 0 to 16777215

Name

maxTac - Max Tracking Area Code

Syntax

--maxTac <integer>

Description

Maximum TAC value in defining set of TACs to update, used with mintac. TAC range can only be up to 16 and the max number of serviceAreaRestrictionItem is 2.

Class

Required key.

Type / Range

Integer in the range of 0 to 16777215

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.21 cmm serviceAreaRestrictionList

Service Area Restriction List. Syntax, description, parameters, and roles for cmm serviceAreaRestrictionList command.

Syntax

Description

Service Area Restriction List

Parameters

Name

name - Service Area Restriction List Name

Syntax

--name <string>

Description

Service Area Restriction List Name. Maximum of 1024 list entries can be created.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.22 cmm servicelp

Service IP. Syntax, description, parameters, and roles for cmm serviceIp command.

Syntax

Description

This command is used to administer a Service IP and includes parameters for defining the pool type, ID, and member ID, service IP address, subnet ID and location, network interface type, and assignment type. For Multiple IPDS, the Service IPs have to be defined on all non 0 IPDS pools for the following interfaces: - S1MME* - Meaning any S1MME Interface 1-9 and Multihoming - M3 - M3INTF1 - M3INTF2 - S11 - S11U For Multiple IPDS the remaining IPDS interfaces can only be defined on IPDS pool 0.

Parameters

Name

poolType - Pool type

Syntax

--poolType <list-item>

Description

Pool (service) type.

Class

Required key.

List

- ALMS - Alarm Service - Alarm Service.
- CPPS - Control plane processing service - Control plane processing service.
- CTCS - Call Trace Collection service - Call Trace Collection service.
- DBS - Database service - Database service.
- EEMS - Event Exposure Management service - Event Exposure Management service.
- IPDS - IP director service - IP director service.

- IPPS - IP packet processing service - IP packet processing service.
- LIHS - Lawful Interception Handling service
- MLIS - Machine Learning Intelligence Service
- NECC - Network element cluster controller - Network element cluster controller.
- NWDS - Network Data Service - NWDAF service
- PAPS - Packet processing service - Packet processing service.
- PCMS - Per Call Measurements Data Service - Per Call Measurements Data (PCMD) Service
- RCIS - Radio Capability Information storage Service
- SFDS - SCTP flow distributor service
- UCDS - User equipment Context Data-dump Service
- UFDS - UDP flow distributor service

Name

poolId - Pool id

Syntax

--poolId <integer>

Description

Pool identification number.

Class

Required key.

Type / Range

Integer in the range of 0 to 63

Name

poolMemberId - Pool member ID

Syntax

--poolMemberId <integer>

Description

Pool member identification number.

Class

Required key.

Type / Range

Integer in the range of 0 to 10

Name

niType - Network interface type

Syntax

--niType <list-item>

Description

Network interface type.

Class

Required key.

List

- AIML_DATA_STREAM - AI ML Data Stream - Specify network interface of AIML_DATA_STREAM interface.
- ALMS_ACCESS - Specify the network interface of the ALMS service.
- AMF - Specify the network interface of AMF on IPDS
- ANALYTICSSRV - Specify network interface ANALYTICSSRV interface on the NECC.
- BFDSVC - Specify the network interface of BFDSVC for the L3NS on IPDS
- BFDTRANS - BFSTRANS - Specify network interface of bidirectional forwarding transmission interface.
- BP - Specify network interface of Bp interface for CDR access by the billing domain.
- CTSTREAM - Specify network interface CTSTREAM on CTCS.
- DEFAULT - Specify network interface of default.
- DNS - Specify network interface of DNS interface to the domain name server.
- GA - Specify network interface of Ga interface to a charging gateway.
- GBAUTO - Specify network interface of gb auto interface to a BSC.
- GBNS - Specify network interface of gb NS interface to a BSC.
- GN - Specify network interface of Gn interface to an SGSN.
- GN_2G_DATA - Specify network interface of Gn interface for 2G data.
- GN_3G_DATA - Specify network interface of Gn interface for 3G data.
- GN_C - Specify network interface of Gn control interface to a GGSN.
- INFRA_ACCESS - Specify network interface of infra access interface.
- IU_DATA - Specify network interface of iu interface for data to an RNC.
- M3 - Specify network interface of M3 interface to an eNodeB.
- M3INTF1 - Specify network interface of M3 multi-homed interface 1 to an eNodeB.
- M3INTF2 - Specify network interface of M3 multi-homed interface 2 to an eNodeB.
- M3UA1MHA - Specify network interface of 1st M3UA multi-homed interface a.

- M3UA1MHB - Specify network interface of 1st M3UA multi-homed interface b.
- M3UA1SH - Specify network interface of 1st M3UA single homed interface.
- M3UA2MHA - Specify network interface of 2nd M3UA multi-homed interface a.
- M3UA2MHB - Specify network interface of 2nd M3UA multi-homed interface b.
- M3UA2SH - Specify network interface of 2nd M3UA single homed interface.
- M3UA3MHA - Specify network interface of 3rd M3UA multi-homed interface a.
- M3UA3MHB - Specify network interface of 3rd M3UA multi-homed interface b.
- M3UA3SH - Specify network interface of 3rd M3UA single homed interface.
- M3UA4MHA - Specify network interface of 4th M3UA multi-homed interface a.
- M3UA4MHB - Specify network interface of 4th M3UA multi-homed interface b.
- M3UA4SH - Specify network interface of 4th M3UA single homed interface.
- M3UA5MHA - Specify network interface of 5th M3UA multi-homed interface a.
- M3UA5MHB - Specify network interface of 5th M3UA multi-homed interface b.
- M3UA5SH - Specify network interface of 5th M3UA single homed interface.
- M3UA6MHA - Specify network interface of 6th M3UA multi-homed interface a.
- M3UA6MHB - Specify network interface of 6th M3UA multi-homed interface b.
- M3UA6SH - Specify network interface of 6th M3UA single homed interface.
- M3UA7MHA - Specify network interface of 7th M3UA multi-homed interface a.
- M3UA7MHB - Specify network interface of 7th M3UA multi-homed interface a.
- M3UA7SH - Specify network interface of 7th M3UA single homed interface.
- M3UA8MHA - Specify network interface of 8th M3UA multi-homed interface a.
- M3UA8MHB - Specify network interface of 8th M3UA multi-homed interface b.
- M3UA8SH - Specify network interface of 8th M3UA single homed interface.
- M3UA9MHA - Specify network interface of 9th M3UA multi-homed interface a.
- M3UA9MHB - Specify network interface of 9th M3UA multi-homed interface b.
- M3UA9SH - Specify network interface of 9th M3UA single homed interface.
- N26 - Specify network interface of N26 between the MME and 5GS AMF.
- NETWORK_TRACE - Specify the network interface of the NETWORK TRACE (tapping) interface on the IPDS.
- OAM_ACCESS - OAM access - Specify network interface of OAM access interface.
- ONAP_DCAE - Specify network interface of ONAP DCAE interface on the NECC.

- PROMETHEUS - Specify network interface PROMETHEUS interface on the NECC.
- RS10 - Specify network interface of RS10 interface to an SRS.
- S10 - Specify network interface of S10 interface to another MME.
- S101 - Specify network interface of S101 interface to an IWF.
- S102 - Specify network interface of S102 interface to an IWF.
- S11 - Specify network interface of S10 interface to another MME.
- S11U - Specify network interface of S11u interface to a SGW.
- S13 - Specify network interface of S13 interface to an EIR.
- S13INTF1 - Specify network interface of S13 multi-homed interface 1 to an EIR.
- S13INTF2 - Specify network interface of S13 multi-homed interface 2 to an EIR.
- S16 - Specify network interface of S16 interface loop around to this SGSN.
- S1MME - Specify network interface of S1-MME single homed interface to an eNodeB.
- S1MME2 - Specify network interface of 2nd S1-MME single homed interface to an eNodeB.
- S1MME2INTF1 - Specify network interface of 2nd S1-MME multi-homed interface 1 to an eNodeB.
- S1MME2INTF2 - Specify network interface of 2nd S1-MME multi-homed interface 2 to an eNodeB.
- S1MME3 - Specify network interface of 3rd S1-MME single homed interface to an eNodeB.
- S1MME3INTF1 - Specify network interface of 3rd S1-MME multi-homed interface 1 to an eNodeB.
- S1MME3INTF2 - Specify network interface of 3rd S1-MME multi-homed interface 2 to an eNodeB.
- S1MME4 - Specify network interface of 4th S1-MME single homed interface to an eNodeB.
- S1MME4INTF1 - Specify network interface of 4th S1-MME multi-homed interface 1 to an eNodeB.
- S1MME4INTF2 - Specify network interface of 4th S1-MME multi-homed interface 2 to an eNodeB.
- S1MME5 - Specify network interface of 5th S1-MME single homed interface to an eNodeB.
- S1MME5INTF1 - Specify network interface of 5th S1-MME multi-homed interface 1 to an eNodeB.

- S1MME5INTF2 - Specify network interface of 5th S1-MME multi-homed interface 2 to an eNodeB.
- S1MME6 - Specify network interface of 6th S1-MME single homed interface to an eNodeB.
- S1MME6INTF1 - Specify network interface of 6th S1-MME multi-homed interface 1 to an eNodeB.
- S1MME6INTF2 - Specify network interface of 6th S1-MME multi-homed interface 2 to an eNodeB.
- S1MME7 - Specify network interface of 7th S1-MME single homed interface to an eNodeB.
- S1MME7INTF1 - Specify network interface of 7th S1-MME multi-homed interface 1 to an eNodeB.
- S1MME7INTF2 - Specify network interface of 7th S1-MME multi-homed interface 2 to an eNodeB.
- S1MME8 - Specify network interface of 8th S1-MME single homed interface to an eNodeB.
- S1MME8INTF1 - Specify network interface of 8th S1-MME multi-homed interface 1 to an eNodeB.
- S1MME8INTF2 - Specify network interface of 8th S1-MME multi-homed interface 2 to an eNodeB.
- S1MME9 - Specify network interface of 9th S1-MME single homed interface to an eNodeB.
- S1MME9INTF1 - Specify network interface of 9th S1-MME multi-homed interface 1 to an eNodeB.
- S1MME9INTF2 - Specify network interface of 9th S1-MME multi-homed interface 2 to an eNodeB.
- S1MMEINTF1 - Specify network interface of S1-MME multi-homed interface 1 to an eNodeB.
- S1MMEINTF2 - Specify network interface of S1-MME multi-homed interface 2 to an eNodeB.
- S3 - Specify network interface of S3 interface from the MME to an SGSN.
- S3SGSN - Specify network interface of S3 interface from the SGSN to an SGSN.
- S4 - Specify network interface of S4 interface to an SGW.
- S4_2G_DATA - Specify network interface of S4 interface for 2G data to an SGW.
- S4_3G_DATA - Specify network interface of S4 interface for 3G data to an SGW.
- S6A - Specify network interface of S6a interface to an HSS.

- S6AINTF1 - Specify network interface of S6a multi-homed interface 1 to an HSS.
- S6AINTF2 - Specify network interface of S6a multi-homed interface 2 to an HSS.
- SBC - Specify network interface of SBc interface to a CBC.
- SBCINTF1 - Specify network interface of SBc multi-homed interface 1 to a CBC.
- SBCINTF2 - Specify network interface of SBc multi-homed interface 2 to a CBC.
- SERVICE_MIRROR - Specify network interface of service mirror interface.
- SGD - Specify network interface of SGd single homed interface to an SMSC/SMS router.
- SGDINTF1 - Specify network interface of SGd multi-homed interface 1 to an SMSC/SMS router.
- SGDINTF2 - Specify network interface of SGd multi-homed interface 2 to an SMSC/SMS router.
- SGS - Specify network interface of SGs interface to a GSM/UMTS MSC.
- SGSINTF1 - Specify network interface of SGs multi-homed interface 1 to a GSM/UMTS MSC.
- SGSINTF2 - Specify network interface of SGs multi-homed interface 2 to a GSM/UMTS MSC.
- SGSLITE - Specify network interface of SGs lite interface to an IWF.
- SLG - Specify network interface of Slg interface to a GMLC.
- SLGINTF1 - Specify network interface of Slg multi-homed interface 1 to a GMLC.
- SLGINTF2 - Specify network interface of Slg multi-homed interface 2 to a GMLC.
- SLS - Specify network interface of Sls interface to an E-SMLC.
- SLSINTF1 - Specify network interface of Sls multi-homed interface 1 to an E-SMLC.
- SLSINTF2 - Specify network interface of Sls multi-homed interface 1 to an E-SMLC.
- SM - Specify network interface of sm interface to an MBMS gateway.
- SV - Specify network interface of Sv interface to a GSM/UMTS MSC.
- T6A - Specify network interface of T6a single homed interface to a SCEF.
- T6AINTF1 - Specify network interface of T6a multi-homed interface 1 to a SCEF.
- T6AINTF2 - Specify network interface of T6a multi-homed interface 2 to a SCEF.
- UNASSIGNED - Specify network interface of unassigned.
- VIA_TRANS - Via_Transport - This NI type is for the Via transport network between the IPDSs and VSRs for Multiple IPDSs.
- X1_1 - Specify network interface of X1_1 interface to an ADMF.
- X1_MME - Specify network interface of X1 interface for MME.

- X1_SGSN - Specify network interface of X1 interface for SGSN.
- X2 - Specify network interface of X2 interface to a DF2.
- X2_MME - Specify network interface of X2 interface for MME.
- X2_SGSN - Specify network interface of X2 interface for SGSN.
- X3_2G - Specify network interface of X3 interface for 2G.
- X3_3G - Specify network interface of X3 interface for 3G.
- X3_IPDS - Specify network interface of X3 interface on the IPDS.
- SBI_NR10 - Specify network interface of SBI_NR10 interface on the IPDS.
- SBI_NR10_2 - Specify network interface of SBI_NR10_2 interface on the IPDS.
- SBI_NR10_3 - Specify network interface of SBI_NR10_3 interface on the IPDS.
- SBI_NR10_4 - Specify network interface of SBI_NR10_4 interface on the IPDS.
- SBI_NR10_5 - Specify network interface of SBI_NR10_5 interface on the IPDS.
- SBI_NR10_6 - Specify network interface of SBI_NR10_6 interface on the IPDS.
- SBI_NR10_7 - Specify network interface of SBI_NR10_7 interface on the IPDS.
- SBI_NR10_8 - Specify network interface of SBI_NR10_8 interface on the IPDS.
- SBI_NR10_9 - Specify network interface of SBI_NR10_9 interface on the IPDS.
- SBI_NR10_10 - Specify network interface of SBI_NR10_10 interface on the IPDS.
- SBI_NR10_11 - Specify network interface of SBI_NR10_11 interface on the IPDS.
- SBI_NR10_12 - Specify network interface of SBI_NR10_12 interface on the IPDS.
- SBI_NR10_13 - Specify network interface of SBI_NR10_13 interface on the IPDS.
- SBI_NR10_14 - Specify network interface of SBI_NR10_14 interface on the IPDS.
- SBI_NR10_15 - Specify network interface of SBI_NR10_15 interface on the IPDS.
- SBI_NR10_16 - Specify network interface of SBI_NR10_16 interface on the IPDS.

Name

svclp - Service IP address

Syntax

--svcIp <string>

Description

Service IP address.

Class

Required key.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

assignmentType - Assignment type

Syntax

--assignmentType <list-item>

Description

Assignment type: fixed or floating.

Class

Optional.

List

- fixed - Fixed.
- floating - Floating.

Default

floating

Name

bfdGatewayLeft - BFD gateway left IP address

Syntax

--bfdGatewayLeft <string>

Description

For a service IP using a BFD service subnet, the gateway of the associated BFD left transport subnet.

Class

Read only.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

bfdGatewayRight - BFD gateway right IP address

Syntax

--bfdGatewayRight <string>

Description

For a service IP using a BFD service subnet, the gateway of the associated BFD right transport subnet.

Class

Read only.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

bfdReachableLeft - BFD reachable via left IP address

Syntax

--bfdReachableLeft <string>

Description

For a service IP using a BFD service subnet, the associated BFD left transport subnet base.

Class

Read only.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

bfdReachableRight - BFD reachable via right IP address

Syntax

--bfdReachableRight <string>

Description

For a service IP using a BFD service subnet, the associated BFD right transport subnet base.

Class

Read only.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

subnetId - Subnet ID

Syntax

--subnetId <integer>

Description

Subnet identification.

Class

Read only.

Type / Range

Integer in the range of 0 to 63

Name

subnetLoc - Subnet location

Syntax

--subnetLoc <list-item>

Description

Subnet location: internal or external.

Class

Optional.

List

- external - External subnet - External subnet.
- internal - Internal subnet - Internal subnet.

Default

external

Name

useFds - Use Flow Distributor

Syntax

--useFds <list-item>

Description

Indicates whether this service IP shall be used as back-end for single-ip flow distributor

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

viaNexthop - VIA Nexthop Left IP address

Syntax

--viaNexthop <string>

Description

For a service IP using a VIA service subnet, the Nexthop of the associated VIA left transport subnet.

Class

Read only.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

viaReachable - VIA reachable by left IP address

Syntax

--viaReachable <string>

Description

For a service IP using a VIA service subnet, the associated VIA left transport subnet base.

Class

Read only.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.23 cmm serviceMirror

Port mirroring capability. Syntax, description, parameters, and roles for cmm serviceMirror command.

Syntax

Description

Port mirroring allowing external traffic to be replicated in real time. Service Mirror is only supported on VNF. Service Mirror is only supported on single IPDS configurations. Service Mirror must be deactivated before executing an SU. Service Mirror will be deactivated when greater than 100,000 packets per second are received by the IPDS. Service Mirror will be deactivated if a single packet from any interface is dropped.

Parameters

Name

hostName - Internal host name

Syntax

--hostName <string>

Description

Customer defined VM identifier.

Class

Required key.

Type / Range

Digit string of length 1 to 255

Name

localMirrorIp - Local mirror IP address

Syntax

--localMirrorIp <string>

Description

Local service mirror IPv4 address.

Class

Required on create, not modifiable.

Type / Range

Digit string of length 0 or 7 to 15 with the allowed pattern of [0-9\.]*

Name

remoteGreTunnelIp - Remote tunnel IP address

Syntax

--remoteGreTunnelIp <string>

Description

Remote GRE tunnel IP address.

Class

Required on create.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

remoteMirrorIp - Remote service mirror IP address

Syntax

--remoteMirrorIp <string>

Description

Remote service mirror IPv4 address.

Class

Required on create, not modifiable.

Type / Range

Digit string of length 0 or 7 to 15 with the allowed pattern of [0-9\.]*

Name

adminState - Admin state

Syntax

--adminState <list-item>

Description

Admin state of the services.

Class

Optional.

List

- locked - Locked.
- unlocked - Unlocked.

Default

unlocked

Name

destinationAddress - Destination address

Syntax

--destinationAddress <string>

Description

Destination IP mask.

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

destinationPorts - Destination ports

Syntax

--destinationPorts <string>

Description

Destination ports.

Class

Optional.

Type / Range

Digit string of length 0 to 11 with the allowed pattern of ((([0-9]{1,4}|[1-5][0-9]{4}|6[0-4][0-9]{3}|65[0-4][0-9]{2}|655[0-2][0-9]|6553[0-5]))|([0-9]{1,4}|[1-5][0-9]{4}|6[0-4][0-9]{3}|

65[0-4][0-9]{2}|655[0-2][0-9]|6553[0-5])-([0-9]{1,4}|[1-5][0-9]{4}|6[0-4][0-9]{3}|65[0-4][0-9]{2}|655[0-2][0-9]|6553[0-5]))))

Name

externalInterface - External interface

Syntax

--externalInterface <string>

Description

External interface - for example eth2 or eth2.<vlan_id>.

Class

Optional.

Type / Range

Digit string of length 0 to 20 with the allowed pattern of ((\w+|\w+\.\d{1,5})|(\w+|\w+\.\d{1,5}),(\w+|\w+\.\d{1,5})))\$

Name

ipv - IP version

Syntax

--ipv <list-item>

Description

Internet protocol version number.

Class

Optional.

List

- " - Blank
- 4 - 4
- 6 - 6

Name

localMirrorIp6 - Local mirror IP6 address

Syntax

--localMirrorIp6 <string>

Description

Local service mirror IPv6 address.

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of '[0-9a-f:]*'

Name

localMirrorIp6PrefixLength - Local service mirror IPv6 prefix length

Syntax

--localMirrorIp6PrefixLength <integer>

Description

Prefix length for local service mirror IPv6.

Class

Optional.

Type / Range

Integer in the range of 0 to 128

Default

128

Name

localMirrorIpPrefixLength - Local service mirror IP prefix length

Syntax

--localMirrorIpPrefixLength <integer>

Description

Prefix length for local service mirror IP.

Class

Optional.

Type / Range

Integer in the range of 0 to 32

Default

32

Name

packetDirection - Packet direction

Syntax

--packetDirection <list-item>

Description

Packet direction - ingress/egress.

Class

Optional.

List

- " - Empty - Packet direction not specified.
- egress - Specify packet direction of egress.

- ingress - Specify packet direction of ingress.

Name

protocol - Protocol

Syntax

--protocol <list-item>

Description

Protocol to monitor.

Class

Optional.

List

- " - Empty - Do not specify a service protocol.
- icmp - Specify service protocol of ICMP.
- icmpv6 - Specify service protocol of ICMPv6.
- sctp - Specify service protocol of SCTP.
- tcp - Specify service protocol of TCP.
- udp - Specify service protocol of UDP.

Name

remoteMirrorIp6 - Remote service mirror IPv6 address

Syntax

--remoteMirrorIp6 <string>

Description

Remote service mirror IPv6 address.

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of '[0-9a-f:]'

Name

sourceAddress - Source address

Syntax

--sourceAddress <string>

Description

Source address IP mask.

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

sourcePorts - Source ports

Syntax

--sourcePorts <string>

Description

Source port(s) (0-65535): <port|port1-port2>

Class

Optional.

Type / Range

Digit string of length 0 to 11 with the allowed pattern of ((([0-9]{1,4}|[1-5][0-9]{4}|6[0-4][0-9]{3}|65[0-4][0-9]{2}|655[0-2][0-9]|6553[0-5])|((([0-9]{1,4}|[1-5][0-9]{4}|6[0-4][0-9]{3}|65[0-4][0-9]{2}|655[0-2][0-9]|6553[0-5])-[0-9]{1,4}|[1-5][0-9]{4}|6[0-4][0-9]{3}|65[0-4][0-9]{2}|655[0-2][0-9]|6553[0-5])))

Name

timeLimit - Time Limit

Syntax

--timeLimit <integer>

Description

A mirror should be able to run indefinitely under normal operation. This parameter is the time limit in hours. Value 0 means unlimited

Class

Optional.

Type / Range

Integer in the range of 0 to 24 in units of Hour

Default

24

Name

timeStamp - Time Stamp

Syntax

--timeStamp <string>

Description

The time when this record was last created or modified.

Class

Read only.

Type / Range

Digit string of length 0 to 26

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.24 cmm serviceMirrorStatus

Port mirroring capability status. Syntax, description, parameters, and roles for cmm serviceMirrorStatus command.

Syntax**Description**

Status of port mirroring allowing external traffic to be replicated in real time. Service Mirror is only supported on VNF. Service Mirror is only supported on single IPDS configurations. Service Mirror must be deactivated before executing an SU. Service Mirror will be deactivated when greater than 100,000 packets per second are received by the IPDS. Service Mirror will be deactivated if a single packet from any interface is dropped.

Parameters**Name**

errorState - Error State

Syntax

--errorState <integer>

Description

Error State

Class

Optional.

Type / Range

Integer in the range of 0 to 499

Name

errorString - Error String

Syntax

--errorString <string>

Description

Error String

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 255

Name

hostName - Internal Host Name

Syntax

--hostName <string>

Description

Customer defined VM identifier

Class

Optional.

Type / Range

Digit string of length 1 to 255

Name

operState - Operation State

Syntax

--operState <integer>

Description

Operation State

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Roles**List**

sysadmin, manage, monitor

7.25 cmm serviceState

Service state. Syntax, description, parameters, and roles for cmm serviceState command.

Syntax

Description

This command is used to administer a service state and includes parameters for defining the pool (service) type, pool ID, and service state. Modification of the service state is allowed on CPPS, IPPS, DBS, PAPS and IPDS (for AMF only). NECC, ALMS, CTCS are Not Applicable.

Parameters

Name

poolType - Pool type

Syntax

--poolType <list-item>

Description

Pool (service) type. Note: Although the syntax allows all poolType values, some might not be modifiable or displayed. See command Description.

Class

Required key.

List

- ALMS - Alarm Service - Alarm Service.
- CPPS - Control plane processing service - Control plane processing service.
- CTCS - Call Trace Collection service - Call Trace Collection service.
- DBS - Database service - Database service.
- EEMS - Event Exposure Management service - Event Exposure Management service.
- IPDS - IP director service - IP director service.
- IPPS - IP packet processing service - IP packet processing service.
- LIHS - Lawful Interception Handling service
- MLIS - Machine Learning Intelligence Service
- NECC - Network element cluster controller - Network element cluster controller.
- NWDS - Network Data Service - NWDAF service

- PAPS - Packet processing service - Packet processing service.
- PCMS - Per Call Measurements Data Service - Per Call Measurements Data (PCMD) Service
- RCIS - Radio Capability Information storage Service
- SFDS - SCTP flow distributor service
- UCDS - User equipment Context Data-dump Service
- UFDS - UDP flow distributor service

Name

poolId - Pool id

Syntax

--poolId <integer>

Description

Pool identification number.

Class

Required key.

Type / Range

Integer in the range of 0 to 63

Name

force - Force

Syntax

--force <list-item>

Description

Bypass crosschecks when locking last service for a pool type.

Restrictions

This field is write only field. Will be set to false when read.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

personality - personality

Syntax

--personality <list-item>

Description

This field indicates the personality of a service. CPPS can have AMMS or EMMS personalities. EMMS indicates the CPPS serves 4G traffic. AMMS indicates the CPPS serves 5G traffic. MLIS can have MLGS (General) or MLUS (UE Group Specific) personalities.

Class

Read only.

List

- - - None Personality
- EMMS - EMMS Personality
- AMMS - AMMS Personality
- MLGS - MLGS Personality
- MLUS - MLUS Personality

Default

-

Name

svcState - Service state

Syntax

--svcState <list-item>

Description

Service state of the pool.

Class

Optional.

List

- blocked - Specify service state of blocked.
- locked - Specify service state of locked.
- shuttingDown - Shutting down - Specify service state of shutting down.
- unknown - Specify service state of unknown.
- unlocked - Specify service state of unlocked.

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

7.26 cmm serviceStatus

Service Status. Syntax, description, parameters, and roles for cmm serviceStatus command.

Syntax**Description**

CMM Service Status

Parameters**Name**

poolType - Pool type

Syntax

--poolType <list-item>

Description

Pool (service) type.

Class

Required key.

List

- ALMS - Alarm Service - Alarm Service.
- CPPS - Control plane processing service - Control plane processing service.
- CTCS - Call Trace Collection service - Call Trace Collection service.
- DBS - Database service - Database service.
- EEMS - Event Exposure Management service - Event Exposure Management service.
- IPDS - IP director service - IP director service.
- IPPS - IP packet processing service - IP packet processing service.
- LIHS - Lawful Interception Handling service
- MLIS - Machine Learning Intelligence Service
- NECC - Network element cluster controller - Network element cluster controller.
- NWDS - Network Data Service - NWDAF service

- PAPS - Packet processing service - Packet processing service.
- PCMS - Per Call Measurements Data Service - Per Call Measurements Data (PCMD) Service
- RCIS - Radio Capability Information storage Service
- SFDS - SCTP flow distributor service
- UCDS - User equipment Context Data-dump Service
- UFDS - UDP flow distributor service

Name

poolId - PoolId

Syntax

--poolId <integer>

Description

Pool id of the service.

Class

Required key.

Type / Range

Integer in the range of 0 to 63

Name

poolMemberId - PoolMember

Syntax

--poolMemberId <integer>

Description

Pool member of the service.

Class

Required key.

Type / Range

Integer in the range of 0 to 10

Name

hostName - Host Name

Syntax

--hostName <string>

Description

Name of host

Class

Optional.

Type / Range

Digit string of length 1 to 63 with the allowed pattern of [0-9a-zA-Z\._:-]*

Name

hostNameLabel - Host name label

Syntax

--hostNameLabel <string>

Description

Host name label.

Class

Optional.

Type / Range

Digit string of length 1 to 128 with the allowed pattern of [0-9a-zA-Z\._:-]*

Name

personality - personality

Syntax

--personality <list-item>

Description

This field indicates the personality of a service. CPPS can have AMMS or EMMS personalities. EMMS indicates the CPPS serves 4G traffic. AMMS indicates the CPPS serves 5G traffic. MLIS can have MLGS (General) or MLUS (UE Group Specific) personalities.

Class

Optional.

List

- - - None Personality
- EMMS - EMMS Personality
- AMMS - AMMS Personality
- MLGS - MLGS Personality
- MLUS - MLUS Personality

Default

-

Name

svclp - Service IpAddress

Syntax

--svcIp <string>

Description

Ip Address of the service.

Class

Optional.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

svcStatus - Service Status

Syntax

--svcStatus <string>

Description

Status of the service.

Class

Optional.

Type / Range

Status of a CMM Service

Name

uuid - UUID

Syntax

--uuid <string>

Description

UUID of the service.

Class

Optional.

Type / Range

Digit string of length 1 to 128 with the allowed pattern of [0-9a-zA-Z\.\-_:]*

Roles**Show**

sysadmin, manage, monitor

List

sysadmin, manage, monitor

7.27 cmm sGdDiamRpMappingProfile

SGd Diameter RP Mapping Profile. Syntax, description, parameters, and roles for cmm sGdDiamRpMappingProfile command.

Syntax

Description

This command is used to administer the SGd Diameter causes to RP causes mapping. The profile is used when an error occurs on native SMS delivery and the SMSC/SMS router reports an error on the delivery towards the MME. The MME uses this mapping profile to provide a corresponding SMS Relay Protocol error further towards the UE. The maximum number of profiles supported is 128.

Parameters

Name

name - SGd Diameter to RP cause code mapping profile name

Syntax

--name <string>

Description

Defines all SGd interface Diameter code to SMS Relay Protocol (RP) cause code mapping.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

default

Name

absentUser - Absent user

Syntax

--absentUser <integer>

Description

Mapping for Diameter Absent user to RP error cause code.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

38

Name

appUnsupported - Application unsupported

Syntax

--appUnsupported <integer>

Description

Mapping for Diameter application unsupported to RP error cause code.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

38

Name

authenticationRejected - Authentication rejected

Syntax

--authenticationRejected <integer>

Description

Mapping for Diameter authentication rejected to RP error cause code.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

38

Name

authorizationRejected - Authorization rejected

Syntax

--authorizationRejected <integer>

Description

Mapping for Diameter authorization rejected to RP error cause code.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

38

Name

avpNotAllowed - AVP not allowed

Syntax

--avpNotAllowed <integer>

Description

Mapping for Diameter AVP not allowed to RP error cause code.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

38

Name

avpOccursTooManyTimes - AVP occurs too many times

Syntax

--avpOccursTooManyTimes <integer>

Description

Mapping for Diameter AVP occurs too many times to RP error cause code.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

38

Name

avpUnsupported - AVP unsupported

Syntax

--avpUnsupported <integer>

Description

Mapping for Diameter AVP unsupported to RP error cause code.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

38

Name

commandUnsupported - Command unsupported

Syntax

--commandUnsupported <integer>

Description

Mapping for Diameter command unsupported to RP error cause code.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

38

Name

contradictingAvps - Contradicting AVPs

Syntax

--contradictingAvps <integer>

Description

Mapping for Diameter contradicting AVPs to RP error cause code.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

38

Name

electionLost - Election lost

Syntax

--electionLost <integer>

Description

Mapping for Diameter election lost to RP error cause code.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

38

Name

facilityNotSupported - Facility not supported

Syntax

--facilityNotSupported <integer>

Description

Mapping for Diameter facility not supported to RP error cause code.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

69

Name

invalidAvpBitCombo - Invalid AVP bit combo

Syntax

--invalidAvpBitCombo <integer>

Description

Mapping for Diameter invalid AVP bit combo to RP error cause code.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

38

Name

invalidAvpBits - Invalid AVP bits

Syntax

--invalidAvpBits <integer>

Description

Mapping for Diameter invalid AVP bits to RP error cause code.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

38

Name

invalidAvpLength - Invalid AVP length

Syntax`--invalidAvpLength <integer>`**Description**

Mapping for Diameter invalid AVP length to RP error cause code.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

38

Name

invalidAvpValue - Invalid AVP value

Syntax`--invalidAvpValue <integer>`**Description**

Mapping for Diameter invalid AVP value to RP error cause code.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

38

Name

invalidBitHeader - Invalid bit in header

Syntax`--invalidBitHeader <integer>`**Description**

Mapping for Diameter invalid bit in header to RP error cause code.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

38

Name

invalidHdrBits - Invalid hearer bits

Syntax`--invalidHdrBits <integer>`**Description**

Mapping for Diameter invalid header bits to RP error cause code.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

38

Name

invalidMessageLength - Invalid message length

Syntax`--invalidMessageLength <integer>`**Description**

Mapping for Diameter invalid message length to RP error cause code.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

38

Name

loopDetected - Loop detected

Syntax`--loopDetected <integer>`**Description**

Mapping for Diameter loop detected to RP error cause code.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

38

Name

missingAvp - Missing AVP

Syntax

--missingAvp <integer>

Description

Mapping for Diameter missing AVP to RP error cause code.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

38

Name

multiRoundAuth - Multi round authorization

Syntax

--multiRoundAuth <integer>

Description

Mapping for Diameter multi round authorization to RP error cause code.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

38

Name

noCommonApplication - No common application

Syntax

--noCommonApplication <integer>

Description

Mapping for Diameter no common application to RP error cause code.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

38

Name

noCommonSecurity - No common security

Syntax

--noCommonSecurity <integer>

Description

Mapping for Diameter no common security to RP error cause code.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

38

Name

noProvisionedSmsc - No provisioned SMSC

Syntax

--noProvisionedSmsc <integer>

Description

Mapping for no provisioned SMSC to RP error cause code.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

38

Name

outOfSpace - Out of space

Syntax

--outOfSpace <integer>

Description

Mapping for Diameter out of space to RP error cause code.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

38

Name

realmNotServed - Realm not served

Syntax

--realmNotServed <integer>

Description

Mapping for Diameter realm not served to RP error cause code.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

38

Name

redirectionIndication - Redirection indication

Syntax

--redirectionIndication <integer>

Description

Mapping for Redirection indication to RP error cause code.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

38

Name

resourcesExceeded - Resources exceeded

Syntax

--resourcesExceeded <integer>

Description

Mapping for Diameter resources exceeded to RP error cause code.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

38

Name

serviceBarred - Service barred

Syntax

--serviceBarred <integer>

Description

Mapping for Diameter Service barred to RP error cause code.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

8

Name

serviceNotSubscribed - Service not subscribed

Syntax

--serviceNotSubscribed <integer>

Description

Mapping for Diameter Service not subscribed to RP error cause code.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

50

Name

smDelFailureInvSmeAddr - SM delivery failure invalid SME address

Syntax

--smDelFailureInvSmeAddr <integer>

Description

Mapping for Experimental-Result-Code AVP indicating SGd specific error
'DIAMETER_ERROR_SM_DELIVERY_FAILURE' and SM-Delivery-Failure-Cause indicating
INVALID_SME-ADDRESS to RP error cause value.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

21

Name

smDelFailureNotScUser - SM delivery failure not SC user

Syntax

--smDelFailureNotScUser <integer>

Description

Mapping for Experimental-Result-Code AVP indicating SGd specific error
'DIAMETER_ERROR_SM_DELIVERY_FAILURE' and SM-Delivery-Failure-Cause indicating
USER_NOT_SC-USER to RP error cause value.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

28

Name

smDelFailureScCongestion - SM delivery failure SC congestion

Syntax

--smDelFailureScCongestion <integer>

Description

Mapping for Experimental-Result-Code AVP indicating SGd specific error
'DIAMETER_ERROR_SM_DELIVERY_FAILURE' and SM-Delivery-Failure-Cause indicating SC-
CONGESTION to RP error cause value.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

42

Name

smDelFailureUnknownSC - SM delivery failure unknown SC

Syntax

--smDelFailureUnknownSC <integer>

Description

Mapping for Experimental-Result-Code AVP indicating SGd specific error
'DIAMETER_ERROR_SM_DELIVERY_FAILURE' and SM-Delivery-Failure-Cause indicating
UNKNOWN_SERVICE_CENTRE to RP error cause value.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

1

Name

smscDown - SMSC down

Syntax

--smscDown <integer>

Description

Mapping for SMSC down to RP error cause code.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

38

Name

unableToComply - Unable to comply

Syntax

--unableToComply <integer>

Description

Mapping for Diameter Unable to comply to RP error cause code.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

38

Name

unableToDeliver - Unable to deliver

Syntax`--unableToDeliver <integer>`**Description**

Mapping for Diameter unable to deliver to RP error cause code.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

38

Name

unknownPeer - Unknown peer

Syntax`--unknownPeer <integer>`**Description**

Mapping for Diameter unknown peer to RP error cause code.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

38

Name

unknownSessionId - Unknown session ID

Syntax`--unknownSessionId <integer>`**Description**

Mapping for Diameter unknown session id to RP error cause code.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

38

Name

unsupportedVersion - Unsupported version

Syntax

--unsupportedVersion <integer>

Description

Mapping for Diameter unsupported version to RP error cause code.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

38

Name

userUnknown - User unknown

Syntax

--userUnknown <integer>

Description

Mapping for Diameter User unknown error to RP error cause code.

Class

Optional.

Type / Range

Integer in the range of 1, 8, 10, 11, 21, 27-30, 38, 41, 42, 47, 50, 69, 81, 95-99, 111, 127

Default

30

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.28 cmm sgnsGsnsLocalEndPtCfg

GBNS SGSN Local Endpoint Configuration. Syntax, description, parameters, and roles for cmm sgnsGsnsLocalEndPtCfg command.

Syntax**Description**

Configure, list and show parameters related to GBNS interface.

Parameters**Name**

id - Interface name

Syntax

<list-item>

Description

Interface name.

Class

Required value.

List

- GBNS - Specify gbns interface.
- GNSGSN - Specify Gn interface to an SGSN.
- Gr - Specify gr interface to an HLR.
- lu - Specify iu interface to an RNC.
- S13 - Specify S13 interface to an EIR.
- S3S16 - Specify S3/S16 interface to an SGSN.
- S4 - Specify S4 interface to an SGW.
- S6d - Specify S6d interface to an HSS.

Name

bvcResetRetries - BVC reset retries

Syntax

--bvcResetRetries <integer>

Description

This parameter defines the number of retries in the BVC reset procedure. the values range from 0 to 255.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 255

Default

3

Name

bvcResetTimer - BVC reset timer

Syntax

--bvcResetTimer <integer>

Description

This parameter defines the BSSGP virtual connection (BVC) reset timer. the parameter guards the reset procedure of the BVC. the values range from 1 to 120 seconds.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 120

Default

3

Name

i6hcack - IPv6 acknowledged data header compression

Syntax

--i6hcack <list-item>

Description

This parameter defines the IPv6 unacknowledged data header compression, true ... IPv6 unacknowledged data header compression is in use false ... IPv6 unacknowledged data header compression is not in use.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Name

i6hcunack - IPv6 unacknowledged data header compression

Syntax

--i6hcunack <list-item>

Description

This parameter defines the IPv6 acknowledged data header compression, true ... IPv6 acknowledged data header compression is in use false ... IPv6 acknowledged data header compression is not in use.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

icp - Gb over IP configuration UDP port

Syntax

--icp <integer>

Description

This parameter defines the pre-configured UDP port number for dynamic gb over IP. the values range from 49999 to 65535.

Class

Output only.

Type / Range

Integer in the range of 49,999 to 65,535

Default

49999

Name

llcRetransmCounter - LLC retransmission count, SGSN proposed value

Syntax

--llcRetransmCounter <integer>

Description

This parameter defines the logical link control (LLC) retransmission count, SGSN proposed value. the values range from 1 to 15.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 15

Default

3

Name

llcRetransmTimer - LLC retransmission timer, SGSN proposed value

Syntax

--llcRetransmTimer <integer>

Description

This parameter defines the logical link control (LLC) retransmission timer, SGSN proposed value. the values range from 0 to 410 seconds.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 410

Default

10

Name

nsAliveRetries - NS alive retries

Syntax

--nsAliveRetries <integer>

Description

This parameter defines the number of retries in the network service (NS) alive procedure. the values range from 0 to 255.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 255

Default

10

Name

nsAliveTimer - NS alive timer

Syntax

--nsAliveTimer <integer>

Description

This parameter defines the network service (NS) alive timer. the parameter guards the test procedure of the network service virtual connections (NS-VC). the values range from 1 to 60 seconds.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 60

Default

3

Name

nsTestTimer - NS test timer

Syntax

--nsTestTimer <integer>

Description

This parameter defines the test procedure of the network service virtual connections (NS-VC). the values range from 1 to 60 seconds.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 60

Default

60

Name

snsAddRetries - SNS add retries

Syntax

--snsAddRetries <integer>

Description

This parameter defines the number of SNS-ADD PDU retries in the SNS add procedure. the values range from 0 to 255.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 255

Default

3

Name

snsChangeWeightRetries - SNS changeweight retries

Syntax

--snsChangeWeightRetries <integer>

Description

This parameter defines the number of SNS-CHANGEWEIGHT PDU retries in the SNS changeweight procedure. the values range from 0 to 255.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 255

Default

3

Name

snsConfigRetries - SNS config retries

Syntax

--snsConfigRetries <integer>

Description

This parameter defines the number of SNS-CONFIG PDU retries in the SNS configuration procedure. the values range from 0 to 255.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 255

Default

3

Name

snsDeleteRetries - SNS delete retries

Syntax

--snsDeleteRetries <integer>

Description

This parameter defines the number of SNS-DELETE PDU retries in the SNS delete procedure. the values range from 0 to 255.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 255

Default

3

Name

snsProvTimer - SNS prov timer

Syntax

--snsProvTimer <integer>

Description

This parameter defines the interval of sub-network service (SNS) prov timer. the SNS prov timer guards the internet protocol (IP) SNS procedures. the values range from 1 to 10 seconds.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

5

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.29 cmm sgsgnGnLocalEndPtCfg

GNSGSN Local Endpoint Configuration. . Syntax, description, parameters, and roles for cmm sgsgnGnLocalEndPtCfg command.

Syntax

Description

Configure, list and show parameters related to GNSGSN interface.

Parameters

Name

id - Interface name

Syntax

<list-item>

Description

Interface name.

Class

Required value.

List

- GBNS - Specify gbns interface.
- GNSGSN - Specify Gn interface to an SGSN.
- Gr - Specify gr interface to an HLR.
- lu - Specify iu interface to an RNC.
- S13 - Specify S13 interface to an EIR.
- S3S16 - Specify S3/S16 interface to an SGSN.
- S4 - Specify S4 interface to an SGW.
- S6d - Specify S6d interface to an HSS.

Name

gni - GGSN IPv4 network identifier

Syntax

--gni <string>

Description

This parameter defines the external network the GGSN is connected to. in the GPRS backbone, the access point name is a reference to the GGSN to be used. this parameter is also applicable for S4.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 to 63 with the allowed pattern of [0-9a-zA-Z\._:]*

Default

""

Name

gni6 - GGSN IPv6 network identifier

Syntax

--gni6 <string>

Description

This parameter defines the external network the GGSN is connected to. in the GPRS backbone, the access point name (APN) is a reference to the GGSN to be used. this parameter is also applicable for S4.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 to 63 with the allowed pattern of [0-9a-zA-Z\._:]*

Default

""

Name

gniv4v6 - GGSN IPv4v6 network identifier

Syntax

--gniv4v6 <string>

Description

This parameter is used to store the primary default gateway network identifier for a dual stack PDP context (IPv4v6). this parameter is also applicable for S4.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 to 63 with the allowed pattern of [0-9a-zA-Z\._:]*

Default

""

Name

goi - GGSN operator identifier

Syntax

--goi <string>

Description

This parameter defines in which PLMN GPRS backbone the GGSN is located. GOI last part has to be .gprs. this parameter is also applicable for S4.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 to 37 with the allowed pattern of [0-9a-zA-Z\.\-_:]*

Default

""

Name

n3CreatePdp - N3-requests counter for create PDP context requests

Syntax

--n3CreatePdp <integer>

Description

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunnelling protocol (GTP) to send a create PDP context request message. the values range from 1 to 7.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

n3DelPdp - N3-requests counter for delete PDP context requests

Syntax

--n3DelPdp <integer>

Description

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunnelling protocol (GTP) to send a delete PDP context request message. the values range from 1 to 7.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

n3EchoUP - N3-requests counter for echo requests of user plane

Syntax

--n3EchoUP <integer>

Description

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunnelling protocol (GTP) to send an echo request message concerning user plane. the values range from 1 to 7.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

n3ForwReloc - N3-requests counter for forward relocation requests

Syntax

--n3ForwReloc <integer>

Description

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunnelling protocol (GTP) to send a forward relocation request message. the values range from 1 to 7.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

n3Id - N3-requests counter for identification requests GTPv1

Syntax

--n3Id <integer>

Description

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunnelling protocol (GTP) to send an identification request message. the values range from 1 to 7.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

n3SgsnCtxt - N3-requests counter for SGSN context requests

Syntax

--n3SgsnCtxt <integer>

Description

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunnelling protocol (GTP) to send an SGSN context request message. the values range from 1 to 7.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

n3UpdPdp - N3-requests counter for update PDP context requests

Syntax

--n3UpdPdp <integer>

Description

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunnelling protocol (GTP) to send an update PDP context request message. the values range from 1 to 7.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

secgni - Secondary GGSN IPv4 network identifier

Syntax

--secgni <string>

Description

This parameter defines which PDP subscription from the HLR is used in the override of requested APN and IMEI based APN override features, remove parameter value using SECGNI=0. this parameter is also applicable for S4.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 to 63 with the allowed pattern of [0-9a-zA-Z\._ :]*

Default

""

Name

secgni6 - Secondary GGSN IPv6 network identifier

Syntax

--secgni6 <string>

Description

This parameter defines which PDP subscription from the HLR is used in the override of requested APN and IMEI based APN override features, remove parameter value using SECGNI6=0. this parameter is also applicable for S4.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 to 63 with the allowed pattern of [0-9a-zA-Z\._ :]*

Default

""

Name

secgniv4v6 - Secondary GGSN IPv4v6 network identifier

Syntax

--secgniv4v6 <string>

Description

This parameter is used to store the secondary default gateway network identifier for a dual stack PDP context (IPv4v6), remove parameter value using SECGNIV4V6=0. this parameter is also applicable for S4.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 to 63 with the allowed pattern of [0-9a-zA-Z\._:\]*

Default

""

Name

t3CreatePdp - T3-response timer of create PDP context response

Syntax

--t3CreatePdp <integer>

Description

This parameter defines the value for the timer that holds the maximum waiting time for a create PDP context response message to a create PDP context request message. the values range from 1 to 10 seconds in 1 second steps.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

t3DelPdp - T3-response timer of delete PDP context response

Syntax

--t3DelPdp <integer>

Description

This parameter defines the value for the timer that holds the maximum waiting time for a delete PDP context response message to a delete PDP context request message. the values range from 1 to 10 seconds in 1 second steps.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

t3EchoUP - T3-response timer of echo response of user plane, from 3 to 15 seconds in 3 second steps

Syntax

--t3EchoUP <integer>

Description

This parameter defines the value for the timer that holds the maximum waiting time for an echo response message to an echo request message concerning user plane. the values range from 3 to 15 seconds.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 3 to 15

Default

3

Name

t3ForwReloc - T3-response timer of forward relocation response

Syntax

--t3ForwReloc <integer>

Description

This parameter defines the value for the timer that holds the maximum waiting time for a forward relocation response message to a forward relocation request message. the values range from 1 to 10 seconds in 1 second steps.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

t3Id - T3-response timer of identification response GTPv1

Syntax

--t3Id <integer>

Description

This parameter defines the value for the timer that holds the maximum waiting time for an identification response message to an identification request message. the values range from 1 to 10 seconds in 1 second steps.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

t3SgsnCtxt - T3-response timer of SGSN context response

Syntax

--t3SgsnCtxt <integer>

Description

This parameter defines the value for the timer that holds the maximum waiting time for an SGSN context response message to an SGSN context request message. the values range from 1 to 10 seconds in 1 second steps.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

t3tunnel - T3 tunnel

Syntax

--t3tunnel <integer>

Description

This parameter defines the T3 tunnel. the T3 tunnel parameter defines the timer that controls the time when protocol data units (PDUs) are forwarded from the old SGSN to the new SGSN as part of the inter-sgsn routing area update procedure. the values range from 20 to 60 seconds in 1 second steps. this parameter is also applicable for S3S16.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 20 to 60

Default

20

Name

t3UpdPdp - T3-response timer of update PDP context response

Syntax

--t3UpdPdp <integer>

Description

This parameter defines the value for the timer that holds the maximum waiting time for an update PDP context response message to an update PDP context request message. the values range from 1 to 10 seconds in 1 second steps.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

techo - Techo

Syntax

--techo <integer>

Description

This parameter defines the timer that holds the time interval for the path validation procedure. the values range from 60 to 600 seconds.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 60 to 600

Default

120

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.30 cmm sgsnGParms

sgsnGParms. Syntax, description, parameters, and roles for **cmm sgsnGParms** command.

Syntax

Description

This command is used to administer SGSN Global Parameters, old profiles

Parameters

Name

sgsnGparmName - sgsnGparmName

Syntax

--sgsnGparmName <list-item>

Description

Name of SGSN parameter, old prfile name

Class

Required key.

List

- **ackDataSupported** - Acknowledged mode PDP context supported - This parameter determines if the acknowledged mode PDP context is supported.
- **aliasApnUsedInConversion** - Alias APN Used In APN Conversion - Alias APN Used in APN Conversion Feature. SGSN feature APN conversion function will be enhanced for Alias APN use case. If this gParm is set, then converted APN will be not checked against HLR subscription. Otherwise (by default), the parameter will not be set, no change to existing APN conversion function
- **allowedApnniEnable** - Active status of APNNI access restriction in SGSN - Active status of APNNI access restriction in SGSN.
- **alternateLineServ** - Alternate line service
- **altImeiChkNasCause** - Alternative NAS cause for IMEI checking in MME/SGSN - Enables or disables the alternative NAS cause for IMEI checking in MME/SGSN.
- **ansiMapSupport** - ANSI MAP allowed or not - ANSI MAP allowed to the operator or not.
- **apnOverrideEnhance** - Apn Override Enhance - Parameter controls feature activation. By default, flag shall be disabled. When enabled, Gn SGSN shall override enhancements to existing APN override feature.

- **apnWithLowestPdpId** - Select APN with lowest PDP id from HLR subscription - Parameter controls whether the APN with lowest PDP id is used from HLR subscription instead of that PDP context is activated with an APN according to the first HLR subscription record.
- **callRedirection** - Call redirection
- **cancelLocationCc** - Cancel location cause code - When HLR sends cancel location message to SGSN, SGSN shall send the detach message without cause code value. in current implementation SGSN uses cause code value 7.
- **cancelLocRes** - Cancel location response - Parameter determines if the cancel location is acknowledged immediately after the cancel location is received.
- **cgfDualStackSupport** - SGSN dual stack PDP context support (3GPP rel 9, gn/gp) for charging gateway - Parameter determines whether the charging gateway is capable of supporting dual stack PDP contexts or not.
- **checkImeiMode** - Check IMEI Mode - This parameter is used to define the check Imei mode. The default value is 0 which means that SGSN will include only IMEI in IMEI check. Possible values (0 to 2), 0 for SGSN will include only IMEI in IMEI check, 1 for SGSN add IMSI in MAP part of IMEI Check, 2 for SGSN add IMSI and MSISDN in TCAP part of IMEI Check.
- **checkImeiSv** - Check IMEI Sv - This parameter is used to define if IMEISV is used for the IMEI check procedure to the EIR over the Gf Interface. When this parameter is set to true SGSN will request from the UE to include the IMEISV in the authentication and ciphering or Identification messages and accordingly use the IMEISV in the IMEI check procedure. Default value is false.
- **delayedIratMod** - Delayed IRAT modification
- **directDelResToGgsn** - Direct delete PDP context response message to GGSN - Parameter controls whether SGSN responds to delete PDP context request from GGSN immediately without waiting for answer from MS.
- **disableSecPdpCtx** - Disable secondary PDP context - Parameter defines if secondary PDP context is supported.
- **dnsFailureCause** - Changed cause code for DNS lookup failure - Parameter controls the cause code for DNS failure, i.e. when a DNS error is detected in internal SGSN RAU the cause code returned to MS is changed from protocol error, unspecified to implicitly detached.
- **dnsFallbackEnabled** - DNS query fallback check - Parameter determines whether a DNS query gets fallback to pre-release 8 in case the post-release 8 DNS query is failed.
- **doubleDnsQuery** - Double DNS query - Double DNS query to determine old node (SGSN or MME) in attach/rau.
- **dtRestrBasedOnTimer** - DT restriction based on timer - Parameter determines whether DT establishment is restricted after consecutive DT establishment/re-establishment happens during the defined time period.

- **dualStackPdpCtxt** - SGSN dual stack PDP context support
- **enforceSubscribedApn** - Enforce subscribed APN - Parameter controls whether the APN override is done in case the subscriber does not have an APN as a 'wild card' (*) defined in the HLR subscription.
- **enhancedPtmsiHandling** - Enhanced P-TMSI handling in SGSN
- **enhOffIdleSubsRate** - Idle subscribers offloading rate - This parameter determines with what rate idle subscribers should be offloaded when subscriber offloading feature is executed in phase 1.
- **enterpriseld** - Enterprise ID - This parameter specifies the enterprise id. all enterprise id's defined in the list.
- **epsBearerMapping** - EPS bearer mapping
- **epsCapability** - EPS capability - Parameter controls whether rel7 or rel8 (if HLR/HSS provides EPS subscription) query will be done.
- **fallbackNormalDns** - Fallback To Normal DNS - With this control parameter set to true, fallback to normal DNS query without tag is supported. Otherwise, if the parameter is not set (default), fallback is not supported
- **followOnProceed** - Support for follow-on proceed information in attach/rau accept - Parameter defines whether follow-on proceed information can be sent to UE.
- **forceAuthentication** - Force Authentication - When true, this parameter forces authentication in all 'SGSN' procedures for 1 hour after PAPS scale out/restart. When false, CMM performs authentication according to 'cmm securityParam' and 'cmm plmnParam' provisioning.
- **forceDnsOverTcp** - Force DNS over TCP - Parameter is used to force DNS queries over TCP instead of UDP.
- **forcefulCtxRem** - Forceful removal of PDP contexts from GGSN - With this parameter, the SGSN will in some abnormal cases send a delete PDP context request to the GGSN in order to remove hanging contexts.
- **ggsnPgwFqdnleCtrl** - PGW node name delivery on SGSN
- **grIfEnhancements** - Gr interface enhancements in SGSN - With this parameter it is possible to command SGSN to perform IMSI analysis for a subscriber before SGSN sends a location update to HLR after a reset indication is received from HLR. it is useful in HLR migration cases to get SGSN quickly to update the new HLR address to SGSN subscriber database and to get SGSN address to the new HLR. with this parameter it is also possible to command SGSN to not send a location update to HLR after a reset indication is received from HLR, until the first routing area update (RAU) request is received from MS. it is useful to decrease SS7 signalling load peak after HLR reset if there are not enough SS7 signalling links between SGSN and HLR.
- **gsIfEnhancements** - Gs interface enhancements in SGSN - This parameter is used to define the support for gs interface enhancement package, which is to avoid paging

problems of combined GPRS and IMSI attached subscribers. Some of the possible allowed values are: 0 Paging requests from VLR are not accepted if subscriber does not have Gs association. Location update to VLR only in case of a combined RA&LA update and LAI changes. 1 Paging requests from VLR are accepted even if subscriber does not have Gs association. 2 Location update to VLR always in case of a combined RA&LA update even if LAI does not change. 3 Location update to VLR always in case of a periodic RA update and when subscriber has Gs association. 4 Location update to VLR always in case of a combined RA&LA update even if LAI does not change or in case of a periodic RA update when subscriber has Gs association. 5 Paging requests from VLR are accepted even if subscriber does not have Gs association. Location update to VLR always in case of a combined RA&LA update even if LAI does not change. 6 Paging requests from VLR are accepted even if subscriber does not have Gs association. Location update to VLR always in case of a periodic RA update and when subscriber has Gs association. 7 Paging requests from VLR are accepted even if subscriber does not have Gs association. Location update to VLR always in case of a combined RA&LA update even if LAI does not change or in case of a periodic RA update and when subscriber has Gs association.

- **gsUpdateLocationThres** - Gs update location threshold - The parameter defines the GS update location overload control limit. Default value is off (10001) and other values define max number of Update Location for nonGPRS services operations on Gs interface that are ongoing in parallel.
- **gtpHopCounter** - Hop counter - Hop counter is used in SGSN pool to control infinite looping. this IE is used in `gtp_sgsn_context_req_s` and `gtp_identification_req_s`. decrease the value of hop counter if the optional parameter hop counter is present in the receiving signal. otherwise, it may include a hop counter with a value of max-1 (where max is the maximum defined value of hop counter) and forward it to other network element. configured value will be included in the very first request b/w SGSN and keeps on decreasing when it moves between SGSN. this decreasing will not result in any change in configured value.
- **gtpTargetIdEncoding** - GTP target identification information element encoding - This parameter controls how the target identification information element is encoded and decoded to the GTP messages. by default, the target identification information element is encoded or decoded to the GTP messages as defined in 3GPP TS 29.060 specification version 6.17.0 or older versions. by changing the parameter, it is possible to encode or decode the target identification information element as defined in 3GPP TS 29.060 change request CR 0668 (i.e. in version 6.18.0 or newer) or change request CR 0808 (i.e. in version 8.14.0 or newer).
- **gtpv1UeNwCapSupport** - UE network capability support - Parameter control whether UE network capability has to be sent in relocation request/context response.
- **hlrQosMbrDLLimit** - HLR QoS maximum bit rate for downlink override limit - This parameter defines bit rate override trigger limit for quality of service (QoS) parameters maximum bit rate (MBR) for downlink (DL) field. if the maximum bit rate (MBR) for downlink (DL) value received from HLR in PDP contexts QoS parameter is equal

or greater than the value of this parameter, the maximum bit rate for downlink value received from HLR is replaced with bit rate value defined in parameter HLR_QOS_MBR_DL_VALUE.

- **hlrQosMbrDlValue** - HLR QoS maximum bit rate for downlink override value - This parameter defines bit rate value for quality of service (QoS) parameters maximum bit rate (MBR) for downlink (DL) field. the given value overrides the maximum bit rate for downlink value received from HLR in PDP contexts QoS parameter, if value received from HLR is equal or greater than value defined in parameter HLR_QOS_MBR_DL_LIMIT.
- **hlrQosMbrUlLimit** - HLR QoS maximum bit rate for uplink override limit - This parameter defines bit rate override trigger limit for quality of service (QoS) parameters maximum bit rate (MBR) for uplink (UL) field. if the maximum bit rate (MBR) for uplink (UL) value received from HLR in PDP contexts QoS parameter is equal or greater than the value of this parameter, the maximum bit rate for uplink value received from HLR is replaced with bit rate value defined in parameter HLR_QOS_MBR_UL_VALUE.
- **hlrQosMbrUlValue** - HLR QoS maximum bit rate for uplink override value - This parameter defines bit rate value for quality of service (QoS) parameters maximum bit rate (MBR) for uplink (UL) field. the given value overrides the maximum bit rate for uplink value received from HLR in PDP contexts QoS parameter, if value received from HLR is equal or greater than value defined in parameter HLR_QOS_MBR_UL_LIMIT.
- **imsApnRouting** - SGW relocation based on IMS or sos substring
- **intCauseMapIllMsMe** - Cause code mapping change in statistics - Cause code mapping change in statistics.
- **ipv6DnsQueriesInUse** - IPv6 address fetching for GSNs in DNS query - Parameter defines, whether SGSN is trying to get IPv6 address for some network element in DNS query or just the IPv4 address.
- **iuQosOverride** - Iu QoS override
- **lboAllowedGnSgsn** - LBO allowed Gn SGSN - Parameter indicates whether Gn SGSN LBO control is allowed or not.
- **ll3MmFilters** - Filtering GPRS layer 3 mobility management messages
- **locationBasedGwSelection** - Location Based Gateway Selection - Based on this control parameter (set to true), location based GW selection feature will be enabled. Gn SGSN will insert a Tag (either zone name or RAC/LAC) to the DNS query for certain APNs. Otherwise (by default), it's not enable.
- **locationInfoInCdr** - LOCATION INFO IN CDR - With this parameter set, UE's location info in RAU will be added to the CDR. If it's not set (by default), UE's location info in RAU will not add to the CDR as existing function.
- **locationRptCtrlFlg** - Location report control - Parameter determines whether SGSN sends the location control messages to RNC.

- **locChangeInfoUpd** - Active status of location change information update to GGSN/PGW
- Parameter controls whether the location change information update to GGSN/PGW feature is active or inactive for subscribers.
- **lrasLoadBalancing** - LRAS load balancing - This parameter determines how load balancing is applied. Default value 0 means is disabled. Other allowed values are: 1 Load Balancing for new subscribers is enabled. 2 Load Balancing for old subscribers RAU request is enabled. 3 Load Balancing for new subscribers and for old subscribers RAU request is enabled.
- **lrasLoadBalancingPerRat** - LRAS load balancing per Radio Access Type - Parameter determines whether load balancing is applied per Radio Access Type or per total of UEs.
- **mapGtNatForm** - MAP use only national GT in own country
- **mapUnhaMsgsLimIn** - Unhandled messages limit for incoming MAP operations - The parameter gives the limit for the unhandled messages. when this limit is exceeded, the outgoing MAP operation start messages to the network are rejected.
- **mapUnhaMsgsLimOut** - Unhandled messages limit for outgoing MAP operations - The parameter gives the limit for unhandled messages. when this limit is exceeded, the MAP operation start messages, coming from the network are rejected.
- **massRabReleaseSpeed** - Speed of controllable mass RAB release procedure - Warning: For Nokia personnel only. Speed of controllable mass RAB release procedure
- **msisdnPgwDnsFallback** - MSISDN PGW DNS fallback - This parameter is used to define the support for fallback to release 8 or 7 DNS without MSISDN. 0 - no fallback 1 - fallback to rel8 DNS without MSISDN 2 - fallback to rel7 DNS fallback without MSISDN.
- **networkNamePlmn** - Network Name Per PLMN - This parameter control network name (short and full) in PLMN level. Normally, They are defined in SGSN level for network operator. With this feature, since MVNO is used, different PLMN with different network names are used. Thus, with this parameter enable, network name can be defined Per PLMN. Otherwise, existing function will be kept (default).
- **nonDrxState** - If the SGSN has valid DRX parameters for a TLLI, the SGSN includes them in the PDU. nevertheless, the SGSN can omit the DRX parameters if the MS identified with the TLLI is in MM non-drx mode period to speed up the transmission of the LLC-PDU on the radio interface. the SGSN does not send a DL-UNITDATA PDU without the DRX parameters IE if the MS identified with the TLLI is not in MM non-drx mode period.
- **oldImsiCoding** - Old IMSI (rel 97) encoding used in RAUs
- **originalCcInCdr** - Original charging characteristics in CDR - This parameter can be used for sending the original charging characteristics received in Mobility Management/Session Management/SMS messages in the corresponding CDRs generated by the SGSN.
- **override2gDowngrade** - QoS downgrade prevention in GERAN access - This parameter determines whether the QoS downgrade in GERAN access is prevented

- **overridePlmn** - Override on PLMN - This parameter will control PLMN (MCCMNC) override function can be applied. If the parameter is set to True, then PLMN level override will apply. If it's false, then no PLMN level control override function. By default, it will be set to false.
- **overrideRoamingApn** - Active status of override requested APN - Active status of override requested APN in SGSN feature for roaming subscribers.
- **overrideWithModApn** - APN override with modified APN - The parameter controls whether the APN is overridden with the modified APN. the modified APN is the same APN as is used for dynamic GGSN selection in the DNS query.in the SG01218 GGSN selection based on charging characteristics feature, this parameter also controls whether subscribed charging characteristics (CC) is required for dynamic GGSN selection (DNS query) and APN override.
- **pagingBufferActive** - Paging Buffer Active - The parameter defines the usage of the SGSN buffer for unacknowledged mode user data during ps-paging procedure
- **papsPortMirror** - PAPS port mirroring - This parameter enables/Disables port mirroring in paps unit.
- **papuBIMassDetachSpd** - Speed of controllable mass detach procedure - This parameter controls the speed of the controllable mass detach procedure, mass detaches per second. the range of the parameter is 1-100. default value is 10.
- **pdpAddrTypeTag** - Intelligent GGSN/PGW selection based on PDP address type - Parameter controls whether the PDN type tag will be included in the DNS query performed from SGSN to select a GGSN/PGW.
- **pdpCntxMassDeactAct** - The active status of controllable PDP context mass deactivation - Parameter controls whether or not the feature controllable PDP context mass deactivation is activated.
- **pdpCntxMassDeactSpd** - The speed of controllable PDP context mass deactivation - This parameter controls the speed of controllable PDP context mass deactivation, deactivations per second. the range of the parameter is 1 - 25. default value is 10.
- **pdpRejIfRabFails** - PDP activation reject if RAB setup procedure fails - Parameter determines whether the PDP activation is rejected if RAB setup procedure fails during PDP activation procedure.
- **poolingCombiSgsnMme** - Pooling SGSN/MME - Parameter enables/disables the support for SGSN/MME in the same hardware and supporting pooling functionality.
- **preemptionSupR8** - Preemption support for R8 subscribers
- **privateNbrPlan** - Private numbering plan
- **pshoBtwSgsnMme** - Support for relocation procedures between S4-SGSN and MME
- **qciMapR8ToR99** - Active status of QCI mapping - Parameter enables or disables the QCI mapping from rel.8 to rel.99 and vice versa.

- qciTrafficClassMap - QCI mapping to traffic class for rel 8 to rel 7 QoS conversion - Parameter determines whether the QCI values 8 and 9 will be mapped to traffic class interactive 2 and interactive 3 respectively in case of rel 8 to rel 7 QoS mapping.
- qosEvolAllRetPrior - Support for QoS parameters evolved allocation/retention priority parameters - Parameter defines whether evolved allocation/retention priority parameters are supported on SGSN for 3G QoS.
- qosHandlingInlsho - Parameter determines whether the QoS profile is renegotiated in case the radio access is changed from GERAN to UTRAN.
- qosSubBr2to3lsho - Negotiated bitrates in case of inter SGSN and inter system RAU - Parameter defines in case of inter SGSN and inter system RAU (2G -> 3G) if HLR max bitrate and guaranteed bitrate are used towards the new SGSN as negotiated bitrates.
- r8DnsEnhancementPool - R8 DNS enhancement pool - Parameter is used to control the 'R8 DNS enhancement in pooling multipoint iu' feature.
- r8DnsR7EpcSubs - r8DnsR7EpcSubs - The parameter is used to control access for R8 DNS Query Support for Gn-SGSN. 1 - roaming subscribers 2 - home subscribers 3 - both home and roaming subscribers
- raileInGtpTunneling - Inclusion of MCC and MNC in the RAI IE of create/update PDP context requests - Parameter defines whether the control of content served to subscribers based on their location is enabled or not.
- randomTlliRemTimer - Random TLLI remove time from SGSN - This parameter determines how long time random format TLLIs are stored in the SGSN after the mobile has completed the attach procedure.
- raStatLevelInUse - RA based statistics reporting resolution - RA based statistics reporting resolution.
- rauCsPsCoordination - CS/PS coordination in RAU procedure
- reAuthActive - Re-Authentication usage - Parameter defines the usage of the re-authentication.
- reAuthSupport - Re-Authentication support
- rel14LrRejectCauses - Release 14 Lr reject causes
- rel8RncQosChgRej1 - REL8 RNC QoS change reject - For Release 8 subscribers, the parameter decides whether to reject the QOS or to accept it when the RNC changes QOS in RAB assignment.
- relAccessBearerCtrl - S4 bearer retainability during iu/rab release procedure triggered from RNC - This parameter provides support for controlling the Release Access Bearer request messages being sent towards SGW during lu/RAB release procedure triggered from RNC, so that the signaling traffic towards SGW interface is not overloaded. If this parameter is enabled, SGSN follows the standard procedure for sending Release Access Bearer request messages towards SGW. If this parameter is disabled, Release Access Bearer request messages will not be sent towards SGW when lu/RAB release happens.

- **restrict11PdpCtx** - Restriction of the maximum allowed PDP contexts - Parameter is used to restrict the maximum allowed PDP contexts from 11 to 4.
- **rfspIndexSelectionOption** - RFSP Index Selection Option - When this parameter is enabled, then gn SGSN enables using rfspIndexSelectionOption option, available under plmn, to decide the way for assigning RFSP Index for the UE
- **rimRel8DnsAllowed** - RIM release 8 DNS allowed - Parameter indicates whether rel8 DNS is allowed or not.
- **s6dRoamingNotAllowedToNasCc11** - S6d CC Roaming Not Allowed Map To CC 11 in NAS - Cause code DIAMETER_ERROR_ROAMING_NOT_ALLOWED (5004) without Error Diagnostic shall be mapped to cause code #11 based on TS29.272 Annex B when sgsnGParms s6dRoamingNotAllowedToNasCc11 is set
- **sgsnNumberInUlr** - Control of SGSN-Number sending to HSS - This parameter controls sending of SGSN-Number AVP to HSS in diameter message update location request (ULR).
- **sgsnSS7Statistics** - SGSN SS7 Statistics
- **sgsnUseCcInApnForR8DnsPgwFallback** - SGSN use CC in APN For R8 DNS PGW Fallback - When this parameter is enabled and MSISDN was not included as an extra tag in parallel with Charging Characteristics, SGSN will retry the R8 DNS query without the CC tag before falling back to pre-R7 DNS queries for P-GW selection.
- **sgwDualStackSupport** - SGW dual stack support
- **signalingIndication** - Support for signaling indication information - Parameter defines whether signaling indication information is set in iu-interface when interactive PDP context is activated.
- **solc** - SGSN overload control - This parameter is used to switch on/off the overload control in paps
- **ss7ResrcStarvCRefCause** - SS7 Refusal cause - Depending on the value of ss7ResrcStarvCRefCause will give operator the following refusal cause options, Default value is 2 (access congestion). Possible values (1 to 4), 1 access failure, 2 access congestion, 3 unqualified, 4 SCCP failure
- **staticGwForBlockService** - Static GW Selection For Block Service - Static GW Selection feature will be used for Blocking Services. Certain IMSIs (configured) access configured APNs, will lead for Gn SGSN to select the configured GW IP address. This will make GGSN/GW to block service for such UE (IMSI) (e.g. what'sApp etc.). With this parameter set, Gn SGSN, will make static GW selection from configured IP addresses for such user(IMSI). Since the IMSIs might be any type of users (home, roamer etc.), only when this parameter is set, new parameter mcc and mnc will be used. Range type will be always IMSI. Otherwise, no change to the existing system.
- **streamingOptAct** - Streaming quality optimization feature - The parameter defines how the streaming quality optimization feature is configured in the SGSN.

- **streamWoPfcSupport** - Streaming PDP supported in gb access without PFC support.
- **supportEpcCapability** - EPC capability check for S/P-GW selection - Parameter determines whether the SGSN supports serving-gw/pdn-gw connectivity for EPC capable mobiles.
- **supportOfImsvops** - Support of IMSVOPS - Parameter is used to support IMS voice flags on the S4-SGSN feature.
- **suspendAcknowledge** - Suspend request acknowledge - When BSS sends a suspend request message to SGSN, this parameter determines whether or not the message is acknowledged at once if the subscriber is not in this PAPU unit.
- **threeDigitMncMcc** - Three digit MNC and MCC usage in DNS query - Parameter controls whether three digit MNC and MCC is used in DNS query.
- **throttleAlarms** - Throttle RAISE and/or CANCEL alarms - Warning: For Nokia personnel only. Throttle RAISE and/or CANCEL alarms of SGSN processes according to the sgsnGparms throttleAlarmsTimeWindow and throttleAlarmsThresholdNumber. value = 0 means the feature is OFF, value = 1 means RAISE alarms are throttled, value = 2 means RAISE and CANCEL alarms are throttled. default value is 1.
- **throttleAlarmsThresholdNumber** - Threshold number for alarm throttling - Warning: For Nokia personnel only. Depending on the value of throttleAlarms sgsnGparmName, if number of alarms with same alarmid that are raised/canceled during the time window configured by throttleAlarmsTimeWindow sgsnGparmName, exceeds this value then throttling starts. throttling stops when the time window ends. default value is 40 alarms.
- **throttleAlarmsTimeWindow** - Time window for alarm throttling - Warning: For Nokia personnel only. Depending on the value of throttleAlarms sgsnGparmName, if number of alarms with same alarmid that are raised/canceled during this time window, exceeds the value of throttleAlarmsThresholdNumber sgsnGparmName, then throttling starts. throttling stops when the time window ends. default value is 60 seconds.
- **timeZoneEncode** - TimeZoneEncode - This parameter controls the encoded time zone format.
- **tripletsToQuintuplets** - Triplets to quintuplets - Parameter defines if new authentication vectors are asked from the HLR when triplets are received from the previous SGSN during inter SGSN RAU in gb or iu access.
- **twoByteChargingChar** - This parameter is used to define the support for two-byte charging characteristics value in SGSN, received from HLR. it is configurable to use four-bit, one-byte, or two-byte charging characteristics in SGSN.
- **ueAmbrDI** - UE AMBR downlink default value - Default value to be used as subscribed UE-AMBR downlink when this QoS parameter is not received from HSS (TS 23.060 par.15.2.2 if no values of UE-AMBR are received from the HSS, the SGSN shall set the UE-AMBR according to implementation specific policies (e.g. a pre-configured maximum UEAMBR).).

- **ueAmbrSupportRanap** - UE-AMBR support - Parameter determines whether SGSN sends UE-AMBR IE in RAB assignment request and relocation request messages to RNC.
- **ueAmbrUI** - ueAmbrUI - Default value to be used as subscribed UE-AMBR uplink when this QoS parameter is not received from HSS (TS 23.060 par.15.2.2 if no values of UE-AMBR are received from the HSS, the SGSN shall set the UE-AMBR according to implementation specific policies (e.g. a pre-configured maximum UEAMBR).).
- **updQosSuppFlg** - Upgrade QoS supported bit in common flag IE for GTPv1 - Parameter is used to control sending the upgrade QoS supported bit in the common flag IE from SGSN to GGSN/PCRF.
- **usedGtpVersion** - Used GTP version - The parameter defines which version of the GTP is used towards the operators own GGSNs.
- **userPlaneEchoSending** - UserPlaneEchoSending - This parameter controls where the GTP user plane echoes are sent. By default (value 4), the echoes are not sent to neither RNC nor GGSN. By changing the parameter, it is possible to enable echo sending to either RNC or GGSN, or enable it for both. Some RNCs do not respond to SGSNs echo requests if the Direct Tunnel is in use. The GTP user plane echo sending is optional in the 29.060 (GTP) specification. Allowed values are: 1 The user plane echoes are sent to RNCs and GGSNs. 2 The user plane echoes are sent only to RNCs. 3 The user plane echoes are sent only to GGSNs. 4 SGSN does not send user plane echoes.
- **zeroNonDrxTimer** - Parameter enables setting the DRX-mode 'ON' to mobiles that send an attach request message with zero non-drx-timer value.
- **zoneSw** - Zone Switch - This parameter controls whether or not the mapping between zone names and RACs/LACs is enabled (switch on). If the parameter is set, zone name will be used as tag in the APN query. Otherwise, RAC+LAC will be used as tag in the APN query.

Name

sgsnGparmValue - sgsnGparmValue

Syntax

- -sgsnGparmValue <string>

Description

Value of SGSN parameter, old prfile value

Class

Optional.

Type / Range

Digit string of length 0 to 64

The sgsnGparmValue string is based on the sgsnGparmName parameter:

Name

ackDataSupported - Acknowledged mode PDP context supported

Syntax

```
--sgsnGparmName ackDataSupported --sgsnGparmValue <string>
```

Description

This parameter determines if the acknowledged mode PDP context is supported.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

aliasApnUsedInConversion - Alias APN Used In APN Conversion

Syntax

```
--sgsnGparmName aliasApnUsedInConversion --sgsnGparmValue <string>
```

Description

Alias APN Used in APN Conversion Feature. SGSN feature APN conversion function will be enhanced for Alias APN use case. If this gParm is set, then converted APN will be not checked against HLR subscription. Otherwise (by default), the parameter will not be set, no change to existing APN conversion function

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

allowedApnniEnable - Active status of APNNI access restriction in SGSN

Syntax

```
--sgsnGparmName allowedApnniEnable --sgsnGparmValue <string>
```

Description

Active status of APNNI access restriction in SGSN.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

alternateLineServ - Alternate line service

Syntax

--sgsnGparmName alternateLineServ --sgsnGparmValue <string>

Description

Alternate line service

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

altImeiChkNasCause - Alternative NAS cause for IMEI checking in MME/SGSN

Syntax

--sgsnGparmName altImeiChkNasCause --sgsnGparmValue <string>

Description

Enables or disables the alternative NAS cause for IMEI checking in MME/SGSN.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ansiMapSupport - ANSI MAP allowed or not

Syntax

--sgsnGparmName ansiMapSupport --sgsnGparmValue <string>

Description

ANSI MAP allowed to the operator or not.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

apnOverrideEnhance - Apn Override Enhance

Syntax

```
--sgsnGparmName apnOverrideEnhance --sgsnGparmValue <string>
```

Description

Parameter controls feature activation. By default, flag shall be disabled. When enabled, Gn SGSN shall override enhancements to existing APN override feature.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

apnWithLowestPdpId - Select APN with lowest PDP id from HLR subscription

Syntax

```
--sgsnGparmName apnWithLowestPdpId --sgsnGparmValue <string>
```

Description

Parameter controls whether the APN with lowest PDP id is used from HLR subscription instead of that PDP context is activated with an APN according to the first HLR subscription record.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

callRedirection - Call redirection

Syntax

```
--sgsnGparmName callRedirection --sgsnGparmValue <string>
```

Description

Call redirection

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

cancelLocationCc - Cancel location cause code

Syntax

```
--sgsnGparmName cancelLocationCc --sgsnGparmValue <string>
```

Description

When HLR sends cancel location message to SGSN, SGSN shall send the detach message without cause code value. in current implementation SGSN uses cause code value 7.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

cancelLocRes - Cancel location response

Syntax

```
--sgsnGparmName cancelLocRes --sgsnGparmValue <string>
```

Description

Parameter determines if the cancel location is acknowledged immediately after the cancel location is received.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

cgfDualStackSupport - SGSN dual stack PDP context support (3GPP rel 9, gn/gp) for charging gateway

Syntax

```
--sgsnGparmName cgfDualStackSupport --sgsnGparmValue <string>
```

Description

Parameter determines whether the charging gateway is capable of supporting dual stack PDP contexts or not.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Name

checkImeiMode - Check IMEI Mode

Syntax

--sgsnGparmName checkImeiMode --sgsnGparmValue <string>

Description

This parameter is used to define the check Imei mode. The default value is 0 which means that SGSN will include only IMEI in IMEI check. Possible values (0 to 2), 0 for SGSN will include only IMEI in IMEI check, 1 for SGSN add IMSI in MAP part of IMEI Check, 2 for SGSN add IMSI and MSISDN in TCAP part of IMEI Check.

Type / Range

Integer in the range of 0 to 2

Default

0

Name

checkImeiSv - Check IMEI Sv

Syntax

--sgsnGparmName checkImeiSv --sgsnGparmValue <string>

Description

This parameter is used to define if IMEISV is used for the IMEI check procedure to the EIR over the Gf Interface. When this parameter is set to true SGSN will request from the UE to include the IMEISV in the authentication and ciphering or Identification messages and accordingly use the IMEISV in the IMEI check procedure. Default value is false.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

delayedIratMod - Delayed IRAT modification

Syntax

--sgsnGparmName delayedIratMod --sgsnGparmValue <string>

Description

Delayed IRAT modification

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

directDelResToGgsn - Direct delete PDP context response message to GGSN

Syntax

--sgsnGparmName directDelResToGgsn --sgsnGparmValue <string>

Description

Parameter controls whether SGSN responds to delete PDP context request from GGSN immediately without waiting for answer from MS.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

disableSecPdpCtx - Disable secondary PDP context

Syntax

--sgsnGparmName disableSecPdpCtx --sgsnGparmValue <string>

Description

Parameter defines if secondary PDP context is supported.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

dnsFailureCause - Changed cause code for DNS lookup failure

Syntax

--sgsnGparmName dnsFailureCause --sgsnGparmValue <string>

Description

Parameter controls the cause code for DNS failure, i.e. when a DNS error is detected in internal SGSN RAU the cause code returned to MS is changed from protocol error, unspecified to implicitly detached.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

dnsFallbackEnabled - DNS query fallback check

Syntax

```
--sgsnGparmName dnsFallbackEnabled --sgsnGparmValue <string>
```

Description

Parameter determines whether a DNS query gets fallback to pre-release 8 in case the post-release 8 DNS query is failed.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

doubleDnsQuery - Double DNS query

Syntax

```
--sgsnGparmName doubleDnsQuery --sgsnGparmValue <string>
```

Description

Double DNS query to determine old node (SGSN or MME) in attach/rau.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

dtRestrBasedOnTimer - DT restriction based on timer

Syntax

```
--sgsnGparmName dtRestrBasedOnTimer --sgsnGparmValue <string>
```

Description

Parameter determines whether DT establishment is restricted after consecutive DT establishment/re-establishment happens during the defined time period.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

dualStackPdpCtxt - SGSN dual stack PDP context support

Syntax

```
--sgsnGparmName dualStackPdpCtxt --sgsnGparmValue <string>
```

Description

SGSN dual stack PDP context support

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

enforceSubscribedApn - Enforce subscribed APN

Syntax

```
--sgsnGparmName enforceSubscribedApn --sgsnGparmValue <string>
```

Description

Parameter controls whether the APN override is done in case the subscriber does not have an APN as a 'wild card' (*) defined in the HLR subscription.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

enhancedPtmsiHandling - Enhanced P-TMSI handling in SGSN

Syntax

```
--sgsnGparmName enhancedPtmsiHandling --sgsnGparmValue <string>
```

Description

Enhanced P-TMSI handling in SGSN

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

enhOfflIdleSubsRate - Idle subscribers offloading rate

Syntax

```
--sgsnGparmName enhOfflIdleSubsRate --sgsnGparmValue <string>
```

Description

This parameter determines with what rate idle subscribers should be offloaded when subscriber offloading feature is executed in phase 1.

Type / Range

Integer in the range of 0 to 50

Default

10

Name

enterpriseId - Enterprise ID

Syntax

```
--sgsnGparmName enterpriseId --sgsnGparmValue <string>
```

Description

This parameter specifies the enterprise id. all enterprise id's defined in the list.

Type / Range

Integer in the range of 1 to 38,101

Default

28458

Name

epsBearerMapping - EPS bearer mapping

Syntax

```
--sgsnGparmName epsBearerMapping --sgsnGparmValue <string>
```

Description

EPS bearer mapping

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

epsCapability - EPS capability

Syntax`--sgsnGparmName epsCapability --sgsnGparmValue <string>`**Description**

Parameter controls whether rel7 or rel8 (if HLR/HSS provides EPS subscription) query will be done.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

fallbackNormalDns - Fallback To Normal DNS

Syntax`--sgsnGparmName fallbackNormalDns --sgsnGparmValue <string>`**Description**

With this control parameter set to true, fallback to normal DNS query without tag is supported. Otherwise, if the parameter is not set (default), fallback is not supported

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

followOnProceed - Support for follow-on proceed information in attach/rau accept

Syntax`--sgsnGparmName followOnProceed --sgsnGparmValue <string>`**Description**

Parameter defines whether follow-on proceed information can be sent to UE.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

forceAuthentication - Force Authentication

Syntax

```
--sgsnGparmName forceAuthentication --sgsnGparmValue <string>
```

Description

When true, this parameter forces authentication in all 'SGSN' procedures for 1 hour after PAPS scale out/restart. When false, CMM performs authentication according to 'cmm securityParam' and 'cmm plmnParam' provisioning.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

forceDnsOverTcp - Force DNS over TCP

Syntax

```
--sgsnGparmName forceDnsOverTcp --sgsnGparmValue <string>
```

Description

Parameter is used to force DNS queries over TCP instead of UDP.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

forcefulCtxRem - Forceful removal of PDP contexts from GGSN

Syntax

```
--sgsnGparmName forcefulCtxRem --sgsnGparmValue <string>
```

Description

With this parameter, the SGSN will in some abnormal cases send a delete PDP context request to the GGSN in order to remove hanging contexts.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ggsnPgwFqdnIeCtrl - PGW node name delivery on SGSN

Syntax

```
--sgsnGparmName ggsnPgwFqdnIeCtrl --sgsnGparmValue <string>
```

Description

PGW node name delivery on SGSN

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

grIfEnhancements - Gr interface enhancements in SGSN

Syntax

```
--sgsnGparmName grIfEnhancements --sgsnGparmValue <string>
```

Description

With this parameter it is possible to command SGSN to perform IMSI analysis for a subscriber before SGSN sends a location update to HLR after a reset indication is received from HLR. it is useful in HLR migration cases to get SGSN quickly to update the new HLR address to SGSN subscriber database and to get SGSN address to the new HLR. with this parameter it is also possible to command SGSN to not send a location update to HLR after a reset indication is received from HLR, until the first routing area update (RAU) request is received from MS. it is useful to decrease SS7 signalling load peak after HLR reset if there are not enough SS7 signalling links between SGSN and HLR.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

gsIfEnhancements - Gs interface enhancements in SGSN

Syntax

--sgsnGparmName gsIfEnhancements --sgsnGparmValue <string>

Description

This parameter is used to define the support for gs interface enhancement package, which is to avoid paging problems of combined GPRS and IMSI attached subscribers. Some of the possible allowed values are: 0 Paging requests from VLR are not accepted if subscriber does not have Gs association. Location update to VLR only in case of a combined RA&LA update and LAI changes. 1 Paging requests from VLR are accepted even if subscriber does not have Gs association. 2 Location update to VLR always in case of a combined RA&LA update even if LAI does not change. 3 Location update to VLR always in case of a periodic RA update and when subscriber has Gs association. 4 Location update to VLR always in case of a combined RA&LA update even if LAI does not change or in case of a periodic RA update when subscriber has Gs association. 5 Paging requests from VLR are accepted even if subscriber does not have Gs association. Location update to VLR always in case of a combined RA&LA update even if LAI does not change. 6 Paging requests from VLR are accepted even if subscriber does not have Gs association. Location update to VLR always in case of a periodic RA update and when subscriber has Gs association. 7 Paging requests from VLR are accepted even if subscriber does not have Gs association. Location update to VLR always in case of a combined RA&LA update even if LAI does not change or in case of a periodic RA update and when subscriber has Gs association.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

gsUpdateLocationThres - Gs update location threshold

Syntax

--sgsnGparmName gsUpdateLocationThres --sgsnGparmValue <string>

Description

The parameter defines the GS update location overload control limit. Default value is off (10001) and other values define max number of Update Location for nonGPRS services operations on Gs interface that are ongoing in parallel.

Type / Range

Integer in the range of 0 to 10,001

Default

10001

Name

gtpHopCounter - Hop counter

Syntax

```
--sgsnGparmName gtpHopCounter --sgsnGparmValue <string>
```

Description

Hop counter is used in SGSN pool to control infinite looping. this IE is used in gtp_sgsn_context_req_s and gtp_identification_req_s. decrease the value of hop counter if the optional parameter hop counter is present in the receiving signal. otherwise, it may include a hop counter with a value of max-1 (where max is the maximum defined value of hop counter) and forward it to other network element. configured value will be included in the very first request b/w SGSN and keeps on decreasing when it moves between SGSN. this decreasing will not result in any change in configured value.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

gtpTargetIdEncoding - GTP target identification information element encoding

Syntax

```
--sgsnGparmName gtpTargetIdEncoding --sgsnGparmValue <string>
```

Description

This parameter controls how the target identification information element is encoded and decoded to the GTP messages. by default, the target identification information element is encoded or decoded to the GTP messages as defined in 3GPP TS 29.060 specification version 6.17.0 or older versions. by changing the parameter, it is possible to encode or decode the target identification information element as defined in 3GPP TS 29.060 change request CR 0668 (i.e. in version 6.18.0 or newer) or change request CR 0808 (i.e. in version 8.14.0 or newer).

List

- 0 - Specify GTP target encoding of 0.
- 1 - Specify GTP target encoding of 1.
- 2 - Specify GTP target encoding of 2.
- 255 - Specify GTP target encoding of 255.

Default

0

Name

gtpv1UeNwCapSupport - UE network capability support

Syntax

```
--sgsnGparmName gtpv1UeNwCapSupport --sgsnGparmValue <string>
```

Description

Parameter control whether UE network capability has to be sent in relocation request/context response.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

h1rQosMbrDLLimit - HLR QoS maximum bit rate for downlink override limit

Syntax

```
--sgsnGparmName h1rQosMbrDLLimit --sgsnGparmValue <string>
```

Description

This parameter defines bit rate override trigger limit for quality of service (QoS) parameters maximum bit rate (MBR) for downlink (DL) field. if the maximum bit rate (MBR) for downlink (DL) value received from HLR in PDP contexts QoS parameter is equal or greater than the value of this parameter, the maximum bit rate for downlink value received from HLR is replaced with bit rate value defined in parameter HLR_QOS_MBR_DL_VALUE.

Type / Range

Integer in the range of 0 to 8

Default

0

Name

h1rQosMbrDLValue - HLR QoS maximum bit rate for downlink override value

Syntax

```
--sgsnGparmName h1rQosMbrDLValue --sgsnGparmValue <string>
```

Description

This parameter defines bit rate value for quality of service (QoS) parameters maximum bit rate (MBR) for downlink (DL) field. the given value overrides the maximum bit rate for downlink value received from HLR in PDP contexts QoS parameter, if value received from HLR is equal or greater than value defined in parameter HLR_QOS_MBR_DL_LIMIT.

Type / Range

Integer in the range of 0 to 168

Default

0

Name

hlrQosMbrULLimit - HLR QoS maximum bit rate for uplink override limit

Syntax

```
--sgsnGparmName hlrQosMbrULLimit --sgsnGparmValue <string>
```

Description

This parameter defines bit rate override trigger limit for quality of service (QoS) parameters maximum bit rate (MBR) for uplink (UL) field. if the maximum bit rate (MBR) for uplink (UL) value received from HLR in PDP contexts QoS parameter is equal or greater than the value of this parameter, the maximum bit rate for uplink value received from HLR is replaced with bit rate value defined in parameter HLR_QOS_MBR_UL_VALUE.

Type / Range

Integer in the range of 0 to 8

Default

0

Name

hlrQosMbrULValue - HLR QoS maximum bit rate for uplink override value

Syntax

```
--sgsnGparmName hlrQosMbrULValue --sgsnGparmValue <string>
```

Description

This parameter defines bit rate value for quality of service (QoS) parameters maximum bit rate (MBR) for uplink (UL) field. the given value overrides the maximum bit rate for uplink value received from HLR in PDP contexts QoS parameter, if value received from HLR is equal or greater than value defined in parameter HLR_QOS_MBR_UL_LIMIT.

Type / Range

Integer in the range of 0 to 46

Default

0

Name

imsApnRouting - SGW relocation based on IMS or sos substring

Syntax

```
--sgsnGparmName imsApnRouting --sgsnGparmValue <string>
```

Description

SGW relocation based on IMS or sos substring

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

intCauseMapIllMsMe - Cause code mapping change in statistics

Syntax

```
--sgsnGparmName intCauseMapIllMsMe --sgsnGparmValue <string>
```

Description

Cause code mapping change in statistics.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ipv6DnsQueriesInUse - IPv6 address fetching for GSNs in DNS query

Syntax

```
--sgsnGparmName ipv6DnsQueriesInUse --sgsnGparmValue <string>
```

Description

Parameter defines, whether SGSN is trying to get IPv6 address for some network element in DNS query or just the IPv4 address.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

iuQosOverride - Iu QoS override

Syntax

```
--sgsnGparmName iuQosOverride --sgsnGparmValue <string>
```

Description

Iu QoS override

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

lboAllowedGnSgsn - LBO allowed Gn SGSN

Syntax

--sgsnGparmName lboAllowedGnSgsn --sgsnGparmValue <string>

Description

Parameter indicates whether Gn SGSN LBO control is allowed or not.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ll3MmFilters - Filtering GPRS layer 3 mobility management messages

Syntax

--sgsnGparmName ll3MmFilters --sgsnGparmValue <string>

Description

Filtering GPRS layer 3 mobility management messages

Type / Range

Integer in the range of 0 to 255

Default

0

Name

locationBasedGwSelection - Location Based Gateway Selection

Syntax

--sgsnGparmName locationBasedGwSelection --sgsnGparmValue <string>

Description

Based on this control parameter (set to true), location based GW selection feature will be enabled. Gn SGSN will insert a Tag (either zone name or RAC/LAC) to the DNS query for certain APNs. Otherwise (by default), it's not enable.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

locationInfoInCdr - LOCATION INFO IN CDR

Syntax

--sgsnGparmName locationInfoInCdr --sgsnGparmValue <string>

Description

With this parameter set, UE's location info in RAU will be added to the CDR. If it's not set (by default), UE's location info in RAU will not add to the CDR as existing function.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

locationRptCtrlFlg - Location report control

Syntax

--sgsnGparmName locationRptCtrlFlg --sgsnGparmValue <string>

Description

Parameter determines whether SGSN sends the location control messages to RNC.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

locChangeInfoUpd - Active status of location change information update to GGSN/PGW

Syntax

--sgsnGparmName locChangeInfoUpd --sgsnGparmValue <string>

Description

Parameter controls whether the location change information update to GGSN/PGW feature is active or inactive for subscribers.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

lrasLoadBalancing - LRAS load balancing

Syntax

--sgsnGparmName lrasLoadBalancing --sgsnGparmValue <string>

Description

This parameter determines how load balancing is applied. Default value 0 means is disabled. Other allowed values are: 1 Load Balancing for new subscribers is enabled. 2 Load Balancing for old subscribers RAU request is enabled. 3 Load Balancing for new subscribers and for old subscribers RAU request is enabled.

Type / Range

Integer in the range of 0 to 3

Default

0

Name

lrasLoadBalancingPerRat - LRAS load balancing per Radio Access Type

Syntax

--sgsnGparmName lrasLoadBalancingPerRat --sgsnGparmValue <string>

Description

Parameter determines whether load balancing is applied per Radio Access Type or per total of UEs.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

mapGtNatForm - MAP use only national GT in own country

Syntax

--sgsnGparmName mapGtNatForm --sgsnGparmValue <string>

Description

MAP use only national GT in own country

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Name

mapUnhaMsgsLimIn - Unhandled messages limit for incoming MAP operations

Syntax

```
--sgsnGparmName mapUnhaMsgsLimIn --sgsnGparmValue <string>
```

Description

The parameter gives the limit for the unhandled messages. when this limit is exceeded, the outgoing MAP operation start messages to the network are rejected.

Type / Range

Integer in the range of 0 to 65,535

Default

300

Name

mapUnhaMsgsLimOut - Unhandled messages limit for outgoing MAP operations

Syntax

```
--sgsnGparmName mapUnhaMsgsLimOut --sgsnGparmValue <string>
```

Description

The parameter gives the limit for unhandled messages. when this limit is exceeded, the MAP operation start messages, coming from the network are rejected.

Type / Range

Integer in the range of 0 to 65,535

Default

300

Name

massRabReleaseSpeed - Speed of controllable mass RAB release procedure

Syntax

```
--sgsnGparmName massRabReleaseSpeed --sgsnGparmValue <string>
```

Description

Warning: For Nokia personnel only. Speed of controllable mass RAB release procedure

Type / Range

Integer in the range of 1 to 100

Default

20

Name

msisdnPgwDnsFallback - MSISDN PGW DNS fallback

Syntax

```
--sgsnGparmName msisdnPgwDnsFallback --sgsnGparmValue <string>
```

Description

This parameter is used to define the support for fallback to release 8 or 7 DNS without MSISDN. 0 - no fallback 1 - fallback to rel8 DNS without MSISDN 2 - fallback to rel7 DNS fallback without MSISDN.

Type / Range

Integer in the range of 0 to 2

Default

0

Name

networkNamePlmn - Network Name Per PLMN

Syntax

```
--sgsnGparmName networkNamePlmn --sgsnGparmValue <string>
```

Description

This parameter control network name (short and full) in PLMN level. Normally, They are defined in SGSN level for network operator. With this feature, since MVNO is used, different PLMN with different network names are used. Thus, with this parameter enable, network name can be defined Per PLMN. Otherwise, existing function will be kept (default).

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

nonDrxState - NonDrxState

Syntax

```
--sgsnGparmName nonDrxState --sgsnGparmValue <string>
```

Description

If the SGSN has valid DRX parameters for a TLLI, the SGSN includes them in the PDU. nevertheless, the SGSN can omit the DRX parameters if the MS identified with the TLLI is in MM non-drx mode period to speed up the transmission of the LLC-PDU on the radio interface. the SGSN does not send a DL-UNITDATA PDU without the DRX parameters IE if the MS identified with the TLLI is not in MM non-drx mode period.

Type / Range

Integer in the range of 0 to 3

Default

0

Name

oldImsiCoding - Old IMSI (rel 97) encoding used in RAUs

Syntax

```
--sgsnGparmName oldImsiCoding --sgsnGparmValue <string>
```

Description

Old IMSI (rel 97) encoding used in RAUs

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

originalCcInCdr - Original charging characteristics in CDR

Syntax

```
--sgsnGparmName originalCcInCdr --sgsnGparmValue <string>
```

Description

This parameter can be used for sending the original charging characteristics received in Mobility Management/Session Management/SMS messages in the corresponding CDRs generated by the SGSN.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

override2gDowngrade - QoS downgrade prevention in GERAN access

Syntax

```
--sgsnGparmName override2gDowngrade --sgsnGparmValue <string>
```

Description

This parameter determines whether the QoS downgrade in GERAN access is prevented

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

overridePlmn - Override on PLMN

Syntax

```
--sgsnGparmName overridePlmn --sgsnGparmValue <string>
```

Description

This parameter will control PLMN (MCCMNC) override function can be applied. If the parameter is set to True, then PLMN level override will apply. If it's false, then no PLMN level control override function. By default, it will be set to false.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

overrideRoamingApn - Active status of override requested APN

Syntax

```
--sgsnGparmName overrideRoamingApn --sgsnGparmValue <string>
```

Description

Active status of override requested APN in SGSN feature for roaming subscribers.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

overrideWithModApn - APN override with modified APN

Syntax

```
--sgsnGparmName overrideWithModApn --sgsnGparmValue <string>
```

Description

The parameter controls whether the APN is overridden with the modified APN. the modified APN is the same APN as is used for dynamic GGSN selection in the DNS query.in the SG01218

GGSN selection based on charging characteristics feature, this parameter also controls whether subscribed charging characteristics (CC) is required for dynamic GGSN selection (DNS query) and APN override.

List

- 0 - Specify override of 0.
- 1 - Specify override of 1.
- 16 - Specify override of 16.
- 17 - Specify override of 17.

Default

0

Name

pagingBufferActive - Paging Buffer Active

Syntax

```
--sgsnGparmName pagingBufferActive --sgsnGparmValue <string>
```

Description

The parameter defines the usage of the SGSN buffer for unacknowledged mode user data during ps-paging procedure

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

papsPortMirror - PAPS port mirroring

Syntax

```
--sgsnGparmName papsPortMirror --sgsnGparmValue <string>
```

Description

This parameter enables/Disables port mirroring in paps unit.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

papuBlMassDetachSpd - Speed of controllable mass detach procedure

Syntax

--sgsnGparmName papuBlMassDetachSpd --sgsnGparmValue <string>

Description

This parameter controls the speed of the controllable mass detach procedure, mass detaches per second. the range of the parameter is 1-100. default value is 10.

Type / Range

Integer in the range of 1 to 100

Default

50

Name

pdpAddrTypeTag - Intelligent GGSN/PGW selection based on PDP address type

Syntax

--sgsnGparmName pdpAddrTypeTag --sgsnGparmValue <string>

Description

Parameter controls whether the PDN type tag will be included in the DNS query performed from SGSN to select a GGSN/PGW.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

pdpCntxMassDeactAct - The active status of controllable PDP context mass deactivation

Syntax

--sgsnGparmName pdpCntxMassDeactAct --sgsnGparmValue <string>

Description

Parameter controls whether or not the feature controllable PDP context mass deactivation is activated.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

pdpCntxMassDeactSpd - The speed of controllable PDP context mass deactivation

Syntax

--sgsnGparmName pdpCntxMassDeactSpd --sgsnGparmValue <string>

Description

This parameter controls the speed of controllable PDP context mass deactivation, deactivations per second. the range of the parameter is 1 - 25. default value is 10.

Type / Range

Integer in the range of 1 to 25

Default

10

Name

pdpRejIfRabFails - PDP activation reject if RAB setup procedure fails

Syntax

--sgsnGparmName pdpRejIfRabFails --sgsnGparmValue <string>

Description

Parameter determines whether the PDP activation is rejected if RAB setup procedure fails during PDP activation procedure.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

poolingCombiSgsnMme - Pooling SGSN/MME

Syntax

--sgsnGparmName poolingCombiSgsnMme --sgsnGparmValue <string>

Description

Parameter enables/disables the support for SGSN/MME in the same hardware and supporting pooling functionality.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

preemptionSupR8 - Preemption support for R8 subscribers

Syntax

```
--sgsnGparmName preemptionSupR8 --sgsnGparmValue <string>
```

Description

Preemption support for R8 subscribers

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

privateNbrPlan - Private numbering plan

Syntax

```
--sgsnGparmName privateNbrPlan --sgsnGparmValue <string>
```

Description

Private numbering plan

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

pshoBtwSgsnMme - Support for relocation procedures between S4-SGSN and MME

Syntax

```
--sgsnGparmName pshoBtwSgsnMme --sgsnGparmValue <string>
```

Description

Support for relocation procedures between S4-SGSN and MME

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

qciMapR8ToR99 - Active status of QCI mapping

Syntax

```
--sgsnGparmName qciMapR8ToR99 --sgsnGparmValue <string>
```

Description

Parameter enables or disables the QCI mapping from rel.8 to rel.99 and vice versa.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

qciTrafficClassMap - QCI mapping to traffic class for rel 8 to rel 7 QoS conversion

Syntax

```
--sgsnGparmName qciTrafficClassMap --sgsnGparmValue <string>
```

Description

Parameter determines whether the QCI values 8 and 9 will be mapped to traffic class interactive 2 and interactive 3 respectively in case of rel 8 to rel 7 QoS mapping.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

qosEvolAllRetPrior - Support for QoS parameters evolved allocation/retention priority parameters

Syntax

```
--sgsnGparmName qosEvolAllRetPrior --sgsnGparmValue <string>
```

Description

Parameter defines whether evolved allocation/retention priority parameters are supported on SGSN for 3G QoS.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

qosHandlingInIsho - qosHandlingInIsho

Syntax

--sgsnGparmName qosHandlingInIsho --sgsnGparmValue <string>

Description

Parameter determines whether the QoS profile is renegotiated in case the radio access is changed from GERAN to UTRAN.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

qosSubBr2to3Isho - Negotiated bitrates in case of inter SGSN and inter system RAU

Syntax

--sgsnGparmName qosSubBr2to3Isho --sgsnGparmValue <string>

Description

Parameter defines in case of inter SGSN and inter system RAU (2G -> 3G) if HLR max bitrate and guaranteed bitrate are used towards the new SGSN as negotiated bitrates.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

r8DnsEnhancementPool - R8 DNS enhancement pool

Syntax

--sgsnGparmName r8DnsEnhancementPool --sgsnGparmValue <string>

Description

Parameter is used to control the 'R8 DNS enhancement in pooling multipoint iu' feature.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

r8DnsR7EpcSubs - r8DnsR7EpcSubs

Syntax

--sgsnGparmName r8DnsR7EpcSubs --sgsnGparmValue <string>

Description

The parameter is used to control access for R8 DNS Query Support for Gn-SGSN. 1 - roaming subscribers 2 - home subscribers 3 - both home and roaming subscribers

Type / Range

Integer in the range of 1 to 3

Default

1

Name

raiInGtpTunneling - Inclusion of MCC and MNC in the RAI IE of create/update PDP context requests

Syntax

--sgsnGparmName raiInGtpTunneling --sgsnGparmValue <string>

Description

Parameter defines whether the control of content served to subscribers based on their location is enabled or not.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

randomTlliRemTimer - Random TLLI remove time from SGSN

Syntax

--sgsnGparmName randomTlliRemTimer --sgsnGparmValue <string>

Description

This parameter determines how long time random format TLLIs are stored in the SGSN after the mobile has completed the attach procedure.

Type / Range

Integer in the range of 0 to 120

Default

0

Name

raStatLevelInUse - RA based statistics reporting resolution

Syntax

--sgsnGparmName raStatLevelInUse --sgsnGparmValue <string>

Description

RA based statistics reporting resolution.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

rauCsPsCoordination - CS/PS coordination in RAU procedure

Syntax

--sgsnGparmName rauCsPsCoordination --sgsnGparmValue <string>

Description

CS/PS coordination in RAU procedure

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

reAuthActive - Re-Authentication usage

Syntax

--sgsnGparmName reAuthActive --sgsnGparmValue <string>

Description

Parameter defines the usage of the re-authentication.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

reAuthSupport - Re-Authentication support

Syntax

--sgsnGparmName reAuthSupport --sgsnGparmValue <string>

Description

Re-Authentication support

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

rel14LrRejectCauses - Release 14 lr reject causes

Syntax

--sgsnGparmName rel14LrRejectCauses --sgsnGparmValue <string>

Description

Release 14 lr reject causes

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

rel8RncQosChgRej1 - REL8 RNC QoS change reject

Syntax

--sgsnGparmName rel8RncQosChgRej1 --sgsnGparmValue <string>

Description

For Release 8 subscribers, the parameter decides whether to reject the QOS or to accept it when the RNC changes QOS in RAB assignment.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

relAccessBearerCtrl - S4 bearer retainability during lu/rab release procedure triggered from RNC

Syntax

```
--sgsnGparmName relAccessBearerCtrl --sgsnGparmValue <string>
```

Description

This parameter provides support for controlling the Release Access Bearer request messages being sent towards SGW during lu/RAB release procedure triggered from RNC, so that the signaling traffic towards SGW interface is not overloaded. If this parameter is enabled, SGSN follows the standard procedure for sending Release Access Bearer request messages towards SGW. If this parameter is disabled, Release Access Bearer request messages will not be sent towards SGW when lu/RAB release happens.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

restrict11PdpCtx - Restriction of the maximum allowed PDP contexts

Syntax

```
--sgsnGparmName restrict11PdpCtx --sgsnGparmValue <string>
```

Description

Parameter is used to restrict the maximum allowed PDP contexts from 11 to 4.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

rfspIndexSelectionOption - RFSP Index Selection Option

Syntax

```
--sgsnGparmName rfspIndexSelectionOption --sgsnGparmValue <string>
```

Description

When this parameter is enabled, then gn SGSN enables using rfspIndexSelectionOption option, available under plmn, to decide the way for assigning RFSP Index for the UE

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

rimRel8DnsAllowed - RIM release 8 DNS allowed

Syntax

```
--sgsnGparmName rimRel8DnsAllowed --sgsnGparmValue <string>
```

Description

Parameter indicates whether rel8 DNS is allowed or not.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

s6dRoamingNotAllowedToNasCc11 - S6d CC Roaming Not Allowed Map To CC 11 in NAS

Syntax

```
--sgsnGparmName s6dRoamingNotAllowedToNasCc11 --sgsnGparmValue  
<string>
```

Description

Cause code DIAMETER_ERROR_ROAMING_NOT_ALLOWED (5004) without Error Diagnostic shall be mapped to cause code #11 based on TS29.272 Annex B when sgsnGParms s6dRoamingNotAllowedToNasCc11 is set

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sgsnNumberInUlr - Control of SGSN-Number sending to HSS

Syntax

```
--sgsnGparmName sgsnNumberInUlr --sgsnGparmValue <string>
```

Description

This parameter controls sending of SGSN-Number AVP to HSS in diameter message update location request (ULR).

Type / Range

Integer in the range of 0 to 2

Default

0

Name

sgsnSS7Statistics - sgsnSS7Statistics

Syntax

```
--sgsnGparmName sgsnSS7Statistics --sgsnGparmValue <string>
```

Description

SGSN SS7 Statistics

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sgsnUseCcInApnForR8DnsPgWFallback - SGSN use CC in APN For R8 DNS PGW Fallback

Syntax

```
--sgsnGparmName sgsnUseCcInApnForR8DnsPgWFallback --sgsnGparmValue  
<string>
```

Description

When this parameter is enabled and MSISDN was not included as an extra tag in parallel with Charging Characteristics, SGSN will retry the R8 DNS query without the CC tag before falling back to pre-R7 DNS queries for P-GW selection.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sgwDualStackSupport - SGW dual stack support

Syntax

```
--sgsnGparmName sgwDualStackSupport --sgsnGparmValue <string>
```

Description

SGW dual stack support

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

signalingIndication - Support for signaling indication information

Syntax

```
--sgsnGparmName signalingIndication --sgsnGparmValue <string>
```

Description

Parameter defines whether signaling indication information is set in iu-interface when interactive PDP context is activated.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

solc - SGSN overload control

Syntax

```
--sgsnGparmName solc --sgsnGparmValue <string>
```

Description

This parameter is used to switch on/off the overload control in paps

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

ss7ResrcStarvCRefCause - SS7 Refusal cause

Syntax

```
--sgsnGparmName ss7ResrcStarvCRefCause --sgsnGparmValue <string>
```

Description

Depending on the value of ss7ResrcStarvCRefCause will give operator the following refusal cause options, Default value is 2 (access congestion). Possible values (1 to 4), 1 access failure, 2 access congestion, 3 unqualified, 4 SCCP failure

Type / Range

Integer in the range of 1 to 4

Default

2

Name

staticGwForBlockService - Static GW Selection For Block Service

Syntax

```
--sgsnGparmName staticGwForBlockService --sgsnGparmValue <string>
```

Description

Static GW Selection feature will be used for Blocking Services. Certain IMSIs (configured) access configured APNs, will lead for Gn SGSN to select the configured GW IP address. This will make GGSN/GW to block service for such UE (IMSI) (e.g. what'sApp etc.). With this parameter set, Gn SGSN, will make static GW selection from configured IP addresses for such user(IMSI). Since the IMSIs might be any type of users (home, roamer etc.), only when this parameter is set, new parameter mcc and mnc will be used. Range type will be always IMSI. Otherwise, no change to the existing system.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

streamingOptAct - Streaming quality optimization feature

Syntax

```
--sgsnGparmName streamingOptAct --sgsnGparmValue <string>
```

Description

The parameter defines how the streaming quality optimization feature is configured in the SGSN.

Type / Range

Integer in the range of 0 to 10 or 255

Default

255

Name

streamWoPfcSupport - Streaming PDP supported in gb access without PFC support.

Syntax

```
--sgsnGparmName streamWoPfcSupport --sgsnGparmValue <string>
```

Description

Streaming PDP supported in gb access without PFC support.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

supportEpcCapability - EPC capability check for S/P-GW selection

Syntax

```
--sgsnGparmName supportEpcCapability --sgsnGparmValue <string>
```

Description

Parameter determines whether the SGSN supports serving-gw/pdn-gw connectivity for EPC capable mobiles.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

supportOfImsvops - Support of IMSVOPS

Syntax

```
--sgsnGparmName supportOfImsvops --sgsnGparmValue <string>
```

Description

Parameter is used to support IMS voice flags on the S4-SGSN feature.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

suspendAcknowledge - Suspend request acknowledge

Syntax

```
--sgsnGparmName suspendAcknowledge --sgsnGparmValue <string>
```

Description

When BSS sends a suspend request message to SGSN, this parameter determines whether or not the message is acknowledged at once if the subscriber is not in this PAPU unit.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

threeDigitMncMcc - Three digit MNC and MCC usage in DNS query

Syntax

```
--sgsnGparmName threeDigitMncMcc --sgsnGparmValue <string>
```

Description

Parameter controls whether three digit MNC and MCC is used in DNS query.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

throttleAlarms - Throttle RAISE and/or CANCEL alarms

Syntax

```
--sgsnGparmName throttleAlarms --sgsnGparmValue <string>
```

Description

Warning: For Nokia personnel only. Throttle RAISE and/or CANCEL alarms of SGSN processes according to the sgsnGParms throttleAlarmsTimeWindow and throttleAlarmsThresholdNumber. value = 0 means the feature is OFF, value = 1 means RAISE alarms are throttled, value = 2 means RAISE and CANCEL alarms are throttled. default value is 1.

Type / Range

Integer in the range of 0 to 2

Default

1

Name

throttleAlarmsThresholdNumber - Threshold number for alarm throttling

Syntax

```
--sgsnGparmName throttleAlarmsThresholdNumber --sgsnGparmValue  
<string>
```

Description

Warning: For Nokia personnel only. Depending on the value of throttleAlarms sgsnGparmName, if number of alarms with same alarmid that are raised/canceled during the time window configured by throttleAlarmsTimeWindow sgsnGparmName, exceeds this value then throttling starts. throttling stops when the time window ends. default value is 40 alarms.

Type / Range

Integer in the range of 1 to 100

Default

40

Name

throttleAlarmsTimeWindow - Time window for alarm throttling

Syntax

```
--sgsnGparmName throttleAlarmsTimeWindow --sgsnGparmValue <string>
```

Description

Warning: For Nokia personnel only. Depending on the value of throttleAlarms sgsnGparmName, if number of alarms with same alarmid that are raised/canceled during this time window, exceeds the value of throttleAlarmsThresholdNumber sgsnGparmName, then throttling starts. throttling stops when the time window ends. default value is 60 seconds.

Type / Range

Integer in the range of 1 to 300

Default

60

Name

timeZoneEncode - TimeZoneEncode

Syntax

```
--sgsnGparmName timeZoneEncode --sgsnGparmValue <string>
```

Description

This parameter controls the encoded time zone format.

Type / Range

Integer in the range of 0 to 1

Default

0

Name

tripletsToQuintuplets - Triplets to quintuplets

Syntax`--sgsnGparmName tripletsToQuintuplets --sgsnGparmValue <string>`**Description**

Parameter defines if new authentication vectors are asked from the HLR when triplets are received from the previous SGSN during inter SGSN RAU in gb or iu access.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

twoByteChargingChar - twoByteChargingChar

Syntax`--sgsnGparmName twoByteChargingChar --sgsnGparmValue <string>`**Description**

This parameter is used to define the support for two-byte charging characteristics value in SGSN, received from HLR. It is configurable to use four-bit, one-byte, or two-byte charging characteristics in SGSN.

Type / Range

Integer in the range of 0 to 2

Default

0

Name

ueAmbrDI - UE AMBR downlink default value

Syntax`--sgsnGparmName ueAmbrDI --sgsnGparmValue <string>`**Description**

Default value to be used as subscribed UE-AMBR downlink when this QoS parameter is not received from HSS (TS 23.060 par.15.2.2 if no values of UE-AMBR are received from the HSS, the SGSN shall set the UE-AMBR according to implementation specific policies (e.g. a pre-configured maximum UEAMBR)).

Type / Range

Integer in the range of 1 to 256

Default

16

Name

ueAmbrSupportRanap - UE-AMBR support

Syntax

```
--sgsnGparmName ueAmbrSupportRanap --sgsnGparmValue <string>
```

Description

Parameter determines whether SGSN sends UE-AMBR IE in RAB assignment request and relocation request messages to RNC.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ueAmbrUI - ueAmbrUI

Syntax

```
--sgsnGparmName ueAmbrUI --sgsnGparmValue <string>
```

Description

Default value to be used as subscribed UE-AMBR uplink when this QoS parameter is not received from HSS (TS 23.060 par.15.2.2 if no values of UE-AMBR are received from the HSS, the SGSN shall set the UE-AMBR according to implementation specific policies (e.g. a pre-configured maximum UEAMBR).).

Type / Range

Integer in the range of 1 to 256

Default

12

Name

updQosSuppFlg - Upgrade QoS supported bit in common flag IE for GTPv1

Syntax

```
--sgsnGparmName updQosSuppFlg --sgsnGparmValue <string>
```

Description

Parameter is used to control sending the upgrade QoS supported bit in the common flag IE from SGSN to GGSN/PCRF.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

usedGtpVersion - Used GTP version

Syntax

```
--sgsnGparmName usedGtpVersion --sgsnGparmValue <string>
```

Description

The parameter defines which version of the GTP is used towards the operators own GGSNs.

List

- 0 - Specify GTP version of 0.
- 1 - Specify GTP version of 1.
- 255 - Specify GTP version of 255.

Default

255

Name

userPlaneEchoSending - UserPlaneEchoSending

Syntax

```
--sgsnGparmName userPlaneEchoSending --sgsnGparmValue <string>
```

Description

This parameter controls where the GTP user plane echoes are sent. By default (value 4), the echoes are not sent to neither RNC nor GGSN. By changing the parameter, it is possible to enable echo sending to either RNC or GGSN, or enable it for both. Some RNCs do not respond to SGSNs echo requests if the Direct Tunnel is in use. The GTP user plane echo sending is optional in the 29.060 (GTP) specification. Allowed values are: 1 The user plane echoes are sent to RNCs and GGSNs. 2 The user plane echoes are sent only to RNCs. 3 The user plane echoes are sent only to GGSNs. 4 SGSN does not send user plane echoes.

Type / Range

Integer in the range of 1 to 4

Default

4

Name

zeroNonDrxTimer - zeroNonDrxTimer

Syntax

```
--sgsnGparmName zeroNonDrxTimer --sgsnGparmValue <string>
```

Description

Parameter enables setting the DRX-mode 'ON' to mobiles that send an attach request message with zero non-drx-timer value.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

zoneSw - Zone Switch

Syntax

```
--sgsnGparmName zoneSw --sgsnGparmValue <string>
```

Description

This parameter controls whether or not the mapping between zone names and RACs/LACs is enabled (switch on). If the parameter is set, zone name will be used as tag in the APN query. Otherwise, RAC+LAC will be used as tag in the APN query.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

7.31 cmm sgsnLai

sgsn location area identifiers. Syntax, description, parameters, and roles for cmm sgsnLai command.

Syntax

Description

Configure, list and show sgsn Location Area Identification.

Parameters

Name

mcc - MCC

Syntax

--mcc <digits>

Description

MCC digits of broadcasted PLMN.

Class

Required key.

Type / Range

Digit string of length 3

Name

mnc - MNC

Syntax

--mnc <digits>

Description

MNC digits of broadcasted PLMN.

Class

Required key.

Type / Range

Digit string of length 2 to 3

Name

lac - LAC

Syntax

--lac <integer>

Description

Location area code.

Class

Required key.

Type / Range

Integer in the range of 1 to 65,533

Name

epind - Epind

Syntax

--epind <integer>

Description

Equivalent PLMN list index the EPIND parameter defines to which location areas (LA) the listed MCC+MNC item is relevant. if the value of the LA's EPIND matches with the EPIND in the 'PLMN parameters' or if the EPIND in the 'PLMN parameters' is 0/all, the listed MCC+MNC item is relevant to the current LA. all relevant MCC+MNC values are collected into the EPLMN list sent to the MS. allowed values 1 ... 5, 0, 255 0 ... all 255 ... remove EPLMN index

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

homeSubscriber - Home subscriber

Syntax

--homeSubscriber <list-item>

Description

Control roaming flag for home subscriber on the LA level.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

visitorSubscriber - Visitor subscriber

Syntax

--visitorSubscriber <list-item>

Description

Control roaming flag for visitor subscriber on the LA level.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.32 cmm sgsnLboAllowedApn

LBO PGW Selection allowed APN. Syntax, description, parameters, and roles for cmm sgsnLboAllowedApn command.

Syntax**Description**

This command is used to provision allowed APNs list for the selection of PGW for a roamer UE LBO PDN connection.

Parameters**Name**

sgsnLboAllowedApnListName - name of sgsn LBO Allowed Apn List

Syntax

--sgsnLboAllowedApnListName <string>

Description

name of sgsn LBO Allowed Apn List

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

apnNi - APN network identifier

Syntax

--apnNi <string>

Description

APN network identifier

Class

Required key.

Type / Range

Digit string of length 1 to 62 with the allowed pattern of [0-9a-zA-Z\._]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.33 cmm sgsnLboAllowedApnList

name. Syntax, description, parameters, and roles for cmm sgsnLboAllowedApnList command.

Syntax**Description**

sgsn LBO Allowed Apn list name

Parameters**Name**

name - name

Syntax

--name <string>

Description

sgsn LBO Allowed Apn List name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.34 cmm sgsnLocalEndPtCfg

SGSN Local End Point Configuration. Syntax, description, parameters, and roles for cmm sgsnLocalEndPtCfg command.

Syntax**Description**

Configure, list and show interface configurations.

Parameters**Name**

bvcResetRetries - BVC reset retries

Syntax

--bvcResetRetries <integer>

Description

This parameter defines the number of retries in the BVC reset procedure. the values range from 0 to 255.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 255

Default

3

Name

bvcResetTimer - BVC reset timer

Syntax

--bvcResetTimer <integer>

Description

This parameter defines the BSSGP virtual connection (BVC) reset timer. the parameter guards the reset procedure of the BVC. the values range from 1 to 120 seconds.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 120

Default

3

Name

gni - GGSN IPv4 network identifier

Syntax

--gni <string>

Description

This parameter defines the external network the GGSN is connected to. in the GPRS backbone, the access point name is a reference to the GGSN to be used. this parameter is also applicable for S4.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 to 63 with the allowed pattern of [0-9a-zA-Z\._:]*

Name

gni6 - GGSN IPv6 network identifier

Syntax

--gni6 <string>

Description

This parameter defines the external network the GGSN is connected to. in the GPRS backbone, the access point name (APN) is a reference to the GGSN to be used. this parameter is also applicable for S4.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 to 63 with the allowed pattern of [0-9a-zA-Z\._:]*

Name

gniv4v6 - GGSN IPv4v6 network identifier

Syntax

--gniv4v6 <string>

Description

This parameter is used to store the primary default gateway network identifier for a dual stack PDP context (IPv4v6). this parameter is also applicable for S4.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 to 63 with the allowed pattern of [0-9a-zA-Z\._:]*

Name

goi - GGSN operator identifier

Syntax

--goi <string>

Description

This parameter defines in which PLMN GPRS backbone the GGSN is located. GOI last part has to be .gprs. this parameter is also applicable for S4.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 to 37 with the allowed pattern of [0-9a-zA-Z\._:]*

Name

i6hcack - IPv6 acknowledged data header compression

Syntax

--i6hcack <list-item>

Description

This parameter defines the IPv6 unacknowledged data header compression, true ... IPv6 unacknowledged data header compression is in use false ... IPv6 unacknowledged data header compression is not in use.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

i6hcunack - IPv6 unacknowledged data header compression

Syntax

--i6hcunack <list-item>

Description

This parameter defines the IPv6 acknowledged data header compression, true ... IPv6 acknowledged data header compression is in use false ... IPv6 acknowledged data header compression is not in use.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

icp - Gb over IP configuration UDP port

Syntax

--icp <integer>

Description

This parameter defines the pre-configured UDP port number for dynamic gb over IP. the values range from 49999 to 65535.

Class

Output only.

Type / Range

Integer in the range of 49,999 to 65,535

Default

49999

Name

IndDataFwdTimer - Timer for indirect data forwarding

Syntax

--IndDataFwdTimer <integer>

Description

This timer is used to determine the duration for which indirect data forwarding tunnel will be established during a handover procedure. the values range from 1 to 10.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

llcRetransmCounter - LLC retransmission count, SGSN proposed value

Syntax

--llcRetransmCounter <integer>

Description

This parameter defines the logical link control (LLC) retransmission count, SGSN proposed value. the values range from 1 to 15.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 15

Default

3

Name

llcRetransmTimer - LLC retransmission timer, SGSN proposed value

Syntax

--llcRetransmTimer <integer>

Description

This parameter defines the logical link control (LLC) retransmission timer, SGSN proposed value. the values range from 0 to 410 seconds.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 410

Default

10

Name

multihoming - Whether multihoming is in use or not

Syntax

--multihoming <list-item>

Description

Whether multihoming is in use or not.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

n3CreatePdp - N3-requests counter for create PDP context requests

Syntax

--n3CreatePdp <integer>

Description

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunnelling protocol (GTP) to send a create PDP context request message. the values range from 1 to 7.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

n3CreIndDataFwdTunnel - N3-request counter for create indirect data fwd tunnel request

Syntax

--n3CreIndDataFwdTunnel <integer>

Description

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunneling protocol v2 (GTPv2) to send a create indirect data forwarding tunnel request message. the values range from 1 to 7.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

n3CreSession - N3-request counter for create session request

Syntax

--n3CreSession <integer>

Description

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunneling protocol V2 (GTPv2) to send a create session request message. the values range from 1 to 7.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

n3Ctxtv2 - N3-request counter for context request, GTPv2

Syntax

--n3Ctxtv2 <integer>

Description

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunneling protocol v2 (GTPv2) to send a context request message. the values range from 1 to 7.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

n3DelIndDataFwdTunnel - N3-request counter for delete indirect data fwd tunnel request

Syntax

--n3DelIndDataFwdTunnel <integer>

Description

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunneling protocol v2 (GTPv2) to send a delete indirect data forwarding tunnel request message. the values range from 1 to 7.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

n3DelPdp - N3-requests counter for delete PDP context requests

Syntax

--n3DelPdp <integer>

Description

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunnelling protocol (GTP) to send a delete PDP context request message. the values range from 1 to 7.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

n3DelSession - N3-request counter for delete session request

Syntax

--n3DelSession <integer>

Description

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunneling protocol v2 (GTPv2) to send a delete session request message. the values range from 1 to 7.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

n3EchoCP - N3-requests counter for echo requests of control plane

Syntax**Description**

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunnelling protocol (GTP) to send an echo request message concerning control plane. the values range from 1 to 7.

Class

Output only.

Type / Range

Digit string of length 1 to 255

Default

5

Name

n3EchoCPv2 - N3-request counter for echo requests of control plane, GTPv2

Syntax

--n3EchoCPv2 <integer>

Description

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunneling protocol v2 (GTPv2) to send an echo request message concerning control plane. the values range from 1 to 7. this parameter is also applicable for S3S16.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

n3EchoUP - N3-requests counter for echo requests of user plane

Syntax

- -n3EchoUP <integer>

Description

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunnelling protocol (GTP) to send an echo request message concerning user plane. the values range from 1 to 7.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

n3ForwReloc - N3-requests counter for forward relocation requests

Syntax

- -n3ForwReloc <integer>

Description

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunnelling protocol (GTP) to send a forward relocation request message. the values range from 1 to 7.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

n3ForwRelocv2 - N3-request counter for fwd relocation request, GTPv2

Syntax

- -n3ForwRelocv2 <integer>

Description

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunneling protocol v2 (GTPv2) to send a forward relocation request message. the values range from 1 to 7.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

n3Id - N3-requests counter for identification requests GTPv1

Syntax

--n3Id <integer>

Description

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunnelling protocol (GTP) to send an identification request message. the values range from 1 to 7.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

n3Idv2 - N3-request counter for identification request, GTPv2

Syntax

--n3Idv2 <integer>

Description

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunneling protocol v2 (GTPv2) to send a identification request message. the values range from 1 to 7.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

n3ModBearer - N3-request counter for modify bearer request

Syntax

--n3ModBearer <integer>

Description

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunneling protocol v2 (GTPv2) to send a modify bearer request message. the values range from 1 to 7.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

n3RelBearer - N3-request counter for release access bearer request

Syntax

--n3RelBearer <integer>

Description

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunneling protocol v2 (GTPv2) to send a release access bearer request message. the values range from 1 to 7.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

n3RelocCancelv2 - N3-request counter for relocation cancel request, GTPv2

Syntax

--n3RelocCancelv2 <integer>

Description

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunneling protocol v2 (GTPv2) to send a relocation cancel request message. the values range from 1 to 7.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

n3SgsnCtxt - N3-requests counter for SGSN context requests

Syntax

--n3SgsnCtxt <integer>

Description

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunnelling protocol (GTP) to send an SGSN context request message. the values range from 1 to 7.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

n3UpdPdp - N3-requests counter for update PDP context requests

Syntax

--n3UpdPdp <integer>

Description

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunnelling protocol (GTP) to send an update PDP context request message. the values range from 1 to 7.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

nsAliveRetries - NS alive retries

Syntax

--nsAliveRetries <integer>

Description

This parameter defines the number of retries in the network service (NS) alive procedure. the values range from 0 to 255.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 255

Default

10

Name

nsAliveTimer - NS alive timer

Syntax

--nsAliveTimer <integer>

Description

This parameter defines the network service (NS) alive timer. the parameter guards the test procedure of the network service virtual connections (NS-VC). the values range from 1 to 60 seconds.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 60

Default

3

Name

nsTestTimer - NS test timer

Syntax

--nsTestTimer <integer>

Description

This parameter defines the test procedure of the network service virtual connections (NS-VC). the values range from 1 to 60 seconds.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 60

Default

60

Name

secgni - Secondary GGSN IPv4 network identifier

Syntax`--secgni <string>`**Description**

This parameter defines which PDP subscription from the HLR is used in the override of requested APN and IMEI based APN override features, remove parameter value using SECGNI=0. this parameter is also applicable for S4.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 to 63 with the allowed pattern of [0-9a-zA-Z\._:]*

Name

secgni6 - Secondary GGSN IPv6 network identifier

Syntax`--secgni6 <string>`**Description**

This parameter defines which PDP subscription from the HLR is used in the override of requested APN and IMEI based APN override features, remove parameter value using SECGNI6=0. this parameter is also applicable for S4.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 to 63 with the allowed pattern of [0-9a-zA-Z\._:]*

Name

secgniv4v6 - Secondary GGSN IPv4v6 network identifier

Syntax`--secgniv4v6 <string>`**Description**

This parameter is used to store the secondary default gateway network identifier for a dual stack PDP context (IPv4v6), remove parameter value using SECGNIV4V6=0. this parameter is also applicable for S4.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 to 63 with the allowed pattern of [0-9a-zA-Z\._:]*

Name

snsAddRetries - SNS add retries

Syntax

--snsAddRetries <integer>

Description

This parameter defines the number of SNS-ADD PDU retries in the SNS add procedure. the values range from 0 to 255.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 255

Default

3

Name

snsChangeWeightRetries - SNS changeweight retries

Syntax

--snsChangeWeightRetries <integer>

Description

This parameter defines the number of SNS-CHANGEWEIGHT PDU retries in the SNS changeweight procedure. the values range from 0 to 255.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 255

Default

3

Name

snsConfigRetries - SNS config retries

Syntax

--snsConfigRetries <integer>

Description

This parameter defines the number of SNS-CONFIG PDU retries in the SNS configuration procedure. the values range from 0 to 255.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 255

Default

3

Name

snsDeleteRetries - SNS delete retries

Syntax

--snsDeleteRetries <integer>

Description

This parameter defines the number of SNS-DELETE PDU retries in the SNS delete procedure. the values range from 0 to 255.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 255

Default

3

Name

snsProvTimer - SNS prov timer

Syntax

--snsProvTimer <integer>

Description

This parameter defines the interval of sub-network service (SNS) prov timer. the SNS prov timer guards the internet protocol (IP) SNS procedures. the values range from 1 to 10 seconds.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

5

Name

t3Cnxtv2 - T3-response timer of context response, GTPv2

Syntax

--t3Cnxtv2 <integer>

Description

This parameter defines the value for the timer that holds the maximum waiting time for a context response message to a context request message (GTPv2). the values range from 1 to 10.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

t3CreatePdp - T3-response timer of create PDP context response

Syntax

--t3CreatePdp <integer>

Description

This parameter defines the value for the timer that holds the maximum waiting time for a create PDP context response message to a create PDP context request message. the values range from 1 to 10 seconds in 1 second steps.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

t3CreIndDataFwdTunnel - T3-response timer of create indirect data fwd tunnel response

Syntax

--t3CreIndDataFwdTunnel <integer>

Description

This parameter defines the value for the timer that holds the maximum waiting time for a create indirect data forwarding tunnel response message to a create indirect data forwarding tunnel request message. the values range from 1 to 10.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

t3CreSession - T3-response timer of create session response in seconds

Syntax`--t3CreSession <integer>`**Description**

This parameter defines the value for the timer that holds the maximum waiting time for a create session response message to a create session request message. the values range from 1 to 10.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

t3DelIndDataFwdTunnel - T3-response timer of delete indirect data fwd tunnel response

Syntax`--t3DelIndDataFwdTunnel <integer>`**Description**

This parameter defines the value for the timer that holds the maximum waiting time for a delete indirect data forwarding tunnel response message to a delete indirect data forwarding tunnel request message. the values range from 1 to 10.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

t3DelPdp - T3-response timer of delete PDP context response

Syntax`--t3DelPdp <integer>`**Description**

This parameter defines the value for the timer that holds the maximum waiting time for a delete PDP context response message to a delete PDP context request message. the values range from 1 to 10 seconds in 1 second steps.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

t3DelSession - T3-response timer of delete session response

Syntax

--t3DelSession <integer>

Description

This parameter defines the value for the timer that holds the maximum waiting time for a delete session response message to a delete session request message. the values range from 1 to 10.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

t3EchoCP - T3-response timer of echo response of control plane

Syntax**Description**

This parameter defines the value for the timer that holds the maximum waiting time for an echo response message to an echo request message concerning control plane. the values range from 3 to 15 seconds.

Class

Output only.

Type / Range

Digit string of length 1 to 255

Default

3

Name

t3EchoCPv2 - T3-response timer of echo response of control plane, GTPv2

Syntax

--t3EchoCPv2 <integer>

Description

This parameter defines the value for the timer that holds the maximum waiting time for an echo response message to an echo request message concerning control plane (GTPv2). the values range from 3 to 15 seconds. this parameter is also applicable for S3S16.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 3 to 15

Default

3

Name

t3EchoUP - T3-response timer of echo response of user plane, from 3 to 15 seconds in 3 second steps

Syntax

--t3EchoUP <integer>

Description

This parameter defines the value for the timer that holds the maximum waiting time for an echo response message to an echo request message concerning user plane. the values range from 3 to 15 seconds.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 3 to 15

Default

3

Name

t3ForwReloc - T3-response timer of forward relocation response

Syntax

--t3ForwReloc <integer>

Description

This parameter defines the value for the timer that holds the maximum waiting time for a forward relocation response message to a forward relocation request message. the values range from 1 to 10 seconds in 1 second steps.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

t3ForwRelocv2 - T3-response timer of fwd relocation response, GTPv2

Syntax

--t3ForwRelocv2 <integer>

Description

This parameter defines the value for the timer that holds the maximum waiting time for a forward relocation request message to a forward relocation response message (GTPv2). the values range from 1 to 10.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

t3Id - T3-response timer of identification response GTPv1

Syntax

--t3Id <integer>

Description

This parameter defines the value for the timer that holds the maximum waiting time for an identification response message to an identification request message. the values range from 1 to 10 seconds in 1 second steps.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

t3Idv2 - T3-response timer of identification response, GTPv2

Syntax

--t3Idv2 <integer>

Description

This parameter defines the value for the timer that holds the maximum waiting time for a identification response message to a identification request message (GTPv2). the values range from 1 to 10.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

t3ModBearer - T3-response timer of modify bearer response

Syntax

--t3ModBearer <integer>

Description

This parameter defines the value for the timer that holds the maximum waiting time for a modify bearer response message to a modify bearer request message. the values range from 1 to 10.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

t3RelBearer - T3-response timer of release access bearer response

Syntax

--t3RelBearer <integer>

Description

This parameter defines the value for the timer that holds the maximum waiting time for a release access bearer response message to a release access bearer request message. the values range from 1 to 10.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

t3RelocCancelv2 - T3-response timer of relocation cancel response, GTPv2

Syntax

--t3RelocCancelv2 <integer>

Description

This parameter defines the value for the timer that holds the maximum waiting time for a relocation cancel response message to a relocation cancel request (GTPv2). the values range from 1 to 10.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

t3SgsnCtxt - T3-response timer of SGSN context response

Syntax

--t3SgsnCtxt <integer>

Description

This parameter defines the value for the timer that holds the maximum waiting time for an SGSN context response message to an SGSN context request message. the values range from 1 to 10 seconds in 1 second steps.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

t3tunnel - T3 tunnel

Syntax

--t3tunnel <integer>

Description

This parameter defines the T3 tunnel. the T3 tunnel parameter defines the timer that controls the time when protocol data units (PDUs) are forwarded from the old SGSN to the new SGSN as part of the inter-sgsn routing area update procedure. the values range from 20 to 60 seconds in 1 second steps. this parameter is also applicable for S3S16.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 20 to 60

Default

20

Name

t3UpdPdp - T3-response timer of update PDP context response

Syntax

--t3UpdPdp <integer>

Description

This parameter defines the value for the timer that holds the maximum waiting time for an update PDP context response message to an update PDP context request message. the values range from 1 to 10 seconds in 1 second steps.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

techo - Techo

Syntax

--techo <integer>

Description

This parameter defines the timer that holds the time interval for the path validation procedure. the values range from 60 to 600 seconds.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 60 to 600

Default

120

Name

techoV2 - Techo, GTPv2

Syntax`--techoV2 <integer>`**Description**

This parameter defines the timer that holds the time interval for the path validation procedure (GTPv2). the values range from 60 to 600 seconds. this parameter is also applicable for S3S16.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 60 to 600

Default

120

Name

id - Interface name

Syntax`<list-item>`**Description**

Interface name.

Class

Required value.

List

- GBNS - Specify gbns interface.
- GNSGSN - Specify Gn interface to an SGSN.
- Gr - Specify gr interface to an HLR.
- lu - Specify iu interface to an RNC.
- S13 - Specify S13 interface to an EIR.
- S3S16 - Specify S3/S16 interface to an SGSN.
- S4 - Specify S4 interface to an SGW.
- S6d - Specify S6d interface to an HSS.

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.35 cmm sgsnPagingGroup

rnc paging group members. Syntax, description, parameters, and roles for cmm sgsnPagingGroup command.

Syntax

Description

This command is used to add RNC members in a paging group. Every group can have up to 32 RNCs.

Parameters

Name

groupId - GroupId of RNC

Syntax

--groupId <integer>

Description

GroupId of RNC.

Class

Required key.

Type / Range

Integer in the range of 1 to 4,900

Name

rncName - RNC name

Syntax

--rncName <string>

Description

RNC name. this parameter should match parameter rncname of table rnc.

Class

Required key.

Type / Range

Digit string of length 1 to 10 with the allowed pattern of [0-9a-zA-Z\._:-]*

Name

rncRole - RNC role

Syntax

--rncRole <integer>

Description

RNC role in group, 1 normal, 2 master, 3 standby.

Class

Optional.

Type / Range

Integer in the range of 1 to 3

Default

1

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.36 cmm sgsnPagingGroupList

paging groups. Syntax, description, parameters, and roles for cmm sgsnPagingGroupList command.

Syntax

Description

This command is used to configure the paging groups under SGSN. The maximum number of paging groups supported is 1000.

Parameters

Name

groupId - GroupId

Syntax

--groupId <integer>

Description

Paging group id.

Class

Required key.

Type / Range

Integer in the range of 1 to 65,535

Name

name - Name

Syntax

--name <string>

Description

Paging group name.

Class

Required on create.

Type / Range

Digit string of length 1 to 10 with the allowed pattern of [0-9a-zA-Z\._ :]*

Name

masterRncName - MasterId

Syntax

--masterRncName <string>

Description

Master id. this parameter should match parameter rncname of table rnc. updated automatically.

Class

Read only.

Type / Range

Digit string of length 0 to 10 with the allowed pattern of [0-9a-zA-Z\.\-:]*

Default

""

Name

standbyRncName - MasterId

Syntax

--standbyRncName <string>

Description

Standby id. this parameter should match parameter rncname of table rnc. updated automatically.

Class

Read only.

Type / Range

Digit string of length 0 to 10 with the allowed pattern of [0-9a-zA-Z\.\-:]*

Default

""

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.37 cmm sgsnPgwSelectionRangeServices

Services based on PGW/GGSN selection range. Syntax, description, parameters, and roles for cmm sgsnPgwSelectionRangeServices command.

Syntax

Description

Services based on PGW/GGSN selection range

Parameters

Name

entryId - Entry Id

Syntax

--entryId <integer>

Description

Entry Id

Class

Required key.

Type / Range

Integer in the range of 1 to 500

Name

apnNi - APN network identifier

Syntax

--apnNi <string>

Description

APN network identifier

Class

Required on create.

Type / Range

Use this parameter to specify the APN Domain Name

Name

ggsnIpAddress - GGSN IP Address

Syntax

--ggsnIpAddress <string>

Description

Use this parameter to specify the GGSN IP Address

Class

Required on create.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

maxMsin - Maximum MSIN

Syntax

--maxMsin <digits>

Description

The maximum IMSI MSIN value or MSISDN. The length 2-15 is only allowed for an MSISDN type. For an IMSI the allowed length is 8-10.

Restrictions

The length 2-15 is only allowed for an MSISDN type. For an IMSI the allowed length is 8-10.

Class

Required on create.

Type / Range

Digit string of length 2 to 15

Name

minMsin - Minimum MSIN

Syntax

--minMsin <digits>

Description

The minimum IMSI MSIN value or MSISDN. The length 2-15 is only allowed for an MSISDN type. For an IMSI the allowed length is 8-10.

Restrictions

The length 2-15 is only allowed for an MSISDN type. For an IMSI the allowed length is 8-10.

Class

Required on create.

Type / Range

Digit string of length 2 to 15

Name

rangeServiceType - Range service type

Syntax

--rangeServiceType <list-item>

Description

Range service type

Class

Required on create, not modifiable.

List

- imsi - Specify range service type of IMSI.

- msisdn - Specify range service type of MSISDN.

Name

mcc - Mobile Country Code

Syntax

--mcc <digits>

Description

Mobile Country Code. Rules including the UE's MCC and MNC will be ignored if sgsnGParms staticGwForBlockService is disabled.

Class

Optional.

Type / Range

Digit string of length 0 or 3

Default

"

Name

mnc - Mobile Network Code

Syntax

--mnc <digits>

Description

Mobile Network Code. Rules including the UE's MCC and MNC will be ignored if sgsnGParms staticGwForBlockService is disabled.

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 3

Default

"

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.38 cmm sgsnS13LocalEndPtCfg

S13 SGSN Local Endpoint Configuration. Syntax, description, parameters, and roles for cmm sgsnS13LocalEndPtCfg command.

Syntax

Description

Configure, list and show parameters related to S13 interface.

Parameters

Name

id - Interface name

Syntax

<list-item>

Description

Interface name.

Class

Required value.

List

- GBNS - Specify gbns interface.
- GNSGSN - Specify Gn interface to an SGSN.
- Gr - Specify gr interface to an HLR.
- lu - Specify iu interface to an RNC.
- S13 - Specify S13 interface to an EIR.
- S3S16 - Specify S3/S16 interface to an SGSN.
- S4 - Specify S4 interface to an SGW.
- S6d - Specify S6d interface to an HSS.

Name

multihoming - Whether multihoming is in use or not

Syntax

--multihoming <list-item>

Description

Whether multihoming is in use or not.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.39 cmm sgsnS3S16LocalEndPtCfg

S13S16 SGSN Local Endpoint Configuration. Syntax, description, parameters, and roles for cmm sgsnS3S16LocalEndPtCfg command.

Syntax**Description**

Configure, list and show parameters related to S13S16 interface.

Parameters**Name**

id - Interface name

Syntax

<list-item>

Description

Interface name.

Class

Required value.

List

- GBNS - Specify gbns interface.
- GNSGSN - Specify Gn interface to an SGSN.
- Gr - Specify gr interface to an HLR.
- lu - Specify iu interface to an RNC.
- S13 - Specify S13 interface to an EIR.
- S3S16 - Specify S3/S16 interface to an SGSN.
- S4 - Specify S4 interface to an SGW.
- S6d - Specify S6d interface to an HSS.

Name

n3Ctxtv2 - N3-request counter for context request, GTPv2

Syntax

--n3Ctxtv2 <integer>

Description

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunneling protocol v2 (GTPv2) to send a context request message. the values range from 1 to 7.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

n3ForwRelocv2 - N3-request counter for fwd relocation request, GTPv2

Syntax

--n3ForwRelocv2 <integer>

Description

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunneling protocol v2 (GTPv2) to send a forward relocation request message. the values range from 1 to 7.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

n3Idv2 - N3-request counter for identification request, GTPv2

Syntax

--n3Idv2 <integer>

Description

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunneling protocol v2 (GTPv2) to send a identification request message. the values range from 1 to 7.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

n3RelocCancelv2 - N3-request counter for relocation cancel request, GTPv2

Syntax

--n3RelocCancelv2 <integer>

Description

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunneling protocol v2 (GTPv2) to send a relocation cancel request message. the values range from 1 to 7.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

t3Cnxtv2 - T3-response timer of context response, GTPv2

Syntax

--t3Cnxtv2 <integer>

Description

This parameter defines the value for the timer that holds the maximum waiting time for a context response message to a context request message (GTPv2). the values range from 1 to 10.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

t3ForwRelocv2 - T3-response timer of fwd relocation response, GTPv2

Syntax

--t3ForwRelocv2 <integer>

Description

This parameter defines the value for the timer that holds the maximum waiting time for a forward relocation request message to a forward relocation response message (GTPv2). the values range from 1 to 10.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

t3Idv2 - T3-response timer of identification response, GTPv2

Syntax

--t3Idv2 <integer>

Description

This parameter defines the value for the timer that holds the maximum waiting time for a identification response message to a identification request message (GTPv2). the values range from 1 to 10.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

t3RelocCancelv2 - T3-response timer of relocation cancel response, GTPv2

Syntax

--t3RelocCancelv2 <integer>

Description

This parameter defines the value for the timer that holds the maximum waiting time for a relocation cancel response message to a relocation cancel request (GTPv2). the values range from 1 to 10.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

3

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.40 cmm sgsnS4LocalEndPtCfg

SGSN S4 Local Endpoint Configuration.. Syntax, description, parameters, and roles for cmm sgsnS4LocalEndPtCfg command.

Syntax

Description

Configure, list and show parameters related to S4 interface.

Parameters

Name

id - Interface name

Syntax

<list-item>

Description

Interface name.

Class

Required value.

List

- GBNS - Specify gbns interface.
- GNSGSN - Specify Gn interface to an SGSN.
- Gr - Specify gr interface to an HLR.
- lu - Specify iu interface to an RNC.
- S13 - Specify S13 interface to an EIR.
- S3S16 - Specify S3/S16 interface to an SGSN.
- S4 - Specify S4 interface to an SGW.
- S6d - Specify S6d interface to an HSS.

Name

IndDataFwdTimer - Timer for indirect data forwarding

Syntax

--IndDataFwdTimer <integer>

Description

This timer is used to determine the duration for which indirect data forwarding tunnel will be established during a handover procedure. the values range from 1 to 10.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

n3CreIndDataFwdTunnel - N3-request counter for create indirect data fwd tunnel request

Syntax`--n3CreIndDataFwdTunnel <integer>`**Description**

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunneling protocol v2 (GTPv2) to send a create indirect data forwarding tunnel request message. the values range from 1 to 7.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

n3CreSession - N3-request counter for create session request

Syntax`--n3CreSession <integer>`**Description**

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunneling protocol V2 (GTPv2) to send a create session request message. the values range from 1 to 7.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

n3DelIndDataFwdTunnel - N3-request counter for delete indirect data fwd tunnel request

Syntax`--n3DelIndDataFwdTunnel <integer>`**Description**

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunneling protocol v2 (GTPv2) to send a delete indirect data forwarding tunnel request message. the values range from 1 to 7.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

n3DelSession - N3-request counter for delete session request

Syntax

--n3DelSession <integer>

Description

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunneling protocol v2 (GTPv2) to send a delete session request message. the values range from 1 to 7.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

n3EchoCPv2 - N3-request counter for echo requests of control plane, GTPv2

Syntax

--n3EchoCPv2 <integer>

Description

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunneling protocol v2 (GTPv2) to send an echo request message concerning control plane. the values range from 1 to 7. this parameter is also applicable for S3S16.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

n3ModBearer - N3-request counter for modify bearer request

Syntax`--n3ModBearer <integer>`**Description**

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunneling protocol v2 (GTPv2) to send a modify bearer request message. the values range from 1 to 7.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

n3RelBearer - N3-request counter for release access bearer request

Syntax`--n3RelBearer <integer>`**Description**

This parameter defines the value for the counter that holds the maximum number of attempts made by the GPRS tunneling protocol v2 (GTPv2) to send a release access bearer request message. the values range from 1 to 7.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 7

Default

5

Name

t3CreIndDataFwdTunnel - T3-response timer of create indirect data fwd tunnel response

Syntax`--t3CreIndDataFwdTunnel <integer>`**Description**

This parameter defines the value for the timer that holds the maximum waiting time for a create indirect data forwarding tunnel response message to a create indirect data forwarding tunnel request message. the values range from 1 to 10.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

t3CreSession - T3-response timer of create session response in seconds

Syntax

--t3CreSession <integer>

Description

This parameter defines the value for the timer that holds the maximum waiting time for a create session response message to a create session request message. the values range from 1 to 10.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

t3DelIndDataFwdTunnel - T3-response timer of delete indirect data fwd tunnel response

Syntax

--t3DelIndDataFwdTunnel <integer>

Description

This parameter defines the value for the timer that holds the maximum waiting time for a delete indirect data forwarding tunnel response message to a delete indirect data forwarding tunnel request message. the values range from 1 to 10.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

t3DelSession - T3-response timer of delete session response

Syntax

--t3DelSession <integer>

Description

This parameter defines the value for the timer that holds the maximum waiting time for a delete session response message to a delete session request message. the values range from 1 to 10.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

t3EchoCPv2 - T3-response timer of echo response of control plane, GTPv2

Syntax

--t3EchoCPv2 <integer>

Description

This parameter defines the value for the timer that holds the maximum waiting time for an echo response message to an echo request message concerning control plane (GTPv2). the values range from 3 to 15 seconds. this parameter is also applicable for S3S16.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 3 to 15

Default

3

Name

t3ModBearer - T3-response timer of modify bearer response

Syntax

--t3ModBearer <integer>

Description

This parameter defines the value for the timer that holds the maximum waiting time for a modify bearer response message to a modify bearer request message. the values range from 1 to 10.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

t3RelBearer - T3-response timer of release access bearer response

Syntax

--t3RelBearer <integer>

Description

This parameter defines the value for the timer that holds the maximum waiting time for a release access bearer response message to a release access bearer request message. the values range from 1 to 10.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 10

Default

3

Name

techoV2 - Techo, GTPv2

Syntax

--techoV2 <integer>

Description

This parameter defines the timer that holds the time interval for the path validation procedure (GTPv2). the values range from 60 to 600 seconds. this parameter is also applicable for S3S16.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 60 to 600

Default

120

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.41 cmm sgsnS6dLocalEndPtCfg

S6d SGSN Local Endpoint Configuration. Syntax, description, parameters, and roles for cmm sgsnS6dLocalEndPtCfg command.

Syntax

Description

Configure, list and show parameters related to S6d interface.

Parameters

Name

id - Interface name

Syntax

<list-item>

Description

Interface name.

Class

Required value.

List

- GBNS - Specify gbns interface.
- GNSGSN - Specify Gn interface to an SGSN.
- Gr - Specify gr interface to an HLR.
- lu - Specify lu interface to an RNC.

- S13 - Specify S13 interface to an EIR.
- S3S16 - Specify S3/S16 interface to an SGSN.
- S4 - Specify S4 interface to an SGW.
- S6d - Specify S6d interface to an HSS.

Name

multihoming - Whether multihoming is in use or not

Syntax

--multihoming <list-item>

Description

Whether multihoming is in use or not.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.42 cmm sgsnTimer

timer configurations. Syntax, description, parameters, and roles for cmm sgsnTimer command.

Syntax

Description

Configure generic timers that are used in SGSN. Interface related timers are defined with the interface group.

Parameters

Name

cert - Cert

Syntax

--cert <integer>

Description

Timeout for diameter base connection establishment in seconds.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 5

Default

3

Name

cit - Cit

Syntax

--cit <integer>

Description

Inactivity timer for diameter connections in seconds.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 5 to 60

Default

30

Name

csPagingTimer - CS paging timer

Syntax

--csPagingTimer <integer>

Description

CS paging sending time to unknown MS after reset in seconds.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 255

Default

20

Name

detachTimer - Detach timer

Syntax

--detachTimer <integer>

Description

Defines the timer that indicates when an inactive subscriber is detached, format <HH>:<MM>

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 1,440

Default

0

Name

diaConnRetryCount - Dia conn retry count

Syntax

--diaConnRetryCount <integer>

Description

Diameter connection retry after connection lost[0 equals infinite retries].

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 100

Default

0

Name

dtEstabTimer - Dt estab timer

Syntax

--dtEstabTimer <integer>

Description

Direct tunnel establishment timer in seconds.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 120

Default

30

Name

dwrr - Dwrr

Syntax

--dwrr <integer>

Description

Number of retries for DWR diameter message if not responded.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 10

Default

2

Name

dwr - Dwrt

Syntax

--dwr <integer>

Description

Timeout for DWR diameter message in seconds.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 5

Default

2

Name

eirResponseTimer - EIR Response Timer

Syntax

--eirResponseTimer <integer>

Description

Time to wait for receiving CheckIMEI response from EIR (Gf)

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 500 to 9000 in units of Millisecond

Default

3000

Name

gsmsRetransmCount - Gsms retransm count

Syntax

--gsmsRetransmCount <integer>

Description

Retransmission.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 3

Default

2

Name

msReachabTimer - Ms reachab timer

Syntax

--msReachabTimer <integer>

Description

Ms reachable timer in seconds.

Class

Output only.

Type / Range

Integer in the range of 0 to 11,160

Default

7200

Name

offloadPrauTimer - Offload prau timer

Syntax

--offloadPrauTimer <integer>

Description

Offload periodic RA update timer in seconds.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 14,400

Default

4

Name

olct - Olct

Syntax

--olct <integer>

Description

Overload control timer for diameter connections in seconds.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 600

Default

0

Name

overwriteMsRdy - Overwrite ms rdy

Syntax

--overwriteMsRdy <list-item>

Description

Override MS ready state timer.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

perRaUpTimer - Per ra up timer

Syntax

--perRaUpTimer <integer>

Description

Periodic RA update timer in seconds.

Class

Output only.

Type / Range

Integer in the range of 0 to 10,800

Default

3240

Name

psPagingInterval2g - PS paging interval 2g

Syntax

--psPagingInterval2g <integer>

Description

Paging interval timer for 2G in seconds.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 or 60 to 11,160

Default

0

Name

psPagingInterval3g - PS paging interval 3g

Syntax

--psPagingInterval3g <integer>

Description

Paging interval timer for 3G in seconds.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 or 60 to 11,160

Default

0

Name

psPagingRaCount2g - PS paging ra count 2g

Syntax

--psPagingRaCount2g <integer>

Description

RA paging area for 2G.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 5

Default

2

Name

psPagingRaCount3g - PS paging ra count 3g

Syntax

--psPagingRaCount3g <integer>

Description

RA paging area for 3G.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 5

Default

2

Name

psPagingSgsnCount2g - PS paging sgsn count 2g

Syntax

--psPagingSgsnCount2g <integer>

Description

SGSN paging area for 2G.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 5

Default

0

Name

psPagingSgsnCount3g - PS paging sgsn count 3g

Syntax

--psPagingSgsnCount3g <integer>

Description

SGSN paging area for 3G.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 5

Default

0

Name

psPagingTimer2g - PS paging timer 2g

Syntax

--psPagingTimer2g <integer>

Description

RA paging repetition for 2G in seconds by 0.1 seconds step.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 25

Default

4

Name

psPagingTimer3g - PS paging timer 3g

Syntax

--psPagingTimer3g <integer>

Description

RA paging repetition for 3G in seconds.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 25

Default

4

Name

readyTimer - Ready timer

Syntax`--readyTimer <integer>`**Description**

Ready state timer in seconds.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 11,160

Default

44

Name

rest - Rest

Syntax`--rest <integer>`**Description**

Resend timer for diameter application interface requests in seconds.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 3

Default

2

Name

standbyTimer - Standby timer

Syntax`--standbyTimer <integer>`**Description**

Standby state timer in seconds.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 11,160

Default

44

Name

tct - Tct

Syntax

--tct <integer>

Description

Transport connection retry timeout for diameter connections in seconds.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 60

Default

30

Name

id - Id

Syntax

config

Description

Automatic.

Class

Required value.

List

- config - Configuration - The id for the configuration list.

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

7.43 cmm sgw

Serving Gateway. Syntax, description, parameters, and roles for cmm sgw command.

Syntax

Description

This command is used to administer a Serving Gateway and includes parameters for defining the serving gateway name, pool ID, IP address, and SGW network protocol.

Parameters

Name

s11Ip - S11 IP

Syntax

--s11Ip <string>

Description

Provisions the MME, for each SGW pool ID, with the S11 IP address of each SGW node that is a member of that SGW pool.

Class

Required key.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

hostName - SGW host name

Syntax

--hostName <string>

Description

SGW host name.

Class

Required on create.

Type / Range

Digit string of length 1 to 255

Name

name - SGW name

Syntax

--name <string>

Description

Serving gateway name.

Class

Required on create.

Type / Range

Digit string of length 1 to 64 with the allowed pattern of [0-9a-zA-Z\.\-_:]*

Name

sgwPoolId - SGW pool ID

Syntax

--sgwPoolId <string>

Description

Serving gateway pool ID this serving gateway is assigned to.

Class

Required on create, not modifiable.

Type / Range

Digit string of length 1 to 20 with the allowed pattern of [0-9a-zA-Z\.\-_:]*

Name

netProtocol - Inter-Gateway protocol

Syntax

--netProtocol <list-item>

Description

Provisions the MME with the specification with the SGW network protocol.

Class

Optional.

List

- BOTH - Specify SGW supports S5/S8 based on GTP and also based on PMIP.
- GTP - Specify SGW supports S5/S8 based on GTP.
- PMIP - Specify SGW supports S5/S8 based on PMIP.

Default

GTP

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.44 cmm sgwPoolTai

Serving Gateway Pool to TAI. Syntax, description, parameters, and roles for `cmm sgwPoolTai` command.

Syntax

Description

This command is used to administer a Serving Gateway Pool to TAI mapping and includes parameters for defining the mobile network name, serving gateway pool ID, and Tracking Area Code (TAC).

Parameters

Name

plmnName - Mobile network name

Syntax

--plmnName <string>

Description

Name that identifies mobile network (alternate key).

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

sgwPoolId - SGW pool ID

Syntax

--sgwPoolId <string>

Description

ID for a serving gateway pool.

Class

Required key.

Type / Range

Digit string of length 1 to 20 with the allowed pattern of [0-9a-zA-Z\._ :]*

Name

tac - Tracking area code

Syntax

--tac <integer>

Description

Tracking area code.

Class

Required key.

Type / Range

Integer in the range of 1 to 65,535

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.45 cmm sgwSelectionRangeServices

Services based on SGW selection range. Syntax, description, parameters, and roles for cmm sgwSelectionRangeServices command.

Syntax**Description**

Services based on SGW selection range.

Parameters**Name**

uePlmnServicesName - Name of services based on UE network and served network

Syntax

--uePlmnServicesName <string>

Description

Name that identifies roaming agreement.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

entryId - Entry ID

Syntax

--entryId <integer>

Description

Entry ID.

Class

Required key.

Type / Range

Integer in the range of 1 to 10,000

Name

gwPoolName - SGW pool name

Syntax

--gwPoolName <string>

Description

Gateway pool name this gateway is assigned to.

Class

Required on create.

Type / Range

Digit string of length 1 to 20 with the allowed pattern of [0-9a-zA-Z\.\-_:]*

Name

maxMsin - Maximum MSIN

Syntax

--maxMsin <digits>

Description

The maximum IMSI MSIN value.

Restrictions

The length 2-15 is only allowed for an MSISDN type. For an IMSI the allowed length is 8-10.

Class

Required on create.

Type / Range

Digit string of length 0 or 2 to 15

Name

minMsin - Minimum MSIN

Syntax

--minMsin <digits>

Description

The minimum IMSI MSIN value.

Restrictions

The length 2-15 is only allowed for an MSISDN type. For an IMSI the allowed length is 8-10.

Class

Required on create.

Type / Range

Digit string of length 0 or 2 to 15

Name

rangeServiceType - Range service type

Syntax

--rangeServiceType <list-item>

Description

Range service type.

Class

Required on create, not modifiable.

List

- imsi - Specify range service type of IMSI.
- msisdn - Specify range service type of MSISDN.

Name

areaListName - Name of TAI list

Syntax

--areaListName <string>

Description

Name of TAI list.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\- _:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.46 cmm sgwSelectionService

Service based on SGW selection. Syntax, description, parameters, and roles for cmm sgwSelectionService command.

Syntax**Description**

Service based on SGW selection.

Parameters**Name**

areaListName - Name of TAI list

Syntax

--areaListName <string>

Description

Name of TAI list.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

gwPoolName - SGW pool name

Syntax

--gwPoolName <string>

Description

Gateway pool name this gateway is assigned to.

Class

Required key.

Type / Range

Digit string of length 1 to 20 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

servedPlmnName - Served PLMN Name

Syntax

--servedPlmnName <string>

Description

The name of the served PLMN.

Class

Required on create, not modifiable.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.47 cmm smfIsolationGeneralSettings

SMF Isolation General Settings. Syntax, description, parameters, and roles for cmm smfIsolationGeneralSettings command.

Syntax

Description

Includes general settings related to SMF Isolation.

Parameters

Name

service - Service

Syntax

--service <list-item>

Description

SMF service for which the settings in the other parameters apply.

Class

Required key.

List

- Nsmf_PDUSession - SMF PDU Session Service - The network function service type of Session Management Function PDU session.

Default

Nsmf_PDUSession

Name

action3xx - Action 3xx

Syntax

--action3xx <list-item>

Description

Default action for 3xx status class responses.

Class

Optional.

List

- Isolate - Isolate the Service Instance
- KPI_Fail - KPI Fail - Increase the Failure Counter used for the Service Instance KPI
- KPI_Success - KPI Success - Increase the Success counter used for the Service Instance KPI

Default

KPI_Fail

Name

action4xx - Action 4xx

Syntax

--action4xx <list-item>

Description

Default action for 4xx status class responses.

Class

Optional.

List

- Isolate - Isolate the Service Instance
- KPI_Fail - KPI Fail - Increase the Failure Counter used for the Service Instance KPI
- KPI_Success - KPI Success - Increase the Success counter used for the Service Instance KPI

Default

KPI_Success

Name

action5xx - Action 5xx

Syntax

--action5xx <list-item>

Description

Default action for 5xx status class responses.

Class

Optional.

List

- Isolate - Isolate the Service Instance
- KPI_Fail - KPI Fail - Increase the Failure Counter used for the Service Instance KPI
- KPI_Success - KPI Success - Increase the Success counter used for the Service Instance KPI

Default

KPI_Fail

Name

actionTimeout - Action Timeout

Syntax

--actionTimeout <list-item>

Description

Action for Timeout.

Class

Optional.

List

- Isolate - Isolate the Service Instance
- KPI_Fail - KPI Fail - Increase the Failure Counter used for the Service Instance KPI
- KPI_Success - KPI Success - Increase the Success counter used for the Service Instance KPI

Default

KPI_Fail

Name

enable - Enable

Syntax

--enable <list-item>

Description

Enables the SMF Service Instance Isolation functionality.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

isolationDuration - Isolation Duration

Syntax

--isolationDuration <integer>

Description

Duration (in seconds) the isolation lasts. 0 effectively means no isolation takes place.

Class

Optional.

Type / Range

Integer in the range of 0 to 3,600 in units of Second

Default

300

Name

kpiIsolationThreshold - KPI Isolation Threshold

Syntax

--kpiIsolationThreshold <integer>

Description

Threshold of success/attempts percent under which a service (set) instance will be isolated.
When value is set to 0 the service instance is not isolated based on KPIs.

Class

Optional.

Type / Range

Integer in the range of 0 to 100

Default

90

Name

kpiMinSamples - KPI Minimum Samples

Syntax

--kpiMinSamples <integer>

Description

Minimum number of requests that are required in order to consider isolation for a service (set) instance based on KPIs.

Class

Optional.

Type / Range

Integer in the range of 1 to 1,000,000

Default

100

Name

kpiSamplingPeriod - KPI Sampling Period

Syntax

--kpiSamplingPeriod <integer>

Description

Sampling period (in seconds) for which response success and failures are aggregated to produce KPI for specific service (set) instance.

Class

Optional.

Type / Range

Integer in the range of 5 to 300 in units of Second

Default

30

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

7.48 cmm smfIsolationPerCauseCodeSettings

SMF Isolation Per Cause Code Settings. Syntax, description, parameters, and roles for cmm smfIsolationPerCauseCodeSettings command.

Syntax**Description**

Includes SMF isolation related rules for specific Cause/Status codes. Used when the action for a Causes/Status codes combination for a service needs to differentiate from the default behaviour provisioned within the SMF Isolation General Settings command.

Parameters**Name**

service - Service

Syntax

--service <list-item>

Description

SMF service for which the settings in the other parameters apply.

Class

Required key.

List

- Nsmf_PDUSession - SMF PDU Session Service - The network function service type of Session Management Function PDU session.

Default

Nsmf_PDUSession

Name

statusCode - Status Code

Syntax

--statusCode <list-item>

Description

HTTP Status Code.

Class

Required key.

List

- 400_Bad_Request - 400 Bad Request
- 401_Unauthorized - 401 Unauthorized
- 403_Forbidden - 403 Forbidden
- 404_Not_Found - 404 Not Found
- 405_Method_Not_Allowed - 405 Method Not Allowed
- 406_Not_Acceptable - 406 Not Acceptable
- 408_Request_Timeout - 408 Request Timeout
- 409_Conflict - 409 Conflict
- 410_Gone - 410 Gone
- 411_Length_Required - 411 Length Required
- 412_Precondition_Failed - 412 Precondition Failed
- 413_Payload_Too_Large - 413 Payload Too Large
- 414_URI_Too_Long - 414 URI Too Long
- 415_Unsupported_Media_Type - 415 Unsupported Media Type
- 429_Too_Many_Requests - 429 Too Many Requests
- 500_Internal_Server_Error - 500 Internal Server Error
- 501_Not_Implemented - 501 Not Implemented
- 502_Bad_Gateway - 502 Bad Gateway
- 503_Service_Unavailable - 503 Service Unavailable
- 504_Gateway_Timeout - 504 Gateway Timeout

Name

errorCause - Error Cause

Syntax

--errorCause <string>

Description

Application Error Cause value. The provided string needs to match with the Cause in the response. 'NULL' string literal has special meaning and is used to match cases where Response did not include Application Error Cause.

Class

Required key.

Type / Range

SBI Application Error Cause value. Digit string of length 1 to 64 with the allowed pattern of [A-Z_]*

Default

"NULL"

Name

action - Action

Syntax

--action <list-item>

Description

Action to be applied for specific status code and error cause.

Class

Optional.

List

- Isolate - Isolate the Service Instance
- KPI_Fail - KPI Fail - Increase the Failure Counter used for the Service Instance KPI
- KPI_Success - KPI Success - Increase the Success counter used for the Service Instance KPI

Default

KPI_Fail

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.49 cmm smscGrp

SMSC Group Profile. Syntax, description, parameters, and roles for cmm smscGrp command.

Syntax

Description

Administer SMSC Group Profile

Parameters

Name

smscGrpListName - SMSC Group List Name

Syntax

--smscGrpListName <string>

Description

SMSC Group List Name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

smscOrderId - SMSC Order Id

Syntax

--smscOrderId <integer>

Description

SMSC order in group.

Class

Required key.

Type / Range

Integer in the range of 1 to 16

Name

smscName - SMSC Name

Syntax

--smscName <string>

Description

SMSC name specified in remote end point configuration.

Class

Required on create, not modifiable.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

lbWeight - Weight for load balancing

Syntax

--lbweight <integer>

Description

Weight for load balancing algorithm.

Class

Optional.

Type / Range

Integer in the range of 0 to 100

Default

0

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.50 cmm smscGrpList

SMSC Group List. Syntax, description, parameters, and roles for cmm smscGrpList command.

Syntax

Description

Administer SMSC Group List.

Parameters

Name

name - SMSC Group Profile Name

Syntax

--name <string>

Description

SMSC Group Profile Name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.51 cmm smscToGrp

SMSC Number to Diameter Destination. Syntax, description, parameters, and roles for cmm smscToGrp command.

Syntax

Description

This command defines for the SMS in MME feature the mapping from a UE provided SMSC address to an SMSC group. The mapping is used for finding the SMSC to which to forward a mobile originated SMS.

Parameters

Name

mappingId - Mapping ID

Syntax

--mappingId <integer>

Description

Identifier for the mapping from SMSC to Group.

Class

Required key.

Type / Range

Integer in the range of 1 to 128

Name

draDestHostFqdnName - DRA destination host name

Syntax

--draDestHostFqdnName <string>

Description

If MME does not have destination host in VLR (UE shows up for the first time), and the destination is a DRA, MME uses the value of this new field to populate destination host AVP in OFR.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

""

Name

draDestRealmFqdnName - DRA destination realm name

Syntax

--draDestRealmFqdnName <string>

Description

If MME does not have destination realm in VLR (UE shows up for the first time), and the destination is a DRA, MME uses the value of this new field to populate destination realm AVP in OFR.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

hssGrpListName - HSS Group List Name

Syntax

--hssGrpListName <string>

Description

HSS group list name for routing SMSC messages when the SGd application is combined with S6a.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

loadBalance - Load balance

Syntax

--loadBalance <list-item>

Description

Indicates load balancing options.

Class

Optional.

List

- all - Balance load on all S6ad links based on weight.
- none - No load balancing between S65ad links.
- priority group - Balance load on all S6ad links based on group priority. the primary group is the set of HSS/DRA that have the same highest weight. the secondary group is the

set of HSS/DRA that have the same second highest weight value, etc. the last N-ary group is always the group with 0 weight (if used).

Default

none

Name

shutdown - Shutdown

Syntax

--shutdown <list-item>

Description

Indicates if the mapping rule is active. if shutdown is true, the rule behaves as if it is deleted.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

smScGrpListName - SMSC Group List Name

Syntax

--smScGrpListName <string>

Description

Name of SMSC group list.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

smScMappingType - SMSC Mapping Type

Syntax

--smScMappingType <list-item>

Description

SMSC Mapping Type.

Class

Optional.

List

- all - SMSC mapping type all
- num - Number - SMSC mapping type Number (mapping for an SMSC number)

Default

all

Name

smscNumber - SMSC Number

Syntax

--smscNumber <digits>

Description

The SMSC number (international number format as described in ITU-T Recommendation E.164) that the UE provides when it sends a mobile originated SMS

Class

Optional.

Type / Range

Digit string of length 0 or 9 to 15

Default

""

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.52 cmm smsProfile

SMS profile for SMS delivery parameters. Syntax, description, parameters, and roles for cmm smsProfile command.

Syntax

Description

This command is used to administer SMS delivery related parameters for the feature SMS in MME.

Parameters

Name

name - SMS profile name

Syntax

--name <string>

Description

The profile name for the profile containing SMS delivery related parameters.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

default

Name

mmelsdn - MME ISDN

Syntax

--mmeIsdn <digits>

Description

MME ISDN for SMS services. When SMS in MME feature is used, the MME registers itself as the SMS service provider towards the HSS for the UE. The MME provides its own number for MT SMS delivery in international ITU-T Rec E-164 format.

Class

Optional.

Type / Range

Digit string of length 9 to 15

Name

mtSmsResendUponCollision - MT SMS Resend Upon Collision

Syntax

--mtSmsResendUponCollision <list-item>

Description

Defines whether the MME attempts to resend the SGd received MT SMS after procedure collision when the MT SMS delivery was aborted by the MME due to the collision.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Recommended

true

Name

nasResends - NAS Resends

Syntax

--nasResends <integer>

Description

The amount of retransmissions sending the DL NAS Transport messages that carry Short Message Control Protocol and Short Message Relay Protocol requests when timer TC1* has expired.

Class

Optional.

Type / Range

Integer in the range of 0 to 3

Default

2

Name

sgdAla - SGd ALA

Syntax

--sgdAla <integer>

Description

Timer to wait for SGd Alert Service Center Answer (ALA) from the SMSC/SMS router when MME has sent SGd Alert Service Center Request (ALR).

Class

Optional.

Type / Range

Integer in the range of 100 to 30,000 in units of Millisecond

Default

5000

Name

sgdOfa - SGd OFA

Syntax`--sgdOfa <integer>`**Description**

Timer to wait for SGd MO Forward Short Message Answer (OFA) from the SMSC when MME has sent SGd MO Forward Short Message Request (OFR).

Class

Optional.

Type / Range

Integer in the range of 100 to 30,000 in units of Millisecond

Default

5000

Name

smsBufferingTime - MT SMS Buffering Time

Syntax`--smsBufferingTime <integer>`**Description**

Time during which the MME buffers an MT SMS. During this timer the MME waits either if the terminating UE is busy with another procedure (e.g delivery of another SMS) or if the UE is currently not reachable due to eDRX but is expected to make contact shortly. This timer should be set lower than the DRA timeout. After this timer expires the MME responds to the TFR. Note: The provisioned retransmission times * 'TC1*' need to be lower than the provisioned MT SMS Buffering Time. Note: the MME does not make use of this timer if the UE is not expected to be reachable before the timer expiry.

Class

Optional.

Type / Range

Integer in the range of 100 to 10,000 in units of Millisecond

Default

5000

Name

smsRegistrationPreference - SMS Registration Preference

Syntax`--smsRegistrationPreference <list-item>`

Description

Defines which interface is used for SMS delivery when both SGd and SGs are possible for a UE.

Class

Optional.

List

- smsRegRequired - SMS Registration Required - SMS registration to HSS is required i.e. SGd interface based SMS delivery is forced when both SGs and SGd interface use is possible.
- smsRegNotPref - SMS Registration Not Preferred - SMS registration to HSS is not preferred i.e. SGs interface based SMS delivery is preferred when both SGs and SGd use is possible.
- noPreference - No Preference - The MME indicates no preference to use either SGd or SGs interface for the SMS delivery. Both options are available from MME perspective and HSS is left to make the choice to register the UE via HSS (use the SGd) or not.

Default

smsRegNotPref

Name

tc1Star - TC1*

Syntax

--tc1Star <integer>

Description

Timer to wait for acknowledgement messages from the UE for MME sent DL NAS Transport messages that carry Short Message Control Protocol and Short Message Relay Protocol requests.

Class

Optional.

Type / Range

Integer in the range of 1 to 5,000,000 in units of Millisecond

Default

8000

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.53 cmm snssai

S-NSSAI. Syntax, description, parameters, and roles for cmm snssai command.

Syntax

Description

Create a S-NSSAI with Name, Service Type and Service Differentiator.

Parameters

Name

name - S-NSSAI Name

Syntax`--name <string>`**Description**

Name of the S-NSSAI.

Class

Required key.

Type / RangeDigit string of length 1 to 31 with the allowed pattern of `[0-9a-zA-Z\.\- _:@]*`**Name**

sliceServiceType - Slice Service Type

Syntax`--sliceServiceType <integer>`**Description**

The S-NSSAI consists of a Slice Service Type (SST) and a Slice Differentiator (SD). Each slice must have SST and SD is optional.

Class

Required on create.

Type / Range

5G Slice Service Type

Name

sliceDifferentiator - Slice Differentiator

Syntax

--sliceDifferentiator <string>

Description

The S-NSSAI consists of a Slice Service Type (SST) and a Slice Differentiator (SD). Each slice must have SST and SD is optional. A SST may have multiple SDs.

Class

Optional.

Type / Range

Identifies Slice Differentiator (SD)

Default

FFFFFF

Name

sliceDifferentiatorRangeEnd - Slice Differentiator Range End

Syntax

--sliceDifferentiatorRangeEnd <string>

Description

This is used when snssai represents an Extended S-NSSAI that consists of a Slice Service Type and a Range of Slice Differentiator (SD). For an Extended S-NSSAI, the Slice Differentiator Range End must be larger than the Slice Differentiator parameter. Otherwise, when the snssai represents a single S-NSSAI, the value must be set to '000000'.

Class

Optional.

Type / Range

Identifies Slice Differentiator (SD)

Default

"000000"

Name

supportedInAllTa - Allow S-NSSAI in all Tracking Areas

Syntax

--supportedInAllTa <list-item>

Description

This parameter is deprecated (was used only for PoC purposes). Should be set to 'false'. If the parameter is set to 'true' then, the S-NSSAI is supported in all TA. If the parameter is set to 'false' then the S-NSSAI is not supported in all TAs served by this AMF but may be supported in a TA or a subset of the available TAs. If S-NSSAI is supported in a TA or subset of the available TAs then use the 'amfTais' command to provision the support.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.54 cmm snssaiList

S-NSSAI list. Syntax, description, parameters, and roles for `cmm snssaiList` command.

Syntax**Description**

Create a list for S-NSSAI.

Parameters**Name**

name - List Name

Syntax

--name <string>

Description

S-NSSAI list name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.55 cmm snssaiListItem

S-NSSAI List Item. Syntax, description, parameters, and roles for `cmm snssaiListItem` command.

Syntax**Description**

Add or delete a S-NSSAI to the named list. The list consists of the name of a S-NSSAI and STS and SD fields of the S-NSSAI.

Parameters**Name**

snssaiListName - S-NSSAI List

Syntax

--snssaiListName <string>

Description

List consisting up to 256 network slice instances IDs.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

snssaiName - S-NSSAI Name

Syntax

--snssaiName <string>

Description

Name of the S-NSSAI.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.56 cmm snssaisOfTaiQuery

Query SNSSAIs of TAI. Syntax, description, parameters, and roles for cmm snssaisOfTaiQuery command.

Syntax

Description

Lists all SNSSAIs supported in a specific TAI. The NG-RAN 'taiMcc', 'taiMnc' and 'tac' are required input filters. If any of these are not given query will return empty list

Parameters

Name

taiMcc - TAI MCC

Syntax

--taiMcc <digits>

Description

3-digit number used to represent the mobile country code for TAI.

Class

Required key.

Type / Range

Digit string of length 3

Name

taiMnc - TAI MNC

Syntax

--taiMnc <digits>

Description

Digits used to represent mobile network code for TAI.

Class

Required key.

Type / Range

Digit string of length 2 to 3

Name

tac - Tracking Area Code

Syntax

--tac <integer>

Description

5G TAC. 5G TAC is 3 octets long.

Class

Required key.

Type / Range

Integer in the range of 0 to 16777215

Name

assignMode - Assignment Mode

Syntax

--assignMode <integer>

Description

This field depicts the assignment mode for the SNPN

Class

Optional.

Type / Range

Integer in the range of 0 to 2

Default

0

Name

nidValue - Nid value

Syntax

--nidValue <integer>

Description

This field depicts the Nid value of the NID for the SNPN

Class

Optional.

Type / Range

Integer in the Range 0 to 1099511627775

Default

0

Name

sliceDifferentiator - Slice Differentiator

Syntax

--sliceDifferentiator <string>

Description

The S-NSSAI consists of a Slice Service Type (SST) and a Slice Differentiator (SD). Each slice must have SST and SD is optional. A SST may have multiple SDs.

Class

Read only.

Type / Range

Identifies Slice Differentiator (SD)

Default

FFFFFF

Name

sliceServiceType - Slice Service Type

Syntax

--sliceServiceType <integer>

Description

The S-NSSAI consists of a Slice Service Type (SST) and a Slice Differentiator (SD). Each slice must have SST and SD is optional.

Class

Optional.

Type / Range

5G Slice Service Type

Roles**List**

sysadmin, manage, monitor

7.57 cmm snssaiToDnn

S-NSSAI to DNN. Syntax, description, parameters, and roles for cmm snssaiToDnn command.

Syntax**Description**

Add or delete a S-NSSAI to the named list. The list consists of the name of a S-NSSAI and STS and SD fields of the S-NSSAI.

Parameters**Name**

snssaiToDnnListName - S-NSSAI to Default DNN List Name

Syntax

--snssaiToDnnListName <string>

Description

S-NSSAI to Default DNN List Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

snssaiName - S-NSSAI Name

Syntax

--snssaiName <string>

Description

Name of the S-NSSAI.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

dataNetworkName - Data Network Name

Syntax

--dataNetworkName <string>

Description

Data Network Name (DNN)

Class

Required on create.

Type / Range

Data Network Name is basically APN Ni.

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.58 cmm snssaiToDnnList

S-NSSAI to Default DNN List. Syntax, description, parameters, and roles for cmm snssaiToDnnList command.

Syntax

Description

List for configuring S-NSSAI to default DNN.

Parameters

Name

name - Name of the List

Syntax

--name <string>

Description

Name of the SNNSAI to default DNN mapping list name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.59 cmm spcParamSet

SCCP signalling point parameter set. Syntax, description, parameters, and roles for **cmm spcParamSet** command.

Syntax

Description

Configure, list and show SCCP signalling point parameter set.

Parameters

Name

name - SCCP signalling point parameter set name

Syntax

--name <string>

Description

SCCP signalling point parameter set name.

Class

Required key.

Type / Range

Digit string of length 1 to 5 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

addDpcIfRiGt - Add dpc if route ind on GT

Syntax

--addDpcIfRiGt <list-item>

Description

21. DPC insertion to called address, when routing indicator is route on GT.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

addDpcIfRiSsn - Add dpc if route ind on ssn

Syntax

--addDpcIfRiSsn <list-item>

Description

19. DPC insertion to called address, when routing indicator is route on SSN.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

addGtIfRiSsn - Add GT if route ind on ssn

Syntax

--addGtIfRiSsn <list-item>

Description

20. GT insertion to called address, when routing indicator is route on SSN.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

aInterface - A interface

Syntax

--aInterface <list-item>

Description

11. interface to signalling point is A-interface.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

allowedGtiValues - Allowed gti values

Syntax

--allowedGtiValues <string>

Description

23. indicates the allowed GTI values of calling address messages with other GTI values are screened. this must be a comma separated list of numbers and ranges of numbers. ITU-T

recommended values 1-4 ANSI recommended values 1-2 ETSI recommended values 4 all values 1-15.

Class

Optional.

Type / Range

Digit string of length 0 to 35 with the allowed pattern of [0-9,-]*

Name

analyseRootOfCallingGt - Analyse root of calling GT

Syntax

--analyseRootOfCallingGt <list-item>

Description

22. indicates whether the translator selector of calling address is analysed.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

coSegmUsed - Co segm used

Syntax

--coSegmUsed <list-item>

Description

27. usage of connection oriented segmentation.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

gtAddrFixedLen - GT addr fixed len

Syntax

--gtAddrFixedLen <list-item>

Description

28. indicates whether the called GT space reservation is optimized in (X)UDT message. if the parameter value is true, no extra space is reserved for GT expansion due to GT modifications in intermediate nodes. the parameter concerns messages coming from a local SCCP user. the parameter value is set in the parameter set of the destination signalling point.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ludtUsed - Ludt used

Syntax

--ludtUsed <list-item>

Description

26. usage of long unitdata (LUDT) messages.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

q714TConnEst - q714TConnEst

Syntax

--q714TConnEst <integer>

Description

1. connection establishment timer, time unit is 1 s.

Class

Optional.

Type / Range

Integer in the range of 30 to 240

Default

90

Name

q714Tlar - q714Tlar

Syntax

--q714TIar <integer>

Description

3. receive inactivity timer, time unit is 1 s.

Class

Optional.

Type / Range

Integer in the range of 150 to 1,260

Default

270

Name

q714Tlas - q714Tlas

Syntax

--q714TIas <integer>

Description

2. send inactivity timer, time unit is 1 s.

Class

Optional.

Type / Range

Integer in the range of 60 to 600

Default

90

Name

q714TInt - q714TInt

Syntax

--q714TInt <integer>

Description

5. connection release timer, time unit is 1 s.

Class

Optional.

Type / Range

Integer in the range of 45 to 90

Default

60

Name

q714TRel - q714TRel

Syntax

--q714TRel <integer>

Description

4. repetition timer of connection release message, time unit is 100 ms.

Class

Optional.

Type / Range

Integer in the range of 100 to 200

Default

150

Name

q714TRepRel - q714TRepRel

Syntax

--q714TRepRel <integer>

Description

7. repetition timer of connection release message, time unit is 100 ms.

Class

Optional.

Type / Range

Integer in the range of 40 to 200

Default

100

Name

q714TRes - q714TRes

Syntax

--q714TRes <integer>

Description

6. signalling connection reset timer, time unit is 1 s.

Class

Optional.

Type / Range

Integer in the range of 10 to 20

Default

15

Name

q714TStatInf1st - q714TStatInf1st

Syntax

--q714TStatInf1st <integer>

Description

8. subsystem status test send interval or time out limit for the first test attempt, time unit is 100 ms.

Class

Optional.

Type / Range

Integer in the range of 50 to 600

Default

100

Name

q714TStatInfInc - q714TStatInfInc

Syntax

--q714TStatInfInc <integer>

Description

9. increment value of subsystem status test, send interval, time unit is 100 ms.

Class

Optional.

Type / Range

Integer in the range of 150 to 3,000

Default

300

Name

q714TStatInfMax - q714TStatInfMax

Syntax

--q714TStatInfMax <integer>

Description

10. maximum value of subsystem status test, send interval, time unit is 100 ms.

Class

Optional.

Type / Range

Integer in the range of 600 to 12,000

Default

9000

Name

segXThres - Seg x threshold

Syntax

--segXThres <integer>

Description

16. lower segmentation threshold.

Class

Optional.

Type / Range

Integer in the range of 1 to 272

Default

272

Name

segYThres - Seg y threshold

Syntax

--segYThres <integer>

Description

17. higher segmentation threshold.

Class

Optional.

Type / Range

Integer in the range of 67 to 272

Default

272

Name

ssaFilterTimer - Ssa filter timer

Syntax

--ssaFilterTimer <integer>

Description

24. delay of local broadcast of subsystem allowed state information, time unit is 100 ms.

Class

Optional.

Type / Range

Integer in the range of 5 to 3,000

Default

10

Name

ssManagementUsed - Ss management used

Syntax

--ssManagementUsed <list-item>

Description

13. usage of subsystem management functions.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

sspFilterTimer - Ssp filter timer

Syntax

--sspFilterTimer <integer>

Description

25. delay of local broadcast of subsystem prohibited state information, time unit is 100 ms.

Class

Optional.

Type / Range

Integer in the range of 5 to 3,000

Default

10

Name

tcapLoadSharingUsed - Tcap load sharing used

Syntax

--tcapLoadSharingUsed <list-item>

Description

18. load sharing used for TCAP messages.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

udtDenied - Udt denied

Syntax

--udtDenied <list-item>

Description

15. deny usage of unitdata (UDT) messages.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

whiteBookMgmtUsed - White book management used

Syntax

--whiteBookMgmtUsed <list-item>

Description

12. usage of white book management procedures.

Class

Optional.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Name

xudtUsed - Xudt used

Syntax

--xudtUsed <list-item>

Description

14. usage of extended unitdata (XUDT) messages.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.60 cmm specialApn

Special Apn. Syntax, description, parameters, and roles for cmm specialApn command.

Syntax

Description

Special Apn handling

Parameters

Name

specialApnListName - Special Apn List Name

Syntax

--specialApnListName <string>

Description

Special Apn List Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

apnNi - APN NI

Syntax

--apnNi <string>

Description

APN NI

Class

Required key.

Type / Range

Digit string of length 1 to 62 with the allowed pattern of [0-9a-zA-Z_\. \-]*

Name

specialTreatmentForUnsubscribedApn - Special Treatment For Unsubscribed Apn

Syntax

--specialTreatmentForUnsubscribedApn <list-item>

Description

Special Treatment handling For Unsubscribed Apn

Class

Optional.

List

- applyGenTreatmentUnsubs - Apply General Treatment For Unsubs - Apply General Treatment For Unsubscribed

- noConvertAndReject - No Convert And Reject - No Convert And Reject PDP

Default

applyGenTreatmentUnsubs

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.61 cmm specialApnList

Special Apn List. Syntax, description, parameters, and roles for `cmm specialApnList` command.

Syntax**Description**

This command shall be used for special Apn handling

Parameters**Name**

name - APNNI

Syntax

--name <string>

Description

special APNNI configured by operator

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.62 cmm ss7DestInfo

SS7 Destination Info. Syntax, description, parameters, and roles for `cmm ss7DestInfo` command.

Syntax

Description

Display the top-down status of a destination in the SS7 network.

Parameters

Name

dpc - Destination point code

Syntax

--dpc <integer>

Description

Use this parameter to specify the destination signalling point code. Please, check that this destination point code has been configured through `cmm muaRoute` list, and that the `destinationSignallingNetwork` is the same as the given network.

Class

Optional.

Type / Range

Integer in the range of 0 to 16,777,215

Name

gt - GT analysis digits identifier

Syntax

--gt <digits>

Description

Use this parameter to specify the global title (GT) analysis digits identifier. Please, check that this GT analysis digits identifier (gtald) has been configured through `cmm gtAnalysis show`.

Class

Optional.

Type / Range

Digit string of length 1 to 24 with the allowed pattern of [0-9]*

Name

network - Destination Signalling Network

Syntax

--network <list-item>

Description

Use this parameter to specify the destination signalling network. Allowed values
IN0,IN1,NA0,NA1

Class

Optional.

List

- IN0 - International network number 0.
- IN1 - International network number 1.
- NA0 - National network number 0.
- NA1 - National network number 1.

Roles**List**

sysadmin, manage, monitor

7.63 cmm ss7DestInfoAll

SS7 Destination Info. Syntax, description, parameters, and roles for `cmm ss7DestInfoAll` command.

Syntax**Description**

Display the top-down status of a destination in the SS7 network.

Parameters**Name**

dpc - Destination point code

Syntax

--dpc <integer>

Description

Use this parameter to specify the destination signalling point code. Please, check that this destination point code has been configured through cmm muaRoute list, and that the destinationSignallingNetwork is the same as the given network.

Class

Optional.

Type / Range

Integer in the range of 0 to 16,777,215

Name

network - Destination Signalling Network

Syntax

--network <list-item>

Description

Use this parameter to specify the destination signalling network. Allowed values
IN0,IN1,NA0,NA1

Class

Optional.

List

- IN0 - International network number 0.
- IN1 - International network number 1.
- NA0 - National network number 0.
- NA1 - National network number 1.

Roles**List**

sysadmin, manage, monitor

7.64 cmm ss7DestInfoGt

SS7 Destination Info With gt. Syntax, description, parameters, and roles for cmm ss7DestInfoGt command.

Syntax

Description

Display the top-down status of a destination in the SS7 network for given gt digits

Parameters

Name

gt - GT analysis digits identifier

Syntax

--gt <digits>

Description

Use this parameter to specify the global title (GT) analysis digits identifier. Please, check that this GT analysis digits identifier (gtald) has been configured through cmm gtAnalysis show.

Class

Optional.

Type / Range

Digit string of length 1 to 24 with the allowed pattern of [0-9]*

Roles

List

sysadmin, manage, monitor

7.65 cmm staticRoute

Static route. Syntax, description, parameters, and roles for cmm staticRoute command.

Syntax

Description

Administer static route

Parameters

Name

poolType - Pool type

Syntax

--poolType <list-item>

Description

Pool (service) type.

Class

Required key.

List

- ALMS - Alarm Service - Alarm Service.
- CPPS - Control plane processing service - Control plane processing service.
- CTCS - Call Trace Collection service - Call Trace Collection service.
- DBS - Database service - Database service.
- EEMS - Event Exposure Management service - Event Exposure Management service.
- IPDS - IP director service - IP director service.
- IPPS - IP packet processing service - IP packet processing service.
- LIHS - Lawful Interception Handling service
- MLIS - Machine Learning Intelligence Service
- NECC - Network element cluster controller - Network element cluster controller.
- NWDS - Network Data Service - NWDAF service
- PAPS - Packet processing service - Packet processing service.
- PCMS - Per Call Measurements Data Service - Per Call Measurements Data (PCMD) Service
- RCIS - Radio Capability Information storage Service
- SFDS - SCTP flow distributor service
- UCDS - User equipment Context Data-dump Service
- UFDS - UDP flow distributor service

Name

poolId - Pool id

Syntax

--poolId <integer>

Description

Pool identification number.

Class

Required key.

Type / Range

Integer in the range of 0 to 39

Name

poolMemberId - Pool member ID

Syntax

--poolMemberId <integer>

Description

Pool member identification number.

Class

Required key.

Type / Range

Integer in the range of 0 to 2

Name

assignmentType - Assignment type

Syntax

--assignmentType <list-item>

Description

Assignment type: fixed or floating

Class

Required key.

List

- fixed - Fixed.
- floating - Floating.

Name

routeId - Route ID

Syntax

--routeId <integer>

Description

Route identification.

Class

Required key.

Type / Range

Integer in the range of 1 to 63

Name

ipv - IP version

Syntax

--ipv <integer>

Description

Internet protocol version number.

Class

Required key.

Type / Range

Integer in the range of 4 or 6

Name

base - Base IP

Syntax

--base <string>

Description

Base IP address for IPv4, or base prefix for IPv6.

Class

Required on create.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

gateway - Gateway IP

Syntax

--gateway <string>

Description

Gateway IP address.

Class

Required on create.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

prefixLen - Base prefix length

Syntax

--prefixLen <integer>

Description

Base prefix length. allowed only for IPv6.

Class

Optional.

Type / Range

Integer in the range of 0 to 128

Name

srcIp - Source IP

Syntax

--srcIp <string>

Description

Source IP address.

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

subnetMask - Subnet mask

Syntax

--subnetMask <string>

Description

Subnet IP mask. allowed only for IPv4.

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.66 cmm subnet

Subnet. Syntax, description, parameters, and roles for cmm subnet command.

Syntax

Description

Administer Subnet. For Multiple IPDS in VNF configurations these subnets are non routable VIA subnets between the IPDSs and the L3NS VNFCs. The Redundancy mode eipm_via must be used for the interfaces (except OAM). There is no Default Gateway IP address. The interfaces and vlans must be aligned to the transport subnets established during Cloud Init. Administer Subnet. For Multiple IPDS in CNF configurations these subnets are non routable subnets between the IPDS PODs and the SFDS/UFDS PODs. The Redundancy mode eipm_arpn dp must be used for these subnets. When multiple interfaces are presented to the SFDS/UFDS pod for interface redundancy, the --fdsMateInterface parameter must be configured to specify the standby interface. The Default Gateway IP address should be configured as an accessible IP, but the IP does not have to have L3 functionality as it is only used as an ARP target for reachability. It is not used for routing purposes. The interfaces and vlans must be aligned to the subnets established during Cloud Init.

Parameters

Name

subnetId - Subnet ID

Syntax

--subnetId <integer>

Description

Subnet identification.

Class

Required key.

Type / Range

Integer in the range of 0 to 63

Name

ipv - IP version

Syntax

--ipv <integer>

Description

Internet protocol version number.

Class

Required key.

Type / Range

Integer in the range of 4 or 6

Name

base - Base IP

Syntax

--base <string>

Description

Base IP address for IPv4, or base prefix for IPv6.

Class

Required on create, not modifiable.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

broadcastIp - Broadcast IP

Syntax

--broadcastIp <string>

Description

Broadcast IP address. allowed only for IPv4.

Class

Read only.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

defaultGateway - Default gateway IP

Syntax

--defaultGateway <string>

Description

Default gateway IP address.

Class

Not modifiable.

Type / Range

Digit string of length 0 or 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Default

"

Name

desiredMinTxInterval - Desired minimum transmission interval

Syntax

--desiredMinTxInterval <integer>

Description

Desired minimum transmission interval, in units of 100 milliseconds.

Class

Optional.

Type / Range

Integer in the range of 0 to 1,000

Name

detectionMultiplier - Detection multiplier

Syntax

--detectionMultiplier <integer>

Description

Detection multiplier.

Class

Optional.

Type / Range

Integer in the range of 0 or 3 to 50

Name

fdsMateInterface - Mate Interface on flow distributor service

Syntax

--fdsMateInterface <string>

Description

Mate interface used for duplex networking on flow distributor service

Restrictions

Applicable to CNF only.

Class

Not modifiable.

Type / Range

Digit string of length 0 to 5 with the allowed pattern of ([a-zA-Z][a-zA-Z0-9]{1,4})|

Default

"

Name

interfaceLabelL - Interface label left

Syntax

--interfaceLabelL <string>

Description

Interface label left.

Class

Not modifiable.

Type / Range

Digit string of length 0 to 5 with the allowed pattern of ([a-zA-Z][a-zA-Z0-9]{1,4})|

Default

"

Name

interfaceLabelR - Interface label right

Syntax

--interfaceLabelR <string>

Description

Interface label right.

Class

Not modifiable.

Type / Range

Digit string of length 0 to 5 with the allowed pattern of ([a-zA-Z][a-zA-Z0-9]{1,4})|

Default

"

Name

prefixLen - Base prefix length

Syntax

--prefixLen <integer>

Description

Base prefix length. allowed only for IPv6.

Class

Not modifiable.

Type / Range

Integer in the range of 0 to 128

Default

0

Name

redundancyMode - Redundancy mode

Syntax

--redundancyMode <list-item>

Description

Redundancy mode.

Class

Not modifiable.

List

- bfd_rsr - Specify redundancy mode of bidirectional forwarding detection transport with reduced state routing.
- bfd_transport - Specify redundancy mode of bidirectional forwarding detection transport.
- eipm_acm - Specify redundancy mode of external internetwork performance monitor with ACM.
- eipm_arpndp - Specify redundancy mode of EIPM with address resolution protocol / network discovery protocol.
- eipm_bfd - Specify redundancy mode of external internetwork performance monitor with bidirectional forwarding detection.
- eipm_via - Specify redundancy mode for Multiple IPDS with VSR routers. This mode is used between the IPDSs and the VSRs. The subnet must not contain a Default Gateway IP address. This mode is applicable for all interface types except OA&M.
- iipm - Specify redundancy mode of internal internetwork performance monitor.
- none - Specify redundancy mode of none.

Default

none

Name

requiredMinRxInterval - Required minimum reception interval

Syntax

--requiredMinRxInterval <integer>

Description

Required minimum reception interval, in units of 100 milliseconds.

Class

Optional.

Type / Range

Integer in the range of 0 to 1,000

Name

rteTableName - Routing table name

Syntax

--rteTableName <string>

Description

The name of the source based routing table if it exists for this subnet.

Class

Read only.

Type / Range

Digit string of length 0 to 20

Name

rteTableNum - Routing table number

Syntax

--rteTableNum <integer>

Description

Table number for source based routing, if applicable. a value of zero implies source based routing is not active for this subnet.

Class

Read only.

Type / Range

Integer in the range of 0 to 255

Name

subnetMask - Subnet mask

Syntax

--subnetMask <string>

Description

Subnet IP mask. allowed only for IPv4.

Class

Not modifiable.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Default

0.0.0.0

Name

subnetName - Subnet name

Syntax

--subnetName <string>

Description

Subnet name.

Class

Not modifiable.

Type / Range

Digit string of length 0 to 20 with the allowed pattern of [0-9a-zA-Z\.\-:]*

Default

"

Name

vlanL - VLAN left

Syntax

--vlanL <integer>

Description

Virtual local area network left.

Class

Not modifiable.

Type / Range

Integer in the range of 0 to 4,094

Default

0

Name

vlanR - VLAN right

Syntax

--vlanR <integer>

Description

Virtual local area network right.

Class

Not modifiable.

Type / Range

Integer in the range of 0 to 4,094

Default

0

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.67 cmm subParamSet

SCCP subsystem parameter set. Syntax, description, parameters, and roles for **cmm subParamSet** command.

Syntax

Description

Configure, list and show SCCP subsystem parameter set.

Parameters

Name

name - SCCP subsystem parameter set name

Syntax

--name <string>

Description

SCCP subsystem parameter set name.

Class

Required key.

Type / Range

Digit string of length 1 to 5 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

addClgGtIfRiSsn - Add clg GT if route ssn

Syntax

--addClgGtIfRiSsn <list-item>

Description

18. indicates whether the GT is left to the calling party address, when the routing indicator of the called party address is route on SSN. the parameter concerns messages coming from a local TC user and destined to network. the parameter value is set in the parameter set of the calling subsystem.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

aInterfApplic - A interf applic

Syntax

--aInterfApplic <list-item>

Description

3. a interface type of SP applicable to SS.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

aInterfErrIgnored - A interf err ignored

Syntax

--aInterfErrIgnored <list-item>

Description

4. error message from a interface is ignored with SS.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

callingAddrModification - Calling addr modification

Syntax

--callingAddrModification <list-item>

Description

6. indicates whether calling address is modified when global title translation result is routed on SSN and message is coming from SCCP user part.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

cscAllowedIn - Csc allowed in

Syntax

--cscAllowedIn <list-item>

Description

8. indicates whether coordinated state change request is delivered to subsystem.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

cscAllowedOut - Csc allowed out

Syntax

--cscAllowedOut <list-item>

Description

7. indicates whether subsystem is allowed to request coordinated state change.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

immediateStateInfo - Immediate state info

Syntax

--immediateStateInfo <list-item>

Description

14. indicates whether subsystem state information is transferred immediately to a local SCCP user.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

keepCldGtIfRiSsn - Keep cld GT if route ssn

Syntax

--keepCldGtIfRiSsn <list-item>

Description

13. indicates if the GT is kept in the called address, even if routing of the called address is based on SSN. the parameter concerns messages going to a local SCCP user.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

q714TCoordChg - q714T coord chg

Syntax

--q714TCoordChg <integer>

Description

1. waiting to subsystem out of service grant, time unit is 1 s.

Class

Optional.

Type / Range

Integer in the range of 30 to 240

Default

90

Name

q714TlgnSst - q714Tl gnsst

Syntax

--q714TlgnSst <integer>

Description

2. subsystem status test ignoring time, time unit is 1 s.

Class

Optional.

Type / Range

Integer in the range of 30 to 300

Default

60

Name

removeSpcClgAddr - Remove spc clg addr

Syntax

--removeSpcClgAddr <list-item>

Description

17. indicates whether the SPC is left out from the calling party address. if the parameter is false, no action is taken. if the parameter value is true, the possible SPC is left out from the calling party address. the parameter concerns messages coming from a local TC user and destined to network. the parameter value is set in the parameter set of the calling subsystem.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sendCallingSpcSsnIfRiGt - Send calling spc ssn if route GT

Syntax

--sendCallingSpcSsnIfRiGt <list-item>

Description

12. indicates whether calling address is modified so that it includes only SSN when the routing of called address is based on GT.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sendCallingSsnIfRiSsn - Send calling ssn if route ssn

Syntax

--sendCallingSsnIfRiSsn <list-item>

Description

11. indicates whether calling address is modified so that it includes only SSN when the routing of called address is based on SSN.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

tcProtocolVersExcluded - Tc protocol vers excluded

Syntax

--tcProtocolVersExcluded <list-item>

Description

15. indicates whether the optional protocol version field is included in the outgoing TCAP message. if the parameter value is true, the protocol version field is not included. the parameter concerns messages coming from a local TC user. the parameter value is set in the parameter set of the calling subsystem.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

tcTransactionIdsThreshold - Tc transaction ids threshold

Syntax

--tcTransactionIdsThreshold <integer>

Description

10. threshold for open TC transaction if the threshold is exceeded a statistics event log is set.

Class

Optional.

Type / Range

Integer in the range of 25 to 90

Default

75

Name

translateAtDpcIfDpcSsnGt - Translate at dpc if dpc ssn GT

Syntax

--translateAtDpcIfDpcSsnGt <list-item>

Description

9. translation selection when DPC, GT and SSN are present and message is coming from SCCP user part.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

translationSelection - Translation selection

Syntax

--translationSelection <list-item>

Description

5. special global title translation method selections to messages from MTP, when translation result is own signalling point and routing indicator in the record is routed on GT.

Class

Optional.

List

- REDIGO - This option is not supported.
- SCCP - Specify translation selection of SCCP.

Default

SCCP

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.68 cmm subsBulkAction

Subscriber Bulk Action. Syntax, description, parameters, and roles for **cmm subsBulkAction** command.

Syntax

Description

Subscriber Bulk Action is a maintenance command to count, deregister/detach or completely purge subscribers bases on subscriber bulk action profile.

Parameters

Name

bulkAction - Bulk Action

Syntax

```
--bulkAction <list-item>
```

Description

Defines Bulk Subscriber Process Action

Class

Required key.

List

- start - start the command
- stop - stop the command

Name

subsBulkActionProfileName - Subscriber Bulk Action Profile Name

Syntax

```
--subsBulkActionProfileName <string>
```

Description

Subscriber Bulk Action Profile Name

Class

Required on create.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Modify

sysadmin, manage

7.69 cmm subsBulkActionProfile

Subscriber Bulk Action Profile. Syntax, description, parameters, and roles for cmm subsBulkActionProfile command.

Syntax

Description

Subscriber Bulk Action Profile. Profile should not be updated or deleted when used by ongoing subsBulkAction Command. The only filter for Bulk Action in this release is all UEs.

Parameters

Name

name - Name

Syntax

--name <string>

Description

Subscriber Bulk Action Profile Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

radioType - Radio Type

Syntax

--radioType <list-item>

Description

Radio Type

Class

Required on create, not modifiable.

List

- 4g - 4g
- 5g - 5g

Name

amfDeleteUeContextCc - AMF Delete UE context cause code

Syntax

--amfDeleteUeContextCc <integer>

Description

NAS Cause code to include in Deregistration Request when UE context is deleted for AMF

Class

Optional.

Type / Range

5GMM Cause Code.

Default

10

Name

amfDeleteUeContextDeregistrationType - AMF Delete UE context deregistration type

Syntax

--amfDeleteUeContextDeregistrationType <list-item>

Description

Deregistration type when UE context is deleted via administrative action e.g. cli command for AMF

Class

Optional.

List

- registrationNotRequired - Registration not required - Specifies registration type of registration not required
- registrationRequired - Registration required - Specifies registration type of registration required

Default

registrationRequired

Name

amfDeleteUeContextSendCc - AMF Delete UE context send cause code

Syntax

--amfDeleteUeContextSendCc <list-item>

Description

Include cause code in deregistration request when deregistration reason is due to deleting UE context for AMF

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

guardTimer - Guard Timer

Syntax

--guardTimer <integer>

Description

Maximum duration once started.

Class

Optional.

Type / Range

Integer in the range of 1 to 1,440 in units of Minute

Default

1440

Name

includeEmergency - Include Emergency

Syntax

--includeEmergency <list-item>

Description

Include Emergency. When the value is set to false, emergency UEs, emergency PDNs and emergency PDUs are not affected

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

mmeDeleteUeContextCc - MME Delete UE context cause code

Syntax

--mmeDeleteUeContextCc <integer>

Description

NAS Cause code to include in Detach Request when UE context is deleted for MME

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

10

Name

mmeDeleteUeContextDetachType - MME Delete UE context detach type

Syntax

- -mmeDeleteUeContextDetachType <list-item>

Description

Detach type when UE context is deleted via administrative action e.g. cli command for MME

Class

Optional.

List

- Reattach not required - Reattached not required - Specifies detach type of reattach not required.
- Reattach required - Specifies detach type of reattach required.

Default

Reattach required

Name

mmeDeleteUeContextSendCc - MME Delete UE context send cause code

Syntax

- -mmeDeleteUeContextSendCc <list-item>

Description

Include cause code in detach request when detach reason is due to deleting UE context for MME

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

purgeContext - Purge Context

Syntax

--purgeContext <list-item>

Description

Detach/De-Register or Detach/De-Register and purge the UE Context from CMM. When set to false only Detach/De-Register is done. When set to true, Detach/De-Register and purge the context from CMM. Also HSS and UDM shall be notified of the context removal.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.70 cmm subsBulkActionStats

Subscriber Bulk Action Status. Syntax, description, parameters, and roles for cmm subsBulkActionStats command.

Syntax

Description

Displays the statistics of the subscriber bulk action command. The stat include radio type, duration, profile name used.

Parameters

Name

commandStatus - Command Status

Syntax

--commandStatus <list-item>

Description

Command Status

Class

Read only.

List

- aborted - Aborted
- timeout - Timeout
- completed - Completed
- failed - Failed
- running - Running
- stopped - Stopped

Name

radioType - Radio Type

Syntax

--radioType <list-item>

Description

Radio Type

Class

Read only.

List

- 4g - 4g
- 5g - 5g

Name

runTime - Run Time

Syntax

--runTime <string>

Description

Run time of the command in hh:mm:ss

Class

Read only.

Type / Range

Digit string of length 0 to 8 with the allowed pattern of [0-9a-zA-Z\.\-_:]*

Name

startDateTime - Start Date and Time

Syntax

--startDateTime <string>

Description

Records the date and time at which the command was entered.

Class

Read only.

Type / Range

Date and Time Digit string of length 0 to 20 with the allowed pattern of '20[0-9][0-9]/[0-1][0-9]/[0-3][0-9] [0-2][0-9]:[0-5][0-9]:[0-5][0-9]'

Name

subsBulkActionProfileName - Subscriber Bulk Action Profile Name

Syntax

--subsBulkActionProfileName <string>

Description

Subscriber Bulk Action Profile Name

Class

Read only.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-:@]*

Roles**List**

sysadmin, manage, monitor

7.71 cmm subsCount

subsCount. Syntax, description, parameters, and roles for cmm subsCount command.

Syntax

Description

Count subscribers per RAT.

Parameters

Name

ecmState - ECM state

Syntax

--ecmState <string>

Description

ECM state.

Class

Optional.

Type / Range

Digit string of length

Name

emmState - EMM state

Syntax

--emmState <string>

Description

EMM state.

Class

Optional.

Type / Range

Digit string of length

Name

filterType - Filter type

Syntax

--filterType <string>

Description

Filter type. Allowed values include: - attachedPerPapsPerRat

Class

Optional.

Type / Range

Digit string of length

Name

papsId - PapsId

Syntax

--papsId <integer>

Description

PAPS index.

Class

Optional.

Type / Range

Integer in the range of 0 to 19

Roles**List**

sysadmin, manage, monitor

7.72 cmm subscriber

Subscriber. Syntax, description, parameters, and roles for **cmm subscriber** command.

Syntax**Description**

This command has three operations: The count operation is used to count subscribers. The delete operation is used to delete subscribers. The show operation is used to show subscribers. The parameter radioType is only applicable on show and delete for imsi or srsImsi. The parameter radioType is only applicable on delete for srsImsi.

Parameters**Name**

amfCmState - AMF CM state

Syntax

--amfCmState <list-item>

Description

AMF Connection Management State

Class

Optional.

List

- connected - AMF Connected - AMF Connected state

- idle - AMF Idle - AMF Idle State

Name

amfNon3gppCmState - AMF Non-3GPP Access CM State

Syntax

--amfNon3gppCmState <list-item>

Description

AMF Non-3GPP Access Connection Management State.

Class

Optional.

List

- connected - AMF Connected - AMF Connected state
- idle - AMF Idle - AMF Idle State

Name

amfNon3gppRmState - AMF Non-3GPP Access RM state

Syntax

--amfNon3gppRmState <list-item>

Description

AMF Non-3GPP Access Registration Management state.

Class

Optional.

List

- deregistered - AMF Deregistered - AMF Deregistered State
- registered - AMF Registered - AMF Registered State

Name

amfRmState - AMF RM state

Syntax

--amfRmState <list-item>

Description

AMF Registration Management state.

Class

Optional.

List

- deregistered - AMF Deregistered - AMF Deregistered State
- registered - AMF Registered - AMF Registered State

Name

bearerImsi - Bearer IMSI

Syntax

--bearerImsi <digits>

Description

Imsi of Bearer.

Class

Optional.

Type / Range

Digit string of length 5 to 15

Name

ecmState - ECM state

Syntax

--ecmState <list-item>

Description

ECM state.

Class

Optional.

List

- connected - ECM state of connected.
- idle - ECM state of idle.

Name

emmState - EMM state

Syntax

--emmState <list-item>

Description

EMM state.

Class

Optional.

List

- deregistered - Specifies de-registered EMM state.
- registered - Specifies registered EMM state.

Name

filterType - Filter type

Syntax

--filterType <list-item>

Description

Count filtering type {attachedperpaps,attachedperpapserrat}.

Class

Optional.

List

- attachedPerPaps - Attached per PAPS - Attached per PAPS count
- attachedPerPapsPerRat - Attached per PAPS per RAT - Attached per PAPS per RAT.
- dualConnUeCount - Dual Connectivity UE Count - Allowed 4G Dual Connectivity UE Count.
- restrictedDualConn5gUeCount - Restricted Dual Connectivity 5G UE Count - Restricted 5G Dual Connectivity UE Count.
- srs - Session Restoration Count - Session Restoration Context UE count

Name

imei - IMEI

Syntax

--imei <digits>

Description

This item is the first 14 digits of the IMEI.

Class

Optional.

Type / Range

Digit string of length 1 to 14

Name

imsi - IMSI

Syntax

--imsi <digits>

Description

IMSI.

Class

Optional.

Type / Range

Digit string of length 5 to 15

Name

msisdn - MSISDN

Syntax

--msisdn <digits>

Description

MSISDN.

Class

Optional.

Type / Range

Digit string of length 5 to 18

Name

radioType - Radio Access Type

Syntax

--radioType <list-item>

Description

This parameter specifies the Radio Access Type of the srsimsi to be shown/deleted. The parameter also specifies the Radio Access Type of the imsi to be shown.

Class

Optional.

List

- 2G3G - SGSN 2G/3G access
- 4G - 4G LTE access
- 5G - 5G access

Default

5G

Name

srsImsi - SRS IMSI

Syntax

--srsImsi <digits>

Description

IMSI of session restoration context UE.

Class

Optional.

Type / Range

Digit string of length 5 to 15

Roles

Count

sysadmin, manage, monitor

Delete

sysadmin, manage

Show

sysadmin, manage, monitor

7.73 cmm subscriberBearerAdmin

Subscriber Bearer Administration. Syntax, description, parameters, and roles for cmm subscriberBearerAdmin command.

Syntax

Description

The subscriber Bearer Admin command is used to display or delete specific subscriber bearers. It may also be used to restart or reset the S1AP connection for a subscriber, Available operations: - S1release - S1AP Release, Release the S1AP connection for the subscriber using detach procedure - deleteBearer - Delete Bearer, Delete a subscriber Bearer - displayBearers - Display Bearers, Display all subscriber Bearers

Parameters

Name

s1operation - Bearer Operation

Syntax

--s1operation <list-item>

Description

The operation to perform on the subscriber: Delete Bearer Display Bearers S1AP Release

Class

Required.

List

- DeleteBearer - Delete Bearer - Delete a subscriber Bearer
- DisplayBearers - Display Bearers - Display all subscriber Bearers
- S1APRelease - S1AP Release - Release the S1AP connection for the subscriber using detach procedure

Name

bearerId - Bearer Id

Syntax

--bearerId <integer>

Description

The bearer ID to delete. This parameter is required with a delete Bearer operation.

Class

Optional.

Type / Range

Integer in the range of 5 to 15

Name

imsi - IMSI

Syntax

--imsi <digits>

Description

International Mobile Subscriber Identity

Class

Optional.

Type / Range

Digit string of length 5 to 15

Roles

sysadmin, manage, monitor

7.74 cmm subscriberDataDump

Subscriber data dump . Syntax, description, parameters, and roles for cmm subscriberDataDump command.

Syntax**Description**

Subscriber data dump parameters.

Parameters**Name**

name - Profile Name

Syntax

--name <string>

Description

Name of the parameter set

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

collect2G3GSubscriberData - Collect 2G3G Subscriber Data

Syntax

--collect2G3GSubscriberData <list-item>

Description

When enabled, periodically collect subscriber data for 2G3G subscribers.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

collect4GSubscriberData - Collect 4G Subscriber Data

Syntax

--collect4GSubscriberData <list-item>

Description

When enabled, periodically collect subscriber data for 4G subscribers.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

collect5GSubscriberData - Collect 5G Subscriber Data

Syntax

--collect5GSubscriberData <list-item>

Description

When enabled, periodically collect subscriber data for 5G subscribers.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

dataRetentionInterval - Subscriber Data Files Retention Interval

Syntax

--dataRetentionInterval <integer>

Description

Time interval for retention of subscriber data dump files.

Class

Optional.

Type / Range

Integer in the range of 1 to 96 in units of Hour

Default

24

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

7.75 cmm subsForceDetach

Subscriber Force Detach. Syntax, description, parameters, and roles for cmm subsForceDetach command.

Syntax

Description

The Subscriber Force Detach command is used to manage the subscriber force detach for the UE PLMN (mcc, mnc and mncLength) entered. When a PLMN is input, will initiate a walk of the UE Contexts in CMM matching UE's PLMN and detach all UEs for that PLMN. A single Subscriber force Detach process may be ongoing at a time. Process to detach is controlled by 'Force Detach Operation' parameter. Force Detach Operation is 'start' restarts the Force Detach process with new parameters, if the previous Start action is performed is already completed or stopped or this the first time it is run. Force Detach Operation is 'stop' stops a currently Started Force Detach process. This command is blocked during ISSU.

Parameters

Name

forceDetachOp - Force Detach Operation

Syntax

--forceDetachOp <list-item>

Description

Force Detach Operation

Class

Required key.

List

- start - start force detach
- stop - stop force detach

Name

mcc - Mobile Country Code

Syntax

--mcc <digits>

Description

3-digit number used to represent the mobile country code.

Class

Required key.

Type / Range

Digit string of length 3

Name

mnc - Mobile Network Code

Syntax

--mnc <digits>

Description

Digits used to represent mobile network code.

Class

Required key.

Type / Range

Digit string of length 2 to 3

Name

mncDigits - Number of MNC Digits

Syntax

--mncDigits <list-item>

Description

Number of digits used to represent the mobile network code.

Class

Required key.

List

- 2/3Digits - 2 or 3 digits - 2 or 3 digits.
- 2Digits - 2 digits - 2 digits.
- 3Digits - 3 digits - 3 digits.

Default

3Digits

Name

detachGuardTimer - Force Detach Guard Timer

Syntax

--detachGuardTimer <integer>

Description

Subscriber Force Detach Guard Timer in unit of minutes

Class

Optional.

Type / Range

Integer in the range of 1 to 10

Default

2

Name

nasCauseCode3g2g - NAS Cause Code for 3g2g

Syntax`--nasCauseCode3g2g <integer>`**Description**

Nas Cause Code for 3g2g

Class

Optional.

Type / Range

Integer in the range of 2-17, 20-23, 25, 28, 32-34, 40, 95- 101, 111

Default

11

Name

nasCauseCode4g - NAS Cause Code for 4g

Syntax`--nasCauseCode4g <integer>`**Description**

Nas Cause Code for 4g UE

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

11

Name

nasCauseCode5g - NAS Cause Code for 5g

Syntax`--nasCauseCode5g <integer>`**Description**

Nas Cause Code for 5g

Class

Optional.

Type / Range

5GMM Cause Code.

Default

11

Name

removeEmergency - Force Remove Emergency UEs

Syntax

--removeEmergency <list-item>

Description

Force Remove Emergency UEs belonging to the PLMN entered

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**Modify**

sysadmin, manage

7.76 cmm subsForceDetachStatus

Subscriber Force Detach Status. Syntax, description, parameters, and roles for cmm subsForceDetachStatus command.

Syntax**Description**

Displays the status of the Subscriber Force Detach Command

Parameters**Name**

detachGuardTimer - Force Detach Guard Timer

Syntax

--detachGuardTimer <integer>

Description

Force Detach Guard Timer

Class

Read only.

Type / Range

Integer in the range of 1 to 10 in units of Minute

Name

forceDetachCmdStatus - Force Detach Command Status

Syntax

--forceDetachCmdStatus <list-item>

Description

Force Detach Command Status

Class

Read only.

List

- completed - Force Detach Command Completed
- failed - Force Detach Command failed with error
- running - Force Detach Command Running
- stopped - Force Detach Command Stopped

Name

forceDetachOp - Force Detach Operation

Syntax

--forceDetachOp <list-item>

Description

Force Detach Operation

Class

Read only.

List

- start - start force detach
- stop - stop force detach

Name

mcc - Mobile Country Code

Syntax

--mcc <digits>

Description

3-digit number used to represent the mobile country code

Class

Optional.

Type / Range

Digit string of length 3

Name

mnc - Mobile Network Code

Syntax

--mnc <digits>

Description

Digits used to represent mobile network code.

Class

Optional.

Type / Range

Digit string of length 2 to 3

Name

mncDigits - Number of MNC Digits

Syntax

--mncDigits <list-item>

Description

Number of digits used to represent the mobile network code.

Class

Optional.

List

- 2/3Digits - 2 or 3 digits - 2 or 3 digits.
- 2Digits - 2 digits - 2 digits.
- 3Digits - 3 digits - 3 digits.

Name

removeEmergency - Force Remove Emergency UEs

Syntax

--removeEmergency <list-item>

Description

Force Remove Emergency UEs belonging to the PLMN entered

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Roles**List**

sysadmin, manage, monitor

7.77 cmm subsImsiDisplay

Subscriber Imsi Display. Syntax, description, parameters, and roles for cmm subsImsiDisplay command.

Syntax**Description**

Display representative IMSIs for an input PLMN.. Operator needs to run subsImsiDisplayAdmin Command before to trigger the IPDS to collect the data.

Parameters**Name**

index - Index

Syntax

--index <integer>

Description

Index

Class

Required key.

Type / Range

Integer in the range of 1 to 100

Name

mcc - Mobile country code

Syntax

--mcc <digits>

Description

3-digit number used to represent the mobile country code.

Class

Required key.

Type / Range

Digit string of length 3

Name

mnc - Mobile network code

Syntax

--mnc <digits>

Description

Digits used to represent mobile network code.

Class

Required key.

Type / Range

Digit string of length 2 to 3

Name

imsi - IMSI

Syntax

--imsi <digits>

Description

IMSI

Class

Read only.

Type / Range

Digit string of length 5 to 15

Name

radioType2g3g - RadioType3g2g

Syntax

--radioType2g3g <list-item>

Description

Indicates if IMSI is radio type is 3g2g

Class

Read only.

List

- false - Boolean value of false.

- true - Boolean value of true.

Name

radioType4g - RadioType4g

Syntax

--radioType4g <list-item>

Description

Indicates if IMSI is radio type 4g

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

radioType5g - RadioType5g

Syntax

--radioType5g <list-item>

Description

Indicates if IMSI is radio type 5g

Class

Read only.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles**List**

sysadmin, manage, monitor

7.78 cmm subslmsiDisplayAdmin

sublmsiDisplayAdmin. Syntax, description, parameters, and roles for cmm subslmsiDisplayAdmin command.

Syntax

Description

SubImsiDisplayAdmin command is to used to start the Display representative IMSIs for an input PLMN

Parameters

Name

mcc - Mobile country code

Syntax

--mcc <digits>

Description

3-digit number used to represent the mobile country code.

Class

Required key.

Type / Range

Digit string of length 3

Name

mnc - Mobile network code

Syntax

--mnc <digits>

Description

Digits used to represent mobile network code

Class

Required key.

Type / Range

Digit string of length 2 to 3

Name

taskOp - Task Operation

Syntax

--taskOp <list-item>

Description

Defines Task Operation

Restrictions

Required for Modify case

Class

Required on create.

List

- clear - Clear Collected IMSIs - Defines Clearing Collected IMSIs operation
- status - Show status of dump - Status query
- dump - Dump IMSIs - Defines Dump IMSIs operation

Name

imsiCount - IMSI Count

Syntax

--imsiCount <integer>

Description

Count of IMSIs requested 1..100. In status message the number of IMSIs found during dump operation.

Class

Optional.

Type / Range

Integer in the range of 0 to 100

Default

20

Name

taskStatus - Task Status

Syntax

--taskStatus <list-item>

Description

Task Status

Class

Read only.

List

- cleared - cleared the collected list. IMSI are not deleted.
- completed - Dump command is completed
- failed - Command failed
- notRunning - Not Running - Command not running
- running - Dump command is running

Roles

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

7.79 cmm subslmsiOffload

subslmsiOffload. Syntax, description, parameters, and roles for cmm subslmsiOffload command.

Syntax

Description

This command is used in order to control the IMSI Offloading Procedure. It will be possible to create/list/show IMSI based offloading parameters like IMSI and NRI.

Parameters

Name

subsnri - NRI of Target SGSN for offloading subscribers

Syntax

--subsnri <integer>

Description

This parameter defines the target SGSN NRI configuration. The value ranges from 0 to 127.

Class

Required on create, not filterable on list.

Type / Range

Integer in the range of 0 to 127

Name

id - IMSI digits of the subscriber to offload

Syntax

<digits>

Description

Use this parameter to indicate the IMSI number of the subscriber to be offloaded

Restrictions

This parameter is also the key for the REDIS DB in order to make sure that it will not be possible to add the same IMSI twice

Class

Required value.

Type / Range

Digit string of length 6 to 15

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.80 cmm subsOffloadCount

subsOffloadCount. Syntax, description, parameters, and roles for cmm subsOffloadCount command.

Syntax**Description**

Show paps offloading subscriber percentages and total counts.

Parameters**Name**

currentCount - Offload rate of target SGSN for offloading subscribers

Syntax

--currentCount <string>

Description

This parameter shows the current number of subscribers offloaded.

Class

Optional.

Type / Range

Digit string of length 1 to 10 with the allowed pattern of [0-9a-zA-Z\._ :]*

Name

nri - Nri

Syntax

--nri <string>

Description

Target SGSN NRI.

Class

Optional.

Type / Range

Digit string of length 1 to 5 with the allowed pattern of [0-9a-zA-Z\._ :]*

Name

percent - Offload count of target SGSN for offloading subscribers

Syntax

--percent <string>

Description

This parameter defines the percentage of subscribers offloaded from source SGSN.

Class

Optional.

Type / Range

Digit string of length 1 to 5 with the allowed pattern of [0-9a-zA-Z\._ :]*

Name

statusCode - Offloading status

Syntax

--statusCode <string>

Description

Status of offloading process.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._ : @]*

Name

targetId - Id of target SGSN for offloading subscribers info

Syntax

--targetId <integer>

Description

This parameter defines the index of the target SGSN configuration record. the values range from 1 to 16.

Class

Optional.

Type / Range

Integer in the range of 1 to 16

Name

totalCount - Total number of subscribers in target SGSN for offloading

Syntax

--totalCount <string>

Description

This parameter shows the total number of subscribers to be offloaded.

Class

Optional.

Type / Range

Digit string of length 1 to 10 with the allowed pattern of [0-9a-zA-Z\._ :]*

Roles**List**

sysadmin, manage, monitor

7.81 cmm subsOffloadTarget

subsOffloadTarget. Syntax, description, parameters, and roles for cmm subsOffloadTarget command.

Syntax**Description**

Configure, list and show Target SGSN for offloading subscriber info.

Parameters**Name**

subsNri - NRI of target SGSN for offloading subscribers

Syntax

--subsNri <integer>

Description

This parameter defines the target SGSN NRI configuration. the value range from 0 to 127.

Class

Required on create, not filterable on list.

Type / Range

Integer in the range of 0 to 127

Name

subsOffPercentage - Offload count of target SGSN for offloading subscribers

Syntax

--subsOffPercentage <integer>

Description

This parameter defines the percentage of subscribers to be offloaded from source SGSN.

Class

Required on create, not filterable on list.

Type / Range

Integer in the range of 0 to 100

Name

subsOffrate - Offload rate of target SGSN for offloading subscribers

Syntax

--subsOffrate <integer>

Description

This parameter defines the number of subscribers to be offloaded per second per PAPS. The values range from 1 to 255

Class

Required on create, not filterable on list.

Type / Range

Integer in the range of 1 to 255

Name

id - Id of target SGSN for offloading subscribers info

Syntax

<integer>

Description

This parameter defines the index of the target SGSN configuration record. the values range from 1 to 16.

Class

Required value.

Type / Range

Integer in the range of 1 to 16

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.82 cmm subsStateInfo

subsStateInfo. Syntax, description, parameters, and roles for **cmm subsStateInfo** command.

Syntax**Description**

Count subscribers per RAT.

Parameters**Name**

imei - IMEI

Syntax

--imei <digits>

Description

IMEI.

Class

Optional.

Type / Range

Digit string of length 1 to 14

Name

imsi - IMSI

Syntax

--imsi <digits>

Description

IMSI.

Class

Optional.

Type / Range

Digit string of length 5 to 15

Name

msisdn - MSISDN

Syntax

--msisdn <digits>

Description

MSISDN.

Class

Optional.

Type / Range

Digit string of length 5 to 18

Name

papsId - PapsId

Syntax

--papsId <integer>

Description

PAPS index.

Class

Optional.

Type / Range

Integer in the range of 0 to 19

Name

ptmsi - PTMSI

Syntax

--ptmsi <digits>

Description

IMSI.

Class

Optional.

Type / Range

Digit string of length 4

Roles**List**

sysadmin, manage, monitor

7.83 cmm suciSchemeAllowedList

UEs SUCI Scheme Output Allowed List. Syntax, description, parameters, and roles for cmm suciSchemeAllowedList command.

Syntax**Description**

Administer the UEs SUCI Scheme Output allowed list.

Parameters**Name**

name - UEs Suci Scheme Output Allow List Name

Syntax

--name <string>

Description

Identifies the name of the UEs SUCI Scheme Output allowed list. The allowed list includes the SUCI Scheme Output of the inbound roaming UEs SUCIs to be allowed in the visited PLMN.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.84 cmm suciSchemeAllowedListItem

UE SUCI Scheme Output Allow List Item. Syntax, description, parameters, and roles for cmm suciSchemeAllowedListItem command.

Syntax

Description

Configures the inbound roaming UEs Scheme Output allow list with each of the UE individual SUCI Scheme Output that is allowed to roam into the visited PLMN network.

Parameters

Name

suciSchemeAllowedListName - UE Suci Scheme Output Allow List Name

Syntax

```
--suciSchemeAllowedListName <string>
```

Description

Identifies the name of the SUCI Scheme Output allow list.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

ueSuciSchemeOutput - UE SUCI Scheme Output

Syntax

```
--ueSuciSchemeOutput <string>
```

Description

Identifies the Scheme Output part of the UE SUCI provided by the UE during UE initial registration.

Class

Required key.

Type / Range

Hexadecimal string of length 1 to 92 with allowed pattern [0-9a-f]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.85 cmm suMaintenance

SU Maintenance. Syntax, description, parameters, and roles for cmm suMaintenance command.

Syntax

Description

Set SU Maintenance State.

Parameters

Name

maintState - Maintenance state

Syntax

--maintState <list-item>

Description

Maintenance state.

Class

Optional.

List

- nodata - No data - Specify maintenance state of no data.
- normal - Specify maintenance state of normal.
- su - Specify maintenance state of software update.

Default

normal

Roles

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

7.86 cmm supiRange

SUPI Range. Syntax, description, parameters, and roles for cmm supiRange command.

Syntax

Description

Add or delete SUPI range to a named list consisting of SUPI range.

Parameters

Name

supiRangeListName - SUPI Range List

Syntax

--supiRangeListName <string>

Description

Name of the list consisting of one more SUPI ranges.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

startSupi - Starting of the SUPI range

Syntax

--startSupi <digits>

Description

Starting of the SUPI range.

Class

Required key.

Type / Range

Digit string of length 9 to 15

Name

endSupi - End of the SUPI range

Syntax

--endSupi <digits>

Description

End of the SUPI range.

Class

Required key.

Type / Range

Digit string of length 9 to 15

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.87 cmm supiRangeList

SUPI Range List. Syntax, description, parameters, and roles for cmm supiRangeList command.

Syntax**Description**

List for SUPI ranges.

Parameters**Name**

name - List Name

Syntax

--name <string>

Description

SUPI ranges list name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

supiRangeListType - AMF Supi Range Type

Syntax

--supiRangeListType <list-item>

Description

Identifies the AMF Supi Range Type associated or linked

Class

Required on create, not modifiable.

List

- amfPolicySelection - Supi Range for AMF Policy Selection - Identifies Supi Range(s) associated or linked to AMF Policy Selection.
- nfnInstance - Supi Range for Network Function Instance(s) - Identifies Supi Range(s) associated or linked to network function instance(s) such as UDM, AUSF and PCF.
- supiRangeServices - Supi Range for Supi Range Services - Identifies Supi Range(s) associated or linked to Supi Range Services.

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.88 cmm sustDelApnList

Sustained or Delayed APN List. Syntax, description, parameters, and roles for cmm sustDelApnList command.

Syntax

Description

Define APN lists that either -disable the UE detaching by regions due to UE connecting to an APN that has to be sustained despite of the mobility, or -delay the UE detaching by regions due to UE connecting to an APN that can only be deleted when dedicated bearers are deleted. An APN list can include max 64 APNs. A single APN NI can only be included in either type of APN list.

Parameters

Name

sustDelApnListName - Sustained or Delayed APN List Name

Syntax

--sustDelApnListName <string>

Description

APN list name for enhanced detach UE move regions functionality.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

listType - List Type

Syntax

--listType <list-item>

Description

Defines the list type of the APN list.

Class

Required on create.

List

- APNLISTTYPE_DELAYED - APN List Type Delayed - APN list of type Delayed.
- APNLISTTYPE_SUSTAINED - APN List Type Sustained - APN list of type Sustained.

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.89 cmm sustDelApnNi

Sustained or Delayed APN Network Identifier. Syntax, description, parameters, and roles for cmm sustDelApnNi command.

Syntax

Description

Define APN that requires enhanced functionality related to detach UE move regions feature Define APN that either -disables the UE detaching by regions due to UE connecting to an APN that has to be sustained despite of the mobility, or -delays the UE detaching by regions due to UE connecting to an APN that can only be deleted when dedicated bearers are deleted. A single APN NI can only be included in either type of APN list.

Parameters

Name

sustDelApnListName - Sustained or Delayed APN List Name

Syntax

--sustDelApnListName <string>

Description

APN list name for enhanced detach UE move regions functionality.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

apnNi - APN Network Identifier

Syntax

--apnNi <string>

Description

APN Network Identifier.

Class

Required key.

Type / Range

Digit string of length 1 to 62 with the allowed pattern of [0-9a-zA-Z\.\-]*

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.90 cmm svcAgreementProfile

UE PLMN and Served PLMN Service Agreement Profile. Syntax, description, parameters, and roles for cmm svcAgreementProfile command.

Syntax**Description**

This command is used to administer a UE PLMN and Served PLMN Service Agreement Profile and includes parameters for defining RAT networks to which handover is not allowed, support for emergency services, and parameters for configuring CSFB, IMS, LPA, MPS, ODB, power save mode, S102, SMS, and VPLMN.

Parameters**Name**

name - Service agreement profile name

Syntax

--name <string>

Description

Defines the PLMN service agreement profile to be used for service.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

accRestCdma2000 - CDMA2000 access restricted

Syntax

--accRestCdma2000 <list-item>

Description

RAT network types to which handover is not allowed from the current (MME home) network. if no options are checked (indicating that handover is not allowed), then the inter-rat handover restrictions received from the HSS/HLR in the UE home network are used.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

accRestEutran - EUTRAN access restricted

Syntax

--accRestEutran <list-item>

Description

RAT network types to which handover is not allowed from the current (MME home) network. if no options are checked (indicating that handover is not allowed), then the inter-rat handover restrictions received from the HSS/HLR in the UE home network are used.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

accRestEutranExceptLtem - Access Restrictions E-UTRAN Except LTE-M

Syntax

--accRestEutranExceptLtem <list-item>

Description

Specifies whether a UE is restricted from accessing LTE but is allowed access to LTE-M RAT. This field applies for home, roamer, shared, and TAHS plmns. If the field allows the access, the MME still checks the UE's access restriction data from the HSS (if any) prior to allowing access to the LTE RAT. The eNB informs the MME whether the UE is using LTE-M via the S1AP UE Capability Info Indication message.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

accRestGeran - GERAN access restricted

Syntax

--accRestGeran <list-item>

Description

RAT network types to which handover is not allowed from the current (MME home) network. if no options are checked (indicating that handover is not allowed), then the inter-rat handover restrictions received from the HSS/HLR in the UE home network are used.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

accRestLtem - Access Restrictions LTE-M

Syntax

--accRestLtem <list-item>

Description

Specifies whether a UE is restricted from accessing LTE-M IoT RAT. This field applies for home, roamer, shared, and TAHS plmns. If the field allows LTE-M RAT, the MME still checks the UE's access restriction data from the HSS (if any) prior to allowing access to LTE-M RAT. The eNB informs the MME whether the UE is using LTE-M via the S1AP UE Capability Info Indication message.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

accRestNbiot - Access Restrictions NB-IoT

Syntax

--accRestNbiot <list-item>

Description

Specifies whether a UE is restricted from accessing NB-IoT RAT. This field applies for home, roamer, shared, and TAHS plmns. If the field allows NB-IoT RAT, the MME still checks the UE's access restriction data from the HSS (if any) prior to allowing access to NB-IoT RAT.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

accRestUtran - UTRAN access restricted

Syntax

--accRestUtran <list-item>

Description

RAT network types to which handover is not allowed from the current (MME home) network. if no options are checked (indicating that handover is not allowed), then the inter-rat handover restrictions received from the HSS/HLR in the UE home network are used.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

checkSrvccCapaForVopsHomogeneous - Check SRVCC Capability for VoPS Homogeneous

Syntax

--checkSrvccCapaForVopsHomogeneous <list-item>

Description

This parameter defines whether the MME checks UE SRVCC capability for indicating homogeneous IMS voice over PS support towards the HSS. When set to 'yes' and the UE supports SRVCC, MME indicates homogeneous support for the UE towards the HSS, if the TAs support voice. When set to 'yes' and the UE does not support SRVCC, MME indicates homogeneous non-support for the UE. When set to 'no', MME indicates homogeneous support based on voice over PS support of the TAs, the UE support for VoPS and the other VoPS impacting parameters.

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

csCapSupported - CS capability supported

Syntax

--csCapSupported <list-item>

Description

The level of CSFB capability supported for this PLMN.

Class

Optional.

List

- CSFB_2G3G - CSFB 2G/3G - Both SMS and CS are supported.

- CSFB_Not_Preferred - CSFB not preferred - This option supports both SMS and CSFB. the MME indicates csfb-not-preferred to the UE, which indicates to the UE to invoke voice services only in the CS domain and use IMS voice, if supported.
- SGS_None - SGS none - All CS based services (CSFB and SMS based on CS) are restricted.
- SMS_Only - SMS only - Only SMS is supported.
- SMS_Only_Faked_Full - SMS only faked full combined - MME fakes towards the UE to support full combined attach i.e. both SMS and CSFB, but locally restricts the service to SMS only. The MME rejects CSFB attempts for regular calls with provisioned causes. The MME follows an additional provisioning parameter to reject or not reject emergency attempts.

Default

SGS_None

Name

csfbDtr - CSFB DTR

Syntax

--csfbDtr <list-item>

Description

Indicates a list of services that UEs can utilize when roaming into the MME home network from the given network.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

csPrefRfsp - CS preferred RFSP

Syntax

--csPrefRfsp <integer>

Description

CS preferred RFSP.

Class

Optional.

Type / Range

Integer in the range of 0 to 256

Default

255

Name

csPrefRfspDataCentricUe - CS preferred RFSP for data centric UE

Syntax

--csPrefRfspDataCentricUe <integer>

Description

CS preferred RFSP for data centric ue.

Class

Optional.

Type / Range

Integer in the range of 0 to 256

Default

256

Name

csRfsp - CS RFSP

Syntax

--csRfsp <integer>

Description

Indicates whether (zero value) or not (non-zero value) the index value for CS RAT/Frequency selection priority (RFSPindex) received from the HSS in the UE home PLMN is to be used. "0" indicates the HSS-provided value should be used. if a non-zero value is provisioned, this numerical value over-rides the value received from the HSS in the home network of the UE.

Class

Optional.

Type / Range

Integer in the range of 0 to 256

Default

255

Name

csRfspDataCentricUe - CS RFSP for data centric UE

Syntax

--csRfspDataCentricUe <integer>

Description

Indicates whether (zero value) or not (non-zero value) the index value for CS RAT/Frequency selection priority (RFSPindex) received from the HSS in the UE home PLMN is to be used. "0" indicates the HSS-provided value should be used. if a non-zero value is provisioned, this numerical value over-rides the value received from the HSS in the home network of the UE.

Class

Optional.

Type / Range

Integer in the range of 0 to 256

Default

256

Name

description - Description

Syntax

--description <string>

Description

Profile description.

Class

Optional.

Type / Range

Digit string of length 0 to 160 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Name

emergencySupported - Emergency supported

Syntax

--emergencySupported <list-item>

Description

This command is used to enable indication of emergency service supported or not to a UE in Attach Accept or TAU Accept message. The no form of the command will result in indicate to a UE that emergency service is not supported. By default the emergency service is disabled. The emergency Service can only be enabled if IMS voice is enabled. However, the emergency service can be enabled irrespective of IMS voice support if the gParm imsEmergencySupportIndication is set to Yes. Based on the emergency bearer provisioning and IMS voice support, MME will set the Emergency bearer services indicator bit (emergency bearer services in S1 mode supported or emergency bearer services in S1 mode not supported) of the EPS network feature support information IE of the Attach Accept or TAU Accept message sent to a UE. UE uses this information to access LTE network when

initiating an emergency call. MME rejects a UE Attach Request for emergency service and PDN Connectivity Request for emergency APN if the emergency service is not enabled.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

fakedFullCombSmsSgsEmerSrvRejCause - Faked Full Combined SMS SGs Emergency Service Reject Cause

Syntax

--fakedFullCombSmsSgsEmerSrvRejCause <integer>

Description

Defines the EMM cause code used to reject Extended Service Request attempts for CSFB for emergency calls when the MME fakes full combined attach/TAU but locally restricts the CS services to SMS only, and MME is provisioned to reject emergency call attempts.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

39

Name

fakedFullCombSmsSgsRejectEmerCalls - Faked Full Combined SMS SGs Reject Emergency Calls

Syntax

--fakedFullCombSmsSgsRejectEmerCalls <list-item>

Description

Defines whether the MME rejects emergency call attempts (Extended Service Requests indicating emergency), when the MME fakes full combined attach/TAU but locally restricts the CS services to SMS only.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

fakedFullCombSmsSgsSgsapPagingRejCause - Faked Full Combined SMS SGs SGsAP Paging Reject Cause

Syntax

--fakedFullCombSmsSgsSgsapPagingRejCause <integer>

Description

The cause the MME uses on SGsAP-PAGING-REJECT to reject MT CSFB attempts, when the MME fakes full combined attach/TAU but locally restricts the CS services to SMS only.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

13

Name

fakedFullCombSmsSgsSrvRejCause - Faked Full Combined SMS SGs Service Reject Cause

Syntax

--fakedFullCombSmsSgsSrvRejCause <integer>

Description

Defines the EMM cause code used to to reject Extended Service Request attempts for CSFB when the MME fakes full combined attach/TAU but locally restricts the CS services to SMS only.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

39

Name

imsOverPs - IMS voice over PS

Syntax

--imsOverPs <list-item>

Description

Indicates a list of services that UEs can utilize when roaming into the MME home network from the given network.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

imsPrefRfsp - IMS preferred RFSP

Syntax

--imsPrefRfsp <integer>

Description

IMS preferred RFSP.

Class

Optional.

Type / Range

Integer in the range of 0 to 256

Default

256

Name

imsPrefRfspDataCentricUe - IMS preferred RFSP for data centric UE

Syntax

--imsPrefRfspDataCentricUe <integer>

Description

IMS preferred RFSP for data centric ue.

Class

Optional.

Type / Range

Integer in the range of 0 to 256

Default

256

Name

imsRfsp - IMS RFSP

Syntax

--imsRfsp <integer>

Description

Indicates whether (zero value) or not (non-zero value) the index value for IMS RAT/Frequency selection priority (RFSPindex) received from the HSS in the UE home PLMN is to be used. "0" indicates the HSS-provided value should be used. if a non-zero value is provisioned, this numerical value over-rides the value received from the HSS in the home network of the UE.

Class

Optional.

Type / Range

Integer in the range of 0 to 256

Default

256

Name

imsRfspDataCentricUe - IMS RFSP for data centric UE

Syntax

--imsRfspDataCentricUe <integer>

Description

Indicates whether (zero value) or not (non-zero value) the index value for IMS RAT/Frequency selection priority (RFSPindex) received from the HSS in the UE home PLMN is to be used. "0" indicates the HSS-provided value should be used. if a non-zero value is provisioned, this numerical value over-rides the value received from the HSS in the home network of the UE.

Class

Optional.

Type / Range

Integer in the range of 0 to 256

Default

256

Name

lowPriorityAccess - Low priority access

Syntax

--lowPriorityAccess <list-item>

Description

Configure to allow a LPA UE to access network.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

lpaMmRejectCc - LPA mobility manager reject cause code

Syntax

--lpaMmRejectCc <integer>

Description

Configure NAS MM CC to be sent to a LPA UE when a LPA mobility management request is rejected due to restricted LPA.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

35

Name

lpaSmRejectCc - LPA session manager reject cause code

Syntax

--lpaSmRejectCc <integer>

Description

Configure NAS SM CC to be sent to a LPA UE when a LPA session management request is rejected due to restricted LPA.

Class

Optional.

Type / Range

Integer in the range of 8, 26-35, 38, 50-51, 53-55, 65-66, 95-112

Default

32

Name

lpaT3412TimerSrc - LPA T3412 timer source

Syntax

```
--lpaT3412TimerSrc <list-item>
```

Description

Configure source of the T3412 timer for LPA UE.

Class

Optional.

List

- HSS - The duration of T3412 is set to the value received from the HSS.
- Local - The duration of T3412 is determined from the MME-provisioned settings.
- Max - Maximum - The duration of T3412 is whatever duration is longer between the value provisioned on the MME or the duration received from the HSS.

Default

Local

Name

mpsCsAllowed - MPS CS allowed

Syntax

```
--mpsCsAllowed <list-item>
```

Description

MPS CS allowed.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

mpsEpsAllowed - MPS EPS allowed

Syntax

```
--mpsEpsAllowed <list-item>
```

Description

MPS EPS allowed.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

networkAccessMode - LTE network access mode

Syntax

--networkAccessMode <list-item>

Description

Determines processing mode for UEs that roam into the MME home network.

Class

Optional.

List

- Packet and Circuit - Specifies network access mode of packet and circuit.
- Packet only - Specifies network access mode of packet only.

Default

Packet and Circuit

Name

odbAllApn - All APN barred

Syntax

--odbAllApn <list-item>

Description

Specifies whether the MME supports receipt of operator determined barring of all packet oriented services (ODB-all-APN) from the HSS. if unchecked (disabled), the MME will not set the corresponding bits in the supported-features IE sent to the HSS in the ULR and IDA message, and the HSS must handle barring locally. if checked, then the MME sets the corresponding bits in the supported-features IE to inform the HSS that the MME will enforce ODB barring restrictions received in a ULA or IDR message.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

odbHplmnApn - HPLMN APN barred

Syntax

--odbHplmnApn <list-item>

Description

Specifies whether the MME supports receipt of operator determined barring of all packet oriented services within the UE home PLMN (ODB-HPLMN-APN) from the HSS. if unchecked (disabled), the MME will not set the corresponding bits in the supported-features IE sent to the HSS in the ULR and IDA message, and the HSS must handle barring locally. if checked, then the MME sets the corresponding bits in the supported-features IE to inform the HSS that the MME will enforce ODB barring restrictions received in a ULA or IDR message.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

odbVplmnApn - VPLMN APN barred

Syntax

--odbVplmnApn <list-item>

Description

Specifies whether the MME supports receipt of all packet oriented services in the MME home PLMN (ODB-VPLMN-APN) from the HSS. if unchecked (disabled), the MME will not set the corresponding bits in the supported-features IE sent to the HSS in the ULR and IDA message, and the HSS must handle barring locally. if checked, then the MME sets the corresponding bits in the supported-features IE to inform the HSS that the MME will enforce ODB barring restrictions received in a ULA or IDR message.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

overwriteRfspIndex - Overwrite RFSP Index

Syntax

--overwriteRfspIndex <integer>

Description

When gParm rfspIndexSelectionOption is enabled and rfspSelOption equals overwriteRfspIndex then the MME sends locally provisioned RFSP value, as defined in current field, ignoring whatever value may have been received from the HSS

Class

Optional.

Type / Range

Integer in the range of 0 to 256

Default

0

Name

pgwRstApnList - PGW restart APN LIST

Syntax

--pgwRstApnList <string>

Description

PGW restart APN NI list name.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

''

Name

psmT3324Timer - Power savings mode T3324 timer

Syntax

--psmT3324Timer <integer>

Description

Configure T3324 timer value to support UE power savings mode.

Class

Optional.

Type / Range

Integer in the range of 0 to 11,160

Default

300

Name

psmT3324TimerSrc - Power savings mode T3324 timer source

Syntax

--psmT3324TimerSrc <list-item>

Description

Configure source of T3324 timer value.

Class

Optional.

List

- HSS - Specify source of T3324 HSS
- Local - Specify source of T3324 timer of local.
- UE - Specify source of T3324 timer of UE.

Default

Local

Name

psmT3412MinTimer - PSM T3412 Minimum Timer Value

Syntax

--psmT3412MinTimer <integer>

Description

This parameter is used to provision T3412 timer to be used for UE supporting PSM. If the provisioned T3412 timer source is UE for UE supporting PSM and T3412 timer received in attach/TAU request from the UE is less than the provisioned minimum T3412 timer then MME sends the provisioned minimum T3412 timer in Attach/TAU accept.

Class

Optional.

Type / Range

Integer in the range of 1 to 31

Default

31

Name

psmT3412MinTimerUnit - Power Savings Mode T3412 Min Timer Unit

Syntax

--psmT3412MinTimerUnit <list-item>

Description

Power savings mode minimum T3412 timer unit.

Class

Optional.

List

- 10Hours - Specify unit of T3412 timer of 10 hours.
- 10Minutes - Specify unit of T3412 timer of 10 minutes.
- 2Seconds - Specify unit of T3412 timer of 2 seconds.
- 30Seconds - Specify unit of T3412 timer of 30 seconds.
- Hour - Specify unit of T3412 timer of hours.
- Minute - Specify unit of T3412 timer of minutes.
- 320Hours - Specify unit of T3412 timer of 320 hours.

Default

Hour

Name

psmT3412Timer - Power savings mode T3412 timer

Syntax

--psmT3412Timer <integer>

Description

Configure T3412 timer to be used for a UE that supports PSM.

Class

Optional.

Type / Range

Integer in the range of 1 to 31

Default

31

Name

psmT3412TimerSrc - Power savings mode T3412 timer source

Syntax

--psmT3412TimerSrc <list-item>

Description

Power savings mode T3412 timer source.

Class

Optional.

List

- HSS - Specify source of T3412 timer of HSS.
- Local - Specify source of T3412 timer of local.
- UE - Specify source of T3412 timer of UE.

Default

Local

Name

psmT3412TimerUnit - Power savings mode T3412 timer unit

Syntax

--psmT3412TimerUnit <list-item>

Description

Power savings mode T3412 timer unit.

Class

Optional.

List

- 10Hours - Specify unit of T3412 timer of 10 hours.
- 10Minutes - Specify unit of T3412 timer of 10 minutes.
- 2Seconds - Specify unit of T3412 timer of 2 seconds.
- 30Seconds - Specify unit of T3412 timer of 30 seconds.
- Hour - Specify unit of T3412 timer of hours.
- Minute - Specify unit of T3412 timer of minutes.
- 320Hours - Specify unit of T3412 timer of 320 hours.

Default

10Minutes

Name

rfspIndex - RFSP index

Syntax

--rfspIndex <integer>

Description

Indicate whether or not the index value for RAT/Frequency selection priority received from the HSS in the UE home PLMN should be used.

Class

Optional.

Type / Range

Integer in the range of 0 to 256

Default

100

Name

rfspIndexVoiceCapableUe - RFSP index for voice capable UEs

Syntax`--rfspIndexVoiceCapableUe <integer>`**Description**

RFSP index for voice capable UEs.

Class

Optional.

Type / Range

Integer in the range of 0 to 128

Default

0

Name

rfspSelOption - RFSP Selection Option

Syntax`--rfspSelOption <list-item>`**Description**

This field defines the way that RFSP Index is selected for the UE. Options are (a) (feature) disabled (b) apply local value as configured in `overwriteRfspIndex` field, (c) use the HSS subscribed value

Class

Optional.

List

- disabled - Feature is disabled at the Service Agreement Profile
- overwriteRfspIndex - Overwrite RFSP Index - MME assigns to the UE the RFSP Index as defined in Overwrite RFSP Index field in Service Agreement Profile.
- sendHssSubRfspIndex - Send HSS Subscription RFSP Index - MME assigns to the UE the RFSP Index as received from the HSS as part of subscription data

Default

disabled

Name

roamerImsPdnConnOption - Roamer IMS PDN connection option

Syntax`--roamerImsPdnConnOption <list-item>`

Description

Roamer IMS PDN connection option. dependency with imsapnroamingcontrol parameter in UE PLMN services.

Class

Optional.

List

- BOTH - Specify roamer IMS PDN connection option of both.
- HRT - Specify roamer IMS PDN connection option of home routed traffic.
- LBO - Specify roamer IMS PDN connection option of local breakout.
- NONE - Specify roamer IMS PDN connection option of none.

Default

BOTH

Name

s102Allowed - S102 services allowed

Syntax

--s102Allowed <list-item>

Description

Indicates whether or not S102 services are allowed.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

s102SrvccAllowed - S102 SRVCC allowed

Syntax

--s102SrvccAllowed <list-item>

Description

This parameter controls activation of S102 1x RTT SRVCC flavor.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sgdAllowed - SGd Allowed

Syntax

--sgdAllowed <list-item>

Description

Defines whether SGd is allowed i.e. whether the MME is allowed to register the user in HSS for SMS delivery.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

smsOnlyLac - LAC of LAI to be used for SMS only

Syntax

--smsOnlyLac <integer>

Description

Location area code, used in combination with smsonlyPlmnname to define the LAI to be used for SMS only.

Class

Optional.

Type / Range

Integer in the range of 0 to 65,535

Default

0

Name

smsOnlyPlmnName - PLMN name of LAI to be used for SMS only

Syntax

--smsOnlyPlmnName <string>

Description

Mobile network name, used in combination with smsonlylac to define the LAI to be used for SMS only.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

srvccPsToCs - SRVCC-Based PS to CS handover

Syntax

--srvccPsToCs <list-item>

Description

Indicates whether handover of a packet switched call to a circuit switched network using the SRVCC protocol is supported.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

t3412Max320HourValue - T3412 Max 320Hour Value

Syntax

--t3412Max320HourValue <integer>

Description

Defines the maximum value for the '320 Hour' Unit for the extended T3412 timer. If a greater value is specified elsewhere in the system (e.g psmAPN), the system will override that value with this provisioned value. Permissible values are 1-31 inclusive. This field is applicable for both lpaT3412 and psmT3412 timers negotiated with the UE.

Class

Optional.

Type / Range

Integer in the range of 1 to 31

Default

1

Name

t3412src - T3412 source

Syntax`--t3412src <list-item>`**Description**

Determines how the T3412 timer is set. this field provides the ability to override the provisioned T3412 timer with subscribed-periodic-rau-tau-timer subscription data or to use the maximum of provisioned and subscription data.

Class

Optional.

List

- MaxProvSub - Maximum of provisioned and subscription - Use maximum value of provisioned T3412 timer and subscribed-periodic-rau-tau-timer subscription data.
- Provisioned - Use provisioned T3412 timer.
- Subscription - Use subscribed-periodic-rau-tau-timer from subscription data.

Default

Provisioned

Name

useLocalRfspHssRfspNotSub - Use Local RFSP if RFSP isn't subscribed at the HSS

Syntax`--useLocalRfspHssRfspNotSub <list-item>`**Description**

Current parameter enables using the locally provisioned RFSP value when local configuration at the uePlmnServices or imsiRangeServices indicates applying the HSS subscribed value but no RFSP value is received from the HSS.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

vplmnAllowed - Honor VPLMN requests

Syntax

--vplmnAllowed <list-item>

Description

Indicates whether a UE is allowed to select a PGW in the VPLMN. if checked, the UE is allowed to use a PGW from the UE HPLMN (home-routed) or from the VPLMN (local-breakout). if unchecked, only home-routed scenario is supported and the roaming UE must use the PGW in their home PLMN.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

vplmnCsgAllowed - CSG allowed

Syntax

--vplmnCsgAllowed <list-item>

Description

CSG allowed.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

vplmnLipaAllowed - LIPA allowed

Syntax

--vplmnLipaAllowed <list-item>

Description

LIPA allowed.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.91 cmm syslogAdmin

Syslog Server Administration. Syntax, description, parameters, and roles for cmm syslogAdmin command.

Syntax

Description

This command will administer the IP address, port and protocol of the remote (r)syslog server

Parameters

Name

syslogConfigName - Syslog Configuration Name

Syntax

--syslogConfigName <string>

Description

Syslog Configuration Name identifies a syslog instance

Class

Required key.

Type / Range

Digit string of length 1 to 50 with the allowed pattern of [0-9a-zA-Z\.\-_:]*

Name

syslogRemoteIp - Syslog server IP Address

Syntax

--syslogRemoteIp <string>

Description

Syslog server IP Address

Class

Required on create.

Type / Range

Digit string of length 2 to 43 with the allowed pattern of [0-9a-fA-F\.:/*]

Name

oamTlsProfileName - OAM TLS Profile

Syntax

--oamTlsProfileName <string>

Description

This parameter contains the profile for TLS. It contains the certificate names needed for establishing secure communication.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Name

permittedPeerName - Name of Remote Peer

Syntax

--permittedPeerName <string>

Description

DNS resolvable remote logging server domain name and this shall be given if --tlsCertificateName has been given.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

syslogProtocol - Syslog Protocol

Syntax

--syslogProtocol <list-item>

Description

Protocol used for streaming transmission of syslog

Class

Optional.

List

- tcp - Specify syslog protocol of TCP.
- udp - Specify syslog protocol of UDP.

Default

tcp

Name

syslogRemotePort - Syslog Remote Port

Syntax

--syslogRemotePort <string>

Description

Syslog Remote Port

Class

Optional.

Type / Range

Digit string of length 0 to 11 with the allowed pattern of ((([0-9]{1,4})[1-5][0-9]{4})6[0-4][0-9]{3})65[0-4][0-9]{2}|655[0-2][0-9]|6553[0-5])|((([0-9]{1,4})[1-5][0-9]{4})6[0-4][0-9]{3}|65[0-4][0-9]{2})655[0-2][0-9]|6553[0-5])-(|[0-9]{1,4})[1-5][0-9]{4}|6[0-4][0-9]{3}|65[0-4][0-9]{2}|655[0-2][0-9]|6553[0-5]))

Default

514

Roles**Create**

sysadmin

Show

sysadmin

Modify

sysadmin

Delete

sysadmin

List

sysadmin

7.92 cmm system

system. Syntax, description, parameters, and roles for **cmm system** command.

Syntax

Description

Configure and show generic system configuration for SGSN.

Parameters

Name

ardMode - Access restriction data

Syntax`--ardMode <list-item>`**Description**

This parameter defines whether the HLR sent access restriction data ARD parameter and the features related to this parameter are used in SGSN or not.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

cc - Country code

Syntax`--cc <digits>`**Description**

Digits characterising the subscribers own country.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 to 18

Default

""

Name

cnId - CN identifier

Syntax

--cnId <integer>

Description

This parameter is core network identifier, CNID. possible values range from 0 to 4095 and 65535 (undefined).

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 4,095 or 65,535

Default

65535

Name

csps2g - CS/PS coordination for 2G

Syntax

--csps2g <list-item>

Description

This parameter defines whether CS/PS coordination is requested from the BSC.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

elddf - Indirect downlink data forwarding value

Syntax

--eIddf <integer>

Description

This parameter is used during the IRAT handovers, so that operator can optimize the user plane route. the possible values are, 1 ... does not apply 2 ... applies always.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 2

Default

1

Name

eir - Equipment identification register ISDN number

Syntax

--eir <digits>

Description

This parameter is an equipment identity register number. when you give the EIR number, give both the numbering plan and the type of number in the second parameter block. the maximum length of the number is 18 digits.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 to 18

Default

""

Name

fullNetwName - Full network name

Syntax

--fullNetwName <string>

Description

This parameter defines the full network name of the network operator.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 to 64

Default

""

Name

gbBufThPri1 - Priority buffer 1 length

Syntax

--gbBufThPri1 <integer>

Description

This parameter defines the buffer length for the scheduling in priority one gb buffer, which is defined as a percent of the total memory. possible values range from 5 to 85.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 5 to 85

Default

10

Name

gbBufThPri2 - Priority buffer 2 length

Syntax

--gbBufThPri2 <integer>

Description

This parameter defines the buffer length for the scheduling in priority two gb buffer, which is defined as a percent of the total memory. possible values range from 5 to 85.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 5 to 85

Default

20

Name

gbBufThPri3 - Priority buffer 3 length

Syntax

--gbBufThPri3 <integer>

Description

This parameter defines the buffer length for the scheduling in priority three gb buffer, which is defined as a percent of the total memory. possible values range from 5 to 85.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 5 to 85

Default

30

Name

gbBufThPri4 - Priority buffer 4 length

Syntax

--gbBufThPri4 <integer>

Description

This parameter defines the buffer length for the scheduling in priority four gb buffer, which is defined as a percent of the total memory. possible values range from 5 to 85.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 5 to 85

Default

40

Name

gsMode - Existence of gs interface

Syntax

--gsMode <list-item>

Description

This parameter defines whether the gs interface exists towards the MSC/VLR.

Class

Optional, not filterable on list.

List

- exists - Specifies the gs interface exists.
- notExists - Not exists - Specifies the gs interface does not exist.

Default

notExists

Name

homeSubs - Home subscriber group controlled roaming

Syntax

--homeSubs <list-item>

Description

This parameter defines whether the home subscriber group controlled roaming is in use in flexi NS SGSN.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

imsiEndDigit - End digit of IMSI digits used in HOME GGSN selection

Syntax

--imsiEndDigit <integer>

Description

This parameter defines the possible range values of IMSI end digits used in HOME GGSN selection. the values range from 1 to 15.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 15

Default

7

Name

imsiStartDigit - Start digit of IMSI digits used in HOME GGSN selection

Syntax

--imsiStartDigit <integer>

Description

This parameter defines the possible range values of IMSI start digits used in HOME GGSN selection. the values range from 1 to 15.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 15

Default

6

Name

ipv4Tag - PDN type tag for IPv4 to be used in DNS query for GGSN/PGW selection

Syntax

--ipv4Tag <string>

Description

PDN type tag for IPv4 to be used in DNS query for GGSN/PGW selection.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 to 10 with the allowed pattern of [0-9a-zA-Z\.\-:]*

Default

""

Name

ipv4v6Tag - PDN type tag for IPv4v6 to be used in DNS query for GGSN/PGW selection

Syntax

--ipv4v6Tag <string>

Description

PDN type tag for IPv4v6 to be used in DNS query for GGSN/PGW selection.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 to 10 with the allowed pattern of [0-9a-zA-Z\.\-:]*

Default

""

Name

ipv6Tag - PDN type tag for IPv6 to be used in DNS query for GGSN/PGW selection

Syntax

--ipv6Tag <string>

Description

PDN type tag for IPv6 to be used in DNS query for GGSN/PGW selection.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 to 10 with the allowed pattern of [0-9a-zA-Z\.\-:]*

Default

""

Name

lboImsi - Local breakout at IMSI NS level

Syntax

--lboImsi <list-item>

Description

Local breakout at IMSI NS level.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

llcMaxIFrame - Maximum number of outstanding i frames in LLC layer

Syntax

--llcMaxIFrame <integer>

Description

This parameter defines the maximum number of outstanding i frames in the logical link control LLC layer. possible values range from 1 to 255.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 255

Default

8

Name

llcMaxN201I - Maximum number of octets in information field N201I

Syntax

--llcMaxN201I <integer>

Description

This parameter defines the maximum number of octets in the information field N201I. possible values range from 140 to 1520.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 140 to 1,520

Default

1250

Name

mcc - Mobile country code, MCC

Syntax

--mcc <digits>

Description

This parameter defines the mobile country code number. the value can be three digits.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 or 3

Default

""

Name

mnc - Mobile network code, MNC

Syntax

--mnc <digits>

Description

This parameter defines the mobile network code number. the value can be two or three digits.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 or 2 to 3

Default

""

Name

msisdnsSubLen - MSISDN subrange starting position for home PGW selection

Syntax

--msisdnsSubLen <integer>

Description

MSISDN subrange length for home PGW selection.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 3 to 7

Default

7

Name

msisdnSubPos - MSISDN subrange starting position for home PGW selection

Syntax

--msisdnSubPos <integer>

Description

MSISDN subrange starting position for home PGW selection.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 3 to 7

Default

3

Name

msStandbyForced - Force to standby

Syntax

--msStandbyForced <list-item>

Description

This parameter defines whether the SGSN forces the MS to the STANDBY state after GMM procedures or not.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

nblac - Non-broadcast location area code, LAC

Syntax

--nblac <integer>

Description

This parameter defines the non-broadcast location area code. possible values range from 1 to 65533. if you enter a value for the nblac parameter, you must enter values also for the nbmcc, nmmnc and nbrac.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 65,535

Default

65535

Name

nbmcc - Non-broadcast mobile country code, MCC

Syntax

--nbmcc <digits>

Description

This parameter defines the non-broadcast mobile country code. if you enter a value for the nbmcc parameter, you must enter values also for the nbmnc, nblac and nbrac.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 or 3

Default

""

Name

nbmnc - Non-broadcast mobile network code, MNC

Syntax

--nbmnc <digits>

Description

This parameter defines the non-broadcast mobile network code. if you enter a value for the nbmnc parameter, you must enter values also for the nbmcc, nblac and nbrac.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 or 2 to 3

Default

""

Name

nbrac - Non-broadcast routing area code, RAC

Syntax

--nbrac <integer>

Description

This parameter defines the non-broadcast routing area code. possible values range from 0 to 255. if you enter a value for the nbrac parameter, you must enter values also for the nbmcc, nbmnc and nblac.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 255

Default

255

Name

ndc - National destination code

Syntax

--ndc <digits>

Description

Digits characterising the called network and/or trunk code.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 or 1 to 18

Default

""

Name

npForCc - Numbering plan indicator for country code

Syntax

--npForCc <list-item>

Description

Numbering plan indicator for country code.

Class

Optional, not filterable on list.

List

- " - No numbering plan - No numbering plan.
- E164 - Numbering plan E164 - Numbering plan E164.
- E212 - Numbering plan E212 - Numbering plan E212.
- E214 - Numbering plan E214 - Numbering plan E214.

Default

"

Name

npForEir - Numbering plan indicator for equipment identity register number

Syntax

--npForEir <list-item>

Description

Numbering plan indicator for equipment identity register number.

Class

Optional, not filterable on list.

List

- " - No numbering plan - No numbering plan.
- E164 - Numbering plan E164 - Numbering plan E164.
- E212 - Numbering plan E212 - Numbering plan E212.
- E214 - Numbering plan E214 - Numbering plan E214.

Default

"

Name

npForNdc - Numbering plan indicator for national destination code

Syntax

--npForNdc <list-item>

Description

Numbering plan indicator for national destination code.

Class

Optional, not filterable on list.

List

- " - No numbering plan - No numbering plan.
- E164 - Numbering plan E164 - Numbering plan E164.

- E212 - Numbering plan E212 - Numbering plan E212.
- E214 - Numbering plan E214 - Numbering plan E214.

Default

"

Name

npForSgsn - Numbering plan indicator for serving GPRS support node

Syntax

--npForSgsn <list-item>

Description

This parameter is the code which refers to the standardised numbering instruction described by the ITU T.

Class

Optional, not filterable on list.

List

- " - No numbering plan - No numbering plan.
- E164 - Numbering plan E164 - Numbering plan E164.
- E212 - Numbering plan E212 - Numbering plan E212.
- E214 - Numbering plan E214 - Numbering plan E214.

Default

"

Name

numbOfProcAttemptsInInterval - Number of procedure attempts allowed within one interval

Syntax

--numbOfProcAttemptsInInterval <integer>

Description

This parameter defines the number of procedure attempts allowed within one interval.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 1000

Default

5

Name

originHost - Origin realm for diameter connections

Syntax

--originHost <string>

Description

Origin host name for diameter connections.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 to 110 with the allowed pattern of [0-9a-zA-Z\.\-_:]*

Default

""

Name

originRealm - Origin realm for diameter connections

Syntax

--originRealm <string>

Description

Origin realm for diameter connections.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 to 49 with the allowed pattern of [0-9a-zA-Z\.\-_:]*

Default

""

Name

pdpActProcMonitoring - PDP activation procedure monitoring

Syntax

--pdpActProcMonitoring <list-item>

Description

This parameter enables and disables the PDP activation procedure monitoring.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

pgwtag - PGW tag name

Syntax

--pgwtag <string>

Description

This is the user defined TAG which is used in pre - release 8 DNS query for selecting PGW. the length of the tag ranges from 1 to 10 alphanumeric characters.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 1 to 10 with the allowed pattern of [0-9a-zA-Z\._ :]*

Default

LTE

Name

procAttemptsInterval - Interval in which the SGSN monitors procedure attempts

Syntax

--procAttemptsInterval <integer>

Description

This parameter defines the interval in which the SGSN monitors procedure attempts.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 3,600

Default

60

Name

restartCount - Restart count

Syntax

--restartCount <integer>

Description

Restart count.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 0 to 255

Default

0

Name

rncIdDigits - Number of RNCID digits used in DNS query

Syntax`--rncIdDigits <integer>`**Description**

This parameter defines the number of RNCID digits used in DNS query. the values range from 1 to 5.

Class

Optional, not filterable on list.

Type / Range

Integer in the range of 1 to 5

Default

2

Name

s6dSelect - S6D or gr mode for roamers

Syntax`--s6dSelect <list-item>`**Description**

This parameter is added with the purpose of selecting either S6d interface or gr interface, y ... S6D n ... gr.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sgsn - Serving GPRS support node, SGSN

Syntax`--sgsn <digits>`**Description**

This parameter is an SGSN number. in the international format the first digits of the number must contain the country code. when you give the SGSN number, give also the numbering plan in the second parameter block. the maximum length of the number is 18 digits.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 or 1 to 18

Default

""

Name

shortNetwName - Short network name

Syntax

--shortNetwName <string>

Description

This parameter defines the short network name of the network operator.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 to 64

Default

""

Name

subsDataKeepMode - Store subscriber data

Syntax

--subsDataKeepMode <list-item>

Description

This parameter defines whether the subscriber data received from the HLR is stored in flexi NS SGSN subscriber database or not, when the subscriber's access to the SGSN is rejected due to access restriction data parameter value received from the HLR.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

tonForEir - Type of number for equipment identity register number

Syntax

--tonForEir <list-item>

Description

Type of number for equipment identity register number.

Class

Optional, not filterable on list.

List

- " - No nature of address - No nature of address.
- INT - Nature of address INT - Nature of address INT.
- NAT - Nature of address NAT - Nature of address NAT.

Default

"

Name

visitorSubs - Visitor subscriber group controlled roaming

Syntax

--visitorSubs <list-item>

Description

This parameter defines whether the visitor subscriber group controlled roaming is in use in flexi NS SGSN.

Class

Optional, not filterable on list.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

vlr - Visitor location register number

Syntax

--vlr <digits>

Description

This parameter is a visitor location register number. the number must be in the E.164 international format and therefore the first digits of the number must contain the country code. the maximum length of the number is 18 digits.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 or 1 to 18

Default

""

Name

id - Id

Syntax

<string>

Description

Just type 'config' here.

Class

Required value.

Type / Range

Digit string of length 6 to 6 with the allowed pattern of [0-9a-zA-Z\.\-_:]*

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

7.93 cmm systemInfo

System Information. Syntax, description, parameters, and roles for cmm systemInfo command.

Syntax**Description**

This command is used to list System Information and includes parameters describing the system switch name, application type, hardware type, time zone, and software release/build information.

Parameters

Name

application - Application type

Syntax

--application <list-item>

Description

Application type that is configured.

Class

Optional.

List

- AMF - Access and Mobility management Function - Access and Mobility management Function supports 5G.
- MME - Mobility management element - Mobility management element supports 4G.
- QA - Quad Access - Quad Access supports 2G/3G/4G/5G or 4G/5G
- SGSN - Serving GPRS support node - Serving GPRS support node supports 2G and 3G.
- TA - Triple access - Triple access supports 2G, 3G, and 4G.

Name

buildNum - Build number

Syntax

--buildNum <string>

Description

Build number for the release.

Class

Optional.

Type / Range

Digit string of length 1 to 20 with the allowed pattern of [0-9a-zA-Z\.\-_:]*

Name

hwType - Hardware type

Syntax

--hwType <list-item>

Description

Hardware type of the CMM.

Class

Optional.

List

- ATE - Specify hardware type ATE.
- CMM - Specify hardware type CMM.
- UnKnown - Specify hardware type unknown.

Name

model - Network Function Model

Syntax

--model <list-item>

Description

This parameter indicates if this CMM is a Virtual Network Function or Container Network Function

Class

Optional.

List

- CNF - Containerized Network Function
- VNF - Virtualized Network Function

Name

releaseName - Release name

Syntax

--releaseName <string>

Description

Software release name.

Class

Optional.

Type / Range

Digit string of length 1 to 20 with the allowed pattern of [0-9a-zA-Z\._:]*

Name

releaseNum - Release number

Syntax

--releaseNum <string>

Description

Software release number.

Class

Optional.

Type / Range

Digit string of length 1 to 20 with the allowed pattern of [0-9a-zA-Z\._:]*

Name

switchName - Switch name

Syntax

--switchName <string>

Description

Name for this CMM.

Class

Optional.

Type / Range

Digit string of length 1 to 40 with the allowed pattern of [0-9a-zA-Z\.-]*

Name

timeZone - Time zone

Syntax

--timeZone <string>

Description

Time zone that this CMM is in.

Class

Optional.

Type / Range

Digit string of length 1 to 16 with the allowed pattern of [0-9a-zA-Z\._:]*

Roles**Show**

sysadmin, manage, monitor

List

sysadmin, manage, monitor

7.94 cmm t6aDiamNasMappingProfile

T6A Diam NAS Mapping Profile. Syntax, description, parameters, and roles for cmm t6aDiamNasMappingProfile command.

Syntax

Description

T6A Diam NAS Mapping Profile.

Parameters

Name

t6aDiamNasMappingProfileName - T6A diam NAS mapping profile name

Syntax

--t6aDiamNasMappingProfileName <string>

Description

T6A diam NAS mapping profile name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

t6aDiamCauseCode - T6A diam cause code

Syntax

--t6aDiamCauseCode <list-item>

Description

T6A diam cause code.

Class

Required key.

List

- avpNotAllowed - AVP not allowed - A message was received with an AVP that MUST NOT be present. the failed-avp AVP MUST be included and contain a copy of the offending AVP.
- avpOccursTooManyTimes - AVP occurs too many times - A message was received that included an AVP that appeared more often than permitted in the message definition. the failed-avp AVP MUST be included and contain a copy of the first instance of the offending AVP that exceeded the maximum number of occurrences.
- avpUnsupported - AVP unsupported - The peer received a message that contained an AVP that is not recognized or supported and was marked with the 'M' (Mandatory) bit. a diameter message with this error MUST contain one or more failed-avp AVPs containing the AVPs that caused the failure.
- commandUnsupported - Command unsupported - This error code is used when a diameter entity receives a message with a command code that it does not support.

- **contradictingAvps** - Contradicting AVPs - The home diameter server has detected AVPs in the request that contradicted each other, and it is not willing to provide service to the user. the failed-avp AVP MUST be present, which contain the AVPs that contradicted each other.
- **electionLost** - Election lost - The peer has determined that it has lost the election process and has therefore disconnected the transport connection.
- **featureUnsupported** - Feature unsupported - The peer received a message that contained an AVP that is not recognized or supported.
- **invalidAvpBitCombination** - Invalid AVP bit combination - The request contained an AVP with which is not allowed to have the given value in the AVP flags field. a diameter message indicating this error MUST include the offending AVPs within a failed-avp AVP.
- **invalidAvpBits** - Invalid AVP bits - A request was received that included an AVP whose flag bits are set to an unrecognized value or that is inconsistent with the AVP's definition.
- **invalidAvpLength** - Invalid AVP length - The request contained an AVP with an invalid length. a diameter message indicating this error MUST include the offending AVPs within a failed-avp AVP. in cases where the erroneous AVP length value exceeds the message length or is less than the minimum AVP header length, it is sufficient to include the offending AVP header and a zero filled payload of the minimum required length for the payloads data type. if the AVP is a grouped AVP, the grouped AVP header with an empty payload would be sufficient to indicate the offending AVP. in the case where the offending AVP header cannot be fully decoded when the AVP length is less than the minimum AVP header length, it is sufficient to include an offending AVP header that is formulated by padding the incomplete AVP header with zero up to the minimum AVP header length.
- **invalidAvpValue** - Invalid AVP value - The request contained an AVP with an invalid value in its data portion. a diameter message indicating this error MUST include the offending AVPs within a failed-avp AVP.
- **invalidBitInHeader** - Invalid bit in header - This error is returned when a reserved bit in the diameter header is set to one (1) or the bits in the diameter header are set incorrectly.
- **invalidEpsBearer** - Invalid EPS bearer - This result code shall be sent by the SCEF or the MME to indicate that there is no existing EPS bearer context for the user.
- **invalidHeaderBits** - Invalid header bits - A request was received whose bits in the diameter header were set either to an invalid combination or to a value that is inconsistent with the command code's definition.
- **invalidMessageLength** - Invalid message length - This error is returned when a request is received with an invalid message length.
- **loopDetected** - Loop detected - An agent detected a loop while trying to get the message to the intended recipient. the message MAY be sent to an alternate peer, if one is available, but the peer reporting the error has identified a configuration problem.

- **missingAvp** - Missing AVP - The request did not contain an AVP that is required by the command code definition. if this value is sent in the result-code AVP, a failed-avp AVP SHOULD be included in the message. the failed-avp AVP MUST contain an example of the missing AVP complete with the vendor-id if applicable. the value field of the missing AVP should be of correct minimum length and contain zeroes.
- **niddConfigurationNotAvailable** - NIDD configuration not available - This result code shall be sent by the SCEF to indicate that there is no valid NIDD configuration available.
- **noCommonApplication** - No common application - This error is returned by a diameter node that receives a CER whereby no applications are common between the CER sending peer and the CER receiving peer.
- **noScefAvailable** - No SCEF available - No SCEF is available to send a request or answer over T6A.
- **operationNotAllowed** - Operation not allowed - This result code shall be sent by the SCEF to indicate that the operation is not allowed when an EPS bearer context exists for the user.
- **outOfSpace** - Out of space - A diameter node received the accounting request but was unable to commit it to stable storage due to a temporary lack of space.
- **redirectionIndication** - Redirection indication - A redirect agent has determined that the request could not be satisfied locally, and the initiator of the request SHOULD direct the request directly to the server.
- **resourcesExceeded** - Resources exceeded - A request was received that cannot be authorized because the user has already expended allowed resources.
- **scefConnectionDown** - SCEF connection down - The target SCEF connection is down while a request or answer is being processed.
- **scefReferenceIdUnknown** - SCEF reference unknown - This result code shall be sent by the SCEF to indicate that the SCEF reference ID is not known by the SCEF.
- **tooBusy** - Too busy - When returned, a diameter node SHOULD attempt to send the message to an alternate peer. this error MUST only be used when a specific server is requested, and it cannot provide the requested service.
- **unableToComply** - Unable to comply - This error is returned when a request is rejected for unspecified reasons.
- **unableToDeliver** - Unable to deliver - This error is given when diameter cannot deliver the message to the destination, either because no host within the realm supporting the required application was available to process the request or because the destination-host AVP was given without the associated destination-realm AVP.
- **unknownPeer** - Unknown peer - A CER was received from an unknown peer.
- **unknownSessionId** - Unknown session ID - The request contained an unknown session-id.

- unsupportedVersion - Unsupported version - This error is returned when a request was received, whose version number is unsupported.
- userUnknown - User unknown - This result code shall be sent by the SCEF or the MME to indicate that the user identified by the IMSI is unknown.

Name

t6aEmmCauseCode - T6A EMM cause code

Syntax

--t6aEmmCauseCode <integer>

Description

T6A EMM cause code.

Class

Optional.

Type / Range

Integer in the range of 2-17, 20-23, 25, 28, 32-34, 40, 95- 101, 111

Name

t6aEsmCauseCode - T6A ESM cause code

Syntax

--t6aEsmCauseCode <integer>

Description

T6A ESM cause code.

Class

Optional.

Type / Range

Integer in the range of 8, 26-35, 38, 50-51, 53-55, 65-66, 95-112

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

7.95 cmm t6aDiamNasMappingProfileList

T6A Diam NAS Mapping Profile List. Syntax, description, parameters, and roles for cmm t6aDiamNasMappingProfileList command.

Syntax

Description

T6A Diam NAS Mapping Profile List.

Parameters

Name

name - T6A diam NAS mapping profile list name

Syntax

--name <string>

Description

T6A diam NAS mapping profile list name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.96 cmm tac

tac. Syntax, description, parameters, and roles for cmm tac command.

Syntax

Description

Configure, list and show TACs.

Parameters

Name

id - ID of TAC

Syntax

<digits>

Description

Id of TAC, which is first eight digits of IMEI.

Class

Required value.

Type / Range

Digit string of length 8

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.97 cmm tacacsAuthRoleMapping

TACACS Authorizaiton Role Mapping. Syntax, description, parameters, and roles for cmm tacacsAuthRoLeMapping command.

Syntax

Description

This command allows mapping of External Authorization Roles to the Internal CMM Roles

Parameters

Name

tacacsMappingName - TACACS Role Mappng Name

Syntax

--tacacsMappingName <string>

Description

Name of this TACACS Mapping

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

tacacsExternalRole - TACACS External Role

Syntax

--tacacsExternalRole <integer>

Description

Value of the external TACACS Defined Role

Class

Required key.

Type / Range

Integer in the range of 0 to 15

Name

rbacRole - RBAC Role

Syntax

--rbacRole <list-item>

Description

The internal RBAC Role Allowed

Class

Required key.

List

- Admin - Admin Role
- Configurator - Configurator Role
- LI - Lawful Intercept - Lawful Intercept Role
- Monitor - Monitor Role
- SecAdmin - Security Admin - Security Admin Role

Roles**Create**

secadmin

Show

secadmin

Delete

secadmin

List

secadmin

7.98 cmm taCountQuery

TA Count Query. Syntax, description, parameters, and roles for cmm taCountQuery command.

Syntax

Description

Show TA Count Query.

Parameters

Name

uePlmnName - UE Mobile Network

Syntax`--uePlmnName <string>`**Description**

Name that identifies roaming network

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

servingPlmnName - Mobile Network Name of TAI

Syntax`--servingPlmnName <string>`**Description**

Name that identifies Mobile Network (alternate key) of the tracking area in this mapping.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

allTac - All Tracking Area Codes

Syntax

--allTac <list-item>

Description

Parameter is used to select all TAC of the serving PLMN. If this parameter is set to 'True' then MME ignores the TAC parameter and shows the total number of registered UEs from all the TAC of the serving PLMN.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

regUeCount - Registered UE Count

Syntax

--regUeCount <integer>

Description

Number of registered users in a serving TA for a roamer PLMN

Class

Read only.

Type / Range

Integer in the range of 0 to 65,535

Default

0

Name

tac - Tracking Area Code

Syntax

--tac <integer>

Description

Tracking Area Code.

Class

Optional.

Type / Range

Integer in the range of 0 to 65,535

Default

0

Roles**List**

sysadmin, manage, monitor

7.99 cmm tai

Tracking Area Identifier. Syntax, description, parameters, and roles for **cmm tai** command.

Syntax**Description**

This command is used to administer an MME Tracking Area and includes parameters for defining the mobile network name, Tracking Area Code, ESMC mapping, whether emergency and IMS services are supported, time zone, and ESMC selection algorithms indicating whether they will load share or function as primary/secondary.

Parameters**Name**

plmnName - Mobile network name

Syntax

--plmnName <string>

Description

Name that identifies the mobile network.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

tac - Tracking area code

Syntax

--tac <integer>

Description

Tracking area code.

Class

Required key.

Type / Range

Integer in the range of 1 to 65,535

Name

aerialUeSupported - Aerial UE Supported

Syntax

--aerialUeSupported <list-item>

Description

This field identifies whether the TAI eNodeBs support Aerial UE operation.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ciotEmmRegisterdWithoutPdn - Enable EMM-REGISTERED without PDN connection for C-IOT UE

Syntax

--ciotEmmRegisterdWithoutPdn <list-item>

Description

Enable EMM-REGISTERED without PDN connection for C-IOT UE.

Restrictions

This element is not applicable when plmnType is Other Network.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

cpCIoT - CP CIOT

Syntax

--cpCIoT <list-item>

Description

Enable CP CIOT optimization.

Restrictions

This element is not applicable when plmnType is Other Network.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

dcnrSupported - DCNR Supported

Syntax

--dcnrSupported <list-item>

Description

This field specifies whether a NSA operation is supported anywhere within this TAI. This field is not automatically provisioned by the system. Instead, it is manually set.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

emergencySupported - Emergency supported

Syntax

--emergencySupported <list-item>

Description

Indicates if emergency is supported. only allowed if IMS supported is true.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

emmInfoOverride - EMM info override

Syntax

--emmInfoOverride <list-item>

Description

Flag defining whether the emmInfo message should be overridden with profile information in the emmInfoOverrideTacSv table.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

esmlcIdentity1 - ESMC identity 1

Syntax

--esmlcIdentity1 <digits>

Description

First TAI to E-SMLC (Evolved serving mobile location center) mapping.

Class

Optional.

Type / Range

Digit string of length 0 to 3

Default

"

Name

esmlcIdentity2 - ESMC identity 2

Syntax

--esmlcIdentity2 <digits>

Description

Second TAI to E-SMLC (Evolved serving mobile location center) mapping.

Class

Optional.

Type / Range

Digit string of length 0 to 3

Default

"

Name

esmlcIdentity3 - ESMC identity 3

Syntax

--esmlcIdentity3 <digits>

Description

Third TAI to E-SMLC (Evolved serving mobile location center) mapping. The E-SMLC is used for emergency services.

Class

Optional.

Type / Range

Digit string of length 0 to 3

Name

esmlcIdentity4 - ESMC identity 4

Syntax

--esmlcIdentity4 <digits>

Description

Fourth TAI to E-SMLC (Evolved serving mobile location center) mapping. The set of 4 ESMC are used for emergency services

Class

Optional.

Type / Range

Digit string of length 0 to 3

Name

forbiddenTailListName - Forbidden TAI list name

Syntax

--forbiddenTailListName <string>

Description

Forbidden TAI list name.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

''

Name

imsSupported - IMS supported

Syntax

--imsSupported <list-item>

Description

Indicates if IMS is supported.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

liSelectionAlgorithm - Lawful Intercept location services selection algorithm

Syntax

--liSelectionAlgorithm <list-item>

Description

Indicates for the set of lawful intercept location services ESMLCs (Evolved Serving Mobile Location Center) whether they will load share or function as primary/secondary.

Class

Optional.

List

- LoadShare - Load share - Load share.

- PrimarySecondary - Primary / secondary - Function as primary / secondary.

Default

PrimarySecondary

Name

mmeRelCapacity - MME relative capacity

Syntax

--mmeRelCapacity <integer>

Description

Provision MME relative capacity per tracking area.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

5

Name

mtEdtCp - mtEdtCp

Syntax

--mtEdtCp <list-item>

Description

Mobile-Terminated Early Data Transfer over Control Plan

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

mtEdtUp - mtEdtUp

Syntax

--mtEdtUp <list-item>

Description

Mobile Terminated Early Data Transfer over User Plane

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

nbiotSmsWithoutCombinedAttach - Enable SMS without combined Attach For NB-IOT UE

Syntax

--nbiotSmsWithoutCombinedAttach <list-item>

Description

Enable SMS without combined Attach For NB-IOT UE

Restrictions

This element is not applicable when plmnType is Other Network.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

operSelectionAlgorithm - Operator location services selection algorithm

Syntax

--operSelectionAlgorithm <list-item>

Description

Indicates for the set of operator location services ESMCs (Evolved serving mobile location center) whether they will load share or function as primary/secondary.

Class

Optional.

List

- LoadShare - Load share - Load share.
- PrimarySecondary - Primary / secondary - Function as primary / secondary.

Default

PrimarySecondary

Name

provisionedStatus - TAI provisioned satuts

Syntax

--provisionedStatus <list-item>

Description

Indicator of whether TAI was provisioned, discovered, or initially discovered with subsequent explicit provisioning changes involving TAI were made.

Class

Read only.

List

- Discovered - TAI was explicitly discovered.
- DiscoveredAndProvisioned - Discovered and provisioned - TAI was initially discovered, but additional provisioning using TAI was done.
- Provisioned - TAI was explicitly provisioned.
- RangeDiscovered - Range Discovered - TAI was explicitly range discovered.
- RangeDiscoveredAndProvisioned - Range Discovered and provisioned - TAI was initially range discovered, but additional provisioning using TAI was done.

Name

s1uDataTrans - S1U data transfer

Syntax

--s1uDataTrans <list-item>

Description

Enable S1-U data transfer.

Restrictions

This element is not applicable when plmnType is Other Network.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

selectionAlgorithm - Selection algorithm

Syntax

--selectionAlgorithm <list-item>

Description

Indicates for the set of ESMLCs, (Evolved serving mobile location center) whether they will load share or function as primary/secondary.

Class

Optional.

List

- LoadShare - Load share - Load share.
- PrimarySecondary - Primary / secondary - Function as primary / secondary.

Default

PrimarySecondary

Name

supportedRatType - Supported RAT type for this TAC

Syntax

--supportedRatType <list-item>

Description

Supported RAT type for this TAC. it can only be wbeutran or nbior.

Restrictions

This element is not applicable when plmnType is Other Network.

Class

Optional.

List

- nbior - Narrow band internet of things - Specify type of narrow band internet of things.
- wbeutran - Wide band EUTRAN - Specify type of wide band EUTRAN.

Default

wbeutran

Name

timeZoneName - Time zone name

Syntax

--timeZoneName <list-item>

Description

Provides time zone names that correspond to the region selected.

Restrictions

See comments on TIME_ZONE type in CfgDataType.xsd.

Class

Optional.

List

- Africa/Abidjan - Timezone for africa/abidjan UTC offset of +00:00.
- Africa/Accra - Timezone for africa/accra UTC offset of +00:00.
- Africa/Addis_Ababa - Timezone for africa/addis_ababa UTC offset of +03:00.
- Africa/Algiers - Timezone for africa/algiers UTC offset of +01:00.
- Africa/Asmara - Timezone for africa/asmara UTC offset of +03:00.
- Africa/Asmera - Timezone for africa/asmera UTC offset of +03:00.
- Africa/Bamako - Timezone for africa/bamako UTC offset of +00:00.
- Africa/Bangui - Timezone for africa/bangui UTC offset of +01:00.
- Africa/Banjul - Timezone for africa/banjul UTC offset of +00:00.
- Africa/Bissau - Timezone for africa/bissau UTC offset of +00:00.
- Africa/Blantyre - Timezone for africa/blantyre UTC offset of +02:00.
- Africa/Brazzaville - Timezone for africa/brazzaville UTC offset of +01:00.
- Africa/Bujumbura - Timezone for africa/bujumbura UTC offset of +02:00.
- Africa/Cairo - Timezone for africa/cairo UTC offset of +02:00.
- Africa/Casablanca - Timezone for africa/casablanca UTC offset of +00:00.
- Africa/Ceuta - Timezone for africa/ceuta UTC offset of +01:00.
- Africa/Conakry - Timezone for africa/conakry UTC offset of +00:00.
- Africa/Dakar - Timezone for africa/dakar UTC offset of +00:00.
- Africa/Dar_es_Salaam - Timezone for africa/dar_es_salaam UTC offset of +03:00.
- Africa/Djibouti - Timezone for africa/djibouti UTC offset of +03:00.
- Africa/Douala - Timezone for africa/douala UTC offset of +01:00.
- Africa/El_Aaiun - Timezone for africa/el_aaiun UTC offset of +00:00.
- Africa/Freetown - Timezone for africa/freetown UTC offset of +00:00.
- Africa/Gaborone - Timezone for africa/gaborone UTC offset of +02:00.
- Africa/Harare - Timezone for africa/harare UTC offset of +02:00.
- Africa/Johannesburg - Timezone for africa/johannesburg UTC offset of +02:00.

- Africa/Juba - Timezone for africa/juba UTC offset of +03:00.
- Africa/Kampala - Timezone for africa/kampala UTC offset of +03:00.
- Africa/Khartoum - Timezone for africa/khartoum UTC offset of +03:00.
- Africa/Kigali - Timezone for africa/kigali UTC offset of +02:00.
- Africa/Kinshasa - Timezone for africa/kinshasa UTC offset of +01:00.
- Africa/Lagos - Timezone for africa/lagos UTC offset of +01:00.
- Africa/Libreville - Timezone for africa/libreville UTC offset of +01:00.
- Africa/Lome - Timezone for africa/lome UTC offset of +00:00.
- Africa/Luanda - Timezone for africa/luanda UTC offset of +01:00.
- Africa/Lubumbashi - Timezone for africa/lubumbashi UTC offset of +02:00.
- Africa/Lusaka - Timezone for africa/lusaka UTC offset of +02:00.
- Africa/Malabo - Timezone for africa/malabo UTC offset of +01:00.
- Africa/Maputo - Timezone for africa/maputo UTC offset of +02:00.
- Africa/Maseru - Timezone for africa/maseru UTC offset of +02:00.
- Africa/Mbabane - Timezone for africa/mbabane UTC offset of +02:00.
- Africa/Mogadishu - Timezone for africa/mogadishu UTC offset of +03:00.
- Africa/Monrovia - Timezone for africa/monrovia UTC offset of +00:00.
- Africa/Nairobi - Timezone for africa/nairobi UTC offset of +03:00.
- Africa/Ndjamena - Timezone for africa/ndjamena UTC offset of +01:00.
- Africa/Niamey - Timezone for africa/niamey UTC offset of +01:00.
- Africa/Nouakchott - Timezone for africa/nouakchott UTC offset of +00:00.
- Africa/Ouagadougou - Timezone for africa/ouagadougou UTC offset of +00:00.
- Africa/Porto-Novo - Timezone for africa/porto-novo UTC offset of +01:00.
- Africa/Sao_Tome - Timezone for africa/sao_tome UTC offset of +00:00.
- Africa/Timbuktu - Timezone for africa/timbuktu UTC offset of +00:00.
- Africa/Tripoli - Timezone for africa/tripoli UTC offset of +02:00.
- Africa/Tunis - Timezone for africa/tunis UTC offset of +01:00.
- Africa/Windhoek - Timezone for africa/windhoek UTC offset of +02:00.
- America/Adak - Timezone for america/adak UTC offset of -10:00.
- America/Anchorage - Timezone for america/anchorage UTC offset of -09:00.
- America/Anguilla - Timezone for america/anguilla UTC offset of -04:00.

- America/Antigua - Timezone for america/antigua UTC offset of -04:00.
- America/Araguaina - Timezone for america/araguaina UTC offset of -03:00.
- America/Argentina/Buenos_Aires - Timezone for america/argentina/buenos_aires UTC offset of -03:00.
- America/Argentina/Catamarca - Timezone for america/argentina/catamarca UTC offset of -03:00.
- America/Argentina/ComodRivadavia - Timezone for america/argentina/comodrivadavia UTC offset of -03:00.
- America/Argentina/Cordoba - Timezone for america/argentina/cordoba UTC offset of -03:00.
- America/Argentina/Jujuy - Timezone for america/argentina/jujuy UTC offset of -03:00.
- America/Argentina/La_Rioja - Timezone for america/argentina/la_rioja UTC offset of -03:00.
- America/Argentina/Mendoza - Timezone for america/argentina/mendoza UTC offset of -03:00.
- America/Argentina/Rio_Gallegos - Timezone for america/argentina/rio_gallegos UTC offset of -03:00.
- America/Argentina/Salta - Timezone for america/argentina/salta UTC offset of -03:00.
- America/Argentina/San_Juan - Timezone for america/argentina/san_juan UTC offset of -03:00.
- America/Argentina/San_Luis - Timezone for america/argentina/san_luis UTC offset of -03:00.
- America/Argentina/Tucuman - Timezone for america/argentina/tucuman UTC offset of -03:00.
- America/Argentina/Ushuaia - Timezone for america/argentina/ushuaia UTC offset of -03:00.
- America/Aruba - Timezone for america/aruba UTC offset of -04:00.
- America/Asuncion - Timezone for america/asuncion UTC offset of -03:00.
- America/Atikokan - Timezone for america/atikokan UTC offset of -05:00.
- America/Atka - Timezone for america/atka UTC offset of -10:00.
- America/Bahia - Timezone for america/bahia UTC offset of -03:00.
- America/Bahia_Banderas - Timezone for america/bahia_banderas UTC offset of -06:00.
- America/Barbados - Timezone for america/barbados UTC offset of -04:00.
- America/Belem - Timezone for america/belem UTC offset of -03:00.

- America/Belize - Timezone for america/belize UTC offset of -06:00.
- America/Blanc-Sablon - Timezone for america/blanc-sablon UTC offset of -04:00.
- America/Boa_Vista - Timezone for america/boa_vista UTC offset of -04:00.
- America/Bogota - Timezone for america/bogota UTC offset of -05:00.
- America/Boise - Timezone for america/boise UTC offset of -07:00.
- America/Buenos_Aires - Timezone for america/buenos_aires UTC offset of -03:00.
- America/Cambridge_Bay - Timezone for america/cambridge_bay UTC offset of -07:00.
- America/Campo_Grande - Timezone for america/campo_grande UTC offset of -03:00.
- America/Cancun - Timezone for america/cancun UTC offset of -05:00.
- America/Caracas - Timezone for america/caracas UTC offset of -04:00.
- America/Catamarca - Timezone for america/catamarca UTC offset of -03:00.
- America/Cayenne - Timezone for america/cayenne UTC offset of -03:00.
- America/Cayman - Timezone for america/cayman UTC offset of -05:00.
- America/Chicago - Timezone for america/chicago UTC offset of -06:00.
- America/Chihuahua - Timezone for america/chihuahua UTC offset of -07:00.
- America/Coral_Harbour - Timezone for america/coral_harbour UTC offset of -05:00.
- America/Cordoba - Timezone for america/cordoba UTC offset of -03:00.
- America/Costa_Rica - Timezone for america/costa_rica UTC offset of -06:00.
- America/Creston - Timezone for america/creston UTC offset of -07:00.
- America/Cuiaba - Timezone for america/cuiaba UTC offset of -03:00.
- America/Curacao - Timezone for america/curacao UTC offset of -04:00.
- America/Danmarkshavn - Timezone for america/danmarkshavn UTC offset of +00:00.
- America/Dawson - Timezone for america/dawson UTC offset of -08:00.
- America/Dawson_Creek - Timezone for america/dawson_creek UTC offset of -07:00.
- America/Denver - Timezone for america/denver UTC offset of -07:00.
- America/Detroit - Timezone for america/detroit UTC offset of -05:00.
- America/Dominica - Timezone for america/dominica UTC offset of -04:00.
- America/Edmonton - Timezone for america/edmonton UTC offset of -07:00.
- America/Eirunepe - Timezone for america/eirunepe UTC offset of -05:00.
- America/El_Salvador - Timezone for america/el_salvador UTC offset of -06:00.
- America/Ensenada - Timezone for america/ensenada UTC offset of -08:00.

- America/Fort_Nelson - Timezone for america/fort_nelson UTC offset of -07:00.
- America/Fort_Wayne - Timezone for america/fort_wayne UTC offset of -05:00.
- America/Fortaleza - Timezone for america/fortaleza UTC offset of -03:00.
- America/Glace_Bay - Timezone for america/glance_bay UTC offset of -04:00.
- America/Godthab - Timezone for america/godthab UTC offset of -03:00.
- America/Goose_Bay - Timezone for america/goose_bay UTC offset of -04:00.
- America/Grand_Turk - Timezone for america/grand_turk UTC offset of -04:00.
- America/Grenada - Timezone for america/grenada UTC offset of -04:00.
- America/Guadeloupe - Timezone for america/guadeloupe UTC offset of -04:00.
- America/Guatemala - Timezone for america/guatemala UTC offset of -06:00.
- America/Guayaquil - Timezone for america/guayaquil UTC offset of -05:00.
- America/Guyana - Timezone for america/guyana UTC offset of -04:00.
- America/Halifax - Timezone for america/halifax UTC offset of -04:00.
- America/Havana - Timezone for america/havana UTC offset of -05:00.
- America/Hermosillo - Timezone for america/hermosillo UTC offset of -07:00.
- America/Indiana/Indianapolis - Timezone for america/indiana/indianapolis UTC offset of -05:00.
- America/Indiana/Knox - Timezone for america/indiana/knox UTC offset of -06:00.
- America/Indiana/Marengo - Timezone for america/indiana/marengo UTC offset of -05:00.
- America/Indiana/Petersburg - Timezone for america/indiana/petersburg UTC offset of -05:00.
- America/Indiana/Tell_City - Timezone for america/indiana/tell_city UTC offset of -06:00.
- America/Indiana/Vevay - Timezone for america/indiana/vevay UTC offset of -05:00.
- America/Indiana/Vincennes - Timezone for america/indiana/vincennes UTC offset of -05:00.
- America/Indiana/Winamac - Timezone for america/indiana/winamac UTC offset of -05:00.
- America/Indianapolis - Timezone for america/indianapolis UTC offset of -05:00.
- America/Inuvik - Timezone for america/inuvik UTC offset of -07:00.
- America/Iqaluit - Timezone for america/iqaluit UTC offset of -05:00.
- America/Jamaica - Timezone for america/jamaica UTC offset of -05:00.

- America/Jujuy - Timezone for america/jujuy UTC offset of -03:00.
- America/Juneau - Timezone for america/juneau UTC offset of -09:00.
- America/Kentucky/Louisville - Timezone for america/kentucky/louisville UTC offset of -05:00.
- America/Kentucky/Monticello - Timezone for america/kentucky/monticello UTC offset of -05:00.
- America/Knox_IN - Timezone for america/knox_in UTC offset of -06:00.
- America/Kralendijk - Timezone for america/kralendijk UTC offset of -04:00.
- America/La_Paz - Timezone for america/la_paz UTC offset of -04:00.
- America/Lima - Timezone for america/lima UTC offset of -05:00.
- America/Los_Angeles - Timezone for america/los_angeles UTC offset of -08:00.
- America/Louisville - Timezone for america/louisville UTC offset of -05:00.
- America/Lower_Princes - Timezone for america/lower_princes UTC offset of -04:00.
- America/Maceio - Timezone for america/maceio UTC offset of -03:00.
- America/Managua - Timezone for america/managua UTC offset of -06:00.
- America/Manaus - Timezone for america/manuel UTC offset of -04:00.
- America/Marigot - Timezone for america/marigot UTC offset of -04:00.
- America/Martinique - Timezone for america/martinique UTC offset of -04:00.
- America/Matamoros - Timezone for america/matamoros UTC offset of -06:00.
- America/Mazatlan - Timezone for america/mazatlan UTC offset of -07:00.
- America/Mendoza - Timezone for america/mendoza UTC offset of -03:00.
- America/Menominee - Timezone for america/menominee UTC offset of -06:00.
- America/Merida - Timezone for america/merida UTC offset of -06:00.
- America/Metlakatla - Timezone for america/metlakatla UTC offset of -09:00.
- America/Mexico_City - Timezone for america/mexico_city UTC offset of -06:00.
- America/Miquelon - Timezone for america/miquelon UTC offset of -03:00.
- America/Moncton - Timezone for america/moncton UTC offset of -04:00.
- America/Monterrey - Timezone for america/monterrey UTC offset of -06:00.
- America/Montevideo - Timezone for america/montevideo UTC offset of -03:00.
- America/Montreal - Timezone for america/montreal UTC offset of -05:00.
- America/Montserrat - Timezone for america/montserrat UTC offset of -04:00.
- America/Nassau - Timezone for america/nassau UTC offset of -05:00.

- America/New_York - Timezone for america/new_york UTC offset of -05:00.
- America/Nipigon - Timezone for america/nipigon UTC offset of -05:00.
- America/Nome - Timezone for america/nome UTC offset of -09:00.
- America/Noronha - Timezone for america/noronha UTC offset of -02:00.
- America/North_Dakota/Beulah - Timezone for america/north_dakota/beulah UTC offset of -06:00.
- America/North_Dakota/Center - Timezone for america/north_dakota/center UTC offset of -06:00.
- America/North_Dakota/New_Salem - Timezone for america/north_dakota/new_salem UTC offset of -06:00.
- America/Ojinaga - Timezone for america/ojinaga UTC offset of -07:00.
- America/Panama - Timezone for america/panama UTC offset of -05:00.
- America/Pangnirtung - Timezone for america/pangnirtung UTC offset of -05:00.
- America/Paramaribo - Timezone for america/paramaribo UTC offset of -03:00.
- America/Phoenix - Timezone for america/phoenix UTC offset of -07:00.
- America/Port-au-Prince - Timezone for america/port-au-prince UTC offset of -05:00.
- America/Port_of_Spain - Timezone for america/port_of_spain UTC offset of -04:00.
- America/Porto_Acre - Timezone for america/porto_acre UTC offset of -05:00.
- America/Porto_Velho - Timezone for america/porto_velho UTC offset of -04:00.
- America/Puerto_Rico - Timezone for america/puerto_rico UTC offset of -04:00.
- America/Rainy_River - Timezone for america/rainy_river UTC offset of -06:00.
- America/Rankin_Inlet - Timezone for america/rankin_inlet UTC offset of -06:00.
- America/Recife - Timezone for america/recife UTC offset of -03:00.
- America/Regina - Timezone for america/regina UTC offset of -06:00.
- America/Resolute - Timezone for america/resolute UTC offset of -06:00.
- America/Rio_Branco - Timezone for america/rio_branco UTC offset of -05:00.
- America/Rosario - Timezone for america/rosario UTC offset of -03:00.
- America/Santa_Isabel - Timezone for america/santa_isabel UTC offset of -08:00.
- America/Santarem - Timezone for america/santarem UTC offset of -03:00.
- America/Santiago - Timezone for america/santiago UTC offset of -03:00.
- America/Santo_Domingo - Timezone for america/santo_domingo UTC offset of -04:00.
- America/Sao_Paulo - Timezone for america/sao_paulo UTC offset of -02:00.

- America/Scoresbysund - Timezone for america/scoresbysund UTC offset of -01:00.
- America/Shiprock - Timezone for america/shiprock UTC offset of -07:00.
- America/Sitka - Timezone for america/sitka UTC offset of -09:00.
- America/St_Barthelemy - Timezone for america/st_barthelemy UTC offset of -04:00.
- America/St_Johns - Timezone for america/st_johns UTC offset of -03:30.
- America/St_Kitts - Timezone for america/st_kitts UTC offset of -04:00.
- America/St_Lucia - Timezone for america/st_lucia UTC offset of -04:00.
- America/St_Thomas - Timezone for america/st_thomas UTC offset of -04:00.
- America/St_Vincent - Timezone for america/st_vincent UTC offset of -04:00.
- America/Swift_Current - Timezone for america/swift_current UTC offset of -06:00.
- America/Tegucigalpa - Timezone for america/tegucigalpa UTC offset of -06:00.
- America/Thule - Timezone for america/thule UTC offset of -04:00.
- America/Thunder_Bay - Timezone for america/thunder_bay UTC offset of -05:00.
- America/Tijuana - Timezone for america/tijuana UTC offset of -08:00.
- America/Toronto - Timezone for america/toronto UTC offset of -05:00.
- America/Tortola - Timezone for america/tortola UTC offset of -04:00.
- America/Vancouver - Timezone for america/vancouver UTC offset of -08:00.
- America/Virgin - Timezone for america/virgin UTC offset of -04:00.
- America/Whitehorse - Timezone for america/whitehorse UTC offset of -08:00.
- America/Winnipeg - Timezone for america/winnipeg UTC offset of -06:00.
- America/Yakutat - Timezone for america/yakutat UTC offset of -09:00.
- America/Yellowknife - Timezone for america/yellowknife UTC offset of -07:00.
- Antarctica/Casey - Timezone for antarctica/casey UTC offset of +11:00.
- Antarctica/Davis - Timezone for antarctica/davis UTC offset of +07:00.
- Antarctica/DumontDURville - Timezone for antarctica/dumontdurville UTC offset of +10:00.
- Antarctica/Macquarie - Timezone for antarctica/macquarie UTC offset of +11:00.
- Antarctica/Mawson - Timezone for antarctica/mawson UTC offset of +05:00.
- Antarctica/McMurdo - Timezone for antarctica/mcmurdo UTC offset of +13:00.
- Antarctica/Palmer - Timezone for antarctica/palmer UTC offset of -03:00.
- Antarctica/Rothera - Timezone for antarctica/rothera UTC offset of -03:00.

- Antarctica/South_Pole - Timezone for antarctica/south_pole UTC offset of +13:00.
- Antarctica/Syowa - Timezone for antarctica/syowa UTC offset of +03:00.
- Antarctica/Troll - Timezone for antarctica/troll UTC offset of +00:00.
- Antarctica/Vostok - Timezone for antarctica/vostok UTC offset of +06:00.
- Arctic/Longyearbyen - Timezone for arctic/longyearbyen UTC offset of +01:00.
- Asia/Aden - Timezone for asia/aden UTC offset of +03:00.
- Asia/Almaty - Timezone for asia/almaty UTC offset of +06:00.
- Asia/Amman - Timezone for asia/amman UTC offset of +02:00.
- Asia/Anadyr - Timezone for asia/anadyr UTC offset of +12:00.
- Asia/Aqtau - Timezone for asia/aqtau UTC offset of +05:00.
- Asia/Aqtobe - Timezone for asia/aqtobe UTC offset of +05:00.
- Asia/Ashgabat - Timezone for asia/ashgabat UTC offset of +05:00.
- Asia/Ashkhabad - Timezone for asia/ashkhabad UTC offset of +05:00.
- Asia/Baghdad - Timezone for asia/baghdad UTC offset of +03:00.
- Asia/Bahrain - Timezone for asia/bahrain UTC offset of +03:00.
- Asia/Baku - Timezone for asia/baku UTC offset of +04:00.
- Asia/Bangkok - Timezone for asia/bangkok UTC offset of +07:00.
- Asia/Barnaul - Timezone for asia/barnaul UTC offset of +07:00.
- Asia/Beirut - Timezone for asia/beirut UTC offset of +02:00.
- Asia/Bishkek - Timezone for asia/bishkek UTC offset of +06:00.
- Asia/Brunei - Timezone for asia/brunei UTC offset of +08:00.
- Asia/Calcutta - Timezone for asia/calcutta UTC offset of +05:30.
- Asia/Chita - Timezone for asia/chita UTC offset of +09:00.
- Asia/Choibalsan - Timezone for asia/choibalsan UTC offset of +08:00.
- Asia/Chongqing - Timezone for asia/chongqing UTC offset of +08:00.
- Asia/Chungking - Timezone for asia/chungking UTC offset of +08:00.
- Asia/Colombo - Timezone for asia/colombo UTC offset of +05:30.
- Asia/Dacca - Timezone for asia/dacca UTC offset of +06:00.
- Asia/Damascus - Timezone for asia/damascus UTC offset of +02:00.
- Asia/Dhaka - Timezone for asia/dhaka UTC offset of +06:00.
- Asia/Dili - Timezone for asia/dili UTC offset of +09:00.

- Asia/Dubai - Timezone for asia/dubai UTC offset of +04:00.
- Asia/Dushanbe - Timezone for asia/dushanbe UTC offset of +05:00.
- Asia/Gaza - Timezone for asia/gaza UTC offset of +02:00.
- Asia/Harbin - Timezone for asia/harbin UTC offset of +08:00.
- Asia/Hebron - Timezone for asia/hebron UTC offset of +02:00.
- Asia/Ho_Chi_Minh - Timezone for asia/ho_chi_minh UTC offset of +07:00.
- Asia/Hong_Kong - Timezone for asia/hong_kong UTC offset of +08:00.
- Asia/Hovd - Timezone for asia/hovd UTC offset of +07:00.
- Asia/Irkutsk - Timezone for asia/irkutsk UTC offset of +08:00.
- Asia/Istanbul - Timezone for asia/istanbul UTC offset of +03:00.
- Asia/Jakarta - Timezone for asia/jakarta UTC offset of +07:00.
- Asia/Jayapura - Timezone for asia/jayapura UTC offset of +09:00.
- Asia/Jerusalem - Timezone for asia/jerusalem UTC offset of +02:00.
- Asia/Kabul - Timezone for asia/kabul UTC offset of +04:30.
- Asia/Kamchatka - Timezone for asia/kamchatka UTC offset of +12:00.
- Asia/Karachi - Timezone for asia/karachi UTC offset of +05:00.
- Asia/Kashgar - Timezone for asia/kashgar UTC offset of +06:00.
- Asia/Kathmandu - Timezone for asia/kathmandu UTC offset of +05:45.
- Asia/Katmandu - Timezone for asia/katmandu UTC offset of +05:45.
- Asia/Khandyga - Timezone for asia/khandyga UTC offset of +09:00.
- Asia/Kolkata - Timezone for asia/kolkata UTC offset of +05:30.
- Asia/Krasnoyarsk - Timezone for asia/krasnoyarsk UTC offset of +07:00.
- Asia/Kuala_Lumpur - Timezone for asia/kuala_lumpur UTC offset of +08:00.
- Asia/Kuching - Timezone for asia/kuching UTC offset of +08:00.
- Asia/Kuwait - Timezone for asia/kuwait UTC offset of +03:00.
- Asia/Macao - Timezone for asia/macao UTC offset of +08:00.
- Asia/Macau - Timezone for asia/macau UTC offset of +08:00.
- Asia/Magadan - Timezone for asia/magadan UTC offset of +11:00.
- Asia/Makassar - Timezone for asia/makassar UTC offset of +08:00.
- Asia/Manila - Timezone for asia/manila UTC offset of +08:00.
- Asia/Muscat - Timezone for asia/muscat UTC offset of +04:00.

- Asia/Nicosia - Timezone for asia/nicosia UTC offset of +02:00.
- Asia/Novokuznetsk - Timezone for asia/novokuznetsk UTC offset of +07:00.
- Asia/Novosibirsk - Timezone for asia/novosibirsk UTC offset of +07:00.
- Asia/Omsk - Timezone for asia/omsk UTC offset of +06:00.
- Asia/Oral - Timezone for asia/oral UTC offset of +05:00.
- Asia/Phnom_Penh - Timezone for asia/phnom_penh UTC offset of +07:00.
- Asia/Pontianak - Timezone for asia/pontianak UTC offset of +07:00.
- Asia/Pyongyang - Timezone for asia/pyongyang UTC offset of +08:30.
- Asia/Qatar - Timezone for asia/qatar UTC offset of +03:00.
- Asia/Qyzylorda - Timezone for asia/qyzylorda UTC offset of +06:00.
- Asia/Rangoon - Timezone for asia/rangoon UTC offset of +06:30.
- Asia/Riyadh - Timezone for asia/riyadh UTC offset of +03:00.
- Asia/Saigon - Timezone for asia/saigon UTC offset of +07:00.
- Asia/Sakhalin - Timezone for asia/sakhalin UTC offset of +11:00.
- Asia/Samarkand - Timezone for asia/samarkand UTC offset of +05:00.
- Asia/Seoul - Timezone for asia/seoul UTC offset of +09:00.
- Asia/Shanghai - Timezone for asia/shanghai UTC offset of +08:00.
- Asia/Singapore - Timezone for asia/singapore UTC offset of +08:00.
- Asia/Srednekolymsk - Timezone for asia/srednekolymsk UTC offset of +11:00.
- Asia/Taipei - Timezone for asia/taipei UTC offset of +08:00.
- Asia/Tashkent - Timezone for asia/tashkent UTC offset of +05:00.
- Asia/Tbilisi - Timezone for asia/tbilisi UTC offset of +04:00.
- Asia/Tehran - Timezone for asia/tehran UTC offset of +03:30.
- Asia/Tel_Aviv - Timezone for asia/tel_aviv UTC offset of +02:00.
- Asia/Thimbu - Timezone for asia/thimbu UTC offset of +06:00.
- Asia/Thimphu - Timezone for asia/thimphu UTC offset of +06:00.
- Asia/Tokyo - Timezone for asia/tokyo UTC offset of +09:00.
- Asia/Tomsk - Timezone for asia/tomsk UTC offset of +07:00.
- Asia/Ujung_Pandang - Timezone for asia/ujung_pandang UTC offset of +08:00.
- Asia/Ulaanbaatar - Timezone for asia/ulaanbaatar UTC offset of +08:00.
- Asia/Ulan_Bator - Timezone for asia/ulan_bator UTC offset of +08:00.

- Asia/Urumqi - Timezone for asia/urumqi UTC offset of +06:00.
- Asia/Ust-Nera - Timezone for asia/ust-nera UTC offset of +10:00.
- Asia/Vientiane - Timezone for asia/vientiane UTC offset of +07:00.
- Asia/Vladivostok - Timezone for asia/vladivostok UTC offset of +10:00.
- Asia/Yakutsk - Timezone for asia/yakutsk UTC offset of +09:00.
- Asia/Yekaterinburg - Timezone for asia/yekaterinburg UTC offset of +05:00.
- Asia/Yerevan - Timezone for asia/yerevan UTC offset of +04:00.
- Atlantic/Azores - Timezone for atlantic/azores UTC offset of -01:00.
- Atlantic/Bermuda - Timezone for atlantic/bermuda UTC offset of -04:00.
- Atlantic/Canary - Timezone for atlantic/canary UTC offset of +00:00.
- Atlantic/Cape_Verde - Timezone for atlantic/cape_verde UTC offset of -01:00.
- Atlantic/Faeroe - Timezone for atlantic/faeroe UTC offset of +00:00.
- Atlantic/Faroe - Timezone for atlantic/faroe UTC offset of +00:00.
- Atlantic/Jan_Mayen - Timezone for atlantic/jan_mayen UTC offset of +01:00.
- Atlantic/Madeira - Timezone for atlantic/madeira UTC offset of +00:00.
- Atlantic/Reykjavik - Timezone for atlantic/reykjavik UTC offset of +00:00.
- Atlantic/South_Georgia - Timezone for atlantic/south_georgia UTC offset of -02:00.
- Atlantic/St_Helena - Timezone for atlantic/st_helena UTC offset of +00:00.
- Atlantic/Stanley - Timezone for atlantic/stanley UTC offset of -03:00.
- Australia/ACT - Timezone for australia/act UTC offset of +11:00.
- Australia/Adelaide - Timezone for australia/adelaide UTC offset of +10:30.
- Australia/Brisbane - Timezone for australia/brisbane UTC offset of +10:00.
- Australia/Broken_Hill - Timezone for australia/broken_hill UTC offset of +10:30.
- Australia/Canberra - Timezone for australia/canberra UTC offset of +11:00.
- Australia/Currie - Timezone for australia/currie UTC offset of +11:00.
- Australia/Darwin - Timezone for australia/darwin UTC offset of +09:30.
- Australia/Eucla - Timezone for australia/eucla UTC offset of +08:45.
- Australia/Hobart - Timezone for australia/hobart UTC offset of +11:00.
- Australia/LHI - Timezone for australia/lhi UTC offset of +11:00.
- Australia/Lindeman - Timezone for australia/lindeman UTC offset of +10:00.
- Australia/Lord_Howe - Timezone for australia/lord_howe UTC offset of +11:00.

- Australia/Melbourne - Timezone for australia/melbourne UTC offset of +11:00.
- Australia/NSW - Timezone for australia/nsw UTC offset of +11:00.
- Australia/North - Timezone for australia/north UTC offset of +09:30.
- Australia/Perth - Timezone for australia/perth UTC offset of +08:00.
- Australia/Queensland - Timezone for australia/queensland UTC offset of +10:00.
- Australia/South - Timezone for australia/south UTC offset of +10:30.
- Australia/Sydney - Timezone for australia/sydney UTC offset of +11:00.
- Australia/Tasmania - Timezone for australia/tasmania UTC offset of +11:00.
- Australia/Victoria - Timezone for australia/victoria UTC offset of +11:00.
- Australia/West - Timezone for australia/west UTC offset of +08:00.
- Australia/Yancowinna - Timezone for australia/yancowinna UTC offset of +10:30.
- CET - Timezone for CET UTC offset of +01:00.
- CST6CDT - Timezone for CST6CDT UTC offset of -06:00.
- Canada/Atlantic - Timezone for canada/atlantic UTC offset of -04:00.
- Canada/Central - Timezone for canada/central UTC offset of -06:00.
- Canada/East-Saskatchewan - Timezone for canada/east-saskatchewan UTC offset of -06:00.
- Canada/Eastern - Timezone for canada/eastern UTC offset of -05:00.
- Canada/Mountain - Timezone for canada/mountain UTC offset of -07:00.
- Canada/Newfoundland - Timezone for canada/newfoundland UTC offset of -03:30.
- Canada/Pacific - Timezone for canada/pacific UTC offset of -08:00.
- Canada/Saskatchewan - Timezone for canada/saskatchewan UTC offset of -06:00.
- Canada/Yukon - Timezone for canada/yukon UTC offset of -08:00.
- Chile/Continental - Timezone for chile/continental UTC offset of -03:00.
- Chile/EasterIsland - Timezone for chile/easterisland UTC offset of -05:00.
- Cuba - Timezone for cuba UTC offset of -05:00.
- Default - Timezone for system UTC (OS time zone)
- EET - Timezone for EET UTC offset of +02:00.
- EST - Timezone for EST UTC offset of -05:00.
- EST5EDT - Timezone for EST5EDT UTC offset of -05:00.
- Egypt - Timezone for egypt UTC offset of +02:00.

- Eire - Timezone for eire UTC offset of +00:00.
- Etc/Greenwich - Timezone for etc/greenwich UTC offset of +00:00.
- Etc/UCT - Timezone for etc/uct UTC offset of +00:00.
- Etc/UTC - Timezone for etc/utc UTC offset of +00:00.
- Etc/Universal - Timezone for etc/universal UTC offset of +00:00.
- Etc/Zulu - Timezone for etc/zulu UTC offset of +00:00.
- Europe/Amsterdam - Timezone for europe/amsterdam UTC offset of +01:00.
- Europe/Andorra - Timezone for europe/andorra UTC offset of +01:00.
- Europe/Astrakhan - Timezone for europe/astrakhan UTC offset of +04:00.
- Europe/Athens - Timezone for europe/athens UTC offset of +02:00.
- Europe/Belfast - Timezone for europe/belfast UTC offset of +00:00.
- Europe/Belgrade - Timezone for europe/belgrade UTC offset of +01:00.
- Europe/Berlin - Timezone for europe/berlin UTC offset of +01:00.
- Europe/Bratislava - Timezone for europe/bratislava UTC offset of +01:00.
- Europe/Brussels - Timezone for europe/brussels UTC offset of +01:00.
- Europe/Bucharest - Timezone for europe/bucharest UTC offset of +02:00.
- Europe/Budapest - Timezone for europe/budapest UTC offset of +01:00.
- Europe/Busingen - Timezone for europe/busingen UTC offset of +01:00.
- Europe/Chisinau - Timezone for europe/chisinau UTC offset of +02:00.
- Europe/Copenhagen - Timezone for europe/copenhagen UTC offset of +01:00.
- Europe/Dublin - Timezone for europe/dublin UTC offset of +00:00.
- Europe/Gibraltar - Timezone for europe/gibraltar UTC offset of +01:00.
- Europe/Guernsey - Timezone for europe/guernsey UTC offset of +00:00.
- Europe/Helsinki - Timezone for europe/helsinki UTC offset of +02:00.
- Europe/Isle_of_Man - Timezone for europe/isle_of_man UTC offset of +00:00.
- Europe/Istanbul - Timezone for europe/istanbul UTC offset of +03:00.
- Europe/Jersey - Timezone for europe/jersey UTC offset of +00:00.
- Europe/Kaliningrad - Timezone for europe/kaliningrad UTC offset of +02:00.
- Europe/Kiev - Timezone for europe/kyiv UTC offset of +02:00.
- Europe/Kirov - Timezone for europe/kirov UTC offset of +03:00.
- Europe/Lisbon - Timezone for europe/lisbon UTC offset of +00:00.

- Europe/Ljubljana - Timezone for europe/ljubljana UTC offset of +01:00.
- Europe/London - Timezone for europe/london UTC offset of +00:00.
- Europe/Luxembourg - Timezone for europe/luxembourg UTC offset of +01:00.
- Europe/Madrid - Timezone for europe/madrid UTC offset of +01:00.
- Europe/Malta - Timezone for europe/malta UTC offset of +01:00.
- Europe/Mariehamn - Timezone for europe/mariehamn UTC offset of +02:00.
- Europe/Minsk - Timezone for europe/minsk UTC offset of +03:00.
- Europe/Monaco - Timezone for europe/monaco UTC offset of +01:00.
- Europe/Moscow - Timezone for europe/moscow UTC offset of +03:00.
- Europe/Nicosia - Timezone for europe/nicosia UTC offset of +02:00.
- Europe/Oslo - Timezone for europe/oslo UTC offset of +01:00.
- Europe/Paris - Timezone for europe/paris UTC offset of +01:00.
- Europe/Podgorica - Timezone for europe/podgorica UTC offset of +01:00.
- Europe/Prague - Timezone for europe/prague UTC offset of +01:00.
- Europe/Riga - Timezone for europe/riga UTC offset of +02:00.
- Europe/Rome - Timezone for europe/rome UTC offset of +01:00.
- Europe/Samara - Timezone for europe/samara UTC offset of +04:00.
- Europe/San_Marino - Timezone for europe/san_marino UTC offset of +01:00.
- Europe/Sarajevo - Timezone for europe/sarajevo UTC offset of +01:00.
- Europe/Simferopol - Timezone for europe/simferopol UTC offset of +03:00.
- Europe/Skopje - Timezone for europe/skopje UTC offset of +01:00.
- Europe/Sofia - Timezone for europe/sofia UTC offset of +02:00.
- Europe/Stockholm - Timezone for europe/stockholm UTC offset of +01:00.
- Europe/Tallinn - Timezone for europe/tallinn UTC offset of +02:00.
- Europe/Tirane - Timezone for europe/tirane UTC offset of +01:00.
- Europe/Tiraspol - Timezone for europe/tiraspol UTC offset of +02:00.
- Europe/Ulyanovsk - Timezone for europe/ulyanovsk UTC offset of +04:00.
- Europe/Uzhgorod - Timezone for europe/uzhgorod UTC offset of +02:00.
- Europe/Vaduz - Timezone for europe/vaduz UTC offset of +01:00.
- Europe/Vatican - Timezone for europe/vatican UTC offset of +01:00.
- Europe/Vienna - Timezone for europe/vienna UTC offset of +01:00.

- Europe/Vilnius - Timezone for europe/vilnius UTC offset of +02:00.
- Europe/Volgograd - Timezone for europe/volgograd UTC offset of +03:00.
- Europe/Warsaw - Timezone for europe/warsaw UTC offset of +01:00.
- Europe/Zagreb - Timezone for europe/zagreb UTC offset of +01:00.
- Europe/Zaporozhye - Timezone for europe/zaporozhye UTC offset of +02:00.
- Europe/Zurich - Timezone for europe/zurich UTC offset of +01:00.
- GB - Timezone for GB UTC offset of +00:00.
- GB-Eire - Timezone for GB-Eire UTC offset of +00:00.
- GMT - Timezone for GMT UTC offset of +00:00.
- GMT0 - Timezone for GMT0 UTC offset of +00:00.
- Greenwich - Timezone for greenwich UTC offset of +00:00.
- HST - Timezone for HST UTC offset of -10:00.
- Hongkong - Timezone for hongkong UTC offset of +08:00.
- Iceland - Timezone for iceland UTC offset of +00:00.
- Indian/Antananarivo - Timezone for indian/antananarivo UTC offset of +03:00.
- Indian/Chagos - Timezone for indian/chagos UTC offset of +06:00.
- Indian/Christmas - Timezone for indian/christmas UTC offset of +07:00.
- Indian/Cocos - Timezone for indian/cocos UTC offset of +06:30.
- Indian/Comoro - Timezone for indian/comoro UTC offset of +03:00.
- Indian/Kerguelen - Timezone for indian/kerguelen UTC offset of +05:00.
- Indian/Mahe - Timezone for indian/mahe UTC offset of +04:00.
- Indian/Maldives - Timezone for indian/maldives UTC offset of +05:00.
- Indian/Mauritius - Timezone for indian/mauritius UTC offset of +04:00.
- Indian/Mayotte - Timezone for indian/mayotte UTC offset of +03:00.
- Indian/Reunion - Timezone for indian/reunion UTC offset of +04:00.
- Iran - Timezone for iran UTC offset of +03:30.
- Israel - Timezone for israel UTC offset of +02:00.
- Jamaica - Timezone for jamaica UTC offset of -05:00.
- Japan - Timezone for japan UTC offset of +09:00.
- Kwajalein - Timezone for kwajalein UTC offset of +12:00.
- Libya - Timezone for libya UTC offset of +02:00.

- MET - Timezone for MET UTC offset of +01:00.
- MST - Timezone for MST UTC offset of -07:00.
- MST7MDT - Timezone for MST7MDT UTC offset of -07:00.
- Mexico/BajaNorte - Timezone for mexico/bajanorte UTC offset of -08:00.
- Mexico/BajaSur - Timezone for mexico/bajasur UTC offset of -07:00.
- Mexico/General - Timezone for mexico/general UTC offset of -06:00.
- NZ - Timezone for NZ UTC offset of +13:00.
- NZ-CHAT - Timezone for NZ-CHAT UTC offset of +13:45.
- Navajo - Timezone for navajo UTC offset of -07:00.
- PRC - Timezone for PRC UTC offset of +08:00.
- PST8PDT - Timezone for PST8PDT UTC offset of -08:00.
- Pacific/Apia - Timezone for pacific/apia UTC offset of +14:00.
- Pacific/Auckland - Timezone for pacific/auckland UTC offset of +13:00.
- Pacific/Bougainville - Timezone for pacific/bougainville UTC offset of +11:00.
- Pacific/Chatham - Timezone for pacific/chatham UTC offset of +13:45.
- Pacific/Chuuk - Timezone for pacific/chuuk UTC offset of +10:00.
- Pacific/Easter - Timezone for pacific/easter UTC offset of -05:00.
- Pacific/Efate - Timezone for pacific/efate UTC offset of +11:00.
- Pacific/Enderbury - Timezone for pacific/enderbury UTC offset of +13:00.
- Pacific/Fakaofu - Timezone for pacific/fakaofu UTC offset of +13:00.
- Pacific/Fiji - Timezone for pacific/fiji UTC offset of +13:00.
- Pacific/Funafuti - Timezone for pacific/funafuti UTC offset of +12:00.
- Pacific/Galapagos - Timezone for pacific/galapagos UTC offset of -06:00.
- Pacific/Gambier - Timezone for pacific/gambier UTC offset of -09:00.
- Pacific/Guadalcanal - Timezone for pacific/guadalcanal UTC offset of +11:00.
- Pacific/Guam - Timezone for pacific/guam UTC offset of +10:00.
- Pacific/Honolulu - Timezone for pacific/honolulu UTC offset of -10:00.
- Pacific/Johnston - Timezone for pacific/johnston UTC offset of -10:00.
- Pacific/Kiritimati - Timezone for pacific/kiritimati UTC offset of +14:00.
- Pacific/Kosrae - Timezone for pacific/kosrae UTC offset of +11:00.
- Pacific/Kwajalein - Timezone for pacific/kwajalein UTC offset of +12:00.

- Pacific/Majuro - Timezone for pacific/majuro UTC offset of +12:00.
- Pacific/Marquesas - Timezone for pacific/marquesas UTC offset of -09:30.
- Pacific/Midway - Timezone for pacific/midway UTC offset of -11:00.
- Pacific/Nauru - Timezone for pacific/nauru UTC offset of +12:00.
- Pacific/Niue - Timezone for pacific/niue UTC offset of -11:00.
- Pacific/Norfolk - Timezone for pacific/norfolk UTC offset of +11:00.
- Pacific/Noumea - Timezone for pacific/noumea UTC offset of +11:00.
- Pacific/Pago_Pago - Timezone for pacific/pago_pago UTC offset of -11:00.
- Pacific/Palau - Timezone for pacific/palau UTC offset of +09:00.
- Pacific/Pitcairn - Timezone for pacific/pitcairn UTC offset of -08:00.
- Pacific/Pohnpei - Timezone for pacific/pohnpei UTC offset of +11:00.
- Pacific/Ponape - Timezone for pacific/ponape UTC offset of +11:00.
- Pacific/Port_Moresby - Timezone for pacific/port_moresby UTC offset of +10:00.
- Pacific/Rarotonga - Timezone for pacific/rarotonga UTC offset of -10:00.
- Pacific/Saipan - Timezone for pacific/saipan UTC offset of +10:00.
- Pacific/Samoa - Timezone for pacific/samoa UTC offset of -11:00.
- Pacific/Tahiti - Timezone for pacific/tahiti UTC offset of -10:00.
- Pacific/Tarawa - Timezone for pacific/tarawa UTC offset of +12:00.
- Pacific/Tongatapu - Timezone for pacific/tongatapu UTC offset of +14:00.
- Pacific/Truk - Timezone for pacific/truk UTC offset of +10:00.
- Pacific/Wake - Timezone for pacific/wake UTC offset of +12:00.
- Pacific/Wallis - Timezone for pacific/wallis UTC offset of +12:00.
- Pacific/Yap - Timezone for pacific/yap UTC offset of +10:00.
- Poland - Timezone for poland UTC offset of +01:00.
- Portugal - Timezone for portugal UTC offset of +00:00.
- ROC - Timezone for ROC UTC offset of +08:00.
- ROK - Timezone for ROK UTC offset of +09:00.
- Singapore - Timezone for singapore UTC offset of +08:00.
- Turkey - Timezone for turkey UTC offset of +03:00.
- UCT - Timezone for UCT UTC offset of +00:00.
- US/Pacific-New - Timezone for US/Pacific-New UTC offset of -08:00.

- Universal - Timezone for universal UTC offset of +00:00.
- W-SU - Timezone for W-SU UTC offset of +03:00.
- WET - Timezone for WET UTC offset of +00:00.
- Zulu - Timezone for zulu UTC offset of +00:00.
- jdj_tz - Timezone for jdj_tz UTC offset of +00:00.

Default

Default

Name

upCloT - UP CloT

Syntax

--upCIoT <list-item>

Description

Enable User Plane Cellular Internet of Things Optimization

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

vaSelectionAlgorithm - Value added location services selection algorithm

Syntax

--vaSelectionAlgorithm <list-item>

Description

Indicates for the set of value added location services ESMLCs (Evolved serving mobile location center) whether they will load share or function as primary/secondary.

Class

Optional.

List

- LoadShare - Load share - Load share.
- PrimarySecondary - Primary / secondary - Function as primary / secondary.

Default

PrimarySecondary

Name

vdpProfileName - VDP profile name

Syntax

--vdpProfileName <string>

Description

Voice domain preference profile name.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

"

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.100 cmm taiDiscoveryDefaults

TAI Discovery Default values. Syntax, description, parameters, and roles for cmm taiDiscoveryDefaults command.

Syntax**Description**

Specifies default values for the TAI when the MME discovers the TAI through the auto-discovery feature.

Parameters**Name**

plmnName - PLMN Name

Syntax

--plmnName <string>

Description

Name that identifies the mobile network

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

aerialUeSupported - Aerial UE Supported

Syntax

--aerialUeSupported <list-item>

Description

This field identifies whether the TAI eNodeBs support Aerial UE operation

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ciotEmmRegisteredWithoutPdn - Enable EMM-REGISTERED without PDN connection for C-IOT UE

Syntax

--ciotEmmRegisteredWithoutPdn <list-item>

Description

Enable EMM-REGISTERED without PDN connection for C-IOT UE

Restrictions

This element is not applicable when plmnType is Other Network

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

cpCIoT - CP CIOT

Syntax

--cpCIoT <list-item>

Description

Enable CP CIOT optimization.

Restrictions

This element is not applicable when plmnType is Other Network

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

emergencySupported - Emergency supported

Syntax

--emergencySupported <list-item>

Description

Indicates if emergency is supported. only allowed if IMS supported is true.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

esmlcIdentity1 - ESMLC identity 1

Syntax

--esmlcIdentity1 <digits>

Description

First TAI to E-SMLC (Evolved serving mobile location center) mapping

Class

Optional.

Type / Range

Digit string of length 0 to 3

Default

"

Name

esmlcIdentity2 - ESMC identity 2

Syntax

--esmlcIdentity2 <digits>

Description

Second TAI to E-SMLC (Evolved serving mobile location center) mapping.

Class

Optional.

Type / Range

Digit string of length 0 to 3

Default

"

Name

esmlcIdentity3 - ESMC identity 3

Syntax

--esmlcIdentity3 <digits>

Description

Third TAI to E-SMLC (Evolved serving mobile location center) mapping. The E-SMLC is used for emergency services.

Class

Optional.

Type / Range

Digit string of length 0 to 3

Default

"

Name

esmlcIdentity4 - ESMC identity 4

Syntax

--esmlcIdentity4 <digits>

Description

Fourth TAI to E-SMLC (Evolved serving mobile location center) mapping. The set of 4 ESMC are used for emergency services

Class

Optional.

Type / Range

Digit string of length 0 to 3

Default

"

Name

forbiddenTailListName - Forbidden TAI list name

Syntax

--forbiddenTailListName <string>

Description

Forbidden TAI list name

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

imsSupported - IMS supported

Syntax

--imsSupported <list-item>

Description

Indicates if IMS is supported.

Class

Optional.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Name

mmeRelCapacity - MME relative capacity

Syntax

--mmeRelCapacity <integer>

Description

Provision MME relative capacity per tracking area.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

5

Name

mtEdtCp - mtEdtCp

Syntax

--mtEdtCp <list-item>

Description

Mobile-Terminated Early Data Transfer over Control Plan

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

mtEdtUp - MT EDT User Plane

Syntax

--mtEdtUp <list-item>

Description

Mobile Terminated Early Data Transfer over User Plane

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

nbiotSmsWithoutCombinedAttach - Enable SMS without combined Attach For NB-IOT UE

Syntax

- nbiotSmsWithoutCombinedAttach <list-item>

Description

Enable SMS without combined Attach For NB-IOT UE

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

s1uDataTrans - S1U data transfer

Syntax

--s1uDataTrans <list-item>

Description

Enable S1-U data transfer.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

selectionAlgorithm - Selection algorithm

Syntax

--selectionAlgorithm <list-item>

Description

Indicates for the set of ESMLCs, (Evolved serving mobile location center) whether they will load share or function as primary/secondary.

Class

Optional.

List

- LoadShare - Load share - Load share.
- PrimarySecondary - Primary / secondary - Function as primary / secondary.

Default

PrimarySecondary

Name

supportedRatType - Supported RAT type for the auto-provisioned TAC

Syntax

--supportedRatType <list-item>

Description

Supported RAT Type for this TAC. It can only be wbeutran or nbio

Restrictions

Supported RAT type for the auto-provisioned TAC. it can only be wbeutran or nbio

Class

Optional.

List

- nbio - Narrow band internet of things - Specify type of narrow band internet of things.
- wbeutran - Wide band EUTRAN - Specify type of wide band EUTRAN.

Default

wbeutran

Name

timeZoneName - Time zone name

Syntax

--timeZoneName <list-item>

Description

Provides time zone names that correspond to the region selected

Restrictions

See comments on TIME_ZONE type in CfgDataType.xsd

Class

Optional.

List

- Africa/Abidjan - Timezone for africa/abidjan UTC offset of +00:00.
- Africa/Accra - Timezone for africa/accra UTC offset of +00:00.
- Africa/Addis_Ababa - Timezone for africa/addis_ababa UTC offset of +03:00.
- Africa/Algiers - Timezone for africa/algiers UTC offset of +01:00.
- Africa/Asmara - Timezone for africa/asmara UTC offset of +03:00.
- Africa/Asmera - Timezone for africa/asmera UTC offset of +03:00.
- Africa/Bamako - Timezone for africa/bamako UTC offset of +00:00.
- Africa/Bangui - Timezone for africa/bangui UTC offset of +01:00.
- Africa/Banjul - Timezone for africa/banjul UTC offset of +00:00.
- Africa/Bissau - Timezone for africa/bissau UTC offset of +00:00.
- Africa/Blantyre - Timezone for africa/blantyre UTC offset of +02:00.
- Africa/Brazzaville - Timezone for africa/brazzaville UTC offset of +01:00.
- Africa/Bujumbura - Timezone for africa/bujumbura UTC offset of +02:00.
- Africa/Cairo - Timezone for africa/cairo UTC offset of +02:00.
- Africa/Casablanca - Timezone for africa/casablanca UTC offset of +00:00.
- Africa/Ceuta - Timezone for africa/ceuta UTC offset of +01:00.
- Africa/Conakry - Timezone for africa/conakry UTC offset of +00:00.
- Africa/Dakar - Timezone for africa/dakar UTC offset of +00:00.
- Africa/Dar_es_Salaam - Timezone for africa/dar_es_salaam UTC offset of +03:00.
- Africa/Djibouti - Timezone for africa/djibouti UTC offset of +03:00.
- Africa/Douala - Timezone for africa/douala UTC offset of +01:00.
- Africa/El_Aaiun - Timezone for africa/el_aaiun UTC offset of +00:00.
- Africa/Freetown - Timezone for africa/freetown UTC offset of +00:00.
- Africa/Gaborone - Timezone for africa/gaborone UTC offset of +02:00.
- Africa/Harare - Timezone for africa/harare UTC offset of +02:00.
- Africa/Johannesburg - Timezone for africa/johannesburg UTC offset of +02:00.
- Africa/Juba - Timezone for africa/juba UTC offset of +03:00.

- Africa/Kampala - Timezone for africa/kampala UTC offset of +03:00.
- Africa/Khartoum - Timezone for africa/khartoum UTC offset of +03:00.
- Africa/Kigali - Timezone for africa/kigali UTC offset of +02:00.
- Africa/Kinshasa - Timezone for africa/kinshasa UTC offset of +01:00.
- Africa/Lagos - Timezone for africa/lagos UTC offset of +01:00.
- Africa/Libreville - Timezone for africa/libreville UTC offset of +01:00.
- Africa/Lome - Timezone for africa/lome UTC offset of +00:00.
- Africa/Luanda - Timezone for africa/luanda UTC offset of +01:00.
- Africa/Lubumbashi - Timezone for africa/lubumbashi UTC offset of +02:00.
- Africa/Lusaka - Timezone for africa/lusaka UTC offset of +02:00.
- Africa/Malabo - Timezone for africa/malabo UTC offset of +01:00.
- Africa/Maputo - Timezone for africa/maputo UTC offset of +02:00.
- Africa/Maseru - Timezone for africa/maseru UTC offset of +02:00.
- Africa/Mbabane - Timezone for africa/mbabane UTC offset of +02:00.
- Africa/Mogadishu - Timezone for africa/mogadishu UTC offset of +03:00.
- Africa/Monrovia - Timezone for africa/monrovia UTC offset of +00:00.
- Africa/Nairobi - Timezone for africa/nairobi UTC offset of +03:00.
- Africa/Ndjamena - Timezone for africa/ndjamena UTC offset of +01:00.
- Africa/Niamey - Timezone for africa/niamey UTC offset of +01:00.
- Africa/Nouakchott - Timezone for africa/nouakchott UTC offset of +00:00.
- Africa/Ouagadougou - Timezone for africa/ouagadougou UTC offset of +00:00.
- Africa/Porto-Novo - Timezone for africa/porto-novo UTC offset of +01:00.
- Africa/Sao_Tome - Timezone for africa/sao_tome UTC offset of +00:00.
- Africa/Timbuktu - Timezone for africa/timbuktu UTC offset of +00:00.
- Africa/Tripoli - Timezone for africa/tripoli UTC offset of +02:00.
- Africa/Tunis - Timezone for africa/tunis UTC offset of +01:00.
- Africa/Windhoek - Timezone for africa/windhoek UTC offset of +02:00.
- America/Adak - Timezone for america/adak UTC offset of -10:00.
- America/Anchorage - Timezone for america/anchorage UTC offset of -09:00.
- America/Anguilla - Timezone for america/anguilla UTC offset of -04:00.
- America/Antigua - Timezone for america/antigua UTC offset of -04:00.

- America/Araguaina - Timezone for america/araguaina UTC offset of -03:00.
- America/Argentina/Buenos_Aires - Timezone for america/argentina/buenos_aires UTC offset of -03:00.
- America/Argentina/Catamarca - Timezone for america/argentina/catamarca UTC offset of -03:00.
- America/Argentina/ComodRivadavia - Timezone for america/argentina/comodrivadavia UTC offset of -03:00.
- America/Argentina/Cordoba - Timezone for america/argentina/cordoba UTC offset of -03:00.
- America/Argentina/Jujuy - Timezone for america/argentina/jujuy UTC offset of -03:00.
- America/Argentina/La_Rioja - Timezone for america/argentina/la_rioja UTC offset of -03:00.
- America/Argentina/Mendoza - Timezone for america/argentina/mendoza UTC offset of -03:00.
- America/Argentina/Rio_Gallegos - Timezone for america/argentina/rio_gallegos UTC offset of -03:00.
- America/Argentina/Salta - Timezone for america/argentina/salta UTC offset of -03:00.
- America/Argentina/San_Juan - Timezone for america/argentina/san_juan UTC offset of -03:00.
- America/Argentina/San_Luis - Timezone for america/argentina/san_luis UTC offset of -03:00.
- America/Argentina/Tucuman - Timezone for america/argentina/tucuman UTC offset of -03:00.
- America/Argentina/Ushuaia - Timezone for america/argentina/ushuaia UTC offset of -03:00.
- America/Aruba - Timezone for america/aruba UTC offset of -04:00.
- America/Asuncion - Timezone for america/asuncion UTC offset of -03:00.
- America/Atikokan - Timezone for america/atikokan UTC offset of -05:00.
- America/Atka - Timezone for america/atka UTC offset of -10:00.
- America/Bahia - Timezone for america/bahia UTC offset of -03:00.
- America/Bahia_Banderas - Timezone for america/bahia_banderas UTC offset of -06:00.
- America/Barbados - Timezone for america/barbados UTC offset of -04:00.
- America/Belem - Timezone for america/belem UTC offset of -03:00.
- America/Belize - Timezone for america/belize UTC offset of -06:00.

- America/Blanc-Sablon - Timezone for america/blanc-sablon UTC offset of -04:00.
- America/Boa_Vista - Timezone for america/boa_vista UTC offset of -04:00.
- America/Bogota - Timezone for america/bogota UTC offset of -05:00.
- America/Boise - Timezone for america/boise UTC offset of -07:00.
- America/Buenos_Aires - Timezone for america/buenos_aires UTC offset of -03:00.
- America/Cambridge_Bay - Timezone for america/cambridge_bay UTC offset of -07:00.
- America/Campo_Grande - Timezone for america/campo_grande UTC offset of -03:00.
- America/Cancun - Timezone for america/cancun UTC offset of -05:00.
- America/Caracas - Timezone for america/caracas UTC offset of -04:00.
- America/Catamarca - Timezone for america/catamarca UTC offset of -03:00.
- America/Cayenne - Timezone for america/cayenne UTC offset of -03:00.
- America/Cayman - Timezone for america/cayman UTC offset of -05:00.
- America/Chicago - Timezone for america/chicago UTC offset of -06:00.
- America/Chihuahua - Timezone for america/chihuahua UTC offset of -07:00.
- America/Coral_Harbour - Timezone for america/coral_harbour UTC offset of -05:00.
- America/Cordoba - Timezone for america/cordoba UTC offset of -03:00.
- America/Costa_Rica - Timezone for america/costa_rica UTC offset of -06:00.
- America/Creston - Timezone for america/creston UTC offset of -07:00.
- America/Cuiaba - Timezone for america/cuiaba UTC offset of -03:00.
- America/Curacao - Timezone for america/curacao UTC offset of -04:00.
- America/Danmarkshavn - Timezone for america/danmarkshavn UTC offset of +00:00.
- America/Dawson - Timezone for america/dawson UTC offset of -08:00.
- America/Dawson_Creek - Timezone for america/dawson_creek UTC offset of -07:00.
- America/Denver - Timezone for america/denver UTC offset of -07:00.
- America/Detroit - Timezone for america/detroit UTC offset of -05:00.
- America/Dominica - Timezone for america/dominica UTC offset of -04:00.
- America/Edmonton - Timezone for america/edmonton UTC offset of -07:00.
- America/Eirunepe - Timezone for america/eirunepe UTC offset of -05:00.
- America/El_Salvador - Timezone for america/el_salvador UTC offset of -06:00.
- America/Ensenada - Timezone for america/ensenada UTC offset of -08:00.
- America/Fort_Nelson - Timezone for america/fort_nelson UTC offset of -07:00.

- America/Fort_Wayne - Timezone for america/fort_wayne UTC offset of -05:00.
- America/Fortaleza - Timezone for america/fortaleza UTC offset of -03:00.
- America/Glace_Bay - Timezone for america/glance_bay UTC offset of -04:00.
- America/Godthab - Timezone for america/godthab UTC offset of -03:00.
- America/Goose_Bay - Timezone for america/goose_bay UTC offset of -04:00.
- America/Grand_Turk - Timezone for america/grand_turk UTC offset of -04:00.
- America/Grenada - Timezone for america/grenada UTC offset of -04:00.
- America/Guadeloupe - Timezone for america/guadeloupe UTC offset of -04:00.
- America/Guatemala - Timezone for america/guatemala UTC offset of -06:00.
- America/Guayaquil - Timezone for america/guayaquil UTC offset of -05:00.
- America/Guyana - Timezone for america/guyana UTC offset of -04:00.
- America/Halifax - Timezone for america/halifax UTC offset of -04:00.
- America/Havana - Timezone for america/havana UTC offset of -05:00.
- America/Hermosillo - Timezone for america/hermosillo UTC offset of -07:00.
- America/Indiana/Indianapolis - Timezone for america/indiana/indianapolis UTC offset of -05:00.
- America/Indiana/Knox - Timezone for america/indiana/knox UTC offset of -06:00.
- America/Indiana/Marengo - Timezone for america/indiana/marengo UTC offset of -05:00.
- America/Indiana/Petersburg - Timezone for america/indiana/petersburg UTC offset of -05:00.
- America/Indiana/Tell_City - Timezone for america/indiana/tell_city UTC offset of -06:00.
- America/Indiana/Vevay - Timezone for america/indiana/vevay UTC offset of -05:00.
- America/Indiana/Vincennes - Timezone for america/indiana/vincennes UTC offset of -05:00.
- America/Indiana/Winamac - Timezone for america/indiana/winamac UTC offset of -05:00.
- America/Indianapolis - Timezone for america/indianapolis UTC offset of -05:00.
- America/Inuvik - Timezone for america/inuvik UTC offset of -07:00.
- America/Iqaluit - Timezone for america/iqaluit UTC offset of -05:00.
- America/Jamaica - Timezone for america/jamaica UTC offset of -05:00.
- America/Jujuy - Timezone for america/jujuy UTC offset of -03:00.

- America/Juneau - Timezone for america/juneau UTC offset of -09:00.
- America/Kentucky/Louisville - Timezone for america/kentucky/louisville UTC offset of -05:00.
- America/Kentucky/Monticello - Timezone for america/kentucky/monticello UTC offset of -05:00.
- America/Knox_IN - Timezone for america/knox_in UTC offset of -06:00.
- America/Kralendijk - Timezone for america/kralendijk UTC offset of -04:00.
- America/La_Paz - Timezone for america/la_paz UTC offset of -04:00.
- America/Lima - Timezone for america/lima UTC offset of -05:00.
- America/Los_Angeles - Timezone for america/los_angeles UTC offset of -08:00.
- America/Louisville - Timezone for america/louisville UTC offset of -05:00.
- America/Lower_Princes - Timezone for america/lower_princes UTC offset of -04:00.
- America/Maceio - Timezone for america/maceio UTC offset of -03:00.
- America/Managua - Timezone for america/managua UTC offset of -06:00.
- America/Manaus - Timezone for america/manuel UTC offset of -04:00.
- America/Marigot - Timezone for america/marigot UTC offset of -04:00.
- America/Martinique - Timezone for america/martinique UTC offset of -04:00.
- America/Matamoros - Timezone for america/matamoros UTC offset of -06:00.
- America/Mazatlan - Timezone for america/mazatlan UTC offset of -07:00.
- America/Mendoza - Timezone for america/mendoza UTC offset of -03:00.
- America/Menominee - Timezone for america/menominee UTC offset of -06:00.
- America/Merida - Timezone for america/merida UTC offset of -06:00.
- America/Metlakatla - Timezone for america/metlakatla UTC offset of -09:00.
- America/Mexico_City - Timezone for america/mexico_city UTC offset of -06:00.
- America/Miquelon - Timezone for america/miquelon UTC offset of -03:00.
- America/Moncton - Timezone for america/moncton UTC offset of -04:00.
- America/Monterrey - Timezone for america/monterrey UTC offset of -06:00.
- America/Montevideo - Timezone for america/montevideo UTC offset of -03:00.
- America/Montreal - Timezone for america/montreal UTC offset of -05:00.
- America/Montserrat - Timezone for america/montserrat UTC offset of -04:00.
- America/Nassau - Timezone for america/nassau UTC offset of -05:00.
- America/New_York - Timezone for america/new_york UTC offset of -05:00.

- America/Nipigon - Timezone for america/nipigon UTC offset of -05:00.
- America/Nome - Timezone for america/nome UTC offset of -09:00.
- America/Noronha - Timezone for america/noronha UTC offset of -02:00.
- America/North_Dakota/Beulah - Timezone for america/north_dakota/beulah UTC offset of -06:00.
- America/North_Dakota/Center - Timezone for america/north_dakota/center UTC offset of -06:00.
- America/North_Dakota/New_Salem - Timezone for america/north_dakota/new_salem UTC offset of -06:00.
- America/Ojinaga - Timezone for america/ojinaga UTC offset of -07:00.
- America/Panama - Timezone for america/panama UTC offset of -05:00.
- America/Pangnirtung - Timezone for america/pangnirtung UTC offset of -05:00.
- America/Paramaribo - Timezone for america/paramaribo UTC offset of -03:00.
- America/Phoenix - Timezone for america/phoenix UTC offset of -07:00.
- America/Port-au-Prince - Timezone for america/port-au-prince UTC offset of -05:00.
- America/Port_of_Spain - Timezone for america/port_of_spain UTC offset of -04:00.
- America/Porto_Acre - Timezone for america/porto_acre UTC offset of -05:00.
- America/Porto_Velho - Timezone for america/porto_velho UTC offset of -04:00.
- America/Puerto_Rico - Timezone for america/puerto_rico UTC offset of -04:00.
- America/Rainy_River - Timezone for america/rainy_river UTC offset of -06:00.
- America/Rankin_Inlet - Timezone for america/rankin_inlet UTC offset of -06:00.
- America/Recife - Timezone for america/recife UTC offset of -03:00.
- America/Regina - Timezone for america/regina UTC offset of -06:00.
- America/Resolute - Timezone for america/resolute UTC offset of -06:00.
- America/Rio_Branco - Timezone for america/rio_branco UTC offset of -05:00.
- America/Rosario - Timezone for america/rosario UTC offset of -03:00.
- America/Santa_Isabel - Timezone for america/santa_isabel UTC offset of -08:00.
- America/Santarem - Timezone for america/santarem UTC offset of -03:00.
- America/Santiago - Timezone for america/santiago UTC offset of -03:00.
- America/Santo_Domingo - Timezone for america/santo_domingo UTC offset of -04:00.
- America/Sao_Paulo - Timezone for america/sao_paulo UTC offset of -02:00.
- America/Scoresbysund - Timezone for america/scoresbysund UTC offset of -01:00.

- America/Shiprock - Timezone for america/shiprock UTC offset of -07:00.
- America/Sitka - Timezone for america/sitka UTC offset of -09:00.
- America/St_Barthelemy - Timezone for america/st_barthelemy UTC offset of -04:00.
- America/St_Johns - Timezone for america/st_johns UTC offset of -03:30.
- America/St_Kitts - Timezone for america/st_kitts UTC offset of -04:00.
- America/St_Lucia - Timezone for america/st_lucia UTC offset of -04:00.
- America/St_Thomas - Timezone for america/st_thomas UTC offset of -04:00.
- America/St_Vincent - Timezone for america/st_vincent UTC offset of -04:00.
- America/Swift_Current - Timezone for america/swift_current UTC offset of -06:00.
- America/Tegucigalpa - Timezone for america/tegucigalpa UTC offset of -06:00.
- America/Thule - Timezone for america/thule UTC offset of -04:00.
- America/Thunder_Bay - Timezone for america/thunder_bay UTC offset of -05:00.
- America/Tijuana - Timezone for america/tijuana UTC offset of -08:00.
- America/Toronto - Timezone for america/toronto UTC offset of -05:00.
- America/Tortola - Timezone for america/tortola UTC offset of -04:00.
- America/Vancouver - Timezone for america/vancouver UTC offset of -08:00.
- America/Virgin - Timezone for america/virgin UTC offset of -04:00.
- America/Whitehorse - Timezone for america/whitehorse UTC offset of -08:00.
- America/Winnipeg - Timezone for america/winnipeg UTC offset of -06:00.
- America/Yakutat - Timezone for america/yakutat UTC offset of -09:00.
- America/Yellowknife - Timezone for america/yellowknife UTC offset of -07:00.
- Antarctica/Casey - Timezone for antarctica/casey UTC offset of +11:00.
- Antarctica/Davis - Timezone for antarctica/davis UTC offset of +07:00.
- Antarctica/DumontDURville - Timezone for antarctica/dumontdurville UTC offset of +10:00.
- Antarctica/Macquarie - Timezone for antarctica/macquarie UTC offset of +11:00.
- Antarctica/Mawson - Timezone for antarctica/mawson UTC offset of +05:00.
- Antarctica/McMurdo - Timezone for antarctica/mcmurdo UTC offset of +13:00.
- Antarctica/Palmer - Timezone for antarctica/palmer UTC offset of -03:00.
- Antarctica/Rothera - Timezone for antarctica/rothera UTC offset of -03:00.
- Antarctica/South_Pole - Timezone for antarctica/south_pole UTC offset of +13:00.

- Antarctica/Syowa - Timezone for antarctica/syowa UTC offset of +03:00.
- Antarctica/Troll - Timezone for antarctica/troll UTC offset of +00:00.
- Antarctica/Vostok - Timezone for antarctica/vostok UTC offset of +06:00.
- Arctic/Longyearbyen - Timezone for arctic/longyearbyen UTC offset of +01:00.
- Asia/Aden - Timezone for asia/aden UTC offset of +03:00.
- Asia/Almaty - Timezone for asia/almaty UTC offset of +06:00.
- Asia/Amman - Timezone for asia/amman UTC offset of +02:00.
- Asia/Anadyr - Timezone for asia/anadyr UTC offset of +12:00.
- Asia/Aqtau - Timezone for asia/aqtau UTC offset of +05:00.
- Asia/Aqtobe - Timezone for asia/aqtobe UTC offset of +05:00.
- Asia/Ashgabat - Timezone for asia/ashgabat UTC offset of +05:00.
- Asia/Ashkhabad - Timezone for asia/ashkhabad UTC offset of +05:00.
- Asia/Baghdad - Timezone for asia/baghdad UTC offset of +03:00.
- Asia/Bahrain - Timezone for asia/bahrain UTC offset of +03:00.
- Asia/Baku - Timezone for asia/baku UTC offset of +04:00.
- Asia/Bangkok - Timezone for asia/bangkok UTC offset of +07:00.
- Asia/Barnaul - Timezone for asia/barnaul UTC offset of +07:00.
- Asia/Beirut - Timezone for asia/beirut UTC offset of +02:00.
- Asia/Bishkek - Timezone for asia/bishkek UTC offset of +06:00.
- Asia/Brunei - Timezone for asia/brunei UTC offset of +08:00.
- Asia/Calcutta - Timezone for asia/calcutta UTC offset of +05:30.
- Asia/Chita - Timezone for asia/chita UTC offset of +09:00.
- Asia/Choibalsan - Timezone for asia/choibalsan UTC offset of +08:00.
- Asia/Chongqing - Timezone for asia/chongqing UTC offset of +08:00.
- Asia/Chungking - Timezone for asia/chungking UTC offset of +08:00.
- Asia/Colombo - Timezone for asia/colombo UTC offset of +05:30.
- Asia/Dacca - Timezone for asia/dacca UTC offset of +06:00.
- Asia/Damascus - Timezone for asia/damascus UTC offset of +02:00.
- Asia/Dhaka - Timezone for asia/dhaka UTC offset of +06:00.
- Asia/Dili - Timezone for asia/dili UTC offset of +09:00.
- Asia/Dubai - Timezone for asia/dubai UTC offset of +04:00.

- Asia/Dushanbe - Timezone for asia/dushanbe UTC offset of +05:00.
- Asia/Gaza - Timezone for asia/gaza UTC offset of +02:00.
- Asia/Harbin - Timezone for asia/harbin UTC offset of +08:00.
- Asia/Hebron - Timezone for asia/hebron UTC offset of +02:00.
- Asia/Ho_Chi_Minh - Timezone for asia/ho_chi_minh UTC offset of +07:00.
- Asia/Hong_Kong - Timezone for asia/hong_kong UTC offset of +08:00.
- Asia/Hovd - Timezone for asia/hovd UTC offset of +07:00.
- Asia/Irkutsk - Timezone for asia/irkutsk UTC offset of +08:00.
- Asia/Istanbul - Timezone for asia/istanbul UTC offset of +03:00.
- Asia/Jakarta - Timezone for asia/jakarta UTC offset of +07:00.
- Asia/Jayapura - Timezone for asia/jayapura UTC offset of +09:00.
- Asia/Jerusalem - Timezone for asia/jerusalem UTC offset of +02:00.
- Asia/Kabul - Timezone for asia/kabul UTC offset of +04:30.
- Asia/Kamchatka - Timezone for asia/kamchatka UTC offset of +12:00.
- Asia/Karachi - Timezone for asia/karachi UTC offset of +05:00.
- Asia/Kashgar - Timezone for asia/kashgar UTC offset of +06:00.
- Asia/Kathmandu - Timezone for asia/kathmandu UTC offset of +05:45.
- Asia/Katmandu - Timezone for asia/katmandu UTC offset of +05:45.
- Asia/Khandyga - Timezone for asia/khandyga UTC offset of +09:00.
- Asia/Kolkata - Timezone for asia/kolkata UTC offset of +05:30.
- Asia/Krasnoyarsk - Timezone for asia/krasnoyarsk UTC offset of +07:00.
- Asia/Kuala_Lumpur - Timezone for asia/kuala_lumpur UTC offset of +08:00.
- Asia/Kuching - Timezone for asia/kuching UTC offset of +08:00.
- Asia/Kuwait - Timezone for asia/kuwait UTC offset of +03:00.
- Asia/Macao - Timezone for asia/macao UTC offset of +08:00.
- Asia/Macau - Timezone for asia/macau UTC offset of +08:00.
- Asia/Magadan - Timezone for asia/magadan UTC offset of +11:00.
- Asia/Makassar - Timezone for asia/makassar UTC offset of +08:00.
- Asia/Manila - Timezone for asia/manila UTC offset of +08:00.
- Asia/Muscat - Timezone for asia/muscat UTC offset of +04:00.
- Asia/Nicosia - Timezone for asia/nicosia UTC offset of +02:00.

- Asia/Novokuznetsk - Timezone for asia/novokuznetsk UTC offset of +07:00.
- Asia/Novosibirsk - Timezone for asia/novosibirsk UTC offset of +07:00.
- Asia/Omsk - Timezone for asia/omsk UTC offset of +06:00.
- Asia/Oral - Timezone for asia/oral UTC offset of +05:00.
- Asia/Phnom_Penh - Timezone for asia/phnom_penh UTC offset of +07:00.
- Asia/Pontianak - Timezone for asia/pontianak UTC offset of +07:00.
- Asia/Pyongyang - Timezone for asia/pyongyang UTC offset of +08:30.
- Asia/Qatar - Timezone for asia/qatar UTC offset of +03:00.
- Asia/Qyzylorda - Timezone for asia/qyzylorda UTC offset of +06:00.
- Asia/Rangoon - Timezone for asia/rangoon UTC offset of +06:30.
- Asia/Riyadh - Timezone for asia/riyadh UTC offset of +03:00.
- Asia/Saigon - Timezone for asia/saigon UTC offset of +07:00.
- Asia/Sakhalin - Timezone for asia/sakhalin UTC offset of +11:00.
- Asia/Samarkand - Timezone for asia/samarkand UTC offset of +05:00.
- Asia/Seoul - Timezone for asia/seoul UTC offset of +09:00.
- Asia/Shanghai - Timezone for asia/shanghai UTC offset of +08:00.
- Asia/Singapore - Timezone for asia/singapore UTC offset of +08:00.
- Asia/Srednekolymsk - Timezone for asia/srednekolymsk UTC offset of +11:00.
- Asia/Taipei - Timezone for asia/taipei UTC offset of +08:00.
- Asia/Tashkent - Timezone for asia/tashkent UTC offset of +05:00.
- Asia/Tbilisi - Timezone for asia/tbilisi UTC offset of +04:00.
- Asia/Tehran - Timezone for asia/tehran UTC offset of +03:30.
- Asia/Tel_Aviv - Timezone for asia/tel_aviv UTC offset of +02:00.
- Asia/Thimbu - Timezone for asia/thimbu UTC offset of +06:00.
- Asia/Thimphu - Timezone for asia/thimphu UTC offset of +06:00.
- Asia/Tokyo - Timezone for asia/tokyo UTC offset of +09:00.
- Asia/Tomsk - Timezone for asia/tomsk UTC offset of +07:00.
- Asia/Ujung_Pandang - Timezone for asia/ujung_pandang UTC offset of +08:00.
- Asia/Ulaanbaatar - Timezone for asia/ulaanbaatar UTC offset of +08:00.
- Asia/Ulan_Bator - Timezone for asia/ulan_bator UTC offset of +08:00.
- Asia/Urumqi - Timezone for asia/urumqi UTC offset of +06:00.

- Asia/Ust-Nera - Timezone for asia/ust-nera UTC offset of +10:00.
- Asia/Vientiane - Timezone for asia/vientiane UTC offset of +07:00.
- Asia/Vladivostok - Timezone for asia/vladivostok UTC offset of +10:00.
- Asia/Yakutsk - Timezone for asia/yakutsk UTC offset of +09:00.
- Asia/Yekaterinburg - Timezone for asia/yekaterinburg UTC offset of +05:00.
- Asia/Yerevan - Timezone for asia/yerevan UTC offset of +04:00.
- Atlantic/Azores - Timezone for atlantic/azores UTC offset of -01:00.
- Atlantic/Bermuda - Timezone for atlantic/bermuda UTC offset of -04:00.
- Atlantic/Canary - Timezone for atlantic/canary UTC offset of +00:00.
- Atlantic/Cape_Verde - Timezone for atlantic/cape_verde UTC offset of -01:00.
- Atlantic/Faeroe - Timezone for atlantic/faeroe UTC offset of +00:00.
- Atlantic/Faroe - Timezone for atlantic/faroe UTC offset of +00:00.
- Atlantic/Jan_Mayen - Timezone for atlantic/jan_mayen UTC offset of +01:00.
- Atlantic/Madeira - Timezone for atlantic/madeira UTC offset of +00:00.
- Atlantic/Reykjavik - Timezone for atlantic/reykjavik UTC offset of +00:00.
- Atlantic/South_Georgia - Timezone for atlantic/south_georgia UTC offset of -02:00.
- Atlantic/St_Helena - Timezone for atlantic/st_helena UTC offset of +00:00.
- Atlantic/Stanley - Timezone for atlantic/stanley UTC offset of -03:00.
- Australia/ACT - Timezone for australia/act UTC offset of +11:00.
- Australia/Adelaide - Timezone for australia/adelaide UTC offset of +10:30.
- Australia/Brisbane - Timezone for australia/brisbane UTC offset of +10:00.
- Australia/Broken_Hill - Timezone for australia/broken_hill UTC offset of +10:30.
- Australia/Canberra - Timezone for australia/canberra UTC offset of +11:00.
- Australia/Currie - Timezone for australia/currie UTC offset of +11:00.
- Australia/Darwin - Timezone for australia/darwin UTC offset of +09:30.
- Australia/Eucla - Timezone for australia/eucla UTC offset of +08:45.
- Australia/Hobart - Timezone for australia/hobart UTC offset of +11:00.
- Australia/LHI - Timezone for australia/lhi UTC offset of +11:00.
- Australia/Lindeman - Timezone for australia/lindeman UTC offset of +10:00.
- Australia/Lord_Howe - Timezone for australia/lord_howe UTC offset of +11:00.
- Australia/Melbourne - Timezone for australia/melbourne UTC offset of +11:00.

- Australia/NSW - Timezone for australia/nsw UTC offset of +11:00.
- Australia/North - Timezone for australia/north UTC offset of +09:30.
- Australia/Perth - Timezone for australia/perth UTC offset of +08:00.
- Australia/Queensland - Timezone for australia/queensland UTC offset of +10:00.
- Australia/South - Timezone for australia/south UTC offset of +10:30.
- Australia/Sydney - Timezone for australia/sydney UTC offset of +11:00.
- Australia/Tasmania - Timezone for australia/tasmania UTC offset of +11:00.
- Australia/Victoria - Timezone for australia/victoria UTC offset of +11:00.
- Australia/West - Timezone for australia/west UTC offset of +08:00.
- Australia/Yancowinna - Timezone for australia/yancowinna UTC offset of +10:30.
- CET - Timezone for CET UTC offset of +01:00.
- CST6CDT - Timezone for CST6CDT UTC offset of -06:00.
- Canada/Atlantic - Timezone for canada/atlantic UTC offset of -04:00.
- Canada/Central - Timezone for canada/central UTC offset of -06:00.
- Canada/East-Saskatchewan - Timezone for canada/east-saskatchewan UTC offset of -06:00.
- Canada/Eastern - Timezone for canada/eastern UTC offset of -05:00.
- Canada/Mountain - Timezone for canada/mountain UTC offset of -07:00.
- Canada/Newfoundland - Timezone for canada/newfoundland UTC offset of -03:30.
- Canada/Pacific - Timezone for canada/pacific UTC offset of -08:00.
- Canada/Saskatchewan - Timezone for canada/saskatchewan UTC offset of -06:00.
- Canada/Yukon - Timezone for canada/yukon UTC offset of -08:00.
- Chile/Continental - Timezone for chile/continental UTC offset of -03:00.
- Chile/EasterIsland - Timezone for chile/easterisland UTC offset of -05:00.
- Cuba - Timezone for cuba UTC offset of -05:00.
- Default - Timezone for system UTC (OS time zone)
- EET - Timezone for EET UTC offset of +02:00.
- EST - Timezone for EST UTC offset of -05:00.
- EST5EDT - Timezone for EST5EDT UTC offset of -05:00.
- Egypt - Timezone for egypt UTC offset of +02:00.
- Eire - Timezone for eire UTC offset of +00:00.

- Etc/Greenwich - Timezone for etc/greenwich UTC offset of +00:00.
- Etc/UCT - Timezone for etc/uct UTC offset of +00:00.
- Etc/UTC - Timezone for etc/utc UTC offset of +00:00.
- Etc/Universal - Timezone for etc/universal UTC offset of +00:00.
- Etc/Zulu - Timezone for etc/zulu UTC offset of +00:00.
- Europe/Amsterdam - Timezone for europe/amsterdam UTC offset of +01:00.
- Europe/Andorra - Timezone for europe/andorra UTC offset of +01:00.
- Europe/Astrakhan - Timezone for europe/astrakhan UTC offset of +04:00.
- Europe/Athens - Timezone for europe/athens UTC offset of +02:00.
- Europe/Belfast - Timezone for europe/belfast UTC offset of +00:00.
- Europe/Belgrade - Timezone for europe/belgrade UTC offset of +01:00.
- Europe/Berlin - Timezone for europe/berlin UTC offset of +01:00.
- Europe/Bratislava - Timezone for europe/bratislava UTC offset of +01:00.
- Europe/Brussels - Timezone for europe/brussels UTC offset of +01:00.
- Europe/Bucharest - Timezone for europe/bucharest UTC offset of +02:00.
- Europe/Budapest - Timezone for europe/budapest UTC offset of +01:00.
- Europe/Busingen - Timezone for europe/busingen UTC offset of +01:00.
- Europe/Chisinau - Timezone for europe/chisinau UTC offset of +02:00.
- Europe/Copenhagen - Timezone for europe/copenhagen UTC offset of +01:00.
- Europe/Dublin - Timezone for europe/dublin UTC offset of +00:00.
- Europe/Gibraltar - Timezone for europe/gibraltar UTC offset of +01:00.
- Europe/Guernsey - Timezone for europe/guernsey UTC offset of +00:00.
- Europe/Helsinki - Timezone for europe/helsinki UTC offset of +02:00.
- Europe/Isle_of_Man - Timezone for europe/isle_of_man UTC offset of +00:00.
- Europe/Istanbul - Timezone for europe/istanbul UTC offset of +03:00.
- Europe/Jersey - Timezone for europe/jersey UTC offset of +00:00.
- Europe/Kaliningrad - Timezone for europe/kaliningrad UTC offset of +02:00.
- Europe/Kiev - Timezone for europe/kiev UTC offset of +02:00.
- Europe/Kirov - Timezone for europe/kirov UTC offset of +03:00.
- Europe/Lisbon - Timezone for europe/lisbon UTC offset of +00:00.
- Europe/Ljubljana - Timezone for europe/ljubljana UTC offset of +01:00.

- Europe/London - Timezone for europe/london UTC offset of +00:00.
- Europe/Luxembourg - Timezone for europe/luxembourg UTC offset of +01:00.
- Europe/Madrid - Timezone for europe/madrid UTC offset of +01:00.
- Europe/Malta - Timezone for europe/malta UTC offset of +01:00.
- Europe/Mariehamn - Timezone for europe/mariehamn UTC offset of +02:00.
- Europe/Minsk - Timezone for europe/minsk UTC offset of +03:00.
- Europe/Monaco - Timezone for europe/monaco UTC offset of +01:00.
- Europe/Moscow - Timezone for europe/moscow UTC offset of +03:00.
- Europe/Nicosia - Timezone for europe/nicosia UTC offset of +02:00.
- Europe/Oslo - Timezone for europe/oslo UTC offset of +01:00.
- Europe/Paris - Timezone for europe/paris UTC offset of +01:00.
- Europe/Podgorica - Timezone for europe/podgorica UTC offset of +01:00.
- Europe/Prague - Timezone for europe/prague UTC offset of +01:00.
- Europe/Riga - Timezone for europe/riga UTC offset of +02:00.
- Europe/Rome - Timezone for europe/rome UTC offset of +01:00.
- Europe/Samara - Timezone for europe/samara UTC offset of +04:00.
- Europe/San_Marino - Timezone for europe/san_marino UTC offset of +01:00.
- Europe/Sarajevo - Timezone for europe/sarajevo UTC offset of +01:00.
- Europe/Simferopol - Timezone for europe/simferopol UTC offset of +03:00.
- Europe/Skopje - Timezone for europe/skopje UTC offset of +01:00.
- Europe/Sofia - Timezone for europe/sofia UTC offset of +02:00.
- Europe/Stockholm - Timezone for europe/stockholm UTC offset of +01:00.
- Europe/Tallinn - Timezone for europe/tallinn UTC offset of +02:00.
- Europe/Tirane - Timezone for europe/tirane UTC offset of +01:00.
- Europe/Tiraspol - Timezone for europe/tiraspol UTC offset of +02:00.
- Europe/Ulyanovsk - Timezone for europe/ulyanovsk UTC offset of +04:00.
- Europe/Uzhgorod - Timezone for europe/uzhgorod UTC offset of +02:00.
- Europe/Vaduz - Timezone for europe/vaduz UTC offset of +01:00.
- Europe/Vatican - Timezone for europe/vatican UTC offset of +01:00.
- Europe/Vienna - Timezone for europe/vienna UTC offset of +01:00.
- Europe/Vilnius - Timezone for europe/vilnius UTC offset of +02:00.

- Europe/Volgograd - Timezone for europe/volgograd UTC offset of +03:00.
- Europe/Warsaw - Timezone for europe/warsaw UTC offset of +01:00.
- Europe/Zagreb - Timezone for europe/zagreb UTC offset of +01:00.
- Europe/Zaporozhye - Timezone for europe/zaporozhye UTC offset of +02:00.
- Europe/Zurich - Timezone for europe/zurich UTC offset of +01:00.
- GB - Timezone for GB UTC offset of +00:00.
- GB-Eire - Timezone for GB-Eire UTC offset of +00:00.
- GMT - Timezone for GMT UTC offset of +00:00.
- GMT0 - Timezone for GMT0 UTC offset of +00:00.
- Greenwich - Timezone for greenwich UTC offset of +00:00.
- HST - Timezone for HST UTC offset of -10:00.
- Hongkong - Timezone for hongkong UTC offset of +08:00.
- Iceland - Timezone for iceland UTC offset of +00:00.
- Indian/Antananarivo - Timezone for indian/antananarivo UTC offset of +03:00.
- Indian/Chagos - Timezone for indian/chagos UTC offset of +06:00.
- Indian/Christmas - Timezone for indian/christmas UTC offset of +07:00.
- Indian/Cocos - Timezone for indian/cocos UTC offset of +06:30.
- Indian/Comoro - Timezone for indian/comoro UTC offset of +03:00.
- Indian/Kerguelen - Timezone for indian/kerguelen UTC offset of +05:00.
- Indian/Mahe - Timezone for indian/mahe UTC offset of +04:00.
- Indian/Maldives - Timezone for indian/maldives UTC offset of +05:00.
- Indian/Mauritius - Timezone for indian/mauritius UTC offset of +04:00.
- Indian/Mayotte - Timezone for indian/mayotte UTC offset of +03:00.
- Indian/Reunion - Timezone for indian/reunion UTC offset of +04:00.
- Iran - Timezone for iran UTC offset of +03:30.
- Israel - Timezone for israel UTC offset of +02:00.
- Jamaica - Timezone for jamaica UTC offset of -05:00.
- Japan - Timezone for japan UTC offset of +09:00.
- Kwajalein - Timezone for kwajalein UTC offset of +12:00.
- Libya - Timezone for libya UTC offset of +02:00.
- MET - Timezone for MET UTC offset of +01:00.

- MST - Timezone for MST UTC offset of -07:00.
- MST7MDT - Timezone for MST7MDT UTC offset of -07:00.
- Mexico/BajaNorte - Timezone for mexico/bajanorte UTC offset of -08:00.
- Mexico/BajaSur - Timezone for mexico/bajasur UTC offset of -07:00.
- Mexico/General - Timezone for mexico/general UTC offset of -06:00.
- NZ - Timezone for NZ UTC offset of +13:00.
- NZ-CHAT - Timezone for NZ-CHAT UTC offset of +13:45.
- Navajo - Timezone for navajo UTC offset of -07:00.
- PRC - Timezone for PRC UTC offset of +08:00.
- PST8PDT - Timezone for PST8PDT UTC offset of -08:00.
- Pacific/Apia - Timezone for pacific/apia UTC offset of +14:00.
- Pacific/Auckland - Timezone for pacific/auckland UTC offset of +13:00.
- Pacific/Bougainville - Timezone for pacific/bougainville UTC offset of +11:00.
- Pacific/Chatham - Timezone for pacific/chatham UTC offset of +13:45.
- Pacific/Chuuk - Timezone for pacific/chuuk UTC offset of +10:00.
- Pacific/Easter - Timezone for pacific/easter UTC offset of -05:00.
- Pacific/Efate - Timezone for pacific/efate UTC offset of +11:00.
- Pacific/Enderbury - Timezone for pacific/enderbury UTC offset of +13:00.
- Pacific/Fakaofo - Timezone for pacific/fakaofo UTC offset of +13:00.
- Pacific/Fiji - Timezone for pacific/fiji UTC offset of +13:00.
- Pacific/Funafuti - Timezone for pacific/funafuti UTC offset of +12:00.
- Pacific/Galapagos - Timezone for pacific/galapagos UTC offset of -06:00.
- Pacific/Gambier - Timezone for pacific/gambier UTC offset of -09:00.
- Pacific/Guadalcanal - Timezone for pacific/guadalcanal UTC offset of +11:00.
- Pacific/Guam - Timezone for pacific/guam UTC offset of +10:00.
- Pacific/Honolulu - Timezone for pacific/honolulu UTC offset of -10:00.
- Pacific/Johnston - Timezone for pacific/johnston UTC offset of -10:00.
- Pacific/Kiritimati - Timezone for pacific/kiritimati UTC offset of +14:00.
- Pacific/Kosrae - Timezone for pacific/kosrae UTC offset of +11:00.
- Pacific/Kwajalein - Timezone for pacific/kwajalein UTC offset of +12:00.
- Pacific/Majuro - Timezone for pacific/majuro UTC offset of +12:00.

- Pacific/Marquesas - Timezone for pacific/marquesas UTC offset of -09:30.
- Pacific/Midway - Timezone for pacific/midway UTC offset of -11:00.
- Pacific/Nauru - Timezone for pacific/nauru UTC offset of +12:00.
- Pacific/Niue - Timezone for pacific/niue UTC offset of -11:00.
- Pacific/Norfolk - Timezone for pacific/norfolk UTC offset of +11:00.
- Pacific/Noumea - Timezone for pacific/noumea UTC offset of +11:00.
- Pacific/Pago_Pago - Timezone for pacific/pago_pago UTC offset of -11:00.
- Pacific/Palau - Timezone for pacific/palau UTC offset of +09:00.
- Pacific/Pitcairn - Timezone for pacific/pitcairn UTC offset of -08:00.
- Pacific/Pohnpei - Timezone for pacific/pohnpei UTC offset of +11:00.
- Pacific/Ponape - Timezone for pacific/ponape UTC offset of +11:00.
- Pacific/Port_Moresby - Timezone for pacific/port_moresby UTC offset of +10:00.
- Pacific/Rarotonga - Timezone for pacific/rarotonga UTC offset of -10:00.
- Pacific/Saipan - Timezone for pacific/saipan UTC offset of +10:00.
- Pacific/Samoa - Timezone for pacific/samoa UTC offset of -11:00.
- Pacific/Tahiti - Timezone for pacific/tahiti UTC offset of -10:00.
- Pacific/Tarawa - Timezone for pacific/tarawa UTC offset of +12:00.
- Pacific/Tongatapu - Timezone for pacific/tongatapu UTC offset of +14:00.
- Pacific/Truk - Timezone for pacific/truk UTC offset of +10:00.
- Pacific/Wake - Timezone for pacific/wake UTC offset of +12:00.
- Pacific/Wallis - Timezone for pacific/wallis UTC offset of +12:00.
- Pacific/Yap - Timezone for pacific/yap UTC offset of +10:00.
- Poland - Timezone for poland UTC offset of +01:00.
- Portugal - Timezone for portugal UTC offset of +00:00.
- ROC - Timezone for ROC UTC offset of +08:00.
- ROK - Timezone for ROK UTC offset of +09:00.
- Singapore - Timezone for singapore UTC offset of +08:00.
- Turkey - Timezone for turkey UTC offset of +03:00.
- UCT - Timezone for UCT UTC offset of +00:00.
- US/Pacific-New - Timezone for US/Pacific-New UTC offset of -08:00.
- Universal - Timezone for universal UTC offset of +00:00.

- W-SU - Timezone for W-SU UTC offset of +03:00.
- WET - Timezone for WET UTC offset of +00:00.
- Zulu - Timezone for zulu UTC offset of +00:00.
- jdj_tz - Timezone for jdj_tz UTC offset of +00:00.

Default

Default

Name

upCloT - UP CloT

Syntax

--upCIoT <list-item>

Description

Enable User Plane Cellular Internet of Things Optimization

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

vdpProfileName - VDP profile name

Syntax

--vdpProfileName <string>

Description

Voice domain preference profile name.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

''

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

7.101 cmm taiEnbQuery

TAI eNB Query. Syntax, description, parameters, and roles for cmm taiEnbQuery command.

Syntax

Description

List eNBs with TAI.

Parameters

Name

taiMcc - TAI MCC

Syntax

--taiMcc <digits>

Description

3-digit number used to represent the mobile country code for TAI.

Class

Required key.

Type / Range

Digit string of length 3

Name

taiMnc - TAI MNC

Syntax

--taiMnc <digits>

Description

Digits used to represent mobile network code for TAI.

Class

Required key.

Type / Range

Digit string of length 2 to 3

Name

tac - Tracking area code

Syntax

--tac <integer>

Description

Tracking Area Code.

Class

Required key.

Type / Range

Integer in the range of 1 to 65,535

Name

enbId - eNB id

Syntax

--enbId <integer>

Description

eNB identifier as specified in TS 36.413 v8.6.1.

Class

Read only.

Type / Range

Integer in the range of 0 to 268,435,455

Name

enbIpAddress - eNB IP address

Syntax

--enbIpAddress <string>

Description

eNB IP address.

Class

Read only.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

enbName - eNB name

Syntax

--enbName <string>

Description

eNB name.

Class

Read only.

Type / Range

Digit string of length 0 or 1 to 255

Name

enbType - EnodeB type

Syntax

--enbType <list-item>

Description

Specifies which enodeb type of the enodeb.

Class

Read only.

List

- home - Specify home eNodeB.
- macro - Specify macro eNodeB.

Name

linkAdminState - Link admin state

Syntax

--linkAdminState <list-item>

Description

Link admin state.

Class

Read only.

List

- locked - Locked.
- unlocked - Unlocked.

Name

linkOperationState - Link operation state

Syntax

--linkOperationState <list-item>

Description

Link operation state.

Class

Read only.

List

- disabled - Disabled state.
- enabled - Enabled state.

Name

maxTac - Maximum Tracking Area Code

Syntax

--maxTac <integer>

Description

Maximum Tracking Area Code. Enter a value equal to minTac to display output for a single TAC or a value greater than 'minTac' to display output for a range of TACs.

Restrictions

The value of this parameter must be greater than or equal to the value of parameter 'minTac'.

Class

Optional.

Type / Range

Integer in the range of 1 to 65,535

Default

65535

Name

mcc - Mobile country code

Syntax

--mcc <digits>

Description

3-digit number used to represent the mobile country code.

Class

Read only.

Type / Range

Digit string of length 3

Name

minTac - Minimum Tracking Area Code

Syntax

--minTac <integer>

Description

Minimum Tracking Area Code. Enter a value equal to maxTac to display output for a single TAC or a value greater than 'maxTac' to display output for a range of TACs.

Restrictions

The value of this parameter must be less than or equal to the value of parameter 'maxTac'.

Class

Optional.

Type / Range

Integer in the range of 1 to 65,535

Default

1

Name

mnc - Mobile network code

Syntax

--mnc <digits>

Description

Digits used to represent mobile network code.

Class

Read only.

Type / Range

Digit string of length 2 to 3

Roles**List**

sysadmin, manage, monitor

7.102 cmm taiEsmcList

TAI E-SMLC List. Syntax, description, parameters, and roles for cmm taiEsmcList command.

Syntax**Description**

Create, delete, list and show TAI E-SMLC List

Parameters**Name**

plmnName - Mobile network name

Syntax

--plmnName <string>

Description

Name that identifies the mobile network.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

tac - Tracking area code

Syntax

--tac <integer>

Description

Tracking area code.

Class

Required key.

Type / Range

Integer in the range of 1 to 65,535

Name

lcsType - Location services type

Syntax

--lcsType <list-item>

Description

Specifies the location services type associated with ESMLC for this tracking area.

Class

Required key.

List

- LawfullIntercept - Lawful interception location services - Lawful interception location services.
- Operator - Operator location services - Operator location services.
- Value Added - Value added location services - Value added location services.

Name

lcsEntryId - Location services entry id

Syntax

--lcsEntryId <integer>

Description

Unique id for this association of E-SMLC to tracking area and location services type. This field is used as the priority when MME TAI selection algorithm matching location services type is primarySecondary.

Class

Required key.

Type / Range

Integer in the range of 1 to 6

Name

esmlcIdentity - ESMC identifier

Syntax

--esmlcIdentity <integer>

Description

Indicates the ESMC being provisioned for a given tracking area and location services type.

Restrictions

esmlcIdentity must be unique for a given plmnName, tac, and lcsType combination.

Class

Required on create, not modifiable.

Type / Range

Integer in the range of 0 to 255

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.103 cmm taiLaiMapping

TAI to LAI Mapping. Syntax, description, parameters, and roles for cmm taiLaiMapping command.

Syntax

Description

This command is used to administer TAI to LAI Mapping and includes parameters for defining the mobile network names of the Tracking Area and Location Areas, the Tracking Area Code and Location Area Code, and record retrieval sort type.

Parameters

Name

taiPlmnName - Mobile network name of TAI

Syntax

--taiPlmnName <string>

Description

Name that identifies mobile network (alternate key) of the tracking area in this mapping.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\- _:@]*

Name

tac - Tracking area code

Syntax

--tac <integer>

Description

Tracking area code.

Class

Required key.

Type / Range

Integer in the range of 1 to 65,535

Name

laiPlmnName - Mobile network name of LAI

Syntax

--laiPlmnName <string>

Description

Name that identifies mobile network (alternate key) of the location area in this mapping.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

lac - Location area code

Syntax

--lac <integer>

Description

Location area code.

Class

Required key.

Type / Range

Integer in the range of 1 to 65,535

Name

sortType - Sort type

Syntax

--sortType <list-item>

Description

Sort type of records retrieved via get-config operation.

Class

Optional.

List

- NoSort - No sort - No sort of records via get-config operation.
- Sort1 - Sort 1 - Sort algorithm 1 applied to records retrieved via get-config operation.

Default

NoSort

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.104 cmm taiList

List of TAI. Syntax, description, parameters, and roles for cmm taiList command.

Syntax

Description

This command is used to administer a list of TAIs and includes parameters for defining the TAI list name, served PLMN name, and Tracking Area Code.

Parameters

Name

areaListName - Name of TAI list

Syntax

--areaListName <string>

Description

Name of TAI list.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

servedPlmnName - Served mobile network name

Syntax

--servedPlmnName <string>

Description

Served mobile network name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

tac - Tracking area code

Syntax

--tac <integer>

Description

Tracking area code.

Class

Required key.

Type / Range

Integer in the range of 1 to 65,535

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.105 cmm taiNb

TAI Neighbor List. Syntax, description, parameters, and roles for cmm taiNb command.

Syntax**Description**

This command is used to administer a TAI Neighbor List and includes parameters for defining the mobile network name, neighbor mobile network name, Tracking Area Code, and neighbor Tracking Area Code. A maximum of 15 tracking area codes from the same mobile network and/or from the equivalent networks can be added as neighbors.

Parameters**Name**

servedPlmnName - Mobile network name

Syntax

--servedPlmnName <string>

Description

Name that identifies mobile network (alternate key).

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

tac - Tracking area code

Syntax

--tac <integer>

Description

Tracking area code.

Class

Required key.

Type / Range

Integer in the range of 1 to 65,535

Name

nbPlmnName - Mobile network name of neighbor

Syntax

--nbPlmnName <string>

Description

Name that identifies mobile network (alternate key) of the neighbor TAI.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

tacNb - Neighbor tracking area code

Syntax

--tacNb <integer>

Description

Neighbor tracking area code.

Class

Required key.

Type / Range

Integer in the range of 1 to 65,535

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.106 cmm taiNgRanQuery

TAI NG-RAN Query. Syntax, description, parameters, and roles for cmm taiNgRanQuery command.

Syntax

Description

Lists NG-RAN (gNB or ng-eNB) with TAI. 'taiMcc', 'taiMnc' and 'tac' are required input filters. If any of these are not given query will return empty list.

Parameters

Name

taiMcc - TAI MCC

Syntax

--taiMcc <digits>

Description

3-digit number used to represent the mobile country code for TAI.

Class

Required key.

Type / Range

Digit string of length 3

Name

taiMnc - TAI MNC

Syntax

--taiMnc <digits>

Description

Digits used to represent mobile network code for TAI.

Class

Required key.

Type / Range

Digit string of length 2 to 3

Name

tac - Tracking area code

Syntax

--tac <integer>

Description

Tracking area code.

Class

Required key.

Type / Range

Integer in the range of 1 to 16,777,215

Name

assignMode - Assignment Mode

Syntax

--assignMode <integer>

Description

This field depicts the assignment mode for the SNPN

Class

Optional.

Type / Range

Integer in the range of 0 to 2

Default

0

Name

mcc - Mobile country code

Syntax

--mcc <digits>

Description

3-digit number used to represent the mobile country code.

Class

Read only.

Type / Range

Digit string of length 3

Name

mnc - Mobile network code

Syntax

--mnc <digits>

Description

Digits used to represent mobile network code.

Class

Read only.

Type / Range

Digit string of length 2 to 3

Name

ngRanId - Global RAN Node ID

Syntax

--ngRanId <integer>

Description

Specifies Global RAN Node identifier as specified in TS 38.413

Class

Read only.

Type / Range

Integer in the range of 0 to 268,435,455

Name

ngRanIpAddress - RAN IP address

Syntax

--ngRanIpAddress <string>

Description

Specifies RAN IP address

Class

Read only.

Type / Range

Digit string of length 2 to 39 with the allowed pattern of [0-9a-f\.:]*

Name

ngRanName - ngRan node name

Syntax

--ngRanName <string>

Description

Specifies RAN node name

Class

Read only.

Type / Range

Digit string of length 0 or 1 to 255

Name

ngRanType - NG-RAN Type

Syntax

--ngRanType <list-item>

Description

Specifies the NG-RAN type.

Class

Optional.

List

- gnb - GNB
- n3iwf - N3IWF

Name

nidValue - Nid value

Syntax

--nidValue <integer>

Description

This field depicts the Nid value of the NID for the SNPN

Class

Optional.

Type / Range

Integer in the Range 0 to 1099511627775

Default

0

Roles**List**

sysadmin, manage, monitor

7.107 cmm taiRangeDiscoveryProfile

TAI Range Discovery Profile Default values. Syntax, description, parameters, and roles for cmm taiRangeDiscoveryProfile command.

Syntax

Description

This command used to define a Profile associated with range of TAI value. The profile specifies default attributes for the TAI when the MME discovers a TAI through the auto-discovery feature.

Parameters

Name

name - TAI Range Discovery Profile name

Syntax

--name <string>

Description

TAI Range Discovery Profile name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

maxTac - Max Tracking Area Code

Syntax

--maxTac <integer>

Description

Ending TAC of a TA range.

Class

Required on create.

Type / Range

Integer in the range of 0 to 65,535

Name

minTac - Min Tracking Area Code

Syntax

--minTac <integer>

Description

The starting Tracking Area Code of a TA range.

Class

Required on create.

Type / Range

Integer in the range of 0 to 65,535

Name

plmnName - Mobile network name

Syntax

--plmnName <string>

Description

Name that identifies the mobile network.

Class

Required on create, not modifiable.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

adminState - Administrative state

Syntax

--adminState <list-item>

Description

Administrative state of the TA Range Profile. The auto-provisioning insertion of new records will not occur for a profile in a locked admin state.

Class

Optional.

List

- locked - Locked.
- unlocked - Unlocked.

Default

locked

Name

aerialUeSupported - Aerial UE Supported

Syntax

--aerialUeSupported <list-item>

Description

This field identifies whether the TAI eNodeBs support Aerial UE operation

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ciotEmmRegisterdWithoutPdn - Enable EMM-REGISTERED without PDN connection for C-IOT UE

Syntax

--ciotEmmRegisterdWithoutPdn <list-item>

Description

Enable EMM-REGISTERED without PDN connection for C-IOT UE

Restrictions

This element is not applicable when plmnType is Other Network

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

cpCIoT - CP CIOT

Syntax

--cpCIoT <list-item>

Description

Enable CP CIOT optimization.

Restrictions

This element is not applicable when plmnType is Other Network

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

createMmeGrpTai - Create mmeGrpTai

Syntax

--createMmeGrpTai <list-item>

Description

Enable the auto-provisioning of a new mmeGrpTai entry using the mmeGi profile attribute.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

emergencySupported - Emergency supported

Syntax

--emergencySupported <list-item>

Description

Indicates if emergency is supported. only allowed if IMS supported is true.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

esmlcIdentity1 - ESMLC identity 1

Syntax

```
--esmlcIdentity1 <digits>
```

Description

First TAI to E-SMLC (Evolved serving mobile location center) mapping

Class

Optional.

Type / Range

Digit string of length 0 to 3

Default

"

Name

esmlcIdentity2 - ESMC identity 2

Syntax

```
--esmlcIdentity2 <digits>
```

Description

Second TAI to E-SMLC (Evolved serving mobile location center) mapping.

Class

Optional.

Type / Range

Digit string of length 0 to 3

Default

"

Name

esmlcIdentity3 - ESMC identity 3

Syntax

```
--esmlcIdentity3 <digits>
```

Description

Third TAI to E-SMLC (Evolved serving mobile location center) mapping. The E-SMLC is used for emergency services.

Class

Optional.

Type / Range

Digit string of length 0 to 3

Default

"

Name

esmlcIdentity4 - ESMC identity 4

Syntax

--esmlcIdentity4 <digits>

Description

Fourth TAI to E-SMLC (Evolved serving mobile location center) mapping. The set of 4 ESMC are used for emergency services

Class

Optional.

Type / Range

Digit string of length 0 to 3

Default

"

Name

forbiddenTailListName - Forbidden TAI list name

Syntax

--forbiddenTailListName <string>

Description

Forbidden TAI list name

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

imsSupported - IMS supported

Syntax

--imsSupported <list-item>

Description

Indicates if IMS is supported.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

liEsmIcIdentity1 - Lawful Intercept ESMC identifier 1

Syntax

--liEsmIcIdentity1 <digits>

Description

Indicates the First TAI to Lawful Intercept E-SMLC (Evolved serving mobile location center) mapping.

Class

Optional.

Type / Range

Digit string of length 0 to 3

Default

"

Name

liEsmIcIdentity2 - Lawful Intercept ESMC identifier 2

Syntax

--liEsmIcIdentity2 <digits>

Description

Indicates the Second TAI to Lawful Intercept E-SMLC (Evolved serving mobile location center) mapping.

Class

Optional.

Type / Range

Digit string of length 0 to 3

Default

"

Name

liSelectionAlgorithm - Lawful Intercept location services selection algorithm

Syntax

--liSelectionAlgorithm <list-item>

Description

Indicates for the set of lawful intercept location services ESMC (Evolved Serving Mobile Location Center) whether they will load share or function as primary/secondary.

Class

Optional.

List

- LoadShare - Load share - Load share.
- PrimarySecondary - Primary / secondary - Function as primary / secondary.

Default

PrimarySecondary

Name

mmeGi - MME group ID

Syntax

--mmeGi <integer>

Description

MME group ID

Class

Optional.

Type / Range

Integer in the range of 0 to 65,535

Default

0

Name

mmeRelCapacity - MME relative capacity

Syntax

--mmeRelCapacity <integer>

Description

Provision MME relative capacity per tracking area.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

5

Name

mtEdtCp - mtEdtCp

Syntax

--mtEdtCp <list-item>

Description

Mobile Terminated Early Data Transfer over Control Plan

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

mtEdtUp - mtEdtUp

Syntax

--mtEdtUp <list-item>

Description

Mobile Terminated Early Data Transfer over User Plane

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

nbiotSmsWithoutCombinedAttach - Enable SMS without combined Attach For NB-IOT UE

Syntax

--nbiotSmsWithoutCombinedAttach <list-item>

Description

Enable SMS without combined Attach For NB-IOT UE

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

operEsmlcIdentity1 - Operator ESMC identifier 1

Syntax

--operEsmlcIdentity1 <digits>

Description

Indicates the First TAI to Operator E-SMLC (Evolved serving mobile location center) mapping.

Class

Optional.

Type / Range

Digit string of length 0 to 3

Default

"

Name

operEsmlcIdentity2 - Operator ESMC identifier 2

Syntax

--operEsmlcIdentity2 <digits>

Description

Indicates the Second TAI to Operator E-SMLC (Evolved serving mobile location center) mapping.

Class

Optional.

Type / Range

Digit string of length 0 to 3

Default

"

Name

operSelectionAlgorithm - Operator location services selection algorithm

Syntax

--operSelectionAlgorithm <list-item>

Description

Indicates for the set of operator location services ESMC (Evolved serving mobile location center) whether they will load share or function as primary/secondary.

Class

Optional.

List

- LoadShare - Load share - Load share.
- PrimarySecondary - Primary / secondary - Function as primary / secondary.

Default

PrimarySecondary

Name

s1uDataTrans - S1U data transfer

Syntax

--s1uDataTrans <list-item>

Description

Enable S1-U data transfer.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

selectionAlgorithm - Selection algorithm

Syntax

--selectionAlgorithm <list-item>

Description

Indicates for the set of ESMCs, (Evolved serving mobile location center) whether they will load share or function as primary/secondary.

Class

Optional.

List

- LoadShare - Load share - Load share.
- PrimarySecondary - Primary / secondary - Function as primary / secondary.

Default

PrimarySecondary

Name

supportedRatType - Supported RAT type for the auto-provisioned TAC

Syntax

--supportedRatType <list-item>

Description

Supported RAT Type for this TAC. It can only be wbeutran or nbiot

Restrictions

Supported RAT type for the auto-provisioned TAC. it can only be wbeutran or nbiot

Class

Optional.

List

- nbiot - Narrow band internet of things - Specify type of narrow band internet of things.
- wbeutran - Wide band EUTRAN - Specify type of wide band EUTRAN.

Default

wbeutran

Name

timeZoneName - Time zone name

Syntax

--timeZoneName <list-item>

Description

Provides time zone names that correspond to the region selected

Restrictions

See comments on TIME_ZONE type in CfgDataType.xsd

Class

Optional.

List

- Africa/Abidjan - Timezone for africa/abidjan UTC offset of +00:00.
- Africa/Accra - Timezone for africa/accra UTC offset of +00:00.
- Africa/Addis_Ababa - Timezone for africa/addis_ababa UTC offset of +03:00.
- Africa/Algiers - Timezone for africa/algiers UTC offset of +01:00.
- Africa/Asmara - Timezone for africa/asmara UTC offset of +03:00.
- Africa/Asmera - Timezone for africa/asmera UTC offset of +03:00.
- Africa/Bamako - Timezone for africa/bamako UTC offset of +00:00.
- Africa/Bangui - Timezone for africa/bangui UTC offset of +01:00.
- Africa/Banjul - Timezone for africa/banjul UTC offset of +00:00.
- Africa/Bissau - Timezone for africa/bissau UTC offset of +00:00.

- Africa/Blantyre - Timezone for africa/blantyre UTC offset of +02:00.
- Africa/Brazzaville - Timezone for africa/brazzaville UTC offset of +01:00.
- Africa/Bujumbura - Timezone for africa/bujumbura UTC offset of +02:00.
- Africa/Cairo - Timezone for africa/cairo UTC offset of +02:00.
- Africa/Casablanca - Timezone for africa/casablanca UTC offset of +00:00.
- Africa/Ceuta - Timezone for africa/ceuta UTC offset of +01:00.
- Africa/Conakry - Timezone for africa/conakry UTC offset of +00:00.
- Africa/Dakar - Timezone for africa/dakar UTC offset of +00:00.
- Africa/Dar_es_Salaam - Timezone for africa/dar_es_salaam UTC offset of +03:00.
- Africa/Djibouti - Timezone for africa/djibouti UTC offset of +03:00.
- Africa/Douala - Timezone for africa/douala UTC offset of +01:00.
- Africa/El_Aaiun - Timezone for africa/el_aaiun UTC offset of +00:00.
- Africa/Freetown - Timezone for africa/freetown UTC offset of +00:00.
- Africa/Gaborone - Timezone for africa/gaborone UTC offset of +02:00.
- Africa/Harare - Timezone for africa/harare UTC offset of +02:00.
- Africa/Johannesburg - Timezone for africa/johannesburg UTC offset of +02:00.
- Africa/Juba - Timezone for africa/juba UTC offset of +03:00.
- Africa/Kampala - Timezone for africa/kampala UTC offset of +03:00.
- Africa/Khartoum - Timezone for africa/khartoum UTC offset of +03:00.
- Africa/Kigali - Timezone for africa/kigali UTC offset of +02:00.
- Africa/Kinshasa - Timezone for africa/kinshasa UTC offset of +01:00.
- Africa/Lagos - Timezone for africa/lagos UTC offset of +01:00.
- Africa/Libreville - Timezone for africa/libreville UTC offset of +01:00.
- Africa/Lome - Timezone for africa/lome UTC offset of +00:00.
- Africa/Luanda - Timezone for africa/luanda UTC offset of +01:00.
- Africa/Lubumbashi - Timezone for africa/lubumbashi UTC offset of +02:00.
- Africa/Lusaka - Timezone for africa/lusaka UTC offset of +02:00.
- Africa/Malabo - Timezone for africa/malabo UTC offset of +01:00.
- Africa/Maputo - Timezone for africa/maputo UTC offset of +02:00.
- Africa/Maseru - Timezone for africa/maseru UTC offset of +02:00.
- Africa/Mbabane - Timezone for africa/mbabane UTC offset of +02:00.

- Africa/Mogadishu - Timezone for africa/mogadishu UTC offset of +03:00.
- Africa/Monrovia - Timezone for africa/monrovia UTC offset of +00:00.
- Africa/Nairobi - Timezone for africa/nairobi UTC offset of +03:00.
- Africa/Ndjamena - Timezone for africa/ndjamena UTC offset of +01:00.
- Africa/Niamey - Timezone for africa/niamey UTC offset of +01:00.
- Africa/Nouakchott - Timezone for africa/nouakchott UTC offset of +00:00.
- Africa/Ouagadougou - Timezone for africa/ouagadougou UTC offset of +00:00.
- Africa/Porto-Novo - Timezone for africa/porto-novo UTC offset of +01:00.
- Africa/Sao_Tome - Timezone for africa/sao_tome UTC offset of +00:00.
- Africa/Timbuktu - Timezone for africa/timbuktu UTC offset of +00:00.
- Africa/Tripoli - Timezone for africa/tripoli UTC offset of +02:00.
- Africa/Tunis - Timezone for africa/tunis UTC offset of +01:00.
- Africa/Windhoek - Timezone for africa/windhoek UTC offset of +02:00.
- America/Adak - Timezone for america/adak UTC offset of -10:00.
- America/Anchorage - Timezone for america/anchorage UTC offset of -09:00.
- America/Anguilla - Timezone for america/anguilla UTC offset of -04:00.
- America/Antigua - Timezone for america/antigua UTC offset of -04:00.
- America/Araguaina - Timezone for america/araguaina UTC offset of -03:00.
- America/Argentina/Buenos_Aires - Timezone for america/argentina/buenos_aires UTC offset of -03:00.
- America/Argentina/Catamarca - Timezone for america/argentina/catamarca UTC offset of -03:00.
- America/Argentina/ComodRivadavia - Timezone for america/argentina/comodrivadavia UTC offset of -03:00.
- America/Argentina/Cordoba - Timezone for america/argentina/cordoba UTC offset of -03:00.
- America/Argentina/Jujuy - Timezone for america/argentina/jujuy UTC offset of -03:00.
- America/Argentina/La_Rioja - Timezone for america/argentina/la_rioja UTC offset of -03:00.
- America/Argentina/Mendoza - Timezone for america/argentina/mendoza UTC offset of -03:00.
- America/Argentina/Rio_Gallegos - Timezone for america/argentina/rio_gallegos UTC offset of -03:00.

- America/Argentina/Salta - Timezone for america/argentina/salta UTC offset of -03:00.
- America/Argentina/San_Juan - Timezone for america/argentina/san_juan UTC offset of -03:00.
- America/Argentina/San_Luis - Timezone for america/argentina/san_luis UTC offset of -03:00.
- America/Argentina/Tucuman - Timezone for america/argentina/tucuman UTC offset of -03:00.
- America/Argentina/Ushuaia - Timezone for america/argentina/ushuaia UTC offset of -03:00.
- America/Aruba - Timezone for america/aruba UTC offset of -04:00.
- America/Asuncion - Timezone for america/asuncion UTC offset of -03:00.
- America/Atikokan - Timezone for america/atikokan UTC offset of -05:00.
- America/Atka - Timezone for america/atka UTC offset of -10:00.
- America/Bahia - Timezone for america/bahia UTC offset of -03:00.
- America/Bahia_Banderas - Timezone for america/bahia_banderas UTC offset of -06:00.
- America/Barbados - Timezone for america/barbados UTC offset of -04:00.
- America/Belem - Timezone for america/belem UTC offset of -03:00.
- America/Belize - Timezone for america/belize UTC offset of -06:00.
- America/Blanc-Sablon - Timezone for america/blanc-sablon UTC offset of -04:00.
- America/Boa_Vista - Timezone for america/boa_vista UTC offset of -04:00.
- America/Bogota - Timezone for america/bogota UTC offset of -05:00.
- America/Boise - Timezone for america/boise UTC offset of -07:00.
- America/Buenos_Aires - Timezone for america/buenos_aires UTC offset of -03:00.
- America/Cambridge_Bay - Timezone for america/cambridge_bay UTC offset of -07:00.
- America/Campo_Grande - Timezone for america/campo_grande UTC offset of -03:00.
- America/Cancun - Timezone for america/cancun UTC offset of -05:00.
- America/Caracas - Timezone for america/caracas UTC offset of -04:00.
- America/Catamarca - Timezone for america/catamarca UTC offset of -03:00.
- America/Cayenne - Timezone for america/cayenne UTC offset of -03:00.
- America/Cayman - Timezone for america/cayman UTC offset of -05:00.
- America/Chicago - Timezone for america/chicago UTC offset of -06:00.
- America/Chihuahua - Timezone for america/chihuahua UTC offset of -07:00.

- America/Coral_Harbour - Timezone for america/coral_harbour UTC offset of -05:00.
- America/Cordoba - Timezone for america/cordoba UTC offset of -03:00.
- America/Costa_Rica - Timezone for america/costa_rica UTC offset of -06:00.
- America/Creston - Timezone for america/creston UTC offset of -07:00.
- America/Cuiaba - Timezone for america/cuiaba UTC offset of -03:00.
- America/Curacao - Timezone for america/curacao UTC offset of -04:00.
- America/Danmarkshavn - Timezone for america/danmarkshavn UTC offset of +00:00.
- America/Dawson - Timezone for america/dawson UTC offset of -08:00.
- America/Dawson_Creek - Timezone for america/dawson_creek UTC offset of -07:00.
- America/Denver - Timezone for america/denver UTC offset of -07:00.
- America/Detroit - Timezone for america/detroit UTC offset of -05:00.
- America/Dominica - Timezone for america/dominica UTC offset of -04:00.
- America/Edmonton - Timezone for america/edmonton UTC offset of -07:00.
- America/Eirunepe - Timezone for america/eirunepe UTC offset of -05:00.
- America/El_Salvador - Timezone for america/el_salvador UTC offset of -06:00.
- America/Ensenada - Timezone for america/ensenada UTC offset of -08:00.
- America/Fort_Nelson - Timezone for america/fort_nelson UTC offset of -07:00.
- America/Fort_Wayne - Timezone for america/fort_wayne UTC offset of -05:00.
- America/Fortaleza - Timezone for america/fortaleza UTC offset of -03:00.
- America/Glace_Bay - Timezone for america/glance_bay UTC offset of -04:00.
- America/Godthab - Timezone for america/godthab UTC offset of -03:00.
- America/Goose_Bay - Timezone for america/goose_bay UTC offset of -04:00.
- America/Grand_Turk - Timezone for america/grand_turk UTC offset of -04:00.
- America/Grenada - Timezone for america/grenada UTC offset of -04:00.
- America/Guadeloupe - Timezone for america/guadeloupe UTC offset of -04:00.
- America/Guatemala - Timezone for america/guatemala UTC offset of -06:00.
- America/Guayaquil - Timezone for america/guayaquil UTC offset of -05:00.
- America/Guyana - Timezone for america/guyana UTC offset of -04:00.
- America/Halifax - Timezone for america/halifax UTC offset of -04:00.
- America/Havana - Timezone for america/havana UTC offset of -05:00.
- America/Hermosillo - Timezone for america/hermosillo UTC offset of -07:00.

- America/Indiana/Indianapolis - Timezone for america/indiana/indianapolis UTC offset of -05:00.
- America/Indiana/Knox - Timezone for america/indiana/knox UTC offset of -06:00.
- America/Indiana/Marengo - Timezone for america/indiana/marengo UTC offset of -05:00.
- America/Indiana/Petersburg - Timezone for america/indiana/petersburg UTC offset of -05:00.
- America/Indiana/Tell_City - Timezone for america/indiana/tell_city UTC offset of -06:00.
- America/Indiana/Vevay - Timezone for america/indiana/vevay UTC offset of -05:00.
- America/Indiana/Vincennes - Timezone for america/indiana/vincennes UTC offset of -05:00.
- America/Indiana/Winamac - Timezone for america/indiana/winamac UTC offset of -05:00.
- America/Indianapolis - Timezone for america/indianapolis UTC offset of -05:00.
- America/Inuvik - Timezone for america/inuvik UTC offset of -07:00.
- America/Iqaluit - Timezone for america/iquait UTC offset of -05:00.
- America/Jamaica - Timezone for america/jamaica UTC offset of -05:00.
- America/Jujuy - Timezone for america/jujuy UTC offset of -03:00.
- America/Juneau - Timezone for america/juneau UTC offset of -09:00.
- America/Kentucky/Louisville - Timezone for america/kentucky/louisville UTC offset of -05:00.
- America/Kentucky/Monticello - Timezone for america/kentucky/monticello UTC offset of -05:00.
- America/Knox_IN - Timezone for america/knox_in UTC offset of -06:00.
- America/Kralendijk - Timezone for america/kralendijk UTC offset of -04:00.
- America/La_Paz - Timezone for america/la_paz UTC offset of -04:00.
- America/Lima - Timezone for america/lima UTC offset of -05:00.
- America/Los_Angeles - Timezone for america/los_angeles UTC offset of -08:00.
- America/Louisville - Timezone for america/louisville UTC offset of -05:00.
- America/Lower_Princes - Timezone for america/lower_princes UTC offset of -04:00.
- America/Maceio - Timezone for america/maceio UTC offset of -03:00.
- America/Managua - Timezone for america/managua UTC offset of -06:00.
- America/Manaus - Timezone for america/manaus UTC offset of -04:00.

- America/Marigot - Timezone for america/marigot UTC offset of -04:00.
- America/Martinique - Timezone for america/martinique UTC offset of -04:00.
- America/Matamoros - Timezone for america/matamoros UTC offset of -06:00.
- America/Mazatlan - Timezone for america/mazatlan UTC offset of -07:00.
- America/Mendoza - Timezone for america/mendoza UTC offset of -03:00.
- America/Menominee - Timezone for america/menominee UTC offset of -06:00.
- America/Merida - Timezone for america/merida UTC offset of -06:00.
- America/Metlakatla - Timezone for america/metlakatla UTC offset of -09:00.
- America/Mexico_City - Timezone for america/mexico_city UTC offset of -06:00.
- America/Miquelon - Timezone for america/miquelon UTC offset of -03:00.
- America/Moncton - Timezone for america/moncton UTC offset of -04:00.
- America/Monterrey - Timezone for america/monterrey UTC offset of -06:00.
- America/Montevideo - Timezone for america/montevideo UTC offset of -03:00.
- America/Montreal - Timezone for america/montreal UTC offset of -05:00.
- America/Montserrat - Timezone for america/montserrat UTC offset of -04:00.
- America/Nassau - Timezone for america/nassau UTC offset of -05:00.
- America/New_York - Timezone for america/new_york UTC offset of -05:00.
- America/Nipigon - Timezone for america/nipigon UTC offset of -05:00.
- America/Nome - Timezone for america/nome UTC offset of -09:00.
- America/Noronha - Timezone for america/noronha UTC offset of -02:00.
- America/North_Dakota/Beulah - Timezone for america/north_dakota/beulah UTC offset of -06:00.
- America/North_Dakota/Center - Timezone for america/north_dakota/center UTC offset of -06:00.
- America/North_Dakota/New_Salem - Timezone for america/north_dakota/new_salem UTC offset of -06:00.
- America/Ojinaga - Timezone for america/ojinaga UTC offset of -07:00.
- America/Panama - Timezone for america/panama UTC offset of -05:00.
- America/Pangnirtung - Timezone for america/pangnirtung UTC offset of -05:00.
- America/Paramaribo - Timezone for america/paramaribo UTC offset of -03:00.
- America/Phoenix - Timezone for america/phoenix UTC offset of -07:00.
- America/Port-au-Prince - Timezone for america/port-au-prince UTC offset of -05:00.

- America/Port_of_Spain - Timezone for america/port_of_spain UTC offset of -04:00.
- America/Porto_Acre - Timezone for america/porto_acre UTC offset of -05:00.
- America/Porto_Velho - Timezone for america/porto_velho UTC offset of -04:00.
- America/Puerto_Rico - Timezone for america/puerto_rico UTC offset of -04:00.
- America/Rainy_River - Timezone for america/rainy_river UTC offset of -06:00.
- America/Rankin_Inlet - Timezone for america/rankin_inlet UTC offset of -06:00.
- America/Recife - Timezone for america/recife UTC offset of -03:00.
- America/Regina - Timezone for america/regina UTC offset of -06:00.
- America/Resolute - Timezone for america/resolute UTC offset of -06:00.
- America/Rio_Branco - Timezone for america/rio_branco UTC offset of -05:00.
- America/Rosario - Timezone for america/rosario UTC offset of -03:00.
- America/Santa_Isabel - Timezone for america/santa_isabel UTC offset of -08:00.
- America/Santarem - Timezone for america/santarem UTC offset of -03:00.
- America/Santiago - Timezone for america/santiago UTC offset of -03:00.
- America/Santo_Domingo - Timezone for america/santo_domingo UTC offset of -04:00.
- America/Sao_Paulo - Timezone for america/sao_paulo UTC offset of -02:00.
- America/Scoresbysund - Timezone for america/scoresbysund UTC offset of -01:00.
- America/Shiprock - Timezone for america/shiprock UTC offset of -07:00.
- America/Sitka - Timezone for america/sitka UTC offset of -09:00.
- America/St_Barthelemy - Timezone for america/st_barthelemy UTC offset of -04:00.
- America/St_Johns - Timezone for america/st_johns UTC offset of -03:30.
- America/St_Kitts - Timezone for america/st_kitts UTC offset of -04:00.
- America/St_Lucia - Timezone for america/st_lucia UTC offset of -04:00.
- America/St_Thomas - Timezone for america/st_thomas UTC offset of -04:00.
- America/St_Vincent - Timezone for america/st_vincent UTC offset of -04:00.
- America/Swift_Current - Timezone for america/swift_current UTC offset of -06:00.
- America/Tegucigalpa - Timezone for america/tegucigalpa UTC offset of -06:00.
- America/Thule - Timezone for america/thule UTC offset of -04:00.
- America/Thunder_Bay - Timezone for america/thunder_bay UTC offset of -05:00.
- America/Tijuana - Timezone for america/tijuana UTC offset of -08:00.
- America/Toronto - Timezone for america/toronto UTC offset of -05:00.

- America/Tortola - Timezone for america/tortola UTC offset of -04:00.
- America/Vancouver - Timezone for america/vancouver UTC offset of -08:00.
- America/Virgin - Timezone for america/virgin UTC offset of -04:00.
- America/Whitehorse - Timezone for america/whitehorse UTC offset of -08:00.
- America/Winnipeg - Timezone for america/winnipeg UTC offset of -06:00.
- America/Yakutat - Timezone for america/yakutat UTC offset of -09:00.
- America/Yellowknife - Timezone for america/yellowknife UTC offset of -07:00.
- Antarctica/Casey - Timezone for antarctica/casey UTC offset of +11:00.
- Antarctica/Davis - Timezone for antarctica/davis UTC offset of +07:00.
- Antarctica/DumontDURville - Timezone for antarctica/dumontdurville UTC offset of +10:00.
- Antarctica/Macquarie - Timezone for antarctica/macquarie UTC offset of +11:00.
- Antarctica/Mawson - Timezone for antarctica/mawson UTC offset of +05:00.
- Antarctica/McMurdo - Timezone for antarctica/mcmurdo UTC offset of +13:00.
- Antarctica/Palmer - Timezone for antarctica/palmer UTC offset of -03:00.
- Antarctica/Rothera - Timezone for antarctica/rothera UTC offset of -03:00.
- Antarctica/South_Pole - Timezone for antarctica/south_pole UTC offset of +13:00.
- Antarctica/Syowa - Timezone for antarctica/syowa UTC offset of +03:00.
- Antarctica/Troll - Timezone for antarctica/troll UTC offset of +00:00.
- Antarctica/Vostok - Timezone for antarctica/vostok UTC offset of +06:00.
- Arctic/Longyearbyen - Timezone for arctic/longyearbyen UTC offset of +01:00.
- Asia/Aden - Timezone for asia/aden UTC offset of +03:00.
- Asia/Almaty - Timezone for asia/almaty UTC offset of +06:00.
- Asia/Amman - Timezone for asia/amman UTC offset of +02:00.
- Asia/Anadyr - Timezone for asia/anadyr UTC offset of +12:00.
- Asia/Aqtau - Timezone for asia/aqtau UTC offset of +05:00.
- Asia/Aqtobe - Timezone for asia/aqtobe UTC offset of +05:00.
- Asia/Ashgabat - Timezone for asia/ashgabat UTC offset of +05:00.
- Asia/Ashkhabad - Timezone for asia/ashkhabad UTC offset of +05:00.
- Asia/Baghdad - Timezone for asia/baghdad UTC offset of +03:00.
- Asia/Bahrain - Timezone for asia/bahrain UTC offset of +03:00.

- Asia/Baku - Timezone for asia/baku UTC offset of +04:00.
- Asia/Bangkok - Timezone for asia/bangkok UTC offset of +07:00.
- Asia/Barnaul - Timezone for asia/barnaul UTC offset of +07:00.
- Asia/Beirut - Timezone for asia/beirut UTC offset of +02:00.
- Asia/Bishkek - Timezone for asia/bishkek UTC offset of +06:00.
- Asia/Brunei - Timezone for asia/brunei UTC offset of +08:00.
- Asia/Calcutta - Timezone for asia/calcutta UTC offset of +05:30.
- Asia/Chita - Timezone for asia/chita UTC offset of +09:00.
- Asia/Choibalsan - Timezone for asia/choibalsan UTC offset of +08:00.
- Asia/Chongqing - Timezone for asia/chongqing UTC offset of +08:00.
- Asia/Chungking - Timezone for asia/chungking UTC offset of +08:00.
- Asia/Colombo - Timezone for asia/colombo UTC offset of +05:30.
- Asia/Dacca - Timezone for asia/dacca UTC offset of +06:00.
- Asia/Damascus - Timezone for asia/damascus UTC offset of +02:00.
- Asia/Dhaka - Timezone for asia/dhaka UTC offset of +06:00.
- Asia/Dili - Timezone for asia/dili UTC offset of +09:00.
- Asia/Dubai - Timezone for asia/dubai UTC offset of +04:00.
- Asia/Dushanbe - Timezone for asia/dushanbe UTC offset of +05:00.
- Asia/Gaza - Timezone for asia/gaza UTC offset of +02:00.
- Asia/Harbin - Timezone for asia/harbin UTC offset of +08:00.
- Asia/Hebron - Timezone for asia/hebron UTC offset of +02:00.
- Asia/Ho_Chi_Minh - Timezone for asia/ho_chi_minh UTC offset of +07:00.
- Asia/Hong_Kong - Timezone for asia/hong_kong UTC offset of +08:00.
- Asia/Hovd - Timezone for asia/hovd UTC offset of +07:00.
- Asia/Irkutsk - Timezone for asia/irkutsk UTC offset of +08:00.
- Asia/Istanbul - Timezone for asia/istanbul UTC offset of +03:00.
- Asia/Jakarta - Timezone for asia/jakarta UTC offset of +07:00.
- Asia/Jayapura - Timezone for asia/jayapura UTC offset of +09:00.
- Asia/Jerusalem - Timezone for asia/jerusalem UTC offset of +02:00.
- Asia/Kabul - Timezone for asia/kabul UTC offset of +04:30.
- Asia/Kamchatka - Timezone for asia/kamchatka UTC offset of +12:00.

- Asia/Karachi - Timezone for asia/karachi UTC offset of +05:00.
- Asia/Kashgar - Timezone for asia/kashgar UTC offset of +06:00.
- Asia/Kathmandu - Timezone for asia/kathmandu UTC offset of +05:45.
- Asia/Katmandu - Timezone for asia/katmandu UTC offset of +05:45.
- Asia/Khandyga - Timezone for asia/khandyga UTC offset of +09:00.
- Asia/Kolkata - Timezone for asia/kolkata UTC offset of +05:30.
- Asia/Krasnoyarsk - Timezone for asia/krasnoyarsk UTC offset of +07:00.
- Asia/Kuala_Lumpur - Timezone for asia/kuala_lumpur UTC offset of +08:00.
- Asia/Kuching - Timezone for asia/kuching UTC offset of +08:00.
- Asia/Kuwait - Timezone for asia/kuwait UTC offset of +03:00.
- Asia/Macao - Timezone for asia/macao UTC offset of +08:00.
- Asia/Macau - Timezone for asia/macau UTC offset of +08:00.
- Asia/Magadan - Timezone for asia/magadan UTC offset of +11:00.
- Asia/Makassar - Timezone for asia/makassar UTC offset of +08:00.
- Asia/Manila - Timezone for asia/manila UTC offset of +08:00.
- Asia/Muscat - Timezone for asia/muscat UTC offset of +04:00.
- Asia/Nicosia - Timezone for asia/nicosia UTC offset of +02:00.
- Asia/Novokuznetsk - Timezone for asia/novokuznetsk UTC offset of +07:00.
- Asia/Novosibirsk - Timezone for asia/novosibirsk UTC offset of +07:00.
- Asia/Omsk - Timezone for asia/omsk UTC offset of +06:00.
- Asia/Oral - Timezone for asia/oral UTC offset of +05:00.
- Asia/Phnom_Penh - Timezone for asia/phnom_penh UTC offset of +07:00.
- Asia/Pontianak - Timezone for asia/pontianak UTC offset of +07:00.
- Asia/Pyongyang - Timezone for asia/pyongyang UTC offset of +08:30.
- Asia/Qatar - Timezone for asia/qatar UTC offset of +03:00.
- Asia/Qyzylorda - Timezone for asia/qyzylorda UTC offset of +06:00.
- Asia/Rangoon - Timezone for asia/rangoon UTC offset of +06:30.
- Asia/Riyadh - Timezone for asia/riyadh UTC offset of +03:00.
- Asia/Saigon - Timezone for asia/saigon UTC offset of +07:00.
- Asia/Sakhalin - Timezone for asia/sakhalin UTC offset of +11:00.
- Asia/Samarkand - Timezone for asia/samarkand UTC offset of +05:00.

- Asia/Seoul - Timezone for asia/seoul UTC offset of +09:00.
- Asia/Shanghai - Timezone for asia/shanghai UTC offset of +08:00.
- Asia/Singapore - Timezone for asia/singapore UTC offset of +08:00.
- Asia/Srednekolymsk - Timezone for asia/srednekolymsk UTC offset of +11:00.
- Asia/Taipei - Timezone for asia/taipei UTC offset of +08:00.
- Asia/Tashkent - Timezone for asia/tashkent UTC offset of +05:00.
- Asia/Tbilisi - Timezone for asia/tbilisi UTC offset of +04:00.
- Asia/Tehran - Timezone for asia/tehran UTC offset of +03:30.
- Asia/Tel_Aviv - Timezone for asia/tel_aviv UTC offset of +02:00.
- Asia/Thimbu - Timezone for asia/thimbu UTC offset of +06:00.
- Asia/Thimphu - Timezone for asia/thimphu UTC offset of +06:00.
- Asia/Tokyo - Timezone for asia/tokyo UTC offset of +09:00.
- Asia/Tomsk - Timezone for asia/tomsk UTC offset of +07:00.
- Asia/Ujung_Pandang - Timezone for asia/ujung_pandang UTC offset of +08:00.
- Asia/Ulaanbaatar - Timezone for asia/ulaanbaatar UTC offset of +08:00.
- Asia/Ulan_Bator - Timezone for asia/ulan_bator UTC offset of +08:00.
- Asia/Urumqi - Timezone for asia/urumqi UTC offset of +06:00.
- Asia/Ust-Nera - Timezone for asia/ust-nera UTC offset of +10:00.
- Asia/Vientiane - Timezone for asia/vientiane UTC offset of +07:00.
- Asia/Vladivostok - Timezone for asia/vladivostok UTC offset of +10:00.
- Asia/Yakutsk - Timezone for asia/yakutsk UTC offset of +09:00.
- Asia/Yekaterinburg - Timezone for asia/yekaterinburg UTC offset of +05:00.
- Asia/Yerevan - Timezone for asia/yerevan UTC offset of +04:00.
- Atlantic/Azores - Timezone for atlantic/azores UTC offset of -01:00.
- Atlantic/Bermuda - Timezone for atlantic/bermuda UTC offset of -04:00.
- Atlantic/Canary - Timezone for atlantic/canary UTC offset of +00:00.
- Atlantic/Cape_Verde - Timezone for atlantic/cape_verde UTC offset of -01:00.
- Atlantic/Faeroe - Timezone for atlantic/faeroe UTC offset of +00:00.
- Atlantic/Faroe - Timezone for atlantic/faroe UTC offset of +00:00.
- Atlantic/Jan_Mayen - Timezone for atlantic/jan_mayen UTC offset of +01:00.
- Atlantic/Madeira - Timezone for atlantic/madeira UTC offset of +00:00.

- Atlantic/Reykjavik - Timezone for atlantic/reykjavik UTC offset of +00:00.
- Atlantic/South_Georgia - Timezone for atlantic/south_georgia UTC offset of -02:00.
- Atlantic/St_Helena - Timezone for atlantic/st_helena UTC offset of +00:00.
- Atlantic/Stanley - Timezone for atlantic/stanley UTC offset of -03:00.
- Australia/ACT - Timezone for australia/act UTC offset of +11:00.
- Australia/Adelaide - Timezone for australia/adelaide UTC offset of +10:30.
- Australia/Brisbane - Timezone for australia/brisbane UTC offset of +10:00.
- Australia/Broken_Hill - Timezone for australia/broken_hill UTC offset of +10:30.
- Australia/Canberra - Timezone for australia/canberra UTC offset of +11:00.
- Australia/Currie - Timezone for australia/currie UTC offset of +11:00.
- Australia/Darwin - Timezone for australia/darwin UTC offset of +09:30.
- Australia/Eucla - Timezone for australia/eucla UTC offset of +08:45.
- Australia/Hobart - Timezone for australia/hobart UTC offset of +11:00.
- Australia/LHI - Timezone for australia/lhi UTC offset of +11:00.
- Australia/Lindeman - Timezone for australia/lindeman UTC offset of +10:00.
- Australia/Lord_Howe - Timezone for australia/lord_howe UTC offset of +11:00.
- Australia/Melbourne - Timezone for australia/melbourne UTC offset of +11:00.
- Australia/NSW - Timezone for australia/nsw UTC offset of +11:00.
- Australia/North - Timezone for australia/north UTC offset of +09:30.
- Australia/Perth - Timezone for australia/perth UTC offset of +08:00.
- Australia/Queensland - Timezone for australia/queensland UTC offset of +10:00.
- Australia/South - Timezone for australia/south UTC offset of +10:30.
- Australia/Sydney - Timezone for australia/sydney UTC offset of +11:00.
- Australia/Tasmania - Timezone for australia/tasmania UTC offset of +11:00.
- Australia/Victoria - Timezone for australia/victoria UTC offset of +11:00.
- Australia/West - Timezone for australia/west UTC offset of +08:00.
- Australia/Yancowinna - Timezone for australia/yancowinna UTC offset of +10:30.
- CET - Timezone for CET UTC offset of +01:00.
- CST6CDT - Timezone for CST6CDT UTC offset of -06:00.
- Canada/Atlantic - Timezone for canada/atlantic UTC offset of -04:00.
- Canada/Central - Timezone for canada/central UTC offset of -06:00.

- Canada/East-Saskatchewan - Timezone for canada/east-saskatchewan UTC offset of -06:00.
- Canada/Eastern - Timezone for canada/eastern UTC offset of -05:00.
- Canada/Mountain - Timezone for canada/mountain UTC offset of -07:00.
- Canada/Newfoundland - Timezone for canada/newfoundland UTC offset of -03:30.
- Canada/Pacific - Timezone for canada/pacific UTC offset of -08:00.
- Canada/Saskatchewan - Timezone for canada/saskatchewan UTC offset of -06:00.
- Canada/Yukon - Timezone for canada/yukon UTC offset of -08:00.
- Chile/Continental - Timezone for chile/continental UTC offset of -03:00.
- Chile/EasterIsland - Timezone for chile/easterisland UTC offset of -05:00.
- Cuba - Timezone for cuba UTC offset of -05:00.
- Default - Timezone for system UTC (OS time zone)
- EET - Timezone for EET UTC offset of +02:00.
- EST - Timezone for EST UTC offset of -05:00.
- EST5EDT - Timezone for EST5EDT UTC offset of -05:00.
- Egypt - Timezone for egypt UTC offset of +02:00.
- Eire - Timezone for eire UTC offset of +00:00.
- Etc/Greenwich - Timezone for etc/greenwich UTC offset of +00:00.
- Etc/UCT - Timezone for etc/uct UTC offset of +00:00.
- Etc/UTC - Timezone for etc/utc UTC offset of +00:00.
- Etc/Universal - Timezone for etc/universal UTC offset of +00:00.
- Etc/Zulu - Timezone for etc/zulu UTC offset of +00:00.
- Europe/Amsterdam - Timezone for europe/amsterdam UTC offset of +01:00.
- Europe/Andorra - Timezone for europe/andorra UTC offset of +01:00.
- Europe/Astrakhan - Timezone for europe/astrakhan UTC offset of +04:00.
- Europe/Athens - Timezone for europe/athens UTC offset of +02:00.
- Europe/Belfast - Timezone for europe/belfast UTC offset of +00:00.
- Europe/Belgrade - Timezone for europe/belgrade UTC offset of +01:00.
- Europe/Berlin - Timezone for europe/berlin UTC offset of +01:00.
- Europe/Bratislava - Timezone for europe/bratislava UTC offset of +01:00.
- Europe/Brussels - Timezone for europe/brussels UTC offset of +01:00.

- Europe/Bucharest - Timezone for europe/bucharest UTC offset of +02:00.
- Europe/Budapest - Timezone for europe/budapest UTC offset of +01:00.
- Europe/Busingen - Timezone for europe/busingen UTC offset of +01:00.
- Europe/Chisinau - Timezone for europe/chisinau UTC offset of +02:00.
- Europe/Copenhagen - Timezone for europe/copenhagen UTC offset of +01:00.
- Europe/Dublin - Timezone for europe/dublin UTC offset of +00:00.
- Europe/Gibraltar - Timezone for europe/gibraltar UTC offset of +01:00.
- Europe/Guernsey - Timezone for europe/guernsey UTC offset of +00:00.
- Europe/Helsinki - Timezone for europe/helsinki UTC offset of +02:00.
- Europe/Isle_of_Man - Timezone for europe/isle_of_man UTC offset of +00:00.
- Europe/Istanbul - Timezone for europe/istanbul UTC offset of +03:00.
- Europe/Jersey - Timezone for europe/jersey UTC offset of +00:00.
- Europe/Kaliningrad - Timezone for europe/kaliningrad UTC offset of +02:00.
- Europe/Kiev - Timezone for europe/kyiv UTC offset of +02:00.
- Europe/Kirov - Timezone for europe/kirov UTC offset of +03:00.
- Europe/Lisbon - Timezone for europe/lisbon UTC offset of +00:00.
- Europe/Ljubljana - Timezone for europe/ljubljana UTC offset of +01:00.
- Europe/London - Timezone for europe/london UTC offset of +00:00.
- Europe/Luxembourg - Timezone for europe/luxembourg UTC offset of +01:00.
- Europe/Madrid - Timezone for europe/madrid UTC offset of +01:00.
- Europe/Malta - Timezone for europe/malta UTC offset of +01:00.
- Europe/Mariehamn - Timezone for europe/mariehamn UTC offset of +02:00.
- Europe/Minsk - Timezone for europe/minsk UTC offset of +03:00.
- Europe/Monaco - Timezone for europe/monaco UTC offset of +01:00.
- Europe/Moscow - Timezone for europe/moscow UTC offset of +03:00.
- Europe/Nicosia - Timezone for europe/nicosia UTC offset of +02:00.
- Europe/Oslo - Timezone for europe/oslo UTC offset of +01:00.
- Europe/Paris - Timezone for europe/paris UTC offset of +01:00.
- Europe/Podgorica - Timezone for europe/podgorica UTC offset of +01:00.
- Europe/Prague - Timezone for europe/prague UTC offset of +01:00.
- Europe/Riga - Timezone for europe/riga UTC offset of +02:00.

- Europe/Rome - Timezone for europe/rome UTC offset of +01:00.
- Europe/Samara - Timezone for europe/samara UTC offset of +04:00.
- Europe/San_Marino - Timezone for europe/san_marino UTC offset of +01:00.
- Europe/Sarajevo - Timezone for europe/sarajevo UTC offset of +01:00.
- Europe/Simferopol - Timezone for europe/simferopol UTC offset of +03:00.
- Europe/Skopje - Timezone for europe/skopje UTC offset of +01:00.
- Europe/Sofia - Timezone for europe/sofia UTC offset of +02:00.
- Europe/Stockholm - Timezone for europe/stockholm UTC offset of +01:00.
- Europe/Tallinn - Timezone for europe/tallinn UTC offset of +02:00.
- Europe/Tirane - Timezone for europe/tirane UTC offset of +01:00.
- Europe/Tiraspol - Timezone for europe/tiraspol UTC offset of +02:00.
- Europe/Ulyanovsk - Timezone for europe/ulyanovsk UTC offset of +04:00.
- Europe/Uzhgorod - Timezone for europe/uzhgorod UTC offset of +02:00.
- Europe/Vaduz - Timezone for europe/vaduz UTC offset of +01:00.
- Europe/Vatican - Timezone for europe/vatican UTC offset of +01:00.
- Europe/Vienna - Timezone for europe/vienna UTC offset of +01:00.
- Europe/Vilnius - Timezone for europe/vilnius UTC offset of +02:00.
- Europe/Volgograd - Timezone for europe/volgograd UTC offset of +03:00.
- Europe/Warsaw - Timezone for europe/warsaw UTC offset of +01:00.
- Europe/Zagreb - Timezone for europe/zagreb UTC offset of +01:00.
- Europe/Zaporozhye - Timezone for europe/zaporozhye UTC offset of +02:00.
- Europe/Zurich - Timezone for europe/zurich UTC offset of +01:00.
- GB - Timezone for GB UTC offset of +00:00.
- GB-Eire - Timezone for GB-Eire UTC offset of +00:00.
- GMT - Timezone for GMT UTC offset of +00:00.
- GMT0 - Timezone for GMT0 UTC offset of +00:00.
- Greenwich - Timezone for greenwich UTC offset of +00:00.
- HST - Timezone for HST UTC offset of -10:00.
- Hongkong - Timezone for hongkong UTC offset of +08:00.
- Iceland - Timezone for iceland UTC offset of +00:00.
- Indian/Antananarivo - Timezone for indian/antananarivo UTC offset of +03:00.

- Indian/Chagos - Timezone for indian/chagos UTC offset of +06:00.
- Indian/Christmas - Timezone for indian/christmas UTC offset of +07:00.
- Indian/Cocos - Timezone for indian/cocos UTC offset of +06:30.
- Indian/Comoro - Timezone for indian/comoro UTC offset of +03:00.
- Indian/Kerguelen - Timezone for indian/kerguelen UTC offset of +05:00.
- Indian/Mahe - Timezone for indian/mahe UTC offset of +04:00.
- Indian/Maldives - Timezone for indian/maldives UTC offset of +05:00.
- Indian/Mauritius - Timezone for indian/mauritius UTC offset of +04:00.
- Indian/Mayotte - Timezone for indian/mayotte UTC offset of +03:00.
- Indian/Reunion - Timezone for indian/reunion UTC offset of +04:00.
- Iran - Timezone for iran UTC offset of +03:30.
- Israel - Timezone for israel UTC offset of +02:00.
- Jamaica - Timezone for jamaica UTC offset of -05:00.
- Japan - Timezone for japan UTC offset of +09:00.
- Kwajalein - Timezone for kwajalein UTC offset of +12:00.
- Libya - Timezone for libya UTC offset of +02:00.
- MET - Timezone for MET UTC offset of +01:00.
- MST - Timezone for MST UTC offset of -07:00.
- MST7MDT - Timezone for MST7MDT UTC offset of -07:00.
- Mexico/BajaNorte - Timezone for mexico/bajanorte UTC offset of -08:00.
- Mexico/BajaSur - Timezone for mexico/bajasur UTC offset of -07:00.
- Mexico/General - Timezone for mexico/general UTC offset of -06:00.
- NZ - Timezone for NZ UTC offset of +13:00.
- NZ-CHAT - Timezone for NZ-CHAT UTC offset of +13:45.
- Navajo - Timezone for navajo UTC offset of -07:00.
- PRC - Timezone for PRC UTC offset of +08:00.
- PST8PDT - Timezone for PST8PDT UTC offset of -08:00.
- Pacific/Apia - Timezone for pacific/apia UTC offset of +14:00.
- Pacific/Auckland - Timezone for pacific/auckland UTC offset of +13:00.
- Pacific/Bougainville - Timezone for pacific/bougainville UTC offset of +11:00.
- Pacific/Chatham - Timezone for pacific/chatham UTC offset of +13:45.

- Pacific/Chuuk - Timezone for pacific/chuuk UTC offset of +10:00.
- Pacific/Easter - Timezone for pacific/easter UTC offset of -05:00.
- Pacific/Efate - Timezone for pacific/efate UTC offset of +11:00.
- Pacific/Enderbury - Timezone for pacific/enderbury UTC offset of +13:00.
- Pacific/Fakaofo - Timezone for pacific/fakaofo UTC offset of +13:00.
- Pacific/Fiji - Timezone for pacific/fiji UTC offset of +13:00.
- Pacific/Funafuti - Timezone for pacific/funafuti UTC offset of +12:00.
- Pacific/Galapagos - Timezone for pacific/galapagos UTC offset of -06:00.
- Pacific/Gambier - Timezone for pacific/gambier UTC offset of -09:00.
- Pacific/Guadalcanal - Timezone for pacific/guadalcanal UTC offset of +11:00.
- Pacific/Guam - Timezone for pacific/guam UTC offset of +10:00.
- Pacific/Honolulu - Timezone for pacific/honolulu UTC offset of -10:00.
- Pacific/Johnston - Timezone for pacific/johnston UTC offset of -10:00.
- Pacific/Kiritimati - Timezone for pacific/kiritimati UTC offset of +14:00.
- Pacific/Kosrae - Timezone for pacific/kosrae UTC offset of +11:00.
- Pacific/Kwajalein - Timezone for pacific/kwajalein UTC offset of +12:00.
- Pacific/Majuro - Timezone for pacific/majuro UTC offset of +12:00.
- Pacific/Marquesas - Timezone for pacific/marquesas UTC offset of -09:30.
- Pacific/Midway - Timezone for pacific/midway UTC offset of -11:00.
- Pacific/Nauru - Timezone for pacific/nauru UTC offset of +12:00.
- Pacific/Niue - Timezone for pacific/niue UTC offset of -11:00.
- Pacific/Norfolk - Timezone for pacific/norfolk UTC offset of +11:00.
- Pacific/Noumea - Timezone for pacific/noumea UTC offset of +11:00.
- Pacific/Pago_Pago - Timezone for pacific/pago_pago UTC offset of -11:00.
- Pacific/Palau - Timezone for pacific/palau UTC offset of +09:00.
- Pacific/Pitcairn - Timezone for pacific/pitcairn UTC offset of -08:00.
- Pacific/Pohnpei - Timezone for pacific/pohnpei UTC offset of +11:00.
- Pacific/Ponape - Timezone for pacific/ponape UTC offset of +11:00.
- Pacific/Port_Moresby - Timezone for pacific/port_moresby UTC offset of +10:00.
- Pacific/Rarotonga - Timezone for pacific/rarotonga UTC offset of -10:00.
- Pacific/Saipan - Timezone for pacific/saipan UTC offset of +10:00.

- Pacific/Samoa - Timezone for pacific/samoa UTC offset of -11:00.
- Pacific/Tahiti - Timezone for pacific/tahiti UTC offset of -10:00.
- Pacific/Tarawa - Timezone for pacific/tarawa UTC offset of +12:00.
- Pacific/Tongatapu - Timezone for pacific/tongatapu UTC offset of +14:00.
- Pacific/Truk - Timezone for pacific/truk UTC offset of +10:00.
- Pacific/Wake - Timezone for pacific/wake UTC offset of +12:00.
- Pacific/Wallis - Timezone for pacific/wallis UTC offset of +12:00.
- Pacific/Yap - Timezone for pacific/yap UTC offset of +10:00.
- Poland - Timezone for poland UTC offset of +01:00.
- Portugal - Timezone for portugal UTC offset of +00:00.
- ROC - Timezone for ROC UTC offset of +08:00.
- ROK - Timezone for ROK UTC offset of +09:00.
- Singapore - Timezone for singapore UTC offset of +08:00.
- Turkey - Timezone for turkey UTC offset of +03:00.
- UCT - Timezone for UCT UTC offset of +00:00.
- US/Pacific-New - Timezone for US/Pacific-New UTC offset of -08:00.
- Universal - Timezone for universal UTC offset of +00:00.
- W-SU - Timezone for W-SU UTC offset of +03:00.
- WET - Timezone for WET UTC offset of +00:00.
- Zulu - Timezone for zulu UTC offset of +00:00.
- jdj_tz - Timezone for jdj_tz UTC offset of +00:00.

Default

Default

Name

upCloT - UP CloT

Syntax

--upCIoT <list-item>

Description

Enable User Plane Cellular Internet of Things Optimization

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

vaEsmIcIdentity1 - Value Added ESMC identifier 1

Syntax

--vaEsmIcIdentity1 <digits>

Description

Indicates the First TAI to Value Added E-SMC (Evolved serving mobile location center) mapping.

Class

Optional.

Type / Range

Digit string of length 0 to 3

Default

"

Name

vaEsmIcIdentity2 - Value Added ESMC identifier 2

Syntax

--vaEsmIcIdentity2 <digits>

Description

Indicates the Second TAI to Value Added E-SMC (Evolved serving mobile location center) mapping.

Class

Optional.

Type / Range

Digit string of length 0 to 3

Default

"

Name

vaSelectionAlgorithm - Value added location services selection algorithm

Syntax

--vaSelectionAlgorithm <list-item>

Description

Indicates for the set of value added location services ESMLCs (Evolved serving mobile location center) whether they will load share or function as primary/secondary.

Class

Optional.

List

- LoadShare - Load share - Load share.
- PrimarySecondary - Primary / secondary - Function as primary / secondary.

Default

PrimarySecondary

Name

vdpProfileName - VDP profile name

Syntax

--vdpProfileName <string>

Description

Voice domain preference profile name.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

"

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.108 cmm taiRangeList

TAI Range List. Syntax, description, parameters, and roles for cmm taiRangeList command.

Syntax

Description

This command administers a list of TAIs. The list is specified via TAI ranges. Up to 16 ranges can be specified. Corresponding TAIs need not be provisioned in the tai translation.

Parameters

Name

servedPlmnName - Served mobile network name

Syntax

--servedPlmnName <string>

Description

Served mobile network name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

areaListName - Name of TAI Range List

Syntax

--areaListName <string>

Description

Name of TAI Range List

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

minTac - Min Tracking Area Code

Syntax

--minTac <integer>

Description

The starting Tracking Area Code of a TA range

Class

Required key.

Type / Range

Integer in the range of 0 to 65,535

Name

maxTac - Max Tracking Area Code

Syntax

--maxTac <integer>

Description

Ending TAC of a TA Range

Class

Required key.

Type / Range

Integer in the range of 0 to 65,535

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.109 cmm taisOfSnssaiQuery

TAs of S-NSSAI. Syntax, description, parameters, and roles for cmm taisOfSnssaiQuery command.

Syntax**Description**

Lists all TAs that based on NG-RAN information can serve a specific S-NSSAI. Slice Service Type is a required input filter. If not given query will return empty list. Slice Differentiator is optional input. If not

given query will return TAs where S-NSSAI SST matches the given Slice Service Type and have any SD (including SST with no SD)

Parameters

Name

sliceServiceType - Slice Service Type

Syntax

--sliceServiceType <integer>

Description

The S-NSSAI consists of a Slice Service Type (SST) and a Slice Differentiator (SD). Each slice must have SST and SD is optional.

Class

Required key.

Type / Range

5G Slice Service Type

Name

assignMode - Assignment Mode

Syntax

--assignMode <integer>

Description

This field depicts the assignment mode for the SNPN

Class

Optional.

Type / Range

Integer in the range of 0 to 2

Default

0

Name

nidValue - Nid value

Syntax

--nidValue <integer>

Description

This field depicts the Nid value of the NID for the SNPN

Class

Optional.

Type / Range

Integer in the Range 0 to 1099511627775

Default

0

Name

sliceDifferentiator - Slice Differentiator

Syntax

--sliceDifferentiator <string>

Description

The S-NSSAI consists of a Slice Service Type (SST) and a Slice Differentiator (SD). Each slice must have SST and SD is optional. A SST may have multiple SDs.

Class

Read only.

Type / Range

Identifies Slice Differentiator (SD)

Default

FFFFFF

Name

tac - Tracking Area Code

Syntax

--tac <integer>

Description

5G TAC. 5G TAC is 3 octets long.

Class

Read only.

Type / Range

Integer in the range of 0 to 16777215

Name

taiMcc - TAI MCC

Syntax

--taiMcc <digits>

Description

3-digit number used to represent the mobile country code for TAI.

Class

Read only.

Type / Range

Digit string of length 3

Name

taiMnc - TAI MNC

Syntax

--taiMnc <digits>

Description

Digits used to represent mobile network code for TAI.

Class

Read only.

Type / Range

Digit string of length 2 to 3

Roles**List**

sysadmin, manage, monitor

7.110 cmm taiToLaiImsiRangeServices

TAI to LAI IMSI Range Service. Syntax, description, parameters, and roles for cmm taiToLaiImsiRangeServices command.

Syntax**Description**

This command is used to administer TAI to LAI Mapping with IMSI range as the key.

Parameters**Name**

uePlmnServicesName - UE PLMN Services Name

Syntax

--uePlmnServicesName <string>

Description

Name that identifies the assigned network.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

entryId - Entry ID

Syntax

--entryId <integer>

Description

Entry ID

Class

Required key.

Type / Range

Integer in the range of 1 to 10,000

Name

lac - Location Area Code

Syntax

--lac <integer>

Description

Location Area Code

Class

Required on create.

Type / Range

Integer in the range of 1 to 65,535

Name

laiPlmnName - Mobile network name of LAI

Syntax

--laiPlmnName <string>

Description

Name that identifies mobile network (alternate key) of the location area in this mapping.

Class

Required on create.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

maxMsin - Maximum MSIN value

Syntax

--maxMsin <digits>

Description

The maximum IMSI MSIN value

Class

Optional.

Type / Range

Digit string of length 0 or 8 to 10

Default

""

Name

minMsin - Minimum MSIN

Syntax

--minMsin <digits>

Description

The minimum IMSI MSIN value.

Class

Optional.

Type / Range

Digit string of length 0 or 8 to 10

Default

""

Name

taiAreaListName - Tracking Area Identity List

Syntax

--taiAreaListName <string>

Description

Tracking Area Identity List

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.111 cmm taiZoneMapping

TAI Zone Mapping. Syntax, description, parameters, and roles for `cmm taiZoneMapping` command.

Syntax

Description

Command maps zone name with a given TAC. It is used to compare with current location of UEs. If current location of UE (TAC) matches with a zone's name then the zone name will be used as a tag with the DNS query.

Parameters

Name

taiZoneMappingListName - TAI Zone Mapping List Name

Syntax`--taiZoneMappingListName <string>`**Description**

TAI Zone Mapping List Name

Class

Required key.

Type / RangeDigit string of length 1 to 31 with the allowed pattern of `[0-9a-zA-Z\.\_:@]*`**Name**

plmnName - PLMN Name

Syntax`--plmnName <string>`**Description**

Identifies the name of the PLMN which can be Home or Roamer PLMN mobile network name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

tac - Tracking Area Code

Syntax

--tac <integer>

Description

Tracking Area Code

Restrictions

Only one to one association between TAC and ZoneName is allowed.

Class

Required key.

Type / Range

Integer in the range of 1 to 65,535

Name

zoneName - Zone Name

Syntax

--zoneName <string>

Description

Name that identifies a Zone.

Class

Required on create.

Type / Range

Digit string of length 1 to 32 with the allowed pattern of [A-Za-z0-9.-]* A Zone name can not start or end with hyphen (-) or period (.)

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.112 cmm taiZoneMappingList

TAI Zone Mapping List. Syntax, description, parameters, and roles for cmm taiZoneMappingList command.

Syntax

Description

TAI Zone Mapping List

Parameters

Name

name - Name

Syntax

--name <string>

Description

Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.113 cmm tcpProfile

TCP Profile. Syntax, description, parameters, and roles for `cmm tcpProfile` command.

Syntax

Description

This command is used to administer a TCP Profile and includes parameters for defining the profile name and keep-alive timer, interval, and maximum number of re-transmissions. A default profile, named 'default', is provided.

Parameters

Name

name - TCP profile name

Syntax

--name <string>

Description

TCP profile name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

keepAliveIdleTimer - Keep alive idle timer

Syntax

--keepAliveIdleTimer <integer>

Description

Keep alive idle timer in units of seconds.

Class

Optional.

Type / Range

Integer in the range of 1 to 300 in units of Second

Default

30

Name

keepAliveInterval - Keep alive interval

Syntax

--keepAliveInterval <integer>

Description

Specify keep alive interval in seconds.

Class

Optional.

Type / Range

Integer in the range of 1 to 3,600 in units of Second

Default

30

Name

keepAliveRetry - Keep alive retry

Syntax

--keepAliveRetry <integer>

Description

Maximum number of keep alive re-transmissions.

Class

Optional.

Type / Range

Integer in the range of 1 to 100

Default

10

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.114 cmm timer

MME Timers. Syntax, description, parameters, and roles for `cmm timer` command.

Syntax

Description

Administer MME Timers

Parameters

Name

timerName - Timer name

Syntax

--timerName <list-item>

Description

Name of timer.

Class

Required key.

List

- alertMessageHoldingTime - Alert message holding time - The time in seconds that a SBC CMAS message will be queued for while waiting for a previously received SBC CMAS message to finish processing.
- connectionManagementAnswer - Connection management answer - Configure timer to wait for connection management answer message after sending connection management request. this timer is also used to respond CMA to CMR received in the case of CMR release request from SCEF.
- controlPlaneUserInactivity - Control plane user inactivity - This timer is used to detect user inactivity and release S1 connection of UE that have user data over NAS only. The cmm subscriber show command may show NB UEs as IDLE if the UE undergoing CP user inactivity timing. This can occur if the inactivity timer has been running for less than 20 seconds when the command is entered. The MME controlPlaneUserInactivity timer is recommended to have a value greater than the the eNB inactivity timers.
- cpiotUndeliveredNasWait - This timer defines the time that the MME will wait for NAS non Delivery Indication msg(s) from source eNB as part of eNB CP Relocation Indication procedure
- cppsRemHeartbeat - CPPS REM heartbeat - Configure REM heartbeat interval for CPPS service. VNF: Default is 200, Range is 200 to 20000 CNF: Default is 600, Range is 600 to 20000

- **critDnsAlarmSustainedInterval** - Critical DNS Alarm Sustained Internal - The interval of time the critical DNS alarm threshold conditions shall be exceeded and sustained before the 42014 CommunicationDegradationOrLost alarm will be raised.
- **csOpsBearerDel** - CS HO procedure delete bearer timer - Used in an EUTRAN to GERAN CS handover procedure to time for the release of UE resources.
- **dbsRemHeartbeat** - DBS REM heartbeat - Configure REM heartbeat interval for DBS service. VNF: Default is 200, Range is 200 to 20000 CNF: Default is 600, Range is 600 to 20000
- **ddnDlBufExtEdrx** - DDN DL Buffer Extend for eDRX - Specifies the amount that the DL Buffer Duration IE in the DDN Ack message should be extended for a UE in eDRX. If this field is set to 0, no additional time is added and the value continues to be set to when the UE will exit its next paging time window.
- **ddnDlBufExtPsm** - DDN DL Buffer Extend for PSM - Specifies the amount that the DL Buffer Duration IE in the DDN Ack message should be extended for a UE in power saving mode. If this field is set to 0, no additional time is added and the value continues to be set to the expected time until the UE performs a TAU.
- **delBearerAtSrcSgw** - Delete bearer at source SGW timer - This timer starts when the MME receives a create session response from the target SGW, and the UE bearers are removed at the source SGW when the timer expires. a value of 0 indicates that the UE bearers at the source SGW are deleted once the bearers are set up at the target SGW.
- **deleteGbr** - Delete guaranteed bit rate - This command is used to configure how long to keep GBR bearers after the completion of an abnormal release (i.e. S1 release cause is other than user inactivity or redirection to UTRAN/GERAN or 1xrtt) of S1 connection of a UE. the no form of the command sets the timer value to zero.
- **dnsApplicationTimer** - DNS application timer - When a DNS query is made, a request is sent to the DNS resolver. the resolver either finds the result in a local cache, or makes a request of a local DNS server. when this occurs, the resolver starts a two-second timer for getting a result. it either returns the result, or no result is returned if the two-second timer expires. this timer allows for changing the 2 second default for proceeding with call processing.
- **eemsRemHeartbeat** - EEMS REM Heartbeat - Configure REM heartbeat interval for EEMS service. VNF: Default is 200, Range is 200 to 20000 CNF: Default is 600, Range is 600 to 20000
- **emergencyMobileReachable** - Emergency mobile reachable - Emergency mobile reachable timer.
- **erabHoFailureTimer** - E-RAB Setup Failure For HO Operations - E-RAB setup failure for HO operations. Includes X2-handover-triggered and dS1-intra-system-handover-triggered.

- **eRabModify** - E-RAB modify - When the MME sends a message to the eNB related to a bearer request, it expects to receive a response. if the timer expires while waiting for this response, the MME considers that this class 1 procedure failed.
- **eRabRelease** - E-RAB release - When the MME sends a message to the eNB related to a bearer request, it expects to receive a response. if the timer expires while waiting for this response, the MME considers that this class 1 procedure failed.
- **eRabSetup** - E-RAB setup - When the MME sends a message to the eNB related to a bearer request, it expects to receive a response. if the timer expires while waiting for this response, the MME considers that this class 1 procedure failed.
- **gutiStalenessTimer** - GUTI Staleness Timer - a GUTI Re-Allocation staleness timer. Timer value range from 0-24 hours. If it's set to 0 (by default), then no timer force GUTI re-allocation (this feature function) would take Place. It means that MME follows existing GUTI allocation function. If it's set to a non-zero value, at timer expiry, MME forces GUTI Re-Allocation at next UE event (e.g. Attach, TAU, Service request etc.).
- **gwtsDelayS1release** - GWTS delay S1 release - Configure delay S1 connection release timer for SDS UE.
- **gwtsTcpLinkFailureDeclaration** - GWTS tcp link failure declaration - Configure TCP link failure declaration timer.
- **handoverRequest** - Hand over request - When the MME sends a handover request message to the eNB, it expects to receive an acknowledgment response message from the eNB. if the timer expires while waiting for this response, the MME considers that this class 1 procedure failed.
- **ho2g3gDeletion** - HO 2G to 3G deletion - Used to handle the deletion of the UE bearers during a handover to a UTRAN or GERAN network. the timer is started when the MME receives the SGSN context request message from the new SGSN. when the timer expires, the MME deletes the UE bearers.
- **hoRcvCsSvcNotifyNasNonDeliveryRadioConnLost** - Handover Received CS Service Notify NAS Non-Delivery Radio Conn Lost - When MME receives NAS Non-Delivery Indication (cause 'radio connection with ue lost') as response to SGs Downlink service operation, it uses this timer to safeguard reception of X2 Ho Request, at expiration of which it will re-attempt sending downlink message
- **hoTauRequestWait** - HO TAU request wait - Configure how long to wait for TAU request after X2 HO or S1 HO.
- **hssUnavailBlacklisting** - HSS unavailable blacklisting - The time to apply extended HSS disabling/blacklisting.
- **hssUnavailFailedPercentCalc** - HSS unavailable failed percentage calculation - Time interval to calculate the failure percentage for HSS blacklisting/disabling detection.
- **imsPdnActivation** - IMS PDN Activation Timer - Timer to wait for IMS PDN activation for IMS supervision functionality. The timer is started after performing Attach and

inter MME/IRAT TAU if the IMS PDN is not already activated and the IMS supervision is configured for the UE.

- **imsRegistration** - IMS Registration Timer - Timer for IMS registration for IMS supervision functionality. The timer is started after IMS PDN activation or after inter MME/IRAT TAU procedures when either the MME does not yet have an STN-SR for the UE, or the STN-SR indicates unsuccessful IMS registration and the IMS supervision is configured for the UE.
- **initContextSetup** - Initial context setup - When the MME sends a message to the eNB for an initial context setup request, it expects to receive a response message from the eNB (Initial context setup response or initial context setup failure). If the timer expires while waiting for this response, the MME considers that this class 1 procedure failed.
- **internalIpIsolationThreshold** - Internal IP isolation threshold - Threshold to be used for determining internal IP isolation of VMs. For VNF and a8 appliance the default is 5. For CNF and a2 appliance the default is 0 and cannot be changed.
- **iotPurgeTimer** - IoT Purge Timer - This timer starts after the MME-initiated detach timer fires. If this timer fires before the IoT UE contacts the MME again, the UE VLR data is purged.
- **ipdsRemHeartbeat** - IPDS REM heartbeat - Configure REM heartbeat interval for IPDS service. VNF: Default is 200, Range is 200 to 20000 CNF: Default is 600, Range is 600 to 20000
- **ippsRemHeartbeat** - IPPS REM heartbeat - Configure REM heartbeat interval for IPPS service. VNF: Default is 200, Range is 200 to 20000 CNF: Default is 600, Range is 600 to 20000
- **iuUeLocationReport** - IU UE location report - Specifies the length of time SGSN waits for the receipt of a location report response from the RNC.
- **keepCtxAfterN26HoFail** - Keep Context After N26 Handover Fail - This timer starts when serving S1 Release by which the ongoing N26 handover is decided as failed at target MME. While the timer is running the MME discards (silent drop) any DDN (or other SM message) received from the SGW. When the timer expires, the MME shall remove resources associated with the N26 handover
- **lihsRemHeartbeat** - LIHS REM Heartbeat - Configure REM heartbeat interval for LIHS service. VNF: Default is 200, Range is 200 to 20000 CNF: Default is 600, Range is 600 to 20000
- **mapM** - Map m to HLR - Defines the map m (medium) protocol when interfacing to an HLR.
- **mbmsResponseFromMce** - MBMS response from MCE - MBMS response from MCE.
- **mBReachable** - Timer beyond T3412 - Minutes beyond T3412 for taking actions when a UE does not send TAU. Timer stops when a NAS message is received.
- **minTimeToMbmsDataTransfer** - Min time to MBMS data transfer - Min time to MBMS data transfer.

- mmeCnfgUpdate - MME config update - When the MME sends a configuration update message to the eNB, it expects to receive an update acknowledgment response message from the eNB. if the timer expires while waiting for this response, the MME considers that this class 1 procedure failed.
- mmelnitDetach - MME Initiated Detach - Timer starts when 'Mobile Reachable' timer expires. Timer stops when a NAS message is received.
- mobileReachableLpa - Mobile reachable LPA - Mobile reachable time for the LPA UE.
- moDataAnswer - MO data answer - Time to wait for ODA after sending ODR (MO-Data-Request).
- monitorArrivalDownlinkData - Monitor arrival downlink data - Monitor arrival of downlink data if downlink data is expected.
- mtDataAnswer - MT data answer - This timer is used for responding to SCEF with TDA when SCEF sends TDR and does not include SCEF-Wait-Time AVP.
- mtDataRequest - MT data request - This timer is started to wait for MT data request before releasing the S1 connection. this timer is started immediately after sending connection-management-request (CMR) with connection-action AVP set to CONNECTION-UPDATE.
- n26rmUeDataAtSrcMme - N26 remove Ue Data at source MME
- nwdsRemHeartbeat - NWDS REM heartbeat - Configure REM heartbeat interval for NWDS service. Applicable only for CNF. Default is 600. Range is 600 to 20000.
- pagingGapTimer - Paging gap timer - This timer indicates the interval in which the MME will suppress paging for a UE after multiple consecutive page response failures have occurred for the UE. a value of 0 indicates that paging should not be suppressed due to previous page response failures.
- papsRemHeartbeat - PAPS REM heartbeat - Configure REM heartbeat interval for PAPS service.
- pcmsRemHeartbeat - PCMS REM Heartbeat - Configure REM heartbeat interval for PCMS service. VNF: Default is 200, Range is 200 to 20000 CNF: Default is 600, Range is 600 to 20000
- pdnConnectAfterEcmlidle - PDN connect after ECM idle - Defines how long the subscriber needs to be in ECM-IDLE state before the PDN re-connection is initiated.
- pgwIsolationDeclarationTimer - PGW isolation declaration timer - If the number of the P-GW faults (remote peer not responding cause received in the S11/S4 create session response) exceed the provisioned threshold in this timer, MME will isolate the S-GW/P-GW pair that reported this fault for new PDN connections for a configurable isolation duration. changes to the timer will take into effect whenever MME checks whether total faults exceeded the fault threshold in the timer that is either triggered by a new fault or some other event.

- **pgwIsolationDuration** - PGW isolation duration - This command is used to provision the duration of PGW isolation once it is declared that the PGW is isolated for any new PDN connections. WMM will start selecting the PGW once the duration expires. This duration can be stopped and PGW selection can be resumed by using the sig-link CLI command
- **pgwRestartRecovery** - PGW restart recovery - It is used to provision a duration in which MME would attempt to re-establish PDN connections associated with a PGW indicated by the PGW restart notification message of the ECM-IDLE UE. MME would start the timer when it starts the recovery. once the timer expires, MME will stop the recovery and deletes the PDN session of the left over UEs.
- **praDcConnectedIn** - PRA Dual Connectivity Connected In - Specifies the timer for Segment Based Dual Connectivity PRA reporting functionality, during which the MME suppresses the PRA status change report when the UE is in ECM-CONNECTED state and the last reported PRA status was 'inside'. The timer helps the MME to skip sequential back and forth status updates related to the segment based reporting. If MME receives new radio transport layer endpoint(s) for user plane, it waits for possible subsequential next update or UE context release before sending the PRA report towards the SGW/PGW.
- **praDcConnectedOut** - PRA Dual Connectivity Connected Out - Specifies the timer for Segment Based Dual Connectivity PRA reporting functionality, during which the MME suppresses the PRA status change report when the UE is in ECM-CONNECTED state and the last reported PRA status was 'outside'. The timer helps the MME to skip sequential back and forth status updates related to the segment based reporting. If MME receives new radio transport layer endpoint(s) for user plane, it waits for possible subsequential next update or UE context release before sending the PRA report towards the SGW/PGW.
- **praDcIdle** - PRA Dual Connectivity Idle - Defines a timer for Presence Reporting Area (PRA) functionality for optimizing Dual Connectivity related reporting when the UE was previously ECM-IDLE. MME starts this timer when the UE was ECM-IDLE previously and the MME receives new RAN endpoints (receives Initial Context Setup Response during Service Request, Control Plane Service Request or Intra MME TAU). MME stops this timer when it receives either new RAN endpoints (Path Switch Request, Handover Request Ack, Initial Context Setup Response or E-RAB modification indication) or UE context release complete. MME sends PRA Information towards the SGW/PGW when timer either expires or was stopped and reporting is required to be done i.e. status differs from previous reported and the UE is still ECM-CONNECTED.
- **praDcSegmentOptServReq** - PRA DC Segment Optimization Service Request - Specifies the timer for IPv6 Address Segment Based Dual Connectivity PRA reporting feature and the Service Request optimization functionality. The timer helps the MME to skip any sequential back and forth status updates related to the segment based reporting related to the idle to connected transferring and the possible subsequent gNB addition. When MME receives a Service Request message for a UE for which address segment based PRA reporting has previously been requested, the MME waits during this timer for a subsequent E-RAB modification indication before sending any PRA status update

towards the SGW/PGW/PCRF, if such would still be needed due to the Service request. If the timer is running and the MME receives an E-RAB Modification Indication, the MME stops the timer. The MME reports the segment based status at either the receipt of the E-RAB modification indication, or the timer expiry (if status at that point is different from the previously sent status).

- **praDcSingleTimer** - PRA DC Single Timer - Specifies the single optimisation timer for Segment Based Dual Connectivity PRA reporting functionality, during which the MME suppresses the PRA status change reports. The MME starts the timer when the timer is not yet running and the MME receives new RAN endpoints that would trigger status update for any PRA id that is reported for the UE. The timer is stopped when the MME receives new RAN endpoints and the PRA status of all UE's PRA ids is back to the previously reported status, or when the UE context is released. The MME restarts the timer when the MME receives new RAN endpoints that cause the status of a different PRA id to be updated to a status that is not yet reported.
- **praStdSuppressionCancel** - PRA Standard Suppression Cancel - PRA Suppression cancel timer for 3GPP standard based Presence Area Reporting. Effective when PRA standard suppression is set to Yes. The functionality includes two time definitions- PRA suppression time and PRA suppression cancel time. MME suppresses a subsequent PRA report after a PRA report (PRA Information) has been sent out towards the SGW/PGW (during the time defined by the PRA suppression time), unless the PRA reporting status stays the same for a suppress cancel defined time period. When more time than the provisioned PRA suppression time has passed since the last delivered PRA report, the MME sends any PRA report to SGW/PGW and stores the timestamp of the report sending. If MME detects next PRA status change before the provisioned suppression time has passed since previous PRA report, MME (suppresses the report for a while and) starts the suppression cancelling timer to wait for another status change. If MME detects another status change before the suppression cancel timer expires, the MME discards the report since the status is back to the same as already reported. If the suppression cancel timer expires, MME sends the latest (suppressed) PRA report towards the SGW/PGW. The timer can be provisioned with 100ms granularity.
- **praStdSuppressionPeriod** - PRA Standard Suppression Period - Defines the time period during which the MME suppresses a subsequent PRA report after a PRA report (PRA Information) has been sent out towards the SGW/PGW, unless the PRA status stays the same during a time period defined by the suppression cancel time. The timer can be provisioned with 100ms granularity.
- **purgeTimer** - Purge timer - This timer starts after the MME-initiated detach timer fires. If this timer fires before the UE contacts the MME again, the UE VLR data is purged.
- **rcisPurgeTimer** - RCIS Purge Timer
- **rcisRemHeartbeat** - RCIS REM Heartbeat - Configure REM heartbeat interval for RCIS service. VNF: Default is 200, Range is 200 to 20000 CNF: Default is 600, Range is 600 to 20000

- **receiveNotificationReturnFromUe** - Receive notification return from UE - Receive notification return from UE.
- **reserveUeSubscriptionIratMob** - Reserve UE Subscription IRAT Mobility - This command is used to provision the duration of preserving UE subscription data. Timer shall be restarted upon UE transitioning back to MME. UE's subscription data shall be kept until the timer expires, however if UE comes back before timer expiration, NOR instead of ULR shall be sent to HSS in TAU procedure.
- **s102tack** - S102 tack - S102 tack.
- **s10rmUeDataAtSrcMme** - S10 remove UE data at source MME - Used in S10 interactions to remove UE data at the source MME after a handover with MME relocation.
- **s1hoComplete** - S1 HO complete - Used to time the receipt of either a handover notify message from the target eNB or a forward relocation complete message from the target MME in the case of MME relocation. the handover is not successful if either of these messages is not received before the expiration of the timer.
- **s1hoCompleteExtended** - Extended S1 HO complete - This timer is introduced to handle extra latency introduced by CIOT and CAT-M1 CE mode b devices in completing S1HO. for S1HO without MME relocation, the timer is used to monitor handover notify reception. for S1HO with MME relocation, this timer covers the extra latency to execute the handover, get the handover notify and receive forward relocation complete notification.
- **s1hoResourceRelease** - S1 HO resource release - During a relocation request (SGW only or MME and SGW), the timer starts upon receipt of a forward relocation complete acknowledge. if the release does not occur before the timer expires, the target MME sends a delete indirect data forwarding message to the target SGW to delete it if the target MME has setup indirect forward paths on the target SGW. the source MME deletes any bearers on the source SGW (with SGW relocation) and deletes the tunnel by sending a delete indirect data forwarding message if the source MME has setup indirect forward paths on the source SGW. the source MME also releases S1 UE connections with the source eNB.
- **s1hoResourceReleaseWithNoMmeSgwRel** - S1 Handover Resource Release With No MME SGW Relocation - During intra-mme intra-sgw s1-based ho request, the timer starts after Handover Notify message and once the timer expires, MME shall send UE Context Release Command to the source eNB and the source eNodeB releases its resources related to the UE and responds with a UE Context Release Complete message. When the timer is set to 0, the S1 Handover Resource Release timer shall be used.
- **s1releaseOnCsgExpiration** - S1 release on CSG expiration - This timer is to specify the time to wait to do an S1 release when CSG registration expires.
- **s1Reset** - S1 reset - When the MME sends a S1 reset message to the eNB, it expects to receive a reset acknowledgment response message from the eNB. if the timer expires while waiting for this response, the MME considers that this class 1 procedure failed.

- **s1UeCapInfoIndForLtemWaitTime - s1UeCapInfoIndWaitTime** - Specifies the timeout for the MME to receive the S1 UE Capability Information Indication message from the eNB. The timer applies if the MME needs to find out from the eNB whether the UE is capable of LTE-M. The timer is started when the MME sends the UE Capability Match Request to the eNB as part of an attach or TAU procedure.
- **s1ueLocationReport - S1 UE location report** - This timer specifies the length of time to wait for the receipt of a location report from the eNB.
- **s3gnHoComplete - S3 Gn HO complete** - Supports inter-rat handover procedures.
- **s3gnIndirectForwarding - S3 Gn indirect forwarding** - Used to time for resource release for indirect forwarding of UE data. when the timer expires, the MME releases the resources that have been allocated for indirect forwarding.
- **scefCmrAnswer - SCEF CMR answer** - This timer is used for responding with connection management answer to SCEF for SCEF sent connection management request (CMR) for an update. the timer applies to SCEF CMR updates other than T6a connection release. CMR for release from SCEF answer uses connectionmanagementanwer timer.
- **scefExpiredTimeReject** - If the MME receives a T6a (SCEF) MT Data Request that contains the Maximum-Retransmission-Time AVP and/or SCEF-Wait-Time AVP, this timer defines how much in the past each AVP value can be in order for the MME to consider it as a still valid AVP. The default is 30 seconds to align with the prior hard-coded value.
- **secondaryRatUsageReportingDelay - Secondary Rat Usage Reporting Delay during Handover** - Timer to delay forwarding the Secondary RAT Usage report to the S/PGW during Handover
- **serviceGapTimerOverrideValue - Service Gap Timer Override Value** - Specifies the timer value to use for the UE's service gap timer
- **sessionOrientedLinkMoDeletionTimer - Session oriented link MO deletion timer** - Session oriented link MO deletion timer for GTP based interfaces. value of 0 means do not do inactivity deletion.
- **sfdsRemHeartbeat - SFDS REM Heartbeat** - Configure REM heartbeat interval for SCTP Flow Distributor Service. VNF: N/A CNF: Default is 600, Range is 600 to 20000
- **sgsLiteDetachIndicationAck - SGs lite detach indication ack** - Configure SGsLite-IMSI-DETACH-ACK timeout in milliseconds.
- **sgsLiteLocationUpdateAck - SGs lite location update ack** - Configure timer for the reception of SGsLite-LOCATION-UPDATE-ACK timer.
- **sgsLiteUplinkDataAck - SGs lite uplink data ack** - Configure SGsLite-UPLINK-ACK timeout.
- **sgsPaging - SGs paging** - Defines the value for receiving responses to a SG paging request. this timer should be set slightly lower than the paging timer value provisioned in the MSC/VLR.

- **sgwIsolationDeclarationTimer** - SGW isolation declaration timer - This command is used to provision SGW isolation declaration timer. If the number of the SGW faults (remote peer not responding cause received in the S11/S4 create session response) exceed the provisioned threshold in this provisioned timer then WMM will isolate the SGW from the selection. A sliding window will be used in counting the total number faults for the isolation declaration duration.
- **sgwIsolationDuration** - SGW isolation duration - This command is used to provision the duration of SGW isolation once it is declared that the SGW is isolated from selection. WMM will start selecting the SGW once the duration expires. This duration can be stopped and PGW selection can be resumed by using the sig-link CLI command
- **sgwRestorationTimer** - SGW restoration timer - This timer is used to guard the MME attempt to assign UE associated with a failed SGW or restarted SGW.
- **srnsCompletion** - SRNS completion - Used to time for receipt of an SRNS completion message from the target SGSN.
- **srnsHoRauRequestWait** - SRNS HO RAU request wait - Specify how long 3G-SGSN waits for a RAU request after SRNS relocation or 4G to 3G HO.
- **srsDataPurge** - SRS Data Purge - SRS Data Purge time
- **srsResponse** - SRS response - SRS response time.
- **srvccPsToCsComplete** - SRVCC PS to CS complete - Time to wait for the SRVCC PS to CS complete message from an MSC.
- **srvccTo1xHoComplete** - SRVCC to 1x HO complete - This is guard timer that is started when S102 signaling with HO status IE (SUCCESS) is received and shall be stopped when S1 UE context release request with cause CIOT cell redirection is received.
- **srvccTo1xHoPreparation** - SRVCC to 1x HO preparation - This is a guard timer that is started when S1 UL CDMA2000 tunneling with HO required IE is received and stopped when S102 Signaling with HO status IE is received.
- **suspendReq** - Suspend request - Used in an EUTRAN to GERAN CS handover procedure to time for the receipt of the suspend request message from the SGSN in the network to which the UE has moved.
- **t3346attachRejectMaximum** - T3346 attach reject maximum - Specifies the maximum value to consider when randomly assigning the T3346 timer value to be used for an attach procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).
- **t3346attachRejectMinimum** - T3346 attach reject minimum - Specifies the minimum value to consider when randomly assigning the T3346 timer value to be used for an attach procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).

- **t3346gtpVolEsrMax** - T3346 GTP VOL ESR max - The maximum value to consider when randomly assigning the T3346 timer value to be used for an ESR procedure rejection due to GTP overload.
- **t3346gtpVolEsrMin** - T3346 GTP VOL ESR min - The minimum value to consider when randomly assigning the T3346 timer value to be used for an SR procedure rejection due to GTP overload.
- **t3346gtpVolSrMax** - T3346 GTP VOL SR max - The maximum value to consider when randomly assigning the T3346 timer value to be used for an SR procedure rejection due to GTP overload.
- **t3346gtpVolSrMin** - T3346 GTP VOL SR min - The minimum value to consider when randomly assigning the T3346 timer value to be used for an SR procedure rejection due to GTP overload.
- **t3346gtpVolTauMax** - T3346 GTP VOL TAU max - The maximum value to consider when randomly assigning the T3346 timer value to be used for a TAU procedure rejection due to GTP overload.
- **t3346gtpVolTauMin** - T3346 GTP VOL TAU min - The minimum value to consider when randomly assigning the T3346 timer value to be used for a TAU procedure rejection due to GTP overload.
- **t3346NblotAttachRejectMaximum** - T3346 NB-IOT Attach Reject Maximum - Specifies the maximum value to consider when randomly assigning the T3346 NB-IOT timer value to be used for an attach procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).
- **t3346NblotAttachRejectMinimum** - T3346 NB-IOT Attach Reject Minimum - Specifies the minimum value to consider when randomly assigning the T3346 NB-IOT timer value to be used for an attach procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).
- **t3346NblotCpsrRejectMaximum** - T3346 NB-IOT Control Plane SR Reject Maximum - Specifies the maximum value to consider when randomly assigning the T3346 NB-IOT timer value to be used for an Control Plane SR procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).
- **t3346NblotCpsrRejectMinimum** - T3346 NB-IOT Control Plane SR Reject Minimum - Specifies the minimum value to consider when randomly assigning the T3346 NB-IOT timer value to be used for an Control Plane SR procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).
- **t3346NblotOdbRejectMaximum** - t3346 NB-IOT ODB Reject Maximum - Specifies the maximum value backoff timer value in Seconds randomly assigned to a NB-IoT UE with ODB trying to access the MME. This timer applies when the MME is provisioned to provide cause 22 to the UE (in the restNasMappingProfile).

- **t3346NblotOdbRejectMinimum** - t3346 NB-IOT ODB Reject Minimum - Specifies the minimum value backoff timer value in Seconds randomly assigned to a NB-IoT UE with ODB trying to access the MME. This timer applies when the MME is provisioned to provide cause 22 to the UE (in the restNasMappingProfile).
- **t3346NblotSrRejectMaximum** - T3346 NB-IOT SR Reject Maximum - Specifies the maximum value to consider when randomly assigning the T3346 NB-IOT timer value to be used for an SR procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).
- **t3346NblotSrRejectMinimum** - T3346 NB-IOT SR Reject Minimum - Specifies the minimum value to consider when randomly assigning the T3346 NB-IOT timer value to be used for an SR procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).
- **t3346NblotTauRejectMaximum** - T3346 NB-IOT TAU Reject Maximum - Specifies the maximum value to consider when randomly assigning the T3346 NB-IOT timer value to be used for a TAU procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).
- **t3346NblotTauRejectMinimum** - T3346 NB-IOT TAU Reject Minimum - Specifies the minimum value to consider when randomly assigning the T3346 NB-IOT timer value to be used for a TAU procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).
- **t3346srRejectMaximum** - T3346 SR reject maximum - Specifies the maximum value to consider when randomly assigning the T3346 timer value to be used for an SR procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).
- **t3346srRejectMinimum** - T3346 SR reject minimum - Specifies the minimum value to consider when randomly assigning the T3346 timer value to be used for an SR procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).
- **t3346tauRejectMaximum** - T3346 TAU reject maximum - Specifies the maximum value to consider when randomly assigning the T3346 timer value to be used for a TAU procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).
- **t3346tauRejectMinimum** - T3346 TAU reject minimum - Specifies the minimum value to consider when randomly assigning the T3346 timer value to be used for a TAU procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).
- **t3396genmax** - T3396 gen max - Parameter t3396genmax indicates maximum value to consider when assigning the T3396gen timer value to be used when MME rejects an IMS PDN Connectivity request (standalone or attach) When the timer has value > 0, then whenever the MME rejects an IMS PDN Connectivity Request (standalone or Attach) due to not subscribed IMS APN (or IMS APN is subscribed only via the wildcard APN

and GPARM is enabled) then the T3396 IE shall be included in PDN Connectivity Reject message to force the UE re-attempt only after the back off timer expires,

- **t3396genmin** - T3396 gen min - Parameter t3396genmin indicates minimum value to consider when assigning the T3396gen timer value to be used when MME rejects an IMS PDN Connectivity request (standalone or attach)
- **t3396gtpVolMax** - T3396 GTP VOL max - The maximum value to consider when randomly assigning the T3396 timer value to be used for an SM request rejection due to GTP overload.
- **t3396gtpVolMin** - T3396 GTP VOL min - The minimum value to consider when randomly assigning the T3396 timer value to be used for an SM request rejection due to GTP overload.
- **t3396max** - T3396 max - The maximum value to consider when randomly assigning the T3396 timer value to be used for an SM request rejection due to overload.
- **t3396min** - T3396 min - The minimum value to consider when randomly assigning the T3396 timer value to be used for an SM request rejection due to overload.
- **t3402** - T3402 reattach request - Part of UE attach accept message. used by UE to initiate the next attach or TAU procedure in case of a failure of a previous attach or TAU procedure.
- **t3402AttachReject** - T3402 Attach Reject - Specifies the wait time for another attach request if the UEs attach counter has reached 5 to the Attach Reject message.
- **t3412** - T3412 periodic TAU - Part of UE attach accept message. used by the UE for periodic TAU. the MME starts this timer when the UE state is changed to the ECM-IDLE state.
- **t3412extendedForLpaUe** - T3412 extended for LPA UE - Periodic TAU timer that MME sends to UE in attach accept message to a UE configured for low priority access.
- **t3412withTauSuppression** - T3412 with TAU suppression - This timer is used instead of the standard T3412 timer when TAU suppression has been triggered by the MME. a value of 0 indicates that the standard T3412 timer should be used even when the MME has triggered TAU suppression.
- **t3422** - T3422 retransmit detach request - Retransmit detach request.
- **t3422Wbs1CeMode** - T3422 for WB-S1 CE mode - T3422 timer value for the maximum interval MME waits after sending a detach request before retry for UE in WB-S1 coverage enhancement (CE) mode.
- **t3442rejectEsrNoSrvcc** - T3442 reject ESR no SRVCC - T3442 timer to be included in the service reject message when ESR is rejected due to UE with IMS voice bearer.
- **t3448Max** - T3448 Maximum - Specifies the maximum value of control plane data back-off timer. If the UE supports the control plane data back-off timer T3448, the UE shall set the CP backoff bit to back-off timer for transport of user data via the control plane supported in the UE network capability IE of the ATTACH REQUEST message.

- **t3448Min** - T3448 Minimum - Specifies the minimum value of control plane data back-off timer. If the UE supports the control plane data back-off timer T3448, the UE shall set the CP backoff bit to back-off timer for transport of user data via the control plane supported in the UE network capability IE of the ATTACH REQUEST message.
- **t3450** - T3450 retransmit attach and TAU accept or GUTI reallocation - Retransmit attach accept, TAU accept, or GUTI reallocation command. timer starts if an attach request is accepted by the network and the MME sends an ATTACH ACCEPT message to the UE. timer stops after receiving an ATTACH COMPLETE message and considers the GUTI sent in the ATTACH ACCEPT message as valid.
- **t3450Wbs1CeMode** - T3450 for WB-S1 CE mode - T3450 timer which is used to supervise the retransmit of attach accept, TAU accept, or GUTI reallocation commands of UE in WB-S1 coverage enhancement (CE) mode.
- **t3460** - T3460 retransmit auth or sec mode command - Retransmit auth request, or security mode command. timer starts when a security mode command is sent. normal stop occurs when security mode complete or security mode reject is received.
- **t3460Wbs1CeMode** - T3460 for WB-S1 CE mode - T3460 timer which is used to supervise the response to the transmission by the MME of the authentication request or security mode command for UE in WB-S1 coverage enhancement(ce) mode.
- **t3470** - T3470 retransmit identity command - Retransmit identity command. timer starts when IDENTITY REQUEST is received. timer stops when IDENTITY REQUEST is received.
- **t3470Wbs1CeMode** - T3470 for WB-S1 CE mode - T3470 timer which is used to supervise the network initiated identification procedure for UE in WB-S1 coverage enhancement (CE) mode.
- **t3485** - T3485 retransmit activate def or ded bearer request - Retransmit activate default bearer request, or activate dedicated bearer request. timer starts when an ACTIVATE DEFAULT/ DEDICATED EPS BEARER CONTEXT REQUEST is sent. timer stops when ACTIVATE DEFAULT EPS BEARER CONTEXT ACCEPT/REJECT or ACTIVATE DEDICATED EPS BEARER CONTEXT ACCEPT/REJECT is received.
- **t3485Wbs1CeMode** - T3485 for WB-S1 CE mode - T3485 timer which is used to supervise the response to the transmission by the MME of the activate default bearer request, or activate dedicated bearer request for UE in WB-S1 coverage enhancement (CE) mode.
- **t3486** - T3486 retransmit mod bearer request - Retransmit modify bearer request.
- **t3486Wbs1CeMode** - T3486 for WB-S1 CE mode - T3486 timer which is used to supervise the response to the transmission by the MME of the modify EPS bearer context request for UE in WB-S1 coverage enhancement (CE) mode.
- **t3489** - T3489 retransmit ESM info request timeout - Timeout before retransmission of ESM information request.

- **t3489Wbs1CeMode** - T3489 for WB-S1 CE mode - T3489 timer which is used to supervise the response to the transmission by the MME of the ESM information request for UE in WB-S1 coverage enhancement (CE) mode.
- **t3495** - T3495 retransmit deactivate bearer request - Retransmit deactivate bearer request. timer starts when DEACTIVATE EPS BEARER CONTEXT REQUEST is sent. timer stops when DEACTIVATE EPS BEARER CONTEXT ACCEPT is received.
- **t3495Wbs1CeMode** - T3495 for WB-S1 CE mode - T3495 timer which is used to supervise the EPS bearer context deactivation procedure for UE in WB-S1 coverage enhancement (CE) mode.
- **t3x01** - T3X01 receive location response from E-SMLC via SLs timer - Timer to receive location response from the E-SMLC via SLs. (timer to transition from location state to LCS-IDLE state for a UE).
- **t3x02** - T3X02 receive reset ack after send reset to E-SMLC timer - Timer to receive reset ack after sending reset to the E-SMLC (SLs).
- **tauAfterHo** - TAU after HO - This command is used to set the value of the tauAfterHo timer which supervises the exchange between the MME and UE after handover occurs from LTE S1, UTRAN or GERAN. The MME detaches the UE if the timer expires before the TAU Request is Received. This timer only applies when a TAU is expected, e.g. if the TAI changed in an LTE-to-LTE HO. Caution! this value should only be set to '0' for troubleshooting purposes. The '0' value is not recommended by Nokia for service operation. If this value is set to '0', the MME does not start the timer, does not detach the UE, and does not start a new AKA as part of the handover procedure. this can generate security vulnerabilities in the EPS that a UE can exploit.
- **tauAfterN26Ho** - TAU after N26 HO - This timer supervises the exchange between the MME and UE after handover occurs from NR. The MME detaches the UE if the timer expires before the TAU Request is Received. Caution! this value should only be set to '0' for troubleshooting purposes. The '0' value is not recommended by Nokia for service operation. If this value is set to '0', the MME does not start the timer, does not detach the UE, and does not start a new AKA as part of the handover procedure. This can generate security vulnerabilities in the EPS that a UE can exploit.
- **tauSuppressionThreshold** - TAU suppression threshold - This timer indicates the maximum interval for cyclic movement between tracking areas that will trigger TAU suppression by MME. a value of 0 indicates that TAU suppression is performed without regard to the interval of the cyclic movement between tracking areas.
- **tDynMO** - Indicates the interval to wait for an interface to return an operational state, before deleting the interface. As of CMM23.2.0, tDynMO is used for the Link Disable timer which only applies to links in DISABLED operational state. S1mme, S11, S10, N26, NR10 and other dynamical MME links except for Gn, S3, and S16 are subject to the tDynMO. Statically provisioned links and administrative locked links are not subject to the Link Disable timer. Disabled N2 links are subject to the tDynMO disable timer. The MME N26 links are subject to the tDynMO timer or Transaction Oriented MO

delete timer depending on the operational state of the links, either being enabled and disabled. Those are exclusive states.

- transactionOrientedLinkMoDeletionTimer - Transaction oriented link MO deletion timer
 - Transaction oriented link MO deletion timer for GTP based interfaces. Value 0 means do not do inactivity deletion. Transaction Oriented Link MO Deletion Time is used for Link Inactivity timer and only applies to operational ENABLED links. Only SGSN Gn, S3, S16, S10 and N26 are subject to this time. GGSN is subjective to Session Oriented Link MO Deletion Time. Administrative locked links are not subject to the inactivity timer.
- tS10 - TS10 implicit IMSI detach from non-eps services procedure guard timer - Guards implicit IMSI detach from non-eps services procedure.
- tS12_1 - TS12-1 MME reset variable guard timer - Controls the resetting of the 'MME-Reset' variable.
- tS12_2 - TS12-2 MME reset procedure guard timer - Guards the MME reset procedure. there is one ts12-2 timer per MME SGs association with a VLR.
- tS13 - T13 SGs MME implicit detach guard timer - Configure SGs TS13 timer for MME implicit detach.
- tS6_1 - T6-1 location update procedure guard timer - Guards the location update procedure.
- tS8 - TS8 explicit IMSI detach from EPS services procedure guard timer - Guards explicit IMSI detach from the EPS services procedure.
- tS9 - TS9 explicit IMSI detach from non-eps services procedure guard timer - Guards explicit IMSI detach from non-eps services procedure.
- tslrAckFromGmlc - TSLR ack from GMLC - TSLR ack from GMLC.
- ucDsRemHeartbeat - UCDS REM Heartbeat - Configure REM heartbeat interval for the UCDS service. VNF: Default is 1400, Range is 1400 to 20000 CNF: Default is 1400, Range is 1400 to 20000
- ueContextModify - UE context modify - When the MME sends a context modification message to the eNB, it expects to receive a modification response message from the eNB. if the timer expires while waiting for this response, the MME considers that this class 1 procedure failed.
- ueContextRelease - UE context release - When the MME sends a context release message to the eNB, it expects to receive a release complete response message from the eNB. if the timer expires while waiting for this response, the MME considers that this class 1 procedure failed.
- ueCtxtEpsMovedTo5gPurgeTimer - 4G Purge Timer for UE Contexts Moved to 5GC - Purge timer starts after the UE moves to 5GC. The UE gets purged either because of the global purge timer or current timer, whichever expires first.
- ueRadioCapabilityMatchRequest - UE radio capability match request - Specifies the time to wait for UE radio capability match request.

- **ufdsRemHeartbeat** - UFDS REM Heartbeat - Configure REM heartbeat interval for UDP Flow Distributor Service. VNF: N/A CNF: Default is 600, Range is 600 to 20000
- **vlrCleanupIratUeCtxt** - VLR Cleanup IRAT UE Contexts - This parameter specifies the time for deleting from the VLR the preserved UE Contexts of the deregistered UEs that have moved to 5GC.
- **warningMessage** - Warning message - This is the timer value that MME waits before re-transmitting the WRWR (Write Replace Warning Request) message to an eNB if the CMAS response is not received from the eNB.

Name

timerUnit - Timer unit

Syntax

--timerUnit <list-item>

Description

Unit of measurement.

Class

Optional.

List

- **320Hour** - 320 hour - Specify timer unit of 320 hours.
- **Millisecond** - Specify timer unit of 1 millisecond.
- **Second** - Specify timer unit of 1 second.
- **2Second** - 2 second - Specify timer unit of 2 seconds.
- **30Second** - 30 second - Specify timer unit of 30 seconds.
- **Minute** - Specify timer unit of 1 minute.
- **DeciHour** - Specify timer unit of 1 decihour (6 minutes).
- **10Minute** - 10 minute - Specify timer unit of 10 minutes.
- **Hour** - Specify timer unit of 1 hour.
- **10Hour** - 10 hour - Specify timer unit of 10 hours.

Default

DeciHour

Name

timerValue - Timer value

Syntax

--timerValue <integer>

Description

Numeric value of timer.

Class

Optional.

Type / Range

Integer in the range of 0 to 4,294,967,295

The timerValue string is based on the timerName parameter:

Name

alertMessageHoldingTime - Alert message holding time

Syntax

```
--timerName alertMessageHoldingTime --timerValue <string> [ --  
timerUnit Second ]
```

Description

The time in seconds that a SBC CMAS message will be queued for while waiting for a previously received SBC CMAS message to finish processing.

Type / Range

Integer in the range of 0 to 300 in units of Second

Default

0

Name

connectionManagementAnswer - Connection management answer

Syntax

```
--timerName connectionManagementAnswer --timerValue <string> [ --  
timerUnit Millisecond ]
```

Description

Configure timer to wait for connection management answer message after sending connection management request. this timer is also used to respond CMA to CMR received in the case of CMR release request from SCEF.

Type / Range

Integer in the range of 100 to 6,000 in units of Millisecond

Default

1000

Name

controlPlaneUserInactivity - Control plane user inactivity

Syntax

```
--timerName controlPlaneUserInactivity --timerValue <string> [ --  
timerUnit Millisecond ]
```

Description

This timer is used to detect user inactivity and release S1 connection of UE that have user data over NAS only. The cmm subscriber show command may show NB UEs as IDLE if the UE undergoing CP user inactivity timing. This can occur if the inactivity timer has been running for less than 20 seconds when the command is entered. The MME controlPlaneUserInactivity timer is recommended to have a value greater than the eNB inactivity timers.

Type / Range

Used for Control Plane User Inactivity Timer in units of Millisecond

Default

3000

Name

cpiotUndeliveredNasWait - cpiotUndeliveredNasWait

Syntax

```
--timerName cpiotUndeliveredNasWait --timerValue <string> [ --  
timerUnit Millisecond ]
```

Description

This timer defines the time that the MME will wait for NAS non Delivery Indication msg(s) from source eNB as part of eNB CP Relocation Indication procedure

Type / Range

Integer in the range of 0 to 3,000 in units of Millisecond

Default

1000

Name

cppsRemHeartbeat - CPPS REM heartbeat

Syntax

```
--timerName cppsRemHeartbeat --timerValue <string> [ --timerUnit  
Millisecond ]
```

Description

Configure REM heartbeat interval for CPPS service. VNF: Default is 200, Range is 200 to 20000 CNF: Default is 600, Range is 600 to 20000

Type / Range

Integer in the range of 200 to 20,000 in units of Millisecond

Default

200

Name

critDnsAlarmSustainedInterval - Critical DNS Alarm Sustained Internal

Syntax

```
--timerName critDnsAlarmSustainedInterval --timerValue <string> [ --  
timerUnit Minute ]
```

Description

The interval of time the critical DNS alarm threshold conditions shall be exceeded and sustained before the 42014 CommunicationDegradationOrLost alarm will be raised.

Type / Range

Integer in the range of 0 to 120 in units of Minute

Default

0

Name

csOpsBearerDel - CS HO procedure delete bearer timer

Syntax

```
--timerName csOpsBearerDel --timerValue <string> [ --timerUnit  
Minute ]
```

Description

Used in an EUTRAN to GERAN CS handover procedure to time for the release of UE resources.

Type / Range

Integer in the range of 10 to 480 in units of Minute

Default

120

Name

dbbRemHeartbeat - DBS REM heartbeat

Syntax

```
--timerName dbbRemHeartbeat --timerValue <string> [ --timerUnit  
Millisecond ]
```

Description

Configure REM heartbeat interval for DBS service. VNF: Default is 200, Range is 200 to 20000 CNF: Default is 600, Range is 600 to 20000

Type / Range

Integer in the range of 200 to 20,000 in units of Millisecond

Default

200

Name

ddnDlBufExtEdrx - DDN DL Buffer Extend for eDRX

Syntax

```
--timerName ddndlBufExtEdrx --timerValue <string> [ --timerUnit  
Second ]
```

Description

Specifies the amount that the DL Buffer Duration IE in the DDN Ack message should be extended for a UE in eDRX. If this field is set to 0, no additional time is added and the value continues to be set to when the UE will exit its next paging time window.

Type / Range

Integer in the range of 0 to 60 in units of Second

Default

1

Name

ddndlBufExtPsm - DDN DL Buffer Extend for PSM

Syntax

```
--timerName ddndlBufExtPsm --timerValue <string> [ --timerUnit  
Second ]
```

Description

Specifies the amount that the DL Buffer Duration IE in the DDN Ack message should be extended for a UE in power saving mode. If this field is set to 0, no additional time is added and the value continues to be set to the expected time until the UE performs a TAU.

Type / Range

Integer in the range of 0 to 60 in units of Second

Default

1

Name

delBearerAtSrcSgw - Delete bearer at source SGW timer

Syntax

```
--timerName delBearerAtSrcSgw --timerValue <string> [ --timerUnit  
Millisecond ]
```

Description

This timer starts when the MME receives a create session response from the target SGW, and the UE bearers are removed at the source SGW when the timer expires. a value of 0 indicates that the UE bearers at the source SGW are deleted once the bearers are set up at the target SGW.

Type / Range

Integer in the range of 0 to 5,000 in units of Millisecond

Default

500

Name

deleteGbr - Delete guaranteed bit rate

Syntax

```
--timerName deleteGbr --timerValue <string> [ --timerUnit Second ]
```

Description

This command is used to configure how long to keep GBR bearers after the completion of an abnormal release (i.e. S1 release cause is other than user inactivity or redirection to UTRAN/ GERAN or 1xrtt) of S1 connection of a UE. the no form of the command sets the timer value to zero.

Type / Range

Integer in the range of 0 to 20 in units of Second

Default

0

Name

dnsApplicationTimer - DNS application timer

Syntax

```
--timerName dnsApplicationTimer --timerValue <string> [ --timerUnit Millisecond ]
```

Description

When a DNS query is made, a request is sent to the DNS resolver. the resolver either finds the result in a local cache, or makes a request of a local DNS server. when this occurs, the resolver starts a two-second timer for getting a result. it either returns the result, or no result is returned if the two-second timer expires. this timer allows for changing the 2 second default for proceeding with call processing.

Type / Range

Integer in the range of 100 to 5,000 in units of Millisecond

Default

5000

Name

eemsRemHeartbeat - EEMS REM Heartbeat

Syntax

```
--timerName eemsRemHeartbeat --timerValue <string> [ --timerUnit Millisecond ]
```

Description

Configure REM heartbeat interval for EEMS service. VNF: Default is 200, Range is 200 to 20000 CNF: Default is 600, Range is 600 to 20000

Type / Range

Integer in the range of 200 to 20,000 in units of Millisecond

Default

200

Name

emergencyMobileReachable - Emergency mobile reachable

Syntax

```
--timerName emergencyMobileReachable --timerValue <string> [ --  
timerUnit Minute ]
```

Description

Emergency mobile reachable timer.

Type / Range

Integer in the range of 1 to 1,500 in units of Minute

Default

124

Name

erabHoFailureTimer - E-RAB Setup Failure For HO Operations

Syntax

```
--timerName erabHoFailureTimer --timerValue <string> [ --timerUnit  
Millisecond ]
```

Description

E-RAB setup failure for HO operations. Includes X2-handover-triggered an dS1-intra-system-handover-triggered.

Type / Range

Integer range from 300 to 2000 in units of Millisecond

Default

300

Name

eRabModify - E-RAB modify

Syntax

```
--timerName eRabModify --timerValue <string> [ --timerUnit Second ]
```

Description

When the MME sends a message to the eNB related to a bearer request, it expects to receive a response. if the timer expires while waiting for this response, the MME considers that this class 1 procedure failed.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

8

Name

eRabRelease - E-RAB release

Syntax

```
--timerName eRabRelease --timerValue <string> [ --timerUnit Second ]
```

Description

When the MME sends a message to the eNB related to a bearer request, it expects to receive a response. if the timer expires while waiting for this response, the MME considers that this class 1 procedure failed.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

8

Name

eRabSetup - E-RAB setup

Syntax

```
--timerName eRabSetup --timerValue <string> [ --timerUnit Second ]
```

Description

When the MME sends a message to the eNB related to a bearer request, it expects to receive a response. if the timer expires while waiting for this response, the MME considers that this class 1 procedure failed.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

8

Name

gutiStalenessTimer - GUTI Staleness Timer

Syntax

```
--timerName gutiStalenessTimer --timerValue <string> [ --timerUnit Minute ]
```

Description

a GUTI Re-Allocation staleness timer. Timer value range from 0-24 hours. If it's set to 0 (by default), then no timer force GUTI re-allocation (this feature function) would take Place. It means that MME follows existing GUTI allocation function. If it's set to a non-zero value, at timer expiry, MME forces GUTI Re-Allocation at next UE event (e.g. Attach, TAU, Service request etc.).

Type / Range

Integer in the range of 5 to 1440, allowed 0 value also. in units of Minute

Default

0

Name

gwtsDelayS1release - GWTS delay S1 release

Syntax

```
--timerName gwtsDelayS1release --timerValue <string> [ --timerUnit  
Millisecond ]
```

Description

Configure delay S1 connection release timer for SDS UE.

Type / Range

Integer in the range of 0 to 10,000 in units of Millisecond

Default

0

Name

gwtsTcpLinkFailureDeclaration - GWTS tcp link failure declaration

Syntax

```
--timerName gwtsTcpLinkFailureDeclaration --timerValue <string> [ --  
timerUnit Second ]
```

Description

Configure TCP link failure declaration timer.

Type / Range

Integer in the range of 1 to 300 in units of Second

Default

2

Name

handoverRequest - Hand over request

Syntax

```
--timerName handoverRequest --timerValue <string> [ --timerUnit  
Second ]
```

Description

When the MME sends a handover request message to the eNB, it expects to receive an acknowledgment response message from the eNB. if the timer expires while waiting for this response, the MME considers that this class 1 procedure failed.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

8

Name

ho2g3gDeletion - HO 2G to 3G deletion

Syntax

```
--timerName ho2g3gDeletion --timerValue <string> [ --timerUnit  
Millisecond ]
```

Description

Used to handle the deletion of the UE bearers during a handover to a UTRAN or GERAN network. the timer is started when the MME receives the SGSN context request message from the new SGSN. when the timer expires, the MME deletes the UE bearers.

Type / Range

Integer in the range of 100 to 5,000 in units of Millisecond

Default

1000

Name

hoRcvCsSvcNotifyNasNonDeliveryRadioConnLost - Handover Received CS Service Notify NAS Non-Delivery Radio Conn Lost

Syntax

```
--timerName hoRcvCsSvcNotifyNasNonDeliveryRadioConnLost --timerValue  
<string> [ --timerUnit Millisecond ]
```

Description

When MME receives NAS Non-Delivery Indication (cause 'radio connection with ue lost') as response to SGs Downlink service operation, it uses this timer to safeguard reception of X2 Ho Request, at expiration of which it will re-attempt sending downlink message

Type / Range

Integer in the range of 100 to 3,000 in units of Millisecond

Default

2000

Name

hoTauRequestWait - HO TAU request wait

Syntax

```
--timerName hoTauRequestWait --timerValue <string> [ --timerUnit  
Millisecond ]
```

Description

Configure how long to wait for TAU request after X2 HO or S1 HO.

Type / Range

Integer in the range of 100 to 10,000 in units of Millisecond

Default

200

Name

hssUnavailBlacklisting - HSS unavailable blacklisting

Syntax

```
--timerName hssUnavailBlacklisting --timerValue <string> [ --  
timerUnit Second ]
```

Description

The time to apply extended HSS disabling/blacklisting.

Type / Range

Integer in the range of 0 to 900 in units of Second

Default

0

Name

hssUnavailFailedPercentCalc - HSS unavailable failed percentage calculation

Syntax

```
--timerName hssUnavailFailedPercentCalc --timerValue <string> [ --  
timerUnit Second ]
```

Description

Time interval to calculate the failure percentage for HSS blacklisting/disabling detection.

Type / Range

Integer in the range of 5 to 900 in units of Second

Default

300

Name

imsPdnActivation - IMS PDN Activation Timer

Syntax

```
--timerName imsPdnActivation --timerValue <string> [ --timerUnit  
Second ]
```

Description

Timer to wait for IMS PDN activation for IMS supervision functionality. The timer is started after performing Attach and inter MME/IRAT TAU if the IMS PDN is not already activated and the IMS supervision is configured for the UE.

Type / Range

Integer in the range of 2 to 60 in units of Second

Default

5

Name

imsRegistration - IMS Registration Timer

Syntax

```
--timerName imsRegistration --timerValue <string> [ --timerUnit  
Second ]
```

Description

Timer for IMS registration for IMS supervision functionality. The timer is started after IMS PDN activation or after inter MME/IRAT TAU procedures when either the MME does not yet have an STN-SR for the UE, or the STN-SR indicates unsuccessful IMS registration and the IMS supervision is configured for the UE.

Type / Range

Integer in the range of 5 to 600 in units of Second

Default

300

Name

initContextSetup - Initial context setup

Syntax

```
--timerName initContextSetup --timerValue <string> [ --timerUnit  
Second ]
```

Description

When the MME sends a message to the eNB for an initial context setup request, it expects to receive a response message from the eNB (Initial context setup response or initial context setup failure). If the timer expires while waiting for this response, the MME considers that this class 1 procedure failed.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

8

Name

internalIpIsolationThreshold - Internal IP isolation threshold

Syntax

```
--timerName internalIpIsolationThreshold --timerValue <string> [ --  
timerUnit Second ]
```

Description

Threshold to be used for determining internal IP isolation of VMs. For VNF and a8 appliance the default is 5. For CNF and a2 appliance the default is 0 and cannot be changed.

Type / Range

Integer in the range of 0 to 60 in units of Second

Default

5

Name

iotPurgeTimer - IoT Purge Timer

Syntax

```
--timerName iotPurgeTimer --timerValue <string> [ --timerUnit Hour ]
```

Description

This timer starts after the MME-initiated detach timer fires. If this timer fires before the IoT UE contacts the MME again, the UE VLR data is purged.

Type / Range

Integer in the range of 0 to 720 in units of Hour

Default

0

Name

ipdsRemHeartbeat - IPDS REM heartbeat

Syntax

```
--timerName ipdsRemHeartbeat --timerValue <string> [ --timerUnit  
Millisecond ]
```

Description

Configure REM heartbeat interval for IPDS service. VNF: Default is 200, Range is 200 to 20000 CNF: Default is 600, Range is 600 to 20000

Type / Range

Integer in the range of 200 to 20,000 in units of Millisecond

Default

200

Name

ippsRemHeartbeat - IPPS REM heartbeat

Syntax

```
--timerName ippsRemHeartbeat --timerValue <string> [ --timerUnit  
Millisecond ]
```

Description

Configure REM heartbeat interval for IPPS service. VNF: Default is 200, Range is 200 to 20000
CNF: Default is 600, Range is 600 to 20000

Type / Range

Integer in the range of 200 to 20,000 in units of Millisecond

Default

200

Name

iuUeLocationReport - IU UE location report

Syntax

```
--timerName iuUeLocationReport --timerValue <string> [ --timerUnit  
Millisecond ]
```

Description

Specifies the length of time SGSN waits for the receipt of a location report response from the RNC.

Type / Range

Integer in the range of 100 to 8,000 in units of Millisecond

Default

2000

Name

keepCtxAfterN26HoFail - Keep Context After N26 Handover Fail

Syntax

```
--timerName keepCtxAfterN26HoFail --timerValue <string> [ --timerUnit  
Second ]
```

Description

This timer starts when serving S1 Release by which the ongoing N26 handover is decided as failed at target MME. While the timer is running the MME discards (silent drop) any DDN (or other SM message) received from the SGW. When the timer expires, the MME shall remove resources associated with the N26 handover

Type / Range

Integer in the range of 0 to 10 in units of Second

Default

0

Name

lihsRemHeartbeat - LIHS REM Heartbeat

Syntax

```
--timerName lihsRemHeartbeat --timerValue <string> [ --timerUnit  
Millisecond ]
```

Description

Configure REM heartbeat interval for LIHS service. VNF: Default is 200, Range is 200 to 20000 CNF: Default is 600, Range is 600 to 20000

Type / Range

Integer in the range of 200 to 20,000 in units of Millisecond

Default

200

Name

mapM - Map m to HLR

Syntax

```
--timerName mapM --timerValue <string> [ --timerUnit Millisecond ]
```

Description

Defines the map m (medium) protocol when interfacing to an HLR.

Type / Range

Integer in the range of 15,000 to 30,000 in units of Millisecond

Default

20000

Name

mbmsResponseFromMce - MBMS response from MCE

Syntax

```
--timerName mbmsResponseFromMce --timerValue <string> [ --timerUnit  
Second ]
```

Description

MBMS response from MCE.

Type / Range

Integer in the range of 1 to 30 in units of Second

Default

5

Name

mBReachable - Timer beyond T3412

Syntax

```
--timerName mBReachable --timerValue <string> [ --timerUnit Minute ]
```

Description

Minutes beyond T3412 for taking actions when a UE does not send TAU. Timer stops when a NAS message is received.

Type / Range

Integer in the range of 1 to 60 in units of Minute

Default

4

Name

minTimeToMbmsDataTransfer - Min time to MBMS data transfer

Syntax

```
--timerName minTimeToMbmsDataTransfer --timerValue <string> [ --  
timerUnit Second ]
```

Description

Min time to MBMS data transfer.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

1

Name

mmeCnfgUpdate - MME config update

Syntax

```
--timerName mmeCnfgUpdate --timerValue <string> [ --timerUnit  
Second ]
```

Description

When the MME sends a configuration update message to the eNB, it expects to receive an update acknowledgment response message from the eNB. If the timer expires while waiting for this response, the MME considers that this class 1 procedure failed.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

8

Name

mmelInitDetach - MME Initiated Detach

Syntax

```
--timerName mmeInitDetach --timerValue <string> [ --timerUnit  
Minute ]
```

Description

Timer starts when 'Mobile Reachable' timer expires. Timer stops when a NAS message is received.

Type / Range

Integer in the range of 0 to 1,500 in units of Minute

Default

240

Name

mobileReachableLpa - Mobile reachable LPA

Syntax

```
--timerName mobileReachableLpa --timerValue <string> [ --timerUnit  
Minute ]
```

Description

Mobile reachable time for the LPA UE.

Type / Range

Integer in the range of 0 to 3,600 in units of Minute

Default

180

Name

moDataAnswer - MO data answer

Syntax

```
--timerName moDataAnswer --timerValue <string> [ --timerUnit  
Millisecond ]
```

Description

Time to wait for ODA after sending ODR (MO-Data-Request).

Type / Range

Integer in the range of 100 to 6,000 in units of Millisecond

Default

1000

Name

monitorArrivalDownlinkData - Monitor arrival downlink data

Syntax

```
--timerName monitorArrivalDownlinkData --timerValue <string> [ --  
timerUnit Millisecond ]
```

Description

Monitor arrival of downlink data if downlink data is expected.

Type / Range

Integer in the range of 200 to 10,000 in units of Millisecond

Default

2000

Name

mtDataAnswer - MT data answer

Syntax

```
--timerName mtDataAnswer --timerValue <string> [ --timerUnit
Millisecond ]
```

Description

This timer is used for responding to SCEF with TDA when SCEF sends TDR and does not include SCEF-Wait-Time AVP.

Type / Range

Integer in the range of 100 to 30,000 in units of Millisecond

Default

12000

Name

mtDataRequest - MT data request

Syntax

```
--timerName mtDataRequest --timerValue <string> [ --timerUnit
Millisecond ]
```

Description

This timer is started to wait for MT data request before releasing the S1 connection. this timer is started immediately after sending connection-management-request (CMR) with connection-action AVP set to CONNECTION-UPDATE.

Type / Range

Integer in the range of 0 to 10,000 in units of Millisecond

Default

1000

Name

n26rmUeDataAtSrcMme - N26 remove Ue Data at source MME

Syntax

```
--timerName n26rmUeDataAtSrcMme --timerValue <string> [ --timerUnit
Millisecond ]
```

Description

N26 remove Ue Data at source MME

Type / Range

Integer in the range of 500 to 5,000 in units of Millisecond

Default

2500

Name

nwdsRemHeartbeat - NWDS REM heartbeat

Syntax

```
--timerName nwdsRemHeartbeat --timerValue <string> [ --timerUnit  
Millisecond ]
```

Description

Configure REM heartbeat interval for NWDS service. Applicable only for CNF. Default is 600.
Range is 600 to 20000.

Type / Range

Integer in the range of 600 to 20000 in units of Millisecond

Default

600

Name

pagingGapTimer - Paging gap timer

Syntax

```
--timerName pagingGapTimer --timerValue <string> [ --timerUnit  
Minute ]
```

Description

This timer indicates the interval in which the MME will suppress paging for a UE after multiple consecutive page response failures have occurred for the UE. a value of 0 indicates that paging should not be suppressed due to previous page response failures.

Type / Range

Integer in the range of 0 to 1,440 in units of Minute

Default

0

Name

papsRemHeartbeat - PAPS REM heartbeat

Syntax

```
--timerName papsRemHeartbeat --timerValue <string> [ --timerUnit  
Millisecond ]
```

Description

Configure REM heartbeat interval for PAPS service.

Type / Range

Integer in the range of 1,000 to 20,000 in units of Millisecond

Default

1000

Name

pcmsRemHeartbeat - PCMS REM Heartbeat

Syntax

```
--timerName pcmsRemHeartbeat --timerValue <string> [ --timerUnit  
Millisecond ]
```

Description

Configure REM heartbeat interval for PCMS service. VNF: Default is 200, Range is 200 to 20000 CNF: Default is 600, Range is 600 to 20000

Type / Range

Integer in the range of 200 to 20,000 in units of Millisecond

Default

200

Name

pdnConnectAfterEcmIdle - PDN connect after ECM idle

Syntax

```
--timerName pdnConnectAfterEcmIdle --timerValue <string> [ --  
timerUnit Second ]
```

Description

Defines how long the subscriber needs to be in ECM-IDLE state before the PDN re-connection is initiated.

Type / Range

Integer in the range of 0 to 120 in units of Second

Default

2

Name

pgwIsolationDeclarationTimer - PGW isolation declaration timer

Syntax

```
--timerName pgwIsolationDeclarationTimer --timerValue <string> [ --  
timerUnit Second ]
```

Description

If the number of the P-GW faults (remote peer not responding cause received in the S11/S4 create session response) exceed the provisioned threshold in this timer, MME will isolate the S-GW/P-GW pair that reported this fault for new PDN connections for a configurable isolation duration. changes to the timer will take into effect whenever MME checks whether total faults exceeded the fault threshold in the timer that is either triggered by a new fault or some other event.

Type / Range

Integer in the range of 1 to 600 in units of Second

Default

3

Name

pgwIsolationDuration - PGW isolation duration

Syntax

```
--timerName pgwIsolationDuration --timerValue <string> [ --timerUnit  
Minute ]
```

Description

This command is used to provision the duration of PGW isolation once it is declared that the PGW is isolated for any new PDN connections. WMM will start selecting the PGW once the duration expires. This duration can be stopped and PGW selection can be resumed by using the sig-link CLI command

Type / Range

Integer in the range of 1 to 3,600 in units of Minute

Default

15

Name

pgwRestartRecovery - PGW restart recovery

Syntax

```
--timerName pgwRestartRecovery --timerValue <string> [ --timerUnit  
Minute ]
```

Description

It is used to provision a duration in which MME would attempt to re-establish PDN connections associated with a PGW indicated by the PGW restart notification message of the ECM-IDLE UE. MME would start the timer when it starts the recovery. once the timer expires, MME will stop the recovery and deletes the PDN session of the left over UEs.

Type / Range

Integer in the range of 30 to 300 in units of Minute

Default

60

Name

praDcConnectedIn - PRA Dual Connectivity Connected In

Syntax

```
--timerName praDcConnectedIn --timerValue <string> [ --timerUnit  
Millisecond ]
```

Description

Specifies the timer for Segment Based Dual Connectivity PRA reporting functionality, during which the MME suppresses the PRA status change report when the UE is in ECM-CONNECTED state and the last reported PRA status was 'inside'. The timer helps the MME to skip sequential back and forth status updates related to the segment based reporting. If MME receives new radio transport layer endpoint(s) for user plane, it waits for possible subsequential next update or UE context release before sending the PRA report towards the SGW/PGW.

Type / Range

Integer in the range of 0 to 60000 in units of Millisecond

Default

0

Name

praDcConnectedOut - PRA Dual Connectivity Connected Out

Syntax

```
--timerName praDcConnectedOut --timerValue <string> [ --timerUnit  
Millisecond ]
```

Description

Specifies the timer for Segment Based Dual Connectivity PRA reporting functionality, during which the MME suppresses the PRA status change report when the UE is in ECM-CONNECTED state and the last reported PRA status was 'outside'. The timer helps the MME to skip sequential back and forth status updates related to the segment based reporting. If MME receives new radio transport layer endpoint(s) for user plane, it waits for possible subsequential next update or UE context release before sending the PRA report towards the SGW/PGW.

Type / Range

Integer in the range of 0 to 60000 in units of Millisecond

Default

0

Name

praDcIdle - PRA Dual Connectivity Idle

Syntax

```
--timerName praDcIdle --timerValue <string> [ --timerUnit  
Millisecond ]
```

Description

Defines a timer for Presence Reporting Area (PRA) functionality for optimizing Dual Connectivity related reporting when the UE was previously ECM-IDLE. MME starts this timer when the UE was ECM-IDLE previously and the MME receives new RAN endpoints (receives Initial Context Setup Response during Service Request, Control Plane Service Request or Intra MME TAU). MME stops this timer when it receives either new RAN endpoints (Path Switch

Request, Handover Request Ack, Initial Context Setup Response or E-RAB modification indication) or UE context release complete. MME sends PRA Information towards the SGW/PGW when timer either expires or was stopped and reporting is required to be done i.e. status differs from previous reported and the UE is still ECM-CONNECTED.

Type / Range

Integer in the range of 0 to 10,000 in units of Millisecond

Default

0

Name

praDcSegmentOptServReq - PRA DC Segment Optimization Service Request

Syntax

```
--timerName praDcSegmentOptServReq --timerValue <string> [ --  
timerUnit Millisecond ]
```

Description

Specifies the timer for IPv6 Address Segment Based Dual Connectivity PRA reporting feature and the Service Request optimization functionality. The timer helps the MME to skip any sequential back and forth status updates related to the segment based reporting related to the idle to connected transferring and the possible subsequent gNB addition. When MME receives a Service Request message for a UE for which address segment based PRA reporting has previously been requested, the MME waits during this timer for a subsequent E-RAB modification indication before sending any PRA status update towards the SGW/PGW/PCRF, if such would still be needed due to the Service request. If the timer is running and the MME receives an E-RAB Modification Indication, the MME stops the timer. The MME reports the segment based status at either the receipt of the E-RAB modification indication, or the timer expiry (if status at that point is different from the previously sent status).

Type / Range

Integer in the range of 0 to 10,000 in units of Millisecond

Default

0

Name

praDcSingleTimer - PRA DC Single Timer

Syntax

```
--timerName praDcSingleTimer --timerValue <string> [ --timerUnit  
Millisecond ]
```

Description

Specifies the single optimisation timer for Segment Based Dual Connectivity PRA reporting functionality, during which the MME suppresses the PRA status change reports. The MME starts the timer when the timer is not yet running and the MME receives new RAN endpoints that would trigger status update for any PRA id that is reported for the UE. The timer is

stopped when the MME receives new RAN endpoints and the PRA status of all UE's PRA ids is back to the previously reported status, or when the UE context is released. The MME restarts the timer when the MME receives new RAN endpoints that cause the status of a different PRA id to be updated to a status that is not yet reported.

Type / Range

Integer in the range of 0 to 60000 in units of Millisecond

Default

0

Name

praStdSuppressionCancel - PRA Standard Suppression Cancel

Syntax

```
--timerName praStdSuppressionCancel --timerValue <string> [ --  
timerUnit Millisecond ]
```

Description

PRA Suppression cancel timer for 3GPP standard based Presence Area Reporting. Effective when PRA standard suppression is set to Yes. The functionality includes two time definitions- PRA suppression time and PRA suppression cancel time. MME suppresses a subsequent PRA report after a PRA report (PRA Information) has been sent out towards the SGW/PGW (during the time defined by the PRA suppression time), unless the PRA reporting status stays the same for a suppress cancel defined time period. When more time than the provisioned PRA suppression time has passed since the last delivered PRA report, the MME sends any PRA report to SGW/PGW and stores the timestamp of the report sending. If MME detects next PRA status change before the provisioned suppression time has passed since previous PRA report, MME (suppresses the report for a while and) starts the suppression cancelling timer to wait for another status change. If MME detects another status change before the suppression cancel timer expires, the MME discards the report since the status is back to the same as already reported. If the suppression cancel timer expires, MME sends the latest (suppressed) PRA report towards the SGW/PGW. The timer can be provisioned with 100ms granularity.

Type / Range

Integer in the range of 100 to 600,000 in units of Millisecond

Default

500

Name

praStdSuppressionPeriod - PRA Standard Suppression Period

Syntax

```
--timerName praStdSuppressionPeriod --timerValue <string> [ --  
timerUnit Millisecond ]
```

Description

Defines the time period during which the MME suppresses a subsequent PRA report after a PRA report (PRA Information) has been sent out towards the SGW/PGW, unless the PRA status stays the same during a time period defined by the suppression cancel time. The timer can be provisioned with 100ms granularity.

Type / Range

Integer in the range of 100 to 600,000 in units of Millisecond

Default

1000

Name

purgeTimer - Purge timer

Syntax

```
--timerName purgeTimer --timerValue <string> [ --timerUnit Hour ]
```

Description

This timer starts after the MME-initiated detach timer fires. If this timer fires before the UE contacts the MME again, the UE VLR data is purged.

Type / Range

Integer in the range of 1 to 120 in units of Hour

Default

48

Name

rcisPurgeTimer - RCIS Purge Timer

Syntax

```
--timerName rcisPurgeTimer --timerValue <string> [ --timerUnit Hour ]
```

Description

RCIS Purge Timer

Type / Range

Integer in the range of 1 to 720 in units of Hour

Default

120

Name

rcisRemHeartbeat - RCIS REM Heartbeat

Syntax

```
--timerName rcisRemHeartbeat --timerValue <string> [ --timerUnit  
Millisecond ]
```

Description

Configure REM heartbeat interval for RCIS service. VNF: Default is 200, Range is 200 to 20000
CNF: Default is 600, Range is 600 to 20000

Type / Range

Integer in the range of 200 to 20,000 in units of Millisecond

Default

200

Name

receiveNotificationReturnFromUe - Receive notification return from UE

Syntax

```
--timerName receiveNotificationReturnFromUe --timerValue <string>
[ --timerUnit Second ]
```

Description

Receive notification return from UE.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

9

Name

reserveUeSubscriptionIratMob - Reserve UE Subscription IRAT Mobility

Syntax

```
--timerName reserveUeSubscriptionIratMob --timerValue <string> [ --
timerUnit Minute ]
```

Description

This command is used to provision the duration of preserving UE subscription data. Timer shall be restarted upon UE transitioning back to MME. UE's subscription data shall be kept until the timer expires, however if UE comes back before timer expiration, NOR instead of ULR shall be sent to HSS in TAU procedure.

Type / Range

Integer in the range of 1 to 1,440 in units of Minute

Default

360

Name

s102tack - S102 tack

Syntax

```
--timerName s102tack --timerValue <string> [ --timerUnit
Millisecond ]
```

Description

S102 tack.

Type / Range

Integer in the range of 100 to 60,000 in units of Millisecond

Default

1000

Name

s10rmUeDataAtSrcMme - S10 remove UE data at source MME

Syntax

```
--timerName s10rmUeDataAtSrcMme --timerValue <string> [ --timerUnit  
Millisecond ]
```

Description

Used in S10 interactions to remove UE data at the source MME after a handover with MME relocation.

Type / Range

Integer in the range of 0 to 5,000 in units of Millisecond

Default

0

Name

s1hoComplete - S1 HO complete

Syntax

```
--timerName s1hoComplete --timerValue <string> [ --timerUnit Second ]
```

Description

Used to time the receipt of either a handover notify message from the target eNB or a forward relocation complete message from the target MME in the case of MME relocation. the handover is not successful if either of these messages is not received before the expiration of the timer.

Type / Range

Integer in the range of 1 to 10 in units of Second

Default

5

Name

s1hoCompleteExtended - Extended S1 HO complete

Syntax

```
--timerName s1hoCompleteExtended --timerValue <string> [ --timerUnit  
Second ]
```

Description

This timer is introduced to handle extra latency introduced by CIOT and CAT-M1 CE mode b devices in completing S1HO. for S1HO without MME relocation, the timer is used to monitor handover notify reception. for S1HO with MME relocation, this timer covers the extra latency to execute the handover, get the handover notify and receive forward relocation complete notification.

Type / Range

Integer in the range of 5 to 100 in units of Second

Default

5

Name

s1hoResourceRelease - S1 HO resource release

Syntax

```
--timerName s1hoResourceRelease --timerValue <string> [ --timerUnit  
Millisecond ]
```

Description

During a relocation request (SGW only or MME and SGW), the timer starts upon receipt of a forward relocation complete acknowledge. if the release does not occur before the timer expires, the target MME sends a delete indirect data forwarding message to the target SGW to delete it if the target MME has setup indirect forward paths on the target SGW. the source MME deletes any bearers on the source SGW (with SGW relocation) and deletes the tunnel by sending a delete indirect data forwarding message if the source MME has setup indirect forward paths on the source SGW. the source MME also releases S1 UE connections with the source eNB.

Type / Range

Integer in the range of 100 to 5,000 in units of Millisecond

Default

2000

Name

s1hoResourceReleaseWithNoMmeSgwRel - S1 Handover Resource Release With No MME SGW Relocation

Syntax

```
--timerName s1hoResourceReleaseWithNoMmeSgwRel --timerValue <string>  
[ --timerUnit Millisecond ]
```

Description

During intra-mme intra-sgw s1-based ho request, the timer starts after Handover Notify message and once the timer expires, MME shall send UE Context Release Command to the source eNB and the source eNodeB releases its resources related to the UE and responds with a UE Context Release Complete message. When the timer is set to 0, the S1 Handover Resource Release timer shall be used.

Type / Range

Integer value in the range of 0 or 200 to 3,000 in units of Millisecond

Default

0

Name

s1releaseOnCsgExpiration - S1 release on CSG expiration

Syntax

```
--timerName s1releaseOnCsgExpiration --timerValue <string> [ --  
timerUnit Millisecond ]
```

Description

This timer is to specify the time to wait to do an S1 release when CSG registration expires.

Type / Range

Integer in the range of 500 to 5,000 in units of Millisecond

Default

2000

Name

s1Reset - S1 reset

Syntax

```
--timerName s1Reset --timerValue <string> [ --timerUnit Second ]
```

Description

When the MME sends a S1 reset message to the eNB, it expects to receive a reset acknowledgment response message from the eNB. If the timer expires while waiting for this response, the MME considers that this class 1 procedure failed.

Type / Range

Integer in the range of 1 to 180 in units of Second

Default

45

Name

s1UeCapInfoIndForLtemWaitTime - s1UeCapInfoIndWaitTime

Syntax

```
--timerName s1UeCapInfoIndForLtemWaitTime --timerValue <string> [ --  
timerUnit Millisecond ]
```

Description

Specifies the timeout for the MME to receive the S1 UE Capability Information Indication message from the eNB. The timer applies if the MME needs to find out from the eNB whether the UE is capable of LTE-M. The timer is started when the MME sends the UE Capability Match Request to the eNB as part of an attach or TAU procedure.

Type / Range

Integer in the range of 100 to 9000 in units of Millisecond

Default

100

Name

s1ueLocationReport - S1 UE location report

Syntax

```
--timerName s1ueLocationReport --timerValue <string> [ --timerUnit  
Millisecond ]
```

Description

This timer specifies the length of time to wait for the receipt of a location report from the eNB.

Type / Range

Integer in the range of 100 to 8,000 in units of Millisecond

Default

2000

Name

s3gnHoComplete - S3 Gn HO complete

Syntax

```
--timerName s3gnHoComplete --timerValue <string> [ --timerUnit  
Millisecond ]
```

Description

Supports inter-rat handover procedures.

Type / Range

Integer in the range of 100 to 10,000 in units of Millisecond

Default

4000

Name

s3gnIndirectForwarding - S3 Gn indirect forwarding

Syntax

```
--timerName s3gnIndirectForwarding --timerValue <string> [ --  
timerUnit Millisecond ]
```

Description

Used to time for resource release for indirect forwarding of UE data. when the timer expires, the MME releases the resources that have been allocated for indirect forwarding.

Type / Range

Integer in the range of 100 to 5,000 in units of Millisecond

Default

4000

Name

scefCmrAnswer - SCEF CMR answer

Syntax

```
--timerName scefCmrAnswer --timerValue <string> [ --timerUnit  
Millisecond ]
```

Description

This timer is used for responding with connection management answer to SCEF for SCEF sent connection management request (CMR) for an update. the timer applies to SCEF CMR updates other than T6a connection release. CMR for release from SCEF answer uses connectionmanagementanwer timer.

Type / Range

Integer in the range of 100 to 30,000 in units of Millisecond

Default

12000

Name

scefExpiredTimeReject - scefExpiredTimeReject

Syntax

```
--timerName scefExpiredTimeReject --timerValue <string> [ --timerUnit  
Second ]
```

Description

If the MME receives a T6a (SCEF) MT Data Request that contains the Maximum-Retransmission-Time AVP and/or SCEF-Wait-Time AVP, this timer defines how much in the past each AVP value can be in order for the MME to consider it as a still valid AVP. The default is 30 seconds to align with the prior hard-coded value.

Type / Range

Integer in the range of 1 to 120 in units of Second

Default

30

Name

secondaryRatUsageReportingDelay - Secondary Rat Usage Reporting Delay during Handover

Syntax

```
--timerName secondaryRatUsageReportingDelay --timerValue <string>  
[ --timerUnit Second ]
```

Description

Timer to delay forwarding the Secondary RAT Usage report to the S/PGW during Handover

Type / Range

Integer in the range of 1 to 30 in units of Second

Default

10

Name

serviceGapTimerOverrideValue - Service Gap Timer Override Value

Syntax

```
--timerName serviceGapTimerOverrideValue --timerValue <string> [ --  
timerUnit Second ]
```

Description

Specifies the timer value to use for the UE's service gap timer

Type / Range

Integer in the range of 0 to 65,535 in units of Second

Default

0

Name

sessionOrientedLinkMoDeletionTimer - Session oriented link MO deletion timer

Syntax

```
--timerName sessionOrientedLinkMoDeletionTimer --timerValue <string>  
[ --timerUnit Hour ]
```

Description

Session oriented link MO deletion timer for GTP based interfaces. value of 0 means do not do inactivity deletion.

Type / Range

Integer in the range of 0 to 168 in units of Hour

Default

24

Name

sfdsRemHeartbeat - SFDS REM Heartbeat

Syntax

```
--timerName sfdsRemHeartbeat --timerValue <string> [ --timerUnit  
Millisecond ]
```

Description

Configure REM heartbeat interval for SCTP Flow Distributor Service. VNF: N/A CNF: Default is 600, Range is 600 to 20000

Type / Range

Integer in the range of 200 to 20,000 in units of Millisecond

Default

600

Name

sgsLiteDetachIndicationAck - SGs lite detach indication ack

Syntax

```
--timerName sgsLiteDetachIndicationAck --timerValue <string> [ --  
timerUnit Millisecond ]
```

Description

Configure SGsLite-IMSI-DETACH-ACK timeout in milliseconds.

Type / Range

Integer in the range of 100 to 60,000 in units of Millisecond

Default

1000

Name

sgsLiteLocationUpdateAck - SGs lite location update ack

Syntax

```
--timerName sgsLiteLocationUpdateAck --timerValue <string> [ --  
timerUnit Millisecond ]
```

Description

Configure timer for the reception of SGsLite-LOCATION-UPDATE-ACK timer.

Type / Range

Integer in the range of 100 to 60,000 in units of Millisecond

Default

1000

Name

sgsLiteUplinkDataAck - SGs lite uplink data ack

Syntax

```
--timerName sgsLiteUplinkDataAck --timerValue <string> [ --timerUnit  
Millisecond ]
```

Description

Configure SGsLite-UPLINK-ACK timeout.

Type / Range

Integer in the range of 100 to 60,000 in units of Millisecond

Default

1000

Name

sgsPaging - SGs paging

Syntax`--timerName sgsPaging --timerValue <string> [ --timerUnit Second ]`**Description**

Defines the value for receiving responses to a SG paging request. this timer should be set slightly lower than the paging timer value provisioned in the MSC/VLR.

Type / Range

Integer in the range of 1 to 25 in units of Second

Default

6

Name

sgwIsolationDeclarationTimer - SGW isolation declaration timer

Syntax`--timerName sgwIsolationDeclarationTimer --timerValue <string> [ --timerUnit Second ]`**Description**

This command is used to provision SGW isolation declaration timer. if the number of the SGW faults (remote peer not responding cause received in the S11/S4 create session response) exceed the provisioned threshold in this provisioned timer then WMM will isolate the SGW from the selection. a sliding window will be used in counting the total number faults for the isolation declaration duration.

Type / Range

Integer in the range of 1 to 600 in units of Second

Default

3

Name

sgwIsolationDuration - SGW isolation duration

Syntax`--timerName sgwIsolationDuration --timerValue <string> [ --timerUnit Minute ]`**Description**

This command is used to provision the duration of SGW isolation once it is declared that the SGW is isolated from selection. WMM will start selecting the SGW once the duration expires. This duration can be stopped and PGW selection can be resumed by using the sig-link CLI command

Type / Range

Integer in the range of 1 to 3,600 in units of Minute

Default

15

Name

sgwRestorationTimer - SGW restoration timer

Syntax

```
--timerName sgwRestorationTimer --timerValue <string> [ --timerUnit  
Minute ]
```

Description

This timer is used to guard the MME attempt to assign UE associated with a failed SGW or restarted SGW.

Type / Range

Integer in the range of 1 to 180 in units of Minute

Default

60

Name

srnsCompletion - SRNS completion

Syntax

```
--timerName srnsCompletion --timerValue <string> [ --timerUnit  
Millisecond ]
```

Description

Used to time for receipt of an SRNS completion message from the target SGSN.

Type / Range

Integer in the range of 100 to 10,000 in units of Millisecond

Default

4000

Name

srnsHoRauRequestWait - SRNS HO RAU request wait

Syntax

```
--timerName srnsHoRauRequestWait --timerValue <string> [ --timerUnit  
Millisecond ]
```

Description

Specify how long 3G-SGSN waits for a RAU request after SRNS relocation or 4G to 3G HO.

Type / Range

Integer in the range of 100 to 10,000 in units of Millisecond

Default

200

Name

srsDataPurge - SRS Data Purge

Syntax`--timerName srsDataPurge --timerValue <string> [ --timerUnit Minute ]`**Description**

SRS Data Purge time

Type / Range

Integer in the range of 30 to 7,200 in units of Minute

Default

2880

Name

srsResponse - SRS response

Syntax`--timerName srsResponse --timerValue <string> [ --timerUnit Second ]`**Description**

SRS response time.

Type / Range

Integer in the range of 1 to 30 in units of Second

Default

6

Name

srvccPsToCsComplete - SRVCC PS to CS complete

Syntax`--timerName srvccPsToCsComplete --timerValue <string> [ --timerUnit Millisecond ]`**Description**

Time to wait for the SRVCC PS to CS complete message from an MSC.

Type / Range

Integer in the range of 500 to 15,000 in units of Millisecond

Default

4000

Name

srvccTo1xHoComplete - SRVCC to 1x HO complete

Syntax

```
--timerName srvccTo1xHoComplete --timerValue <string> [ --timerUnit  
Millisecond ]
```

Description

This is guard timer that is started when S102 signaling with HO status IE (SUCCESS) is received and shall be stopped when S1 UE context release request with cause CIOT cell redirection is received.

Type / Range

Integer in the range of 1 to 17,407 in units of Millisecond

Default

12000

Name

srvccTo1xHoPreparation - SRVCC to 1x HO preparation

Syntax

```
--timerName srvccTo1xHoPreparation --timerValue <string> [ --  
timerUnit Millisecond ]
```

Description

This is a guard timer that is started when S1 UL CDMA2000 tunneling with HO required IE is received and stopped when S102 Signaling with HO status IE is received.

Type / Range

Integer in the range of 1 to 8,192 in units of Millisecond

Default

5000

Name

suspendReq - Suspend request

Syntax

```
--timerName suspendReq --timerValue <string> [ --timerUnit  
Millisecond ]
```

Description

Used in an EUTRAN to GERAN CS handover procedure to time for the receipt of the suspend request message from the SGSN in the network to which the UE has moved.

Type / Range

Integer in the range of 500 to 10,000 in units of Millisecond

Default

2000

Name

t3346attachRejectMaximum - T3346 attach reject maximum

Syntax

```
--timerName t3346attachRejectMaximum --timerValue <string> [ --  
timerUnit Second ]
```

Description

Specifies the maximum value to consider when randomly assigning the T3346 timer value to be used for an attach procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).

Type / Range

Integer in the range of 0 to 600 in units of Second

Default

60

Name

t3346attachRejectMinimum - T3346 attach reject minimum

Syntax

```
--timerName t3346attachRejectMinimum --timerValue <string> [ --  
timerUnit Second ]
```

Description

Specifies the minimum value to consider when randomly assigning the T3346 timer value to be used for an attach procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).

Type / Range

Integer in the range of 0 to 600 in units of Second

Default

15

Name

t3346gtpVolEsrMax - T3346 GTP VOL ESR max

Syntax

```
--timerName t3346gtpVolEsrMax --timerValue <string> [ --timerUnit  
Second ]
```

Description

The maximum value to consider when randomly assigning the T3346 timer value to be used for an ESR procedure rejection due to GTP overload.

Type / Range

Integer in the range of 0 to 600 in units of Second

Default

60

Name

t3346gtpVolEsrMin - T3346 GTP VOL ESR min

Syntax

```
--timerName t3346gtpVolEsrMin --timerValue <string> [ --timerUnit  
Second ]
```

Description

The minimum value to consider when randomly assigning the T3346 timer value to be used for an SR procedure rejection due to GTP overload.

Type / Range

Integer in the range of 0 to 600 in units of Second

Default

15

Name

t3346gtpVolSrMax - T3346 GTP VOL SR max

Syntax

```
--timerName t3346gtpVolSrMax --timerValue <string> [ --timerUnit  
Second ]
```

Description

The maximum value to consider when randomly assigning the T3346 timer value to be used for an SR procedure rejection due to GTP overload.

Type / Range

Integer in the range of 0 to 600 in units of Second

Default

60

Name

t3346gtpVolSrMin - T3346 GTP VOL SR min

Syntax

```
--timerName t3346gtpVolSrMin --timerValue <string> [ --timerUnit  
Second ]
```

Description

The minimum value to consider when randomly assigning the T3346 timer value to be used for an SR procedure rejection due to GTP overload.

Type / Range

Integer in the range of 0 to 600 in units of Second

Default

15

Name

t3346gtpVolTauMax - T3346 GTP VOL TAU max

Syntax

```
--timerName t3346gtpVolTauMax --timerValue <string> [ --timerUnit  
Second ]
```

Description

The maximum value to consider when randomly assigning the T3346 timer value to be used for a TAU procedure rejection due to GTP overload.

Type / Range

Integer in the range of 0 to 600 in units of Second

Default

60

Name

t3346gtpVolTauMin - T3346 GTP VOL TAU min

Syntax

```
--timerName t3346gtpVolTauMin --timerValue <string> [ --timerUnit  
Second ]
```

Description

The minimum value to consider when randomly assigning the T3346 timer value to be used for a TAU procedure rejection due to GTP overload.

Type / Range

Integer in the range of 0 to 600 in units of Second

Default

15

Name

t3346NblotAttachRejectMaximum - T3346 NB-IOT Attach Reject Maximum

Syntax

```
--timerName t3346NbIotAttachRejectMaximum --timerValue <string> [ --  
timerUnit Second ]
```

Description

Specifies the maximum value to consider when randomly assigning the T3346 NB-IOT timer value to be used for an attach procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).

Type / Range

Integer in the range of 0 to 11,160 in units of Second

Default

60

Name

t3346NbIotAttachRejectMinimum - T3346 NB-IOT Attach Reject Minimum**Syntax**

```
--timerName t3346NbIotAttachRejectMinimum --timerValue <string> [ --  
timerUnit Second ]
```

Description

Specifies the minimum value to consider when randomly assigning the T3346 NB-IOT timer value to be used for an attach procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).

Type / Range

Integer in the range of 0 to 11,160 in units of Second

Default

15

Name

t3346NbIotCpsrRejectMaximum - T3346 NB-IOT Control Plane SR Reject Maximum

Syntax

```
--timerName t3346NbIotCpsrRejectMaximum --timerValue <string> [ --  
timerUnit Second ]
```

Description

Specifies the maximum value to consider when randomly assigning the T3346 NB-IOT timer value to be used for an Control Plane SR procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).

Type / Range

Integer in the range of 0 to 11,160 in units of Second

Default

60

Name

t3346NbIotCpsrRejectMinimum - T3346 NB-IOT Control Plane SR Reject Minimum

Syntax

```
--timerName t3346NbIotCpsrRejectMinimum --timerValue <string> [ --  
timerUnit Second ]
```

Description

Specifies the minimum value to consider when randomly assigning the T3346 NB-IOT timer value to be used for an Control Plane SR procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).

Type / Range

Integer in the range of 0 to 11,160 in units of Second

Default

15

Name

t3346NbIotOdbRejectMaximum - t3346 NB-IOT ODB Reject Maximum

Syntax

```
--timerName t3346NbIotOdbRejectMaximum --timerValue <string> [ --  
timerUnit Second ]
```

Description

Specifies the maximum value backoff timer value in Seconds randomly assigned to a NB-IoT UE with ODB trying to access the MME. This timer applies when the MME is provisioned to provide cause 22 to the UE (in the restNasMappingProfile).

Type / Range

Integer in the range of 0 to 11,160 in units of Second

Default

60

Name

t3346NbIotOdbRejectMinimum - t3346 NB-IOT ODB Reject Minimum

Syntax

```
--timerName t3346NbIotOdbRejectMinimum --timerValue <string> [ --  
timerUnit Second ]
```

Description

Specifies the minimum value backoff timer value in Seconds randomly assigned to a NB-IoT UE with ODB trying to access the MME. This timer applies when the MME is provisioned to provide cause 22 to the UE (in the restNasMappingProfile).

Type / Range

Integer in the range of 0 to 11,160 in units of Second

Default

15

Name

t3346NbIotSrRejectMaximum - T3346 NB-IOT SR Reject Maximum

Syntax

```
--timerName t3346NbIotSrRejectMaximum --timerValue <string> [ --  
timerUnit Second ]
```

Description

Specifies the maximum value to consider when randomly assigning the T3346 NB-IOT timer value to be used for an SR procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).

Type / Range

Integer in the range of 0 to 11,160 in units of Second

Default

60

Name

t3346NbIotSrRejectMinimum - T3346 NB-IOT SR Reject Minimum

Syntax

```
--timerName t3346NbIotSrRejectMinimum --timerValue <string> [ --  
timerUnit Second ]
```

Description

Specifies the minimum value to consider when randomly assigning the T3346 NB-IOT timer value to be used for an SR procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).

Type / Range

Integer in the range of 0 to 11,160 in units of Second

Default

15

Name

t3346NbIotTauRejectMaximum - T3346 NB-IOT TAU Reject Maximum

Syntax

```
--timerName t3346NbIotTauRejectMaximum --timerValue <string> [ --  
timerUnit Second ]
```

Description

Specifies the maximum value to consider when randomly assigning the T3346 NB-IOT timer value to be used for a TAU procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).

Type / Range

Integer in the range of 0 to 11,160 in units of Second

Default

60

Name

t3346NbIotTauRejectMinimum - T3346 NB-IOT TAU Reject Minimum

Syntax

```
--timerName t3346NbIotTauRejectMinimum --timerValue <string> [ --  
timerUnit Second ]
```

Description

Specifies the minimum value to consider when randomly assigning the T3346 NB-IOT timer value to be used for a TAU procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).

Type / Range

Integer in the range of 0 to 11,160 in units of Second

Default

15

Name

t3346srRejectMaximum - T3346 SR reject maximum

Syntax

```
--timerName t3346srRejectMaximum --timerValue <string> [ --timerUnit  
Second ]
```

Description

Specifies the maximum value to consider when randomly assigning the T3346 timer value to be used for an SR procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).

Type / Range

Integer in the range of 0 to 600 in units of Second

Default

60

Name

t3346srRejectMinimum - T3346 SR reject minimum

Syntax

```
--timerName t3346srRejectMinimum --timerValue <string> [ --timerUnit  
Second ]
```

Description

Specifies the minimum value to consider when randomly assigning the T3346 timer value to be used for an SR procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).

Type / Range

Integer in the range of 0 to 600 in units of Second

Default

15

Name

t3346tauRejectMaximum - T3346 TAU reject maximum

Syntax

```
--timerName t3346tauRejectMaximum --timerValue <string> [ --timerUnit  
Second ]
```

Description

Specifies the maximum value to consider when randomly assigning the T3346 timer value to be used for a TAU procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).

Type / Range

Integer in the range of 0 to 600 in units of Second

Default

60

Name

t3346tauRejectMinimum - T3346 TAU reject minimum

Syntax

```
--timerName t3346tauRejectMinimum --timerValue <string> [ --timerUnit  
Second ]
```

Description

Specifies the minimum value to consider when randomly assigning the T3346 timer value to be used for a TAU procedure rejection due to IPDS CPU overload, CPPS CPU overload, and CXN object overloads with NAS cause code #22 (Congestion).

Type / Range

Integer in the range of 0 to 600 in units of Second

Default

15

Name

t3396genmax - T3396 gen max

Syntax

```
--timerName t3396genmax --timerValue <string> [ --timerUnit  
{Minute,Second} ]
```

Description

Parameter t3396genmax indicates maximum value to consider when assigning the T3396gen timer value to be used when MME rejects an IMS PDN Connectivity request (standalone or attach) When the timer has value > 0, then whenever the MME rejects an IMS PDN Connectivity Request (standalone or Attach) due to not subscribed IMS APN (or IMS APN is subscribed only via the wildcard APN and GPARM is enabled) then the T3396 IE shall be included in PDN Connectivity Reject message to force the UE re-attempt only after the back off timer expires,

Type / Range

Integer in the range of 0 to 360 in units of Minute,Second

Default

0

Name

t3396genmin - T3396 gen min

Syntax

```
--timerName t3396genmin --timerValue <string> [ --timerUnit  
{Minute,Second} ]
```

Description

Parameter t3396genmin indicates minimum value to consider when assigning the T3396gen timer value to be used when MME rejects an IMS PDN Connectivity request (standalone or attach)

Type / Range

Integer in the range of 0 to 360 in units of Minute,Second

Default

0

Name

t3396gtpVolMax - T3396 GTP VOL max

Syntax

```
--timerName t3396gtpVolMax --timerValue <string> [ --timerUnit  
Second ]
```

Description

The maximum value to consider when randomly assigning the T3396 timer value to be used for an SM request rejection due to GTP overload.

Type / Range

Integer in the range of 0 to 600 in units of Second

Default

60

Name

t3396gtpVolMin - T3396 GTP VOL min

Syntax

```
--timerName t3396gtpVolMin --timerValue <string> [ --timerUnit  
Second ]
```

Description

The minimum value to consider when randomly assigning the T3396 timer value to be used for an SM request rejection due to GTP overload.

Type / Range

Integer in the range of 0 to 600 in units of Second

Default

15

Name

t3396max - T3396 max

Syntax`--timerName t3396max --timerValue <string> [ --timerUnit Second ]`**Description**

The maximum value to consider when randomly assigning the T3396 timer value to be used for an SM request rejection due to overload.

Type / Range

Integer in the range of 0 to 600 in units of Second

Default

60

Name

t3396min - T3396 min

Syntax`--timerName t3396min --timerValue <string> [ --timerUnit Second ]`**Description**

The minimum value to consider when randomly assigning the T3396 timer value to be used for an SM request rejection due to overload.

Type / Range

Integer in the range of 0 to 599 in units of Second

Default

15

Name

t3402 - T3402 reattach request

Syntax`--timerName t3402 --timerValue <string> [ --timerUnit Minute ]`**Description**

Part of UE attach accept message. used by UE to initiate the next attach or TAU procedure in case of a failure of a previous attach or TAU procedure.

Type / Range

Integer in the range of 1 to 31 in units of Minute

Default

12

Name

t3402AttachReject - T3402 Attach Reject**Syntax**

```
--timerName t3402AttachReject --timerValue <string> [ --timerUnit  
{DeciHour,Minute} ]
```

Description

Specifies the wait time for another attach request if the UEs attach counter has reached 5 to the Attach Reject message.

Type / Range

Integer in the range of 0 to 31 in units of DeciHour,Minute

Default

0

Name

t3412 - T3412 periodic TAU

Syntax

```
--timerName t3412 --timerValue <string> [ --timerUnit  
{2Second,DeciHour,Minute} ]
```

Description

Part of UE attach accept message. used by the UE for periodic TAU. the MME starts this timer when the UE state is changed to the ECM-IDLE state.

Type / Range

Integer in the range of 1 to 31 in units of 2Second,DeciHour,Minute

Default

20

Name

t3412extendedForLpaUe - T3412 extended for LPA UE

Syntax

```
--timerName t3412extendedForLpaUe --timerValue <string> [ --timerUnit  
{10Hour,10Minute,2Second,30Second,320Hour,Hour,Minute} ]
```

Description

Periodic TAU timer that MME sends to UE in attach accept message to a UE configured for low priority access.

Type / Range

Integer in the range of 1 to 31 in units of
10Hour,10Minute,2Second,30Second,320Hour,Hour,Minute

Default

18

Name

t3412withTauSuppression - T3412 with TAU suppression

Syntax

```
--timerName t3412withTauSuppression --timerValue <string> [ --  
timerUnit {2Second,DeciHour,Minute} ]
```

Description

This timer is used instead of the standard T3412 timer when TAU suppression has been triggered by the MME. a value of 0 indicates that the standard T3412 timer should be used even when the MME has triggered TAU suppression.

Type / Range

Integer in the range of 0 to 31 in units of 2Second,DeciHour,Minute

Default

0

Name

t3422 - T3422 retransmit detach request

Syntax

```
--timerName t3422 --timerValue <string> [ --timerUnit Second ]
```

Description

Retransmit detach request.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

6

Name

t3422Wbs1CeMode - T3422 for WB-S1 CE mode

Syntax

```
--timerName t3422Wbs1CeMode --timerValue <string> [ --timerUnit  
Second ]
```

Description

T3422 timer value for the maximum interval MME waits after sending a detach request before retry for UE in WB-S1 coverage enhancement (CE) mode.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

24

Name

t3442rejectEsrNoSrvcc - T3442 reject ESR no SRVCC

Syntax

```
--timerName t3442rejectEsrNoSrvcc --timerValue <string> [ --timerUnit  
Second ]
```

Description

T3442 timer to be included in the service reject message when ESR is rejected due to UE with IMS voice bearer.

Type / Range

Integer in the range of 0 to 60 in units of Second

Default

0

Name

t3448Max - T3448 Maximum

Syntax

```
--timerName t3448Max --timerValue <string> [ --timerUnit Minute ]
```

Description

Specifies the maximum value of control plane data back-off timer. If the UE supports the control plane data back-off timer T3448, the UE shall set the CP backoff bit to back-off timer for transport of user data via the control plane supported in the UE network capability IE of the ATTACH REQUEST message.

Type / Range

Integer in the range of 15 to 30 in units of Minute

Default

30

Name

t3448Min - T3448 Minimum

Syntax

```
--timerName t3448Min --timerValue <string> [ --timerUnit Minute ]
```

Description

Specifies the minimum value of control plane data back-off timer. If the UE supports the control plane data back-off timer T3448, the UE shall set the CP backoff bit to back-off timer for transport of user data via the control plane supported in the UE network capability IE of the ATTACH REQUEST message.

Type / Range

Integer in the range of 15 to 30 in units of Minute

Default

15

Name

t3450 - T3450 retransmit attach and TAU accept or GUTI reallocation

Syntax

```
--timerName t3450 --timerValue <string> [ --timerUnit Second ]
```

Description

Retransmit attach accept, TAU accept, or GUTI reallocation command. timer starts if an attach request is accepted by the network and the MME sends an ATTACH ACCEPT message to the UE. timer stops after receiving an ATTACH COMPLETE message and considers the GUTI sent in the ATTACH ACCEPT message as valid.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

6

Name

t3450Wbs1CeMode - T3450 for WB-S1 CE mode

Syntax

```
--timerName t3450Wbs1CeMode --timerValue <string> [ --timerUnit  
Second ]
```

Description

T3450 timer which is used to supervise the retransmit of attach accept, TAU accept, or GUTI reallocation commands of UE in WB-S1 coverage enhancement (CE) mode.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

18

Name

t3460 - T3460 retransmit auth or sec mode command

Syntax

```
--timerName t3460 --timerValue <string> [ --timerUnit Second ]
```

Description

Retransmit auth request, or security mode command. timer starts when a security mode command is sent. normal stop occurs when security mode complete or security mode reject is received.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

6

Name

t3460Wbs1CeMode - T3460 for WB-S1 CE mode

Syntax

```
--timerName t3460Wbs1CeMode --timerValue <string> [ --timerUnit  
Second ]
```

Description

T3460 timer which is used to supervise the response to the transmission by the MME of the authentication request or security mode command for UE in WB-S1 coverage enhancement(ce) mode.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

24

Name

t3470 - T3470 retransmit identity command

Syntax

```
--timerName t3470 --timerValue <string> [ --timerUnit Second ]
```

Description

Retransmit identity command. timer starts when IDENTITY REQUEST is received. timer stops when IDENTITY REQUEST is received.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

6

Name

t3470Wbs1CeMode - T3470 for WB-S1 CE mode

Syntax

```
--timerName t3470Wbs1CeMode --timerValue <string> [ --timerUnit  
Second ]
```

Description

T3470 timer which is used to supervise the network initiated identification procedure for UE in WB-S1 coverage enhancement (CE) mode.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

24

Name

t3485 - T3485 retransmit activate def or ded bearer request

Syntax

```
--timerName t3485 --timerValue <string> [ --timerUnit Second ]
```

Description

Retransmit activate default bearer request, or activate dedicated bearer request. timer starts when an ACTIVATE DEFAULT/ DEDICATED EPS BEARER CONTEXT REQUEST is sent. timer stops when ACTIVATE DEFAULT EPS BEARER CONTEXT ACCEPT/REJECT or ACTIVATE DEDICATED EPS BEARER CONTEXT ACCEPT/REJECT is received.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

6

Name

t3485Wbs1CeMode - T3485 for WB-S1 CE mode

Syntax

```
--timerName t3485Wbs1CeMode --timerValue <string> [ --timerUnit  
Second ]
```

Description

T3485 timer which is used to supervise the response to the transmission by the MME of the activate default bearer request, or activate dedicated bearer request for UE in WB-S1 coverage enhancement (CE) mode.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

16

Name

t3486 - T3486 retransmit mod bearer request

Syntax

```
--timerName t3486 --timerValue <string> [ --timerUnit Second ]
```

Description

Retransmit modify bearer request.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

6

Name

t3486Wbs1CeMode - T3486 for WB-S1 CE mode

Syntax

```
--timerName t3486Wbs1CeMode --timerValue <string> [ --timerUnit
Second ]
```

Description

T3486 timer which is used to supervise the response to the transmission by the MME of the modify EPS bearer context request for UE in WB-S1 coverage enhancement (CE) mode.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

16

Name

t3489 - T3489 retransmit ESM info request timeout

Syntax

```
--timerName t3489 --timerValue <string> [ --timerUnit Second ]
```

Description

Timeout before retransmission of ESM information request.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

4

Name

t3489Wbs1CeMode - T3489 for WB-S1 CE mode

Syntax

```
--timerName t3489Wbs1CeMode --timerValue <string> [ --timerUnit
Second ]
```

Description

T3489 timer which is used to supervise the response to the transmission by the MME of the ESM information request for UE in WB-S1 coverage enhancement (CE) mode.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

12

Name

t3495 - T3495 retransmit deact bearer request

Syntax

```
--timerName t3495 --timerValue <string> [ --timerUnit Second ]
```

Description

Retransmit deactivate bearer request. timer starts when DEACTIVATE EPS BEARER CONTEXT REQUEST is sent. timer stops when DEACTIVATE EPS BEARER CONTEXT ACCEPT is received.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

6

Name

t3495Wbs1CeMode - T3495 for WB-S1 CE mode

Syntax

```
--timerName t3495Wbs1CeMode --timerValue <string> [ --timerUnit  
Second ]
```

Description

T3495 timer which is used to supervise the EPS bearer context deactivation procedure for UE in WB-S1 coverage enhancement (CE) mode.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

16

Name

t3x01 - T3X01 receive location response from E-SMLC via SIs timer

Syntax

```
--timerName t3x01 --timerValue <string> [ --timerUnit Second ]
```

Description

Timer to receive location response from the E-SMLC via SLs. (timer to transition from location state to LCS-IDLE state for a UE).

Type / Range

Integer in the range of 1 to 180 in units of Second

Default

8

Name

t3x02 - T3X02 receive reset ack after send reset to E-SMLC timer

Syntax

```
--timerName t3x02 --timerValue <string> [ --timerUnit Second ]
```

Description

Timer to receive reset ack after sending reset to the E-SMLC (SLs).

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

3

Name

tauAfterHo - TAU after HO

Syntax

```
--timerName tauAfterHo --timerValue <string> [ --timerUnit Second ]
```

Description

This command is used to set the value of the tauAfterHo timer which supervises the exchange between the MME and UE after handover occurs from LTE S1, UTRAN or GERAN. The MME detaches the UE if the timer expires before the TAU Request is Received. This timer only applies when a TAU is expected, e.g. if the TAI changed in an LTE-to-LTE HO. Caution! this value should only be set to '0' for troubleshooting purposes. The '0' value is not recommended by Nokia for service operation. If this value is set to '0', the MME does not start the timer, does not detach the UE, and does not start a new AKA as part of the handover procedure. this can generate security vulnerabilities in the EPS that a UE can exploit.

Type / Range

Integer in the range of 0 to 10 in units of Second

Default

2

Name

tauAfterN26Ho - TAU after N26 HO

Syntax

```
--timerName tauAfterN26Ho --timerValue <string> [ --timerUnit  
Second ]
```

Description

This timer supervises the exchange between the MME and UE after handover occurs from NR. The MME detaches the UE if the timer expires before the TAU Request is Received. Caution! this value should only be set to '0' for troubleshooting purposes. The '0' value is not recommended by Nokia for service operation. If this value is set to '0', the MME does not start the timer, does not detach the UE, and does not start a new AKA as part of the handover procedure. This can generate security vulnerabilities in the EPS that a UE can exploit.

Type / Range

Integer in the range of 0 to 10 in units of Second

Default

2

Name

tauSuppressionThreshold - TAU suppression threshold

Syntax

```
--timerName tauSuppressionThreshold --timerValue <string> [ --  
timerUnit Second ]
```

Description

This timer indicates the maximum interval for cyclic movement between tracking areas that will trigger TAU suppression by MME. a value of 0 indicates that TAU suppression is performed without regard to the interval of the cyclic movement between tracking areas.

Type / Range

Integer in the range of 0 to 300 in units of Second

Default

0

Name

tDynMO - TdynMO

Syntax

```
--timerName tDynMO --timerValue <string> [ --timerUnit Hour ]
```

Description

Indicates the interval to wait for an interface to return an operational state, before deleting the interface. As of CMM23.2.0, tDynMO is used for the Link Disable timer which only applies to links in DISABLED operational state. S1mme, S11, S10, N26, NR10 and other dynamical MME links except for Gn, S3, and S16 are subject to the tDynMO. Statically provisioned links and administrative locked links are not subject to the Link Disable timer. Disabled N2 links are subject to the tDynMO disable timer. The MME N26 links are subject to the tDynMO timer or Transaction Oriented MO delete timer depending on the operational state of the links, either being enabled and disabled. Those are exclusive states.

Type / Range

Integer in the range of 1 to 24 in units of Hour

Default

1

Name

transactionOrientedLinkMoDeletionTimer - Transaction oriented link MO deletion timer

Syntax

```
--timerName transactionOrientedLinkMoDeletionTimer --timerValue  
<string> [ --timerUnit Minute ]
```

Description

Transaction oriented link MO deletion timer for GTP based interfaces. Value 0 means do not do inactivity deletion. Transaction Oriented Link MO Deletion Time is used for Link Inactivity timer and only applies to operational ENABLED links. Only SGSN Gn, S3, S16, S10 and N26 are subject to this time. GGSN is subjective to Session Oriented Link MO Deletion Time. Administrative locked links are not subject to the inactivity timer.

Type / Range

Integer in the range of 0 to 1,440 in units of Minute

Default

30

Name

tS10 - TS10 implicit IMSI detach from non-eps services procedure guard timer

Syntax

```
--timerName tS10 --timerValue <string> [ --timerUnit Second ]
```

Description

Guards implicit IMSI detach from non-eps services procedure.

Type / Range

Integer in the range of 1 to 30 in units of Second

Default

4

Name

tS12_1 - TS12-1 MME reset variable guard timer

Syntax

```
--timerName tS12_1 --timerValue <string> [ --timerUnit Second ]
```

Description

Controls the resetting of the 'MME-Reset' variable.

Type / Range

Integer in the range of 8 to 23,048 in units of Second

Default

11168

Name

tS12_2 - TS12-2 MME reset procedure guard timer

Syntax

```
--timerName tS12_2 --timerValue <string> [ --timerUnit Second ]
```

Description

Guards the MME reset procedure. there is one ts12-2 timer per MME SGs association with a VLR.

Type / Range

Integer in the range of 1 to 120 in units of Second

Default

4

Name

tS13 - T13 SGs MME implicit detach guard timer

Syntax

```
--timerName tS13 --timerValue <string> [ --timerUnit Second ]
```

Description

Configure SGs TS13 timer for MME implicit detach.

Type / Range

Integer in the range of 1 to 32 in units of Second

Default

4

Name

tS6_1 - T6-1 location update procedure guard timer

Syntax

```
--timerName tS6_1 --timerValue <string> [ --timerUnit Second ]
```

Description

Guards the location update procedure.

Type / Range

Integer in the range of 10 to 90 in units of Second

Default

41

Name

tS8 - TS8 explicit IMSI detach from EPS services procedure guard timer

Syntax

```
--timerName tS8 --timerValue <string> [ --timerUnit Second ]
```

Description

Guards explicit IMSI detach from the EPS services procedure.

Type / Range

Integer in the range of 1 to 30 in units of Second

Default

4

Name

ts9 - TS9 explicit IMSI detach from non-eps services procedure guard timer

Syntax

```
--timerName ts9 --timerValue <string> [ --timerUnit Second ]
```

Description

Guards explicit IMSI detach from non-eps services procedure.

Type / Range

Integer in the range of 1 to 30 in units of Second

Default

4

Name

tslrAckFromGmlc - TSLR ack from GMLC

Syntax

```
--timerName tslrAckFromGmlc --timerValue <string> [ --timerUnit  
Second ]
```

Description

TSLR ack from GMLC.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

10

Name

ucdsRemHeartbeat - UCDS REM Heartbeat

Syntax

```
--timerName ucdsRemHeartbeat --timerValue <string> [ --timerUnit  
Millisecond ]
```

Description

Configure REM heartbeat interval for the UCDS service. VNF: Default is 1400, Range is 1400 to 20000 CNF: Default is 1400, Range is 1400 to 20000

Type / Range

Integer in the range of 200 to 20,000 in units of Millisecond

Default

1400

Name

ueContextModify - UE context modify

Syntax

```
--timerName ueContextModify --timerValue <string> [ --timerUnit  
Second ]
```

Description

When the MME sends a context modification message to the eNB, it expects to receive a modification response message from the eNB. if the timer expires while waiting for this response, the MME considers that this class 1 procedure failed.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

8

Name

ueContextRelease - UE context release

Syntax

```
--timerName ueContextRelease --timerValue <string> [ --timerUnit  
Second ]
```

Description

When the MME sends a context release message to the eNB, it expects to receive a release complete response message from the eNB. if the timer expires while waiting for this response, the MME considers that this class 1 procedure failed.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

8

Name

ueCtxtEpsMovedTo5gPurgeTimer - 4G Purge Timer for UE Contexts Moved to 5GC

Syntax

```
--timerName ueCtxtEpsMovedTo5gPurgeTimer --timerValue <string> [ --  
timerUnit Minute ]
```

Description

Purge timer starts after the UE moves to 5GC. The UE gets purged either because of the global purge timer or current timer, whichever expires first.

Type / Range

Integer in the range of 5 to 7,200 in units of Minute

Default

30

Name

ueRadioCapabilityMatchRequest - UE radio capability match request

Syntax

```
--timerName ueRadioCapabilityMatchRequest --timerValue <string> [ --  
timerUnit Millisecond ]
```

Description

Specifies the time to wait for UE radio capability match request.

Type / Range

Integer in the range of 100 to 10,000 in units of Millisecond

Default

8000

Name

ufdsRemHeartbeat - UFDS REM Heartbeat

Syntax

```
--timerName ufdsRemHeartbeat --timerValue <string> [ --timerUnit  
Millisecond ]
```

Description

Configure REM heartbeat interval for UDP Flow Distributor Service. VNF: N/A CNF: Default is 600, Range is 600 to 20000

Type / Range

Integer in the range of 200 to 20,000 in units of Millisecond

Default

600

Name

vlrCleanupIratUeCtxt - VLR Cleanup IRAT UE Contexts

Syntax

```
--timerName vlrCleanupIratUeCtxt --timerValue <string> [ --timerUnit  
Minute ]
```

Description

This parameter specifies the time for deleting from the VLR the preserved UE Contexts of the deregistered UEs that have moved to 5GC.

Type / Range

Integer in the range of 5 to 120, allowed 0 value also. in units of Minute

Default

5

Recommended

5

Name

warningMessage - Warning message

Syntax

```
--timerName warningMessage --timerValue <string> [ --timerUnit  
Second ]
```

Description

This is the timer value that MME waits before re-transmitting the WRWR (Write Replace Warning Request) message to an eNB if the CMAS response is not received from the eNB.

Type / Range

Integer in the range of 1 to 60 in units of Second

Default

8

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

7.115 cmm tlsCAcert

TLS CA Certificate. Syntax, description, parameters, and roles for cmm tlsCAcert command.

Syntax**Description**

This command creates the CA certificate to be loaded into openssl. The Certificate is used to validate certificates received from other tls entities.

Parameters**Name**

name - TLS CA Certificate Name

Syntax

```
--name <string>
```

Description

TLS CA Certificate Name. This is the pem file for the CA authority.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

signedCertFile - Signed Certificate File Name

Syntax

--signedCertFile <string>

Description

Absolute path of the signed certificate file name. Example: /tmp/certificate

Class

Required on create.

Type / Range

Digit string of length 1 to 100 with the allowed pattern of [0-9a-zA-Z\._:@,()/*]*

Name

content - certificate content

Syntax

--content <string>

Description

certificate content

Class

Read only.

Type / Range

String of length 0 to 120

Name

lineNumber - content line number

Syntax

--lineNumber <integer>

Description

line number of the certificate content

Class

Read only.

Type / Range

Integer in the range of 0 to 65,535

Roles

Create

secadmin

Delete

secadmin

List

secadmin

7.116 cmm tlsCAcertList

TLS CA cert List. Syntax, description, parameters, and roles for `cmm tlsCAcertList` command.

Syntax

Description

This command is to create/delete TLS Certified Authority certificate name.

Parameters

Name

name - TLS CA Cert list

Syntax

--name <string>

Description

TLS Certified Authority Certificate

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of `[0-9a-zA-Z\.\-_:@]*`

Roles

Create

secadmin

Show

secadmin

Delete

secadmin

List

secadmin

7.117 cmm tlsCertificate

TLS Certificate. Syntax, description, parameters, and roles for cmm tlsCertificate command.

Syntax**Description**

Manage TLS Certificate. The file pair must be specified together for the create operation

Parameters**Name**

name - TLS Certificate Name

Syntax

--name <string>

Description

TLS Certificate Name. Not applicable for the list operation

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

privateKeyFile - Private key file name

Syntax

--privateKeyFile <string>

Description

Absolute path of the private key file name. Example: /tmp/privkey.pem

Class

Required on create.

Type / Range

Digit string of length 1 to 100 with the allowed pattern of [0-9a-zA-Z\._:@&(),/]*

Name

signedCertFile - Signed Certificate File Name

Syntax

--signedCertFile <string>

Description

Absolute path of the signed certificate file name. Example: /tmp/certificate

Class

Required on create.

Type / Range

Digit string of length 1 to 100 with the allowed pattern of [0-9a-zA-Z\._&,(/]*

Name

content - certificate content

Syntax

--content <string>

Description

certificate content

Class

Optional.

Type / Range

String of length 0 to 120

Name

lineNumber - content line number

Syntax

--lineNumber <integer>

Description

line number of the certificate content

Class

Optional.

Type / Range

Integer in the range of 0 to 65,535

Name

type - certificate type

Syntax

--type <string>

Description

certificate type

Class

Optional.

Type / Range

Digit string of length 0 to 20

Roles**Create**

secadmin

Delete

secadmin

List

secadmin

7.118 cmm tlsCertificateList

TLS Certificate List. Syntax, description, parameters, and roles for cmm tlsCertificateList command.

Syntax**Description**

This command is to create/delete TLS server certificate name.

Parameters**Name**

name - TLS Certificate Name

Syntax

--name <string>

Description

TLS Certificate Name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles**Create**

secadmin

Show

secaadmin

Delete

secaadmin

List

secaadmin

7.119 cmm tlsClientCert

TLS Client Certificate. Syntax, description, parameters, and roles for `cmm tlsClientCert` command.

Syntax

Description

Manage TLS Client Certificate. The file pair must be specified together for the create operation

Parameters

Name

name - TLS Client Certificate Name

Syntax

--name <string>

Description

TLS Client Certificate Name. Not applicable with the list operation

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of `[0-9a-zA-Z\.\_:@]*`

Name

privateKeyFile - Client private key file name

Syntax

--privateKeyFile <string>

Description

Absolute path of the client private key file name. Example: `/tmp/clientprivkey.pem`

Class

Required on create.

Type / Range

Digit string of length 1 to 100 with the allowed pattern of `[0-9a-zA-Z\.\_:@,()/*]*`

Name

signedCertFile - Signed Client Certificate File Name

Syntax

--signedCertFile <string>

Description

Absolute path of the signed client certificate file name. Example: /tmp/clientcert

Class

Required on create.

Type / Range

Digit string of length 1 to 100 with the allowed pattern of [0-9a-zA-Z\.\-_:&(),/]*

Name

content - client certificate content

Syntax

--content <string>

Description

client certificate content

Class

Optional.

Type / Range

String of length 0 to 120

Name

lineNumber - client certificate content line number

Syntax

--lineNumber <integer>

Description

line number of the client certificate content

Class

Optional.

Type / Range

Integer in the range of 0 to 65,535

Name

type - client certificate type

Syntax

--type <string>

Description

client certificate type

Class

Optional.

Type / Range

Digit string of length 0 to 20

Roles**Create**

secadmin

Delete

secadmin

List

secadmin

7.120 cmm tlsClientCertList

TLS Client Certificate List. Syntax, description, parameters, and roles for `cmm tlsClientCertList` command.

Syntax**Description**

This command is to create/delete TLS client certificate name.

Parameters**Name**

name - TLS Client Certificate Name

Syntax

--name <string>

Description

TLS Client Certificate Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of `[0-9a-zA-Z\.\-_\:@]*`

Roles

Create

secadmin

Show

secadmin

Delete

secadmin

List

secadmin

7.121 cmm tlsProfile

TLS Profile. Syntax, description, parameters, and roles for `cmm tlsProfile` command.

Syntax

Description

Create , modify and delete a Transport Layer Security profile. A TLS profile will be provisioned to a SBI.

Parameters

Name

name - TLS Profile Name

Syntax

--name <string>

Description

TLS profile name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

mutualTlsAuth - Mutual TLS Authentication

Syntax

--mutualTlsAuth <list-item>

Description

This parameter indicates if the Producer Service should ask to Authenticate the Client through mutual Authentication. If the EndPoint associated with this Profile is a Client this parameter is ignored.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sendSni - Send SNI

Syntax

```
--sendSni <list-item>
```

Description

When this parameter is set to true CMM will send the Server Name Indication (SNI) extension when setting up a TLS connection as a HTTP/2 client. The SNI is set to the FQDN of the Server that terminates the connection.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

tlsCertificateName - TLS Certificate Name for SBI

Syntax

```
--tlsCertificateName <string>
```

Description

Configure TLS certificate used in establishing secure communication.

Restrictions

This element is only applicable when interfaceName is a SBI interface.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

tlsEncrypted - TLS Encrypted for SBI

Syntax

--tlsEncrypted <list-item>

Description

If this parameter is set to 'True' then AMF uses https. If not AMF uses http.

Restrictions

This element is only applicable when interfaceName is a SBI interface.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

tlsVersion - TLS Version

Syntax

--tlsVersion <list-item>

Description

TLS version used to set up secure TCP connection for HTTPS.

Class

Optional.

List

- 1.2 - Version 1.2
- 1.3 - Version 1.3
- negotiation - Negotiation of TLS 1.3 OR 1.2

Default

1.2

Name

validateSni - Validate SNI

Syntax

--validateSni <list-item>

Description

When this parameter is set to true CMM will validate the Server Name Indication (SNI) extension when setting up a TLS connection as a HTTP/2 server. The SNI must be set to one of the FQDNs associated with the IP endpoint where the connection is terminated.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Roles

Create

secadmin

Show

sysadmin, manage, secadmin

Modify

secadmin

Delete

secadmin

List

sysadmin, manage, secadmin

7.122 cmm tlsRevocation

TLS Revocation CRL. Syntax, description, parameters, and roles for `cmm tlsRevocation` command.

Syntax

Description

TLS Revocation CRL - A file of Revoke Certificate Serial Numbers. This is a complete list of Revoked Certificates not a delta

Parameters

Name

name - TLS CA Certificate Name

Syntax

--name <string>

Description

TLS CA Certificate Name. This is the pem file for the CA authority.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

revocationListFile - Certificate Revocation List

Syntax

--revocationListFile <string>

Description

Absolute path of the CRL file

Class

Required on create.

Type / Range

Digit string of length 1 to 100 with the allowed pattern of [0-9a-zA-Z\._&(),/]*

Name

content - certificate content

Syntax

--content <string>

Description

certificate content

Class

Read only.

Type / Range

String of length 0 to 120

Name

lineNumber - content line number

Syntax

--lineNumber <integer>

Description

line number of the certificate content

Class

Read only.

Type / Range

Integer in the range of 0 to 65,535

Roles**Create**

secadmin

Delete

secadmin

List

secadmin

7.123 cmm tlsRevocationList

TLS Revocation List. Syntax, description, parameters, and roles for cmm tlsRevocationList command.

Syntax**Description**

This command is to create/delete TLS Revocation List name.

Parameters**Name**

name - TLS Revocation List

Syntax

--name <string>

Description

TLS Revocation List Name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles**Create**

secadmin

Show

secadmin

Delete

secadmin

List

secadmin

7.124 cmm tnlaProfile

TNLA Profile. Syntax, description, parameters, and roles for `cmm tnlaProfile` command.

Syntax

Description

Administer TNLA Profile.

Parameters

Name

name - TNLA Profile Name

Syntax

--name <string>

Description

TNLA profile name

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

numberTnlaPerGnb - Number of TNLAs to establish per gNB

Syntax

--numberTnlaPerGnb <integer>

Description

Parameter specifies the number of TNLAs to an AMF establishes with per gNB. The parameter can be set from 1 to N where N is the number of maximum number of AMF N2 local endpoints that can be provisioned.

Class

Optional.

Type / Range

Integer in the range of 1 to 16

Default

1

Name

tnlaDistributorEnable - TNLA Distributor Enable

Syntax

--tnlaDistributorEnable <list-item>

Description

Parameter enables and disables TNLA distributor functionality. When the parameter is set to 'true', the ability of the AMF to add and remove TNLAs for a gNB is enabled. When the parameter is set to 'false', the ability of the AMF to add and remove TNLAs for a gNB is disabled.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

tnlaWeightFactor - Weight factor of each TNLA of the AMF

Syntax

--tnlaWeightFactor <integer>

Description

Parameter specifies the weight factor of each TNLA of the AMF.

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

255

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.125 cmm traceData

Tracedata. Syntax, description, parameters, and roles for cmm traceData command.

Syntax

Description

Create, delete, list and show traces.

Parameters

Name

end - End

Syntax

--end <string>

Description

End time of the trace, format <YYYY>--<MM>--<DD>T<HH>:<mm>:<ss>.<SSS>

Class

Optional.

Type / Range

Digit string of length

Name

file - File

Syntax

--file <string>

Description

File name of the trace.

Class

Optional.

Type / Range

Digit string of length

Name

fileFolder - File folder

Syntax

--fileFolder <string>

Description

Trace file folder.

Class

Optional.

Type / Range

Digit string of length

Name

fileLocation - File location

Syntax

--fileLocation <string>

Description

Trace file location.

Class

Optional.

Type / Range

Digit string of length

Name

fileName - File name

Syntax

--fileName <string>

Description

Trace file name.

Class

Optional.

Type / Range

Digit string of length

Name

imsi - IMSI

Syntax

--imsi <digits>

Description

IMSI.

Class

Optional.

Type / Range

Digit string of length 5 to 15

Name

limit - Limit

Syntax

--limit <list-item>

Description

Delete limit of the trace.

Class

Optional.

List

- all - The specified call trace job packets will be deleted from elasticsearch DB and CT *.pcap file(s) has been created (per CRAFT request) and stored at ACTIVE NECC /shared/restfm/traces.
- db - The specified call trace job packets stored in elasticsearch DB will be deleted.

Name

msisdn - MSISDN

Syntax

--msisdn <digits>

Description

MSISDN.

Class

Optional.

Type / Range

Digit string of length 5 to 18

Name

packetCount - Packet count

Syntax

--packetCount <string>

Description

Packet count.

Class

Optional.

Type / Range

Digit string of length

Name

start - Start

Syntax

--start <string>

Description

Start time of the trace, format <YYYY>--<MM>--<DD>T<HH>:<mm>:<ss>.<SSS>

Class

Optional.

Type / Range

Digit string of length

Name

status - Status

Syntax

--status <string>

Description

Status of the operation.

Class

Optional.

Type / Range

Digit string of length

Name

traceID - Trace id

Syntax

--traceID <integer>

Description

Id of the trace.

Class

Optional.

Type / Range

Integer in the range of 1 to 100

Name

traceOrigin - Trace origin

Syntax

--traceOrigin <list-item>

Description

Type of the trace.

Class

Optional.

List

- cli - cli
- hss - hss
- udm - udm

Roles**Create**

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.126 cmm traceDataStatus

trace Data Status. Syntax, description, parameters, and roles for **cmm traceDataStatus** command.

Syntax**Description**

List and show available trace data.

Parameters**Name**

end - End

Syntax

--end <string>

Description

Stop date of the trace.

Class

Optional.

Type / Range

Digit string of length

Name

first - First

Syntax

--first <string>

Description

First trace.

Class

Optional.

Type / Range

Digit string of length

Name

for - For

Syntax

--for <string>

Description

Who the trace is for.

Class

Optional.

Type / Range

Digit string of length

Name

hits - Hits

Syntax

--hits <integer>

Description

How many trace hits.

Class

Optional.

Type / Range

Integer in the range of 1 to 65,535

Name

imsi - IMSI

Syntax

--imsi <digits>

Description

IMSI.

Class

Optional.

Type / Range

Digit string of length 5 to 15

Name

last - Last

Syntax

--last <string>

Description

Last trace.

Class

Optional.

Type / Range

Digit string of length

Name

msisdn - MSISDN

Syntax

--msisdn <digits>

Description

MSISDN.

Class

Optional.

Type / Range

Digit string of length 5 to 18

Name

start - Start

Syntax

--start <string>

Description

Start date of the trace.

Class

Optional.

Type / Range

Digit string of length

Name

traceID - Trace id

Syntax

--traceID <integer>

Description

Id of the trace.

Class

Optional.

Type / Range

Integer in the range of 1 to 100

Name

traceOrigin - Trace origin

Syntax

--traceOrigin <list-item>

Description

Type of the trace.

Class

Optional.

List

- cli - cli
- hss - hss
- udm - udm

Roles**Show**

sysadmin, manage, monitor

List

sysadmin, manage, monitor

7.127 cmm ueCounts

ueCounts. Syntax, description, parameters, and roles for cmm ueCounts command.

Syntax

Description

ueCounts from cppsId.

Parameters

Name

cppsId - Cpps identifier

Syntax

--cppsId <integer>

Description

Cpps identifier.

Class

Required key.

Type / Range

Integer in the range of 0 to 63

Name

countName - Count name

Syntax

--countName <string>

Description

Count name.

Class

Read only.

Type / Range

Digit string of length

Name

countValue - Count value

Syntax

--countValue <integer>

Description

Count value.

Class

Read only.

Type / Range

Integer in the range of 0 to 2,147,483,647

Roles**List**

sysadmin, manage, monitor

7.128 cmm ueFailopenProfile

UE Failopen Profile. Syntax, description, parameters, and roles for `cmm ueFailopenProfile` command.

Syntax**Description**

UE failopen profile for HSS unavailability. At maximum 50 UE failopen profiles can be created.

Parameters**Name**

name - UE Failopen Profile Name

Syntax

--name <string>

Description

The name of the UE failopen profile.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

chargingCharacteristics - Charging Characteristics

Syntax

--chargingCharacteristics <string>

Description

Charging characteristics.

Class

Required on create.

Type / Range

Digit string of length 1 to 4 with the allowed pattern of [0-9a-fA-F]*

Name

apnOverrideWithFailopen - APN Override with Failopen

Syntax

--apnOverrideWithFailopen <list-item>

Description

Controls whether the MME uses the UE failopen profile to to override UE requested APN, when the HSS default APN configuration is not available due to HSS unavailability, and the MME is provisioned to both use the UE failopen profile and override/correct/replace the requested APN based on HSS subscription. The parameter is applied with any of the below settings: 1) 'Support APN correction' global parameter is set to 'Yes' 2) Per UE PLMN/serving PLMN 'APN correction during attach' and/or 'APN correction during standalone PDN connect' is/are defined as 'Received from HSS', 3) Support IMSI IMEI APN conversion is set to 'IMEI priority' or 'IMSI priority', and on 'APN IMSI Range Services' command the 'Replacement action' for the IMSI/IMEI/APN defines 'Use default' 4) 'IMSI IMEI lookup fail' global parameter is set to 'Use Default' 5) 'Support IMS IIMEI APN enhancement' global parameter is set to 'Enable and use default APN context ID'

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

defaultApnNi - Default APN NI

Syntax

--defaultApnNi <string>

Description

The default Access Point Name Network Indicator.

Class

Optional.

Type / Range

Digit string of length 0 to 62 with the allowed pattern of ([0-9a-zA-Z\.\-]*|*)

Default

"

Name

failopenProfileApnListName - Failopen Profile APN List Name

Syntax

--failopenProfileApnListName <string>

Description

The name of the APN list associated with the UE failopen profile.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\- _:@]*

Default

"

Name

networkAccessMode - Network Access Mode

Syntax

--networkAccessMode <list-item>

Description

Network access mode.

Class

Optional.

List

- Packet and Circuit - Specifies network access mode of packet and circuit.
- Packet only - Specifies network access mode of packet only.

Default

Packet only

Name

pdnDeactivationCc - PDN Deactivation Cause Code

Syntax

--pdnDeactivationCc <integer>

Description

ESM cause that the MME sends if the failopen profile based PDN needs to be deactivated due to HSS unavailability recovery.

Class

Optional.

Type / Range

Integer in the range of 8, 26-39, 41-47, 49-60, 65-66, 81, 95-101, 111-113

Default

36

Name

restorationDeactNotifyIdleUe - Restoration Deactivation Notify Idle UE

Syntax

--restorationDeactNotifyIdleUe <list-item>

Description

Define whether an ECM-IDLE UE will be paged for PDN deactivation during manual HSS restoration process.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

rfspId - RFSP ID

Syntax

--rfspId <integer>

Description

RAT/frequency selection priority identification.

Class

Optional.

Type / Range

Integer in the range of 1 to 256

Default

256

Name

serviceRejectCode - Service Reject Code

Syntax

--serviceRejectCode <integer>

Description

The Service reject cause code when automatic HSS restoration is ongoing and the UE needs to reattach.

Class

Optional.

Type / Range

Integer in the range of 3,6-12,35,40,42

Default

10

Name

tauRejectCauseCode - TAU Reject Cause Code

Syntax

--tauRejectCauseCode <integer>

Description

The TAU reject cause code when automatic HSS restoration is ongoing and the UE needs to reattach.

Class

Optional.

Type / Range

Integer in the range of 3,6,7,9,10-12,14,15,35,40,42

Default

10

Name

ueAmbrDL - UE-AMBR DL

Syntax

--ueAmbrDL <integer>

Description

User equipment aggregate maximum bit rate downlink.

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64- 568, 576-8640, 8700-16000, 17000-128000,
130000-4000000000

Default

128000

Recommended

The recommended value is 100000000.

Name

ueAmbrUI - UE-AMBR UL

Syntax

--ueAmbrUl <integer>

Description

User equipment aggregate maximum bit rate uplink.

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64- 568, 576-8640, 8700-16000, 17000-128000, 130000-4000000000

Default

128000

Recommended

The recommended value is 100000000.

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.129 cmm ueLbStatistics

UE Load Balance Statistics. Syntax, description, parameters, and roles for cmm ueLbStatistics command.

Syntax

Description

This command shows the UE Load Balance Statistics.

Parameters

Name

imsiRangeFrom - IMSI range from

Syntax

--imsiRangeFrom <digits>

Description

IMSI range from.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 15

Name

imsiRangeTo - IMSI range to

Syntax

--imsiRangeTo <digits>

Description

IMSI range to.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 15

Name

lbAbsolute - Number of UEs to load balance

Syntax

--lbAbsolute <integer>

Description

This field specifies the absolute number of UEs to be load balanced. changing this setting can be done only when ue-load balancing is administratively shutdown.

Class

Optional.

Type / Range

Integer in the range of 1 to 5,000,000

Name

lbcmmLocked - Locked status

Syntax

```
--lbcmmLocked <list-item>
```

Description

Locked status.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Name

lbmmeQuarantine - MME quarantined

Syntax

```
--lbmmeQuarantine <list-item>
```

Description

MME quarantined.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Name

lbMoreStatus - More status

Syntax

```
--lbMoreStatus <string>
```

Description

Additional status.

Class

Optional.

Type / Range

Digit string of length

Name

lbNumConUesRelocated4G - Connected ues relocated in LTE

Syntax

```
--lbNumConUesRelocated4G <integer>
```

Description

Connected UEs relocated in LTE.

Class

Optional.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

lbNumConUesToReloc4G - Connected ues to relocate in LTE

Syntax

-- lbNumConUesToReloc4G <integer>

Description

Connected UEs to relocate in LTE.

Class

Optional.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

lbNumIdleUesRelocated4G - Idle ues relocated in LTE

Syntax

-- lbNumIdleUesRelocated4G <integer>

Description

Idle UEs relocated in LTE.

Class

Optional.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

lbNumIdleUesToReloc4G - Idle ues to relocate in LTE

Syntax

-- lbNumIdleUesToReloc4G <integer>

Description

Idle UEs to relocate in LTE.

Class

Optional.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

lbNumUesNotPaged4G - Number of UEs not paged

Syntax

--lbNumUesNotPaged4G <integer>

Description

UEs not paged in LTE.

Class

Optional.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

lbNumUesRelocated4G - Ues relocated in LTE

Syntax

--lbNumUesRelocated4G <integer>

Description

UEs relocated in LTE.

Class

Optional.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

lbNumUesToReloc4G - Ues to relocate in LTE

Syntax

--lbNumUesToReloc4G <integer>

Description

UEs to relocate in LTE.

Class

Optional.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

lbPercentage - Percentage of UEs to load balance

Syntax

--lbPercentage <integer>

Description

This field specifies the percentage of UEs to be load balanced. changing this setting can be done only when ue-load balancing is administratively shutdown.

Class

Optional.

Type / Range

Integer in the range of 1 to 100

Name

lbPhase - Phase

Syntax

--lbPhase <list-item>

Description

Phase of load balance.

Class

Optional.

List

- AGGR - AGGR phase - AGGR phase.
- Campon - Campon phase - Campon phase.
- Drain - Drain phase - Drain phase.
- FetchFromDbbs - FetchFromDbbs.
- NotRunning - Not running - Not running.
- WaitLock - Wait lock - Wait lock phase.

Name

lbStatus - Status

Syntax

--lbStatus <list-item>

Description

Load balance status.

Class

Optional.

List

- Aborted - Aborted state.
- CapReached - Completed capacity reached - Completed capacity reached state.
- CompleteLocked - Complete locked - Complete locked.
- CompleteTimeoutLocked - Complete timeout locked - Complete timeout locked.

- Completed - Completed state.
- DInterMME - Denied inter MME reloaction - Denied inter MME reloaction inprogress state.
- DIntraMME - Denied intra MME reloaction - Denied intra MME reloaction inprogress state.
- DMMEAGGR - Denied MME AGGR locked - Denied MME AGGR locked state.
- DNoActS10L - Denied noactive S10 link - Denied noactive S10 link state.
- DeniedMMENotConfigured - Denied MME not configured - Denied MME not configured.
- DeniedSgsnAggrLocked - Denied SGSN aggr locked - Denied SGSN aggr locked.
- DeniedSgsnNotConfigured - Denied SGSN not configured - Denied SGSN not configured.
- Failed - Failed state.
- Inprogress - Inprogress state.
- MMECompleteSgsnAbort - MME complete SGSN abort - MME complete SGSN abort.
- MMECompleteSgsnFailed - MME complete SGSN failed - MME complete SGSN failed.
- NotRunning - Not running - Not running state.
- OverLoad - Aborted over loaded - Aborted due to over loaded state.
- Requested - Requested state.
- SgsnCompleteMMEAbort - SGSN complete MME abort - SGSN complete MME abort.
- SgsnCompleteMMEFailed - SGSN complete MME failed - SGSN complete MME failed.
- Stop4GNoS10SgsnComplete - Stop 4G no S10 SGSN complete - Stop 4G no S10 SGSN complete.
- Stop4GNoS10SgsnFailed - Stop 4G no S10 SGSN failed - Stop 4G no S10 SGSN failed.
- TimeOut - Completed time out - Completed time out state.
- Unlocked - Unlocked state.

Name

lbTimeRemainInPhase - lbTimeRemainInPhase

Syntax

--lbTimeRemainInPhase <integer>

Description

Time remaining in phase.

Class

Optional.

Type / Range

Integer in the range of 0 to 4,294,967,295

Name

lbTotalTimeInterval - LBTotalTimeInterval

Syntax

--lbTotalTimeInterval <integer>

Description

Total time interval.

Class

Optional.

Type / Range

Integer in the range of 1 to 54,000

Name

maxDuration - MaxDuration requested

Syntax

--maxDuration <integer>

Description

Maximum duration requested.

Class

Optional.

Type / Range

Integer in the range of 900 to 54,000

Name

maxOffLoadRate - Maximum Off Load Rate

Syntax

--maxOffLoadRate <integer>

Description

The CMM shall limit the off-load rate to a --maxOffLoadRate per CPPS (UEs/sec). The rate is a function the VLR Scan rate and the impact of --SkipPostProcLb. If --maxOffLoadRate is not supplied the CMM instance off-loads at the default rate. The default value is 150 UEs/Sec per CPPS. --maxOffLoadRate range allowed is 1 to 600 UEs/sec.

Class

Optional.

Type / Range

Maximum offload rate for UE load balance.

Default

150

Name

mmeCode - MmeCode

Syntax

--mmeCode <digits>

Description

MME code.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 3

Name

msisdnRangeFrom - MSISDN range from

Syntax

--msisdnRangeFrom <digits>

Description

MSISDN range from.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 15

Name

msisdnRangeTo - MSISDN range to

Syntax

--msisdnRangeTo <digits>

Description

MSISDN range to.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 15

Name

nonBroadcastTac - NonBroadcast TAC

Syntax

--nonBroadcastTac <integer>

Description

NonBroadcast TAC.

Class

Optional.

Type / Range

Integer in the range of 0 to 65,535

Name

nonBroadcastTaMcc - NonBroadcast TAC MCC

Syntax

--nonBroadcastTaMcc <digits>

Description

NonBroadcast TAC MCC.

Class

Optional.

Type / Range

Digit string of length 3

Name

nonBroadcastTaMnc - NonBroadcast TAC MNC

Syntax

--nonBroadcastTaMnc <digits>

Description

NonBroadcast TAC MNC.

Class

Optional.

Type / Range

Digit string of length 2 to 3

Name

offloadRate - Offload Rate

Syntax

--offloadRate <integer>

Description

Calculated offload rate from $(lbNumUesRelocated4G)/(maxDuration-lbTimeRemainInPhase)$.

Class

Optional.

Type / Range

Rate of CPPS off-load for UELB show commands. Range is 0 to 1000 UEs/Sec.

Default

0

Name

offloadTaiAreaListName - Offload TAI Area List Name

Syntax

--offloadTaiAreaListName <string>

Description

This is the first list of TACs in a PLMN. When specified all users in the TACs specified will be moved by the UELB command.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

offloadTaiAreaListName2 - Offload TAI Area List Name 2

Syntax

--offloadTaiAreaListName2 <string>

Description

This is a second list of TACs in a PLMN. When specified all users in the TACs specified will be moved by the UELB command.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

offloadTaiAreaListName3 - Offload TAI Area List Name 3

Syntax

--offloadTaiAreaListName3 <string>

Description

This is the third list of TACs in a PLMN. When specified all users in the TACs specified will be moved by the UELB command.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

offloadTaiAreaListPlmnName - Offload TAI Area List Plmn Name

Syntax

--offloadTaiAreaListPlmnName <string>

Description

This specifies the PLMN name for the first TAI Area List.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

offloadTaiAreaListPlmnName2 - Offload TAI Area List Plmn Name 2

Syntax

--offloadTaiAreaListPlmnName2 <string>

Description

This specifies the PLMN name for the second TAI Area List.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

offloadTaiAreaListPlmnName3 - Offload TAI Area List Plmn Name 3

Syntax

--offloadTaiAreaListPlmnName3 <string>

Description

This specifies the PLMN name for the third TAI Area List.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

quarantineMode - QuarantineMode

Syntax

--quarantineMode <list-item>

Description

Quarantine mode.

Class

Optional.

List

- all - Specify quarantine mode of all.
- lte-only - LTE only - Specify quarantine mode of LTE only.
- not-used - Not used - Specify quarantine mode of not used.

Name

radioType - Radio type

Syntax

--radioType <list-item>

Description

Radio type.

Class

Optional.

List

- all - Specify radio type of all.
- lte - Specify radio type of LTE.

Name

serviceLock - ServiceLock

Syntax

--serviceLock <list-item>

Description

Service lock requested.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Name

shutdownFlag - ShutDown state requested

Syntax

--shutdownFlag <list-item>

Description

The load balancing process has to be explicitly started using no shutdown command. once started, none of the parameters can be changed. the shutdown command stops load balancing process. if this is done before the completion of the load balancing process the system will continue its normal operation. if the shutdown command has been issued after the job was completed and the system was in quarantine or service-lock mode, this action will effectively reset the corresponding mode and the system will continue its normal operation.

Class

Optional.

List

- noshutdown - Execute - Execute load balance on the WMM.
- shutdown - Abort-Stop - Aborted\t load balance on the WMM if one is in progress. if loadbalance\ \ is already completed, shutdown will let you to change the state to start\ \ a new one.

Name

skipPostProcLb - Skip Load Balancing after Procedure

Syntax

--skipPostProcLb <list-item>

Description

The GPARAM skipPostSrLbDuringUelb shall be deprecated and replaced by the - - skipPostProcLb option. skipPostSrLbDuringUelb shall not be used together with the -- soft option. Possible values are True/False. Default is True

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

soft - Soft UELB Option

Syntax

--soft <list-item>

Description

If the option 'soft' is set to true, the CMM makes two passes (Pass1, Pass2) in the CAMPON phase. The following actions take place during Pass1: For ECM-CONNECTED UEs (including emergency, Volte, high priority, non-emergency) the CMM shall issue a UE CONTEXT RELEASE

COMMAND with cause 'Loadbalancing TAU required' upon receiving a UE CONTEXT RELEASE REQUEST with cause 'User inactivity'. If the option soft is not set then the current behaviour applied. Soft off-load cannot be used with Directed loadbalancing as directed loadbalancing uses GUTI reallocation. For ECM-IDLE UEs(including emergency, Volte, high priority, non-emergency) the CMM shall page the UE. Once UE sends SR/CPSR and becomes ECM-CONNECTED then CMM shall use the procedure specified above for the ECM-CONNECTED UE to offload the UE

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

softSelectedProcs - Soft UELB for selected call processing procedures

Syntax

--softSelectedProcs <list-item>

Description

The -- softSelectedProcs option is introduced to enable soft handling of selected procedures with the current off-loading mechanism. If the softSelectedProcs option is set to True the CMM shall send UE CONTEXT RELEASE COMMAND with cause 'load balancing TAU required' when: CMM determines that it can initiate UE context release or a Volte UE releases its QCI1 bearer. Note1: softSelectedProcs cannot be used with the soft option because the treatment of Volte UEs would be inconsistent with the treatment of other UEs

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

srsUsed - SRS Used for UELB

Syntax

--srsUsed <list-item>

Description

When the CMM is configured to support in-instance session restoration (SRS) the -- sRsUsed option can be set to speed up the Database phase. The Database phase deals with unclaimed and aged-out UEs. During UELB Cache-Age is suspended for the UE types. The selected sRsUsed option value determines which UE types are to be moved with the help of the in-instance service restoration and therefore excluded from the Database phase.

Class

Optional.

List

- all - All UE Types - All UE Types are selected
- mbb - 4G Mobile Broadband UEs - 4G Mobile Broadband UEs Selection
- nbioT - NBioT UEs - NBioT UEs selection
- none - No UE Type is selected - No UE Type is selected. This is the default Value

Default

none

Name

ueType - Ue type

Syntax

--ueType <list-item>

Description

UE type.

Class

Optional.

List

- active - Specify active UEs.
- all - Specify all UEs.
- idle - Specify idle UEs.

Roles**Show**

sysadmin, manage, monitor

List

sysadmin, manage, monitor

7.130 cmm ueLoadBalance

UE Load Balance. Syntax, description, parameters, and roles for cmm ueLoadBalance command.

Syntax

Description

UE Load Balance. In case of TA/QA configuration, SGSN PAPSS should be drained (using the cmm maintenance command) and must be locked (using cmm serviceState modify command) before ueLoadBalance with serviceLock=true is entered.

Parameters

Name

dcnBasedOffload - Load balancing using DCN ID of the UE

Syntax

--dcnBasedOffload <list-item>

Description

Parameter is used to enable load balancing of UE supporting a DCN ID

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

dcnId - Dedicated Core Network Identifier

Syntax

--dcnId <integer>

Description

This parameter used to offload UEs supporting a DCN ID. MME determines DNC ID of UE by using the DNC ID to UE Usage Type provisioning

Class

Optional.

Type / Range

Integer in the range of 1 to 126 or 128 to 65,535

Default

65535

Name

directedMoveMode - Directed Move Mode

Syntax

--directedMoveMode <list-item>

Description

Parameter to select mode of UE move to a targeted MME in the same MME pool consisting of CMM and FNS. Two modes are supported: CMM - uses the CMM method of directive move method, FNS - uses the FNS method of directive move. Both the methods use GUTI reallocation to move to the target MME within the same pool. However, for FNS method, the target MME forces the UE to reattach to the network where as the CMM method gets UE context from the old MME thus eliminating need to force the UE to reattach.

Class

Optional.

List

- CMM - Mode 1 - CMM supported UE directed move.
- FNS - FNS supported UE directed move.

Default

CMM

Name

dynUeLbDrainIntv - Dynamic UELB drain interval

Syntax

--dynUeLbDrainIntv <list-item>

Description

Enables quick (dynamically calculated) UE-LB drain interval.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

imsiRangeFrom - IMSI range from

Syntax

--imsiRangeFrom <digits>

Description

Specifies which imsi-range start for UEs will be subject to load balancing process. there can be only single imsi-range specific data this command is mutually exclusive with mobilenetworkname, msisdn-range, location and ran data. imsirangefrom and imsirango must be specified together. changing this setting can be done only when ue-load balancing is administratively shutdown.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 15

Default

''

Name

imsiRangeTo - IMSI range to

Syntax

--imsiRangeTo <digits>

Description

Specifies which imsi-range end for UEs will be subject to load balancing process. there can be only single imsi-range specific data this command is mutually exclusive with mobilenetworkname, msisdn-range, location and ran data. imsirangefrom and imsirango must be specified together. changing this setting can be done only when ue-load balancing is administratively shutdown.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 15

Default

''

Name

lbAbsolute - Number of UEs to load balance

Syntax

--lbAbsolute <integer>

Description

This field specifies the absolute number of UEs to be load balanced. changing this setting can be done only when ue-load balancing is administratively shutdown.

Class

Optional.

Type / Range

Integer in the range of 1 to 5,000,000

Name

lbPercentage - Percentage of UEs to load balance

Syntax

--lbPercentage <integer>

Description

This field specifies the percentage of UEs to be load balanced. changing this setting can be done only when ue-load balancing is administratively shutdown.

Class

Optional.

Type / Range

Integer in the range of 1 to 100

Name

maxDuration - Maximum duration

Syntax

--maxDuration <integer>

Description

Specifies the maximum time in seconds load balancing process will be active. after reaching the maximum time before reaching target number of UEs the load balancing effort will stop and the system will behave in the same way as if the target number of UEs would have been reached. changing this setting can be done only when ue-load balancing is administratively shutdown.

Class

Optional.

Type / Range

Integer in the range of 900 to 54,000

Default

5400

Name

maxOffLoadRate - Maximum Off Load Rate

Syntax

--maxOffLoadRate <integer>

Description

The CMM shall limit the off-load rate to a --maxOffLoadRate per CPPS (UEs/sec). The rate is a function the VLR Scan rate and the impact of --SkipPostProcLb. If --maxOffLoadRate is not supplied the CMM instance off-loads at the default rate. The default value is 150 UEs/Sec per CPPS. --maxOffLoadRate range allowed is 1 to 600 UEs/sec.

Class

Optional.

Type / Range

Maximum offload rate for UE load balance.

Default

150

Name

mmeCode - MmeCode

Syntax

--mmeCode <digits>

Description

The target MME node in the pool the UEs should be moved to. the target MME is specified in the terms of mme-code. changing this setting can be done only when ue-load balancing is administratively shutdown.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 3

Name

msisdnRangeFrom - MSISDN range from

Syntax

--msisdnRangeFrom <digits>

Description

Specifies which msisdn-range start for UEs will be subject to load balancing process. there can be only single imsi-range specific data this command is mutually exclusive with mobilenetworkname, imsi-range, location and ran data. msisdnrangefrom and msisdnrangeto must be specified together. changing this setting can be done only when ue-load balancing is administratively shutdown.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 15

Default

''

Name

msisdnRangeTo - MSISDN range to

Syntax

--msisdnRangeTo <digits>

Description

Specifies which msisdn-range end for UEs will be subject to load balancing process. there can be only single imsi-range specific data this command is mutually exclusive with mobilenetworkname, imsi-range, location and ran data. msisdnrangefrom and msisdnrangeto must be specified together. changing this setting can be done only when ue-load balancing is administratively shutdown.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 15

Default

''

Name

nonBroadcastTac - NonBroadcastTAC

Syntax

--nonBroadcastTac <integer>

Description

Non broadcast TAC.

Class

Optional.

Type / Range

Integer in the range of 0 to 65,535

Default

0

Name

nonBroadcastTaMcc - NonBroadcastTaMCC

Syntax

--nonBroadcastTaMcc <digits>

Description

Non broadcast ta MCC.

Class

Optional.

Type / Range

Digit string of length 3

Default

000

Name

nonBroadcastTaMnc - NonBroadcastTaMNC

Syntax

--nonBroadcastTaMnc <digits>

Description

Non broadcast ta MNC.

Class

Optional.

Type / Range

Digit string of length 2 to 3

Default

000

Name

offloadTaiAreaListName - Offload TAI Area List Name

Syntax

--offloadTaiAreaListName <string>

Description

This is the first list of TACs in a PLMN. When specified all users in the TACs specified will be moved by the UELB command.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Default

"

Name

offloadTaiAreaListName2 - Offload TAI Area List Name 2

Syntax

--offloadTaiAreaListName2 <string>

Description

This is a second list of TACs in a PLMN. When specified all users in the TACs specified will be moved by the UELB command.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

offloadTaiAreaListName3 - Offload TAI Area List Name 3

Syntax

--offloadTaiAreaListName3 <string>

Description

This is a third list of TACs in a PLMN. When specified all users in the TACs specified will be moved by the UELB command.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

offloadTaiAreaListPlmnName - Offload TAI Area List Plmn Name

Syntax

--offloadTaiAreaListPlmnName <string>

Description

This specifies the PLMN name for the first TAI Area List.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

offloadTaiAreaListPlmnName2 - Offload TAI Area List Plmn Name 2

Syntax

--offloadTaiAreaListPlmnName2 <string>

Description

This specifies the PLMN name for the second TAI Area List.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

offloadTaiAreaListPlmnName3 - Offload TAI Area List Plmn Name 3

Syntax

--offloadTaiAreaListPlmnName3 <string>

Description

This specifies the PLMN name for the third TAI Area List.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

quarantineMode - Quarantine mode

Syntax

--quarantineMode <list-item>

Description

Specifies the action the system will take after load balancing process is completed (or in other words the target number of moved UEs has been reached). this is mutually exclusive with service-lock command. quarantine-mode means that the system will not accept any new UEs after load balancing process completion and it will continue to serve all remaining UEs attached to the system. in combination with ue-filter criteria, the selection of 'non acceptance' of new UEs is the same as the selection of UEs to be moved. changing this setting can be done only when ue-load balancing is administratively shutdown.

Class

Optional.

List

- all - Specify quarantine mode of all.
- lte-only - LTE only - Specify quarantine mode of LTE only.
- not-used - Not used - Specify quarantine mode of not used.

Default

not-used

Name

radioType - Radio type

Syntax

--radioType <list-item>

Description

Specifies which radio-type connected UEs will be subject to load-balancing process. changing this setting can be done only when ue-load balancing is administratively shutdown.

Class

Optional.

List

- all - Specify radio type of all.
- lte - Specify radio type of LTE.

Default

all

Name

serviceLock - Service lock

Syntax

--serviceLock <list-item>

Description

Specifies the action the system will take after load balancing process is completed (or in other words the target number of moved UEs has been reached). this is mutually exclusive with quarantine-mode command. service-lock means that the system will block all its interfaces after load balancing process completion and it will not serve any UEs. this type of command is mainly meant to be used with moving all UEs from the system to allow system maintenance operation. the service-lock works regardless the UE-filter criteria, which means that after load balancing process no UEs will be served. using radio-type lte with service-lock flag allows to apply service-lock mode to MME changing this setting can be done only when ue-load balancing is administratively shutdown.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

shutDownFlag - ShutDown flag

Syntax

--shutDownFlag <list-item>

Description

The load balancing process has to be explicitly started using no shutdown command. once started, none of the parameters can be changed. the shutdown command stops load balancing process. if this is done before the completion of the load balancing process the system will continue its normal operation. if the shutdown command has been issued after the job was completed and the system was in quarantine or service-lock mode, this action will effectively reset the corresponding mode and the system will continue its normal operation. In Release 20.0BR the UELB command auto-shutdown if the --quarantine and --serviceLock options and False. All parameters are returned to their specified default values.

Class

Optional.

List

- noshutdown - Execute - Execute load balance on the WMM.
- shutdown - Abort-Stop - Aborted\t load balance on the WMM if one is in progress. if loadbalance\ \ is already completed, shutdown will let you to change the state to start\ \ a new one.

Default

shutdown

Name

skipPostProcLb - Skip Load Balancing after Procedure

Syntax

--skipPostProcLb <list-item>

Description

The GPARM skipPostSrLbDuringUelb shall be deprecated and replaced by the --skipPostProcLb option. skipPostSrLbDuringUelb shall not be used together with the --soft option. Possible values are True/False. Default is True

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

soft - Soft UELB Option

Syntax

--soft <list-item>

Description

If the option 'soft' is set to true, the CMM makes two passes (Pass1, Pass2) in the CAMPON phase. The following actions take place during Pass1: For ECM-CONNECTED UEs (including emergency, Volte, high priority, non-emergency) the CMM shall issue a UE CONTEXT RELEASE COMMAND with cause 'Loadbalancing TAU required' upon receiving a UE CONTEXT RELEASE REQUEST with cause 'User inactivity'. If the option soft is not set then the current behaviour applied. Soft off-load cannot be used with Directed loadbalancing as directed loadbalancing uses GUTI reallocation. For ECM-IDLE UEs(including emergency, Volte, high priority, non-emergency) the CMM shall page the UE. Once UE sends SR/CPSR and becomes ECM-CONNECTED then CMM shall use the procedure specified above for the ECM-CONNECTED UE to offload the UE

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

softSelectedProcs - Soft UELB for selected call processing procedures

Syntax

--softSelectedProcs <list-item>

Description

The -- softSelectedProcs option is introduced to enable soft handling of selected procedures with the current off-loading mechanism. If the softSelectedProcs option is set to True the CMM shall send UE CONTEXT RELEASE COMMAND with cause 'load balancing TAU required' when: CMM determines that it can initiate UE context release or a Volte UE releases its QCI1 bearer. Note1: softSelectedProcs cannot be used with the soft option because the treatment of Volte UEs would be inconsistent with the treatment of other UEs

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

srsUsed - SRS Used for UELB

Syntax

--srsUsed <list-item>

Description

When the CMM is configured to support in-instance session restoration (SRS) the -- sRsUsed option can be set to speed up the Database phase. The Database phase deals with unclaimed and aged-out UEs. During UELB Cache-Age is suspended for the UE types. The selected sRsUsed option value determines which UE types are to be moved with the help of the in-instance service restoration and therefore excluded from the Database phase.

Class

Optional.

List

- all - All UE Types - All UE Types are selected
- mbb - 4G Mobile Broadband UEs - 4G Mobile Broadband UEs Selection
- nbiot - NBloT UEs - NBloT UEs selection
- none - No UE Type is selected - No UE Type is selected. This is the default Value

Default

none

Name

suppressRcZero - Suppress RC Zero

Syntax

--suppressRcZero <list-item>

Description

When this parameter is disabled the MME updates the eNBs by setting the relative capacity to zero when the Directed ue-load balancing procedure is initiated and restores the value when the procedure is stopped. When this parameter is enabled the relative capacity is not updated. Changing this setting can be done only when ue-load balancing is administratively shutdown.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ueType - Ue type

Syntax

--ueType <list-item>

Description

Specifies which UEs will be subject to load-balancing process. changing this setting can be done only when ue-load balancing is administratively shutdown.

Class

Optional.

List

- active - Specify active UEs.
- all - Specify all UEs.
- idle - Specify idle UEs.

Default

all

Roles**Show**

sysadmin, manage, monitor

Modify

sysadmin, manage

List

sysadmin, manage, monitor

7.131 cmm ueMonitoringProfile

UE Monitoring Profile. Syntax, description, parameters, and roles for cmm ueMonitoringProfile command.

Syntax**Description**

Administer UE Monitoring Profile.

Parameters

Name

name - UE monitoring profile name

Syntax

--name <string>

Description

UE monitoring profile name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\.\-:@]*

Name

allowExtScefRefId - Allow Extended SCEF Reference ID

Syntax

--allowExtScefRefId <list-item>

Description

Specifies whether the MME is to allow negotiation of the use of extended SCEF reference IDs for monitoring Events. This field applies only for home/shared/TAHS UEs. The default is false.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Recommended

true

Name

allowExtScefRefIdRoamers - Allow Extended SCEF Reference ID for Roamers

Syntax

--allowExtScefRefIdRoamers <list-item>

Description

Specifies whether the MME is to allow negotiation of the use of extended SCEF reference IDs for monitoring Events. This field applies only for inbound roaming UEs. The default is false.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Recommended

true

Name

description - Description

Syntax

--description <string>

Description

UE monitoring profile description.

Class

Optional.

Type / Range

Digit string of length 0 to 160 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Name

eventReportPreferIda - monitoring report via IDA

Syntax

--eventReportPreferIda <list-item>

Description

Current in MME, monitoring reports are with RIR only. With this parameter set, MME can report via IDA for any of the requested Monitoring events are available to be reported. Default is no set-> MME keep the same behavior as current.

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

includeReportingTimeStamp - include Reporting Time Stamp

Syntax

--includeReportingTimeStamp <list-item>

Description

Specifies whether the Reporting Time Stamp shall included in Monitoring Event Reports generated by the MME. The report can appear in S6a:IDA or T6a:RIR messages, depending on the event scenario.

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

minPerLocRptTimeOverS10 - Min Per Loc Rpt Time Over S10

Syntax

--minPerLocRptTimeOverS10 <list-item>

Description

Specifies whether the MME shall send and receive the Minimum Periodic Location Reporting Time in S10 messaging for inter-MME relocation.

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

minPerLocTimer - minPerLocTimer

Syntax

--minPerLocTimer <integer>

Description

Specifies the minimum value of the Periodic Timer. This floor value applies if the periodic timer for Current Location Event Monitoring is received from the HSS over S6a. If the S6a value is lower than the provisioned value on the MME, the MME will use the provisioned value as the minimum reporting interval for current location event reporting.

Class

Optional.

Type / Range

Integer in the range of 3 to 50

Default

3

Name

monAvailAfterDdnFail - Monitor availability after DDN failure

Syntax

--monAvailAfterDdnFail <list-item>

Description

Allowability of UE monitoring type availability after DDN failure.

Class

Optional.

List

- none - No monitoring of availability after DDN failure.
- all - Allow monitoring of availability after DDN failure when UE moves to connected mode.
- allEdrx - Allow monitoring of availability after DDN failure to be reported at the beginning of an eDRX cycle in addition to the 'All' case.

Default

none

Name

monAvailAfterDdnFailWithIdleStatus - Mon Avail After DDN Fail With Idle Status

Syntax

--monAvailAfterDdnFailWithIdleStatus <list-item>

Description

Specifies whether the MME shall generate the Available After DDN Failure Monitoring Event for a UE if that UE is configured with the AVAILABILITY_AFTER_DDN_FAILURE_AND_IDLE_STATUS_INDICATION (9) monitoring event.

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

monCommFailure - monCommFailure

Syntax

--monCommFailure <list-item>

Description

Communication Failure MONTE Event.

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

monIdleStatusIndication - monitor idle status indication

Syntax

--monIdleStatusIndication <list-item>

Description

Allowability of UE monitoring type idle status indication

Class

Optional.

List

- all - support this event
- availAfterDdn - available After DDN failure - Support for monitor event available after DDN failure and idle states ind.
- none - not support this event
- ueReach - Ue Reachability - Support for monitor event UE reachability and idle states ind.

Default

none

Name

monIdleStatusRptAtIdr - monIdleStatusRptAtIdr

Syntax

--monIdleStatusRptAtIdr <list-item>

Description

Specifies whether the MME is to sent the Idle Status Monitoring Report if the UE is already idle when the event is configured via S6a:IDR message.

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

monLocation - Monitor location

Syntax

--monLocation <list-item>

Description

Allowability of UE monitoring type location.

Class

Optional.

List

- all - Allow monitoring of location including ecgi, enodeb and TAI.
- enodeb/tai - Enodeb and TAI - Allow monitoring of location limiting to enodeb and TAI.
- none - No monitoring of location.

Default

none

Name

monLossOfConnectivity - Monitor loss of connectivity

Syntax

--monLossOfConnectivity <list-item>

Description

Allowability of UE monitoring type loss of connectivity.

Class

Optional.

List

- all - Allow monitoring of loss of connectivity including uedetach, maxtimedetach and uepurge.
- none - No monitoring of loss of connectivity.
- allPsm - Allow generation of loss of connectivity reporting event when UE enters PSM mode (after T3324 expiration) in addition to the 'All' cases.
- allEdrx - Allow generation of loss of connectivity reporting event when UE exits a paging time window within an eDRX cycle in addition to the 'All' case.
- allEdrxPsm - Allow generation of loss of connectivity reporting event for eDRX, PSM, and All cases.

Default

none

Name

monPdnConnectivityStatus - PDN Connectivity Status Event

Syntax

--monPdnConnectivityStatus <list-item>

Description

Defines whether the MME is to support the PDN Connectivity Status Event

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

monReachability - Monitor reachability

Syntax

--monReachability <list-item>

Description

Allowability of UE monitoring type UE reachability.

Class

Optional.

List

- all - Allow monitoring of reachability including data and sms.
- none - No monitoring of reachability.

Default

none

Name

monReachabilityWithIdleStatus - Mon Reachability With Idle Status

Syntax

--monReachabilityWithIdleStatus <list-item>

Description

Specifies whether the MME shall generate the Reachability Monitoring Event for a UE if that UE is configured with the UE_REACHABILITY_AND_IDLE_STATUS_INDICATION (8) monitoring event.

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

nsurNsuiOverS10 - NSUR NSUI Over S10

Syntax

--nsurNsuiOverS10 <list-item>

Description

Specifies whether the MME shall send and receive the NSUR (Notify SCEF when UE becomes Reachable) and NSUI (Notify SCEF when UE becomes Idle) flags in S10 messaging for inter-MME relocation.

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

pageIdleUe4Location - page IdleUe For Location

Syntax

--pageIdleUe4Location <list-item>

Description

With this parameter set, MME will page UE in idle for Location info if IDR comes with current location for monitor type location. if paging failed, then last known location should be reported to SCEF. If this parameter does not set, MME keep the current behave, no RIR towards SCEF for current location request.

Restrictions

Note: pageIdleUe4Location parameter is only for idle UE which is not in eDRX sleep mode (i.e. unreachable windows) and not in PSM sleep mode.

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

pageIdleUeForLocImmediateIda - Page Idle UE for Location Immediate IDA

Syntax

--pageIdleUeForLocImmediateIda <list-item>

Description

Specifies whether the MME, when receiving an S6a:IDR message that configures a current location monitoring event, will send the S6a:IDA response prior to paging for the location monitoring event. pageIdleUeForLocImmediateIda applies only if pageIdleUe4Location is also set true.

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

pdnConnBearerIdentityIncluded - PDN Connectivity Bearer Identity Included

Syntax

--pdnConnBearerIdentityIncluded <list-item>

Description

Specifies whether to include the Bearer Identity as a non-standardized AVP in the PDN Connectivity Status Report

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

pdnConnMip6AgentInfoIncluded - PDN Connectivity MIPv6AgentInfo Included

Syntax

--pdnConnMip6AgentInfoIncluded <list-item>

Description

Specifies whether to include the MIPv6AgentInfo AVP as a non-standardized parameter in the PDN Connectivity Status Report.

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

pdnConnRatTypeAvpIncluded - PDN Connectivity RAT Type AVP Included

Syntax

--pdnConnRatTypeAvpIncluded <list-item>

Description

Specifies whether the RAT Type AVP is to be included as a non-standardized parameter in the PDN Connectivity Status Report.

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

reportOldLastKnownLoc4Current - report Old LastKnown Location for Current location request

Syntax

--reportOldLastKnownLoc4Current <list-item>

Description

If current-location is not available, report last known location even if last-known-location is old (e.g. older than 1 minute).

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

reportUeLocationInPsmEdrx - report Ue last known Location in PsmEdrx

Syntax

--reportUeLocationInPsmEdrx <list-item>

Description

with this parameter set, MME will report UE last known location towards SCEF when Ue is in either Psm or eDrx slepp mode if MME receives current location request via IDR. If the parameter does not set, then MME keeps current behave, no report towards SCEF if request is current location.

Restrictions

Note 1: When current location report is requested in IDR, in all cases (even for an existing continuous report update) and even if the location has not changed, always send the first report (one-time or continuous) for current-location. Note 2: For an eDRX UE, pagIdleUe4Location parameter is impacted during eDRX UE reachable window.

reportUeLocationInPsmEdrx parameter is impacted during eDRX sleep mode (i.e. unreachable window of eDRX cycle).

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

supportPeriodicLocRpt - Support Periodic LocRpt

Syntax

--supportPeriodicLocRpt <list-item>

Description

Specifies whether the MME is to support a minimum reporting interval between successive MONTE Location Event Reports for the same UE.

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

suppressAaddnfForSr - Suppress AADDNF Report for Service Request

Syntax

--suppressAaddnfForSr <list-item>

Description

Specifies whether the AADDNF (availability after DDN failure) report is to be suppressed for a UE that is not assigned eDRX and/or is not assigned PSM currently. The suppression applies only when the UE performs a Service Request. TAU and Attach continue to generate the reachability report. Events (6) and (9) from the UE subscription apply.

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

suppressReachForSr - Suppress Reachability Report for Service Request

Syntax

--suppressReachForSr <list-item>

Description

Specifies whether the UE reachability Report is to be suppressed for a UE that is not assigned eDRX and/or is not assigned PSM currently. The suppression applies only when the UE performs a Service Request. TAU and Attach continue to generate the reachability report. Events (1) and (8) from the UE subscription apply.

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

t6aExternalIdSupported - T6a External ID Supported

Syntax

--t6aExternalIdSupported <list-item>

Description

Specifies whether the External ID can be included in the T6a:RIR message.

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

ueReachReportWithCauseCode - UE Reachability Report with Cause Code

Syntax

--ueReachReportWithCauseCode <list-item>

Description

Specifies whether the Reachability-Cause AVP is to be included in the UE Reachability Report.

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.132 cmm uePlmnServices

Services based on UE Network and Served Network. Syntax, description, parameters, and roles for cmm uePlmnServices command.

Syntax**Description**

This command is used to administer services based on the UE Network and Served Network.

Parameters**Name**

name - Name of services based on UE network and served network

Syntax

--name <string>

Description

Name that identifies roaming agreement.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

servedPlmnName - Mobile network name

Syntax

--servedPlmnName <string>

Description

Name that identifies mobile network.

Class

Required on create, not modifiable.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

uePlmnName - UE mobile network

Syntax

--uePlmnName <string>

Description

Name that identifies roaming network.

Class

Required on create, not modifiable.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

acceptDab - Accept DAB

Syntax

--acceptDab <list-item>

Description

Indicates whether PLMN accepts DAB (Dual access bearer).

Restrictions

This element is not applicable when the serving and UE PLMNs are the same.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

allowEthernetPdnType - Allow Ethernet PDN Type

Syntax

--allowEthernetPdnType <list-item>

Description

Parameter controls MME support for allowing Ethernet PDN Type. By default this shall be set to false.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

allowNonIpOverS1u - allow non IP over S1

Syntax

--allowNonIpOverS1u <list-item>

Description

with allowNonIpOverS1u parameter under uePlmnServices set, MME will allow an UE has non IP connection (with other condition, HSS allow non IP APN etc) under this PLMN.

Restrictions

This element is applicable when the serving and UE PLMNs are same or when lteTreatAsHomeSubscriber set to true

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

apnCorrectionAttach - APN correction during attach

Syntax

--apnCorrectionAttach <list-item>

Description

APN correction that is done during attach.

Class

Optional.

List

- featureDisabled - Feature disabled - This feature is disabled.
- locallyConfigured - Locally configured - Locally configured.
- receivedFromHss - Received from HSS - Received from HSS.

Default

featureDisabled

Name

apnCorrectionStandalone - APN correction during standalone PDN connect

Syntax

--apnCorrectionStandalone <list-item>

Description

APN correction that is done during standalone PDN connectivity request.

Class

Optional.

List

- featureDisabled - Feature disabled - This feature is disabled.
- locallyConfigured - Locally configured - Locally configured.
- receivedFromHss - Received from HSS - Received from HSS.

Default

featureDisabled

Name

apnPriorityListName - APN priority list name

Syntax

--apnPriorityListName <string>

Description

APN priority list name.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

apnWtLstLte - LTE APN white list

Syntax

--apnWtLstLte <string>

Description

It is used to assign a local breakout (LBO) APN white list to a roaming network partner for LBO PDN connections.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

apn-all

Name

applProtocolFallbackN26ToS10 - Application Protocol Fallback from N26 to S10

Syntax

--applProtocolFallbackN26ToS10 <list-item>

Description

Current parameter enables fallback to search for 's10' candidate when DNS Query to find target MME during S10 handover using 'n26' fails. This parameter is expected to be used together with parameter Use N26 Service Parameter in S10 Handover

Class

Optional.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Name

applyApnWtLstLte - Apply LTE APN white list

Syntax

--applyApnWtLstLte <list-item>

Description

It is used to enable APN white list checks on the roamer UE in the LTE network.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

camelAllowed - CAMEL subscriber roaming allowed

Syntax

--camelAllowed <list-item>

Description

Indicates whether camel subscriber roaming in mobile network is allowed or not.

Restrictions

This element is not applicable when the serving and UE PLMNs are the same. This element only applies to 2G/3G.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

chargingCharAccessRestrictionProfileName - Charging Characteristics Access Restriction Profile Name

Syntax

--chargingCharAccessRestrictionProfileName <string>

Description

Defines the name of the charging characteristics access restriction profile.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

ciotEmmRegisterdWithoutPdn - Enable EMM-REGISTERED without PDN connection for C-IOT UE

Syntax

--ciotEmmRegisterdWithoutPdn <list-item>

Description

Enable EMM-REGISTERED without PDN connection for C-IOT UE.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

combinedAttachWithoutCsProfileName - Combined Attach without CS Profile Name

Syntax

--combinedAttachWithoutCsProfileName <string>

Description

The profile name to include parameters for combined attach without CS domain.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

..

Name

cpCIoT - CP CIOT

Syntax

--cpCIoT <list-item>

Description

Enable CP CIOT optimization.

Restrictions

This element is applicable when the serving and UE PLMNs are same or when lteTreatAsHomeSubscriber set to true.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

cpCIoTDeliveryMechanism - CP CIOT delivery mechanism

Syntax

--cpCIoTDeliveryMechanism <list-item>

Description

CP CIOT delivery mechanism.

Restrictions

This element is only applicable when plmn is home/shared.

Class

Optional.

List

- all - Delivery using both S11-u and T6a is supported.
- scef - Delivery using T6a interface.
- sgi - Delivery using S11-u interface.

Default

sgi

Name

csfbPagingPriProfileName - CSFB paging priority profile name

Syntax

--csfbPagingPriProfileName <string>

Description

Identifies the CSFB paging priority profile.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

custApn1QosprofileName - Custom APN QoS profile-1 for roamers

Syntax

--custApn1QosprofileName <string>

Description

Custom APN QoS profile-1 for roamers.

Restrictions

This element is not applicable when the serving and UE PLMNs are the same.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

custApn1WpsQosprofileName - Custom APN WPS QoS profile for roamers - 1

Syntax

--custApn1WpsQosprofileName <string>

Description

Points to the first custom APN WPS QoS profile for roamers.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

custApn2QosprofileName - Custom APN QoS profile 2 for roamers

Syntax

--custApn2QosprofileName <string>

Description

Custom APN QoS profile-2 for roamers.

Restrictions

This element is not applicable when the serving and UE PLMNs are the same.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

''

Name

custApn2WpsQosprofileName - Custom APN WPS QoS profile for roamers - 2

Syntax

--custApn2WpsQosprofileName <string>

Description

Points to the second custom APN WPS QoS profile for roamers.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

''

Name

custApn3QosprofileName - Custom APN QoS profile-3 for roamers

Syntax

--custApn3QosprofileName <string>

Description

Custom APN QoS profile-3 for roamers.

Restrictions

This element is not applicable when the serving and UE PLMNs are the same.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Name

custApn3WpsQosprofileName - Custom APN WPS QoS profile for roamers - 3

Syntax

--custApn3WpsQosprofileName <string>

Description

Points to the third custom APN WPS QoS profile for roamers.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Name

dedicatedCoreNetwork - Dedicated Core Network Support

Syntax

--dedicatedCoreNetwork <list-item>

Description

Parameter indicates whether DCN feature is activated. By default the parameter is set to 'False' indicating that the DCN feature is not activated.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

description - Description

Syntax

--description <string>

Description

Roaming agreement description.

Class

Optional.

Type / Range

Digit string of length 0 to 160 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

detachBehaviorMappingProfileName - Detach behavior profile name

Syntax

--detachBehaviorMappingProfileName <string>

Description

Identifies the detach behavior profile to use.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\.\-:@]*

Default

default

Name

diamApplProfileListName - Diameter Application Profile List Name

Syntax

--diamApplProfileListName <string>

Description

Diameter Application Profile List Name

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

""

Name

diamNasMappingProfileName - Diameter to NAS cause code profile name

Syntax

--diamNasMappingProfileName <string>

Description

Defines all MME diameter code to NAS cause code mapping.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

default

Name

dlRateLimit - Downlink Rate Limit

Syntax

--dlRateLimit <integer>

Description

The parameter defined maximum value MME allows an UE to transfer data over CPO

Class

Optional.

Type / Range

Integer in the range of 0 or 10 to 10000 in units of DeciHour

Default

20

Name

drmpCiotProfileName - DRMP CIOT profile name

Syntax

--drmpCiotProfileName <string>

Description

DRMP CIOT profile name.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

default

Name

drmpProfileName - DRMP profile name

Syntax

--drmpProfileName <string>

Description

DRMP profile name.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

default

Name

edrxApnListName - eDRX APN list name

Syntax

--edrxApnListName <string>

Description

eDRX APN list name.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

"

Name

edrxEnabled - eDRX enabled

Syntax

--edrxEnabled <list-item>

Description

eDRX enabled.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

edrxHssPtwProfileName - eDRX HSS PTW profile name

Syntax

--edrxHssPtwProfileName <string>

Description

eDRX HSS PTW profile name.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

default

Name

edrxModeWithPsm - eDRX mode with PSM

Syntax

--edrxModeWithPsm <list-item>

Description

eDRX mode with PSM.

Class

Optional.

List

- Both - Specifies both eDRX and PSM.
- PSM - Specifies PSM.
- eDRX - Specifies eDRX.

Default

eDRX

Name

edrxProfileName - eDRX profile name

Syntax

--edrxProfileName <string>

Description

eDRX profile name.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

default

Name

edrxSource - eDRX source

Syntax

--edrxSource <list-item>

Description

eDRX source.

Class

Optional.

List

- HSS - Specifies eDRX source as HSS.
- Local - Specifies eDRX source as local.

Default

Local

Name

emmInfoOverride - EMM info override

Syntax

--emmInfoOverride <list-item>

Description

Flag defining whether the emmInfo message should be overridden with profile information in the emmInfoOverrideTacSv table.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

emmInfoOverrideProfileName - EMM Info Override Profile Name

Syntax

--emmInfoOverrideProfileName <string>

Description

EMM Info Override Profile to be used for this PLMN

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Default

"

Name

enableRszcForbiddenPlmn - Enable RSZC forbidden PLMN

Syntax

--enableRszcForbiddenPlmn <list-item>

Description

Enable forbidden PLMN list using regional subscription zone code.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

enhancedDualConnectivity - 5G NSA Option 3X

Syntax

--enhancedDualConnectivity <list-item>

Description

This parameter enables 5G NSA Option 3x with enhanced capabilities which include higher bit rates, NR access restriction and S/PGW selection with NR capability

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

enhancedDualConnectivityProfileName - Enhanced Dual Connectivity Profile Name

Syntax

--enhancedDualConnectivityProfileName <string>

Description

Identifies the Enhanced Dual Connectivity Profile for NSA Option 3A/3X

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

ePcoSupport - Extended protocol configuration option support

Syntax

--ePcoSupport <list-item>

Description

Extended protocol configuration support.

Restrictions

This element is only applicable when plmn is home/shared.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ePlmnListName - EPLMN list name

Syntax

--ePlmnListName <string>

Description

EPLMN list name

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

ePlmnListNameEnb - EPLMN List Name for eNB

Syntax

--ePlmnListNameEnb <string>

Description

Identifies Equivalent PLMN list name to be sent to eNB

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

eps5gslwkWithoutN26 - EPS-5GS Interworking Without N26

Syntax

--eps5gsIwkWithoutN26 <list-item>

Description

Enables the EPS-5GS interworking without N26 interface support for the UE on MME side.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

epsUpip - EPS UPIP

Syntax

--epsUpip <list-item>

Description

Current parameter enables support for the EPS-UIPI (User Plane Integrity Protection)

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

extendedEmergencyNumListName - Extended Emergency Number List Name

Syntax

--extendedEmergencyNumListName <string>

Description

Extended Emergency Number List Name

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\- _:@]*

Default

"

Name

externalIdSupported - External ID Supported

Syntax

--externalIdSupported <list-item>

Description

Specifies whether the MME advertises support of the External ID to the HSS.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

freqCountMonitoringProfileName - Frequency Count Monitoring Profile

Syntax

--freqCountMonitoringProfileName <string>

Description

Defines the frequency count monitoring profile name applied for the UE.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

fullPgwApnGtpOvlControlSupport - Full PGW APN GTP Overload Control Support

Syntax

--fullPgwApnGtpOvlControlSupport <list-item>

Description

Defines whether the MME supports PGW indicated overload for the UE: -PGW Overload Control Information IE containing APN list on GTPv2 messages, when APN is congested on a PGW, -Back-off Time IE on Create Session Response when PGW rejects the request due to APN congestion.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

grpSvcId - Group service identification

Syntax

--grpSvcId <integer>

Description

Group service identification.

Restrictions

This element can not be Zero if supportGrpSvcProvision is true.

Class

Optional.

Type / Range

Integer in the range of 0 or 1,001 to 4,294,967,295

Default

0

Name

gtpProfileName - GTP profile name

Syntax

--gtpProfileName <string>

Description

GTP profile to be assigned for the roamers.

Restrictions

1. When not empty string, refers valid gtpProfile.name 2. This element is not applicable when the Served and UE PLMNs are the same.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

hlcomDdnAckWithDlBuffering - HLCOM ddn ack with downlink dl buffering

Syntax

--hlcomDdnAckWithDlBuffering <list-item>

Description

DDN ACK with downlink buffering for HLCOM.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

hlcomEnabled - HLCOM enabled

Syntax

--hlcomEnabled <list-item>

Description

HLCOM enabled.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

hlcomPsmDlBufferingPacketCount - HLCOM/PSM downlink buffering packet count

Syntax

--hlcomPsmDlBufferingPacketCount <integer>

Description

PSM downlink buffering packet count for HLCOM and PSM.

Class

Optional.

Type / Range

Integer in the range of 1 to 65,535

Default

1

Name

hlcomUseLocalDlBufferingPacketCount - HLCOM use local downlink buffering packet count

Syntax

--hlcomUseLocalDlBufferingPacketCount <list-item>

Description

Use local downlink buffering packet count for HLCOM.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

hoCellIdRestrictionListNameN26 - NR Cells List for Handover Restriction over N26

Syntax

--hoCellIdRestrictionListNameN26 <string>

Description

Identifies the list of source NR Cells that the UE is either allowed or restricted from being handover over from 5G to EPS. The MME uses the list to either receive or reject the Forward Relocation Request from the AMF.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

""

Name

homerUeTzInCreateAndUpdateBearerResp - Homer UE Time Zone in Create And Update Bearer Response

Syntax

--homerUeTzInCreateAndUpdateBearerResp <list-item>

Description

Parameter controls inclusion and sending of UE time zone IE in Update and Create Bearer Response for Homers. By default, flag shall be set to false. When enabled, MME will include and send UE time zone IE in Update and Create Bearer Response messages towards SGW. Definition of homers is subscribers that are using operator EPC network. In other words, this means UEs with IMSIs PLMN ID same as Serving PLMN ID, UEs with IMSIs PLMN ID different from Serving PLMN ID but treat as home subscriber, and multiple Home PLMN UEs.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

hoRestrictAreaListName - Handover restrict area list name

Syntax

--hoRestrictAreaListName <string>

Description

Defines the UE roaming restrict profile to be used for the service in 4G.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

imeiTacSvRejectionProfileName - IMEI TAC SV Rejection Profile Name

Syntax

```
--imeiTacSvRejectionProfileName <string>
```

Description

The name of the IMEI-TAC SV rejection profile.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

default

Name

imsApnQosprofileName - Default IMS APN QoS profile for roamers

Syntax

```
--imsApnQosprofileName <string>
```

Description

Default IMS APN QoS profile for roamers. IMS APN QoS profile name is not used when served PLMN and UE PLMN are the same. Otherwise, the default is Def-Roam-IMS-APN-QoS-Prfl.

Restrictions

This element is not applicable when the serving and UE PLMNs are the same.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

imsApnRoamingControl - IMS APN roaming control allowed

Syntax

```
--imsApnRoamingControl <list-item>
```

Description

IMS APN roaming control allowed.

Restrictions

This element is only applicable when plmn is roamer.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

imsApnWpsQosprofileName - Default IMS APN WPS QoS profile for roamers

Syntax

--imsApnWpsQosprofileName <string>

Description

Points to the roaming WPS QoS profile for IMS APNs. IMS APN WPS QoS profile name is not used when Serving PLMN equals UE PLMN or when Treat As Home Subscriber is true.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

imsLboProfileName - IMS LBO Profile Name

Syntax

--imsLboProfileName <string>

Description

The name of the IMS LBO profile.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

imsRegistrationFailureStnSr - IMS Registration Failure STN-SR

Syntax

--imsRegistrationFailureStnSr <digits>

Description

The STN-SR value indicating IMS registration failure. HSS indicates the MME the IMS registration success of a UE with special STN-SR parameter values. When IMS registration for a UE has failed, the HSS indicates the MME the IMS registration failure STN-SR as the subscribed STN-SR of the UE.

Class

Optional.

Type / Range

Digit string of length 9 to 15

Default

11111111111

Name

imsRegistrationSuccessStnSr - IMS Registration Success STN-SR

Syntax

--imsRegistrationSuccessStnSr <digits>

Description

The STN-SR value indicating IMS registration success. HSS indicates the MME the IMS registration success of a UE with special STN-SR parameter values. When IMS registration for a UE has succeeded, the HSS indicates the MME the IMS registration success STN-SR (or the real STN-SR of the UE) as the subscribed STN-SR of the UE.

Class

Optional.

Type / Range

Digit string of length 9 to 15

Default

22222222222

Name

includeNwPolicyInAttachAndTauAccept - Include Network Policy in Attach and TAU Accept

Syntax

--includeNwPolicyInAttachAndTauAccept <list-item>

Description

Parameter controls inclusion of Network Policy IE in Attach Accept and TAU Accept messages. By default, it shall be disabled. When enabled, MME shall include Network Policy IE in Attach Accept and TAU accept messages sent to the UE.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

initialGutiAttachEirPercentage - Initial GUTI Attach EIR Percentage

Syntax

--initialGutiAttachEirPercentage <integer>

Description

Parameter controls percentage EIR checking for Initial GUTI Attach Procedure. Default value of 0 indicates no EIR checking. This field is valid when the Attach EIR Percentage Checking General Global Parameter is enabled.

Class

Optional.

Type / Range

Integer in the range of 0 to 100

Default

0

Name

initialImsiAttachEirPercentage - Initial IMSI Attach ECR Percentage

Syntax

--initialImsiAttachEirPercentage <integer>

Description

Parameter controls percentage EIR checking for Initial IMSI Attach Procedure. Default value of 0 indicates no EIR checking. This field is valid when the Attach EIR Percentage Checking General Global Parameter is enabled.

Class

Optional.

Type / Range

Integer in the range of 0 to 100

Default

0

Name

interPlmn5gsToEpsAttach - Inter PLMN 5GS to EPS Attach

Syntax`--interPlmn5gsToEpsAttach <list-item>`**Description**

Allows or restricts inbound inter-plmn Attach from 5G to 4G based on UE GUTI PLMN ID. When restricted, MME invokes NAS Identity Request to retrieve IMSI and proceed with the Attach Request.

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

interPlmn5gsToEpsTau - Inter PLMN Tracking Area Update

Syntax`--interPlmn5gsToEpsTau <list-item>`**Description**

Allows or restricts inbound inter-plmn idle mode Tracking Area Update from 5G to 4G based on UE GUTI PLMN ID. When restricted, MME rejects the TAU with NAS cause code #9, which triggers UE to initiate Attach with IMSI.

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

interPlmnHandoverEpsTo5gs - Inter PLMN EPS to 5GS Handover

Syntax

```
--interPlmnHandoverEpsTo5gs <list-item>
```

Description

Allows or restricts eNB initiated EPS to 5GS Handover at the source MME.

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

interPlmnHandoverTargetNodeBehavior - Inter PLMN Handover Target Node Behavior

Syntax

```
--interPlmnHandoverTargetNodeBehavior <list-item>
```

Description

Sets target MME behaviour over S10 or N26 during inter-plmn Handover.

Class

Optional.

List

- both - S10 and N26 Interfaces - The target MME allows inter-plmn Forward Relocation Request procedure from source MME or from source AMF to proceed.
- n26 - N26 Interface - The target MME allows inter-plmn Forward Relocation Request procedure from source AMF to proceed.
- none - The target MME rejects inter-plmn Forward Relocation Request procedure from source MME or from source AMF due to Handover not allowed.
- s10 - S10 Interface - The target MME allows inter-plmn Forward Relocation Request procedure from source MME to proceed.

Default

both

Name

interPlmnIdleModeMobSrcNodeBehavior - Inter PLMN Idle Mode Mobility Source Node Behavior

Syntax

```
--interPlmnIdleModeMobSrcNodeBehavior <list-item>
```

Description

Sets source or old MME behaviour over S10 or N26 during inter-plmn UE Idle Mode Mobility.

Class

Optional.

List

- both - Both S10 and N26 - Source or old MME allows Context Request procedure from new or target MME or UE Context Transfer procedure from new or target AMF to proceed based on Serving-Target PLMN ID combination during inter-plmn UE Idle Mode Mobility.
- n26 - N26 Interface - Source or old MME allows Context Request procedure from new or target AMF to proceed based on Serving-Target PLMN ID combination during inter-plmn UE Mobility Registration Update.
- none - Source or old MME rejects Context Request procedure from new or target MME or UE Context Transfer procedure from new or target AMF based on Serving-Target PLMN ID combination during inter-plmn UE Idle Mode Mobility.
- s10 - S10 Interface - Source or old MME allows Context Request procedure from new or target MME to proceed based on Serving-Target PLMN ID combination during inter-plmn UE Tracking Area Update.

Default

both

Name

interPlmnInitAttachRegSrcNodeBehavior - Inter PLMN Initial Attach or Registration Source Node Behavior

Syntax

--interPlmnInitAttachRegSrcNodeBehavior <list-item>

Description

Sets source or old MME behaviour over S10 or N26 during inter-plmn UE Initial Attach or Registration.

Class

Optional.

List

- both - Both S10 and N26 Interfaces - Source or old MME allows UE Identification Request procedure from new or target MME or Identification Request procedure from new or target AMF to proceed based on Serving-Target PLMN ID combination during inter-plmn UE Initial Attach or Initial Registration procedure.
- none - Source or old MME rejects UE Identification Request procedure from new or target MME or Identification Request procedure from new or target AMF based on Serving-Target PLMN ID combination during inter-plmn UE Initial Attach or Initial Registration procedure.

Default

both

Name

interPlmnN26EmergencyHoSourceNode - Inter Plmn N26 Emergency Call Handover Source Node

Syntax

--interPlmnN26EmergencyHoSourceNode <list-item>

Description

Identifies MME source node behaviour (allow or reject) for inter plmn 4G to 5G handover of emergency calls triggered by the source eNB.

Class

Optional.

List

- both - Both LBO and Home Routed emergency calls are allowed by the source node to be handed over.
- hrt - Home Routed Traffic - when UE originates emergency call in UE H-PLMN, the call is allowed by source node to be handed over to V-PLMN as home routed call.
- lbo - Local Break Out - When roaming UE originates LBO emergency call in V-PLMN, the call is allowed by source node to be handed over to another PLMN (another V-PLMN or UE H-PLMN).
- none - Neither LBO nor Home Routed emergency calls are allowed by the source node to be handed over.

Default

both

Name

interPlmnN26EmergencyHoTargetNode - Inter PLMN N26 Emergency Call Handover Target Node

Syntax

--interPlmnN26EmergencyHoTargetNode <list-item>

Description

Identifies MME target node behaviour (allow or reject) for inter plmn 5G to 4G handover of emergency calls from AMF source node.

Class

Optional.

List

- both - LBO and HRT - Both LBO and Home Routed emergency calls are allowed by the target node to be handed in.

- hrt - Home Routed Traffic - When UE originates emergency call in UE H-PLMN, the call is allowed by V-PLMN target node to be handed in.
- lbo - Local Break Out - When roaming UE originates emergency call in V-PLMN, the call is allowed by target node in another Plmn (V-PLMN or H-PLMN) to be handed in.
- none - Neither LBO nor Home Routed emergency calls are allowed by the target node to be handed in.

Default

both

Name

interPlmnS10EmergencyHoSourceNode - Inter Plmn S10 Emergency Call Handover Source Node

Syntax

--interPlmnS10EmergencyHoSourceNode <list-item>

Description

Identifies MME source node behaviour (allow or reject) for inter plmn 4G to 4G handover of emergency calls triggered by the source eNB.

Class

Optional.

List

- both - Both LBO and Home Routed emergency calls are allowed by the source node to be handed over.
- hrt - Home Routed Traffic - when UE originates emergency call in UE H-PLMN, the call is allowed by source node to be handed over to V-PLMN as home routed call.
- lbo - Local Break Out - When roaming UE originates LBO emergency call in V-PLMN, the call is allowed by source node to be handed over to another PLMN (another V-PLMN or UE H-PLMN).
- none - Neither LBO nor Home Routed emergency calls are allowed by the source node to be handed over.

Default

both

Name

interPlmnS10EmergencyHoTargetNode - Inter PLMN S10 Emergency Call Handover Target Node

Syntax

--interPlmnS10EmergencyHoTargetNode <list-item>

Description

Identifies MME target node behaviour (allow or reject) for inter plmn 4G to 4G handover of emergency calls from MME source node.

Class

Optional.

List

- both - LBO and HRT - Both LBO and Home Routed emergency calls are allowed by the target node to be handed in.
- hrt - Home Routed Traffic - When UE originates emergency call in UE H-PLMN, the call is allowed by V-PLMN target node to be handed in.
- lbo - Local Break Out - When roaming UE originates emergency call in V-PLMN, the call is allowed by target node in another PLMN (V-PLMN or H-PLMN) to be handed in.
- none - Neither LBO nor Home Routed emergency calls are allowed by the target node to be handed in.

Default

both

Name

interPlmnTauAttachEnable - Inter PLMN TAU attach enable

Syntax

--interPlmnTauAttachEnable <list-item>

Description

This field when set to yes, enables serving network on MME to send S10/S3/Gn messages to the old node for TAU request and attach request with a foreign GUTI. this enable is based on the global parameter inter PLMN TAU attach. if global parameter inter PLMN TAU attach is disabled, this field is ignored. this field when set to yes enables inter-plmn S10/S3/Gn messages to be sent to the old MME/SGSN by the new MME in case of TAU request and attach request with foreign GUTI to get UE context from the old MME/SGSN in a different network.

Restrictions

1. This element is not applicable when the serving and UE PLMNs are the same. 2. This element applies only to 4G.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ipV6SupportForNoDab - IPv6 PDN type used when DAB is not supported

Syntax

--ipV6SupportForNoDab <list-item>

Description

Indicates whether IPv6 PDN type is used when PLMN does not support DAB (Dual access bearer) otherwise IPv4 PDN type is used.

Restrictions

This element is not applicable when the serving and UE PLMNs are the same.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

lapPagingPriProfileName - LAP paging priority profile name

Syntax

--lapPagingPriProfileName <string>

Description

Identifies the LAP paging priority profile.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

lcsOptionsProfileName - Location services options profile name

Syntax

--lcsOptionsProfileName <string>

Description

Defines location services options.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

default

Name

locationBasedGatewaySelection - Location Based Gateway Selection.

Syntax

--locationBasedGatewaySelection <list-item>

Description

Specifies whether location based gateway selection is allowed under UePlmnServices.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ltePgwDomainMccOverride - LTE PGW domain name MCC override

Syntax

--ltePgwDomainMccOverride <digits>

Description

Override of normal PGW DNS domain derivation by using this mobility country code and the mobility network code represented by ltepgwdomainmncoveride.

Restrictions

1. This element is not applicable when the serving and UE PLMNs are the same. 2. If this element is present so must be ltePgwDomainMncOverride

Class

Optional.

Type / Range

Digit string of length 0 or 3

Name

ltePgwDomainMncOverride - LTE PGW domain name MNC override

Syntax

--ltePgwDomainMncOverride <digits>

Description

Override of normal PGW DNS domain derivation by using this mobility network code and the mobility country code represented by ltepgwdomainmccoverride.

Restrictions

1. This element is not applicable when the serving and UE PLMNs are the same. 2. If this element is present so must be ltePgwDomainMccOverride

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 3

Name

lteRoamingAllowed - LTE roaming allowed

Syntax

--lteRoamingAllowed <list-item>

Description

Flag indicating if LTE roaming is allowed.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

lteTo2G3GQoSMappingName - LTE to 2G3G QoS mapping profile name

Syntax

--lteTo2G3GQoSMappingName <string>

Description

Name that identifies the LTE-2G3G QoS mapping profile, whose parameters are used during handover between 2G/3G and 4G networks.

Restrictions

This element is not applicable when the serving and UE PLMNs are the same.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

default

Name

lteTreatAsHomeSubscriber - Treat roamer as home subscriber

Syntax

--lteTreatAsHomeSubscriber <list-item>

Description

When set, treat a roaming UE as if it were a home subscriber of the serving PLMN. If LTE Treat As Home Subscriber is true for a roaming UE, the IMS APN QoS Profile and Non-IMS APN QoS Profile fields do not take effect even if populated.

Restrictions

This element is not applicable when the serving and UE PLMNs are the same.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

lteUeAmbrUseHssValue - LTE UE AMBR use HSS value

Syntax

--lteUeAmbrUseHssValue <list-item>

Description

Indicates for LTE to enable/disable use of HSS subscribed UL UE-AMBR and DL UE-AMBR values at roaming PLMN level.

Restrictions

1. This element applies only to 4G. 2. This element is not applicable when the serving and UE PLMNs are the same.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

maxUeAllowedInAllTa - Maximum UE Allowed in All TA

Syntax

--maxUeAllowedInAllTa <integer>

Description

Maximum number of UE allowed in all the TA of the serving PLMN. This maximum number of allowed UE for TA list overrides this provisioning

Class

Optional.

Type / Range

Integer in the range of 0 to 4,294,967,294

Default

0

Name

maxWildcardApns - Maximum APNs allowed for wildcard APN

Syntax

--maxWildcardApns <integer>

Description

Maximum number of APNs allowed for a wildcard APN in UE subscription data. a UE subscription data may contain a wild card APN if the HPLMN operator allows subscriber to access any network. if MME has received such a wild card APN, MME can allow an APN NI received from the UE even if the APN is not in the UE subscription. the wild card APN is coded as an APN with "*" as its single label, (i.e. a length octet with value one, followed by the ASCII code for the asterisk) in UE subscription.

Class

Optional.

Type / Range

Integer in the range of 0 to 8

Default

1

Name

mccLteHmRtPgWdnsDmOvrd - MCC LTE home routed PGW DNS domain override

Syntax

--mccLteHmRtPgWdnsDmOvrd <digits>

Description

Override the normal PGW domain derivation by using the given MCC and MNC values for a home routed APN.

Class

Optional.

Type / Range

Digit string of length 0 or 3

Name

mmeNbIotCptoUpSwitch - Enable MME triggered control plane bearers to user plane bearers switch for CIOT UE

Syntax

--mmeNbIotCptoUpSwitch <list-item>

Description

This parameters enables MME triggered control plane (CP) bearers to use user plane (UP) i.e. S1-U bearers for CIOT UE. if this parameter is set then MME triggers CP bearers to UP. By default the parameter is not set.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

mmeWbEutranCpToUpSwitch - Enable MME triggered control plane bearers to user plane bearers switch for WB-EUTRAN UE

Syntax

--mmeWbEutranCpToUpSwitch <list-item>

Description

This parameters enables MME triggered control plane (CP) bearers to use user plane (UP) i.e. S1-U bearers for WB-EUTRAN UE. if this parameter is set then MME triggers CP bearers to UP. By default the parameter is not set.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

mncLteHmRtPgWdnsDmOvrd - MNC LTE home routed PGW DNS domain override

Syntax

--mncLteHmRtPgWdnsDmOvrd <digits>

Description

Override the normal PGW domain derivation by using the given MCC and MNC values for a home routed APN.

Class

Optional.

Type / Range

Digit string of length 0 or 2 to 3

Default

"

Name

moEdtCp - moEdtCp

Syntax

--moEdtCp <list-item>

Description

Mobile Originated Early Data Transfer for Control Plane for S11u and T6a

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

mpdnconn - Multiple PDN connection to same APN

Syntax

--mpdnconn <list-item>

Description

If true, MME shall allow and accept multiple PDN connection to the same APN with the same PDN type.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

mtEdtCp - mtEdtCp

Syntax

--mtEdtCp <list-item>

Description

Mobile-Terminated Early Data Transfer over Control Plan

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

mtEdtT6a - mtEdtT6a

Syntax

--mtEdtT6a <list-item>

Description

Mobile Terminated Early Data Transfer over T6a

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

mtEdtUp - mtEdtUp

Syntax

--mtEdtUp <list-item>

Description

Mobile Terminated Early Data Transfer User Plane

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

mtSmsOpt2ForUeInPowerSavings - MT SMS option 2 deployment for UE in power savings

Syntax

--mtSmsOpt2ForUeInPowerSavings <list-item>

Description

Use deployment option 2 specified in TS 23.272 for handling mobile terminated SMS for UE in power savings (idle mode DRX and power savings mode). if this parameter is false then MME uses the option 1 specified in TS 23.272.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

n26IwkMme - n26 interworking on MME

Syntax

--n26IwkMme <list-item>

Description

Controlling enable of N26 interworking on MME

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

n26IwkMmeRestrictionTaiListName - MME N26 IWK Restriction TAI List

Syntax

--n26IwkMmeRestrictionTaiListName <string>

Description

N26 interworking restriction TAI list used by MME to restrict 4G service when UE(s) move from 5G into the respective TACs in the TAI list.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 31 with the allowed pattern of [0-9a-zA-Z_\.\\-:@]*

Default

""

Name

nblotDlDataSizeForCpToUpSwitch - Downlink data packet size threshold for CP bearers to UP bearers switch

Syntax

--nbIotDlDataSizeForCpToUpSwitch <integer>

Description

Provision the size of CIOT UE downlink control plane data packet size threshold in octets for MME to trigger control plane bearers to switch to use user plane bearers i.e. S1-U bearers. By default the data packet size is set to 120 octets. the range of allowed values are 20 to 1024 octets.

Class

Optional.

Type / Range

Integer in the range of 20 to 1,024

Default

120

Name

nbiotSmsWithoutCombinedAttach - Enable SMS without combined Attach For NB-IOT UE

Syntax

--nbiotSmsWithoutCombinedAttach <list-item>

Description

Enable SMS without combined Attach For NB-IOT UE

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

nBlotUeCpToUpSwitch - CIOT UE control plane bearer to user plane bearer switching

Syntax

--nBIotUeCpToUpSwitch <list-item>

Description

Parameter to allow/not allow CIOT UE control plane traffic to user plane i.e. setup radio bearers and transport user data over S1-U.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

nblotUlDataSizeForCpToUpSwitch - Upnlk data packet size threshold for CP bearers to UP bearers switch

Syntax

--nBIotUlDataSizeForCpToUpSwitch <integer>

Description

Provision the size of CIOT UE uplink control plane data packet size threshold in octets for MME to trigger control plane bearers to switch to use user plane bearers i.e. S1-U bearers. By default the data packet size is set to 120 octets. the range of allowed values are 20 to 1024 octets.

Class

Optional.

Type / Range

Integer in the range of 20 to 1,024

Default

120

Name

nonImsApnQosprofileName - Default non-ims APN QoS profile for roamers

Syntax

--nonImsApnQosprofileName <string>

Description

Default non-IMS APN QoS profile for roamers. Non IMS APN QoS profile name is not used when served PLMN and UE PLMN are the same. Otherwise, the default is Def-Roam-NonIMS-APN-QoS-Prfl.

Restrictions

This element is not applicable when the serving and UE PLMNs are the same.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

nonImsApnWpsQosprofileName - Default non-ims APN WPS QoS profile for roamers

Syntax

--nonImsApnWpsQosprofileName <string>

Description

Points to the roaming QoS profile for non IMS APNs. Non IMS APN WPS QoS profile name is not used when Serving PLMN equals UE PLMN or when Treat As Home Subscriber is true.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

nonIpDataTrans - Non-IP data transport

Syntax

--nonIpDataTrans <list-item>

Description

Enable/disable non-ip data transport.

Restrictions

This element is applicable when the serving and UE PLMNs are same or when lteTreatAsHomeSubscriber set to true.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

numberOfAvRequested - Number of Authentication Vectors Requested

Syntax

--numberOfAvRequested <integer>

Description

Defines the number of authentication vectors requested. This parameter, when defined to be used, takes precedence over the Local endpoint parameter. Value 0 for this parameter stands for 'not used', and when that is set, the Local endpoint parameter is applied.

Class

Optional.

Type / Range

Integer in the range of 0 to 5

Default

0

Name

pagingPriProfileListName - Paging priority profile name

Syntax

--pagingPriProfileListName <string>

Description

Identifies the paging priority profile.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

pdnReactivateN1Enable - PDN Reactivation when N1 mode gets enabled

Syntax

--pdnReactivateN1Enable <list-item>

Description

Current parameter enables the MME performing PDN deactivation (reactivation requested) for the non-emergency PDNs after TAU completes for which the UE re-enabled N1 mode for those PDN sessions that are established with standalone PGWs but the associated APNs allow 5GS IWK and UE is not 5G Restricted.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

pdnTypeIpv6Restriction - PDN Type IPv6 Restriction

Syntax

--pdnTypeIpv6Restriction <list-item>

Description

Restricts the use of PDN Type IPv6 or IPv4v6 for the UE.

Class

Optional.

List

- pdnTypeIpv6RestrictionDisabled - PDN Type IPv6 Restriction Disabled - MME functionality to restrict PDN Connectivity procedure attempts with PDN Type IPv6 or IPv4v6 is disabled.
- pdnTypeIpv6RestrictionModify - PDN Type IPv6 Restriction Modify - MME restricts any PDN Connectivity procedure attempts with PDN Type IPv6 or IPv4v6 that the UE performs by continuing the PDN connectivity request procedure using the IPv4 PDN Type.

- pdnTypeIpv6RestrictionReject - PDN Type IPv6 Restriction Reject - MME restricts any PDN Connectivity procedure attempts with PDN Type IPv6 or IPv4v6 that the UE performs by directly rejecting the UE request.

Default

pdnTypeIpv6RestrictionDisabled

Name

pdnTypeIpv6RestrictionPdnRejectionCause - PDN Type IPv6 Restriction PDN Rejection Cause

Syntax

--pdnTypeIpv6RestrictionPdnRejectionCause <integer>

Description

Defines the ESM cause value when MME rejects the PDN connectivity request of the UE due to PDN Type IPv6 being restricted.

Class

Optional.

Type / Range

Integer in the range of 8, 26-39, 41-47, 49-60, 65-66, 81, 95-101, 111-113

Default

50

Name

pgwSelectionCcApnName - P-GW selection by charging characteristics APN name

Syntax

--pgwSelectionCcApnName <string>

Description

Identifies the table to be used for P-Gateway selection by charging characteristics using APN.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

provisionedDefaultApn - Provisioned default APN

Syntax

--provisionedDefaultApn <string>

Description

Access point name, the corrected value.

Class

Optional.

Type / Range

Digit string of length 0 to 62 with the allowed pattern of [0-9a-zA-Z\.\-]*

Name

psmApnListName - PSM APN list name

Syntax

--psmApnListName <string>

Description

PSM APN list name.

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 31 with the allowed pattern of [0-9a-zA-Z_\.\-:@]*

Default

"

Name

psmEnabled - PSM enabled

Syntax

--psmEnabled <list-item>

Description

PSM enabled.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

reAuthTimePeriod - Reauthentication Time Period

Syntax

--reAuthTimePeriod <integer>

Description

This field defines the time period between successive UE authentications forced by the MME.
When value equals zero then the feature is disabled

Class

Optional.

Type / Range

Integer in the range of 60 to 1440 including value zero in units of Minute

Default

0

Name

registeredUePerTaListName - Registered UE per TA List Name

Syntax

--registeredUePerTaListName <string>

Description

This list consists of TA and maximum number of registered UE allowed for each TA in the list. The list is created using the following commands: cmm registeredUePerTa create --name <string> cmm registeredUePerTaList create --registeredUePerTaListName <string> --servingPlmnName <integer> --tac <integer> [--maxUeAllowedPerTac <integer>

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

rejectGrpSrvAttachPdnRestricted - Reject group service attach PDN restricted

Syntax

--rejectGrpSrvAttachPdnRestricted <list-item>

Description

Selection of option to reject or accept a UE attach request if UE does not support ERw/oPDN and bit 2 of local-group-id is set to indicate attach without PDN connection restricted.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

rejectUeTauMaxUeExceed - Reject UE TAU request if Maximum Number Exceeded

Syntax

--rejectUeTauMaxUeExceed <list-item>

Description

If the parameter is set then MME rejects the UE TAU request if the allowed UEs exceed the max provisioned for the tracking area.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

restNasMappingProfileName - Access restriction to NAS cause code profile name

Syntax

--restNasMappingProfileName <string>

Description

Defines all MME access restriction to NAS cause code mapping.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\.\-:@]*

Default

default

Name

rfspAccessRestrictionListName - Access Restriction for RFSP List Name

Syntax

--rfspAccessRestrictionListName <string>

Description

A reference to a list of RFSP values for which access is to be restricted for the UE.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z_\.\-:@]*

Default

""

Name

roamerUeTzInCreateAndUpdateBearerResp - Roamer UE Time Zone in Create And Update Bearer Response

Syntax

--roamerUeTzInCreateAndUpdateBearerResp <list-item>

Description

Parameter controls inclusion and sending of UE time zone IE in Update and Create Bearer Response for Roamers. By default, flag shall be set to false. When enabled, MME will include and send UE time zone IE in Update and Create Bearer Response messages towards SGW. Definition of Roamers for eg: is subscriber roaming into 'Operator A' network but using their own PGW, PLMN not owned by 'Operator A'. In other words, HRT, meaning the roamer is using PGW of the roamer network (e.g., 'Operator B' UE roaming in 'Operator A' network but using 'Operator B' PGW. However LBO case doesn't apply for this feature.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

rohc - ROHC

Syntax

--rohc <list-item>

Description

ROHC.

Restrictions

This element is applicable when the serving and UE PLMNs are same or when lteTreatAsHomeSubscriber set to true.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

rszcForbiddenNasCauseCode - NAS cause code for RSZC PLMN

Syntax

--rszcForbiddenNasCauseCode <integer>

Description

NAS cause to be sent to UE if UE MM request is rejected due to regional subscription zone code PLMN checks.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

11

Name

s102PagingPriProfileName - S102 paging priority profile name

Syntax

--s102PagingPriProfileName <string>

Description

Identifies the S102 paging priority profile.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

s1uDataTrans - S1U data transfer

Syntax

--s1uDataTrans <list-item>

Description

Enable S1-U data transfer.

Restrictions

This element is applicable when the serving and UE PLMNs are same or when lteTreatAsHomeSubscriber set to true.

Class

Optional.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

false

Name

sendDcnrRestrictionsIndToUe - Send DC-NR Restriction Indication to UE

Syntax

--sendDcnrRestrictionsIndToUe <list-item>

Description

This parameter applies when the 'useNrSecondaryArdAsDcnrIndication' parameter is set and the UE does not provide a DC-NR indication to the MME. In this case, the setting of sendDcnrRestrictionsIndToUe defines whether the MME is to send DC-NR restrictions to the UE in NAS messaging.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

sendLtemRatToPgw - Send LTE-M RAT to PGW

Syntax

--sendLtemRatToPgw <list-item>

Description

Specifies whether the MME should send an indication to the PGW (via S11 messaging to the SGW) when the UE is using the LTE-M RAT Type. The PGWs would then include this in subsequent charging records.

Class

Optional.

List

- No - Boolean value of no.
- Yes - Boolean value of yes.

Default

No

Name

separateEplmnListNasS1ap - Separate EPLMN List on NAS and S1AP

Syntax

--separateEplmnListNasS1ap <list-item>

Description

Enable sending separate Equivalent PLMN Lists to NAS and S1AP. The EPLMN Lists applied to NAS and S1AP are copied from EPLMN List Name and EPLMN List Name for eNB, respectively.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

serviceGapControlActive - Service Gap Control Active

Syntax

--serviceGapControlActive <list-item>

Description

Service Gap Control is active for UEs from the PLMN

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

serviceGapTimerSource - Service Gap Timer Source

Syntax

--serviceGapTimerSource <list-item>

Description

Specifies whether to use Service Gap Timer from HSS or Local provisioning

Class

Optional.

List

- Local - Use locally-provisioned Service Gap Timer
- HSS - Use HSS-provided Service Gap Timer

Default

HSS

Name

sGdDiamRpMappingProfileName - SGd Diameter RP Mapping Profile Name

Syntax

--sGdDiamRpMappingProfileName <string>

Description

Defines all SGd interface Diameter code to SMS Relay Protocol (RP) cause code mapping.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\.\\-:@]*

Default

default

Name

sgwRelocationAllow - SGW Relocation Allow

Syntax

--sgwRelocationAllow <list-item>

Description

Parameter 'sgwRelocationAllow' will override the existing 'sgwRelocationRestriction' global gparm 'on per UEPLMN/serving PLMN combination. If the flag is set to true, SGW will be relocated at TAU procedure where the current SGW is not in the candidate list of the new TAI. This applies to gateway selection mode 2. By default this flag is disabled.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

smsProfileName - SMS profile name

Syntax

--smsProfileName <string>

Description

The profile containing the SMS delivery related parameters for the SMS in MME feature.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\.\-:@]*

Default

default

Name

smsS102PagingPriProfileName - S102 SMS paging priority profile name

Syntax

--smsS102PagingPriProfileName <string>

Description

Identifies the S102 SMS paging priority profile.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z_\.\-:@]*

Name

srvcc5gMscServerAddrToAmf - 5G-SRVCC MSC Server Address to AMF

Syntax

--srvcc5gMscServerAddrToAmf <list-item>

Description

Enables MME (acting as the MME_SRVCC) to deliver the MSC server address to the AMF over N26 interface.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

support5gSrvcc - Support 5G-SRVCC

Syntax

--support5gSrvcc <list-item>

Description

Enables the intermediary role of the MME_SRVCC (MME supporting 5G-SRVCC) for the UE for regular voice calls. The MME_SRVCC passes messages between the AMF and MSC server during 5G-SRVCC handover of voice calls from NG-RAN/5GS to 3GPP UTRAN/CS domain.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

supportGrpSvcProvision - Support group service provision

Syntax

--supportGrpSvcProvision <list-item>

Description

Support group service provision.

Restrictions

This element is only applicable when plmn is home/shared.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

svcAgreementProfileName - Service agreement profile name

Syntax

--svcAgreementProfileName <string>

Description

Defines the MME PLMN service agreement profile to be used for service.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

default

Name

t6aDiamToNasProfileName - T6A diam to NAS profile name

Syntax

--t6aDiamToNasProfileName <string>

Description

Configure T6a diameter error to NAS cause code profile.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

default

Name

taiZoneMappingListName - TAI Zone Mapping List Name

Syntax

--taiZoneMappingListName <string>

Description

TAI Zone Mapping List Name

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

""

Name

tauDataForwarding - TAU data forwarding

Syntax

--tauDataForwarding <list-item>

Description

TAU data forwarding.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ueAmbrDownLink - UE AMBR downlink bitrate

Syntax

--ueAmbrDownLink <integer>

Description

LTE UE aggregate maximum downlink bitrate.

Restrictions

This element applies only to LTE. This element is not applicable when the serving and UE PLMNs are the same. The granularity is 8 when the value is between 64 and 568. The granularity is 64 when the value is between 576 and 8640. The granularity is 100 when the value is between 8700 and 16000. The granularity is 1000 when the value is between 17000 and 128000. The granularity is 2000 when the value is between 130000 and 256000.

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64- 568, 576-8640, 8700-16000, 17000-128000, 130000-4000000000 in units of kbps

Default

0

Name

ueAmbrUpLink - UE AMBR uplink bitrate

Syntax

--ueAmbrUpLink <integer>

Description

LTE UE aggregate maximum uplink bitrate.

Restrictions

This element applies only to LTE. This element is not applicable when the serving and UE PLMNs are the same. The granularity is 8 when the value is between 64 and 568. The granularity is 64 when the value is between 576 and 8640. The granularity is 100 when the value is between 8700 and 16000. The granularity is 1000 when the value is between 17000 and 128000. The granularity is 2000 when the value is between 130000 and 256000.

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64- 568, 576-8640, 8700-16000, 17000-128000, 130000-4000000000 in units of kbps

Default

0

Name

ueFailopenProfileName - MME Failopen Profile Name

Syntax

--ueFailopenProfileName <string>

Description

The name of the failopen profile. A failopen profile as part of uePlmnServices is only applied if an APN configuration profile list is defined as part of it.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

""

Name

ueMonitoringEnabled - UE monitoring enabled

Syntax

--ueMonitoringEnabled <list-item>

Description

Indicate if UE monitoring is enabled.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ueMonitoringProfileName - UE monitoring profile name

Syntax

--ueMonitoringProfileName <string>

Description

Reference to a UE monitoring profile in uemonitoringprofile.

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

default

Name

ueUsagePgwFallbackSelection - UE Usage PGW Fallback Selection

Syntax

--ueUsagePgwFallbackSelection <list-item>

Description

If the parameter is set to 'True' then MME attempts to select a PGW that does not support the UE Usage Type if MME failed to select a PGW supporting the UE Usage Type.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ueUsagePgwRelocation - PGW Relocation on UE Usage Type change

Syntax

--ueUsagePgwRelocation <list-item>

Description

This parameter is used by a new MME for TAU with MME relocation. The new MME checks whether the UE usage Type is supported by the PGW associated with the active PDN

connection. If the PGW does not support the UE Usage Type and if the parameter is set then the new MME will deactivate the PDN connection so that a PGW supporting the UE Usage Type can be selected. If the PDN connection is the last PDN connection then the UE is detached. if the parameter is set to false then the MME will not take check UE Usage Type of the PGW.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ueUsagePgwSelection - PGW Selection using UE Usage Type

Syntax

--ueUsagePgwSelection <list-item>

Description

Parameter enables the use of UE Usage Type to select a PGW if the FID f11601-01 is enabled.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ueUsageSgwFallbackSelection - UE Usage SGW Fallback Selection

Syntax

--ueUsageSgwFallbackSelection <list-item>

Description

If the parameter is set to 'True' then MME will select a SGW that does not support the UE Usage Type if MME fails to select a SGW supporting the UE Usage Type.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ueUsageSgwRelocation - UE Usage SGW Relocation

Syntax

--ueUsageSgwRelocation <list-item>

Description

This parameter applies to target MME when there is MME relocation. In the case of MME relocation with SGW relocation then the target MME will select a SGW supporting the UE Usage Type. If SGW relocation is not required, then the target MME checks to see whether the SGW supports the UE Usage Type. If the SGW does not support the UE Usage type and if this parameters is set to 'Yes' then MME initiates SGW relocation to select a SGW supporting the UE Usage Type.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ueUsageSgwSelection - UE Usage SGW Selection

Syntax

--ueUsageSgwSelection <list-item>

Description

If this parameter is set to to 'True', then MME selects SGW supporting the UE Usage Type.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

ueUsageSourceImsiRange - ueUsageSourceImsiRange

Syntax

--ueUsageSourceImsiRange <list-item>

Description

if ueUsageSourceImsiRange is true and if the UE matches an imsiRangeServices record, the CMM will assign the ueUsageType from the imsiRangeServices to the UE (or blank if no UUT is defined for that IMSI Range). Otherwise, the CMM will assign to the UE the ueUsageType from the HSS (or blank if not present).

Restrictions

This element is applicable when the serving and UE PLMNs are same or when lteTreatAsHomeSubscriber set to true.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ulRateLimit - uplink rate limit

Syntax

--ulRateLimit <integer>

Description

MME allows an UE to transfer up to the maximum data over CPO with the parameter limited value for uplink data.

Class

Optional.

Type / Range

Integer in the range of 0 or 10 to 10000 in units of DeciHour

Default

20

Name

upCloT - UP CloT

Syntax

--upCIoT <list-item>

Description

Enable CP CIOT optimization.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

useApnOiToDetermineHrt - Use APN-OI to Determine HRT

Syntax

--useApnOiToDetermineHrt <list-item>

Description

Current parameter enables using the APN-OI instead of the PGW FQDN to determine if a PDN Session is HRT for a roaming subscriber when applying QoS modification for a PDN session in accordance to local configuration and when sending the UE Time Zone IE in UBR and CBR Response messages. The parameter is effective only when the gParm useApnOiToDetermineHrt is also enabled.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

useLocallyProvQoS - Use Locally Provisioned QoS

Syntax

--useLocallyProvQoS <list-item>

Description

Parameter controls whether to use locally configured QoS values all the time. By default this shall be disabled, when enabled MME shall always use the locally configured QoS values, HSS QoS subscription values shall be ignored.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

useLocallyProvQoSForHomers - Use Locally Provisioned QoS for Homers

Syntax

--useLocallyProvQoSForHomers <list-item>

Description

Parameter controls whether to use locally configured QoS values all the time. By default this shall be disabled, when enabled MME shall always use the locally configured QoS values for homers. HSS QoS subscription values shall be ignored.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

useN26SvcParamS10Ho - Use N26 Service Parameter in S10 Handover

Syntax

--useN26SvcParamS10Ho <list-item>

Description

Current parameter enables using N26 service parameter as part of DNS query for target MME selection during S10 handover

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

useNrSecondaryArdAsDcnrIndication - Use NR Secondary ARD As DC-NR Indication

Syntax

--useNrSecondaryArdAsDcnrIndication <list-item>

Description

MME preferentially selects an S/PGW with 'nc-nr' in the app-protocol name of the DNS record based solely on the Access-Restriction-Data (NR as secondary RAT in E-UTRAN not allowed bit).

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

uutOverrideProfileName - UE Usage Type Override Profile Name

Syntax

--uutOverrideProfileName <string>

Description

Specifies profile name for assigning local UE Usage Type override

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 31 with the allowed pattern of [0-9a-zA-Z_\.\\-:@]*

Default

"

Name

wbEutranDlDataSizeForCpToUpSwitch - Downlink data packet size threshold for CP bearers to UP bearers switch

Syntax

--wbEutranDlDataSizeForCpToUpSwitch <integer>

Description

Provision the size of WB-EUTRAN UE downlink control plane data packet size threshold in octets for MME to trigger control plane bearers to switch to use user plane bearers i.e. S1-U bearers. By default the data packet size is set to 120 octets. The range of allowed values are 20 to 1024 octets.

Class

Optional.

Type / Range

Integer in the range of 20 to 1,024

Default

120

Name

wbEutranUeCpToUpSwitch - WB-EUTRAN UE control plane bearers to user plane switching

Syntax

--wbEutranUeCpToUpSwitch <list-item>

Description

Parameter to allow WB-EUTRAN UE control plane bearers switching to user plane i.e. set up radio bearers and transport user data over S1-U.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

wbEutranUlDataSizeForCpToUpSwitch - Upnlink data packet size threshold for CP bearers to UP bearers switch

Syntax

--wbEutranUlDataSizeForCpToUpSwitch <integer>

Description

Provision the size of WB-EUTRAN UE uplink control plane data packet size threshold in octets for MME to trigger control plane bearers to switch to use user plane bearers i.e. S1-U bearers. By default the data packet size is set to 120 octets. The range of allowed values are 20 to 1024 octets.

Class

Optional.

Type / Range

Integer in the range of 20 to 1,024

Default

120

Name

wusProfileName - Wake Up Signal Profile Name

Syntax

--wusProfileName <string>

Description

Specifies the applicable profile for Wake Up Signaling for UEs from this PLMN

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

zoneCodeExtendedEmergencyNumListName - Zone Code Extended Emergency Number List Name

Syntax

--zoneCodeExtendedEmergencyNumListName <string>

Description

Zone Code Extended Emergency Number List Name

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.133 cmm ueRoamRestrictProfile

UE Roaming TAI and LAI Restriction Profile. Syntax, description, parameters, and roles for cmm ueRoamRestrictProfile command.

Syntax

Description

This command is used to administer the UE Roaming TAI and LAI Restriction Profile and includes parameters for defining the list type, serving or equivalent network name, LAI or TAI network name, and Location Area Code or Tracking Area Code.

Parameters

Name

areaListName - TAI and LAI area list name

Syntax

--areaListName <string>

Description

Name that identifies roaming restriction profile.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

servedPlmnName - Mobile network name

Syntax

--servedPlmnName <string>

Description

Name that identifies the serving or equivalent network containing the restricted TAI or LAI.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

listType - List type

Syntax

--listType <list-item>

Description

Indicates the type of list being provisioned.

Class

Required key.

List

- LAI List - Specify location area identifier list.
- TAI List - Specify tracking area identifier list.

Name

taiLaiPlmnName - LAI/TAI mobile network name

Syntax

--taiLaiPlmnName <string>

Description

Name that identifies LAI/TAI network.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

tacLac - Restricted TAC or restricted LAC

Syntax

--tacLac <integer>

Description

Restricted tracking area code or location area code.

Class

Required key.

Type / Range

Integer in the range of 1 to 65,535

Name

ueRoamRestrictProfileId - UE roaming restrict profile ID

Syntax

--ueRoamRestrictProfileId <integer>

Description

Defines the UE roaming restrict profile to be used for the service.

Class

Read only.

Type / Range

Integer in the range of 1 to 5,000

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.134 cmm uutOverrideProfile

UUT Override Profile. Syntax, description, parameters, and roles for cmm uutOverrideProfile command.

Syntax**Description**

Specifies conditions under which the MME is to assign a local value for UE Usage Type. This local value is then used to determine whether to reroute or keep the UE on the MME. If the UE is kept, the value is used for SGW and PGW DNS selection.

Parameters**Name**

name - UE Usage Type Override Profile Name

Syntax

--name <string>

Description

Name of the override profile

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

dcnrPrioritizedOverN1 - DCNR Prioritized

Syntax

--dcnrPrioritizedOverN1 <list-item>

Description

Specifies the relative priority of local UUT assignment. For this priority field to be used, the DC-NR and N1 overrides (UUTs) must be provisioned in the profile. Otherwise, this field is ignored. If this field is set to true, DC-NR has priority over N1. Otherwise, N1 has priority. The default is false (N1 priority).

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ignoreArdDcnrDisallowed - UE DC NR Ignore ARD

Syntax

--ignoreArdDcnrDisallowed <list-item>

Description

Specifies whether to ignore ARD from HSS that restricts UE the use of DC-NR. If ignore is false, the MME will not assign a local UUT override for DC-NR regardless of UE capability. The default is false.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

ue5gcRestricted - UE 5G Core Restricted

Syntax

--ue5gcRestricted <integer>

Description

UUT Override value to apply to UEs whose HSS restricts 5G core use as indicated in subscription profile in S6a:ULA and S6a:IDR messaging. If a value of 0 is specified, there is no UUT override assigned.

Class

Optional.

Type / Range

UE Usage type in the range of 1 to 126 or 128 to 255, or 0 if not used.

Default

0

Name

ue5gNasNotSupported - UE 5G NAS Not Supported

Syntax

--ue5gNasNotSupported <integer>

Description

UUT Override value to apply to UEs not capable of 5G NAS operation as indicated by the 'N1 mode' bit in the UE network capability IE that the UE provides in Attach and TAU messaging. If a value of 0 is specified, there is no UUT override assigned.

Class

Optional.

Type / Range

UE Usage type in the range of 1 to 126 or 128 to 255, or 0 if not used.

Default

0

Name

ue5gNasSupported - UE 5G NAS Supported

Syntax

--ue5gNasSupported <integer>

Description

UUT Override value to apply to UEs capable of 5G NAS operation as indicated by the 'N1 mode' bit in the UE network capability IE that the UE provides in Attach and TAU messaging. If a value of 0 is specified, there is no UUT override assigned.

Class

Optional.

Type / Range

UE Usage type in the range of 1 to 126 or 128 to 255, or 0 if not used.

Default

0

Name

ueDcnrNotSupportedUut - UE DC-NR Not Supported

Syntax

--ueDcnrNotSupportedUut <integer>

Description

UUT Override value to assign to UEs that do not indicate support of DC-NR operation via the 'DCNR' bit (octet 9, bit 5) in the UE network capability IE that the UE provides in Attach and TAU messaging. If an override value of 0 is specified, there is no UUT override to be assigned to the UE.

Class

Optional.

Type / Range

UE Usage type in the range of 1 to 126 or 128 to 255, or 0 if not used.

Default

0

Name

ueDcnrSupportedUut - UE DC-NR Supported

Syntax

--ueDcnrSupportedUut <integer>

Description

UUT Override value to assign to UEs that indicate support of DC-NR operation via the 'DCNR' bit (octet 9, bit 5) in the UE network capability IE that the UE provides in Attach and TAU messaging. If an override value of 0 is specified, there is no UUT override to be assigned to the UE.

Class

Optional.

Type / Range

UE Usage type in the range of 1 to 126 or 128 to 255, or 0 if not used.

Default

0

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.135 cmm vdpProfile

VDP profile. Syntax, description, parameters, and roles for `cmm vdpProfile` command.

Syntax

Description

Voice Domain Preference profile.

Parameters

Name

plmnName - Mobile network name

Syntax`--plmnName <string>`**Description**

Name that identifies serving network.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of `[0-9a-zA-Z\.\-_:@]*`

Name

name - VDP profile name

Syntax`--name <string>`**Description**

Voice domain preference profile name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

dCsOnlyCombAttachTreatment - Data centric CS only combined Attach treatment

Syntax

--dCsOnlyCombAttachTreatment <list-item>

Description

Data centric circuit switched only treatment for combined attach scenarios. If DEFAULT, the behavior will be to use the value from dCsOnlyTreatment. If not DEFAULT, override dCsOnlyTreatment for Combined Attach scenarios with the specified value.

Class

Optional.

List

- ALW - Allow - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are allowed.
- CS_D_NOT_AVAIL - CS domain not available - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. in attach accept/tau accept emmcause portion send cause "CS domain not available" for the noneps part.
- CSFB_NOT_PREF - CSFB not preferred - Anytype of noneps procedure (i.e. combined-attach or combined-ta) is allowed. in the attach accept/tau accept message we will indicate to the UE that "CSFB is not preferred".
- NAS - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are not allowed. use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure.
- SMS_ONLY - SMS only - NonEps procedure allowed however indicate to the UE in attach accept/tau accept that they are only allowed to have SMS service.
- EPS_ONLY_NAS - EPS only NAS - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. Use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure for the noneps part.
- DEFAULT - Use the treatment provisioned commonly for all Attach and TAU procedures.

Default

DEFAULT

Name

dCsOnlyCombTauTreatment - Data centric CS only combined TAU treatment

Syntax

--dCsOnlyCombTauTreatment <list-item>

Description

Data centric circuit switched only treatment for combined TAU scenarios. If DEFAULT, the behavior will be to use the value from dCsOnlyTreatment. If not DEFAULT, override dCsOnlyTreatment for Combined TAU scenarios with the specified value.

Class

Optional.

List

- ALW - Allow - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are allowed.
- CS_D_NOT_AVAIL - CS domain not available - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. in attach accept/tau accept emmcause portion send cause "CS domain not available" for the noneps part.
- CSFB_NOT_PREF - CSFB not preferred - Anytype of noneps procedure (i.e. combined-attach or combined-ta) is allowed. in the attach accept/tau accept message we will indicate to the UE that "CSFB is not preferred".
- NAS - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are not allowed. use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure.
- SMS_ONLY - SMS only - NonEps procedure allowed however indicate to the UE in attach accept/tau accept that they are only allowed to have SMS service.
- EPS_ONLY_NAS - EPS only NAS - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. Use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure for the noneps part.
- DEFAULT - Use the treatment provisioned commonly for all Attach and TAU procedures.

Default

DEFAULT

Name

dCsOnlyImsSupervision - Data centric CS only IMS supervision

Syntax

--dCsOnlyImsSupervision <list-item>

Description

Data centric circuit switched only IMS supervision flag. The IMS supervision functionality of the MME checks that the UE activates the IMS PDN connection and performs successful IMS registration. Otherwise the MME detaches the UE.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

dCsOnlyNasCause - Data centric CS only NAS cause code

Syntax

--dCsOnlyNasCause <integer>

Description

Data centric circuit switched only cause code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

15

Name

dCsOnlyNonEpsNasCause - Data centric CS only Non EPS NAS cause

Syntax

--dCsOnlyNonEpsNasCause <integer>

Description

The NAS cause used when non-EPS procedure is not allowed for Data centric CS only UE.

Class

Optional.

Type / Range

Non EPS rejection NAS cause

Default

18

Name

dCsOnlySendForbiddenTaiList - Data Centric CS Voice Only Send Forbidden TAI List

Syntax

--dCsOnlySendForbiddenTaiList <list-item>

Description

Controls whether the forbidden TAI list is sent for Data centric, CS voice only UEs. Applied when global parameter VDP Forbidden TAI List on VDP Profile is set to true.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

dCsOnlyTreatment - Data centric CS only treatment

Syntax

--dCsOnlyTreatment <list-item>

Description

Data centric circuit switched only treatment.

Class

Optional.

List

- ALW - Allow - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are allowed.
- CSFB_NOT_PREF - CSFB not preferred - Anytype of noneps procedure (i.e. combined-attach or combined-ta) is allowed. in the attach accept/tau accept message we will indicate to the UE that "CSFB is not preferred".
- CS_D_NOT_AVAIL - CS domain not available - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. in attach accept/tau accept emmcause portion send cause "CS domain not available" for the noneps part.
- EPS_ONLY_NAS - EPS only NAS - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. Use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure for the noneps part.
- NAS - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are not allowed. use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure.
- SMS_ONLY - SMS only - NonEps procedure allowed however indicate to the UE in attach accept/tau accept that they are only allowed to have SMS service.

Default

ALW

Name

dCsPreferCombAttachTreatment - Data centric CS preferred Combined Attach treatment

Syntax

--dCsPreferCombAttachTreatment <list-item>

Description

Data centric circuit switched preferred, IMS secondary treatment for combined attach scenarios. If DEFAULT, the behavior will be to use the value from dCsPreferTreatment. If not DEFAULT, override dCsPreferTreatment for Combined Attach scenarios with the specified value.

Class

Optional.

List

- ALW - Allow - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are allowed.
- CS_D_NOT_AVAIL - CS domain not available - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. in attach accept/tau accept emmcause portion send cause "CS domain not available" for the noneps part.
- CSFB_NOT_PREF - CSFB not preferred - Anytype of noneps procedure (i.e. combined-attach or combined-ta) is allowed. in the attach accept/tau accept message we will indicate to the UE that "CSFB is not preferred".
- NAS - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are not allowed. use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure.
- SMS_ONLY - SMS only - NonEps procedure allowed however indicate to the UE in attach accept/tau accept that they are only allowed to have SMS service.
- EPS_ONLY_NAS - EPS only NAS - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. Use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure for the noneps part.
- DEFAULT - Use the treatment provisioned commonly for all Attach and TAU procedures.

Default

DEFAULT

Name

dCsPreferCombTauTreatment - Data centric CS preferred combined TAU treatment

Syntax

--dCsPreferCombTauTreatment <list-item>

Description

Data centric circuit switched preferred, IMS secondary treatment for combined TAU scenarios. If DEFAULT, the behavior will be to use the value from dCsPreferTreatment. If not DEFAULT, override dCsPreferTreatment for Combined TAU scenarios with the specified value.

Class

Optional.

List

- ALW - Allow - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are allowed.
- CS_D_NOT_AVAIL - CS domain not available - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. in attach accept/tau accept emmcause portion send cause "CS domain not available" for the noneps part.
- CSFB_NOT_PREF - CSFB not preferred - Anytype of noneps procedure (i.e. combined-attach or combined-ta) is allowed. in the attach accept/tau accept message we will indicate to the UE that "CSFB is not preferred".
- NAS - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are not allowed. use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure.
- SMS_ONLY - SMS only - NonEps procedure allowed however indicate to the UE in attach accept/tau accept that they are only allowed to have SMS service.
- EPS_ONLY_NAS - EPS only NAS - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. Use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure for the noneps part.
- DEFAULT - Use the treatment provisioned commonly for all Attach and TAU procedures.

Default

DEFAULT

Name

dCsPreferImsSupervision - Data centric CS preferred IMS supervision

Syntax

--dCsPreferImsSupervision <list-item>

Description

Data centric CS preferred IMS secondary IMS supervision flag. The IMS supervision functionality of the MME checks that the UE activates the IMS PDN connection and performs successful IMS registration. Otherwise the MME detaches the UE.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

dCsPreferNasCause - Data centric CS preferred NAS cause code

Syntax

--dCsPreferNasCause <integer>

Description

Data centric circuit switched preferred IMS secondary cause code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

15

Name

dCsPreferNonEpsNasCause - Data centric CS preferred Non EPS NAS cause

Syntax

--dCsPreferNonEpsNasCause <integer>

Description

The NAS cause used when non-EPS procedure is not allowed for Data centric circuit switched preferred, IMS secondary UE.

Class

Optional.

Type / Range

Non EPS rejection NAS cause

Default

18

Name

dCsPreferSendForbiddenTaiList - Data Centric CS Voice Preferred Send Forbidden TAI List

Syntax

--dCsPreferSendForbiddenTaiList <list-item>

Description

Controls whether the forbidden TAI list is sent for Data centric, CS voice preferred UEs.
Applied when global parameter VDP Forbidden TAI List on VDP Profile is set to true.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

dCsPreferTreatment - Data centric CS preferred treatment

Syntax

--dCsPreferTreatment <list-item>

Description

Data centric circuit switched preferred IMS secondary treatment.

Class

Optional.

List

- ALW - Allow - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are allowed.
- CSFB_NOT_PREF - CSFB not preferred - Anytype of noneps procedure (i.e. combined-attach or combined-ta) is allowed. in the attach accept/tau accept message we will indicate to the UE that "CSFB is not preferred".
- CS_D_NOT_AVAIL - CS domain not available - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. in attach accept/tau accept emmcause portion send cause "CS domain not available" for the noneps part.
- EPS_ONLY_NAS - EPS only NAS - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. Use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure for the noneps part.
- NAS - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are not allowed. use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure.
- SMS_ONLY - SMS only - NonEps procedure allowed however indicate to the UE in attach accept/tau accept that they are only allowed to have SMS service.

Default

ALW

Name

description - Description

Syntax

--description <string>

Description

Profile description.

Class

Optional.

Type / Range

Digit string of length 0 to 160 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Name

dlmsOnlyCombAttachTreatment - Data centric IMS only Combined Attach treatment

Syntax

-dImsOnlyCombAttachTreatment <list-item>

Description

Data centric IMS only treatment for combined attach scenarios. If DEFAULT, the behavior will be to use the value from dlmsOnlyTreatment. If not DEFAULT, override dlmsOnlyTreatment for Combined Attach scenarios with the specified value.

Class

Optional.

List

- ALW - Allow - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are allowed.
- CS_D_NOT_AVAIL - CS domain not available - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. in attach accept/tau accept emmcause portion send cause "CS domain not available" for the noneps part.
- CSFB_NOT_PREF - CSFB not preferred - Anytype of noneps procedure (i.e. combined-attach or combined-ta) is allowed. in the attach accept/tau accept message we will indicate to the UE that "CSFB is not preferred".
- NAS - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are not allowed. use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure.
- SMS_ONLY - SMS only - NonEps procedure allowed however indicate to the UE in attach accept/tau accept that they are only allowed to have SMS service.
- EPS_ONLY_NAS - EPS only NAS - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. Use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure for the noneps part.
- DEFAULT - Use the treatment provisioned commonly for all Attach and TAU procedures.

Default

DEFAULT

Name

dlmsOnlyCombTauTreatment - Data centric IMS only Combined TAU treatment

Syntax

--dlmsOnlyCombTauTreatment <list-item>

Description

Data centric IMS only treatment for combined TAU scenarios. If DEFAULT, the behavior will be to use the value from dlmsOnlyTreatment. If not DEFAULT, override dlmsOnlyTreatment for Combined TAU scenarios with the specified value.

Class

Optional.

List

- ALW - Allow - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are allowed.
- CS_D_NOT_AVAIL - CS domain not available - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. in attach accept/tau accept emmcause portion send cause "CS domain not available" for the noneps part.
- CSFB_NOT_PREF - CSFB not preferred - Anytype of noneps procedure (i.e. combined-attach or combined-ta) is allowed. in the attach accept/tau accept message we will indicate to the UE that "CSFB is not preferred".
- NAS - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are not allowed. use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure.
- SMS_ONLY - SMS only - NonEps procedure allowed however indicate to the UE in attach accept/tau accept that they are only allowed to have SMS service.
- EPS_ONLY_NAS - EPS only NAS - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. Use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure for the noneps part.
- DEFAULT - Use the treatment provisioned commonly for all Attach and TAU procedures.

Default

DEFAULT

Name

dlmsOnlyImsSupervision - Data centric IMS only IMS supervision

Syntax

--dlmsOnlyImsSupervision <list-item>

Description

Data centric IMS only IMS supervision flag. The IMS supervision functionality of the MME checks that the UE activates the IMS PDN connection and performs successful IMS registration. Otherwise the MME detaches the UE.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

dlmsOnlyNasCause - Data centric IMS only NAS cause code

Syntax

--dlmsOnlyNasCause <integer>

Description

Data centric IMS only cause code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

15

Name

dlmsOnlyNonEpsNasCause - Data centric IMS only Non EPS NAS cause

Syntax

--dlmsOnlyNonEpsNasCause <integer>

Description

The NAS cause used when non-EPS procedure is not allowed for Data centric IMS only UE.

Class

Optional.

Type / Range

Non EPS rejection NAS cause

Default

18

Name

dlmsOnlySendForbiddenTailList - Data Centric IMS PS voice only Send Forbidden TAI List

Syntax

--dlmsOnlySendForbiddenTailList <list-item>

Description

Controls whether the forbidden TAI list is sent for Data centric, IMS PS voice only UEs.
Applied when global parameter VDP Forbidden TAI List on VDP Profile is set to true.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

dlmsOnlyTreatment - Data centric IMS only treatment

Syntax

--dlmsOnlyTreatment <list-item>

Description

Data centric IMS only treatment.

Class

Optional.

List

- ALW - Allow - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are allowed.
- CSFB_NOT_PREF - CSFB not preferred - Anytype of noneps procedure (i.e. combined-attach or combined-ta) is allowed. in the attach accept/tau accept message we will indicate to the UE that "CSFB is not preferred".
- CS_D_NOT_AVAIL - CS domain not available - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. in attach accept/tau accept emmcause portion send cause "CS domain not available" for the noneps part.
- EPS_ONLY_NAS - EPS only NAS - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. Use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure for the noneps part.
- NAS - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are not allowed. use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure.

- SMS_ONLY - SMS only - NonEps procedure allowed however indicate to the UE in attach accept/tau accept that they are only allowed to have SMS service.

Default

ALW

Name

dlmsPreferCombAttachTreatment - Data centric IMS preferred Combined Attach treatment

Syntax

- -dlmsPreferCombAttachTreatment <list-item>

Description

Data centric IMS preferred, circuit switched secondary treatment for combined attach scenarios. If DEFAULT, the behavior will be to use the value from dlmsPreferTreatment. If not DEFAULT, override dlmsPreferTreatment for Combined Attach scenarios with the specified value.

Class

Optional.

List

- ALW - Allow - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are allowed.
- CS_D_NOT_AVAIL - CS domain not available - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. in attach accept/tau accept emmcause portion send cause "CS domain not available" for the noneps part.
- CSFB_NOT_PREF - CSFB not preferred - Anytype of noneps procedure (i.e. combined-attach or combined-ta) is allowed. in the attach accept/tau accept message we will indicate to the UE that "CSFB is not preferred".
- NAS - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are not allowed. use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure.
- SMS_ONLY - SMS only - NonEps procedure allowed however indicate to the UE in attach accept/tau accept that they are only allowed to have SMS service.
- EPS_ONLY_NAS - EPS only NAS - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. Use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure for the noneps part.
- DEFAULT - Use the treatment provisioned commonly for all Attach and TAU procedures.

Default

DEFAULT

Name

dlmsPreferCombTauTreatment - Data centric IMS preferred Combined TAU treatment

Syntax

--dIm PreferCombTauTreatment <list-item>

Description

Data centric IMS preferred, circuit switched secondary treatment for combined TAU scenarios. If DEFAULT, the behavior will be to use the value from dlmsPreferTreatment. If not DEFAULT, override dlmsPreferTreatment for Combined TAU scenarios with the specified value.

Class

Optional.

List

- ALW - Allow - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are allowed.
- CS_D_NOT_AVAIL - CS domain not available - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. in attach accept/tau accept emmcause portion send cause "CS domain not available" for the noneps part.
- CSFB_NOT_PREF - CSFB not preferred - Anytype of noneps procedure (i.e. combined-attach or combined-ta) is allowed. in the attach accept/tau accept message we will indicate to the UE that "CSFB is not preferred".
- NAS - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are not allowed. use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure.
- SMS_ONLY - SMS only - NonEps procedure allowed however indicate to the UE in attach accept/tau accept that they are only allowed to have SMS service.
- EPS_ONLY_NAS - EPS only NAS - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. Use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure for the noneps part.
- DEFAULT - Use the treatment provisioned commonly for all Attach and TAU procedures.

Default

DEFAULT

Name

dlmsPreferImSupervision - Data centric IMS preferred IMS supervision

Syntax

--dIm PreferImSupervision <list-item>

Description

Data centric IMS preferred IMS supervision flag. The IMS supervision functionality of the MME checks that the UE activates the IMS PDN connection and performs successful IMS registration. Otherwise the MME detaches the UE.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

dlmsPreferNasCause - Data centric IMS preferred NAS cause code

Syntax

--dlmsPreferNasCause <integer>

Description

Data centric IMS preferred circuit switched secondary cause code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

15

Name

dlmsPreferNonEpsNasCause - Data centric IMS preferred Non EPS NAS cause

Syntax

--dlmsPreferNonEpsNasCause <integer>

Description

The NAS cause used when non-EPS procedure is not allowed for Data centric IMS preferred, circuit switched secondary UE.

Class

Optional.

Type / Range

Non EPS rejection NAS cause

Default

18

Name

dlmsPreferSendForbiddenTailList - Data Centric IMS PS Voice Preferred Send Forbidden TAI List

Syntax

--dImsPreferSendForbiddenTaiList <list-item>

Description

Controls whether the forbidden TAI list is sent for Data centric, IMS PS voice preferred UEs. Applied when global parameter VDP Forbidden TAI List on VDP Profile is set to true.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

dlImsPreferTreatment - Data centric IMS preferred treatment

Syntax

--dImsPreferTreatment <list-item>

Description

Data centric IMS preferred circuit switched secondary treatment.

Class

Optional.

List

- ALW - Allow - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are allowed.
- CSFB_NOT_PREF - CSFB not preferred - Anytype of noneps procedure (i.e. combined-attach or combined-ta) is allowed. in the attach accept/tau accept message we will indicate to the UE that "CSFB is not preferred".
- CS_D_NOT_AVAIL - CS domain not available - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. in attach accept/tau accept emmcause portion send cause "CS domain not available" for the noneps part.
- EPS_ONLY_NAS - EPS only NAS - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. Use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure for the noneps part.
- NAS - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are not allowed. use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure.

- SMS_ONLY - SMS only - NonEps procedure allowed however indicate to the UE in attach accept/tau accept that they are only allowed to have SMS service.

Default

ALW

Name

noneCombAttachTreatment - None Combined Attach treatment

Syntax

--noneCombAttachTreatment <list-item>

Description

Combined Attach procedure treatment when UE usage or VDP is not given. If DEFAULT, the behavior will be to use the value from noneTreatment. If not DEFAULT, override noneTreatment for Combined Attach scenarios with the specified value.

Class

Optional.

List

- ALW - Allow - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are allowed.
- CS_D_NOT_AVAIL - CS domain not available - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. in attach accept/tau accept emmcause portion send cause "CS domain not available" for the noneps part.
- CSFB_NOT_PREF - CSFB not preferred - Anytype of noneps procedure (i.e. combined-attach or combined-ta) is allowed. in the attach accept/tau accept message we will indicate to the UE that "CSFB is not preferred".
- NAS - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are not allowed. use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure.
- SMS_ONLY - SMS only - NonEps procedure allowed however indicate to the UE in attach accept/tau accept that they are only allowed to have SMS service.
- EPS_ONLY_NAS - EPS only NAS - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. Use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure for the noneps part.
- DEFAULT - Use the treatment provisioned commonly for all Attach and TAU procedures.

Default

DEFAULT

Name

noneCombTauTreatment - None Combined TAU treatment

Syntax

--noneCombTauTreatment <list-item>

Description

Combined TAU procedure treatment when UE usage or VDP is not given. If DEFAULT, the behavior will be to use the value from noneTreatment. If not DEFAULT, override noneTreatment for Combined TAU scenarios with the specified value.

Class

Optional.

List

- ALW - Allow - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are allowed.
- CS_D_NOT_AVAIL - CS domain not available - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. in attach accept/tau accept emmcause portion send cause "CS domain not available" for the noneps part.
- CSFB_NOT_PREF - CSFB not preferred - Anytype of noneps procedure (i.e. combined-attach or combined-ta) is allowed. in the attach accept/tau accept message we will indicate to the UE that "CSFB is not preferred".
- NAS - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are not allowed. use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure.
- SMS_ONLY - SMS only - NonEps procedure allowed however indicate to the UE in attach accept/tau accept that they are only allowed to have SMS service.
- EPS_ONLY_NAS - EPS only NAS - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. Use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure for the noneps part.
- DEFAULT - Use the treatment provisioned commonly for all Attach and TAU procedures.

Default

DEFAULT

Name

noneImsSupervision - None IMS supervision

Syntax

--noneImsSupervision <list-item>

Description

IMS supervision flag when UE usage or VDP is not given. The IMS supervision functionality of the MME checks that the UE activates the IMS PDN connection and performs successful IMS registration. Otherwise the MME detaches the UE.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

noneNasCause - None NAS cause code

Syntax

--noneNasCause <integer>

Description

Cause code when UE usage or VDP is not given.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

15

Name

noneNonEpsNasCause - None Non EPS NAS cause

Syntax

--noneNonEpsNasCause <integer>

Description

The NAS cause used when non-EPS procedure is not allowed for UE that does not provide any UE usage type or VDP.

Class

Optional.

Type / Range

Non EPS rejection NAS cause

Default

18

Name

noneSendForbiddenTaiList - None Send Forbidden TAI List

Syntax

--noneSendForbiddenTaiList <list-item>

Description

Controls whether the forbidden TAI list is sent for UEs not indicating UE's usage setting and voice domain preference. Applied when global parameter VDP Forbidden TAI List on VDP Profile is set to true.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

noneTreatment - None treatment

Syntax

--noneTreatment <list-item>

Description

Treatment when UE usage or VDP is not given.

Class

Optional.

List

- ALW - Allow - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are allowed.
- CSFB_NOT_PREF - CSFB not preferred - Anytype of noneps procedure (i.e. combined-attach or combined-ta) is allowed. in the attach accept/tau accept message we will indicate to the UE that "CSFB is not preferred".
- CS_D_NOT_AVAIL - CS domain not available - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. in attach accept/tau accept emmcause portion send cause "CS domain not available" for the noneps part.
- EPS_ONLY_NAS - EPS only NAS - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. Use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure for the noneps part.
- NAS - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are not allowed. use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure.
- SMS_ONLY - SMS only - NonEps procedure allowed however indicate to the UE in attach accept/tau accept that they are only allowed to have SMS service.

Default

ALW

Name

vCsOnlyCombAttachTreatment - Voice centric CS only Combined Attach treatment

Syntax`--vCsOnlyCombAttachTreatment <list-item>`**Description**

Voice centric circuit switched only treatment for combined attach scenarios. If DEFAULT, the behavior will be to use the value from vCsOnlyTreatment. If not DEFAULT, override vCsOnlyTreatment for Combined Attach scenarios with the specified value.

Class

Optional.

List

- ALW - Allow - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are allowed.
- CS_D_NOT_AVAIL - CS domain not available - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. in attach accept/tau accept emmcause portion send cause "CS domain not available" for the noneps part.
- CSFB_NOT_PREF - CSFB not preferred - Anytype of noneps procedure (i.e. combined-attach or combined-ta) is allowed. in the attach accept/tau accept message we will indicate to the UE that "CSFB is not preferred".
- NAS - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are not allowed. use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure.
- SMS_ONLY - SMS only - NonEps procedure allowed however indicate to the UE in attach accept/tau accept that they are only allowed to have SMS service.
- EPS_ONLY_NAS - EPS only NAS - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. Use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure for the noneps part.
- DEFAULT - Use the treatment provisioned commonly for all Attach and TAU procedures.

Default

DEFAULT

Name

vCsOnlyCombTauTreatment - Voice Centric CS only Combined TAU treatment

Syntax`--vCsOnlyCombTauTreatment <list-item>`**Description**

Voice Centric Circuit Switched only Combined TAU procedure treatment. Voice centric circuit switched only treatment for combined TAU scenarios. If DEFAULT, the behavior will be to use the value from vCsOnlyTreatment. If not DEFAULT, override vCsOnlyTreatment for Combined TAU scenarios with the specified value.

Class

Optional.

List

- ALW - Allow - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are allowed.
- CS_D_NOT_AVAIL - CS domain not available - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. in attach accept/tau accept emmcause portion send cause "CS domain not available" for the noneps part.
- CSFB_NOT_PREF - CSFB not preferred - Anytype of noneps procedure (i.e. combined-attach or combined-ta) is allowed. in the attach accept/tau accept message we will indicate to the UE that "CSFB is not preferred".
- NAS - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are not allowed. use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure.
- SMS_ONLY - SMS only - NonEps procedure allowed however indicate to the UE in attach accept/tau accept that they are only allowed to have SMS service.
- EPS_ONLY_NAS - EPS only NAS - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. Use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure for the noneps part.
- DEFAULT - Use the treatment provisioned commonly for all Attach and TAU procedures.

Default

DEFAULT

Name

vCsOnlyImsSupervision - Voice centric CS only IMS supervision

Syntax

--vCsOnlyImsSupervision <list-item>

Description

Voice centric Circuit Switched only IMS supervision flag. The IMS supervision functionality of the MME checks that the UE activates the IMS PDN connection and performs successful IMS registration. Otherwise the MME detaches the UE.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

vCsOnlyNasCause - Voice centric CS only NAS cause code

Syntax

--vCsOnlyNasCause <integer>

Description

Voice centric circuit switched only cause code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

15

Name

vCsOnlyNonEpsNasCause - Voice centric CS only Non EPS NAS cause

Syntax

--vCsOnlyNonEpsNasCause <integer>

Description

The NAS cause used when non-EPS procedure is not allowed for voice centric CS only UE.

Class

Optional.

Type / Range

Non EPS rejection NAS cause

Default

18

Name

vCsOnlySendForbiddenTaiList - Voice Centric CS Voice Only Send Forbidden TAI List

Syntax

--vCsOnlySendForbiddenTaiList <list-item>

Description

Controls whether the forbidden TAI list is sent for Voice centric, CS voice only UEs. Applied when global parameter VDP Forbidden TAI List on VDP Profile is set to true.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

vCsOnlyTreatment - Voice centric CS only treatment

Syntax

-vCsOnlyTreatment <list-item>

Description

Voice centric circuit switched only treatment.

Class

Optional.

List

- ALW - Allow - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are allowed.
- CSFB_NOT_PREF - CSFB not preferred - Anytype of noneps procedure (i.e. combined-attach or combined-ta) is allowed. in the attach accept/tau accept message we will indicate to the UE that "CSFB is not preferred".
- CS_D_NOT_AVAIL - CS domain not available - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. in attach accept/tau accept emmcause portion send cause "CS domain not available" for the noneps part.
- EPS_ONLY_NAS - EPS only NAS - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. Use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure for the noneps part.
- NAS - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are not allowed. use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure.
- SMS_ONLY - SMS only - NonEps procedure allowed however indicate to the UE in attach accept/tau accept that they are only allowed to have SMS service.

Default

ALW

Name

vCsPreferCombAttachTreatment - Voice centric CS preferred Combined Attach treatment

Syntax

--vCsPreferCombAttachTreatment <list-item>

Description

Voice centric circuit switched preferred, IMS secondary treatment for combined attach scenarios. If DEFAULT, the behavior will be to use the value from vCsPreferTreatment. If not DEFAULT, override vCsPreferTreatment for Combined Attach scenarios with the specified value.

Class

Optional.

List

- ALW - Allow - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are allowed.
- CS_D_NOT_AVAIL - CS domain not available - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. in attach accept/tau accept emmcause portion send cause "CS domain not available" for the noneps part.
- CSFB_NOT_PREF - CSFB not preferred - Anytype of noneps procedure (i.e. combined-attach or combined-ta) is allowed. in the attach accept/tau accept message we will indicate to the UE that "CSFB is not preferred".
- NAS - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are not allowed. use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure.
- SMS_ONLY - SMS only - NonEps procedure allowed however indicate to the UE in attach accept/tau accept that they are only allowed to have SMS service.
- EPS_ONLY_NAS - EPS only NAS - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. Use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure for the noneps part.
- DEFAULT - Use the treatment provisioned commonly for all Attach and TAU procedures.

Default

DEFAULT

Name

vCsPreferCombTauTreatment - Voice centric CS preferred Combined TAU treatment

Syntax

--vCsPreferCombTauTreatment <list-item>

Description

Voice centric circuit switched preferred, IMS secondary treatment for combined TAU scenarios. If DEFAULT, the behavior will be to use the value from vCsPreferTreatment. If not DEFAULT, override vCsPreferTreatment for Combined TAU scenarios with the specified value.

Class

Optional.

List

- ALW - Allow - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are allowed.
- CS_D_NOT_AVAIL - CS domain not available - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. in attach accept/tau accept emmcause portion send cause "CS domain not available" for the noneps part.
- CSFB_NOT_PREF - CSFB not preferred - Anytype of noneps procedure (i.e. combined-attach or combined-ta) is allowed. in the attach accept/tau accept message we will indicate to the UE that "CSFB is not preferred".
- NAS - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are not allowed. use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure.
- SMS_ONLY - SMS only - NonEps procedure allowed however indicate to the UE in attach accept/tau accept that they are only allowed to have SMS service.
- EPS_ONLY_NAS - EPS only NAS - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. Use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure for the noneps part.
- DEFAULT - Use the treatment provisioned commonly for all Attach and TAU procedures.

Default

DEFAULT

Name

vCsPreferImsSupervision - Voice centric CS preferred IMS supervision

Syntax

--vCsPreferImsSupervision <list-item>

Description

Voice centric Circuit Switched preferred IMS secondary IMS supervision flag. The IMS supervision functionality of the MME checks that the UE activates the IMS PDN connection and performs successful IMS registration. Otherwise the MME detaches the UE.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

vCsPreferNasCause - Voice centric CS preferred NAS cause code

Syntax

--vCsPreferNasCause <integer>

Description

Voice centric circuit switched preferred IMS secondary cause code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

15

Name

vCsPreferNonEpsNasCause - Voice centric CS preferred Non EPS NAS cause

Syntax

--vCsPreferNonEpsNasCause <integer>

Description

The NAS cause used when non-EPS procedure is not allowed for voice centric CS preferred, IMS secondary UE.

Class

Optional.

Type / Range

Non EPS rejection NAS cause

Default

18

Name

vCsPreferSendForbiddenTaiList - Voice Centric CS Voice Preferred Send Forbidden TAI List

Syntax

--vCsPreferSendForbiddenTaiList <list-item>

Description

Controls whether the forbidden TAI list is sent for Voice centric, CS voice preferred UEs. Applied when global parameter VDP Forbidden TAI List on VDP Profile is set to true.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

vCsPreferTreatment - Voice centric CS preferred treatment

Syntax

--vCsPreferTreatment <list-item>

Description

Voice centric circuit switched preferred IMS secondary treatment.

Class

Optional.

List

- ALW - Allow - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are allowed.
- CSFB_NOT_PREF - CSFB not preferred - Anytype of noneps procedure (i.e. combined-attach or combined-ta) is allowed. in the attach accept/tau accept message we will indicate to the UE that "CSFB is not preferred".
- CS_D_NOT_AVAIL - CS domain not available - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. in attach accept/tau accept emmcause portion send cause "CS domain not available" for the noneps part.
- EPS_ONLY_NAS - EPS only NAS - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. Use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure for the noneps part.
- NAS - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are not allowed. use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure.
- SMS_ONLY - SMS only - NonEps procedure allowed however indicate to the UE in attach accept/tau accept that they are only allowed to have SMS service.

Default

ALW

Name

vlmsOnlyCombAttachTreatment - Voice centric IMS only Combined Attach treatment

Syntax

--vImsOnlyCombAttachTreatment <list-item>

Description

Voice centric IMS only treatment for combined attach scenarios. If DEFAULT, the behavior will be to use the value from vlmsOnlyTreatment. If not DEFAULT, override vlmsOnlyTreatment for Combined Attach scenarios with the specified value.

Class

Optional.

List

- ALW - Allow - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are allowed.
- CS_D_NOT_AVAIL - CS domain not available - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. in attach accept/tau accept emmcause portion send cause "CS domain not available" for the noneps part.
- CSFB_NOT_PREF - CSFB not preferred - Anytype of noneps procedure (i.e. combined-attach or combined-ta) is allowed. in the attach accept/tau accept message we will indicate to the UE that "CSFB is not preferred".
- NAS - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are not allowed. use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure.
- SMS_ONLY - SMS only - NonEps procedure allowed however indicate to the UE in attach accept/tau accept that they are only allowed to have SMS service.
- EPS_ONLY_NAS - EPS only NAS - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. Use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure for the noneps part.
- DEFAULT - Use the treatment provisioned commonly for all Attach and TAU procedures.

Default

DEFAULT

Name

vlmsOnlyCombTauTreatment - Voice centric IMS only Combined TAU treatment

Syntax

--vImOnlyCombTauTreatment <list-item>

Description

Voice centric IMS only treatment for combined TAU scenarios. If DEFAULT, the behavior will be to use the value from vlmsOnlyTreatment. If not DEFAULT, override vlmsOnlyTreatment for Combined TAU scenarios with the specified value.

Class

Optional.

List

- ALW - Allow - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are allowed.
- CS_D_NOT_AVAIL - CS domain not available - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. in attach accept/tau accept emmcause portion send cause “CS domain not available” for the noneps part.
- CSFB_NOT_PREF - CSFB not preferred - Anytype of noneps procedure (i.e. combined-attach or combined-ta) is allowed. in the attach accept/tau accept message we will indicate to the UE that “CSFB is not preferred”.
- NAS - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are not allowed. use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure.
- SMS_ONLY - SMS only - NonEps procedure allowed however indicate to the UE in attach accept/tau accept that they are only allowed to have SMS service.
- EPS_ONLY_NAS - EPS only NAS - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. Use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure for the noneps part.
- DEFAULT - Use the treatment provisioned commonly for all Attach and TAU procedures.

Default

DEFAULT

Name

vlmsOnlyImsSupervision - Voice centric IMS only IMS supervision

Syntax

--vImsOnlyImsSupervision <list-item>

Description

Voice centric IMS only IMS supervision flag. The IMS supervision functionality of the MME checks that the UE activates the IMS PDN connection and performs successful IMS registration. Otherwise the MME detaches the UE.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

vlmsOnlyNasCause - Voice centric IMS only NAS cause code

Syntax

--vImsOnlyNasCause <integer>

Description

Voice centric IMS only cause code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

15

Name

vlmsOnlyNonEpsNasCause - Voice centric IMS only Non EPS NAS cause

Syntax

--vImsOnlyNonEpsNasCause <integer>

Description

The NAS cause used when non-EPS procedure is not allowed for Voice centric IMS only UE.

Class

Optional.

Type / Range

Non EPS rejection NAS cause

Default

18

Name

vlmsOnlySendForbiddenTailList - Voice Centric IMS PS Voice Only Send Forbidden TAI List

Syntax

--vImsOnlySendForbiddenTailList <list-item>

Description

Controls whether the forbidden TAI list is sent for Voice centric, IMS PS voice only UEs.
Applied when global parameter VDP Forbidden TAI List on VDP Profile is set to true.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

vlmsOnlyTreatment - Voice centric IMS only treatment

Syntax

--vImOnlyTreatment <list-item>

Description

Voice centric IMS only treatment.

Class

Optional.

List

- ALW - Allow - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are allowed.
- CSFB_NOT_PREF - CSFB not preferred - Anytype of noneps procedure (i.e. combined-attach or combined-ta) is allowed. in the attach accept/tau accept message we will indicate to the UE that "CSFB is not preferred".
- CS_D_NOT_AVAIL - CS domain not available - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. in attach accept/tau accept emmcause portion send cause "CS domain not available" for the noneps part.
- EPS_ONLY_NAS - EPS only NAS - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. Use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure for the noneps part.
- NAS - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are not allowed. use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure.
- SMS_ONLY - SMS only - NonEps procedure allowed however indicate to the UE in attach accept/tau accept that they are only allowed to have SMS service.

Default

ALW

Name

vlmsPreferCombAttachTreatment - Voice centric IMS preferred Combined Attach treatment

Syntax

--vImPreferCombAttachTreatment <list-item>

Description

Voice centric IMS preferred, circuit switched secondary treatment for combined attach scenarios. If DEFAULT, the behavior will be to use the value from vlmsPreferTreatment. If not DEFAULT, override vlmsPreferTreatment for Combined Attach scenarios with the specified value.

Class

Optional.

List

- ALW - Allow - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are allowed.
- CS_D_NOT_AVAIL - CS domain not available - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. in attach accept/tau accept emmcause portion send cause "CS domain not available" for the noneps part.
- CSFB_NOT_PREF - CSFB not preferred - Anytype of noneps procedure (i.e. combined-attach or combined-ta) is allowed. in the attach accept/tau accept message we will indicate to the UE that "CSFB is not preferred".
- NAS - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are not allowed. use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure.
- SMS_ONLY - SMS only - NonEps procedure allowed however indicate to the UE in attach accept/tau accept that they are only allowed to have SMS service.
- EPS_ONLY_NAS - EPS only NAS - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. Use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure for the noneps part.
- DEFAULT - Use the treatment provisioned commonly for all Attach and TAU procedures.

Default

DEFAULT

Name

vlmsPreferCombTauTreatment - Voice centric IMS preferred Combined TAU treatment

Syntax

--vlmsPreferCombTauTreatment <list-item>

Description

Voice centric IMS preferred, circuit switched secondary treatment for combined TAU scenarios. If DEFAULT, the behavior will be to use the value from vlmsPreferTreatment. If not DEFAULT, override vlmsPreferTreatment for Combined TAU scenarios with the specified value.

Class

Optional.

List

- ALW - Allow - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are allowed.

- CS_D_NOT_AVAIL - CS domain not available - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. in attach accept/tau accept emmcause portion send cause “CS domain not available” for the noneps part.
- CSFB_NOT_PREF - CSFB not preferred - Anytype of noneps procedure (i.e. combined-attach or combined-ta) is allowed. in the attach accept/tau accept message we will indicate to the UE that “CSFB is not preferred”.
- NAS - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are not allowed. use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure.
- SMS_ONLY - SMS only - NonEps procedure allowed however indicate to the UE in attach accept/tau accept that they are only allowed to have SMS service.
- EPS_ONLY_NAS - EPS only NAS - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. Use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure for the noneps part.
- DEFAULT - Use the treatment provisioned commonly for all Attach and TAU procedures.

Default

DEFAULT

Name

vlmsPreferImsSupervision - Voice centric IMS preferred IMS supervision

Syntax

--vlmsPreferImsSupervision <list-item>

Description

Voice centric IMS preferred Circuit Switched secondary IMS supervision flag. The IMS supervision functionality of the MME checks that the UE activates the IMS PDN connection and performs successful IMS registration. Otherwise the MME detaches the UE.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

vlmsPreferNasCause - Voice centric IMS preferred NAS cause code

Syntax

--vlmsPreferNasCause <integer>

Description

Voice centric IMS preferred circuit switched secondary cause code.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

15

Name

vImsPreferNonEpsNasCause - Voice centric IMS preferred Non EPS NAS cause

Syntax

--vImsPreferNonEpsNasCause <integer>

Description

The NAS cause used when non-EPS procedure is not allowed for voice centric IMS preferred, circuit switched secondary UE.

Class

Optional.

Type / Range

Non EPS rejection NAS cause

Default

18

Name

vImsPreferSendForbiddenTaiList - Voice Centric IMS PS Voice Preferred Send Forbidden TAI List

Syntax

--vImsPreferSendForbiddenTaiList <list-item>

Description

Controls whether the forbidden TAI list is sent for Voice centric, IMS PS voice preferred UEs. Applied when global parameter VDP Forbidden TAI List on VDP Profile is set to true.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

vImSPreferTreatment - Voice centric IMS preferred treatment

Syntax

--vImSPreferTreatment <list-item>

Description

Voice centric IMS preferred circuit switched secondary treatment.

Class

Optional.

List

- ALW - Allow - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are allowed.
- CSFB_NOT_PREF - CSFB not preferred - Anytype of noneps procedure (i.e. combined-attach or combined-ta) is allowed. in the attach accept/tau accept message we will indicate to the UE that "CSFB is not preferred".
- CS_D_NOT_AVAIL - CS domain not available - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. in attach accept/tau accept emmcause portion send cause "CS domain not available" for the noneps part.
- EPS_ONLY_NAS - EPS only NAS - NonEps procedure (i.e. combined-attach or combined-ta) is not allowed. Use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure for the noneps part.
- NAS - NonEps procedure (i.e. combined-attach or combined-ta) and Eps procedures are not allowed. use the provisioned NAS cause code in the attach accept/tau accept emmcause portion to indicate the failure.
- SMS_ONLY - SMS only - NonEps procedure allowed however indicate to the UE in attach accept/tau accept that they are only allowed to have SMS service.

Default

ALW

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.136 cmm vdpRangeServices

VDP Range Services. Syntax, description, parameters, and roles for **cmm vdpRangeServices** command.

Syntax

Description

Administer Voice Domain Preference based upon IMSI, IMEI, and TAC. If imeiTacSvListName is defined, the IMEI range parameters (minImeiTac/ maxImeiTac/ minImeiSv/ maxImeiSv/ inclusive) are not used when finding a match for the IMEI.

Parameters

Name

uePlmnServicesName - Name of Services based on UE Network and Served Network

Syntax

--uePlmnServicesName <string>

Description

Name that identifies roaming agreement

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Name

entryId - Entry ID

Syntax

--entryId <integer>

Description

Entry ID.

Class

Required key.

Type / Range

Integer in the range of 1 to 10,000

Name

description - Entry description

Syntax

--description <string>

Description

Description for VDP range services entry.

Class

Optional.

Type / Range

Digit string of length 0 to 160 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

fakedFullCombSmsSgd - Faked Full Combined SMS SGd

Syntax

--fakedFullCombSmsSgd <list-item>

Description

Defines whether faked full combined attach shall be granted for the UE with SMS service provided over SGd. MME locally restricts the service to SMS only but accepts full combined attach towards the UE. The MME provides SMS service via SGd interface, but rejects any Extended Service Request from the UE with provisioned EMM cause code. This parameter applies only when SMS in MME is enabled for the UE.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

fakedFullCombSmsSgdSrvRejCause - Faked Full Combined SMS SGd Service Reject Cause

Syntax

--fakedFullCombSmsSgdSrvRejCause <integer>

Description

Defines the EMM cause code used to reject Extended Service Request CSFB call attempts when faked combined attach is enabled with SMS over SGd.

Class

Optional.

Type / Range

Integer in the range of 2-3, 5-26, 35, 39-40, 42, 95-101, 111

Default

39

Name

forbiddenListPref - Forbidden List Preference

Syntax

--forbiddenListPref <list-item>

Description

Specifies which ForbiddenTAI list to use for the VPD Range Services record

Class

Optional.

List

- TAI List - TAI List.
- TAI Range List - TAI Range List

Default

TAI List

Name

forbiddenTailistName - Forbidden TAI List Name

Syntax

--forbiddenTailistName <string>

Description

Forbidden TAI List Name

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

"

Name

forbiddenTaiRangeListName - Forbidden TAI Range List Name

Syntax

--forbiddenTaiRangeListName <string>

Description

Reference to a Forbidden TAI Range List (as created in the taiRangeList translation).

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

imeiTacSvListName - imeiTacSvListName

Syntax

--imeiTacSvListName <string>

Description

Defines IMEI TAC SV list for which the rule applies.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

imsApnSubscrOverride - IMS APN Subscription Override

Syntax

--imsApnSubscrOverride <list-item>

Description

Controls override functionality related to whether IMS APN is subscribed. If enabled, and the UE does NOT have IMS APN subscribed, the MME -accepts any Combined Attach/TAU attempt from the UE (i.e. does NOT apply Combined Attach Attach/TAU restriction for this UE, if such would be resulted based on other parameters of the restriction profile) -sets IMS voice over PS session indicator (IMS VoPS) to IMS voice over PS session in S1 mode not supported on every Attach Accept/TAU Accept for the UE. If disabled, any restriction otherwise resulted as part of the restriction profile is applicable.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

inclusive - Inclusive IMEI

Syntax

--inclusive <list-item>

Description

Specify if the IMEI range is inclusive or exclusive.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

maxImeiSv - Maximum IMEI SV

Syntax

--maxImeiSv <digits>

Description

The maximum IMEI SV in range.

Class

Optional.

Type / Range

Digit string of length 2 to 2

Default

99

Name

maxImeiTac - Maximum IMEI TAC

Syntax

--maxImeiTac <digits>

Description

The maximum IMEI TAC in range.

Class

Optional.

Type / Range

Digit string of length 8 to 8

Default

99999999

Name

maxMsin - Maximum MSIN

Syntax

--maxMsin <digits>

Description

The maximum IMSI MSIN value.

Class

Optional.

Type / Range

Digit string of length 8 to 10

Default

999999999

Name

minImeiSv - Minimum IMEI SV

Syntax

--minImeiSv <digits>

Description

The minimum IMEI SV in range.

Class

Optional.

Type / Range

Digit string of length 2 to 2

Default

00

Name

minImeiTac - Minimum IMEI TAC

Syntax

--minImeiTac <digits>

Description

The minimum IMEI TAC in range.

Class

Optional.

Type / Range

Digit string of length 8 to 8

Default

00000000

Name

minMsin - Minimum MSIN

Syntax

--minMsin <digits>

Description

The minimum IMSI MSIN value.

Class

Optional.

Type / Range

Digit string of length 8 to 10

Default

000000000

Name

shutdown - Shutdown

Syntax

--shutdown <list-item>

Description

When shutdown is true, the rule is not active.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

taiAreaListName - TAI Area List Name

Syntax

--taiAreaListName <string>

Description

Defines TAI restrictions

Class

Optional.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

all

Name

taiListPref - TAI List Preference

Syntax

--taiListPref <list-item>

Description

Specifies which TAI list to use for the imsi Range Services record match

Class

Optional.

List

- TAI List - TAI List.
- TAI Range List - TAI Range List

Default

TAI List

Name

taiRangeListName - TAI Range List Name

Syntax

--taiRangeListName <string>

Description

Reference to a TAI Range List

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_:@]*

Default

""

Name

vdpProfileName - VDP profile name

Syntax

--vdpProfileName <string>

Description

Voice domain preference profile name

Class

Optional.

Type / Range

Digit string of length 0 or 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Default

""

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.137 cmm virtualSgsnAddress

virtualSgsnAddress. Syntax, description, parameters, and roles for cmm virtualSgsnAddress command.

Syntax**Description**

Configure, list and show virtual SGSN address.

Parameters**Name**

mcc - Mobile country code for radio network PLMN

Syntax

--mcc <digits>

Description

Mobile country code for radio network PLMN. in the SGSN/MME, the value of this parameter is in sync with the value of the parameter MCC of the supported PLMN in index 0 in the MME.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 or 3

Default

""

Name

mnc - Mobile network code for radio network PLMN

Syntax

--mnc <digits>

Description

Mobile network code for radio network PLMN. in the SGSN/MME, the value of this parameter is in sync with the value of the parameter MCC of the supported PLMN in index 0 in the MME.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 or 2 to 3

Default

""

Name

np - Numbering plan indicator E164

Syntax

--np <list-item>

Description

Numbering plan indicator E164. allowed values are E164, E212 and E214.

Class

Optional, not filterable on list.

List

- E164 - E.164 - Specify numbering plan of E.164.
- E212 - E.212 - Specify numbering plan of E.212.
- E214 - E.214 - Specify numbering plan of E.214.

Default

E212

Name

ton - Nature of address indicator

Syntax

--ton <list-item>

Description

Nature of address indicator. allowed values are INT (international) and NAT (national).

Class

Optional, not filterable on list.

List

- INT - International - Specify address format as international.
- NAT - National - Specify address format as national.

Default

INT

Name

vsa - Virtual SGSN address for radio network PLMN

Syntax

--vsa <digits>

Description

This parameter is a virtual SGSN address number. the maximum length of the number is 18 digits.

Class

Optional, not filterable on list.

Type / Range

Digit string of length 0 or 1 to 18

Default

""

Name

id - Id of virtual SGSN address

Syntax

<integer>

Description

Id of virtual SGSN address.

Class

Required value.

Type / Range

Integer in the range of 1 to 10

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.138 cmm visitingSubscriberGroup

visiting subscriber group. Syntax, description, parameters, and roles for cmm visitingSubscriberGroup command.

Syntax**Description**

Configure, list and show SGSN visiting subscribers.

Parameters**Name**

visitingSubscriberGroupId - Visiting subs group id

Syntax

--visitingSubscriberGroupId <integer>

Description

Visiting subscriber group identifier.

Class

Required key.

Type / Range

Integer in the range of 1 to 5,000

Name

visitorAccessListId - Visitor access list id

Syntax

--visitorAccessListId <integer>

Description

Visitor access list identifier. this parameter should match parameter visitoraccesslistid of table visitoraccesslist.

Class

Required on create, not modifiable.

Type / Range

Integer in the range of 1 to 1000

Name

accessRights - Access rights

Syntax

--accessRights <list-item>

Description

Access rights of subscriber group. possible values, norights ... no rights
2groamgeranandutran ... 2G roaming allowed with GERAN and UTRAN
3groamgeranandutran ... 3G roaming allowed with GERAN and UTRAN
2gand3groamgeranandutran ... 2G and 3G roaming allowed with GERAN and UTRAN
2groamgeran ... 2G roaming allowed with GERAN 3groamgeran ... 3G roaming allowed with GERAN
2groamutran ... 2G roaming allowed with UTRAN 3groamutran ... 3G roaming allowed with UTRAN
2gand3groamgeran ... 2G and 3G roaming allowed with GERAN
2gand3groamutran ... 2G and 3G roaming allowed with UTRAN.

Class

Optional.

List

- 2gAnd3gRoamGeran - 2G and 3G roaming allowed with GERAN - 2G and 3G roaming allowed with GERAN.
- 2gAnd3gRoamGeranAndUtran - 2G and 3G roaming allowed with GERAN and UTRAN - 2G and 3G roaming allowed with GERAN and UTRAN.
- 2gAnd3gRoamUtran - 2G and 3G roaming allowed with UTRAN - 2G and 3G roaming allowed with UTRAN.
- 2gRoamGeran - 2G roaming allowed with GERAN - 2G roaming allowed with GERAN.
- 2gRoamGeranAndUtran - 2G roaming allowed with GERAN and UTRAN - 2G roaming allowed with GERAN and UTRAN.
- 2gRoamUtran - 2G roaming allowed with UTRAN - 2G roaming allowed with UTRAN.
- 3gRoamGeran - 3G roaming allowed with GERAN - 3G roaming allowed with GERAN.

- 3gRoamGeranAndUtran - 3G roaming allowed with GERAN and UTRAN - 3G roaming allowed with GERAN and UTRAN.
- 3gRoamUtran - 3G roaming allowed with UTRAN - 3G roaming allowed with UTRAN.
- noRights - No rights - No rights.

Default

noRights

Name

imsi - IMSI

Syntax

--imsi <digits>

Description

Imsi.

Class

Optional.

Type / Range

Digit string of length 0 to 15

Default

""

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.139 cmm visitorAccessList

visitor access list. Syntax, description, parameters, and roles for cmm visitorAccessList command.

Syntax

Description

Configure, list and show visitor access list.

Parameters

Name

visitorAccessListId - Visitor access list id

Syntax

--visitorAccessListId <integer>

Description

Visitor access list identifier.

Class

Required key.

Type / Range

Integer in the range of 1 to 1000

Name

alName - AI name

Syntax

--alName <string>

Description

Access list name.

Class

Optional.

Type / Range

Digit string of length 0 to 15 with the allowed pattern of [0-9a-zA-Z\.\-:]*

Default

""

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.140 cmm vlr

VLR. Syntax, description, parameters, and roles for cmm vlr command.

Syntax

Description

VLR.

Parameters

Name

vlrId - VlrId

Syntax

--vlrId <integer>

Description

VLR index.

Class

Required key.

Type / Range

Integer in the range of 0 to 199

Name

toa - Type of VLR address

Syntax

--toa <list-item>

Description

Identify type of VLR address. allowed values. GT ... global title address. SPC ... destination point code (ni + spc) if GT is given, the dig parameter should be given. if SPC is given, the ni and the spc parameters should be given.

Class

Required on create.

List

- GT - Global title address.

- SPC - Destination point code (ni + spc).

Name

vlrDigits - VLR address

Syntax

--vlrDigits <digits>

Description

Use this parameter to specify the virtual VLR address in E164 format.

Class

Required on create.

Type / Range

Digit string of length 1 to 18

Name

ni - Network indicator

Syntax

--ni <list-item>

Description

Network indicator. this parameter defines the value that represents the type of network where the VLR is located. allowed values, IN0 ... international network IN1 ... reserved for international use NA0 ... national network NA1 ... reserved for national use.

Class

Optional.

List

- IN0 - International network number 0.
- IN1 - International network number 1.
- NA0 - National network number 0.
- NA1 - National network number 1.

Default

NA0

Name

spc - Spc

Syntax

--spc <integer>

Description

Signalling point code.

Class

Optional.

Type / Range

Integer in the range of 0 to 16,777,215

Default

0

Name

tom - Tunneling of messages

Syntax

--tom <list-item>

Description

Tunneling of messages.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

false

Name

vlrTon - Type of VLR address

Syntax

--vlrTon <list-item>

Description

Use this parameter to specify the country format of virtual VLR address.

Class

Optional.

List

- INT - International - Specify address format as international.
- NAT - National - Specify address format as national.

Default

INT

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.141 cmm voiceTrafficList

Voice Traffic List. Syntax, description, parameters, and roles for cmm voiceTrafficList command.

Syntax

Description

This command is used to administer a Voice Traffic List.

Parameters

Name

name - Voice traffic list name

Syntax

--name <string>

Description

Voice traffic list name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.142 cmm voiceTrafficListNi

Voice Traffic List NI. Syntax, description, parameters, and roles for cmm voiceTrafficListNi command.

Syntax**Description**

This command is used to administer a Voice Traffic List Access Point Name Network Indicator.

Parameters**Name**

voiceTrafficListName - Voice traffic list name

Syntax

--voiceTrafficListName <string>

Description

Voice traffic list name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

apnNi - Access point name network indicator

Syntax

--apnNi <string>

Description

Access point name - network indicator.

Class

Required key.

Type / Range

Digit string of length 1 to 62 with the allowed pattern of [0-9a-zA-Z\._]*

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.143 cmm wpsQosProfile

WPS QoS Profile. Syntax, description, parameters, and roles for `cmm wpsQosProfile` command.

Syntax

Description

This command is used to administer an Wireless Priority Services QoS Profile for Roaming Subscribers and includes parameters for defining the APN-NI, APN AMBR bitrates, default bearer ARP, QCI, PCI and PVI values, and guaranteed and non-guaranteed dedicated bearer ARP, bitrate, QCI, PCI and PVI values.

Parameters

Name

name - WPS Roaming QoS Profile

Syntax

--name <string>

Description

A name that identifies this roaming QoS profile.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z_\. \-:@]*

Name

profileType - WPS roaming QoS profile type

Syntax

--profileType <list-item>

Description

Profile Type. Uniquely identifies the various types of profiles we support: IMS, Non-IMS, Custom APN 1, 2 or 3

Class

Required on create, not modifiable.

List

- CustomAPNWpsPrfl1 - Custom APN Wps profile - 1 - Roaming WPS QoS profile for custom APN WPS 1.
- CustomAPNWpsPrfl2 - Custom APN Wps profile - 2 - Roaming WPS QoS profile for custom APN WPS 2.
- CustomAPNWpsPrfl3 - Custom APN Wps profile - 3 - Roaming WPS QoS profile for custom APN WPS 3
- IMSAPNWpsPrfl - IMS APN WPS profile - Roaming WPS QoS profile for IMS APNs.
- NonIMSAPNWpsPrfl - Non-IMS APN WPS profile - Roaming WPS QoS profile for non-ims APNs

Name

apnAmbrDownLink - APN AMBR downlink for default bearer

Syntax

--apnAmbrDownLink <integer>

Description

APN aggregated maximum bitrate downlink for default bearer.

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64- 568, 576-8640, 8700-16000, 17000-128000, 130000-4000000000 in units of Kbps

Default

128000

Name

apnAmbrUpLink - APN AMBR uplink for default bearer

Syntax

--apnAmbrUpLink <integer>

Description

APN aggregated maximum bitrate uplink for default bearer.

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64- 568, 576-8640, 8700-16000, 17000-128000, 130000-4000000000 in units of Kbps

Default

128000

Name

apnNi - Access point name - network indicator

Syntax

--apnNi <string>

Description

APN NI for this roaming WPS QoS profile.

Class

Optional.

Type / Range

Digit string of length 0 to 62 with the allowed pattern of [0-9a-zA-Z\.\-]*

Default

""

Name

defaultBrrArp - Default bearer ARP

Syntax

--defaultBrrArp <integer>

Description

Default bearer allocation and retention priority.

Class

Optional.

Type / Range

Integer in the range of 1 to 15

Default

15

Name

defaultBrrQci - Default bearer QCI

Syntax

--defaultBrrQci <integer>

Description

QoS class identifier for default bearer.

Class

Optional.

Type / Range

Integer in the range of 5 to 9

Default

9

Name

defBrrPci - Default bearer PCI

Syntax

--defBrrPci <list-item>

Description

Default bearer pre-emption capability indicator.

Class

Optional.

List

- Disabled - PCI disabled - Pre-emption capability disabled.
- Enabled - PCI enabled - Pre-emption capability enabled.

Default

Enabled

Name

defBrrPvi - Default bearer PVI

Syntax

--defBrrPvi <list-item>

Description

Default bearer pre-emption vulnerability indicator.

Class

Optional.

List

- Disabled - PVI disabled - Pre-emption vulnerability disabled.
- Enabled - PVI enabled - Pre-emption vulnerability enabled.

Default

Enabled

Name

description - Description

Syntax

--description <string>

Description

Profile description.

Class

Optional.

Type / Range

Digit string of length 0 to 160 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

gbrDedBrrArp - Guaranteed dedicated bearer ARP

Syntax

--gbrDedBrrArp <integer>

Description

Guaranteed dedicated bearer allocation and retention priority.

Class

Optional.

Type / Range

Integer in the range of 0 to 15

Default

15

Name

gbrDedBrrBitrateDownlink - Guaranteed dedicated bearer downlink bitrate

Syntax

--gbrDedBrrBitrateDownlink <integer>

Description

Guaranteed dedicated bearer downlink bitrate.

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64-568, 576-8640, 8700-16000, 17000-128000, 130000-256000, 260000-500000, 510000-1500000, 1600000-4000000000 in units of Kbps

Default

128000

Name

gbrDedBrrBitrateUplink - Guaranteed dedicated bearer uplink bitrate

Syntax

--gbrDedBrrBitrateUplink <integer>

Description

Guaranteed dedicated bearer uplink bitrate.

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64-568, 576-8640, 8700-16000, 17000-128000, 130000-256000, 260000-500000, 510000-1500000, 1600000-4000000000 in units of Kbps

Default

128000

Name

gbrDedBrrMaxBitrateDownlink - Guaranteed dedicated bearer maximum downlink bitrate

Syntax

--gbrDedBrrMaxBitrateDownlink <integer>

Description

Guaranteed dedicated bearer maximum downlink bitrate.

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64-568, 576-8640, 8700-16000, 17000-128000, 130000-256000, 260000-500000, 510000-1500000, 1600000-4000000000 in units of Kbps

Default

128000

Name

gbrDedBrrMaxBitrateUplink - Guaranteed dedicated bearer maximum uplink bitrate

Syntax

--gbrDedBrrMaxBitrateUplink <integer>

Description

Guaranteed dedicated bearer maximum uplink bitrate.

Class

Optional.

Type / Range

Integer in the range of 0, 1-63, 64-568, 576-8640, 8700-16000, 17000-128000, 130000-256000, 260000-500000, 510000-1500000, 1600000-4000000000 in units of Kbps

Default

128000

Name

gbrDedBrrPci - Guaranteed dedicated bearer PCI

Syntax

--gbrDedBrrPci <list-item>

Description

Guaranteed dedicated bearer pre-emption capability indicator.

Class

Optional.

List

- Disabled - PCI disabled - Pre-emption capability disabled.
- Enabled - PCI enabled - Pre-emption capability enabled.

Default

Disabled

Name

gbrDedBrrPvi - Guaranteed dedicated bearer PVI

Syntax

--gbrDedBrrPvi <list-item>

Description

Guaranteed dedicated bearer pre-emption vulnerability indicator.

Class

Optional.

List

- Disabled - PVI disabled - Pre-emption vulnerability disabled.
- Enabled - PVI enabled - Pre-emption vulnerability enabled.

Default

Disabled

Name

gbrDedBrrQci - Guaranteed dedicated bearer QCI

Syntax

--gbrDedBrrQci <integer>

Description

Guaranteed dedicated bearer QCI

Class

Optional.

Type / Range

Integer in the range of 0 to 4

Default

4

Name

nonGbrDedBrrArp - Non guaranteed dedicated bearer ARP

Syntax

--nonGbrDedBrrArp <integer>

Description

Non guaranteed dedicated bearer allocation and retention priority.

Class

Optional.

Type / Range

Integer in the range of 0 to 15

Default

15

Name

nonGbrDedBrrPci - Non guaranteed dedicated bearer PCI

Syntax

--nonGbrDedBrrPci <list-item>

Description

Non guaranteed dedicated bearer pre-emption capability indicator.

Class

Optional.

List

- Disabled - PCI disabled - Pre-emption capability disabled.
- Enabled - PCI enabled - Pre-emption capability enabled.

Default

Disabled

Name

nonGbrDedBrrPvi - Non guaranteed dedicated bearer PVI

Syntax

--nonGbrDedBrrPvi <list-item>

Description

Non guaranteed dedicated bearer pre-emption vulnerability indicator.

Class

Optional.

List

- Disabled - PVI disabled - Pre-emption vulnerability disabled.
- Enabled - PVI enabled - Pre-emption vulnerability enabled.

Default

Disabled

Name

nonGbrDedBrrQci - Non guaranteed dedicated bearer QCI

Syntax

--nonGbrDedBrrQci <integer>

Description

Non guaranteed dedicated bearer QoS class identifier.

Class

Optional.

Type / Range

Integer in the range of 0 or 6 to 9

Default

9

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.144 cmm wusProfile

Wake Up Signal Profile. Syntax, description, parameters, and roles for `cmm wusProfile` command.

Syntax

Description

This profile specifies the wake up signal configuration to be used for the uePlmnServices and imsiRangeServices.

Parameters

Name

name - Wake Up Signal Profile Name

Syntax

--name <string>

Description

Name of the WUS profile.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Name

localWusAiValue - Local Wake Up Signal Assistance Information Value

Syntax

--localWusAiValue <list-item>

Description

This field specifies the paging probability Assistance Information value that is to be assigned to a UE that supports Group WUS functionality. This value will override any recommended value from the UE.

Class

Optional.

List

- p00 - 0% paging probability value
- p05 - 5% paging probability value
- p10 - 10% paging probability value
- p15 - 15% paging probability value

- p20 - 20% paging probability value
- p25 - 25% paging probability value
- p30 - 30% paging probability value
- p35 - 35% paging probability value
- p40 - 40% paging probability value
- p45 - 45% paging probability value
- p50 - 50% paging probability value
- p55 - 55% paging probability value
- p60 - 60% paging probability value
- p65 - 65% paging probability value
- p70 - 70% paging probability value
- p75 - 75% paging probability value
- p80 - 80% paging probability value
- p85 - 85% paging probability value
- p90 - 90% paging probability value
- p95 - 95% paging probability value
- p100 - 100% paging probability value

Default

p100

Name

ueWusAiAdjustment - UE Wake Up Signal Assistance Information Adjustment

Syntax

- ueWusAiAdjustment <integer>

Description

This field specifies the adjustment to be applied to the UE recommended value for Assistance information used for Group WUS. The specified value is an integer in the range [-20, 20] and defines the adjustment in 5% increments or decrements. If the adjustment takes the value to over 100%, the adjustment is changed to 100%. If the adjustment takes the value to less than 0%, the adjustment is changed to 0%. As an example, assume the field is provisioned to 10 (i.e. 50%). If the UE recommended value is 30%, the MME will adjust to 80%. If the UE recommended value is 75%, the MME will adjust to 100%.

Class

Optional.

Type / Range

Integer in the range of [-20, 20].

Default

0

Name

useLocalIfNoRecommendedAi - Use Local if No Recommended Assistance Information

Syntax

--useLocalIfNoRecommendedAi <list-item>

Description

Specifies whether the MME is to negotiate the use of Group WUS if the UE indicates support of Group WUS (WUSA bit in UE capabilities) but the UE does not provide a recommended assistance value. If the value is true, the MME will negotiate using the local provisioned value.

Class

Optional.

List

- false - Boolean value of false.
- true - Boolean value of true.

Default

true

Name

wusSource - Wake Up Signal Source

Syntax

--wusSource <list-item>

Description

Specifies whether the MME is to use the UE recommended value or the MME local value for the negotiation of Group WUS with the UE.

Class

Optional.

List

- UE - Specifies to use the UE provided recommended assistance information for Group WUS negotiation. if the UE does not provide a recommended value, the local value will be used.
- Local - Specifies to use the locally provisioned value for assistance information for Group WUS negotiation.

Default

UE

Roles

Create

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.145 cmm zone

Zone Name. Syntax, description, parameters, and roles for cmm zone command.

Syntax

Description

This command intends to map zone name with a given RAC and LAC. It is used to compare with current location of UEs. if current location of UE (RAC+LAC) is matching with a zone's name, then the zone name will be used as a Tag to the DNS query.

Parameters

Name

lac - LAC

Syntax

--lac <integer>

Description

Location area code.

Restrictions

configuration should be in Hexadecimal format no matter if there a prefix 0x is added.

Class

Required key.

Type / Range

Integer in the range of 1 to 65,533

Name

rac - RAC

Syntax

--rac <integer>

Description

Routing area code

Restrictions

configuration should be in Hexadecimal format no matter if there a prefix 0x is added.

Class

Required key.

Type / Range

Integer in the range of 0 to 255

Name

name - Zone Table Name

Syntax

--name <string>

Description

Name that identifies a Zone

Restrictions

configuration should not allow case like below: rac1 + lac1 --> zone name1 rac2 + lac2 --> zone name1

Class

Required on create.

Type / Range

Digit string of length 1 to 32 with the allowed pattern of [A-Za-z0-9.-]* A Zone name can not start or end with hyphen (-) or period (.).

Name

racRange - Rac Range

Syntax

--racRange <integer>

Description

This parameter indicates range of RAC. The value of the parameter must be greater or equal to RAC parameter. If this is not configured, then single RAC will be used.

Restrictions

configuration should be in Hexadecimal format no matter if there a prefix 0x is added

Class

Optional.

Type / Range

Integer in the range of 0 to 255

Default

0

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor

7.146 cmm zoneCode

Zone Code. Syntax, description, parameters, and roles for cmm zoneCode command.

Syntax

Description

This command is used to administer a Zone Code in support of regional subscription and includes parameters for defining the mobile network name, technology type, zone code list type, and zone code. The 'shutdown' parameter is used to enable/shutdown the zone code.

Parameters

Name

name - Zone code name

Syntax

--name <string>

Description

Zone code name.

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

plmnName - Mobile network name

Syntax

--plmnName <string>

Description

Name that identifies mobile network (alternate key).

Class

Required key.

Type / Range

Digit string of length 1 to 31 with the allowed pattern of [0-9a-zA-Z\.\-_\:@]*

Name

techType - Network technology type

Syntax

--techType <list-item>

Description

Network technology type.

Class

Required on create, not modifiable.

List

- 2g3g - Zone code for 2G/3G (SGSN).
- 4g - Zone code for 4G (MME).

Name

zcId - Zone code

Syntax

--zcId <integer>

Description

Indicates one, of up to 128, unique zone codes.

Class

Required on create, not modifiable.

Type / Range

Integer in the range of 0 to 65,535

Name

limitedList - Limited list name

Syntax

--limitedList <string>

Description

Name of the LA list or TA list for a zone code.

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

"

Name

rszListPref - Regional Zone List Preference

Syntax

--rszListPref <list-item>

Description

Specifies the list to use for zone code decision

Class

Optional.

List

- Limited List - The Limited List
- TAI Range List - The TAI Range List

Default

Limited List

Name

shutdown - Shutdown

Syntax

--shutdown <list-item>

Description

When it is true, indicates that zone code is shutdown.

Class

Optional.

List

- false - Boolean value of false.

- true - Boolean value of true.

Default

true

Name

taiRangeListName - TAI Range List Name

Syntax

--taiRangeListName <string>

Description

Reference to a TAI Range List

Class

Optional.

Type / Range

Digit string of length 0 to 31 with the allowed pattern of [0-9a-zA-Z\._:@]*

Default

""

Name

wmmZcType - Zone code type

Syntax

--wmmZcType <list-item>

Description

Indicates zone code list type of LAs or TAs.

Class

Optional.

List

- all - Regional subscription allowed in all LAs or all TAs.
- exclusiveLimited - Exclusive limited - Regional subscription not allowed in LAs or TAs in the list.
- inclusiveLimited - Inclusive limited - Regional subscription allowed in LAs or TAs in the list.
- no - Regional subscription not allowed in all LAs or all TAs.

Default

all

Roles**Create**

sysadmin, manage

Show

sysadmin, manage, monitor

Modify

sysadmin, manage

Delete

sysadmin, manage

List

sysadmin, manage, monitor